

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Steady State Hydraulic Analysis-HYSYS

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R01	16.06.22	First Issue for Client's Review
A01	23.01.23	Approved for Design

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED- PSE-RPT-013

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
API	American Petroleum Institute
BFPCL	Brass Fertilizer and Petrochemical Company Limited
BOD	Basis Of Design
DEP	Design and Engineering Practice
DPR (now NUPRC)	Department for Petroleum Resources
FEED	Front End Engineering Design
GPP	Gas Processing Plant
MMSCFD	Metric Million Standard Cubic Feet Per Day
MTPD	Metric Tonnes Per Day
NUPRC (formerly DPR)	Nigerian Upstream Petroleum Regulatory Commission
OML	Oil Mining License
PIG	Pipeline Inspection Gauge
PMC	Project Management Consultant
RTU	Remote Telemetry Unit
SPDC	Shell Petroleum Development Company

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opubene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

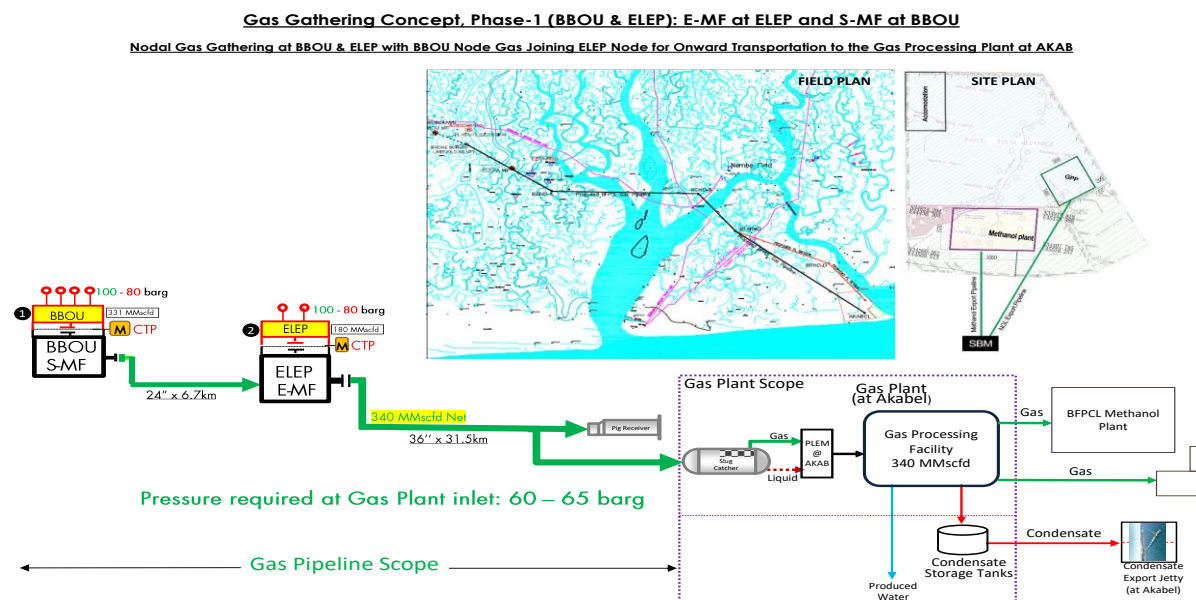


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The objective of this report is to document the simulation results of the gas pipelines from BBOU to Elepa and Elepa to Akabel. It details the line sizing, flow assurance, and gas deliverability to Akabel Gas plant at required pressure.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
Petroleum Act, CAP P10, Laws of the Federation of Nigeria, LFN 2004, amended.	This is the principal statute regulating the oil industry. The following regulation relevant to this project is issued pursuant to the Act.
Mineral Oil (Safety) Regulations 1963, as amended.	These regulations, in so far as they apply to pipelines, require compliance with relevant IP, API and ASME codes and standards.
Oil Pipelines Act, 1956. CAP. 145 (Amended 1965, CAP. 338, LFN 1990)	Provides for licenses to be granted for the establishment and maintenance of pipelines incidental and supplementary to oilfields and oil mining, and for purposes ancillary to such pipelines.
Oil and Gas Pipelines Regulations, 1995	Details the regulations for the design, construction and maintenance of oil and gas pipelines.
Environmental Impact Assessment Act, CAP E12, LFN 2004.	Makes the conduct of an EIA mandatory prior to the development of any project or activity likely to have a significant effect on the environment
Federal Environment Protection Agency Act, CAP F13, LFN 2004.	Provides for the protection of the environment and sustainable development of Nigeria's natural resources.
Environmental Guidelines and Standards For The Petroleum Industry In Nigeria, 1991.	DPR (NUPRC) issued document providing guidelines on pollution abatement and relevant legislation.
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA/SPDC-BFPC UGS-GEP/SVY-RPT-001	Pipeline Route Selection Study
EA/SPDC-BFPC UGS-GEP/PRC-	Process Simulation Study – HYSYS, PIPESIM, & OLGA

RPT-001- 3	
EA/SPDC-BFPC UGS-GEP/MAT-RPT-001	Pipeline Materials Selection & Corrosion Analysis Study
EA/SPDC-BFPC UGS-GEP/ELE-RPT-001	Manifold Platforms Power Supply Options
EA/SPDC-BFPC UGS-GEP/MEC-RPT-001	Manifold Platforms Concept & Integration with SPDC
EA/SPDC-BFPC UGS-GEP/PRC-RPT-002	Gas Processing Technology Option
6064/PLRS/001 α-PRP	Onshore Pipeline Route Survey for BFCL; Topographical Surveys
6064/PLRS/001 PRP	Onshore Pipeline Route Survey for BFCL; Soil Investigation Surveys
40km Gas Supply Pipeline Project ESIA	Environmental and Social Impact Assessment Report prepared by Messrs. Fundamental Integrated Site Appraisal Services Limited - Draft Report November 2020.

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. SIMULATION METHODOLOGY

3.1. Simulation Scope

- i. Generate mass and material balances for flow sources in-situ the pipelines and at the delivery outlets.
- ii. Establish appropriate line sizes suitable for the anticipated peak flow from the fields.
- iii. Flow assurance (parameter checks; which includes pressure, temperature, velocity and hydrate formation).
- iv. Develop a suitable operating flow conditions (pressure and temperature).

3.2. Simulation Basis and Assumptions

3.2.1. Base Assumption

The pipeline temperature ambient medium was taken as 25 OC. Also, it is assumed for this conceptual study, that there are no elevation changes across the pipeline route.

3.2.2. Software

The hydraulics process simulation study was undertaken using Aspen Hysys v7.3.

Water saturation extension was explored to determine the saturated water in the gas.

3.2.3. Simulation Property Package

The thermodynamic equation of state (EOS) used for the process simulation is; Peng-Robinson (PR) fluid property package.

3.3. Well Fluid Simulation

The composition of the well fluids were simulated using the following conditions:

- v. Dry gas composition at each Well
- vi. Reservoir pressure and temperature conditions at each Well
- vii. Estimated flow at from each Well

3.3.1. Dry gas composition at each Well

The gas composition for each field was culled from the Basis of Design Document, EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002-R02. Each of the well fluid composition was assumed to be given on a "dry basis", therefore they were saturated with water before free water and condensate are added in order to arrive at a mixed fluid flow that was used in the simulation of the manifolds and pipelines.

The flashed gas and liquid composition are presented below:

	BBOU		ELEP		NUNR		SEIB		OPUG	
Component	Gas	Liq.	Gas	Liq.	Gas	Liq.	Gas	Liq.	Gas	Liq.
N2	0.07	0.00	0.00	0.00	0.00	0.00	0.07	0.00	0.12	0.00
CO2	0.59	0.00	2.30	0.00	0.00	0.00	1.63	0.00	2.53	0.00
H2S	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
C1	93.26	0.00	84.63	0.00	0.00	0.00	83.24	0.00	77.45	0.00
C2	3.17	0.01	6.35	0.03	0.00	0.00	6.27	0.28	9.06	0.13
C3	1.01	0.03	2.67	0.27	0.00	0.00	3.49	1.88	5.40	0.53
iC4	0.61	0.07	0.76	0.37	0.00	0.00	1.20	1.55	2.13	0.76
nC4	0.44	0.07	0.82	0.54	0.00	0.00	1.31	2.36	1.00	1.59
iC5	0.25	0.16	0.43	0.87	0.00	0.00	0.73	2.34	1.07	1.79
nC5	0.19	0.16	0.32	0.68	0.00	0.00	0.50	1.96	0.53	1.97
C6	0.22	1.88	0.59	2.26	0.00	0.00	0.68	4.76	0.53	5.23
C7	0.17	6.21	0.69	6.75	0.00	0.00	0.62	9.71	0.17	8.16
C8	0.02	6.61	0.44	12.76	0.00	0.00	0.28	15.65	0.05	16.75
C9	0.00	7.13	0.00	10.87	0.00	0.00	0.00	11.71	0.00	8.80
C10	0.00	5.70	0.00	8.62	0.00	0.00	0.00	9.14	0.00	5.87
C11	0.00	7.83	0.00	6.64	0.00	0.00	0.00	5.68	0.00	6.44
C12+	0.00	64.14	0.00	49.34	0.00	0.00	0.00	33.00	0.00	42.00
Total	100.00	100.00	100.00	100.00	0.00	0.00	100.00	100.00	100.00	100.00

The Peng Robinson EOS was selected for the VLE calculations because it is the generally recommended model For Oil, Gas, or Petrochemical applications. The PR property package rigorously solves any single-, two-, or three-phase system with a high degree of efficiency and reliability and is applicable over a wide range of conditions:

- Temperature Range > -271 °C or -456 °F
- Pressure Range < 100,000 kPa or 15,000 psia

The well fluids compositions fall within these range.

3.3.2. Reservoir pressure and temperature conditions at each Well

The table below shows the reservoir pressure and temperature used for each Well.

	BBOU	ELEP	NUNR	SEIB	OPUG
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Reservoir Pressure, psia	3,246	6,910	7,259	6,533	5,176
Reservoir Temp, deg. C	78	123	102	117	117

3.3.3. Flow Simulation

The two Pipelines (BBOU-to-Elepa and Elepa-to-Akabel) were simulated using the maximum combined flow of gas, condensate and water that will be handled by each of the Pipeline – BBOU and Elepa.

BBOU Manifold. This is a very conservative assumptions made since the actual forecast are not available at the time of preparing this report.

3.3.4. Selected Flow for BBOU-to-Elepa Pipeline

The maximum flow used for gas, condensate and water for the BBOU-to-Elepa Pipeline occurred in Years 9, 18 and 7 respectively as highlighted in the table below. Maximum combined flow are 331 MMSCFD gas, 16226 bpd of condensate and 9002 bpd of water.

Year	TOTAL FLUID RATE			BUBOUWE BOU			OPUGBENE		
	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd
	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)
1	181	944	907	181	944	907	0	0	0
2	181	878	907	181	878	907	0	0	0
3	181	788	907	181	788	907	0	0	0
4	181	742	7,855	181	742	7,855	0	0	0
5	181	707	7,547	181	707	7,547	0	0	0
6	215	840	8,491	215	840	8,491	0	0	0
7	305	1,217	9,002	305	1,217	9,002	0	0	0
8	327	1,323	8,193	327	1,323	8,193	0	0	0
9	331	1,338	8,038	331	1,338	8,038	0	0	0
10	93	1,763	2,527	67	239	2,397	26	1,525	129
11	75	3,242	1,301	21	59	1,031	54	3,183	270
12	74	3,457	1,074	16	42	784	58	3,416	290
13	71	4,084	1,513	2	4	85	69	4,080	1,429
14	71	4,161	3,048	0	0	0	71	4,161	3,048
15	94	5,541	3,748	0	0	0	94	5,541	3,748
16	154	9,082	3,859	0	0	0	154	9,082	3,859
17	170	12,115	3,148	0	0	0	170	12,115	3,148
18	174	16,226	1,667	0	0	0	174	16,226	1,667
19	177	15,812	1,558	0	0	0	177	15,812	1,558
20	182	15,606	2,775	0	0	0	182	15,606	2,775

3.3.4.1. Selected Flow for Elepa-to-Akabel Pipeline

year	TOTAL FLUID RATE			BUBOUWE BOU			ELEPA			NUN RIVER			SEIB			OPUG		
	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd
	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)
1	361	11,032	1,807	181	944	907	180	10,088	900	0	0	0	0	0	0	0	0	0
2	361	10,965	1,807	181	878	907	180	10,088	900	0	0	0	0	0	0	0	0	0
3	361	10,876	1,807	181	788	907	180	10,088	900	0	0	0	0	0	0	0	0	0
4	361	10,830	15,835	181	742	7,855	180	10,088	7,980	0	0	0	0	0	0	0	0	0
5	361	10,794	15,213	181	707	7,547	180	10,088	7,666	0	0	0	0	0	0	0	0	0
6	360	8,985	13,393	215	840	8,491	145	8,146	4,902	0	0	0	0	0	0	0	0	0
7	357	4,128	11,599	305	1,217	9,002	52	2,911	2,597	0	0	0	0	0	0	0	0	0
8	356	2,953	9,647	327	1,323	8,193	29	1,630	1,454	0	0	0	0	0	0	0	0	0
9	356	2,724	9,275	331	1,338	8,038	25	1,386	1,236	0	0	0	0	0	0	0	0	0
10	382	18,828	4,837	67	239	2,397	19	1,076	960	244	14,464	1,221	26	1,525	129	26	1,525	129
11	385	21,637	3,357	21	59	1,031	11	628	561	245	14,583	1,225	54	3,183	270	54	3,183	270
12	385	21,933	3,015	16	42	784	9	478	426	245	14,583	1,225	58	3,416	290	58	3,416	290
13	386	22,799	13,832	2	4	85	1	52	46	245	14,583	10,844	69	4,080	1,429	69	4,080	1,429
14	386	22,905	16,513	0	0	0	0	0	0	245	14,583	10,417	71	4,161	3,048	71	4,161	3,048
15	384	21,996	14,149	0	0	0	0	0	0	196	10,914	6,653	94	5,541	3,748	94	5,541	3,748
16	379	22,323	11,246	0	0	0	0	0	0	71	4,159	3,528	154	9,082	3,859	154	9,082	3,859
17	380	26,558	8,272	0	0	0	0	0	0	40	2,329	1,976	170	12,115	3,148	170	12,115	3,148
18	382	34,433	5,014	0	0	0	0	0	0	34	1,980	1,680	174	16,226	1,667	174	16,226	1,667
19	380	33,161	4,421	0	0	0	0	0	0	26	1,538	1,304	177	15,812	1,558	177	15,812	1,558

20	379	32,111	6,313	0	0	0	0	0	0	15	898	762	182	15,606	2,775	182	15,606	2,775
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4. Simulation Results

For phase-1 gas production, Two (2) scenarios were explored for each pipeline:

4.1. Scenario Definitions

4.1.1. BBOU-Elepa Pipeline

- Scenario-1 is minimum supply pressure of 80 barg by SPDC at BBOU Manifold
- Scenario-2 is maximum supply pressure of 100 barg by SPDC at BBOU Manifold

4.1.2. Elepa-Akabel Pipeline

- Scenario-3 is minimum supply pressure of 80 barg by SPDC at Elepa Manifold
- Scenario-4 is maximum supply pressure of 100 barg by SPDC at Elepa Manifold

4.2. Phase-1/Scenario-1 Gas Production

This scenario shows the deliverability of maximum flow of gas, condensate and water from both BBOU and Opugbene Fields to the 24 inch BBOU-Elepa Pipeline when supply pressure at BBOU Manifold is 80 barg. The maximum flows occur in Years 9, 18 and 7 respectively as shown in the production forecast. Maximum combined flows are 331 MMSCFD gas, 16226 bpd of condensate and 9002 bpd of water.

4.2.1. Pressure and Temperature Regime

Source Conditions at the CTP; 100 barg pressure and corresponding temperature of 62.14 °C. Cumulative molar flow of gas, condensate and water at this condition is 346 MMSCFD.

Delivery Pressure; at ELEP E-MF is 97.49 bar and the corresponding temperature is 58.10 °C.

The Line Size; which also supported 346 MMscfd cumulative molar flow of gas, condensate and water of gas flow across the 6.7 km BBOU-ELEP E-MF bulk-line with minimal pressure loss and with a velocity below erosional velocity mark is 24". And the pressure drop is 0.44 bar/km.

See Appendix 1 for detailed simulation results for more details

Flow Velocity Profile

Minimum Fluid Flow Velocity; that will ensure the cleanliness of the pipe at the inlet and outlet of the bulk-line is 3.5 m/s and 3.9 m/s respectively.

Prevailing Fluid Flow Velocity; that supports fluid delivery is 5.36 m/s and 5.48 m/s at both inlet and outlet of the Pipeline. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at inlet and outlet of the Pipeline is 13.41 m/s and 13.56 m/s respectively.

The indication is that, prevailing velocity is below erosional velocity and pipe wall erosion cannot set-in.

See Appendix 1 for detailed simulation results for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 80.0 bar, the formation temperature is 16.31 °C. While the prevailing flow conditions at the inlet is 80.0 bar pressure and 57.94 °C temperature.

Similarly, for outlet stream, at stream pressure of 77.05 bar, the formation temperature is 16.07 °C. While the prevailing flow conditions at the outlet is, pressure 77.05 bar and temperature 54.12 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See Appendix 1 for detailed simulation results for more details

4.3. Phase-1/Scenario-2 Gas Production

This scenario shows the deliverability of maximum flow of gas, condensate and water from both BBOU and Opugbene Fields to the 24 inch BBOU-Elepa Pipeline when supply pressure at BBOU Manifold is 100 barg. The maximum flows occur in Years 9, 18 and 7 respectively as shown in the production forecast. Maximum combined flows are 331 MMSCFD gas, 16226 bpd of condensate and 9002 bpd of water.

4.3.1. Pressure and Temperature Regime

Source Conditions at the CTP; 80 barg pressure and corresponding temperature of 62.14 °C. Cumulative molar flow of gas, condensate and water at this condition is 346 MMSCFD.

Delivery Pressure; at ELEM E-MF is 77.04 bar and the corresponding temperature is 58.10 °C.

The Line Size; which also supported 346 MMscfd cumulative molar flow of gas, condensate and water of gas flow across the 6.7 km BBOU-ELEM E-MF bulk-line with minimal pressure loss and with a velocity below erosional velocity mark is 24". And the pressure drop is 0.37 bar/km.

See Appendix 2 for detailed simulation results for more details

Flow Velocity Profile

Minimum Fluid Flow Velocity; that will ensure the cleanliness of the pipe at the inlet and outlet of the bulk-line is 3.5 m/s and 3.9 m/s respectively.

Prevailing Fluid Flow Velocity; that supports fluid delivery is 4.31 m/s and 4.34 m/s at both inlet and outlet of the Pipeline. This indicates fluid velocity is above the minimum required for efficient

flow.

Erosional Velocity; at inlet and outlet of the Pipeline is 12.02 m/s and 12.07 m/s respectively. The indication is that, prevailing velocity is below erosional velocity and pipe wall erosion cannot set-in.

See Appendix 2 for detailed simulation results for more details.

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 100 bar, the hydrate formation temperature is 17.68 °C. While the prevailing flow conditions at the inlet is 100 bar pressure and 62.14°C temperature.

Similarly, for outlet stream, at stream pressure of 97.49 bar, the hydrate formation temperature is 17.53 °C. While the prevailing flow conditions at the outlet is, pressure 97.49 bar and temperature 58.1 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See Appendix 2 for detailed simulation results for the prevailing conditions.

4.4. Phase-1/Scenario-3 Gas Production

This scenario shows the deliverability of maximum flow of gas, condensate and water from both Seibou, Nun River, BBOU, Opugbene and Elepa Fields to the 36 inch Elepa-Akabel Pipeline when supply pressure at Elepa Manifold is 80 barg. The maximum flows occur in Years 13, 14 and 18 respectively as shown in the production forecast. Maximum combined flows are 386 MMSCFD gas, 34,433 bpd of condensate and 16,513 bpd of water.

4.4.1. Pressure and Temperature Regime

Source Conditions at the CTP; 78.57 barg pressure and corresponding temperature of 59.75 OC. Cumulative molar flow of gas, condensate and water at this condition is 414 MMSCFD.

Delivery Pressure; at Akabel Slug Catcher Tie-in is 66.77 bar and the corresponding temperature is 45.35 OC.

The Line Size; which also supported 414 MMscfd cumulative molar flow of gas, condensate and water of gas flow across the 35.7 km ELEP-AKABEL bulk-line with minimal pressure loss and with a velocity below erosional velocity mark is 36". And the pressure drop is 0.33 bar/km.

See Appendix 3 for detailed simulation results for more details.

Flow Velocity Profile

Minimum Fluid Flow Velocity; that will ensure the cleanliness of the pipe at the inlet and outlet of

the bulk-line is 3.5 m/s and 3.9 m/s respectively.

Prevailing Fluid Flow Velocity; that supports fluid delivery is 2.83 m/s and 3.15 m/s at both inlet and outlet of the Pipeline. This indicates fluid velocity is below the minimum required for efficient flow.

Erosional Velocity; at inlet and outlet of the Pipeline is 11.96 m/s and 12.60 m/s respectively. The indication is that, prevailing velocity is below erosional velocity and pipe wall erosion cannot set-in.

See Appendix 3 for detailed simulation results for more details.

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 78.57 bar, the formation temperature is 17.06 °C. While the prevailing flow conditions at the inlet is 78.57 bar pressure and 59.75 °C temperature.

Similarly, for outlet stream, at stream pressure of 66.77 bar, the formation temperature is 16.03 °C. While the prevailing flow conditions at the outlet is, pressure 66.77 bar and temperature 45.35 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See Appendix 3 for detailed simulation results for more details.

4.5. Phase-1/Scenario-4 Gas Production

This scenario shows the deliverability of maximum flow of gas, condensate and water from both Seibou, Nun River, BBOU, Opugbene and Elepa Fields to the 36 inch Elepa-Akabel Pipeline when supply pressure at Elepa Manifold is 80 barg. The maximum flows occur in Years 13, 14 and 18 respectively as shown in the production forecast. Maximum combined flows are 386 MMSCFD gas, 34,433 bpd of condensate and 16,513 bpd of water.

4.5.1. Pressure and Temperature Regime

Source Conditions at the CTP; 98.66 barg pressure and corresponding temperature of 63.19 °C. Cumulative molar flow of gas, condensate and water at this condition is 414 MMSCFD.

Delivery Pressure; at Akabel Slug Catcher Tie-in is 87.77 bar and the corresponding temperature is 47.89 °C.

The Line Size; which also supported 414 MMscfd cumulative molar flow of gas, condensate and water of gas flow across the 35.7 km ELEM-AKABEL bulk-line with minimal pressure loss and with a velocity below erosional velocity mark is 36". And the pressure drop is 0.31 bar/km.

See Appendix 4 for detailed simulation results for more details.

Flow Velocity Profile

Minimum Fluid Flow Velocity; that will ensure the cleanliness of the pipe at the inlet and outlet of the bulk-line is 3.5 m/s and 3.9 m/s respectively.

Prevailing Fluid Flow Velocity; that supports fluid delivery is 2.26 m/s and 2.37 m/s at both inlet and outlet of the Pipeline. This indicates fluid velocity is below the minimum required for efficient flow.

Erosional Velocity; at inlet and outlet of the Pipeline is 10.69 m/s and 10.94 m/s respectively. The indication is that, prevailing velocity is below erosional velocity and pipe wall erosion cannot set-in.

See Appendix 4 for detailed simulation results for more details.

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 98.66 bar, the hydrate formation temperature is 18.4 OC. While the prevailing flow conditions at the inlet is 98.66 bar pressure and 63.19 OC temperature.

Similarly, for outlet stream, at stream pressure of 87.77 bar, the hydrate formation temperature is 17.73 OC. While the prevailing flow conditions at the outlet is, pressure 87.77 bar and temperature 47.89 OC.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See Appendix 4 for detailed simulation results for the prevailing conditions.

5. Conclusion

Pressure deliverability:

All scenarios studied gives satisfactory results that guarantees flow assurance and gas deliverability at required 60 – 65 barg pressure required at Akabel for gas plant operations.

Pipeline Velocity:

Notably, at the declined stage of the fields when low volume flow sets-in, the prevailing fluid velocity is below minimum velocity required to sweep the pipe clean.

Line Sizes:

All the line were sized to flow at velocities below erosional velocity and also, to achieve a minimal pressure losses. This criterion guarantees that, the pipeline wall will not be eroded and also ensures adequate pressure deliverability.

Hydrate Formation:

For the entire pipeline examined; at the flow conditions (pressure & temperature regimes) no indication that hydrates will be formed on the lines.

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:09:19 2022		
5					
6					
7	Pipe Segment: BBOU-ELEP Pipeline				
8					
9					
10	CONDITIONS				
11	Name	BB-Elepa Pipeline Inlet	BB-Elepa Pipeline Outlet	e	
12	Vapour	0.9346	0.9340	---	
13	Temperature (C)	57.9405	54.1186	---	
14	Pressure (bar)	80.0000	77.0470	---	
15	Molar Flow (MMSCFD)	345.5645	345.5645	---	
16	Mass Flow (kg/h)	419320.3253	419320.3253	---	
17	Std Ideal Liq Vol Flow (m3/h)	1085.9821	1085.9821	---	
18	Molar Enthalpy (kJ/kgmole)	-9.031e+004	-9.053e+004	---	
19	Molar Entropy (kJ/kgmole-C)	157.9	157.5	---	
20	Heat Flow (kJ/h)	-1.5543e+09	-1.5581e+09	3.7786e+06	
21					
22	PROPERTIES				
23	Name	BB-Elepa Pipeline Inlet	BB-Elepa Pipeline Outlet		
24	Molecular Weight	24.36	24.36		
25	Molar Density (kgmole/m3)	3.396	3.321		
26	Mass Density (kg/m3)	82.74	80.90		
27	Act. Volume Flow (m3/h)	5068	5183		
28	Mass Enthalpy (kJ/kg)	-3707	-3716		
29	Mass Entropy (kJ/kg-C)	6.481	6.465		
30	Heat Capacity (kJ/kgmole-C)	62.55	62.23		
31	Mass Heat Capacity (kJ/kg-C)	2.568	2.554		
32	LHV Molar Basis (Std) (kJ/kgmole)	1.153e+006	1.153e+006		
33	HHV Molar Basis (Std) (kJ/kgmole)	1.258e+006	1.258e+006		
34	HHV Mass Basis (Std) (kJ/kg)	5.165e+004	5.165e+004		
35	CO2 Loading	---	---		
36	CO2 Apparent Mole Conc. (kgmole/m3)	---	---		
37	CO2 Apparent Wt. Conc. (kgmol/kg)	---	---		
38	LHV Mass Basis (Std) (kJ/kg)	4.733e+004	4.733e+004		
39	Phase Fraction [Vol. Basis]	0.8234	0.8224		
40	Phase Fraction [Mass Basis]	0.6781	0.6763		
41	Phase Fraction [Act. Vol. Basis]	0.9611	0.9620		
42	Mass Exergy (kJ/kg)	413.2	409.0		
43	Partial Pressure of CO2 (bar)	0.4645	0.4476		
44	Cost Based on Flow (Cost/s)	0.0000	0.0000		
45	Act. Gas Flow (ACT_m3/h)	4871	4986		
46	Avg. Liq. Density (kgmole/m3)	15.85	15.85		
47	Specific Heat (kJ/kgmole-C)	62.55	62.23		
48	Std. Gas Flow (MMSCFD)	344.9	344.9		
49	Std. Ideal Liq. Mass Density (kg/m3)	386.1	386.1		
50	Act. Liq. Flow (m3/s)	5.469e-002	5.471e-002		
51	Z Factor	---	---		
52	Watson K	16.58	16.58		
53	User Property	---	---		
54	Partial Pressure of H2S (bar)	0.0000	0.0000		
55	Cp/(Cp - R)	1.153	1.154		
56	Cp/Cv	1.318	1.318		
57	Heat of Vap. (kJ/kgmole)	2.752e+004	2.757e+004		
58	Kinematic Viscosity (cSt)	---	---		
59	Liq. Mass Density (Std. Cond) (kg/m3)	1.036	1.036		
60	Liq. Vol. Flow (Std. Cond) (m3/h)	4.048e+005	4.048e+005		
61	Liquid Fraction	6.537e-002	6.597e-002		
62	Molar Volume (m3/kgmole)	0.2944	0.3012		
63	Mass Heat of Vap. (kJ/kg)	1130	1132		
64	Phase Fraction [Molar Basis]	0.9346	0.9340		
65	Surface Tension (dyne/cm)	---	---		
66	Thermal Conductivity (W/m-K)	---	---		
67	Viscosity (lbm/s-ft)	---	---		
68	Cv (Semi-Ideal) (kJ/kgmole-C)	54.24	53.91		
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 1 of 116				

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5				Date/Time: Mon Jun 13 15:09:19 2022		
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	PROPERTIES					
10						
11	Name	BB-Elepa Pipeline Inlet	BB-Elepa Pipeline Outlet			
12	Mass Cv (Semi-Ideal) (kJ/kg-C)	2.226	2.213			
13	Cv (kJ/kgmole-C)	47.47	47.20			
14	Mass Cv (kJ/kg-C)	1.948	1.937			
15	Cv (Ent. Method) (kJ/kgmole-C)	---	---			
16	Mass Cv (Ent. Method) (kJ/kg-C)	---	---			
17	Cp/Cv (Ent. Method)	---	---			
18	Reid VP at 37.8 C (bar)	---	---			
19	True VP at 37.8 C (bar)	---	---			
20	Liq. Vol. Flow - Sum(Std. Cond) (m3/h)	3.794e+005	3.792e+005			
21	Viscosity Index	---	---			
22	Ideal Gas Cp/Cv	1.194	1.195			
23	Ideal Gas Cp (kJ/kgmole-C)	51.25	50.94			
24	Mass Ideal Gas Cp (kJ/kg-C)	2.104	2.091			
25	Bubble Point Pressure (bar)	---	---			
26	PROFILES					
27						
28	Data					
29						
30	Distance	Elevation		Increments		
31	(m)	(m)				
32	0.0000	0.0000		5		
33	2546	0.0000		5		
34	2571	-0.1600		5		
35	2596	4.000e-002		5		
36	2621	-1.000e-002		5		
37	2636	-8.000e-002		5		
38	2656	1.000e-002		5		
39	2681	-7.000e-002		5		
40	2706	-0.1000		5		
41	2731	-0.1600		5		
42	2756	-0.1000		5		
43	2781	-0.1600		5		
44	2806	-0.2000		5		
45	2831	-0.2100		5		
46	2856	-0.2100		5		
47	2881	-0.5700		5		
48	2906	-0.2200		5		
49	2931	0.2000		5		
50	2956	8.000e-002		5		
51	2981	3.000e-002		5		
52	3006	6.000e-002		5		
53	3031	6.000e-002		5		
54	3056	5.000e-002		5		
55	3081	6.000e-002		5		
56	3106	8.000e-002		5		
57	3131	7.000e-002		5		
58	3156	1.000e-001		5		
59	3181	0.1300		5		
60	3206	0.1500		5		
61	3236	0.1700		5		
62	3251	9.000e-002		5		
63	3281	0.2200		5		
64	3306	0.1500		5		
65	3331	1.000e-001		5		
66	3356	8.000e-002		5		
67	3381	1.000e-001		5		
68	3406	1.000e-001		5		
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1	 Epic Atlantic Limited		Case Name:	Simulation - 331MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:09:19 2022
4				
5				
6	Pipe Segment: BBOU-ELEP Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	3431	-0.2700	5	
16	3456	5.000e-002	5	
17	3481	1.000e-002	5	
18	3506	-2.000e-002	5	
19	3531	1.000e-002	5	
20	3556	1.000e-002	5	
21	3577	-1.232e-016	5	
22	3602	0.1800	5	
23	3627	-0.2300	5	
24	3652	-5.000e-002	5	
25	3677	2.000e-002	5	
26	3702	7.000e-002	5	
27	3727	0.1400	5	
28	3752	1.000e-001	5	
29	3775	0.1500	5	
30	3800	7.000e-002	5	
31	3825	-9.714e-017	5	
32	3850	-9.714e-017	5	
33	3875	3.000e-002	5	
34	3900	9.000e-002	5	
35	3925	0.2200	5	
36	3935	-0.3700	5	
37	3991	-3.600	5	
38	4047	-5.300	5	
39	4103	-5.900	5	
40	4158	-5.300	5	
41	4214	-6.300	5	
42	4270	-11.60	5	
43	4325	0.4400	5	
44	4340	1.150	5	
45	4355	0.8400	5	
46	4365	0.5100	5	
47	4375	0.6200	5	
48	4400	0.5900	5	
49	4425	0.2200	5	
50	4451	0.1600	5	
51	4476	0.1700	5	
52	4501	0.2400	5	
53	4526	0.3400	5	
54	4551	0.4800	5	
55	4576	0.5700	5	
56	4601	0.6300	5	
57	4626	0.6700	5	
58	4651	0.6700	5	
59	4676	0.7000	5	
60	4701	0.7000	5	
61	4726	0.5200	5	
62	4751	0.5200	5	
63	4776	0.6800	5	
64	4801	0.6300	5	
65	4826	0.6500	5	
66	4851	0.7000	5	
67	4876	0.7300	5	
68	4901	0.6800	5	
69	Aspen Technology Inc.			Aspen HYSYS Version 10
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1	 Epic Atlantic Limited		Case Name:	Simulation - 331MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:09:19 2022
4				
5				
6	Pipe Segment: BBOU-ELEP Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	4926	0.6700	5	
16	4951	0.6600	5	
17	4976	0.5900	5	
18	5001	0.6300	5	
19	5026	0.6800	5	
20	5052	0.6500	5	
21	5077	0.6200	5	
22	5102	0.6100	5	
23	5127	0.6600	5	
24	5152	0.7500	5	
25	5162	0.6600	5	
26	5192	0.7100	5	
27	5202	0.7600	5	
28	5227	0.6500	5	
29	5252	0.4700	5	
30	5277	0.5100	5	
31	5302	0.5500	5	
32	5327	0.5400	5	
33	5352	0.5800	5	
34	5377	0.5800	5	
35	5402	0.6000	5	
36	5427	0.6000	5	
37	5453	0.6000	5	
38	5478	0.6500	5	
39	5503	0.4800	5	
40	5528	0.5700	5	
41	5553	0.6500	5	
42	5578	0.7200	5	
43	5588	0.4000	5	
44	5598	0.2900	5	
45	5628	0.6500	5	
46	5658	-0.7100	5	
47	5725	-0.6000	5	
48	5792	-8.000	5	
49	5859	-10.40	5	
50	5926	-11.60	5	
51	5993	2.000e-002	5	
52	6013	0.5200	5	
53	6032	0.3200	5	
54	6037	0.2100	5	
55	6062	0.4800	5	
56	6087	0.4100	5	
57	6112	0.1700	5	
58	6137	0.1600	5	
59	6162	0.1500	5	
60	6187	0.2300	5	
61	6212	0.3200	5	
62	6238	0.3700	5	
63	6263	0.3900	5	
64	6288	0.4000	5	
65	6313	0.3800	5	
66	6338	0.3800	5	
67	6363	0.3200	5	
68	6388	0.3200	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	
			Page 4 of 116	

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:09:19 2022		
4					
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Data				
12					
13	Distance	Elevation	Increments		
14	(m)	(m)			
15	6413	0.2800	5		
16	6443	0.1300	5		
17	6473	0.1600	5		
18	6488	0.1900	5		
19	6513	0.3100	5		
20	6523	0.2400	5		
21	6538	0.1300	5		
22	6563	0.1600	5		
23	6588	0.2000	5		
24	6613	0.1200	5		
25	6638	7.000e-002	5		
26	6663	6.000e-002			
27					
28	Pipe Table				
29	Length (m)	0.0000	509.2	1018	1528
30	Elevation (m)	0.0000	0.0000	0.0000	0.0000
31	Pressure (bar)	80.00	79.82	79.64	79.47
32	Temperature (C)	57.94	57.65	57.35	57.06
33	Vapour Fraction	0.9346	0.9346	0.9345	0.9344
34	Heat Transferred (kJ/h-m)	---	605.4	599.0	592.7
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1678	0.1677	0.1676	0.1674
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	3.505e-002	3.509e-002	3.513e-002	3.517e-002
40	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
41	Accel. Gradient (kPa/m)	1.002e-005	1.006e-005	1.010e-005	1.015e-005
42	Liq. Reynolds	1.372e+005 *	1.369e+005 *	1.366e+005 *	1.362e+005 *
43	Vap. Reynolds	1.210e+007 *	1.210e+007 *	1.211e+007 *	1.212e+007 *
44	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
45	Vapour Velocity (m/s)	5.152	5.158	5.163	5.169
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	2037	2546	2551	2556
49	Elevation (m)	0.0000	0.0000	-3.200e-002	-6.400e-002
50	Pressure (bar)	79.29	79.11	79.11	79.10
51	Temperature (C)	56.77	56.49	56.49	56.48
52	Vapour Fraction	0.9344	0.9343	0.9343	0.9343
53	Heat Transferred (kJ/h-m)	586.4	580.1	577.0	577.0
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.1673	0.1672	0.1120	0.1120
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	3.521e-002	3.525e-002	3.870e-002	3.870e-002
59	Static Gradient (kPa/m)	0.0000	0.0000	-7.997e-003	-7.997e-003
60	Accel. Gradient (kPa/m)	1.019e-005	1.024e-005	8.922e-006	8.922e-006
61	Liq. Reynolds	1.359e+005 *	1.356e+005 *	1.356e+005 *	1.356e+005 *
62	Vap. Reynolds	1.213e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
63	Liquid Velocity (m/s)	0.2083	0.2084	0.2084	0.2084
64	Vapour Velocity (m/s)	5.175	5.180	5.181	5.181
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	2561	2566	2571	2576
68	Elevation (m)	-9.600e-002	-0.1280	-0.1600	-0.1200
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:09:19 2022		
4					
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Pressure (bar)	79.10	79.10	79.10	79.10
14	Temperature (C)	56.48	56.48	56.47	56.47
15	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
16	Heat Transferred (kJ/h-m)	576.9	576.8	576.8	576.6
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.1120	0.1120	0.1120	0.1662
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	3.870e-002	3.870e-002	3.870e-002	3.527e-002
22	Static Gradient (kPa/m)	-7.997e-003	-7.997e-003	-7.997e-003	1.275e-002
23	Accel. Gradient (kPa/m)	8.922e-006	8.923e-006	8.923e-006	1.395e-005
24	Liq. Reynolds	1.356e+005 *	1.356e+005 *	1.356e+005 *	1.356e+005 *
25	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
26	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
27	Vapour Velocity (m/s)	5.181	5.181	5.181	5.181
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	2581	2586	2591	2596
31	Elevation (m)	-8.000e-002	-4.000e-002	2.776e-017	4.000e-002
32	Pressure (bar)	79.09	79.09	79.09	79.09
33	Temperature (C)	56.47	56.47	56.46	56.46
34	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
35	Heat Transferred (kJ/h-m)	576.6	576.5	576.4	576.4
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.1662	0.1662	0.1662	0.1662
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	3.527e-002	3.527e-002	3.527e-002	3.528e-002
41	Static Gradient (kPa/m)	1.275e-002	1.275e-002	1.275e-002	1.275e-002
42	Accel. Gradient (kPa/m)	1.396e-005	1.396e-005	1.396e-005	1.396e-005
43	Liq. Reynolds	1.356e+005 *	1.356e+005 *	1.355e+005 *	1.355e+005 *
44	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
45	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
46	Vapour Velocity (m/s)	5.181	5.181	5.181	5.181
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	2601	2606	2611	2616
50	Elevation (m)	3.000e-002	2.000e-002	1.000e-002	0.0000
51	Pressure (bar)	79.09	79.08	79.08	79.08
52	Temperature (C)	56.46	56.45	56.45	56.45
53	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
54	Heat Transferred (kJ/h-m)	576.4	576.3	576.3	576.2
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.1137	0.1137	0.1137	0.1137
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	3.852e-002	3.852e-002	3.852e-002	3.852e-002
60	Static Gradient (kPa/m)	-2.540e-003	-2.540e-003	-2.540e-003	-2.540e-003
61	Accel. Gradient (kPa/m)	1.046e-005	1.046e-005	1.046e-005	1.046e-005
62	Liq. Reynolds	1.355e+005 *	1.355e+005 *	1.355e+005 *	1.355e+005 *
63	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
64	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
65	Vapour Velocity (m/s)	5.181	5.181	5.181	5.181
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	2621	2624	2627	2630
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:09:19 2022				
5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Elevation (m)	-1.000e-002	-2.400e-002	-3.800e-002	-5.200e-002		
14	Pressure (bar)	79.08	79.08	79.08	79.08		
15	Temperature (C)	56.45	56.44	56.44	56.44		
16	Vapour Fraction	0.9343	0.9343	0.9343	0.9343		
17	Heat Transferred (kJ/h-m)	576.1	576.1	576.1	576.0		
18	Flow Regime	Transition	Transition	Transition	Transition		
19	Water Flow Regime						
20	Liquid HoldUp	0.1137	0.1127	0.1127	0.1127		
21	Water HoldUp	---	---	---	---		
22	Friction Gradient (kPa/m)	3.852e-002	3.864e-002	3.864e-002	3.864e-002		
23	Static Gradient (kPa/m)	-2.540e-003	-5.897e-003	-5.896e-003	-5.896e-003		
24	Accel. Gradient (kPa/m)	1.046e-005	9.518e-006	9.519e-006	9.519e-006		
25	Liq. Reynolds	1.355e+005 *	1.355e+005 *	1.355e+005 *	1.355e+005 *		
26	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *		
27	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084		
28	Vapour Velocity (m/s)	5.182	5.182	5.182	5.182		
29	Deposit Thickness (mm)	---	---	---	---		
30	Deposit Volume (m3)	---	---	---	---		
31	Length (m)	2633	2636	2640	2644		
32	Elevation (m)	-6.600e-002	-8.000e-002	-6.200e-002	-4.400e-002		
33	Pressure (bar)	79.07	79.07	79.07	79.07		
34	Temperature (C)	56.44	56.44	56.43	56.43		
35	Vapour Fraction	0.9343	0.9343	0.9343	0.9343		
36	Heat Transferred (kJ/h-m)	576.0	576.0	575.8	575.8		
37	Flow Regime	Transition	Transition	Transition	Transition		
38	Water Flow Regime						
39	Liquid HoldUp	0.1127	0.1127	0.1610	0.1610		
40	Water HoldUp	---	---	---	---		
41	Friction Gradient (kPa/m)	3.864e-002	3.864e-002	3.542e-002	3.542e-002		
42	Static Gradient (kPa/m)	-5.896e-003	-5.896e-003	7.028e-003	7.028e-003		
43	Accel. Gradient (kPa/m)	9.519e-006	9.519e-006	1.234e-005	1.234e-005		
44	Liq. Reynolds	1.355e+005 *	1.355e+005 *	1.355e+005 *	1.355e+005 *		
45	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *		
46	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084		
47	Vapour Velocity (m/s)	5.182	5.182	5.182	5.182		
48	Deposit Thickness (mm)	---	---	---	---		
49	Deposit Volume (m3)	---	---	---	---		
50	Length (m)	2648	2652	2656	2661		
51	Elevation (m)	-2.600e-002	-8.000e-003	1.000e-002	-6.000e-003		
52	Pressure (bar)	79.07	79.07	79.06	79.06		
53	Temperature (C)	56.43	56.43	56.43	56.42		
54	Vapour Fraction	0.9343	0.9343	0.9343	0.9343		
55	Heat Transferred (kJ/h-m)	575.7	575.7	575.6	575.6		
56	Flow Regime	Transition	Transition	Transition	Transition		
57	Water Flow Regime						
58	Liquid HoldUp	0.1610	0.1610	0.1610	0.1133		
59	Water HoldUp	---	---	---	---		
60	Friction Gradient (kPa/m)	3.542e-002	3.542e-002	3.542e-002	3.858e-002		
61	Static Gradient (kPa/m)	7.028e-003	7.028e-003	7.028e-003	-4.055e-003		
62	Accel. Gradient (kPa/m)	1.234e-005	1.234e-005	1.234e-005	1.004e-005		
63	Liq. Reynolds	1.355e+005 *	1.355e+005 *	1.355e+005 *	1.355e+005 *		
64	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *		
65	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084		
66	Vapour Velocity (m/s)	5.182	5.182	5.182	5.182		
67	Deposit Thickness (mm)	---	---	---	---		
68	Deposit Volume (m3)	---	---	---	---		
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1			Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:09:19 2022		
6					
7	Pipe Segment: BBOU-ELEP Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	2666	2671	2676	2681
14	Elevation (m)	-2.200e-002	-3.800e-002	-5.400e-002	-7.000e-002
15	Pressure (bar)	79.06	79.06	79.06	79.06
16	Temperature (C)	56.42	56.42	56.41	56.41
17	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
18	Heat Transferred (kJ/h-m)	575.6	575.5	575.5	575.4
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.1133	0.1133	0.1133	0.1132
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	3.858e-002	3.858e-002	3.858e-002	3.858e-002
24	Static Gradient (kPa/m)	-4.054e-003	-4.054e-003	-4.054e-003	-4.054e-003
25	Accel. Gradient (kPa/m)	1.004e-005	1.004e-005	1.004e-005	1.004e-005
26	Liq. Reynolds	1.355e+005 *	1.355e+005 *	1.355e+005 *	1.355e+005 *
27	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
28	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
29	Vapour Velocity (m/s)	5.182	5.182	5.182	5.182
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	2686	2691	2696	2701
33	Elevation (m)	-7.600e-002	-8.200e-002	-8.800e-002	-9.400e-002
34	Pressure (bar)	79.05	79.05	79.05	79.05
35	Temperature (C)	56.41	56.41	56.40	56.40
36	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
37	Heat Transferred (kJ/h-m)	575.3	575.3	575.2	575.2
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.1140	0.1140	0.1140	0.1140
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	3.850e-002	3.850e-002	3.850e-002	3.850e-002
43	Static Gradient (kPa/m)	-1.526e-003	-1.526e-003	-1.526e-003	-1.526e-003
44	Accel. Gradient (kPa/m)	1.075e-005	1.075e-005	1.075e-005	1.075e-005
45	Liq. Reynolds	1.355e+005 *	1.355e+005 *	1.355e+005 *	1.355e+005 *
46	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
47	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
48	Vapour Velocity (m/s)	5.182	5.182	5.182	5.183
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	2706	2711	2716	2721
52	Elevation (m)	-0.1000	-0.1120	-0.1240	-0.1360
53	Pressure (bar)	79.05	79.04	79.04	79.04
54	Temperature (C)	56.40	56.39	56.39	56.39
55	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
56	Heat Transferred (kJ/h-m)	575.1	575.0	575.0	574.9
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.1140	0.1136	0.1136	0.1136
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	3.850e-002	3.855e-002	3.855e-002	3.855e-002
62	Static Gradient (kPa/m)	-1.526e-003	-3.045e-003	-3.045e-003	-3.045e-003
63	Accel. Gradient (kPa/m)	1.075e-005	1.033e-005	1.033e-005	1.033e-005
64	Liq. Reynolds	1.355e+005 *	1.355e+005 *	1.355e+005 *	1.355e+005 *
65	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
66	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
67	Vapour Velocity (m/s)	5.183	5.183	5.183	5.183
68	Deposit Thickness (mm)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:09:19 2022
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	2726	2731	2736	2741
15	Elevation (m)	-0.1480	-0.1600	-0.1480	-0.1360
16	Pressure (bar)	79.04	79.04	79.04	79.03
17	Temperature (C)	56.39	56.38	56.38	56.38
18	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
19	Heat Transferred (kJ/h-m)	574.9	574.8	574.7	574.6
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.1136	0.1136	0.1580	0.1580
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	3.855e-002	3.855e-002	3.554e-002	3.554e-002
25	Static Gradient (kPa/m)	-3.045e-003	-3.045e-003	3.702e-003	3.702e-003
26	Accel. Gradient (kPa/m)	1.033e-005	1.033e-005	1.142e-005	1.142e-005
27	Liq. Reynolds	1.355e+005 *	1.355e+005 *	1.355e+005 *	1.355e+005 *
28	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
29	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
30	Vapour Velocity (m/s)	5.183	5.183	5.183	5.183
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	2746	2751	2756	2761
34	Elevation (m)	-0.1240	-0.1120	-0.1000	-0.1120
35	Pressure (bar)	79.03	79.03	79.03	79.03
36	Temperature (C)	56.38	56.37	56.37	56.37
37	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
38	Heat Transferred (kJ/h-m)	574.6	574.5	574.4	574.4
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.1580	0.1580	0.1579	0.1135
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	3.554e-002	3.554e-002	3.554e-002	3.856e-002
44	Static Gradient (kPa/m)	3.702e-003	3.702e-003	3.702e-003	-3.072e-003
45	Accel. Gradient (kPa/m)	1.142e-005	1.142e-005	1.142e-005	1.033e-005
46	Liq. Reynolds	1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *
47	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
48	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
49	Vapour Velocity (m/s)	5.183	5.183	5.183	5.183
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	2766	2771	2776	2781
53	Elevation (m)	-0.1240	-0.1360	-0.1480	-0.1600
54	Pressure (bar)	79.02	79.02	79.02	79.02
55	Temperature (C)	56.36	56.36	56.36	56.36
56	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
57	Heat Transferred (kJ/h-m)	574.4	574.3	574.3	574.2
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.1135	0.1135	0.1135	0.1135
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	3.856e-002	3.856e-002	3.856e-002	3.856e-002
63	Static Gradient (kPa/m)	-3.072e-003	-3.072e-003	-3.072e-003	-3.072e-003
64	Accel. Gradient (kPa/m)	1.033e-005	1.033e-005	1.033e-005	1.033e-005
65	Liq. Reynolds	1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *
66	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
67	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
68	Vapour Velocity (m/s)	5.183	5.183	5.183	5.184
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:09:19 2022		
4					
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	2786	2791	2796	2801
16	Elevation (m)	-0.1680	-0.1760	-0.1840	-0.1920
17	Pressure (bar)	79.02	79.02	79.01	79.01
18	Temperature (C)	56.35	56.35	56.35	56.34
19	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
20	Heat Transferred (kJ/h-m)	574.1	574.1	574.0	574.0
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.1139	0.1139	0.1139	0.1139
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	3.852e-002	3.852e-002	3.852e-002	3.852e-002
26	Static Gradient (kPa/m)	-2.033e-003	-2.033e-003	-2.033e-003	-2.033e-003
27	Accel. Gradient (kPa/m)	1.062e-005	1.062e-005	1.062e-005	1.062e-005
28	Liq. Reynolds	1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *
29	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
30	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
31	Vapour Velocity (m/s)	5.184	5.184	5.184	5.184
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	2806	2811	2816	2821
35	Elevation (m)	-0.2000	-0.2020	-0.2040	-0.2060
36	Pressure (bar)	79.01	79.01	79.01	79.00
37	Temperature (C)	56.34	56.34	56.34	56.33
38	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
39	Heat Transferred (kJ/h-m)	573.9	573.8	573.8	573.7
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.1139	0.1143	0.1143	0.1143
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	3.853e-002	3.848e-002	3.848e-002	3.848e-002
45	Static Gradient (kPa/m)	-2.033e-003	-5.094e-004	-5.094e-004	-5.094e-004
46	Accel. Gradient (kPa/m)	1.062e-005	1.105e-005	1.105e-005	1.105e-005
47	Liq. Reynolds	1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *
48	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
49	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
50	Vapour Velocity (m/s)	5.184	5.184	5.184	5.184
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	2826	2831	2836	2841
54	Elevation (m)	-0.2080	-0.2100	-0.2100	-0.2100
55	Pressure (bar)	79.00	79.00	79.00	79.00
56	Temperature (C)	56.33	56.33	56.33	56.32
57	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
58	Heat Transferred (kJ/h-m)	573.6	573.6	573.4	573.4
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.1143	0.1143	0.1671	0.1671
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	3.848e-002	3.848e-002	3.527e-002	3.527e-002
64	Static Gradient (kPa/m)	-5.094e-004	-5.094e-004	0.0000	0.0000
65	Accel. Gradient (kPa/m)	1.105e-005	1.105e-005	1.027e-005	1.027e-005
66	Liq. Reynolds	1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *
67	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
68	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:09:19 2022			
5										
6	Pipe Segment: BBOU-ELEP Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Vapour Velocity	(m/s)	5.184	5.184	5.184	5.184				
14	Deposit Thickness	(mm)	---	---	---	---				
15	Deposit Volume	(m3)	---	---	---	---				
16	Length	(m)	2846	2851	2856	2861				
17	Elevation	(m)	-0.2100	-0.2100	-0.2100	-0.2820				
18	Pressure	(bar)	79.00	78.99	78.99	78.99				
19	Temperature	(C)	56.32	56.32	56.31	56.31				
20	Vapour Fraction		0.9343	0.9343	0.9343	0.9343				
21	Heat Transferred	(kJ/h-m)	573.3	573.3	573.2	573.2				
22	Flow Regime		Transition	Transition	Transition	Transition				
23	Water Flow Regime									
24	Liquid HoldUp		0.1671	0.1671	0.1671	0.1088				
25	Water HoldUp		---	---	---	---				
26	Friction Gradient	(kPa/m)	3.527e-002	3.527e-002	3.527e-002	3.910e-002				
27	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	-1.785e-002				
28	Accel. Gradient	(kPa/m)	1.027e-005	1.027e-005	1.027e-005	6.188e-006				
29	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *				
30	Vap. Reynolds		1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *				
31	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084				
32	Vapour Velocity	(m/s)	5.184	5.184	5.185	5.185				
33	Deposit Thickness	(mm)	---	---	---	---				
34	Deposit Volume	(m3)	---	---	---	---				
35	Length	(m)	2866	2871	2876	2881				
36	Elevation	(m)	-0.3540	-0.4260	-0.4980	-0.5700				
37	Pressure	(bar)	78.99	78.99	78.99	78.99				
38	Temperature	(C)	56.31	56.31	56.30	56.30				
39	Vapour Fraction		0.9343	0.9343	0.9343	0.9343				
40	Heat Transferred	(kJ/h-m)	573.2	573.1	573.1	573.0				
41	Flow Regime		Transition	Transition	Transition	Transition				
42	Water Flow Regime									
43	Liquid HoldUp		0.1088	0.1088	0.1088	0.1088				
44	Water HoldUp		---	---	---	---				
45	Friction Gradient	(kPa/m)	3.910e-002	3.910e-002	3.910e-002	3.910e-002				
46	Static Gradient	(kPa/m)	-1.785e-002	-1.785e-002	-1.785e-002	-1.785e-002				
47	Accel. Gradient	(kPa/m)	6.189e-006	6.189e-006	6.189e-006	6.189e-006				
48	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *				
49	Vap. Reynolds		1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *				
50	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084				
51	Vapour Velocity	(m/s)	5.185	5.185	5.185	5.185				
52	Deposit Thickness	(mm)	---	---	---	---				
53	Deposit Volume	(m3)	---	---	---	---				
54	Length	(m)	2886	2891	2896	2901				
55	Elevation	(m)	-0.5000	-0.4300	-0.3600	-0.2900				
56	Pressure	(bar)	78.98	78.98	78.98	78.97				
57	Temperature	(C)	56.30	56.30	56.29	56.29				
58	Vapour Fraction		0.9343	0.9343	0.9343	0.9343				
59	Heat Transferred	(kJ/h-m)	572.8	572.8	572.7	572.6				
60	Flow Regime		Transition	Transition	Transition	Transition				
61	Water Flow Regime									
62	Liquid HoldUp		0.1748	0.1748	0.1748	0.1748				
63	Water HoldUp		---	---	---	---				
64	Friction Gradient	(kPa/m)	3.520e-002	3.520e-002	3.520e-002	3.520e-002				
65	Static Gradient	(kPa/m)	2.305e-002	2.305e-002	2.305e-002	2.305e-002				
66	Accel. Gradient	(kPa/m)	1.697e-005	1.697e-005	1.697e-005	1.697e-005				
67	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *				
68	Vap. Reynolds		1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *				
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3							
4							
5			Unit Set: Mixed				
6			Date/Time: Mon Jun 13 15:09:19 2022				
7	Pipe Segment: BBOU-ELEP Pipeline (continued)						
8							
9							
10	PROFILES						
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
14	Vapour Velocity	(m/s)	5.185	5.185	5.185	5.185	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	2906	2911	2916	2921	
18	Elevation	(m)	-0.2200	-0.1360	-5.200e-002	3.200e-002	
19	Pressure	(bar)	78.97	78.97	78.97	78.96	
20	Temperature	(C)	56.29	56.28	56.28	56.28	
21	Vapour Fraction		0.9343	0.9343	0.9343	0.9343	
22	Heat Transferred	(kJ/h-m)	572.6	572.5	572.4	572.3	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.1748	0.1789	0.1789	0.1789	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	3.520e-002	3.527e-002	3.527e-002	3.527e-002	
28	Static Gradient	(kPa/m)	2.305e-002	2.809e-002	2.808e-002	2.808e-002	
29	Accel. Gradient	(kPa/m)	1.697e-005	1.846e-005	1.846e-005	1.846e-005	
30	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *	
31	Vap. Reynolds		1.214e+007 *	1.214e+007 *	1.215e+007 *	1.215e+007 *	
32	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
33	Vapour Velocity	(m/s)	5.185	5.185	5.186	5.186	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	2926	2931	2936	2941	
37	Elevation	(m)	0.1160	0.2000	0.1760	0.1520	
38	Pressure	(bar)	78.96	78.96	78.95	78.95	
39	Temperature	(C)	56.27	56.27	56.27	56.26	
40	Vapour Fraction		0.9343	0.9343	0.9343	0.9343	
41	Heat Transferred	(kJ/h-m)	572.3	572.2	572.2	572.2	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.1789	0.1789	0.1126	0.1126	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	3.528e-002	3.528e-002	3.868e-002	3.868e-002	
47	Static Gradient	(kPa/m)	2.808e-002	2.808e-002	-6.060e-003	-6.060e-003	
48	Accel. Gradient	(kPa/m)	1.846e-005	1.846e-005	9.505e-006	9.506e-006	
49	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *	
50	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *	
51	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
52	Vapour Velocity	(m/s)	5.186	5.186	5.186	5.186	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	2946	2951	2956	2961	
56	Elevation	(m)	0.1280	0.1040	8.000e-002	7.000e-002	
57	Pressure	(bar)	78.95	78.95	78.95	78.95	
58	Temperature	(C)	56.26	56.26	56.26	56.25	
59	Vapour Fraction		0.9343	0.9343	0.9343	0.9343	
60	Heat Transferred	(kJ/h-m)	572.1	572.1	572.0	571.9	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.1126	0.1126	0.1126	0.1137	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	3.868e-002	3.868e-002	3.868e-002	3.856e-002	
66	Static Gradient	(kPa/m)	-6.060e-003	-6.060e-003	-6.060e-003	-2.512e-003	
67	Accel. Gradient	(kPa/m)	9.506e-006	9.507e-006	9.507e-006	1.051e-005	
68	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:09:19 2022		
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
14	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
15	Vapour Velocity (m/s)	5.186	5.186	5.186	5.186
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	2966	2971	2976	2981
19	Elevation (m)	6.000e-002	5.000e-002	4.000e-002	3.000e-002
20	Pressure (bar)	78.94	78.94	78.94	78.94
21	Temperature (C)	56.25	56.25	56.24	56.24
22	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
23	Heat Transferred (kJ/h-m)	571.9	571.8	571.8	571.7
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1137	0.1137	0.1137	0.1137
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	3.856e-002	3.856e-002	3.856e-002	3.856e-002
29	Static Gradient (kPa/m)	-2.512e-003	-2.512e-003	-2.512e-003	-2.512e-003
30	Accel. Gradient (kPa/m)	1.051e-005	1.051e-005	1.051e-005	1.051e-005
31	Liq. Reynolds	1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *
32	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
33	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
34	Vapour Velocity (m/s)	5.186	5.186	5.187	5.187
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	2986	2991	2996	3001
38	Elevation (m)	3.600e-002	4.200e-002	4.800e-002	5.400e-002
39	Pressure (bar)	78.94	78.94	78.93	78.93
40	Temperature (C)	56.24	56.24	56.23	56.23
41	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
42	Heat Transferred (kJ/h-m)	571.6	571.5	571.5	571.4
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1561	0.1561	0.1561	0.1561
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	3.563e-002	3.563e-002	3.563e-002	3.563e-002
48	Static Gradient (kPa/m)	1.838e-003	1.838e-003	1.838e-003	1.838e-003
49	Accel. Gradient (kPa/m)	1.092e-005	1.092e-005	1.092e-005	1.092e-005
50	Liq. Reynolds	1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *
51	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
52	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
53	Vapour Velocity (m/s)	5.187	5.187	5.187	5.187
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	3006	3011	3016	3021
57	Elevation (m)	6.000e-002	6.000e-002	6.000e-002	6.000e-002
58	Pressure (bar)	78.93	78.93	78.93	78.92
59	Temperature (C)	56.23	56.22	56.22	56.22
60	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
61	Heat Transferred (kJ/h-m)	571.3	571.2	571.2	571.1
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1561	0.1671	0.1671	0.1671
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	3.563e-002	3.529e-002	3.529e-002	3.529e-002
67	Static Gradient (kPa/m)	1.838e-003	0.0000	0.0000	0.0000
68	Accel. Gradient (kPa/m)	1.092e-005	1.029e-005	1.029e-005	1.029e-005
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:09:19 2022				
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *	
14	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *	
15	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084	0.2084	
16	Vapour Velocity (m/s)	5.187	5.187	5.187	5.187	5.187	
17	Deposit Thickness (mm)	---	---	---	---	---	
18	Deposit Volume (m3)	---	---	---	---	---	
19	Length (m)	3026	3031	3036	3041		
20	Elevation (m)	6.000e-002	6.000e-002	5.800e-002	5.600e-002		
21	Pressure (bar)	78.92	78.92	78.92	78.92		
22	Temperature (C)	56.22	56.21	56.21	56.21		
23	Vapour Fraction	0.9343	0.9343	0.9343	0.9343		
24	Heat Transferred (kJ/h-m)	571.1	571.0	571.0	571.0		
25	Flow Regime	Transition	Transition	Transition	Transition		
26	Water Flow Regime						
27	Liquid HoldUp	0.1671	0.1671	0.1143	0.1143		
28	Water HoldUp	---	---	---	---		
29	Friction Gradient (kPa/m)	3.529e-002	3.529e-002	3.850e-002	3.850e-002		
30	Static Gradient (kPa/m)	0.0000	0.0000	-5.092e-004	-5.092e-004		
31	Accel. Gradient (kPa/m)	1.029e-005	1.029e-005	1.108e-005	1.108e-005		
32	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *	
33	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *	
34	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084	0.2084	
35	Vapour Velocity (m/s)	5.187	5.187	5.187	5.187	5.187	
36	Deposit Thickness (mm)	---	---	---	---	---	
37	Deposit Volume (m3)	---	---	---	---	---	
38	Length (m)	3046	3051	3056	3061		
39	Elevation (m)	5.400e-002	5.200e-002	5.000e-002	5.200e-002		
40	Pressure (bar)	78.91	78.91	78.91	78.91		
41	Temperature (C)	56.21	56.20	56.20	56.20		
42	Vapour Fraction	0.9343	0.9343	0.9343	0.9343		
43	Heat Transferred (kJ/h-m)	570.9	570.8	570.8	570.7		
44	Flow Regime	Transition	Transition	Transition	Transition		
45	Water Flow Regime						
46	Liquid HoldUp	0.1143	0.1143	0.1143	0.1550		
47	Water HoldUp	---	---	---	---		
48	Friction Gradient (kPa/m)	3.850e-002	3.850e-002	3.850e-002	3.568e-002		
49	Static Gradient (kPa/m)	-5.092e-004	-5.092e-004	-5.092e-004	6.096e-004		
50	Accel. Gradient (kPa/m)	1.108e-005	1.108e-005	1.108e-005	1.058e-005		
51	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *	
52	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *	
53	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084	0.2084	
54	Vapour Velocity (m/s)	5.187	5.187	5.188	5.188	5.188	
55	Deposit Thickness (mm)	---	---	---	---	---	
56	Deposit Volume (m3)	---	---	---	---	---	
57	Length (m)	3066	3071	3076	3081		
58	Elevation (m)	5.400e-002	5.600e-002	5.800e-002	6.000e-002		
59	Pressure (bar)	78.91	78.91	78.90	78.90		
60	Temperature (C)	56.19	56.19	56.19	56.19		
61	Vapour Fraction	0.9343	0.9343	0.9343	0.9343		
62	Heat Transferred (kJ/h-m)	570.6	570.5	570.5	570.4		
63	Flow Regime	Transition	Transition	Transition	Transition		
64	Water Flow Regime						
65	Liquid HoldUp	0.1550	0.1550	0.1549	0.1549		
66	Water HoldUp	---	---	---	---		
67	Friction Gradient (kPa/m)	3.568e-002	3.568e-002	3.568e-002	3.568e-002		
68	Static Gradient (kPa/m)	6.096e-004	6.096e-004	6.096e-004	6.096e-004		
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:09:19 2022				
5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient (kPa/m)	1.059e-005	1.059e-005	1.059e-005	1.059e-005		
14	Liq. Reynolds	1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *		
15	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *		
16	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084		
17	Vapour Velocity (m/s)	5.188	5.188	5.188	5.188		
18	Deposit Thickness (mm)	---	---	---	---		
19	Deposit Volume (m3)	---	---	---	---		
20	Length (m)	3086	3091	3096	3101		
21	Elevation (m)	6.400e-002	6.800e-002	7.200e-002	7.600e-002		
22	Pressure (bar)	78.90	78.90	78.90	78.89		
23	Temperature (C)	56.18	56.18	56.18	56.17		
24	Vapour Fraction	0.9343	0.9343	0.9343	0.9343		
25	Heat Transferred (kJ/h-m)	570.4	570.3	570.2	570.2		
26	Flow Regime	Transition	Transition	Transition	Transition		
27	Water Flow Regime						
28	Liquid HoldUp	0.1555	0.1555	0.1555	0.1555		
29	Water HoldUp	---	---	---	---		
30	Friction Gradient (kPa/m)	3.566e-002	3.566e-002	3.566e-002	3.566e-002		
31	Static Gradient (kPa/m)	1.222e-003	1.222e-003	1.222e-003	1.222e-003		
32	Accel. Gradient (kPa/m)	1.076e-005	1.076e-005	1.076e-005	1.076e-005		
33	Liq. Reynolds	1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *		
34	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *		
35	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084		
36	Vapour Velocity (m/s)	5.188	5.188	5.188	5.188		
37	Deposit Thickness (mm)	---	---	---	---		
38	Deposit Volume (m3)	---	---	---	---		
39	Length (m)	3106	3111	3116	3121		
40	Elevation (m)	8.000e-002	7.800e-002	7.600e-002	7.400e-002		
41	Pressure (bar)	78.89	78.89	78.89	78.89		
42	Temperature (C)	56.17	56.17	56.17	56.16		
43	Vapour Fraction	0.9343	0.9343	0.9343	0.9343		
44	Heat Transferred (kJ/h-m)	570.1	570.1	570.1	570.0		
45	Flow Regime	Transition	Transition	Transition	Transition		
46	Water Flow Regime						
47	Liquid HoldUp	0.1555	0.1143	0.1143	0.1143		
48	Water HoldUp	---	---	---	---		
49	Friction Gradient (kPa/m)	3.566e-002	3.851e-002	3.851e-002	3.851e-002		
50	Static Gradient (kPa/m)	1.222e-003	-5.092e-004	-5.092e-004	-5.092e-004		
51	Accel. Gradient (kPa/m)	1.076e-005	1.109e-005	1.109e-005	1.109e-005		
52	Liq. Reynolds	1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *		
53	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *		
54	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084		
55	Vapour Velocity (m/s)	5.188	5.188	5.188	5.188		
56	Deposit Thickness (mm)	---	---	---	---		
57	Deposit Volume (m3)	---	---	---	---		
58	Length (m)	3126	3131	3136	3141		
59	Elevation (m)	7.200e-002	7.000e-002	7.600e-002	8.200e-002		
60	Pressure (bar)	78.89	78.88	78.88	78.88		
61	Temperature (C)	56.16	56.16	56.15	56.15		
62	Vapour Fraction	0.9343	0.9343	0.9343	0.9343		
63	Heat Transferred (kJ/h-m)	569.9	569.9	569.8	569.7		
64	Flow Regime	Transition	Transition	Transition	Transition		
65	Water Flow Regime						
66	Liquid HoldUp	0.1143	0.1143	0.1561	0.1561		
67	Water HoldUp	---	---	---	---		
68	Friction Gradient (kPa/m)	3.851e-002	3.851e-002	3.564e-002	3.564e-002		
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1				Case Name: Simulation - 331MMSCFD_80barg.hsc		
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3	 Epic Atlantic Limited			Unit Set: Mixed		
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5				Date/Time: Mon Jun 13 15:09:19 2022		
6						
7	Pipe Segment: BBOU-ELEP Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	-5.091e-004	-5.091e-004	1.837e-003	1.837e-003
14	Accel. Gradient	(kPa/m)	1.109e-005	1.109e-005	1.094e-005	1.094e-005
15	Liq. Reynolds		1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
16	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
17	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084
18	Vapour Velocity	(m/s)	5.188	5.189	5.189	5.189
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	3146	3151	3156	3161
22	Elevation	(m)	8.800e-002	9.400e-002	1.000e-001	0.1060
23	Pressure	(bar)	78.88	78.88	78.87	78.87
24	Temperature	(C)	56.15	56.15	56.14	56.14
25	Vapour Fraction		0.9343	0.9343	0.9343	0.9343
26	Heat Transferred	(kJ/h-m)	569.6	569.6	569.5	569.4
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.1561	0.1561	0.1561	0.1561
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	3.564e-002	3.564e-002	3.564e-002	3.564e-002
32	Static Gradient	(kPa/m)	1.837e-003	1.837e-003	1.837e-003	1.827e-003
33	Accel. Gradient	(kPa/m)	1.094e-005	1.094e-005	1.094e-005	1.094e-005
34	Liq. Reynolds		1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
35	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
36	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084
37	Vapour Velocity	(m/s)	5.189	5.189	5.189	5.189
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	3166	3171	3176	3181
41	Elevation	(m)	0.1120	0.1180	0.1240	0.1300
42	Pressure	(bar)	78.87	78.87	78.87	78.86
43	Temperature	(C)	56.14	56.13	56.13	56.13
44	Vapour Fraction		0.9343	0.9343	0.9343	0.9343
45	Heat Transferred	(kJ/h-m)	569.4	569.3	569.3	569.2
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.1561	0.1561	0.1561	0.1561
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	3.564e-002	3.564e-002	3.564e-002	3.564e-002
51	Static Gradient	(kPa/m)	1.827e-003	1.827e-003	1.827e-003	1.827e-003
52	Accel. Gradient	(kPa/m)	1.094e-005	1.094e-005	1.094e-005	1.094e-005
53	Liq. Reynolds		1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
54	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
55	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084
56	Vapour Velocity	(m/s)	5.189	5.189	5.189	5.189
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	3186	3191	3196	3201
60	Elevation	(m)	0.1340	0.1380	0.1420	0.1460
61	Pressure	(bar)	78.86	78.86	78.86	78.86
62	Temperature	(C)	56.13	56.12	56.12	56.12
63	Vapour Fraction		0.9343	0.9342	0.9342	0.9342
64	Heat Transferred	(kJ/h-m)	569.1	569.1	569.0	569.0
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.1555	0.1555	0.1555	0.1555
68	Water HoldUp		---	---	---	---
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3	 Epic Atlantic Limited		Unit Set: Mixed				
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11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	3.567e-002	3.567e-002	3.567e-002	3.567e-002	
14	Static Gradient	(kPa/m)	1.222e-003	1.222e-003	1.222e-003	1.222e-003	
15	Accel. Gradient	(kPa/m)	1.077e-005	1.077e-005	1.077e-005	1.077e-005	
16	Liq. Reynolds		1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *	
17	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *	
18	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
19	Vapour Velocity	(m/s)	5.189	5.189	5.189	5.189	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	3206	3212	3218	3224	
23	Elevation	(m)	0.1500	0.1540	0.1580	0.1620	
24	Pressure	(bar)	78.86	78.85	78.85	78.85	
25	Temperature	(C)	56.12	56.11	56.11	56.11	
26	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
27	Heat Transferred	(kJ/h-m)	568.9	568.8	568.7	568.7	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.1555	0.1553	0.1553	0.1553	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	3.567e-002	3.568e-002	3.568e-002	3.568e-002	
33	Static Gradient	(kPa/m)	1.222e-003	1.017e-003	1.017e-003	1.017e-003	
34	Accel. Gradient	(kPa/m)	1.077e-005	1.071e-005	1.072e-005	1.072e-005	
35	Liq. Reynolds		1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *	
36	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *	
37	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
38	Vapour Velocity	(m/s)	5.190	5.190	5.190	5.190	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	3230	3236	3239	3242	
42	Elevation	(m)	0.1660	0.1700	0.1540	0.1380	
43	Pressure	(bar)	78.85	78.84	78.84	78.84	
44	Temperature	(C)	56.10	56.10	56.10	56.10	
45	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
46	Heat Transferred	(kJ/h-m)	568.6	568.5	568.6	568.5	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.1553	0.1553	0.1123	0.1123	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	3.568e-002	3.568e-002	3.873e-002	3.873e-002	
52	Static Gradient	(kPa/m)	1.017e-003	1.017e-003	-6.723e-003	-6.723e-003	
53	Accel. Gradient	(kPa/m)	1.072e-005	1.072e-005	9.347e-006	9.348e-006	
54	Liq. Reynolds		1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *	
55	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *	
56	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
57	Vapour Velocity	(m/s)	5.190	5.190	5.190	5.190	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	3245	3248	3251	3257	
61	Elevation	(m)	0.1220	0.1060	9.000e-002	0.1160	
62	Pressure	(bar)	78.84	78.84	78.84	78.84	
63	Temperature	(C)	56.09	56.09	56.09	56.09	
64	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
65	Heat Transferred	(kJ/h-m)	568.5	568.5	568.4	568.3	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.1123	0.1123	0.1123	0.1606	
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
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11	Pipe Table				
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13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	3.873e-002	3.873e-002	3.873e-002	3.549e-002
15	Static Gradient (kPa/m)	-6.723e-003	-6.723e-003	-6.723e-003	6.755e-003
16	Accel. Gradient (kPa/m)	9.348e-006	9.348e-006	9.348e-006	1.234e-005
17	Liq. Reynolds	1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
18	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
19	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
20	Vapour Velocity (m/s)	5.190	5.190	5.190	5.190
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	3263	3269	3275	3281
24	Elevation (m)	0.1420	0.1680	0.1940	0.2200
25	Pressure (bar)	78.83	78.83	78.83	78.83
26	Temperature (C)	56.08	56.08	56.08	56.07
27	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
28	Heat Transferred (kJ/h-m)	568.2	568.1	568.0	568.0
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.1606	0.1606	0.1606	0.1606
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	3.549e-002	3.549e-002	3.549e-002	3.549e-002
34	Static Gradient (kPa/m)	6.755e-003	6.755e-003	6.755e-003	6.755e-003
35	Accel. Gradient (kPa/m)	1.234e-005	1.234e-005	1.234e-005	1.234e-005
36	Liq. Reynolds	1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
37	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
38	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
39	Vapour Velocity (m/s)	5.190	5.190	5.190	5.191
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	3286	3291	3296	3301
43	Elevation (m)	0.2060	0.1920	0.1780	0.1640
44	Pressure (bar)	78.83	78.82	78.82	78.82
45	Temperature (C)	56.07	56.07	56.06	56.06
46	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
47	Heat Transferred (kJ/h-m)	568.0	567.9	567.9	567.8
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.1133	0.1133	0.1133	0.1133
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	3.863e-002	3.863e-002	3.863e-002	3.863e-002
53	Static Gradient (kPa/m)	-3.547e-003	-3.547e-003	-3.547e-003	-3.547e-003
54	Accel. Gradient (kPa/m)	1.025e-005	1.025e-005	1.025e-005	1.025e-005
55	Liq. Reynolds	1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
56	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
57	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
58	Vapour Velocity (m/s)	5.191	5.191	5.191	5.191
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	3306	3311	3316	3321
62	Elevation (m)	0.1500	0.1400	0.1300	0.1200
63	Pressure (bar)	78.82	78.82	78.81	78.81
64	Temperature (C)	56.06	56.06	56.05	56.05
65	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
66	Heat Transferred (kJ/h-m)	567.7	567.7	567.6	567.6
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
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13	Liquid HoldUp	0.1133	0.1136	0.1136	0.1136
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	3.863e-002	3.859e-002	3.860e-002	3.860e-002
16	Static Gradient (kPa/m)	-3.547e-003	-2.537e-003	-2.537e-003	-2.537e-003
17	Accel. Gradient (kPa/m)	1.025e-005	1.054e-005	1.054e-005	1.054e-005
18	Liq. Reynolds	1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
19	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
20	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
21	Vapour Velocity (m/s)	5.191	5.191	5.191	5.191
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	3326	3331	3336	3341
25	Elevation (m)	0.1100	1.000e-001	9.600e-002	9.200e-002
26	Pressure (bar)	78.81	78.81	78.81	78.81
27	Temperature (C)	56.05	56.05	56.04	56.04
28	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
29	Heat Transferred (kJ/h-m)	567.5	567.4	567.4	567.3
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1136	0.1136	0.1141	0.1141
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	3.860e-002	3.860e-002	3.855e-002	3.855e-002
35	Static Gradient (kPa/m)	-2.537e-003	-2.537e-003	-1.017e-003	-1.017e-003
36	Accel. Gradient (kPa/m)	1.054e-005	1.054e-005	1.097e-005	1.097e-005
37	Liq. Reynolds	1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
38	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
39	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
40	Vapour Velocity (m/s)	5.191	5.191	5.191	5.191
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	3346	3351	3356	3361
44	Elevation (m)	8.800e-002	8.400e-002	8.000e-002	8.400e-002
45	Pressure (bar)	78.80	78.80	78.80	78.80
46	Temperature (C)	56.04	56.03	56.03	56.03
47	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
48	Heat Transferred (kJ/h-m)	567.3	567.2	567.1	567.0
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1141	0.1141	0.1141	0.1555
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	3.855e-002	3.855e-002	3.855e-002	3.568e-002
54	Static Gradient (kPa/m)	-1.017e-003	-1.017e-003	-1.017e-003	1.221e-003
55	Accel. Gradient (kPa/m)	1.097e-005	1.097e-005	1.097e-005	1.079e-005
56	Liq. Reynolds	1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
57	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
58	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
59	Vapour Velocity (m/s)	5.191	5.191	5.191	5.192
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	3366	3371	3376	3381
63	Elevation (m)	8.800e-002	9.200e-002	9.600e-002	1.000e-001
64	Pressure (bar)	78.80	78.79	78.79	78.79
65	Temperature (C)	56.03	56.02	56.02	56.02
66	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
67	Heat Transferred (kJ/h-m)	567.0	566.9	566.8	566.8
68	Flow Regime	Transition	Transition	Transition	Transition
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2			Unit Set: Mixed		
3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:09:19 2022		
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11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1555	0.1555	0.1555	0.1555
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	3.568e-002	3.568e-002	3.568e-002	3.569e-002
17	Static Gradient (kPa/m)	1.221e-003	1.221e-003	1.221e-003	1.221e-003
18	Accel. Gradient (kPa/m)	1.079e-005	1.079e-005	1.079e-005	1.079e-005
19	Liq. Reynolds	1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
20	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
21	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
22	Vapour Velocity (m/s)	5.192	5.192	5.192	5.192
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	3386	3391	3396	3401
26	Elevation (m)	1.000e-001	1.000e-001	1.000e-001	1.000e-001
27	Pressure (bar)	78.79	78.79	78.79	78.78
28	Temperature (C)	56.02	56.01	56.01	56.01
29	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
30	Heat Transferred (kJ/h-m)	566.7	566.6	566.6	566.5
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1670	0.1670	0.1670	0.1670
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	3.532e-002	3.532e-002	3.532e-002	3.532e-002
36	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
37	Accel. Gradient (kPa/m)	1.033e-005	1.033e-005	1.033e-005	1.033e-005
38	Liq. Reynolds	1.353e+005 *	1.353e+005 *	1.352e+005 *	1.352e+005 *
39	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
40	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
41	Vapour Velocity (m/s)	5.192	5.192	5.192	5.192
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	3406	3411	3416	3421
45	Elevation (m)	1.000e-001	2.600e-002	-4.800e-002	-0.1220
46	Pressure (bar)	78.78	78.78	78.78	78.78
47	Temperature (C)	56.00	56.00	56.00	56.00
48	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
49	Heat Transferred (kJ/h-m)	566.4	566.5	566.4	566.4
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.1670	0.1085	0.1085	0.1085
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	3.533e-002	3.918e-002	3.918e-002	3.918e-002
55	Static Gradient (kPa/m)	0.0000	-1.831e-002	-1.831e-002	-1.831e-002
56	Accel. Gradient (kPa/m)	1.033e-005	6.101e-006	6.101e-006	6.101e-006
57	Liq. Reynolds	1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *
58	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
59	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
60	Vapour Velocity (m/s)	5.192	5.192	5.192	5.192
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	3426	3431	3436	3441
64	Elevation (m)	-0.1960	-0.2700	-0.2060	-0.1420
65	Pressure (bar)	78.78	78.78	78.77	78.77
66	Temperature (C)	55.99	55.99	55.99	55.99
67	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
68	Heat Transferred (kJ/h-m)	566.3	566.3	566.1	566.0
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:09:19 2022
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.1085	0.1085	0.1729	0.1729
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	3.918e-002	3.918e-002	3.525e-002	3.525e-002
18	Static Gradient (kPa/m)	-1.831e-002	-1.831e-002	2.092e-002	2.092e-002
19	Accel. Gradient (kPa/m)	6.101e-006	6.101e-006	1.643e-005	1.643e-005
20	Liq. Reynolds	1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *
21	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
22	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
23	Vapour Velocity (m/s)	5.192	5.192	5.192	5.192
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	3446	3451	3456	3461
27	Elevation (m)	-7.800e-002	-1.400e-002	5.000e-002	4.200e-002
28	Pressure (bar)	78.77	78.77	78.76	78.76
29	Temperature (C)	55.98	55.98	55.98	55.97
30	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
31	Heat Transferred (kJ/h-m)	566.0	565.9	565.8	565.9
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1729	0.1729	0.1729	0.1137
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	3.525e-002	3.525e-002	3.525e-002	3.859e-002
37	Static Gradient (kPa/m)	2.092e-002	2.092e-002	2.092e-002	-2.031e-003
38	Accel. Gradient (kPa/m)	1.643e-005	1.643e-005	1.643e-005	1.070e-005
39	Liq. Reynolds	1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *
40	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
41	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
42	Vapour Velocity (m/s)	5.193	5.193	5.193	5.193
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	3466	3471	3476	3481
46	Elevation (m)	3.400e-002	2.600e-002	1.800e-002	1.000e-002
47	Pressure (bar)	78.76	78.76	78.76	78.75
48	Temperature (C)	55.97	55.97	55.97	55.96
49	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
50	Heat Transferred (kJ/h-m)	565.8	565.7	565.7	565.6
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1137	0.1137	0.1137	0.1137
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	3.859e-002	3.859e-002	3.859e-002	3.859e-002
56	Static Gradient (kPa/m)	-2.031e-003	-2.031e-003	-2.031e-003	-2.031e-003
57	Accel. Gradient (kPa/m)	1.070e-005	1.070e-005	1.070e-005	1.070e-005
58	Liq. Reynolds	1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *
59	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
60	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
61	Vapour Velocity (m/s)	5.193	5.193	5.193	5.193
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	3486	3491	3496	3501
65	Elevation (m)	4.000e-003	-2.000e-003	-8.000e-003	-1.400e-002
66	Pressure (bar)	78.75	78.75	78.75	78.75
67	Temperature (C)	55.96	55.96	55.95	55.95
68	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:09:19 2022				
5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Heat Transferred	(kJ/h-m)	565.6	565.5	565.4	565.4	
14	Flow Regime		Transition	Transition	Transition	Transition	
15	Water Flow Regime						
16	Liquid HoldUp		0.1139	0.1139	0.1139	0.1139	
17	Water HoldUp		---	---	---	---	
18	Friction Gradient	(kPa/m)	3.858e-002	3.858e-002	3.858e-002	3.858e-002	
19	Static Gradient	(kPa/m)	-1.524e-003	-1.524e-003	-1.524e-003	-1.524e-003	
20	Accel. Gradient	(kPa/m)	1.084e-005	1.084e-005	1.084e-005	1.084e-005	
21	Liq. Reynolds		1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *	
22	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *	
23	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
24	Vapour Velocity	(m/s)	5.193	5.193	5.193	5.193	
25	Deposit Thickness	(mm)	---	---	---	---	
26	Deposit Volume	(m3)	---	---	---	---	
27	Length	(m)	3506	3511	3516	3521	
28	Elevation	(m)	-2.000e-002	-1.400e-002	-8.000e-003	-2.000e-003	
29	Pressure	(bar)	78.74	78.74	78.74	78.74	
30	Temperature	(C)	55.95	55.95	55.94	55.94	
31	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
32	Heat Transferred	(kJ/h-m)	565.3	565.2	565.1	565.1	
33	Flow Regime		Transition	Transition	Transition	Transition	
34	Water Flow Regime						
35	Liquid HoldUp		0.1139	0.1560	0.1560	0.1560	
36	Water HoldUp		---	---	---	---	
37	Friction Gradient	(kPa/m)	3.858e-002	3.567e-002	3.567e-002	3.567e-002	
38	Static Gradient	(kPa/m)	-1.524e-003	1.836e-003	1.836e-003	1.836e-003	
39	Accel. Gradient	(kPa/m)	1.084e-005	1.098e-005	1.098e-005	1.098e-005	
40	Liq. Reynolds		1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *	
41	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *	
42	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
43	Vapour Velocity	(m/s)	5.193	5.194	5.194	5.194	
44	Deposit Thickness	(mm)	---	---	---	---	
45	Deposit Volume	(m3)	---	---	---	---	
46	Length	(m)	3526	3531	3536	3541	
47	Elevation	(m)	4.000e-003	1.000e-002	1.000e-002	1.000e-002	
48	Pressure	(bar)	78.74	78.73	78.73	78.73	
49	Temperature	(C)	55.94	55.93	55.93	55.93	
50	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
51	Heat Transferred	(kJ/h-m)	565.0	565.0	564.9	564.8	
52	Flow Regime		Transition	Transition	Transition	Transition	
53	Water Flow Regime						
54	Liquid HoldUp		0.1560	0.1560	0.1669	0.1669	
55	Water HoldUp		---	---	---	---	
56	Friction Gradient	(kPa/m)	3.568e-002	3.568e-002	3.534e-002	3.534e-002	
57	Static Gradient	(kPa/m)	1.836e-003	1.836e-003	0.0000	0.0000	
58	Accel. Gradient	(kPa/m)	1.098e-005	1.098e-005	1.034e-005	1.034e-005	
59	Liq. Reynolds		1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *	
60	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *	
61	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
62	Vapour Velocity	(m/s)	5.194	5.194	5.194	5.194	
63	Deposit Thickness	(mm)	---	---	---	---	
64	Deposit Volume	(m3)	---	---	---	---	
65	Length	(m)	3546	3551	3556	3561	
66	Elevation	(m)	1.000e-002	1.000e-002	1.000e-002	8.000e-003	
67	Pressure	(bar)	78.73	78.73	78.73	78.72	
68	Temperature	(C)	55.93	55.92	55.92	55.92	
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:09:19 2022		
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
14	Heat Transferred (kJ/h-m)	564.8	564.7	564.6	564.7
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.1669	0.1669	0.1669	0.1142
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	3.534e-002	3.534e-002	3.534e-002	3.856e-002
20	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-6.207e-004
21	Accel. Gradient (kPa/m)	1.034e-005	1.034e-005	1.034e-005	1.111e-005
22	Liq. Reynolds	1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *
23	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
24	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
25	Vapour Velocity (m/s)	5.194	5.194	5.194	5.194
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	3565	3569	3573	3577
29	Elevation (m)	6.000e-003	4.000e-003	2.000e-003	-1.232e-016
30	Pressure (bar)	78.72	78.72	78.72	78.72
31	Temperature (C)	55.92	55.91	55.91	55.91
32	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
33	Heat Transferred (kJ/h-m)	564.6	564.6	564.5	564.5
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1142	0.1142	0.1142	0.1142
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	3.856e-002	3.856e-002	3.856e-002	3.856e-002
39	Static Gradient (kPa/m)	-6.207e-004	-6.207e-004	-6.207e-004	-6.207e-004
40	Accel. Gradient (kPa/m)	1.111e-005	1.111e-005	1.111e-005	1.111e-005
41	Liq. Reynolds	1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *
42	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
43	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
44	Vapour Velocity (m/s)	5.194	5.194	5.194	5.194
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	3582	3587	3592	3597
48	Elevation (m)	3.600e-002	7.200e-002	0.1080	0.1440
49	Pressure (bar)	78.72	78.71	78.71	78.71
50	Temperature (C)	55.91	55.90	55.90	55.90
51	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
52	Heat Transferred (kJ/h-m)	564.3	564.3	564.2	564.1
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.1647	0.1647	0.1647	0.1647
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	3.539e-002	3.539e-002	3.539e-002	3.539e-002
58	Static Gradient (kPa/m)	1.140e-002	1.140e-002	1.140e-002	1.140e-002
59	Accel. Gradient (kPa/m)	1.370e-005	1.370e-005	1.370e-005	1.370e-005
60	Liq. Reynolds	1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *
61	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
62	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
63	Vapour Velocity (m/s)	5.195	5.195	5.195	5.195
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	3602	3607	3612	3617
67	Elevation (m)	0.1800	9.800e-002	1.600e-002	-6.600e-002
68	Pressure (bar)	78.71	78.71	78.70	78.70
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:09:19 2022		
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	55.89	55.89	55.89	55.89
14	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
15	Heat Transferred (kJ/h-m)	564.1	564.1	564.1	564.0
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.1647	0.1079	0.1079	0.1079
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	3.540e-002	3.927e-002	3.927e-002	3.927e-002
21	Static Gradient (kPa/m)	1.140e-002	-2.022e-002	-2.022e-002	-2.022e-002
22	Accel. Gradient (kPa/m)	1.370e-005	5.579e-006	5.579e-006	5.580e-006
23	Liq. Reynolds	1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *
24	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
25	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
26	Vapour Velocity (m/s)	5.195	5.195	5.195	5.195
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	3622	3627	3632	3637
30	Elevation (m)	-0.1480	-0.2300	-0.1940	-0.1580
31	Pressure (bar)	78.70	78.70	78.70	78.70
32	Temperature (C)	55.89	55.88	55.88	55.88
33	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
34	Heat Transferred (kJ/h-m)	564.0	563.9	563.8	563.7
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1079	0.1079	0.1647	0.1647
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	3.927e-002	3.927e-002	3.540e-002	3.540e-002
40	Static Gradient (kPa/m)	-2.022e-002	-2.022e-002	1.140e-002	1.140e-002
41	Accel. Gradient (kPa/m)	5.580e-006	5.580e-006	1.371e-005	1.371e-005
42	Liq. Reynolds	1.352e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *
43	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
44	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
45	Vapour Velocity (m/s)	5.195	5.195	5.195	5.195
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	3642	3647	3652	3657
49	Elevation (m)	-0.1220	-8.600e-002	-5.000e-002	-3.600e-002
50	Pressure (bar)	78.69	78.69	78.69	78.69
51	Temperature (C)	55.87	55.87	55.87	55.87
52	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
53	Heat Transferred (kJ/h-m)	563.6	563.6	563.5	563.4
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.1647	0.1647	0.1647	0.1583
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	3.540e-002	3.540e-002	3.540e-002	3.560e-002
59	Static Gradient (kPa/m)	1.140e-002	1.140e-002	1.140e-002	4.324e-003
60	Accel. Gradient (kPa/m)	1.371e-005	1.371e-005	1.371e-005	1.170e-005
61	Liq. Reynolds	1.351e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *
62	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
63	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
64	Vapour Velocity (m/s)	5.195	5.195	5.195	5.196
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	3662	3667	3672	3677
68	Elevation (m)	-2.200e-002	-8.000e-003	6.000e-003	2.000e-002
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
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10					
11	Pipe Table				
12					
13	Pressure (bar)	78.69	78.68	78.68	78.68
14	Temperature (C)	55.86	55.86	55.86	55.85
15	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
16	Heat Transferred (kJ/h-m)	563.4	563.3	563.3	563.2
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.1583	0.1583	0.1583	0.1583
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	3.560e-002	3.560e-002	3.560e-002	3.560e-002
22	Static Gradient (kPa/m)	4.324e-003	4.324e-003	4.324e-003	4.324e-003
23	Accel. Gradient (kPa/m)	1.170e-005	1.170e-005	1.170e-005	1.170e-005
24	Liq. Reynolds	1.351e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *
25	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
26	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
27	Vapour Velocity (m/s)	5.196	5.196	5.196	5.196
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	3682	3687	3692	3697
31	Elevation (m)	3.000e-002	4.000e-002	5.000e-002	6.000e-002
32	Pressure (bar)	78.68	78.68	78.67	78.67
33	Temperature (C)	55.85	55.85	55.85	55.84
34	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
35	Heat Transferred (kJ/h-m)	563.1	563.1	563.0	562.9
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.1571	0.1571	0.1571	0.1571
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	3.564e-002	3.565e-002	3.565e-002	3.565e-002
41	Static Gradient (kPa/m)	3.074e-003	3.074e-003	3.074e-003	3.074e-003
42	Accel. Gradient (kPa/m)	1.134e-005	1.135e-005	1.135e-005	1.135e-005
43	Liq. Reynolds	1.351e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *
44	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
45	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
46	Vapour Velocity (m/s)	5.196	5.196	5.196	5.196
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	3702	3707	3712	3717
50	Elevation (m)	7.000e-002	8.400e-002	9.800e-002	0.1120
51	Pressure (bar)	78.67	78.67	78.67	78.66
52	Temperature (C)	55.84	55.84	55.83	55.83
53	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
54	Heat Transferred (kJ/h-m)	562.9	562.8	562.8	562.7
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.1571	0.1583	0.1583	0.1583
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	3.565e-002	3.560e-002	3.560e-002	3.561e-002
60	Static Gradient (kPa/m)	3.074e-003	4.324e-003	4.323e-003	4.323e-003
61	Accel. Gradient (kPa/m)	1.135e-005	1.170e-005	1.170e-005	1.170e-005
62	Liq. Reynolds	1.351e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *
63	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
64	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
65	Vapour Velocity (m/s)	5.196	5.196	5.196	5.196
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	3722	3727	3732	3737
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13	Elevation (m)	0.1260	0.1400	0.1320	0.1240
14	Pressure (bar)	78.66	78.66	78.66	78.66
15	Temperature (C)	55.83	55.83	55.82	55.82
16	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
17	Heat Transferred (kJ/h-m)	562.6	562.6	562.6	562.5
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.1583	0.1583	0.1137	0.1137
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	3.561e-002	3.561e-002	3.862e-002	3.862e-002
23	Static Gradient (kPa/m)	4.323e-003	4.323e-003	-2.030e-003	-2.030e-003
24	Accel. Gradient (kPa/m)	1.170e-005	1.170e-005	1.073e-005	1.073e-005
25	Liq. Reynolds	1.351e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *
26	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
27	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
28	Vapour Velocity (m/s)	5.196	5.197	5.197	5.197
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	3742	3747	3752	3757
32	Elevation (m)	0.1160	0.1080	1.000e-001	0.1100
33	Pressure (bar)	78.65	78.65	78.65	78.65
34	Temperature (C)	55.82	55.81	55.81	55.81
35	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
36	Heat Transferred (kJ/h-m)	562.5	562.4	562.3	562.2
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.1137	0.1137	0.1137	0.1573
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	3.862e-002	3.862e-002	3.862e-002	3.564e-002
42	Static Gradient (kPa/m)	-2.030e-003	-2.030e-003	-2.030e-003	3.331e-003
43	Accel. Gradient (kPa/m)	1.073e-005	1.073e-005	1.073e-005	1.143e-005
44	Liq. Reynolds	1.351e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *
45	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
46	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
47	Vapour Velocity (m/s)	5.197	5.197	5.197	5.197
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	3761	3766	3770	3775
51	Elevation (m)	0.1200	0.1300	0.1400	0.1500
52	Pressure (bar)	78.65	78.65	78.64	78.64
53	Temperature (C)	55.81	55.80	55.80	55.80
54	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
55	Heat Transferred (kJ/h-m)	562.2	562.1	562.1	562.0
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.1573	0.1573	0.1573	0.1573
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	3.564e-002	3.564e-002	3.564e-002	3.564e-002
61	Static Gradient (kPa/m)	3.331e-003	3.331e-003	3.331e-003	3.331e-003
62	Accel. Gradient (kPa/m)	1.143e-005	1.143e-005	1.143e-005	1.143e-005
63	Liq. Reynolds	1.351e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *
64	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
65	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
66	Vapour Velocity (m/s)	5.197	5.197	5.197	5.197
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
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13	Length (m)	3780	3785	3790	3795
14	Elevation (m)	0.1340	0.1180	0.1020	8.600e-002
15	Pressure (bar)	78.64	78.64	78.64	78.64
16	Temperature (C)	55.80	55.79	55.79	55.79
17	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
18	Heat Transferred (kJ/h-m)	562.0	561.9	561.9	561.8
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.1130	0.1130	0.1130	0.1130
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	3.869e-002	3.869e-002	3.869e-002	3.870e-002
24	Static Gradient (kPa/m)	-4.047e-003	-4.047e-003	-4.047e-003	-4.047e-003
25	Accel. Gradient (kPa/m)	1.016e-005	1.016e-005	1.016e-005	1.016e-005
26	Liq. Reynolds	1.351e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *
27	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
28	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
29	Vapour Velocity (m/s)	5.197	5.197	5.197	5.197
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	3800	3805	3810	3815
33	Elevation (m)	7.000e-002	5.600e-002	4.200e-002	2.800e-002
34	Pressure (bar)	78.63	78.63	78.63	78.63
35	Temperature (C)	55.79	55.78	55.78	55.78
36	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
37	Heat Transferred (kJ/h-m)	561.8	561.7	561.6	561.6
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.1130	0.1132	0.1132	0.1132
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	3.870e-002	3.868e-002	3.868e-002	3.868e-002
43	Static Gradient (kPa/m)	-4.047e-003	-3.544e-003	-3.544e-003	-3.544e-003
44	Accel. Gradient (kPa/m)	1.016e-005	1.030e-005	1.030e-005	1.031e-005
45	Liq. Reynolds	1.351e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *
46	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
47	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
48	Vapour Velocity (m/s)	5.198	5.198	5.198	5.198
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	3820	3825	3830	3835
52	Elevation (m)	1.400e-002	-9.714e-017	-9.714e-017	-9.714e-017
53	Pressure (bar)	78.63	78.62	78.62	78.62
54	Temperature (C)	55.77	55.77	55.77	55.77
55	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
56	Heat Transferred (kJ/h-m)	561.5	561.5	561.3	561.3
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.1132	0.1132	0.1669	0.1669
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	3.868e-002	3.868e-002	3.536e-002	3.536e-002
62	Static Gradient (kPa/m)	-3.544e-003	-3.544e-003	0.0000	0.0000
63	Accel. Gradient (kPa/m)	1.031e-005	1.031e-005	1.037e-005	1.037e-005
64	Liq. Reynolds	1.351e+005 *	1.351e+005 *	1.351e+005 *	1.350e+005 *
65	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
66	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
67	Vapour Velocity (m/s)	5.198	5.198	5.198	5.198
68	Deposit Thickness (mm)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
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13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	3840	3845	3850	3855
15	Elevation (m)	-9.714e-017	-9.714e-017	-9.714e-017	6.000e-003
16	Pressure (bar)	78.62	78.62	78.62	78.61
17	Temperature (C)	55.76	55.76	55.76	55.76
18	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
19	Heat Transferred (kJ/h-m)	561.2	561.1	561.1	561.1
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.1669	0.1669	0.1669	0.1559
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	3.536e-002	3.537e-002	3.537e-002	3.571e-002
25	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	1.835e-003
26	Accel. Gradient (kPa/m)	1.037e-005	1.037e-005	1.037e-005	1.101e-005
27	Liq. Reynolds	1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *
28	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
29	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
30	Vapour Velocity (m/s)	5.198	5.198	5.198	5.198
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	3860	3865	3870	3875
34	Elevation (m)	1.200e-002	1.800e-002	2.400e-002	3.000e-002
35	Pressure (bar)	78.61	78.61	78.61	78.61
36	Temperature (C)	55.75	55.75	55.75	55.74
37	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
38	Heat Transferred (kJ/h-m)	561.0	560.9	560.9	560.8
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.1559	0.1559	0.1559	0.1559
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	3.571e-002	3.571e-002	3.571e-002	3.571e-002
44	Static Gradient (kPa/m)	1.835e-003	1.835e-003	1.835e-003	1.835e-003
45	Accel. Gradient (kPa/m)	1.101e-005	1.101e-005	1.102e-005	1.102e-005
46	Liq. Reynolds	1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *
47	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
48	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
49	Vapour Velocity (m/s)	5.198	5.198	5.198	5.198
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	3880	3885	3890	3895
53	Elevation (m)	4.200e-002	5.400e-002	6.600e-002	7.800e-002
54	Pressure (bar)	78.60	78.60	78.60	78.60
55	Temperature (C)	55.74	55.74	55.74	55.73
56	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
57	Heat Transferred (kJ/h-m)	560.7	560.7	560.6	560.6
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.1577	0.1577	0.1577	0.1577
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	3.564e-002	3.564e-002	3.564e-002	3.564e-002
63	Static Gradient (kPa/m)	3.696e-003	3.696e-003	3.696e-003	3.696e-003
64	Accel. Gradient (kPa/m)	1.154e-005	1.154e-005	1.154e-005	1.154e-005
65	Liq. Reynolds	1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *
66	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
67	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
68	Vapour Velocity (m/s)	5.199	5.199	5.199	5.199
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11	Pipe Table				
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13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	3900	3905	3910	3915
16	Elevation (m)	9.000e-002	0.1160	0.1420	0.1680
17	Pressure (bar)	78.60	78.59	78.59	78.59
18	Temperature (C)	55.73	55.73	55.72	55.72
19	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
20	Heat Transferred (kJ/h-m)	560.5	560.4	560.4	560.3
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.1577	0.1617	0.1617	0.1617
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	3.564e-002	3.551e-002	3.551e-002	3.551e-002
26	Static Gradient (kPa/m)	3.696e-003	8.139e-003	8.139e-003	8.139e-003
27	Accel. Gradient (kPa/m)	1.155e-005	1.281e-005	1.281e-005	1.281e-005
28	Liq. Reynolds	1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *
29	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
30	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
31	Vapour Velocity (m/s)	5.199	5.199	5.199	5.199
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	3920	3925	3927	3929
35	Elevation (m)	0.1940	0.2200	0.1020	-1.600e-002
36	Pressure (bar)	78.59	78.59	78.59	78.59
37	Temperature (C)	55.72	55.72	55.72	55.71
38	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
39	Heat Transferred (kJ/h-m)	560.2	560.2	560.3	560.3
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.1617	0.1617	9.223e-002	9.224e-002
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	3.551e-002	3.551e-002	4.163e-002	4.163e-002
45	Static Gradient (kPa/m)	8.139e-003	8.139e-003	-6.389e-002	-6.389e-002
46	Accel. Gradient (kPa/m)	1.281e-005	1.281e-005	-6.534e-006	-6.534e-006
47	Liq. Reynolds	1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *
48	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
49	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
50	Vapour Velocity (m/s)	5.199	5.199	5.199	5.199
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	3931	3933	3935	3947
54	Elevation (m)	-0.1340	-0.2520	-0.3700	-1.016
55	Pressure (bar)	78.59	78.59	78.59	78.59
56	Temperature (C)	55.71	55.71	55.71	55.71
57	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
58	Heat Transferred (kJ/h-m)	560.3	560.2	560.2	560.2
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	9.224e-002	9.224e-002	9.224e-002	9.157e-002
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	4.163e-002	4.163e-002	4.163e-002	4.175e-002
64	Static Gradient (kPa/m)	-6.390e-002	-6.390e-002	-6.390e-002	-6.562e-002
65	Accel. Gradient (kPa/m)	-6.534e-006	-6.534e-006	-6.535e-006	-7.005e-006
66	Liq. Reynolds	1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *
67	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
68	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2			Pipe Segment: BBOU-ELEP Pipeline (continued)				
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:09:19 2022		
6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	5.199	5.199	5.199	5.199	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	3958	3969	3980	3991	
17	Elevation	(m)	-1.662	-2.308	-2.954	-3.600	
18	Pressure	(bar)	78.59	78.60	78.60	78.60	
19	Temperature	(C)	55.71	55.71	55.70	55.70	
20	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
21	Heat Transferred	(kJ/h-m)	560.1	560.1	560.0	560.0	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		9.157e-002	9.157e-002	9.158e-002	9.158e-002	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	4.175e-002	4.175e-002	4.175e-002	4.175e-002	
27	Static Gradient	(kPa/m)	-6.563e-002	-6.563e-002	-6.563e-002	-6.563e-002	
28	Accel. Gradient	(kPa/m)	-7.006e-006	-7.007e-006	-7.007e-006	-7.008e-006	
29	Liq. Reynolds		1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *	
30	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *	
31	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
32	Vapour Velocity	(m/s)	5.199	5.198	5.198	5.198	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	4002	4013	4025	4036	
36	Elevation	(m)	-3.940	-4.280	-4.620	-4.960	
37	Pressure	(bar)	78.60	78.60	78.60	78.60	
38	Temperature	(C)	55.70	55.69	55.69	55.68	
39	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
40	Heat Transferred	(kJ/h-m)	559.9	559.8	559.7	559.6	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.1023	0.1023	0.1023	0.1023	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	4.002e-002	4.002e-002	4.002e-002	4.002e-002	
46	Static Gradient	(kPa/m)	-3.656e-002	-3.657e-002	-3.657e-002	-3.657e-002	
47	Accel. Gradient	(kPa/m)	1.013e-006	1.013e-006	1.013e-006	1.012e-006	
48	Liq. Reynolds		1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *	
49	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *	
50	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
51	Vapour Velocity	(m/s)	5.198	5.198	5.198	5.197	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	4047	4058	4069	4080	
55	Elevation	(m)	-5.300	-5.420	-5.540	-5.660	
56	Pressure	(bar)	78.60	78.60	78.59	78.59	
57	Temperature	(C)	55.68	55.68	55.67	55.66	
58	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
59	Heat Transferred	(kJ/h-m)	559.5	559.4	559.3	559.1	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.1023	0.1101	0.1101	0.1101	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	4.002e-002	3.903e-002	3.903e-002	3.903e-002	
65	Static Gradient	(kPa/m)	-3.657e-002	-1.342e-002	-1.342e-002	-1.342e-002	
66	Accel. Gradient	(kPa/m)	1.012e-006	7.513e-006	7.514e-006	7.514e-006	
67	Liq. Reynolds		1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *	
68	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *	
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1			Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:09:19 2022			
6						
7	Pipe Segment: BBOU-ELEP Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084
14	Vapour Velocity	(m/s)	5.197	5.197	5.198	5.198
15	Deposit Thickness	(mm)	---	---	---	---
16	Deposit Volume	(m3)	---	---	---	---
17	Length	(m)	4091	4103	4114	4125
18	Elevation	(m)	-5.780	-5.900	-5.780	-5.660
19	Pressure	(bar)	78.59	78.58	78.58	78.57
20	Temperature	(C)	55.66	55.65	55.65	55.64
21	Vapour Fraction		0.9341	0.9341	0.9341	0.9341
22	Heat Transferred	(kJ/h-m)	559.0	558.9	558.7	558.5
23	Flow Regime		Transition	Transition	Transition	Transition
24	Water Flow Regime					
25	Liquid HoldUp		0.1101	0.1101	0.1698	0.1698
26	Water HoldUp		---	---	---	---
27	Friction Gradient	(kPa/m)	3.903e-002	3.903e-002	3.531e-002	3.531e-002
28	Static Gradient	(kPa/m)	-1.342e-002	-1.342e-002	1.740e-002	1.740e-002
29	Accel. Gradient	(kPa/m)	7.515e-006	7.515e-006	1.547e-005	1.547e-005
30	Liq. Reynolds		1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *
31	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
32	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084
33	Vapour Velocity	(m/s)	5.198	5.198	5.198	5.198
34	Deposit Thickness	(mm)	---	---	---	---
35	Deposit Volume	(m3)	---	---	---	---
36	Length	(m)	4136	4147	4158	4169
37	Elevation	(m)	-5.540	-5.420	-5.300	-5.500
38	Pressure	(bar)	78.57	78.56	78.56	78.55
39	Temperature	(C)	55.63	55.63	55.62	55.61
40	Vapour Fraction		0.9341	0.9341	0.9341	0.9341
41	Heat Transferred	(kJ/h-m)	558.3	558.2	558.0	558.0
42	Flow Regime		Transition	Transition	Transition	Transition
43	Water Flow Regime					
44	Liquid HoldUp		0.1698	0.1698	0.1698	0.1072
45	Water HoldUp		---	---	---	---
46	Friction Gradient	(kPa/m)	3.532e-002	3.532e-002	3.532e-002	3.938e-002
47	Static Gradient	(kPa/m)	1.740e-002	1.740e-002	1.740e-002	-2.205e-002
48	Accel. Gradient	(kPa/m)	1.547e-005	1.548e-005	1.548e-005	5.088e-006
49	Liq. Reynolds		1.350e+005 *	1.349e+005 *	1.349e+005 *	1.349e+005 *
50	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
51	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084
52	Vapour Velocity	(m/s)	5.199	5.199	5.199	5.199
53	Deposit Thickness	(mm)	---	---	---	---
54	Deposit Volume	(m3)	---	---	---	---
55	Length	(m)	4181	4192	4203	4214
56	Elevation	(m)	-5.700	-5.900	-6.100	-6.300
57	Pressure	(bar)	78.55	78.55	78.55	78.55
58	Temperature	(C)	55.61	55.60	55.60	55.59
59	Vapour Fraction		0.9341	0.9341	0.9341	0.9341
60	Heat Transferred	(kJ/h-m)	557.9	557.8	557.7	557.6
61	Flow Regime		Transition	Transition	Transition	Transition
62	Water Flow Regime					
63	Liquid HoldUp		0.1072	0.1072	0.1072	0.1072
64	Water HoldUp		---	---	---	---
65	Friction Gradient	(kPa/m)	3.938e-002	3.938e-002	3.938e-002	3.938e-002
66	Static Gradient	(kPa/m)	-2.205e-002	-2.205e-002	-2.205e-002	-2.205e-002
67	Accel. Gradient	(kPa/m)	5.088e-006	5.088e-006	5.089e-006	5.089e-006
68	Liq. Reynolds		1.349e+005 *	1.349e+005 *	1.349e+005 *	1.349e+005 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:09:19 2022				
5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.217e+007 *	1.217e+007 *	
14	Liquid Velocity (m/s)		0.2084	0.2084	0.2084	0.2084	
15	Vapour Velocity (m/s)		5.199	5.199	5.199	5.199	
16	Deposit Thickness (mm)		---	---	---	---	
17	Deposit Volume (m3)		---	---	---	---	
18	Length (m)		4225	4236	4247	4259	
19	Elevation (m)		-7.360	-8.420	-9.480	-10.54	
20	Pressure (bar)		78.55	78.56	78.56	78.57	
21	Temperature (C)		55.59	55.59	55.59	55.59	
22	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
23	Heat Transferred (kJ/h-m)		557.7	557.7	557.7	557.7	
24	Flow Regime		Transition	Transition	Transition	Transition	
25	Water Flow Regime						
26	Liquid HoldUp		7.732e-002	7.732e-002	7.732e-002	7.733e-002	
27	Water HoldUp		---	---	---	---	
28	Friction Gradient (kPa/m)		4.504e-002	4.503e-002	4.503e-002	4.503e-002	
29	Static Gradient (kPa/m)		-9.931e-002	-9.932e-002	-9.932e-002	-9.933e-002	
30	Accel. Gradient (kPa/m)		-1.594e-005	-1.594e-005	-1.594e-005	-1.594e-005	
31	Liq. Reynolds		1.349e+005 *	1.349e+005 *	1.349e+005 *	1.349e+005 *	
32	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *	
33	Liquid Velocity (m/s)		0.2084	0.2084	0.2084	0.2084	
34	Vapour Velocity (m/s)		5.199	5.198	5.198	5.197	
35	Deposit Thickness (mm)		---	---	---	---	
36	Deposit Volume (m3)		---	---	---	---	
37	Length (m)		4270	4281	4292	4303	
38	Elevation (m)		-11.60	-9.192	-6.784	-4.376	
39	Pressure (bar)		78.58	78.49	78.41	78.33	
40	Temperature (C)		55.59	55.56	55.53	55.50	
41	Vapour Fraction		0.9341	0.9341	0.9342	0.9342	
42	Heat Transferred (kJ/h-m)		557.7	557.4	556.7	556.0	
43	Flow Regime		Transition	Transition	Transition	Transition	
44	Water Flow Regime						
45	Liquid HoldUp		7.733e-002	0.4477	0.4474	0.4472	
46	Water HoldUp		---	---	---	---	
47	Friction Gradient (kPa/m)		4.502e-002	3.057e-002	3.060e-002	3.063e-002	
48	Static Gradient (kPa/m)		-9.933e-002	0.7201	0.7197	0.7193	
49	Accel. Gradient (kPa/m)		-1.594e-005	2.207e-004	2.211e-004	2.214e-004	
50	Liq. Reynolds		1.349e+005 *	1.349e+005 *	1.348e+005 *	1.347e+005 *	
51	Vap. Reynolds		1.216e+007 *	1.217e+007 *	1.217e+007 *	1.217e+007 *	
52	Liquid Velocity (m/s)		0.2084	0.2084	0.2084	0.2084	
53	Vapour Velocity (m/s)		5.197	5.202	5.208	5.213	
54	Deposit Thickness (mm)		---	---	---	---	
55	Deposit Volume (m3)		---	---	---	---	
56	Length (m)		4314	4325	4328	4331	
57	Elevation (m)		-1.968	0.4400	0.5820	0.7240	
58	Pressure (bar)		78.24	78.16	78.15	78.15	
59	Temperature (C)		55.46	55.43	55.43	55.42	
60	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
61	Heat Transferred (kJ/h-m)		555.2	554.5	553.6	553.5	
62	Flow Regime		Transition	Transition	Transition	Transition	
63	Water Flow Regime						
64	Liquid HoldUp		0.4469	0.4466	0.2221	0.2221	
65	Water HoldUp		---	---	---	---	
66	Friction Gradient (kPa/m)		3.066e-002	3.069e-002	3.254e-002	3.254e-002	
67	Static Gradient (kPa/m)		0.7189	0.7185	9.168e-002	9.168e-002	
68	Accel. Gradient (kPa/m)		2.218e-004	2.222e-004	3.684e-005	3.684e-005	
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1			Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:09:19 2022		
6					
7	Pipe Segment: BBOU-ELEP Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	1.347e+005 *	1.346e+005 *	1.346e+005 *	1.346e+005 *
14	Vap. Reynolds	1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
15	Liquid Velocity (m/s)	0.2084	0.2083	0.2083	0.2083
16	Vapour Velocity (m/s)	5.218	5.224	5.224	5.224
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	4334	4337	4340	4343
20	Elevation (m)	0.8660	1.008	1.150	1.088
21	Pressure (bar)	78.15	78.14	78.14	78.14
22	Temperature (C)	55.42	55.42	55.42	55.41
23	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
24	Heat Transferred (kJ/h-m)	553.5	553.4	553.4	553.7
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2221	0.2221	0.2221	0.1058
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	3.254e-002	3.254e-002	3.254e-002	3.971e-002
30	Static Gradient (kPa/m)	9.167e-002	9.167e-002	9.167e-002	-2.517e-002
31	Accel. Gradient (kPa/m)	3.684e-005	3.685e-005	3.685e-005	4.314e-006
32	Liq. Reynolds	1.346e+005 *	1.346e+005 *	1.346e+005 *	1.346e+005 *
33	Vap. Reynolds	1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
34	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
35	Vapour Velocity (m/s)	5.225	5.225	5.225	5.225
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	4346	4349	4352	4355
39	Elevation (m)	1.026	0.9640	0.9020	0.8400
40	Pressure (bar)	78.14	78.14	78.14	78.14
41	Temperature (C)	55.41	55.41	55.41	55.41
42	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
43	Heat Transferred (kJ/h-m)	553.7	553.6	553.6	553.6
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.1058	0.1058	0.1058	0.1058
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	3.971e-002	3.971e-002	3.971e-002	3.971e-002
49	Static Gradient (kPa/m)	-2.517e-002	-2.517e-002	-2.517e-002	-2.517e-002
50	Accel. Gradient (kPa/m)	4.314e-006	4.314e-006	4.314e-006	4.314e-006
51	Liq. Reynolds	1.346e+005 *	1.346e+005 *	1.346e+005 *	1.346e+005 *
52	Vap. Reynolds	1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
53	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
54	Vapour Velocity (m/s)	5.225	5.225	5.225	5.225
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	4357	4359	4361	4363
58	Elevation (m)	0.7740	0.7080	0.6420	0.5760
59	Pressure (bar)	78.14	78.14	78.14	78.14
60	Temperature (C)	55.41	55.41	55.41	55.41
61	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
62	Heat Transferred (kJ/h-m)	553.6	553.6	553.6	553.5
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.1010	0.1010	0.1010	0.1010
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	4.036e-002	4.036e-002	4.036e-002	4.036e-002
68	Static Gradient (kPa/m)	-3.920e-002	-3.920e-002	-3.920e-002	-3.920e-002
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3						
4			Unit Set: Mixed			
5	Date/Time: Mon Jun 13 15:09:19 2022					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	PROFILES					
10	Pipe Table					
11						
12						
13	Accel. Gradient (kPa/m)	3.444e-007	3.443e-007	3.442e-007	3.442e-007	
14	Liq. Reynolds	1.346e+005 *	1.346e+005 *	1.346e+005 *	1.346e+005 *	
15	Vap. Reynolds	1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *	
16	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083	
17	Vapour Velocity (m/s)	5.225	5.225	5.225	5.225	
18	Deposit Thickness (mm)	---	---	---	---	
19	Deposit Volume (m3)	---	---	---	---	
20	Length (m)	4365	4367	4369	4371	
21	Elevation (m)	0.5100	0.5320	0.5540	0.5760	
22	Pressure (bar)	78.14	78.14	78.14	78.13	
23	Temperature (C)	55.41	55.40	55.40	55.40	
24	Vapour Fraction	0.9342	0.9342	0.9342	0.9342	
25	Heat Transferred (kJ/h-m)	553.5	553.4	553.4	553.3	
26	Flow Regime	Transition	Transition	Transition	Transition	
27	Water Flow Regime					
28	Liquid HoldUp	0.1010	0.1696	0.1696	0.1696	
29	Water HoldUp	---	---	---	---	
30	Friction Gradient (kPa/m)	4.036e-002	3.548e-002	3.548e-002	3.548e-002	
31	Static Gradient (kPa/m)	-3.920e-002	1.774e-002	1.773e-002	1.773e-002	
32	Accel. Gradient (kPa/m)	3.441e-007	1.579e-005	1.579e-005	1.579e-005	
33	Liq. Reynolds	1.346e+005 *	1.346e+005 *	1.346e+005 *	1.346e+005 *	
34	Vap. Reynolds	1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *	
35	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083	
36	Vapour Velocity (m/s)	5.225	5.225	5.225	5.225	
37	Deposit Thickness (mm)	---	---	---	---	
38	Deposit Volume (m3)	---	---	---	---	
39	Length (m)	4373	4375	4380	4385	
40	Elevation (m)	0.5980	0.6200	0.6140	0.6080	
41	Pressure (bar)	78.13	78.13	78.13	78.13	
42	Temperature (C)	55.40	55.40	55.40	55.39	
43	Vapour Fraction	0.9342	0.9342	0.9342	0.9342	
44	Heat Transferred (kJ/h-m)	553.3	553.3	553.3	553.2	
45	Flow Regime	Transition	Transition	Transition	Transition	
46	Water Flow Regime					
47	Liquid HoldUp	0.1696	0.1696	0.1134	0.1134	
48	Water HoldUp	---	---	---	---	
49	Friction Gradient (kPa/m)	3.549e-002	3.549e-002	3.882e-002	3.882e-002	
50	Static Gradient (kPa/m)	1.773e-002	1.773e-002	-1.518e-003	-1.518e-003	
51	Accel. Gradient (kPa/m)	1.579e-005	1.579e-005	1.107e-005	1.107e-005	
52	Liq. Reynolds	1.346e+005 *	1.346e+005 *	1.345e+005 *	1.345e+005 *	
53	Vap. Reynolds	1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *	
54	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083	
55	Vapour Velocity (m/s)	5.225	5.225	5.225	5.225	
56	Deposit Thickness (mm)	---	---	---	---	
57	Deposit Volume (m3)	---	---	---	---	
58	Length (m)	4390	4395	4400	4405	
59	Elevation (m)	0.6020	0.5960	0.5900	0.5160	
60	Pressure (bar)	78.13	78.12	78.12	78.12	
61	Temperature (C)	55.39	55.39	55.39	55.38	
62	Vapour Fraction	0.9342	0.9342	0.9342	0.9342	
63	Heat Transferred (kJ/h-m)	553.2	553.1	553.1	553.0	
64	Flow Regime	Transition	Transition	Transition	Transition	
65	Water Flow Regime					
66	Liquid HoldUp	0.1134	0.1134	0.1134	0.1081	
67	Water HoldUp	---	---	---	---	
68	Friction Gradient (kPa/m)	3.882e-002	3.882e-002	3.882e-002	3.943e-002	
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1				Case Name: Simulation - 331MMSCFD_80barg.hsc		
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
Date/Time: Mon Jun 13 15:09:19 2022						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	-1.518e-003	-1.518e-003	-1.518e-003	-1.823e-002
14	Accel. Gradient	(kPa/m)	1.107e-005	1.107e-005	1.107e-005	6.290e-006
15	Liq. Reynolds		1.345e+005 *	1.345e+005 *	1.345e+005 *	1.345e+005 *
16	Vap. Reynolds		1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
17	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083
18	Vapour Velocity	(m/s)	5.225	5.225	5.225	5.226
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	4410	4415	4420	4425
22	Elevation	(m)	0.4420	0.3680	0.2940	0.2200
23	Pressure	(bar)	78.12	78.12	78.12	78.12
24	Temperature	(C)	55.38	55.38	55.38	55.37
25	Vapour Fraction		0.9342	0.9342	0.9342	0.9342
26	Heat Transferred	(kJ/h-m)	553.0	552.9	552.9	552.8
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.1081	0.1081	0.1081	0.1081
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	3.943e-002	3.943e-002	3.943e-002	3.943e-002
32	Static Gradient	(kPa/m)	-1.823e-002	-1.823e-002	-1.823e-002	-1.823e-002
33	Accel. Gradient	(kPa/m)	6.290e-006	6.290e-006	6.290e-006	6.290e-006
34	Liq. Reynolds		1.345e+005 *	1.345e+005 *	1.345e+005 *	1.345e+005 *
35	Vap. Reynolds		1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
36	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083
37	Vapour Velocity	(m/s)	5.226	5.226	5.226	5.226
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	4431	4436	4441	4446
41	Elevation	(m)	0.2080	0.1960	0.1840	0.1720
42	Pressure	(bar)	78.12	78.11	78.11	78.11
43	Temperature	(C)	55.37	55.37	55.37	55.36
44	Vapour Fraction		0.9342	0.9342	0.9342	0.9342
45	Heat Transferred	(kJ/h-m)	552.8	552.7	552.6	552.6
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.1130	0.1130	0.1130	0.1130
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	3.887e-002	3.887e-002	3.887e-002	3.887e-002
51	Static Gradient	(kPa/m)	-2.998e-003	-2.998e-003	-2.998e-003	-2.998e-003
52	Accel. Gradient	(kPa/m)	1.065e-005	1.065e-005	1.065e-005	1.065e-005
53	Liq. Reynolds		1.345e+005 *	1.345e+005 *	1.345e+005 *	1.345e+005 *
54	Vap. Reynolds		1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
55	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083
56	Vapour Velocity	(m/s)	5.226	5.226	5.226	5.226
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	4451	4456	4461	4466
60	Elevation	(m)	0.1600	0.1620	0.1640	0.1660
61	Pressure	(bar)	78.11	78.11	78.10	78.10
62	Temperature	(C)	55.36	55.36	55.36	55.35
63	Vapour Fraction		0.9342	0.9342	0.9342	0.9342
64	Heat Transferred	(kJ/h-m)	552.5	552.4	552.4	552.3
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.1130	0.1542	0.1542	0.1542
68	Water HoldUp		---	---	---	---
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1				Case Name: Simulation - 331MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:09:19 2022		
6						
7	Pipe Segment: BBOU-ELEP Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	3.887e-002	3.594e-002	3.594e-002	3.594e-002
14	Static Gradient	(kPa/m)	-2.998e-003	6.068e-004	6.068e-004	6.068e-004
15	Accel. Gradient	(kPa/m)	1.065e-005	1.085e-005	1.085e-005	1.085e-005
16	Liq. Reynolds		1.345e+005 *	1.345e+005 *	1.345e+005 *	1.345e+005 *
17	Vap. Reynolds		1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
18	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083
19	Vapour Velocity	(m/s)	5.226	5.226	5.226	5.226
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	4471	4476	4481	4486
23	Elevation	(m)	0.1680	0.1700	0.1840	0.1980
24	Pressure	(bar)	78.10	78.10	78.10	78.10
25	Temperature	(C)	55.35	55.35	55.34	55.34
26	Vapour Fraction		0.9342	0.9342	0.9342	0.9342
27	Heat Transferred	(kJ/h-m)	552.2	552.2	552.1	552.0
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.1542	0.1542	0.1577	0.1577
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	3.594e-002	3.594e-002	3.581e-002	3.581e-002
33	Static Gradient	(kPa/m)	6.068e-004	6.068e-004	4.308e-003	4.308e-003
34	Accel. Gradient	(kPa/m)	1.085e-005	1.085e-005	1.191e-005	1.191e-005
35	Liq. Reynolds		1.345e+005 *	1.345e+005 *	1.345e+005 *	1.345e+005 *
36	Vap. Reynolds		1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
37	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083
38	Vapour Velocity	(m/s)	5.226	5.226	5.226	5.226
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	4491	4496	4501	4506
42	Elevation	(m)	0.2120	0.2260	0.2400	0.2600
43	Pressure	(bar)	78.09	78.09	78.09	78.09
44	Temperature	(C)	55.34	55.34	55.33	55.33
45	Vapour Fraction		0.9342	0.9342	0.9342	0.9342
46	Heat Transferred	(kJ/h-m)	552.0	551.9	551.9	551.8
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.1577	0.1577	0.1577	0.1594
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	3.581e-002	3.581e-002	3.581e-002	3.575e-002
52	Static Gradient	(kPa/m)	4.308e-003	4.308e-003	4.308e-003	6.197e-003
53	Accel. Gradient	(kPa/m)	1.191e-005	1.191e-005	1.191e-005	1.246e-005
54	Liq. Reynolds		1.345e+005 *	1.345e+005 *	1.345e+005 *	1.345e+005 *
55	Vap. Reynolds		1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
56	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083
57	Vapour Velocity	(m/s)	5.226	5.227	5.227	5.227
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	4511	4516	4521	4526
61	Elevation	(m)	0.2800	0.3000	0.3200	0.3400
62	Pressure	(bar)	78.08	78.08	78.08	78.08
63	Temperature	(C)	55.33	55.33	55.32	55.32
64	Vapour Fraction		0.9342	0.9342	0.9342	0.9342
65	Heat Transferred	(kJ/h-m)	551.7	551.7	551.6	551.5
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.1594	0.1594	0.1594	0.1594
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:09:19 2022
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	3.575e-002	3.575e-002	3.575e-002	3.575e-002
15	Static Gradient (kPa/m)	6.197e-003	6.197e-003	6.197e-003	6.197e-003
16	Accel. Gradient (kPa/m)	1.246e-005	1.246e-005	1.246e-005	1.246e-005
17	Liq. Reynolds	1.345e+005 *	1.345e+005 *	1.345e+005 *	1.345e+005 *
18	Vap. Reynolds	1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
19	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
20	Vapour Velocity (m/s)	5.227	5.227	5.227	5.227
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	4531	4536	4541	4546
24	Elevation (m)	0.3680	0.3960	0.4240	0.4520
25	Pressure (bar)	78.08	78.07	78.07	78.07
26	Temperature (C)	55.32	55.31	55.31	55.31
27	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
28	Heat Transferred (kJ/h-m)	551.5	551.4	551.3	551.3
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.1617	0.1617	0.1617	0.1617
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	3.567e-002	3.568e-002	3.568e-002	3.568e-002
34	Static Gradient (kPa/m)	8.756e-003	8.756e-003	8.755e-003	8.755e-003
35	Accel. Gradient (kPa/m)	1.320e-005	1.320e-005	1.320e-005	1.320e-005
36	Liq. Reynolds	1.345e+005 *	1.345e+005 *	1.345e+005 *	1.345e+005 *
37	Vap. Reynolds	1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
38	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
39	Vapour Velocity (m/s)	5.227	5.227	5.227	5.227
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	4551	4556	4561	4566
43	Elevation (m)	0.4800	0.4980	0.5160	0.5340
44	Pressure (bar)	78.07	78.07	78.06	78.06
45	Temperature (C)	55.30	55.30	55.30	55.30
46	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
47	Heat Transferred (kJ/h-m)	551.2	551.2	551.1	551.0
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.1617	0.1588	0.1588	0.1588
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	3.568e-002	3.577e-002	3.577e-002	3.577e-002
53	Static Gradient (kPa/m)	8.755e-003	5.564e-003	5.564e-003	5.564e-003
54	Accel. Gradient (kPa/m)	1.320e-005	1.228e-005	1.228e-005	1.228e-005
55	Liq. Reynolds	1.345e+005 *	1.345e+005 *	1.345e+005 *	1.345e+005 *
56	Vap. Reynolds	1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
57	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
58	Vapour Velocity (m/s)	5.228	5.228	5.228	5.228
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	4571	4576	4581	4586
62	Elevation (m)	0.5520	0.5700	0.5820	0.5940
63	Pressure (bar)	78.06	78.06	78.06	78.05
64	Temperature (C)	55.29	55.29	55.29	55.29
65	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
66	Heat Transferred (kJ/h-m)	551.0	550.9	550.9	550.8
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:09:19 2022
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.1588	0.1588	0.1571	0.1571
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	3.577e-002	3.577e-002	3.584e-002	3.584e-002
16	Static Gradient (kPa/m)	5.564e-003	5.564e-003	3.683e-003	3.683e-003
17	Accel. Gradient (kPa/m)	1.228e-005	1.228e-005	1.174e-005	1.175e-005
18	Liq. Reynolds	1.344e+005 *	1.344e+005 *	1.344e+005 *	1.344e+005 *
19	Vap. Reynolds	1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
20	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
21	Vapour Velocity (m/s)	5.228	5.228	5.228	5.228
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	4591	4596	4601	4606
25	Elevation (m)	0.6060	0.6180	0.6300	0.6380
26	Pressure (bar)	78.05	78.05	78.05	78.05
27	Temperature (C)	55.28	55.28	55.28	55.27
28	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
29	Heat Transferred (kJ/h-m)	550.7	550.7	550.6	550.6
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1571	0.1571	0.1571	0.1559
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	3.584e-002	3.584e-002	3.584e-002	3.589e-002
35	Static Gradient (kPa/m)	3.683e-003	3.683e-003	3.683e-003	2.444e-003
36	Accel. Gradient (kPa/m)	1.175e-005	1.175e-005	1.175e-005	1.139e-005
37	Liq. Reynolds	1.344e+005 *	1.344e+005 *	1.344e+005 *	1.344e+005 *
38	Vap. Reynolds	1.218e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
39	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
40	Vapour Velocity (m/s)	5.228	5.228	5.228	5.228
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	4611	4616	4621	4626
44	Elevation (m)	0.6460	0.6540	0.6620	0.6700
45	Pressure (bar)	78.04	78.04	78.04	78.04
46	Temperature (C)	55.27	55.27	55.27	55.26
47	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
48	Heat Transferred (kJ/h-m)	550.5	550.4	550.4	550.3
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1559	0.1559	0.1559	0.1559
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	3.589e-002	3.589e-002	3.589e-002	3.589e-002
54	Static Gradient (kPa/m)	2.444e-003	2.444e-003	2.444e-003	2.444e-003
55	Accel. Gradient (kPa/m)	1.139e-005	1.139e-005	1.139e-005	1.139e-005
56	Liq. Reynolds	1.344e+005 *	1.344e+005 *	1.344e+005 *	1.344e+005 *
57	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
58	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
59	Vapour Velocity (m/s)	5.228	5.229	5.229	5.229
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	4631	4636	4641	4646
63	Elevation (m)	0.6700	0.6700	0.6700	0.6700
64	Pressure (bar)	78.04	78.03	78.03	78.03
65	Temperature (C)	55.26	55.26	55.25	55.25
66	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
67	Heat Transferred (kJ/h-m)	550.2	550.2	550.1	550.1
68	Flow Regime	Transition	Transition	Transition	Transition
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:09:19 2022		
4					
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1662	0.1662	0.1662	0.1662
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	3.557e-002	3.557e-002	3.557e-002	3.557e-002
17	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
18	Accel. Gradient (kPa/m)	1.057e-005	1.057e-005	1.057e-005	1.058e-005
19	Liq. Reynolds	1.344e+005 *	1.344e+005 *	1.344e+005 *	1.344e+005 *
20	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
21	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
22	Vapour Velocity (m/s)	5.229	5.229	5.229	5.229
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	4651	4656	4661	4666
26	Elevation (m)	0.6700	0.6760	0.6820	0.6880
27	Pressure (bar)	78.03	78.03	78.02	78.02
28	Temperature (C)	55.25	55.25	55.24	55.24
29	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
30	Heat Transferred (kJ/h-m)	550.0	550.0	549.9	549.8
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1662	0.1553	0.1553	0.1553
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	3.557e-002	3.591e-002	3.591e-002	3.591e-002
36	Static Gradient (kPa/m)	0.0000	1.828e-003	1.828e-003	1.828e-003
37	Accel. Gradient (kPa/m)	1.058e-005	1.122e-005	1.122e-005	1.122e-005
38	Liq. Reynolds	1.344e+005 *	1.344e+005 *	1.344e+005 *	1.344e+005 *
39	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
40	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
41	Vapour Velocity (m/s)	5.229	5.229	5.229	5.229
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	4671	4676	4681	4686
45	Elevation (m)	0.6940	0.7000	0.7000	0.7000
46	Pressure (bar)	78.02	78.02	78.02	78.02
47	Temperature (C)	55.24	55.24	55.23	55.23
48	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
49	Heat Transferred (kJ/h-m)	549.8	549.7	549.6	549.6
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.1553	0.1553	0.1662	0.1662
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	3.591e-002	3.592e-002	3.557e-002	3.557e-002
55	Static Gradient (kPa/m)	1.828e-003	1.828e-003	0.0000	0.0000
56	Accel. Gradient (kPa/m)	1.122e-005	1.122e-005	1.058e-005	1.058e-005
57	Liq. Reynolds	1.344e+005 *	1.344e+005 *	1.344e+005 *	1.344e+005 *
58	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
59	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
60	Vapour Velocity (m/s)	5.229	5.229	5.229	5.230
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	4691	4696	4701	4706
64	Elevation (m)	0.7000	0.7000	0.7000	0.6640
65	Pressure (bar)	78.01	78.01	78.01	78.01
66	Temperature (C)	55.23	55.22	55.22	55.22
67	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
68	Heat Transferred (kJ/h-m)	549.5	549.5	549.4	549.4
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
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5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.1662	0.1662	0.1662	0.1110
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	3.557e-002	3.557e-002	3.557e-002	3.911e-002
18	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-8.998e-003
19	Accel. Gradient (kPa/m)	1.058e-005	1.058e-005	1.058e-005	8.957e-006
20	Liq. Reynolds	1.344e+005 *	1.344e+005 *	1.344e+005 *	1.344e+005 *
21	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
22	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
23	Vapour Velocity (m/s)	5.230	5.230	5.230	5.230
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	4711	4716	4721	4726
27	Elevation (m)	0.6280	0.5920	0.5560	0.5200
28	Pressure (bar)	78.01	78.01	78.00	78.00
29	Temperature (C)	55.22	55.21	55.21	55.21
30	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
31	Heat Transferred (kJ/h-m)	549.4	549.3	549.3	549.2
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1110	0.1110	0.1110	0.1110
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	3.911e-002	3.911e-002	3.911e-002	3.911e-002
37	Static Gradient (kPa/m)	-8.998e-003	-8.998e-003	-8.998e-003	-8.998e-003
38	Accel. Gradient (kPa/m)	8.957e-006	8.957e-006	8.958e-006	8.958e-006
39	Liq. Reynolds	1.344e+005 *	1.344e+005 *	1.344e+005 *	1.344e+005 *
40	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
41	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
42	Vapour Velocity (m/s)	5.230	5.230	5.230	5.230
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	4731	4736	4741	4746
46	Elevation (m)	0.5200	0.5200	0.5200	0.5200
47	Pressure (bar)	78.00	78.00	78.00	78.00
48	Temperature (C)	55.21	55.20	55.20	55.20
49	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
50	Heat Transferred (kJ/h-m)	549.1	549.0	549.0	548.9
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1662	0.1662	0.1662	0.1662
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	3.558e-002	3.558e-002	3.558e-002	3.558e-002
56	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
57	Accel. Gradient (kPa/m)	1.058e-005	1.058e-005	1.058e-005	1.058e-005
58	Liq. Reynolds	1.344e+005 *	1.344e+005 *	1.344e+005 *	1.344e+005 *
59	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
60	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
61	Vapour Velocity (m/s)	5.230	5.230	5.230	5.230
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	4751	4756	4761	4766
65	Elevation (m)	0.5200	0.5520	0.5840	0.6160
66	Pressure (bar)	77.99	77.99	77.99	77.99
67	Temperature (C)	55.20	55.19	55.19	55.19
68	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
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5										
6	Pipe Segment: BBOU-ELEP Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Heat Transferred (kJ/h-m)	548.8	548.8	548.7	548.7					
14	Flow Regime	Transition	Transition	Transition	Transition					
15	Water Flow Regime									
16	Liquid HoldUp	0.1662	0.1628	0.1628	0.1628					
17	Water HoldUp	---	---	---	---					
18	Friction Gradient (kPa/m)	3.558e-002	3.566e-002	3.566e-002	3.566e-002					
19	Static Gradient (kPa/m)	0.0000	1.005e-002	1.005e-002	1.005e-002					
20	Accel. Gradient (kPa/m)	1.059e-005	1.360e-005	1.360e-005	1.360e-005					
21	Liq. Reynolds	1.344e+005 *	1.343e+005 *	1.343e+005 *	1.343e+005 *					
22	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *					
23	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083					
24	Vapour Velocity (m/s)	5.230	5.230	5.230	5.231					
25	Deposit Thickness (mm)	---	---	---	---					
26	Deposit Volume (m3)	---	---	---	---					
27	Length (m)	4771	4776	4781	4786					
28	Elevation (m)	0.6480	0.6800	0.6700	0.6600					
29	Pressure (bar)	77.98	77.98	77.98	77.98					
30	Temperature (C)	55.18	55.18	55.18	55.18					
31	Vapour Fraction	0.9342	0.9342	0.9342	0.9341					
32	Heat Transferred (kJ/h-m)	548.6	548.5	548.5	548.5					
33	Flow Regime	Transition	Transition	Transition	Transition					
34	Water Flow Regime									
35	Liquid HoldUp	0.1628	0.1628	0.1131	0.1131					
36	Water HoldUp	---	---	---	---					
37	Friction Gradient (kPa/m)	3.566e-002	3.566e-002	3.889e-002	3.889e-002					
38	Static Gradient (kPa/m)	1.005e-002	1.005e-002	-2.524e-003	-2.524e-003					
39	Accel. Gradient (kPa/m)	1.360e-005	1.360e-005	1.082e-005	1.082e-005					
40	Liq. Reynolds	1.343e+005 *	1.343e+005 *	1.343e+005 *	1.343e+005 *					
41	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *					
42	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083					
43	Vapour Velocity (m/s)	5.231	5.231	5.231	5.231					
44	Deposit Thickness (mm)	---	---	---	---					
45	Deposit Volume (m3)	---	---	---	---					
46	Length (m)	4791	4796	4801	4806					
47	Elevation (m)	0.6500	0.6400	0.6300	0.6340					
48	Pressure (bar)	77.98	77.98	77.97	77.97					
49	Temperature (C)	55.17	55.17	55.17	55.16					
50	Vapour Fraction	0.9341	0.9341	0.9341	0.9341					
51	Heat Transferred (kJ/h-m)	548.4	548.4	548.3	548.2					
52	Flow Regime	Transition	Transition	Transition	Transition					
53	Water Flow Regime									
54	Liquid HoldUp	0.1131	0.1131	0.1131	0.1547					
55	Water HoldUp	---	---	---	---					
56	Friction Gradient (kPa/m)	3.889e-002	3.889e-002	3.889e-002	3.595e-002					
57	Static Gradient (kPa/m)	-2.524e-003	-2.524e-003	-2.524e-003	1.216e-003					
58	Accel. Gradient (kPa/m)	1.082e-005	1.083e-005	1.083e-005	1.106e-005					
59	Liq. Reynolds	1.343e+005 *	1.343e+005 *	1.343e+005 *	1.343e+005 *					
60	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *					
61	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083					
62	Vapour Velocity (m/s)	5.231	5.231	5.231	5.231					
63	Deposit Thickness (mm)	---	---	---	---					
64	Deposit Volume (m3)	---	---	---	---					
65	Length (m)	4811	4816	4821	4826					
66	Elevation (m)	0.6380	0.6420	0.6460	0.6500					
67	Pressure (bar)	77.97	77.97	77.97	77.96					
68	Temperature (C)	55.16	55.16	55.16	55.15					
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
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4					
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
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9	PROFILES				
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11	Pipe Table				
12					
13	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
14	Heat Transferred (kJ/h-m)	548.1	548.1	548.0	548.0
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.1547	0.1547	0.1547	0.1547
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	3.595e-002	3.595e-002	3.595e-002	3.595e-002
20	Static Gradient (kPa/m)	1.216e-003	1.216e-003	1.216e-003	1.216e-003
21	Accel. Gradient (kPa/m)	1.106e-005	1.106e-005	1.106e-005	1.107e-005
22	Liq. Reynolds	1.343e+005 *	1.343e+005 *	1.343e+005 *	1.343e+005 *
23	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
24	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
25	Vapour Velocity (m/s)	5.231	5.231	5.231	5.231
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	4831	4836	4841	4846
29	Elevation (m)	0.6600	0.6700	0.6800	0.6900
30	Pressure (bar)	77.96	77.96	77.96	77.96
31	Temperature (C)	55.15	55.15	55.15	55.14
32	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
33	Heat Transferred (kJ/h-m)	547.9	547.8	547.8	547.7
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1564	0.1564	0.1564	0.1564
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	3.589e-002	3.589e-002	3.589e-002	3.589e-002
39	Static Gradient (kPa/m)	3.019e-003	3.019e-003	3.019e-003	3.019e-003
40	Accel. Gradient (kPa/m)	1.158e-005	1.158e-005	1.158e-005	1.158e-005
41	Liq. Reynolds	1.343e+005 *	1.343e+005 *	1.343e+005 *	1.343e+005 *
42	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
43	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
44	Vapour Velocity (m/s)	5.232	5.232	5.232	5.232
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	4851	4856	4861	4866
48	Elevation (m)	0.7000	0.7060	0.7120	0.7180
49	Pressure (bar)	77.95	77.95	77.95	77.95
50	Temperature (C)	55.14	55.14	55.13	55.13
51	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
52	Heat Transferred (kJ/h-m)	547.6	547.6	547.5	547.5
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.1564	0.1553	0.1553	0.1553
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	3.589e-002	3.593e-002	3.593e-002	3.593e-002
58	Static Gradient (kPa/m)	3.019e-003	1.828e-003	1.828e-003	1.828e-003
59	Accel. Gradient (kPa/m)	1.159e-005	1.124e-005	1.124e-005	1.125e-005
60	Liq. Reynolds	1.343e+005 *	1.343e+005 *	1.343e+005 *	1.343e+005 *
61	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
62	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
63	Vapour Velocity (m/s)	5.232	5.232	5.232	5.232
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	4871	4876	4881	4886
67	Elevation (m)	0.7240	0.7300	0.7200	0.7100
68	Pressure (bar)	77.95	77.94	77.94	77.94
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1			Case Name: Simulation - 331MMSCFD_80barg.hsc		
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3		Epic Atlantic Limited	Unit Set: Mixed		
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5			Date/Time: Mon Jun 13 15:09:19 2022		
6					
7	Pipe Segment: BBOU-ELEP Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	55.13	55.13	55.12	55.12
14	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
15	Heat Transferred (kJ/h-m)	547.4	547.3	547.3	547.3
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.1553	0.1553	0.1130	0.1130
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	3.593e-002	3.593e-002	3.890e-002	3.890e-002
21	Static Gradient (kPa/m)	1.828e-003	1.828e-003	-2.524e-003	-2.524e-003
22	Accel. Gradient (kPa/m)	1.125e-005	1.125e-005	1.084e-005	1.084e-005
23	Liq. Reynolds	1.343e+005 *	1.343e+005 *	1.343e+005 *	1.343e+005 *
24	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
25	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
26	Vapour Velocity (m/s)	5.232	5.232	5.232	5.232
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	4891	4896	4901	4906
30	Elevation (m)	0.7000	0.6900	0.6800	0.6780
31	Pressure (bar)	77.94	77.94	77.94	77.93
32	Temperature (C)	55.12	55.12	55.11	55.11
33	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
34	Heat Transferred (kJ/h-m)	547.2	547.2	547.1	547.1
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1130	0.1130	0.1130	0.1137
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	3.890e-002	3.890e-002	3.890e-002	3.884e-002
40	Static Gradient (kPa/m)	-2.524e-003	-2.524e-003	-2.524e-003	-5.063e-004
41	Accel. Gradient (kPa/m)	1.084e-005	1.084e-005	1.084e-005	1.142e-005
42	Liq. Reynolds	1.343e+005 *	1.343e+005 *	1.343e+005 *	1.343e+005 *
43	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
44	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
45	Vapour Velocity (m/s)	5.232	5.232	5.233	5.233
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	4911	4916	4921	4926
49	Elevation (m)	0.6760	0.6740	0.6720	0.6700
50	Pressure (bar)	77.93	77.93	77.93	77.93
51	Temperature (C)	55.11	55.10	55.10	55.10
52	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
53	Heat Transferred (kJ/h-m)	547.0	546.9	546.9	546.8
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.1137	0.1137	0.1137	0.1137
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	3.884e-002	3.884e-002	3.884e-002	3.884e-002
59	Static Gradient (kPa/m)	-5.063e-004	-5.063e-004	-5.063e-004	-5.063e-004
60	Accel. Gradient (kPa/m)	1.142e-005	1.142e-005	1.142e-005	1.142e-005
61	Liq. Reynolds	1.343e+005 *	1.343e+005 *	1.343e+005 *	1.343e+005 *
62	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
63	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
64	Vapour Velocity (m/s)	5.233	5.233	5.233	5.233
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	4931	4936	4941	4946
68	Elevation (m)	0.6680	0.6660	0.6640	0.6620
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:09:19 2022				
5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	77.92	77.92	77.92	77.92		
14	Temperature (C)	55.10	55.09	55.09	55.09		
15	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
16	Heat Transferred (kJ/h-m)	546.8	546.7	546.6	546.6		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.1137	0.1137	0.1137	0.1137		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	3.884e-002	3.884e-002	3.884e-002	3.884e-002		
22	Static Gradient (kPa/m)	-5.063e-004	-5.063e-004	-5.063e-004	-5.063e-004		
23	Accel. Gradient (kPa/m)	1.142e-005	1.142e-005	1.142e-005	1.142e-005		
24	Liq. Reynolds	1.343e+005 *	1.343e+005 *	1.342e+005 *	1.342e+005 *		
25	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
26	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
27	Vapour Velocity (m/s)	5.233	5.233	5.233	5.233		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	4951	4956	4961	4966		
31	Elevation (m)	0.6600	0.6460	0.6320	0.6180		
32	Pressure (bar)	77.92	77.91	77.91	77.91		
33	Temperature (C)	55.09	55.08	55.08	55.08		
34	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
35	Heat Transferred (kJ/h-m)	546.5	546.5	546.4	546.3		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.1137	0.1127	0.1127	0.1127		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	3.884e-002	3.895e-002	3.895e-002	3.895e-002		
41	Static Gradient (kPa/m)	-5.063e-004	-3.527e-003	-3.527e-003	-3.527e-003		
42	Accel. Gradient (kPa/m)	1.142e-005	1.055e-005	1.056e-005	1.056e-005		
43	Liq. Reynolds	1.342e+005 *	1.342e+005 *	1.342e+005 *	1.342e+005 *		
44	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
45	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
46	Vapour Velocity (m/s)	5.233	5.233	5.233	5.233		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	4971	4976	4981	4986		
50	Elevation (m)	0.6040	0.5900	0.5980	0.6060		
51	Pressure (bar)	77.91	77.91	77.91	77.90		
52	Temperature (C)	55.08	55.07	55.07	55.07		
53	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
54	Heat Transferred (kJ/h-m)	546.3	546.2	546.1	546.1		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.1127	0.1127	0.1558	0.1558		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	3.895e-002	3.895e-002	3.592e-002	3.592e-002		
60	Static Gradient (kPa/m)	-3.527e-003	-3.527e-003	2.443e-003	2.442e-003		
61	Accel. Gradient (kPa/m)	1.056e-005	1.056e-005	1.143e-005	1.144e-005		
62	Liq. Reynolds	1.342e+005 *	1.342e+005 *	1.342e+005 *	1.342e+005 *		
63	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
64	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
65	Vapour Velocity (m/s)	5.234	5.234	5.234	5.234		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	4991	4996	5001	5006		
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3							
4							
5			Unit Set: Mixed				
6			Date/Time: Mon Jun 13 15:09:19 2022				
7	Pipe Segment: BBOU-ELEP Pipeline (continued)						
8							
9							
10	PROFILES						
11	Pipe Table						
12							
13	Elevation (m)	0.6140	0.6220	0.6300	0.6400		
14	Pressure (bar)	77.90	77.90	77.90	77.90		
15	Temperature (C)	55.06	55.06	55.06	55.06		
16	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
17	Heat Transferred (kJ/h-m)	546.0	545.9	545.9	545.8		
18	Flow Regime	Transition	Transition	Transition	Transition		
19	Water Flow Regime						
20	Liquid HoldUp	0.1558	0.1558	0.1558	0.1564		
21	Water HoldUp	---	---	---	---		
22	Friction Gradient (kPa/m)	3.592e-002	3.592e-002	3.592e-002	3.590e-002		
23	Static Gradient (kPa/m)	2.442e-003	2.442e-003	2.442e-003	3.060e-003		
24	Accel. Gradient (kPa/m)	1.144e-005	1.144e-005	1.144e-005	1.162e-005		
25	Liq. Reynolds	1.342e+005 *	1.342e+005 *	1.342e+005 *	1.342e+005 *		
26	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
27	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
28	Vapour Velocity (m/s)	5.234	5.234	5.234	5.234		
29	Deposit Thickness (mm)	---	---	---	---		
30	Deposit Volume (m3)	---	---	---	---		
31	Length (m)	5011	5016	5021	5026		
32	Elevation (m)	0.6500	0.6600	0.6700	0.6800		
33	Pressure (bar)	77.89	77.89	77.89	77.89		
34	Temperature (C)	55.05	55.05	55.05	55.04		
35	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
36	Heat Transferred (kJ/h-m)	545.8	545.7	545.6	545.6		
37	Flow Regime	Transition	Transition	Transition	Transition		
38	Water Flow Regime						
39	Liquid HoldUp	0.1564	0.1564	0.1564	0.1564		
40	Water HoldUp	---	---	---	---		
41	Friction Gradient (kPa/m)	3.590e-002	3.590e-002	3.590e-002	3.590e-002		
42	Static Gradient (kPa/m)	3.060e-003	3.060e-003	3.060e-003	3.060e-003		
43	Accel. Gradient (kPa/m)	1.162e-005	1.162e-005	1.162e-005	1.162e-005		
44	Liq. Reynolds	1.342e+005 *	1.342e+005 *	1.342e+005 *	1.342e+005 *		
45	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
46	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
47	Vapour Velocity (m/s)	5.234	5.234	5.234	5.234		
48	Deposit Thickness (mm)	---	---	---	---		
49	Deposit Volume (m3)	---	---	---	---		
50	Length (m)	5032	5037	5042	5047		
51	Elevation (m)	0.6740	0.6680	0.6620	0.6560		
52	Pressure (bar)	77.89	77.88	77.88	77.88		
53	Temperature (C)	55.04	55.04	55.04	55.03		
54	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
55	Heat Transferred (kJ/h-m)	545.6	545.5	545.5	545.4		
56	Flow Regime	Transition	Transition	Transition	Transition		
57	Water Flow Regime						
58	Liquid HoldUp	0.1133	0.1133	0.1133	0.1133		
59	Water HoldUp	---	---	---	---		
60	Friction Gradient (kPa/m)	3.888e-002	3.889e-002	3.889e-002	3.889e-002		
61	Static Gradient (kPa/m)	-1.495e-003	-1.495e-003	-1.495e-003	-1.495e-003		
62	Accel. Gradient (kPa/m)	1.115e-005	1.115e-005	1.115e-005	1.115e-005		
63	Liq. Reynolds	1.342e+005 *	1.342e+005 *	1.342e+005 *	1.342e+005 *		
64	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
65	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
66	Vapour Velocity (m/s)	5.234	5.234	5.235	5.235		
67	Deposit Thickness (mm)	---	---	---	---		
68	Deposit Volume (m3)	---	---	---	---		
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
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5										
6	Pipe Segment: BBOU-ELEP Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Length	(m)	5052	5057	5062	5067				
14	Elevation	(m)	0.6500	0.6440	0.6380	0.6320				
15	Pressure	(bar)	77.88	77.88	77.87	77.87				
16	Temperature	(C)	55.03	55.03	55.03	55.02				
17	Vapour Fraction		0.9341	0.9341	0.9341	0.9341				
18	Heat Transferred	(kJ/h-m)	545.3	545.3	545.2	545.2				
19	Flow Regime		Transition	Transition	Transition	Transition				
20	Water Flow Regime									
21	Liquid HoldUp		0.1133	0.1133	0.1133	0.1133				
22	Water HoldUp		---	---	---	---				
23	Friction Gradient	(kPa/m)	3.889e-002	3.889e-002	3.889e-002	3.889e-002				
24	Static Gradient	(kPa/m)	-1.495e-003	-1.516e-003	-1.516e-003	-1.516e-003				
25	Accel. Gradient	(kPa/m)	1.115e-005	1.115e-005	1.115e-005	1.115e-005				
26	Liq. Reynolds		1.342e+005 *	1.342e+005 *	1.342e+005 *	1.342e+005 *				
27	Vap. Reynolds		1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *				
28	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083				
29	Vapour Velocity	(m/s)	5.235	5.235	5.235	5.235				
30	Deposit Thickness	(mm)	---	---	---	---				
31	Deposit Volume	(m3)	---	---	---	---				
32	Length	(m)	5072	5077	5082	5087				
33	Elevation	(m)	0.6260	0.6200	0.6180	0.6160				
34	Pressure	(bar)	77.87	77.87	77.87	77.87				
35	Temperature	(C)	55.02	55.02	55.02	55.01				
36	Vapour Fraction		0.9341	0.9341	0.9341	0.9341				
37	Heat Transferred	(kJ/h-m)	545.1	545.0	545.0	544.9				
38	Flow Regime		Transition	Transition	Transition	Transition				
39	Water Flow Regime									
40	Liquid HoldUp		0.1133	0.1133	0.1136	0.1136				
41	Water HoldUp		---	---	---	---				
42	Friction Gradient	(kPa/m)	3.889e-002	3.889e-002	3.886e-002	3.886e-002				
43	Static Gradient	(kPa/m)	-1.516e-003	-1.516e-003	-5.061e-004	-5.061e-004				
44	Accel. Gradient	(kPa/m)	1.115e-005	1.115e-005	1.144e-005	1.144e-005				
45	Liq. Reynolds		1.342e+005 *	1.342e+005 *	1.342e+005 *	1.342e+005 *				
46	Vap. Reynolds		1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *				
47	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083				
48	Vapour Velocity	(m/s)	5.235	5.235	5.235	5.235				
49	Deposit Thickness	(mm)	---	---	---	---				
50	Deposit Volume	(m3)	---	---	---	---				
51	Length	(m)	5092	5097	5102	5107				
52	Elevation	(m)	0.6140	0.6120	0.6100	0.6200				
53	Pressure	(bar)	77.86	77.86	77.86	77.86				
54	Temperature	(C)	55.01	55.01	55.00	55.00				
55	Vapour Fraction		0.9341	0.9341	0.9341	0.9341				
56	Heat Transferred	(kJ/h-m)	544.9	544.8	544.7	544.6				
57	Flow Regime		Transition	Transition	Transition	Transition				
58	Water Flow Regime									
59	Liquid HoldUp		0.1136	0.1136	0.1136	0.1564				
60	Water HoldUp		---	---	---	---				
61	Friction Gradient	(kPa/m)	3.886e-002	3.886e-002	3.886e-002	3.591e-002				
62	Static Gradient	(kPa/m)	-5.061e-004	-5.061e-004	-5.061e-004	3.060e-003				
63	Accel. Gradient	(kPa/m)	1.144e-005	1.144e-005	1.144e-005	1.163e-005				
64	Liq. Reynolds		1.342e+005 *	1.342e+005 *	1.342e+005 *	1.342e+005 *				
65	Vap. Reynolds		1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *				
66	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083				
67	Vapour Velocity	(m/s)	5.235	5.235	5.235	5.236				
68	Deposit Thickness	(mm)	---	---	---	---				
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:09:19 2022				
5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	5112	5117	5122	5127		
15	Elevation (m)	0.6300	0.6400	0.6500	0.6600		
16	Pressure (bar)	77.86	77.85	77.85	77.85		
17	Temperature (C)	55.00	55.00	54.99	54.99		
18	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
19	Heat Transferred (kJ/h-m)	544.6	544.5	544.4	544.4		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	0.1564	0.1564	0.1564	0.1564		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	3.591e-002	3.591e-002	3.591e-002	3.591e-002		
25	Static Gradient (kPa/m)	3.060e-003	3.060e-003	3.060e-003	3.060e-003		
26	Accel. Gradient (kPa/m)	1.163e-005	1.163e-005	1.163e-005	1.163e-005		
27	Liq. Reynolds	1.342e+005 *	1.342e+005 *	1.341e+005 *	1.341e+005 *		
28	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
29	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
30	Vapour Velocity (m/s)	5.236	5.236	5.236	5.236		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	5132	5137	5142	5147		
34	Elevation (m)	0.6780	0.6960	0.7140	0.7320		
35	Pressure (bar)	77.85	77.85	77.84	77.84		
36	Temperature (C)	54.99	54.98	54.98	54.98		
37	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
38	Heat Transferred (kJ/h-m)	544.3	544.3	544.2	544.1		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.1587	0.1587	0.1587	0.1587		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	3.583e-002	3.583e-002	3.583e-002	3.583e-002		
44	Static Gradient (kPa/m)	5.559e-003	5.559e-003	5.559e-003	5.559e-003		
45	Accel. Gradient (kPa/m)	1.235e-005	1.235e-005	1.235e-005	1.235e-005		
46	Liq. Reynolds	1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *		
47	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
48	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
49	Vapour Velocity (m/s)	5.236	5.236	5.236	5.236		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	5152	5154	5156	5158		
53	Elevation (m)	0.7500	0.7320	0.7140	0.6960		
54	Pressure (bar)	77.84	77.84	77.84	77.84		
55	Temperature (C)	54.98	54.98	54.97	54.97		
56	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
57	Heat Transferred (kJ/h-m)	544.1	544.1	544.1	544.1		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.1587	0.1102	0.1102	0.1102		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	3.583e-002	3.924e-002	3.924e-002	3.924e-002		
63	Static Gradient (kPa/m)	5.559e-003	-1.120e-002	-1.120e-002	-1.120e-002		
64	Accel. Gradient (kPa/m)	1.235e-005	8.369e-006	8.369e-006	8.369e-006		
65	Liq. Reynolds	1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *		
66	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
67	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
68	Vapour Velocity (m/s)	5.236	5.236	5.236	5.236		
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
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5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	5160	5162	5168	5174
16	Elevation (m)	0.6780	0.6600	0.6700	0.6800
17	Pressure (bar)	77.84	77.84	77.83	77.83
18	Temperature (C)	54.97	54.97	54.97	54.97
19	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
20	Heat Transferred (kJ/h-m)	544.1	544.0	543.9	543.8
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.1102	0.1102	0.1559	0.1559
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	3.924e-002	3.924e-002	3.594e-002	3.594e-002
26	Static Gradient (kPa/m)	-1.120e-002	-1.120e-002	2.545e-003	2.545e-003
27	Accel. Gradient (kPa/m)	8.369e-006	8.370e-006	1.149e-005	1.149e-005
28	Liq. Reynolds	1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *
29	Vap. Reynolds	1.219e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
30	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
31	Vapour Velocity (m/s)	5.236	5.236	5.236	5.237
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	5180	5186	5192	5194
35	Elevation (m)	0.6900	0.7000	0.7100	0.7200
36	Pressure (bar)	77.83	77.83	77.83	77.82
37	Temperature (C)	54.96	54.96	54.96	54.95
38	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
39	Heat Transferred (kJ/h-m)	543.8	543.7	543.6	543.6
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.1559	0.1559	0.1559	0.1607
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	3.594e-002	3.594e-002	3.594e-002	3.577e-002
45	Static Gradient (kPa/m)	2.544e-003	2.544e-003	2.544e-003	7.783e-003
46	Accel. Gradient (kPa/m)	1.149e-005	1.149e-005	1.149e-005	1.300e-005
47	Liq. Reynolds	1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *
48	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
49	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
50	Vapour Velocity (m/s)	5.237	5.237	5.237	5.237
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	5196	5198	5200	5202
54	Elevation (m)	0.7300	0.7400	0.7500	0.7600
55	Pressure (bar)	77.82	77.82	77.82	77.82
56	Temperature (C)	54.95	54.95	54.95	54.95
57	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
58	Heat Transferred (kJ/h-m)	543.6	543.5	543.5	543.5
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.1607	0.1607	0.1607	0.1607
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	3.577e-002	3.577e-002	3.577e-002	3.577e-002
64	Static Gradient (kPa/m)	7.783e-003	7.783e-003	7.782e-003	7.782e-003
65	Accel. Gradient (kPa/m)	1.300e-005	1.300e-005	1.300e-005	1.300e-005
66	Liq. Reynolds	1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *
67	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
68	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
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3			Unit Set: Mixed				
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
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9	PROFILES						
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11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	5.237	5.237	5.237	5.237	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	5207	5212	5217	5222	
17	Elevation	(m)	0.7380	0.7160	0.6940	0.6720	
18	Pressure	(bar)	77.82	77.82	77.82	77.81	
19	Temperature	(C)	54.95	54.94	54.94	54.94	
20	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
21	Heat Transferred	(kJ/h-m)	543.5	543.4	543.4	543.3	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.1120	0.1120	0.1120	0.1120	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	3.904e-002	3.904e-002	3.904e-002	3.904e-002	
27	Static Gradient	(kPa/m)	-5.524e-003	-5.524e-003	-5.524e-003	-5.524e-003	
28	Accel. Gradient	(kPa/m)	1.001e-005	1.001e-005	1.001e-005	1.001e-005	
29	Liq. Reynolds		1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *	
30	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
31	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083	
32	Vapour Velocity	(m/s)	5.237	5.237	5.237	5.237	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	5227	5232	5237	5242	
36	Elevation	(m)	0.6500	0.6140	0.5780	0.5420	
37	Pressure	(bar)	77.81	77.81	77.81	77.81	
38	Temperature	(C)	54.94	54.93	54.93	54.93	
39	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
40	Heat Transferred	(kJ/h-m)	543.3	543.2	543.2	543.1	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.1120	0.1110	0.1110	0.1110	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	3.904e-002	3.916e-002	3.916e-002	3.916e-002	
46	Static Gradient	(kPa/m)	-5.524e-003	-8.836e-003	-8.836e-003	-8.836e-003	
47	Accel. Gradient	(kPa/m)	1.001e-005	9.056e-006	9.056e-006	9.057e-006	
48	Liq. Reynolds		1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *	
49	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
50	Liquid Velocity	(m/s)	0.2083	0.2083	0.2084	0.2084	
51	Vapour Velocity	(m/s)	5.237	5.237	5.237	5.237	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	5247	5252	5257	5262	
55	Elevation	(m)	0.5060	0.4700	0.4780	0.4860	
56	Pressure	(bar)	77.81	77.80	77.80	77.80	
57	Temperature	(C)	54.93	54.92	54.92	54.92	
58	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
59	Heat Transferred	(kJ/h-m)	543.1	543.0	542.9	542.8	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.1110	0.1110	0.1558	0.1558	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	3.916e-002	3.916e-002	3.595e-002	3.595e-002	
65	Static Gradient	(kPa/m)	-8.836e-003	-8.836e-003	2.441e-003	2.441e-003	
66	Accel. Gradient	(kPa/m)	9.057e-006	9.058e-006	1.147e-005	1.147e-005	
67	Liq. Reynolds		1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *	
68	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
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1			Case Name: Simulation - 331MMSCFD_80barg.hsc				
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3	 Epic Atlantic Limited		Unit Set: Mixed				
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6							
7	Pipe Segment: BBOU-ELEP Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
14	Vapour Velocity	(m/s)	5.237	5.238	5.238	5.238	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	5267	5272	5277	5282	
18	Elevation	(m)	0.4940	0.5020	0.5100	0.5180	
19	Pressure	(bar)	77.80	77.80	77.80	77.79	
20	Temperature	(C)	54.92	54.91	54.91	54.91	
21	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
22	Heat Transferred	(kJ/h-m)	542.8	542.7	542.6	542.6	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.1558	0.1558	0.1557	0.1557	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	3.595e-002	3.595e-002	3.595e-002	3.595e-002	
28	Static Gradient	(kPa/m)	2.441e-003	2.441e-003	2.441e-003	2.441e-003	
29	Accel. Gradient	(kPa/m)	1.147e-005	1.147e-005	1.147e-005	1.147e-005	
30	Liq. Reynolds		1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *	
31	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
32	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
33	Vapour Velocity	(m/s)	5.238	5.238	5.238	5.238	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	5287	5292	5297	5302	
37	Elevation	(m)	0.5260	0.5340	0.5420	0.5500	
38	Pressure	(bar)	77.79	77.79	77.79	77.79	
39	Temperature	(C)	54.90	54.90	54.90	54.90	
40	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
41	Heat Transferred	(kJ/h-m)	542.5	542.5	542.4	542.3	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.1557	0.1557	0.1557	0.1557	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	3.595e-002	3.595e-002	3.595e-002	3.595e-002	
47	Static Gradient	(kPa/m)	2.441e-003	2.441e-003	2.441e-003	2.441e-003	
48	Accel. Gradient	(kPa/m)	1.147e-005	1.147e-005	1.147e-005	1.147e-005	
49	Liq. Reynolds		1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *	
50	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
51	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
52	Vapour Velocity	(m/s)	5.238	5.238	5.238	5.238	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	5307	5312	5317	5322	
56	Elevation	(m)	0.5480	0.5460	0.5440	0.5420	
57	Pressure	(bar)	77.78	77.78	77.78	77.78	
58	Temperature	(C)	54.89	54.89	54.89	54.89	
59	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
60	Heat Transferred	(kJ/h-m)	542.3	542.3	542.2	542.2	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.1136	0.1136	0.1136	0.1136	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	3.888e-002	3.888e-002	3.888e-002	3.888e-002	
66	Static Gradient	(kPa/m)	-5.059e-004	-5.059e-004	-5.059e-004	-5.059e-004	
67	Accel. Gradient	(kPa/m)	1.147e-005	1.147e-005	1.147e-005	1.147e-005	
68	Liq. Reynolds		1.340e+005 *	1.340e+005 *	1.340e+005 *	1.340e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
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13	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
14	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
15	Vapour Velocity (m/s)	5.238	5.238	5.238	5.239
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	5327	5332	5337	5342
19	Elevation (m)	0.5400	0.5480	0.5560	0.5640
20	Pressure (bar)	77.78	77.77	77.77	77.77
21	Temperature (C)	54.88	54.88	54.88	54.87
22	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
23	Heat Transferred (kJ/h-m)	542.1	542.0	541.9	541.9
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1136	0.1557	0.1557	0.1557
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	3.888e-002	3.596e-002	3.596e-002	3.596e-002
29	Static Gradient (kPa/m)	-5.059e-004	2.441e-003	2.441e-003	2.441e-003
30	Accel. Gradient (kPa/m)	1.147e-005	1.148e-005	1.148e-005	1.148e-005
31	Liq. Reynolds	1.340e+005 *	1.340e+005 *	1.340e+005 *	1.340e+005 *
32	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
33	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
34	Vapour Velocity (m/s)	5.239	5.239	5.239	5.239
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	5347	5352	5357	5362
38	Elevation (m)	0.5720	0.5800	0.5800	0.5800
39	Pressure (bar)	77.77	77.77	77.76	77.76
40	Temperature (C)	54.87	54.87	54.87	54.86
41	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
42	Heat Transferred (kJ/h-m)	541.8	541.7	541.7	541.6
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1557	0.1557	0.1660	0.1660
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	3.596e-002	3.596e-002	3.564e-002	3.564e-002
48	Static Gradient (kPa/m)	2.441e-003	2.441e-003	0.0000	0.0000
49	Accel. Gradient (kPa/m)	1.148e-005	1.148e-005	1.065e-005	1.065e-005
50	Liq. Reynolds	1.340e+005 *	1.340e+005 *	1.340e+005 *	1.340e+005 *
51	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
52	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
53	Vapour Velocity (m/s)	5.239	5.239	5.239	5.239
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	5367	5372	5377	5382
57	Elevation (m)	0.5800	0.5800	0.5800	0.5840
58	Pressure (bar)	77.76	77.76	77.76	77.76
59	Temperature (C)	54.86	54.86	54.86	54.85
60	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
61	Heat Transferred (kJ/h-m)	541.5	541.5	541.4	541.4
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1660	0.1660	0.1660	0.1546
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	3.564e-002	3.564e-002	3.564e-002	3.601e-002
67	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	1.215e-003
68	Accel. Gradient (kPa/m)	1.065e-005	1.065e-005	1.065e-005	1.113e-005
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		1.340e+005 *	1.340e+005 *	1.340e+005 *	1.340e+005 *	
14	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
15	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084		
16	Vapour Velocity (m/s)	5.239	5.239	5.239	5.239		
17	Deposit Thickness (mm)	---	---	---	---		
18	Deposit Volume (m3)	---	---	---	---		
19	Length (m)	5387	5392	5397	5402		
20	Elevation (m)	0.5880	0.5920	0.5960	0.6000		
21	Pressure (bar)	77.75	77.75	77.75	77.75		
22	Temperature (C)	54.85	54.85	54.85	54.84		
23	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
24	Heat Transferred (kJ/h-m)	541.3	541.3	541.2	541.2		
25	Flow Regime	Transition	Transition	Transition	Transition		
26	Water Flow Regime						
27	Liquid HoldUp	0.1546	0.1546	0.1546	0.1546		
28	Water HoldUp	---	---	---	---		
29	Friction Gradient (kPa/m)	3.601e-002	3.601e-002	3.601e-002	3.601e-002		
30	Static Gradient (kPa/m)	1.215e-003	1.215e-003	1.215e-003	1.215e-003		
31	Accel. Gradient (kPa/m)	1.113e-005	1.113e-005	1.113e-005	1.113e-005		
32	Liq. Reynolds		1.340e+005 *	1.340e+005 *	1.340e+005 *		
33	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *		
34	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084		
35	Vapour Velocity (m/s)	5.239	5.240	5.240	5.240		
36	Deposit Thickness (mm)	---	---	---	---		
37	Deposit Volume (m3)	---	---	---	---		
38	Length (m)	5407	5412	5417	5422		
39	Elevation (m)	0.6000	0.6000	0.6000	0.6000		
40	Pressure (bar)	77.75	77.74	77.74	77.74		
41	Temperature (C)	54.84	54.84	54.83	54.83		
42	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
43	Heat Transferred (kJ/h-m)	541.1	541.0	541.0	540.9		
44	Flow Regime	Transition	Transition	Transition	Transition		
45	Water Flow Regime						
46	Liquid HoldUp	0.1660	0.1660	0.1660	0.1660		
47	Water HoldUp	---	---	---	---		
48	Friction Gradient (kPa/m)	3.564e-002	3.564e-002	3.564e-002	3.564e-002		
49	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000		
50	Accel. Gradient (kPa/m)	1.066e-005	1.066e-005	1.066e-005	1.066e-005		
51	Liq. Reynolds		1.340e+005 *	1.340e+005 *	1.340e+005 *		
52	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *		
53	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084		
54	Vapour Velocity (m/s)	5.240	5.240	5.240	5.240		
55	Deposit Thickness (mm)	---	---	---	---		
56	Deposit Volume (m3)	---	---	---	---		
57	Length (m)	5427	5432	5437	5443		
58	Elevation (m)	0.6000	0.6000	0.6000	0.6000		
59	Pressure (bar)	77.74	77.74	77.74	77.73		
60	Temperature (C)	54.83	54.83	54.82	54.82		
61	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
62	Heat Transferred (kJ/h-m)	540.8	540.8	540.7	540.7		
63	Flow Regime	Transition	Transition	Transition	Transition		
64	Water Flow Regime						
65	Liquid HoldUp	0.1660	0.1660	0.1660	0.1660		
66	Water HoldUp	---	---	---	---		
67	Friction Gradient (kPa/m)	3.564e-002	3.564e-002	3.564e-002	3.564e-002		
68	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000		
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
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5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient	(kPa/m)	1.066e-005	1.066e-005	1.066e-005	1.066e-005	
14	Liq. Reynolds		1.340e+005 *	1.340e+005 *	1.340e+005 *	1.340e+005 *	
15	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
16	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
17	Vapour Velocity	(m/s)	5.240	5.240	5.240	5.240	
18	Deposit Thickness	(mm)	---	---	---	---	
19	Deposit Volume	(m3)	---	---	---	---	
20	Length	(m)	5448	5453	5458	5463	
21	Elevation	(m)	0.6000	0.6000	0.6100	0.6200	
22	Pressure	(bar)	77.73	77.73	77.73	77.73	
23	Temperature	(C)	54.82	54.82	54.81	54.81	
24	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
25	Heat Transferred	(kJ/h-m)	540.6	540.5	540.5	540.5	
26	Flow Regime		Transition	Transition	Transition	Transition	
27	Water Flow Regime						
28	Liquid HoldUp		0.1660	0.1660	0.1563	0.1563	
29	Water HoldUp		---	---	---	---	
30	Friction Gradient	(kPa/m)	3.564e-002	3.565e-002	3.595e-002	3.595e-002	
31	Static Gradient	(kPa/m)	0.0000	0.0000	3.058e-003	3.058e-003	
32	Accel. Gradient	(kPa/m)	1.066e-005	1.066e-005	1.167e-005	1.167e-005	
33	Liq. Reynolds		1.340e+005 *	1.340e+005 *	1.340e+005 *	1.340e+005 *	
34	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
35	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
36	Vapour Velocity	(m/s)	5.240	5.240	5.240	5.241	
37	Deposit Thickness	(mm)	---	---	---	---	
38	Deposit Volume	(m3)	---	---	---	---	
39	Length	(m)	5468	5473	5478	5483	
40	Elevation	(m)	0.6300	0.6400	0.6500	0.6160	
41	Pressure	(bar)	77.72	77.72	77.72	77.72	
42	Temperature	(C)	54.81	54.80	54.80	54.80	
43	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
44	Heat Transferred	(kJ/h-m)	540.4	540.3	540.3	540.3	
45	Flow Regime		Transition	Transition	Transition	Transition	
46	Water Flow Regime						
47	Liquid HoldUp		0.1563	0.1563	0.1563	0.1110	
48	Water HoldUp		---	---	---	---	
49	Friction Gradient	(kPa/m)	3.595e-002	3.595e-002	3.595e-002	3.918e-002	
50	Static Gradient	(kPa/m)	3.058e-003	3.058e-003	3.058e-003	-8.493e-003	
51	Accel. Gradient	(kPa/m)	1.167e-005	1.167e-005	1.167e-005	9.180e-006	
52	Liq. Reynolds		1.340e+005 *	1.340e+005 *	1.340e+005 *	1.340e+005 *	
53	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
54	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
55	Vapour Velocity	(m/s)	5.241	5.241	5.241	5.241	
56	Deposit Thickness	(mm)	---	---	---	---	
57	Deposit Volume	(m3)	---	---	---	---	
58	Length	(m)	5488	5493	5498	5503	
59	Elevation	(m)	0.5820	0.5480	0.5140	0.4800	
60	Pressure	(bar)	77.72	77.72	77.71	77.71	
61	Temperature	(C)	54.80	54.79	54.79	54.79	
62	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
63	Heat Transferred	(kJ/h-m)	540.2	540.2	540.1	540.1	
64	Flow Regime		Transition	Transition	Transition	Transition	
65	Water Flow Regime						
66	Liquid HoldUp		0.1110	0.1110	0.1110	0.1110	
67	Water HoldUp		---	---	---	---	
68	Friction Gradient	(kPa/m)	3.918e-002	3.918e-002	3.918e-002	3.918e-002	
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2										
3							Unit Set: Mixed			
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5										
6	Pipe Segment: BBOU-ELEP Pipeline (continued)									
7										
8										
9	PROFILES									
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11	Pipe Table									
12										
13	Static Gradient	(kPa/m)	-8.493e-003	-8.493e-003	-8.493e-003	-8.493e-003				
14	Accel. Gradient	(kPa/m)	9.180e-006	9.181e-006	9.181e-006	9.181e-006				
15	Liq. Reynolds		1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *				
16	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *				
17	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084				
18	Vapour Velocity	(m/s)	5.241	5.241	5.241	5.241				
19	Deposit Thickness	(mm)	---	---	---	---				
20	Deposit Volume	(m3)	---	---	---	---				
21	Length	(m)	5508	5513	5518	5523				
22	Elevation	(m)	0.4980	0.5160	0.5340	0.5520				
23	Pressure	(bar)	77.71	77.71	77.71	77.70				
24	Temperature	(C)	54.79	54.78	54.78	54.78				
25	Vapour Fraction		0.9341	0.9341	0.9341	0.9341				
26	Heat Transferred	(kJ/h-m)	539.9	539.9	539.8	539.8				
27	Flow Regime		Transition	Transition	Transition	Transition				
28	Water Flow Regime									
29	Liquid HoldUp		0.1586	0.1586	0.1586	0.1586				
30	Water HoldUp		---	---	---	---				
31	Friction Gradient	(kPa/m)	3.586e-002	3.586e-002	3.587e-002	3.587e-002				
32	Static Gradient	(kPa/m)	5.556e-003	5.556e-003	5.556e-003	5.556e-003				
33	Accel. Gradient	(kPa/m)	1.239e-005	1.239e-005	1.240e-005	1.240e-005				
34	Liq. Reynolds		1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *				
35	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *				
36	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084				
37	Vapour Velocity	(m/s)	5.241	5.241	5.241	5.241				
38	Deposit Thickness	(mm)	---	---	---	---				
39	Deposit Volume	(m3)	---	---	---	---				
40	Length	(m)	5528	5533	5538	5543				
41	Elevation	(m)	0.5700	0.5860	0.6020	0.6180				
42	Pressure	(bar)	77.70	77.70	77.70	77.70				
43	Temperature	(C)	54.78	54.77	54.77	54.77				
44	Vapour Fraction		0.9341	0.9341	0.9341	0.9341				
45	Heat Transferred	(kJ/h-m)	539.7	539.6	539.6	539.5				
46	Flow Regime		Transition	Transition	Transition	Transition				
47	Water Flow Regime									
48	Liquid HoldUp		0.1586	0.1580	0.1580	0.1580				
49	Water HoldUp		---	---	---	---				
50	Friction Gradient	(kPa/m)	3.587e-002	3.589e-002	3.589e-002	3.589e-002				
51	Static Gradient	(kPa/m)	5.556e-003	4.927e-003	4.927e-003	4.927e-003				
52	Accel. Gradient	(kPa/m)	1.240e-005	1.222e-005	1.222e-005	1.222e-005				
53	Liq. Reynolds		1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *				
54	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *				
55	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084				
56	Vapour Velocity	(m/s)	5.241	5.242	5.242	5.242				
57	Deposit Thickness	(mm)	---	---	---	---				
58	Deposit Volume	(m3)	---	---	---	---				
59	Length	(m)	5548	5553	5558	5563				
60	Elevation	(m)	0.6340	0.6500	0.6640	0.6780				
61	Pressure	(bar)	77.69	77.69	77.69	77.69				
62	Temperature	(C)	54.76	54.76	54.76	54.76				
63	Vapour Fraction		0.9341	0.9341	0.9341	0.9341				
64	Heat Transferred	(kJ/h-m)	539.4	539.4	539.3	539.3				
65	Flow Regime		Transition	Transition	Transition	Transition				
66	Water Flow Regime									
67	Liquid HoldUp		0.1580	0.1580	0.1574	0.1574				
68	Water HoldUp		---	---	---	---				
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1				Case Name: Simulation - 331MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
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6						
7	Pipe Segment: BBOU-ELEP Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	3.589e-002	3.589e-002	3.591e-002	3.591e-002
14	Static Gradient	(kPa/m)	4.927e-003	4.927e-003	4.301e-003	4.301e-003
15	Accel. Gradient	(kPa/m)	1.222e-005	1.222e-005	1.204e-005	1.204e-005
16	Liq. Reynolds		1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *
17	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
18	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084
19	Vapour Velocity	(m/s)	5.242	5.242	5.242	5.242
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	5568	5573	5578	5580
23	Elevation	(m)	0.6920	0.7060	0.7200	0.6560
24	Pressure	(bar)	77.69	77.68	77.68	77.68
25	Temperature	(C)	54.75	54.75	54.75	54.75
26	Vapour Fraction		0.9341	0.9341	0.9341	0.9341
27	Heat Transferred	(kJ/h-m)	539.2	539.1	539.1	539.2
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.1574	0.1574	0.1574	0.1012
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	3.591e-002	3.591e-002	3.591e-002	4.044e-002
33	Static Gradient	(kPa/m)	4.301e-003	4.301e-003	4.301e-003	-3.801e-002
34	Accel. Gradient	(kPa/m)	1.204e-005	1.204e-005	1.204e-005	7.277e-007
35	Liq. Reynolds		1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *
36	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
37	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084
38	Vapour Velocity	(m/s)	5.242	5.242	5.242	5.242
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	5582	5584	5586	5588
42	Elevation	(m)	0.5920	0.5280	0.4640	0.4000
43	Pressure	(bar)	77.68	77.68	77.68	77.68
44	Temperature	(C)	54.75	54.75	54.74	54.74
45	Vapour Fraction		0.9341	0.9341	0.9341	0.9341
46	Heat Transferred	(kJ/h-m)	539.1	539.1	539.1	539.1
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.1012	0.1012	0.1012	0.1012
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	4.044e-002	4.044e-002	4.044e-002	4.044e-002
52	Static Gradient	(kPa/m)	-3.801e-002	-3.801e-002	-3.801e-002	-3.801e-002
53	Accel. Gradient	(kPa/m)	7.277e-007	7.276e-007	7.276e-007	7.275e-007
54	Liq. Reynolds		1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *
55	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
56	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084
57	Vapour Velocity	(m/s)	5.242	5.242	5.242	5.242
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	5590	5592	5594	5596
61	Elevation	(m)	0.3780	0.3560	0.3340	0.3120
62	Pressure	(bar)	77.68	77.68	77.68	77.68
63	Temperature	(C)	54.74	54.74	54.74	54.74
64	Vapour Fraction		0.9341	0.9341	0.9341	0.9341
65	Heat Transferred	(kJ/h-m)	539.1	539.0	539.0	539.0
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.1094	0.1094	0.1094	0.1094
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
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5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Water HoldUp		---	---	---	---	
14	Friction Gradient (kPa/m)		3.938e-002	3.938e-002	3.938e-002	3.938e-002	
15	Static Gradient (kPa/m)		-1.362e-002	-1.362e-002	-1.362e-002	-1.362e-002	
16	Accel. Gradient (kPa/m)		7.710e-006	7.710e-006	7.710e-006	7.710e-006	
17	Liq. Reynolds		1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *	
18	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
19	Liquid Velocity (m/s)		0.2084	0.2084	0.2084	0.2084	
20	Vapour Velocity (m/s)		5.242	5.242	5.242	5.242	
21	Deposit Thickness (mm)		---	---	---	---	
22	Deposit Volume (m3)		---	---	---	---	
23	Length (m)		5598	5604	5610	5616	
24	Elevation (m)		0.2900	0.3620	0.4340	0.5060	
25	Pressure (bar)		77.68	77.68	77.67	77.67	
26	Temperature (C)		54.74	54.74	54.73	54.73	
27	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
28	Heat Transferred (kJ/h-m)		539.0	538.8	538.7	538.6	
29	Flow Regime		Transition	Transition	Transition	Transition	
30	Water Flow Regime						
31	Liquid HoldUp		0.1094	0.1707	0.1707	0.1707	
32	Water HoldUp		---	---	---	---	
33	Friction Gradient (kPa/m)		3.938e-002	3.558e-002	3.559e-002	3.559e-002	
34	Static Gradient (kPa/m)		-1.362e-002	1.942e-002	1.942e-002	1.942e-002	
35	Accel. Gradient (kPa/m)		7.711e-006	1.647e-005	1.647e-005	1.647e-005	
36	Liq. Reynolds		1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *	
37	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
38	Liquid Velocity (m/s)		0.2084	0.2084	0.2084	0.2084	
39	Vapour Velocity (m/s)		5.242	5.242	5.243	5.243	
40	Deposit Thickness (mm)		---	---	---	---	
41	Deposit Volume (m3)		---	---	---	---	
42	Length (m)		5622	5628	5634	5640	
43	Elevation (m)		0.5780	0.6500	0.3780	0.1060	
44	Pressure (bar)		77.67	77.66	77.66	77.66	
45	Temperature (C)		54.72	54.72	54.72	54.72	
46	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
47	Heat Transferred (kJ/h-m)		538.6	538.5	538.6	538.5	
48	Flow Regime		Transition	Transition	Transition	Transition	
49	Water Flow Regime						
50	Liquid HoldUp		0.1707	0.1707	9.599e-002	9.599e-002	
51	Water HoldUp		---	---	---	---	
52	Friction Gradient (kPa/m)		3.559e-002	3.559e-002	4.125e-002	4.125e-002	
53	Static Gradient (kPa/m)		1.942e-002	1.942e-002	-5.246e-002	-5.246e-002	
54	Accel. Gradient (kPa/m)		1.647e-005	1.648e-005	-3.357e-006	-3.357e-006	
55	Liq. Reynolds		1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *	
56	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
57	Liquid Velocity (m/s)		0.2084	0.2084	0.2084	0.2084	
58	Vapour Velocity (m/s)		5.243	5.243	5.243	5.243	
59	Deposit Thickness (mm)		---	---	---	---	
60	Deposit Volume (m3)		---	---	---	---	
61	Length (m)		5646	5652	5658	5671	
62	Elevation (m)		-0.1660	-0.4380	-0.7100	-0.6880	
63	Pressure (bar)		77.66	77.67	77.67	77.66	
64	Temperature (C)		54.72	54.71	54.71	54.71	
65	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
66	Heat Transferred (kJ/h-m)		538.5	538.5	538.4	538.2	
67	Flow Regime		Transition	Transition	Transition	Transition	
68	Water Flow Regime						
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 56 of 116	

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:09:19 2022
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	9.599e-002	9.599e-002	9.599e-002	0.1557
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	4.125e-002	4.125e-002	4.125e-002	3.598e-002
16	Static Gradient (kPa/m)	-5.246e-002	-5.246e-002	-5.246e-002	2.505e-003
17	Accel. Gradient (kPa/m)	-3.358e-006	-3.358e-006	-3.358e-006	1.153e-005
18	Liq. Reynolds	1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *
19	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
20	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
21	Vapour Velocity (m/s)	5.243	5.243	5.243	5.243
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	5684	5698	5711	5725
25	Elevation (m)	-0.6660	-0.6440	-0.6220	-0.6000
26	Pressure (bar)	77.66	77.65	77.65	77.64
27	Temperature (C)	54.70	54.69	54.68	54.68
28	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
29	Heat Transferred (kJ/h-m)	538.0	537.9	537.7	537.5
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1557	0.1557	0.1557	0.1557
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	3.598e-002	3.598e-002	3.599e-002	3.599e-002
35	Static Gradient (kPa/m)	2.504e-003	2.504e-003	2.504e-003	2.504e-003
36	Accel. Gradient (kPa/m)	1.153e-005	1.153e-005	1.153e-005	1.153e-005
37	Liq. Reynolds	1.339e+005 *	1.338e+005 *	1.338e+005 *	1.338e+005 *
38	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
39	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
40	Vapour Velocity (m/s)	5.243	5.243	5.243	5.244
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	5738	5751	5765	5778
44	Elevation (m)	-2.080	-3.560	-5.040	-6.520
45	Pressure (bar)	77.65	77.66	77.67	77.67
46	Temperature (C)	54.68	54.68	54.68	54.68
47	Vapour Fraction	0.9341	0.9341	0.9341	0.9340
48	Heat Transferred (kJ/h-m)	537.7	537.7	537.8	537.8
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	7.127e-002	7.128e-002	7.128e-002	7.128e-002
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	4.723e-002	4.722e-002	4.722e-002	4.721e-002
54	Static Gradient (kPa/m)	-0.1107	-0.1107	-0.1107	-0.1108
55	Accel. Gradient (kPa/m)	-1.902e-005	-1.902e-005	-1.902e-005	-1.902e-005
56	Liq. Reynolds	1.338e+005 *	1.338e+005 *	1.338e+005 *	1.339e+005 *
57	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
58	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
59	Vapour Velocity (m/s)	5.243	5.242	5.242	5.241
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	5792	5805	5818	5832
63	Elevation (m)	-8.000	-8.480	-8.960	-9.440
64	Pressure (bar)	77.68	77.68	77.68	77.68
65	Temperature (C)	54.68	54.68	54.67	54.67
66	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
67	Heat Transferred (kJ/h-m)	537.8	537.6	537.5	537.5
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 57 of 116

1			Case Name:	Simulation - 331MMSCFD_80barg.hsc								
2			Unit Set:	Mixed								
3	 Epic Atlantic Limited		Date/Time:	Mon Jun 13 15:09:19 2022								
4												
5												
6	Pipe Segment: BBOU-ELEP Pipeline (continued)											
7												
8												
9	PROFILES											
10												
11	Pipe Table											
12												
13	Water Flow Regime											
14	Liquid HoldUp	7.129e-002	9.974e-002	9.974e-002	9.974e-002							
15	Water HoldUp	---	---	---	---							
16	Friction Gradient (kPa/m)	4.721e-002	4.065e-002	4.065e-002	4.065e-002							
17	Static Gradient (kPa/m)	-0.1108	-4.223e-002	-4.224e-002	-4.224e-002							
18	Accel. Gradient (kPa/m)	-1.902e-005	-4.748e-007	-4.754e-007	-4.759e-007							
19	Liq. Reynolds	1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *							
20	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *							
21	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084							
22	Vapour Velocity (m/s)	5.240	5.240	5.240	5.240							
23	Deposit Thickness (mm)	---	---	---	---							
24	Deposit Volume (m3)	---	---	---	---							
25	Length (m)	5845	5859	5872	5885							
26	Elevation (m)	-9.920	-10.40	-10.64	-10.88							
27	Pressure (bar)	77.68	77.68	77.68	77.68							
28	Temperature (C)	54.66	54.66	54.65	54.65							
29	Vapour Fraction	0.9340	0.9340	0.9340	0.9340							
30	Heat Transferred (kJ/h-m)	537.4	537.3	537.1	537.0							
31	Flow Regime	Transition	Transition	Transition	Transition							
32	Water Flow Regime											
33	Liquid HoldUp	9.974e-002	9.974e-002	0.1067	0.1067							
34	Water HoldUp	---	---	---	---							
35	Friction Gradient (kPa/m)	4.065e-002	4.064e-002	3.969e-002	3.969e-002							
36	Static Gradient (kPa/m)	-4.224e-002	-4.224e-002	-2.189e-002	-2.189e-002							
37	Accel. Gradient (kPa/m)	-4.764e-007	-4.769e-007	5.326e-006	5.326e-006							
38	Liq. Reynolds	1.339e+005 *	1.338e+005 *	1.338e+005 *	1.338e+005 *							
39	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *							
40	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084							
41	Vapour Velocity (m/s)	5.240	5.240	5.240	5.240							
42	Deposit Thickness (mm)	---	---	---	---							
43	Deposit Volume (m3)	---	---	---	---							
44	Length (m)	5899	5912	5926	5939							
45	Elevation (m)	-11.12	-11.36	-11.60	-9.276							
46	Pressure (bar)	77.68	77.67	77.67	77.60							
47	Temperature (C)	54.64	54.64	54.63	54.60							
48	Vapour Fraction	0.9340	0.9340	0.9340	0.9340							
49	Heat Transferred (kJ/h-m)	536.9	536.8	536.6	532.0							
50	Flow Regime	Transition	Transition	Transition	Transition							
51	Water Flow Regime											
52	Liquid HoldUp	0.1067	0.1067	0.1067	0.3934							
53	Water HoldUp	---	---	---	---							
54	Friction Gradient (kPa/m)	3.969e-002	3.969e-002	3.969e-002	3.109e-002							
55	Static Gradient (kPa/m)	-2.189e-002	-2.189e-002	-2.189e-002	0.5198							
56	Accel. Gradient (kPa/m)	5.326e-006	5.326e-006	5.327e-006	1.652e-004							
57	Liq. Reynolds	1.338e+005 *	1.338e+005 *	1.338e+005 *	1.338e+005 *							
58	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.221e+007 *							
59	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084							
60	Vapour Velocity (m/s)	5.240	5.240	5.240	5.245							
61	Deposit Thickness (mm)	---	---	---	---							
62	Deposit Volume (m3)	---	---	---	---							
63	Length (m)	5952	5966	5979	5993							
64	Elevation (m)	-6.952	-4.628	-2.304	2.000e-002							
65	Pressure (bar)	77.52	77.45	77.38	77.30							
66	Temperature (C)	54.57	54.54	54.51	54.48							
67	Vapour Fraction	0.9341	0.9341	0.9341	0.9341							
68	Heat Transferred (kJ/h-m)	531.4	530.7	530.0	529.3							
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 58 of 116							

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:09:19 2022		
4					
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.3932	0.3929	0.3927	0.3925
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	3.112e-002	3.115e-002	3.117e-002	3.120e-002
18	Static Gradient (kPa/m)	0.5196	0.5193	0.5191	0.5188
19	Accel. Gradient (kPa/m)	1.654e-004	1.657e-004	1.659e-004	1.662e-004
20	Liq. Reynolds	1.337e+005 *	1.336e+005 *	1.336e+005 *	1.335e+005 *
21	Vap. Reynolds	1.221e+007 *	1.221e+007 *	1.222e+007 *	1.222e+007 *
22	Liquid Velocity (m/s)	0.2084	0.2083	0.2083	0.2083
23	Vapour Velocity (m/s)	5.250	5.254	5.259	5.264
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	5997	6001	6005	6009
27	Elevation (m)	0.1200	0.2200	0.3200	0.4200
28	Pressure (bar)	77.30	77.30	77.29	77.29
29	Temperature (C)	54.47	54.47	54.47	54.46
30	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
31	Heat Transferred (kJ/h-m)	533.0	532.9	532.9	532.8
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1890	0.1890	0.1890	0.1890
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	2.844e-002	2.844e-002	2.844e-002	2.844e-002
37	Static Gradient (kPa/m)	4.325e-002	4.325e-002	4.325e-002	4.325e-002
38	Accel. Gradient (kPa/m)	2.166e-005	2.166e-005	2.167e-005	2.167e-005
39	Liq. Reynolds	1.335e+005 *	1.335e+005 *	1.335e+005 *	1.335e+005 *
40	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
41	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
42	Vapour Velocity (m/s)	5.264	5.264	5.265	5.265
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	6013	6017	6021	6024
46	Elevation (m)	0.5200	0.4800	0.4400	0.4000
47	Pressure (bar)	77.29	77.29	77.29	77.29
48	Temperature (C)	54.46	54.46	54.46	54.46
49	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
50	Heat Transferred (kJ/h-m)	532.8	532.9	532.8	532.8
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1890	0.1094	0.1094	0.1094
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	2.845e-002	3.951e-002	3.951e-002	3.951e-002
56	Static Gradient (kPa/m)	4.325e-002	-1.256e-002	-1.256e-002	-1.256e-002
57	Accel. Gradient (kPa/m)	2.167e-005	8.143e-006	8.143e-006	8.143e-006
58	Liq. Reynolds	1.335e+005 *	1.335e+005 *	1.335e+005 *	1.335e+005 *
59	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
60	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
61	Vapour Velocity (m/s)	5.265	5.265	5.265	5.265
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	6028	6032	6033	6034
65	Elevation (m)	0.3600	0.3200	0.2980	0.2760
66	Pressure (bar)	77.28	77.28	77.28	77.28
67	Temperature (C)	54.45	54.45	54.45	54.45
68	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:09:19 2022				
5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Heat Transferred (kJ/h-m)	532.8	532.7	532.7	532.7		
14	Flow Regime	Transition	Transition	Transition	Transition		
15	Water Flow Regime						
16	Liquid HoldUp	0.1094	0.1094	0.1048	0.1048		
17	Water HoldUp	---	---	---	---		
18	Friction Gradient (kPa/m)	3.951e-002	3.951e-002	4.008e-002	4.008e-002		
19	Static Gradient (kPa/m)	-1.256e-002	-1.256e-002	-2.659e-002	-2.659e-002		
20	Accel. Gradient (kPa/m)	8.144e-006	8.144e-006	4.079e-006	4.079e-006		
21	Liq. Reynolds	1.335e+005 *	1.335e+005 *	1.335e+005 *	1.335e+005 *		
22	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *		
23	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
24	Vapour Velocity (m/s)	5.265	5.265	5.265	5.265		
25	Deposit Thickness (mm)	---	---	---	---		
26	Deposit Volume (m3)	---	---	---	---		
27	Length (m)	6035	6036	6037	6042		
28	Elevation (m)	0.2540	0.2320	0.2100	0.2640		
29	Pressure (bar)	77.28	77.28	77.28	77.28		
30	Temperature (C)	54.45	54.45	54.45	54.45		
31	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
32	Heat Transferred (kJ/h-m)	532.7	532.7	532.7	532.5		
33	Flow Regime	Transition	Transition	Transition	Transition		
34	Water Flow Regime						
35	Liquid HoldUp	0.1048	0.1048	0.1048	0.1685		
36	Water HoldUp	---	---	---	---		
37	Friction Gradient (kPa/m)	4.008e-002	4.008e-002	4.008e-002	3.575e-002		
38	Static Gradient (kPa/m)	-2.659e-002	-2.659e-002	-2.659e-002	1.731e-002		
39	Accel. Gradient (kPa/m)	4.079e-006	4.079e-006	4.079e-006	1.604e-005		
40	Liq. Reynolds	1.335e+005 *	1.335e+005 *	1.335e+005 *	1.335e+005 *		
41	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *		
42	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
43	Vapour Velocity (m/s)	5.265	5.265	5.265	5.265		
44	Deposit Thickness (mm)	---	---	---	---		
45	Deposit Volume (m3)	---	---	---	---		
46	Length (m)	6047	6052	6057	6062		
47	Elevation (m)	0.3180	0.3720	0.4260	0.4800		
48	Pressure (bar)	77.28	77.27	77.27	77.27		
49	Temperature (C)	54.44	54.44	54.44	54.43		
50	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
51	Heat Transferred (kJ/h-m)	532.4	532.4	532.3	532.3		
52	Flow Regime	Transition	Transition	Transition	Transition		
53	Water Flow Regime						
54	Liquid HoldUp	0.1685	0.1685	0.1685	0.1685		
55	Water HoldUp	---	---	---	---		
56	Friction Gradient (kPa/m)	3.575e-002	3.575e-002	3.576e-002	3.576e-002		
57	Static Gradient (kPa/m)	1.731e-002	1.731e-002	1.731e-002	1.731e-002		
58	Accel. Gradient (kPa/m)	1.604e-005	1.604e-005	1.604e-005	1.605e-005		
59	Liq. Reynolds	1.335e+005 *	1.335e+005 *	1.334e+005 *	1.334e+005 *		
60	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *		
61	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
62	Vapour Velocity (m/s)	5.265	5.265	5.266	5.266		
63	Deposit Thickness (mm)	---	---	---	---		
64	Deposit Volume (m3)	---	---	---	---		
65	Length (m)	6067	6072	6077	6082		
66	Elevation (m)	0.4660	0.4520	0.4380	0.4240		
67	Pressure (bar)	77.27	77.27	77.26	77.26		
68	Temperature (C)	54.43	54.43	54.43	54.42		
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 60 of 116		

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:09:19 2022
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
14	Heat Transferred (kJ/h-m)	532.3	532.2	532.2	532.1
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.1123	0.1123	0.1123	0.1123
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	3.919e-002	3.919e-002	3.919e-002	3.919e-002
20	Static Gradient (kPa/m)	-3.513e-003	-3.513e-003	-3.513e-003	-3.513e-003
21	Accel. Gradient (kPa/m)	1.079e-005	1.079e-005	1.079e-005	1.079e-005
22	Liq. Reynolds	1.334e+005 *	1.334e+005 *	1.334e+005 *	1.334e+005 *
23	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
24	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
25	Vapour Velocity (m/s)	5.266	5.266	5.266	5.266
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	6087	6092	6097	6102
29	Elevation (m)	0.4100	0.3620	0.3140	0.2660
30	Pressure (bar)	77.26	77.26	77.26	77.26
31	Temperature (C)	54.42	54.42	54.42	54.41
32	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
33	Heat Transferred (kJ/h-m)	532.1	532.0	532.0	531.9
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1123	0.1096	0.1096	0.1096
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	3.919e-002	3.949e-002	3.949e-002	3.949e-002
39	Static Gradient (kPa/m)	-3.513e-003	-1.189e-002	-1.189e-002	-1.189e-002
40	Accel. Gradient (kPa/m)	1.079e-005	8.348e-006	8.348e-006	8.348e-006
41	Liq. Reynolds	1.334e+005 *	1.334e+005 *	1.334e+005 *	1.334e+005 *
42	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
43	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
44	Vapour Velocity (m/s)	5.266	5.266	5.266	5.266
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	6107	6112	6117	6122
48	Elevation (m)	0.2180	0.1700	0.1680	0.1660
49	Pressure (bar)	77.26	77.25	77.25	77.25
50	Temperature (C)	54.41	54.41	54.41	54.40
51	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
52	Heat Transferred (kJ/h-m)	531.9	531.8	531.7	531.7
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.1096	0.1096	0.1132	0.1132
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	3.949e-002	3.949e-002	3.909e-002	3.909e-002
58	Static Gradient (kPa/m)	-1.189e-002	-1.189e-002	-5.041e-004	-5.041e-004
59	Accel. Gradient (kPa/m)	8.349e-006	8.349e-006	1.167e-005	1.167e-005
60	Liq. Reynolds	1.334e+005 *	1.334e+005 *	1.334e+005 *	1.334e+005 *
61	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
62	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
63	Vapour Velocity (m/s)	5.266	5.266	5.266	5.266
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	6127	6132	6137	6142
67	Elevation (m)	0.1640	0.1620	0.1600	0.1580
68	Pressure (bar)	77.25	77.25	77.24	77.24
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:09:19 2022		
4					
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	54.40	54.40	54.40	54.39
14	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
15	Heat Transferred (kJ/h-m)	531.6	531.6	531.5	531.4
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.1132	0.1132	0.1132	0.1132
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	3.909e-002	3.909e-002	3.909e-002	3.909e-002
21	Static Gradient (kPa/m)	-5.041e-004	-5.041e-004	-5.041e-004	-5.041e-004
22	Accel. Gradient (kPa/m)	1.167e-005	1.167e-005	1.167e-005	1.167e-005
23	Liq. Reynolds	1.334e+005 *	1.334e+005 *	1.334e+005 *	1.334e+005 *
24	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
25	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
26	Vapour Velocity (m/s)	5.266	5.267	5.267	5.267
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	6147	6152	6157	6162
30	Elevation (m)	0.1560	0.1540	0.1520	0.1500
31	Pressure (bar)	77.24	77.24	77.24	77.23
32	Temperature (C)	54.39	54.39	54.39	54.38
33	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
34	Heat Transferred (kJ/h-m)	531.4	531.3	531.3	531.2
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1132	0.1132	0.1132	0.1132
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	3.909e-002	3.909e-002	3.909e-002	3.909e-002
40	Static Gradient (kPa/m)	-5.041e-004	-5.041e-004	-5.041e-004	-5.041e-004
41	Accel. Gradient (kPa/m)	1.167e-005	1.167e-005	1.168e-005	1.168e-005
42	Liq. Reynolds	1.334e+005 *	1.334e+005 *	1.334e+005 *	1.334e+005 *
43	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
44	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
45	Vapour Velocity (m/s)	5.267	5.267	5.267	5.267
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	6167	6172	6177	6182
49	Elevation (m)	0.1660	0.1820	0.1980	0.2140
50	Pressure (bar)	77.23	77.23	77.23	77.23
51	Temperature (C)	54.38	54.38	54.37	54.37
52	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
53	Heat Transferred (kJ/h-m)	531.1	531.0	531.0	530.9
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.1575	0.1575	0.1575	0.1575
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	3.606e-002	3.606e-002	3.606e-002	3.606e-002
59	Static Gradient (kPa/m)	4.912e-003	4.912e-003	4.912e-003	4.912e-003
60	Accel. Gradient (kPa/m)	1.240e-005	1.240e-005	1.240e-005	1.240e-005
61	Liq. Reynolds	1.334e+005 *	1.334e+005 *	1.334e+005 *	1.334e+005 *
62	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
63	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
64	Vapour Velocity (m/s)	5.267	5.267	5.267	5.267
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	6187	6192	6197	6202
68	Elevation (m)	0.2300	0.2480	0.2660	0.2840
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:09:19 2022			
5										
6	Pipe Segment: BBOU-ELEP Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Pressure (bar)	77.22	77.22	77.22	77.22					
14	Temperature (C)	54.37	54.37	54.36	54.36					
15	Vapour Fraction	0.9341	0.9341	0.9341	0.9341					
16	Heat Transferred (kJ/h-m)	530.9	530.8	530.7	530.7					
17	Flow Regime	Transition	Transition	Transition	Transition					
18	Water Flow Regime									
19	Liquid HoldUp	0.1575	0.1581	0.1581	0.1581					
20	Water HoldUp	---	---	---	---					
21	Friction Gradient (kPa/m)	3.606e-002	3.604e-002	3.604e-002	3.604e-002					
22	Static Gradient (kPa/m)	4.912e-003	5.538e-003	5.538e-003	5.538e-003					
23	Accel. Gradient (kPa/m)	1.240e-005	1.258e-005	1.258e-005	1.259e-005					
24	Liq. Reynolds	1.334e+005 *	1.334e+005 *	1.334e+005 *	1.334e+005 *					
25	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *					
26	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083					
27	Vapour Velocity (m/s)	5.267	5.268	5.268	5.268					
28	Deposit Thickness (mm)	---	---	---	---					
29	Deposit Volume (m3)	---	---	---	---					
30	Length (m)	6207	6212	6217	6222					
31	Elevation (m)	0.3020	0.3200	0.3300	0.3400					
32	Pressure (bar)	77.22	77.21	77.21	77.21					
33	Temperature (C)	54.36	54.36	54.35	54.35					
34	Vapour Fraction	0.9341	0.9341	0.9341	0.9341					
35	Heat Transferred (kJ/h-m)	530.6	530.5	530.5	530.4					
36	Flow Regime	Transition	Transition	Transition	Transition					
37	Water Flow Regime									
38	Liquid HoldUp	0.1581	0.1581	0.1557	0.1557					
39	Water HoldUp	---	---	---	---					
40	Friction Gradient (kPa/m)	3.604e-002	3.604e-002	3.613e-002	3.613e-002					
41	Static Gradient (kPa/m)	5.538e-003	5.538e-003	3.009e-003	3.009e-003					
42	Accel. Gradient (kPa/m)	1.259e-005	1.259e-005	1.185e-005	1.185e-005					
43	Liq. Reynolds	1.334e+005 *	1.334e+005 *	1.334e+005 *	1.334e+005 *					
44	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *					
45	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083					
46	Vapour Velocity (m/s)	5.268	5.268	5.268	5.268					
47	Deposit Thickness (mm)	---	---	---	---					
48	Deposit Volume (m3)	---	---	---	---					
49	Length (m)	6228	6233	6238	6243					
50	Elevation (m)	0.3500	0.3600	0.3700	0.3740					
51	Pressure (bar)	77.21	77.21	77.20	77.20					
52	Temperature (C)	54.35	54.34	54.34	54.34					
53	Vapour Fraction	0.9341	0.9341	0.9341	0.9341					
54	Heat Transferred (kJ/h-m)	530.4	530.3	530.3	530.2					
55	Flow Regime	Transition	Transition	Transition	Transition					
56	Water Flow Regime									
57	Liquid HoldUp	0.1557	0.1557	0.1557	0.1540					
58	Water HoldUp	---	---	---	---					
59	Friction Gradient (kPa/m)	3.613e-002	3.613e-002	3.613e-002	3.620e-002					
60	Static Gradient (kPa/m)	3.009e-003	3.009e-003	3.009e-003	1.211e-003					
61	Accel. Gradient (kPa/m)	1.185e-005	1.185e-005	1.185e-005	1.133e-005					
62	Liq. Reynolds	1.334e+005 *	1.334e+005 *	1.333e+005 *	1.333e+005 *					
63	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *					
64	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083					
65	Vapour Velocity (m/s)	5.268	5.268	5.268	5.268					
66	Deposit Thickness (mm)	---	---	---	---					
67	Deposit Volume (m3)	---	---	---	---					
68	Length (m)	6248	6253	6258	6263					
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:09:19 2022
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Elevation (m)	0.3780	0.3820	0.3860	0.3900
14	Pressure (bar)	77.20	77.20	77.20	77.19
15	Temperature (C)	54.34	54.33	54.33	54.33
16	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
17	Heat Transferred (kJ/h-m)	530.1	530.1	530.0	530.0
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.1540	0.1540	0.1540	0.1540
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	3.620e-002	3.620e-002	3.620e-002	3.620e-002
23	Static Gradient (kPa/m)	1.211e-003	1.211e-003	1.211e-003	1.211e-003
24	Accel. Gradient (kPa/m)	1.133e-005	1.133e-005	1.133e-005	1.133e-005
25	Liq. Reynolds	1.333e+005 *	1.333e+005 *	1.333e+005 *	1.333e+005 *
26	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
27	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
28	Vapour Velocity (m/s)	5.268	5.268	5.269	5.269
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	6268	6273	6278	6283
32	Elevation (m)	0.3920	0.3940	0.3960	0.3980
33	Pressure (bar)	77.19	77.19	77.19	77.19
34	Temperature (C)	54.33	54.32	54.32	54.32
35	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
36	Heat Transferred (kJ/h-m)	529.9	529.9	529.8	529.7
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.1534	0.1534	0.1534	0.1534
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	3.623e-002	3.623e-002	3.623e-002	3.623e-002
42	Static Gradient (kPa/m)	6.038e-004	6.038e-004	6.038e-004	6.038e-004
43	Accel. Gradient (kPa/m)	1.115e-005	1.115e-005	1.116e-005	1.116e-005
44	Liq. Reynolds	1.333e+005 *	1.333e+005 *	1.333e+005 *	1.333e+005 *
45	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
46	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
47	Vapour Velocity (m/s)	5.269	5.269	5.269	5.269
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	6288	6293	6298	6303
51	Elevation (m)	0.4000	0.3960	0.3920	0.3880
52	Pressure (bar)	77.19	77.18	77.18	77.18
53	Temperature (C)	54.32	54.31	54.31	54.31
54	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
55	Heat Transferred (kJ/h-m)	529.7	529.7	529.6	529.6
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.1534	0.1130	0.1130	0.1130
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	3.623e-002	3.913e-002	3.913e-002	3.913e-002
61	Static Gradient (kPa/m)	6.038e-004	-1.007e-003	-1.007e-003	-1.007e-003
62	Accel. Gradient (kPa/m)	1.116e-005	1.155e-005	1.155e-005	1.155e-005
63	Liq. Reynolds	1.333e+005 *	1.333e+005 *	1.333e+005 *	1.333e+005 *
64	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
65	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
66	Vapour Velocity (m/s)	5.269	5.269	5.269	5.269
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1			Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:09:19 2022				
6							
7	Pipe Segment: BBOU-ELEP Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length	(m)	6308	6313	6318	6323	
14	Elevation	(m)	0.3840	0.3800	0.3800	0.3800	
15	Pressure	(bar)	77.18	77.18	77.17	77.17	
16	Temperature	(C)	54.30	54.30	54.30	54.30	
17	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
18	Heat Transferred	(kJ/h-m)	529.5	529.4	529.3	529.3	
19	Flow Regime		Transition	Transition	Transition	Transition	
20	Water Flow Regime						
21	Liquid HoldUp		0.1130	0.1130	0.1654	0.1654	
22	Water HoldUp		---	---	---	---	
23	Friction Gradient	(kPa/m)	3.913e-002	3.913e-002	3.584e-002	3.584e-002	
24	Static Gradient	(kPa/m)	-1.007e-003	-1.007e-003	0.0000	0.0000	
25	Accel. Gradient	(kPa/m)	1.155e-005	1.155e-005	1.086e-005	1.086e-005	
26	Liq. Reynolds		1.333e+005 *	1.333e+005 *	1.333e+005 *	1.333e+005 *	
27	Vap. Reynolds		1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *	
28	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083	
29	Vapour Velocity	(m/s)	5.269	5.269	5.269	5.270	
30	Deposit Thickness	(mm)	---	---	---	---	
31	Deposit Volume	(m3)	---	---	---	---	
32	Length	(m)	6328	6333	6338	6343	
33	Elevation	(m)	0.3800	0.3800	0.3800	0.3680	
34	Pressure	(bar)	77.17	77.17	77.17	77.17	
35	Temperature	(C)	54.29	54.29	54.29	54.29	
36	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
37	Heat Transferred	(kJ/h-m)	529.2	529.1	529.1	529.1	
38	Flow Regime		Transition	Transition	Transition	Transition	
39	Water Flow Regime						
40	Liquid HoldUp		0.1654	0.1654	0.1654	0.1124	
41	Water HoldUp		---	---	---	---	
42	Friction Gradient	(kPa/m)	3.584e-002	3.584e-002	3.584e-002	3.920e-002	
43	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	-3.012e-003	
44	Accel. Gradient	(kPa/m)	1.086e-005	1.086e-005	1.086e-005	1.097e-005	
45	Liq. Reynolds		1.333e+005 *	1.333e+005 *	1.333e+005 *	1.333e+005 *	
46	Vap. Reynolds		1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *	
47	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083	
48	Vapour Velocity	(m/s)	5.270	5.270	5.270	5.270	
49	Deposit Thickness	(mm)	---	---	---	---	
50	Deposit Volume	(m3)	---	---	---	---	
51	Length	(m)	6348	6353	6358	6363	
52	Elevation	(m)	0.3560	0.3440	0.3320	0.3200	
53	Pressure	(bar)	77.16	77.16	77.16	77.16	
54	Temperature	(C)	54.28	54.28	54.28	54.28	
55	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
56	Heat Transferred	(kJ/h-m)	529.0	529.0	528.9	528.9	
57	Flow Regime		Transition	Transition	Transition	Transition	
58	Water Flow Regime						
59	Liquid HoldUp		0.1124	0.1124	0.1124	0.1124	
60	Water HoldUp		---	---	---	---	
61	Friction Gradient	(kPa/m)	3.920e-002	3.920e-002	3.920e-002	3.920e-002	
62	Static Gradient	(kPa/m)	-3.012e-003	-3.012e-003	-3.012e-003	-3.012e-003	
63	Accel. Gradient	(kPa/m)	1.097e-005	1.097e-005	1.097e-005	1.097e-005	
64	Liq. Reynolds		1.333e+005 *	1.333e+005 *	1.333e+005 *	1.333e+005 *	
65	Vap. Reynolds		1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *	
66	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083	
67	Vapour Velocity	(m/s)	5.270	5.270	5.270	5.270	
68	Deposit Thickness	(mm)	---	---	---	---	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 65 of 116	

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:09:19 2022		
4					
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	6368	6373	6378	6383
15	Elevation (m)	0.3200	0.3200	0.3200	0.3200
16	Pressure (bar)	77.16	77.15	77.15	77.15
17	Temperature (C)	54.27	54.27	54.27	54.27
18	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
19	Heat Transferred (kJ/h-m)	528.7	528.7	528.6	528.6
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.1654	0.1654	0.1654	0.1654
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	3.584e-002	3.584e-002	3.584e-002	3.584e-002
25	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
26	Accel. Gradient (kPa/m)	1.086e-005	1.086e-005	1.086e-005	1.086e-005
27	Liq. Reynolds	1.333e+005 *	1.333e+005 *	1.333e+005 *	1.333e+005 *
28	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.223e+007 *
29	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
30	Vapour Velocity (m/s)	5.270	5.270	5.270	5.270
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	6388	6393	6398	6403
34	Elevation (m)	0.3200	0.3120	0.3040	0.2960
35	Pressure (bar)	77.15	77.15	77.15	77.14
36	Temperature (C)	54.26	54.26	54.26	54.25
37	Vapour Fraction	0.9341	0.9341	0.9340	0.9340
38	Heat Transferred (kJ/h-m)	528.5	528.5	528.5	528.4
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.1654	0.1127	0.1127	0.1127
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	3.584e-002	3.917e-002	3.917e-002	3.917e-002
44	Static Gradient (kPa/m)	0.0000	-2.011e-003	-2.011e-003	-2.011e-003
45	Accel. Gradient (kPa/m)	1.086e-005	1.126e-005	1.126e-005	1.127e-005
46	Liq. Reynolds	1.333e+005 *	1.333e+005 *	1.333e+005 *	1.333e+005 *
47	Vap. Reynolds	1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *
48	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
49	Vapour Velocity (m/s)	5.270	5.271	5.271	5.271
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	6408	6413	6419	6425
53	Elevation (m)	0.2880	0.2800	0.2500	0.2200
54	Pressure (bar)	77.14	77.14	77.14	77.14
55	Temperature (C)	54.25	54.25	54.25	54.24
56	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
57	Heat Transferred (kJ/h-m)	528.4	528.3	528.2	528.2
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.1127	0.1127	0.1114	0.1114
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	3.917e-002	3.917e-002	3.932e-002	3.932e-002
63	Static Gradient (kPa/m)	-2.011e-003	-2.011e-003	-6.230e-003	-6.230e-003
64	Accel. Gradient (kPa/m)	1.127e-005	1.127e-005	1.003e-005	1.003e-005
65	Liq. Reynolds	1.333e+005 *	1.333e+005 *	1.332e+005 *	1.332e+005 *
66	Vap. Reynolds	1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *
67	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
68	Vapour Velocity (m/s)	5.271	5.271	5.271	5.271
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1			Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:09:19 2022		
6					
7	Pipe Segment: BBOU-ELEP Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	6431	6437	6443	6449
16	Elevation (m)	0.1900	0.1600	0.1300	0.1360
17	Pressure (bar)	77.13	77.13	77.13	77.13
18	Temperature (C)	54.24	54.24	54.23	54.23
19	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
20	Heat Transferred (kJ/h-m)	528.1	528.0	528.0	527.8
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.1114	0.1114	0.1113	0.1543
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	3.932e-002	3.932e-002	3.932e-002	3.621e-002
26	Static Gradient (kPa/m)	-6.230e-003	-6.230e-003	-6.230e-003	1.515e-003
27	Accel. Gradient (kPa/m)	1.003e-005	1.003e-005	1.003e-005	1.144e-005
28	Liq. Reynolds	1.332e+005 *	1.332e+005 *	1.332e+005 *	1.332e+005 *
29	Vap. Reynolds	1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *
30	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
31	Vapour Velocity (m/s)	5.271	5.271	5.271	5.271
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	6455	6461	6467	6473
35	Elevation (m)	0.1420	0.1480	0.1540	0.1600
36	Pressure (bar)	77.13	77.12	77.12	77.12
37	Temperature (C)	54.23	54.22	54.22	54.22
38	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
39	Heat Transferred (kJ/h-m)	527.8	527.7	527.6	527.6
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.1543	0.1543	0.1543	0.1543
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	3.621e-002	3.621e-002	3.621e-002	3.621e-002
45	Static Gradient (kPa/m)	1.515e-003	1.515e-003	1.515e-003	1.514e-003
46	Accel. Gradient (kPa/m)	1.144e-005	1.144e-005	1.144e-005	1.144e-005
47	Liq. Reynolds	1.332e+005 *	1.332e+005 *	1.332e+005 *	1.332e+005 *
48	Vap. Reynolds	1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *
49	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
50	Vapour Velocity (m/s)	5.271	5.271	5.272	5.272
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	6476	6479	6482	6485
54	Elevation (m)	0.1660	0.1720	0.1780	0.1840
55	Pressure (bar)	77.12	77.12	77.11	77.11
56	Temperature (C)	54.22	54.22	54.21	54.21
57	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
58	Heat Transferred (kJ/h-m)	527.5	527.5	527.5	527.4
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.1557	0.1557	0.1557	0.1557
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	3.616e-002	3.616e-002	3.616e-002	3.616e-002
64	Static Gradient (kPa/m)	3.047e-003	3.047e-003	3.047e-003	3.047e-003
65	Accel. Gradient (kPa/m)	1.189e-005	1.189e-005	1.189e-005	1.189e-005
66	Liq. Reynolds	1.332e+005 *	1.332e+005 *	1.332e+005 *	1.332e+005 *
67	Vap. Reynolds	1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *
68	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
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1			Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:09:19 2022		
6					
7	Pipe Segment: BBOU-ELEP Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Velocity (m/s)	5.272	5.272	5.272	5.272
14	Deposit Thickness (mm)	---	---	---	---
15	Deposit Volume (m3)	---	---	---	---
16	Length (m)	6488	6493	6498	6503
17	Elevation (m)	0.1900	0.2140	0.2380	0.2620
18	Pressure (bar)	77.11	77.11	77.11	77.11
19	Temperature (C)	54.21	54.21	54.20	54.20
20	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
21	Heat Transferred (kJ/h-m)	527.4	527.3	527.3	527.2
22	Flow Regime	Transition	Transition	Transition	Transition
23	Water Flow Regime				
24	Liquid HoldUp	0.1557	0.1597	0.1597	0.1597
25	Water HoldUp	---	---	---	---
26	Friction Gradient (kPa/m)	3.616e-002	3.601e-002	3.601e-002	3.601e-002
27	Static Gradient (kPa/m)	3.047e-003	7.432e-003	7.432e-003	7.432e-003
28	Accel. Gradient (kPa/m)	1.189e-005	1.318e-005	1.318e-005	1.318e-005
29	Liq. Reynolds	1.332e+005 *	1.332e+005 *	1.332e+005 *	1.332e+005 *
30	Vap. Reynolds	1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *
31	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
32	Vapour Velocity (m/s)	5.272	5.272	5.272	5.272
33	Deposit Thickness (mm)	---	---	---	---
34	Deposit Volume (m3)	---	---	---	---
35	Length (m)	6508	6513	6515	6517
36	Elevation (m)	0.2860	0.3100	0.2960	0.2820
37	Pressure (bar)	77.10	77.10	77.10	77.10
38	Temperature (C)	54.20	54.20	54.20	54.19
39	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
40	Heat Transferred (kJ/h-m)	527.1	527.1	527.1	527.1
41	Flow Regime	Transition	Transition	Transition	Transition
42	Water Flow Regime				
43	Liquid HoldUp	0.1597	0.1597	0.1106	0.1106
44	Water HoldUp	---	---	---	---
45	Friction Gradient (kPa/m)	3.601e-002	3.601e-002	3.942e-002	3.942e-002
46	Static Gradient (kPa/m)	7.432e-003	7.432e-003	-8.704e-003	-8.704e-003
47	Accel. Gradient (kPa/m)	1.318e-005	1.318e-005	9.319e-006	9.320e-006
48	Liq. Reynolds	1.332e+005 *	1.332e+005 *	1.332e+005 *	1.332e+005 *
49	Vap. Reynolds	1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *
50	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
51	Vapour Velocity (m/s)	5.272	5.272	5.272	5.272
52	Deposit Thickness (mm)	---	---	---	---
53	Deposit Volume (m3)	---	---	---	---
54	Length (m)	6519	6521	6523	6526
55	Elevation (m)	0.2680	0.2540	0.2400	0.2180
56	Pressure (bar)	77.10	77.10	77.10	77.10
57	Temperature (C)	54.19	54.19	54.19	54.19
58	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
59	Heat Transferred (kJ/h-m)	527.1	527.1	527.0	527.0
60	Flow Regime	Transition	Transition	Transition	Transition
61	Water Flow Regime				
62	Liquid HoldUp	0.1106	0.1106	0.1106	0.1104
63	Water HoldUp	---	---	---	---
64	Friction Gradient (kPa/m)	3.942e-002	3.942e-002	3.942e-002	3.944e-002
65	Static Gradient (kPa/m)	-8.704e-003	-8.704e-003	-8.704e-003	-9.113e-003
66	Accel. Gradient (kPa/m)	9.320e-006	9.320e-006	9.320e-006	9.201e-006
67	Liq. Reynolds	1.332e+005 *	1.332e+005 *	1.332e+005 *	1.332e+005 *
68	Vap. Reynolds	1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *
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1			Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:09:19 2022				
6							
7	Pipe Segment: BBOU-ELEP Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083	
14	Vapour Velocity	(m/s)	5.272	5.272	5.272	5.272	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	6529	6532	6535	6538	
18	Elevation	(m)	0.1960	0.1740	0.1520	0.1300	
19	Pressure	(bar)	77.10	77.10	77.09	77.09	
20	Temperature	(C)	54.19	54.19	54.19	54.18	
21	Vapour Fraction		0.9340	0.9340	0.9340	0.9340	
22	Heat Transferred	(kJ/h-m)	527.0	526.9	526.9	526.9	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.1104	0.1104	0.1104	0.1104	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	3.944e-002	3.944e-002	3.944e-002	3.944e-002	
28	Static Gradient	(kPa/m)	-9.113e-003	-9.113e-003	-9.113e-003	-9.113e-003	
29	Accel. Gradient	(kPa/m)	9.201e-006	9.201e-006	9.202e-006	9.202e-006	
30	Liq. Reynolds		1.332e+005 *	1.332e+005 *	1.332e+005 *	1.332e+005 *	
31	Vap. Reynolds		1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *	
32	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083	
33	Vapour Velocity	(m/s)	5.273	5.273	5.273	5.273	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	6543	6548	6553	6558	
37	Elevation	(m)	0.1360	0.1420	0.1480	0.1540	
38	Pressure	(bar)	77.09	77.09	77.09	77.09	
39	Temperature	(C)	54.18	54.18	54.18	54.17	
40	Vapour Fraction		0.9340	0.9340	0.9340	0.9340	
41	Heat Transferred	(kJ/h-m)	526.8	526.7	526.6	526.6	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.1545	0.1545	0.1545	0.1545	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	3.621e-002	3.621e-002	3.621e-002	3.621e-002	
47	Static Gradient	(kPa/m)	1.819e-003	1.819e-003	1.819e-003	1.819e-003	
48	Accel. Gradient	(kPa/m)	1.154e-005	1.154e-005	1.154e-005	1.154e-005	
49	Liq. Reynolds		1.332e+005 *	1.332e+005 *	1.332e+005 *	1.332e+005 *	
50	Vap. Reynolds		1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *	
51	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083	
52	Vapour Velocity	(m/s)	5.273	5.273	5.273	5.273	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	6563	6568	6573	6578	
56	Elevation	(m)	0.1600	0.1680	0.1760	0.1840	
57	Pressure	(bar)	77.08	77.08	77.08	77.08	
58	Temperature	(C)	54.17	54.17	54.17	54.16	
59	Vapour Fraction		0.9340	0.9340	0.9340	0.9340	
60	Heat Transferred	(kJ/h-m)	526.5	526.5	526.4	526.4	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.1545	0.1551	0.1551	0.1551	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	3.621e-002	3.619e-002	3.619e-002	3.619e-002	
66	Static Gradient	(kPa/m)	1.819e-003	2.431e-003	2.431e-003	2.431e-003	
67	Accel. Gradient	(kPa/m)	1.154e-005	1.172e-005	1.172e-005	1.172e-005	
68	Liq. Reynolds		1.332e+005 *	1.332e+005 *	1.332e+005 *	1.332e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:09:19 2022		
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *
14	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
15	Vapour Velocity (m/s)	5.273	5.273	5.273	5.273
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	6583	6588	6593	6598
19	Elevation (m)	0.1920	0.2000	0.1840	0.1680
20	Pressure (bar)	77.08	77.07	77.07	77.07
21	Temperature (C)	54.16	54.16	54.15	54.15
22	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
23	Heat Transferred (kJ/h-m)	526.3	526.2	526.2	526.2
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1551	0.1551	0.1120	0.1120
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	3.619e-002	3.619e-002	3.926e-002	3.926e-002
29	Static Gradient (kPa/m)	2.431e-003	2.431e-003	-4.008e-003	-4.008e-003
30	Accel. Gradient (kPa/m)	1.172e-005	1.172e-005	1.070e-005	1.070e-005
31	Liq. Reynolds	1.332e+005 *	1.332e+005 *	1.332e+005 *	1.331e+005 *
32	Vap. Reynolds	1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *
33	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
34	Vapour Velocity (m/s)	5.273	5.273	5.273	5.274
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	6603	6608	6613	6618
38	Elevation (m)	0.1520	0.1360	0.1200	0.1100
39	Pressure (bar)	77.07	77.07	77.07	77.06
40	Temperature (C)	54.15	54.15	54.14	54.14
41	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
42	Heat Transferred (kJ/h-m)	526.1	526.1	526.0	526.0
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1120	0.1120	0.1120	0.1125
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	3.926e-002	3.926e-002	3.926e-002	3.921e-002
48	Static Gradient (kPa/m)	-4.008e-003	-4.008e-003	-4.008e-003	-2.511e-003
49	Accel. Gradient (kPa/m)	1.070e-005	1.070e-005	1.070e-005	1.114e-005
50	Liq. Reynolds	1.331e+005 *	1.331e+005 *	1.331e+005 *	1.331e+005 *
51	Vap. Reynolds	1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *
52	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
53	Vapour Velocity (m/s)	5.274	5.274	5.274	5.274
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	6623	6628	6633	6638
57	Elevation (m)	1.000e-001	9.000e-002	8.000e-002	7.000e-002
58	Pressure (bar)	77.06	77.06	77.06	77.06
59	Temperature (C)	54.14	54.14	54.13	54.13
60	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
61	Heat Transferred (kJ/h-m)	525.9	525.8	525.8	525.7
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1125	0.1125	0.1125	0.1125
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	3.921e-002	3.921e-002	3.922e-002	3.922e-002
67	Static Gradient (kPa/m)	-2.511e-003	-2.511e-003	-2.511e-003	-2.511e-003
68	Accel. Gradient (kPa/m)	1.114e-005	1.115e-005	1.115e-005	1.115e-005
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:09:19 2022			
5										
6	Pipe Segment: BBOU-ELEP Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Liq. Reynolds		1.331e+005 *	1.331e+005 *	1.331e+005 *	1.331e+005 *				
14	Vap. Reynolds		1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *				
15	Liquid Velocity (m/s)		0.2083	0.2083	0.2083	0.2083				
16	Vapour Velocity (m/s)		5.274	5.274	5.274	5.274				
17	Deposit Thickness (mm)		---	---	---	---				
18	Deposit Volume (m3)		---	---	---	---				
19	Length (m)		6643	6648	6653	6658				
20	Elevation (m)		6.800e-002	6.600e-002	6.400e-002	6.200e-002				
21	Pressure (bar)		77.05	77.05	77.05	77.05				
22	Temperature (C)		54.13	54.13	54.12	54.12				
23	Vapour Fraction		0.9340	0.9340	0.9340	0.9340				
24	Heat Transferred (kJ/h-m)		525.7	525.6	525.6	525.5				
25	Flow Regime		Transition	Transition	Transition	Transition				
26	Water Flow Regime									
27	Liquid HoldUp		0.1131	0.1131	0.1131	0.1131				
28	Water HoldUp		---	---	---	---				
29	Friction Gradient (kPa/m)		3.915e-002	3.915e-002	3.915e-002	3.915e-002				
30	Static Gradient (kPa/m)		-5.036e-004	-5.036e-004	-5.036e-004	-5.036e-004				
31	Accel. Gradient (kPa/m)		1.174e-005	1.174e-005	1.174e-005	1.174e-005				
32	Liq. Reynolds		1.331e+005 *	1.331e+005 *	1.331e+005 *	1.331e+005 *				
33	Vap. Reynolds		1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *				
34	Liquid Velocity (m/s)		0.2083	0.2083	0.2083	0.2083				
35	Vapour Velocity (m/s)		5.274	5.274	5.274	5.274				
36	Deposit Thickness (mm)		---	---	---	---				
37	Deposit Volume (m3)		---	---	---	---				
38	Length (m)		6663							
39	Elevation (m)		6.000e-002							
40	Pressure (bar)		77.05							
41	Temperature (C)		54.12							
42	Vapour Fraction		0.9340							
43	Heat Transferred (kJ/h-m)		525.4							
44	Flow Regime		Transition							
45	Water Flow Regime									
46	Liquid HoldUp		0.1131							
47	Water HoldUp		---							
48	Friction Gradient (kPa/m)		3.915e-002							
49	Static Gradient (kPa/m)		-5.036e-004							
50	Accel. Gradient (kPa/m)		1.174e-005							
51	Liq. Reynolds		1.331e+005 *							
52	Vap. Reynolds		1.223e+007 *							
53	Liquid Velocity (m/s)		0.2083							
54	Vapour Velocity (m/s)		5.274							
55	Deposit Thickness (mm)		---							
56	Deposit Volume (m3)		---							
57	Erosion									
58										
59	Model									
60										
61	Erosion Model:		API-RP-14E Continuous service		Empirical Constant: 100.0					
62										
63	Erosion Table									
64	Length (m)		0.0000	509.2	1018	1528	2037			
65	Bulk Density (kg/m3)		82.74	82.66	82.57	82.49	82.40			
66	Bulk Velocity (m/s)		5.360	5.366	5.371	5.377	5.383			
67	Erosion Velocity (m/s)		13.41	13.42	13.42	13.43	13.44			
68	Length (m)		2546	2551	2556	2561	2566			
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2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:09:19 2022		
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Bulk Density	(kg/m3)	82.31	82.31	82.31	82.31
14	Bulk Velocity	(m/s)	5.389	5.389	5.389	5.389
15	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
16	Length	(m)	2571	2576	2581	2591
17	Bulk Density	(kg/m3)	82.30	82.30	82.30	82.30
18	Bulk Velocity	(m/s)	5.389	5.389	5.389	5.389
19	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
20	Length	(m)	2596	2601	2606	2611
21	Bulk Density	(kg/m3)	82.30	82.30	82.30	82.29
22	Bulk Velocity	(m/s)	5.390	5.390	5.390	5.390
23	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
24	Length	(m)	2621	2624	2627	2633
25	Bulk Density	(kg/m3)	82.29	82.29	82.29	82.29
26	Bulk Velocity	(m/s)	5.390	5.390	5.390	5.390
27	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
28	Length	(m)	2636	2640	2644	2652
29	Bulk Density	(kg/m3)	82.29	82.29	82.29	82.29
30	Bulk Velocity	(m/s)	5.390	5.390	5.390	5.390
31	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
32	Length	(m)	2656	2661	2666	2671
33	Bulk Density	(kg/m3)	82.29	82.28	82.28	82.28
34	Bulk Velocity	(m/s)	5.390	5.390	5.390	5.391
35	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
36	Length	(m)	2681	2686	2691	2701
37	Bulk Density	(kg/m3)	82.28	82.28	82.28	82.28
38	Bulk Velocity	(m/s)	5.391	5.391	5.391	5.391
39	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
40	Length	(m)	2706	2711	2716	2726
41	Bulk Density	(kg/m3)	82.28	82.27	82.27	82.27
42	Bulk Velocity	(m/s)	5.391	5.391	5.391	5.391
43	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
44	Length	(m)	2731	2736	2741	2751
45	Bulk Density	(kg/m3)	82.27	82.27	82.27	82.27
46	Bulk Velocity	(m/s)	5.391	5.391	5.391	5.392
47	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
48	Length	(m)	2756	2761	2766	2771
49	Bulk Density	(kg/m3)	82.27	82.26	82.26	82.26
50	Bulk Velocity	(m/s)	5.392	5.392	5.392	5.392
51	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
52	Length	(m)	2781	2786	2791	2796
53	Bulk Density	(kg/m3)	82.26	82.26	82.26	82.26
54	Bulk Velocity	(m/s)	5.392	5.392	5.392	5.392
55	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
56	Length	(m)	2806	2811	2816	2826
57	Bulk Density	(kg/m3)	82.26	82.26	82.25	82.25
58	Bulk Velocity	(m/s)	5.392	5.392	5.392	5.393
59	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
60	Length	(m)	2831	2836	2841	2851
61	Bulk Density	(kg/m3)	82.25	82.25	82.25	82.25
62	Bulk Velocity	(m/s)	5.393	5.393	5.393	5.393
63	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
64	Length	(m)	2856	2861	2866	2876
65	Bulk Density	(kg/m3)	82.25	82.25	82.25	82.25
66	Bulk Velocity	(m/s)	5.393	5.393	5.393	5.393
67	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
68	Length	(m)	2881	2886	2891	2901
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5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Bulk Density (kg/m3)	82.25	82.24	82.24	82.24	82.24
14	Bulk Velocity (m/s)	5.393	5.393	5.393	5.393	5.393
15	Erosion Velocity (m/s)	13.45	13.45	13.45	13.45	13.45
16	Length (m)	2906	2911	2916	2921	2926
17	Bulk Density (kg/m3)	82.24	82.23	82.23	82.23	82.23
18	Bulk Velocity (m/s)	5.394	5.394	5.394	5.394	5.394
19	Erosion Velocity (m/s)	13.45	13.45	13.45	13.45	13.45
20	Length (m)	2931	2936	2941	2946	2951
21	Bulk Density (kg/m3)	82.22	82.22	82.22	82.22	82.22
22	Bulk Velocity (m/s)	5.394	5.394	5.394	5.395	5.395
23	Erosion Velocity (m/s)	13.45	13.45	13.45	13.45	13.45
24	Length (m)	2956	2961	2966	2971	2976
25	Bulk Density (kg/m3)	82.22	82.22	82.22	82.22	82.22
26	Bulk Velocity (m/s)	5.395	5.395	5.395	5.395	5.395
27	Erosion Velocity (m/s)	13.45	13.45	13.45	13.45	13.45
28	Length (m)	2981	2986	2991	2996	3001
29	Bulk Density (kg/m3)	82.22	82.21	82.21	82.21	82.21
30	Bulk Velocity (m/s)	5.395	5.395	5.395	5.395	5.395
31	Erosion Velocity (m/s)	13.45	13.45	13.45	13.45	13.45
32	Length (m)	3006	3011	3016	3021	3026
33	Bulk Density (kg/m3)	82.21	82.21	82.21	82.21	82.21
34	Bulk Velocity (m/s)	5.395	5.395	5.395	5.395	5.396
35	Erosion Velocity (m/s)	13.45	13.45	13.45	13.45	13.45
36	Length (m)	3031	3036	3041	3046	3051
37	Bulk Density (kg/m3)	82.21	82.20	82.20	82.20	82.20
38	Bulk Velocity (m/s)	5.396	5.396	5.396	5.396	5.396
39	Erosion Velocity (m/s)	13.45	13.45	13.45	13.45	13.46
40	Length (m)	3056	3061	3066	3071	3076
41	Bulk Density (kg/m3)	82.20	82.20	82.20	82.20	82.20
42	Bulk Velocity (m/s)	5.396	5.396	5.396	5.396	5.396
43	Erosion Velocity (m/s)	13.46	13.46	13.46	13.46	13.46
44	Length (m)	3081	3086	3091	3096	3101
45	Bulk Density (kg/m3)	82.20	82.19	82.19	82.19	82.19
46	Bulk Velocity (m/s)	5.396	5.396	5.396	5.396	5.396
47	Erosion Velocity (m/s)	13.46	13.46	13.46	13.46	13.46
48	Length (m)	3106	3111	3116	3121	3126
49	Bulk Density (kg/m3)	82.19	82.19	82.19	82.19	82.19
50	Bulk Velocity (m/s)	5.397	5.397	5.397	5.397	5.397
51	Erosion Velocity (m/s)	13.46	13.46	13.46	13.46	13.46
52	Length (m)	3131	3136	3141	3146	3151
53	Bulk Density (kg/m3)	82.19	82.18	82.18	82.18	82.18
54	Bulk Velocity (m/s)	5.397	5.397	5.397	5.397	5.397
55	Erosion Velocity (m/s)	13.46	13.46	13.46	13.46	13.46
56	Length (m)	3156	3161	3166	3171	3176
57	Bulk Density (kg/m3)	82.18	82.18	82.18	82.18	82.18
58	Bulk Velocity (m/s)	5.397	5.397	5.397	5.397	5.397
59	Erosion Velocity (m/s)	13.46	13.46	13.46	13.46	13.46
60	Length (m)	3181	3186	3191	3196	3201
61	Bulk Density (kg/m3)	82.18	82.17	82.17	82.17	82.17
62	Bulk Velocity (m/s)	5.398	5.398	5.398	5.398	5.398
63	Erosion Velocity (m/s)	13.46	13.46	13.46	13.46	13.46
64	Length (m)	3206	3212	3218	3224	3230
65	Bulk Density (kg/m3)	82.17	82.17	82.17	82.17	82.17
66	Bulk Velocity (m/s)	5.398	5.398	5.398	5.398	5.398
67	Erosion Velocity (m/s)	13.46	13.46	13.46	13.46	13.46
68	Length (m)	3236	3239	3242	3245	3248
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Bulk Density	(kg/m3)	82.16	82.16	82.16	82.16
14	Bulk Velocity	(m/s)	5.398	5.398	5.398	5.398
15	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
16	Length	(m)	3251	3257	3263	3275
17	Bulk Density	(kg/m3)	82.16	82.16	82.16	82.16
18	Bulk Velocity	(m/s)	5.398	5.399	5.399	5.399
19	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
20	Length	(m)	3281	3286	3291	3301
21	Bulk Density	(kg/m3)	82.15	82.15	82.15	82.15
22	Bulk Velocity	(m/s)	5.399	5.399	5.399	5.399
23	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
24	Length	(m)	3306	3311	3316	3326
25	Bulk Density	(kg/m3)	82.15	82.15	82.15	82.15
26	Bulk Velocity	(m/s)	5.399	5.399	5.399	5.399
27	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
28	Length	(m)	3331	3336	3341	3351
29	Bulk Density	(kg/m3)	82.15	82.14	82.14	82.14
30	Bulk Velocity	(m/s)	5.400	5.400	5.400	5.400
31	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
32	Length	(m)	3356	3361	3366	3376
33	Bulk Density	(kg/m3)	82.14	82.14	82.14	82.14
34	Bulk Velocity	(m/s)	5.400	5.400	5.400	5.400
35	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
36	Length	(m)	3381	3386	3391	3401
37	Bulk Density	(kg/m3)	82.13	82.13	82.13	82.13
38	Bulk Velocity	(m/s)	5.400	5.400	5.400	5.400
39	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
40	Length	(m)	3406	3411	3416	3426
41	Bulk Density	(kg/m3)	82.13	82.13	82.13	82.13
42	Bulk Velocity	(m/s)	5.400	5.401	5.401	5.401
43	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
44	Length	(m)	3431	3436	3441	3451
45	Bulk Density	(kg/m3)	82.13	82.13	82.13	82.12
46	Bulk Velocity	(m/s)	5.401	5.401	5.401	5.401
47	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
48	Length	(m)	3456	3461	3466	3476
49	Bulk Density	(kg/m3)	82.12	82.12	82.12	82.12
50	Bulk Velocity	(m/s)	5.401	5.401	5.401	5.401
51	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
52	Length	(m)	3481	3486	3491	3501
53	Bulk Density	(kg/m3)	82.11	82.11	82.11	82.11
54	Bulk Velocity	(m/s)	5.402	5.402	5.402	5.402
55	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
56	Length	(m)	3506	3511	3516	3526
57	Bulk Density	(kg/m3)	82.11	82.11	82.11	82.11
58	Bulk Velocity	(m/s)	5.402	5.402	5.402	5.402
59	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
60	Length	(m)	3531	3536	3541	3551
61	Bulk Density	(kg/m3)	82.10	82.10	82.10	82.10
62	Bulk Velocity	(m/s)	5.402	5.402	5.402	5.402
63	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
64	Length	(m)	3556	3561	3565	3573
65	Bulk Density	(kg/m3)	82.10	82.10	82.10	82.10
66	Bulk Velocity	(m/s)	5.403	5.403	5.403	5.403
67	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
68	Length	(m)	3577	3582	3587	3597
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1	Epic Atlantic Limited			Case Name: Simulation - 331MMSCFD_80barg.hsc			
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3				Unit Set: Mixed			
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	Erosion						
10							
11	Erosion Table						
12							
13	Bulk Density	(kg/m3)	82.10	82.09	82.09	82.09	82.09
14	Bulk Velocity	(m/s)	5.403	5.403	5.403	5.403	5.403
15	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46	13.46
16	Length	(m)	3602	3607	3612	3617	3622
17	Bulk Density	(kg/m3)	82.09	82.09	82.09	82.09	82.09
18	Bulk Velocity	(m/s)	5.403	5.403	5.403	5.403	5.403
19	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46	13.46
20	Length	(m)	3627	3632	3637	3642	3647
21	Bulk Density	(kg/m3)	82.09	82.09	82.08	82.08	82.08
22	Bulk Velocity	(m/s)	5.403	5.403	5.404	5.404	5.404
23	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46	13.46
24	Length	(m)	3652	3657	3662	3667	3672
25	Bulk Density	(kg/m3)	82.08	82.08	82.08	82.08	82.08
26	Bulk Velocity	(m/s)	5.404	5.404	5.404	5.404	5.404
27	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47	13.47
28	Length	(m)	3677	3682	3687	3692	3697
29	Bulk Density	(kg/m3)	82.07	82.07	82.07	82.07	82.07
30	Bulk Velocity	(m/s)	5.404	5.404	5.404	5.404	5.404
31	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47	13.47
32	Length	(m)	3702	3707	3712	3717	3722
33	Bulk Density	(kg/m3)	82.07	82.07	82.07	82.07	82.06
34	Bulk Velocity	(m/s)	5.405	5.405	5.405	5.405	5.405
35	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47	13.47
36	Length	(m)	3727	3732	3737	3742	3747
37	Bulk Density	(kg/m3)	82.06	82.06	82.06	82.06	82.06
38	Bulk Velocity	(m/s)	5.405	5.405	5.405	5.405	5.405
39	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47	13.47
40	Length	(m)	3752	3757	3761	3766	3770
41	Bulk Density	(kg/m3)	82.06	82.06	82.06	82.05	82.05
42	Bulk Velocity	(m/s)	5.405	5.405	5.405	5.405	5.406
43	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47	13.47
44	Length	(m)	3775	3780	3785	3790	3795
45	Bulk Density	(kg/m3)	82.05	82.05	82.05	82.05	82.05
46	Bulk Velocity	(m/s)	5.406	5.406	5.406	5.406	5.406
47	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47	13.47
48	Length	(m)	3800	3805	3810	3815	3820
49	Bulk Density	(kg/m3)	82.05	82.05	82.05	82.05	82.04
50	Bulk Velocity	(m/s)	5.406	5.406	5.406	5.406	5.406
51	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47	13.47
52	Length	(m)	3825	3830	3835	3840	3845
53	Bulk Density	(kg/m3)	82.04	82.04	82.04	82.04	82.04
54	Bulk Velocity	(m/s)	5.406	5.406	5.406	5.406	5.406
55	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47	13.47
56	Length	(m)	3850	3855	3860	3865	3870
57	Bulk Density	(kg/m3)	82.04	82.04	82.04	82.04	82.03
58	Bulk Velocity	(m/s)	5.407	5.407	5.407	5.407	5.407
59	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47	13.47
60	Length	(m)	3875	3880	3885	3890	3895
61	Bulk Density	(kg/m3)	82.03	82.03	82.03	82.03	82.03
62	Bulk Velocity	(m/s)	5.407	5.407	5.407	5.407	5.407
63	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47	13.47
64	Length	(m)	3900	3905	3910	3915	3920
65	Bulk Density	(kg/m3)	82.03	82.03	82.03	82.02	82.02
66	Bulk Velocity	(m/s)	5.407	5.407	5.407	5.407	5.408
67	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47	13.47
68	Length	(m)	3925	3927	3929	3931	3933
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Bulk Density	(kg/m3)	82.02	82.02	82.02	82.02
14	Bulk Velocity	(m/s)	5.408	5.408	5.408	5.407
15	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
16	Length	(m)	3935	3947	3958	3980
17	Bulk Density	(kg/m3)	82.02	82.03	82.03	82.04
18	Bulk Velocity	(m/s)	5.407	5.407	5.407	5.406
19	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
20	Length	(m)	3991	4002	4013	4036
21	Bulk Density	(kg/m3)	82.04	82.05	82.05	82.05
22	Bulk Velocity	(m/s)	5.406	5.406	5.406	5.406
23	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
24	Length	(m)	4047	4058	4069	4091
25	Bulk Density	(kg/m3)	82.05	82.05	82.05	82.05
26	Bulk Velocity	(m/s)	5.406	5.406	5.406	5.406
27	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
28	Length	(m)	4103	4114	4125	4147
29	Bulk Density	(kg/m3)	82.04	82.04	82.04	82.03
30	Bulk Velocity	(m/s)	5.406	5.406	5.407	5.407
31	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
32	Length	(m)	4158	4169	4181	4203
33	Bulk Density	(kg/m3)	82.03	82.02	82.02	82.02
34	Bulk Velocity	(m/s)	5.407	5.407	5.407	5.407
35	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
36	Length	(m)	4214	4225	4236	4259
37	Bulk Density	(kg/m3)	82.02	82.03	82.04	82.05
38	Bulk Velocity	(m/s)	5.408	5.407	5.407	5.406
39	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
40	Length	(m)	4270	4281	4292	4314
41	Bulk Density	(kg/m3)	82.06	81.98	81.89	81.73
42	Bulk Velocity	(m/s)	5.405	5.411	5.416	5.427
43	Erosion Velocity	(m/s)	13.47	13.47	13.48	13.49
44	Length	(m)	4325	4328	4331	4337
45	Bulk Density	(kg/m3)	81.65	81.65	81.64	81.64
46	Bulk Velocity	(m/s)	5.432	5.432	5.433	5.433
47	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
48	Length	(m)	4340	4343	4346	4352
49	Bulk Density	(kg/m3)	81.63	81.63	81.63	81.63
50	Bulk Velocity	(m/s)	5.433	5.433	5.433	5.433
51	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
52	Length	(m)	4355	4357	4359	4363
53	Bulk Density	(kg/m3)	81.63	81.63	81.63	81.64
54	Bulk Velocity	(m/s)	5.433	5.433	5.433	5.433
55	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
56	Length	(m)	4365	4367	4369	4373
57	Bulk Density	(kg/m3)	81.64	81.63	81.63	81.63
58	Bulk Velocity	(m/s)	5.433	5.433	5.433	5.433
59	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
60	Length	(m)	4375	4380	4385	4395
61	Bulk Density	(kg/m3)	81.63	81.63	81.63	81.63
62	Bulk Velocity	(m/s)	5.433	5.434	5.434	5.434
63	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
64	Length	(m)	4400	4405	4410	4420
65	Bulk Density	(kg/m3)	81.63	81.63	81.63	81.63
66	Bulk Velocity	(m/s)	5.434	5.434	5.434	5.434
67	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
68	Length	(m)	4425	4431	4436	4446
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13	Bulk Density	(kg/m3)	81.63	81.62	81.62	81.62
14	Bulk Velocity	(m/s)	5.434	5.434	5.434	5.434
15	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
16	Length	(m)	4451	4456	4461	4471
17	Bulk Density	(kg/m3)	81.62	81.62	81.62	81.62
18	Bulk Velocity	(m/s)	5.434	5.434	5.434	5.435
19	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
20	Length	(m)	4476	4481	4486	4496
21	Bulk Density	(kg/m3)	81.62	81.61	81.61	81.61
22	Bulk Velocity	(m/s)	5.435	5.435	5.435	5.435
23	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
24	Length	(m)	4501	4506	4511	4521
25	Bulk Density	(kg/m3)	81.61	81.61	81.61	81.60
26	Bulk Velocity	(m/s)	5.435	5.435	5.435	5.435
27	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
28	Length	(m)	4526	4531	4536	4546
29	Bulk Density	(kg/m3)	81.60	81.60	81.60	81.60
30	Bulk Velocity	(m/s)	5.435	5.436	5.436	5.436
31	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
32	Length	(m)	4551	4556	4561	4571
33	Bulk Density	(kg/m3)	81.60	81.59	81.59	81.59
34	Bulk Velocity	(m/s)	5.436	5.436	5.436	5.436
35	Erosion Velocity	(m/s)	13.50	13.50	13.51	13.51
36	Length	(m)	4576	4581	4586	4596
37	Bulk Density	(kg/m3)	81.59	81.59	81.59	81.59
38	Bulk Velocity	(m/s)	5.436	5.436	5.436	5.437
39	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51
40	Length	(m)	4601	4606	4611	4621
41	Bulk Density	(kg/m3)	81.58	81.58	81.58	81.58
42	Bulk Velocity	(m/s)	5.437	5.437	5.437	5.437
43	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51
44	Length	(m)	4626	4631	4636	4646
45	Bulk Density	(kg/m3)	81.58	81.58	81.58	81.57
46	Bulk Velocity	(m/s)	5.437	5.437	5.437	5.437
47	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51
48	Length	(m)	4651	4656	4661	4671
49	Bulk Density	(kg/m3)	81.57	81.57	81.57	81.57
50	Bulk Velocity	(m/s)	5.437	5.437	5.438	5.438
51	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51
52	Length	(m)	4676	4681	4686	4696
53	Bulk Density	(kg/m3)	81.57	81.57	81.57	81.56
54	Bulk Velocity	(m/s)	5.438	5.438	5.438	5.438
55	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51
56	Length	(m)	4701	4706	4711	4721
57	Bulk Density	(kg/m3)	81.56	81.56	81.56	81.56
58	Bulk Velocity	(m/s)	5.438	5.438	5.438	5.438
59	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51
60	Length	(m)	4726	4731	4736	4746
61	Bulk Density	(kg/m3)	81.56	81.56	81.56	81.56
62	Bulk Velocity	(m/s)	5.438	5.438	5.438	5.439
63	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51
64	Length	(m)	4751	4756	4761	4771
65	Bulk Density	(kg/m3)	81.55	81.55	81.55	81.55
66	Bulk Velocity	(m/s)	5.439	5.439	5.439	5.439
67	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51
68	Length	(m)	4776	4781	4786	4796
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	Erosion						
10							
11	Erosion Table						
12							
13	Bulk Density	(kg/m3)	81.55	81.55	81.55	81.54	81.54
14	Bulk Velocity	(m/s)	5.439	5.439	5.439	5.439	5.439
15	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51	13.51
16	Length	(m)	4801	4806	4811	4816	4821
17	Bulk Density	(kg/m3)	81.54	81.54	81.54	81.54	81.54
18	Bulk Velocity	(m/s)	5.439	5.439	5.440	5.440	5.440
19	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51	13.51
20	Length	(m)	4826	4831	4836	4841	4846
21	Bulk Density	(kg/m3)	81.54	81.54	81.53	81.53	81.53
22	Bulk Velocity	(m/s)	5.440	5.440	5.440	5.440	5.440
23	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51	13.51
24	Length	(m)	4851	4856	4861	4866	4871
25	Bulk Density	(kg/m3)	81.53	81.53	81.53	81.53	81.53
26	Bulk Velocity	(m/s)	5.440	5.440	5.440	5.440	5.440
27	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51	13.51
28	Length	(m)	4876	4881	4886	4891	4896
29	Bulk Density	(kg/m3)	81.53	81.53	81.52	81.52	81.52
30	Bulk Velocity	(m/s)	5.441	5.441	5.441	5.441	5.441
31	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51	13.51
32	Length	(m)	4901	4906	4911	4916	4921
33	Bulk Density	(kg/m3)	81.52	81.52	81.52	81.52	81.52
34	Bulk Velocity	(m/s)	5.441	5.441	5.441	5.441	5.441
35	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51	13.51
36	Length	(m)	4926	4931	4936	4941	4946
37	Bulk Density	(kg/m3)	81.52	81.51	81.51	81.51	81.51
38	Bulk Velocity	(m/s)	5.441	5.441	5.441	5.441	5.442
39	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51	13.51
40	Length	(m)	4951	4956	4961	4966	4971
41	Bulk Density	(kg/m3)	81.51	81.51	81.51	81.51	81.51
42	Bulk Velocity	(m/s)	5.442	5.442	5.442	5.442	5.442
43	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51	13.51
44	Length	(m)	4976	4981	4986	4991	4996
45	Bulk Density	(kg/m3)	81.50	81.50	81.50	81.50	81.50
46	Bulk Velocity	(m/s)	5.442	5.442	5.442	5.442	5.442
47	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51	13.51
48	Length	(m)	5001	5006	5011	5016	5021
49	Bulk Density	(kg/m3)	81.50	81.50	81.50	81.50	81.49
50	Bulk Velocity	(m/s)	5.442	5.442	5.442	5.443	5.443
51	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51	13.51
52	Length	(m)	5026	5032	5037	5042	5047
53	Bulk Density	(kg/m3)	81.49	81.49	81.49	81.49	81.49
54	Bulk Velocity	(m/s)	5.443	5.443	5.443	5.443	5.443
55	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51	13.51
56	Length	(m)	5052	5057	5062	5067	5072
57	Bulk Density	(kg/m3)	81.49	81.49	81.49	81.49	81.48
58	Bulk Velocity	(m/s)	5.443	5.443	5.443	5.443	5.443
59	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51	13.51
60	Length	(m)	5077	5082	5087	5092	5097
61	Bulk Density	(kg/m3)	81.48	81.48	81.48	81.48	81.48
62	Bulk Velocity	(m/s)	5.443	5.443	5.444	5.444	5.444
63	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51	13.51
64	Length	(m)	5102	5107	5112	5117	5122
65	Bulk Density	(kg/m3)	81.48	81.48	81.47	81.47	81.47
66	Bulk Velocity	(m/s)	5.444	5.444	5.444	5.444	5.444
67	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.52	13.52
68	Length	(m)	5127	5132	5137	5142	5147
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3				Unit Set: Mixed		
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
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8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Bulk Density	(kg/m3)	81.47	81.47	81.47	81.47
14	Bulk Velocity	(m/s)	5.444	5.444	5.444	5.445
15	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
16	Length	(m)	5152	5154	5156	5160
17	Bulk Density	(kg/m3)	81.47	81.47	81.46	81.46
18	Bulk Velocity	(m/s)	5.445	5.445	5.445	5.445
19	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
20	Length	(m)	5162	5168	5174	5186
21	Bulk Density	(kg/m3)	81.46	81.46	81.46	81.46
22	Bulk Velocity	(m/s)	5.445	5.445	5.445	5.445
23	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
24	Length	(m)	5192	5194	5196	5200
25	Bulk Density	(kg/m3)	81.46	81.46	81.46	81.46
26	Bulk Velocity	(m/s)	5.445	5.445	5.445	5.445
27	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
28	Length	(m)	5202	5207	5212	5222
29	Bulk Density	(kg/m3)	81.45	81.45	81.45	81.45
30	Bulk Velocity	(m/s)	5.445	5.445	5.445	5.446
31	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
32	Length	(m)	5227	5232	5237	5247
33	Bulk Density	(kg/m3)	81.45	81.45	81.45	81.45
34	Bulk Velocity	(m/s)	5.446	5.446	5.446	5.446
35	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
36	Length	(m)	5252	5257	5262	5272
37	Bulk Density	(kg/m3)	81.45	81.45	81.44	81.44
38	Bulk Velocity	(m/s)	5.446	5.446	5.446	5.446
39	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
40	Length	(m)	5277	5282	5287	5297
41	Bulk Density	(kg/m3)	81.44	81.44	81.44	81.44
42	Bulk Velocity	(m/s)	5.446	5.446	5.446	5.447
43	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
44	Length	(m)	5302	5307	5312	5322
45	Bulk Density	(kg/m3)	81.44	81.43	81.43	81.43
46	Bulk Velocity	(m/s)	5.447	5.447	5.447	5.447
47	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
48	Length	(m)	5327	5332	5337	5347
49	Bulk Density	(kg/m3)	81.43	81.43	81.43	81.42
50	Bulk Velocity	(m/s)	5.447	5.447	5.447	5.447
51	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
52	Length	(m)	5352	5357	5362	5372
53	Bulk Density	(kg/m3)	81.42	81.42	81.42	81.42
54	Bulk Velocity	(m/s)	5.447	5.447	5.447	5.448
55	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
56	Length	(m)	5377	5382	5387	5397
57	Bulk Density	(kg/m3)	81.42	81.42	81.42	81.41
58	Bulk Velocity	(m/s)	5.448	5.448	5.448	5.448
59	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
60	Length	(m)	5402	5407	5412	5422
61	Bulk Density	(kg/m3)	81.41	81.41	81.41	81.41
62	Bulk Velocity	(m/s)	5.448	5.448	5.448	5.448
63	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
64	Length	(m)	5427	5432	5437	5448
65	Bulk Density	(kg/m3)	81.41	81.41	81.41	81.40
66	Bulk Velocity	(m/s)	5.448	5.448	5.449	5.449
67	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
68	Length	(m)	5453	5458	5463	5473
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1	Epic Atlantic Limited			Case Name: Simulation - 331MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Erosion					
10	Erosion Table					
11						
12						
13	Bulk Density	(kg/m3)	81.40	81.40	81.40	81.40
14	Bulk Velocity	(m/s)	5.449	5.449	5.449	5.449
15	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
16	Length	(m)	5478	5483	5488	5493
17	Bulk Density	(kg/m3)	81.40	81.40	81.40	81.39
18	Bulk Velocity	(m/s)	5.449	5.449	5.449	5.449
19	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
20	Length	(m)	5503	5508	5513	5518
21	Bulk Density	(kg/m3)	81.39	81.39	81.39	81.39
22	Bulk Velocity	(m/s)	5.449	5.449	5.450	5.450
23	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
24	Length	(m)	5528	5533	5538	5543
25	Bulk Density	(kg/m3)	81.39	81.39	81.39	81.38
26	Bulk Velocity	(m/s)	5.450	5.450	5.450	5.450
27	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
28	Length	(m)	5553	5558	5563	5568
29	Bulk Density	(kg/m3)	81.38	81.38	81.38	81.38
30	Bulk Velocity	(m/s)	5.450	5.450	5.450	5.451
31	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
32	Length	(m)	5578	5580	5582	5584
33	Bulk Density	(kg/m3)	81.38	81.38	81.38	81.38
34	Bulk Velocity	(m/s)	5.451	5.451	5.451	5.451
35	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
36	Length	(m)	5588	5590	5592	5594
37	Bulk Density	(kg/m3)	81.38	81.38	81.38	81.38
38	Bulk Velocity	(m/s)	5.451	5.451	5.451	5.451
39	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
40	Length	(m)	5598	5604	5610	5616
41	Bulk Density	(kg/m3)	81.38	81.37	81.37	81.37
42	Bulk Velocity	(m/s)	5.451	5.451	5.451	5.451
43	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
44	Length	(m)	5628	5634	5640	5646
45	Bulk Density	(kg/m3)	81.36	81.37	81.37	81.37
46	Bulk Velocity	(m/s)	5.451	5.451	5.451	5.451
47	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
48	Length	(m)	5658	5671	5684	5698
49	Bulk Density	(kg/m3)	81.37	81.37	81.36	81.36
50	Bulk Velocity	(m/s)	5.451	5.451	5.451	5.452
51	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
52	Length	(m)	5725	5738	5751	5778
53	Bulk Density	(kg/m3)	81.36	81.36	81.37	81.38
54	Bulk Velocity	(m/s)	5.452	5.451	5.451	5.450
55	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
56	Length	(m)	5792	5805	5818	5832
57	Bulk Density	(kg/m3)	81.40	81.40	81.41	81.41
58	Bulk Velocity	(m/s)	5.449	5.449	5.449	5.448
59	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
60	Length	(m)	5859	5872	5885	5899
61	Bulk Density	(kg/m3)	81.41	81.41	81.41	81.41
62	Bulk Velocity	(m/s)	5.448	5.448	5.448	5.448
63	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
64	Length	(m)	5926	5939	5952	5979
65	Bulk Density	(kg/m3)	81.41	81.34	81.27	81.19
66	Bulk Velocity	(m/s)	5.448	5.453	5.458	5.468
67	Erosion Velocity	(m/s)	13.52	13.53	13.53	13.54
68	Length	(m)	5993	5997	6001	6009
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1	Epic Atlantic Limited			Case Name: Simulation - 331MMSCFD_80barg.hsc			
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3				Unit Set: Mixed			
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5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	Erosion						
10							
11	Erosion Table						
12							
13	Bulk Density	(kg/m3)	81.05	81.05	81.05	81.04	81.04
14	Bulk Velocity	(m/s)	5.472	5.473	5.473	5.473	5.473
15	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
16	Length	(m)	6013	6017	6021	6024	6028
17	Bulk Density	(kg/m3)	81.04	81.04	81.04	81.04	81.04
18	Bulk Velocity	(m/s)	5.473	5.473	5.473	5.473	5.473
19	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
20	Length	(m)	6032	6033	6034	6035	6036
21	Bulk Density	(kg/m3)	81.04	81.04	81.04	81.04	81.04
22	Bulk Velocity	(m/s)	5.473	5.473	5.473	5.473	5.473
23	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
24	Length	(m)	6037	6042	6047	6052	6057
25	Bulk Density	(kg/m3)	81.04	81.04	81.03	81.03	81.03
26	Bulk Velocity	(m/s)	5.473	5.473	5.474	5.474	5.474
27	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
28	Length	(m)	6062	6067	6072	6077	6082
29	Bulk Density	(kg/m3)	81.03	81.03	81.03	81.03	81.02
30	Bulk Velocity	(m/s)	5.474	5.474	5.474	5.474	5.474
31	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
32	Length	(m)	6087	6092	6097	6102	6107
33	Bulk Density	(kg/m3)	81.02	81.02	81.02	81.02	81.02
34	Bulk Velocity	(m/s)	5.474	5.474	5.474	5.474	5.474
35	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
36	Length	(m)	6112	6117	6122	6127	6132
37	Bulk Density	(kg/m3)	81.02	81.02	81.02	81.02	81.02
38	Bulk Velocity	(m/s)	5.475	5.475	5.475	5.475	5.475
39	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
40	Length	(m)	6137	6142	6147	6152	6157
41	Bulk Density	(kg/m3)	81.01	81.01	81.01	81.01	81.01
42	Bulk Velocity	(m/s)	5.475	5.475	5.475	5.475	5.475
43	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
44	Length	(m)	6162	6167	6172	6177	6182
45	Bulk Density	(kg/m3)	81.01	81.01	81.01	81.00	81.00
46	Bulk Velocity	(m/s)	5.475	5.475	5.475	5.476	5.476
47	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
48	Length	(m)	6187	6192	6197	6202	6207
49	Bulk Density	(kg/m3)	81.00	81.00	81.00	81.00	81.00
50	Bulk Velocity	(m/s)	5.476	5.476	5.476	5.476	5.476
51	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
52	Length	(m)	6212	6217	6222	6228	6233
53	Bulk Density	(kg/m3)	81.00	80.99	80.99	80.99	80.99
54	Bulk Velocity	(m/s)	5.476	5.476	5.476	5.476	5.476
55	Erosion Velocity	(m/s)	13.55	13.55	13.56	13.56	13.56
56	Length	(m)	6238	6243	6248	6253	6258
57	Bulk Density	(kg/m3)	80.99	80.99	80.99	80.99	80.99
58	Bulk Velocity	(m/s)	5.477	5.477	5.477	5.477	5.477
59	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56	13.56
60	Length	(m)	6263	6268	6273	6278	6283
61	Bulk Density	(kg/m3)	80.98	80.98	80.98	80.98	80.98
62	Bulk Velocity	(m/s)	5.477	5.477	5.477	5.477	5.477
63	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56	13.56
64	Length	(m)	6288	6293	6298	6303	6308
65	Bulk Density	(kg/m3)	80.98	80.98	80.98	80.98	80.97
66	Bulk Velocity	(m/s)	5.477	5.477	5.477	5.478	5.478
67	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56	13.56
68	Length	(m)	6313	6318	6323	6328	6333
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2						
3				Unit Set: Mixed		
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Bulk Density	(kg/m3)	80.97	80.97	80.97	80.97
14	Bulk Velocity	(m/s)	5.478	5.478	5.478	5.478
15	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
16	Length	(m)	6338	6343	6348	6358
17	Bulk Density	(kg/m3)	80.97	80.97	80.97	80.96
18	Bulk Velocity	(m/s)	5.478	5.478	5.478	5.478
19	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
20	Length	(m)	6363	6368	6373	6383
21	Bulk Density	(kg/m3)	80.96	80.96	80.96	80.96
22	Bulk Velocity	(m/s)	5.478	5.478	5.479	5.479
23	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
24	Length	(m)	6388	6393	6398	6408
25	Bulk Density	(kg/m3)	80.96	80.96	80.96	80.95
26	Bulk Velocity	(m/s)	5.479	5.479	5.479	5.479
27	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
28	Length	(m)	6413	6419	6425	6437
29	Bulk Density	(kg/m3)	80.95	80.95	80.95	80.95
30	Bulk Velocity	(m/s)	5.479	5.479	5.479	5.479
31	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
32	Length	(m)	6443	6449	6455	6467
33	Bulk Density	(kg/m3)	80.95	80.95	80.94	80.94
34	Bulk Velocity	(m/s)	5.479	5.480	5.480	5.480
35	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
36	Length	(m)	6473	6476	6479	6485
37	Bulk Density	(kg/m3)	80.94	80.94	80.94	80.94
38	Bulk Velocity	(m/s)	5.480	5.480	5.480	5.480
39	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
40	Length	(m)	6488	6493	6498	6508
41	Bulk Density	(kg/m3)	80.94	80.93	80.93	80.93
42	Bulk Velocity	(m/s)	5.480	5.480	5.480	5.481
43	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
44	Length	(m)	6513	6515	6517	6521
45	Bulk Density	(kg/m3)	80.93	80.93	80.93	80.93
46	Bulk Velocity	(m/s)	5.481	5.481	5.481	5.481
47	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
48	Length	(m)	6523	6526	6529	6535
49	Bulk Density	(kg/m3)	80.93	80.93	80.93	80.93
50	Bulk Velocity	(m/s)	5.481	5.481	5.481	5.481
51	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
52	Length	(m)	6538	6543	6548	6558
53	Bulk Density	(kg/m3)	80.93	80.92	80.92	80.92
54	Bulk Velocity	(m/s)	5.481	5.481	5.481	5.481
55	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
56	Length	(m)	6563	6568	6573	6583
57	Bulk Density	(kg/m3)	80.92	80.92	80.92	80.91
58	Bulk Velocity	(m/s)	5.481	5.481	5.481	5.482
59	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
60	Length	(m)	6588	6593	6598	6608
61	Bulk Density	(kg/m3)	80.91	80.91	80.91	80.91
62	Bulk Velocity	(m/s)	5.482	5.482	5.482	5.482
63	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
64	Length	(m)	6613	6618	6623	6633
65	Bulk Density	(kg/m3)	80.91	80.91	80.91	80.90
66	Bulk Velocity	(m/s)	5.482	5.482	5.482	5.482
67	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
68	Length	(m)	6638	6643	6648	6658
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2						
3				Unit Set: Mixed		
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Bulk Density (kg/m3)	80.90	80.90	80.90	80.90	80.90
14	Bulk Velocity (m/s)	5.482	5.482	5.483	5.483	5.483
15	Erosion Velocity (m/s)	13.56	13.56	13.56	13.56	13.56
16	Length (m)	6663				
17	Bulk Density (kg/m3)	80.90				
18	Bulk Velocity (m/s)	5.483				
19	Erosion Velocity (m/s)	13.56				
20	Hydrates					
21						
22	Model Options					
23						
24	Hydrate Model:	Ng & Robinson				
25	Override Ng and Robinson:	False				
26	Initial Pressure:	From a Correlation				
27	Initial Pressure User Input:	---				
28	Hydrates Table					
29						
30	Distance (m)	0.0000	509.2	1018	1528	2037
31	Pressure (bar)	80.00	79.82	79.64	79.47	79.29
32	Temperature (C)	57.94	57.65	57.35	57.06	56.77
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.31	16.30	16.28	16.27	16.26
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	2546	2551	2556	2561	2566
43	Pressure (bar)	79.11	79.11	79.10	79.10	79.10
44	Temperature (C)	56.49	56.49	56.48	56.48	56.48
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	2571	2576	2581	2586	2591
55	Pressure (bar)	79.10	79.10	79.09	79.09	79.09
56	Temperature (C)	56.47	56.47	56.47	56.47	56.46
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	2596	2601	2606	2611	2616
67	Pressure (bar)	79.09	79.09	79.08	79.08	79.08
68	Temperature (C)	56.46	56.46	56.45	56.45	56.45
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2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:09:19 2022					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	2621	2624	2627	2630	2633
23	Pressure (bar)	79.08	79.08	79.08	79.08	79.07
24	Temperature (C)	56.45	56.44	56.44	56.44	56.44
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	2636	2640	2644	2648	2652
35	Pressure (bar)	79.07	79.07	79.07	79.07	79.07
36	Temperature (C)	56.44	56.43	56.43	56.43	56.43
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	2656	2661	2666	2671	2676
47	Pressure (bar)	79.06	79.06	79.06	79.06	79.06
48	Temperature (C)	56.43	56.42	56.42	56.42	56.41
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	2681	2686	2691	2696	2701
59	Pressure (bar)	79.06	79.05	79.05	79.05	79.05
60	Temperature (C)	56.41	56.41	56.41	56.40	56.40
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2			Unit Set: Mixed			
3			Date/Time: Mon Jun 13 15:09:19 2022			
4						
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	2706	2711	2716	2721	2726
15	Pressure (bar)	79.05	79.04	79.04	79.04	79.04
16	Temperature (C)	56.40	56.39	56.39	56.39	56.39
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	2731	2736	2741	2746	2751
27	Pressure (bar)	79.04	79.04	79.03	79.03	79.03
28	Temperature (C)	56.38	56.38	56.38	56.38	56.37
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	2756	2761	2766	2771	2776
39	Pressure (bar)	79.03	79.03	79.02	79.02	79.02
40	Temperature (C)	56.37	56.37	56.36	56.36	56.36
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.23
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	2781	2786	2791	2796	2801
51	Pressure (bar)	79.02	79.02	79.02	79.01	79.01
52	Temperature (C)	56.36	56.35	56.35	56.35	56.34
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	2806	2811	2816	2821	2826
63	Pressure (bar)	79.01	79.01	79.01	79.00	79.00
64	Temperature (C)	56.34	56.34	56.34	56.33	56.33
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	2831	2836	2841	2846	2851
19	Pressure (bar)	79.00	79.00	79.00	79.00	78.99
20	Temperature (C)	56.33	56.33	56.32	56.32	56.32
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	2856	2861	2866	2871	2876
31	Pressure (bar)	78.99	78.99	78.99	78.99	78.99
32	Temperature (C)	56.31	56.31	56.31	56.31	56.30
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	2881	2886	2891	2896	2901
43	Pressure (bar)	78.99	78.98	78.98	78.98	78.97
44	Temperature (C)	56.30	56.30	56.30	56.29	56.29
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	2906	2911	2916	2921	2926
55	Pressure (bar)	78.97	78.97	78.97	78.96	78.96
56	Temperature (C)	56.29	56.28	56.28	56.28	56.27
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	2931	2936	2941	2946	2951
67	Pressure (bar)	78.96	78.95	78.95	78.95	78.95
68	Temperature (C)	56.27	56.27	56.26	56.26	56.26
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	2956	2961	2966	2971	2976
23	Pressure (bar)	78.95	78.95	78.94	78.94	78.94
24	Temperature (C)	56.26	56.25	56.25	56.25	56.24
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	2981	2986	2991	2996	3001
35	Pressure (bar)	78.94	78.94	78.94	78.93	78.93
36	Temperature (C)	56.24	56.24	56.24	56.23	56.23
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	3006	3011	3016	3021	3026
47	Pressure (bar)	78.93	78.93	78.93	78.92	78.92
48	Temperature (C)	56.23	56.22	56.22	56.22	56.22
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	3031	3036	3041	3046	3051
59	Pressure (bar)	78.92	78.92	78.92	78.91	78.91
60	Temperature (C)	56.21	56.21	56.21	56.21	56.20
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	3056	3061	3066	3071	3076
15	Pressure (bar)	78.91	78.91	78.91	78.91	78.90
16	Temperature (C)	56.20	56.20	56.19	56.19	56.19
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	3081	3086	3091	3096	3101
27	Pressure (bar)	78.90	78.90	78.90	78.90	78.89
28	Temperature (C)	56.19	56.18	56.18	56.18	56.17
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.23	16.23	16.23	16.22	16.22
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	3106	3111	3116	3121	3126
39	Pressure (bar)	78.89	78.89	78.89	78.89	78.89
40	Temperature (C)	56.17	56.17	56.17	56.16	56.16
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	3131	3136	3141	3146	3151
51	Pressure (bar)	78.88	78.88	78.88	78.88	78.88
52	Temperature (C)	56.16	56.15	56.15	56.15	56.15
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	3156	3161	3166	3171	3176
63	Pressure (bar)	78.87	78.87	78.87	78.87	78.87
64	Temperature (C)	56.14	56.14	56.14	56.13	56.13
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	3181	3186	3191	3196	3201
19	Pressure (bar)	78.86	78.86	78.86	78.86	78.86
20	Temperature (C)	56.13	56.13	56.12	56.12	56.12
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	3206	3212	3218	3224	3230
31	Pressure (bar)	78.86	78.85	78.85	78.85	78.85
32	Temperature (C)	56.12	56.11	56.11	56.11	56.10
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	3236	3239	3242	3245	3248
43	Pressure (bar)	78.84	78.84	78.84	78.84	78.84
44	Temperature (C)	56.10	56.10	56.10	56.09	56.09
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	3251	3257	3263	3269	3275
55	Pressure (bar)	78.84	78.84	78.83	78.83	78.83
56	Temperature (C)	56.09	56.09	56.08	56.08	56.08
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	3281	3286	3291	3296	3301
67	Pressure (bar)	78.83	78.83	78.82	78.82	78.82
68	Temperature (C)	56.07	56.07	56.07	56.06	56.06
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2			Unit Set: Mixed			
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5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	3306	3311	3316	3321	3326
23	Pressure (bar)	78.82	78.82	78.81	78.81	78.81
24	Temperature (C)	56.06	56.06	56.05	56.05	56.05
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	3331	3336	3341	3346	3351
35	Pressure (bar)	78.81	78.81	78.81	78.80	78.80
36	Temperature (C)	56.05	56.04	56.04	56.04	56.03
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	3356	3361	3366	3371	3376
47	Pressure (bar)	78.80	78.80	78.80	78.79	78.79
48	Temperature (C)	56.03	56.03	56.03	56.02	56.02
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	3381	3386	3391	3396	3401
59	Pressure (bar)	78.79	78.79	78.79	78.79	78.78
60	Temperature (C)	56.02	56.02	56.01	56.01	56.01
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	3406	3411	3416	3421	3426
15	Pressure (bar)	78.78	78.78	78.78	78.78	78.78
16	Temperature (C)	56.00	56.00	56.00	56.00	55.99
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	3431	3436	3441	3446	3451
27	Pressure (bar)	78.78	78.77	78.77	78.77	78.77
28	Temperature (C)	55.99	55.99	55.99	55.98	55.98
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.22	16.21	16.21	16.21	16.21
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	3456	3461	3466	3471	3476
39	Pressure (bar)	78.76	78.76	78.76	78.76	78.76
40	Temperature (C)	55.98	55.97	55.97	55.97	55.97
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	3481	3486	3491	3496	3501
51	Pressure (bar)	78.75	78.75	78.75	78.75	78.75
52	Temperature (C)	55.96	55.96	55.96	55.95	55.95
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	3506	3511	3516	3521	3526
63	Pressure (bar)	78.74	78.74	78.74	78.74	78.74
64	Temperature (C)	55.95	55.95	55.94	55.94	55.94
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	3531	3536	3541	3546	3551
19	Pressure (bar)	78.73	78.73	78.73	78.73	78.73
20	Temperature (C)	55.93	55.93	55.93	55.93	55.92
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	3556	3561	3565	3569	3573
31	Pressure (bar)	78.73	78.72	78.72	78.72	78.72
32	Temperature (C)	55.92	55.92	55.92	55.91	55.91
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	3577	3582	3587	3592	3597
43	Pressure (bar)	78.72	78.72	78.71	78.71	78.71
44	Temperature (C)	55.91	55.91	55.90	55.90	55.90
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	3602	3607	3612	3617	3622
55	Pressure (bar)	78.71	78.71	78.70	78.70	78.70
56	Temperature (C)	55.89	55.89	55.89	55.89	55.89
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	3627	3632	3637	3642	3647
67	Pressure (bar)	78.70	78.70	78.70	78.69	78.69
68	Temperature (C)	55.88	55.88	55.88	55.87	55.87
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	3652	3657	3662	3667	3672
23	Pressure (bar)	78.69	78.69	78.69	78.68	78.68
24	Temperature (C)	55.87	55.87	55.86	55.86	55.86
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	3677	3682	3687	3692	3697
35	Pressure (bar)	78.68	78.68	78.68	78.67	78.67
36	Temperature (C)	55.85	55.85	55.85	55.85	55.84
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	3702	3707	3712	3717	3722
47	Pressure (bar)	78.67	78.67	78.67	78.66	78.66
48	Temperature (C)	55.84	55.84	55.83	55.83	55.83
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	3727	3732	3737	3742	3747
59	Pressure (bar)	78.66	78.66	78.66	78.65	78.65
60	Temperature (C)	55.83	55.82	55.82	55.82	55.81
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	3752	3757	3761	3766	3770
15	Pressure (bar)	78.65	78.65	78.65	78.65	78.64
16	Temperature (C)	55.81	55.81	55.81	55.80	55.80
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.21	16.20	16.20	16.20	16.20
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	3775	3780	3785	3790	3795
27	Pressure (bar)	78.64	78.64	78.64	78.64	78.64
28	Temperature (C)	55.80	55.80	55.79	55.79	55.79
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	3800	3805	3810	3815	3820
39	Pressure (bar)	78.63	78.63	78.63	78.63	78.63
40	Temperature (C)	55.79	55.78	55.78	55.78	55.77
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	3825	3830	3835	3840	3845
51	Pressure (bar)	78.62	78.62	78.62	78.62	78.62
52	Temperature (C)	55.77	55.77	55.77	55.76	55.76
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	3850	3855	3860	3865	3870
63	Pressure (bar)	78.62	78.61	78.61	78.61	78.61
64	Temperature (C)	55.76	55.76	55.75	55.75	55.75
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	3875	3880	3885	3890	3895
19	Pressure (bar)	78.61	78.60	78.60	78.60	78.60
20	Temperature (C)	55.74	55.74	55.74	55.74	55.73
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	3900	3905	3910	3915	3920
31	Pressure (bar)	78.60	78.59	78.59	78.59	78.59
32	Temperature (C)	55.73	55.73	55.72	55.72	55.72
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	3925	3927	3929	3931	3933
43	Pressure (bar)	78.59	78.59	78.59	78.59	78.59
44	Temperature (C)	55.72	55.72	55.71	55.71	55.71
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	3935	3947	3958	3969	3980
55	Pressure (bar)	78.59	78.59	78.59	78.60	78.60
56	Temperature (C)	55.71	55.71	55.71	55.71	55.70
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	3991	4002	4013	4025	4036
67	Pressure (bar)	78.60	78.60	78.60	78.60	78.60
68	Temperature (C)	55.70	55.70	55.69	55.69	55.68
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	4047	4058	4069	4080	4091
23	Pressure (bar)	78.60	78.60	78.59	78.59	78.59
24	Temperature (C)	55.68	55.68	55.67	55.66	55.66
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	4103	4114	4125	4136	4147
35	Pressure (bar)	78.58	78.58	78.57	78.57	78.56
36	Temperature (C)	55.65	55.65	55.64	55.63	55.63
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	4158	4169	4181	4192	4203
47	Pressure (bar)	78.56	78.55	78.55	78.55	78.55
48	Temperature (C)	55.62	55.61	55.61	55.60	55.60
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	4214	4225	4236	4247	4259
59	Pressure (bar)	78.55	78.55	78.56	78.56	78.57
60	Temperature (C)	55.59	55.59	55.59	55.59	55.59
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:09:19 2022					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	4270	4281	4292	4303	4314
15	Pressure (bar)	78.58	78.49	78.41	78.33	78.24
16	Temperature (C)	55.59	55.56	55.53	55.50	55.46
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.20	16.19	16.19	16.18	16.17
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	4325	4328	4331	4334	4337
27	Pressure (bar)	78.16	78.15	78.15	78.15	78.14
28	Temperature (C)	55.43	55.43	55.42	55.42	55.42
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	4340	4343	4346	4349	4352
39	Pressure (bar)	78.14	78.14	78.14	78.14	78.14
40	Temperature (C)	55.42	55.41	55.41	55.41	55.41
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	4355	4357	4359	4361	4363
51	Pressure (bar)	78.14	78.14	78.14	78.14	78.14
52	Temperature (C)	55.41	55.41	55.41	55.41	55.41
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	4365	4367	4369	4371	4373
63	Pressure (bar)	78.14	78.14	78.14	78.13	78.13
64	Temperature (C)	55.41	55.40	55.40	55.40	55.40
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	4375	4380	4385	4390	4395
19	Pressure (bar)	78.13	78.13	78.13	78.13	78.12
20	Temperature (C)	55.40	55.40	55.39	55.39	55.39
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	4400	4405	4410	4415	4420
31	Pressure (bar)	78.12	78.12	78.12	78.12	78.12
32	Temperature (C)	55.39	55.38	55.38	55.38	55.38
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	4425	4431	4436	4441	4446
43	Pressure (bar)	78.12	78.12	78.11	78.11	78.11
44	Temperature (C)	55.37	55.37	55.37	55.37	55.36
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	4451	4456	4461	4466	4471
55	Pressure (bar)	78.11	78.11	78.10	78.10	78.10
56	Temperature (C)	55.36	55.36	55.36	55.35	55.35
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	4476	4481	4486	4491	4496
67	Pressure (bar)	78.10	78.10	78.10	78.09	78.09
68	Temperature (C)	55.35	55.34	55.34	55.34	55.34
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	4501	4506	4511	4516	4521
23	Pressure (bar)	78.09	78.09	78.08	78.08	78.08
24	Temperature (C)	55.33	55.33	55.33	55.33	55.32
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	4526	4531	4536	4541	4546
35	Pressure (bar)	78.08	78.08	78.07	78.07	78.07
36	Temperature (C)	55.32	55.32	55.31	55.31	55.31
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	4551	4556	4561	4566	4571
47	Pressure (bar)	78.07	78.07	78.06	78.06	78.06
48	Temperature (C)	55.30	55.30	55.30	55.30	55.29
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	4576	4581	4586	4591	4596
59	Pressure (bar)	78.06	78.06	78.05	78.05	78.05
60	Temperature (C)	55.29	55.29	55.29	55.28	55.28
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	4601	4606	4611	4616	4621
15	Pressure (bar)	78.05	78.05	78.04	78.04	78.04
16	Temperature (C)	55.28	55.27	55.27	55.27	55.27
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	4626	4631	4636	4641	4646
27	Pressure (bar)	78.04	78.04	78.03	78.03	78.03
28	Temperature (C)	55.26	55.26	55.26	55.25	55.25
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.16	16.15	16.15	16.15	16.15
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	4651	4656	4661	4666	4671
39	Pressure (bar)	78.03	78.03	78.02	78.02	78.02
40	Temperature (C)	55.25	55.25	55.24	55.24	55.24
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	4676	4681	4686	4691	4696
51	Pressure (bar)	78.02	78.02	78.02	78.01	78.01
52	Temperature (C)	55.24	55.23	55.23	55.23	55.22
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	4701	4706	4711	4716	4721
63	Pressure (bar)	78.01	78.01	78.01	78.01	78.00
64	Temperature (C)	55.22	55.22	55.22	55.21	55.21
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 100 of 116

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	4726	4731	4736	4741	4746
19	Pressure (bar)	78.00	78.00	78.00	78.00	78.00
20	Temperature (C)	55.21	55.21	55.20	55.20	55.20
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	4751	4756	4761	4766	4771
31	Pressure (bar)	77.99	77.99	77.99	77.99	77.98
32	Temperature (C)	55.20	55.19	55.19	55.19	55.18
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	4776	4781	4786	4791	4796
43	Pressure (bar)	77.98	77.98	77.98	77.98	77.98
44	Temperature (C)	55.18	55.18	55.18	55.17	55.17
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	4801	4806	4811	4816	4821
55	Pressure (bar)	77.97	77.97	77.97	77.97	77.97
56	Temperature (C)	55.17	55.16	55.16	55.16	55.16
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	4826	4831	4836	4841	4846
67	Pressure (bar)	77.96	77.96	77.96	77.96	77.96
68	Temperature (C)	55.15	55.15	55.15	55.15	55.14
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 101 of 116

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:09:19 2022					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	4851	4856	4861	4866	4871
23	Pressure (bar)	77.95	77.95	77.95	77.95	77.95
24	Temperature (C)	55.14	55.14	55.13	55.13	55.13
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	4876	4881	4886	4891	4896
35	Pressure (bar)	77.94	77.94	77.94	77.94	77.94
36	Temperature (C)	55.13	55.12	55.12	55.12	55.12
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	4901	4906	4911	4916	4921
47	Pressure (bar)	77.94	77.93	77.93	77.93	77.93
48	Temperature (C)	55.11	55.11	55.11	55.10	55.10
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	4926	4931	4936	4941	4946
59	Pressure (bar)	77.93	77.92	77.92	77.92	77.92
60	Temperature (C)	55.10	55.10	55.09	55.09	55.09
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	4951	4956	4961	4966	4971
15	Pressure (bar)	77.92	77.91	77.91	77.91	77.91
16	Temperature (C)	55.09	55.08	55.08	55.08	55.08
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.15	16.14	16.14	16.14	16.14
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	4976	4981	4986	4991	4996
27	Pressure (bar)	77.91	77.91	77.90	77.90	77.90
28	Temperature (C)	55.07	55.07	55.07	55.06	55.06
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	5001	5006	5011	5016	5021
39	Pressure (bar)	77.90	77.90	77.89	77.89	77.89
40	Temperature (C)	55.06	55.06	55.05	55.05	55.05
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	5026	5032	5037	5042	5047
51	Pressure (bar)	77.89	77.89	77.88	77.88	77.88
52	Temperature (C)	55.04	55.04	55.04	55.04	55.03
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	5052	5057	5062	5067	5072
63	Pressure (bar)	77.88	77.88	77.87	77.87	77.87
64	Temperature (C)	55.03	55.03	55.03	55.02	55.02
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 103 of 116

1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	5077	5082	5087	5092	5097
19	Pressure (bar)	77.87	77.87	77.87	77.86	77.86
20	Temperature (C)	55.02	55.02	55.01	55.01	55.01
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	5102	5107	5112	5117	5122
31	Pressure (bar)	77.86	77.86	77.86	77.85	77.85
32	Temperature (C)	55.00	55.00	55.00	55.00	54.99
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	5127	5132	5137	5142	5147
43	Pressure (bar)	77.85	77.85	77.85	77.84	77.84
44	Temperature (C)	54.99	54.99	54.98	54.98	54.98
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	5152	5154	5156	5158	5160
55	Pressure (bar)	77.84	77.84	77.84	77.84	77.84
56	Temperature (C)	54.98	54.98	54.97	54.97	54.97
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	5162	5168	5174	5180	5186
67	Pressure (bar)	77.84	77.83	77.83	77.83	77.83
68	Temperature (C)	54.97	54.97	54.97	54.96	54.96
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:09:19 2022			
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	5192	5194	5196	5198	5200
23	Pressure (bar)	77.83	77.82	77.82	77.82	77.82
24	Temperature (C)	54.96	54.95	54.95	54.95	54.95
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	5202	5207	5212	5217	5222
35	Pressure (bar)	77.82	77.82	77.82	77.82	77.81
36	Temperature (C)	54.95	54.95	54.94	54.94	54.94
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	5227	5232	5237	5242	5247
47	Pressure (bar)	77.81	77.81	77.81	77.81	77.81
48	Temperature (C)	54.94	54.93	54.93	54.93	54.93
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	5252	5257	5262	5267	5272
59	Pressure (bar)	77.80	77.80	77.80	77.80	77.80
60	Temperature (C)	54.92	54.92	54.92	54.92	54.91
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	5277	5282	5287	5292	5297
15	Pressure (bar)	77.80	77.79	77.79	77.79	77.79
16	Temperature (C)	54.91	54.91	54.90	54.90	54.90
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.14	16.13	16.13	16.13	16.13
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	5302	5307	5312	5317	5322
27	Pressure (bar)	77.79	77.78	77.78	77.78	77.78
28	Temperature (C)	54.90	54.89	54.89	54.89	54.89
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	5327	5332	5337	5342	5347
39	Pressure (bar)	77.78	77.77	77.77	77.77	77.77
40	Temperature (C)	54.88	54.88	54.88	54.87	54.87
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	5352	5357	5362	5367	5372
51	Pressure (bar)	77.77	77.76	77.76	77.76	77.76
52	Temperature (C)	54.87	54.87	54.86	54.86	54.86
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	5377	5382	5387	5392	5397
63	Pressure (bar)	77.76	77.76	77.75	77.75	77.75
64	Temperature (C)	54.86	54.85	54.85	54.85	54.85
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	5402	5407	5412	5417	5422
19	Pressure (bar)	77.75	77.75	77.74	77.74	77.74
20	Temperature (C)	54.84	54.84	54.84	54.83	54.83
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	5427	5432	5437	5443	5448
31	Pressure (bar)	77.74	77.74	77.74	77.73	77.73
32	Temperature (C)	54.83	54.83	54.82	54.82	54.82
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	5453	5458	5463	5468	5473
43	Pressure (bar)	77.73	77.73	77.73	77.72	77.72
44	Temperature (C)	54.82	54.81	54.81	54.81	54.80
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	5478	5483	5488	5493	5498
55	Pressure (bar)	77.72	77.72	77.72	77.72	77.71
56	Temperature (C)	54.80	54.80	54.80	54.79	54.79
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	5503	5508	5513	5518	5523
67	Pressure (bar)	77.71	77.71	77.71	77.71	77.70
68	Temperature (C)	54.79	54.79	54.78	54.78	54.78
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:09:19 2022					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	5528	5533	5538	5543	5548
23	Pressure (bar)	77.70	77.70	77.70	77.70	77.69
24	Temperature (C)	54.78	54.77	54.77	54.77	54.76
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	5553	5558	5563	5568	5573
35	Pressure (bar)	77.69	77.69	77.69	77.69	77.68
36	Temperature (C)	54.76	54.76	54.76	54.75	54.75
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	5578	5580	5582	5584	5586
47	Pressure (bar)	77.68	77.68	77.68	77.68	77.68
48	Temperature (C)	54.75	54.75	54.75	54.75	54.74
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	5588	5590	5592	5594	5596
59	Pressure (bar)	77.68	77.68	77.68	77.68	77.68
60	Temperature (C)	54.74	54.74	54.74	54.74	54.74
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	5598	5604	5610	5616	5622
15	Pressure (bar)	77.68	77.68	77.67	77.67	77.67
16	Temperature (C)	54.74	54.74	54.73	54.73	54.72
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.13	16.13	16.13	16.12	16.12
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	5628	5634	5640	5646	5652
27	Pressure (bar)	77.66	77.66	77.66	77.66	77.67
28	Temperature (C)	54.72	54.72	54.72	54.72	54.71
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	5658	5671	5684	5698	5711
39	Pressure (bar)	77.67	77.66	77.66	77.65	77.65
40	Temperature (C)	54.71	54.71	54.70	54.69	54.68
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	5725	5738	5751	5765	5778
51	Pressure (bar)	77.64	77.65	77.66	77.67	77.67
52	Temperature (C)	54.68	54.68	54.68	54.68	54.68
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.13
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	5792	5805	5818	5832	5845
63	Pressure (bar)	77.68	77.68	77.68	77.68	77.68
64	Temperature (C)	54.68	54.68	54.67	54.67	54.66
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 109 of 116

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	5859	5872	5885	5899	5912
19	Pressure (bar)	77.68	77.68	77.68	77.68	77.67
20	Temperature (C)	54.66	54.65	54.65	54.64	54.64
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	5926	5939	5952	5966	5979
31	Pressure (bar)	77.67	77.60	77.52	77.45	77.38
32	Temperature (C)	54.63	54.60	54.57	54.54	54.51
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.13	16.12	16.11	16.11	16.10
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	5993	5997	6001	6005	6009
43	Pressure (bar)	77.30	77.30	77.30	77.29	77.29
44	Temperature (C)	54.48	54.47	54.47	54.47	54.46
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	6013	6017	6021	6024	6028
55	Pressure (bar)	77.29	77.29	77.29	77.29	77.28
56	Temperature (C)	54.46	54.46	54.46	54.46	54.45
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	6032	6033	6034	6035	6036
67	Pressure (bar)	77.28	77.28	77.28	77.28	77.28
68	Temperature (C)	54.45	54.45	54.45	54.45	54.45
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	6037	6042	6047	6052	6057
23	Pressure (bar)	77.28	77.28	77.28	77.27	77.27
24	Temperature (C)	54.45	54.45	54.44	54.44	54.44
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	6062	6067	6072	6077	6082
35	Pressure (bar)	77.27	77.27	77.27	77.26	77.26
36	Temperature (C)	54.43	54.43	54.43	54.43	54.42
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	6087	6092	6097	6102	6107
47	Pressure (bar)	77.26	77.26	77.26	77.26	77.26
48	Temperature (C)	54.42	54.42	54.42	54.41	54.41
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	6112	6117	6122	6127	6132
59	Pressure (bar)	77.25	77.25	77.25	77.25	77.25
60	Temperature (C)	54.41	54.41	54.40	54.40	54.40
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	6137	6142	6147	6152	6157
15	Pressure (bar)	77.24	77.24	77.24	77.24	77.24
16	Temperature (C)	54.40	54.39	54.39	54.39	54.39
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	6162	6167	6172	6177	6182
27	Pressure (bar)	77.23	77.23	77.23	77.23	77.23
28	Temperature (C)	54.38	54.38	54.38	54.37	54.37
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	6187	6192	6197	6202	6207
39	Pressure (bar)	77.22	77.22	77.22	77.22	77.22
40	Temperature (C)	54.37	54.37	54.36	54.36	54.36
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	6212	6217	6222	6228	6233
51	Pressure (bar)	77.21	77.21	77.21	77.21	77.21
52	Temperature (C)	54.36	54.35	54.35	54.35	54.34
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	6238	6243	6248	6253	6258
63	Pressure (bar)	77.20	77.20	77.20	77.20	77.20
64	Temperature (C)	54.34	54.34	54.34	54.33	54.33
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	6263	6268	6273	6278	6283
19	Pressure (bar)	77.19	77.19	77.19	77.19	77.19
20	Temperature (C)	54.33	54.33	54.32	54.32	54.32
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.08
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	6288	6293	6298	6303	6308
31	Pressure (bar)	77.19	77.18	77.18	77.18	77.18
32	Temperature (C)	54.32	54.31	54.31	54.31	54.30
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	6313	6318	6323	6328	6333
43	Pressure (bar)	77.18	77.17	77.17	77.17	77.17
44	Temperature (C)	54.30	54.30	54.30	54.29	54.29
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	6338	6343	6348	6353	6358
55	Pressure (bar)	77.17	77.17	77.16	77.16	77.16
56	Temperature (C)	54.29	54.29	54.28	54.28	54.28
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	6363	6368	6373	6378	6383
67	Pressure (bar)	77.16	77.16	77.15	77.15	77.15
68	Temperature (C)	54.28	54.27	54.27	54.27	54.27
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:09:19 2022					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	6388	6393	6398	6403	6408
23	Pressure (bar)	77.15	77.15	77.15	77.14	77.14
24	Temperature (C)	54.26	54.26	54.26	54.25	54.25
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	6413	6419	6425	6431	6437
35	Pressure (bar)	77.14	77.14	77.14	77.13	77.13
36	Temperature (C)	54.25	54.25	54.24	54.24	54.24
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	6443	6449	6455	6461	6467
47	Pressure (bar)	77.13	77.13	77.13	77.12	77.12
48	Temperature (C)	54.23	54.23	54.23	54.22	54.22
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	6473	6476	6479	6482	6485
59	Pressure (bar)	77.12	77.12	77.12	77.11	77.11
60	Temperature (C)	54.22	54.22	54.22	54.21	54.21
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 114 of 116

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:09:19 2022			
5										
6	Pipe Segment: BBOU-ELEP Pipeline (continued)									
7										
8										
9	Hydrates									
10										
11	Hydrates Table									
12										
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included				
14	Distance (m)	6488	6493	6498	6503	6508				
15	Pressure (bar)	77.11	77.11	77.11	77.11	77.10				
16	Temperature (C)	54.21	54.21	54.20	54.20	54.20				
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form				
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types				
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model				
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included				
21	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08				
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II				
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model				
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H				
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included				
26	Distance (m)	6513	6515	6517	6519	6521				
27	Pressure (bar)	77.10	77.10	77.10	77.10	77.10				
28	Temperature (C)	54.20	54.20	54.19	54.19	54.19				
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form				
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types				
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model				
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included				
33	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08				
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II				
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model				
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H				
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included				
38	Distance (m)	6523	6526	6529	6532	6535				
39	Pressure (bar)	77.10	77.10	77.10	77.10	77.09				
40	Temperature (C)	54.19	54.19	54.19	54.19	54.19				
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form				
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types				
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model				
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included				
45	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08				
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II				
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model				
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H				
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included				
50	Distance (m)	6538	6543	6548	6553	6558				
51	Pressure (bar)	77.09	77.09	77.09	77.09	77.09				
52	Temperature (C)	54.18	54.18	54.18	54.18	54.17				
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form				
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types				
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model				
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included				
57	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08				
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II				
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model				
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H				
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included				
62	Distance (m)	6563	6568	6573	6578	6583				
63	Pressure (bar)	77.08	77.08	77.08	77.08	77.08				
64	Temperature (C)	54.17	54.17	54.17	54.16	54.16				
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form				
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types				
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model				
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included				
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 115 of 116				

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	6588	6593	6598	6603	6608
19	Pressure (bar)	77.07	77.07	77.07	77.07	77.07
20	Temperature (C)	54.16	54.15	54.15	54.15	54.15
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.07
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	6613	6618	6623	6628	6633
31	Pressure (bar)	77.07	77.06	77.06	77.06	77.06
32	Temperature (C)	54.14	54.14	54.14	54.14	54.13
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	6638	6643	6648	6653	6658
43	Pressure (bar)	77.06	77.05	77.05	77.05	77.05
44	Temperature (C)	54.13	54.13	54.13	54.12	54.12
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	6663				
55	Pressure (bar)	77.05				
56	Temperature (C)	54.12				
57	Formation	Will NOT Form				
58	Hydrate Type	No Types				
59	Mode	Use 3-Phase Model				
60	Inhibitors	Not Included				
61	Formation Temperature (C)	16.07				
62	Type at Formation Temperature	Type II				
63	Mode at Formation Temperature	Use 3-Phase Model				
64	Phases at Formation Temperature	V - Aq - L - H				
65	Inhibitors at Formation Temperature	Not Included				
66						
67						
68						
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 116 of 116	

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:09:19 2022		
5					
6					
7	Pipe Segment: BBOU-ELEP Pipeline				
8					
9					
10	CONDITIONS				
11	Name	BB-Elepa Pipeline Inlet	BB-Elepa Pipeline Outlet	e	
12	Vapour	0.9346	0.9340	---	
13	Temperature (C)	57.9405	54.1186	---	
14	Pressure (bar)	80.0000	77.0470	---	
15	Molar Flow (MMSCFD)	345.5645	345.5645	---	
16	Mass Flow (kg/h)	419320.3253	419320.3253	---	
17	Std Ideal Liq Vol Flow (m3/h)	1085.9821	1085.9821	---	
18	Molar Enthalpy (kJ/kgmole)	-9.031e+004	-9.053e+004	---	
19	Molar Entropy (kJ/kgmole-C)	157.9	157.5	---	
20	Heat Flow (kJ/h)	-1.5543e+09	-1.5581e+09	3.7786e+06	
21					
22	PROPERTIES				
23	Name	BB-Elepa Pipeline Inlet	BB-Elepa Pipeline Outlet		
24	Molecular Weight	24.36	24.36		
25	Molar Density (kgmole/m3)	3.396	3.321		
26	Mass Density (kg/m3)	82.74	80.90		
27	Act. Volume Flow (m3/h)	5068	5183		
28	Mass Enthalpy (kJ/kg)	-3707	-3716		
29	Mass Entropy (kJ/kg-C)	6.481	6.465		
30	Heat Capacity (kJ/kgmole-C)	62.55	62.23		
31	Mass Heat Capacity (kJ/kg-C)	2.568	2.554		
32	LHV Molar Basis (Std) (kJ/kgmole)	1.153e+006	1.153e+006		
33	HHV Molar Basis (Std) (kJ/kgmole)	1.258e+006	1.258e+006		
34	HHV Mass Basis (Std) (kJ/kg)	5.165e+004	5.165e+004		
35	CO2 Loading	---	---		
36	CO2 Apparent Mole Conc. (kgmole/m3)	---	---		
37	CO2 Apparent Wt. Conc. (kgmol/kg)	---	---		
38	LHV Mass Basis (Std) (kJ/kg)	4.733e+004	4.733e+004		
39	Phase Fraction [Vol. Basis]	0.8234	0.8224		
40	Phase Fraction [Mass Basis]	0.6781	0.6763		
41	Phase Fraction [Act. Vol. Basis]	0.9611	0.9620		
42	Mass Exergy (kJ/kg)	413.2	409.0		
43	Partial Pressure of CO2 (bar)	0.4645	0.4476		
44	Cost Based on Flow (Cost/s)	0.0000	0.0000		
45	Act. Gas Flow (ACT_m3/h)	4871	4986		
46	Avg. Liq. Density (kgmole/m3)	15.85	15.85		
47	Specific Heat (kJ/kgmole-C)	62.55	62.23		
48	Std. Gas Flow (MMSCFD)	344.9	344.9		
49	Std. Ideal Liq. Mass Density (kg/m3)	386.1	386.1		
50	Act. Liq. Flow (m3/s)	5.469e-002	5.471e-002		
51	Z Factor	---	---		
52	Watson K	16.58	16.58		
53	User Property	---	---		
54	Partial Pressure of H2S (bar)	0.0000	0.0000		
55	Cp/(Cp - R)	1.153	1.154		
56	Cp/Cv	1.318	1.318		
57	Heat of Vap. (kJ/kgmole)	2.752e+004	2.757e+004		
58	Kinematic Viscosity (cSt)	---	---		
59	Liq. Mass Density (Std. Cond) (kg/m3)	1.036	1.036		
60	Liq. Vol. Flow (Std. Cond) (m3/h)	4.048e+005	4.048e+005		
61	Liquid Fraction	6.537e-002	6.597e-002		
62	Molar Volume (m3/kgmole)	0.2944	0.3012		
63	Mass Heat of Vap. (kJ/kg)	1130	1132		
64	Phase Fraction [Molar Basis]	0.9346	0.9340		
65	Surface Tension (dyne/cm)	---	---		
66	Thermal Conductivity (W/m-K)	---	---		
67	Viscosity (lbm/s-ft)	---	---		
68	Cv (Semi-Ideal) (kJ/kgmole-C)	54.24	53.91		
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 1 of 116				

1				Case Name: Simulation - 331MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:09:19 2022		
6						
7	Pipe Segment: BBOU-ELEP Pipeline (continued)					
8						
9	PROPERTIES					
10						
11	Name	BB-Elepa Pipeline Inlet	BB-Elepa Pipeline Outlet			
12	Mass Cv (Semi-Ideal) (kJ/kg-C)	2.226	2.213			
13	Cv (kJ/kgmole-C)	47.47	47.20			
14	Mass Cv (kJ/kg-C)	1.948	1.937			
15	Cv (Ent. Method) (kJ/kgmole-C)	---	---			
16	Mass Cv (Ent. Method) (kJ/kg-C)	---	---			
17	Cp/Cv (Ent. Method)	---	---			
18	Reid VP at 37.8 C (bar)	---	---			
19	True VP at 37.8 C (bar)	---	---			
20	Liq. Vol. Flow - Sum(Std. Cond) (m3/h)	3.794e+005	3.792e+005			
21	Viscosity Index	---	---			
22	Ideal Gas Cp/Cv	1.194	1.195			
23	Ideal Gas Cp (kJ/kgmole-C)	51.25	50.94			
24	Mass Ideal Gas Cp (kJ/kg-C)	2.104	2.091			
25	Bubble Point Pressure (bar)	---	---			
26	PROFILES					
27						
28	Data					
29						
30	Distance	Elevation		Increments		
31	(m)	(m)				
32	0.0000	0.0000		5		
33	2546	0.0000		5		
34	2571	-0.1600		5		
35	2596	4.000e-002		5		
36	2621	-1.000e-002		5		
37	2636	-8.000e-002		5		
38	2656	1.000e-002		5		
39	2681	-7.000e-002		5		
40	2706	-0.1000		5		
41	2731	-0.1600		5		
42	2756	-0.1000		5		
43	2781	-0.1600		5		
44	2806	-0.2000		5		
45	2831	-0.2100		5		
46	2856	-0.2100		5		
47	2881	-0.5700		5		
48	2906	-0.2200		5		
49	2931	0.2000		5		
50	2956	8.000e-002		5		
51	2981	3.000e-002		5		
52	3006	6.000e-002		5		
53	3031	6.000e-002		5		
54	3056	5.000e-002		5		
55	3081	6.000e-002		5		
56	3106	8.000e-002		5		
57	3131	7.000e-002		5		
58	3156	1.000e-001		5		
59	3181	0.1300		5		
60	3206	0.1500		5		
61	3236	0.1700		5		
62	3251	9.000e-002		5		
63	3281	0.2200		5		
64	3306	0.1500		5		
65	3331	1.000e-001		5		
66	3356	8.000e-002		5		
67	3381	1.000e-001		5		
68	3406	1.000e-001		5		
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1	 Epic Atlantic Limited		Case Name:	Simulation - 331MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:09:19 2022
4				
5				
6	Pipe Segment: BBOU-ELEP Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	3431	-0.2700	5	
16	3456	5.000e-002	5	
17	3481	1.000e-002	5	
18	3506	-2.000e-002	5	
19	3531	1.000e-002	5	
20	3556	1.000e-002	5	
21	3577	-1.232e-016	5	
22	3602	0.1800	5	
23	3627	-0.2300	5	
24	3652	-5.000e-002	5	
25	3677	2.000e-002	5	
26	3702	7.000e-002	5	
27	3727	0.1400	5	
28	3752	1.000e-001	5	
29	3775	0.1500	5	
30	3800	7.000e-002	5	
31	3825	-9.714e-017	5	
32	3850	-9.714e-017	5	
33	3875	3.000e-002	5	
34	3900	9.000e-002	5	
35	3925	0.2200	5	
36	3935	-0.3700	5	
37	3991	-3.600	5	
38	4047	-5.300	5	
39	4103	-5.900	5	
40	4158	-5.300	5	
41	4214	-6.300	5	
42	4270	-11.60	5	
43	4325	0.4400	5	
44	4340	1.150	5	
45	4355	0.8400	5	
46	4365	0.5100	5	
47	4375	0.6200	5	
48	4400	0.5900	5	
49	4425	0.2200	5	
50	4451	0.1600	5	
51	4476	0.1700	5	
52	4501	0.2400	5	
53	4526	0.3400	5	
54	4551	0.4800	5	
55	4576	0.5700	5	
56	4601	0.6300	5	
57	4626	0.6700	5	
58	4651	0.6700	5	
59	4676	0.7000	5	
60	4701	0.7000	5	
61	4726	0.5200	5	
62	4751	0.5200	5	
63	4776	0.6800	5	
64	4801	0.6300	5	
65	4826	0.6500	5	
66	4851	0.7000	5	
67	4876	0.7300	5	
68	4901	0.6800	5	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 331MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:09:19 2022
4				
5				
6	Pipe Segment: BBOU-ELEP Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	4926	0.6700	5	
16	4951	0.6600	5	
17	4976	0.5900	5	
18	5001	0.6300	5	
19	5026	0.6800	5	
20	5052	0.6500	5	
21	5077	0.6200	5	
22	5102	0.6100	5	
23	5127	0.6600	5	
24	5152	0.7500	5	
25	5162	0.6600	5	
26	5192	0.7100	5	
27	5202	0.7600	5	
28	5227	0.6500	5	
29	5252	0.4700	5	
30	5277	0.5100	5	
31	5302	0.5500	5	
32	5327	0.5400	5	
33	5352	0.5800	5	
34	5377	0.5800	5	
35	5402	0.6000	5	
36	5427	0.6000	5	
37	5453	0.6000	5	
38	5478	0.6500	5	
39	5503	0.4800	5	
40	5528	0.5700	5	
41	5553	0.6500	5	
42	5578	0.7200	5	
43	5588	0.4000	5	
44	5598	0.2900	5	
45	5628	0.6500	5	
46	5658	-0.7100	5	
47	5725	-0.6000	5	
48	5792	-8.000	5	
49	5859	-10.40	5	
50	5926	-11.60	5	
51	5993	2.000e-002	5	
52	6013	0.5200	5	
53	6032	0.3200	5	
54	6037	0.2100	5	
55	6062	0.4800	5	
56	6087	0.4100	5	
57	6112	0.1700	5	
58	6137	0.1600	5	
59	6162	0.1500	5	
60	6187	0.2300	5	
61	6212	0.3200	5	
62	6238	0.3700	5	
63	6263	0.3900	5	
64	6288	0.4000	5	
65	6313	0.3800	5	
66	6338	0.3800	5	
67	6363	0.3200	5	
68	6388	0.3200	5	
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:09:19 2022		
4					
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Pressure (bar)	79.10	79.10	79.10	79.10
14	Temperature (C)	56.48	56.48	56.47	56.47
15	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
16	Heat Transferred (kJ/h-m)	576.9	576.8	576.8	576.6
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.1120	0.1120	0.1120	0.1662
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	3.870e-002	3.870e-002	3.870e-002	3.527e-002
22	Static Gradient (kPa/m)	-7.997e-003	-7.997e-003	-7.997e-003	1.275e-002
23	Accel. Gradient (kPa/m)	8.922e-006	8.923e-006	8.923e-006	1.395e-005
24	Liq. Reynolds	1.356e+005 *	1.356e+005 *	1.356e+005 *	1.356e+005 *
25	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
26	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
27	Vapour Velocity (m/s)	5.181	5.181	5.181	5.181
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	2581	2586	2591	2596
31	Elevation (m)	-8.000e-002	-4.000e-002	2.776e-017	4.000e-002
32	Pressure (bar)	79.09	79.09	79.09	79.09
33	Temperature (C)	56.47	56.47	56.46	56.46
34	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
35	Heat Transferred (kJ/h-m)	576.6	576.5	576.4	576.4
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.1662	0.1662	0.1662	0.1662
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	3.527e-002	3.527e-002	3.527e-002	3.528e-002
41	Static Gradient (kPa/m)	1.275e-002	1.275e-002	1.275e-002	1.275e-002
42	Accel. Gradient (kPa/m)	1.396e-005	1.396e-005	1.396e-005	1.396e-005
43	Liq. Reynolds	1.356e+005 *	1.356e+005 *	1.355e+005 *	1.355e+005 *
44	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
45	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
46	Vapour Velocity (m/s)	5.181	5.181	5.181	5.181
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	2601	2606	2611	2616
50	Elevation (m)	3.000e-002	2.000e-002	1.000e-002	0.0000
51	Pressure (bar)	79.09	79.08	79.08	79.08
52	Temperature (C)	56.46	56.45	56.45	56.45
53	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
54	Heat Transferred (kJ/h-m)	576.4	576.3	576.3	576.2
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.1137	0.1137	0.1137	0.1137
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	3.852e-002	3.852e-002	3.852e-002	3.852e-002
60	Static Gradient (kPa/m)	-2.540e-003	-2.540e-003	-2.540e-003	-2.540e-003
61	Accel. Gradient (kPa/m)	1.046e-005	1.046e-005	1.046e-005	1.046e-005
62	Liq. Reynolds	1.355e+005 *	1.355e+005 *	1.355e+005 *	1.355e+005 *
63	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
64	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
65	Vapour Velocity (m/s)	5.181	5.181	5.181	5.181
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	2621	2624	2627	2630
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:09:19 2022				
5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Elevation (m)	-1.000e-002	-2.400e-002	-3.800e-002	-5.200e-002		
14	Pressure (bar)	79.08	79.08	79.08	79.08		
15	Temperature (C)	56.45	56.44	56.44	56.44		
16	Vapour Fraction	0.9343	0.9343	0.9343	0.9343		
17	Heat Transferred (kJ/h-m)	576.1	576.1	576.1	576.0		
18	Flow Regime	Transition	Transition	Transition	Transition		
19	Water Flow Regime						
20	Liquid HoldUp	0.1137	0.1127	0.1127	0.1127		
21	Water HoldUp	---	---	---	---		
22	Friction Gradient (kPa/m)	3.852e-002	3.864e-002	3.864e-002	3.864e-002		
23	Static Gradient (kPa/m)	-2.540e-003	-5.897e-003	-5.896e-003	-5.896e-003		
24	Accel. Gradient (kPa/m)	1.046e-005	9.518e-006	9.519e-006	9.519e-006		
25	Liq. Reynolds	1.355e+005 *	1.355e+005 *	1.355e+005 *	1.355e+005 *		
26	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *		
27	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084		
28	Vapour Velocity (m/s)	5.182	5.182	5.182	5.182		
29	Deposit Thickness (mm)	---	---	---	---		
30	Deposit Volume (m3)	---	---	---	---		
31	Length (m)	2633	2636	2640	2644		
32	Elevation (m)	-6.600e-002	-8.000e-002	-6.200e-002	-4.400e-002		
33	Pressure (bar)	79.07	79.07	79.07	79.07		
34	Temperature (C)	56.44	56.44	56.43	56.43		
35	Vapour Fraction	0.9343	0.9343	0.9343	0.9343		
36	Heat Transferred (kJ/h-m)	576.0	576.0	575.8	575.8		
37	Flow Regime	Transition	Transition	Transition	Transition		
38	Water Flow Regime						
39	Liquid HoldUp	0.1127	0.1127	0.1610	0.1610		
40	Water HoldUp	---	---	---	---		
41	Friction Gradient (kPa/m)	3.864e-002	3.864e-002	3.542e-002	3.542e-002		
42	Static Gradient (kPa/m)	-5.896e-003	-5.896e-003	7.028e-003	7.028e-003		
43	Accel. Gradient (kPa/m)	9.519e-006	9.519e-006	1.234e-005	1.234e-005		
44	Liq. Reynolds	1.355e+005 *	1.355e+005 *	1.355e+005 *	1.355e+005 *		
45	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *		
46	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084		
47	Vapour Velocity (m/s)	5.182	5.182	5.182	5.182		
48	Deposit Thickness (mm)	---	---	---	---		
49	Deposit Volume (m3)	---	---	---	---		
50	Length (m)	2648	2652	2656	2661		
51	Elevation (m)	-2.600e-002	-8.000e-003	1.000e-002	-6.000e-003		
52	Pressure (bar)	79.07	79.07	79.06	79.06		
53	Temperature (C)	56.43	56.43	56.43	56.42		
54	Vapour Fraction	0.9343	0.9343	0.9343	0.9343		
55	Heat Transferred (kJ/h-m)	575.7	575.7	575.6	575.6		
56	Flow Regime	Transition	Transition	Transition	Transition		
57	Water Flow Regime						
58	Liquid HoldUp	0.1610	0.1610	0.1610	0.1133		
59	Water HoldUp	---	---	---	---		
60	Friction Gradient (kPa/m)	3.542e-002	3.542e-002	3.542e-002	3.858e-002		
61	Static Gradient (kPa/m)	7.028e-003	7.028e-003	7.028e-003	-4.055e-003		
62	Accel. Gradient (kPa/m)	1.234e-005	1.234e-005	1.234e-005	1.004e-005		
63	Liq. Reynolds	1.355e+005 *	1.355e+005 *	1.355e+005 *	1.355e+005 *		
64	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *		
65	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084		
66	Vapour Velocity (m/s)	5.182	5.182	5.182	5.182		
67	Deposit Thickness (mm)	---	---	---	---		
68	Deposit Volume (m3)	---	---	---	---		
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1			Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:09:19 2022		
6					
7	Pipe Segment: BBOU-ELEP Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	2666	2671	2676	2681
14	Elevation (m)	-2.200e-002	-3.800e-002	-5.400e-002	-7.000e-002
15	Pressure (bar)	79.06	79.06	79.06	79.06
16	Temperature (C)	56.42	56.42	56.41	56.41
17	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
18	Heat Transferred (kJ/h-m)	575.6	575.5	575.5	575.4
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.1133	0.1133	0.1133	0.1132
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	3.858e-002	3.858e-002	3.858e-002	3.858e-002
24	Static Gradient (kPa/m)	-4.054e-003	-4.054e-003	-4.054e-003	-4.054e-003
25	Accel. Gradient (kPa/m)	1.004e-005	1.004e-005	1.004e-005	1.004e-005
26	Liq. Reynolds	1.355e+005 *	1.355e+005 *	1.355e+005 *	1.355e+005 *
27	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
28	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
29	Vapour Velocity (m/s)	5.182	5.182	5.182	5.182
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	2686	2691	2696	2701
33	Elevation (m)	-7.600e-002	-8.200e-002	-8.800e-002	-9.400e-002
34	Pressure (bar)	79.05	79.05	79.05	79.05
35	Temperature (C)	56.41	56.41	56.40	56.40
36	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
37	Heat Transferred (kJ/h-m)	575.3	575.3	575.2	575.2
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.1140	0.1140	0.1140	0.1140
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	3.850e-002	3.850e-002	3.850e-002	3.850e-002
43	Static Gradient (kPa/m)	-1.526e-003	-1.526e-003	-1.526e-003	-1.526e-003
44	Accel. Gradient (kPa/m)	1.075e-005	1.075e-005	1.075e-005	1.075e-005
45	Liq. Reynolds	1.355e+005 *	1.355e+005 *	1.355e+005 *	1.355e+005 *
46	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
47	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
48	Vapour Velocity (m/s)	5.182	5.182	5.182	5.183
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	2706	2711	2716	2721
52	Elevation (m)	-0.1000	-0.1120	-0.1240	-0.1360
53	Pressure (bar)	79.05	79.04	79.04	79.04
54	Temperature (C)	56.40	56.39	56.39	56.39
55	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
56	Heat Transferred (kJ/h-m)	575.1	575.0	575.0	574.9
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.1140	0.1136	0.1136	0.1136
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	3.850e-002	3.855e-002	3.855e-002	3.855e-002
62	Static Gradient (kPa/m)	-1.526e-003	-3.045e-003	-3.045e-003	-3.045e-003
63	Accel. Gradient (kPa/m)	1.075e-005	1.033e-005	1.033e-005	1.033e-005
64	Liq. Reynolds	1.355e+005 *	1.355e+005 *	1.355e+005 *	1.355e+005 *
65	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
66	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
67	Vapour Velocity (m/s)	5.183	5.183	5.183	5.183
68	Deposit Thickness (mm)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:09:19 2022		
4					
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	2726	2731	2736	2741
15	Elevation (m)	-0.1480	-0.1600	-0.1480	-0.1360
16	Pressure (bar)	79.04	79.04	79.04	79.03
17	Temperature (C)	56.39	56.38	56.38	56.38
18	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
19	Heat Transferred (kJ/h-m)	574.9	574.8	574.7	574.6
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.1136	0.1136	0.1580	0.1580
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	3.855e-002	3.855e-002	3.554e-002	3.554e-002
25	Static Gradient (kPa/m)	-3.045e-003	-3.045e-003	3.702e-003	3.702e-003
26	Accel. Gradient (kPa/m)	1.033e-005	1.033e-005	1.142e-005	1.142e-005
27	Liq. Reynolds	1.355e+005 *	1.355e+005 *	1.355e+005 *	1.355e+005 *
28	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
29	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
30	Vapour Velocity (m/s)	5.183	5.183	5.183	5.183
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	2746	2751	2756	2761
34	Elevation (m)	-0.1240	-0.1120	-0.1000	-0.1120
35	Pressure (bar)	79.03	79.03	79.03	79.03
36	Temperature (C)	56.38	56.37	56.37	56.37
37	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
38	Heat Transferred (kJ/h-m)	574.6	574.5	574.4	574.4
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.1580	0.1580	0.1579	0.1135
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	3.554e-002	3.554e-002	3.554e-002	3.856e-002
44	Static Gradient (kPa/m)	3.702e-003	3.702e-003	3.702e-003	-3.072e-003
45	Accel. Gradient (kPa/m)	1.142e-005	1.142e-005	1.142e-005	1.033e-005
46	Liq. Reynolds	1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *
47	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
48	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
49	Vapour Velocity (m/s)	5.183	5.183	5.183	5.183
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	2766	2771	2776	2781
53	Elevation (m)	-0.1240	-0.1360	-0.1480	-0.1600
54	Pressure (bar)	79.02	79.02	79.02	79.02
55	Temperature (C)	56.36	56.36	56.36	56.36
56	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
57	Heat Transferred (kJ/h-m)	574.4	574.3	574.3	574.2
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.1135	0.1135	0.1135	0.1135
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	3.856e-002	3.856e-002	3.856e-002	3.856e-002
63	Static Gradient (kPa/m)	-3.072e-003	-3.072e-003	-3.072e-003	-3.072e-003
64	Accel. Gradient (kPa/m)	1.033e-005	1.033e-005	1.033e-005	1.033e-005
65	Liq. Reynolds	1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *
66	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
67	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
68	Vapour Velocity (m/s)	5.183	5.183	5.183	5.184
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 9 of 116

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:09:19 2022		
4					
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	2786	2791	2796	2801
16	Elevation (m)	-0.1680	-0.1760	-0.1840	-0.1920
17	Pressure (bar)	79.02	79.02	79.01	79.01
18	Temperature (C)	56.35	56.35	56.35	56.34
19	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
20	Heat Transferred (kJ/h-m)	574.1	574.1	574.0	574.0
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.1139	0.1139	0.1139	0.1139
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	3.852e-002	3.852e-002	3.852e-002	3.852e-002
26	Static Gradient (kPa/m)	-2.033e-003	-2.033e-003	-2.033e-003	-2.033e-003
27	Accel. Gradient (kPa/m)	1.062e-005	1.062e-005	1.062e-005	1.062e-005
28	Liq. Reynolds	1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *
29	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
30	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
31	Vapour Velocity (m/s)	5.184	5.184	5.184	5.184
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	2806	2811	2816	2821
35	Elevation (m)	-0.2000	-0.2020	-0.2040	-0.2060
36	Pressure (bar)	79.01	79.01	79.01	79.00
37	Temperature (C)	56.34	56.34	56.34	56.33
38	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
39	Heat Transferred (kJ/h-m)	573.9	573.8	573.8	573.7
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.1139	0.1143	0.1143	0.1143
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	3.853e-002	3.848e-002	3.848e-002	3.848e-002
45	Static Gradient (kPa/m)	-2.033e-003	-5.094e-004	-5.094e-004	-5.094e-004
46	Accel. Gradient (kPa/m)	1.062e-005	1.105e-005	1.105e-005	1.105e-005
47	Liq. Reynolds	1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *
48	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
49	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
50	Vapour Velocity (m/s)	5.184	5.184	5.184	5.184
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	2826	2831	2836	2841
54	Elevation (m)	-0.2080	-0.2100	-0.2100	-0.2100
55	Pressure (bar)	79.00	79.00	79.00	79.00
56	Temperature (C)	56.33	56.33	56.33	56.32
57	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
58	Heat Transferred (kJ/h-m)	573.6	573.6	573.4	573.4
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.1143	0.1143	0.1671	0.1671
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	3.848e-002	3.848e-002	3.527e-002	3.527e-002
64	Static Gradient (kPa/m)	-5.094e-004	-5.094e-004	0.0000	0.0000
65	Accel. Gradient (kPa/m)	1.105e-005	1.105e-005	1.027e-005	1.027e-005
66	Liq. Reynolds	1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *
67	Vap. Reynolds	1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *
68	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2			Pipe Segment: BBOU-ELEP Pipeline (continued)				
3							
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6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	5.184	5.184	5.184	5.184	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	2846	2851	2856	2861	
17	Elevation	(m)	-0.2100	-0.2100	-0.2100	-0.2820	
18	Pressure	(bar)	79.00	78.99	78.99	78.99	
19	Temperature	(C)	56.32	56.32	56.31	56.31	
20	Vapour Fraction		0.9343	0.9343	0.9343	0.9343	
21	Heat Transferred	(kJ/h-m)	573.3	573.3	573.2	573.2	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.1671	0.1671	0.1671	0.1088	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	3.527e-002	3.527e-002	3.527e-002	3.910e-002	
27	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	-1.785e-002	
28	Accel. Gradient	(kPa/m)	1.027e-005	1.027e-005	1.027e-005	6.188e-006	
29	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *	
30	Vap. Reynolds		1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *	
31	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
32	Vapour Velocity	(m/s)	5.184	5.184	5.185	5.185	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	2866	2871	2876	2881	
36	Elevation	(m)	-0.3540	-0.4260	-0.4980	-0.5700	
37	Pressure	(bar)	78.99	78.99	78.99	78.99	
38	Temperature	(C)	56.31	56.31	56.30	56.30	
39	Vapour Fraction		0.9343	0.9343	0.9343	0.9343	
40	Heat Transferred	(kJ/h-m)	573.2	573.1	573.1	573.0	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.1088	0.1088	0.1088	0.1088	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	3.910e-002	3.910e-002	3.910e-002	3.910e-002	
46	Static Gradient	(kPa/m)	-1.785e-002	-1.785e-002	-1.785e-002	-1.785e-002	
47	Accel. Gradient	(kPa/m)	6.189e-006	6.189e-006	6.189e-006	6.189e-006	
48	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *	
49	Vap. Reynolds		1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *	
50	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
51	Vapour Velocity	(m/s)	5.185	5.185	5.185	5.185	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	2886	2891	2896	2901	
55	Elevation	(m)	-0.5000	-0.4300	-0.3600	-0.2900	
56	Pressure	(bar)	78.98	78.98	78.98	78.97	
57	Temperature	(C)	56.30	56.30	56.29	56.29	
58	Vapour Fraction		0.9343	0.9343	0.9343	0.9343	
59	Heat Transferred	(kJ/h-m)	572.8	572.8	572.7	572.6	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.1748	0.1748	0.1748	0.1748	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	3.520e-002	3.520e-002	3.520e-002	3.520e-002	
65	Static Gradient	(kPa/m)	2.305e-002	2.305e-002	2.305e-002	2.305e-002	
66	Accel. Gradient	(kPa/m)	1.697e-005	1.697e-005	1.697e-005	1.697e-005	
67	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *	
68	Vap. Reynolds		1.214e+007 *	1.214e+007 *	1.214e+007 *	1.214e+007 *	
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:09:19 2022				
5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
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9	PROFILES						
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11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
14	Vapour Velocity	(m/s)	5.185	5.185	5.185	5.185	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	2906	2911	2916	2921	
18	Elevation	(m)	-0.2200	-0.1360	-5.200e-002	3.200e-002	
19	Pressure	(bar)	78.97	78.97	78.97	78.96	
20	Temperature	(C)	56.29	56.28	56.28	56.28	
21	Vapour Fraction		0.9343	0.9343	0.9343	0.9343	
22	Heat Transferred	(kJ/h-m)	572.6	572.5	572.4	572.3	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.1748	0.1789	0.1789	0.1789	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	3.520e-002	3.527e-002	3.527e-002	3.527e-002	
28	Static Gradient	(kPa/m)	2.305e-002	2.809e-002	2.808e-002	2.808e-002	
29	Accel. Gradient	(kPa/m)	1.697e-005	1.846e-005	1.846e-005	1.846e-005	
30	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *	
31	Vap. Reynolds		1.214e+007 *	1.214e+007 *	1.215e+007 *	1.215e+007 *	
32	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
33	Vapour Velocity	(m/s)	5.185	5.185	5.186	5.186	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	2926	2931	2936	2941	
37	Elevation	(m)	0.1160	0.2000	0.1760	0.1520	
38	Pressure	(bar)	78.96	78.96	78.95	78.95	
39	Temperature	(C)	56.27	56.27	56.27	56.26	
40	Vapour Fraction		0.9343	0.9343	0.9343	0.9343	
41	Heat Transferred	(kJ/h-m)	572.3	572.2	572.2	572.2	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.1789	0.1789	0.1126	0.1126	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	3.528e-002	3.528e-002	3.868e-002	3.868e-002	
47	Static Gradient	(kPa/m)	2.808e-002	2.808e-002	-6.060e-003	-6.060e-003	
48	Accel. Gradient	(kPa/m)	1.846e-005	1.846e-005	9.505e-006	9.506e-006	
49	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *	
50	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *	
51	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
52	Vapour Velocity	(m/s)	5.186	5.186	5.186	5.186	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	2946	2951	2956	2961	
56	Elevation	(m)	0.1280	0.1040	8.000e-002	7.000e-002	
57	Pressure	(bar)	78.95	78.95	78.95	78.95	
58	Temperature	(C)	56.26	56.26	56.26	56.25	
59	Vapour Fraction		0.9343	0.9343	0.9343	0.9343	
60	Heat Transferred	(kJ/h-m)	572.1	572.1	572.0	571.9	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.1126	0.1126	0.1126	0.1137	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	3.868e-002	3.868e-002	3.868e-002	3.856e-002	
66	Static Gradient	(kPa/m)	-6.060e-003	-6.060e-003	-6.060e-003	-2.512e-003	
67	Accel. Gradient	(kPa/m)	9.506e-006	9.507e-006	9.507e-006	1.051e-005	
68	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *	
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
Unit Set: Mixed					
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7	Pipe Segment: BBOU-ELEP Pipeline (continued)				
8					
9	PROFILES				
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11	Pipe Table				
12					
13	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
14	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
15	Vapour Velocity (m/s)	5.186	5.186	5.186	5.186
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	2966	2971	2976	2981
19	Elevation (m)	6.000e-002	5.000e-002	4.000e-002	3.000e-002
20	Pressure (bar)	78.94	78.94	78.94	78.94
21	Temperature (C)	56.25	56.25	56.24	56.24
22	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
23	Heat Transferred (kJ/h-m)	571.9	571.8	571.8	571.7
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1137	0.1137	0.1137	0.1137
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	3.856e-002	3.856e-002	3.856e-002	3.856e-002
29	Static Gradient (kPa/m)	-2.512e-003	-2.512e-003	-2.512e-003	-2.512e-003
30	Accel. Gradient (kPa/m)	1.051e-005	1.051e-005	1.051e-005	1.051e-005
31	Liq. Reynolds	1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *
32	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
33	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
34	Vapour Velocity (m/s)	5.186	5.186	5.187	5.187
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	2986	2991	2996	3001
38	Elevation (m)	3.600e-002	4.200e-002	4.800e-002	5.400e-002
39	Pressure (bar)	78.94	78.94	78.93	78.93
40	Temperature (C)	56.24	56.24	56.23	56.23
41	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
42	Heat Transferred (kJ/h-m)	571.6	571.5	571.5	571.4
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1561	0.1561	0.1561	0.1561
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	3.563e-002	3.563e-002	3.563e-002	3.563e-002
48	Static Gradient (kPa/m)	1.838e-003	1.838e-003	1.838e-003	1.838e-003
49	Accel. Gradient (kPa/m)	1.092e-005	1.092e-005	1.092e-005	1.092e-005
50	Liq. Reynolds	1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *
51	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
52	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
53	Vapour Velocity (m/s)	5.187	5.187	5.187	5.187
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	3006	3011	3016	3021
57	Elevation (m)	6.000e-002	6.000e-002	6.000e-002	6.000e-002
58	Pressure (bar)	78.93	78.93	78.93	78.92
59	Temperature (C)	56.23	56.22	56.22	56.22
60	Vapour Fraction	0.9343	0.9343	0.9343	0.9343
61	Heat Transferred (kJ/h-m)	571.3	571.2	571.2	571.1
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1561	0.1671	0.1671	0.1671
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	3.563e-002	3.529e-002	3.529e-002	3.529e-002
67	Static Gradient (kPa/m)	1.838e-003	0.0000	0.0000	0.0000
68	Accel. Gradient (kPa/m)	1.092e-005	1.029e-005	1.029e-005	1.029e-005
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:09:19 2022				
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
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11	Pipe Table						
12							
13	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *	
14	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *	
15	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084		
16	Vapour Velocity (m/s)	5.187	5.187	5.187	5.187		
17	Deposit Thickness (mm)	---	---	---	---		
18	Deposit Volume (m3)	---	---	---	---		
19	Length (m)	3026	3031	3036	3041		
20	Elevation (m)	6.000e-002	6.000e-002	5.800e-002	5.600e-002		
21	Pressure (bar)	78.92	78.92	78.92	78.92		
22	Temperature (C)	56.22	56.21	56.21	56.21		
23	Vapour Fraction	0.9343	0.9343	0.9343	0.9343		
24	Heat Transferred (kJ/h-m)	571.1	571.0	571.0	571.0		
25	Flow Regime	Transition	Transition	Transition	Transition		
26	Water Flow Regime						
27	Liquid HoldUp	0.1671	0.1671	0.1143	0.1143		
28	Water HoldUp	---	---	---	---		
29	Friction Gradient (kPa/m)	3.529e-002	3.529e-002	3.850e-002	3.850e-002		
30	Static Gradient (kPa/m)	0.0000	0.0000	-5.092e-004	-5.092e-004		
31	Accel. Gradient (kPa/m)	1.029e-005	1.029e-005	1.108e-005	1.108e-005		
32	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *		
33	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *		
34	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084		
35	Vapour Velocity (m/s)	5.187	5.187	5.187	5.187		
36	Deposit Thickness (mm)	---	---	---	---		
37	Deposit Volume (m3)	---	---	---	---		
38	Length (m)	3046	3051	3056	3061		
39	Elevation (m)	5.400e-002	5.200e-002	5.000e-002	5.200e-002		
40	Pressure (bar)	78.91	78.91	78.91	78.91		
41	Temperature (C)	56.21	56.20	56.20	56.20		
42	Vapour Fraction	0.9343	0.9343	0.9343	0.9343		
43	Heat Transferred (kJ/h-m)	570.9	570.8	570.8	570.7		
44	Flow Regime	Transition	Transition	Transition	Transition		
45	Water Flow Regime						
46	Liquid HoldUp	0.1143	0.1143	0.1143	0.1550		
47	Water HoldUp	---	---	---	---		
48	Friction Gradient (kPa/m)	3.850e-002	3.850e-002	3.850e-002	3.568e-002		
49	Static Gradient (kPa/m)	-5.092e-004	-5.092e-004	-5.092e-004	6.096e-004		
50	Accel. Gradient (kPa/m)	1.108e-005	1.108e-005	1.108e-005	1.058e-005		
51	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *		
52	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *		
53	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084		
54	Vapour Velocity (m/s)	5.187	5.187	5.188	5.188		
55	Deposit Thickness (mm)	---	---	---	---		
56	Deposit Volume (m3)	---	---	---	---		
57	Length (m)	3066	3071	3076	3081		
58	Elevation (m)	5.400e-002	5.600e-002	5.800e-002	6.000e-002		
59	Pressure (bar)	78.91	78.91	78.90	78.90		
60	Temperature (C)	56.19	56.19	56.19	56.19		
61	Vapour Fraction	0.9343	0.9343	0.9343	0.9343		
62	Heat Transferred (kJ/h-m)	570.6	570.5	570.5	570.4		
63	Flow Regime	Transition	Transition	Transition	Transition		
64	Water Flow Regime						
65	Liquid HoldUp	0.1550	0.1550	0.1549	0.1549		
66	Water HoldUp	---	---	---	---		
67	Friction Gradient (kPa/m)	3.568e-002	3.568e-002	3.568e-002	3.568e-002		
68	Static Gradient (kPa/m)	6.096e-004	6.096e-004	6.096e-004	6.096e-004		
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:09:19 2022				
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient	(kPa/m)	1.059e-005	1.059e-005	1.059e-005	1.059e-005	
14	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *	
15	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *	
16	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
17	Vapour Velocity	(m/s)	5.188	5.188	5.188	5.188	
18	Deposit Thickness	(mm)	---	---	---	---	
19	Deposit Volume	(m3)	---	---	---	---	
20	Length	(m)	3086	3091	3096	3101	
21	Elevation	(m)	6.400e-002	6.800e-002	7.200e-002	7.600e-002	
22	Pressure	(bar)	78.90	78.90	78.90	78.89	
23	Temperature	(C)	56.18	56.18	56.18	56.17	
24	Vapour Fraction		0.9343	0.9343	0.9343	0.9343	
25	Heat Transferred	(kJ/h-m)	570.4	570.3	570.2	570.2	
26	Flow Regime		Transition	Transition	Transition	Transition	
27	Water Flow Regime						
28	Liquid HoldUp		0.1555	0.1555	0.1555	0.1555	
29	Water HoldUp		---	---	---	---	
30	Friction Gradient	(kPa/m)	3.566e-002	3.566e-002	3.566e-002	3.566e-002	
31	Static Gradient	(kPa/m)	1.222e-003	1.222e-003	1.222e-003	1.222e-003	
32	Accel. Gradient	(kPa/m)	1.076e-005	1.076e-005	1.076e-005	1.076e-005	
33	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *	
34	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *	
35	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
36	Vapour Velocity	(m/s)	5.188	5.188	5.188	5.188	
37	Deposit Thickness	(mm)	---	---	---	---	
38	Deposit Volume	(m3)	---	---	---	---	
39	Length	(m)	3106	3111	3116	3121	
40	Elevation	(m)	8.000e-002	7.800e-002	7.600e-002	7.400e-002	
41	Pressure	(bar)	78.89	78.89	78.89	78.89	
42	Temperature	(C)	56.17	56.17	56.17	56.16	
43	Vapour Fraction		0.9343	0.9343	0.9343	0.9343	
44	Heat Transferred	(kJ/h-m)	570.1	570.1	570.1	570.0	
45	Flow Regime		Transition	Transition	Transition	Transition	
46	Water Flow Regime						
47	Liquid HoldUp		0.1555	0.1143	0.1143	0.1143	
48	Water HoldUp		---	---	---	---	
49	Friction Gradient	(kPa/m)	3.566e-002	3.851e-002	3.851e-002	3.851e-002	
50	Static Gradient	(kPa/m)	1.222e-003	-5.092e-004	-5.092e-004	-5.092e-004	
51	Accel. Gradient	(kPa/m)	1.076e-005	1.109e-005	1.109e-005	1.109e-005	
52	Liq. Reynolds		1.354e+005 *	1.354e+005 *	1.354e+005 *	1.354e+005 *	
53	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *	
54	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
55	Vapour Velocity	(m/s)	5.188	5.188	5.188	5.188	
56	Deposit Thickness	(mm)	---	---	---	---	
57	Deposit Volume	(m3)	---	---	---	---	
58	Length	(m)	3126	3131	3136	3141	
59	Elevation	(m)	7.200e-002	7.000e-002	7.600e-002	8.200e-002	
60	Pressure	(bar)	78.89	78.88	78.88	78.88	
61	Temperature	(C)	56.16	56.16	56.15	56.15	
62	Vapour Fraction		0.9343	0.9343	0.9343	0.9343	
63	Heat Transferred	(kJ/h-m)	569.9	569.9	569.8	569.7	
64	Flow Regime		Transition	Transition	Transition	Transition	
65	Water Flow Regime						
66	Liquid HoldUp		0.1143	0.1143	0.1561	0.1561	
67	Water HoldUp		---	---	---	---	
68	Friction Gradient	(kPa/m)	3.851e-002	3.851e-002	3.564e-002	3.564e-002	
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1				Case Name: Simulation - 331MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:09:19 2022		
6						
7	Pipe Segment: BBOU-ELEP Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	-5.091e-004	-5.091e-004	1.837e-003	1.837e-003
14	Accel. Gradient	(kPa/m)	1.109e-005	1.109e-005	1.094e-005	1.094e-005
15	Liq. Reynolds		1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
16	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
17	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084
18	Vapour Velocity	(m/s)	5.188	5.189	5.189	5.189
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	3146	3151	3156	3161
22	Elevation	(m)	8.800e-002	9.400e-002	1.000e-001	0.1060
23	Pressure	(bar)	78.88	78.88	78.87	78.87
24	Temperature	(C)	56.15	56.15	56.14	56.14
25	Vapour Fraction		0.9343	0.9343	0.9343	0.9343
26	Heat Transferred	(kJ/h-m)	569.6	569.6	569.5	569.4
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.1561	0.1561	0.1561	0.1561
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	3.564e-002	3.564e-002	3.564e-002	3.564e-002
32	Static Gradient	(kPa/m)	1.837e-003	1.837e-003	1.837e-003	1.827e-003
33	Accel. Gradient	(kPa/m)	1.094e-005	1.094e-005	1.094e-005	1.094e-005
34	Liq. Reynolds		1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
35	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
36	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084
37	Vapour Velocity	(m/s)	5.189	5.189	5.189	5.189
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	3166	3171	3176	3181
41	Elevation	(m)	0.1120	0.1180	0.1240	0.1300
42	Pressure	(bar)	78.87	78.87	78.87	78.86
43	Temperature	(C)	56.14	56.13	56.13	56.13
44	Vapour Fraction		0.9343	0.9343	0.9343	0.9343
45	Heat Transferred	(kJ/h-m)	569.4	569.3	569.3	569.2
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.1561	0.1561	0.1561	0.1561
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	3.564e-002	3.564e-002	3.564e-002	3.564e-002
51	Static Gradient	(kPa/m)	1.827e-003	1.827e-003	1.827e-003	1.827e-003
52	Accel. Gradient	(kPa/m)	1.094e-005	1.094e-005	1.094e-005	1.094e-005
53	Liq. Reynolds		1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
54	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
55	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084
56	Vapour Velocity	(m/s)	5.189	5.189	5.189	5.189
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	3186	3191	3196	3201
60	Elevation	(m)	0.1340	0.1380	0.1420	0.1460
61	Pressure	(bar)	78.86	78.86	78.86	78.86
62	Temperature	(C)	56.13	56.12	56.12	56.12
63	Vapour Fraction		0.9343	0.9342	0.9342	0.9342
64	Heat Transferred	(kJ/h-m)	569.1	569.1	569.0	569.0
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.1555	0.1555	0.1555	0.1555
68	Water HoldUp		---	---	---	---
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1			Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
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5			Date/Time: Mon Jun 13 15:09:19 2022				
6							
7	Pipe Segment: BBOU-ELEP Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	3.567e-002	3.567e-002	3.567e-002	3.567e-002	
14	Static Gradient	(kPa/m)	1.222e-003	1.222e-003	1.222e-003	1.222e-003	
15	Accel. Gradient	(kPa/m)	1.077e-005	1.077e-005	1.077e-005	1.077e-005	
16	Liq. Reynolds		1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *	
17	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *	
18	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
19	Vapour Velocity	(m/s)	5.189	5.189	5.189	5.189	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	3206	3212	3218	3224	
23	Elevation	(m)	0.1500	0.1540	0.1580	0.1620	
24	Pressure	(bar)	78.86	78.85	78.85	78.85	
25	Temperature	(C)	56.12	56.11	56.11	56.11	
26	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
27	Heat Transferred	(kJ/h-m)	568.9	568.8	568.7	568.7	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.1555	0.1553	0.1553	0.1553	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	3.567e-002	3.568e-002	3.568e-002	3.568e-002	
33	Static Gradient	(kPa/m)	1.222e-003	1.017e-003	1.017e-003	1.017e-003	
34	Accel. Gradient	(kPa/m)	1.077e-005	1.071e-005	1.072e-005	1.072e-005	
35	Liq. Reynolds		1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *	
36	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *	
37	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
38	Vapour Velocity	(m/s)	5.190	5.190	5.190	5.190	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	3230	3236	3239	3242	
42	Elevation	(m)	0.1660	0.1700	0.1540	0.1380	
43	Pressure	(bar)	78.85	78.84	78.84	78.84	
44	Temperature	(C)	56.10	56.10	56.10	56.10	
45	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
46	Heat Transferred	(kJ/h-m)	568.6	568.5	568.6	568.5	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.1553	0.1553	0.1123	0.1123	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	3.568e-002	3.568e-002	3.873e-002	3.873e-002	
52	Static Gradient	(kPa/m)	1.017e-003	1.017e-003	-6.723e-003	-6.723e-003	
53	Accel. Gradient	(kPa/m)	1.072e-005	1.072e-005	9.347e-006	9.348e-006	
54	Liq. Reynolds		1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *	
55	Vap. Reynolds		1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *	
56	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
57	Vapour Velocity	(m/s)	5.190	5.190	5.190	5.190	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	3245	3248	3251	3257	
61	Elevation	(m)	0.1220	0.1060	9.000e-002	0.1160	
62	Pressure	(bar)	78.84	78.84	78.84	78.84	
63	Temperature	(C)	56.09	56.09	56.09	56.09	
64	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
65	Heat Transferred	(kJ/h-m)	568.5	568.5	568.4	568.3	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.1123	0.1123	0.1123	0.1606	
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:09:19 2022		
4					
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	3.873e-002	3.873e-002	3.873e-002	3.549e-002
15	Static Gradient (kPa/m)	-6.723e-003	-6.723e-003	-6.723e-003	6.755e-003
16	Accel. Gradient (kPa/m)	9.348e-006	9.348e-006	9.348e-006	1.234e-005
17	Liq. Reynolds	1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
18	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
19	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
20	Vapour Velocity (m/s)	5.190	5.190	5.190	5.190
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	3263	3269	3275	3281
24	Elevation (m)	0.1420	0.1680	0.1940	0.2200
25	Pressure (bar)	78.83	78.83	78.83	78.83
26	Temperature (C)	56.08	56.08	56.08	56.07
27	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
28	Heat Transferred (kJ/h-m)	568.2	568.1	568.0	568.0
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.1606	0.1606	0.1606	0.1606
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	3.549e-002	3.549e-002	3.549e-002	3.549e-002
34	Static Gradient (kPa/m)	6.755e-003	6.755e-003	6.755e-003	6.755e-003
35	Accel. Gradient (kPa/m)	1.234e-005	1.234e-005	1.234e-005	1.234e-005
36	Liq. Reynolds	1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
37	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
38	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
39	Vapour Velocity (m/s)	5.190	5.190	5.190	5.191
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	3286	3291	3296	3301
43	Elevation (m)	0.2060	0.1920	0.1780	0.1640
44	Pressure (bar)	78.83	78.82	78.82	78.82
45	Temperature (C)	56.07	56.07	56.06	56.06
46	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
47	Heat Transferred (kJ/h-m)	568.0	567.9	567.9	567.8
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.1133	0.1133	0.1133	0.1133
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	3.863e-002	3.863e-002	3.863e-002	3.863e-002
53	Static Gradient (kPa/m)	-3.547e-003	-3.547e-003	-3.547e-003	-3.547e-003
54	Accel. Gradient (kPa/m)	1.025e-005	1.025e-005	1.025e-005	1.025e-005
55	Liq. Reynolds	1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
56	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
57	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
58	Vapour Velocity (m/s)	5.191	5.191	5.191	5.191
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	3306	3311	3316	3321
62	Elevation (m)	0.1500	0.1400	0.1300	0.1200
63	Pressure (bar)	78.82	78.82	78.81	78.81
64	Temperature (C)	56.06	56.06	56.05	56.05
65	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
66	Heat Transferred (kJ/h-m)	567.7	567.7	567.6	567.6
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:09:19 2022
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.1133	0.1136	0.1136	0.1136
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	3.863e-002	3.859e-002	3.860e-002	3.860e-002
16	Static Gradient (kPa/m)	-3.547e-003	-2.537e-003	-2.537e-003	-2.537e-003
17	Accel. Gradient (kPa/m)	1.025e-005	1.054e-005	1.054e-005	1.054e-005
18	Liq. Reynolds	1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
19	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
20	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
21	Vapour Velocity (m/s)	5.191	5.191	5.191	5.191
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	3326	3331	3336	3341
25	Elevation (m)	0.1100	1.000e-001	9.600e-002	9.200e-002
26	Pressure (bar)	78.81	78.81	78.81	78.81
27	Temperature (C)	56.05	56.05	56.04	56.04
28	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
29	Heat Transferred (kJ/h-m)	567.5	567.4	567.4	567.3
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1136	0.1136	0.1141	0.1141
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	3.860e-002	3.860e-002	3.855e-002	3.855e-002
35	Static Gradient (kPa/m)	-2.537e-003	-2.537e-003	-1.017e-003	-1.017e-003
36	Accel. Gradient (kPa/m)	1.054e-005	1.054e-005	1.097e-005	1.097e-005
37	Liq. Reynolds	1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
38	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
39	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
40	Vapour Velocity (m/s)	5.191	5.191	5.191	5.191
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	3346	3351	3356	3361
44	Elevation (m)	8.800e-002	8.400e-002	8.000e-002	8.400e-002
45	Pressure (bar)	78.80	78.80	78.80	78.80
46	Temperature (C)	56.04	56.03	56.03	56.03
47	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
48	Heat Transferred (kJ/h-m)	567.3	567.2	567.1	567.0
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1141	0.1141	0.1141	0.1555
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	3.855e-002	3.855e-002	3.855e-002	3.568e-002
54	Static Gradient (kPa/m)	-1.017e-003	-1.017e-003	-1.017e-003	1.221e-003
55	Accel. Gradient (kPa/m)	1.097e-005	1.097e-005	1.097e-005	1.079e-005
56	Liq. Reynolds	1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
57	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
58	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
59	Vapour Velocity (m/s)	5.191	5.191	5.191	5.192
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	3366	3371	3376	3381
63	Elevation (m)	8.800e-002	9.200e-002	9.600e-002	1.000e-001
64	Pressure (bar)	78.80	78.79	78.79	78.79
65	Temperature (C)	56.03	56.02	56.02	56.02
66	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
67	Heat Transferred (kJ/h-m)	567.0	566.9	566.8	566.8
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 19 of 116

1			Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:09:19 2022		
4					
5					
6					
7	Pipe Segment: BBOU-ELEP Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1555	0.1555	0.1555	0.1555
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	3.568e-002	3.568e-002	3.568e-002	3.569e-002
17	Static Gradient (kPa/m)	1.221e-003	1.221e-003	1.221e-003	1.221e-003
18	Accel. Gradient (kPa/m)	1.079e-005	1.079e-005	1.079e-005	1.079e-005
19	Liq. Reynolds	1.353e+005 *	1.353e+005 *	1.353e+005 *	1.353e+005 *
20	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
21	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
22	Vapour Velocity (m/s)	5.192	5.192	5.192	5.192
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	3386	3391	3396	3401
26	Elevation (m)	1.000e-001	1.000e-001	1.000e-001	1.000e-001
27	Pressure (bar)	78.79	78.79	78.79	78.78
28	Temperature (C)	56.02	56.01	56.01	56.01
29	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
30	Heat Transferred (kJ/h-m)	566.7	566.6	566.6	566.5
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1670	0.1670	0.1670	0.1670
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	3.532e-002	3.532e-002	3.532e-002	3.532e-002
36	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
37	Accel. Gradient (kPa/m)	1.033e-005	1.033e-005	1.033e-005	1.033e-005
38	Liq. Reynolds	1.353e+005 *	1.353e+005 *	1.352e+005 *	1.352e+005 *
39	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
40	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
41	Vapour Velocity (m/s)	5.192	5.192	5.192	5.192
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	3406	3411	3416	3421
45	Elevation (m)	1.000e-001	2.600e-002	-4.800e-002	-0.1220
46	Pressure (bar)	78.78	78.78	78.78	78.78
47	Temperature (C)	56.00	56.00	56.00	56.00
48	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
49	Heat Transferred (kJ/h-m)	566.4	566.5	566.4	566.4
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.1670	0.1085	0.1085	0.1085
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	3.533e-002	3.918e-002	3.918e-002	3.918e-002
55	Static Gradient (kPa/m)	0.0000	-1.831e-002	-1.831e-002	-1.831e-002
56	Accel. Gradient (kPa/m)	1.033e-005	6.101e-006	6.101e-006	6.101e-006
57	Liq. Reynolds	1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *
58	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
59	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
60	Vapour Velocity (m/s)	5.192	5.192	5.192	5.192
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	3426	3431	3436	3441
64	Elevation (m)	-0.1960	-0.2700	-0.2060	-0.1420
65	Pressure (bar)	78.78	78.78	78.77	78.77
66	Temperature (C)	55.99	55.99	55.99	55.99
67	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
68	Heat Transferred (kJ/h-m)	566.3	566.3	566.1	566.0
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 20 of 116

1			Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:09:19 2022		
6					
7	Pipe Segment: BBOU-ELEP Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.1085	0.1085	0.1729	0.1729
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	3.918e-002	3.918e-002	3.525e-002	3.525e-002
18	Static Gradient (kPa/m)	-1.831e-002	-1.831e-002	2.092e-002	2.092e-002
19	Accel. Gradient (kPa/m)	6.101e-006	6.101e-006	1.643e-005	1.643e-005
20	Liq. Reynolds	1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *
21	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
22	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
23	Vapour Velocity (m/s)	5.192	5.192	5.192	5.192
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	3446	3451	3456	3461
27	Elevation (m)	-7.800e-002	-1.400e-002	5.000e-002	4.200e-002
28	Pressure (bar)	78.77	78.77	78.76	78.76
29	Temperature (C)	55.98	55.98	55.98	55.97
30	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
31	Heat Transferred (kJ/h-m)	566.0	565.9	565.8	565.9
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1729	0.1729	0.1729	0.1137
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	3.525e-002	3.525e-002	3.525e-002	3.859e-002
37	Static Gradient (kPa/m)	2.092e-002	2.092e-002	2.092e-002	-2.031e-003
38	Accel. Gradient (kPa/m)	1.643e-005	1.643e-005	1.643e-005	1.070e-005
39	Liq. Reynolds	1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *
40	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
41	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
42	Vapour Velocity (m/s)	5.193	5.193	5.193	5.193
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	3466	3471	3476	3481
46	Elevation (m)	3.400e-002	2.600e-002	1.800e-002	1.000e-002
47	Pressure (bar)	78.76	78.76	78.76	78.75
48	Temperature (C)	55.97	55.97	55.97	55.96
49	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
50	Heat Transferred (kJ/h-m)	565.8	565.7	565.7	565.6
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1137	0.1137	0.1137	0.1137
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	3.859e-002	3.859e-002	3.859e-002	3.859e-002
56	Static Gradient (kPa/m)	-2.031e-003	-2.031e-003	-2.031e-003	-2.031e-003
57	Accel. Gradient (kPa/m)	1.070e-005	1.070e-005	1.070e-005	1.070e-005
58	Liq. Reynolds	1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *
59	Vap. Reynolds	1.215e+007 *	1.215e+007 *	1.215e+007 *	1.215e+007 *
60	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
61	Vapour Velocity (m/s)	5.193	5.193	5.193	5.193
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	3486	3491	3496	3501
65	Elevation (m)	4.000e-003	-2.000e-003	-8.000e-003	-1.400e-002
66	Pressure (bar)	78.75	78.75	78.75	78.75
67	Temperature (C)	55.96	55.96	55.95	55.95
68	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:09:19 2022			
5										
6	Pipe Segment: BBOU-ELEP Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Heat Transferred (kJ/h-m)	565.6	565.5	565.4	565.4					
14	Flow Regime	Transition	Transition	Transition	Transition					
15	Water Flow Regime									
16	Liquid HoldUp	0.1139	0.1139	0.1139	0.1139					
17	Water HoldUp	---	---	---	---					
18	Friction Gradient (kPa/m)	3.858e-002	3.858e-002	3.858e-002	3.858e-002					
19	Static Gradient (kPa/m)	-1.524e-003	-1.524e-003	-1.524e-003	-1.524e-003					
20	Accel. Gradient (kPa/m)	1.084e-005	1.084e-005	1.084e-005	1.084e-005					
21	Liq. Reynolds	1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *					
22	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *					
23	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084					
24	Vapour Velocity (m/s)	5.193	5.193	5.193	5.193					
25	Deposit Thickness (mm)	---	---	---	---					
26	Deposit Volume (m3)	---	---	---	---					
27	Length (m)	3506	3511	3516	3521					
28	Elevation (m)	-2.000e-002	-1.400e-002	-8.000e-003	-2.000e-003					
29	Pressure (bar)	78.74	78.74	78.74	78.74					
30	Temperature (C)	55.95	55.95	55.94	55.94					
31	Vapour Fraction	0.9342	0.9342	0.9342	0.9342					
32	Heat Transferred (kJ/h-m)	565.3	565.2	565.1	565.1					
33	Flow Regime	Transition	Transition	Transition	Transition					
34	Water Flow Regime									
35	Liquid HoldUp	0.1139	0.1560	0.1560	0.1560					
36	Water HoldUp	---	---	---	---					
37	Friction Gradient (kPa/m)	3.858e-002	3.567e-002	3.567e-002	3.567e-002					
38	Static Gradient (kPa/m)	-1.524e-003	1.836e-003	1.836e-003	1.836e-003					
39	Accel. Gradient (kPa/m)	1.084e-005	1.098e-005	1.098e-005	1.098e-005					
40	Liq. Reynolds	1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *					
41	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *					
42	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084					
43	Vapour Velocity (m/s)	5.193	5.194	5.194	5.194					
44	Deposit Thickness (mm)	---	---	---	---					
45	Deposit Volume (m3)	---	---	---	---					
46	Length (m)	3526	3531	3536	3541					
47	Elevation (m)	4.000e-003	1.000e-002	1.000e-002	1.000e-002					
48	Pressure (bar)	78.74	78.73	78.73	78.73					
49	Temperature (C)	55.94	55.93	55.93	55.93					
50	Vapour Fraction	0.9342	0.9342	0.9342	0.9342					
51	Heat Transferred (kJ/h-m)	565.0	565.0	564.9	564.8					
52	Flow Regime	Transition	Transition	Transition	Transition					
53	Water Flow Regime									
54	Liquid HoldUp	0.1560	0.1560	0.1669	0.1669					
55	Water HoldUp	---	---	---	---					
56	Friction Gradient (kPa/m)	3.568e-002	3.568e-002	3.534e-002	3.534e-002					
57	Static Gradient (kPa/m)	1.836e-003	1.836e-003	0.0000	0.0000					
58	Accel. Gradient (kPa/m)	1.098e-005	1.098e-005	1.034e-005	1.034e-005					
59	Liq. Reynolds	1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *					
60	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *					
61	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084					
62	Vapour Velocity (m/s)	5.194	5.194	5.194	5.194					
63	Deposit Thickness (mm)	---	---	---	---					
64	Deposit Volume (m3)	---	---	---	---					
65	Length (m)	3546	3551	3556	3561					
66	Elevation (m)	1.000e-002	1.000e-002	1.000e-002	8.000e-003					
67	Pressure (bar)	78.73	78.73	78.73	78.72					
68	Temperature (C)	55.93	55.92	55.92	55.92					
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 22 of 116					

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:09:19 2022		
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
14	Heat Transferred (kJ/h-m)	564.8	564.7	564.6	564.7
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.1669	0.1669	0.1669	0.1142
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	3.534e-002	3.534e-002	3.534e-002	3.856e-002
20	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-6.207e-004
21	Accel. Gradient (kPa/m)	1.034e-005	1.034e-005	1.034e-005	1.111e-005
22	Liq. Reynolds	1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *
23	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
24	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
25	Vapour Velocity (m/s)	5.194	5.194	5.194	5.194
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	3565	3569	3573	3577
29	Elevation (m)	6.000e-003	4.000e-003	2.000e-003	-1.232e-016
30	Pressure (bar)	78.72	78.72	78.72	78.72
31	Temperature (C)	55.92	55.91	55.91	55.91
32	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
33	Heat Transferred (kJ/h-m)	564.6	564.6	564.5	564.5
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1142	0.1142	0.1142	0.1142
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	3.856e-002	3.856e-002	3.856e-002	3.856e-002
39	Static Gradient (kPa/m)	-6.207e-004	-6.207e-004	-6.207e-004	-6.207e-004
40	Accel. Gradient (kPa/m)	1.111e-005	1.111e-005	1.111e-005	1.111e-005
41	Liq. Reynolds	1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *
42	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
43	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
44	Vapour Velocity (m/s)	5.194	5.194	5.194	5.194
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	3582	3587	3592	3597
48	Elevation (m)	3.600e-002	7.200e-002	0.1080	0.1440
49	Pressure (bar)	78.72	78.71	78.71	78.71
50	Temperature (C)	55.91	55.90	55.90	55.90
51	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
52	Heat Transferred (kJ/h-m)	564.3	564.3	564.2	564.1
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.1647	0.1647	0.1647	0.1647
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	3.539e-002	3.539e-002	3.539e-002	3.539e-002
58	Static Gradient (kPa/m)	1.140e-002	1.140e-002	1.140e-002	1.140e-002
59	Accel. Gradient (kPa/m)	1.370e-005	1.370e-005	1.370e-005	1.370e-005
60	Liq. Reynolds	1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *
61	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
62	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
63	Vapour Velocity (m/s)	5.195	5.195	5.195	5.195
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	3602	3607	3612	3617
67	Elevation (m)	0.1800	9.800e-002	1.600e-002	-6.600e-002
68	Pressure (bar)	78.71	78.71	78.70	78.70
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:09:19 2022		
4					
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	55.89	55.89	55.89	55.89
14	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
15	Heat Transferred (kJ/h-m)	564.1	564.1	564.1	564.0
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.1647	0.1079	0.1079	0.1079
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	3.540e-002	3.927e-002	3.927e-002	3.927e-002
21	Static Gradient (kPa/m)	1.140e-002	-2.022e-002	-2.022e-002	-2.022e-002
22	Accel. Gradient (kPa/m)	1.370e-005	5.579e-006	5.579e-006	5.580e-006
23	Liq. Reynolds	1.352e+005 *	1.352e+005 *	1.352e+005 *	1.352e+005 *
24	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
25	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
26	Vapour Velocity (m/s)	5.195	5.195	5.195	5.195
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	3622	3627	3632	3637
30	Elevation (m)	-0.1480	-0.2300	-0.1940	-0.1580
31	Pressure (bar)	78.70	78.70	78.70	78.70
32	Temperature (C)	55.89	55.88	55.88	55.88
33	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
34	Heat Transferred (kJ/h-m)	564.0	563.9	563.8	563.7
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1079	0.1079	0.1647	0.1647
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	3.927e-002	3.927e-002	3.540e-002	3.540e-002
40	Static Gradient (kPa/m)	-2.022e-002	-2.022e-002	1.140e-002	1.140e-002
41	Accel. Gradient (kPa/m)	5.580e-006	5.580e-006	1.371e-005	1.371e-005
42	Liq. Reynolds	1.352e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *
43	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
44	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
45	Vapour Velocity (m/s)	5.195	5.195	5.195	5.195
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	3642	3647	3652	3657
49	Elevation (m)	-0.1220	-8.600e-002	-5.000e-002	-3.600e-002
50	Pressure (bar)	78.69	78.69	78.69	78.69
51	Temperature (C)	55.87	55.87	55.87	55.87
52	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
53	Heat Transferred (kJ/h-m)	563.6	563.6	563.5	563.4
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.1647	0.1647	0.1647	0.1583
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	3.540e-002	3.540e-002	3.540e-002	3.560e-002
59	Static Gradient (kPa/m)	1.140e-002	1.140e-002	1.140e-002	4.324e-003
60	Accel. Gradient (kPa/m)	1.371e-005	1.371e-005	1.371e-005	1.170e-005
61	Liq. Reynolds	1.351e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *
62	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
63	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
64	Vapour Velocity (m/s)	5.195	5.195	5.195	5.196
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	3662	3667	3672	3677
68	Elevation (m)	-2.200e-002	-8.000e-003	6.000e-003	2.000e-002
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
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5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Pressure (bar)	78.69	78.68	78.68	78.68
14	Temperature (C)	55.86	55.86	55.86	55.85
15	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
16	Heat Transferred (kJ/h-m)	563.4	563.3	563.3	563.2
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.1583	0.1583	0.1583	0.1583
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	3.560e-002	3.560e-002	3.560e-002	3.560e-002
22	Static Gradient (kPa/m)	4.324e-003	4.324e-003	4.324e-003	4.324e-003
23	Accel. Gradient (kPa/m)	1.170e-005	1.170e-005	1.170e-005	1.170e-005
24	Liq. Reynolds	1.351e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *
25	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
26	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
27	Vapour Velocity (m/s)	5.196	5.196	5.196	5.196
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	3682	3687	3692	3697
31	Elevation (m)	3.000e-002	4.000e-002	5.000e-002	6.000e-002
32	Pressure (bar)	78.68	78.68	78.67	78.67
33	Temperature (C)	55.85	55.85	55.85	55.84
34	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
35	Heat Transferred (kJ/h-m)	563.1	563.1	563.0	562.9
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.1571	0.1571	0.1571	0.1571
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	3.564e-002	3.565e-002	3.565e-002	3.565e-002
41	Static Gradient (kPa/m)	3.074e-003	3.074e-003	3.074e-003	3.074e-003
42	Accel. Gradient (kPa/m)	1.134e-005	1.135e-005	1.135e-005	1.135e-005
43	Liq. Reynolds	1.351e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *
44	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
45	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
46	Vapour Velocity (m/s)	5.196	5.196	5.196	5.196
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	3702	3707	3712	3717
50	Elevation (m)	7.000e-002	8.400e-002	9.800e-002	0.1120
51	Pressure (bar)	78.67	78.67	78.67	78.66
52	Temperature (C)	55.84	55.84	55.83	55.83
53	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
54	Heat Transferred (kJ/h-m)	562.9	562.8	562.8	562.7
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.1571	0.1583	0.1583	0.1583
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	3.565e-002	3.560e-002	3.560e-002	3.561e-002
60	Static Gradient (kPa/m)	3.074e-003	4.324e-003	4.323e-003	4.323e-003
61	Accel. Gradient (kPa/m)	1.135e-005	1.170e-005	1.170e-005	1.170e-005
62	Liq. Reynolds	1.351e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *
63	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
64	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
65	Vapour Velocity (m/s)	5.196	5.196	5.196	5.196
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	3722	3727	3732	3737
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
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3					Unit Set: Mixed
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
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11	Pipe Table				
12					
13	Elevation (m)	0.1260	0.1400	0.1320	0.1240
14	Pressure (bar)	78.66	78.66	78.66	78.66
15	Temperature (C)	55.83	55.83	55.82	55.82
16	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
17	Heat Transferred (kJ/h-m)	562.6	562.6	562.6	562.5
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.1583	0.1583	0.1137	0.1137
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	3.561e-002	3.561e-002	3.862e-002	3.862e-002
23	Static Gradient (kPa/m)	4.323e-003	4.323e-003	-2.030e-003	-2.030e-003
24	Accel. Gradient (kPa/m)	1.170e-005	1.170e-005	1.073e-005	1.073e-005
25	Liq. Reynolds	1.351e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *
26	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
27	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
28	Vapour Velocity (m/s)	5.196	5.197	5.197	5.197
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	3742	3747	3752	3757
32	Elevation (m)	0.1160	0.1080	1.000e-001	0.1100
33	Pressure (bar)	78.65	78.65	78.65	78.65
34	Temperature (C)	55.82	55.81	55.81	55.81
35	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
36	Heat Transferred (kJ/h-m)	562.5	562.4	562.3	562.2
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.1137	0.1137	0.1137	0.1573
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	3.862e-002	3.862e-002	3.862e-002	3.564e-002
42	Static Gradient (kPa/m)	-2.030e-003	-2.030e-003	-2.030e-003	3.331e-003
43	Accel. Gradient (kPa/m)	1.073e-005	1.073e-005	1.073e-005	1.143e-005
44	Liq. Reynolds	1.351e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *
45	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
46	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
47	Vapour Velocity (m/s)	5.197	5.197	5.197	5.197
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	3761	3766	3770	3775
51	Elevation (m)	0.1200	0.1300	0.1400	0.1500
52	Pressure (bar)	78.65	78.65	78.64	78.64
53	Temperature (C)	55.81	55.80	55.80	55.80
54	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
55	Heat Transferred (kJ/h-m)	562.2	562.1	562.1	562.0
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.1573	0.1573	0.1573	0.1573
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	3.564e-002	3.564e-002	3.564e-002	3.564e-002
61	Static Gradient (kPa/m)	3.331e-003	3.331e-003	3.331e-003	3.331e-003
62	Accel. Gradient (kPa/m)	1.143e-005	1.143e-005	1.143e-005	1.143e-005
63	Liq. Reynolds	1.351e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *
64	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
65	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
66	Vapour Velocity (m/s)	5.197	5.197	5.197	5.197
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
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3			Unit Set: Mixed				
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length	(m)	3780	3785	3790	3795	
14	Elevation	(m)	0.1340	0.1180	0.1020	8.600e-002	
15	Pressure	(bar)	78.64	78.64	78.64	78.64	
16	Temperature	(C)	55.80	55.79	55.79	55.79	
17	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
18	Heat Transferred	(kJ/h-m)	562.0	561.9	561.9	561.8	
19	Flow Regime		Transition	Transition	Transition	Transition	
20	Water Flow Regime						
21	Liquid HoldUp		0.1130	0.1130	0.1130	0.1130	
22	Water HoldUp		---	---	---	---	
23	Friction Gradient	(kPa/m)	3.869e-002	3.869e-002	3.869e-002	3.870e-002	
24	Static Gradient	(kPa/m)	-4.047e-003	-4.047e-003	-4.047e-003	-4.047e-003	
25	Accel. Gradient	(kPa/m)	1.016e-005	1.016e-005	1.016e-005	1.016e-005	
26	Liq. Reynolds		1.351e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *	
27	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *	
28	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
29	Vapour Velocity	(m/s)	5.197	5.197	5.197	5.197	
30	Deposit Thickness	(mm)	---	---	---	---	
31	Deposit Volume	(m3)	---	---	---	---	
32	Length	(m)	3800	3805	3810	3815	
33	Elevation	(m)	7.000e-002	5.600e-002	4.200e-002	2.800e-002	
34	Pressure	(bar)	78.63	78.63	78.63	78.63	
35	Temperature	(C)	55.79	55.78	55.78	55.78	
36	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
37	Heat Transferred	(kJ/h-m)	561.8	561.7	561.6	561.6	
38	Flow Regime		Transition	Transition	Transition	Transition	
39	Water Flow Regime						
40	Liquid HoldUp		0.1130	0.1132	0.1132	0.1132	
41	Water HoldUp		---	---	---	---	
42	Friction Gradient	(kPa/m)	3.870e-002	3.868e-002	3.868e-002	3.868e-002	
43	Static Gradient	(kPa/m)	-4.047e-003	-3.544e-003	-3.544e-003	-3.544e-003	
44	Accel. Gradient	(kPa/m)	1.016e-005	1.030e-005	1.030e-005	1.031e-005	
45	Liq. Reynolds		1.351e+005 *	1.351e+005 *	1.351e+005 *	1.351e+005 *	
46	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *	
47	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
48	Vapour Velocity	(m/s)	5.198	5.198	5.198	5.198	
49	Deposit Thickness	(mm)	---	---	---	---	
50	Deposit Volume	(m3)	---	---	---	---	
51	Length	(m)	3820	3825	3830	3835	
52	Elevation	(m)	1.400e-002	-9.714e-017	-9.714e-017	-9.714e-017	
53	Pressure	(bar)	78.63	78.62	78.62	78.62	
54	Temperature	(C)	55.77	55.77	55.77	55.77	
55	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
56	Heat Transferred	(kJ/h-m)	561.5	561.5	561.3	561.3	
57	Flow Regime		Transition	Transition	Transition	Transition	
58	Water Flow Regime						
59	Liquid HoldUp		0.1132	0.1132	0.1669	0.1669	
60	Water HoldUp		---	---	---	---	
61	Friction Gradient	(kPa/m)	3.868e-002	3.868e-002	3.536e-002	3.536e-002	
62	Static Gradient	(kPa/m)	-3.544e-003	-3.544e-003	0.0000	0.0000	
63	Accel. Gradient	(kPa/m)	1.031e-005	1.031e-005	1.037e-005	1.037e-005	
64	Liq. Reynolds		1.351e+005 *	1.351e+005 *	1.351e+005 *	1.350e+005 *	
65	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *	
66	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
67	Vapour Velocity	(m/s)	5.198	5.198	5.198	5.198	
68	Deposit Thickness	(mm)	---	---	---	---	
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
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3			Unit Set: Mixed		
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	3840	3845	3850	3855
15	Elevation (m)	-9.714e-017	-9.714e-017	-9.714e-017	6.000e-003
16	Pressure (bar)	78.62	78.62	78.62	78.61
17	Temperature (C)	55.76	55.76	55.76	55.76
18	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
19	Heat Transferred (kJ/h-m)	561.2	561.1	561.1	561.1
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.1669	0.1669	0.1669	0.1559
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	3.536e-002	3.537e-002	3.537e-002	3.571e-002
25	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	1.835e-003
26	Accel. Gradient (kPa/m)	1.037e-005	1.037e-005	1.037e-005	1.101e-005
27	Liq. Reynolds	1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *
28	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
29	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
30	Vapour Velocity (m/s)	5.198	5.198	5.198	5.198
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	3860	3865	3870	3875
34	Elevation (m)	1.200e-002	1.800e-002	2.400e-002	3.000e-002
35	Pressure (bar)	78.61	78.61	78.61	78.61
36	Temperature (C)	55.75	55.75	55.75	55.74
37	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
38	Heat Transferred (kJ/h-m)	561.0	560.9	560.9	560.8
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.1559	0.1559	0.1559	0.1559
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	3.571e-002	3.571e-002	3.571e-002	3.571e-002
44	Static Gradient (kPa/m)	1.835e-003	1.835e-003	1.835e-003	1.835e-003
45	Accel. Gradient (kPa/m)	1.101e-005	1.101e-005	1.102e-005	1.102e-005
46	Liq. Reynolds	1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *
47	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
48	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
49	Vapour Velocity (m/s)	5.198	5.198	5.198	5.198
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	3880	3885	3890	3895
53	Elevation (m)	4.200e-002	5.400e-002	6.600e-002	7.800e-002
54	Pressure (bar)	78.60	78.60	78.60	78.60
55	Temperature (C)	55.74	55.74	55.74	55.73
56	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
57	Heat Transferred (kJ/h-m)	560.7	560.7	560.6	560.6
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.1577	0.1577	0.1577	0.1577
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	3.564e-002	3.564e-002	3.564e-002	3.564e-002
63	Static Gradient (kPa/m)	3.696e-003	3.696e-003	3.696e-003	3.696e-003
64	Accel. Gradient (kPa/m)	1.154e-005	1.154e-005	1.154e-005	1.154e-005
65	Liq. Reynolds	1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *
66	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
67	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
68	Vapour Velocity (m/s)	5.199	5.199	5.199	5.199
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:09:19 2022		
4					
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	3900	3905	3910	3915
16	Elevation (m)	9.000e-002	0.1160	0.1420	0.1680
17	Pressure (bar)	78.60	78.59	78.59	78.59
18	Temperature (C)	55.73	55.73	55.72	55.72
19	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
20	Heat Transferred (kJ/h-m)	560.5	560.4	560.4	560.3
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.1577	0.1617	0.1617	0.1617
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	3.564e-002	3.551e-002	3.551e-002	3.551e-002
26	Static Gradient (kPa/m)	3.696e-003	8.139e-003	8.139e-003	8.139e-003
27	Accel. Gradient (kPa/m)	1.155e-005	1.281e-005	1.281e-005	1.281e-005
28	Liq. Reynolds	1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *
29	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
30	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
31	Vapour Velocity (m/s)	5.199	5.199	5.199	5.199
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	3920	3925	3927	3929
35	Elevation (m)	0.1940	0.2200	0.1020	-1.600e-002
36	Pressure (bar)	78.59	78.59	78.59	78.59
37	Temperature (C)	55.72	55.72	55.72	55.71
38	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
39	Heat Transferred (kJ/h-m)	560.2	560.2	560.3	560.3
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.1617	0.1617	9.223e-002	9.224e-002
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	3.551e-002	3.551e-002	4.163e-002	4.163e-002
45	Static Gradient (kPa/m)	8.139e-003	8.139e-003	-6.389e-002	-6.389e-002
46	Accel. Gradient (kPa/m)	1.281e-005	1.281e-005	-6.534e-006	-6.534e-006
47	Liq. Reynolds	1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *
48	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
49	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
50	Vapour Velocity (m/s)	5.199	5.199	5.199	5.199
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	3931	3933	3935	3947
54	Elevation (m)	-0.1340	-0.2520	-0.3700	-1.016
55	Pressure (bar)	78.59	78.59	78.59	78.59
56	Temperature (C)	55.71	55.71	55.71	55.71
57	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
58	Heat Transferred (kJ/h-m)	560.3	560.2	560.2	560.2
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	9.224e-002	9.224e-002	9.224e-002	9.157e-002
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	4.163e-002	4.163e-002	4.163e-002	4.175e-002
64	Static Gradient (kPa/m)	-6.390e-002	-6.390e-002	-6.390e-002	-6.562e-002
65	Accel. Gradient (kPa/m)	-6.534e-006	-6.534e-006	-6.535e-006	-7.005e-006
66	Liq. Reynolds	1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *
67	Vap. Reynolds	1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *
68	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3							
4							
5			Unit Set: Mixed				
6			Date/Time: Mon Jun 13 15:09:19 2022				
7	Pipe Segment: BBOU-ELEP Pipeline (continued)						
8							
9							
10	PROFILES						
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	5.199	5.199	5.199	5.199	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	3958	3969	3980	3991	
17	Elevation	(m)	-1.662	-2.308	-2.954	-3.600	
18	Pressure	(bar)	78.59	78.60	78.60	78.60	
19	Temperature	(C)	55.71	55.71	55.70	55.70	
20	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
21	Heat Transferred	(kJ/h-m)	560.1	560.1	560.0	560.0	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		9.157e-002	9.157e-002	9.158e-002	9.158e-002	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	4.175e-002	4.175e-002	4.175e-002	4.175e-002	
27	Static Gradient	(kPa/m)	-6.563e-002	-6.563e-002	-6.563e-002	-6.563e-002	
28	Accel. Gradient	(kPa/m)	-7.006e-006	-7.007e-006	-7.007e-006	-7.008e-006	
29	Liq. Reynolds		1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *	
30	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *	
31	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
32	Vapour Velocity	(m/s)	5.199	5.198	5.198	5.198	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	4002	4013	4025	4036	
36	Elevation	(m)	-3.940	-4.280	-4.620	-4.960	
37	Pressure	(bar)	78.60	78.60	78.60	78.60	
38	Temperature	(C)	55.70	55.69	55.69	55.68	
39	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
40	Heat Transferred	(kJ/h-m)	559.9	559.8	559.7	559.6	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.1023	0.1023	0.1023	0.1023	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	4.002e-002	4.002e-002	4.002e-002	4.002e-002	
46	Static Gradient	(kPa/m)	-3.656e-002	-3.657e-002	-3.657e-002	-3.657e-002	
47	Accel. Gradient	(kPa/m)	1.013e-006	1.013e-006	1.013e-006	1.012e-006	
48	Liq. Reynolds		1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *	
49	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *	
50	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
51	Vapour Velocity	(m/s)	5.198	5.198	5.198	5.197	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	4047	4058	4069	4080	
55	Elevation	(m)	-5.300	-5.420	-5.540	-5.660	
56	Pressure	(bar)	78.60	78.60	78.59	78.59	
57	Temperature	(C)	55.68	55.68	55.67	55.66	
58	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
59	Heat Transferred	(kJ/h-m)	559.5	559.4	559.3	559.1	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.1023	0.1101	0.1101	0.1101	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	4.002e-002	3.903e-002	3.903e-002	3.903e-002	
65	Static Gradient	(kPa/m)	-3.657e-002	-1.342e-002	-1.342e-002	-1.342e-002	
66	Accel. Gradient	(kPa/m)	1.012e-006	7.513e-006	7.514e-006	7.514e-006	
67	Liq. Reynolds		1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *	
68	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:09:19 2022				
5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
14	Vapour Velocity	(m/s)	5.197	5.197	5.198	5.198	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	4091	4103	4114	4125	
18	Elevation	(m)	-5.780	-5.900	-5.780	-5.660	
19	Pressure	(bar)	78.59	78.58	78.58	78.57	
20	Temperature	(C)	55.66	55.65	55.65	55.64	
21	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
22	Heat Transferred	(kJ/h-m)	559.0	558.9	558.7	558.5	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.1101	0.1101	0.1698	0.1698	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	3.903e-002	3.903e-002	3.531e-002	3.531e-002	
28	Static Gradient	(kPa/m)	-1.342e-002	-1.342e-002	1.740e-002	1.740e-002	
29	Accel. Gradient	(kPa/m)	7.515e-006	7.515e-006	1.547e-005	1.547e-005	
30	Liq. Reynolds		1.350e+005 *	1.350e+005 *	1.350e+005 *	1.350e+005 *	
31	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *	
32	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
33	Vapour Velocity	(m/s)	5.198	5.198	5.198	5.198	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	4136	4147	4158	4169	
37	Elevation	(m)	-5.540	-5.420	-5.300	-5.500	
38	Pressure	(bar)	78.57	78.56	78.56	78.55	
39	Temperature	(C)	55.63	55.63	55.62	55.61	
40	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
41	Heat Transferred	(kJ/h-m)	558.3	558.2	558.0	558.0	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.1698	0.1698	0.1698	0.1072	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	3.532e-002	3.532e-002	3.532e-002	3.938e-002	
47	Static Gradient	(kPa/m)	1.740e-002	1.740e-002	1.740e-002	-2.205e-002	
48	Accel. Gradient	(kPa/m)	1.547e-005	1.548e-005	1.548e-005	5.088e-006	
49	Liq. Reynolds		1.350e+005 *	1.349e+005 *	1.349e+005 *	1.349e+005 *	
50	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *	
51	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
52	Vapour Velocity	(m/s)	5.199	5.199	5.199	5.199	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	4181	4192	4203	4214	
56	Elevation	(m)	-5.700	-5.900	-6.100	-6.300	
57	Pressure	(bar)	78.55	78.55	78.55	78.55	
58	Temperature	(C)	55.61	55.60	55.60	55.59	
59	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
60	Heat Transferred	(kJ/h-m)	557.9	557.8	557.7	557.6	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.1072	0.1072	0.1072	0.1072	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	3.938e-002	3.938e-002	3.938e-002	3.938e-002	
66	Static Gradient	(kPa/m)	-2.205e-002	-2.205e-002	-2.205e-002	-2.205e-002	
67	Accel. Gradient	(kPa/m)	5.088e-006	5.088e-006	5.089e-006	5.089e-006	
68	Liq. Reynolds		1.349e+005 *	1.349e+005 *	1.349e+005 *	1.349e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:09:19 2022				
5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.217e+007 *	1.217e+007 *	
14	Liquid Velocity (m/s)		0.2084	0.2084	0.2084	0.2084	
15	Vapour Velocity (m/s)		5.199	5.199	5.199	5.199	
16	Deposit Thickness (mm)		---	---	---	---	
17	Deposit Volume (m3)		---	---	---	---	
18	Length (m)		4225	4236	4247	4259	
19	Elevation (m)		-7.360	-8.420	-9.480	-10.54	
20	Pressure (bar)		78.55	78.56	78.56	78.57	
21	Temperature (C)		55.59	55.59	55.59	55.59	
22	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
23	Heat Transferred (kJ/h-m)		557.7	557.7	557.7	557.7	
24	Flow Regime		Transition	Transition	Transition	Transition	
25	Water Flow Regime						
26	Liquid HoldUp		7.732e-002	7.732e-002	7.732e-002	7.733e-002	
27	Water HoldUp		---	---	---	---	
28	Friction Gradient (kPa/m)		4.504e-002	4.503e-002	4.503e-002	4.503e-002	
29	Static Gradient (kPa/m)		-9.931e-002	-9.932e-002	-9.932e-002	-9.933e-002	
30	Accel. Gradient (kPa/m)		-1.594e-005	-1.594e-005	-1.594e-005	-1.594e-005	
31	Liq. Reynolds		1.349e+005 *	1.349e+005 *	1.349e+005 *	1.349e+005 *	
32	Vap. Reynolds		1.216e+007 *	1.216e+007 *	1.216e+007 *	1.216e+007 *	
33	Liquid Velocity (m/s)		0.2084	0.2084	0.2084	0.2084	
34	Vapour Velocity (m/s)		5.199	5.198	5.198	5.197	
35	Deposit Thickness (mm)		---	---	---	---	
36	Deposit Volume (m3)		---	---	---	---	
37	Length (m)		4270	4281	4292	4303	
38	Elevation (m)		-11.60	-9.192	-6.784	-4.376	
39	Pressure (bar)		78.58	78.49	78.41	78.33	
40	Temperature (C)		55.59	55.56	55.53	55.50	
41	Vapour Fraction		0.9341	0.9341	0.9342	0.9342	
42	Heat Transferred (kJ/h-m)		557.7	557.4	556.7	556.0	
43	Flow Regime		Transition	Transition	Transition	Transition	
44	Water Flow Regime						
45	Liquid HoldUp		7.733e-002	0.4477	0.4474	0.4472	
46	Water HoldUp		---	---	---	---	
47	Friction Gradient (kPa/m)		4.502e-002	3.057e-002	3.060e-002	3.063e-002	
48	Static Gradient (kPa/m)		-9.933e-002	0.7201	0.7197	0.7193	
49	Accel. Gradient (kPa/m)		-1.594e-005	2.207e-004	2.211e-004	2.214e-004	
50	Liq. Reynolds		1.349e+005 *	1.349e+005 *	1.348e+005 *	1.347e+005 *	
51	Vap. Reynolds		1.216e+007 *	1.217e+007 *	1.217e+007 *	1.217e+007 *	
52	Liquid Velocity (m/s)		0.2084	0.2084	0.2084	0.2084	
53	Vapour Velocity (m/s)		5.197	5.202	5.208	5.213	
54	Deposit Thickness (mm)		---	---	---	---	
55	Deposit Volume (m3)		---	---	---	---	
56	Length (m)		4314	4325	4328	4331	
57	Elevation (m)		-1.968	0.4400	0.5820	0.7240	
58	Pressure (bar)		78.24	78.16	78.15	78.15	
59	Temperature (C)		55.46	55.43	55.43	55.42	
60	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
61	Heat Transferred (kJ/h-m)		555.2	554.5	553.6	553.5	
62	Flow Regime		Transition	Transition	Transition	Transition	
63	Water Flow Regime						
64	Liquid HoldUp		0.4469	0.4466	0.2221	0.2221	
65	Water HoldUp		---	---	---	---	
66	Friction Gradient (kPa/m)		3.066e-002	3.069e-002	3.254e-002	3.254e-002	
67	Static Gradient (kPa/m)		0.7189	0.7185	9.168e-002	9.168e-002	
68	Accel. Gradient (kPa/m)		2.218e-004	2.222e-004	3.684e-005	3.684e-005	
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:09:19 2022				
5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		1.347e+005 *	1.346e+005 *	1.346e+005 *	1.346e+005 *	
14	Vap. Reynolds		1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *	
15	Liquid Velocity (m/s)		0.2084	0.2083	0.2083	0.2083	
16	Vapour Velocity (m/s)		5.218	5.224	5.224	5.224	
17	Deposit Thickness (mm)		---	---	---	---	
18	Deposit Volume (m3)		---	---	---	---	
19	Length (m)		4334	4337	4340	4343	
20	Elevation (m)		0.8660	1.008	1.150	1.088	
21	Pressure (bar)		78.15	78.14	78.14	78.14	
22	Temperature (C)		55.42	55.42	55.42	55.41	
23	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
24	Heat Transferred (kJ/h-m)		553.5	553.4	553.4	553.7	
25	Flow Regime		Transition	Transition	Transition	Transition	
26	Water Flow Regime						
27	Liquid HoldUp		0.2221	0.2221	0.2221	0.1058	
28	Water HoldUp		---	---	---	---	
29	Friction Gradient (kPa/m)		3.254e-002	3.254e-002	3.254e-002	3.971e-002	
30	Static Gradient (kPa/m)		9.167e-002	9.167e-002	9.167e-002	-2.517e-002	
31	Accel. Gradient (kPa/m)		3.684e-005	3.685e-005	3.685e-005	4.314e-006	
32	Liq. Reynolds		1.346e+005 *	1.346e+005 *	1.346e+005 *	1.346e+005 *	
33	Vap. Reynolds		1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *	
34	Liquid Velocity (m/s)		0.2083	0.2083	0.2083	0.2083	
35	Vapour Velocity (m/s)		5.225	5.225	5.225	5.225	
36	Deposit Thickness (mm)		---	---	---	---	
37	Deposit Volume (m3)		---	---	---	---	
38	Length (m)		4346	4349	4352	4355	
39	Elevation (m)		1.026	0.9640	0.9020	0.8400	
40	Pressure (bar)		78.14	78.14	78.14	78.14	
41	Temperature (C)		55.41	55.41	55.41	55.41	
42	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
43	Heat Transferred (kJ/h-m)		553.7	553.6	553.6	553.6	
44	Flow Regime		Transition	Transition	Transition	Transition	
45	Water Flow Regime						
46	Liquid HoldUp		0.1058	0.1058	0.1058	0.1058	
47	Water HoldUp		---	---	---	---	
48	Friction Gradient (kPa/m)		3.971e-002	3.971e-002	3.971e-002	3.971e-002	
49	Static Gradient (kPa/m)		-2.517e-002	-2.517e-002	-2.517e-002	-2.517e-002	
50	Accel. Gradient (kPa/m)		4.314e-006	4.314e-006	4.314e-006	4.314e-006	
51	Liq. Reynolds		1.346e+005 *	1.346e+005 *	1.346e+005 *	1.346e+005 *	
52	Vap. Reynolds		1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *	
53	Liquid Velocity (m/s)		0.2083	0.2083	0.2083	0.2083	
54	Vapour Velocity (m/s)		5.225	5.225	5.225	5.225	
55	Deposit Thickness (mm)		---	---	---	---	
56	Deposit Volume (m3)		---	---	---	---	
57	Length (m)		4357	4359	4361	4363	
58	Elevation (m)		0.7740	0.7080	0.6420	0.5760	
59	Pressure (bar)		78.14	78.14	78.14	78.14	
60	Temperature (C)		55.41	55.41	55.41	55.41	
61	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
62	Heat Transferred (kJ/h-m)		553.6	553.6	553.6	553.5	
63	Flow Regime		Transition	Transition	Transition	Transition	
64	Water Flow Regime						
65	Liquid HoldUp		0.1010	0.1010	0.1010	0.1010	
66	Water HoldUp		---	---	---	---	
67	Friction Gradient (kPa/m)		4.036e-002	4.036e-002	4.036e-002	4.036e-002	
68	Static Gradient (kPa/m)		-3.920e-002	-3.920e-002	-3.920e-002	-3.920e-002	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 33 of 116	

1			Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:09:19 2022		
6					
7	Pipe Segment: BBOU-ELEP Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	3.444e-007	3.443e-007	3.442e-007	3.442e-007
14	Liq. Reynolds	1.346e+005 *	1.346e+005 *	1.346e+005 *	1.346e+005 *
15	Vap. Reynolds	1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
16	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
17	Vapour Velocity (m/s)	5.225	5.225	5.225	5.225
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	4365	4367	4369	4371
21	Elevation (m)	0.5100	0.5320	0.5540	0.5760
22	Pressure (bar)	78.14	78.14	78.14	78.13
23	Temperature (C)	55.41	55.40	55.40	55.40
24	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
25	Heat Transferred (kJ/h-m)	553.5	553.4	553.4	553.3
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.1010	0.1696	0.1696	0.1696
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	4.036e-002	3.548e-002	3.548e-002	3.548e-002
31	Static Gradient (kPa/m)	-3.920e-002	1.774e-002	1.773e-002	1.773e-002
32	Accel. Gradient (kPa/m)	3.441e-007	1.579e-005	1.579e-005	1.579e-005
33	Liq. Reynolds	1.346e+005 *	1.346e+005 *	1.346e+005 *	1.346e+005 *
34	Vap. Reynolds	1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
35	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
36	Vapour Velocity (m/s)	5.225	5.225	5.225	5.225
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	4373	4375	4380	4385
40	Elevation (m)	0.5980	0.6200	0.6140	0.6080
41	Pressure (bar)	78.13	78.13	78.13	78.13
42	Temperature (C)	55.40	55.40	55.40	55.39
43	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
44	Heat Transferred (kJ/h-m)	553.3	553.3	553.3	553.2
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.1696	0.1696	0.1134	0.1134
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	3.549e-002	3.549e-002	3.882e-002	3.882e-002
50	Static Gradient (kPa/m)	1.773e-002	1.773e-002	-1.518e-003	-1.518e-003
51	Accel. Gradient (kPa/m)	1.579e-005	1.579e-005	1.107e-005	1.107e-005
52	Liq. Reynolds	1.346e+005 *	1.346e+005 *	1.345e+005 *	1.345e+005 *
53	Vap. Reynolds	1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
54	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
55	Vapour Velocity (m/s)	5.225	5.225	5.225	5.225
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	4390	4395	4400	4405
59	Elevation (m)	0.6020	0.5960	0.5900	0.5160
60	Pressure (bar)	78.13	78.12	78.12	78.12
61	Temperature (C)	55.39	55.39	55.39	55.38
62	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
63	Heat Transferred (kJ/h-m)	553.2	553.1	553.1	553.0
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.1134	0.1134	0.1134	0.1081
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	3.882e-002	3.882e-002	3.882e-002	3.943e-002
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1			Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:09:19 2022				
6							
7	Pipe Segment: BBOU-ELEP Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Static Gradient	(kPa/m)	-1.518e-003	-1.518e-003	-1.518e-003	-1.823e-002	
14	Accel. Gradient	(kPa/m)	1.107e-005	1.107e-005	1.107e-005	6.290e-006	
15	Liq. Reynolds		1.345e+005 *	1.345e+005 *	1.345e+005 *	1.345e+005 *	
16	Vap. Reynolds		1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *	
17	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083	
18	Vapour Velocity	(m/s)	5.225	5.225	5.225	5.226	
19	Deposit Thickness	(mm)	---	---	---	---	
20	Deposit Volume	(m3)	---	---	---	---	
21	Length	(m)	4410	4415	4420	4425	
22	Elevation	(m)	0.4420	0.3680	0.2940	0.2200	
23	Pressure	(bar)	78.12	78.12	78.12	78.12	
24	Temperature	(C)	55.38	55.38	55.38	55.37	
25	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
26	Heat Transferred	(kJ/h-m)	553.0	552.9	552.9	552.8	
27	Flow Regime		Transition	Transition	Transition	Transition	
28	Water Flow Regime						
29	Liquid HoldUp		0.1081	0.1081	0.1081	0.1081	
30	Water HoldUp		---	---	---	---	
31	Friction Gradient	(kPa/m)	3.943e-002	3.943e-002	3.943e-002	3.943e-002	
32	Static Gradient	(kPa/m)	-1.823e-002	-1.823e-002	-1.823e-002	-1.823e-002	
33	Accel. Gradient	(kPa/m)	6.290e-006	6.290e-006	6.290e-006	6.290e-006	
34	Liq. Reynolds		1.345e+005 *	1.345e+005 *	1.345e+005 *	1.345e+005 *	
35	Vap. Reynolds		1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *	
36	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083	
37	Vapour Velocity	(m/s)	5.226	5.226	5.226	5.226	
38	Deposit Thickness	(mm)	---	---	---	---	
39	Deposit Volume	(m3)	---	---	---	---	
40	Length	(m)	4431	4436	4441	4446	
41	Elevation	(m)	0.2080	0.1960	0.1840	0.1720	
42	Pressure	(bar)	78.12	78.11	78.11	78.11	
43	Temperature	(C)	55.37	55.37	55.37	55.36	
44	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
45	Heat Transferred	(kJ/h-m)	552.8	552.7	552.6	552.6	
46	Flow Regime		Transition	Transition	Transition	Transition	
47	Water Flow Regime						
48	Liquid HoldUp		0.1130	0.1130	0.1130	0.1130	
49	Water HoldUp		---	---	---	---	
50	Friction Gradient	(kPa/m)	3.887e-002	3.887e-002	3.887e-002	3.887e-002	
51	Static Gradient	(kPa/m)	-2.998e-003	-2.998e-003	-2.998e-003	-2.998e-003	
52	Accel. Gradient	(kPa/m)	1.065e-005	1.065e-005	1.065e-005	1.065e-005	
53	Liq. Reynolds		1.345e+005 *	1.345e+005 *	1.345e+005 *	1.345e+005 *	
54	Vap. Reynolds		1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *	
55	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083	
56	Vapour Velocity	(m/s)	5.226	5.226	5.226	5.226	
57	Deposit Thickness	(mm)	---	---	---	---	
58	Deposit Volume	(m3)	---	---	---	---	
59	Length	(m)	4451	4456	4461	4466	
60	Elevation	(m)	0.1600	0.1620	0.1640	0.1660	
61	Pressure	(bar)	78.11	78.11	78.10	78.10	
62	Temperature	(C)	55.36	55.36	55.36	55.35	
63	Vapour Fraction		0.9342	0.9342	0.9342	0.9342	
64	Heat Transferred	(kJ/h-m)	552.5	552.4	552.4	552.3	
65	Flow Regime		Transition	Transition	Transition	Transition	
66	Water Flow Regime						
67	Liquid HoldUp		0.1130	0.1542	0.1542	0.1542	
68	Water HoldUp		---	---	---	---	
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1				Case Name: Simulation - 331MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:09:19 2022		
6						
7	Pipe Segment: BBOU-ELEP Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	3.887e-002	3.594e-002	3.594e-002	3.594e-002
14	Static Gradient	(kPa/m)	-2.998e-003	6.068e-004	6.068e-004	6.068e-004
15	Accel. Gradient	(kPa/m)	1.065e-005	1.085e-005	1.085e-005	1.085e-005
16	Liq. Reynolds		1.345e+005 *	1.345e+005 *	1.345e+005 *	1.345e+005 *
17	Vap. Reynolds		1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
18	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083
19	Vapour Velocity	(m/s)	5.226	5.226	5.226	5.226
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	4471	4476	4481	4486
23	Elevation	(m)	0.1680	0.1700	0.1840	0.1980
24	Pressure	(bar)	78.10	78.10	78.10	78.10
25	Temperature	(C)	55.35	55.35	55.34	55.34
26	Vapour Fraction		0.9342	0.9342	0.9342	0.9342
27	Heat Transferred	(kJ/h-m)	552.2	552.2	552.1	552.0
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.1542	0.1542	0.1577	0.1577
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	3.594e-002	3.594e-002	3.581e-002	3.581e-002
33	Static Gradient	(kPa/m)	6.068e-004	6.068e-004	4.308e-003	4.308e-003
34	Accel. Gradient	(kPa/m)	1.085e-005	1.085e-005	1.191e-005	1.191e-005
35	Liq. Reynolds		1.345e+005 *	1.345e+005 *	1.345e+005 *	1.345e+005 *
36	Vap. Reynolds		1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
37	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083
38	Vapour Velocity	(m/s)	5.226	5.226	5.226	5.226
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	4491	4496	4501	4506
42	Elevation	(m)	0.2120	0.2260	0.2400	0.2600
43	Pressure	(bar)	78.09	78.09	78.09	78.09
44	Temperature	(C)	55.34	55.34	55.33	55.33
45	Vapour Fraction		0.9342	0.9342	0.9342	0.9342
46	Heat Transferred	(kJ/h-m)	552.0	551.9	551.9	551.8
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.1577	0.1577	0.1577	0.1594
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	3.581e-002	3.581e-002	3.581e-002	3.575e-002
52	Static Gradient	(kPa/m)	4.308e-003	4.308e-003	4.308e-003	6.197e-003
53	Accel. Gradient	(kPa/m)	1.191e-005	1.191e-005	1.191e-005	1.246e-005
54	Liq. Reynolds		1.345e+005 *	1.345e+005 *	1.345e+005 *	1.345e+005 *
55	Vap. Reynolds		1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
56	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083
57	Vapour Velocity	(m/s)	5.226	5.227	5.227	5.227
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	4511	4516	4521	4526
61	Elevation	(m)	0.2800	0.3000	0.3200	0.3400
62	Pressure	(bar)	78.08	78.08	78.08	78.08
63	Temperature	(C)	55.33	55.33	55.32	55.32
64	Vapour Fraction		0.9342	0.9342	0.9342	0.9342
65	Heat Transferred	(kJ/h-m)	551.7	551.7	551.6	551.5
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.1594	0.1594	0.1594	0.1594
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:09:19 2022
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	3.575e-002	3.575e-002	3.575e-002	3.575e-002
15	Static Gradient (kPa/m)	6.197e-003	6.197e-003	6.197e-003	6.197e-003
16	Accel. Gradient (kPa/m)	1.246e-005	1.246e-005	1.246e-005	1.246e-005
17	Liq. Reynolds	1.345e+005 *	1.345e+005 *	1.345e+005 *	1.345e+005 *
18	Vap. Reynolds	1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
19	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
20	Vapour Velocity (m/s)	5.227	5.227	5.227	5.227
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	4531	4536	4541	4546
24	Elevation (m)	0.3680	0.3960	0.4240	0.4520
25	Pressure (bar)	78.08	78.07	78.07	78.07
26	Temperature (C)	55.32	55.31	55.31	55.31
27	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
28	Heat Transferred (kJ/h-m)	551.5	551.4	551.3	551.3
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.1617	0.1617	0.1617	0.1617
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	3.567e-002	3.568e-002	3.568e-002	3.568e-002
34	Static Gradient (kPa/m)	8.756e-003	8.756e-003	8.755e-003	8.755e-003
35	Accel. Gradient (kPa/m)	1.320e-005	1.320e-005	1.320e-005	1.320e-005
36	Liq. Reynolds	1.345e+005 *	1.345e+005 *	1.345e+005 *	1.345e+005 *
37	Vap. Reynolds	1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
38	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
39	Vapour Velocity (m/s)	5.227	5.227	5.227	5.227
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	4551	4556	4561	4566
43	Elevation (m)	0.4800	0.4980	0.5160	0.5340
44	Pressure (bar)	78.07	78.07	78.06	78.06
45	Temperature (C)	55.30	55.30	55.30	55.30
46	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
47	Heat Transferred (kJ/h-m)	551.2	551.2	551.1	551.0
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.1617	0.1588	0.1588	0.1588
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	3.568e-002	3.577e-002	3.577e-002	3.577e-002
53	Static Gradient (kPa/m)	8.755e-003	5.564e-003	5.564e-003	5.564e-003
54	Accel. Gradient (kPa/m)	1.320e-005	1.228e-005	1.228e-005	1.228e-005
55	Liq. Reynolds	1.345e+005 *	1.345e+005 *	1.345e+005 *	1.345e+005 *
56	Vap. Reynolds	1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
57	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
58	Vapour Velocity (m/s)	5.228	5.228	5.228	5.228
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	4571	4576	4581	4586
62	Elevation (m)	0.5520	0.5700	0.5820	0.5940
63	Pressure (bar)	78.06	78.06	78.06	78.05
64	Temperature (C)	55.29	55.29	55.29	55.29
65	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
66	Heat Transferred (kJ/h-m)	551.0	550.9	550.9	550.8
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 37 of 116

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:09:19 2022
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.1588	0.1588	0.1571	0.1571
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	3.577e-002	3.577e-002	3.584e-002	3.584e-002
16	Static Gradient (kPa/m)	5.564e-003	5.564e-003	3.683e-003	3.683e-003
17	Accel. Gradient (kPa/m)	1.228e-005	1.228e-005	1.174e-005	1.175e-005
18	Liq. Reynolds	1.344e+005 *	1.344e+005 *	1.344e+005 *	1.344e+005 *
19	Vap. Reynolds	1.218e+007 *	1.218e+007 *	1.218e+007 *	1.218e+007 *
20	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
21	Vapour Velocity (m/s)	5.228	5.228	5.228	5.228
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	4591	4596	4601	4606
25	Elevation (m)	0.6060	0.6180	0.6300	0.6380
26	Pressure (bar)	78.05	78.05	78.05	78.05
27	Temperature (C)	55.28	55.28	55.28	55.27
28	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
29	Heat Transferred (kJ/h-m)	550.7	550.7	550.6	550.6
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1571	0.1571	0.1571	0.1559
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	3.584e-002	3.584e-002	3.584e-002	3.589e-002
35	Static Gradient (kPa/m)	3.683e-003	3.683e-003	3.683e-003	2.444e-003
36	Accel. Gradient (kPa/m)	1.175e-005	1.175e-005	1.175e-005	1.139e-005
37	Liq. Reynolds	1.344e+005 *	1.344e+005 *	1.344e+005 *	1.344e+005 *
38	Vap. Reynolds	1.218e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
39	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
40	Vapour Velocity (m/s)	5.228	5.228	5.228	5.228
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	4611	4616	4621	4626
44	Elevation (m)	0.6460	0.6540	0.6620	0.6700
45	Pressure (bar)	78.04	78.04	78.04	78.04
46	Temperature (C)	55.27	55.27	55.27	55.26
47	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
48	Heat Transferred (kJ/h-m)	550.5	550.4	550.4	550.3
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1559	0.1559	0.1559	0.1559
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	3.589e-002	3.589e-002	3.589e-002	3.589e-002
54	Static Gradient (kPa/m)	2.444e-003	2.444e-003	2.444e-003	2.444e-003
55	Accel. Gradient (kPa/m)	1.139e-005	1.139e-005	1.139e-005	1.139e-005
56	Liq. Reynolds	1.344e+005 *	1.344e+005 *	1.344e+005 *	1.344e+005 *
57	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
58	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
59	Vapour Velocity (m/s)	5.228	5.229	5.229	5.229
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	4631	4636	4641	4646
63	Elevation (m)	0.6700	0.6700	0.6700	0.6700
64	Pressure (bar)	78.04	78.03	78.03	78.03
65	Temperature (C)	55.26	55.26	55.25	55.25
66	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
67	Heat Transferred (kJ/h-m)	550.2	550.2	550.1	550.1
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 38 of 116				

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:09:19 2022		
4					
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1662	0.1662	0.1662	0.1662
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	3.557e-002	3.557e-002	3.557e-002	3.557e-002
17	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
18	Accel. Gradient (kPa/m)	1.057e-005	1.057e-005	1.057e-005	1.058e-005
19	Liq. Reynolds	1.344e+005 *	1.344e+005 *	1.344e+005 *	1.344e+005 *
20	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
21	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
22	Vapour Velocity (m/s)	5.229	5.229	5.229	5.229
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	4651	4656	4661	4666
26	Elevation (m)	0.6700	0.6760	0.6820	0.6880
27	Pressure (bar)	78.03	78.03	78.02	78.02
28	Temperature (C)	55.25	55.25	55.24	55.24
29	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
30	Heat Transferred (kJ/h-m)	550.0	550.0	549.9	549.8
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1662	0.1553	0.1553	0.1553
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	3.557e-002	3.591e-002	3.591e-002	3.591e-002
36	Static Gradient (kPa/m)	0.0000	1.828e-003	1.828e-003	1.828e-003
37	Accel. Gradient (kPa/m)	1.058e-005	1.122e-005	1.122e-005	1.122e-005
38	Liq. Reynolds	1.344e+005 *	1.344e+005 *	1.344e+005 *	1.344e+005 *
39	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
40	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
41	Vapour Velocity (m/s)	5.229	5.229	5.229	5.229
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	4671	4676	4681	4686
45	Elevation (m)	0.6940	0.7000	0.7000	0.7000
46	Pressure (bar)	78.02	78.02	78.02	78.02
47	Temperature (C)	55.24	55.24	55.23	55.23
48	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
49	Heat Transferred (kJ/h-m)	549.8	549.7	549.6	549.6
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.1553	0.1553	0.1662	0.1662
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	3.591e-002	3.592e-002	3.557e-002	3.557e-002
55	Static Gradient (kPa/m)	1.828e-003	1.828e-003	0.0000	0.0000
56	Accel. Gradient (kPa/m)	1.122e-005	1.122e-005	1.058e-005	1.058e-005
57	Liq. Reynolds	1.344e+005 *	1.344e+005 *	1.344e+005 *	1.344e+005 *
58	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
59	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
60	Vapour Velocity (m/s)	5.229	5.229	5.229	5.230
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	4691	4696	4701	4706
64	Elevation (m)	0.7000	0.7000	0.7000	0.6640
65	Pressure (bar)	78.01	78.01	78.01	78.01
66	Temperature (C)	55.23	55.22	55.22	55.22
67	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
68	Heat Transferred (kJ/h-m)	549.5	549.5	549.4	549.4
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 39 of 116

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
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4					
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.1662	0.1662	0.1662	0.1110
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	3.557e-002	3.557e-002	3.557e-002	3.911e-002
18	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-8.998e-003
19	Accel. Gradient (kPa/m)	1.058e-005	1.058e-005	1.058e-005	8.957e-006
20	Liq. Reynolds	1.344e+005 *	1.344e+005 *	1.344e+005 *	1.344e+005 *
21	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
22	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
23	Vapour Velocity (m/s)	5.230	5.230	5.230	5.230
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	4711	4716	4721	4726
27	Elevation (m)	0.6280	0.5920	0.5560	0.5200
28	Pressure (bar)	78.01	78.01	78.00	78.00
29	Temperature (C)	55.22	55.21	55.21	55.21
30	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
31	Heat Transferred (kJ/h-m)	549.4	549.3	549.3	549.2
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1110	0.1110	0.1110	0.1110
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	3.911e-002	3.911e-002	3.911e-002	3.911e-002
37	Static Gradient (kPa/m)	-8.998e-003	-8.998e-003	-8.998e-003	-8.998e-003
38	Accel. Gradient (kPa/m)	8.957e-006	8.957e-006	8.958e-006	8.958e-006
39	Liq. Reynolds	1.344e+005 *	1.344e+005 *	1.344e+005 *	1.344e+005 *
40	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
41	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
42	Vapour Velocity (m/s)	5.230	5.230	5.230	5.230
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	4731	4736	4741	4746
46	Elevation (m)	0.5200	0.5200	0.5200	0.5200
47	Pressure (bar)	78.00	78.00	78.00	78.00
48	Temperature (C)	55.21	55.20	55.20	55.20
49	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
50	Heat Transferred (kJ/h-m)	549.1	549.0	549.0	548.9
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1662	0.1662	0.1662	0.1662
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	3.558e-002	3.558e-002	3.558e-002	3.558e-002
56	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
57	Accel. Gradient (kPa/m)	1.058e-005	1.058e-005	1.058e-005	1.058e-005
58	Liq. Reynolds	1.344e+005 *	1.344e+005 *	1.344e+005 *	1.344e+005 *
59	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
60	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
61	Vapour Velocity (m/s)	5.230	5.230	5.230	5.230
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	4751	4756	4761	4766
65	Elevation (m)	0.5200	0.5520	0.5840	0.6160
66	Pressure (bar)	77.99	77.99	77.99	77.99
67	Temperature (C)	55.20	55.19	55.19	55.19
68	Vapour Fraction	0.9342	0.9342	0.9342	0.9342
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:09:19 2022				
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Heat Transferred (kJ/h-m)	548.8	548.8	548.7	548.7		
14	Flow Regime	Transition	Transition	Transition	Transition		
15	Water Flow Regime						
16	Liquid HoldUp	0.1662	0.1628	0.1628	0.1628		
17	Water HoldUp	---	---	---	---		
18	Friction Gradient (kPa/m)	3.558e-002	3.566e-002	3.566e-002	3.566e-002		
19	Static Gradient (kPa/m)	0.0000	1.005e-002	1.005e-002	1.005e-002		
20	Accel. Gradient (kPa/m)	1.059e-005	1.360e-005	1.360e-005	1.360e-005		
21	Liq. Reynolds	1.344e+005 *	1.343e+005 *	1.343e+005 *	1.343e+005 *		
22	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
23	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
24	Vapour Velocity (m/s)	5.230	5.230	5.230	5.231		
25	Deposit Thickness (mm)	---	---	---	---		
26	Deposit Volume (m3)	---	---	---	---		
27	Length (m)	4771	4776	4781	4786		
28	Elevation (m)	0.6480	0.6800	0.6700	0.6600		
29	Pressure (bar)	77.98	77.98	77.98	77.98		
30	Temperature (C)	55.18	55.18	55.18	55.18		
31	Vapour Fraction	0.9342	0.9342	0.9342	0.9341		
32	Heat Transferred (kJ/h-m)	548.6	548.5	548.5	548.5		
33	Flow Regime	Transition	Transition	Transition	Transition		
34	Water Flow Regime						
35	Liquid HoldUp	0.1628	0.1628	0.1131	0.1131		
36	Water HoldUp	---	---	---	---		
37	Friction Gradient (kPa/m)	3.566e-002	3.566e-002	3.889e-002	3.889e-002		
38	Static Gradient (kPa/m)	1.005e-002	1.005e-002	-2.524e-003	-2.524e-003		
39	Accel. Gradient (kPa/m)	1.360e-005	1.360e-005	1.082e-005	1.082e-005		
40	Liq. Reynolds	1.343e+005 *	1.343e+005 *	1.343e+005 *	1.343e+005 *		
41	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
42	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
43	Vapour Velocity (m/s)	5.231	5.231	5.231	5.231		
44	Deposit Thickness (mm)	---	---	---	---		
45	Deposit Volume (m3)	---	---	---	---		
46	Length (m)	4791	4796	4801	4806		
47	Elevation (m)	0.6500	0.6400	0.6300	0.6340		
48	Pressure (bar)	77.98	77.98	77.97	77.97		
49	Temperature (C)	55.17	55.17	55.17	55.16		
50	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
51	Heat Transferred (kJ/h-m)	548.4	548.4	548.3	548.2		
52	Flow Regime	Transition	Transition	Transition	Transition		
53	Water Flow Regime						
54	Liquid HoldUp	0.1131	0.1131	0.1131	0.1547		
55	Water HoldUp	---	---	---	---		
56	Friction Gradient (kPa/m)	3.889e-002	3.889e-002	3.889e-002	3.595e-002		
57	Static Gradient (kPa/m)	-2.524e-003	-2.524e-003	-2.524e-003	1.216e-003		
58	Accel. Gradient (kPa/m)	1.082e-005	1.083e-005	1.083e-005	1.106e-005		
59	Liq. Reynolds	1.343e+005 *	1.343e+005 *	1.343e+005 *	1.343e+005 *		
60	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
61	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
62	Vapour Velocity (m/s)	5.231	5.231	5.231	5.231		
63	Deposit Thickness (mm)	---	---	---	---		
64	Deposit Volume (m3)	---	---	---	---		
65	Length (m)	4811	4816	4821	4826		
66	Elevation (m)	0.6380	0.6420	0.6460	0.6500		
67	Pressure (bar)	77.97	77.97	77.97	77.96		
68	Temperature (C)	55.16	55.16	55.16	55.15		
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:09:19 2022		
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
14	Heat Transferred (kJ/h-m)	548.1	548.1	548.0	548.0
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.1547	0.1547	0.1547	0.1547
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	3.595e-002	3.595e-002	3.595e-002	3.595e-002
20	Static Gradient (kPa/m)	1.216e-003	1.216e-003	1.216e-003	1.216e-003
21	Accel. Gradient (kPa/m)	1.106e-005	1.106e-005	1.106e-005	1.107e-005
22	Liq. Reynolds	1.343e+005 *	1.343e+005 *	1.343e+005 *	1.343e+005 *
23	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
24	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
25	Vapour Velocity (m/s)	5.231	5.231	5.231	5.231
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	4831	4836	4841	4846
29	Elevation (m)	0.6600	0.6700	0.6800	0.6900
30	Pressure (bar)	77.96	77.96	77.96	77.96
31	Temperature (C)	55.15	55.15	55.15	55.14
32	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
33	Heat Transferred (kJ/h-m)	547.9	547.8	547.8	547.7
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1564	0.1564	0.1564	0.1564
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	3.589e-002	3.589e-002	3.589e-002	3.589e-002
39	Static Gradient (kPa/m)	3.019e-003	3.019e-003	3.019e-003	3.019e-003
40	Accel. Gradient (kPa/m)	1.158e-005	1.158e-005	1.158e-005	1.158e-005
41	Liq. Reynolds	1.343e+005 *	1.343e+005 *	1.343e+005 *	1.343e+005 *
42	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
43	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
44	Vapour Velocity (m/s)	5.232	5.232	5.232	5.232
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	4851	4856	4861	4866
48	Elevation (m)	0.7000	0.7060	0.7120	0.7180
49	Pressure (bar)	77.95	77.95	77.95	77.95
50	Temperature (C)	55.14	55.14	55.13	55.13
51	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
52	Heat Transferred (kJ/h-m)	547.6	547.6	547.5	547.5
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.1564	0.1553	0.1553	0.1553
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	3.589e-002	3.593e-002	3.593e-002	3.593e-002
58	Static Gradient (kPa/m)	3.019e-003	1.828e-003	1.828e-003	1.828e-003
59	Accel. Gradient (kPa/m)	1.159e-005	1.124e-005	1.124e-005	1.125e-005
60	Liq. Reynolds	1.343e+005 *	1.343e+005 *	1.343e+005 *	1.343e+005 *
61	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
62	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
63	Vapour Velocity (m/s)	5.232	5.232	5.232	5.232
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	4871	4876	4881	4886
67	Elevation (m)	0.7240	0.7300	0.7200	0.7100
68	Pressure (bar)	77.95	77.94	77.94	77.94
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:09:19 2022
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	55.13	55.13	55.12	55.12
14	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
15	Heat Transferred (kJ/h-m)	547.4	547.3	547.3	547.3
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.1553	0.1553	0.1130	0.1130
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	3.593e-002	3.593e-002	3.890e-002	3.890e-002
21	Static Gradient (kPa/m)	1.828e-003	1.828e-003	-2.524e-003	-2.524e-003
22	Accel. Gradient (kPa/m)	1.125e-005	1.125e-005	1.084e-005	1.084e-005
23	Liq. Reynolds	1.343e+005 *	1.343e+005 *	1.343e+005 *	1.343e+005 *
24	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
25	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
26	Vapour Velocity (m/s)	5.232	5.232	5.232	5.232
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	4891	4896	4901	4906
30	Elevation (m)	0.7000	0.6900	0.6800	0.6780
31	Pressure (bar)	77.94	77.94	77.94	77.93
32	Temperature (C)	55.12	55.12	55.11	55.11
33	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
34	Heat Transferred (kJ/h-m)	547.2	547.2	547.1	547.1
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1130	0.1130	0.1130	0.1137
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	3.890e-002	3.890e-002	3.890e-002	3.884e-002
40	Static Gradient (kPa/m)	-2.524e-003	-2.524e-003	-2.524e-003	-5.063e-004
41	Accel. Gradient (kPa/m)	1.084e-005	1.084e-005	1.084e-005	1.142e-005
42	Liq. Reynolds	1.343e+005 *	1.343e+005 *	1.343e+005 *	1.343e+005 *
43	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
44	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
45	Vapour Velocity (m/s)	5.232	5.232	5.233	5.233
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	4911	4916	4921	4926
49	Elevation (m)	0.6760	0.6740	0.6720	0.6700
50	Pressure (bar)	77.93	77.93	77.93	77.93
51	Temperature (C)	55.11	55.10	55.10	55.10
52	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
53	Heat Transferred (kJ/h-m)	547.0	546.9	546.9	546.8
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.1137	0.1137	0.1137	0.1137
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	3.884e-002	3.884e-002	3.884e-002	3.884e-002
59	Static Gradient (kPa/m)	-5.063e-004	-5.063e-004	-5.063e-004	-5.063e-004
60	Accel. Gradient (kPa/m)	1.142e-005	1.142e-005	1.142e-005	1.142e-005
61	Liq. Reynolds	1.343e+005 *	1.343e+005 *	1.343e+005 *	1.343e+005 *
62	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *
63	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
64	Vapour Velocity (m/s)	5.233	5.233	5.233	5.233
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	4931	4936	4941	4946
68	Elevation (m)	0.6680	0.6660	0.6640	0.6620
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:09:19 2022				
5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
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9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	77.92	77.92	77.92	77.92		
14	Temperature (C)	55.10	55.09	55.09	55.09		
15	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
16	Heat Transferred (kJ/h-m)	546.8	546.7	546.6	546.6		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.1137	0.1137	0.1137	0.1137		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	3.884e-002	3.884e-002	3.884e-002	3.884e-002		
22	Static Gradient (kPa/m)	-5.063e-004	-5.063e-004	-5.063e-004	-5.063e-004		
23	Accel. Gradient (kPa/m)	1.142e-005	1.142e-005	1.142e-005	1.142e-005		
24	Liq. Reynolds	1.343e+005 *	1.343e+005 *	1.342e+005 *	1.342e+005 *		
25	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
26	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
27	Vapour Velocity (m/s)	5.233	5.233	5.233	5.233		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	4951	4956	4961	4966		
31	Elevation (m)	0.6600	0.6460	0.6320	0.6180		
32	Pressure (bar)	77.92	77.91	77.91	77.91		
33	Temperature (C)	55.09	55.08	55.08	55.08		
34	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
35	Heat Transferred (kJ/h-m)	546.5	546.5	546.4	546.3		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.1137	0.1127	0.1127	0.1127		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	3.884e-002	3.895e-002	3.895e-002	3.895e-002		
41	Static Gradient (kPa/m)	-5.063e-004	-3.527e-003	-3.527e-003	-3.527e-003		
42	Accel. Gradient (kPa/m)	1.142e-005	1.055e-005	1.056e-005	1.056e-005		
43	Liq. Reynolds	1.342e+005 *	1.342e+005 *	1.342e+005 *	1.342e+005 *		
44	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
45	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
46	Vapour Velocity (m/s)	5.233	5.233	5.233	5.233		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	4971	4976	4981	4986		
50	Elevation (m)	0.6040	0.5900	0.5980	0.6060		
51	Pressure (bar)	77.91	77.91	77.91	77.90		
52	Temperature (C)	55.08	55.07	55.07	55.07		
53	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
54	Heat Transferred (kJ/h-m)	546.3	546.2	546.1	546.1		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.1127	0.1127	0.1558	0.1558		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	3.895e-002	3.895e-002	3.592e-002	3.592e-002		
60	Static Gradient (kPa/m)	-3.527e-003	-3.527e-003	2.443e-003	2.442e-003		
61	Accel. Gradient (kPa/m)	1.056e-005	1.056e-005	1.143e-005	1.144e-005		
62	Liq. Reynolds	1.342e+005 *	1.342e+005 *	1.342e+005 *	1.342e+005 *		
63	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
64	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
65	Vapour Velocity (m/s)	5.234	5.234	5.234	5.234		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	4991	4996	5001	5006		
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
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5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Elevation (m)	0.6140	0.6220	0.6300	0.6400		
14	Pressure (bar)	77.90	77.90	77.90	77.90		
15	Temperature (C)	55.06	55.06	55.06	55.06		
16	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
17	Heat Transferred (kJ/h-m)	546.0	545.9	545.9	545.8		
18	Flow Regime	Transition	Transition	Transition	Transition		
19	Water Flow Regime						
20	Liquid HoldUp	0.1558	0.1558	0.1558	0.1564		
21	Water HoldUp	---	---	---	---		
22	Friction Gradient (kPa/m)	3.592e-002	3.592e-002	3.592e-002	3.590e-002		
23	Static Gradient (kPa/m)	2.442e-003	2.442e-003	2.442e-003	3.060e-003		
24	Accel. Gradient (kPa/m)	1.144e-005	1.144e-005	1.144e-005	1.162e-005		
25	Liq. Reynolds	1.342e+005 *	1.342e+005 *	1.342e+005 *	1.342e+005 *		
26	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
27	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
28	Vapour Velocity (m/s)	5.234	5.234	5.234	5.234		
29	Deposit Thickness (mm)	---	---	---	---		
30	Deposit Volume (m3)	---	---	---	---		
31	Length (m)	5011	5016	5021	5026		
32	Elevation (m)	0.6500	0.6600	0.6700	0.6800		
33	Pressure (bar)	77.89	77.89	77.89	77.89		
34	Temperature (C)	55.05	55.05	55.05	55.04		
35	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
36	Heat Transferred (kJ/h-m)	545.8	545.7	545.6	545.6		
37	Flow Regime	Transition	Transition	Transition	Transition		
38	Water Flow Regime						
39	Liquid HoldUp	0.1564	0.1564	0.1564	0.1564		
40	Water HoldUp	---	---	---	---		
41	Friction Gradient (kPa/m)	3.590e-002	3.590e-002	3.590e-002	3.590e-002		
42	Static Gradient (kPa/m)	3.060e-003	3.060e-003	3.060e-003	3.060e-003		
43	Accel. Gradient (kPa/m)	1.162e-005	1.162e-005	1.162e-005	1.162e-005		
44	Liq. Reynolds	1.342e+005 *	1.342e+005 *	1.342e+005 *	1.342e+005 *		
45	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
46	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
47	Vapour Velocity (m/s)	5.234	5.234	5.234	5.234		
48	Deposit Thickness (mm)	---	---	---	---		
49	Deposit Volume (m3)	---	---	---	---		
50	Length (m)	5032	5037	5042	5047		
51	Elevation (m)	0.6740	0.6680	0.6620	0.6560		
52	Pressure (bar)	77.89	77.88	77.88	77.88		
53	Temperature (C)	55.04	55.04	55.04	55.03		
54	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
55	Heat Transferred (kJ/h-m)	545.6	545.5	545.5	545.4		
56	Flow Regime	Transition	Transition	Transition	Transition		
57	Water Flow Regime						
58	Liquid HoldUp	0.1133	0.1133	0.1133	0.1133		
59	Water HoldUp	---	---	---	---		
60	Friction Gradient (kPa/m)	3.888e-002	3.889e-002	3.889e-002	3.889e-002		
61	Static Gradient (kPa/m)	-1.495e-003	-1.495e-003	-1.495e-003	-1.495e-003		
62	Accel. Gradient (kPa/m)	1.115e-005	1.115e-005	1.115e-005	1.115e-005		
63	Liq. Reynolds	1.342e+005 *	1.342e+005 *	1.342e+005 *	1.342e+005 *		
64	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
65	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
66	Vapour Velocity (m/s)	5.234	5.234	5.235	5.235		
67	Deposit Thickness (mm)	---	---	---	---		
68	Deposit Volume (m3)	---	---	---	---		
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
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5										
6	Pipe Segment: BBOU-ELEP Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Length	(m)	5052	5057	5062	5067				
14	Elevation	(m)	0.6500	0.6440	0.6380	0.6320				
15	Pressure	(bar)	77.88	77.88	77.87	77.87				
16	Temperature	(C)	55.03	55.03	55.03	55.02				
17	Vapour Fraction		0.9341	0.9341	0.9341	0.9341				
18	Heat Transferred	(kJ/h-m)	545.3	545.3	545.2	545.2				
19	Flow Regime		Transition	Transition	Transition	Transition				
20	Water Flow Regime									
21	Liquid HoldUp		0.1133	0.1133	0.1133	0.1133				
22	Water HoldUp		---	---	---	---				
23	Friction Gradient	(kPa/m)	3.889e-002	3.889e-002	3.889e-002	3.889e-002				
24	Static Gradient	(kPa/m)	-1.495e-003	-1.516e-003	-1.516e-003	-1.516e-003				
25	Accel. Gradient	(kPa/m)	1.115e-005	1.115e-005	1.115e-005	1.115e-005				
26	Liq. Reynolds		1.342e+005 *	1.342e+005 *	1.342e+005 *	1.342e+005 *				
27	Vap. Reynolds		1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *				
28	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083				
29	Vapour Velocity	(m/s)	5.235	5.235	5.235	5.235				
30	Deposit Thickness	(mm)	---	---	---	---				
31	Deposit Volume	(m3)	---	---	---	---				
32	Length	(m)	5072	5077	5082	5087				
33	Elevation	(m)	0.6260	0.6200	0.6180	0.6160				
34	Pressure	(bar)	77.87	77.87	77.87	77.87				
35	Temperature	(C)	55.02	55.02	55.02	55.01				
36	Vapour Fraction		0.9341	0.9341	0.9341	0.9341				
37	Heat Transferred	(kJ/h-m)	545.1	545.0	545.0	544.9				
38	Flow Regime		Transition	Transition	Transition	Transition				
39	Water Flow Regime									
40	Liquid HoldUp		0.1133	0.1133	0.1136	0.1136				
41	Water HoldUp		---	---	---	---				
42	Friction Gradient	(kPa/m)	3.889e-002	3.889e-002	3.886e-002	3.886e-002				
43	Static Gradient	(kPa/m)	-1.516e-003	-1.516e-003	-5.061e-004	-5.061e-004				
44	Accel. Gradient	(kPa/m)	1.115e-005	1.115e-005	1.144e-005	1.144e-005				
45	Liq. Reynolds		1.342e+005 *	1.342e+005 *	1.342e+005 *	1.342e+005 *				
46	Vap. Reynolds		1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *				
47	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083				
48	Vapour Velocity	(m/s)	5.235	5.235	5.235	5.235				
49	Deposit Thickness	(mm)	---	---	---	---				
50	Deposit Volume	(m3)	---	---	---	---				
51	Length	(m)	5092	5097	5102	5107				
52	Elevation	(m)	0.6140	0.6120	0.6100	0.6200				
53	Pressure	(bar)	77.86	77.86	77.86	77.86				
54	Temperature	(C)	55.01	55.01	55.00	55.00				
55	Vapour Fraction		0.9341	0.9341	0.9341	0.9341				
56	Heat Transferred	(kJ/h-m)	544.9	544.8	544.7	544.6				
57	Flow Regime		Transition	Transition	Transition	Transition				
58	Water Flow Regime									
59	Liquid HoldUp		0.1136	0.1136	0.1136	0.1564				
60	Water HoldUp		---	---	---	---				
61	Friction Gradient	(kPa/m)	3.886e-002	3.886e-002	3.886e-002	3.591e-002				
62	Static Gradient	(kPa/m)	-5.061e-004	-5.061e-004	-5.061e-004	3.060e-003				
63	Accel. Gradient	(kPa/m)	1.144e-005	1.144e-005	1.144e-005	1.163e-005				
64	Liq. Reynolds		1.342e+005 *	1.342e+005 *	1.342e+005 *	1.342e+005 *				
65	Vap. Reynolds		1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *				
66	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083				
67	Vapour Velocity	(m/s)	5.235	5.235	5.235	5.236				
68	Deposit Thickness	(mm)	---	---	---	---				
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:09:19 2022				
5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	5112	5117	5122	5127		
15	Elevation (m)	0.6300	0.6400	0.6500	0.6600		
16	Pressure (bar)	77.86	77.85	77.85	77.85		
17	Temperature (C)	55.00	55.00	54.99	54.99		
18	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
19	Heat Transferred (kJ/h-m)	544.6	544.5	544.4	544.4		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	0.1564	0.1564	0.1564	0.1564		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	3.591e-002	3.591e-002	3.591e-002	3.591e-002		
25	Static Gradient (kPa/m)	3.060e-003	3.060e-003	3.060e-003	3.060e-003		
26	Accel. Gradient (kPa/m)	1.163e-005	1.163e-005	1.163e-005	1.163e-005		
27	Liq. Reynolds	1.342e+005 *	1.342e+005 *	1.341e+005 *	1.341e+005 *		
28	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
29	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
30	Vapour Velocity (m/s)	5.236	5.236	5.236	5.236		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	5132	5137	5142	5147		
34	Elevation (m)	0.6780	0.6960	0.7140	0.7320		
35	Pressure (bar)	77.85	77.85	77.84	77.84		
36	Temperature (C)	54.99	54.98	54.98	54.98		
37	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
38	Heat Transferred (kJ/h-m)	544.3	544.3	544.2	544.1		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.1587	0.1587	0.1587	0.1587		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	3.583e-002	3.583e-002	3.583e-002	3.583e-002		
44	Static Gradient (kPa/m)	5.559e-003	5.559e-003	5.559e-003	5.559e-003		
45	Accel. Gradient (kPa/m)	1.235e-005	1.235e-005	1.235e-005	1.235e-005		
46	Liq. Reynolds	1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *		
47	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
48	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
49	Vapour Velocity (m/s)	5.236	5.236	5.236	5.236		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	5152	5154	5156	5158		
53	Elevation (m)	0.7500	0.7320	0.7140	0.6960		
54	Pressure (bar)	77.84	77.84	77.84	77.84		
55	Temperature (C)	54.98	54.98	54.97	54.97		
56	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
57	Heat Transferred (kJ/h-m)	544.1	544.1	544.1	544.1		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.1587	0.1102	0.1102	0.1102		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	3.583e-002	3.924e-002	3.924e-002	3.924e-002		
63	Static Gradient (kPa/m)	5.559e-003	-1.120e-002	-1.120e-002	-1.120e-002		
64	Accel. Gradient (kPa/m)	1.235e-005	8.369e-006	8.369e-006	8.369e-006		
65	Liq. Reynolds	1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *		
66	Vap. Reynolds	1.219e+007 *	1.219e+007 *	1.219e+007 *	1.219e+007 *		
67	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
68	Vapour Velocity (m/s)	5.236	5.236	5.236	5.236		
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:09:19 2022		
4					
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	5160	5162	5168	5174
16	Elevation (m)	0.6780	0.6600	0.6700	0.6800
17	Pressure (bar)	77.84	77.84	77.83	77.83
18	Temperature (C)	54.97	54.97	54.97	54.97
19	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
20	Heat Transferred (kJ/h-m)	544.1	544.0	543.9	543.8
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.1102	0.1102	0.1559	0.1559
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	3.924e-002	3.924e-002	3.594e-002	3.594e-002
26	Static Gradient (kPa/m)	-1.120e-002	-1.120e-002	2.545e-003	2.545e-003
27	Accel. Gradient (kPa/m)	8.369e-006	8.370e-006	1.149e-005	1.149e-005
28	Liq. Reynolds	1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *
29	Vap. Reynolds	1.219e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
30	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
31	Vapour Velocity (m/s)	5.236	5.236	5.236	5.237
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	5180	5186	5192	5194
35	Elevation (m)	0.6900	0.7000	0.7100	0.7200
36	Pressure (bar)	77.83	77.83	77.83	77.82
37	Temperature (C)	54.96	54.96	54.96	54.95
38	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
39	Heat Transferred (kJ/h-m)	543.8	543.7	543.6	543.6
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.1559	0.1559	0.1559	0.1607
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	3.594e-002	3.594e-002	3.594e-002	3.577e-002
45	Static Gradient (kPa/m)	2.544e-003	2.544e-003	2.544e-003	7.783e-003
46	Accel. Gradient (kPa/m)	1.149e-005	1.149e-005	1.149e-005	1.300e-005
47	Liq. Reynolds	1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *
48	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
49	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
50	Vapour Velocity (m/s)	5.237	5.237	5.237	5.237
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	5196	5198	5200	5202
54	Elevation (m)	0.7300	0.7400	0.7500	0.7600
55	Pressure (bar)	77.82	77.82	77.82	77.82
56	Temperature (C)	54.95	54.95	54.95	54.95
57	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
58	Heat Transferred (kJ/h-m)	543.6	543.5	543.5	543.5
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.1607	0.1607	0.1607	0.1607
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	3.577e-002	3.577e-002	3.577e-002	3.577e-002
64	Static Gradient (kPa/m)	7.783e-003	7.783e-003	7.782e-003	7.782e-003
65	Accel. Gradient (kPa/m)	1.300e-005	1.300e-005	1.300e-005	1.300e-005
66	Liq. Reynolds	1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *
67	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
68	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3							
4							
5	Unit Set: Mixed						
6	Date/Time: Mon Jun 13 15:09:19 2022						
7	Pipe Segment: BBOU-ELEP Pipeline (continued)						
8							
9							
10	PROFILES						
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	5.237	5.237	5.237	5.237	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	5207	5212	5217	5222	
17	Elevation	(m)	0.7380	0.7160	0.6940	0.6720	
18	Pressure	(bar)	77.82	77.82	77.82	77.81	
19	Temperature	(C)	54.95	54.94	54.94	54.94	
20	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
21	Heat Transferred	(kJ/h-m)	543.5	543.4	543.4	543.3	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.1120	0.1120	0.1120	0.1120	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	3.904e-002	3.904e-002	3.904e-002	3.904e-002	
27	Static Gradient	(kPa/m)	-5.524e-003	-5.524e-003	-5.524e-003	-5.524e-003	
28	Accel. Gradient	(kPa/m)	1.001e-005	1.001e-005	1.001e-005	1.001e-005	
29	Liq. Reynolds		1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *	
30	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
31	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083	
32	Vapour Velocity	(m/s)	5.237	5.237	5.237	5.237	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	5227	5232	5237	5242	
36	Elevation	(m)	0.6500	0.6140	0.5780	0.5420	
37	Pressure	(bar)	77.81	77.81	77.81	77.81	
38	Temperature	(C)	54.94	54.93	54.93	54.93	
39	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
40	Heat Transferred	(kJ/h-m)	543.3	543.2	543.2	543.1	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.1120	0.1110	0.1110	0.1110	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	3.904e-002	3.916e-002	3.916e-002	3.916e-002	
46	Static Gradient	(kPa/m)	-5.524e-003	-8.836e-003	-8.836e-003	-8.836e-003	
47	Accel. Gradient	(kPa/m)	1.001e-005	9.056e-006	9.056e-006	9.057e-006	
48	Liq. Reynolds		1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *	
49	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
50	Liquid Velocity	(m/s)	0.2083	0.2083	0.2084	0.2084	
51	Vapour Velocity	(m/s)	5.237	5.237	5.237	5.237	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	5247	5252	5257	5262	
55	Elevation	(m)	0.5060	0.4700	0.4780	0.4860	
56	Pressure	(bar)	77.81	77.80	77.80	77.80	
57	Temperature	(C)	54.93	54.92	54.92	54.92	
58	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
59	Heat Transferred	(kJ/h-m)	543.1	543.0	542.9	542.8	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.1110	0.1110	0.1558	0.1558	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	3.916e-002	3.916e-002	3.595e-002	3.595e-002	
65	Static Gradient	(kPa/m)	-8.836e-003	-8.836e-003	2.441e-003	2.441e-003	
66	Accel. Gradient	(kPa/m)	9.057e-006	9.058e-006	1.147e-005	1.147e-005	
67	Liq. Reynolds		1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *	
68	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
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1			Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:09:19 2022				
6							
7	Pipe Segment: BBOU-ELEP Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
14	Vapour Velocity	(m/s)	5.237	5.238	5.238	5.238	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	5267	5272	5277	5282	
18	Elevation	(m)	0.4940	0.5020	0.5100	0.5180	
19	Pressure	(bar)	77.80	77.80	77.80	77.79	
20	Temperature	(C)	54.92	54.91	54.91	54.91	
21	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
22	Heat Transferred	(kJ/h-m)	542.8	542.7	542.6	542.6	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.1558	0.1558	0.1557	0.1557	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	3.595e-002	3.595e-002	3.595e-002	3.595e-002	
28	Static Gradient	(kPa/m)	2.441e-003	2.441e-003	2.441e-003	2.441e-003	
29	Accel. Gradient	(kPa/m)	1.147e-005	1.147e-005	1.147e-005	1.147e-005	
30	Liq. Reynolds		1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *	
31	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
32	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
33	Vapour Velocity	(m/s)	5.238	5.238	5.238	5.238	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	5287	5292	5297	5302	
37	Elevation	(m)	0.5260	0.5340	0.5420	0.5500	
38	Pressure	(bar)	77.79	77.79	77.79	77.79	
39	Temperature	(C)	54.90	54.90	54.90	54.90	
40	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
41	Heat Transferred	(kJ/h-m)	542.5	542.5	542.4	542.3	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.1557	0.1557	0.1557	0.1557	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	3.595e-002	3.595e-002	3.595e-002	3.595e-002	
47	Static Gradient	(kPa/m)	2.441e-003	2.441e-003	2.441e-003	2.441e-003	
48	Accel. Gradient	(kPa/m)	1.147e-005	1.147e-005	1.147e-005	1.147e-005	
49	Liq. Reynolds		1.341e+005 *	1.341e+005 *	1.341e+005 *	1.341e+005 *	
50	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
51	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
52	Vapour Velocity	(m/s)	5.238	5.238	5.238	5.238	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	5307	5312	5317	5322	
56	Elevation	(m)	0.5480	0.5460	0.5440	0.5420	
57	Pressure	(bar)	77.78	77.78	77.78	77.78	
58	Temperature	(C)	54.89	54.89	54.89	54.89	
59	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
60	Heat Transferred	(kJ/h-m)	542.3	542.3	542.2	542.2	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.1136	0.1136	0.1136	0.1136	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	3.888e-002	3.888e-002	3.888e-002	3.888e-002	
66	Static Gradient	(kPa/m)	-5.059e-004	-5.059e-004	-5.059e-004	-5.059e-004	
67	Accel. Gradient	(kPa/m)	1.147e-005	1.147e-005	1.147e-005	1.147e-005	
68	Liq. Reynolds		1.340e+005 *	1.340e+005 *	1.340e+005 *	1.340e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:09:19 2022		
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
14	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
15	Vapour Velocity (m/s)	5.238	5.238	5.238	5.239
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	5327	5332	5337	5342
19	Elevation (m)	0.5400	0.5480	0.5560	0.5640
20	Pressure (bar)	77.78	77.77	77.77	77.77
21	Temperature (C)	54.88	54.88	54.88	54.87
22	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
23	Heat Transferred (kJ/h-m)	542.1	542.0	541.9	541.9
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1136	0.1557	0.1557	0.1557
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	3.888e-002	3.596e-002	3.596e-002	3.596e-002
29	Static Gradient (kPa/m)	-5.059e-004	2.441e-003	2.441e-003	2.441e-003
30	Accel. Gradient (kPa/m)	1.147e-005	1.148e-005	1.148e-005	1.148e-005
31	Liq. Reynolds	1.340e+005 *	1.340e+005 *	1.340e+005 *	1.340e+005 *
32	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
33	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
34	Vapour Velocity (m/s)	5.239	5.239	5.239	5.239
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	5347	5352	5357	5362
38	Elevation (m)	0.5720	0.5800	0.5800	0.5800
39	Pressure (bar)	77.77	77.77	77.76	77.76
40	Temperature (C)	54.87	54.87	54.87	54.86
41	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
42	Heat Transferred (kJ/h-m)	541.8	541.7	541.7	541.6
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1557	0.1557	0.1660	0.1660
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	3.596e-002	3.596e-002	3.564e-002	3.564e-002
48	Static Gradient (kPa/m)	2.441e-003	2.441e-003	0.0000	0.0000
49	Accel. Gradient (kPa/m)	1.148e-005	1.148e-005	1.065e-005	1.065e-005
50	Liq. Reynolds	1.340e+005 *	1.340e+005 *	1.340e+005 *	1.340e+005 *
51	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
52	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
53	Vapour Velocity (m/s)	5.239	5.239	5.239	5.239
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	5367	5372	5377	5382
57	Elevation (m)	0.5800	0.5800	0.5800	0.5840
58	Pressure (bar)	77.76	77.76	77.76	77.76
59	Temperature (C)	54.86	54.86	54.86	54.85
60	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
61	Heat Transferred (kJ/h-m)	541.5	541.5	541.4	541.4
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1660	0.1660	0.1660	0.1546
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	3.564e-002	3.564e-002	3.564e-002	3.601e-002
67	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	1.215e-003
68	Accel. Gradient (kPa/m)	1.065e-005	1.065e-005	1.065e-005	1.113e-005
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:09:19 2022		
4					
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	1.340e+005 *	1.340e+005 *	1.340e+005 *	1.340e+005 *
14	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
15	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
16	Vapour Velocity (m/s)	5.239	5.239	5.239	5.239
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	5387	5392	5397	5402
20	Elevation (m)	0.5880	0.5920	0.5960	0.6000
21	Pressure (bar)	77.75	77.75	77.75	77.75
22	Temperature (C)	54.85	54.85	54.85	54.84
23	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
24	Heat Transferred (kJ/h-m)	541.3	541.3	541.2	541.2
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.1546	0.1546	0.1546	0.1546
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	3.601e-002	3.601e-002	3.601e-002	3.601e-002
30	Static Gradient (kPa/m)	1.215e-003	1.215e-003	1.215e-003	1.215e-003
31	Accel. Gradient (kPa/m)	1.113e-005	1.113e-005	1.113e-005	1.113e-005
32	Liq. Reynolds	1.340e+005 *	1.340e+005 *	1.340e+005 *	1.340e+005 *
33	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
34	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
35	Vapour Velocity (m/s)	5.239	5.240	5.240	5.240
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	5407	5412	5417	5422
39	Elevation (m)	0.6000	0.6000	0.6000	0.6000
40	Pressure (bar)	77.75	77.74	77.74	77.74
41	Temperature (C)	54.84	54.84	54.83	54.83
42	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
43	Heat Transferred (kJ/h-m)	541.1	541.0	541.0	540.9
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.1660	0.1660	0.1660	0.1660
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	3.564e-002	3.564e-002	3.564e-002	3.564e-002
49	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
50	Accel. Gradient (kPa/m)	1.066e-005	1.066e-005	1.066e-005	1.066e-005
51	Liq. Reynolds	1.340e+005 *	1.340e+005 *	1.340e+005 *	1.340e+005 *
52	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
53	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
54	Vapour Velocity (m/s)	5.240	5.240	5.240	5.240
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	5427	5432	5437	5443
58	Elevation (m)	0.6000	0.6000	0.6000	0.6000
59	Pressure (bar)	77.74	77.74	77.74	77.73
60	Temperature (C)	54.83	54.83	54.82	54.82
61	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
62	Heat Transferred (kJ/h-m)	540.8	540.8	540.7	540.7
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.1660	0.1660	0.1660	0.1660
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	3.564e-002	3.564e-002	3.564e-002	3.564e-002
68	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
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5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient (kPa/m)	1.066e-005	1.066e-005	1.066e-005	1.066e-005		
14	Liq. Reynolds	1.340e+005 *	1.340e+005 *	1.340e+005 *	1.340e+005 *		
15	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *		
16	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084		
17	Vapour Velocity (m/s)	5.240	5.240	5.240	5.240		
18	Deposit Thickness (mm)	---	---	---	---		
19	Deposit Volume (m3)	---	---	---	---		
20	Length (m)	5448	5453	5458	5463		
21	Elevation (m)	0.6000	0.6000	0.6100	0.6200		
22	Pressure (bar)	77.73	77.73	77.73	77.73		
23	Temperature (C)	54.82	54.82	54.81	54.81		
24	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
25	Heat Transferred (kJ/h-m)	540.6	540.5	540.5	540.5		
26	Flow Regime	Transition	Transition	Transition	Transition		
27	Water Flow Regime						
28	Liquid HoldUp	0.1660	0.1660	0.1563	0.1563		
29	Water HoldUp	---	---	---	---		
30	Friction Gradient (kPa/m)	3.564e-002	3.565e-002	3.595e-002	3.595e-002		
31	Static Gradient (kPa/m)	0.0000	0.0000	3.058e-003	3.058e-003		
32	Accel. Gradient (kPa/m)	1.066e-005	1.066e-005	1.167e-005	1.167e-005		
33	Liq. Reynolds	1.340e+005 *	1.340e+005 *	1.340e+005 *	1.340e+005 *		
34	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *		
35	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084		
36	Vapour Velocity (m/s)	5.240	5.240	5.240	5.241		
37	Deposit Thickness (mm)	---	---	---	---		
38	Deposit Volume (m3)	---	---	---	---		
39	Length (m)	5468	5473	5478	5483		
40	Elevation (m)	0.6300	0.6400	0.6500	0.6160		
41	Pressure (bar)	77.72	77.72	77.72	77.72		
42	Temperature (C)	54.81	54.80	54.80	54.80		
43	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
44	Heat Transferred (kJ/h-m)	540.4	540.3	540.3	540.3		
45	Flow Regime	Transition	Transition	Transition	Transition		
46	Water Flow Regime						
47	Liquid HoldUp	0.1563	0.1563	0.1563	0.1110		
48	Water HoldUp	---	---	---	---		
49	Friction Gradient (kPa/m)	3.595e-002	3.595e-002	3.595e-002	3.918e-002		
50	Static Gradient (kPa/m)	3.058e-003	3.058e-003	3.058e-003	-8.493e-003		
51	Accel. Gradient (kPa/m)	1.167e-005	1.167e-005	1.167e-005	9.180e-006		
52	Liq. Reynolds	1.340e+005 *	1.340e+005 *	1.340e+005 *	1.340e+005 *		
53	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *		
54	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084		
55	Vapour Velocity (m/s)	5.241	5.241	5.241	5.241		
56	Deposit Thickness (mm)	---	---	---	---		
57	Deposit Volume (m3)	---	---	---	---		
58	Length (m)	5488	5493	5498	5503		
59	Elevation (m)	0.5820	0.5480	0.5140	0.4800		
60	Pressure (bar)	77.72	77.72	77.71	77.71		
61	Temperature (C)	54.80	54.79	54.79	54.79		
62	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
63	Heat Transferred (kJ/h-m)	540.2	540.2	540.1	540.1		
64	Flow Regime	Transition	Transition	Transition	Transition		
65	Water Flow Regime						
66	Liquid HoldUp	0.1110	0.1110	0.1110	0.1110		
67	Water HoldUp	---	---	---	---		
68	Friction Gradient (kPa/m)	3.918e-002	3.918e-002	3.918e-002	3.918e-002		
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1			Case Name: Simulation - 331MMSCFD_80barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
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6							
7	Pipe Segment: BBOU-ELEP Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Static Gradient	(kPa/m)	-8.493e-003	-8.493e-003	-8.493e-003	-8.493e-003	
14	Accel. Gradient	(kPa/m)	9.180e-006	9.181e-006	9.181e-006	9.181e-006	
15	Liq. Reynolds		1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *	
16	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
17	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
18	Vapour Velocity	(m/s)	5.241	5.241	5.241	5.241	
19	Deposit Thickness	(mm)	---	---	---	---	
20	Deposit Volume	(m3)	---	---	---	---	
21	Length	(m)	5508	5513	5518	5523	
22	Elevation	(m)	0.4980	0.5160	0.5340	0.5520	
23	Pressure	(bar)	77.71	77.71	77.71	77.70	
24	Temperature	(C)	54.79	54.78	54.78	54.78	
25	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
26	Heat Transferred	(kJ/h-m)	539.9	539.9	539.8	539.8	
27	Flow Regime		Transition	Transition	Transition	Transition	
28	Water Flow Regime						
29	Liquid HoldUp		0.1586	0.1586	0.1586	0.1586	
30	Water HoldUp		---	---	---	---	
31	Friction Gradient	(kPa/m)	3.586e-002	3.586e-002	3.587e-002	3.587e-002	
32	Static Gradient	(kPa/m)	5.556e-003	5.556e-003	5.556e-003	5.556e-003	
33	Accel. Gradient	(kPa/m)	1.239e-005	1.239e-005	1.240e-005	1.240e-005	
34	Liq. Reynolds		1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *	
35	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
36	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
37	Vapour Velocity	(m/s)	5.241	5.241	5.241	5.241	
38	Deposit Thickness	(mm)	---	---	---	---	
39	Deposit Volume	(m3)	---	---	---	---	
40	Length	(m)	5528	5533	5538	5543	
41	Elevation	(m)	0.5700	0.5860	0.6020	0.6180	
42	Pressure	(bar)	77.70	77.70	77.70	77.70	
43	Temperature	(C)	54.78	54.77	54.77	54.77	
44	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
45	Heat Transferred	(kJ/h-m)	539.7	539.6	539.6	539.5	
46	Flow Regime		Transition	Transition	Transition	Transition	
47	Water Flow Regime						
48	Liquid HoldUp		0.1586	0.1580	0.1580	0.1580	
49	Water HoldUp		---	---	---	---	
50	Friction Gradient	(kPa/m)	3.587e-002	3.589e-002	3.589e-002	3.589e-002	
51	Static Gradient	(kPa/m)	5.556e-003	4.927e-003	4.927e-003	4.927e-003	
52	Accel. Gradient	(kPa/m)	1.240e-005	1.222e-005	1.222e-005	1.222e-005	
53	Liq. Reynolds		1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *	
54	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
55	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084	
56	Vapour Velocity	(m/s)	5.241	5.242	5.242	5.242	
57	Deposit Thickness	(mm)	---	---	---	---	
58	Deposit Volume	(m3)	---	---	---	---	
59	Length	(m)	5548	5553	5558	5563	
60	Elevation	(m)	0.6340	0.6500	0.6640	0.6780	
61	Pressure	(bar)	77.69	77.69	77.69	77.69	
62	Temperature	(C)	54.76	54.76	54.76	54.76	
63	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
64	Heat Transferred	(kJ/h-m)	539.4	539.4	539.3	539.3	
65	Flow Regime		Transition	Transition	Transition	Transition	
66	Water Flow Regime						
67	Liquid HoldUp		0.1580	0.1580	0.1574	0.1574	
68	Water HoldUp		---	---	---	---	
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1				Case Name: Simulation - 331MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:09:19 2022		
6						
7	Pipe Segment: BBOU-ELEP Pipeline (continued)					
8						
9	PROFILES					
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11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	3.589e-002	3.589e-002	3.591e-002	3.591e-002
14	Static Gradient	(kPa/m)	4.927e-003	4.927e-003	4.301e-003	4.301e-003
15	Accel. Gradient	(kPa/m)	1.222e-005	1.222e-005	1.204e-005	1.204e-005
16	Liq. Reynolds		1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *
17	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
18	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084
19	Vapour Velocity	(m/s)	5.242	5.242	5.242	5.242
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	5568	5573	5578	5580
23	Elevation	(m)	0.6920	0.7060	0.7200	0.6560
24	Pressure	(bar)	77.69	77.68	77.68	77.68
25	Temperature	(C)	54.75	54.75	54.75	54.75
26	Vapour Fraction		0.9341	0.9341	0.9341	0.9341
27	Heat Transferred	(kJ/h-m)	539.2	539.1	539.1	539.2
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.1574	0.1574	0.1574	0.1012
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	3.591e-002	3.591e-002	3.591e-002	4.044e-002
33	Static Gradient	(kPa/m)	4.301e-003	4.301e-003	4.301e-003	-3.801e-002
34	Accel. Gradient	(kPa/m)	1.204e-005	1.204e-005	1.204e-005	7.277e-007
35	Liq. Reynolds		1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *
36	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
37	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084
38	Vapour Velocity	(m/s)	5.242	5.242	5.242	5.242
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	5582	5584	5586	5588
42	Elevation	(m)	0.5920	0.5280	0.4640	0.4000
43	Pressure	(bar)	77.68	77.68	77.68	77.68
44	Temperature	(C)	54.75	54.75	54.74	54.74
45	Vapour Fraction		0.9341	0.9341	0.9341	0.9341
46	Heat Transferred	(kJ/h-m)	539.1	539.1	539.1	539.1
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.1012	0.1012	0.1012	0.1012
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	4.044e-002	4.044e-002	4.044e-002	4.044e-002
52	Static Gradient	(kPa/m)	-3.801e-002	-3.801e-002	-3.801e-002	-3.801e-002
53	Accel. Gradient	(kPa/m)	7.277e-007	7.276e-007	7.276e-007	7.275e-007
54	Liq. Reynolds		1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *
55	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
56	Liquid Velocity	(m/s)	0.2084	0.2084	0.2084	0.2084
57	Vapour Velocity	(m/s)	5.242	5.242	5.242	5.242
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	5590	5592	5594	5596
61	Elevation	(m)	0.3780	0.3560	0.3340	0.3120
62	Pressure	(bar)	77.68	77.68	77.68	77.68
63	Temperature	(C)	54.74	54.74	54.74	54.74
64	Vapour Fraction		0.9341	0.9341	0.9341	0.9341
65	Heat Transferred	(kJ/h-m)	539.1	539.0	539.0	539.0
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.1094	0.1094	0.1094	0.1094
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Water HoldUp		---	---	---	---	
14	Friction Gradient (kPa/m)		3.938e-002	3.938e-002	3.938e-002	3.938e-002	
15	Static Gradient (kPa/m)		-1.362e-002	-1.362e-002	-1.362e-002	-1.362e-002	
16	Accel. Gradient (kPa/m)		7.710e-006	7.710e-006	7.710e-006	7.710e-006	
17	Liq. Reynolds		1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *	
18	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
19	Liquid Velocity (m/s)		0.2084	0.2084	0.2084	0.2084	
20	Vapour Velocity (m/s)		5.242	5.242	5.242	5.242	
21	Deposit Thickness (mm)		---	---	---	---	
22	Deposit Volume (m3)		---	---	---	---	
23	Length (m)		5598	5604	5610	5616	
24	Elevation (m)		0.2900	0.3620	0.4340	0.5060	
25	Pressure (bar)		77.68	77.68	77.67	77.67	
26	Temperature (C)		54.74	54.74	54.73	54.73	
27	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
28	Heat Transferred (kJ/h-m)		539.0	538.8	538.7	538.6	
29	Flow Regime		Transition	Transition	Transition	Transition	
30	Water Flow Regime						
31	Liquid HoldUp		0.1094	0.1707	0.1707	0.1707	
32	Water HoldUp		---	---	---	---	
33	Friction Gradient (kPa/m)		3.938e-002	3.558e-002	3.559e-002	3.559e-002	
34	Static Gradient (kPa/m)		-1.362e-002	1.942e-002	1.942e-002	1.942e-002	
35	Accel. Gradient (kPa/m)		7.711e-006	1.647e-005	1.647e-005	1.647e-005	
36	Liq. Reynolds		1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *	
37	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
38	Liquid Velocity (m/s)		0.2084	0.2084	0.2084	0.2084	
39	Vapour Velocity (m/s)		5.242	5.242	5.243	5.243	
40	Deposit Thickness (mm)		---	---	---	---	
41	Deposit Volume (m3)		---	---	---	---	
42	Length (m)		5622	5628	5634	5640	
43	Elevation (m)		0.5780	0.6500	0.3780	0.1060	
44	Pressure (bar)		77.67	77.66	77.66	77.66	
45	Temperature (C)		54.72	54.72	54.72	54.72	
46	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
47	Heat Transferred (kJ/h-m)		538.6	538.5	538.6	538.5	
48	Flow Regime		Transition	Transition	Transition	Transition	
49	Water Flow Regime						
50	Liquid HoldUp		0.1707	0.1707	9.599e-002	9.599e-002	
51	Water HoldUp		---	---	---	---	
52	Friction Gradient (kPa/m)		3.559e-002	3.559e-002	4.125e-002	4.125e-002	
53	Static Gradient (kPa/m)		1.942e-002	1.942e-002	-5.246e-002	-5.246e-002	
54	Accel. Gradient (kPa/m)		1.647e-005	1.648e-005	-3.357e-006	-3.357e-006	
55	Liq. Reynolds		1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *	
56	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
57	Liquid Velocity (m/s)		0.2084	0.2084	0.2084	0.2084	
58	Vapour Velocity (m/s)		5.243	5.243	5.243	5.243	
59	Deposit Thickness (mm)		---	---	---	---	
60	Deposit Volume (m3)		---	---	---	---	
61	Length (m)		5646	5652	5658	5671	
62	Elevation (m)		-0.1660	-0.4380	-0.7100	-0.6880	
63	Pressure (bar)		77.66	77.67	77.67	77.66	
64	Temperature (C)		54.72	54.71	54.71	54.71	
65	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
66	Heat Transferred (kJ/h-m)		538.5	538.5	538.4	538.2	
67	Flow Regime		Transition	Transition	Transition	Transition	
68	Water Flow Regime						
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:09:19 2022				
5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid HoldUp		9.599e-002	9.599e-002	9.599e-002	0.1557	
14	Water HoldUp		---	---	---	---	
15	Friction Gradient (kPa/m)		4.125e-002	4.125e-002	4.125e-002	3.598e-002	
16	Static Gradient (kPa/m)		-5.246e-002	-5.246e-002	-5.246e-002	2.505e-003	
17	Accel. Gradient (kPa/m)		-3.358e-006	-3.358e-006	-3.358e-006	1.153e-005	
18	Liq. Reynolds		1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *	
19	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
20	Liquid Velocity (m/s)		0.2084	0.2084	0.2084	0.2084	
21	Vapour Velocity (m/s)		5.243	5.243	5.243	5.243	
22	Deposit Thickness (mm)		---	---	---	---	
23	Deposit Volume (m3)		---	---	---	---	
24	Length (m)		5684	5698	5711	5725	
25	Elevation (m)		-0.6660	-0.6440	-0.6220	-0.6000	
26	Pressure (bar)		77.66	77.65	77.65	77.64	
27	Temperature (C)		54.70	54.69	54.68	54.68	
28	Vapour Fraction		0.9341	0.9341	0.9341	0.9341	
29	Heat Transferred (kJ/h-m)		538.0	537.9	537.7	537.5	
30	Flow Regime		Transition	Transition	Transition	Transition	
31	Water Flow Regime						
32	Liquid HoldUp		0.1557	0.1557	0.1557	0.1557	
33	Water HoldUp		---	---	---	---	
34	Friction Gradient (kPa/m)		3.598e-002	3.598e-002	3.599e-002	3.599e-002	
35	Static Gradient (kPa/m)		2.504e-003	2.504e-003	2.504e-003	2.504e-003	
36	Accel. Gradient (kPa/m)		1.153e-005	1.153e-005	1.153e-005	1.153e-005	
37	Liq. Reynolds		1.339e+005 *	1.338e+005 *	1.338e+005 *	1.338e+005 *	
38	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
39	Liquid Velocity (m/s)		0.2084	0.2084	0.2084	0.2084	
40	Vapour Velocity (m/s)		5.243	5.243	5.243	5.244	
41	Deposit Thickness (mm)		---	---	---	---	
42	Deposit Volume (m3)		---	---	---	---	
43	Length (m)		5738	5751	5765	5778	
44	Elevation (m)		-2.080	-3.560	-5.040	-6.520	
45	Pressure (bar)		77.65	77.66	77.67	77.67	
46	Temperature (C)		54.68	54.68	54.68	54.68	
47	Vapour Fraction		0.9341	0.9341	0.9341	0.9340	
48	Heat Transferred (kJ/h-m)		537.7	537.7	537.8	537.8	
49	Flow Regime		Transition	Transition	Transition	Transition	
50	Water Flow Regime						
51	Liquid HoldUp		7.127e-002	7.128e-002	7.128e-002	7.128e-002	
52	Water HoldUp		---	---	---	---	
53	Friction Gradient (kPa/m)		4.723e-002	4.722e-002	4.722e-002	4.721e-002	
54	Static Gradient (kPa/m)		-0.1107	-0.1107	-0.1107	-0.1108	
55	Accel. Gradient (kPa/m)		-1.902e-005	-1.902e-005	-1.902e-005	-1.902e-005	
56	Liq. Reynolds		1.338e+005 *	1.338e+005 *	1.338e+005 *	1.339e+005 *	
57	Vap. Reynolds		1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *	
58	Liquid Velocity (m/s)		0.2084	0.2084	0.2084	0.2084	
59	Vapour Velocity (m/s)		5.243	5.242	5.242	5.241	
60	Deposit Thickness (mm)		---	---	---	---	
61	Deposit Volume (m3)		---	---	---	---	
62	Length (m)		5792	5805	5818	5832	
63	Elevation (m)		-8.000	-8.480	-8.960	-9.440	
64	Pressure (bar)		77.68	77.68	77.68	77.68	
65	Temperature (C)		54.68	54.68	54.67	54.67	
66	Vapour Fraction		0.9340	0.9340	0.9340	0.9340	
67	Heat Transferred (kJ/h-m)		537.8	537.6	537.5	537.5	
68	Flow Regime		Transition	Transition	Transition	Transition	
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1			Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:09:19 2022		
4					
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6					
7	Pipe Segment: BBOU-ELEP Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	7.129e-002	9.974e-002	9.974e-002	9.974e-002
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	4.721e-002	4.065e-002	4.065e-002	4.065e-002
17	Static Gradient (kPa/m)	-0.1108	-4.223e-002	-4.224e-002	-4.224e-002
18	Accel. Gradient (kPa/m)	-1.902e-005	-4.748e-007	-4.754e-007	-4.759e-007
19	Liq. Reynolds	1.339e+005 *	1.339e+005 *	1.339e+005 *	1.339e+005 *
20	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
21	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
22	Vapour Velocity (m/s)	5.240	5.240	5.240	5.240
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	5845	5859	5872	5885
26	Elevation (m)	-9.920	-10.40	-10.64	-10.88
27	Pressure (bar)	77.68	77.68	77.68	77.68
28	Temperature (C)	54.66	54.66	54.65	54.65
29	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
30	Heat Transferred (kJ/h-m)	537.4	537.3	537.1	537.0
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	9.974e-002	9.974e-002	0.1067	0.1067
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	4.065e-002	4.064e-002	3.969e-002	3.969e-002
36	Static Gradient (kPa/m)	-4.224e-002	-4.224e-002	-2.189e-002	-2.189e-002
37	Accel. Gradient (kPa/m)	-4.764e-007	-4.769e-007	5.326e-006	5.326e-006
38	Liq. Reynolds	1.339e+005 *	1.338e+005 *	1.338e+005 *	1.338e+005 *
39	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.220e+007 *
40	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
41	Vapour Velocity (m/s)	5.240	5.240	5.240	5.240
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	5899	5912	5926	5939
45	Elevation (m)	-11.12	-11.36	-11.60	-9.276
46	Pressure (bar)	77.68	77.67	77.67	77.60
47	Temperature (C)	54.64	54.64	54.63	54.60
48	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
49	Heat Transferred (kJ/h-m)	536.9	536.8	536.6	532.0
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.1067	0.1067	0.1067	0.3934
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	3.969e-002	3.969e-002	3.969e-002	3.109e-002
55	Static Gradient (kPa/m)	-2.189e-002	-2.189e-002	-2.189e-002	0.5198
56	Accel. Gradient (kPa/m)	5.326e-006	5.326e-006	5.327e-006	1.652e-004
57	Liq. Reynolds	1.338e+005 *	1.338e+005 *	1.338e+005 *	1.338e+005 *
58	Vap. Reynolds	1.220e+007 *	1.220e+007 *	1.220e+007 *	1.221e+007 *
59	Liquid Velocity (m/s)	0.2084	0.2084	0.2084	0.2084
60	Vapour Velocity (m/s)	5.240	5.240	5.240	5.245
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	5952	5966	5979	5993
64	Elevation (m)	-6.952	-4.628	-2.304	2.000e-002
65	Pressure (bar)	77.52	77.45	77.38	77.30
66	Temperature (C)	54.57	54.54	54.51	54.48
67	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
68	Heat Transferred (kJ/h-m)	531.4	530.7	530.0	529.3
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:09:19 2022		
4					
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.3932	0.3929	0.3927	0.3925
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	3.112e-002	3.115e-002	3.117e-002	3.120e-002
18	Static Gradient (kPa/m)	0.5196	0.5193	0.5191	0.5188
19	Accel. Gradient (kPa/m)	1.654e-004	1.657e-004	1.659e-004	1.662e-004
20	Liq. Reynolds	1.337e+005 *	1.336e+005 *	1.336e+005 *	1.335e+005 *
21	Vap. Reynolds	1.221e+007 *	1.221e+007 *	1.222e+007 *	1.222e+007 *
22	Liquid Velocity (m/s)	0.2084	0.2083	0.2083	0.2083
23	Vapour Velocity (m/s)	5.250	5.254	5.259	5.264
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	5997	6001	6005	6009
27	Elevation (m)	0.1200	0.2200	0.3200	0.4200
28	Pressure (bar)	77.30	77.30	77.29	77.29
29	Temperature (C)	54.47	54.47	54.47	54.46
30	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
31	Heat Transferred (kJ/h-m)	533.0	532.9	532.9	532.8
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1890	0.1890	0.1890	0.1890
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	2.844e-002	2.844e-002	2.844e-002	2.844e-002
37	Static Gradient (kPa/m)	4.325e-002	4.325e-002	4.325e-002	4.325e-002
38	Accel. Gradient (kPa/m)	2.166e-005	2.166e-005	2.167e-005	2.167e-005
39	Liq. Reynolds	1.335e+005 *	1.335e+005 *	1.335e+005 *	1.335e+005 *
40	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
41	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
42	Vapour Velocity (m/s)	5.264	5.264	5.265	5.265
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	6013	6017	6021	6024
46	Elevation (m)	0.5200	0.4800	0.4400	0.4000
47	Pressure (bar)	77.29	77.29	77.29	77.29
48	Temperature (C)	54.46	54.46	54.46	54.46
49	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
50	Heat Transferred (kJ/h-m)	532.8	532.9	532.8	532.8
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1890	0.1094	0.1094	0.1094
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	2.845e-002	3.951e-002	3.951e-002	3.951e-002
56	Static Gradient (kPa/m)	4.325e-002	-1.256e-002	-1.256e-002	-1.256e-002
57	Accel. Gradient (kPa/m)	2.167e-005	8.143e-006	8.143e-006	8.143e-006
58	Liq. Reynolds	1.335e+005 *	1.335e+005 *	1.335e+005 *	1.335e+005 *
59	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
60	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
61	Vapour Velocity (m/s)	5.265	5.265	5.265	5.265
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	6028	6032	6033	6034
65	Elevation (m)	0.3600	0.3200	0.2980	0.2760
66	Pressure (bar)	77.28	77.28	77.28	77.28
67	Temperature (C)	54.45	54.45	54.45	54.45
68	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:09:19 2022				
5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Heat Transferred (kJ/h-m)	532.8	532.7	532.7	532.7		
14	Flow Regime	Transition	Transition	Transition	Transition		
15	Water Flow Regime						
16	Liquid HoldUp	0.1094	0.1094	0.1048	0.1048		
17	Water HoldUp	---	---	---	---		
18	Friction Gradient (kPa/m)	3.951e-002	3.951e-002	4.008e-002	4.008e-002		
19	Static Gradient (kPa/m)	-1.256e-002	-1.256e-002	-2.659e-002	-2.659e-002		
20	Accel. Gradient (kPa/m)	8.144e-006	8.144e-006	4.079e-006	4.079e-006		
21	Liq. Reynolds	1.335e+005 *	1.335e+005 *	1.335e+005 *	1.335e+005 *		
22	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *		
23	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
24	Vapour Velocity (m/s)	5.265	5.265	5.265	5.265		
25	Deposit Thickness (mm)	---	---	---	---		
26	Deposit Volume (m3)	---	---	---	---		
27	Length (m)	6035	6036	6037	6042		
28	Elevation (m)	0.2540	0.2320	0.2100	0.2640		
29	Pressure (bar)	77.28	77.28	77.28	77.28		
30	Temperature (C)	54.45	54.45	54.45	54.45		
31	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
32	Heat Transferred (kJ/h-m)	532.7	532.7	532.7	532.5		
33	Flow Regime	Transition	Transition	Transition	Transition		
34	Water Flow Regime						
35	Liquid HoldUp	0.1048	0.1048	0.1048	0.1685		
36	Water HoldUp	---	---	---	---		
37	Friction Gradient (kPa/m)	4.008e-002	4.008e-002	4.008e-002	3.575e-002		
38	Static Gradient (kPa/m)	-2.659e-002	-2.659e-002	-2.659e-002	1.731e-002		
39	Accel. Gradient (kPa/m)	4.079e-006	4.079e-006	4.079e-006	1.604e-005		
40	Liq. Reynolds	1.335e+005 *	1.335e+005 *	1.335e+005 *	1.335e+005 *		
41	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *		
42	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
43	Vapour Velocity (m/s)	5.265	5.265	5.265	5.265		
44	Deposit Thickness (mm)	---	---	---	---		
45	Deposit Volume (m3)	---	---	---	---		
46	Length (m)	6047	6052	6057	6062		
47	Elevation (m)	0.3180	0.3720	0.4260	0.4800		
48	Pressure (bar)	77.28	77.27	77.27	77.27		
49	Temperature (C)	54.44	54.44	54.44	54.43		
50	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
51	Heat Transferred (kJ/h-m)	532.4	532.4	532.3	532.3		
52	Flow Regime	Transition	Transition	Transition	Transition		
53	Water Flow Regime						
54	Liquid HoldUp	0.1685	0.1685	0.1685	0.1685		
55	Water HoldUp	---	---	---	---		
56	Friction Gradient (kPa/m)	3.575e-002	3.575e-002	3.576e-002	3.576e-002		
57	Static Gradient (kPa/m)	1.731e-002	1.731e-002	1.731e-002	1.731e-002		
58	Accel. Gradient (kPa/m)	1.604e-005	1.604e-005	1.604e-005	1.605e-005		
59	Liq. Reynolds	1.335e+005 *	1.335e+005 *	1.334e+005 *	1.334e+005 *		
60	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *		
61	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
62	Vapour Velocity (m/s)	5.265	5.265	5.266	5.266		
63	Deposit Thickness (mm)	---	---	---	---		
64	Deposit Volume (m3)	---	---	---	---		
65	Length (m)	6067	6072	6077	6082		
66	Elevation (m)	0.4660	0.4520	0.4380	0.4240		
67	Pressure (bar)	77.27	77.27	77.26	77.26		
68	Temperature (C)	54.43	54.43	54.43	54.42		
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:09:19 2022
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
14	Heat Transferred (kJ/h-m)	532.3	532.2	532.2	532.1
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.1123	0.1123	0.1123	0.1123
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	3.919e-002	3.919e-002	3.919e-002	3.919e-002
20	Static Gradient (kPa/m)	-3.513e-003	-3.513e-003	-3.513e-003	-3.513e-003
21	Accel. Gradient (kPa/m)	1.079e-005	1.079e-005	1.079e-005	1.079e-005
22	Liq. Reynolds	1.334e+005 *	1.334e+005 *	1.334e+005 *	1.334e+005 *
23	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
24	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
25	Vapour Velocity (m/s)	5.266	5.266	5.266	5.266
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	6087	6092	6097	6102
29	Elevation (m)	0.4100	0.3620	0.3140	0.2660
30	Pressure (bar)	77.26	77.26	77.26	77.26
31	Temperature (C)	54.42	54.42	54.42	54.41
32	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
33	Heat Transferred (kJ/h-m)	532.1	532.0	532.0	531.9
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1123	0.1096	0.1096	0.1096
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	3.919e-002	3.949e-002	3.949e-002	3.949e-002
39	Static Gradient (kPa/m)	-3.513e-003	-1.189e-002	-1.189e-002	-1.189e-002
40	Accel. Gradient (kPa/m)	1.079e-005	8.348e-006	8.348e-006	8.348e-006
41	Liq. Reynolds	1.334e+005 *	1.334e+005 *	1.334e+005 *	1.334e+005 *
42	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
43	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
44	Vapour Velocity (m/s)	5.266	5.266	5.266	5.266
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	6107	6112	6117	6122
48	Elevation (m)	0.2180	0.1700	0.1680	0.1660
49	Pressure (bar)	77.26	77.25	77.25	77.25
50	Temperature (C)	54.41	54.41	54.41	54.40
51	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
52	Heat Transferred (kJ/h-m)	531.9	531.8	531.7	531.7
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.1096	0.1096	0.1132	0.1132
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	3.949e-002	3.949e-002	3.909e-002	3.909e-002
58	Static Gradient (kPa/m)	-1.189e-002	-1.189e-002	-5.041e-004	-5.041e-004
59	Accel. Gradient (kPa/m)	8.349e-006	8.349e-006	1.167e-005	1.167e-005
60	Liq. Reynolds	1.334e+005 *	1.334e+005 *	1.334e+005 *	1.334e+005 *
61	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
62	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
63	Vapour Velocity (m/s)	5.266	5.266	5.266	5.266
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	6127	6132	6137	6142
67	Elevation (m)	0.1640	0.1620	0.1600	0.1580
68	Pressure (bar)	77.25	77.25	77.24	77.24
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:09:19 2022
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	54.40	54.40	54.40	54.39
14	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
15	Heat Transferred (kJ/h-m)	531.6	531.6	531.5	531.4
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.1132	0.1132	0.1132	0.1132
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	3.909e-002	3.909e-002	3.909e-002	3.909e-002
21	Static Gradient (kPa/m)	-5.041e-004	-5.041e-004	-5.041e-004	-5.041e-004
22	Accel. Gradient (kPa/m)	1.167e-005	1.167e-005	1.167e-005	1.167e-005
23	Liq. Reynolds	1.334e+005 *	1.334e+005 *	1.334e+005 *	1.334e+005 *
24	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
25	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
26	Vapour Velocity (m/s)	5.266	5.267	5.267	5.267
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	6147	6152	6157	6162
30	Elevation (m)	0.1560	0.1540	0.1520	0.1500
31	Pressure (bar)	77.24	77.24	77.24	77.23
32	Temperature (C)	54.39	54.39	54.39	54.38
33	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
34	Heat Transferred (kJ/h-m)	531.4	531.3	531.3	531.2
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1132	0.1132	0.1132	0.1132
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	3.909e-002	3.909e-002	3.909e-002	3.909e-002
40	Static Gradient (kPa/m)	-5.041e-004	-5.041e-004	-5.041e-004	-5.041e-004
41	Accel. Gradient (kPa/m)	1.167e-005	1.167e-005	1.168e-005	1.168e-005
42	Liq. Reynolds	1.334e+005 *	1.334e+005 *	1.334e+005 *	1.334e+005 *
43	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
44	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
45	Vapour Velocity (m/s)	5.267	5.267	5.267	5.267
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	6167	6172	6177	6182
49	Elevation (m)	0.1660	0.1820	0.1980	0.2140
50	Pressure (bar)	77.23	77.23	77.23	77.23
51	Temperature (C)	54.38	54.38	54.37	54.37
52	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
53	Heat Transferred (kJ/h-m)	531.1	531.0	531.0	530.9
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.1575	0.1575	0.1575	0.1575
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	3.606e-002	3.606e-002	3.606e-002	3.606e-002
59	Static Gradient (kPa/m)	4.912e-003	4.912e-003	4.912e-003	4.912e-003
60	Accel. Gradient (kPa/m)	1.240e-005	1.240e-005	1.240e-005	1.240e-005
61	Liq. Reynolds	1.334e+005 *	1.334e+005 *	1.334e+005 *	1.334e+005 *
62	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
63	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
64	Vapour Velocity (m/s)	5.267	5.267	5.267	5.267
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	6187	6192	6197	6202
68	Elevation (m)	0.2300	0.2480	0.2660	0.2840
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:09:19 2022				
5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	77.22	77.22	77.22	77.22		
14	Temperature (C)	54.37	54.37	54.36	54.36		
15	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
16	Heat Transferred (kJ/h-m)	530.9	530.8	530.7	530.7		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.1575	0.1581	0.1581	0.1581		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	3.606e-002	3.604e-002	3.604e-002	3.604e-002		
22	Static Gradient (kPa/m)	4.912e-003	5.538e-003	5.538e-003	5.538e-003		
23	Accel. Gradient (kPa/m)	1.240e-005	1.258e-005	1.258e-005	1.259e-005		
24	Liq. Reynolds	1.334e+005 *	1.334e+005 *	1.334e+005 *	1.334e+005 *		
25	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *		
26	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
27	Vapour Velocity (m/s)	5.267	5.268	5.268	5.268		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	6207	6212	6217	6222		
31	Elevation (m)	0.3020	0.3200	0.3300	0.3400		
32	Pressure (bar)	77.22	77.21	77.21	77.21		
33	Temperature (C)	54.36	54.36	54.35	54.35		
34	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
35	Heat Transferred (kJ/h-m)	530.6	530.5	530.5	530.4		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.1581	0.1581	0.1557	0.1557		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	3.604e-002	3.604e-002	3.613e-002	3.613e-002		
41	Static Gradient (kPa/m)	5.538e-003	5.538e-003	3.009e-003	3.009e-003		
42	Accel. Gradient (kPa/m)	1.259e-005	1.259e-005	1.185e-005	1.185e-005		
43	Liq. Reynolds	1.334e+005 *	1.334e+005 *	1.334e+005 *	1.334e+005 *		
44	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *		
45	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
46	Vapour Velocity (m/s)	5.268	5.268	5.268	5.268		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	6228	6233	6238	6243		
50	Elevation (m)	0.3500	0.3600	0.3700	0.3740		
51	Pressure (bar)	77.21	77.21	77.20	77.20		
52	Temperature (C)	54.35	54.34	54.34	54.34		
53	Vapour Fraction	0.9341	0.9341	0.9341	0.9341		
54	Heat Transferred (kJ/h-m)	530.4	530.3	530.3	530.2		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.1557	0.1557	0.1557	0.1540		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	3.613e-002	3.613e-002	3.613e-002	3.620e-002		
60	Static Gradient (kPa/m)	3.009e-003	3.009e-003	3.009e-003	1.211e-003		
61	Accel. Gradient (kPa/m)	1.185e-005	1.185e-005	1.185e-005	1.133e-005		
62	Liq. Reynolds	1.334e+005 *	1.334e+005 *	1.333e+005 *	1.333e+005 *		
63	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *		
64	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083		
65	Vapour Velocity (m/s)	5.268	5.268	5.268	5.268		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	6248	6253	6258	6263		
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:09:19 2022		
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5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	0.3780	0.3820	0.3860	0.3900
14	Pressure (bar)	77.20	77.20	77.20	77.19
15	Temperature (C)	54.34	54.33	54.33	54.33
16	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
17	Heat Transferred (kJ/h-m)	530.1	530.1	530.0	530.0
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.1540	0.1540	0.1540	0.1540
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	3.620e-002	3.620e-002	3.620e-002	3.620e-002
23	Static Gradient (kPa/m)	1.211e-003	1.211e-003	1.211e-003	1.211e-003
24	Accel. Gradient (kPa/m)	1.133e-005	1.133e-005	1.133e-005	1.133e-005
25	Liq. Reynolds	1.333e+005 *	1.333e+005 *	1.333e+005 *	1.333e+005 *
26	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
27	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
28	Vapour Velocity (m/s)	5.268	5.268	5.269	5.269
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	6268	6273	6278	6283
32	Elevation (m)	0.3920	0.3940	0.3960	0.3980
33	Pressure (bar)	77.19	77.19	77.19	77.19
34	Temperature (C)	54.33	54.32	54.32	54.32
35	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
36	Heat Transferred (kJ/h-m)	529.9	529.9	529.8	529.7
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.1534	0.1534	0.1534	0.1534
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	3.623e-002	3.623e-002	3.623e-002	3.623e-002
42	Static Gradient (kPa/m)	6.038e-004	6.038e-004	6.038e-004	6.038e-004
43	Accel. Gradient (kPa/m)	1.115e-005	1.115e-005	1.116e-005	1.116e-005
44	Liq. Reynolds	1.333e+005 *	1.333e+005 *	1.333e+005 *	1.333e+005 *
45	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
46	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
47	Vapour Velocity (m/s)	5.269	5.269	5.269	5.269
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	6288	6293	6298	6303
51	Elevation (m)	0.4000	0.3960	0.3920	0.3880
52	Pressure (bar)	77.19	77.18	77.18	77.18
53	Temperature (C)	54.32	54.31	54.31	54.31
54	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
55	Heat Transferred (kJ/h-m)	529.7	529.7	529.6	529.6
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.1534	0.1130	0.1130	0.1130
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	3.623e-002	3.913e-002	3.913e-002	3.913e-002
61	Static Gradient (kPa/m)	6.038e-004	-1.007e-003	-1.007e-003	-1.007e-003
62	Accel. Gradient (kPa/m)	1.116e-005	1.155e-005	1.155e-005	1.155e-005
63	Liq. Reynolds	1.333e+005 *	1.333e+005 *	1.333e+005 *	1.333e+005 *
64	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
65	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
66	Vapour Velocity (m/s)	5.269	5.269	5.269	5.269
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1			Case Name: Simulation - 331MMSCFD_80barg.hsc		
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3		Epic Atlantic Limited	Unit Set: Mixed		
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5			Date/Time: Mon Jun 13 15:09:19 2022		
6					
7	Pipe Segment: BBOU-ELEP Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	6308	6313	6318	6323
14	Elevation (m)	0.3840	0.3800	0.3800	0.3800
15	Pressure (bar)	77.18	77.18	77.17	77.17
16	Temperature (C)	54.30	54.30	54.30	54.30
17	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
18	Heat Transferred (kJ/h-m)	529.5	529.4	529.3	529.3
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.1130	0.1130	0.1654	0.1654
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	3.913e-002	3.913e-002	3.584e-002	3.584e-002
24	Static Gradient (kPa/m)	-1.007e-003	-1.007e-003	0.0000	0.0000
25	Accel. Gradient (kPa/m)	1.155e-005	1.155e-005	1.086e-005	1.086e-005
26	Liq. Reynolds	1.333e+005 *	1.333e+005 *	1.333e+005 *	1.333e+005 *
27	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
28	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
29	Vapour Velocity (m/s)	5.269	5.269	5.269	5.270
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	6328	6333	6338	6343
33	Elevation (m)	0.3800	0.3800	0.3800	0.3680
34	Pressure (bar)	77.17	77.17	77.17	77.17
35	Temperature (C)	54.29	54.29	54.29	54.29
36	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
37	Heat Transferred (kJ/h-m)	529.2	529.1	529.1	529.1
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.1654	0.1654	0.1654	0.1124
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	3.584e-002	3.584e-002	3.584e-002	3.920e-002
43	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-3.012e-003
44	Accel. Gradient (kPa/m)	1.086e-005	1.086e-005	1.086e-005	1.097e-005
45	Liq. Reynolds	1.333e+005 *	1.333e+005 *	1.333e+005 *	1.333e+005 *
46	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
47	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
48	Vapour Velocity (m/s)	5.270	5.270	5.270	5.270
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	6348	6353	6358	6363
52	Elevation (m)	0.3560	0.3440	0.3320	0.3200
53	Pressure (bar)	77.16	77.16	77.16	77.16
54	Temperature (C)	54.28	54.28	54.28	54.28
55	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
56	Heat Transferred (kJ/h-m)	529.0	529.0	528.9	528.9
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.1124	0.1124	0.1124	0.1124
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	3.920e-002	3.920e-002	3.920e-002	3.920e-002
62	Static Gradient (kPa/m)	-3.012e-003	-3.012e-003	-3.012e-003	-3.012e-003
63	Accel. Gradient (kPa/m)	1.097e-005	1.097e-005	1.097e-005	1.097e-005
64	Liq. Reynolds	1.333e+005 *	1.333e+005 *	1.333e+005 *	1.333e+005 *
65	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.222e+007 *
66	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
67	Vapour Velocity (m/s)	5.270	5.270	5.270	5.270
68	Deposit Thickness (mm)	---	---	---	---
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2					
3					Unit Set: Mixed
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
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10	Pipe Table				
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12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	6368	6373	6378	6383
15	Elevation (m)	0.3200	0.3200	0.3200	0.3200
16	Pressure (bar)	77.16	77.15	77.15	77.15
17	Temperature (C)	54.27	54.27	54.27	54.27
18	Vapour Fraction	0.9341	0.9341	0.9341	0.9341
19	Heat Transferred (kJ/h-m)	528.7	528.7	528.6	528.6
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.1654	0.1654	0.1654	0.1654
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	3.584e-002	3.584e-002	3.584e-002	3.584e-002
25	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
26	Accel. Gradient (kPa/m)	1.086e-005	1.086e-005	1.086e-005	1.086e-005
27	Liq. Reynolds	1.333e+005 *	1.333e+005 *	1.333e+005 *	1.333e+005 *
28	Vap. Reynolds	1.222e+007 *	1.222e+007 *	1.222e+007 *	1.223e+007 *
29	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
30	Vapour Velocity (m/s)	5.270	5.270	5.270	5.270
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	6388	6393	6398	6403
34	Elevation (m)	0.3200	0.3120	0.3040	0.2960
35	Pressure (bar)	77.15	77.15	77.15	77.14
36	Temperature (C)	54.26	54.26	54.26	54.25
37	Vapour Fraction	0.9341	0.9341	0.9340	0.9340
38	Heat Transferred (kJ/h-m)	528.5	528.5	528.5	528.4
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.1654	0.1127	0.1127	0.1127
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	3.584e-002	3.917e-002	3.917e-002	3.917e-002
44	Static Gradient (kPa/m)	0.0000	-2.011e-003	-2.011e-003	-2.011e-003
45	Accel. Gradient (kPa/m)	1.086e-005	1.126e-005	1.126e-005	1.127e-005
46	Liq. Reynolds	1.333e+005 *	1.333e+005 *	1.333e+005 *	1.333e+005 *
47	Vap. Reynolds	1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *
48	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
49	Vapour Velocity (m/s)	5.270	5.271	5.271	5.271
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	6408	6413	6419	6425
53	Elevation (m)	0.2880	0.2800	0.2500	0.2200
54	Pressure (bar)	77.14	77.14	77.14	77.14
55	Temperature (C)	54.25	54.25	54.25	54.24
56	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
57	Heat Transferred (kJ/h-m)	528.4	528.3	528.2	528.2
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.1127	0.1127	0.1114	0.1114
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	3.917e-002	3.917e-002	3.932e-002	3.932e-002
63	Static Gradient (kPa/m)	-2.011e-003	-2.011e-003	-6.230e-003	-6.230e-003
64	Accel. Gradient (kPa/m)	1.127e-005	1.127e-005	1.003e-005	1.003e-005
65	Liq. Reynolds	1.333e+005 *	1.333e+005 *	1.332e+005 *	1.332e+005 *
66	Vap. Reynolds	1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *
67	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
68	Vapour Velocity (m/s)	5.271	5.271	5.271	5.271
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
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13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	6431	6437	6443	6449
16	Elevation (m)	0.1900	0.1600	0.1300	0.1360
17	Pressure (bar)	77.13	77.13	77.13	77.13
18	Temperature (C)	54.24	54.24	54.23	54.23
19	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
20	Heat Transferred (kJ/h-m)	528.1	528.0	528.0	527.8
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.1114	0.1114	0.1113	0.1543
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	3.932e-002	3.932e-002	3.932e-002	3.621e-002
26	Static Gradient (kPa/m)	-6.230e-003	-6.230e-003	-6.230e-003	1.515e-003
27	Accel. Gradient (kPa/m)	1.003e-005	1.003e-005	1.003e-005	1.144e-005
28	Liq. Reynolds	1.332e+005 *	1.332e+005 *	1.332e+005 *	1.332e+005 *
29	Vap. Reynolds	1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *
30	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
31	Vapour Velocity (m/s)	5.271	5.271	5.271	5.271
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	6455	6461	6467	6473
35	Elevation (m)	0.1420	0.1480	0.1540	0.1600
36	Pressure (bar)	77.13	77.12	77.12	77.12
37	Temperature (C)	54.23	54.22	54.22	54.22
38	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
39	Heat Transferred (kJ/h-m)	527.8	527.7	527.6	527.6
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.1543	0.1543	0.1543	0.1543
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	3.621e-002	3.621e-002	3.621e-002	3.621e-002
45	Static Gradient (kPa/m)	1.515e-003	1.515e-003	1.515e-003	1.514e-003
46	Accel. Gradient (kPa/m)	1.144e-005	1.144e-005	1.144e-005	1.144e-005
47	Liq. Reynolds	1.332e+005 *	1.332e+005 *	1.332e+005 *	1.332e+005 *
48	Vap. Reynolds	1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *
49	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
50	Vapour Velocity (m/s)	5.271	5.271	5.272	5.272
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	6476	6479	6482	6485
54	Elevation (m)	0.1660	0.1720	0.1780	0.1840
55	Pressure (bar)	77.12	77.12	77.11	77.11
56	Temperature (C)	54.22	54.22	54.21	54.21
57	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
58	Heat Transferred (kJ/h-m)	527.5	527.5	527.5	527.4
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.1557	0.1557	0.1557	0.1557
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	3.616e-002	3.616e-002	3.616e-002	3.616e-002
64	Static Gradient (kPa/m)	3.047e-003	3.047e-003	3.047e-003	3.047e-003
65	Accel. Gradient (kPa/m)	1.189e-005	1.189e-005	1.189e-005	1.189e-005
66	Liq. Reynolds	1.332e+005 *	1.332e+005 *	1.332e+005 *	1.332e+005 *
67	Vap. Reynolds	1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *
68	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
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5	Unit Set: Mixed						
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7	Pipe Segment: BBOU-ELEP Pipeline (continued)						
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10	PROFILES						
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	5.272	5.272	5.272	5.272	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	6488	6493	6498	6503	
17	Elevation	(m)	0.1900	0.2140	0.2380	0.2620	
18	Pressure	(bar)	77.11	77.11	77.11	77.11	
19	Temperature	(C)	54.21	54.21	54.20	54.20	
20	Vapour Fraction		0.9340	0.9340	0.9340	0.9340	
21	Heat Transferred	(kJ/h-m)	527.4	527.3	527.3	527.2	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.1557	0.1597	0.1597	0.1597	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	3.616e-002	3.601e-002	3.601e-002	3.601e-002	
27	Static Gradient	(kPa/m)	3.047e-003	7.432e-003	7.432e-003	7.432e-003	
28	Accel. Gradient	(kPa/m)	1.189e-005	1.318e-005	1.318e-005	1.318e-005	
29	Liq. Reynolds		1.332e+005 *	1.332e+005 *	1.332e+005 *	1.332e+005 *	
30	Vap. Reynolds		1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *	
31	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083	
32	Vapour Velocity	(m/s)	5.272	5.272	5.272	5.272	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	6508	6513	6515	6517	
36	Elevation	(m)	0.2860	0.3100	0.2960	0.2820	
37	Pressure	(bar)	77.10	77.10	77.10	77.10	
38	Temperature	(C)	54.20	54.20	54.20	54.19	
39	Vapour Fraction		0.9340	0.9340	0.9340	0.9340	
40	Heat Transferred	(kJ/h-m)	527.1	527.1	527.1	527.1	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.1597	0.1597	0.1106	0.1106	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	3.601e-002	3.601e-002	3.942e-002	3.942e-002	
46	Static Gradient	(kPa/m)	7.432e-003	7.432e-003	-8.704e-003	-8.704e-003	
47	Accel. Gradient	(kPa/m)	1.318e-005	1.318e-005	9.319e-006	9.320e-006	
48	Liq. Reynolds		1.332e+005 *	1.332e+005 *	1.332e+005 *	1.332e+005 *	
49	Vap. Reynolds		1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *	
50	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083	
51	Vapour Velocity	(m/s)	5.272	5.272	5.272	5.272	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	6519	6521	6523	6526	
55	Elevation	(m)	0.2680	0.2540	0.2400	0.2180	
56	Pressure	(bar)	77.10	77.10	77.10	77.10	
57	Temperature	(C)	54.19	54.19	54.19	54.19	
58	Vapour Fraction		0.9340	0.9340	0.9340	0.9340	
59	Heat Transferred	(kJ/h-m)	527.1	527.1	527.0	527.0	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.1106	0.1106	0.1106	0.1104	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	3.942e-002	3.942e-002	3.942e-002	3.944e-002	
65	Static Gradient	(kPa/m)	-8.704e-003	-8.704e-003	-8.704e-003	-9.113e-003	
66	Accel. Gradient	(kPa/m)	9.320e-006	9.320e-006	9.320e-006	9.201e-006	
67	Liq. Reynolds		1.332e+005 *	1.332e+005 *	1.332e+005 *	1.332e+005 *	
68	Vap. Reynolds		1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	PROFILES						
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11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083	
14	Vapour Velocity	(m/s)	5.272	5.272	5.272	5.272	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	6529	6532	6535	6538	
18	Elevation	(m)	0.1960	0.1740	0.1520	0.1300	
19	Pressure	(bar)	77.10	77.10	77.09	77.09	
20	Temperature	(C)	54.19	54.19	54.19	54.18	
21	Vapour Fraction		0.9340	0.9340	0.9340	0.9340	
22	Heat Transferred	(kJ/h-m)	527.0	526.9	526.9	526.9	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.1104	0.1104	0.1104	0.1104	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	3.944e-002	3.944e-002	3.944e-002	3.944e-002	
28	Static Gradient	(kPa/m)	-9.113e-003	-9.113e-003	-9.113e-003	-9.113e-003	
29	Accel. Gradient	(kPa/m)	9.201e-006	9.201e-006	9.202e-006	9.202e-006	
30	Liq. Reynolds		1.332e+005 *	1.332e+005 *	1.332e+005 *	1.332e+005 *	
31	Vap. Reynolds		1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *	
32	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083	
33	Vapour Velocity	(m/s)	5.273	5.273	5.273	5.273	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	6543	6548	6553	6558	
37	Elevation	(m)	0.1360	0.1420	0.1480	0.1540	
38	Pressure	(bar)	77.09	77.09	77.09	77.09	
39	Temperature	(C)	54.18	54.18	54.18	54.17	
40	Vapour Fraction		0.9340	0.9340	0.9340	0.9340	
41	Heat Transferred	(kJ/h-m)	526.8	526.7	526.6	526.6	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.1545	0.1545	0.1545	0.1545	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	3.621e-002	3.621e-002	3.621e-002	3.621e-002	
47	Static Gradient	(kPa/m)	1.819e-003	1.819e-003	1.819e-003	1.819e-003	
48	Accel. Gradient	(kPa/m)	1.154e-005	1.154e-005	1.154e-005	1.154e-005	
49	Liq. Reynolds		1.332e+005 *	1.332e+005 *	1.332e+005 *	1.332e+005 *	
50	Vap. Reynolds		1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *	
51	Liquid Velocity	(m/s)	0.2083	0.2083	0.2083	0.2083	
52	Vapour Velocity	(m/s)	5.273	5.273	5.273	5.273	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	6563	6568	6573	6578	
56	Elevation	(m)	0.1600	0.1680	0.1760	0.1840	
57	Pressure	(bar)	77.08	77.08	77.08	77.08	
58	Temperature	(C)	54.17	54.17	54.17	54.16	
59	Vapour Fraction		0.9340	0.9340	0.9340	0.9340	
60	Heat Transferred	(kJ/h-m)	526.5	526.5	526.4	526.4	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.1545	0.1551	0.1551	0.1551	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	3.621e-002	3.619e-002	3.619e-002	3.619e-002	
66	Static Gradient	(kPa/m)	1.819e-003	2.431e-003	2.431e-003	2.431e-003	
67	Accel. Gradient	(kPa/m)	1.154e-005	1.172e-005	1.172e-005	1.172e-005	
68	Liq. Reynolds		1.332e+005 *	1.332e+005 *	1.332e+005 *	1.332e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:09:19 2022		
5					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *
14	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
15	Vapour Velocity (m/s)	5.273	5.273	5.273	5.273
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	6583	6588	6593	6598
19	Elevation (m)	0.1920	0.2000	0.1840	0.1680
20	Pressure (bar)	77.08	77.07	77.07	77.07
21	Temperature (C)	54.16	54.16	54.15	54.15
22	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
23	Heat Transferred (kJ/h-m)	526.3	526.2	526.2	526.2
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1551	0.1551	0.1120	0.1120
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	3.619e-002	3.619e-002	3.926e-002	3.926e-002
29	Static Gradient (kPa/m)	2.431e-003	2.431e-003	-4.008e-003	-4.008e-003
30	Accel. Gradient (kPa/m)	1.172e-005	1.172e-005	1.070e-005	1.070e-005
31	Liq. Reynolds	1.332e+005 *	1.332e+005 *	1.332e+005 *	1.331e+005 *
32	Vap. Reynolds	1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *
33	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
34	Vapour Velocity (m/s)	5.273	5.273	5.273	5.274
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	6603	6608	6613	6618
38	Elevation (m)	0.1520	0.1360	0.1200	0.1100
39	Pressure (bar)	77.07	77.07	77.07	77.06
40	Temperature (C)	54.15	54.15	54.14	54.14
41	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
42	Heat Transferred (kJ/h-m)	526.1	526.1	526.0	526.0
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1120	0.1120	0.1120	0.1125
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	3.926e-002	3.926e-002	3.926e-002	3.921e-002
48	Static Gradient (kPa/m)	-4.008e-003	-4.008e-003	-4.008e-003	-2.511e-003
49	Accel. Gradient (kPa/m)	1.070e-005	1.070e-005	1.070e-005	1.114e-005
50	Liq. Reynolds	1.331e+005 *	1.331e+005 *	1.331e+005 *	1.331e+005 *
51	Vap. Reynolds	1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *
52	Liquid Velocity (m/s)	0.2083	0.2083	0.2083	0.2083
53	Vapour Velocity (m/s)	5.274	5.274	5.274	5.274
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	6623	6628	6633	6638
57	Elevation (m)	1.000e-001	9.000e-002	8.000e-002	7.000e-002
58	Pressure (bar)	77.06	77.06	77.06	77.06
59	Temperature (C)	54.14	54.14	54.13	54.13
60	Vapour Fraction	0.9340	0.9340	0.9340	0.9340
61	Heat Transferred (kJ/h-m)	525.9	525.8	525.8	525.7
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1125	0.1125	0.1125	0.1125
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	3.921e-002	3.921e-002	3.922e-002	3.922e-002
67	Static Gradient (kPa/m)	-2.511e-003	-2.511e-003	-2.511e-003	-2.511e-003
68	Accel. Gradient (kPa/m)	1.114e-005	1.115e-005	1.115e-005	1.115e-005
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:09:19 2022			
5										
6	Pipe Segment: BBOU-ELEP Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Liq. Reynolds		1.331e+005 *	1.331e+005 *	1.331e+005 *	1.331e+005 *				
14	Vap. Reynolds		1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *				
15	Liquid Velocity (m/s)		0.2083	0.2083	0.2083	0.2083				
16	Vapour Velocity (m/s)		5.274	5.274	5.274	5.274				
17	Deposit Thickness (mm)		---	---	---	---				
18	Deposit Volume (m3)		---	---	---	---				
19	Length (m)		6643	6648	6653	6658				
20	Elevation (m)		6.800e-002	6.600e-002	6.400e-002	6.200e-002				
21	Pressure (bar)		77.05	77.05	77.05	77.05				
22	Temperature (C)		54.13	54.13	54.12	54.12				
23	Vapour Fraction		0.9340	0.9340	0.9340	0.9340				
24	Heat Transferred (kJ/h-m)		525.7	525.6	525.6	525.5				
25	Flow Regime		Transition	Transition	Transition	Transition				
26	Water Flow Regime									
27	Liquid HoldUp		0.1131	0.1131	0.1131	0.1131				
28	Water HoldUp		---	---	---	---				
29	Friction Gradient (kPa/m)		3.915e-002	3.915e-002	3.915e-002	3.915e-002				
30	Static Gradient (kPa/m)		-5.036e-004	-5.036e-004	-5.036e-004	-5.036e-004				
31	Accel. Gradient (kPa/m)		1.174e-005	1.174e-005	1.174e-005	1.174e-005				
32	Liq. Reynolds		1.331e+005 *	1.331e+005 *	1.331e+005 *	1.331e+005 *				
33	Vap. Reynolds		1.223e+007 *	1.223e+007 *	1.223e+007 *	1.223e+007 *				
34	Liquid Velocity (m/s)		0.2083	0.2083	0.2083	0.2083				
35	Vapour Velocity (m/s)		5.274	5.274	5.274	5.274				
36	Deposit Thickness (mm)		---	---	---	---				
37	Deposit Volume (m3)		---	---	---	---				
38	Length (m)		6663							
39	Elevation (m)		6.000e-002							
40	Pressure (bar)		77.05							
41	Temperature (C)		54.12							
42	Vapour Fraction		0.9340							
43	Heat Transferred (kJ/h-m)		525.4							
44	Flow Regime		Transition							
45	Water Flow Regime									
46	Liquid HoldUp		0.1131							
47	Water HoldUp		---							
48	Friction Gradient (kPa/m)		3.915e-002							
49	Static Gradient (kPa/m)		-5.036e-004							
50	Accel. Gradient (kPa/m)		1.174e-005							
51	Liq. Reynolds		1.331e+005 *							
52	Vap. Reynolds		1.223e+007 *							
53	Liquid Velocity (m/s)		0.2083							
54	Vapour Velocity (m/s)		5.274							
55	Deposit Thickness (mm)		---							
56	Deposit Volume (m3)		---							
57	Erosion									
58										
59	Model									
60										
61	Erosion Model:		API-RP-14E Continuous service		Empirical Constant:		100.0			
62										
63	Erosion Table									
64	Length (m)		0.0000	509.2	1018	1528	2037			
65	Bulk Density (kg/m3)		82.74	82.66	82.57	82.49	82.40			
66	Bulk Velocity (m/s)		5.360	5.366	5.371	5.377	5.383			
67	Erosion Velocity (m/s)		13.41	13.42	13.42	13.43	13.44			
68	Length (m)		2546	2551	2556	2561	2566			
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1	Epic Atlantic Limited			Case Name: Simulation - 331MMSCFD_80barg.hsc		
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3				Unit Set: Mixed		
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Bulk Density	(kg/m3)	82.31	82.31	82.31	82.31
14	Bulk Velocity	(m/s)	5.389	5.389	5.389	5.389
15	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
16	Length	(m)	2571	2576	2581	2591
17	Bulk Density	(kg/m3)	82.30	82.30	82.30	82.30
18	Bulk Velocity	(m/s)	5.389	5.389	5.389	5.389
19	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
20	Length	(m)	2596	2601	2606	2611
21	Bulk Density	(kg/m3)	82.30	82.30	82.30	82.29
22	Bulk Velocity	(m/s)	5.390	5.390	5.390	5.390
23	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
24	Length	(m)	2621	2624	2627	2633
25	Bulk Density	(kg/m3)	82.29	82.29	82.29	82.29
26	Bulk Velocity	(m/s)	5.390	5.390	5.390	5.390
27	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
28	Length	(m)	2636	2640	2644	2652
29	Bulk Density	(kg/m3)	82.29	82.29	82.29	82.29
30	Bulk Velocity	(m/s)	5.390	5.390	5.390	5.390
31	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
32	Length	(m)	2656	2661	2666	2671
33	Bulk Density	(kg/m3)	82.29	82.28	82.28	82.28
34	Bulk Velocity	(m/s)	5.390	5.390	5.390	5.391
35	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
36	Length	(m)	2681	2686	2691	2701
37	Bulk Density	(kg/m3)	82.28	82.28	82.28	82.28
38	Bulk Velocity	(m/s)	5.391	5.391	5.391	5.391
39	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
40	Length	(m)	2706	2711	2716	2726
41	Bulk Density	(kg/m3)	82.28	82.27	82.27	82.27
42	Bulk Velocity	(m/s)	5.391	5.391	5.391	5.391
43	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
44	Length	(m)	2731	2736	2741	2751
45	Bulk Density	(kg/m3)	82.27	82.27	82.27	82.27
46	Bulk Velocity	(m/s)	5.391	5.391	5.391	5.392
47	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
48	Length	(m)	2756	2761	2766	2771
49	Bulk Density	(kg/m3)	82.27	82.26	82.26	82.26
50	Bulk Velocity	(m/s)	5.392	5.392	5.392	5.392
51	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
52	Length	(m)	2781	2786	2791	2801
53	Bulk Density	(kg/m3)	82.26	82.26	82.26	82.26
54	Bulk Velocity	(m/s)	5.392	5.392	5.392	5.392
55	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
56	Length	(m)	2806	2811	2816	2826
57	Bulk Density	(kg/m3)	82.26	82.26	82.25	82.25
58	Bulk Velocity	(m/s)	5.392	5.392	5.392	5.393
59	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
60	Length	(m)	2831	2836	2841	2851
61	Bulk Density	(kg/m3)	82.25	82.25	82.25	82.25
62	Bulk Velocity	(m/s)	5.393	5.393	5.393	5.393
63	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
64	Length	(m)	2856	2861	2866	2876
65	Bulk Density	(kg/m3)	82.25	82.25	82.25	82.25
66	Bulk Velocity	(m/s)	5.393	5.393	5.393	5.393
67	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
68	Length	(m)	2881	2886	2891	2901
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3				Unit Set: Mixed		
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Bulk Density	(kg/m3)	82.25	82.24	82.24	82.24
14	Bulk Velocity	(m/s)	5.393	5.393	5.393	5.393
15	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
16	Length	(m)	2906	2911	2916	2926
17	Bulk Density	(kg/m3)	82.24	82.23	82.23	82.23
18	Bulk Velocity	(m/s)	5.394	5.394	5.394	5.394
19	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
20	Length	(m)	2931	2936	2941	2951
21	Bulk Density	(kg/m3)	82.22	82.22	82.22	82.22
22	Bulk Velocity	(m/s)	5.394	5.394	5.394	5.395
23	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
24	Length	(m)	2956	2961	2966	2976
25	Bulk Density	(kg/m3)	82.22	82.22	82.22	82.22
26	Bulk Velocity	(m/s)	5.395	5.395	5.395	5.395
27	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
28	Length	(m)	2981	2986	2991	3001
29	Bulk Density	(kg/m3)	82.22	82.21	82.21	82.21
30	Bulk Velocity	(m/s)	5.395	5.395	5.395	5.395
31	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
32	Length	(m)	3006	3011	3016	3026
33	Bulk Density	(kg/m3)	82.21	82.21	82.21	82.21
34	Bulk Velocity	(m/s)	5.395	5.395	5.395	5.396
35	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.45
36	Length	(m)	3031	3036	3041	3051
37	Bulk Density	(kg/m3)	82.21	82.20	82.20	82.20
38	Bulk Velocity	(m/s)	5.396	5.396	5.396	5.396
39	Erosion Velocity	(m/s)	13.45	13.45	13.45	13.46
40	Length	(m)	3056	3061	3066	3076
41	Bulk Density	(kg/m3)	82.20	82.20	82.20	82.20
42	Bulk Velocity	(m/s)	5.396	5.396	5.396	5.396
43	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
44	Length	(m)	3081	3086	3091	3101
45	Bulk Density	(kg/m3)	82.20	82.19	82.19	82.19
46	Bulk Velocity	(m/s)	5.396	5.396	5.396	5.396
47	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
48	Length	(m)	3106	3111	3116	3126
49	Bulk Density	(kg/m3)	82.19	82.19	82.19	82.19
50	Bulk Velocity	(m/s)	5.397	5.397	5.397	5.397
51	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
52	Length	(m)	3131	3136	3141	3151
53	Bulk Density	(kg/m3)	82.19	82.18	82.18	82.18
54	Bulk Velocity	(m/s)	5.397	5.397	5.397	5.397
55	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
56	Length	(m)	3156	3161	3166	3176
57	Bulk Density	(kg/m3)	82.18	82.18	82.18	82.18
58	Bulk Velocity	(m/s)	5.397	5.397	5.397	5.397
59	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
60	Length	(m)	3181	3186	3191	3201
61	Bulk Density	(kg/m3)	82.18	82.17	82.17	82.17
62	Bulk Velocity	(m/s)	5.398	5.398	5.398	5.398
63	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
64	Length	(m)	3206	3212	3218	3230
65	Bulk Density	(kg/m3)	82.17	82.17	82.17	82.17
66	Bulk Velocity	(m/s)	5.398	5.398	5.398	5.398
67	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
68	Length	(m)	3236	3239	3242	3248
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1	Epic Atlantic Limited			Case Name: Simulation - 331MMSCFD_80barg.hsc		
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3				Unit Set: Mixed		
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Bulk Density	(kg/m3)	82.16	82.16	82.16	82.16
14	Bulk Velocity	(m/s)	5.398	5.398	5.398	5.398
15	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
16	Length	(m)	3251	3257	3263	3275
17	Bulk Density	(kg/m3)	82.16	82.16	82.16	82.16
18	Bulk Velocity	(m/s)	5.398	5.399	5.399	5.399
19	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
20	Length	(m)	3281	3286	3291	3301
21	Bulk Density	(kg/m3)	82.15	82.15	82.15	82.15
22	Bulk Velocity	(m/s)	5.399	5.399	5.399	5.399
23	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
24	Length	(m)	3306	3311	3316	3326
25	Bulk Density	(kg/m3)	82.15	82.15	82.15	82.15
26	Bulk Velocity	(m/s)	5.399	5.399	5.399	5.399
27	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
28	Length	(m)	3331	3336	3341	3351
29	Bulk Density	(kg/m3)	82.15	82.14	82.14	82.14
30	Bulk Velocity	(m/s)	5.400	5.400	5.400	5.400
31	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
32	Length	(m)	3356	3361	3366	3376
33	Bulk Density	(kg/m3)	82.14	82.14	82.14	82.14
34	Bulk Velocity	(m/s)	5.400	5.400	5.400	5.400
35	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
36	Length	(m)	3381	3386	3391	3401
37	Bulk Density	(kg/m3)	82.13	82.13	82.13	82.13
38	Bulk Velocity	(m/s)	5.400	5.400	5.400	5.400
39	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
40	Length	(m)	3406	3411	3416	3426
41	Bulk Density	(kg/m3)	82.13	82.13	82.13	82.13
42	Bulk Velocity	(m/s)	5.400	5.401	5.401	5.401
43	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
44	Length	(m)	3431	3436	3441	3451
45	Bulk Density	(kg/m3)	82.13	82.13	82.13	82.12
46	Bulk Velocity	(m/s)	5.401	5.401	5.401	5.401
47	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
48	Length	(m)	3456	3461	3466	3476
49	Bulk Density	(kg/m3)	82.12	82.12	82.12	82.12
50	Bulk Velocity	(m/s)	5.401	5.401	5.401	5.401
51	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
52	Length	(m)	3481	3486	3491	3501
53	Bulk Density	(kg/m3)	82.11	82.11	82.11	82.11
54	Bulk Velocity	(m/s)	5.402	5.402	5.402	5.402
55	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
56	Length	(m)	3506	3511	3516	3526
57	Bulk Density	(kg/m3)	82.11	82.11	82.11	82.11
58	Bulk Velocity	(m/s)	5.402	5.402	5.402	5.402
59	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
60	Length	(m)	3531	3536	3541	3551
61	Bulk Density	(kg/m3)	82.10	82.10	82.10	82.10
62	Bulk Velocity	(m/s)	5.402	5.402	5.402	5.402
63	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
64	Length	(m)	3556	3561	3565	3573
65	Bulk Density	(kg/m3)	82.10	82.10	82.10	82.10
66	Bulk Velocity	(m/s)	5.403	5.403	5.403	5.403
67	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
68	Length	(m)	3577	3582	3587	3597
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1	 Epic Atlantic Limited			Case Name: Simulation - 331MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
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8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Bulk Density	(kg/m3)	82.10	82.09	82.09	82.09
14	Bulk Velocity	(m/s)	5.403	5.403	5.403	5.403
15	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
16	Length	(m)	3602	3607	3612	3622
17	Bulk Density	(kg/m3)	82.09	82.09	82.09	82.09
18	Bulk Velocity	(m/s)	5.403	5.403	5.403	5.403
19	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
20	Length	(m)	3627	3632	3637	3647
21	Bulk Density	(kg/m3)	82.09	82.09	82.08	82.08
22	Bulk Velocity	(m/s)	5.403	5.403	5.404	5.404
23	Erosion Velocity	(m/s)	13.46	13.46	13.46	13.46
24	Length	(m)	3652	3657	3662	3672
25	Bulk Density	(kg/m3)	82.08	82.08	82.08	82.08
26	Bulk Velocity	(m/s)	5.404	5.404	5.404	5.404
27	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
28	Length	(m)	3677	3682	3687	3697
29	Bulk Density	(kg/m3)	82.07	82.07	82.07	82.07
30	Bulk Velocity	(m/s)	5.404	5.404	5.404	5.404
31	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
32	Length	(m)	3702	3707	3712	3722
33	Bulk Density	(kg/m3)	82.07	82.07	82.07	82.06
34	Bulk Velocity	(m/s)	5.405	5.405	5.405	5.405
35	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
36	Length	(m)	3727	3732	3737	3747
37	Bulk Density	(kg/m3)	82.06	82.06	82.06	82.06
38	Bulk Velocity	(m/s)	5.405	5.405	5.405	5.405
39	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
40	Length	(m)	3752	3757	3761	3770
41	Bulk Density	(kg/m3)	82.06	82.06	82.06	82.05
42	Bulk Velocity	(m/s)	5.405	5.405	5.405	5.406
43	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
44	Length	(m)	3775	3780	3785	3795
45	Bulk Density	(kg/m3)	82.05	82.05	82.05	82.05
46	Bulk Velocity	(m/s)	5.406	5.406	5.406	5.406
47	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
48	Length	(m)	3800	3805	3810	3820
49	Bulk Density	(kg/m3)	82.05	82.05	82.05	82.04
50	Bulk Velocity	(m/s)	5.406	5.406	5.406	5.406
51	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
52	Length	(m)	3825	3830	3835	3845
53	Bulk Density	(kg/m3)	82.04	82.04	82.04	82.04
54	Bulk Velocity	(m/s)	5.406	5.406	5.406	5.406
55	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
56	Length	(m)	3850	3855	3860	3870
57	Bulk Density	(kg/m3)	82.04	82.04	82.04	82.03
58	Bulk Velocity	(m/s)	5.407	5.407	5.407	5.407
59	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
60	Length	(m)	3875	3880	3885	3895
61	Bulk Density	(kg/m3)	82.03	82.03	82.03	82.03
62	Bulk Velocity	(m/s)	5.407	5.407	5.407	5.407
63	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
64	Length	(m)	3900	3905	3910	3920
65	Bulk Density	(kg/m3)	82.03	82.03	82.03	82.02
66	Bulk Velocity	(m/s)	5.407	5.407	5.407	5.408
67	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
68	Length	(m)	3925	3927	3929	3933
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
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9	Erosion					
10						
11	Erosion Table					
12						
13	Bulk Density	(kg/m3)	82.02	82.02	82.02	82.02
14	Bulk Velocity	(m/s)	5.408	5.408	5.408	5.407
15	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
16	Length	(m)	3935	3947	3958	3980
17	Bulk Density	(kg/m3)	82.02	82.03	82.03	82.04
18	Bulk Velocity	(m/s)	5.407	5.407	5.407	5.406
19	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
20	Length	(m)	3991	4002	4013	4036
21	Bulk Density	(kg/m3)	82.04	82.05	82.05	82.05
22	Bulk Velocity	(m/s)	5.406	5.406	5.406	5.406
23	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
24	Length	(m)	4047	4058	4069	4091
25	Bulk Density	(kg/m3)	82.05	82.05	82.05	82.05
26	Bulk Velocity	(m/s)	5.406	5.406	5.406	5.406
27	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
28	Length	(m)	4103	4114	4125	4147
29	Bulk Density	(kg/m3)	82.04	82.04	82.04	82.03
30	Bulk Velocity	(m/s)	5.406	5.406	5.407	5.407
31	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
32	Length	(m)	4158	4169	4181	4203
33	Bulk Density	(kg/m3)	82.03	82.02	82.02	82.02
34	Bulk Velocity	(m/s)	5.407	5.407	5.407	5.407
35	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
36	Length	(m)	4214	4225	4236	4259
37	Bulk Density	(kg/m3)	82.02	82.03	82.04	82.05
38	Bulk Velocity	(m/s)	5.408	5.407	5.407	5.406
39	Erosion Velocity	(m/s)	13.47	13.47	13.47	13.47
40	Length	(m)	4270	4281	4292	4314
41	Bulk Density	(kg/m3)	82.06	81.98	81.89	81.73
42	Bulk Velocity	(m/s)	5.405	5.411	5.416	5.427
43	Erosion Velocity	(m/s)	13.47	13.47	13.48	13.49
44	Length	(m)	4325	4328	4331	4337
45	Bulk Density	(kg/m3)	81.65	81.65	81.64	81.64
46	Bulk Velocity	(m/s)	5.432	5.432	5.433	5.433
47	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
48	Length	(m)	4340	4343	4346	4352
49	Bulk Density	(kg/m3)	81.63	81.63	81.63	81.63
50	Bulk Velocity	(m/s)	5.433	5.433	5.433	5.433
51	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
52	Length	(m)	4355	4357	4359	4363
53	Bulk Density	(kg/m3)	81.63	81.63	81.63	81.64
54	Bulk Velocity	(m/s)	5.433	5.433	5.433	5.433
55	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
56	Length	(m)	4365	4367	4369	4373
57	Bulk Density	(kg/m3)	81.64	81.63	81.63	81.63
58	Bulk Velocity	(m/s)	5.433	5.433	5.433	5.433
59	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
60	Length	(m)	4375	4380	4385	4395
61	Bulk Density	(kg/m3)	81.63	81.63	81.63	81.63
62	Bulk Velocity	(m/s)	5.433	5.434	5.434	5.434
63	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
64	Length	(m)	4400	4405	4410	4420
65	Bulk Density	(kg/m3)	81.63	81.63	81.63	81.63
66	Bulk Velocity	(m/s)	5.434	5.434	5.434	5.434
67	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
68	Length	(m)	4425	4431	4436	4446
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8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Bulk Density	(kg/m3)	81.63	81.62	81.62	81.62
14	Bulk Velocity	(m/s)	5.434	5.434	5.434	5.434
15	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
16	Length	(m)	4451	4456	4461	4471
17	Bulk Density	(kg/m3)	81.62	81.62	81.62	81.62
18	Bulk Velocity	(m/s)	5.434	5.434	5.434	5.435
19	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
20	Length	(m)	4476	4481	4486	4496
21	Bulk Density	(kg/m3)	81.62	81.61	81.61	81.61
22	Bulk Velocity	(m/s)	5.435	5.435	5.435	5.435
23	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
24	Length	(m)	4501	4506	4511	4521
25	Bulk Density	(kg/m3)	81.61	81.61	81.61	81.60
26	Bulk Velocity	(m/s)	5.435	5.435	5.435	5.435
27	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
28	Length	(m)	4526	4531	4536	4546
29	Bulk Density	(kg/m3)	81.60	81.60	81.60	81.60
30	Bulk Velocity	(m/s)	5.435	5.436	5.436	5.436
31	Erosion Velocity	(m/s)	13.50	13.50	13.50	13.50
32	Length	(m)	4551	4556	4561	4571
33	Bulk Density	(kg/m3)	81.60	81.59	81.59	81.59
34	Bulk Velocity	(m/s)	5.436	5.436	5.436	5.436
35	Erosion Velocity	(m/s)	13.50	13.50	13.51	13.51
36	Length	(m)	4576	4581	4586	4596
37	Bulk Density	(kg/m3)	81.59	81.59	81.59	81.59
38	Bulk Velocity	(m/s)	5.436	5.436	5.436	5.437
39	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51
40	Length	(m)	4601	4606	4611	4621
41	Bulk Density	(kg/m3)	81.58	81.58	81.58	81.58
42	Bulk Velocity	(m/s)	5.437	5.437	5.437	5.437
43	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51
44	Length	(m)	4626	4631	4636	4646
45	Bulk Density	(kg/m3)	81.58	81.58	81.58	81.57
46	Bulk Velocity	(m/s)	5.437	5.437	5.437	5.437
47	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51
48	Length	(m)	4651	4656	4661	4671
49	Bulk Density	(kg/m3)	81.57	81.57	81.57	81.57
50	Bulk Velocity	(m/s)	5.437	5.437	5.438	5.438
51	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51
52	Length	(m)	4676	4681	4686	4696
53	Bulk Density	(kg/m3)	81.57	81.57	81.57	81.56
54	Bulk Velocity	(m/s)	5.438	5.438	5.438	5.438
55	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51
56	Length	(m)	4701	4706	4711	4721
57	Bulk Density	(kg/m3)	81.56	81.56	81.56	81.56
58	Bulk Velocity	(m/s)	5.438	5.438	5.438	5.438
59	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51
60	Length	(m)	4726	4731	4736	4746
61	Bulk Density	(kg/m3)	81.56	81.56	81.56	81.56
62	Bulk Velocity	(m/s)	5.438	5.438	5.438	5.439
63	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51
64	Length	(m)	4751	4756	4761	4771
65	Bulk Density	(kg/m3)	81.55	81.55	81.55	81.55
66	Bulk Velocity	(m/s)	5.439	5.439	5.439	5.439
67	Erosion Velocity	(m/s)	13.51	13.51	13.51	13.51
68	Length	(m)	4776	4781	4786	4796
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3				Unit Set: Mixed		
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Erosion					
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11	Erosion Table					
12						
13	Bulk Density (kg/m3)	81.55	81.55	81.55	81.54	81.54
14	Bulk Velocity (m/s)	5.439	5.439	5.439	5.439	5.439
15	Erosion Velocity (m/s)	13.51	13.51	13.51	13.51	13.51
16	Length (m)	4801	4806	4811	4816	4821
17	Bulk Density (kg/m3)	81.54	81.54	81.54	81.54	81.54
18	Bulk Velocity (m/s)	5.439	5.439	5.440	5.440	5.440
19	Erosion Velocity (m/s)	13.51	13.51	13.51	13.51	13.51
20	Length (m)	4826	4831	4836	4841	4846
21	Bulk Density (kg/m3)	81.54	81.54	81.53	81.53	81.53
22	Bulk Velocity (m/s)	5.440	5.440	5.440	5.440	5.440
23	Erosion Velocity (m/s)	13.51	13.51	13.51	13.51	13.51
24	Length (m)	4851	4856	4861	4866	4871
25	Bulk Density (kg/m3)	81.53	81.53	81.53	81.53	81.53
26	Bulk Velocity (m/s)	5.440	5.440	5.440	5.440	5.440
27	Erosion Velocity (m/s)	13.51	13.51	13.51	13.51	13.51
28	Length (m)	4876	4881	4886	4891	4896
29	Bulk Density (kg/m3)	81.53	81.53	81.52	81.52	81.52
30	Bulk Velocity (m/s)	5.441	5.441	5.441	5.441	5.441
31	Erosion Velocity (m/s)	13.51	13.51	13.51	13.51	13.51
32	Length (m)	4901	4906	4911	4916	4921
33	Bulk Density (kg/m3)	81.52	81.52	81.52	81.52	81.52
34	Bulk Velocity (m/s)	5.441	5.441	5.441	5.441	5.441
35	Erosion Velocity (m/s)	13.51	13.51	13.51	13.51	13.51
36	Length (m)	4926	4931	4936	4941	4946
37	Bulk Density (kg/m3)	81.52	81.51	81.51	81.51	81.51
38	Bulk Velocity (m/s)	5.441	5.441	5.441	5.441	5.442
39	Erosion Velocity (m/s)	13.51	13.51	13.51	13.51	13.51
40	Length (m)	4951	4956	4961	4966	4971
41	Bulk Density (kg/m3)	81.51	81.51	81.51	81.51	81.51
42	Bulk Velocity (m/s)	5.442	5.442	5.442	5.442	5.442
43	Erosion Velocity (m/s)	13.51	13.51	13.51	13.51	13.51
44	Length (m)	4976	4981	4986	4991	4996
45	Bulk Density (kg/m3)	81.50	81.50	81.50	81.50	81.50
46	Bulk Velocity (m/s)	5.442	5.442	5.442	5.442	5.442
47	Erosion Velocity (m/s)	13.51	13.51	13.51	13.51	13.51
48	Length (m)	5001	5006	5011	5016	5021
49	Bulk Density (kg/m3)	81.50	81.50	81.50	81.50	81.49
50	Bulk Velocity (m/s)	5.442	5.442	5.442	5.443	5.443
51	Erosion Velocity (m/s)	13.51	13.51	13.51	13.51	13.51
52	Length (m)	5026	5032	5037	5042	5047
53	Bulk Density (kg/m3)	81.49	81.49	81.49	81.49	81.49
54	Bulk Velocity (m/s)	5.443	5.443	5.443	5.443	5.443
55	Erosion Velocity (m/s)	13.51	13.51	13.51	13.51	13.51
56	Length (m)	5052	5057	5062	5067	5072
57	Bulk Density (kg/m3)	81.49	81.49	81.49	81.49	81.48
58	Bulk Velocity (m/s)	5.443	5.443	5.443	5.443	5.443
59	Erosion Velocity (m/s)	13.51	13.51	13.51	13.51	13.51
60	Length (m)	5077	5082	5087	5092	5097
61	Bulk Density (kg/m3)	81.48	81.48	81.48	81.48	81.48
62	Bulk Velocity (m/s)	5.443	5.443	5.444	5.444	5.444
63	Erosion Velocity (m/s)	13.51	13.51	13.51	13.51	13.51
64	Length (m)	5102	5107	5112	5117	5122
65	Bulk Density (kg/m3)	81.48	81.48	81.47	81.47	81.47
66	Bulk Velocity (m/s)	5.444	5.444	5.444	5.444	5.444
67	Erosion Velocity (m/s)	13.51	13.51	13.51	13.52	13.52
68	Length (m)	5127	5132	5137	5142	5147
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
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9	Erosion					
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11	Erosion Table					
12						
13	Bulk Density	(kg/m3)	81.47	81.47	81.47	81.47
14	Bulk Velocity	(m/s)	5.444	5.444	5.444	5.445
15	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
16	Length	(m)	5152	5154	5156	5160
17	Bulk Density	(kg/m3)	81.47	81.47	81.46	81.46
18	Bulk Velocity	(m/s)	5.445	5.445	5.445	5.445
19	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
20	Length	(m)	5162	5168	5174	5186
21	Bulk Density	(kg/m3)	81.46	81.46	81.46	81.46
22	Bulk Velocity	(m/s)	5.445	5.445	5.445	5.445
23	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
24	Length	(m)	5192	5194	5196	5200
25	Bulk Density	(kg/m3)	81.46	81.46	81.46	81.46
26	Bulk Velocity	(m/s)	5.445	5.445	5.445	5.445
27	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
28	Length	(m)	5202	5207	5212	5222
29	Bulk Density	(kg/m3)	81.45	81.45	81.45	81.45
30	Bulk Velocity	(m/s)	5.445	5.445	5.445	5.446
31	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
32	Length	(m)	5227	5232	5237	5247
33	Bulk Density	(kg/m3)	81.45	81.45	81.45	81.45
34	Bulk Velocity	(m/s)	5.446	5.446	5.446	5.446
35	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
36	Length	(m)	5252	5257	5262	5272
37	Bulk Density	(kg/m3)	81.45	81.45	81.44	81.44
38	Bulk Velocity	(m/s)	5.446	5.446	5.446	5.446
39	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
40	Length	(m)	5277	5282	5287	5297
41	Bulk Density	(kg/m3)	81.44	81.44	81.44	81.44
42	Bulk Velocity	(m/s)	5.446	5.446	5.446	5.447
43	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
44	Length	(m)	5302	5307	5312	5322
45	Bulk Density	(kg/m3)	81.44	81.43	81.43	81.43
46	Bulk Velocity	(m/s)	5.447	5.447	5.447	5.447
47	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
48	Length	(m)	5327	5332	5337	5347
49	Bulk Density	(kg/m3)	81.43	81.43	81.43	81.42
50	Bulk Velocity	(m/s)	5.447	5.447	5.447	5.447
51	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
52	Length	(m)	5352	5357	5362	5372
53	Bulk Density	(kg/m3)	81.42	81.42	81.42	81.42
54	Bulk Velocity	(m/s)	5.447	5.447	5.447	5.448
55	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
56	Length	(m)	5377	5382	5387	5397
57	Bulk Density	(kg/m3)	81.42	81.42	81.42	81.41
58	Bulk Velocity	(m/s)	5.448	5.448	5.448	5.448
59	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
60	Length	(m)	5402	5407	5412	5422
61	Bulk Density	(kg/m3)	81.41	81.41	81.41	81.41
62	Bulk Velocity	(m/s)	5.448	5.448	5.448	5.448
63	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
64	Length	(m)	5427	5432	5437	5448
65	Bulk Density	(kg/m3)	81.41	81.41	81.41	81.40
66	Bulk Velocity	(m/s)	5.448	5.448	5.449	5.449
67	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
68	Length	(m)	5453	5458	5463	5473
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1	Epic Atlantic Limited			Case Name: Simulation - 331MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
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5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Bulk Density	(kg/m3)	81.40	81.40	81.40	81.40
14	Bulk Velocity	(m/s)	5.449	5.449	5.449	5.449
15	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
16	Length	(m)	5478	5483	5488	5493
17	Bulk Density	(kg/m3)	81.40	81.40	81.40	81.39
18	Bulk Velocity	(m/s)	5.449	5.449	5.449	5.449
19	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
20	Length	(m)	5503	5508	5513	5518
21	Bulk Density	(kg/m3)	81.39	81.39	81.39	81.39
22	Bulk Velocity	(m/s)	5.449	5.449	5.450	5.450
23	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
24	Length	(m)	5528	5533	5538	5543
25	Bulk Density	(kg/m3)	81.39	81.39	81.39	81.38
26	Bulk Velocity	(m/s)	5.450	5.450	5.450	5.450
27	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
28	Length	(m)	5553	5558	5563	5568
29	Bulk Density	(kg/m3)	81.38	81.38	81.38	81.38
30	Bulk Velocity	(m/s)	5.450	5.450	5.450	5.451
31	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
32	Length	(m)	5578	5580	5582	5584
33	Bulk Density	(kg/m3)	81.38	81.38	81.38	81.38
34	Bulk Velocity	(m/s)	5.451	5.451	5.451	5.451
35	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
36	Length	(m)	5588	5590	5592	5594
37	Bulk Density	(kg/m3)	81.38	81.38	81.38	81.38
38	Bulk Velocity	(m/s)	5.451	5.451	5.451	5.451
39	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
40	Length	(m)	5598	5604	5610	5616
41	Bulk Density	(kg/m3)	81.38	81.37	81.37	81.37
42	Bulk Velocity	(m/s)	5.451	5.451	5.451	5.451
43	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
44	Length	(m)	5628	5634	5640	5646
45	Bulk Density	(kg/m3)	81.36	81.37	81.37	81.37
46	Bulk Velocity	(m/s)	5.451	5.451	5.451	5.451
47	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
48	Length	(m)	5658	5671	5684	5698
49	Bulk Density	(kg/m3)	81.37	81.37	81.36	81.36
50	Bulk Velocity	(m/s)	5.451	5.451	5.451	5.452
51	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
52	Length	(m)	5725	5738	5751	5765
53	Bulk Density	(kg/m3)	81.36	81.36	81.37	81.38
54	Bulk Velocity	(m/s)	5.452	5.451	5.451	5.450
55	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
56	Length	(m)	5792	5805	5818	5832
57	Bulk Density	(kg/m3)	81.40	81.40	81.41	81.41
58	Bulk Velocity	(m/s)	5.449	5.449	5.449	5.448
59	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
60	Length	(m)	5859	5872	5885	5899
61	Bulk Density	(kg/m3)	81.41	81.41	81.41	81.41
62	Bulk Velocity	(m/s)	5.448	5.448	5.448	5.448
63	Erosion Velocity	(m/s)	13.52	13.52	13.52	13.52
64	Length	(m)	5926	5939	5952	5966
65	Bulk Density	(kg/m3)	81.41	81.34	81.27	81.19
66	Bulk Velocity	(m/s)	5.448	5.453	5.458	5.463
67	Erosion Velocity	(m/s)	13.52	13.53	13.53	13.54
68	Length	(m)	5993	5997	6001	6005
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2							
3				Unit Set: Mixed			
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5							
6	Pipe Segment: BBOU-ELEP Pipeline (continued)						
7							
8							
9	Erosion						
10							
11	Erosion Table						
12							
13	Bulk Density	(kg/m3)	81.05	81.05	81.05	81.04	81.04
14	Bulk Velocity	(m/s)	5.472	5.473	5.473	5.473	5.473
15	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
16	Length	(m)	6013	6017	6021	6024	6028
17	Bulk Density	(kg/m3)	81.04	81.04	81.04	81.04	81.04
18	Bulk Velocity	(m/s)	5.473	5.473	5.473	5.473	5.473
19	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
20	Length	(m)	6032	6033	6034	6035	6036
21	Bulk Density	(kg/m3)	81.04	81.04	81.04	81.04	81.04
22	Bulk Velocity	(m/s)	5.473	5.473	5.473	5.473	5.473
23	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
24	Length	(m)	6037	6042	6047	6052	6057
25	Bulk Density	(kg/m3)	81.04	81.04	81.03	81.03	81.03
26	Bulk Velocity	(m/s)	5.473	5.473	5.474	5.474	5.474
27	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
28	Length	(m)	6062	6067	6072	6077	6082
29	Bulk Density	(kg/m3)	81.03	81.03	81.03	81.03	81.02
30	Bulk Velocity	(m/s)	5.474	5.474	5.474	5.474	5.474
31	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
32	Length	(m)	6087	6092	6097	6102	6107
33	Bulk Density	(kg/m3)	81.02	81.02	81.02	81.02	81.02
34	Bulk Velocity	(m/s)	5.474	5.474	5.474	5.474	5.474
35	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
36	Length	(m)	6112	6117	6122	6127	6132
37	Bulk Density	(kg/m3)	81.02	81.02	81.02	81.02	81.02
38	Bulk Velocity	(m/s)	5.475	5.475	5.475	5.475	5.475
39	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
40	Length	(m)	6137	6142	6147	6152	6157
41	Bulk Density	(kg/m3)	81.01	81.01	81.01	81.01	81.01
42	Bulk Velocity	(m/s)	5.475	5.475	5.475	5.475	5.475
43	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
44	Length	(m)	6162	6167	6172	6177	6182
45	Bulk Density	(kg/m3)	81.01	81.01	81.01	81.00	81.00
46	Bulk Velocity	(m/s)	5.475	5.475	5.475	5.476	5.476
47	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
48	Length	(m)	6187	6192	6197	6202	6207
49	Bulk Density	(kg/m3)	81.00	81.00	81.00	81.00	81.00
50	Bulk Velocity	(m/s)	5.476	5.476	5.476	5.476	5.476
51	Erosion Velocity	(m/s)	13.55	13.55	13.55	13.55	13.55
52	Length	(m)	6212	6217	6222	6228	6233
53	Bulk Density	(kg/m3)	81.00	80.99	80.99	80.99	80.99
54	Bulk Velocity	(m/s)	5.476	5.476	5.476	5.476	5.476
55	Erosion Velocity	(m/s)	13.55	13.55	13.56	13.56	13.56
56	Length	(m)	6238	6243	6248	6253	6258
57	Bulk Density	(kg/m3)	80.99	80.99	80.99	80.99	80.99
58	Bulk Velocity	(m/s)	5.477	5.477	5.477	5.477	5.477
59	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56	13.56
60	Length	(m)	6263	6268	6273	6278	6283
61	Bulk Density	(kg/m3)	80.98	80.98	80.98	80.98	80.98
62	Bulk Velocity	(m/s)	5.477	5.477	5.477	5.477	5.477
63	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56	13.56
64	Length	(m)	6288	6293	6298	6303	6308
65	Bulk Density	(kg/m3)	80.98	80.98	80.98	80.98	80.97
66	Bulk Velocity	(m/s)	5.477	5.477	5.477	5.478	5.478
67	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56	13.56
68	Length	(m)	6313	6318	6323	6328	6333
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1	Epic Atlantic Limited			Case Name: Simulation - 331MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
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5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Bulk Density	(kg/m3)	80.97	80.97	80.97	80.97
14	Bulk Velocity	(m/s)	5.478	5.478	5.478	5.478
15	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
16	Length	(m)	6338	6343	6348	6358
17	Bulk Density	(kg/m3)	80.97	80.97	80.97	80.96
18	Bulk Velocity	(m/s)	5.478	5.478	5.478	5.478
19	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
20	Length	(m)	6363	6368	6373	6383
21	Bulk Density	(kg/m3)	80.96	80.96	80.96	80.96
22	Bulk Velocity	(m/s)	5.478	5.478	5.479	5.479
23	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
24	Length	(m)	6388	6393	6398	6408
25	Bulk Density	(kg/m3)	80.96	80.96	80.96	80.95
26	Bulk Velocity	(m/s)	5.479	5.479	5.479	5.479
27	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
28	Length	(m)	6413	6419	6425	6437
29	Bulk Density	(kg/m3)	80.95	80.95	80.95	80.95
30	Bulk Velocity	(m/s)	5.479	5.479	5.479	5.479
31	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
32	Length	(m)	6443	6449	6455	6467
33	Bulk Density	(kg/m3)	80.95	80.95	80.94	80.94
34	Bulk Velocity	(m/s)	5.479	5.480	5.480	5.480
35	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
36	Length	(m)	6473	6476	6479	6485
37	Bulk Density	(kg/m3)	80.94	80.94	80.94	80.94
38	Bulk Velocity	(m/s)	5.480	5.480	5.480	5.480
39	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
40	Length	(m)	6488	6493	6498	6508
41	Bulk Density	(kg/m3)	80.94	80.93	80.93	80.93
42	Bulk Velocity	(m/s)	5.480	5.480	5.480	5.481
43	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
44	Length	(m)	6513	6515	6517	6521
45	Bulk Density	(kg/m3)	80.93	80.93	80.93	80.93
46	Bulk Velocity	(m/s)	5.481	5.481	5.481	5.481
47	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
48	Length	(m)	6523	6526	6529	6535
49	Bulk Density	(kg/m3)	80.93	80.93	80.93	80.93
50	Bulk Velocity	(m/s)	5.481	5.481	5.481	5.481
51	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
52	Length	(m)	6538	6543	6548	6558
53	Bulk Density	(kg/m3)	80.93	80.92	80.92	80.92
54	Bulk Velocity	(m/s)	5.481	5.481	5.481	5.481
55	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
56	Length	(m)	6563	6568	6573	6583
57	Bulk Density	(kg/m3)	80.92	80.92	80.92	80.91
58	Bulk Velocity	(m/s)	5.481	5.481	5.481	5.482
59	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
60	Length	(m)	6588	6593	6598	6608
61	Bulk Density	(kg/m3)	80.91	80.91	80.91	80.91
62	Bulk Velocity	(m/s)	5.482	5.482	5.482	5.482
63	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
64	Length	(m)	6613	6618	6623	6633
65	Bulk Density	(kg/m3)	80.91	80.91	80.91	80.90
66	Bulk Velocity	(m/s)	5.482	5.482	5.482	5.482
67	Erosion Velocity	(m/s)	13.56	13.56	13.56	13.56
68	Length	(m)	6638	6643	6648	6658
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2						
3				Unit Set: Mixed		
4						
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6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Bulk Density (kg/m3)	80.90	80.90	80.90	80.90	80.90
14	Bulk Velocity (m/s)	5.482	5.482	5.483	5.483	5.483
15	Erosion Velocity (m/s)	13.56	13.56	13.56	13.56	13.56
16	Length (m)	6663				
17	Bulk Density (kg/m3)	80.90				
18	Bulk Velocity (m/s)	5.483				
19	Erosion Velocity (m/s)	13.56				
20	Hydrates					
21						
22	Model Options					
23						
24	Hydrate Model:	Ng & Robinson				
25	Override Ng and Robinson:	False				
26	Initial Pressure:	From a Correlation				
27	Initial Pressure User Input:	---				
28	Hydrates Table					
29						
30	Distance (m)	0.0000	509.2	1018	1528	2037
31	Pressure (bar)	80.00	79.82	79.64	79.47	79.29
32	Temperature (C)	57.94	57.65	57.35	57.06	56.77
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.31	16.30	16.28	16.27	16.26
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	2546	2551	2556	2561	2566
43	Pressure (bar)	79.11	79.11	79.10	79.10	79.10
44	Temperature (C)	56.49	56.49	56.48	56.48	56.48
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	2571	2576	2581	2586	2591
55	Pressure (bar)	79.10	79.10	79.09	79.09	79.09
56	Temperature (C)	56.47	56.47	56.47	56.47	56.46
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	2596	2601	2606	2611	2616
67	Pressure (bar)	79.09	79.09	79.08	79.08	79.08
68	Temperature (C)	56.46	56.46	56.45	56.45	56.45
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:09:19 2022					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	2621	2624	2627	2630	2633
23	Pressure (bar)	79.08	79.08	79.08	79.08	79.07
24	Temperature (C)	56.45	56.44	56.44	56.44	56.44
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	2636	2640	2644	2648	2652
35	Pressure (bar)	79.07	79.07	79.07	79.07	79.07
36	Temperature (C)	56.44	56.43	56.43	56.43	56.43
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	2656	2661	2666	2671	2676
47	Pressure (bar)	79.06	79.06	79.06	79.06	79.06
48	Temperature (C)	56.43	56.42	56.42	56.42	56.41
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	2681	2686	2691	2696	2701
59	Pressure (bar)	79.06	79.05	79.05	79.05	79.05
60	Temperature (C)	56.41	56.41	56.41	56.40	56.40
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2			Unit Set: Mixed			
3			Date/Time: Mon Jun 13 15:09:19 2022			
4						
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	2706	2711	2716	2721	2726
15	Pressure (bar)	79.05	79.04	79.04	79.04	79.04
16	Temperature (C)	56.40	56.39	56.39	56.39	56.39
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	2731	2736	2741	2746	2751
27	Pressure (bar)	79.04	79.04	79.03	79.03	79.03
28	Temperature (C)	56.38	56.38	56.38	56.38	56.37
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	2756	2761	2766	2771	2776
39	Pressure (bar)	79.03	79.03	79.02	79.02	79.02
40	Temperature (C)	56.37	56.37	56.36	56.36	56.36
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.23
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	2781	2786	2791	2796	2801
51	Pressure (bar)	79.02	79.02	79.02	79.01	79.01
52	Temperature (C)	56.36	56.35	56.35	56.35	56.34
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	2806	2811	2816	2821	2826
63	Pressure (bar)	79.01	79.01	79.01	79.00	79.00
64	Temperature (C)	56.34	56.34	56.34	56.33	56.33
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	2831	2836	2841	2846	2851
19	Pressure (bar)	79.00	79.00	79.00	79.00	78.99
20	Temperature (C)	56.33	56.33	56.32	56.32	56.32
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	2856	2861	2866	2871	2876
31	Pressure (bar)	78.99	78.99	78.99	78.99	78.99
32	Temperature (C)	56.31	56.31	56.31	56.31	56.30
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	2881	2886	2891	2896	2901
43	Pressure (bar)	78.99	78.98	78.98	78.98	78.97
44	Temperature (C)	56.30	56.30	56.30	56.29	56.29
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	2906	2911	2916	2921	2926
55	Pressure (bar)	78.97	78.97	78.97	78.96	78.96
56	Temperature (C)	56.29	56.28	56.28	56.28	56.27
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	2931	2936	2941	2946	2951
67	Pressure (bar)	78.96	78.95	78.95	78.95	78.95
68	Temperature (C)	56.27	56.27	56.26	56.26	56.26
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	2956	2961	2966	2971	2976
23	Pressure (bar)	78.95	78.95	78.94	78.94	78.94
24	Temperature (C)	56.26	56.25	56.25	56.25	56.24
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	2981	2986	2991	2996	3001
35	Pressure (bar)	78.94	78.94	78.94	78.93	78.93
36	Temperature (C)	56.24	56.24	56.24	56.23	56.23
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	3006	3011	3016	3021	3026
47	Pressure (bar)	78.93	78.93	78.93	78.92	78.92
48	Temperature (C)	56.23	56.22	56.22	56.22	56.22
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	3031	3036	3041	3046	3051
59	Pressure (bar)	78.92	78.92	78.92	78.91	78.91
60	Temperature (C)	56.21	56.21	56.21	56.21	56.20
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	3056	3061	3066	3071	3076
15	Pressure (bar)	78.91	78.91	78.91	78.91	78.90
16	Temperature (C)	56.20	56.20	56.19	56.19	56.19
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.23	16.23	16.23	16.23	16.23
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	3081	3086	3091	3096	3101
27	Pressure (bar)	78.90	78.90	78.90	78.90	78.89
28	Temperature (C)	56.19	56.18	56.18	56.18	56.17
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.23	16.23	16.23	16.22	16.22
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	3106	3111	3116	3121	3126
39	Pressure (bar)	78.89	78.89	78.89	78.89	78.89
40	Temperature (C)	56.17	56.17	56.17	56.16	56.16
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	3131	3136	3141	3146	3151
51	Pressure (bar)	78.88	78.88	78.88	78.88	78.88
52	Temperature (C)	56.16	56.15	56.15	56.15	56.15
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	3156	3161	3166	3171	3176
63	Pressure (bar)	78.87	78.87	78.87	78.87	78.87
64	Temperature (C)	56.14	56.14	56.14	56.13	56.13
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	3181	3186	3191	3196	3201
19	Pressure (bar)	78.86	78.86	78.86	78.86	78.86
20	Temperature (C)	56.13	56.13	56.12	56.12	56.12
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	3206	3212	3218	3224	3230
31	Pressure (bar)	78.86	78.85	78.85	78.85	78.85
32	Temperature (C)	56.12	56.11	56.11	56.11	56.10
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	3236	3239	3242	3245	3248
43	Pressure (bar)	78.84	78.84	78.84	78.84	78.84
44	Temperature (C)	56.10	56.10	56.10	56.09	56.09
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	3251	3257	3263	3269	3275
55	Pressure (bar)	78.84	78.84	78.83	78.83	78.83
56	Temperature (C)	56.09	56.09	56.08	56.08	56.08
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	3281	3286	3291	3296	3301
67	Pressure (bar)	78.83	78.83	78.82	78.82	78.82
68	Temperature (C)	56.07	56.07	56.07	56.06	56.06
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	3306	3311	3316	3321	3326
23	Pressure (bar)	78.82	78.82	78.81	78.81	78.81
24	Temperature (C)	56.06	56.06	56.05	56.05	56.05
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	3331	3336	3341	3346	3351
35	Pressure (bar)	78.81	78.81	78.81	78.80	78.80
36	Temperature (C)	56.05	56.04	56.04	56.04	56.03
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	3356	3361	3366	3371	3376
47	Pressure (bar)	78.80	78.80	78.80	78.79	78.79
48	Temperature (C)	56.03	56.03	56.03	56.02	56.02
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	3381	3386	3391	3396	3401
59	Pressure (bar)	78.79	78.79	78.79	78.79	78.78
60	Temperature (C)	56.02	56.02	56.01	56.01	56.01
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	3406	3411	3416	3421	3426
15	Pressure (bar)	78.78	78.78	78.78	78.78	78.78
16	Temperature (C)	56.00	56.00	56.00	56.00	55.99
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.22	16.22	16.22	16.22	16.22
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	3431	3436	3441	3446	3451
27	Pressure (bar)	78.78	78.77	78.77	78.77	78.77
28	Temperature (C)	55.99	55.99	55.99	55.98	55.98
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.22	16.21	16.21	16.21	16.21
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	3456	3461	3466	3471	3476
39	Pressure (bar)	78.76	78.76	78.76	78.76	78.76
40	Temperature (C)	55.98	55.97	55.97	55.97	55.97
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	3481	3486	3491	3496	3501
51	Pressure (bar)	78.75	78.75	78.75	78.75	78.75
52	Temperature (C)	55.96	55.96	55.96	55.95	55.95
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	3506	3511	3516	3521	3526
63	Pressure (bar)	78.74	78.74	78.74	78.74	78.74
64	Temperature (C)	55.95	55.95	55.94	55.94	55.94
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	3531	3536	3541	3546	3551
19	Pressure (bar)	78.73	78.73	78.73	78.73	78.73
20	Temperature (C)	55.93	55.93	55.93	55.93	55.92
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	3556	3561	3565	3569	3573
31	Pressure (bar)	78.73	78.72	78.72	78.72	78.72
32	Temperature (C)	55.92	55.92	55.92	55.91	55.91
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	3577	3582	3587	3592	3597
43	Pressure (bar)	78.72	78.72	78.71	78.71	78.71
44	Temperature (C)	55.91	55.91	55.90	55.90	55.90
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	3602	3607	3612	3617	3622
55	Pressure (bar)	78.71	78.71	78.70	78.70	78.70
56	Temperature (C)	55.89	55.89	55.89	55.89	55.89
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	3627	3632	3637	3642	3647
67	Pressure (bar)	78.70	78.70	78.70	78.69	78.69
68	Temperature (C)	55.88	55.88	55.88	55.87	55.87
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	3652	3657	3662	3667	3672
23	Pressure (bar)	78.69	78.69	78.69	78.68	78.68
24	Temperature (C)	55.87	55.87	55.86	55.86	55.86
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	3677	3682	3687	3692	3697
35	Pressure (bar)	78.68	78.68	78.68	78.67	78.67
36	Temperature (C)	55.85	55.85	55.85	55.85	55.84
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	3702	3707	3712	3717	3722
47	Pressure (bar)	78.67	78.67	78.67	78.66	78.66
48	Temperature (C)	55.84	55.84	55.83	55.83	55.83
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	3727	3732	3737	3742	3747
59	Pressure (bar)	78.66	78.66	78.66	78.65	78.65
60	Temperature (C)	55.83	55.82	55.82	55.82	55.81
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	3752	3757	3761	3766	3770
15	Pressure (bar)	78.65	78.65	78.65	78.65	78.64
16	Temperature (C)	55.81	55.81	55.81	55.80	55.80
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.21	16.20	16.20	16.20	16.20
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	3775	3780	3785	3790	3795
27	Pressure (bar)	78.64	78.64	78.64	78.64	78.64
28	Temperature (C)	55.80	55.80	55.79	55.79	55.79
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	3800	3805	3810	3815	3820
39	Pressure (bar)	78.63	78.63	78.63	78.63	78.63
40	Temperature (C)	55.79	55.78	55.78	55.78	55.77
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	3825	3830	3835	3840	3845
51	Pressure (bar)	78.62	78.62	78.62	78.62	78.62
52	Temperature (C)	55.77	55.77	55.77	55.76	55.76
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	3850	3855	3860	3865	3870
63	Pressure (bar)	78.62	78.61	78.61	78.61	78.61
64	Temperature (C)	55.76	55.76	55.75	55.75	55.75
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	3875	3880	3885	3890	3895
19	Pressure (bar)	78.61	78.60	78.60	78.60	78.60
20	Temperature (C)	55.74	55.74	55.74	55.74	55.73
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	3900	3905	3910	3915	3920
31	Pressure (bar)	78.60	78.59	78.59	78.59	78.59
32	Temperature (C)	55.73	55.73	55.72	55.72	55.72
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	3925	3927	3929	3931	3933
43	Pressure (bar)	78.59	78.59	78.59	78.59	78.59
44	Temperature (C)	55.72	55.72	55.71	55.71	55.71
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	3935	3947	3958	3969	3980
55	Pressure (bar)	78.59	78.59	78.59	78.60	78.60
56	Temperature (C)	55.71	55.71	55.71	55.71	55.70
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	3991	4002	4013	4025	4036
67	Pressure (bar)	78.60	78.60	78.60	78.60	78.60
68	Temperature (C)	55.70	55.70	55.69	55.69	55.68
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:09:19 2022					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	4047	4058	4069	4080	4091
23	Pressure (bar)	78.60	78.60	78.59	78.59	78.59
24	Temperature (C)	55.68	55.68	55.67	55.66	55.66
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	4103	4114	4125	4136	4147
35	Pressure (bar)	78.58	78.58	78.57	78.57	78.56
36	Temperature (C)	55.65	55.65	55.64	55.63	55.63
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	4158	4169	4181	4192	4203
47	Pressure (bar)	78.56	78.55	78.55	78.55	78.55
48	Temperature (C)	55.62	55.61	55.61	55.60	55.60
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	4214	4225	4236	4247	4259
59	Pressure (bar)	78.55	78.55	78.56	78.56	78.57
60	Temperature (C)	55.59	55.59	55.59	55.59	55.59
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:09:19 2022			
5										
6	Pipe Segment: BBOU-ELEP Pipeline (continued)									
7										
8										
9	Hydrates									
10										
11	Hydrates Table									
12										
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included				
14	Distance (m)	4270	4281	4292	4303	4314				
15	Pressure (bar)	78.58	78.49	78.41	78.33	78.24				
16	Temperature (C)	55.59	55.56	55.53	55.50	55.46				
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form				
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types				
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model				
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included				
21	Formation Temperature (C)	16.20	16.19	16.19	16.18	16.17				
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II				
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model				
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H				
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included				
26	Distance (m)	4325	4328	4331	4334	4337				
27	Pressure (bar)	78.16	78.15	78.15	78.15	78.14				
28	Temperature (C)	55.43	55.43	55.42	55.42	55.42				
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form				
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types				
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model				
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included				
33	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16				
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II				
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model				
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H				
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included				
38	Distance (m)	4340	4343	4346	4349	4352				
39	Pressure (bar)	78.14	78.14	78.14	78.14	78.14				
40	Temperature (C)	55.42	55.41	55.41	55.41	55.41				
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form				
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types				
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model				
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included				
45	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16				
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II				
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model				
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H				
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included				
50	Distance (m)	4355	4357	4359	4361	4363				
51	Pressure (bar)	78.14	78.14	78.14	78.14	78.14				
52	Temperature (C)	55.41	55.41	55.41	55.41	55.41				
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form				
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types				
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model				
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included				
57	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16				
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II				
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model				
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H				
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included				
62	Distance (m)	4365	4367	4369	4371	4373				
63	Pressure (bar)	78.14	78.14	78.14	78.13	78.13				
64	Temperature (C)	55.41	55.40	55.40	55.40	55.40				
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form				
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types				
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model				
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included				
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	4375	4380	4385	4390	4395
19	Pressure (bar)	78.13	78.13	78.13	78.13	78.12
20	Temperature (C)	55.40	55.40	55.39	55.39	55.39
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	4400	4405	4410	4415	4420
31	Pressure (bar)	78.12	78.12	78.12	78.12	78.12
32	Temperature (C)	55.39	55.38	55.38	55.38	55.38
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	4425	4431	4436	4441	4446
43	Pressure (bar)	78.12	78.12	78.11	78.11	78.11
44	Temperature (C)	55.37	55.37	55.37	55.37	55.36
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	4451	4456	4461	4466	4471
55	Pressure (bar)	78.11	78.11	78.10	78.10	78.10
56	Temperature (C)	55.36	55.36	55.36	55.35	55.35
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	4476	4481	4486	4491	4496
67	Pressure (bar)	78.10	78.10	78.10	78.09	78.09
68	Temperature (C)	55.35	55.34	55.34	55.34	55.34
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	4501	4506	4511	4516	4521
23	Pressure (bar)	78.09	78.09	78.08	78.08	78.08
24	Temperature (C)	55.33	55.33	55.33	55.33	55.32
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	4526	4531	4536	4541	4546
35	Pressure (bar)	78.08	78.08	78.07	78.07	78.07
36	Temperature (C)	55.32	55.32	55.31	55.31	55.31
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	4551	4556	4561	4566	4571
47	Pressure (bar)	78.07	78.07	78.06	78.06	78.06
48	Temperature (C)	55.30	55.30	55.30	55.30	55.29
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	4576	4581	4586	4591	4596
59	Pressure (bar)	78.06	78.06	78.05	78.05	78.05
60	Temperature (C)	55.29	55.29	55.29	55.28	55.28
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	4601	4606	4611	4616	4621
15	Pressure (bar)	78.05	78.05	78.04	78.04	78.04
16	Temperature (C)	55.28	55.27	55.27	55.27	55.27
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.16	16.16	16.16	16.16	16.16
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	4626	4631	4636	4641	4646
27	Pressure (bar)	78.04	78.04	78.03	78.03	78.03
28	Temperature (C)	55.26	55.26	55.26	55.25	55.25
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.16	16.15	16.15	16.15	16.15
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	4651	4656	4661	4666	4671
39	Pressure (bar)	78.03	78.03	78.02	78.02	78.02
40	Temperature (C)	55.25	55.25	55.24	55.24	55.24
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	4676	4681	4686	4691	4696
51	Pressure (bar)	78.02	78.02	78.02	78.01	78.01
52	Temperature (C)	55.24	55.23	55.23	55.23	55.22
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	4701	4706	4711	4716	4721
63	Pressure (bar)	78.01	78.01	78.01	78.01	78.00
64	Temperature (C)	55.22	55.22	55.22	55.21	55.21
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	4726	4731	4736	4741	4746
19	Pressure (bar)	78.00	78.00	78.00	78.00	78.00
20	Temperature (C)	55.21	55.21	55.20	55.20	55.20
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	4751	4756	4761	4766	4771
31	Pressure (bar)	77.99	77.99	77.99	77.99	77.98
32	Temperature (C)	55.20	55.19	55.19	55.19	55.18
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	4776	4781	4786	4791	4796
43	Pressure (bar)	77.98	77.98	77.98	77.98	77.98
44	Temperature (C)	55.18	55.18	55.18	55.17	55.17
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	4801	4806	4811	4816	4821
55	Pressure (bar)	77.97	77.97	77.97	77.97	77.97
56	Temperature (C)	55.17	55.16	55.16	55.16	55.16
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	4826	4831	4836	4841	4846
67	Pressure (bar)	77.96	77.96	77.96	77.96	77.96
68	Temperature (C)	55.15	55.15	55.15	55.15	55.14
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 101 of 116

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:09:19 2022					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	4851	4856	4861	4866	4871
23	Pressure (bar)	77.95	77.95	77.95	77.95	77.95
24	Temperature (C)	55.14	55.14	55.13	55.13	55.13
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	4876	4881	4886	4891	4896
35	Pressure (bar)	77.94	77.94	77.94	77.94	77.94
36	Temperature (C)	55.13	55.12	55.12	55.12	55.12
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	4901	4906	4911	4916	4921
47	Pressure (bar)	77.94	77.93	77.93	77.93	77.93
48	Temperature (C)	55.11	55.11	55.11	55.10	55.10
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	4926	4931	4936	4941	4946
59	Pressure (bar)	77.93	77.92	77.92	77.92	77.92
60	Temperature (C)	55.10	55.10	55.09	55.09	55.09
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	4951	4956	4961	4966	4971
15	Pressure (bar)	77.92	77.91	77.91	77.91	77.91
16	Temperature (C)	55.09	55.08	55.08	55.08	55.08
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.15	16.14	16.14	16.14	16.14
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	4976	4981	4986	4991	4996
27	Pressure (bar)	77.91	77.91	77.90	77.90	77.90
28	Temperature (C)	55.07	55.07	55.07	55.06	55.06
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	5001	5006	5011	5016	5021
39	Pressure (bar)	77.90	77.90	77.89	77.89	77.89
40	Temperature (C)	55.06	55.06	55.05	55.05	55.05
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	5026	5032	5037	5042	5047
51	Pressure (bar)	77.89	77.89	77.88	77.88	77.88
52	Temperature (C)	55.04	55.04	55.04	55.04	55.03
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	5052	5057	5062	5067	5072
63	Pressure (bar)	77.88	77.88	77.87	77.87	77.87
64	Temperature (C)	55.03	55.03	55.03	55.02	55.02
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 103 of 116

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	5077	5082	5087	5092	5097
19	Pressure (bar)	77.87	77.87	77.87	77.86	77.86
20	Temperature (C)	55.02	55.02	55.01	55.01	55.01
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	5102	5107	5112	5117	5122
31	Pressure (bar)	77.86	77.86	77.86	77.85	77.85
32	Temperature (C)	55.00	55.00	55.00	55.00	54.99
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	5127	5132	5137	5142	5147
43	Pressure (bar)	77.85	77.85	77.85	77.84	77.84
44	Temperature (C)	54.99	54.99	54.98	54.98	54.98
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	5152	5154	5156	5158	5160
55	Pressure (bar)	77.84	77.84	77.84	77.84	77.84
56	Temperature (C)	54.98	54.98	54.97	54.97	54.97
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	5162	5168	5174	5180	5186
67	Pressure (bar)	77.84	77.83	77.83	77.83	77.83
68	Temperature (C)	54.97	54.97	54.97	54.96	54.96
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:09:19 2022					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	5192	5194	5196	5198	5200
23	Pressure (bar)	77.83	77.82	77.82	77.82	77.82
24	Temperature (C)	54.96	54.95	54.95	54.95	54.95
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	5202	5207	5212	5217	5222
35	Pressure (bar)	77.82	77.82	77.82	77.82	77.81
36	Temperature (C)	54.95	54.95	54.94	54.94	54.94
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	5227	5232	5237	5242	5247
47	Pressure (bar)	77.81	77.81	77.81	77.81	77.81
48	Temperature (C)	54.94	54.93	54.93	54.93	54.93
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	5252	5257	5262	5267	5272
59	Pressure (bar)	77.80	77.80	77.80	77.80	77.80
60	Temperature (C)	54.92	54.92	54.92	54.92	54.91
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 105 of 116

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	5277	5282	5287	5292	5297
15	Pressure (bar)	77.80	77.79	77.79	77.79	77.79
16	Temperature (C)	54.91	54.91	54.90	54.90	54.90
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.14	16.13	16.13	16.13	16.13
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	5302	5307	5312	5317	5322
27	Pressure (bar)	77.79	77.78	77.78	77.78	77.78
28	Temperature (C)	54.90	54.89	54.89	54.89	54.89
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	5327	5332	5337	5342	5347
39	Pressure (bar)	77.78	77.77	77.77	77.77	77.77
40	Temperature (C)	54.88	54.88	54.88	54.87	54.87
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	5352	5357	5362	5367	5372
51	Pressure (bar)	77.77	77.76	77.76	77.76	77.76
52	Temperature (C)	54.87	54.87	54.86	54.86	54.86
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	5377	5382	5387	5392	5397
63	Pressure (bar)	77.76	77.76	77.75	77.75	77.75
64	Temperature (C)	54.86	54.85	54.85	54.85	54.85
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	5402	5407	5412	5417	5422
19	Pressure (bar)	77.75	77.75	77.74	77.74	77.74
20	Temperature (C)	54.84	54.84	54.84	54.83	54.83
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	5427	5432	5437	5443	5448
31	Pressure (bar)	77.74	77.74	77.74	77.73	77.73
32	Temperature (C)	54.83	54.83	54.82	54.82	54.82
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	5453	5458	5463	5468	5473
43	Pressure (bar)	77.73	77.73	77.73	77.72	77.72
44	Temperature (C)	54.82	54.81	54.81	54.81	54.80
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	5478	5483	5488	5493	5498
55	Pressure (bar)	77.72	77.72	77.72	77.72	77.71
56	Temperature (C)	54.80	54.80	54.80	54.79	54.79
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	5503	5508	5513	5518	5523
67	Pressure (bar)	77.71	77.71	77.71	77.71	77.70
68	Temperature (C)	54.79	54.79	54.78	54.78	54.78
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:09:19 2022					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	5528	5533	5538	5543	5548
23	Pressure (bar)	77.70	77.70	77.70	77.70	77.69
24	Temperature (C)	54.78	54.77	54.77	54.77	54.76
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	5553	5558	5563	5568	5573
35	Pressure (bar)	77.69	77.69	77.69	77.69	77.68
36	Temperature (C)	54.76	54.76	54.76	54.75	54.75
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	5578	5580	5582	5584	5586
47	Pressure (bar)	77.68	77.68	77.68	77.68	77.68
48	Temperature (C)	54.75	54.75	54.75	54.75	54.74
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	5588	5590	5592	5594	5596
59	Pressure (bar)	77.68	77.68	77.68	77.68	77.68
60	Temperature (C)	54.74	54.74	54.74	54.74	54.74
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	5598	5604	5610	5616	5622
15	Pressure (bar)	77.68	77.68	77.67	77.67	77.67
16	Temperature (C)	54.74	54.74	54.73	54.73	54.72
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.13	16.13	16.13	16.12	16.12
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	5628	5634	5640	5646	5652
27	Pressure (bar)	77.66	77.66	77.66	77.66	77.67
28	Temperature (C)	54.72	54.72	54.72	54.72	54.71
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	5658	5671	5684	5698	5711
39	Pressure (bar)	77.67	77.66	77.66	77.65	77.65
40	Temperature (C)	54.71	54.71	54.70	54.69	54.68
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	5725	5738	5751	5765	5778
51	Pressure (bar)	77.64	77.65	77.66	77.67	77.67
52	Temperature (C)	54.68	54.68	54.68	54.68	54.68
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.13
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	5792	5805	5818	5832	5845
63	Pressure (bar)	77.68	77.68	77.68	77.68	77.68
64	Temperature (C)	54.68	54.68	54.67	54.67	54.66
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 109 of 116

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	5859	5872	5885	5899	5912
19	Pressure (bar)	77.68	77.68	77.68	77.68	77.67
20	Temperature (C)	54.66	54.65	54.65	54.64	54.64
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	5926	5939	5952	5966	5979
31	Pressure (bar)	77.67	77.60	77.52	77.45	77.38
32	Temperature (C)	54.63	54.60	54.57	54.54	54.51
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.13	16.12	16.11	16.11	16.10
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	5993	5997	6001	6005	6009
43	Pressure (bar)	77.30	77.30	77.30	77.29	77.29
44	Temperature (C)	54.48	54.47	54.47	54.47	54.46
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	6013	6017	6021	6024	6028
55	Pressure (bar)	77.29	77.29	77.29	77.29	77.28
56	Temperature (C)	54.46	54.46	54.46	54.46	54.45
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	6032	6033	6034	6035	6036
67	Pressure (bar)	77.28	77.28	77.28	77.28	77.28
68	Temperature (C)	54.45	54.45	54.45	54.45	54.45
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	6037	6042	6047	6052	6057
23	Pressure (bar)	77.28	77.28	77.28	77.27	77.27
24	Temperature (C)	54.45	54.45	54.44	54.44	54.44
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	6062	6067	6072	6077	6082
35	Pressure (bar)	77.27	77.27	77.27	77.26	77.26
36	Temperature (C)	54.43	54.43	54.43	54.43	54.42
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	6087	6092	6097	6102	6107
47	Pressure (bar)	77.26	77.26	77.26	77.26	77.26
48	Temperature (C)	54.42	54.42	54.42	54.41	54.41
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	6112	6117	6122	6127	6132
59	Pressure (bar)	77.25	77.25	77.25	77.25	77.25
60	Temperature (C)	54.41	54.41	54.40	54.40	54.40
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	6137	6142	6147	6152	6157
15	Pressure (bar)	77.24	77.24	77.24	77.24	77.24
16	Temperature (C)	54.40	54.39	54.39	54.39	54.39
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	6162	6167	6172	6177	6182
27	Pressure (bar)	77.23	77.23	77.23	77.23	77.23
28	Temperature (C)	54.38	54.38	54.38	54.37	54.37
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	6187	6192	6197	6202	6207
39	Pressure (bar)	77.22	77.22	77.22	77.22	77.22
40	Temperature (C)	54.37	54.37	54.36	54.36	54.36
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	6212	6217	6222	6228	6233
51	Pressure (bar)	77.21	77.21	77.21	77.21	77.21
52	Temperature (C)	54.36	54.35	54.35	54.35	54.34
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	6238	6243	6248	6253	6258
63	Pressure (bar)	77.20	77.20	77.20	77.20	77.20
64	Temperature (C)	54.34	54.34	54.34	54.33	54.33
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
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1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.09
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	6263	6268	6273	6278	6283
19	Pressure (bar)	77.19	77.19	77.19	77.19	77.19
20	Temperature (C)	54.33	54.33	54.32	54.32	54.32
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.09	16.09	16.09	16.09	16.08
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	6288	6293	6298	6303	6308
31	Pressure (bar)	77.19	77.18	77.18	77.18	77.18
32	Temperature (C)	54.32	54.31	54.31	54.31	54.30
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	6313	6318	6323	6328	6333
43	Pressure (bar)	77.18	77.17	77.17	77.17	77.17
44	Temperature (C)	54.30	54.30	54.30	54.29	54.29
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	6338	6343	6348	6353	6358
55	Pressure (bar)	77.17	77.17	77.16	77.16	77.16
56	Temperature (C)	54.29	54.29	54.28	54.28	54.28
57	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
58	Hydrate Type	No Types	No Types	No Types	No Types	No Types
59	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
61	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
62	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
63	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
65	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
66	Distance (m)	6363	6368	6373	6378	6383
67	Pressure (bar)	77.16	77.16	77.15	77.15	77.15
68	Temperature (C)	54.28	54.27	54.27	54.27	54.27
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1	Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:09:19 2022					
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
14	Hydrate Type	No Types	No Types	No Types	No Types	No Types
15	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
17	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
18	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
19	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
21	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
22	Distance (m)	6388	6393	6398	6403	6408
23	Pressure (bar)	77.15	77.15	77.15	77.14	77.14
24	Temperature (C)	54.26	54.26	54.26	54.25	54.25
25	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
26	Hydrate Type	No Types	No Types	No Types	No Types	No Types
27	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
29	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
30	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
31	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
33	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
34	Distance (m)	6413	6419	6425	6431	6437
35	Pressure (bar)	77.14	77.14	77.14	77.13	77.13
36	Temperature (C)	54.25	54.25	54.24	54.24	54.24
37	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
38	Hydrate Type	No Types	No Types	No Types	No Types	No Types
39	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
41	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
42	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
43	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
45	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
46	Distance (m)	6443	6449	6455	6461	6467
47	Pressure (bar)	77.13	77.13	77.13	77.12	77.12
48	Temperature (C)	54.23	54.23	54.23	54.22	54.22
49	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
50	Hydrate Type	No Types	No Types	No Types	No Types	No Types
51	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
53	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
54	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
55	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
57	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
58	Distance (m)	6473	6476	6479	6482	6485
59	Pressure (bar)	77.12	77.12	77.12	77.11	77.11
60	Temperature (C)	54.22	54.22	54.22	54.21	54.21
61	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
62	Hydrate Type	No Types	No Types	No Types	No Types	No Types
63	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
64	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
65	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
66	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
67	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 114 of 116

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
14	Distance (m)	6488	6493	6498	6503	6508
15	Pressure (bar)	77.11	77.11	77.11	77.11	77.10
16	Temperature (C)	54.21	54.21	54.20	54.20	54.20
17	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
18	Hydrate Type	No Types	No Types	No Types	No Types	No Types
19	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
20	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
21	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
22	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
23	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
25	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
26	Distance (m)	6513	6515	6517	6519	6521
27	Pressure (bar)	77.10	77.10	77.10	77.10	77.10
28	Temperature (C)	54.20	54.20	54.19	54.19	54.19
29	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
30	Hydrate Type	No Types	No Types	No Types	No Types	No Types
31	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
32	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
33	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
34	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
35	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
37	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
38	Distance (m)	6523	6526	6529	6532	6535
39	Pressure (bar)	77.10	77.10	77.10	77.10	77.09
40	Temperature (C)	54.19	54.19	54.19	54.19	54.19
41	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
42	Hydrate Type	No Types	No Types	No Types	No Types	No Types
43	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
44	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
45	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
46	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
47	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
49	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
50	Distance (m)	6538	6543	6548	6553	6558
51	Pressure (bar)	77.09	77.09	77.09	77.09	77.09
52	Temperature (C)	54.18	54.18	54.18	54.18	54.17
53	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
54	Hydrate Type	No Types	No Types	No Types	No Types	No Types
55	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
56	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
57	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
58	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
59	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
60	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
61	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
62	Distance (m)	6563	6568	6573	6578	6583
63	Pressure (bar)	77.08	77.08	77.08	77.08	77.08
64	Temperature (C)	54.17	54.17	54.17	54.16	54.16
65	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
66	Hydrate Type	No Types	No Types	No Types	No Types	No Types
67	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
68	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 115 of 116

1	 Epic Atlantic Limited		Case Name: Simulation - 331MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:09:19 2022			
5						
6	Pipe Segment: BBOU-ELEP Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.08
14	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
15	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
16	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
17	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
18	Distance (m)	6588	6593	6598	6603	6608
19	Pressure (bar)	77.07	77.07	77.07	77.07	77.07
20	Temperature (C)	54.16	54.15	54.15	54.15	54.15
21	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
22	Hydrate Type	No Types	No Types	No Types	No Types	No Types
23	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
24	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
25	Formation Temperature (C)	16.08	16.08	16.08	16.08	16.07
26	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
27	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
28	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
29	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
30	Distance (m)	6613	6618	6623	6628	6633
31	Pressure (bar)	77.07	77.06	77.06	77.06	77.06
32	Temperature (C)	54.14	54.14	54.14	54.14	54.13
33	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
34	Hydrate Type	No Types	No Types	No Types	No Types	No Types
35	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
36	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
37	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
38	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
39	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
40	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
41	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
42	Distance (m)	6638	6643	6648	6653	6658
43	Pressure (bar)	77.06	77.05	77.05	77.05	77.05
44	Temperature (C)	54.13	54.13	54.13	54.12	54.12
45	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
46	Hydrate Type	No Types	No Types	No Types	No Types	No Types
47	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
48	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
49	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
50	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
51	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
52	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
53	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
54	Distance (m)	6663				
55	Pressure (bar)	77.05				
56	Temperature (C)	54.12				
57	Formation	Will NOT Form				
58	Hydrate Type	No Types				
59	Mode	Use 3-Phase Model				
60	Inhibitors	Not Included				
61	Formation Temperature (C)	16.07				
62	Type at Formation Temperature	Type II				
63	Mode at Formation Temperature	Use 3-Phase Model				
64	Phases at Formation Temperature	V - Aq - L - H				
65	Inhibitors at Formation Temperature	Not Included				
66						
67						
68						
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 116 of 116	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline				
7					
8					
9	CONDITIONS				
10					
11	Name	To Elep-Akabel Pipeline	Slug Catcher Tie-In	Per Pipeline Energy Stream	
12	Vapour	0.8815	0.8799	---	
13	Temperature (C)	59.7451	45.3486	---	
14	Pressure (bar)	78.5704	66.7720	---	
15	Molar Flow (MMSCFD)	413.8643	413.8643	---	
16	Mass Flow (kg/h)	625694.7736	625694.7736	---	
17	Std Ideal Liq Vol Flow (m3/h)	1439.3914	1439.3914	---	
18	Molar Enthalpy (kJ/kgmole)	-1.003e+005	-1.013e+005	---	
19	Molar Entropy (kJ/kgmole-C)	164.7	162.8	---	
20	Heat Flow (kJ/h)	-2.0673e+09	-2.0878e+09	2.0459e+07	
21	PROPERTIES				
22					
23	Name	To Elep-Akabel Pipeline	Slug Catcher Tie-In		
24	Molecular Weight	30.35	30.35		
25	Molar Density (kgmole/m3)	3.429	3.087		
26	Mass Density (kg/m3)	104.1	93.70		
27	Act. Volume Flow (m3/h)	6011	6678		
28	Mass Enthalpy (kJ/kg)	-3304	-3337		
29	Mass Entropy (kJ/kg-C)	5.427	5.364		
30	Heat Capacity (kJ/kgmole-C)	75.38	73.53		
31	Mass Heat Capacity (kJ/kg-C)	2.483	2.422		
32	LHV Molar Basis (Std) (kJ/kgmole)	1.398e+006	1.398e+006		
33	HHV Molar Basis (Std) (kJ/kgmole)	1.519e+006	1.519e+006		
34	HHV Mass Basis (Std) (kJ/kg)	5.005e+004	5.005e+004		
35	CO2 Loading	---	---		
36	CO2 Apparent Mole Conc. (kgmole/m3)	---	---		
37	CO2 Apparent Wt. Conc. (kgmol/kg)	---	---		
38	LHV Mass Basis (Std) (kJ/kg)	4.606e+004	4.606e+004		
39	Phase Fraction [Vol. Basis]	0.7142	0.7107		
40	Phase Fraction [Mass Basis]	0.5403	0.5337		
41	Phase Fraction [Act. Vol. Basis]	0.9296	0.9371		
42	Mass Exergy (kJ/kg)	320.2	306.4		
43	Partial Pressure of CO2 (bar)	0.8074	0.6873		
44	Cost Based on Flow (Cost/s)	0.0000	0.0000		
45	Act. Gas Flow (ACT_m3/h)	5587	6258		
46	Avg. Liq. Density (kgmole/m3)	14.32	14.32		
47	Specific Heat (kJ/kgmole-C)	75.38	73.53		
48	Std. Gas Flow (MMSCFD)	413.1	413.1		
49	Std. Ideal Liq. Mass Density (kg/m3)	434.7	434.7		
50	Act. Liq. Flow (m3/s)	0.1175	0.1167		
51	Z Factor	---	---		
52	Watson K	15.46	15.46		
53	User Property	---	---		
54	Partial Pressure of H2S (bar)	0.0000	0.0000		
55	Cp/(Cp - R)	1.124	1.127		
56	Cp/Cv	1.247	1.245		
57	Heat of Vap. (kJ/kgmole)	3.581e+004	3.602e+004		
58	Kinematic Viscosity (cSt)	---	---		
59	Liq. Mass Density (Std. Cond) (kg/m3)	1.294	1.294		
60	Liq. Vol. Flow (Std. Cond) (m3/h)	4.834e+005	4.834e+005		
61	Liquid Fraction	0.1185	0.1201		
62	Molar Volume (m3/kgmole)	0.2916	0.3240		
63	Mass Heat of Vap. (kJ/kg)	1180	1187		
64	Phase Fraction [Molar Basis]	0.8815	0.8799		
65	Surface Tension (dyne/cm)	---	---		
66	Thermal Conductivity (W/m-K)	---	---		
67	Viscosity (lbm/s-ft)	---	---		
68	Cv (Semi-Ideal) (kJ/kgmole-C)	67.06	65.21		
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 1 of 937				

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROPERTIES					
10						
11	Name	To Elep-Akabel Pipeline	Slug Catcher Tie-In			
12	Mass Cv (Semi-Ideal) (kJ/kg-C)	2.209	2.148			
13	Cv (kJ/kgmole-C)	60.45	59.08			
14	Mass Cv (kJ/kg-C)	1.992	1.946			
15	Cv (Ent. Method) (kJ/kgmole-C)	---	---			
16	Mass Cv (Ent. Method) (kJ/kg-C)	---	---			
17	Cp/Cv (Ent. Method)	---	---			
18	Reid VP at 37.8 C (bar)	---	---			
19	True VP at 37.8 C (bar)	---	---			
20	Liq. Vol. Flow - Sum(Std. Cond) (m3/h)	4.287e+005	4.279e+005			
21	Viscosity Index	---	---			
22	Ideal Gas Cp/Cv	1.157	1.162			
23	Ideal Gas Cp (kJ/kgmole-C)	61.31	59.73			
24	Mass Ideal Gas Cp (kJ/kg-C)	2.020	1.968			
25	Bubble Point Pressure (bar)	---	---			
26						
27	PROFILES					
28						
29	Data					
30	Distance	Elevation		Increments		
31	(m)	(m)				
32	0.0000	0.0000		5		
33	25.00	-0.1600		5		
34	50.00	4.000e-002		5		
35	75.00	-1.000e-002		5		
36	100.0	-8.000e-002		5		
37	125.0	1.000e-002		5		
38	150.0	-7.000e-002		5		
39	175.0	-0.1000		5		
40	200.1	-0.1600		5		
41	225.1	-0.1000		5		
42	250.1	-0.1600		5		
43	275.1	-0.2000		5		
44	300.1	-0.2100		5		
45	325.1	-0.2100		5		
46	350.1	-0.5700		5		
47	375.1	-0.2200		5		
48	400.0	0.2000		5		
49	425.0	8.000e-002		5		
50	450.0	3.000e-002		5		
51	475.0	6.000e-002		5		
52	500.0	6.000e-002		5		
53	525.0	5.000e-002		5		
54	550.0	6.000e-002		5		
55	575.0	8.000e-002		5		
56	600.0	7.000e-002		5		
57	625.0	1.000e-001		5		
58	650.0	0.1300		5		
59	675.0	0.1500		5		
60	710.0	0.1700		5		
61	720.0	9.000e-002		5		
62	730.0	0.2200		5		
63	740.0	0.1500		5		
64	750.0	1.000e-001		5		
65	775.0	8.000e-002		5		
66	800.1	1.000e-001		5		
67	825.1	1.000e-001		5		
68	850.1	-0.2700		5		
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 2 of 937					

1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	875.1	5.000e-002	5	
16	900.1	1.000e-002	5	
17	925.1	-2.000e-002	5	
18	950.1	1.000e-002	5	
19	975.1	1.000e-002	5	
20	1000	-1.232e-016	5	
21	1025	0.1800	5	
22	1050	-0.2300	5	
23	1075	-5.000e-002	5	
24	1100	2.000e-002	5	
25	1125	7.000e-002	5	
26	1150	0.1400	5	
27	1175	1.000e-001	5	
28	1200	0.1500	5	
29	1225	7.000e-002	5	
30	1250	-9.714e-017	5	
31	1275	-9.714e-017	5	
32	1300	3.000e-002	5	
33	1325	9.000e-002	5	
34	1350	0.2200	5	
35	1375	-0.3700	5	
36	1400	-3.600	5	
37	1425	-5.300	5	
38	1450	-5.900	5	
39	1475	-5.300	5	
40	1500	-6.300	5	
41	1525	-11.60	5	
42	1550	0.4400	5	
43	1575	1.150	5	
44	1600	0.8400	5	
45	1625	0.5100	5	
46	1650	0.6200	5	
47	1675	0.5900	5	
48	1700	0.2200	5	
49	1725	0.1600	5	
50	1750	0.1700	5	
51	1775	0.2400	5	
52	1800	0.3400	5	
53	1825	0.4800	5	
54	1850	0.5700	5	
55	1875	0.6300	5	
56	1900	0.6700	5	
57	1925	0.6700	5	
58	1950	0.7000	5	
59	1975	0.7000	5	
60	2000	0.5200	5	
61	2025	0.5200	5	
62	2050	0.6800	5	
63	2075	0.6300	5	
64	2100	0.6500	5	
65	2125	0.7000	5	
66	2150	0.7300	5	
67	2175	0.6800	5	
68	2201	0.6700	5	
69	Aspen Technology Inc.			Aspen HYSYS Version 10
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	2226	0.6600	5	
16	2251	0.5900	5	
17	2276	0.6300	5	
18	2301	0.6800	5	
19	2326	0.6500	5	
20	2351	0.6200	5	
21	2376	0.6100	5	
22	2401	0.6600	5	
23	2426	0.7500	5	
24	2451	0.6600	5	
25	2476	0.7100	5	
26	2501	0.7600	5	
27	2526	0.6500	5	
28	2551	0.4700	5	
29	2576	0.5100	5	
30	2601	0.5500	5	
31	2626	0.5400	5	
32	2651	0.5800	5	
33	2676	0.5800	5	
34	2701	0.6000	5	
35	2726	0.6000	5	
36	2751	0.6000	5	
37	2776	0.6500	5	
38	2801	0.4800	5	
39	2826	0.5700	5	
40	2851	0.6500	5	
41	2876	0.7200	5	
42	2901	0.4000	5	
43	2926	0.2900	5	
44	2951	0.6500	5	
45	2976	-0.7100	5	
46	3001	-0.6000	5	
47	3026	-8.000	5	
48	3051	-10.40	5	
49	3076	-11.60	5	
50	3101	2.000e-002	5	
51	3126	0.5200	5	
52	3151	0.3200	5	
53	3176	0.2100	5	
54	3202	0.4800	5	
55	3227	0.4100	5	
56	3252	0.1700	5	
57	3277	0.1600	5	
58	3302	0.1500	5	
59	3327	0.2300	5	
60	3352	0.3200	5	
61	3377	0.3700	5	
62	3401	0.3900	5	
63	3426	0.4000	5	
64	3451	0.3800	5	
65	3476	0.3800	5	
66	3501	0.3200	5	
67	3526	0.3200	5	
68	3551	0.2800	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	3576	0.1300	5	
16	3601	0.1600	5	
17	3626	0.1900	5	
18	3651	0.3100	5	
19	3676	0.2400	5	
20	3701	0.1300	5	
21	3726	0.1600	5	
22	3751	0.2000	5	
23	3776	0.1200	5	
24	3802	7.000e-002	5	
25	3827	6.000e-002	5	
26	3852	0.1600	5	
27	3877	0.1900	5	
28	3902	0.2600	5	
29	3927	0.3000	5	
30	3952	0.3600	5	
31	3977	0.3800	5	
32	4002	0.3800	5	
33	4027	0.2800	5	
34	4052	0.2600	5	
35	4077	0.1600	5	
36	4112	0.1500	5	
37	4150	0.1300	5	
38	4188	5.000e-002	5	
39	4226	-0.1300	5	
40	4264	0.1100	5	
41	4302	0.1900	5	
42	4309	0.1200	5	
43	4334	-0.1300	5	
44	4359	-0.2700	5	
45	4384	-8.000e-002	5	
46	4409	-3.000e-002	5	
47	4434	0.1600	5	
48	4459	0.1300	5	
49	4484	0.1900	5	
50	4510	0.1600	5	
51	4535	0.1800	5	
52	4560	0.1300	5	
53	4585	0.1600	5	
54	4610	0.2700	5	
55	4675	0.1200	5	
56	4685	0.1600	5	
57	4710	0.2600	5	
58	4735	0.2700	5	
59	4760	0.2800	5	
60	4785	0.2700	5	
61	4810	0.2200	5	
62	4835	0.1900	5	
63	4860	0.1400	5	
64	4885	0.1300	5	
65	4910	0.1300	5	
66	4935	2.000e-002	5	
67	4960	-2.000e-002	5	
68	4985	-2.217e-015	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	Page 5 of 937

1	Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	5010	7.000e-002	5	
16	5035	-2.220e-015	5	
17	5060	0.1300	5	
18	5085	1.000e-001	5	
19	5110	0.1900	5	
20	5135	0.1700	5	
21	5160	0.1400	5	
22	5185	0.1800	5	
23	5210	0.1200	5	
24	5235	9.000e-002	5	
25	5260	8.000e-002	5	
26	5285	2.000e-002	5	
27	5310	-7.000e-002	5	
28	5335	-0.2000	5	
29	5360	-0.3200	5	
30	5385	-0.2600	5	
31	5410	-0.6900	5	
32	5435	-0.3800	5	
33	5460	-2.000e-002	5	
34	5485	-5.000e-002	5	
35	5510	-0.1400	5	
36	5535	-0.2000	5	
37	5570	-0.3200	5	
38	5580	-0.4600	5	
39	5605	-0.3900	5	
40	5630	-0.4300	5	
41	5655	-0.7200	5	
42	5680	-0.7100	5	
43	5705	-0.2900	5	
44	5730	-0.3900	5	
45	5755	-0.2500	5	
46	5780	-5.000e-002	5	
47	5805	-0.3500	5	
48	5830	-0.3900	5	
49	5855	1.000e-001	5	
50	5880	0.2400	5	
51	5905	-7.800	5	
52	5930	-16.10	5	
53	5955	-4.300	5	
54	5980	-0.3100	5	
55	6005	-14.00	5	
56	6030	-0.3600	5	
57	6055	-3.000e-002	5	
58	6080	-0.1400	5	
59	6106	-10.60	5	
60	6131	-0.3100	5	
61	6156	-0.2000	5	
62	6181	-0.1700	5	
63	6206	-7.000e-002	5	
64	6231	-6.000e-002	5	
65	6256	-0.1800	5	
66	6281	-0.1400	5	
67	6303	-10.90	5	
68	6328	-6.800	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	6353	-10.60	5	
16	6378	-17.30	5	
17	6403	-13.80	5	
18	6428	-0.5700	5	
19	6453	-0.1400	5	
20	6483	-0.1000	5	
21	6503	-9.000e-002	5	
22	6528	4.000e-002	5	
23	6553	3.000e-002	5	
24	6578	2.000e-002	5	
25	6603	8.000e-002	5	
26	6628	4.000e-002	5	
27	6653	-2.000e-002	5	
28	6678	7.000e-002	5	
29	6703	-8.000e-002	5	
30	6728	-0.2500	5	
31	6754	-13.00	5	
32	6779	-8.100	5	
33	6804	-6.300	5	
34	6829	-0.2500	5	
35	6854	-17.80	5	
36	6879	-0.3500	5	
37	6904	-1.000e-002	5	
38	6929	-4.000e-002	5	
39	6954	-8.000e-002	5	
40	6979	-4.000e-002	5	
41	7004	-1.402e-015	5	
42	7029	5.000e-002	5	
43	7054	-0.1200	5	
44	7079	5.000e-002	5	
45	7089	3.000e-002	5	
46	7129	-1.398e-015	5	
47	7154	2.000e-002	5	
48	7179	7.000e-002	5	
49	7204	0.1700	5	
50	7229	8.000e-002	5	
51	7254	6.000e-002	5	
52	7279	0.1600	5	
53	7304	3.000e-002	5	
54	7329	7.000e-002	5	
55	7354	0.1600	5	
56	7379	0.2000	5	
57	7404	0.2200	5	
58	7429	0.2300	5	
59	7454	0.2400	5	
60	7479	0.2100	5	
61	7504	0.1200	5	
62	7529	0.1800	5	
63	7553	0.1200	5	
64	7578	0.1800	5	
65	7603	0.1800	5	
66	7628	0.1100	5	
67	7653	0.1400	5	
68	7678	7.000e-002	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	Page 7 of 937

1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	7703	0.1600	5	
16	7728	-0.3700	5	
17	7753	-0.4300	5	
18	7778	7.000e-002	5	
19	7803	0.2000	5	
20	7828	0.2000	5	
21	7853	0.2200	5	
22	7878	0.2600	5	
23	7903	0.2400	5	
24	7928	0.2300	5	
25	7953	0.3400	5	
26	7978	0.1500	5	
27	8003	8.000e-002	5	
28	8028	1.000e-002	5	
29	8053	-0.3100	5	
30	8078	-4.600	5	
31	8103	-0.6800	5	
32	8128	0.2000	5	
33	8153	0.2500	5	
34	8178	0.2400	5	
35	8203	0.2300	5	
36	8228	0.3000	5	
37	8253	0.3000	5	
38	8278	0.3400	5	
39	8303	0.3100	5	
40	8328	0.3700	5	
41	8353	6.000e-002	5	
42	8378	6.000e-002	5	
43	8403	-4.000e-002	5	
44	8428	-0.1600	5	
45	8453	-0.2000	5	
46	8478	-2.000e-002	5	
47	8503	-1.000e-002	5	
48	8528	-1.000e-002	5	
49	8553	-3.000e-002	5	
50	8578	-2.000e-002	5	
51	8603	-3.000e-002	5	
52	8628	-1.000e-002	5	
53	8653	-1.518e-015	5	
54	8678	1.000e-002	5	
55	8703	1.000e-002	5	
56	8728	2.000e-002	5	
57	8753	-1.516e-015	5	
58	8778	-3.000e-002	5	
59	8803	-7.000e-002	5	
60	8813	-2.000e-002	5	
61	8948	-5.000e-002	5	
62	8953	-3.000e-002	5	
63	8978	-1.000e-002	5	
64	9003	8.000e-002	5	
65	9028	-1.513e-015	5	
66	9053	-4.000e-002	5	
67	9058	-8.000e-002	5	
68	9091	-0.1400	5	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	9125	-0.2500	5	
16	9158	-0.3300	5	
17	9178	-0.3000	5	
18	9203	-0.4100	5	
19	9228	-0.4400	5	
20	9253	-0.4500	5	
21	9278	-0.2000	5	
22	9303	-0.1500	5	
23	9328	-0.1600	5	
24	9353	-3.000e-002	5	
25	9378	1.000e-002	5	
26	9403	-0.1500	5	
27	9428	-0.4400	5	
28	9453	-0.6100	5	
29	9468	-9.000e-002	5	
30	9491	-9.000e-002	5	
31	9515	-0.5500	5	
32	9538	-0.4500	5	
33	9561	-0.2800	5	
34	9585	-9.000e-002	5	
35	9608	-2.000e-002	5	
36	9628	0.2300	5	
37	9653	-1.582e-015	5	
38	9678	-7.000e-002	5	
39	9703	0.1700	5	
40	9728	0.1800	5	
41	9753	0.1700	5	
42	9778	0.1500	5	
43	9803	0.1800	5	
44	9828	0.1100	5	
45	9853	0.1800	5	
46	9878	0.1400	5	
47	9903	0.1600	5	
48	9913	0.1400	5	
49	1.001e+004	1.000e-001	5	
50	1.003e+004	-0.5000	5	
51	1.005e+004	-0.4700	5	
52	1.008e+004	-4.000e-002	5	
53	1.010e+004	0.1600	5	
54	1.013e+004	0.1800	5	
55	1.015e+004	0.2100	5	
56	1.018e+004	0.2200	5	
57	1.020e+004	0.1900	5	
58	1.023e+004	0.2200	5	
59	1.025e+004	0.1800	5	
60	1.028e+004	-0.2200	5	
61	1.030e+004	0.1800	5	
62	1.033e+004	1.000e-001	5	
63	1.035e+004	3.000e-002	5	
64	1.038e+004	2.000e-002	5	
65	1.040e+004	-0.1200	5	
66	1.043e+004	-1.554e-015	5	
67	1.045e+004	-0.2900	5	
68	1.048e+004	-0.4800	5	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	1.050e+004	-0.5800		5
16	1.053e+004	-8.000e-002		5
17	1.055e+004	-6.000e-002		5
18	1.058e+004	-1.513e-015		5
19	1.060e+004	9.000e-002		5
20	1.063e+004	0.1600		5
21	1.065e+004	0.1800		5
22	1.068e+004	0.2000		5
23	1.070e+004	0.2500		5
24	1.073e+004	0.2600		5
25	1.075e+004	0.2900		5
26	1.078e+004	0.3100		5
27	1.080e+004	0.2900		5
28	1.083e+004	0.3100		5
29	1.085e+004	0.3300		5
30	1.088e+004	0.3200		5
31	1.090e+004	0.3400		5
32	1.093e+004	0.3600		5
33	1.095e+004	0.4100		5
34	1.098e+004	0.3600		5
35	1.100e+004	0.3600		5
36	1.103e+004	0.3400		5
37	1.105e+004	0.3300		5
38	1.108e+004	0.3400		5
39	1.110e+004	0.3500		5
40	1.113e+004	0.3200		5
41	1.115e+004	0.3400		5
42	1.118e+004	0.3100		5
43	1.120e+004	0.3000		5
44	1.123e+004	0.3000		5
45	1.125e+004	0.3300		5
46	1.128e+004	0.3200		5
47	1.130e+004	0.3200		5
48	1.133e+004	0.1300		5
49	1.135e+004	0.1700		5
50	1.138e+004	0.2000		5
51	1.140e+004	1.000e-001		5
52	1.143e+004	-0.5100		5
53	1.145e+004	-0.3200		5
54	1.148e+004	0.2400		5
55	1.150e+004	0.2800		5
56	1.153e+004	0.3000		5
57	1.155e+004	0.3400		5
58	1.158e+004	0.3300		5
59	1.160e+004	0.3000		5
60	1.163e+004	0.2900		5
61	1.165e+004	0.2200		5
62	1.168e+004	0.2600		5
63	1.170e+004	0.3200		5
64	1.173e+004	0.3200		5
65	1.175e+004	0.3300		5
66	1.178e+004	0.3300		5
67	1.180e+004	0.3000		5
68	1.183e+004	0.2800		5
69	Aspen Technology Inc.		Aspen HYSYS Version 10	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	1.185e+004	0.3000	5	
16	1.188e+004	0.2900	5	
17	1.190e+004	0.3100	5	
18	1.193e+004	0.3000	5	
19	1.195e+004	0.3000	5	
20	1.198e+004	0.2900	5	
21	1.202e+004	0.2900	5	
22	1.205e+004	0.2500	5	
23	1.209e+004	0.8000	5	
24	1.212e+004	0.3000	5	
25	1.216e+004	0.3000	5	
26	1.219e+004	0.3000	5	
27	1.223e+004	0.2800	5	
28	1.226e+004	0.2500	5	
29	1.228e+004	0.1100	5	
30	1.231e+004	0.2200	5	
31	1.233e+004	0.2000	5	
32	1.236e+004	0.2000	5	
33	1.239e+004	0.2100	5	
34	1.242e+004	0.2800	5	
35	1.244e+004	0.2200	5	
36	1.247e+004	0.2800	5	
37	1.249e+004	0.2100	5	
38	1.252e+004	-0.2300	5	
39	1.254e+004	-0.1700	5	
40	1.257e+004	0.1600	5	
41	1.259e+004	0.2300	5	
42	1.262e+004	0.4400	5	
43	1.265e+004	0.6200	5	
44	1.268e+004	0.6000	5	
45	1.270e+004	0.2800	5	
46	1.273e+004	0.2600	5	
47	1.275e+004	0.2500	5	
48	1.277e+004	3.000e-002	5	
49	1.279e+004	1.000e-002	5	
50	1.280e+004	0.2000	5	
51	1.283e+004	-0.1600	5	
52	1.285e+004	0.1800	5	
53	1.288e+004	0.2000	5	
54	1.290e+004	0.2200	5	
55	1.293e+004	0.2400	5	
56	1.295e+004	0.2500	5	
57	1.298e+004	0.2200	5	
58	1.300e+004	0.2700	5	
59	1.303e+004	0.2800	5	
60	1.305e+004	0.2900	5	
61	1.308e+004	0.3000	5	
62	1.310e+004	0.2700	5	
63	1.313e+004	0.3300	5	
64	1.315e+004	0.3400	5	
65	1.318e+004	0.3500	5	
66	1.320e+004	0.3400	5	
67	1.323e+004	-0.8100	5	
68	1.325e+004	-1.440	5	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	1.328e+004	-5.200	5	
16	1.330e+004	-6.300	5	
17	1.333e+004	-7.800	5	
18	1.335e+004	-11.30	5	
19	1.338e+004	-13.55	5	
20	1.340e+004	-13.55	5	
21	1.343e+004	-11.30	5	
22	1.345e+004	-9.650	5	
23	1.348e+004	-0.6500	5	
24	1.350e+004	0.1800	5	
25	1.353e+004	0.2500	5	
26	1.354e+004	0.2500	5	
27	1.355e+004	0.2600	5	
28	1.357e+004	0.2700	5	
29	1.359e+004	0.2800	5	
30	1.360e+004	0.2600	5	
31	1.363e+004	0.2700	5	
32	1.365e+004	0.3100	5	
33	1.368e+004	0.2900	5	
34	1.370e+004	0.3000	5	
35	1.373e+004	0.2000	5	
36	1.375e+004	9.000e-002	5	
37	1.378e+004	-7.000e-002	5	
38	1.380e+004	-2.000e-002	5	
39	1.383e+004	0.2600	5	
40	1.385e+004	-0.3600	5	
41	1.388e+004	-0.2000	5	
42	1.390e+004	-8.000e-002	5	
43	1.393e+004	-4.000e-002	5	
44	1.395e+004	-0.1100	5	
45	1.398e+004	8.000e-002	5	
46	1.400e+004	1.000e-001	5	
47	1.403e+004	-0.2100	5	
48	1.405e+004	1.000e-001	5	
49	1.406e+004	0.1400	5	
50	1.408e+004	9.000e-002	5	
51	1.409e+004	-5.000e-002	5	
52	1.412e+004	-0.3800	5	
53	1.415e+004	-0.4700	5	
54	1.418e+004	-4.000e-002	5	
55	1.420e+004	-0.1000	5	
56	1.423e+004	2.000e-002	5	
57	1.425e+004	0.1500	5	
58	1.428e+004	0.1700	5	
59	1.430e+004	0.1900	5	
60	1.433e+004	0.2000	5	
61	1.435e+004	0.1100	5	
62	1.438e+004	0.1400	5	
63	1.440e+004	0.1400	5	
64	1.443e+004	0.2000	5	
65	1.445e+004	0.2000	5	
66	1.448e+004	0.2000	5	
67	1.450e+004	0.2000	5	
68	1.453e+004	0.2100	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	1.455e+004	0.2000	5	
16	1.458e+004	0.1700	5	
17	1.460e+004	0.1600	5	
18	1.463e+004	0.1700	5	
19	1.465e+004	0.1400	5	
20	1.468e+004	0.1800	5	
21	1.470e+004	0.1100	5	
22	1.473e+004	0.1200	5	
23	1.475e+004	9.000e-002	5	
24	1.478e+004	7.000e-002	5	
25	1.480e+004	7.000e-002	5	
26	1.483e+004	5.000e-002	5	
27	1.485e+004	3.000e-002	5	
28	1.486e+004	2.000e-002	5	
29	1.496e+004	3.000e-002	5	
30	1.506e+004	-4.000e-002	5	
31	1.516e+004	-1.000e-002	5	
32	1.526e+004	-6.000e-002	5	
33	1.536e+004	-0.3800	5	
34	1.546e+004	-3.550	5	
35	1.556e+004	-0.3700	5	
36	1.566e+004	2.000e-002	5	
37	1.576e+004	-9.000e-002	5	
38	1.579e+004	-9.000e-002	5	
39	1.589e+004	2.000e-002	5	
40	1.591e+004	1.000e-001	5	
41	1.593e+004	0.1100	5	
42	1.594e+004	0.1700	5	
43	1.596e+004	0.1800	5	
44	1.598e+004	0.2000	5	
45	1.601e+004	0.2300	5	
46	1.603e+004	0.1400	5	
47	1.606e+004	7.000e-002	5	
48	1.608e+004	-2.000e-002	5	
49	1.611e+004	-9.000e-002	5	
50	1.613e+004	-0.1700	5	
51	1.616e+004	-0.2800	5	
52	1.618e+004	-0.3400	5	
53	1.621e+004	-0.4600	5	
54	1.623e+004	0.1600	5	
55	1.626e+004	0.1400	5	
56	1.628e+004	0.1200	5	
57	1.631e+004	5.000e-002	5	
58	1.632e+004	-0.4100	5	
59	1.633e+004	-1.110e-016	5	
60	1.636e+004	4.000e-002	5	
61	1.638e+004	7.000e-002	5	
62	1.641e+004	1.000e-001	5	
63	1.643e+004	0.1400	5	
64	1.646e+004	1.000e-001	5	
65	1.648e+004	9.000e-002	5	
66	1.651e+004	7.000e-002	5	
67	1.653e+004	0.1200	5	
68	1.656e+004	0.1200	5	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	1.658e+004	0.1500	5	
16	1.661e+004	1.000e-002	5	
17	1.663e+004	2.000e-002	5	
18	1.666e+004	-6.000e-002	5	
19	1.668e+004	-2.000e-002	5	
20	1.671e+004	-1.006e-016	5	
21	1.673e+004	-2.000e-002	5	
22	1.676e+004	-7.000e-002	5	
23	1.678e+004	-0.1400	5	
24	1.681e+004	-0.1000	5	
25	1.683e+004	-0.1300	5	
26	1.686e+004	-0.1700	5	
27	1.688e+004	-0.1700	5	
28	1.690e+004	-0.2600	5	
29	1.693e+004	-0.3500	5	
30	1.695e+004	-0.3600	5	
31	1.698e+004	-0.3500	5	
32	1.700e+004	-0.2400	5	
33	1.703e+004	-0.2700	5	
34	1.705e+004	-0.3400	5	
35	1.708e+004	-0.2500	5	
36	1.710e+004	-0.2100	5	
37	1.713e+004	-0.1500	5	
38	1.715e+004	-8.000e-002	5	
39	1.718e+004	-3.000e-002	5	
40	1.720e+004	-1.000e-002	5	
41	1.723e+004	6.000e-002	5	
42	1.725e+004	8.000e-002	5	
43	1.728e+004	0.1200	5	
44	1.730e+004	0.1200	5	
45	1.733e+004	0.1400	5	
46	1.735e+004	0.1400	5	
47	1.738e+004	0.1600	5	
48	1.740e+004	0.1600	5	
49	1.743e+004	0.1400	5	
50	1.745e+004	0.1100	5	
51	1.748e+004	0.1100	5	
52	1.750e+004	5.000e-002	5	
53	1.753e+004	1.000e-001	5	
54	1.755e+004	-9.714e-017	5	
55	1.758e+004	-0.1700	5	
56	1.760e+004	9.000e-002	5	
57	1.763e+004	1.000e-002	5	
58	1.765e+004	-1.000e-002	5	
59	1.768e+004	-5.000e-002	5	
60	1.770e+004	-5.000e-002	5	
61	1.773e+004	-6.000e-002	5	
62	1.775e+004	-0.1800	5	
63	1.778e+004	-0.1800	5	
64	1.780e+004	-0.3800	5	
65	1.783e+004	-0.1700	5	
66	1.785e+004	-0.8700	5	
67	1.787e+004	-0.8700	5	
68	1.788e+004	-1.300	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	Page 14 of 937

1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	1.790e+004	-4.300	5	
16	1.791e+004	-5.800	5	
17	1.793e+004	-6.800	5	
18	1.795e+004	-8.300	5	
19	1.798e+004	-9.800	5	
20	1.800e+004	-12.30	5	
21	1.803e+004	-14.80	5	
22	1.805e+004	-13.30	5	
23	1.808e+004	-0.7000	5	
24	1.810e+004	-0.5000	5	
25	1.813e+004	-0.3000	5	
26	1.815e+004	-0.2600	5	
27	1.818e+004	-0.2800	5	
28	1.820e+004	-0.2400	5	
29	1.823e+004	-0.2100	5	
30	1.825e+004	-0.2200	5	
31	1.828e+004	-0.2300	5	
32	1.830e+004	-0.2500	5	
33	1.833e+004	-0.3000	5	
34	1.835e+004	-0.2800	5	
35	1.838e+004	-0.3000	5	
36	1.840e+004	-0.2800	5	
37	1.843e+004	-0.2700	5	
38	1.845e+004	-0.3500	5	
39	1.848e+004	-0.3000	5	
40	1.850e+004	-0.2900	5	
41	1.853e+004	-0.2600	5	
42	1.855e+004	-0.3400	5	
43	1.858e+004	-0.3300	5	
44	1.860e+004	-0.3800	5	
45	1.863e+004	-0.4800	5	
46	1.865e+004	-0.5400	5	
47	1.868e+004	-0.5200	5	
48	1.870e+004	-0.8000	5	
49	1.873e+004	-0.5500	5	
50	1.875e+004	-0.4800	5	
51	1.878e+004	-0.5900	5	
52	1.880e+004	-0.6800	5	
53	1.882e+004	-0.5400	5	
54	1.883e+004	-0.8600	5	
55	1.885e+004	-0.3700	5	
56	1.888e+004	-0.6800	5	
57	1.890e+004	-0.5500	5	
58	1.891e+004	-0.6000	5	
59	1.893e+004	-0.5600	5	
60	1.895e+004	-0.7000	5	
61	1.898e+004	-0.9800	5	
62	1.900e+004	-0.7100	5	
63	1.903e+004	-0.6100	5	
64	1.905e+004	-0.5400	5	
65	1.908e+004	-0.5700	5	
66	1.910e+004	-0.4500	5	
67	1.913e+004	-0.4400	5	
68	1.915e+004	-0.4500	5	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	1.918e+004	-0.4800	5	
16	1.920e+004	-0.5000	5	
17	1.923e+004	-0.5900	5	
18	1.925e+004	-0.4800	5	
19	1.928e+004	-0.5500	5	
20	1.930e+004	-0.7000	5	
21	1.933e+004	-0.7100	5	
22	1.935e+004	-0.6500	5	
23	1.938e+004	-4.600	5	
24	1.940e+004	-3.550	5	
25	1.943e+004	-1.090	5	
26	1.945e+004	-1.120	5	
27	1.947e+004	-0.7500	5	
28	1.950e+004	-0.6200	5	
29	1.961e+004	-0.4800	5	
30	1.973e+004	-0.4700	5	
31	1.985e+004	-0.4500	5	
32	1.996e+004	-0.4400	5	
33	2.008e+004	-0.4300	5	
34	2.020e+004	-0.7000	5	
35	2.031e+004	-0.3900	5	
36	2.043e+004	-0.4000	5	
37	2.055e+004	-0.3800	5	
38	2.069e+004	-0.4900	5	
39	2.070e+004	-0.5400	5	
40	2.073e+004	-0.5200	5	
41	2.075e+004	-0.4900	5	
42	2.078e+004	-0.4900	5	
43	2.080e+004	-0.4900	5	
44	2.083e+004	-0.5400	5	
45	2.085e+004	-0.5700	5	
46	2.088e+004	-0.5100	5	
47	2.090e+004	-0.6900	5	
48	2.093e+004	-0.6100	5	
49	2.095e+004	-0.5400	5	
50	2.098e+004	-0.5900	5	
51	2.100e+004	-0.7200	5	
52	2.103e+004	-0.7000	5	
53	2.105e+004	-0.8000	5	
54	2.108e+004	-0.6300	5	
55	2.110e+004	-0.7700	5	
56	2.112e+004	-1.230	5	
57	2.114e+004	-0.9300	5	
58	2.115e+004	-0.7900	5	
59	2.118e+004	-0.5700	5	
60	2.120e+004	-0.4900	5	
61	2.123e+004	-0.5000	5	
62	2.125e+004	-0.4100	5	
63	2.128e+004	-0.5100	5	
64	2.130e+004	-0.6100	5	
65	2.133e+004	-0.6500	5	
66	2.134e+004	-0.6600	5	
67	2.135e+004	-0.5100	5	
68	2.138e+004	-0.4900	5	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	2.140e+004	-0.4800	5	
16	2.143e+004	-0.2400	5	
17	2.145e+004	-0.2800	5	
18	2.148e+004	-0.2600	5	
19	2.150e+004	-0.2700	5	
20	2.152e+004	-0.3400	5	
21	2.153e+004	-0.4700	5	
22	2.155e+004	-0.2800	5	
23	2.158e+004	-4.000e-002	5	
24	2.160e+004	-0.5700	5	
25	2.163e+004	-2.920	5	
26	2.165e+004	-5.420	5	
27	2.168e+004	-7.300	5	
28	2.170e+004	-7.300	5	
29	2.173e+004	-6.800	5	
30	2.175e+004	-6.300	5	
31	2.178e+004	-5.800	5	
32	2.180e+004	-0.3600	5	
33	2.183e+004	-0.1800	5	
34	2.185e+004	-0.1700	5	
35	2.188e+004	-0.1400	5	
36	2.190e+004	-9.000e-002	5	
37	2.193e+004	-9.000e-002	5	
38	2.195e+004	-9.000e-002	5	
39	2.198e+004	-8.000e-002	5	
40	2.200e+004	-7.000e-002	5	
41	2.203e+004	-9.000e-002	5	
42	2.205e+004	-0.1000	5	
43	2.208e+004	-0.2300	5	
44	2.210e+004	-0.3600	5	
45	2.213e+004	-0.3600	5	
46	2.215e+004	-0.3300	5	
47	2.218e+004	-0.1900	5	
48	2.220e+004	-0.3000	5	
49	2.223e+004	-0.7500	5	
50	2.225e+004	-0.3200	5	
51	2.228e+004	-0.1600	5	
52	2.233e+004	-0.1200	5	
53	2.345e+004	-0.1000	5	
54	2.349e+004	-0.1200	5	
55	2.352e+004	-0.1100	5	
56	2.354e+004	-0.1300	5	
57	2.357e+004	-0.1300	5	
58	2.359e+004	0.6800	5	
59	2.362e+004	-0.1600	5	
60	2.364e+004	-0.2100	5	
61	2.367e+004	-0.1000	5	
62	2.369e+004	-0.1500	5	
63	2.372e+004	-0.1900	5	
64	2.374e+004	-0.2500	5	
65	2.377e+004	-0.4500	5	
66	2.379e+004	-0.4100	5	
67	2.382e+004	-0.3800	5	
68	2.384e+004	-0.1700	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	Page 17 of 937

1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	2.387e+004	-0.2000	5	
16	2.389e+004	-0.6700	5	
17	2.392e+004	-0.2400	5	
18	2.394e+004	-0.1300	5	
19	2.397e+004	-0.1200	5	
20	2.399e+004	-0.1600	5	
21	2.402e+004	-0.3200	5	
22	2.404e+004	-0.5000	5	
23	2.407e+004	-0.2700	5	
24	2.409e+004	-0.2100	5	
25	2.412e+004	-0.1500	5	
26	2.414e+004	-0.1000	5	
27	2.417e+004	-9.000e-002	5	
28	2.419e+004	-0.1300	5	
29	2.422e+004	-0.1700	5	
30	2.424e+004	-0.1100	5	
31	2.427e+004	-0.2200	5	
32	2.429e+004	-0.1400	5	
33	2.432e+004	-9.000e-002	5	
34	2.434e+004	-0.1200	5	
35	2.437e+004	-0.1500	5	
36	2.438e+004	-0.1100	5	
37	2.439e+004	-0.1300	5	
38	2.442e+004	-0.1300	5	
39	2.444e+004	-0.1600	5	
40	2.447e+004	-0.2400	5	
41	2.449e+004	-0.2900	5	
42	2.452e+004	-0.2000	5	
43	2.454e+004	-0.2600	5	
44	2.457e+004	-0.2200	5	
45	2.459e+004	-0.1800	5	
46	2.462e+004	-0.1400	5	
47	2.464e+004	-9.000e-002	5	
48	2.467e+004	-0.2200	5	
49	2.469e+004	-0.5800	5	
50	2.472e+004	-0.2300	5	
51	2.474e+004	-7.000e-002	5	
52	2.477e+004	-7.000e-002	5	
53	2.479e+004	-0.1000	5	
54	2.482e+004	-0.1700	5	
55	2.484e+004	-0.3500	5	
56	2.487e+004	-0.7300	5	
57	2.489e+004	-0.7300	5	
58	2.492e+004	-0.1500	5	
59	2.494e+004	-7.000e-002	5	
60	2.497e+004	-0.1000	5	
61	2.499e+004	-0.1300	5	
62	2.502e+004	-0.1400	5	
63	2.504e+004	-0.1600	5	
64	2.507e+004	-0.2100	5	
65	2.509e+004	-0.1300	5	
66	2.512e+004	-0.1700	5	
67	2.514e+004	-0.2000	5	
68	2.517e+004	-0.2900	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	2.519e+004	-0.4200	5	
16	2.522e+004	-0.5200	5	
17	2.525e+004	-3.900	5	
18	2.527e+004	-5.300	5	
19	2.529e+004	-3.900	5	
20	2.539e+004	-3.300	5	
21	2.540e+004	-3.100	5	
22	2.543e+004	-0.4400	5	
23	2.545e+004	-0.2400	5	
24	2.548e+004	-0.1200	5	
25	2.550e+004	-0.1300	5	
26	2.553e+004	-9.000e-002	5	
27	2.555e+004	-9.000e-002	5	
28	2.558e+004	-5.000e-002	5	
29	2.560e+004	-3.000e-002	5	
30	2.563e+004	-5.000e-002	5	
31	2.566e+004	-6.000e-002	5	
32	2.568e+004	-7.000e-002	5	
33	2.571e+004	-0.1000	5	
34	2.571e+004	-0.2000	5	
35	2.572e+004	-0.8000	5	
36	2.573e+004	-3.500	5	
37	2.576e+004	-4.800	5	
38	2.578e+004	-0.7000	5	
39	2.581e+004	-0.3000	5	
40	2.583e+004	-0.2200	5	
41	2.586e+004	-6.000e-002	5	
42	2.588e+004	-7.000e-002	5	
43	2.591e+004	-0.2700	5	
44	2.593e+004	-0.6200	5	
45	2.596e+004	-0.7700	5	
46	2.598e+004	-0.1200	5	
47	2.601e+004	-0.1000	5	
48	2.603e+004	-0.2300	5	
49	2.606e+004	-4.000e-002	5	
50	2.608e+004	7.000e-002	5	
51	2.611e+004	-8.000e-002	5	
52	2.613e+004	-7.000e-002	5	
53	2.616e+004	-0.1400	5	
54	2.618e+004	-0.2000	5	
55	2.621e+004	-0.3800	5	
56	2.623e+004	-0.2000	5	
57	2.626e+004	-0.2300	5	
58	2.628e+004	-0.1000	5	
59	2.631e+004	-0.1700	5	
60	2.633e+004	-0.1700	5	
61	2.636e+004	-0.2400	5	
62	2.638e+004	-0.3100	5	
63	2.641e+004	-0.1800	5	
64	2.644e+004	-8.000e-002	5	
65	2.646e+004	-0.3700	5	
66	2.649e+004	-0.2900	5	
67	2.651e+004	-0.3900	5	
68	2.654e+004	-0.4100	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	Page 19 of 937

1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	2.656e+004	-0.3800	5	
16	2.659e+004	-0.4600	5	
17	2.661e+004	-0.4400	5	
18	2.664e+004	-0.5800	5	
19	2.666e+004	-0.5600	5	
20	2.669e+004	-0.4300	5	
21	2.671e+004	-0.2000	5	
22	2.674e+004	-0.1200	5	
23	2.676e+004	-3.000e-002	5	
24	2.679e+004	-1.000e-002	5	
25	2.681e+004	1.000e-002	5	
26	2.684e+004	-4.000e-002	5	
27	2.686e+004	3.000e-002	5	
28	2.689e+004	1.000e-002	5	
29	2.691e+004	-4.000e-002	5	
30	2.694e+004	-0.1400	5	
31	2.696e+004	-0.2600	5	
32	2.699e+004	-0.3600	5	
33	2.701e+004	-0.4500	5	
34	2.704e+004	-0.5500	5	
35	2.705e+004	-0.4400	5	
36	2.706e+004	-0.4900	5	
37	2.709e+004	-0.6500	5	
38	2.711e+004	-0.6000	5	
39	2.714e+004	-0.3200	5	
40	2.716e+004	-0.1300	5	
41	2.719e+004	-4.000e-002	5	
42	2.721e+004	4.000e-002	5	
43	2.724e+004	3.000e-002	5	
44	2.726e+004	8.000e-002	5	
45	2.729e+004	8.000e-002	5	
46	2.731e+004	0.1100	5	
47	2.734e+004	0.1100	5	
48	2.736e+004	0.1300	5	
49	2.739e+004	0.1500	5	
50	2.741e+004	0.1400	5	
51	2.744e+004	0.1500	5	
52	2.746e+004	0.1600	5	
53	2.749e+004	0.1600	5	
54	2.750e+004	0.1600	5	
55	2.753e+004	0.1500	5	
56	2.754e+004	0.1800	5	
57	2.756e+004	0.1800	5	
58	2.759e+004	0.8000	5	
59	2.761e+004	-0.3300	5	
60	2.764e+004	0.4500	5	
61	2.766e+004	0.7700	5	
62	2.769e+004	0.1800	5	
63	2.771e+004	0.1800	5	
64	2.774e+004	0.1900	5	
65	2.776e+004	0.1600	5	
66	2.779e+004	0.1700	5	
67	2.781e+004	0.1600	5	
68	2.784e+004	0.1500	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	2.786e+004	0.1600	5	
16	2.795e+004	0.1800	5	
17	2.804e+004	0.1500	5	
18	2.814e+004	0.1400	5	
19	2.823e+004	1.000e-001	5	
20	2.832e+004	8.000e-002	5	
21	2.841e+004	9.000e-002	5	
22	2.845e+004	8.000e-002	5	
23	2.847e+004	7.000e-002	5	
24	2.850e+004	7.000e-002	5	
25	2.852e+004	2.000e-002	5	
26	2.855e+004	5.000e-002	5	
27	2.857e+004	2.000e-002	5	
28	2.860e+004	-0.1600	5	
29	2.862e+004	-0.5000	5	
30	2.865e+004	-0.6800	5	
31	2.867e+004	-0.5300	5	
32	2.870e+004	-3.000e-002	5	
33	2.872e+004	-1.000e-002	5	
34	2.877e+004	2.000e-002	5	
35	2.883e+004	-1.000e-002	5	
36	2.888e+004	-2.000e-002	5	
37	2.893e+004	-9.000e-002	5	
38	2.898e+004	-0.1100	5	
39	2.903e+004	-0.1500	5	
40	2.905e+004	-0.1200	5	
41	2.907e+004	-0.1700	5	
42	2.910e+004	-0.2800	5	
43	2.912e+004	-0.3200	5	
44	2.915e+004	-0.1400	5	
45	2.917e+004	-8.000e-002	5	
46	2.920e+004	2.000e-002	5	
47	2.922e+004	-5.655e-016	5	
48	2.925e+004	3.000e-002	5	
49	2.926e+004	7.000e-002	5	
50	2.931e+004	0.1100	5	
51	2.936e+004	8.000e-002	5	
52	2.940e+004	9.000e-002	5	
53	2.945e+004	8.000e-002	5	
54	2.950e+004	2.000e-002	5	
55	2.955e+004	2.000e-002	5	
56	2.955e+004	3.000e-002	5	
57	2.958e+004	4.000e-002	5	
58	2.960e+004	6.000e-002	5	
59	2.963e+004	4.000e-002	5	
60	2.965e+004	3.000e-002	5	
61	2.966e+004	-5.621e-016	5	
62	2.967e+004	-3.000e-002	5	
63	2.969e+004	-4.000e-002	5	
64	2.971e+004	-3.000e-002	5	
65	2.973e+004	-3.000e-002	5	
66	2.975e+004	-4.000e-002	5	
67	2.978e+004	-5.000e-002	5	
68	2.980e+004	-3.000e-002	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	Page 21 of 937

1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	2.983e+004	1.000e-002	5	
16	2.985e+004	2.000e-002	5	
17	2.988e+004	7.000e-002	5	
18	2.990e+004	5.000e-002	5	
19	2.991e+004	6.000e-002	5	
20	2.994e+004	5.000e-002	5	
21	2.996e+004	5.000e-002	5	
22	2.998e+004	7.000e-002	5	
23	3.001e+004	6.000e-002	5	
24	3.003e+004	5.000e-002	5	
25	3.006e+004	3.000e-002	5	
26	3.008e+004	5.000e-002	5	
27	3.011e+004	3.000e-002	5	
28	3.013e+004	1.000e-002	5	
29	3.016e+004	3.000e-002	5	
30	3.018e+004	4.000e-002	5	
31	3.021e+004	4.000e-002	5	
32	3.023e+004	6.000e-002	5	
33	3.026e+004	4.000e-002	5	
34	3.028e+004	7.000e-002	5	
35	3.031e+004	1.000e-001	5	
36	3.033e+004	0.2000	5	
37	3.037e+004	0.1800	5	
38	3.044e+004	0.1900	5	
39	3.046e+004	0.1400	5	
40	3.048e+004	0.1400	5	
41	3.051e+004	0.1300	5	
42	3.053e+004	1.000e-001	5	
43	3.056e+004	8.000e-002	5	
44	3.058e+004	7.000e-002	5	
45	3.061e+004	0.1100	5	
46	3.063e+004	0.1400	5	
47	3.066e+004	0.1700	5	
48	3.068e+004	0.2600	5	
49	3.071e+004	-4.000e-002	5	
50	3.073e+004	-8.000e-002	5	
51	3.076e+004	-1.000e-002	5	
52	3.078e+004	-5.360e-016	5	
53	3.081e+004	-3.000e-002	5	
54	3.083e+004	2.000e-002	5	
55	3.086e+004	5.000e-002	5	
56	3.088e+004	0.1900	5	
57	3.091e+004	0.2100	5	
58	3.093e+004	0.2200	5	
59	3.096e+004	0.2300	5	
60	3.098e+004	0.2300	5	
61	3.101e+004	0.2200	5	
62	3.111e+004	0.2400	5	
63	3.113e+004	0.2300	5	
64	3.116e+004	0.8100	5	
65	3.118e+004	-8.300	5	
66	3.120e+004	-7.900	5	
67	3.124e+004	-6.300	5	
68	3.126e+004	-4.800	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	
			Page 22 of 937	

1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5	Pipe Segment: ELEP-Akabel Pipeline (continued)			
6	PROFILES			
7	Data			
8				
9	Distance	Elevation	Increments	
10	(m)	(m)		
11	3.128e+004	0.1500	5	
12	3.131e+004	-5.000e-002	5	
13	3.133e+004	4.000e-002	5	
14	3.136e+004	-2.000e-002	5	
15	3.138e+004	-5.447e-016	5	
16	3.141e+004	0.1200	5	
17	3.143e+004	6.000e-002	5	
18	3.146e+004	2.000e-002	5	
19	3.148e+004	1.000e-002	5	
20	3.151e+004	6.000e-002	5	
21	3.153e+004	3.000e-002	5	
22	3.156e+004	0.1300	5	
23	3.159e+004	1.000e-002	5	
24	3.161e+004	0.1300	5	
25	3.164e+004	6.000e-002	5	
26	3.166e+004	1.000e-001	5	
27	3.169e+004	0.1400	5	
28	3.170e+004	0.1200	5	
29	3.171e+004	1.000e-001	5	
30	3.174e+004	-6.000e-002	5	
31	3.175e+004	5.000e-002	5	
32	3.178e+004	-1.000e-002	5	
33	3.181e+004	7.000e-002	5	
34	3.184e+004	3.000e-002	5	
35	3.186e+004	1.000e-001	5	
36	3.189e+004	0.1100	5	
37	3.191e+004	9.000e-002	5	
38	3.194e+004	0.3800	5	
39	3.196e+004	0.2400	5	
40	3.199e+004	0.2400	5	
41	3.201e+004	0.2100	5	
42	3.204e+004	0.1800	5	
43	3.206e+004	0.1600	5	
44	3.209e+004	0.1200	5	
45	3.211e+004	0.1500	5	
46	3.214e+004	0.1700	5	
47	3.216e+004	0.4000	5	
48	3.219e+004	0.4200	5	
49	3.219e+004	0.7600	5	
50	3.221e+004	1.190	5	
51	3.223e+004	1.320	5	
52	3.230e+004	1.330	5	
53	3.236e+004	1.280	5	
54	3.243e+004	1.340	5	
55	3.250e+004	1.380	5	
56	3.257e+004	1.180	5	
57	3.260e+004	1.090	5	
58	3.263e+004	1.030	5	
59	3.264e+004	1.010	5	
60	3.265e+004	0.9700	5	
61	3.267e+004	0.8800	5	
62	3.270e+004	0.8200	5	
63	3.272e+004	0.9000	5	
64	3.275e+004	0.8300	5	
65	Aspen Technology Inc.		Aspen HYSYS Version 10	
66	Licensed to: Company Name Not Available		Page 23 of 937	
67			* Specified by user.	

1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	3.277e+004	0.9900	5	
16	3.280e+004	0.8500	5	
17	3.282e+004	0.7400	5	
18	3.285e+004	0.5100	5	
19	3.287e+004	0.4200	5	
20	3.290e+004	0.2100	5	
21	3.292e+004	0.1800	5	
22	3.295e+004	0.1800	5	
23	3.297e+004	0.1600	5	
24	3.300e+004	0.1700	5	
25	3.302e+004	0.1500	5	
26	3.303e+004	0.1500	5	
27	3.306e+004	0.1900	5	
28	3.307e+004	0.1700	5	
29	3.310e+004	0.1600	5	
30	3.312e+004	0.1900	5	
31	3.315e+004	0.1600	5	
32	3.317e+004	0.1400	5	
33	3.320e+004	0.2000	5	
34	3.322e+004	0.2000	5	
35	3.325e+004	0.2000	5	
36	3.327e+004	0.1900	5	
37	3.330e+004	0.2100	5	
38	3.332e+004	0.2000	5	
39	3.335e+004	0.2000	5	
40	3.337e+004	0.2300	5	
41	3.340e+004	0.2100	5	
42	3.342e+004	0.2000	5	
43	3.345e+004	0.2200	5	
44	3.347e+004	0.2100	5	
45	3.350e+004	0.2100	5	
46	3.352e+004	0.2200	5	
47	3.355e+004	0.1900	5	
48	3.357e+004	0.2100	5	
49	3.360e+004	0.1600	5	
50	3.362e+004	0.2900	5	
51	3.365e+004	0.3300	5	
52	3.367e+004	0.4100	5	
53	3.370e+004	0.2200	5	
54	3.372e+004	0.2700	5	
55	3.375e+004	0.3200	5	
56	3.377e+004	0.3600	5	
57	3.380e+004	0.1700	5	
58	3.382e+004	0.2000	5	
59	3.385e+004	0.2500	5	
60	3.386e+004	0.3000	5	
61	3.387e+004	0.3000	5	
62	3.388e+004	0.3000	5	
63	3.390e+004	0.3100	5	
64	3.395e+004	0.3100	5	
65	3.401e+004	0.3000	5	
66	3.406e+004	0.2900	5	
67	3.412e+004	0.2800	5	
68	3.418e+004	0.3000	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	Page 24 of 937

1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_80barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:13:43 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	3.423e+004	0.4500	5	
16	3.429e+004	0.2900	5	
17	3.430e+004	0.3200	5	
18	3.432e+004	0.3400	5	
19	3.435e+004	0.3200	5	
20	3.437e+004	0.3300	5	
21	3.440e+004	0.3000	5	
22	3.442e+004	0.3000	5	
23	3.445e+004	0.3900	5	
24	3.447e+004	0.3000	5	
25	3.450e+004	0.3000	5	
26	3.452e+004	0.2900	5	
27	3.455e+004	0.3400	5	
28	3.457e+004	0.3000	5	
29	3.460e+004	0.3200	5	
30	3.462e+004	0.3100	5	
31	3.465e+004	0.3100	5	
32	3.467e+004	0.3900	5	
33	3.469e+004	0.3900	5	
34	3.472e+004	0.3500	5	
35	3.474e+004	0.3000	5	
36	3.477e+004	0.2400	5	
37	3.479e+004	0.1400	5	
38	3.482e+004	8.000e-002	5	
39	3.484e+004	0.1400	5	
40	3.487e+004	0.1100	5	
41	3.489e+004	0.4000	5	
42	3.492e+004	-0.3200	5	
43	3.494e+004	-0.8000	5	
44	3.497e+004	-0.3000	5	
45	3.499e+004	-2.000e-002	5	
46	3.502e+004	0.1500	5	
47	3.504e+004	0.1800	5	
48	3.507e+004	0.1600	5	
49	3.509e+004	0.2200	5	
50	3.512e+004	0.2200	5	
51	3.514e+004	0.2100	5	
52	3.517e+004	0.2200	5	
53	3.519e+004	0.2200	5	
54	3.522e+004	0.2400	5	
55	3.524e+004	0.2500	5	
56	3.527e+004	0.2400	5	
57	3.529e+004	0.2000	5	
58	3.532e+004	0.2100	5	
59	3.534e+004	0.2200	5	
60	3.537e+004	-0.7500	5	
61	3.539e+004	0.2400	5	
62	3.542e+004	0.3300	5	
63	3.544e+004	0.1500	5	
64	3.547e+004	0.1300	5	
65	3.549e+004	0.1400	5	
66	3.552e+004	0.1200	5	
67	3.554e+004	0.1300	5	
68	3.557e+004	0.1100	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	
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1				Case Name: Simulation - 386MMSCFD_80barg.hsc		
2				Unit Set: Mixed		
3	 Epic Atlantic Limited			Date/Time: Mon Jun 13 15:13:43 2022		
4						
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Data					
12						
13	Distance		Elevation		Increments	
14	(m)		(m)			
15	3.559e+004		0.1100		5	
16	3.560e+004		0.1400		5	
17	3.563e+004		0.2100		5	
18	3.565e+004		0.1300		5	
19	3.568e+004		0.4000			
20						
21	Pipe Table					
22	Length	(m)	0.0000	5.000	10.00	15.00
23	Elevation	(m)	0.0000	-3.200e-002	-6.400e-002	-9.600e-002
24	Pressure	(bar)	78.57	78.57	78.57	78.57
25	Temperature	(C)	59.75	59.74	59.74	59.74
26	Vapour Fraction		0.8815	0.8815	0.8815	0.8815
27	Heat Transferred	(kJ/h-m)	---	792.4	792.4	792.3
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.1779	0.1779	0.1779	0.1779
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	8.228e-003	8.228e-003	8.228e-003	8.228e-003
33	Static Gradient	(kPa/m)	-1.071e-002	-1.071e-002	-1.071e-002	-1.071e-002
34	Accel. Gradient	(kPa/m)	-2.455e-007	-2.455e-007	-2.455e-007	-2.455e-007
35	Liq. Reynolds		2.043e+005 *	2.043e+005 *	2.043e+005 *	2.043e+005 *
36	Vap. Reynolds		9.499e+006 *	9.499e+006 *	9.499e+006 *	9.499e+006 *
37	Liquid Velocity	(m/s)	0.1994	0.1994	0.1994	0.1994
38	Vapour Velocity	(m/s)	2.633	2.633	2.633	2.633
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	20.00	25.00	30.00	35.00
42	Elevation	(m)	-0.1280	-0.1600	-0.1200	-8.000e-002
43	Pressure	(bar)	78.57	78.57	78.57	78.57
44	Temperature	(C)	59.74	59.73	59.73	59.73
45	Vapour Fraction		0.8815	0.8815	0.8815	0.8815
46	Heat Transferred	(kJ/h-m)	792.3	792.2	792.1	792.0
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.1779	0.1779	0.2703	0.2703
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	8.228e-003	8.228e-003	6.767e-003	6.767e-003
52	Static Gradient	(kPa/m)	-1.071e-002	-1.071e-002	1.788e-002	1.788e-002
53	Accel. Gradient	(kPa/m)	-2.456e-007	-2.456e-007	2.435e-006	2.435e-006
54	Liq. Reynolds		2.043e+005 *	2.042e+005 *	2.042e+005 *	2.042e+005 *
55	Vap. Reynolds		9.499e+006 *	9.499e+006 *	9.499e+006 *	9.499e+006 *
56	Liquid Velocity	(m/s)	0.1994	0.1994	0.1994	0.1994
57	Vapour Velocity	(m/s)	2.633	2.633	2.633	2.633
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	40.00	45.00	50.00	55.00
61	Elevation	(m)	-4.000e-002	2.776e-017	4.000e-002	3.000e-002
62	Pressure	(bar)	78.57	78.57	78.56	78.56
63	Temperature	(C)	59.73	59.72	59.72	59.72
64	Vapour Fraction		0.8815	0.8815	0.8815	0.8815
65	Heat Transferred	(kJ/h-m)	792.0	791.9	791.8	791.8
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2703	0.2703	0.2703	0.1815
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	6.767e-003	6.767e-003	6.767e-003	8.187e-003
15	Static Gradient (kPa/m)	1.788e-002	1.788e-002	1.788e-002	-3.391e-003
16	Accel. Gradient (kPa/m)	2.435e-006	2.435e-006	2.435e-006	4.740e-007
17	Liq. Reynolds	2.042e+005 *	2.042e+005 *	2.042e+005 *	2.042e+005 *
18	Vap. Reynolds	9.499e+006 *	9.499e+006 *	9.499e+006 *	9.499e+006 *
19	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
20	Vapour Velocity (m/s)	2.633	2.633	2.633	2.633
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	60.00	65.00	70.00	75.00
24	Elevation (m)	2.000e-002	1.000e-002	0.0000	-1.000e-002
25	Pressure (bar)	78.56	78.56	78.56	78.56
26	Temperature (C)	59.72	59.71	59.71	59.71
27	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
28	Heat Transferred (kJ/h-m)	791.7	791.7	791.6	791.6
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.1815	0.1815	0.1815	0.1815
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	8.187e-003	8.187e-003	8.187e-003	8.187e-003
34	Static Gradient (kPa/m)	-3.391e-003	-3.391e-003	-3.391e-003	-3.391e-003
35	Accel. Gradient (kPa/m)	4.740e-007	4.740e-007	4.740e-007	4.739e-007
36	Liq. Reynolds	2.042e+005 *	2.042e+005 *	2.042e+005 *	2.042e+005 *
37	Vap. Reynolds	9.499e+006 *	9.499e+006 *	9.499e+006 *	9.499e+006 *
38	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
39	Vapour Velocity (m/s)	2.633	2.633	2.633	2.633
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	80.00	85.00	90.00	95.00
43	Elevation (m)	-2.400e-002	-3.800e-002	-5.200e-002	-6.600e-002
44	Pressure (bar)	78.56	78.56	78.56	78.56
45	Temperature (C)	59.71	59.71	59.70	59.70
46	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
47	Heat Transferred (kJ/h-m)	791.5	791.4	791.4	791.3
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.1808	0.1808	0.1808	0.1808
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	8.194e-003	8.194e-003	8.194e-003	8.194e-003
53	Static Gradient (kPa/m)	-4.736e-003	-4.736e-003	-4.737e-003	-4.737e-003
54	Accel. Gradient (kPa/m)	3.417e-007	3.417e-007	3.417e-007	3.417e-007
55	Liq. Reynolds	2.042e+005 *	2.042e+005 *	2.042e+005 *	2.042e+005 *
56	Vap. Reynolds	9.499e+006 *	9.499e+006 *	9.499e+006 *	9.499e+006 *
57	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
58	Vapour Velocity (m/s)	2.633	2.633	2.633	2.633
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	100.0	105.0	110.0	115.0
62	Elevation (m)	-8.000e-002	-6.200e-002	-4.400e-002	-2.600e-002
63	Pressure (bar)	78.56	78.56	78.56	78.56
64	Temperature (C)	59.70	59.70	59.69	59.69
65	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
66	Heat Transferred (kJ/h-m)	791.3	791.1	791.1	791.0
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.1808	0.2575	0.2575	0.2575
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	8.194e-003	6.195e-003	6.195e-003	6.195e-003
16	Static Gradient (kPa/m)	-4.737e-003	7.765e-003	7.765e-003	7.765e-003
17	Accel. Gradient (kPa/m)	3.416e-007	1.379e-006	1.379e-006	1.379e-006
18	Liq. Reynolds	2.042e+005 *	2.042e+005 *	2.042e+005 *	2.042e+005 *
19	Vap. Reynolds	9.499e+006 *	9.499e+006 *	9.499e+006 *	9.499e+006 *
20	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
21	Vapour Velocity (m/s)	2.633	2.633	2.633	2.633
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	120.0	125.0	130.0	135.0
25	Elevation (m)	-8.000e-003	1.000e-002	-6.000e-003	-2.200e-002
26	Pressure (bar)	78.56	78.56	78.56	78.56
27	Temperature (C)	59.69	59.69	59.68	59.68
28	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
29	Heat Transferred (kJ/h-m)	790.9	790.9	790.9	790.8
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2575	0.2575	0.1805	0.1805
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.195e-003	6.195e-003	8.198e-003	8.198e-003
35	Static Gradient (kPa/m)	7.765e-003	7.765e-003	-5.407e-003	-5.407e-003
36	Accel. Gradient (kPa/m)	1.379e-006	1.379e-006	2.757e-007	2.757e-007
37	Liq. Reynolds	2.042e+005 *	2.042e+005 *	2.042e+005 *	2.042e+005 *
38	Vap. Reynolds	9.499e+006 *	9.499e+006 *	9.499e+006 *	9.499e+006 *
39	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
40	Vapour Velocity (m/s)	2.633	2.633	2.633	2.633
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	140.0	145.0	150.0	155.0
44	Elevation (m)	-3.800e-002	-5.400e-002	-7.000e-002	-7.600e-002
45	Pressure (bar)	78.56	78.56	78.56	78.56
46	Temperature (C)	59.68	59.68	59.68	59.67
47	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
48	Heat Transferred (kJ/h-m)	790.8	790.7	790.6	790.6
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1805	0.1805	0.1805	0.1821
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	8.198e-003	8.197e-003	8.197e-003	8.179e-003
54	Static Gradient (kPa/m)	-5.407e-003	-5.407e-003	-5.407e-003	-2.039e-003
55	Accel. Gradient (kPa/m)	2.757e-007	2.757e-007	2.757e-007	6.067e-007
56	Liq. Reynolds	2.042e+005 *	2.042e+005 *	2.042e+005 *	2.042e+005 *
57	Vap. Reynolds	9.499e+006 *	9.500e+006 *	9.500e+006 *	9.500e+006 *
58	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
59	Vapour Velocity (m/s)	2.633	2.633	2.632	2.632
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	160.0	165.0	170.0	175.0
63	Elevation (m)	-8.200e-002	-8.800e-002	-9.400e-002	-0.1000
64	Pressure (bar)	78.56	78.56	78.56	78.56
65	Temperature (C)	59.67	59.67	59.67	59.66
66	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
67	Heat Transferred (kJ/h-m)	790.5	790.5	790.4	790.3
68	Flow Regime	Transition	Transition	Transition	Transition
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1821	0.1821	0.1821	0.1821
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	8.179e-003	8.179e-003	8.179e-003	8.179e-003
17	Static Gradient (kPa/m)	-2.039e-003	-2.039e-003	-2.039e-003	-2.039e-003
18	Accel. Gradient (kPa/m)	6.067e-007	6.067e-007	6.067e-007	6.067e-007
19	Liq. Reynolds	2.042e+005 *	2.042e+005 *	2.042e+005 *	2.042e+005 *
20	Vap. Reynolds	9.500e+006 *	9.500e+006 *	9.500e+006 *	9.500e+006 *
21	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
22	Vapour Velocity (m/s)	2.632	2.632	2.632	2.632
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	180.0	185.0	190.0	195.0
26	Elevation (m)	-0.1120	-0.1240	-0.1360	-0.1480
27	Pressure (bar)	78.56	78.56	78.56	78.56
28	Temperature (C)	59.66	59.66	59.66	59.65
29	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
30	Heat Transferred (kJ/h-m)	790.3	790.2	790.2	790.1
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1812	0.1812	0.1812	0.1812
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	8.190e-003	8.190e-003	8.190e-003	8.190e-003
36	Static Gradient (kPa/m)	-4.057e-003	-4.057e-003	-4.057e-003	-4.057e-003
37	Accel. Gradient (kPa/m)	4.084e-007	4.084e-007	4.083e-007	4.083e-007
38	Liq. Reynolds	2.042e+005 *	2.042e+005 *	2.042e+005 *	2.042e+005 *
39	Vap. Reynolds	9.500e+006 *	9.500e+006 *	9.500e+006 *	9.500e+006 *
40	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
41	Vapour Velocity (m/s)	2.632	2.632	2.632	2.632
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	200.1	205.1	210.1	215.1
45	Elevation (m)	-0.1600	-0.1480	-0.1360	-0.1240
46	Pressure (bar)	78.56	78.56	78.55	78.55
47	Temperature (C)	59.65	59.65	59.65	59.65
48	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
49	Heat Transferred (kJ/h-m)	790.0	789.9	789.9	789.8
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.1812	0.2540	0.2540	0.2540
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	8.190e-003	6.547e-003	6.547e-003	6.547e-003
55	Static Gradient (kPa/m)	-4.057e-003	5.126e-003	5.126e-003	5.126e-003
56	Accel. Gradient (kPa/m)	4.083e-007	1.153e-006	1.153e-006	1.153e-006
57	Liq. Reynolds	2.041e+005 *	2.041e+005 *	2.041e+005 *	2.041e+005 *
58	Vap. Reynolds	9.500e+006 *	9.500e+006 *	9.500e+006 *	9.500e+006 *
59	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
60	Vapour Velocity (m/s)	2.632	2.632	2.632	2.632
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	220.1	225.1	230.1	235.1
64	Elevation (m)	-0.1120	-0.1000	-0.1120	-0.1240
65	Pressure (bar)	78.55	78.55	78.55	78.55
66	Temperature (C)	59.64	59.64	59.64	59.64
67	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
68	Heat Transferred (kJ/h-m)	789.7	789.7	789.7	789.6
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2540	0.2540	0.1812	0.1812
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	6.547e-003	6.547e-003	8.189e-003	8.189e-003
18	Static Gradient (kPa/m)	5.127e-003	5.127e-003	-4.065e-003	-4.065e-003
19	Accel. Gradient (kPa/m)	1.153e-006	1.153e-006	4.075e-007	4.075e-007
20	Liq. Reynolds	2.041e+005 *	2.041e+005 *	2.041e+005 *	2.041e+005 *
21	Vap. Reynolds	9.500e+006 *	9.500e+006 *	9.500e+006 *	9.500e+006 *
22	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
23	Vapour Velocity (m/s)	2.632	2.632	2.632	2.632
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	240.1	245.1	250.1	255.1
27	Elevation (m)	-0.1360	-0.1480	-0.1600	-0.1680
28	Pressure (bar)	78.55	78.55	78.55	78.55
29	Temperature (C)	59.63	59.63	59.63	59.63
30	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
31	Heat Transferred (kJ/h-m)	789.5	789.5	789.4	789.4
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1812	0.1812	0.1812	0.1818
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	8.189e-003	8.189e-003	8.189e-003	8.182e-003
37	Static Gradient (kPa/m)	-4.065e-003	-4.065e-003	-4.065e-003	-2.716e-003
38	Accel. Gradient (kPa/m)	4.075e-007	4.075e-007	4.075e-007	5.400e-007
39	Liq. Reynolds	2.041e+005 *	2.041e+005 *	2.041e+005 *	2.041e+005 *
40	Vap. Reynolds	9.500e+006 *	9.500e+006 *	9.500e+006 *	9.500e+006 *
41	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
42	Vapour Velocity (m/s)	2.632	2.632	2.632	2.632
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	260.1	265.1	270.1	275.1
46	Elevation (m)	-0.1760	-0.1840	-0.1920	-0.2000
47	Pressure (bar)	78.55	78.55	78.55	78.55
48	Temperature (C)	59.62	59.62	59.62	59.62
49	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
50	Heat Transferred (kJ/h-m)	789.3	789.2	789.2	789.1
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1818	0.1818	0.1818	0.1818
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	8.182e-003	8.182e-003	8.182e-003	8.182e-003
56	Static Gradient (kPa/m)	-2.716e-003	-2.716e-003	-2.717e-003	-2.717e-003
57	Accel. Gradient (kPa/m)	5.400e-007	5.400e-007	5.400e-007	5.400e-007
58	Liq. Reynolds	2.041e+005 *	2.041e+005 *	2.041e+005 *	2.041e+005 *
59	Vap. Reynolds	9.500e+006 *	9.500e+006 *	9.500e+006 *	9.500e+006 *
60	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
61	Vapour Velocity (m/s)	2.632	2.632	2.632	2.632
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	280.1	285.1	290.1	295.1
65	Elevation (m)	-0.2020	-0.2040	-0.2060	-0.2080
66	Pressure (bar)	78.55	78.55	78.55	78.55
67	Temperature (C)	59.62	59.61	59.61	59.61
68	Vapour Fraction	0.8815	0.8815	0.8814	0.8814
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 30 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	789.0	789.0	788.9	788.9
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.1828	0.1828	0.1828	0.1828
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	8.171e-003	8.171e-003	8.171e-003	8.171e-003
19	Static Gradient (kPa/m)	-6.815e-004	-6.815e-004	-6.815e-004	-6.815e-004
20	Accel. Gradient (kPa/m)	7.400e-007	7.400e-007	7.400e-007	7.400e-007
21	Liq. Reynolds	2.041e+005 *	2.041e+005 *	2.041e+005 *	2.041e+005 *
22	Vap. Reynolds	9.500e+006 *	9.500e+006 *	9.500e+006 *	9.500e+006 *
23	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
24	Vapour Velocity (m/s)	2.632	2.632	2.632	2.632
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	300.1	305.1	310.1	315.1
28	Elevation (m)	-0.2100	-0.2100	-0.2100	-0.2100
29	Pressure (bar)	78.55	78.55	78.55	78.55
30	Temperature (C)	59.61	59.60	59.60	59.60
31	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
32	Heat Transferred (kJ/h-m)	788.8	788.7	788.6	788.6
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.1828	0.2673	0.2673	0.2673
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	8.171e-003	6.318e-003	6.318e-003	6.318e-003
38	Static Gradient (kPa/m)	-6.815e-004	0.0000	0.0000	0.0000
39	Accel. Gradient (kPa/m)	7.400e-007	6.243e-007	6.243e-007	6.243e-007
40	Liq. Reynolds	2.041e+005 *	2.041e+005 *	2.041e+005 *	2.041e+005 *
41	Vap. Reynolds	9.500e+006 *	9.500e+006 *	9.500e+006 *	9.500e+006 *
42	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
43	Vapour Velocity (m/s)	2.632	2.632	2.632	2.632
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	320.1	325.1	330.1	335.1
47	Elevation (m)	-0.2100	-0.2100	-0.2820	-0.3540
48	Pressure (bar)	78.55	78.55	78.55	78.55
49	Temperature (C)	59.60	59.59	59.59	59.59
50	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
51	Heat Transferred (kJ/h-m)	788.5	788.4	788.5	788.4
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.2673	0.2673	0.1716	0.1716
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	6.318e-003	6.318e-003	8.305e-003	8.305e-003
57	Static Gradient (kPa/m)	0.0000	0.0000	-2.355e-002	-2.355e-002
58	Accel. Gradient (kPa/m)	6.243e-007	6.243e-007	-1.506e-006	-1.506e-006
59	Liq. Reynolds	2.041e+005 *	2.041e+005 *	2.041e+005 *	2.041e+005 *
60	Vap. Reynolds	9.500e+006 *	9.500e+006 *	9.500e+006 *	9.500e+006 *
61	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
62	Vapour Velocity (m/s)	2.632	2.632	2.632	2.632
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	340.1	345.1	350.1	355.1
66	Elevation (m)	-0.4260	-0.4980	-0.5700	-0.5000
67	Pressure (bar)	78.55	78.55	78.55	78.55
68	Temperature (C)	59.59	59.59	59.59	59.58
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
14	Heat Transferred (kJ/h-m)		788.4	788.3	788.3	788.1	
15	Flow Regime		Transition	Transition	Transition	Transition	
16	Water Flow Regime						
17	Liquid HoldUp		0.1716	0.1716	0.1716	0.2879	
18	Water HoldUp		---	---	---	---	
19	Friction Gradient (kPa/m)		8.305e-003	8.305e-003	8.305e-003	7.169e-003	
20	Static Gradient (kPa/m)		-2.355e-002	-2.355e-002	-2.355e-002	3.279e-002	
21	Accel. Gradient (kPa/m)		-1.506e-006	-1.506e-006	-1.506e-006	3.947e-006	
22	Liq. Reynolds		2.041e+005 *	2.041e+005 *	2.041e+005 *	2.041e+005 *	
23	Vap. Reynolds		9.500e+006 *	9.500e+006 *	9.500e+006 *	9.500e+006 *	
24	Liquid Velocity (m/s)		0.1994	0.1994	0.1994	0.1994	
25	Vapour Velocity (m/s)		2.632	2.632	2.632	2.632	
26	Deposit Thickness (mm)		---	---	---	---	
27	Deposit Volume (m3)		---	---	---	---	
28	Length (m)		360.1	365.1	370.1	375.1	
29	Elevation (m)		-0.4300	-0.3600	-0.2900	-0.2200	
30	Pressure (bar)		78.55	78.55	78.54	78.54	
31	Temperature (C)		59.58	59.58	59.57	59.57	
32	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
33	Heat Transferred (kJ/h-m)		788.0	788.0	787.9	787.8	
34	Flow Regime		Transition	Transition	Transition	Transition	
35	Water Flow Regime						
36	Liquid HoldUp		0.2879	0.2879	0.2879	0.2879	
37	Water HoldUp		---	---	---	---	
38	Friction Gradient (kPa/m)		7.169e-003	7.169e-003	7.169e-003	7.169e-003	
39	Static Gradient (kPa/m)		3.279e-002	3.279e-002	3.279e-002	3.279e-002	
40	Accel. Gradient (kPa/m)		3.947e-006	3.948e-006	3.948e-006	3.948e-006	
41	Liq. Reynolds		2.041e+005 *	2.041e+005 *	2.041e+005 *	2.041e+005 *	
42	Vap. Reynolds		9.500e+006 *	9.500e+006 *	9.500e+006 *	9.500e+006 *	
43	Liquid Velocity (m/s)		0.1994	0.1994	0.1994	0.1994	
44	Vapour Velocity (m/s)		2.632	2.632	2.632	2.632	
45	Deposit Thickness (mm)		---	---	---	---	
46	Deposit Volume (m3)		---	---	---	---	
47	Length (m)		380.0	385.0	390.0	395.0	
48	Elevation (m)		-0.1360	-5.200e-002	3.200e-002	0.1160	
49	Pressure (bar)		78.54	78.54	78.53	78.53	
50	Temperature (C)		59.57	59.57	59.56	59.56	
51	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
52	Heat Transferred (kJ/h-m)		787.7	787.7	787.6	787.5	
53	Flow Regime		Transition	Transition	Transition	Transition	
54	Water Flow Regime						
55	Liquid HoldUp		0.2961	0.2961	0.2961	0.2960	
56	Water HoldUp		---	---	---	---	
57	Friction Gradient (kPa/m)		7.188e-003	7.188e-003	7.188e-003	7.188e-003	
58	Static Gradient (kPa/m)		4.021e-002	4.021e-002	4.021e-002	4.021e-002	
59	Accel. Gradient (kPa/m)		4.684e-006	4.684e-006	4.684e-006	4.684e-006	
60	Liq. Reynolds		2.040e+005 *	2.040e+005 *	2.040e+005 *	2.040e+005 *	
61	Vap. Reynolds		9.500e+006 *	9.500e+006 *	9.500e+006 *	9.501e+006 *	
62	Liquid Velocity (m/s)		0.1994	0.1994	0.1994	0.1994	
63	Vapour Velocity (m/s)		2.632	2.632	2.632	2.632	
64	Deposit Thickness (mm)		---	---	---	---	
65	Deposit Volume (m3)		---	---	---	---	
66	Length (m)		400.0	405.0	410.0	415.0	
67	Elevation (m)		0.2000	0.1760	0.1520	0.1280	
68	Pressure (bar)		78.53	78.53	78.53	78.53	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	59.56	59.55	59.55	59.55
14	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
15	Heat Transferred (kJ/h-m)	787.4	787.4	787.3	787.3
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2960	0.1793	0.1793	0.1793
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	7.188e-003	8.211e-003	8.211e-003	8.211e-003
21	Static Gradient (kPa/m)	4.021e-002	-8.075e-003	-8.075e-003	-8.075e-003
22	Accel. Gradient (kPa/m)	4.684e-006	1.344e-008	1.343e-008	1.342e-008
23	Liq. Reynolds	2.040e+005 *	2.040e+005 *	2.040e+005 *	2.040e+005 *
24	Vap. Reynolds	9.501e+006 *	9.501e+006 *	9.501e+006 *	9.501e+006 *
25	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
26	Vapour Velocity (m/s)	2.632	2.632	2.632	2.632
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	420.0	425.0	430.0	435.0
30	Elevation (m)	0.1040	8.000e-002	7.000e-002	6.000e-002
31	Pressure (bar)	78.53	78.53	78.53	78.53
32	Temperature (C)	59.55	59.55	59.54	59.54
33	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
34	Heat Transferred (kJ/h-m)	787.2	787.2	787.1	787.1
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1793	0.1793	0.1815	0.1815
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	8.211e-003	8.211e-003	8.185e-003	8.185e-003
40	Static Gradient (kPa/m)	-8.075e-003	-8.075e-003	-3.392e-003	-3.392e-003
41	Accel. Gradient (kPa/m)	1.340e-008	1.339e-008	4.737e-007	4.736e-007
42	Liq. Reynolds	2.040e+005 *	2.040e+005 *	2.040e+005 *	2.040e+005 *
43	Vap. Reynolds	9.501e+006 *	9.501e+006 *	9.501e+006 *	9.501e+006 *
44	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
45	Vapour Velocity (m/s)	2.632	2.632	2.632	2.632
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	440.0	445.0	450.0	455.0
49	Elevation (m)	5.000e-002	4.000e-002	3.000e-002	3.600e-002
50	Pressure (bar)	78.53	78.53	78.53	78.53
51	Temperature (C)	59.54	59.54	59.53	59.53
52	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
53	Heat Transferred (kJ/h-m)	787.0	786.9	786.9	786.8
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.1815	0.1815	0.1815	0.2505
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	8.185e-003	8.185e-003	8.185e-003	6.914e-003
59	Static Gradient (kPa/m)	-3.392e-003	-3.392e-003	-3.392e-003	2.538e-003
60	Accel. Gradient (kPa/m)	4.736e-007	4.736e-007	4.736e-007	9.341e-007
61	Liq. Reynolds	2.040e+005 *	2.040e+005 *	2.040e+005 *	2.040e+005 *
62	Vap. Reynolds	9.501e+006 *	9.501e+006 *	9.501e+006 *	9.501e+006 *
63	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
64	Vapour Velocity (m/s)	2.632	2.632	2.632	2.632
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	460.0	465.0	470.0	475.0
68	Elevation (m)	4.200e-002	4.800e-002	5.400e-002	6.000e-002
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	78.53	78.53	78.53	78.53		
14	Temperature (C)	59.53	59.53	59.52	59.52		
15	Vapour Fraction	0.8814	0.8814	0.8814	0.8814		
16	Heat Transferred (kJ/h-m)	786.7	786.6	786.6	786.5		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.2505	0.2505	0.2505	0.2505		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	6.914e-003	6.914e-003	6.914e-003	6.914e-003		
22	Static Gradient (kPa/m)	2.538e-003	2.538e-003	2.538e-003	2.538e-003		
23	Accel. Gradient (kPa/m)	9.341e-007	9.341e-007	9.341e-007	9.341e-007		
24	Liq. Reynolds	2.040e+005 *	2.040e+005 *	2.040e+005 *	2.040e+005 *		
25	Vap. Reynolds	9.501e+006 *	9.501e+006 *	9.501e+006 *	9.501e+006 *		
26	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994		
27	Vapour Velocity (m/s)	2.632	2.632	2.632	2.632		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	480.0	485.0	490.0	495.0		
31	Elevation (m)	6.000e-002	6.000e-002	6.000e-002	6.000e-002		
32	Pressure (bar)	78.53	78.53	78.52	78.52		
33	Temperature (C)	59.52	59.52	59.52	59.51		
34	Vapour Fraction	0.8814	0.8814	0.8814	0.8814		
35	Heat Transferred (kJ/h-m)	786.4	786.4	786.3	786.3		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.2674	0.2674	0.2674	0.2674		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	6.319e-003	6.319e-003	6.319e-003	6.319e-003		
41	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000		
42	Accel. Gradient (kPa/m)	6.244e-007	6.244e-007	6.244e-007	6.244e-007		
43	Liq. Reynolds	2.040e+005 *	2.040e+005 *	2.040e+005 *	2.040e+005 *		
44	Vap. Reynolds	9.501e+006 *	9.501e+006 *	9.501e+006 *	9.501e+006 *		
45	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994		
46	Vapour Velocity (m/s)	2.632	2.632	2.632	2.632		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	500.0	505.0	510.0	515.0		
50	Elevation (m)	6.000e-002	5.800e-002	5.600e-002	5.400e-002		
51	Pressure (bar)	78.52	78.52	78.52	78.52		
52	Temperature (C)	59.51	59.51	59.51	59.50		
53	Vapour Fraction	0.8814	0.8814	0.8814	0.8814		
54	Heat Transferred (kJ/h-m)	786.2	786.2	786.1	786.1		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.2674	0.1828	0.1828	0.1828		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	6.319e-003	8.170e-003	8.170e-003	8.170e-003		
60	Static Gradient (kPa/m)	0.0000	-6.816e-004	-6.816e-004	-6.816e-004		
61	Accel. Gradient (kPa/m)	6.244e-007	7.400e-007	7.400e-007	7.400e-007		
62	Liq. Reynolds	2.040e+005 *	2.040e+005 *	2.040e+005 *	2.040e+005 *		
63	Vap. Reynolds	9.501e+006 *	9.501e+006 *	9.501e+006 *	9.501e+006 *		
64	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994		
65	Vapour Velocity (m/s)	2.632	2.632	2.632	2.632		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	520.0	525.0	530.0	535.0		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	5.200e-002	5.000e-002	5.200e-002	5.400e-002
14	Pressure (bar)	78.52	78.52	78.52	78.52
15	Temperature (C)	59.50	59.50	59.50	59.49
16	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
17	Heat Transferred (kJ/h-m)	786.0	785.9	785.8	785.8
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.1828	0.1828	0.2482	0.2482
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	8.170e-003	8.170e-003	7.168e-003	7.168e-003
23	Static Gradient (kPa/m)	-6.816e-004	-6.816e-004	8.405e-004	8.405e-004
24	Accel. Gradient (kPa/m)	7.400e-007	7.400e-007	7.914e-007	7.913e-007
25	Liq. Reynolds	2.040e+005 *	2.040e+005 *	2.040e+005 *	2.040e+005 *
26	Vap. Reynolds	9.501e+006 *	9.501e+006 *	9.501e+006 *	9.501e+006 *
27	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
28	Vapour Velocity (m/s)	2.631	2.631	2.631	2.631
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	540.0	545.0	550.0	555.0
32	Elevation (m)	5.600e-002	5.800e-002	6.000e-002	6.400e-002
33	Pressure (bar)	78.52	78.52	78.52	78.52
34	Temperature (C)	59.49	59.49	59.49	59.49
35	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
36	Heat Transferred (kJ/h-m)	785.7	785.6	785.6	785.5
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2482	0.2482	0.2482	0.2494
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	7.167e-003	7.167e-003	7.167e-003	7.039e-003
42	Static Gradient (kPa/m)	8.405e-004	8.405e-004	8.405e-004	1.687e-003
43	Accel. Gradient (kPa/m)	7.913e-007	7.913e-007	7.913e-007	8.623e-007
44	Liq. Reynolds	2.039e+005 *	2.039e+005 *	2.039e+005 *	2.039e+005 *
45	Vap. Reynolds	9.501e+006 *	9.501e+006 *	9.501e+006 *	9.501e+006 *
46	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
47	Vapour Velocity (m/s)	2.631	2.631	2.631	2.631
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	560.0	565.0	570.0	575.0
51	Elevation (m)	6.800e-002	7.200e-002	7.600e-002	8.000e-002
52	Pressure (bar)	78.52	78.52	78.52	78.52
53	Temperature (C)	59.48	59.48	59.48	59.48
54	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
55	Heat Transferred (kJ/h-m)	785.5	785.4	785.3	785.3
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2494	0.2494	0.2494	0.2494
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	7.039e-003	7.039e-003	7.039e-003	7.039e-003
61	Static Gradient (kPa/m)	1.687e-003	1.687e-003	1.687e-003	1.687e-003
62	Accel. Gradient (kPa/m)	8.623e-007	8.623e-007	8.623e-007	8.623e-007
63	Liq. Reynolds	2.039e+005 *	2.039e+005 *	2.039e+005 *	2.039e+005 *
64	Vap. Reynolds	9.501e+006 *	9.501e+006 *	9.501e+006 *	9.501e+006 *
65	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
66	Vapour Velocity (m/s)	2.631	2.631	2.631	2.631
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	580.0	585.0	590.0	595.0
14	Elevation (m)	7.800e-002	7.600e-002	7.400e-002	7.200e-002
15	Pressure (bar)	78.52	78.52	78.52	78.52
16	Temperature (C)	59.47	59.47	59.47	59.47
17	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
18	Heat Transferred (kJ/h-m)	785.3	785.2	785.1	785.1
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.1828	0.1828	0.1828	0.1828
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	8.169e-003	8.169e-003	8.169e-003	8.169e-003
24	Static Gradient (kPa/m)	-6.833e-004	-6.833e-004	-6.833e-004	-6.833e-004
25	Accel. Gradient (kPa/m)	7.398e-007	7.398e-007	7.398e-007	7.398e-007
26	Liq. Reynolds	2.039e+005 *	2.039e+005 *	2.039e+005 *	2.039e+005 *
27	Vap. Reynolds	9.501e+006 *	9.501e+006 *	9.501e+006 *	9.501e+006 *
28	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
29	Vapour Velocity (m/s)	2.631	2.631	2.631	2.631
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	600.0	605.0	610.0	615.0
33	Elevation (m)	7.000e-002	7.600e-002	8.200e-002	8.800e-002
34	Pressure (bar)	78.52	78.52	78.52	78.51
35	Temperature (C)	59.46	59.46	59.46	59.46
36	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
37	Heat Transferred (kJ/h-m)	785.0	784.9	784.8	784.8
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.1828	0.2506	0.2506	0.2506
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	8.169e-003	6.913e-003	6.913e-003	6.913e-003
43	Static Gradient (kPa/m)	-6.833e-004	2.539e-003	2.539e-003	2.539e-003
44	Accel. Gradient (kPa/m)	7.398e-007	9.340e-007	9.340e-007	9.340e-007
45	Liq. Reynolds	2.039e+005 *	2.039e+005 *	2.039e+005 *	2.039e+005 *
46	Vap. Reynolds	9.501e+006 *	9.501e+006 *	9.501e+006 *	9.501e+006 *
47	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
48	Vapour Velocity (m/s)	2.631	2.631	2.631	2.631
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	620.0	625.0	630.0	635.0
52	Elevation (m)	9.400e-002	1.000e-001	0.1060	0.1120
53	Pressure (bar)	78.51	78.51	78.51	78.51
54	Temperature (C)	59.45	59.45	59.45	59.45
55	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
56	Heat Transferred (kJ/h-m)	784.7	784.6	784.6	784.5
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.2506	0.2506	0.2506	0.2506
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	6.913e-003	6.912e-003	6.912e-003	6.912e-003
62	Static Gradient (kPa/m)	2.539e-003	2.539e-003	2.539e-003	2.539e-003
63	Accel. Gradient (kPa/m)	9.340e-007	9.340e-007	9.340e-007	9.340e-007
64	Liq. Reynolds	2.039e+005 *	2.039e+005 *	2.039e+005 *	2.039e+005 *
65	Vap. Reynolds	9.501e+006 *	9.501e+006 *	9.501e+006 *	9.501e+006 *
66	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
67	Vapour Velocity (m/s)	2.631	2.631	2.631	2.631
68	Deposit Thickness (mm)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
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11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	640.0	645.0	650.0	655.0
15	Elevation (m)	0.1180	0.1240	0.1300	0.1340
16	Pressure (bar)	78.51	78.51	78.51	78.51
17	Temperature (C)	59.45	59.44	59.44	59.44
18	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
19	Heat Transferred (kJ/h-m)	784.5	784.4	784.3	784.3
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2506	0.2506	0.2506	0.2494
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	6.912e-003	6.912e-003	6.912e-003	7.038e-003
25	Static Gradient (kPa/m)	2.539e-003	2.539e-003	2.539e-003	1.687e-003
26	Accel. Gradient (kPa/m)	9.339e-007	9.339e-007	9.339e-007	8.622e-007
27	Liq. Reynolds	2.039e+005 *	2.039e+005 *	2.039e+005 *	2.039e+005 *
28	Vap. Reynolds	9.501e+006 *	9.501e+006 *	9.501e+006 *	9.501e+006 *
29	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
30	Vapour Velocity (m/s)	2.631	2.631	2.631	2.631
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	660.0	665.0	670.0	675.0
34	Elevation (m)	0.1380	0.1420	0.1460	0.1500
35	Pressure (bar)	78.51	78.51	78.51	78.51
36	Temperature (C)	59.44	59.43	59.43	59.43
37	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
38	Heat Transferred (kJ/h-m)	784.2	784.1	784.1	784.0
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2494	0.2494	0.2494	0.2494
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	7.038e-003	7.038e-003	7.038e-003	7.038e-003
44	Static Gradient (kPa/m)	1.687e-003	1.687e-003	1.687e-003	1.687e-003
45	Accel. Gradient (kPa/m)	8.622e-007	8.622e-007	8.622e-007	8.622e-007
46	Liq. Reynolds	2.039e+005 *	2.039e+005 *	2.039e+005 *	2.039e+005 *
47	Vap. Reynolds	9.501e+006 *	9.501e+006 *	9.501e+006 *	9.501e+006 *
48	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
49	Vapour Velocity (m/s)	2.631	2.631	2.631	2.631
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	682.0	689.0	696.0	703.0
53	Elevation (m)	0.1540	0.1580	0.1620	0.1660
54	Pressure (bar)	78.51	78.51	78.51	78.51
55	Temperature (C)	59.43	59.42	59.42	59.42
56	Vapour Fraction	0.8814	0.8814	0.8814	0.8813
57	Heat Transferred (kJ/h-m)	783.9	783.9	783.8	783.7
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2487	0.2487	0.2487	0.2488
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	7.111e-003	7.111e-003	7.111e-003	7.111e-003
63	Static Gradient (kPa/m)	1.203e-003	1.203e-003	1.203e-003	1.203e-003
64	Accel. Gradient (kPa/m)	8.215e-007	8.215e-007	8.215e-007	8.215e-007
65	Liq. Reynolds	2.039e+005 *	2.039e+005 *	2.039e+005 *	2.039e+005 *
66	Vap. Reynolds	9.501e+006 *	9.501e+006 *	9.501e+006 *	9.502e+006 *
67	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
68	Vapour Velocity (m/s)	2.631	2.631	2.631	2.631
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	710.0	712.0	714.0	716.0
16	Elevation (m)	0.1700	0.1540	0.1380	0.1220
17	Pressure (bar)	78.51	78.51	78.51	78.51
18	Temperature (C)	59.41	59.41	59.41	59.41
19	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
20	Heat Transferred (kJ/h-m)	783.6	783.6	783.6	783.6
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2488	0.1767	0.1767	0.1767
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	7.111e-003	8.239e-003	8.239e-003	8.239e-003
26	Static Gradient (kPa/m)	1.203e-003	-1.334e-002	-1.334e-002	-1.334e-002
27	Accel. Gradient (kPa/m)	8.215e-007	-5.039e-007	-5.039e-007	-5.039e-007
28	Liq. Reynolds	2.039e+005 *	2.039e+005 *	2.039e+005 *	2.039e+005 *
29	Vap. Reynolds	9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *
30	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
31	Vapour Velocity (m/s)	2.631	2.631	2.631	2.631
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	718.0	720.0	722.0	724.0
35	Elevation (m)	0.1060	9.000e-002	0.1160	0.1420
36	Pressure (bar)	78.51	78.51	78.51	78.51
37	Temperature (C)	59.41	59.41	59.41	59.41
38	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
39	Heat Transferred (kJ/h-m)	783.6	783.5	783.5	783.4
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.1767	0.1767	0.2850	0.2850
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	8.239e-003	8.239e-003	7.152e-003	7.153e-003
45	Static Gradient (kPa/m)	-1.334e-002	-1.334e-002	3.022e-002	3.022e-002
46	Accel. Gradient (kPa/m)	-5.039e-007	-5.039e-007	3.693e-006	3.693e-006
47	Liq. Reynolds	2.038e+005 *	2.038e+005 *	2.038e+005 *	2.038e+005 *
48	Vap. Reynolds	9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *
49	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
50	Vapour Velocity (m/s)	2.631	2.631	2.631	2.631
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	726.0	728.0	730.0	732.0
54	Elevation (m)	0.1680	0.1940	0.2200	0.2060
55	Pressure (bar)	78.50	78.50	78.50	78.50
56	Temperature (C)	59.41	59.40	59.40	59.40
57	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
58	Heat Transferred (kJ/h-m)	783.4	783.4	783.3	783.4
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2850	0.2850	0.2850	0.1775
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	7.153e-003	7.153e-003	7.153e-003	8.229e-003
64	Static Gradient (kPa/m)	3.022e-002	3.022e-002	3.022e-002	-1.170e-002
65	Accel. Gradient (kPa/m)	3.693e-006	3.694e-006	3.694e-006	-3.435e-007
66	Liq. Reynolds	2.038e+005 *	2.038e+005 *	2.038e+005 *	2.038e+005 *
67	Vap. Reynolds	9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *
68	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3							
4							
5			Unit Set: Mixed				
6			Date/Time: Mon Jun 13 15:13:43 2022				
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9							
10	PROFILES						
11	Pipe Table						
12							
13	Vapour Velocity (m/s)	2.631	2.631	2.631	2.631		
14	Deposit Thickness (mm)	---	---	---	---		
15	Deposit Volume (m3)	---	---	---	---		
16	Length (m)	734.0	736.0	738.0	740.0		
17	Elevation (m)	0.1920	0.1780	0.1640	0.1500		
18	Pressure (bar)	78.50	78.50	78.50	78.50		
19	Temperature (C)	59.40	59.40	59.40	59.40		
20	Vapour Fraction	0.8813	0.8813	0.8813	0.8813		
21	Heat Transferred (kJ/h-m)	783.3	783.3	783.3	783.3		
22	Flow Regime	Transition	Transition	Transition	Transition		
23	Water Flow Regime						
24	Liquid HoldUp	0.1775	0.1775	0.1775	0.1776		
25	Water HoldUp	---	---	---	---		
26	Friction Gradient (kPa/m)	8.229e-003	8.229e-003	8.229e-003	8.229e-003		
27	Static Gradient (kPa/m)	-1.170e-002	-1.170e-002	-1.170e-002	-1.170e-002		
28	Accel. Gradient (kPa/m)	-3.435e-007	-3.435e-007	-3.435e-007	-3.435e-007		
29	Liq. Reynolds	2.038e+005 *	2.038e+005 *	2.038e+005 *	2.038e+005 *		
30	Vap. Reynolds	9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *		
31	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994		
32	Vapour Velocity (m/s)	2.631	2.631	2.631	2.631		
33	Deposit Thickness (mm)	---	---	---	---		
34	Deposit Volume (m3)	---	---	---	---		
35	Length (m)	742.0	744.0	746.0	748.0		
36	Elevation (m)	0.1400	0.1300	0.1200	0.1100		
37	Pressure (bar)	78.50	78.50	78.50	78.50		
38	Temperature (C)	59.40	59.40	59.40	59.40		
39	Vapour Fraction	0.8813	0.8813	0.8813	0.8813		
40	Heat Transferred (kJ/h-m)	783.3	783.2	783.2	783.2		
41	Flow Regime	Transition	Transition	Transition	Transition		
42	Water Flow Regime						
43	Liquid HoldUp	0.1792	0.1792	0.1792	0.1792		
44	Water HoldUp	---	---	---	---		
45	Friction Gradient (kPa/m)	8.210e-003	8.210e-003	8.210e-003	8.210e-003		
46	Static Gradient (kPa/m)	-8.410e-003	-8.410e-003	-8.410e-003	-8.410e-003		
47	Accel. Gradient (kPa/m)	-1.972e-008	-1.972e-008	-1.973e-008	-1.973e-008		
48	Liq. Reynolds	2.038e+005 *	2.038e+005 *	2.038e+005 *	2.038e+005 *		
49	Vap. Reynolds	9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *		
50	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994		
51	Vapour Velocity (m/s)	2.631	2.631	2.631	2.631		
52	Deposit Thickness (mm)	---	---	---	---		
53	Deposit Volume (m3)	---	---	---	---		
54	Length (m)	750.0	755.0	760.0	765.0		
55	Elevation (m)	1.000e-001	9.600e-002	9.200e-002	8.800e-002		
56	Pressure (bar)	78.50	78.50	78.50	78.50		
57	Temperature (C)	59.39	59.39	59.39	59.39		
58	Vapour Fraction	0.8813	0.8813	0.8813	0.8813		
59	Heat Transferred (kJ/h-m)	783.2	783.1	783.0	783.0		
60	Flow Regime	Transition	Transition	Transition	Transition		
61	Water Flow Regime						
62	Liquid HoldUp	0.1792	0.1825	0.1825	0.1825		
63	Water HoldUp	---	---	---	---		
64	Friction Gradient (kPa/m)	8.210e-003	8.172e-003	8.172e-003	8.172e-003		
65	Static Gradient (kPa/m)	-8.410e-003	-1.362e-003	-1.362e-003	-1.362e-003		
66	Accel. Gradient (kPa/m)	-1.974e-008	6.729e-007	6.729e-007	6.729e-007		
67	Liq. Reynolds	2.038e+005 *	2.038e+005 *	2.038e+005 *	2.038e+005 *		
68	Vap. Reynolds	9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Liquid Velocity	(m/s)	0.1994	0.1994	0.1994	0.1994
14	Vapour Velocity	(m/s)	2.631	2.631	2.631	2.631
15	Deposit Thickness	(mm)	---	---	---	---
16	Deposit Volume	(m3)	---	---	---	---
17	Length	(m)	770.0	775.0	780.0	785.0
18	Elevation	(m)	8.400e-002	8.000e-002	8.400e-002	8.800e-002
19	Pressure	(bar)	78.50	78.50	78.50	78.50
20	Temperature	(C)	59.39	59.38	59.38	59.38
21	Vapour Fraction		0.8813	0.8813	0.8813	0.8813
22	Heat Transferred	(kJ/h-m)	782.9	782.8	782.7	782.7
23	Flow Regime		Transition	Transition	Transition	Transition
24	Water Flow Regime					
25	Liquid HoldUp		0.1825	0.1825	0.2494	0.2494
26	Water HoldUp		---	---	---	---
27	Friction Gradient	(kPa/m)	8.171e-003	8.171e-003	7.039e-003	7.039e-003
28	Static Gradient	(kPa/m)	-1.362e-003	-1.362e-003	1.675e-003	1.675e-003
29	Accel. Gradient	(kPa/m)	6.729e-007	6.729e-007	8.611e-007	8.611e-007
30	Liq. Reynolds		2.038e+005 *	2.038e+005 *	2.038e+005 *	2.038e+005 *
31	Vap. Reynolds		9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *
32	Liquid Velocity	(m/s)	0.1994	0.1994	0.1994	0.1994
33	Vapour Velocity	(m/s)	2.631	2.631	2.631	2.631
34	Deposit Thickness	(mm)	---	---	---	---
35	Deposit Volume	(m3)	---	---	---	---
36	Length	(m)	790.1	795.1	800.1	805.1
37	Elevation	(m)	9.200e-002	9.600e-002	1.000e-001	1.000e-001
38	Pressure	(bar)	78.50	78.50	78.50	78.50
39	Temperature	(C)	59.38	59.37	59.37	59.37
40	Vapour Fraction		0.8813	0.8813	0.8813	0.8813
41	Heat Transferred	(kJ/h-m)	782.6	782.5	782.5	782.4
42	Flow Regime		Transition	Transition	Transition	Transition
43	Water Flow Regime					
44	Liquid HoldUp		0.2494	0.2494	0.2494	0.2674
45	Water HoldUp		---	---	---	---
46	Friction Gradient	(kPa/m)	7.039e-003	7.039e-003	7.039e-003	6.319e-003
47	Static Gradient	(kPa/m)	1.675e-003	1.675e-003	1.675e-003	0.0000
48	Accel. Gradient	(kPa/m)	8.611e-007	8.611e-007	8.611e-007	6.244e-007
49	Liq. Reynolds		2.038e+005 *	2.038e+005 *	2.038e+005 *	2.038e+005 *
50	Vap. Reynolds		9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *
51	Liquid Velocity	(m/s)	0.1994	0.1994	0.1994	0.1994
52	Vapour Velocity	(m/s)	2.631	2.631	2.631	2.631
53	Deposit Thickness	(mm)	---	---	---	---
54	Deposit Volume	(m3)	---	---	---	---
55	Length	(m)	810.1	815.1	820.1	825.1
56	Elevation	(m)	1.000e-001	1.000e-001	1.000e-001	1.000e-001
57	Pressure	(bar)	78.50	78.50	78.50	78.50
58	Temperature	(C)	59.37	59.36	59.36	59.36
59	Vapour Fraction		0.8813	0.8813	0.8813	0.8813
60	Heat Transferred	(kJ/h-m)	782.4	782.3	782.2	782.2
61	Flow Regime		Transition	Transition	Transition	Transition
62	Water Flow Regime					
63	Liquid HoldUp		0.2674	0.2674	0.2674	0.2674
64	Water HoldUp		---	---	---	---
65	Friction Gradient	(kPa/m)	6.319e-003	6.319e-003	6.319e-003	6.319e-003
66	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000
67	Accel. Gradient	(kPa/m)	6.244e-007	6.244e-007	6.244e-007	6.244e-007
68	Liq. Reynolds		2.038e+005 *	2.038e+005 *	2.038e+005 *	2.038e+005 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *
14	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
15	Vapour Velocity (m/s)	2.631	2.631	2.631	2.631
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	830.1	835.1	840.1	845.1
19	Elevation (m)	2.600e-002	-4.800e-002	-0.1220	-0.1960
20	Pressure (bar)	78.50	78.50	78.50	78.50
21	Temperature (C)	59.36	59.36	59.35	59.35
22	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
23	Heat Transferred (kJ/h-m)	782.2	782.2	782.1	782.1
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1713	0.1713	0.1713	0.1713
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	8.306e-003	8.306e-003	8.306e-003	8.306e-003
29	Static Gradient (kPa/m)	-2.419e-002	-2.419e-002	-2.419e-002	-2.419e-002
30	Accel. Gradient (kPa/m)	-1.569e-006	-1.569e-006	-1.569e-006	-1.569e-006
31	Liq. Reynolds	2.038e+005 *	2.038e+005 *	2.038e+005 *	2.038e+005 *
32	Vap. Reynolds	9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *
33	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
34	Vapour Velocity (m/s)	2.630	2.630	2.630	2.630
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	850.1	855.1	860.1	865.1
38	Elevation (m)	-0.2700	-0.2060	-0.1420	-7.800e-002
39	Pressure (bar)	78.50	78.50	78.50	78.50
40	Temperature (C)	59.35	59.35	59.35	59.34
41	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
42	Heat Transferred (kJ/h-m)	782.0	781.9	781.8	781.7
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1713	0.2845	0.2845	0.2845
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	8.306e-003	7.147e-003	7.147e-003	7.147e-003
48	Static Gradient (kPa/m)	-2.419e-002	2.972e-002	2.972e-002	2.972e-002
49	Accel. Gradient (kPa/m)	-1.569e-006	3.642e-006	3.643e-006	3.643e-006
50	Liq. Reynolds	2.038e+005 *	2.038e+005 *	2.038e+005 *	2.038e+005 *
51	Vap. Reynolds	9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *
52	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
53	Vapour Velocity (m/s)	2.630	2.630	2.630	2.630
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	870.1	875.1	880.1	885.1
57	Elevation (m)	-1.400e-002	5.000e-002	4.200e-002	3.400e-002
58	Pressure (bar)	78.49	78.49	78.49	78.49
59	Temperature (C)	59.34	59.34	59.33	59.33
60	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
61	Heat Transferred (kJ/h-m)	781.6	781.6	781.6	781.5
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2845	0.2845	0.1819	0.1819
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	7.148e-003	7.148e-003	8.178e-003	8.178e-003
67	Static Gradient (kPa/m)	2.972e-002	2.972e-002	-2.718e-003	-2.718e-003
68	Accel. Gradient (kPa/m)	3.643e-006	3.643e-006	5.395e-007	5.395e-007
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Liq. Reynolds		2.038e+005 *	2.038e+005 *	2.038e+005 *	2.038e+005 *				
14	Vap. Reynolds		9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *				
15	Liquid Velocity	(m/s)	0.1994	0.1994	0.1994	0.1994				
16	Vapour Velocity	(m/s)	2.630	2.630	2.630	2.630				
17	Deposit Thickness	(mm)	---	---	---	---				
18	Deposit Volume	(m3)	---	---	---	---				
19	Length	(m)	890.1	895.1	900.1	905.1				
20	Elevation	(m)	2.600e-002	1.800e-002	1.000e-002	4.000e-003				
21	Pressure	(bar)	78.49	78.49	78.49	78.49				
22	Temperature	(C)	59.33	59.33	59.33	59.32				
23	Vapour Fraction		0.8813	0.8813	0.8813	0.8813				
24	Heat Transferred	(kJ/h-m)	781.4	781.4	781.3	781.3				
25	Flow Regime		Transition	Transition	Transition	Transition				
26	Water Flow Regime									
27	Liquid HoldUp		0.1819	0.1819	0.1819	0.1822				
28	Water HoldUp		---	---	---	---				
29	Friction Gradient	(kPa/m)	8.178e-003	8.178e-003	8.178e-003	8.174e-003				
30	Static Gradient	(kPa/m)	-2.718e-003	-2.718e-003	-2.718e-003	-2.041e-003				
31	Accel. Gradient	(kPa/m)	5.395e-007	5.395e-007	5.395e-007	6.060e-007				
32	Liq. Reynolds		2.038e+005 *	2.038e+005 *	2.037e+005 *	2.037e+005 *				
33	Vap. Reynolds		9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *				
34	Liquid Velocity	(m/s)	0.1994	0.1994	0.1994	0.1994				
35	Vapour Velocity	(m/s)	2.630	2.630	2.630	2.630				
36	Deposit Thickness	(mm)	---	---	---	---				
37	Deposit Volume	(m3)	---	---	---	---				
38	Length	(m)	910.1	915.1	920.1	925.1				
39	Elevation	(m)	-2.000e-003	-8.000e-003	-1.400e-002	-2.000e-002				
40	Pressure	(bar)	78.49	78.49	78.49	78.49				
41	Temperature	(C)	59.32	59.32	59.32	59.31				
42	Vapour Fraction		0.8813	0.8813	0.8813	0.8813				
43	Heat Transferred	(kJ/h-m)	781.2	781.1	781.1	781.0				
44	Flow Regime		Transition	Transition	Transition	Transition				
45	Water Flow Regime									
46	Liquid HoldUp		0.1822	0.1822	0.1822	0.1822				
47	Water HoldUp		---	---	---	---				
48	Friction Gradient	(kPa/m)	8.174e-003	8.174e-003	8.174e-003	8.174e-003				
49	Static Gradient	(kPa/m)	-2.041e-003	-2.041e-003	-2.041e-003	-2.041e-003				
50	Accel. Gradient	(kPa/m)	6.060e-007	6.060e-007	6.060e-007	6.060e-007				
51	Liq. Reynolds		2.037e+005 *	2.037e+005 *	2.037e+005 *	2.037e+005 *				
52	Vap. Reynolds		9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *				
53	Liquid Velocity	(m/s)	0.1994	0.1994	0.1994	0.1994				
54	Vapour Velocity	(m/s)	2.630	2.630	2.630	2.630				
55	Deposit Thickness	(mm)	---	---	---	---				
56	Deposit Volume	(m3)	---	---	---	---				
57	Length	(m)	930.1	935.1	940.1	945.1				
58	Elevation	(m)	-1.400e-002	-8.000e-003	-2.000e-003	4.000e-003				
59	Pressure	(bar)	78.49	78.49	78.49	78.49				
60	Temperature	(C)	59.31	59.31	59.31	59.30				
61	Vapour Fraction		0.8813	0.8813	0.8813	0.8813				
62	Heat Transferred	(kJ/h-m)	780.9	780.8	780.8	780.7				
63	Flow Regime		Transition	Transition	Transition	Transition				
64	Water Flow Regime									
65	Liquid HoldUp		0.2506	0.2506	0.2506	0.2506				
66	Water HoldUp		---	---	---	---				
67	Friction Gradient	(kPa/m)	6.910e-003	6.909e-003	6.909e-003	6.909e-003				
68	Static Gradient	(kPa/m)	2.540e-003	2.540e-003	2.540e-003	2.540e-003				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Accel. Gradient	(kPa/m)	9.337e-007	9.337e-007	9.337e-007	9.337e-007				
14	Liq. Reynolds		2.037e+005 *	2.037e+005 *	2.037e+005 *	2.037e+005 *				
15	Vap. Reynolds		9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *				
16	Liquid Velocity	(m/s)	0.1994	0.1994	0.1994	0.1994				
17	Vapour Velocity	(m/s)	2.630	2.630	2.630	2.630				
18	Deposit Thickness	(mm)	---	---	---	---				
19	Deposit Volume	(m3)	---	---	---	---				
20	Length	(m)	950.1	955.1	960.1	965.1				
21	Elevation	(m)	1.000e-002	1.000e-002	1.000e-002	1.000e-002				
22	Pressure	(bar)	78.49	78.49	78.49	78.49				
23	Temperature	(C)	59.30	59.30	59.30	59.30				
24	Vapour Fraction		0.8813	0.8813	0.8813	0.8813				
25	Heat Transferred	(kJ/h-m)	780.6	780.6	780.5	780.5				
26	Flow Regime		Transition	Transition	Transition	Transition				
27	Water Flow Regime									
28	Liquid HoldUp		0.2506	0.2675	0.2675	0.2675				
29	Water HoldUp		---	---	---	---				
30	Friction Gradient	(kPa/m)	6.909e-003	6.320e-003	6.320e-003	6.320e-003				
31	Static Gradient	(kPa/m)	2.540e-003	0.0000	0.0000	0.0000				
32	Accel. Gradient	(kPa/m)	9.337e-007	6.245e-007	6.245e-007	6.245e-007				
33	Liq. Reynolds		2.037e+005 *	2.037e+005 *	2.037e+005 *	2.037e+005 *				
34	Vap. Reynolds		9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *				
35	Liquid Velocity	(m/s)	0.1994	0.1994	0.1994	0.1994				
36	Vapour Velocity	(m/s)	2.630	2.630	2.630	2.630				
37	Deposit Thickness	(mm)	---	---	---	---				
38	Deposit Volume	(m3)	---	---	---	---				
39	Length	(m)	970.1	975.1	980.1	985.1				
40	Elevation	(m)	1.000e-002	1.000e-002	8.000e-003	6.000e-003				
41	Pressure	(bar)	78.49	78.49	78.49	78.49				
42	Temperature	(C)	59.29	59.29	59.29	59.29				
43	Vapour Fraction		0.8813	0.8813	0.8813	0.8813				
44	Heat Transferred	(kJ/h-m)	780.4	780.3	780.3	780.3				
45	Flow Regime		Transition	Transition	Transition	Transition				
46	Water Flow Regime									
47	Liquid HoldUp		0.2675	0.2675	0.1829	0.1829				
48	Water HoldUp		---	---	---	---				
49	Friction Gradient	(kPa/m)	6.320e-003	6.320e-003	8.166e-003	8.166e-003				
50	Static Gradient	(kPa/m)	0.0000	0.0000	-6.827e-004	-6.827e-004				
51	Accel. Gradient	(kPa/m)	6.245e-007	6.245e-007	7.395e-007	7.394e-007				
52	Liq. Reynolds		2.037e+005 *	2.037e+005 *	2.037e+005 *	2.037e+005 *				
53	Vap. Reynolds		9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *				
54	Liquid Velocity	(m/s)	0.1994	0.1994	0.1994	0.1994				
55	Vapour Velocity	(m/s)	2.630	2.630	2.630	2.630				
56	Deposit Thickness	(mm)	---	---	---	---				
57	Deposit Volume	(m3)	---	---	---	---				
58	Length	(m)	990.1	995.1	1000	1005				
59	Elevation	(m)	4.000e-003	2.000e-003	-1.232e-016	3.600e-002				
60	Pressure	(bar)	78.48	78.48	78.48	78.48				
61	Temperature	(C)	59.28	59.28	59.28	59.28				
62	Vapour Fraction		0.8813	0.8813	0.8813	0.8813				
63	Heat Transferred	(kJ/h-m)	780.2	780.1	780.1	780.0				
64	Flow Regime		Transition	Transition	Transition	Transition				
65	Water Flow Regime									
66	Liquid HoldUp		0.1829	0.1829	0.1829	0.2682				
67	Water HoldUp		---	---	---	---				
68	Friction Gradient	(kPa/m)	8.166e-003	8.166e-003	8.166e-003	6.466e-003				
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1				Case Name:	Simulation - 386MMSCFD_80barg.hsc		
2				Unit Set:	Mixed		
3	 Epic Atlantic Limited			Date/Time:	Mon Jun 13 15:13:43 2022		
4							
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Static Gradient	(kPa/m)	-6.827e-004	-6.827e-004	-6.828e-004	1.600e-002	
14	Accel. Gradient	(kPa/m)	7.394e-007	7.394e-007	7.394e-007	2.220e-006	
15	Liq. Reynolds		2.037e+005 *	2.037e+005 *	2.037e+005 *	2.037e+005 *	
16	Vap. Reynolds		9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *	
17	Liquid Velocity	(m/s)	0.1994	0.1994	0.1994	0.1994	
18	Vapour Velocity	(m/s)	2.630	2.630	2.630	2.630	
19	Deposit Thickness	(mm)	---	---	---	---	
20	Deposit Volume	(m3)	---	---	---	---	
21	Length	(m)	1010	1015	1020	1025	
22	Elevation	(m)	7.200e-002	0.1080	0.1440	0.1800	
23	Pressure	(bar)	78.48	78.48	78.48	78.48	
24	Temperature	(C)	59.27	59.27	59.27	59.27	
25	Vapour Fraction		0.8813	0.8813	0.8813	0.8813	
26	Heat Transferred	(kJ/h-m)	779.9	779.8	779.8	779.7	
27	Flow Regime		Transition	Transition	Transition	Transition	
28	Water Flow Regime						
29	Liquid HoldUp		0.2682	0.2682	0.2682	0.2682	
30	Water HoldUp		---	---	---	---	
31	Friction Gradient	(kPa/m)	6.466e-003	6.466e-003	6.466e-003	6.466e-003	
32	Static Gradient	(kPa/m)	1.600e-002	1.600e-002	1.600e-002	1.600e-002	
33	Accel. Gradient	(kPa/m)	2.220e-006	2.220e-006	2.220e-006	2.220e-006	
34	Liq. Reynolds		2.037e+005 *	2.037e+005 *	2.037e+005 *	2.037e+005 *	
35	Vap. Reynolds		9.503e+006 *	9.503e+006 *	9.503e+006 *	9.503e+006 *	
36	Liquid Velocity	(m/s)	0.1994	0.1994	0.1994	0.1994	
37	Vapour Velocity	(m/s)	2.630	2.630	2.630	2.630	
38	Deposit Thickness	(mm)	---	---	---	---	
39	Deposit Volume	(m3)	---	---	---	---	
40	Length	(m)	1030	1035	1040	1045	
41	Elevation	(m)	9.800e-002	1.600e-002	-6.600e-002	-0.1480	
42	Pressure	(bar)	78.48	78.48	78.48	78.48	
43	Temperature	(C)	59.26	59.26	59.26	59.26	
44	Vapour Fraction		0.8813	0.8813	0.8813	0.8813	
45	Heat Transferred	(kJ/h-m)	779.7	779.7	779.6	779.6	
46	Flow Regime		Transition	Transition	Transition	Transition	
47	Water Flow Regime						
48	Liquid HoldUp		0.1700	0.1700	0.1700	0.1700	
49	Water HoldUp		---	---	---	---	
50	Friction Gradient	(kPa/m)	8.322e-003	8.322e-003	8.322e-003	8.322e-003	
51	Static Gradient	(kPa/m)	-2.668e-002	-2.668e-002	-2.668e-002	-2.668e-002	
52	Accel. Gradient	(kPa/m)	-1.814e-006	-1.814e-006	-1.814e-006	-1.814e-006	
53	Liq. Reynolds		2.037e+005 *	2.037e+005 *	2.037e+005 *	2.037e+005 *	
54	Vap. Reynolds		9.503e+006 *	9.503e+006 *	9.503e+006 *	9.503e+006 *	
55	Liquid Velocity	(m/s)	0.1994	0.1994	0.1994	0.1994	
56	Vapour Velocity	(m/s)	2.630	2.630	2.630	2.630	
57	Deposit Thickness	(mm)	---	---	---	---	
58	Deposit Volume	(m3)	---	---	---	---	
59	Length	(m)	1050	1055	1060	1065	
60	Elevation	(m)	-0.2300	-0.1940	-0.1580	-0.1220	
61	Pressure	(bar)	78.48	78.48	78.48	78.48	
62	Temperature	(C)	59.26	59.26	59.25	59.25	
63	Vapour Fraction		0.8813	0.8813	0.8813	0.8813	
64	Heat Transferred	(kJ/h-m)	779.5	779.4	779.3	779.3	
65	Flow Regime		Transition	Transition	Transition	Transition	
66	Water Flow Regime						
67	Liquid HoldUp		0.1700	0.2682	0.2682	0.2682	
68	Water HoldUp		---	---	---	---	
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1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	8.322e-003	6.466e-003	6.466e-003	6.466e-003
14	Static Gradient	(kPa/m)	-2.668e-002	1.601e-002	1.601e-002	1.601e-002
15	Accel. Gradient	(kPa/m)	-1.814e-006	2.220e-006	2.220e-006	2.220e-006
16	Liq. Reynolds		2.037e+005 *	2.037e+005 *	2.037e+005 *	2.037e+005 *
17	Vap. Reynolds		9.503e+006 *	9.503e+006 *	9.503e+006 *	9.503e+006 *
18	Liquid Velocity	(m/s)	0.1994	0.1995	0.1995	0.1995
19	Vapour Velocity	(m/s)	2.630	2.630	2.630	2.630
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	1070	1075	1080	1085
23	Elevation	(m)	-8.600e-002	-5.000e-002	-3.600e-002	-2.200e-002
24	Pressure	(bar)	78.48	78.48	78.48	78.48
25	Temperature	(C)	59.25	59.24	59.24	59.24
26	Vapour Fraction		0.8813	0.8813	0.8813	0.8813
27	Heat Transferred	(kJ/h-m)	779.2	779.1	779.1	779.0
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2682	0.2682	0.2553	0.2553
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	6.466e-003	6.466e-003	6.421e-003	6.421e-003
33	Static Gradient	(kPa/m)	1.601e-002	1.601e-002	6.006e-003	6.006e-003
34	Accel. Gradient	(kPa/m)	2.220e-006	2.220e-006	1.228e-006	1.228e-006
35	Liq. Reynolds		2.037e+005 *	2.037e+005 *	2.036e+005 *	2.036e+005 *
36	Vap. Reynolds		9.503e+006 *	9.503e+006 *	9.503e+006 *	9.503e+006 *
37	Liquid Velocity	(m/s)	0.1995	0.1995	0.1995	0.1995
38	Vapour Velocity	(m/s)	2.630	2.630	2.630	2.630
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	1090	1095	1100	1105
42	Elevation	(m)	-8.000e-003	6.000e-003	2.000e-002	3.000e-002
43	Pressure	(bar)	78.48	78.47	78.47	78.47
44	Temperature	(C)	59.24	59.24	59.23	59.23
45	Vapour Fraction		0.8813	0.8813	0.8813	0.8813
46	Heat Transferred	(kJ/h-m)	778.9	778.9	778.8	778.7
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2553	0.2553	0.2553	0.2530
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	6.421e-003	6.421e-003	6.421e-003	6.661e-003
52	Static Gradient	(kPa/m)	6.006e-003	6.006e-003	6.006e-003	4.262e-003
53	Accel. Gradient	(kPa/m)	1.228e-006	1.228e-006	1.228e-006	1.079e-006
54	Liq. Reynolds		2.036e+005 *	2.036e+005 *	2.036e+005 *	2.036e+005 *
55	Vap. Reynolds		9.503e+006 *	9.503e+006 *	9.503e+006 *	9.503e+006 *
56	Liquid Velocity	(m/s)	0.1995	0.1995	0.1995	0.1995
57	Vapour Velocity	(m/s)	2.630	2.630	2.630	2.630
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	1110	1115	1120	1125
61	Elevation	(m)	4.000e-002	5.000e-002	6.000e-002	7.000e-002
62	Pressure	(bar)	78.47	78.47	78.47	78.47
63	Temperature	(C)	59.23	59.23	59.22	59.22
64	Vapour Fraction		0.8812	0.8812	0.8812	0.8812
65	Heat Transferred	(kJ/h-m)	778.7	778.6	778.5	778.5
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2530	0.2530	0.2530	0.2530
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2										
3							Unit Set: Mixed			
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5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Water HoldUp		---	---	---	---				
14	Friction Gradient (kPa/m)		6.661e-003	6.661e-003	6.661e-003	6.661e-003				
15	Static Gradient (kPa/m)		4.262e-003	4.262e-003	4.262e-003	4.262e-003				
16	Accel. Gradient (kPa/m)		1.079e-006	1.079e-006	1.079e-006	1.079e-006				
17	Liq. Reynolds		2.036e+005 *	2.036e+005 *	2.036e+005 *	2.036e+005 *				
18	Vap. Reynolds		9.503e+006 *	9.503e+006 *	9.503e+006 *	9.503e+006 *				
19	Liquid Velocity (m/s)		0.1995	0.1995	0.1995	0.1995				
20	Vapour Velocity (m/s)		2.630	2.630	2.630	2.630				
21	Deposit Thickness (mm)		---	---	---	---				
22	Deposit Volume (m3)		---	---	---	---				
23	Length (m)		1130	1135	1140	1145				
24	Elevation (m)		8.400e-002	9.800e-002	0.1120	0.1260				
25	Pressure (bar)		78.47	78.47	78.47	78.47				
26	Temperature (C)		59.22	59.22	59.21	59.21				
27	Vapour Fraction		0.8812	0.8812	0.8812	0.8812				
28	Heat Transferred (kJ/h-m)		778.4	778.4	778.3	778.2				
29	Flow Regime		Transition	Transition	Transition	Transition				
30	Water Flow Regime									
31	Liquid HoldUp		0.2553	0.2553	0.2553	0.2553				
32	Water HoldUp		---	---	---	---				
33	Friction Gradient (kPa/m)		6.421e-003	6.421e-003	6.421e-003	6.421e-003				
34	Static Gradient (kPa/m)		6.006e-003	6.006e-003	6.006e-003	6.006e-003				
35	Accel. Gradient (kPa/m)		1.228e-006	1.228e-006	1.228e-006	1.228e-006				
36	Liq. Reynolds		2.036e+005 *	2.036e+005 *	2.036e+005 *	2.036e+005 *				
37	Vap. Reynolds		9.503e+006 *	9.503e+006 *	9.503e+006 *	9.503e+006 *				
38	Liquid Velocity (m/s)		0.1995	0.1995	0.1995	0.1995				
39	Vapour Velocity (m/s)		2.630	2.630	2.630	2.630				
40	Deposit Thickness (mm)		---	---	---	---				
41	Deposit Volume (m3)		---	---	---	---				
42	Length (m)		1150	1155	1160	1165				
43	Elevation (m)		0.1400	0.1320	0.1240	0.1160				
44	Pressure (bar)		78.47	78.47	78.47	78.47				
45	Temperature (C)		59.21	59.21	59.20	59.20				
46	Vapour Fraction		0.8812	0.8812	0.8812	0.8812				
47	Heat Transferred (kJ/h-m)		778.2	778.2	778.1	778.0				
48	Flow Regime		Transition	Transition	Transition	Transition				
49	Water Flow Regime									
50	Liquid HoldUp		0.2553	0.1820	0.1820	0.1820				
51	Water HoldUp		---	---	---	---				
52	Friction Gradient (kPa/m)		6.421e-003	8.176e-003	8.176e-003	8.176e-003				
53	Static Gradient (kPa/m)		6.006e-003	-2.719e-003	-2.719e-003	-2.719e-003				
54	Accel. Gradient (kPa/m)		1.228e-006	5.393e-007	5.392e-007	5.392e-007				
55	Liq. Reynolds		2.036e+005 *	2.036e+005 *	2.036e+005 *	2.036e+005 *				
56	Vap. Reynolds		9.503e+006 *	9.503e+006 *	9.503e+006 *	9.503e+006 *				
57	Liquid Velocity (m/s)		0.1995	0.1995	0.1995	0.1995				
58	Vapour Velocity (m/s)		2.630	2.630	2.630	2.630				
59	Deposit Thickness (mm)		---	---	---	---				
60	Deposit Volume (m3)		---	---	---	---				
61	Length (m)		1170	1175	1180	1185				
62	Elevation (m)		0.1080	1.000e-001	0.1100	0.1200				
63	Pressure (bar)		78.47	78.47	78.47	78.47				
64	Temperature (C)		59.20	59.20	59.20	59.19				
65	Vapour Fraction		0.8812	0.8812	0.8812	0.8812				
66	Heat Transferred (kJ/h-m)		778.0	777.9	777.8	777.7				
67	Flow Regime		Transition	Transition	Transition	Transition				
68	Water Flow Regime									
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.1820	0.1820	0.2530	0.2530
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	8.176e-003	8.176e-003	6.664e-003	6.664e-003
16	Static Gradient (kPa/m)	-2.719e-003	-2.719e-003	4.240e-003	4.240e-003
17	Accel. Gradient (kPa/m)	5.392e-007	5.392e-007	1.077e-006	1.077e-006
18	Liq. Reynolds	2.036e+005 *	2.036e+005 *	2.036e+005 *	2.036e+005 *
19	Vap. Reynolds	9.503e+006 *	9.503e+006 *	9.503e+006 *	9.503e+006 *
20	Liquid Velocity (m/s)	0.1995	0.1995	0.1995	0.1995
21	Vapour Velocity (m/s)	2.630	2.630	2.629	2.629
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	1190	1195	1200	1205
25	Elevation (m)	0.1300	0.1400	0.1500	0.1340
26	Pressure (bar)	78.47	78.46	78.46	78.46
27	Temperature (C)	59.19	59.19	59.19	59.18
28	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
29	Heat Transferred (kJ/h-m)	777.7	777.6	777.5	777.5
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2530	0.2530	0.2530	0.1807
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.664e-003	6.664e-003	6.664e-003	8.190e-003
35	Static Gradient (kPa/m)	4.240e-003	4.240e-003	4.240e-003	-5.413e-003
36	Accel. Gradient (kPa/m)	1.077e-006	1.077e-006	1.077e-006	2.744e-007
37	Liq. Reynolds	2.036e+005 *	2.036e+005 *	2.036e+005 *	2.036e+005 *
38	Vap. Reynolds	9.503e+006 *	9.503e+006 *	9.503e+006 *	9.503e+006 *
39	Liquid Velocity (m/s)	0.1995	0.1995	0.1995	0.1995
40	Vapour Velocity (m/s)	2.629	2.629	2.629	2.629
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	1210	1215	1220	1225
44	Elevation (m)	0.1180	0.1020	8.600e-002	7.000e-002
45	Pressure (bar)	78.46	78.46	78.46	78.46
46	Temperature (C)	59.18	59.18	59.18	59.18
47	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
48	Heat Transferred (kJ/h-m)	777.5	777.4	777.4	777.3
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1807	0.1807	0.1807	0.1807
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	8.190e-003	8.190e-003	8.190e-003	8.190e-003
54	Static Gradient (kPa/m)	-5.413e-003	-5.413e-003	-5.413e-003	-5.413e-003
55	Accel. Gradient (kPa/m)	2.744e-007	2.744e-007	2.744e-007	2.744e-007
56	Liq. Reynolds	2.036e+005 *	2.036e+005 *	2.036e+005 *	2.036e+005 *
57	Vap. Reynolds	9.503e+006 *	9.503e+006 *	9.503e+006 *	9.503e+006 *
58	Liquid Velocity (m/s)	0.1995	0.1995	0.1995	0.1995
59	Vapour Velocity (m/s)	2.629	2.629	2.629	2.629
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	1230	1235	1240	1245
63	Elevation (m)	5.600e-002	4.200e-002	2.800e-002	1.400e-002
64	Pressure (bar)	78.46	78.46	78.46	78.46
65	Temperature (C)	59.17	59.17	59.17	59.17
66	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
67	Heat Transferred (kJ/h-m)	777.2	777.2	777.1	777.1
68	Flow Regime	Transition	Transition	Transition	Transition
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1810	0.1810	0.1810	0.1810
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	8.186e-003	8.186e-003	8.186e-003	8.186e-003
17	Static Gradient (kPa/m)	-4.742e-003	-4.742e-003	-4.742e-003	-4.742e-003
18	Accel. Gradient (kPa/m)	3.403e-007	3.403e-007	3.403e-007	3.403e-007
19	Liq. Reynolds	2.036e+005 *	2.036e+005 *	2.036e+005 *	2.036e+005 *
20	Vap. Reynolds	9.503e+006 *	9.503e+006 *	9.503e+006 *	9.503e+006 *
21	Liquid Velocity (m/s)	0.1995	0.1995	0.1995	0.1995
22	Vapour Velocity (m/s)	2.629	2.629	2.629	2.629
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	1250	1255	1260	1265
26	Elevation (m)	-9.714e-017	-9.714e-017	-9.714e-017	-9.714e-017
27	Pressure (bar)	78.46	78.46	78.46	78.46
28	Temperature (C)	59.16	59.16	59.16	59.16
29	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
30	Heat Transferred (kJ/h-m)	777.0	776.9	776.8	776.8
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1810	0.2675	0.2675	0.2675
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	8.186e-003	6.320e-003	6.320e-003	6.320e-003
36	Static Gradient (kPa/m)	-4.742e-003	0.0000	0.0000	0.0000
37	Accel. Gradient (kPa/m)	3.403e-007	6.245e-007	6.245e-007	6.245e-007
38	Liq. Reynolds	2.036e+005 *	2.036e+005 *	2.035e+005 *	2.035e+005 *
39	Vap. Reynolds	9.503e+006 *	9.503e+006 *	9.503e+006 *	9.503e+006 *
40	Liquid Velocity (m/s)	0.1995	0.1995	0.1995	0.1995
41	Vapour Velocity (m/s)	2.629	2.629	2.629	2.629
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	1270	1275	1280	1285
45	Elevation (m)	-9.714e-017	-9.714e-017	6.000e-003	1.200e-002
46	Pressure (bar)	78.46	78.46	78.46	78.46
47	Temperature (C)	59.15	59.15	59.15	59.15
48	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
49	Heat Transferred (kJ/h-m)	776.7	776.7	776.6	776.5
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2675	0.2675	0.2507	0.2507
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	6.320e-003	6.320e-003	6.906e-003	6.906e-003
55	Static Gradient (kPa/m)	0.0000	0.0000	2.540e-003	2.540e-003
56	Accel. Gradient (kPa/m)	6.245e-007	6.245e-007	9.334e-007	9.334e-007
57	Liq. Reynolds	2.035e+005 *	2.035e+005 *	2.035e+005 *	2.035e+005 *
58	Vap. Reynolds	9.503e+006 *	9.503e+006 *	9.503e+006 *	9.503e+006 *
59	Liquid Velocity (m/s)	0.1995	0.1995	0.1995	0.1995
60	Vapour Velocity (m/s)	2.629	2.629	2.629	2.629
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	1290	1295	1300	1305
64	Elevation (m)	1.800e-002	2.400e-002	3.000e-002	4.200e-002
65	Pressure (bar)	78.46	78.46	78.46	78.46
66	Temperature (C)	59.15	59.14	59.14	59.14
67	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
68	Heat Transferred (kJ/h-m)	776.5	776.4	776.3	776.3
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2507	0.2507	0.2507	0.2542
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	6.906e-003	6.906e-003	6.906e-003	6.538e-003
18	Static Gradient (kPa/m)	2.540e-003	2.540e-003	2.541e-003	5.132e-003
19	Accel. Gradient (kPa/m)	9.334e-007	9.334e-007	9.334e-007	1.153e-006
20	Liq. Reynolds	2.035e+005 *	2.035e+005 *	2.035e+005 *	2.035e+005 *
21	Vap. Reynolds	9.503e+006 *	9.503e+006 *	9.503e+006 *	9.503e+006 *
22	Liquid Velocity (m/s)	0.1995	0.1995	0.1995	0.1995
23	Vapour Velocity (m/s)	2.629	2.629	2.629	2.629
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	1310	1315	1320	1325
27	Elevation (m)	5.400e-002	6.600e-002	7.800e-002	9.000e-002
28	Pressure (bar)	78.46	78.46	78.46	78.46
29	Temperature (C)	59.14	59.13	59.13	59.13
30	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
31	Heat Transferred (kJ/h-m)	776.2	776.2	776.1	776.0
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2542	0.2542	0.2542	0.2542
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	6.538e-003	6.538e-003	6.538e-003	6.538e-003
37	Static Gradient (kPa/m)	5.132e-003	5.132e-003	5.132e-003	5.132e-003
38	Accel. Gradient (kPa/m)	1.153e-006	1.153e-006	1.153e-006	1.153e-006
39	Liq. Reynolds	2.035e+005 *	2.035e+005 *	2.035e+005 *	2.035e+005 *
40	Vap. Reynolds	9.503e+006 *	9.504e+006 *	9.504e+006 *	9.504e+006 *
41	Liquid Velocity (m/s)	0.1995	0.1995	0.1995	0.1995
42	Vapour Velocity (m/s)	2.629	2.629	2.629	2.629
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	1330	1335	1340	1345
46	Elevation (m)	0.1160	0.1420	0.1680	0.1940
47	Pressure (bar)	78.46	78.45	78.45	78.45
48	Temperature (C)	59.13	59.12	59.12	59.12
49	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
50	Heat Transferred (kJ/h-m)	776.0	775.9	775.8	775.8
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2624	0.2624	0.2624	0.2624
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	5.737e-003	5.737e-003	5.737e-003	5.737e-003
56	Static Gradient (kPa/m)	1.138e-002	1.138e-002	1.138e-002	1.138e-002
57	Accel. Gradient (kPa/m)	1.691e-006	1.691e-006	1.691e-006	1.691e-006
58	Liq. Reynolds	2.035e+005 *	2.035e+005 *	2.035e+005 *	2.035e+005 *
59	Vap. Reynolds	9.504e+006 *	9.504e+006 *	9.504e+006 *	9.504e+006 *
60	Liquid Velocity (m/s)	0.1995	0.1995	0.1995	0.1995
61	Vapour Velocity (m/s)	2.629	2.629	2.629	2.629
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	1350	1355	1360	1365
65	Elevation (m)	0.2200	0.1020	-1.600e-002	-0.1340
66	Pressure (bar)	78.45	78.45	78.45	78.46
67	Temperature (C)	59.12	59.12	59.11	59.11
68	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Heat Transferred (kJ/h-m)	775.7	775.7	775.7	775.7					
14	Flow Regime	Transition	Transition	Transition	Transition					
15	Water Flow Regime									
16	Liquid HoldUp	0.2624	0.1643	0.1643	0.1643					
17	Water HoldUp	---	---	---	---					
18	Friction Gradient (kPa/m)	5.737e-003	8.401e-003	8.401e-003	8.400e-003					
19	Static Gradient (kPa/m)	1.138e-002	-3.757e-002	-3.757e-002	-3.758e-002					
20	Accel. Gradient (kPa/m)	1.691e-006	-2.883e-006	-2.883e-006	-2.882e-006					
21	Liq. Reynolds	2.035e+005 *	2.035e+005 *	2.035e+005 *	2.035e+005 *					
22	Vap. Reynolds	9.504e+006 *	9.504e+006 *	9.504e+006 *	9.504e+006 *					
23	Liquid Velocity (m/s)	0.1995	0.1995	0.1995	0.1995					
24	Vapour Velocity (m/s)	2.629	2.629	2.629	2.629					
25	Deposit Thickness (mm)	---	---	---	---					
26	Deposit Volume (m3)	---	---	---	---					
27	Length (m)	1370	1375	1380	1385					
28	Elevation (m)	-0.2520	-0.3700	-1.016	-1.662					
29	Pressure (bar)	78.46	78.46	78.47	78.47					
30	Temperature (C)	59.11	59.11	59.11	59.11					
31	Vapour Fraction	0.8812	0.8812	0.8812	0.8812					
32	Heat Transferred (kJ/h-m)	775.6	775.6	776.0	776.0					
33	Flow Regime	Transition	Transition	Transition	Transition					
34	Water Flow Regime									
35	Liquid HoldUp	0.1643	0.1643	8.190e-002	8.191e-002					
36	Water HoldUp	---	---	---	---					
37	Friction Gradient (kPa/m)	8.400e-003	8.400e-003	1.143e-002	1.143e-002					
38	Static Gradient (kPa/m)	-3.758e-002	-3.758e-002	-0.1409	-0.1409					
39	Accel. Gradient (kPa/m)	-2.882e-006	-2.882e-006	-1.279e-005	-1.279e-005					
40	Liq. Reynolds	2.035e+005 *	2.035e+005 *	2.035e+005 *	2.035e+005 *					
41	Vap. Reynolds	9.504e+006 *	9.503e+006 *	9.503e+006 *	9.503e+006 *					
42	Liquid Velocity (m/s)	0.1995	0.1995	0.1995	0.1995					
43	Vapour Velocity (m/s)	2.629	2.629	2.628	2.628					
44	Deposit Thickness (mm)	---	---	---	---					
45	Deposit Volume (m3)	---	---	---	---					
46	Length (m)	1390	1395	1400	1405					
47	Elevation (m)	-2.308	-2.954	-3.600	-3.940					
48	Pressure (bar)	78.48	78.48	78.49	78.50					
49	Temperature (C)	59.11	59.11	59.12	59.12					
50	Vapour Fraction	0.8812	0.8812	0.8812	0.8812					
51	Heat Transferred (kJ/h-m)	776.0	776.1	776.1	775.9					
52	Flow Regime	Transition	Transition	Transition	Transition					
53	Water Flow Regime									
54	Liquid HoldUp	8.191e-002	8.191e-002	8.192e-002	0.1290					
55	Water HoldUp	---	---	---	---					
56	Friction Gradient (kPa/m)	1.143e-002	1.143e-002	1.143e-002	9.110e-003					
57	Static Gradient (kPa/m)	-0.1409	-0.1409	-0.1409	-9.369e-002					
58	Accel. Gradient (kPa/m)	-1.279e-005	-1.279e-005	-1.278e-005	-8.347e-006					
59	Liq. Reynolds	2.035e+005 *	2.035e+005 *	2.035e+005 *	2.035e+005 *					
60	Vap. Reynolds	9.503e+006 *	9.503e+006 *	9.502e+006 *	9.502e+006 *					
61	Liquid Velocity (m/s)	0.1995	0.1995	0.1995	0.1995					
62	Vapour Velocity (m/s)	2.628	2.628	2.627	2.627					
63	Deposit Thickness (mm)	---	---	---	---					
64	Deposit Volume (m3)	---	---	---	---					
65	Length (m)	1410	1415	1420	1425					
66	Elevation (m)	-4.280	-4.620	-4.960	-5.300					
67	Pressure (bar)	78.50	78.50	78.51	78.51					
68	Temperature (C)	59.12	59.12	59.11	59.11					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
14	Heat Transferred (kJ/h-m)	775.9	775.8	775.8	775.8
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.1290	0.1290	0.1290	0.1290
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	9.109e-003	9.109e-003	9.108e-003	9.108e-003
20	Static Gradient (kPa/m)	-9.369e-002	-9.370e-002	-9.370e-002	-9.370e-002
21	Accel. Gradient (kPa/m)	-8.347e-006	-8.346e-006	-8.346e-006	-8.345e-006
22	Liq. Reynolds	2.035e+005 *	2.035e+005 *	2.036e+005 *	2.036e+005 *
23	Vap. Reynolds	9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *
24	Liquid Velocity (m/s)	0.1995	0.1995	0.1995	0.1995
25	Vapour Velocity (m/s)	2.627	2.627	2.627	2.627
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	1430	1435	1440	1445
29	Elevation (m)	-5.420	-5.540	-5.660	-5.780
30	Pressure (bar)	78.51	78.52	78.52	78.52
31	Temperature (C)	59.11	59.11	59.11	59.11
32	Vapour Fraction	0.8812	0.8811	0.8811	0.8811
33	Heat Transferred (kJ/h-m)	775.7	775.6	775.6	775.6
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1641	0.1641	0.1641	0.1641
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	8.398e-003	8.398e-003	8.398e-003	8.398e-003
39	Static Gradient (kPa/m)	-3.819e-002	-3.819e-002	-3.819e-002	-3.819e-002
40	Accel. Gradient (kPa/m)	-2.938e-006	-2.938e-006	-2.938e-006	-2.938e-006
41	Liq. Reynolds	2.036e+005 *	2.036e+005 *	2.036e+005 *	2.036e+005 *
42	Vap. Reynolds	9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *
43	Liquid Velocity (m/s)	0.1995	0.1995	0.1995	0.1995
44	Vapour Velocity (m/s)	2.627	2.626	2.626	2.626
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	1450	1455	1460	1465
48	Elevation (m)	-5.900	-5.780	-5.660	-5.540
49	Pressure (bar)	78.52	78.52	78.51	78.51
50	Temperature (C)	59.11	59.10	59.10	59.10
51	Vapour Fraction	0.8811	0.8811	0.8811	0.8811
52	Heat Transferred (kJ/h-m)	775.5	775.4	775.3	775.2
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.1641	0.3174	0.3174	0.3174
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	8.398e-003	7.161e-003	7.161e-003	7.161e-003
58	Static Gradient (kPa/m)	-3.819e-002	6.054e-002	6.054e-002	6.054e-002
59	Accel. Gradient (kPa/m)	-2.938e-006	6.678e-006	6.678e-006	6.679e-006
60	Liq. Reynolds	2.036e+005 *	2.036e+005 *	2.035e+005 *	2.035e+005 *
61	Vap. Reynolds	9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *
62	Liquid Velocity (m/s)	0.1995	0.1995	0.1995	0.1995
63	Vapour Velocity (m/s)	2.626	2.626	2.626	2.626
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	1470	1475	1480	1485
67	Elevation (m)	-5.420	-5.300	-5.500	-5.700
68	Pressure (bar)	78.51	78.50	78.51	78.51
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	59.09	59.09	59.09	59.09
14	Vapour Fraction	0.8811	0.8811	0.8811	0.8811
15	Heat Transferred (kJ/h-m)	775.1	775.0	775.1	775.1
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.3174	0.3174	0.1512	0.1513
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	7.162e-003	7.162e-003	8.609e-003	8.609e-003
21	Static Gradient (kPa/m)	6.054e-002	6.054e-002	-6.053e-002	-6.053e-002
22	Accel. Gradient (kPa/m)	6.679e-006	6.679e-006	-5.122e-006	-5.122e-006
23	Liq. Reynolds	2.035e+005 *	2.035e+005 *	2.035e+005 *	2.035e+005 *
24	Vap. Reynolds	9.502e+006 *	9.502e+006 *	9.502e+006 *	9.502e+006 *
25	Liquid Velocity (m/s)	0.1995	0.1995	0.1995	0.1995
26	Vapour Velocity (m/s)	2.627	2.627	2.627	2.626
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	1490	1495	1500	1505
30	Elevation (m)	-5.900	-6.100	-6.300	-7.360
31	Pressure (bar)	78.51	78.51	78.52	78.52
32	Temperature (C)	59.09	59.09	59.09	59.09
33	Vapour Fraction	0.8811	0.8811	0.8811	0.8811
34	Heat Transferred (kJ/h-m)	775.0	775.0	775.0	0.0000
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1513	0.1513	0.1513	2.347e-002
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	8.608e-003	8.608e-003	8.608e-003	1.403e-002
40	Static Gradient (kPa/m)	-6.053e-002	-6.053e-002	-6.053e-002	-0.1562
41	Accel. Gradient (kPa/m)	-5.122e-006	-5.121e-006	-5.121e-006	-1.402e-005
42	Liq. Reynolds	2.035e+005 *	2.035e+005 *	2.035e+005 *	2.035e+005 *
43	Vap. Reynolds	9.502e+006 *	9.502e+006 *	9.502e+006 *	9.501e+006 *
44	Liquid Velocity (m/s)	0.1995	0.1995	0.1995	0.1995
45	Vapour Velocity (m/s)	2.626	2.626	2.626	2.626
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	1510	1515	1520	1525
49	Elevation (m)	-8.420	-9.480	-10.54	-11.60
50	Pressure (bar)	78.53	78.54	78.54	78.55
51	Temperature (C)	59.10	59.10	59.11	59.11
52	Vapour Fraction	0.8811	0.8811	0.8811	0.8811
53	Heat Transferred (kJ/h-m)	0.0000	0.0000	0.0000	0.0000
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	2.347e-002	2.347e-002	2.347e-002	2.347e-002
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	1.403e-002	1.403e-002	1.403e-002	1.403e-002
59	Static Gradient (kPa/m)	-0.1563	-0.1563	-0.1563	-0.1563
60	Accel. Gradient (kPa/m)	-1.402e-005	-1.402e-005	-1.402e-005	-1.402e-005
61	Liq. Reynolds	2.036e+005 *	2.036e+005 *	2.036e+005 *	2.036e+005 *
62	Vap. Reynolds	9.501e+006 *	9.501e+006 *	9.501e+006 *	9.501e+006 *
63	Liquid Velocity (m/s)	0.1995	0.1995	0.1995	0.1995
64	Vapour Velocity (m/s)	2.626	2.625	2.625	2.625
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	1530	1535	1540	1545
68	Elevation (m)	-9.192	-6.784	-4.376	-1.968
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Pressure (bar)	78.39	78.23	78.07	77.91
14	Temperature (C)	59.07	59.03	58.99	58.94
15	Vapour Fraction	0.8812	0.8813	0.8813	0.8814
16	Heat Transferred (kJ/h-m)	767.4	766.3	765.2	764.1
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	1.000	1.000	1.000	1.000
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	6.611e-003	6.624e-003	6.636e-003	6.649e-003
22	Static Gradient (kPa/m)	3.213	3.213	3.214	3.214
23	Accel. Gradient (kPa/m)	3.186e-004	3.200e-004	3.214e-004	3.228e-004
24	Liq. Reynolds	2.034e+005 *	2.032e+005 *	2.029e+005 *	2.027e+005 *
25	Vap. Reynolds	9.506e+006 *	9.511e+006 *	9.516e+006 *	9.521e+006 *
26	Liquid Velocity (m/s)	0.1995	0.1994	0.1994	0.1993
27	Vapour Velocity (m/s)	2.631	2.636	2.642	2.648
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	1550	1555	1560	1565
31	Elevation (m)	0.4400	0.5820	0.7240	0.8660
32	Pressure (bar)	77.75	77.74	77.74	77.73
33	Temperature (C)	58.90	58.90	58.89	58.89
34	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
35	Heat Transferred (kJ/h-m)	763.0	769.9	769.8	769.7
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	1.000	0.3282	0.3282	0.3281
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	6.662e-003	7.207e-003	7.207e-003	7.207e-003
41	Static Gradient (kPa/m)	3.215	7.344e-002	7.344e-002	7.343e-002
42	Accel. Gradient (kPa/m)	3.242e-004	8.117e-006	8.118e-006	8.119e-006
43	Liq. Reynolds	2.025e+005 *	2.025e+005 *	2.025e+005 *	2.025e+005 *
44	Vap. Reynolds	9.526e+006 *	9.526e+006 *	9.527e+006 *	9.527e+006 *
45	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
46	Vapour Velocity (m/s)	2.654	2.654	2.654	2.654
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	1570	1575	1580	1585
50	Elevation (m)	1.008	1.150	1.088	1.026
51	Pressure (bar)	77.73	77.73	77.73	77.73
52	Temperature (C)	58.89	58.88	58.88	58.88
53	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
54	Heat Transferred (kJ/h-m)	769.6	769.5	769.5	769.5
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.3281	0.3281	0.1723	0.1723
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	7.207e-003	7.208e-003	8.352e-003	8.352e-003
60	Static Gradient (kPa/m)	7.343e-002	7.343e-002	-2.028e-002	-2.028e-002
61	Accel. Gradient (kPa/m)	8.119e-006	8.120e-006	-1.201e-006	-1.201e-006
62	Liq. Reynolds	2.025e+005 *	2.025e+005 *	2.025e+005 *	2.025e+005 *
63	Vap. Reynolds	9.527e+006 *	9.527e+006 *	9.527e+006 *	9.527e+006 *
64	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
65	Vapour Velocity (m/s)	2.654	2.654	2.654	2.654
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	1590	1595	1600	1605
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Elevation (m)	0.9640	0.9020	0.8400	0.7740
14	Pressure (bar)	77.73	77.73	77.73	77.73
15	Temperature (C)	58.88	58.88	58.87	58.87
16	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
17	Heat Transferred (kJ/h-m)	769.4	769.4	769.3	769.3
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.1723	0.1723	0.1723	0.1716
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	8.351e-003	8.351e-003	8.351e-003	8.359e-003
23	Static Gradient (kPa/m)	-2.028e-002	-2.028e-002	-2.028e-002	-2.156e-002
24	Accel. Gradient (kPa/m)	-1.201e-006	-1.201e-006	-1.201e-006	-1.329e-006
25	Liq. Reynolds	2.025e+005 *	2.025e+005 *	2.025e+005 *	2.025e+005 *
26	Vap. Reynolds	9.527e+006 *	9.527e+006 *	9.527e+006 *	9.527e+006 *
27	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
28	Vapour Velocity (m/s)	2.654	2.654	2.654	2.654
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	1610	1615	1620	1625
32	Elevation (m)	0.7080	0.6420	0.5760	0.5100
33	Pressure (bar)	77.73	77.73	77.73	77.73
34	Temperature (C)	58.87	58.87	58.87	58.86
35	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
36	Heat Transferred (kJ/h-m)	769.2	769.2	769.1	769.1
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.1716	0.1716	0.1716	0.1716
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	8.359e-003	8.359e-003	8.359e-003	8.359e-003
42	Static Gradient (kPa/m)	-2.156e-002	-2.156e-002	-2.156e-002	-2.156e-002
43	Accel. Gradient (kPa/m)	-1.329e-006	-1.329e-006	-1.329e-006	-1.329e-006
44	Liq. Reynolds	2.025e+005 *	2.025e+005 *	2.025e+005 *	2.025e+005 *
45	Vap. Reynolds	9.527e+006 *	9.527e+006 *	9.527e+006 *	9.527e+006 *
46	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
47	Vapour Velocity (m/s)	2.654	2.654	2.654	2.654
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	1630	1635	1640	1645
51	Elevation (m)	0.5320	0.5540	0.5760	0.5980
52	Pressure (bar)	77.73	77.73	77.73	77.73
53	Temperature (C)	58.86	58.86	58.86	58.86
54	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
55	Heat Transferred (kJ/h-m)	768.9	768.9	768.8	768.7
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2585	0.2585	0.2585	0.2585
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	6.036e-003	6.036e-003	6.036e-003	6.036e-003
61	Static Gradient (kPa/m)	9.513e-003	9.513e-003	9.513e-003	9.513e-003
62	Accel. Gradient (kPa/m)	1.565e-006	1.565e-006	1.565e-006	1.565e-006
63	Liq. Reynolds	2.025e+005 *	2.025e+005 *	2.025e+005 *	2.024e+005 *
64	Vap. Reynolds	9.527e+006 *	9.527e+006 *	9.527e+006 *	9.527e+006 *
65	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
66	Vapour Velocity (m/s)	2.654	2.654	2.654	2.654
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
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9	PROFILES									
10										
11	Pipe Table									
12										
13	Length	(m)	1650	1655	1660	1665				
14	Elevation	(m)	0.6200	0.6140	0.6080	0.6020				
15	Pressure	(bar)	77.73	77.73	77.73	77.73				
16	Temperature	(C)	58.85	58.85	58.85	58.85				
17	Vapour Fraction		0.8815	0.8815	0.8815	0.8815				
18	Heat Transferred	(kJ/h-m)	768.7	768.7	768.6	768.5				
19	Flow Regime		Transition	Transition	Transition	Transition				
20	Water Flow Regime									
21	Liquid HoldUp		0.2585	0.1812	0.1812	0.1812				
22	Water HoldUp		---	---	---	---				
23	Friction Gradient	(kPa/m)	6.036e-003	8.242e-003	8.242e-003	8.242e-003				
24	Static Gradient	(kPa/m)	9.513e-003	-2.030e-003	-2.030e-003	-2.030e-003				
25	Accel. Gradient	(kPa/m)	1.565e-006	6.254e-007	6.254e-007	6.254e-007				
26	Liq. Reynolds		2.024e+005 *	2.024e+005 *	2.024e+005 *	2.024e+005 *				
27	Vap. Reynolds		9.527e+006 *	9.527e+006 *	9.527e+006 *	9.527e+006 *				
28	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993				
29	Vapour Velocity	(m/s)	2.654	2.654	2.654	2.654				
30	Deposit Thickness	(mm)	---	---	---	---				
31	Deposit Volume	(m3)	---	---	---	---				
32	Length	(m)	1670	1675	1680	1685				
33	Elevation	(m)	0.5960	0.5900	0.5160	0.4420				
34	Pressure	(bar)	77.73	77.73	77.73	77.73				
35	Temperature	(C)	58.84	58.84	58.84	58.84				
36	Vapour Fraction		0.8815	0.8815	0.8815	0.8814				
37	Heat Transferred	(kJ/h-m)	768.5	768.4	768.4	768.3				
38	Flow Regime		Transition	Transition	Transition	Transition				
39	Water Flow Regime									
40	Liquid HoldUp		0.1812	0.1812	0.1704	0.1704				
41	Water HoldUp		---	---	---	---				
42	Friction Gradient	(kPa/m)	8.242e-003	8.242e-003	8.375e-003	8.375e-003				
43	Static Gradient	(kPa/m)	-2.030e-003	-2.030e-003	-2.406e-002	-2.406e-002				
44	Accel. Gradient	(kPa/m)	6.254e-007	6.254e-007	-1.579e-006	-1.579e-006				
45	Liq. Reynolds		2.024e+005 *	2.024e+005 *	2.024e+005 *	2.024e+005 *				
46	Vap. Reynolds		9.527e+006 *	9.527e+006 *	9.527e+006 *	9.527e+006 *				
47	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993				
48	Vapour Velocity	(m/s)	2.654	2.654	2.654	2.654				
49	Deposit Thickness	(mm)	---	---	---	---				
50	Deposit Volume	(m3)	---	---	---	---				
51	Length	(m)	1690	1695	1700	1705				
52	Elevation	(m)	0.3680	0.2940	0.2200	0.2080				
53	Pressure	(bar)	77.73	77.73	77.73	77.73				
54	Temperature	(C)	58.84	58.83	58.83	58.83				
55	Vapour Fraction		0.8814	0.8814	0.8814	0.8814				
56	Heat Transferred	(kJ/h-m)	768.3	768.2	768.2	768.1				
57	Flow Regime		Transition	Transition	Transition	Transition				
58	Water Flow Regime									
59	Liquid HoldUp		0.1704	0.1704	0.1704	0.1803				
60	Water HoldUp		---	---	---	---				
61	Friction Gradient	(kPa/m)	8.375e-003	8.375e-003	8.375e-003	8.252e-003				
62	Static Gradient	(kPa/m)	-2.406e-002	-2.406e-002	-2.406e-002	-4.046e-003				
63	Accel. Gradient	(kPa/m)	-1.579e-006	-1.579e-006	-1.579e-006	4.234e-007				
64	Liq. Reynolds		2.024e+005 *	2.024e+005 *	2.024e+005 *	2.024e+005 *				
65	Vap. Reynolds		9.527e+006 *	9.527e+006 *	9.527e+006 *	9.527e+006 *				
66	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993				
67	Vapour Velocity	(m/s)	2.653	2.653	2.653	2.653				
68	Deposit Thickness	(mm)	---	---	---	---				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	1710	1715	1720	1725
15	Elevation (m)	0.1960	0.1840	0.1720	0.1600
16	Pressure (bar)	77.73	77.73	77.73	77.73
17	Temperature (C)	58.83	58.83	58.82	58.82
18	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
19	Heat Transferred (kJ/h-m)	768.1	768.0	767.9	767.9
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.1803	0.1803	0.1803	0.1803
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	8.252e-003	8.252e-003	8.252e-003	8.252e-003
25	Static Gradient (kPa/m)	-4.046e-003	-4.046e-003	-4.046e-003	-4.046e-003
26	Accel. Gradient (kPa/m)	4.234e-007	4.234e-007	4.234e-007	4.233e-007
27	Liq. Reynolds	2.024e+005 *	2.024e+005 *	2.024e+005 *	2.024e+005 *
28	Vap. Reynolds	9.527e+006 *	9.527e+006 *	9.527e+006 *	9.527e+006 *
29	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
30	Vapour Velocity (m/s)	2.653	2.653	2.653	2.653
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	1730	1735	1740	1745
34	Elevation (m)	0.1620	0.1640	0.1660	0.1680
35	Pressure (bar)	77.73	77.73	77.73	77.73
36	Temperature (C)	58.82	58.82	58.81	58.81
37	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
38	Heat Transferred (kJ/h-m)	767.8	767.7	767.7	767.6
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2469	0.2469	0.2469	0.2469
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	7.250e-003	7.250e-003	7.250e-003	7.250e-003
44	Static Gradient (kPa/m)	8.366e-004	8.366e-004	8.366e-004	8.366e-004
45	Accel. Gradient (kPa/m)	8.140e-007	8.140e-007	8.140e-007	8.140e-007
46	Liq. Reynolds	2.024e+005 *	2.024e+005 *	2.024e+005 *	2.024e+005 *
47	Vap. Reynolds	9.527e+006 *	9.527e+006 *	9.527e+006 *	9.527e+006 *
48	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
49	Vapour Velocity (m/s)	2.653	2.653	2.653	2.653
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	1750	1755	1760	1765
53	Elevation (m)	0.1700	0.1840	0.1980	0.2120
54	Pressure (bar)	77.73	77.73	77.73	77.73
55	Temperature (C)	58.81	58.81	58.81	58.80
56	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
57	Heat Transferred (kJ/h-m)	767.5	767.5	767.4	767.3
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2469	0.2539	0.2539	0.2539
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	7.250e-003	6.500e-003	6.500e-003	6.500e-003
63	Static Gradient (kPa/m)	8.367e-004	5.975e-003	5.975e-003	5.976e-003
64	Accel. Gradient (kPa/m)	8.140e-007	1.256e-006	1.256e-006	1.256e-006
65	Liq. Reynolds	2.024e+005 *	2.024e+005 *	2.024e+005 *	2.024e+005 *
66	Vap. Reynolds	9.527e+006 *	9.527e+006 *	9.527e+006 *	9.527e+006 *
67	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
68	Vapour Velocity (m/s)	2.653	2.653	2.653	2.653
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
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5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
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9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	1770	1775	1780	1785
16	Elevation (m)	0.2260	0.2400	0.2600	0.2800
17	Pressure (bar)	77.73	77.72	77.72	77.72
18	Temperature (C)	58.80	58.80	58.80	58.79
19	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
20	Heat Transferred (kJ/h-m)	767.3	767.2	767.2	767.1
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2539	0.2539	0.2574	0.2574
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	6.500e-003	6.500e-003	6.153e-003	6.153e-003
26	Static Gradient (kPa/m)	5.976e-003	5.976e-003	8.586e-003	8.586e-003
27	Accel. Gradient (kPa/m)	1.256e-006	1.256e-006	1.484e-006	1.484e-006
28	Liq. Reynolds	2.024e+005 *	2.024e+005 *	2.024e+005 *	2.024e+005 *
29	Vap. Reynolds	9.527e+006 *	9.527e+006 *	9.527e+006 *	9.527e+006 *
30	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
31	Vapour Velocity (m/s)	2.653	2.653	2.653	2.653
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	1790	1795	1800	1805
35	Elevation (m)	0.3000	0.3200	0.3400	0.3680
36	Pressure (bar)	77.72	77.72	77.72	77.72
37	Temperature (C)	58.79	58.79	58.79	58.78
38	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
39	Heat Transferred (kJ/h-m)	767.0	767.0	766.9	766.8
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2574	0.2574	0.2574	0.2621
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	6.153e-003	6.153e-003	6.153e-003	5.700e-003
45	Static Gradient (kPa/m)	8.586e-003	8.586e-003	8.586e-003	1.223e-002
46	Accel. Gradient (kPa/m)	1.484e-006	1.484e-006	1.484e-006	1.805e-006
47	Liq. Reynolds	2.024e+005 *	2.024e+005 *	2.024e+005 *	2.024e+005 *
48	Vap. Reynolds	9.527e+006 *	9.527e+006 *	9.527e+006 *	9.527e+006 *
49	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
50	Vapour Velocity (m/s)	2.653	2.653	2.653	2.653
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	1810	1815	1820	1825
54	Elevation (m)	0.3960	0.4240	0.4520	0.4800
55	Pressure (bar)	77.72	77.72	77.72	77.72
56	Temperature (C)	58.78	58.78	58.78	58.77
57	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
58	Heat Transferred (kJ/h-m)	766.8	766.7	766.6	766.6
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2621	0.2621	0.2621	0.2621
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	5.700e-003	5.700e-003	5.700e-003	5.700e-003
64	Static Gradient (kPa/m)	1.223e-002	1.223e-002	1.223e-002	1.223e-002
65	Accel. Gradient (kPa/m)	1.805e-006	1.805e-006	1.805e-006	1.805e-006
66	Liq. Reynolds	2.024e+005 *	2.024e+005 *	2.024e+005 *	2.024e+005 *
67	Vap. Reynolds	9.527e+006 *	9.527e+006 *	9.527e+006 *	9.527e+006 *
68	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.653	2.653	2.653	2.653	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	1830	1835	1840	1845	
17	Elevation	(m)	0.4980	0.5160	0.5340	0.5520	
18	Pressure	(bar)	77.72	77.72	77.71	77.71	
19	Temperature	(C)	58.77	58.77	58.77	58.76	
20	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
21	Heat Transferred	(kJ/h-m)	766.5	766.4	766.4	766.3	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2562	0.2562	0.2562	0.2562	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	6.264e-003	6.264e-003	6.264e-003	6.264e-003	
27	Static Gradient	(kPa/m)	7.734e-003	7.734e-003	7.734e-003	7.734e-003	
28	Accel. Gradient	(kPa/m)	1.409e-006	1.409e-006	1.409e-006	1.409e-006	
29	Liq. Reynolds		2.023e+005 *	2.023e+005 *	2.023e+005 *	2.023e+005 *	
30	Vap. Reynolds		9.527e+006 *	9.528e+006 *	9.528e+006 *	9.528e+006 *	
31	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993	
32	Vapour Velocity	(m/s)	2.653	2.653	2.653	2.653	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	1850	1855	1860	1865	
36	Elevation	(m)	0.5700	0.5820	0.5940	0.6060	
37	Pressure	(bar)	77.71	77.71	77.71	77.71	
38	Temperature	(C)	58.76	58.76	58.76	58.76	
39	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
40	Heat Transferred	(kJ/h-m)	766.2	766.2	766.1	766.1	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2562	0.2528	0.2528	0.2528	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	6.264e-003	6.621e-003	6.621e-003	6.621e-003	
46	Static Gradient	(kPa/m)	7.734e-003	5.105e-003	5.105e-003	5.105e-003	
47	Accel. Gradient	(kPa/m)	1.409e-006	1.180e-006	1.180e-006	1.180e-006	
48	Liq. Reynolds		2.023e+005 *	2.023e+005 *	2.023e+005 *	2.023e+005 *	
49	Vap. Reynolds		9.528e+006 *	9.528e+006 *	9.528e+006 *	9.528e+006 *	
50	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993	
51	Vapour Velocity	(m/s)	2.653	2.653	2.653	2.653	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	1870	1875	1880	1885	
55	Elevation	(m)	0.6180	0.6300	0.6380	0.6460	
56	Pressure	(bar)	77.71	77.71	77.71	77.71	
57	Temperature	(C)	58.75	58.75	58.75	58.75	
58	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
59	Heat Transferred	(kJ/h-m)	766.0	765.9	765.9	765.8	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.2528	0.2528	0.2504	0.2504	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	6.621e-003	6.621e-003	6.867e-003	6.867e-003	
65	Static Gradient	(kPa/m)	5.105e-003	5.105e-003	3.381e-003	3.381e-003	
66	Accel. Gradient	(kPa/m)	1.180e-006	1.180e-006	1.032e-006	1.032e-006	
67	Liq. Reynolds		2.023e+005 *	2.023e+005 *	2.023e+005 *	2.023e+005 *	
68	Vap. Reynolds		9.528e+006 *	9.528e+006 *	9.528e+006 *	9.528e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
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11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993	
14	Vapour Velocity	(m/s)	2.653	2.653	2.653	2.653	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	1890	1895	1900	1905	
18	Elevation	(m)	0.6540	0.6620	0.6700	0.6700	
19	Pressure	(bar)	77.71	77.71	77.71	77.71	
20	Temperature	(C)	58.74	58.74	58.74	58.74	
21	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
22	Heat Transferred	(kJ/h-m)	765.8	765.7	765.6	765.6	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.2504	0.2504	0.2504	0.2660	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	6.867e-003	6.867e-003	6.867e-003	6.293e-003	
28	Static Gradient	(kPa/m)	3.381e-003	3.381e-003	3.381e-003	0.0000	
29	Accel. Gradient	(kPa/m)	1.032e-006	1.032e-006	1.032e-006	6.335e-007	
30	Liq. Reynolds		2.023e+005 *	2.023e+005 *	2.023e+005 *	2.023e+005 *	
31	Vap. Reynolds		9.528e+006 *	9.528e+006 *	9.528e+006 *	9.528e+006 *	
32	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993	
33	Vapour Velocity	(m/s)	2.653	2.653	2.653	2.653	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	1910	1915	1920	1925	
37	Elevation	(m)	0.6700	0.6700	0.6700	0.6700	
38	Pressure	(bar)	77.71	77.71	77.71	77.71	
39	Temperature	(C)	58.73	58.73	58.73	58.73	
40	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
41	Heat Transferred	(kJ/h-m)	765.5	765.4	765.4	765.3	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2660	0.2660	0.2660	0.2660	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	6.293e-003	6.293e-003	6.293e-003	6.293e-003	
47	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000	
48	Accel. Gradient	(kPa/m)	6.335e-007	6.335e-007	6.335e-007	6.335e-007	
49	Liq. Reynolds		2.023e+005 *	2.023e+005 *	2.023e+005 *	2.023e+005 *	
50	Vap. Reynolds		9.528e+006 *	9.528e+006 *	9.528e+006 *	9.528e+006 *	
51	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993	
52	Vapour Velocity	(m/s)	2.653	2.653	2.653	2.653	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	1930	1935	1940	1945	
56	Elevation	(m)	0.6760	0.6820	0.6880	0.6940	
57	Pressure	(bar)	77.71	77.71	77.70	77.70	
58	Temperature	(C)	58.73	58.72	58.72	58.72	
59	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
60	Heat Transferred	(kJ/h-m)	765.3	765.2	765.1	765.1	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2493	0.2493	0.2493	0.2493	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	6.993e-003	6.992e-003	6.992e-003	6.992e-003	
66	Static Gradient	(kPa/m)	2.527e-003	2.527e-003	2.527e-003	2.527e-003	
67	Accel. Gradient	(kPa/m)	9.584e-007	9.584e-007	9.584e-007	9.584e-007	
68	Liq. Reynolds		2.023e+005 *	2.023e+005 *	2.023e+005 *	2.023e+005 *	
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3			Unit Set: Mixed		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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13	Vap. Reynolds	9.528e+006 *	9.528e+006 *	9.528e+006 *	9.528e+006 *
14	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
15	Vapour Velocity (m/s)	2.653	2.653	2.653	2.653
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	1950	1955	1960	1965
19	Elevation (m)	0.7000	0.7000	0.7000	0.7000
20	Pressure (bar)	77.70	77.70	77.70	77.70
21	Temperature (C)	58.72	58.71	58.71	58.71
22	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
23	Heat Transferred (kJ/h-m)	765.0	765.0	764.9	764.8
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2493	0.2660	0.2660	0.2660
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.992e-003	6.293e-003	6.293e-003	6.293e-003
29	Static Gradient (kPa/m)	2.527e-003	0.0000	0.0000	0.0000
30	Accel. Gradient (kPa/m)	9.584e-007	6.335e-007	6.335e-007	6.335e-007
31	Liq. Reynolds	2.023e+005 *	2.023e+005 *	2.023e+005 *	2.023e+005 *
32	Vap. Reynolds	9.528e+006 *	9.528e+006 *	9.528e+006 *	9.528e+006 *
33	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
34	Vapour Velocity (m/s)	2.653	2.653	2.653	2.653
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	1970	1975	1980	1985
38	Elevation (m)	0.7000	0.7000	0.6640	0.6280
39	Pressure (bar)	77.70	77.70	77.70	77.70
40	Temperature (C)	58.71	58.70	58.70	58.70
41	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
42	Heat Transferred (kJ/h-m)	764.8	764.7	764.7	764.7
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2660	0.2660	0.1765	0.1765
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.293e-003	6.293e-003	8.296e-003	8.296e-003
48	Static Gradient (kPa/m)	0.0000	0.0000	-1.198e-002	-1.198e-002
49	Accel. Gradient (kPa/m)	6.335e-007	6.335e-007	-3.710e-007	-3.711e-007
50	Liq. Reynolds	2.023e+005 *	2.023e+005 *	2.023e+005 *	2.023e+005 *
51	Vap. Reynolds	9.528e+006 *	9.528e+006 *	9.528e+006 *	9.528e+006 *
52	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
53	Vapour Velocity (m/s)	2.653	2.653	2.653	2.653
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1990	1995	2000	2005
57	Elevation (m)	0.5920	0.5560	0.5200	0.5200
58	Pressure (bar)	77.70	77.70	77.70	77.70
59	Temperature (C)	58.70	58.70	58.69	58.69
60	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
61	Heat Transferred (kJ/h-m)	764.6	764.6	764.5	764.4
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1765	0.1765	0.1765	0.2660
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	8.296e-003	8.296e-003	8.296e-003	6.293e-003
67	Static Gradient (kPa/m)	-1.198e-002	-1.198e-002	-1.198e-002	0.0000
68	Accel. Gradient (kPa/m)	-3.711e-007	-3.711e-007	-3.711e-007	6.335e-007
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2					
3			Unit Set: Mixed		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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12					
13	Liq. Reynolds	2.023e+005 *	2.023e+005 *	2.023e+005 *	2.023e+005 *
14	Vap. Reynolds	9.528e+006 *	9.528e+006 *	9.528e+006 *	9.528e+006 *
15	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
16	Vapour Velocity (m/s)	2.653	2.653	2.653	2.653
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	2010	2015	2020	2025
20	Elevation (m)	0.5200	0.5200	0.5200	0.5200
21	Pressure (bar)	77.70	77.70	77.70	77.70
22	Temperature (C)	58.69	58.69	58.69	58.68
23	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
24	Heat Transferred (kJ/h-m)	764.3	764.3	764.2	764.2
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2660	0.2660	0.2660	0.2660
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	6.293e-003	6.294e-003	6.294e-003	6.294e-003
30	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
31	Accel. Gradient (kPa/m)	6.335e-007	6.335e-007	6.335e-007	6.335e-007
32	Liq. Reynolds	2.023e+005 *	2.022e+005 *	2.022e+005 *	2.022e+005 *
33	Vap. Reynolds	9.528e+006 *	9.528e+006 *	9.528e+006 *	9.528e+006 *
34	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
35	Vapour Velocity (m/s)	2.653	2.652	2.652	2.652
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	2030	2035	2040	2045
39	Elevation (m)	0.5520	0.5840	0.6160	0.6480
40	Pressure (bar)	77.70	77.70	77.70	77.70
41	Temperature (C)	58.68	58.68	58.68	58.67
42	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
43	Heat Transferred (kJ/h-m)	764.1	764.0	764.0	763.9
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2644	0.2644	0.2644	0.2644
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	5.537e-003	5.537e-003	5.537e-003	5.537e-003
49	Static Gradient (kPa/m)	1.407e-002	1.407e-002	1.407e-002	1.407e-002
50	Accel. Gradient (kPa/m)	1.974e-006	1.974e-006	1.974e-006	1.974e-006
51	Liq. Reynolds	2.022e+005 *	2.022e+005 *	2.022e+005 *	2.022e+005 *
52	Vap. Reynolds	9.528e+006 *	9.528e+006 *	9.528e+006 *	9.528e+006 *
53	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
54	Vapour Velocity (m/s)	2.652	2.652	2.652	2.652
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	2050	2055	2060	2065
58	Elevation (m)	0.6800	0.6700	0.6600	0.6500
59	Pressure (bar)	77.70	77.70	77.70	77.70
60	Temperature (C)	58.67	58.67	58.67	58.66
61	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
62	Heat Transferred (kJ/h-m)	763.8	763.8	763.8	763.7
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2644	0.1806	0.1806	0.1806
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	5.537e-003	8.247e-003	8.247e-003	8.246e-003
68	Static Gradient (kPa/m)	1.407e-002	-3.376e-003	-3.377e-003	-3.377e-003
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
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8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient	(kPa/m)	1.974e-006	4.903e-007	4.902e-007	4.902e-007	
14	Liq. Reynolds		2.022e+005 *	2.022e+005 *	2.022e+005 *	2.022e+005 *	
15	Vap. Reynolds		9.528e+006 *	9.528e+006 *	9.528e+006 *	9.528e+006 *	
16	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993	
17	Vapour Velocity	(m/s)	2.652	2.652	2.652	2.652	
18	Deposit Thickness	(mm)	---	---	---	---	
19	Deposit Volume	(m3)	---	---	---	---	
20	Length	(m)	2070	2075	2080	2085	
21	Elevation	(m)	0.6400	0.6300	0.6340	0.6380	
22	Pressure	(bar)	77.70	77.70	77.70	77.69	
23	Temperature	(C)	58.66	58.66	58.66	58.66	
24	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
25	Heat Transferred	(kJ/h-m)	763.6	763.6	763.5	763.4	
26	Flow Regime		Transition	Transition	Transition	Transition	
27	Water Flow Regime						
28	Liquid HoldUp		0.1806	0.1806	0.2482	0.2482	
29	Water HoldUp		---	---	---	---	
30	Friction Gradient	(kPa/m)	8.246e-003	8.246e-003	7.118e-003	7.118e-003	
31	Static Gradient	(kPa/m)	-3.377e-003	-3.377e-003	1.679e-003	1.679e-003	
32	Accel. Gradient	(kPa/m)	4.902e-007	4.902e-007	8.856e-007	8.856e-007	
33	Liq. Reynolds		2.022e+005 *	2.022e+005 *	2.022e+005 *	2.022e+005 *	
34	Vap. Reynolds		9.528e+006 *	9.528e+006 *	9.528e+006 *	9.528e+006 *	
35	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993	
36	Vapour Velocity	(m/s)	2.652	2.652	2.652	2.652	
37	Deposit Thickness	(mm)	---	---	---	---	
38	Deposit Volume	(m3)	---	---	---	---	
39	Length	(m)	2090	2095	2100	2105	
40	Elevation	(m)	0.6420	0.6460	0.6500	0.6600	
41	Pressure	(bar)	77.69	77.69	77.69	77.69	
42	Temperature	(C)	58.65	58.65	58.65	58.65	
43	Vapour Fraction		0.8814	0.8813	0.8813	0.8813	
44	Heat Transferred	(kJ/h-m)	763.3	763.3	763.2	763.2	
45	Flow Regime		Transition	Transition	Transition	Transition	
46	Water Flow Regime						
47	Liquid HoldUp		0.2482	0.2482	0.2482	0.2516	
48	Water HoldUp		---	---	---	---	
49	Friction Gradient	(kPa/m)	7.118e-003	7.118e-003	7.118e-003	6.741e-003	
50	Static Gradient	(kPa/m)	1.679e-003	1.679e-003	1.679e-003	4.241e-003	
51	Accel. Gradient	(kPa/m)	8.856e-007	8.856e-007	8.856e-007	1.106e-006	
52	Liq. Reynolds		2.022e+005 *	2.022e+005 *	2.022e+005 *	2.022e+005 *	
53	Vap. Reynolds		9.528e+006 *	9.528e+006 *	9.528e+006 *	9.528e+006 *	
54	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993	
55	Vapour Velocity	(m/s)	2.652	2.652	2.652	2.652	
56	Deposit Thickness	(mm)	---	---	---	---	
57	Deposit Volume	(m3)	---	---	---	---	
58	Length	(m)	2110	2115	2120	2125	
59	Elevation	(m)	0.6700	0.6800	0.6900	0.7000	
60	Pressure	(bar)	77.69	77.69	77.69	77.69	
61	Temperature	(C)	58.64	58.64	58.64	58.64	
62	Vapour Fraction		0.8813	0.8813	0.8813	0.8813	
63	Heat Transferred	(kJ/h-m)	763.1	763.0	763.0	762.9	
64	Flow Regime		Transition	Transition	Transition	Transition	
65	Water Flow Regime						
66	Liquid HoldUp		0.2516	0.2516	0.2516	0.2516	
67	Water HoldUp		---	---	---	---	
68	Friction Gradient	(kPa/m)	6.741e-003	6.741e-003	6.741e-003	6.741e-003	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 62 of 937	

1			Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	4.241e-003	4.241e-003	4.241e-003	4.241e-003
14	Accel. Gradient	(kPa/m)	1.106e-006	1.106e-006	1.106e-006	1.106e-006
15	Liq. Reynolds		2.022e+005 *	2.022e+005 *	2.022e+005 *	2.022e+005 *
16	Vap. Reynolds		9.528e+006 *	9.528e+006 *	9.528e+006 *	9.529e+006 *
17	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993
18	Vapour Velocity	(m/s)	2.652	2.652	2.652	2.652
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	2130	2135	2140	2145
22	Elevation	(m)	0.7060	0.7120	0.7180	0.7240
23	Pressure	(bar)	77.69	77.69	77.69	77.69
24	Temperature	(C)	58.63	58.63	58.63	58.63
25	Vapour Fraction		0.8813	0.8813	0.8813	0.8813
26	Heat Transferred	(kJ/h-m)	762.9	762.8	762.7	762.7
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2493	0.2493	0.2493	0.2493
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	6.991e-003	6.991e-003	6.991e-003	6.990e-003
32	Static Gradient	(kPa/m)	2.528e-003	2.528e-003	2.528e-003	2.528e-003
33	Accel. Gradient	(kPa/m)	9.582e-007	9.582e-007	9.582e-007	9.582e-007
34	Liq. Reynolds		2.022e+005 *	2.022e+005 *	2.022e+005 *	2.022e+005 *
35	Vap. Reynolds		9.529e+006 *	9.529e+006 *	9.529e+006 *	9.529e+006 *
36	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993
37	Vapour Velocity	(m/s)	2.652	2.652	2.652	2.652
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	2150	2155	2160	2165
41	Elevation	(m)	0.7300	0.7200	0.7100	0.7000
42	Pressure	(bar)	77.69	77.69	77.69	77.69
43	Temperature	(C)	58.63	58.62	58.62	58.62
44	Vapour Fraction		0.8813	0.8813	0.8813	0.8813
45	Heat Transferred	(kJ/h-m)	762.6	762.6	762.5	762.5
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2493	0.1807	0.1807	0.1807
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	6.990e-003	8.246e-003	8.246e-003	8.246e-003
51	Static Gradient	(kPa/m)	2.528e-003	-3.377e-003	-3.377e-003	-3.377e-003
52	Accel. Gradient	(kPa/m)	9.582e-007	4.902e-007	4.901e-007	4.901e-007
53	Liq. Reynolds		2.022e+005 *	2.022e+005 *	2.022e+005 *	2.022e+005 *
54	Vap. Reynolds		9.529e+006 *	9.529e+006 *	9.529e+006 *	9.529e+006 *
55	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993
56	Vapour Velocity	(m/s)	2.652	2.652	2.652	2.652
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	2170	2175	2180	2185
60	Elevation	(m)	0.6900	0.6800	0.6780	0.6760
61	Pressure	(bar)	77.69	77.69	77.69	77.69
62	Temperature	(C)	58.62	58.61	58.61	58.61
63	Vapour Fraction		0.8813	0.8813	0.8813	0.8813
64	Heat Transferred	(kJ/h-m)	762.4	762.4	762.3	762.2
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.1807	0.1807	0.1819	0.1819
68	Water HoldUp		---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 63 of 937	

1				Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	8.246e-003	8.246e-003	8.231e-003	8.231e-003
14	Static Gradient	(kPa/m)	-3.377e-003	-3.377e-003	-6.720e-004	-6.721e-004
15	Accel. Gradient	(kPa/m)	4.901e-007	4.901e-007	7.609e-007	7.609e-007
16	Liq. Reynolds		2.022e+005 *	2.022e+005 *	2.022e+005 *	2.022e+005 *
17	Vap. Reynolds		9.529e+006 *	9.529e+006 *	9.529e+006 *	9.529e+006 *
18	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993
19	Vapour Velocity	(m/s)	2.652	2.652	2.652	2.652
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	2191	2196	2201	2206
23	Elevation	(m)	0.6740	0.6720	0.6700	0.6680
24	Pressure	(bar)	77.69	77.69	77.68	77.68
25	Temperature	(C)	58.61	58.61	58.60	58.60
26	Vapour Fraction		0.8813	0.8813	0.8813	0.8813
27	Heat Transferred	(kJ/h-m)	762.2	762.1	762.1	762.0
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.1819	0.1819	0.1819	0.1819
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	8.231e-003	8.231e-003	8.231e-003	8.231e-003
33	Static Gradient	(kPa/m)	-6.721e-004	-6.721e-004	-6.721e-004	-6.785e-004
34	Accel. Gradient	(kPa/m)	7.609e-007	7.609e-007	7.609e-007	7.603e-007
35	Liq. Reynolds		2.021e+005 *	2.021e+005 *	2.021e+005 *	2.021e+005 *
36	Vap. Reynolds		9.529e+006 *	9.529e+006 *	9.529e+006 *	9.529e+006 *
37	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993
38	Vapour Velocity	(m/s)	2.652	2.652	2.652	2.652
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	2211	2216	2221	2226
42	Elevation	(m)	0.6660	0.6640	0.6620	0.6600
43	Pressure	(bar)	77.68	77.68	77.68	77.68
44	Temperature	(C)	58.60	58.60	58.59	58.59
45	Vapour Fraction		0.8813	0.8813	0.8813	0.8813
46	Heat Transferred	(kJ/h-m)	761.9	761.9	761.8	761.8
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.1819	0.1819	0.1819	0.1819
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	8.231e-003	8.231e-003	8.231e-003	8.231e-003
52	Static Gradient	(kPa/m)	-6.785e-004	-6.785e-004	-6.785e-004	-6.785e-004
53	Accel. Gradient	(kPa/m)	7.603e-007	7.603e-007	7.603e-007	7.602e-007
54	Liq. Reynolds		2.021e+005 *	2.021e+005 *	2.021e+005 *	2.021e+005 *
55	Vap. Reynolds		9.529e+006 *	9.529e+006 *	9.529e+006 *	9.529e+006 *
56	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993
57	Vapour Velocity	(m/s)	2.652	2.652	2.652	2.652
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	2231	2236	2241	2246
61	Elevation	(m)	0.6460	0.6320	0.6180	0.6040
62	Pressure	(bar)	77.68	77.68	77.68	77.68
63	Temperature	(C)	58.59	58.59	58.59	58.58
64	Vapour Fraction		0.8813	0.8813	0.8813	0.8813
65	Heat Transferred	(kJ/h-m)	761.7	761.6	761.6	761.5
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.1800	0.1800	0.1800	0.1800
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Water HoldUp		---	---	---	---	
14	Friction Gradient (kPa/m)		8.252e-003	8.252e-003	8.252e-003	8.252e-003	
15	Static Gradient (kPa/m)		-4.717e-003	-4.717e-003	-4.717e-003	-4.717e-003	
16	Accel. Gradient (kPa/m)		3.559e-007	3.559e-007	3.559e-007	3.559e-007	
17	Liq. Reynolds		2.021e+005 *	2.021e+005 *	2.021e+005 *	2.021e+005 *	
18	Vap. Reynolds		9.529e+006 *	9.529e+006 *	9.529e+006 *	9.529e+006 *	
19	Liquid Velocity (m/s)		0.1993	0.1993	0.1993	0.1993	
20	Vapour Velocity (m/s)		2.652	2.652	2.652	2.652	
21	Deposit Thickness (mm)		---	---	---	---	
22	Deposit Volume (m3)		---	---	---	---	
23	Length (m)		2251	2256	2261	2266	
24	Elevation (m)		0.5900	0.5980	0.6060	0.6140	
25	Pressure (bar)		77.68	77.68	77.68	77.68	
26	Temperature (C)		58.58	58.58	58.58	58.57	
27	Vapour Fraction		0.8813	0.8813	0.8813	0.8813	
28	Heat Transferred (kJ/h-m)		761.5	761.4	761.3	761.2	
29	Flow Regime		Transition	Transition	Transition	Transition	
30	Water Flow Regime						
31	Liquid HoldUp		0.1800	0.2505	0.2505	0.2505	
32	Water HoldUp		---	---	---	---	
33	Friction Gradient (kPa/m)		8.252e-003	6.864e-003	6.864e-003	6.863e-003	
34	Static Gradient (kPa/m)		-4.717e-003	3.382e-003	3.382e-003	3.382e-003	
35	Accel. Gradient (kPa/m)		3.558e-007	1.031e-006	1.031e-006	1.031e-006	
36	Liq. Reynolds		2.021e+005 *	2.021e+005 *	2.021e+005 *	2.021e+005 *	
37	Vap. Reynolds		9.529e+006 *	9.529e+006 *	9.529e+006 *	9.529e+006 *	
38	Liquid Velocity (m/s)		0.1993	0.1993	0.1993	0.1993	
39	Vapour Velocity (m/s)		2.652	2.652	2.652	2.652	
40	Deposit Thickness (mm)		---	---	---	---	
41	Deposit Volume (m3)		---	---	---	---	
42	Length (m)		2271	2276	2281	2286	
43	Elevation (m)		0.6220	0.6300	0.6400	0.6500	
44	Pressure (bar)		77.68	77.68	77.68	77.68	
45	Temperature (C)		58.57	58.57	58.57	58.56	
46	Vapour Fraction		0.8813	0.8813	0.8813	0.8813	
47	Heat Transferred (kJ/h-m)		761.2	761.1	761.1	761.0	
48	Flow Regime		Transition	Transition	Transition	Transition	
49	Water Flow Regime						
50	Liquid HoldUp		0.2505	0.2505	0.2517	0.2517	
51	Water HoldUp		---	---	---	---	
52	Friction Gradient (kPa/m)		6.863e-003	6.863e-003	6.739e-003	6.739e-003	
53	Static Gradient (kPa/m)		3.382e-003	3.382e-003	4.242e-003	4.242e-003	
54	Accel. Gradient (kPa/m)		1.031e-006	1.031e-006	1.105e-006	1.105e-006	
55	Liq. Reynolds		2.021e+005 *	2.021e+005 *	2.021e+005 *	2.021e+005 *	
56	Vap. Reynolds		9.529e+006 *	9.529e+006 *	9.529e+006 *	9.529e+006 *	
57	Liquid Velocity (m/s)		0.1993	0.1993	0.1993	0.1993	
58	Vapour Velocity (m/s)		2.652	2.652	2.652	2.652	
59	Deposit Thickness (mm)		---	---	---	---	
60	Deposit Volume (m3)		---	---	---	---	
61	Length (m)		2291	2296	2301	2306	
62	Elevation (m)		0.6600	0.6700	0.6800	0.6740	
63	Pressure (bar)		77.68	77.68	77.68	77.68	
64	Temperature (C)		58.56	58.56	58.56	58.56	
65	Vapour Fraction		0.8813	0.8813	0.8813	0.8813	
66	Heat Transferred (kJ/h-m)		760.9	760.9	760.8	760.8	
67	Flow Regime		Transition	Transition	Transition	Transition	
68	Water Flow Regime						
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 65 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.2517	0.2517	0.2517	0.1813
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	6.739e-003	6.739e-003	6.739e-003	8.237e-003
16	Static Gradient (kPa/m)	4.242e-003	4.242e-003	4.242e-003	-2.031e-003
17	Accel. Gradient (kPa/m)	1.105e-006	1.105e-006	1.105e-006	6.247e-007
18	Liq. Reynolds	2.021e+005 *	2.021e+005 *	2.021e+005 *	2.021e+005 *
19	Vap. Reynolds	9.529e+006 *	9.529e+006 *	9.529e+006 *	9.529e+006 *
20	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
21	Vapour Velocity (m/s)	2.652	2.652	2.652	2.652
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	2311	2316	2321	2326
25	Elevation (m)	0.6680	0.6620	0.6560	0.6500
26	Pressure (bar)	77.68	77.68	77.68	77.68
27	Temperature (C)	58.55	58.55	58.55	58.55
28	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
29	Heat Transferred (kJ/h-m)	760.7	760.7	760.6	760.6
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1813	0.1813	0.1813	0.1813
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	8.237e-003	8.237e-003	8.237e-003	8.237e-003
35	Static Gradient (kPa/m)	-2.031e-003	-2.031e-003	-2.031e-003	-2.031e-003
36	Accel. Gradient (kPa/m)	6.247e-007	6.247e-007	6.247e-007	6.247e-007
37	Liq. Reynolds	2.021e+005 *	2.021e+005 *	2.021e+005 *	2.021e+005 *
38	Vap. Reynolds	9.529e+006 *	9.529e+006 *	9.529e+006 *	9.529e+006 *
39	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
40	Vapour Velocity (m/s)	2.652	2.652	2.652	2.652
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	2331	2336	2341	2346
44	Elevation (m)	0.6440	0.6380	0.6320	0.6260
45	Pressure (bar)	77.68	77.67	77.67	77.67
46	Temperature (C)	58.54	58.54	58.54	58.54
47	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
48	Heat Transferred (kJ/h-m)	760.5	760.4	760.4	760.3
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1813	0.1813	0.1813	0.1813
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	8.237e-003	8.237e-003	8.237e-003	8.237e-003
54	Static Gradient (kPa/m)	-2.031e-003	-2.031e-003	-2.031e-003	-2.031e-003
55	Accel. Gradient (kPa/m)	6.247e-007	6.247e-007	6.247e-007	6.247e-007
56	Liq. Reynolds	2.021e+005 *	2.021e+005 *	2.021e+005 *	2.021e+005 *
57	Vap. Reynolds	9.529e+006 *	9.529e+006 *	9.529e+006 *	9.529e+006 *
58	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
59	Vapour Velocity (m/s)	2.652	2.652	2.652	2.652
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	2351	2356	2361	2366
63	Elevation (m)	0.6200	0.6180	0.6160	0.6140
64	Pressure (bar)	77.67	77.67	77.67	77.67
65	Temperature (C)	58.54	58.53	58.53	58.53
66	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
67	Heat Transferred (kJ/h-m)	760.3	760.2	760.1	760.1
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 66 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1813	0.1820	0.1820	0.1820
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	8.237e-003	8.230e-003	8.230e-003	8.230e-003
17	Static Gradient (kPa/m)	-2.031e-003	-6.786e-004	-6.786e-004	-6.786e-004
18	Accel. Gradient (kPa/m)	6.247e-007	7.601e-007	7.601e-007	7.601e-007
19	Liq. Reynolds	2.021e+005 *	2.021e+005 *	2.021e+005 *	2.021e+005 *
20	Vap. Reynolds	9.529e+006 *	9.529e+006 *	9.529e+006 *	9.529e+006 *
21	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
22	Vapour Velocity (m/s)	2.652	2.652	2.651	2.651
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	2371	2376	2381	2386
26	Elevation (m)	0.6120	0.6100	0.6200	0.6300
27	Pressure (bar)	77.67	77.67	77.67	77.67
28	Temperature (C)	58.53	58.52	58.52	58.52
29	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
30	Heat Transferred (kJ/h-m)	760.0	760.0	759.9	759.8
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1820	0.1820	0.2517	0.2517
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	8.230e-003	8.230e-003	6.740e-003	6.740e-003
36	Static Gradient (kPa/m)	-6.786e-004	-6.786e-004	4.230e-003	4.230e-003
37	Accel. Gradient (kPa/m)	7.601e-007	7.601e-007	1.104e-006	1.104e-006
38	Liq. Reynolds	2.021e+005 *	2.021e+005 *	2.020e+005 *	2.020e+005 *
39	Vap. Reynolds	9.529e+006 *	9.529e+006 *	9.529e+006 *	9.529e+006 *
40	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
41	Vapour Velocity (m/s)	2.651	2.651	2.651	2.651
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	2391	2396	2401	2406
45	Elevation (m)	0.6400	0.6500	0.6600	0.6780
46	Pressure (bar)	77.67	77.67	77.67	77.67
47	Temperature (C)	58.52	58.52	58.51	58.51
48	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
49	Heat Transferred (kJ/h-m)	759.7	759.7	759.6	759.5
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2517	0.2517	0.2517	0.2564
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	6.740e-003	6.740e-003	6.740e-003	6.259e-003
55	Static Gradient (kPa/m)	4.230e-003	4.230e-003	4.230e-003	7.738e-003
56	Accel. Gradient (kPa/m)	1.104e-006	1.104e-006	1.104e-006	1.409e-006
57	Liq. Reynolds	2.020e+005 *	2.020e+005 *	2.020e+005 *	2.020e+005 *
58	Vap. Reynolds	9.529e+006 *	9.529e+006 *	9.529e+006 *	9.529e+006 *
59	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
60	Vapour Velocity (m/s)	2.651	2.651	2.651	2.651
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	2411	2416	2421	2426
64	Elevation (m)	0.6960	0.7140	0.7320	0.7500
65	Pressure (bar)	77.67	77.67	77.67	77.67
66	Temperature (C)	58.51	58.51	58.50	58.50
67	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
68	Heat Transferred (kJ/h-m)	759.5	759.4	759.4	759.3
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2564	0.2564	0.2564	0.2564
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	6.259e-003	6.259e-003	6.259e-003	6.259e-003
18	Static Gradient (kPa/m)	7.738e-003	7.738e-003	7.738e-003	7.739e-003
19	Accel. Gradient (kPa/m)	1.409e-006	1.409e-006	1.409e-006	1.409e-006
20	Liq. Reynolds	2.020e+005 *	2.020e+005 *	2.020e+005 *	2.020e+005 *
21	Vap. Reynolds	9.529e+006 *	9.529e+006 *	9.529e+006 *	9.530e+006 *
22	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
23	Vapour Velocity (m/s)	2.651	2.651	2.651	2.651
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	2431	2436	2441	2446
27	Elevation (m)	0.7320	0.7140	0.6960	0.6780
28	Pressure (bar)	77.67	77.67	77.67	77.67
29	Temperature (C)	58.50	58.50	58.49	58.49
30	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
31	Heat Transferred (kJ/h-m)	759.3	759.2	759.2	759.1
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1794	0.1794	0.1794	0.1794
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	8.258e-003	8.258e-003	8.258e-003	8.258e-003
37	Static Gradient (kPa/m)	-6.052e-003	-6.052e-003	-6.052e-003	-6.052e-003
38	Accel. Gradient (kPa/m)	2.221e-007	2.221e-007	2.221e-007	2.221e-007
39	Liq. Reynolds	2.020e+005 *	2.020e+005 *	2.020e+005 *	2.020e+005 *
40	Vap. Reynolds	9.530e+006 *	9.530e+006 *	9.530e+006 *	9.530e+006 *
41	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
42	Vapour Velocity (m/s)	2.651	2.651	2.651	2.651
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	2451	2456	2461	2466
46	Elevation (m)	0.6600	0.6700	0.6800	0.6900
47	Pressure (bar)	77.67	77.66	77.66	77.66
48	Temperature (C)	58.49	58.49	58.49	58.48
49	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
50	Heat Transferred (kJ/h-m)	759.1	758.9	758.9	758.8
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1794	0.2517	0.2517	0.2517
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	8.258e-003	6.738e-003	6.738e-003	6.738e-003
56	Static Gradient (kPa/m)	-6.052e-003	4.243e-003	4.243e-003	4.243e-003
57	Accel. Gradient (kPa/m)	2.221e-007	1.105e-006	1.105e-006	1.105e-006
58	Liq. Reynolds	2.020e+005 *	2.020e+005 *	2.020e+005 *	2.020e+005 *
59	Vap. Reynolds	9.530e+006 *	9.530e+006 *	9.530e+006 *	9.530e+006 *
60	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
61	Vapour Velocity (m/s)	2.651	2.651	2.651	2.651
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	2471	2476	2481	2486
65	Elevation (m)	0.7000	0.7100	0.7200	0.7300
66	Pressure (bar)	77.66	77.66	77.66	77.66
67	Temperature (C)	58.48	58.48	58.48	58.47
68	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Heat Transferred (kJ/h-m)	758.8	758.7	758.6	758.6					
14	Flow Regime	Transition	Transition	Transition	Transition					
15	Water Flow Regime									
16	Liquid HoldUp	0.2517	0.2517	0.2517	0.2517					
17	Water HoldUp	---	---	---	---					
18	Friction Gradient (kPa/m)	6.738e-003	6.738e-003	6.738e-003	6.738e-003					
19	Static Gradient (kPa/m)	4.243e-003	4.243e-003	4.243e-003	4.243e-003					
20	Accel. Gradient (kPa/m)	1.105e-006	1.105e-006	1.105e-006	1.105e-006					
21	Liq. Reynolds	2.020e+005 *	2.020e+005 *	2.020e+005 *	2.020e+005 *					
22	Vap. Reynolds	9.530e+006 *	9.530e+006 *	9.530e+006 *	9.530e+006 *					
23	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993					
24	Vapour Velocity (m/s)	2.651	2.651	2.651	2.651					
25	Deposit Thickness (mm)	---	---	---	---					
26	Deposit Volume (m3)	---	---	---	---					
27	Length (m)	2491	2496	2501	2506					
28	Elevation (m)	0.7400	0.7500	0.7600	0.7380					
29	Pressure (bar)	77.66	77.66	77.66	77.66					
30	Temperature (C)	58.47	58.47	58.47	58.46					
31	Vapour Fraction	0.8813	0.8813	0.8813	0.8813					
32	Heat Transferred (kJ/h-m)	758.5	758.4	758.4	758.4					
33	Flow Regime	Transition	Transition	Transition	Transition					
34	Water Flow Regime									
35	Liquid HoldUp	0.2517	0.2517	0.2517	0.1788					
36	Water HoldUp	---	---	---	---					
37	Friction Gradient (kPa/m)	6.738e-003	6.738e-003	6.738e-003	8.265e-003					
38	Static Gradient (kPa/m)	4.243e-003	4.243e-003	4.243e-003	-7.380e-003					
39	Accel. Gradient (kPa/m)	1.105e-006	1.105e-006	1.105e-006	8.911e-008					
40	Liq. Reynolds	2.020e+005 *	2.020e+005 *	2.020e+005 *	2.020e+005 *					
41	Vap. Reynolds	9.530e+006 *	9.530e+006 *	9.530e+006 *	9.530e+006 *					
42	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993					
43	Vapour Velocity (m/s)	2.651	2.651	2.651	2.651					
44	Deposit Thickness (mm)	---	---	---	---					
45	Deposit Volume (m3)	---	---	---	---					
46	Length (m)	2511	2516	2521	2526					
47	Elevation (m)	0.7160	0.6940	0.6720	0.6500					
48	Pressure (bar)	77.66	77.66	77.66	77.66					
49	Temperature (C)	58.46	58.46	58.46	58.46					
50	Vapour Fraction	0.8812	0.8812	0.8812	0.8812					
51	Heat Transferred (kJ/h-m)	758.3	758.3	758.2	758.2					
52	Flow Regime	Transition	Transition	Transition	Transition					
53	Water Flow Regime									
54	Liquid HoldUp	0.1788	0.1788	0.1788	0.1788					
55	Water HoldUp	---	---	---	---					
56	Friction Gradient (kPa/m)	8.265e-003	8.265e-003	8.265e-003	8.265e-003					
57	Static Gradient (kPa/m)	-7.380e-003	-7.380e-003	-7.380e-003	-7.380e-003					
58	Accel. Gradient (kPa/m)	8.909e-008	8.908e-008	8.907e-008	8.905e-008					
59	Liq. Reynolds	2.020e+005 *	2.020e+005 *	2.020e+005 *	2.020e+005 *					
60	Vap. Reynolds	9.530e+006 *	9.530e+006 *	9.530e+006 *	9.530e+006 *					
61	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993					
62	Vapour Velocity (m/s)	2.651	2.651	2.651	2.651					
63	Deposit Thickness (mm)	---	---	---	---					
64	Deposit Volume (m3)	---	---	---	---					
65	Length (m)	2531	2536	2541	2546					
66	Elevation (m)	0.6140	0.5780	0.5420	0.5060					
67	Pressure (bar)	77.66	77.66	77.66	77.66					
68	Temperature (C)	58.45	58.45	58.45	58.45					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
14	Heat Transferred (kJ/h-m)	758.1	758.1	758.0	758.0
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.1766	0.1766	0.1766	0.1766
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	8.292e-003	8.291e-003	8.291e-003	8.291e-003
20	Static Gradient (kPa/m)	-1.198e-002	-1.198e-002	-1.198e-002	-1.198e-002
21	Accel. Gradient (kPa/m)	-3.712e-007	-3.712e-007	-3.712e-007	-3.712e-007
22	Liq. Reynolds	2.020e+005 *	2.020e+005 *	2.020e+005 *	2.020e+005 *
23	Vap. Reynolds	9.530e+006 *	9.530e+006 *	9.530e+006 *	9.530e+006 *
24	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
25	Vapour Velocity (m/s)	2.651	2.651	2.651	2.651
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	2551	2556	2561	2566
29	Elevation (m)	0.4700	0.4780	0.4860	0.4940
30	Pressure (bar)	77.66	77.66	77.66	77.66
31	Temperature (C)	58.45	58.44	58.44	58.44
32	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
33	Heat Transferred (kJ/h-m)	757.9	757.8	757.7	757.7
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1766	0.2506	0.2506	0.2506
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	8.291e-003	6.861e-003	6.861e-003	6.860e-003
39	Static Gradient (kPa/m)	-1.198e-002	3.383e-003	3.383e-003	3.383e-003
40	Accel. Gradient (kPa/m)	-3.712e-007	1.031e-006	1.031e-006	1.031e-006
41	Liq. Reynolds	2.020e+005 *	2.020e+005 *	2.020e+005 *	2.020e+005 *
42	Vap. Reynolds	9.530e+006 *	9.530e+006 *	9.530e+006 *	9.530e+006 *
43	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
44	Vapour Velocity (m/s)	2.651	2.651	2.651	2.651
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	2571	2576	2581	2586
48	Elevation (m)	0.5020	0.5100	0.5180	0.5260
49	Pressure (bar)	77.66	77.66	77.66	77.66
50	Temperature (C)	58.44	58.44	58.43	58.43
51	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
52	Heat Transferred (kJ/h-m)	757.6	757.5	757.5	757.4
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2506	0.2506	0.2506	0.2506
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	6.860e-003	6.860e-003	6.862e-003	6.862e-003
58	Static Gradient (kPa/m)	3.383e-003	3.383e-003	3.371e-003	3.371e-003
59	Accel. Gradient (kPa/m)	1.031e-006	1.031e-006	1.030e-006	1.030e-006
60	Liq. Reynolds	2.019e+005 *	2.019e+005 *	2.019e+005 *	2.019e+005 *
61	Vap. Reynolds	9.530e+006 *	9.530e+006 *	9.530e+006 *	9.530e+006 *
62	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
63	Vapour Velocity (m/s)	2.651	2.651	2.651	2.651
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	2591	2596	2601	2606
67	Elevation (m)	0.5340	0.5420	0.5500	0.5480
68	Pressure (bar)	77.66	77.66	77.66	77.65
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	58.43	58.43	58.42	58.42
14	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
15	Heat Transferred (kJ/h-m)	757.3	757.3	757.2	757.2
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2506	0.2506	0.2506	0.1820
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	6.862e-003	6.862e-003	6.862e-003	8.228e-003
21	Static Gradient (kPa/m)	3.371e-003	3.371e-003	3.371e-003	-6.788e-004
22	Accel. Gradient (kPa/m)	1.030e-006	1.030e-006	1.030e-006	7.598e-007
23	Liq. Reynolds	2.019e+005 *	2.019e+005 *	2.019e+005 *	2.019e+005 *
24	Vap. Reynolds	9.530e+006 *	9.530e+006 *	9.530e+006 *	9.530e+006 *
25	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
26	Vapour Velocity (m/s)	2.651	2.651	2.651	2.651
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	2611	2616	2621	2626
30	Elevation (m)	0.5460	0.5440	0.5420	0.5400
31	Pressure (bar)	77.65	77.65	77.65	77.65
32	Temperature (C)	58.42	58.42	58.41	58.41
33	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
34	Heat Transferred (kJ/h-m)	757.2	757.1	757.0	757.0
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1820	0.1820	0.1820	0.1820
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	8.228e-003	8.228e-003	8.228e-003	8.228e-003
40	Static Gradient (kPa/m)	-6.788e-004	-6.788e-004	-6.788e-004	-6.788e-004
41	Accel. Gradient (kPa/m)	7.598e-007	7.598e-007	7.598e-007	7.598e-007
42	Liq. Reynolds	2.019e+005 *	2.019e+005 *	2.019e+005 *	2.019e+005 *
43	Vap. Reynolds	9.530e+006 *	9.530e+006 *	9.530e+006 *	9.530e+006 *
44	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
45	Vapour Velocity (m/s)	2.651	2.651	2.651	2.651
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	2631	2636	2641	2646
49	Elevation (m)	0.5480	0.5560	0.5640	0.5720
50	Pressure (bar)	77.65	77.65	77.65	77.65
51	Temperature (C)	58.41	58.41	58.41	58.40
52	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
53	Heat Transferred (kJ/h-m)	756.9	756.8	756.7	756.7
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2506	0.2506	0.2506	0.2506
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	6.860e-003	6.860e-003	6.860e-003	6.860e-003
59	Static Gradient (kPa/m)	3.383e-003	3.384e-003	3.384e-003	3.384e-003
60	Accel. Gradient (kPa/m)	1.031e-006	1.031e-006	1.031e-006	1.031e-006
61	Liq. Reynolds	2.019e+005 *	2.019e+005 *	2.019e+005 *	2.019e+005 *
62	Vap. Reynolds	9.530e+006 *	9.530e+006 *	9.530e+006 *	9.530e+006 *
63	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
64	Vapour Velocity (m/s)	2.651	2.651	2.651	2.651
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	2651	2656	2661	2666
68	Elevation (m)	0.5800	0.5800	0.5800	0.5800
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 71 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Pressure (bar)	77.65	77.65	77.65	77.65
14	Temperature (C)	58.40	58.40	58.40	58.39
15	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
16	Heat Transferred (kJ/h-m)	756.6	756.6	756.5	756.4
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.2506	0.2662	0.2662	0.2662
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	6.860e-003	6.296e-003	6.296e-003	6.296e-003
22	Static Gradient (kPa/m)	3.384e-003	0.0000	0.0000	0.0000
23	Accel. Gradient (kPa/m)	1.031e-006	6.337e-007	6.337e-007	6.337e-007
24	Liq. Reynolds	2.019e+005 *	2.019e+005 *	2.019e+005 *	2.019e+005 *
25	Vap. Reynolds	9.530e+006 *	9.530e+006 *	9.530e+006 *	9.530e+006 *
26	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
27	Vapour Velocity (m/s)	2.651	2.651	2.651	2.651
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	2671	2676	2681	2686
31	Elevation (m)	0.5800	0.5800	0.5840	0.5880
32	Pressure (bar)	77.65	77.65	77.65	77.65
33	Temperature (C)	58.39	58.39	58.39	58.39
34	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
35	Heat Transferred (kJ/h-m)	756.4	756.3	756.3	756.2
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.2662	0.2662	0.2483	0.2483
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	6.296e-003	6.296e-003	7.113e-003	7.113e-003
41	Static Gradient (kPa/m)	0.0000	0.0000	1.681e-003	1.681e-003
42	Accel. Gradient (kPa/m)	6.337e-007	6.337e-007	8.850e-007	8.850e-007
43	Liq. Reynolds	2.019e+005 *	2.019e+005 *	2.019e+005 *	2.019e+005 *
44	Vap. Reynolds	9.530e+006 *	9.530e+006 *	9.530e+006 *	9.530e+006 *
45	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
46	Vapour Velocity (m/s)	2.650	2.650	2.650	2.650
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	2691	2696	2701	2706
50	Elevation (m)	0.5920	0.5960	0.6000	0.6000
51	Pressure (bar)	77.65	77.65	77.65	77.65
52	Temperature (C)	58.38	58.38	58.38	58.38
53	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
54	Heat Transferred (kJ/h-m)	756.1	756.1	756.0	756.0
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.2483	0.2483	0.2483	0.2662
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	7.112e-003	7.112e-003	7.112e-003	6.296e-003
60	Static Gradient (kPa/m)	1.681e-003	1.681e-003	1.681e-003	0.0000
61	Accel. Gradient (kPa/m)	8.850e-007	8.850e-007	8.850e-007	6.337e-007
62	Liq. Reynolds	2.019e+005 *	2.019e+005 *	2.019e+005 *	2.019e+005 *
63	Vap. Reynolds	9.530e+006 *	9.530e+006 *	9.530e+006 *	9.530e+006 *
64	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
65	Vapour Velocity (m/s)	2.650	2.650	2.650	2.650
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	2711	2716	2721	2726
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Elevation (m)	0.6000	0.6000	0.6000	0.6000
14	Pressure (bar)	77.65	77.65	77.65	77.65
15	Temperature (C)	58.37	58.37	58.37	58.37
16	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
17	Heat Transferred (kJ/h-m)	755.9	755.8	755.8	755.7
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2662	0.2662	0.2662	0.2662
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	6.296e-003	6.296e-003	6.296e-003	6.296e-003
23	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
24	Accel. Gradient (kPa/m)	6.337e-007	6.337e-007	6.337e-007	6.337e-007
25	Liq. Reynolds	2.019e+005 *	2.019e+005 *	2.019e+005 *	2.019e+005 *
26	Vap. Reynolds	9.530e+006 *	9.530e+006 *	9.530e+006 *	9.530e+006 *
27	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
28	Vapour Velocity (m/s)	2.650	2.650	2.650	2.650
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	2731	2736	2741	2746
32	Elevation (m)	0.6000	0.6000	0.6000	0.6000
33	Pressure (bar)	77.65	77.64	77.64	77.64
34	Temperature (C)	58.36	58.36	58.36	58.36
35	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
36	Heat Transferred (kJ/h-m)	755.7	755.6	755.5	755.5
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2662	0.2662	0.2662	0.2662
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	6.296e-003	6.296e-003	6.296e-003	6.296e-003
42	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
43	Accel. Gradient (kPa/m)	6.337e-007	6.337e-007	6.337e-007	6.337e-007
44	Liq. Reynolds	2.019e+005 *	2.019e+005 *	2.019e+005 *	2.019e+005 *
45	Vap. Reynolds	9.530e+006 *	9.530e+006 *	9.530e+006 *	9.530e+006 *
46	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
47	Vapour Velocity (m/s)	2.650	2.650	2.650	2.650
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	2751	2756	2761	2766
51	Elevation (m)	0.6000	0.6100	0.6200	0.6300
52	Pressure (bar)	77.64	77.64	77.64	77.64
53	Temperature (C)	58.36	58.35	58.35	58.35
54	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
55	Heat Transferred (kJ/h-m)	755.4	755.4	755.3	755.2
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2662	0.2518	0.2518	0.2518
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	6.296e-003	6.735e-003	6.735e-003	6.735e-003
61	Static Gradient (kPa/m)	0.0000	4.244e-003	4.244e-003	4.244e-003
62	Accel. Gradient (kPa/m)	6.337e-007	1.105e-006	1.105e-006	1.105e-006
63	Liq. Reynolds	2.019e+005 *	2.018e+005 *	2.018e+005 *	2.018e+005 *
64	Vap. Reynolds	9.530e+006 *	9.530e+006 *	9.531e+006 *	9.531e+006 *
65	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
66	Vapour Velocity (m/s)	2.650	2.650	2.650	2.650
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	2771	2776	2781	2786
14	Elevation (m)	0.6400	0.6500	0.6160	0.5820
15	Pressure (bar)	77.64	77.64	77.64	77.64
16	Temperature (C)	58.35	58.34	58.34	58.34
17	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
18	Heat Transferred (kJ/h-m)	755.2	755.1	755.1	755.1
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.2518	0.2518	0.1769	0.1769
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	6.735e-003	6.735e-003	8.286e-003	8.286e-003
24	Static Gradient (kPa/m)	4.244e-003	4.244e-003	-1.145e-002	-1.145e-002
25	Accel. Gradient (kPa/m)	1.105e-006	1.105e-006	-3.185e-007	-3.185e-007
26	Liq. Reynolds	2.018e+005 *	2.018e+005 *	2.018e+005 *	2.018e+005 *
27	Vap. Reynolds	9.531e+006 *	9.531e+006 *	9.531e+006 *	9.531e+006 *
28	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
29	Vapour Velocity (m/s)	2.650	2.650	2.650	2.650
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	2791	2796	2801	2806
33	Elevation (m)	0.5480	0.5140	0.4800	0.4980
34	Pressure (bar)	77.64	77.64	77.64	77.64
35	Temperature (C)	58.34	58.34	58.33	58.33
36	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
37	Heat Transferred (kJ/h-m)	755.0	755.0	754.9	754.8
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.1769	0.1769	0.1769	0.2564
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	8.286e-003	8.286e-003	8.286e-003	6.256e-003
43	Static Gradient (kPa/m)	-1.145e-002	-1.145e-002	-1.145e-002	7.742e-003
44	Accel. Gradient (kPa/m)	-3.185e-007	-3.185e-007	-3.186e-007	1.409e-006
45	Liq. Reynolds	2.018e+005 *	2.018e+005 *	2.018e+005 *	2.018e+005 *
46	Vap. Reynolds	9.531e+006 *	9.531e+006 *	9.531e+006 *	9.531e+006 *
47	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
48	Vapour Velocity (m/s)	2.650	2.650	2.650	2.650
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	2811	2816	2821	2826
52	Elevation (m)	0.5160	0.5340	0.5520	0.5700
53	Pressure (bar)	77.64	77.64	77.64	77.64
54	Temperature (C)	58.33	58.33	58.33	58.32
55	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
56	Heat Transferred (kJ/h-m)	754.7	754.7	754.6	754.5
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.2564	0.2564	0.2564	0.2564
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	6.256e-003	6.256e-003	6.256e-003	6.256e-003
62	Static Gradient (kPa/m)	7.742e-003	7.742e-003	7.742e-003	7.742e-003
63	Accel. Gradient (kPa/m)	1.409e-006	1.409e-006	1.409e-006	1.409e-006
64	Liq. Reynolds	2.018e+005 *	2.018e+005 *	2.018e+005 *	2.018e+005 *
65	Vap. Reynolds	9.531e+006 *	9.531e+006 *	9.531e+006 *	9.531e+006 *
66	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
67	Vapour Velocity (m/s)	2.650	2.650	2.650	2.650
68	Deposit Thickness (mm)	---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 74 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	2831	2836	2841	2846
15	Elevation (m)	0.5860	0.6020	0.6180	0.6340
16	Pressure (bar)	77.64	77.64	77.64	77.64
17	Temperature (C)	58.32	58.32	58.32	58.31
18	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
19	Heat Transferred (kJ/h-m)	754.5	754.4	754.4	754.3
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2553	0.2553	0.2553	0.2553
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	6.373e-003	6.373e-003	6.373e-003	6.373e-003
25	Static Gradient (kPa/m)	6.859e-003	6.859e-003	6.859e-003	6.859e-003
26	Accel. Gradient (kPa/m)	1.332e-006	1.332e-006	1.332e-006	1.332e-006
27	Liq. Reynolds	2.018e+005 *	2.018e+005 *	2.018e+005 *	2.018e+005 *
28	Vap. Reynolds	9.531e+006 *	9.531e+006 *	9.531e+006 *	9.531e+006 *
29	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
30	Vapour Velocity (m/s)	2.650	2.650	2.650	2.650
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	2851	2856	2861	2866
34	Elevation (m)	0.6500	0.6640	0.6780	0.6920
35	Pressure (bar)	77.64	77.63	77.63	77.63
36	Temperature (C)	58.31	58.31	58.31	58.30
37	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
38	Heat Transferred (kJ/h-m)	754.2	754.2	754.1	754.0
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2553	0.2541	0.2541	0.2541
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	6.373e-003	6.492e-003	6.492e-003	6.492e-003
44	Static Gradient (kPa/m)	6.859e-003	5.982e-003	5.982e-003	5.982e-003
45	Accel. Gradient (kPa/m)	1.332e-006	1.255e-006	1.255e-006	1.255e-006
46	Liq. Reynolds	2.018e+005 *	2.018e+005 *	2.018e+005 *	2.018e+005 *
47	Vap. Reynolds	9.531e+006 *	9.531e+006 *	9.531e+006 *	9.531e+006 *
48	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
49	Vapour Velocity (m/s)	2.650	2.650	2.650	2.650
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	2871	2876	2881	2886
53	Elevation (m)	0.7060	0.7200	0.6560	0.5920
54	Pressure (bar)	77.63	77.63	77.63	77.63
55	Temperature (C)	58.30	58.30	58.30	58.30
56	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
57	Heat Transferred (kJ/h-m)	754.0	753.9	753.9	753.9
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2541	0.2541	0.1721	0.1721
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	6.491e-003	6.491e-003	8.345e-003	8.345e-003
63	Static Gradient (kPa/m)	5.982e-003	5.982e-003	-2.095e-002	-2.095e-002
64	Accel. Gradient (kPa/m)	1.255e-006	1.255e-006	-1.269e-006	-1.269e-006
65	Liq. Reynolds	2.018e+005 *	2.018e+005 *	2.018e+005 *	2.018e+005 *
66	Vap. Reynolds	9.531e+006 *	9.531e+006 *	9.531e+006 *	9.531e+006 *
67	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
68	Vapour Velocity (m/s)	2.650	2.650	2.650	2.650
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	2891	2896	2901	2906
16	Elevation (m)	0.5280	0.4640	0.4000	0.3780
17	Pressure (bar)	77.63	77.63	77.64	77.64
18	Temperature (C)	58.29	58.29	58.29	58.29
19	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
20	Heat Transferred (kJ/h-m)	753.8	753.8	753.7	753.7
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.1721	0.1721	0.1721	0.1789
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	8.345e-003	8.344e-003	8.344e-003	8.262e-003
26	Static Gradient (kPa/m)	-2.095e-002	-2.096e-002	-2.096e-002	-7.383e-003
27	Accel. Gradient (kPa/m)	-1.269e-006	-1.269e-006	-1.269e-006	8.837e-008
28	Liq. Reynolds	2.018e+005 *	2.018e+005 *	2.018e+005 *	2.018e+005 *
29	Vap. Reynolds	9.531e+006 *	9.531e+006 *	9.531e+006 *	9.531e+006 *
30	Liquid Velocity (m/s)	0.1993	0.1993	0.1994	0.1994
31	Vapour Velocity (m/s)	2.650	2.650	2.650	2.650
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	2911	2916	2921	2926
35	Elevation (m)	0.3560	0.3340	0.3120	0.2900
36	Pressure (bar)	77.64	77.63	77.63	77.63
37	Temperature (C)	58.29	58.28	58.28	58.28
38	Vapour Fraction	0.8812	0.8812	0.8811	0.8811
39	Heat Transferred (kJ/h-m)	753.6	753.6	753.5	753.5
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.1789	0.1789	0.1789	0.1789
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	8.261e-003	8.261e-003	8.261e-003	8.261e-003
45	Static Gradient (kPa/m)	-7.383e-003	-7.383e-003	-7.384e-003	-7.384e-003
46	Accel. Gradient (kPa/m)	8.836e-008	8.835e-008	8.833e-008	8.832e-008
47	Liq. Reynolds	2.018e+005 *	2.018e+005 *	2.018e+005 *	2.018e+005 *
48	Vap. Reynolds	9.531e+006 *	9.531e+006 *	9.531e+006 *	9.531e+006 *
49	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
50	Vapour Velocity (m/s)	2.650	2.650	2.650	2.650
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	2931	2936	2941	2946
54	Elevation (m)	0.3620	0.4340	0.5060	0.5780
55	Pressure (bar)	77.63	77.63	77.63	77.63
56	Temperature (C)	58.28	58.27	58.27	58.27
57	Vapour Fraction	0.8811	0.8811	0.8811	0.8811
58	Heat Transferred (kJ/h-m)	753.3	753.3	753.2	753.1
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2879	0.2879	0.2879	0.2878
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	7.219e-003	7.219e-003	7.220e-003	7.220e-003
64	Static Gradient (kPa/m)	3.372e-002	3.372e-002	3.372e-002	3.372e-002
65	Accel. Gradient (kPa/m)	4.121e-006	4.121e-006	4.121e-006	4.121e-006
66	Liq. Reynolds	2.018e+005 *	2.018e+005 *	2.017e+005 *	2.017e+005 *
67	Vap. Reynolds	9.531e+006 *	9.531e+006 *	9.531e+006 *	9.531e+006 *
68	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Vapour Velocity	(m/s)	2.650	2.650	2.650	2.650				
14	Deposit Thickness	(mm)	---	---	---	---				
15	Deposit Volume	(m3)	---	---	---	---				
16	Length	(m)	2951	2956	2961	2966				
17	Elevation	(m)	0.6500	0.3780	0.1060	-0.1660				
18	Pressure	(bar)	77.62	77.63	77.63	77.64				
19	Temperature	(C)	58.27	58.27	58.26	58.26				
20	Vapour Fraction		0.8811	0.8811	0.8811	0.8811				
21	Heat Transferred	(kJ/h-m)	753.0	753.2	753.1	753.1				
22	Flow Regime		Transition	Transition	Transition	Transition				
23	Water Flow Regime									
24	Liquid HoldUp		0.2878	0.1391	0.1391	0.1391				
25	Water HoldUp		---	---	---	---				
26	Friction Gradient	(kPa/m)	7.220e-003	8.915e-003	8.914e-003	8.914e-003				
27	Static Gradient	(kPa/m)	3.372e-002	-7.810e-002	-7.811e-002	-7.811e-002				
28	Accel. Gradient	(kPa/m)	4.121e-006	-6.963e-006	-6.963e-006	-6.963e-006				
29	Liq. Reynolds		2.017e+005 *	2.017e+005 *	2.017e+005 *	2.017e+005 *				
30	Vap. Reynolds		9.531e+006 *	9.531e+006 *	9.531e+006 *	9.531e+006 *				
31	Liquid Velocity	(m/s)	0.1994	0.1994	0.1994	0.1994				
32	Vapour Velocity	(m/s)	2.650	2.650	2.649	2.649				
33	Deposit Thickness	(mm)	---	---	---	---				
34	Deposit Volume	(m3)	---	---	---	---				
35	Length	(m)	2971	2976	2981	2986				
36	Elevation	(m)	-0.4380	-0.7100	-0.6880	-0.6660				
37	Pressure	(bar)	77.64	77.64	77.64	77.64				
38	Temperature	(C)	58.26	58.26	58.26	58.26				
39	Vapour Fraction		0.8811	0.8811	0.8811	0.8811				
40	Heat Transferred	(kJ/h-m)	753.1	753.1	752.9	752.8				
41	Flow Regime		Transition	Transition	Transition	Transition				
42	Water Flow Regime									
43	Liquid HoldUp		0.1391	0.1391	0.2585	0.2585				
44	Water HoldUp		---	---	---	---				
45	Friction Gradient	(kPa/m)	8.913e-003	8.913e-003	6.051e-003	6.051e-003				
46	Static Gradient	(kPa/m)	-7.811e-002	-7.811e-002	9.307e-003	9.307e-003				
47	Accel. Gradient	(kPa/m)	-6.962e-006	-6.962e-006	1.545e-006	1.545e-006				
48	Liq. Reynolds		2.018e+005 *	2.018e+005 *	2.018e+005 *	2.017e+005 *				
49	Vap. Reynolds		9.531e+006 *	9.531e+006 *	9.531e+006 *	9.531e+006 *				
50	Liquid Velocity	(m/s)	0.1994	0.1994	0.1994	0.1994				
51	Vapour Velocity	(m/s)	2.649	2.649	2.649	2.649				
52	Deposit Thickness	(mm)	---	---	---	---				
53	Deposit Volume	(m3)	---	---	---	---				
54	Length	(m)	2991	2996	3001	3006				
55	Elevation	(m)	-0.6440	-0.6220	-0.6000	-2.080				
56	Pressure	(bar)	77.64	77.64	77.64	77.65				
57	Temperature	(C)	58.26	58.25	58.25	58.26				
58	Vapour Fraction		0.8811	0.8811	0.8811	0.8811				
59	Heat Transferred	(kJ/h-m)	752.8	752.7	752.6	754.3				
60	Flow Regime		Transition	Transition	Transition	Transition				
61	Water Flow Regime									
62	Liquid HoldUp		0.2585	0.2585	0.2585	7.000e-002				
63	Water HoldUp		---	---	---	---				
64	Friction Gradient	(kPa/m)	6.051e-003	6.051e-003	6.051e-003	1.270e-002				
65	Static Gradient	(kPa/m)	9.307e-003	9.307e-003	9.307e-003	-0.3005				
66	Accel. Gradient	(kPa/m)	1.545e-006	1.545e-006	1.545e-006	-2.895e-005				
67	Liq. Reynolds		2.017e+005 *	2.017e+005 *	2.017e+005 *	2.018e+005 *				
68	Vap. Reynolds		9.531e+006 *	9.531e+006 *	9.531e+006 *	9.530e+006 *				
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:13:43 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.1994	0.1994	0.1994	0.1994	
14	Vapour Velocity	(m/s)	2.649	2.649	2.649	2.649	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	3011	3016	3021	3026	
18	Elevation	(m)	-3.560	-5.040	-6.520	-8.000	
19	Pressure	(bar)	77.67	77.68	77.70	77.71	
20	Temperature	(C)	58.26	58.27	58.27	58.28	
21	Vapour Fraction		0.8811	0.8811	0.8811	0.8811	
22	Heat Transferred	(kJ/h-m)	754.4	754.6	754.7	754.9	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		7.002e-002	7.003e-002	7.004e-002	7.006e-002	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	1.270e-002	1.269e-002	1.269e-002	1.269e-002	
28	Static Gradient	(kPa/m)	-0.3006	-0.3006	-0.3007	-0.3007	
29	Accel. Gradient	(kPa/m)	-2.894e-005	-2.894e-005	-2.893e-005	-2.893e-005	
30	Liq. Reynolds		2.018e+005 *	2.018e+005 *	2.018e+005 *	2.018e+005 *	
31	Vap. Reynolds		9.530e+006 *	9.529e+006 *	9.529e+006 *	9.528e+006 *	
32	Liquid Velocity	(m/s)	0.1994	0.1994	0.1994	0.1994	
33	Vapour Velocity	(m/s)	2.648	2.648	2.647	2.647	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	3031	3036	3041	3046	
37	Elevation	(m)	-8.480	-8.960	-9.440	-9.920	
38	Pressure	(bar)	77.72	77.72	77.73	77.73	
39	Temperature	(C)	58.28	58.28	58.28	58.28	
40	Vapour Fraction		0.8811	0.8811	0.8811	0.8811	
41	Heat Transferred	(kJ/h-m)	753.7	753.7	753.7	753.8	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.1067	0.1067	0.1067	0.1067	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	9.939e-003	9.938e-003	9.937e-003	9.937e-003	
47	Static Gradient	(kPa/m)	-0.1190	-0.1190	-0.1190	-0.1190	
48	Accel. Gradient	(kPa/m)	-1.095e-005	-1.095e-005	-1.094e-005	-1.094e-005	
49	Liq. Reynolds		2.019e+005 *	2.019e+005 *	2.019e+005 *	2.019e+005 *	
50	Vap. Reynolds		9.528e+006 *	9.528e+006 *	9.528e+006 *	9.528e+006 *	
51	Liquid Velocity	(m/s)	0.1994	0.1994	0.1994	0.1994	
52	Vapour Velocity	(m/s)	2.646	2.646	2.646	2.646	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	3051	3056	3061	3066	
56	Elevation	(m)	-10.40	-10.64	-10.88	-11.12	
57	Pressure	(bar)	77.74	77.74	77.74	77.75	
58	Temperature	(C)	58.28	58.28	58.28	58.28	
59	Vapour Fraction		0.8811	0.8811	0.8811	0.8811	
60	Heat Transferred	(kJ/h-m)	753.8	753.6	753.6	753.6	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.1067	0.1443	0.1443	0.1443	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	9.936e-003	8.792e-003	8.792e-003	8.791e-003	
66	Static Gradient	(kPa/m)	-0.1190	-7.046e-002	-7.047e-002	-7.047e-002	
67	Accel. Gradient	(kPa/m)	-1.094e-005	-6.188e-006	-6.188e-006	-6.188e-006	
68	Liq. Reynolds		2.019e+005 *	2.019e+005 *	2.019e+005 *	2.019e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.528e+006 *	9.528e+006 *	9.527e+006 *	9.527e+006 *
14	Liquid Velocity (m/s)	0.1994	0.1994	0.1994	0.1994
15	Vapour Velocity (m/s)	2.646	2.645	2.645	2.645
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	3071	3076	3081	3086
19	Elevation (m)	-11.36	-11.60	-9.276	-6.952
20	Pressure (bar)	77.75	77.75	77.60	77.44
21	Temperature (C)	58.28	58.28	58.24	58.20
22	Vapour Fraction	0.8811	0.8811	0.8811	0.8812
23	Heat Transferred (kJ/h-m)	753.5	753.5	745.5	744.4
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1443	0.1443	1.000	1.000
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	8.791e-003	8.791e-003	6.656e-003	6.669e-003
29	Static Gradient (kPa/m)	-7.047e-002	-7.047e-002	3.105	3.105
30	Accel. Gradient (kPa/m)	-6.187e-006	-6.187e-006	3.134e-004	3.147e-004
31	Liq. Reynolds	2.019e+005 *	2.019e+005 *	2.017e+005 *	2.015e+005 *
32	Vap. Reynolds	9.527e+006 *	9.527e+006 *	9.532e+006 *	9.537e+006 *
33	Liquid Velocity (m/s)	0.1994	0.1994	0.1993	0.1993
34	Vapour Velocity (m/s)	2.645	2.645	2.650	2.656
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	3091	3096	3101	3106
38	Elevation (m)	-4.628	-2.304	2.000e-002	0.1200
39	Pressure (bar)	77.29	77.13	76.97	76.97
40	Temperature (C)	58.16	58.12	58.08	58.07
41	Vapour Fraction	0.8813	0.8813	0.8814	0.8814
42	Heat Transferred (kJ/h-m)	743.3	742.2	741.1	747.9
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	1.000	1.000	1.000	0.3025
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.681e-003	6.694e-003	6.706e-003	7.292e-003
48	Static Gradient (kPa/m)	3.106	3.106	3.107	4.858e-002
49	Accel. Gradient (kPa/m)	3.161e-004	3.174e-004	3.188e-004	5.722e-006
50	Liq. Reynolds	2.013e+005 *	2.011e+005 *	2.008e+005 *	2.008e+005 *
51	Vap. Reynolds	9.542e+006 *	9.547e+006 *	9.552e+006 *	9.552e+006 *
52	Liquid Velocity (m/s)	0.1993	0.1992	0.1992	0.1992
53	Vapour Velocity (m/s)	2.662	2.667	2.673	2.673
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	3111	3116	3121	3126
57	Elevation (m)	0.2200	0.3200	0.4200	0.5200
58	Pressure (bar)	76.97	76.97	76.96	76.96
59	Temperature (C)	58.07	58.07	58.06	58.06
60	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
61	Heat Transferred (kJ/h-m)	747.8	747.7	747.6	747.5
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.3025	0.3025	0.3025	0.3025
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	7.292e-003	7.292e-003	7.293e-003	7.293e-003
67	Static Gradient (kPa/m)	4.858e-002	4.858e-002	4.858e-002	4.858e-002
68	Accel. Gradient (kPa/m)	5.722e-006	5.722e-006	5.723e-006	5.723e-006
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.008e+005 *	2.008e+005 *	2.008e+005 *	2.008e+005 *
14	Vap. Reynolds	9.552e+006 *	9.552e+006 *	9.553e+006 *	9.553e+006 *
15	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
16	Vapour Velocity (m/s)	2.673	2.673	2.673	2.673
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	3131	3136	3141	3146
20	Elevation (m)	0.4800	0.4400	0.4000	0.3600
21	Pressure (bar)	76.96	76.96	76.96	76.96
22	Temperature (C)	58.06	58.06	58.05	58.05
23	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
24	Heat Transferred (kJ/h-m)	747.5	747.5	747.4	747.4
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.1750	0.1750	0.1750	0.1750
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	8.365e-003	8.365e-003	8.365e-003	8.365e-003
30	Static Gradient (kPa/m)	-1.321e-002	-1.321e-002	-1.321e-002	-1.321e-002
31	Accel. Gradient (kPa/m)	-4.962e-007	-4.962e-007	-4.962e-007	-4.962e-007
32	Liq. Reynolds	2.008e+005 *	2.008e+005 *	2.008e+005 *	2.008e+005 *
33	Vap. Reynolds	9.553e+006 *	9.553e+006 *	9.553e+006 *	9.553e+006 *
34	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
35	Vapour Velocity (m/s)	2.673	2.673	2.673	2.673
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	3151	3156	3161	3166
39	Elevation (m)	0.3200	0.2980	0.2760	0.2540
40	Pressure (bar)	76.96	76.96	76.96	76.96
41	Temperature (C)	58.05	58.05	58.05	58.04
42	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
43	Heat Transferred (kJ/h-m)	747.3	747.3	747.2	747.2
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.1750	0.1779	0.1779	0.1779
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	8.365e-003	8.330e-003	8.330e-003	8.330e-003
49	Static Gradient (kPa/m)	-1.321e-002	-7.342e-003	-7.342e-003	-7.342e-003
50	Accel. Gradient (kPa/m)	-4.962e-007	1.012e-007	1.012e-007	1.012e-007
51	Liq. Reynolds	2.008e+005 *	2.008e+005 *	2.008e+005 *	2.008e+005 *
52	Vap. Reynolds	9.553e+006 *	9.553e+006 *	9.553e+006 *	9.553e+006 *
53	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
54	Vapour Velocity (m/s)	2.673	2.673	2.673	2.673
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	3171	3176	3181	3186
58	Elevation (m)	0.2320	0.2100	0.2640	0.3180
59	Pressure (bar)	76.96	76.96	76.96	76.96
60	Temperature (C)	58.04	58.04	58.04	58.03
61	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
62	Heat Transferred (kJ/h-m)	747.1	747.1	746.9	746.9
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.1779	0.1779	0.2748	0.2748
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	8.330e-003	8.330e-003	7.158e-003	7.158e-003
68	Static Gradient (kPa/m)	-7.342e-003	-7.342e-003	2.353e-002	2.353e-002
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	1.011e-007	1.011e-007	3.143e-006	3.143e-006
14	Liq. Reynolds	2.008e+005 *	2.008e+005 *	2.008e+005 *	2.008e+005 *
15	Vap. Reynolds	9.553e+006 *	9.553e+006 *	9.553e+006 *	9.553e+006 *
16	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
17	Vapour Velocity (m/s)	2.673	2.673	2.673	2.673
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	3192	3197	3202	3207
21	Elevation (m)	0.3720	0.4260	0.4800	0.4660
22	Pressure (bar)	76.96	76.96	76.95	76.95
23	Temperature (C)	58.03	58.03	58.03	58.02
24	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
25	Heat Transferred (kJ/h-m)	746.8	746.7	746.6	746.6
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.2748	0.2748	0.2748	0.1791
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	7.158e-003	7.158e-003	7.158e-003	8.315e-003
31	Static Gradient (kPa/m)	2.353e-002	2.353e-002	2.353e-002	-4.694e-003
32	Accel. Gradient (kPa/m)	3.144e-006	3.144e-006	3.144e-006	3.709e-007
33	Liq. Reynolds	2.008e+005 *	2.008e+005 *	2.008e+005 *	2.008e+005 *
34	Vap. Reynolds	9.553e+006 *	9.553e+006 *	9.553e+006 *	9.553e+006 *
35	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
36	Vapour Velocity (m/s)	2.673	2.673	2.673	2.673
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	3212	3217	3222	3227
40	Elevation (m)	0.4520	0.4380	0.4240	0.4100
41	Pressure (bar)	76.95	76.95	76.95	76.95
42	Temperature (C)	58.02	58.02	58.02	58.02
43	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
44	Heat Transferred (kJ/h-m)	746.6	746.5	746.5	746.4
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.1791	0.1791	0.1791	0.1791
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	8.315e-003	8.315e-003	8.315e-003	8.315e-003
50	Static Gradient (kPa/m)	-4.694e-003	-4.694e-003	-4.694e-003	-4.694e-003
51	Accel. Gradient (kPa/m)	3.709e-007	3.709e-007	3.709e-007	3.709e-007
52	Liq. Reynolds	2.008e+005 *	2.008e+005 *	2.008e+005 *	2.008e+005 *
53	Vap. Reynolds	9.553e+006 *	9.553e+006 *	9.553e+006 *	9.553e+006 *
54	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
55	Vapour Velocity (m/s)	2.673	2.673	2.673	2.673
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	3232	3237	3242	3247
59	Elevation (m)	0.3620	0.3140	0.2660	0.2180
60	Pressure (bar)	76.95	76.95	76.95	76.95
61	Temperature (C)	58.01	58.01	58.01	58.01
62	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
63	Heat Transferred (kJ/h-m)	746.4	746.3	746.3	746.2
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.1737	0.1737	0.1737	0.1737
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	8.380e-003	8.380e-003	8.380e-003	8.380e-003
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1				Case Name:	Simulation - 386MMSCFD_80barg.hsc	
2				Unit Set:	Mixed	
3	 Epic Atlantic Limited			Date/Time:	Mon Jun 13 15:13:43 2022	
4						
5						
6						
7						
8						
9						
10						
11						
12						
13	Static Gradient	(kPa/m)	-1.578e-002	-1.578e-002	-1.578e-002	-1.578e-002
14	Accel. Gradient	(kPa/m)	-7.578e-007	-7.578e-007	-7.578e-007	-7.578e-007
15	Liq. Reynolds		2.008e+005 *	2.008e+005 *	2.008e+005 *	2.008e+005 *
16	Vap. Reynolds		9.553e+006 *	9.553e+006 *	9.553e+006 *	9.553e+006 *
17	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992
18	Vapour Velocity	(m/s)	2.673	2.673	2.673	2.673
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	3252	3257	3262	3267
22	Elevation	(m)	0.1700	0.1680	0.1660	0.1640
23	Pressure	(bar)	76.95	76.95	76.95	76.95
24	Temperature	(C)	58.01	58.00	58.00	58.00
25	Vapour Fraction		0.8814	0.8814	0.8814	0.8814
26	Heat Transferred	(kJ/h-m)	746.2	746.1	746.0	746.0
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.1737	0.1810	0.1810	0.1810
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	8.380e-003	8.293e-003	8.293e-003	8.293e-003
32	Static Gradient	(kPa/m)	-1.578e-002	-6.753e-004	-6.753e-004	-6.753e-004
33	Accel. Gradient	(kPa/m)	-7.578e-007	7.802e-007	7.802e-007	7.802e-007
34	Liq. Reynolds		2.008e+005 *	2.007e+005 *	2.007e+005 *	2.007e+005 *
35	Vap. Reynolds		9.553e+006 *	9.553e+006 *	9.553e+006 *	9.553e+006 *
36	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992
37	Vapour Velocity	(m/s)	2.673	2.673	2.673	2.673
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	3272	3277	3282	3287
41	Elevation	(m)	0.1620	0.1600	0.1580	0.1560
42	Pressure	(bar)	76.95	76.95	76.95	76.95
43	Temperature	(C)	58.00	57.99	57.99	57.99
44	Vapour Fraction		0.8814	0.8814	0.8814	0.8814
45	Heat Transferred	(kJ/h-m)	745.9	745.9	745.8	745.7
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.1810	0.1811	0.1811	0.1811
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	8.292e-003	8.292e-003	8.292e-003	8.292e-003
51	Static Gradient	(kPa/m)	-6.753e-004	-6.753e-004	-6.753e-004	-6.753e-004
52	Accel. Gradient	(kPa/m)	7.802e-007	7.801e-007	7.801e-007	7.801e-007
53	Liq. Reynolds		2.007e+005 *	2.007e+005 *	2.007e+005 *	2.007e+005 *
54	Vap. Reynolds		9.553e+006 *	9.553e+006 *	9.553e+006 *	9.553e+006 *
55	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992
56	Vapour Velocity	(m/s)	2.673	2.673	2.673	2.673
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	3292	3297	3302	3307
60	Elevation	(m)	0.1540	0.1520	0.1500	0.1480
61	Pressure	(bar)	76.95	76.95	76.95	76.95
62	Temperature	(C)	57.99	57.99	57.98	57.98
63	Vapour Fraction		0.8814	0.8814	0.8814	0.8814
64	Heat Transferred	(kJ/h-m)	745.7	745.6	745.6	745.5
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.1811	0.1811	0.1811	0.2539
68	Water HoldUp		---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:13:43 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	8.292e-003	8.292e-003	8.292e-003	6.449e-003	
14	Static Gradient	(kPa/m)	-6.753e-004	-6.753e-004	-6.753e-004	6.825e-003	
15	Accel. Gradient	(kPa/m)	7.801e-007	7.801e-007	7.801e-007	1.360e-006	
16	Liq. Reynolds		2.007e+005 *	2.007e+005 *	2.007e+005 *	2.007e+005 *	
17	Vap. Reynolds		9.553e+006 *	9.553e+006 *	9.553e+006 *	9.553e+006 *	
18	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992	
19	Vapour Velocity	(m/s)	2.673	2.673	2.673	2.673	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	3312	3317	3322	3327	
23	Elevation	(m)	0.1820	0.1980	0.2140	0.2300	
24	Pressure	(bar)	76.95	76.95	76.95	76.95	
25	Temperature	(C)	57.98	57.98	57.97	57.97	
26	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
27	Heat Transferred	(kJ/h-m)	745.4	745.3	745.3	745.2	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.2539	0.2539	0.2539	0.2539	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	6.449e-003	6.449e-003	6.449e-003	6.449e-003	
33	Static Gradient	(kPa/m)	6.825e-003	6.825e-003	6.825e-003	6.825e-003	
34	Accel. Gradient	(kPa/m)	1.360e-006	1.360e-006	1.360e-006	1.360e-006	
35	Liq. Reynolds		2.007e+005 *	2.007e+005 *	2.007e+005 *	2.007e+005 *	
36	Vap. Reynolds		9.553e+006 *	9.553e+006 *	9.553e+006 *	9.553e+006 *	
37	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992	
38	Vapour Velocity	(m/s)	2.673	2.673	2.673	2.673	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	3332	3337	3342	3347	
42	Elevation	(m)	0.2480	0.2660	0.2840	0.3020	
43	Pressure	(bar)	76.95	76.95	76.95	76.94	
44	Temperature	(C)	57.97	57.97	57.97	57.96	
45	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
46	Heat Transferred	(kJ/h-m)	745.1	745.1	745.0	745.0	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.2551	0.2551	0.2551	0.2551	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	6.330e-003	6.330e-003	6.330e-003	6.330e-003	
52	Static Gradient	(kPa/m)	7.704e-003	7.704e-003	7.704e-003	7.704e-003	
53	Accel. Gradient	(kPa/m)	1.437e-006	1.437e-006	1.437e-006	1.437e-006	
54	Liq. Reynolds		2.007e+005 *	2.007e+005 *	2.007e+005 *	2.007e+005 *	
55	Vap. Reynolds		9.553e+006 *	9.553e+006 *	9.553e+006 *	9.553e+006 *	
56	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992	
57	Vapour Velocity	(m/s)	2.673	2.673	2.673	2.673	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	3352	3357	3362	3367	
61	Elevation	(m)	0.3200	0.3300	0.3400	0.3500	
62	Pressure	(bar)	76.94	76.94	76.94	76.94	
63	Temperature	(C)	57.96	57.96	57.96	57.95	
64	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
65	Heat Transferred	(kJ/h-m)	744.9	744.8	744.8	744.7	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.2551	0.2504	0.2504	0.2504	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	6.330e-003	6.814e-003	6.814e-003	6.814e-003
15	Static Gradient (kPa/m)	7.704e-003	4.223e-003	4.223e-003	4.223e-003
16	Accel. Gradient (kPa/m)	1.437e-006	1.131e-006	1.131e-006	1.131e-006
17	Liq. Reynolds	2.007e+005 *	2.007e+005 *	2.007e+005 *	2.007e+005 *
18	Vap. Reynolds	9.553e+006 *	9.553e+006 *	9.553e+006 *	9.553e+006 *
19	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
20	Vapour Velocity (m/s)	2.673	2.673	2.673	2.673
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	3372	3377	3382	3387
24	Elevation (m)	0.3600	0.3700	0.3740	0.3780
25	Pressure (bar)	76.94	76.94	76.94	76.94
26	Temperature (C)	57.95	57.95	57.95	57.94
27	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
28	Heat Transferred (kJ/h-m)	744.6	744.6	744.5	744.5
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2504	0.2504	0.2471	0.2471
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	6.814e-003	6.814e-003	7.187e-003	7.186e-003
34	Static Gradient (kPa/m)	4.223e-003	4.223e-003	1.727e-003	1.727e-003
35	Accel. Gradient (kPa/m)	1.131e-006	1.131e-006	9.130e-007	9.130e-007
36	Liq. Reynolds	2.007e+005 *	2.007e+005 *	2.007e+005 *	2.007e+005 *
37	Vap. Reynolds	9.553e+006 *	9.553e+006 *	9.554e+006 *	9.554e+006 *
38	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
39	Vapour Velocity (m/s)	2.673	2.673	2.673	2.673
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	3392	3396	3401	3406
43	Elevation (m)	0.3820	0.3860	0.3900	0.3920
44	Pressure (bar)	76.94	76.94	76.94	76.94
45	Temperature (C)	57.94	57.94	57.94	57.94
46	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
47	Heat Transferred (kJ/h-m)	744.4	744.4	744.3	744.2
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2471	0.2471	0.2471	0.2458
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	7.186e-003	7.186e-003	7.186e-003	7.325e-003
53	Static Gradient (kPa/m)	1.727e-003	1.727e-003	1.727e-003	8.334e-004
54	Accel. Gradient (kPa/m)	9.130e-007	9.130e-007	9.130e-007	8.357e-007
55	Liq. Reynolds	2.007e+005 *	2.007e+005 *	2.007e+005 *	2.007e+005 *
56	Vap. Reynolds	9.554e+006 *	9.554e+006 *	9.554e+006 *	9.554e+006 *
57	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
58	Vapour Velocity (m/s)	2.673	2.673	2.673	2.673
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	3411	3416	3421	3426
62	Elevation (m)	0.3940	0.3960	0.3980	0.4000
63	Pressure (bar)	76.94	76.94	76.94	76.94
64	Temperature (C)	57.93	57.93	57.93	57.93
65	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
66	Heat Transferred (kJ/h-m)	744.2	744.1	744.1	744.0
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 84 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.2458	0.2458	0.2458	0.2458
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	7.325e-003	7.325e-003	7.325e-003	7.325e-003
16	Static Gradient (kPa/m)	8.334e-004	8.334e-004	8.334e-004	8.334e-004
17	Accel. Gradient (kPa/m)	8.357e-007	8.357e-007	8.357e-007	8.357e-007
18	Liq. Reynolds	2.007e+005 *	2.007e+005 *	2.007e+005 *	2.007e+005 *
19	Vap. Reynolds	9.554e+006 *	9.554e+006 *	9.554e+006 *	9.554e+006 *
20	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
21	Vapour Velocity (m/s)	2.673	2.673	2.673	2.673
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	3431	3436	3441	3446
25	Elevation (m)	0.3960	0.3920	0.3880	0.3840
26	Pressure (bar)	76.94	76.94	76.94	76.94
27	Temperature (C)	57.93	57.92	57.92	57.92
28	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
29	Heat Transferred (kJ/h-m)	744.0	743.9	743.9	743.8
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1807	0.1807	0.1807	0.1808
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	8.295e-003	8.295e-003	8.295e-003	8.295e-003
35	Static Gradient (kPa/m)	-1.349e-003	-1.349e-003	-1.349e-003	-1.349e-003
36	Accel. Gradient (kPa/m)	7.115e-007	7.115e-007	7.115e-007	7.115e-007
37	Liq. Reynolds	2.007e+005 *	2.006e+005 *	2.006e+005 *	2.006e+005 *
38	Vap. Reynolds	9.554e+006 *	9.554e+006 *	9.554e+006 *	9.554e+006 *
39	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
40	Vapour Velocity (m/s)	2.672	2.672	2.672	2.672
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	3451	3456	3461	3466
44	Elevation (m)	0.3800	0.3800	0.3800	0.3800
45	Pressure (bar)	76.94	76.93	76.93	76.93
46	Temperature (C)	57.92	57.91	57.91	57.91
47	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
48	Heat Transferred (kJ/h-m)	743.8	743.7	743.6	743.5
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1808	0.2648	0.2648	0.2648
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	8.295e-003	6.260e-003	6.260e-003	6.260e-003
54	Static Gradient (kPa/m)	-1.349e-003	0.0000	0.0000	0.0000
55	Accel. Gradient (kPa/m)	7.115e-007	6.412e-007	6.412e-007	6.412e-007
56	Liq. Reynolds	2.006e+005 *	2.006e+005 *	2.006e+005 *	2.006e+005 *
57	Vap. Reynolds	9.554e+006 *	9.554e+006 *	9.554e+006 *	9.554e+006 *
58	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
59	Vapour Velocity (m/s)	2.672	2.672	2.672	2.672
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	3471	3476	3481	3486
63	Elevation (m)	0.3800	0.3800	0.3680	0.3560
64	Pressure (bar)	76.93	76.93	76.93	76.93
65	Temperature (C)	57.91	57.91	57.90	57.90
66	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
67	Heat Transferred (kJ/h-m)	743.5	743.4	743.4	743.4
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 85 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2648	0.2648	0.1795	0.1795
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.260e-003	6.260e-003	8.309e-003	8.309e-003
17	Static Gradient (kPa/m)	0.0000	0.0000	-4.029e-003	-4.029e-003
18	Accel. Gradient (kPa/m)	6.412e-007	6.412e-007	4.384e-007	4.384e-007
19	Liq. Reynolds	2.006e+005 *	2.006e+005 *	2.006e+005 *	2.006e+005 *
20	Vap. Reynolds	9.554e+006 *	9.554e+006 *	9.554e+006 *	9.554e+006 *
21	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
22	Vapour Velocity (m/s)	2.672	2.672	2.672	2.672
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	3491	3496	3501	3506
26	Elevation (m)	0.3440	0.3320	0.3200	0.3200
27	Pressure (bar)	76.93	76.93	76.93	76.93
28	Temperature (C)	57.90	57.90	57.89	57.89
29	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
30	Heat Transferred (kJ/h-m)	743.3	743.2	743.2	743.1
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1795	0.1795	0.1795	0.2648
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	8.309e-003	8.309e-003	8.309e-003	6.261e-003
36	Static Gradient (kPa/m)	-4.029e-003	-4.029e-003	-4.029e-003	0.0000
37	Accel. Gradient (kPa/m)	4.384e-007	4.384e-007	4.384e-007	6.412e-007
38	Liq. Reynolds	2.006e+005 *	2.006e+005 *	2.006e+005 *	2.006e+005 *
39	Vap. Reynolds	9.554e+006 *	9.554e+006 *	9.554e+006 *	9.554e+006 *
40	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
41	Vapour Velocity (m/s)	2.672	2.672	2.672	2.672
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	3511	3516	3521	3526
45	Elevation (m)	0.3200	0.3200	0.3200	0.3200
46	Pressure (bar)	76.93	76.93	76.93	76.93
47	Temperature (C)	57.89	57.89	57.89	57.88
48	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
49	Heat Transferred (kJ/h-m)	743.0	743.0	742.9	742.8
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2648	0.2648	0.2648	0.2648
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	6.261e-003	6.261e-003	6.261e-003	6.261e-003
55	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
56	Accel. Gradient (kPa/m)	6.412e-007	6.412e-007	6.412e-007	6.412e-007
57	Liq. Reynolds	2.006e+005 *	2.006e+005 *	2.006e+005 *	2.006e+005 *
58	Vap. Reynolds	9.554e+006 *	9.554e+006 *	9.554e+006 *	9.554e+006 *
59	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
60	Vapour Velocity (m/s)	2.672	2.672	2.672	2.672
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	3531	3536	3541	3546
64	Elevation (m)	0.3120	0.3040	0.2960	0.2880
65	Pressure (bar)	76.93	76.93	76.93	76.93
66	Temperature (C)	57.88	57.88	57.88	57.88
67	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
68	Heat Transferred (kJ/h-m)	742.8	742.8	742.7	742.7
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 86 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.1801	0.1801	0.1801	0.1801
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	8.302e-003	8.302e-003	8.301e-003	8.301e-003
18	Static Gradient (kPa/m)	-2.693e-003	-2.693e-003	-2.693e-003	-2.693e-003
19	Accel. Gradient (kPa/m)	5.745e-007	5.745e-007	5.745e-007	5.745e-007
20	Liq. Reynolds	2.006e+005 *	2.006e+005 *	2.006e+005 *	2.006e+005 *
21	Vap. Reynolds	9.554e+006 *	9.554e+006 *	9.554e+006 *	9.554e+006 *
22	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
23	Vapour Velocity (m/s)	2.672	2.672	2.672	2.672
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	3551	3556	3561	3566
27	Elevation (m)	0.2800	0.2500	0.2200	0.1900
28	Pressure (bar)	76.93	76.93	76.93	76.93
29	Temperature (C)	57.87	57.87	57.87	57.87
30	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
31	Heat Transferred (kJ/h-m)	742.6	742.6	742.5	742.4
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1801	0.1766	0.1766	0.1766
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	8.301e-003	8.342e-003	8.342e-003	8.342e-003
37	Static Gradient (kPa/m)	-2.693e-003	-9.969e-003	-9.969e-003	-9.969e-003
38	Accel. Gradient (kPa/m)	5.745e-007	-1.666e-007	-1.666e-007	-1.667e-007
39	Liq. Reynolds	2.006e+005 *	2.006e+005 *	2.006e+005 *	2.006e+005 *
40	Vap. Reynolds	9.554e+006 *	9.554e+006 *	9.554e+006 *	9.554e+006 *
41	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
42	Vapour Velocity (m/s)	2.672	2.672	2.672	2.672
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	3571	3576	3581	3586
46	Elevation (m)	0.1600	0.1300	0.1360	0.1420
47	Pressure (bar)	76.93	76.93	76.93	76.93
48	Temperature (C)	57.86	57.86	57.86	57.86
49	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
50	Heat Transferred (kJ/h-m)	742.4	742.3	742.2	742.2
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1766	0.1766	0.2482	0.2482
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	8.342e-003	8.342e-003	7.065e-003	7.065e-003
56	Static Gradient (kPa/m)	-9.970e-003	-9.970e-003	2.507e-003	2.507e-003
57	Accel. Gradient (kPa/m)	-1.667e-007	-1.667e-007	9.803e-007	9.803e-007
58	Liq. Reynolds	2.006e+005 *	2.006e+005 *	2.006e+005 *	2.006e+005 *
59	Vap. Reynolds	9.554e+006 *	9.554e+006 *	9.554e+006 *	9.554e+006 *
60	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
61	Vapour Velocity (m/s)	2.672	2.672	2.672	2.672
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	3591	3596	3601	3606
65	Elevation (m)	0.1480	0.1540	0.1600	0.1660
66	Pressure (bar)	76.93	76.93	76.93	76.93
67	Temperature (C)	57.86	57.85	57.85	57.85
68	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 87 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Heat Transferred (kJ/h-m)	742.1	742.0	742.0	741.9					
14	Flow Regime	Transition	Transition	Transition	Transition					
15	Water Flow Regime									
16	Liquid HoldUp	0.2482	0.2482	0.2482	0.2482					
17	Water HoldUp	---	---	---	---					
18	Friction Gradient (kPa/m)	7.065e-003	7.065e-003	7.065e-003	7.064e-003					
19	Static Gradient (kPa/m)	2.507e-003	2.507e-003	2.507e-003	2.518e-003					
20	Accel. Gradient (kPa/m)	9.803e-007	9.803e-007	9.803e-007	9.813e-007					
21	Liq. Reynolds	2.006e+005 *	2.006e+005 *	2.006e+005 *	2.006e+005 *					
22	Vap. Reynolds	9.554e+006 *	9.554e+006 *	9.554e+006 *	9.554e+006 *					
23	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992					
24	Vapour Velocity (m/s)	2.672	2.672	2.672	2.672					
25	Deposit Thickness (mm)	---	---	---	---					
26	Deposit Volume (m3)	---	---	---	---					
27	Length (m)	3611	3616	3621	3626					
28	Elevation (m)	0.1720	0.1780	0.1840	0.1900					
29	Pressure (bar)	76.93	76.93	76.93	76.93					
30	Temperature (C)	57.85	57.84	57.84	57.84					
31	Vapour Fraction	0.8813	0.8813	0.8813	0.8813					
32	Heat Transferred (kJ/h-m)	741.9	741.8	741.7	741.7					
33	Flow Regime	Transition	Transition	Transition	Transition					
34	Water Flow Regime									
35	Liquid HoldUp	0.2482	0.2482	0.2482	0.2482					
36	Water HoldUp	---	---	---	---					
37	Friction Gradient (kPa/m)	7.064e-003	7.063e-003	7.063e-003	7.063e-003					
38	Static Gradient (kPa/m)	2.518e-003	2.518e-003	2.518e-003	2.518e-003					
39	Accel. Gradient (kPa/m)	9.813e-007	9.813e-007	9.813e-007	9.813e-007					
40	Liq. Reynolds	2.006e+005 *	2.006e+005 *	2.006e+005 *	2.006e+005 *					
41	Vap. Reynolds	9.554e+006 *	9.554e+006 *	9.554e+006 *	9.554e+006 *					
42	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992					
43	Vapour Velocity (m/s)	2.672	2.672	2.672	2.672					
44	Deposit Thickness (mm)	---	---	---	---					
45	Deposit Volume (m3)	---	---	---	---					
46	Length (m)	3631	3636	3641	3646					
47	Elevation (m)	0.2140	0.2380	0.2620	0.2860					
48	Pressure (bar)	76.92	76.92	76.92	76.92					
49	Temperature (C)	57.84	57.84	57.83	57.83					
50	Vapour Fraction	0.8813	0.8813	0.8813	0.8813					
51	Heat Transferred (kJ/h-m)	741.6	741.6	741.5	741.4					
52	Flow Regime	Transition	Transition	Transition	Transition					
53	Water Flow Regime									
54	Liquid HoldUp	0.2586	0.2586	0.2586	0.2586					
55	Water HoldUp	---	---	---	---					
56	Friction Gradient (kPa/m)	5.982e-003	5.982e-003	5.982e-003	5.982e-003					
57	Static Gradient (kPa/m)	1.038e-002	1.038e-002	1.038e-002	1.038e-002					
58	Accel. Gradient (kPa/m)	1.675e-006	1.675e-006	1.675e-006	1.675e-006					
59	Liq. Reynolds	2.006e+005 *	2.005e+005 *	2.005e+005 *	2.005e+005 *					
60	Vap. Reynolds	9.554e+006 *	9.554e+006 *	9.554e+006 *	9.554e+006 *					
61	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992					
62	Vapour Velocity (m/s)	2.672	2.672	2.672	2.672					
63	Deposit Thickness (mm)	---	---	---	---					
64	Deposit Volume (m3)	---	---	---	---					
65	Length (m)	3651	3656	3661	3666					
66	Elevation (m)	0.3100	0.2960	0.2820	0.2680					
67	Pressure (bar)	76.92	76.92	76.92	76.92					
68	Temperature (C)	57.83	57.83	57.82	57.82					
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 88 of 937					

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
14	Heat Transferred (kJ/h-m)	741.4	741.4	741.3	741.3
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2586	0.1792	0.1792	0.1792
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	5.982e-003	8.312e-003	8.312e-003	8.311e-003
20	Static Gradient (kPa/m)	1.038e-002	-4.696e-003	-4.696e-003	-4.696e-003
21	Accel. Gradient (kPa/m)	1.675e-006	3.703e-007	3.703e-007	3.703e-007
22	Liq. Reynolds	2.005e+005 *	2.005e+005 *	2.005e+005 *	2.005e+005 *
23	Vap. Reynolds	9.554e+006 *	9.554e+006 *	9.554e+006 *	9.554e+006 *
24	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
25	Vapour Velocity (m/s)	2.672	2.672	2.672	2.672
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	3671	3676	3681	3686
29	Elevation (m)	0.2540	0.2400	0.2180	0.1960
30	Pressure (bar)	76.92	76.92	76.92	76.92
31	Temperature (C)	57.82	57.82	57.82	57.81
32	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
33	Heat Transferred (kJ/h-m)	741.2	741.1	741.1	741.0
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1792	0.1792	0.1779	0.1779
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	8.311e-003	8.311e-003	8.326e-003	8.326e-003
39	Static Gradient (kPa/m)	-4.696e-003	-4.696e-003	-7.346e-003	-7.346e-003
40	Accel. Gradient (kPa/m)	3.703e-007	3.703e-007	1.004e-007	1.004e-007
41	Liq. Reynolds	2.005e+005 *	2.005e+005 *	2.005e+005 *	2.005e+005 *
42	Vap. Reynolds	9.554e+006 *	9.554e+006 *	9.554e+006 *	9.554e+006 *
43	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
44	Vapour Velocity (m/s)	2.672	2.672	2.672	2.672
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	3691	3696	3701	3706
48	Elevation (m)	0.1740	0.1520	0.1300	0.1360
49	Pressure (bar)	76.92	76.92	76.92	76.92
50	Temperature (C)	57.81	57.81	57.81	57.81
51	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
52	Heat Transferred (kJ/h-m)	741.0	740.9	740.9	740.8
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.1779	0.1779	0.1779	0.2482
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	8.326e-003	8.326e-003	8.326e-003	7.063e-003
58	Static Gradient (kPa/m)	-7.346e-003	-7.346e-003	-7.346e-003	2.518e-003
59	Accel. Gradient (kPa/m)	1.004e-007	1.004e-007	1.004e-007	9.812e-007
60	Liq. Reynolds	2.005e+005 *	2.005e+005 *	2.005e+005 *	2.005e+005 *
61	Vap. Reynolds	9.554e+006 *	9.554e+006 *	9.554e+006 *	9.554e+006 *
62	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
63	Vapour Velocity (m/s)	2.672	2.672	2.672	2.672
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	3711	3716	3721	3726
67	Elevation (m)	0.1420	0.1480	0.1540	0.1600
68	Pressure (bar)	76.92	76.92	76.92	76.92
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 89 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Temperature (C)	57.80	57.80	57.80	57.80		
14	Vapour Fraction	0.8813	0.8813	0.8813	0.8813		
15	Heat Transferred (kJ/h-m)	740.7	740.6	740.6	740.5		
16	Flow Regime	Transition	Transition	Transition	Transition		
17	Water Flow Regime						
18	Liquid HoldUp	0.2482	0.2482	0.2482	0.2482		
19	Water HoldUp	---	---	---	---		
20	Friction Gradient (kPa/m)	7.063e-003	7.063e-003	7.063e-003	7.062e-003		
21	Static Gradient (kPa/m)	2.518e-003	2.518e-003	2.518e-003	2.518e-003		
22	Accel. Gradient (kPa/m)	9.812e-007	9.812e-007	9.812e-007	9.812e-007		
23	Liq. Reynolds	2.005e+005 *	2.005e+005 *	2.005e+005 *	2.005e+005 *		
24	Vap. Reynolds	9.554e+006 *	9.554e+006 *	9.554e+006 *	9.555e+006 *		
25	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992		
26	Vapour Velocity (m/s)	2.672	2.671	2.671	2.671		
27	Deposit Thickness (mm)	---	---	---	---		
28	Deposit Volume (m3)	---	---	---	---		
29	Length (m)	3731	3736	3741	3746		
30	Elevation (m)	0.1680	0.1760	0.1840	0.1920		
31	Pressure (bar)	76.92	76.92	76.92	76.92		
32	Temperature (C)	57.79	57.79	57.79	57.79		
33	Vapour Fraction	0.8813	0.8813	0.8813	0.8813		
34	Heat Transferred (kJ/h-m)	740.5	740.4	740.3	740.3		
35	Flow Regime	Transition	Transition	Transition	Transition		
36	Water Flow Regime						
37	Liquid HoldUp	0.2494	0.2494	0.2494	0.2494		
38	Water HoldUp	---	---	---	---		
39	Friction Gradient (kPa/m)	6.936e-003	6.935e-003	6.935e-003	6.935e-003		
40	Static Gradient (kPa/m)	3.369e-003	3.369e-003	3.369e-003	3.369e-003		
41	Accel. Gradient (kPa/m)	1.055e-006	1.055e-006	1.055e-006	1.055e-006		
42	Liq. Reynolds	2.005e+005 *	2.005e+005 *	2.005e+005 *	2.005e+005 *		
43	Vap. Reynolds	9.555e+006 *	9.555e+006 *	9.555e+006 *	9.555e+006 *		
44	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992		
45	Vapour Velocity (m/s)	2.671	2.671	2.671	2.671		
46	Deposit Thickness (mm)	---	---	---	---		
47	Deposit Volume (m3)	---	---	---	---		
48	Length (m)	3751	3756	3761	3766		
49	Elevation (m)	0.2000	0.1840	0.1680	0.1520		
50	Pressure (bar)	76.91	76.91	76.91	76.91		
51	Temperature (C)	57.79	57.78	57.78	57.78		
52	Vapour Fraction	0.8813	0.8813	0.8813	0.8813		
53	Heat Transferred (kJ/h-m)	740.2	740.2	740.2	740.1		
54	Flow Regime	Transition	Transition	Transition	Transition		
55	Water Flow Regime						
56	Liquid HoldUp	0.2494	0.1789	0.1789	0.1789		
57	Water HoldUp	---	---	---	---		
58	Friction Gradient (kPa/m)	6.935e-003	8.314e-003	8.314e-003	8.314e-003		
59	Static Gradient (kPa/m)	3.369e-003	-5.361e-003	-5.361e-003	-5.361e-003		
60	Accel. Gradient (kPa/m)	1.055e-006	3.024e-007	3.024e-007	3.024e-007		
61	Liq. Reynolds	2.005e+005 *	2.005e+005 *	2.005e+005 *	2.005e+005 *		
62	Vap. Reynolds	9.555e+006 *	9.555e+006 *	9.555e+006 *	9.555e+006 *		
63	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992		
64	Vapour Velocity (m/s)	2.671	2.671	2.671	2.671		
65	Deposit Thickness (mm)	---	---	---	---		
66	Deposit Volume (m3)	---	---	---	---		
67	Length (m)	3771	3776	3781	3787		
68	Elevation (m)	0.1360	0.1200	0.1100	1.000e-001		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Pressure (bar)	76.91	76.91	76.91	76.91					
14	Temperature (C)	57.78	57.77	57.77	57.77					
15	Vapour Fraction	0.8813	0.8813	0.8813	0.8813					
16	Heat Transferred (kJ/h-m)	740.0	740.0	739.9	739.9					
17	Flow Regime	Transition	Transition	Transition	Transition					
18	Water Flow Regime									
19	Liquid HoldUp	0.1789	0.1789	0.1799	0.1799					
20	Water HoldUp	---	---	---	---					
21	Friction Gradient (kPa/m)	8.314e-003	8.314e-003	8.303e-003	8.302e-003					
22	Static Gradient (kPa/m)	-5.362e-003	-5.362e-003	-3.267e-003	-3.267e-003					
23	Accel. Gradient (kPa/m)	3.024e-007	3.024e-007	5.158e-007	5.157e-007					
24	Liq. Reynolds	2.005e+005 *	2.005e+005 *	2.005e+005 *	2.005e+005 *					
25	Vap. Reynolds	9.555e+006 *	9.555e+006 *	9.555e+006 *	9.555e+006 *					
26	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992					
27	Vapour Velocity (m/s)	2.671	2.671	2.671	2.671					
28	Deposit Thickness (mm)	---	---	---	---					
29	Deposit Volume (m3)	---	---	---	---					
30	Length (m)	3792	3797	3802	3807					
31	Elevation (m)	9.000e-002	8.000e-002	7.000e-002	6.800e-002					
32	Pressure (bar)	76.91	76.91	76.91	76.91					
33	Temperature (C)	57.77	57.77	57.76	57.76					
34	Vapour Fraction	0.8813	0.8813	0.8813	0.8813					
35	Heat Transferred (kJ/h-m)	739.8	739.8	739.7	739.6					
36	Flow Regime	Transition	Transition	Transition	Transition					
37	Water Flow Regime									
38	Liquid HoldUp	0.1799	0.1799	0.1799	0.1811					
39	Water HoldUp	---	---	---	---					
40	Friction Gradient (kPa/m)	8.302e-003	8.302e-003	8.302e-003	8.288e-003					
41	Static Gradient (kPa/m)	-3.267e-003	-3.267e-003	-3.267e-003	-6.756e-004					
42	Accel. Gradient (kPa/m)	5.157e-007	5.157e-007	5.157e-007	7.796e-007					
43	Liq. Reynolds	2.005e+005 *	2.005e+005 *	2.005e+005 *	2.005e+005 *					
44	Vap. Reynolds	9.555e+006 *	9.555e+006 *	9.555e+006 *	9.555e+006 *					
45	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992					
46	Vapour Velocity (m/s)	2.671	2.671	2.671	2.671					
47	Deposit Thickness (mm)	---	---	---	---					
48	Deposit Volume (m3)	---	---	---	---					
49	Length (m)	3812	3817	3822	3827					
50	Elevation (m)	6.600e-002	6.400e-002	6.200e-002	6.000e-002					
51	Pressure (bar)	76.91	76.91	76.91	76.91					
52	Temperature (C)	57.76	57.76	57.75	57.75					
53	Vapour Fraction	0.8813	0.8813	0.8813	0.8812					
54	Heat Transferred (kJ/h-m)	739.6	739.5	739.5	739.4					
55	Flow Regime	Transition	Transition	Transition	Transition					
56	Water Flow Regime									
57	Liquid HoldUp	0.1811	0.1811	0.1811	0.1811					
58	Water HoldUp	---	---	---	---					
59	Friction Gradient (kPa/m)	8.288e-003	8.288e-003	8.288e-003	8.288e-003					
60	Static Gradient (kPa/m)	-6.757e-004	-6.757e-004	-6.757e-004	-6.757e-004					
61	Accel. Gradient (kPa/m)	7.796e-007	7.796e-007	7.796e-007	7.796e-007					
62	Liq. Reynolds	2.005e+005 *	2.005e+005 *	2.005e+005 *	2.005e+005 *					
63	Vap. Reynolds	9.555e+006 *	9.555e+006 *	9.555e+006 *	9.555e+006 *					
64	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992					
65	Vapour Velocity (m/s)	2.671	2.671	2.671	2.671					
66	Deposit Thickness (mm)	---	---	---	---					
67	Deposit Volume (m3)	---	---	---	---					
68	Length (m)	3832	3837	3842	3847					
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	8.000e-002	1.000e-001	0.1200	0.1400
14	Pressure (bar)	76.91	76.91	76.91	76.91
15	Temperature (C)	57.75	57.75	57.75	57.74
16	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
17	Heat Transferred (kJ/h-m)	739.3	739.2	739.2	739.1
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2563	0.2563	0.2563	0.2563
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	6.209e-003	6.209e-003	6.209e-003	6.209e-003
23	Static Gradient (kPa/m)	8.592e-003	8.592e-003	8.593e-003	8.593e-003
24	Accel. Gradient (kPa/m)	1.516e-006	1.516e-006	1.516e-006	1.516e-006
25	Liq. Reynolds	2.004e+005 *	2.004e+005 *	2.004e+005 *	2.004e+005 *
26	Vap. Reynolds	9.555e+006 *	9.555e+006 *	9.555e+006 *	9.555e+006 *
27	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
28	Vapour Velocity (m/s)	2.671	2.671	2.671	2.671
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	3852	3857	3862	3867
32	Elevation (m)	0.1600	0.1660	0.1720	0.1780
33	Pressure (bar)	76.91	76.91	76.91	76.91
34	Temperature (C)	57.74	57.74	57.74	57.73
35	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
36	Heat Transferred (kJ/h-m)	739.0	739.0	738.9	738.9
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2563	0.2482	0.2482	0.2482
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	6.209e-003	7.061e-003	7.061e-003	7.061e-003
42	Static Gradient (kPa/m)	8.593e-003	2.518e-003	2.518e-003	2.518e-003
43	Accel. Gradient (kPa/m)	1.516e-006	9.811e-007	9.811e-007	9.811e-007
44	Liq. Reynolds	2.004e+005 *	2.004e+005 *	2.004e+005 *	2.004e+005 *
45	Vap. Reynolds	9.555e+006 *	9.555e+006 *	9.555e+006 *	9.555e+006 *
46	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
47	Vapour Velocity (m/s)	2.671	2.671	2.671	2.671
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	3872	3877	3882	3887
51	Elevation (m)	0.1840	0.1900	0.2040	0.2180
52	Pressure (bar)	76.91	76.90	76.90	76.90
53	Temperature (C)	57.73	57.73	57.73	57.73
54	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
55	Heat Transferred (kJ/h-m)	738.8	738.7	738.7	738.6
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2482	0.2482	0.2529	0.2529
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	7.061e-003	7.061e-003	6.564e-003	6.564e-003
61	Static Gradient (kPa/m)	2.518e-003	2.518e-003	5.956e-003	5.956e-003
62	Accel. Gradient (kPa/m)	9.811e-007	9.811e-007	1.282e-006	1.282e-006
63	Liq. Reynolds	2.004e+005 *	2.004e+005 *	2.004e+005 *	2.004e+005 *
64	Vap. Reynolds	9.555e+006 *	9.555e+006 *	9.555e+006 *	9.555e+006 *
65	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
66	Vapour Velocity (m/s)	2.671	2.671	2.671	2.671
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 92 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	3892	3897	3902	3907
14	Elevation (m)	0.2320	0.2460	0.2600	0.2680
15	Pressure (bar)	76.90	76.90	76.90	76.90
16	Temperature (C)	57.72	57.72	57.72	57.72
17	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
18	Heat Transferred (kJ/h-m)	738.6	738.5	738.4	738.4
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.2529	0.2529	0.2529	0.2494
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	6.564e-003	6.564e-003	6.564e-003	6.934e-003
24	Static Gradient (kPa/m)	5.956e-003	5.956e-003	5.956e-003	3.369e-003
25	Accel. Gradient (kPa/m)	1.282e-006	1.282e-006	1.282e-006	1.055e-006
26	Liq. Reynolds	2.004e+005 *	2.004e+005 *	2.004e+005 *	2.004e+005 *
27	Vap. Reynolds	9.555e+006 *	9.555e+006 *	9.555e+006 *	9.555e+006 *
28	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
29	Vapour Velocity (m/s)	2.671	2.671	2.671	2.671
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	3912	3917	3922	3927
33	Elevation (m)	0.2760	0.2840	0.2920	0.3000
34	Pressure (bar)	76.90	76.90	76.90	76.90
35	Temperature (C)	57.71	57.71	57.71	57.71
36	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
37	Heat Transferred (kJ/h-m)	738.3	738.3	738.2	738.1
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.2494	0.2494	0.2494	0.2494
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	6.934e-003	6.934e-003	6.934e-003	6.934e-003
43	Static Gradient (kPa/m)	3.369e-003	3.369e-003	3.369e-003	3.369e-003
44	Accel. Gradient (kPa/m)	1.055e-006	1.055e-006	1.055e-006	1.055e-006
45	Liq. Reynolds	2.004e+005 *	2.004e+005 *	2.004e+005 *	2.004e+005 *
46	Vap. Reynolds	9.555e+006 *	9.555e+006 *	9.555e+006 *	9.555e+006 *
47	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
48	Vapour Velocity (m/s)	2.671	2.671	2.671	2.671
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	3932	3937	3942	3947
52	Elevation (m)	0.3120	0.3240	0.3360	0.3480
53	Pressure (bar)	76.90	76.90	76.90	76.90
54	Temperature (C)	57.70	57.70	57.70	57.70
55	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
56	Heat Transferred (kJ/h-m)	738.1	738.0	738.0	737.9
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.2517	0.2517	0.2517	0.2517
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	6.686e-003	6.686e-003	6.686e-003	6.685e-003
62	Static Gradient (kPa/m)	5.088e-003	5.088e-003	5.088e-003	5.088e-003
63	Accel. Gradient (kPa/m)	1.206e-006	1.206e-006	1.206e-006	1.206e-006
64	Liq. Reynolds	2.004e+005 *	2.004e+005 *	2.004e+005 *	2.004e+005 *
65	Vap. Reynolds	9.555e+006 *	9.555e+006 *	9.555e+006 *	9.555e+006 *
66	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
67	Vapour Velocity (m/s)	2.671	2.671	2.671	2.671
68	Deposit Thickness (mm)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	3952	3957	3962	3967
15	Elevation (m)	0.3600	0.3640	0.3680	0.3720
16	Pressure (bar)	76.90	76.90	76.90	76.89
17	Temperature (C)	57.70	57.69	57.69	57.69
18	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
19	Heat Transferred (kJ/h-m)	737.8	737.8	737.7	737.7
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2517	0.2471	0.2471	0.2471
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	6.685e-003	7.190e-003	7.189e-003	7.189e-003
25	Static Gradient (kPa/m)	5.088e-003	1.673e-003	1.673e-003	1.673e-003
26	Accel. Gradient (kPa/m)	1.206e-006	9.078e-007	9.078e-007	9.078e-007
27	Liq. Reynolds	2.004e+005 *	2.004e+005 *	2.004e+005 *	2.004e+005 *
28	Vap. Reynolds	9.555e+006 *	9.555e+006 *	9.555e+006 *	9.555e+006 *
29	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
30	Vapour Velocity (m/s)	2.671	2.671	2.671	2.671
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	3972	3977	3982	3987
34	Elevation (m)	0.3760	0.3800	0.3800	0.3800
35	Pressure (bar)	76.89	76.89	76.89	76.89
36	Temperature (C)	57.69	57.68	57.68	57.68
37	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
38	Heat Transferred (kJ/h-m)	737.6	737.5	737.5	737.4
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2471	0.2471	0.2649	0.2649
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	7.189e-003	7.189e-003	6.263e-003	6.263e-003
44	Static Gradient (kPa/m)	1.673e-003	1.673e-003	0.0000	0.0000
45	Accel. Gradient (kPa/m)	9.078e-007	9.078e-007	6.414e-007	6.414e-007
46	Liq. Reynolds	2.004e+005 *	2.004e+005 *	2.004e+005 *	2.004e+005 *
47	Vap. Reynolds	9.555e+006 *	9.555e+006 *	9.555e+006 *	9.556e+006 *
48	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
49	Vapour Velocity (m/s)	2.671	2.671	2.671	2.671
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	3992	3997	4002	4007
53	Elevation (m)	0.3800	0.3800	0.3800	0.3600
54	Pressure (bar)	76.89	76.89	76.89	76.89
55	Temperature (C)	57.68	57.68	57.67	57.67
56	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
57	Heat Transferred (kJ/h-m)	737.4	737.3	737.2	737.2
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2649	0.2649	0.2649	0.1783
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	6.263e-003	6.263e-003	6.263e-003	8.320e-003
63	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-6.688e-003
64	Accel. Gradient (kPa/m)	6.414e-007	6.414e-007	6.414e-007	1.672e-007
65	Liq. Reynolds	2.004e+005 *	2.004e+005 *	2.004e+005 *	2.003e+005 *
66	Vap. Reynolds	9.556e+006 *	9.556e+006 *	9.556e+006 *	9.556e+006 *
67	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
68	Vapour Velocity (m/s)	2.671	2.671	2.671	2.671
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Thickness	(mm)	---	---	---	---	
14	Deposit Volume	(m3)	---	---	---	---	
15	Length	(m)	4012	4017	4022	4027	
16	Elevation	(m)	0.3400	0.3200	0.3000	0.2800	
17	Pressure	(bar)	76.89	76.89	76.89	76.89	
18	Temperature	(C)	57.67	57.67	57.67	57.66	
19	Vapour Fraction		0.8812	0.8812	0.8812	0.8812	
20	Heat Transferred	(kJ/h-m)	737.2	737.1	737.1	737.0	
21	Flow Regime		Transition	Transition	Transition	Transition	
22	Water Flow Regime						
23	Liquid HoldUp		0.1783	0.1783	0.1783	0.1783	
24	Water HoldUp		---	---	---	---	
25	Friction Gradient	(kPa/m)	8.320e-003	8.320e-003	8.320e-003	8.320e-003	
26	Static Gradient	(kPa/m)	-6.688e-003	-6.688e-003	-6.688e-003	-6.688e-003	
27	Accel. Gradient	(kPa/m)	1.672e-007	1.672e-007	1.672e-007	1.672e-007	
28	Liq. Reynolds		2.003e+005 *	2.003e+005 *	2.003e+005 *	2.003e+005 *	
29	Vap. Reynolds		9.556e+006 *	9.556e+006 *	9.556e+006 *	9.556e+006 *	
30	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992	
31	Vapour Velocity	(m/s)	2.671	2.671	2.671	2.671	
32	Deposit Thickness	(mm)	---	---	---	---	
33	Deposit Volume	(m3)	---	---	---	---	
34	Length	(m)	4032	4037	4042	4047	
35	Elevation	(m)	0.2760	0.2720	0.2680	0.2640	
36	Pressure	(bar)	76.89	76.89	76.89	76.89	
37	Temperature	(C)	57.66	57.66	57.66	57.65	
38	Vapour Fraction		0.8812	0.8812	0.8812	0.8812	
39	Heat Transferred	(kJ/h-m)	737.0	736.9	736.8	736.8	
40	Flow Regime		Transition	Transition	Transition	Transition	
41	Water Flow Regime						
42	Liquid HoldUp		0.1808	0.1808	0.1808	0.1808	
43	Water HoldUp		---	---	---	---	
44	Friction Gradient	(kPa/m)	8.291e-003	8.291e-003	8.291e-003	8.290e-003	
45	Static Gradient	(kPa/m)	-1.350e-003	-1.350e-003	-1.350e-003	-1.350e-003	
46	Accel. Gradient	(kPa/m)	7.108e-007	7.108e-007	7.108e-007	7.108e-007	
47	Liq. Reynolds		2.003e+005 *	2.003e+005 *	2.003e+005 *	2.003e+005 *	
48	Vap. Reynolds		9.556e+006 *	9.556e+006 *	9.556e+006 *	9.556e+006 *	
49	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992	
50	Vapour Velocity	(m/s)	2.671	2.671	2.671	2.671	
51	Deposit Thickness	(mm)	---	---	---	---	
52	Deposit Volume	(m3)	---	---	---	---	
53	Length	(m)	4052	4057	4062	4067	
54	Elevation	(m)	0.2600	0.2400	0.2200	0.2000	
55	Pressure	(bar)	76.89	76.89	76.89	76.89	
56	Temperature	(C)	57.65	57.65	57.65	57.65	
57	Vapour Fraction		0.8812	0.8812	0.8812	0.8812	
58	Heat Transferred	(kJ/h-m)	736.7	736.7	736.6	736.6	
59	Flow Regime		Transition	Transition	Transition	Transition	
60	Water Flow Regime						
61	Liquid HoldUp		0.1808	0.1783	0.1783	0.1783	
62	Water HoldUp		---	---	---	---	
63	Friction Gradient	(kPa/m)	8.290e-003	8.320e-003	8.320e-003	8.320e-003	
64	Static Gradient	(kPa/m)	-1.350e-003	-6.688e-003	-6.688e-003	-6.688e-003	
65	Accel. Gradient	(kPa/m)	7.108e-007	1.671e-007	1.671e-007	1.671e-007	
66	Liq. Reynolds		2.003e+005 *	2.003e+005 *	2.003e+005 *	2.003e+005 *	
67	Vap. Reynolds		9.556e+006 *	9.556e+006 *	9.556e+006 *	9.556e+006 *	
68	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.671	2.671	2.671	2.670	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	4072	4077	4084	4091	
17	Elevation	(m)	0.1800	0.1600	0.1580	0.1560	
18	Pressure	(bar)	76.89	76.89	76.89	76.89	
19	Temperature	(C)	57.64	57.64	57.64	57.64	
20	Vapour Fraction		0.8812	0.8812	0.8812	0.8812	
21	Heat Transferred	(kJ/h-m)	736.5	736.5	736.4	736.3	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.1783	0.1783	0.1813	0.1813	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	8.320e-003	8.319e-003	8.285e-003	8.285e-003	
27	Static Gradient	(kPa/m)	-6.688e-003	-6.688e-003	-4.829e-004	-4.829e-004	
28	Accel. Gradient	(kPa/m)	1.671e-007	1.671e-007	7.991e-007	7.991e-007	
29	Liq. Reynolds		2.003e+005 *	2.003e+005 *	2.003e+005 *	2.003e+005 *	
30	Vap. Reynolds		9.556e+006 *	9.556e+006 *	9.556e+006 *	9.556e+006 *	
31	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992	
32	Vapour Velocity	(m/s)	2.670	2.670	2.670	2.670	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	4098	4105	4112	4120	
36	Elevation	(m)	0.1540	0.1520	0.1500	0.1460	
37	Pressure	(bar)	76.89	76.89	76.89	76.89	
38	Temperature	(C)	57.63	57.63	57.63	57.62	
39	Vapour Fraction		0.8812	0.8812	0.8812	0.8812	
40	Heat Transferred	(kJ/h-m)	736.2	736.1	736.0	736.0	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.1813	0.1813	0.1813	0.1811	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	8.285e-003	8.285e-003	8.285e-003	8.287e-003	
46	Static Gradient	(kPa/m)	-4.829e-004	-4.829e-004	-4.829e-004	-8.889e-004	
47	Accel. Gradient	(kPa/m)	7.991e-007	7.991e-007	7.990e-007	7.577e-007	
48	Liq. Reynolds		2.003e+005 *	2.003e+005 *	2.003e+005 *	2.003e+005 *	
49	Vap. Reynolds		9.556e+006 *	9.556e+006 *	9.556e+006 *	9.556e+006 *	
50	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992	
51	Vapour Velocity	(m/s)	2.670	2.670	2.670	2.670	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	4127	4135	4143	4150	
55	Elevation	(m)	0.1420	0.1380	0.1340	0.1300	
56	Pressure	(bar)	76.89	76.89	76.88	76.88	
57	Temperature	(C)	57.62	57.62	57.61	57.61	
58	Vapour Fraction		0.8812	0.8812	0.8812	0.8812	
59	Heat Transferred	(kJ/h-m)	735.9	735.8	735.7	735.6	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.1811	0.1811	0.1811	0.1811	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	8.287e-003	8.287e-003	8.287e-003	8.287e-003	
65	Static Gradient	(kPa/m)	-8.889e-004	-8.890e-004	-8.890e-004	-8.890e-004	
66	Accel. Gradient	(kPa/m)	7.577e-007	7.577e-007	7.577e-007	7.576e-007	
67	Liq. Reynolds		2.003e+005 *	2.003e+005 *	2.003e+005 *	2.003e+005 *	
68	Vap. Reynolds		9.556e+006 *	9.556e+006 *	9.556e+006 *	9.556e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992
14	Vapour Velocity	(m/s)	2.670	2.670	2.670	2.670
15	Deposit Thickness	(mm)	---	---	---	---
16	Deposit Volume	(m3)	---	---	---	---
17	Length	(m)	4158	4165	4173	4181
18	Elevation	(m)	0.1140	9.800e-002	8.200e-002	6.600e-002
19	Pressure	(bar)	76.88	76.88	76.88	76.88
20	Temperature	(C)	57.61	57.60	57.60	57.60
21	Vapour Fraction		0.8812	0.8812	0.8812	0.8812
22	Heat Transferred	(kJ/h-m)	735.5	735.4	735.3	735.3
23	Flow Regime		Transition	Transition	Transition	Transition
24	Water Flow Regime					
25	Liquid HoldUp		0.1798	0.1798	0.1798	0.1798
26	Water HoldUp		---	---	---	---
27	Friction Gradient	(kPa/m)	8.301e-003	8.301e-003	8.301e-003	8.301e-003
28	Static Gradient	(kPa/m)	-3.540e-003	-3.540e-003	-3.540e-003	-3.540e-003
29	Accel. Gradient	(kPa/m)	4.876e-007	4.876e-007	4.876e-007	4.876e-007
30	Liq. Reynolds		2.003e+005 *	2.003e+005 *	2.003e+005 *	2.003e+005 *
31	Vap. Reynolds		9.556e+006 *	9.556e+006 *	9.556e+006 *	9.556e+006 *
32	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992
33	Vapour Velocity	(m/s)	2.670	2.670	2.670	2.670
34	Deposit Thickness	(mm)	---	---	---	---
35	Deposit Volume	(m3)	---	---	---	---
36	Length	(m)	4188	4196	4203	4211
37	Elevation	(m)	5.000e-002	1.400e-002	-2.200e-002	-5.800e-002
38	Pressure	(bar)	76.88	76.88	76.88	76.88
39	Temperature	(C)	57.59	57.59	57.59	57.58
40	Vapour Fraction		0.8812	0.8812	0.8812	0.8812
41	Heat Transferred	(kJ/h-m)	735.2	735.1	735.0	734.9
42	Flow Regime		Transition	Transition	Transition	Transition
43	Water Flow Regime					
44	Liquid HoldUp		0.1798	0.1777	0.1777	0.1777
45	Water HoldUp		---	---	---	---
46	Friction Gradient	(kPa/m)	8.301e-003	8.325e-003	8.325e-003	8.325e-003
47	Static Gradient	(kPa/m)	-3.540e-003	-7.904e-003	-7.904e-003	-7.905e-003
48	Accel. Gradient	(kPa/m)	4.876e-007	4.310e-008	4.308e-008	4.306e-008
49	Liq. Reynolds		2.003e+005 *	2.003e+005 *	2.003e+005 *	2.003e+005 *
50	Vap. Reynolds		9.556e+006 *	9.556e+006 *	9.556e+006 *	9.556e+006 *
51	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992
52	Vapour Velocity	(m/s)	2.670	2.670	2.670	2.670
53	Deposit Thickness	(mm)	---	---	---	---
54	Deposit Volume	(m3)	---	---	---	---
55	Length	(m)	4219	4226	4234	4241
56	Elevation	(m)	-9.400e-002	-0.1300	-8.200e-002	-3.400e-002
57	Pressure	(bar)	76.88	76.88	76.88	76.88
58	Temperature	(C)	57.58	57.58	57.58	57.57
59	Vapour Fraction		0.8812	0.8812	0.8812	0.8812
60	Heat Transferred	(kJ/h-m)	734.9	734.8	734.6	734.5
61	Flow Regime		Transition	Transition	Transition	Transition
62	Water Flow Regime					
63	Liquid HoldUp		0.1777	0.1777	0.2631	0.2631
64	Water HoldUp		---	---	---	---
65	Friction Gradient	(kPa/m)	8.325e-003	8.325e-003	5.560e-003	5.560e-003
66	Static Gradient	(kPa/m)	-7.905e-003	-7.905e-003	1.383e-002	1.383e-002
67	Accel. Gradient	(kPa/m)	4.304e-008	4.302e-008	1.986e-006	1.986e-006
68	Liq. Reynolds		2.002e+005 *	2.002e+005 *	2.002e+005 *	2.002e+005 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.556e+006 *	9.556e+006 *	9.556e+006 *	9.556e+006 *
14	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
15	Vapour Velocity (m/s)	2.670	2.670	2.670	2.670
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	4249	4257	4264	4272
19	Elevation (m)	1.400e-002	6.200e-002	0.1100	0.1260
20	Pressure (bar)	76.88	76.88	76.88	76.87
21	Temperature (C)	57.57	57.56	57.56	57.56
22	Vapour Fraction	0.8812	0.8812	0.8811	0.8811
23	Heat Transferred (kJ/h-m)	734.4	734.3	734.2	734.1
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2631	0.2631	0.2631	0.2509
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	5.560e-003	5.560e-003	5.560e-003	6.773e-003
29	Static Gradient (kPa/m)	1.383e-002	1.383e-002	1.383e-002	4.454e-003
30	Accel. Gradient (kPa/m)	1.986e-006	1.986e-006	1.986e-006	1.150e-006
31	Liq. Reynolds	2.002e+005 *	2.002e+005 *	2.002e+005 *	2.002e+005 *
32	Vap. Reynolds	9.556e+006 *	9.556e+006 *	9.556e+006 *	9.556e+006 *
33	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
34	Vapour Velocity (m/s)	2.670	2.670	2.670	2.670
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	4279	4287	4295	4302
38	Elevation (m)	0.1420	0.1580	0.1740	0.1900
39	Pressure (bar)	76.87	76.87	76.87	76.87
40	Temperature (C)	57.55	57.55	57.55	57.54
41	Vapour Fraction	0.8811	0.8811	0.8811	0.8811
42	Heat Transferred (kJ/h-m)	734.0	734.0	733.9	733.8
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2509	0.2509	0.2509	0.2509
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.773e-003	6.773e-003	6.773e-003	6.773e-003
48	Static Gradient (kPa/m)	4.454e-003	4.454e-003	4.454e-003	4.454e-003
49	Accel. Gradient (kPa/m)	1.150e-006	1.150e-006	1.150e-006	1.150e-006
50	Liq. Reynolds	2.002e+005 *	2.002e+005 *	2.002e+005 *	2.002e+005 *
51	Vap. Reynolds	9.556e+006 *	9.556e+006 *	9.556e+006 *	9.556e+006 *
52	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
53	Vapour Velocity (m/s)	2.670	2.670	2.670	2.670
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	4304	4305	4306	4308
57	Elevation (m)	0.1760	0.1620	0.1480	0.1340
58	Pressure (bar)	76.87	76.87	76.87	76.87
59	Temperature (C)	57.54	57.54	57.54	57.54
60	Vapour Fraction	0.8811	0.8811	0.8811	0.8811
61	Heat Transferred (kJ/h-m)	733.8	733.8	733.8	733.8
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1737	0.1737	0.1737	0.1737
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	8.374e-003	8.374e-003	8.374e-003	8.374e-003
67	Static Gradient (kPa/m)	-1.614e-002	-1.614e-002	-1.614e-002	-1.614e-002
68	Accel. Gradient (kPa/m)	-7.955e-007	-7.955e-007	-7.955e-007	-7.955e-007
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		2.002e+005 *	2.002e+005 *	2.002e+005 *	2.002e+005 *	
14	Vap. Reynolds		9.556e+006 *	9.556e+006 *	9.556e+006 *	9.556e+006 *	
15	Liquid Velocity (m/s)		0.1992	0.1992	0.1992	0.1992	
16	Vapour Velocity (m/s)		2.670	2.670	2.670	2.670	
17	Deposit Thickness (mm)		---	---	---	---	
18	Deposit Volume (m3)		---	---	---	---	
19	Length (m)		4309	4314	4319	4324	
20	Elevation (m)		0.1200	7.000e-002	2.000e-002	-3.000e-002	
21	Pressure (bar)		76.87	76.87	76.87	76.87	
22	Temperature (C)		57.54	57.54	57.54	57.54	
23	Vapour Fraction		0.8811	0.8811	0.8811	0.8811	
24	Heat Transferred (kJ/h-m)		733.8	733.7	733.7	733.6	
25	Flow Regime		Transition	Transition	Transition	Transition	
26	Water Flow Regime						
27	Liquid HoldUp		0.1737	0.1736	0.1736	0.1736	
28	Water HoldUp		---	---	---	---	
29	Friction Gradient (kPa/m)		8.374e-003	8.376e-003	8.376e-003	8.376e-003	
30	Static Gradient (kPa/m)		-1.614e-002	-1.643e-002	-1.643e-002	-1.643e-002	
31	Accel. Gradient (kPa/m)		-7.955e-007	-8.251e-007	-8.252e-007	-8.252e-007	
32	Liq. Reynolds		2.002e+005 *	2.002e+005 *	2.002e+005 *	2.002e+005 *	
33	Vap. Reynolds		9.556e+006 *	9.556e+006 *	9.556e+006 *	9.556e+006 *	
34	Liquid Velocity (m/s)		0.1992	0.1992	0.1992	0.1992	
35	Vapour Velocity (m/s)		2.670	2.670	2.670	2.670	
36	Deposit Thickness (mm)		---	---	---	---	
37	Deposit Volume (m3)		---	---	---	---	
38	Length (m)		4329	4334	4339	4344	
39	Elevation (m)		-8.000e-002	-0.1300	-0.1580	-0.1860	
40	Pressure (bar)		76.87	76.87	76.87	76.87	
41	Temperature (C)		57.53	57.53	57.53	57.53	
42	Vapour Fraction		0.8811	0.8811	0.8811	0.8811	
43	Heat Transferred (kJ/h-m)		733.6	733.5	733.5	733.4	
44	Flow Regime		Transition	Transition	Transition	Transition	
45	Water Flow Regime						
46	Liquid HoldUp		0.1736	0.1736	0.1771	0.1771	
47	Water HoldUp		---	---	---	---	
48	Friction Gradient (kPa/m)		8.376e-003	8.376e-003	8.332e-003	8.332e-003	
49	Static Gradient (kPa/m)		-1.643e-002	-1.643e-002	-9.323e-003	-9.323e-003	
50	Accel. Gradient (kPa/m)		-8.252e-007	-8.252e-007	-1.014e-007	-1.014e-007	
51	Liq. Reynolds		2.002e+005 *	2.002e+005 *	2.002e+005 *	2.002e+005 *	
52	Vap. Reynolds		9.556e+006 *	9.556e+006 *	9.556e+006 *	9.556e+006 *	
53	Liquid Velocity (m/s)		0.1992	0.1992	0.1992	0.1992	
54	Vapour Velocity (m/s)		2.670	2.670	2.670	2.669	
55	Deposit Thickness (mm)		---	---	---	---	
56	Deposit Volume (m3)		---	---	---	---	
57	Length (m)		4349	4354	4359	4364	
58	Elevation (m)		-0.2140	-0.2420	-0.2700	-0.2320	
59	Pressure (bar)		76.87	76.87	76.87	76.87	
60	Temperature (C)		57.53	57.52	57.52	57.52	
61	Vapour Fraction		0.8811	0.8811	0.8811	0.8811	
62	Heat Transferred (kJ/h-m)		733.3	733.3	733.2	733.1	
63	Flow Regime		Transition	Transition	Transition	Transition	
64	Water Flow Regime						
65	Liquid HoldUp		0.1771	0.1771	0.1771	0.2669	
66	Water HoldUp		---	---	---	---	
67	Friction Gradient (kPa/m)		8.332e-003	8.332e-003	8.332e-003	6.673e-003	
68	Static Gradient (kPa/m)		-9.323e-003	-9.323e-003	-9.323e-003	1.682e-002	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient	(kPa/m)	-1.015e-007	-1.015e-007	-1.015e-007	2.405e-006	
14	Liq. Reynolds		2.002e+005 *	2.002e+005 *	2.002e+005 *	2.002e+005 *	
15	Vap. Reynolds		9.556e+006 *	9.556e+006 *	9.556e+006 *	9.556e+006 *	
16	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992	
17	Vapour Velocity	(m/s)	2.669	2.669	2.669	2.669	
18	Deposit Thickness	(mm)	---	---	---	---	
19	Deposit Volume	(m3)	---	---	---	---	
20	Length	(m)	4369	4374	4379	4384	
21	Elevation	(m)	-0.1940	-0.1560	-0.1180	-8.000e-002	
22	Pressure	(bar)	76.87	76.87	76.87	76.87	
23	Temperature	(C)	57.52	57.51	57.51	57.51	
24	Vapour Fraction		0.8811	0.8811	0.8811	0.8811	
25	Heat Transferred	(kJ/h-m)	733.1	733.0	732.9	732.9	
26	Flow Regime		Transition	Transition	Transition	Transition	
27	Water Flow Regime						
28	Liquid HoldUp		0.2669	0.2669	0.2669	0.2669	
29	Water HoldUp		---	---	---	---	
30	Friction Gradient	(kPa/m)	6.673e-003	6.673e-003	6.673e-003	6.673e-003	
31	Static Gradient	(kPa/m)	1.682e-002	1.682e-002	1.682e-002	1.682e-002	
32	Accel. Gradient	(kPa/m)	2.405e-006	2.405e-006	2.406e-006	2.406e-006	
33	Liq. Reynolds		2.002e+005 *	2.002e+005 *	2.002e+005 *	2.002e+005 *	
34	Vap. Reynolds		9.557e+006 *	9.557e+006 *	9.557e+006 *	9.557e+006 *	
35	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992	
36	Vapour Velocity	(m/s)	2.669	2.669	2.669	2.669	
37	Deposit Thickness	(mm)	---	---	---	---	
38	Deposit Volume	(m3)	---	---	---	---	
39	Length	(m)	4389	4394	4399	4404	
40	Elevation	(m)	-7.000e-002	-6.000e-002	-5.000e-002	-4.000e-002	
41	Pressure	(bar)	76.87	76.87	76.87	76.87	
42	Temperature	(C)	57.51	57.50	57.50	57.50	
43	Vapour Fraction		0.8811	0.8811	0.8811	0.8811	
44	Heat Transferred	(kJ/h-m)	732.8	732.7	732.7	732.6	
45	Flow Regime		Transition	Transition	Transition	Transition	
46	Water Flow Regime						
47	Liquid HoldUp		0.2507	0.2507	0.2507	0.2507	
48	Water HoldUp		---	---	---	---	
49	Friction Gradient	(kPa/m)	6.805e-003	6.804e-003	6.804e-003	6.804e-003	
50	Static Gradient	(kPa/m)	4.228e-003	4.228e-003	4.228e-003	4.228e-003	
51	Accel. Gradient	(kPa/m)	1.130e-006	1.130e-006	1.130e-006	1.130e-006	
52	Liq. Reynolds		2.002e+005 *	2.002e+005 *	2.002e+005 *	2.001e+005 *	
53	Vap. Reynolds		9.557e+006 *	9.557e+006 *	9.557e+006 *	9.557e+006 *	
54	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992	
55	Vapour Velocity	(m/s)	2.669	2.669	2.669	2.669	
56	Deposit Thickness	(mm)	---	---	---	---	
57	Deposit Volume	(m3)	---	---	---	---	
58	Length	(m)	4409	4414	4419	4424	
59	Elevation	(m)	-3.000e-002	8.000e-003	4.600e-002	8.400e-002	
60	Pressure	(bar)	76.87	76.86	76.86	76.86	
61	Temperature	(C)	57.50	57.50	57.49	57.49	
62	Vapour Fraction		0.8811	0.8811	0.8811	0.8811	
63	Heat Transferred	(kJ/h-m)	732.6	732.5	732.4	732.4	
64	Flow Regime		Transition	Transition	Transition	Transition	
65	Water Flow Regime						
66	Liquid HoldUp		0.2507	0.2669	0.2669	0.2669	
67	Water HoldUp		---	---	---	---	
68	Friction Gradient	(kPa/m)	6.804e-003	6.673e-003	6.673e-003	6.673e-003	
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1				Case Name: Simulation - 386MMSCFD_80barg.hsc	
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3	 Epic Atlantic Limited			Unit Set: Mixed	
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6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
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11	Pipe Table				
12					
13	Static Gradient	(kPa/m)	4.228e-003	1.682e-002	1.682e-002
14	Accel. Gradient	(kPa/m)	1.130e-006	2.406e-006	2.406e-006
15	Liq. Reynolds		2.001e+005 *	2.001e+005 *	2.001e+005 *
16	Vap. Reynolds		9.557e+006 *	9.557e+006 *	9.557e+006 *
17	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992
18	Vapour Velocity	(m/s)	2.669	2.669	2.669
19	Deposit Thickness	(mm)	---	---	---
20	Deposit Volume	(m3)	---	---	---
21	Length	(m)	4429	4434	4439
22	Elevation	(m)	0.1220	0.1600	0.1540
23	Pressure	(bar)	76.86	76.86	76.86
24	Temperature	(C)	57.49	57.49	57.48
25	Vapour Fraction		0.8811	0.8811	0.8811
26	Heat Transferred	(kJ/h-m)	732.3	732.2	732.2
27	Flow Regime		Transition	Transition	Transition
28	Water Flow Regime				
29	Liquid HoldUp		0.2669	0.2669	0.1806
30	Water HoldUp		---	---	---
31	Friction Gradient	(kPa/m)	6.673e-003	6.673e-003	8.291e-003
32	Static Gradient	(kPa/m)	1.682e-002	1.682e-002	-2.023e-003
33	Accel. Gradient	(kPa/m)	2.406e-006	2.406e-006	6.419e-007
34	Liq. Reynolds		2.001e+005 *	2.001e+005 *	2.001e+005 *
35	Vap. Reynolds		9.557e+006 *	9.557e+006 *	9.557e+006 *
36	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992
37	Vapour Velocity	(m/s)	2.669	2.669	2.669
38	Deposit Thickness	(mm)	---	---	---
39	Deposit Volume	(m3)	---	---	---
40	Length	(m)	4449	4454	4459
41	Elevation	(m)	0.1420	0.1360	0.1300
42	Pressure	(bar)	76.86	76.86	76.86
43	Temperature	(C)	57.48	57.48	57.47
44	Vapour Fraction		0.8811	0.8811	0.8811
45	Heat Transferred	(kJ/h-m)	732.1	732.0	732.0
46	Flow Regime		Transition	Transition	Transition
47	Water Flow Regime				
48	Liquid HoldUp		0.1806	0.1806	0.1806
49	Water HoldUp		---	---	---
50	Friction Gradient	(kPa/m)	8.291e-003	8.291e-003	8.291e-003
51	Static Gradient	(kPa/m)	-2.023e-003	-2.023e-003	-2.023e-003
52	Accel. Gradient	(kPa/m)	6.419e-007	6.419e-007	6.419e-007
53	Liq. Reynolds		2.001e+005 *	2.001e+005 *	2.001e+005 *
54	Vap. Reynolds		9.557e+006 *	9.557e+006 *	9.557e+006 *
55	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992
56	Vapour Velocity	(m/s)	2.669	2.669	2.669
57	Deposit Thickness	(mm)	---	---	---
58	Deposit Volume	(m3)	---	---	---
59	Length	(m)	4469	4474	4479
60	Elevation	(m)	0.1540	0.1660	0.1780
61	Pressure	(bar)	76.86	76.86	76.86
62	Temperature	(C)	57.47	57.47	57.47
63	Vapour Fraction		0.8811	0.8811	0.8811
64	Heat Transferred	(kJ/h-m)	731.8	731.8	731.7
65	Flow Regime		Transition	Transition	Transition
66	Water Flow Regime				
67	Liquid HoldUp		0.2518	0.2518	0.2518
68	Water HoldUp		---	---	---
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2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
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6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
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11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	6.681e-003	6.681e-003	6.681e-003	6.681e-003
14	Static Gradient	(kPa/m)	5.091e-003	5.091e-003	5.091e-003	5.091e-003
15	Accel. Gradient	(kPa/m)	1.206e-006	1.206e-006	1.206e-006	1.206e-006
16	Liq. Reynolds		2.001e+005 *	2.001e+005 *	2.001e+005 *	2.001e+005 *
17	Vap. Reynolds		9.557e+006 *	9.557e+006 *	9.557e+006 *	9.557e+006 *
18	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992
19	Vapour Velocity	(m/s)	2.669	2.669	2.669	2.669
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	4489	4494	4499	4504
23	Elevation	(m)	0.1840	0.1780	0.1720	0.1660
24	Pressure	(bar)	76.85	76.85	76.85	76.85
25	Temperature	(C)	57.46	57.46	57.46	57.45
26	Vapour Fraction		0.8811	0.8811	0.8811	0.8811
27	Heat Transferred	(kJ/h-m)	731.6	731.6	731.5	731.5
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.1806	0.1806	0.1806	0.1806
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	8.291e-003	8.291e-003	8.291e-003	8.291e-003
33	Static Gradient	(kPa/m)	-2.001e-003	-2.001e-003	-2.001e-003	-2.001e-003
34	Accel. Gradient	(kPa/m)	6.442e-007	6.442e-007	6.442e-007	6.442e-007
35	Liq. Reynolds		2.001e+005 *	2.001e+005 *	2.001e+005 *	2.001e+005 *
36	Vap. Reynolds		9.557e+006 *	9.557e+006 *	9.557e+006 *	9.557e+006 *
37	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992
38	Vapour Velocity	(m/s)	2.669	2.669	2.669	2.669
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	4510	4515	4520	4525
42	Elevation	(m)	0.1600	0.1640	0.1680	0.1720
43	Pressure	(bar)	76.85	76.85	76.85	76.85
44	Temperature	(C)	57.45	57.45	57.45	57.45
45	Vapour Fraction		0.8811	0.8811	0.8811	0.8811
46	Heat Transferred	(kJ/h-m)	731.4	731.3	731.2	731.2
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.1806	0.2472	0.2472	0.2472
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	8.291e-003	7.184e-003	7.184e-003	7.184e-003
52	Static Gradient	(kPa/m)	-2.001e-003	1.674e-003	1.674e-003	1.674e-003
53	Accel. Gradient	(kPa/m)	6.442e-007	9.073e-007	9.073e-007	9.073e-007
54	Liq. Reynolds		2.001e+005 *	2.001e+005 *	2.001e+005 *	2.001e+005 *
55	Vap. Reynolds		9.557e+006 *	9.557e+006 *	9.557e+006 *	9.557e+006 *
56	Liquid Velocity	(m/s)	0.1992	0.1992	0.1992	0.1992
57	Vapour Velocity	(m/s)	2.669	2.669	2.669	2.669
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	4530	4535	4540	4545
61	Elevation	(m)	0.1760	0.1800	0.1700	0.1600
62	Pressure	(bar)	76.85	76.85	76.85	76.85
63	Temperature	(C)	57.44	57.44	57.44	57.44
64	Vapour Fraction		0.8811	0.8811	0.8811	0.8811
65	Heat Transferred	(kJ/h-m)	731.1	731.1	731.1	731.0
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2472	0.2472	0.1799	0.1799
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
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6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
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10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	7.184e-003	7.184e-003	8.298e-003	8.298e-003
15	Static Gradient (kPa/m)	1.674e-003	1.674e-003	-3.365e-003	-3.365e-003
16	Accel. Gradient (kPa/m)	9.073e-007	9.073e-007	5.052e-007	5.052e-007
17	Liq. Reynolds	2.001e+005 *	2.001e+005 *	2.001e+005 *	2.001e+005 *
18	Vap. Reynolds	9.557e+006 *	9.557e+006 *	9.557e+006 *	9.557e+006 *
19	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
20	Vapour Velocity (m/s)	2.669	2.669	2.669	2.669
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	4550	4555	4560	4565
24	Elevation (m)	0.1500	0.1400	0.1300	0.1360
25	Pressure (bar)	76.85	76.85	76.85	76.85
26	Temperature (C)	57.44	57.43	57.43	57.43
27	Vapour Fraction	0.8811	0.8811	0.8811	0.8811
28	Heat Transferred (kJ/h-m)	730.9	730.9	730.8	730.7
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.1799	0.1799	0.1799	0.2484
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	8.298e-003	8.298e-003	8.298e-003	7.055e-003
34	Static Gradient (kPa/m)	-3.365e-003	-3.365e-003	-3.365e-003	2.520e-003
35	Accel. Gradient (kPa/m)	5.052e-007	5.052e-007	5.052e-007	9.807e-007
36	Liq. Reynolds	2.001e+005 *	2.001e+005 *	2.001e+005 *	2.001e+005 *
37	Vap. Reynolds	9.557e+006 *	9.557e+006 *	9.557e+006 *	9.557e+006 *
38	Liquid Velocity (m/s)	0.1992	0.1992	0.1992	0.1992
39	Vapour Velocity (m/s)	2.669	2.669	2.669	2.669
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	4570	4575	4580	4585
43	Elevation (m)	0.1420	0.1480	0.1540	0.1600
44	Pressure (bar)	76.85	76.85	76.85	76.85
45	Temperature (C)	57.43	57.42	57.42	57.42
46	Vapour Fraction	0.8811	0.8811	0.8811	0.8811
47	Heat Transferred (kJ/h-m)	730.7	730.6	730.5	730.5
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2484	0.2484	0.2484	0.2484
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	7.055e-003	7.055e-003	7.055e-003	7.055e-003
53	Static Gradient (kPa/m)	2.520e-003	2.520e-003	2.520e-003	2.520e-003
54	Accel. Gradient (kPa/m)	9.807e-007	9.807e-007	9.807e-007	9.807e-007
55	Liq. Reynolds	2.001e+005 *	2.001e+005 *	2.001e+005 *	2.000e+005 *
56	Vap. Reynolds	9.557e+006 *	9.557e+006 *	9.557e+006 *	9.557e+006 *
57	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
58	Vapour Velocity (m/s)	2.669	2.669	2.669	2.669
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	4590	4595	4600	4605
62	Elevation (m)	0.1820	0.2040	0.2260	0.2480
63	Pressure (bar)	76.85	76.85	76.84	76.84
64	Temperature (C)	57.42	57.42	57.41	57.41
65	Vapour Fraction	0.8811	0.8811	0.8811	0.8811
66	Heat Transferred (kJ/h-m)	730.4	730.4	730.3	730.2
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
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11	Pipe Table				
12					
13	Liquid HoldUp	0.2576	0.2576	0.2576	0.2576
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	6.088e-003	6.088e-003	6.088e-003	6.088e-003
16	Static Gradient (kPa/m)	9.489e-003	9.489e-003	9.489e-003	9.490e-003
17	Accel. Gradient (kPa/m)	1.595e-006	1.595e-006	1.595e-006	1.595e-006
18	Liq. Reynolds	2.000e+005 *	2.000e+005 *	2.000e+005 *	2.000e+005 *
19	Vap. Reynolds	9.557e+006 *	9.558e+006 *	9.558e+006 *	9.558e+006 *
20	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
21	Vapour Velocity (m/s)	2.669	2.669	2.669	2.669
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	4610	4623	4636	4649
25	Elevation (m)	0.2700	0.2400	0.2100	0.1800
26	Pressure (bar)	76.84	76.84	76.84	76.84
27	Temperature (C)	57.41	57.40	57.40	57.39
28	Vapour Fraction	0.8811	0.8811	0.8811	0.8811
29	Heat Transferred (kJ/h-m)	730.2	730.1	729.9	729.8
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2576	0.1797	0.1797	0.1797
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.088e-003	8.300e-003	8.300e-003	8.300e-003
35	Static Gradient (kPa/m)	9.490e-003	-3.879e-003	-3.879e-003	-3.879e-003
36	Accel. Gradient (kPa/m)	1.596e-006	4.528e-007	4.528e-007	4.527e-007
37	Liq. Reynolds	2.000e+005 *	2.000e+005 *	2.000e+005 *	2.000e+005 *
38	Vap. Reynolds	9.558e+006 *	9.558e+006 *	9.558e+006 *	9.558e+006 *
39	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
40	Vapour Velocity (m/s)	2.669	2.669	2.669	2.669
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	4662	4675	4677	4679
44	Elevation (m)	0.1500	0.1200	0.1280	0.1360
45	Pressure (bar)	76.84	76.84	76.84	76.84
46	Temperature (C)	57.39	57.38	57.38	57.38
47	Vapour Fraction	0.8811	0.8811	0.8811	0.8811
48	Heat Transferred (kJ/h-m)	729.6	729.5	729.4	729.4
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1797	0.1797	0.2565	0.2565
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	8.300e-003	8.300e-003	6.203e-003	6.203e-003
54	Static Gradient (kPa/m)	-3.880e-003	-3.880e-003	8.599e-003	8.599e-003
55	Accel. Gradient (kPa/m)	4.527e-007	4.527e-007	1.516e-006	1.516e-006
56	Liq. Reynolds	2.000e+005 *	2.000e+005 *	2.000e+005 *	2.000e+005 *
57	Vap. Reynolds	9.558e+006 *	9.558e+006 *	9.558e+006 *	9.558e+006 *
58	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
59	Vapour Velocity (m/s)	2.669	2.669	2.669	2.669
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	4681	4683	4685	4690
63	Elevation (m)	0.1440	0.1520	0.1600	0.1800
64	Pressure (bar)	76.84	76.84	76.84	76.84
65	Temperature (C)	57.38	57.38	57.38	57.37
66	Vapour Fraction	0.8811	0.8811	0.8811	0.8811
67	Heat Transferred (kJ/h-m)	729.4	729.3	729.3	729.3
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 104 of 937

1			Case Name:	Simulation - 386MMSCFD_80barg.hsc	
2			Unit Set:	Mixed	
3	 Epic Atlantic Limited		Date/Time:	Mon Jun 13 15:13:43 2022	
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2565	0.2565	0.2565	0.2563
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.203e-003	6.203e-003	6.203e-003	6.222e-003
17	Static Gradient (kPa/m)	8.599e-003	8.599e-003	8.599e-003	8.453e-003
18	Accel. Gradient (kPa/m)	1.516e-006	1.516e-006	1.516e-006	1.503e-006
19	Liq. Reynolds	2.000e+005 *	2.000e+005 *	2.000e+005 *	2.000e+005 *
20	Vap. Reynolds	9.558e+006 *	9.558e+006 *	9.558e+006 *	9.558e+006 *
21	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
22	Vapour Velocity (m/s)	2.669	2.669	2.669	2.669
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	4695	4700	4705	4710
26	Elevation (m)	0.2000	0.2200	0.2400	0.2600
27	Pressure (bar)	76.84	76.84	76.84	76.84
28	Temperature (C)	57.37	57.37	57.37	57.36
29	Vapour Fraction	0.8811	0.8811	0.8810	0.8810
30	Heat Transferred (kJ/h-m)	729.2	729.1	729.1	729.0
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2563	0.2563	0.2563	0.2563
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	6.222e-003	6.222e-003	6.222e-003	6.222e-003
36	Static Gradient (kPa/m)	8.453e-003	8.453e-003	8.453e-003	8.453e-003
37	Accel. Gradient (kPa/m)	1.503e-006	1.503e-006	1.503e-006	1.503e-006
38	Liq. Reynolds	2.000e+005 *	2.000e+005 *	2.000e+005 *	2.000e+005 *
39	Vap. Reynolds	9.558e+006 *	9.558e+006 *	9.558e+006 *	9.558e+006 *
40	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
41	Vapour Velocity (m/s)	2.669	2.669	2.669	2.669
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	4715	4720	4725	4730
45	Elevation (m)	0.2620	0.2640	0.2660	0.2680
46	Pressure (bar)	76.83	76.83	76.83	76.83
47	Temperature (C)	57.36	57.36	57.36	57.36
48	Vapour Fraction	0.8810	0.8810	0.8810	0.8810
49	Heat Transferred (kJ/h-m)	729.0	728.9	728.8	728.8
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2461	0.2461	0.2461	0.2461
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	7.313e-003	7.313e-003	7.313e-003	7.313e-003
55	Static Gradient (kPa/m)	8.345e-004	8.345e-004	8.345e-004	8.345e-004
56	Accel. Gradient (kPa/m)	8.345e-007	8.345e-007	8.345e-007	8.345e-007
57	Liq. Reynolds	2.000e+005 *	2.000e+005 *	2.000e+005 *	2.000e+005 *
58	Vap. Reynolds	9.558e+006 *	9.558e+006 *	9.558e+006 *	9.558e+006 *
59	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
60	Vapour Velocity (m/s)	2.669	2.669	2.669	2.669
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	4735	4740	4745	4750
64	Elevation (m)	0.2700	0.2720	0.2740	0.2760
65	Pressure (bar)	76.83	76.83	76.83	76.83
66	Temperature (C)	57.35	57.35	57.35	57.35
67	Vapour Fraction	0.8810	0.8810	0.8810	0.8810
68	Heat Transferred (kJ/h-m)	728.7	728.7	728.6	728.5
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2461	0.2461	0.2461	0.2461
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	7.313e-003	7.313e-003	7.313e-003	7.313e-003
18	Static Gradient (kPa/m)	8.345e-004	8.345e-004	8.345e-004	8.345e-004
19	Accel. Gradient (kPa/m)	8.345e-007	8.345e-007	8.345e-007	8.345e-007
20	Liq. Reynolds	2.000e+005 *	2.000e+005 *	2.000e+005 *	2.000e+005 *
21	Vap. Reynolds	9.558e+006 *	9.558e+006 *	9.558e+006 *	9.558e+006 *
22	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
23	Vapour Velocity (m/s)	2.669	2.669	2.669	2.669
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	4755	4760	4765	4770
27	Elevation (m)	0.2780	0.2800	0.2780	0.2760
28	Pressure (bar)	76.83	76.83	76.83	76.83
29	Temperature (C)	57.34	57.34	57.34	57.34
30	Vapour Fraction	0.8810	0.8810	0.8810	0.8810
31	Heat Transferred (kJ/h-m)	728.5	728.4	728.4	728.4
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2461	0.2461	0.1812	0.1812
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	7.313e-003	7.313e-003	8.282e-003	8.282e-003
37	Static Gradient (kPa/m)	8.345e-004	8.345e-004	-6.762e-004	-6.762e-004
38	Accel. Gradient (kPa/m)	8.345e-007	8.345e-007	7.790e-007	7.790e-007
39	Liq. Reynolds	2.000e+005 *	2.000e+005 *	2.000e+005 *	1.999e+005 *
40	Vap. Reynolds	9.558e+006 *	9.558e+006 *	9.558e+006 *	9.558e+006 *
41	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
42	Vapour Velocity (m/s)	2.669	2.669	2.669	2.669
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	4775	4780	4785	4790
46	Elevation (m)	0.2740	0.2720	0.2700	0.2600
47	Pressure (bar)	76.83	76.83	76.83	76.83
48	Temperature (C)	57.34	57.33	57.33	57.33
49	Vapour Fraction	0.8810	0.8810	0.8810	0.8810
50	Heat Transferred (kJ/h-m)	728.3	728.2	728.2	728.1
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1812	0.1812	0.1812	0.1800
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	8.282e-003	8.282e-003	8.282e-003	8.296e-003
56	Static Gradient (kPa/m)	-6.762e-004	-6.762e-004	-6.762e-004	-3.366e-003
57	Accel. Gradient (kPa/m)	7.790e-007	7.790e-007	7.790e-007	5.050e-007
58	Liq. Reynolds	1.999e+005 *	1.999e+005 *	1.999e+005 *	1.999e+005 *
59	Vap. Reynolds	9.558e+006 *	9.558e+006 *	9.558e+006 *	9.558e+006 *
60	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
61	Vapour Velocity (m/s)	2.669	2.669	2.669	2.669
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	4795	4800	4805	4810
65	Elevation (m)	0.2500	0.2400	0.2300	0.2200
66	Pressure (bar)	76.83	76.83	76.83	76.83
67	Temperature (C)	57.33	57.33	57.32	57.32
68	Vapour Fraction	0.8810	0.8810	0.8810	0.8810
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 106 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Heat Transferred (kJ/h-m)	728.1	728.0	728.0	727.9					
14	Flow Regime	Transition	Transition	Transition	Transition					
15	Water Flow Regime									
16	Liquid HoldUp	0.1800	0.1800	0.1800	0.1800					
17	Water HoldUp	---	---	---	---					
18	Friction Gradient (kPa/m)	8.296e-003	8.296e-003	8.296e-003	8.296e-003					
19	Static Gradient (kPa/m)	-3.366e-003	-3.366e-003	-3.366e-003	-3.366e-003					
20	Accel. Gradient (kPa/m)	5.050e-007	5.050e-007	5.050e-007	5.050e-007					
21	Liq. Reynolds	1.999e+005 *	1.999e+005 *	1.999e+005 *	1.999e+005 *					
22	Vap. Reynolds	9.558e+006 *	9.558e+006 *	9.558e+006 *	9.558e+006 *					
23	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993					
24	Vapour Velocity (m/s)	2.669	2.668	2.668	2.668					
25	Deposit Thickness (mm)	---	---	---	---					
26	Deposit Volume (m3)	---	---	---	---					
27	Length (m)	4815	4820	4825	4830					
28	Elevation (m)	0.2140	0.2080	0.2020	0.1960					
29	Pressure (bar)	76.83	76.83	76.83	76.83					
30	Temperature (C)	57.32	57.32	57.32	57.31					
31	Vapour Fraction	0.8810	0.8810	0.8810	0.8810					
32	Heat Transferred (kJ/h-m)	727.9	727.8	727.7	727.7					
33	Flow Regime	Transition	Transition	Transition	Transition					
34	Water Flow Regime									
35	Liquid HoldUp	0.1806	0.1806	0.1806	0.1806					
36	Water HoldUp	---	---	---	---					
37	Friction Gradient (kPa/m)	8.289e-003	8.289e-003	8.289e-003	8.289e-003					
38	Static Gradient (kPa/m)	-2.024e-003	-2.024e-003	-2.024e-003	-2.024e-003					
39	Accel. Gradient (kPa/m)	6.416e-007	6.416e-007	6.416e-007	6.416e-007					
40	Liq. Reynolds	1.999e+005 *	1.999e+005 *	1.999e+005 *	1.999e+005 *					
41	Vap. Reynolds	9.558e+006 *	9.558e+006 *	9.558e+006 *	9.558e+006 *					
42	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993					
43	Vapour Velocity (m/s)	2.668	2.668	2.668	2.668					
44	Deposit Thickness (mm)	---	---	---	---					
45	Deposit Volume (m3)	---	---	---	---					
46	Length (m)	4835	4840	4845	4850					
47	Elevation (m)	0.1900	0.1800	0.1700	0.1600					
48	Pressure (bar)	76.83	76.83	76.83	76.83					
49	Temperature (C)	57.31	57.31	57.31	57.30					
50	Vapour Fraction	0.8810	0.8810	0.8810	0.8810					
51	Heat Transferred (kJ/h-m)	727.6	727.6	727.5	727.5					
52	Flow Regime	Transition	Transition	Transition	Transition					
53	Water Flow Regime									
54	Liquid HoldUp	0.1806	0.1800	0.1800	0.1800					
55	Water HoldUp	---	---	---	---					
56	Friction Gradient (kPa/m)	8.289e-003	8.296e-003	8.296e-003	8.296e-003					
57	Static Gradient (kPa/m)	-2.024e-003	-3.366e-003	-3.366e-003	-3.366e-003					
58	Accel. Gradient (kPa/m)	6.416e-007	5.049e-007	5.049e-007	5.049e-007					
59	Liq. Reynolds	1.999e+005 *	1.999e+005 *	1.999e+005 *	1.999e+005 *					
60	Vap. Reynolds	9.558e+006 *	9.558e+006 *	9.558e+006 *	9.558e+006 *					
61	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993					
62	Vapour Velocity (m/s)	2.668	2.668	2.668	2.668					
63	Deposit Thickness (mm)	---	---	---	---					
64	Deposit Volume (m3)	---	---	---	---					
65	Length (m)	4855	4860	4865	4870					
66	Elevation (m)	0.1500	0.1400	0.1380	0.1360					
67	Pressure (bar)	76.83	76.83	76.82	76.82					
68	Temperature (C)	57.30	57.30	57.30	57.30					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:13:43 2022		
6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Fraction		0.8810	0.8810	0.8810	0.8810	
14	Heat Transferred (kJ/h-m)		727.4	727.3	727.3	727.2	
15	Flow Regime		Transition	Transition	Transition	Transition	
16	Water Flow Regime						
17	Liquid HoldUp		0.1800	0.1800	0.1813	0.1813	
18	Water HoldUp		---	---	---	---	
19	Friction Gradient (kPa/m)		8.296e-003	8.296e-003	8.281e-003	8.281e-003	
20	Static Gradient (kPa/m)		-3.366e-003	-3.366e-003	-6.763e-004	-6.763e-004	
21	Accel. Gradient (kPa/m)		5.049e-007	5.049e-007	7.789e-007	7.789e-007	
22	Liq. Reynolds		1.999e+005 *	1.999e+005 *	1.999e+005 *	1.999e+005 *	
23	Vap. Reynolds		9.558e+006 *	9.558e+006 *	9.558e+006 *	9.558e+006 *	
24	Liquid Velocity (m/s)		0.1993	0.1993	0.1993	0.1993	
25	Vapour Velocity (m/s)		2.668	2.668	2.668	2.668	
26	Deposit Thickness (mm)		---	---	---	---	
27	Deposit Volume (m3)		---	---	---	---	
28	Length (m)		4875	4880	4885	4890	
29	Elevation (m)		0.1340	0.1320	0.1300	0.1300	
30	Pressure (bar)		76.82	76.82	76.82	76.82	
31	Temperature (C)		57.29	57.29	57.29	57.29	
32	Vapour Fraction		0.8810	0.8810	0.8810	0.8810	
33	Heat Transferred (kJ/h-m)		727.2	727.1	727.1	726.9	
34	Flow Regime		Transition	Transition	Transition	Transition	
35	Water Flow Regime						
36	Liquid HoldUp		0.1813	0.1813	0.1813	0.2651	
37	Water HoldUp		---	---	---	---	
38	Friction Gradient (kPa/m)		8.281e-003	8.281e-003	8.281e-003	6.266e-003	
39	Static Gradient (kPa/m)		-6.763e-004	-6.763e-004	-6.763e-004	0.0000	
40	Accel. Gradient (kPa/m)		7.788e-007	7.788e-007	7.788e-007	6.418e-007	
41	Liq. Reynolds		1.999e+005 *	1.999e+005 *	1.999e+005 *	1.999e+005 *	
42	Vap. Reynolds		9.558e+006 *	9.558e+006 *	9.558e+006 *	9.558e+006 *	
43	Liquid Velocity (m/s)		0.1993	0.1993	0.1993	0.1993	
44	Vapour Velocity (m/s)		2.668	2.668	2.668	2.668	
45	Deposit Thickness (mm)		---	---	---	---	
46	Deposit Volume (m3)		---	---	---	---	
47	Length (m)		4895	4900	4905	4910	
48	Elevation (m)		0.1300	0.1300	0.1300	0.1300	
49	Pressure (bar)		76.82	76.82	76.82	76.82	
50	Temperature (C)		57.29	57.28	57.28	57.28	
51	Vapour Fraction		0.8810	0.8810	0.8810	0.8810	
52	Heat Transferred (kJ/h-m)		726.9	726.8	726.8	726.7	
53	Flow Regime		Transition	Transition	Transition	Transition	
54	Water Flow Regime						
55	Liquid HoldUp		0.2651	0.2651	0.2651	0.2651	
56	Water HoldUp		---	---	---	---	
57	Friction Gradient (kPa/m)		6.266e-003	6.266e-003	6.266e-003	6.266e-003	
58	Static Gradient (kPa/m)		0.0000	0.0000	0.0000	0.0000	
59	Accel. Gradient (kPa/m)		6.418e-007	6.418e-007	6.418e-007	6.418e-007	
60	Liq. Reynolds		1.999e+005 *	1.999e+005 *	1.999e+005 *	1.999e+005 *	
61	Vap. Reynolds		9.558e+006 *	9.559e+006 *	9.559e+006 *	9.559e+006 *	
62	Liquid Velocity (m/s)		0.1993	0.1993	0.1993	0.1993	
63	Vapour Velocity (m/s)		2.668	2.668	2.668	2.668	
64	Deposit Thickness (mm)		---	---	---	---	
65	Deposit Volume (m3)		---	---	---	---	
66	Length (m)		4915	4920	4925	4930	
67	Elevation (m)		0.1080	8.600e-002	6.400e-002	4.200e-002	
68	Pressure (bar)		76.82	76.82	76.82	76.82	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 108 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Temperature (C)	57.28	57.27	57.27	57.27		
14	Vapour Fraction	0.8810	0.8810	0.8810	0.8810		
15	Heat Transferred (kJ/h-m)	726.7	726.7	726.6	726.6		
16	Flow Regime	Transition	Transition	Transition	Transition		
17	Water Flow Regime						
18	Liquid HoldUp	0.1781	0.1781	0.1781	0.1781		
19	Water HoldUp	---	---	---	---		
20	Friction Gradient (kPa/m)	8.317e-003	8.317e-003	8.317e-003	8.317e-003		
21	Static Gradient (kPa/m)	-7.354e-003	-7.354e-003	-7.354e-003	-7.354e-003		
22	Accel. Gradient (kPa/m)	9.865e-008	9.864e-008	9.863e-008	9.861e-008		
23	Liq. Reynolds	1.999e+005 *	1.999e+005 *	1.999e+005 *	1.999e+005 *		
24	Vap. Reynolds	9.559e+006 *	9.559e+006 *	9.559e+006 *	9.559e+006 *		
25	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993		
26	Vapour Velocity (m/s)	2.668	2.668	2.668	2.668		
27	Deposit Thickness (mm)	---	---	---	---		
28	Deposit Volume (m3)	---	---	---	---		
29	Length (m)	4935	4940	4945	4950		
30	Elevation (m)	2.000e-002	1.200e-002	4.000e-003	-4.000e-003		
31	Pressure (bar)	76.82	76.82	76.82	76.82		
32	Temperature (C)	57.27	57.27	57.26	57.26		
33	Vapour Fraction	0.8810	0.8810	0.8810	0.8810		
34	Heat Transferred (kJ/h-m)	726.5	726.4	726.4	726.3		
35	Flow Regime	Transition	Transition	Transition	Transition		
36	Water Flow Regime						
37	Liquid HoldUp	0.1781	0.1803	0.1803	0.1803		
38	Water HoldUp	---	---	---	---		
39	Friction Gradient (kPa/m)	8.317e-003	8.291e-003	8.291e-003	8.291e-003		
40	Static Gradient (kPa/m)	-7.354e-003	-2.696e-003	-2.696e-003	-2.696e-003		
41	Accel. Gradient (kPa/m)	9.860e-008	5.730e-007	5.730e-007	5.730e-007		
42	Liq. Reynolds	1.999e+005 *	1.999e+005 *	1.999e+005 *	1.999e+005 *		
43	Vap. Reynolds	9.559e+006 *	9.559e+006 *	9.559e+006 *	9.559e+006 *		
44	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993		
45	Vapour Velocity (m/s)	2.668	2.668	2.668	2.668		
46	Deposit Thickness (mm)	---	---	---	---		
47	Deposit Volume (m3)	---	---	---	---		
48	Length (m)	4955	4960	4965	4970		
49	Elevation (m)	-1.200e-002	-2.000e-002	-1.600e-002	-1.200e-002		
50	Pressure (bar)	76.82	76.82	76.82	76.82		
51	Temperature (C)	57.26	57.26	57.26	57.25		
52	Vapour Fraction	0.8810	0.8810	0.8810	0.8810		
53	Heat Transferred (kJ/h-m)	726.3	726.2	726.1	726.1		
54	Flow Regime	Transition	Transition	Transition	Transition		
55	Water Flow Regime						
56	Liquid HoldUp	0.1803	0.1803	0.2473	0.2473		
57	Water HoldUp	---	---	---	---		
58	Friction Gradient (kPa/m)	8.291e-003	8.291e-003	7.180e-003	7.180e-003		
59	Static Gradient (kPa/m)	-2.696e-003	-2.696e-003	1.675e-003	1.675e-003		
60	Accel. Gradient (kPa/m)	5.730e-007	5.730e-007	9.068e-007	9.068e-007		
61	Liq. Reynolds	1.999e+005 *	1.999e+005 *	1.999e+005 *	1.999e+005 *		
62	Vap. Reynolds	9.559e+006 *	9.559e+006 *	9.559e+006 *	9.559e+006 *		
63	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993		
64	Vapour Velocity (m/s)	2.668	2.668	2.668	2.668		
65	Deposit Thickness (mm)	---	---	---	---		
66	Deposit Volume (m3)	---	---	---	---		
67	Length (m)	4975	4980	4985	4990		
68	Elevation (m)	-8.000e-003	-4.000e-003	-2.217e-015	1.400e-002		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Pressure (bar)	76.82	76.82	76.82	76.82					
14	Temperature (C)	57.25	57.25	57.25	57.24					
15	Vapour Fraction	0.8810	0.8810	0.8810	0.8810					
16	Heat Transferred (kJ/h-m)	726.0	725.9	725.9	725.8					
17	Flow Regime	Transition	Transition	Transition	Transition					
18	Water Flow Regime									
19	Liquid HoldUp	0.2473	0.2473	0.2473	0.2531					
20	Water HoldUp	---	---	---	---					
21	Friction Gradient (kPa/m)	7.180e-003	7.180e-003	7.180e-003	6.555e-003					
22	Static Gradient (kPa/m)	1.675e-003	1.675e-003	1.675e-003	5.962e-003					
23	Accel. Gradient (kPa/m)	9.068e-007	9.068e-007	9.068e-007	1.282e-006					
24	Liq. Reynolds	1.998e+005 *	1.998e+005 *	1.998e+005 *	1.998e+005 *					
25	Vap. Reynolds	9.559e+006 *	9.559e+006 *	9.559e+006 *	9.559e+006 *					
26	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993					
27	Vapour Velocity (m/s)	2.668	2.668	2.668	2.668					
28	Deposit Thickness (mm)	---	---	---	---					
29	Deposit Volume (m3)	---	---	---	---					
30	Length (m)	4995	5000	5005	5010					
31	Elevation (m)	2.800e-002	4.200e-002	5.600e-002	7.000e-002					
32	Pressure (bar)	76.82	76.82	76.82	76.81					
33	Temperature (C)	57.24	57.24	57.24	57.24					
34	Vapour Fraction	0.8810	0.8810	0.8810	0.8810					
35	Heat Transferred (kJ/h-m)	725.8	725.7	725.6	725.6					
36	Flow Regime	Transition	Transition	Transition	Transition					
37	Water Flow Regime									
38	Liquid HoldUp	0.2531	0.2531	0.2531	0.2531					
39	Water HoldUp	---	---	---	---					
40	Friction Gradient (kPa/m)	6.555e-003	6.555e-003	6.555e-003	6.555e-003					
41	Static Gradient (kPa/m)	5.962e-003	5.962e-003	5.962e-003	5.962e-003					
42	Accel. Gradient (kPa/m)	1.282e-006	1.282e-006	1.282e-006	1.282e-006					
43	Liq. Reynolds	1.998e+005 *	1.998e+005 *	1.998e+005 *	1.998e+005 *					
44	Vap. Reynolds	9.559e+006 *	9.559e+006 *	9.559e+006 *	9.559e+006 *					
45	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993					
46	Vapour Velocity (m/s)	2.668	2.668	2.668	2.668					
47	Deposit Thickness (mm)	---	---	---	---					
48	Deposit Volume (m3)	---	---	---	---					
49	Length (m)	5015	5020	5025	5030					
50	Elevation (m)	5.600e-002	4.200e-002	2.800e-002	1.400e-002					
51	Pressure (bar)	76.81	76.81	76.81	76.81					
52	Temperature (C)	57.23	57.23	57.23	57.23					
53	Vapour Fraction	0.8810	0.8810	0.8810	0.8810					
54	Heat Transferred (kJ/h-m)	725.6	725.5	725.5	725.4					
55	Flow Regime	Transition	Transition	Transition	Transition					
56	Water Flow Regime									
57	Liquid HoldUp	0.1794	0.1794	0.1794	0.1794					
58	Water HoldUp	---	---	---	---					
59	Friction Gradient (kPa/m)	8.302e-003	8.302e-003	8.302e-003	8.302e-003					
60	Static Gradient (kPa/m)	-4.702e-003	-4.702e-003	-4.702e-003	-4.702e-003					
61	Accel. Gradient (kPa/m)	3.686e-007	3.686e-007	3.686e-007	3.686e-007					
62	Liq. Reynolds	1.998e+005 *	1.998e+005 *	1.998e+005 *	1.998e+005 *					
63	Vap. Reynolds	9.559e+006 *	9.559e+006 *	9.559e+006 *	9.559e+006 *					
64	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993					
65	Vapour Velocity (m/s)	2.668	2.668	2.668	2.668					
66	Deposit Thickness (mm)	---	---	---	---					
67	Deposit Volume (m3)	---	---	---	---					
68	Length (m)	5035	5040	5045	5050					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11										
12										
13	Pipe Table									
13	Elevation	(m)	-2.220e-015	2.600e-002	5.200e-002	7.800e-002				
14	Pressure	(bar)	76.81	76.81	76.81	76.81				
15	Temperature	(C)	57.23	57.22	57.22	57.22				
16	Vapour Fraction		0.8810	0.8810	0.8810	0.8810				
17	Heat Transferred	(kJ/h-m)	725.3	725.2	725.2	725.1				
18	Flow Regime		Transition	Transition	Transition	Transition				
19	Water Flow Regime									
20	Liquid HoldUp		0.1794	0.2600	0.2600	0.2600				
21	Water HoldUp		---	---	---	---				
22	Friction Gradient	(kPa/m)	8.301e-003	5.859e-003	5.859e-003	5.859e-003				
23	Static Gradient	(kPa/m)	-4.702e-003	1.129e-002	1.129e-002	1.129e-002				
24	Accel. Gradient	(kPa/m)	3.686e-007	1.757e-006	1.757e-006	1.757e-006				
25	Liq. Reynolds		1.998e+005 *	1.998e+005 *	1.998e+005 *	1.998e+005 *				
26	Vap. Reynolds		9.559e+006 *	9.559e+006 *	9.559e+006 *	9.559e+006 *				
27	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993				
28	Vapour Velocity	(m/s)	2.668	2.668	2.668	2.668				
29	Deposit Thickness	(mm)	---	---	---	---				
30	Deposit Volume	(m3)	---	---	---	---				
31	Length	(m)	5055	5060	5065	5070				
32	Elevation	(m)	0.1040	0.1300	0.1240	0.1180				
33	Pressure	(bar)	76.81	76.81	76.81	76.81				
34	Temperature	(C)	57.22	57.21	57.21	57.21				
35	Vapour Fraction		0.8810	0.8810	0.8810	0.8810				
36	Heat Transferred	(kJ/h-m)	725.0	725.0	725.0	724.9				
37	Flow Regime		Transition	Transition	Transition	Transition				
38	Water Flow Regime									
39	Liquid HoldUp		0.2600	0.2600	0.1807	0.1807				
40	Water HoldUp		---	---	---	---				
41	Friction Gradient	(kPa/m)	5.859e-003	5.859e-003	8.287e-003	8.287e-003				
42	Static Gradient	(kPa/m)	1.129e-002	1.129e-002	-2.025e-003	-2.025e-003				
43	Accel. Gradient	(kPa/m)	1.757e-006	1.757e-006	6.413e-007	6.413e-007				
44	Liq. Reynolds		1.998e+005 *	1.998e+005 *	1.998e+005 *	1.998e+005 *				
45	Vap. Reynolds		9.559e+006 *	9.559e+006 *	9.559e+006 *	9.559e+006 *				
46	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993				
47	Vapour Velocity	(m/s)	2.668	2.668	2.668	2.668				
48	Deposit Thickness	(mm)	---	---	---	---				
49	Deposit Volume	(m3)	---	---	---	---				
50	Length	(m)	5075	5080	5085	5090				
51	Elevation	(m)	0.1120	0.1060	1.000e-001	0.1180				
52	Pressure	(bar)	76.81	76.81	76.81	76.81				
53	Temperature	(C)	57.21	57.21	57.20	57.20				
54	Vapour Fraction		0.8810	0.8810	0.8810	0.8810				
55	Heat Transferred	(kJ/h-m)	724.9	724.8	724.8	724.6				
56	Flow Regime		Transition	Transition	Transition	Transition				
57	Water Flow Regime									
58	Liquid HoldUp		0.1807	0.1807	0.1807	0.2553				
59	Water HoldUp		---	---	---	---				
60	Friction Gradient	(kPa/m)	8.287e-003	8.287e-003	8.287e-003	6.323e-003				
61	Static Gradient	(kPa/m)	-2.025e-003	-2.025e-003	-2.025e-003	7.666e-003				
62	Accel. Gradient	(kPa/m)	6.413e-007	6.413e-007	6.413e-007	1.433e-006				
63	Liq. Reynolds		1.998e+005 *	1.998e+005 *	1.998e+005 *	1.998e+005 *				
64	Vap. Reynolds		9.559e+006 *	9.559e+006 *	9.559e+006 *	9.559e+006 *				
65	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993				
66	Vapour Velocity	(m/s)	2.668	2.668	2.668	2.668				
67	Deposit Thickness	(mm)	---	---	---	---				
68	Deposit Volume	(m3)	---	---	---	---				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:13:43 2022		
6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length	(m)	5095	5100	5105	5110	
14	Elevation	(m)	0.1360	0.1540	0.1720	0.1900	
15	Pressure	(bar)	76.81	76.81	76.81	76.80	
16	Temperature	(C)	57.20	57.20	57.19	57.19	
17	Vapour Fraction		0.8810	0.8810	0.8810	0.8810	
18	Heat Transferred	(kJ/h-m)	724.6	724.5	724.5	724.4	
19	Flow Regime		Transition	Transition	Transition	Transition	
20	Water Flow Regime						
21	Liquid HoldUp		0.2553	0.2553	0.2553	0.2553	
22	Water HoldUp		---	---	---	---	
23	Friction Gradient	(kPa/m)	6.323e-003	6.323e-003	6.323e-003	6.323e-003	
24	Static Gradient	(kPa/m)	7.666e-003	7.666e-003	7.666e-003	7.666e-003	
25	Accel. Gradient	(kPa/m)	1.433e-006	1.433e-006	1.433e-006	1.433e-006	
26	Liq. Reynolds		1.998e+005 *	1.998e+005 *	1.998e+005 *	1.998e+005 *	
27	Vap. Reynolds		9.559e+006 *	9.559e+006 *	9.559e+006 *	9.559e+006 *	
28	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993	
29	Vapour Velocity	(m/s)	2.668	2.668	2.668	2.668	
30	Deposit Thickness	(mm)	---	---	---	---	
31	Deposit Volume	(m3)	---	---	---	---	
32	Length	(m)	5115	5120	5125	5130	
33	Elevation	(m)	0.1860	0.1820	0.1780	0.1740	
34	Pressure	(bar)	76.80	76.80	76.80	76.80	
35	Temperature	(C)	57.19	57.19	57.19	57.18	
36	Vapour Fraction		0.8810	0.8810	0.8810	0.8810	
37	Heat Transferred	(kJ/h-m)	724.4	724.3	724.3	724.2	
38	Flow Regime		Transition	Transition	Transition	Transition	
39	Water Flow Regime						
40	Liquid HoldUp		0.1810	0.1810	0.1810	0.1810	
41	Water HoldUp		---	---	---	---	
42	Friction Gradient	(kPa/m)	8.283e-003	8.283e-003	8.283e-003	8.283e-003	
43	Static Gradient	(kPa/m)	-1.351e-003	-1.351e-003	-1.351e-003	-1.351e-003	
44	Accel. Gradient	(kPa/m)	7.099e-007	7.099e-007	7.099e-007	7.099e-007	
45	Liq. Reynolds		1.998e+005 *	1.998e+005 *	1.998e+005 *	1.998e+005 *	
46	Vap. Reynolds		9.559e+006 *	9.559e+006 *	9.559e+006 *	9.559e+006 *	
47	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993	
48	Vapour Velocity	(m/s)	2.668	2.668	2.668	2.668	
49	Deposit Thickness	(mm)	---	---	---	---	
50	Deposit Volume	(m3)	---	---	---	---	
51	Length	(m)	5135	5140	5145	5150	
52	Elevation	(m)	0.1700	0.1640	0.1580	0.1520	
53	Pressure	(bar)	76.80	76.80	76.80	76.80	
54	Temperature	(C)	57.18	57.18	57.18	57.17	
55	Vapour Fraction		0.8810	0.8809	0.8809	0.8809	
56	Heat Transferred	(kJ/h-m)	724.2	724.1	724.0	724.0	
57	Flow Regime		Transition	Transition	Transition	Transition	
58	Water Flow Regime						
59	Liquid HoldUp		0.1810	0.1807	0.1807	0.1807	
60	Water HoldUp		---	---	---	---	
61	Friction Gradient	(kPa/m)	8.283e-003	8.286e-003	8.286e-003	8.286e-003	
62	Static Gradient	(kPa/m)	-1.351e-003	-2.025e-003	-2.025e-003	-2.025e-003	
63	Accel. Gradient	(kPa/m)	7.099e-007	6.413e-007	6.413e-007	6.413e-007	
64	Liq. Reynolds		1.998e+005 *	1.998e+005 *	1.998e+005 *	1.998e+005 *	
65	Vap. Reynolds		9.559e+006 *	9.559e+006 *	9.559e+006 *	9.559e+006 *	
66	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993	
67	Vapour Velocity	(m/s)	2.668	2.667	2.667	2.667	
68	Deposit Thickness	(mm)	---	---	---	---	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 112 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	5155	5160	5165	5170
15	Elevation (m)	0.1460	0.1400	0.1480	0.1560
16	Pressure (bar)	76.80	76.80	76.80	76.80
17	Temperature (C)	57.17	57.17	57.17	57.17
18	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
19	Heat Transferred (kJ/h-m)	723.9	723.9	723.8	723.7
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.1807	0.1807	0.2496	0.2496
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	8.286e-003	8.286e-003	6.924e-003	6.924e-003
25	Static Gradient (kPa/m)	-2.025e-003	-2.025e-003	3.373e-003	3.373e-003
26	Accel. Gradient (kPa/m)	6.413e-007	6.413e-007	1.055e-006	1.055e-006
27	Liq. Reynolds	1.998e+005 *	1.997e+005 *	1.997e+005 *	1.997e+005 *
28	Vap. Reynolds	9.559e+006 *	9.559e+006 *	9.559e+006 *	9.559e+006 *
29	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
30	Vapour Velocity (m/s)	2.667	2.667	2.667	2.667
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	5175	5180	5185	5190
34	Elevation (m)	0.1640	0.1720	0.1800	0.1680
35	Pressure (bar)	76.80	76.80	76.80	76.80
36	Temperature (C)	57.16	57.16	57.16	57.16
37	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
38	Heat Transferred (kJ/h-m)	723.7	723.6	723.5	723.5
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2496	0.2496	0.2496	0.1797
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	6.924e-003	6.924e-003	6.924e-003	8.297e-003
44	Static Gradient (kPa/m)	3.373e-003	3.373e-003	3.373e-003	-4.036e-003
45	Accel. Gradient (kPa/m)	1.055e-006	1.055e-006	1.055e-006	4.364e-007
46	Liq. Reynolds	1.997e+005 *	1.997e+005 *	1.997e+005 *	1.997e+005 *
47	Vap. Reynolds	9.559e+006 *	9.559e+006 *	9.559e+006 *	9.560e+006 *
48	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
49	Vapour Velocity (m/s)	2.667	2.667	2.667	2.667
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	5195	5200	5205	5210
53	Elevation (m)	0.1560	0.1440	0.1320	0.1200
54	Pressure (bar)	76.80	76.80	76.80	76.80
55	Temperature (C)	57.15	57.15	57.15	57.15
56	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
57	Heat Transferred (kJ/h-m)	723.5	723.4	723.4	723.3
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.1797	0.1797	0.1797	0.1797
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	8.297e-003	8.297e-003	8.297e-003	8.297e-003
63	Static Gradient (kPa/m)	-4.036e-003	-4.036e-003	-4.036e-003	-4.036e-003
64	Accel. Gradient (kPa/m)	4.364e-007	4.364e-007	4.364e-007	4.364e-007
65	Liq. Reynolds	1.997e+005 *	1.997e+005 *	1.997e+005 *	1.997e+005 *
66	Vap. Reynolds	9.560e+006 *	9.560e+006 *	9.560e+006 *	9.560e+006 *
67	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
68	Vapour Velocity (m/s)	2.667	2.667	2.667	2.667
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
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9	PROFILES				
10	Pipe Table				
11					
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13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	5215	5220	5225	5230
16	Elevation (m)	0.1140	0.1080	0.1020	9.600e-002
17	Pressure (bar)	76.80	76.80	76.80	76.80
18	Temperature (C)	57.15	57.14	57.14	57.14
19	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
20	Heat Transferred (kJ/h-m)	723.2	723.2	723.1	723.1
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.1807	0.1807	0.1807	0.1807
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	8.286e-003	8.286e-003	8.286e-003	8.286e-003
26	Static Gradient (kPa/m)	-2.025e-003	-2.025e-003	-2.025e-003	-2.025e-003
27	Accel. Gradient (kPa/m)	6.412e-007	6.412e-007	6.412e-007	6.412e-007
28	Liq. Reynolds	1.997e+005 *	1.997e+005 *	1.997e+005 *	1.997e+005 *
29	Vap. Reynolds	9.560e+006 *	9.560e+006 *	9.560e+006 *	9.560e+006 *
30	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
31	Vapour Velocity (m/s)	2.667	2.667	2.667	2.667
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	5235	5240	5245	5250
35	Elevation (m)	9.000e-002	8.800e-002	8.600e-002	8.400e-002
36	Pressure (bar)	76.80	76.80	76.80	76.79
37	Temperature (C)	57.14	57.14	57.13	57.13
38	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
39	Heat Transferred (kJ/h-m)	723.0	723.0	722.9	722.9
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.1807	0.1813	0.1813	0.1813
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	8.286e-003	8.278e-003	8.278e-003	8.278e-003
45	Static Gradient (kPa/m)	-2.025e-003	-6.765e-004	-6.765e-004	-6.765e-004
46	Accel. Gradient (kPa/m)	6.412e-007	7.785e-007	7.785e-007	7.785e-007
47	Liq. Reynolds	1.997e+005 *	1.997e+005 *	1.997e+005 *	1.997e+005 *
48	Vap. Reynolds	9.560e+006 *	9.560e+006 *	9.560e+006 *	9.560e+006 *
49	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
50	Vapour Velocity (m/s)	2.667	2.667	2.667	2.667
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	5255	5260	5265	5270
54	Elevation (m)	8.200e-002	8.000e-002	6.800e-002	5.600e-002
55	Pressure (bar)	76.79	76.79	76.79	76.79
56	Temperature (C)	57.13	57.13	57.13	57.12
57	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
58	Heat Transferred (kJ/h-m)	722.8	722.7	722.7	722.6
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.1813	0.1813	0.1797	0.1797
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	8.278e-003	8.278e-003	8.296e-003	8.296e-003
64	Static Gradient (kPa/m)	-6.765e-004	-6.765e-004	-4.036e-003	-4.036e-003
65	Accel. Gradient (kPa/m)	7.785e-007	7.785e-007	4.363e-007	4.363e-007
66	Liq. Reynolds	1.997e+005 *	1.997e+005 *	1.997e+005 *	1.997e+005 *
67	Vap. Reynolds	9.560e+006 *	9.560e+006 *	9.560e+006 *	9.560e+006 *
68	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
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13	Vapour Velocity (m/s)	2.667	2.667	2.667	2.667
14	Deposit Thickness (mm)	---	---	---	---
15	Deposit Volume (m3)	---	---	---	---
16	Length (m)	5275	5280	5285	5290
17	Elevation (m)	4.400e-002	3.200e-002	2.000e-002	2.000e-003
18	Pressure (bar)	76.79	76.79	76.79	76.79
19	Temperature (C)	57.12	57.12	57.12	57.12
20	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
21	Heat Transferred (kJ/h-m)	722.6	722.5	722.5	722.4
22	Flow Regime	Transition	Transition	Transition	Transition
23	Water Flow Regime				
24	Liquid HoldUp	0.1797	0.1797	0.1797	0.1788
25	Water HoldUp	---	---	---	---
26	Friction Gradient (kPa/m)	8.296e-003	8.296e-003	8.296e-003	8.307e-003
27	Static Gradient (kPa/m)	-4.036e-003	-4.036e-003	-4.036e-003	-6.050e-003
28	Accel. Gradient (kPa/m)	4.363e-007	4.363e-007	4.363e-007	2.312e-007
29	Liq. Reynolds	1.997e+005 *	1.997e+005 *	1.997e+005 *	1.997e+005 *
30	Vap. Reynolds	9.560e+006 *	9.560e+006 *	9.560e+006 *	9.560e+006 *
31	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
32	Vapour Velocity (m/s)	2.667	2.667	2.667	2.667
33	Deposit Thickness (mm)	---	---	---	---
34	Deposit Volume (m3)	---	---	---	---
35	Length (m)	5295	5300	5305	5310
36	Elevation (m)	-1.600e-002	-3.400e-002	-5.200e-002	-7.000e-002
37	Pressure (bar)	76.79	76.79	76.79	76.79
38	Temperature (C)	57.11	57.11	57.11	57.11
39	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
40	Heat Transferred (kJ/h-m)	722.4	722.3	722.2	722.2
41	Flow Regime	Transition	Transition	Transition	Transition
42	Water Flow Regime				
43	Liquid HoldUp	0.1788	0.1788	0.1788	0.1788
44	Water HoldUp	---	---	---	---
45	Friction Gradient (kPa/m)	8.307e-003	8.307e-003	8.307e-003	8.307e-003
46	Static Gradient (kPa/m)	-6.050e-003	-6.050e-003	-6.050e-003	-6.050e-003
47	Accel. Gradient (kPa/m)	2.312e-007	2.312e-007	2.311e-007	2.311e-007
48	Liq. Reynolds	1.997e+005 *	1.997e+005 *	1.997e+005 *	1.997e+005 *
49	Vap. Reynolds	9.560e+006 *	9.560e+006 *	9.560e+006 *	9.560e+006 *
50	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
51	Vapour Velocity (m/s)	2.667	2.667	2.667	2.667
52	Deposit Thickness (mm)	---	---	---	---
53	Deposit Volume (m3)	---	---	---	---
54	Length (m)	5315	5320	5325	5330
55	Elevation (m)	-9.600e-002	-0.1220	-0.1480	-0.1740
56	Pressure (bar)	76.79	76.79	76.79	76.79
57	Temperature (C)	57.10	57.10	57.10	57.10
58	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
59	Heat Transferred (kJ/h-m)	722.1	722.1	722.0	722.0
60	Flow Regime	Transition	Transition	Transition	Transition
61	Water Flow Regime				
62	Liquid HoldUp	0.1775	0.1775	0.1775	0.1775
63	Water HoldUp	---	---	---	---
64	Friction Gradient (kPa/m)	8.322e-003	8.322e-003	8.322e-003	8.322e-003
65	Static Gradient (kPa/m)	-8.674e-003	-8.674e-003	-8.674e-003	-8.674e-003
66	Accel. Gradient (kPa/m)	-3.609e-008	-3.610e-008	-3.612e-008	-3.613e-008
67	Liq. Reynolds	1.997e+005 *	1.997e+005 *	1.997e+005 *	1.997e+005 *
68	Vap. Reynolds	9.560e+006 *	9.560e+006 *	9.560e+006 *	9.560e+006 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993	
14	Vapour Velocity	(m/s)	2.667	2.667	2.667	2.667	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	5335	5340	5345	5350	
18	Elevation	(m)	-0.2000	-0.2240	-0.2480	-0.2720	
19	Pressure	(bar)	76.79	76.79	76.79	76.79	
20	Temperature	(C)	57.10	57.09	57.09	57.09	
21	Vapour Fraction		0.8809	0.8809	0.8809	0.8809	
22	Heat Transferred	(kJ/h-m)	721.9	721.9	721.8	721.8	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.1775	0.1778	0.1778	0.1778	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	8.322e-003	8.318e-003	8.318e-003	8.318e-003	
28	Static Gradient	(kPa/m)	-8.675e-003	-8.017e-003	-8.017e-003	-8.017e-003	
29	Accel. Gradient	(kPa/m)	-3.615e-008	3.083e-008	3.081e-008	3.080e-008	
30	Liq. Reynolds		1.997e+005 *	1.997e+005 *	1.997e+005 *	1.997e+005 *	
31	Vap. Reynolds		9.560e+006 *	9.560e+006 *	9.560e+006 *	9.560e+006 *	
32	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993	
33	Vapour Velocity	(m/s)	2.667	2.667	2.667	2.667	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	5355	5360	5365	5370	
37	Elevation	(m)	-0.2960	-0.3200	-0.3080	-0.2960	
38	Pressure	(bar)	76.79	76.79	76.79	76.79	
39	Temperature	(C)	57.09	57.09	57.08	57.08	
40	Vapour Fraction		0.8809	0.8809	0.8809	0.8809	
41	Heat Transferred	(kJ/h-m)	721.7	721.7	721.6	721.5	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.1778	0.1778	0.2520	0.2520	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	8.317e-003	8.317e-003	6.673e-003	6.673e-003	
47	Static Gradient	(kPa/m)	-8.017e-003	-8.017e-003	5.095e-003	5.095e-003	
48	Accel. Gradient	(kPa/m)	3.078e-008	3.077e-008	1.205e-006	1.205e-006	
49	Liq. Reynolds		1.997e+005 *	1.997e+005 *	1.997e+005 *	1.997e+005 *	
50	Vap. Reynolds		9.560e+006 *	9.560e+006 *	9.560e+006 *	9.560e+006 *	
51	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993	
52	Vapour Velocity	(m/s)	2.667	2.667	2.667	2.667	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	5375	5380	5385	5390	
56	Elevation	(m)	-0.2840	-0.2720	-0.2600	-0.3460	
57	Pressure	(bar)	76.79	76.79	76.79	76.79	
58	Temperature	(C)	57.08	57.08	57.08	57.07	
59	Vapour Fraction		0.8809	0.8809	0.8809	0.8809	
60	Heat Transferred	(kJ/h-m)	721.4	721.4	721.3	721.3	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2520	0.2520	0.2520	0.1680	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	6.673e-003	6.673e-003	6.673e-003	8.444e-003	
66	Static Gradient	(kPa/m)	5.095e-003	5.095e-003	5.095e-003	-2.769e-002	
67	Accel. Gradient	(kPa/m)	1.205e-006	1.205e-006	1.205e-006	-1.971e-006	
68	Liq. Reynolds		1.996e+005 *	1.996e+005 *	1.996e+005 *	1.996e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
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5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
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12					
13	Vap. Reynolds	9.560e+006 *	9.560e+006 *	9.560e+006 *	9.560e+006 *
14	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
15	Vapour Velocity (m/s)	2.667	2.667	2.667	2.667
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	5395	5400	5405	5410
19	Elevation (m)	-0.4320	-0.5180	-0.6040	-0.6900
20	Pressure (bar)	76.79	76.79	76.79	76.79
21	Temperature (C)	57.07	57.07	57.07	57.07
22	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
23	Heat Transferred (kJ/h-m)	721.3	721.3	721.2	721.2
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1680	0.1680	0.1680	0.1680
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	8.444e-003	8.444e-003	8.443e-003	8.443e-003
29	Static Gradient (kPa/m)	-2.769e-002	-2.769e-002	-2.769e-002	-2.769e-002
30	Accel. Gradient (kPa/m)	-1.971e-006	-1.971e-006	-1.971e-006	-1.971e-006
31	Liq. Reynolds	1.996e+005 *	1.996e+005 *	1.996e+005 *	1.996e+005 *
32	Vap. Reynolds	9.560e+006 *	9.560e+006 *	9.560e+006 *	9.560e+006 *
33	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
34	Vapour Velocity (m/s)	2.666	2.666	2.666	2.666
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	5415	5420	5425	5430
38	Elevation (m)	-0.6280	-0.5660	-0.5040	-0.4420
39	Pressure (bar)	76.79	76.79	76.79	76.79
40	Temperature (C)	57.07	57.06	57.06	57.06
41	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
42	Heat Transferred (kJ/h-m)	721.0	721.0	720.9	720.8
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2810	0.2810	0.2810	0.2810
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	7.227e-003	7.227e-003	7.227e-003	7.227e-003
48	Static Gradient (kPa/m)	2.852e-002	2.852e-002	2.852e-002	2.852e-002
49	Accel. Gradient (kPa/m)	3.660e-006	3.660e-006	3.660e-006	3.660e-006
50	Liq. Reynolds	1.996e+005 *	1.996e+005 *	1.996e+005 *	1.996e+005 *
51	Vap. Reynolds	9.560e+006 *	9.560e+006 *	9.560e+006 *	9.560e+006 *
52	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
53	Vapour Velocity (m/s)	2.666	2.666	2.666	2.666
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	5435	5440	5445	5450
57	Elevation (m)	-0.3800	-0.3080	-0.2360	-0.1640
58	Pressure (bar)	76.79	76.78	76.78	76.78
59	Temperature (C)	57.05	57.05	57.05	57.05
60	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
61	Heat Transferred (kJ/h-m)	720.8	720.7	720.6	720.5
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2810	0.2868	0.2868	0.2868
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	7.228e-003	7.263e-003	7.263e-003	7.263e-003
67	Static Gradient (kPa/m)	2.852e-002	3.363e-002	3.363e-002	3.363e-002
68	Accel. Gradient (kPa/m)	3.661e-006	4.187e-006	4.188e-006	4.188e-006
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3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:13:43 2022	
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10	Pipe Table					
11						
12						
13	Liq. Reynolds	1.996e+005 *	1.996e+005 *	1.996e+005 *	1.996e+005 *	
14	Vap. Reynolds	9.560e+006 *	9.560e+006 *	9.560e+006 *	9.560e+006 *	
15	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993	
16	Vapour Velocity (m/s)	2.666	2.666	2.667	2.667	
17	Deposit Thickness (mm)	---	---	---	---	
18	Deposit Volume (m3)	---	---	---	---	
19	Length (m)	5455	5460	5465	5470	
20	Elevation (m)	-9.200e-002	-2.000e-002	-2.600e-002	-3.200e-002	
21	Pressure (bar)	76.78	76.78	76.77	76.77	
22	Temperature (C)	57.04	57.04	57.04	57.04	
23	Vapour Fraction	0.8809	0.8809	0.8809	0.8809	
24	Heat Transferred (kJ/h-m)	720.5	720.4	720.4	720.3	
25	Flow Regime	Transition	Transition	Transition	Transition	
26	Water Flow Regime					
27	Liquid HoldUp	0.2867	0.2867	0.1807	0.1807	
28	Water HoldUp	---	---	---	---	
29	Friction Gradient (kPa/m)	7.263e-003	7.263e-003	8.284e-003	8.284e-003	
30	Static Gradient (kPa/m)	3.363e-002	3.363e-002	-2.025e-003	-2.025e-003	
31	Accel. Gradient (kPa/m)	4.188e-006	4.188e-006	6.411e-007	6.411e-007	
32	Liq. Reynolds	1.996e+005 *	1.996e+005 *	1.996e+005 *	1.996e+005 *	
33	Vap. Reynolds	9.560e+006 *	9.560e+006 *	9.560e+006 *	9.560e+006 *	
34	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993	
35	Vapour Velocity (m/s)	2.667	2.667	2.667	2.667	
36	Deposit Thickness (mm)	---	---	---	---	
37	Deposit Volume (m3)	---	---	---	---	
38	Length (m)	5475	5480	5485	5490	
39	Elevation (m)	-3.800e-002	-4.400e-002	-5.000e-002	-6.800e-002	
40	Pressure (bar)	76.77	76.77	76.77	76.77	
41	Temperature (C)	57.03	57.03	57.03	57.03	
42	Vapour Fraction	0.8809	0.8809	0.8809	0.8809	
43	Heat Transferred (kJ/h-m)	720.3	720.2	720.2	720.1	
44	Flow Regime	Transition	Transition	Transition	Transition	
45	Water Flow Regime					
46	Liquid HoldUp	0.1807	0.1807	0.1807	0.1788	
47	Water HoldUp	---	---	---	---	
48	Friction Gradient (kPa/m)	8.284e-003	8.284e-003	8.284e-003	8.306e-003	
49	Static Gradient (kPa/m)	-2.025e-003	-2.025e-003	-2.025e-003	-5.982e-003	
50	Accel. Gradient (kPa/m)	6.411e-007	6.410e-007	6.410e-007	2.380e-007	
51	Liq. Reynolds	1.996e+005 *	1.996e+005 *	1.996e+005 *	1.996e+005 *	
52	Vap. Reynolds	9.561e+006 *	9.561e+006 *	9.561e+006 *	9.561e+006 *	
53	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993	
54	Vapour Velocity (m/s)	2.667	2.667	2.667	2.667	
55	Deposit Thickness (mm)	---	---	---	---	
56	Deposit Volume (m3)	---	---	---	---	
57	Length (m)	5495	5500	5505	5510	
58	Elevation (m)	-8.600e-002	-0.1040	-0.1220	-0.1400	
59	Pressure (bar)	76.77	76.77	76.77	76.77	
60	Temperature (C)	57.03	57.02	57.02	57.02	
61	Vapour Fraction	0.8809	0.8809	0.8809	0.8809	
62	Heat Transferred (kJ/h-m)	720.0	720.0	719.9	719.9	
63	Flow Regime	Transition	Transition	Transition	Transition	
64	Water Flow Regime					
65	Liquid HoldUp	0.1788	0.1788	0.1788	0.1788	
66	Water HoldUp	---	---	---	---	
67	Friction Gradient (kPa/m)	8.306e-003	8.306e-003	8.306e-003	8.306e-003	
68	Static Gradient (kPa/m)	-5.982e-003	-5.982e-003	-5.982e-003	-5.982e-003	
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:13:43 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient	(kPa/m)	2.380e-007	2.380e-007	2.380e-007	2.380e-007	
14	Liq. Reynolds		1.996e+005 *	1.996e+005 *	1.996e+005 *	1.996e+005 *	
15	Vap. Reynolds		9.561e+006 *	9.561e+006 *	9.561e+006 *	9.561e+006 *	
16	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993	
17	Vapour Velocity	(m/s)	2.667	2.667	2.666	2.666	
18	Deposit Thickness	(mm)	---	---	---	---	
19	Deposit Volume	(m3)	---	---	---	---	
20	Length	(m)	5515	5520	5525	5530	
21	Elevation	(m)	-0.1520	-0.1640	-0.1760	-0.1880	
22	Pressure	(bar)	76.77	76.77	76.77	76.77	
23	Temperature	(C)	57.02	57.02	57.01	57.01	
24	Vapour Fraction		0.8809	0.8809	0.8809	0.8809	
25	Heat Transferred	(kJ/h-m)	719.8	719.8	719.7	719.7	
26	Flow Regime		Transition	Transition	Transition	Transition	
27	Water Flow Regime						
28	Liquid HoldUp		0.1798	0.1798	0.1798	0.1798	
29	Water HoldUp		---	---	---	---	
30	Friction Gradient	(kPa/m)	8.295e-003	8.295e-003	8.295e-003	8.295e-003	
31	Static Gradient	(kPa/m)	-4.037e-003	-4.037e-003	-4.037e-003	-4.037e-003	
32	Accel. Gradient	(kPa/m)	4.361e-007	4.361e-007	4.361e-007	4.361e-007	
33	Liq. Reynolds		1.996e+005 *	1.996e+005 *	1.996e+005 *	1.996e+005 *	
34	Vap. Reynolds		9.561e+006 *	9.561e+006 *	9.561e+006 *	9.561e+006 *	
35	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993	
36	Vapour Velocity	(m/s)	2.666	2.666	2.666	2.666	
37	Deposit Thickness	(mm)	---	---	---	---	
38	Deposit Volume	(m3)	---	---	---	---	
39	Length	(m)	5535	5542	5549	5556	
40	Elevation	(m)	-0.2000	-0.2240	-0.2480	-0.2720	
41	Pressure	(bar)	76.77	76.77	76.77	76.77	
42	Temperature	(C)	57.01	57.01	57.00	57.00	
43	Vapour Fraction		0.8809	0.8809	0.8809	0.8809	
44	Heat Transferred	(kJ/h-m)	719.6	719.5	719.5	719.4	
45	Flow Regime		Transition	Transition	Transition	Transition	
46	Water Flow Regime						
47	Liquid HoldUp		0.1798	0.1789	0.1789	0.1789	
48	Water HoldUp		---	---	---	---	
49	Friction Gradient	(kPa/m)	8.295e-003	8.304e-003	8.304e-003	8.304e-003	
50	Static Gradient	(kPa/m)	-4.037e-003	-5.750e-003	-5.750e-003	-5.750e-003	
51	Accel. Gradient	(kPa/m)	4.360e-007	2.616e-007	2.616e-007	2.615e-007	
52	Liq. Reynolds		1.996e+005 *	1.996e+005 *	1.996e+005 *	1.995e+005 *	
53	Vap. Reynolds		9.561e+006 *	9.561e+006 *	9.561e+006 *	9.561e+006 *	
54	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993	
55	Vapour Velocity	(m/s)	2.666	2.666	2.666	2.666	
56	Deposit Thickness	(mm)	---	---	---	---	
57	Deposit Volume	(m3)	---	---	---	---	
58	Length	(m)	5563	5570	5572	5574	
59	Elevation	(m)	-0.2960	-0.3200	-0.3480	-0.3760	
60	Pressure	(bar)	76.77	76.77	76.77	76.77	
61	Temperature	(C)	57.00	57.00	56.99	56.99	
62	Vapour Fraction		0.8809	0.8809	0.8809	0.8809	
63	Heat Transferred	(kJ/h-m)	719.3	719.2	719.2	719.2	
64	Flow Regime		Transition	Transition	Transition	Transition	
65	Water Flow Regime						
66	Liquid HoldUp		0.1789	0.1789	0.1705	0.1705	
67	Water HoldUp		---	---	---	---	
68	Friction Gradient	(kPa/m)	8.304e-003	8.304e-003	8.408e-003	8.408e-003	
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1				Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	-5.750e-003	-5.750e-003	-2.276e-002	-2.276e-002
14	Accel. Gradient	(kPa/m)	2.615e-007	2.615e-007	-1.470e-006	-1.470e-006
15	Liq. Reynolds		1.995e+005 *	1.995e+005 *	1.995e+005 *	1.995e+005 *
16	Vap. Reynolds		9.561e+006 *	9.561e+006 *	9.561e+006 *	9.561e+006 *
17	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993
18	Vapour Velocity	(m/s)	2.666	2.666	2.666	2.666
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	5576	5578	5580	5585
22	Elevation	(m)	-0.4040	-0.4320	-0.4600	-0.4460
23	Pressure	(bar)	76.77	76.77	76.77	76.77
24	Temperature	(C)	56.99	56.99	56.99	56.99
25	Vapour Fraction		0.8808	0.8808	0.8808	0.8808
26	Heat Transferred	(kJ/h-m)	719.2	719.2	719.2	719.0
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.1705	0.1705	0.1705	0.2532
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	8.408e-003	8.408e-003	8.408e-003	6.550e-003
32	Static Gradient	(kPa/m)	-2.276e-002	-2.276e-002	-2.276e-002	5.965e-003
33	Accel. Gradient	(kPa/m)	-1.470e-006	-1.470e-006	-1.470e-006	1.282e-006
34	Liq. Reynolds		1.995e+005 *	1.995e+005 *	1.995e+005 *	1.995e+005 *
35	Vap. Reynolds		9.561e+006 *	9.561e+006 *	9.561e+006 *	9.561e+006 *
36	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993
37	Vapour Velocity	(m/s)	2.666	2.666	2.666	2.666
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	5590	5595	5600	5605
41	Elevation	(m)	-0.4320	-0.4180	-0.4040	-0.3900
42	Pressure	(bar)	76.77	76.77	76.77	76.77
43	Temperature	(C)	56.99	56.99	56.98	56.98
44	Vapour Fraction		0.8808	0.8808	0.8808	0.8808
45	Heat Transferred	(kJ/h-m)	719.0	718.9	718.8	718.8
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2532	0.2532	0.2532	0.2532
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	6.550e-003	6.550e-003	6.550e-003	6.550e-003
51	Static Gradient	(kPa/m)	5.965e-003	5.965e-003	5.965e-003	5.965e-003
52	Accel. Gradient	(kPa/m)	1.282e-006	1.282e-006	1.282e-006	1.282e-006
53	Liq. Reynolds		1.995e+005 *	1.995e+005 *	1.995e+005 *	1.995e+005 *
54	Vap. Reynolds		9.561e+006 *	9.561e+006 *	9.561e+006 *	9.561e+006 *
55	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993
56	Vapour Velocity	(m/s)	2.666	2.666	2.666	2.666
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	5610	5615	5620	5625
60	Elevation	(m)	-0.3980	-0.4060	-0.4140	-0.4220
61	Pressure	(bar)	76.77	76.77	76.77	76.77
62	Temperature	(C)	56.98	56.98	56.97	56.97
63	Vapour Fraction		0.8808	0.8808	0.8808	0.8808
64	Heat Transferred	(kJ/h-m)	718.8	718.7	718.7	718.6
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.1804	0.1804	0.1804	0.1804
68	Water HoldUp		---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3						
4						
5			Unit Set: Mixed			
6			Date/Time: Mon Jun 13 15:13:43 2022			
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	PROFILES					
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	8.286e-003	8.286e-003	8.286e-003	8.286e-003
14	Static Gradient	(kPa/m)	-2.698e-003	-2.698e-003	-2.698e-003	-2.698e-003
15	Accel. Gradient	(kPa/m)	5.723e-007	5.723e-007	5.723e-007	5.723e-007
16	Liq. Reynolds		1.995e+005 *	1.995e+005 *	1.995e+005 *	1.995e+005 *
17	Vap. Reynolds		9.561e+006 *	9.561e+006 *	9.561e+006 *	9.561e+006 *
18	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993
19	Vapour Velocity	(m/s)	2.666	2.666	2.666	2.666
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	5630	5635	5640	5645
23	Elevation	(m)	-0.4300	-0.4880	-0.5460	-0.6040
24	Pressure	(bar)	76.77	76.77	76.77	76.77
25	Temperature	(C)	56.97	56.97	56.97	56.97
26	Vapour Fraction		0.8808	0.8808	0.8808	0.8808
27	Heat Transferred	(kJ/h-m)	718.6	718.5	718.5	718.4
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.1804	0.1725	0.1725	0.1725
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	8.286e-003	8.383e-003	8.383e-003	8.383e-003
33	Static Gradient	(kPa/m)	-2.698e-003	-1.900e-002	-1.900e-002	-1.900e-002
34	Accel. Gradient	(kPa/m)	5.723e-007	-1.087e-006	-1.087e-006	-1.087e-006
35	Liq. Reynolds		1.995e+005 *	1.995e+005 *	1.995e+005 *	1.995e+005 *
36	Vap. Reynolds		9.561e+006 *	9.561e+006 *	9.561e+006 *	9.561e+006 *
37	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993
38	Vapour Velocity	(m/s)	2.666	2.666	2.666	2.666
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	5650	5655	5660	5665
42	Elevation	(m)	-0.6620	-0.7200	-0.7180	-0.7160
43	Pressure	(bar)	76.77	76.77	76.77	76.77
44	Temperature	(C)	56.96	56.96	56.96	56.96
45	Vapour Fraction		0.8808	0.8808	0.8808	0.8808
46	Heat Transferred	(kJ/h-m)	718.4	718.3	718.2	718.2
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.1725	0.1725	0.2462	0.2463
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	8.382e-003	8.382e-003	7.304e-003	7.304e-003
52	Static Gradient	(kPa/m)	-1.900e-002	-1.900e-002	8.353e-004	8.353e-004
53	Accel. Gradient	(kPa/m)	-1.087e-006	-1.087e-006	8.333e-007	8.333e-007
54	Liq. Reynolds		1.995e+005 *	1.995e+005 *	1.995e+005 *	1.995e+005 *
55	Vap. Reynolds		9.561e+006 *	9.561e+006 *	9.561e+006 *	9.561e+006 *
56	Liquid Velocity	(m/s)	0.1993	0.1993	0.1993	0.1993
57	Vapour Velocity	(m/s)	2.666	2.666	2.666	2.666
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	5670	5675	5680	5685
61	Elevation	(m)	-0.7140	-0.7120	-0.7100	-0.6260
62	Pressure	(bar)	76.77	76.77	76.77	76.77
63	Temperature	(C)	56.96	56.95	56.95	56.95
64	Vapour Fraction		0.8808	0.8808	0.8808	0.8808
65	Heat Transferred	(kJ/h-m)	718.1	718.0	718.0	717.9
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2463	0.2463	0.2463	0.2941
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	7.303e-003	7.303e-003	7.303e-003	7.277e-003
15	Static Gradient (kPa/m)	8.353e-004	8.353e-004	8.353e-004	4.030e-002
16	Accel. Gradient (kPa/m)	8.333e-007	8.333e-007	8.333e-007	4.872e-006
17	Liq. Reynolds	1.995e+005 *	1.995e+005 *	1.995e+005 *	1.995e+005 *
18	Vap. Reynolds	9.561e+006 *	9.561e+006 *	9.561e+006 *	9.561e+006 *
19	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
20	Vapour Velocity (m/s)	2.666	2.666	2.666	2.666
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	5690	5695	5700	5705
24	Elevation (m)	-0.5420	-0.4580	-0.3740	-0.2900
25	Pressure (bar)	76.76	76.76	76.76	76.76
26	Temperature (C)	56.95	56.94	56.94	56.94
27	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
28	Heat Transferred (kJ/h-m)	717.8	717.8	717.7	717.6
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2941	0.2941	0.2941	0.2941
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	7.277e-003	7.277e-003	7.277e-003	7.277e-003
34	Static Gradient (kPa/m)	4.030e-002	4.030e-002	4.030e-002	4.030e-002
35	Accel. Gradient (kPa/m)	4.872e-006	4.872e-006	4.872e-006	4.873e-006
36	Liq. Reynolds	1.995e+005 *	1.995e+005 *	1.995e+005 *	1.995e+005 *
37	Vap. Reynolds	9.561e+006 *	9.561e+006 *	9.561e+006 *	9.561e+006 *
38	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
39	Vapour Velocity (m/s)	2.666	2.666	2.666	2.666
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	5710	5715	5720	5725
43	Elevation (m)	-0.3100	-0.3300	-0.3500	-0.3700
44	Pressure (bar)	76.76	76.76	76.76	76.76
45	Temperature (C)	56.93	56.93	56.93	56.93
46	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
47	Heat Transferred (kJ/h-m)	717.6	717.6	717.5	717.5
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.1785	0.1785	0.1785	0.1785
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	8.308e-003	8.308e-003	8.308e-003	8.308e-003
53	Static Gradient (kPa/m)	-6.698e-003	-6.698e-003	-6.698e-003	-6.698e-003
54	Accel. Gradient (kPa/m)	1.649e-007	1.649e-007	1.649e-007	1.649e-007
55	Liq. Reynolds	1.995e+005 *	1.995e+005 *	1.995e+005 *	1.995e+005 *
56	Vap. Reynolds	9.561e+006 *	9.561e+006 *	9.561e+006 *	9.561e+006 *
57	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
58	Vapour Velocity (m/s)	2.666	2.666	2.666	2.666
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	5730	5735	5740	5745
62	Elevation (m)	-0.3900	-0.3620	-0.3340	-0.3060
63	Pressure (bar)	76.76	76.76	76.75	76.75
64	Temperature (C)	56.93	56.92	56.92	56.92
65	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
66	Heat Transferred (kJ/h-m)	717.4	717.3	717.2	717.2
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 122 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.1785	0.2613	0.2613	0.2613
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	8.308e-003	5.745e-003	5.745e-003	5.745e-003
16	Static Gradient (kPa/m)	-6.698e-003	1.221e-002	1.221e-002	1.221e-002
17	Accel. Gradient (kPa/m)	1.649e-007	1.839e-006	1.839e-006	1.839e-006
18	Liq. Reynolds	1.995e+005 *	1.995e+005 *	1.995e+005 *	1.994e+005 *
19	Vap. Reynolds	9.561e+006 *	9.561e+006 *	9.561e+006 *	9.561e+006 *
20	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
21	Vapour Velocity (m/s)	2.666	2.666	2.666	2.666
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	5750	5755	5760	5765
25	Elevation (m)	-0.2780	-0.2500	-0.2100	-0.1700
26	Pressure (bar)	76.75	76.75	76.75	76.75
27	Temperature (C)	56.92	56.92	56.91	56.91
28	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
29	Heat Transferred (kJ/h-m)	717.1	717.0	717.0	716.9
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2613	0.2613	0.2683	0.2683
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	5.745e-003	5.745e-003	6.809e-003	6.809e-003
35	Static Gradient (kPa/m)	1.221e-002	1.221e-002	1.778e-002	1.778e-002
36	Accel. Gradient (kPa/m)	1.839e-006	1.839e-006	2.519e-006	2.519e-006
37	Liq. Reynolds	1.994e+005 *	1.994e+005 *	1.994e+005 *	1.994e+005 *
38	Vap. Reynolds	9.561e+006 *	9.561e+006 *	9.561e+006 *	9.562e+006 *
39	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
40	Vapour Velocity (m/s)	2.666	2.666	2.666	2.666
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	5770	5775	5780	5785
44	Elevation (m)	-0.1300	-9.000e-002	-5.000e-002	-0.1100
45	Pressure (bar)	76.75	76.75	76.75	76.75
46	Temperature (C)	56.91	56.91	56.90	56.90
47	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
48	Heat Transferred (kJ/h-m)	716.8	716.8	716.7	716.7
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2683	0.2683	0.2683	0.1721
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.809e-003	6.809e-003	6.809e-003	8.387e-003
54	Static Gradient (kPa/m)	1.778e-002	1.778e-002	1.778e-002	-1.963e-002
55	Accel. Gradient (kPa/m)	2.519e-006	2.519e-006	2.519e-006	-1.151e-006
56	Liq. Reynolds	1.994e+005 *	1.994e+005 *	1.994e+005 *	1.994e+005 *
57	Vap. Reynolds	9.562e+006 *	9.562e+006 *	9.562e+006 *	9.562e+006 *
58	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
59	Vapour Velocity (m/s)	2.666	2.666	2.666	2.666
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	5790	5795	5800	5805
63	Elevation (m)	-0.1700	-0.2300	-0.2900	-0.3500
64	Pressure (bar)	76.75	76.75	76.75	76.75
65	Temperature (C)	56.90	56.90	56.90	56.89
66	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
67	Heat Transferred (kJ/h-m)	716.7	716.6	716.6	716.5
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 123 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1721	0.1721	0.1721	0.1722
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	8.387e-003	8.387e-003	8.387e-003	8.386e-003
17	Static Gradient (kPa/m)	-1.963e-002	-1.963e-002	-1.963e-002	-1.963e-002
18	Accel. Gradient (kPa/m)	-1.151e-006	-1.151e-006	-1.151e-006	-1.151e-006
19	Liq. Reynolds	1.994e+005 *	1.994e+005 *	1.994e+005 *	1.994e+005 *
20	Vap. Reynolds	9.562e+006 *	9.562e+006 *	9.562e+006 *	9.562e+006 *
21	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
22	Vapour Velocity (m/s)	2.666	2.666	2.666	2.666
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	5810	5815	5820	5825
26	Elevation (m)	-0.3580	-0.3660	-0.3740	-0.3820
27	Pressure (bar)	76.75	76.75	76.75	76.75
28	Temperature (C)	56.89	56.89	56.89	56.89
29	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
30	Heat Transferred (kJ/h-m)	716.5	716.4	716.4	716.3
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1804	0.1804	0.1804	0.1804
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	8.286e-003	8.286e-003	8.286e-003	8.285e-003
36	Static Gradient (kPa/m)	-2.698e-003	-2.698e-003	-2.698e-003	-2.698e-003
37	Accel. Gradient (kPa/m)	5.723e-007	5.723e-007	5.723e-007	5.722e-007
38	Liq. Reynolds	1.994e+005 *	1.994e+005 *	1.994e+005 *	1.994e+005 *
39	Vap. Reynolds	9.562e+006 *	9.562e+006 *	9.562e+006 *	9.562e+006 *
40	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
41	Vapour Velocity (m/s)	2.666	2.666	2.666	2.666
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	5830	5835	5840	5845
45	Elevation (m)	-0.3900	-0.2920	-0.1940	-9.600e-002
46	Pressure (bar)	76.75	76.74	76.74	76.74
47	Temperature (C)	56.88	56.88	56.88	56.87
48	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
49	Heat Transferred (kJ/h-m)	716.2	716.1	716.1	716.0
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.1804	0.3019	0.3019	0.3019
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	8.285e-003	7.277e-003	7.277e-003	7.277e-003
55	Static Gradient (kPa/m)	-2.698e-003	4.759e-002	4.759e-002	4.759e-002
56	Accel. Gradient (kPa/m)	5.722e-007	5.620e-006	5.620e-006	5.620e-006
57	Liq. Reynolds	1.994e+005 *	1.994e+005 *	1.994e+005 *	1.994e+005 *
58	Vap. Reynolds	9.562e+006 *	9.562e+006 *	9.562e+006 *	9.562e+006 *
59	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
60	Vapour Velocity (m/s)	2.666	2.666	2.666	2.666
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	5850	5855	5860	5865
64	Elevation (m)	2.000e-003	1.000e-001	0.1280	0.1560
65	Pressure (bar)	76.74	76.73	76.73	76.73
66	Temperature (C)	56.87	56.87	56.87	56.86
67	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
68	Heat Transferred (kJ/h-m)	715.9	715.8	715.7	715.7
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 124 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.3018	0.3018	0.2613	0.2613
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	7.278e-003	7.278e-003	5.745e-003	5.745e-003
18	Static Gradient (kPa/m)	4.759e-002	4.759e-002	1.221e-002	1.221e-002
19	Accel. Gradient (kPa/m)	5.620e-006	5.621e-006	1.839e-006	1.839e-006
20	Liq. Reynolds	1.994e+005 *	1.994e+005 *	1.994e+005 *	1.994e+005 *
21	Vap. Reynolds	9.562e+006 *	9.562e+006 *	9.562e+006 *	9.562e+006 *
22	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
23	Vapour Velocity (m/s)	2.666	2.666	2.666	2.666
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	5870	5875	5880	5885
27	Elevation (m)	0.1840	0.2120	0.2400	-1.368
28	Pressure (bar)	76.73	76.73	76.73	76.74
29	Temperature (C)	56.86	56.86	56.86	56.86
30	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
31	Heat Transferred (kJ/h-m)	715.6	715.6	715.5	717.1
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2613	0.2613	0.2613	6.958e-002
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	5.745e-003	5.745e-003	5.745e-003	1.280e-002
37	Static Gradient (kPa/m)	1.221e-002	1.221e-002	1.221e-002	-0.3224
38	Accel. Gradient (kPa/m)	1.839e-006	1.840e-006	1.840e-006	-3.171e-005
39	Liq. Reynolds	1.994e+005 *	1.994e+005 *	1.994e+005 *	1.994e+005 *
40	Vap. Reynolds	9.562e+006 *	9.562e+006 *	9.562e+006 *	9.562e+006 *
41	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
42	Vapour Velocity (m/s)	2.666	2.666	2.666	2.665
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	5890	5895	5900	5905
46	Elevation (m)	-2.976	-4.584	-6.192	-7.800
47	Pressure (bar)	76.76	76.78	76.79	76.81
48	Temperature (C)	56.87	56.88	56.88	56.89
49	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
50	Heat Transferred (kJ/h-m)	717.3	717.4	717.6	717.8
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	6.959e-002	6.960e-002	6.962e-002	6.963e-002
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	1.280e-002	1.280e-002	1.279e-002	1.279e-002
56	Static Gradient (kPa/m)	-0.3225	-0.3225	-0.3226	-0.3226
57	Accel. Gradient (kPa/m)	-3.170e-005	-3.169e-005	-3.169e-005	-3.168e-005
58	Liq. Reynolds	1.994e+005 *	1.994e+005 *	1.995e+005 *	1.995e+005 *
59	Vap. Reynolds	9.561e+006 *	9.561e+006 *	9.560e+006 *	9.560e+006 *
60	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1993
61	Vapour Velocity (m/s)	2.665	2.664	2.664	2.663
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	5910	5915	5920	5925
65	Elevation (m)	-9.460	-11.12	-12.78	-14.44
66	Pressure (bar)	76.82	76.84	76.86	76.87
67	Temperature (C)	56.90	56.90	56.91	56.92
68	Vapour Fraction	0.8808	0.8808	0.8807	0.8807
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
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11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	718.0	718.2	718.4	718.5
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	6.965e-002	6.966e-002	6.968e-002	6.969e-002
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	1.279e-002	1.278e-002	1.278e-002	1.278e-002
19	Static Gradient (kPa/m)	-0.3354	-0.3355	-0.3355	-0.3356
20	Accel. Gradient (kPa/m)	-3.297e-005	-3.297e-005	-3.296e-005	-3.295e-005
21	Liq. Reynolds	1.995e+005 *	1.995e+005 *	1.996e+005 *	1.996e+005 *
22	Vap. Reynolds	9.559e+006 *	9.559e+006 *	9.558e+006 *	9.558e+006 *
23	Liquid Velocity (m/s)	0.1993	0.1993	0.1993	0.1994
24	Vapour Velocity (m/s)	2.663	2.662	2.662	2.661
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	5930	5935	5940	5945
28	Elevation (m)	-16.10	-13.74	-11.38	-9.020
29	Pressure (bar)	76.89	76.73	76.57	76.41
30	Temperature (C)	56.93	56.88	56.84	56.80
31	Vapour Fraction	0.8807	0.8808	0.8809	0.8809
32	Heat Transferred (kJ/h-m)	718.7	709.6	708.5	707.4
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	6.971e-002	1.000	1.000	1.000
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	1.277e-002	6.694e-003	6.707e-003	6.720e-003
38	Static Gradient (kPa/m)	-0.3357	3.159	3.159	3.160
39	Accel. Gradient (kPa/m)	-3.295e-005	3.243e-004	3.258e-004	3.272e-004
40	Liq. Reynolds	1.996e+005 *	1.994e+005 *	1.992e+005 *	1.990e+005 *
41	Vap. Reynolds	9.557e+006 *	9.562e+006 *	9.567e+006 *	9.573e+006 *
42	Liquid Velocity (m/s)	0.1994	0.1993	0.1993	0.1992
43	Vapour Velocity (m/s)	2.661	2.666	2.672	2.678
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	5950	5955	5960	5965
47	Elevation (m)	-6.660	-4.300	-3.502	-2.704
48	Pressure (bar)	76.25	76.10	76.06	76.02
49	Temperature (C)	56.76	56.72	56.70	56.69
50	Vapour Fraction	0.8810	0.8811	0.8811	0.8811
51	Heat Transferred (kJ/h-m)	706.3	705.2	712.9	712.5
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	1.000	1.000	0.6874	0.6872
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	6.733e-003	6.746e-003	6.866e-003	6.869e-003
57	Static Gradient (kPa/m)	3.160	3.161	0.7635	0.7634
58	Accel. Gradient (kPa/m)	3.287e-004	3.302e-004	8.038e-005	8.045e-005
59	Liq. Reynolds	1.987e+005 *	1.985e+005 *	1.985e+005 *	1.984e+005 *
60	Vap. Reynolds	9.578e+006 *	9.583e+006 *	9.584e+006 *	9.585e+006 *
61	Liquid Velocity (m/s)	0.1992	0.1991	0.1991	0.1991
62	Vapour Velocity (m/s)	2.684	2.690	2.691	2.693
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	5970	5975	5980	5985
66	Elevation (m)	-1.906	-1.108	-0.3100	-3.048
67	Pressure (bar)	75.98	75.94	75.90	75.93
68	Temperature (C)	56.68	56.67	56.65	56.67
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
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5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
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11	Pipe Table				
12					
13	Vapour Fraction	0.8811	0.8811	0.8812	0.8812
14	Heat Transferred (kJ/h-m)	712.2	711.9	711.5	711.8
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.6870	0.6868	0.6866	6.877e-002
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	6.873e-003	6.876e-003	6.879e-003	1.298e-002
20	Static Gradient (kPa/m)	0.7632	0.7630	0.7629	-0.5469
21	Accel. Gradient (kPa/m)	8.051e-005	8.058e-005	8.064e-005	-5.590e-005
22	Liq. Reynolds	1.984e+005 *	1.983e+005 *	1.983e+005 *	1.983e+005 *
23	Vap. Reynolds	9.587e+006 *	9.588e+006 *	9.589e+006 *	9.588e+006 *
24	Liquid Velocity (m/s)	0.1991	0.1991	0.1991	0.1991
25	Vapour Velocity (m/s)	2.694	2.695	2.697	2.696
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	5990	5995	6000	6005
29	Elevation (m)	-5.786	-8.524	-11.26	-14.00
30	Pressure (bar)	75.96	75.98	76.01	76.04
31	Temperature (C)	56.68	56.69	56.71	56.72
32	Vapour Fraction	0.8811	0.8811	0.8811	0.8811
33	Heat Transferred (kJ/h-m)	712.2	712.5	712.9	713.2
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	6.879e-002	6.882e-002	6.884e-002	6.886e-002
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	1.298e-002	1.297e-002	1.297e-002	1.296e-002
39	Static Gradient (kPa/m)	-0.5471	-0.5473	-0.5474	-0.5476
40	Accel. Gradient (kPa/m)	-5.588e-005	-5.586e-005	-5.584e-005	-5.582e-005
41	Liq. Reynolds	1.983e+005 *	1.984e+005 *	1.984e+005 *	1.985e+005 *
42	Vap. Reynolds	9.587e+006 *	9.587e+006 *	9.586e+006 *	9.585e+006 *
43	Liquid Velocity (m/s)	0.1991	0.1991	0.1991	0.1991
44	Vapour Velocity (m/s)	2.695	2.694	2.693	2.692
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	6010	6015	6020	6025
48	Elevation (m)	-11.27	-8.544	-5.816	-3.088
49	Pressure (bar)	75.85	75.67	75.49	75.30
50	Temperature (C)	56.67	56.62	56.57	56.52
51	Vapour Fraction	0.8812	0.8813	0.8814	0.8814
52	Heat Transferred (kJ/h-m)	704.0	702.7	701.4	700.1
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	1.000	1.000	1.000	1.000
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	6.766e-003	6.782e-003	6.797e-003	6.813e-003
58	Static Gradient (kPa/m)	3.655	3.655	3.656	3.657
59	Accel. Gradient (kPa/m)	3.842e-004	3.862e-004	3.882e-004	3.902e-004
60	Liq. Reynolds	1.982e+005 *	1.980e+005 *	1.977e+005 *	1.975e+005 *
61	Vap. Reynolds	9.591e+006 *	9.597e+006 *	9.603e+006 *	9.609e+006 *
62	Liquid Velocity (m/s)	0.1991	0.1990	0.1990	0.1989
63	Vapour Velocity (m/s)	2.699	2.706	2.713	2.720
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	6030	6035	6040	6045
67	Elevation (m)	-0.3600	-0.2940	-0.2280	-0.1620
68	Pressure (bar)	75.12	75.12	75.12	75.12
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2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	56.48	56.47	56.47	56.47
14	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
15	Heat Transferred (kJ/h-m)	698.8	705.3	705.2	705.1
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	1.000	0.2794	0.2794	0.2794
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	6.828e-003	7.389e-003	7.389e-003	7.389e-003
21	Static Gradient (kPa/m)	3.657	3.016e-002	3.016e-002	3.016e-002
22	Accel. Gradient (kPa/m)	3.922e-004	4.020e-006	4.020e-006	4.020e-006
23	Liq. Reynolds	1.972e+005 *	1.972e+005 *	1.972e+005 *	1.972e+005 *
24	Vap. Reynolds	9.615e+006 *	9.615e+006 *	9.615e+006 *	9.615e+006 *
25	Liquid Velocity (m/s)	0.1989	0.1988	0.1988	0.1988
26	Vapour Velocity (m/s)	2.727	2.727	2.727	2.727
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	6050	6055	6060	6065
30	Elevation (m)	-9.600e-002	-3.000e-002	-5.200e-002	-7.400e-002
31	Pressure (bar)	75.11	75.11	75.11	75.11
32	Temperature (C)	56.47	56.46	56.46	56.46
33	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
34	Heat Transferred (kJ/h-m)	705.0	705.0	705.0	704.9
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2794	0.2794	0.1757	0.1757
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	7.390e-003	7.390e-003	8.490e-003	8.490e-003
40	Static Gradient (kPa/m)	3.016e-002	3.016e-002	-7.254e-003	-7.254e-003
41	Accel. Gradient (kPa/m)	4.020e-006	4.020e-006	1.323e-007	1.323e-007
42	Liq. Reynolds	1.972e+005 *	1.972e+005 *	1.972e+005 *	1.972e+005 *
43	Vap. Reynolds	9.615e+006 *	9.615e+006 *	9.615e+006 *	9.615e+006 *
44	Liquid Velocity (m/s)	0.1988	0.1988	0.1988	0.1989
45	Vapour Velocity (m/s)	2.727	2.727	2.727	2.727
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	6070	6075	6080	6085
49	Elevation (m)	-9.600e-002	-0.1180	-0.1400	-2.232
50	Pressure (bar)	75.11	75.11	75.11	75.13
51	Temperature (C)	56.46	56.45	56.45	56.46
52	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
53	Heat Transferred (kJ/h-m)	704.9	704.8	704.8	706.4
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.1757	0.1757	0.1757	6.798e-002
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	8.490e-003	8.490e-003	8.490e-003	1.317e-002
59	Static Gradient (kPa/m)	-7.254e-003	-7.254e-003	-7.254e-003	-0.4090
60	Accel. Gradient (kPa/m)	1.323e-007	1.323e-007	1.323e-007	-4.235e-005
61	Liq. Reynolds	1.972e+005 *	1.972e+005 *	1.972e+005 *	1.972e+005 *
62	Vap. Reynolds	9.615e+006 *	9.615e+006 *	9.615e+006 *	9.614e+006 *
63	Liquid Velocity (m/s)	0.1989	0.1989	0.1989	0.1989
64	Vapour Velocity (m/s)	2.727	2.727	2.727	2.726
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	6090	6095	6101	6106
68	Elevation (m)	-4.324	-6.416	-8.508	-10.60
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2										
3							Unit Set: Mixed			
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5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Pressure (bar)	75.15	75.17	75.19	75.21					
14	Temperature (C)	56.47	56.48	56.49	56.50					
15	Vapour Fraction	0.8815	0.8815	0.8815	0.8815					
16	Heat Transferred (kJ/h-m)	706.6	706.9	707.1	707.4					
17	Flow Regime	Transition	Transition	Transition	Transition					
18	Water Flow Regime									
19	Liquid HoldUp	6.800e-002	6.802e-002	6.804e-002	6.805e-002					
20	Water HoldUp	---	---	---	---					
21	Friction Gradient (kPa/m)	1.316e-002	1.316e-002	1.315e-002	1.315e-002					
22	Static Gradient (kPa/m)	-0.4091	-0.4092	-0.4093	-0.4094					
23	Accel. Gradient (kPa/m)	-4.234e-005	-4.233e-005	-4.232e-005	-4.231e-005					
24	Liq. Reynolds	1.973e+005 *	1.973e+005 *	1.973e+005 *	1.974e+005 *					
25	Vap. Reynolds	9.614e+006 *	9.613e+006 *	9.612e+006 *	9.612e+006 *					
26	Liquid Velocity (m/s)	0.1989	0.1989	0.1989	0.1989					
27	Vapour Velocity (m/s)	2.726	2.725	2.724	2.723					
28	Deposit Thickness (mm)	---	---	---	---					
29	Deposit Volume (m3)	---	---	---	---					
30	Length (m)	6111	6116	6121	6126					
31	Elevation (m)	-8.542	-6.484	-4.426	-2.368					
32	Pressure (bar)	75.07	74.94	74.80	74.66					
33	Temperature (C)	56.46	56.43	56.39	56.35					
34	Vapour Fraction	0.8816	0.8816	0.8817	0.8817					
35	Heat Transferred (kJ/h-m)	698.5	697.5	696.5	695.5					
36	Flow Regime	Transition	Transition	Transition	Transition					
37	Water Flow Regime									
38	Liquid HoldUp	1.000	1.000	1.000	1.000					
39	Water HoldUp	---	---	---	---					
40	Friction Gradient (kPa/m)	6.832e-003	6.844e-003	6.856e-003	6.868e-003					
41	Static Gradient (kPa/m)	2.759	2.760	2.760	2.761					
42	Accel. Gradient (kPa/m)	2.965e-004	2.976e-004	2.988e-004	3.000e-004					
43	Liq. Reynolds	1.972e+005 *	1.970e+005 *	1.968e+005 *	1.966e+005 *					
44	Vap. Reynolds	9.616e+006 *	9.621e+006 *	9.625e+006 *	9.630e+006 *					
45	Liquid Velocity (m/s)	0.1988	0.1988	0.1988	0.1987					
46	Vapour Velocity (m/s)	2.729	2.734	2.739	2.745					
47	Deposit Thickness (mm)	---	---	---	---					
48	Deposit Volume (m3)	---	---	---	---					
49	Length (m)	6131	6136	6141	6146					
50	Elevation (m)	-0.3100	-0.2880	-0.2660	-0.2440					
51	Pressure (bar)	74.52	74.52	74.52	74.52					
52	Temperature (C)	56.31	56.31	56.31	56.31					
53	Vapour Fraction	0.8818	0.8818	0.8818	0.8818					
54	Heat Transferred (kJ/h-m)	694.5	700.9	700.9	700.8					
55	Flow Regime	Transition	Transition	Transition	Transition					
56	Water Flow Regime									
57	Liquid HoldUp	1.000	0.2529	0.2529	0.2529					
58	Water HoldUp	---	---	---	---					
59	Friction Gradient (kPa/m)	6.880e-003	6.346e-003	6.346e-003	6.346e-003					
60	Static Gradient (kPa/m)	2.761	9.326e-003	9.326e-003	9.326e-003					
61	Accel. Gradient (kPa/m)	3.012e-004	1.706e-006	1.706e-006	1.706e-006					
62	Liq. Reynolds	1.965e+005 *	1.965e+005 *	1.965e+005 *	1.965e+005 *					
63	Vap. Reynolds	9.634e+006 *	9.635e+006 *	9.635e+006 *	9.635e+006 *					
64	Liquid Velocity (m/s)	0.1987	0.1987	0.1987	0.1987					
65	Vapour Velocity (m/s)	2.750	2.750	2.750	2.750					
66	Deposit Thickness (mm)	---	---	---	---					
67	Deposit Volume (m3)	---	---	---	---					
68	Length (m)	6151	6156	6161	6166					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	-0.2220	-0.2000	-0.1940	-0.1880
14	Pressure (bar)	74.52	74.52	74.52	74.52
15	Temperature (C)	56.30	56.30	56.30	56.30
16	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
17	Heat Transferred (kJ/h-m)	700.8	700.7	700.6	700.6
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2529	0.2529	0.2437	0.2437
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	6.346e-003	6.346e-003	7.351e-003	7.351e-003
23	Static Gradient (kPa/m)	9.326e-003	9.326e-003	2.476e-003	2.476e-003
24	Accel. Gradient (kPa/m)	1.706e-006	1.706e-006	1.069e-006	1.069e-006
25	Liq. Reynolds	1.965e+005 *	1.965e+005 *	1.965e+005 *	1.965e+005 *
26	Vap. Reynolds	9.635e+006 *	9.635e+006 *	9.635e+006 *	9.635e+006 *
27	Liquid Velocity (m/s)	0.1987	0.1987	0.1987	0.1987
28	Vapour Velocity (m/s)	2.750	2.750	2.750	2.750
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	6171	6176	6181	6186
32	Elevation (m)	-0.1820	-0.1760	-0.1700	-0.1500
33	Pressure (bar)	74.51	74.51	74.51	74.51
34	Temperature (C)	56.30	56.29	56.29	56.29
35	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
36	Heat Transferred (kJ/h-m)	700.5	700.5	700.4	700.4
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2437	0.2437	0.2437	0.2517
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	7.351e-003	7.351e-003	7.351e-003	6.465e-003
42	Static Gradient (kPa/m)	2.476e-003	2.476e-003	2.476e-003	8.451e-003
43	Accel. Gradient (kPa/m)	1.069e-006	1.069e-006	1.069e-006	1.623e-006
44	Liq. Reynolds	1.965e+005 *	1.965e+005 *	1.965e+005 *	1.965e+005 *
45	Vap. Reynolds	9.635e+006 *	9.635e+006 *	9.635e+006 *	9.635e+006 *
46	Liquid Velocity (m/s)	0.1987	0.1987	0.1987	0.1987
47	Vapour Velocity (m/s)	2.750	2.750	2.750	2.750
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	6191	6196	6201	6206
51	Elevation (m)	-0.1300	-0.1100	-9.000e-002	-7.000e-002
52	Pressure (bar)	74.51	74.51	74.51	74.51
53	Temperature (C)	56.29	56.28	56.28	56.28
54	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
55	Heat Transferred (kJ/h-m)	700.3	700.2	700.2	700.1
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2517	0.2517	0.2517	0.2517
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	6.465e-003	6.465e-003	6.465e-003	6.465e-003
61	Static Gradient (kPa/m)	8.451e-003	8.451e-003	8.451e-003	8.451e-003
62	Accel. Gradient (kPa/m)	1.623e-006	1.623e-006	1.623e-006	1.623e-006
63	Liq. Reynolds	1.965e+005 *	1.965e+005 *	1.965e+005 *	1.965e+005 *
64	Vap. Reynolds	9.635e+006 *	9.635e+006 *	9.635e+006 *	9.635e+006 *
65	Liquid Velocity (m/s)	0.1987	0.1987	0.1987	0.1987
66	Vapour Velocity (m/s)	2.750	2.750	2.750	2.750
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
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5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	6211	6216	6221	6226
14	Elevation (m)	-6.800e-002	-6.600e-002	-6.400e-002	-6.200e-002
15	Pressure (bar)	74.51	74.51	74.51	74.51
16	Temperature (C)	56.28	56.28	56.27	56.27
17	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
18	Heat Transferred (kJ/h-m)	700.1	700.0	699.9	699.9
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.2414	0.2414	0.2414	0.2414
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	7.620e-003	7.620e-003	7.620e-003	7.620e-003
24	Static Gradient (kPa/m)	8.198e-004	8.198e-004	8.198e-004	8.198e-004
25	Accel. Gradient (kPa/m)	9.186e-007	9.186e-007	9.186e-007	9.185e-007
26	Liq. Reynolds	1.965e+005 *	1.965e+005 *	1.965e+005 *	1.965e+005 *
27	Vap. Reynolds	9.635e+006 *	9.635e+006 *	9.635e+006 *	9.635e+006 *
28	Liquid Velocity (m/s)	0.1987	0.1987	0.1987	0.1987
29	Vapour Velocity (m/s)	2.750	2.750	2.750	2.750
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	6231	6236	6241	6246
33	Elevation (m)	-6.000e-002	-8.400e-002	-0.1080	-0.1320
34	Pressure (bar)	74.51	74.51	74.51	74.51
35	Temperature (C)	56.27	56.27	56.27	56.26
36	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
37	Heat Transferred (kJ/h-m)	699.8	699.8	699.8	699.7
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.2414	0.1744	0.1744	0.1744
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	7.620e-003	8.560e-003	8.560e-003	8.560e-003
43	Static Gradient (kPa/m)	8.198e-004	-7.862e-003	-7.862e-003	-7.862e-003
44	Accel. Gradient (kPa/m)	9.185e-007	7.596e-008	7.595e-008	7.593e-008
45	Liq. Reynolds	1.965e+005 *	1.965e+005 *	1.965e+005 *	1.966e+005 *
46	Vap. Reynolds	9.635e+006 *	9.635e+006 *	9.635e+006 *	9.635e+006 *
47	Liquid Velocity (m/s)	0.1987	0.1987	0.1987	0.1987
48	Vapour Velocity (m/s)	2.750	2.750	2.750	2.750
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	6251	6256	6261	6266
52	Elevation (m)	-0.1560	-0.1800	-0.1720	-0.1640
53	Pressure (bar)	74.51	74.51	74.51	74.51
54	Temperature (C)	56.26	56.26	56.26	56.26
55	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
56	Heat Transferred (kJ/h-m)	699.7	699.6	699.5	699.5
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.1744	0.1744	0.2449	0.2449
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	8.560e-003	8.560e-003	7.218e-003	7.218e-003
62	Static Gradient (kPa/m)	-7.862e-003	-7.862e-003	3.313e-003	3.313e-003
63	Accel. Gradient (kPa/m)	7.592e-008	7.590e-008	1.146e-006	1.146e-006
64	Liq. Reynolds	1.966e+005 *	1.966e+005 *	1.966e+005 *	1.966e+005 *
65	Vap. Reynolds	9.635e+006 *	9.635e+006 *	9.635e+006 *	9.635e+006 *
66	Liquid Velocity (m/s)	0.1987	0.1987	0.1987	0.1987
67	Vapour Velocity (m/s)	2.750	2.750	2.750	2.750
68	Deposit Thickness (mm)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	6271	6276	6281	6285
15	Elevation (m)	-0.1560	-0.1480	-0.1400	-2.292
16	Pressure (bar)	74.51	74.51	74.50	74.53
17	Temperature (C)	56.25	56.25	56.25	56.26
18	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
19	Heat Transferred (kJ/h-m)	699.4	699.3	699.3	701.0
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2449	0.2449	0.2449	6.740e-002
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	7.218e-003	7.218e-003	7.218e-003	1.330e-002
25	Static Gradient (kPa/m)	3.313e-003	3.313e-003	3.313e-003	-0.4647
26	Accel. Gradient (kPa/m)	1.146e-006	1.146e-006	1.146e-006	-4.910e-005
27	Liq. Reynolds	1.966e+005 *	1.966e+005 *	1.966e+005 *	1.966e+005 *
28	Vap. Reynolds	9.635e+006 *	9.635e+006 *	9.635e+006 *	9.634e+006 *
29	Liquid Velocity (m/s)	0.1987	0.1987	0.1987	0.1987
30	Vapour Velocity (m/s)	2.750	2.750	2.750	2.749
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	6290	6294	6299	6303
34	Elevation (m)	-4.444	-6.596	-8.748	-10.90
35	Pressure (bar)	74.55	74.57	74.59	74.61
36	Temperature (C)	56.27	56.28	56.29	56.30
37	Vapour Fraction	0.8818	0.8817	0.8817	0.8817
38	Heat Transferred (kJ/h-m)	701.2	701.5	701.8	702.0
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	6.742e-002	6.744e-002	6.746e-002	6.748e-002
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	1.330e-002	1.330e-002	1.329e-002	1.329e-002
44	Static Gradient (kPa/m)	-0.4649	-0.4650	-0.4651	-0.4652
45	Accel. Gradient (kPa/m)	-4.909e-005	-4.908e-005	-4.906e-005	-4.905e-005
46	Liq. Reynolds	1.966e+005 *	1.966e+005 *	1.966e+005 *	1.966e+005 *
47	Vap. Reynolds	9.634e+006 *	9.633e+006 *	9.632e+006 *	9.632e+006 *
48	Liquid Velocity (m/s)	0.1987	0.1987	0.1987	0.1987
49	Vapour Velocity (m/s)	2.748	2.748	2.747	2.746
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	6308	6313	6318	6323
53	Elevation (m)	-10.08	-9.260	-8.440	-7.620
54	Pressure (bar)	74.57	74.53	74.49	74.45
55	Temperature (C)	56.29	56.27	56.26	56.25
56	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
57	Heat Transferred (kJ/h-m)	701.7	701.4	701.0	700.7
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.6910	0.6908	0.6906	0.6904
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	6.990e-003	6.993e-003	6.997e-003	7.000e-003
63	Static Gradient (kPa/m)	0.7889	0.7888	0.7886	0.7884
64	Accel. Gradient (kPa/m)	8.648e-005	8.656e-005	8.663e-005	8.671e-005
65	Liq. Reynolds	1.966e+005 *	1.966e+005 *	1.965e+005 *	1.965e+005 *
66	Vap. Reynolds	9.633e+006 *	9.634e+006 *	9.636e+006 *	9.637e+006 *
67	Liquid Velocity (m/s)	0.1987	0.1987	0.1987	0.1987
68	Vapour Velocity (m/s)	2.748	2.749	2.751	2.752
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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9	PROFILES				
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11	Pipe Table				
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13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	6328	6333	6338	6343
16	Elevation (m)	-6.800	-7.560	-8.320	-9.080
17	Pressure (bar)	74.41	74.42	74.42	74.43
18	Temperature (C)	56.23	56.24	56.24	56.24
19	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
20	Heat Transferred (kJ/h-m)	700.3	699.5	699.5	699.6
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.6902	6.394e-002	6.394e-002	6.394e-002
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	7.003e-003	1.380e-002	1.380e-002	1.380e-002
26	Static Gradient (kPa/m)	0.7882	-0.1457	-0.1458	-0.1458
27	Accel. Gradient (kPa/m)	8.678e-005	-1.440e-005	-1.440e-005	-1.439e-005
28	Liq. Reynolds	1.965e+005 *	1.965e+005 *	1.965e+005 *	1.965e+005 *
29	Vap. Reynolds	9.638e+006 *	9.638e+006 *	9.638e+006 *	9.638e+006 *
30	Liquid Velocity (m/s)	0.1987	0.1987	0.1987	0.1987
31	Vapour Velocity (m/s)	2.754	2.753	2.753	2.753
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	6348	6353	6358	6363
35	Elevation (m)	-9.840	-10.60	-11.94	-13.28
36	Pressure (bar)	74.43	74.44	74.45	74.47
37	Temperature (C)	56.24	56.24	56.25	56.25
38	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
39	Heat Transferred (kJ/h-m)	699.6	699.7	700.7	700.8
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	6.395e-002	6.395e-002	6.733e-002	6.734e-002
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	1.380e-002	1.379e-002	1.332e-002	1.332e-002
45	Static Gradient (kPa/m)	-0.1458	-0.1458	-0.2626	-0.2627
46	Accel. Gradient (kPa/m)	-1.439e-005	-1.439e-005	-2.717e-005	-2.717e-005
47	Liq. Reynolds	1.965e+005 *	1.965e+005 *	1.965e+005 *	1.965e+005 *
48	Vap. Reynolds	9.637e+006 *	9.637e+006 *	9.637e+006 *	9.636e+006 *
49	Liquid Velocity (m/s)	0.1987	0.1987	0.1987	0.1987
50	Vapour Velocity (m/s)	2.753	2.752	2.752	2.752
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	6368	6373	6378	6383
54	Elevation (m)	-14.62	-15.96	-17.30	-16.60
55	Pressure (bar)	74.48	74.49	74.50	74.47
56	Temperature (C)	56.26	56.27	56.27	56.26
57	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
58	Heat Transferred (kJ/h-m)	701.0	701.1	701.3	701.0
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	6.735e-002	6.736e-002	6.738e-002	0.6287
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	1.332e-002	1.331e-002	1.331e-002	7.037e-003
64	Static Gradient (kPa/m)	-0.2627	-0.2627	-0.2628	0.6200
65	Accel. Gradient (kPa/m)	-2.716e-005	-2.716e-005	-2.715e-005	6.831e-005
66	Liq. Reynolds	1.965e+005 *	1.965e+005 *	1.965e+005 *	1.965e+005 *
67	Vap. Reynolds	9.636e+006 *	9.636e+006 *	9.635e+006 *	9.636e+006 *
68	Liquid Velocity (m/s)	0.1987	0.1987	0.1987	0.1987
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
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10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.751	2.751	2.750	2.751	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	6388	6393	6398	6403	
17	Elevation	(m)	-15.90	-15.20	-14.50	-13.80	
18	Pressure	(bar)	74.44	74.41	74.38	74.35	
19	Temperature	(C)	56.25	56.24	56.23	56.22	
20	Vapour Fraction		0.8818	0.8818	0.8818	0.8818	
21	Heat Transferred	(kJ/h-m)	700.7	700.4	700.1	699.8	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.6286	0.6285	0.6283	0.6282	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	7.040e-003	7.043e-003	7.046e-003	7.048e-003	
27	Static Gradient	(kPa/m)	0.6198	0.6197	0.6196	0.6195	
28	Accel. Gradient	(kPa/m)	6.835e-005	6.840e-005	6.844e-005	6.849e-005	
29	Liq. Reynolds		1.965e+005 *	1.965e+005 *	1.965e+005 *	1.964e+005 *	
30	Vap. Reynolds		9.637e+006 *	9.638e+006 *	9.639e+006 *	9.640e+006 *	
31	Liquid Velocity	(m/s)	0.1987	0.1987	0.1986	0.1986	
32	Vapour Velocity	(m/s)	2.753	2.754	2.755	2.756	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	6408	6413	6418	6423	
36	Elevation	(m)	-11.15	-8.508	-5.862	-3.216	
37	Pressure	(bar)	74.17	73.99	73.81	73.64	
38	Temperature	(C)	56.17	56.12	56.07	56.03	
39	Vapour Fraction		0.8819	0.8820	0.8821	0.8822	
40	Heat Transferred	(kJ/h-m)	690.7	689.4	688.2	686.9	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		1.000	1.000	1.000	1.000	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	6.908e-003	6.924e-003	6.939e-003	6.955e-003	
46	Static Gradient	(kPa/m)	3.551	3.552	3.553	3.553	
47	Accel. Gradient	(kPa/m)	3.909e-004	3.929e-004	3.949e-004	3.970e-004	
48	Liq. Reynolds		1.963e+005 *	1.961e+005 *	1.959e+005 *	1.957e+005 *	
49	Vap. Reynolds		9.646e+006 *	9.652e+006 *	9.658e+006 *	9.664e+006 *	
50	Liquid Velocity	(m/s)	0.1986	0.1985	0.1985	0.1984	
51	Vapour Velocity	(m/s)	2.763	2.770	2.777	2.784	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	6428	6433	6438	6443	
55	Elevation	(m)	-0.5700	-0.4840	-0.3980	-0.3120	
56	Pressure	(bar)	73.46	73.45	73.45	73.45	
57	Temperature	(C)	55.98	55.98	55.97	55.97	
58	Vapour Fraction		0.8822	0.8822	0.8822	0.8822	
59	Heat Transferred	(kJ/h-m)	685.6	692.0	691.9	691.8	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		1.000	0.2868	0.2868	0.2868	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	6.971e-003	7.585e-003	7.585e-003	7.585e-003	
65	Static Gradient	(kPa/m)	3.554	3.998e-002	3.998e-002	3.998e-002	
66	Accel. Gradient	(kPa/m)	3.990e-004	5.330e-006	5.330e-006	5.331e-006	
67	Liq. Reynolds		1.955e+005 *	1.955e+005 *	1.955e+005 *	1.955e+005 *	
68	Vap. Reynolds		9.670e+006 *	9.670e+006 *	9.670e+006 *	9.670e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
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4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
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9	PROFILES						
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11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.1984	0.1984	0.1984	0.1984	
14	Vapour Velocity	(m/s)	2.791	2.791	2.792	2.792	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	6448	6453	6459	6465	
18	Elevation	(m)	-0.2260	-0.1400	-0.1320	-0.1240	
19	Pressure	(bar)	73.45	73.45	73.44	73.44	
20	Temperature	(C)	55.97	55.96	55.96	55.96	
21	Vapour Fraction		0.8822	0.8822	0.8822	0.8822	
22	Heat Transferred	(kJ/h-m)	691.8	691.7	691.6	691.5	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.2868	0.2868	0.2418	0.2418	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	7.585e-003	7.585e-003	7.457e-003	7.457e-003	
28	Static Gradient	(kPa/m)	3.998e-002	3.998e-002	2.730e-003	2.730e-003	
29	Accel. Gradient	(kPa/m)	5.331e-006	5.331e-006	1.142e-006	1.142e-006	
30	Liq. Reynolds		1.955e+005 *	1.955e+005 *	1.955e+005 *	1.955e+005 *	
31	Vap. Reynolds		9.670e+006 *	9.670e+006 *	9.670e+006 *	9.670e+006 *	
32	Liquid Velocity	(m/s)	0.1984	0.1984	0.1984	0.1984	
33	Vapour Velocity	(m/s)	2.792	2.792	2.792	2.792	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	6471	6477	6483	6487	
37	Elevation	(m)	-0.1160	-0.1080	-0.1000	-9.800e-002	
38	Pressure	(bar)	73.44	73.44	73.44	73.44	
39	Temperature	(C)	55.96	55.95	55.95	55.95	
40	Vapour Fraction		0.8822	0.8822	0.8822	0.8822	
41	Heat Transferred	(kJ/h-m)	691.5	691.4	691.3	691.3	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2418	0.2418	0.2418	0.2394	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	7.457e-003	7.457e-003	7.457e-003	7.742e-003	
47	Static Gradient	(kPa/m)	2.730e-003	2.730e-003	2.730e-003	1.016e-003	
48	Accel. Gradient	(kPa/m)	1.142e-006	1.142e-006	1.142e-006	9.817e-007	
49	Liq. Reynolds		1.955e+005 *	1.955e+005 *	1.955e+005 *	1.955e+005 *	
50	Vap. Reynolds		9.670e+006 *	9.670e+006 *	9.670e+006 *	9.670e+006 *	
51	Liquid Velocity	(m/s)	0.1984	0.1984	0.1984	0.1984	
52	Vapour Velocity	(m/s)	2.792	2.792	2.792	2.792	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	6491	6495	6499	6503	
56	Elevation	(m)	-9.600e-002	-9.400e-002	-9.200e-002	-9.000e-002	
57	Pressure	(bar)	73.44	73.44	73.44	73.44	
58	Temperature	(C)	55.95	55.95	55.94	55.94	
59	Vapour Fraction		0.8822	0.8822	0.8822	0.8822	
60	Heat Transferred	(kJ/h-m)	691.3	691.2	691.2	691.1	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2394	0.2394	0.2394	0.2394	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	7.742e-003	7.742e-003	7.742e-003	7.742e-003	
66	Static Gradient	(kPa/m)	1.016e-003	1.016e-003	1.016e-003	1.016e-003	
67	Accel. Gradient	(kPa/m)	9.817e-007	9.817e-007	9.817e-007	9.817e-007	
68	Liq. Reynolds		1.955e+005 *	1.955e+005 *	1.955e+005 *	1.955e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
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5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.670e+006 *	9.670e+006 *	9.670e+006 *	9.670e+006 *
14	Liquid Velocity (m/s)	0.1984	0.1984	0.1984	0.1984
15	Vapour Velocity (m/s)	2.792	2.792	2.792	2.792
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	6508	6513	6518	6523
19	Elevation (m)	-6.400e-002	-3.800e-002	-1.200e-002	1.400e-002
20	Pressure (bar)	73.44	73.44	73.44	73.44
21	Temperature (C)	55.94	55.94	55.94	55.93
22	Vapour Fraction	0.8822	0.8822	0.8822	0.8822
23	Heat Transferred (kJ/h-m)	691.1	691.0	690.9	690.9
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2528	0.2528	0.2528	0.2528
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.238e-003	6.238e-003	6.238e-003	6.238e-003
29	Static Gradient (kPa/m)	1.100e-002	1.100e-002	1.100e-002	1.100e-002
30	Accel. Gradient (kPa/m)	1.932e-006	1.932e-006	1.932e-006	1.932e-006
31	Liq. Reynolds	1.955e+005 *	1.955e+005 *	1.955e+005 *	1.955e+005 *
32	Vap. Reynolds	9.670e+006 *	9.670e+006 *	9.670e+006 *	9.670e+006 *
33	Liquid Velocity (m/s)	0.1984	0.1984	0.1984	0.1984
34	Vapour Velocity (m/s)	2.792	2.792	2.792	2.792
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	6528	6533	6538	6543
38	Elevation (m)	4.000e-002	3.800e-002	3.600e-002	3.400e-002
39	Pressure (bar)	73.44	73.44	73.44	73.44
40	Temperature (C)	55.93	55.93	55.93	55.93
41	Vapour Fraction	0.8822	0.8822	0.8822	0.8822
42	Heat Transferred (kJ/h-m)	690.8	690.8	690.8	690.7
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2528	0.1761	0.1761	0.1761
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.239e-003	8.638e-003	8.638e-003	8.638e-003
48	Static Gradient (kPa/m)	1.100e-002	-6.588e-004	-6.588e-004	-6.588e-004
49	Accel. Gradient (kPa/m)	1.932e-006	8.945e-007	8.945e-007	8.945e-007
50	Liq. Reynolds	1.955e+005 *	1.955e+005 *	1.955e+005 *	1.955e+005 *
51	Vap. Reynolds	9.670e+006 *	9.670e+006 *	9.670e+006 *	9.671e+006 *
52	Liquid Velocity (m/s)	0.1984	0.1984	0.1984	0.1984
53	Vapour Velocity (m/s)	2.792	2.792	2.792	2.792
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	6548	6553	6558	6563
57	Elevation (m)	3.200e-002	3.000e-002	2.800e-002	2.600e-002
58	Pressure (bar)	73.43	73.43	73.43	73.43
59	Temperature (C)	55.92	55.92	55.92	55.92
60	Vapour Fraction	0.8822	0.8822	0.8822	0.8822
61	Heat Transferred (kJ/h-m)	690.6	690.6	690.5	690.5
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1761	0.1761	0.1761	0.1761
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	8.638e-003	8.638e-003	8.638e-003	8.638e-003
67	Static Gradient (kPa/m)	-6.588e-004	-6.588e-004	-6.573e-004	-6.573e-004
68	Accel. Gradient (kPa/m)	8.945e-007	8.945e-007	8.947e-007	8.947e-007
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		1.955e+005 *	1.955e+005 *	1.955e+005 *	1.955e+005 *	
14	Vap. Reynolds		9.671e+006 *	9.671e+006 *	9.671e+006 *	9.671e+006 *	
15	Liquid Velocity (m/s)	0.1984	0.1984	0.1984	0.1984		
16	Vapour Velocity (m/s)	2.792	2.792	2.792	2.792		
17	Deposit Thickness (mm)	---	---	---	---		
18	Deposit Volume (m3)	---	---	---	---		
19	Length (m)	6568	6573	6578	6583		
20	Elevation (m)	2.400e-002	2.200e-002	2.000e-002	3.200e-002		
21	Pressure (bar)	73.43	73.43	73.43	73.43		
22	Temperature (C)	55.91	55.91	55.91	55.91		
23	Vapour Fraction	0.8822	0.8822	0.8822	0.8822		
24	Heat Transferred (kJ/h-m)	690.4	690.4	690.3	690.2		
25	Flow Regime	Transition	Transition	Transition	Transition		
26	Water Flow Regime						
27	Liquid HoldUp	0.1761	0.1761	0.1761	0.2448		
28	Water HoldUp	---	---	---	---		
29	Friction Gradient (kPa/m)	8.638e-003	8.638e-003	8.638e-003	7.104e-003		
30	Static Gradient (kPa/m)	-6.573e-004	-6.573e-004	-6.573e-004	4.959e-003		
31	Accel. Gradient (kPa/m)	8.946e-007	8.946e-007	8.946e-007	1.352e-006		
32	Liq. Reynolds		1.955e+005 *	1.955e+005 *	1.955e+005 *		
33	Vap. Reynolds		9.671e+006 *	9.671e+006 *	9.671e+006 *		
34	Liquid Velocity (m/s)	0.1984	0.1984	0.1984	0.1984		
35	Vapour Velocity (m/s)	2.792	2.792	2.792	2.792		
36	Deposit Thickness (mm)	---	---	---	---		
37	Deposit Volume (m3)	---	---	---	---		
38	Length (m)	6588	6593	6598	6603		
39	Elevation (m)	4.400e-002	5.600e-002	6.800e-002	8.000e-002		
40	Pressure (bar)	73.43	73.43	73.43	73.43		
41	Temperature (C)	55.91	55.90	55.90	55.90		
42	Vapour Fraction	0.8822	0.8822	0.8822	0.8822		
43	Heat Transferred (kJ/h-m)	690.2	690.1	690.0	690.0		
44	Flow Regime	Transition	Transition	Transition	Transition		
45	Water Flow Regime						
46	Liquid HoldUp	0.2448	0.2448	0.2448	0.2448		
47	Water HoldUp	---	---	---	---		
48	Friction Gradient (kPa/m)	7.104e-003	7.104e-003	7.104e-003	7.104e-003		
49	Static Gradient (kPa/m)	4.959e-003	4.959e-003	4.959e-003	4.959e-003		
50	Accel. Gradient (kPa/m)	1.352e-006	1.352e-006	1.352e-006	1.352e-006		
51	Liq. Reynolds		1.955e+005 *	1.955e+005 *	1.955e+005 *		
52	Vap. Reynolds		9.671e+006 *	9.671e+006 *	9.671e+006 *		
53	Liquid Velocity (m/s)	0.1984	0.1984	0.1984	0.1984		
54	Vapour Velocity (m/s)	2.792	2.792	2.792	2.792		
55	Deposit Thickness (mm)	---	---	---	---		
56	Deposit Volume (m3)	---	---	---	---		
57	Length (m)	6608	6613	6618	6623		
58	Elevation (m)	7.200e-002	6.400e-002	5.600e-002	4.800e-002		
59	Pressure (bar)	73.43	73.43	73.43	73.43		
60	Temperature (C)	55.90	55.90	55.89	55.89		
61	Vapour Fraction	0.8822	0.8822	0.8822	0.8822		
62	Heat Transferred (kJ/h-m)	690.0	689.9	689.9	689.8		
63	Flow Regime	Transition	Transition	Transition	Transition		
64	Water Flow Regime						
65	Liquid HoldUp	0.1752	0.1752	0.1752	0.1752		
66	Water HoldUp	---	---	---	---		
67	Friction Gradient (kPa/m)	8.649e-003	8.649e-003	8.649e-003	8.649e-003		
68	Static Gradient (kPa/m)	-2.620e-003	-2.620e-003	-2.620e-003	-2.620e-003		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Accel. Gradient	(kPa/m)	6.759e-007	6.759e-007	6.759e-007	6.759e-007				
14	Liq. Reynolds		1.955e+005 *	1.955e+005 *	1.955e+005 *	1.955e+005 *				
15	Vap. Reynolds		9.671e+006 *	9.671e+006 *	9.671e+006 *	9.671e+006 *				
16	Liquid Velocity	(m/s)	0.1984	0.1984	0.1984	0.1984				
17	Vapour Velocity	(m/s)	2.792	2.791	2.791	2.791				
18	Deposit Thickness	(mm)	---	---	---	---				
19	Deposit Volume	(m3)	---	---	---	---				
20	Length	(m)	6628	6633	6638	6643				
21	Elevation	(m)	4.000e-002	2.800e-002	1.600e-002	4.000e-003				
22	Pressure	(bar)	73.43	73.43	73.43	73.43				
23	Temperature	(C)	55.89	55.89	55.89	55.88				
24	Vapour Fraction		0.8822	0.8822	0.8822	0.8822				
25	Heat Transferred	(kJ/h-m)	689.8	689.7	689.7	689.6				
26	Flow Regime		Transition	Transition	Transition	Transition				
27	Water Flow Regime									
28	Liquid HoldUp		0.1752	0.1746	0.1746	0.1746				
29	Water HoldUp		---	---	---	---				
30	Friction Gradient	(kPa/m)	8.649e-003	8.657e-003	8.657e-003	8.657e-003				
31	Static Gradient	(kPa/m)	-2.620e-003	-3.921e-003	-3.921e-003	-3.921e-003				
32	Accel. Gradient	(kPa/m)	6.759e-007	5.309e-007	5.309e-007	5.308e-007				
33	Liq. Reynolds		1.955e+005 *	1.955e+005 *	1.955e+005 *	1.955e+005 *				
34	Vap. Reynolds		9.671e+006 *	9.671e+006 *	9.671e+006 *	9.671e+006 *				
35	Liquid Velocity	(m/s)	0.1984	0.1984	0.1984	0.1984				
36	Vapour Velocity	(m/s)	2.791	2.791	2.791	2.791				
37	Deposit Thickness	(mm)	---	---	---	---				
38	Deposit Volume	(m3)	---	---	---	---				
39	Length	(m)	6648	6653	6658	6663				
40	Elevation	(m)	-8.000e-003	-2.000e-002	-2.000e-003	1.600e-002				
41	Pressure	(bar)	73.43	73.43	73.43	73.43				
42	Temperature	(C)	55.88	55.88	55.88	55.88				
43	Vapour Fraction		0.8822	0.8822	0.8822	0.8822				
44	Heat Transferred	(kJ/h-m)	689.6	689.5	689.4	689.3				
45	Flow Regime		Transition	Transition	Transition	Transition				
46	Water Flow Regime									
47	Liquid HoldUp		0.1746	0.1746	0.2482	0.2482				
48	Water HoldUp		---	---	---	---				
49	Friction Gradient	(kPa/m)	8.657e-003	8.657e-003	6.722e-003	6.722e-003				
50	Static Gradient	(kPa/m)	-3.921e-003	-3.921e-003	7.514e-003	7.514e-003				
51	Accel. Gradient	(kPa/m)	5.308e-007	5.308e-007	1.596e-006	1.596e-006				
52	Liq. Reynolds		1.955e+005 *	1.955e+005 *	1.955e+005 *	1.955e+005 *				
53	Vap. Reynolds		9.671e+006 *	9.671e+006 *	9.671e+006 *	9.671e+006 *				
54	Liquid Velocity	(m/s)	0.1984	0.1984	0.1984	0.1984				
55	Vapour Velocity	(m/s)	2.791	2.791	2.791	2.791				
56	Deposit Thickness	(mm)	---	---	---	---				
57	Deposit Volume	(m3)	---	---	---	---				
58	Length	(m)	6668	6673	6678	6683				
59	Elevation	(m)	3.400e-002	5.200e-002	7.000e-002	4.000e-002				
60	Pressure	(bar)	73.42	73.42	73.42	73.42				
61	Temperature	(C)	55.87	55.87	55.87	55.87				
62	Vapour Fraction		0.8822	0.8822	0.8822	0.8822				
63	Heat Transferred	(kJ/h-m)	689.3	689.2	689.1	689.2				
64	Flow Regime		Transition	Transition	Transition	Transition				
65	Water Flow Regime									
66	Liquid HoldUp		0.2482	0.2482	0.2482	0.1719				
67	Water HoldUp		---	---	---	---				
68	Friction Gradient	(kPa/m)	6.722e-003	6.722e-003	6.722e-003	8.692e-003				
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 138 of 937				

1			Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	7.514e-003	7.514e-003	7.514e-003	-9.701e-003
14	Accel. Gradient	(kPa/m)	1.596e-006	1.596e-006	1.596e-006	-1.131e-007
15	Liq. Reynolds		1.955e+005 *	1.955e+005 *	1.955e+005 *	1.955e+005 *
16	Vap. Reynolds		9.671e+006 *	9.671e+006 *	9.671e+006 *	9.671e+006 *
17	Liquid Velocity	(m/s)	0.1984	0.1984	0.1984	0.1984
18	Vapour Velocity	(m/s)	2.791	2.791	2.791	2.791
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	6688	6693	6698	6703
22	Elevation	(m)	1.000e-002	-2.000e-002	-5.000e-002	-8.000e-002
23	Pressure	(bar)	73.42	73.42	73.42	73.42
24	Temperature	(C)	55.87	55.86	55.86	55.86
25	Vapour Fraction		0.8822	0.8822	0.8822	0.8822
26	Heat Transferred	(kJ/h-m)	689.1	689.1	689.0	689.0
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.1719	0.1719	0.1719	0.1719
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	8.692e-003	8.692e-003	8.691e-003	8.691e-003
32	Static Gradient	(kPa/m)	-9.701e-003	-9.701e-003	-9.701e-003	-9.701e-003
33	Accel. Gradient	(kPa/m)	-1.132e-007	-1.132e-007	-1.132e-007	-1.132e-007
34	Liq. Reynolds		1.955e+005 *	1.955e+005 *	1.955e+005 *	1.955e+005 *
35	Vap. Reynolds		9.671e+006 *	9.671e+006 *	9.671e+006 *	9.671e+006 *
36	Liquid Velocity	(m/s)	0.1984	0.1984	0.1984	0.1984
37	Vapour Velocity	(m/s)	2.791	2.791	2.791	2.791
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	6708	6713	6718	6723
41	Elevation	(m)	-0.1140	-0.1480	-0.1820	-0.2160
42	Pressure	(bar)	73.42	73.42	73.42	73.42
43	Temperature	(C)	55.86	55.86	55.85	55.85
44	Vapour Fraction		0.8822	0.8822	0.8822	0.8822
45	Heat Transferred	(kJ/h-m)	688.9	688.9	688.8	688.8
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.1713	0.1713	0.1713	0.1713
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	8.699e-003	8.699e-003	8.699e-003	8.699e-003
51	Static Gradient	(kPa/m)	-1.097e-002	-1.097e-002	-1.097e-002	-1.097e-002
52	Accel. Gradient	(kPa/m)	-2.544e-007	-2.545e-007	-2.545e-007	-2.545e-007
53	Liq. Reynolds		1.955e+005 *	1.955e+005 *	1.955e+005 *	1.955e+005 *
54	Vap. Reynolds		9.671e+006 *	9.671e+006 *	9.671e+006 *	9.671e+006 *
55	Liquid Velocity	(m/s)	0.1984	0.1984	0.1984	0.1984
56	Vapour Velocity	(m/s)	2.791	2.791	2.791	2.791
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	6728	6733	6738	6743
60	Elevation	(m)	-0.2500	-2.800	-5.350	-7.900
61	Pressure	(bar)	73.42	73.45	73.47	73.50
62	Temperature	(C)	55.85	55.86	55.87	55.89
63	Vapour Fraction		0.8822	0.8822	0.8822	0.8822
64	Heat Transferred	(kJ/h-m)	688.7	690.4	690.7	691.0
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.1713	6.639e-002	6.641e-002	6.643e-002
68	Water HoldUp		---	---	---	---
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1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	8.699e-003	1.355e-002	1.355e-002	1.354e-002
14	Static Gradient	(kPa/m)	-1.097e-002	-0.4883	-0.4885	-0.4886
15	Accel. Gradient	(kPa/m)	-2.545e-007	-5.319e-005	-5.317e-005	-5.315e-005
16	Liq. Reynolds		1.955e+005 *	1.955e+005 *	1.956e+005 *	1.956e+005 *
17	Vap. Reynolds		9.671e+006 *	9.670e+006 *	9.669e+006 *	9.669e+006 *
18	Liquid Velocity	(m/s)	0.1984	0.1984	0.1984	0.1984
19	Vapour Velocity	(m/s)	2.791	2.790	2.789	2.788
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	6748	6754	6759	6764
23	Elevation	(m)	-10.45	-13.00	-12.02	-11.04
24	Pressure	(bar)	73.52	73.54	73.49	73.44
25	Temperature	(C)	55.90	55.91	55.89	55.88
26	Vapour Fraction		0.8822	0.8821	0.8822	0.8822
27	Heat Transferred	(kJ/h-m)	691.3	691.7	691.3	690.8
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		6.645e-002	6.647e-002	0.7641	0.7639
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	1.354e-002	1.353e-002	7.043e-003	7.048e-003
33	Static Gradient	(kPa/m)	-0.4888	-0.4889	1.032	1.031
34	Accel. Gradient	(kPa/m)	-5.313e-005	-5.312e-005	1.162e-004	1.164e-004
35	Liq. Reynolds		1.956e+005 *	1.956e+005 *	1.956e+005 *	1.955e+005 *
36	Vap. Reynolds		9.668e+006 *	9.667e+006 *	9.669e+006 *	9.670e+006 *
37	Liquid Velocity	(m/s)	0.1984	0.1984	0.1984	0.1984
38	Vapour Velocity	(m/s)	2.788	2.787	2.789	2.791
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	6769	6774	6779	6784
42	Elevation	(m)	-10.06	-9.080	-8.100	-7.740
43	Pressure	(bar)	73.39	73.34	73.28	73.27
44	Temperature	(C)	55.86	55.85	55.83	55.82
45	Vapour Fraction		0.8822	0.8822	0.8823	0.8823
46	Heat Transferred	(kJ/h-m)	690.4	690.0	689.5	687.6
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.7636	0.7633	0.7630	0.4402
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	7.052e-003	7.057e-003	7.062e-003	7.334e-003
52	Static Gradient	(kPa/m)	1.031	1.031	1.030	0.2354
53	Accel. Gradient	(kPa/m)	1.165e-004	1.166e-004	1.168e-004	2.732e-005
54	Liq. Reynolds		1.955e+005 *	1.954e+005 *	1.953e+005 *	1.953e+005 *
55	Vap. Reynolds		9.672e+006 *	9.674e+006 *	9.676e+006 *	9.676e+006 *
56	Liquid Velocity	(m/s)	0.1984	0.1984	0.1984	0.1983
57	Vapour Velocity	(m/s)	2.793	2.795	2.797	2.797
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	6789	6794	6799	6804
61	Elevation	(m)	-7.380	-7.020	-6.660	-6.300
62	Pressure	(bar)	73.26	73.25	73.24	73.22
63	Temperature	(C)	55.82	55.81	55.81	55.80
64	Vapour Fraction		0.8823	0.8823	0.8823	0.8823
65	Heat Transferred	(kJ/h-m)	687.4	687.2	687.1	686.9
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.4401	0.4401	0.4400	0.4400
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	7.335e-003	7.336e-003	7.337e-003	7.338e-003
15	Static Gradient (kPa/m)	0.2353	0.2353	0.2353	0.2353
16	Accel. Gradient (kPa/m)	2.733e-005	2.734e-005	2.734e-005	2.735e-005
17	Liq. Reynolds	1.953e+005 *	1.953e+005 *	1.953e+005 *	1.953e+005 *
18	Vap. Reynolds	9.676e+006 *	9.677e+006 *	9.677e+006 *	9.678e+006 *
19	Liquid Velocity (m/s)	0.1983	0.1983	0.1983	0.1983
20	Vapour Velocity (m/s)	2.798	2.798	2.799	2.799
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	6809	6814	6819	6824
24	Elevation (m)	-5.090	-3.880	-2.670	-1.460
25	Pressure (bar)	73.15	73.08	73.01	72.94
26	Temperature (C)	55.78	55.76	55.74	55.72
27	Vapour Fraction	0.8823	0.8823	0.8824	0.8824
28	Heat Transferred (kJ/h-m)	688.2	687.7	687.1	686.5
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.8668	0.8664	0.8659	0.8654
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	7.032e-003	7.039e-003	7.045e-003	7.052e-003
34	Static Gradient (kPa/m)	1.427	1.427	1.426	1.426
35	Accel. Gradient (kPa/m)	1.620e-004	1.623e-004	1.625e-004	1.628e-004
36	Liq. Reynolds	1.952e+005 *	1.951e+005 *	1.950e+005 *	1.949e+005 *
37	Vap. Reynolds	9.680e+006 *	9.682e+006 *	9.685e+006 *	9.687e+006 *
38	Liquid Velocity (m/s)	0.1983	0.1983	0.1983	0.1983
39	Vapour Velocity (m/s)	2.802	2.805	2.808	2.811
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	6829	6834	6839	6844
43	Elevation (m)	-0.2500	-3.760	-7.270	-10.78
44	Pressure (bar)	72.87	72.90	72.93	72.96
45	Temperature (C)	55.70	55.71	55.73	55.75
46	Vapour Fraction	0.8824	0.8824	0.8824	0.8824
47	Heat Transferred (kJ/h-m)	686.0	686.4	686.9	687.3
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.8650	6.585e-002	6.588e-002	6.591e-002
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	7.058e-003	1.369e-002	1.368e-002	1.367e-002
53	Static Gradient (kPa/m)	1.425	-0.6742	-0.6745	-0.6748
54	Accel. Gradient (kPa/m)	1.631e-004	-7.514e-005	-7.511e-005	-7.507e-005
55	Liq. Reynolds	1.948e+005 *	1.949e+005 *	1.949e+005 *	1.950e+005 *
56	Vap. Reynolds	9.690e+006 *	9.688e+006 *	9.687e+006 *	9.686e+006 *
57	Liquid Velocity (m/s)	0.1982	0.1982	0.1983	0.1983
58	Vapour Velocity (m/s)	2.814	2.812	2.811	2.810
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	6849	6854	6859	6864
62	Elevation (m)	-14.29	-17.80	-14.31	-10.82
63	Pressure (bar)	73.00	73.03	72.80	72.56
64	Temperature (C)	55.76	55.78	55.72	55.66
65	Vapour Fraction	0.8824	0.8824	0.8825	0.8826
66	Heat Transferred (kJ/h-m)	687.8	688.2	678.8	677.1
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 141 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	6.594e-002	6.597e-002	1.000	1.000
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	1.367e-002	1.366e-002	7.027e-003	7.049e-003
16	Static Gradient (kPa/m)	-0.6750	-0.6753	4.691	4.693
17	Accel. Gradient (kPa/m)	-7.504e-005	-7.500e-005	5.362e-004	5.399e-004
18	Liq. Reynolds	1.950e+005 *	1.951e+005 *	1.948e+005 *	1.945e+005 *
19	Vap. Reynolds	9.685e+006 *	9.684e+006 *	9.692e+006 *	9.700e+006 *
20	Liquid Velocity (m/s)	0.1983	0.1983	0.1982	0.1981
21	Vapour Velocity (m/s)	2.809	2.807	2.817	2.827
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	6869	6874	6879	6884
25	Elevation (m)	-7.330	-3.840	-0.3500	-0.2820
26	Pressure (bar)	72.33	72.09	71.86	71.85
27	Temperature (C)	55.59	55.53	55.47	55.47
28	Vapour Fraction	0.8827	0.8828	0.8829	0.8829
29	Heat Transferred (kJ/h-m)	675.5	673.8	672.1	678.4
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	1.000	1.000	1.000	0.2727
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	7.070e-003	7.092e-003	7.114e-003	7.703e-003
35	Static Gradient (kPa/m)	4.694	4.695	4.696	3.036e-002
36	Accel. Gradient (kPa/m)	5.436e-004	5.474e-004	5.513e-004	4.461e-006
37	Liq. Reynolds	1.942e+005 *	1.939e+005 *	1.937e+005 *	1.937e+005 *
38	Vap. Reynolds	9.707e+006 *	9.715e+006 *	9.723e+006 *	9.723e+006 *
39	Liquid Velocity (m/s)	0.1981	0.1980	0.1979	0.1979
40	Vapour Velocity (m/s)	2.836	2.846	2.856	2.856
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	6889	6894	6899	6904
44	Elevation (m)	-0.2140	-0.1460	-7.800e-002	-1.000e-002
45	Pressure (bar)	71.85	71.85	71.85	71.85
46	Temperature (C)	55.46	55.46	55.46	55.45
47	Vapour Fraction	0.8829	0.8829	0.8829	0.8829
48	Heat Transferred (kJ/h-m)	678.3	678.3	678.2	678.1
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2727	0.2727	0.2727	0.2727
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	7.704e-003	7.704e-003	7.704e-003	7.704e-003
54	Static Gradient (kPa/m)	3.036e-002	3.036e-002	3.036e-002	3.036e-002
55	Accel. Gradient (kPa/m)	4.461e-006	4.462e-006	4.462e-006	4.462e-006
56	Liq. Reynolds	1.937e+005 *	1.936e+005 *	1.936e+005 *	1.936e+005 *
57	Vap. Reynolds	9.723e+006 *	9.723e+006 *	9.723e+006 *	9.723e+006 *
58	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
59	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	6909	6914	6919	6924
63	Elevation (m)	-1.600e-002	-2.200e-002	-2.800e-002	-3.400e-002
64	Pressure (bar)	71.85	71.84	71.84	71.84
65	Temperature (C)	55.45	55.45	55.45	55.45
66	Vapour Fraction	0.8829	0.8829	0.8829	0.8829
67	Heat Transferred (kJ/h-m)	678.1	678.1	678.0	678.0
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 142 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1730	0.1730	0.1730	0.1730
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	8.833e-003	8.833e-003	8.833e-003	8.833e-003
17	Static Gradient (kPa/m)	-1.939e-003	-1.939e-003	-1.939e-003	-1.939e-003
18	Accel. Gradient (kPa/m)	8.082e-007	8.082e-007	8.082e-007	8.082e-007
19	Liq. Reynolds	1.936e+005 *	1.936e+005 *	1.936e+005 *	1.936e+005 *
20	Vap. Reynolds	9.723e+006 *	9.723e+006 *	9.723e+006 *	9.723e+006 *
21	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
22	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	6929	6934	6939	6944
26	Elevation (m)	-4.000e-002	-4.800e-002	-5.600e-002	-6.400e-002
27	Pressure (bar)	71.84	71.84	71.84	71.84
28	Temperature (C)	55.44	55.44	55.44	55.44
29	Vapour Fraction	0.8829	0.8829	0.8829	0.8829
30	Heat Transferred (kJ/h-m)	677.9	677.8	677.8	677.7
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1730	0.1727	0.1727	0.1727
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	8.833e-003	8.837e-003	8.837e-003	8.836e-003
36	Static Gradient (kPa/m)	-1.939e-003	-2.575e-003	-2.575e-003	-2.575e-003
37	Accel. Gradient (kPa/m)	8.082e-007	7.341e-007	7.341e-007	7.341e-007
38	Liq. Reynolds	1.936e+005 *	1.936e+005 *	1.936e+005 *	1.936e+005 *
39	Vap. Reynolds	9.723e+006 *	9.724e+006 *	9.724e+006 *	9.724e+006 *
40	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
41	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	6949	6954	6959	6964
45	Elevation (m)	-7.200e-002	-8.000e-002	-7.200e-002	-6.400e-002
46	Pressure (bar)	71.84	71.84	71.84	71.84
47	Temperature (C)	55.44	55.43	55.43	55.43
48	Vapour Fraction	0.8829	0.8829	0.8829	0.8829
49	Heat Transferred (kJ/h-m)	677.7	677.6	677.5	677.5
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.1727	0.1727	0.2391	0.2391
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	8.836e-003	8.836e-003	7.601e-003	7.601e-003
55	Static Gradient (kPa/m)	-2.575e-003	-2.575e-003	3.239e-003	3.239e-003
56	Accel. Gradient (kPa/m)	7.341e-007	7.341e-007	1.271e-006	1.271e-006
57	Liq. Reynolds	1.936e+005 *	1.936e+005 *	1.936e+005 *	1.936e+005 *
58	Vap. Reynolds	9.724e+006 *	9.724e+006 *	9.724e+006 *	9.724e+006 *
59	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
60	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	6969	6974	6979	6984
64	Elevation (m)	-5.600e-002	-4.800e-002	-4.000e-002	-3.200e-002
65	Pressure (bar)	71.84	71.84	71.84	71.84
66	Temperature (C)	55.43	55.43	55.42	55.42
67	Vapour Fraction	0.8829	0.8829	0.8829	0.8829
68	Heat Transferred (kJ/h-m)	677.4	677.4	677.3	677.2
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2391	0.2391	0.2391	0.2391
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	7.601e-003	7.600e-003	7.600e-003	7.600e-003
18	Static Gradient (kPa/m)	3.239e-003	3.239e-003	3.239e-003	3.239e-003
19	Accel. Gradient (kPa/m)	1.271e-006	1.271e-006	1.271e-006	1.271e-006
20	Liq. Reynolds	1.936e+005 *	1.936e+005 *	1.936e+005 *	1.936e+005 *
21	Vap. Reynolds	9.724e+006 *	9.724e+006 *	9.724e+006 *	9.724e+006 *
22	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
23	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	6989	6994	6999	7004
27	Elevation (m)	-2.400e-002	-1.600e-002	-8.000e-003	-1.402e-015
28	Pressure (bar)	71.84	71.84	71.84	71.84
29	Temperature (C)	55.42	55.42	55.42	55.41
30	Vapour Fraction	0.8829	0.8829	0.8829	0.8829
31	Heat Transferred (kJ/h-m)	677.2	677.1	677.1	677.0
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2391	0.2391	0.2391	0.2391
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	7.600e-003	7.600e-003	7.600e-003	7.600e-003
37	Static Gradient (kPa/m)	3.239e-003	3.239e-003	3.239e-003	3.239e-003
38	Accel. Gradient (kPa/m)	1.271e-006	1.271e-006	1.271e-006	1.271e-006
39	Liq. Reynolds	1.936e+005 *	1.936e+005 *	1.936e+005 *	1.936e+005 *
40	Vap. Reynolds	9.724e+006 *	9.724e+006 *	9.724e+006 *	9.724e+006 *
41	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
42	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	7009	7014	7019	7024
46	Elevation (m)	1.000e-002	2.000e-002	3.000e-002	4.000e-002
47	Pressure (bar)	71.84	71.84	71.84	71.83
48	Temperature (C)	55.41	55.41	55.41	55.41
49	Vapour Fraction	0.8829	0.8829	0.8829	0.8829
50	Heat Transferred (kJ/h-m)	677.0	676.9	676.8	676.8
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2402	0.2402	0.2402	0.2402
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	7.464e-003	7.464e-003	7.464e-003	7.464e-003
56	Static Gradient (kPa/m)	4.062e-003	4.062e-003	4.062e-003	4.062e-003
57	Accel. Gradient (kPa/m)	1.351e-006	1.351e-006	1.351e-006	1.351e-006
58	Liq. Reynolds	1.936e+005 *	1.936e+005 *	1.936e+005 *	1.936e+005 *
59	Vap. Reynolds	9.724e+006 *	9.724e+006 *	9.724e+006 *	9.724e+006 *
60	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
61	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	7029	7034	7039	7044
65	Elevation (m)	5.000e-002	1.600e-002	-1.800e-002	-5.200e-002
66	Pressure (bar)	71.83	71.83	71.83	71.83
67	Temperature (C)	55.40	55.40	55.40	55.40
68	Vapour Fraction	0.8829	0.8829	0.8829	0.8829
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:13:43 2022		
6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Heat Transferred (kJ/h-m)	676.7	676.7	676.7	676.7		
14	Flow Regime	Transition	Transition	Transition	Transition		
15	Water Flow Regime						
16	Liquid HoldUp	0.2402	0.1688	0.1688	0.1688		
17	Water HoldUp	---	---	---	---		
18	Friction Gradient (kPa/m)	7.464e-003	8.888e-003	8.888e-003	8.888e-003		
19	Static Gradient (kPa/m)	4.062e-003	-1.081e-002	-1.081e-002	-1.081e-002		
20	Accel. Gradient (kPa/m)	1.351e-006	-2.252e-007	-2.252e-007	-2.252e-007		
21	Liq. Reynolds	1.936e+005 *	1.936e+005 *	1.936e+005 *	1.936e+005 *		
22	Vap. Reynolds	9.724e+006 *	9.724e+006 *	9.724e+006 *	9.724e+006 *		
23	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
24	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856		
25	Deposit Thickness (mm)	---	---	---	---		
26	Deposit Volume (m3)	---	---	---	---		
27	Length (m)	7049	7054	7059	7064		
28	Elevation (m)	-8.600e-002	-0.1200	-8.600e-002	-5.200e-002		
29	Pressure (bar)	71.83	71.83	71.83	71.83		
30	Temperature (C)	55.40	55.39	55.39	55.39		
31	Vapour Fraction	0.8829	0.8829	0.8829	0.8829		
32	Heat Transferred (kJ/h-m)	676.6	676.6	676.4	676.4		
33	Flow Regime	Transition	Transition	Transition	Transition		
34	Water Flow Regime						
35	Liquid HoldUp	0.1688	0.1688	0.2537	0.2537		
36	Water HoldUp	---	---	---	---		
37	Friction Gradient (kPa/m)	8.888e-003	8.888e-003	5.964e-003	5.964e-003		
38	Static Gradient (kPa/m)	-1.081e-002	-1.081e-002	1.438e-002	1.438e-002		
39	Accel. Gradient (kPa/m)	-2.253e-007	-2.253e-007	2.385e-006	2.385e-006		
40	Liq. Reynolds	1.936e+005 *	1.936e+005 *	1.936e+005 *	1.936e+005 *		
41	Vap. Reynolds	9.724e+006 *	9.724e+006 *	9.724e+006 *	9.724e+006 *		
42	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
43	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856		
44	Deposit Thickness (mm)	---	---	---	---		
45	Deposit Volume (m3)	---	---	---	---		
46	Length (m)	7069	7074	7079	7081		
47	Elevation (m)	-1.800e-002	1.600e-002	5.000e-002	4.600e-002		
48	Pressure (bar)	71.83	71.83	71.83	71.83		
49	Temperature (C)	55.39	55.38	55.38	55.38		
50	Vapour Fraction	0.8829	0.8829	0.8829	0.8829		
51	Heat Transferred (kJ/h-m)	676.3	676.2	676.2	676.2		
52	Flow Regime	Transition	Transition	Transition	Transition		
53	Water Flow Regime						
54	Liquid HoldUp	0.2537	0.2537	0.2537	0.1724		
55	Water HoldUp	---	---	---	---		
56	Friction Gradient (kPa/m)	5.964e-003	5.964e-003	5.964e-003	8.840e-003		
57	Static Gradient (kPa/m)	1.438e-002	1.438e-002	1.438e-002	-3.224e-003		
58	Accel. Gradient (kPa/m)	2.385e-006	2.385e-006	2.385e-006	6.585e-007		
59	Liq. Reynolds	1.936e+005 *	1.936e+005 *	1.936e+005 *	1.936e+005 *		
60	Vap. Reynolds	9.724e+006 *	9.724e+006 *	9.724e+006 *	9.724e+006 *		
61	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
62	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856		
63	Deposit Thickness (mm)	---	---	---	---		
64	Deposit Volume (m3)	---	---	---	---		
65	Length (m)	7083	7085	7087	7089		
66	Elevation (m)	4.200e-002	3.800e-002	3.400e-002	3.000e-002		
67	Pressure (bar)	71.83	71.83	71.83	71.83		
68	Temperature (C)	55.38	55.38	55.38	55.38		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8829	0.8829	0.8829	0.8829
14	Heat Transferred (kJ/h-m)	676.2	676.2	676.2	676.1
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.1724	0.1724	0.1724	0.1724
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	8.840e-003	8.840e-003	8.840e-003	8.840e-003
20	Static Gradient (kPa/m)	-3.224e-003	-3.224e-003	-3.224e-003	-3.224e-003
21	Accel. Gradient (kPa/m)	6.584e-007	6.584e-007	6.584e-007	6.584e-007
22	Liq. Reynolds	1.936e+005 *	1.936e+005 *	1.936e+005 *	1.936e+005 *
23	Vap. Reynolds	9.724e+006 *	9.724e+006 *	9.724e+006 *	9.724e+006 *
24	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
25	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	7097	7105	7113	7121
29	Elevation (m)	2.400e-002	1.800e-002	1.200e-002	6.000e-003
30	Pressure (bar)	71.83	71.83	71.83	71.83
31	Temperature (C)	55.38	55.37	55.37	55.37
32	Vapour Fraction	0.8829	0.8829	0.8829	0.8829
33	Heat Transferred (kJ/h-m)	676.0	676.0	675.9	675.8
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1734	0.1734	0.1734	0.1734
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	8.828e-003	8.828e-003	8.828e-003	8.828e-003
39	Static Gradient (kPa/m)	-1.213e-003	-1.213e-003	-1.213e-003	-1.213e-003
40	Accel. Gradient (kPa/m)	8.928e-007	8.928e-007	8.927e-007	8.927e-007
41	Liq. Reynolds	1.936e+005 *	1.936e+005 *	1.936e+005 *	1.936e+005 *
42	Vap. Reynolds	9.724e+006 *	9.724e+006 *	9.724e+006 *	9.724e+006 *
43	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
44	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	7129	7134	7139	7144
48	Elevation (m)	-1.398e-015	4.000e-003	8.000e-003	1.200e-002
49	Pressure (bar)	71.83	71.83	71.82	71.82
50	Temperature (C)	55.36	55.36	55.36	55.36
51	Vapour Fraction	0.8829	0.8829	0.8829	0.8829
52	Heat Transferred (kJ/h-m)	675.7	675.6	675.5	675.5
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.1734	0.2369	0.2369	0.2369
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	8.827e-003	7.879e-003	7.879e-003	7.879e-003
58	Static Gradient (kPa/m)	-1.213e-003	1.608e-003	1.608e-003	1.608e-003
59	Accel. Gradient (kPa/m)	8.927e-007	1.112e-006	1.112e-006	1.112e-006
60	Liq. Reynolds	1.936e+005 *	1.936e+005 *	1.936e+005 *	1.936e+005 *
61	Vap. Reynolds	9.724e+006 *	9.724e+006 *	9.724e+006 *	9.724e+006 *
62	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
63	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	7149	7154	7159	7164
67	Elevation (m)	1.600e-002	2.000e-002	3.000e-002	4.000e-002
68	Pressure (bar)	71.82	71.82	71.82	71.82
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	55.35	55.35	55.35	55.35
14	Vapour Fraction	0.8829	0.8829	0.8829	0.8829
15	Heat Transferred (kJ/h-m)	675.4	675.4	675.3	675.3
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2369	0.2369	0.2402	0.2402
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	7.878e-003	7.878e-003	7.463e-003	7.463e-003
21	Static Gradient (kPa/m)	1.608e-003	1.608e-003	4.063e-003	4.063e-003
22	Accel. Gradient (kPa/m)	1.112e-006	1.112e-006	1.351e-006	1.351e-006
23	Liq. Reynolds	1.936e+005 *	1.936e+005 *	1.936e+005 *	1.936e+005 *
24	Vap. Reynolds	9.724e+006 *	9.724e+006 *	9.724e+006 *	9.724e+006 *
25	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
26	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	7169	7174	7179	7184
30	Elevation (m)	5.000e-002	6.000e-002	7.000e-002	9.000e-002
31	Pressure (bar)	71.82	71.82	71.82	71.82
32	Temperature (C)	55.35	55.34	55.34	55.34
33	Vapour Fraction	0.8829	0.8829	0.8829	0.8829
34	Heat Transferred (kJ/h-m)	675.2	675.1	675.1	675.0
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2402	0.2402	0.2402	0.2458
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	7.463e-003	7.463e-003	7.463e-003	6.808e-003
40	Static Gradient (kPa/m)	4.063e-003	4.063e-003	4.063e-003	8.264e-003
41	Accel. Gradient (kPa/m)	1.351e-006	1.351e-006	1.351e-006	1.767e-006
42	Liq. Reynolds	1.936e+005 *	1.936e+005 *	1.936e+005 *	1.936e+005 *
43	Vap. Reynolds	9.725e+006 *	9.725e+006 *	9.725e+006 *	9.725e+006 *
44	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
45	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	7189	7194	7199	7204
49	Elevation (m)	0.1100	0.1300	0.1500	0.1700
50	Pressure (bar)	71.82	71.82	71.82	71.82
51	Temperature (C)	55.34	55.33	55.33	55.33
52	Vapour Fraction	0.8829	0.8829	0.8829	0.8829
53	Heat Transferred (kJ/h-m)	675.0	674.9	674.9	674.8
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2458	0.2458	0.2458	0.2458
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	6.808e-003	6.808e-003	6.808e-003	6.808e-003
59	Static Gradient (kPa/m)	8.264e-003	8.264e-003	8.264e-003	8.264e-003
60	Accel. Gradient (kPa/m)	1.767e-006	1.767e-006	1.767e-006	1.767e-006
61	Liq. Reynolds	1.936e+005 *	1.936e+005 *	1.936e+005 *	1.936e+005 *
62	Vap. Reynolds	9.725e+006 *	9.725e+006 *	9.725e+006 *	9.725e+006 *
63	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
64	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	7209	7214	7219	7224
68	Elevation (m)	0.1520	0.1340	0.1160	9.800e-002
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
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5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Pressure (bar)	71.82	71.82	71.82	71.82					
14	Temperature (C)	55.33	55.33	55.32	55.32					
15	Vapour Fraction	0.8828	0.8828	0.8828	0.8828					
16	Heat Transferred (kJ/h-m)	674.8	674.8	674.7	674.7					
17	Flow Regime	Transition	Transition	Transition	Transition					
18	Water Flow Regime									
19	Liquid HoldUp	0.1712	0.1712	0.1712	0.1712					
20	Water HoldUp	---	---	---	---					
21	Friction Gradient (kPa/m)	8.855e-003	8.855e-003	8.855e-003	8.855e-003					
22	Static Gradient (kPa/m)	-5.777e-003	-5.777e-003	-5.777e-003	-5.777e-003					
23	Accel. Gradient (kPa/m)	3.610e-007	3.610e-007	3.610e-007	3.609e-007					
24	Liq. Reynolds	1.936e+005 *	1.936e+005 *	1.936e+005 *	1.935e+005 *					
25	Vap. Reynolds	9.725e+006 *	9.725e+006 *	9.725e+006 *	9.725e+006 *					
26	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
27	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856					
28	Deposit Thickness (mm)	---	---	---	---					
29	Deposit Volume (m3)	---	---	---	---					
30	Length (m)	7229	7234	7239	7244					
31	Elevation (m)	8.000e-002	7.600e-002	7.200e-002	6.800e-002					
32	Pressure (bar)	71.82	71.82	71.82	71.81					
33	Temperature (C)	55.32	55.32	55.32	55.31					
34	Vapour Fraction	0.8828	0.8828	0.8828	0.8828					
35	Heat Transferred (kJ/h-m)	674.6	674.5	674.5	674.4					
36	Flow Regime	Transition	Transition	Transition	Transition					
37	Water Flow Regime									
38	Liquid HoldUp	0.1712	0.1733	0.1733	0.1733					
39	Water HoldUp	---	---	---	---					
40	Friction Gradient (kPa/m)	8.855e-003	8.828e-003	8.828e-003	8.827e-003					
41	Static Gradient (kPa/m)	-5.777e-003	-1.294e-003	-1.294e-003	-1.294e-003					
42	Accel. Gradient (kPa/m)	3.609e-007	8.833e-007	8.833e-007	8.833e-007					
43	Liq. Reynolds	1.935e+005 *	1.935e+005 *	1.935e+005 *	1.935e+005 *					
44	Vap. Reynolds	9.725e+006 *	9.725e+006 *	9.725e+006 *	9.725e+006 *					
45	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
46	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856					
47	Deposit Thickness (mm)	---	---	---	---					
48	Deposit Volume (m3)	---	---	---	---					
49	Length (m)	7249	7254	7259	7264					
50	Elevation (m)	6.400e-002	6.000e-002	8.000e-002	1.000e-001					
51	Pressure (bar)	71.81	71.81	71.81	71.81					
52	Temperature (C)	55.31	55.31	55.31	55.31					
53	Vapour Fraction	0.8828	0.8828	0.8828	0.8828					
54	Heat Transferred (kJ/h-m)	674.4	674.3	674.2	674.2					
55	Flow Regime	Transition	Transition	Transition	Transition					
56	Water Flow Regime									
57	Liquid HoldUp	0.1733	0.1733	0.2458	0.2458					
58	Water HoldUp	---	---	---	---					
59	Friction Gradient (kPa/m)	8.827e-003	8.827e-003	6.808e-003	6.808e-003					
60	Static Gradient (kPa/m)	-1.294e-003	-1.294e-003	8.264e-003	8.265e-003					
61	Accel. Gradient (kPa/m)	8.833e-007	8.833e-007	1.767e-006	1.767e-006					
62	Liq. Reynolds	1.935e+005 *	1.935e+005 *	1.935e+005 *	1.935e+005 *					
63	Vap. Reynolds	9.725e+006 *	9.725e+006 *	9.725e+006 *	9.725e+006 *					
64	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1980					
65	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856					
66	Deposit Thickness (mm)	---	---	---	---					
67	Deposit Volume (m3)	---	---	---	---					
68	Length (m)	7269	7274	7279	7284					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	0.1200	0.1400	0.1600	0.1340
14	Pressure (bar)	71.81	71.81	71.81	71.81
15	Temperature (C)	55.30	55.30	55.30	55.30
16	Vapour Fraction	0.8828	0.8828	0.8828	0.8828
17	Heat Transferred (kJ/h-m)	674.1	674.0	674.0	674.0
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2458	0.2458	0.2458	0.1700
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	6.808e-003	6.808e-003	6.808e-003	8.871e-003
23	Static Gradient (kPa/m)	8.265e-003	8.265e-003	8.265e-003	-8.306e-003
24	Accel. Gradient (kPa/m)	1.767e-006	1.767e-006	1.767e-006	6.625e-008
25	Liq. Reynolds	1.935e+005 *	1.935e+005 *	1.935e+005 *	1.935e+005 *
26	Vap. Reynolds	9.725e+006 *	9.725e+006 *	9.725e+006 *	9.725e+006 *
27	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
28	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	7289	7294	7299	7304
32	Elevation (m)	0.1080	8.200e-002	5.600e-002	3.000e-002
33	Pressure (bar)	71.81	71.81	71.81	71.81
34	Temperature (C)	55.30	55.29	55.29	55.29
35	Vapour Fraction	0.8828	0.8828	0.8828	0.8828
36	Heat Transferred (kJ/h-m)	673.9	673.9	673.8	673.8
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.1700	0.1700	0.1700	0.1700
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	8.871e-003	8.871e-003	8.871e-003	8.871e-003
42	Static Gradient (kPa/m)	-8.306e-003	-8.306e-003	-8.306e-003	-8.306e-003
43	Accel. Gradient (kPa/m)	6.624e-008	6.622e-008	6.621e-008	6.619e-008
44	Liq. Reynolds	1.935e+005 *	1.935e+005 *	1.935e+005 *	1.935e+005 *
45	Vap. Reynolds	9.725e+006 *	9.725e+006 *	9.725e+006 *	9.725e+006 *
46	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
47	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	7309	7314	7319	7324
51	Elevation (m)	3.800e-002	4.600e-002	5.400e-002	6.200e-002
52	Pressure (bar)	71.81	71.81	71.81	71.81
53	Temperature (C)	55.29	55.29	55.28	55.28
54	Vapour Fraction	0.8828	0.8828	0.8828	0.8828
55	Heat Transferred (kJ/h-m)	673.7	673.6	673.6	673.5
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2391	0.2391	0.2391	0.2391
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	7.599e-003	7.599e-003	7.599e-003	7.599e-003
61	Static Gradient (kPa/m)	3.239e-003	3.239e-003	3.239e-003	3.239e-003
62	Accel. Gradient (kPa/m)	1.271e-006	1.271e-006	1.271e-006	1.271e-006
63	Liq. Reynolds	1.935e+005 *	1.935e+005 *	1.935e+005 *	1.935e+005 *
64	Vap. Reynolds	9.725e+006 *	9.725e+006 *	9.725e+006 *	9.725e+006 *
65	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
66	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
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6						
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Length (m)	7329	7334	7339	7344	
14	Elevation (m)	7.000e-002	8.800e-002	0.1060	0.1240	
15	Pressure (bar)	71.81	71.81	71.81	71.81	
16	Temperature (C)	55.28	55.28	55.28	55.27	
17	Vapour Fraction	0.8828	0.8828	0.8828	0.8828	
18	Heat Transferred (kJ/h-m)	673.5	673.4	673.3	673.3	
19	Flow Regime	Transition	Transition	Transition	Transition	
20	Water Flow Regime					
21	Liquid HoldUp	0.2391	0.2447	0.2447	0.2447	
22	Water HoldUp	---	---	---	---	
23	Friction Gradient (kPa/m)	7.599e-003	6.932e-003	6.932e-003	6.932e-003	
24	Static Gradient (kPa/m)	3.239e-003	7.429e-003	7.429e-003	7.429e-003	
25	Accel. Gradient (kPa/m)	1.271e-006	1.684e-006	1.684e-006	1.684e-006	
26	Liq. Reynolds	1.935e+005 *	1.935e+005 *	1.935e+005 *	1.935e+005 *	
27	Vap. Reynolds	9.725e+006 *	9.725e+006 *	9.725e+006 *	9.725e+006 *	
28	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	
29	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856	
30	Deposit Thickness (mm)	---	---	---	---	
31	Deposit Volume (m3)	---	---	---	---	
32	Length (m)	7349	7354	7359	7364	
33	Elevation (m)	0.1420	0.1600	0.1680	0.1760	
34	Pressure (bar)	71.80	71.80	71.80	71.80	
35	Temperature (C)	55.27	55.27	55.27	55.27	
36	Vapour Fraction	0.8828	0.8828	0.8828	0.8828	
37	Heat Transferred (kJ/h-m)	673.2	673.2	673.1	673.1	
38	Flow Regime	Transition	Transition	Transition	Transition	
39	Water Flow Regime					
40	Liquid HoldUp	0.2447	0.2447	0.2391	0.2391	
41	Water HoldUp	---	---	---	---	
42	Friction Gradient (kPa/m)	6.932e-003	6.933e-003	7.599e-003	7.599e-003	
43	Static Gradient (kPa/m)	7.429e-003	7.429e-003	3.239e-003	3.239e-003	
44	Accel. Gradient (kPa/m)	1.684e-006	1.684e-006	1.271e-006	1.271e-006	
45	Liq. Reynolds	1.935e+005 *	1.935e+005 *	1.935e+005 *	1.935e+005 *	
46	Vap. Reynolds	9.725e+006 *	9.725e+006 *	9.725e+006 *	9.725e+006 *	
47	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	
48	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856	
49	Deposit Thickness (mm)	---	---	---	---	
50	Deposit Volume (m3)	---	---	---	---	
51	Length (m)	7369	7374	7379	7384	
52	Elevation (m)	0.1840	0.1920	0.2000	0.2040	
53	Pressure (bar)	71.80	71.80	71.80	71.80	
54	Temperature (C)	55.26	55.26	55.26	55.26	
55	Vapour Fraction	0.8828	0.8828	0.8828	0.8828	
56	Heat Transferred (kJ/h-m)	673.0	672.9	672.9	672.8	
57	Flow Regime	Transition	Transition	Transition	Transition	
58	Water Flow Regime					
59	Liquid HoldUp	0.2391	0.2391	0.2391	0.2369	
60	Water HoldUp	---	---	---	---	
61	Friction Gradient (kPa/m)	7.599e-003	7.599e-003	7.599e-003	7.877e-003	
62	Static Gradient (kPa/m)	3.239e-003	3.239e-003	3.239e-003	1.609e-003	
63	Accel. Gradient (kPa/m)	1.271e-006	1.271e-006	1.271e-006	1.112e-006	
64	Liq. Reynolds	1.935e+005 *	1.935e+005 *	1.935e+005 *	1.935e+005 *	
65	Vap. Reynolds	9.725e+006 *	9.725e+006 *	9.725e+006 *	9.725e+006 *	
66	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	
67	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855	
68	Deposit Thickness (mm)	---	---	---	---	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	7389	7394	7399	7404
15	Elevation (m)	0.2080	0.2120	0.2160	0.2200
16	Pressure (bar)	71.80	71.80	71.80	71.80
17	Temperature (C)	55.25	55.25	55.25	55.25
18	Vapour Fraction	0.8828	0.8828	0.8828	0.8828
19	Heat Transferred (kJ/h-m)	672.8	672.7	672.7	672.6
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2369	0.2369	0.2369	0.2369
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	7.877e-003	7.877e-003	7.877e-003	7.877e-003
25	Static Gradient (kPa/m)	1.609e-003	1.609e-003	1.609e-003	1.609e-003
26	Accel. Gradient (kPa/m)	1.112e-006	1.112e-006	1.112e-006	1.112e-006
27	Liq. Reynolds	1.935e+005 *	1.935e+005 *	1.935e+005 *	1.935e+005 *
28	Vap. Reynolds	9.725e+006 *	9.726e+006 *	9.726e+006 *	9.726e+006 *
29	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
30	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	7409	7414	7419	7424
34	Elevation (m)	0.2220	0.2240	0.2260	0.2280
35	Pressure (bar)	71.80	71.80	71.80	71.80
36	Temperature (C)	55.25	55.24	55.24	55.24
37	Vapour Fraction	0.8828	0.8828	0.8828	0.8828
38	Heat Transferred (kJ/h-m)	672.6	672.5	672.5	672.4
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2358	0.2358	0.2358	0.2358
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	8.020e-003	8.020e-003	8.020e-003	8.020e-003
44	Static Gradient (kPa/m)	8.015e-004	8.015e-004	8.015e-004	8.015e-004
45	Accel. Gradient (kPa/m)	1.035e-006	1.035e-006	1.035e-006	1.035e-006
46	Liq. Reynolds	1.935e+005 *	1.935e+005 *	1.935e+005 *	1.935e+005 *
47	Vap. Reynolds	9.726e+006 *	9.726e+006 *	9.726e+006 *	9.726e+006 *
48	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
49	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	7429	7434	7439	7444
53	Elevation (m)	0.2300	0.2320	0.2340	0.2360
54	Pressure (bar)	71.80	71.80	71.80	71.80
55	Temperature (C)	55.24	55.24	55.23	55.23
56	Vapour Fraction	0.8828	0.8828	0.8828	0.8828
57	Heat Transferred (kJ/h-m)	672.3	672.3	672.2	672.2
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2358	0.2358	0.2358	0.2358
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	8.020e-003	8.020e-003	8.020e-003	8.020e-003
63	Static Gradient (kPa/m)	8.016e-004	8.016e-004	8.016e-004	8.016e-004
64	Accel. Gradient (kPa/m)	1.035e-006	1.035e-006	1.035e-006	1.035e-006
65	Liq. Reynolds	1.935e+005 *	1.935e+005 *	1.935e+005 *	1.935e+005 *
66	Vap. Reynolds	9.726e+006 *	9.726e+006 *	9.726e+006 *	9.726e+006 *
67	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
68	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
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3					Unit Set: Mixed
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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9	PROFILES				
10	Pipe Table				
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13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	7449	7454	7459	7464
16	Elevation (m)	0.2380	0.2400	0.2340	0.2280
17	Pressure (bar)	71.79	71.79	71.79	71.79
18	Temperature (C)	55.23	55.23	55.23	55.22
19	Vapour Fraction	0.8828	0.8828	0.8828	0.8828
20	Heat Transferred (kJ/h-m)	672.1	672.1	672.1	672.0
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2358	0.2358	0.1730	0.1730
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	8.019e-003	8.019e-003	8.830e-003	8.830e-003
26	Static Gradient (kPa/m)	8.016e-004	8.016e-004	-1.939e-003	-1.939e-003
27	Accel. Gradient (kPa/m)	1.035e-006	1.034e-006	8.082e-007	8.082e-007
28	Liq. Reynolds	1.935e+005 *	1.935e+005 *	1.935e+005 *	1.935e+005 *
29	Vap. Reynolds	9.726e+006 *	9.726e+006 *	9.726e+006 *	9.726e+006 *
30	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
31	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	7469	7474	7479	7484
35	Elevation (m)	0.2220	0.2160	0.2100	0.1920
36	Pressure (bar)	71.79	71.79	71.79	71.79
37	Temperature (C)	55.22	55.22	55.22	55.22
38	Vapour Fraction	0.8828	0.8828	0.8828	0.8828
39	Heat Transferred (kJ/h-m)	672.0	671.9	671.9	671.8
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.1730	0.1730	0.1730	0.1712
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	8.830e-003	8.830e-003	8.830e-003	8.854e-003
45	Static Gradient (kPa/m)	-1.939e-003	-1.939e-003	-1.939e-003	-5.778e-003
46	Accel. Gradient (kPa/m)	8.082e-007	8.082e-007	8.082e-007	3.607e-007
47	Liq. Reynolds	1.935e+005 *	1.935e+005 *	1.935e+005 *	1.935e+005 *
48	Vap. Reynolds	9.726e+006 *	9.726e+006 *	9.726e+006 *	9.726e+006 *
49	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
50	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	7489	7494	7499	7504
54	Elevation (m)	0.1740	0.1560	0.1380	0.1200
55	Pressure (bar)	71.79	71.79	71.79	71.79
56	Temperature (C)	55.21	55.21	55.21	55.21
57	Vapour Fraction	0.8828	0.8828	0.8828	0.8828
58	Heat Transferred (kJ/h-m)	671.8	671.7	671.7	671.6
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.1712	0.1712	0.1712	0.1712
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	8.854e-003	8.854e-003	8.854e-003	8.854e-003
64	Static Gradient (kPa/m)	-5.778e-003	-5.778e-003	-5.778e-003	-5.778e-003
65	Accel. Gradient (kPa/m)	3.607e-007	3.607e-007	3.607e-007	3.607e-007
66	Liq. Reynolds	1.935e+005 *	1.935e+005 *	1.935e+005 *	1.935e+005 *
67	Vap. Reynolds	9.726e+006 *	9.726e+006 *	9.726e+006 *	9.726e+006 *
68	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
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5			Unit Set: Mixed			
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7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	PROFILES					
11	Pipe Table					
12						
13	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855	
14	Deposit Thickness (mm)	---	---	---	---	
15	Deposit Volume (m3)	---	---	---	---	
16	Length (m)	7509	7514	7519	7524	
17	Elevation (m)	0.1320	0.1440	0.1560	0.1680	
18	Pressure (bar)	71.79	71.79	71.79	71.79	
19	Temperature (C)	55.21	55.20	55.20	55.20	
20	Vapour Fraction	0.8828	0.8828	0.8828	0.8828	
21	Heat Transferred (kJ/h-m)	671.5	671.4	671.4	671.3	
22	Flow Regime	Transition	Transition	Transition	Transition	
23	Water Flow Regime					
24	Liquid HoldUp	0.2414	0.2414	0.2414	0.2414	
25	Water HoldUp	---	---	---	---	
26	Friction Gradient (kPa/m)	7.327e-003	7.327e-003	7.327e-003	7.327e-003	
27	Static Gradient (kPa/m)	4.893e-003	4.893e-003	4.893e-003	4.893e-003	
28	Accel. Gradient (kPa/m)	1.433e-006	1.433e-006	1.433e-006	1.433e-006	
29	Liq. Reynolds	1.935e+005 *	1.935e+005 *	1.935e+005 *	1.935e+005 *	
30	Vap. Reynolds	9.726e+006 *	9.726e+006 *	9.726e+006 *	9.726e+006 *	
31	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	
32	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855	
33	Deposit Thickness (mm)	---	---	---	---	
34	Deposit Volume (m3)	---	---	---	---	
35	Length (m)	7529	7533	7538	7543	
36	Elevation (m)	0.1800	0.1680	0.1560	0.1440	
37	Pressure (bar)	71.79	71.79	71.79	71.79	
38	Temperature (C)	55.20	55.20	55.19	55.19	
39	Vapour Fraction	0.8828	0.8828	0.8828	0.8828	
40	Heat Transferred (kJ/h-m)	671.3	671.3	671.2	671.2	
41	Flow Regime	Transition	Transition	Transition	Transition	
42	Water Flow Regime					
43	Liquid HoldUp	0.2414	0.1721	0.1721	0.1721	
44	Water HoldUp	---	---	---	---	
45	Friction Gradient (kPa/m)	7.327e-003	8.842e-003	8.842e-003	8.842e-003	
46	Static Gradient (kPa/m)	4.893e-003	-3.931e-003	-3.931e-003	-3.931e-003	
47	Accel. Gradient (kPa/m)	1.433e-006	5.760e-007	5.759e-007	5.759e-007	
48	Liq. Reynolds	1.935e+005 *	1.935e+005 *	1.935e+005 *	1.935e+005 *	
49	Vap. Reynolds	9.726e+006 *	9.726e+006 *	9.726e+006 *	9.726e+006 *	
50	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	
51	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855	
52	Deposit Thickness (mm)	---	---	---	---	
53	Deposit Volume (m3)	---	---	---	---	
54	Length (m)	7548	7553	7558	7563	
55	Elevation (m)	0.1320	0.1200	0.1320	0.1440	
56	Pressure (bar)	71.79	71.79	71.79	71.79	
57	Temperature (C)	55.19	55.19	55.19	55.18	
58	Vapour Fraction	0.8828	0.8828	0.8828	0.8828	
59	Heat Transferred (kJ/h-m)	671.1	671.1	671.0	670.9	
60	Flow Regime	Transition	Transition	Transition	Transition	
61	Water Flow Regime					
62	Liquid HoldUp	0.1721	0.1721	0.2414	0.2414	
63	Water HoldUp	---	---	---	---	
64	Friction Gradient (kPa/m)	8.842e-003	8.842e-003	7.326e-003	7.326e-003	
65	Static Gradient (kPa/m)	-3.931e-003	-3.931e-003	4.893e-003	4.893e-003	
66	Accel. Gradient (kPa/m)	5.759e-007	5.759e-007	1.433e-006	1.433e-006	
67	Liq. Reynolds	1.935e+005 *	1.935e+005 *	1.934e+005 *	1.934e+005 *	
68	Vap. Reynolds	9.726e+006 *	9.726e+006 *	9.726e+006 *	9.726e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
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5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
14	Vapour Velocity	(m/s)	2.855	2.855	2.855	2.855				
15	Deposit Thickness	(mm)	---	---	---	---				
16	Deposit Volume	(m3)	---	---	---	---				
17	Length	(m)	7568	7573	7578	7583				
18	Elevation	(m)	0.1560	0.1680	0.1800	0.1800				
19	Pressure	(bar)	71.79	71.79	71.78	71.78				
20	Temperature	(C)	55.18	55.18	55.18	55.18				
21	Vapour Fraction		0.8828	0.8828	0.8828	0.8828				
22	Heat Transferred	(kJ/h-m)	670.8	670.8	670.7	670.7				
23	Flow Regime		Transition	Transition	Transition	Transition				
24	Water Flow Regime									
25	Liquid HoldUp		0.2414	0.2414	0.2414	0.2540				
26	Water HoldUp		---	---	---	---				
27	Friction Gradient	(kPa/m)	7.326e-003	7.326e-003	7.326e-003	5.939e-003				
28	Static Gradient	(kPa/m)	4.893e-003	4.893e-003	4.893e-003	0.0000				
29	Accel. Gradient	(kPa/m)	1.433e-006	1.433e-006	1.433e-006	6.966e-007				
30	Liq. Reynolds		1.934e+005 *	1.934e+005 *	1.934e+005 *	1.934e+005 *				
31	Vap. Reynolds		9.726e+006 *	9.726e+006 *	9.726e+006 *	9.726e+006 *				
32	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
33	Vapour Velocity	(m/s)	2.855	2.855	2.855	2.855				
34	Deposit Thickness	(mm)	---	---	---	---				
35	Deposit Volume	(m3)	---	---	---	---				
36	Length	(m)	7588	7593	7598	7603				
37	Elevation	(m)	0.1800	0.1800	0.1800	0.1800				
38	Pressure	(bar)	71.78	71.78	71.78	71.78				
39	Temperature	(C)	55.17	55.17	55.17	55.17				
40	Vapour Fraction		0.8828	0.8828	0.8828	0.8828				
41	Heat Transferred	(kJ/h-m)	670.6	670.6	670.5	670.5				
42	Flow Regime		Transition	Transition	Transition	Transition				
43	Water Flow Regime									
44	Liquid HoldUp		0.2540	0.2540	0.2540	0.2540				
45	Water HoldUp		---	---	---	---				
46	Friction Gradient	(kPa/m)	5.939e-003	5.939e-003	5.939e-003	5.939e-003				
47	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000				
48	Accel. Gradient	(kPa/m)	6.966e-007	6.966e-007	6.966e-007	6.965e-007				
49	Liq. Reynolds		1.934e+005 *	1.934e+005 *	1.934e+005 *	1.934e+005 *				
50	Vap. Reynolds		9.726e+006 *	9.726e+006 *	9.726e+006 *	9.726e+006 *				
51	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
52	Vapour Velocity	(m/s)	2.855	2.855	2.855	2.855				
53	Deposit Thickness	(mm)	---	---	---	---				
54	Deposit Volume	(m3)	---	---	---	---				
55	Length	(m)	7608	7613	7618	7623				
56	Elevation	(m)	0.1660	0.1520	0.1380	0.1240				
57	Pressure	(bar)	71.78	71.78	71.78	71.78				
58	Temperature	(C)	55.17	55.16	55.16	55.16				
59	Vapour Fraction		0.8828	0.8828	0.8828	0.8828				
60	Heat Transferred	(kJ/h-m)	670.5	670.4	670.4	670.3				
61	Flow Regime		Transition	Transition	Transition	Transition				
62	Water Flow Regime									
63	Liquid HoldUp		0.1719	0.1719	0.1719	0.1719				
64	Water HoldUp		---	---	---	---				
65	Friction Gradient	(kPa/m)	8.845e-003	8.845e-003	8.845e-003	8.845e-003				
66	Static Gradient	(kPa/m)	-4.505e-003	-4.505e-003	-4.505e-003	-4.505e-003				
67	Accel. Gradient	(kPa/m)	5.090e-007	5.090e-007	5.090e-007	5.090e-007				
68	Liq. Reynolds		1.934e+005 *	1.934e+005 *	1.934e+005 *	1.934e+005 *				
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 154 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.726e+006 *	9.726e+006 *	9.726e+006 *	9.726e+006 *
14	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
15	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	7628	7633	7638	7643
19	Elevation (m)	0.1100	0.1160	0.1220	0.1280
20	Pressure (bar)	71.78	71.78	71.78	71.78
21	Temperature (C)	55.16	55.16	55.15	55.15
22	Vapour Fraction	0.8828	0.8828	0.8828	0.8828
23	Heat Transferred (kJ/h-m)	670.3	670.2	670.1	670.0
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1719	0.2380	0.2380	0.2380
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	8.845e-003	7.735e-003	7.735e-003	7.735e-003
29	Static Gradient (kPa/m)	-4.505e-003	2.422e-003	2.422e-003	2.422e-003
30	Accel. Gradient (kPa/m)	5.090e-007	1.191e-006	1.191e-006	1.191e-006
31	Liq. Reynolds	1.934e+005 *	1.934e+005 *	1.934e+005 *	1.934e+005 *
32	Vap. Reynolds	9.726e+006 *	9.726e+006 *	9.726e+006 *	9.726e+006 *
33	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
34	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	7648	7653	7658	7663
38	Elevation (m)	0.1340	0.1400	0.1260	0.1120
39	Pressure (bar)	71.78	71.78	71.78	71.78
40	Temperature (C)	55.15	55.15	55.15	55.14
41	Vapour Fraction	0.8828	0.8828	0.8828	0.8828
42	Heat Transferred (kJ/h-m)	670.0	669.9	669.9	669.9
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2380	0.2380	0.1719	0.1719
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	7.735e-003	7.735e-003	8.845e-003	8.845e-003
48	Static Gradient (kPa/m)	2.422e-003	2.422e-003	-4.505e-003	-4.505e-003
49	Accel. Gradient (kPa/m)	1.191e-006	1.191e-006	5.090e-007	5.090e-007
50	Liq. Reynolds	1.934e+005 *	1.934e+005 *	1.934e+005 *	1.934e+005 *
51	Vap. Reynolds	9.726e+006 *	9.726e+006 *	9.726e+006 *	9.727e+006 *
52	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
53	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	7668	7673	7678	7683
57	Elevation (m)	9.800e-002	8.400e-002	7.000e-002	8.800e-002
58	Pressure (bar)	71.78	71.78	71.78	71.78
59	Temperature (C)	55.14	55.14	55.14	55.14
60	Vapour Fraction	0.8828	0.8828	0.8828	0.8828
61	Heat Transferred (kJ/h-m)	669.8	669.8	669.7	669.6
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1719	0.1719	0.1719	0.2448
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	8.845e-003	8.845e-003	8.845e-003	6.933e-003
67	Static Gradient (kPa/m)	-4.505e-003	-4.505e-003	-4.505e-003	7.415e-003
68	Accel. Gradient (kPa/m)	5.090e-007	5.090e-007	5.089e-007	1.683e-006
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		1.934e+005 *	1.934e+005 *	1.934e+005 *	1.934e+005 *	
14	Vap. Reynolds		9.727e+006 *	9.727e+006 *	9.727e+006 *	9.727e+006 *	
15	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
16	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855		
17	Deposit Thickness (mm)	---	---	---	---		
18	Deposit Volume (m3)	---	---	---	---		
19	Length (m)	7688	7693	7698	7703		
20	Elevation (m)	0.1060	0.1240	0.1420	0.1600		
21	Pressure (bar)	71.78	71.78	71.78	71.77		
22	Temperature (C)	55.13	55.13	55.13	55.13		
23	Vapour Fraction	0.8828	0.8828	0.8828	0.8827		
24	Heat Transferred (kJ/h-m)	669.6	669.5	669.4	669.4		
25	Flow Regime	Transition	Transition	Transition	Transition		
26	Water Flow Regime						
27	Liquid HoldUp	0.2448	0.2448	0.2448	0.2448		
28	Water HoldUp	---	---	---	---		
29	Friction Gradient (kPa/m)	6.933e-003	6.933e-003	6.933e-003	6.933e-003		
30	Static Gradient (kPa/m)	7.415e-003	7.415e-003	7.415e-003	7.415e-003		
31	Accel. Gradient (kPa/m)	1.683e-006	1.683e-006	1.683e-006	1.683e-006		
32	Liq. Reynolds		1.934e+005 *	1.934e+005 *	1.934e+005 *	1.934e+005 *	
33	Vap. Reynolds		9.727e+006 *	9.727e+006 *	9.727e+006 *	9.727e+006 *	
34	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
35	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855		
36	Deposit Thickness (mm)	---	---	---	---		
37	Deposit Volume (m3)	---	---	---	---		
38	Length (m)	7708	7713	7718	7723		
39	Elevation (m)	5.400e-002	-5.200e-002	-0.1580	-0.2640		
40	Pressure (bar)	71.78	71.78	71.78	71.78		
41	Temperature (C)	55.13	55.12	55.12	55.12		
42	Vapour Fraction	0.8827	0.8827	0.8827	0.8827		
43	Heat Transferred (kJ/h-m)	669.4	669.4	669.4	669.3		
44	Flow Regime	Transition	Transition	Transition	Transition		
45	Water Flow Regime						
46	Liquid HoldUp	0.1580	0.1580	0.1580	0.1580		
47	Water HoldUp	---	---	---	---		
48	Friction Gradient (kPa/m)	9.048e-003	9.047e-003	9.047e-003	9.047e-003		
49	Static Gradient (kPa/m)	-3.229e-002	-3.229e-002	-3.229e-002	-3.229e-002		
50	Accel. Gradient (kPa/m)	-2.726e-006	-2.726e-006	-2.726e-006	-2.726e-006		
51	Liq. Reynolds		1.934e+005 *	1.934e+005 *	1.934e+005 *	1.934e+005 *	
52	Vap. Reynolds		9.727e+006 *	9.727e+006 *	9.727e+006 *	9.727e+006 *	
53	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
54	Vapour Velocity (m/s)	2.855	2.855	2.855	2.854		
55	Deposit Thickness (mm)	---	---	---	---		
56	Deposit Volume (m3)	---	---	---	---		
57	Length (m)	7728	7733	7738	7743		
58	Elevation (m)	-0.3700	-0.3820	-0.3940	-0.4060		
59	Pressure (bar)	71.78	71.78	71.78	71.78		
60	Temperature (C)	55.12	55.12	55.12	55.11		
61	Vapour Fraction	0.8827	0.8827	0.8827	0.8827		
62	Heat Transferred (kJ/h-m)	669.3	669.2	669.2	669.1		
63	Flow Regime	Transition	Transition	Transition	Transition		
64	Water Flow Regime						
65	Liquid HoldUp	0.1580	0.1722	0.1722	0.1722		
66	Water HoldUp	---	---	---	---		
67	Friction Gradient (kPa/m)	9.047e-003	8.840e-003	8.840e-003	8.840e-003		
68	Static Gradient (kPa/m)	-3.229e-002	-3.872e-003	-3.872e-003	-3.872e-003		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3						
4						
5			Unit Set: Mixed			
6			Date/Time: Mon Jun 13 15:13:43 2022			
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	PROFILES					
11	Pipe Table					
12						
13	Accel. Gradient	(kPa/m)	-2.726e-006	5.825e-007	5.824e-007	5.824e-007
14	Liq. Reynolds		1.934e+005 *	1.934e+005 *	1.934e+005 *	1.934e+005 *
15	Vap. Reynolds		9.727e+006 *	9.727e+006 *	9.727e+006 *	9.727e+006 *
16	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
17	Vapour Velocity	(m/s)	2.854	2.854	2.854	2.854
18	Deposit Thickness	(mm)	---	---	---	---
19	Deposit Volume	(m3)	---	---	---	---
20	Length	(m)	7748	7753	7758	7763
21	Elevation	(m)	-0.4180	-0.4300	-0.3300	-0.2300
22	Pressure	(bar)	71.78	71.78	71.78	71.77
23	Temperature	(C)	55.11	55.11	55.11	55.11
24	Vapour Fraction		0.8827	0.8827	0.8827	0.8827
25	Heat Transferred	(kJ/h-m)	669.1	669.0	668.9	668.8
26	Flow Regime		Transition	Transition	Transition	Transition
27	Water Flow Regime					
28	Liquid HoldUp		0.1722	0.1722	0.2908	0.2908
29	Water HoldUp		---	---	---	---
30	Friction Gradient	(kPa/m)	8.840e-003	8.839e-003	7.741e-003	7.741e-003
31	Static Gradient	(kPa/m)	-3.872e-003	-3.872e-003	4.689e-002	4.689e-002
32	Accel. Gradient	(kPa/m)	5.824e-007	5.824e-007	6.406e-006	6.407e-006
33	Liq. Reynolds		1.934e+005 *	1.934e+005 *	1.934e+005 *	1.934e+005 *
34	Vap. Reynolds		9.727e+006 *	9.727e+006 *	9.727e+006 *	9.727e+006 *
35	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
36	Vapour Velocity	(m/s)	2.854	2.854	2.854	2.854
37	Deposit Thickness	(mm)	---	---	---	---
38	Deposit Volume	(m3)	---	---	---	---
39	Length	(m)	7768	7773	7778	7783
40	Elevation	(m)	-0.1300	-3.000e-002	7.000e-002	9.600e-002
41	Pressure	(bar)	71.77	71.77	71.77	71.76
42	Temperature	(C)	55.10	55.10	55.10	55.09
43	Vapour Fraction		0.8827	0.8827	0.8827	0.8827
44	Heat Transferred	(kJ/h-m)	668.7	668.7	668.6	668.5
45	Flow Regime		Transition	Transition	Transition	Transition
46	Water Flow Regime					
47	Liquid HoldUp		0.2908	0.2908	0.2908	0.2493
48	Water HoldUp		---	---	---	---
49	Friction Gradient	(kPa/m)	7.742e-003	7.742e-003	7.742e-003	6.434e-003
50	Static Gradient	(kPa/m)	4.689e-002	4.689e-002	4.689e-002	1.086e-002
51	Accel. Gradient	(kPa/m)	6.407e-006	6.407e-006	6.408e-006	2.028e-006
52	Liq. Reynolds		1.934e+005 *	1.934e+005 *	1.934e+005 *	1.934e+005 *
53	Vap. Reynolds		9.727e+006 *	9.727e+006 *	9.727e+006 *	9.727e+006 *
54	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
55	Vapour Velocity	(m/s)	2.855	2.855	2.855	2.855
56	Deposit Thickness	(mm)	---	---	---	---
57	Deposit Volume	(m3)	---	---	---	---
58	Length	(m)	7788	7793	7798	7803
59	Elevation	(m)	0.1220	0.1480	0.1740	0.2000
60	Pressure	(bar)	71.76	71.76	71.76	71.76
61	Temperature	(C)	55.09	55.09	55.09	55.09
62	Vapour Fraction		0.8827	0.8827	0.8827	0.8827
63	Heat Transferred	(kJ/h-m)	668.5	668.4	668.3	668.3
64	Flow Regime		Transition	Transition	Transition	Transition
65	Water Flow Regime					
66	Liquid HoldUp		0.2493	0.2493	0.2493	0.2493
67	Water HoldUp		---	---	---	---
68	Friction Gradient	(kPa/m)	6.434e-003	6.434e-003	6.434e-003	6.434e-003
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1				Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	1.086e-002	1.086e-002	1.086e-002	1.086e-002
14	Accel. Gradient	(kPa/m)	2.028e-006	2.028e-006	2.028e-006	2.028e-006
15	Liq. Reynolds		1.934e+005 *	1.934e+005 *	1.934e+005 *	1.934e+005 *
16	Vap. Reynolds		9.727e+006 *	9.727e+006 *	9.727e+006 *	9.727e+006 *
17	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
18	Vapour Velocity	(m/s)	2.855	2.855	2.855	2.855
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	7808	7813	7818	7823
22	Elevation	(m)	0.2000	0.2000	0.2000	0.2000
23	Pressure	(bar)	71.76	71.76	71.76	71.76
24	Temperature	(C)	55.08	55.08	55.08	55.08
25	Vapour Fraction		0.8827	0.8827	0.8827	0.8827
26	Heat Transferred	(kJ/h-m)	668.2	668.2	668.1	668.1
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2540	0.2540	0.2540	0.2540
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	5.939e-003	5.939e-003	5.939e-003	5.939e-003
32	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000
33	Accel. Gradient	(kPa/m)	6.967e-007	6.966e-007	6.966e-007	6.966e-007
34	Liq. Reynolds		1.934e+005 *	1.934e+005 *	1.934e+005 *	1.934e+005 *
35	Vap. Reynolds		9.727e+006 *	9.727e+006 *	9.727e+006 *	9.727e+006 *
36	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
37	Vapour Velocity	(m/s)	2.855	2.855	2.855	2.855
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	7828	7833	7838	7843
41	Elevation	(m)	0.2000	0.2040	0.2080	0.2120
42	Pressure	(bar)	71.76	71.76	71.76	71.76
43	Temperature	(C)	55.08	55.07	55.07	55.07
44	Vapour Fraction		0.8827	0.8827	0.8827	0.8827
45	Heat Transferred	(kJ/h-m)	668.0	668.0	667.9	667.8
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2540	0.2369	0.2369	0.2369
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	5.939e-003	7.875e-003	7.875e-003	7.875e-003
51	Static Gradient	(kPa/m)	0.0000	1.609e-003	1.609e-003	1.609e-003
52	Accel. Gradient	(kPa/m)	6.966e-007	1.113e-006	1.113e-006	1.113e-006
53	Liq. Reynolds		1.934e+005 *	1.934e+005 *	1.934e+005 *	1.934e+005 *
54	Vap. Reynolds		9.727e+006 *	9.727e+006 *	9.727e+006 *	9.727e+006 *
55	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
56	Vapour Velocity	(m/s)	2.855	2.855	2.855	2.855
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	7848	7853	7858	7863
60	Elevation	(m)	0.2160	0.2200	0.2280	0.2360
61	Pressure	(bar)	71.76	71.76	71.76	71.76
62	Temperature	(C)	55.07	55.07	55.06	55.06
63	Vapour Fraction		0.8827	0.8827	0.8827	0.8827
64	Heat Transferred	(kJ/h-m)	667.8	667.7	667.7	667.6
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2369	0.2369	0.2392	0.2392
68	Water HoldUp		---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
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5			Date/Time: Mon Jun 13 15:13:43 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	7.875e-003	7.875e-003	7.596e-003	7.596e-003
14	Static Gradient	(kPa/m)	1.609e-003	1.609e-003	3.240e-003	3.240e-003
15	Accel. Gradient	(kPa/m)	1.113e-006	1.113e-006	1.271e-006	1.271e-006
16	Liq. Reynolds		1.934e+005 *	1.934e+005 *	1.934e+005 *	1.933e+005 *
17	Vap. Reynolds		9.727e+006 *	9.727e+006 *	9.727e+006 *	9.727e+006 *
18	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
19	Vapour Velocity	(m/s)	2.855	2.855	2.855	2.855
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	7868	7873	7878	7883
23	Elevation	(m)	0.2440	0.2520	0.2600	0.2560
24	Pressure	(bar)	71.76	71.76	71.75	71.75
25	Temperature	(C)	55.06	55.06	55.06	55.05
26	Vapour Fraction		0.8827	0.8827	0.8827	0.8827
27	Heat Transferred	(kJ/h-m)	667.6	667.5	667.5	667.5
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2392	0.2392	0.2392	0.1734
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	7.596e-003	7.596e-003	7.596e-003	8.825e-003
33	Static Gradient	(kPa/m)	3.240e-003	3.240e-003	3.240e-003	-1.295e-003
34	Accel. Gradient	(kPa/m)	1.271e-006	1.271e-006	1.271e-006	8.834e-007
35	Liq. Reynolds		1.933e+005 *	1.933e+005 *	1.933e+005 *	1.933e+005 *
36	Vap. Reynolds		9.728e+006 *	9.728e+006 *	9.728e+006 *	9.728e+006 *
37	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
38	Vapour Velocity	(m/s)	2.855	2.855	2.855	2.855
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	7888	7893	7898	7903
42	Elevation	(m)	0.2520	0.2480	0.2440	0.2400
43	Pressure	(bar)	71.75	71.75	71.75	71.75
44	Temperature	(C)	55.05	55.05	55.05	55.05
45	Vapour Fraction		0.8827	0.8827	0.8827	0.8827
46	Heat Transferred	(kJ/h-m)	667.4	667.4	667.3	667.3
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.1734	0.1734	0.1734	0.1734
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	8.825e-003	8.825e-003	8.825e-003	8.825e-003
52	Static Gradient	(kPa/m)	-1.295e-003	-1.295e-003	-1.295e-003	-1.295e-003
53	Accel. Gradient	(kPa/m)	8.834e-007	8.834e-007	8.834e-007	8.834e-007
54	Liq. Reynolds		1.933e+005 *	1.933e+005 *	1.933e+005 *	1.933e+005 *
55	Vap. Reynolds		9.728e+006 *	9.728e+006 *	9.728e+006 *	9.728e+006 *
56	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
57	Vapour Velocity	(m/s)	2.855	2.855	2.855	2.855
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	7908	7913	7918	7923
61	Elevation	(m)	0.2380	0.2360	0.2340	0.2320
62	Pressure	(bar)	71.75	71.75	71.75	71.75
63	Temperature	(C)	55.04	55.04	55.04	55.04
64	Vapour Fraction		0.8827	0.8827	0.8827	0.8827
65	Heat Transferred	(kJ/h-m)	667.2	667.1	667.1	667.0
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.1737	0.1737	0.1737	0.1737
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	8.821e-003	8.821e-003	8.821e-003	8.821e-003
15	Static Gradient (kPa/m)	-6.481e-004	-6.481e-004	-6.481e-004	-6.481e-004
16	Accel. Gradient (kPa/m)	9.588e-007	9.588e-007	9.588e-007	9.588e-007
17	Liq. Reynolds	1.933e+005 *	1.933e+005 *	1.933e+005 *	1.933e+005 *
18	Vap. Reynolds	9.728e+006 *	9.728e+006 *	9.728e+006 *	9.728e+006 *
19	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
20	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	7928	7933	7938	7943
24	Elevation (m)	0.2300	0.2520	0.2740	0.2960
25	Pressure (bar)	71.75	71.75	71.75	71.75
26	Temperature (C)	55.03	55.03	55.03	55.03
27	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
28	Heat Transferred (kJ/h-m)	667.0	666.9	666.8	666.8
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.1737	0.2471	0.2471	0.2471
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	8.821e-003	6.671e-003	6.671e-003	6.671e-003
34	Static Gradient (kPa/m)	-6.481e-004	9.186e-003	9.186e-003	9.186e-003
35	Accel. Gradient (kPa/m)	9.588e-007	1.860e-006	1.860e-006	1.860e-006
36	Liq. Reynolds	1.933e+005 *	1.933e+005 *	1.933e+005 *	1.933e+005 *
37	Vap. Reynolds	9.728e+006 *	9.728e+006 *	9.728e+006 *	9.728e+006 *
38	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
39	Vapour Velocity (m/s)	2.854	2.854	2.854	2.855
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	7948	7953	7958	7963
43	Elevation (m)	0.3180	0.3400	0.3020	0.2640
44	Pressure (bar)	71.75	71.75	71.75	71.75
45	Temperature (C)	55.03	55.02	55.02	55.02
46	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
47	Heat Transferred (kJ/h-m)	666.7	666.6	666.7	666.6
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2471	0.2471	0.1683	0.1683
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	6.671e-003	6.671e-003	8.893e-003	8.893e-003
53	Static Gradient (kPa/m)	9.186e-003	9.186e-003	-1.206e-002	-1.206e-002
54	Accel. Gradient (kPa/m)	1.860e-006	1.860e-006	-3.715e-007	-3.715e-007
55	Liq. Reynolds	1.933e+005 *	1.933e+005 *	1.933e+005 *	1.933e+005 *
56	Vap. Reynolds	9.728e+006 *	9.728e+006 *	9.728e+006 *	9.728e+006 *
57	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
58	Vapour Velocity (m/s)	2.855	2.855	2.854	2.854
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	7968	7973	7978	7983
62	Elevation (m)	0.2260	0.1880	0.1500	0.1360
63	Pressure (bar)	71.75	71.75	71.75	71.75
64	Temperature (C)	55.02	55.02	55.02	55.01
65	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
66	Heat Transferred (kJ/h-m)	666.6	666.5	666.5	666.4
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.1683	0.1683	0.1683	0.1719
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	8.893e-003	8.892e-003	8.892e-003	8.844e-003
16	Static Gradient (kPa/m)	-1.206e-002	-1.206e-002	-1.206e-002	-4.506e-003
17	Accel. Gradient (kPa/m)	-3.715e-007	-3.715e-007	-3.715e-007	5.089e-007
18	Liq. Reynolds	1.933e+005 *	1.933e+005 *	1.933e+005 *	1.933e+005 *
19	Vap. Reynolds	9.728e+006 *	9.728e+006 *	9.728e+006 *	9.728e+006 *
20	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
21	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	7988	7993	7998	8003
25	Elevation (m)	0.1220	0.1080	9.400e-002	8.000e-002
26	Pressure (bar)	71.75	71.75	71.75	71.75
27	Temperature (C)	55.01	55.01	55.01	55.01
28	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
29	Heat Transferred (kJ/h-m)	666.4	666.3	666.3	666.2
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1719	0.1719	0.1719	0.1719
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	8.844e-003	8.844e-003	8.843e-003	8.843e-003
35	Static Gradient (kPa/m)	-4.506e-003	-4.506e-003	-4.506e-003	-4.506e-003
36	Accel. Gradient (kPa/m)	5.089e-007	5.089e-007	5.089e-007	5.088e-007
37	Liq. Reynolds	1.933e+005 *	1.933e+005 *	1.933e+005 *	1.933e+005 *
38	Vap. Reynolds	9.728e+006 *	9.728e+006 *	9.728e+006 *	9.728e+006 *
39	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
40	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	8008	8013	8018	8023
44	Elevation (m)	6.600e-002	5.200e-002	3.800e-002	2.400e-002
45	Pressure (bar)	71.75	71.75	71.75	71.75
46	Temperature (C)	55.00	55.00	55.00	55.00
47	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
48	Heat Transferred (kJ/h-m)	666.2	666.1	666.1	666.0
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1719	0.1719	0.1719	0.1719
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	8.843e-003	8.843e-003	8.843e-003	8.843e-003
54	Static Gradient (kPa/m)	-4.506e-003	-4.506e-003	-4.506e-003	-4.506e-003
55	Accel. Gradient (kPa/m)	5.088e-007	5.088e-007	5.088e-007	5.088e-007
56	Liq. Reynolds	1.933e+005 *	1.933e+005 *	1.933e+005 *	1.933e+005 *
57	Vap. Reynolds	9.728e+006 *	9.728e+006 *	9.728e+006 *	9.728e+006 *
58	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
59	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	8028	8033	8038	8043
63	Elevation (m)	1.000e-002	-5.400e-002	-0.1180	-0.1820
64	Pressure (bar)	71.75	71.75	71.75	71.75
65	Temperature (C)	55.00	54.99	54.99	54.99
66	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
67	Heat Transferred (kJ/h-m)	665.9	665.9	665.9	665.8
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 161 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1719	0.1643	0.1643	0.1643
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	8.843e-003	8.948e-003	8.948e-003	8.947e-003
17	Static Gradient (kPa/m)	-4.506e-003	-2.000e-002	-2.000e-002	-2.000e-002
18	Accel. Gradient (kPa/m)	5.088e-007	-1.297e-006	-1.297e-006	-1.297e-006
19	Liq. Reynolds	1.933e+005 *	1.933e+005 *	1.933e+005 *	1.933e+005 *
20	Vap. Reynolds	9.728e+006 *	9.728e+006 *	9.728e+006 *	9.728e+006 *
21	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
22	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	8048	8053	8058	8063
26	Elevation (m)	-0.2460	-0.3100	-1.168	-2.026
27	Pressure (bar)	71.75	71.75	71.75	71.76
28	Temperature (C)	54.99	54.99	54.99	54.99
29	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
30	Heat Transferred (kJ/h-m)	665.8	665.8	666.4	666.4
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1644	0.1644	4.996e-002	4.996e-002
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	8.947e-003	8.947e-003	1.655e-002	1.655e-002
36	Static Gradient (kPa/m)	-2.000e-002	-2.000e-002	-0.1468	-0.1468
37	Accel. Gradient (kPa/m)	-1.297e-006	-1.297e-006	-1.527e-005	-1.527e-005
38	Liq. Reynolds	1.933e+005 *	1.933e+005 *	1.933e+005 *	1.933e+005 *
39	Vap. Reynolds	9.728e+006 *	9.728e+006 *	9.728e+006 *	9.728e+006 *
40	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
41	Vapour Velocity (m/s)	2.854	2.854	2.854	2.853
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	8068	8073	8078	8083
45	Elevation (m)	-2.884	-3.742	-4.600	-3.816
46	Pressure (bar)	71.77	71.77	71.78	71.74
47	Temperature (C)	55.00	55.00	55.00	54.99
48	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
49	Heat Transferred (kJ/h-m)	666.5	666.6	666.6	667.1
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	4.997e-002	4.997e-002	4.997e-002	0.6593
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	1.655e-002	1.655e-002	1.655e-002	7.246e-003
55	Static Gradient (kPa/m)	-0.1468	-0.1468	-0.1468	0.7251
56	Accel. Gradient (kPa/m)	-1.527e-005	-1.527e-005	-1.526e-005	8.591e-005
57	Liq. Reynolds	1.933e+005 *	1.933e+005 *	1.933e+005 *	1.933e+005 *
58	Vap. Reynolds	9.727e+006 *	9.727e+006 *	9.727e+006 *	9.728e+006 *
59	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
60	Vapour Velocity (m/s)	2.853	2.853	2.853	2.854
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	8088	8093	8098	8103
64	Elevation (m)	-3.032	-2.248	-1.464	-0.6800
65	Pressure (bar)	71.71	71.67	71.63	71.60
66	Temperature (C)	54.98	54.96	54.95	54.94
67	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
68	Heat Transferred (kJ/h-m)	666.7	666.4	666.1	665.8
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 162 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.6591	0.6590	0.6588	0.6586
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	7.249e-003	7.253e-003	7.256e-003	7.260e-003
18	Static Gradient (kPa/m)	0.7249	0.7247	0.7246	0.7244
19	Accel. Gradient (kPa/m)	8.598e-005	8.605e-005	8.612e-005	8.619e-005
20	Liq. Reynolds	1.932e+005 *	1.932e+005 *	1.932e+005 *	1.931e+005 *
21	Vap. Reynolds	9.729e+006 *	9.731e+006 *	9.732e+006 *	9.733e+006 *
22	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1979
23	Vapour Velocity (m/s)	2.856	2.857	2.859	2.860
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	8108	8113	8118	8123
27	Elevation (m)	-0.5040	-0.3280	-0.1520	2.400e-002
28	Pressure (bar)	71.59	71.59	71.58	71.58
29	Temperature (C)	54.94	54.93	54.93	54.92
30	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
31	Heat Transferred (kJ/h-m)	664.3	664.2	664.1	664.0
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.3329	0.3329	0.3329	0.3329
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	7.679e-003	7.679e-003	7.680e-003	7.680e-003
37	Static Gradient (kPa/m)	9.169e-002	9.168e-002	9.168e-002	9.168e-002
38	Accel. Gradient (kPa/m)	1.171e-005	1.171e-005	1.171e-005	1.171e-005
39	Liq. Reynolds	1.931e+005 *	1.931e+005 *	1.931e+005 *	1.931e+005 *
40	Vap. Reynolds	9.733e+006 *	9.733e+006 *	9.734e+006 *	9.734e+006 *
41	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
42	Vapour Velocity (m/s)	2.860	2.860	2.861	2.861
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	8128	8133	8138	8143
46	Elevation (m)	0.2000	0.2100	0.2200	0.2300
47	Pressure (bar)	71.57	71.57	71.57	71.57
48	Temperature (C)	54.92	54.92	54.92	54.91
49	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
50	Heat Transferred (kJ/h-m)	663.9	663.8	663.8	663.7
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.3328	0.2400	0.2400	0.2400
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	7.681e-003	7.480e-003	7.480e-003	7.480e-003
56	Static Gradient (kPa/m)	9.168e-002	4.076e-003	4.076e-003	4.076e-003
57	Accel. Gradient (kPa/m)	1.171e-005	1.362e-006	1.362e-006	1.362e-006
58	Liq. Reynolds	1.931e+005 *	1.931e+005 *	1.931e+005 *	1.931e+005 *
59	Vap. Reynolds	9.734e+006 *	9.734e+006 *	9.734e+006 *	9.734e+006 *
60	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
61	Vapour Velocity (m/s)	2.861	2.861	2.861	2.861
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	8148	8153	8158	8163
65	Elevation (m)	0.2400	0.2500	0.2480	0.2460
66	Pressure (bar)	71.57	71.57	71.57	71.57
67	Temperature (C)	54.91	54.91	54.91	54.91
68	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 163 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Heat Transferred (kJ/h-m)	663.7	663.6	663.6	663.6					
14	Flow Regime	Transition	Transition	Transition	Transition					
15	Water Flow Regime									
16	Liquid HoldUp	0.2400	0.2400	0.1734	0.1734					
17	Water HoldUp	---	---	---	---					
18	Friction Gradient (kPa/m)	7.480e-003	7.480e-003	8.840e-003	8.840e-003					
19	Static Gradient (kPa/m)	4.076e-003	4.076e-003	-6.472e-004	-6.472e-004					
20	Accel. Gradient (kPa/m)	1.362e-006	1.362e-006	9.657e-007	9.657e-007					
21	Liq. Reynolds	1.931e+005 *	1.931e+005 *	1.931e+005 *	1.931e+005 *					
22	Vap. Reynolds	9.734e+006 *	9.734e+006 *	9.734e+006 *	9.734e+006 *					
23	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
24	Vapour Velocity (m/s)	2.861	2.861	2.861	2.861					
25	Deposit Thickness (mm)	---	---	---	---					
26	Deposit Volume (m3)	---	---	---	---					
27	Length (m)	8168	8173	8178	8183					
28	Elevation (m)	0.2440	0.2420	0.2400	0.2380					
29	Pressure (bar)	71.57	71.57	71.57	71.57					
30	Temperature (C)	54.90	54.90	54.90	54.90					
31	Vapour Fraction	0.8827	0.8827	0.8827	0.8827					
32	Heat Transferred (kJ/h-m)	663.5	663.5	663.4	663.3					
33	Flow Regime	Transition	Transition	Transition	Transition					
34	Water Flow Regime									
35	Liquid HoldUp	0.1734	0.1734	0.1734	0.1734					
36	Water HoldUp	---	---	---	---					
37	Friction Gradient (kPa/m)	8.840e-003	8.840e-003	8.840e-003	8.840e-003					
38	Static Gradient (kPa/m)	-6.472e-004	-6.472e-004	-6.472e-004	-6.472e-004					
39	Accel. Gradient (kPa/m)	9.657e-007	9.657e-007	9.657e-007	9.657e-007					
40	Liq. Reynolds	1.931e+005 *	1.931e+005 *	1.931e+005 *	1.931e+005 *					
41	Vap. Reynolds	9.734e+006 *	9.734e+006 *	9.734e+006 *	9.734e+006 *					
42	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
43	Vapour Velocity (m/s)	2.861	2.861	2.861	2.861					
44	Deposit Thickness (mm)	---	---	---	---					
45	Deposit Volume (m3)	---	---	---	---					
46	Length (m)	8188	8193	8198	8203					
47	Elevation (m)	0.2360	0.2340	0.2320	0.2300					
48	Pressure (bar)	71.57	71.57	71.57	71.57					
49	Temperature (C)	54.90	54.89	54.89	54.89					
50	Vapour Fraction	0.8827	0.8827	0.8827	0.8827					
51	Heat Transferred (kJ/h-m)	663.3	663.2	663.2	663.1					
52	Flow Regime	Transition	Transition	Transition	Transition					
53	Water Flow Regime									
54	Liquid HoldUp	0.1734	0.1734	0.1734	0.1734					
55	Water HoldUp	---	---	---	---					
56	Friction Gradient (kPa/m)	8.840e-003	8.840e-003	8.840e-003	8.839e-003					
57	Static Gradient (kPa/m)	-6.472e-004	-6.472e-004	-6.472e-004	-6.472e-004					
58	Accel. Gradient (kPa/m)	9.657e-007	9.657e-007	9.657e-007	9.657e-007					
59	Liq. Reynolds	1.930e+005 *	1.930e+005 *	1.930e+005 *	1.930e+005 *					
60	Vap. Reynolds	9.734e+006 *	9.734e+006 *	9.734e+006 *	9.734e+006 *					
61	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
62	Vapour Velocity (m/s)	2.861	2.861	2.861	2.861					
63	Deposit Thickness (mm)	---	---	---	---					
64	Deposit Volume (m3)	---	---	---	---					
65	Length (m)	8208	8213	8218	8223					
66	Elevation (m)	0.2440	0.2580	0.2720	0.2860					
67	Pressure (bar)	71.57	71.56	71.56	71.56					
68	Temperature (C)	54.89	54.89	54.88	54.88					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
14	Heat Transferred (kJ/h-m)	663.0	663.0	662.9	662.9
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2422	0.2422	0.2422	0.2422
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	7.214e-003	7.214e-003	7.214e-003	7.214e-003
20	Static Gradient (kPa/m)	5.722e-003	5.722e-003	5.722e-003	5.722e-003
21	Accel. Gradient (kPa/m)	1.525e-006	1.525e-006	1.525e-006	1.525e-006
22	Liq. Reynolds	1.930e+005 *	1.930e+005 *	1.930e+005 *	1.930e+005 *
23	Vap. Reynolds	9.734e+006 *	9.734e+006 *	9.734e+006 *	9.734e+006 *
24	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
25	Vapour Velocity (m/s)	2.861	2.861	2.861	2.861
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	8228	8233	8238	8243
29	Elevation (m)	0.3000	0.3000	0.3000	0.3000
30	Pressure (bar)	71.56	71.56	71.56	71.56
31	Temperature (C)	54.88	54.88	54.88	54.87
32	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
33	Heat Transferred (kJ/h-m)	662.8	662.7	662.7	662.6
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2422	0.2536	0.2536	0.2536
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	7.214e-003	5.957e-003	5.957e-003	5.957e-003
39	Static Gradient (kPa/m)	5.722e-003	0.0000	0.0000	0.0000
40	Accel. Gradient (kPa/m)	1.525e-006	7.023e-007	7.023e-007	7.023e-007
41	Liq. Reynolds	1.930e+005 *	1.930e+005 *	1.930e+005 *	1.930e+005 *
42	Vap. Reynolds	9.734e+006 *	9.734e+006 *	9.734e+006 *	9.734e+006 *
43	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
44	Vapour Velocity (m/s)	2.861	2.861	2.861	2.861
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	8248	8253	8258	8263
48	Elevation (m)	0.3000	0.3000	0.3080	0.3160
49	Pressure (bar)	71.56	71.56	71.56	71.56
50	Temperature (C)	54.87	54.87	54.87	54.87
51	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
52	Heat Transferred (kJ/h-m)	662.6	662.5	662.5	662.4
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2536	0.2536	0.2388	0.2388
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	5.957e-003	5.957e-003	7.619e-003	7.619e-003
58	Static Gradient (kPa/m)	0.0000	0.0000	3.237e-003	3.237e-003
59	Accel. Gradient (kPa/m)	7.023e-007	7.023e-007	1.280e-006	1.280e-006
60	Liq. Reynolds	1.930e+005 *	1.930e+005 *	1.930e+005 *	1.930e+005 *
61	Vap. Reynolds	9.734e+006 *	9.734e+006 *	9.734e+006 *	9.734e+006 *
62	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
63	Vapour Velocity (m/s)	2.861	2.861	2.861	2.861
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	8268	8273	8278	8283
67	Elevation (m)	0.3240	0.3320	0.3400	0.3340
68	Pressure (bar)	71.56	71.56	71.56	71.56
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	54.86	54.86	54.86	54.86
14	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
15	Heat Transferred (kJ/h-m)	662.4	662.3	662.3	662.3
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2388	0.2388	0.2388	0.1728
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	7.619e-003	7.619e-003	7.619e-003	8.847e-003
21	Static Gradient (kPa/m)	3.237e-003	3.237e-003	3.237e-003	-1.937e-003
22	Accel. Gradient (kPa/m)	1.280e-006	1.280e-006	1.280e-006	8.146e-007
23	Liq. Reynolds	1.930e+005 *	1.930e+005 *	1.930e+005 *	1.930e+005 *
24	Vap. Reynolds	9.735e+006 *	9.735e+006 *	9.735e+006 *	9.735e+006 *
25	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
26	Vapour Velocity (m/s)	2.861	2.861	2.861	2.861
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	8288	8293	8298	8303
30	Elevation (m)	0.3280	0.3220	0.3160	0.3100
31	Pressure (bar)	71.56	71.56	71.56	71.56
32	Temperature (C)	54.86	54.85	54.85	54.85
33	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
34	Heat Transferred (kJ/h-m)	662.2	662.2	662.1	662.1
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1728	0.1728	0.1728	0.1728
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	8.847e-003	8.847e-003	8.847e-003	8.847e-003
40	Static Gradient (kPa/m)	-1.937e-003	-1.937e-003	-1.937e-003	-1.937e-003
41	Accel. Gradient (kPa/m)	8.145e-007	8.145e-007	8.145e-007	8.145e-007
42	Liq. Reynolds	1.930e+005 *	1.930e+005 *	1.930e+005 *	1.930e+005 *
43	Vap. Reynolds	9.735e+006 *	9.735e+006 *	9.735e+006 *	9.735e+006 *
44	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
45	Vapour Velocity (m/s)	2.861	2.861	2.861	2.861
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	8308	8313	8318	8323
49	Elevation (m)	0.3220	0.3340	0.3460	0.3580
50	Pressure (bar)	71.56	71.56	71.56	71.55
51	Temperature (C)	54.85	54.85	54.84	54.84
52	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
53	Heat Transferred (kJ/h-m)	661.9	661.9	661.8	661.8
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2411	0.2411	0.2411	0.2411
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	7.347e-003	7.347e-003	7.347e-003	7.347e-003
59	Static Gradient (kPa/m)	4.888e-003	4.888e-003	4.888e-003	4.888e-003
60	Accel. Gradient (kPa/m)	1.442e-006	1.442e-006	1.442e-006	1.442e-006
61	Liq. Reynolds	1.930e+005 *	1.930e+005 *	1.930e+005 *	1.930e+005 *
62	Vap. Reynolds	9.735e+006 *	9.735e+006 *	9.735e+006 *	9.735e+006 *
63	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
64	Vapour Velocity (m/s)	2.861	2.861	2.861	2.861
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	8328	8333	8338	8343
68	Elevation (m)	0.3700	0.3080	0.2460	0.1840
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Pressure (bar)	71.55	71.55	71.55	71.56					
14	Temperature (C)	54.84	54.84	54.84	54.83					
15	Vapour Fraction	0.8827	0.8827	0.8827	0.8827					
16	Heat Transferred (kJ/h-m)	661.7	661.7	661.7	661.7					
17	Flow Regime	Transition	Transition	Transition	Transition					
18	Water Flow Regime									
19	Liquid HoldUp	0.2411	0.1644	0.1644	0.1644					
20	Water HoldUp	---	---	---	---					
21	Friction Gradient (kPa/m)	7.347e-003	8.963e-003	8.963e-003	8.963e-003					
22	Static Gradient (kPa/m)	4.888e-003	-1.939e-002	-1.939e-002	-1.940e-002					
23	Accel. Gradient (kPa/m)	1.442e-006	-1.230e-006	-1.230e-006	-1.230e-006					
24	Liq. Reynolds	1.930e+005 *	1.930e+005 *	1.930e+005 *	1.930e+005 *					
25	Vap. Reynolds	9.735e+006 *	9.735e+006 *	9.735e+006 *	9.735e+006 *					
26	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
27	Vapour Velocity (m/s)	2.861	2.861	2.861	2.861					
28	Deposit Thickness (mm)	---	---	---	---					
29	Deposit Volume (m3)	---	---	---	---					
30	Length (m)	8348	8353	8358	8363					
31	Elevation (m)	0.1220	6.000e-002	6.000e-002	6.000e-002					
32	Pressure (bar)	71.56	71.56	71.56	71.56					
33	Temperature (C)	54.83	54.83	54.83	54.83					
34	Vapour Fraction	0.8827	0.8827	0.8827	0.8827					
35	Heat Transferred (kJ/h-m)	661.6	661.6	661.5	661.4					
36	Flow Regime	Transition	Transition	Transition	Transition					
37	Water Flow Regime									
38	Liquid HoldUp	0.1644	0.1644	0.2537	0.2537					
39	Water HoldUp	---	---	---	---					
40	Friction Gradient (kPa/m)	8.962e-003	8.962e-003	5.956e-003	5.956e-003					
41	Static Gradient (kPa/m)	-1.940e-002	-1.940e-002	0.0000	0.0000					
42	Accel. Gradient (kPa/m)	-1.230e-006	-1.230e-006	7.021e-007	7.021e-007					
43	Liq. Reynolds	1.930e+005 *	1.930e+005 *	1.930e+005 *	1.930e+005 *					
44	Vap. Reynolds	9.735e+006 *	9.735e+006 *	9.735e+006 *	9.735e+006 *					
45	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
46	Vapour Velocity (m/s)	2.861	2.860	2.860	2.860					
47	Deposit Thickness (mm)	---	---	---	---					
48	Deposit Volume (m3)	---	---	---	---					
49	Length (m)	8368	8373	8378	8383					
50	Elevation (m)	6.000e-002	6.000e-002	6.000e-002	4.000e-002					
51	Pressure (bar)	71.56	71.56	71.56	71.55					
52	Temperature (C)	54.83	54.82	54.82	54.82					
53	Vapour Fraction	0.8827	0.8827	0.8827	0.8827					
54	Heat Transferred (kJ/h-m)	661.3	661.3	661.2	661.3					
55	Flow Regime	Transition	Transition	Transition	Transition					
56	Water Flow Regime									
57	Liquid HoldUp	0.2537	0.2537	0.2537	0.1707					
58	Water HoldUp	---	---	---	---					
59	Friction Gradient (kPa/m)	5.956e-003	5.956e-003	5.956e-003	8.873e-003					
60	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-6.406e-003					
61	Accel. Gradient (kPa/m)	7.021e-007	7.021e-007	7.021e-007	2.908e-007					
62	Liq. Reynolds	1.930e+005 *	1.930e+005 *	1.930e+005 *	1.930e+005 *					
63	Vap. Reynolds	9.735e+006 *	9.735e+006 *	9.735e+006 *	9.735e+006 *					
64	Liquid Velocity (m/s)	0.1979	0.1979	0.1980	0.1980					
65	Vapour Velocity (m/s)	2.860	2.860	2.860	2.860					
66	Deposit Thickness (mm)	---	---	---	---					
67	Deposit Volume (m3)	---	---	---	---					
68	Length (m)	8388	8393	8398	8403					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Elevation (m)	2.000e-002	-1.499e-015	-2.000e-002	-4.000e-002		
14	Pressure (bar)	71.55	71.55	71.55	71.55		
15	Temperature (C)	54.82	54.82	54.81	54.81		
16	Vapour Fraction	0.8827	0.8827	0.8827	0.8827		
17	Heat Transferred (kJ/h-m)	661.2	661.2	661.1	661.1		
18	Flow Regime	Transition	Transition	Transition	Transition		
19	Water Flow Regime						
20	Liquid HoldUp	0.1707	0.1707	0.1708	0.1708		
21	Water HoldUp	---	---	---	---		
22	Friction Gradient (kPa/m)	8.873e-003	8.873e-003	8.873e-003	8.873e-003		
23	Static Gradient (kPa/m)	-6.406e-003	-6.406e-003	-6.406e-003	-6.406e-003		
24	Accel. Gradient (kPa/m)	2.908e-007	2.908e-007	2.908e-007	2.908e-007		
25	Liq. Reynolds	1.930e+005 *	1.930e+005 *	1.930e+005 *	1.930e+005 *		
26	Vap. Reynolds	9.735e+006 *	9.735e+006 *	9.735e+006 *	9.735e+006 *		
27	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
28	Vapour Velocity (m/s)	2.860	2.860	2.860	2.860		
29	Deposit Thickness (mm)	---	---	---	---		
30	Deposit Volume (m3)	---	---	---	---		
31	Length (m)	8408	8413	8418	8423		
32	Elevation (m)	-6.400e-002	-8.800e-002	-0.1120	-0.1360		
33	Pressure (bar)	71.55	71.55	71.55	71.55		
34	Temperature (C)	54.81	54.81	54.81	54.80		
35	Vapour Fraction	0.8827	0.8827	0.8827	0.8827		
36	Heat Transferred (kJ/h-m)	661.0	661.0	660.9	660.9		
37	Flow Regime	Transition	Transition	Transition	Transition		
38	Water Flow Regime						
39	Liquid HoldUp	0.1701	0.1702	0.1702	0.1702		
40	Water HoldUp	---	---	---	---		
41	Friction Gradient (kPa/m)	8.881e-003	8.881e-003	8.881e-003	8.881e-003		
42	Static Gradient (kPa/m)	-7.670e-003	-7.670e-003	-7.670e-003	-7.670e-003		
43	Accel. Gradient (kPa/m)	1.428e-007	1.428e-007	1.428e-007	1.427e-007		
44	Liq. Reynolds	1.930e+005 *	1.930e+005 *	1.930e+005 *	1.930e+005 *		
45	Vap. Reynolds	9.735e+006 *	9.735e+006 *	9.735e+006 *	9.735e+006 *		
46	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
47	Vapour Velocity (m/s)	2.860	2.860	2.860	2.860		
48	Deposit Thickness (mm)	---	---	---	---		
49	Deposit Volume (m3)	---	---	---	---		
50	Length (m)	8428	8433	8438	8443		
51	Elevation (m)	-0.1600	-0.1680	-0.1760	-0.1840		
52	Pressure (bar)	71.55	71.55	71.55	71.55		
53	Temperature (C)	54.80	54.80	54.80	54.80		
54	Vapour Fraction	0.8827	0.8827	0.8827	0.8827		
55	Heat Transferred (kJ/h-m)	660.8	660.8	660.7	660.6		
56	Flow Regime	Transition	Transition	Transition	Transition		
57	Water Flow Regime						
58	Liquid HoldUp	0.1702	0.1726	0.1726	0.1726		
59	Water HoldUp	---	---	---	---		
60	Friction Gradient (kPa/m)	8.881e-003	8.849e-003	8.849e-003	8.849e-003		
61	Static Gradient (kPa/m)	-7.670e-003	-2.581e-003	-2.581e-003	-2.581e-003		
62	Accel. Gradient (kPa/m)	1.427e-007	7.389e-007	7.389e-007	7.389e-007		
63	Liq. Reynolds	1.930e+005 *	1.930e+005 *	1.930e+005 *	1.930e+005 *		
64	Vap. Reynolds	9.735e+006 *	9.735e+006 *	9.735e+006 *	9.735e+006 *		
65	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
66	Vapour Velocity (m/s)	2.860	2.860	2.860	2.860		
67	Deposit Thickness (mm)	---	---	---	---		
68	Deposit Volume (m3)	---	---	---	---		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Length (m)	8448	8453	8458	8463					
14	Elevation (m)	-0.1920	-0.2000	-0.1640	-0.1280					
15	Pressure (bar)	71.55	71.55	71.55	71.55					
16	Temperature (C)	54.80	54.79	54.79	54.79					
17	Vapour Fraction	0.8827	0.8827	0.8827	0.8827					
18	Heat Transferred (kJ/h-m)	660.6	660.5	660.4	660.4					
19	Flow Regime	Transition	Transition	Transition	Transition					
20	Water Flow Regime									
21	Liquid HoldUp	0.1726	0.1726	0.2546	0.2546					
22	Water HoldUp	---	---	---	---					
23	Friction Gradient (kPa/m)	8.849e-003	8.849e-003	6.012e-003	6.012e-003					
24	Static Gradient (kPa/m)	-2.581e-003	-2.581e-003	1.527e-002	1.527e-002					
25	Accel. Gradient (kPa/m)	7.388e-007	7.388e-007	2.508e-006	2.508e-006					
26	Liq. Reynolds	1.930e+005 *	1.930e+005 *	1.930e+005 *	1.930e+005 *					
27	Vap. Reynolds	9.735e+006 *	9.735e+006 *	9.735e+006 *	9.735e+006 *					
28	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
29	Vapour Velocity (m/s)	2.860	2.860	2.860	2.860					
30	Deposit Thickness (mm)	---	---	---	---					
31	Deposit Volume (m3)	---	---	---	---					
32	Length (m)	8468	8473	8478	8483					
33	Elevation (m)	-9.200e-002	-5.600e-002	-2.000e-002	-1.800e-002					
34	Pressure (bar)	71.55	71.55	71.55	71.55					
35	Temperature (C)	54.79	54.78	54.78	54.78					
36	Vapour Fraction	0.8827	0.8827	0.8827	0.8827					
37	Heat Transferred (kJ/h-m)	660.3	660.2	660.2	660.1					
38	Flow Regime	Transition	Transition	Transition	Transition					
39	Water Flow Regime									
40	Liquid HoldUp	0.2546	0.2546	0.2546	0.2355					
41	Water HoldUp	---	---	---	---					
42	Friction Gradient (kPa/m)	6.012e-003	6.012e-003	6.012e-003	8.039e-003					
43	Static Gradient (kPa/m)	1.527e-002	1.527e-002	1.527e-002	8.010e-004					
44	Accel. Gradient (kPa/m)	2.508e-006	2.508e-006	2.508e-006	1.042e-006					
45	Liq. Reynolds	1.930e+005 *	1.930e+005 *	1.930e+005 *	1.930e+005 *					
46	Vap. Reynolds	9.735e+006 *	9.735e+006 *	9.735e+006 *	9.735e+006 *					
47	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
48	Vapour Velocity (m/s)	2.860	2.860	2.860	2.860					
49	Deposit Thickness (mm)	---	---	---	---					
50	Deposit Volume (m3)	---	---	---	---					
51	Length (m)	8488	8493	8498	8503					
52	Elevation (m)	-1.600e-002	-1.400e-002	-1.200e-002	-1.000e-002					
53	Pressure (bar)	71.55	71.55	71.55	71.55					
54	Temperature (C)	54.78	54.78	54.77	54.77					
55	Vapour Fraction	0.8827	0.8827	0.8827	0.8827					
56	Heat Transferred (kJ/h-m)	660.1	660.0	660.0	659.9					
57	Flow Regime	Transition	Transition	Transition	Transition					
58	Water Flow Regime									
59	Liquid HoldUp	0.2355	0.2355	0.2355	0.2355					
60	Water HoldUp	---	---	---	---					
61	Friction Gradient (kPa/m)	8.039e-003	8.039e-003	8.039e-003	8.039e-003					
62	Static Gradient (kPa/m)	8.010e-004	8.010e-004	8.010e-004	8.010e-004					
63	Accel. Gradient (kPa/m)	1.042e-006	1.042e-006	1.042e-006	1.042e-006					
64	Liq. Reynolds	1.930e+005 *	1.930e+005 *	1.930e+005 *	1.930e+005 *					
65	Vap. Reynolds	9.735e+006 *	9.735e+006 *	9.735e+006 *	9.735e+006 *					
66	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
67	Vapour Velocity (m/s)	2.860	2.860	2.860	2.860					
68	Deposit Thickness (mm)	---	---	---	---					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	8508	8513	8518	8523
15	Elevation (m)	-1.000e-002	-1.000e-002	-1.000e-002	-1.000e-002
16	Pressure (bar)	71.54	71.54	71.54	71.54
17	Temperature (C)	54.77	54.77	54.77	54.76
18	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
19	Heat Transferred (kJ/h-m)	659.9	659.8	659.8	659.7
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2537	0.2537	0.2537	0.2537
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	5.955e-003	5.955e-003	5.955e-003	5.955e-003
25	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
26	Accel. Gradient (kPa/m)	7.020e-007	7.020e-007	7.020e-007	7.020e-007
27	Liq. Reynolds	1.929e+005 *	1.929e+005 *	1.929e+005 *	1.929e+005 *
28	Vap. Reynolds	9.735e+006 *	9.735e+006 *	9.735e+006 *	9.735e+006 *
29	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
30	Vapour Velocity (m/s)	2.860	2.860	2.860	2.860
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	8528	8533	8538	8543
34	Elevation (m)	-1.000e-002	-1.400e-002	-1.800e-002	-2.200e-002
35	Pressure (bar)	71.54	71.54	71.54	71.54
36	Temperature (C)	54.76	54.76	54.76	54.76
37	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
38	Heat Transferred (kJ/h-m)	659.7	659.7	659.6	659.5
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2537	0.1732	0.1732	0.1732
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	5.955e-003	8.841e-003	8.841e-003	8.841e-003
44	Static Gradient (kPa/m)	0.0000	-1.293e-003	-1.293e-003	-1.293e-003
45	Accel. Gradient (kPa/m)	7.020e-007	8.897e-007	8.897e-007	8.897e-007
46	Liq. Reynolds	1.929e+005 *	1.929e+005 *	1.929e+005 *	1.929e+005 *
47	Vap. Reynolds	9.735e+006 *	9.735e+006 *	9.735e+006 *	9.735e+006 *
48	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
49	Vapour Velocity (m/s)	2.860	2.860	2.860	2.860
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	8548	8553	8558	8563
53	Elevation (m)	-2.600e-002	-3.000e-002	-2.800e-002	-2.600e-002
54	Pressure (bar)	71.54	71.54	71.54	71.54
55	Temperature (C)	54.75	54.75	54.75	54.75
56	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
57	Heat Transferred (kJ/h-m)	659.5	659.4	659.3	659.3
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.1732	0.1732	0.2355	0.2355
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	8.841e-003	8.841e-003	8.039e-003	8.039e-003
63	Static Gradient (kPa/m)	-1.293e-003	-1.293e-003	7.981e-004	7.981e-004
64	Accel. Gradient (kPa/m)	8.897e-007	8.897e-007	1.042e-006	1.042e-006
65	Liq. Reynolds	1.929e+005 *	1.929e+005 *	1.929e+005 *	1.929e+005 *
66	Vap. Reynolds	9.735e+006 *	9.735e+006 *	9.735e+006 *	9.735e+006 *
67	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
68	Vapour Velocity (m/s)	2.860	2.860	2.860	2.860
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	8568	8573	8578	8583
16	Elevation (m)	-2.400e-002	-2.200e-002	-2.000e-002	-2.200e-002
17	Pressure (bar)	71.54	71.54	71.54	71.54
18	Temperature (C)	54.75	54.74	54.74	54.74
19	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
20	Heat Transferred (kJ/h-m)	659.2	659.2	659.1	659.1
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2355	0.2355	0.2355	0.1735
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	8.039e-003	8.039e-003	8.039e-003	8.837e-003
26	Static Gradient (kPa/m)	7.981e-004	7.981e-004	7.981e-004	-6.474e-004
27	Accel. Gradient (kPa/m)	1.042e-006	1.042e-006	1.042e-006	9.654e-007
28	Liq. Reynolds	1.929e+005 *	1.929e+005 *	1.929e+005 *	1.929e+005 *
29	Vap. Reynolds	9.736e+006 *	9.736e+006 *	9.736e+006 *	9.736e+006 *
30	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
31	Vapour Velocity (m/s)	2.860	2.860	2.860	2.860
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	8588	8593	8598	8603
35	Elevation (m)	-2.400e-002	-2.600e-002	-2.800e-002	-3.000e-002
36	Pressure (bar)	71.54	71.54	71.54	71.54
37	Temperature (C)	54.74	54.74	54.73	54.73
38	Vapour Fraction	0.8827	0.8827	0.8827	0.8827
39	Heat Transferred (kJ/h-m)	659.1	659.0	659.0	658.9
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.1735	0.1735	0.1735	0.1735
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	8.837e-003	8.837e-003	8.837e-003	8.837e-003
45	Static Gradient (kPa/m)	-6.474e-004	-6.474e-004	-6.474e-004	-6.474e-004
46	Accel. Gradient (kPa/m)	9.654e-007	9.654e-007	9.654e-007	9.654e-007
47	Liq. Reynolds	1.929e+005 *	1.929e+005 *	1.929e+005 *	1.929e+005 *
48	Vap. Reynolds	9.736e+006 *	9.736e+006 *	9.736e+006 *	9.736e+006 *
49	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
50	Vapour Velocity (m/s)	2.860	2.860	2.860	2.860
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	8608	8613	8618	8623
54	Elevation (m)	-2.600e-002	-2.200e-002	-1.800e-002	-1.400e-002
55	Pressure (bar)	71.54	71.54	71.54	71.54
56	Temperature (C)	54.73	54.73	54.73	54.72
57	Vapour Fraction	0.8826	0.8826	0.8826	0.8826
58	Heat Transferred (kJ/h-m)	658.8	658.7	658.7	658.6
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2367	0.2367	0.2367	0.2367
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	7.896e-003	7.896e-003	7.896e-003	7.896e-003
64	Static Gradient (kPa/m)	1.608e-003	1.608e-003	1.608e-003	1.608e-003
65	Accel. Gradient (kPa/m)	1.120e-006	1.120e-006	1.120e-006	1.120e-006
66	Liq. Reynolds	1.929e+005 *	1.929e+005 *	1.929e+005 *	1.929e+005 *
67	Vap. Reynolds	9.736e+006 *	9.736e+006 *	9.736e+006 *	9.736e+006 *
68	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.860	2.860	2.860	2.860	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	8628	8633	8638	8643	
17	Elevation	(m)	-1.000e-002	-8.000e-003	-6.000e-003	-4.000e-003	
18	Pressure	(bar)	71.54	71.53	71.53	71.53	
19	Temperature	(C)	54.72	54.72	54.72	54.72	
20	Vapour Fraction		0.8826	0.8826	0.8826	0.8826	
21	Heat Transferred	(kJ/h-m)	658.6	658.5	658.5	658.4	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2367	0.2355	0.2355	0.2355	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	7.895e-003	8.038e-003	8.038e-003	8.038e-003	
27	Static Gradient	(kPa/m)	1.608e-003	8.011e-004	8.011e-004	8.011e-004	
28	Accel. Gradient	(kPa/m)	1.120e-006	1.042e-006	1.042e-006	1.042e-006	
29	Liq. Reynolds		1.929e+005 *	1.929e+005 *	1.929e+005 *	1.929e+005 *	
30	Vap. Reynolds		9.736e+006 *	9.736e+006 *	9.736e+006 *	9.736e+006 *	
31	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
32	Vapour Velocity	(m/s)	2.860	2.860	2.860	2.860	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	8648	8653	8658	8663	
36	Elevation	(m)	-2.000e-003	-1.518e-015	2.000e-003	4.000e-003	
37	Pressure	(bar)	71.53	71.53	71.53	71.53	
38	Temperature	(C)	54.71	54.71	54.71	54.71	
39	Vapour Fraction		0.8826	0.8826	0.8826	0.8826	
40	Heat Transferred	(kJ/h-m)	658.4	658.3	658.3	658.2	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2355	0.2355	0.2355	0.2355	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	8.038e-003	8.038e-003	8.038e-003	8.038e-003	
46	Static Gradient	(kPa/m)	8.011e-004	8.011e-004	8.011e-004	8.011e-004	
47	Accel. Gradient	(kPa/m)	1.042e-006	1.042e-006	1.042e-006	1.042e-006	
48	Liq. Reynolds		1.929e+005 *	1.929e+005 *	1.929e+005 *	1.929e+005 *	
49	Vap. Reynolds		9.736e+006 *	9.736e+006 *	9.736e+006 *	9.736e+006 *	
50	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
51	Vapour Velocity	(m/s)	2.860	2.860	2.860	2.860	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	8668	8673	8678	8683	
55	Elevation	(m)	6.000e-003	8.000e-003	1.000e-002	1.000e-002	
56	Pressure	(bar)	71.53	71.53	71.53	71.53	
57	Temperature	(C)	54.71	54.70	54.70	54.70	
58	Vapour Fraction		0.8826	0.8826	0.8826	0.8826	
59	Heat Transferred	(kJ/h-m)	658.2	658.1	658.1	658.0	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.2355	0.2355	0.2355	0.2537	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	8.038e-003	8.038e-003	8.038e-003	5.955e-003	
65	Static Gradient	(kPa/m)	8.011e-004	8.011e-004	8.011e-004	0.0000	
66	Accel. Gradient	(kPa/m)	1.042e-006	1.042e-006	1.042e-006	7.020e-007	
67	Liq. Reynolds		1.929e+005 *	1.929e+005 *	1.929e+005 *	1.929e+005 *	
68	Vap. Reynolds		9.736e+006 *	9.736e+006 *	9.736e+006 *	9.736e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:13:43 2022	
6						
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
14	Vapour Velocity	(m/s)	2.860	2.860	2.860	2.860
15	Deposit Thickness	(mm)	---	---	---	---
16	Deposit Volume	(m3)	---	---	---	---
17	Length	(m)	8688	8693	8698	8703
18	Elevation	(m)	1.000e-002	1.000e-002	1.000e-002	1.000e-002
19	Pressure	(bar)	71.53	71.53	71.53	71.53
20	Temperature	(C)	54.70	54.70	54.69	54.69
21	Vapour Fraction		0.8826	0.8826	0.8826	0.8826
22	Heat Transferred	(kJ/h-m)	657.9	657.9	657.8	657.8
23	Flow Regime		Transition	Transition	Transition	Transition
24	Water Flow Regime					
25	Liquid HoldUp		0.2537	0.2537	0.2537	0.2537
26	Water HoldUp		---	---	---	---
27	Friction Gradient	(kPa/m)	5.955e-003	5.954e-003	5.954e-003	5.954e-003
28	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000
29	Accel. Gradient	(kPa/m)	7.020e-007	7.020e-007	7.020e-007	7.020e-007
30	Liq. Reynolds		1.929e+005 *	1.929e+005 *	1.929e+005 *	1.929e+005 *
31	Vap. Reynolds		9.736e+006 *	9.736e+006 *	9.736e+006 *	9.736e+006 *
32	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
33	Vapour Velocity	(m/s)	2.860	2.860	2.860	2.860
34	Deposit Thickness	(mm)	---	---	---	---
35	Deposit Volume	(m3)	---	---	---	---
36	Length	(m)	8708	8713	8718	8723
37	Elevation	(m)	1.200e-002	1.400e-002	1.600e-002	1.800e-002
38	Pressure	(bar)	71.53	71.53	71.53	71.53
39	Temperature	(C)	54.69	54.69	54.69	54.68
40	Vapour Fraction		0.8826	0.8826	0.8826	0.8826
41	Heat Transferred	(kJ/h-m)	657.7	657.7	657.6	657.6
42	Flow Regime		Transition	Transition	Transition	Transition
43	Water Flow Regime					
44	Liquid HoldUp		0.2355	0.2355	0.2355	0.2355
45	Water HoldUp		---	---	---	---
46	Friction Gradient	(kPa/m)	8.038e-003	8.038e-003	8.038e-003	8.038e-003
47	Static Gradient	(kPa/m)	8.011e-004	8.011e-004	8.011e-004	8.011e-004
48	Accel. Gradient	(kPa/m)	1.042e-006	1.042e-006	1.042e-006	1.042e-006
49	Liq. Reynolds		1.929e+005 *	1.929e+005 *	1.929e+005 *	1.929e+005 *
50	Vap. Reynolds		9.736e+006 *	9.736e+006 *	9.736e+006 *	9.736e+006 *
51	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
52	Vapour Velocity	(m/s)	2.860	2.860	2.860	2.860
53	Deposit Thickness	(mm)	---	---	---	---
54	Deposit Volume	(m3)	---	---	---	---
55	Length	(m)	8728	8733	8738	8743
56	Elevation	(m)	2.000e-002	1.600e-002	1.200e-002	8.000e-003
57	Pressure	(bar)	71.53	71.53	71.53	71.53
58	Temperature	(C)	54.68	54.68	54.68	54.68
59	Vapour Fraction		0.8826	0.8826	0.8826	0.8826
60	Heat Transferred	(kJ/h-m)	657.5	657.5	657.5	657.4
61	Flow Regime		Transition	Transition	Transition	Transition
62	Water Flow Regime					
63	Liquid HoldUp		0.2356	0.1732	0.1732	0.1732
64	Water HoldUp		---	---	---	---
65	Friction Gradient	(kPa/m)	8.038e-003	8.840e-003	8.840e-003	8.840e-003
66	Static Gradient	(kPa/m)	8.011e-004	-1.293e-003	-1.293e-003	-1.293e-003
67	Accel. Gradient	(kPa/m)	1.042e-006	8.897e-007	8.897e-007	8.897e-007
68	Liq. Reynolds		1.929e+005 *	1.929e+005 *	1.929e+005 *	1.929e+005 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.736e+006 *	9.736e+006 *	9.736e+006 *	9.736e+006 *
14	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
15	Vapour Velocity (m/s)	2.860	2.860	2.860	2.860
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	8748	8753	8758	8763
19	Elevation (m)	4.000e-003	-1.516e-015	-6.000e-003	-1.200e-002
20	Pressure (bar)	71.53	71.52	71.52	71.52
21	Temperature (C)	54.67	54.67	54.67	54.67
22	Vapour Fraction	0.8826	0.8826	0.8826	0.8826
23	Heat Transferred (kJ/h-m)	657.4	657.3	657.3	657.2
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1732	0.1732	0.1729	0.1729
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	8.840e-003	8.840e-003	8.844e-003	8.844e-003
29	Static Gradient (kPa/m)	-1.293e-003	-1.293e-003	-1.939e-003	-1.939e-003
30	Accel. Gradient (kPa/m)	8.897e-007	8.897e-007	8.140e-007	8.140e-007
31	Liq. Reynolds	1.929e+005 *	1.929e+005 *	1.929e+005 *	1.929e+005 *
32	Vap. Reynolds	9.736e+006 *	9.736e+006 *	9.736e+006 *	9.736e+006 *
33	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
34	Vapour Velocity (m/s)	2.860	2.860	2.860	2.860
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	8768	8773	8778	8783
38	Elevation (m)	-1.800e-002	-2.400e-002	-3.000e-002	-3.800e-002
39	Pressure (bar)	71.52	71.52	71.52	71.52
40	Temperature (C)	54.67	54.66	54.66	54.66
41	Vapour Fraction	0.8826	0.8826	0.8826	0.8826
42	Heat Transferred (kJ/h-m)	657.2	657.1	657.1	657.0
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1729	0.1729	0.1729	0.1726
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	8.844e-003	8.844e-003	8.844e-003	8.847e-003
48	Static Gradient (kPa/m)	-1.939e-003	-1.939e-003	-1.939e-003	-2.581e-003
49	Accel. Gradient (kPa/m)	8.140e-007	8.140e-007	8.140e-007	7.388e-007
50	Liq. Reynolds	1.929e+005 *	1.929e+005 *	1.929e+005 *	1.929e+005 *
51	Vap. Reynolds	9.736e+006 *	9.736e+006 *	9.736e+006 *	9.736e+006 *
52	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
53	Vapour Velocity (m/s)	2.860	2.860	2.859	2.859
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	8788	8793	8798	8803
57	Elevation (m)	-4.600e-002	-5.400e-002	-6.200e-002	-7.000e-002
58	Pressure (bar)	71.52	71.52	71.52	71.52
59	Temperature (C)	54.66	54.66	54.65	54.65
60	Vapour Fraction	0.8826	0.8826	0.8826	0.8826
61	Heat Transferred (kJ/h-m)	656.9	656.9	656.8	656.8
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1726	0.1726	0.1726	0.1726
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	8.847e-003	8.847e-003	8.847e-003	8.847e-003
67	Static Gradient (kPa/m)	-2.581e-003	-2.581e-003	-2.581e-003	-2.581e-003
68	Accel. Gradient (kPa/m)	7.388e-007	7.387e-007	7.387e-007	7.387e-007
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:13:43 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		1.929e+005 *	1.929e+005 *	1.929e+005 *	1.929e+005 *	
14	Vap. Reynolds		9.736e+006 *	9.736e+006 *	9.736e+006 *	9.736e+006 *	
15	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	0.1980	
16	Vapour Velocity (m/s)	2.859	2.859	2.859	2.859	2.859	
17	Deposit Thickness (mm)	---	---	---	---	---	
18	Deposit Volume (m3)	---	---	---	---	---	
19	Length (m)	8805	8807	8809	8811		
20	Elevation (m)	-6.000e-002	-5.000e-002	-4.000e-002	-3.000e-002		
21	Pressure (bar)	71.52	71.52	71.52	71.52		
22	Temperature (C)	54.65	54.65	54.65	54.65		
23	Vapour Fraction	0.8826	0.8826	0.8826	0.8826		
24	Heat Transferred (kJ/h-m)	656.7	656.7	656.7	656.6		
25	Flow Regime	Transition	Transition	Transition	Transition		
26	Water Flow Regime						
27	Liquid HoldUp	0.2485	0.2485	0.2485	0.2485		
28	Water HoldUp	---	---	---	---		
29	Friction Gradient (kPa/m)	6.511e-003	6.511e-003	6.511e-003	6.511e-003		
30	Static Gradient (kPa/m)	1.041e-002	1.041e-002	1.041e-002	1.041e-002		
31	Accel. Gradient (kPa/m)	1.995e-006	1.995e-006	1.995e-006	1.995e-006		
32	Liq. Reynolds		1.929e+005 *	1.929e+005 *	1.929e+005 *	1.928e+005 *	
33	Vap. Reynolds		9.736e+006 *	9.736e+006 *	9.736e+006 *	9.736e+006 *	
34	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	0.1980	
35	Vapour Velocity (m/s)	2.859	2.859	2.859	2.859	2.859	
36	Deposit Thickness (mm)	---	---	---	---	---	
37	Deposit Volume (m3)	---	---	---	---	---	
38	Length (m)	8813	8840	8867	8894		
39	Elevation (m)	-2.000e-002	-2.600e-002	-3.200e-002	-3.800e-002		
40	Pressure (bar)	71.52	71.52	71.52	71.51		
41	Temperature (C)	54.65	54.64	54.63	54.62		
42	Vapour Fraction	0.8826	0.8826	0.8826	0.8826		
43	Heat Transferred (kJ/h-m)	656.6	656.4	656.1	655.8		
44	Flow Regime	Transition	Transition	Transition	Transition		
45	Water Flow Regime						
46	Liquid HoldUp	0.2485	0.1736	0.1736	0.1736		
47	Water HoldUp	---	---	---	---		
48	Friction Gradient (kPa/m)	6.511e-003	8.834e-003	8.834e-003	8.834e-003		
49	Static Gradient (kPa/m)	1.041e-002	-3.599e-004	-3.599e-004	-3.599e-004		
50	Accel. Gradient (kPa/m)	1.995e-006	9.991e-007	9.991e-007	9.991e-007		
51	Liq. Reynolds		1.928e+005 *	1.928e+005 *	1.928e+005 *	1.928e+005 *	
52	Vap. Reynolds		9.736e+006 *	9.737e+006 *	9.737e+006 *	9.737e+006 *	
53	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	0.1980	
54	Vapour Velocity (m/s)	2.859	2.859	2.859	2.859	2.859	
55	Deposit Thickness (mm)	---	---	---	---	---	
56	Deposit Volume (m3)	---	---	---	---	---	
57	Length (m)	8921	8948	8949	8950		
58	Elevation (m)	-4.400e-002	-5.000e-002	-4.600e-002	-4.200e-002		
59	Pressure (bar)	71.51	71.51	71.51	71.51		
60	Temperature (C)	54.61	54.59	54.59	54.59		
61	Vapour Fraction	0.8826	0.8826	0.8826	0.8826		
62	Heat Transferred (kJ/h-m)	655.5	655.2	655.2	655.2		
63	Flow Regime	Transition	Transition	Transition	Transition		
64	Water Flow Regime						
65	Liquid HoldUp	0.1736	0.1736	0.2457	0.2457		
66	Water HoldUp	---	---	---	---		
67	Friction Gradient (kPa/m)	8.834e-003	8.833e-003	6.822e-003	6.822e-003		
68	Static Gradient (kPa/m)	-3.599e-004	-3.599e-004	8.262e-003	8.262e-003		
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	9.991e-007	9.991e-007	1.778e-006	1.778e-006
14	Liq. Reynolds	1.928e+005 *	1.928e+005 *	1.928e+005 *	1.928e+005 *
15	Vap. Reynolds	9.737e+006 *	9.737e+006 *	9.737e+006 *	9.737e+006 *
16	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
17	Vapour Velocity (m/s)	2.859	2.859	2.859	2.859
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	8951	8952	8953	8958
21	Elevation (m)	-3.800e-002	-3.400e-002	-3.000e-002	-2.600e-002
22	Pressure (bar)	71.51	71.51	71.51	71.51
23	Temperature (C)	54.59	54.59	54.59	54.59
24	Vapour Fraction	0.8826	0.8826	0.8826	0.8826
25	Heat Transferred (kJ/h-m)	655.2	655.1	655.1	655.1
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.2457	0.2457	0.2457	0.2367
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	6.822e-003	6.822e-003	6.822e-003	7.893e-003
31	Static Gradient (kPa/m)	8.262e-003	8.262e-003	8.262e-003	1.609e-003
32	Accel. Gradient (kPa/m)	1.778e-006	1.778e-006	1.778e-006	1.120e-006
33	Liq. Reynolds	1.928e+005 *	1.928e+005 *	1.928e+005 *	1.928e+005 *
34	Vap. Reynolds	9.737e+006 *	9.737e+006 *	9.737e+006 *	9.737e+006 *
35	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
36	Vapour Velocity (m/s)	2.859	2.859	2.859	2.859
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	8963	8968	8973	8978
40	Elevation (m)	-2.200e-002	-1.800e-002	-1.400e-002	-1.000e-002
41	Pressure (bar)	71.51	71.51	71.51	71.51
42	Temperature (C)	54.59	54.59	54.58	54.58
43	Vapour Fraction	0.8826	0.8826	0.8826	0.8826
44	Heat Transferred (kJ/h-m)	655.0	655.0	654.9	654.9
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.2367	0.2367	0.2367	0.2367
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	7.893e-003	7.893e-003	7.893e-003	7.893e-003
50	Static Gradient (kPa/m)	1.609e-003	1.609e-003	1.609e-003	1.609e-003
51	Accel. Gradient (kPa/m)	1.120e-006	1.120e-006	1.120e-006	1.120e-006
52	Liq. Reynolds	1.928e+005 *	1.928e+005 *	1.928e+005 *	1.928e+005 *
53	Vap. Reynolds	9.737e+006 *	9.737e+006 *	9.737e+006 *	9.737e+006 *
54	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
55	Vapour Velocity (m/s)	2.859	2.859	2.859	2.859
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	8983	8988	8993	8998
59	Elevation (m)	8.000e-003	2.600e-002	4.400e-002	6.200e-002
60	Pressure (bar)	71.50	71.50	71.50	71.50
61	Temperature (C)	54.58	54.58	54.58	54.57
62	Vapour Fraction	0.8826	0.8826	0.8826	0.8826
63	Heat Transferred (kJ/h-m)	654.8	654.8	654.7	654.6
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.2446	0.2446	0.2446	0.2446
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	6.949e-003	6.949e-003	6.949e-003	6.949e-003
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1				Case Name:	Simulation - 386MMSCFD_80barg.hsc	
2				Unit Set:	Mixed	
3		Epic Atlantic Limited				
4				Date/Time:	Mon Jun 13 15:13:43 2022	
5						
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	7.411e-003	7.411e-003	7.411e-003	7.411e-003
14	Accel. Gradient	(kPa/m)	1.693e-006	1.693e-006	1.693e-006	1.693e-006
15	Liq. Reynolds		1.928e+005 *	1.928e+005 *	1.928e+005 *	1.928e+005 *
16	Vap. Reynolds		9.737e+006 *	9.737e+006 *	9.737e+006 *	9.737e+006 *
17	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
18	Vapour Velocity	(m/s)	2.859	2.859	2.859	2.859
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	9003	9008	9013	9018
22	Elevation	(m)	8.000e-002	6.400e-002	4.800e-002	3.200e-002
23	Pressure	(bar)	71.50	71.50	71.50	71.50
24	Temperature	(C)	54.57	54.57	54.57	54.57
25	Vapour Fraction		0.8826	0.8826	0.8826	0.8826
26	Heat Transferred	(kJ/h-m)	654.6	654.6	654.5	654.5
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2446	0.1714	0.1714	0.1714
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	6.949e-003	8.862e-003	8.862e-003	8.862e-003
32	Static Gradient	(kPa/m)	7.411e-003	-5.139e-003	-5.139e-003	-5.139e-003
33	Accel. Gradient	(kPa/m)	1.693e-006	4.390e-007	4.390e-007	4.390e-007
34	Liq. Reynolds		1.928e+005 *	1.928e+005 *	1.928e+005 *	1.928e+005 *
35	Vap. Reynolds		9.737e+006 *	9.737e+006 *	9.737e+006 *	9.737e+006 *
36	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
37	Vapour Velocity	(m/s)	2.859	2.859	2.859	2.859
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	9023	9028	9033	9038
41	Elevation	(m)	1.600e-002	-1.513e-015	-8.000e-003	-1.600e-002
42	Pressure	(bar)	71.50	71.50	71.50	71.50
43	Temperature	(C)	54.56	54.56	54.56	54.56
44	Vapour Fraction		0.8826	0.8826	0.8826	0.8826
45	Heat Transferred	(kJ/h-m)	654.4	654.4	654.3	654.3
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.1714	0.1714	0.1726	0.1726
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	8.862e-003	8.862e-003	8.846e-003	8.846e-003
51	Static Gradient	(kPa/m)	-5.139e-003	-5.139e-003	-2.581e-003	-2.581e-003
52	Accel. Gradient	(kPa/m)	4.390e-007	4.390e-007	7.387e-007	7.387e-007
53	Liq. Reynolds		1.928e+005 *	1.928e+005 *	1.928e+005 *	1.928e+005 *
54	Vap. Reynolds		9.737e+006 *	9.737e+006 *	9.737e+006 *	9.737e+006 *
55	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
56	Vapour Velocity	(m/s)	2.859	2.859	2.859	2.859
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	9043	9048	9053	9054
60	Elevation	(m)	-2.400e-002	-3.200e-002	-4.000e-002	-4.800e-002
61	Pressure	(bar)	71.50	71.50	71.50	71.50
62	Temperature	(C)	54.56	54.55	54.55	54.55
63	Vapour Fraction		0.8826	0.8826	0.8826	0.8826
64	Heat Transferred	(kJ/h-m)	654.2	654.2	654.1	654.1
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.1726	0.1726	0.1726	0.1678
68	Water HoldUp		---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4							
5	Date/Time: Mon Jun 13 15:13:43 2022						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	8.846e-003	8.846e-003	8.846e-003	8.911e-003	
14	Static Gradient	(kPa/m)	-2.581e-003	-2.581e-003	-2.582e-003	-1.267e-002	
15	Accel. Gradient	(kPa/m)	7.387e-007	7.387e-007	7.387e-007	-4.431e-007	
16	Liq. Reynolds		1.928e+005 *	1.928e+005 *	1.928e+005 *	1.928e+005 *	
17	Vap. Reynolds		9.737e+006 *	9.737e+006 *	9.737e+006 *	9.737e+006 *	
18	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
19	Vapour Velocity	(m/s)	2.859	2.859	2.859	2.859	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	9055	9056	9057	9058	
23	Elevation	(m)	-5.600e-002	-6.400e-002	-7.200e-002	-8.000e-002	
24	Pressure	(bar)	71.50	71.50	71.50	71.50	
25	Temperature	(C)	54.55	54.55	54.55	54.55	
26	Vapour Fraction		0.8826	0.8826	0.8826	0.8826	
27	Heat Transferred	(kJ/h-m)	654.1	654.1	654.1	654.1	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.1678	0.1678	0.1678	0.1678	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	8.911e-003	8.911e-003	8.911e-003	8.911e-003	
33	Static Gradient	(kPa/m)	-1.267e-002	-1.267e-002	-1.267e-002	-1.267e-002	
34	Accel. Gradient	(kPa/m)	-4.431e-007	-4.431e-007	-4.431e-007	-4.431e-007	
35	Liq. Reynolds		1.928e+005 *	1.928e+005 *	1.928e+005 *	1.928e+005 *	
36	Vap. Reynolds		9.737e+006 *	9.737e+006 *	9.737e+006 *	9.737e+006 *	
37	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
38	Vapour Velocity	(m/s)	2.859	2.859	2.859	2.859	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	9065	9071	9078	9085	
42	Elevation	(m)	-9.200e-002	-0.1040	-0.1160	-0.1280	
43	Pressure	(bar)	71.50	71.50	71.50	71.50	
44	Temperature	(C)	54.55	54.55	54.54	54.54	
45	Vapour Fraction		0.8826	0.8826	0.8826	0.8826	
46	Heat Transferred	(kJ/h-m)	654.0	654.0	653.9	653.8	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.1725	0.1725	0.1725	0.1725	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	8.848e-003	8.848e-003	8.848e-003	8.848e-003	
52	Static Gradient	(kPa/m)	-2.903e-003	-2.903e-003	-2.903e-003	-2.903e-003	
53	Accel. Gradient	(kPa/m)	7.010e-007	7.010e-007	7.010e-007	7.010e-007	
54	Liq. Reynolds		1.928e+005 *	1.928e+005 *	1.928e+005 *	1.928e+005 *	
55	Vap. Reynolds		9.737e+006 *	9.737e+006 *	9.738e+006 *	9.738e+006 *	
56	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
57	Vapour Velocity	(m/s)	2.859	2.859	2.859	2.859	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	9091	9098	9105	9111	
61	Elevation	(m)	-0.1400	-0.1620	-0.1840	-0.2060	
62	Pressure	(bar)	71.50	71.50	71.50	71.50	
63	Temperature	(C)	54.54	54.54	54.53	54.53	
64	Vapour Fraction		0.8826	0.8826	0.8826	0.8825	
65	Heat Transferred	(kJ/h-m)	653.7	653.7	653.6	653.5	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.1725	0.1713	0.1713	0.1713	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Water HoldUp		---	---	---	---	
14	Friction Gradient (kPa/m)		8.848e-003	8.863e-003	8.862e-003	8.862e-003	
15	Static Gradient (kPa/m)		-2.903e-003	-5.299e-003	-5.299e-003	-5.299e-003	
16	Accel. Gradient (kPa/m)		7.010e-007	4.202e-007	4.202e-007	4.201e-007	
17	Liq. Reynolds		1.928e+005 *	1.928e+005 *	1.928e+005 *	1.928e+005 *	
18	Vap. Reynolds		9.738e+006 *	9.738e+006 *	9.738e+006 *	9.738e+006 *	
19	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
20	Vapour Velocity (m/s)		2.859	2.859	2.859	2.859	
21	Deposit Thickness (mm)		---	---	---	---	
22	Deposit Volume (m3)		---	---	---	---	
23	Length (m)		9118	9125	9131	9138	
24	Elevation (m)		-0.2280	-0.2500	-0.2660	-0.2820	
25	Pressure (bar)		71.50	71.50	71.50	71.50	
26	Temperature (C)		54.53	54.53	54.52	54.52	
27	Vapour Fraction		0.8825	0.8825	0.8825	0.8825	
28	Heat Transferred (kJ/h-m)		653.5	653.4	653.3	653.3	
29	Flow Regime		Transition	Transition	Transition	Transition	
30	Water Flow Regime						
31	Liquid HoldUp		0.1713	0.1713	0.1720	0.1720	
32	Water HoldUp		---	---	---	---	
33	Friction Gradient (kPa/m)		8.862e-003	8.862e-003	8.853e-003	8.853e-003	
34	Static Gradient (kPa/m)		-5.299e-003	-5.299e-003	-3.863e-003	-3.863e-003	
35	Accel. Gradient (kPa/m)		4.201e-007	4.201e-007	5.884e-007	5.884e-007	
36	Liq. Reynolds		1.927e+005 *	1.927e+005 *	1.927e+005 *	1.927e+005 *	
37	Vap. Reynolds		9.738e+006 *	9.738e+006 *	9.738e+006 *	9.738e+006 *	
38	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
39	Vapour Velocity (m/s)		2.859	2.859	2.859	2.859	
40	Deposit Thickness (mm)		---	---	---	---	
41	Deposit Volume (m3)		---	---	---	---	
42	Length (m)		9145	9151	9158	9162	
43	Elevation (m)		-0.2980	-0.3140	-0.3300	-0.3240	
44	Pressure (bar)		71.50	71.50	71.49	71.49	
45	Temperature (C)		54.52	54.52	54.51	54.51	
46	Vapour Fraction		0.8825	0.8825	0.8825	0.8825	
47	Heat Transferred (kJ/h-m)		653.2	653.1	653.1	653.0	
48	Flow Regime		Transition	Transition	Transition	Transition	
49	Water Flow Regime						
50	Liquid HoldUp		0.1720	0.1720	0.1720	0.2387	
51	Water HoldUp		---	---	---	---	
52	Friction Gradient (kPa/m)		8.853e-003	8.853e-003	8.853e-003	7.648e-003	
53	Static Gradient (kPa/m)		-3.863e-003	-3.863e-003	-3.863e-003	3.028e-003	
54	Accel. Gradient (kPa/m)		5.884e-007	5.884e-007	5.883e-007	1.259e-006	
55	Liq. Reynolds		1.927e+005 *	1.927e+005 *	1.927e+005 *	1.927e+005 *	
56	Vap. Reynolds		9.738e+006 *	9.738e+006 *	9.738e+006 *	9.738e+006 *	
57	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
58	Vapour Velocity (m/s)		2.859	2.859	2.859	2.859	
59	Deposit Thickness (mm)		---	---	---	---	
60	Deposit Volume (m3)		---	---	---	---	
61	Length (m)		9166	9170	9174	9178	
62	Elevation (m)		-0.3180	-0.3120	-0.3060	-0.3000	
63	Pressure (bar)		71.49	71.49	71.49	71.49	
64	Temperature (C)		54.51	54.51	54.51	54.50	
65	Vapour Fraction		0.8825	0.8825	0.8825	0.8825	
66	Heat Transferred (kJ/h-m)		652.9	652.9	652.8	652.8	
67	Flow Regime		Transition	Transition	Transition	Transition	
68	Water Flow Regime						
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.2387	0.2387	0.2387	0.2387
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	7.648e-003	7.648e-003	7.648e-003	7.648e-003
16	Static Gradient (kPa/m)	3.028e-003	3.028e-003	3.028e-003	3.028e-003
17	Accel. Gradient (kPa/m)	1.259e-006	1.259e-006	1.259e-006	1.259e-006
18	Liq. Reynolds	1.927e+005 *	1.927e+005 *	1.927e+005 *	1.927e+005 *
19	Vap. Reynolds	9.738e+006 *	9.738e+006 *	9.738e+006 *	9.738e+006 *
20	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
21	Vapour Velocity (m/s)	2.859	2.859	2.859	2.859
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	9183	9188	9193	9198
25	Elevation (m)	-0.3220	-0.3440	-0.3660	-0.3880
26	Pressure (bar)	71.49	71.49	71.49	71.49
27	Temperature (C)	54.50	54.50	54.50	54.50
28	Vapour Fraction	0.8825	0.8825	0.8825	0.8825
29	Heat Transferred (kJ/h-m)	652.8	652.8	652.7	652.7
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1705	0.1705	0.1705	0.1705
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	8.873e-003	8.873e-003	8.873e-003	8.873e-003
35	Static Gradient (kPa/m)	-7.043e-003	-7.043e-003	-7.043e-003	-7.043e-003
36	Accel. Gradient (kPa/m)	2.158e-007	2.158e-007	2.158e-007	2.158e-007
37	Liq. Reynolds	1.927e+005 *	1.927e+005 *	1.927e+005 *	1.927e+005 *
38	Vap. Reynolds	9.738e+006 *	9.738e+006 *	9.738e+006 *	9.738e+006 *
39	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
40	Vapour Velocity (m/s)	2.859	2.859	2.859	2.858
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	9203	9208	9213	9218
44	Elevation (m)	-0.4100	-0.4160	-0.4220	-0.4280
45	Pressure (bar)	71.49	71.49	71.49	71.49
46	Temperature (C)	54.50	54.49	54.49	54.49
47	Vapour Fraction	0.8825	0.8825	0.8825	0.8825
48	Heat Transferred (kJ/h-m)	652.6	652.6	652.5	652.5
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1705	0.1729	0.1729	0.1729
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	8.873e-003	8.841e-003	8.841e-003	8.841e-003
54	Static Gradient (kPa/m)	-7.043e-003	-1.939e-003	-1.939e-003	-1.939e-003
55	Accel. Gradient (kPa/m)	2.157e-007	8.138e-007	8.138e-007	8.138e-007
56	Liq. Reynolds	1.927e+005 *	1.927e+005 *	1.927e+005 *	1.927e+005 *
57	Vap. Reynolds	9.738e+006 *	9.738e+006 *	9.738e+006 *	9.738e+006 *
58	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
59	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	9223	9228	9233	9238
63	Elevation (m)	-0.4340	-0.4400	-0.4420	-0.4440
64	Pressure (bar)	71.49	71.49	71.49	71.49
65	Temperature (C)	54.49	54.49	54.48	54.48
66	Vapour Fraction	0.8825	0.8825	0.8825	0.8825
67	Heat Transferred (kJ/h-m)	652.4	652.4	652.3	652.2
68	Flow Regime	Transition	Transition	Transition	Transition
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1729	0.1729	0.1735	0.1735
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	8.841e-003	8.841e-003	8.833e-003	8.833e-003
17	Static Gradient (kPa/m)	-1.939e-003	-1.939e-003	-6.477e-004	-6.477e-004
18	Accel. Gradient (kPa/m)	8.138e-007	8.138e-007	9.651e-007	9.651e-007
19	Liq. Reynolds	1.927e+005 *	1.927e+005 *	1.927e+005 *	1.927e+005 *
20	Vap. Reynolds	9.738e+006 *	9.738e+006 *	9.738e+006 *	9.738e+006 *
21	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
22	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	9243	9248	9253	9258
26	Elevation (m)	-0.4460	-0.4480	-0.4500	-0.4000
27	Pressure (bar)	71.49	71.49	71.49	71.49
28	Temperature (C)	54.48	54.48	54.48	54.47
29	Vapour Fraction	0.8825	0.8825	0.8825	0.8825
30	Heat Transferred (kJ/h-m)	652.2	652.1	652.1	652.0
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1735	0.1735	0.1735	0.2625
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	8.833e-003	8.833e-003	8.833e-003	7.509e-003
36	Static Gradient (kPa/m)	-6.477e-004	-6.477e-004	-6.477e-004	2.170e-002
37	Accel. Gradient (kPa/m)	9.651e-007	9.651e-007	9.651e-007	3.444e-006
38	Liq. Reynolds	1.927e+005 *	1.927e+005 *	1.927e+005 *	1.927e+005 *
39	Vap. Reynolds	9.738e+006 *	9.738e+006 *	9.738e+006 *	9.738e+006 *
40	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
41	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	9263	9268	9273	9278
45	Elevation (m)	-0.3500	-0.3000	-0.2500	-0.2000
46	Pressure (bar)	71.49	71.48	71.48	71.48
47	Temperature (C)	54.47	54.47	54.47	54.46
48	Vapour Fraction	0.8825	0.8825	0.8825	0.8825
49	Heat Transferred (kJ/h-m)	651.9	651.8	651.8	651.7
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2625	0.2625	0.2625	0.2625
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	7.509e-003	7.509e-003	7.509e-003	7.509e-003
55	Static Gradient (kPa/m)	2.170e-002	2.170e-002	2.170e-002	2.170e-002
56	Accel. Gradient (kPa/m)	3.444e-006	3.444e-006	3.445e-006	3.445e-006
57	Liq. Reynolds	1.927e+005 *	1.927e+005 *	1.927e+005 *	1.927e+005 *
58	Vap. Reynolds	9.738e+006 *	9.738e+006 *	9.738e+006 *	9.738e+006 *
59	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
60	Vapour Velocity (m/s)	2.858	2.858	2.858	2.859
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	9283	9288	9293	9298
64	Elevation (m)	-0.1900	-0.1800	-0.1700	-0.1600
65	Pressure (bar)	71.48	71.48	71.48	71.48
66	Temperature (C)	54.46	54.46	54.46	54.46
67	Vapour Fraction	0.8825	0.8825	0.8825	0.8825
68	Heat Transferred (kJ/h-m)	651.7	651.6	651.6	651.5
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2401	0.2401	0.2401	0.2401
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	7.476e-003	7.476e-003	7.476e-003	7.476e-003
18	Static Gradient (kPa/m)	4.063e-003	4.063e-003	4.063e-003	4.063e-003
19	Accel. Gradient (kPa/m)	1.361e-006	1.361e-006	1.361e-006	1.361e-006
20	Liq. Reynolds	1.927e+005 *	1.927e+005 *	1.927e+005 *	1.927e+005 *
21	Vap. Reynolds	9.738e+006 *	9.738e+006 *	9.738e+006 *	9.738e+006 *
22	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
23	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	9303	9308	9313	9318
27	Elevation (m)	-0.1500	-0.1520	-0.1540	-0.1560
28	Pressure (bar)	71.48	71.48	71.48	71.48
29	Temperature (C)	54.45	54.45	54.45	54.45
30	Vapour Fraction	0.8825	0.8825	0.8825	0.8825
31	Heat Transferred (kJ/h-m)	651.4	651.4	651.4	651.3
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2401	0.1735	0.1735	0.1735
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	7.476e-003	8.833e-003	8.833e-003	8.833e-003
37	Static Gradient (kPa/m)	4.063e-003	-6.477e-004	-6.477e-004	-6.477e-004
38	Accel. Gradient (kPa/m)	1.361e-006	9.653e-007	9.653e-007	9.653e-007
39	Liq. Reynolds	1.927e+005 *	1.927e+005 *	1.927e+005 *	1.927e+005 *
40	Vap. Reynolds	9.738e+006 *	9.738e+006 *	9.738e+006 *	9.738e+006 *
41	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
42	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	9323	9328	9333	9338
46	Elevation (m)	-0.1580	-0.1600	-0.1340	-0.1080
47	Pressure (bar)	71.48	71.48	71.48	71.47
48	Temperature (C)	54.45	54.44	54.44	54.44
49	Vapour Fraction	0.8825	0.8825	0.8825	0.8825
50	Heat Transferred (kJ/h-m)	651.3	651.2	651.1	651.1
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1735	0.1735	0.2491	0.2491
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	8.833e-003	8.833e-003	6.448e-003	6.448e-003
56	Static Gradient (kPa/m)	-6.477e-004	-6.477e-004	1.085e-002	1.085e-002
57	Accel. Gradient (kPa/m)	9.653e-007	9.653e-007	2.040e-006	2.040e-006
58	Liq. Reynolds	1.927e+005 *	1.927e+005 *	1.927e+005 *	1.927e+005 *
59	Vap. Reynolds	9.739e+006 *	9.739e+006 *	9.739e+006 *	9.739e+006 *
60	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
61	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	9343	9348	9353	9358
65	Elevation (m)	-8.200e-002	-5.600e-002	-3.000e-002	-2.200e-002
66	Pressure (bar)	71.47	71.47	71.47	71.47
67	Temperature (C)	54.44	54.44	54.43	54.43
68	Vapour Fraction	0.8825	0.8825	0.8825	0.8825
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Heat Transferred (kJ/h-m)	651.0	651.0	650.9	650.8					
14	Flow Regime	Transition	Transition	Transition	Transition					
15	Water Flow Regime									
16	Liquid HoldUp	0.2491	0.2491	0.2491	0.2390					
17	Water HoldUp	---	---	---	---					
18	Friction Gradient (kPa/m)	6.448e-003	6.448e-003	6.448e-003	7.613e-003					
19	Static Gradient (kPa/m)	1.085e-002	1.085e-002	1.085e-002	3.235e-003					
20	Accel. Gradient (kPa/m)	2.040e-006	2.040e-006	2.040e-006	1.279e-006					
21	Liq. Reynolds	1.927e+005 *	1.927e+005 *	1.927e+005 *	1.927e+005 *					
22	Vap. Reynolds	9.739e+006 *	9.739e+006 *	9.739e+006 *	9.739e+006 *					
23	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
24	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858					
25	Deposit Thickness (mm)	---	---	---	---					
26	Deposit Volume (m3)	---	---	---	---					
27	Length (m)	9363	9368	9373	9378					
28	Elevation (m)	-1.400e-002	-6.000e-003	2.000e-003	1.000e-002					
29	Pressure (bar)	71.47	71.47	71.47	71.47					
30	Temperature (C)	54.43	54.43	54.43	54.42					
31	Vapour Fraction	0.8825	0.8825	0.8825	0.8825					
32	Heat Transferred (kJ/h-m)	650.8	650.7	650.7	650.6					
33	Flow Regime	Transition	Transition	Transition	Transition					
34	Water Flow Regime									
35	Liquid HoldUp	0.2390	0.2390	0.2390	0.2390					
36	Water HoldUp	---	---	---	---					
37	Friction Gradient (kPa/m)	7.613e-003	7.613e-003	7.613e-003	7.613e-003					
38	Static Gradient (kPa/m)	3.235e-003	3.235e-003	3.235e-003	3.235e-003					
39	Accel. Gradient (kPa/m)	1.279e-006	1.279e-006	1.279e-006	1.279e-006					
40	Liq. Reynolds	1.927e+005 *	1.927e+005 *	1.927e+005 *	1.927e+005 *					
41	Vap. Reynolds	9.739e+006 *	9.739e+006 *	9.739e+006 *	9.739e+006 *					
42	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
43	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858					
44	Deposit Thickness (mm)	---	---	---	---					
45	Deposit Volume (m3)	---	---	---	---					
46	Length (m)	9383	9388	9393	9398					
47	Elevation (m)	-2.200e-002	-5.400e-002	-8.600e-002	-0.1180					
48	Pressure (bar)	71.47	71.47	71.47	71.47					
49	Temperature (C)	54.42	54.42	54.42	54.42					
50	Vapour Fraction	0.8825	0.8825	0.8825	0.8825					
51	Heat Transferred (kJ/h-m)	650.6	650.6	650.5	650.5					
52	Flow Regime	Transition	Transition	Transition	Transition					
53	Water Flow Regime									
54	Liquid HoldUp	0.1690	0.1690	0.1690	0.1690					
55	Water HoldUp	---	---	---	---					
56	Friction Gradient (kPa/m)	8.893e-003	8.893e-003	8.893e-003	8.893e-003					
57	Static Gradient (kPa/m)	-1.018e-002	-1.018e-002	-1.018e-002	-1.018e-002					
58	Accel. Gradient (kPa/m)	-1.523e-007	-1.524e-007	-1.524e-007	-1.524e-007					
59	Liq. Reynolds	1.927e+005 *	1.926e+005 *	1.926e+005 *	1.926e+005 *					
60	Vap. Reynolds	9.739e+006 *	9.739e+006 *	9.739e+006 *	9.739e+006 *					
61	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
62	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858					
63	Deposit Thickness (mm)	---	---	---	---					
64	Deposit Volume (m3)	---	---	---	---					
65	Length (m)	9403	9408	9413	9418					
66	Elevation (m)	-0.1500	-0.2080	-0.2660	-0.3240					
67	Pressure (bar)	71.47	71.47	71.47	71.47					
68	Temperature (C)	54.41	54.41	54.41	54.41					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8825	0.8825	0.8825	0.8825
14	Heat Transferred (kJ/h-m)	650.4	650.4	650.4	650.3
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.1690	0.1651	0.1651	0.1651
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	8.893e-003	8.948e-003	8.948e-003	8.948e-003
20	Static Gradient (kPa/m)	-1.019e-002	-1.818e-002	-1.818e-002	-1.818e-002
21	Accel. Gradient (kPa/m)	-1.524e-007	-1.089e-006	-1.089e-006	-1.089e-006
22	Liq. Reynolds	1.926e+005 *	1.926e+005 *	1.926e+005 *	1.926e+005 *
23	Vap. Reynolds	9.739e+006 *	9.739e+006 *	9.739e+006 *	9.739e+006 *
24	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
25	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	9423	9428	9433	9438
29	Elevation (m)	-0.3820	-0.4400	-0.4740	-0.5080
30	Pressure (bar)	71.47	71.47	71.47	71.47
31	Temperature (C)	54.41	54.41	54.40	54.40
32	Vapour Fraction	0.8825	0.8825	0.8825	0.8825
33	Heat Transferred (kJ/h-m)	650.3	650.2	650.2	650.1
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1651	0.1651	0.1687	0.1687
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	8.947e-003	8.947e-003	8.896e-003	8.896e-003
39	Static Gradient (kPa/m)	-1.818e-002	-1.818e-002	-1.081e-002	-1.081e-002
40	Accel. Gradient (kPa/m)	-1.089e-006	-1.089e-006	-2.256e-007	-2.257e-007
41	Liq. Reynolds	1.926e+005 *	1.926e+005 *	1.926e+005 *	1.926e+005 *
42	Vap. Reynolds	9.739e+006 *	9.739e+006 *	9.739e+006 *	9.739e+006 *
43	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
44	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	9443	9448	9453	9456
48	Elevation (m)	-0.5420	-0.5760	-0.6100	-0.5060
49	Pressure (bar)	71.47	71.47	71.47	71.47
50	Temperature (C)	54.40	54.40	54.40	54.40
51	Vapour Fraction	0.8825	0.8825	0.8825	0.8825
52	Heat Transferred (kJ/h-m)	650.1	650.1	650.0	649.9
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.1687	0.1687	0.1687	0.3316
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	8.896e-003	8.896e-003	8.896e-003	7.678e-003
58	Static Gradient (kPa/m)	-1.081e-002	-1.081e-002	-1.081e-002	9.005e-002
59	Accel. Gradient (kPa/m)	-2.257e-007	-2.257e-007	-2.257e-007	1.152e-005
60	Liq. Reynolds	1.926e+005 *	1.926e+005 *	1.926e+005 *	1.926e+005 *
61	Vap. Reynolds	9.739e+006 *	9.739e+006 *	9.739e+006 *	9.739e+006 *
62	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
63	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	9459	9462	9465	9468
67	Elevation (m)	-0.4020	-0.2980	-0.1940	-9.000e-002
68	Pressure (bar)	71.47	71.46	71.46	71.46
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Temperature (C)	54.39	54.39	54.39	54.39					
14	Vapour Fraction	0.8825	0.8825	0.8825	0.8825					
15	Heat Transferred (kJ/h-m)	649.9	649.8	649.7	649.7					
16	Flow Regime	Transition	Transition	Transition	Transition					
17	Water Flow Regime									
18	Liquid HoldUp	0.3316	0.3315	0.3315	0.3315					
19	Water HoldUp	---	---	---	---					
20	Friction Gradient (kPa/m)	7.678e-003	7.678e-003	7.678e-003	7.679e-003					
21	Static Gradient (kPa/m)	9.005e-002	9.005e-002	9.005e-002	9.005e-002					
22	Accel. Gradient (kPa/m)	1.153e-005	1.153e-005	1.153e-005	1.153e-005					
23	Liq. Reynolds	1.926e+005 *	1.926e+005 *	1.926e+005 *	1.926e+005 *					
24	Vap. Reynolds	9.739e+006 *	9.739e+006 *	9.739e+006 *	9.739e+006 *					
25	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
26	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858					
27	Deposit Thickness (mm)	---	---	---	---					
28	Deposit Volume (m3)	---	---	---	---					
29	Length (m)	9473	9477	9482	9487					
30	Elevation (m)	-9.000e-002	-9.000e-002	-9.000e-002	-9.000e-002					
31	Pressure (bar)	71.46	71.46	71.46	71.46					
32	Temperature (C)	54.39	54.38	54.38	54.38					
33	Vapour Fraction	0.8825	0.8825	0.8825	0.8825					
34	Heat Transferred (kJ/h-m)	649.6	649.6	649.5	649.5					
35	Flow Regime	Transition	Transition	Transition	Transition					
36	Water Flow Regime									
37	Liquid HoldUp	0.2538	0.2538	0.2538	0.2538					
38	Water HoldUp	---	---	---	---					
39	Friction Gradient (kPa/m)	5.952e-003	5.952e-003	5.952e-003	5.952e-003					
40	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000					
41	Accel. Gradient (kPa/m)	7.020e-007	7.020e-007	7.020e-007	7.020e-007					
42	Liq. Reynolds	1.926e+005 *	1.926e+005 *	1.926e+005 *	1.926e+005 *					
43	Vap. Reynolds	9.739e+006 *	9.739e+006 *	9.739e+006 *	9.739e+006 *					
44	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
45	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858					
46	Deposit Thickness (mm)	---	---	---	---					
47	Deposit Volume (m3)	---	---	---	---					
48	Length (m)	9491	9496	9501	9505					
49	Elevation (m)	-9.000e-002	-0.1820	-0.2740	-0.3660					
50	Pressure (bar)	71.46	71.46	71.46	71.46					
51	Temperature (C)	54.38	54.38	54.38	54.37					
52	Vapour Fraction	0.8825	0.8825	0.8825	0.8825					
53	Heat Transferred (kJ/h-m)	649.4	649.5	649.4	649.4					
54	Flow Regime	Transition	Transition	Transition	Transition					
55	Water Flow Regime									
56	Liquid HoldUp	0.2538	0.1590	0.1590	0.1590					
57	Water HoldUp	---	---	---	---					
58	Friction Gradient (kPa/m)	5.951e-003	9.042e-003	9.042e-003	9.041e-003					
59	Static Gradient (kPa/m)	0.0000	-3.014e-002	-3.014e-002	-3.014e-002					
60	Accel. Gradient (kPa/m)	7.020e-007	-2.489e-006	-2.489e-006	-2.489e-006					
61	Liq. Reynolds	1.926e+005 *	1.926e+005 *	1.926e+005 *	1.926e+005 *					
62	Vap. Reynolds	9.739e+006 *	9.739e+006 *	9.739e+006 *	9.739e+006 *					
63	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
64	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858					
65	Deposit Thickness (mm)	---	---	---	---					
66	Deposit Volume (m3)	---	---	---	---					
67	Length (m)	9510	9515	9519	9524					
68	Elevation (m)	-0.4580	-0.5500	-0.5300	-0.5100					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	71.46	71.46	71.46	71.46		
14	Temperature (C)	54.37	54.37	54.37	54.37		
15	Vapour Fraction	0.8825	0.8825	0.8825	0.8825		
16	Heat Transferred (kJ/h-m)	649.4	649.3	649.2	649.2		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.1590	0.1590	0.2465	0.2465		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	9.041e-003	9.041e-003	6.729e-003	6.729e-003		
22	Static Gradient (kPa/m)	-3.014e-002	-3.014e-002	8.878e-003	8.878e-003		
23	Accel. Gradient (kPa/m)	-2.489e-006	-2.489e-006	1.841e-006	1.841e-006		
24	Liq. Reynolds	1.926e+005 *	1.926e+005 *	1.926e+005 *	1.926e+005 *		
25	Vap. Reynolds	9.739e+006 *	9.739e+006 *	9.739e+006 *	9.739e+006 *		
26	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
27	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	9529	9533	9538	9543		
31	Elevation (m)	-0.4900	-0.4700	-0.4500	-0.4160		
32	Pressure (bar)	71.46	71.46	71.46	71.46		
33	Temperature (C)	54.37	54.36	54.36	54.36		
34	Vapour Fraction	0.8825	0.8825	0.8825	0.8825		
35	Heat Transferred (kJ/h-m)	649.1	649.0	649.0	648.9		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.2465	0.2465	0.2465	0.2549		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	6.729e-003	6.729e-003	6.729e-003	6.188e-003		
41	Static Gradient (kPa/m)	8.879e-003	8.879e-003	8.879e-003	1.547e-002		
42	Accel. Gradient (kPa/m)	1.841e-006	1.841e-006	1.841e-006	2.555e-006		
43	Liq. Reynolds	1.926e+005 *	1.926e+005 *	1.926e+005 *	1.926e+005 *		
44	Vap. Reynolds	9.739e+006 *	9.739e+006 *	9.739e+006 *	9.739e+006 *		
45	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
46	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	9547	9552	9557	9561		
50	Elevation (m)	-0.3820	-0.3480	-0.3140	-0.2800		
51	Pressure (bar)	71.46	71.45	71.45	71.45		
52	Temperature (C)	54.36	54.36	54.35	54.35		
53	Vapour Fraction	0.8825	0.8825	0.8825	0.8825		
54	Heat Transferred (kJ/h-m)	648.9	648.8	648.8	648.7		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.2549	0.2549	0.2549	0.2549		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	6.188e-003	6.188e-003	6.188e-003	6.188e-003		
60	Static Gradient (kPa/m)	1.547e-002	1.547e-002	1.547e-002	1.547e-002		
61	Accel. Gradient (kPa/m)	2.555e-006	2.555e-006	2.555e-006	2.555e-006		
62	Liq. Reynolds	1.926e+005 *	1.926e+005 *	1.926e+005 *	1.926e+005 *		
63	Vap. Reynolds	9.739e+006 *	9.740e+006 *	9.740e+006 *	9.740e+006 *		
64	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
65	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	9566	9571	9575	9580		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	-0.2420	-0.2040	-0.1660	-0.1280
14	Pressure (bar)	71.45	71.45	71.45	71.45
15	Temperature (C)	54.35	54.35	54.35	54.34
16	Vapour Fraction	0.8825	0.8825	0.8825	0.8825
17	Heat Transferred (kJ/h-m)	648.7	648.6	648.5	648.5
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2573	0.2573	0.2573	0.2573
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	7.028e-003	7.028e-003	7.028e-003	7.028e-003
23	Static Gradient (kPa/m)	1.741e-002	1.741e-002	1.741e-002	1.741e-002
24	Accel. Gradient (kPa/m)	2.882e-006	2.882e-006	2.882e-006	2.882e-006
25	Liq. Reynolds	1.926e+005 *	1.926e+005 *	1.926e+005 *	1.926e+005 *
26	Vap. Reynolds	9.740e+006 *	9.740e+006 *	9.740e+006 *	9.740e+006 *
27	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
28	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	9585	9589	9594	9599
32	Elevation (m)	-9.000e-002	-7.600e-002	-6.200e-002	-4.800e-002
33	Pressure (bar)	71.45	71.45	71.45	71.44
34	Temperature (C)	54.34	54.34	54.34	54.33
35	Vapour Fraction	0.8825	0.8825	0.8825	0.8825
36	Heat Transferred (kJ/h-m)	648.4	648.4	648.3	648.3
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2573	0.2429	0.2429	0.2429
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	7.028e-003	7.142e-003	7.142e-003	7.142e-003
42	Static Gradient (kPa/m)	1.741e-002	6.148e-003	6.148e-003	6.148e-003
43	Accel. Gradient (kPa/m)	2.882e-006	1.568e-006	1.568e-006	1.568e-006
44	Liq. Reynolds	1.926e+005 *	1.926e+005 *	1.926e+005 *	1.926e+005 *
45	Vap. Reynolds	9.740e+006 *	9.740e+006 *	9.740e+006 *	9.740e+006 *
46	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
47	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	9603	9608	9612	9616
51	Elevation (m)	-3.400e-002	-2.000e-002	3.000e-002	8.000e-002
52	Pressure (bar)	71.44	71.44	71.44	71.44
53	Temperature (C)	54.33	54.33	54.33	54.33
54	Vapour Fraction	0.8825	0.8825	0.8825	0.8825
55	Heat Transferred (kJ/h-m)	648.2	648.2	648.1	648.1
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2429	0.2429	0.2694	0.2694
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	7.142e-003	7.142e-003	7.677e-003	7.677e-003
61	Static Gradient (kPa/m)	6.148e-003	6.148e-003	2.755e-002	2.755e-002
62	Accel. Gradient (kPa/m)	1.568e-006	1.568e-006	4.156e-006	4.156e-006
63	Liq. Reynolds	1.926e+005 *	1.926e+005 *	1.926e+005 *	1.926e+005 *
64	Vap. Reynolds	9.740e+006 *	9.740e+006 *	9.740e+006 *	9.740e+006 *
65	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
66	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
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11	Pipe Table					
12						
13	Length (m)	9620	9624	9628	9633	
14	Elevation (m)	0.1300	0.1800	0.2300	0.1840	
15	Pressure (bar)	71.44	71.44	71.44	71.44	
16	Temperature (C)	54.33	54.32	54.32	54.32	
17	Vapour Fraction	0.8825	0.8825	0.8825	0.8825	
18	Heat Transferred (kJ/h-m)	648.0	648.0	647.9	647.9	
19	Flow Regime	Transition	Transition	Transition	Transition	
20	Water Flow Regime					
21	Liquid HoldUp	0.2694	0.2694	0.2694	0.1669	
22	Water HoldUp	---	---	---	---	
23	Friction Gradient (kPa/m)	7.678e-003	7.678e-003	7.678e-003	8.922e-003	
24	Static Gradient (kPa/m)	2.755e-002	2.755e-002	2.755e-002	-1.452e-002	
25	Accel. Gradient (kPa/m)	4.157e-006	4.157e-006	4.157e-006	-6.606e-007	
26	Liq. Reynolds	1.926e+005 *	1.926e+005 *	1.925e+005 *	1.925e+005 *	
27	Vap. Reynolds	9.740e+006 *	9.740e+006 *	9.740e+006 *	9.740e+006 *	
28	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	
29	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858	
30	Deposit Thickness (mm)	---	---	---	---	
31	Deposit Volume (m3)	---	---	---	---	
32	Length (m)	9638	9643	9648	9653	
33	Elevation (m)	0.1380	9.200e-002	4.600e-002	-1.582e-015	
34	Pressure (bar)	71.44	71.44	71.44	71.44	
35	Temperature (C)	54.32	54.32	54.31	54.31	
36	Vapour Fraction	0.8825	0.8825	0.8825	0.8825	
37	Heat Transferred (kJ/h-m)	647.9	647.8	647.8	647.8	
38	Flow Regime	Transition	Transition	Transition	Transition	
39	Water Flow Regime					
40	Liquid HoldUp	0.1669	0.1669	0.1669	0.1669	
41	Water HoldUp	---	---	---	---	
42	Friction Gradient (kPa/m)	8.922e-003	8.922e-003	8.922e-003	8.922e-003	
43	Static Gradient (kPa/m)	-1.452e-002	-1.452e-002	-1.452e-002	-1.452e-002	
44	Accel. Gradient (kPa/m)	-6.606e-007	-6.606e-007	-6.606e-007	-6.607e-007	
45	Liq. Reynolds	1.925e+005 *	1.925e+005 *	1.925e+005 *	1.925e+005 *	
46	Vap. Reynolds	9.740e+006 *	9.740e+006 *	9.740e+006 *	9.740e+006 *	
47	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	
48	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858	
49	Deposit Thickness (mm)	---	---	---	---	
50	Deposit Volume (m3)	---	---	---	---	
51	Length (m)	9658	9663	9668	9673	
52	Elevation (m)	-1.400e-002	-2.800e-002	-4.200e-002	-5.600e-002	
53	Pressure (bar)	71.44	71.44	71.44	71.44	
54	Temperature (C)	54.31	54.31	54.31	54.31	
55	Vapour Fraction	0.8824	0.8824	0.8824	0.8824	
56	Heat Transferred (kJ/h-m)	647.7	647.6	647.6	647.5	
57	Flow Regime	Transition	Transition	Transition	Transition	
58	Water Flow Regime					
59	Liquid HoldUp	0.1717	0.1717	0.1717	0.1717	
60	Water HoldUp	---	---	---	---	
61	Friction Gradient (kPa/m)	8.856e-003	8.856e-003	8.856e-003	8.856e-003	
62	Static Gradient (kPa/m)	-4.503e-003	-4.503e-003	-4.503e-003	-4.503e-003	
63	Accel. Gradient (kPa/m)	5.136e-007	5.136e-007	5.136e-007	5.136e-007	
64	Liq. Reynolds	1.925e+005 *	1.925e+005 *	1.925e+005 *	1.925e+005 *	
65	Vap. Reynolds	9.740e+006 *	9.740e+006 *	9.740e+006 *	9.740e+006 *	
66	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	
67	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858	
68	Deposit Thickness (mm)	---	---	---	---	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
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9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	9678	9683	9688	9693
15	Elevation (m)	-7.000e-002	-2.200e-002	2.600e-002	7.400e-002
16	Pressure (bar)	71.44	71.44	71.43	71.43
17	Temperature (C)	54.30	54.30	54.30	54.30
18	Vapour Fraction	0.8824	0.8824	0.8824	0.8824
19	Heat Transferred (kJ/h-m)	647.5	647.4	647.3	647.3
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.1717	0.2614	0.2614	0.2614
23	Water HoldUp				
24	Friction Gradient (kPa/m)	8.856e-003	7.453e-003	7.453e-003	7.453e-003
25	Static Gradient (kPa/m)	-4.503e-003	2.077e-002	2.077e-002	2.077e-002
26	Accel. Gradient (kPa/m)	5.136e-007	3.330e-006	3.330e-006	3.330e-006
27	Liq. Reynolds	1.925e+005 *	1.925e+005 *	1.925e+005 *	1.925e+005 *
28	Vap. Reynolds	9.740e+006 *	9.740e+006 *	9.740e+006 *	9.740e+006 *
29	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
30	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	9698	9703	9708	9713
34	Elevation (m)	0.1220	0.1700	0.1720	0.1740
35	Pressure (bar)	71.43	71.43	71.43	71.43
36	Temperature (C)	54.29	54.29	54.29	54.29
37	Vapour Fraction	0.8824	0.8824	0.8824	0.8824
38	Heat Transferred (kJ/h-m)	647.2	647.1	647.1	647.0
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2614	0.2614	0.2356	0.2356
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	7.454e-003	7.454e-003	8.034e-003	8.034e-003
44	Static Gradient (kPa/m)	2.077e-002	2.077e-002	8.016e-004	8.016e-004
45	Accel. Gradient (kPa/m)	3.330e-006	3.330e-006	1.043e-006	1.043e-006
46	Liq. Reynolds	1.925e+005 *	1.925e+005 *	1.925e+005 *	1.925e+005 *
47	Vap. Reynolds	9.740e+006 *	9.741e+006 *	9.741e+006 *	9.741e+006 *
48	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
49	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	9718	9723	9728	9733
53	Elevation (m)	0.1760	0.1780	0.1800	0.1780
54	Pressure (bar)	71.43	71.43	71.43	71.43
55	Temperature (C)	54.29	54.28	54.28	54.28
56	Vapour Fraction	0.8824	0.8824	0.8824	0.8824
57	Heat Transferred (kJ/h-m)	647.0	646.9	646.9	646.9
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2356	0.2356	0.2356	0.1736
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	8.034e-003	8.034e-003	8.034e-003	8.833e-003
63	Static Gradient (kPa/m)	8.016e-004	8.016e-004	8.016e-004	-6.478e-004
64	Accel. Gradient (kPa/m)	1.043e-006	1.043e-006	1.043e-006	9.659e-007
65	Liq. Reynolds	1.925e+005 *	1.925e+005 *	1.925e+005 *	1.925e+005 *
66	Vap. Reynolds	9.741e+006 *	9.741e+006 *	9.741e+006 *	9.741e+006 *
67	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
68	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
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9	PROFILES				
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11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	9738	9743	9748	9753
16	Elevation (m)	0.1760	0.1740	0.1720	0.1700
17	Pressure (bar)	71.43	71.43	71.43	71.43
18	Temperature (C)	54.28	54.28	54.27	54.27
19	Vapour Fraction	0.8824	0.8824	0.8824	0.8824
20	Heat Transferred (kJ/h-m)	646.8	646.8	646.7	646.7
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.1736	0.1736	0.1736	0.1736
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	8.833e-003	8.833e-003	8.833e-003	8.833e-003
26	Static Gradient (kPa/m)	-6.478e-004	-6.478e-004	-6.478e-004	-6.478e-004
27	Accel. Gradient (kPa/m)	9.659e-007	9.659e-007	9.659e-007	9.659e-007
28	Liq. Reynolds	1.925e+005 *	1.925e+005 *	1.925e+005 *	1.925e+005 *
29	Vap. Reynolds	9.741e+006 *	9.741e+006 *	9.741e+006 *	9.741e+006 *
30	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
31	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	9758	9763	9768	9773
35	Elevation (m)	0.1660	0.1620	0.1580	0.1540
36	Pressure (bar)	71.43	71.42	71.42	71.42
37	Temperature (C)	54.27	54.27	54.27	54.26
38	Vapour Fraction	0.8824	0.8824	0.8824	0.8824
39	Heat Transferred (kJ/h-m)	646.6	646.6	646.5	646.4
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.1733	0.1733	0.1733	0.1733
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	8.837e-003	8.837e-003	8.837e-003	8.837e-003
45	Static Gradient (kPa/m)	-1.294e-003	-1.294e-003	-1.294e-003	-1.294e-003
46	Accel. Gradient (kPa/m)	8.901e-007	8.901e-007	8.901e-007	8.901e-007
47	Liq. Reynolds	1.925e+005 *	1.925e+005 *	1.925e+005 *	1.925e+005 *
48	Vap. Reynolds	9.741e+006 *	9.741e+006 *	9.741e+006 *	9.741e+006 *
49	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
50	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	9778	9783	9788	9793
54	Elevation (m)	0.1500	0.1560	0.1620	0.1680
55	Pressure (bar)	71.42	71.42	71.42	71.42
56	Temperature (C)	54.26	54.26	54.26	54.26
57	Vapour Fraction	0.8824	0.8824	0.8824	0.8824
58	Heat Transferred (kJ/h-m)	646.4	646.3	646.2	646.2
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.1733	0.2379	0.2379	0.2379
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	8.837e-003	7.750e-003	7.750e-003	7.750e-003
64	Static Gradient (kPa/m)	-1.294e-003	2.421e-003	2.421e-003	2.421e-003
65	Accel. Gradient (kPa/m)	8.901e-007	1.200e-006	1.200e-006	1.200e-006
66	Liq. Reynolds	1.925e+005 *	1.925e+005 *	1.925e+005 *	1.925e+005 *
67	Vap. Reynolds	9.741e+006 *	9.741e+006 *	9.741e+006 *	9.741e+006 *
68	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
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5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Vapour Velocity	(m/s)	2.858	2.858	2.858	2.858				
14	Deposit Thickness	(mm)	---	---	---	---				
15	Deposit Volume	(m3)	---	---	---	---				
16	Length	(m)	9798	9803	9808	9813				
17	Elevation	(m)	0.1740	0.1800	0.1660	0.1520				
18	Pressure	(bar)	71.42	71.42	71.42	71.42				
19	Temperature	(C)	54.25	54.25	54.25	54.25				
20	Vapour Fraction		0.8824	0.8824	0.8824	0.8824				
21	Heat Transferred	(kJ/h-m)	646.1	646.1	646.1	646.0				
22	Flow Regime		Transition	Transition	Transition	Transition				
23	Water Flow Regime									
24	Liquid HoldUp		0.2379	0.2379	0.1717	0.1717				
25	Water HoldUp		---	---	---	---				
26	Friction Gradient	(kPa/m)	7.750e-003	7.750e-003	8.856e-003	8.856e-003				
27	Static Gradient	(kPa/m)	2.421e-003	2.421e-003	-4.516e-003	-4.516e-003				
28	Accel. Gradient	(kPa/m)	1.200e-006	1.200e-006	5.122e-007	5.122e-007				
29	Liq. Reynolds		1.925e+005 *	1.925e+005 *	1.925e+005 *	1.925e+005 *				
30	Vap. Reynolds		9.741e+006 *	9.741e+006 *	9.741e+006 *	9.741e+006 *				
31	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
32	Vapour Velocity	(m/s)	2.858	2.858	2.858	2.858				
33	Deposit Thickness	(mm)	---	---	---	---				
34	Deposit Volume	(m3)	---	---	---	---				
35	Length	(m)	9818	9823	9828	9833				
36	Elevation	(m)	0.1380	0.1240	0.1100	0.1240				
37	Pressure	(bar)	71.42	71.42	71.42	71.42				
38	Temperature	(C)	54.25	54.24	54.24	54.24				
39	Vapour Fraction		0.8824	0.8824	0.8824	0.8824				
40	Heat Transferred	(kJ/h-m)	646.0	645.9	645.9	645.8				
41	Flow Regime		Transition	Transition	Transition	Transition				
42	Water Flow Regime									
43	Liquid HoldUp		0.1717	0.1717	0.1717	0.2424				
44	Water HoldUp		---	---	---	---				
45	Friction Gradient	(kPa/m)	8.856e-003	8.856e-003	8.856e-003	7.207e-003				
46	Static Gradient	(kPa/m)	-4.516e-003	-4.516e-003	-4.516e-003	5.728e-003				
47	Accel. Gradient	(kPa/m)	5.121e-007	5.121e-007	5.121e-007	1.526e-006				
48	Liq. Reynolds		1.925e+005 *	1.925e+005 *	1.925e+005 *	1.925e+005 *				
49	Vap. Reynolds		9.741e+006 *	9.741e+006 *	9.741e+006 *	9.741e+006 *				
50	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
51	Vapour Velocity	(m/s)	2.858	2.858	2.858	2.858				
52	Deposit Thickness	(mm)	---	---	---	---				
53	Deposit Volume	(m3)	---	---	---	---				
54	Length	(m)	9838	9843	9848	9853				
55	Elevation	(m)	0.1380	0.1520	0.1660	0.1800				
56	Pressure	(bar)	71.42	71.42	71.42	71.42				
57	Temperature	(C)	54.24	54.24	54.23	54.23				
58	Vapour Fraction		0.8824	0.8824	0.8824	0.8824				
59	Heat Transferred	(kJ/h-m)	645.7	645.7	645.6	645.6				
60	Flow Regime		Transition	Transition	Transition	Transition				
61	Water Flow Regime									
62	Liquid HoldUp		0.2424	0.2424	0.2424	0.2424				
63	Water HoldUp		---	---	---	---				
64	Friction Gradient	(kPa/m)	7.207e-003	7.207e-003	7.207e-003	7.207e-003				
65	Static Gradient	(kPa/m)	5.728e-003	5.728e-003	5.728e-003	5.728e-003				
66	Accel. Gradient	(kPa/m)	1.526e-006	1.526e-006	1.526e-006	1.526e-006				
67	Liq. Reynolds		1.925e+005 *	1.925e+005 *	1.925e+005 *	1.925e+005 *				
68	Vap. Reynolds		9.741e+006 *	9.741e+006 *	9.741e+006 *	9.741e+006 *				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
5			Date/Time: Mon Jun 13 15:13:43 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
14	Vapour Velocity	(m/s)	2.858	2.858	2.858	2.858	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	9858	9863	9868	9873	
18	Elevation	(m)	0.1720	0.1640	0.1560	0.1480	
19	Pressure	(bar)	71.42	71.42	71.42	71.42	
20	Temperature	(C)	54.23	54.23	54.23	54.23	
21	Vapour Fraction		0.8824	0.8824	0.8824	0.8824	
22	Heat Transferred	(kJ/h-m)	645.6	645.5	645.5	645.4	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.1727	0.1727	0.1727	0.1727	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	8.844e-003	8.844e-003	8.844e-003	8.844e-003	
28	Static Gradient	(kPa/m)	-2.582e-003	-2.582e-003	-2.582e-003	-2.583e-003	
29	Accel. Gradient	(kPa/m)	7.389e-007	7.389e-007	7.389e-007	7.389e-007	
30	Liq. Reynolds		1.925e+005 *	1.925e+005 *	1.925e+005 *	1.925e+005 *	
31	Vap. Reynolds		9.741e+006 *	9.741e+006 *	9.741e+006 *	9.741e+006 *	
32	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
33	Vapour Velocity	(m/s)	2.858	2.858	2.858	2.858	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	9878	9883	9888	9893	
37	Elevation	(m)	0.1400	0.1440	0.1480	0.1520	
38	Pressure	(bar)	71.42	71.41	71.41	71.41	
39	Temperature	(C)	54.22	54.22	54.22	54.22	
40	Vapour Fraction		0.8824	0.8824	0.8824	0.8824	
41	Heat Transferred	(kJ/h-m)	645.4	645.2	645.2	645.1	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.1727	0.2368	0.2368	0.2368	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	8.844e-003	7.890e-003	7.890e-003	7.890e-003	
47	Static Gradient	(kPa/m)	-2.583e-003	1.609e-003	1.609e-003	1.609e-003	
48	Accel. Gradient	(kPa/m)	7.389e-007	1.121e-006	1.121e-006	1.121e-006	
49	Liq. Reynolds		1.925e+005 *	1.925e+005 *	1.925e+005 *	1.925e+005 *	
50	Vap. Reynolds		9.741e+006 *	9.741e+006 *	9.741e+006 *	9.741e+006 *	
51	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
52	Vapour Velocity	(m/s)	2.858	2.858	2.858	2.858	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	9898	9903	9905	9907	
56	Elevation	(m)	0.1560	0.1600	0.1560	0.1520	
57	Pressure	(bar)	71.41	71.41	71.41	71.41	
58	Temperature	(C)	54.22	54.21	54.21	54.21	
59	Vapour Fraction		0.8824	0.8824	0.8824	0.8824	
60	Heat Transferred	(kJ/h-m)	645.1	645.0	645.1	645.0	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2368	0.2368	0.1724	0.1724	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	7.890e-003	7.890e-003	8.848e-003	8.848e-003	
66	Static Gradient	(kPa/m)	1.609e-003	1.609e-003	-3.224e-003	-3.224e-003	
67	Accel. Gradient	(kPa/m)	1.121e-006	1.121e-006	6.636e-007	6.636e-007	
68	Liq. Reynolds		1.925e+005 *	1.925e+005 *	1.925e+005 *	1.925e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.741e+006 *	9.741e+006 *	9.741e+006 *	9.741e+006 *
14	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
15	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	9909	9911	9913	9933
19	Elevation (m)	0.1480	0.1440	0.1400	0.1320
20	Pressure (bar)	71.41	71.41	71.41	71.41
21	Temperature (C)	54.21	54.21	54.21	54.20
22	Vapour Fraction	0.8824	0.8824	0.8824	0.8824
23	Heat Transferred (kJ/h-m)	645.0	645.0	645.0	644.8
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1724	0.1724	0.1724	0.1736
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	8.848e-003	8.848e-003	8.848e-003	8.832e-003
29	Static Gradient (kPa/m)	-3.224e-003	-3.224e-003	-3.224e-003	-6.479e-004
30	Accel. Gradient (kPa/m)	6.636e-007	6.636e-007	6.636e-007	9.659e-007
31	Liq. Reynolds	1.925e+005 *	1.925e+005 *	1.924e+005 *	1.924e+005 *
32	Vap. Reynolds	9.741e+006 *	9.741e+006 *	9.741e+006 *	9.741e+006 *
33	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
34	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	9953	9973	9993	1.001e+004
38	Elevation (m)	0.1240	0.1160	0.1080	1.000e-001
39	Pressure (bar)	71.41	71.41	71.41	71.40
40	Temperature (C)	54.19	54.19	54.18	54.17
41	Vapour Fraction	0.8824	0.8824	0.8824	0.8824
42	Heat Transferred (kJ/h-m)	644.6	644.4	644.2	643.9
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1736	0.1736	0.1736	0.1736
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	8.832e-003	8.832e-003	8.832e-003	8.832e-003
48	Static Gradient (kPa/m)	-6.479e-004	-6.479e-004	-6.479e-004	-6.479e-004
49	Accel. Gradient (kPa/m)	9.658e-007	9.658e-007	9.658e-007	9.658e-007
50	Liq. Reynolds	1.924e+005 *	1.924e+005 *	1.924e+005 *	1.924e+005 *
51	Vap. Reynolds	9.742e+006 *	9.742e+006 *	9.742e+006 *	9.742e+006 *
52	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
53	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1.002e+004	1.002e+004	1.002e+004	1.003e+004
57	Elevation (m)	-2.000e-002	-0.1400	-0.2600	-0.3800
58	Pressure (bar)	71.41	71.41	71.41	71.41
59	Temperature (C)	54.17	54.17	54.17	54.17
60	Vapour Fraction	0.8824	0.8824	0.8824	0.8824
61	Heat Transferred (kJ/h-m)	644.0	644.0	644.0	644.0
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1440	0.1440	0.1440	0.1440
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	9.317e-003	9.317e-003	9.317e-003	9.316e-003
67	Static Gradient (kPa/m)	-5.712e-002	-5.712e-002	-5.712e-002	-5.712e-002
68	Accel. Gradient (kPa/m)	-5.641e-006	-5.641e-006	-5.641e-006	-5.641e-006
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Liq. Reynolds		1.924e+005 *	1.924e+005 *	1.924e+005 *	1.924e+005 *				
14	Vap. Reynolds		9.742e+006 *	9.742e+006 *	9.742e+006 *	9.742e+006 *				
15	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980				
16	Vapour Velocity (m/s)		2.858	2.858	2.857	2.857				
17	Deposit Thickness (mm)		---	---	---	---				
18	Deposit Volume (m3)		---	---	---	---				
19	Length (m)		1.003e+004	1.003e+004	1.004e+004	1.004e+004				
20	Elevation (m)		-0.5000	-0.4940	-0.4880	-0.4820				
21	Pressure (bar)		71.41	71.41	71.41	71.41				
22	Temperature (C)		54.17	54.17	54.16	54.16				
23	Vapour Fraction		0.8824	0.8824	0.8824	0.8824				
24	Heat Transferred (kJ/h-m)		644.0	643.8	643.7	643.7				
25	Flow Regime		Transition	Transition	Transition	Transition				
26	Water Flow Regime									
27	Liquid HoldUp		0.1440	0.2379	0.2379	0.2379				
28	Water HoldUp		---	---	---	---				
29	Friction Gradient (kPa/m)		9.316e-003	7.748e-003	7.748e-003	7.748e-003				
30	Static Gradient (kPa/m)		-5.712e-002	2.422e-003	2.422e-003	2.422e-003				
31	Accel. Gradient (kPa/m)		-5.641e-006	1.200e-006	1.200e-006	1.200e-006				
32	Liq. Reynolds		1.924e+005 *	1.924e+005 *	1.924e+005 *	1.924e+005 *				
33	Vap. Reynolds		9.742e+006 *	9.742e+006 *	9.742e+006 *	9.742e+006 *				
34	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980				
35	Vapour Velocity (m/s)		2.857	2.857	2.857	2.857				
36	Deposit Thickness (mm)		---	---	---	---				
37	Deposit Volume (m3)		---	---	---	---				
38	Length (m)		1.005e+004	1.005e+004	1.006e+004	1.006e+004				
39	Elevation (m)		-0.4760	-0.4700	-0.3840	-0.2980				
40	Pressure (bar)		71.41	71.41	71.41	71.40				
41	Temperature (C)		54.16	54.16	54.16	54.15				
42	Vapour Fraction		0.8824	0.8824	0.8824	0.8824				
43	Heat Transferred (kJ/h-m)		643.6	643.6	643.5	643.4				
44	Flow Regime		Transition	Transition	Transition	Transition				
45	Water Flow Regime									
46	Liquid HoldUp		0.2379	0.2379	0.2828	0.2828				
47	Water HoldUp		---	---	---	---				
48	Friction Gradient (kPa/m)		7.748e-003	7.748e-003	7.748e-003	7.748e-003				
49	Static Gradient (kPa/m)		2.422e-003	2.422e-003	3.949e-002	3.949e-002				
50	Accel. Gradient (kPa/m)		1.200e-006	1.200e-006	5.574e-006	5.575e-006				
51	Liq. Reynolds		1.924e+005 *	1.924e+005 *	1.924e+005 *	1.924e+005 *				
52	Vap. Reynolds		9.742e+006 *	9.742e+006 *	9.742e+006 *	9.742e+006 *				
53	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980				
54	Vapour Velocity (m/s)		2.857	2.857	2.857	2.857				
55	Deposit Thickness (mm)		---	---	---	---				
56	Deposit Volume (m3)		---	---	---	---				
57	Length (m)		1.007e+004	1.007e+004	1.008e+004	1.008e+004				
58	Elevation (m)		-0.2120	-0.1260	-4.000e-002	-1.527e-015				
59	Pressure (bar)		71.40	71.40	71.40	71.40				
60	Temperature (C)		54.15	54.15	54.15	54.14				
61	Vapour Fraction		0.8824	0.8824	0.8824	0.8824				
62	Heat Transferred (kJ/h-m)		643.4	643.3	643.2	643.2				
63	Flow Regime		Transition	Transition	Transition	Transition				
64	Water Flow Regime									
65	Liquid HoldUp		0.2828	0.2828	0.2828	0.2570				
66	Water HoldUp		---	---	---	---				
67	Friction Gradient (kPa/m)		7.748e-003	7.749e-003	7.749e-003	6.947e-003				
68	Static Gradient (kPa/m)		3.949e-002	3.949e-002	3.949e-002	1.709e-002				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	5.575e-006	5.575e-006	5.575e-006	2.837e-006
14	Liq. Reynolds	1.924e+005 *	1.924e+005 *	1.924e+005 *	1.924e+005 *
15	Vap. Reynolds	9.742e+006 *	9.742e+006 *	9.742e+006 *	9.742e+006 *
16	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
17	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	1.009e+004	1.009e+004	1.010e+004	1.010e+004
21	Elevation (m)	4.000e-002	8.000e-002	0.1200	0.1600
22	Pressure (bar)	71.39	71.39	71.39	71.39
23	Temperature (C)	54.14	54.14	54.14	54.13
24	Vapour Fraction	0.8824	0.8824	0.8824	0.8824
25	Heat Transferred (kJ/h-m)	643.1	643.0	643.0	642.9
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.2570	0.2570	0.2570	0.2570
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	6.947e-003	6.947e-003	6.947e-003	6.947e-003
31	Static Gradient (kPa/m)	1.709e-002	1.709e-002	1.709e-002	1.709e-002
32	Accel. Gradient (kPa/m)	2.837e-006	2.837e-006	2.837e-006	2.837e-006
33	Liq. Reynolds	1.924e+005 *	1.924e+005 *	1.924e+005 *	1.924e+005 *
34	Vap. Reynolds	9.742e+006 *	9.742e+006 *	9.742e+006 *	9.742e+006 *
35	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
36	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	1.011e+004	1.011e+004	1.012e+004	1.012e+004
40	Elevation (m)	0.1640	0.1680	0.1720	0.1760
41	Pressure (bar)	71.39	71.39	71.39	71.39
42	Temperature (C)	54.13	54.13	54.13	54.13
43	Vapour Fraction	0.8824	0.8824	0.8824	0.8824
44	Heat Transferred (kJ/h-m)	642.9	642.8	642.8	642.7
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.2368	0.2368	0.2368	0.2368
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	7.890e-003	7.890e-003	7.890e-003	7.890e-003
50	Static Gradient (kPa/m)	1.609e-003	1.609e-003	1.609e-003	1.609e-003
51	Accel. Gradient (kPa/m)	1.121e-006	1.121e-006	1.121e-006	1.121e-006
52	Liq. Reynolds	1.924e+005 *	1.924e+005 *	1.924e+005 *	1.924e+005 *
53	Vap. Reynolds	9.742e+006 *	9.742e+006 *	9.742e+006 *	9.742e+006 *
54	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
55	Vapour Velocity (m/s)	2.858	2.858	2.858	2.858
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	1.013e+004	1.013e+004	1.014e+004	1.014e+004
59	Elevation (m)	0.1800	0.1860	0.1920	0.1980
60	Pressure (bar)	71.39	71.39	71.39	71.39
61	Temperature (C)	54.12	54.12	54.12	54.12
62	Vapour Fraction	0.8824	0.8824	0.8824	0.8824
63	Heat Transferred (kJ/h-m)	642.7	642.6	642.5	642.5
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.2368	0.2379	0.2379	0.2379
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	7.890e-003	7.749e-003	7.749e-003	7.749e-003
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	1.609e-003	2.422e-003	2.422e-003	2.422e-003
14	Accel. Gradient	(kPa/m)	1.121e-006	1.201e-006	1.201e-006	1.201e-006
15	Liq. Reynolds		1.924e+005 *	1.924e+005 *	1.924e+005 *	1.924e+005 *
16	Vap. Reynolds		9.742e+006 *	9.742e+006 *	9.742e+006 *	9.742e+006 *
17	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
18	Vapour Velocity	(m/s)	2.858	2.858	2.858	2.858
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	1.015e+004	1.015e+004	1.016e+004	1.016e+004
22	Elevation	(m)	0.2040	0.2100	0.2120	0.2140
23	Pressure	(bar)	71.39	71.39	71.39	71.39
24	Temperature	(C)	54.12	54.11	54.11	54.11
25	Vapour Fraction		0.8824	0.8824	0.8824	0.8824
26	Heat Transferred	(kJ/h-m)	642.4	642.4	642.3	642.3
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2379	0.2379	0.2357	0.2357
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	7.749e-003	7.749e-003	8.032e-003	8.032e-003
32	Static Gradient	(kPa/m)	2.422e-003	2.422e-003	8.018e-004	8.018e-004
33	Accel. Gradient	(kPa/m)	1.201e-006	1.201e-006	1.043e-006	1.043e-006
34	Liq. Reynolds		1.924e+005 *	1.924e+005 *	1.924e+005 *	1.924e+005 *
35	Vap. Reynolds		9.742e+006 *	9.743e+006 *	9.743e+006 *	9.743e+006 *
36	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
37	Vapour Velocity	(m/s)	2.858	2.858	2.858	2.858
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	1.017e+004	1.017e+004	1.018e+004	1.018e+004
41	Elevation	(m)	0.2160	0.2180	0.2200	0.2140
42	Pressure	(bar)	71.38	71.38	71.38	71.38
43	Temperature	(C)	54.11	54.11	54.10	54.10
44	Vapour Fraction		0.8824	0.8824	0.8824	0.8824
45	Heat Transferred	(kJ/h-m)	642.2	642.2	642.1	642.1
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2357	0.2357	0.2357	0.1730
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	8.032e-003	8.032e-003	8.032e-003	8.840e-003
51	Static Gradient	(kPa/m)	8.018e-004	8.018e-004	8.018e-004	-1.939e-003
52	Accel. Gradient	(kPa/m)	1.043e-006	1.043e-006	1.043e-006	8.145e-007
53	Liq. Reynolds		1.924e+005 *	1.924e+005 *	1.923e+005 *	1.923e+005 *
54	Vap. Reynolds		9.743e+006 *	9.743e+006 *	9.743e+006 *	9.743e+006 *
55	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
56	Vapour Velocity	(m/s)	2.858	2.858	2.858	2.858
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	1.019e+004	1.019e+004	1.020e+004	1.020e+004
60	Elevation	(m)	0.2080	0.2020	0.1960	0.1900
61	Pressure	(bar)	71.38	71.38	71.38	71.38
62	Temperature	(C)	54.10	54.10	54.10	54.09
63	Vapour Fraction		0.8824	0.8824	0.8823	0.8823
64	Heat Transferred	(kJ/h-m)	642.1	642.0	642.0	641.9
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.1730	0.1730	0.1730	0.1730
68	Water HoldUp		---	---	---	---
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1				Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	8.840e-003	8.839e-003	8.839e-003	8.839e-003
14	Static Gradient	(kPa/m)	-1.939e-003	-1.939e-003	-1.939e-003	-1.939e-003
15	Accel. Gradient	(kPa/m)	8.145e-007	8.145e-007	8.145e-007	8.145e-007
16	Liq. Reynolds		1.923e+005 *	1.923e+005 *	1.923e+005 *	1.923e+005 *
17	Vap. Reynolds		9.743e+006 *	9.743e+006 *	9.743e+006 *	9.743e+006 *
18	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
19	Vapour Velocity	(m/s)	2.858	2.858	2.858	2.858
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	1.021e+004	1.021e+004	1.022e+004	1.022e+004
23	Elevation	(m)	0.1960	0.2020	0.2080	0.2140
24	Pressure	(bar)	71.38	71.38	71.38	71.38
25	Temperature	(C)	54.09	54.09	54.09	54.09
26	Vapour Fraction		0.8823	0.8823	0.8823	0.8823
27	Heat Transferred	(kJ/h-m)	641.8	641.8	641.7	641.7
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2379	0.2379	0.2379	0.2379
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	7.748e-003	7.748e-003	7.748e-003	7.748e-003
33	Static Gradient	(kPa/m)	2.429e-003	2.429e-003	2.429e-003	2.429e-003
34	Accel. Gradient	(kPa/m)	1.201e-006	1.201e-006	1.201e-006	1.201e-006
35	Liq. Reynolds		1.923e+005 *	1.923e+005 *	1.923e+005 *	1.923e+005 *
36	Vap. Reynolds		9.743e+006 *	9.743e+006 *	9.743e+006 *	9.743e+006 *
37	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
38	Vapour Velocity	(m/s)	2.858	2.858	2.858	2.858
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	1.023e+004	1.023e+004	1.024e+004	1.024e+004
42	Elevation	(m)	0.2200	0.2120	0.2040	0.1960
43	Pressure	(bar)	71.38	71.38	71.38	71.38
44	Temperature	(C)	54.08	54.08	54.08	54.08
45	Vapour Fraction		0.8823	0.8823	0.8823	0.8823
46	Heat Transferred	(kJ/h-m)	641.6	641.6	641.6	641.5
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2379	0.1727	0.1727	0.1727
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	7.748e-003	8.843e-003	8.843e-003	8.843e-003
52	Static Gradient	(kPa/m)	2.429e-003	-2.583e-003	-2.583e-003	-2.583e-003
53	Accel. Gradient	(kPa/m)	1.201e-006	7.390e-007	7.390e-007	7.390e-007
54	Liq. Reynolds		1.923e+005 *	1.923e+005 *	1.923e+005 *	1.923e+005 *
55	Vap. Reynolds		9.743e+006 *	9.743e+006 *	9.743e+006 *	9.743e+006 *
56	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
57	Vapour Velocity	(m/s)	2.858	2.858	2.858	2.858
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	1.025e+004	1.025e+004	1.026e+004	1.026e+004
61	Elevation	(m)	0.1880	0.1800	1.000e-001	2.000e-002
62	Pressure	(bar)	71.38	71.38	71.38	71.38
63	Temperature	(C)	54.08	54.08	54.07	54.07
64	Vapour Fraction		0.8823	0.8823	0.8823	0.8823
65	Heat Transferred	(kJ/h-m)	641.5	641.4	641.4	641.3
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.1727	0.1727	0.1618	0.1618
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	8.843e-003	8.843e-003	8.996e-003	8.996e-003
15	Static Gradient (kPa/m)	-2.583e-003	-2.583e-003	-2.476e-002	-2.476e-002
16	Accel. Gradient (kPa/m)	7.390e-007	7.390e-007	-1.860e-006	-1.860e-006
17	Liq. Reynolds	1.923e+005 *	1.923e+005 *	1.923e+005 *	1.923e+005 *
18	Vap. Reynolds	9.743e+006 *	9.743e+006 *	9.743e+006 *	9.743e+006 *
19	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
20	Vapour Velocity (m/s)	2.858	2.857	2.857	2.857
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	1.027e+004	1.027e+004	1.028e+004	1.028e+004
24	Elevation (m)	-6.000e-002	-0.1400	-0.2200	-0.1400
25	Pressure (bar)	71.38	71.38	71.38	71.38
26	Temperature (C)	54.07	54.07	54.07	54.07
27	Vapour Fraction	0.8823	0.8823	0.8823	0.8823
28	Heat Transferred (kJ/h-m)	641.3	641.3	641.2	641.1
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.1618	0.1618	0.1618	0.2794
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	8.996e-003	8.995e-003	8.995e-003	7.741e-003
34	Static Gradient (kPa/m)	-2.476e-002	-2.476e-002	-2.476e-002	3.641e-002
35	Accel. Gradient (kPa/m)	-1.860e-006	-1.860e-006	-1.860e-006	5.211e-006
36	Liq. Reynolds	1.923e+005 *	1.923e+005 *	1.923e+005 *	1.923e+005 *
37	Vap. Reynolds	9.743e+006 *	9.743e+006 *	9.743e+006 *	9.743e+006 *
38	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
39	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	1.029e+004	1.029e+004	1.030e+004	1.030e+004
43	Elevation (m)	-6.000e-002	2.000e-002	1.000e-001	0.1800
44	Pressure (bar)	71.38	71.38	71.37	71.37
45	Temperature (C)	54.06	54.06	54.06	54.05
46	Vapour Fraction	0.8823	0.8823	0.8823	0.8823
47	Heat Transferred (kJ/h-m)	641.0	641.0	640.9	640.8
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2794	0.2794	0.2794	0.2794
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	7.741e-003	7.742e-003	7.742e-003	7.742e-003
53	Static Gradient (kPa/m)	3.641e-002	3.641e-002	3.641e-002	3.641e-002
54	Accel. Gradient (kPa/m)	5.211e-006	5.212e-006	5.212e-006	5.212e-006
55	Liq. Reynolds	1.923e+005 *	1.923e+005 *	1.923e+005 *	1.923e+005 *
56	Vap. Reynolds	9.743e+006 *	9.743e+006 *	9.743e+006 *	9.743e+006 *
57	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
58	Vapour Velocity (m/s)	2.857	2.857	2.857	2.858
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	1.031e+004	1.031e+004	1.032e+004	1.032e+004
62	Elevation (m)	0.1640	0.1480	0.1320	0.1160
63	Pressure (bar)	71.37	71.37	71.37	71.37
64	Temperature (C)	54.05	54.05	54.05	54.05
65	Vapour Fraction	0.8823	0.8823	0.8823	0.8823
66	Heat Transferred (kJ/h-m)	640.8	640.8	640.7	640.7
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.1715	0.1715	0.1715	0.1715
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	8.859e-003	8.859e-003	8.859e-003	8.859e-003
16	Static Gradient (kPa/m)	-5.142e-003	-5.142e-003	-5.142e-003	-5.142e-003
17	Accel. Gradient (kPa/m)	4.388e-007	4.388e-007	4.388e-007	4.388e-007
18	Liq. Reynolds	1.923e+005 *	1.923e+005 *	1.923e+005 *	1.923e+005 *
19	Vap. Reynolds	9.743e+006 *	9.743e+006 *	9.743e+006 *	9.743e+006 *
20	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
21	Vapour Velocity (m/s)	2.858	2.857	2.857	2.857
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	1.033e+004	1.033e+004	1.034e+004	1.034e+004
25	Elevation (m)	1.000e-001	8.600e-002	7.200e-002	5.800e-002
26	Pressure (bar)	71.37	71.37	71.37	71.37
27	Temperature (C)	54.05	54.04	54.04	54.04
28	Vapour Fraction	0.8823	0.8823	0.8823	0.8823
29	Heat Transferred (kJ/h-m)	640.6	640.6	640.5	640.5
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1715	0.1718	0.1718	0.1718
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	8.859e-003	8.855e-003	8.855e-003	8.855e-003
35	Static Gradient (kPa/m)	-5.142e-003	-4.505e-003	-4.505e-003	-4.505e-003
36	Accel. Gradient (kPa/m)	4.388e-007	5.136e-007	5.135e-007	5.135e-007
37	Liq. Reynolds	1.923e+005 *	1.923e+005 *	1.923e+005 *	1.923e+005 *
38	Vap. Reynolds	9.743e+006 *	9.743e+006 *	9.743e+006 *	9.743e+006 *
39	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
40	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	1.035e+004	1.035e+004	1.036e+004	1.036e+004
44	Elevation (m)	4.400e-002	3.000e-002	2.800e-002	2.600e-002
45	Pressure (bar)	71.37	71.37	71.37	71.37
46	Temperature (C)	54.04	54.04	54.03	54.03
47	Vapour Fraction	0.8823	0.8823	0.8823	0.8823
48	Heat Transferred (kJ/h-m)	640.4	640.4	640.3	640.3
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1718	0.1718	0.1736	0.1736
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	8.854e-003	8.854e-003	8.831e-003	8.831e-003
54	Static Gradient (kPa/m)	-4.505e-003	-4.505e-003	-6.480e-004	-6.480e-004
55	Accel. Gradient (kPa/m)	5.135e-007	5.135e-007	9.661e-007	9.661e-007
56	Liq. Reynolds	1.923e+005 *	1.923e+005 *	1.923e+005 *	1.923e+005 *
57	Vap. Reynolds	9.743e+006 *	9.743e+006 *	9.743e+006 *	9.743e+006 *
58	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
59	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	1.037e+004	1.037e+004	1.038e+004	1.038e+004
63	Elevation (m)	2.400e-002	2.200e-002	2.000e-002	-8.000e-003
64	Pressure (bar)	71.37	71.37	71.37	71.37
65	Temperature (C)	54.03	54.03	54.03	54.02
66	Vapour Fraction	0.8823	0.8823	0.8823	0.8823
67	Heat Transferred (kJ/h-m)	640.2	640.2	640.1	640.1
68	Flow Regime	Transition	Transition	Transition	Transition
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1736	0.1736	0.1736	0.1697
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	8.831e-003	8.831e-003	8.831e-003	8.882e-003
17	Static Gradient (kPa/m)	-6.480e-004	-6.480e-004	-6.480e-004	-8.937e-003
18	Accel. Gradient (kPa/m)	9.661e-007	9.661e-007	9.661e-007	-6.406e-009
19	Liq. Reynolds	1.923e+005 *	1.923e+005 *	1.923e+005 *	1.923e+005 *
20	Vap. Reynolds	9.743e+006 *	9.743e+006 *	9.743e+006 *	9.743e+006 *
21	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
22	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	1.039e+004	1.039e+004	1.040e+004	1.040e+004
26	Elevation (m)	-3.600e-002	-6.400e-002	-9.200e-002	-0.1200
27	Pressure (bar)	71.37	71.37	71.37	71.37
28	Temperature (C)	54.02	54.02	54.02	54.02
29	Vapour Fraction	0.8823	0.8823	0.8823	0.8823
30	Heat Transferred (kJ/h-m)	640.0	640.0	639.9	639.9
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1697	0.1697	0.1697	0.1697
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	8.882e-003	8.882e-003	8.882e-003	8.882e-003
36	Static Gradient (kPa/m)	-8.937e-003	-8.937e-003	-8.937e-003	-8.937e-003
37	Accel. Gradient (kPa/m)	-6.422e-009	-6.437e-009	-6.453e-009	-6.469e-009
38	Liq. Reynolds	1.923e+005 *	1.923e+005 *	1.923e+005 *	1.923e+005 *
39	Vap. Reynolds	9.743e+006 *	9.743e+006 *	9.743e+006 *	9.743e+006 *
40	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
41	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	1.041e+004	1.041e+004	1.042e+004	1.042e+004
45	Elevation (m)	-9.600e-002	-7.200e-002	-4.800e-002	-2.400e-002
46	Pressure (bar)	71.37	71.37	71.36	71.36
47	Temperature (C)	54.02	54.01	54.01	54.01
48	Vapour Fraction	0.8823	0.8823	0.8823	0.8823
49	Heat Transferred (kJ/h-m)	639.8	639.7	639.7	639.6
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2480	0.2480	0.2480	0.2480
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	6.567e-003	6.567e-003	6.567e-003	6.567e-003
55	Static Gradient (kPa/m)	9.994e-003	9.994e-003	9.994e-003	9.994e-003
56	Accel. Gradient (kPa/m)	1.955e-006	1.955e-006	1.955e-006	1.955e-006
57	Liq. Reynolds	1.923e+005 *	1.923e+005 *	1.923e+005 *	1.923e+005 *
58	Vap. Reynolds	9.743e+006 *	9.744e+006 *	9.744e+006 *	9.744e+006 *
59	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
60	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	1.043e+004	1.043e+004	1.044e+004	1.044e+004
64	Elevation (m)	-1.554e-015	-5.800e-002	-0.1160	-0.1740
65	Pressure (bar)	71.36	71.36	71.36	71.36
66	Temperature (C)	54.01	54.01	54.00	54.00
67	Vapour Fraction	0.8823	0.8823	0.8823	0.8823
68	Heat Transferred (kJ/h-m)	639.5	639.6	639.5	639.5
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2480	0.1652	0.1652	0.1652
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	6.567e-003	8.946e-003	8.945e-003	8.945e-003
18	Static Gradient (kPa/m)	9.994e-003	-1.819e-002	-1.819e-002	-1.819e-002
19	Accel. Gradient (kPa/m)	1.955e-006	-1.091e-006	-1.091e-006	-1.091e-006
20	Liq. Reynolds	1.923e+005 *	1.923e+005 *	1.923e+005 *	1.923e+005 *
21	Vap. Reynolds	9.744e+006 *	9.744e+006 *	9.744e+006 *	9.744e+006 *
22	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
23	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	1.045e+004	1.045e+004	1.046e+004	1.046e+004
27	Elevation (m)	-0.2320	-0.2900	-0.3280	-0.3660
28	Pressure (bar)	71.36	71.36	71.37	71.37
29	Temperature (C)	54.00	54.00	54.00	54.00
30	Vapour Fraction	0.8823	0.8823	0.8823	0.8823
31	Heat Transferred (kJ/h-m)	639.4	639.4	639.4	639.3
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1652	0.1652	0.1682	0.1682
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	8.945e-003	8.945e-003	8.902e-003	8.902e-003
37	Static Gradient (kPa/m)	-1.819e-002	-1.819e-002	-1.206e-002	-1.206e-002
38	Accel. Gradient (kPa/m)	-1.091e-006	-1.091e-006	-3.726e-007	-3.726e-007
39	Liq. Reynolds	1.923e+005 *	1.923e+005 *	1.923e+005 *	1.923e+005 *
40	Vap. Reynolds	9.744e+006 *	9.744e+006 *	9.744e+006 *	9.744e+006 *
41	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
42	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	1.047e+004	1.047e+004	1.048e+004	1.048e+004
46	Elevation (m)	-0.4040	-0.4420	-0.4800	-0.5000
47	Pressure (bar)	71.37	71.37	71.37	71.37
48	Temperature (C)	53.99	53.99	53.99	53.99
49	Vapour Fraction	0.8823	0.8823	0.8823	0.8823
50	Heat Transferred (kJ/h-m)	639.3	639.2	639.2	639.1
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1682	0.1682	0.1682	0.1709
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	8.902e-003	8.902e-003	8.902e-003	8.865e-003
56	Static Gradient (kPa/m)	-1.206e-002	-1.206e-002	-1.206e-002	-6.414e-003
57	Accel. Gradient (kPa/m)	-3.726e-007	-3.726e-007	-3.727e-007	2.893e-007
58	Liq. Reynolds	1.923e+005 *	1.923e+005 *	1.923e+005 *	1.923e+005 *
59	Vap. Reynolds	9.744e+006 *	9.744e+006 *	9.744e+006 *	9.744e+006 *
60	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
61	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	1.049e+004	1.049e+004	1.050e+004	1.050e+004
65	Elevation (m)	-0.5200	-0.5400	-0.5600	-0.5800
66	Pressure (bar)	71.37	71.37	71.37	71.37
67	Temperature (C)	53.99	53.99	53.98	53.98
68	Vapour Fraction	0.8823	0.8823	0.8823	0.8823
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:13:43 2022		
6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Heat Transferred (kJ/h-m)	639.1	639.0	639.0	638.9		
14	Flow Regime	Transition	Transition	Transition	Transition		
15	Water Flow Regime						
16	Liquid HoldUp	0.1709	0.1709	0.1709	0.1709		
17	Water HoldUp	---	---	---	---		
18	Friction Gradient (kPa/m)	8.865e-003	8.865e-003	8.865e-003	8.865e-003		
19	Static Gradient (kPa/m)	-6.414e-003	-6.414e-003	-6.414e-003	-6.414e-003		
20	Accel. Gradient (kPa/m)	2.893e-007	2.893e-007	2.893e-007	2.893e-007		
21	Liq. Reynolds	1.923e+005 *	1.922e+005 *	1.922e+005 *	1.922e+005 *		
22	Vap. Reynolds	9.744e+006 *	9.744e+006 *	9.744e+006 *	9.744e+006 *		
23	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
24	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857		
25	Deposit Thickness (mm)	---	---	---	---		
26	Deposit Volume (m3)	---	---	---	---		
27	Length (m)	1.051e+004	1.051e+004	1.052e+004	1.052e+004		
28	Elevation (m)	-0.4800	-0.3800	-0.2800	-0.1800		
29	Pressure (bar)	71.36	71.36	71.36	71.35		
30	Temperature (C)	53.98	53.98	53.97	53.97		
31	Vapour Fraction	0.8823	0.8823	0.8823	0.8823		
32	Heat Transferred (kJ/h-m)	638.8	638.7	638.7	638.6		
33	Flow Regime	Transition	Transition	Transition	Transition		
34	Water Flow Regime						
35	Liquid HoldUp	0.2907	0.2906	0.2906	0.2906		
36	Water HoldUp	---	---	---	---		
37	Friction Gradient (kPa/m)	7.749e-003	7.749e-003	7.750e-003	7.750e-003		
38	Static Gradient (kPa/m)	4.690e-002	4.690e-002	4.690e-002	4.690e-002		
39	Accel. Gradient (kPa/m)	6.452e-006	6.452e-006	6.452e-006	6.453e-006		
40	Liq. Reynolds	1.922e+005 *	1.922e+005 *	1.922e+005 *	1.922e+005 *		
41	Vap. Reynolds	9.744e+006 *	9.744e+006 *	9.744e+006 *	9.744e+006 *		
42	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
43	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857		
44	Deposit Thickness (mm)	---	---	---	---		
45	Deposit Volume (m3)	---	---	---	---		
46	Length (m)	1.053e+004	1.053e+004	1.054e+004	1.054e+004		
47	Elevation (m)	-8.000e-002	-7.600e-002	-7.200e-002	-6.800e-002		
48	Pressure (bar)	71.35	71.35	71.35	71.35		
49	Temperature (C)	53.97	53.97	53.96	53.96		
50	Vapour Fraction	0.8823	0.8823	0.8823	0.8823		
51	Heat Transferred (kJ/h-m)	638.5	638.4	638.4	638.3		
52	Flow Regime	Transition	Transition	Transition	Transition		
53	Water Flow Regime						
54	Liquid HoldUp	0.2906	0.2368	0.2368	0.2368		
55	Water HoldUp	---	---	---	---		
56	Friction Gradient (kPa/m)	7.750e-003	7.888e-003	7.888e-003	7.888e-003		
57	Static Gradient (kPa/m)	4.690e-002	1.609e-003	1.609e-003	1.609e-003		
58	Accel. Gradient (kPa/m)	6.453e-006	1.121e-006	1.121e-006	1.121e-006		
59	Liq. Reynolds	1.922e+005 *	1.922e+005 *	1.922e+005 *	1.922e+005 *		
60	Vap. Reynolds	9.744e+006 *	9.744e+006 *	9.744e+006 *	9.744e+006 *		
61	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
62	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857		
63	Deposit Thickness (mm)	---	---	---	---		
64	Deposit Volume (m3)	---	---	---	---		
65	Length (m)	1.055e+004	1.055e+004	1.056e+004	1.056e+004		
66	Elevation (m)	-6.400e-002	-6.000e-002	-4.800e-002	-3.600e-002		
67	Pressure (bar)	71.35	71.35	71.35	71.35		
68	Temperature (C)	53.96	53.96	53.96	53.95		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Fraction		0.8823	0.8823	0.8823	0.8823	
14	Heat Transferred (kJ/h-m)		638.3	638.2	638.2	638.1	
15	Flow Regime		Transition	Transition	Transition	Transition	
16	Water Flow Regime						
17	Liquid HoldUp		0.2368	0.2368	0.2413	0.2413	
18	Water HoldUp		---	---	---	---	
19	Friction Gradient (kPa/m)		7.888e-003	7.888e-003	7.338e-003	7.338e-003	
20	Static Gradient (kPa/m)		1.609e-003	1.609e-003	4.895e-003	4.895e-003	
21	Accel. Gradient (kPa/m)		1.121e-006	1.121e-006	1.444e-006	1.444e-006	
22	Liq. Reynolds		1.922e+005 *	1.922e+005 *	1.922e+005 *	1.922e+005 *	
23	Vap. Reynolds		9.744e+006 *	9.744e+006 *	9.744e+006 *	9.744e+006 *	
24	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
25	Vapour Velocity (m/s)		2.857	2.857	2.857	2.857	
26	Deposit Thickness (mm)		---	---	---	---	
27	Deposit Volume (m3)		---	---	---	---	
28	Length (m)		1.057e+004	1.057e+004	1.058e+004	1.058e+004	
29	Elevation (m)		-2.400e-002	-1.200e-002	-1.513e-015	1.800e-002	
30	Pressure (bar)		71.35	71.35	71.35	71.35	
31	Temperature (C)		53.95	53.95	53.95	53.95	
32	Vapour Fraction		0.8823	0.8823	0.8823	0.8823	
33	Heat Transferred (kJ/h-m)		638.1	638.0	638.0	637.9	
34	Flow Regime		Transition	Transition	Transition	Transition	
35	Water Flow Regime						
36	Liquid HoldUp		0.2413	0.2413	0.2413	0.2447	
37	Water HoldUp		---	---	---	---	
38	Friction Gradient (kPa/m)		7.338e-003	7.338e-003	7.338e-003	6.945e-003	
39	Static Gradient (kPa/m)		4.895e-003	4.895e-003	4.895e-003	7.418e-003	
40	Accel. Gradient (kPa/m)		1.444e-006	1.445e-006	1.445e-006	1.696e-006	
41	Liq. Reynolds		1.922e+005 *	1.922e+005 *	1.922e+005 *	1.922e+005 *	
42	Vap. Reynolds		9.744e+006 *	9.744e+006 *	9.744e+006 *	9.744e+006 *	
43	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
44	Vapour Velocity (m/s)		2.857	2.857	2.857	2.857	
45	Deposit Thickness (mm)		---	---	---	---	
46	Deposit Volume (m3)		---	---	---	---	
47	Length (m)		1.059e+004	1.059e+004	1.060e+004	1.060e+004	
48	Elevation (m)		3.600e-002	5.400e-002	7.200e-002	9.000e-002	
49	Pressure (bar)		71.34	71.34	71.34	71.34	
50	Temperature (C)		53.94	53.94	53.94	53.94	
51	Vapour Fraction		0.8823	0.8823	0.8823	0.8823	
52	Heat Transferred (kJ/h-m)		637.9	637.8	637.7	637.7	
53	Flow Regime		Transition	Transition	Transition	Transition	
54	Water Flow Regime						
55	Liquid HoldUp		0.2447	0.2447	0.2447	0.2447	
56	Water HoldUp		---	---	---	---	
57	Friction Gradient (kPa/m)		6.945e-003	6.945e-003	6.945e-003	6.945e-003	
58	Static Gradient (kPa/m)		7.418e-003	7.418e-003	7.418e-003	7.418e-003	
59	Accel. Gradient (kPa/m)		1.696e-006	1.696e-006	1.696e-006	1.696e-006	
60	Liq. Reynolds		1.922e+005 *	1.922e+005 *	1.922e+005 *	1.922e+005 *	
61	Vap. Reynolds		9.744e+006 *	9.744e+006 *	9.744e+006 *	9.744e+006 *	
62	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
63	Vapour Velocity (m/s)		2.857	2.857	2.857	2.857	
64	Deposit Thickness (mm)		---	---	---	---	
65	Deposit Volume (m3)		---	---	---	---	
66	Length (m)		1.061e+004	1.061e+004	1.062e+004	1.062e+004	
67	Elevation (m)		0.1040	0.1180	0.1320	0.1460	
68	Pressure (bar)		71.34	71.34	71.34	71.34	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	53.94	53.93	53.93	53.93
14	Vapour Fraction	0.8823	0.8823	0.8823	0.8823
15	Heat Transferred (kJ/h-m)	637.6	637.6	637.5	637.5
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2424	0.2424	0.2424	0.2424
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	7.208e-003	7.208e-003	7.208e-003	7.208e-003
21	Static Gradient (kPa/m)	5.712e-003	5.712e-003	5.712e-003	5.712e-003
22	Accel. Gradient (kPa/m)	1.526e-006	1.526e-006	1.526e-006	1.526e-006
23	Liq. Reynolds	1.922e+005 *	1.922e+005 *	1.922e+005 *	1.922e+005 *
24	Vap. Reynolds	9.745e+006 *	9.745e+006 *	9.745e+006 *	9.745e+006 *
25	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
26	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	1.063e+004	1.063e+004	1.064e+004	1.064e+004
30	Elevation (m)	0.1600	0.1640	0.1680	0.1720
31	Pressure (bar)	71.34	71.34	71.34	71.34
32	Temperature (C)	53.93	53.93	53.92	53.92
33	Vapour Fraction	0.8823	0.8823	0.8823	0.8823
34	Heat Transferred (kJ/h-m)	637.4	637.4	637.3	637.3
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2424	0.2368	0.2368	0.2368
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	7.208e-003	7.888e-003	7.888e-003	7.888e-003
40	Static Gradient (kPa/m)	5.712e-003	1.609e-003	1.609e-003	1.609e-003
41	Accel. Gradient (kPa/m)	1.526e-006	1.122e-006	1.122e-006	1.122e-006
42	Liq. Reynolds	1.922e+005 *	1.922e+005 *	1.922e+005 *	1.922e+005 *
43	Vap. Reynolds	9.745e+006 *	9.745e+006 *	9.745e+006 *	9.745e+006 *
44	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
45	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	1.065e+004	1.065e+004	1.066e+004	1.066e+004
49	Elevation (m)	0.1760	0.1800	0.1840	0.1880
50	Pressure (bar)	71.34	71.34	71.34	71.34
51	Temperature (C)	53.92	53.92	53.92	53.91
52	Vapour Fraction	0.8823	0.8823	0.8823	0.8823
53	Heat Transferred (kJ/h-m)	637.2	637.2	637.1	637.1
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2368	0.2368	0.2368	0.2368
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	7.888e-003	7.888e-003	7.888e-003	7.888e-003
59	Static Gradient (kPa/m)	1.609e-003	1.609e-003	1.609e-003	1.610e-003
60	Accel. Gradient (kPa/m)	1.122e-006	1.122e-006	1.122e-006	1.122e-006
61	Liq. Reynolds	1.922e+005 *	1.922e+005 *	1.922e+005 *	1.922e+005 *
62	Vap. Reynolds	9.745e+006 *	9.745e+006 *	9.745e+006 *	9.745e+006 *
63	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
64	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	1.067e+004	1.067e+004	1.068e+004	1.068e+004
68	Elevation (m)	0.1920	0.1960	0.2000	0.2100
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:13:43 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	71.34	71.33	71.33	71.33		
14	Temperature (C)	53.91	53.91	53.91	53.91		
15	Vapour Fraction	0.8823	0.8823	0.8823	0.8823		
16	Heat Transferred (kJ/h-m)	637.0	637.0	636.9	636.8		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.2368	0.2368	0.2368	0.2402		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	7.888e-003	7.888e-003	7.888e-003	7.472e-003		
22	Static Gradient (kPa/m)	1.610e-003	1.610e-003	1.610e-003	4.065e-003		
23	Accel. Gradient (kPa/m)	1.122e-006	1.122e-006	1.122e-006	1.363e-006		
24	Liq. Reynolds	1.922e+005 *	1.922e+005 *	1.922e+005 *	1.922e+005 *		
25	Vap. Reynolds	9.745e+006 *	9.745e+006 *	9.745e+006 *	9.745e+006 *		
26	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
27	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	1.069e+004	1.069e+004	1.070e+004	1.070e+004		
31	Elevation (m)	0.2200	0.2300	0.2400	0.2500		
32	Pressure (bar)	71.33	71.33	71.33	71.33		
33	Temperature (C)	53.90	53.90	53.90	53.90		
34	Vapour Fraction	0.8823	0.8823	0.8823	0.8823		
35	Heat Transferred (kJ/h-m)	636.8	636.7	636.7	636.6		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.2402	0.2402	0.2402	0.2402		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	7.472e-003	7.472e-003	7.472e-003	7.472e-003		
41	Static Gradient (kPa/m)	4.066e-003	4.066e-003	4.066e-003	4.066e-003		
42	Accel. Gradient (kPa/m)	1.363e-006	1.363e-006	1.363e-006	1.363e-006		
43	Liq. Reynolds	1.922e+005 *	1.922e+005 *	1.922e+005 *	1.922e+005 *		
44	Vap. Reynolds	9.745e+006 *	9.745e+006 *	9.745e+006 *	9.745e+006 *		
45	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
46	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	1.071e+004	1.071e+004	1.072e+004	1.072e+004		
50	Elevation (m)	0.2520	0.2540	0.2560	0.2580		
51	Pressure (bar)	71.33	71.33	71.33	71.33		
52	Temperature (C)	53.90	53.89	53.89	53.89		
53	Vapour Fraction	0.8823	0.8823	0.8823	0.8823		
54	Heat Transferred (kJ/h-m)	636.6	636.5	636.5	636.4		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.2357	0.2357	0.2357	0.2357		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	8.030e-003	8.030e-003	8.030e-003	8.030e-003		
60	Static Gradient (kPa/m)	8.020e-004	8.020e-004	8.020e-004	8.020e-004		
61	Accel. Gradient (kPa/m)	1.043e-006	1.043e-006	1.043e-006	1.043e-006		
62	Liq. Reynolds	1.921e+005 *	1.921e+005 *	1.921e+005 *	1.921e+005 *		
63	Vap. Reynolds	9.745e+006 *	9.745e+006 *	9.745e+006 *	9.745e+006 *		
64	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
65	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	1.073e+004	1.073e+004	1.074e+004	1.074e+004		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Elevation (m)	0.2600	0.2660	0.2720	0.2780
14	Pressure (bar)	71.33	71.33	71.33	71.33
15	Temperature (C)	53.89	53.89	53.88	53.88
16	Vapour Fraction	0.8823	0.8823	0.8823	0.8822
17	Heat Transferred (kJ/h-m)	636.4	636.3	636.3	636.2
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2357	0.2380	0.2380	0.2380
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	8.030e-003	7.747e-003	7.747e-003	7.747e-003
23	Static Gradient (kPa/m)	8.020e-004	2.423e-003	2.423e-003	2.423e-003
24	Accel. Gradient (kPa/m)	1.043e-006	1.201e-006	1.201e-006	1.201e-006
25	Liq. Reynolds	1.921e+005 *	1.921e+005 *	1.921e+005 *	1.921e+005 *
26	Vap. Reynolds	9.745e+006 *	9.745e+006 *	9.745e+006 *	9.745e+006 *
27	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
28	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	1.075e+004	1.075e+004	1.076e+004	1.076e+004
32	Elevation (m)	0.2840	0.2900	0.2940	0.2980
33	Pressure (bar)	71.33	71.33	71.33	71.33
34	Temperature (C)	53.88	53.88	53.88	53.87
35	Vapour Fraction	0.8822	0.8822	0.8822	0.8822
36	Heat Transferred (kJ/h-m)	636.2	636.1	636.1	636.0
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2380	0.2380	0.2368	0.2368
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	7.747e-003	7.747e-003	7.888e-003	7.888e-003
42	Static Gradient (kPa/m)	2.423e-003	2.423e-003	1.610e-003	1.610e-003
43	Accel. Gradient (kPa/m)	1.201e-006	1.201e-006	1.122e-006	1.122e-006
44	Liq. Reynolds	1.921e+005 *	1.921e+005 *	1.921e+005 *	1.921e+005 *
45	Vap. Reynolds	9.745e+006 *	9.745e+006 *	9.745e+006 *	9.745e+006 *
46	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
47	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	1.077e+004	1.077e+004	1.078e+004	1.078e+004
51	Elevation (m)	0.3020	0.3060	0.3100	0.3060
52	Pressure (bar)	71.33	71.32	71.32	71.32
53	Temperature (C)	53.87	53.87	53.87	53.87
54	Vapour Fraction	0.8822	0.8822	0.8822	0.8822
55	Heat Transferred (kJ/h-m)	636.0	635.9	635.8	635.8
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2368	0.2368	0.2368	0.1733
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	7.888e-003	7.888e-003	7.888e-003	8.834e-003
61	Static Gradient (kPa/m)	1.610e-003	1.610e-003	1.610e-003	-1.295e-003
62	Accel. Gradient (kPa/m)	1.122e-006	1.122e-006	1.122e-006	8.905e-007
63	Liq. Reynolds	1.921e+005 *	1.921e+005 *	1.921e+005 *	1.921e+005 *
64	Vap. Reynolds	9.745e+006 *	9.745e+006 *	9.745e+006 *	9.745e+006 *
65	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
66	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Length	(m)	1.079e+004	1.079e+004	1.080e+004	1.080e+004				
14	Elevation	(m)	0.3020	0.2980	0.2940	0.2900				
15	Pressure	(bar)	71.32	71.32	71.32	71.32				
16	Temperature	(C)	53.86	53.86	53.86	53.86				
17	Vapour Fraction		0.8822	0.8822	0.8822	0.8822				
18	Heat Transferred	(kJ/h-m)	635.8	635.7	635.7	635.6				
19	Flow Regime		Transition	Transition	Transition	Transition				
20	Water Flow Regime									
21	Liquid HoldUp		0.1733	0.1733	0.1733	0.1733				
22	Water HoldUp		---	---	---	---				
23	Friction Gradient	(kPa/m)	8.834e-003	8.834e-003	8.834e-003	8.834e-003				
24	Static Gradient	(kPa/m)	-1.295e-003	-1.295e-003	-1.295e-003	-1.295e-003				
25	Accel. Gradient	(kPa/m)	8.905e-007	8.905e-007	8.905e-007	8.905e-007				
26	Liq. Reynolds		1.921e+005 *	1.921e+005 *	1.921e+005 *	1.921e+005 *				
27	Vap. Reynolds		9.745e+006 *	9.745e+006 *	9.745e+006 *	9.745e+006 *				
28	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
29	Vapour Velocity	(m/s)	2.857	2.857	2.857	2.857				
30	Deposit Thickness	(mm)	---	---	---	---				
31	Deposit Volume	(m3)	---	---	---	---				
32	Length	(m)	1.081e+004	1.081e+004	1.082e+004	1.082e+004				
33	Elevation	(m)	0.2940	0.2980	0.3020	0.3060				
34	Pressure	(bar)	71.32	71.32	71.32	71.32				
35	Temperature	(C)	53.86	53.85	53.85	53.85				
36	Vapour Fraction		0.8822	0.8822	0.8822	0.8822				
37	Heat Transferred	(kJ/h-m)	635.5	635.5	635.4	635.4				
38	Flow Regime		Transition	Transition	Transition	Transition				
39	Water Flow Regime									
40	Liquid HoldUp		0.2368	0.2368	0.2368	0.2368				
41	Water HoldUp		---	---	---	---				
42	Friction Gradient	(kPa/m)	7.887e-003	7.887e-003	7.887e-003	7.887e-003				
43	Static Gradient	(kPa/m)	1.611e-003	1.611e-003	1.611e-003	1.611e-003				
44	Accel. Gradient	(kPa/m)	1.122e-006	1.122e-006	1.122e-006	1.122e-006				
45	Liq. Reynolds		1.921e+005 *	1.921e+005 *	1.921e+005 *	1.921e+005 *				
46	Vap. Reynolds		9.745e+006 *	9.745e+006 *	9.745e+006 *	9.745e+006 *				
47	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
48	Vapour Velocity	(m/s)	2.857	2.857	2.857	2.857				
49	Deposit Thickness	(mm)	---	---	---	---				
50	Deposit Volume	(m3)	---	---	---	---				
51	Length	(m)	1.083e+004	1.083e+004	1.084e+004	1.084e+004				
52	Elevation	(m)	0.3100	0.3140	0.3180	0.3220				
53	Pressure	(bar)	71.32	71.32	71.32	71.32				
54	Temperature	(C)	53.85	53.85	53.85	53.84				
55	Vapour Fraction		0.8822	0.8822	0.8822	0.8822				
56	Heat Transferred	(kJ/h-m)	635.3	635.3	635.2	635.2				
57	Flow Regime		Transition	Transition	Transition	Transition				
58	Water Flow Regime									
59	Liquid HoldUp		0.2368	0.2368	0.2368	0.2368				
60	Water HoldUp		---	---	---	---				
61	Friction Gradient	(kPa/m)	7.887e-003	7.887e-003	7.887e-003	7.887e-003				
62	Static Gradient	(kPa/m)	1.611e-003	1.610e-003	1.610e-003	1.610e-003				
63	Accel. Gradient	(kPa/m)	1.122e-006	1.122e-006	1.122e-006	1.122e-006				
64	Liq. Reynolds		1.921e+005 *	1.921e+005 *	1.921e+005 *	1.921e+005 *				
65	Vap. Reynolds		9.745e+006 *	9.746e+006 *	9.746e+006 *	9.746e+006 *				
66	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
67	Vapour Velocity	(m/s)	2.857	2.857	2.857	2.857				
68	Deposit Thickness	(mm)	---	---	---	---				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---	---	---
14	Length (m)	1.085e+004	1.085e+004	1.086e+004	1.086e+004	1.086e+004	1.086e+004
15	Elevation (m)	0.3260	0.3300	0.3280	0.3280	0.3260	0.3260
16	Pressure (bar)	71.32	71.32	71.32	71.32	71.32	71.32
17	Temperature (C)	53.84	53.84	53.84	53.84	53.84	53.84
18	Vapour Fraction	0.8822	0.8822	0.8822	0.8822	0.8822	0.8822
19	Heat Transferred (kJ/h-m)	635.1	635.1	635.1	635.1	635.0	635.0
20	Flow Regime	Transition	Transition	Transition	Transition	Transition	Transition
21	Water Flow Regime						
22	Liquid HoldUp	0.2368	0.2368	0.1736	0.1736	0.1736	0.1736
23	Water HoldUp	---	---	---	---	---	---
24	Friction Gradient (kPa/m)	7.887e-003	7.887e-003	8.830e-003	8.830e-003	8.830e-003	8.830e-003
25	Static Gradient (kPa/m)	1.610e-003	1.610e-003	-6.482e-004	-6.482e-004	-6.482e-004	-6.482e-004
26	Accel. Gradient (kPa/m)	1.122e-006	1.122e-006	9.664e-007	9.664e-007	9.664e-007	9.664e-007
27	Liq. Reynolds	1.921e+005 *	1.921e+005 *	1.921e+005 *	1.921e+005 *	1.921e+005 *	1.921e+005 *
28	Vap. Reynolds	9.746e+006 *	9.746e+006 *	9.746e+006 *	9.746e+006 *	9.746e+006 *	9.746e+006 *
29	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	0.1980	0.1980
30	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857	2.857	2.857
31	Deposit Thickness (mm)	---	---	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---	---	---
33	Length (m)	1.087e+004	1.087e+004	1.088e+004	1.088e+004	1.088e+004	1.088e+004
34	Elevation (m)	0.3240	0.3220	0.3200	0.3200	0.3240	0.3240
35	Pressure (bar)	71.32	71.32	71.32	71.32	71.32	71.32
36	Temperature (C)	53.83	53.83	53.83	53.83	53.83	53.83
37	Vapour Fraction	0.8822	0.8822	0.8822	0.8822	0.8822	0.8822
38	Heat Transferred (kJ/h-m)	635.0	634.9	634.9	634.9	634.8	634.8
39	Flow Regime	Transition	Transition	Transition	Transition	Transition	Transition
40	Water Flow Regime						
41	Liquid HoldUp	0.1736	0.1736	0.1736	0.1736	0.2368	0.2368
42	Water HoldUp	---	---	---	---	---	---
43	Friction Gradient (kPa/m)	8.830e-003	8.830e-003	8.830e-003	8.830e-003	7.887e-003	7.887e-003
44	Static Gradient (kPa/m)	-6.482e-004	-6.482e-004	-6.482e-004	-6.482e-004	1.610e-003	1.610e-003
45	Accel. Gradient (kPa/m)	9.664e-007	9.664e-007	9.664e-007	9.664e-007	1.122e-006	1.122e-006
46	Liq. Reynolds	1.921e+005 *	1.921e+005 *	1.921e+005 *	1.921e+005 *	1.921e+005 *	1.921e+005 *
47	Vap. Reynolds	9.746e+006 *	9.746e+006 *	9.746e+006 *	9.746e+006 *	9.746e+006 *	9.746e+006 *
48	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	0.1980	0.1980
49	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857	2.857	2.857
50	Deposit Thickness (mm)	---	---	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---	---	---
52	Length (m)	1.089e+004	1.089e+004	1.090e+004	1.090e+004	1.090e+004	1.090e+004
53	Elevation (m)	0.3280	0.3320	0.3360	0.3360	0.3400	0.3400
54	Pressure (bar)	71.31	71.31	71.31	71.31	71.31	71.31
55	Temperature (C)	53.83	53.82	53.82	53.82	53.82	53.82
56	Vapour Fraction	0.8822	0.8822	0.8822	0.8822	0.8822	0.8822
57	Heat Transferred (kJ/h-m)	634.7	634.7	634.6	634.6	634.6	634.6
58	Flow Regime	Transition	Transition	Transition	Transition	Transition	Transition
59	Water Flow Regime						
60	Liquid HoldUp	0.2368	0.2368	0.2368	0.2368	0.2368	0.2368
61	Water HoldUp	---	---	---	---	---	---
62	Friction Gradient (kPa/m)	7.887e-003	7.887e-003	7.887e-003	7.887e-003	7.887e-003	7.887e-003
63	Static Gradient (kPa/m)	1.610e-003	1.610e-003	1.610e-003	1.610e-003	1.610e-003	1.610e-003
64	Accel. Gradient (kPa/m)	1.122e-006	1.122e-006	1.122e-006	1.122e-006	1.122e-006	1.122e-006
65	Liq. Reynolds	1.921e+005 *	1.921e+005 *	1.921e+005 *	1.921e+005 *	1.921e+005 *	1.921e+005 *
66	Vap. Reynolds	9.746e+006 *	9.746e+006 *	9.746e+006 *	9.746e+006 *	9.746e+006 *	9.746e+006 *
67	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	0.1980	0.1980
68	Vapour Velocity (m/s)	2.857	2.857	2.857	2.857	2.857	2.857
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 208 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Thickness	(mm)	---	---	---	---	
14	Deposit Volume	(m3)	---	---	---	---	
15	Length	(m)	1.091e+004	1.091e+004	1.092e+004	1.092e+004	
16	Elevation	(m)	0.3440	0.3480	0.3520	0.3560	
17	Pressure	(bar)	71.31	71.31	71.31	71.31	
18	Temperature	(C)	53.82	53.82	53.81	53.81	
19	Vapour Fraction		0.8822	0.8822	0.8822	0.8822	
20	Heat Transferred	(kJ/h-m)	634.5	634.4	634.4	634.3	
21	Flow Regime		Transition	Transition	Transition	Transition	
22	Water Flow Regime						
23	Liquid HoldUp		0.2368	0.2368	0.2368	0.2368	
24	Water HoldUp		---	---	---	---	
25	Friction Gradient	(kPa/m)	7.887e-003	7.887e-003	7.887e-003	7.887e-003	
26	Static Gradient	(kPa/m)	1.610e-003	1.610e-003	1.610e-003	1.610e-003	
27	Accel. Gradient	(kPa/m)	1.122e-006	1.122e-006	1.122e-006	1.122e-006	
28	Liq. Reynolds		1.921e+005 *	1.921e+005 *	1.921e+005 *	1.921e+005 *	
29	Vap. Reynolds		9.746e+006 *	9.746e+006 *	9.746e+006 *	9.746e+006 *	
30	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
31	Vapour Velocity	(m/s)	2.857	2.857	2.857	2.857	
32	Deposit Thickness	(mm)	---	---	---	---	
33	Deposit Volume	(m3)	---	---	---	---	
34	Length	(m)	1.093e+004	1.093e+004	1.094e+004	1.094e+004	
35	Elevation	(m)	0.3600	0.3700	0.3800	0.3900	
36	Pressure	(bar)	71.31	71.31	71.31	71.31	
37	Temperature	(C)	53.81	53.81	53.81	53.80	
38	Vapour Fraction		0.8822	0.8822	0.8822	0.8822	
39	Heat Transferred	(kJ/h-m)	634.3	634.2	634.2	634.1	
40	Flow Regime		Transition	Transition	Transition	Transition	
41	Water Flow Regime						
42	Liquid HoldUp		0.2368	0.2402	0.2402	0.2402	
43	Water HoldUp		---	---	---	---	
44	Friction Gradient	(kPa/m)	7.887e-003	7.471e-003	7.471e-003	7.471e-003	
45	Static Gradient	(kPa/m)	1.610e-003	4.066e-003	4.066e-003	4.066e-003	
46	Accel. Gradient	(kPa/m)	1.122e-006	1.363e-006	1.363e-006	1.363e-006	
47	Liq. Reynolds		1.921e+005 *	1.921e+005 *	1.921e+005 *	1.921e+005 *	
48	Vap. Reynolds		9.746e+006 *	9.746e+006 *	9.746e+006 *	9.746e+006 *	
49	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
50	Vapour Velocity	(m/s)	2.857	2.857	2.857	2.857	
51	Deposit Thickness	(mm)	---	---	---	---	
52	Deposit Volume	(m3)	---	---	---	---	
53	Length	(m)	1.095e+004	1.095e+004	1.096e+004	1.096e+004	
54	Elevation	(m)	0.4000	0.4100	0.4000	0.3900	
55	Pressure	(bar)	71.31	71.31	71.31	71.31	
56	Temperature	(C)	53.80	53.80	53.80	53.80	
57	Vapour Fraction		0.8822	0.8822	0.8822	0.8822	
58	Heat Transferred	(kJ/h-m)	634.1	634.0	634.0	634.0	
59	Flow Regime		Transition	Transition	Transition	Transition	
60	Water Flow Regime						
61	Liquid HoldUp		0.2402	0.2402	0.1724	0.1724	
62	Water HoldUp		---	---	---	---	
63	Friction Gradient	(kPa/m)	7.471e-003	7.471e-003	8.845e-003	8.845e-003	
64	Static Gradient	(kPa/m)	4.066e-003	4.066e-003	-3.226e-003	-3.226e-003	
65	Accel. Gradient	(kPa/m)	1.363e-006	1.363e-006	6.638e-007	6.638e-007	
66	Liq. Reynolds		1.921e+005 *	1.921e+005 *	1.921e+005 *	1.921e+005 *	
67	Vap. Reynolds		9.746e+006 *	9.746e+006 *	9.746e+006 *	9.746e+006 *	
68	Liquid Velocity	(m/s)	0.1980	0.1981	0.1981	0.1981	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Vapour Velocity	(m/s)	2.857	2.857	2.857	2.857				
14	Deposit Thickness	(mm)	---	---	---	---				
15	Deposit Volume	(m3)	---	---	---	---				
16	Length	(m)	1.097e+004	1.097e+004	1.098e+004	1.098e+004				
17	Elevation	(m)	0.3800	0.3700	0.3600	0.3600				
18	Pressure	(bar)	71.31	71.31	71.31	71.31				
19	Temperature	(C)	53.79	53.79	53.79	53.79				
20	Vapour Fraction		0.8822	0.8822	0.8822	0.8822				
21	Heat Transferred	(kJ/h-m)	633.9	633.9	633.8	633.7				
22	Flow Regime		Transition	Transition	Transition	Transition				
23	Water Flow Regime									
24	Liquid HoldUp		0.1724	0.1724	0.1724	0.2539				
25	Water HoldUp		---	---	---	---				
26	Friction Gradient	(kPa/m)	8.845e-003	8.845e-003	8.845e-003	5.948e-003				
27	Static Gradient	(kPa/m)	-3.226e-003	-3.226e-003	-3.226e-003	0.0000				
28	Accel. Gradient	(kPa/m)	6.638e-007	6.638e-007	6.637e-007	7.026e-007				
29	Liq. Reynolds		1.921e+005 *	1.921e+005 *	1.921e+005 *	1.920e+005 *				
30	Vap. Reynolds		9.746e+006 *	9.746e+006 *	9.746e+006 *	9.746e+006 *				
31	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981				
32	Vapour Velocity	(m/s)	2.857	2.857	2.857	2.857				
33	Deposit Thickness	(mm)	---	---	---	---				
34	Deposit Volume	(m3)	---	---	---	---				
35	Length	(m)	1.099e+004	1.099e+004	1.099e+004	1.100e+004				
36	Elevation	(m)	0.3600	0.3600	0.3600	0.3600				
37	Pressure	(bar)	71.31	71.31	71.31	71.31				
38	Temperature	(C)	53.79	53.79	53.78	53.78				
39	Vapour Fraction		0.8822	0.8822	0.8822	0.8822				
40	Heat Transferred	(kJ/h-m)	633.7	633.7	633.6	633.6				
41	Flow Regime		Transition	Transition	Transition	Transition				
42	Water Flow Regime									
43	Liquid HoldUp		0.2539	0.2539	0.2539	0.2539				
44	Water HoldUp		---	---	---	---				
45	Friction Gradient	(kPa/m)	5.948e-003	5.948e-003	5.948e-003	5.948e-003				
46	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000				
47	Accel. Gradient	(kPa/m)	7.026e-007	7.026e-007	7.026e-007	7.026e-007				
48	Liq. Reynolds		1.920e+005 *	1.920e+005 *	1.920e+005 *	1.920e+005 *				
49	Vap. Reynolds		9.746e+006 *	9.746e+006 *	9.746e+006 *	9.746e+006 *				
50	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981				
51	Vapour Velocity	(m/s)	2.857	2.857	2.857	2.857				
52	Deposit Thickness	(mm)	---	---	---	---				
53	Deposit Volume	(m3)	---	---	---	---				
54	Length	(m)	1.100e+004	1.101e+004	1.102e+004	1.102e+004				
55	Elevation	(m)	0.3560	0.3520	0.3480	0.3440				
56	Pressure	(bar)	71.31	71.30	71.30	71.30				
57	Temperature	(C)	53.78	53.78	53.78	53.77				
58	Vapour Fraction		0.8822	0.8822	0.8822	0.8822				
59	Heat Transferred	(kJ/h-m)	633.6	633.5	633.4	633.4				
60	Flow Regime		Transition	Transition	Transition	Transition				
61	Water Flow Regime									
62	Liquid HoldUp		0.1734	0.1734	0.1734	0.1734				
63	Water HoldUp		---	---	---	---				
64	Friction Gradient	(kPa/m)	8.832e-003	8.832e-003	8.832e-003	8.832e-003				
65	Static Gradient	(kPa/m)	-1.077e-003	-1.077e-003	-1.077e-003	-1.077e-003				
66	Accel. Gradient	(kPa/m)	9.160e-007	9.160e-007	9.160e-007	9.160e-007				
67	Liq. Reynolds		1.920e+005 *	1.920e+005 *	1.920e+005 *	1.920e+005 *				
68	Vap. Reynolds		9.746e+006 *	9.746e+006 *	9.746e+006 *	9.746e+006 *				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
14	Vapour Velocity	(m/s)	2.857	2.856	2.856	2.856	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	1.103e+004	1.103e+004	1.104e+004	1.104e+004	
18	Elevation	(m)	0.3400	0.3380	0.3360	0.3340	
19	Pressure	(bar)	71.30	71.30	71.30	71.30	
20	Temperature	(C)	53.77	53.77	53.77	53.77	
21	Vapour Fraction		0.8822	0.8822	0.8822	0.8822	
22	Heat Transferred	(kJ/h-m)	633.3	633.3	633.2	633.2	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.1734	0.1736	0.1736	0.1736	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	8.832e-003	8.829e-003	8.829e-003	8.829e-003	
28	Static Gradient	(kPa/m)	-1.077e-003	-6.482e-004	-6.482e-004	-6.482e-004	
29	Accel. Gradient	(kPa/m)	9.160e-007	9.664e-007	9.664e-007	9.664e-007	
30	Liq. Reynolds		1.920e+005 *	1.920e+005 *	1.920e+005 *	1.920e+005 *	
31	Vap. Reynolds		9.746e+006 *	9.746e+006 *	9.746e+006 *	9.746e+006 *	
32	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
33	Vapour Velocity	(m/s)	2.856	2.856	2.856	2.856	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	1.105e+004	1.105e+004	1.106e+004	1.106e+004	
37	Elevation	(m)	0.3320	0.3300	0.3320	0.3340	
38	Pressure	(bar)	71.30	71.30	71.30	71.30	
39	Temperature	(C)	53.76	53.76	53.76	53.76	
40	Vapour Fraction		0.8822	0.8822	0.8822	0.8822	
41	Heat Transferred	(kJ/h-m)	633.1	633.1	633.0	632.9	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.1736	0.1736	0.2357	0.2357	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	8.829e-003	8.829e-003	8.029e-003	8.029e-003	
47	Static Gradient	(kPa/m)	-6.482e-004	-6.483e-004	8.022e-004	8.022e-004	
48	Accel. Gradient	(kPa/m)	9.664e-007	9.664e-007	1.043e-006	1.043e-006	
49	Liq. Reynolds		1.920e+005 *	1.920e+005 *	1.920e+005 *	1.920e+005 *	
50	Vap. Reynolds		9.746e+006 *	9.746e+006 *	9.746e+006 *	9.746e+006 *	
51	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
52	Vapour Velocity	(m/s)	2.856	2.856	2.856	2.856	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	1.107e+004	1.107e+004	1.108e+004	1.108e+004	
56	Elevation	(m)	0.3360	0.3380	0.3400	0.3420	
57	Pressure	(bar)	71.30	71.30	71.30	71.30	
58	Temperature	(C)	53.76	53.75	53.75	53.75	
59	Vapour Fraction		0.8822	0.8822	0.8822	0.8822	
60	Heat Transferred	(kJ/h-m)	632.9	632.8	632.8	632.7	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2357	0.2357	0.2357	0.2357	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	8.029e-003	8.029e-003	8.029e-003	8.029e-003	
66	Static Gradient	(kPa/m)	8.022e-004	8.022e-004	8.022e-004	8.022e-004	
67	Accel. Gradient	(kPa/m)	1.043e-006	1.043e-006	1.043e-006	1.043e-006	
68	Liq. Reynolds		1.920e+005 *	1.920e+005 *	1.920e+005 *	1.920e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.746e+006 *	9.746e+006 *	9.746e+006 *	9.747e+006 *
14	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
15	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	1.109e+004	1.109e+004	1.110e+004	1.110e+004
19	Elevation (m)	0.3440	0.3460	0.3480	0.3500
20	Pressure (bar)	71.30	71.30	71.30	71.30
21	Temperature (C)	53.75	53.75	53.74	53.74
22	Vapour Fraction	0.8822	0.8822	0.8822	0.8822
23	Heat Transferred (kJ/h-m)	632.6	632.6	632.5	632.5
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2357	0.2357	0.2357	0.2357
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	8.028e-003	8.028e-003	8.028e-003	8.028e-003
29	Static Gradient (kPa/m)	8.022e-004	8.022e-004	8.022e-004	8.022e-004
30	Accel. Gradient (kPa/m)	1.043e-006	1.043e-006	1.043e-006	1.043e-006
31	Liq. Reynolds	1.920e+005 *	1.920e+005 *	1.920e+005 *	1.920e+005 *
32	Vap. Reynolds	9.747e+006 *	9.747e+006 *	9.747e+006 *	9.747e+006 *
33	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
34	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	1.111e+004	1.111e+004	1.112e+004	1.112e+004
38	Elevation (m)	0.3440	0.3380	0.3320	0.3260
39	Pressure (bar)	71.30	71.30	71.30	71.30
40	Temperature (C)	53.74	53.74	53.74	53.73
41	Vapour Fraction	0.8822	0.8822	0.8822	0.8822
42	Heat Transferred (kJ/h-m)	632.5	632.4	632.4	632.3
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1730	0.1730	0.1730	0.1730
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	8.837e-003	8.836e-003	8.836e-003	8.836e-003
48	Static Gradient (kPa/m)	-1.940e-003	-1.940e-003	-1.940e-003	-1.940e-003
49	Accel. Gradient (kPa/m)	8.147e-007	8.147e-007	8.147e-007	8.147e-007
50	Liq. Reynolds	1.920e+005 *	1.920e+005 *	1.920e+005 *	1.920e+005 *
51	Vap. Reynolds	9.747e+006 *	9.747e+006 *	9.747e+006 *	9.747e+006 *
52	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
53	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1.113e+004	1.113e+004	1.114e+004	1.114e+004
57	Elevation (m)	0.3200	0.3240	0.3280	0.3320
58	Pressure (bar)	71.30	71.29	71.29	71.29
59	Temperature (C)	53.73	53.73	53.73	53.73
60	Vapour Fraction	0.8822	0.8822	0.8822	0.8822
61	Heat Transferred (kJ/h-m)	632.3	632.2	632.1	632.1
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1730	0.2369	0.2369	0.2369
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	8.836e-003	7.886e-003	7.886e-003	7.886e-003
67	Static Gradient (kPa/m)	-1.940e-003	1.610e-003	1.610e-003	1.610e-003
68	Accel. Gradient (kPa/m)	8.147e-007	1.122e-006	1.122e-006	1.122e-006
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 212 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	1.920e+005 *	1.920e+005 *	1.920e+005 *	1.920e+005 *
14	Vap. Reynolds	9.747e+006 *	9.747e+006 *	9.747e+006 *	9.747e+006 *
15	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
16	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	1.115e+004	1.115e+004	1.116e+004	1.116e+004
20	Elevation (m)	0.3360	0.3400	0.3340	0.3280
21	Pressure (bar)	71.29	71.29	71.29	71.29
22	Temperature (C)	53.73	53.72	53.72	53.72
23	Vapour Fraction	0.8822	0.8822	0.8822	0.8822
24	Heat Transferred (kJ/h-m)	632.0	632.0	632.0	631.9
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2369	0.2369	0.1730	0.1730
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	7.886e-003	7.886e-003	8.836e-003	8.836e-003
30	Static Gradient (kPa/m)	1.610e-003	1.610e-003	-1.940e-003	-1.940e-003
31	Accel. Gradient (kPa/m)	1.122e-006	1.122e-006	8.147e-007	8.147e-007
32	Liq. Reynolds	1.920e+005 *	1.920e+005 *	1.920e+005 *	1.920e+005 *
33	Vap. Reynolds	9.747e+006 *	9.747e+006 *	9.747e+006 *	9.747e+006 *
34	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
35	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	1.117e+004	1.117e+004	1.118e+004	1.118e+004
39	Elevation (m)	0.3220	0.3160	0.3100	0.3080
40	Pressure (bar)	71.29	71.29	71.29	71.29
41	Temperature (C)	53.72	53.72	53.71	53.71
42	Vapour Fraction	0.8822	0.8822	0.8822	0.8822
43	Heat Transferred (kJ/h-m)	631.9	631.8	631.8	631.7
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.1730	0.1730	0.1730	0.1736
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	8.836e-003	8.836e-003	8.836e-003	8.828e-003
49	Static Gradient (kPa/m)	-1.940e-003	-1.940e-003	-1.940e-003	-6.483e-004
50	Accel. Gradient (kPa/m)	8.147e-007	8.147e-007	8.147e-007	9.664e-007
51	Liq. Reynolds	1.920e+005 *	1.920e+005 *	1.920e+005 *	1.920e+005 *
52	Vap. Reynolds	9.747e+006 *	9.747e+006 *	9.747e+006 *	9.747e+006 *
53	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
54	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	1.119e+004	1.119e+004	1.120e+004	1.120e+004
58	Elevation (m)	0.3060	0.3040	0.3020	0.3000
59	Pressure (bar)	71.29	71.29	71.29	71.29
60	Temperature (C)	53.71	53.71	53.71	53.70
61	Vapour Fraction	0.8822	0.8822	0.8822	0.8822
62	Heat Transferred (kJ/h-m)	631.7	631.6	631.6	631.5
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.1736	0.1736	0.1736	0.1736
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	8.828e-003	8.828e-003	8.828e-003	8.828e-003
68	Static Gradient (kPa/m)	-6.483e-004	-6.483e-004	-6.483e-004	-6.483e-004
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Accel. Gradient	(kPa/m)	9.664e-007	9.664e-007	9.664e-007	9.664e-007				
14	Liq. Reynolds		1.920e+005 *	1.920e+005 *	1.920e+005 *	1.920e+005 *				
15	Vap. Reynolds		9.747e+006 *	9.747e+006 *	9.747e+006 *	9.747e+006 *				
16	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981				
17	Vapour Velocity	(m/s)	2.856	2.856	2.856	2.856				
18	Deposit Thickness	(mm)	---	---	---	---				
19	Deposit Volume	(m3)	---	---	---	---				
20	Length	(m)	1.121e+004	1.121e+004	1.122e+004	1.122e+004				
21	Elevation	(m)	0.3000	0.3000	0.3000	0.3000				
22	Pressure	(bar)	71.29	71.29	71.29	71.29				
23	Temperature	(C)	53.70	53.70	53.70	53.70				
24	Vapour Fraction		0.8822	0.8822	0.8822	0.8822				
25	Heat Transferred	(kJ/h-m)	631.4	631.4	631.3	631.3				
26	Flow Regime		Transition	Transition	Transition	Transition				
27	Water Flow Regime									
28	Liquid HoldUp		0.2539	0.2539	0.2539	0.2539				
29	Water HoldUp		---	---	---	---				
30	Friction Gradient	(kPa/m)	5.947e-003	5.947e-003	5.947e-003	5.947e-003				
31	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000				
32	Accel. Gradient	(kPa/m)	7.025e-007	7.025e-007	7.025e-007	7.025e-007				
33	Liq. Reynolds		1.920e+005 *	1.920e+005 *	1.920e+005 *	1.920e+005 *				
34	Vap. Reynolds		9.747e+006 *	9.747e+006 *	9.747e+006 *	9.747e+006 *				
35	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981				
36	Vapour Velocity	(m/s)	2.856	2.856	2.856	2.856				
37	Deposit Thickness	(mm)	---	---	---	---				
38	Deposit Volume	(m3)	---	---	---	---				
39	Length	(m)	1.123e+004	1.123e+004	1.124e+004	1.124e+004				
40	Elevation	(m)	0.3000	0.3060	0.3120	0.3180				
41	Pressure	(bar)	71.29	71.29	71.29	71.29				
42	Temperature	(C)	53.69	53.69	53.69	53.69				
43	Vapour Fraction		0.8822	0.8822	0.8822	0.8822				
44	Heat Transferred	(kJ/h-m)	631.2	631.2	631.1	631.1				
45	Flow Regime		Transition	Transition	Transition	Transition				
46	Water Flow Regime									
47	Liquid HoldUp		0.2539	0.2380	0.2380	0.2380				
48	Water HoldUp		---	---	---	---				
49	Friction Gradient	(kPa/m)	5.946e-003	7.745e-003	7.745e-003	7.745e-003				
50	Static Gradient	(kPa/m)	0.0000	2.423e-003	2.423e-003	2.423e-003				
51	Accel. Gradient	(kPa/m)	7.025e-007	1.201e-006	1.201e-006	1.201e-006				
52	Liq. Reynolds		1.920e+005 *	1.920e+005 *	1.920e+005 *	1.920e+005 *				
53	Vap. Reynolds		9.747e+006 *	9.747e+006 *	9.747e+006 *	9.747e+006 *				
54	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981				
55	Vapour Velocity	(m/s)	2.856	2.856	2.856	2.856				
56	Deposit Thickness	(mm)	---	---	---	---				
57	Deposit Volume	(m3)	---	---	---	---				
58	Length	(m)	1.125e+004	1.125e+004	1.126e+004	1.126e+004				
59	Elevation	(m)	0.3240	0.3300	0.3280	0.3260				
60	Pressure	(bar)	71.29	71.28	71.28	71.28				
61	Temperature	(C)	53.69	53.68	53.68	53.68				
62	Vapour Fraction		0.8822	0.8822	0.8822	0.8822				
63	Heat Transferred	(kJ/h-m)	631.0	631.0	631.0	630.9				
64	Flow Regime		Transition	Transition	Transition	Transition				
65	Water Flow Regime									
66	Liquid HoldUp		0.2380	0.2380	0.1736	0.1737				
67	Water HoldUp		---	---	---	---				
68	Friction Gradient	(kPa/m)	7.745e-003	7.745e-003	8.828e-003	8.828e-003				
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1				Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	2.424e-003	2.424e-003	-6.483e-004	-6.483e-004
14	Accel. Gradient	(kPa/m)	1.201e-006	1.201e-006	9.664e-007	9.664e-007
15	Liq. Reynolds		1.920e+005 *	1.920e+005 *	1.920e+005 *	1.919e+005 *
16	Vap. Reynolds		9.747e+006 *	9.747e+006 *	9.747e+006 *	9.747e+006 *
17	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981
18	Vapour Velocity	(m/s)	2.856	2.856	2.856	2.856
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	1.127e+004	1.127e+004	1.128e+004	1.128e+004
22	Elevation	(m)	0.3240	0.3220	0.3200	0.3200
23	Pressure	(bar)	71.28	71.28	71.28	71.28
24	Temperature	(C)	53.68	53.68	53.68	53.67
25	Vapour Fraction		0.8822	0.8821	0.8821	0.8821
26	Heat Transferred	(kJ/h-m)	630.9	630.8	630.8	630.6
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.1737	0.1737	0.1737	0.2539
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	8.828e-003	8.828e-003	8.828e-003	5.946e-003
32	Static Gradient	(kPa/m)	-6.483e-004	-6.483e-004	-6.483e-004	0.0000
33	Accel. Gradient	(kPa/m)	9.664e-007	9.664e-007	9.664e-007	7.025e-007
34	Liq. Reynolds		1.919e+005 *	1.919e+005 *	1.919e+005 *	1.919e+005 *
35	Vap. Reynolds		9.747e+006 *	9.747e+006 *	9.747e+006 *	9.747e+006 *
36	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981
37	Vapour Velocity	(m/s)	2.856	2.856	2.856	2.856
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	1.129e+004	1.129e+004	1.130e+004	1.130e+004
41	Elevation	(m)	0.3200	0.3200	0.3200	0.3200
42	Pressure	(bar)	71.28	71.28	71.28	71.28
43	Temperature	(C)	53.67	53.67	53.67	53.67
44	Vapour Fraction		0.8821	0.8821	0.8821	0.8821
45	Heat Transferred	(kJ/h-m)	630.6	630.5	630.5	630.4
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2539	0.2539	0.2539	0.2540
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	5.946e-003	5.946e-003	5.946e-003	5.946e-003
51	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000
52	Accel. Gradient	(kPa/m)	7.025e-007	7.025e-007	7.025e-007	7.025e-007
53	Liq. Reynolds		1.919e+005 *	1.919e+005 *	1.919e+005 *	1.919e+005 *
54	Vap. Reynolds		9.747e+006 *	9.747e+006 *	9.747e+006 *	9.747e+006 *
55	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981
56	Vapour Velocity	(m/s)	2.856	2.856	2.856	2.856
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	1.131e+004	1.131e+004	1.132e+004	1.132e+004
60	Elevation	(m)	0.2820	0.2440	0.2060	0.1680
61	Pressure	(bar)	71.28	71.28	71.28	71.28
62	Temperature	(C)	53.66	53.66	53.66	53.66
63	Vapour Fraction		0.8821	0.8821	0.8821	0.8821
64	Heat Transferred	(kJ/h-m)	630.5	630.4	630.4	630.3
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.1682	0.1682	0.1682	0.1682
68	Water HoldUp		---	---	---	---
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1				Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	8.900e-003	8.900e-003	8.900e-003	8.899e-003
14	Static Gradient	(kPa/m)	-1.206e-002	-1.206e-002	-1.206e-002	-1.206e-002
15	Accel. Gradient	(kPa/m)	-3.737e-007	-3.738e-007	-3.738e-007	-3.738e-007
16	Liq. Reynolds		1.919e+005 *	1.919e+005 *	1.919e+005 *	1.919e+005 *
17	Vap. Reynolds		9.747e+006 *	9.747e+006 *	9.747e+006 *	9.747e+006 *
18	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981
19	Vapour Velocity	(m/s)	2.856	2.856	2.856	2.856
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	1.133e+004	1.133e+004	1.134e+004	1.134e+004
23	Elevation	(m)	0.1300	0.1380	0.1460	0.1540
24	Pressure	(bar)	71.28	71.28	71.28	71.28
25	Temperature	(C)	53.66	53.66	53.65	53.65
26	Vapour Fraction		0.8821	0.8821	0.8821	0.8821
27	Heat Transferred	(kJ/h-m)	630.3	630.2	630.1	630.1
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.1682	0.2391	0.2391	0.2391
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	8.899e-003	7.605e-003	7.605e-003	7.605e-003
33	Static Gradient	(kPa/m)	-1.206e-002	3.243e-003	3.243e-003	3.243e-003
34	Accel. Gradient	(kPa/m)	-3.738e-007	1.282e-006	1.282e-006	1.282e-006
35	Liq. Reynolds		1.919e+005 *	1.919e+005 *	1.919e+005 *	1.919e+005 *
36	Vap. Reynolds		9.747e+006 *	9.747e+006 *	9.747e+006 *	9.747e+006 *
37	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981
38	Vapour Velocity	(m/s)	2.856	2.856	2.856	2.856
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	1.135e+004	1.135e+004	1.136e+004	1.136e+004
42	Elevation	(m)	0.1620	0.1700	0.1760	0.1820
43	Pressure	(bar)	71.28	71.28	71.28	71.28
44	Temperature	(C)	53.65	53.65	53.65	53.64
45	Vapour Fraction		0.8821	0.8821	0.8821	0.8821
46	Heat Transferred	(kJ/h-m)	630.0	630.0	629.9	629.9
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2391	0.2391	0.2380	0.2380
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	7.605e-003	7.605e-003	7.744e-003	7.744e-003
52	Static Gradient	(kPa/m)	3.243e-003	3.243e-003	2.424e-003	2.424e-003
53	Accel. Gradient	(kPa/m)	1.282e-006	1.282e-006	1.201e-006	1.201e-006
54	Liq. Reynolds		1.919e+005 *	1.919e+005 *	1.919e+005 *	1.919e+005 *
55	Vap. Reynolds		9.747e+006 *	9.747e+006 *	9.747e+006 *	9.748e+006 *
56	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981
57	Vapour Velocity	(m/s)	2.856	2.856	2.856	2.856
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	1.137e+004	1.137e+004	1.138e+004	1.138e+004
61	Elevation	(m)	0.1880	0.1940	0.2000	0.1800
62	Pressure	(bar)	71.28	71.28	71.28	71.28
63	Temperature	(C)	53.64	53.64	53.64	53.64
64	Vapour Fraction		0.8821	0.8821	0.8821	0.8821
65	Heat Transferred	(kJ/h-m)	629.8	629.8	629.7	629.7
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2380	0.2380	0.2380	0.1709
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	7.744e-003	7.744e-003	7.744e-003	8.862e-003
15	Static Gradient (kPa/m)	2.424e-003	2.424e-003	2.424e-003	-6.417e-003
16	Accel. Gradient (kPa/m)	1.201e-006	1.201e-006	1.201e-006	2.889e-007
17	Liq. Reynolds	1.919e+005 *	1.919e+005 *	1.919e+005 *	1.919e+005 *
18	Vap. Reynolds	9.748e+006 *	9.748e+006 *	9.748e+006 *	9.748e+006 *
19	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
20	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	1.139e+004	1.139e+004	1.140e+004	1.140e+004
24	Elevation (m)	0.1600	0.1400	0.1200	1.000e-001
25	Pressure (bar)	71.28	71.28	71.28	71.28
26	Temperature (C)	53.63	53.63	53.63	53.63
27	Vapour Fraction	0.8821	0.8821	0.8821	0.8821
28	Heat Transferred (kJ/h-m)	629.7	629.6	629.6	629.5
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.1710	0.1710	0.1710	0.1710
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	8.862e-003	8.862e-003	8.862e-003	8.862e-003
34	Static Gradient (kPa/m)	-6.417e-003	-6.417e-003	-6.417e-003	-6.417e-003
35	Accel. Gradient (kPa/m)	2.889e-007	2.889e-007	2.889e-007	2.889e-007
36	Liq. Reynolds	1.919e+005 *	1.919e+005 *	1.919e+005 *	1.919e+005 *
37	Vap. Reynolds	9.748e+006 *	9.748e+006 *	9.748e+006 *	9.748e+006 *
38	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
39	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	1.141e+004	1.141e+004	1.142e+004	1.142e+004
43	Elevation (m)	-2.200e-002	-0.1440	-0.2660	-0.3880
44	Pressure (bar)	71.28	71.28	71.28	71.28
45	Temperature (C)	53.63	53.63	53.63	53.62
46	Vapour Fraction	0.8821	0.8821	0.8821	0.8821
47	Heat Transferred (kJ/h-m)	629.5	629.5	629.5	629.4
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.1556	0.1556	0.1556	0.1556
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	9.093e-003	9.093e-003	9.093e-003	9.093e-003
53	Static Gradient (kPa/m)	-3.677e-002	-3.677e-002	-3.677e-002	-3.677e-002
54	Accel. Gradient (kPa/m)	-3.269e-006	-3.269e-006	-3.269e-006	-3.269e-006
55	Liq. Reynolds	1.919e+005 *	1.919e+005 *	1.919e+005 *	1.919e+005 *
56	Vap. Reynolds	9.748e+006 *	9.748e+006 *	9.748e+006 *	9.747e+006 *
57	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
58	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	1.143e+004	1.143e+004	1.144e+004	1.144e+004
62	Elevation (m)	-0.5100	-0.4720	-0.4340	-0.3960
63	Pressure (bar)	71.28	71.28	71.28	71.28
64	Temperature (C)	53.62	53.62	53.62	53.62
65	Vapour Fraction	0.8821	0.8821	0.8821	0.8821
66	Heat Transferred (kJ/h-m)	629.4	629.3	629.2	629.1
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.1556	0.2560	0.2560	0.2560
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	9.092e-003	6.622e-003	6.622e-003	6.622e-003
16	Static Gradient (kPa/m)	-3.677e-002	1.620e-002	1.620e-002	1.620e-002
17	Accel. Gradient (kPa/m)	-3.269e-006	2.695e-006	2.695e-006	2.695e-006
18	Liq. Reynolds	1.919e+005 *	1.919e+005 *	1.919e+005 *	1.919e+005 *
19	Vap. Reynolds	9.747e+006 *	9.747e+006 *	9.748e+006 *	9.748e+006 *
20	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
21	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	1.145e+004	1.145e+004	1.146e+004	1.146e+004
25	Elevation (m)	-0.3580	-0.3200	-0.2080	-9.600e-002
26	Pressure (bar)	71.28	71.28	71.27	71.27
27	Temperature (C)	53.61	53.61	53.61	53.61
28	Vapour Fraction	0.8821	0.8821	0.8821	0.8821
29	Heat Transferred (kJ/h-m)	629.1	629.0	629.0	628.9
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2560	0.2560	0.2975	0.2975
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.622e-003	6.622e-003	7.740e-003	7.740e-003
35	Static Gradient (kPa/m)	1.620e-002	1.620e-002	5.349e-002	5.349e-002
36	Accel. Gradient (kPa/m)	2.695e-006	2.696e-006	7.234e-006	7.234e-006
37	Liq. Reynolds	1.919e+005 *	1.919e+005 *	1.919e+005 *	1.919e+005 *
38	Vap. Reynolds	9.748e+006 *	9.748e+006 *	9.748e+006 *	9.748e+006 *
39	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
40	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	1.147e+004	1.147e+004	1.148e+004	1.148e+004
44	Elevation (m)	1.600e-002	0.1280	0.2400	0.2480
45	Pressure (bar)	71.27	71.27	71.26	71.26
46	Temperature (C)	53.60	53.60	53.60	53.60
47	Vapour Fraction	0.8821	0.8821	0.8821	0.8821
48	Heat Transferred (kJ/h-m)	628.8	628.7	628.6	628.6
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2974	0.2974	0.2974	0.2391
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	7.741e-003	7.741e-003	7.741e-003	7.606e-003
54	Static Gradient (kPa/m)	5.349e-002	5.349e-002	5.349e-002	3.243e-003
55	Accel. Gradient (kPa/m)	7.235e-006	7.235e-006	7.236e-006	1.282e-006
56	Liq. Reynolds	1.919e+005 *	1.919e+005 *	1.919e+005 *	1.919e+005 *
57	Vap. Reynolds	9.748e+006 *	9.748e+006 *	9.748e+006 *	9.748e+006 *
58	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
59	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	1.149e+004	1.149e+004	1.150e+004	1.150e+004
63	Elevation (m)	0.2560	0.2640	0.2720	0.2800
64	Pressure (bar)	71.26	71.26	71.26	71.26
65	Temperature (C)	53.59	53.59	53.59	53.59
66	Vapour Fraction	0.8821	0.8821	0.8821	0.8821
67	Heat Transferred (kJ/h-m)	628.5	628.5	628.4	628.4
68	Flow Regime	Transition	Transition	Transition	Transition
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2391	0.2391	0.2391	
15	Water HoldUp	---	---	---	
16	Friction Gradient (kPa/m)	7.606e-003	7.606e-003	7.606e-003	
17	Static Gradient (kPa/m)	3.243e-003	3.243e-003	3.243e-003	
18	Accel. Gradient (kPa/m)	1.282e-006	1.282e-006	1.282e-006	
19	Liq. Reynolds	1.919e+005 *	1.919e+005 *	1.919e+005 *	
20	Vap. Reynolds	9.748e+006 *	9.748e+006 *	9.748e+006 *	
21	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	
22	Vapour Velocity (m/s)	2.856	2.856	2.856	
23	Deposit Thickness (mm)	---	---	---	
24	Deposit Volume (m3)	---	---	---	
25	Length (m)	1.151e+004	1.151e+004	1.152e+004	
26	Elevation (m)	0.2840	0.2880	0.2920	
27	Pressure (bar)	71.26	71.26	71.26	
28	Temperature (C)	53.59	53.58	53.58	
29	Vapour Fraction	0.8821	0.8821	0.8821	
30	Heat Transferred (kJ/h-m)	628.3	628.3	628.2	
31	Flow Regime	Transition	Transition	Transition	
32	Water Flow Regime				
33	Liquid HoldUp	0.2369	0.2369	0.2369	
34	Water HoldUp	---	---	---	
35	Friction Gradient (kPa/m)	7.884e-003	7.884e-003	7.884e-003	
36	Static Gradient (kPa/m)	1.610e-003	1.610e-003	1.610e-003	
37	Accel. Gradient (kPa/m)	1.122e-006	1.122e-006	1.122e-006	
38	Liq. Reynolds	1.919e+005 *	1.919e+005 *	1.919e+005 *	
39	Vap. Reynolds	9.748e+006 *	9.748e+006 *	9.748e+006 *	
40	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	
41	Vapour Velocity (m/s)	2.856	2.856	2.856	
42	Deposit Thickness (mm)	---	---	---	
43	Deposit Volume (m3)	---	---	---	
44	Length (m)	1.153e+004	1.153e+004	1.154e+004	
45	Elevation (m)	0.3000	0.3080	0.3160	
46	Pressure (bar)	71.26	71.26	71.26	
47	Temperature (C)	53.58	53.58	53.57	
48	Vapour Fraction	0.8821	0.8821	0.8821	
49	Heat Transferred (kJ/h-m)	628.1	628.1	628.0	
50	Flow Regime	Transition	Transition	Transition	
51	Water Flow Regime				
52	Liquid HoldUp	0.2369	0.2391	0.2391	
53	Water HoldUp	---	---	---	
54	Friction Gradient (kPa/m)	7.884e-003	7.605e-003	7.605e-003	
55	Static Gradient (kPa/m)	1.610e-003	3.243e-003	3.243e-003	
56	Accel. Gradient (kPa/m)	1.122e-006	1.282e-006	1.282e-006	
57	Liq. Reynolds	1.918e+005 *	1.918e+005 *	1.918e+005 *	
58	Vap. Reynolds	9.748e+006 *	9.748e+006 *	9.748e+006 *	
59	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	
60	Vapour Velocity (m/s)	2.856	2.856	2.856	
61	Deposit Thickness (mm)	---	---	---	
62	Deposit Volume (m3)	---	---	---	
63	Length (m)	1.155e+004	1.155e+004	1.156e+004	
64	Elevation (m)	0.3320	0.3400	0.3380	
65	Pressure (bar)	71.25	71.25	71.25	
66	Temperature (C)	53.57	53.57	53.57	
67	Vapour Fraction	0.8821	0.8821	0.8821	
68	Heat Transferred (kJ/h-m)	627.9	627.9	627.9	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2392	0.2392	0.1737	0.1737
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	7.605e-003	7.605e-003	8.827e-003	8.827e-003
18	Static Gradient (kPa/m)	3.243e-003	3.243e-003	-6.484e-004	-6.484e-004
19	Accel. Gradient (kPa/m)	1.282e-006	1.282e-006	9.666e-007	9.666e-007
20	Liq. Reynolds	1.918e+005 *	1.918e+005 *	1.918e+005 *	1.918e+005 *
21	Vap. Reynolds	9.749e+006 *	9.749e+006 *	9.749e+006 *	9.749e+006 *
22	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
23	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	1.157e+004	1.157e+004	1.158e+004	1.158e+004
27	Elevation (m)	0.3340	0.3320	0.3300	0.3240
28	Pressure (bar)	71.25	71.25	71.25	71.25
29	Temperature (C)	53.56	53.56	53.56	53.56
30	Vapour Fraction	0.8821	0.8821	0.8821	0.8821
31	Heat Transferred (kJ/h-m)	627.8	627.7	627.7	627.6
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1737	0.1737	0.1737	0.1731
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	8.827e-003	8.827e-003	8.827e-003	8.835e-003
37	Static Gradient (kPa/m)	-6.484e-004	-6.484e-004	-6.484e-004	-1.941e-003
38	Accel. Gradient (kPa/m)	9.666e-007	9.666e-007	9.666e-007	8.148e-007
39	Liq. Reynolds	1.918e+005 *	1.918e+005 *	1.918e+005 *	1.918e+005 *
40	Vap. Reynolds	9.749e+006 *	9.749e+006 *	9.749e+006 *	9.749e+006 *
41	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
42	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	1.159e+004	1.159e+004	1.160e+004	1.160e+004
46	Elevation (m)	0.3180	0.3120	0.3060	0.3000
47	Pressure (bar)	71.25	71.25	71.25	71.25
48	Temperature (C)	53.56	53.55	53.55	53.55
49	Vapour Fraction	0.8821	0.8821	0.8821	0.8821
50	Heat Transferred (kJ/h-m)	627.6	627.5	627.5	627.4
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1731	0.1731	0.1731	0.1731
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	8.835e-003	8.835e-003	8.835e-003	8.835e-003
56	Static Gradient (kPa/m)	-1.941e-003	-1.941e-003	-1.941e-003	-1.941e-003
57	Accel. Gradient (kPa/m)	8.148e-007	8.148e-007	8.148e-007	8.148e-007
58	Liq. Reynolds	1.918e+005 *	1.918e+005 *	1.918e+005 *	1.918e+005 *
59	Vap. Reynolds	9.749e+006 *	9.749e+006 *	9.749e+006 *	9.749e+006 *
60	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
61	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	1.161e+004	1.161e+004	1.162e+004	1.162e+004
65	Elevation (m)	0.2980	0.2960	0.2940	0.2920
66	Pressure (bar)	71.25	71.25	71.25	71.25
67	Temperature (C)	53.55	53.55	53.54	53.54
68	Vapour Fraction	0.8821	0.8821	0.8821	0.8821
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:13:43 2022		
6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Heat Transferred (kJ/h-m)	627.4	627.3	627.3	627.2		
14	Flow Regime	Transition	Transition	Transition	Transition		
15	Water Flow Regime						
16	Liquid HoldUp	0.1737	0.1737	0.1737	0.1737		
17	Water HoldUp	---	---	---	---		
18	Friction Gradient (kPa/m)	8.827e-003	8.827e-003	8.827e-003	8.827e-003		
19	Static Gradient (kPa/m)	-6.497e-004	-6.497e-004	-6.497e-004	-6.497e-004		
20	Accel. Gradient (kPa/m)	9.664e-007	9.664e-007	9.664e-007	9.664e-007		
21	Liq. Reynolds	1.918e+005 *	1.918e+005 *	1.918e+005 *	1.918e+005 *		
22	Vap. Reynolds	9.749e+006 *	9.749e+006 *	9.749e+006 *	9.749e+006 *		
23	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
24	Vapour Velocity (m/s)	2.856	2.856	2.856	2.856		
25	Deposit Thickness (mm)	---	---	---	---		
26	Deposit Volume (m3)	---	---	---	---		
27	Length (m)	1.163e+004	1.163e+004	1.164e+004	1.164e+004		
28	Elevation (m)	0.2900	0.2760	0.2620	0.2480		
29	Pressure (bar)	71.25	71.25	71.25	71.25		
30	Temperature (C)	53.54	53.54	53.54	53.53		
31	Vapour Fraction	0.8821	0.8821	0.8821	0.8821		
32	Heat Transferred (kJ/h-m)	627.1	627.1	627.1	627.0		
33	Flow Regime	Transition	Transition	Transition	Transition		
34	Water Flow Regime						
35	Liquid HoldUp	0.1737	0.1719	0.1719	0.1719		
36	Water HoldUp	---	---	---	---		
37	Friction Gradient (kPa/m)	8.827e-003	8.850e-003	8.850e-003	8.850e-003		
38	Static Gradient (kPa/m)	-6.497e-004	-4.508e-003	-4.508e-003	-4.508e-003		
39	Accel. Gradient (kPa/m)	9.664e-007	5.132e-007	5.132e-007	5.132e-007		
40	Liq. Reynolds	1.918e+005 *	1.918e+005 *	1.918e+005 *	1.918e+005 *		
41	Vap. Reynolds	9.749e+006 *	9.749e+006 *	9.749e+006 *	9.749e+006 *		
42	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
43	Vapour Velocity (m/s)	2.856	2.856	2.856	2.855		
44	Deposit Thickness (mm)	---	---	---	---		
45	Deposit Volume (m3)	---	---	---	---		
46	Length (m)	1.165e+004	1.165e+004	1.166e+004	1.166e+004		
47	Elevation (m)	0.2340	0.2200	0.2280	0.2360		
48	Pressure (bar)	71.25	71.25	71.25	71.25		
49	Temperature (C)	53.53	53.53	53.53	53.53		
50	Vapour Fraction	0.8821	0.8821	0.8821	0.8821		
51	Heat Transferred (kJ/h-m)	627.0	626.9	626.8	626.8		
52	Flow Regime	Transition	Transition	Transition	Transition		
53	Water Flow Regime						
54	Liquid HoldUp	0.1719	0.1719	0.2392	0.2392		
55	Water HoldUp	---	---	---	---		
56	Friction Gradient (kPa/m)	8.850e-003	8.850e-003	7.605e-003	7.605e-003		
57	Static Gradient (kPa/m)	-4.508e-003	-4.508e-003	3.243e-003	3.243e-003		
58	Accel. Gradient (kPa/m)	5.132e-007	5.132e-007	1.282e-006	1.282e-006		
59	Liq. Reynolds	1.918e+005 *	1.918e+005 *	1.918e+005 *	1.918e+005 *		
60	Vap. Reynolds	9.749e+006 *	9.749e+006 *	9.749e+006 *	9.749e+006 *		
61	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
62	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855		
63	Deposit Thickness (mm)	---	---	---	---		
64	Deposit Volume (m3)	---	---	---	---		
65	Length (m)	1.167e+004	1.167e+004	1.168e+004	1.168e+004		
66	Elevation (m)	0.2440	0.2520	0.2600	0.2720		
67	Pressure (bar)	71.25	71.25	71.24	71.24		
68	Temperature (C)	53.53	53.52	53.52	53.52		
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 221 of 937		

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Vapour Fraction		0.8821	0.8821	0.8821	0.8821				
14	Heat Transferred (kJ/h-m)		626.7	626.6	626.6	626.5				
15	Flow Regime		Transition	Transition	Transition	Transition				
16	Water Flow Regime									
17	Liquid HoldUp		0.2392	0.2392	0.2392	0.2414				
18	Water HoldUp		---	---	---	---				
19	Friction Gradient (kPa/m)		7.605e-003	7.605e-003	7.605e-003	7.334e-003				
20	Static Gradient (kPa/m)		3.243e-003	3.243e-003	3.243e-003	4.898e-003				
21	Accel. Gradient (kPa/m)		1.282e-006	1.282e-006	1.282e-006	1.446e-006				
22	Liq. Reynolds		1.918e+005 *	1.918e+005 *	1.918e+005 *	1.918e+005 *				
23	Vap. Reynolds		9.749e+006 *	9.749e+006 *	9.749e+006 *	9.749e+006 *				
24	Liquid Velocity (m/s)		0.1981	0.1981	0.1981	0.1981				
25	Vapour Velocity (m/s)		2.855	2.855	2.855	2.855				
26	Deposit Thickness (mm)		---	---	---	---				
27	Deposit Volume (m3)		---	---	---	---				
28	Length (m)		1.169e+004	1.169e+004	1.170e+004	1.170e+004				
29	Elevation (m)		0.2840	0.2960	0.3080	0.3200				
30	Pressure (bar)		71.24	71.24	71.24	71.24				
31	Temperature (C)		53.52	53.52	53.51	53.51				
32	Vapour Fraction		0.8821	0.8821	0.8821	0.8821				
33	Heat Transferred (kJ/h-m)		626.5	626.4	626.4	626.3				
34	Flow Regime		Transition	Transition	Transition	Transition				
35	Water Flow Regime									
36	Liquid HoldUp		0.2414	0.2414	0.2414	0.2414				
37	Water HoldUp		---	---	---	---				
38	Friction Gradient (kPa/m)		7.334e-003	7.334e-003	7.334e-003	7.334e-003				
39	Static Gradient (kPa/m)		4.898e-003	4.898e-003	4.898e-003	4.898e-003				
40	Accel. Gradient (kPa/m)		1.446e-006	1.446e-006	1.446e-006	1.446e-006				
41	Liq. Reynolds		1.918e+005 *	1.918e+005 *	1.918e+005 *	1.918e+005 *				
42	Vap. Reynolds		9.749e+006 *	9.749e+006 *	9.749e+006 *	9.749e+006 *				
43	Liquid Velocity (m/s)		0.1981	0.1981	0.1981	0.1981				
44	Vapour Velocity (m/s)		2.855	2.855	2.855	2.855				
45	Deposit Thickness (mm)		---	---	---	---				
46	Deposit Volume (m3)		---	---	---	---				
47	Length (m)		1.171e+004	1.171e+004	1.172e+004	1.172e+004				
48	Elevation (m)		0.3200	0.3200	0.3200	0.3200				
49	Pressure (bar)		71.24	71.24	71.24	71.24				
50	Temperature (C)		53.51	53.51	53.51	53.50				
51	Vapour Fraction		0.8821	0.8821	0.8821	0.8821				
52	Heat Transferred (kJ/h-m)		626.3	626.2	626.2	626.1				
53	Flow Regime		Transition	Transition	Transition	Transition				
54	Water Flow Regime									
55	Liquid HoldUp		0.2540	0.2540	0.2540	0.2540				
56	Water HoldUp		---	---	---	---				
57	Friction Gradient (kPa/m)		5.945e-003	5.945e-003	5.945e-003	5.945e-003				
58	Static Gradient (kPa/m)		0.0000	0.0000	0.0000	0.0000				
59	Accel. Gradient (kPa/m)		7.027e-007	7.027e-007	7.027e-007	7.026e-007				
60	Liq. Reynolds		1.918e+005 *	1.918e+005 *	1.918e+005 *	1.918e+005 *				
61	Vap. Reynolds		9.749e+006 *	9.749e+006 *	9.749e+006 *	9.749e+006 *				
62	Liquid Velocity (m/s)		0.1981	0.1981	0.1981	0.1981				
63	Vapour Velocity (m/s)		2.855	2.855	2.855	2.855				
64	Deposit Thickness (mm)		---	---	---	---				
65	Deposit Volume (m3)		---	---	---	---				
66	Length (m)		1.173e+004	1.173e+004	1.174e+004	1.174e+004				
67	Elevation (m)		0.3200	0.3220	0.3240	0.3260				
68	Pressure (bar)		71.24	71.24	71.24	71.24				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Temperature (C)	53.50	53.50	53.50	53.50
14	Vapour Fraction	0.8821	0.8821	0.8821	0.8821
15	Heat Transferred (kJ/h-m)	626.1	626.0	626.0	625.9
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2540	0.2358	0.2358	0.2358
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	5.945e-003	8.026e-003	8.026e-003	8.026e-003
21	Static Gradient (kPa/m)	0.0000	8.025e-004	8.025e-004	8.025e-004
22	Accel. Gradient (kPa/m)	7.026e-007	1.043e-006	1.043e-006	1.043e-006
23	Liq. Reynolds	1.918e+005 *	1.918e+005 *	1.918e+005 *	1.918e+005 *
24	Vap. Reynolds	9.749e+006 *	9.749e+006 *	9.749e+006 *	9.749e+006 *
25	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
26	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	1.175e+004	1.175e+004	1.176e+004	1.176e+004
30	Elevation (m)	0.3280	0.3300	0.3300	0.3300
31	Pressure (bar)	71.24	71.24	71.24	71.24
32	Temperature (C)	53.49	53.49	53.49	53.49
33	Vapour Fraction	0.8821	0.8821	0.8821	0.8821
34	Heat Transferred (kJ/h-m)	625.9	625.8	625.8	625.7
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2358	0.2358	0.2540	0.2540
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	8.026e-003	8.026e-003	5.945e-003	5.945e-003
40	Static Gradient (kPa/m)	8.025e-004	8.025e-004	0.0000	0.0000
41	Accel. Gradient (kPa/m)	1.043e-006	1.043e-006	7.026e-007	7.026e-007
42	Liq. Reynolds	1.918e+005 *	1.918e+005 *	1.918e+005 *	1.918e+005 *
43	Vap. Reynolds	9.749e+006 *	9.749e+006 *	9.749e+006 *	9.749e+006 *
44	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
45	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	1.177e+004	1.177e+004	1.178e+004	1.178e+004
49	Elevation (m)	0.3300	0.3300	0.3300	0.3240
50	Pressure (bar)	71.24	71.24	71.24	71.24
51	Temperature (C)	53.49	53.48	53.48	53.48
52	Vapour Fraction	0.8821	0.8821	0.8821	0.8821
53	Heat Transferred (kJ/h-m)	625.7	625.6	625.6	625.6
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2540	0.2540	0.2540	0.1731
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	5.945e-003	5.945e-003	5.945e-003	8.834e-003
59	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-1.941e-003
60	Accel. Gradient (kPa/m)	7.026e-007	7.026e-007	7.026e-007	8.147e-007
61	Liq. Reynolds	1.918e+005 *	1.918e+005 *	1.918e+005 *	1.918e+005 *
62	Vap. Reynolds	9.749e+006 *	9.749e+006 *	9.749e+006 *	9.749e+006 *
63	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
64	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	1.179e+004	1.179e+004	1.180e+004	1.180e+004
68	Elevation (m)	0.3180	0.3120	0.3060	0.3000
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 223 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:13:43 2022		
6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	71.24	71.24	71.24	71.23		
14	Temperature (C)	53.48	53.48	53.48	53.47		
15	Vapour Fraction	0.8821	0.8821	0.8821	0.8821		
16	Heat Transferred (kJ/h-m)	625.5	625.5	625.4	625.4		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.1731	0.1731	0.1731	0.1731		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	8.834e-003	8.834e-003	8.834e-003	8.834e-003		
22	Static Gradient (kPa/m)	-1.941e-003	-1.941e-003	-1.941e-003	-1.941e-003		
23	Accel. Gradient (kPa/m)	8.147e-007	8.147e-007	8.147e-007	8.147e-007		
24	Liq. Reynolds	1.918e+005 *	1.918e+005 *	1.918e+005 *	1.917e+005 *		
25	Vap. Reynolds	9.749e+006 *	9.749e+006 *	9.749e+006 *	9.749e+006 *		
26	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
27	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	1.181e+004	1.181e+004	1.182e+004	1.182e+004		
31	Elevation (m)	0.2960	0.2920	0.2880	0.2840		
32	Pressure (bar)	71.23	71.23	71.23	71.23		
33	Temperature (C)	53.47	53.47	53.47	53.47		
34	Vapour Fraction	0.8821	0.8821	0.8820	0.8820		
35	Heat Transferred (kJ/h-m)	625.3	625.3	625.2	625.2		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.1734	0.1734	0.1734	0.1734		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	8.830e-003	8.830e-003	8.830e-003	8.830e-003		
41	Static Gradient (kPa/m)	-1.299e-003	-1.299e-003	-1.299e-003	-1.299e-003		
42	Accel. Gradient (kPa/m)	8.901e-007	8.901e-007	8.901e-007	8.901e-007		
43	Liq. Reynolds	1.917e+005 *	1.917e+005 *	1.917e+005 *	1.917e+005 *		
44	Vap. Reynolds	9.750e+006 *	9.750e+006 *	9.750e+006 *	9.750e+006 *		
45	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
46	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	1.183e+004	1.183e+004	1.184e+004	1.184e+004		
50	Elevation (m)	0.2800	0.2840	0.2880	0.2920		
51	Pressure (bar)	71.23	71.23	71.23	71.23		
52	Temperature (C)	53.46	53.46	53.46	53.46		
53	Vapour Fraction	0.8820	0.8820	0.8820	0.8820		
54	Heat Transferred (kJ/h-m)	625.1	625.0	625.0	624.9		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.1734	0.2369	0.2369	0.2369		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	8.830e-003	7.883e-003	7.883e-003	7.883e-003		
60	Static Gradient (kPa/m)	-1.299e-003	1.611e-003	1.611e-003	1.611e-003		
61	Accel. Gradient (kPa/m)	8.901e-007	1.122e-006	1.122e-006	1.122e-006		
62	Liq. Reynolds	1.917e+005 *	1.917e+005 *	1.917e+005 *	1.917e+005 *		
63	Vap. Reynolds	9.750e+006 *	9.750e+006 *	9.750e+006 *	9.750e+006 *		
64	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
65	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	1.185e+004	1.185e+004	1.186e+004	1.186e+004		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Elevation (m)	0.2960	0.3000	0.2980	0.2960
14	Pressure (bar)	71.23	71.23	71.23	71.23
15	Temperature (C)	53.46	53.45	53.45	53.45
16	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
17	Heat Transferred (kJ/h-m)	624.9	624.8	624.8	624.8
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2369	0.2369	0.1737	0.1737
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	7.882e-003	7.882e-003	8.826e-003	8.826e-003
23	Static Gradient (kPa/m)	1.611e-003	1.611e-003	-6.485e-004	-6.485e-004
24	Accel. Gradient (kPa/m)	1.122e-006	1.122e-006	9.665e-007	9.665e-007
25	Liq. Reynolds	1.917e+005 *	1.917e+005 *	1.917e+005 *	1.917e+005 *
26	Vap. Reynolds	9.750e+006 *	9.750e+006 *	9.750e+006 *	9.750e+006 *
27	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
28	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	1.187e+004	1.187e+004	1.188e+004	1.188e+004
32	Elevation (m)	0.2940	0.2920	0.2900	0.2940
33	Pressure (bar)	71.23	71.23	71.23	71.23
34	Temperature (C)	53.45	53.45	53.44	53.44
35	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
36	Heat Transferred (kJ/h-m)	624.7	624.7	624.6	624.5
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.1737	0.1737	0.1737	0.2369
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	8.826e-003	8.826e-003	8.826e-003	7.882e-003
42	Static Gradient (kPa/m)	-6.485e-004	-6.485e-004	-6.486e-004	1.611e-003
43	Accel. Gradient (kPa/m)	9.665e-007	9.665e-007	9.665e-007	1.122e-006
44	Liq. Reynolds	1.917e+005 *	1.917e+005 *	1.917e+005 *	1.917e+005 *
45	Vap. Reynolds	9.750e+006 *	9.750e+006 *	9.750e+006 *	9.750e+006 *
46	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
47	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	1.189e+004	1.189e+004	1.190e+004	1.190e+004
51	Elevation (m)	0.2980	0.3020	0.3060	0.3100
52	Pressure (bar)	71.23	71.23	71.23	71.23
53	Temperature (C)	53.44	53.44	53.44	53.44
54	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
55	Heat Transferred (kJ/h-m)	624.5	624.4	624.4	624.3
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2369	0.2369	0.2369	0.2369
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	7.882e-003	7.882e-003	7.882e-003	7.882e-003
61	Static Gradient (kPa/m)	1.611e-003	1.611e-003	1.611e-003	1.611e-003
62	Accel. Gradient (kPa/m)	1.122e-006	1.122e-006	1.122e-006	1.122e-006
63	Liq. Reynolds	1.917e+005 *	1.917e+005 *	1.917e+005 *	1.917e+005 *
64	Vap. Reynolds	9.750e+006 *	9.750e+006 *	9.750e+006 *	9.750e+006 *
65	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
66	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 225 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:13:43 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length	(m)	1.191e+004	1.191e+004	1.192e+004	1.192e+004	
14	Elevation	(m)	0.3080	0.3060	0.3040	0.3020	
15	Pressure	(bar)	71.23	71.23	71.22	71.22	
16	Temperature	(C)	53.43	53.43	53.43	53.43	
17	Vapour Fraction		0.8820	0.8820	0.8820	0.8820	
18	Heat Transferred	(kJ/h-m)	624.3	624.3	624.2	624.1	
19	Flow Regime		Transition	Transition	Transition	Transition	
20	Water Flow Regime						
21	Liquid HoldUp		0.1737	0.1737	0.1737	0.1737	
22	Water HoldUp		---	---	---	---	
23	Friction Gradient	(kPa/m)	8.826e-003	8.826e-003	8.826e-003	8.826e-003	
24	Static Gradient	(kPa/m)	-6.447e-004	-6.447e-004	-6.447e-004	-6.447e-004	
25	Accel. Gradient	(kPa/m)	9.670e-007	9.670e-007	9.670e-007	9.670e-007	
26	Liq. Reynolds		1.917e+005 *	1.917e+005 *	1.917e+005 *	1.917e+005 *	
27	Vap. Reynolds		9.750e+006 *	9.750e+006 *	9.750e+006 *	9.750e+006 *	
28	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
29	Vapour Velocity	(m/s)	2.855	2.855	2.855	2.855	
30	Deposit Thickness	(mm)	---	---	---	---	
31	Deposit Volume	(m3)	---	---	---	---	
32	Length	(m)	1.193e+004	1.193e+004	1.194e+004	1.194e+004	
33	Elevation	(m)	0.3000	0.3000	0.3000	0.3000	
34	Pressure	(bar)	71.22	71.22	71.22	71.22	
35	Temperature	(C)	53.43	53.42	53.42	53.42	
36	Vapour Fraction		0.8820	0.8820	0.8820	0.8820	
37	Heat Transferred	(kJ/h-m)	624.1	624.0	624.0	623.9	
38	Flow Regime		Transition	Transition	Transition	Transition	
39	Water Flow Regime						
40	Liquid HoldUp		0.1737	0.2540	0.2540	0.2540	
41	Water HoldUp		---	---	---	---	
42	Friction Gradient	(kPa/m)	8.826e-003	5.944e-003	5.944e-003	5.944e-003	
43	Static Gradient	(kPa/m)	-6.447e-004	0.0000	0.0000	0.0000	
44	Accel. Gradient	(kPa/m)	9.670e-007	7.026e-007	7.026e-007	7.026e-007	
45	Liq. Reynolds		1.917e+005 *	1.917e+005 *	1.917e+005 *	1.917e+005 *	
46	Vap. Reynolds		9.750e+006 *	9.750e+006 *	9.750e+006 *	9.750e+006 *	
47	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
48	Vapour Velocity	(m/s)	2.855	2.855	2.855	2.855	
49	Deposit Thickness	(mm)	---	---	---	---	
50	Deposit Volume	(m3)	---	---	---	---	
51	Length	(m)	1.194e+004	1.195e+004	1.196e+004	1.196e+004	
52	Elevation	(m)	0.3000	0.3000	0.2980	0.2960	
53	Pressure	(bar)	71.22	71.22	71.22	71.22	
54	Temperature	(C)	53.42	53.42	53.42	53.41	
55	Vapour Fraction		0.8820	0.8820	0.8820	0.8820	
56	Heat Transferred	(kJ/h-m)	623.9	623.8	623.8	623.8	
57	Flow Regime		Transition	Transition	Transition	Transition	
58	Water Flow Regime						
59	Liquid HoldUp		0.2540	0.2540	0.1738	0.1738	
60	Water HoldUp		---	---	---	---	
61	Friction Gradient	(kPa/m)	5.944e-003	5.944e-003	8.824e-003	8.824e-003	
62	Static Gradient	(kPa/m)	0.0000	0.0000	-4.634e-004	-4.634e-004	
63	Accel. Gradient	(kPa/m)	7.026e-007	7.026e-007	9.883e-007	9.883e-007	
64	Liq. Reynolds		1.917e+005 *	1.917e+005 *	1.917e+005 *	1.917e+005 *	
65	Vap. Reynolds		9.750e+006 *	9.750e+006 *	9.750e+006 *	9.750e+006 *	
66	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
67	Vapour Velocity	(m/s)	2.855	2.855	2.855	2.855	
68	Deposit Thickness	(mm)	---	---	---	---	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 226 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	1.197e+004	1.198e+004	1.198e+004	1.199e+004		
15	Elevation (m)	0.2940	0.2920	0.2900	0.2900		
16	Pressure (bar)	71.22	71.22	71.22	71.22		
17	Temperature (C)	53.41	53.41	53.40	53.40		
18	Vapour Fraction	0.8820	0.8820	0.8820	0.8820		
19	Heat Transferred (kJ/h-m)	623.7	623.6	623.5	623.4		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	0.1738	0.1738	0.1738	0.2540		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	8.824e-003	8.824e-003	8.824e-003	5.944e-003		
25	Static Gradient (kPa/m)	-4.634e-004	-4.634e-004	-4.634e-004	0.0000		
26	Accel. Gradient (kPa/m)	9.883e-007	9.883e-007	9.883e-007	7.026e-007		
27	Liq. Reynolds	1.917e+005 *	1.917e+005 *	1.917e+005 *	1.917e+005 *		
28	Vap. Reynolds	9.750e+006 *	9.750e+006 *	9.750e+006 *	9.750e+006 *		
29	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
30	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	1.200e+004	1.200e+004	1.201e+004	1.202e+004		
34	Elevation (m)	0.2900	0.2900	0.2900	0.2900		
35	Pressure (bar)	71.22	71.22	71.22	71.22		
36	Temperature (C)	53.40	53.40	53.39	53.39		
37	Vapour Fraction	0.8820	0.8820	0.8820	0.8820		
38	Heat Transferred (kJ/h-m)	623.4	623.3	623.2	623.1		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.2540	0.2540	0.2540	0.2540		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	5.944e-003	5.944e-003	5.944e-003	5.944e-003		
44	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000		
45	Accel. Gradient (kPa/m)	7.026e-007	7.026e-007	7.025e-007	7.025e-007		
46	Liq. Reynolds	1.917e+005 *	1.917e+005 *	1.917e+005 *	1.917e+005 *		
47	Vap. Reynolds	9.750e+006 *	9.750e+006 *	9.750e+006 *	9.750e+006 *		
48	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
49	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	1.203e+004	1.203e+004	1.204e+004	1.205e+004		
53	Elevation (m)	0.2820	0.2740	0.2660	0.2580		
54	Pressure (bar)	71.22	71.22	71.22	71.22		
55	Temperature (C)	53.39	53.39	53.38	53.38		
56	Vapour Fraction	0.8820	0.8820	0.8820	0.8820		
57	Heat Transferred (kJ/h-m)	623.1	623.1	623.0	622.9		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.1731	0.1731	0.1731	0.1731		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	8.832e-003	8.832e-003	8.832e-003	8.832e-003		
63	Static Gradient (kPa/m)	-1.849e-003	-1.849e-003	-1.849e-003	-1.849e-003		
64	Accel. Gradient (kPa/m)	8.254e-007	8.254e-007	8.254e-007	8.254e-007		
65	Liq. Reynolds	1.917e+005 *	1.917e+005 *	1.917e+005 *	1.917e+005 *		
66	Vap. Reynolds	9.750e+006 *	9.750e+006 *	9.750e+006 *	9.750e+006 *		
67	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
68	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Deposit Thickness (mm)	---	---	---	---					
14	Deposit Volume (m3)	---	---	---	---					
15	Length (m)	1.205e+004	1.206e+004	1.207e+004	1.207e+004					
16	Elevation (m)	0.2500	0.3600	0.4700	0.5800					
17	Pressure (bar)	71.22	71.21	71.21	71.21					
18	Temperature (C)	53.38	53.38	53.37	53.37					
19	Vapour Fraction	0.8820	0.8820	0.8820	0.8820					
20	Heat Transferred (kJ/h-m)	622.8	622.7	622.6	622.5					
21	Flow Regime	Transition	Transition	Transition	Transition					
22	Water Flow Regime									
23	Liquid HoldUp	0.1731	0.2788	0.2788	0.2788					
24	Water HoldUp	---	---	---	---					
25	Friction Gradient (kPa/m)	8.832e-003	7.734e-003	7.734e-003	7.735e-003					
26	Static Gradient (kPa/m)	-1.849e-003	3.571e-002	3.571e-002	3.571e-002					
27	Accel. Gradient (kPa/m)	8.254e-007	5.136e-006	5.137e-006	5.137e-006					
28	Liq. Reynolds	1.917e+005 *	1.917e+005 *	1.916e+005 *	1.916e+005 *					
29	Vap. Reynolds	9.750e+006 *	9.751e+006 *	9.751e+006 *	9.751e+006 *					
30	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981					
31	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855					
32	Deposit Thickness (mm)	---	---	---	---					
33	Deposit Volume (m3)	---	---	---	---					
34	Length (m)	1.208e+004	1.209e+004	1.210e+004	1.210e+004					
35	Elevation (m)	0.6900	0.8000	0.7000	0.6000					
36	Pressure (bar)	71.20	71.20	71.20	71.20					
37	Temperature (C)	53.36	53.36	53.36	53.36					
38	Vapour Fraction	0.8820	0.8820	0.8820	0.8820					
39	Heat Transferred (kJ/h-m)	622.4	622.3	622.3	622.3					
40	Flow Regime	Transition	Transition	Transition	Transition					
41	Water Flow Regime									
42	Liquid HoldUp	0.2788	0.2788	0.1632	0.1632					
43	Water HoldUp	---	---	---	---					
44	Friction Gradient (kPa/m)	7.735e-003	7.735e-003	8.971e-003	8.971e-003					
45	Static Gradient (kPa/m)	3.571e-002	3.571e-002	-2.224e-002	-2.224e-002					
46	Accel. Gradient (kPa/m)	5.137e-006	5.138e-006	-1.569e-006	-1.569e-006					
47	Liq. Reynolds	1.916e+005 *	1.916e+005 *	1.916e+005 *	1.916e+005 *					
48	Vap. Reynolds	9.751e+006 *	9.751e+006 *	9.751e+006 *	9.751e+006 *					
49	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981					
50	Vapour Velocity (m/s)	2.855	2.855	2.855	2.855					
51	Deposit Thickness (mm)	---	---	---	---					
52	Deposit Volume (m3)	---	---	---	---					
53	Length (m)	1.211e+004	1.212e+004	1.212e+004	1.213e+004					
54	Elevation (m)	0.5000	0.4000	0.3000	0.3000					
55	Pressure (bar)	71.20	71.20	71.20	71.20					
56	Temperature (C)	53.35	53.35	53.35	53.35					
57	Vapour Fraction	0.8820	0.8820	0.8820	0.8820					
58	Heat Transferred (kJ/h-m)	622.2	622.2	622.1	622.0					
59	Flow Regime	Transition	Transition	Transition	Transition					
60	Water Flow Regime									
61	Liquid HoldUp	0.1632	0.1632	0.1632	0.2540					
62	Water HoldUp	---	---	---	---					
63	Friction Gradient (kPa/m)	8.970e-003	8.970e-003	8.970e-003	5.944e-003					
64	Static Gradient (kPa/m)	-2.224e-002	-2.224e-002	-2.224e-002	0.0000					
65	Accel. Gradient (kPa/m)	-1.569e-006	-1.569e-006	-1.569e-006	7.027e-007					
66	Liq. Reynolds	1.916e+005 *	1.916e+005 *	1.916e+005 *	1.916e+005 *					
67	Vap. Reynolds	9.751e+006 *	9.751e+006 *	9.751e+006 *	9.751e+006 *					
68	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Vapour Velocity	(m/s)	2.855	2.855	2.855	2.855				
14	Deposit Thickness	(mm)	---	---	---	---				
15	Deposit Volume	(m3)	---	---	---	---				
16	Length	(m)	1.214e+004	1.214e+004	1.215e+004	1.216e+004				
17	Elevation	(m)	0.3000	0.3000	0.3000	0.3000				
18	Pressure	(bar)	71.20	71.20	71.20	71.20				
19	Temperature	(C)	53.35	53.34	53.34	53.34				
20	Vapour Fraction		0.8820	0.8820	0.8820	0.8820				
21	Heat Transferred	(kJ/h-m)	621.9	621.8	621.8	621.7				
22	Flow Regime		Transition	Transition	Transition	Transition				
23	Water Flow Regime									
24	Liquid HoldUp		0.2540	0.2540	0.2540	0.2540				
25	Water HoldUp		---	---	---	---				
26	Friction Gradient	(kPa/m)	5.944e-003	5.944e-003	5.943e-003	5.943e-003				
27	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000				
28	Accel. Gradient	(kPa/m)	7.027e-007	7.027e-007	7.027e-007	7.026e-007				
29	Liq. Reynolds		1.916e+005 *	1.916e+005 *	1.916e+005 *	1.916e+005 *				
30	Vap. Reynolds		9.751e+006 *	9.751e+006 *	9.751e+006 *	9.751e+006 *				
31	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981				
32	Vapour Velocity	(m/s)	2.855	2.855	2.855	2.855				
33	Deposit Thickness	(mm)	---	---	---	---				
34	Deposit Volume	(m3)	---	---	---	---				
35	Length	(m)	1.217e+004	1.217e+004	1.218e+004	1.219e+004				
36	Elevation	(m)	0.3000	0.3000	0.3000	0.3000				
37	Pressure	(bar)	71.20	71.20	71.20	71.20				
38	Temperature	(C)	53.34	53.33	53.33	53.33				
39	Vapour Fraction		0.8820	0.8820	0.8820	0.8820				
40	Heat Transferred	(kJ/h-m)	621.6	621.6	621.5	621.4				
41	Flow Regime		Transition	Transition	Transition	Transition				
42	Water Flow Regime									
43	Liquid HoldUp		0.2540	0.2540	0.2540	0.2540				
44	Water HoldUp		---	---	---	---				
45	Friction Gradient	(kPa/m)	5.943e-003	5.943e-003	5.943e-003	5.943e-003				
46	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000				
47	Accel. Gradient	(kPa/m)	7.026e-007	7.026e-007	7.026e-007	7.026e-007				
48	Liq. Reynolds		1.916e+005 *	1.916e+005 *	1.916e+005 *	1.916e+005 *				
49	Vap. Reynolds		9.751e+006 *	9.751e+006 *	9.751e+006 *	9.751e+006 *				
50	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981				
51	Vapour Velocity	(m/s)	2.855	2.855	2.855	2.855				
52	Deposit Thickness	(mm)	---	---	---	---				
53	Deposit Volume	(m3)	---	---	---	---				
54	Length	(m)	1.219e+004	1.220e+004	1.221e+004	1.221e+004				
55	Elevation	(m)	0.3000	0.2960	0.2920	0.2880				
56	Pressure	(bar)	71.20	71.20	71.20	71.20				
57	Temperature	(C)	53.32	53.32	53.32	53.32				
58	Vapour Fraction		0.8820	0.8820	0.8820	0.8820				
59	Heat Transferred	(kJ/h-m)	621.4	621.3	621.3	621.2				
60	Flow Regime		Transition	Transition	Transition	Transition				
61	Water Flow Regime									
62	Liquid HoldUp		0.2540	0.1736	0.1736	0.1736				
63	Water HoldUp		---	---	---	---				
64	Friction Gradient	(kPa/m)	5.943e-003	8.826e-003	8.826e-003	8.826e-003				
65	Static Gradient	(kPa/m)	0.0000	-9.262e-004	-9.262e-004	-9.262e-004				
66	Accel. Gradient	(kPa/m)	7.026e-007	9.340e-007	9.340e-007	9.340e-007				
67	Liq. Reynolds		1.916e+005 *	1.916e+005 *	1.916e+005 *	1.916e+005 *				
68	Vap. Reynolds		9.751e+006 *	9.751e+006 *	9.751e+006 *	9.751e+006 *				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
14	Vapour Velocity	(m/s)	2.855	2.855	2.855	2.855	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	1.222e+004	1.223e+004	1.223e+004	1.224e+004	
18	Elevation	(m)	0.2840	0.2800	0.2740	0.2680	
19	Pressure	(bar)	71.20	71.20	71.20	71.20	
20	Temperature	(C)	53.31	53.31	53.31	53.31	
21	Vapour Fraction		0.8820	0.8820	0.8820	0.8820	
22	Heat Transferred	(kJ/h-m)	621.1	621.1	621.0	620.9	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.1736	0.1736	0.1733	0.1733	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	8.826e-003	8.826e-003	8.830e-003	8.830e-003	
28	Static Gradient	(kPa/m)	-9.262e-004	-9.262e-004	-1.624e-003	-1.624e-003	
29	Accel. Gradient	(kPa/m)	9.340e-007	9.340e-007	8.520e-007	8.520e-007	
30	Liq. Reynolds		1.916e+005 *	1.916e+005 *	1.916e+005 *	1.916e+005 *	
31	Vap. Reynolds		9.751e+006 *	9.751e+006 *	9.751e+006 *	9.751e+006 *	
32	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
33	Vapour Velocity	(m/s)	2.855	2.854	2.854	2.854	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	1.225e+004	1.225e+004	1.226e+004	1.226e+004	
37	Elevation	(m)	0.2620	0.2560	0.2500	0.2220	
38	Pressure	(bar)	71.20	71.20	71.20	71.20	
39	Temperature	(C)	53.30	53.30	53.30	53.30	
40	Vapour Fraction		0.8820	0.8820	0.8820	0.8820	
41	Heat Transferred	(kJ/h-m)	620.9	620.8	620.8	620.7	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.1733	0.1733	0.1733	0.1698	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	8.830e-003	8.830e-003	8.830e-003	8.876e-003	
47	Static Gradient	(kPa/m)	-1.624e-003	-1.624e-003	-1.624e-003	-8.946e-003	
48	Accel. Gradient	(kPa/m)	8.520e-007	8.520e-007	8.520e-007	-8.267e-009	
49	Liq. Reynolds		1.916e+005 *	1.916e+005 *	1.916e+005 *	1.916e+005 *	
50	Vap. Reynolds		9.751e+006 *	9.751e+006 *	9.751e+006 *	9.751e+006 *	
51	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
52	Vapour Velocity	(m/s)	2.854	2.854	2.854	2.854	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	1.227e+004	1.227e+004	1.228e+004	1.228e+004	
56	Elevation	(m)	0.1940	0.1660	0.1380	0.1100	
57	Pressure	(bar)	71.20	71.20	71.20	71.20	
58	Temperature	(C)	53.30	53.30	53.29	53.29	
59	Vapour Fraction		0.8820	0.8820	0.8820	0.8820	
60	Heat Transferred	(kJ/h-m)	620.7	620.6	620.6	620.5	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.1698	0.1698	0.1698	0.1698	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	8.876e-003	8.876e-003	8.875e-003	8.875e-003	
66	Static Gradient	(kPa/m)	-8.946e-003	-8.946e-003	-8.946e-003	-8.946e-003	
67	Accel. Gradient	(kPa/m)	-8.283e-009	-8.298e-009	-8.313e-009	-8.329e-009	
68	Liq. Reynolds		1.916e+005 *	1.916e+005 *	1.916e+005 *	1.916e+005 *	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 230 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.751e+006 *	9.751e+006 *	9.751e+006 *	9.751e+006 *
14	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
15	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	1.229e+004	1.229e+004	1.230e+004	1.230e+004
19	Elevation (m)	0.1320	0.1540	0.1760	0.1980
20	Pressure (bar)	71.19	71.19	71.19	71.19
21	Temperature (C)	53.29	53.29	53.29	53.28
22	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
23	Heat Transferred (kJ/h-m)	620.4	620.4	620.3	620.3
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2471	0.2471	0.2471	0.2471
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.685e-003	6.685e-003	6.685e-003	6.685e-003
29	Static Gradient (kPa/m)	9.137e-003	9.137e-003	9.137e-003	9.137e-003
30	Accel. Gradient (kPa/m)	1.870e-006	1.870e-006	1.871e-006	1.871e-006
31	Liq. Reynolds	1.916e+005 *	1.916e+005 *	1.916e+005 *	1.916e+005 *
32	Vap. Reynolds	9.751e+006 *	9.751e+006 *	9.751e+006 *	9.751e+006 *
33	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
34	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	1.231e+004	1.231e+004	1.232e+004	1.232e+004
38	Elevation (m)	0.2200	0.2160	0.2120	0.2080
39	Pressure (bar)	71.19	71.19	71.19	71.19
40	Temperature (C)	53.28	53.28	53.28	53.28
41	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
42	Heat Transferred (kJ/h-m)	620.2	620.2	620.2	620.1
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2471	0.1734	0.1734	0.1734
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.685e-003	8.828e-003	8.828e-003	8.828e-003
48	Static Gradient (kPa/m)	9.137e-003	-1.296e-003	-1.296e-003	-1.296e-003
49	Accel. Gradient (kPa/m)	1.871e-006	8.905e-007	8.905e-007	8.905e-007
50	Liq. Reynolds	1.916e+005 *	1.916e+005 *	1.916e+005 *	1.916e+005 *
51	Vap. Reynolds	9.752e+006 *	9.752e+006 *	9.752e+006 *	9.752e+006 *
52	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
53	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1.233e+004	1.233e+004	1.234e+004	1.234e+004
57	Elevation (m)	0.2040	0.2000	0.2000	0.2000
58	Pressure (bar)	71.19	71.19	71.19	71.19
59	Temperature (C)	53.27	53.27	53.27	53.27
60	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
61	Heat Transferred (kJ/h-m)	620.1	620.0	619.9	619.9
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1734	0.1734	0.2541	0.2541
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	8.828e-003	8.828e-003	5.942e-003	5.942e-003
67	Static Gradient (kPa/m)	-1.296e-003	-1.296e-003	0.0000	0.0000
68	Accel. Gradient (kPa/m)	8.905e-007	8.905e-007	7.026e-007	7.026e-007
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 231 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	1.916e+005 *	1.916e+005 *	1.916e+005 *	1.916e+005 *
14	Vap. Reynolds	9.752e+006 *	9.752e+006 *	9.752e+006 *	9.752e+006 *
15	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
16	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	1.235e+004	1.235e+004	1.236e+004	1.237e+004
20	Elevation (m)	0.2000	0.2000	0.2000	0.2020
21	Pressure (bar)	71.19	71.19	71.19	71.19
22	Temperature (C)	53.27	53.26	53.26	53.26
23	Vapour Fraction	0.8820	0.8820	0.8819	0.8819
24	Heat Transferred (kJ/h-m)	619.8	619.8	619.7	619.6
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2541	0.2541	0.2541	0.2356
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	5.942e-003	5.942e-003	5.942e-003	8.060e-003
30	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	5.896e-004
31	Accel. Gradient (kPa/m)	7.026e-007	7.025e-007	7.025e-007	1.023e-006
32	Liq. Reynolds	1.916e+005 *	1.915e+005 *	1.915e+005 *	1.915e+005 *
33	Vap. Reynolds	9.752e+006 *	9.752e+006 *	9.752e+006 *	9.752e+006 *
34	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
35	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	1.237e+004	1.238e+004	1.239e+004	1.239e+004
39	Elevation (m)	0.2040	0.2060	0.2080	0.2100
40	Pressure (bar)	71.19	71.19	71.19	71.19
41	Temperature (C)	53.26	53.26	53.25	53.25
42	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
43	Heat Transferred (kJ/h-m)	619.6	619.5	619.4	619.4
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2356	0.2356	0.2356	0.2356
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	8.060e-003	8.060e-003	8.060e-003	8.060e-003
49	Static Gradient (kPa/m)	5.896e-004	5.896e-004	5.896e-004	5.896e-004
50	Accel. Gradient (kPa/m)	1.023e-006	1.023e-006	1.023e-006	1.023e-006
51	Liq. Reynolds	1.915e+005 *	1.915e+005 *	1.915e+005 *	1.915e+005 *
52	Vap. Reynolds	9.752e+006 *	9.752e+006 *	9.752e+006 *	9.752e+006 *
53	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
54	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	1.240e+004	1.240e+004	1.241e+004	1.241e+004
58	Elevation (m)	0.2240	0.2380	0.2520	0.2660
59	Pressure (bar)	71.18	71.18	71.18	71.18
60	Temperature (C)	53.25	53.25	53.24	53.24
61	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
62	Heat Transferred (kJ/h-m)	619.3	619.3	619.2	619.2
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2426	0.2426	0.2426	0.2426
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	7.197e-003	7.197e-003	7.197e-003	7.197e-003
68	Static Gradient (kPa/m)	5.737e-003	5.737e-003	5.737e-003	5.737e-003
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 232 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient	(kPa/m)	1.529e-006	1.529e-006	1.529e-006	1.529e-006	
14	Liq. Reynolds		1.915e+005 *	1.915e+005 *	1.915e+005 *	1.915e+005 *	
15	Vap. Reynolds		9.752e+006 *	9.752e+006 *	9.752e+006 *	9.752e+006 *	
16	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
17	Vapour Velocity	(m/s)	2.854	2.854	2.854	2.854	
18	Deposit Thickness	(mm)	---	---	---	---	
19	Deposit Volume	(m3)	---	---	---	---	
20	Length	(m)	1.242e+004	1.242e+004	1.243e+004	1.243e+004	
21	Elevation	(m)	0.2800	0.2680	0.2560	0.2440	
22	Pressure	(bar)	71.18	71.18	71.18	71.18	
23	Temperature	(C)	53.24	53.24	53.24	53.23	
24	Vapour Fraction		0.8819	0.8819	0.8819	0.8819	
25	Heat Transferred	(kJ/h-m)	619.1	619.1	619.1	619.0	
26	Flow Regime		Transition	Transition	Transition	Transition	
27	Water Flow Regime						
28	Liquid HoldUp		0.2426	0.1722	0.1722	0.1722	
29	Water HoldUp		---	---	---	---	
30	Friction Gradient	(kPa/m)	7.197e-003	8.843e-003	8.843e-003	8.843e-003	
31	Static Gradient	(kPa/m)	5.737e-003	-3.870e-003	-3.870e-003	-3.870e-003	
32	Accel. Gradient	(kPa/m)	1.529e-006	5.880e-007	5.880e-007	5.880e-007	
33	Liq. Reynolds		1.915e+005 *	1.915e+005 *	1.915e+005 *	1.915e+005 *	
34	Vap. Reynolds		9.752e+006 *	9.752e+006 *	9.752e+006 *	9.752e+006 *	
35	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
36	Vapour Velocity	(m/s)	2.854	2.854	2.854	2.854	
37	Deposit Thickness	(mm)	---	---	---	---	
38	Deposit Volume	(m3)	---	---	---	---	
39	Length	(m)	1.244e+004	1.244e+004	1.245e+004	1.245e+004	
40	Elevation	(m)	0.2320	0.2200	0.2320	0.2440	
41	Pressure	(bar)	71.18	71.18	71.18	71.18	
42	Temperature	(C)	53.23	53.23	53.23	53.23	
43	Vapour Fraction		0.8819	0.8819	0.8819	0.8819	
44	Heat Transferred	(kJ/h-m)	619.0	618.9	618.8	618.8	
45	Flow Regime		Transition	Transition	Transition	Transition	
46	Water Flow Regime						
47	Liquid HoldUp		0.1722	0.1722	0.2415	0.2415	
48	Water HoldUp		---	---	---	---	
49	Friction Gradient	(kPa/m)	8.843e-003	8.843e-003	7.330e-003	7.330e-003	
50	Static Gradient	(kPa/m)	-3.870e-003	-3.870e-003	4.901e-003	4.901e-003	
51	Accel. Gradient	(kPa/m)	5.880e-007	5.880e-007	1.446e-006	1.446e-006	
52	Liq. Reynolds		1.915e+005 *	1.915e+005 *	1.915e+005 *	1.915e+005 *	
53	Vap. Reynolds		9.752e+006 *	9.752e+006 *	9.752e+006 *	9.752e+006 *	
54	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
55	Vapour Velocity	(m/s)	2.854	2.854	2.854	2.854	
56	Deposit Thickness	(mm)	---	---	---	---	
57	Deposit Volume	(m3)	---	---	---	---	
58	Length	(m)	1.246e+004	1.246e+004	1.247e+004	1.247e+004	
59	Elevation	(m)	0.2560	0.2680	0.2800	0.2660	
60	Pressure	(bar)	71.18	71.18	71.18	71.18	
61	Temperature	(C)	53.22	53.22	53.22	53.22	
62	Vapour Fraction		0.8819	0.8819	0.8819	0.8819	
63	Heat Transferred	(kJ/h-m)	618.7	618.6	618.6	618.6	
64	Flow Regime		Transition	Transition	Transition	Transition	
65	Water Flow Regime						
66	Liquid HoldUp		0.2415	0.2415	0.2415	0.1719	
67	Water HoldUp		---	---	---	---	
68	Friction Gradient	(kPa/m)	7.330e-003	7.330e-003	7.330e-003	8.847e-003	
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	4.901e-003	4.901e-003	4.901e-003	-4.510e-003
14	Accel. Gradient	(kPa/m)	1.446e-006	1.446e-006	1.446e-006	5.128e-007
15	Liq. Reynolds		1.915e+005 *	1.915e+005 *	1.915e+005 *	1.915e+005 *
16	Vap. Reynolds		9.752e+006 *	9.752e+006 *	9.752e+006 *	9.752e+006 *
17	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981
18	Vapour Velocity	(m/s)	2.854	2.854	2.854	2.854
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	1.248e+004	1.248e+004	1.249e+004	1.249e+004
22	Elevation	(m)	0.2520	0.2380	0.2240	0.2100
23	Pressure	(bar)	71.18	71.18	71.18	71.18
24	Temperature	(C)	53.22	53.22	53.21	53.21
25	Vapour Fraction		0.8819	0.8819	0.8819	0.8819
26	Heat Transferred	(kJ/h-m)	618.6	618.5	618.5	618.4
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.1719	0.1719	0.1719	0.1719
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	8.847e-003	8.847e-003	8.847e-003	8.847e-003
32	Static Gradient	(kPa/m)	-4.510e-003	-4.510e-003	-4.510e-003	-4.510e-003
33	Accel. Gradient	(kPa/m)	5.128e-007	5.128e-007	5.128e-007	5.127e-007
34	Liq. Reynolds		1.915e+005 *	1.915e+005 *	1.915e+005 *	1.915e+005 *
35	Vap. Reynolds		9.752e+006 *	9.752e+006 *	9.752e+006 *	9.752e+006 *
36	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981
37	Vapour Velocity	(m/s)	2.854	2.854	2.854	2.854
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	1.250e+004	1.250e+004	1.251e+004	1.251e+004
41	Elevation	(m)	0.1220	3.400e-002	-5.400e-002	-0.1420
42	Pressure	(bar)	71.18	71.18	71.18	71.18
43	Temperature	(C)	53.21	53.21	53.21	53.21
44	Vapour Fraction		0.8819	0.8819	0.8819	0.8819
45	Heat Transferred	(kJ/h-m)	618.4	618.4	618.3	618.3
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.1608	0.1608	0.1608	0.1608
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	9.006e-003	9.006e-003	9.006e-003	9.006e-003
51	Static Gradient	(kPa/m)	-2.713e-002	-2.713e-002	-2.713e-002	-2.713e-002
52	Accel. Gradient	(kPa/m)	-2.143e-006	-2.143e-006	-2.143e-006	-2.143e-006
53	Liq. Reynolds		1.915e+005 *	1.915e+005 *	1.915e+005 *	1.915e+005 *
54	Vap. Reynolds		9.752e+006 *	9.752e+006 *	9.752e+006 *	9.752e+006 *
55	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981
56	Vapour Velocity	(m/s)	2.854	2.854	2.854	2.854
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	1.252e+004	1.252e+004	1.253e+004	1.253e+004
60	Elevation	(m)	-0.2300	-0.2180	-0.2060	-0.1940
61	Pressure	(bar)	71.18	71.18	71.18	71.18
62	Temperature	(C)	53.21	53.20	53.20	53.20
63	Vapour Fraction		0.8819	0.8819	0.8819	0.8819
64	Heat Transferred	(kJ/h-m)	618.3	618.1	618.1	618.0
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.1608	0.2415	0.2415	0.2415
68	Water HoldUp		---	---	---	---
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1				Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	9.006e-003	7.329e-003	7.329e-003	7.329e-003
14	Static Gradient	(kPa/m)	-2.713e-002	4.901e-003	4.901e-003	4.901e-003
15	Accel. Gradient	(kPa/m)	-2.143e-006	1.446e-006	1.446e-006	1.446e-006
16	Liq. Reynolds		1.915e+005 *	1.915e+005 *	1.915e+005 *	1.915e+005 *
17	Vap. Reynolds		9.752e+006 *	9.752e+006 *	9.752e+006 *	9.752e+006 *
18	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981
19	Vapour Velocity	(m/s)	2.854	2.854	2.854	2.854
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	1.254e+004	1.254e+004	1.255e+004	1.255e+004
23	Elevation	(m)	-0.1820	-0.1700	-0.1040	-3.800e-002
24	Pressure	(bar)	71.18	71.18	71.18	71.17
25	Temperature	(C)	53.20	53.20	53.19	53.19
26	Vapour Fraction		0.8819	0.8819	0.8819	0.8819
27	Heat Transferred	(kJ/h-m)	618.0	617.9	617.9	617.8
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2415	0.2415	0.2718	0.2718
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	7.329e-003	7.329e-003	7.692e-003	7.692e-003
33	Static Gradient	(kPa/m)	4.901e-003	4.901e-003	2.943e-002	2.943e-002
34	Accel. Gradient	(kPa/m)	1.446e-006	1.446e-006	4.389e-006	4.389e-006
35	Liq. Reynolds		1.915e+005 *	1.915e+005 *	1.915e+005 *	1.915e+005 *
36	Vap. Reynolds		9.752e+006 *	9.752e+006 *	9.752e+006 *	9.752e+006 *
37	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981
38	Vapour Velocity	(m/s)	2.854	2.854	2.854	2.854
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	1.256e+004	1.256e+004	1.257e+004	1.257e+004
42	Elevation	(m)	2.800e-002	9.400e-002	0.1600	0.1740
43	Pressure	(bar)	71.17	71.17	71.17	71.17
44	Temperature	(C)	53.19	53.19	53.18	53.18
45	Vapour Fraction		0.8819	0.8819	0.8819	0.8819
46	Heat Transferred	(kJ/h-m)	617.7	617.7	617.6	617.5
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2718	0.2718	0.2718	0.2426
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	7.692e-003	7.692e-003	7.692e-003	7.197e-003
52	Static Gradient	(kPa/m)	2.943e-002	2.943e-002	2.943e-002	5.737e-003
53	Accel. Gradient	(kPa/m)	4.390e-006	4.390e-006	4.390e-006	1.529e-006
54	Liq. Reynolds		1.915e+005 *	1.915e+005 *	1.915e+005 *	1.915e+005 *
55	Vap. Reynolds		9.752e+006 *	9.752e+006 *	9.753e+006 *	9.753e+006 *
56	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981
57	Vapour Velocity	(m/s)	2.854	2.854	2.854	2.854
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	1.258e+004	1.258e+004	1.259e+004	1.259e+004
61	Elevation	(m)	0.1880	0.2020	0.2160	0.2300
62	Pressure	(bar)	71.17	71.17	71.17	71.17
63	Temperature	(C)	53.18	53.18	53.18	53.17
64	Vapour Fraction		0.8819	0.8819	0.8819	0.8819
65	Heat Transferred	(kJ/h-m)	617.5	617.4	617.4	617.3
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2426	0.2426	0.2426	0.2426
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	7.197e-003	7.197e-003	7.197e-003	7.197e-003
15	Static Gradient (kPa/m)	5.737e-003	5.737e-003	5.737e-003	5.737e-003
16	Accel. Gradient (kPa/m)	1.529e-006	1.529e-006	1.530e-006	1.530e-006
17	Liq. Reynolds	1.915e+005 *	1.915e+005 *	1.915e+005 *	1.915e+005 *
18	Vap. Reynolds	9.753e+006 *	9.753e+006 *	9.753e+006 *	9.753e+006 *
19	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
20	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	1.260e+004	1.260e+004	1.261e+004	1.261e+004
24	Elevation (m)	0.2720	0.3140	0.3560	0.3980
25	Pressure (bar)	71.16	71.16	71.16	71.16
26	Temperature (C)	53.17	53.17	53.17	53.16
27	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
28	Heat Transferred (kJ/h-m)	617.3	617.2	617.2	617.1
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2583	0.2583	0.2583	0.2583
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	7.143e-003	7.143e-003	7.143e-003	7.143e-003
34	Static Gradient (kPa/m)	1.803e-002	1.803e-002	1.803e-002	1.803e-002
35	Accel. Gradient (kPa/m)	2.977e-006	2.977e-006	2.977e-006	2.977e-006
36	Liq. Reynolds	1.915e+005 *	1.915e+005 *	1.914e+005 *	1.914e+005 *
37	Vap. Reynolds	9.753e+006 *	9.753e+006 *	9.753e+006 *	9.753e+006 *
38	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
39	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	1.262e+004	1.262e+004	1.263e+004	1.264e+004
43	Elevation (m)	0.4400	0.4760	0.5120	0.5480
44	Pressure (bar)	71.16	71.16	71.16	71.16
45	Temperature (C)	53.16	53.16	53.16	53.15
46	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
47	Heat Transferred (kJ/h-m)	617.0	617.0	616.9	616.8
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2583	0.2491	0.2491	0.2491
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	7.143e-003	6.461e-003	6.461e-003	6.461e-003
53	Static Gradient (kPa/m)	1.803e-002	1.072e-002	1.072e-002	1.072e-002
54	Accel. Gradient (kPa/m)	2.977e-006	2.031e-006	2.031e-006	2.032e-006
55	Liq. Reynolds	1.914e+005 *	1.914e+005 *	1.914e+005 *	1.914e+005 *
56	Vap. Reynolds	9.753e+006 *	9.753e+006 *	9.753e+006 *	9.753e+006 *
57	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
58	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	1.265e+004	1.265e+004	1.266e+004	1.266e+004
62	Elevation (m)	0.5840	0.6200	0.6160	0.6120
63	Pressure (bar)	71.15	71.15	71.15	71.15
64	Temperature (C)	53.15	53.15	53.15	53.14
65	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
66	Heat Transferred (kJ/h-m)	616.7	616.7	616.7	616.6
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.2491	0.2491	0.1734	0.1734
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	6.461e-003	6.461e-003	8.828e-003	8.828e-003
16	Static Gradient (kPa/m)	1.072e-002	1.072e-002	-1.296e-003	-1.296e-003
17	Accel. Gradient (kPa/m)	2.032e-006	2.032e-006	8.909e-007	8.909e-007
18	Liq. Reynolds	1.914e+005 *	1.914e+005 *	1.914e+005 *	1.914e+005 *
19	Vap. Reynolds	9.753e+006 *	9.753e+006 *	9.753e+006 *	9.753e+006 *
20	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
21	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	1.267e+004	1.267e+004	1.268e+004	1.268e+004
25	Elevation (m)	0.6080	0.6040	0.6000	0.5360
26	Pressure (bar)	71.15	71.15	71.15	71.15
27	Temperature (C)	53.14	53.14	53.14	53.14
28	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
29	Heat Transferred (kJ/h-m)	616.6	616.5	616.5	616.4
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1734	0.1734	0.1734	0.1644
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	8.828e-003	8.828e-003	8.828e-003	8.952e-003
35	Static Gradient (kPa/m)	-1.296e-003	-1.296e-003	-1.296e-003	-2.002e-002
36	Accel. Gradient (kPa/m)	8.909e-007	8.909e-007	8.909e-007	-1.309e-006
37	Liq. Reynolds	1.914e+005 *	1.914e+005 *	1.914e+005 *	1.914e+005 *
38	Vap. Reynolds	9.753e+006 *	9.753e+006 *	9.753e+006 *	9.753e+006 *
39	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
40	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	1.269e+004	1.269e+004	1.270e+004	1.270e+004
44	Elevation (m)	0.4720	0.4080	0.3440	0.2800
45	Pressure (bar)	71.15	71.15	71.15	71.15
46	Temperature (C)	53.14	53.13	53.13	53.13
47	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
48	Heat Transferred (kJ/h-m)	616.4	616.4	616.3	616.3
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1644	0.1644	0.1644	0.1644
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	8.952e-003	8.951e-003	8.951e-003	8.951e-003
54	Static Gradient (kPa/m)	-2.002e-002	-2.002e-002	-2.002e-002	-2.002e-002
55	Accel. Gradient (kPa/m)	-1.309e-006	-1.309e-006	-1.309e-006	-1.309e-006
56	Liq. Reynolds	1.914e+005 *	1.914e+005 *	1.914e+005 *	1.914e+005 *
57	Vap. Reynolds	9.753e+006 *	9.753e+006 *	9.753e+006 *	9.753e+006 *
58	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
59	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	1.271e+004	1.271e+004	1.272e+004	1.272e+004
63	Elevation (m)	0.2760	0.2720	0.2680	0.2640
64	Pressure (bar)	71.15	71.15	71.15	71.15
65	Temperature (C)	53.13	53.13	53.13	53.12
66	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
67	Heat Transferred (kJ/h-m)	616.2	616.2	616.1	616.1
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 237 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1734	0.1734	0.1734	0.1734
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	8.827e-003	8.827e-003	8.827e-003	8.827e-003
17	Static Gradient (kPa/m)	-1.296e-003	-1.296e-003	-1.296e-003	-1.296e-003
18	Accel. Gradient (kPa/m)	8.907e-007	8.907e-007	8.906e-007	8.906e-007
19	Liq. Reynolds	1.914e+005 *	1.914e+005 *	1.914e+005 *	1.914e+005 *
20	Vap. Reynolds	9.753e+006 *	9.753e+006 *	9.753e+006 *	9.753e+006 *
21	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
22	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	1.273e+004	1.273e+004	1.274e+004	1.274e+004
26	Elevation (m)	0.2600	0.2580	0.2560	0.2540
27	Pressure (bar)	71.15	71.15	71.15	71.15
28	Temperature (C)	53.12	53.12	53.12	53.12
29	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
30	Heat Transferred (kJ/h-m)	616.0	616.0	615.9	615.9
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1734	0.1737	0.1737	0.1737
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	8.827e-003	8.823e-003	8.823e-003	8.823e-003
36	Static Gradient (kPa/m)	-1.296e-003	-6.488e-004	-6.488e-004	-6.489e-004
37	Accel. Gradient (kPa/m)	8.906e-007	9.668e-007	9.668e-007	9.668e-007
38	Liq. Reynolds	1.914e+005 *	1.914e+005 *	1.914e+005 *	1.914e+005 *
39	Vap. Reynolds	9.753e+006 *	9.753e+006 *	9.753e+006 *	9.753e+006 *
40	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
41	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	1.275e+004	1.275e+004	1.276e+004	1.276e+004
45	Elevation (m)	0.2520	0.2500	0.2060	0.1620
46	Pressure (bar)	71.15	71.15	71.15	71.15
47	Temperature (C)	53.11	53.11	53.11	53.11
48	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
49	Heat Transferred (kJ/h-m)	615.8	615.8	615.7	615.7
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.1737	0.1737	0.1658	0.1658
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	8.823e-003	8.823e-003	8.931e-003	8.931e-003
55	Static Gradient (kPa/m)	-6.489e-004	-6.489e-004	-1.730e-002	-1.730e-002
56	Accel. Gradient (kPa/m)	9.668e-007	9.668e-007	-9.896e-007	-9.896e-007
57	Liq. Reynolds	1.914e+005 *	1.914e+005 *	1.914e+005 *	1.914e+005 *
58	Vap. Reynolds	9.753e+006 *	9.753e+006 *	9.753e+006 *	9.753e+006 *
59	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
60	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	1.276e+004	1.277e+004	1.277e+004	1.278e+004
64	Elevation (m)	0.1180	7.400e-002	3.000e-002	2.600e-002
65	Pressure (bar)	71.15	71.15	71.15	71.15
66	Temperature (C)	53.11	53.11	53.11	53.11
67	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
68	Heat Transferred (kJ/h-m)	615.7	615.6	615.6	615.6
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 238 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc	
Unit Set: Mixed				
Date/Time: Mon Jun 13 15:13:43 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Pipe Table			
12				
13	Flow Regime	Transition	Transition	Transition
14	Water Flow Regime			
15	Liquid HoldUp	0.1658	0.1658	0.1658
16	Water HoldUp	---	---	---
17	Friction Gradient (kPa/m)	8.931e-003	8.931e-003	8.931e-003
18	Static Gradient (kPa/m)	-1.730e-002	-1.730e-002	-1.730e-002
19	Accel. Gradient (kPa/m)	-9.896e-007	-9.896e-007	-9.897e-007
20	Liq. Reynolds	1.914e+005 *	1.914e+005 *	1.914e+005 *
21	Vap. Reynolds	9.753e+006 *	9.753e+006 *	9.753e+006 *
22	Liquid Velocity (m/s)	0.1981	0.1981	0.1981
23	Vapour Velocity (m/s)	2.854	2.854	2.854
24	Deposit Thickness (mm)	---	---	---
25	Deposit Volume (m3)	---	---	---
26	Length (m)	1.278e+004	1.278e+004	1.278e+004
27	Elevation (m)	2.200e-002	1.800e-002	1.400e-002
28	Pressure (bar)	71.15	71.15	71.15
29	Temperature (C)	53.10	53.10	53.10
30	Vapour Fraction	0.8819	0.8819	0.8819
31	Heat Transferred (kJ/h-m)	615.5	615.5	615.5
32	Flow Regime	Transition	Transition	Transition
33	Water Flow Regime			
34	Liquid HoldUp	0.1731	0.1731	0.1731
35	Water HoldUp	---	---	---
36	Friction Gradient (kPa/m)	8.831e-003	8.831e-003	8.831e-003
37	Static Gradient (kPa/m)	-2.157e-003	-2.157e-003	-2.157e-003
38	Accel. Gradient (kPa/m)	7.893e-007	7.893e-007	7.893e-007
39	Liq. Reynolds	1.914e+005 *	1.914e+005 *	1.914e+005 *
40	Vap. Reynolds	9.753e+006 *	9.753e+006 *	9.753e+006 *
41	Liquid Velocity (m/s)	0.1981	0.1981	0.1981
42	Vapour Velocity (m/s)	2.854	2.854	2.854
43	Deposit Thickness (mm)	---	---	---
44	Deposit Volume (m3)	---	---	---
45	Length (m)	1.279e+004	1.279e+004	1.280e+004
46	Elevation (m)	4.800e-002	8.600e-002	0.1240
47	Pressure (bar)	71.15	71.15	71.15
48	Temperature (C)	53.10	53.10	53.10
49	Vapour Fraction	0.8819	0.8819	0.8819
50	Heat Transferred (kJ/h-m)	615.4	615.3	615.3
51	Flow Regime	Transition	Transition	Transition
52	Water Flow Regime			
53	Liquid HoldUp	0.2703	0.2703	0.2703
54	Water HoldUp	---	---	---
55	Friction Gradient (kPa/m)	7.676e-003	7.676e-003	7.676e-003
56	Static Gradient (kPa/m)	2.813e-002	2.813e-002	2.813e-002
57	Accel. Gradient (kPa/m)	4.235e-006	4.235e-006	4.235e-006
58	Liq. Reynolds	1.914e+005 *	1.914e+005 *	1.914e+005 *
59	Vap. Reynolds	9.753e+006 *	9.753e+006 *	9.754e+006 *
60	Liquid Velocity (m/s)	0.1981	0.1981	0.1981
61	Vapour Velocity (m/s)	2.854	2.854	2.854
62	Deposit Thickness (mm)	---	---	---
63	Deposit Volume (m3)	---	---	---
64	Length (m)	1.280e+004	1.281e+004	1.281e+004
65	Elevation (m)	0.2000	0.1280	5.600e-002
66	Pressure (bar)	71.15	71.15	71.15
67	Temperature (C)	53.09	53.09	53.09
68	Vapour Fraction	0.8819	0.8819	0.8819
69	Aspen Technology Inc.		Aspen HYSYS Version 10	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Heat Transferred (kJ/h-m)	615.2	615.2	615.2	615.2					
14	Flow Regime	Transition	Transition	Transition	Transition					
15	Water Flow Regime									
16	Liquid HoldUp	0.2703	0.1632	0.1632	0.1632					
17	Water HoldUp	---	---	---	---					
18	Friction Gradient (kPa/m)	7.676e-003	8.969e-003	8.968e-003	8.968e-003					
19	Static Gradient (kPa/m)	2.813e-002	-2.242e-002	-2.242e-002	-2.242e-002					
20	Accel. Gradient (kPa/m)	4.235e-006	-1.591e-006	-1.591e-006	-1.591e-006					
21	Liq. Reynolds	1.914e+005 *	1.914e+005 *	1.914e+005 *	1.914e+005 *					
22	Vap. Reynolds	9.754e+006 *	9.754e+006 *	9.754e+006 *	9.754e+006 *					
23	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981					
24	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854					
25	Deposit Thickness (mm)	---	---	---	---					
26	Deposit Volume (m3)	---	---	---	---					
27	Length (m)	1.282e+004	1.283e+004	1.283e+004	1.284e+004					
28	Elevation (m)	-8.800e-002	-0.1600	-9.200e-002	-2.400e-002					
29	Pressure (bar)	71.15	71.15	71.15	71.15					
30	Temperature (C)	53.09	53.09	53.08	53.08					
31	Vapour Fraction	0.8819	0.8819	0.8819	0.8819					
32	Heat Transferred (kJ/h-m)	615.1	615.1	615.0	614.9					
33	Flow Regime	Transition	Transition	Transition	Transition					
34	Water Flow Regime									
35	Liquid HoldUp	0.1632	0.1632	0.2727	0.2727					
36	Water HoldUp	---	---	---	---					
37	Friction Gradient (kPa/m)	8.968e-003	8.968e-003	7.699e-003	7.699e-003					
38	Static Gradient (kPa/m)	-2.242e-002	-2.242e-002	3.022e-002	3.022e-002					
39	Accel. Gradient (kPa/m)	-1.591e-006	-1.591e-006	4.485e-006	4.485e-006					
40	Liq. Reynolds	1.914e+005 *	1.914e+005 *	1.914e+005 *	1.914e+005 *					
41	Vap. Reynolds	9.754e+006 *	9.754e+006 *	9.754e+006 *	9.754e+006 *					
42	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981					
43	Vapour Velocity (m/s)	2.853	2.853	2.853	2.854					
44	Deposit Thickness (mm)	---	---	---	---					
45	Deposit Volume (m3)	---	---	---	---					
46	Length (m)	1.284e+004	1.285e+004	1.285e+004	1.286e+004					
47	Elevation (m)	4.400e-002	0.1120	0.1800	0.1840					
48	Pressure (bar)	71.14	71.14	71.14	71.14					
49	Temperature (C)	53.08	53.08	53.07	53.07					
50	Vapour Fraction	0.8819	0.8819	0.8819	0.8819					
51	Heat Transferred (kJ/h-m)	614.8	614.8	614.7	614.7					
52	Flow Regime	Transition	Transition	Transition	Transition					
53	Water Flow Regime									
54	Liquid HoldUp	0.2727	0.2727	0.2727	0.2370					
55	Water HoldUp	---	---	---	---					
56	Friction Gradient (kPa/m)	7.699e-003	7.699e-003	7.700e-003	7.879e-003					
57	Static Gradient (kPa/m)	3.022e-002	3.022e-002	3.022e-002	1.612e-003					
58	Accel. Gradient (kPa/m)	4.485e-006	4.486e-006	4.486e-006	1.123e-006					
59	Liq. Reynolds	1.914e+005 *	1.914e+005 *	1.914e+005 *	1.914e+005 *					
60	Vap. Reynolds	9.754e+006 *	9.754e+006 *	9.754e+006 *	9.754e+006 *					
61	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981					
62	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854					
63	Deposit Thickness (mm)	---	---	---	---					
64	Deposit Volume (m3)	---	---	---	---					
65	Length (m)	1.286e+004	1.287e+004	1.287e+004	1.288e+004					
66	Elevation (m)	0.1880	0.1920	0.1960	0.2000					
67	Pressure (bar)	71.14	71.14	71.14	71.14					
68	Temperature (C)	53.07	53.07	53.07	53.07					
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Fraction		0.8819	0.8819	0.8819	0.8819	
14	Heat Transferred (kJ/h-m)		614.6	614.5	614.5	614.4	
15	Flow Regime		Transition	Transition	Transition	Transition	
16	Water Flow Regime						
17	Liquid HoldUp		0.2370	0.2370	0.2370	0.2370	
18	Water HoldUp		---	---	---	---	
19	Friction Gradient (kPa/m)		7.879e-003	7.878e-003	7.878e-003	7.878e-003	
20	Static Gradient (kPa/m)		1.612e-003	1.612e-003	1.612e-003	1.612e-003	
21	Accel. Gradient (kPa/m)		1.123e-006	1.123e-006	1.123e-006	1.123e-006	
22	Liq. Reynolds		1.914e+005 *	1.914e+005 *	1.914e+005 *	1.913e+005 *	
23	Vap. Reynolds		9.754e+006 *	9.754e+006 *	9.754e+006 *	9.754e+006 *	
24	Liquid Velocity (m/s)		0.1981	0.1981	0.1981	0.1981	
25	Vapour Velocity (m/s)		2.854	2.854	2.854	2.854	
26	Deposit Thickness (mm)		---	---	---	---	
27	Deposit Volume (m3)		---	---	---	---	
28	Length (m)		1.288e+004	1.289e+004	1.289e+004	1.290e+004	
29	Elevation (m)		0.2040	0.2080	0.2120	0.2160	
30	Pressure (bar)		71.14	71.14	71.14	71.14	
31	Temperature (C)		53.06	53.06	53.06	53.06	
32	Vapour Fraction		0.8819	0.8819	0.8819	0.8819	
33	Heat Transferred (kJ/h-m)		614.4	614.3	614.3	614.2	
34	Flow Regime		Transition	Transition	Transition	Transition	
35	Water Flow Regime						
36	Liquid HoldUp		0.2370	0.2370	0.2370	0.2370	
37	Water HoldUp		---	---	---	---	
38	Friction Gradient (kPa/m)		7.878e-003	7.878e-003	7.878e-003	7.878e-003	
39	Static Gradient (kPa/m)		1.612e-003	1.612e-003	1.612e-003	1.612e-003	
40	Accel. Gradient (kPa/m)		1.123e-006	1.123e-006	1.123e-006	1.123e-006	
41	Liq. Reynolds		1.913e+005 *	1.913e+005 *	1.913e+005 *	1.913e+005 *	
42	Vap. Reynolds		9.754e+006 *	9.754e+006 *	9.754e+006 *	9.754e+006 *	
43	Liquid Velocity (m/s)		0.1981	0.1981	0.1981	0.1981	
44	Vapour Velocity (m/s)		2.854	2.854	2.854	2.854	
45	Deposit Thickness (mm)		---	---	---	---	
46	Deposit Volume (m3)		---	---	---	---	
47	Length (m)		1.290e+004	1.291e+004	1.291e+004	1.292e+004	
48	Elevation (m)		0.2200	0.2240	0.2280	0.2320	
49	Pressure (bar)		71.13	71.13	71.13	71.13	
50	Temperature (C)		53.06	53.05	53.05	53.05	
51	Vapour Fraction		0.8819	0.8819	0.8819	0.8819	
52	Heat Transferred (kJ/h-m)		614.2	614.1	614.1	614.0	
53	Flow Regime		Transition	Transition	Transition	Transition	
54	Water Flow Regime						
55	Liquid HoldUp		0.2370	0.2370	0.2370	0.2370	
56	Water HoldUp		---	---	---	---	
57	Friction Gradient (kPa/m)		7.878e-003	7.878e-003	7.878e-003	7.878e-003	
58	Static Gradient (kPa/m)		1.612e-003	1.612e-003	1.612e-003	1.612e-003	
59	Accel. Gradient (kPa/m)		1.123e-006	1.123e-006	1.123e-006	1.123e-006	
60	Liq. Reynolds		1.913e+005 *	1.913e+005 *	1.913e+005 *	1.913e+005 *	
61	Vap. Reynolds		9.754e+006 *	9.754e+006 *	9.754e+006 *	9.754e+006 *	
62	Liquid Velocity (m/s)		0.1981	0.1981	0.1981	0.1981	
63	Vapour Velocity (m/s)		2.854	2.854	2.854	2.854	
64	Deposit Thickness (mm)		---	---	---	---	
65	Deposit Volume (m3)		---	---	---	---	
66	Length (m)		1.292e+004	1.293e+004	1.293e+004	1.294e+004	
67	Elevation (m)		0.2360	0.2400	0.2420	0.2440	
68	Pressure (bar)		71.13	71.13	71.13	71.13	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 241 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	53.05	53.05	53.04	53.04
14	Vapour Fraction	0.8819	0.8818	0.8818	0.8818
15	Heat Transferred (kJ/h-m)	614.0	613.9	613.9	613.8
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2370	0.2370	0.2359	0.2359
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	7.878e-003	7.878e-003	8.021e-003	8.021e-003
21	Static Gradient (kPa/m)	1.612e-003	1.612e-003	8.030e-004	8.030e-004
22	Accel. Gradient (kPa/m)	1.123e-006	1.123e-006	1.044e-006	1.044e-006
23	Liq. Reynolds	1.913e+005 *	1.913e+005 *	1.913e+005 *	1.913e+005 *
24	Vap. Reynolds	9.754e+006 *	9.754e+006 *	9.754e+006 *	9.754e+006 *
25	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
26	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	1.294e+004	1.295e+004	1.295e+004	1.296e+004
30	Elevation (m)	0.2460	0.2480	0.2500	0.2440
31	Pressure (bar)	71.13	71.13	71.13	71.13
32	Temperature (C)	53.04	53.04	53.04	53.04
33	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
34	Heat Transferred (kJ/h-m)	613.8	613.7	613.7	613.7
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2359	0.2359	0.2359	0.1732
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	8.021e-003	8.020e-003	8.020e-003	8.830e-003
40	Static Gradient (kPa/m)	8.030e-004	8.030e-004	8.031e-004	-1.942e-003
41	Accel. Gradient (kPa/m)	1.044e-006	1.044e-006	1.044e-006	8.148e-007
42	Liq. Reynolds	1.913e+005 *	1.913e+005 *	1.913e+005 *	1.913e+005 *
43	Vap. Reynolds	9.754e+006 *	9.754e+006 *	9.754e+006 *	9.754e+006 *
44	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
45	Vapour Velocity (m/s)	2.854	2.854	2.854	2.854
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	1.296e+004	1.297e+004	1.297e+004	1.298e+004
49	Elevation (m)	0.2380	0.2320	0.2260	0.2200
50	Pressure (bar)	71.13	71.13	71.13	71.13
51	Temperature (C)	53.03	53.03	53.03	53.03
52	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
53	Heat Transferred (kJ/h-m)	613.6	613.6	613.5	613.5
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.1732	0.1732	0.1732	0.1732
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	8.830e-003	8.830e-003	8.830e-003	8.830e-003
59	Static Gradient (kPa/m)	-1.942e-003	-1.942e-003	-1.942e-003	-1.942e-003
60	Accel. Gradient (kPa/m)	8.148e-007	8.148e-007	8.148e-007	8.148e-007
61	Liq. Reynolds	1.913e+005 *	1.913e+005 *	1.913e+005 *	1.913e+005 *
62	Vap. Reynolds	9.754e+006 *	9.754e+006 *	9.754e+006 *	9.754e+006 *
63	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
64	Vapour Velocity (m/s)	2.854	2.854	2.853	2.853
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	1.298e+004	1.299e+004	1.299e+004	1.300e+004
68	Elevation (m)	0.2300	0.2400	0.2500	0.2600
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Pressure (bar)	71.13	71.13	71.13	71.13					
14	Temperature (C)	53.03	53.02	53.02	53.02					
15	Vapour Fraction	0.8818	0.8818	0.8818	0.8818					
16	Heat Transferred (kJ/h-m)	613.4	613.3	613.3	613.2					
17	Flow Regime	Transition	Transition	Transition	Transition					
18	Water Flow Regime									
19	Liquid HoldUp	0.2404	0.2404	0.2404	0.2404					
20	Water HoldUp	---	---	---	---					
21	Friction Gradient (kPa/m)	7.463e-003	7.463e-003	7.463e-003	7.463e-003					
22	Static Gradient (kPa/m)	4.071e-003	4.071e-003	4.071e-003	4.071e-003					
23	Accel. Gradient (kPa/m)	1.364e-006	1.364e-006	1.364e-006	1.364e-006					
24	Liq. Reynolds	1.913e+005 *	1.913e+005 *	1.913e+005 *	1.913e+005 *					
25	Vap. Reynolds	9.754e+006 *	9.754e+006 *	9.754e+006 *	9.754e+006 *					
26	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981					
27	Vapour Velocity (m/s)	2.853	2.853	2.853	2.853					
28	Deposit Thickness (mm)	---	---	---	---					
29	Deposit Volume (m3)	---	---	---	---					
30	Length (m)	1.300e+004	1.301e+004	1.301e+004	1.302e+004					
31	Elevation (m)	0.2700	0.2720	0.2740	0.2760					
32	Pressure (bar)	71.13	71.13	71.12	71.12					
33	Temperature (C)	53.02	53.02	53.01	53.01					
34	Vapour Fraction	0.8818	0.8818	0.8818	0.8818					
35	Heat Transferred (kJ/h-m)	613.2	613.1	613.1	613.0					
36	Flow Regime	Transition	Transition	Transition	Transition					
37	Water Flow Regime									
38	Liquid HoldUp	0.2404	0.2359	0.2359	0.2359					
39	Water HoldUp	---	---	---	---					
40	Friction Gradient (kPa/m)	7.463e-003	8.020e-003	8.020e-003	8.020e-003					
41	Static Gradient (kPa/m)	4.071e-003	8.031e-004	8.031e-004	8.031e-004					
42	Accel. Gradient (kPa/m)	1.365e-006	1.044e-006	1.044e-006	1.044e-006					
43	Liq. Reynolds	1.913e+005 *	1.913e+005 *	1.913e+005 *	1.913e+005 *					
44	Vap. Reynolds	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.755e+006 *					
45	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981					
46	Vapour Velocity (m/s)	2.853	2.853	2.853	2.853					
47	Deposit Thickness (mm)	---	---	---	---					
48	Deposit Volume (m3)	---	---	---	---					
49	Length (m)	1.302e+004	1.303e+004	1.303e+004	1.304e+004					
50	Elevation (m)	0.2780	0.2800	0.2820	0.2840					
51	Pressure (bar)	71.12	71.12	71.12	71.12					
52	Temperature (C)	53.01	53.01	53.01	53.00					
53	Vapour Fraction	0.8818	0.8818	0.8818	0.8818					
54	Heat Transferred (kJ/h-m)	613.0	612.9	612.9	612.8					
55	Flow Regime	Transition	Transition	Transition	Transition					
56	Water Flow Regime									
57	Liquid HoldUp	0.2359	0.2359	0.2359	0.2359					
58	Water HoldUp	---	---	---	---					
59	Friction Gradient (kPa/m)	8.020e-003	8.020e-003	8.021e-003	8.021e-003					
60	Static Gradient (kPa/m)	8.031e-004	8.031e-004	7.980e-004	7.980e-004					
61	Accel. Gradient (kPa/m)	1.044e-006	1.044e-006	1.043e-006	1.043e-006					
62	Liq. Reynolds	1.913e+005 *	1.913e+005 *	1.913e+005 *	1.913e+005 *					
63	Vap. Reynolds	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.755e+006 *					
64	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981					
65	Vapour Velocity (m/s)	2.853	2.853	2.853	2.853					
66	Deposit Thickness (mm)	---	---	---	---					
67	Deposit Volume (m3)	---	---	---	---					
68	Length (m)	1.304e+004	1.305e+004	1.305e+004	1.306e+004					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	0.2860	0.2880	0.2900	0.2920
14	Pressure (bar)	71.12	71.12	71.12	71.12
15	Temperature (C)	53.00	53.00	53.00	53.00
16	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
17	Heat Transferred (kJ/h-m)	612.8	612.7	612.7	612.6
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2359	0.2359	0.2359	0.2359
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	8.021e-003	8.021e-003	8.021e-003	8.020e-003
23	Static Gradient (kPa/m)	7.980e-004	7.980e-004	7.980e-004	8.031e-004
24	Accel. Gradient (kPa/m)	1.043e-006	1.043e-006	1.043e-006	1.044e-006
25	Liq. Reynolds	1.913e+005 *	1.913e+005 *	1.913e+005 *	1.913e+005 *
26	Vap. Reynolds	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.755e+006 *
27	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
28	Vapour Velocity (m/s)	2.853	2.853	2.853	2.853
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	1.306e+004	1.307e+004	1.307e+004	1.308e+004
32	Elevation (m)	0.2940	0.2960	0.2980	0.3000
33	Pressure (bar)	71.12	71.12	71.12	71.12
34	Temperature (C)	53.00	52.99	52.99	52.99
35	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
36	Heat Transferred (kJ/h-m)	612.6	612.5	612.5	612.4
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2359	0.2359	0.2359	0.2359
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	8.020e-003	8.020e-003	8.020e-003	8.020e-003
42	Static Gradient (kPa/m)	8.031e-004	8.031e-004	8.031e-004	8.031e-004
43	Accel. Gradient (kPa/m)	1.044e-006	1.044e-006	1.044e-006	1.044e-006
44	Liq. Reynolds	1.913e+005 *	1.913e+005 *	1.913e+005 *	1.913e+005 *
45	Vap. Reynolds	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.755e+006 *
46	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
47	Vapour Velocity (m/s)	2.853	2.853	2.853	2.853
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	1.308e+004	1.309e+004	1.309e+004	1.310e+004
51	Elevation (m)	0.2940	0.2880	0.2820	0.2760
52	Pressure (bar)	71.12	71.12	71.12	71.12
53	Temperature (C)	52.99	52.99	52.98	52.98
54	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
55	Heat Transferred (kJ/h-m)	612.4	612.4	612.3	612.3
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.1732	0.1732	0.1732	0.1732
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	8.830e-003	8.830e-003	8.830e-003	8.830e-003
61	Static Gradient (kPa/m)	-1.942e-003	-1.942e-003	-1.942e-003	-1.942e-003
62	Accel. Gradient (kPa/m)	8.148e-007	8.148e-007	8.148e-007	8.148e-007
63	Liq. Reynolds	1.913e+005 *	1.913e+005 *	1.913e+005 *	1.913e+005 *
64	Vap. Reynolds	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.755e+006 *
65	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
66	Vapour Velocity (m/s)	2.853	2.853	2.853	2.853
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 244 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc					
2								
3							Unit Set: Mixed	
4							Date/Time: Mon Jun 13 15:13:43 2022	
5								
6	Pipe Segment: ELEP-Akabel Pipeline (continued)							
7								
8								
9	PROFILES							
10	Pipe Table							
11								
12								
13	Length (m)	1.310e+004	1.311e+004	1.311e+004	1.312e+004			
14	Elevation (m)	0.2700	0.2820	0.2940	0.3060			
15	Pressure (bar)	71.12	71.12	71.12	71.12			
16	Temperature (C)	52.98	52.98	52.98	52.97			
17	Vapour Fraction	0.8818	0.8818	0.8818	0.8818			
18	Heat Transferred (kJ/h-m)	612.2	612.1	612.1	612.0			
19	Flow Regime	Transition	Transition	Transition	Transition			
20	Water Flow Regime							
21	Liquid HoldUp	0.1732	0.2415	0.2415	0.2415			
22	Water HoldUp	---	---	---	---			
23	Friction Gradient (kPa/m)	8.830e-003	7.328e-003	7.328e-003	7.328e-003			
24	Static Gradient (kPa/m)	-1.942e-003	4.902e-003	4.902e-003	4.902e-003			
25	Accel. Gradient (kPa/m)	8.148e-007	1.447e-006	1.447e-006	1.447e-006			
26	Liq. Reynolds	1.913e+005 *	1.913e+005 *	1.913e+005 *	1.913e+005 *			
27	Vap. Reynolds	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.755e+006 *			
28	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981			
29	Vapour Velocity (m/s)	2.853	2.853	2.853	2.853			
30	Deposit Thickness (mm)	---	---	---	---			
31	Deposit Volume (m3)	---	---	---	---			
32	Length (m)	1.312e+004	1.313e+004	1.313e+004	1.314e+004			
33	Elevation (m)	0.3180	0.3300	0.3320	0.3340			
34	Pressure (bar)	71.11	71.11	71.11	71.11			
35	Temperature (C)	52.97	52.97	52.97	52.97			
36	Vapour Fraction	0.8818	0.8818	0.8818	0.8818			
37	Heat Transferred (kJ/h-m)	612.0	611.9	611.9	611.8			
38	Flow Regime	Transition	Transition	Transition	Transition			
39	Water Flow Regime							
40	Liquid HoldUp	0.2415	0.2415	0.2359	0.2359			
41	Water HoldUp	---	---	---	---			
42	Friction Gradient (kPa/m)	7.328e-003	7.328e-003	8.020e-003	8.020e-003			
43	Static Gradient (kPa/m)	4.902e-003	4.902e-003	8.031e-004	8.031e-004			
44	Accel. Gradient (kPa/m)	1.447e-006	1.447e-006	1.044e-006	1.044e-006			
45	Liq. Reynolds	1.913e+005 *	1.913e+005 *	1.913e+005 *	1.913e+005 *			
46	Vap. Reynolds	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.755e+006 *			
47	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981			
48	Vapour Velocity (m/s)	2.853	2.853	2.853	2.853			
49	Deposit Thickness (mm)	---	---	---	---			
50	Deposit Volume (m3)	---	---	---	---			
51	Length (m)	1.314e+004	1.315e+004	1.315e+004	1.316e+004			
52	Elevation (m)	0.3360	0.3380	0.3400	0.3420			
53	Pressure (bar)	71.11	71.11	71.11	71.11			
54	Temperature (C)	52.96	52.96	52.96	52.96			
55	Vapour Fraction	0.8818	0.8818	0.8818	0.8818			
56	Heat Transferred (kJ/h-m)	611.8	611.7	611.7	611.6			
57	Flow Regime	Transition	Transition	Transition	Transition			
58	Water Flow Regime							
59	Liquid HoldUp	0.2359	0.2359	0.2359	0.2359			
60	Water HoldUp	---	---	---	---			
61	Friction Gradient (kPa/m)	8.020e-003	8.020e-003	8.020e-003	8.020e-003			
62	Static Gradient (kPa/m)	8.031e-004	8.031e-004	8.031e-004	8.031e-004			
63	Accel. Gradient (kPa/m)	1.044e-006	1.044e-006	1.044e-006	1.044e-006			
64	Liq. Reynolds	1.912e+005 *	1.912e+005 *	1.912e+005 *	1.912e+005 *			
65	Vap. Reynolds	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.755e+006 *			
66	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981			
67	Vapour Velocity (m/s)	2.853	2.853	2.853	2.853			
68	Deposit Thickness (mm)	---	---	---	---			
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 245 of 937			

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---	---	---
14	Length (m)	1.316e+004	1.317e+004	1.317e+004	1.317e+004	1.318e+004	1.318e+004
15	Elevation (m)	0.3440	0.3460	0.3480	0.3480	0.3500	0.3500
16	Pressure (bar)	71.11	71.11	71.11	71.11	71.11	71.11
17	Temperature (C)	52.96	52.96	52.95	52.95	52.95	52.95
18	Vapour Fraction	0.8818	0.8818	0.8818	0.8818	0.8818	0.8818
19	Heat Transferred (kJ/h-m)	611.6	611.5	611.5	611.5	611.4	611.4
20	Flow Regime	Transition	Transition	Transition	Transition	Transition	Transition
21	Water Flow Regime						
22	Liquid HoldUp	0.2359	0.2359	0.2359	0.2359	0.2359	0.2359
23	Water HoldUp	---	---	---	---	---	---
24	Friction Gradient (kPa/m)	8.020e-003	8.020e-003	8.020e-003	8.020e-003	8.020e-003	8.020e-003
25	Static Gradient (kPa/m)	8.031e-004	8.031e-004	8.032e-004	8.032e-004	8.032e-004	8.032e-004
26	Accel. Gradient (kPa/m)	1.044e-006	1.044e-006	1.044e-006	1.044e-006	1.044e-006	1.044e-006
27	Liq. Reynolds	1.912e+005 *	1.912e+005 *	1.912e+005 *	1.912e+005 *	1.912e+005 *	1.912e+005 *
28	Vap. Reynolds	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.755e+006 *
29	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981	0.1981	0.1981
30	Vapour Velocity (m/s)	2.853	2.853	2.853	2.853	2.853	2.853
31	Deposit Thickness (mm)	---	---	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---	---	---
33	Length (m)	1.318e+004	1.319e+004	1.319e+004	1.319e+004	1.320e+004	1.320e+004
34	Elevation (m)	0.3480	0.3460	0.3440	0.3440	0.3420	0.3420
35	Pressure (bar)	71.11	71.11	71.11	71.11	71.11	71.11
36	Temperature (C)	52.95	52.95	52.95	52.95	52.94	52.94
37	Vapour Fraction	0.8818	0.8818	0.8818	0.8818	0.8818	0.8818
38	Heat Transferred (kJ/h-m)	611.4	611.4	611.3	611.3	611.3	611.3
39	Flow Regime	Transition	Transition	Transition	Transition	Transition	Transition
40	Water Flow Regime						
41	Liquid HoldUp	0.1738	0.1738	0.1738	0.1738	0.1738	0.1738
42	Water HoldUp	---	---	---	---	---	---
43	Friction Gradient (kPa/m)	8.822e-003	8.822e-003	8.822e-003	8.822e-003	8.822e-003	8.822e-003
44	Static Gradient (kPa/m)	-6.490e-004	-6.490e-004	-6.490e-004	-6.490e-004	-6.490e-004	-6.490e-004
45	Accel. Gradient (kPa/m)	9.670e-007	9.670e-007	9.670e-007	9.670e-007	9.670e-007	9.670e-007
46	Liq. Reynolds	1.912e+005 *	1.912e+005 *	1.912e+005 *	1.912e+005 *	1.912e+005 *	1.912e+005 *
47	Vap. Reynolds	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.755e+006 *
48	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981	0.1981	0.1981
49	Vapour Velocity (m/s)	2.853	2.853	2.853	2.853	2.853	2.853
50	Deposit Thickness (mm)	---	---	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---	---	---
52	Length (m)	1.320e+004	1.321e+004	1.321e+004	1.321e+004	1.322e+004	1.322e+004
53	Elevation (m)	0.3400	0.1100	-0.1200	-0.1200	-0.3500	-0.3500
54	Pressure (bar)	71.11	71.11	71.11	71.11	71.12	71.12
55	Temperature (C)	52.94	52.94	52.94	52.94	52.94	52.94
56	Vapour Fraction	0.8818	0.8818	0.8818	0.8818	0.8818	0.8818
57	Heat Transferred (kJ/h-m)	611.2	611.3	611.3	611.3	611.3	611.3
58	Flow Regime	Transition	Transition	Transition	Transition	Transition	Transition
59	Water Flow Regime						
60	Liquid HoldUp	0.1738	0.1395	0.1395	0.1395	0.1395	0.1395
61	Water HoldUp	---	---	---	---	---	---
62	Friction Gradient (kPa/m)	8.822e-003	9.408e-003	9.407e-003	9.407e-003	9.407e-003	9.407e-003
63	Static Gradient (kPa/m)	-6.490e-004	-6.485e-002	-6.485e-002	-6.485e-002	-6.486e-002	-6.486e-002
64	Accel. Gradient (kPa/m)	9.670e-007	-6.559e-006	-6.559e-006	-6.559e-006	-6.559e-006	-6.559e-006
65	Liq. Reynolds	1.912e+005 *	1.912e+005 *	1.912e+005 *	1.912e+005 *	1.912e+005 *	1.912e+005 *
66	Vap. Reynolds	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.755e+006 *
67	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981	0.1981	0.1981
68	Vapour Velocity (m/s)	2.853	2.853	2.853	2.853	2.853	2.853
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 246 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	1.322e+004	1.323e+004	1.323e+004	1.324e+004
16	Elevation (m)	-0.5800	-0.8100	-0.9360	-1.062
17	Pressure (bar)	71.12	71.12	71.12	71.12
18	Temperature (C)	52.94	52.94	52.94	52.94
19	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
20	Heat Transferred (kJ/h-m)	611.3	611.2	611.2	611.2
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.1395	0.1395	0.1552	0.1552
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	9.406e-003	9.406e-003	9.095e-003	9.095e-003
26	Static Gradient (kPa/m)	-6.486e-002	-6.486e-002	-3.778e-002	-3.778e-002
27	Accel. Gradient (kPa/m)	-6.559e-006	-6.558e-006	-3.392e-006	-3.392e-006
28	Liq. Reynolds	1.912e+005 *	1.912e+005 *	1.912e+005 *	1.912e+005 *
29	Vap. Reynolds	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.755e+006 *
30	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
31	Vapour Velocity (m/s)	2.853	2.852	2.852	2.852
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	1.324e+004	1.325e+004	1.325e+004	1.326e+004
35	Elevation (m)	-1.188	-1.314	-1.440	-2.192
36	Pressure (bar)	71.13	71.13	71.13	71.14
37	Temperature (C)	52.94	52.94	52.94	52.94
38	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
39	Heat Transferred (kJ/h-m)	611.1	611.1	611.1	611.5
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.1552	0.1552	0.1552	6.426e-002
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	9.094e-003	9.094e-003	9.094e-003	1.404e-002
45	Static Gradient (kPa/m)	-3.778e-002	-3.778e-002	-3.778e-002	-0.1419
46	Accel. Gradient (kPa/m)	-3.392e-006	-3.392e-006	-3.392e-006	-1.512e-005
47	Liq. Reynolds	1.912e+005 *	1.912e+005 *	1.912e+005 *	1.913e+005 *
48	Vap. Reynolds	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.754e+006 *
49	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
50	Vapour Velocity (m/s)	2.852	2.852	2.852	2.852
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	1.326e+004	1.327e+004	1.327e+004	1.328e+004
54	Elevation (m)	-2.944	-3.696	-4.448	-5.200
55	Pressure (bar)	71.14	71.15	71.15	71.16
56	Temperature (C)	52.94	52.94	52.94	52.95
57	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
58	Heat Transferred (kJ/h-m)	611.6	611.6	611.7	611.8
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	6.426e-002	6.426e-002	6.426e-002	6.427e-002
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	1.403e-002	1.403e-002	1.403e-002	1.403e-002
64	Static Gradient (kPa/m)	-0.1419	-0.1419	-0.1419	-0.1420
65	Accel. Gradient (kPa/m)	-1.512e-005	-1.512e-005	-1.511e-005	-1.511e-005
66	Liq. Reynolds	1.913e+005 *	1.913e+005 *	1.913e+005 *	1.913e+005 *
67	Vap. Reynolds	9.754e+006 *	9.754e+006 *	9.754e+006 *	9.754e+006 *
68	Liquid Velocity (m/s)	0.1981	0.1981	0.1982	0.1982
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 247 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Velocity (m/s)	2.852	2.851	2.851	2.851
14	Deposit Thickness (mm)	---	---	---	---
15	Deposit Volume (m3)	---	---	---	---
16	Length (m)	1.328e+004	1.329e+004	1.329e+004	1.330e+004
17	Elevation (m)	-5.420	-5.640	-5.860	-6.080
18	Pressure (bar)	71.16	71.17	71.17	71.17
19	Temperature (C)	52.95	52.95	52.95	52.94
20	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
21	Heat Transferred (kJ/h-m)	611.4	611.4	611.4	611.4
22	Flow Regime	Transition	Transition	Transition	Transition
23	Water Flow Regime				
24	Liquid HoldUp	0.1410	0.1410	0.1410	0.1411
25	Water HoldUp	---	---	---	---
26	Friction Gradient (kPa/m)	9.367e-003	9.366e-003	9.366e-003	9.366e-003
27	Static Gradient (kPa/m)	-6.247e-002	-6.248e-002	-6.248e-002	-6.248e-002
28	Accel. Gradient (kPa/m)	-6.273e-006	-6.273e-006	-6.273e-006	-6.272e-006
29	Liq. Reynolds	1.913e+005 *	1.913e+005 *	1.913e+005 *	1.913e+005 *
30	Vap. Reynolds	9.753e+006 *	9.753e+006 *	9.753e+006 *	9.753e+006 *
31	Liquid Velocity (m/s)	0.1982	0.1982	0.1982	0.1982
32	Vapour Velocity (m/s)	2.851	2.851	2.850	2.850
33	Deposit Thickness (mm)	---	---	---	---
34	Deposit Volume (m3)	---	---	---	---
35	Length (m)	1.330e+004	1.331e+004	1.331e+004	1.332e+004
36	Elevation (m)	-6.300	-6.600	-6.900	-7.200
37	Pressure (bar)	71.17	71.18	71.18	71.18
38	Temperature (C)	52.94	52.94	52.94	52.94
39	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
40	Heat Transferred (kJ/h-m)	611.3	611.4	611.4	611.4
41	Flow Regime	Transition	Transition	Transition	Transition
42	Water Flow Regime				
43	Liquid HoldUp	0.1411	0.1291	0.1291	0.1291
44	Water HoldUp	---	---	---	---
45	Friction Gradient (kPa/m)	9.365e-003	9.660e-003	9.659e-003	9.659e-003
46	Static Gradient (kPa/m)	-6.248e-002	-8.076e-002	-8.077e-002	-8.077e-002
47	Accel. Gradient (kPa/m)	-6.272e-006	-8.395e-006	-8.395e-006	-8.394e-006
48	Liq. Reynolds	1.913e+005 *	1.913e+005 *	1.913e+005 *	1.913e+005 *
49	Vap. Reynolds	9.753e+006 *	9.753e+006 *	9.753e+006 *	9.753e+006 *
50	Liquid Velocity (m/s)	0.1982	0.1982	0.1982	0.1982
51	Vapour Velocity (m/s)	2.850	2.850	2.850	2.850
52	Deposit Thickness (mm)	---	---	---	---
53	Deposit Volume (m3)	---	---	---	---
54	Length (m)	1.332e+004	1.333e+004	1.333e+004	1.334e+004
55	Elevation (m)	-7.500	-7.800	-8.500	-9.200
56	Pressure (bar)	71.19	71.19	71.20	71.20
57	Temperature (C)	52.94	52.94	52.95	52.95
58	Vapour Fraction	0.8818	0.8817	0.8817	0.8817
59	Heat Transferred (kJ/h-m)	611.4	611.4	611.7	611.8
60	Flow Regime	Transition	Transition	Transition	Transition
61	Water Flow Regime				
62	Liquid HoldUp	0.1291	0.1291	7.143e-002	7.144e-002
63	Water HoldUp	---	---	---	---
64	Friction Gradient (kPa/m)	9.658e-003	9.658e-003	1.308e-002	1.307e-002
65	Static Gradient (kPa/m)	-8.077e-002	-8.078e-002	-0.1384	-0.1384
66	Accel. Gradient (kPa/m)	-8.394e-006	-8.393e-006	-1.479e-005	-1.479e-005
67	Liq. Reynolds	1.913e+005 *	1.913e+005 *	1.913e+005 *	1.913e+005 *
68	Vap. Reynolds	9.753e+006 *	9.753e+006 *	9.752e+006 *	9.752e+006 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2			Unit Set: Mixed			
3			Date/Time: Mon Jun 13 15:13:43 2022			
4						
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Liquid Velocity	(m/s)	0.1982	0.1982	0.1982	0.1982
14	Vapour Velocity	(m/s)	2.849	2.849	2.849	2.849
15	Deposit Thickness	(mm)	---	---	---	---
16	Deposit Volume	(m3)	---	---	---	---
17	Length	(m)	1.334e+004	1.335e+004	1.335e+004	1.336e+004
18	Elevation	(m)	-9.900	-10.60	-11.30	-11.75
19	Pressure	(bar)	71.21	71.22	71.22	71.23
20	Temperature	(C)	52.95	52.95	52.95	52.96
21	Vapour Fraction		0.8817	0.8817	0.8817	0.8817
22	Heat Transferred	(kJ/h-m)	611.8	611.9	611.9	611.8
23	Flow Regime		Transition	Transition	Transition	Transition
24	Water Flow Regime					
25	Liquid HoldUp		7.144e-002	7.144e-002	7.145e-002	0.1071
26	Water HoldUp		---	---	---	---
27	Friction Gradient	(kPa/m)	1.307e-002	1.307e-002	1.307e-002	1.043e-002
28	Static Gradient	(kPa/m)	-0.1384	-0.1384	-0.1384	-0.1089
29	Accel. Gradient	(kPa/m)	-1.478e-005	-1.478e-005	-1.478e-005	-1.161e-005
30	Liq. Reynolds		1.913e+005 *	1.914e+005 *	1.914e+005 *	1.914e+005 *
31	Vap. Reynolds		9.752e+006 *	9.752e+006 *	9.751e+006 *	9.751e+006 *
32	Liquid Velocity	(m/s)	0.1982	0.1982	0.1982	0.1982
33	Vapour Velocity	(m/s)	2.849	2.848	2.848	2.848
34	Deposit Thickness	(mm)	---	---	---	---
35	Deposit Volume	(m3)	---	---	---	---
36	Length	(m)	1.336e+004	1.337e+004	1.337e+004	1.338e+004
37	Elevation	(m)	-12.20	-12.65	-13.10	-13.55
38	Pressure	(bar)	71.23	71.24	71.24	71.25
39	Temperature	(C)	52.96	52.96	52.96	52.96
40	Vapour Fraction		0.8817	0.8817	0.8817	0.8817
41	Heat Transferred	(kJ/h-m)	611.8	611.8	611.8	611.8
42	Flow Regime		Transition	Transition	Transition	Transition
43	Water Flow Regime					
44	Liquid HoldUp		0.1071	0.1071	0.1071	0.1071
45	Water HoldUp		---	---	---	---
46	Friction Gradient	(kPa/m)	1.043e-002	1.043e-002	1.043e-002	1.043e-002
47	Static Gradient	(kPa/m)	-0.1089	-0.1089	-0.1089	-0.1089
48	Accel. Gradient	(kPa/m)	-1.160e-005	-1.160e-005	-1.160e-005	-1.160e-005
49	Liq. Reynolds		1.914e+005 *	1.914e+005 *	1.914e+005 *	1.914e+005 *
50	Vap. Reynolds		9.751e+006 *	9.751e+006 *	9.751e+006 *	9.751e+006 *
51	Liquid Velocity	(m/s)	0.1982	0.1982	0.1982	0.1982
52	Vapour Velocity	(m/s)	2.848	2.847	2.847	2.847
53	Deposit Thickness	(mm)	---	---	---	---
54	Deposit Volume	(m3)	---	---	---	---
55	Length	(m)	1.338e+004	1.339e+004	1.339e+004	1.340e+004
56	Elevation	(m)	-13.55	-13.55	-13.55	-13.55
57	Pressure	(bar)	71.25	71.25	71.25	71.25
58	Temperature	(C)	52.96	52.96	52.95	52.95
59	Vapour Fraction		0.8817	0.8817	0.8817	0.8817
60	Heat Transferred	(kJ/h-m)	611.6	611.5	611.5	611.4
61	Flow Regime		Transition	Transition	Transition	Transition
62	Water Flow Regime					
63	Liquid HoldUp		0.2545	0.2545	0.2545	0.2545
64	Water HoldUp		---	---	---	---
65	Friction Gradient	(kPa/m)	5.921e-003	5.921e-003	5.921e-003	5.921e-003
66	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000
67	Accel. Gradient	(kPa/m)	6.977e-007	6.977e-007	6.977e-007	6.977e-007
68	Liq. Reynolds		1.914e+005 *	1.914e+005 *	1.914e+005 *	1.914e+005 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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11	Pipe Table				
12					
13	Vap. Reynolds	9.751e+006 *	9.751e+006 *	9.751e+006 *	9.751e+006 *
14	Liquid Velocity (m/s)	0.1982	0.1982	0.1982	0.1982
15	Vapour Velocity (m/s)	2.847	2.847	2.847	2.847
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	1.340e+004	1.341e+004	1.341e+004	1.342e+004
19	Elevation (m)	-13.55	-13.10	-12.65	-12.20
20	Pressure (bar)	71.25	71.23	71.21	71.20
21	Temperature (C)	52.95	52.94	52.94	52.93
22	Vapour Fraction	0.8817	0.8817	0.8817	0.8817
23	Heat Transferred (kJ/h-m)	611.4	611.0	610.8	610.6
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2545	0.4849	0.4848	0.4847
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	5.921e-003	7.391e-003	7.392e-003	7.394e-003
29	Static Gradient (kPa/m)	0.0000	0.3196	0.3196	0.3196
30	Accel. Gradient (kPa/m)	6.977e-007	3.855e-005	3.857e-005	3.858e-005
31	Liq. Reynolds	1.914e+005 *	1.914e+005 *	1.913e+005 *	1.913e+005 *
32	Vap. Reynolds	9.751e+006 *	9.751e+006 *	9.752e+006 *	9.752e+006 *
33	Liquid Velocity (m/s)	0.1982	0.1982	0.1982	0.1982
34	Vapour Velocity (m/s)	2.847	2.847	2.848	2.849
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	1.342e+004	1.343e+004	1.343e+004	1.344e+004
38	Elevation (m)	-11.75	-11.30	-10.97	-10.64
39	Pressure (bar)	71.18	71.16	71.15	71.14
40	Temperature (C)	52.92	52.92	52.91	52.91
41	Vapour Fraction	0.8817	0.8817	0.8818	0.8818
42	Heat Transferred (kJ/h-m)	610.4	610.3	610.2	610.0
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.4847	0.4846	0.4175	0.4175
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	7.396e-003	7.397e-003	7.494e-003	7.495e-003
48	Static Gradient (kPa/m)	0.3195	0.3195	0.2051	0.2051
49	Accel. Gradient (kPa/m)	3.859e-005	3.861e-005	2.512e-005	2.512e-005
50	Liq. Reynolds	1.913e+005 *	1.913e+005 *	1.913e+005 *	1.912e+005 *
51	Vap. Reynolds	9.753e+006 *	9.754e+006 *	9.754e+006 *	9.754e+006 *
52	Liquid Velocity (m/s)	0.1982	0.1982	0.1982	0.1982
53	Vapour Velocity (m/s)	2.849	2.850	2.851	2.851
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1.344e+004	1.345e+004	1.345e+004	1.346e+004
57	Elevation (m)	-10.31	-9.980	-9.650	-7.850
58	Pressure (bar)	71.13	71.12	71.11	70.99
59	Temperature (C)	52.90	52.89	52.89	52.86
60	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
61	Heat Transferred (kJ/h-m)	609.9	609.8	609.6	603.1
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.4174	0.4174	0.4174	1.000
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	7.496e-003	7.497e-003	7.498e-003	7.121e-003
67	Static Gradient (kPa/m)	0.2051	0.2051	0.2051	2.429
68	Accel. Gradient (kPa/m)	2.513e-005	2.513e-005	2.514e-005	2.891e-004
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	1.912e+005 *	1.912e+005 *	1.912e+005 *	1.910e+005 *
14	Vap. Reynolds	9.755e+006 *	9.755e+006 *	9.755e+006 *	9.760e+006 *
15	Liquid Velocity (m/s)	0.1982	0.1981	0.1981	0.1981
16	Vapour Velocity (m/s)	2.851	2.852	2.852	2.857
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	1.346e+004	1.347e+004	1.347e+004	1.348e+004
20	Elevation (m)	-6.050	-4.250	-2.450	-0.6500
21	Pressure (bar)	70.87	70.75	70.62	70.50
22	Temperature (C)	52.82	52.79	52.75	52.72
23	Vapour Fraction	0.8819	0.8819	0.8820	0.8820
24	Heat Transferred (kJ/h-m)	602.2	601.3	600.4	599.5
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	1.000	1.000	1.000	1.000
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	7.133e-003	7.144e-003	7.156e-003	7.167e-003
30	Static Gradient (kPa/m)	2.429	2.429	2.430	2.430
31	Accel. Gradient (kPa/m)	2.902e-004	2.913e-004	2.923e-004	2.934e-004
32	Liq. Reynolds	1.909e+005 *	1.907e+005 *	1.906e+005 *	1.904e+005 *
33	Vap. Reynolds	9.764e+006 *	9.768e+006 *	9.772e+006 *	9.776e+006 *
34	Liquid Velocity (m/s)	0.1981	0.1980	0.1980	0.1980
35	Vapour Velocity (m/s)	2.863	2.868	2.873	2.878
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	1.348e+004	1.349e+004	1.349e+004	1.350e+004
39	Elevation (m)	-0.4840	-0.3180	-0.1520	1.400e-002
40	Pressure (bar)	70.50	70.49	70.49	70.48
41	Temperature (C)	52.72	52.71	52.71	52.71
42	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
43	Heat Transferred (kJ/h-m)	605.2	605.1	605.0	604.9
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.3261	0.3261	0.3261	0.3261
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	7.740e-003	7.740e-003	7.741e-003	7.741e-003
49	Static Gradient (kPa/m)	8.517e-002	8.517e-002	8.517e-002	8.516e-002
50	Accel. Gradient (kPa/m)	1.119e-005	1.119e-005	1.119e-005	1.119e-005
51	Liq. Reynolds	1.904e+005 *	1.904e+005 *	1.904e+005 *	1.904e+005 *
52	Vap. Reynolds	9.776e+006 *	9.777e+006 *	9.777e+006 *	9.777e+006 *
53	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
54	Vapour Velocity (m/s)	2.878	2.879	2.879	2.879
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	1.350e+004	1.351e+004	1.351e+004	1.352e+004
58	Elevation (m)	0.1800	0.1940	0.2080	0.2220
59	Pressure (bar)	70.48	70.48	70.48	70.48
60	Temperature (C)	52.70	52.70	52.70	52.70
61	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
62	Heat Transferred (kJ/h-m)	604.8	604.7	604.7	604.6
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.3261	0.2413	0.2413	0.2413
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	7.742e-003	7.285e-003	7.285e-003	7.285e-003
68	Static Gradient (kPa/m)	8.516e-002	5.708e-003	5.708e-003	5.708e-003
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
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11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	1.119e-005	1.565e-006	1.565e-006	1.565e-006
14	Liq. Reynolds	1.904e+005 *	1.904e+005 *	1.904e+005 *	1.903e+005 *
15	Vap. Reynolds	9.777e+006 *	9.777e+006 *	9.777e+006 *	9.777e+006 *
16	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
17	Vapour Velocity (m/s)	2.879	2.879	2.879	2.879
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	1.352e+004	1.353e+004	1.353e+004	1.353e+004
21	Elevation (m)	0.2360	0.2500	0.2500	0.2500
22	Pressure (bar)	70.48	70.48	70.48	70.48
23	Temperature (C)	52.70	52.69	52.69	52.69
24	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
25	Heat Transferred (kJ/h-m)	604.6	604.5	604.5	604.5
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.2413	0.2413	0.2527	0.2527
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	7.285e-003	7.285e-003	6.018e-003	6.018e-003
31	Static Gradient (kPa/m)	5.708e-003	5.708e-003	0.0000	0.0000
32	Accel. Gradient (kPa/m)	1.565e-006	1.565e-006	7.250e-007	7.250e-007
33	Liq. Reynolds	1.903e+005 *	1.903e+005 *	1.903e+005 *	1.903e+005 *
34	Vap. Reynolds	9.777e+006 *	9.777e+006 *	9.777e+006 *	9.777e+006 *
35	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
36	Vapour Velocity (m/s)	2.879	2.879	2.879	2.879
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	1.353e+004	1.354e+004	1.354e+004	1.354e+004
40	Elevation (m)	0.2500	0.2500	0.2500	0.2520
41	Pressure (bar)	70.48	70.47	70.47	70.47
42	Temperature (C)	52.69	52.69	52.69	52.69
43	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
44	Heat Transferred (kJ/h-m)	604.5	604.5	604.4	604.4
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.2527	0.2527	0.2527	0.2351
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	6.018e-003	6.018e-003	6.018e-003	8.047e-003
50	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	1.200e-003
51	Accel. Gradient (kPa/m)	7.250e-007	7.250e-007	7.250e-007	1.114e-006
52	Liq. Reynolds	1.903e+005 *	1.903e+005 *	1.903e+005 *	1.903e+005 *
53	Vap. Reynolds	9.777e+006 *	9.777e+006 *	9.777e+006 *	9.777e+006 *
54	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
55	Vapour Velocity (m/s)	2.879	2.879	2.879	2.879
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	1.354e+004	1.355e+004	1.355e+004	1.355e+004
59	Elevation (m)	0.2540	0.2560	0.2580	0.2600
60	Pressure (bar)	70.47	70.47	70.47	70.47
61	Temperature (C)	52.69	52.69	52.68	52.68
62	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
63	Heat Transferred (kJ/h-m)	604.4	604.3	604.3	604.3
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.2351	0.2351	0.2351	0.2351
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	8.047e-003	8.047e-003	8.047e-003	8.047e-003
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1				Case Name: Simulation - 386MMSCFD_80barg.hsc	
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
Date/Time: Mon Jun 13 15:13:43 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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11	Pipe Table				
12					
13	Static Gradient	(kPa/m)	1.200e-003	1.200e-003	1.200e-003
14	Accel. Gradient	(kPa/m)	1.114e-006	1.114e-006	1.114e-006
15	Liq. Reynolds		1.903e+005 *	1.903e+005 *	1.903e+005 *
16	Vap. Reynolds		9.777e+006 *	9.777e+006 *	9.777e+006 *
17	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980
18	Vapour Velocity	(m/s)	2.879	2.879	2.879
19	Deposit Thickness	(mm)	---	---	---
20	Deposit Volume	(m3)	---	---	---
21	Length	(m)	1.356e+004	1.356e+004	1.357e+004
22	Elevation	(m)	0.2620	0.2640	0.2660
23	Pressure	(bar)	70.47	70.47	70.47
24	Temperature	(C)	52.68	52.68	52.68
25	Vapour Fraction		0.8820	0.8820	0.8820
26	Heat Transferred	(kJ/h-m)	604.2	604.2	604.1
27	Flow Regime		Transition	Transition	Transition
28	Water Flow Regime				
29	Liquid HoldUp		0.2351	0.2351	0.2351
30	Water HoldUp		---	---	---
31	Friction Gradient	(kPa/m)	8.047e-003	8.047e-003	8.047e-003
32	Static Gradient	(kPa/m)	1.201e-003	1.201e-003	1.201e-003
33	Accel. Gradient	(kPa/m)	1.114e-006	1.114e-006	1.114e-006
34	Liq. Reynolds		1.903e+005 *	1.903e+005 *	1.903e+005 *
35	Vap. Reynolds		9.777e+006 *	9.777e+006 *	9.777e+006 *
36	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980
37	Vapour Velocity	(m/s)	2.879	2.879	2.879
38	Deposit Thickness	(mm)	---	---	---
39	Deposit Volume	(m3)	---	---	---
40	Length	(m)	1.357e+004	1.357e+004	1.358e+004
41	Elevation	(m)	0.2700	0.2720	0.2740
42	Pressure	(bar)	70.47	70.47	70.47
43	Temperature	(C)	52.68	52.68	52.67
44	Vapour Fraction		0.8820	0.8820	0.8820
45	Heat Transferred	(kJ/h-m)	604.1	604.1	604.0
46	Flow Regime		Transition	Transition	Transition
47	Water Flow Regime				
48	Liquid HoldUp		0.2351	0.2351	0.2351
49	Water HoldUp		---	---	---
50	Friction Gradient	(kPa/m)	8.047e-003	8.047e-003	8.047e-003
51	Static Gradient	(kPa/m)	1.201e-003	1.200e-003	1.200e-003
52	Accel. Gradient	(kPa/m)	1.114e-006	1.114e-006	1.114e-006
53	Liq. Reynolds		1.903e+005 *	1.903e+005 *	1.903e+005 *
54	Vap. Reynolds		9.777e+006 *	9.777e+006 *	9.777e+006 *
55	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980
56	Vapour Velocity	(m/s)	2.879	2.879	2.879
57	Deposit Thickness	(mm)	---	---	---
58	Deposit Volume	(m3)	---	---	---
59	Length	(m)	1.358e+004	1.359e+004	1.359e+004
60	Elevation	(m)	0.2780	0.2800	0.2760
61	Pressure	(bar)	70.47	70.47	70.47
62	Temperature	(C)	52.67	52.67	52.67
63	Vapour Fraction		0.8820	0.8820	0.8820
64	Heat Transferred	(kJ/h-m)	604.0	603.9	604.0
65	Flow Regime		Transition	Transition	Transition
66	Water Flow Regime				
67	Liquid HoldUp		0.2351	0.2351	0.1721
68	Water HoldUp		---	---	---
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1				Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	8.047e-003	8.047e-003	8.906e-003	8.906e-003
14	Static Gradient	(kPa/m)	1.200e-003	1.200e-003	-2.145e-003	-2.145e-003
15	Accel. Gradient	(kPa/m)	1.114e-006	1.114e-006	8.145e-007	8.145e-007
16	Liq. Reynolds		1.903e+005 *	1.903e+005 *	1.903e+005 *	1.903e+005 *
17	Vap. Reynolds		9.777e+006 *	9.777e+006 *	9.777e+006 *	9.777e+006 *
18	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
19	Vapour Velocity	(m/s)	2.879	2.879	2.879	2.879
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	1.360e+004	1.360e+004	1.360e+004	1.361e+004
23	Elevation	(m)	0.2680	0.2640	0.2600	0.2620
24	Pressure	(bar)	70.47	70.47	70.47	70.47
25	Temperature	(C)	52.67	52.67	52.67	52.66
26	Vapour Fraction		0.8820	0.8820	0.8820	0.8820
27	Heat Transferred	(kJ/h-m)	603.9	603.9	603.8	603.7
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.1721	0.1721	0.1721	0.2346
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	8.906e-003	8.906e-003	8.906e-003	8.119e-003
33	Static Gradient	(kPa/m)	-2.145e-003	-2.145e-003	-2.145e-003	7.988e-004
34	Accel. Gradient	(kPa/m)	8.145e-007	8.145e-007	8.145e-007	1.074e-006
35	Liq. Reynolds		1.903e+005 *	1.903e+005 *	1.903e+005 *	1.903e+005 *
36	Vap. Reynolds		9.777e+006 *	9.777e+006 *	9.777e+006 *	9.777e+006 *
37	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
38	Vapour Velocity	(m/s)	2.879	2.879	2.879	2.879
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	1.361e+004	1.362e+004	1.362e+004	1.363e+004
42	Elevation	(m)	0.2640	0.2660	0.2680	0.2700
43	Pressure	(bar)	70.47	70.47	70.47	70.47
44	Temperature	(C)	52.66	52.66	52.66	52.66
45	Vapour Fraction		0.8820	0.8820	0.8820	0.8820
46	Heat Transferred	(kJ/h-m)	603.7	603.6	603.6	603.5
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2346	0.2346	0.2346	0.2346
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	8.119e-003	8.119e-003	8.119e-003	8.119e-003
52	Static Gradient	(kPa/m)	7.988e-004	7.988e-004	7.988e-004	7.988e-004
53	Accel. Gradient	(kPa/m)	1.074e-006	1.074e-006	1.074e-006	1.074e-006
54	Liq. Reynolds		1.903e+005 *	1.903e+005 *	1.903e+005 *	1.903e+005 *
55	Vap. Reynolds		9.777e+006 *	9.778e+006 *	9.778e+006 *	9.778e+006 *
56	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
57	Vapour Velocity	(m/s)	2.879	2.879	2.879	2.879
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	1.363e+004	1.364e+004	1.364e+004	1.365e+004
61	Elevation	(m)	0.2780	0.2860	0.2940	0.3020
62	Pressure	(bar)	70.47	70.47	70.47	70.46
63	Temperature	(C)	52.65	52.65	52.65	52.65
64	Vapour Fraction		0.8820	0.8820	0.8820	0.8820
65	Heat Transferred	(kJ/h-m)	603.5	603.4	603.4	603.3
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2379	0.2379	0.2379	0.2379
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	7.697e-003	7.697e-003	7.697e-003	7.697e-003
15	Static Gradient (kPa/m)	3.201e-003	3.201e-003	3.201e-003	3.201e-003
16	Accel. Gradient (kPa/m)	1.313e-006	1.313e-006	1.313e-006	1.313e-006
17	Liq. Reynolds	1.903e+005 *	1.903e+005 *	1.903e+005 *	1.903e+005 *
18	Vap. Reynolds	9.778e+006 *	9.778e+006 *	9.778e+006 *	9.778e+006 *
19	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
20	Vapour Velocity (m/s)	2.879	2.879	2.879	2.879
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	1.365e+004	1.366e+004	1.366e+004	1.367e+004
24	Elevation (m)	0.3100	0.3060	0.3020	0.2980
25	Pressure (bar)	70.46	70.46	70.46	70.46
26	Temperature (C)	52.65	52.64	52.64	52.64
27	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
28	Heat Transferred (kJ/h-m)	603.3	603.3	603.2	603.2
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2379	0.1725	0.1725	0.1725
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	7.697e-003	8.901e-003	8.901e-003	8.901e-003
34	Static Gradient (kPa/m)	3.201e-003	-1.289e-003	-1.289e-003	-1.289e-003
35	Accel. Gradient (kPa/m)	1.313e-006	9.170e-007	9.170e-007	9.170e-007
36	Liq. Reynolds	1.903e+005 *	1.903e+005 *	1.903e+005 *	1.903e+005 *
37	Vap. Reynolds	9.778e+006 *	9.778e+006 *	9.778e+006 *	9.778e+006 *
38	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
39	Vapour Velocity (m/s)	2.879	2.879	2.879	2.879
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	1.367e+004	1.368e+004	1.368e+004	1.369e+004
43	Elevation (m)	0.2940	0.2900	0.2920	0.2940
44	Pressure (bar)	70.46	70.46	70.46	70.46
45	Temperature (C)	52.64	52.64	52.64	52.63
46	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
47	Heat Transferred (kJ/h-m)	603.1	603.1	603.0	602.9
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.1725	0.1725	0.2346	0.2346
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	8.900e-003	8.900e-003	8.118e-003	8.118e-003
53	Static Gradient (kPa/m)	-1.289e-003	-1.289e-003	7.989e-004	7.989e-004
54	Accel. Gradient (kPa/m)	9.170e-007	9.170e-007	1.074e-006	1.074e-006
55	Liq. Reynolds	1.903e+005 *	1.903e+005 *	1.903e+005 *	1.903e+005 *
56	Vap. Reynolds	9.778e+006 *	9.778e+006 *	9.778e+006 *	9.778e+006 *
57	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
58	Vapour Velocity (m/s)	2.879	2.879	2.879	2.879
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	1.369e+004	1.370e+004	1.370e+004	1.371e+004
62	Elevation (m)	0.2960	0.2980	0.3000	0.2800
63	Pressure (bar)	70.46	70.46	70.46	70.46
64	Temperature (C)	52.63	52.63	52.63	52.63
65	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
66	Heat Transferred (kJ/h-m)	602.9	602.8	602.8	602.8
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.2346	0.2346	0.2346	0.1701
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	8.118e-003	8.118e-003	8.118e-003	8.932e-003
16	Static Gradient (kPa/m)	7.989e-004	7.989e-004	7.989e-004	-6.386e-003
17	Accel. Gradient (kPa/m)	1.074e-006	1.074e-006	1.074e-006	3.067e-007
18	Liq. Reynolds	1.903e+005 *	1.903e+005 *	1.903e+005 *	1.903e+005 *
19	Vap. Reynolds	9.778e+006 *	9.778e+006 *	9.778e+006 *	9.778e+006 *
20	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
21	Vapour Velocity (m/s)	2.879	2.879	2.879	2.879
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	1.371e+004	1.372e+004	1.372e+004	1.373e+004
25	Elevation (m)	0.2600	0.2400	0.2200	0.2000
26	Pressure (bar)	70.46	70.46	70.46	70.46
27	Temperature (C)	52.62	52.62	52.62	52.62
28	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
29	Heat Transferred (kJ/h-m)	602.8	602.7	602.7	602.6
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1701	0.1701	0.1701	0.1701
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	8.932e-003	8.932e-003	8.932e-003	8.932e-003
35	Static Gradient (kPa/m)	-6.386e-003	-6.386e-003	-6.386e-003	-6.386e-003
36	Accel. Gradient (kPa/m)	3.067e-007	3.067e-007	3.067e-007	3.067e-007
37	Liq. Reynolds	1.903e+005 *	1.903e+005 *	1.903e+005 *	1.903e+005 *
38	Vap. Reynolds	9.778e+006 *	9.778e+006 *	9.778e+006 *	9.778e+006 *
39	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
40	Vapour Velocity (m/s)	2.879	2.879	2.879	2.879
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	1.373e+004	1.374e+004	1.374e+004	1.375e+004
44	Elevation (m)	0.1780	0.1560	0.1340	0.1120
45	Pressure (bar)	70.46	70.46	70.46	70.46
46	Temperature (C)	52.62	52.62	52.61	52.61
47	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
48	Heat Transferred (kJ/h-m)	602.6	602.5	602.5	602.4
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1698	0.1698	0.1698	0.1698
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	8.936e-003	8.936e-003	8.936e-003	8.936e-003
54	Static Gradient (kPa/m)	-7.017e-003	-7.017e-003	-7.017e-003	-7.017e-003
55	Accel. Gradient (kPa/m)	2.312e-007	2.312e-007	2.312e-007	2.312e-007
56	Liq. Reynolds	1.903e+005 *	1.903e+005 *	1.903e+005 *	1.903e+005 *
57	Vap. Reynolds	9.778e+006 *	9.778e+006 *	9.778e+006 *	9.778e+006 *
58	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
59	Vapour Velocity (m/s)	2.879	2.879	2.879	2.879
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	1.375e+004	1.376e+004	1.376e+004	1.377e+004
63	Elevation (m)	9.000e-002	5.800e-002	2.600e-002	-6.000e-003
64	Pressure (bar)	70.46	70.46	70.46	70.46
65	Temperature (C)	52.61	52.61	52.61	52.61
66	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
67	Heat Transferred (kJ/h-m)	602.4	602.4	602.3	602.3
68	Flow Regime	Transition	Transition	Transition	Transition
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1698	0.1683	0.1683	0.1683
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	8.935e-003	8.956e-003	8.956e-003	8.956e-003
17	Static Gradient (kPa/m)	-7.017e-003	-1.015e-002	-1.015e-002	-1.015e-002
18	Accel. Gradient (kPa/m)	2.311e-007	-1.435e-007	-1.435e-007	-1.435e-007
19	Liq. Reynolds	1.903e+005 *	1.903e+005 *	1.903e+005 *	1.903e+005 *
20	Vap. Reynolds	9.778e+006 *	9.778e+006 *	9.778e+006 *	9.778e+006 *
21	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
22	Vapour Velocity (m/s)	2.879	2.879	2.879	2.879
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	1.377e+004	1.378e+004	1.378e+004	1.379e+004
26	Elevation (m)	-3.800e-002	-7.000e-002	-6.000e-002	-5.000e-002
27	Pressure (bar)	70.46	70.46	70.46	70.46
28	Temperature (C)	52.60	52.60	52.60	52.60
29	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
30	Heat Transferred (kJ/h-m)	602.2	602.2	602.1	602.0
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1683	0.1683	0.2391	0.2391
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	8.956e-003	8.956e-003	7.553e-003	7.553e-003
36	Static Gradient (kPa/m)	-1.015e-002	-1.015e-002	4.050e-003	4.050e-003
37	Accel. Gradient (kPa/m)	-1.435e-007	-1.436e-007	1.398e-006	1.398e-006
38	Liq. Reynolds	1.903e+005 *	1.903e+005 *	1.903e+005 *	1.903e+005 *
39	Vap. Reynolds	9.778e+006 *	9.778e+006 *	9.778e+006 *	9.778e+006 *
40	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
41	Vapour Velocity (m/s)	2.879	2.879	2.879	2.879
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	1.379e+004	1.380e+004	1.380e+004	1.381e+004
45	Elevation (m)	-4.000e-002	-3.000e-002	-2.000e-002	3.600e-002
46	Pressure (bar)	70.46	70.46	70.46	70.45
47	Temperature (C)	52.60	52.60	52.59	52.59
48	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
49	Heat Transferred (kJ/h-m)	602.0	601.9	601.9	601.8
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2391	0.2391	0.2391	0.2648
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	7.553e-003	7.553e-003	7.553e-003	7.663e-003
55	Static Gradient (kPa/m)	4.050e-003	4.050e-003	4.050e-003	2.447e-002
56	Accel. Gradient (kPa/m)	1.398e-006	1.398e-006	1.398e-006	3.871e-006
57	Liq. Reynolds	1.903e+005 *	1.903e+005 *	1.902e+005 *	1.902e+005 *
58	Vap. Reynolds	9.778e+006 *	9.778e+006 *	9.778e+006 *	9.778e+006 *
59	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
60	Vapour Velocity (m/s)	2.879	2.879	2.879	2.879
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	1.381e+004	1.382e+004	1.382e+004	1.383e+004
64	Elevation (m)	9.200e-002	0.1480	0.2040	0.2600
65	Pressure (bar)	70.45	70.45	70.45	70.45
66	Temperature (C)	52.59	52.59	52.58	52.58
67	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
68	Heat Transferred (kJ/h-m)	601.7	601.7	601.6	601.6
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 257 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2648	0.2648	0.2648	0.2648
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	7.663e-003	7.663e-003	7.663e-003	7.663e-003
18	Static Gradient (kPa/m)	2.447e-002	2.447e-002	2.447e-002	2.447e-002
19	Accel. Gradient (kPa/m)	3.871e-006	3.872e-006	3.872e-006	3.872e-006
20	Liq. Reynolds	1.902e+005 *	1.902e+005 *	1.902e+005 *	1.902e+005 *
21	Vap. Reynolds	9.778e+006 *	9.778e+006 *	9.778e+006 *	9.778e+006 *
22	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
23	Vapour Velocity (m/s)	2.879	2.879	2.879	2.879
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	1.383e+004	1.384e+004	1.384e+004	1.385e+004
27	Elevation (m)	0.1360	1.200e-002	-0.1120	-0.2360
28	Pressure (bar)	70.45	70.45	70.45	70.45
29	Temperature (C)	52.58	52.58	52.58	52.58
30	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
31	Heat Transferred (kJ/h-m)	601.6	601.6	601.6	601.5
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1547	0.1547	0.1547	0.1547
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	9.167e-003	9.167e-003	9.167e-003	9.167e-003
37	Static Gradient (kPa/m)	-3.689e-002	-3.689e-002	-3.689e-002	-3.689e-002
38	Accel. Gradient (kPa/m)	-3.340e-006	-3.340e-006	-3.340e-006	-3.340e-006
39	Liq. Reynolds	1.902e+005 *	1.902e+005 *	1.902e+005 *	1.902e+005 *
40	Vap. Reynolds	9.778e+006 *	9.778e+006 *	9.778e+006 *	9.778e+006 *
41	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
42	Vapour Velocity (m/s)	2.879	2.879	2.879	2.878
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	1.385e+004	1.386e+004	1.386e+004	1.387e+004
46	Elevation (m)	-0.3600	-0.3280	-0.2960	-0.2640
47	Pressure (bar)	70.46	70.45	70.45	70.45
48	Temperature (C)	52.58	52.57	52.57	52.57
49	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
50	Heat Transferred (kJ/h-m)	601.5	601.4	601.3	601.3
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1547	0.2514	0.2514	0.2514
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	9.166e-003	6.155e-003	6.155e-003	6.155e-003
56	Static Gradient (kPa/m)	-3.689e-002	1.345e-002	1.345e-002	1.345e-002
57	Accel. Gradient (kPa/m)	-3.340e-006	2.362e-006	2.362e-006	2.362e-006
58	Liq. Reynolds	1.902e+005 *	1.902e+005 *	1.902e+005 *	1.902e+005 *
59	Vap. Reynolds	9.778e+006 *	9.778e+006 *	9.778e+006 *	9.778e+006 *
60	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
61	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	1.387e+004	1.388e+004	1.388e+004	1.389e+004
65	Elevation (m)	-0.2320	-0.2000	-0.1760	-0.1520
66	Pressure (bar)	70.45	70.45	70.45	70.45
67	Temperature (C)	52.57	52.57	52.56	52.56
68	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Heat Transferred (kJ/h-m)	601.2	601.1	601.1	601.0					
14	Flow Regime	Transition	Transition	Transition	Transition					
15	Water Flow Regime									
16	Liquid HoldUp	0.2514	0.2514	0.2469	0.2469					
17	Water HoldUp	---	---	---	---					
18	Friction Gradient (kPa/m)	6.155e-003	6.155e-003	6.640e-003	6.640e-003					
19	Static Gradient (kPa/m)	1.345e-002	1.345e-002	9.954e-003	9.954e-003					
20	Accel. Gradient (kPa/m)	2.362e-006	2.362e-006	1.999e-006	1.999e-006					
21	Liq. Reynolds	1.902e+005 *	1.902e+005 *	1.902e+005 *	1.902e+005 *					
22	Vap. Reynolds	9.778e+006 *	9.778e+006 *	9.778e+006 *	9.778e+006 *					
23	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
24	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878					
25	Deposit Thickness (mm)	---	---	---	---					
26	Deposit Volume (m3)	---	---	---	---					
27	Length (m)	1.389e+004	1.390e+004	1.390e+004	1.391e+004					
28	Elevation (m)	-0.1280	-0.1040	-8.000e-002	-7.200e-002					
29	Pressure (bar)	70.45	70.45	70.45	70.45					
30	Temperature (C)	52.56	52.56	52.56	52.55					
31	Vapour Fraction	0.8820	0.8820	0.8820	0.8820					
32	Heat Transferred (kJ/h-m)	601.0	600.9	600.9	600.8					
33	Flow Regime	Transition	Transition	Transition	Transition					
34	Water Flow Regime									
35	Liquid HoldUp	0.2469	0.2469	0.2469	0.2380					
36	Water HoldUp	---	---	---	---					
37	Friction Gradient (kPa/m)	6.640e-003	6.640e-003	6.640e-003	7.691e-003					
38	Static Gradient (kPa/m)	9.954e-003	9.954e-003	9.954e-003	3.229e-003					
39	Accel. Gradient (kPa/m)	1.999e-006	1.999e-006	1.999e-006	1.316e-006					
40	Liq. Reynolds	1.902e+005 *	1.902e+005 *	1.902e+005 *	1.902e+005 *					
41	Vap. Reynolds	9.778e+006 *	9.779e+006 *	9.779e+006 *	9.779e+006 *					
42	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
43	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878					
44	Deposit Thickness (mm)	---	---	---	---					
45	Deposit Volume (m3)	---	---	---	---					
46	Length (m)	1.391e+004	1.392e+004	1.392e+004	1.393e+004					
47	Elevation (m)	-6.400e-002	-5.600e-002	-4.800e-002	-4.000e-002					
48	Pressure (bar)	70.45	70.44	70.44	70.44					
49	Temperature (C)	52.55	52.55	52.55	52.55					
50	Vapour Fraction	0.8820	0.8820	0.8820	0.8820					
51	Heat Transferred (kJ/h-m)	600.8	600.7	600.7	600.6					
52	Flow Regime	Transition	Transition	Transition	Transition					
53	Water Flow Regime									
54	Liquid HoldUp	0.2380	0.2380	0.2380	0.2380					
55	Water HoldUp	---	---	---	---					
56	Friction Gradient (kPa/m)	7.691e-003	7.691e-003	7.691e-003	7.691e-003					
57	Static Gradient (kPa/m)	3.229e-003	3.229e-003	3.229e-003	3.229e-003					
58	Accel. Gradient (kPa/m)	1.316e-006	1.316e-006	1.316e-006	1.316e-006					
59	Liq. Reynolds	1.902e+005 *	1.902e+005 *	1.902e+005 *	1.902e+005 *					
60	Vap. Reynolds	9.779e+006 *	9.779e+006 *	9.779e+006 *	9.779e+006 *					
61	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
62	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878					
63	Deposit Thickness (mm)	---	---	---	---					
64	Deposit Volume (m3)	---	---	---	---					
65	Length (m)	1.393e+004	1.394e+004	1.394e+004	1.395e+004					
66	Elevation (m)	-5.400e-002	-6.800e-002	-8.200e-002	-9.600e-002					
67	Pressure (bar)	70.44	70.44	70.44	70.44					
68	Temperature (C)	52.54	52.54	52.54	52.54					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Fraction		0.8820	0.8820	0.8820	0.8820	
14	Heat Transferred (kJ/h-m)		600.6	600.6	600.5	600.5	
15	Flow Regime		Transition	Transition	Transition	Transition	
16	Water Flow Regime						
17	Liquid HoldUp		0.1710	0.1710	0.1710	0.1710	
18	Water HoldUp		---	---	---	---	
19	Friction Gradient (kPa/m)		8.919e-003	8.919e-003	8.919e-003	8.919e-003	
20	Static Gradient (kPa/m)		-4.487e-003	-4.487e-003	-4.487e-003	-4.487e-003	
21	Accel. Gradient (kPa/m)		5.341e-007	5.341e-007	5.340e-007	5.340e-007	
22	Liq. Reynolds		1.902e+005 *	1.902e+005 *	1.902e+005 *	1.902e+005 *	
23	Vap. Reynolds		9.779e+006 *	9.779e+006 *	9.779e+006 *	9.779e+006 *	
24	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
25	Vapour Velocity (m/s)		2.878	2.878	2.878	2.878	
26	Deposit Thickness (mm)		---	---	---	---	
27	Deposit Volume (m3)		---	---	---	---	
28	Length (m)		1.395e+004	1.396e+004	1.396e+004	1.397e+004	
29	Elevation (m)		-0.1100	-7.200e-002	-3.400e-002	4.000e-003	
30	Pressure (bar)		70.44	70.44	70.44	70.44	
31	Temperature (C)		52.54	52.54	52.53	52.53	
32	Vapour Fraction		0.8820	0.8820	0.8820	0.8820	
33	Heat Transferred (kJ/h-m)		600.4	600.3	600.3	600.2	
34	Flow Regime		Transition	Transition	Transition	Transition	
35	Water Flow Regime						
36	Liquid HoldUp		0.1710	0.2547	0.2547	0.2547	
37	Water HoldUp		---	---	---	---	
38	Friction Gradient (kPa/m)		8.919e-003	6.575e-003	6.575e-003	6.574e-003	
39	Static Gradient (kPa/m)		-4.487e-003	1.613e-002	1.613e-002	1.613e-002	
40	Accel. Gradient (kPa/m)		5.340e-007	2.736e-006	2.736e-006	2.736e-006	
41	Liq. Reynolds		1.902e+005 *	1.902e+005 *	1.902e+005 *	1.902e+005 *	
42	Vap. Reynolds		9.779e+006 *	9.779e+006 *	9.779e+006 *	9.779e+006 *	
43	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
44	Vapour Velocity (m/s)		2.878	2.878	2.878	2.878	
45	Deposit Thickness (mm)		---	---	---	---	
46	Deposit Volume (m3)		---	---	---	---	
47	Length (m)		1.397e+004	1.398e+004	1.398e+004	1.399e+004	
48	Elevation (m)		4.200e-002	8.000e-002	8.400e-002	8.800e-002	
49	Pressure (bar)		70.44	70.44	70.44	70.44	
50	Temperature (C)		52.53	52.53	52.53	52.52	
51	Vapour Fraction		0.8820	0.8820	0.8820	0.8820	
52	Heat Transferred (kJ/h-m)		600.2	600.1	600.1	600.0	
53	Flow Regime		Transition	Transition	Transition	Transition	
54	Water Flow Regime						
55	Liquid HoldUp		0.2547	0.2547	0.2357	0.2357	
56	Water HoldUp		---	---	---	---	
57	Friction Gradient (kPa/m)		6.574e-003	6.574e-003	7.973e-003	7.973e-003	
58	Static Gradient (kPa/m)		1.613e-002	1.613e-002	1.604e-003	1.604e-003	
59	Accel. Gradient (kPa/m)		2.736e-006	2.736e-006	1.154e-006	1.154e-006	
60	Liq. Reynolds		1.902e+005 *	1.902e+005 *	1.902e+005 *	1.902e+005 *	
61	Vap. Reynolds		9.779e+006 *	9.779e+006 *	9.779e+006 *	9.779e+006 *	
62	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
63	Vapour Velocity (m/s)		2.878	2.878	2.878	2.878	
64	Deposit Thickness (mm)		---	---	---	---	
65	Deposit Volume (m3)		---	---	---	---	
66	Length (m)		1.399e+004	1.400e+004	1.400e+004	1.401e+004	
67	Elevation (m)		9.200e-002	9.600e-002	1.000e-001	3.800e-002	
68	Pressure (bar)		70.44	70.43	70.43	70.43	
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	52.52	52.52	52.52	52.52
14	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
15	Heat Transferred (kJ/h-m)	600.0	599.9	599.9	599.9
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2357	0.2357	0.2357	0.1638
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	7.973e-003	7.973e-003	7.973e-003	9.020e-003
21	Static Gradient (kPa/m)	1.604e-003	1.604e-003	1.604e-003	-1.932e-002
22	Accel. Gradient (kPa/m)	1.154e-006	1.154e-006	1.154e-006	-1.241e-006
23	Liq. Reynolds	1.902e+005 *	1.902e+005 *	1.902e+005 *	1.902e+005 *
24	Vap. Reynolds	9.779e+006 *	9.779e+006 *	9.779e+006 *	9.779e+006 *
25	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
26	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	1.401e+004	1.402e+004	1.402e+004	1.403e+004
30	Elevation (m)	-2.400e-002	-8.600e-002	-0.1480	-0.2100
31	Pressure (bar)	70.44	70.44	70.44	70.44
32	Temperature (C)	52.52	52.51	52.51	52.51
33	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
34	Heat Transferred (kJ/h-m)	599.8	599.8	599.8	599.7
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1638	0.1638	0.1638	0.1638
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	9.020e-003	9.020e-003	9.020e-003	9.019e-003
40	Static Gradient (kPa/m)	-1.932e-002	-1.932e-002	-1.932e-002	-1.932e-002
41	Accel. Gradient (kPa/m)	-1.241e-006	-1.241e-006	-1.241e-006	-1.241e-006
42	Liq. Reynolds	1.902e+005 *	1.902e+005 *	1.902e+005 *	1.902e+005 *
43	Vap. Reynolds	9.779e+006 *	9.779e+006 *	9.779e+006 *	9.779e+006 *
44	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
45	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	1.403e+004	1.404e+004	1.404e+004	1.405e+004
49	Elevation (m)	-0.1480	-8.600e-002	-2.400e-002	3.800e-002
50	Pressure (bar)	70.44	70.43	70.43	70.43
51	Temperature (C)	52.51	52.51	52.50	52.50
52	Vapour Fraction	0.8820	0.8820	0.8820	0.8820
53	Heat Transferred (kJ/h-m)	599.6	599.5	599.5	599.4
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2680	0.2680	0.2680	0.2679
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	7.722e-003	7.722e-003	7.722e-003	7.722e-003
59	Static Gradient (kPa/m)	2.718e-002	2.718e-002	2.718e-002	2.718e-002
60	Accel. Gradient (kPa/m)	4.206e-006	4.206e-006	4.207e-006	4.207e-006
61	Liq. Reynolds	1.902e+005 *	1.902e+005 *	1.902e+005 *	1.902e+005 *
62	Vap. Reynolds	9.779e+006 *	9.779e+006 *	9.779e+006 *	9.779e+006 *
63	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
64	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	1.405e+004	1.405e+004	1.406e+004	1.406e+004
68	Elevation (m)	1.000e-001	0.1080	0.1160	0.1240
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Pressure (bar)	70.43	70.43	70.43	70.43					
14	Temperature (C)	52.50	52.50	52.50	52.50					
15	Vapour Fraction	0.8820	0.8820	0.8820	0.8820					
16	Heat Transferred (kJ/h-m)	599.4	599.3	599.3	599.3					
17	Flow Regime	Transition	Transition	Transition	Transition					
18	Water Flow Regime									
19	Liquid HoldUp	0.2679	0.2559	0.2559	0.2559					
20	Water HoldUp	---	---	---	---					
21	Friction Gradient (kPa/m)	7.722e-003	6.941e-003	6.941e-003	6.941e-003					
22	Static Gradient (kPa/m)	2.718e-002	1.704e-002	1.704e-002	1.704e-002					
23	Accel. Gradient (kPa/m)	4.207e-006	2.890e-006	2.890e-006	2.890e-006					
24	Liq. Reynolds	1.901e+005 *	1.901e+005 *	1.901e+005 *	1.901e+005 *					
25	Vap. Reynolds	9.779e+006 *	9.779e+006 *	9.779e+006 *	9.779e+006 *					
26	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
27	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878					
28	Deposit Thickness (mm)	---	---	---	---					
29	Deposit Volume (m3)	---	---	---	---					
30	Length (m)	1.406e+004	1.406e+004	1.406e+004	1.407e+004					
31	Elevation (m)	0.1320	0.1400	0.1300	0.1200					
32	Pressure (bar)	70.43	70.43	70.43	70.43					
33	Temperature (C)	52.50	52.50	52.50	52.49					
34	Vapour Fraction	0.8820	0.8820	0.8819	0.8819					
35	Heat Transferred (kJ/h-m)	599.3	599.3	599.3	599.3					
36	Flow Regime	Transition	Transition	Transition	Transition					
37	Water Flow Regime									
38	Liquid HoldUp	0.2559	0.2559	0.1716	0.1716					
39	Water HoldUp	---	---	---	---					
40	Friction Gradient (kPa/m)	6.941e-003	6.941e-003	8.911e-003	8.911e-003					
41	Static Gradient (kPa/m)	1.704e-002	1.704e-002	-3.212e-003	-3.212e-003					
42	Accel. Gradient (kPa/m)	2.890e-006	2.890e-006	6.869e-007	6.869e-007					
43	Liq. Reynolds	1.901e+005 *	1.901e+005 *	1.901e+005 *	1.901e+005 *					
44	Vap. Reynolds	9.779e+006 *	9.779e+006 *	9.779e+006 *	9.779e+006 *					
45	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
46	Vapour Velocity (m/s)	2.879	2.879	2.878	2.878					
47	Deposit Thickness (mm)	---	---	---	---					
48	Deposit Volume (m3)	---	---	---	---					
49	Length (m)	1.407e+004	1.408e+004	1.408e+004	1.408e+004					
50	Elevation (m)	0.1100	1.000e-001	9.000e-002	6.200e-002					
51	Pressure (bar)	70.43	70.43	70.43	70.43					
52	Temperature (C)	52.49	52.49	52.49	52.49					
53	Vapour Fraction	0.8819	0.8819	0.8819	0.8819					
54	Heat Transferred (kJ/h-m)	599.2	599.2	599.1	599.1					
55	Flow Regime	Transition	Transition	Transition	Transition					
56	Water Flow Regime									
57	Liquid HoldUp	0.1716	0.1716	0.1716	0.1522					
58	Water HoldUp	---	---	---	---					
59	Friction Gradient (kPa/m)	8.911e-003	8.911e-003	8.911e-003	9.212e-003					
60	Static Gradient (kPa/m)	-3.212e-003	-3.212e-003	-3.212e-003	-4.159e-002					
61	Accel. Gradient (kPa/m)	6.869e-007	6.868e-007	6.868e-007	-3.902e-006					
62	Liq. Reynolds	1.901e+005 *	1.901e+005 *	1.901e+005 *	1.901e+005 *					
63	Vap. Reynolds	9.779e+006 *	9.779e+006 *	9.779e+006 *	9.779e+006 *					
64	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
65	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878					
66	Deposit Thickness (mm)	---	---	---	---					
67	Deposit Volume (m3)	---	---	---	---					
68	Length (m)	1.409e+004	1.409e+004	1.409e+004	1.409e+004					
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 262 of 937					

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Elevation (m)	3.400e-002	6.000e-003	-2.200e-002	-5.000e-002		
14	Pressure (bar)	70.43	70.43	70.43	70.43		
15	Temperature (C)	52.49	52.49	52.49	52.49		
16	Vapour Fraction	0.8819	0.8819	0.8819	0.8819		
17	Heat Transferred (kJ/h-m)	599.1	599.1	599.1	599.1		
18	Flow Regime	Transition	Transition	Transition	Transition		
19	Water Flow Regime						
20	Liquid HoldUp	0.1522	0.1522	0.1522	0.1522		
21	Water HoldUp	---	---	---	---		
22	Friction Gradient (kPa/m)	9.212e-003	9.212e-003	9.212e-003	9.212e-003		
23	Static Gradient (kPa/m)	-4.159e-002	-4.159e-002	-4.159e-002	-4.159e-002		
24	Accel. Gradient (kPa/m)	-3.902e-006	-3.902e-006	-3.902e-006	-3.902e-006		
25	Liq. Reynolds	1.901e+005 *	1.901e+005 *	1.901e+005 *	1.901e+005 *		
26	Vap. Reynolds	9.779e+006 *	9.779e+006 *	9.779e+006 *	9.779e+006 *		
27	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
28	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878		
29	Deposit Thickness (mm)	---	---	---	---		
30	Deposit Volume (m3)	---	---	---	---		
31	Length (m)	1.409e+004	1.410e+004	1.411e+004	1.411e+004		
32	Elevation (m)	-0.1160	-0.1820	-0.2480	-0.3140		
33	Pressure (bar)	70.43	70.43	70.43	70.43		
34	Temperature (C)	52.49	52.48	52.48	52.48		
35	Vapour Fraction	0.8819	0.8819	0.8819	0.8819		
36	Heat Transferred (kJ/h-m)	599.1	599.0	599.0	598.9		
37	Flow Regime	Transition	Transition	Transition	Transition		
38	Water Flow Regime						
39	Liquid HoldUp	0.1649	0.1649	0.1649	0.1649		
40	Water HoldUp	---	---	---	---		
41	Friction Gradient (kPa/m)	9.004e-003	9.004e-003	9.004e-003	9.004e-003		
42	Static Gradient (kPa/m)	-1.721e-002	-1.721e-002	-1.721e-002	-1.721e-002		
43	Accel. Gradient (kPa/m)	-9.886e-007	-9.886e-007	-9.886e-007	-9.886e-007		
44	Liq. Reynolds	1.901e+005 *	1.901e+005 *	1.901e+005 *	1.901e+005 *		
45	Vap. Reynolds	9.779e+006 *	9.779e+006 *	9.779e+006 *	9.779e+006 *		
46	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
47	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878		
48	Deposit Thickness (mm)	---	---	---	---		
49	Deposit Volume (m3)	---	---	---	---		
50	Length (m)	1.412e+004	1.413e+004	1.413e+004	1.414e+004		
51	Elevation (m)	-0.3800	-0.3980	-0.4160	-0.4340		
52	Pressure (bar)	70.43	70.43	70.43	70.43		
53	Temperature (C)	52.48	52.48	52.47	52.47		
54	Vapour Fraction	0.8819	0.8819	0.8819	0.8819		
55	Heat Transferred (kJ/h-m)	598.9	598.8	598.7	598.7		
56	Flow Regime	Transition	Transition	Transition	Transition		
57	Water Flow Regime						
58	Liquid HoldUp	0.1649	0.1712	0.1712	0.1712		
59	Water HoldUp	---	---	---	---		
60	Friction Gradient (kPa/m)	9.004e-003	8.916e-003	8.916e-003	8.916e-003		
61	Static Gradient (kPa/m)	-1.721e-002	-4.124e-003	-4.124e-003	-4.124e-003		
62	Accel. Gradient (kPa/m)	-9.887e-007	5.775e-007	5.775e-007	5.774e-007		
63	Liq. Reynolds	1.901e+005 *	1.901e+005 *	1.901e+005 *	1.901e+005 *		
64	Vap. Reynolds	9.779e+006 *	9.779e+006 *	9.779e+006 *	9.779e+006 *		
65	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
66	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878		
67	Deposit Thickness (mm)	---	---	---	---		
68	Deposit Volume (m3)	---	---	---	---		
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 263 of 937		

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	1.415e+004	1.415e+004	1.416e+004	1.416e+004
14	Elevation (m)	-0.4520	-0.4700	-0.3840	-0.2980
15	Pressure (bar)	70.43	70.43	70.43	70.42
16	Temperature (C)	52.47	52.47	52.46	52.46
17	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
18	Heat Transferred (kJ/h-m)	598.6	598.5	598.4	598.3
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.1712	0.1712	0.2816	0.2816
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	8.916e-003	8.915e-003	7.802e-003	7.802e-003
24	Static Gradient (kPa/m)	-4.124e-003	-4.124e-003	3.937e-002	3.937e-002
25	Accel. Gradient (kPa/m)	5.774e-007	5.774e-007	5.685e-006	5.685e-006
26	Liq. Reynolds	1.901e+005 *	1.901e+005 *	1.901e+005 *	1.901e+005 *
27	Vap. Reynolds	9.779e+006 *	9.779e+006 *	9.780e+006 *	9.780e+006 *
28	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
29	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	1.417e+004	1.417e+004	1.418e+004	1.418e+004
33	Elevation (m)	-0.2120	-0.1260	-4.000e-002	-5.200e-002
34	Pressure (bar)	70.42	70.42	70.42	70.42
35	Temperature (C)	52.46	52.46	52.45	52.45
36	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
37	Heat Transferred (kJ/h-m)	598.3	598.2	598.1	598.1
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.2816	0.2815	0.2815	0.1713
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	7.802e-003	7.802e-003	7.802e-003	8.915e-003
43	Static Gradient (kPa/m)	3.937e-002	3.937e-002	3.937e-002	-3.850e-003
44	Accel. Gradient (kPa/m)	5.686e-006	5.686e-006	5.686e-006	6.104e-007
45	Liq. Reynolds	1.901e+005 *	1.901e+005 *	1.901e+005 *	1.901e+005 *
46	Vap. Reynolds	9.780e+006 *	9.780e+006 *	9.780e+006 *	9.780e+006 *
47	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
48	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	1.419e+004	1.419e+004	1.420e+004	1.420e+004
52	Elevation (m)	-6.400e-002	-7.600e-002	-8.800e-002	-0.1000
53	Pressure (bar)	70.42	70.42	70.42	70.41
54	Temperature (C)	52.45	52.45	52.45	52.44
55	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
56	Heat Transferred (kJ/h-m)	598.1	598.0	598.0	598.0
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.1713	0.1713	0.1713	0.1713
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	8.915e-003	8.915e-003	8.915e-003	8.915e-003
62	Static Gradient (kPa/m)	-3.850e-003	-3.850e-003	-3.850e-003	-3.850e-003
63	Accel. Gradient (kPa/m)	6.104e-007	6.104e-007	6.104e-007	6.104e-007
64	Liq. Reynolds	1.901e+005 *	1.901e+005 *	1.901e+005 *	1.901e+005 *
65	Vap. Reynolds	9.780e+006 *	9.780e+006 *	9.780e+006 *	9.780e+006 *
66	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
67	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
68	Deposit Thickness (mm)	---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 264 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	1.421e+004	1.421e+004	1.422e+004	1.422e+004		
15	Elevation (m)	-7.600e-002	-5.200e-002	-2.800e-002	-4.000e-003		
16	Pressure (bar)	70.41	70.41	70.41	70.41		
17	Temperature (C)	52.44	52.44	52.44	52.44		
18	Vapour Fraction	0.8819	0.8819	0.8819	0.8819		
19	Heat Transferred (kJ/h-m)	597.8	597.8	597.7	597.7		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	0.2469	0.2469	0.2469	0.2469		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	6.640e-003	6.640e-003	6.640e-003	6.640e-003		
25	Static Gradient (kPa/m)	9.955e-003	9.955e-003	9.955e-003	9.955e-003		
26	Accel. Gradient (kPa/m)	2.000e-006	2.000e-006	2.000e-006	2.000e-006		
27	Liq. Reynolds	1.901e+005 *	1.901e+005 *	1.901e+005 *	1.901e+005 *		
28	Vap. Reynolds	9.780e+006 *	9.780e+006 *	9.780e+006 *	9.780e+006 *		
29	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
30	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	1.423e+004	1.423e+004	1.424e+004	1.424e+004		
34	Elevation (m)	2.000e-002	4.600e-002	7.200e-002	9.800e-002		
35	Pressure (bar)	70.41	70.41	70.41	70.41		
36	Temperature (C)	52.43	52.43	52.43	52.43		
37	Vapour Fraction	0.8819	0.8819	0.8819	0.8819		
38	Heat Transferred (kJ/h-m)	597.6	597.6	597.5	597.5		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.2469	0.2480	0.2480	0.2480		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	6.640e-003	6.523e-003	6.523e-003	6.523e-003		
44	Static Gradient (kPa/m)	9.955e-003	1.078e-002	1.078e-002	1.078e-002		
45	Accel. Gradient (kPa/m)	2.000e-006	2.085e-006	2.085e-006	2.085e-006		
46	Liq. Reynolds	1.901e+005 *	1.901e+005 *	1.901e+005 *	1.901e+005 *		
47	Vap. Reynolds	9.780e+006 *	9.780e+006 *	9.780e+006 *	9.780e+006 *		
48	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
49	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	1.425e+004	1.425e+004	1.426e+004	1.426e+004		
53	Elevation (m)	0.1240	0.1500	0.1540	0.1580		
54	Pressure (bar)	70.41	70.41	70.41	70.41		
55	Temperature (C)	52.43	52.42	52.42	52.42		
56	Vapour Fraction	0.8819	0.8819	0.8819	0.8819		
57	Heat Transferred (kJ/h-m)	597.4	597.4	597.3	597.3		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.2480	0.2480	0.2357	0.2357		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	6.523e-003	6.523e-003	7.973e-003	7.973e-003		
63	Static Gradient (kPa/m)	1.078e-002	1.078e-002	1.604e-003	1.604e-003		
64	Accel. Gradient (kPa/m)	2.085e-006	2.085e-006	1.155e-006	1.155e-006		
65	Liq. Reynolds	1.901e+005 *	1.901e+005 *	1.901e+005 *	1.901e+005 *		
66	Vap. Reynolds	9.780e+006 *	9.780e+006 *	9.780e+006 *	9.780e+006 *		
67	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
68	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878		
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 265 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	1.427e+004	1.427e+004	1.428e+004	1.428e+004
16	Elevation (m)	0.1620	0.1660	0.1700	0.1740
17	Pressure (bar)	70.40	70.40	70.40	70.40
18	Temperature (C)	52.42	52.42	52.41	52.41
19	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
20	Heat Transferred (kJ/h-m)	597.2	597.2	597.1	597.1
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2357	0.2357	0.2357	0.2358
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	7.973e-003	7.973e-003	7.973e-003	7.973e-003
26	Static Gradient (kPa/m)	1.604e-003	1.604e-003	1.604e-003	1.604e-003
27	Accel. Gradient (kPa/m)	1.155e-006	1.155e-006	1.155e-006	1.155e-006
28	Liq. Reynolds	1.901e+005 *	1.901e+005 *	1.901e+005 *	1.901e+005 *
29	Vap. Reynolds	9.780e+006 *	9.780e+006 *	9.780e+006 *	9.780e+006 *
30	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
31	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	1.429e+004	1.429e+004	1.430e+004	1.430e+004
35	Elevation (m)	0.1780	0.1820	0.1860	0.1900
36	Pressure (bar)	70.40	70.40	70.40	70.40
37	Temperature (C)	52.41	52.41	52.41	52.41
38	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
39	Heat Transferred (kJ/h-m)	597.0	597.0	596.9	596.9
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2358	0.2358	0.2358	0.2358
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	7.973e-003	7.973e-003	7.973e-003	7.973e-003
45	Static Gradient (kPa/m)	1.604e-003	1.604e-003	1.604e-003	1.604e-003
46	Accel. Gradient (kPa/m)	1.155e-006	1.155e-006	1.155e-006	1.155e-006
47	Liq. Reynolds	1.901e+005 *	1.901e+005 *	1.900e+005 *	1.900e+005 *
48	Vap. Reynolds	9.780e+006 *	9.781e+006 *	9.781e+006 *	9.781e+006 *
49	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
50	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	1.431e+004	1.431e+004	1.432e+004	1.432e+004
54	Elevation (m)	0.1920	0.1940	0.1960	0.1980
55	Pressure (bar)	70.40	70.40	70.40	70.40
56	Temperature (C)	52.40	52.40	52.40	52.40
57	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
58	Heat Transferred (kJ/h-m)	596.8	596.8	596.7	596.7
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2346	0.2346	0.2346	0.2346
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	8.117e-003	8.117e-003	8.117e-003	8.117e-003
64	Static Gradient (kPa/m)	7.991e-004	7.991e-004	7.991e-004	7.991e-004
65	Accel. Gradient (kPa/m)	1.075e-006	1.075e-006	1.075e-006	1.075e-006
66	Liq. Reynolds	1.900e+005 *	1.900e+005 *	1.900e+005 *	1.900e+005 *
67	Vap. Reynolds	9.781e+006 *	9.781e+006 *	9.781e+006 *	9.781e+006 *
68	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 266 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
14	Deposit Thickness (mm)	---	---	---	---
15	Deposit Volume (m3)	---	---	---	---
16	Length (m)	1.433e+004	1.433e+004	1.434e+004	1.434e+004
17	Elevation (m)	0.2000	0.1820	0.1640	0.1460
18	Pressure (bar)	70.40	70.40	70.40	70.40
19	Temperature (C)	52.40	52.39	52.39	52.39
20	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
21	Heat Transferred (kJ/h-m)	596.6	596.6	596.6	596.5
22	Flow Regime	Transition	Transition	Transition	Transition
23	Water Flow Regime				
24	Liquid HoldUp	0.2346	0.1704	0.1704	0.1704
25	Water HoldUp	---	---	---	---
26	Friction Gradient (kPa/m)	8.117e-003	8.927e-003	8.927e-003	8.927e-003
27	Static Gradient (kPa/m)	7.991e-004	-5.756e-003	-5.756e-003	-5.756e-003
28	Accel. Gradient (kPa/m)	1.075e-006	3.823e-007	3.823e-007	3.823e-007
29	Liq. Reynolds	1.900e+005 *	1.900e+005 *	1.900e+005 *	1.900e+005 *
30	Vap. Reynolds	9.781e+006 *	9.781e+006 *	9.781e+006 *	9.781e+006 *
31	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
32	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
33	Deposit Thickness (mm)	---	---	---	---
34	Deposit Volume (m3)	---	---	---	---
35	Length (m)	1.435e+004	1.435e+004	1.436e+004	1.436e+004
36	Elevation (m)	0.1280	0.1100	0.1160	0.1220
37	Pressure (bar)	70.40	70.40	70.40	70.40
38	Temperature (C)	52.39	52.39	52.39	52.38
39	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
40	Heat Transferred (kJ/h-m)	596.5	596.4	596.3	596.3
41	Flow Regime	Transition	Transition	Transition	Transition
42	Water Flow Regime				
43	Liquid HoldUp	0.1704	0.1704	0.2369	0.2369
44	Water HoldUp	---	---	---	---
45	Friction Gradient (kPa/m)	8.927e-003	8.927e-003	7.831e-003	7.831e-003
46	Static Gradient (kPa/m)	-5.756e-003	-5.756e-003	2.414e-003	2.414e-003
47	Accel. Gradient (kPa/m)	3.823e-007	3.823e-007	1.235e-006	1.235e-006
48	Liq. Reynolds	1.900e+005 *	1.900e+005 *	1.900e+005 *	1.900e+005 *
49	Vap. Reynolds	9.781e+006 *	9.781e+006 *	9.781e+006 *	9.781e+006 *
50	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
51	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
52	Deposit Thickness (mm)	---	---	---	---
53	Deposit Volume (m3)	---	---	---	---
54	Length (m)	1.437e+004	1.437e+004	1.438e+004	1.438e+004
55	Elevation (m)	0.1280	0.1340	0.1400	0.1400
56	Pressure (bar)	70.40	70.40	70.40	70.40
57	Temperature (C)	52.38	52.38	52.38	52.38
58	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
59	Heat Transferred (kJ/h-m)	596.2	596.2	596.1	596.1
60	Flow Regime	Transition	Transition	Transition	Transition
61	Water Flow Regime				
62	Liquid HoldUp	0.2369	0.2369	0.2369	0.2527
63	Water HoldUp	---	---	---	---
64	Friction Gradient (kPa/m)	7.831e-003	7.831e-003	7.831e-003	6.016e-003
65	Static Gradient (kPa/m)	2.414e-003	2.414e-003	2.414e-003	0.0000
66	Accel. Gradient (kPa/m)	1.235e-006	1.235e-006	1.235e-006	7.253e-007
67	Liq. Reynolds	1.900e+005 *	1.900e+005 *	1.900e+005 *	1.900e+005 *
68	Vap. Reynolds	9.781e+006 *	9.781e+006 *	9.781e+006 *	9.781e+006 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
14	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
15	Deposit Thickness (mm)	---	---	---	---
16	Deposit Volume (m3)	---	---	---	---
17	Length (m)	1.439e+004	1.439e+004	1.440e+004	1.440e+004
18	Elevation (m)	0.1400	0.1400	0.1400	0.1400
19	Pressure (bar)	70.40	70.40	70.39	70.39
20	Temperature (C)	52.37	52.37	52.37	52.37
21	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
22	Heat Transferred (kJ/h-m)	596.1	596.0	596.0	595.9
23	Flow Regime	Transition	Transition	Transition	Transition
24	Water Flow Regime				
25	Liquid HoldUp	0.2527	0.2527	0.2527	0.2527
26	Water HoldUp	---	---	---	---
27	Friction Gradient (kPa/m)	6.016e-003	6.016e-003	6.016e-003	6.016e-003
28	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
29	Accel. Gradient (kPa/m)	7.253e-007	7.253e-007	7.253e-007	7.253e-007
30	Liq. Reynolds	1.900e+005 *	1.900e+005 *	1.900e+005 *	1.900e+005 *
31	Vap. Reynolds	9.781e+006 *	9.781e+006 *	9.781e+006 *	9.781e+006 *
32	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
33	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
34	Deposit Thickness (mm)	---	---	---	---
35	Deposit Volume (m3)	---	---	---	---
36	Length (m)	1.441e+004	1.441e+004	1.442e+004	1.442e+004
37	Elevation (m)	0.1520	0.1640	0.1760	0.1880
38	Pressure (bar)	70.39	70.39	70.39	70.39
39	Temperature (C)	52.37	52.37	52.36	52.36
40	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
41	Heat Transferred (kJ/h-m)	595.9	595.8	595.8	595.7
42	Flow Regime	Transition	Transition	Transition	Transition
43	Water Flow Regime				
44	Liquid HoldUp	0.2402	0.2402	0.2402	0.2402
45	Water HoldUp	---	---	---	---
46	Friction Gradient (kPa/m)	7.417e-003	7.417e-003	7.417e-003	7.417e-003
47	Static Gradient (kPa/m)	4.878e-003	4.878e-003	4.878e-003	4.878e-003
48	Accel. Gradient (kPa/m)	1.482e-006	1.482e-006	1.482e-006	1.482e-006
49	Liq. Reynolds	1.900e+005 *	1.900e+005 *	1.900e+005 *	1.900e+005 *
50	Vap. Reynolds	9.781e+006 *	9.781e+006 *	9.781e+006 *	9.781e+006 *
51	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
52	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
53	Deposit Thickness (mm)	---	---	---	---
54	Deposit Volume (m3)	---	---	---	---
55	Length (m)	1.443e+004	1.443e+004	1.444e+004	1.444e+004
56	Elevation (m)	0.2000	0.2000	0.2000	0.2000
57	Pressure (bar)	70.39	70.39	70.39	70.39
58	Temperature (C)	52.36	52.36	52.36	52.35
59	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
60	Heat Transferred (kJ/h-m)	595.7	595.6	595.6	595.5
61	Flow Regime	Transition	Transition	Transition	Transition
62	Water Flow Regime				
63	Liquid HoldUp	0.2402	0.2527	0.2527	0.2527
64	Water HoldUp	---	---	---	---
65	Friction Gradient (kPa/m)	7.417e-003	6.016e-003	6.016e-003	6.016e-003
66	Static Gradient (kPa/m)	4.878e-003	0.0000	0.0000	0.0000
67	Accel. Gradient (kPa/m)	1.482e-006	7.253e-007	7.253e-007	7.253e-007
68	Liq. Reynolds	1.900e+005 *	1.900e+005 *	1.900e+005 *	1.900e+005 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.781e+006 *	9.781e+006 *	9.781e+006 *	9.781e+006 *
14	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
15	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	1.445e+004	1.445e+004	1.446e+004	1.446e+004
19	Elevation (m)	0.2000	0.2000	0.2000	0.2000
20	Pressure (bar)	70.39	70.39	70.39	70.39
21	Temperature (C)	52.35	52.35	52.35	52.35
22	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
23	Heat Transferred (kJ/h-m)	595.5	595.4	595.4	595.3
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2527	0.2527	0.2527	0.2527
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.016e-003	6.016e-003	6.016e-003	6.016e-003
29	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
30	Accel. Gradient (kPa/m)	7.253e-007	7.253e-007	7.253e-007	7.253e-007
31	Liq. Reynolds	1.900e+005 *	1.900e+005 *	1.900e+005 *	1.900e+005 *
32	Vap. Reynolds	9.781e+006 *	9.781e+006 *	9.781e+006 *	9.781e+006 *
33	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
34	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	1.447e+004	1.447e+004	1.448e+004	1.448e+004
38	Elevation (m)	0.2000	0.2000	0.2000	0.2000
39	Pressure (bar)	70.39	70.39	70.39	70.39
40	Temperature (C)	52.35	52.34	52.34	52.34
41	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
42	Heat Transferred (kJ/h-m)	595.3	595.2	595.2	595.1
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2527	0.2527	0.2527	0.2527
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.016e-003	6.016e-003	6.016e-003	6.016e-003
48	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
49	Accel. Gradient (kPa/m)	7.253e-007	7.253e-007	7.253e-007	7.253e-007
50	Liq. Reynolds	1.900e+005 *	1.900e+005 *	1.900e+005 *	1.900e+005 *
51	Vap. Reynolds	9.781e+006 *	9.781e+006 *	9.781e+006 *	9.781e+006 *
52	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
53	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1.449e+004	1.449e+004	1.450e+004	1.450e+004
57	Elevation (m)	0.2000	0.2000	0.2000	0.2000
58	Pressure (bar)	70.39	70.39	70.39	70.39
59	Temperature (C)	52.34	52.34	52.33	52.33
60	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
61	Heat Transferred (kJ/h-m)	595.1	595.0	595.0	594.9
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2527	0.2527	0.2527	0.2527
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	6.016e-003	6.016e-003	6.016e-003	6.016e-003
67	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
68	Accel. Gradient (kPa/m)	7.253e-007	7.253e-007	7.253e-007	7.253e-007
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	1.900e+005 *	1.900e+005 *	1.900e+005 *	1.900e+005 *
14	Vap. Reynolds	9.781e+006 *	9.781e+006 *	9.781e+006 *	9.781e+006 *
15	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
16	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	1.451e+004	1.451e+004	1.452e+004	1.452e+004
20	Elevation (m)	0.2020	0.2040	0.2060	0.2080
21	Pressure (bar)	70.39	70.39	70.39	70.39
22	Temperature (C)	52.33	52.33	52.33	52.33
23	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
24	Heat Transferred (kJ/h-m)	594.9	594.8	594.8	594.7
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2347	0.2347	0.2347	0.2347
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	8.116e-003	8.116e-003	8.116e-003	8.116e-003
30	Static Gradient (kPa/m)	7.992e-004	7.992e-004	7.992e-004	7.992e-004
31	Accel. Gradient (kPa/m)	1.075e-006	1.075e-006	1.075e-006	1.075e-006
32	Liq. Reynolds	1.900e+005 *	1.900e+005 *	1.900e+005 *	1.900e+005 *
33	Vap. Reynolds	9.781e+006 *	9.781e+006 *	9.781e+006 *	9.781e+006 *
34	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
35	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	1.453e+004	1.453e+004	1.454e+004	1.454e+004
39	Elevation (m)	0.2100	0.2080	0.2060	0.2040
40	Pressure (bar)	70.38	70.38	70.38	70.38
41	Temperature (C)	52.32	52.32	52.32	52.32
42	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
43	Heat Transferred (kJ/h-m)	594.7	594.7	594.6	594.6
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2347	0.1728	0.1728	0.1728
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	8.116e-003	8.894e-003	8.894e-003	8.894e-003
49	Static Gradient (kPa/m)	7.992e-004	-6.456e-004	-6.456e-004	-6.456e-004
50	Accel. Gradient (kPa/m)	1.075e-006	9.946e-007	9.946e-007	9.946e-007
51	Liq. Reynolds	1.900e+005 *	1.900e+005 *	1.900e+005 *	1.900e+005 *
52	Vap. Reynolds	9.781e+006 *	9.781e+006 *	9.781e+006 *	9.781e+006 *
53	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
54	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	1.455e+004	1.455e+004	1.456e+004	1.456e+004
58	Elevation (m)	0.2020	0.2000	0.1940	0.1880
59	Pressure (bar)	70.38	70.38	70.38	70.38
60	Temperature (C)	52.32	52.31	52.31	52.31
61	Vapour Fraction	0.8819	0.8819	0.8819	0.8819
62	Heat Transferred (kJ/h-m)	594.5	594.5	594.4	594.4
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.1728	0.1728	0.1722	0.1722
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	8.894e-003	8.894e-003	8.902e-003	8.902e-003
68	Static Gradient (kPa/m)	-6.456e-004	-6.456e-004	-1.932e-003	-1.932e-003
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Accel. Gradient	(kPa/m)	9.946e-007	9.946e-007	8.404e-007	8.404e-007				
14	Liq. Reynolds		1.900e+005 *	1.900e+005 *	1.900e+005 *	1.900e+005 *				
15	Vap. Reynolds		9.781e+006 *	9.781e+006 *	9.782e+006 *	9.782e+006 *				
16	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
17	Vapour Velocity	(m/s)	2.878	2.878	2.878	2.878				
18	Deposit Thickness	(mm)	---	---	---	---				
19	Deposit Volume	(m3)	---	---	---	---				
20	Length	(m)	1.457e+004	1.457e+004	1.458e+004	1.458e+004				
21	Elevation	(m)	0.1820	0.1760	0.1700	0.1680				
22	Pressure	(bar)	70.38	70.38	70.38	70.38				
23	Temperature	(C)	52.31	52.31	52.31	52.30				
24	Vapour Fraction		0.8819	0.8819	0.8819	0.8819				
25	Heat Transferred	(kJ/h-m)	594.3	594.3	594.2	594.2				
26	Flow Regime		Transition	Transition	Transition	Transition				
27	Water Flow Regime									
28	Liquid HoldUp		0.1722	0.1722	0.1722	0.1728				
29	Water HoldUp		---	---	---	---				
30	Friction Gradient	(kPa/m)	8.902e-003	8.902e-003	8.902e-003	8.894e-003				
31	Static Gradient	(kPa/m)	-1.932e-003	-1.932e-003	-1.932e-003	-6.456e-004				
32	Accel. Gradient	(kPa/m)	8.404e-007	8.404e-007	8.404e-007	9.946e-007				
33	Liq. Reynolds		1.900e+005 *	1.900e+005 *	1.899e+005 *	1.899e+005 *				
34	Vap. Reynolds		9.782e+006 *	9.782e+006 *	9.782e+006 *	9.782e+006 *				
35	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
36	Vapour Velocity	(m/s)	2.878	2.878	2.878	2.878				
37	Deposit Thickness	(mm)	---	---	---	---				
38	Deposit Volume	(m3)	---	---	---	---				
39	Length	(m)	1.459e+004	1.459e+004	1.460e+004	1.460e+004				
40	Elevation	(m)	0.1660	0.1640	0.1620	0.1600				
41	Pressure	(bar)	70.38	70.38	70.38	70.38				
42	Temperature	(C)	52.30	52.30	52.30	52.30				
43	Vapour Fraction		0.8819	0.8819	0.8819	0.8819				
44	Heat Transferred	(kJ/h-m)	594.2	594.1	594.1	594.0				
45	Flow Regime		Transition	Transition	Transition	Transition				
46	Water Flow Regime									
47	Liquid HoldUp		0.1728	0.1728	0.1728	0.1728				
48	Water HoldUp		---	---	---	---				
49	Friction Gradient	(kPa/m)	8.894e-003	8.894e-003	8.894e-003	8.894e-003				
50	Static Gradient	(kPa/m)	-6.456e-004	-6.456e-004	-6.456e-004	-6.456e-004				
51	Accel. Gradient	(kPa/m)	9.946e-007	9.946e-007	9.946e-007	9.946e-007				
52	Liq. Reynolds		1.899e+005 *	1.899e+005 *	1.899e+005 *	1.899e+005 *				
53	Vap. Reynolds		9.782e+006 *	9.782e+006 *	9.782e+006 *	9.782e+006 *				
54	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
55	Vapour Velocity	(m/s)	2.878	2.878	2.878	2.878				
56	Deposit Thickness	(mm)	---	---	---	---				
57	Deposit Volume	(m3)	---	---	---	---				
58	Length	(m)	1.461e+004	1.461e+004	1.462e+004	1.462e+004				
59	Elevation	(m)	0.1620	0.1640	0.1660	0.1680				
60	Pressure	(bar)	70.38	70.38	70.38	70.38				
61	Temperature	(C)	52.29	52.29	52.29	52.29				
62	Vapour Fraction		0.8819	0.8819	0.8819	0.8819				
63	Heat Transferred	(kJ/h-m)	593.9	593.9	593.8	593.8				
64	Flow Regime		Transition	Transition	Transition	Transition				
65	Water Flow Regime									
66	Liquid HoldUp		0.2347	0.2347	0.2347	0.2347				
67	Water HoldUp		---	---	---	---				
68	Friction Gradient	(kPa/m)	8.115e-003	8.115e-003	8.115e-003	8.115e-003				
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 271 of 937				

1				Case Name:	Simulation - 386MMSCFD_80barg.hsc	
2				Unit Set:	Mixed	
3		Epic Atlantic Limited				
4				Date/Time:	Mon Jun 13 15:13:43 2022	
5						
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	7.992e-004	7.993e-004	7.993e-004	7.993e-004
14	Accel. Gradient	(kPa/m)	1.075e-006	1.075e-006	1.075e-006	1.075e-006
15	Liq. Reynolds		1.899e+005 *	1.899e+005 *	1.899e+005 *	1.899e+005 *
16	Vap. Reynolds		9.782e+006 *	9.782e+006 *	9.782e+006 *	9.782e+006 *
17	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
18	Vapour Velocity	(m/s)	2.878	2.878	2.878	2.878
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	1.463e+004	1.463e+004	1.464e+004	1.464e+004
22	Elevation	(m)	0.1700	0.1640	0.1580	0.1520
23	Pressure	(bar)	70.38	70.38	70.38	70.38
24	Temperature	(C)	52.29	52.29	52.28	52.28
25	Vapour Fraction		0.8819	0.8819	0.8818	0.8818
26	Heat Transferred	(kJ/h-m)	593.7	593.7	593.7	593.6
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2347	0.1722	0.1722	0.1722
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	8.115e-003	8.902e-003	8.902e-003	8.902e-003
32	Static Gradient	(kPa/m)	7.993e-004	-1.969e-003	-1.969e-003	-1.969e-003
33	Accel. Gradient	(kPa/m)	1.075e-006	8.359e-007	8.359e-007	8.359e-007
34	Liq. Reynolds		1.899e+005 *	1.899e+005 *	1.899e+005 *	1.899e+005 *
35	Vap. Reynolds		9.782e+006 *	9.782e+006 *	9.782e+006 *	9.782e+006 *
36	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
37	Vapour Velocity	(m/s)	2.878	2.878	2.878	2.878
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	1.465e+004	1.465e+004	1.466e+004	1.466e+004
41	Elevation	(m)	0.1460	0.1400	0.1480	0.1560
42	Pressure	(bar)	70.38	70.37	70.37	70.37
43	Temperature	(C)	52.28	52.28	52.28	52.27
44	Vapour Fraction		0.8818	0.8818	0.8818	0.8818
45	Heat Transferred	(kJ/h-m)	593.6	593.5	593.4	593.4
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.1722	0.1722	0.2380	0.2380
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	8.902e-003	8.902e-003	7.689e-003	7.689e-003
51	Static Gradient	(kPa/m)	-1.969e-003	-1.969e-003	3.230e-003	3.230e-003
52	Accel. Gradient	(kPa/m)	8.359e-007	8.359e-007	1.317e-006	1.317e-006
53	Liq. Reynolds		1.899e+005 *	1.899e+005 *	1.899e+005 *	1.899e+005 *
54	Vap. Reynolds		9.782e+006 *	9.782e+006 *	9.782e+006 *	9.782e+006 *
55	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
56	Vapour Velocity	(m/s)	2.878	2.878	2.878	2.878
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	1.467e+004	1.467e+004	1.468e+004	1.468e+004
60	Elevation	(m)	0.1640	0.1720	0.1800	0.1660
61	Pressure	(bar)	70.37	70.37	70.37	70.37
62	Temperature	(C)	52.27	52.27	52.27	52.27
63	Vapour Fraction		0.8818	0.8818	0.8818	0.8818
64	Heat Transferred	(kJ/h-m)	593.3	593.3	593.2	593.2
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2380	0.2380	0.2380	0.1710
68	Water HoldUp		---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 272 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	7.689e-003	7.689e-003	7.689e-003	8.917e-003	
14	Static Gradient	(kPa/m)	3.230e-003	3.230e-003	3.230e-003	-4.488e-003	
15	Accel. Gradient	(kPa/m)	1.317e-006	1.317e-006	1.317e-006	5.341e-007	
16	Liq. Reynolds		1.899e+005 *	1.899e+005 *	1.899e+005 *	1.899e+005 *	
17	Vap. Reynolds		9.782e+006 *	9.782e+006 *	9.782e+006 *	9.782e+006 *	
18	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
19	Vapour Velocity	(m/s)	2.878	2.878	2.878	2.878	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	1.469e+004	1.469e+004	1.470e+004	1.470e+004	
23	Elevation	(m)	0.1520	0.1380	0.1240	0.1100	
24	Pressure	(bar)	70.37	70.37	70.37	70.37	
25	Temperature	(C)	52.27	52.26	52.26	52.26	
26	Vapour Fraction		0.8818	0.8818	0.8818	0.8818	
27	Heat Transferred	(kJ/h-m)	593.2	593.1	593.1	593.0	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.1710	0.1710	0.1710	0.1710	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	8.917e-003	8.917e-003	8.917e-003	8.917e-003	
33	Static Gradient	(kPa/m)	-4.488e-003	-4.488e-003	-4.488e-003	-4.488e-003	
34	Accel. Gradient	(kPa/m)	5.341e-007	5.341e-007	5.341e-007	5.341e-007	
35	Liq. Reynolds		1.899e+005 *	1.899e+005 *	1.899e+005 *	1.899e+005 *	
36	Vap. Reynolds		9.782e+006 *	9.782e+006 *	9.782e+006 *	9.782e+006 *	
37	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
38	Vapour Velocity	(m/s)	2.878	2.878	2.878	2.878	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	1.471e+004	1.471e+004	1.472e+004	1.472e+004	
42	Elevation	(m)	0.1120	0.1140	0.1160	0.1180	
43	Pressure	(bar)	70.37	70.37	70.37	70.37	
44	Temperature	(C)	52.26	52.26	52.25	52.25	
45	Vapour Fraction		0.8818	0.8818	0.8818	0.8818	
46	Heat Transferred	(kJ/h-m)	592.9	592.9	592.9	592.8	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.2347	0.2347	0.2347	0.2347	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	8.115e-003	8.115e-003	8.115e-003	8.115e-003	
52	Static Gradient	(kPa/m)	7.993e-004	7.993e-004	7.993e-004	7.993e-004	
53	Accel. Gradient	(kPa/m)	1.075e-006	1.075e-006	1.075e-006	1.075e-006	
54	Liq. Reynolds		1.899e+005 *	1.899e+005 *	1.899e+005 *	1.899e+005 *	
55	Vap. Reynolds		9.782e+006 *	9.782e+006 *	9.782e+006 *	9.782e+006 *	
56	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
57	Vapour Velocity	(m/s)	2.878	2.878	2.878	2.878	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	1.473e+004	1.473e+004	1.474e+004	1.474e+004	
61	Elevation	(m)	0.1200	0.1140	0.1080	0.1020	
62	Pressure	(bar)	70.37	70.37	70.37	70.37	
63	Temperature	(C)	52.25	52.25	52.25	52.25	
64	Vapour Fraction		0.8818	0.8818	0.8818	0.8818	
65	Heat Transferred	(kJ/h-m)	592.8	592.8	592.7	592.7	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.2347	0.1722	0.1722	0.1723	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 273 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	8.115e-003	8.901e-003	8.901e-003	8.901e-003
15	Static Gradient (kPa/m)	7.993e-004	-1.932e-003	-1.932e-003	-1.932e-003
16	Accel. Gradient (kPa/m)	1.075e-006	8.403e-007	8.403e-007	8.403e-007
17	Liq. Reynolds	1.899e+005 *	1.899e+005 *	1.899e+005 *	1.899e+005 *
18	Vap. Reynolds	9.782e+006 *	9.782e+006 *	9.782e+006 *	9.782e+006 *
19	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
20	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	1.475e+004	1.475e+004	1.476e+004	1.476e+004
24	Elevation (m)	9.600e-002	9.000e-002	8.600e-002	8.200e-002
25	Pressure (bar)	70.37	70.37	70.37	70.37
26	Temperature (C)	52.24	52.24	52.24	52.24
27	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
28	Heat Transferred (kJ/h-m)	592.6	592.6	592.5	592.5
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.1723	0.1723	0.1726	0.1726
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	8.901e-003	8.901e-003	8.897e-003	8.897e-003
34	Static Gradient (kPa/m)	-1.932e-003	-1.932e-003	-1.290e-003	-1.290e-003
35	Accel. Gradient (kPa/m)	8.403e-007	8.403e-007	9.173e-007	9.173e-007
36	Liq. Reynolds	1.899e+005 *	1.899e+005 *	1.899e+005 *	1.899e+005 *
37	Vap. Reynolds	9.782e+006 *	9.782e+006 *	9.782e+006 *	9.782e+006 *
38	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
39	Vapour Velocity (m/s)	2.878	2.878	2.878	2.877
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	1.477e+004	1.477e+004	1.478e+004	1.478e+004
43	Elevation (m)	7.800e-002	7.400e-002	7.000e-002	7.000e-002
44	Pressure (bar)	70.37	70.37	70.37	70.36
45	Temperature (C)	52.24	52.24	52.23	52.23
46	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
47	Heat Transferred (kJ/h-m)	592.4	592.4	592.3	592.2
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.1726	0.1726	0.1726	0.2528
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	8.897e-003	8.897e-003	8.897e-003	6.014e-003
53	Static Gradient (kPa/m)	-1.290e-003	-1.290e-003	-1.290e-003	0.0000
54	Accel. Gradient (kPa/m)	9.173e-007	9.173e-007	9.173e-007	7.253e-007
55	Liq. Reynolds	1.899e+005 *	1.899e+005 *	1.899e+005 *	1.899e+005 *
56	Vap. Reynolds	9.782e+006 *	9.782e+006 *	9.782e+006 *	9.782e+006 *
57	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
58	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	1.479e+004	1.479e+004	1.480e+004	1.480e+004
62	Elevation (m)	7.000e-002	7.000e-002	7.000e-002	7.000e-002
63	Pressure (bar)	70.36	70.36	70.36	70.36
64	Temperature (C)	52.23	52.23	52.23	52.22
65	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
66	Heat Transferred (kJ/h-m)	592.2	592.1	592.1	592.0
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 274 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.2528	0.2528	0.2528	0.2528
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	6.014e-003	6.014e-003	6.014e-003	6.014e-003
16	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
17	Accel. Gradient (kPa/m)	7.253e-007	7.253e-007	7.252e-007	7.252e-007
18	Liq. Reynolds	1.899e+005 *	1.899e+005 *	1.899e+005 *	1.899e+005 *
19	Vap. Reynolds	9.782e+006 *	9.782e+006 *	9.782e+006 *	9.782e+006 *
20	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
21	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	1.481e+004	1.481e+004	1.482e+004	1.482e+004
25	Elevation (m)	6.600e-002	6.200e-002	5.800e-002	5.400e-002
26	Pressure (bar)	70.36	70.36	70.36	70.36
27	Temperature (C)	52.22	52.22	52.22	52.22
28	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
29	Heat Transferred (kJ/h-m)	592.0	592.0	591.9	591.9
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1726	0.1726	0.1726	0.1726
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	8.897e-003	8.897e-003	8.897e-003	8.897e-003
35	Static Gradient (kPa/m)	-1.290e-003	-1.290e-003	-1.290e-003	-1.290e-003
36	Accel. Gradient (kPa/m)	9.173e-007	9.173e-007	9.173e-007	9.173e-007
37	Liq. Reynolds	1.899e+005 *	1.899e+005 *	1.899e+005 *	1.899e+005 *
38	Vap. Reynolds	9.782e+006 *	9.782e+006 *	9.782e+006 *	9.782e+006 *
39	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
40	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	1.483e+004	1.483e+004	1.484e+004	1.484e+004
44	Elevation (m)	5.000e-002	4.600e-002	4.200e-002	3.800e-002
45	Pressure (bar)	70.36	70.36	70.36	70.36
46	Temperature (C)	52.22	52.21	52.21	52.21
47	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
48	Heat Transferred (kJ/h-m)	591.8	591.8	591.8	591.7
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1726	0.1726	0.1726	0.1726
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	8.897e-003	8.897e-003	8.897e-003	8.897e-003
54	Static Gradient (kPa/m)	-1.290e-003	-1.281e-003	-1.281e-003	-1.281e-003
55	Accel. Gradient (kPa/m)	9.173e-007	9.184e-007	9.184e-007	9.184e-007
56	Liq. Reynolds	1.899e+005 *	1.899e+005 *	1.899e+005 *	1.899e+005 *
57	Vap. Reynolds	9.783e+006 *	9.783e+006 *	9.783e+006 *	9.783e+006 *
58	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
59	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	1.485e+004	1.485e+004	1.486e+004	1.486e+004
63	Elevation (m)	3.400e-002	3.000e-002	2.800e-002	2.600e-002
64	Pressure (bar)	70.36	70.36	70.36	70.36
65	Temperature (C)	52.21	52.21	52.21	52.20
66	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
67	Heat Transferred (kJ/h-m)	591.7	591.6	591.6	591.6
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 275 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1726	0.1726	0.1724	0.1724
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	8.897e-003	8.897e-003	8.899e-003	8.899e-003
17	Static Gradient (kPa/m)	-1.281e-003	-1.281e-003	-1.611e-003	-1.611e-003
18	Accel. Gradient (kPa/m)	9.184e-007	9.184e-007	8.788e-007	8.788e-007
19	Liq. Reynolds	1.899e+005 *	1.899e+005 *	1.899e+005 *	1.899e+005 *
20	Vap. Reynolds	9.783e+006 *	9.783e+006 *	9.783e+006 *	9.783e+006 *
21	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
22	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	1.486e+004	1.486e+004	1.486e+004	1.488e+004
26	Elevation (m)	2.400e-002	2.200e-002	2.000e-002	2.200e-002
27	Pressure (bar)	70.36	70.36	70.36	70.36
28	Temperature (C)	52.20	52.20	52.20	52.20
29	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
30	Heat Transferred (kJ/h-m)	591.5	591.5	591.5	591.3
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1724	0.1724	0.1724	0.2339
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	8.899e-003	8.899e-003	8.899e-003	8.223e-003
36	Static Gradient (kPa/m)	-1.611e-003	-1.611e-003	-1.611e-003	1.993e-004
37	Accel. Gradient (kPa/m)	8.788e-007	8.788e-007	8.788e-007	1.016e-006
38	Liq. Reynolds	1.898e+005 *	1.898e+005 *	1.898e+005 *	1.898e+005 *
39	Vap. Reynolds	9.783e+006 *	9.783e+006 *	9.783e+006 *	9.783e+006 *
40	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
41	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	1.490e+004	1.492e+004	1.494e+004	1.496e+004
45	Elevation (m)	2.400e-002	2.600e-002	2.800e-002	3.000e-002
46	Pressure (bar)	70.36	70.35	70.35	70.35
47	Temperature (C)	52.19	52.18	52.17	52.17
48	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
49	Heat Transferred (kJ/h-m)	591.1	590.9	590.7	590.5
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2339	0.2339	0.2339	0.2339
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	8.223e-003	8.223e-003	8.223e-003	8.223e-003
55	Static Gradient (kPa/m)	1.993e-004	1.993e-004	1.993e-004	1.993e-004
56	Accel. Gradient (kPa/m)	1.016e-006	1.016e-006	1.016e-006	1.016e-006
57	Liq. Reynolds	1.898e+005 *	1.898e+005 *	1.898e+005 *	1.898e+005 *
58	Vap. Reynolds	9.783e+006 *	9.783e+006 *	9.783e+006 *	9.783e+006 *
59	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
60	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	1.498e+004	1.500e+004	1.502e+004	1.504e+004
64	Elevation (m)	1.600e-002	2.000e-003	-1.200e-002	-2.600e-002
65	Pressure (bar)	70.35	70.35	70.35	70.34
66	Temperature (C)	52.16	52.15	52.14	52.14
67	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
68	Heat Transferred (kJ/h-m)	590.4	590.2	590.0	589.8
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.1726	0.1726	0.1726	0.1726
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	8.895e-003	8.895e-003	8.895e-003	8.895e-003
18	Static Gradient (kPa/m)	-1.129e-003	-1.129e-003	-1.129e-003	-1.129e-003
19	Accel. Gradient (kPa/m)	9.366e-007	9.366e-007	9.366e-007	9.366e-007
20	Liq. Reynolds	1.898e+005 *	1.898e+005 *	1.898e+005 *	1.898e+005 *
21	Vap. Reynolds	9.783e+006 *	9.783e+006 *	9.783e+006 *	9.783e+006 *
22	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
23	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	1.506e+004	1.508e+004	1.510e+004	1.512e+004
27	Elevation (m)	-4.000e-002	-3.400e-002	-2.800e-002	-2.200e-002
28	Pressure (bar)	70.34	70.34	70.34	70.34
29	Temperature (C)	52.13	52.12	52.12	52.11
30	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
31	Heat Transferred (kJ/h-m)	589.6	589.3	589.2	589.0
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1726	0.2344	0.2344	0.2344
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	8.895e-003	8.149e-003	8.149e-003	8.149e-003
37	Static Gradient (kPa/m)	-1.129e-003	5.991e-004	5.991e-004	5.991e-004
38	Accel. Gradient (kPa/m)	9.366e-007	1.055e-006	1.055e-006	1.055e-006
39	Liq. Reynolds	1.898e+005 *	1.898e+005 *	1.898e+005 *	1.898e+005 *
40	Vap. Reynolds	9.783e+006 *	9.784e+006 *	9.784e+006 *	9.784e+006 *
41	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
42	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	1.514e+004	1.516e+004	1.518e+004	1.520e+004
46	Elevation (m)	-1.600e-002	-1.000e-002	-2.000e-002	-3.000e-002
47	Pressure (bar)	70.34	70.33	70.33	70.33
48	Temperature (C)	52.10	52.09	52.09	52.08
49	Vapour Fraction	0.8818	0.8818	0.8818	0.8818
50	Heat Transferred (kJ/h-m)	588.8	588.6	588.4	588.2
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2344	0.2344	0.1728	0.1728
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	8.149e-003	8.149e-003	8.893e-003	8.893e-003
56	Static Gradient (kPa/m)	5.991e-004	5.991e-004	-8.070e-004	-8.070e-004
57	Accel. Gradient (kPa/m)	1.055e-006	1.055e-006	9.753e-007	9.753e-007
58	Liq. Reynolds	1.897e+005 *	1.897e+005 *	1.897e+005 *	1.897e+005 *
59	Vap. Reynolds	9.784e+006 *	9.784e+006 *	9.784e+006 *	9.784e+006 *
60	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
61	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	1.522e+004	1.524e+004	1.526e+004	1.528e+004
65	Elevation (m)	-4.000e-002	-5.000e-002	-6.000e-002	-0.1240
66	Pressure (bar)	70.33	70.33	70.33	70.33
67	Temperature (C)	52.07	52.07	52.06	52.05
68	Vapour Fraction	0.8817	0.8817	0.8817	0.8817
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	588.0	587.9	587.7	587.5
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.1728	0.1728	0.1728	0.1708
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	8.893e-003	8.893e-003	8.892e-003	8.919e-003
19	Static Gradient (kPa/m)	-8.070e-004	-8.070e-004	-8.070e-004	-5.125e-003
20	Accel. Gradient (kPa/m)	9.753e-007	9.753e-007	9.753e-007	4.576e-007
21	Liq. Reynolds	1.897e+005 *	1.897e+005 *	1.897e+005 *	1.897e+005 *
22	Vap. Reynolds	9.784e+006 *	9.784e+006 *	9.784e+006 *	9.784e+006 *
23	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
24	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	1.530e+004	1.532e+004	1.534e+004	1.536e+004
28	Elevation (m)	-0.1880	-0.2520	-0.3160	-0.3800
29	Pressure (bar)	70.32	70.32	70.32	70.32
30	Temperature (C)	52.04	52.04	52.03	52.02
31	Vapour Fraction	0.8817	0.8817	0.8817	0.8817
32	Heat Transferred (kJ/h-m)	587.3	587.1	586.9	586.8
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.1708	0.1708	0.1708	0.1708
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	8.919e-003	8.918e-003	8.918e-003	8.918e-003
38	Static Gradient (kPa/m)	-5.125e-003	-5.125e-003	-5.125e-003	-5.125e-003
39	Accel. Gradient (kPa/m)	4.576e-007	4.575e-007	4.575e-007	4.574e-007
40	Liq. Reynolds	1.897e+005 *	1.897e+005 *	1.897e+005 *	1.897e+005 *
41	Vap. Reynolds	9.784e+006 *	9.784e+006 *	9.784e+006 *	9.784e+006 *
42	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
43	Vapour Velocity (m/s)	2.877	2.876	2.876	2.876
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	1.538e+004	1.540e+004	1.542e+004	1.544e+004
47	Elevation (m)	-1.014	-1.648	-2.282	-2.916
48	Pressure (bar)	70.33	70.34	70.34	70.35
49	Temperature (C)	52.02	52.02	52.02	52.01
50	Vapour Fraction	0.8817	0.8817	0.8817	0.8817
51	Heat Transferred (kJ/h-m)	586.7	586.6	586.6	586.5
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.1495	0.1495	0.1495	0.1495
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	9.257e-003	9.256e-003	9.255e-003	9.254e-003
57	Static Gradient (kPa/m)	-4.657e-002	-4.658e-002	-4.658e-002	-4.659e-002
58	Accel. Gradient (kPa/m)	-4.500e-006	-4.499e-006	-4.499e-006	-4.498e-006
59	Liq. Reynolds	1.897e+005 *	1.897e+005 *	1.897e+005 *	1.897e+005 *
60	Vap. Reynolds	9.784e+006 *	9.784e+006 *	9.784e+006 *	9.784e+006 *
61	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
62	Vapour Velocity (m/s)	2.876	2.876	2.875	2.875
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	1.546e+004	1.548e+004	1.550e+004	1.552e+004
66	Elevation (m)	-3.550	-2.914	-2.278	-1.642
67	Pressure (bar)	70.36	70.34	70.32	70.31
68	Temperature (C)	52.01	52.00	51.98	51.97
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8817	0.8817	0.8817	0.8817
14	Heat Transferred (kJ/h-m)	586.4	586.0	585.6	585.3
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.1495	0.3224	0.3224	0.3224
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	9.253e-003	7.742e-003	7.743e-003	7.745e-003
20	Static Gradient (kPa/m)	-4.659e-002	8.091e-002	8.090e-002	8.089e-002
21	Accel. Gradient (kPa/m)	-4.498e-006	1.068e-005	1.069e-005	1.069e-005
22	Liq. Reynolds	1.897e+005 *	1.897e+005 *	1.896e+005 *	1.896e+005 *
23	Vap. Reynolds	9.783e+006 *	9.784e+006 *	9.785e+006 *	9.785e+006 *
24	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
25	Vapour Velocity (m/s)	2.874	2.875	2.876	2.876
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	1.554e+004	1.556e+004	1.558e+004	1.560e+004
29	Elevation (m)	-1.006	-0.3700	-0.2920	-0.2140
30	Pressure (bar)	70.29	70.27	70.27	70.26
31	Temperature (C)	51.96	51.94	51.94	51.93
32	Vapour Fraction	0.8817	0.8817	0.8817	0.8817
33	Heat Transferred (kJ/h-m)	585.0	584.6	584.4	584.2
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.3223	0.3223	0.2445	0.2445
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	7.746e-003	7.748e-003	6.923e-003	6.923e-003
39	Static Gradient (kPa/m)	8.088e-002	8.087e-002	8.032e-003	8.032e-003
40	Accel. Gradient (kPa/m)	1.070e-005	1.070e-005	1.806e-006	1.806e-006
41	Liq. Reynolds	1.896e+005 *	1.896e+005 *	1.895e+005 *	1.895e+005 *
42	Vap. Reynolds	9.786e+006 *	9.786e+006 *	9.787e+006 *	9.787e+006 *
43	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
44	Vapour Velocity (m/s)	2.877	2.878	2.878	2.878
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	1.562e+004	1.564e+004	1.566e+004	1.568e+004
48	Elevation (m)	-0.1360	-5.800e-002	2.000e-002	-2.000e-003
49	Pressure (bar)	70.26	70.26	70.26	70.25
50	Temperature (C)	51.92	51.91	51.91	51.90
51	Vapour Fraction	0.8817	0.8817	0.8817	0.8817
52	Heat Transferred (kJ/h-m)	584.0	583.8	583.6	583.4
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2445	0.2445	0.2445	0.1723
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	6.924e-003	6.924e-003	6.924e-003	8.901e-003
58	Static Gradient (kPa/m)	8.032e-003	8.032e-003	8.032e-003	-1.772e-003
59	Accel. Gradient (kPa/m)	1.806e-006	1.806e-006	1.806e-006	8.611e-007
60	Liq. Reynolds	1.895e+005 *	1.895e+005 *	1.895e+005 *	1.895e+005 *
61	Vap. Reynolds	9.787e+006 *	9.787e+006 *	9.787e+006 *	9.787e+006 *
62	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
63	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	1.570e+004	1.572e+004	1.574e+004	1.576e+004
67	Elevation (m)	-2.400e-002	-4.600e-002	-6.800e-002	-9.000e-002
68	Pressure (bar)	70.25	70.25	70.25	70.25
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	51.89	51.88	51.88	51.87
14	Vapour Fraction	0.8817	0.8817	0.8817	0.8817
15	Heat Transferred (kJ/h-m)	583.2	583.0	582.9	582.7
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.1723	0.1723	0.1723	0.1723
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	8.901e-003	8.901e-003	8.901e-003	8.901e-003
21	Static Gradient (kPa/m)	-1.772e-003	-1.772e-003	-1.772e-003	-1.772e-003
22	Accel. Gradient (kPa/m)	8.611e-007	8.611e-007	8.611e-007	8.611e-007
23	Liq. Reynolds	1.895e+005 *	1.895e+005 *	1.895e+005 *	1.895e+005 *
24	Vap. Reynolds	9.787e+006 *	9.787e+006 *	9.787e+006 *	9.787e+006 *
25	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
26	Vapour Velocity (m/s)	2.878	2.878	2.878	2.878
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	1.577e+004	1.577e+004	1.578e+004	1.578e+004
30	Elevation (m)	-9.000e-002	-9.000e-002	-9.000e-002	-9.000e-002
31	Pressure (bar)	70.25	70.25	70.25	70.25
32	Temperature (C)	51.87	51.87	51.87	51.86
33	Vapour Fraction	0.8817	0.8817	0.8817	0.8817
34	Heat Transferred (kJ/h-m)	582.6	582.5	582.5	582.4
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2528	0.2528	0.2528	0.2528
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	6.016e-003	6.016e-003	6.016e-003	6.016e-003
40	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
41	Accel. Gradient (kPa/m)	7.266e-007	7.266e-007	7.266e-007	7.266e-007
42	Liq. Reynolds	1.895e+005 *	1.895e+005 *	1.895e+005 *	1.895e+005 *
43	Vap. Reynolds	9.787e+006 *	9.787e+006 *	9.787e+006 *	9.787e+006 *
44	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
45	Vapour Velocity (m/s)	2.878	2.877	2.877	2.877
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	1.579e+004	1.581e+004	1.583e+004	1.585e+004
49	Elevation (m)	-9.000e-002	-6.800e-002	-4.600e-002	-2.400e-002
50	Pressure (bar)	70.25	70.25	70.24	70.24
51	Temperature (C)	51.86	51.85	51.85	51.84
52	Vapour Fraction	0.8817	0.8817	0.8817	0.8817
53	Heat Transferred (kJ/h-m)	582.4	582.2	582.0	581.8
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2528	0.2367	0.2367	0.2367
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	6.016e-003	7.865e-003	7.865e-003	7.865e-003
59	Static Gradient (kPa/m)	0.0000	2.212e-003	2.212e-003	2.212e-003
60	Accel. Gradient (kPa/m)	7.266e-007	1.217e-006	1.217e-006	1.217e-006
61	Liq. Reynolds	1.895e+005 *	1.895e+005 *	1.895e+005 *	1.894e+005 *
62	Vap. Reynolds	9.788e+006 *	9.788e+006 *	9.788e+006 *	9.788e+006 *
63	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
64	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	1.587e+004	1.589e+004	1.589e+004	1.590e+004
68	Elevation (m)	-2.000e-003	2.000e-002	3.600e-002	5.200e-002
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 280 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Pressure (bar)	70.24	70.24	70.24	70.24
14	Temperature (C)	51.83	51.83	51.82	51.82
15	Vapour Fraction	0.8817	0.8817	0.8817	0.8817
16	Heat Transferred (kJ/h-m)	581.6	581.4	581.4	581.3
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.2367	0.2367	0.2461	0.2461
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	7.865e-003	7.865e-003	6.742e-003	6.742e-003
22	Static Gradient (kPa/m)	2.212e-003	2.212e-003	9.255e-003	9.255e-003
23	Accel. Gradient (kPa/m)	1.217e-006	1.217e-006	1.932e-006	1.932e-006
24	Liq. Reynolds	1.894e+005 *	1.894e+005 *	1.894e+005 *	1.894e+005 *
25	Vap. Reynolds	9.788e+006 *	9.788e+006 *	9.788e+006 *	9.788e+006 *
26	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
27	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	1.590e+004	1.590e+004	1.591e+004	1.591e+004
31	Elevation (m)	6.800e-002	8.400e-002	1.000e-001	0.1020
32	Pressure (bar)	70.24	70.23	70.23	70.23
33	Temperature (C)	51.82	51.82	51.82	51.82
34	Vapour Fraction	0.8816	0.8816	0.8816	0.8816
35	Heat Transferred (kJ/h-m)	581.3	581.3	581.2	581.2
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.2461	0.2461	0.2461	0.2350
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	6.742e-003	6.742e-003	6.742e-003	8.080e-003
41	Static Gradient (kPa/m)	9.255e-003	9.255e-003	9.255e-003	1.000e-003
42	Accel. Gradient (kPa/m)	1.932e-006	1.932e-006	1.933e-006	1.097e-006
43	Liq. Reynolds	1.894e+005 *	1.894e+005 *	1.894e+005 *	1.894e+005 *
44	Vap. Reynolds	9.788e+006 *	9.788e+006 *	9.788e+006 *	9.788e+006 *
45	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
46	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	1.592e+004	1.592e+004	1.592e+004	1.593e+004
50	Elevation (m)	0.1040	0.1060	0.1080	0.1100
51	Pressure (bar)	70.23	70.23	70.23	70.23
52	Temperature (C)	51.82	51.81	51.81	51.81
53	Vapour Fraction	0.8816	0.8816	0.8816	0.8816
54	Heat Transferred (kJ/h-m)	581.1	581.1	581.1	581.0
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.2350	0.2350	0.2350	0.2350
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	8.080e-003	8.080e-003	8.080e-003	8.080e-003
60	Static Gradient (kPa/m)	1.000e-003	1.000e-003	1.000e-003	1.000e-003
61	Accel. Gradient (kPa/m)	1.097e-006	1.097e-006	1.097e-006	1.097e-006
62	Liq. Reynolds	1.894e+005 *	1.894e+005 *	1.894e+005 *	1.894e+005 *
63	Vap. Reynolds	9.788e+006 *	9.788e+006 *	9.788e+006 *	9.788e+006 *
64	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
65	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	1.593e+004	1.593e+004	1.593e+004	1.594e+004
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Elevation (m)	0.1220	0.1340	0.1460	0.1580
14	Pressure (bar)	70.23	70.23	70.23	70.23
15	Temperature (C)	51.81	51.81	51.81	51.81
16	Vapour Fraction	0.8816	0.8816	0.8816	0.8816
17	Heat Transferred (kJ/h-m)	581.0	581.0	581.0	580.9
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2503	0.2503	0.2503	0.2503
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	6.274e-003	6.274e-003	6.274e-003	6.274e-003
23	Static Gradient (kPa/m)	1.258e-002	1.258e-002	1.258e-002	1.258e-002
24	Accel. Gradient (kPa/m)	2.277e-006	2.277e-006	2.277e-006	2.277e-006
25	Liq. Reynolds	1.894e+005 *	1.894e+005 *	1.894e+005 *	1.894e+005 *
26	Vap. Reynolds	9.788e+006 *	9.788e+006 *	9.788e+006 *	9.788e+006 *
27	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
28	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	1.594e+004	1.594e+004	1.595e+004	1.595e+004
32	Elevation (m)	0.1700	0.1720	0.1740	0.1760
33	Pressure (bar)	70.23	70.23	70.23	70.23
34	Temperature (C)	51.81	51.81	51.80	51.80
35	Vapour Fraction	0.8816	0.8816	0.8816	0.8816
36	Heat Transferred (kJ/h-m)	580.9	580.9	580.8	580.8
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2503	0.2350	0.2350	0.2350
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	6.274e-003	8.080e-003	8.080e-003	8.080e-003
42	Static Gradient (kPa/m)	1.258e-002	1.000e-003	1.000e-003	1.000e-003
43	Accel. Gradient (kPa/m)	2.277e-006	1.097e-006	1.097e-006	1.097e-006
44	Liq. Reynolds	1.894e+005 *	1.894e+005 *	1.894e+005 *	1.894e+005 *
45	Vap. Reynolds	9.788e+006 *	9.788e+006 *	9.788e+006 *	9.788e+006 *
46	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
47	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	1.595e+004	1.596e+004	1.596e+004	1.597e+004
51	Elevation (m)	0.1780	0.1800	0.1840	0.1880
52	Pressure (bar)	70.23	70.23	70.23	70.23
53	Temperature (C)	51.80	51.80	51.80	51.80
54	Vapour Fraction	0.8816	0.8816	0.8816	0.8816
55	Heat Transferred (kJ/h-m)	580.8	580.7	580.7	580.6
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2350	0.2350	0.2358	0.2358
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	8.080e-003	8.080e-003	7.972e-003	7.972e-003
61	Static Gradient (kPa/m)	1.000e-003	1.000e-003	1.605e-003	1.605e-003
62	Accel. Gradient (kPa/m)	1.097e-006	1.097e-006	1.157e-006	1.157e-006
63	Liq. Reynolds	1.894e+005 *	1.894e+005 *	1.894e+005 *	1.894e+005 *
64	Vap. Reynolds	9.788e+006 *	9.788e+006 *	9.788e+006 *	9.788e+006 *
65	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
66	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 282 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3							
4			Unit Set: Mixed				
5			Date/Time: Mon Jun 13 15:13:43 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Length (m)	1.597e+004	1.598e+004	1.598e+004	1.599e+004		
14	Elevation (m)	0.1920	0.1960	0.2000	0.2060		
15	Pressure (bar)	70.23	70.23	70.23	70.23		
16	Temperature (C)	51.79	51.79	51.79	51.79		
17	Vapour Fraction	0.8816	0.8816	0.8816	0.8816		
18	Heat Transferred (kJ/h-m)	580.6	580.5	580.5	580.4		
19	Flow Regime	Transition	Transition	Transition	Transition		
20	Water Flow Regime						
21	Liquid HoldUp	0.2358	0.2358	0.2358	0.2369		
22	Water HoldUp	---	---	---	---		
23	Friction Gradient (kPa/m)	7.972e-003	7.972e-003	7.972e-003	7.830e-003		
24	Static Gradient (kPa/m)	1.605e-003	1.605e-003	1.605e-003	2.415e-003		
25	Accel. Gradient (kPa/m)	1.157e-006	1.157e-006	1.157e-006	1.238e-006		
26	Liq. Reynolds	1.894e+005 *	1.894e+005 *	1.894e+005 *	1.894e+005 *		
27	Vap. Reynolds	9.788e+006 *	9.788e+006 *	9.788e+006 *	9.788e+006 *		
28	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
29	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877		
30	Deposit Thickness (mm)	---	---	---	---		
31	Deposit Volume (m3)	---	---	---	---		
32	Length (m)	1.599e+004	1.600e+004	1.600e+004	1.601e+004		
33	Elevation (m)	0.2120	0.2180	0.2240	0.2300		
34	Pressure (bar)	70.23	70.22	70.22	70.22		
35	Temperature (C)	51.79	51.79	51.78	51.78		
36	Vapour Fraction	0.8816	0.8816	0.8816	0.8816		
37	Heat Transferred (kJ/h-m)	580.4	580.3	580.3	580.2		
38	Flow Regime	Transition	Transition	Transition	Transition		
39	Water Flow Regime						
40	Liquid HoldUp	0.2369	0.2369	0.2369	0.2369		
41	Water HoldUp	---	---	---	---		
42	Friction Gradient (kPa/m)	7.830e-003	7.830e-003	7.830e-003	7.830e-003		
43	Static Gradient (kPa/m)	2.415e-003	2.415e-003	2.415e-003	2.415e-003		
44	Accel. Gradient (kPa/m)	1.238e-006	1.238e-006	1.238e-006	1.238e-006		
45	Liq. Reynolds	1.894e+005 *	1.894e+005 *	1.894e+005 *	1.894e+005 *		
46	Vap. Reynolds	9.788e+006 *	9.789e+006 *	9.789e+006 *	9.789e+006 *		
47	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
48	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877		
49	Deposit Thickness (mm)	---	---	---	---		
50	Deposit Volume (m3)	---	---	---	---		
51	Length (m)	1.601e+004	1.602e+004	1.602e+004	1.603e+004		
52	Elevation (m)	0.2120	0.1940	0.1760	0.1580		
53	Pressure (bar)	70.22	70.22	70.22	70.22		
54	Temperature (C)	51.78	51.78	51.78	51.77		
55	Vapour Fraction	0.8816	0.8816	0.8816	0.8816		
56	Heat Transferred (kJ/h-m)	580.3	580.2	580.2	580.1		
57	Flow Regime	Transition	Transition	Transition	Transition		
58	Water Flow Regime						
59	Liquid HoldUp	0.1705	0.1705	0.1705	0.1705		
60	Water HoldUp	---	---	---	---		
61	Friction Gradient (kPa/m)	8.926e-003	8.925e-003	8.925e-003	8.925e-003		
62	Static Gradient (kPa/m)	-5.759e-003	-5.759e-003	-5.759e-003	-5.759e-003		
63	Accel. Gradient (kPa/m)	3.826e-007	3.826e-007	3.826e-007	3.826e-007		
64	Liq. Reynolds	1.894e+005 *	1.894e+005 *	1.894e+005 *	1.894e+005 *		
65	Vap. Reynolds	9.789e+006 *	9.789e+006 *	9.789e+006 *	9.789e+006 *		
66	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
67	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877		
68	Deposit Thickness (mm)	---	---	---	---		
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 283 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	1.603e+004	1.604e+004	1.604e+004	1.605e+004		
15	Elevation (m)	0.1400	0.1260	0.1120	9.800e-002		
16	Pressure (bar)	70.22	70.22	70.22	70.22		
17	Temperature (C)	51.77	51.77	51.77	51.77		
18	Vapour Fraction	0.8816	0.8816	0.8816	0.8816		
19	Heat Transferred (kJ/h-m)	580.1	580.0	580.0	579.9		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	0.1705	0.1711	0.1711	0.1711		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	8.925e-003	8.917e-003	8.917e-003	8.917e-003		
25	Static Gradient (kPa/m)	-5.759e-003	-4.490e-003	-4.490e-003	-4.490e-003		
26	Accel. Gradient (kPa/m)	3.826e-007	5.350e-007	5.350e-007	5.350e-007		
27	Liq. Reynolds	1.894e+005 *	1.894e+005 *	1.894e+005 *	1.894e+005 *		
28	Vap. Reynolds	9.789e+006 *	9.789e+006 *	9.789e+006 *	9.789e+006 *		
29	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
30	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	1.605e+004	1.606e+004	1.606e+004	1.607e+004		
34	Elevation (m)	8.400e-002	7.000e-002	5.200e-002	3.400e-002		
35	Pressure (bar)	70.22	70.22	70.22	70.22		
36	Temperature (C)	51.77	51.76	51.76	51.76		
37	Vapour Fraction	0.8816	0.8816	0.8816	0.8816		
38	Heat Transferred (kJ/h-m)	579.9	579.8	579.8	579.8		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.1711	0.1711	0.1705	0.1705		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	8.917e-003	8.917e-003	8.925e-003	8.925e-003		
44	Static Gradient (kPa/m)	-4.490e-003	-4.490e-003	-5.759e-003	-5.759e-003		
45	Accel. Gradient (kPa/m)	5.349e-007	5.349e-007	3.825e-007	3.825e-007		
46	Liq. Reynolds	1.894e+005 *	1.894e+005 *	1.894e+005 *	1.894e+005 *		
47	Vap. Reynolds	9.789e+006 *	9.789e+006 *	9.789e+006 *	9.789e+006 *		
48	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
49	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	1.607e+004	1.608e+004	1.608e+004	1.609e+004		
53	Elevation (m)	1.600e-002	-2.000e-003	-2.000e-002	-3.400e-002		
54	Pressure (bar)	70.22	70.22	70.22	70.22		
55	Temperature (C)	51.76	51.76	51.76	51.75		
56	Vapour Fraction	0.8816	0.8816	0.8816	0.8816		
57	Heat Transferred (kJ/h-m)	579.7	579.7	579.6	579.6		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.1705	0.1705	0.1705	0.1711		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	8.925e-003	8.925e-003	8.925e-003	8.917e-003		
63	Static Gradient (kPa/m)	-5.759e-003	-5.759e-003	-5.759e-003	-4.476e-003		
64	Accel. Gradient (kPa/m)	3.825e-007	3.825e-007	3.825e-007	5.366e-007		
65	Liq. Reynolds	1.894e+005 *	1.894e+005 *	1.894e+005 *	1.894e+005 *		
66	Vap. Reynolds	9.789e+006 *	9.789e+006 *	9.789e+006 *	9.789e+006 *		
67	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
68	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877		
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 284 of 937		

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Thickness	(mm)	---	---	---	---	
14	Deposit Volume	(m3)	---	---	---	---	
15	Length	(m)	1.609e+004	1.610e+004	1.610e+004	1.611e+004	
16	Elevation	(m)	-4.800e-002	-6.200e-002	-7.600e-002	-9.000e-002	
17	Pressure	(bar)	70.22	70.22	70.22	70.22	
18	Temperature	(C)	51.75	51.75	51.75	51.75	
19	Vapour Fraction		0.8816	0.8816	0.8816	0.8816	
20	Heat Transferred	(kJ/h-m)	579.5	579.5	579.4	579.4	
21	Flow Regime		Transition	Transition	Transition	Transition	
22	Water Flow Regime						
23	Liquid HoldUp		0.1711	0.1711	0.1711	0.1711	
24	Water HoldUp		---	---	---	---	
25	Friction Gradient	(kPa/m)	8.917e-003	8.917e-003	8.917e-003	8.917e-003	
26	Static Gradient	(kPa/m)	-4.476e-003	-4.476e-003	-4.476e-003	-4.476e-003	
27	Accel. Gradient	(kPa/m)	5.366e-007	5.366e-007	5.366e-007	5.366e-007	
28	Liq. Reynolds		1.894e+005 *	1.894e+005 *	1.894e+005 *	1.893e+005 *	
29	Vap. Reynolds		9.789e+006 *	9.789e+006 *	9.789e+006 *	9.789e+006 *	
30	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
31	Vapour Velocity	(m/s)	2.877	2.877	2.877	2.877	
32	Deposit Thickness	(mm)	---	---	---	---	
33	Deposit Volume	(m3)	---	---	---	---	
34	Length	(m)	1.611e+004	1.612e+004	1.612e+004	1.613e+004	
35	Elevation	(m)	-0.1060	-0.1220	-0.1380	-0.1540	
36	Pressure	(bar)	70.22	70.22	70.22	70.22	
37	Temperature	(C)	51.75	51.74	51.74	51.74	
38	Vapour Fraction		0.8816	0.8816	0.8816	0.8816	
39	Heat Transferred	(kJ/h-m)	579.4	579.3	579.3	579.2	
40	Flow Regime		Transition	Transition	Transition	Transition	
41	Water Flow Regime						
42	Liquid HoldUp		0.1708	0.1708	0.1708	0.1708	
43	Water HoldUp		---	---	---	---	
44	Friction Gradient	(kPa/m)	8.921e-003	8.921e-003	8.921e-003	8.921e-003	
45	Static Gradient	(kPa/m)	-5.125e-003	-5.125e-003	-5.125e-003	-5.125e-003	
46	Accel. Gradient	(kPa/m)	4.585e-007	4.585e-007	4.585e-007	4.585e-007	
47	Liq. Reynolds		1.893e+005 *	1.893e+005 *	1.893e+005 *	1.893e+005 *	
48	Vap. Reynolds		9.789e+006 *	9.789e+006 *	9.789e+006 *	9.789e+006 *	
49	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
50	Vapour Velocity	(m/s)	2.877	2.877	2.877	2.877	
51	Deposit Thickness	(mm)	---	---	---	---	
52	Deposit Volume	(m3)	---	---	---	---	
53	Length	(m)	1.613e+004	1.614e+004	1.614e+004	1.615e+004	
54	Elevation	(m)	-0.1700	-0.1920	-0.2140	-0.2360	
55	Pressure	(bar)	70.22	70.22	70.22	70.22	
56	Temperature	(C)	51.74	51.74	51.74	51.73	
57	Vapour Fraction		0.8816	0.8816	0.8816	0.8816	
58	Heat Transferred	(kJ/h-m)	579.2	579.1	579.1	579.0	
59	Flow Regime		Transition	Transition	Transition	Transition	
60	Water Flow Regime						
61	Liquid HoldUp		0.1708	0.1699	0.1699	0.1699	
62	Water HoldUp		---	---	---	---	
63	Friction Gradient	(kPa/m)	8.920e-003	8.932e-003	8.932e-003	8.932e-003	
64	Static Gradient	(kPa/m)	-5.125e-003	-7.023e-003	-7.023e-003	-7.023e-003	
65	Accel. Gradient	(kPa/m)	4.585e-007	2.307e-007	2.307e-007	2.307e-007	
66	Liq. Reynolds		1.893e+005 *	1.893e+005 *	1.893e+005 *	1.893e+005 *	
67	Vap. Reynolds		9.789e+006 *	9.789e+006 *	9.789e+006 *	9.789e+006 *	
68	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.877	2.877	2.877	2.877	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	1.615e+004	1.616e+004	1.616e+004	1.617e+004	
17	Elevation	(m)	-0.2580	-0.2800	-0.2920	-0.3040	
18	Pressure	(bar)	70.22	70.22	70.22	70.22	
19	Temperature	(C)	51.73	51.73	51.73	51.73	
20	Vapour Fraction		0.8816	0.8816	0.8816	0.8816	
21	Heat Transferred	(kJ/h-m)	579.0	579.0	578.9	578.9	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.1699	0.1699	0.1714	0.1714	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	8.932e-003	8.932e-003	8.912e-003	8.912e-003	
27	Static Gradient	(kPa/m)	-7.023e-003	-7.023e-003	-3.853e-003	-3.853e-003	
28	Accel. Gradient	(kPa/m)	2.307e-007	2.307e-007	6.112e-007	6.112e-007	
29	Liq. Reynolds		1.893e+005 *	1.893e+005 *	1.893e+005 *	1.893e+005 *	
30	Vap. Reynolds		9.789e+006 *	9.789e+006 *	9.789e+006 *	9.789e+006 *	
31	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
32	Vapour Velocity	(m/s)	2.877	2.877	2.877	2.877	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	1.617e+004	1.618e+004	1.618e+004	1.619e+004	
36	Elevation	(m)	-0.3160	-0.3280	-0.3400	-0.3640	
37	Pressure	(bar)	70.22	70.22	70.22	70.22	
38	Temperature	(C)	51.73	51.72	51.72	51.72	
39	Vapour Fraction		0.8816	0.8816	0.8816	0.8816	
40	Heat Transferred	(kJ/h-m)	578.8	578.8	578.7	578.7	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.1714	0.1714	0.1714	0.1696	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	8.912e-003	8.912e-003	8.912e-003	8.936e-003	
46	Static Gradient	(kPa/m)	-3.853e-003	-3.853e-003	-3.853e-003	-7.653e-003	
47	Accel. Gradient	(kPa/m)	6.112e-007	6.112e-007	6.112e-007	1.551e-007	
48	Liq. Reynolds		1.893e+005 *	1.893e+005 *	1.893e+005 *	1.893e+005 *	
49	Vap. Reynolds		9.789e+006 *	9.789e+006 *	9.789e+006 *	9.789e+006 *	
50	Liquid Velocity	(m/s)	0.1980	0.1981	0.1981	0.1981	
51	Vapour Velocity	(m/s)	2.877	2.877	2.877	2.877	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	1.619e+004	1.620e+004	1.620e+004	1.621e+004	
55	Elevation	(m)	-0.3880	-0.4120	-0.4360	-0.4600	
56	Pressure	(bar)	70.22	70.22	70.22	70.22	
57	Temperature	(C)	51.72	51.72	51.72	51.71	
58	Vapour Fraction		0.8816	0.8816	0.8816	0.8816	
59	Heat Transferred	(kJ/h-m)	578.6	578.6	578.6	578.5	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.1696	0.1696	0.1696	0.1696	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	8.936e-003	8.936e-003	8.936e-003	8.936e-003	
65	Static Gradient	(kPa/m)	-7.653e-003	-7.653e-003	-7.653e-003	-7.653e-003	
66	Accel. Gradient	(kPa/m)	1.550e-007	1.550e-007	1.550e-007	1.550e-007	
67	Liq. Reynolds		1.893e+005 *	1.893e+005 *	1.893e+005 *	1.893e+005 *	
68	Vap. Reynolds		9.789e+006 *	9.789e+006 *	9.789e+006 *	9.789e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981
14	Vapour Velocity	(m/s)	2.877	2.877	2.877	2.877
15	Deposit Thickness	(mm)	---	---	---	---
16	Deposit Volume	(m3)	---	---	---	---
17	Length	(m)	1.621e+004	1.622e+004	1.622e+004	1.623e+004
18	Elevation	(m)	-0.3360	-0.2120	-8.800e-002	3.600e-002
19	Pressure	(bar)	70.21	70.21	70.21	70.20
20	Temperature	(C)	51.71	51.71	51.71	51.70
21	Vapour Fraction		0.8816	0.8816	0.8816	0.8816
22	Heat Transferred	(kJ/h-m)	578.4	578.3	578.2	578.2
23	Flow Regime		Transition	Transition	Transition	Transition
24	Water Flow Regime					
25	Liquid HoldUp		0.3028	0.3028	0.3028	0.3028
26	Water HoldUp		---	---	---	---
27	Friction Gradient	(kPa/m)	7.786e-003	7.787e-003	7.787e-003	7.787e-003
28	Static Gradient	(kPa/m)	6.008e-002	6.008e-002	6.008e-002	6.008e-002
29	Accel. Gradient	(kPa/m)	8.200e-006	8.201e-006	8.201e-006	8.202e-006
30	Liq. Reynolds		1.893e+005 *	1.893e+005 *	1.893e+005 *	1.893e+005 *
31	Vap. Reynolds		9.789e+006 *	9.789e+006 *	9.789e+006 *	9.790e+006 *
32	Liquid Velocity	(m/s)	0.1981	0.1981	0.1980	0.1980
33	Vapour Velocity	(m/s)	2.877	2.877	2.877	2.877
34	Deposit Thickness	(mm)	---	---	---	---
35	Deposit Volume	(m3)	---	---	---	---
36	Length	(m)	1.623e+004	1.624e+004	1.624e+004	1.625e+004
37	Elevation	(m)	0.1600	0.1560	0.1520	0.1480
38	Pressure	(bar)	70.20	70.20	70.20	70.20
39	Temperature	(C)	51.70	51.70	51.70	51.70
40	Vapour Fraction		0.8816	0.8816	0.8816	0.8816
41	Heat Transferred	(kJ/h-m)	578.1	578.1	578.0	578.0
42	Flow Regime		Transition	Transition	Transition	Transition
43	Water Flow Regime					
44	Liquid HoldUp		0.3028	0.1726	0.1726	0.1726
45	Water HoldUp		---	---	---	---
46	Friction Gradient	(kPa/m)	7.788e-003	8.898e-003	8.898e-003	8.898e-003
47	Static Gradient	(kPa/m)	6.008e-002	-1.290e-003	-1.290e-003	-1.290e-003
48	Accel. Gradient	(kPa/m)	8.202e-006	9.195e-007	9.195e-007	9.195e-007
49	Liq. Reynolds		1.893e+005 *	1.893e+005 *	1.893e+005 *	1.893e+005 *
50	Vap. Reynolds		9.790e+006 *	9.790e+006 *	9.790e+006 *	9.790e+006 *
51	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
52	Vapour Velocity	(m/s)	2.877	2.877	2.877	2.877
53	Deposit Thickness	(mm)	---	---	---	---
54	Deposit Volume	(m3)	---	---	---	---
55	Length	(m)	1.625e+004	1.626e+004	1.626e+004	1.627e+004
56	Elevation	(m)	0.1440	0.1400	0.1360	0.1320
57	Pressure	(bar)	70.20	70.20	70.20	70.20
58	Temperature	(C)	51.69	51.69	51.69	51.69
59	Vapour Fraction		0.8816	0.8816	0.8816	0.8816
60	Heat Transferred	(kJ/h-m)	577.9	577.9	577.8	577.8
61	Flow Regime		Transition	Transition	Transition	Transition
62	Water Flow Regime					
63	Liquid HoldUp		0.1726	0.1726	0.1726	0.1726
64	Water HoldUp		---	---	---	---
65	Friction Gradient	(kPa/m)	8.898e-003	8.898e-003	8.898e-003	8.898e-003
66	Static Gradient	(kPa/m)	-1.290e-003	-1.290e-003	-1.290e-003	-1.290e-003
67	Accel. Gradient	(kPa/m)	9.195e-007	9.195e-007	9.195e-007	9.195e-007
68	Liq. Reynolds		1.893e+005 *	1.893e+005 *	1.893e+005 *	1.893e+005 *
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.790e+006 *	9.790e+006 *	9.790e+006 *	9.790e+006 *
14	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
15	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	1.627e+004	1.628e+004	1.628e+004	1.629e+004
19	Elevation (m)	0.1280	0.1240	0.1200	0.1060
20	Pressure (bar)	70.20	70.20	70.20	70.20
21	Temperature (C)	51.69	51.68	51.68	51.68
22	Vapour Fraction	0.8816	0.8816	0.8816	0.8816
23	Heat Transferred (kJ/h-m)	577.8	577.7	577.7	577.6
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1726	0.1726	0.1726	0.1711
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	8.898e-003	8.898e-003	8.898e-003	8.917e-003
29	Static Gradient (kPa/m)	-1.290e-003	-1.290e-003	-1.290e-003	-4.490e-003
30	Accel. Gradient (kPa/m)	9.195e-007	9.195e-007	9.195e-007	5.351e-007
31	Liq. Reynolds	1.893e+005 *	1.893e+005 *	1.893e+005 *	1.893e+005 *
32	Vap. Reynolds	9.790e+006 *	9.790e+006 *	9.790e+006 *	9.790e+006 *
33	Liquid Velocity (m/s)	0.1980	0.1981	0.1981	0.1981
34	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	1.629e+004	1.630e+004	1.630e+004	1.631e+004
38	Elevation (m)	9.200e-002	7.800e-002	6.400e-002	5.000e-002
39	Pressure (bar)	70.20	70.20	70.20	70.20
40	Temperature (C)	51.68	51.68	51.68	51.67
41	Vapour Fraction	0.8816	0.8816	0.8816	0.8816
42	Heat Transferred (kJ/h-m)	577.6	577.5	577.5	577.4
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1711	0.1711	0.1711	0.1711
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	8.917e-003	8.917e-003	8.917e-003	8.917e-003
48	Static Gradient (kPa/m)	-4.490e-003	-4.490e-003	-4.490e-003	-4.490e-003
49	Accel. Gradient (kPa/m)	5.351e-007	5.351e-007	5.351e-007	5.351e-007
50	Liq. Reynolds	1.893e+005 *	1.893e+005 *	1.893e+005 *	1.893e+005 *
51	Vap. Reynolds	9.790e+006 *	9.790e+006 *	9.790e+006 *	9.790e+006 *
52	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
53	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1.631e+004	1.631e+004	1.631e+004	1.632e+004
57	Elevation (m)	-4.200e-002	-0.1340	-0.2260	-0.3180
58	Pressure (bar)	70.20	70.20	70.20	70.20
59	Temperature (C)	51.67	51.67	51.67	51.67
60	Vapour Fraction	0.8816	0.8816	0.8816	0.8816
61	Heat Transferred (kJ/h-m)	577.5	577.5	577.5	577.5
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1388	0.1388	0.1388	0.1388
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	9.484e-003	9.484e-003	9.484e-003	9.484e-003
67	Static Gradient (kPa/m)	-6.453e-002	-6.453e-002	-6.453e-002	-6.453e-002
68	Accel. Gradient (kPa/m)	-6.653e-006	-6.653e-006	-6.653e-006	-6.652e-006
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 288 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		1.893e+005 *	1.893e+005 *	1.893e+005 *	1.893e+005 *	
14	Vap. Reynolds		9.790e+006 *	9.790e+006 *	9.790e+006 *	9.790e+006 *	
15	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
16	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877		
17	Deposit Thickness (mm)	---	---	---	---		
18	Deposit Volume (m3)	---	---	---	---		
19	Length (m)	1.632e+004	1.632e+004	1.632e+004	1.633e+004		
20	Elevation (m)	-0.4100	-0.3280	-0.2460	-0.1640		
21	Pressure (bar)	70.20	70.20	70.20	70.19		
22	Temperature (C)	51.67	51.67	51.67	51.67		
23	Vapour Fraction	0.8816	0.8816	0.8816	0.8816		
24	Heat Transferred (kJ/h-m)	577.5	577.3	577.3	577.2		
25	Flow Regime	Transition	Transition	Transition	Transition		
26	Water Flow Regime						
27	Liquid HoldUp	0.1388	0.3099	0.3099	0.3099		
28	Water HoldUp	---	---	---	---		
29	Friction Gradient (kPa/m)	9.484e-003	7.773e-003	7.774e-003	7.774e-003		
30	Static Gradient (kPa/m)	-6.453e-002	6.742e-002	6.742e-002	6.742e-002		
31	Accel. Gradient (kPa/m)	-6.652e-006	9.087e-006	9.088e-006	9.088e-006		
32	Liq. Reynolds		1.893e+005 *	1.893e+005 *	1.893e+005 *		
33	Vap. Reynolds		9.790e+006 *	9.790e+006 *	9.790e+006 *		
34	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
35	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877		
36	Deposit Thickness (mm)	---	---	---	---		
37	Deposit Volume (m3)	---	---	---	---		
38	Length (m)	1.633e+004	1.633e+004	1.634e+004	1.634e+004		
39	Elevation (m)	-8.200e-002	-1.110e-016	8.000e-003	1.600e-002		
40	Pressure (bar)	70.19	70.19	70.19	70.19		
41	Temperature (C)	51.67	51.67	51.66	51.66		
42	Vapour Fraction	0.8816	0.8816	0.8816	0.8816		
43	Heat Transferred (kJ/h-m)	577.2	577.1	577.1	577.0		
44	Flow Regime	Transition	Transition	Transition	Transition		
45	Water Flow Regime						
46	Liquid HoldUp	0.3099	0.3099	0.2381	0.2381		
47	Water HoldUp	---	---	---	---		
48	Friction Gradient (kPa/m)	7.774e-003	7.774e-003	7.690e-003	7.690e-003		
49	Static Gradient (kPa/m)	6.742e-002	6.742e-002	3.232e-003	3.232e-003		
50	Accel. Gradient (kPa/m)	9.089e-006	9.089e-006	1.320e-006	1.320e-006		
51	Liq. Reynolds		1.893e+005 *	1.893e+005 *	1.892e+005 *		
52	Vap. Reynolds		9.790e+006 *	9.790e+006 *	9.790e+006 *		
53	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
54	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877		
55	Deposit Thickness (mm)	---	---	---	---		
56	Deposit Volume (m3)	---	---	---	---		
57	Length (m)	1.635e+004	1.635e+004	1.636e+004	1.636e+004		
58	Elevation (m)	2.400e-002	3.200e-002	4.000e-002	4.600e-002		
59	Pressure (bar)	70.19	70.19	70.19	70.19		
60	Temperature (C)	51.66	51.66	51.66	51.65		
61	Vapour Fraction	0.8816	0.8816	0.8816	0.8816		
62	Heat Transferred (kJ/h-m)	577.0	576.9	576.9	576.8		
63	Flow Regime	Transition	Transition	Transition	Transition		
64	Water Flow Regime						
65	Liquid HoldUp	0.2381	0.2381	0.2381	0.2369		
66	Water HoldUp	---	---	---	---		
67	Friction Gradient (kPa/m)	7.690e-003	7.690e-003	7.690e-003	7.830e-003		
68	Static Gradient (kPa/m)	3.232e-003	3.232e-003	3.232e-003	2.416e-003		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3							
4			Unit Set: Mixed				
5			Date/Time: Mon Jun 13 15:13:43 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Accel. Gradient (kPa/m)	1.320e-006	1.320e-006	1.320e-006	1.239e-006		
14	Liq. Reynolds	1.892e+005 *	1.892e+005 *	1.892e+005 *	1.892e+005 *		
15	Vap. Reynolds	9.790e+006 *	9.790e+006 *	9.790e+006 *	9.790e+006 *		
16	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
17	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877		
18	Deposit Thickness (mm)	---	---	---	---		
19	Deposit Volume (m3)	---	---	---	---		
20	Length (m)	1.637e+004	1.637e+004	1.638e+004	1.638e+004		
21	Elevation (m)	5.200e-002	5.800e-002	6.400e-002	7.000e-002		
22	Pressure (bar)	70.19	70.19	70.18	70.18		
23	Temperature (C)	51.65	51.65	51.65	51.65		
24	Vapour Fraction	0.8816	0.8816	0.8816	0.8816		
25	Heat Transferred (kJ/h-m)	576.8	576.7	576.7	576.7		
26	Flow Regime	Transition	Transition	Transition	Transition		
27	Water Flow Regime						
28	Liquid HoldUp	0.2369	0.2369	0.2369	0.2369		
29	Water HoldUp	---	---	---	---		
30	Friction Gradient (kPa/m)	7.830e-003	7.830e-003	7.830e-003	7.830e-003		
31	Static Gradient (kPa/m)	2.416e-003	2.416e-003	2.416e-003	2.416e-003		
32	Accel. Gradient (kPa/m)	1.239e-006	1.239e-006	1.239e-006	1.239e-006		
33	Liq. Reynolds	1.892e+005 *	1.892e+005 *	1.892e+005 *	1.892e+005 *		
34	Vap. Reynolds	9.790e+006 *	9.790e+006 *	9.790e+006 *	9.790e+006 *		
35	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
36	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877		
37	Deposit Thickness (mm)	---	---	---	---		
38	Deposit Volume (m3)	---	---	---	---		
39	Length (m)	1.639e+004	1.639e+004	1.640e+004	1.640e+004		
40	Elevation (m)	7.600e-002	8.200e-002	8.800e-002	9.400e-002		
41	Pressure (bar)	70.18	70.18	70.18	70.18		
42	Temperature (C)	51.64	51.64	51.64	51.64		
43	Vapour Fraction	0.8816	0.8816	0.8816	0.8816		
44	Heat Transferred (kJ/h-m)	576.6	576.6	576.5	576.5		
45	Flow Regime	Transition	Transition	Transition	Transition		
46	Water Flow Regime						
47	Liquid HoldUp	0.2369	0.2369	0.2369	0.2369		
48	Water HoldUp	---	---	---	---		
49	Friction Gradient (kPa/m)	7.830e-003	7.830e-003	7.830e-003	7.830e-003		
50	Static Gradient (kPa/m)	2.416e-003	2.416e-003	2.416e-003	2.416e-003		
51	Accel. Gradient (kPa/m)	1.239e-006	1.239e-006	1.239e-006	1.239e-006		
52	Liq. Reynolds	1.892e+005 *	1.892e+005 *	1.892e+005 *	1.892e+005 *		
53	Vap. Reynolds	9.790e+006 *	9.790e+006 *	9.790e+006 *	9.790e+006 *		
54	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
55	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877		
56	Deposit Thickness (mm)	---	---	---	---		
57	Deposit Volume (m3)	---	---	---	---		
58	Length (m)	1.641e+004	1.641e+004	1.642e+004	1.642e+004		
59	Elevation (m)	1.000e-001	0.1080	0.1160	0.1240		
60	Pressure (bar)	70.18	70.18	70.18	70.18		
61	Temperature (C)	51.64	51.64	51.63	51.63		
62	Vapour Fraction	0.8816	0.8816	0.8816	0.8816		
63	Heat Transferred (kJ/h-m)	576.4	576.4	576.3	576.3		
64	Flow Regime	Transition	Transition	Transition	Transition		
65	Water Flow Regime						
66	Liquid HoldUp	0.2369	0.2381	0.2381	0.2381		
67	Water HoldUp	---	---	---	---		
68	Friction Gradient (kPa/m)	7.830e-003	7.690e-003	7.690e-003	7.690e-003		
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1				Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	2.416e-003	3.232e-003	3.232e-003	3.232e-003
14	Accel. Gradient	(kPa/m)	1.239e-006	1.320e-006	1.320e-006	1.320e-006
15	Liq. Reynolds		1.892e+005 *	1.892e+005 *	1.892e+005 *	1.892e+005 *
16	Vap. Reynolds		9.790e+006 *	9.790e+006 *	9.791e+006 *	9.791e+006 *
17	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981
18	Vapour Velocity	(m/s)	2.877	2.877	2.877	2.877
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	1.643e+004	1.643e+004	1.644e+004	1.644e+004
22	Elevation	(m)	0.1320	0.1400	0.1320	0.1240
23	Pressure	(bar)	70.18	70.18	70.18	70.18
24	Temperature	(C)	51.63	51.63	51.63	51.63
25	Vapour Fraction		0.8816	0.8816	0.8816	0.8816
26	Heat Transferred	(kJ/h-m)	576.2	576.2	576.2	576.1
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2381	0.2381	0.1720	0.1720
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	7.690e-003	7.690e-003	8.906e-003	8.906e-003
32	Static Gradient	(kPa/m)	3.232e-003	3.232e-003	-2.575e-003	-2.575e-003
33	Accel. Gradient	(kPa/m)	1.320e-006	1.320e-006	7.654e-007	7.654e-007
34	Liq. Reynolds		1.892e+005 *	1.892e+005 *	1.892e+005 *	1.892e+005 *
35	Vap. Reynolds		9.791e+006 *	9.791e+006 *	9.791e+006 *	9.791e+006 *
36	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981
37	Vapour Velocity	(m/s)	2.877	2.877	2.877	2.877
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	1.645e+004	1.645e+004	1.646e+004	1.646e+004
41	Elevation	(m)	0.1160	0.1080	1.000e-001	9.800e-002
42	Pressure	(bar)	70.18	70.18	70.18	70.18
43	Temperature	(C)	51.62	51.62	51.62	51.62
44	Vapour Fraction		0.8816	0.8816	0.8816	0.8816
45	Heat Transferred	(kJ/h-m)	576.1	576.0	576.0	575.9
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.1720	0.1720	0.1720	0.1729
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	8.906e-003	8.906e-003	8.906e-003	8.894e-003
51	Static Gradient	(kPa/m)	-2.575e-003	-2.575e-003	-2.575e-003	-6.459e-004
52	Accel. Gradient	(kPa/m)	7.654e-007	7.654e-007	7.654e-007	9.971e-007
53	Liq. Reynolds		1.892e+005 *	1.892e+005 *	1.892e+005 *	1.892e+005 *
54	Vap. Reynolds		9.791e+006 *	9.791e+006 *	9.791e+006 *	9.791e+006 *
55	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981
56	Vapour Velocity	(m/s)	2.877	2.877	2.877	2.877
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	1.647e+004	1.647e+004	1.648e+004	1.648e+004
60	Elevation	(m)	9.600e-002	9.400e-002	9.200e-002	9.000e-002
61	Pressure	(bar)	70.18	70.18	70.18	70.18
62	Temperature	(C)	51.62	51.61	51.61	51.61
63	Vapour Fraction		0.8816	0.8816	0.8816	0.8816
64	Heat Transferred	(kJ/h-m)	575.9	575.8	575.8	575.7
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.1729	0.1729	0.1729	0.1729
68	Water HoldUp		---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Friction Gradient	(kPa/m)	8.894e-003	8.894e-003	8.894e-003	8.894e-003				
14	Static Gradient	(kPa/m)	-6.459e-004	-6.459e-004	-6.459e-004	-6.459e-004				
15	Accel. Gradient	(kPa/m)	9.971e-007	9.971e-007	9.971e-007	9.971e-007				
16	Liq. Reynolds		1.892e+005 *	1.892e+005 *	1.892e+005 *	1.892e+005 *				
17	Vap. Reynolds		9.791e+006 *	9.791e+006 *	9.791e+006 *	9.791e+006 *				
18	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981				
19	Vapour Velocity	(m/s)	2.877	2.877	2.877	2.877				
20	Deposit Thickness	(mm)	---	---	---	---				
21	Deposit Volume	(m3)	---	---	---	---				
22	Length	(m)	1.649e+004	1.649e+004	1.650e+004	1.650e+004				
23	Elevation	(m)	8.600e-002	8.200e-002	7.800e-002	7.400e-002				
24	Pressure	(bar)	70.17	70.17	70.17	70.17				
25	Temperature	(C)	51.61	51.61	51.61	51.60				
26	Vapour Fraction		0.8816	0.8816	0.8816	0.8816				
27	Heat Transferred	(kJ/h-m)	575.7	575.7	575.6	575.6				
28	Flow Regime		Transition	Transition	Transition	Transition				
29	Water Flow Regime									
30	Liquid HoldUp		0.1726	0.1726	0.1726	0.1726				
31	Water HoldUp		---	---	---	---				
32	Friction Gradient	(kPa/m)	8.898e-003	8.898e-003	8.898e-003	8.898e-003				
33	Static Gradient	(kPa/m)	-1.298e-003	-1.298e-003	-1.298e-003	-1.298e-003				
34	Accel. Gradient	(kPa/m)	9.188e-007	9.188e-007	9.188e-007	9.188e-007				
35	Liq. Reynolds		1.892e+005 *	1.892e+005 *	1.892e+005 *	1.892e+005 *				
36	Vap. Reynolds		9.791e+006 *	9.791e+006 *	9.791e+006 *	9.791e+006 *				
37	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981				
38	Vapour Velocity	(m/s)	2.877	2.877	2.877	2.877				
39	Deposit Thickness	(mm)	---	---	---	---				
40	Deposit Volume	(m3)	---	---	---	---				
41	Length	(m)	1.651e+004	1.651e+004	1.652e+004	1.652e+004				
42	Elevation	(m)	7.000e-002	8.000e-002	9.000e-002	1.000e-001				
43	Pressure	(bar)	70.17	70.17	70.17	70.17				
44	Temperature	(C)	51.60	51.60	51.60	51.60				
45	Vapour Fraction		0.8816	0.8816	0.8816	0.8816				
46	Heat Transferred	(kJ/h-m)	575.5	575.4	575.4	575.3				
47	Flow Regime		Transition	Transition	Transition	Transition				
48	Water Flow Regime									
49	Liquid HoldUp		0.1726	0.2392	0.2392	0.2392				
50	Water HoldUp		---	---	---	---				
51	Friction Gradient	(kPa/m)	8.898e-003	7.552e-003	7.552e-003	7.552e-003				
52	Static Gradient	(kPa/m)	-1.298e-003	4.054e-003	4.054e-003	4.054e-003				
53	Accel. Gradient	(kPa/m)	9.188e-007	1.403e-006	1.403e-006	1.403e-006				
54	Liq. Reynolds		1.892e+005 *	1.892e+005 *	1.892e+005 *	1.892e+005 *				
55	Vap. Reynolds		9.791e+006 *	9.791e+006 *	9.791e+006 *	9.791e+006 *				
56	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981				
57	Vapour Velocity	(m/s)	2.877	2.877	2.877	2.877				
58	Deposit Thickness	(mm)	---	---	---	---				
59	Deposit Volume	(m3)	---	---	---	---				
60	Length	(m)	1.653e+004	1.653e+004	1.654e+004	1.654e+004				
61	Elevation	(m)	0.1100	0.1200	0.1200	0.1200				
62	Pressure	(bar)	70.17	70.17	70.17	70.17				
63	Temperature	(C)	51.59	51.59	51.59	51.59				
64	Vapour Fraction		0.8815	0.8815	0.8815	0.8815				
65	Heat Transferred	(kJ/h-m)	575.3	575.2	575.2	575.1				
66	Flow Regime		Transition	Transition	Transition	Transition				
67	Water Flow Regime									
68	Liquid HoldUp		0.2392	0.2392	0.2528	0.2528				
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 292 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	7.552e-003	7.552e-003	6.015e-003	6.015e-003
15	Static Gradient (kPa/m)	4.054e-003	4.054e-003	0.0000	0.0000
16	Accel. Gradient (kPa/m)	1.403e-006	1.403e-006	7.273e-007	7.273e-007
17	Liq. Reynolds	1.892e+005 *	1.892e+005 *	1.892e+005 *	1.892e+005 *
18	Vap. Reynolds	9.791e+006 *	9.791e+006 *	9.791e+006 *	9.791e+006 *
19	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
20	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	1.655e+004	1.655e+004	1.656e+004	1.656e+004
24	Elevation (m)	0.1200	0.1200	0.1200	0.1260
25	Pressure (bar)	70.17	70.17	70.17	70.17
26	Temperature (C)	51.59	51.59	51.58	51.58
27	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
28	Heat Transferred (kJ/h-m)	575.1	575.0	575.0	574.9
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2528	0.2528	0.2528	0.2370
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	6.015e-003	6.015e-003	6.015e-003	7.829e-003
34	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	2.416e-003
35	Accel. Gradient (kPa/m)	7.272e-007	7.272e-007	7.272e-007	1.239e-006
36	Liq. Reynolds	1.892e+005 *	1.892e+005 *	1.892e+005 *	1.892e+005 *
37	Vap. Reynolds	9.791e+006 *	9.791e+006 *	9.791e+006 *	9.791e+006 *
38	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
39	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	1.657e+004	1.657e+004	1.658e+004	1.658e+004
43	Elevation (m)	0.1320	0.1380	0.1440	0.1500
44	Pressure (bar)	70.17	70.17	70.17	70.17
45	Temperature (C)	51.58	51.58	51.58	51.58
46	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
47	Heat Transferred (kJ/h-m)	574.9	574.8	574.8	574.7
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2370	0.2370	0.2370	0.2370
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	7.829e-003	7.829e-003	7.829e-003	7.829e-003
53	Static Gradient (kPa/m)	2.416e-003	2.416e-003	2.416e-003	2.416e-003
54	Accel. Gradient (kPa/m)	1.239e-006	1.239e-006	1.239e-006	1.239e-006
55	Liq. Reynolds	1.892e+005 *	1.892e+005 *	1.892e+005 *	1.892e+005 *
56	Vap. Reynolds	9.791e+006 *	9.791e+006 *	9.791e+006 *	9.791e+006 *
57	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
58	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	1.659e+004	1.659e+004	1.660e+004	1.660e+004
62	Elevation (m)	0.1220	9.400e-002	6.600e-002	3.800e-002
63	Pressure (bar)	70.17	70.17	70.17	70.17
64	Temperature (C)	51.57	51.57	51.57	51.57
65	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
66	Heat Transferred (kJ/h-m)	574.8	574.7	574.7	574.6
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 293 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.1690	0.1690	0.1690	0.1690
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	8.945e-003	8.945e-003	8.945e-003	8.945e-003
16	Static Gradient (kPa/m)	-8.907e-003	-8.907e-003	-8.908e-003	-8.908e-003
17	Accel. Gradient (kPa/m)	4.568e-009	4.554e-009	4.539e-009	4.525e-009
18	Liq. Reynolds	1.892e+005 *	1.892e+005 *	1.892e+005 *	1.892e+005 *
19	Vap. Reynolds	9.791e+006 *	9.791e+006 *	9.791e+006 *	9.791e+006 *
20	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
21	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	1.661e+004	1.661e+004	1.662e+004	1.662e+004
25	Elevation (m)	1.000e-002	1.200e-002	1.400e-002	1.600e-002
26	Pressure (bar)	70.17	70.17	70.17	70.16
27	Temperature (C)	51.57	51.57	51.56	51.56
28	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
29	Heat Transferred (kJ/h-m)	574.6	574.5	574.4	574.4
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1690	0.2347	0.2347	0.2347
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	8.945e-003	8.115e-003	8.115e-003	8.115e-003
35	Static Gradient (kPa/m)	-8.908e-003	7.998e-004	7.998e-004	7.998e-004
36	Accel. Gradient (kPa/m)	4.510e-009	1.078e-006	1.078e-006	1.078e-006
37	Liq. Reynolds	1.892e+005 *	1.892e+005 *	1.891e+005 *	1.891e+005 *
38	Vap. Reynolds	9.791e+006 *	9.791e+006 *	9.791e+006 *	9.791e+006 *
39	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
40	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	1.663e+004	1.663e+004	1.664e+004	1.664e+004
44	Elevation (m)	1.800e-002	2.000e-002	4.000e-003	-1.200e-002
45	Pressure (bar)	70.16	70.16	70.16	70.16
46	Temperature (C)	51.56	51.56	51.56	51.56
47	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
48	Heat Transferred (kJ/h-m)	574.3	574.3	574.3	574.3
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2347	0.2347	0.1708	0.1708
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	8.115e-003	8.115e-003	8.921e-003	8.920e-003
54	Static Gradient (kPa/m)	7.998e-004	7.998e-004	-5.126e-003	-5.126e-003
55	Accel. Gradient (kPa/m)	1.078e-006	1.078e-006	4.588e-007	4.588e-007
56	Liq. Reynolds	1.891e+005 *	1.891e+005 *	1.891e+005 *	1.891e+005 *
57	Vap. Reynolds	9.791e+006 *	9.791e+006 *	9.791e+006 *	9.791e+006 *
58	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
59	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	1.665e+004	1.665e+004	1.666e+004	1.666e+004
63	Elevation (m)	-2.800e-002	-4.400e-002	-6.000e-002	-5.200e-002
64	Pressure (bar)	70.16	70.16	70.16	70.16
65	Temperature (C)	51.55	51.55	51.55	51.55
66	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
67	Heat Transferred (kJ/h-m)	574.2	574.2	574.1	574.0
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 294 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1708	0.1708	0.1708	0.2381
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	8.920e-003	8.920e-003	8.920e-003	7.689e-003
17	Static Gradient (kPa/m)	-5.126e-003	-5.126e-003	-5.126e-003	3.233e-003
18	Accel. Gradient (kPa/m)	4.588e-007	4.588e-007	4.587e-007	1.320e-006
19	Liq. Reynolds	1.891e+005 *	1.891e+005 *	1.891e+005 *	1.891e+005 *
20	Vap. Reynolds	9.791e+006 *	9.791e+006 *	9.791e+006 *	9.791e+006 *
21	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
22	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	1.667e+004	1.667e+004	1.668e+004	1.668e+004
26	Elevation (m)	-4.400e-002	-3.600e-002	-2.800e-002	-2.000e-002
27	Pressure (bar)	70.16	70.16	70.16	70.16
28	Temperature (C)	51.55	51.54	51.54	51.54
29	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
30	Heat Transferred (kJ/h-m)	574.0	573.9	573.9	573.8
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2381	0.2381	0.2381	0.2381
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	7.689e-003	7.689e-003	7.689e-003	7.689e-003
36	Static Gradient (kPa/m)	3.233e-003	3.233e-003	3.233e-003	3.233e-003
37	Accel. Gradient (kPa/m)	1.320e-006	1.320e-006	1.320e-006	1.321e-006
38	Liq. Reynolds	1.891e+005 *	1.891e+005 *	1.891e+005 *	1.891e+005 *
39	Vap. Reynolds	9.791e+006 *	9.791e+006 *	9.791e+006 *	9.792e+006 *
40	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
41	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	1.669e+004	1.669e+004	1.670e+004	1.670e+004
45	Elevation (m)	-1.600e-002	-1.200e-002	-8.000e-003	-4.000e-003
46	Pressure (bar)	70.16	70.16	70.16	70.16
47	Temperature (C)	51.54	51.54	51.54	51.53
48	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
49	Heat Transferred (kJ/h-m)	573.8	573.7	573.7	573.6
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2359	0.2359	0.2359	0.2359
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	7.971e-003	7.971e-003	7.971e-003	7.971e-003
55	Static Gradient (kPa/m)	1.601e-003	1.601e-003	1.601e-003	1.601e-003
56	Accel. Gradient (kPa/m)	1.157e-006	1.157e-006	1.157e-006	1.157e-006
57	Liq. Reynolds	1.891e+005 *	1.891e+005 *	1.891e+005 *	1.891e+005 *
58	Vap. Reynolds	9.792e+006 *	9.792e+006 *	9.792e+006 *	9.792e+006 *
59	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
60	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	1.671e+004	1.671e+004	1.672e+004	1.672e+004
64	Elevation (m)	-1.006e-016	-4.000e-003	-8.000e-003	-1.200e-002
65	Pressure (bar)	70.16	70.16	70.16	70.16
66	Temperature (C)	51.53	51.53	51.53	51.53
67	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
68	Heat Transferred (kJ/h-m)	573.6	573.6	573.5	573.5
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 295 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2359	0.1726	0.1726	0.1726
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	7.971e-003	8.897e-003	8.897e-003	8.897e-003
18	Static Gradient (kPa/m)	1.601e-003	-1.290e-003	-1.290e-003	-1.290e-003
19	Accel. Gradient (kPa/m)	1.157e-006	9.197e-007	9.197e-007	9.197e-007
20	Liq. Reynolds	1.891e+005 *	1.891e+005 *	1.891e+005 *	1.891e+005 *
21	Vap. Reynolds	9.792e+006 *	9.792e+006 *	9.792e+006 *	9.792e+006 *
22	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
23	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	1.673e+004	1.673e+004	1.674e+004	1.674e+004
27	Elevation (m)	-1.600e-002	-2.000e-002	-3.000e-002	-4.000e-002
28	Pressure (bar)	70.16	70.16	70.16	70.16
29	Temperature (C)	51.53	51.52	51.52	51.52
30	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
31	Heat Transferred (kJ/h-m)	573.5	573.4	573.4	573.3
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1726	0.1726	0.1717	0.1717
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	8.897e-003	8.897e-003	8.908e-003	8.908e-003
37	Static Gradient (kPa/m)	-1.290e-003	-1.291e-003	-3.215e-003	-3.215e-003
38	Accel. Gradient (kPa/m)	9.197e-007	9.197e-007	6.884e-007	6.884e-007
39	Liq. Reynolds	1.891e+005 *	1.891e+005 *	1.891e+005 *	1.891e+005 *
40	Vap. Reynolds	9.792e+006 *	9.792e+006 *	9.792e+006 *	9.792e+006 *
41	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
42	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	1.675e+004	1.675e+004	1.676e+004	1.676e+004
46	Elevation (m)	-5.000e-002	-6.000e-002	-7.000e-002	-8.400e-002
47	Pressure (bar)	70.16	70.15	70.15	70.15
48	Temperature (C)	51.52	51.52	51.51	51.51
49	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
50	Heat Transferred (kJ/h-m)	573.3	573.2	573.2	573.1
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1717	0.1717	0.1717	0.1711
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	8.908e-003	8.908e-003	8.908e-003	8.916e-003
56	Static Gradient (kPa/m)	-3.215e-003	-3.215e-003	-3.215e-003	-4.491e-003
57	Accel. Gradient (kPa/m)	6.884e-007	6.884e-007	6.884e-007	5.351e-007
58	Liq. Reynolds	1.891e+005 *	1.891e+005 *	1.891e+005 *	1.891e+005 *
59	Vap. Reynolds	9.792e+006 *	9.792e+006 *	9.792e+006 *	9.792e+006 *
60	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
61	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	1.677e+004	1.677e+004	1.678e+004	1.678e+004
65	Elevation (m)	-9.800e-002	-0.1120	-0.1260	-0.1400
66	Pressure (bar)	70.15	70.15	70.15	70.15
67	Temperature (C)	51.51	51.51	51.51	51.51
68	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 296 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Heat Transferred (kJ/h-m)	573.1	573.0	573.0	573.0					
14	Flow Regime	Transition	Transition	Transition	Transition					
15	Water Flow Regime									
16	Liquid HoldUp	0.1711	0.1711	0.1711	0.1711					
17	Water HoldUp	---	---	---	---					
18	Friction Gradient (kPa/m)	8.916e-003	8.916e-003	8.916e-003	8.916e-003					
19	Static Gradient (kPa/m)	-4.491e-003	-4.491e-003	-4.491e-003	-4.491e-003					
20	Accel. Gradient (kPa/m)	5.351e-007	5.351e-007	5.351e-007	5.350e-007					
21	Liq. Reynolds	1.891e+005 *	1.891e+005 *	1.891e+005 *	1.891e+005 *					
22	Vap. Reynolds	9.792e+006 *	9.792e+006 *	9.792e+006 *	9.792e+006 *					
23	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981					
24	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877					
25	Deposit Thickness (mm)	---	---	---	---					
26	Deposit Volume (m3)	---	---	---	---					
27	Length (m)	1.679e+004	1.679e+004	1.680e+004	1.680e+004					
28	Elevation (m)	-0.1320	-0.1240	-0.1160	-0.1080					
29	Pressure (bar)	70.15	70.15	70.15	70.15					
30	Temperature (C)	51.50	51.50	51.50	51.50					
31	Vapour Fraction	0.8815	0.8815	0.8815	0.8815					
32	Heat Transferred (kJ/h-m)	572.9	572.8	572.8	572.7					
33	Flow Regime	Transition	Transition	Transition	Transition					
34	Water Flow Regime									
35	Liquid HoldUp	0.2381	0.2381	0.2381	0.2381					
36	Water HoldUp	---	---	---	---					
37	Friction Gradient (kPa/m)	7.688e-003	7.688e-003	7.688e-003	7.688e-003					
38	Static Gradient (kPa/m)	3.233e-003	3.233e-003	3.233e-003	3.233e-003					
39	Accel. Gradient (kPa/m)	1.321e-006	1.321e-006	1.321e-006	1.321e-006					
40	Liq. Reynolds	1.891e+005 *	1.891e+005 *	1.891e+005 *	1.891e+005 *					
41	Vap. Reynolds	9.792e+006 *	9.792e+006 *	9.792e+006 *	9.792e+006 *					
42	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981					
43	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877					
44	Deposit Thickness (mm)	---	---	---	---					
45	Deposit Volume (m3)	---	---	---	---					
46	Length (m)	1.681e+004	1.681e+004	1.682e+004	1.682e+004					
47	Elevation (m)	-0.1000	-0.1060	-0.1120	-0.1180					
48	Pressure (bar)	70.15	70.15	70.15	70.15					
49	Temperature (C)	51.50	51.50	51.49	51.49					
50	Vapour Fraction	0.8815	0.8815	0.8815	0.8815					
51	Heat Transferred (kJ/h-m)	572.7	572.7	572.6	572.6					
52	Flow Regime	Transition	Transition	Transition	Transition					
53	Water Flow Regime									
54	Liquid HoldUp	0.2381	0.1723	0.1723	0.1723					
55	Water HoldUp	---	---	---	---					
56	Friction Gradient (kPa/m)	7.688e-003	8.900e-003	8.900e-003	8.900e-003					
57	Static Gradient (kPa/m)	3.233e-003	-1.934e-003	-1.934e-003	-1.934e-003					
58	Accel. Gradient (kPa/m)	1.321e-006	8.424e-007	8.424e-007	8.424e-007					
59	Liq. Reynolds	1.891e+005 *	1.891e+005 *	1.891e+005 *	1.891e+005 *					
60	Vap. Reynolds	9.792e+006 *	9.792e+006 *	9.792e+006 *	9.792e+006 *					
61	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981					
62	Vapour Velocity (m/s)	2.877	2.877	2.877	2.877					
63	Deposit Thickness (mm)	---	---	---	---					
64	Deposit Volume (m3)	---	---	---	---					
65	Length (m)	1.683e+004	1.683e+004	1.684e+004	1.684e+004					
66	Elevation (m)	-0.1240	-0.1300	-0.1380	-0.1460					
67	Pressure (bar)	70.15	70.15	70.15	70.15					
68	Temperature (C)	51.49	51.49	51.49	51.49					
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 297 of 937					

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
14	Heat Transferred (kJ/h-m)	572.5	572.5	572.4	572.4
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.1723	0.1723	0.1720	0.1720
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	8.900e-003	8.900e-003	8.904e-003	8.904e-003
20	Static Gradient (kPa/m)	-1.934e-003	-1.934e-003	-2.575e-003	-2.575e-003
21	Accel. Gradient (kPa/m)	8.424e-007	8.424e-007	7.653e-007	7.652e-007
22	Liq. Reynolds	1.891e+005 *	1.891e+005 *	1.891e+005 *	1.891e+005 *
23	Vap. Reynolds	9.792e+006 *	9.792e+006 *	9.792e+006 *	9.792e+006 *
24	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
25	Vapour Velocity (m/s)	2.877	2.876	2.876	2.876
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	1.685e+004	1.685e+004	1.686e+004	1.686e+004
29	Elevation (m)	-0.1540	-0.1620	-0.1700	-0.1700
30	Pressure (bar)	70.15	70.15	70.15	70.15
31	Temperature (C)	51.48	51.48	51.48	51.48
32	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
33	Heat Transferred (kJ/h-m)	572.3	572.3	572.3	572.2
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1720	0.1720	0.1720	0.2529
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	8.904e-003	8.904e-003	8.904e-003	6.014e-003
39	Static Gradient (kPa/m)	-2.575e-003	-2.575e-003	-2.575e-003	0.0000
40	Accel. Gradient (kPa/m)	7.652e-007	7.652e-007	7.652e-007	7.272e-007
41	Liq. Reynolds	1.891e+005 *	1.891e+005 *	1.891e+005 *	1.891e+005 *
42	Vap. Reynolds	9.792e+006 *	9.792e+006 *	9.792e+006 *	9.792e+006 *
43	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
44	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	1.687e+004	1.687e+004	1.688e+004	1.688e+004
48	Elevation (m)	-0.1700	-0.1700	-0.1700	-0.1700
49	Pressure (bar)	70.15	70.15	70.15	70.15
50	Temperature (C)	51.48	51.47	51.47	51.47
51	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
52	Heat Transferred (kJ/h-m)	572.1	572.1	572.0	572.0
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2529	0.2529	0.2529	0.2529
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	6.014e-003	6.014e-003	6.014e-003	6.014e-003
58	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
59	Accel. Gradient (kPa/m)	7.271e-007	7.271e-007	7.271e-007	7.271e-007
60	Liq. Reynolds	1.891e+005 *	1.891e+005 *	1.891e+005 *	1.891e+005 *
61	Vap. Reynolds	9.792e+006 *	9.792e+006 *	9.792e+006 *	9.792e+006 *
62	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
63	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	1.689e+004	1.689e+004	1.690e+004	1.690e+004
67	Elevation (m)	-0.1880	-0.2060	-0.2240	-0.2420
68	Pressure (bar)	70.15	70.15	70.15	70.15
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 298 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	51.47	51.47	51.47	51.47
14	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
15	Heat Transferred (kJ/h-m)	572.0	572.0	571.9	571.9
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.1702	0.1702	0.1702	0.1702
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	8.928e-003	8.928e-003	8.928e-003	8.928e-003
21	Static Gradient (kPa/m)	-6.442e-003	-6.442e-003	-6.442e-003	-6.442e-003
22	Accel. Gradient (kPa/m)	3.006e-007	3.006e-007	3.006e-007	3.006e-007
23	Liq. Reynolds	1.891e+005 *	1.891e+005 *	1.891e+005 *	1.891e+005 *
24	Vap. Reynolds	9.792e+006 *	9.792e+006 *	9.792e+006 *	9.792e+006 *
25	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
26	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	1.690e+004	1.691e+004	1.691e+004	1.692e+004
30	Elevation (m)	-0.2600	-0.2780	-0.2960	-0.3140
31	Pressure (bar)	70.15	70.15	70.15	70.14
32	Temperature (C)	51.46	51.46	51.46	51.46
33	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
34	Heat Transferred (kJ/h-m)	571.8	571.8	571.7	571.7
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1702	0.1705	0.1705	0.1705
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	8.928e-003	8.923e-003	8.923e-003	8.923e-003
40	Static Gradient (kPa/m)	-6.442e-003	-5.761e-003	-5.761e-003	-5.761e-003
41	Accel. Gradient (kPa/m)	3.006e-007	3.824e-007	3.824e-007	3.823e-007
42	Liq. Reynolds	1.891e+005 *	1.890e+005 *	1.890e+005 *	1.890e+005 *
43	Vap. Reynolds	9.792e+006 *	9.792e+006 *	9.792e+006 *	9.792e+006 *
44	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
45	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	1.692e+004	1.693e+004	1.693e+004	1.694e+004
49	Elevation (m)	-0.3320	-0.3500	-0.3520	-0.3540
50	Pressure (bar)	70.14	70.14	70.14	70.14
51	Temperature (C)	51.46	51.46	51.45	51.45
52	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
53	Heat Transferred (kJ/h-m)	571.7	571.6	571.6	571.5
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.1705	0.1705	0.1729	0.1729
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	8.923e-003	8.923e-003	8.892e-003	8.892e-003
59	Static Gradient (kPa/m)	-5.761e-003	-5.761e-003	-6.461e-004	-6.461e-004
60	Accel. Gradient (kPa/m)	3.823e-007	3.823e-007	9.970e-007	9.970e-007
61	Liq. Reynolds	1.890e+005 *	1.890e+005 *	1.890e+005 *	1.890e+005 *
62	Vap. Reynolds	9.792e+006 *	9.792e+006 *	9.792e+006 *	9.792e+006 *
63	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
64	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	1.694e+004	1.695e+004	1.695e+004	1.696e+004
68	Elevation (m)	-0.3560	-0.3580	-0.3600	-0.3580
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 299 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:13:43 2022	
6						
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Pressure (bar)	70.14	70.14	70.14	70.14	
14	Temperature (C)	51.45	51.45	51.45	51.45	
15	Vapour Fraction	0.8815	0.8815	0.8815	0.8815	
16	Heat Transferred (kJ/h-m)	571.5	571.4	571.4	571.3	
17	Flow Regime	Transition	Transition	Transition	Transition	
18	Water Flow Regime					
19	Liquid HoldUp	0.1729	0.1729	0.1729	0.2348	
20	Water HoldUp	---	---	---	---	
21	Friction Gradient (kPa/m)	8.892e-003	8.892e-003	8.892e-003	8.112e-003	
22	Static Gradient (kPa/m)	-6.461e-004	-6.461e-004	-6.461e-004	8.000e-004	
23	Accel. Gradient (kPa/m)	9.970e-007	9.970e-007	9.970e-007	1.078e-006	
24	Liq. Reynolds	1.890e+005 *	1.890e+005 *	1.890e+005 *	1.890e+005 *	
25	Vap. Reynolds	9.792e+006 *	9.792e+006 *	9.792e+006 *	9.792e+006 *	
26	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981	
27	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876	
28	Deposit Thickness (mm)	---	---	---	---	
29	Deposit Volume (m3)	---	---	---	---	
30	Length (m)	1.696e+004	1.697e+004	1.697e+004	1.698e+004	
31	Elevation (m)	-0.3560	-0.3540	-0.3520	-0.3500	
32	Pressure (bar)	70.14	70.14	70.14	70.14	
33	Temperature (C)	51.44	51.44	51.44	51.44	
34	Vapour Fraction	0.8815	0.8815	0.8815	0.8815	
35	Heat Transferred (kJ/h-m)	571.2	571.2	571.1	571.1	
36	Flow Regime	Transition	Transition	Transition	Transition	
37	Water Flow Regime					
38	Liquid HoldUp	0.2348	0.2348	0.2348	0.2348	
39	Water HoldUp	---	---	---	---	
40	Friction Gradient (kPa/m)	8.112e-003	8.112e-003	8.112e-003	8.112e-003	
41	Static Gradient (kPa/m)	8.000e-004	8.000e-004	8.000e-004	8.000e-004	
42	Accel. Gradient (kPa/m)	1.078e-006	1.078e-006	1.078e-006	1.078e-006	
43	Liq. Reynolds	1.890e+005 *	1.890e+005 *	1.890e+005 *	1.890e+005 *	
44	Vap. Reynolds	9.792e+006 *	9.792e+006 *	9.792e+006 *	9.793e+006 *	
45	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981	
46	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876	
47	Deposit Thickness (mm)	---	---	---	---	
48	Deposit Volume (m3)	---	---	---	---	
49	Length (m)	1.698e+004	1.699e+004	1.699e+004	1.700e+004	
50	Elevation (m)	-0.3280	-0.3060	-0.2840	-0.2620	
51	Pressure (bar)	70.14	70.14	70.14	70.14	
52	Temperature (C)	51.44	51.43	51.43	51.43	
53	Vapour Fraction	0.8815	0.8815	0.8815	0.8815	
54	Heat Transferred (kJ/h-m)	571.0	571.0	570.9	570.9	
55	Flow Regime	Transition	Transition	Transition	Transition	
56	Water Flow Regime					
57	Liquid HoldUp	0.2459	0.2459	0.2459	0.2459	
58	Water HoldUp	---	---	---	---	
59	Friction Gradient (kPa/m)	6.762e-003	6.762e-003	6.762e-003	6.762e-003	
60	Static Gradient (kPa/m)	9.106e-003	9.106e-003	9.106e-003	9.106e-003	
61	Accel. Gradient (kPa/m)	1.919e-006	1.919e-006	1.919e-006	1.919e-006	
62	Liq. Reynolds	1.890e+005 *	1.890e+005 *	1.890e+005 *	1.890e+005 *	
63	Vap. Reynolds	9.793e+006 *	9.793e+006 *	9.793e+006 *	9.793e+006 *	
64	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981	
65	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876	
66	Deposit Thickness (mm)	---	---	---	---	
67	Deposit Volume (m3)	---	---	---	---	
68	Length (m)	1.700e+004	1.701e+004	1.701e+004	1.702e+004	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	-0.2400	-0.2460	-0.2520	-0.2580
14	Pressure (bar)	70.14	70.14	70.14	70.14
15	Temperature (C)	51.43	51.43	51.43	51.42
16	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
17	Heat Transferred (kJ/h-m)	570.8	570.8	570.8	570.7
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2459	0.1723	0.1723	0.1723
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	6.762e-003	8.899e-003	8.899e-003	8.899e-003
23	Static Gradient (kPa/m)	9.106e-003	-1.934e-003	-1.934e-003	-1.934e-003
24	Accel. Gradient (kPa/m)	1.919e-006	8.423e-007	8.423e-007	8.423e-007
25	Liq. Reynolds	1.890e+005 *	1.890e+005 *	1.890e+005 *	1.890e+005 *
26	Vap. Reynolds	9.793e+006 *	9.793e+006 *	9.793e+006 *	9.793e+006 *
27	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
28	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	1.702e+004	1.703e+004	1.703e+004	1.704e+004
32	Elevation (m)	-0.2640	-0.2700	-0.2840	-0.2980
33	Pressure (bar)	70.14	70.13	70.13	70.13
34	Temperature (C)	51.42	51.42	51.42	51.42
35	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
36	Heat Transferred (kJ/h-m)	570.7	570.7	570.6	570.6
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.1723	0.1723	0.1711	0.1711
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	8.899e-003	8.899e-003	8.915e-003	8.915e-003
42	Static Gradient (kPa/m)	-1.934e-003	-1.934e-003	-4.491e-003	-4.491e-003
43	Accel. Gradient (kPa/m)	8.423e-007	8.423e-007	5.349e-007	5.349e-007
44	Liq. Reynolds	1.890e+005 *	1.890e+005 *	1.890e+005 *	1.890e+005 *
45	Vap. Reynolds	9.793e+006 *	9.793e+006 *	9.793e+006 *	9.793e+006 *
46	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
47	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	1.704e+004	1.705e+004	1.705e+004	1.706e+004
51	Elevation (m)	-0.3120	-0.3260	-0.3400	-0.3220
52	Pressure (bar)	70.13	70.13	70.13	70.13
53	Temperature (C)	51.42	51.41	51.41	51.41
54	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
55	Heat Transferred (kJ/h-m)	570.5	570.5	570.4	570.3
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.1711	0.1711	0.1711	0.2437
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	8.915e-003	8.915e-003	8.915e-003	7.017e-003
61	Static Gradient (kPa/m)	-4.491e-003	-4.491e-003	-4.492e-003	7.400e-003
62	Accel. Gradient (kPa/m)	5.349e-007	5.349e-007	5.349e-007	1.743e-006
63	Liq. Reynolds	1.890e+005 *	1.890e+005 *	1.890e+005 *	1.890e+005 *
64	Vap. Reynolds	9.793e+006 *	9.793e+006 *	9.793e+006 *	9.793e+006 *
65	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
66	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 301 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length (m)	1.706e+004	1.707e+004	1.707e+004	1.708e+004		
14	Elevation (m)	-0.3040	-0.2860	-0.2680	-0.2500		
15	Pressure (bar)	70.13	70.13	70.13	70.13		
16	Temperature (C)	51.41	51.41	51.40	51.40		
17	Vapour Fraction	0.8815	0.8815	0.8815	0.8815		
18	Heat Transferred (kJ/h-m)	570.3	570.2	570.2	570.1		
19	Flow Regime	Transition	Transition	Transition	Transition		
20	Water Flow Regime						
21	Liquid HoldUp	0.2437	0.2437	0.2437	0.2437		
22	Water HoldUp	---	---	---	---		
23	Friction Gradient (kPa/m)	7.017e-003	7.017e-003	7.017e-003	7.017e-003		
24	Static Gradient (kPa/m)	7.400e-003	7.400e-003	7.400e-003	7.400e-003		
25	Accel. Gradient (kPa/m)	1.743e-006	1.743e-006	1.743e-006	1.744e-006		
26	Liq. Reynolds	1.890e+005 *	1.890e+005 *	1.890e+005 *	1.890e+005 *		
27	Vap. Reynolds	9.793e+006 *	9.793e+006 *	9.793e+006 *	9.793e+006 *		
28	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
29	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876		
30	Deposit Thickness (mm)	---	---	---	---		
31	Deposit Volume (m3)	---	---	---	---		
32	Length (m)	1.708e+004	1.709e+004	1.709e+004	1.710e+004		
33	Elevation (m)	-0.2420	-0.2340	-0.2260	-0.2180		
34	Pressure (bar)	70.13	70.13	70.13	70.13		
35	Temperature (C)	51.40	51.40	51.40	51.40		
36	Vapour Fraction	0.8815	0.8815	0.8815	0.8815		
37	Heat Transferred (kJ/h-m)	570.1	570.0	570.0	569.9		
38	Flow Regime	Transition	Transition	Transition	Transition		
39	Water Flow Regime						
40	Liquid HoldUp	0.2383	0.2383	0.2383	0.2383		
41	Water HoldUp	---	---	---	---		
42	Friction Gradient (kPa/m)	7.669e-003	7.669e-003	7.669e-003	7.669e-003		
43	Static Gradient (kPa/m)	3.340e-003	3.340e-003	3.340e-003	3.340e-003		
44	Accel. Gradient (kPa/m)	1.331e-006	1.331e-006	1.331e-006	1.331e-006		
45	Liq. Reynolds	1.890e+005 *	1.890e+005 *	1.890e+005 *	1.890e+005 *		
46	Vap. Reynolds	9.793e+006 *	9.793e+006 *	9.793e+006 *	9.793e+006 *		
47	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
48	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876		
49	Deposit Thickness (mm)	---	---	---	---		
50	Deposit Volume (m3)	---	---	---	---		
51	Length (m)	1.710e+004	1.711e+004	1.711e+004	1.712e+004		
52	Elevation (m)	-0.2100	-0.1980	-0.1860	-0.1740		
53	Pressure (bar)	70.13	70.13	70.13	70.13		
54	Temperature (C)	51.39	51.39	51.39	51.39		
55	Vapour Fraction	0.8815	0.8815	0.8815	0.8814		
56	Heat Transferred (kJ/h-m)	569.9	569.9	569.8	569.8		
57	Flow Regime	Transition	Transition	Transition	Transition		
58	Water Flow Regime						
59	Liquid HoldUp	0.2383	0.2404	0.2404	0.2404		
60	Water HoldUp	---	---	---	---		
61	Friction Gradient (kPa/m)	7.669e-003	7.413e-003	7.413e-003	7.413e-003		
62	Static Gradient (kPa/m)	3.340e-003	4.883e-003	4.883e-003	4.883e-003		
63	Accel. Gradient (kPa/m)	1.331e-006	1.487e-006	1.487e-006	1.487e-006		
64	Liq. Reynolds	1.890e+005 *	1.890e+005 *	1.890e+005 *	1.890e+005 *		
65	Vap. Reynolds	9.793e+006 *	9.793e+006 *	9.793e+006 *	9.793e+006 *		
66	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
67	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876		
68	Deposit Thickness (mm)	---	---	---	---		
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 302 of 937		

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	1.712e+004	1.713e+004	1.713e+004	1.714e+004
15	Elevation (m)	-0.1620	-0.1500	-0.1360	-0.1220
16	Pressure (bar)	70.12	70.12	70.12	70.12
17	Temperature (C)	51.39	51.38	51.38	51.38
18	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
19	Heat Transferred (kJ/h-m)	569.7	569.7	569.6	569.6
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2404	0.2404	0.2415	0.2415
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	7.413e-003	7.414e-003	7.280e-003	7.280e-003
25	Static Gradient (kPa/m)	4.883e-003	4.883e-003	5.717e-003	5.717e-003
26	Accel. Gradient (kPa/m)	1.487e-006	1.487e-006	1.572e-006	1.572e-006
27	Liq. Reynolds	1.890e+005 *	1.890e+005 *	1.890e+005 *	1.890e+005 *
28	Vap. Reynolds	9.793e+006 *	9.793e+006 *	9.793e+006 *	9.793e+006 *
29	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
30	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	1.714e+004	1.715e+004	1.715e+004	1.716e+004
34	Elevation (m)	-0.1080	-9.400e-002	-8.000e-002	-7.000e-002
35	Pressure (bar)	70.12	70.12	70.12	70.12
36	Temperature (C)	51.38	51.38	51.37	51.37
37	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
38	Heat Transferred (kJ/h-m)	569.5	569.5	569.4	569.4
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2415	0.2415	0.2415	0.2392
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	7.280e-003	7.280e-003	7.280e-003	7.549e-003
44	Static Gradient (kPa/m)	5.717e-003	5.717e-003	5.717e-003	4.056e-003
45	Accel. Gradient (kPa/m)	1.572e-006	1.572e-006	1.572e-006	1.404e-006
46	Liq. Reynolds	1.890e+005 *	1.890e+005 *	1.890e+005 *	1.890e+005 *
47	Vap. Reynolds	9.793e+006 *	9.793e+006 *	9.793e+006 *	9.793e+006 *
48	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
49	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	1.716e+004	1.717e+004	1.717e+004	1.718e+004
53	Elevation (m)	-6.000e-002	-5.000e-002	-4.000e-002	-3.000e-002
54	Pressure (bar)	70.12	70.12	70.12	70.12
55	Temperature (C)	51.37	51.37	51.37	51.37
56	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
57	Heat Transferred (kJ/h-m)	569.3	569.3	569.2	569.2
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2392	0.2392	0.2392	0.2392
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	7.549e-003	7.549e-003	7.549e-003	7.549e-003
63	Static Gradient (kPa/m)	4.056e-003	4.056e-003	4.056e-003	4.056e-003
64	Accel. Gradient (kPa/m)	1.404e-006	1.404e-006	1.404e-006	1.404e-006
65	Liq. Reynolds	1.890e+005 *	1.889e+005 *	1.889e+005 *	1.889e+005 *
66	Vap. Reynolds	9.793e+006 *	9.793e+006 *	9.793e+006 *	9.794e+006 *
67	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
68	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 303 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Thickness	(mm)	---	---	---	---	
14	Deposit Volume	(m3)	---	---	---	---	
15	Length	(m)	1.718e+004	1.719e+004	1.719e+004	1.720e+004	
16	Elevation	(m)	-2.600e-002	-2.200e-002	-1.800e-002	-1.400e-002	
17	Pressure	(bar)	70.12	70.12	70.12	70.12	
18	Temperature	(C)	51.36	51.36	51.36	51.36	
19	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
20	Heat Transferred	(kJ/h-m)	569.1	569.1	569.0	569.0	
21	Flow Regime		Transition	Transition	Transition	Transition	
22	Water Flow Regime						
23	Liquid HoldUp		0.2359	0.2359	0.2359	0.2359	
24	Water HoldUp		---	---	---	---	
25	Friction Gradient	(kPa/m)	7.969e-003	7.969e-003	7.969e-003	7.969e-003	
26	Static Gradient	(kPa/m)	1.606e-003	1.606e-003	1.606e-003	1.606e-003	
27	Accel. Gradient	(kPa/m)	1.158e-006	1.158e-006	1.158e-006	1.158e-006	
28	Liq. Reynolds		1.889e+005 *	1.889e+005 *	1.889e+005 *	1.889e+005 *	
29	Vap. Reynolds		9.794e+006 *	9.794e+006 *	9.794e+006 *	9.794e+006 *	
30	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
31	Vapour Velocity	(m/s)	2.876	2.876	2.876	2.876	
32	Deposit Thickness	(mm)	---	---	---	---	
33	Deposit Volume	(m3)	---	---	---	---	
34	Length	(m)	1.720e+004	1.721e+004	1.721e+004	1.722e+004	
35	Elevation	(m)	-1.000e-002	4.000e-003	1.800e-002	3.200e-002	
36	Pressure	(bar)	70.12	70.12	70.11	70.11	
37	Temperature	(C)	51.36	51.36	51.35	51.35	
38	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
39	Heat Transferred	(kJ/h-m)	568.9	568.9	568.8	568.8	
40	Flow Regime		Transition	Transition	Transition	Transition	
41	Water Flow Regime						
42	Liquid HoldUp		0.2359	0.2415	0.2415	0.2415	
43	Water HoldUp		---	---	---	---	
44	Friction Gradient	(kPa/m)	7.969e-003	7.280e-003	7.280e-003	7.280e-003	
45	Static Gradient	(kPa/m)	1.606e-003	5.717e-003	5.717e-003	5.717e-003	
46	Accel. Gradient	(kPa/m)	1.158e-006	1.572e-006	1.572e-006	1.572e-006	
47	Liq. Reynolds		1.889e+005 *	1.889e+005 *	1.889e+005 *	1.889e+005 *	
48	Vap. Reynolds		9.794e+006 *	9.794e+006 *	9.794e+006 *	9.794e+006 *	
49	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
50	Vapour Velocity	(m/s)	2.876	2.876	2.876	2.876	
51	Deposit Thickness	(mm)	---	---	---	---	
52	Deposit Volume	(m3)	---	---	---	---	
53	Length	(m)	1.722e+004	1.723e+004	1.723e+004	1.724e+004	
54	Elevation	(m)	4.600e-002	6.000e-002	6.400e-002	6.800e-002	
55	Pressure	(bar)	70.11	70.11	70.11	70.11	
56	Temperature	(C)	51.35	51.35	51.35	51.34	
57	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
58	Heat Transferred	(kJ/h-m)	568.7	568.7	568.6	568.6	
59	Flow Regime		Transition	Transition	Transition	Transition	
60	Water Flow Regime						
61	Liquid HoldUp		0.2415	0.2415	0.2359	0.2359	
62	Water HoldUp		---	---	---	---	
63	Friction Gradient	(kPa/m)	7.280e-003	7.280e-003	7.969e-003	7.969e-003	
64	Static Gradient	(kPa/m)	5.717e-003	5.717e-003	1.606e-003	1.606e-003	
65	Accel. Gradient	(kPa/m)	1.572e-006	1.572e-006	1.158e-006	1.158e-006	
66	Liq. Reynolds		1.889e+005 *	1.889e+005 *	1.889e+005 *	1.889e+005 *	
67	Vap. Reynolds		9.794e+006 *	9.794e+006 *	9.794e+006 *	9.794e+006 *	
68	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
5			Date/Time: Mon Jun 13 15:13:43 2022				
6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.876	2.876	2.876	2.876	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	1.724e+004	1.725e+004	1.725e+004	1.726e+004	
17	Elevation	(m)	7.200e-002	7.600e-002	8.000e-002	8.800e-002	
18	Pressure	(bar)	70.11	70.11	70.11	70.11	
19	Temperature	(C)	51.34	51.34	51.34	51.34	
20	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
21	Heat Transferred	(kJ/h-m)	568.5	568.5	568.4	568.4	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2359	0.2359	0.2359	0.2381	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	7.969e-003	7.969e-003	7.969e-003	7.687e-003	
27	Static Gradient	(kPa/m)	1.606e-003	1.606e-003	1.606e-003	3.234e-003	
28	Accel. Gradient	(kPa/m)	1.158e-006	1.158e-006	1.158e-006	1.321e-006	
29	Liq. Reynolds		1.889e+005 *	1.889e+005 *	1.889e+005 *	1.889e+005 *	
30	Vap. Reynolds		9.794e+006 *	9.794e+006 *	9.794e+006 *	9.794e+006 *	
31	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
32	Vapour Velocity	(m/s)	2.876	2.876	2.876	2.876	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	1.726e+004	1.727e+004	1.727e+004	1.728e+004	
36	Elevation	(m)	9.600e-002	0.1040	0.1120	0.1200	
37	Pressure	(bar)	70.11	70.11	70.11	70.11	
38	Temperature	(C)	51.34	51.33	51.33	51.33	
39	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
40	Heat Transferred	(kJ/h-m)	568.3	568.3	568.3	568.2	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2381	0.2381	0.2381	0.2381	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	7.687e-003	7.687e-003	7.687e-003	7.687e-003	
46	Static Gradient	(kPa/m)	3.234e-003	3.234e-003	3.234e-003	3.234e-003	
47	Accel. Gradient	(kPa/m)	1.321e-006	1.321e-006	1.321e-006	1.321e-006	
48	Liq. Reynolds		1.889e+005 *	1.889e+005 *	1.889e+005 *	1.889e+005 *	
49	Vap. Reynolds		9.794e+006 *	9.794e+006 *	9.794e+006 *	9.794e+006 *	
50	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
51	Vapour Velocity	(m/s)	2.876	2.876	2.876	2.876	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	1.728e+004	1.729e+004	1.729e+004	1.730e+004	
55	Elevation	(m)	0.1200	0.1200	0.1200	0.1200	
56	Pressure	(bar)	70.11	70.11	70.11	70.11	
57	Temperature	(C)	51.33	51.33	51.32	51.32	
58	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
59	Heat Transferred	(kJ/h-m)	568.2	568.1	568.1	568.0	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.2529	0.2529	0.2529	0.2529	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	6.013e-003	6.013e-003	6.013e-003	6.013e-003	
65	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000	
66	Accel. Gradient	(kPa/m)	7.274e-007	7.274e-007	7.274e-007	7.274e-007	
67	Liq. Reynolds		1.889e+005 *	1.889e+005 *	1.889e+005 *	1.889e+005 *	
68	Vap. Reynolds		9.794e+006 *	9.794e+006 *	9.794e+006 *	9.794e+006 *	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 305 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
14	Vapour Velocity	(m/s)	2.876	2.876	2.876	2.876	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	1.730e+004	1.731e+004	1.731e+004	1.732e+004	
18	Elevation	(m)	0.1200	0.1240	0.1280	0.1320	
19	Pressure	(bar)	70.11	70.11	70.10	70.10	
20	Temperature	(C)	51.32	51.32	51.32	51.32	
21	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
22	Heat Transferred	(kJ/h-m)	568.0	567.9	567.9	567.8	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.2529	0.2359	0.2359	0.2359	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	6.013e-003	7.969e-003	7.969e-003	7.969e-003	
28	Static Gradient	(kPa/m)	0.0000	1.606e-003	1.606e-003	1.606e-003	
29	Accel. Gradient	(kPa/m)	7.274e-007	1.158e-006	1.158e-006	1.158e-006	
30	Liq. Reynolds		1.889e+005 *	1.889e+005 *	1.889e+005 *	1.889e+005 *	
31	Vap. Reynolds		9.794e+006 *	9.794e+006 *	9.794e+006 *	9.794e+006 *	
32	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
33	Vapour Velocity	(m/s)	2.876	2.876	2.876	2.876	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	1.732e+004	1.733e+004	1.733e+004	1.734e+004	
37	Elevation	(m)	0.1360	0.1400	0.1400	0.1400	
38	Pressure	(bar)	70.10	70.10	70.10	70.10	
39	Temperature	(C)	51.31	51.31	51.31	51.31	
40	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
41	Heat Transferred	(kJ/h-m)	567.8	567.7	567.7	567.6	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2359	0.2359	0.2529	0.2529	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	7.969e-003	7.969e-003	6.013e-003	6.013e-003	
47	Static Gradient	(kPa/m)	1.606e-003	1.606e-003	0.0000	0.0000	
48	Accel. Gradient	(kPa/m)	1.158e-006	1.158e-006	7.274e-007	7.274e-007	
49	Liq. Reynolds		1.889e+005 *	1.889e+005 *	1.889e+005 *	1.889e+005 *	
50	Vap. Reynolds		9.794e+006 *	9.794e+006 *	9.794e+006 *	9.794e+006 *	
51	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
52	Vapour Velocity	(m/s)	2.876	2.876	2.876	2.876	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	1.734e+004	1.735e+004	1.735e+004	1.736e+004	
56	Elevation	(m)	0.1400	0.1400	0.1400	0.1440	
57	Pressure	(bar)	70.10	70.10	70.10	70.10	
58	Temperature	(C)	51.31	51.31	51.30	51.30	
59	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
60	Heat Transferred	(kJ/h-m)	567.6	567.5	567.5	567.5	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2529	0.2529	0.2529	0.2359	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	6.013e-003	6.013e-003	6.013e-003	7.968e-003	
66	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	1.606e-003	
67	Accel. Gradient	(kPa/m)	7.274e-007	7.274e-007	7.274e-007	1.158e-006	
68	Liq. Reynolds		1.889e+005 *	1.889e+005 *	1.889e+005 *	1.889e+005 *	
69	Aspen Technology Inc.		Aspen HYSYS Version 10				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.794e+006 *	9.794e+006 *	9.794e+006 *	9.794e+006 *
14	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
15	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	1.736e+004	1.737e+004	1.737e+004	1.738e+004
19	Elevation (m)	0.1480	0.1520	0.1560	0.1600
20	Pressure (bar)	70.10	70.10	70.10	70.10
21	Temperature (C)	51.30	51.30	51.30	51.29
22	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
23	Heat Transferred (kJ/h-m)	567.4	567.4	567.3	567.3
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2359	0.2359	0.2359	0.2359
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	7.968e-003	7.968e-003	7.968e-003	7.968e-003
29	Static Gradient (kPa/m)	1.606e-003	1.606e-003	1.606e-003	1.606e-003
30	Accel. Gradient (kPa/m)	1.158e-006	1.158e-006	1.158e-006	1.158e-006
31	Liq. Reynolds	1.889e+005 *	1.889e+005 *	1.889e+005 *	1.889e+005 *
32	Vap. Reynolds	9.794e+006 *	9.794e+006 *	9.794e+006 *	9.794e+006 *
33	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
34	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	1.738e+004	1.739e+004	1.739e+004	1.740e+004
38	Elevation (m)	0.1600	0.1600	0.1600	0.1600
39	Pressure (bar)	70.10	70.10	70.10	70.10
40	Temperature (C)	51.29	51.29	51.29	51.29
41	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
42	Heat Transferred (kJ/h-m)	567.2	567.2	567.1	567.1
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2529	0.2529	0.2529	0.2529
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.013e-003	6.013e-003	6.013e-003	6.013e-003
48	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
49	Accel. Gradient (kPa/m)	7.274e-007	7.274e-007	7.274e-007	7.274e-007
50	Liq. Reynolds	1.889e+005 *	1.889e+005 *	1.889e+005 *	1.889e+005 *
51	Vap. Reynolds	9.794e+006 *	9.794e+006 *	9.794e+006 *	9.794e+006 *
52	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
53	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1.740e+004	1.741e+004	1.741e+004	1.742e+004
57	Elevation (m)	0.1600	0.1560	0.1520	0.1480
58	Pressure (bar)	70.10	70.10	70.10	70.10
59	Temperature (C)	51.29	51.28	51.28	51.28
60	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
61	Heat Transferred (kJ/h-m)	567.0	567.0	567.0	566.9
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2529	0.1726	0.1726	0.1726
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	6.013e-003	8.895e-003	8.895e-003	8.895e-003
67	Static Gradient (kPa/m)	0.0000	-1.291e-003	-1.291e-003	-1.291e-003
68	Accel. Gradient (kPa/m)	7.274e-007	9.199e-007	9.199e-007	9.199e-007
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Liq. Reynolds		1.889e+005 *	1.889e+005 *	1.889e+005 *	1.889e+005 *				
14	Vap. Reynolds		9.794e+006 *	9.794e+006 *	9.794e+006 *	9.795e+006 *				
15	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981					
16	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876					
17	Deposit Thickness (mm)	---	---	---	---					
18	Deposit Volume (m3)	---	---	---	---					
19	Length (m)	1.742e+004	1.743e+004	1.743e+004	1.744e+004					
20	Elevation (m)	0.1440	0.1400	0.1340	0.1280					
21	Pressure (bar)	70.10	70.10	70.10	70.10					
22	Temperature (C)	51.28	51.28	51.28	51.27					
23	Vapour Fraction	0.8814	0.8814	0.8814	0.8814					
24	Heat Transferred (kJ/h-m)	566.9	566.9	566.8	566.8					
25	Flow Regime	Transition	Transition	Transition	Transition					
26	Water Flow Regime									
27	Liquid HoldUp	0.1726	0.1726	0.1723	0.1723					
28	Water HoldUp	---	---	---	---					
29	Friction Gradient (kPa/m)	8.895e-003	8.895e-003	8.899e-003	8.899e-003					
30	Static Gradient (kPa/m)	-1.291e-003	-1.291e-003	-1.934e-003	-1.934e-003					
31	Accel. Gradient (kPa/m)	9.199e-007	9.199e-007	8.426e-007	8.426e-007					
32	Liq. Reynolds	1.889e+005 *	1.889e+005 *	1.888e+005 *	1.888e+005 *					
33	Vap. Reynolds	9.795e+006 *	9.795e+006 *	9.795e+006 *	9.795e+006 *					
34	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981					
35	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876					
36	Deposit Thickness (mm)	---	---	---	---					
37	Deposit Volume (m3)	---	---	---	---					
38	Length (m)	1.744e+004	1.745e+004	1.745e+004	1.746e+004					
39	Elevation (m)	0.1220	0.1160	0.1100	0.1100					
40	Pressure (bar)	70.10	70.09	70.09	70.09					
41	Temperature (C)	51.27	51.27	51.27	51.27					
42	Vapour Fraction	0.8814	0.8814	0.8814	0.8814					
43	Heat Transferred (kJ/h-m)	566.7	566.7	566.6	566.5					
44	Flow Regime	Transition	Transition	Transition	Transition					
45	Water Flow Regime									
46	Liquid HoldUp	0.1723	0.1723	0.1723	0.2529					
47	Water HoldUp	---	---	---	---					
48	Friction Gradient (kPa/m)	8.899e-003	8.899e-003	8.899e-003	6.012e-003					
49	Static Gradient (kPa/m)	-1.934e-003	-1.934e-003	-1.934e-003	0.0000					
50	Accel. Gradient (kPa/m)	8.426e-007	8.426e-007	8.426e-007	7.274e-007					
51	Liq. Reynolds	1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *					
52	Vap. Reynolds	9.795e+006 *	9.795e+006 *	9.795e+006 *	9.795e+006 *					
53	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981					
54	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876					
55	Deposit Thickness (mm)	---	---	---	---					
56	Deposit Volume (m3)	---	---	---	---					
57	Length (m)	1.746e+004	1.747e+004	1.747e+004	1.748e+004					
58	Elevation (m)	0.1100	0.1100	0.1100	0.1100					
59	Pressure (bar)	70.09	70.09	70.09	70.09					
60	Temperature (C)	51.27	51.26	51.26	51.26					
61	Vapour Fraction	0.8814	0.8814	0.8814	0.8814					
62	Heat Transferred (kJ/h-m)	566.5	566.4	566.4	566.3					
63	Flow Regime	Transition	Transition	Transition	Transition					
64	Water Flow Regime									
65	Liquid HoldUp	0.2529	0.2529	0.2529	0.2529					
66	Water HoldUp	---	---	---	---					
67	Friction Gradient (kPa/m)	6.012e-003	6.012e-003	6.012e-003	6.012e-003					
68	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000					
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	7.274e-007	7.274e-007	7.274e-007	7.274e-007
14	Liq. Reynolds	1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *
15	Vap. Reynolds	9.795e+006 *	9.795e+006 *	9.795e+006 *	9.795e+006 *
16	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
17	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	1.748e+004	1.749e+004	1.749e+004	1.750e+004
21	Elevation (m)	9.800e-002	8.600e-002	7.400e-002	6.200e-002
22	Pressure (bar)	70.09	70.09	70.09	70.09
23	Temperature (C)	51.26	51.26	51.26	51.25
24	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
25	Heat Transferred (kJ/h-m)	566.4	566.3	566.3	566.2
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.1713	0.1713	0.1713	0.1713
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	8.912e-003	8.912e-003	8.912e-003	8.912e-003
31	Static Gradient (kPa/m)	-4.184e-003	-4.184e-003	-4.184e-003	-4.184e-003
32	Accel. Gradient (kPa/m)	5.720e-007	5.720e-007	5.720e-007	5.720e-007
33	Liq. Reynolds	1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *
34	Vap. Reynolds	9.795e+006 *	9.795e+006 *	9.795e+006 *	9.795e+006 *
35	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
36	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	1.750e+004	1.751e+004	1.751e+004	1.752e+004
40	Elevation (m)	5.000e-002	6.000e-002	7.000e-002	8.000e-002
41	Pressure (bar)	70.09	70.09	70.09	70.09
42	Temperature (C)	51.25	51.25	51.25	51.25
43	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
44	Heat Transferred (kJ/h-m)	566.2	566.1	566.0	566.0
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.1713	0.2393	0.2393	0.2393
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	8.912e-003	7.548e-003	7.548e-003	7.548e-003
50	Static Gradient (kPa/m)	-4.184e-003	4.056e-003	4.056e-003	4.056e-003
51	Accel. Gradient (kPa/m)	5.720e-007	1.404e-006	1.404e-006	1.404e-006
52	Liq. Reynolds	1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *
53	Vap. Reynolds	9.795e+006 *	9.795e+006 *	9.795e+006 *	9.795e+006 *
54	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
55	Vapour Velocity (m/s)	2.876	2.876	2.876	2.876
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	1.752e+004	1.753e+004	1.753e+004	1.754e+004
59	Elevation (m)	9.000e-002	1.000e-001	8.000e-002	6.000e-002
60	Pressure (bar)	70.09	70.09	70.09	70.09
61	Temperature (C)	51.25	51.24	51.24	51.24
62	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
63	Heat Transferred (kJ/h-m)	566.0	565.9	565.9	565.9
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.2393	0.2393	0.1702	0.1702
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	7.548e-003	7.548e-003	8.926e-003	8.926e-003
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1				Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	4.056e-003	4.056e-003	-6.395e-003	-6.395e-003
14	Accel. Gradient	(kPa/m)	1.404e-006	1.404e-006	3.062e-007	3.062e-007
15	Liq. Reynolds		1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *
16	Vap. Reynolds		9.795e+006 *	9.795e+006 *	9.795e+006 *	9.795e+006 *
17	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981
18	Vapour Velocity	(m/s)	2.876	2.876	2.876	2.876
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	1.754e+004	1.755e+004	1.755e+004	1.756e+004
22	Elevation	(m)	4.000e-002	2.000e-002	-9.714e-017	-3.400e-002
23	Pressure	(bar)	70.09	70.09	70.09	70.09
24	Temperature	(C)	51.24	51.24	51.24	51.23
25	Vapour Fraction		0.8814	0.8814	0.8814	0.8814
26	Heat Transferred	(kJ/h-m)	565.8	565.8	565.7	565.7
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.1702	0.1702	0.1702	0.1681
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	8.926e-003	8.926e-003	8.926e-003	8.954e-003
32	Static Gradient	(kPa/m)	-6.395e-003	-6.395e-003	-6.395e-003	-1.078e-002
33	Accel. Gradient	(kPa/m)	3.062e-007	3.061e-007	3.061e-007	-2.213e-007
34	Liq. Reynolds		1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *
35	Vap. Reynolds		9.795e+006 *	9.795e+006 *	9.795e+006 *	9.795e+006 *
36	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981
37	Vapour Velocity	(m/s)	2.876	2.876	2.876	2.876
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	1.756e+004	1.757e+004	1.757e+004	1.758e+004
41	Elevation	(m)	-6.800e-002	-0.1020	-0.1360	-0.1700
42	Pressure	(bar)	70.09	70.09	70.09	70.09
43	Temperature	(C)	51.23	51.23	51.23	51.23
44	Vapour Fraction		0.8814	0.8814	0.8814	0.8814
45	Heat Transferred	(kJ/h-m)	565.7	565.6	565.6	565.5
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.1681	0.1681	0.1681	0.1682
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	8.954e-003	8.954e-003	8.954e-003	8.954e-003
51	Static Gradient	(kPa/m)	-1.078e-002	-1.078e-002	-1.078e-002	-1.078e-002
52	Accel. Gradient	(kPa/m)	-2.213e-007	-2.213e-007	-2.213e-007	-2.213e-007
53	Liq. Reynolds		1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *
54	Vap. Reynolds		9.795e+006 *	9.795e+006 *	9.795e+006 *	9.795e+006 *
55	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981
56	Vapour Velocity	(m/s)	2.876	2.875	2.875	2.875
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	1.758e+004	1.759e+004	1.759e+004	1.760e+004
60	Elevation	(m)	-0.1180	-6.600e-002	-1.400e-002	3.800e-002
61	Pressure	(bar)	70.09	70.09	70.08	70.08
62	Temperature	(C)	51.23	51.22	51.22	51.22
63	Vapour Fraction		0.8814	0.8814	0.8814	0.8814
64	Heat Transferred	(kJ/h-m)	565.4	565.4	565.3	565.3
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2627	0.2627	0.2627	0.2627
68	Water HoldUp		---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:13:43 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	7.591e-003	7.591e-003	7.591e-003	7.591e-003	
14	Static Gradient	(kPa/m)	2.261e-002	2.261e-002	2.261e-002	2.261e-002	
15	Accel. Gradient	(kPa/m)	3.654e-006	3.654e-006	3.654e-006	3.654e-006	
16	Liq. Reynolds		1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *	
17	Vap. Reynolds		9.795e+006 *	9.795e+006 *	9.795e+006 *	9.795e+006 *	
18	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
19	Vapour Velocity	(m/s)	2.875	2.876	2.876	2.876	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	1.760e+004	1.761e+004	1.761e+004	1.762e+004	
23	Elevation	(m)	9.000e-002	7.400e-002	5.800e-002	4.200e-002	
24	Pressure	(bar)	70.08	70.08	70.08	70.08	
25	Temperature	(C)	51.22	51.22	51.21	51.21	
26	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
27	Heat Transferred	(kJ/h-m)	565.2	565.2	565.2	565.1	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.2627	0.1708	0.1708	0.1708	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	7.592e-003	8.918e-003	8.918e-003	8.918e-003	
33	Static Gradient	(kPa/m)	2.261e-002	-5.128e-003	-5.128e-003	-5.128e-003	
34	Accel. Gradient	(kPa/m)	3.655e-006	4.585e-007	4.585e-007	4.585e-007	
35	Liq. Reynolds		1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *	
36	Vap. Reynolds		9.795e+006 *	9.795e+006 *	9.795e+006 *	9.795e+006 *	
37	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
38	Vapour Velocity	(m/s)	2.876	2.876	2.876	2.876	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	1.762e+004	1.763e+004	1.763e+004	1.764e+004	
42	Elevation	(m)	2.600e-002	1.000e-002	6.000e-003	2.000e-003	
43	Pressure	(bar)	70.08	70.08	70.08	70.08	
44	Temperature	(C)	51.21	51.21	51.21	51.21	
45	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
46	Heat Transferred	(kJ/h-m)	565.1	565.0	565.0	564.9	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.1708	0.1708	0.1726	0.1726	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	8.918e-003	8.918e-003	8.894e-003	8.894e-003	
52	Static Gradient	(kPa/m)	-5.128e-003	-5.128e-003	-1.291e-003	-1.291e-003	
53	Accel. Gradient	(kPa/m)	4.585e-007	4.585e-007	9.199e-007	9.199e-007	
54	Liq. Reynolds		1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *	
55	Vap. Reynolds		9.795e+006 *	9.795e+006 *	9.795e+006 *	9.795e+006 *	
56	Liquid Velocity	(m/s)	0.1981	0.1981	0.1981	0.1981	
57	Vapour Velocity	(m/s)	2.876	2.876	2.876	2.876	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	1.764e+004	1.765e+004	1.765e+004	1.766e+004	
61	Elevation	(m)	-2.000e-003	-6.000e-003	-1.000e-002	-1.800e-002	
62	Pressure	(bar)	70.08	70.08	70.08	70.08	
63	Temperature	(C)	51.20	51.20	51.20	51.20	
64	Vapour Fraction		0.8814	0.8814	0.8814	0.8814	
65	Heat Transferred	(kJ/h-m)	564.9	564.8	564.8	564.8	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.1726	0.1726	0.1726	0.1720	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 311 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	8.894e-003	8.894e-003	8.894e-003	8.902e-003
15	Static Gradient (kPa/m)	-1.291e-003	-1.291e-003	-1.291e-003	-2.576e-003
16	Accel. Gradient (kPa/m)	9.199e-007	9.199e-007	9.199e-007	7.654e-007
17	Liq. Reynolds	1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *
18	Vap. Reynolds	9.795e+006 *	9.795e+006 *	9.795e+006 *	9.795e+006 *
19	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
20	Vapour Velocity (m/s)	2.876	2.876	2.876	2.875
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	1.766e+004	1.767e+004	1.767e+004	1.768e+004
24	Elevation (m)	-2.600e-002	-3.400e-002	-4.200e-002	-5.000e-002
25	Pressure (bar)	70.08	70.08	70.08	70.08
26	Temperature (C)	51.20	51.20	51.19	51.19
27	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
28	Heat Transferred (kJ/h-m)	564.7	564.7	564.6	564.6
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.1720	0.1720	0.1720	0.1720
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	8.902e-003	8.902e-003	8.902e-003	8.902e-003
34	Static Gradient (kPa/m)	-2.576e-003	-2.576e-003	-2.576e-003	-2.576e-003
35	Accel. Gradient (kPa/m)	7.654e-007	7.654e-007	7.654e-007	7.654e-007
36	Liq. Reynolds	1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *
37	Vap. Reynolds	9.795e+006 *	9.795e+006 *	9.795e+006 *	9.795e+006 *
38	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
39	Vapour Velocity (m/s)	2.875	2.875	2.875	2.875
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	1.768e+004	1.769e+004	1.769e+004	1.770e+004
43	Elevation (m)	-5.000e-002	-5.000e-002	-5.000e-002	-5.000e-002
44	Pressure (bar)	70.08	70.08	70.08	70.08
45	Temperature (C)	51.19	51.19	51.19	51.18
46	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
47	Heat Transferred (kJ/h-m)	564.5	564.4	564.4	564.3
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2529	0.2529	0.2529	0.2529
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	6.012e-003	6.012e-003	6.011e-003	6.011e-003
53	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
54	Accel. Gradient (kPa/m)	7.274e-007	7.274e-007	7.274e-007	7.274e-007
55	Liq. Reynolds	1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *
56	Vap. Reynolds	9.796e+006 *	9.796e+006 *	9.796e+006 *	9.796e+006 *
57	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
58	Vapour Velocity (m/s)	2.875	2.875	2.875	2.875
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	1.770e+004	1.771e+004	1.771e+004	1.772e+004
62	Elevation (m)	-5.000e-002	-5.200e-002	-5.400e-002	-5.600e-002
63	Pressure (bar)	70.08	70.07	70.07	70.07
64	Temperature (C)	51.18	51.18	51.18	51.18
65	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
66	Heat Transferred (kJ/h-m)	564.3	564.3	564.3	564.2
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.2529	0.1730	0.1730	0.1730
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	6.011e-003	8.890e-003	8.890e-003	8.890e-003
16	Static Gradient (kPa/m)	0.0000	-6.463e-004	-6.463e-004	-6.463e-004
17	Accel. Gradient (kPa/m)	7.274e-007	9.975e-007	9.975e-007	9.975e-007
18	Liq. Reynolds	1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *
19	Vap. Reynolds	9.796e+006 *	9.796e+006 *	9.796e+006 *	9.796e+006 *
20	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
21	Vapour Velocity (m/s)	2.875	2.875	2.875	2.875
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	1.772e+004	1.773e+004	1.773e+004	1.774e+004
25	Elevation (m)	-5.800e-002	-6.000e-002	-8.400e-002	-0.1080
26	Pressure (bar)	70.07	70.07	70.07	70.07
27	Temperature (C)	51.18	51.17	51.17	51.17
28	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
29	Heat Transferred (kJ/h-m)	564.2	564.1	564.1	564.0
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1730	0.1730	0.1697	0.1697
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	8.890e-003	8.890e-003	8.933e-003	8.933e-003
35	Static Gradient (kPa/m)	-6.463e-004	-6.463e-004	-7.657e-003	-7.657e-003
36	Accel. Gradient (kPa/m)	9.975e-007	9.975e-007	1.544e-007	1.544e-007
37	Liq. Reynolds	1.887e+005 *	1.887e+005 *	1.887e+005 *	1.887e+005 *
38	Vap. Reynolds	9.796e+006 *	9.796e+006 *	9.796e+006 *	9.796e+006 *
39	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
40	Vapour Velocity (m/s)	2.875	2.875	2.875	2.875
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	1.774e+004	1.775e+004	1.775e+004	1.776e+004
44	Elevation (m)	-0.1320	-0.1560	-0.1800	-0.1800
45	Pressure (bar)	70.07	70.07	70.07	70.07
46	Temperature (C)	51.17	51.17	51.17	51.16
47	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
48	Heat Transferred (kJ/h-m)	564.0	563.9	563.9	563.8
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1697	0.1697	0.1697	0.2529
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	8.933e-003	8.933e-003	8.933e-003	6.011e-003
54	Static Gradient (kPa/m)	-7.657e-003	-7.657e-003	-7.657e-003	0.0000
55	Accel. Gradient (kPa/m)	1.544e-007	1.544e-007	1.544e-007	7.273e-007
56	Liq. Reynolds	1.887e+005 *	1.887e+005 *	1.887e+005 *	1.887e+005 *
57	Vap. Reynolds	9.796e+006 *	9.796e+006 *	9.796e+006 *	9.796e+006 *
58	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
59	Vapour Velocity (m/s)	2.875	2.875	2.875	2.875
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	1.776e+004	1.777e+004	1.777e+004	1.778e+004
63	Elevation (m)	-0.1800	-0.1800	-0.1800	-0.1800
64	Pressure (bar)	70.07	70.07	70.07	70.07
65	Temperature (C)	51.16	51.16	51.16	51.16
66	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
67	Heat Transferred (kJ/h-m)	563.8	563.7	563.7	563.6
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 313 of 937				

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2529	0.2529	0.2529	0.2529
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.011e-003	6.011e-003	6.011e-003	6.011e-003
17	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
18	Accel. Gradient (kPa/m)	7.273e-007	7.273e-007	7.273e-007	7.273e-007
19	Liq. Reynolds	1.887e+005 *	1.887e+005 *	1.887e+005 *	1.887e+005 *
20	Vap. Reynolds	9.796e+006 *	9.796e+006 *	9.796e+006 *	9.796e+006 *
21	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
22	Vapour Velocity (m/s)	2.875	2.875	2.875	2.875
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	1.778e+004	1.779e+004	1.779e+004	1.780e+004
26	Elevation (m)	-0.2200	-0.2600	-0.3000	-0.3400
27	Pressure (bar)	70.07	70.07	70.07	70.07
28	Temperature (C)	51.16	51.16	51.15	51.15
29	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
30	Heat Transferred (kJ/h-m)	563.6	563.6	563.6	563.5
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1673	0.1673	0.1673	0.1673
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	8.966e-003	8.966e-003	8.966e-003	8.966e-003
36	Static Gradient (kPa/m)	-1.264e-002	-1.264e-002	-1.264e-002	-1.264e-002
37	Accel. Gradient (kPa/m)	-4.449e-007	-4.449e-007	-4.449e-007	-4.449e-007
38	Liq. Reynolds	1.887e+005 *	1.887e+005 *	1.887e+005 *	1.887e+005 *
39	Vap. Reynolds	9.796e+006 *	9.796e+006 *	9.796e+006 *	9.796e+006 *
40	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
41	Vapour Velocity (m/s)	2.875	2.875	2.875	2.875
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	1.780e+004	1.781e+004	1.781e+004	1.782e+004
45	Elevation (m)	-0.3800	-0.3380	-0.2960	-0.2540
46	Pressure (bar)	70.07	70.07	70.07	70.07
47	Temperature (C)	51.15	51.15	51.15	51.14
48	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
49	Heat Transferred (kJ/h-m)	563.5	563.4	563.3	563.3
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.1673	0.2572	0.2572	0.2572
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	8.966e-003	7.157e-003	7.157e-003	7.157e-003
55	Static Gradient (kPa/m)	-1.264e-002	1.797e-002	1.797e-002	1.797e-002
56	Accel. Gradient (kPa/m)	-4.450e-007	3.041e-006	3.041e-006	3.041e-006
57	Liq. Reynolds	1.887e+005 *	1.887e+005 *	1.887e+005 *	1.887e+005 *
58	Vap. Reynolds	9.796e+006 *	9.796e+006 *	9.796e+006 *	9.796e+006 *
59	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
60	Vapour Velocity (m/s)	2.875	2.875	2.875	2.875
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	1.782e+004	1.783e+004	1.783e+004	1.784e+004
64	Elevation (m)	-0.2120	-0.1700	-0.3100	-0.4500
65	Pressure (bar)	70.07	70.07	70.07	70.07
66	Temperature (C)	51.14	51.14	51.14	51.14
67	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
68	Heat Transferred (kJ/h-m)	563.2	563.2	563.2	563.2
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 314 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2572	0.2572	0.1523	0.1523
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	7.157e-003	7.157e-003	9.206e-003	9.205e-003
18	Static Gradient (kPa/m)	1.797e-002	1.797e-002	-4.164e-002	-4.164e-002
19	Accel. Gradient (kPa/m)	3.041e-006	3.041e-006	-3.925e-006	-3.925e-006
20	Liq. Reynolds	1.887e+005 *	1.887e+005 *	1.887e+005 *	1.887e+005 *
21	Vap. Reynolds	9.796e+006 *	9.796e+006 *	9.796e+006 *	9.796e+006 *
22	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
23	Vapour Velocity (m/s)	2.875	2.875	2.875	2.875
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	1.784e+004	1.785e+004	1.785e+004	1.785e+004
27	Elevation (m)	-0.5900	-0.7300	-0.8700	-0.8700
28	Pressure (bar)	70.07	70.07	70.07	70.07
29	Temperature (C)	51.14	51.14	51.14	51.14
30	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
31	Heat Transferred (kJ/h-m)	563.2	563.2	563.1	563.0
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1523	0.1523	0.1523	0.2530
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	9.205e-003	9.205e-003	9.205e-003	6.010e-003
37	Static Gradient (kPa/m)	-4.164e-002	-4.164e-002	-4.165e-002	0.0000
38	Accel. Gradient (kPa/m)	-3.925e-006	-3.925e-006	-3.924e-006	7.270e-007
39	Liq. Reynolds	1.887e+005 *	1.887e+005 *	1.887e+005 *	1.887e+005 *
40	Vap. Reynolds	9.796e+006 *	9.796e+006 *	9.796e+006 *	9.796e+006 *
41	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
42	Vapour Velocity (m/s)	2.875	2.875	2.875	2.875
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	1.786e+004	1.786e+004	1.786e+004	1.787e+004
46	Elevation (m)	-0.8700	-0.8700	-0.8700	-0.8700
47	Pressure (bar)	70.07	70.07	70.07	70.07
48	Temperature (C)	51.13	51.13	51.13	51.13
49	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
50	Heat Transferred (kJ/h-m)	563.0	563.0	562.9	562.9
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2530	0.2530	0.2530	0.2530
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	6.010e-003	6.010e-003	6.010e-003	6.010e-003
56	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
57	Accel. Gradient (kPa/m)	7.270e-007	7.270e-007	7.270e-007	7.270e-007
58	Liq. Reynolds	1.887e+005 *	1.887e+005 *	1.887e+005 *	1.887e+005 *
59	Vap. Reynolds	9.796e+006 *	9.796e+006 *	9.796e+006 *	9.796e+006 *
60	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
61	Vapour Velocity (m/s)	2.875	2.875	2.875	2.875
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	1.787e+004	1.787e+004	1.787e+004	1.787e+004
65	Elevation (m)	-0.9560	-1.042	-1.128	-1.214
66	Pressure (bar)	70.07	70.08	70.08	70.08
67	Temperature (C)	51.13	51.13	51.13	51.13
68	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 315 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Heat Transferred (kJ/h-m)	563.0	563.0	563.0	563.0		
14	Flow Regime	Transition	Transition	Transition	Transition		
15	Water Flow Regime						
16	Liquid HoldUp	0.1411	0.1411	0.1411	0.1411		
17	Water HoldUp	---	---	---	---		
18	Friction Gradient (kPa/m)	9.428e-003	9.428e-003	9.428e-003	9.428e-003		
19	Static Gradient (kPa/m)	-6.096e-002	-6.096e-002	-6.097e-002	-6.097e-002		
20	Accel. Gradient (kPa/m)	-6.234e-006	-6.234e-006	-6.234e-006	-6.234e-006		
21	Liq. Reynolds	1.887e+005 *	1.887e+005 *	1.887e+005 *	1.887e+005 *		
22	Vap. Reynolds	9.796e+006 *	9.796e+006 *	9.796e+006 *	9.796e+006 *		
23	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
24	Vapour Velocity (m/s)	2.875	2.875	2.875	2.874		
25	Deposit Thickness (mm)	---	---	---	---		
26	Deposit Volume (m3)	---	---	---	---		
27	Length (m)	1.788e+004	1.788e+004	1.789e+004	1.789e+004		
28	Elevation (m)	-1.300	-1.900	-2.500	-3.100		
29	Pressure (bar)	70.08	70.08	70.09	70.10		
30	Temperature (C)	51.13	51.13	51.13	51.14		
31	Vapour Fraction	0.8813	0.8813	0.8813	0.8813		
32	Heat Transferred (kJ/h-m)	563.0	563.2	563.3	563.3		
33	Flow Regime	Transition	Transition	Transition	Transition		
34	Water Flow Regime						
35	Liquid HoldUp	0.1411	8.513e-002	8.513e-002	8.513e-002		
36	Water HoldUp	---	---	---	---		
37	Friction Gradient (kPa/m)	9.428e-003	1.181e-002	1.181e-002	1.181e-002		
38	Static Gradient (kPa/m)	-6.097e-002	-0.1283	-0.1284	-0.1284		
39	Accel. Gradient (kPa/m)	-6.234e-006	-1.409e-005	-1.409e-005	-1.409e-005		
40	Liq. Reynolds	1.887e+005 *	1.887e+005 *	1.887e+005 *	1.887e+005 *		
41	Vap. Reynolds	9.796e+006 *	9.795e+006 *	9.795e+006 *	9.795e+006 *		
42	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
43	Vapour Velocity (m/s)	2.874	2.874	2.874	2.874		
44	Deposit Thickness (mm)	---	---	---	---		
45	Deposit Volume (m3)	---	---	---	---		
46	Length (m)	1.790e+004	1.790e+004	1.790e+004	1.791e+004		
47	Elevation (m)	-3.700	-4.300	-4.600	-4.900		
48	Pressure (bar)	70.10	70.11	70.11	70.11		
49	Temperature (C)	51.14	51.14	51.14	51.14		
50	Vapour Fraction	0.8813	0.8813	0.8813	0.8813		
51	Heat Transferred (kJ/h-m)	563.4	563.4	563.6	563.6		
52	Flow Regime	Transition	Transition	Transition	Transition		
53	Water Flow Regime						
54	Liquid HoldUp	8.514e-002	8.514e-002	6.438e-002	6.438e-002		
55	Water HoldUp	---	---	---	---		
56	Friction Gradient (kPa/m)	1.181e-002	1.181e-002	1.408e-002	1.408e-002		
57	Static Gradient (kPa/m)	-0.1284	-0.1284	-0.1411	-0.1411		
58	Accel. Gradient (kPa/m)	-1.409e-005	-1.409e-005	-1.535e-005	-1.535e-005		
59	Liq. Reynolds	1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *		
60	Vap. Reynolds	9.795e+006 *	9.795e+006 *	9.795e+006 *	9.794e+006 *		
61	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981		
62	Vapour Velocity (m/s)	2.873	2.873	2.873	2.873		
63	Deposit Thickness (mm)	---	---	---	---		
64	Deposit Volume (m3)	---	---	---	---		
65	Length (m)	1.791e+004	1.791e+004	1.791e+004	1.792e+004		
66	Elevation (m)	-5.200	-5.500	-5.800	-6.000		
67	Pressure (bar)	70.12	70.12	70.12	70.12		
68	Temperature (C)	51.14	51.14	51.14	51.14		
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
14	Heat Transferred (kJ/h-m)	563.6	563.7	563.7	563.4
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	6.438e-002	6.438e-002	6.438e-002	0.1307
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	1.408e-002	1.408e-002	1.408e-002	9.677e-003
20	Static Gradient (kPa/m)	-0.1411	-0.1411	-0.1411	-7.732e-002
21	Accel. Gradient (kPa/m)	-1.535e-005	-1.535e-005	-1.535e-005	-8.171e-006
22	Liq. Reynolds	1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *
23	Vap. Reynolds	9.794e+006 *	9.794e+006 *	9.794e+006 *	9.794e+006 *
24	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
25	Vapour Velocity (m/s)	2.873	2.873	2.873	2.873
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	1.792e+004	1.792e+004	1.793e+004	1.793e+004
29	Elevation (m)	-6.200	-6.400	-6.600	-6.800
30	Pressure (bar)	70.12	70.13	70.13	70.13
31	Temperature (C)	51.14	51.14	51.14	51.14
32	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
33	Heat Transferred (kJ/h-m)	563.4	563.4	563.4	563.4
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1307	0.1307	0.1307	0.1307
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	9.677e-003	9.677e-003	9.676e-003	9.676e-003
39	Static Gradient (kPa/m)	-7.732e-002	-7.732e-002	-7.733e-002	-7.733e-002
40	Accel. Gradient (kPa/m)	-8.171e-006	-8.170e-006	-8.170e-006	-8.170e-006
41	Liq. Reynolds	1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *
42	Vap. Reynolds	9.794e+006 *	9.794e+006 *	9.794e+006 *	9.794e+006 *
43	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
44	Vapour Velocity (m/s)	2.872	2.872	2.872	2.872
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	1.793e+004	1.794e+004	1.794e+004	1.794e+004
48	Elevation (m)	-7.100	-7.400	-7.700	-8.000
49	Pressure (bar)	70.14	70.14	70.14	70.15
50	Temperature (C)	51.15	51.15	51.15	51.15
51	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
52	Heat Transferred (kJ/h-m)	563.5	563.5	563.5	563.5
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.1097	0.1097	0.1097	0.1097
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	1.038e-002	1.038e-002	1.038e-002	1.038e-002
58	Static Gradient (kPa/m)	-0.1048	-0.1048	-0.1048	-0.1048
59	Accel. Gradient (kPa/m)	-1.140e-005	-1.140e-005	-1.140e-005	-1.140e-005
60	Liq. Reynolds	1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *
61	Vap. Reynolds	9.794e+006 *	9.794e+006 *	9.793e+006 *	9.793e+006 *
62	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
63	Vapour Velocity (m/s)	2.872	2.872	2.872	2.872
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	1.795e+004	1.795e+004	1.796e+004	1.796e+004
67	Elevation (m)	-8.300	-8.600	-8.900	-9.200
68	Pressure (bar)	70.15	70.15	70.16	70.16
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	51.15	51.15	51.15	51.15
14	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
15	Heat Transferred (kJ/h-m)	563.5	563.5	563.5	563.5
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.1097	0.1360	0.1360	0.1360
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	1.038e-002	9.538e-003	9.537e-003	9.537e-003
21	Static Gradient (kPa/m)	-0.1048	-6.932e-002	-6.933e-002	-6.933e-002
22	Accel. Gradient (kPa/m)	-1.140e-005	-7.215e-006	-7.215e-006	-7.215e-006
23	Liq. Reynolds	1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *
24	Vap. Reynolds	9.793e+006 *	9.793e+006 *	9.793e+006 *	9.793e+006 *
25	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
26	Vapour Velocity (m/s)	2.871	2.871	2.871	2.871
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	1.797e+004	1.798e+004	1.798e+004	1.799e+004
30	Elevation (m)	-9.500	-9.800	-10.30	-10.80
31	Pressure (bar)	70.16	70.17	70.17	70.18
32	Temperature (C)	51.15	51.15	51.15	51.15
33	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
34	Heat Transferred (kJ/h-m)	563.5	563.5	563.6	563.6
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1360	0.1360	9.942e-002	9.943e-002
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	9.536e-003	9.536e-003	1.087e-002	1.087e-002
40	Static Gradient (kPa/m)	-6.933e-002	-6.933e-002	-0.1159	-0.1159
41	Accel. Gradient (kPa/m)	-7.214e-006	-7.214e-006	-1.267e-005	-1.267e-005
42	Liq. Reynolds	1.888e+005 *	1.888e+005 *	1.888e+005 *	1.888e+005 *
43	Vap. Reynolds	9.793e+006 *	9.793e+006 *	9.792e+006 *	9.792e+006 *
44	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
45	Vapour Velocity (m/s)	2.871	2.870	2.870	2.870
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	1.799e+004	1.800e+004	1.800e+004	1.801e+004
49	Elevation (m)	-11.30	-11.80	-12.30	-12.80
50	Pressure (bar)	70.18	70.19	70.19	70.20
51	Temperature (C)	51.15	51.15	51.15	51.15
52	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
53	Heat Transferred (kJ/h-m)	563.7	563.7	563.7	563.8
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	9.943e-002	9.943e-002	9.944e-002	9.944e-002
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	1.086e-002	1.086e-002	1.086e-002	1.086e-002
59	Static Gradient (kPa/m)	-0.1159	-0.1159	-0.1159	-0.1159
60	Accel. Gradient (kPa/m)	-1.267e-005	-1.267e-005	-1.267e-005	-1.266e-005
61	Liq. Reynolds	1.888e+005 *	1.889e+005 *	1.889e+005 *	1.889e+005 *
62	Vap. Reynolds	9.792e+006 *	9.792e+006 *	9.792e+006 *	9.792e+006 *
63	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981
64	Vapour Velocity (m/s)	2.870	2.870	2.869	2.869
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	1.801e+004	1.802e+004	1.802e+004	1.803e+004
68	Elevation (m)	-13.30	-13.80	-14.30	-14.80
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Pressure (bar)	70.20	70.21	70.21	70.22					
14	Temperature (C)	51.16	51.16	51.16	51.16					
15	Vapour Fraction	0.8813	0.8813	0.8812	0.8812					
16	Heat Transferred (kJ/h-m)	563.8	563.8	563.9	563.9					
17	Flow Regime	Transition	Transition	Transition	Transition					
18	Water Flow Regime									
19	Liquid HoldUp	9.945e-002	9.945e-002	9.945e-002	9.946e-002					
20	Water HoldUp	---	---	---	---					
21	Friction Gradient (kPa/m)	1.086e-002	1.086e-002	1.086e-002	1.086e-002					
22	Static Gradient (kPa/m)	-0.1159	-0.1159	-0.1160	-0.1160					
23	Accel. Gradient (kPa/m)	-1.266e-005	-1.266e-005	-1.266e-005	-1.266e-005					
24	Liq. Reynolds	1.889e+005 *	1.889e+005 *	1.889e+005 *	1.889e+005 *					
25	Vap. Reynolds	9.791e+006 *	9.791e+006 *	9.791e+006 *	9.791e+006 *					
26	Liquid Velocity (m/s)	0.1981	0.1981	0.1982	0.1982					
27	Vapour Velocity (m/s)	2.869	2.869	2.868	2.868					
28	Deposit Thickness (mm)	---	---	---	---					
29	Deposit Volume (m3)	---	---	---	---					
30	Length (m)	1.803e+004	1.804e+004	1.804e+004	1.805e+004					
31	Elevation (m)	-14.50	-14.20	-13.90	-13.60					
32	Pressure (bar)	70.21	70.20	70.19	70.18					
33	Temperature (C)	51.15	51.15	51.15	51.14					
34	Vapour Fraction	0.8813	0.8813	0.8813	0.8813					
35	Heat Transferred (kJ/h-m)	563.5	563.4	563.3	563.1					
36	Flow Regime	Transition	Transition	Transition	Transition					
37	Water Flow Regime									
38	Liquid HoldUp	0.4010	0.4010	0.4009	0.4009					
39	Water HoldUp	---	---	---	---					
40	Friction Gradient (kPa/m)	7.567e-003	7.567e-003	7.568e-003	7.569e-003					
41	Static Gradient (kPa/m)	0.1821	0.1821	0.1821	0.1821					
42	Accel. Gradient (kPa/m)	2.285e-005	2.285e-005	2.286e-005	2.286e-005					
43	Liq. Reynolds	1.889e+005 *	1.889e+005 *	1.889e+005 *	1.888e+005 *					
44	Vap. Reynolds	9.791e+006 *	9.791e+006 *	9.792e+006 *	9.792e+006 *					
45	Liquid Velocity (m/s)	0.1981	0.1981	0.1981	0.1981					
46	Vapour Velocity (m/s)	2.869	2.869	2.869	2.870					
47	Deposit Thickness (mm)	---	---	---	---					
48	Deposit Volume (m3)	---	---	---	---					
49	Length (m)	1.805e+004	1.806e+004	1.806e+004	1.807e+004					
50	Elevation (m)	-13.30	-10.78	-8.260	-5.740					
51	Pressure (bar)	70.17	70.00	69.83	69.66					
52	Temperature (C)	51.14	51.09	51.04	50.99					
53	Vapour Fraction	0.8813	0.8813	0.8814	0.8815					
54	Heat Transferred (kJ/h-m)	563.0	556.5	555.2	553.9					
55	Flow Regime	Transition	Transition	Transition	Transition					
56	Water Flow Regime									
57	Liquid HoldUp	0.4009	1.000	1.000	1.000					
58	Water HoldUp	---	---	---	---					
59	Friction Gradient (kPa/m)	7.570e-003	7.169e-003	7.185e-003	7.202e-003					
60	Static Gradient (kPa/m)	0.1820	3.408	3.409	3.409					
61	Accel. Gradient (kPa/m)	2.287e-005	4.140e-004	4.161e-004	4.183e-004					
62	Liq. Reynolds	1.888e+005 *	1.886e+005 *	1.884e+005 *	1.881e+005 *					
63	Vap. Reynolds	9.792e+006 *	9.798e+006 *	9.804e+006 *	9.810e+006 *					
64	Liquid Velocity (m/s)	0.1981	0.1981	0.1980	0.1980					
65	Vapour Velocity (m/s)	2.870	2.878	2.885	2.892					
66	Deposit Thickness (mm)	---	---	---	---					
67	Deposit Volume (m3)	---	---	---	---					
68	Length (m)	1.807e+004	1.808e+004	1.808e+004	1.809e+004					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Elevation	(m)	-3.220	-0.7000	-0.6600	-0.6200	
14	Pressure	(bar)	69.49	69.32	69.32	69.32	
15	Temperature	(C)	50.95	50.90	50.90	50.89	
16	Vapour Fraction		0.8816	0.8817	0.8817	0.8817	
17	Heat Transferred	(kJ/h-m)	552.7	551.4	556.7	556.6	
18	Flow Regime		Transition	Transition	Transition	Transition	
19	Water Flow Regime						
20	Liquid HoldUp		1.000	1.000	0.2543	0.2543	
21	Water HoldUp		---	---	---	---	
22	Friction Gradient	(kPa/m)	7.219e-003	7.235e-003	6.925e-003	6.925e-003	
23	Static Gradient	(kPa/m)	3.410	3.410	1.695e-002	1.695e-002	
24	Accel. Gradient	(kPa/m)	4.205e-004	4.227e-004	2.953e-006	2.953e-006	
25	Liq. Reynolds		1.879e+005 *	1.877e+005 *	1.877e+005 *	1.877e+005 *	
26	Vap. Reynolds		9.816e+006 *	9.822e+006 *	9.822e+006 *	9.822e+006 *	
27	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
28	Vapour Velocity	(m/s)	2.900	2.908	2.908	2.908	
29	Deposit Thickness	(mm)	---	---	---	---	
30	Deposit Volume	(m3)	---	---	---	---	
31	Length	(m)	1.809e+004	1.810e+004	1.810e+004	1.811e+004	
32	Elevation	(m)	-0.5800	-0.5400	-0.5000	-0.4600	
33	Pressure	(bar)	69.31	69.31	69.31	69.31	
34	Temperature	(C)	50.89	50.89	50.89	50.89	
35	Vapour Fraction		0.8817	0.8817	0.8817	0.8817	
36	Heat Transferred	(kJ/h-m)	556.5	556.5	556.4	556.4	
37	Flow Regime		Transition	Transition	Transition	Transition	
38	Water Flow Regime						
39	Liquid HoldUp		0.2543	0.2543	0.2543	0.2543	
40	Water HoldUp		---	---	---	---	
41	Friction Gradient	(kPa/m)	6.925e-003	6.925e-003	6.925e-003	6.925e-003	
42	Static Gradient	(kPa/m)	1.695e-002	1.695e-002	1.695e-002	1.695e-002	
43	Accel. Gradient	(kPa/m)	2.953e-006	2.954e-006	2.954e-006	2.954e-006	
44	Liq. Reynolds		1.877e+005 *	1.877e+005 *	1.877e+005 *	1.877e+005 *	
45	Vap. Reynolds		9.822e+006 *	9.822e+006 *	9.822e+006 *	9.822e+006 *	
46	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
47	Vapour Velocity	(m/s)	2.908	2.908	2.908	2.908	
48	Deposit Thickness	(mm)	---	---	---	---	
49	Deposit Volume	(m3)	---	---	---	---	
50	Length	(m)	1.811e+004	1.812e+004	1.812e+004	1.813e+004	
51	Elevation	(m)	-0.4200	-0.3800	-0.3400	-0.3000	
52	Pressure	(bar)	69.31	69.31	69.31	69.31	
53	Temperature	(C)	50.88	50.88	50.88	50.88	
54	Vapour Fraction		0.8817	0.8817	0.8817	0.8817	
55	Heat Transferred	(kJ/h-m)	556.3	556.3	556.2	556.2	
56	Flow Regime		Transition	Transition	Transition	Transition	
57	Water Flow Regime						
58	Liquid HoldUp		0.2543	0.2543	0.2543	0.2543	
59	Water HoldUp		---	---	---	---	
60	Friction Gradient	(kPa/m)	6.925e-003	6.925e-003	6.925e-003	6.925e-003	
61	Static Gradient	(kPa/m)	1.695e-002	1.695e-002	1.695e-002	1.695e-002	
62	Accel. Gradient	(kPa/m)	2.954e-006	2.954e-006	2.954e-006	2.954e-006	
63	Liq. Reynolds		1.877e+005 *	1.877e+005 *	1.877e+005 *	1.877e+005 *	
64	Vap. Reynolds		9.822e+006 *	9.822e+006 *	9.822e+006 *	9.823e+006 *	
65	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
66	Vapour Velocity	(m/s)	2.908	2.908	2.908	2.908	
67	Deposit Thickness	(mm)	---	---	---	---	
68	Deposit Volume	(m3)	---	---	---	---	
69	Aspen Technology Inc.		Aspen HYSYS Version 10				Page 320 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length	(m)	1.813e+004	1.814e+004	1.814e+004	1.815e+004	
14	Elevation	(m)	-0.2920	-0.2840	-0.2760	-0.2680	
15	Pressure	(bar)	69.30	69.30	69.30	69.30	
16	Temperature	(C)	50.88	50.87	50.87	50.87	
17	Vapour Fraction		0.8817	0.8817	0.8817	0.8817	
18	Heat Transferred	(kJ/h-m)	556.1	556.1	556.0	556.0	
19	Flow Regime		Transition	Transition	Transition	Transition	
20	Water Flow Regime						
21	Liquid HoldUp		0.2365	0.2365	0.2365	0.2365	
22	Water HoldUp		---	---	---	---	
23	Friction Gradient	(kPa/m)	7.804e-003	7.804e-003	7.804e-003	7.804e-003	
24	Static Gradient	(kPa/m)	3.213e-003	3.213e-003	3.213e-003	3.213e-003	
25	Accel. Gradient	(kPa/m)	1.363e-006	1.363e-006	1.363e-006	1.363e-006	
26	Liq. Reynolds		1.877e+005 *	1.877e+005 *	1.877e+005 *	1.877e+005 *	
27	Vap. Reynolds		9.823e+006 *	9.823e+006 *	9.823e+006 *	9.823e+006 *	
28	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
29	Vapour Velocity	(m/s)	2.908	2.908	2.908	2.908	
30	Deposit Thickness	(mm)	---	---	---	---	
31	Deposit Volume	(m3)	---	---	---	---	
32	Length	(m)	1.815e+004	1.816e+004	1.816e+004	1.817e+004	
33	Elevation	(m)	-0.2600	-0.2640	-0.2680	-0.2720	
34	Pressure	(bar)	69.30	69.30	69.30	69.30	
35	Temperature	(C)	50.87	50.87	50.87	50.86	
36	Vapour Fraction		0.8816	0.8816	0.8816	0.8816	
37	Heat Transferred	(kJ/h-m)	555.9	555.9	555.9	555.8	
38	Flow Regime		Transition	Transition	Transition	Transition	
39	Water Flow Regime						
40	Liquid HoldUp		0.2365	0.1714	0.1714	0.1714	
41	Water HoldUp		---	---	---	---	
42	Friction Gradient	(kPa/m)	7.804e-003	8.988e-003	8.988e-003	8.988e-003	
43	Static Gradient	(kPa/m)	3.213e-003	-1.282e-003	-1.282e-003	-1.282e-003	
44	Accel. Gradient	(kPa/m)	1.363e-006	9.535e-007	9.535e-007	9.535e-007	
45	Liq. Reynolds		1.877e+005 *	1.877e+005 *	1.877e+005 *	1.876e+005 *	
46	Vap. Reynolds		9.823e+006 *	9.823e+006 *	9.823e+006 *	9.823e+006 *	
47	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
48	Vapour Velocity	(m/s)	2.908	2.908	2.908	2.908	
49	Deposit Thickness	(mm)	---	---	---	---	
50	Deposit Volume	(m3)	---	---	---	---	
51	Length	(m)	1.817e+004	1.818e+004	1.818e+004	1.819e+004	
52	Elevation	(m)	-0.2760	-0.2800	-0.2720	-0.2640	
53	Pressure	(bar)	69.30	69.30	69.30	69.30	
54	Temperature	(C)	50.86	50.86	50.86	50.86	
55	Vapour Fraction		0.8816	0.8816	0.8816	0.8816	
56	Heat Transferred	(kJ/h-m)	555.8	555.8	555.7	555.6	
57	Flow Regime		Transition	Transition	Transition	Transition	
58	Water Flow Regime						
59	Liquid HoldUp		0.1714	0.1714	0.2365	0.2365	
60	Water HoldUp		---	---	---	---	
61	Friction Gradient	(kPa/m)	8.988e-003	8.988e-003	7.804e-003	7.804e-003	
62	Static Gradient	(kPa/m)	-1.282e-003	-1.282e-003	3.213e-003	3.213e-003	
63	Accel. Gradient	(kPa/m)	9.535e-007	9.535e-007	1.363e-006	1.363e-006	
64	Liq. Reynolds		1.876e+005 *	1.876e+005 *	1.876e+005 *	1.876e+005 *	
65	Vap. Reynolds		9.823e+006 *	9.823e+006 *	9.823e+006 *	9.823e+006 *	
66	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
67	Vapour Velocity	(m/s)	2.908	2.908	2.908	2.908	
68	Deposit Thickness	(mm)	---	---	---	---	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	1.819e+004	1.820e+004	1.820e+004	1.821e+004		
15	Elevation (m)	-0.2560	-0.2480	-0.2400	-0.2340		
16	Pressure (bar)	69.30	69.30	69.30	69.30		
17	Temperature (C)	50.86	50.85	50.85	50.85		
18	Vapour Fraction	0.8816	0.8816	0.8816	0.8816		
19	Heat Transferred (kJ/h-m)	555.6	555.5	555.5	555.4		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	0.2365	0.2365	0.2365	0.2354		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	7.804e-003	7.804e-003	7.804e-003	7.946e-003		
25	Static Gradient (kPa/m)	3.213e-003	3.213e-003	3.213e-003	2.401e-003		
26	Accel. Gradient (kPa/m)	1.363e-006	1.363e-006	1.363e-006	1.280e-006		
27	Liq. Reynolds	1.876e+005 *	1.876e+005 *	1.876e+005 *	1.876e+005 *		
28	Vap. Reynolds	9.823e+006 *	9.823e+006 *	9.823e+006 *	9.823e+006 *		
29	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
30	Vapour Velocity (m/s)	2.908	2.908	2.908	2.908		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	1.821e+004	1.822e+004	1.822e+004	1.823e+004		
34	Elevation (m)	-0.2280	-0.2220	-0.2160	-0.2100		
35	Pressure (bar)	69.30	69.30	69.30	69.30		
36	Temperature (C)	50.85	50.85	50.84	50.84		
37	Vapour Fraction	0.8816	0.8816	0.8816	0.8816		
38	Heat Transferred (kJ/h-m)	555.4	555.3	555.3	555.2		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.2354	0.2354	0.2354	0.2354		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	7.946e-003	7.946e-003	7.946e-003	7.946e-003		
44	Static Gradient (kPa/m)	2.401e-003	2.401e-003	2.401e-003	2.401e-003		
45	Accel. Gradient (kPa/m)	1.280e-006	1.280e-006	1.280e-006	1.280e-006		
46	Liq. Reynolds	1.876e+005 *	1.876e+005 *	1.876e+005 *	1.876e+005 *		
47	Vap. Reynolds	9.823e+006 *	9.823e+006 *	9.823e+006 *	9.823e+006 *		
48	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
49	Vapour Velocity (m/s)	2.908	2.908	2.908	2.908		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	1.823e+004	1.824e+004	1.824e+004	1.825e+004		
53	Elevation (m)	-0.2120	-0.2140	-0.2160	-0.2180		
54	Pressure (bar)	69.30	69.29	69.29	69.29		
55	Temperature (C)	50.84	50.84	50.84	50.84		
56	Vapour Fraction	0.8816	0.8816	0.8816	0.8816		
57	Heat Transferred (kJ/h-m)	555.2	555.2	555.1	555.1		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.1717	0.1717	0.1717	0.1717		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	8.984e-003	8.984e-003	8.984e-003	8.984e-003		
63	Static Gradient (kPa/m)	-6.417e-004	-6.417e-004	-6.417e-004	-6.417e-004		
64	Accel. Gradient (kPa/m)	1.032e-006	1.032e-006	1.032e-006	1.032e-006		
65	Liq. Reynolds	1.876e+005 *	1.876e+005 *	1.876e+005 *	1.876e+005 *		
66	Vap. Reynolds	9.823e+006 *	9.823e+006 *	9.823e+006 *	9.823e+006 *		
67	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
68	Vapour Velocity (m/s)	2.908	2.908	2.908	2.908		
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 322 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Deposit Thickness (mm)	---	---	---	---					
14	Deposit Volume (m3)	---	---	---	---					
15	Length (m)	1.825e+004	1.826e+004	1.826e+004	1.827e+004					
16	Elevation (m)	-0.2200	-0.2220	-0.2240	-0.2260					
17	Pressure (bar)	69.29	69.29	69.29	69.29					
18	Temperature (C)	50.83	50.83	50.83	50.83					
19	Vapour Fraction	0.8816	0.8816	0.8816	0.8816					
20	Heat Transferred (kJ/h-m)	555.1	555.0	555.0	554.9					
21	Flow Regime	Transition	Transition	Transition	Transition					
22	Water Flow Regime									
23	Liquid HoldUp	0.1717	0.1717	0.1717	0.1717					
24	Water HoldUp	---	---	---	---					
25	Friction Gradient (kPa/m)	8.984e-003	8.984e-003	8.984e-003	8.984e-003					
26	Static Gradient (kPa/m)	-6.417e-004	-6.417e-004	-6.417e-004	-6.417e-004					
27	Accel. Gradient (kPa/m)	1.032e-006	1.032e-006	1.032e-006	1.032e-006					
28	Liq. Reynolds	1.876e+005 *	1.876e+005 *	1.876e+005 *	1.876e+005 *					
29	Vap. Reynolds	9.823e+006 *	9.823e+006 *	9.823e+006 *	9.823e+006 *					
30	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
31	Vapour Velocity (m/s)	2.908	2.908	2.908	2.908					
32	Deposit Thickness (mm)	---	---	---	---					
33	Deposit Volume (m3)	---	---	---	---					
34	Length (m)	1.827e+004	1.828e+004	1.828e+004	1.829e+004					
35	Elevation (m)	-0.2280	-0.2300	-0.2340	-0.2380					
36	Pressure (bar)	69.29	69.29	69.29	69.29					
37	Temperature (C)	50.83	50.83	50.82	50.82					
38	Vapour Fraction	0.8816	0.8816	0.8816	0.8816					
39	Heat Transferred (kJ/h-m)	554.9	554.8	554.8	554.7					
40	Flow Regime	Transition	Transition	Transition	Transition					
41	Water Flow Regime									
42	Liquid HoldUp	0.1717	0.1717	0.1714	0.1714					
43	Water HoldUp	---	---	---	---					
44	Friction Gradient (kPa/m)	8.984e-003	8.984e-003	8.988e-003	8.988e-003					
45	Static Gradient (kPa/m)	-6.417e-004	-6.417e-004	-1.290e-003	-1.290e-003					
46	Accel. Gradient (kPa/m)	1.032e-006	1.032e-006	9.526e-007	9.526e-007					
47	Liq. Reynolds	1.876e+005 *	1.876e+005 *	1.876e+005 *	1.876e+005 *					
48	Vap. Reynolds	9.823e+006 *	9.823e+006 *	9.823e+006 *	9.823e+006 *					
49	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
50	Vapour Velocity (m/s)	2.908	2.908	2.908	2.908					
51	Deposit Thickness (mm)	---	---	---	---					
52	Deposit Volume (m3)	---	---	---	---					
53	Length (m)	1.829e+004	1.830e+004	1.830e+004	1.831e+004					
54	Elevation (m)	-0.2420	-0.2460	-0.2500	-0.2600					
55	Pressure (bar)	69.29	69.29	69.29	69.29					
56	Temperature (C)	50.82	50.82	50.82	50.82					
57	Vapour Fraction	0.8816	0.8816	0.8816	0.8816					
58	Heat Transferred (kJ/h-m)	554.7	554.6	554.6	554.6					
59	Flow Regime	Transition	Transition	Transition	Transition					
60	Water Flow Regime									
61	Liquid HoldUp	0.1714	0.1714	0.1714	0.1705					
62	Water HoldUp	---	---	---	---					
63	Friction Gradient (kPa/m)	8.988e-003	8.988e-003	8.988e-003	9.000e-003					
64	Static Gradient (kPa/m)	-1.290e-003	-1.290e-003	-1.290e-003	-3.194e-003					
65	Accel. Gradient (kPa/m)	9.526e-007	9.526e-007	9.526e-007	7.185e-007					
66	Liq. Reynolds	1.876e+005 *	1.876e+005 *	1.876e+005 *	1.876e+005 *					
67	Vap. Reynolds	9.823e+006 *	9.823e+006 *	9.823e+006 *	9.823e+006 *					
68	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 323 of 937					

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3						
4						
5			Unit Set: Mixed			
6			Date/Time: Mon Jun 13 15:13:43 2022			
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	PROFILES					
11	Pipe Table					
12						
13	Vapour Velocity (m/s)	2.908	2.908	2.908	2.908	
14	Deposit Thickness (mm)	---	---	---	---	
15	Deposit Volume (m3)	---	---	---	---	
16	Length (m)	1.831e+004	1.832e+004	1.832e+004	1.833e+004	
17	Elevation (m)	-0.2700	-0.2800	-0.2900	-0.3000	
18	Pressure (bar)	69.29	69.29	69.29	69.29	
19	Temperature (C)	50.81	50.81	50.81	50.81	
20	Vapour Fraction	0.8816	0.8816	0.8816	0.8816	
21	Heat Transferred (kJ/h-m)	554.5	554.5	554.4	554.4	
22	Flow Regime	Transition	Transition	Transition	Transition	
23	Water Flow Regime					
24	Liquid HoldUp	0.1705	0.1705	0.1705	0.1705	
25	Water HoldUp	---	---	---	---	
26	Friction Gradient (kPa/m)	9.000e-003	9.000e-003	9.000e-003	9.000e-003	
27	Static Gradient (kPa/m)	-3.194e-003	-3.194e-003	-3.194e-003	-3.194e-003	
28	Accel. Gradient (kPa/m)	7.185e-007	7.184e-007	7.184e-007	7.184e-007	
29	Liq. Reynolds	1.876e+005 *	1.876e+005 *	1.876e+005 *	1.876e+005 *	
30	Vap. Reynolds	9.823e+006 *	9.823e+006 *	9.823e+006 *	9.823e+006 *	
31	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979	
32	Vapour Velocity (m/s)	2.908	2.908	2.908	2.908	
33	Deposit Thickness (mm)	---	---	---	---	
34	Deposit Volume (m3)	---	---	---	---	
35	Length (m)	1.833e+004	1.834e+004	1.834e+004	1.835e+004	
36	Elevation (m)	-0.2960	-0.2920	-0.2880	-0.2840	
37	Pressure (bar)	69.29	69.29	69.29	69.29	
38	Temperature (C)	50.81	50.81	50.80	50.80	
39	Vapour Fraction	0.8816	0.8816	0.8816	0.8816	
40	Heat Transferred (kJ/h-m)	554.3	554.2	554.2	554.1	
41	Flow Regime	Transition	Transition	Transition	Transition	
42	Water Flow Regime					
43	Liquid HoldUp	0.2343	0.2343	0.2343	0.2343	
44	Water HoldUp	---	---	---	---	
45	Friction Gradient (kPa/m)	8.089e-003	8.089e-003	8.089e-003	8.089e-003	
46	Static Gradient (kPa/m)	1.595e-003	1.595e-003	1.595e-003	1.595e-003	
47	Accel. Gradient (kPa/m)	1.198e-006	1.198e-006	1.198e-006	1.198e-006	
48	Liq. Reynolds	1.876e+005 *	1.876e+005 *	1.876e+005 *	1.876e+005 *	
49	Vap. Reynolds	9.823e+006 *	9.823e+006 *	9.823e+006 *	9.823e+006 *	
50	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979	
51	Vapour Velocity (m/s)	2.908	2.908	2.908	2.908	
52	Deposit Thickness (mm)	---	---	---	---	
53	Deposit Volume (m3)	---	---	---	---	
54	Length (m)	1.835e+004	1.836e+004	1.836e+004	1.837e+004	
55	Elevation (m)	-0.2800	-0.2840	-0.2880	-0.2920	
56	Pressure (bar)	69.29	69.29	69.28	69.28	
57	Temperature (C)	50.80	50.80	50.80	50.80	
58	Vapour Fraction	0.8816	0.8816	0.8816	0.8816	
59	Heat Transferred (kJ/h-m)	554.1	554.1	554.1	554.0	
60	Flow Regime	Transition	Transition	Transition	Transition	
61	Water Flow Regime					
62	Liquid HoldUp	0.2343	0.1714	0.1714	0.1714	
63	Water HoldUp	---	---	---	---	
64	Friction Gradient (kPa/m)	8.089e-003	8.988e-003	8.988e-003	8.988e-003	
65	Static Gradient (kPa/m)	1.595e-003	-1.282e-003	-1.282e-003	-1.282e-003	
66	Accel. Gradient (kPa/m)	1.198e-006	9.536e-007	9.536e-007	9.536e-007	
67	Liq. Reynolds	1.876e+005 *	1.876e+005 *	1.876e+005 *	1.876e+005 *	
68	Vap. Reynolds	9.823e+006 *	9.824e+006 *	9.824e+006 *	9.824e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
14	Vapour Velocity	(m/s)	2.908	2.908	2.908	2.908	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	1.837e+004	1.838e+004	1.838e+004	1.839e+004	
18	Elevation	(m)	-0.2960	-0.3000	-0.2960	-0.2920	
19	Pressure	(bar)	69.28	69.28	69.28	69.28	
20	Temperature	(C)	50.79	50.79	50.79	50.79	
21	Vapour Fraction		0.8816	0.8816	0.8816	0.8816	
22	Heat Transferred	(kJ/h-m)	554.0	553.9	553.8	553.8	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.1714	0.1714	0.2343	0.2343	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	8.988e-003	8.988e-003	8.089e-003	8.089e-003	
28	Static Gradient	(kPa/m)	-1.282e-003	-1.282e-003	1.595e-003	1.595e-003	
29	Accel. Gradient	(kPa/m)	9.536e-007	9.536e-007	1.198e-006	1.198e-006	
30	Liq. Reynolds		1.876e+005 *	1.876e+005 *	1.876e+005 *	1.876e+005 *	
31	Vap. Reynolds		9.824e+006 *	9.824e+006 *	9.824e+006 *	9.824e+006 *	
32	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
33	Vapour Velocity	(m/s)	2.908	2.908	2.908	2.908	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	1.839e+004	1.840e+004	1.840e+004	1.841e+004	
37	Elevation	(m)	-0.2880	-0.2840	-0.2800	-0.2780	
38	Pressure	(bar)	69.28	69.28	69.28	69.28	
39	Temperature	(C)	50.79	50.79	50.78	50.78	
40	Vapour Fraction		0.8816	0.8816	0.8816	0.8816	
41	Heat Transferred	(kJ/h-m)	553.7	553.7	553.6	553.6	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2343	0.2343	0.2343	0.2332	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	8.089e-003	8.089e-003	8.089e-003	8.235e-003	
47	Static Gradient	(kPa/m)	1.595e-003	1.595e-003	1.595e-003	7.949e-004	
48	Accel. Gradient	(kPa/m)	1.198e-006	1.198e-006	1.199e-006	1.117e-006	
49	Liq. Reynolds		1.876e+005 *	1.876e+005 *	1.876e+005 *	1.876e+005 *	
50	Vap. Reynolds		9.824e+006 *	9.824e+006 *	9.824e+006 *	9.824e+006 *	
51	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
52	Vapour Velocity	(m/s)	2.908	2.908	2.908	2.908	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	1.841e+004	1.842e+004	1.842e+004	1.843e+004	
56	Elevation	(m)	-0.2760	-0.2740	-0.2720	-0.2700	
57	Pressure	(bar)	69.28	69.28	69.28	69.28	
58	Temperature	(C)	50.78	50.78	50.78	50.77	
59	Vapour Fraction		0.8816	0.8816	0.8816	0.8816	
60	Heat Transferred	(kJ/h-m)	553.6	553.5	553.5	553.4	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2332	0.2332	0.2332	0.2332	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	8.235e-003	8.235e-003	8.235e-003	8.235e-003	
66	Static Gradient	(kPa/m)	7.949e-004	7.949e-004	7.949e-004	7.949e-004	
67	Accel. Gradient	(kPa/m)	1.117e-006	1.118e-006	1.118e-006	1.118e-006	
68	Liq. Reynolds		1.876e+005 *	1.876e+005 *	1.876e+005 *	1.876e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.824e+006 *	9.824e+006 *	9.824e+006 *	9.824e+006 *
14	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
15	Vapour Velocity (m/s)	2.908	2.908	2.908	2.908
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	1.843e+004	1.844e+004	1.844e+004	1.845e+004
19	Elevation (m)	-0.2860	-0.3020	-0.3180	-0.3340
20	Pressure (bar)	69.28	69.28	69.28	69.28
21	Temperature (C)	50.77	50.77	50.77	50.77
22	Vapour Fraction	0.8816	0.8816	0.8816	0.8816
23	Heat Transferred (kJ/h-m)	553.4	553.4	553.3	553.3
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1697	0.1697	0.1697	0.1697
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	9.011e-003	9.011e-003	9.011e-003	9.011e-003
29	Static Gradient (kPa/m)	-5.093e-003	-5.093e-003	-5.093e-003	-5.093e-003
30	Accel. Gradient (kPa/m)	4.850e-007	4.850e-007	4.850e-007	4.850e-007
31	Liq. Reynolds	1.876e+005 *	1.876e+005 *	1.875e+005 *	1.875e+005 *
32	Vap. Reynolds	9.824e+006 *	9.824e+006 *	9.824e+006 *	9.824e+006 *
33	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
34	Vapour Velocity (m/s)	2.908	2.908	2.908	2.908
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	1.845e+004	1.846e+004	1.846e+004	1.847e+004
38	Elevation (m)	-0.3500	-0.3400	-0.3300	-0.3200
39	Pressure (bar)	69.28	69.28	69.28	69.28
40	Temperature (C)	50.77	50.76	50.76	50.76
41	Vapour Fraction	0.8816	0.8816	0.8816	0.8816
42	Heat Transferred (kJ/h-m)	553.2	553.2	553.1	553.1
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1697	0.2376	0.2376	0.2376
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	9.011e-003	7.663e-003	7.663e-003	7.663e-003
48	Static Gradient (kPa/m)	-5.093e-003	4.030e-003	4.030e-003	4.030e-003
49	Accel. Gradient (kPa/m)	4.850e-007	1.447e-006	1.447e-006	1.447e-006
50	Liq. Reynolds	1.875e+005 *	1.875e+005 *	1.875e+005 *	1.875e+005 *
51	Vap. Reynolds	9.824e+006 *	9.824e+006 *	9.824e+006 *	9.824e+006 *
52	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
53	Vapour Velocity (m/s)	2.908	2.908	2.908	2.908
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1.847e+004	1.848e+004	1.848e+004	1.849e+004
57	Elevation (m)	-0.3100	-0.3000	-0.2980	-0.2960
58	Pressure (bar)	69.28	69.28	69.27	69.27
59	Temperature (C)	50.76	50.76	50.76	50.75
60	Vapour Fraction	0.8816	0.8816	0.8816	0.8816
61	Heat Transferred (kJ/h-m)	553.0	553.0	552.9	552.9
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2376	0.2376	0.2332	0.2332
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	7.663e-003	7.663e-003	8.235e-003	8.235e-003
67	Static Gradient (kPa/m)	4.030e-003	4.030e-003	7.937e-004	7.937e-004
68	Accel. Gradient (kPa/m)	1.447e-006	1.447e-006	1.117e-006	1.117e-006
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	1.875e+005 *	1.875e+005 *	1.875e+005 *	1.875e+005 *
14	Vap. Reynolds	9.824e+006 *	9.824e+006 *	9.824e+006 *	9.824e+006 *
15	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
16	Vapour Velocity (m/s)	2.908	2.908	2.908	2.908
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	1.849e+004	1.850e+004	1.850e+004	1.851e+004
20	Elevation (m)	-0.2940	-0.2920	-0.2900	-0.2840
21	Pressure (bar)	69.27	69.27	69.27	69.27
22	Temperature (C)	50.75	50.75	50.75	50.75
23	Vapour Fraction	0.8816	0.8816	0.8816	0.8816
24	Heat Transferred (kJ/h-m)	552.8	552.8	552.7	552.7
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2332	0.2332	0.2332	0.2354
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	8.235e-003	8.235e-003	8.235e-003	7.945e-003
30	Static Gradient (kPa/m)	7.937e-004	7.937e-004	7.937e-004	2.402e-003
31	Accel. Gradient (kPa/m)	1.117e-006	1.117e-006	1.117e-006	1.281e-006
32	Liq. Reynolds	1.875e+005 *	1.875e+005 *	1.875e+005 *	1.875e+005 *
33	Vap. Reynolds	9.824e+006 *	9.824e+006 *	9.824e+006 *	9.824e+006 *
34	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
35	Vapour Velocity (m/s)	2.908	2.908	2.907	2.907
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	1.851e+004	1.852e+004	1.852e+004	1.853e+004
39	Elevation (m)	-0.2780	-0.2720	-0.2660	-0.2600
40	Pressure (bar)	69.27	69.27	69.27	69.27
41	Temperature (C)	50.75	50.74	50.74	50.74
42	Vapour Fraction	0.8816	0.8816	0.8816	0.8816
43	Heat Transferred (kJ/h-m)	552.6	552.6	552.5	552.5
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2354	0.2354	0.2354	0.2354
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	7.945e-003	7.945e-003	7.945e-003	7.945e-003
49	Static Gradient (kPa/m)	2.402e-003	2.402e-003	2.402e-003	2.402e-003
50	Accel. Gradient (kPa/m)	1.281e-006	1.281e-006	1.281e-006	1.281e-006
51	Liq. Reynolds	1.875e+005 *	1.875e+005 *	1.875e+005 *	1.875e+005 *
52	Vap. Reynolds	9.824e+006 *	9.824e+006 *	9.824e+006 *	9.824e+006 *
53	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
54	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	1.853e+004	1.854e+004	1.854e+004	1.855e+004
58	Elevation (m)	-0.2760	-0.2920	-0.3080	-0.3240
59	Pressure (bar)	69.27	69.27	69.27	69.27
60	Temperature (C)	50.74	50.74	50.74	50.73
61	Vapour Fraction	0.8816	0.8816	0.8816	0.8816
62	Heat Transferred (kJ/h-m)	552.5	552.5	552.4	552.4
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.1697	0.1697	0.1697	0.1697
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	9.011e-003	9.011e-003	9.011e-003	9.011e-003
68	Static Gradient (kPa/m)	-5.093e-003	-5.093e-003	-5.093e-003	-5.093e-003
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	4.850e-007	4.850e-007	4.850e-007	4.850e-007
14	Liq. Reynolds	1.875e+005 *	1.875e+005 *	1.875e+005 *	1.875e+005 *
15	Vap. Reynolds	9.824e+006 *	9.824e+006 *	9.824e+006 *	9.824e+006 *
16	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
17	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	1.855e+004	1.856e+004	1.856e+004	1.857e+004
21	Elevation (m)	-0.3400	-0.3380	-0.3360	-0.3340
22	Pressure (bar)	69.27	69.27	69.27	69.27
23	Temperature (C)	50.73	50.73	50.73	50.73
24	Vapour Fraction	0.8816	0.8816	0.8816	0.8816
25	Heat Transferred (kJ/h-m)	552.3	552.2	552.2	552.1
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.1697	0.2332	0.2332	0.2332
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	9.011e-003	8.234e-003	8.234e-003	8.234e-003
31	Static Gradient (kPa/m)	-5.093e-003	7.950e-004	7.950e-004	7.950e-004
32	Accel. Gradient (kPa/m)	4.849e-007	1.118e-006	1.118e-006	1.118e-006
33	Liq. Reynolds	1.875e+005 *	1.875e+005 *	1.875e+005 *	1.875e+005 *
34	Vap. Reynolds	9.824e+006 *	9.824e+006 *	9.824e+006 *	9.824e+006 *
35	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
36	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	1.857e+004	1.858e+004	1.858e+004	1.859e+004
40	Elevation (m)	-0.3320	-0.3300	-0.3400	-0.3500
41	Pressure (bar)	69.27	69.27	69.27	69.27
42	Temperature (C)	50.73	50.72	50.72	50.72
43	Vapour Fraction	0.8816	0.8816	0.8816	0.8816
44	Heat Transferred (kJ/h-m)	552.1	552.1	552.1	552.0
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.2332	0.2332	0.1706	0.1706
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	8.234e-003	8.234e-003	8.999e-003	8.999e-003
50	Static Gradient (kPa/m)	7.950e-004	7.950e-004	-3.194e-003	-3.194e-003
51	Accel. Gradient (kPa/m)	1.118e-006	1.118e-006	7.185e-007	7.185e-007
52	Liq. Reynolds	1.875e+005 *	1.875e+005 *	1.875e+005 *	1.875e+005 *
53	Vap. Reynolds	9.824e+006 *	9.824e+006 *	9.824e+006 *	9.824e+006 *
54	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
55	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	1.859e+004	1.860e+004	1.860e+004	1.861e+004
59	Elevation (m)	-0.3600	-0.3700	-0.3800	-0.4000
60	Pressure (bar)	69.27	69.27	69.27	69.27
61	Temperature (C)	50.72	50.72	50.72	50.71
62	Vapour Fraction	0.8816	0.8816	0.8816	0.8816
63	Heat Transferred (kJ/h-m)	552.0	551.9	551.9	551.8
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.1706	0.1706	0.1706	0.1691
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	8.999e-003	8.999e-003	8.999e-003	9.019e-003
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:13:43 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Static Gradient	(kPa/m)	-3.194e-003	-3.194e-003	-3.194e-003	-6.351e-003	
14	Accel. Gradient	(kPa/m)	7.185e-007	7.185e-007	7.184e-007	3.302e-007	
15	Liq. Reynolds		1.875e+005 *	1.875e+005 *	1.875e+005 *	1.875e+005 *	
16	Vap. Reynolds		9.824e+006 *	9.824e+006 *	9.824e+006 *	9.824e+006 *	
17	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
18	Vapour Velocity	(m/s)	2.907	2.907	2.907	2.907	
19	Deposit Thickness	(mm)	---	---	---	---	
20	Deposit Volume	(m3)	---	---	---	---	
21	Length	(m)	1.861e+004	1.862e+004	1.862e+004	1.863e+004	
22	Elevation	(m)	-0.4200	-0.4400	-0.4600	-0.4800	
23	Pressure	(bar)	69.27	69.27	69.26	69.26	
24	Temperature	(C)	50.71	50.71	50.71	50.71	
25	Vapour Fraction		0.8816	0.8816	0.8816	0.8816	
26	Heat Transferred	(kJ/h-m)	551.8	551.8	551.7	551.7	
27	Flow Regime		Transition	Transition	Transition	Transition	
28	Water Flow Regime						
29	Liquid HoldUp		0.1691	0.1691	0.1691	0.1691	
30	Water HoldUp		---	---	---	---	
31	Friction Gradient	(kPa/m)	9.019e-003	9.019e-003	9.019e-003	9.019e-003	
32	Static Gradient	(kPa/m)	-6.351e-003	-6.352e-003	-6.352e-003	-6.352e-003	
33	Accel. Gradient	(kPa/m)	3.301e-007	3.301e-007	3.301e-007	3.301e-007	
34	Liq. Reynolds		1.875e+005 *	1.875e+005 *	1.875e+005 *	1.875e+005 *	
35	Vap. Reynolds		9.824e+006 *	9.825e+006 *	9.825e+006 *	9.825e+006 *	
36	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
37	Vapour Velocity	(m/s)	2.907	2.907	2.907	2.907	
38	Deposit Thickness	(mm)	---	---	---	---	
39	Deposit Volume	(m3)	---	---	---	---	
40	Length	(m)	1.863e+004	1.864e+004	1.864e+004	1.865e+004	
41	Elevation	(m)	-0.4920	-0.5040	-0.5160	-0.5280	
42	Pressure	(bar)	69.26	69.26	69.26	69.26	
43	Temperature	(C)	50.71	50.70	50.70	50.70	
44	Vapour Fraction		0.8816	0.8816	0.8816	0.8816	
45	Heat Transferred	(kJ/h-m)	551.6	551.6	551.5	551.5	
46	Flow Regime		Transition	Transition	Transition	Transition	
47	Water Flow Regime						
48	Liquid HoldUp		0.1703	0.1703	0.1703	0.1703	
49	Water HoldUp		---	---	---	---	
50	Friction Gradient	(kPa/m)	9.003e-003	9.003e-003	9.002e-003	9.002e-003	
51	Static Gradient	(kPa/m)	-3.829e-003	-3.829e-003	-3.829e-003	-3.829e-003	
52	Accel. Gradient	(kPa/m)	6.404e-007	6.404e-007	6.403e-007	6.403e-007	
53	Liq. Reynolds		1.875e+005 *	1.875e+005 *	1.875e+005 *	1.875e+005 *	
54	Vap. Reynolds		9.825e+006 *	9.825e+006 *	9.825e+006 *	9.825e+006 *	
55	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
56	Vapour Velocity	(m/s)	2.907	2.907	2.907	2.907	
57	Deposit Thickness	(mm)	---	---	---	---	
58	Deposit Volume	(m3)	---	---	---	---	
59	Length	(m)	1.865e+004	1.866e+004	1.866e+004	1.867e+004	
60	Elevation	(m)	-0.5400	-0.5360	-0.5320	-0.5280	
61	Pressure	(bar)	69.26	69.26	69.26	69.26	
62	Temperature	(C)	50.70	50.70	50.70	50.69	
63	Vapour Fraction		0.8816	0.8816	0.8816	0.8816	
64	Heat Transferred	(kJ/h-m)	551.5	551.4	551.3	551.3	
65	Flow Regime		Transition	Transition	Transition	Transition	
66	Water Flow Regime						
67	Liquid HoldUp		0.1703	0.2343	0.2343	0.2343	
68	Water HoldUp		---	---	---	---	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 329 of 937	

1				Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	9.002e-003	8.088e-003	8.088e-003	8.088e-003
14	Static Gradient	(kPa/m)	-3.829e-003	1.596e-003	1.596e-003	1.596e-003
15	Accel. Gradient	(kPa/m)	6.403e-007	1.199e-006	1.199e-006	1.199e-006
16	Liq. Reynolds		1.875e+005 *	1.875e+005 *	1.875e+005 *	1.875e+005 *
17	Vap. Reynolds		9.825e+006 *	9.825e+006 *	9.825e+006 *	9.825e+006 *
18	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
19	Vapour Velocity	(m/s)	2.907	2.907	2.907	2.907
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	1.867e+004	1.868e+004	1.868e+004	1.869e+004
23	Elevation	(m)	-0.5240	-0.5200	-0.5760	-0.6320
24	Pressure	(bar)	69.26	69.26	69.26	69.26
25	Temperature	(C)	50.69	50.69	50.69	50.69
26	Vapour Fraction		0.8816	0.8816	0.8816	0.8816
27	Heat Transferred	(kJ/h-m)	551.2	551.2	551.2	551.2
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2343	0.2343	0.1637	0.1637
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	8.088e-003	8.088e-003	9.095e-003	9.095e-003
33	Static Gradient	(kPa/m)	1.596e-003	1.596e-003	-1.740e-002	-1.740e-002
34	Accel. Gradient	(kPa/m)	1.199e-006	1.199e-006	-1.028e-006	-1.029e-006
35	Liq. Reynolds		1.875e+005 *	1.875e+005 *	1.875e+005 *	1.875e+005 *
36	Vap. Reynolds		9.825e+006 *	9.825e+006 *	9.825e+006 *	9.825e+006 *
37	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
38	Vapour Velocity	(m/s)	2.907	2.907	2.907	2.907
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	1.869e+004	1.870e+004	1.870e+004	1.871e+004
42	Elevation	(m)	-0.6880	-0.7440	-0.8000	-0.7500
43	Pressure	(bar)	69.26	69.26	69.26	69.26
44	Temperature	(C)	50.69	50.69	50.68	50.68
45	Vapour Fraction		0.8816	0.8816	0.8816	0.8816
46	Heat Transferred	(kJ/h-m)	551.1	551.1	551.1	550.9
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.1637	0.1637	0.1637	0.2599
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	9.095e-003	9.095e-003	9.094e-003	7.606e-003
52	Static Gradient	(kPa/m)	-1.740e-002	-1.741e-002	-1.741e-002	2.154e-002
53	Accel. Gradient	(kPa/m)	-1.029e-006	-1.029e-006	-1.029e-006	3.607e-006
54	Liq. Reynolds		1.875e+005 *	1.875e+005 *	1.875e+005 *	1.875e+005 *
55	Vap. Reynolds		9.825e+006 *	9.825e+006 *	9.825e+006 *	9.825e+006 *
56	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
57	Vapour Velocity	(m/s)	2.907	2.907	2.907	2.907
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	1.871e+004	1.872e+004	1.872e+004	1.873e+004
61	Elevation	(m)	-0.7000	-0.6500	-0.6000	-0.5500
62	Pressure	(bar)	69.26	69.26	69.26	69.26
63	Temperature	(C)	50.68	50.68	50.68	50.67
64	Vapour Fraction		0.8816	0.8816	0.8816	0.8816
65	Heat Transferred	(kJ/h-m)	550.9	550.8	550.8	550.7
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2599	0.2599	0.2599	0.2599
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	7.606e-003	7.606e-003	7.606e-003	7.606e-003
15	Static Gradient (kPa/m)	2.154e-002	2.154e-002	2.154e-002	2.154e-002
16	Accel. Gradient (kPa/m)	3.607e-006	3.608e-006	3.608e-006	3.608e-006
17	Liq. Reynolds	1.875e+005 *	1.875e+005 *	1.875e+005 *	1.874e+005 *
18	Vap. Reynolds	9.825e+006 *	9.825e+006 *	9.825e+006 *	9.825e+006 *
19	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
20	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	1.873e+004	1.874e+004	1.874e+004	1.875e+004
24	Elevation (m)	-0.5360	-0.5220	-0.5080	-0.4940
25	Pressure (bar)	69.26	69.25	69.25	69.25
26	Temperature (C)	50.67	50.67	50.67	50.67
27	Vapour Fraction	0.8816	0.8816	0.8816	0.8816
28	Heat Transferred (kJ/h-m)	550.7	550.6	550.6	550.5
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2399	0.2399	0.2399	0.2399
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	7.389e-003	7.389e-003	7.389e-003	7.389e-003
34	Static Gradient (kPa/m)	5.682e-003	5.682e-003	5.682e-003	5.682e-003
35	Accel. Gradient (kPa/m)	1.618e-006	1.618e-006	1.618e-006	1.618e-006
36	Liq. Reynolds	1.874e+005 *	1.874e+005 *	1.874e+005 *	1.874e+005 *
37	Vap. Reynolds	9.825e+006 *	9.825e+006 *	9.825e+006 *	9.825e+006 *
38	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
39	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	1.875e+004	1.876e+004	1.876e+004	1.877e+004
43	Elevation (m)	-0.4800	-0.5020	-0.5240	-0.5460
44	Pressure (bar)	69.25	69.25	69.25	69.25
45	Temperature (C)	50.66	50.66	50.66	50.66
46	Vapour Fraction	0.8816	0.8816	0.8816	0.8816
47	Heat Transferred (kJ/h-m)	550.5	550.5	550.4	550.4
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2399	0.1688	0.1688	0.1688
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	7.389e-003	9.023e-003	9.023e-003	9.023e-003
53	Static Gradient (kPa/m)	5.682e-003	-6.979e-003	-6.979e-003	-6.979e-003
54	Accel. Gradient (kPa/m)	1.618e-006	2.530e-007	2.530e-007	2.530e-007
55	Liq. Reynolds	1.874e+005 *	1.874e+005 *	1.874e+005 *	1.874e+005 *
56	Vap. Reynolds	9.825e+006 *	9.825e+006 *	9.825e+006 *	9.825e+006 *
57	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
58	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	1.877e+004	1.878e+004	1.878e+004	1.879e+004
62	Elevation (m)	-0.5680	-0.5900	-0.6080	-0.6260
63	Pressure (bar)	69.25	69.25	69.25	69.25
64	Temperature (C)	50.66	50.66	50.66	50.65
65	Vapour Fraction	0.8816	0.8815	0.8815	0.8815
66	Heat Transferred (kJ/h-m)	550.4	550.3	550.3	550.2
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 331 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.1688	0.1688	0.1694	0.1694
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	9.023e-003	9.022e-003	9.014e-003	9.014e-003
16	Static Gradient (kPa/m)	-6.979e-003	-6.979e-003	-5.723e-003	-5.723e-003
17	Accel. Gradient (kPa/m)	2.530e-007	2.530e-007	4.074e-007	4.073e-007
18	Liq. Reynolds	1.874e+005 *	1.874e+005 *	1.874e+005 *	1.874e+005 *
19	Vap. Reynolds	9.825e+006 *	9.825e+006 *	9.825e+006 *	9.825e+006 *
20	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
21	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	1.879e+004	1.880e+004	1.880e+004	1.880e+004
25	Elevation (m)	-0.6440	-0.6620	-0.6800	-0.6520
26	Pressure (bar)	69.25	69.25	69.25	69.25
27	Temperature (C)	50.65	50.65	50.65	50.65
28	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
29	Heat Transferred (kJ/h-m)	550.2	550.2	550.1	550.0
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1694	0.1694	0.1694	0.2580
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	9.014e-003	9.014e-003	9.014e-003	7.489e-003
35	Static Gradient (kPa/m)	-5.724e-003	-5.724e-003	-5.724e-003	2.000e-002
36	Accel. Gradient (kPa/m)	4.073e-007	4.073e-007	4.073e-007	3.402e-006
37	Liq. Reynolds	1.874e+005 *	1.874e+005 *	1.874e+005 *	1.874e+005 *
38	Vap. Reynolds	9.825e+006 *	9.825e+006 *	9.825e+006 *	9.825e+006 *
39	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
40	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	1.881e+004	1.881e+004	1.881e+004	1.882e+004
44	Elevation (m)	-0.6240	-0.5960	-0.5680	-0.5400
45	Pressure (bar)	69.25	69.25	69.25	69.25
46	Temperature (C)	50.65	50.65	50.64	50.64
47	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
48	Heat Transferred (kJ/h-m)	550.0	550.0	549.9	549.9
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2580	0.2580	0.2580	0.2580
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	7.489e-003	7.489e-003	7.489e-003	7.489e-003
54	Static Gradient (kPa/m)	2.000e-002	2.000e-002	2.000e-002	2.000e-002
55	Accel. Gradient (kPa/m)	3.402e-006	3.403e-006	3.403e-006	3.403e-006
56	Liq. Reynolds	1.874e+005 *	1.874e+005 *	1.874e+005 *	1.874e+005 *
57	Vap. Reynolds	9.825e+006 *	9.825e+006 *	9.825e+006 *	9.825e+006 *
58	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
59	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	1.882e+004	1.882e+004	1.882e+004	1.882e+004
63	Elevation (m)	-0.6040	-0.6680	-0.7320	-0.7960
64	Pressure (bar)	69.25	69.25	69.25	69.25
65	Temperature (C)	50.64	50.64	50.64	50.64
66	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
67	Heat Transferred (kJ/h-m)	550.0	550.0	550.0	550.0
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 332 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1483	0.1483	0.1483	0.1483
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	9.358e-003	9.358e-003	9.358e-003	9.358e-003
17	Static Gradient (kPa/m)	-4.667e-002	-4.667e-002	-4.667e-002	-4.667e-002
18	Accel. Gradient (kPa/m)	-4.619e-006	-4.619e-006	-4.619e-006	-4.619e-006
19	Liq. Reynolds	1.874e+005 *	1.874e+005 *	1.874e+005 *	1.874e+005 *
20	Vap. Reynolds	9.825e+006 *	9.825e+006 *	9.825e+006 *	9.825e+006 *
21	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
22	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	1.883e+004	1.883e+004	1.884e+004	1.884e+004
26	Elevation (m)	-0.8600	-0.7620	-0.6640	-0.5660
27	Pressure (bar)	69.25	69.25	69.25	69.24
28	Temperature (C)	50.64	50.64	50.64	50.63
29	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
30	Heat Transferred (kJ/h-m)	549.9	549.8	549.7	549.7
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1483	0.2865	0.2865	0.2865
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	9.358e-003	7.879e-003	7.880e-003	7.880e-003
36	Static Gradient (kPa/m)	-4.667e-002	4.548e-002	4.548e-002	4.548e-002
37	Accel. Gradient (kPa/m)	-4.619e-006	6.605e-006	6.606e-006	6.606e-006
38	Liq. Reynolds	1.874e+005 *	1.874e+005 *	1.874e+005 *	1.874e+005 *
39	Vap. Reynolds	9.825e+006 *	9.825e+006 *	9.825e+006 *	9.826e+006 *
40	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
41	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	1.885e+004	1.885e+004	1.886e+004	1.886e+004
45	Elevation (m)	-0.4680	-0.3700	-0.4320	-0.4940
46	Pressure (bar)	69.24	69.24	69.24	69.24
47	Temperature (C)	50.63	50.63	50.63	50.63
48	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
49	Heat Transferred (kJ/h-m)	549.6	549.5	549.5	549.5
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2865	0.2865	0.1628	0.1628
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	7.880e-003	7.880e-003	9.109e-003	9.109e-003
55	Static Gradient (kPa/m)	4.548e-002	4.548e-002	-1.921e-002	-1.921e-002
56	Accel. Gradient (kPa/m)	6.606e-006	6.607e-006	-1.250e-006	-1.250e-006
57	Liq. Reynolds	1.874e+005 *	1.874e+005 *	1.874e+005 *	1.874e+005 *
58	Vap. Reynolds	9.826e+006 *	9.826e+006 *	9.826e+006 *	9.826e+006 *
59	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
60	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	1.887e+004	1.887e+004	1.888e+004	1.888e+004
64	Elevation (m)	-0.5560	-0.6180	-0.6800	-0.6540
65	Pressure (bar)	69.24	69.24	69.24	69.24
66	Temperature (C)	50.62	50.62	50.62	50.62
67	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
68	Heat Transferred (kJ/h-m)	549.5	549.4	549.4	549.3
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.1628	0.1628	0.1628	0.2466
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	9.109e-003	9.109e-003	9.109e-003	6.611e-003
18	Static Gradient (kPa/m)	-1.921e-002	-1.921e-002	-1.921e-002	1.078e-002
19	Accel. Gradient (kPa/m)	-1.250e-006	-1.250e-006	-1.250e-006	2.153e-006
20	Liq. Reynolds	1.874e+005 *	1.874e+005 *	1.874e+005 *	1.874e+005 *
21	Vap. Reynolds	9.826e+006 *	9.826e+006 *	9.826e+006 *	9.826e+006 *
22	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
23	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	1.889e+004	1.889e+004	1.890e+004	1.890e+004
27	Elevation (m)	-0.6280	-0.6020	-0.5760	-0.5500
28	Pressure (bar)	69.24	69.24	69.24	69.24
29	Temperature (C)	50.62	50.62	50.61	50.61
30	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
31	Heat Transferred (kJ/h-m)	549.2	549.2	549.1	549.1
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2465	0.2465	0.2465	0.2465
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	6.611e-003	6.611e-003	6.611e-003	6.611e-003
37	Static Gradient (kPa/m)	1.078e-002	1.078e-002	1.078e-002	1.078e-002
38	Accel. Gradient (kPa/m)	2.153e-006	2.153e-006	2.153e-006	2.153e-006
39	Liq. Reynolds	1.874e+005 *	1.874e+005 *	1.874e+005 *	1.874e+005 *
40	Vap. Reynolds	9.826e+006 *	9.826e+006 *	9.826e+006 *	9.826e+006 *
41	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
42	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	1.890e+004	1.891e+004	1.891e+004	1.891e+004
46	Elevation (m)	-0.5600	-0.5700	-0.5800	-0.5900
47	Pressure (bar)	69.24	69.24	69.24	69.24
48	Temperature (C)	50.61	50.61	50.61	50.61
49	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
50	Heat Transferred (kJ/h-m)	549.1	549.1	549.1	549.1
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1683	0.1683	0.1683	0.1683
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	9.029e-003	9.029e-003	9.029e-003	9.029e-003
56	Static Gradient (kPa/m)	-7.917e-003	-7.917e-003	-7.917e-003	-7.917e-003
57	Accel. Gradient (kPa/m)	1.377e-007	1.377e-007	1.377e-007	1.377e-007
58	Liq. Reynolds	1.874e+005 *	1.874e+005 *	1.874e+005 *	1.874e+005 *
59	Vap. Reynolds	9.826e+006 *	9.826e+006 *	9.826e+006 *	9.826e+006 *
60	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
61	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	1.891e+004	1.891e+004	1.892e+004	1.892e+004
65	Elevation (m)	-0.6000	-0.5920	-0.5840	-0.5760
66	Pressure (bar)	69.24	69.24	69.23	69.23
67	Temperature (C)	50.61	50.61	50.61	50.61
68	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Heat Transferred (kJ/h-m)	549.1	549.0	549.0	548.9					
14	Flow Regime	Transition	Transition	Transition	Transition					
15	Water Flow Regime									
16	Liquid HoldUp	0.1683	0.2395	0.2395	0.2395					
17	Water HoldUp	---	---	---	---					
18	Friction Gradient (kPa/m)	9.029e-003	7.434e-003	7.434e-003	7.434e-003					
19	Static Gradient (kPa/m)	-7.917e-003	5.405e-003	5.405e-003	5.405e-003					
20	Accel. Gradient (kPa/m)	1.377e-007	1.590e-006	1.590e-006	1.590e-006					
21	Liq. Reynolds	1.874e+005 *	1.874e+005 *	1.874e+005 *	1.874e+005 *					
22	Vap. Reynolds	9.826e+006 *	9.826e+006 *	9.826e+006 *	9.826e+006 *					
23	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
24	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907					
25	Deposit Thickness (mm)	---	---	---	---					
26	Deposit Volume (m3)	---	---	---	---					
27	Length (m)	1.892e+004	1.893e+004	1.893e+004	1.894e+004					
28	Elevation (m)	-0.5680	-0.5600	-0.5880	-0.6160					
29	Pressure (bar)	69.23	69.23	69.23	69.23					
30	Temperature (C)	50.61	50.60	50.60	50.60					
31	Vapour Fraction	0.8815	0.8815	0.8815	0.8815					
32	Heat Transferred (kJ/h-m)	548.9	548.9	548.9	548.8					
33	Flow Regime	Transition	Transition	Transition	Transition					
34	Water Flow Regime									
35	Liquid HoldUp	0.2395	0.2395	0.1679	0.1679					
36	Water HoldUp	---	---	---	---					
37	Friction Gradient (kPa/m)	7.434e-003	7.434e-003	9.035e-003	9.035e-003					
38	Static Gradient (kPa/m)	5.405e-003	5.405e-003	-8.851e-003	-8.851e-003					
39	Accel. Gradient (kPa/m)	1.590e-006	1.590e-006	2.277e-008	2.275e-008					
40	Liq. Reynolds	1.874e+005 *	1.874e+005 *	1.874e+005 *	1.874e+005 *					
41	Vap. Reynolds	9.826e+006 *	9.826e+006 *	9.826e+006 *	9.826e+006 *					
42	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
43	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907					
44	Deposit Thickness (mm)	---	---	---	---					
45	Deposit Volume (m3)	---	---	---	---					
46	Length (m)	1.894e+004	1.895e+004	1.895e+004	1.896e+004					
47	Elevation (m)	-0.6440	-0.6720	-0.7000	-0.7560					
48	Pressure (bar)	69.23	69.23	69.23	69.23					
49	Temperature (C)	50.60	50.60	50.60	50.60					
50	Vapour Fraction	0.8815	0.8815	0.8815	0.8815					
51	Heat Transferred (kJ/h-m)	548.8	548.8	548.7	548.7					
52	Flow Regime	Transition	Transition	Transition	Transition					
53	Water Flow Regime									
54	Liquid HoldUp	0.1679	0.1679	0.1679	0.1637					
55	Water HoldUp	---	---	---	---					
56	Friction Gradient (kPa/m)	9.035e-003	9.035e-003	9.035e-003	9.095e-003					
57	Static Gradient (kPa/m)	-8.851e-003	-8.851e-003	-8.851e-003	-1.741e-002					
58	Accel. Gradient (kPa/m)	2.274e-008	2.272e-008	2.271e-008	-1.030e-006					
59	Liq. Reynolds	1.874e+005 *	1.874e+005 *	1.874e+005 *	1.874e+005 *					
60	Vap. Reynolds	9.826e+006 *	9.826e+006 *	9.826e+006 *	9.826e+006 *					
61	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
62	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907					
63	Deposit Thickness (mm)	---	---	---	---					
64	Deposit Volume (m3)	---	---	---	---					
65	Length (m)	1.896e+004	1.897e+004	1.897e+004	1.898e+004					
66	Elevation (m)	-0.8120	-0.8680	-0.9240	-0.9800					
67	Pressure (bar)	69.23	69.23	69.24	69.24					
68	Temperature (C)	50.59	50.59	50.59	50.59					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Fraction		0.8815	0.8815	0.8815	0.8815	
14	Heat Transferred (kJ/h-m)		548.7	548.6	548.6	548.6	
15	Flow Regime		Transition	Transition	Transition	Transition	
16	Water Flow Regime						
17	Liquid HoldUp		0.1637	0.1637	0.1637	0.1637	
18	Water HoldUp		---	---	---	---	
19	Friction Gradient (kPa/m)		9.095e-003	9.095e-003	9.095e-003	9.094e-003	
20	Static Gradient (kPa/m)		-1.741e-002	-1.741e-002	-1.741e-002	-1.741e-002	
21	Accel. Gradient (kPa/m)		-1.030e-006	-1.030e-006	-1.030e-006	-1.030e-006	
22	Liq. Reynolds		1.874e+005 *	1.874e+005 *	1.874e+005 *	1.874e+005 *	
23	Vap. Reynolds		9.826e+006 *	9.826e+006 *	9.826e+006 *	9.826e+006 *	
24	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979	
25	Vapour Velocity (m/s)		2.907	2.907	2.907	2.907	
26	Deposit Thickness (mm)		---	---	---	---	
27	Deposit Volume (m3)		---	---	---	---	
28	Length (m)		1.898e+004	1.899e+004	1.899e+004	1.900e+004	
29	Elevation (m)		-0.9260	-0.8720	-0.8180	-0.7640	
30	Pressure (bar)		69.23	69.23	69.23	69.23	
31	Temperature (C)		50.59	50.59	50.58	50.58	
32	Vapour Fraction		0.8815	0.8815	0.8815	0.8815	
33	Heat Transferred (kJ/h-m)		548.4	548.4	548.3	548.3	
34	Flow Regime		Transition	Transition	Transition	Transition	
35	Water Flow Regime						
36	Liquid HoldUp		0.2621	0.2621	0.2621	0.2621	
37	Water HoldUp		---	---	---	---	
38	Friction Gradient (kPa/m)		7.695e-003	7.695e-003	7.696e-003	7.696e-003	
39	Static Gradient (kPa/m)		2.342e-002	2.342e-002	2.342e-002	2.341e-002	
40	Accel. Gradient (kPa/m)		3.852e-006	3.852e-006	3.852e-006	3.852e-006	
41	Liq. Reynolds		1.874e+005 *	1.874e+005 *	1.874e+005 *	1.873e+005 *	
42	Vap. Reynolds		9.826e+006 *	9.826e+006 *	9.826e+006 *	9.826e+006 *	
43	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979	
44	Vapour Velocity (m/s)		2.907	2.907	2.907	2.907	
45	Deposit Thickness (mm)		---	---	---	---	
46	Deposit Volume (m3)		---	---	---	---	
47	Length (m)		1.900e+004	1.901e+004	1.901e+004	1.902e+004	
48	Elevation (m)		-0.7100	-0.6900	-0.6700	-0.6500	
49	Pressure (bar)		69.23	69.23	69.23	69.23	
50	Temperature (C)		50.58	50.58	50.58	50.57	
51	Vapour Fraction		0.8815	0.8815	0.8815	0.8815	
52	Heat Transferred (kJ/h-m)		548.2	548.2	548.1	548.1	
53	Flow Regime		Transition	Transition	Transition	Transition	
54	Water Flow Regime						
55	Liquid HoldUp		0.2621	0.2432	0.2432	0.2432	
56	Water HoldUp		---	---	---	---	
57	Friction Gradient (kPa/m)		7.696e-003	6.992e-003	6.993e-003	6.993e-003	
58	Static Gradient (kPa/m)		2.341e-002	8.201e-003	8.201e-003	8.201e-003	
59	Accel. Gradient (kPa/m)		3.852e-006	1.881e-006	1.881e-006	1.881e-006	
60	Liq. Reynolds		1.873e+005 *	1.873e+005 *	1.873e+005 *	1.873e+005 *	
61	Vap. Reynolds		9.826e+006 *	9.826e+006 *	9.826e+006 *	9.826e+006 *	
62	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979	
63	Vapour Velocity (m/s)		2.907	2.907	2.907	2.907	
64	Deposit Thickness (mm)		---	---	---	---	
65	Deposit Volume (m3)		---	---	---	---	
66	Length (m)		1.902e+004	1.903e+004	1.903e+004	1.904e+004	
67	Elevation (m)		-0.6300	-0.6100	-0.5960	-0.5820	
68	Pressure (bar)		69.22	69.22	69.22	69.22	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	50.57	50.57	50.57	50.57
14	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
15	Heat Transferred (kJ/h-m)	548.0	548.0	547.9	547.9
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2432	0.2432	0.2399	0.2399
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	6.993e-003	6.993e-003	7.389e-003	7.389e-003
21	Static Gradient (kPa/m)	8.201e-003	8.201e-003	5.682e-003	5.682e-003
22	Accel. Gradient (kPa/m)	1.881e-006	1.882e-006	1.619e-006	1.619e-006
23	Liq. Reynolds	1.873e+005 *	1.873e+005 *	1.873e+005 *	1.873e+005 *
24	Vap. Reynolds	9.826e+006 *	9.826e+006 *	9.826e+006 *	9.826e+006 *
25	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
26	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	1.904e+004	1.905e+004	1.905e+004	1.906e+004
30	Elevation (m)	-0.5680	-0.5540	-0.5400	-0.5460
31	Pressure (bar)	69.22	69.22	69.22	69.22
32	Temperature (C)	50.56	50.56	50.56	50.56
33	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
34	Heat Transferred (kJ/h-m)	547.8	547.8	547.7	547.7
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2399	0.2399	0.2399	0.1712
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	7.389e-003	7.389e-003	7.389e-003	8.991e-003
40	Static Gradient (kPa/m)	5.682e-003	5.682e-003	5.682e-003	-1.921e-003
41	Accel. Gradient (kPa/m)	1.619e-006	1.619e-006	1.619e-006	8.755e-007
42	Liq. Reynolds	1.873e+005 *	1.873e+005 *	1.873e+005 *	1.873e+005 *
43	Vap. Reynolds	9.826e+006 *	9.827e+006 *	9.827e+006 *	9.827e+006 *
44	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
45	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	1.906e+004	1.907e+004	1.907e+004	1.908e+004
49	Elevation (m)	-0.5520	-0.5580	-0.5640	-0.5700
50	Pressure (bar)	69.22	69.22	69.22	69.22
51	Temperature (C)	50.56	50.56	50.55	50.55
52	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
53	Heat Transferred (kJ/h-m)	547.7	547.6	547.6	547.5
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.1712	0.1712	0.1712	0.1712
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	8.991e-003	8.991e-003	8.991e-003	8.991e-003
59	Static Gradient (kPa/m)	-1.921e-003	-1.921e-003	-1.921e-003	-1.921e-003
60	Accel. Gradient (kPa/m)	8.755e-007	8.755e-007	8.755e-007	8.755e-007
61	Liq. Reynolds	1.873e+005 *	1.873e+005 *	1.873e+005 *	1.873e+005 *
62	Vap. Reynolds	9.827e+006 *	9.827e+006 *	9.827e+006 *	9.827e+006 *
63	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
64	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	1.908e+004	1.909e+004	1.909e+004	1.910e+004
68	Elevation (m)	-0.5460	-0.5220	-0.4980	-0.4740
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Pressure (bar)	69.22	69.22	69.22	69.22					
14	Temperature (C)	50.55	50.55	50.55	50.55					
15	Vapour Fraction	0.8815	0.8815	0.8815	0.8815					
16	Heat Transferred (kJ/h-m)	547.5	547.4	547.4	547.3					
17	Flow Regime	Transition	Transition	Transition	Transition					
18	Water Flow Regime									
19	Liquid HoldUp	0.2455	0.2455	0.2455	0.2455					
20	Water HoldUp	---	---	---	---					
21	Friction Gradient (kPa/m)	6.728e-003	6.728e-003	6.728e-003	6.728e-003					
22	Static Gradient (kPa/m)	9.974e-003	9.974e-003	9.974e-003	9.974e-003					
23	Accel. Gradient (kPa/m)	2.068e-006	2.069e-006	2.069e-006	2.069e-006					
24	Liq. Reynolds	1.873e+005 *	1.873e+005 *	1.873e+005 *	1.873e+005 *					
25	Vap. Reynolds	9.827e+006 *	9.827e+006 *	9.827e+006 *	9.827e+006 *					
26	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
27	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907					
28	Deposit Thickness (mm)	---	---	---	---					
29	Deposit Volume (m3)	---	---	---	---					
30	Length (m)	1.910e+004	1.911e+004	1.911e+004	1.912e+004					
31	Elevation (m)	-0.4500	-0.4480	-0.4460	-0.4440					
32	Pressure (bar)	69.22	69.21	69.21	69.21					
33	Temperature (C)	50.54	50.54	50.54	50.54					
34	Vapour Fraction	0.8815	0.8815	0.8815	0.8815					
35	Heat Transferred (kJ/h-m)	547.3	547.2	547.2	547.1					
36	Flow Regime	Transition	Transition	Transition	Transition					
37	Water Flow Regime									
38	Liquid HoldUp	0.2455	0.2332	0.2332	0.2332					
39	Water HoldUp	---	---	---	---					
40	Friction Gradient (kPa/m)	6.728e-003	8.234e-003	8.234e-003	8.234e-003					
41	Static Gradient (kPa/m)	9.974e-003	7.951e-004	7.951e-004	7.951e-004					
42	Accel. Gradient (kPa/m)	2.069e-006	1.118e-006	1.118e-006	1.118e-006					
43	Liq. Reynolds	1.873e+005 *	1.873e+005 *	1.873e+005 *	1.873e+005 *					
44	Vap. Reynolds	9.827e+006 *	9.827e+006 *	9.827e+006 *	9.827e+006 *					
45	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
46	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907					
47	Deposit Thickness (mm)	---	---	---	---					
48	Deposit Volume (m3)	---	---	---	---					
49	Length (m)	1.912e+004	1.913e+004	1.913e+004	1.914e+004					
50	Elevation (m)	-0.4420	-0.4400	-0.4420	-0.4440					
51	Pressure (bar)	69.21	69.21	69.21	69.21					
52	Temperature (C)	50.54	50.53	50.53	50.53					
53	Vapour Fraction	0.8815	0.8815	0.8815	0.8815					
54	Heat Transferred (kJ/h-m)	547.1	547.0	547.0	547.0					
55	Flow Regime	Transition	Transition	Transition	Transition					
56	Water Flow Regime									
57	Liquid HoldUp	0.2332	0.2332	0.1718	0.1718					
58	Water HoldUp	---	---	---	---					
59	Friction Gradient (kPa/m)	8.234e-003	8.234e-003	8.983e-003	8.983e-003					
60	Static Gradient (kPa/m)	7.951e-004	7.951e-004	-6.419e-004	-6.419e-004					
61	Accel. Gradient (kPa/m)	1.118e-006	1.118e-006	1.033e-006	1.033e-006					
62	Liq. Reynolds	1.873e+005 *	1.873e+005 *	1.873e+005 *	1.873e+005 *					
63	Vap. Reynolds	9.827e+006 *	9.827e+006 *	9.827e+006 *	9.827e+006 *					
64	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
65	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907					
66	Deposit Thickness (mm)	---	---	---	---					
67	Deposit Volume (m3)	---	---	---	---					
68	Length (m)	1.914e+004	1.915e+004	1.915e+004	1.916e+004					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Elevation (m)	-0.4460	-0.4480	-0.4500	-0.4560					
14	Pressure (bar)	69.21	69.21	69.21	69.21					
15	Temperature (C)	50.53	50.53	50.53	50.52					
16	Vapour Fraction	0.8815	0.8815	0.8815	0.8815					
17	Heat Transferred (kJ/h-m)	546.9	546.9	546.8	546.8					
18	Flow Regime	Transition	Transition	Transition	Transition					
19	Water Flow Regime									
20	Liquid HoldUp	0.1718	0.1718	0.1718	0.1712					
21	Water HoldUp	---	---	---	---					
22	Friction Gradient (kPa/m)	8.983e-003	8.983e-003	8.983e-003	8.991e-003					
23	Static Gradient (kPa/m)	-6.419e-004	-6.419e-004	-6.419e-004	-1.921e-003					
24	Accel. Gradient (kPa/m)	1.033e-006	1.033e-006	1.033e-006	8.756e-007					
25	Liq. Reynolds	1.873e+005 *	1.873e+005 *	1.873e+005 *	1.873e+005 *					
26	Vap. Reynolds	9.827e+006 *	9.827e+006 *	9.827e+006 *	9.827e+006 *					
27	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
28	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907					
29	Deposit Thickness (mm)	---	---	---	---					
30	Deposit Volume (m3)	---	---	---	---					
31	Length (m)	1.916e+004	1.917e+004	1.917e+004	1.918e+004					
32	Elevation (m)	-0.4620	-0.4680	-0.4740	-0.4800					
33	Pressure (bar)	69.21	69.21	69.21	69.21					
34	Temperature (C)	50.52	50.52	50.52	50.52					
35	Vapour Fraction	0.8815	0.8815	0.8815	0.8815					
36	Heat Transferred (kJ/h-m)	546.8	546.7	546.7	546.6					
37	Flow Regime	Transition	Transition	Transition	Transition					
38	Water Flow Regime									
39	Liquid HoldUp	0.1712	0.1712	0.1712	0.1712					
40	Water HoldUp	---	---	---	---					
41	Friction Gradient (kPa/m)	8.991e-003	8.991e-003	8.991e-003	8.991e-003					
42	Static Gradient (kPa/m)	-1.921e-003	-1.921e-003	-1.921e-003	-1.921e-003					
43	Accel. Gradient (kPa/m)	8.756e-007	8.756e-007	8.756e-007	8.756e-007					
44	Liq. Reynolds	1.873e+005 *	1.873e+005 *	1.873e+005 *	1.873e+005 *					
45	Vap. Reynolds	9.827e+006 *	9.827e+006 *	9.827e+006 *	9.827e+006 *					
46	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
47	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907					
48	Deposit Thickness (mm)	---	---	---	---					
49	Deposit Volume (m3)	---	---	---	---					
50	Length (m)	1.918e+004	1.919e+004	1.919e+004	1.920e+004					
51	Elevation (m)	-0.4840	-0.4880	-0.4920	-0.4960					
52	Pressure (bar)	69.21	69.21	69.21	69.21					
53	Temperature (C)	50.52	50.51	50.51	50.51					
54	Vapour Fraction	0.8815	0.8815	0.8815	0.8815					
55	Heat Transferred (kJ/h-m)	546.6	546.5	546.5	546.4					
56	Flow Regime	Transition	Transition	Transition	Transition					
57	Water Flow Regime									
58	Liquid HoldUp	0.1715	0.1715	0.1715	0.1715					
59	Water HoldUp	---	---	---	---					
60	Friction Gradient (kPa/m)	8.987e-003	8.987e-003	8.987e-003	8.987e-003					
61	Static Gradient (kPa/m)	-1.282e-003	-1.282e-003	-1.282e-003	-1.282e-003					
62	Accel. Gradient (kPa/m)	9.543e-007	9.543e-007	9.543e-007	9.543e-007					
63	Liq. Reynolds	1.873e+005 *	1.873e+005 *	1.873e+005 *	1.873e+005 *					
64	Vap. Reynolds	9.827e+006 *	9.827e+006 *	9.827e+006 *	9.827e+006 *					
65	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
66	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907					
67	Deposit Thickness (mm)	---	---	---	---					
68	Deposit Volume (m3)	---	---	---	---					
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 339 of 937					

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length	(m)	1.920e+004	1.921e+004	1.921e+004	1.922e+004	
14	Elevation	(m)	-0.5000	-0.5180	-0.5360	-0.5540	
15	Pressure	(bar)	69.21	69.21	69.21	69.21	
16	Temperature	(C)	50.51	50.51	50.51	50.51	
17	Vapour Fraction		0.8815	0.8815	0.8815	0.8815	
18	Heat Transferred	(kJ/h-m)	546.4	546.4	546.3	546.3	
19	Flow Regime		Transition	Transition	Transition	Transition	
20	Water Flow Regime						
21	Liquid HoldUp		0.1715	0.1694	0.1694	0.1694	
22	Water HoldUp		---	---	---	---	
23	Friction Gradient	(kPa/m)	8.987e-003	9.015e-003	9.014e-003	9.014e-003	
24	Static Gradient	(kPa/m)	-1.282e-003	-5.724e-003	-5.724e-003	-5.724e-003	
25	Accel. Gradient	(kPa/m)	9.543e-007	4.076e-007	4.076e-007	4.076e-007	
26	Liq. Reynolds		1.873e+005 *	1.873e+005 *	1.873e+005 *	1.873e+005 *	
27	Vap. Reynolds		9.827e+006 *	9.827e+006 *	9.827e+006 *	9.827e+006 *	
28	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
29	Vapour Velocity	(m/s)	2.907	2.907	2.907	2.907	
30	Deposit Thickness	(mm)	---	---	---	---	
31	Deposit Volume	(m3)	---	---	---	---	
32	Length	(m)	1.922e+004	1.923e+004	1.923e+004	1.924e+004	
33	Elevation	(m)	-0.5720	-0.5900	-0.5680	-0.5460	
34	Pressure	(bar)	69.21	69.21	69.21	69.20	
35	Temperature	(C)	50.50	50.50	50.50	50.50	
36	Vapour Fraction		0.8815	0.8815	0.8815	0.8815	
37	Heat Transferred	(kJ/h-m)	546.2	546.2	546.1	546.0	
38	Flow Regime		Transition	Transition	Transition	Transition	
39	Water Flow Regime						
40	Liquid HoldUp		0.1694	0.1694	0.2443	0.2443	
41	Water HoldUp		---	---	---	---	
42	Friction Gradient	(kPa/m)	9.014e-003	9.014e-003	6.864e-003	6.864e-003	
43	Static Gradient	(kPa/m)	-5.724e-003	-5.724e-003	9.052e-003	9.052e-003	
44	Accel. Gradient	(kPa/m)	4.076e-007	4.075e-007	1.971e-006	1.971e-006	
45	Liq. Reynolds		1.873e+005 *	1.873e+005 *	1.873e+005 *	1.873e+005 *	
46	Vap. Reynolds		9.827e+006 *	9.827e+006 *	9.827e+006 *	9.827e+006 *	
47	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
48	Vapour Velocity	(m/s)	2.907	2.907	2.907	2.907	
49	Deposit Thickness	(mm)	---	---	---	---	
50	Deposit Volume	(m3)	---	---	---	---	
51	Length	(m)	1.924e+004	1.925e+004	1.925e+004	1.926e+004	
52	Elevation	(m)	-0.5240	-0.5020	-0.4800	-0.4940	
53	Pressure	(bar)	69.20	69.20	69.20	69.20	
54	Temperature	(C)	50.50	50.49	50.49	50.49	
55	Vapour Fraction		0.8815	0.8815	0.8815	0.8815	
56	Heat Transferred	(kJ/h-m)	546.0	545.9	545.9	545.9	
57	Flow Regime		Transition	Transition	Transition	Transition	
58	Water Flow Regime						
59	Liquid HoldUp		0.2443	0.2443	0.2443	0.1700	
60	Water HoldUp		---	---	---	---	
61	Friction Gradient	(kPa/m)	6.864e-003	6.864e-003	6.864e-003	9.006e-003	
62	Static Gradient	(kPa/m)	9.052e-003	9.052e-003	9.052e-003	-4.462e-003	
63	Accel. Gradient	(kPa/m)	1.972e-006	1.972e-006	1.972e-006	5.629e-007	
64	Liq. Reynolds		1.873e+005 *	1.873e+005 *	1.872e+005 *	1.872e+005 *	
65	Vap. Reynolds		9.827e+006 *	9.827e+006 *	9.827e+006 *	9.827e+006 *	
66	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
67	Vapour Velocity	(m/s)	2.907	2.907	2.907	2.907	
68	Deposit Thickness	(mm)	---	---	---	---	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---	---	
14	Length (m)	1.926e+004	1.927e+004	1.927e+004	1.928e+004	1.928e+004	
15	Elevation (m)	-0.5080	-0.5220	-0.5360	-0.5500	-0.5500	
16	Pressure (bar)	69.20	69.20	69.20	69.20	69.20	
17	Temperature (C)	50.49	50.49	50.49	50.48	50.48	
18	Vapour Fraction	0.8815	0.8815	0.8815	0.8815	0.8815	
19	Heat Transferred (kJ/h-m)	545.9	545.8	545.8	545.7	545.7	
20	Flow Regime	Transition	Transition	Transition	Transition	Transition	
21	Water Flow Regime						
22	Liquid HoldUp	0.1700	0.1700	0.1700	0.1700	0.1700	
23	Water HoldUp	---	---	---	---	---	
24	Friction Gradient (kPa/m)	9.006e-003	9.006e-003	9.006e-003	9.006e-003	9.006e-003	
25	Static Gradient (kPa/m)	-4.462e-003	-4.462e-003	-4.462e-003	-4.462e-003	-4.462e-003	
26	Accel. Gradient (kPa/m)	5.629e-007	5.629e-007	5.629e-007	5.629e-007	5.629e-007	
27	Liq. Reynolds	1.872e+005 *	1.872e+005 *	1.872e+005 *	1.872e+005 *	1.872e+005 *	
28	Vap. Reynolds	9.827e+006 *	9.827e+006 *	9.827e+006 *	9.827e+006 *	9.827e+006 *	
29	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979	0.1979	
30	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907	2.907	
31	Deposit Thickness (mm)	---	---	---	---	---	
32	Deposit Volume (m3)	---	---	---	---	---	
33	Length (m)	1.928e+004	1.929e+004	1.929e+004	1.930e+004	1.930e+004	
34	Elevation (m)	-0.5800	-0.6100	-0.6400	-0.6700	-0.6700	
35	Pressure (bar)	69.20	69.20	69.20	69.20	69.20	
36	Temperature (C)	50.48	50.48	50.48	50.48	50.48	
37	Vapour Fraction	0.8815	0.8815	0.8815	0.8815	0.8815	
38	Heat Transferred (kJ/h-m)	545.7	545.7	545.6	545.6	545.6	
39	Flow Regime	Transition	Transition	Transition	Transition	Transition	
40	Water Flow Regime						
41	Liquid HoldUp	0.1676	0.1676	0.1676	0.1676	0.1676	
42	Water HoldUp	---	---	---	---	---	
43	Friction Gradient (kPa/m)	9.038e-003	9.038e-003	9.038e-003	9.038e-003	9.038e-003	
44	Static Gradient (kPa/m)	-9.407e-003	-9.407e-003	-9.407e-003	-9.407e-003	-9.407e-003	
45	Accel. Gradient (kPa/m)	-4.564e-008	-4.565e-008	-4.567e-008	-4.568e-008	-4.568e-008	
46	Liq. Reynolds	1.872e+005 *	1.872e+005 *	1.872e+005 *	1.872e+005 *	1.872e+005 *	
47	Vap. Reynolds	9.827e+006 *	9.827e+006 *	9.827e+006 *	9.827e+006 *	9.827e+006 *	
48	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979	0.1979	
49	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907	2.907	
50	Deposit Thickness (mm)	---	---	---	---	---	
51	Deposit Volume (m3)	---	---	---	---	---	
52	Length (m)	1.930e+004	1.931e+004	1.931e+004	1.932e+004	1.932e+004	
53	Elevation (m)	-0.7000	-0.7020	-0.7040	-0.7060	-0.7060	
54	Pressure (bar)	69.20	69.20	69.20	69.20	69.20	
55	Temperature (C)	50.48	50.48	50.47	50.47	50.47	
56	Vapour Fraction	0.8815	0.8815	0.8815	0.8815	0.8815	
57	Heat Transferred (kJ/h-m)	545.5	545.5	545.4	545.4	545.4	
58	Flow Regime	Transition	Transition	Transition	Transition	Transition	
59	Water Flow Regime						
60	Liquid HoldUp	0.1676	0.1718	0.1718	0.1718	0.1718	
61	Water HoldUp	---	---	---	---	---	
62	Friction Gradient (kPa/m)	9.038e-003	8.982e-003	8.982e-003	8.982e-003	8.982e-003	
63	Static Gradient (kPa/m)	-9.407e-003	-6.420e-004	-6.420e-004	-6.420e-004	-6.420e-004	
64	Accel. Gradient (kPa/m)	-4.569e-008	1.033e-006	1.033e-006	1.033e-006	1.033e-006	
65	Liq. Reynolds	1.872e+005 *	1.872e+005 *	1.872e+005 *	1.872e+005 *	1.872e+005 *	
66	Vap. Reynolds	9.828e+006 *	9.828e+006 *	9.828e+006 *	9.828e+006 *	9.828e+006 *	
67	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979	0.1979	
68	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907	2.907	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Deposit Thickness (mm)	---	---	---	---					
14	Deposit Volume (m3)	---	---	---	---					
15	Length (m)	1.932e+004	1.933e+004	1.933e+004	1.934e+004					
16	Elevation (m)	-0.7080	-0.7100	-0.6980	-0.6860					
17	Pressure (bar)	69.20	69.20	69.20	69.20					
18	Temperature (C)	50.47	50.47	50.47	50.47					
19	Vapour Fraction	0.8815	0.8815	0.8815	0.8815					
20	Heat Transferred (kJ/h-m)	545.4	545.3	545.2	545.2					
21	Flow Regime	Transition	Transition	Transition	Transition					
22	Water Flow Regime									
23	Liquid HoldUp	0.1718	0.1718	0.2388	0.2388					
24	Water HoldUp	---	---	---	---					
25	Friction Gradient (kPa/m)	8.982e-003	8.982e-003	7.524e-003	7.524e-003					
26	Static Gradient (kPa/m)	-6.420e-004	-6.420e-004	4.855e-003	4.855e-003					
27	Accel. Gradient (kPa/m)	1.033e-006	1.033e-006	1.533e-006	1.533e-006					
28	Liq. Reynolds	1.872e+005 *	1.872e+005 *	1.872e+005 *	1.872e+005 *					
29	Vap. Reynolds	9.828e+006 *	9.828e+006 *	9.828e+006 *	9.828e+006 *					
30	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
31	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907					
32	Deposit Thickness (mm)	---	---	---	---					
33	Deposit Volume (m3)	---	---	---	---					
34	Length (m)	1.934e+004	1.935e+004	1.935e+004	1.936e+004					
35	Elevation (m)	-0.6740	-0.6620	-0.6500	-1.440					
36	Pressure (bar)	69.20	69.20	69.20	69.20					
37	Temperature (C)	50.46	50.46	50.46	50.46					
38	Vapour Fraction	0.8815	0.8815	0.8815	0.8815					
39	Heat Transferred (kJ/h-m)	545.1	545.1	545.0	545.5					
40	Flow Regime	Transition	Transition	Transition	Transition					
41	Water Flow Regime									
42	Liquid HoldUp	0.2388	0.2388	0.2388	5.876e-002					
43	Water HoldUp	---	---	---	---					
44	Friction Gradient (kPa/m)	7.524e-003	7.524e-003	7.524e-003	1.508e-002					
45	Static Gradient (kPa/m)	4.855e-003	4.855e-003	4.855e-003	-0.1422					
46	Accel. Gradient (kPa/m)	1.533e-006	1.533e-006	1.533e-006	-1.575e-005					
47	Liq. Reynolds	1.872e+005 *	1.872e+005 *	1.872e+005 *	1.872e+005 *					
48	Vap. Reynolds	9.828e+006 *	9.828e+006 *	9.828e+006 *	9.828e+006 *					
49	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
50	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907					
51	Deposit Thickness (mm)	---	---	---	---					
52	Deposit Volume (m3)	---	---	---	---					
53	Length (m)	1.936e+004	1.937e+004	1.937e+004	1.938e+004					
54	Elevation (m)	-2.230	-3.020	-3.810	-4.600					
55	Pressure (bar)	69.21	69.22	69.22	69.23					
56	Temperature (C)	50.47	50.47	50.47	50.47					
57	Vapour Fraction	0.8815	0.8815	0.8815	0.8815					
58	Heat Transferred (kJ/h-m)	545.6	545.7	545.8	545.8					
59	Flow Regime	Transition	Transition	Transition	Transition					
60	Water Flow Regime									
61	Liquid HoldUp	5.876e-002	5.876e-002	5.876e-002	5.877e-002					
62	Water HoldUp	---	---	---	---					
63	Friction Gradient (kPa/m)	1.508e-002	1.508e-002	1.508e-002	1.508e-002					
64	Static Gradient (kPa/m)	-0.1422	-0.1423	-0.1423	-0.1423					
65	Accel. Gradient (kPa/m)	-1.575e-005	-1.574e-005	-1.574e-005	-1.574e-005					
66	Liq. Reynolds	1.872e+005 *	1.872e+005 *	1.873e+005 *	1.873e+005 *					
67	Vap. Reynolds	9.827e+006 *	9.827e+006 *	9.827e+006 *	9.827e+006 *					
68	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Velocity (m/s)	2.906	2.906	2.906	2.905
14	Deposit Thickness (mm)	---	---	---	---
15	Deposit Volume (m3)	---	---	---	---
16	Length (m)	1.938e+004	1.939e+004	1.939e+004	1.940e+004
17	Elevation (m)	-4.390	-4.180	-3.970	-3.760
18	Pressure (bar)	69.22	69.22	69.21	69.20
19	Temperature (C)	50.47	50.47	50.46	50.46
20	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
21	Heat Transferred (kJ/h-m)	545.3	545.2	545.1	545.0
22	Flow Regime	Transition	Transition	Transition	Transition
23	Water Flow Regime				
24	Liquid HoldUp	0.3486	0.3486	0.3486	0.3486
25	Water HoldUp	---	---	---	---
26	Friction Gradient (kPa/m)	7.757e-003	7.757e-003	7.758e-003	7.758e-003
27	Static Gradient (kPa/m)	0.1137	0.1137	0.1137	0.1137
28	Accel. Gradient (kPa/m)	1.504e-005	1.504e-005	1.504e-005	1.504e-005
29	Liq. Reynolds	1.873e+005 *	1.872e+005 *	1.872e+005 *	1.872e+005 *
30	Vap. Reynolds	9.827e+006 *	9.827e+006 *	9.827e+006 *	9.827e+006 *
31	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
32	Vapour Velocity (m/s)	2.906	2.906	2.906	2.906
33	Deposit Thickness (mm)	---	---	---	---
34	Deposit Volume (m3)	---	---	---	---
35	Length (m)	1.940e+004	1.941e+004	1.941e+004	1.942e+004
36	Elevation (m)	-3.550	-3.058	-2.566	-2.074
37	Pressure (bar)	69.20	69.18	69.16	69.14
38	Temperature (C)	50.45	50.45	50.44	50.43
39	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
40	Heat Transferred (kJ/h-m)	544.9	544.9	544.7	544.5
41	Flow Regime	Transition	Transition	Transition	Transition
42	Water Flow Regime				
43	Liquid HoldUp	0.3486	0.5017	0.5017	0.5016
44	Water HoldUp	---	---	---	---
45	Friction Gradient (kPa/m)	7.759e-003	7.509e-003	7.511e-003	7.513e-003
46	Static Gradient (kPa/m)	0.1137	0.3604	0.3603	0.3603
47	Accel. Gradient (kPa/m)	1.505e-005	4.559e-005	4.561e-005	4.563e-005
48	Liq. Reynolds	1.872e+005 *	1.872e+005 *	1.872e+005 *	1.871e+005 *
49	Vap. Reynolds	9.828e+006 *	9.828e+006 *	9.829e+006 *	9.830e+006 *
50	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
51	Vapour Velocity (m/s)	2.907	2.907	2.908	2.909
52	Deposit Thickness (mm)	---	---	---	---
53	Deposit Volume (m3)	---	---	---	---
54	Length (m)	1.942e+004	1.943e+004	1.943e+004	1.944e+004
55	Elevation (m)	-1.582	-1.090	-1.096	-1.102
56	Pressure (bar)	69.12	69.11	69.11	69.10
57	Temperature (C)	50.43	50.42	50.42	50.41
58	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
59	Heat Transferred (kJ/h-m)	544.3	544.1	543.9	543.9
60	Flow Regime	Transition	Transition	Transition	Transition
61	Water Flow Regime				
62	Liquid HoldUp	0.5015	0.5014	0.1710	0.1710
63	Water HoldUp	---	---	---	---
64	Friction Gradient (kPa/m)	7.515e-003	7.517e-003	9.001e-003	9.001e-003
65	Static Gradient (kPa/m)	0.3603	0.3602	-1.920e-003	-1.920e-003
66	Accel. Gradient (kPa/m)	4.565e-005	4.567e-005	8.795e-007	8.795e-007
67	Liq. Reynolds	1.871e+005 *	1.871e+005 *	1.871e+005 *	1.871e+005 *
68	Vap. Reynolds	9.830e+006 *	9.831e+006 *	9.831e+006 *	9.831e+006 *
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
14	Vapour Velocity	(m/s)	2.910	2.911	2.911	2.911
15	Deposit Thickness	(mm)	---	---	---	---
16	Deposit Volume	(m3)	---	---	---	---
17	Length	(m)	1.944e+004	1.945e+004	1.945e+004	1.946e+004
18	Elevation	(m)	-1.108	-1.114	-1.120	-1.046
19	Pressure	(bar)	69.10	69.10	69.10	69.10
20	Temperature	(C)	50.41	50.41	50.41	50.41
21	Vapour Fraction		0.8815	0.8815	0.8815	0.8815
22	Heat Transferred	(kJ/h-m)	543.8	543.8	543.7	543.6
23	Flow Regime		Transition	Transition	Transition	Transition
24	Water Flow Regime					
25	Liquid HoldUp		0.1710	0.1710	0.1710	0.2831
26	Water HoldUp		---	---	---	---
27	Friction Gradient	(kPa/m)	9.001e-003	9.001e-003	9.001e-003	7.888e-003
28	Static Gradient	(kPa/m)	-1.920e-003	-1.920e-003	-1.920e-003	4.238e-002
29	Accel. Gradient	(kPa/m)	8.795e-007	8.795e-007	8.795e-007	6.244e-006
30	Liq. Reynolds		1.871e+005 *	1.871e+005 *	1.871e+005 *	1.871e+005 *
31	Vap. Reynolds		9.831e+006 *	9.831e+006 *	9.831e+006 *	9.831e+006 *
32	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
33	Vapour Velocity	(m/s)	2.911	2.911	2.911	2.911
34	Deposit Thickness	(mm)	---	---	---	---
35	Deposit Volume	(m3)	---	---	---	---
36	Length	(m)	1.946e+004	1.946e+004	1.947e+004	1.947e+004
37	Elevation	(m)	-0.9720	-0.8980	-0.8240	-0.7500
38	Pressure	(bar)	69.10	69.10	69.10	69.09
39	Temperature	(C)	50.41	50.40	50.40	50.40
40	Vapour Fraction		0.8815	0.8815	0.8815	0.8815
41	Heat Transferred	(kJ/h-m)	543.6	543.5	543.5	543.4
42	Flow Regime		Transition	Transition	Transition	Transition
43	Water Flow Regime					
44	Liquid HoldUp		0.2831	0.2831	0.2831	0.2831
45	Water HoldUp		---	---	---	---
46	Friction Gradient	(kPa/m)	7.888e-003	7.889e-003	7.889e-003	7.889e-003
47	Static Gradient	(kPa/m)	4.238e-002	4.238e-002	4.238e-002	4.238e-002
48	Accel. Gradient	(kPa/m)	6.244e-006	6.244e-006	6.245e-006	6.245e-006
49	Liq. Reynolds		1.871e+005 *	1.871e+005 *	1.871e+005 *	1.871e+005 *
50	Vap. Reynolds		9.831e+006 *	9.831e+006 *	9.831e+006 *	9.831e+006 *
51	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
52	Vapour Velocity	(m/s)	2.911	2.911	2.911	2.911
53	Deposit Thickness	(mm)	---	---	---	---
54	Deposit Volume	(m3)	---	---	---	---
55	Length	(m)	1.948e+004	1.948e+004	1.949e+004	1.949e+004
56	Elevation	(m)	-0.7240	-0.6980	-0.6720	-0.6460
57	Pressure	(bar)	69.09	69.09	69.09	69.09
58	Temperature	(C)	50.40	50.40	50.39	50.39
59	Vapour Fraction		0.8815	0.8815	0.8815	0.8815
60	Heat Transferred	(kJ/h-m)	543.4	543.3	543.3	543.2
61	Flow Regime		Transition	Transition	Transition	Transition
62	Water Flow Regime					
63	Liquid HoldUp		0.2463	0.2463	0.2463	0.2463
64	Water HoldUp		---	---	---	---
65	Friction Gradient	(kPa/m)	6.625e-003	6.625e-003	6.625e-003	6.625e-003
66	Static Gradient	(kPa/m)	1.076e-002	1.076e-002	1.076e-002	1.076e-002
67	Accel. Gradient	(kPa/m)	2.160e-006	2.160e-006	2.160e-006	2.160e-006
68	Liq. Reynolds		1.871e+005 *	1.871e+005 *	1.870e+005 *	1.870e+005 *
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.831e+006 *	9.831e+006 *	9.832e+006 *	9.832e+006 *
14	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
15	Vapour Velocity (m/s)	2.911	2.911	2.911	2.911
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	1.950e+004	1.952e+004	1.954e+004	1.957e+004
19	Elevation (m)	-0.6200	-0.5920	-0.5640	-0.5360
20	Pressure (bar)	69.09	69.09	69.08	69.08
21	Temperature (C)	50.39	50.38	50.37	50.37
22	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
23	Heat Transferred (kJ/h-m)	543.2	543.0	542.7	542.5
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2463	0.2352	0.2352	0.2352
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.625e-003	7.959e-003	7.959e-003	7.959e-003
29	Static Gradient (kPa/m)	1.076e-002	2.400e-003	2.400e-003	2.400e-003
30	Accel. Gradient (kPa/m)	2.160e-006	1.287e-006	1.287e-006	1.287e-006
31	Liq. Reynolds	1.870e+005 *	1.870e+005 *	1.870e+005 *	1.870e+005 *
32	Vap. Reynolds	9.832e+006 *	9.832e+006 *	9.832e+006 *	9.832e+006 *
33	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
34	Vapour Velocity (m/s)	2.911	2.911	2.911	2.911
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	1.959e+004	1.961e+004	1.964e+004	1.966e+004
38	Elevation (m)	-0.5080	-0.4800	-0.4780	-0.4760
39	Pressure (bar)	69.08	69.08	69.08	69.07
40	Temperature (C)	50.36	50.35	50.34	50.33
41	Vapour Fraction	0.8815	0.8815	0.8815	0.8815
42	Heat Transferred (kJ/h-m)	542.3	542.1	541.9	541.7
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2352	0.2352	0.2321	0.2321
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	7.959e-003	7.959e-003	8.365e-003	8.365e-003
48	Static Gradient (kPa/m)	2.400e-003	2.400e-003	1.698e-004	1.698e-004
49	Accel. Gradient (kPa/m)	1.287e-006	1.287e-006	1.061e-006	1.061e-006
50	Liq. Reynolds	1.870e+005 *	1.870e+005 *	1.870e+005 *	1.870e+005 *
51	Vap. Reynolds	9.832e+006 *	9.832e+006 *	9.832e+006 *	9.832e+006 *
52	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
53	Vapour Velocity (m/s)	2.911	2.911	2.911	2.911
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1.968e+004	1.971e+004	1.973e+004	1.975e+004
57	Elevation (m)	-0.4740	-0.4720	-0.4700	-0.4660
58	Pressure (bar)	69.07	69.07	69.07	69.07
59	Temperature (C)	50.33	50.32	50.31	50.30
60	Vapour Fraction	0.8815	0.8815	0.8815	0.8814
61	Heat Transferred (kJ/h-m)	541.5	541.3	541.1	540.8
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2321	0.2321	0.2321	0.2324
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	8.365e-003	8.365e-003	8.365e-003	8.333e-003
67	Static Gradient (kPa/m)	1.698e-004	1.698e-004	1.698e-004	3.399e-004
68	Accel. Gradient (kPa/m)	1.061e-006	1.061e-006	1.061e-006	1.078e-006
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	1.870e+005 *	1.870e+005 *	1.870e+005 *	1.869e+005 *
14	Vap. Reynolds	9.832e+006 *	9.833e+006 *	9.833e+006 *	9.833e+006 *
15	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
16	Vapour Velocity (m/s)	2.911	2.911	2.911	2.911
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	1.978e+004	1.980e+004	1.982e+004	1.985e+004
20	Elevation (m)	-0.4620	-0.4580	-0.4540	-0.4500
21	Pressure (bar)	69.06	69.06	69.06	69.06
22	Temperature (C)	50.29	50.29	50.28	50.27
23	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
24	Heat Transferred (kJ/h-m)	540.6	540.4	540.2	540.0
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2324	0.2324	0.2324	0.2324
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	8.333e-003	8.333e-003	8.333e-003	8.333e-003
30	Static Gradient (kPa/m)	3.399e-004	3.399e-004	3.399e-004	3.399e-004
31	Accel. Gradient (kPa/m)	1.078e-006	1.078e-006	1.078e-006	1.078e-006
32	Liq. Reynolds	1.869e+005 *	1.869e+005 *	1.869e+005 *	1.869e+005 *
33	Vap. Reynolds	9.833e+006 *	9.833e+006 *	9.833e+006 *	9.833e+006 *
34	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
35	Vapour Velocity (m/s)	2.911	2.911	2.911	2.911
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	1.987e+004	1.989e+004	1.992e+004	1.994e+004
39	Elevation (m)	-0.4480	-0.4460	-0.4440	-0.4420
40	Pressure (bar)	69.06	69.05	69.05	69.05
41	Temperature (C)	50.26	50.26	50.25	50.24
42	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
43	Heat Transferred (kJ/h-m)	539.8	539.6	539.4	539.2
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2322	0.2322	0.2322	0.2322
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	8.365e-003	8.365e-003	8.365e-003	8.365e-003
49	Static Gradient (kPa/m)	1.698e-004	1.698e-004	1.698e-004	1.698e-004
50	Accel. Gradient (kPa/m)	1.061e-006	1.061e-006	1.061e-006	1.061e-006
51	Liq. Reynolds	1.869e+005 *	1.869e+005 *	1.869e+005 *	1.869e+005 *
52	Vap. Reynolds	9.833e+006 *	9.833e+006 *	9.833e+006 *	9.834e+006 *
53	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
54	Vapour Velocity (m/s)	2.911	2.911	2.911	2.911
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	1.996e+004	1.999e+004	2.001e+004	2.003e+004
58	Elevation (m)	-0.4400	-0.4380	-0.4360	-0.4340
59	Pressure (bar)	69.05	69.05	69.04	69.04
60	Temperature (C)	50.23	50.22	50.22	50.21
61	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
62	Heat Transferred (kJ/h-m)	539.0	538.8	538.5	538.3
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2322	0.2322	0.2322	0.2322
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	8.365e-003	8.365e-003	8.365e-003	8.364e-003
68	Static Gradient (kPa/m)	1.698e-004	1.698e-004	1.698e-004	1.698e-004
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	1.061e-006	1.061e-006	1.061e-006	1.061e-006
14	Liq. Reynolds	1.869e+005 *	1.869e+005 *	1.869e+005 *	1.868e+005 *
15	Vap. Reynolds	9.834e+006 *	9.834e+006 *	9.834e+006 *	9.834e+006 *
16	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
17	Vapour Velocity (m/s)	2.911	2.911	2.911	2.911
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	2.006e+004	2.008e+004	2.010e+004	2.013e+004
21	Elevation (m)	-0.4320	-0.4300	-0.4840	-0.5380
22	Pressure (bar)	69.04	69.04	69.04	69.03
23	Temperature (C)	50.20	50.19	50.19	50.18
24	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
25	Heat Transferred (kJ/h-m)	538.1	537.9	537.8	537.6
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.2322	0.2322	0.1702	0.1702
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	8.364e-003	8.364e-003	9.012e-003	9.012e-003
31	Static Gradient (kPa/m)	1.698e-004	1.698e-004	-3.691e-003	-3.691e-003
32	Accel. Gradient (kPa/m)	1.061e-006	1.061e-006	6.615e-007	6.615e-007
33	Liq. Reynolds	1.868e+005 *	1.868e+005 *	1.868e+005 *	1.868e+005 *
34	Vap. Reynolds	9.834e+006 *	9.834e+006 *	9.834e+006 *	9.834e+006 *
35	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
36	Vapour Velocity (m/s)	2.911	2.910	2.910	2.910
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	2.015e+004	2.017e+004	2.020e+004	2.022e+004
40	Elevation (m)	-0.5920	-0.6460	-0.7000	-0.6380
41	Pressure (bar)	69.03	69.03	69.03	69.03
42	Temperature (C)	50.17	50.16	50.16	50.15
43	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
44	Heat Transferred (kJ/h-m)	537.4	537.2	537.0	536.7
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.1702	0.1702	0.1702	0.2393
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	9.012e-003	9.012e-003	9.012e-003	7.450e-003
50	Static Gradient (kPa/m)	-3.691e-003	-3.691e-003	-3.691e-003	5.384e-003
51	Accel. Gradient (kPa/m)	6.615e-007	6.614e-007	6.614e-007	1.596e-006
52	Liq. Reynolds	1.868e+005 *	1.868e+005 *	1.868e+005 *	1.868e+005 *
53	Vap. Reynolds	9.834e+006 *	9.834e+006 *	9.834e+006 *	9.835e+006 *
54	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
55	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	2.024e+004	2.027e+004	2.029e+004	2.031e+004
59	Elevation (m)	-0.5760	-0.5140	-0.4520	-0.3900
60	Pressure (bar)	69.02	69.02	69.02	69.02
61	Temperature (C)	50.14	50.13	50.12	50.11
62	Vapour Fraction	0.8814	0.8814	0.8814	0.8814
63	Heat Transferred (kJ/h-m)	536.5	536.3	536.0	535.8
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.2393	0.2393	0.2393	0.2393
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	7.450e-003	7.450e-003	7.450e-003	7.450e-003
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:13:43 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Static Gradient	(kPa/m)	5.384e-003	5.384e-003	5.384e-003	5.384e-003	
14	Accel. Gradient	(kPa/m)	1.596e-006	1.596e-006	1.596e-006	1.596e-006	
15	Liq. Reynolds		1.868e+005 *	1.868e+005 *	1.868e+005 *	1.867e+005 *	
16	Vap. Reynolds		9.835e+006 *	9.835e+006 *	9.835e+006 *	9.835e+006 *	
17	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
18	Vapour Velocity	(m/s)	2.910	2.910	2.910	2.910	
19	Deposit Thickness	(mm)	---	---	---	---	
20	Deposit Volume	(m3)	---	---	---	---	
21	Length	(m)	2.034e+004	2.036e+004	2.038e+004	2.041e+004	
22	Elevation	(m)	-0.3920	-0.3940	-0.3960	-0.3980	
23	Pressure	(bar)	69.01	69.01	69.01	69.01	
24	Temperature	(C)	50.11	50.10	50.09	50.08	
25	Vapour Fraction		0.8814	0.8814	0.8814	0.8813	
26	Heat Transferred	(kJ/h-m)	535.7	535.5	535.3	535.0	
27	Flow Regime		Transition	Transition	Transition	Transition	
28	Water Flow Regime						
29	Liquid HoldUp		0.1719	0.1719	0.1719	0.1719	
30	Water HoldUp		---	---	---	---	
31	Friction Gradient	(kPa/m)	8.990e-003	8.990e-003	8.990e-003	8.990e-003	
32	Static Gradient	(kPa/m)	-1.376e-004	-1.376e-004	-1.376e-004	-1.376e-004	
33	Accel. Gradient	(kPa/m)	1.101e-006	1.101e-006	1.101e-006	1.101e-006	
34	Liq. Reynolds		1.867e+005 *	1.867e+005 *	1.867e+005 *	1.867e+005 *	
35	Vap. Reynolds		9.835e+006 *	9.835e+006 *	9.835e+006 *	9.835e+006 *	
36	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
37	Vapour Velocity	(m/s)	2.910	2.910	2.910	2.910	
38	Deposit Thickness	(mm)	---	---	---	---	
39	Deposit Volume	(m3)	---	---	---	---	
40	Length	(m)	2.043e+004	2.045e+004	2.048e+004	2.050e+004	
41	Elevation	(m)	-0.4000	-0.3960	-0.3920	-0.3880	
42	Pressure	(bar)	69.01	69.00	69.00	69.00	
43	Temperature	(C)	50.08	50.07	50.06	50.05	
44	Vapour Fraction		0.8813	0.8813	0.8813	0.8813	
45	Heat Transferred	(kJ/h-m)	534.8	534.6	534.4	534.2	
46	Flow Regime		Transition	Transition	Transition	Transition	
47	Water Flow Regime						
48	Liquid HoldUp		0.1719	0.2324	0.2324	0.2324	
49	Water HoldUp		---	---	---	---	
50	Friction Gradient	(kPa/m)	8.990e-003	8.332e-003	8.332e-003	8.332e-003	
51	Static Gradient	(kPa/m)	-1.376e-004	3.400e-004	3.400e-004	3.400e-004	
52	Accel. Gradient	(kPa/m)	1.101e-006	1.078e-006	1.079e-006	1.079e-006	
53	Liq. Reynolds		1.867e+005 *	1.867e+005 *	1.867e+005 *	1.867e+005 *	
54	Vap. Reynolds		9.836e+006 *	9.836e+006 *	9.836e+006 *	9.836e+006 *	
55	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
56	Vapour Velocity	(m/s)	2.910	2.910	2.910	2.910	
57	Deposit Thickness	(mm)	---	---	---	---	
58	Deposit Volume	(m3)	---	---	---	---	
59	Length	(m)	2.052e+004	2.055e+004	2.058e+004	2.060e+004	
60	Elevation	(m)	-0.3840	-0.3800	-0.4020	-0.4240	
61	Pressure	(bar)	69.00	69.00	68.99	68.99	
62	Temperature	(C)	50.04	50.04	50.03	50.02	
63	Vapour Fraction		0.8813	0.8813	0.8813	0.8813	
64	Heat Transferred	(kJ/h-m)	534.0	533.8	533.5	533.3	
65	Flow Regime		Transition	Transition	Transition	Transition	
66	Water Flow Regime						
67	Liquid HoldUp		0.2324	0.2324	0.1714	0.1714	
68	Water HoldUp		---	---	---	---	
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	8.332e-003	8.332e-003	8.996e-003	8.996e-003
14	Static Gradient	(kPa/m)	3.400e-004	3.400e-004	-1.216e-003	-1.216e-003
15	Accel. Gradient	(kPa/m)	1.079e-006	1.079e-006	9.678e-007	9.678e-007
16	Liq. Reynolds		1.867e+005 *	1.867e+005 *	1.867e+005 *	1.866e+005 *
17	Vap. Reynolds		9.836e+006 *	9.836e+006 *	9.836e+006 *	9.836e+006 *
18	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
19	Vapour Velocity	(m/s)	2.910	2.910	2.910	2.910
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	2.063e+004	2.066e+004	2.069e+004	2.069e+004
23	Elevation	(m)	-0.4460	-0.4680	-0.4900	-0.5000
24	Pressure	(bar)	68.99	68.99	68.98	68.98
25	Temperature	(C)	50.01	50.00	49.99	49.99
26	Vapour Fraction		0.8813	0.8813	0.8813	0.8813
27	Heat Transferred	(kJ/h-m)	533.0	532.8	532.5	532.5
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.1714	0.1714	0.1714	0.1677
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	8.996e-003	8.996e-003	8.996e-003	9.046e-003
33	Static Gradient	(kPa/m)	-1.216e-003	-1.216e-003	-1.216e-003	-8.975e-003
34	Accel. Gradient	(kPa/m)	9.678e-007	9.678e-007	9.678e-007	8.731e-009
35	Liq. Reynolds		1.866e+005 *	1.866e+005 *	1.866e+005 *	1.866e+005 *
36	Vap. Reynolds		9.836e+006 *	9.837e+006 *	9.837e+006 *	9.837e+006 *
37	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
38	Vapour Velocity	(m/s)	2.910	2.910	2.910	2.910
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	2.070e+004	2.070e+004	2.070e+004	2.070e+004
42	Elevation	(m)	-0.5100	-0.5200	-0.5300	-0.5400
43	Pressure	(bar)	68.98	68.98	68.98	68.98
44	Temperature	(C)	49.99	49.99	49.99	49.99
45	Vapour Fraction		0.8813	0.8813	0.8813	0.8813
46	Heat Transferred	(kJ/h-m)	532.5	532.5	532.5	532.5
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.1677	0.1677	0.1677	0.1677
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	9.046e-003	9.046e-003	9.046e-003	9.046e-003
52	Static Gradient	(kPa/m)	-8.975e-003	-8.976e-003	-8.976e-003	-8.976e-003
53	Accel. Gradient	(kPa/m)	8.726e-009	8.721e-009	8.716e-009	8.711e-009
54	Liq. Reynolds		1.866e+005 *	1.866e+005 *	1.866e+005 *	1.866e+005 *
55	Vap. Reynolds		9.837e+006 *	9.837e+006 *	9.837e+006 *	9.837e+006 *
56	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
57	Vapour Velocity	(m/s)	2.910	2.910	2.910	2.910
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	2.071e+004	2.071e+004	2.072e+004	2.072e+004
61	Elevation	(m)	-0.5360	-0.5320	-0.5280	-0.5240
62	Pressure	(bar)	68.98	68.98	68.98	68.98
63	Temperature	(C)	49.99	49.98	49.98	49.98
64	Vapour Fraction		0.8813	0.8813	0.8813	0.8813
65	Heat Transferred	(kJ/h-m)	532.4	532.3	532.3	532.2
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2342	0.2342	0.2342	0.2342
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	8.101e-003	8.101e-003	8.101e-003	8.101e-003
15	Static Gradient (kPa/m)	1.595e-003	1.595e-003	1.595e-003	1.595e-003
16	Accel. Gradient (kPa/m)	1.206e-006	1.206e-006	1.206e-006	1.206e-006
17	Liq. Reynolds	1.866e+005 *	1.866e+005 *	1.866e+005 *	1.866e+005 *
18	Vap. Reynolds	9.837e+006 *	9.837e+006 *	9.837e+006 *	9.837e+006 *
19	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
20	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	2.073e+004	2.073e+004	2.074e+004	2.074e+004
24	Elevation (m)	-0.5200	-0.5140	-0.5080	-0.5020
25	Pressure (bar)	68.98	68.98	68.98	68.98
26	Temperature (C)	49.98	49.98	49.97	49.97
27	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
28	Heat Transferred (kJ/h-m)	532.2	532.2	532.1	532.1
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2342	0.2353	0.2353	0.2353
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	8.101e-003	7.957e-003	7.957e-003	7.957e-003
34	Static Gradient (kPa/m)	1.595e-003	2.402e-003	2.402e-003	2.402e-003
35	Accel. Gradient (kPa/m)	1.206e-006	1.289e-006	1.289e-006	1.289e-006
36	Liq. Reynolds	1.866e+005 *	1.866e+005 *	1.866e+005 *	1.866e+005 *
37	Vap. Reynolds	9.837e+006 *	9.837e+006 *	9.837e+006 *	9.837e+006 *
38	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
39	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	2.075e+004	2.075e+004	2.076e+004	2.076e+004
43	Elevation (m)	-0.4960	-0.4900	-0.4900	-0.4900
44	Pressure (bar)	68.98	68.98	68.98	68.98
45	Temperature (C)	49.97	49.97	49.97	49.97
46	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
47	Heat Transferred (kJ/h-m)	532.0	532.0	531.9	531.9
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2353	0.2353	0.2510	0.2510
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	7.957e-003	7.957e-003	6.118e-003	6.118e-003
53	Static Gradient (kPa/m)	2.402e-003	2.402e-003	0.0000	0.0000
54	Accel. Gradient (kPa/m)	1.289e-006	1.289e-006	7.611e-007	7.611e-007
55	Liq. Reynolds	1.866e+005 *	1.866e+005 *	1.866e+005 *	1.866e+005 *
56	Vap. Reynolds	9.837e+006 *	9.837e+006 *	9.837e+006 *	9.837e+006 *
57	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
58	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	2.077e+004	2.077e+004	2.078e+004	2.078e+004
62	Elevation (m)	-0.4900	-0.4900	-0.4900	-0.4900
63	Pressure (bar)	68.98	68.98	68.98	68.98
64	Temperature (C)	49.96	49.96	49.96	49.96
65	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
66	Heat Transferred (kJ/h-m)	531.8	531.8	531.8	531.7
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 350 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.2510	0.2510	0.2510	0.2510
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	6.118e-003	6.118e-003	6.118e-003	6.118e-003
16	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
17	Accel. Gradient (kPa/m)	7.611e-007	7.611e-007	7.611e-007	7.611e-007
18	Liq. Reynolds	1.866e+005 *	1.866e+005 *	1.866e+005 *	1.866e+005 *
19	Vap. Reynolds	9.837e+006 *	9.837e+006 *	9.837e+006 *	9.837e+006 *
20	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
21	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	2.079e+004	2.079e+004	2.080e+004	2.080e+004
25	Elevation (m)	-0.4900	-0.4900	-0.4900	-0.4900
26	Pressure (bar)	68.98	68.98	68.98	68.98
27	Temperature (C)	49.96	49.96	49.96	49.95
28	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
29	Heat Transferred (kJ/h-m)	531.7	531.6	531.6	531.5
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2510	0.2510	0.2510	0.2510
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.118e-003	6.118e-003	6.118e-003	6.118e-003
35	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
36	Accel. Gradient (kPa/m)	7.611e-007	7.611e-007	7.611e-007	7.611e-007
37	Liq. Reynolds	1.866e+005 *	1.866e+005 *	1.866e+005 *	1.866e+005 *
38	Vap. Reynolds	9.837e+006 *	9.837e+006 *	9.837e+006 *	9.837e+006 *
39	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
40	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	2.081e+004	2.081e+004	2.082e+004	2.082e+004
44	Elevation (m)	-0.5000	-0.5100	-0.5200	-0.5300
45	Pressure (bar)	68.98	68.98	68.98	68.97
46	Temperature (C)	49.95	49.95	49.95	49.95
47	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
48	Heat Transferred (kJ/h-m)	531.5	531.5	531.5	531.4
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1705	0.1705	0.1705	0.1705
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	9.008e-003	9.008e-003	9.008e-003	9.008e-003
54	Static Gradient (kPa/m)	-3.194e-003	-3.194e-003	-3.194e-003	-3.194e-003
55	Accel. Gradient (kPa/m)	7.233e-007	7.233e-007	7.233e-007	7.233e-007
56	Liq. Reynolds	1.866e+005 *	1.866e+005 *	1.866e+005 *	1.866e+005 *
57	Vap. Reynolds	9.837e+006 *	9.837e+006 *	9.837e+006 *	9.837e+006 *
58	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
59	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	2.083e+004	2.083e+004	2.084e+004	2.084e+004
63	Elevation (m)	-0.5400	-0.5460	-0.5520	-0.5580
64	Pressure (bar)	68.97	68.97	68.97	68.97
65	Temperature (C)	49.95	49.94	49.94	49.94
66	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
67	Heat Transferred (kJ/h-m)	531.4	531.3	531.3	531.2
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 351 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:13:43 2022		
6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Water Flow Regime						
14	Liquid HoldUp		0.1705	0.1711	0.1711	0.1711	
15	Water HoldUp		---	---	---	---	
16	Friction Gradient (kPa/m)		9.008e-003	9.000e-003	9.000e-003	9.000e-003	
17	Static Gradient (kPa/m)		-3.194e-003	-1.921e-003	-1.921e-003	-1.921e-003	
18	Accel. Gradient (kPa/m)		7.233e-007	8.806e-007	8.806e-007	8.806e-007	
19	Liq. Reynolds		1.866e+005 *	1.866e+005 *	1.866e+005 *	1.866e+005 *	
20	Vap. Reynolds		9.837e+006 *	9.837e+006 *	9.837e+006 *	9.837e+006 *	
21	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979	
22	Vapour Velocity (m/s)		2.910	2.910	2.910	2.910	
23	Deposit Thickness (mm)		---	---	---	---	
24	Deposit Volume (m3)		---	---	---	---	
25	Length (m)		2.085e+004	2.085e+004	2.086e+004	2.086e+004	
26	Elevation (m)		-0.5640	-0.5700	-0.5580	-0.5460	
27	Pressure (bar)		68.97	68.97	68.97	68.97	
28	Temperature (C)		49.94	49.94	49.94	49.93	
29	Vapour Fraction		0.8813	0.8813	0.8813	0.8813	
30	Heat Transferred (kJ/h-m)		531.2	531.2	531.1	531.0	
31	Flow Regime		Transition	Transition	Transition	Transition	
32	Water Flow Regime						
33	Liquid HoldUp		0.1711	0.1711	0.2386	0.2386	
34	Water HoldUp		---	---	---	---	
35	Friction Gradient (kPa/m)		9.000e-003	9.000e-003	7.537e-003	7.537e-003	
36	Static Gradient (kPa/m)		-1.921e-003	-1.921e-003	4.853e-003	4.853e-003	
37	Accel. Gradient (kPa/m)		8.806e-007	8.806e-007	1.541e-006	1.541e-006	
38	Liq. Reynolds		1.866e+005 *	1.866e+005 *	1.866e+005 *	1.866e+005 *	
39	Vap. Reynolds		9.837e+006 *	9.837e+006 *	9.837e+006 *	9.837e+006 *	
40	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979	
41	Vapour Velocity (m/s)		2.910	2.910	2.910	2.910	
42	Deposit Thickness (mm)		---	---	---	---	
43	Deposit Volume (m3)		---	---	---	---	
44	Length (m)		2.087e+004	2.087e+004	2.088e+004	2.088e+004	
45	Elevation (m)		-0.5340	-0.5220	-0.5100	-0.5460	
46	Pressure (bar)		68.97	68.97	68.97	68.97	
47	Temperature (C)		49.93	49.93	49.93	49.93	
48	Vapour Fraction		0.8813	0.8813	0.8813	0.8813	
49	Heat Transferred (kJ/h-m)		531.0	530.9	530.9	530.9	
50	Flow Regime		Transition	Transition	Transition	Transition	
51	Water Flow Regime						
52	Liquid HoldUp		0.2386	0.2386	0.2386	0.1667	
53	Water HoldUp		---	---	---	---	
54	Friction Gradient (kPa/m)		7.537e-003	7.537e-003	7.537e-003	9.060e-003	
55	Static Gradient (kPa/m)		4.853e-003	4.853e-003	4.853e-003	-1.121e-002	
56	Accel. Gradient (kPa/m)		1.541e-006	1.541e-006	1.541e-006	-2.680e-007	
57	Liq. Reynolds		1.866e+005 *	1.866e+005 *	1.865e+005 *	1.865e+005 *	
58	Vap. Reynolds		9.837e+006 *	9.837e+006 *	9.837e+006 *	9.837e+006 *	
59	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979	
60	Vapour Velocity (m/s)		2.910	2.910	2.910	2.910	
61	Deposit Thickness (mm)		---	---	---	---	
62	Deposit Volume (m3)		---	---	---	---	
63	Length (m)		2.089e+004	2.089e+004	2.090e+004	2.090e+004	
64	Elevation (m)		-0.5820	-0.6180	-0.6540	-0.6900	
65	Pressure (bar)		68.97	68.97	68.97	68.97	
66	Temperature (C)		49.93	49.92	49.92	49.92	
67	Vapour Fraction		0.8813	0.8813	0.8813	0.8813	
68	Heat Transferred (kJ/h-m)		530.9	530.8	530.8	530.7	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 352 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.1667	0.1667	0.1667	0.1667
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	9.060e-003	9.060e-003	9.060e-003	9.060e-003
18	Static Gradient (kPa/m)	-1.121e-002	-1.121e-002	-1.121e-002	-1.121e-002
19	Accel. Gradient (kPa/m)	-2.680e-007	-2.680e-007	-2.680e-007	-2.680e-007
20	Liq. Reynolds	1.865e+005 *	1.865e+005 *	1.865e+005 *	1.865e+005 *
21	Vap. Reynolds	9.837e+006 *	9.837e+006 *	9.837e+006 *	9.837e+006 *
22	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
23	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	2.091e+004	2.091e+004	2.092e+004	2.092e+004
27	Elevation (m)	-0.6740	-0.6580	-0.6420	-0.6260
28	Pressure (bar)	68.97	68.97	68.97	68.97
29	Temperature (C)	49.92	49.92	49.92	49.91
30	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
31	Heat Transferred (kJ/h-m)	530.6	530.6	530.5	530.5
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2409	0.2409	0.2409	0.2409
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	7.266e-003	7.266e-003	7.266e-003	7.266e-003
37	Static Gradient (kPa/m)	6.516e-003	6.516e-003	6.516e-003	6.516e-003
38	Accel. Gradient (kPa/m)	1.714e-006	1.714e-006	1.715e-006	1.715e-006
39	Liq. Reynolds	1.865e+005 *	1.865e+005 *	1.865e+005 *	1.865e+005 *
40	Vap. Reynolds	9.837e+006 *	9.837e+006 *	9.837e+006 *	9.837e+006 *
41	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
42	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	2.093e+004	2.093e+004	2.094e+004	2.094e+004
46	Elevation (m)	-0.6100	-0.5960	-0.5820	-0.5680
47	Pressure (bar)	68.97	68.97	68.97	68.96
48	Temperature (C)	49.91	49.91	49.91	49.91
49	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
50	Heat Transferred (kJ/h-m)	530.5	530.4	530.4	530.3
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2409	0.2398	0.2398	0.2398
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	7.266e-003	7.400e-003	7.400e-003	7.400e-003
56	Static Gradient (kPa/m)	6.516e-003	5.682e-003	5.682e-003	5.682e-003
57	Accel. Gradient (kPa/m)	1.715e-006	1.628e-006	1.628e-006	1.628e-006
58	Liq. Reynolds	1.865e+005 *	1.865e+005 *	1.865e+005 *	1.865e+005 *
59	Vap. Reynolds	9.837e+006 *	9.838e+006 *	9.838e+006 *	9.838e+006 *
60	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
61	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	2.095e+004	2.095e+004	2.096e+004	2.096e+004
65	Elevation (m)	-0.5540	-0.5400	-0.5500	-0.5600
66	Pressure (bar)	68.96	68.96	68.96	68.96
67	Temperature (C)	49.91	49.90	49.90	49.90
68	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Heat Transferred (kJ/h-m)	530.3	530.2	530.2	530.2					
14	Flow Regime	Transition	Transition	Transition	Transition					
15	Water Flow Regime									
16	Liquid HoldUp	0.2398	0.2398	0.1705	0.1705					
17	Water HoldUp	---	---	---	---					
18	Friction Gradient (kPa/m)	7.400e-003	7.400e-003	9.008e-003	9.008e-003					
19	Static Gradient (kPa/m)	5.682e-003	5.682e-003	-3.194e-003	-3.194e-003					
20	Accel. Gradient (kPa/m)	1.628e-006	1.628e-006	7.233e-007	7.233e-007					
21	Liq. Reynolds	1.865e+005 *	1.865e+005 *	1.865e+005 *	1.865e+005 *					
22	Vap. Reynolds	9.838e+006 *	9.838e+006 *	9.838e+006 *	9.838e+006 *					
23	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
24	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910					
25	Deposit Thickness (mm)	---	---	---	---					
26	Deposit Volume (m3)	---	---	---	---					
27	Length (m)	2.097e+004	2.097e+004	2.098e+004	2.098e+004					
28	Elevation (m)	-0.5700	-0.5800	-0.5900	-0.6160					
29	Pressure (bar)	68.96	68.96	68.96	68.96					
30	Temperature (C)	49.90	49.90	49.90	49.89					
31	Vapour Fraction	0.8813	0.8813	0.8813	0.8813					
32	Heat Transferred (kJ/h-m)	530.1	530.1	530.1	530.0					
33	Flow Regime	Transition	Transition	Transition	Transition					
34	Water Flow Regime									
35	Liquid HoldUp	0.1705	0.1705	0.1705	0.1681					
36	Water HoldUp	---	---	---	---					
37	Friction Gradient (kPa/m)	9.008e-003	9.008e-003	9.007e-003	9.040e-003					
38	Static Gradient (kPa/m)	-3.194e-003	-3.194e-003	-3.194e-003	-8.227e-003					
39	Accel. Gradient (kPa/m)	7.233e-007	7.233e-007	7.233e-007	1.011e-007					
40	Liq. Reynolds	1.865e+005 *	1.865e+005 *	1.865e+005 *	1.865e+005 *					
41	Vap. Reynolds	9.838e+006 *	9.838e+006 *	9.838e+006 *	9.838e+006 *					
42	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
43	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910					
44	Deposit Thickness (mm)	---	---	---	---					
45	Deposit Volume (m3)	---	---	---	---					
46	Length (m)	2.099e+004	2.099e+004	2.100e+004	2.100e+004					
47	Elevation (m)	-0.6420	-0.6680	-0.6940	-0.7200					
48	Pressure (bar)	68.96	68.96	68.96	68.96					
49	Temperature (C)	49.89	49.89	49.89	49.89					
50	Vapour Fraction	0.8813	0.8813	0.8813	0.8813					
51	Heat Transferred (kJ/h-m)	530.0	529.9	529.9	529.9					
52	Flow Regime	Transition	Transition	Transition	Transition					
53	Water Flow Regime									
54	Liquid HoldUp	0.1681	0.1681	0.1681	0.1681					
55	Water HoldUp	---	---	---	---					
56	Friction Gradient (kPa/m)	9.040e-003	9.040e-003	9.040e-003	9.040e-003					
57	Static Gradient (kPa/m)	-8.227e-003	-8.227e-003	-8.227e-003	-8.227e-003					
58	Accel. Gradient (kPa/m)	1.011e-007	1.011e-007	1.011e-007	1.011e-007					
59	Liq. Reynolds	1.865e+005 *	1.865e+005 *	1.865e+005 *	1.865e+005 *					
60	Vap. Reynolds	9.838e+006 *	9.838e+006 *	9.838e+006 *	9.838e+006 *					
61	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
62	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910					
63	Deposit Thickness (mm)	---	---	---	---					
64	Deposit Volume (m3)	---	---	---	---					
65	Length (m)	2.101e+004	2.101e+004	2.102e+004	2.102e+004					
66	Elevation (m)	-0.7160	-0.7120	-0.7080	-0.7040					
67	Pressure (bar)	68.96	68.96	68.96	68.96					
68	Temperature (C)	49.89	49.89	49.88	49.88					
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 354 of 937					

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Vapour Fraction		0.8813	0.8813	0.8813	0.8813				
14	Heat Transferred (kJ/h-m)		529.8	529.7	529.7	529.6				
15	Flow Regime		Transition	Transition	Transition	Transition				
16	Water Flow Regime									
17	Liquid HoldUp		0.2342	0.2342	0.2342	0.2342				
18	Water HoldUp		---	---	---	---				
19	Friction Gradient (kPa/m)		8.100e-003	8.100e-003	8.100e-003	8.100e-003				
20	Static Gradient (kPa/m)		1.596e-003	1.596e-003	1.596e-003	1.596e-003				
21	Accel. Gradient (kPa/m)		1.206e-006	1.206e-006	1.206e-006	1.206e-006				
22	Liq. Reynolds		1.865e+005 *	1.865e+005 *	1.865e+005 *	1.865e+005 *				
23	Vap. Reynolds		9.838e+006 *	9.838e+006 *	9.838e+006 *	9.838e+006 *				
24	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979				
25	Vapour Velocity (m/s)		2.910	2.910	2.910	2.910				
26	Deposit Thickness (mm)		---	---	---	---				
27	Deposit Volume (m3)		---	---	---	---				
28	Length (m)		2.103e+004	2.103e+004	2.104e+004	2.104e+004				
29	Elevation (m)		-0.7000	-0.7200	-0.7400	-0.7600				
30	Pressure (bar)		68.96	68.96	68.96	68.96				
31	Temperature (C)		49.88	49.88	49.88	49.88				
32	Vapour Fraction		0.8813	0.8813	0.8813	0.8813				
33	Heat Transferred (kJ/h-m)		529.6	529.6	529.6	529.5				
34	Flow Regime		Transition	Transition	Transition	Transition				
35	Water Flow Regime									
36	Liquid HoldUp		0.2342	0.1690	0.1690	0.1690				
37	Water HoldUp		---	---	---	---				
38	Friction Gradient (kPa/m)		8.100e-003	9.027e-003	9.027e-003	9.027e-003				
39	Static Gradient (kPa/m)		1.596e-003	-6.351e-003	-6.351e-003	-6.351e-003				
40	Accel. Gradient (kPa/m)		1.206e-006	3.330e-007	3.330e-007	3.330e-007				
41	Liq. Reynolds		1.865e+005 *	1.865e+005 *	1.865e+005 *	1.865e+005 *				
42	Vap. Reynolds		9.838e+006 *	9.838e+006 *	9.838e+006 *	9.838e+006 *				
43	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979				
44	Vapour Velocity (m/s)		2.910	2.909	2.909	2.909				
45	Deposit Thickness (mm)		---	---	---	---				
46	Deposit Volume (m3)		---	---	---	---				
47	Length (m)		2.105e+004	2.105e+004	2.106e+004	2.106e+004				
48	Elevation (m)		-0.7800	-0.8000	-0.7660	-0.7320				
49	Pressure (bar)		68.96	68.96	68.96	68.96				
50	Temperature (C)		49.87	49.87	49.87	49.87				
51	Vapour Fraction		0.8813	0.8812	0.8812	0.8812				
52	Heat Transferred (kJ/h-m)		529.5	529.4	529.3	529.3				
53	Flow Regime		Transition	Transition	Transition	Transition				
54	Water Flow Regime									
55	Liquid HoldUp		0.1690	0.1690	0.2509	0.2509				
56	Water HoldUp		---	---	---	---				
57	Friction Gradient (kPa/m)		9.027e-003	9.027e-003	6.138e-003	6.138e-003				
58	Static Gradient (kPa/m)		-6.351e-003	-6.351e-003	1.427e-002	1.427e-002				
59	Accel. Gradient (kPa/m)		3.329e-007	3.329e-007	2.539e-006	2.539e-006				
60	Liq. Reynolds		1.865e+005 *	1.865e+005 *	1.865e+005 *	1.865e+005 *				
61	Vap. Reynolds		9.838e+006 *	9.838e+006 *	9.838e+006 *	9.838e+006 *				
62	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979				
63	Vapour Velocity (m/s)		2.909	2.909	2.909	2.909				
64	Deposit Thickness (mm)		---	---	---	---				
65	Deposit Volume (m3)		---	---	---	---				
66	Length (m)		2.107e+004	2.107e+004	2.108e+004	2.108e+004				
67	Elevation (m)		-0.6980	-0.6640	-0.6300	-0.6580				
68	Pressure (bar)		68.96	68.95	68.95	68.95				
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 355 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	49.87	49.87	49.86	49.86
14	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
15	Heat Transferred (kJ/h-m)	529.2	529.2	529.1	529.1
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2509	0.2509	0.2509	0.1679
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	6.138e-003	6.138e-003	6.138e-003	9.043e-003
21	Static Gradient (kPa/m)	1.427e-002	1.427e-002	1.427e-002	-8.716e-003
22	Accel. Gradient (kPa/m)	2.539e-006	2.539e-006	2.539e-006	4.067e-008
23	Liq. Reynolds	1.865e+005 *	1.865e+005 *	1.865e+005 *	1.865e+005 *
24	Vap. Reynolds	9.838e+006 *	9.838e+006 *	9.838e+006 *	9.838e+006 *
25	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
26	Vapour Velocity (m/s)	2.909	2.910	2.910	2.910
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	2.109e+004	2.109e+004	2.110e+004	2.110e+004
30	Elevation (m)	-0.6860	-0.7140	-0.7420	-0.7700
31	Pressure (bar)	68.95	68.95	68.95	68.95
32	Temperature (C)	49.86	49.86	49.86	49.86
33	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
34	Heat Transferred (kJ/h-m)	529.1	529.1	529.0	529.0
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1679	0.1679	0.1679	0.1679
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	9.043e-003	9.043e-003	9.043e-003	9.043e-003
40	Static Gradient (kPa/m)	-8.716e-003	-8.716e-003	-8.716e-003	-8.716e-003
41	Accel. Gradient (kPa/m)	4.066e-008	4.064e-008	4.063e-008	4.061e-008
42	Liq. Reynolds	1.865e+005 *	1.865e+005 *	1.865e+005 *	1.865e+005 *
43	Vap. Reynolds	9.838e+006 *	9.838e+006 *	9.838e+006 *	9.838e+006 *
44	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
45	Vapour Velocity (m/s)	2.909	2.909	2.909	2.909
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	2.111e+004	2.111e+004	2.111e+004	2.112e+004
49	Elevation (m)	-0.8620	-0.9540	-1.046	-1.138
50	Pressure (bar)	68.95	68.96	68.96	68.96
51	Temperature (C)	49.86	49.85	49.85	49.85
52	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
53	Heat Transferred (kJ/h-m)	529.0	529.0	529.0	528.9
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.1549	0.1549	0.1549	0.1549
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	9.245e-003	9.245e-003	9.244e-003	9.244e-003
59	Static Gradient (kPa/m)	-3.449e-002	-3.449e-002	-3.450e-002	-3.450e-002
60	Accel. Gradient (kPa/m)	-3.141e-006	-3.141e-006	-3.141e-006	-3.141e-006
61	Liq. Reynolds	1.865e+005 *	1.865e+005 *	1.865e+005 *	1.865e+005 *
62	Vap. Reynolds	9.838e+006 *	9.838e+006 *	9.838e+006 *	9.838e+006 *
63	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1980
64	Vapour Velocity (m/s)	2.909	2.909	2.909	2.909
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	2.112e+004	2.113e+004	2.113e+004	2.113e+004
68	Elevation (m)	-1.230	-1.170	-1.110	-1.050
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	68.96	68.96	68.96	68.95		
14	Temperature (C)	49.85	49.85	49.85	49.85		
15	Vapour Fraction	0.8812	0.8812	0.8812	0.8812		
16	Heat Transferred (kJ/h-m)	528.9	528.8	528.7	528.7		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.1549	0.2737	0.2737	0.2737		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	9.244e-003	7.864e-003	7.864e-003	7.864e-003		
22	Static Gradient (kPa/m)	-3.450e-002	3.361e-002	3.361e-002	3.361e-002		
23	Accel. Gradient (kPa/m)	-3.141e-006	5.160e-006	5.160e-006	5.160e-006		
24	Liq. Reynolds	1.865e+005 *	1.865e+005 *	1.865e+005 *	1.865e+005 *		
25	Vap. Reynolds	9.838e+006 *	9.838e+006 *	9.838e+006 *	9.838e+006 *		
26	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
27	Vapour Velocity (m/s)	2.909	2.909	2.909	2.909		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	2.114e+004	2.114e+004	2.114e+004	2.115e+004		
31	Elevation (m)	-0.9900	-0.9300	-0.9020	-0.8740		
32	Pressure (bar)	68.95	68.95	68.95	68.95		
33	Temperature (C)	49.85	49.84	49.84	49.84		
34	Vapour Fraction	0.8812	0.8812	0.8812	0.8812		
35	Heat Transferred (kJ/h-m)	528.7	528.6	528.6	528.6		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.2737	0.2736	0.2709	0.2709		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	7.864e-003	7.865e-003	7.848e-003	7.848e-003		
41	Static Gradient (kPa/m)	3.361e-002	3.361e-002	3.113e-002	3.113e-002		
42	Accel. Gradient (kPa/m)	5.161e-006	5.161e-006	4.850e-006	4.850e-006		
43	Liq. Reynolds	1.865e+005 *	1.865e+005 *	1.865e+005 *	1.865e+005 *		
44	Vap. Reynolds	9.838e+006 *	9.838e+006 *	9.838e+006 *	9.838e+006 *		
45	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
46	Vapour Velocity (m/s)	2.909	2.909	2.909	2.909		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	2.115e+004	2.115e+004	2.115e+004	2.116e+004		
50	Elevation (m)	-0.8460	-0.8180	-0.7900	-0.7460		
51	Pressure (bar)	68.95	68.95	68.95	68.95		
52	Temperature (C)	49.84	49.84	49.84	49.84		
53	Vapour Fraction	0.8812	0.8812	0.8812	0.8812		
54	Heat Transferred (kJ/h-m)	528.5	528.5	528.5	528.4		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.2709	0.2709	0.2709	0.2564		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	7.848e-003	7.848e-003	7.848e-003	7.345e-003		
60	Static Gradient (kPa/m)	3.113e-002	3.113e-002	3.113e-002	1.877e-002		
61	Accel. Gradient (kPa/m)	4.850e-006	4.850e-006	4.850e-006	3.250e-006		
62	Liq. Reynolds	1.865e+005 *	1.865e+005 *	1.865e+005 *	1.865e+005 *		
63	Vap. Reynolds	9.838e+006 *	9.838e+006 *	9.838e+006 *	9.838e+006 *		
64	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
65	Vapour Velocity (m/s)	2.909	2.909	2.910	2.910		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	2.116e+004	2.117e+004	2.117e+004	2.118e+004		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Elevation (m)	-0.7020	-0.6580	-0.6140	-0.5700		
14	Pressure (bar)	68.94	68.94	68.94	68.94		
15	Temperature (C)	49.83	49.83	49.83	49.83		
16	Vapour Fraction	0.8812	0.8812	0.8812	0.8812		
17	Heat Transferred (kJ/h-m)	528.4	528.3	528.3	528.2		
18	Flow Regime	Transition	Transition	Transition	Transition		
19	Water Flow Regime						
20	Liquid HoldUp	0.2564	0.2564	0.2564	0.2564		
21	Water HoldUp	---	---	---	---		
22	Friction Gradient (kPa/m)	7.345e-003	7.345e-003	7.345e-003	7.345e-003		
23	Static Gradient (kPa/m)	1.877e-002	1.877e-002	1.877e-002	1.877e-002		
24	Accel. Gradient (kPa/m)	3.250e-006	3.250e-006	3.251e-006	3.251e-006		
25	Liq. Reynolds	1.864e+005 *	1.864e+005 *	1.864e+005 *	1.864e+005 *		
26	Vap. Reynolds	9.839e+006 *	9.839e+006 *	9.839e+006 *	9.839e+006 *		
27	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
28	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910		
29	Deposit Thickness (mm)	---	---	---	---		
30	Deposit Volume (m3)	---	---	---	---		
31	Length (m)	2.118e+004	2.119e+004	2.119e+004	2.120e+004		
32	Elevation (m)	-0.5540	-0.5380	-0.5220	-0.5060		
33	Pressure (bar)	68.94	68.94	68.94	68.94		
34	Temperature (C)	49.83	49.83	49.82	49.82		
35	Vapour Fraction	0.8812	0.8812	0.8812	0.8812		
36	Heat Transferred (kJ/h-m)	528.2	528.1	528.1	528.0		
37	Flow Regime	Transition	Transition	Transition	Transition		
38	Water Flow Regime						
39	Liquid HoldUp	0.2409	0.2409	0.2409	0.2409		
40	Water HoldUp	---	---	---	---		
41	Friction Gradient (kPa/m)	7.267e-003	7.267e-003	7.267e-003	7.267e-003		
42	Static Gradient (kPa/m)	6.516e-003	6.516e-003	6.516e-003	6.516e-003		
43	Accel. Gradient (kPa/m)	1.715e-006	1.715e-006	1.715e-006	1.715e-006		
44	Liq. Reynolds	1.864e+005 *	1.864e+005 *	1.864e+005 *	1.864e+005 *		
45	Vap. Reynolds	9.839e+006 *	9.839e+006 *	9.839e+006 *	9.839e+006 *		
46	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
47	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910		
48	Deposit Thickness (mm)	---	---	---	---		
49	Deposit Volume (m3)	---	---	---	---		
50	Length (m)	2.120e+004	2.121e+004	2.121e+004	2.122e+004		
51	Elevation (m)	-0.4900	-0.4920	-0.4940	-0.4960		
52	Pressure (bar)	68.94	68.94	68.94	68.94		
53	Temperature (C)	49.82	49.82	49.82	49.82		
54	Vapour Fraction	0.8812	0.8812	0.8812	0.8812		
55	Heat Transferred (kJ/h-m)	528.0	528.0	527.9	527.9		
56	Flow Regime	Transition	Transition	Transition	Transition		
57	Water Flow Regime						
58	Liquid HoldUp	0.2409	0.1717	0.1717	0.1717		
59	Water HoldUp	---	---	---	---		
60	Friction Gradient (kPa/m)	7.267e-003	8.992e-003	8.992e-003	8.992e-003		
61	Static Gradient (kPa/m)	6.516e-003	-6.418e-004	-6.418e-004	-6.418e-004		
62	Accel. Gradient (kPa/m)	1.715e-006	1.039e-006	1.039e-006	1.039e-006		
63	Liq. Reynolds	1.864e+005 *	1.864e+005 *	1.864e+005 *	1.864e+005 *		
64	Vap. Reynolds	9.839e+006 *	9.839e+006 *	9.839e+006 *	9.839e+006 *		
65	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
66	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910		
67	Deposit Thickness (mm)	---	---	---	---		
68	Deposit Volume (m3)	---	---	---	---		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3						
4						
5			Unit Set: Mixed			
6			Date/Time: Mon Jun 13 15:13:43 2022			
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	PROFILES					
11	Pipe Table					
12						
13	Length (m)	2.122e+004	2.123e+004	2.123e+004	2.124e+004	
14	Elevation (m)	-0.4980	-0.5000	-0.4820	-0.4640	
15	Pressure (bar)	68.93	68.93	68.93	68.93	
16	Temperature (C)	49.81	49.81	49.81	49.81	
17	Vapour Fraction	0.8812	0.8812	0.8812	0.8812	
18	Heat Transferred (kJ/h-m)	527.9	527.8	527.7	527.7	
19	Flow Regime	Transition	Transition	Transition	Transition	
20	Water Flow Regime					
21	Liquid HoldUp	0.1717	0.1717	0.2420	0.2420	
22	Water HoldUp	---	---	---	---	
23	Friction Gradient (kPa/m)	8.992e-003	8.992e-003	7.134e-003	7.134e-003	
24	Static Gradient (kPa/m)	-6.418e-004	-6.418e-004	7.356e-003	7.356e-003	
25	Accel. Gradient (kPa/m)	1.039e-006	1.039e-006	1.804e-006	1.804e-006	
26	Liq. Reynolds	1.864e+005 *	1.864e+005 *	1.864e+005 *	1.864e+005 *	
27	Vap. Reynolds	9.839e+006 *	9.839e+006 *	9.839e+006 *	9.839e+006 *	
28	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979	
29	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910	
30	Deposit Thickness (mm)	---	---	---	---	
31	Deposit Volume (m3)	---	---	---	---	
32	Length (m)	2.124e+004	2.125e+004	2.125e+004	2.126e+004	
33	Elevation (m)	-0.4460	-0.4280	-0.4100	-0.4300	
34	Pressure (bar)	68.93	68.93	68.93	68.93	
35	Temperature (C)	49.81	49.80	49.80	49.80	
36	Vapour Fraction	0.8812	0.8812	0.8812	0.8812	
37	Heat Transferred (kJ/h-m)	527.6	527.6	527.5	527.5	
38	Flow Regime	Transition	Transition	Transition	Transition	
39	Water Flow Regime					
40	Liquid HoldUp	0.2420	0.2420	0.2420	0.1690	
41	Water HoldUp	---	---	---	---	
42	Friction Gradient (kPa/m)	7.135e-003	7.135e-003	7.135e-003	9.028e-003	
43	Static Gradient (kPa/m)	7.356e-003	7.356e-003	7.356e-003	-6.351e-003	
44	Accel. Gradient (kPa/m)	1.804e-006	1.804e-006	1.804e-006	3.332e-007	
45	Liq. Reynolds	1.864e+005 *	1.864e+005 *	1.864e+005 *	1.864e+005 *	
46	Vap. Reynolds	9.839e+006 *	9.839e+006 *	9.839e+006 *	9.839e+006 *	
47	Liquid Velocity (m/s)	0.1979	0.1980	0.1980	0.1980	
48	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910	
49	Deposit Thickness (mm)	---	---	---	---	
50	Deposit Volume (m3)	---	---	---	---	
51	Length (m)	2.126e+004	2.127e+004	2.127e+004	2.128e+004	
52	Elevation (m)	-0.4500	-0.4700	-0.4900	-0.5100	
53	Pressure (bar)	68.93	68.93	68.93	68.93	
54	Temperature (C)	49.80	49.80	49.80	49.80	
55	Vapour Fraction	0.8812	0.8812	0.8812	0.8812	
56	Heat Transferred (kJ/h-m)	527.5	527.5	527.4	527.4	
57	Flow Regime	Transition	Transition	Transition	Transition	
58	Water Flow Regime					
59	Liquid HoldUp	0.1690	0.1690	0.1690	0.1690	
60	Water HoldUp	---	---	---	---	
61	Friction Gradient (kPa/m)	9.028e-003	9.028e-003	9.028e-003	9.028e-003	
62	Static Gradient (kPa/m)	-6.351e-003	-6.351e-003	-6.351e-003	-6.351e-003	
63	Accel. Gradient (kPa/m)	3.332e-007	3.332e-007	3.332e-007	3.332e-007	
64	Liq. Reynolds	1.864e+005 *	1.864e+005 *	1.864e+005 *	1.864e+005 *	
65	Vap. Reynolds	9.839e+006 *	9.839e+006 *	9.839e+006 *	9.839e+006 *	
66	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	
67	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910	
68	Deposit Thickness (mm)	---	---	---	---	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	2.128e+004	2.129e+004	2.129e+004	2.130e+004		
15	Elevation (m)	-0.5300	-0.5500	-0.5700	-0.5900		
16	Pressure (bar)	68.93	68.93	68.93	68.93		
17	Temperature (C)	49.79	49.79	49.79	49.79		
18	Vapour Fraction	0.8812	0.8812	0.8812	0.8812		
19	Heat Transferred (kJ/h-m)	527.3	527.3	527.3	527.2		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	0.1690	0.1690	0.1690	0.1690		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	9.028e-003	9.028e-003	9.028e-003	9.028e-003		
25	Static Gradient (kPa/m)	-6.399e-003	-6.399e-003	-6.399e-003	-6.399e-003		
26	Accel. Gradient (kPa/m)	3.273e-007	3.272e-007	3.272e-007	3.272e-007		
27	Liq. Reynolds	1.864e+005 *	1.864e+005 *	1.864e+005 *	1.864e+005 *		
28	Vap. Reynolds	9.839e+006 *	9.839e+006 *	9.839e+006 *	9.839e+006 *		
29	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
30	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	2.130e+004	2.131e+004	2.131e+004	2.132e+004		
34	Elevation (m)	-0.6100	-0.6180	-0.6260	-0.6340		
35	Pressure (bar)	68.93	68.93	68.93	68.93		
36	Temperature (C)	49.79	49.79	49.78	49.78		
37	Vapour Fraction	0.8812	0.8812	0.8812	0.8812		
38	Heat Transferred (kJ/h-m)	527.2	527.1	527.1	527.0		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.1690	0.1708	0.1708	0.1708		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	9.028e-003	9.004e-003	9.004e-003	9.004e-003		
44	Static Gradient (kPa/m)	-6.399e-003	-2.558e-003	-2.558e-003	-2.558e-003		
45	Accel. Gradient (kPa/m)	3.272e-007	8.023e-007	8.023e-007	8.023e-007		
46	Liq. Reynolds	1.864e+005 *	1.864e+005 *	1.864e+005 *	1.864e+005 *		
47	Vap. Reynolds	9.839e+006 *	9.839e+006 *	9.839e+006 *	9.839e+006 *		
48	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
49	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	2.132e+004	2.133e+004	2.133e+004	2.133e+004		
53	Elevation (m)	-0.6420	-0.6500	-0.6520	-0.6540		
54	Pressure (bar)	68.93	68.93	68.93	68.93		
55	Temperature (C)	49.78	49.78	49.78	49.78		
56	Vapour Fraction	0.8812	0.8812	0.8812	0.8812		
57	Heat Transferred (kJ/h-m)	527.0	527.0	526.9	526.9		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.1708	0.1708	0.1712	0.1712		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	9.004e-003	9.004e-003	8.998e-003	8.998e-003		
63	Static Gradient (kPa/m)	-2.558e-003	-2.558e-003	-1.602e-003	-1.602e-003		
64	Accel. Gradient (kPa/m)	8.023e-007	8.023e-007	9.206e-007	9.206e-007		
65	Liq. Reynolds	1.864e+005 *	1.864e+005 *	1.864e+005 *	1.864e+005 *		
66	Vap. Reynolds	9.839e+006 *	9.839e+006 *	9.839e+006 *	9.839e+006 *		
67	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
68	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
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8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Deposit Thickness	(mm)	---	---	---	---				
14	Deposit Volume	(m3)	---	---	---	---				
15	Length	(m)	2.133e+004	2.133e+004	2.134e+004	2.134e+004				
16	Elevation	(m)	-0.6560	-0.6580	-0.6600	-0.6300				
17	Pressure	(bar)	68.93	68.93	68.93	68.93				
18	Temperature	(C)	49.78	49.78	49.78	49.78				
19	Vapour Fraction		0.8812	0.8812	0.8812	0.8812				
20	Heat Transferred	(kJ/h-m)	526.9	526.9	526.9	526.8				
21	Flow Regime		Transition	Transition	Transition	Transition				
22	Water Flow Regime									
23	Liquid HoldUp		0.1712	0.1712	0.1712	0.2598				
24	Water HoldUp		---	---	---	---				
25	Friction Gradient	(kPa/m)	8.998e-003	8.998e-003	8.998e-003	7.612e-003				
26	Static Gradient	(kPa/m)	-1.602e-003	-1.602e-003	-1.602e-003	2.154e-002				
27	Accel. Gradient	(kPa/m)	9.206e-007	9.206e-007	9.206e-007	3.629e-006				
28	Liq. Reynolds		1.864e+005 *	1.864e+005 *	1.864e+005 *	1.864e+005 *				
29	Vap. Reynolds		9.839e+006 *	9.839e+006 *	9.839e+006 *	9.839e+006 *				
30	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
31	Vapour Velocity	(m/s)	2.910	2.910	2.909	2.910				
32	Deposit Thickness	(mm)	---	---	---	---				
33	Deposit Volume	(m3)	---	---	---	---				
34	Length	(m)	2.134e+004	2.134e+004	2.135e+004	2.135e+004				
35	Elevation	(m)	-0.6000	-0.5700	-0.5400	-0.5100				
36	Pressure	(bar)	68.93	68.92	68.92	68.92				
37	Temperature	(C)	49.77	49.77	49.77	49.77				
38	Vapour Fraction		0.8812	0.8812	0.8812	0.8812				
39	Heat Transferred	(kJ/h-m)	526.8	526.7	526.7	526.7				
40	Flow Regime		Transition	Transition	Transition	Transition				
41	Water Flow Regime									
42	Liquid HoldUp		0.2598	0.2598	0.2598	0.2598				
43	Water HoldUp		---	---	---	---				
44	Friction Gradient	(kPa/m)	7.612e-003	7.612e-003	7.612e-003	7.612e-003				
45	Static Gradient	(kPa/m)	2.154e-002	2.154e-002	2.154e-002	2.154e-002				
46	Accel. Gradient	(kPa/m)	3.629e-006	3.629e-006	3.629e-006	3.629e-006				
47	Liq. Reynolds		1.864e+005 *	1.864e+005 *	1.864e+005 *	1.864e+005 *				
48	Vap. Reynolds		9.839e+006 *	9.839e+006 *	9.839e+006 *	9.840e+006 *				
49	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
50	Vapour Velocity	(m/s)	2.910	2.910	2.910	2.910				
51	Deposit Thickness	(mm)	---	---	---	---				
52	Deposit Volume	(m3)	---	---	---	---				
53	Length	(m)	2.136e+004	2.136e+004	2.137e+004	2.137e+004				
54	Elevation	(m)	-0.5060	-0.5020	-0.4980	-0.4940				
55	Pressure	(bar)	68.92	68.92	68.92	68.92				
56	Temperature	(C)	49.77	49.77	49.77	49.76				
57	Vapour Fraction		0.8812	0.8812	0.8812	0.8812				
58	Heat Transferred	(kJ/h-m)	526.6	526.6	526.5	526.5				
59	Flow Regime		Transition	Transition	Transition	Transition				
60	Water Flow Regime									
61	Liquid HoldUp		0.2342	0.2342	0.2342	0.2342				
62	Water HoldUp		---	---	---	---				
63	Friction Gradient	(kPa/m)	8.100e-003	8.100e-003	8.100e-003	8.100e-003				
64	Static Gradient	(kPa/m)	1.596e-003	1.596e-003	1.596e-003	1.596e-003				
65	Accel. Gradient	(kPa/m)	1.207e-006	1.207e-006	1.207e-006	1.207e-006				
66	Liq. Reynolds		1.864e+005 *	1.864e+005 *	1.864e+005 *	1.864e+005 *				
67	Vap. Reynolds		9.840e+006 *	9.840e+006 *	9.840e+006 *	9.840e+006 *				
68	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.910	2.910	2.910	2.910	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	2.138e+004	2.138e+004	2.139e+004	2.139e+004	
17	Elevation	(m)	-0.4900	-0.4880	-0.4860	-0.4840	
18	Pressure	(bar)	68.92	68.92	68.92	68.92	
19	Temperature	(C)	49.76	49.76	49.76	49.76	
20	Vapour Fraction		0.8812	0.8812	0.8812	0.8812	
21	Heat Transferred	(kJ/h-m)	526.4	526.4	526.4	526.3	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2342	0.2331	0.2331	0.2331	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	8.100e-003	8.246e-003	8.246e-003	8.246e-003	
27	Static Gradient	(kPa/m)	1.596e-003	7.951e-004	7.951e-004	7.951e-004	
28	Accel. Gradient	(kPa/m)	1.207e-006	1.126e-006	1.126e-006	1.126e-006	
29	Liq. Reynolds		1.864e+005 *	1.864e+005 *	1.864e+005 *	1.864e+005 *	
30	Vap. Reynolds		9.840e+006 *	9.840e+006 *	9.840e+006 *	9.840e+006 *	
31	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
32	Vapour Velocity	(m/s)	2.910	2.910	2.910	2.910	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	2.140e+004	2.140e+004	2.141e+004	2.141e+004	
36	Elevation	(m)	-0.4820	-0.4800	-0.4320	-0.3840	
37	Pressure	(bar)	68.92	68.92	68.92	68.92	
38	Temperature	(C)	49.76	49.75	49.75	49.75	
39	Vapour Fraction		0.8812	0.8812	0.8812	0.8812	
40	Heat Transferred	(kJ/h-m)	526.3	526.2	526.2	526.1	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2331	0.2331	0.2586	0.2586	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	8.246e-003	8.246e-003	7.547e-003	7.547e-003	
46	Static Gradient	(kPa/m)	7.951e-004	7.951e-004	2.061e-002	2.061e-002	
47	Accel. Gradient	(kPa/m)	1.126e-006	1.126e-006	3.506e-006	3.506e-006	
48	Liq. Reynolds		1.864e+005 *	1.864e+005 *	1.864e+005 *	1.863e+005 *	
49	Vap. Reynolds		9.840e+006 *	9.840e+006 *	9.840e+006 *	9.840e+006 *	
50	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
51	Vapour Velocity	(m/s)	2.910	2.910	2.910	2.910	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	2.142e+004	2.142e+004	2.143e+004	2.143e+004	
55	Elevation	(m)	-0.3360	-0.2880	-0.2400	-0.2480	
56	Pressure	(bar)	68.91	68.91	68.91	68.91	
57	Temperature	(C)	49.75	49.75	49.74	49.74	
58	Vapour Fraction		0.8812	0.8812	0.8812	0.8812	
59	Heat Transferred	(kJ/h-m)	526.1	526.0	526.0	526.0	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.2586	0.2586	0.2586	0.1708	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	7.547e-003	7.547e-003	7.547e-003	9.004e-003	
65	Static Gradient	(kPa/m)	2.061e-002	2.061e-002	2.061e-002	-2.558e-003	
66	Accel. Gradient	(kPa/m)	3.506e-006	3.506e-006	3.506e-006	8.026e-007	
67	Liq. Reynolds		1.863e+005 *	1.863e+005 *	1.863e+005 *	1.863e+005 *	
68	Vap. Reynolds		9.840e+006 *	9.840e+006 *	9.840e+006 *	9.840e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
14	Vapour Velocity	(m/s)	2.910	2.910	2.910	2.910	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	2.144e+004	2.144e+004	2.145e+004	2.145e+004	
18	Elevation	(m)	-0.2560	-0.2640	-0.2720	-0.2800	
19	Pressure	(bar)	68.91	68.91	68.91	68.91	
20	Temperature	(C)	49.74	49.74	49.74	49.74	
21	Vapour Fraction		0.8812	0.8812	0.8812	0.8812	
22	Heat Transferred	(kJ/h-m)	525.9	525.9	525.8	525.8	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.1708	0.1708	0.1708	0.1708	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	9.004e-003	9.004e-003	9.004e-003	9.004e-003	
28	Static Gradient	(kPa/m)	-2.558e-003	-2.558e-003	-2.558e-003	-2.558e-003	
29	Accel. Gradient	(kPa/m)	8.026e-007	8.026e-007	8.026e-007	8.026e-007	
30	Liq. Reynolds		1.863e+005 *	1.863e+005 *	1.863e+005 *	1.863e+005 *	
31	Vap. Reynolds		9.840e+006 *	9.840e+006 *	9.840e+006 *	9.840e+006 *	
32	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
33	Vapour Velocity	(m/s)	2.910	2.910	2.910	2.910	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	2.146e+004	2.146e+004	2.147e+004	2.147e+004	
37	Elevation	(m)	-0.2760	-0.2720	-0.2680	-0.2640	
38	Pressure	(bar)	68.91	68.91	68.91	68.91	
39	Temperature	(C)	49.73	49.73	49.73	49.73	
40	Vapour Fraction		0.8812	0.8812	0.8812	0.8812	
41	Heat Transferred	(kJ/h-m)	525.7	525.7	525.6	525.6	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2342	0.2342	0.2342	0.2342	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	8.101e-003	8.101e-003	8.101e-003	8.101e-003	
47	Static Gradient	(kPa/m)	1.596e-003	1.596e-003	1.596e-003	1.596e-003	
48	Accel. Gradient	(kPa/m)	1.207e-006	1.207e-006	1.207e-006	1.207e-006	
49	Liq. Reynolds		1.863e+005 *	1.863e+005 *	1.863e+005 *	1.863e+005 *	
50	Vap. Reynolds		9.840e+006 *	9.840e+006 *	9.840e+006 *	9.840e+006 *	
51	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
52	Vapour Velocity	(m/s)	2.910	2.910	2.910	2.910	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	2.148e+004	2.148e+004	2.149e+004	2.149e+004	
56	Elevation	(m)	-0.2600	-0.2620	-0.2640	-0.2660	
57	Pressure	(bar)	68.91	68.91	68.91	68.91	
58	Temperature	(C)	49.73	49.73	49.72	49.72	
59	Vapour Fraction		0.8812	0.8812	0.8812	0.8812	
60	Heat Transferred	(kJ/h-m)	525.5	525.5	525.5	525.4	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2342	0.1717	0.1717	0.1717	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	8.101e-003	8.992e-003	8.992e-003	8.992e-003	
66	Static Gradient	(kPa/m)	1.596e-003	-6.418e-004	-6.418e-004	-6.418e-004	
67	Accel. Gradient	(kPa/m)	1.207e-006	1.040e-006	1.040e-006	1.040e-006	
68	Liq. Reynolds		1.863e+005 *	1.863e+005 *	1.863e+005 *	1.863e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
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11	Pipe Table				
12					
13	Vap. Reynolds	9.840e+006 *	9.840e+006 *	9.840e+006 *	9.840e+006 *
14	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
15	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	2.150e+004	2.150e+004	2.150e+004	2.151e+004
19	Elevation (m)	-0.2680	-0.2700	-0.2840	-0.2980
20	Pressure (bar)	68.91	68.90	68.90	68.90
21	Temperature (C)	49.72	49.72	49.72	49.72
22	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
23	Heat Transferred (kJ/h-m)	525.4	525.3	525.3	525.3
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1717	0.1717	0.1694	0.1694
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	8.992e-003	8.992e-003	9.023e-003	9.023e-003
29	Static Gradient (kPa/m)	-6.418e-004	-6.418e-004	-5.565e-003	-5.565e-003
30	Accel. Gradient (kPa/m)	1.040e-006	1.040e-006	4.306e-007	4.306e-007
31	Liq. Reynolds	1.863e+005 *	1.863e+005 *	1.863e+005 *	1.863e+005 *
32	Vap. Reynolds	9.840e+006 *	9.840e+006 *	9.840e+006 *	9.840e+006 *
33	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
34	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	2.151e+004	2.152e+004	2.152e+004	2.152e+004
38	Elevation (m)	-0.3120	-0.3260	-0.3400	-0.3660
39	Pressure (bar)	68.90	68.90	68.90	68.90
40	Temperature (C)	49.72	49.71	49.71	49.71
41	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
42	Heat Transferred (kJ/h-m)	525.3	525.2	525.2	525.2
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1694	0.1694	0.1694	0.1523
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	9.023e-003	9.023e-003	9.023e-003	9.292e-003
48	Static Gradient (kPa/m)	-5.565e-003	-5.565e-003	-5.565e-003	-3.936e-002
49	Accel. Gradient (kPa/m)	4.306e-007	4.306e-007	4.306e-007	-3.744e-006
50	Liq. Reynolds	1.863e+005 *	1.863e+005 *	1.863e+005 *	1.863e+005 *
51	Vap. Reynolds	9.840e+006 *	9.840e+006 *	9.840e+006 *	9.840e+006 *
52	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
53	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	2.152e+004	2.152e+004	2.152e+004	2.153e+004
57	Elevation (m)	-0.3920	-0.4180	-0.4440	-0.4700
58	Pressure (bar)	68.90	68.91	68.91	68.91
59	Temperature (C)	49.71	49.71	49.71	49.71
60	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
61	Heat Transferred (kJ/h-m)	525.2	525.2	525.2	525.2
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1523	0.1523	0.1523	0.1523
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	9.292e-003	9.292e-003	9.292e-003	9.292e-003
67	Static Gradient (kPa/m)	-3.936e-002	-3.936e-002	-3.936e-002	-3.936e-002
68	Accel. Gradient (kPa/m)	-3.743e-006	-3.743e-006	-3.743e-006	-3.743e-006
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		1.863e+005 *	1.863e+005 *	1.863e+005 *	1.863e+005 *	
14	Vap. Reynolds		9.840e+006 *	9.840e+006 *	9.840e+006 *	9.840e+006 *	
15	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
16	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910		
17	Deposit Thickness (mm)	---	---	---	---		
18	Deposit Volume (m3)	---	---	---	---		
19	Length (m)	2.153e+004	2.154e+004	2.154e+004	2.155e+004		
20	Elevation (m)	-0.4320	-0.3940	-0.3560	-0.3180		
21	Pressure (bar)	68.90	68.90	68.90	68.90		
22	Temperature (C)	49.71	49.71	49.71	49.71		
23	Vapour Fraction	0.8812	0.8812	0.8812	0.8812		
24	Heat Transferred (kJ/h-m)	525.1	525.0	525.0	524.9		
25	Flow Regime	Transition	Transition	Transition	Transition		
26	Water Flow Regime						
27	Liquid HoldUp	0.2531	0.2531	0.2531	0.2531		
28	Water HoldUp	---	---	---	---		
29	Friction Gradient (kPa/m)	6.473e-003	6.473e-003	6.472e-003	6.472e-003		
30	Static Gradient (kPa/m)	1.606e-002	1.606e-002	1.606e-002	1.606e-002		
31	Accel. Gradient (kPa/m)	2.805e-006	2.805e-006	2.805e-006	2.805e-006		
32	Liq. Reynolds		1.863e+005 *	1.863e+005 *	1.863e+005 *		
33	Vap. Reynolds		9.840e+006 *	9.840e+006 *	9.840e+006 *		
34	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
35	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910		
36	Deposit Thickness (mm)	---	---	---	---		
37	Deposit Volume (m3)	---	---	---	---		
38	Length (m)	2.155e+004	2.156e+004	2.156e+004	2.157e+004		
39	Elevation (m)	-0.2800	-0.2320	-0.1840	-0.1360		
40	Pressure (bar)	68.90	68.90	68.90	68.90		
41	Temperature (C)	49.70	49.70	49.70	49.70		
42	Vapour Fraction	0.8812	0.8812	0.8812	0.8812		
43	Heat Transferred (kJ/h-m)	524.9	524.8	524.8	524.7		
44	Flow Regime	Transition	Transition	Transition	Transition		
45	Water Flow Regime						
46	Liquid HoldUp	0.2531	0.2586	0.2586	0.2586		
47	Water HoldUp	---	---	---	---		
48	Friction Gradient (kPa/m)	6.472e-003	7.547e-003	7.547e-003	7.548e-003		
49	Static Gradient (kPa/m)	1.606e-002	2.061e-002	2.061e-002	2.061e-002		
50	Accel. Gradient (kPa/m)	2.806e-006	3.507e-006	3.507e-006	3.507e-006		
51	Liq. Reynolds		1.863e+005 *	1.863e+005 *	1.863e+005 *		
52	Vap. Reynolds		9.841e+006 *	9.841e+006 *	9.841e+006 *		
53	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
54	Vapour Velocity (m/s)	2.910	2.910	2.910	2.910		
55	Deposit Thickness (mm)	---	---	---	---		
56	Deposit Volume (m3)	---	---	---	---		
57	Length (m)	2.157e+004	2.158e+004	2.158e+004	2.159e+004		
58	Elevation (m)	-8.800e-002	-4.000e-002	-0.1460	-0.2520		
59	Pressure (bar)	68.89	68.89	68.89	68.90		
60	Temperature (C)	49.70	49.69	49.69	49.69		
61	Vapour Fraction	0.8812	0.8812	0.8812	0.8812		
62	Heat Transferred (kJ/h-m)	524.6	524.6	524.6	524.6		
63	Flow Regime	Transition	Transition	Transition	Transition		
64	Water Flow Regime						
65	Liquid HoldUp	0.2586	0.2586	0.1562	0.1562		
66	Water HoldUp	---	---	---	---		
67	Friction Gradient (kPa/m)	7.548e-003	7.548e-003	9.224e-003	9.224e-003		
68	Static Gradient (kPa/m)	2.061e-002	2.061e-002	-3.197e-002	-3.197e-002		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient	(kPa/m)	3.507e-006	3.508e-006	-2.833e-006	-2.833e-006	
14	Liq. Reynolds		1.863e+005 *	1.863e+005 *	1.863e+005 *	1.863e+005 *	
15	Vap. Reynolds		9.841e+006 *	9.841e+006 *	9.841e+006 *	9.841e+006 *	
16	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
17	Vapour Velocity	(m/s)	2.910	2.910	2.910	2.910	
18	Deposit Thickness	(mm)	---	---	---	---	
19	Deposit Volume	(m3)	---	---	---	---	
20	Length	(m)	2.159e+004	2.160e+004	2.160e+004	2.161e+004	
21	Elevation	(m)	-0.3580	-0.4640	-0.5700	-1.040	
22	Pressure	(bar)	68.90	68.90	68.90	68.90	
23	Temperature	(C)	49.69	49.69	49.69	49.69	
24	Vapour Fraction		0.8812	0.8812	0.8812	0.8812	
25	Heat Transferred	(kJ/h-m)	524.6	524.6	524.5	524.7	
26	Flow Regime		Transition	Transition	Transition	Transition	
27	Water Flow Regime						
28	Liquid HoldUp		0.1562	0.1562	0.1562	0.1030	
29	Water HoldUp		---	---	---	---	
30	Friction Gradient	(kPa/m)	9.223e-003	9.223e-003	9.223e-003	1.078e-002	
31	Static Gradient	(kPa/m)	-3.197e-002	-3.197e-002	-3.197e-002	-0.1105	
32	Accel. Gradient	(kPa/m)	-2.833e-006	-2.833e-006	-2.833e-006	-1.242e-005	
33	Liq. Reynolds		1.863e+005 *	1.863e+005 *	1.863e+005 *	1.863e+005 *	
34	Vap. Reynolds		9.841e+006 *	9.841e+006 *	9.841e+006 *	9.840e+006 *	
35	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
36	Vapour Velocity	(m/s)	2.910	2.910	2.910	2.909	
37	Deposit Thickness	(mm)	---	---	---	---	
38	Deposit Volume	(m3)	---	---	---	---	
39	Length	(m)	2.161e+004	2.162e+004	2.162e+004	2.163e+004	
40	Elevation	(m)	-1.510	-1.980	-2.450	-2.920	
41	Pressure	(bar)	68.91	68.91	68.92	68.92	
42	Temperature	(C)	49.69	49.69	49.69	49.69	
43	Vapour Fraction		0.8812	0.8812	0.8812	0.8812	
44	Heat Transferred	(kJ/h-m)	524.7	524.8	524.8	524.8	
45	Flow Regime		Transition	Transition	Transition	Transition	
46	Water Flow Regime						
47	Liquid HoldUp		0.1030	0.1030	0.1030	0.1030	
48	Water HoldUp		---	---	---	---	
49	Friction Gradient	(kPa/m)	1.078e-002	1.078e-002	1.078e-002	1.078e-002	
50	Static Gradient	(kPa/m)	-0.1105	-0.1106	-0.1106	-0.1106	
51	Accel. Gradient	(kPa/m)	-1.242e-005	-1.242e-005	-1.242e-005	-1.242e-005	
52	Liq. Reynolds		1.863e+005 *	1.863e+005 *	1.863e+005 *	1.863e+005 *	
53	Vap. Reynolds		9.840e+006 *	9.840e+006 *	9.840e+006 *	9.840e+006 *	
54	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
55	Vapour Velocity	(m/s)	2.909	2.909	2.909	2.908	
56	Deposit Thickness	(mm)	---	---	---	---	
57	Deposit Volume	(m3)	---	---	---	---	
58	Length	(m)	2.163e+004	2.164e+004	2.164e+004	2.165e+004	
59	Elevation	(m)	-3.420	-3.920	-4.420	-4.920	
60	Pressure	(bar)	68.93	68.93	68.94	68.94	
61	Temperature	(C)	49.70	49.70	49.70	49.70	
62	Vapour Fraction		0.8812	0.8812	0.8812	0.8812	
63	Heat Transferred	(kJ/h-m)	524.9	524.9	525.0	525.0	
64	Flow Regime		Transition	Transition	Transition	Transition	
65	Water Flow Regime						
66	Liquid HoldUp		9.877e-002	9.877e-002	9.878e-002	9.878e-002	
67	Water HoldUp		---	---	---	---	
68	Friction Gradient	(kPa/m)	1.100e-002	1.100e-002	1.100e-002	1.100e-002	
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	-0.1150	-0.1150	-0.1150	-0.1150
14	Accel. Gradient	(kPa/m)	-1.293e-005	-1.293e-005	-1.293e-005	-1.293e-005
15	Liq. Reynolds		1.863e+005 *	1.863e+005 *	1.863e+005 *	1.863e+005 *
16	Vap. Reynolds		9.840e+006 *	9.839e+006 *	9.839e+006 *	9.839e+006 *
17	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
18	Vapour Velocity	(m/s)	2.908	2.908	2.908	2.907
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	2.165e+004	2.166e+004	2.166e+004	2.167e+004
22	Elevation	(m)	-5.420	-5.796	-6.172	-6.548
23	Pressure	(bar)	68.95	68.95	68.96	68.96
24	Temperature	(C)	49.70	49.70	49.70	49.70
25	Vapour Fraction		0.8811	0.8811	0.8811	0.8811
26	Heat Transferred	(kJ/h-m)	525.0	525.0	525.0	525.0
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		9.878e-002	0.1166	0.1166	0.1166
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	1.100e-002	1.020e-002	1.020e-002	1.020e-002
32	Static Gradient	(kPa/m)	-0.1150	-9.484e-002	-9.484e-002	-9.484e-002
33	Accel. Gradient	(kPa/m)	-1.293e-005	-1.052e-005	-1.052e-005	-1.052e-005
34	Liq. Reynolds		1.863e+005 *	1.864e+005 *	1.864e+005 *	1.864e+005 *
35	Vap. Reynolds		9.839e+006 *	9.839e+006 *	9.839e+006 *	9.838e+006 *
36	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
37	Vapour Velocity	(m/s)	2.907	2.907	2.907	2.907
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	2.167e+004	2.168e+004	2.168e+004	2.169e+004
41	Elevation	(m)	-6.924	-7.300	-7.300	-7.300
42	Pressure	(bar)	68.97	68.97	68.97	68.97
43	Temperature	(C)	49.70	49.71	49.70	49.70
44	Vapour Fraction		0.8811	0.8811	0.8811	0.8811
45	Heat Transferred	(kJ/h-m)	525.1	525.1	524.9	524.8
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.1166	0.1166	0.2513	0.2513
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	1.020e-002	1.020e-002	6.108e-003	6.108e-003
51	Static Gradient	(kPa/m)	-9.485e-002	-9.485e-002	0.0000	0.0000
52	Accel. Gradient	(kPa/m)	-1.052e-005	-1.052e-005	7.590e-007	7.590e-007
53	Liq. Reynolds		1.864e+005 *	1.864e+005 *	1.864e+005 *	1.864e+005 *
54	Vap. Reynolds		9.838e+006 *	9.838e+006 *	9.838e+006 *	9.838e+006 *
55	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
56	Vapour Velocity	(m/s)	2.906	2.906	2.906	2.906
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	2.169e+004	2.170e+004	2.170e+004	2.171e+004
60	Elevation	(m)	-7.300	-7.300	-7.300	-7.200
61	Pressure	(bar)	68.97	68.97	68.97	68.97
62	Temperature	(C)	49.70	49.70	49.70	49.69
63	Vapour Fraction		0.8811	0.8811	0.8811	0.8811
64	Heat Transferred	(kJ/h-m)	524.8	524.7	524.7	524.6
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2513	0.2513	0.2513	0.2876
68	Water HoldUp		---	---	---	---
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1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	6.108e-003	6.108e-003	6.108e-003	7.879e-003
14	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	4.650e-002
15	Accel. Gradient	(kPa/m)	7.590e-007	7.590e-007	7.590e-007	6.757e-006
16	Liq. Reynolds		1.864e+005 *	1.864e+005 *	1.864e+005 *	1.864e+005 *
17	Vap. Reynolds		9.838e+006 *	9.838e+006 *	9.838e+006 *	9.838e+006 *
18	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
19	Vapour Velocity	(m/s)	2.906	2.906	2.906	2.906
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	2.171e+004	2.172e+004	2.172e+004	2.173e+004
23	Elevation	(m)	-7.100	-7.000	-6.900	-6.800
24	Pressure	(bar)	68.96	68.96	68.96	68.96
25	Temperature	(C)	49.69	49.69	49.69	49.68
26	Vapour Fraction		0.8811	0.8811	0.8811	0.8811
27	Heat Transferred	(kJ/h-m)	524.6	524.5	524.4	524.4
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2876	0.2876	0.2876	0.2876
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	7.880e-003	7.880e-003	7.880e-003	7.880e-003
33	Static Gradient	(kPa/m)	4.650e-002	4.649e-002	4.649e-002	4.649e-002
34	Accel. Gradient	(kPa/m)	6.757e-006	6.758e-006	6.758e-006	6.759e-006
35	Liq. Reynolds		1.864e+005 *	1.864e+005 *	1.863e+005 *	1.863e+005 *
36	Vap. Reynolds		9.838e+006 *	9.838e+006 *	9.839e+006 *	9.839e+006 *
37	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
38	Vapour Velocity	(m/s)	2.906	2.906	2.907	2.907
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	2.173e+004	2.174e+004	2.174e+004	2.175e+004
42	Elevation	(m)	-6.700	-6.600	-6.500	-6.400
43	Pressure	(bar)	68.95	68.95	68.95	68.94
44	Temperature	(C)	49.68	49.68	49.68	49.67
45	Vapour Fraction		0.8811	0.8811	0.8811	0.8811
46	Heat Transferred	(kJ/h-m)	524.3	524.2	524.2	524.1
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2877	0.2877	0.2877	0.2877
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	7.880e-003	7.881e-003	7.881e-003	7.881e-003
52	Static Gradient	(kPa/m)	4.658e-002	4.658e-002	4.658e-002	4.658e-002
53	Accel. Gradient	(kPa/m)	6.770e-006	6.770e-006	6.771e-006	6.771e-006
54	Liq. Reynolds		1.863e+005 *	1.863e+005 *	1.863e+005 *	1.863e+005 *
55	Vap. Reynolds		9.839e+006 *	9.839e+006 *	9.839e+006 *	9.839e+006 *
56	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
57	Vapour Velocity	(m/s)	2.907	2.907	2.907	2.907
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	2.175e+004	2.176e+004	2.176e+004	2.177e+004
61	Elevation	(m)	-6.300	-6.200	-6.100	-6.000
62	Pressure	(bar)	68.94	68.94	68.94	68.93
63	Temperature	(C)	49.67	49.67	49.67	49.66
64	Vapour Fraction		0.8811	0.8811	0.8811	0.8811
65	Heat Transferred	(kJ/h-m)	524.0	524.0	523.9	523.8
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2877	0.2876	0.2876	0.2876
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	7.881e-003	7.882e-003	7.882e-003	7.882e-003
15	Static Gradient (kPa/m)	4.657e-002	4.657e-002	4.657e-002	4.657e-002
16	Accel. Gradient (kPa/m)	6.771e-006	6.772e-006	6.772e-006	6.773e-006
17	Liq. Reynolds	1.863e+005 *	1.863e+005 *	1.863e+005 *	1.863e+005 *
18	Vap. Reynolds	9.839e+006 *	9.839e+006 *	9.839e+006 *	9.840e+006 *
19	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
20	Vapour Velocity (m/s)	2.907	2.907	2.907	2.907
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	2.177e+004	2.178e+004	2.178e+004	2.179e+004
24	Elevation (m)	-5.900	-5.800	-4.712	-3.624
25	Pressure (bar)	68.93	68.93	68.87	68.81
26	Temperature (C)	49.66	49.66	49.64	49.62
27	Vapour Fraction	0.8811	0.8811	0.8812	0.8812
28	Heat Transferred (kJ/h-m)	523.8	523.7	524.4	523.9
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2876	0.2876	0.7962	0.7958
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	7.882e-003	7.883e-003	7.309e-003	7.315e-003
34	Static Gradient (kPa/m)	4.657e-002	4.657e-002	1.198	1.197
35	Accel. Gradient (kPa/m)	6.773e-006	6.773e-006	1.501e-004	1.504e-004
36	Liq. Reynolds	1.863e+005 *	1.863e+005 *	1.862e+005 *	1.861e+005 *
37	Vap. Reynolds	9.840e+006 *	9.840e+006 *	9.842e+006 *	9.844e+006 *
38	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1979
39	Vapour Velocity (m/s)	2.908	2.908	2.910	2.913
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	2.179e+004	2.180e+004	2.180e+004	2.181e+004
43	Elevation (m)	-2.536	-1.448	-0.3600	-0.3240
44	Pressure (bar)	68.75	68.69	68.63	68.63
45	Temperature (C)	49.60	49.59	49.57	49.56
46	Vapour Fraction	0.8812	0.8812	0.8813	0.8813
47	Heat Transferred (kJ/h-m)	523.4	522.9	522.4	521.2
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.7954	0.7951	0.7947	0.2514
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	7.321e-003	7.327e-003	7.333e-003	6.055e-003
53	Static Gradient (kPa/m)	1.197	1.196	1.196	1.513e-002
54	Accel. Gradient (kPa/m)	1.506e-004	1.508e-004	1.510e-004	2.659e-006
55	Liq. Reynolds	1.860e+005 *	1.860e+005 *	1.859e+005 *	1.859e+005 *
56	Vap. Reynolds	9.846e+006 *	9.848e+006 *	9.850e+006 *	9.850e+006 *
57	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
58	Vapour Velocity (m/s)	2.916	2.918	2.921	2.921
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	2.181e+004	2.182e+004	2.182e+004	2.183e+004
62	Elevation (m)	-0.2880	-0.2520	-0.2160	-0.1800
63	Pressure (bar)	68.63	68.62	68.62	68.62
64	Temperature (C)	49.56	49.56	49.56	49.56
65	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
66	Heat Transferred (kJ/h-m)	521.1	521.1	521.0	521.0
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.2514	0.2514	0.2514	0.2514
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	6.055e-003	6.055e-003	6.055e-003	6.055e-003
16	Static Gradient (kPa/m)	1.513e-002	1.513e-002	1.513e-002	1.513e-002
17	Accel. Gradient (kPa/m)	2.659e-006	2.659e-006	2.659e-006	2.659e-006
18	Liq. Reynolds	1.859e+005 *	1.859e+005 *	1.859e+005 *	1.859e+005 *
19	Vap. Reynolds	9.850e+006 *	9.850e+006 *	9.850e+006 *	9.851e+006 *
20	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
21	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	2.183e+004	2.184e+004	2.184e+004	2.185e+004
25	Elevation (m)	-0.1780	-0.1760	-0.1740	-0.1720
26	Pressure (bar)	68.62	68.62	68.62	68.62
27	Temperature (C)	49.56	49.55	49.55	49.55
28	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
29	Heat Transferred (kJ/h-m)	520.9	520.9	520.8	520.8
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2325	0.2325	0.2325	0.2325
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	8.291e-003	8.291e-003	8.291e-003	8.291e-003
35	Static Gradient (kPa/m)	7.933e-004	7.933e-004	7.933e-004	7.933e-004
36	Accel. Gradient (kPa/m)	1.140e-006	1.140e-006	1.140e-006	1.140e-006
37	Liq. Reynolds	1.859e+005 *	1.859e+005 *	1.859e+005 *	1.859e+005 *
38	Vap. Reynolds	9.851e+006 *	9.851e+006 *	9.851e+006 *	9.851e+006 *
39	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
40	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	2.185e+004	2.186e+004	2.186e+004	2.187e+004
44	Elevation (m)	-0.1700	-0.1640	-0.1580	-0.1520
45	Pressure (bar)	68.62	68.62	68.62	68.62
46	Temperature (C)	49.55	49.55	49.55	49.54
47	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
48	Heat Transferred (kJ/h-m)	520.8	520.7	520.7	520.6
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2325	0.2347	0.2347	0.2347
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	8.291e-003	8.000e-003	8.000e-003	8.000e-003
54	Static Gradient (kPa/m)	7.933e-004	2.396e-003	2.396e-003	2.396e-003
55	Accel. Gradient (kPa/m)	1.140e-006	1.305e-006	1.305e-006	1.305e-006
56	Liq. Reynolds	1.859e+005 *	1.859e+005 *	1.859e+005 *	1.859e+005 *
57	Vap. Reynolds	9.851e+006 *	9.851e+006 *	9.851e+006 *	9.851e+006 *
58	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
59	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	2.187e+004	2.188e+004	2.188e+004	2.189e+004
63	Elevation (m)	-0.1460	-0.1400	-0.1300	-0.1200
64	Pressure (bar)	68.62	68.62	68.62	68.62
65	Temperature (C)	49.54	49.54	49.54	49.54
66	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
67	Heat Transferred (kJ/h-m)	520.6	520.5	520.5	520.4
68	Flow Regime	Transition	Transition	Transition	Transition
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2347	0.2347	0.2370	0.2370
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	8.000e-003	8.000e-003	7.717e-003	7.717e-003
17	Static Gradient (kPa/m)	2.396e-003	2.396e-003	4.022e-003	4.022e-003
18	Accel. Gradient (kPa/m)	1.305e-006	1.305e-006	1.474e-006	1.474e-006
19	Liq. Reynolds	1.859e+005 *	1.859e+005 *	1.859e+005 *	1.858e+005 *
20	Vap. Reynolds	9.851e+006 *	9.851e+006 *	9.851e+006 *	9.851e+006 *
21	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
22	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	2.189e+004	2.190e+004	2.190e+004	2.191e+004
26	Elevation (m)	-0.1100	-0.1000	-9.000e-002	-9.000e-002
27	Pressure (bar)	68.62	68.61	68.61	68.61
28	Temperature (C)	49.54	49.53	49.53	49.53
29	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
30	Heat Transferred (kJ/h-m)	520.4	520.4	520.3	520.3
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2370	0.2370	0.2370	0.2504
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	7.717e-003	7.717e-003	7.717e-003	6.153e-003
36	Static Gradient (kPa/m)	4.022e-003	4.022e-003	4.022e-003	0.0000
37	Accel. Gradient (kPa/m)	1.474e-006	1.474e-006	1.474e-006	7.725e-007
38	Liq. Reynolds	1.858e+005 *	1.858e+005 *	1.858e+005 *	1.858e+005 *
39	Vap. Reynolds	9.851e+006 *	9.851e+006 *	9.851e+006 *	9.851e+006 *
40	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
41	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	2.191e+004	2.192e+004	2.192e+004	2.193e+004
45	Elevation (m)	-9.000e-002	-9.000e-002	-9.000e-002	-9.000e-002
46	Pressure (bar)	68.61	68.61	68.61	68.61
47	Temperature (C)	49.53	49.53	49.53	49.52
48	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
49	Heat Transferred (kJ/h-m)	520.2	520.2	520.1	520.1
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2504	0.2504	0.2504	0.2504
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	6.153e-003	6.153e-003	6.153e-003	6.153e-003
55	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
56	Accel. Gradient (kPa/m)	7.725e-007	7.725e-007	7.725e-007	7.725e-007
57	Liq. Reynolds	1.858e+005 *	1.858e+005 *	1.858e+005 *	1.858e+005 *
58	Vap. Reynolds	9.851e+006 *	9.851e+006 *	9.851e+006 *	9.851e+006 *
59	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
60	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	2.193e+004	2.194e+004	2.194e+004	2.195e+004
64	Elevation (m)	-9.000e-002	-9.000e-002	-9.000e-002	-9.000e-002
65	Pressure (bar)	68.61	68.61	68.61	68.61
66	Temperature (C)	49.52	49.52	49.52	49.52
67	Vapour Fraction	0.8813	0.8813	0.8813	0.8813
68	Heat Transferred (kJ/h-m)	520.1	520.0	520.0	519.9
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2504	0.2504	0.2504	0.2504
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	6.153e-003	6.153e-003	6.153e-003	6.153e-003
18	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
19	Accel. Gradient (kPa/m)	7.725e-007	7.725e-007	7.725e-007	7.725e-007
20	Liq. Reynolds	1.858e+005 *	1.858e+005 *	1.858e+005 *	1.858e+005 *
21	Vap. Reynolds	9.851e+006 *	9.851e+006 *	9.851e+006 *	9.851e+006 *
22	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
23	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	2.195e+004	2.196e+004	2.196e+004	2.197e+004
27	Elevation (m)	-9.000e-002	-8.800e-002	-8.600e-002	-8.400e-002
28	Pressure (bar)	68.61	68.61	68.61	68.61
29	Temperature (C)	49.52	49.51	49.51	49.51
30	Vapour Fraction	0.8813	0.8813	0.8812	0.8812
31	Heat Transferred (kJ/h-m)	519.9	519.8	519.8	519.8
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2504	0.2325	0.2325	0.2325
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	6.153e-003	8.291e-003	8.291e-003	8.291e-003
37	Static Gradient (kPa/m)	0.0000	7.933e-004	7.933e-004	7.933e-004
38	Accel. Gradient (kPa/m)	7.725e-007	1.141e-006	1.141e-006	1.141e-006
39	Liq. Reynolds	1.858e+005 *	1.858e+005 *	1.858e+005 *	1.858e+005 *
40	Vap. Reynolds	9.851e+006 *	9.851e+006 *	9.851e+006 *	9.851e+006 *
41	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
42	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	2.197e+004	2.198e+004	2.198e+004	2.199e+004
46	Elevation (m)	-8.200e-002	-8.000e-002	-7.800e-002	-7.600e-002
47	Pressure (bar)	68.61	68.61	68.61	68.61
48	Temperature (C)	49.51	49.51	49.51	49.50
49	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
50	Heat Transferred (kJ/h-m)	519.7	519.7	519.6	519.6
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2325	0.2325	0.2325	0.2325
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	8.291e-003	8.291e-003	8.291e-003	8.291e-003
56	Static Gradient (kPa/m)	7.933e-004	7.933e-004	7.933e-004	7.933e-004
57	Accel. Gradient (kPa/m)	1.141e-006	1.141e-006	1.141e-006	1.141e-006
58	Liq. Reynolds	1.858e+005 *	1.858e+005 *	1.858e+005 *	1.858e+005 *
59	Vap. Reynolds	9.851e+006 *	9.851e+006 *	9.851e+006 *	9.851e+006 *
60	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
61	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	2.199e+004	2.200e+004	2.200e+004	2.201e+004
65	Elevation (m)	-7.400e-002	-7.200e-002	-7.000e-002	-7.400e-002
66	Pressure (bar)	68.61	68.61	68.61	68.61
67	Temperature (C)	49.50	49.50	49.50	49.50
68	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3							Unit Set: Mixed
4							Date/Time: Mon Jun 13 15:13:43 2022
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Heat Transferred (kJ/h-m)	519.5	519.5	519.4	519.4		
14	Flow Regime	Transition	Transition	Transition	Transition		
15	Water Flow Regime						
16	Liquid HoldUp	0.2325	0.2325	0.2325	0.1710		
17	Water HoldUp	---	---	---	---		
18	Friction Gradient (kPa/m)	8.291e-003	8.291e-003	8.291e-003	9.030e-003		
19	Static Gradient (kPa/m)	7.933e-004	7.933e-004	7.933e-004	-1.279e-003		
20	Accel. Gradient (kPa/m)	1.141e-006	1.141e-006	1.141e-006	9.731e-007		
21	Liq. Reynolds	1.858e+005 *	1.858e+005 *	1.858e+005 *	1.858e+005 *		
22	Vap. Reynolds	9.851e+006 *	9.851e+006 *	9.851e+006 *	9.851e+006 *		
23	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
24	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921		
25	Deposit Thickness (mm)	---	---	---	---		
26	Deposit Volume (m3)	---	---	---	---		
27	Length (m)	2.201e+004	2.202e+004	2.202e+004	2.203e+004		
28	Elevation (m)	-7.800e-002	-8.200e-002	-8.600e-002	-9.000e-002		
29	Pressure (bar)	68.61	68.61	68.61	68.60		
30	Temperature (C)	49.50	49.50	49.49	49.49		
31	Vapour Fraction	0.8812	0.8812	0.8812	0.8812		
32	Heat Transferred (kJ/h-m)	519.4	519.4	519.3	519.3		
33	Flow Regime	Transition	Transition	Transition	Transition		
34	Water Flow Regime						
35	Liquid HoldUp	0.1710	0.1710	0.1710	0.1710		
36	Water HoldUp	---	---	---	---		
37	Friction Gradient (kPa/m)	9.030e-003	9.030e-003	9.030e-003	9.030e-003		
38	Static Gradient (kPa/m)	-1.279e-003	-1.279e-003	-1.279e-003	-1.279e-003		
39	Accel. Gradient (kPa/m)	9.731e-007	9.731e-007	9.731e-007	9.731e-007		
40	Liq. Reynolds	1.858e+005 *	1.858e+005 *	1.858e+005 *	1.858e+005 *		
41	Vap. Reynolds	9.851e+006 *	9.851e+006 *	9.851e+006 *	9.851e+006 *		
42	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
43	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921		
44	Deposit Thickness (mm)	---	---	---	---		
45	Deposit Volume (m3)	---	---	---	---		
46	Length (m)	2.203e+004	2.204e+004	2.204e+004	2.205e+004		
47	Elevation (m)	-9.200e-002	-9.400e-002	-9.600e-002	-9.800e-002		
48	Pressure (bar)	68.60	68.60	68.60	68.60		
49	Temperature (C)	49.49	49.49	49.49	49.49		
50	Vapour Fraction	0.8812	0.8812	0.8812	0.8812		
51	Heat Transferred (kJ/h-m)	519.2	519.2	519.2	519.1		
52	Flow Regime	Transition	Transition	Transition	Transition		
53	Water Flow Regime						
54	Liquid HoldUp	0.1713	0.1713	0.1713	0.1713		
55	Water HoldUp	---	---	---	---		
56	Friction Gradient (kPa/m)	9.026e-003	9.026e-003	9.026e-003	9.026e-003		
57	Static Gradient (kPa/m)	-6.402e-004	-6.402e-004	-6.402e-004	-6.402e-004		
58	Accel. Gradient (kPa/m)	1.053e-006	1.053e-006	1.053e-006	1.053e-006		
59	Liq. Reynolds	1.858e+005 *	1.858e+005 *	1.858e+005 *	1.858e+005 *		
60	Vap. Reynolds	9.851e+006 *	9.851e+006 *	9.851e+006 *	9.851e+006 *		
61	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
62	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921		
63	Deposit Thickness (mm)	---	---	---	---		
64	Deposit Volume (m3)	---	---	---	---		
65	Length (m)	2.205e+004	2.206e+004	2.206e+004	2.207e+004		
66	Elevation (m)	-0.1000	-0.1260	-0.1520	-0.1780		
67	Pressure (bar)	68.60	68.60	68.60	68.60		
68	Temperature (C)	49.48	49.48	49.48	49.48		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Fraction		0.8812	0.8812	0.8812	0.8812	
14	Heat Transferred (kJ/h-m)		519.1	519.0	519.0	519.0	
15	Flow Regime		Transition	Transition	Transition	Transition	
16	Water Flow Regime						
17	Liquid HoldUp		0.1713	0.1677	0.1677	0.1677	
18	Water HoldUp		---	---	---	---	
19	Friction Gradient (kPa/m)		9.026e-003	9.074e-003	9.074e-003	9.074e-003	
20	Static Gradient (kPa/m)		-6.403e-004	-8.208e-003	-8.208e-003	-8.208e-003	
21	Accel. Gradient (kPa/m)		1.053e-006	1.087e-007	1.087e-007	1.087e-007	
22	Liq. Reynolds		1.858e+005 *	1.858e+005 *	1.858e+005 *	1.858e+005 *	
23	Vap. Reynolds		9.851e+006 *	9.851e+006 *	9.851e+006 *	9.851e+006 *	
24	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979	
25	Vapour Velocity (m/s)		2.921	2.921	2.921	2.921	
26	Deposit Thickness (mm)		---	---	---	---	
27	Deposit Volume (m3)		---	---	---	---	
28	Length (m)		2.207e+004	2.208e+004	2.208e+004	2.209e+004	
29	Elevation (m)		-0.2040	-0.2300	-0.2560	-0.2820	
30	Pressure (bar)		68.60	68.60	68.60	68.60	
31	Temperature (C)		49.48	49.48	49.48	49.47	
32	Vapour Fraction		0.8812	0.8812	0.8812	0.8812	
33	Heat Transferred (kJ/h-m)		518.9	518.9	518.8	518.8	
34	Flow Regime		Transition	Transition	Transition	Transition	
35	Water Flow Regime						
36	Liquid HoldUp		0.1677	0.1677	0.1677	0.1677	
37	Water HoldUp		---	---	---	---	
38	Friction Gradient (kPa/m)		9.074e-003	9.074e-003	9.074e-003	9.074e-003	
39	Static Gradient (kPa/m)		-8.208e-003	-8.208e-003	-8.208e-003	-8.208e-003	
40	Accel. Gradient (kPa/m)		1.087e-007	1.087e-007	1.087e-007	1.087e-007	
41	Liq. Reynolds		1.858e+005 *	1.858e+005 *	1.858e+005 *	1.858e+005 *	
42	Vap. Reynolds		9.851e+006 *	9.851e+006 *	9.852e+006 *	9.852e+006 *	
43	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979	
44	Vapour Velocity (m/s)		2.921	2.921	2.921	2.921	
45	Deposit Thickness (mm)		---	---	---	---	
46	Deposit Volume (m3)		---	---	---	---	
47	Length (m)		2.209e+004	2.210e+004	2.210e+004	2.211e+004	
48	Elevation (m)		-0.3080	-0.3340	-0.3600	-0.3600	
49	Pressure (bar)		68.60	68.60	68.60	68.60	
50	Temperature (C)		49.47	49.47	49.47	49.47	
51	Vapour Fraction		0.8812	0.8812	0.8812	0.8812	
52	Heat Transferred (kJ/h-m)		518.8	518.7	518.7	518.6	
53	Flow Regime		Transition	Transition	Transition	Transition	
54	Water Flow Regime						
55	Liquid HoldUp		0.1677	0.1677	0.1677	0.2505	
56	Water HoldUp		---	---	---	---	
57	Friction Gradient (kPa/m)		9.074e-003	9.073e-003	9.073e-003	6.152e-003	
58	Static Gradient (kPa/m)		-8.208e-003	-8.208e-003	-8.208e-003	0.0000	
59	Accel. Gradient (kPa/m)		1.087e-007	1.086e-007	1.086e-007	7.724e-007	
60	Liq. Reynolds		1.858e+005 *	1.858e+005 *	1.858e+005 *	1.858e+005 *	
61	Vap. Reynolds		9.852e+006 *	9.852e+006 *	9.852e+006 *	9.852e+006 *	
62	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979	
63	Vapour Velocity (m/s)		2.921	2.921	2.921	2.921	
64	Deposit Thickness (mm)		---	---	---	---	
65	Deposit Volume (m3)		---	---	---	---	
66	Length (m)		2.211e+004	2.212e+004	2.212e+004	2.213e+004	
67	Elevation (m)		-0.3600	-0.3600	-0.3600	-0.3600	
68	Pressure (bar)		68.60	68.60	68.60	68.60	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	49.47	49.46	49.46	49.46
14	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
15	Heat Transferred (kJ/h-m)	518.5	518.5	518.4	518.4
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2505	0.2505	0.2505	0.2505
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	6.152e-003	6.152e-003	6.152e-003	6.152e-003
21	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
22	Accel. Gradient (kPa/m)	7.724e-007	7.724e-007	7.724e-007	7.724e-007
23	Liq. Reynolds	1.858e+005 *	1.858e+005 *	1.858e+005 *	1.858e+005 *
24	Vap. Reynolds	9.852e+006 *	9.852e+006 *	9.852e+006 *	9.852e+006 *
25	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
26	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	2.213e+004	2.214e+004	2.214e+004	2.215e+004
30	Elevation (m)	-0.3540	-0.3480	-0.3420	-0.3360
31	Pressure (bar)	68.60	68.60	68.60	68.60
32	Temperature (C)	49.46	49.46	49.46	49.45
33	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
34	Heat Transferred (kJ/h-m)	518.4	518.3	518.3	518.2
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2348	0.2348	0.2348	0.2348
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	7.999e-003	7.999e-003	7.999e-003	7.999e-003
40	Static Gradient (kPa/m)	2.397e-003	2.397e-003	2.397e-003	2.397e-003
41	Accel. Gradient (kPa/m)	1.305e-006	1.305e-006	1.305e-006	1.305e-006
42	Liq. Reynolds	1.858e+005 *	1.858e+005 *	1.858e+005 *	1.858e+005 *
43	Vap. Reynolds	9.852e+006 *	9.852e+006 *	9.852e+006 *	9.852e+006 *
44	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
45	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	2.215e+004	2.216e+004	2.216e+004	2.217e+004
49	Elevation (m)	-0.3300	-0.3020	-0.2740	-0.2460
50	Pressure (bar)	68.60	68.60	68.60	68.60
51	Temperature (C)	49.45	49.45	49.45	49.45
52	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
53	Heat Transferred (kJ/h-m)	518.2	518.1	518.1	518.0
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2348	0.2470	0.2470	0.2470
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	7.999e-003	6.534e-003	6.535e-003	6.535e-003
59	Static Gradient (kPa/m)	2.397e-003	1.161e-002	1.161e-002	1.161e-002
60	Accel. Gradient (kPa/m)	1.305e-006	2.278e-006	2.278e-006	2.279e-006
61	Liq. Reynolds	1.858e+005 *	1.858e+005 *	1.858e+005 *	1.858e+005 *
62	Vap. Reynolds	9.852e+006 *	9.852e+006 *	9.852e+006 *	9.852e+006 *
63	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
64	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	2.217e+004	2.218e+004	2.218e+004	2.219e+004
68	Elevation (m)	-0.2180	-0.1900	-0.2120	-0.2340
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	68.59	68.59	68.59	68.59		
14	Temperature (C)	49.44	49.44	49.44	49.44		
15	Vapour Fraction	0.8812	0.8812	0.8812	0.8812		
16	Heat Transferred (kJ/h-m)	518.0	517.9	517.9	517.9		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.2470	0.2470	0.1683	0.1683		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	6.535e-003	6.535e-003	9.065e-003	9.065e-003		
22	Static Gradient (kPa/m)	1.161e-002	1.161e-002	-6.962e-003	-6.962e-003		
23	Accel. Gradient (kPa/m)	2.279e-006	2.279e-006	2.641e-007	2.641e-007		
24	Liq. Reynolds	1.858e+005 *	1.857e+005 *	1.857e+005 *	1.857e+005 *		
25	Vap. Reynolds	9.852e+006 *	9.852e+006 *	9.852e+006 *	9.852e+006 *		
26	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
27	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	2.219e+004	2.220e+004	2.220e+004	2.221e+004		
31	Elevation (m)	-0.2560	-0.2780	-0.3000	-0.3900		
32	Pressure (bar)	68.59	68.59	68.59	68.59		
33	Temperature (C)	49.44	49.44	49.44	49.43		
34	Vapour Fraction	0.8812	0.8812	0.8812	0.8812		
35	Heat Transferred (kJ/h-m)	517.9	517.8	517.8	517.8		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.1683	0.1683	0.1683	0.1582		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	9.065e-003	9.065e-003	9.065e-003	9.217e-003		
41	Static Gradient (kPa/m)	-6.962e-003	-6.962e-003	-6.962e-003	-2.735e-002		
42	Accel. Gradient (kPa/m)	2.641e-007	2.641e-007	2.641e-007	-2.277e-006		
43	Liq. Reynolds	1.857e+005 *	1.857e+005 *	1.857e+005 *	1.857e+005 *		
44	Vap. Reynolds	9.852e+006 *	9.852e+006 *	9.852e+006 *	9.852e+006 *		
45	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
46	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	2.221e+004	2.222e+004	2.222e+004	2.223e+004		
50	Elevation (m)	-0.4800	-0.5700	-0.6600	-0.7500		
51	Pressure (bar)	68.59	68.60	68.60	68.60		
52	Temperature (C)	49.43	49.43	49.43	49.43		
53	Vapour Fraction	0.8812	0.8812	0.8812	0.8812		
54	Heat Transferred (kJ/h-m)	517.7	517.7	517.7	517.7		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.1582	0.1582	0.1582	0.1582		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	9.217e-003	9.217e-003	9.216e-003	9.216e-003		
60	Static Gradient (kPa/m)	-2.735e-002	-2.735e-002	-2.735e-002	-2.735e-002		
61	Accel. Gradient (kPa/m)	-2.277e-006	-2.277e-006	-2.277e-006	-2.277e-006		
62	Liq. Reynolds	1.857e+005 *	1.857e+005 *	1.857e+005 *	1.857e+005 *		
63	Vap. Reynolds	9.852e+006 *	9.852e+006 *	9.852e+006 *	9.852e+006 *		
64	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
65	Vapour Velocity (m/s)	2.921	2.921	2.920	2.920		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	2.223e+004	2.224e+004	2.224e+004	2.225e+004		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	-0.6640	-0.5780	-0.4920	-0.4060
14	Pressure (bar)	68.59	68.59	68.59	68.59
15	Temperature (C)	49.43	49.43	49.42	49.42
16	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
17	Heat Transferred (kJ/h-m)	517.5	517.5	517.4	517.4
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2791	0.2791	0.2791	0.2791
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	7.911e-003	7.911e-003	7.911e-003	7.912e-003
23	Static Gradient (kPa/m)	3.912e-002	3.912e-002	3.912e-002	3.912e-002
24	Accel. Gradient (kPa/m)	5.905e-006	5.905e-006	5.906e-006	5.906e-006
25	Liq. Reynolds	1.857e+005 *	1.857e+005 *	1.857e+005 *	1.857e+005 *
26	Vap. Reynolds	9.852e+006 *	9.852e+006 *	9.852e+006 *	9.852e+006 *
27	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
28	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	2.225e+004	2.226e+004	2.226e+004	2.227e+004
32	Elevation (m)	-0.3200	-0.2880	-0.2560	-0.2240
33	Pressure (bar)	68.59	68.58	68.58	68.58
34	Temperature (C)	49.42	49.42	49.42	49.41
35	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
36	Heat Transferred (kJ/h-m)	517.3	517.2	517.2	517.1
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2791	0.2492	0.2492	0.2492
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	7.912e-003	6.283e-003	6.283e-003	6.283e-003
42	Static Gradient (kPa/m)	3.912e-002	1.342e-002	1.342e-002	1.342e-002
43	Accel. Gradient (kPa/m)	5.906e-006	2.475e-006	2.475e-006	2.475e-006
44	Liq. Reynolds	1.857e+005 *	1.857e+005 *	1.857e+005 *	1.857e+005 *
45	Vap. Reynolds	9.852e+006 *	9.852e+006 *	9.852e+006 *	9.852e+006 *
46	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
47	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	2.227e+004	2.228e+004	2.229e+004	2.230e+004
51	Elevation (m)	-0.1920	-0.1600	-0.1520	-0.1440
52	Pressure (bar)	68.58	68.58	68.58	68.58
53	Temperature (C)	49.41	49.41	49.41	49.40
54	Vapour Fraction	0.8812	0.8812	0.8812	0.8812
55	Heat Transferred (kJ/h-m)	517.1	517.0	517.0	516.9
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2492	0.2492	0.2336	0.2336
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	6.283e-003	6.283e-003	8.144e-003	8.144e-003
61	Static Gradient (kPa/m)	1.342e-002	1.342e-002	1.592e-003	1.592e-003
62	Accel. Gradient (kPa/m)	2.475e-006	2.475e-006	1.223e-006	1.223e-006
63	Liq. Reynolds	1.857e+005 *	1.857e+005 *	1.857e+005 *	1.857e+005 *
64	Vap. Reynolds	9.852e+006 *	9.852e+006 *	9.853e+006 *	9.853e+006 *
65	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
66	Vapour Velocity (m/s)	2.921	2.921	2.921	2.921
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length	(m)	2.231e+004	2.232e+004	2.233e+004	2.255e+004	
14	Elevation	(m)	-0.1360	-0.1280	-0.1200	-0.1160	
15	Pressure	(bar)	68.58	68.58	68.58	68.56	
16	Temperature	(C)	49.40	49.40	49.39	49.32	
17	Vapour Fraction		0.8812	0.8812	0.8812	0.8812	
18	Heat Transferred	(kJ/h-m)	516.8	516.7	516.6	514.7	
19	Flow Regime		Transition	Transition	Transition	Transition	
20	Water Flow Regime						
21	Liquid HoldUp		0.2336	0.2336	0.2336	0.2315	
22	Water HoldUp		---	---	---	---	
23	Friction Gradient	(kPa/m)	8.144e-003	8.144e-003	8.144e-003	8.433e-003	
24	Static Gradient	(kPa/m)	1.592e-003	1.592e-003	1.592e-003	3.515e-005	
25	Accel. Gradient	(kPa/m)	1.223e-006	1.223e-006	1.223e-006	1.064e-006	
26	Liq. Reynolds		1.857e+005 *	1.857e+005 *	1.857e+005 *	1.856e+005 *	
27	Vap. Reynolds		9.853e+006 *	9.853e+006 *	9.853e+006 *	9.854e+006 *	
28	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
29	Vapour Velocity	(m/s)	2.921	2.921	2.921	2.921	
30	Deposit Thickness	(mm)	---	---	---	---	
31	Deposit Volume	(m3)	---	---	---	---	
32	Length	(m)	2.278e+004	2.300e+004	2.323e+004	2.345e+004	
33	Elevation	(m)	-0.1120	-0.1080	-0.1040	-0.1000	
34	Pressure	(bar)	68.54	68.52	68.50	68.48	
35	Temperature	(C)	49.25	49.18	49.11	49.03	
36	Vapour Fraction		0.8811	0.8811	0.8811	0.8810	
37	Heat Transferred	(kJ/h-m)	512.8	510.8	508.9	507.0	
38	Flow Regime		Transition	Transition	Transition	Transition	
39	Water Flow Regime						
40	Liquid HoldUp		0.2315	0.2315	0.2315	0.2315	
41	Water HoldUp		---	---	---	---	
42	Friction Gradient	(kPa/m)	8.433e-003	8.432e-003	8.432e-003	8.432e-003	
43	Static Gradient	(kPa/m)	3.515e-005	3.515e-005	3.516e-005	3.516e-005	
44	Accel. Gradient	(kPa/m)	1.064e-006	1.064e-006	1.064e-006	1.065e-006	
45	Liq. Reynolds		1.855e+005 *	1.855e+005 *	1.854e+005 *	1.853e+005 *	
46	Vap. Reynolds		9.855e+006 *	9.855e+006 *	9.856e+006 *	9.857e+006 *	
47	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
48	Vapour Velocity	(m/s)	2.921	2.920	2.920	2.920	
49	Deposit Thickness	(mm)	---	---	---	---	
50	Deposit Volume	(m3)	---	---	---	---	
51	Length	(m)	2.346e+004	2.347e+004	2.348e+004	2.348e+004	
52	Elevation	(m)	-0.1040	-0.1080	-0.1120	-0.1160	
53	Pressure	(bar)	68.48	68.48	68.48	68.48	
54	Temperature	(C)	49.03	49.03	49.03	49.02	
55	Vapour Fraction		0.8810	0.8810	0.8810	0.8810	
56	Heat Transferred	(kJ/h-m)	507.0	506.9	506.9	506.8	
57	Flow Regime		Transition	Transition	Transition	Transition	
58	Water Flow Regime						
59	Liquid HoldUp		0.1712	0.1712	0.1712	0.1712	
60	Water HoldUp		---	---	---	---	
61	Friction Gradient	(kPa/m)	9.026e-003	9.026e-003	9.026e-003	9.025e-003	
62	Static Gradient	(kPa/m)	-8.709e-004	-8.709e-004	-8.709e-004	-8.709e-004	
63	Accel. Gradient	(kPa/m)	1.025e-006	1.025e-006	1.025e-006	1.025e-006	
64	Liq. Reynolds		1.853e+005 *	1.853e+005 *	1.853e+005 *	1.853e+005 *	
65	Vap. Reynolds		9.857e+006 *	9.857e+006 *	9.857e+006 *	9.857e+006 *	
66	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
67	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
68	Deposit Thickness	(mm)	---	---	---	---	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---	---	---
14	Length (m)	2.349e+004	2.350e+004	2.350e+004	2.351e+004	2.351e+004	2.351e+004
15	Elevation (m)	-0.1200	-0.1180	-0.1160	-0.1140	-0.1140	-0.1140
16	Pressure (bar)	68.48	68.48	68.48	68.48	68.48	68.48
17	Temperature (C)	49.02	49.02	49.02	49.02	49.02	49.02
18	Vapour Fraction	0.8810	0.8810	0.8810	0.8810	0.8810	0.8810
19	Heat Transferred (kJ/h-m)	506.8	506.7	506.6	506.6	506.6	506.6
20	Flow Regime	Transition	Transition	Transition	Transition	Transition	Transition
21	Water Flow Regime						
22	Liquid HoldUp	0.1712	0.2326	0.2326	0.2326	0.2326	0.2326
23	Water HoldUp	---	---	---	---	---	---
24	Friction Gradient (kPa/m)	9.025e-003	8.289e-003	8.289e-003	8.289e-003	8.289e-003	8.289e-003
25	Static Gradient (kPa/m)	-8.709e-004	7.937e-004	7.937e-004	7.937e-004	7.937e-004	7.937e-004
26	Accel. Gradient (kPa/m)	1.025e-006	1.142e-006	1.142e-006	1.142e-006	1.142e-006	1.142e-006
27	Liq. Reynolds	1.853e+005 *	1.853e+005 *	1.853e+005 *	1.853e+005 *	1.853e+005 *	1.853e+005 *
28	Vap. Reynolds	9.857e+006 *	9.857e+006 *	9.858e+006 *	9.858e+006 *	9.858e+006 *	9.858e+006 *
29	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979	0.1979	0.1979
30	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920	2.920	2.920
31	Deposit Thickness (mm)	---	---	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---	---	---
33	Length (m)	2.351e+004	2.352e+004	2.352e+004	2.353e+004	2.353e+004	2.353e+004
34	Elevation (m)	-0.1120	-0.1100	-0.1140	-0.1180	-0.1180	-0.1180
35	Pressure (bar)	68.48	68.48	68.47	68.47	68.47	68.47
36	Temperature (C)	49.02	49.01	49.01	49.01	49.01	49.01
37	Vapour Fraction	0.8810	0.8810	0.8810	0.8810	0.8810	0.8810
38	Heat Transferred (kJ/h-m)	506.5	506.5	506.5	506.5	506.5	506.5
39	Flow Regime	Transition	Transition	Transition	Transition	Transition	Transition
40	Water Flow Regime						
41	Liquid HoldUp	0.2326	0.2326	0.1710	0.1710	0.1710	0.1710
42	Water HoldUp	---	---	---	---	---	---
43	Friction Gradient (kPa/m)	8.289e-003	8.289e-003	9.028e-003	9.028e-003	9.028e-003	9.028e-003
44	Static Gradient (kPa/m)	7.937e-004	7.937e-004	-1.280e-003	-1.280e-003	-1.280e-003	-1.280e-003
45	Accel. Gradient (kPa/m)	1.142e-006	1.142e-006	9.744e-007	9.744e-007	9.744e-007	9.744e-007
46	Liq. Reynolds	1.853e+005 *	1.853e+005 *	1.853e+005 *	1.853e+005 *	1.853e+005 *	1.853e+005 *
47	Vap. Reynolds	9.858e+006 *	9.858e+006 *	9.858e+006 *	9.858e+006 *	9.858e+006 *	9.858e+006 *
48	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979	0.1979	0.1979
49	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920	2.920	2.920
50	Deposit Thickness (mm)	---	---	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---	---	---
52	Length (m)	2.353e+004	2.354e+004	2.354e+004	2.355e+004	2.355e+004	2.355e+004
53	Elevation (m)	-0.1220	-0.1260	-0.1300	-0.1300	-0.1300	-0.1300
54	Pressure (bar)	68.47	68.47	68.47	68.47	68.47	68.47
55	Temperature (C)	49.01	49.01	49.01	49.01	49.01	49.00
56	Vapour Fraction	0.8810	0.8810	0.8810	0.8810	0.8810	0.8810
57	Heat Transferred (kJ/h-m)	506.4	506.4	506.3	506.3	506.3	506.3
58	Flow Regime	Transition	Transition	Transition	Transition	Transition	Transition
59	Water Flow Regime						
60	Liquid HoldUp	0.1710	0.1710	0.1710	0.1710	0.1710	0.2505
61	Water HoldUp	---	---	---	---	---	---
62	Friction Gradient (kPa/m)	9.028e-003	9.028e-003	9.028e-003	9.028e-003	9.028e-003	6.151e-003
63	Static Gradient (kPa/m)	-1.280e-003	-1.280e-003	-1.280e-003	-1.280e-003	-1.280e-003	0.0000
64	Accel. Gradient (kPa/m)	9.744e-007	9.744e-007	9.744e-007	9.744e-007	9.744e-007	7.736e-007
65	Liq. Reynolds	1.853e+005 *	1.853e+005 *	1.853e+005 *	1.853e+005 *	1.853e+005 *	1.853e+005 *
66	Vap. Reynolds	9.858e+006 *	9.858e+006 *	9.858e+006 *	9.858e+006 *	9.858e+006 *	9.858e+006 *
67	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979	0.1979	0.1979
68	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920	2.920	2.920
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Thickness	(mm)	---	---	---	---	
14	Deposit Volume	(m3)	---	---	---	---	
15	Length	(m)	2.355e+004	2.356e+004	2.356e+004	2.357e+004	
16	Elevation	(m)	-0.1300	-0.1300	-0.1300	-0.1300	
17	Pressure	(bar)	68.47	68.47	68.47	68.47	
18	Temperature	(C)	49.00	49.00	49.00	49.00	
19	Vapour Fraction		0.8810	0.8810	0.8810	0.8810	
20	Heat Transferred	(kJ/h-m)	506.2	506.2	506.1	506.1	
21	Flow Regime		Transition	Transition	Transition	Transition	
22	Water Flow Regime						
23	Liquid HoldUp		0.2505	0.2505	0.2505	0.2505	
24	Water HoldUp		---	---	---	---	
25	Friction Gradient	(kPa/m)	6.151e-003	6.151e-003	6.151e-003	6.151e-003	
26	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000	
27	Accel. Gradient	(kPa/m)	7.736e-007	7.736e-007	7.735e-007	7.735e-007	
28	Liq. Reynolds		1.853e+005 *	1.853e+005 *	1.853e+005 *	1.853e+005 *	
29	Vap. Reynolds		9.858e+006 *	9.858e+006 *	9.858e+006 *	9.858e+006 *	
30	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
31	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
32	Deposit Thickness	(mm)	---	---	---	---	
33	Deposit Volume	(m3)	---	---	---	---	
34	Length	(m)	2.357e+004	2.358e+004	2.358e+004	2.359e+004	
35	Elevation	(m)	3.200e-002	0.1940	0.3560	0.5180	
36	Pressure	(bar)	68.47	68.46	68.46	68.45	
37	Temperature	(C)	49.00	48.99	48.99	48.99	
38	Vapour Fraction		0.8810	0.8810	0.8810	0.8810	
39	Heat Transferred	(kJ/h-m)	506.0	505.9	505.9	505.8	
40	Flow Regime		Transition	Transition	Transition	Transition	
41	Water Flow Regime						
42	Liquid HoldUp		0.3212	0.3212	0.3211	0.3211	
43	Water HoldUp		---	---	---	---	
44	Friction Gradient	(kPa/m)	7.855e-003	7.855e-003	7.856e-003	7.856e-003	
45	Static Gradient	(kPa/m)	8.223e-002	8.223e-002	8.223e-002	8.223e-002	
46	Accel. Gradient	(kPa/m)	1.133e-005	1.133e-005	1.133e-005	1.133e-005	
47	Liq. Reynolds		1.853e+005 *	1.852e+005 *	1.852e+005 *	1.852e+005 *	
48	Vap. Reynolds		9.858e+006 *	9.858e+006 *	9.858e+006 *	9.858e+006 *	
49	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
50	Vapour Velocity	(m/s)	2.920	2.920	2.921	2.921	
51	Deposit Thickness	(mm)	---	---	---	---	
52	Deposit Volume	(m3)	---	---	---	---	
53	Length	(m)	2.359e+004	2.360e+004	2.360e+004	2.361e+004	
54	Elevation	(m)	0.6800	0.5120	0.3440	0.1760	
55	Pressure	(bar)	68.45	68.45	68.45	68.46	
56	Temperature	(C)	48.98	48.98	48.98	48.98	
57	Vapour Fraction		0.8810	0.8810	0.8810	0.8810	
58	Heat Transferred	(kJ/h-m)	505.7	505.8	505.7	505.7	
59	Flow Regime		Transition	Transition	Transition	Transition	
60	Water Flow Regime						
61	Liquid HoldUp		0.3211	0.1467	0.1467	0.1467	
62	Water HoldUp		---	---	---	---	
63	Friction Gradient	(kPa/m)	7.857e-003	9.427e-003	9.427e-003	9.427e-003	
64	Static Gradient	(kPa/m)	8.222e-002	-4.864e-002	-4.864e-002	-4.864e-002	
65	Accel. Gradient	(kPa/m)	1.133e-005	-4.934e-006	-4.934e-006	-4.934e-006	
66	Liq. Reynolds		1.852e+005 *	1.852e+005 *	1.852e+005 *	1.852e+005 *	
67	Vap. Reynolds		9.859e+006 *	9.859e+006 *	9.858e+006 *	9.858e+006 *	
68	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3							
4			Unit Set: Mixed				
5			Date/Time: Mon Jun 13 15:13:43 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Vapour Velocity	(m/s)	2.921	2.921	2.921	2.921	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	2.361e+004	2.362e+004	2.362e+004	2.363e+004	
17	Elevation	(m)	8.000e-003	-0.1600	-0.1700	-0.1800	
18	Pressure	(bar)	68.46	68.46	68.46	68.46	
19	Temperature	(C)	48.98	48.98	48.98	48.98	
20	Vapour Fraction		0.8810	0.8810	0.8810	0.8810	
21	Heat Transferred	(kJ/h-m)	505.7	505.7	505.6	505.6	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.1467	0.1467	0.1701	0.1701	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	9.426e-003	9.426e-003	9.041e-003	9.041e-003	
27	Static Gradient	(kPa/m)	-4.864e-002	-4.864e-002	-3.188e-003	-3.188e-003	
28	Accel. Gradient	(kPa/m)	-4.933e-006	-4.933e-006	7.363e-007	7.363e-007	
29	Liq. Reynolds		1.852e+005 *	1.852e+005 *	1.852e+005 *	1.852e+005 *	
30	Vap. Reynolds		9.858e+006 *	9.858e+006 *	9.858e+006 *	9.858e+006 *	
31	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
32	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	2.363e+004	2.364e+004	2.364e+004	2.365e+004	
36	Elevation	(m)	-0.1900	-0.2000	-0.2100	-0.1880	
37	Pressure	(bar)	68.46	68.46	68.46	68.46	
38	Temperature	(C)	48.98	48.97	48.97	48.97	
39	Vapour Fraction		0.8810	0.8810	0.8810	0.8810	
40	Heat Transferred	(kJ/h-m)	505.5	505.5	505.5	505.4	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.1701	0.1701	0.1701	0.2437	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	9.041e-003	9.041e-003	9.041e-003	6.913e-003	
46	Static Gradient	(kPa/m)	-3.188e-003	-3.188e-003	-3.188e-003	9.036e-003	
47	Accel. Gradient	(kPa/m)	7.363e-007	7.363e-007	7.363e-007	2.006e-006	
48	Liq. Reynolds		1.852e+005 *	1.852e+005 *	1.852e+005 *	1.852e+005 *	
49	Vap. Reynolds		9.858e+006 *	9.858e+006 *	9.858e+006 *	9.858e+006 *	
50	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
51	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	2.365e+004	2.366e+004	2.366e+004	2.367e+004	
55	Elevation	(m)	-0.1660	-0.1440	-0.1220	-0.1000	
56	Pressure	(bar)	68.46	68.46	68.45	68.45	
57	Temperature	(C)	48.97	48.97	48.97	48.96	
58	Vapour Fraction		0.8810	0.8810	0.8810	0.8810	
59	Heat Transferred	(kJ/h-m)	505.3	505.3	505.2	505.2	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.2437	0.2437	0.2437	0.2437	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	6.913e-003	6.913e-003	6.913e-003	6.913e-003	
65	Static Gradient	(kPa/m)	9.036e-003	9.036e-003	9.036e-003	9.036e-003	
66	Accel. Gradient	(kPa/m)	2.006e-006	2.006e-006	2.006e-006	2.006e-006	
67	Liq. Reynolds		1.852e+005 *	1.852e+005 *	1.852e+005 *	1.852e+005 *	
68	Vap. Reynolds		9.858e+006 *	9.858e+006 *	9.858e+006 *	9.858e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
14	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	2.367e+004	2.368e+004	2.368e+004	2.369e+004	
18	Elevation	(m)	-0.1100	-0.1200	-0.1300	-0.1400	
19	Pressure	(bar)	68.45	68.45	68.45	68.45	
20	Temperature	(C)	48.96	48.96	48.96	48.96	
21	Vapour Fraction		0.8810	0.8810	0.8810	0.8810	
22	Heat Transferred	(kJ/h-m)	505.2	505.1	505.1	505.1	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.1701	0.1701	0.1701	0.1701	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	9.041e-003	9.041e-003	9.041e-003	9.041e-003	
28	Static Gradient	(kPa/m)	-3.185e-003	-3.185e-003	-3.185e-003	-3.185e-003	
29	Accel. Gradient	(kPa/m)	7.367e-007	7.367e-007	7.366e-007	7.366e-007	
30	Liq. Reynolds		1.852e+005 *	1.852e+005 *	1.852e+005 *	1.852e+005 *	
31	Vap. Reynolds		9.859e+006 *	9.859e+006 *	9.859e+006 *	9.859e+006 *	
32	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
33	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	2.369e+004	2.370e+004	2.370e+004	2.371e+004	
37	Elevation	(m)	-0.1500	-0.1580	-0.1660	-0.1740	
38	Pressure	(bar)	68.45	68.45	68.45	68.45	
39	Temperature	(C)	48.96	48.96	48.95	48.95	
40	Vapour Fraction		0.8810	0.8810	0.8810	0.8810	
41	Heat Transferred	(kJ/h-m)	505.0	505.0	504.9	504.9	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.1701	0.1704	0.1704	0.1704	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	9.041e-003	9.037e-003	9.037e-003	9.037e-003	
47	Static Gradient	(kPa/m)	-3.185e-003	-2.553e-003	-2.553e-003	-2.553e-003	
48	Accel. Gradient	(kPa/m)	7.366e-007	8.157e-007	8.157e-007	8.157e-007	
49	Liq. Reynolds		1.852e+005 *	1.852e+005 *	1.852e+005 *	1.852e+005 *	
50	Vap. Reynolds		9.859e+006 *	9.859e+006 *	9.859e+006 *	9.859e+006 *	
51	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
52	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	2.371e+004	2.372e+004	2.372e+004	2.373e+004	
56	Elevation	(m)	-0.1820	-0.1900	-0.2020	-0.2140	
57	Pressure	(bar)	68.45	68.45	68.45	68.45	
58	Temperature	(C)	48.95	48.95	48.95	48.95	
59	Vapour Fraction		0.8810	0.8810	0.8810	0.8810	
60	Heat Transferred	(kJ/h-m)	504.9	504.8	504.8	504.7	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.1704	0.1704	0.1698	0.1698	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	9.037e-003	9.037e-003	9.045e-003	9.045e-003	
66	Static Gradient	(kPa/m)	-2.553e-003	-2.553e-003	-3.821e-003	-3.821e-003	
67	Accel. Gradient	(kPa/m)	8.157e-007	8.157e-007	6.572e-007	6.572e-007	
68	Liq. Reynolds		1.852e+005 *	1.852e+005 *	1.852e+005 *	1.852e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.859e+006 *	9.859e+006 *	9.859e+006 *	9.859e+006 *
14	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
15	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	2.373e+004	2.374e+004	2.374e+004	2.375e+004
19	Elevation (m)	-0.2260	-0.2380	-0.2500	-0.2900
20	Pressure (bar)	68.45	68.45	68.45	68.45
21	Temperature (C)	48.94	48.94	48.94	48.94
22	Vapour Fraction	0.8810	0.8810	0.8810	0.8810
23	Heat Transferred (kJ/h-m)	504.7	504.7	504.6	504.6
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1698	0.1698	0.1698	0.1657
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	9.045e-003	9.045e-003	9.045e-003	9.103e-003
29	Static Gradient (kPa/m)	-3.821e-003	-3.821e-003	-3.821e-003	-1.253e-002
30	Accel. Gradient (kPa/m)	6.572e-007	6.572e-007	6.572e-007	-4.311e-007
31	Liq. Reynolds	1.852e+005 *	1.852e+005 *	1.852e+005 *	1.852e+005 *
32	Vap. Reynolds	9.859e+006 *	9.859e+006 *	9.859e+006 *	9.859e+006 *
33	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
34	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	2.375e+004	2.376e+004	2.376e+004	2.377e+004
38	Elevation (m)	-0.3300	-0.3700	-0.4100	-0.4500
39	Pressure (bar)	68.45	68.45	68.45	68.45
40	Temperature (C)	48.94	48.94	48.94	48.94
41	Vapour Fraction	0.8810	0.8810	0.8810	0.8810
42	Heat Transferred (kJ/h-m)	504.6	504.5	504.5	504.5
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1657	0.1657	0.1657	0.1657
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	9.102e-003	9.102e-003	9.102e-003	9.102e-003
48	Static Gradient (kPa/m)	-1.253e-002	-1.253e-002	-1.253e-002	-1.253e-002
49	Accel. Gradient (kPa/m)	-4.311e-007	-4.311e-007	-4.311e-007	-4.311e-007
50	Liq. Reynolds	1.852e+005 *	1.852e+005 *	1.852e+005 *	1.852e+005 *
51	Vap. Reynolds	9.859e+006 *	9.859e+006 *	9.859e+006 *	9.859e+006 *
52	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
53	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	2.377e+004	2.378e+004	2.378e+004	2.379e+004
57	Elevation (m)	-0.4420	-0.4340	-0.4260	-0.4180
58	Pressure (bar)	68.45	68.45	68.45	68.45
59	Temperature (C)	48.93	48.93	48.93	48.93
60	Vapour Fraction	0.8810	0.8810	0.8810	0.8810
61	Heat Transferred (kJ/h-m)	504.4	504.3	504.3	504.2
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2359	0.2359	0.2359	0.2359
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	7.856e-003	7.856e-003	7.856e-003	7.856e-003
67	Static Gradient (kPa/m)	3.208e-003	3.208e-003	3.208e-003	3.208e-003
68	Accel. Gradient (kPa/m)	1.392e-006	1.392e-006	1.392e-006	1.392e-006
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		1.852e+005 *	1.852e+005 *	1.852e+005 *	1.852e+005 *	
14	Vap. Reynolds		9.859e+006 *	9.859e+006 *	9.859e+006 *	9.859e+006 *	
15	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979	0.1979	
16	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920	2.920	
17	Deposit Thickness (mm)	---	---	---	---	---	
18	Deposit Volume (m3)	---	---	---	---	---	
19	Length (m)	2.379e+004	2.380e+004	2.380e+004	2.381e+004		
20	Elevation (m)	-0.4100	-0.4040	-0.3980	-0.3920		
21	Pressure (bar)	68.45	68.45	68.45	68.45		
22	Temperature (C)	48.93	48.93	48.92	48.92		
23	Vapour Fraction	0.8810	0.8810	0.8810	0.8810		
24	Heat Transferred (kJ/h-m)	504.2	504.1	504.1	504.1		
25	Flow Regime	Transition	Transition	Transition	Transition		
26	Water Flow Regime						
27	Liquid HoldUp	0.2359	0.2348	0.2348	0.2348		
28	Water HoldUp	---	---	---	---		
29	Friction Gradient (kPa/m)	7.856e-003	7.999e-003	7.999e-003	7.999e-003		
30	Static Gradient (kPa/m)	3.208e-003	2.398e-003	2.398e-003	2.398e-003		
31	Accel. Gradient (kPa/m)	1.392e-006	1.308e-006	1.308e-006	1.308e-006		
32	Liq. Reynolds		1.852e+005 *	1.852e+005 *	1.852e+005 *	1.852e+005 *	
33	Vap. Reynolds		9.859e+006 *	9.859e+006 *	9.859e+006 *	9.859e+006 *	
34	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979	0.1979	
35	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920	2.920	
36	Deposit Thickness (mm)	---	---	---	---	---	
37	Deposit Volume (m3)	---	---	---	---	---	
38	Length (m)	2.381e+004	2.382e+004	2.382e+004	2.383e+004		
39	Elevation (m)	-0.3860	-0.3800	-0.3380	-0.2960		
40	Pressure (bar)	68.45	68.44	68.44	68.44		
41	Temperature (C)	48.92	48.92	48.92	48.92		
42	Vapour Fraction	0.8810	0.8810	0.8810	0.8810		
43	Heat Transferred (kJ/h-m)	504.0	504.0	503.9	503.9		
44	Flow Regime	Transition	Transition	Transition	Transition		
45	Water Flow Regime						
46	Liquid HoldUp	0.2348	0.2348	0.2548	0.2548		
47	Water HoldUp	---	---	---	---		
48	Friction Gradient (kPa/m)	7.999e-003	7.999e-003	7.181e-003	7.181e-003		
49	Static Gradient (kPa/m)	2.398e-003	2.398e-003	1.783e-002	1.783e-002		
50	Accel. Gradient (kPa/m)	1.308e-006	1.308e-006	3.147e-006	3.147e-006		
51	Liq. Reynolds		1.852e+005 *	1.852e+005 *	1.852e+005 *	1.852e+005 *	
52	Vap. Reynolds		9.859e+006 *	9.859e+006 *	9.859e+006 *	9.859e+006 *	
53	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979	0.1979	
54	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920	2.920	
55	Deposit Thickness (mm)	---	---	---	---	---	
56	Deposit Volume (m3)	---	---	---	---	---	
57	Length (m)	2.383e+004	2.384e+004	2.384e+004	2.385e+004		
58	Elevation (m)	-0.2540	-0.2120	-0.1700	-0.1760		
59	Pressure (bar)	68.44	68.44	68.44	68.44		
60	Temperature (C)	48.91	48.91	48.91	48.91		
61	Vapour Fraction	0.8810	0.8810	0.8810	0.8810		
62	Heat Transferred (kJ/h-m)	503.8	503.8	503.7	503.7		
63	Flow Regime	Transition	Transition	Transition	Transition		
64	Water Flow Regime						
65	Liquid HoldUp	0.2548	0.2548	0.2548	0.1707		
66	Water HoldUp	---	---	---	---		
67	Friction Gradient (kPa/m)	7.181e-003	7.181e-003	7.181e-003	9.033e-003		
68	Static Gradient (kPa/m)	1.783e-002	1.783e-002	1.783e-002	-1.917e-003		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Accel. Gradient	(kPa/m)	3.147e-006	3.147e-006	3.147e-006	8.953e-007				
14	Liq. Reynolds		1.852e+005 *	1.852e+005 *	1.852e+005 *	1.852e+005 *				
15	Vap. Reynolds		9.859e+006 *	9.859e+006 *	9.859e+006 *	9.859e+006 *				
16	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979				
17	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920				
18	Deposit Thickness	(mm)	---	---	---	---				
19	Deposit Volume	(m3)	---	---	---	---				
20	Length	(m)	2.385e+004	2.386e+004	2.386e+004	2.387e+004				
21	Elevation	(m)	-0.1820	-0.1880	-0.1940	-0.2000				
22	Pressure	(bar)	68.44	68.44	68.44	68.44				
23	Temperature	(C)	48.91	48.90	48.90	48.90				
24	Vapour Fraction		0.8810	0.8810	0.8810	0.8810				
25	Heat Transferred	(kJ/h-m)	503.7	503.6	503.6	503.6				
26	Flow Regime		Transition	Transition	Transition	Transition				
27	Water Flow Regime									
28	Liquid HoldUp		0.1707	0.1707	0.1707	0.1707				
29	Water HoldUp		---	---	---	---				
30	Friction Gradient	(kPa/m)	9.033e-003	9.033e-003	9.033e-003	9.033e-003				
31	Static Gradient	(kPa/m)	-1.917e-003	-1.917e-003	-1.917e-003	-1.917e-003				
32	Accel. Gradient	(kPa/m)	8.953e-007	8.953e-007	8.953e-007	8.953e-007				
33	Liq. Reynolds		1.852e+005 *	1.852e+005 *	1.852e+005 *	1.851e+005 *				
34	Vap. Reynolds		9.859e+006 *	9.859e+006 *	9.859e+006 *	9.859e+006 *				
35	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979				
36	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920				
37	Deposit Thickness	(mm)	---	---	---	---				
38	Deposit Volume	(m3)	---	---	---	---				
39	Length	(m)	2.387e+004	2.388e+004	2.388e+004	2.389e+004				
40	Elevation	(m)	-0.2940	-0.3880	-0.4820	-0.5760				
41	Pressure	(bar)	68.44	68.44	68.44	68.44				
42	Temperature	(C)	48.90	48.90	48.90	48.90				
43	Vapour Fraction		0.8810	0.8810	0.8810	0.8810				
44	Heat Transferred	(kJ/h-m)	503.5	503.5	503.5	503.5				
45	Flow Regime		Transition	Transition	Transition	Transition				
46	Water Flow Regime									
47	Liquid HoldUp		0.1577	0.1577	0.1577	0.1577				
48	Water HoldUp		---	---	---	---				
49	Friction Gradient	(kPa/m)	9.226e-003	9.226e-003	9.226e-003	9.226e-003				
50	Static Gradient	(kPa/m)	-2.849e-002	-2.849e-002	-2.849e-002	-2.849e-002				
51	Accel. Gradient	(kPa/m)	-2.424e-006	-2.424e-006	-2.424e-006	-2.424e-006				
52	Liq. Reynolds		1.851e+005 *	1.852e+005 *	1.852e+005 *	1.852e+005 *				
53	Vap. Reynolds		9.859e+006 *	9.859e+006 *	9.859e+006 *	9.859e+006 *				
54	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979				
55	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920				
56	Deposit Thickness	(mm)	---	---	---	---				
57	Deposit Volume	(m3)	---	---	---	---				
58	Length	(m)	2.389e+004	2.390e+004	2.390e+004	2.391e+004				
59	Elevation	(m)	-0.6700	-0.5840	-0.4980	-0.4120				
60	Pressure	(bar)	68.44	68.44	68.44	68.43				
61	Temperature	(C)	48.90	48.89	48.89	48.89				
62	Vapour Fraction		0.8810	0.8810	0.8810	0.8810				
63	Heat Transferred	(kJ/h-m)	503.5	503.3	503.3	503.2				
64	Flow Regime		Transition	Transition	Transition	Transition				
65	Water Flow Regime									
66	Liquid HoldUp		0.1577	0.2791	0.2791	0.2791				
67	Water HoldUp		---	---	---	---				
68	Friction Gradient	(kPa/m)	9.226e-003	7.911e-003	7.911e-003	7.911e-003				
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 385 of 937				

1				Case Name:	Simulation - 386MMSCFD_80barg.hsc		
2				Unit Set:	Mixed		
3	 Epic Atlantic Limited			Date/Time:	Mon Jun 13 15:13:43 2022		
4							
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Static Gradient	(kPa/m)	-2.850e-002	3.914e-002	3.914e-002	3.914e-002	
14	Accel. Gradient	(kPa/m)	-2.424e-006	5.920e-006	5.920e-006	5.920e-006	
15	Liq. Reynolds		1.852e+005 *	1.851e+005 *	1.851e+005 *	1.851e+005 *	
16	Vap. Reynolds		9.859e+006 *	9.859e+006 *	9.859e+006 *	9.859e+006 *	
17	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
18	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
19	Deposit Thickness	(mm)	---	---	---	---	
20	Deposit Volume	(m3)	---	---	---	---	
21	Length	(m)	2.391e+004	2.392e+004	2.392e+004	2.393e+004	
22	Elevation	(m)	-0.3260	-0.2400	-0.2180	-0.1960	
23	Pressure	(bar)	68.43	68.43	68.43	68.43	
24	Temperature	(C)	48.89	48.89	48.88	48.88	
25	Vapour Fraction		0.8810	0.8810	0.8810	0.8810	
26	Heat Transferred	(kJ/h-m)	503.1	503.1	503.0	503.0	
27	Flow Regime		Transition	Transition	Transition	Transition	
28	Water Flow Regime						
29	Liquid HoldUp		0.2791	0.2791	0.2437	0.2437	
30	Water HoldUp		---	---	---	---	
31	Friction Gradient	(kPa/m)	7.911e-003	7.912e-003	6.913e-003	6.913e-003	
32	Static Gradient	(kPa/m)	3.914e-002	3.914e-002	9.036e-003	9.036e-003	
33	Accel. Gradient	(kPa/m)	5.921e-006	5.921e-006	2.007e-006	2.007e-006	
34	Liq. Reynolds		1.851e+005 *	1.851e+005 *	1.851e+005 *	1.851e+005 *	
35	Vap. Reynolds		9.860e+006 *	9.860e+006 *	9.860e+006 *	9.860e+006 *	
36	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
37	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
38	Deposit Thickness	(mm)	---	---	---	---	
39	Deposit Volume	(m3)	---	---	---	---	
40	Length	(m)	2.393e+004	2.394e+004	2.394e+004	2.395e+004	
41	Elevation	(m)	-0.1740	-0.1520	-0.1300	-0.1280	
42	Pressure	(bar)	68.43	68.43	68.43	68.43	
43	Temperature	(C)	48.88	48.88	48.88	48.88	
44	Vapour Fraction		0.8810	0.8810	0.8810	0.8810	
45	Heat Transferred	(kJ/h-m)	502.9	502.9	502.8	502.8	
46	Flow Regime		Transition	Transition	Transition	Transition	
47	Water Flow Regime						
48	Liquid HoldUp		0.2437	0.2437	0.2437	0.2326	
49	Water HoldUp		---	---	---	---	
50	Friction Gradient	(kPa/m)	6.913e-003	6.913e-003	6.913e-003	8.291e-003	
51	Static Gradient	(kPa/m)	9.036e-003	9.036e-003	9.036e-003	7.938e-004	
52	Accel. Gradient	(kPa/m)	2.007e-006	2.007e-006	2.007e-006	1.143e-006	
53	Liq. Reynolds		1.851e+005 *	1.851e+005 *	1.851e+005 *	1.851e+005 *	
54	Vap. Reynolds		9.860e+006 *	9.860e+006 *	9.860e+006 *	9.860e+006 *	
55	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
56	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
57	Deposit Thickness	(mm)	---	---	---	---	
58	Deposit Volume	(m3)	---	---	---	---	
59	Length	(m)	2.395e+004	2.396e+004	2.396e+004	2.397e+004	
60	Elevation	(m)	-0.1260	-0.1240	-0.1220	-0.1200	
61	Pressure	(bar)	68.42	68.42	68.42	68.42	
62	Temperature	(C)	48.87	48.87	48.87	48.87	
63	Vapour Fraction		0.8810	0.8810	0.8810	0.8810	
64	Heat Transferred	(kJ/h-m)	502.8	502.7	502.7	502.6	
65	Flow Regime		Transition	Transition	Transition	Transition	
66	Water Flow Regime						
67	Liquid HoldUp		0.2326	0.2326	0.2326	0.2326	
68	Water HoldUp		---	---	---	---	
69	Aspen Technology Inc.			Aspen HYSYS Version 10		Page 386 of 937	

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	8.291e-003	8.291e-003	8.291e-003	8.291e-003
14	Static Gradient	(kPa/m)	7.938e-004	7.938e-004	7.938e-004	7.938e-004
15	Accel. Gradient	(kPa/m)	1.143e-006	1.143e-006	1.143e-006	1.143e-006
16	Liq. Reynolds		1.851e+005 *	1.851e+005 *	1.851e+005 *	1.851e+005 *
17	Vap. Reynolds		9.860e+006 *	9.860e+006 *	9.860e+006 *	9.860e+006 *
18	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
19	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	2.397e+004	2.398e+004	2.398e+004	2.399e+004
23	Elevation	(m)	-0.1280	-0.1360	-0.1440	-0.1520
24	Pressure	(bar)	68.42	68.42	68.42	68.42
25	Temperature	(C)	48.87	48.87	48.86	48.86
26	Vapour Fraction		0.8810	0.8810	0.8810	0.8810
27	Heat Transferred	(kJ/h-m)	502.6	502.6	502.6	502.5
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.1704	0.1704	0.1704	0.1704
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	9.037e-003	9.037e-003	9.037e-003	9.037e-003
33	Static Gradient	(kPa/m)	-2.553e-003	-2.553e-003	-2.553e-003	-2.553e-003
34	Accel. Gradient	(kPa/m)	8.161e-007	8.161e-007	8.161e-007	8.161e-007
35	Liq. Reynolds		1.851e+005 *	1.851e+005 *	1.851e+005 *	1.851e+005 *
36	Vap. Reynolds		9.860e+006 *	9.860e+006 *	9.860e+006 *	9.860e+006 *
37	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
38	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	2.399e+004	2.400e+004	2.400e+004	2.401e+004
42	Elevation	(m)	-0.1600	-0.1920	-0.2240	-0.2560
43	Pressure	(bar)	68.42	68.42	68.42	68.42
44	Temperature	(C)	48.86	48.86	48.86	48.86
45	Vapour Fraction		0.8810	0.8810	0.8810	0.8810
46	Heat Transferred	(kJ/h-m)	502.5	502.4	502.4	502.4
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.1704	0.1669	0.1669	0.1669
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	9.037e-003	9.086e-003	9.086e-003	9.086e-003
52	Static Gradient	(kPa/m)	-2.553e-003	-1.007e-002	-1.007e-002	-1.007e-002
53	Accel. Gradient	(kPa/m)	8.161e-007	-1.239e-007	-1.239e-007	-1.240e-007
54	Liq. Reynolds		1.851e+005 *	1.851e+005 *	1.851e+005 *	1.851e+005 *
55	Vap. Reynolds		9.860e+006 *	9.860e+006 *	9.860e+006 *	9.860e+006 *
56	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
57	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	2.401e+004	2.402e+004	2.402e+004	2.403e+004
61	Elevation	(m)	-0.2880	-0.3200	-0.3560	-0.3920
62	Pressure	(bar)	68.42	68.42	68.42	68.42
63	Temperature	(C)	48.86	48.85	48.85	48.85
64	Vapour Fraction		0.8810	0.8810	0.8810	0.8810
65	Heat Transferred	(kJ/h-m)	502.3	502.3	502.3	502.2
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.1669	0.1669	0.1663	0.1663
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 387 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	9.086e-003	9.086e-003	9.094e-003	9.094e-003
15	Static Gradient (kPa/m)	-1.007e-002	-1.007e-002	-1.130e-002	-1.130e-002
16	Accel. Gradient (kPa/m)	-1.240e-007	-1.240e-007	-2.780e-007	-2.780e-007
17	Liq. Reynolds	1.851e+005 *	1.851e+005 *	1.851e+005 *	1.851e+005 *
18	Vap. Reynolds	9.860e+006 *	9.860e+006 *	9.860e+006 *	9.860e+006 *
19	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
20	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	2.403e+004	2.404e+004	2.404e+004	2.405e+004
24	Elevation (m)	-0.4280	-0.4640	-0.5000	-0.4540
25	Pressure (bar)	68.42	68.42	68.42	68.42
26	Temperature (C)	48.85	48.85	48.85	48.85
27	Vapour Fraction	0.8810	0.8810	0.8810	0.8810
28	Heat Transferred (kJ/h-m)	502.2	502.2	502.1	502.0
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.1663	0.1663	0.1663	0.2570
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	9.094e-003	9.094e-003	9.094e-003	7.477e-003
34	Static Gradient (kPa/m)	-1.130e-002	-1.130e-002	-1.130e-002	1.966e-002
35	Accel. Gradient (kPa/m)	-2.780e-007	-2.781e-007	-2.781e-007	3.415e-006
36	Liq. Reynolds	1.851e+005 *	1.851e+005 *	1.851e+005 *	1.851e+005 *
37	Vap. Reynolds	9.860e+006 *	9.860e+006 *	9.860e+006 *	9.860e+006 *
38	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
39	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	2.405e+004	2.406e+004	2.406e+004	2.407e+004
43	Elevation (m)	-0.4080	-0.3620	-0.3160	-0.2700
44	Pressure (bar)	68.42	68.42	68.42	68.42
45	Temperature (C)	48.84	48.84	48.84	48.84
46	Vapour Fraction	0.8810	0.8810	0.8810	0.8810
47	Heat Transferred (kJ/h-m)	502.0	501.9	501.9	501.8
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2570	0.2570	0.2570	0.2570
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	7.478e-003	7.478e-003	7.478e-003	7.478e-003
53	Static Gradient (kPa/m)	1.966e-002	1.966e-002	1.966e-002	1.966e-002
54	Accel. Gradient (kPa/m)	3.416e-006	3.416e-006	3.416e-006	3.416e-006
55	Liq. Reynolds	1.851e+005 *	1.851e+005 *	1.851e+005 *	1.851e+005 *
56	Vap. Reynolds	9.860e+006 *	9.860e+006 *	9.860e+006 *	9.860e+006 *
57	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
58	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	2.407e+004	2.408e+004	2.408e+004	2.409e+004
62	Elevation (m)	-0.2580	-0.2460	-0.2340	-0.2220
63	Pressure (bar)	68.42	68.41	68.41	68.41
64	Temperature (C)	48.84	48.83	48.83	48.83
65	Vapour Fraction	0.8810	0.8810	0.8810	0.8810
66	Heat Transferred (kJ/h-m)	501.8	501.7	501.7	501.6
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 388 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.2381	0.2381	0.2381	0.2381
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	7.577e-003	7.577e-003	7.577e-003	7.577e-003
16	Static Gradient (kPa/m)	4.854e-003	4.854e-003	4.854e-003	4.854e-003
17	Accel. Gradient (kPa/m)	1.565e-006	1.565e-006	1.565e-006	1.565e-006
18	Liq. Reynolds	1.851e+005 *	1.851e+005 *	1.851e+005 *	1.851e+005 *
19	Vap. Reynolds	9.860e+006 *	9.860e+006 *	9.860e+006 *	9.860e+006 *
20	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
21	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	2.409e+004	2.410e+004	2.410e+004	2.411e+004
25	Elevation (m)	-0.2100	-0.1980	-0.1860	-0.1740
26	Pressure (bar)	68.41	68.41	68.41	68.41
27	Temperature (C)	48.83	48.83	48.83	48.82
28	Vapour Fraction	0.8810	0.8810	0.8810	0.8809
29	Heat Transferred (kJ/h-m)	501.6	501.6	501.5	501.5
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2381	0.2381	0.2381	0.2381
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	7.577e-003	7.578e-003	7.578e-003	7.578e-003
35	Static Gradient (kPa/m)	4.854e-003	4.846e-003	4.846e-003	4.846e-003
36	Accel. Gradient (kPa/m)	1.565e-006	1.564e-006	1.564e-006	1.564e-006
37	Liq. Reynolds	1.851e+005 *	1.851e+005 *	1.851e+005 *	1.851e+005 *
38	Vap. Reynolds	9.860e+006 *	9.860e+006 *	9.860e+006 *	9.860e+006 *
39	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
40	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	2.411e+004	2.412e+004	2.412e+004	2.413e+004
44	Elevation (m)	-0.1620	-0.1500	-0.1400	-0.1300
45	Pressure (bar)	68.41	68.41	68.41	68.41
46	Temperature (C)	48.82	48.82	48.82	48.82
47	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
48	Heat Transferred (kJ/h-m)	501.4	501.4	501.3	501.3
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2381	0.2381	0.2370	0.2370
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	7.578e-003	7.578e-003	7.717e-003	7.717e-003
54	Static Gradient (kPa/m)	4.846e-003	4.846e-003	4.024e-003	4.024e-003
55	Accel. Gradient (kPa/m)	1.564e-006	1.564e-006	1.478e-006	1.478e-006
56	Liq. Reynolds	1.851e+005 *	1.851e+005 *	1.851e+005 *	1.851e+005 *
57	Vap. Reynolds	9.861e+006 *	9.861e+006 *	9.861e+006 *	9.861e+006 *
58	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
59	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	2.413e+004	2.414e+004	2.414e+004	2.415e+004
63	Elevation (m)	-0.1200	-0.1100	-0.1000	-9.800e-002
64	Pressure (bar)	68.41	68.41	68.41	68.41
65	Temperature (C)	48.82	48.81	48.81	48.81
66	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
67	Heat Transferred (kJ/h-m)	501.2	501.2	501.2	501.1
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 389 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2370	0.2370	0.2370	0.2326
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	7.717e-003	7.717e-003	7.717e-003	8.291e-003
17	Static Gradient (kPa/m)	4.024e-003	4.024e-003	4.024e-003	7.938e-004
18	Accel. Gradient (kPa/m)	1.478e-006	1.478e-006	1.478e-006	1.144e-006
19	Liq. Reynolds	1.851e+005 *	1.850e+005 *	1.850e+005 *	1.850e+005 *
20	Vap. Reynolds	9.861e+006 *	9.861e+006 *	9.861e+006 *	9.861e+006 *
21	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
22	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	2.415e+004	2.416e+004	2.416e+004	2.417e+004
26	Elevation (m)	-9.600e-002	-9.400e-002	-9.200e-002	-9.000e-002
27	Pressure (bar)	68.41	68.41	68.40	68.40
28	Temperature (C)	48.81	48.81	48.81	48.81
29	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
30	Heat Transferred (kJ/h-m)	501.1	501.0	501.0	500.9
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2326	0.2326	0.2326	0.2326
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	8.291e-003	8.291e-003	8.291e-003	8.291e-003
36	Static Gradient (kPa/m)	7.938e-004	7.938e-004	7.938e-004	7.938e-004
37	Accel. Gradient (kPa/m)	1.144e-006	1.144e-006	1.144e-006	1.144e-006
38	Liq. Reynolds	1.850e+005 *	1.850e+005 *	1.850e+005 *	1.850e+005 *
39	Vap. Reynolds	9.861e+006 *	9.861e+006 *	9.861e+006 *	9.861e+006 *
40	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
41	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	2.417e+004	2.418e+004	2.418e+004	2.419e+004
45	Elevation (m)	-9.800e-002	-0.1060	-0.1140	-0.1220
46	Pressure (bar)	68.40	68.40	68.40	68.40
47	Temperature (C)	48.80	48.80	48.80	48.80
48	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
49	Heat Transferred (kJ/h-m)	500.9	500.9	500.9	500.8
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.1704	0.1704	0.1704	0.1704
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	9.037e-003	9.037e-003	9.037e-003	9.037e-003
55	Static Gradient (kPa/m)	-2.553e-003	-2.553e-003	-2.553e-003	-2.553e-003
56	Accel. Gradient (kPa/m)	8.163e-007	8.163e-007	8.163e-007	8.163e-007
57	Liq. Reynolds	1.850e+005 *	1.850e+005 *	1.850e+005 *	1.850e+005 *
58	Vap. Reynolds	9.861e+006 *	9.861e+006 *	9.861e+006 *	9.861e+006 *
59	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
60	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	2.419e+004	2.420e+004	2.420e+004	2.421e+004
64	Elevation (m)	-0.1300	-0.1380	-0.1460	-0.1540
65	Pressure (bar)	68.40	68.40	68.40	68.40
66	Temperature (C)	48.80	48.80	48.79	48.79
67	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
68	Heat Transferred (kJ/h-m)	500.8	500.7	500.7	500.7
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 390 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.1704	0.1704	0.1704	0.1704
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	9.037e-003	9.037e-003	9.037e-003	9.037e-003
18	Static Gradient (kPa/m)	-2.553e-003	-2.553e-003	-2.553e-003	-2.553e-003
19	Accel. Gradient (kPa/m)	8.163e-007	8.163e-007	8.163e-007	8.163e-007
20	Liq. Reynolds	1.850e+005 *	1.850e+005 *	1.850e+005 *	1.850e+005 *
21	Vap. Reynolds	9.861e+006 *	9.861e+006 *	9.861e+006 *	9.861e+006 *
22	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
23	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	2.421e+004	2.422e+004	2.422e+004	2.423e+004
27	Elevation (m)	-0.1620	-0.1700	-0.1580	-0.1460
28	Pressure (bar)	68.40	68.40	68.40	68.40
29	Temperature (C)	48.79	48.79	48.79	48.79
30	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
31	Heat Transferred (kJ/h-m)	500.6	500.6	500.5	500.5
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1704	0.1704	0.2381	0.2381
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	9.037e-003	9.037e-003	7.578e-003	7.578e-003
37	Static Gradient (kPa/m)	-2.553e-003	-2.553e-003	4.846e-003	4.846e-003
38	Accel. Gradient (kPa/m)	8.163e-007	8.163e-007	1.564e-006	1.564e-006
39	Liq. Reynolds	1.850e+005 *	1.850e+005 *	1.850e+005 *	1.850e+005 *
40	Vap. Reynolds	9.861e+006 *	9.861e+006 *	9.861e+006 *	9.861e+006 *
41	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
42	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	2.423e+004	2.424e+004	2.424e+004	2.425e+004
46	Elevation (m)	-0.1340	-0.1220	-0.1100	-0.1320
47	Pressure (bar)	68.40	68.40	68.40	68.40
48	Temperature (C)	48.79	48.78	48.78	48.78
49	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
50	Heat Transferred (kJ/h-m)	500.4	500.4	500.3	500.3
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2381	0.2381	0.2381	0.1683
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	7.578e-003	7.578e-003	7.578e-003	9.065e-003
56	Static Gradient (kPa/m)	4.846e-003	4.846e-003	4.846e-003	-6.965e-003
57	Accel. Gradient (kPa/m)	1.564e-006	1.564e-006	1.564e-006	2.645e-007
58	Liq. Reynolds	1.850e+005 *	1.850e+005 *	1.850e+005 *	1.850e+005 *
59	Vap. Reynolds	9.861e+006 *	9.861e+006 *	9.861e+006 *	9.861e+006 *
60	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
61	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	2.425e+004	2.426e+004	2.426e+004	2.427e+004
65	Elevation (m)	-0.1540	-0.1760	-0.1980	-0.2200
66	Pressure (bar)	68.40	68.40	68.40	68.40
67	Temperature (C)	48.78	48.78	48.78	48.77
68	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Heat Transferred (kJ/h-m)	500.3	500.3	500.2	500.2					
14	Flow Regime	Transition	Transition	Transition	Transition					
15	Water Flow Regime									
16	Liquid HoldUp	0.1683	0.1683	0.1683	0.1683					
17	Water HoldUp	---	---	---	---					
18	Friction Gradient (kPa/m)	9.065e-003	9.065e-003	9.065e-003	9.065e-003					
19	Static Gradient (kPa/m)	-6.965e-003	-6.965e-003	-6.965e-003	-6.965e-003					
20	Accel. Gradient (kPa/m)	2.645e-007	2.644e-007	2.644e-007	2.644e-007					
21	Liq. Reynolds	1.850e+005 *	1.850e+005 *	1.850e+005 *	1.850e+005 *					
22	Vap. Reynolds	9.861e+006 *	9.861e+006 *	9.861e+006 *	9.861e+006 *					
23	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
24	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920					
25	Deposit Thickness (mm)	---	---	---	---					
26	Deposit Volume (m3)	---	---	---	---					
27	Length (m)	2.427e+004	2.428e+004	2.428e+004	2.429e+004					
28	Elevation (m)	-0.2040	-0.1880	-0.1720	-0.1560					
29	Pressure (bar)	68.40	68.40	68.40	68.39					
30	Temperature (C)	48.77	48.77	48.77	48.77					
31	Vapour Fraction	0.8809	0.8809	0.8809	0.8809					
32	Heat Transferred (kJ/h-m)	500.1	500.0	500.0	500.0					
33	Flow Regime	Transition	Transition	Transition	Transition					
34	Water Flow Regime									
35	Liquid HoldUp	0.2403	0.2403	0.2403	0.2403					
36	Water HoldUp	---	---	---	---					
37	Friction Gradient (kPa/m)	7.311e-003	7.311e-003	7.311e-003	7.311e-003					
38	Static Gradient (kPa/m)	6.477e-003	6.477e-003	6.477e-003	6.477e-003					
39	Accel. Gradient (kPa/m)	1.736e-006	1.736e-006	1.736e-006	1.736e-006					
40	Liq. Reynolds	1.850e+005 *	1.850e+005 *	1.850e+005 *	1.850e+005 *					
41	Vap. Reynolds	9.861e+006 *	9.861e+006 *	9.861e+006 *	9.861e+006 *					
42	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
43	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920					
44	Deposit Thickness (mm)	---	---	---	---					
45	Deposit Volume (m3)	---	---	---	---					
46	Length (m)	2.429e+004	2.430e+004	2.430e+004	2.431e+004					
47	Elevation (m)	-0.1400	-0.1300	-0.1200	-0.1100					
48	Pressure (bar)	68.39	68.39	68.39	68.39					
49	Temperature (C)	48.77	48.76	48.76	48.76					
50	Vapour Fraction	0.8809	0.8809	0.8809	0.8809					
51	Heat Transferred (kJ/h-m)	499.9	499.9	499.8	499.8					
52	Flow Regime	Transition	Transition	Transition	Transition					
53	Water Flow Regime									
54	Liquid HoldUp	0.2403	0.2370	0.2370	0.2370					
55	Water HoldUp	---	---	---	---					
56	Friction Gradient (kPa/m)	7.311e-003	7.717e-003	7.717e-003	7.717e-003					
57	Static Gradient (kPa/m)	6.477e-003	4.025e-003	4.025e-003	4.025e-003					
58	Accel. Gradient (kPa/m)	1.736e-006	1.478e-006	1.478e-006	1.478e-006					
59	Liq. Reynolds	1.850e+005 *	1.850e+005 *	1.850e+005 *	1.850e+005 *					
60	Vap. Reynolds	9.861e+006 *	9.861e+006 *	9.861e+006 *	9.861e+006 *					
61	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
62	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920					
63	Deposit Thickness (mm)	---	---	---	---					
64	Deposit Volume (m3)	---	---	---	---					
65	Length (m)	2.431e+004	2.432e+004	2.432e+004	2.433e+004					
66	Elevation (m)	-0.1000	-9.000e-002	-9.600e-002	-0.1020					
67	Pressure (bar)	68.39	68.39	68.39	68.39					
68	Temperature (C)	48.76	48.76	48.76	48.76					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
14	Heat Transferred (kJ/h-m)	499.7	499.7	499.7	499.6
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2370	0.2370	0.1707	0.1707
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	7.717e-003	7.717e-003	9.033e-003	9.033e-003
20	Static Gradient (kPa/m)	4.025e-003	4.025e-003	-1.917e-003	-1.917e-003
21	Accel. Gradient (kPa/m)	1.478e-006	1.478e-006	8.960e-007	8.960e-007
22	Liq. Reynolds	1.850e+005 *	1.850e+005 *	1.850e+005 *	1.850e+005 *
23	Vap. Reynolds	9.861e+006 *	9.861e+006 *	9.861e+006 *	9.861e+006 *
24	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
25	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	2.433e+004	2.434e+004	2.434e+004	2.435e+004
29	Elevation (m)	-0.1080	-0.1140	-0.1200	-0.1260
30	Pressure (bar)	68.39	68.39	68.39	68.39
31	Temperature (C)	48.75	48.75	48.75	48.75
32	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
33	Heat Transferred (kJ/h-m)	499.6	499.6	499.5	499.5
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1707	0.1707	0.1707	0.1707
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	9.033e-003	9.033e-003	9.033e-003	9.033e-003
39	Static Gradient (kPa/m)	-1.917e-003	-1.917e-003	-1.917e-003	-1.917e-003
40	Accel. Gradient (kPa/m)	8.960e-007	8.960e-007	8.960e-007	8.960e-007
41	Liq. Reynolds	1.850e+005 *	1.850e+005 *	1.850e+005 *	1.850e+005 *
42	Vap. Reynolds	9.861e+006 *	9.861e+006 *	9.861e+006 *	9.862e+006 *
43	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
44	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	2.435e+004	2.436e+004	2.436e+004	2.437e+004
48	Elevation (m)	-0.1320	-0.1380	-0.1440	-0.1500
49	Pressure (bar)	68.39	68.39	68.39	68.39
50	Temperature (C)	48.75	48.75	48.74	48.74
51	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
52	Heat Transferred (kJ/h-m)	499.4	499.4	499.4	499.3
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.1707	0.1707	0.1707	0.1707
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	9.033e-003	9.033e-003	9.033e-003	9.033e-003
58	Static Gradient (kPa/m)	-1.917e-003	-1.917e-003	-1.917e-003	-1.917e-003
59	Accel. Gradient (kPa/m)	8.960e-007	8.960e-007	8.960e-007	8.960e-007
60	Liq. Reynolds	1.850e+005 *	1.850e+005 *	1.850e+005 *	1.850e+005 *
61	Vap. Reynolds	9.862e+006 *	9.862e+006 *	9.862e+006 *	9.862e+006 *
62	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
63	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	2.437e+004	2.437e+004	2.437e+004	2.437e+004
67	Elevation (m)	-0.1420	-0.1340	-0.1260	-0.1180
68	Pressure (bar)	68.39	68.39	68.39	68.39
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 393 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Temperature (C)	48.74	48.74	48.74	48.74		
14	Vapour Fraction	0.8809	0.8809	0.8809	0.8809		
15	Heat Transferred (kJ/h-m)	499.3	499.2	499.2	499.2		
16	Flow Regime	Transition	Transition	Transition	Transition		
17	Water Flow Regime						
18	Liquid HoldUp	0.2426	0.2426	0.2426	0.2426		
19	Water HoldUp	---	---	---	---		
20	Friction Gradient (kPa/m)	7.042e-003	7.043e-003	7.043e-003	7.043e-003		
21	Static Gradient (kPa/m)	8.188e-003	8.188e-003	8.188e-003	8.188e-003		
22	Accel. Gradient (kPa/m)	1.918e-006	1.918e-006	1.918e-006	1.918e-006		
23	Liq. Reynolds	1.850e+005 *	1.850e+005 *	1.850e+005 *	1.850e+005 *		
24	Vap. Reynolds	9.862e+006 *	9.862e+006 *	9.862e+006 *	9.862e+006 *		
25	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
26	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920		
27	Deposit Thickness (mm)	---	---	---	---		
28	Deposit Volume (m3)	---	---	---	---		
29	Length (m)	2.438e+004	2.438e+004	2.438e+004	2.438e+004		
30	Elevation (m)	-0.1100	-0.1140	-0.1180	-0.1220		
31	Pressure (bar)	68.39	68.39	68.39	68.39		
32	Temperature (C)	48.74	48.74	48.74	48.74		
33	Vapour Fraction	0.8809	0.8809	0.8809	0.8809		
34	Heat Transferred (kJ/h-m)	499.2	499.2	499.2	499.2		
35	Flow Regime	Transition	Transition	Transition	Transition		
36	Water Flow Regime						
37	Liquid HoldUp	0.2426	0.1701	0.1701	0.1701		
38	Water HoldUp	---	---	---	---		
39	Friction Gradient (kPa/m)	7.043e-003	9.041e-003	9.041e-003	9.041e-003		
40	Static Gradient (kPa/m)	8.188e-003	-3.188e-003	-3.188e-003	-3.188e-003		
41	Accel. Gradient (kPa/m)	1.918e-006	7.370e-007	7.370e-007	7.370e-007		
42	Liq. Reynolds	1.850e+005 *	1.850e+005 *	1.850e+005 *	1.850e+005 *		
43	Vap. Reynolds	9.862e+006 *	9.862e+006 *	9.862e+006 *	9.862e+006 *		
44	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
45	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920		
46	Deposit Thickness (mm)	---	---	---	---		
47	Deposit Volume (m3)	---	---	---	---		
48	Length (m)	2.438e+004	2.439e+004	2.439e+004	2.440e+004		
49	Elevation (m)	-0.1260	-0.1300	-0.1300	-0.1300		
50	Pressure (bar)	68.39	68.39	68.39	68.38		
51	Temperature (C)	48.74	48.74	48.73	48.73		
52	Vapour Fraction	0.8809	0.8809	0.8809	0.8809		
53	Heat Transferred (kJ/h-m)	499.2	499.2	499.1	499.0		
54	Flow Regime	Transition	Transition	Transition	Transition		
55	Water Flow Regime						
56	Liquid HoldUp	0.1701	0.1701	0.2505	0.2505		
57	Water HoldUp	---	---	---	---		
58	Friction Gradient (kPa/m)	9.041e-003	9.041e-003	6.153e-003	6.153e-003		
59	Static Gradient (kPa/m)	-3.188e-003	-3.188e-003	0.0000	0.0000		
60	Accel. Gradient (kPa/m)	7.370e-007	7.370e-007	7.748e-007	7.748e-007		
61	Liq. Reynolds	1.850e+005 *	1.850e+005 *	1.850e+005 *	1.850e+005 *		
62	Vap. Reynolds	9.862e+006 *	9.862e+006 *	9.862e+006 *	9.862e+006 *		
63	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
64	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920		
65	Deposit Thickness (mm)	---	---	---	---		
66	Deposit Volume (m3)	---	---	---	---		
67	Length (m)	2.440e+004	2.441e+004	2.442e+004	2.442e+004		
68	Elevation (m)	-0.1300	-0.1300	-0.1300	-0.1360		
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 394 of 937		

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Pressure (bar)	68.38	68.38	68.38	68.38					
14	Temperature (C)	48.73	48.73	48.73	48.73					
15	Vapour Fraction	0.8809	0.8809	0.8809	0.8809					
16	Heat Transferred (kJ/h-m)	499.0	498.9	498.9	498.9					
17	Flow Regime	Transition	Transition	Transition	Transition					
18	Water Flow Regime									
19	Liquid HoldUp	0.2505	0.2505	0.2505	0.1707					
20	Water HoldUp	---	---	---	---					
21	Friction Gradient (kPa/m)	6.153e-003	6.153e-003	6.153e-003	9.033e-003					
22	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-1.917e-003					
23	Accel. Gradient (kPa/m)	7.748e-007	7.748e-007	7.748e-007	8.960e-007					
24	Liq. Reynolds	1.850e+005 *	1.850e+005 *	1.850e+005 *	1.850e+005 *					
25	Vap. Reynolds	9.862e+006 *	9.862e+006 *	9.862e+006 *	9.862e+006 *					
26	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
27	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920					
28	Deposit Thickness (mm)	---	---	---	---					
29	Deposit Volume (m3)	---	---	---	---					
30	Length (m)	2.443e+004	2.443e+004	2.444e+004	2.444e+004					
31	Elevation (m)	-0.1420	-0.1480	-0.1540	-0.1600					
32	Pressure (bar)	68.38	68.38	68.38	68.38					
33	Temperature (C)	48.72	48.72	48.72	48.72					
34	Vapour Fraction	0.8809	0.8809	0.8809	0.8809					
35	Heat Transferred (kJ/h-m)	498.8	498.8	498.7	498.7					
36	Flow Regime	Transition	Transition	Transition	Transition					
37	Water Flow Regime									
38	Liquid HoldUp	0.1707	0.1707	0.1707	0.1707					
39	Water HoldUp	---	---	---	---					
40	Friction Gradient (kPa/m)	9.033e-003	9.033e-003	9.033e-003	9.033e-003					
41	Static Gradient (kPa/m)	-1.917e-003	-1.917e-003	-1.917e-003	-1.917e-003					
42	Accel. Gradient (kPa/m)	8.960e-007	8.960e-007	8.960e-007	8.960e-007					
43	Liq. Reynolds	1.849e+005 *	1.849e+005 *	1.849e+005 *	1.849e+005 *					
44	Vap. Reynolds	9.862e+006 *	9.862e+006 *	9.862e+006 *	9.862e+006 *					
45	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
46	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920					
47	Deposit Thickness (mm)	---	---	---	---					
48	Deposit Volume (m3)	---	---	---	---					
49	Length (m)	2.445e+004	2.445e+004	2.446e+004	2.446e+004					
50	Elevation (m)	-0.1760	-0.1920	-0.2080	-0.2240					
51	Pressure (bar)	68.38	68.38	68.38	68.38					
52	Temperature (C)	48.72	48.72	48.72	48.71					
53	Vapour Fraction	0.8809	0.8809	0.8809	0.8809					
54	Heat Transferred (kJ/h-m)	498.7	498.6	498.6	498.5					
55	Flow Regime	Transition	Transition	Transition	Transition					
56	Water Flow Regime									
57	Liquid HoldUp	0.1692	0.1692	0.1692	0.1692					
58	Water HoldUp	---	---	---	---					
59	Friction Gradient (kPa/m)	9.053e-003	9.053e-003	9.053e-003	9.053e-003					
60	Static Gradient (kPa/m)	-5.083e-003	-5.083e-003	-5.083e-003	-5.083e-003					
61	Accel. Gradient (kPa/m)	4.998e-007	4.998e-007	4.998e-007	4.998e-007					
62	Liq. Reynolds	1.849e+005 *	1.849e+005 *	1.849e+005 *	1.849e+005 *					
63	Vap. Reynolds	9.862e+006 *	9.862e+006 *	9.862e+006 *	9.862e+006 *					
64	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
65	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920					
66	Deposit Thickness (mm)	---	---	---	---					
67	Deposit Volume (m3)	---	---	---	---					
68	Length (m)	2.447e+004	2.447e+004	2.448e+004	2.448e+004					
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 395 of 937					

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Elevation	(m)	-0.2400	-0.2500	-0.2600	-0.2700	
14	Pressure	(bar)	68.38	68.38	68.38	68.38	
15	Temperature	(C)	48.71	48.71	48.71	48.71	
16	Vapour Fraction		0.8809	0.8809	0.8809	0.8809	
17	Heat Transferred	(kJ/h-m)	498.5	498.5	498.4	498.4	
18	Flow Regime		Transition	Transition	Transition	Transition	
19	Water Flow Regime						
20	Liquid HoldUp		0.1692	0.1701	0.1701	0.1701	
21	Water HoldUp		---	---	---	---	
22	Friction Gradient	(kPa/m)	9.053e-003	9.041e-003	9.041e-003	9.041e-003	
23	Static Gradient	(kPa/m)	-5.083e-003	-3.186e-003	-3.186e-003	-3.186e-003	
24	Accel. Gradient	(kPa/m)	4.998e-007	7.373e-007	7.373e-007	7.373e-007	
25	Liq. Reynolds		1.849e+005 *	1.849e+005 *	1.849e+005 *	1.849e+005 *	
26	Vap. Reynolds		9.862e+006 *	9.862e+006 *	9.862e+006 *	9.862e+006 *	
27	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
28	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
29	Deposit Thickness	(mm)	---	---	---	---	
30	Deposit Volume	(m3)	---	---	---	---	
31	Length	(m)	2.449e+004	2.449e+004	2.450e+004	2.450e+004	
32	Elevation	(m)	-0.2800	-0.2900	-0.2720	-0.2540	
33	Pressure	(bar)	68.38	68.38	68.38	68.38	
34	Temperature	(C)	48.71	48.70	48.70	48.70	
35	Vapour Fraction		0.8809	0.8809	0.8809	0.8809	
36	Heat Transferred	(kJ/h-m)	498.3	498.3	498.2	498.2	
37	Flow Regime		Transition	Transition	Transition	Transition	
38	Water Flow Regime						
39	Liquid HoldUp		0.1701	0.1701	0.2415	0.2415	
40	Water HoldUp		---	---	---	---	
41	Friction Gradient	(kPa/m)	9.041e-003	9.041e-003	7.173e-003	7.173e-003	
42	Static Gradient	(kPa/m)	-3.186e-003	-3.186e-003	7.345e-003	7.345e-003	
43	Accel. Gradient	(kPa/m)	7.373e-007	7.373e-007	1.828e-006	1.828e-006	
44	Liq. Reynolds		1.849e+005 *	1.849e+005 *	1.849e+005 *	1.849e+005 *	
45	Vap. Reynolds		9.862e+006 *	9.862e+006 *	9.862e+006 *	9.862e+006 *	
46	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
47	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
48	Deposit Thickness	(mm)	---	---	---	---	
49	Deposit Volume	(m3)	---	---	---	---	
50	Length	(m)	2.451e+004	2.451e+004	2.452e+004	2.452e+004	
51	Elevation	(m)	-0.2360	-0.2180	-0.2000	-0.2120	
52	Pressure	(bar)	68.38	68.38	68.38	68.38	
53	Temperature	(C)	48.70	48.70	48.70	48.69	
54	Vapour Fraction		0.8809	0.8809	0.8809	0.8809	
55	Heat Transferred	(kJ/h-m)	498.1	498.1	498.0	498.0	
56	Flow Regime		Transition	Transition	Transition	Transition	
57	Water Flow Regime						
58	Liquid HoldUp		0.2415	0.2415	0.2415	0.1698	
59	Water HoldUp		---	---	---	---	
60	Friction Gradient	(kPa/m)	7.173e-003	7.173e-003	7.173e-003	9.045e-003	
61	Static Gradient	(kPa/m)	7.345e-003	7.345e-003	7.345e-003	-3.822e-003	
62	Accel. Gradient	(kPa/m)	1.828e-006	1.828e-006	1.828e-006	6.578e-007	
63	Liq. Reynolds		1.849e+005 *	1.849e+005 *	1.849e+005 *	1.849e+005 *	
64	Vap. Reynolds		9.862e+006 *	9.862e+006 *	9.862e+006 *	9.862e+006 *	
65	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
66	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
67	Deposit Thickness	(mm)	---	---	---	---	
68	Deposit Volume	(m3)	---	---	---	---	
69	Aspen Technology Inc.		Aspen HYSYS Version 10				Page 396 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	2.453e+004	2.453e+004	2.454e+004	2.454e+004
14	Elevation (m)	-0.2240	-0.2360	-0.2480	-0.2600
15	Pressure (bar)	68.38	68.38	68.37	68.37
16	Temperature (C)	48.69	48.69	48.69	48.69
17	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
18	Heat Transferred (kJ/h-m)	498.0	498.0	497.9	497.9
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.1698	0.1698	0.1698	0.1698
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	9.045e-003	9.045e-003	9.045e-003	9.045e-003
24	Static Gradient (kPa/m)	-3.822e-003	-3.822e-003	-3.822e-003	-3.822e-003
25	Accel. Gradient (kPa/m)	6.578e-007	6.578e-007	6.578e-007	6.578e-007
26	Liq. Reynolds	1.849e+005 *	1.849e+005 *	1.849e+005 *	1.849e+005 *
27	Vap. Reynolds	9.862e+006 *	9.862e+006 *	9.862e+006 *	9.862e+006 *
28	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
29	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	2.455e+004	2.455e+004	2.456e+004	2.456e+004
33	Elevation (m)	-0.2520	-0.2440	-0.2360	-0.2280
34	Pressure (bar)	68.37	68.37	68.37	68.37
35	Temperature (C)	48.69	48.69	48.68	48.68
36	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
37	Heat Transferred (kJ/h-m)	497.8	497.8	497.7	497.7
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.2359	0.2359	0.2359	0.2359
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	7.857e-003	7.857e-003	7.857e-003	7.857e-003
43	Static Gradient (kPa/m)	3.209e-003	3.209e-003	3.209e-003	3.209e-003
44	Accel. Gradient (kPa/m)	1.394e-006	1.394e-006	1.394e-006	1.394e-006
45	Liq. Reynolds	1.849e+005 *	1.849e+005 *	1.849e+005 *	1.849e+005 *
46	Vap. Reynolds	9.862e+006 *	9.862e+006 *	9.862e+006 *	9.862e+006 *
47	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
48	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	2.457e+004	2.457e+004	2.458e+004	2.458e+004
52	Elevation (m)	-0.2200	-0.2120	-0.2040	-0.1960
53	Pressure (bar)	68.37	68.37	68.37	68.37
54	Temperature (C)	48.68	48.68	48.68	48.68
55	Vapour Fraction	0.8809	0.8809	0.8809	0.8809
56	Heat Transferred (kJ/h-m)	497.6	497.6	497.5	497.5
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.2359	0.2359	0.2359	0.2359
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	7.857e-003	7.857e-003	7.857e-003	7.857e-003
62	Static Gradient (kPa/m)	3.209e-003	3.209e-003	3.209e-003	3.209e-003
63	Accel. Gradient (kPa/m)	1.394e-006	1.394e-006	1.394e-006	1.394e-006
64	Liq. Reynolds	1.849e+005 *	1.849e+005 *	1.849e+005 *	1.849e+005 *
65	Vap. Reynolds	9.862e+006 *	9.862e+006 *	9.862e+006 *	9.862e+006 *
66	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
67	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
68	Deposit Thickness (mm)	---	---	---	---
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 397 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	2.459e+004	2.459e+004	2.460e+004	2.460e+004		
15	Elevation (m)	-0.1880	-0.1800	-0.1720	-0.1640		
16	Pressure (bar)	68.37	68.37	68.37	68.37		
17	Temperature (C)	48.67	48.67	48.67	48.67		
18	Vapour Fraction	0.8809	0.8809	0.8809	0.8809		
19	Heat Transferred (kJ/h-m)	497.5	497.4	497.4	497.3		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	0.2359	0.2359	0.2359	0.2359		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	7.857e-003	7.857e-003	7.857e-003	7.857e-003		
25	Static Gradient (kPa/m)	3.209e-003	3.209e-003	3.209e-003	3.209e-003		
26	Accel. Gradient (kPa/m)	1.394e-006	1.394e-006	1.394e-006	1.394e-006		
27	Liq. Reynolds	1.849e+005 *	1.849e+005 *	1.849e+005 *	1.849e+005 *		
28	Vap. Reynolds	9.862e+006 *	9.862e+006 *	9.863e+006 *	9.863e+006 *		
29	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
30	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	2.461e+004	2.461e+004	2.462e+004	2.462e+004		
34	Elevation (m)	-0.1560	-0.1480	-0.1400	-0.1300		
35	Pressure (bar)	68.37	68.37	68.37	68.37		
36	Temperature (C)	48.67	48.67	48.66	48.66		
37	Vapour Fraction	0.8809	0.8809	0.8809	0.8809		
38	Heat Transferred (kJ/h-m)	497.3	497.2	497.2	497.2		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.2359	0.2359	0.2359	0.2370		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	7.857e-003	7.857e-003	7.857e-003	7.716e-003		
44	Static Gradient (kPa/m)	3.209e-003	3.209e-003	3.209e-003	4.025e-003		
45	Accel. Gradient (kPa/m)	1.394e-006	1.394e-006	1.394e-006	1.479e-006		
46	Liq. Reynolds	1.849e+005 *	1.849e+005 *	1.849e+005 *	1.849e+005 *		
47	Vap. Reynolds	9.863e+006 *	9.863e+006 *	9.863e+006 *	9.863e+006 *		
48	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
49	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	2.463e+004	2.463e+004	2.464e+004	2.464e+004		
53	Elevation (m)	-0.1200	-0.1100	-0.1000	-9.000e-002		
54	Pressure (bar)	68.37	68.36	68.36	68.36		
55	Temperature (C)	48.66	48.66	48.66	48.66		
56	Vapour Fraction	0.8809	0.8809	0.8809	0.8809		
57	Heat Transferred (kJ/h-m)	497.1	497.1	497.0	497.0		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.2370	0.2370	0.2370	0.2370		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	7.716e-003	7.716e-003	7.716e-003	7.716e-003		
63	Static Gradient (kPa/m)	4.025e-003	4.025e-003	4.025e-003	4.025e-003		
64	Accel. Gradient (kPa/m)	1.479e-006	1.479e-006	1.479e-006	1.479e-006		
65	Liq. Reynolds	1.849e+005 *	1.849e+005 *	1.849e+005 *	1.849e+005 *		
66	Vap. Reynolds	9.863e+006 *	9.863e+006 *	9.863e+006 *	9.863e+006 *		
67	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
68	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920		
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 398 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Deposit Thickness (mm)	---	---	---	---					
14	Deposit Volume (m3)	---	---	---	---					
15	Length (m)	2.465e+004	2.465e+004	2.466e+004	2.466e+004					
16	Elevation (m)	-0.1160	-0.1420	-0.1680	-0.1940					
17	Pressure (bar)	68.36	68.36	68.36	68.36					
18	Temperature (C)	48.66	48.65	48.65	48.65					
19	Vapour Fraction	0.8809	0.8809	0.8809	0.8809					
20	Heat Transferred (kJ/h-m)	497.0	497.0	496.9	496.9					
21	Flow Regime	Transition	Transition	Transition	Transition					
22	Water Flow Regime									
23	Liquid HoldUp	0.1678	0.1678	0.1678	0.1678					
24	Water HoldUp	---	---	---	---					
25	Friction Gradient (kPa/m)	9.073e-003	9.073e-003	9.073e-003	9.073e-003					
26	Static Gradient (kPa/m)	-8.213e-003	-8.213e-003	-8.213e-003	-8.213e-003					
27	Accel. Gradient (kPa/m)	1.084e-007	1.084e-007	1.084e-007	1.084e-007					
28	Liq. Reynolds	1.849e+005 *	1.849e+005 *	1.849e+005 *	1.849e+005 *					
29	Vap. Reynolds	9.863e+006 *	9.863e+006 *	9.863e+006 *	9.863e+006 *					
30	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
31	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920					
32	Deposit Thickness (mm)	---	---	---	---					
33	Deposit Volume (m3)	---	---	---	---					
34	Length (m)	2.467e+004	2.467e+004	2.468e+004	2.468e+004					
35	Elevation (m)	-0.2200	-0.2920	-0.3640	-0.4360					
36	Pressure (bar)	68.36	68.36	68.36	68.36					
37	Temperature (C)	48.65	48.65	48.65	48.65					
38	Vapour Fraction	0.8809	0.8809	0.8809	0.8809					
39	Heat Transferred (kJ/h-m)	496.8	496.8	496.8	496.8					
40	Flow Regime	Transition	Transition	Transition	Transition					
41	Water Flow Regime									
42	Liquid HoldUp	0.1678	0.1609	0.1609	0.1609					
43	Water HoldUp	---	---	---	---					
44	Friction Gradient (kPa/m)	9.073e-003	9.174e-003	9.174e-003	9.174e-003					
45	Static Gradient (kPa/m)	-8.213e-003	-2.213e-002	-2.213e-002	-2.213e-002					
46	Accel. Gradient (kPa/m)	1.084e-007	-1.632e-006	-1.632e-006	-1.632e-006					
47	Liq. Reynolds	1.849e+005 *	1.849e+005 *	1.849e+005 *	1.849e+005 *					
48	Vap. Reynolds	9.863e+006 *	9.863e+006 *	9.863e+006 *	9.863e+006 *					
49	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
50	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920					
51	Deposit Thickness (mm)	---	---	---	---					
52	Deposit Volume (m3)	---	---	---	---					
53	Length (m)	2.469e+004	2.469e+004	2.470e+004	2.470e+004					
54	Elevation (m)	-0.5080	-0.5800	-0.5100	-0.4400					
55	Pressure (bar)	68.37	68.37	68.36	68.36					
56	Temperature (C)	48.65	48.64	48.64	48.64					
57	Vapour Fraction	0.8809	0.8809	0.8809	0.8809					
58	Heat Transferred (kJ/h-m)	496.7	496.7	496.6	496.5					
59	Flow Regime	Transition	Transition	Transition	Transition					
60	Water Flow Regime									
61	Liquid HoldUp	0.1610	0.1610	0.2703	0.2703					
62	Water HoldUp	---	---	---	---					
63	Friction Gradient (kPa/m)	9.174e-003	9.173e-003	7.875e-003	7.875e-003					
64	Static Gradient (kPa/m)	-2.213e-002	-2.213e-002	3.109e-002	3.109e-002					
65	Accel. Gradient (kPa/m)	-1.632e-006	-1.632e-006	4.907e-006	4.907e-006					
66	Liq. Reynolds	1.849e+005 *	1.849e+005 *	1.849e+005 *	1.849e+005 *					
67	Vap. Reynolds	9.863e+006 *	9.863e+006 *	9.863e+006 *	9.863e+006 *					
68	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	2.471e+004	2.471e+004	2.472e+004	2.472e+004	
17	Elevation	(m)	-0.3700	-0.3000	-0.2300	-0.1980	
18	Pressure	(bar)	68.36	68.36	68.36	68.36	
19	Temperature	(C)	48.64	48.64	48.63	48.63	
20	Vapour Fraction		0.8809	0.8809	0.8809	0.8809	
21	Heat Transferred	(kJ/h-m)	496.5	496.4	496.4	496.3	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2703	0.2703	0.2703	0.2492	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	7.875e-003	7.875e-003	7.875e-003	6.292e-003	
27	Static Gradient	(kPa/m)	3.109e-002	3.109e-002	3.109e-002	1.337e-002	
28	Accel. Gradient	(kPa/m)	4.908e-006	4.908e-006	4.908e-006	2.477e-006	
29	Liq. Reynolds		1.849e+005 *	1.849e+005 *	1.848e+005 *	1.848e+005 *	
30	Vap. Reynolds		9.863e+006 *	9.863e+006 *	9.863e+006 *	9.863e+006 *	
31	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
32	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	2.473e+004	2.473e+004	2.474e+004	2.474e+004	
36	Elevation	(m)	-0.1660	-0.1340	-0.1020	-7.000e-002	
37	Pressure	(bar)	68.35	68.35	68.35	68.35	
38	Temperature	(C)	48.63	48.63	48.63	48.62	
39	Vapour Fraction		0.8809	0.8809	0.8809	0.8809	
40	Heat Transferred	(kJ/h-m)	496.3	496.2	496.2	496.1	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2492	0.2492	0.2492	0.2492	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	6.292e-003	6.292e-003	6.292e-003	6.293e-003	
46	Static Gradient	(kPa/m)	1.337e-002	1.337e-002	1.337e-002	1.337e-002	
47	Accel. Gradient	(kPa/m)	2.477e-006	2.477e-006	2.477e-006	2.477e-006	
48	Liq. Reynolds		1.848e+005 *	1.848e+005 *	1.848e+005 *	1.848e+005 *	
49	Vap. Reynolds		9.863e+006 *	9.863e+006 *	9.863e+006 *	9.863e+006 *	
50	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
51	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	2.475e+004	2.475e+004	2.476e+004	2.476e+004	
55	Elevation	(m)	-7.000e-002	-7.000e-002	-7.000e-002	-7.000e-002	
56	Pressure	(bar)	68.35	68.35	68.35	68.35	
57	Temperature	(C)	48.62	48.62	48.62	48.62	
58	Vapour Fraction		0.8809	0.8809	0.8809	0.8809	
59	Heat Transferred	(kJ/h-m)	496.1	496.0	496.0	496.0	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.2505	0.2505	0.2505	0.2505	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	6.153e-003	6.153e-003	6.153e-003	6.153e-003	
65	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000	
66	Accel. Gradient	(kPa/m)	7.752e-007	7.752e-007	7.752e-007	7.752e-007	
67	Liq. Reynolds		1.848e+005 *	1.848e+005 *	1.848e+005 *	1.848e+005 *	
68	Vap. Reynolds		9.863e+006 *	9.863e+006 *	9.863e+006 *	9.863e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
14	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	2.477e+004	2.477e+004	2.478e+004	2.478e+004	
18	Elevation	(m)	-7.000e-002	-7.600e-002	-8.200e-002	-8.800e-002	
19	Pressure	(bar)	68.35	68.35	68.35	68.35	
20	Temperature	(C)	48.62	48.62	48.61	48.61	
21	Vapour Fraction		0.8809	0.8809	0.8809	0.8809	
22	Heat Transferred	(kJ/h-m)	495.9	495.9	495.9	495.8	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.2505	0.1707	0.1707	0.1707	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	6.153e-003	9.033e-003	9.033e-003	9.033e-003	
28	Static Gradient	(kPa/m)	0.0000	-1.917e-003	-1.917e-003	-1.918e-003	
29	Accel. Gradient	(kPa/m)	7.752e-007	8.964e-007	8.964e-007	8.964e-007	
30	Liq. Reynolds		1.848e+005 *	1.848e+005 *	1.848e+005 *	1.848e+005 *	
31	Vap. Reynolds		9.863e+006 *	9.863e+006 *	9.863e+006 *	9.863e+006 *	
32	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
33	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	2.479e+004	2.479e+004	2.480e+004	2.480e+004	
37	Elevation	(m)	-9.400e-002	-0.1000	-0.1140	-0.1280	
38	Pressure	(bar)	68.35	68.35	68.35	68.35	
39	Temperature	(C)	48.61	48.61	48.61	48.61	
40	Vapour Fraction		0.8809	0.8809	0.8809	0.8809	
41	Heat Transferred	(kJ/h-m)	495.8	495.8	495.7	495.7	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.1707	0.1707	0.1695	0.1695	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	9.033e-003	9.033e-003	9.049e-003	9.049e-003	
47	Static Gradient	(kPa/m)	-1.918e-003	-1.918e-003	-4.453e-003	-4.453e-003	
48	Accel. Gradient	(kPa/m)	8.964e-007	8.964e-007	5.790e-007	5.790e-007	
49	Liq. Reynolds		1.848e+005 *	1.848e+005 *	1.848e+005 *	1.848e+005 *	
50	Vap. Reynolds		9.863e+006 *	9.863e+006 *	9.863e+006 *	9.863e+006 *	
51	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
52	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	2.481e+004	2.481e+004	2.482e+004	2.482e+004	
56	Elevation	(m)	-0.1420	-0.1560	-0.1700	-0.2060	
57	Pressure	(bar)	68.35	68.35	68.35	68.35	
58	Temperature	(C)	48.60	48.60	48.60	48.60	
59	Vapour Fraction		0.8809	0.8809	0.8809	0.8809	
60	Heat Transferred	(kJ/h-m)	495.6	495.6	495.6	495.5	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.1695	0.1695	0.1695	0.1663	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	9.049e-003	9.049e-003	9.049e-003	9.094e-003	
66	Static Gradient	(kPa/m)	-4.453e-003	-4.453e-003	-4.453e-003	-1.131e-002	
67	Accel. Gradient	(kPa/m)	5.790e-007	5.789e-007	5.789e-007	-2.785e-007	
68	Liq. Reynolds		1.848e+005 *	1.848e+005 *	1.848e+005 *	1.848e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.863e+006 *	9.863e+006 *	9.864e+006 *	9.864e+006 *
14	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
15	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	2.483e+004	2.483e+004	2.484e+004	2.484e+004
19	Elevation (m)	-0.2420	-0.2780	-0.3140	-0.3500
20	Pressure (bar)	68.35	68.35	68.35	68.35
21	Temperature (C)	48.60	48.60	48.60	48.60
22	Vapour Fraction	0.8809	0.8808	0.8808	0.8808
23	Heat Transferred (kJ/h-m)	495.5	495.5	495.4	495.4
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1663	0.1663	0.1663	0.1663
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	9.094e-003	9.094e-003	9.094e-003	9.094e-003
29	Static Gradient (kPa/m)	-1.131e-002	-1.131e-002	-1.131e-002	-1.131e-002
30	Accel. Gradient (kPa/m)	-2.785e-007	-2.785e-007	-2.786e-007	-2.786e-007
31	Liq. Reynolds	1.848e+005 *	1.848e+005 *	1.848e+005 *	1.848e+005 *
32	Vap. Reynolds	9.864e+006 *	9.864e+006 *	9.864e+006 *	9.864e+006 *
33	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
34	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	2.485e+004	2.485e+004	2.486e+004	2.486e+004
38	Elevation (m)	-0.4260	-0.5020	-0.5780	-0.6540
39	Pressure (bar)	68.35	68.35	68.35	68.35
40	Temperature (C)	48.59	48.59	48.59	48.59
41	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
42	Heat Transferred (kJ/h-m)	495.4	495.4	495.3	495.3
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1604	0.1604	0.1604	0.1604
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	9.183e-003	9.183e-003	9.183e-003	9.183e-003
48	Static Gradient (kPa/m)	-2.330e-002	-2.330e-002	-2.330e-002	-2.330e-002
49	Accel. Gradient (kPa/m)	-1.779e-006	-1.779e-006	-1.779e-006	-1.779e-006
50	Liq. Reynolds	1.848e+005 *	1.848e+005 *	1.848e+005 *	1.848e+005 *
51	Vap. Reynolds	9.863e+006 *	9.863e+006 *	9.863e+006 *	9.863e+006 *
52	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
53	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	2.487e+004	2.487e+004	2.488e+004	2.488e+004
57	Elevation (m)	-0.7300	-0.7300	-0.7300	-0.7300
58	Pressure (bar)	68.35	68.35	68.35	68.35
59	Temperature (C)	48.59	48.59	48.59	48.59
60	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
61	Heat Transferred (kJ/h-m)	495.3	495.2	495.1	495.1
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1604	0.2505	0.2505	0.2506
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	9.183e-003	6.152e-003	6.152e-003	6.152e-003
67	Static Gradient (kPa/m)	-2.331e-002	0.0000	0.0000	0.0000
68	Accel. Gradient (kPa/m)	-1.779e-006	7.749e-007	7.749e-007	7.749e-007
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		1.848e+005 *	1.848e+005 *	1.848e+005 *	1.848e+005 *	
14	Vap. Reynolds		9.863e+006 *	9.863e+006 *	9.863e+006 *	9.863e+006 *	
15	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
16	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920		
17	Deposit Thickness (mm)	---	---	---	---		
18	Deposit Volume (m3)	---	---	---	---		
19	Length (m)	2.489e+004	2.489e+004	2.490e+004	2.490e+004		
20	Elevation (m)	-0.7300	-0.7300	-0.6140	-0.4980		
21	Pressure (bar)	68.35	68.35	68.35	68.34		
22	Temperature (C)	48.58	48.58	48.58	48.58		
23	Vapour Fraction	0.8808	0.8808	0.8808	0.8808		
24	Heat Transferred (kJ/h-m)	495.0	495.0	494.9	494.9		
25	Flow Regime	Transition	Transition	Transition	Transition		
26	Water Flow Regime						
27	Liquid HoldUp	0.2506	0.2506	0.2958	0.2958		
28	Water HoldUp	---	---	---	---		
29	Friction Gradient (kPa/m)	6.152e-003	6.152e-003	7.906e-003	7.906e-003		
30	Static Gradient (kPa/m)	0.0000	0.0000	5.522e-002	5.522e-002		
31	Accel. Gradient (kPa/m)	7.749e-007	7.749e-007	7.952e-006	7.952e-006		
32	Liq. Reynolds		1.848e+005 *	1.848e+005 *	1.848e+005 *		
33	Vap. Reynolds		9.863e+006 *	9.863e+006 *	9.864e+006 *		
34	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
35	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920		
36	Deposit Thickness (mm)	---	---	---	---		
37	Deposit Volume (m3)	---	---	---	---		
38	Length (m)	2.491e+004	2.491e+004	2.492e+004	2.492e+004		
39	Elevation (m)	-0.3820	-0.2660	-0.1500	-0.1340		
40	Pressure (bar)	68.34	68.34	68.33	68.33		
41	Temperature (C)	48.57	48.57	48.57	48.57		
42	Vapour Fraction	0.8808	0.8808	0.8808	0.8808		
43	Heat Transferred (kJ/h-m)	494.8	494.7	494.7	494.6		
44	Flow Regime	Transition	Transition	Transition	Transition		
45	Water Flow Regime						
46	Liquid HoldUp	0.2958	0.2958	0.2957	0.2404		
47	Water HoldUp	---	---	---	---		
48	Friction Gradient (kPa/m)	7.906e-003	7.907e-003	7.907e-003	7.307e-003		
49	Static Gradient (kPa/m)	5.522e-002	5.521e-002	5.521e-002	6.507e-003		
50	Accel. Gradient (kPa/m)	7.953e-006	7.953e-006	7.954e-006	1.741e-006		
51	Liq. Reynolds		1.848e+005 *	1.848e+005 *	1.848e+005 *		
52	Vap. Reynolds		9.864e+006 *	9.864e+006 *	9.864e+006 *		
53	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
54	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920		
55	Deposit Thickness (mm)	---	---	---	---		
56	Deposit Volume (m3)	---	---	---	---		
57	Length (m)	2.493e+004	2.493e+004	2.494e+004	2.494e+004		
58	Elevation (m)	-0.1180	-0.1020	-8.600e-002	-7.000e-002		
59	Pressure (bar)	68.33	68.33	68.33	68.33		
60	Temperature (C)	48.57	48.56	48.56	48.56		
61	Vapour Fraction	0.8808	0.8808	0.8808	0.8808		
62	Heat Transferred (kJ/h-m)	494.6	494.5	494.5	494.4		
63	Flow Regime	Transition	Transition	Transition	Transition		
64	Water Flow Regime						
65	Liquid HoldUp	0.2404	0.2403	0.2403	0.2403		
66	Water HoldUp	---	---	---	---		
67	Friction Gradient (kPa/m)	7.307e-003	7.307e-003	7.307e-003	7.307e-003		
68	Static Gradient (kPa/m)	6.507e-003	6.507e-003	6.507e-003	6.507e-003		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient	(kPa/m)	1.741e-006	1.741e-006	1.741e-006	1.741e-006	
14	Liq. Reynolds		1.848e+005 *	1.848e+005 *	1.848e+005 *	1.848e+005 *	
15	Vap. Reynolds		9.864e+006 *	9.864e+006 *	9.864e+006 *	9.864e+006 *	
16	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
17	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
18	Deposit Thickness	(mm)	---	---	---	---	
19	Deposit Volume	(m3)	---	---	---	---	
20	Length	(m)	2.495e+004	2.495e+004	2.496e+004	2.496e+004	
21	Elevation	(m)	-7.600e-002	-8.200e-002	-8.800e-002	-9.400e-002	
22	Pressure	(bar)	68.33	68.33	68.33	68.33	
23	Temperature	(C)	48.56	48.56	48.56	48.56	
24	Vapour Fraction		0.8808	0.8808	0.8808	0.8808	
25	Heat Transferred	(kJ/h-m)	494.4	494.4	494.4	494.3	
26	Flow Regime		Transition	Transition	Transition	Transition	
27	Water Flow Regime						
28	Liquid HoldUp		0.1707	0.1707	0.1707	0.1707	
29	Water HoldUp		---	---	---	---	
30	Friction Gradient	(kPa/m)	9.034e-003	9.034e-003	9.034e-003	9.034e-003	
31	Static Gradient	(kPa/m)	-1.917e-003	-1.917e-003	-1.918e-003	-1.918e-003	
32	Accel. Gradient	(kPa/m)	8.968e-007	8.968e-007	8.968e-007	8.968e-007	
33	Liq. Reynolds		1.848e+005 *	1.848e+005 *	1.848e+005 *	1.848e+005 *	
34	Vap. Reynolds		9.864e+006 *	9.864e+006 *	9.864e+006 *	9.864e+006 *	
35	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
36	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
37	Deposit Thickness	(mm)	---	---	---	---	
38	Deposit Volume	(m3)	---	---	---	---	
39	Length	(m)	2.497e+004	2.497e+004	2.498e+004	2.498e+004	
40	Elevation	(m)	-0.1000	-0.1060	-0.1120	-0.1180	
41	Pressure	(bar)	68.33	68.33	68.33	68.33	
42	Temperature	(C)	48.55	48.55	48.55	48.55	
43	Vapour Fraction		0.8808	0.8808	0.8808	0.8808	
44	Heat Transferred	(kJ/h-m)	494.3	494.2	494.2	494.2	
45	Flow Regime		Transition	Transition	Transition	Transition	
46	Water Flow Regime						
47	Liquid HoldUp		0.1707	0.1707	0.1707	0.1707	
48	Water HoldUp		---	---	---	---	
49	Friction Gradient	(kPa/m)	9.034e-003	9.034e-003	9.034e-003	9.034e-003	
50	Static Gradient	(kPa/m)	-1.918e-003	-1.918e-003	-1.918e-003	-1.918e-003	
51	Accel. Gradient	(kPa/m)	8.968e-007	8.968e-007	8.968e-007	8.968e-007	
52	Liq. Reynolds		1.848e+005 *	1.848e+005 *	1.848e+005 *	1.848e+005 *	
53	Vap. Reynolds		9.864e+006 *	9.864e+006 *	9.864e+006 *	9.864e+006 *	
54	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
55	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920	
56	Deposit Thickness	(mm)	---	---	---	---	
57	Deposit Volume	(m3)	---	---	---	---	
58	Length	(m)	2.499e+004	2.499e+004	2.500e+004	2.500e+004	
59	Elevation	(m)	-0.1240	-0.1300	-0.1320	-0.1340	
60	Pressure	(bar)	68.33	68.33	68.33	68.33	
61	Temperature	(C)	48.55	48.55	48.54	48.54	
62	Vapour Fraction		0.8808	0.8808	0.8808	0.8808	
63	Heat Transferred	(kJ/h-m)	494.1	494.1	494.0	494.0	
64	Flow Regime		Transition	Transition	Transition	Transition	
65	Water Flow Regime						
66	Liquid HoldUp		0.1707	0.1707	0.1713	0.1713	
67	Water HoldUp		---	---	---	---	
68	Friction Gradient	(kPa/m)	9.034e-003	9.033e-003	9.026e-003	9.026e-003	
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1				Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
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11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	-1.918e-003	-1.918e-003	-6.407e-004	-6.407e-004
14	Accel. Gradient	(kPa/m)	8.968e-007	8.968e-007	1.057e-006	1.057e-006
15	Liq. Reynolds		1.847e+005 *	1.847e+005 *	1.847e+005 *	1.847e+005 *
16	Vap. Reynolds		9.864e+006 *	9.864e+006 *	9.864e+006 *	9.864e+006 *
17	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
18	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	2.501e+004	2.501e+004	2.502e+004	2.502e+004
22	Elevation	(m)	-0.1360	-0.1380	-0.1400	-0.1440
23	Pressure	(bar)	68.33	68.33	68.32	68.32
24	Temperature	(C)	48.54	48.54	48.54	48.54
25	Vapour Fraction		0.8808	0.8808	0.8808	0.8808
26	Heat Transferred	(kJ/h-m)	494.0	493.9	493.9	493.8
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.1713	0.1713	0.1713	0.1710
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	9.026e-003	9.026e-003	9.026e-003	9.030e-003
32	Static Gradient	(kPa/m)	-6.407e-004	-6.407e-004	-6.407e-004	-1.280e-003
33	Accel. Gradient	(kPa/m)	1.057e-006	1.057e-006	1.057e-006	9.767e-007
34	Liq. Reynolds		1.847e+005 *	1.847e+005 *	1.847e+005 *	1.847e+005 *
35	Vap. Reynolds		9.864e+006 *	9.864e+006 *	9.865e+006 *	9.865e+006 *
36	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
37	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	2.503e+004	2.503e+004	2.504e+004	2.504e+004
41	Elevation	(m)	-0.1480	-0.1520	-0.1560	-0.1600
42	Pressure	(bar)	68.32	68.32	68.32	68.32
43	Temperature	(C)	48.54	48.53	48.53	48.53
44	Vapour Fraction		0.8808	0.8808	0.8808	0.8808
45	Heat Transferred	(kJ/h-m)	493.8	493.8	493.7	493.7
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.1710	0.1710	0.1710	0.1710
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	9.030e-003	9.029e-003	9.029e-003	9.029e-003
51	Static Gradient	(kPa/m)	-1.280e-003	-1.280e-003	-1.280e-003	-1.280e-003
52	Accel. Gradient	(kPa/m)	9.767e-007	9.767e-007	9.767e-007	9.767e-007
53	Liq. Reynolds		1.847e+005 *	1.847e+005 *	1.847e+005 *	1.847e+005 *
54	Vap. Reynolds		9.865e+006 *	9.865e+006 *	9.865e+006 *	9.865e+006 *
55	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
56	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	2.505e+004	2.505e+004	2.506e+004	2.506e+004
60	Elevation	(m)	-0.1700	-0.1800	-0.1900	-0.2000
61	Pressure	(bar)	68.32	68.32	68.32	68.32
62	Temperature	(C)	48.53	48.53	48.53	48.52
63	Vapour Fraction		0.8808	0.8808	0.8808	0.8808
64	Heat Transferred	(kJ/h-m)	493.6	493.6	493.6	493.5
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.1701	0.1701	0.1701	0.1701
68	Water HoldUp		---	---	---	---
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1				Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	9.041e-003	9.041e-003	9.041e-003	9.041e-003
14	Static Gradient	(kPa/m)	-3.189e-003	-3.189e-003	-3.189e-003	-3.189e-003
15	Accel. Gradient	(kPa/m)	7.376e-007	7.376e-007	7.376e-007	7.376e-007
16	Liq. Reynolds		1.847e+005 *	1.847e+005 *	1.847e+005 *	1.847e+005 *
17	Vap. Reynolds		9.865e+006 *	9.865e+006 *	9.865e+006 *	9.865e+006 *
18	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
19	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	2.507e+004	2.507e+004	2.508e+004	2.508e+004
23	Elevation	(m)	-0.2100	-0.1940	-0.1780	-0.1620
24	Pressure	(bar)	68.32	68.32	68.32	68.32
25	Temperature	(C)	48.52	48.52	48.52	48.52
26	Vapour Fraction		0.8808	0.8808	0.8808	0.8808
27	Heat Transferred	(kJ/h-m)	493.5	493.4	493.3	493.3
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.1701	0.2403	0.2403	0.2403
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	9.041e-003	7.309e-003	7.309e-003	7.309e-003
33	Static Gradient	(kPa/m)	-3.189e-003	6.497e-003	6.497e-003	6.497e-003
34	Accel. Gradient	(kPa/m)	7.376e-007	1.740e-006	1.740e-006	1.740e-006
35	Liq. Reynolds		1.847e+005 *	1.847e+005 *	1.847e+005 *	1.847e+005 *
36	Vap. Reynolds		9.865e+006 *	9.865e+006 *	9.865e+006 *	9.865e+006 *
37	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
38	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	2.509e+004	2.509e+004	2.510e+004	2.510e+004
42	Elevation	(m)	-0.1460	-0.1300	-0.1380	-0.1460
43	Pressure	(bar)	68.32	68.32	68.32	68.32
44	Temperature	(C)	48.52	48.52	48.51	48.51
45	Vapour Fraction		0.8808	0.8808	0.8808	0.8808
46	Heat Transferred	(kJ/h-m)	493.3	493.2	493.2	493.2
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2403	0.2403	0.1704	0.1704
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	7.309e-003	7.309e-003	9.037e-003	9.037e-003
52	Static Gradient	(kPa/m)	6.497e-003	6.497e-003	-2.554e-003	-2.554e-003
53	Accel. Gradient	(kPa/m)	1.740e-006	1.740e-006	8.172e-007	8.172e-007
54	Liq. Reynolds		1.847e+005 *	1.847e+005 *	1.847e+005 *	1.847e+005 *
55	Vap. Reynolds		9.865e+006 *	9.865e+006 *	9.865e+006 *	9.865e+006 *
56	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
57	Vapour Velocity	(m/s)	2.920	2.920	2.920	2.920
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	2.511e+004	2.511e+004	2.512e+004	2.512e+004
61	Elevation	(m)	-0.1540	-0.1620	-0.1700	-0.1760
62	Pressure	(bar)	68.32	68.32	68.32	68.32
63	Temperature	(C)	48.51	48.51	48.51	48.51
64	Vapour Fraction		0.8808	0.8808	0.8808	0.8808
65	Heat Transferred	(kJ/h-m)	493.1	493.1	493.1	493.0
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.1704	0.1704	0.1704	0.1707
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	9.037e-003	9.037e-003	9.037e-003	9.033e-003
15	Static Gradient (kPa/m)	-2.554e-003	-2.554e-003	-2.554e-003	-1.918e-003
16	Accel. Gradient (kPa/m)	8.172e-007	8.172e-007	8.172e-007	8.969e-007
17	Liq. Reynolds	1.847e+005 *	1.847e+005 *	1.847e+005 *	1.847e+005 *
18	Vap. Reynolds	9.865e+006 *	9.865e+006 *	9.865e+006 *	9.865e+006 *
19	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
20	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	2.513e+004	2.513e+004	2.514e+004	2.514e+004
24	Elevation (m)	-0.1820	-0.1880	-0.1940	-0.2000
25	Pressure (bar)	68.32	68.32	68.31	68.31
26	Temperature (C)	48.50	48.50	48.50	48.50
27	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
28	Heat Transferred (kJ/h-m)	493.0	492.9	492.9	492.9
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.1707	0.1707	0.1707	0.1707
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	9.033e-003	9.033e-003	9.033e-003	9.033e-003
34	Static Gradient (kPa/m)	-1.918e-003	-1.918e-003	-1.918e-003	-1.918e-003
35	Accel. Gradient (kPa/m)	8.969e-007	8.969e-007	8.969e-007	8.969e-007
36	Liq. Reynolds	1.847e+005 *	1.847e+005 *	1.847e+005 *	1.847e+005 *
37	Vap. Reynolds	9.865e+006 *	9.865e+006 *	9.865e+006 *	9.865e+006 *
38	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
39	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	2.515e+004	2.515e+004	2.516e+004	2.516e+004
43	Elevation (m)	-0.2180	-0.2360	-0.2540	-0.2720
44	Pressure (bar)	68.31	68.31	68.31	68.31
45	Temperature (C)	48.50	48.50	48.50	48.49
46	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
47	Heat Transferred (kJ/h-m)	492.8	492.8	492.7	492.7
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.1689	0.1689	0.1689	0.1689
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	9.057e-003	9.057e-003	9.057e-003	9.057e-003
53	Static Gradient (kPa/m)	-5.713e-003	-5.713e-003	-5.713e-003	-5.713e-003
54	Accel. Gradient (kPa/m)	4.216e-007	4.216e-007	4.215e-007	4.215e-007
55	Liq. Reynolds	1.847e+005 *	1.847e+005 *	1.847e+005 *	1.847e+005 *
56	Vap. Reynolds	9.865e+006 *	9.865e+006 *	9.865e+006 *	9.865e+006 *
57	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
58	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	2.517e+004	2.517e+004	2.518e+004	2.518e+004
62	Elevation (m)	-0.2900	-0.3160	-0.3420	-0.3680
63	Pressure (bar)	68.31	68.31	68.31	68.31
64	Temperature (C)	48.49	48.49	48.49	48.49
65	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
66	Heat Transferred (kJ/h-m)	492.7	492.6	492.6	492.6
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.1689	0.1678	0.1678	0.1678
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	9.057e-003	9.074e-003	9.074e-003	9.073e-003
16	Static Gradient (kPa/m)	-5.713e-003	-8.213e-003	-8.214e-003	-8.214e-003
17	Accel. Gradient (kPa/m)	4.215e-007	1.084e-007	1.084e-007	1.084e-007
18	Liq. Reynolds	1.847e+005 *	1.847e+005 *	1.847e+005 *	1.847e+005 *
19	Vap. Reynolds	9.865e+006 *	9.865e+006 *	9.865e+006 *	9.865e+006 *
20	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
21	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	2.519e+004	2.519e+004	2.520e+004	2.520e+004
25	Elevation (m)	-0.3940	-0.4200	-0.4400	-0.4600
26	Pressure (bar)	68.31	68.31	68.31	68.31
27	Temperature (C)	48.49	48.49	48.48	48.48
28	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
29	Heat Transferred (kJ/h-m)	492.5	492.5	492.4	492.4
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1678	0.1678	0.1693	0.1693
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	9.073e-003	9.073e-003	9.051e-003	9.051e-003
35	Static Gradient (kPa/m)	-8.214e-003	-8.214e-003	-4.870e-003	-4.870e-003
36	Accel. Gradient (kPa/m)	1.084e-007	1.084e-007	5.271e-007	5.271e-007
37	Liq. Reynolds	1.847e+005 *	1.847e+005 *	1.847e+005 *	1.847e+005 *
38	Vap. Reynolds	9.865e+006 *	9.865e+006 *	9.865e+006 *	9.865e+006 *
39	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
40	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	2.521e+004	2.522e+004	2.522e+004	2.523e+004
44	Elevation (m)	-0.4800	-0.5000	-0.5200	-1.196
45	Pressure (bar)	68.31	68.31	68.31	68.32
46	Temperature (C)	48.48	48.48	48.48	48.48
47	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
48	Heat Transferred (kJ/h-m)	492.3	492.3	492.2	492.6
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1693	0.1694	0.1694	7.406e-002
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	9.051e-003	9.051e-003	9.051e-003	1.298e-002
54	Static Gradient (kPa/m)	-4.870e-003	-4.870e-003	-4.870e-003	-0.1343
55	Accel. Gradient (kPa/m)	5.271e-007	5.271e-007	5.270e-007	-1.529e-005
56	Liq. Reynolds	1.847e+005 *	1.847e+005 *	1.847e+005 *	1.847e+005 *
57	Vap. Reynolds	9.865e+006 *	9.865e+006 *	9.865e+006 *	9.865e+006 *
58	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
59	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	2.523e+004	2.524e+004	2.524e+004	2.525e+004
63	Elevation (m)	-1.872	-2.548	-3.224	-3.900
64	Pressure (bar)	68.32	68.33	68.34	68.34
65	Temperature (C)	48.48	48.48	48.49	48.49
66	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
67	Heat Transferred (kJ/h-m)	492.6	492.7	492.8	492.8
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 408 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	7.406e-002	7.406e-002	7.406e-002	7.407e-002
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	1.298e-002	1.298e-002	1.298e-002	1.297e-002
17	Static Gradient (kPa/m)	-0.1343	-0.1343	-0.1343	-0.1343
18	Accel. Gradient (kPa/m)	-1.529e-005	-1.529e-005	-1.529e-005	-1.528e-005
19	Liq. Reynolds	1.847e+005 *	1.847e+005 *	1.847e+005 *	1.847e+005 *
20	Vap. Reynolds	9.865e+006 *	9.865e+006 *	9.864e+006 *	9.864e+006 *
21	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
22	Vapour Velocity (m/s)	2.919	2.919	2.919	2.918
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	2.525e+004	2.526e+004	2.526e+004	2.527e+004
26	Elevation (m)	-4.180	-4.460	-4.740	-5.020
27	Pressure (bar)	68.35	68.35	68.35	68.36
28	Temperature (C)	48.49	48.49	48.49	48.49
29	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
30	Heat Transferred (kJ/h-m)	492.6	492.6	492.6	492.6
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1303	0.1304	0.1304	0.1304
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	9.805e-003	9.804e-003	9.804e-003	9.803e-003
36	Static Gradient (kPa/m)	-7.536e-002	-7.536e-002	-7.537e-002	-7.537e-002
37	Accel. Gradient (kPa/m)	-8.254e-006	-8.254e-006	-8.253e-006	-8.253e-006
38	Liq. Reynolds	1.847e+005 *	1.847e+005 *	1.847e+005 *	1.847e+005 *
39	Vap. Reynolds	9.864e+006 *	9.864e+006 *	9.864e+006 *	9.864e+006 *
40	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
41	Vapour Velocity (m/s)	2.918	2.918	2.918	2.918
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	2.527e+004	2.528e+004	2.528e+004	2.529e+004
45	Elevation (m)	-5.300	-5.020	-4.740	-4.460
46	Pressure (bar)	68.36	68.35	68.34	68.33
47	Temperature (C)	48.49	48.49	48.48	48.48
48	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
49	Heat Transferred (kJ/h-m)	492.7	492.2	492.1	492.0
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.1304	0.4243	0.4243	0.4243
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	9.803e-003	7.644e-003	7.645e-003	7.646e-003
55	Static Gradient (kPa/m)	-7.537e-002	0.2229	0.2229	0.2229
56	Accel. Gradient (kPa/m)	-8.253e-006	2.902e-005	2.903e-005	2.904e-005
57	Liq. Reynolds	1.847e+005 *	1.847e+005 *	1.847e+005 *	1.847e+005 *
58	Vap. Reynolds	9.864e+006 *	9.864e+006 *	9.864e+006 *	9.865e+006 *
59	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
60	Vapour Velocity (m/s)	2.918	2.918	2.918	2.919
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	2.529e+004	2.529e+004	2.531e+004	2.533e+004
64	Elevation (m)	-4.180	-3.900	-3.780	-3.660
65	Pressure (bar)	68.32	68.31	68.31	68.31
66	Temperature (C)	48.47	48.47	48.46	48.46
67	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
68	Heat Transferred (kJ/h-m)	491.9	491.8	491.8	491.6
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.4242	0.4242	0.2490	0.2490
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	7.647e-003	7.648e-003	6.317e-003	6.318e-003
18	Static Gradient (kPa/m)	0.2229	0.2228	1.319e-002	1.319e-002
19	Accel. Gradient (kPa/m)	2.904e-005	2.905e-005	2.458e-006	2.458e-006
20	Liq. Reynolds	1.847e+005 *	1.847e+005 *	1.847e+005 *	1.847e+005 *
21	Vap. Reynolds	9.865e+006 *	9.865e+006 *	9.865e+006 *	9.865e+006 *
22	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
23	Vapour Velocity (m/s)	2.919	2.920	2.920	2.920
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	2.535e+004	2.537e+004	2.539e+004	2.539e+004
27	Elevation (m)	-3.540	-3.420	-3.300	-3.260
28	Pressure (bar)	68.30	68.30	68.29	68.29
29	Temperature (C)	48.45	48.44	48.43	48.43
30	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
31	Heat Transferred (kJ/h-m)	491.4	491.3	491.1	491.1
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2490	0.2490	0.2490	0.2869
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	6.318e-003	6.318e-003	6.318e-003	7.916e-003
37	Static Gradient (kPa/m)	1.319e-002	1.319e-002	1.319e-002	4.649e-002
38	Accel. Gradient (kPa/m)	2.459e-006	2.459e-006	2.459e-006	6.860e-006
39	Liq. Reynolds	1.846e+005 *	1.846e+005 *	1.846e+005 *	1.846e+005 *
40	Vap. Reynolds	9.866e+006 *	9.866e+006 *	9.866e+006 *	9.866e+006 *
41	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
42	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	2.539e+004	2.539e+004	2.540e+004	2.540e+004
46	Elevation (m)	-3.220	-3.180	-3.140	-3.100
47	Pressure (bar)	68.29	68.29	68.29	68.29
48	Temperature (C)	48.43	48.43	48.43	48.43
49	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
50	Heat Transferred (kJ/h-m)	491.0	491.0	491.0	490.9
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2869	0.2869	0.2869	0.2869
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	7.916e-003	7.916e-003	7.916e-003	7.916e-003
56	Static Gradient (kPa/m)	4.649e-002	4.649e-002	4.649e-002	4.649e-002
57	Accel. Gradient (kPa/m)	6.861e-006	6.861e-006	6.861e-006	6.861e-006
58	Liq. Reynolds	1.846e+005 *	1.846e+005 *	1.846e+005 *	1.846e+005 *
59	Vap. Reynolds	9.866e+006 *	9.866e+006 *	9.866e+006 *	9.866e+006 *
60	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
61	Vapour Velocity (m/s)	2.920	2.920	2.920	2.920
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	2.540e+004	2.541e+004	2.541e+004	2.542e+004
65	Elevation (m)	-2.568	-2.036	-1.504	-0.9720
66	Pressure (bar)	68.27	68.25	68.23	68.21
67	Temperature (C)	48.42	48.41	48.41	48.40
68	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	490.9	490.7	490.5	490.3
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.5022	0.5021	0.5021	0.5020
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	7.544e-003	7.546e-003	7.548e-003	7.550e-003
19	Static Gradient (kPa/m)	0.3639	0.3638	0.3638	0.3637
20	Accel. Gradient (kPa/m)	4.687e-005	4.689e-005	4.691e-005	4.693e-005
21	Liq. Reynolds	1.846e+005 *	1.846e+005 *	1.845e+005 *	1.845e+005 *
22	Vap. Reynolds	9.867e+006 *	9.868e+006 *	9.868e+006 *	9.869e+006 *
23	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
24	Vapour Velocity (m/s)	2.921	2.922	2.923	2.924
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	2.543e+004	2.543e+004	2.544e+004	2.544e+004
28	Elevation (m)	-0.4400	-0.4000	-0.3600	-0.3200
29	Pressure (bar)	68.19	68.19	68.19	68.19
30	Temperature (C)	48.39	48.39	48.39	48.39
31	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
32	Heat Transferred (kJ/h-m)	490.1	489.8	489.8	489.7
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.5019	0.2534	0.2534	0.2534
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	7.552e-003	6.902e-003	6.902e-003	6.902e-003
38	Static Gradient (kPa/m)	0.3637	1.692e-002	1.692e-002	1.692e-002
39	Accel. Gradient (kPa/m)	4.695e-005	3.013e-006	3.013e-006	3.013e-006
40	Liq. Reynolds	1.845e+005 *	1.845e+005 *	1.845e+005 *	1.845e+005 *
41	Vap. Reynolds	9.870e+006 *	9.870e+006 *	9.870e+006 *	9.870e+006 *
42	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
43	Vapour Velocity (m/s)	2.925	2.925	2.925	2.925
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	2.545e+004	2.545e+004	2.546e+004	2.546e+004
47	Elevation (m)	-0.2800	-0.2400	-0.2160	-0.1920
48	Pressure (bar)	68.18	68.18	68.18	68.18
49	Temperature (C)	48.38	48.38	48.38	48.38
50	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
51	Heat Transferred (kJ/h-m)	489.7	489.6	489.6	489.5
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.2534	0.2534	0.2423	0.2423
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	6.902e-003	6.902e-003	7.059e-003	7.059e-003
57	Static Gradient (kPa/m)	1.692e-002	1.692e-002	8.182e-003	8.182e-003
58	Accel. Gradient (kPa/m)	3.013e-006	3.013e-006	1.928e-006	1.928e-006
59	Liq. Reynolds	1.845e+005 *	1.845e+005 *	1.845e+005 *	1.845e+005 *
60	Vap. Reynolds	9.870e+006 *	9.870e+006 *	9.870e+006 *	9.870e+006 *
61	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
62	Vapour Velocity (m/s)	2.925	2.925	2.925	2.925
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	2.547e+004	2.547e+004	2.548e+004	2.548e+004
66	Elevation (m)	-0.1680	-0.1440	-0.1200	-0.1220
67	Pressure (bar)	68.18	68.18	68.18	68.18
68	Temperature (C)	48.38	48.37	48.37	48.37
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 411 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
14	Heat Transferred (kJ/h-m)	489.5	489.4	489.4	489.4
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2423	0.2423	0.2423	0.1711
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	7.059e-003	7.059e-003	7.059e-003	9.041e-003
20	Static Gradient (kPa/m)	8.182e-003	8.182e-003	8.182e-003	-7.998e-004
21	Accel. Gradient (kPa/m)	1.928e-006	1.928e-006	1.928e-006	1.042e-006
22	Liq. Reynolds	1.844e+005 *	1.844e+005 *	1.844e+005 *	1.844e+005 *
23	Vap. Reynolds	9.870e+006 *	9.870e+006 *	9.870e+006 *	9.870e+006 *
24	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
25	Vapour Velocity (m/s)	2.925	2.925	2.925	2.925
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	2.549e+004	2.549e+004	2.550e+004	2.550e+004
29	Elevation (m)	-0.1240	-0.1260	-0.1280	-0.1300
30	Pressure (bar)	68.18	68.18	68.18	68.18
31	Temperature (C)	48.37	48.37	48.37	48.37
32	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
33	Heat Transferred (kJ/h-m)	489.4	489.3	489.3	489.3
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1711	0.1711	0.1711	0.1711
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	9.041e-003	9.041e-003	9.041e-003	9.041e-003
39	Static Gradient (kPa/m)	-7.998e-004	-7.998e-004	-7.998e-004	-7.998e-004
40	Accel. Gradient (kPa/m)	1.042e-006	1.042e-006	1.042e-006	1.043e-006
41	Liq. Reynolds	1.844e+005 *	1.844e+005 *	1.844e+005 *	1.844e+005 *
42	Vap. Reynolds	9.870e+006 *	9.870e+006 *	9.870e+006 *	9.870e+006 *
43	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
44	Vapour Velocity (m/s)	2.925	2.925	2.925	2.925
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	2.551e+004	2.551e+004	2.552e+004	2.552e+004
48	Elevation (m)	-0.1220	-0.1140	-0.1060	-9.800e-002
49	Pressure (bar)	68.18	68.18	68.18	68.17
50	Temperature (C)	48.36	48.36	48.36	48.36
51	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
52	Heat Transferred (kJ/h-m)	489.2	489.1	489.1	489.0
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2357	0.2357	0.2357	0.2357
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	7.876e-003	7.876e-003	7.876e-003	7.876e-003
58	Static Gradient (kPa/m)	3.206e-003	3.206e-003	3.206e-003	3.206e-003
59	Accel. Gradient (kPa/m)	1.402e-006	1.402e-006	1.402e-006	1.402e-006
60	Liq. Reynolds	1.844e+005 *	1.844e+005 *	1.844e+005 *	1.844e+005 *
61	Vap. Reynolds	9.870e+006 *	9.870e+006 *	9.870e+006 *	9.870e+006 *
62	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
63	Vapour Velocity (m/s)	2.925	2.925	2.925	2.925
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	2.553e+004	2.553e+004	2.554e+004	2.554e+004
67	Elevation (m)	-9.000e-002	-9.000e-002	-9.000e-002	-9.000e-002
68	Pressure (bar)	68.17	68.17	68.17	68.17
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	48.36	48.36	48.35	48.35
14	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
15	Heat Transferred (kJ/h-m)	489.0	489.0	488.9	488.9
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2357	0.2503	0.2503	0.2503
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	7.876e-003	6.168e-003	6.168e-003	6.168e-003
21	Static Gradient (kPa/m)	3.206e-003	0.0000	0.0000	0.0000
22	Accel. Gradient (kPa/m)	1.402e-006	7.803e-007	7.803e-007	7.803e-007
23	Liq. Reynolds	1.844e+005 *	1.844e+005 *	1.844e+005 *	1.844e+005 *
24	Vap. Reynolds	9.870e+006 *	9.870e+006 *	9.870e+006 *	9.870e+006 *
25	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
26	Vapour Velocity (m/s)	2.925	2.925	2.925	2.925
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	2.555e+004	2.555e+004	2.556e+004	2.556e+004
30	Elevation (m)	-9.000e-002	-9.000e-002	-8.200e-002	-7.400e-002
31	Pressure (bar)	68.17	68.17	68.17	68.17
32	Temperature (C)	48.35	48.35	48.35	48.35
33	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
34	Heat Transferred (kJ/h-m)	488.8	488.8	488.8	488.7
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2503	0.2503	0.2357	0.2357
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	6.168e-003	6.168e-003	7.876e-003	7.876e-003
40	Static Gradient (kPa/m)	0.0000	0.0000	3.206e-003	3.206e-003
41	Accel. Gradient (kPa/m)	7.803e-007	7.803e-007	1.402e-006	1.402e-006
42	Liq. Reynolds	1.844e+005 *	1.844e+005 *	1.844e+005 *	1.844e+005 *
43	Vap. Reynolds	9.870e+006 *	9.870e+006 *	9.870e+006 *	9.871e+006 *
44	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
45	Vapour Velocity (m/s)	2.925	2.925	2.925	2.925
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	2.557e+004	2.557e+004	2.558e+004	2.558e+004
49	Elevation (m)	-6.600e-002	-5.800e-002	-5.000e-002	-4.600e-002
50	Pressure (bar)	68.17	68.17	68.17	68.17
51	Temperature (C)	48.34	48.34	48.34	48.34
52	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
53	Heat Transferred (kJ/h-m)	488.7	488.6	488.6	488.5
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2357	0.2357	0.2357	0.2335
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	7.876e-003	7.876e-003	7.876e-003	8.163e-003
59	Static Gradient (kPa/m)	3.206e-003	3.206e-003	3.206e-003	1.592e-003
60	Accel. Gradient (kPa/m)	1.402e-006	1.402e-006	1.402e-006	1.234e-006
61	Liq. Reynolds	1.844e+005 *	1.844e+005 *	1.844e+005 *	1.844e+005 *
62	Vap. Reynolds	9.871e+006 *	9.871e+006 *	9.871e+006 *	9.871e+006 *
63	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
64	Vapour Velocity (m/s)	2.925	2.925	2.925	2.925
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	2.559e+004	2.559e+004	2.560e+004	2.560e+004
68	Elevation (m)	-4.200e-002	-3.800e-002	-3.400e-002	-3.000e-002
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Pressure (bar)	68.17	68.17	68.17	68.17					
14	Temperature (C)	48.34	48.34	48.34	48.33					
15	Vapour Fraction	0.8808	0.8808	0.8808	0.8808					
16	Heat Transferred (kJ/h-m)	488.5	488.5	488.4	488.4					
17	Flow Regime	Transition	Transition	Transition	Transition					
18	Water Flow Regime									
19	Liquid HoldUp	0.2335	0.2335	0.2335	0.2335					
20	Water HoldUp	---	---	---	---					
21	Friction Gradient (kPa/m)	8.163e-003	8.163e-003	8.163e-003	8.163e-003					
22	Static Gradient (kPa/m)	1.592e-003	1.592e-003	1.592e-003	1.592e-003					
23	Accel. Gradient (kPa/m)	1.234e-006	1.234e-006	1.234e-006	1.234e-006					
24	Liq. Reynolds	1.844e+005 *	1.844e+005 *	1.844e+005 *	1.844e+005 *					
25	Vap. Reynolds	9.871e+006 *	9.871e+006 *	9.871e+006 *	9.871e+006 *					
26	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
27	Vapour Velocity (m/s)	2.925	2.925	2.925	2.925					
28	Deposit Thickness (mm)	---	---	---	---					
29	Deposit Volume (m3)	---	---	---	---					
30	Length (m)	2.561e+004	2.561e+004	2.562e+004	2.563e+004					
31	Elevation (m)	-3.400e-002	-3.800e-002	-4.200e-002	-4.600e-002					
32	Pressure (bar)	68.17	68.17	68.17	68.17					
33	Temperature (C)	48.33	48.33	48.33	48.33					
34	Vapour Fraction	0.8808	0.8808	0.8808	0.8808					
35	Heat Transferred (kJ/h-m)	488.4	488.3	488.3	488.2					
36	Flow Regime	Transition	Transition	Transition	Transition					
37	Water Flow Regime									
38	Liquid HoldUp	0.1710	0.1710	0.1710	0.1710					
39	Water HoldUp	---	---	---	---					
40	Friction Gradient (kPa/m)	9.041e-003	9.041e-003	9.041e-003	9.041e-003					
41	Static Gradient (kPa/m)	-9.280e-004	-9.280e-004	-9.280e-004	-9.280e-004					
42	Accel. Gradient (kPa/m)	1.027e-006	1.027e-006	1.027e-006	1.027e-006					
43	Liq. Reynolds	1.844e+005 *	1.844e+005 *	1.844e+005 *	1.844e+005 *					
44	Vap. Reynolds	9.871e+006 *	9.871e+006 *	9.871e+006 *	9.871e+006 *					
45	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
46	Vapour Velocity (m/s)	2.925	2.925	2.925	2.925					
47	Deposit Thickness (mm)	---	---	---	---					
48	Deposit Volume (m3)	---	---	---	---					
49	Length (m)	2.563e+004	2.564e+004	2.564e+004	2.565e+004					
50	Elevation (m)	-5.000e-002	-5.200e-002	-5.400e-002	-5.600e-002					
51	Pressure (bar)	68.16	68.16	68.16	68.16					
52	Temperature (C)	48.32	48.32	48.32	48.32					
53	Vapour Fraction	0.8808	0.8808	0.8808	0.8808					
54	Heat Transferred (kJ/h-m)	488.1	488.1	488.1	488.0					
55	Flow Regime	Transition	Transition	Transition	Transition					
56	Water Flow Regime									
57	Liquid HoldUp	0.1710	0.1711	0.1711	0.1711					
58	Water HoldUp	---	---	---	---					
59	Friction Gradient (kPa/m)	9.041e-003	9.040e-003	9.040e-003	9.040e-003					
60	Static Gradient (kPa/m)	-9.280e-004	-6.401e-004	-6.401e-004	-6.401e-004					
61	Accel. Gradient (kPa/m)	1.027e-006	1.063e-006	1.063e-006	1.063e-006					
62	Liq. Reynolds	1.844e+005 *	1.844e+005 *	1.844e+005 *	1.844e+005 *					
63	Vap. Reynolds	9.871e+006 *	9.871e+006 *	9.871e+006 *	9.871e+006 *					
64	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
65	Vapour Velocity (m/s)	2.925	2.925	2.925	2.925					
66	Deposit Thickness (mm)	---	---	---	---					
67	Deposit Volume (m3)	---	---	---	---					
68	Length (m)	2.565e+004	2.566e+004	2.566e+004	2.567e+004					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:13:43 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Elevation (m)	-5.800e-002	-6.000e-002	-6.200e-002	-6.400e-002		
14	Pressure (bar)	68.16	68.16	68.16	68.16		
15	Temperature (C)	48.32	48.32	48.31	48.31		
16	Vapour Fraction	0.8808	0.8808	0.8808	0.8808		
17	Heat Transferred (kJ/h-m)	488.0	487.9	487.9	487.9		
18	Flow Regime	Transition	Transition	Transition	Transition		
19	Water Flow Regime						
20	Liquid HoldUp	0.1711	0.1711	0.1711	0.1711		
21	Water HoldUp	---	---	---	---		
22	Friction Gradient (kPa/m)	9.040e-003	9.040e-003	9.040e-003	9.040e-003		
23	Static Gradient (kPa/m)	-6.401e-004	-6.401e-004	-6.401e-004	-6.401e-004		
24	Accel. Gradient (kPa/m)	1.063e-006	1.063e-006	1.063e-006	1.063e-006		
25	Liq. Reynolds	1.844e+005 *	1.844e+005 *	1.844e+005 *	1.844e+005 *		
26	Vap. Reynolds	9.871e+006 *	9.871e+006 *	9.871e+006 *	9.871e+006 *		
27	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
28	Vapour Velocity (m/s)	2.925	2.925	2.925	2.925		
29	Deposit Thickness (mm)	---	---	---	---		
30	Deposit Volume (m3)	---	---	---	---		
31	Length (m)	2.567e+004	2.568e+004	2.568e+004	2.569e+004		
32	Elevation (m)	-6.600e-002	-6.800e-002	-7.000e-002	-7.600e-002		
33	Pressure (bar)	68.16	68.16	68.16	68.16		
34	Temperature (C)	48.31	48.31	48.31	48.31		
35	Vapour Fraction	0.8808	0.8808	0.8808	0.8808		
36	Heat Transferred (kJ/h-m)	487.8	487.8	487.7	487.7		
37	Flow Regime	Transition	Transition	Transition	Transition		
38	Water Flow Regime						
39	Liquid HoldUp	0.1711	0.1711	0.1711	0.1706		
40	Water HoldUp	---	---	---	---		
41	Friction Gradient (kPa/m)	9.040e-003	9.040e-003	9.040e-003	9.047e-003		
42	Static Gradient (kPa/m)	-6.401e-004	-6.401e-004	-6.401e-004	-1.916e-003		
43	Accel. Gradient (kPa/m)	1.063e-006	1.063e-006	1.063e-006	9.024e-007		
44	Liq. Reynolds	1.844e+005 *	1.844e+005 *	1.844e+005 *	1.844e+005 *		
45	Vap. Reynolds	9.871e+006 *	9.871e+006 *	9.871e+006 *	9.871e+006 *		
46	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
47	Vapour Velocity (m/s)	2.925	2.925	2.925	2.925		
48	Deposit Thickness (mm)	---	---	---	---		
49	Deposit Volume (m3)	---	---	---	---		
50	Length (m)	2.569e+004	2.570e+004	2.570e+004	2.571e+004		
51	Elevation (m)	-8.200e-002	-8.800e-002	-9.400e-002	-0.1000		
52	Pressure (bar)	68.16	68.16	68.16	68.16		
53	Temperature (C)	48.31	48.30	48.30	48.30		
54	Vapour Fraction	0.8808	0.8808	0.8808	0.8808		
55	Heat Transferred (kJ/h-m)	487.7	487.6	487.6	487.5		
56	Flow Regime	Transition	Transition	Transition	Transition		
57	Water Flow Regime						
58	Liquid HoldUp	0.1706	0.1706	0.1706	0.1706		
59	Water HoldUp	---	---	---	---		
60	Friction Gradient (kPa/m)	9.047e-003	9.047e-003	9.047e-003	9.047e-003		
61	Static Gradient (kPa/m)	-1.916e-003	-1.916e-003	-1.916e-003	-1.916e-003		
62	Accel. Gradient (kPa/m)	9.024e-007	9.024e-007	9.024e-007	9.024e-007		
63	Liq. Reynolds	1.844e+005 *	1.844e+005 *	1.844e+005 *	1.844e+005 *		
64	Vap. Reynolds	9.871e+006 *	9.871e+006 *	9.871e+006 *	9.871e+006 *		
65	Liquid Velocity (m/s)	0.1979	0.1980	0.1980	0.1980		
66	Vapour Velocity (m/s)	2.925	2.925	2.925	2.925		
67	Deposit Thickness (mm)	---	---	---	---		
68	Deposit Volume (m3)	---	---	---	---		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Length (m)	2.571e+004	2.571e+004	2.571e+004	2.571e+004					
14	Elevation (m)	-0.1200	-0.1400	-0.1600	-0.1800					
15	Pressure (bar)	68.16	68.16	68.16	68.16					
16	Temperature (C)	48.30	48.30	48.30	48.30					
17	Vapour Fraction	0.8808	0.8808	0.8808	0.8808					
18	Heat Transferred (kJ/h-m)	487.6	487.5	487.5	487.5					
19	Flow Regime	Transition	Transition	Transition	Transition					
20	Water Flow Regime									
21	Liquid HoldUp	0.1566	0.1566	0.1566	0.1566					
22	Water HoldUp	---	---	---	---					
23	Friction Gradient (kPa/m)	9.256e-003	9.256e-003	9.256e-003	9.256e-003					
24	Static Gradient (kPa/m)	-3.019e-002	-3.019e-002	-3.019e-002	-3.019e-002					
25	Accel. Gradient (kPa/m)	-2.649e-006	-2.649e-006	-2.649e-006	-2.649e-006					
26	Liq. Reynolds	1.844e+005 *	1.844e+005 *	1.844e+005 *	1.844e+005 *					
27	Vap. Reynolds	9.871e+006 *	9.871e+006 *	9.871e+006 *	9.871e+006 *					
28	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
29	Vapour Velocity (m/s)	2.925	2.925	2.925	2.925					
30	Deposit Thickness (mm)	---	---	---	---					
31	Deposit Volume (m3)	---	---	---	---					
32	Length (m)	2.571e+004	2.572e+004	2.572e+004	2.572e+004					
33	Elevation (m)	-0.2000	-0.3200	-0.4400	-0.5600					
34	Pressure (bar)	68.16	68.16	68.16	68.16					
35	Temperature (C)	48.30	48.30	48.30	48.30					
36	Vapour Fraction	0.8808	0.8808	0.8808	0.8808					
37	Heat Transferred (kJ/h-m)	487.5	487.8	487.8	487.8					
38	Flow Regime	Transition	Transition	Transition	Transition					
39	Water Flow Regime									
40	Liquid HoldUp	0.1566	8.447e-002	8.447e-002	8.447e-002					
41	Water HoldUp	---	---	---	---					
42	Friction Gradient (kPa/m)	9.256e-003	1.200e-002	1.200e-002	1.200e-002					
43	Static Gradient (kPa/m)	-3.019e-002	-0.1269	-0.1269	-0.1269					
44	Accel. Gradient (kPa/m)	-2.649e-006	-1.454e-005	-1.454e-005	-1.454e-005					
45	Liq. Reynolds	1.844e+005 *	1.844e+005 *	1.844e+005 *	1.844e+005 *					
46	Vap. Reynolds	9.871e+006 *	9.871e+006 *	9.871e+006 *	9.871e+006 *					
47	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
48	Vapour Velocity (m/s)	2.925	2.925	2.925	2.925					
49	Deposit Thickness (mm)	---	---	---	---					
50	Deposit Volume (m3)	---	---	---	---					
51	Length (m)	2.572e+004	2.572e+004	2.572e+004	2.573e+004					
52	Elevation (m)	-0.6800	-0.8000	-1.340	-1.880					
53	Pressure (bar)	68.16	68.17	68.17	68.17					
54	Temperature (C)	48.30	48.30	48.30	48.31					
55	Vapour Fraction	0.8808	0.8808	0.8808	0.8808					
56	Heat Transferred (kJ/h-m)	487.8	487.8	488.9	489.0					
57	Flow Regime	Transition	Transition	Transition	Transition					
58	Water Flow Regime									
59	Liquid HoldUp	8.447e-002	8.447e-002	4.435e-002	4.435e-002					
60	Water HoldUp	---	---	---	---					
61	Friction Gradient (kPa/m)	1.200e-002	1.200e-002	1.797e-002	1.797e-002					
62	Static Gradient (kPa/m)	-0.1269	-0.1269	-0.1452	-0.1452					
63	Accel. Gradient (kPa/m)	-1.454e-005	-1.454e-005	-1.610e-005	-1.610e-005					
64	Liq. Reynolds	1.844e+005 *	1.844e+005 *	1.844e+005 *	1.844e+005 *					
65	Vap. Reynolds	9.871e+006 *	9.871e+006 *	9.871e+006 *	9.871e+006 *					
66	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
67	Vapour Velocity (m/s)	2.924	2.924	2.924	2.924					
68	Deposit Thickness (mm)	---	---	---	---					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3							
4			Unit Set: Mixed				
5			Date/Time: Mon Jun 13 15:13:43 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	2.573e+004	2.573e+004	2.573e+004	2.574e+004		
15	Elevation (m)	-2.420	-2.960	-3.500	-3.760		
16	Pressure (bar)	68.18	68.18	68.18	68.19		
17	Temperature (C)	48.31	48.31	48.31	48.31		
18	Vapour Fraction	0.8808	0.8808	0.8808	0.8808		
19	Heat Transferred (kJ/h-m)	489.0	489.1	489.1	487.9		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	4.435e-002	4.435e-002	4.435e-002	0.1331		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	1.797e-002	1.796e-002	1.796e-002	9.744e-003		
25	Static Gradient (kPa/m)	-0.1452	-0.1452	-0.1453	-7.085e-002		
26	Accel. Gradient (kPa/m)	-1.610e-005	-1.610e-005	-1.609e-005	-7.726e-006		
27	Liq. Reynolds	1.844e+005 *	1.844e+005 *	1.844e+005 *	1.844e+005 *		
28	Vap. Reynolds	9.870e+006 *	9.870e+006 *	9.870e+006 *	9.870e+006 *		
29	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
30	Vapour Velocity (m/s)	2.924	2.924	2.924	2.923		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	2.574e+004	2.575e+004	2.575e+004	2.576e+004		
34	Elevation (m)	-4.020	-4.280	-4.540	-4.800		
35	Pressure (bar)	68.19	68.19	68.20	68.20		
36	Temperature (C)	48.31	48.31	48.31	48.31		
37	Vapour Fraction	0.8808	0.8808	0.8808	0.8808		
38	Heat Transferred (kJ/h-m)	487.9	487.9	487.9	487.9		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.1331	0.1331	0.1332	0.1332		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	9.744e-003	9.743e-003	9.743e-003	9.742e-003		
44	Static Gradient (kPa/m)	-7.086e-002	-7.086e-002	-7.086e-002	-7.086e-002		
45	Accel. Gradient (kPa/m)	-7.726e-006	-7.725e-006	-7.725e-006	-7.724e-006		
46	Liq. Reynolds	1.844e+005 *	1.844e+005 *	1.844e+005 *	1.844e+005 *		
47	Vap. Reynolds	9.870e+006 *	9.870e+006 *	9.870e+006 *	9.870e+006 *		
48	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
49	Vapour Velocity (m/s)	2.923	2.923	2.923	2.923		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	2.576e+004	2.577e+004	2.577e+004	2.578e+004		
53	Elevation (m)	-3.980	-3.160	-2.340	-1.520		
54	Pressure (bar)	68.16	68.12	68.08	68.04		
55	Temperature (C)	48.30	48.29	48.27	48.26		
56	Vapour Fraction	0.8808	0.8808	0.8808	0.8808		
57	Heat Transferred (kJ/h-m)	488.5	488.2	487.8	487.5		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.6682	0.6679	0.6677	0.6675		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	7.409e-003	7.413e-003	7.417e-003	7.421e-003		
63	Static Gradient (kPa/m)	0.7725	0.7723	0.7721	0.7719		
64	Accel. Gradient (kPa/m)	9.868e-005	9.877e-005	9.886e-005	9.895e-005		
65	Liq. Reynolds	1.844e+005 *	1.843e+005 *	1.843e+005 *	1.842e+005 *		
66	Vap. Reynolds	9.871e+006 *	9.872e+006 *	9.874e+006 *	9.875e+006 *		
67	Liquid Velocity (m/s)	0.1980	0.1979	0.1979	0.1979		
68	Vapour Velocity (m/s)	2.925	2.926	2.928	2.930		
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
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5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Thickness	(mm)	---	---	---	---	
14	Deposit Volume	(m3)	---	---	---	---	
15	Length	(m)	2.578e+004	2.579e+004	2.579e+004	2.580e+004	
16	Elevation	(m)	-0.7000	-0.6200	-0.5400	-0.4600	
17	Pressure	(bar)	68.00	68.00	68.00	68.00	
18	Temperature	(C)	48.25	48.24	48.24	48.24	
19	Vapour Fraction		0.8809	0.8809	0.8809	0.8809	
20	Heat Transferred	(kJ/h-m)	487.1	486.0	486.0	485.9	
21	Flow Regime		Transition	Transition	Transition	Transition	
22	Water Flow Regime						
23	Liquid HoldUp		0.6673	0.2752	0.2752	0.2752	
24	Water HoldUp		---	---	---	---	
25	Friction Gradient	(kPa/m)	7.425e-003	7.933e-003	7.933e-003	7.933e-003	
26	Static Gradient	(kPa/m)	0.7717	3.601e-002	3.601e-002	3.601e-002	
27	Accel. Gradient	(kPa/m)	9.904e-005	5.586e-006	5.586e-006	5.587e-006	
28	Liq. Reynolds		1.842e+005 *	1.841e+005 *	1.841e+005 *	1.841e+005 *	
29	Vap. Reynolds		9.877e+006 *	9.877e+006 *	9.877e+006 *	9.877e+006 *	
30	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
31	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932	
32	Deposit Thickness	(mm)	---	---	---	---	
33	Deposit Volume	(m3)	---	---	---	---	
34	Length	(m)	2.580e+004	2.581e+004	2.581e+004	2.582e+004	
35	Elevation	(m)	-0.3800	-0.3000	-0.2840	-0.2680	
36	Pressure	(bar)	68.00	67.99	67.99	67.99	
37	Temperature	(C)	48.24	48.24	48.23	48.23	
38	Vapour Fraction		0.8809	0.8809	0.8809	0.8809	
39	Heat Transferred	(kJ/h-m)	485.8	485.8	485.7	485.7	
40	Flow Regime		Transition	Transition	Transition	Transition	
41	Water Flow Regime						
42	Liquid HoldUp		0.2752	0.2751	0.2399	0.2399	
43	Water HoldUp		---	---	---	---	
44	Friction Gradient	(kPa/m)	7.933e-003	7.933e-003	7.332e-003	7.332e-003	
45	Static Gradient	(kPa/m)	3.601e-002	3.601e-002	6.592e-003	6.592e-003	
46	Accel. Gradient	(kPa/m)	5.587e-006	5.587e-006	1.771e-006	1.771e-006	
47	Liq. Reynolds		1.841e+005 *	1.841e+005 *	1.841e+005 *	1.841e+005 *	
48	Vap. Reynolds		9.877e+006 *	9.877e+006 *	9.877e+006 *	9.877e+006 *	
49	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
50	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932	
51	Deposit Thickness	(mm)	---	---	---	---	
52	Deposit Volume	(m3)	---	---	---	---	
53	Length	(m)	2.582e+004	2.583e+004	2.583e+004	2.584e+004	
54	Elevation	(m)	-0.2520	-0.2360	-0.2200	-0.1880	
55	Pressure	(bar)	67.99	67.99	67.99	67.99	
56	Temperature	(C)	48.23	48.23	48.23	48.23	
57	Vapour Fraction		0.8809	0.8809	0.8809	0.8808	
58	Heat Transferred	(kJ/h-m)	485.6	485.6	485.6	485.5	
59	Flow Regime		Transition	Transition	Transition	Transition	
60	Water Flow Regime						
61	Liquid HoldUp		0.2399	0.2399	0.2399	0.2486	
62	Water HoldUp		---	---	---	---	
63	Friction Gradient	(kPa/m)	7.332e-003	7.333e-003	7.333e-003	6.329e-003	
64	Static Gradient	(kPa/m)	6.592e-003	6.592e-003	6.592e-003	1.334e-002	
65	Accel. Gradient	(kPa/m)	1.771e-006	1.771e-006	1.771e-006	2.501e-006	
66	Liq. Reynolds		1.841e+005 *	1.841e+005 *	1.841e+005 *	1.841e+005 *	
67	Vap. Reynolds		9.877e+006 *	9.877e+006 *	9.877e+006 *	9.877e+006 *	
68	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
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5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	2.584e+004	2.585e+004	2.585e+004	2.586e+004	
17	Elevation	(m)	-0.1560	-0.1240	-9.200e-002	-6.000e-002	
18	Pressure	(bar)	67.99	67.99	67.99	67.99	
19	Temperature	(C)	48.22	48.22	48.22	48.22	
20	Vapour Fraction		0.8808	0.8808	0.8808	0.8808	
21	Heat Transferred	(kJ/h-m)	485.5	485.4	485.4	485.3	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2486	0.2486	0.2486	0.2486	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	6.329e-003	6.329e-003	6.329e-003	6.329e-003	
27	Static Gradient	(kPa/m)	1.334e-002	1.334e-002	1.334e-002	1.334e-002	
28	Accel. Gradient	(kPa/m)	2.501e-006	2.501e-006	2.501e-006	2.501e-006	
29	Liq. Reynolds		1.841e+005 *	1.841e+005 *	1.841e+005 *	1.841e+005 *	
30	Vap. Reynolds		9.877e+006 *	9.877e+006 *	9.877e+006 *	9.877e+006 *	
31	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
32	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	2.586e+004	2.587e+004	2.587e+004	2.588e+004	
36	Elevation	(m)	-6.200e-002	-6.400e-002	-6.600e-002	-6.800e-002	
37	Pressure	(bar)	67.99	67.98	67.98	67.98	
38	Temperature	(C)	48.22	48.22	48.21	48.21	
39	Vapour Fraction		0.8808	0.8808	0.8808	0.8808	
40	Heat Transferred	(kJ/h-m)	485.3	485.3	485.2	485.2	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.1709	0.1709	0.1709	0.1709	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	9.061e-003	9.061e-003	9.061e-003	9.061e-003	
46	Static Gradient	(kPa/m)	-6.391e-004	-6.391e-004	-6.391e-004	-6.391e-004	
47	Accel. Gradient	(kPa/m)	1.071e-006	1.071e-006	1.071e-006	1.071e-006	
48	Liq. Reynolds		1.841e+005 *	1.841e+005 *	1.841e+005 *	1.841e+005 *	
49	Vap. Reynolds		9.877e+006 *	9.877e+006 *	9.877e+006 *	9.877e+006 *	
50	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
51	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	2.588e+004	2.589e+004	2.589e+004	2.590e+004	
55	Elevation	(m)	-7.000e-002	-0.1100	-0.1500	-0.1900	
56	Pressure	(bar)	67.98	67.98	67.98	67.98	
57	Temperature	(C)	48.21	48.21	48.21	48.21	
58	Vapour Fraction		0.8808	0.8808	0.8808	0.8808	
59	Heat Transferred	(kJ/h-m)	485.2	485.1	485.1	485.1	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.1709	0.1653	0.1653	0.1653	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	9.061e-003	9.139e-003	9.139e-003	9.139e-003	
65	Static Gradient	(kPa/m)	-6.391e-004	-1.250e-002	-1.250e-002	-1.250e-002	
66	Accel. Gradient	(kPa/m)	1.071e-006	-4.275e-007	-4.275e-007	-4.275e-007	
67	Liq. Reynolds		1.841e+005 *	1.841e+005 *	1.841e+005 *	1.841e+005 *	
68	Vap. Reynolds		9.878e+006 *	9.87e+006 *	9.877e+006 *	9.877e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3							
4							
5			Unit Set: Mixed				
6			Date/Time: Mon Jun 13 15:13:43 2022				
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9							
10	PROFILES						
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
14	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	2.590e+004	2.591e+004	2.591e+004	2.592e+004	
18	Elevation	(m)	-0.2300	-0.2700	-0.3400	-0.4100	
19	Pressure	(bar)	67.98	67.98	67.98	67.99	
20	Temperature	(C)	48.21	48.21	48.20	48.20	
21	Vapour Fraction		0.8808	0.8808	0.8808	0.8808	
22	Heat Transferred	(kJ/h-m)	485.0	485.0	485.0	485.0	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.1653	0.1653	0.1608	0.1608	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	9.139e-003	9.139e-003	9.206e-003	9.206e-003	
28	Static Gradient	(kPa/m)	-1.250e-002	-1.250e-002	-2.149e-002	-2.149e-002	
29	Accel. Gradient	(kPa/m)	-4.275e-007	-4.275e-007	-1.562e-006	-1.562e-006	
30	Liq. Reynolds		1.841e+005 *	1.841e+005 *	1.841e+005 *	1.841e+005 *	
31	Vap. Reynolds		9.877e+006 *	9.877e+006 *	9.877e+006 *	9.877e+006 *	
32	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
33	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	2.592e+004	2.593e+004	2.593e+004	2.594e+004	
37	Elevation	(m)	-0.4800	-0.5500	-0.6200	-0.6500	
38	Pressure	(bar)	67.99	67.99	67.99	67.99	
39	Temperature	(C)	48.20	48.20	48.20	48.20	
40	Vapour Fraction		0.8808	0.8808	0.8808	0.8808	
41	Heat Transferred	(kJ/h-m)	484.9	484.9	484.9	484.8	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.1608	0.1608	0.1608	0.1667	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	9.206e-003	9.205e-003	9.205e-003	9.117e-003	
47	Static Gradient	(kPa/m)	-2.149e-002	-2.149e-002	-2.149e-002	-9.431e-003	
48	Accel. Gradient	(kPa/m)	-1.562e-006	-1.562e-006	-1.562e-006	-4.001e-008	
49	Liq. Reynolds		1.841e+005 *	1.841e+005 *	1.841e+005 *	1.841e+005 *	
50	Vap. Reynolds		9.877e+006 *	9.877e+006 *	9.877e+006 *	9.877e+006 *	
51	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
52	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	2.594e+004	2.595e+004	2.595e+004	2.596e+004	
56	Elevation	(m)	-0.6800	-0.7100	-0.7400	-0.7700	
57	Pressure	(bar)	67.99	67.99	67.99	67.99	
58	Temperature	(C)	48.20	48.20	48.19	48.19	
59	Vapour Fraction		0.8808	0.8808	0.8808	0.8808	
60	Heat Transferred	(kJ/h-m)	484.8	484.8	484.7	484.7	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.1668	0.1668	0.1668	0.1668	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	9.117e-003	9.117e-003	9.117e-003	9.117e-003	
66	Static Gradient	(kPa/m)	-9.431e-003	-9.432e-003	-9.432e-003	-9.432e-003	
67	Accel. Gradient	(kPa/m)	-4.002e-008	-4.003e-008	-4.005e-008	-4.006e-008	
68	Liq. Reynolds		1.841e+005 *	1.841e+005 *	1.841e+005 *	1.841e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.877e+006 *	9.877e+006 *	9.877e+006 *	9.877e+006 *
14	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
15	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	2.596e+004	2.597e+004	2.597e+004	2.598e+004
19	Elevation (m)	-0.6400	-0.5100	-0.3800	-0.2500
20	Pressure (bar)	67.98	67.98	67.98	67.97
21	Temperature (C)	48.19	48.19	48.19	48.18
22	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
23	Heat Transferred (kJ/h-m)	484.6	484.5	484.4	484.4
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.3028	0.3028	0.3028	0.3027
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	7.923e-003	7.923e-003	7.924e-003	7.924e-003
29	Static Gradient (kPa/m)	6.301e-002	6.301e-002	6.301e-002	6.301e-002
30	Accel. Gradient (kPa/m)	9.020e-006	9.021e-006	9.022e-006	9.023e-006
31	Liq. Reynolds	1.841e+005 *	1.841e+005 *	1.841e+005 *	1.841e+005 *
32	Vap. Reynolds	9.878e+006 *	9.878e+006 *	9.878e+006 *	9.878e+006 *
33	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
34	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	2.598e+004	2.599e+004	2.599e+004	2.600e+004
38	Elevation (m)	-0.1200	-0.1160	-0.1120	-0.1080
39	Pressure (bar)	67.97	67.97	67.97	67.97
40	Temperature (C)	48.18	48.18	48.18	48.18
41	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
42	Heat Transferred (kJ/h-m)	484.3	484.2	484.2	484.2
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.3027	0.2331	0.2331	0.2331
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	7.924e-003	8.192e-003	8.192e-003	8.192e-003
48	Static Gradient (kPa/m)	6.300e-002	1.590e-003	1.590e-003	1.590e-003
49	Accel. Gradient (kPa/m)	9.023e-006	1.244e-006	1.244e-006	1.244e-006
50	Liq. Reynolds	1.841e+005 *	1.841e+005 *	1.841e+005 *	1.841e+005 *
51	Vap. Reynolds	9.878e+006 *	9.878e+006 *	9.878e+006 *	9.878e+006 *
52	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
53	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	2.600e+004	2.601e+004	2.601e+004	2.602e+004
57	Elevation (m)	-0.1040	-0.1000	-0.1260	-0.1520
58	Pressure (bar)	67.97	67.97	67.97	67.97
59	Temperature (C)	48.17	48.17	48.17	48.17
60	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
61	Heat Transferred (kJ/h-m)	484.1	484.1	484.1	484.1
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2331	0.2331	0.1673	0.1673
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	8.192e-003	8.192e-003	9.111e-003	9.111e-003
67	Static Gradient (kPa/m)	1.590e-003	1.590e-003	-8.323e-003	-8.323e-003
68	Accel. Gradient (kPa/m)	1.244e-006	1.245e-006	1.002e-007	1.002e-007
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		1.841e+005 *	1.840e+005 *	1.840e+005 *	1.840e+005 *	
14	Vap. Reynolds		9.878e+006 *	9.878e+006 *	9.878e+006 *	9.878e+006 *	
15	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979	
16	Vapour Velocity (m/s)		2.932	2.932	2.932	2.932	
17	Deposit Thickness (mm)		---	---	---	---	
18	Deposit Volume (m3)		---	---	---	---	
19	Length (m)		2.602e+004	2.603e+004	2.603e+004	2.604e+004	
20	Elevation (m)		-0.1780	-0.2040	-0.2300	-0.1920	
21	Pressure (bar)		67.97	67.97	67.97	67.97	
22	Temperature (C)		48.17	48.17	48.17	48.16	
23	Vapour Fraction		0.8808	0.8808	0.8808	0.8808	
24	Heat Transferred (kJ/h-m)		484.0	484.0	483.9	483.9	
25	Flow Regime		Transition	Transition	Transition	Transition	
26	Water Flow Regime						
27	Liquid HoldUp		0.1673	0.1673	0.1673	0.2519	
28	Water HoldUp		---	---	---	---	
29	Friction Gradient (kPa/m)		9.111e-003	9.111e-003	9.111e-003	6.378e-003	
30	Static Gradient (kPa/m)		-8.323e-003	-8.323e-003	-8.324e-003	1.600e-002	
31	Accel. Gradient (kPa/m)		1.002e-007	1.002e-007	1.002e-007	2.847e-006	
32	Liq. Reynolds		1.840e+005 *	1.840e+005 *	1.840e+005 *	1.840e+005 *	
33	Vap. Reynolds		9.878e+006 *	9.878e+006 *	9.878e+006 *	9.878e+006 *	
34	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979	
35	Vapour Velocity (m/s)		2.932	2.932	2.932	2.932	
36	Deposit Thickness (mm)		---	---	---	---	
37	Deposit Volume (m3)		---	---	---	---	
38	Length (m)		2.604e+004	2.605e+004	2.605e+004	2.606e+004	
39	Elevation (m)		-0.1540	-0.1160	-7.800e-002	-4.000e-002	
40	Pressure (bar)		67.96	67.96	67.96	67.96	
41	Temperature (C)		48.16	48.16	48.16	48.16	
42	Vapour Fraction		0.8808	0.8808	0.8808	0.8808	
43	Heat Transferred (kJ/h-m)		483.8	483.8	483.7	483.7	
44	Flow Regime		Transition	Transition	Transition	Transition	
45	Water Flow Regime						
46	Liquid HoldUp		0.2519	0.2519	0.2519	0.2519	
47	Water HoldUp		---	---	---	---	
48	Friction Gradient (kPa/m)		6.378e-003	6.378e-003	6.378e-003	6.377e-003	
49	Static Gradient (kPa/m)		1.600e-002	1.600e-002	1.600e-002	1.600e-002	
50	Accel. Gradient (kPa/m)		2.847e-006	2.847e-006	2.847e-006	2.847e-006	
51	Liq. Reynolds		1.840e+005 *	1.840e+005 *	1.840e+005 *	1.840e+005 *	
52	Vap. Reynolds		9.878e+006 *	9.878e+006 *	9.878e+006 *	9.878e+006 *	
53	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979	
54	Vapour Velocity (m/s)		2.932	2.932	2.932	2.932	
55	Deposit Thickness (mm)		---	---	---	---	
56	Deposit Volume (m3)		---	---	---	---	
57	Length (m)		2.606e+004	2.607e+004	2.607e+004	2.608e+004	
58	Elevation (m)		-1.800e-002	4.000e-003	2.600e-002	4.800e-002	
59	Pressure (bar)		67.96	67.96	67.96	67.96	
60	Temperature (C)		48.15	48.15	48.15	48.15	
61	Vapour Fraction		0.8808	0.8808	0.8808	0.8808	
62	Heat Transferred (kJ/h-m)		483.6	483.6	483.5	483.5	
63	Flow Regime		Transition	Transition	Transition	Transition	
64	Water Flow Regime						
65	Liquid HoldUp		0.2430	0.2430	0.2430	0.2430	
66	Water HoldUp		---	---	---	---	
67	Friction Gradient (kPa/m)		6.955e-003	6.955e-003	6.955e-003	6.955e-003	
68	Static Gradient (kPa/m)		9.018e-003	9.018e-003	9.018e-003	9.018e-003	
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	2.032e-006	2.032e-006	2.032e-006	2.032e-006
14	Liq. Reynolds	1.840e+005 *	1.840e+005 *	1.840e+005 *	1.840e+005 *
15	Vap. Reynolds	9.879e+006 *	9.879e+006 *	9.879e+006 *	9.879e+006 *
16	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
17	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	2.608e+004	2.609e+004	2.609e+004	2.610e+004
21	Elevation (m)	7.000e-002	4.000e-002	1.000e-002	-2.000e-002
22	Pressure (bar)	67.96	67.96	67.96	67.96
23	Temperature (C)	48.15	48.15	48.15	48.14
24	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
25	Heat Transferred (kJ/h-m)	483.4	483.4	483.4	483.4
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.2430	0.1667	0.1667	0.1667
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	6.955e-003	9.119e-003	9.119e-003	9.119e-003
31	Static Gradient (kPa/m)	9.018e-003	-9.430e-003	-9.430e-003	-9.430e-003
32	Accel. Gradient (kPa/m)	2.032e-006	-3.960e-008	-3.962e-008	-3.963e-008
33	Liq. Reynolds	1.840e+005 *	1.840e+005 *	1.840e+005 *	1.840e+005 *
34	Vap. Reynolds	9.879e+006 *	9.879e+006 *	9.879e+006 *	9.879e+006 *
35	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
36	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	2.610e+004	2.611e+004	2.611e+004	2.612e+004
40	Elevation (m)	-5.000e-002	-8.000e-002	-7.800e-002	-7.600e-002
41	Pressure (bar)	67.96	67.96	67.96	67.96
42	Temperature (C)	48.14	48.14	48.14	48.14
43	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
44	Heat Transferred (kJ/h-m)	483.3	483.3	483.2	483.2
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.1667	0.1667	0.2320	0.2320
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	9.119e-003	9.119e-003	8.340e-003	8.340e-003
50	Static Gradient (kPa/m)	-9.430e-003	-9.430e-003	7.921e-004	7.921e-004
51	Accel. Gradient (kPa/m)	-3.965e-008	-3.966e-008	1.162e-006	1.162e-006
52	Liq. Reynolds	1.840e+005 *	1.840e+005 *	1.840e+005 *	1.840e+005 *
53	Vap. Reynolds	9.879e+006 *	9.879e+006 *	9.879e+006 *	9.879e+006 *
54	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
55	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	2.612e+004	2.613e+004	2.613e+004	2.614e+004
59	Elevation (m)	-7.400e-002	-7.200e-002	-7.000e-002	-8.400e-002
60	Pressure (bar)	67.96	67.96	67.96	67.96
61	Temperature (C)	48.14	48.14	48.13	48.13
62	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
63	Heat Transferred (kJ/h-m)	483.1	483.1	483.1	483.1
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.2320	0.2320	0.2320	0.1691
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	8.340e-003	8.340e-003	8.340e-003	9.086e-003
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3	 Epic Atlantic Limited		Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	7.921e-004	7.921e-004	7.921e-004	-4.442e-003
14	Accel. Gradient	(kPa/m)	1.162e-006	1.162e-006	1.162e-006	5.908e-007
15	Liq. Reynolds		1.840e+005 *	1.840e+005 *	1.840e+005 *	1.840e+005 *
16	Vap. Reynolds		9.879e+006 *	9.879e+006 *	9.879e+006 *	9.879e+006 *
17	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
18	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	2.614e+004	2.615e+004	2.615e+004	2.616e+004
22	Elevation	(m)	-9.800e-002	-0.1120	-0.1260	-0.1400
23	Pressure	(bar)	67.95	67.95	67.95	67.95
24	Temperature	(C)	48.13	48.13	48.13	48.13
25	Vapour Fraction		0.8808	0.8808	0.8808	0.8808
26	Heat Transferred	(kJ/h-m)	483.0	483.0	482.9	482.9
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.1691	0.1691	0.1691	0.1691
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	9.086e-003	9.086e-003	9.086e-003	9.086e-003
32	Static Gradient	(kPa/m)	-4.442e-003	-4.442e-003	-4.442e-003	-4.442e-003
33	Accel. Gradient	(kPa/m)	5.908e-007	5.908e-007	5.908e-007	5.908e-007
34	Liq. Reynolds		1.840e+005 *	1.840e+005 *	1.840e+005 *	1.840e+005 *
35	Vap. Reynolds		9.879e+006 *	9.879e+006 *	9.879e+006 *	9.879e+006 *
36	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
37	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	2.616e+004	2.617e+004	2.617e+004	2.618e+004
41	Elevation	(m)	-0.1520	-0.1640	-0.1760	-0.1880
42	Pressure	(bar)	67.95	67.95	67.95	67.95
43	Temperature	(C)	48.13	48.12	48.12	48.12
44	Vapour Fraction		0.8808	0.8808	0.8808	0.8808
45	Heat Transferred	(kJ/h-m)	482.9	482.8	482.8	482.8
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.1694	0.1694	0.1694	0.1694
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	9.082e-003	9.082e-003	9.081e-003	9.081e-003
51	Static Gradient	(kPa/m)	-3.812e-003	-3.812e-003	-3.812e-003	-3.812e-003
52	Accel. Gradient	(kPa/m)	6.705e-007	6.705e-007	6.705e-007	6.705e-007
53	Liq. Reynolds		1.840e+005 *	1.840e+005 *	1.840e+005 *	1.840e+005 *
54	Vap. Reynolds		9.879e+006 *	9.879e+006 *	9.879e+006 *	9.879e+006 *
55	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
56	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	2.618e+004	2.619e+004	2.619e+004	2.620e+004
60	Elevation	(m)	-0.2000	-0.2360	-0.2720	-0.3080
61	Pressure	(bar)	67.95	67.95	67.95	67.95
62	Temperature	(C)	48.12	48.12	48.12	48.12
63	Vapour Fraction		0.8808	0.8808	0.8808	0.8808
64	Heat Transferred	(kJ/h-m)	482.7	482.7	482.7	482.6
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.1694	0.1658	0.1658	0.1658
68	Water HoldUp		---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:13:43 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	9.081e-003	9.131e-003	9.131e-003	9.131e-003	
14	Static Gradient	(kPa/m)	-3.812e-003	-1.128e-002	-1.128e-002	-1.128e-002	
15	Accel. Gradient	(kPa/m)	6.705e-007	-2.730e-007	-2.731e-007	-2.731e-007	
16	Liq. Reynolds		1.840e+005 *	1.840e+005 *	1.840e+005 *	1.840e+005 *	
17	Vap. Reynolds		9.879e+006 *	9.879e+006 *	9.879e+006 *	9.879e+006 *	
18	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
19	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	2.620e+004	2.621e+004	2.621e+004	2.622e+004	
23	Elevation	(m)	-0.3440	-0.3800	-0.3440	-0.3080	
24	Pressure	(bar)	67.95	67.95	67.95	67.95	
25	Temperature	(C)	48.11	48.11	48.11	48.11	
26	Vapour Fraction		0.8808	0.8808	0.8808	0.8808	
27	Heat Transferred	(kJ/h-m)	482.6	482.6	482.5	482.4	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.1659	0.1659	0.2511	0.2511	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	9.131e-003	9.131e-003	6.063e-003	6.063e-003	
33	Static Gradient	(kPa/m)	-1.128e-002	-1.128e-002	1.532e-002	1.532e-002	
34	Accel. Gradient	(kPa/m)	-2.731e-007	-2.731e-007	2.721e-006	2.721e-006	
35	Liq. Reynolds		1.840e+005 *	1.840e+005 *	1.840e+005 *	1.840e+005 *	
36	Vap. Reynolds		9.879e+006 *	9.879e+006 *	9.879e+006 *	9.879e+006 *	
37	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
38	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	2.622e+004	2.623e+004	2.623e+004	2.624e+004	
42	Elevation	(m)	-0.2720	-0.2360	-0.2000	-0.2060	
43	Pressure	(bar)	67.95	67.95	67.95	67.95	
44	Temperature	(C)	48.11	48.11	48.10	48.10	
45	Vapour Fraction		0.8808	0.8808	0.8808	0.8808	
46	Heat Transferred	(kJ/h-m)	482.4	482.3	482.3	482.3	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.2511	0.2511	0.2511	0.1703	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	6.064e-003	6.064e-003	6.064e-003	9.069e-003	
52	Static Gradient	(kPa/m)	1.532e-002	1.532e-002	1.532e-002	-1.913e-003	
53	Accel. Gradient	(kPa/m)	2.721e-006	2.721e-006	2.721e-006	9.107e-007	
54	Liq. Reynolds		1.840e+005 *	1.840e+005 *	1.840e+005 *	1.840e+005 *	
55	Vap. Reynolds		9.879e+006 *	9.879e+006 *	9.879e+006 *	9.879e+006 *	
56	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
57	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	2.624e+004	2.625e+004	2.625e+004	2.626e+004	
61	Elevation	(m)	-0.2120	-0.2180	-0.2240	-0.2300	
62	Pressure	(bar)	67.95	67.95	67.95	67.95	
63	Temperature	(C)	48.10	48.10	48.10	48.10	
64	Vapour Fraction		0.8808	0.8808	0.8808	0.8808	
65	Heat Transferred	(kJ/h-m)	482.2	482.2	482.2	482.1	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.1703	0.1703	0.1703	0.1703	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	9.069e-003	9.069e-003	9.069e-003	9.069e-003
15	Static Gradient (kPa/m)	-1.913e-003	-1.913e-003	-1.913e-003	-1.913e-003
16	Accel. Gradient (kPa/m)	9.107e-007	9.107e-007	9.107e-007	9.107e-007
17	Liq. Reynolds	1.840e+005 *	1.840e+005 *	1.840e+005 *	1.840e+005 *
18	Vap. Reynolds	9.879e+006 *	9.879e+006 *	9.879e+006 *	9.879e+006 *
19	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
20	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	2.626e+004	2.627e+004	2.627e+004	2.628e+004
24	Elevation (m)	-0.2040	-0.1780	-0.1520	-0.1260
25	Pressure (bar)	67.95	67.94	67.94	67.94
26	Temperature (C)	48.10	48.09	48.09	48.09
27	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
28	Heat Transferred (kJ/h-m)	482.0	482.0	481.9	481.9
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2453	0.2453	0.2453	0.2453
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	6.700e-003	6.700e-003	6.700e-003	6.700e-003
34	Static Gradient (kPa/m)	1.073e-002	1.073e-002	1.073e-002	1.073e-002
35	Accel. Gradient (kPa/m)	2.218e-006	2.218e-006	2.218e-006	2.218e-006
36	Liq. Reynolds	1.840e+005 *	1.840e+005 *	1.840e+005 *	1.840e+005 *
37	Vap. Reynolds	9.879e+006 *	9.879e+006 *	9.879e+006 *	9.879e+006 *
38	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
39	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	2.628e+004	2.629e+004	2.629e+004	2.630e+004
43	Elevation (m)	-0.1000	-0.1140	-0.1280	-0.1420
44	Pressure (bar)	67.94	67.94	67.94	67.94
45	Temperature (C)	48.09	48.09	48.09	48.08
46	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
47	Heat Transferred (kJ/h-m)	481.8	481.8	481.8	481.8
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2453	0.1691	0.1691	0.1691
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	6.700e-003	9.086e-003	9.086e-003	9.086e-003
53	Static Gradient (kPa/m)	1.073e-002	-4.442e-003	-4.442e-003	-4.442e-003
54	Accel. Gradient (kPa/m)	2.218e-006	5.909e-007	5.909e-007	5.909e-007
55	Liq. Reynolds	1.840e+005 *	1.840e+005 *	1.840e+005 *	1.840e+005 *
56	Vap. Reynolds	9.879e+006 *	9.879e+006 *	9.879e+006 *	9.879e+006 *
57	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
58	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	2.630e+004	2.631e+004	2.631e+004	2.632e+004
62	Elevation (m)	-0.1560	-0.1700	-0.1700	-0.1700
63	Pressure (bar)	67.94	67.94	67.94	67.94
64	Temperature (C)	48.08	48.08	48.08	48.08
65	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
66	Heat Transferred (kJ/h-m)	481.7	481.7	481.6	481.6
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.1691	0.1691	0.2499	0.2499
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	9.086e-003	9.086e-003	6.191e-003	6.191e-003
16	Static Gradient (kPa/m)	-4.443e-003	-4.443e-003	0.0000	0.0000
17	Accel. Gradient (kPa/m)	5.909e-007	5.909e-007	7.879e-007	7.879e-007
18	Liq. Reynolds	1.839e+005 *	1.839e+005 *	1.839e+005 *	1.839e+005 *
19	Vap. Reynolds	9.879e+006 *	9.879e+006 *	9.879e+006 *	9.880e+006 *
20	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
21	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	2.632e+004	2.633e+004	2.633e+004	2.634e+004
25	Elevation (m)	-0.1700	-0.1700	-0.1700	-0.1840
26	Pressure (bar)	67.94	67.94	67.94	67.94
27	Temperature (C)	48.08	48.08	48.07	48.07
28	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
29	Heat Transferred (kJ/h-m)	481.5	481.5	481.5	481.5
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2499	0.2499	0.2499	0.1691
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.191e-003	6.191e-003	6.191e-003	9.085e-003
35	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-4.443e-003
36	Accel. Gradient (kPa/m)	7.879e-007	7.879e-007	7.879e-007	5.909e-007
37	Liq. Reynolds	1.839e+005 *	1.839e+005 *	1.839e+005 *	1.839e+005 *
38	Vap. Reynolds	9.880e+006 *	9.880e+006 *	9.880e+006 *	9.880e+006 *
39	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
40	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	2.634e+004	2.635e+004	2.635e+004	2.636e+004
44	Elevation (m)	-0.1980	-0.2120	-0.2260	-0.2400
45	Pressure (bar)	67.94	67.94	67.94	67.94
46	Temperature (C)	48.07	48.07	48.07	48.07
47	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
48	Heat Transferred (kJ/h-m)	481.4	481.4	481.3	481.3
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1691	0.1691	0.1691	0.1691
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	9.085e-003	9.085e-003	9.085e-003	9.085e-003
54	Static Gradient (kPa/m)	-4.443e-003	-4.443e-003	-4.443e-003	-4.443e-003
55	Accel. Gradient (kPa/m)	5.909e-007	5.909e-007	5.909e-007	5.909e-007
56	Liq. Reynolds	1.839e+005 *	1.839e+005 *	1.839e+005 *	1.839e+005 *
57	Vap. Reynolds	9.880e+006 *	9.880e+006 *	9.880e+006 *	9.880e+006 *
58	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
59	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	2.636e+004	2.637e+004	2.637e+004	2.638e+004
63	Elevation (m)	-0.2540	-0.2680	-0.2820	-0.2960
64	Pressure (bar)	67.94	67.94	67.94	67.94
65	Temperature (C)	48.07	48.06	48.06	48.06
66	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
67	Heat Transferred (kJ/h-m)	481.3	481.2	481.2	481.2
68	Flow Regime	Transition	Transition	Transition	Transition
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1691	0.1691	0.1691	0.1691
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	9.085e-003	9.085e-003	9.085e-003	9.085e-003
17	Static Gradient (kPa/m)	-4.443e-003	-4.443e-003	-4.443e-003	-4.443e-003
18	Accel. Gradient (kPa/m)	5.909e-007	5.909e-007	5.909e-007	5.909e-007
19	Liq. Reynolds	1.839e+005 *	1.839e+005 *	1.839e+005 *	1.839e+005 *
20	Vap. Reynolds	9.880e+006 *	9.880e+006 *	9.880e+006 *	9.880e+006 *
21	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
22	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	2.638e+004	2.639e+004	2.639e+004	2.640e+004
26	Elevation (m)	-0.3100	-0.2840	-0.2580	-0.2320
27	Pressure (bar)	67.94	67.94	67.94	67.93
28	Temperature (C)	48.06	48.06	48.06	48.05
29	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
30	Heat Transferred (kJ/h-m)	481.1	481.0	481.0	480.9
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1691	0.2453	0.2453	0.2453
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	9.085e-003	6.699e-003	6.700e-003	6.700e-003
36	Static Gradient (kPa/m)	-4.443e-003	1.073e-002	1.073e-002	1.073e-002
37	Accel. Gradient (kPa/m)	5.909e-007	2.218e-006	2.218e-006	2.218e-006
38	Liq. Reynolds	1.839e+005 *	1.839e+005 *	1.839e+005 *	1.839e+005 *
39	Vap. Reynolds	9.880e+006 *	9.880e+006 *	9.880e+006 *	9.880e+006 *
40	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
41	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	2.640e+004	2.641e+004	2.641e+004	2.642e+004
45	Elevation (m)	-0.2060	-0.1800	-0.1600	-0.1400
46	Pressure (bar)	67.93	67.93	67.93	67.93
47	Temperature (C)	48.05	48.05	48.05	48.05
48	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
49	Heat Transferred (kJ/h-m)	480.9	480.8	480.8	480.8
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2453	0.2453	0.2412	0.2412
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	6.700e-003	6.700e-003	7.176e-003	7.176e-003
55	Static Gradient (kPa/m)	1.073e-002	1.073e-002	7.585e-003	7.585e-003
56	Accel. Gradient (kPa/m)	2.219e-006	2.219e-006	1.879e-006	1.879e-006
57	Liq. Reynolds	1.839e+005 *	1.839e+005 *	1.839e+005 *	1.839e+005 *
58	Vap. Reynolds	9.880e+006 *	9.880e+006 *	9.880e+006 *	9.880e+006 *
59	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
60	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	2.642e+004	2.643e+004	2.644e+004	2.644e+004
64	Elevation (m)	-0.1200	-0.1000	-8.000e-002	-0.1380
65	Pressure (bar)	67.93	67.93	67.93	67.93
66	Temperature (C)	48.05	48.04	48.04	48.04
67	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
68	Heat Transferred (kJ/h-m)	480.7	480.7	480.6	480.6
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2412	0.2412	0.2412	0.1626
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	7.176e-003	7.176e-003	7.176e-003	9.180e-003
18	Static Gradient (kPa/m)	7.585e-003	7.585e-003	7.585e-003	-1.793e-002
19	Accel. Gradient (kPa/m)	1.879e-006	1.879e-006	1.879e-006	-1.114e-006
20	Liq. Reynolds	1.839e+005 *	1.839e+005 *	1.839e+005 *	1.839e+005 *
21	Vap. Reynolds	9.880e+006 *	9.880e+006 *	9.880e+006 *	9.880e+006 *
22	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
23	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	2.645e+004	2.645e+004	2.646e+004	2.646e+004
27	Elevation (m)	-0.1960	-0.2540	-0.3120	-0.3700
28	Pressure (bar)	67.93	67.93	67.93	67.93
29	Temperature (C)	48.04	48.04	48.04	48.04
30	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
31	Heat Transferred (kJ/h-m)	480.6	480.6	480.5	480.5
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1626	0.1626	0.1626	0.1626
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	9.179e-003	9.179e-003	9.179e-003	9.179e-003
37	Static Gradient (kPa/m)	-1.793e-002	-1.793e-002	-1.793e-002	-1.793e-002
38	Accel. Gradient (kPa/m)	-1.114e-006	-1.114e-006	-1.114e-006	-1.114e-006
39	Liq. Reynolds	1.839e+005 *	1.839e+005 *	1.839e+005 *	1.839e+005 *
40	Vap. Reynolds	9.880e+006 *	9.880e+006 *	9.880e+006 *	9.880e+006 *
41	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
42	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	2.647e+004	2.647e+004	2.648e+004	2.648e+004
46	Elevation (m)	-0.3540	-0.3380	-0.3220	-0.3060
47	Pressure (bar)	67.93	67.93	67.93	67.93
48	Temperature (C)	48.04	48.03	48.03	48.03
49	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
50	Heat Transferred (kJ/h-m)	480.4	480.4	480.3	480.3
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2397	0.2397	0.2397	0.2397
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	7.350e-003	7.350e-003	7.350e-003	7.350e-003
56	Static Gradient (kPa/m)	6.493e-003	6.493e-003	6.493e-003	6.493e-003
57	Accel. Gradient (kPa/m)	1.762e-006	1.762e-006	1.762e-006	1.762e-006
58	Liq. Reynolds	1.839e+005 *	1.839e+005 *	1.839e+005 *	1.839e+005 *
59	Vap. Reynolds	9.880e+006 *	9.880e+006 *	9.880e+006 *	9.880e+006 *
60	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
61	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	2.649e+004	2.649e+004	2.650e+004	2.650e+004
65	Elevation (m)	-0.2900	-0.3100	-0.3300	-0.3500
66	Pressure (bar)	67.93	67.93	67.93	67.93
67	Temperature (C)	48.03	48.03	48.03	48.02
68	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Heat Transferred	(kJ/h-m)	480.2	480.3	480.2	480.2				
14	Flow Regime		Transition	Transition	Transition	Transition				
15	Water Flow Regime									
16	Liquid HoldUp		0.2397	0.1682	0.1682	0.1682				
17	Water HoldUp		---	---	---	---				
18	Friction Gradient	(kPa/m)	7.350e-003	9.098e-003	9.097e-003	9.097e-003				
19	Static Gradient	(kPa/m)	6.493e-003	-6.325e-003	-6.325e-003	-6.325e-003				
20	Accel. Gradient	(kPa/m)	1.762e-006	3.530e-007	3.529e-007	3.529e-007				
21	Liq. Reynolds		1.839e+005 *	1.839e+005 *	1.839e+005 *	1.839e+005 *				
22	Vap. Reynolds		9.880e+006 *	9.880e+006 *	9.880e+006 *	9.880e+006 *				
23	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979				
24	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932				
25	Deposit Thickness	(mm)	---	---	---	---				
26	Deposit Volume	(m3)	---	---	---	---				
27	Length	(m)	2.651e+004	2.651e+004	2.652e+004	2.652e+004				
28	Elevation	(m)	-0.3700	-0.3900	-0.3940	-0.3980				
29	Pressure	(bar)	67.93	67.93	67.93	67.93				
30	Temperature	(C)	48.02	48.02	48.02	48.02				
31	Vapour Fraction		0.8808	0.8808	0.8808	0.8808				
32	Heat Transferred	(kJ/h-m)	480.1	480.1	480.1	480.0				
33	Flow Regime		Transition	Transition	Transition	Transition				
34	Water Flow Regime									
35	Liquid HoldUp		0.1682	0.1682	0.1706	0.1706				
36	Water HoldUp		---	---	---	---				
37	Friction Gradient	(kPa/m)	9.097e-003	9.097e-003	9.065e-003	9.065e-003				
38	Static Gradient	(kPa/m)	-6.325e-003	-6.325e-003	-1.277e-003	-1.277e-003				
39	Accel. Gradient	(kPa/m)	3.529e-007	3.529e-007	9.913e-007	9.914e-007				
40	Liq. Reynolds		1.839e+005 *	1.839e+005 *	1.839e+005 *	1.839e+005 *				
41	Vap. Reynolds		9.880e+006 *	9.880e+006 *	9.880e+006 *	9.880e+006 *				
42	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979				
43	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932				
44	Deposit Thickness	(mm)	---	---	---	---				
45	Deposit Volume	(m3)	---	---	---	---				
46	Length	(m)	2.653e+004	2.653e+004	2.654e+004	2.654e+004				
47	Elevation	(m)	-0.4020	-0.4060	-0.4100	-0.4040				
48	Pressure	(bar)	67.93	67.93	67.92	67.92				
49	Temperature	(C)	48.02	48.02	48.01	48.01				
50	Vapour Fraction		0.8808	0.8808	0.8808	0.8808				
51	Heat Transferred	(kJ/h-m)	480.0	479.9	479.9	479.8				
52	Flow Regime		Transition	Transition	Transition	Transition				
53	Water Flow Regime									
54	Liquid HoldUp		0.1706	0.1706	0.1706	0.2342				
55	Water HoldUp		---	---	---	---				
56	Friction Gradient	(kPa/m)	9.065e-003	9.065e-003	9.065e-003	8.046e-003				
57	Static Gradient	(kPa/m)	-1.277e-003	-1.277e-003	-1.277e-003	2.393e-003				
58	Accel. Gradient	(kPa/m)	9.914e-007	9.914e-007	9.914e-007	1.329e-006				
59	Liq. Reynolds		1.839e+005 *	1.839e+005 *	1.839e+005 *	1.839e+005 *				
60	Vap. Reynolds		9.880e+006 *	9.880e+006 *	9.880e+006 *	9.880e+006 *				
61	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979				
62	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932				
63	Deposit Thickness	(mm)	---	---	---	---				
64	Deposit Volume	(m3)	---	---	---	---				
65	Length	(m)	2.655e+004	2.655e+004	2.656e+004	2.656e+004				
66	Elevation	(m)	-0.3980	-0.3920	-0.3860	-0.3800				
67	Pressure	(bar)	67.92	67.92	67.92	67.92				
68	Temperature	(C)	48.01	48.01	48.01	48.01				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
14	Heat Transferred (kJ/h-m)	479.8	479.7	479.7	479.7
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2342	0.2342	0.2342	0.2342
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	8.046e-003	8.046e-003	8.046e-003	8.046e-003
20	Static Gradient (kPa/m)	2.393e-003	2.393e-003	2.393e-003	2.393e-003
21	Accel. Gradient (kPa/m)	1.329e-006	1.329e-006	1.329e-006	1.329e-006
22	Liq. Reynolds	1.839e+005 *	1.839e+005 *	1.839e+005 *	1.839e+005 *
23	Vap. Reynolds	9.880e+006 *	9.880e+006 *	9.880e+006 *	9.880e+006 *
24	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
25	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	2.657e+004	2.657e+004	2.658e+004	2.658e+004
29	Elevation (m)	-0.3960	-0.4120	-0.4280	-0.4440
30	Pressure (bar)	67.92	67.92	67.92	67.92
31	Temperature (C)	48.00	48.00	48.00	48.00
32	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
33	Heat Transferred (kJ/h-m)	479.7	479.6	479.6	479.6
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1688	0.1688	0.1688	0.1688
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	9.089e-003	9.089e-003	9.089e-003	9.089e-003
39	Static Gradient (kPa/m)	-5.072e-003	-5.072e-003	-5.072e-003	-5.072e-003
40	Accel. Gradient (kPa/m)	5.114e-007	5.114e-007	5.114e-007	5.114e-007
41	Liq. Reynolds	1.839e+005 *	1.839e+005 *	1.839e+005 *	1.839e+005 *
42	Vap. Reynolds	9.880e+006 *	9.880e+006 *	9.880e+006 *	9.880e+006 *
43	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
44	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	2.659e+004	2.659e+004	2.660e+004	2.660e+004
48	Elevation (m)	-0.4600	-0.4560	-0.4520	-0.4480
49	Pressure (bar)	67.92	67.92	67.92	67.92
50	Temperature (C)	48.00	48.00	48.00	47.99
51	Vapour Fraction	0.8808	0.8808	0.8808	0.8808
52	Heat Transferred (kJ/h-m)	479.5	479.4	479.4	479.3
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.1688	0.2331	0.2331	0.2331
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	9.089e-003	8.192e-003	8.192e-003	8.192e-003
58	Static Gradient (kPa/m)	-5.072e-003	1.590e-003	1.590e-003	1.590e-003
59	Accel. Gradient (kPa/m)	5.114e-007	1.245e-006	1.245e-006	1.245e-006
60	Liq. Reynolds	1.839e+005 *	1.839e+005 *	1.839e+005 *	1.839e+005 *
61	Vap. Reynolds	9.880e+006 *	9.880e+006 *	9.881e+006 *	9.881e+006 *
62	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
63	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	2.661e+004	2.661e+004	2.662e+004	2.662e+004
67	Elevation (m)	-0.4440	-0.4400	-0.4680	-0.4960
68	Pressure (bar)	67.92	67.92	67.92	67.92
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	47.99	47.99	47.99	47.99
14	Vapour Fraction	0.8808	0.8807	0.8807	0.8807
15	Heat Transferred (kJ/h-m)	479.3	479.3	479.3	479.2
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2331	0.2331	0.1670	0.1670
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	8.192e-003	8.192e-003	9.114e-003	9.114e-003
21	Static Gradient (kPa/m)	1.590e-003	1.590e-003	-8.813e-003	-8.814e-003
22	Accel. Gradient (kPa/m)	1.245e-006	1.245e-006	3.823e-008	3.821e-008
23	Liq. Reynolds	1.839e+005 *	1.839e+005 *	1.839e+005 *	1.839e+005 *
24	Vap. Reynolds	9.881e+006 *	9.881e+006 *	9.881e+006 *	9.881e+006 *
25	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
26	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	2.663e+004	2.663e+004	2.664e+004	2.664e+004
30	Elevation (m)	-0.5240	-0.5520	-0.5800	-0.5760
31	Pressure (bar)	67.92	67.92	67.92	67.92
32	Temperature (C)	47.99	47.99	47.99	47.98
33	Vapour Fraction	0.8807	0.8807	0.8807	0.8807
34	Heat Transferred (kJ/h-m)	479.2	479.2	479.1	479.0
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1670	0.1670	0.1670	0.2331
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	9.114e-003	9.114e-003	9.114e-003	8.191e-003
40	Static Gradient (kPa/m)	-8.814e-003	-8.814e-003	-8.814e-003	1.590e-003
41	Accel. Gradient (kPa/m)	3.820e-008	3.819e-008	3.817e-008	1.245e-006
42	Liq. Reynolds	1.838e+005 *	1.838e+005 *	1.838e+005 *	1.838e+005 *
43	Vap. Reynolds	9.881e+006 *	9.881e+006 *	9.881e+006 *	9.881e+006 *
44	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
45	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	2.665e+004	2.665e+004	2.666e+004	2.666e+004
49	Elevation (m)	-0.5720	-0.5680	-0.5640	-0.5600
50	Pressure (bar)	67.92	67.92	67.92	67.92
51	Temperature (C)	47.98	47.98	47.98	47.98
52	Vapour Fraction	0.8807	0.8807	0.8807	0.8807
53	Heat Transferred (kJ/h-m)	479.0	479.0	478.9	478.9
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2331	0.2331	0.2331	0.2331
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	8.191e-003	8.191e-003	8.191e-003	8.191e-003
59	Static Gradient (kPa/m)	1.590e-003	1.590e-003	1.590e-003	1.590e-003
60	Accel. Gradient (kPa/m)	1.245e-006	1.245e-006	1.245e-006	1.245e-006
61	Liq. Reynolds	1.838e+005 *	1.838e+005 *	1.838e+005 *	1.838e+005 *
62	Vap. Reynolds	9.881e+006 *	9.881e+006 *	9.881e+006 *	9.881e+006 *
63	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
64	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	2.667e+004	2.667e+004	2.668e+004	2.668e+004
68	Elevation (m)	-0.5340	-0.5080	-0.4820	-0.4560
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Pressure (bar)	67.92	67.91	67.91	67.91
14	Temperature (C)	47.98	47.97	47.97	47.97
15	Vapour Fraction	0.8807	0.8807	0.8807	0.8807
16	Heat Transferred (kJ/h-m)	478.8	478.8	478.7	478.7
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.2453	0.2453	0.2453	0.2453
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	6.699e-003	6.699e-003	6.699e-003	6.699e-003
22	Static Gradient (kPa/m)	1.073e-002	1.073e-002	1.073e-002	1.073e-002
23	Accel. Gradient (kPa/m)	2.219e-006	2.219e-006	2.219e-006	2.219e-006
24	Liq. Reynolds	1.838e+005 *	1.838e+005 *	1.838e+005 *	1.838e+005 *
25	Vap. Reynolds	9.881e+006 *	9.881e+006 *	9.881e+006 *	9.881e+006 *
26	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
27	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	2.669e+004	2.669e+004	2.670e+004	2.670e+004
31	Elevation (m)	-0.4300	-0.3840	-0.3380	-0.2920
32	Pressure (bar)	67.91	67.91	67.91	67.91
33	Temperature (C)	47.97	47.97	47.96	47.96
34	Vapour Fraction	0.8807	0.8807	0.8807	0.8807
35	Heat Transferred (kJ/h-m)	478.7	478.6	478.5	478.5
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.2453	0.2540	0.2540	0.2540
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	6.699e-003	7.163e-003	7.163e-003	7.163e-003
41	Static Gradient (kPa/m)	1.073e-002	1.772e-002	1.772e-002	1.772e-002
42	Accel. Gradient (kPa/m)	2.219e-006	3.168e-006	3.168e-006	3.168e-006
43	Liq. Reynolds	1.838e+005 *	1.838e+005 *	1.838e+005 *	1.838e+005 *
44	Vap. Reynolds	9.881e+006 *	9.881e+006 *	9.881e+006 *	9.881e+006 *
45	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
46	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	2.671e+004	2.671e+004	2.672e+004	2.672e+004
50	Elevation (m)	-0.2460	-0.2000	-0.1840	-0.1680
51	Pressure (bar)	67.91	67.90	67.90	67.90
52	Temperature (C)	47.96	47.96	47.96	47.96
53	Vapour Fraction	0.8807	0.8807	0.8807	0.8807
54	Heat Transferred (kJ/h-m)	478.4	478.4	478.3	478.3
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.2540	0.2540	0.2389	0.2389
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	7.163e-003	7.163e-003	7.449e-003	7.449e-003
60	Static Gradient (kPa/m)	1.772e-002	1.772e-002	5.888e-003	5.888e-003
61	Accel. Gradient (kPa/m)	3.168e-006	3.168e-006	1.698e-006	1.698e-006
62	Liq. Reynolds	1.838e+005 *	1.838e+005 *	1.838e+005 *	1.838e+005 *
63	Vap. Reynolds	9.881e+006 *	9.881e+006 *	9.881e+006 *	9.881e+006 *
64	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
65	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	2.673e+004	2.674e+004	2.674e+004	2.674e+004
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Elevation (m)	-0.1520	-0.1360	-0.1200	-0.1020		
14	Pressure (bar)	67.90	67.90	67.90	67.90		
15	Temperature (C)	47.95	47.95	47.95	47.95		
16	Vapour Fraction	0.8807	0.8807	0.8807	0.8807		
17	Heat Transferred (kJ/h-m)	478.2	478.2	478.2	478.1		
18	Flow Regime	Transition	Transition	Transition	Transition		
19	Water Flow Regime						
20	Liquid HoldUp	0.2389	0.2389	0.2389	0.2433		
21	Water HoldUp	---	---	---	---		
22	Friction Gradient (kPa/m)	7.449e-003	7.449e-003	7.449e-003	6.923e-003		
23	Static Gradient (kPa/m)	5.888e-003	5.888e-003	5.888e-003	9.232e-003		
24	Accel. Gradient (kPa/m)	1.698e-006	1.698e-006	1.698e-006	2.057e-006		
25	Liq. Reynolds	1.838e+005 *	1.838e+005 *	1.838e+005 *	1.838e+005 *		
26	Vap. Reynolds	9.881e+006 *	9.881e+006 *	9.881e+006 *	9.881e+006 *		
27	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
28	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932		
29	Deposit Thickness (mm)	---	---	---	---		
30	Deposit Volume (m3)	---	---	---	---		
31	Length (m)	2.675e+004	2.675e+004	2.676e+004	2.676e+004		
32	Elevation (m)	-8.400e-002	-6.600e-002	-4.800e-002	-3.000e-002		
33	Pressure (bar)	67.90	67.90	67.90	67.90		
34	Temperature (C)	47.95	47.95	47.94	47.94		
35	Vapour Fraction	0.8807	0.8807	0.8807	0.8807		
36	Heat Transferred (kJ/h-m)	478.1	478.0	478.0	478.0		
37	Flow Regime	Transition	Transition	Transition	Transition		
38	Water Flow Regime						
39	Liquid HoldUp	0.2433	0.2433	0.2433	0.2433		
40	Water HoldUp	---	---	---	---		
41	Friction Gradient (kPa/m)	6.923e-003	6.923e-003	6.923e-003	6.923e-003		
42	Static Gradient (kPa/m)	9.232e-003	9.232e-003	9.232e-003	9.232e-003		
43	Accel. Gradient (kPa/m)	2.057e-006	2.057e-006	2.057e-006	2.057e-006		
44	Liq. Reynolds	1.838e+005 *	1.838e+005 *	1.838e+005 *	1.838e+005 *		
45	Vap. Reynolds	9.881e+006 *	9.881e+006 *	9.881e+006 *	9.881e+006 *		
46	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
47	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932		
48	Deposit Thickness (mm)	---	---	---	---		
49	Deposit Volume (m3)	---	---	---	---		
50	Length (m)	2.677e+004	2.677e+004	2.678e+004	2.678e+004		
51	Elevation (m)	-2.600e-002	-2.200e-002	-1.800e-002	-1.400e-002		
52	Pressure (bar)	67.90	67.90	67.90	67.90		
53	Temperature (C)	47.94	47.94	47.94	47.94		
54	Vapour Fraction	0.8807	0.8807	0.8807	0.8807		
55	Heat Transferred (kJ/h-m)	477.9	477.9	477.9	477.8		
56	Flow Regime	Transition	Transition	Transition	Transition		
57	Water Flow Regime						
58	Liquid HoldUp	0.2331	0.2331	0.2331	0.2331		
59	Water HoldUp	---	---	---	---		
60	Friction Gradient (kPa/m)	8.193e-003	8.193e-003	8.193e-003	8.193e-003		
61	Static Gradient (kPa/m)	1.590e-003	1.590e-003	1.590e-003	1.590e-003		
62	Accel. Gradient (kPa/m)	1.246e-006	1.246e-006	1.246e-006	1.246e-006		
63	Liq. Reynolds	1.838e+005 *	1.838e+005 *	1.838e+005 *	1.838e+005 *		
64	Vap. Reynolds	9.882e+006 *	9.882e+006 *	9.882e+006 *	9.882e+006 *		
65	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
66	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932		
67	Deposit Thickness (mm)	---	---	---	---		
68	Deposit Volume (m3)	---	---	---	---		
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	2.679e+004	2.679e+004	2.680e+004	2.680e+004
14	Elevation (m)	-1.000e-002	-6.000e-003	-2.000e-003	2.000e-003
15	Pressure (bar)	67.90	67.90	67.89	67.89
16	Temperature (C)	47.94	47.93	47.93	47.93
17	Vapour Fraction	0.8807	0.8807	0.8807	0.8807
18	Heat Transferred (kJ/h-m)	477.8	477.7	477.7	477.6
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.2331	0.2331	0.2331	0.2331
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	8.193e-003	8.193e-003	8.193e-003	8.193e-003
24	Static Gradient (kPa/m)	1.590e-003	1.590e-003	1.590e-003	1.590e-003
25	Accel. Gradient (kPa/m)	1.246e-006	1.246e-006	1.246e-006	1.246e-006
26	Liq. Reynolds	1.838e+005 *	1.838e+005 *	1.838e+005 *	1.838e+005 *
27	Vap. Reynolds	9.882e+006 *	9.882e+006 *	9.882e+006 *	9.882e+006 *
28	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
29	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	2.681e+004	2.681e+004	2.682e+004	2.682e+004
33	Elevation (m)	6.000e-003	1.000e-002	-9.298e-016	-1.000e-002
34	Pressure (bar)	67.89	67.89	67.89	67.89
35	Temperature (C)	47.93	47.93	47.93	47.92
36	Vapour Fraction	0.8807	0.8807	0.8807	0.8807
37	Heat Transferred (kJ/h-m)	477.6	477.6	477.6	477.5
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.2331	0.2331	0.1697	0.1697
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	8.193e-003	8.193e-003	9.078e-003	9.078e-003
43	Static Gradient (kPa/m)	1.590e-003	1.590e-003	-3.145e-003	-3.145e-003
44	Accel. Gradient (kPa/m)	1.246e-006	1.246e-006	7.556e-007	7.556e-007
45	Liq. Reynolds	1.838e+005 *	1.838e+005 *	1.838e+005 *	1.838e+005 *
46	Vap. Reynolds	9.882e+006 *	9.882e+006 *	9.882e+006 *	9.882e+006 *
47	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
48	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	2.683e+004	2.683e+004	2.684e+004	2.684e+004
52	Elevation (m)	-2.000e-002	-3.000e-002	-4.000e-002	-2.600e-002
53	Pressure (bar)	67.89	67.89	67.89	67.89
54	Temperature (C)	47.92	47.92	47.92	47.92
55	Vapour Fraction	0.8807	0.8807	0.8807	0.8807
56	Heat Transferred (kJ/h-m)	477.5	477.5	477.4	477.3
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.1697	0.1697	0.1697	0.2386
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	9.078e-003	9.078e-003	9.078e-003	7.486e-003
62	Static Gradient (kPa/m)	-3.145e-003	-3.145e-003	-3.145e-003	5.662e-003
63	Accel. Gradient (kPa/m)	7.556e-007	7.556e-007	7.556e-007	1.675e-006
64	Liq. Reynolds	1.838e+005 *	1.838e+005 *	1.838e+005 *	1.838e+005 *
65	Vap. Reynolds	9.882e+006 *	9.882e+006 *	9.882e+006 *	9.882e+006 *
66	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
67	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
68	Deposit Thickness (mm)	---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	2.685e+004	2.685e+004	2.686e+004	2.686e+004
15	Elevation (m)	-1.200e-002	2.000e-003	1.600e-002	3.000e-002
16	Pressure (bar)	67.89	67.89	67.89	67.89
17	Temperature (C)	47.92	47.92	47.91	47.91
18	Vapour Fraction	0.8807	0.8807	0.8807	0.8807
19	Heat Transferred (kJ/h-m)	477.3	477.2	477.2	477.2
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2386	0.2386	0.2386	0.2386
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	7.486e-003	7.486e-003	7.486e-003	7.486e-003
25	Static Gradient (kPa/m)	5.662e-003	5.662e-003	5.662e-003	5.662e-003
26	Accel. Gradient (kPa/m)	1.675e-006	1.675e-006	1.675e-006	1.675e-006
27	Liq. Reynolds	1.838e+005 *	1.838e+005 *	1.838e+005 *	1.838e+005 *
28	Vap. Reynolds	9.882e+006 *	9.882e+006 *	9.882e+006 *	9.882e+006 *
29	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
30	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	2.687e+004	2.687e+004	2.688e+004	2.688e+004
34	Elevation (m)	2.600e-002	2.200e-002	1.800e-002	1.400e-002
35	Pressure (bar)	67.89	67.89	67.89	67.89
36	Temperature (C)	47.91	47.91	47.91	47.91
37	Vapour Fraction	0.8807	0.8807	0.8807	0.8807
38	Heat Transferred (kJ/h-m)	477.2	477.1	477.1	477.0
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.1706	0.1706	0.1706	0.1706
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	9.066e-003	9.066e-003	9.066e-003	9.066e-003
44	Static Gradient (kPa/m)	-1.277e-003	-1.277e-003	-1.277e-003	-1.277e-003
45	Accel. Gradient (kPa/m)	9.921e-007	9.921e-007	9.921e-007	9.921e-007
46	Liq. Reynolds	1.838e+005 *	1.838e+005 *	1.838e+005 *	1.837e+005 *
47	Vap. Reynolds	9.882e+006 *	9.882e+006 *	9.882e+006 *	9.882e+006 *
48	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
49	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	2.689e+004	2.689e+004	2.690e+004	2.690e+004
53	Elevation (m)	1.000e-002	-9.263e-016	-1.000e-002	-2.000e-002
54	Pressure (bar)	67.89	67.89	67.89	67.89
55	Temperature (C)	47.91	47.90	47.90	47.90
56	Vapour Fraction	0.8807	0.8807	0.8807	0.8807
57	Heat Transferred (kJ/h-m)	477.0	477.0	476.9	476.9
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.1706	0.1697	0.1697	0.1697
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	9.066e-003	9.078e-003	9.078e-003	9.078e-003
63	Static Gradient (kPa/m)	-1.277e-003	-3.181e-003	-3.181e-003	-3.181e-003
64	Accel. Gradient (kPa/m)	9.921e-007	7.511e-007	7.511e-007	7.511e-007
65	Liq. Reynolds	1.837e+005 *	1.837e+005 *	1.837e+005 *	1.837e+005 *
66	Vap. Reynolds	9.882e+006 *	9.882e+006 *	9.882e+006 *	9.882e+006 *
67	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
68	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Thickness	(mm)	---	---	---	---	
14	Deposit Volume	(m3)	---	---	---	---	
15	Length	(m)	2.691e+004	2.691e+004	2.692e+004	2.692e+004	
16	Elevation	(m)	-3.000e-002	-4.000e-002	-6.000e-002	-8.000e-002	
17	Pressure	(bar)	67.89	67.88	67.88	67.88	
18	Temperature	(C)	47.90	47.90	47.90	47.90	
19	Vapour Fraction		0.8807	0.8807	0.8807	0.8807	
20	Heat Transferred	(kJ/h-m)	476.8	476.8	476.8	476.7	
21	Flow Regime		Transition	Transition	Transition	Transition	
22	Water Flow Regime						
23	Liquid HoldUp		0.1697	0.1697	0.1682	0.1682	
24	Water HoldUp		---	---	---	---	
25	Friction Gradient	(kPa/m)	9.078e-003	9.078e-003	9.098e-003	9.098e-003	
26	Static Gradient	(kPa/m)	-3.181e-003	-3.181e-003	-6.325e-003	-6.325e-003	
27	Accel. Gradient	(kPa/m)	7.511e-007	7.511e-007	3.532e-007	3.532e-007	
28	Liq. Reynolds		1.837e+005 *	1.837e+005 *	1.837e+005 *	1.837e+005 *	
29	Vap. Reynolds		9.882e+006 *	9.882e+006 *	9.882e+006 *	9.882e+006 *	
30	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
31	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932	
32	Deposit Thickness	(mm)	---	---	---	---	
33	Deposit Volume	(m3)	---	---	---	---	
34	Length	(m)	2.693e+004	2.693e+004	2.694e+004	2.694e+004	
35	Elevation	(m)	-0.1000	-0.1200	-0.1400	-0.1640	
36	Pressure	(bar)	67.88	67.88	67.88	67.88	
37	Temperature	(C)	47.89	47.89	47.89	47.89	
38	Vapour Fraction		0.8807	0.8807	0.8807	0.8807	
39	Heat Transferred	(kJ/h-m)	476.7	476.7	476.6	476.6	
40	Flow Regime		Transition	Transition	Transition	Transition	
41	Water Flow Regime						
42	Liquid HoldUp		0.1682	0.1682	0.1682	0.1676	
43	Water HoldUp		---	---	---	---	
44	Friction Gradient	(kPa/m)	9.098e-003	9.098e-003	9.098e-003	9.106e-003	
45	Static Gradient	(kPa/m)	-6.325e-003	-6.325e-003	-6.325e-003	-7.572e-003	
46	Accel. Gradient	(kPa/m)	3.532e-007	3.532e-007	3.532e-007	1.954e-007	
47	Liq. Reynolds		1.837e+005 *	1.837e+005 *	1.837e+005 *	1.837e+005 *	
48	Vap. Reynolds		9.882e+006 *	9.882e+006 *	9.882e+006 *	9.882e+006 *	
49	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
50	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932	
51	Deposit Thickness	(mm)	---	---	---	---	
52	Deposit Volume	(m3)	---	---	---	---	
53	Length	(m)	2.695e+004	2.695e+004	2.696e+004	2.696e+004	
54	Elevation	(m)	-0.1880	-0.2120	-0.2360	-0.2600	
55	Pressure	(bar)	67.88	67.88	67.88	67.88	
56	Temperature	(C)	47.89	47.89	47.89	47.88	
57	Vapour Fraction		0.8807	0.8807	0.8807	0.8807	
58	Heat Transferred	(kJ/h-m)	476.6	476.5	476.5	476.5	
59	Flow Regime		Transition	Transition	Transition	Transition	
60	Water Flow Regime						
61	Liquid HoldUp		0.1676	0.1676	0.1676	0.1676	
62	Water HoldUp		---	---	---	---	
63	Friction Gradient	(kPa/m)	9.106e-003	9.106e-003	9.106e-003	9.106e-003	
64	Static Gradient	(kPa/m)	-7.572e-003	-7.572e-003	-7.572e-003	-7.572e-003	
65	Accel. Gradient	(kPa/m)	1.954e-007	1.954e-007	1.953e-007	1.953e-007	
66	Liq. Reynolds		1.837e+005 *	1.837e+005 *	1.837e+005 *	1.837e+005 *	
67	Vap. Reynolds		9.882e+006 *	9.882e+006 *	9.882e+006 *	9.882e+006 *	
68	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
69	Aspen Technology Inc.		Aspen HYSYS Version 10				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	2.697e+004	2.697e+004	2.698e+004	2.698e+004	
17	Elevation	(m)	-0.2800	-0.3000	-0.3200	-0.3400	
18	Pressure	(bar)	67.88	67.88	67.88	67.88	
19	Temperature	(C)	47.88	47.88	47.88	47.88	
20	Vapour Fraction		0.8807	0.8807	0.8807	0.8807	
21	Heat Transferred	(kJ/h-m)	476.4	476.4	476.3	476.3	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.1682	0.1682	0.1682	0.1682	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	9.098e-003	9.098e-003	9.098e-003	9.098e-003	
27	Static Gradient	(kPa/m)	-6.340e-003	-6.340e-003	-6.340e-003	-6.340e-003	
28	Accel. Gradient	(kPa/m)	3.512e-007	3.512e-007	3.512e-007	3.512e-007	
29	Liq. Reynolds		1.837e+005 *	1.837e+005 *	1.837e+005 *	1.837e+005 *	
30	Vap. Reynolds		9.882e+006 *	9.882e+006 *	9.882e+006 *	9.882e+006 *	
31	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
32	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	2.699e+004	2.699e+004	2.700e+004	2.700e+004	
36	Elevation	(m)	-0.3600	-0.3780	-0.3960	-0.4140	
37	Pressure	(bar)	67.88	67.88	67.88	67.88	
38	Temperature	(C)	47.88	47.88	47.87	47.87	
39	Vapour Fraction		0.8807	0.8807	0.8807	0.8807	
40	Heat Transferred	(kJ/h-m)	476.3	476.2	476.2	476.2	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.1682	0.1685	0.1685	0.1685	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	9.098e-003	9.093e-003	9.093e-003	9.093e-003	
46	Static Gradient	(kPa/m)	-6.340e-003	-5.699e-003	-5.699e-003	-5.699e-003	
47	Accel. Gradient	(kPa/m)	3.512e-007	4.323e-007	4.323e-007	4.323e-007	
48	Liq. Reynolds		1.837e+005 *	1.837e+005 *	1.837e+005 *	1.837e+005 *	
49	Vap. Reynolds		9.882e+006 *	9.882e+006 *	9.882e+006 *	9.882e+006 *	
50	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
51	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	2.701e+004	2.701e+004	2.702e+004	2.702e+004	
55	Elevation	(m)	-0.4320	-0.4500	-0.4700	-0.4900	
56	Pressure	(bar)	67.88	67.88	67.88	67.88	
57	Temperature	(C)	47.87	47.87	47.87	47.87	
58	Vapour Fraction		0.8807	0.8807	0.8807	0.8807	
59	Heat Transferred	(kJ/h-m)	476.1	476.1	476.0	476.0	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.1685	0.1685	0.1682	0.1682	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	9.093e-003	9.093e-003	9.097e-003	9.097e-003	
65	Static Gradient	(kPa/m)	-5.699e-003	-5.699e-003	-6.325e-003	-6.325e-003	
66	Accel. Gradient	(kPa/m)	4.323e-007	4.322e-007	3.530e-007	3.530e-007	
67	Liq. Reynolds		1.837e+005 *	1.837e+005 *	1.837e+005 *	1.837e+005 *	
68	Vap. Reynolds		9.882e+006 *	9.882e+006 *	9.882e+006 *	9.882e+006 *	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 438 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
14	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	2.703e+004	2.703e+004	2.704e+004	2.704e+004	
18	Elevation	(m)	-0.5100	-0.5300	-0.5500	-0.5280	
19	Pressure	(bar)	67.88	67.88	67.88	67.88	
20	Temperature	(C)	47.87	47.87	47.86	47.86	
21	Vapour Fraction		0.8807	0.8807	0.8807	0.8807	
22	Heat Transferred	(kJ/h-m)	476.0	475.9	475.9	475.8	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.1682	0.1682	0.1682	0.2613	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	9.097e-003	9.097e-003	9.097e-003	7.769e-003	
28	Static Gradient	(kPa/m)	-6.325e-003	-6.325e-003	-6.325e-003	2.381e-002	
29	Accel. Gradient	(kPa/m)	3.530e-007	3.530e-007	3.530e-007	4.022e-006	
30	Liq. Reynolds		1.837e+005 *	1.837e+005 *	1.837e+005 *	1.837e+005 *	
31	Vap. Reynolds		9.882e+006 *	9.882e+006 *	9.882e+006 *	9.882e+006 *	
32	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
33	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	2.704e+004	2.704e+004	2.704e+004	2.705e+004	
37	Elevation	(m)	-0.5060	-0.4840	-0.4620	-0.4400	
38	Pressure	(bar)	67.88	67.88	67.88	67.88	
39	Temperature	(C)	47.86	47.86	47.86	47.86	
40	Vapour Fraction		0.8807	0.8807	0.8807	0.8807	
41	Heat Transferred	(kJ/h-m)	475.8	475.8	475.8	475.8	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2613	0.2613	0.2613	0.2613	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	7.769e-003	7.769e-003	7.769e-003	7.769e-003	
47	Static Gradient	(kPa/m)	2.381e-002	2.381e-002	2.381e-002	2.381e-002	
48	Accel. Gradient	(kPa/m)	4.022e-006	4.022e-006	4.022e-006	4.022e-006	
49	Liq. Reynolds		1.837e+005 *	1.837e+005 *	1.837e+005 *	1.837e+005 *	
50	Vap. Reynolds		9.882e+006 *	9.882e+006 *	9.882e+006 *	9.882e+006 *	
51	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
52	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	2.705e+004	2.705e+004	2.705e+004	2.706e+004	
56	Elevation	(m)	-0.4500	-0.4600	-0.4700	-0.4800	
57	Pressure	(bar)	67.88	67.88	67.88	67.88	
58	Temperature	(C)	47.86	47.86	47.86	47.86	
59	Vapour Fraction		0.8807	0.8807	0.8807	0.8807	
60	Heat Transferred	(kJ/h-m)	475.8	475.8	475.7	475.7	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.1687	0.1687	0.1687	0.1687	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	9.091e-003	9.091e-003	9.091e-003	9.091e-003	
66	Static Gradient	(kPa/m)	-5.281e-003	-5.281e-003	-5.281e-003	-5.281e-003	
67	Accel. Gradient	(kPa/m)	4.852e-007	4.852e-007	4.852e-007	4.852e-007	
68	Liq. Reynolds		1.837e+005 *	1.837e+005 *	1.837e+005 *	1.837e+005 *	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 439 of 937	

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.882e+006 *	9.883e+006 *	9.883e+006 *	9.883e+006 *
14	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
15	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	2.706e+004	2.707e+004	2.707e+004	2.708e+004
19	Elevation (m)	-0.4900	-0.5220	-0.5540	-0.5860
20	Pressure (bar)	67.88	67.88	67.88	67.88
21	Temperature (C)	47.86	47.85	47.85	47.85
22	Vapour Fraction	0.8807	0.8807	0.8807	0.8807
23	Heat Transferred (kJ/h-m)	475.7	475.7	475.6	475.6
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1687	0.1665	0.1665	0.1665
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	9.091e-003	9.122e-003	9.122e-003	9.122e-003
29	Static Gradient (kPa/m)	-5.281e-003	-1.005e-002	-1.005e-002	-1.005e-002
30	Accel. Gradient (kPa/m)	4.852e-007	-1.181e-007	-1.181e-007	-1.181e-007
31	Liq. Reynolds	1.837e+005 *	1.837e+005 *	1.837e+005 *	1.837e+005 *
32	Vap. Reynolds	9.883e+006 *	9.883e+006 *	9.883e+006 *	9.883e+006 *
33	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
34	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	2.708e+004	2.709e+004	2.709e+004	2.710e+004
38	Elevation (m)	-0.6180	-0.6500	-0.6400	-0.6300
39	Pressure (bar)	67.88	67.88	67.88	67.88
40	Temperature (C)	47.85	47.85	47.85	47.85
41	Vapour Fraction	0.8807	0.8807	0.8807	0.8807
42	Heat Transferred (kJ/h-m)	475.6	475.5	475.4	475.4
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1665	0.1665	0.2364	0.2364
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	9.122e-003	9.122e-003	7.762e-003	7.762e-003
48	Static Gradient (kPa/m)	-1.005e-002	-1.005e-002	4.017e-003	4.017e-003
49	Accel. Gradient (kPa/m)	-1.182e-007	-1.182e-007	1.500e-006	1.500e-006
50	Liq. Reynolds	1.837e+005 *	1.837e+005 *	1.837e+005 *	1.837e+005 *
51	Vap. Reynolds	9.883e+006 *	9.883e+006 *	9.883e+006 *	9.883e+006 *
52	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
53	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	2.710e+004	2.711e+004	2.711e+004	2.712e+004
57	Elevation (m)	-0.6200	-0.6100	-0.6000	-0.5440
58	Pressure (bar)	67.88	67.88	67.88	67.87
59	Temperature (C)	47.84	47.84	47.84	47.84
60	Vapour Fraction	0.8807	0.8807	0.8807	0.8807
61	Heat Transferred (kJ/h-m)	475.3	475.3	475.3	475.2
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2364	0.2364	0.2364	0.2619
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	7.762e-003	7.762e-003	7.762e-003	7.785e-003
67	Static Gradient (kPa/m)	4.017e-003	4.017e-003	4.017e-003	2.428e-002
68	Accel. Gradient (kPa/m)	1.500e-006	1.500e-006	1.500e-006	4.085e-006
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 440 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3							Unit Set: Mixed
4							Date/Time: Mon Jun 13 15:13:43 2022
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Liq. Reynolds		1.837e+005 *	1.837e+005 *	1.837e+005 *	1.837e+005 *	
14	Vap. Reynolds		9.883e+006 *	9.883e+006 *	9.883e+006 *	9.883e+006 *	
15	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979	
16	Vapour Velocity (m/s)		2.932	2.932	2.932	2.932	
17	Deposit Thickness (mm)		---	---	---	---	
18	Deposit Volume (m3)		---	---	---	---	
19	Length (m)		2.712e+004	2.713e+004	2.713e+004	2.714e+004	
20	Elevation (m)		-0.4880	-0.4320	-0.3760	-0.3200	
21	Pressure (bar)		67.87	67.87	67.87	67.87	
22	Temperature (C)		47.84	47.84	47.83	47.83	
23	Vapour Fraction		0.8807	0.8807	0.8807	0.8807	
24	Heat Transferred (kJ/h-m)		475.2	475.1	475.1	475.0	
25	Flow Regime		Transition	Transition	Transition	Transition	
26	Water Flow Regime						
27	Liquid HoldUp		0.2619	0.2619	0.2619	0.2619	
28	Water HoldUp		---	---	---	---	
29	Friction Gradient (kPa/m)		7.785e-003	7.785e-003	7.785e-003	7.785e-003	
30	Static Gradient (kPa/m)		2.428e-002	2.428e-002	2.428e-002	2.428e-002	
31	Accel. Gradient (kPa/m)		4.085e-006	4.085e-006	4.085e-006	4.085e-006	
32	Liq. Reynolds		1.837e+005 *	1.837e+005 *	1.837e+005 *	1.837e+005 *	
33	Vap. Reynolds		9.883e+006 *	9.883e+006 *	9.883e+006 *	9.883e+006 *	
34	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979	
35	Vapour Velocity (m/s)		2.932	2.932	2.932	2.932	
36	Deposit Thickness (mm)		---	---	---	---	
37	Deposit Volume (m3)		---	---	---	---	
38	Length (m)		2.714e+004	2.715e+004	2.715e+004	2.716e+004	
39	Elevation (m)		-0.2820	-0.2440	-0.2060	-0.1680	
40	Pressure (bar)		67.87	67.86	67.86	67.86	
41	Temperature (C)		47.83	47.83	47.83	47.82	
42	Vapour Fraction		0.8807	0.8807	0.8807	0.8807	
43	Heat Transferred (kJ/h-m)		475.0	474.9	474.9	474.8	
44	Flow Regime		Transition	Transition	Transition	Transition	
45	Water Flow Regime						
46	Liquid HoldUp		0.2519	0.2519	0.2519	0.2519	
47	Water HoldUp		---	---	---	---	
48	Friction Gradient (kPa/m)		6.374e-003	6.374e-003	6.374e-003	6.374e-003	
49	Static Gradient (kPa/m)		1.600e-002	1.600e-002	1.600e-002	1.600e-002	
50	Accel. Gradient (kPa/m)		2.851e-006	2.851e-006	2.851e-006	2.851e-006	
51	Liq. Reynolds		1.837e+005 *	1.837e+005 *	1.837e+005 *	1.837e+005 *	
52	Vap. Reynolds		9.883e+006 *	9.883e+006 *	9.883e+006 *	9.883e+006 *	
53	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979	
54	Vapour Velocity (m/s)		2.932	2.932	2.932	2.932	
55	Deposit Thickness (mm)		---	---	---	---	
56	Deposit Volume (m3)		---	---	---	---	
57	Length (m)		2.716e+004	2.717e+004	2.717e+004	2.718e+004	
58	Elevation (m)		-0.1300	-0.1120	-9.400e-002	-7.600e-002	
59	Pressure (bar)		67.86	67.86	67.86	67.86	
60	Temperature (C)		47.82	47.82	47.82	47.82	
61	Vapour Fraction		0.8807	0.8807	0.8807	0.8807	
62	Heat Transferred (kJ/h-m)		474.8	474.7	474.7	474.6	
63	Flow Regime		Transition	Transition	Transition	Transition	
64	Water Flow Regime						
65	Liquid HoldUp		0.2519	0.2409	0.2409	0.2409	
66	Water HoldUp		---	---	---	---	
67	Friction Gradient (kPa/m)		6.374e-003	7.217e-003	7.217e-003	7.217e-003	
68	Static Gradient (kPa/m)		1.600e-002	7.330e-003	7.330e-003	7.330e-003	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 441 of 937	

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	2.851e-006	1.854e-006	1.854e-006	1.854e-006
14	Liq. Reynolds	1.837e+005 *	1.837e+005 *	1.837e+005 *	1.836e+005 *
15	Vap. Reynolds	9.883e+006 *	9.883e+006 *	9.883e+006 *	9.883e+006 *
16	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
17	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	2.718e+004	2.719e+004	2.719e+004	2.720e+004
21	Elevation (m)	-5.800e-002	-4.000e-002	-2.400e-002	-8.000e-003
22	Pressure (bar)	67.86	67.86	67.86	67.86
23	Temperature (C)	47.82	47.81	47.81	47.81
24	Vapour Fraction	0.8807	0.8807	0.8807	0.8807
25	Heat Transferred (kJ/h-m)	474.6	474.5	474.5	474.5
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.2409	0.2409	0.2397	0.2397
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	7.217e-003	7.217e-003	7.351e-003	7.351e-003
31	Static Gradient (kPa/m)	7.330e-003	7.330e-003	6.494e-003	6.494e-003
32	Accel. Gradient (kPa/m)	1.854e-006	1.854e-006	1.764e-006	1.764e-006
33	Liq. Reynolds	1.836e+005 *	1.836e+005 *	1.836e+005 *	1.836e+005 *
34	Vap. Reynolds	9.883e+006 *	9.883e+006 *	9.883e+006 *	9.883e+006 *
35	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
36	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	2.720e+004	2.721e+004	2.721e+004	2.722e+004
40	Elevation (m)	8.000e-003	2.400e-002	4.000e-002	3.800e-002
41	Pressure (bar)	67.86	67.86	67.85	67.85
42	Temperature (C)	47.81	47.81	47.81	47.80
43	Vapour Fraction	0.8807	0.8807	0.8807	0.8807
44	Heat Transferred (kJ/h-m)	474.4	474.4	474.3	474.3
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.2397	0.2397	0.2397	0.1709
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	7.351e-003	7.351e-003	7.351e-003	9.063e-003
50	Static Gradient (kPa/m)	6.494e-003	6.494e-003	6.494e-003	-6.366e-004
51	Accel. Gradient (kPa/m)	1.764e-006	1.764e-006	1.764e-006	1.074e-006
52	Liq. Reynolds	1.836e+005 *	1.836e+005 *	1.836e+005 *	1.836e+005 *
53	Vap. Reynolds	9.883e+006 *	9.883e+006 *	9.884e+006 *	9.884e+006 *
54	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
55	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	2.722e+004	2.723e+004	2.723e+004	2.724e+004
59	Elevation (m)	3.600e-002	3.400e-002	3.200e-002	3.000e-002
60	Pressure (bar)	67.85	67.85	67.85	67.85
61	Temperature (C)	47.80	47.80	47.80	47.80
62	Vapour Fraction	0.8807	0.8807	0.8807	0.8807
63	Heat Transferred (kJ/h-m)	474.3	474.3	474.2	474.2
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.1709	0.1709	0.1709	0.1709
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	9.063e-003	9.063e-003	9.063e-003	9.063e-003
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 442 of 937

1				Case Name: Simulation - 386MMSCFD_80barg.hsc	
2					
3	 Epic Atlantic Limited			Unit Set: Mixed	
4					
5				Date/Time: Mon Jun 13 15:13:43 2022	
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Static Gradient	(kPa/m)	-6.366e-004	-6.366e-004	-6.366e-004
14	Accel. Gradient	(kPa/m)	1.074e-006	1.074e-006	1.074e-006
15	Liq. Reynolds		1.836e+005 *	1.836e+005 *	1.836e+005 *
16	Vap. Reynolds		9.884e+006 *	9.884e+006 *	9.884e+006 *
17	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979
18	Vapour Velocity	(m/s)	2.932	2.932	2.932
19	Deposit Thickness	(mm)	---	---	---
20	Deposit Volume	(m3)	---	---	---
21	Length	(m)	2.724e+004	2.725e+004	2.725e+004
22	Elevation	(m)	4.000e-002	5.000e-002	6.000e-002
23	Pressure	(bar)	67.85	67.85	67.85
24	Temperature	(C)	47.80	47.80	47.79
25	Vapour Fraction		0.8807	0.8807	0.8807
26	Heat Transferred	(kJ/h-m)	474.1	474.0	474.0
27	Flow Regime		Transition	Transition	Transition
28	Water Flow Regime				
29	Liquid HoldUp		0.2364	0.2364	0.2364
30	Water HoldUp		---	---	---
31	Friction Gradient	(kPa/m)	7.764e-003	7.764e-003	7.764e-003
32	Static Gradient	(kPa/m)	4.017e-003	4.017e-003	4.017e-003
33	Accel. Gradient	(kPa/m)	1.501e-006	1.501e-006	1.501e-006
34	Liq. Reynolds		1.836e+005 *	1.836e+005 *	1.836e+005 *
35	Vap. Reynolds		9.884e+006 *	9.884e+006 *	9.884e+006 *
36	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979
37	Vapour Velocity	(m/s)	2.932	2.932	2.932
38	Deposit Thickness	(mm)	---	---	---
39	Deposit Volume	(m3)	---	---	---
40	Length	(m)	2.726e+004	2.727e+004	2.727e+004
41	Elevation	(m)	8.000e-002	8.000e-002	8.000e-002
42	Pressure	(bar)	67.85	67.85	67.85
43	Temperature	(C)	47.79	47.79	47.79
44	Vapour Fraction		0.8807	0.8807	0.8807
45	Heat Transferred	(kJ/h-m)	473.9	473.9	473.8
46	Flow Regime		Transition	Transition	Transition
47	Water Flow Regime				
48	Liquid HoldUp		0.2364	0.2499	0.2499
49	Water HoldUp		---	---	---
50	Friction Gradient	(kPa/m)	7.764e-003	6.192e-003	6.192e-003
51	Static Gradient	(kPa/m)	4.017e-003	0.0000	0.0000
52	Accel. Gradient	(kPa/m)	1.501e-006	7.892e-007	7.892e-007
53	Liq. Reynolds		1.836e+005 *	1.836e+005 *	1.836e+005 *
54	Vap. Reynolds		9.884e+006 *	9.884e+006 *	9.884e+006 *
55	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979
56	Vapour Velocity	(m/s)	2.932	2.932	2.932
57	Deposit Thickness	(mm)	---	---	---
58	Deposit Volume	(m3)	---	---	---
59	Length	(m)	2.728e+004	2.729e+004	2.729e+004
60	Elevation	(m)	8.000e-002	8.000e-002	8.600e-002
61	Pressure	(bar)	67.85	67.85	67.85
62	Temperature	(C)	47.79	47.78	47.78
63	Vapour Fraction		0.8807	0.8807	0.8807
64	Heat Transferred	(kJ/h-m)	473.8	473.7	473.6
65	Flow Regime		Transition	Transition	Transition
66	Water Flow Regime				
67	Liquid HoldUp		0.2499	0.2499	0.2342
68	Water HoldUp		---	---	---
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 443 of 937				

1				Case Name:	Simulation - 386MMSCFD_80barg.hsc	
2				Unit Set:	Mixed	
3		Epic Atlantic Limited				
Date/Time:			Mon Jun 13 15:13:43 2022			
4						
5						
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	6.192e-003	6.192e-003	8.048e-003	8.048e-003
14	Static Gradient	(kPa/m)	0.0000	0.0000	2.394e-003	2.394e-003
15	Accel. Gradient	(kPa/m)	7.892e-007	7.892e-007	1.331e-006	1.331e-006
16	Liq. Reynolds		1.836e+005 *	1.836e+005 *	1.836e+005 *	1.836e+005 *
17	Vap. Reynolds		9.884e+006 *	9.884e+006 *	9.884e+006 *	9.884e+006 *
18	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
19	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	2.730e+004	2.731e+004	2.731e+004	2.732e+004
23	Elevation	(m)	9.800e-002	0.1040	0.1100	0.1100
24	Pressure	(bar)	67.85	67.85	67.85	67.84
25	Temperature	(C)	47.78	47.78	47.78	47.77
26	Vapour Fraction		0.8807	0.8807	0.8807	0.8807
27	Heat Transferred	(kJ/h-m)	473.6	473.6	473.5	473.5
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2342	0.2342	0.2342	0.2499
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	8.048e-003	8.048e-003	8.048e-003	6.192e-003
33	Static Gradient	(kPa/m)	2.394e-003	2.394e-003	2.394e-003	0.0000
34	Accel. Gradient	(kPa/m)	1.331e-006	1.331e-006	1.331e-006	7.892e-007
35	Liq. Reynolds		1.836e+005 *	1.836e+005 *	1.836e+005 *	1.836e+005 *
36	Vap. Reynolds		9.884e+006 *	9.884e+006 *	9.884e+006 *	9.884e+006 *
37	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
38	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	2.732e+004	2.733e+004	2.733e+004	2.734e+004
42	Elevation	(m)	0.1100	0.1100	0.1100	0.1100
43	Pressure	(bar)	67.84	67.84	67.84	67.84
44	Temperature	(C)	47.77	47.77	47.77	47.77
45	Vapour Fraction		0.8807	0.8807	0.8807	0.8807
46	Heat Transferred	(kJ/h-m)	473.4	473.4	473.4	473.3
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2499	0.2499	0.2499	0.2499
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	6.192e-003	6.192e-003	6.192e-003	6.192e-003
52	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000
53	Accel. Gradient	(kPa/m)	7.892e-007	7.892e-007	7.892e-007	7.892e-007
54	Liq. Reynolds		1.836e+005 *	1.836e+005 *	1.836e+005 *	1.836e+005 *
55	Vap. Reynolds		9.884e+006 *	9.884e+006 *	9.884e+006 *	9.884e+006 *
56	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
57	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	2.734e+004	2.735e+004	2.735e+004	2.736e+004
61	Elevation	(m)	0.1140	0.1180	0.1220	0.1260
62	Pressure	(bar)	67.84	67.84	67.84	67.84
63	Temperature	(C)	47.77	47.77	47.76	47.76
64	Vapour Fraction		0.8807	0.8807	0.8807	0.8807
65	Heat Transferred	(kJ/h-m)	473.3	473.2	473.2	473.2
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2331	0.2331	0.2331	0.2331
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 444 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	8.194e-003	8.194e-003	8.194e-003	8.194e-003
15	Static Gradient (kPa/m)	1.590e-003	1.590e-003	1.590e-003	1.590e-003
16	Accel. Gradient (kPa/m)	1.247e-006	1.247e-006	1.247e-006	1.247e-006
17	Liq. Reynolds	1.836e+005 *	1.836e+005 *	1.836e+005 *	1.836e+005 *
18	Vap. Reynolds	9.884e+006 *	9.884e+006 *	9.884e+006 *	9.884e+006 *
19	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
20	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	2.736e+004	2.737e+004	2.737e+004	2.738e+004
24	Elevation (m)	0.1300	0.1340	0.1380	0.1420
25	Pressure (bar)	67.84	67.84	67.84	67.84
26	Temperature (C)	47.76	47.76	47.76	47.76
27	Vapour Fraction	0.8807	0.8807	0.8807	0.8807
28	Heat Transferred (kJ/h-m)	473.1	473.1	473.0	473.0
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2331	0.2331	0.2331	0.2331
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	8.194e-003	8.194e-003	8.194e-003	8.194e-003
34	Static Gradient (kPa/m)	1.590e-003	1.590e-003	1.590e-003	1.590e-003
35	Accel. Gradient (kPa/m)	1.247e-006	1.247e-006	1.247e-006	1.247e-006
36	Liq. Reynolds	1.836e+005 *	1.836e+005 *	1.836e+005 *	1.836e+005 *
37	Vap. Reynolds	9.884e+006 *	9.884e+006 *	9.884e+006 *	9.884e+006 *
38	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
39	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	2.738e+004	2.739e+004	2.739e+004	2.740e+004
43	Elevation (m)	0.1460	0.1500	0.1480	0.1460
44	Pressure (bar)	67.84	67.84	67.84	67.84
45	Temperature (C)	47.76	47.75	47.75	47.75
46	Vapour Fraction	0.8807	0.8807	0.8807	0.8807
47	Heat Transferred (kJ/h-m)	473.0	472.9	472.9	472.9
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2331	0.2331	0.1709	0.1709
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	8.194e-003	8.194e-003	9.063e-003	9.063e-003
53	Static Gradient (kPa/m)	1.590e-003	1.590e-003	-6.392e-004	-6.392e-004
54	Accel. Gradient (kPa/m)	1.247e-006	1.247e-006	1.074e-006	1.074e-006
55	Liq. Reynolds	1.836e+005 *	1.836e+005 *	1.836e+005 *	1.836e+005 *
56	Vap. Reynolds	9.884e+006 *	9.884e+006 *	9.884e+006 *	9.884e+006 *
57	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
58	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	2.740e+004	2.741e+004	2.741e+004	2.742e+004
62	Elevation (m)	0.1440	0.1420	0.1400	0.1420
63	Pressure (bar)	67.84	67.84	67.84	67.84
64	Temperature (C)	47.75	47.75	47.75	47.74
65	Vapour Fraction	0.8807	0.8807	0.8807	0.8806
66	Heat Transferred (kJ/h-m)	472.8	472.8	472.8	472.7
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 445 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.1709	0.1709	0.1709	0.2320
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	9.063e-003	9.063e-003	9.063e-003	8.344e-003
16	Static Gradient (kPa/m)	-6.392e-004	-6.392e-004	-6.392e-004	7.767e-004
17	Accel. Gradient (kPa/m)	1.074e-006	1.074e-006	1.074e-006	1.163e-006
18	Liq. Reynolds	1.836e+005 *	1.836e+005 *	1.836e+005 *	1.836e+005 *
19	Vap. Reynolds	9.884e+006 *	9.884e+006 *	9.884e+006 *	9.884e+006 *
20	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
21	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	2.742e+004	2.743e+004	2.743e+004	2.744e+004
25	Elevation (m)	0.1440	0.1460	0.1480	0.1500
26	Pressure (bar)	67.84	67.84	67.83	67.83
27	Temperature (C)	47.74	47.74	47.74	47.74
28	Vapour Fraction	0.8806	0.8806	0.8806	0.8806
29	Heat Transferred (kJ/h-m)	472.6	472.6	472.6	472.5
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2320	0.2320	0.2320	0.2320
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	8.344e-003	8.344e-003	8.344e-003	8.344e-003
35	Static Gradient (kPa/m)	7.767e-004	7.767e-004	7.768e-004	7.768e-004
36	Accel. Gradient (kPa/m)	1.163e-006	1.163e-006	1.163e-006	1.163e-006
37	Liq. Reynolds	1.836e+005 *	1.836e+005 *	1.836e+005 *	1.836e+005 *
38	Vap. Reynolds	9.884e+006 *	9.884e+006 *	9.884e+006 *	9.884e+006 *
39	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
40	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	2.744e+004	2.745e+004	2.745e+004	2.746e+004
44	Elevation (m)	0.1520	0.1540	0.1560	0.1580
45	Pressure (bar)	67.83	67.83	67.83	67.83
46	Temperature (C)	47.74	47.74	47.73	47.73
47	Vapour Fraction	0.8806	0.8806	0.8806	0.8806
48	Heat Transferred (kJ/h-m)	472.5	472.4	472.4	472.4
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2320	0.2320	0.2320	0.2320
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	8.341e-003	8.341e-003	8.341e-003	8.341e-003
54	Static Gradient (kPa/m)	7.923e-004	7.923e-004	7.923e-004	7.923e-004
55	Accel. Gradient (kPa/m)	1.164e-006	1.164e-006	1.164e-006	1.164e-006
56	Liq. Reynolds	1.836e+005 *	1.836e+005 *	1.836e+005 *	1.836e+005 *
57	Vap. Reynolds	9.885e+006 *	9.885e+006 *	9.885e+006 *	9.885e+006 *
58	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
59	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	2.746e+004	2.747e+004	2.747e+004	2.748e+004
63	Elevation (m)	0.1600	0.1600	0.1600	0.1600
64	Pressure (bar)	67.83	67.83	67.83	67.83
65	Temperature (C)	47.73	47.73	47.73	47.73
66	Vapour Fraction	0.8806	0.8806	0.8806	0.8806
67	Heat Transferred (kJ/h-m)	472.3	472.3	472.2	472.2
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 446 of 937

1			Case Name:	Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set:	Mixed		
3	 Epic Atlantic Limited		Date/Time:	Mon Jun 13 15:13:43 2022		
4						
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Water Flow Regime					
14	Liquid HoldUp	0.2320	0.2499	0.2499	0.2499	
15	Water HoldUp	---	---	---	---	
16	Friction Gradient (kPa/m)	8.341e-003	6.192e-003	6.192e-003	6.192e-003	
17	Static Gradient (kPa/m)	7.923e-004	0.0000	0.0000	0.0000	
18	Accel. Gradient (kPa/m)	1.164e-006	7.893e-007	7.893e-007	7.893e-007	
19	Liq. Reynolds	1.835e+005 *	1.835e+005 *	1.835e+005 *	1.835e+005 *	
20	Vap. Reynolds	9.885e+006 *	9.885e+006 *	9.885e+006 *	9.885e+006 *	
21	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979	
22	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932	
23	Deposit Thickness (mm)	---	---	---	---	
24	Deposit Volume (m3)	---	---	---	---	
25	Length (m)	2.748e+004	2.749e+004	2.749e+004	2.749e+004	
26	Elevation (m)	0.1600	0.1600	0.1600	0.1600	
27	Pressure (bar)	67.83	67.83	67.83	67.83	
28	Temperature (C)	47.73	47.72	47.72	47.72	
29	Vapour Fraction	0.8806	0.8806	0.8806	0.8806	
30	Heat Transferred (kJ/h-m)	472.2	472.1	472.1	472.1	
31	Flow Regime	Transition	Transition	Transition	Transition	
32	Water Flow Regime					
33	Liquid HoldUp	0.2499	0.2499	0.2499	0.2499	
34	Water HoldUp	---	---	---	---	
35	Friction Gradient (kPa/m)	6.192e-003	6.192e-003	6.192e-003	6.192e-003	
36	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000	
37	Accel. Gradient (kPa/m)	7.893e-007	7.893e-007	7.893e-007	7.893e-007	
38	Liq. Reynolds	1.835e+005 *	1.835e+005 *	1.835e+005 *	1.835e+005 *	
39	Vap. Reynolds	9.885e+006 *	9.885e+006 *	9.885e+006 *	9.885e+006 *	
40	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979	
41	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932	
42	Deposit Thickness (mm)	---	---	---	---	
43	Deposit Volume (m3)	---	---	---	---	
44	Length (m)	2.749e+004	2.749e+004	2.750e+004	2.750e+004	
45	Elevation (m)	0.1600	0.1600	0.1600	0.1580	
46	Pressure (bar)	67.83	67.83	67.83	67.83	
47	Temperature (C)	47.72	47.72	47.72	47.72	
48	Vapour Fraction	0.8806	0.8806	0.8806	0.8806	
49	Heat Transferred (kJ/h-m)	472.1	472.1	472.0	472.0	
50	Flow Regime	Transition	Transition	Transition	Transition	
51	Water Flow Regime					
52	Liquid HoldUp	0.2499	0.2499	0.2499	0.1709	
53	Water HoldUp	---	---	---	---	
54	Friction Gradient (kPa/m)	6.192e-003	6.192e-003	6.192e-003	9.062e-003	
55	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-5.328e-004	
56	Accel. Gradient (kPa/m)	7.893e-007	7.893e-007	7.893e-007	1.087e-006	
57	Liq. Reynolds	1.835e+005 *	1.835e+005 *	1.835e+005 *	1.835e+005 *	
58	Vap. Reynolds	9.885e+006 *	9.885e+006 *	9.885e+006 *	9.885e+006 *	
59	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979	
60	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932	
61	Deposit Thickness (mm)	---	---	---	---	
62	Deposit Volume (m3)	---	---	---	---	
63	Length (m)	2.751e+004	2.751e+004	2.752e+004	2.753e+004	
64	Elevation (m)	0.1560	0.1540	0.1520	0.1500	
65	Pressure (bar)	67.83	67.83	67.83	67.83	
66	Temperature (C)	47.72	47.72	47.71	47.71	
67	Vapour Fraction	0.8806	0.8806	0.8806	0.8806	
68	Heat Transferred (kJ/h-m)	472.0	471.9	471.9	471.8	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.1709	0.1709	0.1709	0.1709
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	9.062e-003	9.062e-003	9.062e-003	9.062e-003
18	Static Gradient (kPa/m)	-5.328e-004	-5.328e-004	-5.328e-004	-5.328e-004
19	Accel. Gradient (kPa/m)	1.087e-006	1.087e-006	1.087e-006	1.087e-006
20	Liq. Reynolds	1.835e+005 *	1.835e+005 *	1.835e+005 *	1.835e+005 *
21	Vap. Reynolds	9.885e+006 *	9.885e+006 *	9.885e+006 *	9.885e+006 *
22	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
23	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	2.753e+004	2.753e+004	2.753e+004	2.753e+004
27	Elevation (m)	0.1560	0.1620	0.1680	0.1740
28	Pressure (bar)	67.83	67.83	67.83	67.83
29	Temperature (C)	47.71	47.71	47.71	47.71
30	Vapour Fraction	0.8806	0.8806	0.8806	0.8806
31	Heat Transferred (kJ/h-m)	471.8	471.8	471.8	471.7
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2392	0.2392	0.2392	0.2392
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	7.419e-003	7.419e-003	7.419e-003	7.419e-003
37	Static Gradient (kPa/m)	6.078e-003	6.078e-003	6.078e-003	6.078e-003
38	Accel. Gradient (kPa/m)	1.721e-006	1.721e-006	1.721e-006	1.721e-006
39	Liq. Reynolds	1.835e+005 *	1.835e+005 *	1.835e+005 *	1.835e+005 *
40	Vap. Reynolds	9.885e+006 *	9.885e+006 *	9.885e+006 *	9.885e+006 *
41	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
42	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	2.754e+004	2.754e+004	2.755e+004	2.755e+004
46	Elevation (m)	0.1800	0.1800	0.1800	0.1800
47	Pressure (bar)	67.83	67.83	67.83	67.82
48	Temperature (C)	47.71	47.71	47.71	47.70
49	Vapour Fraction	0.8806	0.8806	0.8806	0.8806
50	Heat Transferred (kJ/h-m)	471.7	471.7	471.6	471.6
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2392	0.2499	0.2499	0.2499
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	7.419e-003	6.192e-003	6.192e-003	6.192e-003
56	Static Gradient (kPa/m)	6.078e-003	0.0000	0.0000	0.0000
57	Accel. Gradient (kPa/m)	1.721e-006	7.894e-007	7.894e-007	7.894e-007
58	Liq. Reynolds	1.835e+005 *	1.835e+005 *	1.835e+005 *	1.835e+005 *
59	Vap. Reynolds	9.885e+006 *	9.885e+006 *	9.885e+006 *	9.885e+006 *
60	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
61	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	2.756e+004	2.756e+004	2.757e+004	2.757e+004
65	Elevation (m)	0.1800	0.1800	0.3040	0.4280
66	Pressure (bar)	67.82	67.82	67.82	67.82
67	Temperature (C)	47.70	47.70	47.70	47.70
68	Vapour Fraction	0.8806	0.8806	0.8806	0.8806
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:13:43 2022		
6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Heat Transferred (kJ/h-m)	471.6	471.5	471.5	471.4		
14	Flow Regime	Transition	Transition	Transition	Transition		
15	Water Flow Regime						
16	Liquid HoldUp	0.2499	0.2499	0.2994	0.2994		
17	Water HoldUp	---	---	---	---		
18	Friction Gradient (kPa/m)	6.192e-003	6.192e-003	7.932e-003	7.932e-003		
19	Static Gradient (kPa/m)	0.0000	0.0000	5.960e-002	5.960e-002		
20	Accel. Gradient (kPa/m)	7.894e-007	7.894e-007	8.610e-006	8.611e-006		
21	Liq. Reynolds	1.835e+005 *	1.835e+005 *	1.835e+005 *	1.835e+005 *		
22	Vap. Reynolds	9.885e+006 *	9.885e+006 *	9.885e+006 *	9.885e+006 *		
23	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
24	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932		
25	Deposit Thickness (mm)	---	---	---	---		
26	Deposit Volume (m3)	---	---	---	---		
27	Length (m)	2.758e+004	2.758e+004	2.759e+004	2.759e+004		
28	Elevation (m)	0.5520	0.6760	0.8000	0.5740		
29	Pressure (bar)	67.81	67.81	67.81	67.81		
30	Temperature (C)	47.69	47.69	47.69	47.69		
31	Vapour Fraction	0.8806	0.8806	0.8806	0.8806		
32	Heat Transferred (kJ/h-m)	471.3	471.3	471.2	471.3		
33	Flow Regime	Transition	Transition	Transition	Transition		
34	Water Flow Regime						
35	Liquid HoldUp	0.2994	0.2994	0.2994	0.1379		
36	Water HoldUp	---	---	---	---		
37	Friction Gradient (kPa/m)	7.932e-003	7.933e-003	7.933e-003	9.652e-003		
38	Static Gradient (kPa/m)	5.960e-002	5.960e-002	5.960e-002	-6.288e-002		
39	Accel. Gradient (kPa/m)	8.612e-006	8.612e-006	8.613e-006	-6.789e-006		
40	Liq. Reynolds	1.835e+005 *	1.835e+005 *	1.835e+005 *	1.835e+005 *		
41	Vap. Reynolds	9.885e+006 *	9.885e+006 *	9.886e+006 *	9.886e+006 *		
42	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
43	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933		
44	Deposit Thickness (mm)	---	---	---	---		
45	Deposit Volume (m3)	---	---	---	---		
46	Length (m)	2.760e+004	2.760e+004	2.761e+004	2.761e+004		
47	Elevation (m)	0.3480	0.1220	-0.1040	-0.3300		
48	Pressure (bar)	67.81	67.82	67.82	67.82		
49	Temperature (C)	47.69	47.69	47.69	47.69		
50	Vapour Fraction	0.8806	0.8806	0.8806	0.8806		
51	Heat Transferred (kJ/h-m)	471.3	471.3	471.3	471.3		
52	Flow Regime	Transition	Transition	Transition	Transition		
53	Water Flow Regime						
54	Liquid HoldUp	0.1379	0.1379	0.1379	0.1379		
55	Water HoldUp	---	---	---	---		
56	Friction Gradient (kPa/m)	9.651e-003	9.651e-003	9.651e-003	9.650e-003		
57	Static Gradient (kPa/m)	-6.289e-002	-6.289e-002	-6.289e-002	-6.289e-002		
58	Accel. Gradient (kPa/m)	-6.788e-006	-6.788e-006	-6.788e-006	-6.788e-006		
59	Liq. Reynolds	1.835e+005 *	1.835e+005 *	1.835e+005 *	1.835e+005 *		
60	Vap. Reynolds	9.885e+006 *	9.885e+006 *	9.885e+006 *	9.885e+006 *		
61	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
62	Vapour Velocity (m/s)	2.933	2.932	2.932	2.932		
63	Deposit Thickness (mm)	---	---	---	---		
64	Deposit Volume (m3)	---	---	---	---		
65	Length (m)	2.762e+004	2.762e+004	2.763e+004	2.763e+004		
66	Elevation (m)	-0.1740	-1.800e-002	0.1380	0.2940		
67	Pressure (bar)	67.82	67.81	67.81	67.80		
68	Temperature (C)	47.69	47.68	47.68	47.68		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Vapour Fraction		0.8806	0.8806	0.8806	0.8806				
14	Heat Transferred (kJ/h-m)		471.1	471.0	471.0	470.9				
15	Flow Regime		Transition	Transition	Transition	Transition				
16	Water Flow Regime									
17	Liquid HoldUp		0.3188	0.3188	0.3188	0.3188				
18	Water HoldUp		---	---	---	---				
19	Friction Gradient (kPa/m)		7.891e-003	7.892e-003	7.892e-003	7.893e-003				
20	Static Gradient (kPa/m)		8.034e-002	8.034e-002	8.034e-002	8.033e-002				
21	Accel. Gradient (kPa/m)		1.125e-005	1.125e-005	1.125e-005	1.125e-005				
22	Liq. Reynolds		1.835e+005 *	1.835e+005 *	1.835e+005 *	1.835e+005 *				
23	Vap. Reynolds		9.885e+006 *	9.885e+006 *	9.886e+006 *	9.886e+006 *				
24	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979				
25	Vapour Velocity (m/s)		2.932	2.933	2.933	2.933				
26	Deposit Thickness (mm)		---	---	---	---				
27	Deposit Volume (m3)		---	---	---	---				
28	Length (m)		2.764e+004	2.764e+004	2.765e+004	2.765e+004				
29	Elevation (m)		0.4500	0.5140	0.5780	0.6420				
30	Pressure (bar)		67.80	67.80	67.80	67.79				
31	Temperature (C)		47.67	47.67	47.67	47.67				
32	Vapour Fraction		0.8806	0.8806	0.8806	0.8806				
33	Heat Transferred (kJ/h-m)		470.8	470.7	470.7	470.6				
34	Flow Regime		Transition	Transition	Transition	Transition				
35	Water Flow Regime									
36	Liquid HoldUp		0.3188	0.2662	0.2662	0.2662				
37	Water HoldUp		---	---	---	---				
38	Friction Gradient (kPa/m)		7.893e-003	7.874e-003	7.874e-003	7.874e-003				
39	Static Gradient (kPa/m)		8.033e-002	2.810e-002	2.810e-002	2.810e-002				
40	Accel. Gradient (kPa/m)		1.125e-005	4.589e-006	4.590e-006	4.590e-006				
41	Liq. Reynolds		1.835e+005 *	1.835e+005 *	1.835e+005 *	1.835e+005 *				
42	Vap. Reynolds		9.886e+006 *	9.886e+006 *	9.886e+006 *	9.886e+006 *				
43	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979				
44	Vapour Velocity (m/s)		2.933	2.933	2.933	2.933				
45	Deposit Thickness (mm)		---	---	---	---				
46	Deposit Volume (m3)		---	---	---	---				
47	Length (m)		2.766e+004	2.766e+004	2.767e+004	2.767e+004				
48	Elevation (m)		0.7060	0.7700	0.6520	0.5340				
49	Pressure (bar)		67.79	67.79	67.79	67.79				
50	Temperature (C)		47.67	47.66	47.66	47.66				
51	Vapour Fraction		0.8806	0.8806	0.8806	0.8806				
52	Heat Transferred (kJ/h-m)		470.6	470.5	470.6	470.6				
53	Flow Regime		Transition	Transition	Transition	Transition				
54	Water Flow Regime									
55	Liquid HoldUp		0.2662	0.2662	0.1537	0.1537				
56	Water HoldUp		---	---	---	---				
57	Friction Gradient (kPa/m)		7.874e-003	7.874e-003	9.329e-003	9.329e-003				
58	Static Gradient (kPa/m)		2.810e-002	2.810e-002	-3.517e-002	-3.518e-002				
59	Accel. Gradient (kPa/m)		4.590e-006	4.590e-006	-3.298e-006	-3.298e-006				
60	Liq. Reynolds		1.835e+005 *	1.835e+005 *	1.835e+005 *	1.835e+005 *				
61	Vap. Reynolds		9.886e+006 *	9.886e+006 *	9.886e+006 *	9.886e+006 *				
62	Liquid Velocity (m/s)		0.1979	0.1979	0.1979	0.1979				
63	Vapour Velocity (m/s)		2.933	2.933	2.933	2.933				
64	Deposit Thickness (mm)		---	---	---	---				
65	Deposit Volume (m3)		---	---	---	---				
66	Length (m)		2.768e+004	2.768e+004	2.769e+004	2.769e+004				
67	Elevation (m)		0.4160	0.2980	0.1800	0.1800				
68	Pressure (bar)		67.79	67.80	67.80	67.80				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	47.66	47.66	47.66	47.66
14	Vapour Fraction	0.8806	0.8806	0.8806	0.8806
15	Heat Transferred (kJ/h-m)	470.5	470.5	470.5	470.4
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.1537	0.1537	0.1537	0.2499
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	9.329e-003	9.329e-003	9.328e-003	6.194e-003
21	Static Gradient (kPa/m)	-3.518e-002	-3.518e-002	-3.518e-002	0.0000
22	Accel. Gradient (kPa/m)	-3.298e-006	-3.298e-006	-3.297e-006	7.902e-007
23	Liq. Reynolds	1.835e+005 *	1.835e+005 *	1.835e+005 *	1.835e+005 *
24	Vap. Reynolds	9.886e+006 *	9.886e+006 *	9.886e+006 *	9.886e+006 *
25	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
26	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	2.770e+004	2.770e+004	2.771e+004	2.771e+004
30	Elevation (m)	0.1800	0.1800	0.1800	0.1800
31	Pressure (bar)	67.80	67.80	67.80	67.80
32	Temperature (C)	47.66	47.66	47.65	47.65
33	Vapour Fraction	0.8806	0.8806	0.8806	0.8806
34	Heat Transferred (kJ/h-m)	470.4	470.3	470.3	470.2
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2499	0.2499	0.2499	0.2499
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	6.194e-003	6.194e-003	6.194e-003	6.194e-003
40	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
41	Accel. Gradient (kPa/m)	7.902e-007	7.902e-007	7.902e-007	7.902e-007
42	Liq. Reynolds	1.835e+005 *	1.835e+005 *	1.834e+005 *	1.834e+005 *
43	Vap. Reynolds	9.886e+006 *	9.886e+006 *	9.886e+006 *	9.886e+006 *
44	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
45	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	2.772e+004	2.772e+004	2.773e+004	2.773e+004
49	Elevation (m)	0.1820	0.1840	0.1860	0.1880
50	Pressure (bar)	67.79	67.79	67.79	67.79
51	Temperature (C)	47.65	47.65	47.65	47.65
52	Vapour Fraction	0.8806	0.8806	0.8806	0.8806
53	Heat Transferred (kJ/h-m)	470.2	470.2	470.1	470.1
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2320	0.2320	0.2320	0.2320
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	8.344e-003	8.344e-003	8.344e-003	8.344e-003
59	Static Gradient (kPa/m)	7.923e-004	7.923e-004	7.923e-004	7.923e-004
60	Accel. Gradient (kPa/m)	1.166e-006	1.166e-006	1.166e-006	1.166e-006
61	Liq. Reynolds	1.834e+005 *	1.834e+005 *	1.834e+005 *	1.834e+005 *
62	Vap. Reynolds	9.886e+006 *	9.886e+006 *	9.886e+006 *	9.886e+006 *
63	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
64	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	2.774e+004	2.774e+004	2.775e+004	2.775e+004
68	Elevation (m)	0.1900	0.1840	0.1780	0.1720
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Pressure (bar)	67.79	67.79	67.79	67.79					
14	Temperature (C)	47.65	47.64	47.64	47.64					
15	Vapour Fraction	0.8806	0.8806	0.8806	0.8806					
16	Heat Transferred (kJ/h-m)	470.0	470.0	470.0	470.0					
17	Flow Regime	Transition	Transition	Transition	Transition					
18	Water Flow Regime									
19	Liquid HoldUp	0.2320	0.1703	0.1703	0.1703					
20	Water HoldUp	---	---	---	---					
21	Friction Gradient (kPa/m)	8.344e-003	9.073e-003	9.073e-003	9.073e-003					
22	Static Gradient (kPa/m)	7.923e-004	-1.913e-003	-1.913e-003	-1.913e-003					
23	Accel. Gradient (kPa/m)	1.166e-006	9.134e-007	9.134e-007	9.134e-007					
24	Liq. Reynolds	1.834e+005 *	1.834e+005 *	1.834e+005 *	1.834e+005 *					
25	Vap. Reynolds	9.886e+006 *	9.886e+006 *	9.886e+006 *	9.886e+006 *					
26	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
27	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933					
28	Deposit Thickness (mm)	---	---	---	---					
29	Deposit Volume (m3)	---	---	---	---					
30	Length (m)	2.776e+004	2.776e+004	2.777e+004	2.777e+004					
31	Elevation (m)	0.1660	0.1600	0.1620	0.1640					
32	Pressure (bar)	67.79	67.79	67.79	67.79					
33	Temperature (C)	47.64	47.64	47.64	47.64					
34	Vapour Fraction	0.8806	0.8806	0.8806	0.8806					
35	Heat Transferred (kJ/h-m)	469.9	469.9	469.8	469.8					
36	Flow Regime	Transition	Transition	Transition	Transition					
37	Water Flow Regime									
38	Liquid HoldUp	0.1703	0.1703	0.2320	0.2320					
39	Water HoldUp	---	---	---	---					
40	Friction Gradient (kPa/m)	9.073e-003	9.073e-003	8.344e-003	8.344e-003					
41	Static Gradient (kPa/m)	-1.913e-003	-1.913e-003	7.923e-004	7.923e-004					
42	Accel. Gradient (kPa/m)	9.134e-007	9.134e-007	1.166e-006	1.166e-006					
43	Liq. Reynolds	1.834e+005 *	1.834e+005 *	1.834e+005 *	1.834e+005 *					
44	Vap. Reynolds	9.886e+006 *	9.886e+006 *	9.886e+006 *	9.886e+006 *					
45	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
46	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933					
47	Deposit Thickness (mm)	---	---	---	---					
48	Deposit Volume (m3)	---	---	---	---					
49	Length (m)	2.778e+004	2.778e+004	2.779e+004	2.779e+004					
50	Elevation (m)	0.1660	0.1680	0.1700	0.1680					
51	Pressure (bar)	67.79	67.79	67.79	67.79					
52	Temperature (C)	47.63	47.63	47.63	47.63					
53	Vapour Fraction	0.8806	0.8806	0.8806	0.8806					
54	Heat Transferred (kJ/h-m)	469.7	469.7	469.7	469.7					
55	Flow Regime	Transition	Transition	Transition	Transition					
56	Water Flow Regime									
57	Liquid HoldUp	0.2320	0.2320	0.2320	0.1709					
58	Water HoldUp	---	---	---	---					
59	Friction Gradient (kPa/m)	8.344e-003	8.344e-003	8.344e-003	9.065e-003					
60	Static Gradient (kPa/m)	7.923e-004	7.923e-004	7.923e-004	-6.391e-004					
61	Accel. Gradient (kPa/m)	1.166e-006	1.166e-006	1.166e-006	1.075e-006					
62	Liq. Reynolds	1.834e+005 *	1.834e+005 *	1.834e+005 *	1.834e+005 *					
63	Vap. Reynolds	9.886e+006 *	9.886e+006 *	9.886e+006 *	9.886e+006 *					
64	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
65	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933					
66	Deposit Thickness (mm)	---	---	---	---					
67	Deposit Volume (m3)	---	---	---	---					
68	Length (m)	2.780e+004	2.780e+004	2.781e+004	2.781e+004					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	0.1660	0.1640	0.1620	0.1600
14	Pressure (bar)	67.79	67.79	67.79	67.79
15	Temperature (C)	47.63	47.63	47.62	47.62
16	Vapour Fraction	0.8806	0.8806	0.8806	0.8806
17	Heat Transferred (kJ/h-m)	469.6	469.6	469.5	469.5
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.1709	0.1709	0.1709	0.1709
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	9.065e-003	9.065e-003	9.065e-003	9.065e-003
23	Static Gradient (kPa/m)	-6.391e-004	-6.391e-004	-6.391e-004	-6.391e-004
24	Accel. Gradient (kPa/m)	1.075e-006	1.075e-006	1.075e-006	1.075e-006
25	Liq. Reynolds	1.834e+005 *	1.834e+005 *	1.834e+005 *	1.834e+005 *
26	Vap. Reynolds	9.887e+006 *	9.887e+006 *	9.887e+006 *	9.887e+006 *
27	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
28	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	2.782e+004	2.782e+004	2.783e+004	2.783e+004
32	Elevation (m)	0.1580	0.1560	0.1540	0.1520
33	Pressure (bar)	67.79	67.79	67.79	67.78
34	Temperature (C)	47.62	47.62	47.62	47.62
35	Vapour Fraction	0.8806	0.8806	0.8806	0.8806
36	Heat Transferred (kJ/h-m)	469.5	469.4	469.4	469.3
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.1709	0.1709	0.1709	0.1709
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	9.064e-003	9.064e-003	9.064e-003	9.064e-003
42	Static Gradient (kPa/m)	-5.860e-004	-5.860e-004	-5.860e-004	-5.860e-004
43	Accel. Gradient (kPa/m)	1.082e-006	1.082e-006	1.082e-006	1.082e-006
44	Liq. Reynolds	1.834e+005 *	1.834e+005 *	1.834e+005 *	1.834e+005 *
45	Vap. Reynolds	9.887e+006 *	9.887e+006 *	9.887e+006 *	9.887e+006 *
46	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
47	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	2.784e+004	2.784e+004	2.785e+004	2.785e+004
51	Elevation (m)	0.1500	0.1520	0.1540	0.1560
52	Pressure (bar)	67.78	67.78	67.78	67.78
53	Temperature (C)	47.62	47.61	47.61	47.61
54	Vapour Fraction	0.8806	0.8806	0.8806	0.8806
55	Heat Transferred (kJ/h-m)	469.3	469.2	469.2	469.1
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.1709	0.2323	0.2323	0.2323
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	9.064e-003	8.307e-003	8.307e-003	8.307e-003
61	Static Gradient (kPa/m)	-5.860e-004	9.912e-004	9.912e-004	9.912e-004
62	Accel. Gradient (kPa/m)	1.082e-006	1.186e-006	1.186e-006	1.186e-006
63	Liq. Reynolds	1.834e+005 *	1.834e+005 *	1.834e+005 *	1.834e+005 *
64	Vap. Reynolds	9.887e+006 *	9.887e+006 *	9.887e+006 *	9.887e+006 *
65	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
66	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	2.785e+004	2.786e+004	2.788e+004	2.790e+004
14	Elevation (m)	0.1580	0.1600	0.1640	0.1680
15	Pressure (bar)	67.78	67.78	67.78	67.78
16	Temperature (C)	47.61	47.61	47.60	47.60
17	Vapour Fraction	0.8806	0.8806	0.8806	0.8806
18	Heat Transferred (kJ/h-m)	469.1	469.1	468.9	468.8
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.2323	0.2323	0.2315	0.2315
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	8.307e-003	8.307e-003	8.413e-003	8.413e-003
24	Static Gradient (kPa/m)	9.912e-004	9.912e-004	4.276e-004	4.276e-004
25	Accel. Gradient (kPa/m)	1.186e-006	1.186e-006	1.128e-006	1.128e-006
26	Liq. Reynolds	1.834e+005 *	1.834e+005 *	1.834e+005 *	1.834e+005 *
27	Vap. Reynolds	9.887e+006 *	9.887e+006 *	9.887e+006 *	9.887e+006 *
28	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
29	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	2.791e+004	2.793e+004	2.795e+004	2.797e+004
33	Elevation (m)	0.1720	0.1760	0.1800	0.1740
34	Pressure (bar)	67.78	67.78	67.77	67.77
35	Temperature (C)	47.59	47.59	47.58	47.58
36	Vapour Fraction	0.8806	0.8806	0.8806	0.8806
37	Heat Transferred (kJ/h-m)	468.6	468.5	468.3	468.2
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.2315	0.2315	0.2315	0.1709
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	8.413e-003	8.413e-003	8.413e-003	9.064e-003
43	Static Gradient (kPa/m)	4.276e-004	4.276e-004	4.276e-004	-5.184e-004
44	Accel. Gradient (kPa/m)	1.128e-006	1.128e-006	1.128e-006	1.090e-006
45	Liq. Reynolds	1.834e+005 *	1.834e+005 *	1.834e+005 *	1.834e+005 *
46	Vap. Reynolds	9.887e+006 *	9.887e+006 *	9.887e+006 *	9.887e+006 *
47	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
48	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	2.799e+004	2.801e+004	2.802e+004	2.804e+004
52	Elevation (m)	0.1680	0.1620	0.1560	0.1500
53	Pressure (bar)	67.77	67.77	67.77	67.77
54	Temperature (C)	47.57	47.57	47.56	47.55
55	Vapour Fraction	0.8806	0.8806	0.8806	0.8806
56	Heat Transferred (kJ/h-m)	468.1	468.0	467.8	467.7
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.1709	0.1709	0.1709	0.1709
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	9.064e-003	9.064e-003	9.064e-003	9.064e-003
62	Static Gradient (kPa/m)	-5.184e-004	-5.184e-004	-5.184e-004	-5.184e-004
63	Accel. Gradient (kPa/m)	1.091e-006	1.091e-006	1.091e-006	1.091e-006
64	Liq. Reynolds	1.834e+005 *	1.833e+005 *	1.833e+005 *	1.833e+005 *
65	Vap. Reynolds	9.887e+006 *	9.887e+006 *	9.887e+006 *	9.888e+006 *
66	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
67	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
68	Deposit Thickness (mm)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Deposit Volume (m3)	---	---	---	---					
14	Length (m)	2.806e+004	2.808e+004	2.810e+004	2.812e+004					
15	Elevation (m)	0.1480	0.1460	0.1440	0.1420					
16	Pressure (bar)	67.76	67.76	67.76	67.76					
17	Temperature (C)	47.55	47.54	47.54	47.53					
18	Vapour Fraction	0.8806	0.8806	0.8806	0.8806					
19	Heat Transferred (kJ/h-m)	467.5	467.4	467.2	467.1					
20	Flow Regime	Transition	Transition	Transition	Transition					
21	Water Flow Regime									
22	Liquid HoldUp	0.1711	0.1711	0.1711	0.1711					
23	Water HoldUp	---	---	---	---					
24	Friction Gradient (kPa/m)	9.062e-003	9.062e-003	9.062e-003	9.062e-003					
25	Static Gradient (kPa/m)	-1.729e-004	-1.729e-004	-1.729e-004	-1.729e-004					
26	Accel. Gradient (kPa/m)	1.134e-006	1.134e-006	1.134e-006	1.135e-006					
27	Liq. Reynolds	1.833e+005 *	1.833e+005 *	1.833e+005 *	1.833e+005 *					
28	Vap. Reynolds	9.888e+006 *	9.888e+006 *	9.888e+006 *	9.888e+006 *					
29	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
30	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933					
31	Deposit Thickness (mm)	---	---	---	---					
32	Deposit Volume (m3)	---	---	---	---					
33	Length (m)	2.814e+004	2.815e+004	2.817e+004	2.819e+004					
34	Elevation (m)	0.1400	0.1320	0.1240	0.1160					
35	Pressure (bar)	67.76	67.76	67.76	67.75					
36	Temperature (C)	47.53	47.52	47.52	47.51					
37	Vapour Fraction	0.8806	0.8806	0.8806	0.8806					
38	Heat Transferred (kJ/h-m)	466.9	466.8	466.6	466.5					
39	Flow Regime	Transition	Transition	Transition	Transition					
40	Water Flow Regime									
41	Liquid HoldUp	0.1711	0.1708	0.1708	0.1708					
42	Water HoldUp	---	---	---	---					
43	Friction Gradient (kPa/m)	9.062e-003	9.065e-003	9.065e-003	9.065e-003					
44	Static Gradient (kPa/m)	-1.729e-004	-6.909e-004	-6.909e-004	-6.909e-004					
45	Accel. Gradient (kPa/m)	1.135e-006	1.069e-006	1.069e-006	1.069e-006					
46	Liq. Reynolds	1.833e+005 *	1.833e+005 *	1.833e+005 *	1.833e+005 *					
47	Vap. Reynolds	9.888e+006 *	9.888e+006 *	9.888e+006 *	9.888e+006 *					
48	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1980					
49	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933					
50	Deposit Thickness (mm)	---	---	---	---					
51	Deposit Volume (m3)	---	---	---	---					
52	Length (m)	2.821e+004	2.823e+004	2.825e+004	2.827e+004					
53	Elevation (m)	0.1080	1.000e-001	9.600e-002	9.200e-002					
54	Pressure (bar)	67.75	67.75	67.75	67.75					
55	Temperature (C)	47.51	47.50	47.49	47.49					
56	Vapour Fraction	0.8806	0.8806	0.8806	0.8805					
57	Heat Transferred (kJ/h-m)	466.4	466.2	466.1	465.9					
58	Flow Regime	Transition	Transition	Transition	Transition					
59	Water Flow Regime									
60	Liquid HoldUp	0.1708	0.1708	0.1710	0.1710					
61	Water HoldUp	---	---	---	---					
62	Friction Gradient (kPa/m)	9.065e-003	9.065e-003	9.063e-003	9.063e-003					
63	Static Gradient (kPa/m)	-6.910e-004	-6.910e-004	-3.457e-004	-3.457e-004					
64	Accel. Gradient (kPa/m)	1.069e-006	1.069e-006	1.113e-006	1.113e-006					
65	Liq. Reynolds	1.833e+005 *	1.833e+005 *	1.833e+005 *	1.833e+005 *					
66	Vap. Reynolds	9.888e+006 *	9.888e+006 *	9.888e+006 *	9.888e+006 *					
67	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
68	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Thickness	(mm)	---	---	---	---	
14	Deposit Volume	(m3)	---	---	---	---	
15	Length	(m)	2.828e+004	2.830e+004	2.832e+004	2.834e+004	
16	Elevation	(m)	8.800e-002	8.400e-002	8.000e-002	8.200e-002	
17	Pressure	(bar)	67.75	67.74	67.74	67.74	
18	Temperature	(C)	47.48	47.48	47.47	47.47	
19	Vapour Fraction		0.8805	0.8805	0.8805	0.8805	
20	Heat Transferred	(kJ/h-m)	465.8	465.6	465.5	465.3	
21	Flow Regime		Transition	Transition	Transition	Transition	
22	Water Flow Regime						
23	Liquid HoldUp		0.1710	0.1710	0.1710	0.2312	
24	Water HoldUp		---	---	---	---	
25	Friction Gradient	(kPa/m)	9.063e-003	9.063e-003	9.063e-003	8.453e-003	
26	Static Gradient	(kPa/m)	-3.457e-004	-3.457e-004	-3.457e-004	2.136e-004	
27	Accel. Gradient	(kPa/m)	1.113e-006	1.113e-006	1.113e-006	1.106e-006	
28	Liq. Reynolds		1.833e+005 *	1.833e+005 *	1.832e+005 *	1.832e+005 *	
29	Vap. Reynolds		9.889e+006 *	9.889e+006 *	9.889e+006 *	9.889e+006 *	
30	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
31	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
32	Deposit Thickness	(mm)	---	---	---	---	
33	Deposit Volume	(m3)	---	---	---	---	
34	Length	(m)	2.836e+004	2.838e+004	2.839e+004	2.841e+004	
35	Elevation	(m)	8.400e-002	8.600e-002	8.800e-002	9.000e-002	
36	Pressure	(bar)	67.74	67.74	67.74	67.73	
37	Temperature	(C)	47.46	47.46	47.45	47.45	
38	Vapour Fraction		0.8805	0.8805	0.8805	0.8805	
39	Heat Transferred	(kJ/h-m)	465.2	465.0	464.9	464.7	
40	Flow Regime		Transition	Transition	Transition	Transition	
41	Water Flow Regime						
42	Liquid HoldUp		0.2312	0.2312	0.2312	0.2312	
43	Water HoldUp		---	---	---	---	
44	Friction Gradient	(kPa/m)	8.453e-003	8.453e-003	8.453e-003	8.453e-003	
45	Static Gradient	(kPa/m)	2.136e-004	2.136e-004	2.136e-004	2.136e-004	
46	Accel. Gradient	(kPa/m)	1.106e-006	1.106e-006	1.107e-006	1.107e-006	
47	Liq. Reynolds		1.832e+005 *	1.832e+005 *	1.832e+005 *	1.832e+005 *	
48	Vap. Reynolds		9.889e+006 *	9.889e+006 *	9.889e+006 *	9.889e+006 *	
49	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
50	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
51	Deposit Thickness	(mm)	---	---	---	---	
52	Deposit Volume	(m3)	---	---	---	---	
53	Length	(m)	2.842e+004	2.843e+004	2.843e+004	2.844e+004	
54	Elevation	(m)	8.800e-002	8.600e-002	8.400e-002	8.200e-002	
55	Pressure	(bar)	67.73	67.73	67.73	67.73	
56	Temperature	(C)	47.44	47.44	47.44	47.44	
57	Vapour Fraction		0.8805	0.8805	0.8805	0.8805	
58	Heat Transferred	(kJ/h-m)	464.7	464.7	464.6	464.6	
59	Flow Regime		Transition	Transition	Transition	Transition	
60	Water Flow Regime						
61	Liquid HoldUp		0.1709	0.1709	0.1709	0.1709	
62	Water HoldUp		---	---	---	---	
63	Friction Gradient	(kPa/m)	9.064e-003	9.064e-003	9.064e-003	9.064e-003	
64	Static Gradient	(kPa/m)	-4.773e-004	-4.773e-004	-4.773e-004	-4.773e-004	
65	Accel. Gradient	(kPa/m)	1.096e-006	1.096e-006	1.096e-006	1.096e-006	
66	Liq. Reynolds		1.832e+005 *	1.832e+005 *	1.832e+005 *	1.832e+005 *	
67	Vap. Reynolds		9.889e+006 *	9.889e+006 *	9.889e+006 *	9.889e+006 *	
68	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	2.845e+004	2.845e+004	2.846e+004	2.846e+004	
17	Elevation	(m)	8.000e-002	7.800e-002	7.600e-002	7.400e-002	
18	Pressure	(bar)	67.73	67.73	67.73	67.73	
19	Temperature	(C)	47.44	47.43	47.43	47.43	
20	Vapour Fraction		0.8805	0.8805	0.8805	0.8805	
21	Heat Transferred	(kJ/h-m)	464.5	464.5	464.4	464.4	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.1709	0.1709	0.1709	0.1709	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	9.064e-003	9.065e-003	9.065e-003	9.065e-003	
27	Static Gradient	(kPa/m)	-4.773e-004	-6.392e-004	-6.392e-004	-6.392e-004	
28	Accel. Gradient	(kPa/m)	1.096e-006	1.076e-006	1.076e-006	1.076e-006	
29	Liq. Reynolds		1.832e+005 *	1.832e+005 *	1.832e+005 *	1.832e+005 *	
30	Vap. Reynolds		9.889e+006 *	9.889e+006 *	9.889e+006 *	9.889e+006 *	
31	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
32	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	2.847e+004	2.847e+004	2.848e+004	2.848e+004	
36	Elevation	(m)	7.200e-002	7.000e-002	7.000e-002	7.000e-002	
37	Pressure	(bar)	67.73	67.73	67.73	67.73	
38	Temperature	(C)	47.43	47.43	47.43	47.43	
39	Vapour Fraction		0.8805	0.8805	0.8805	0.8805	
40	Heat Transferred	(kJ/h-m)	464.3	464.3	464.2	464.2	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.1709	0.1709	0.2499	0.2499	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	9.065e-003	9.065e-003	6.195e-003	6.195e-003	
46	Static Gradient	(kPa/m)	-6.392e-004	-6.392e-004	0.0000	0.0000	
47	Accel. Gradient	(kPa/m)	1.076e-006	1.076e-006	7.910e-007	7.910e-007	
48	Liq. Reynolds		1.832e+005 *	1.832e+005 *	1.832e+005 *	1.832e+005 *	
49	Vap. Reynolds		9.889e+006 *	9.889e+006 *	9.889e+006 *	9.889e+006 *	
50	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
51	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	2.849e+004	2.849e+004	2.850e+004	2.850e+004	
55	Elevation	(m)	7.000e-002	7.000e-002	7.000e-002	6.000e-002	
56	Pressure	(bar)	67.73	67.73	67.73	67.73	
57	Temperature	(C)	47.42	47.42	47.42	47.42	
58	Vapour Fraction		0.8805	0.8805	0.8805	0.8805	
59	Heat Transferred	(kJ/h-m)	464.1	464.1	464.1	464.1	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.2499	0.2499	0.2499	0.1697	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	6.195e-003	6.194e-003	6.194e-003	9.080e-003	
65	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	-3.181e-003	
66	Accel. Gradient	(kPa/m)	7.910e-007	7.910e-007	7.910e-007	7.532e-007	
67	Liq. Reynolds		1.832e+005 *	1.832e+005 *	1.832e+005 *	1.832e+005 *	
68	Vap. Reynolds		9.889e+006 *	9.889e+006 *	9.889e+006 *	9.889e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
14	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	2.851e+004	2.851e+004	2.852e+004	2.852e+004	
18	Elevation	(m)	5.000e-002	4.000e-002	3.000e-002	2.000e-002	
19	Pressure	(bar)	67.73	67.73	67.73	67.73	
20	Temperature	(C)	47.42	47.42	47.42	47.41	
21	Vapour Fraction		0.8805	0.8805	0.8805	0.8805	
22	Heat Transferred	(kJ/h-m)	464.0	464.0	464.0	463.9	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.1697	0.1697	0.1697	0.1697	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	9.080e-003	9.080e-003	9.080e-003	9.080e-003	
28	Static Gradient	(kPa/m)	-3.181e-003	-3.181e-003	-3.181e-003	-3.181e-003	
29	Accel. Gradient	(kPa/m)	7.532e-007	7.532e-007	7.532e-007	7.532e-007	
30	Liq. Reynolds		1.832e+005 *	1.832e+005 *	1.832e+005 *	1.832e+005 *	
31	Vap. Reynolds		9.889e+006 *	9.889e+006 *	9.889e+006 *	9.889e+006 *	
32	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
33	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	2.853e+004	2.853e+004	2.854e+004	2.854e+004	
37	Elevation	(m)	2.600e-002	3.200e-002	3.800e-002	4.400e-002	
38	Pressure	(bar)	67.73	67.73	67.72	67.72	
39	Temperature	(C)	47.41	47.41	47.41	47.41	
40	Vapour Fraction		0.8805	0.8805	0.8805	0.8805	
41	Heat Transferred	(kJ/h-m)	463.8	463.8	463.8	463.7	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2342	0.2342	0.2342	0.2342	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	8.051e-003	8.051e-003	8.051e-003	8.051e-003	
47	Static Gradient	(kPa/m)	2.394e-003	2.394e-003	2.394e-003	2.394e-003	
48	Accel. Gradient	(kPa/m)	1.334e-006	1.334e-006	1.334e-006	1.334e-006	
49	Liq. Reynolds		1.832e+005 *	1.832e+005 *	1.832e+005 *	1.832e+005 *	
50	Vap. Reynolds		9.890e+006 *	9.890e+006 *	9.890e+006 *	9.890e+006 *	
51	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
52	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	2.855e+004	2.855e+004	2.856e+004	2.856e+004	
56	Elevation	(m)	5.000e-002	4.400e-002	3.800e-002	3.200e-002	
57	Pressure	(bar)	67.72	67.72	67.72	67.72	
58	Temperature	(C)	47.41	47.41	47.40	47.40	
59	Vapour Fraction		0.8805	0.8805	0.8805	0.8805	
60	Heat Transferred	(kJ/h-m)	463.7	463.7	463.6	463.6	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2342	0.1703	0.1703	0.1703	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	8.051e-003	9.072e-003	9.072e-003	9.072e-003	
66	Static Gradient	(kPa/m)	2.394e-003	-1.913e-003	-1.913e-003	-1.913e-003	
67	Accel. Gradient	(kPa/m)	1.334e-006	9.142e-007	9.142e-007	9.142e-007	
68	Liq. Reynolds		1.832e+005 *	1.832e+005 *	1.832e+005 *	1.832e+005 *	
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.890e+006 *	9.890e+006 *	9.890e+006 *	9.890e+006 *
14	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
15	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	2.857e+004	2.857e+004	2.858e+004	2.858e+004
19	Elevation (m)	2.600e-002	2.000e-002	-1.600e-002	-5.200e-002
20	Pressure (bar)	67.72	67.72	67.72	67.72
21	Temperature (C)	47.40	47.40	47.40	47.40
22	Vapour Fraction	0.8805	0.8805	0.8805	0.8805
23	Heat Transferred (kJ/h-m)	463.6	463.5	463.5	463.5
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1703	0.1703	0.1658	0.1658
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	9.072e-003	9.072e-003	9.134e-003	9.134e-003
29	Static Gradient (kPa/m)	-1.913e-003	-1.913e-003	-1.128e-002	-1.128e-002
30	Accel. Gradient (kPa/m)	9.142e-007	9.142e-007	-2.740e-007	-2.740e-007
31	Liq. Reynolds	1.832e+005 *	1.832e+005 *	1.832e+005 *	1.832e+005 *
32	Vap. Reynolds	9.890e+006 *	9.890e+006 *	9.890e+006 *	9.890e+006 *
33	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
34	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	2.859e+004	2.859e+004	2.860e+004	2.860e+004
38	Elevation (m)	-8.800e-002	-0.1240	-0.1600	-0.2280
39	Pressure (bar)	67.72	67.72	67.72	67.72
40	Temperature (C)	47.40	47.39	47.39	47.39
41	Vapour Fraction	0.8805	0.8805	0.8805	0.8805
42	Heat Transferred (kJ/h-m)	463.4	463.4	463.4	463.4
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1658	0.1659	0.1659	0.1611
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	9.134e-003	9.134e-003	9.134e-003	9.205e-003
48	Static Gradient (kPa/m)	-1.128e-002	-1.128e-002	-1.128e-002	-2.090e-002
49	Accel. Gradient (kPa/m)	-2.740e-007	-2.740e-007	-2.741e-007	-1.494e-006
50	Liq. Reynolds	1.832e+005 *	1.832e+005 *	1.832e+005 *	1.832e+005 *
51	Vap. Reynolds	9.890e+006 *	9.890e+006 *	9.890e+006 *	9.890e+006 *
52	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
53	Vapour Velocity (m/s)	2.933	2.933	2.933	2.932
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	2.861e+004	2.861e+004	2.862e+004	2.862e+004
57	Elevation (m)	-0.2960	-0.3640	-0.4320	-0.5000
58	Pressure (bar)	67.72	67.72	67.72	67.73
59	Temperature (C)	47.39	47.39	47.39	47.39
60	Vapour Fraction	0.8805	0.8805	0.8805	0.8805
61	Heat Transferred (kJ/h-m)	463.3	463.3	463.3	463.3
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1611	0.1611	0.1611	0.1611
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	9.205e-003	9.205e-003	9.204e-003	9.204e-003
67	Static Gradient (kPa/m)	-2.090e-002	-2.090e-002	-2.091e-002	-2.091e-002
68	Accel. Gradient (kPa/m)	-1.494e-006	-1.494e-006	-1.494e-006	-1.494e-006
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		1.832e+005 *	1.832e+005 *	1.832e+005 *	1.832e+005 *	
14	Vap. Reynolds		9.890e+006 *	9.890e+006 *	9.890e+006 *	9.890e+006 *	
15	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	0.1980	
16	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932	2.932	
17	Deposit Thickness (mm)	---	---	---	---	---	
18	Deposit Volume (m3)	---	---	---	---	---	
19	Length (m)	2.863e+004	2.863e+004	2.864e+004	2.864e+004	2.864e+004	
20	Elevation (m)	-0.5360	-0.5720	-0.6080	-0.6440	-0.6440	
21	Pressure (bar)	67.73	67.73	67.73	67.73	67.73	
22	Temperature (C)	47.39	47.39	47.38	47.38	47.38	
23	Vapour Fraction	0.8805	0.8805	0.8805	0.8805	0.8805	
24	Heat Transferred (kJ/h-m)	463.2	463.2	463.1	463.1	463.1	
25	Flow Regime	Transition	Transition	Transition	Transition	Transition	
26	Water Flow Regime						
27	Liquid HoldUp	0.1659	0.1659	0.1659	0.1659	0.1659	
28	Water HoldUp	---	---	---	---	---	
29	Friction Gradient (kPa/m)	9.133e-003	9.133e-003	9.133e-003	9.133e-003	9.133e-003	
30	Static Gradient (kPa/m)	-1.129e-002	-1.129e-002	-1.129e-002	-1.129e-002	-1.129e-002	
31	Accel. Gradient (kPa/m)	-2.753e-007	-2.753e-007	-2.753e-007	-2.754e-007	-2.754e-007	
32	Liq. Reynolds		1.832e+005 *	1.832e+005 *	1.832e+005 *	1.832e+005 *	
33	Vap. Reynolds		9.890e+006 *	9.890e+006 *	9.890e+006 *	9.890e+006 *	
34	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	0.1980	
35	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932	2.932	
36	Deposit Thickness (mm)	---	---	---	---	---	
37	Deposit Volume (m3)	---	---	---	---	---	
38	Length (m)	2.865e+004	2.865e+004	2.866e+004	2.866e+004	2.866e+004	
39	Elevation (m)	-0.6800	-0.6500	-0.6200	-0.5900	-0.5900	
40	Pressure (bar)	67.73	67.72	67.72	67.72	67.72	
41	Temperature (C)	47.38	47.38	47.38	47.38	47.38	
42	Vapour Fraction	0.8805	0.8805	0.8805	0.8805	0.8805	
43	Heat Transferred (kJ/h-m)	463.1	463.0	462.9	462.9	462.9	
44	Flow Regime	Transition	Transition	Transition	Transition	Transition	
45	Water Flow Regime						
46	Liquid HoldUp	0.1659	0.2475	0.2475	0.2475	0.2475	
47	Water HoldUp	---	---	---	---	---	
48	Friction Gradient (kPa/m)	9.133e-003	6.454e-003	6.454e-003	6.454e-003	6.454e-003	
49	Static Gradient (kPa/m)	-1.129e-002	1.247e-002	1.247e-002	1.247e-002	1.247e-002	
50	Accel. Gradient (kPa/m)	-2.754e-007	2.416e-006	2.416e-006	2.416e-006	2.416e-006	
51	Liq. Reynolds		1.832e+005 *	1.832e+005 *	1.832e+005 *	1.831e+005 *	
52	Vap. Reynolds		9.890e+006 *	9.890e+006 *	9.890e+006 *	9.890e+006 *	
53	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	0.1980	
54	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932	2.932	
55	Deposit Thickness (mm)	---	---	---	---	---	
56	Deposit Volume (m3)	---	---	---	---	---	
57	Length (m)	2.867e+004	2.867e+004	2.868e+004	2.868e+004	2.868e+004	
58	Elevation (m)	-0.5600	-0.5300	-0.4300	-0.3300	-0.3300	
59	Pressure (bar)	67.72	67.72	67.72	67.72	67.72	
60	Temperature (C)	47.38	47.37	47.37	47.37	47.37	
61	Vapour Fraction	0.8805	0.8805	0.8805	0.8805	0.8805	
62	Heat Transferred (kJ/h-m)	462.9	462.8	462.8	462.8	462.7	
63	Flow Regime	Transition	Transition	Transition	Transition	Transition	
64	Water Flow Regime						
65	Liquid HoldUp	0.2475	0.2475	0.2862	0.2862	0.2862	
66	Water HoldUp	---	---	---	---	---	
67	Friction Gradient (kPa/m)	6.455e-003	6.455e-003	7.948e-003	7.948e-003	7.948e-003	
68	Static Gradient (kPa/m)	1.247e-002	1.247e-002	4.642e-002	4.642e-002	4.642e-002	
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 460 of 937		

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Accel. Gradient	(kPa/m)	2.416e-006	2.416e-006	6.942e-006	6.942e-006				
14	Liq. Reynolds		1.831e+005 *	1.831e+005 *	1.831e+005 *	1.831e+005 *				
15	Vap. Reynolds		9.890e+006 *	9.890e+006 *	9.890e+006 *	9.890e+006 *				
16	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
17	Vapour Velocity	(m/s)	2.932	2.932	2.932	2.932				
18	Deposit Thickness	(mm)	---	---	---	---				
19	Deposit Volume	(m3)	---	---	---	---				
20	Length	(m)	2.869e+004	2.869e+004	2.870e+004	2.870e+004				
21	Elevation	(m)	-0.2300	-0.1300	-3.000e-002	-2.600e-002				
22	Pressure	(bar)	67.71	67.71	67.71	67.71				
23	Temperature	(C)	47.37	47.36	47.36	47.36				
24	Vapour Fraction		0.8805	0.8805	0.8805	0.8805				
25	Heat Transferred	(kJ/h-m)	462.6	462.6	462.5	462.5				
26	Flow Regime		Transition	Transition	Transition	Transition				
27	Water Flow Regime									
28	Liquid HoldUp		0.2862	0.2862	0.2862	0.2331				
29	Water HoldUp		---	---	---	---				
30	Friction Gradient	(kPa/m)	7.949e-003	7.949e-003	7.949e-003	8.197e-003				
31	Static Gradient	(kPa/m)	4.641e-002	4.641e-002	4.641e-002	1.590e-003				
32	Accel. Gradient	(kPa/m)	6.943e-006	6.943e-006	6.944e-006	1.250e-006				
33	Liq. Reynolds		1.831e+005 *	1.831e+005 *	1.831e+005 *	1.831e+005 *				
34	Vap. Reynolds		9.890e+006 *	9.890e+006 *	9.890e+006 *	9.890e+006 *				
35	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
36	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933				
37	Deposit Thickness	(mm)	---	---	---	---				
38	Deposit Volume	(m3)	---	---	---	---				
39	Length	(m)	2.871e+004	2.871e+004	2.872e+004	2.872e+004				
40	Elevation	(m)	-2.200e-002	-1.800e-002	-1.400e-002	-1.000e-002				
41	Pressure	(bar)	67.71	67.71	67.71	67.71				
42	Temperature	(C)	47.36	47.36	47.36	47.35				
43	Vapour Fraction		0.8805	0.8805	0.8805	0.8805				
44	Heat Transferred	(kJ/h-m)	462.4	462.4	462.3	462.3				
45	Flow Regime		Transition	Transition	Transition	Transition				
46	Water Flow Regime									
47	Liquid HoldUp		0.2331	0.2331	0.2331	0.2331				
48	Water HoldUp		---	---	---	---				
49	Friction Gradient	(kPa/m)	8.197e-003	8.197e-003	8.197e-003	8.197e-003				
50	Static Gradient	(kPa/m)	1.590e-003	1.590e-003	1.590e-003	1.590e-003				
51	Accel. Gradient	(kPa/m)	1.250e-006	1.250e-006	1.250e-006	1.250e-006				
52	Liq. Reynolds		1.831e+005 *	1.831e+005 *	1.831e+005 *	1.831e+005 *				
53	Vap. Reynolds		9.890e+006 *	9.890e+006 *	9.890e+006 *	9.890e+006 *				
54	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
55	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933				
56	Deposit Thickness	(mm)	---	---	---	---				
57	Deposit Volume	(m3)	---	---	---	---				
58	Length	(m)	2.873e+004	2.874e+004	2.875e+004	2.876e+004				
59	Elevation	(m)	-4.000e-003	2.000e-003	8.000e-003	1.400e-002				
60	Pressure	(bar)	67.70	67.70	67.70	67.70				
61	Temperature	(C)	47.35	47.35	47.35	47.34				
62	Vapour Fraction		0.8805	0.8805	0.8805	0.8805				
63	Heat Transferred	(kJ/h-m)	462.2	462.1	462.1	462.0				
64	Flow Regime		Transition	Transition	Transition	Transition				
65	Water Flow Regime									
66	Liquid HoldUp		0.2325	0.2325	0.2325	0.2325				
67	Water HoldUp		---	---	---	---				
68	Friction Gradient	(kPa/m)	8.278e-003	8.278e-003	8.278e-003	8.278e-003				
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 461 of 937				

1			Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	1.152e-003	1.152e-003	1.152e-003	1.152e-003
14	Accel. Gradient	(kPa/m)	1.205e-006	1.205e-006	1.205e-006	1.205e-006
15	Liq. Reynolds		1.831e+005 *	1.831e+005 *	1.831e+005 *	1.831e+005 *
16	Vap. Reynolds		9.890e+006 *	9.891e+006 *	9.891e+006 *	9.891e+006 *
17	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
18	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	2.877e+004	2.878e+004	2.879e+004	2.880e+004
22	Elevation	(m)	2.000e-002	1.400e-002	8.000e-003	2.000e-003
23	Pressure	(bar)	67.70	67.70	67.70	67.70
24	Temperature	(C)	47.34	47.34	47.33	47.33
25	Vapour Fraction		0.8805	0.8805	0.8805	0.8805
26	Heat Transferred	(kJ/h-m)	461.9	461.9	461.8	461.7
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2325	0.1707	0.1707	0.1707
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	8.278e-003	9.067e-003	9.067e-003	9.067e-003
32	Static Gradient	(kPa/m)	1.152e-003	-9.274e-004	-9.274e-004	-9.274e-004
33	Accel. Gradient	(kPa/m)	1.205e-006	1.040e-006	1.040e-006	1.040e-006
34	Liq. Reynolds		1.831e+005 *	1.831e+005 *	1.831e+005 *	1.831e+005 *
35	Vap. Reynolds		9.891e+006 *	9.891e+006 *	9.891e+006 *	9.891e+006 *
36	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
37	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	2.881e+004	2.883e+004	2.884e+004	2.885e+004
41	Elevation	(m)	-4.000e-003	-1.000e-002	-1.200e-002	-1.400e-002
42	Pressure	(bar)	67.70	67.70	67.70	67.69
43	Temperature	(C)	47.33	47.32	47.32	47.32
44	Vapour Fraction		0.8805	0.8805	0.8805	0.8805
45	Heat Transferred	(kJ/h-m)	461.6	461.5	461.5	461.4
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.1707	0.1707	0.1710	0.1710
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	9.067e-003	9.067e-003	9.063e-003	9.063e-003
51	Static Gradient	(kPa/m)	-9.275e-004	-9.275e-004	-3.095e-004	-3.095e-004
52	Accel. Gradient	(kPa/m)	1.040e-006	1.040e-006	1.118e-006	1.118e-006
53	Liq. Reynolds		1.831e+005 *	1.831e+005 *	1.831e+005 *	1.831e+005 *
54	Vap. Reynolds		9.891e+006 *	9.891e+006 *	9.891e+006 *	9.891e+006 *
55	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
56	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	2.886e+004	2.887e+004	2.888e+004	2.889e+004
60	Elevation	(m)	-1.600e-002	-1.800e-002	-2.000e-002	-3.400e-002
61	Pressure	(bar)	67.69	67.69	67.69	67.69
62	Temperature	(C)	47.32	47.31	47.31	47.31
63	Vapour Fraction		0.8805	0.8805	0.8805	0.8805
64	Heat Transferred	(kJ/h-m)	461.3	461.2	461.1	461.1
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.1710	0.1710	0.1710	0.1702
68	Water HoldUp		---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 462 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	9.063e-003	9.063e-003	9.063e-003	9.075e-003	
14	Static Gradient	(kPa/m)	-3.095e-004	-3.095e-004	-3.095e-004	-2.159e-003	
15	Accel. Gradient	(kPa/m)	1.118e-006	1.118e-006	1.118e-006	8.835e-007	
16	Liq. Reynolds		1.831e+005 *	1.831e+005 *	1.831e+005 *	1.831e+005 *	
17	Vap. Reynolds		9.891e+006 *	9.891e+006 *	9.891e+006 *	9.891e+006 *	
18	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
19	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	2.890e+004	2.891e+004	2.892e+004	2.893e+004	
23	Elevation	(m)	-4.800e-002	-6.200e-002	-7.600e-002	-9.000e-002	
24	Pressure	(bar)	67.69	67.69	67.69	67.69	
25	Temperature	(C)	47.30	47.30	47.30	47.29	
26	Vapour Fraction		0.8805	0.8805	0.8805	0.8805	
27	Heat Transferred	(kJ/h-m)	461.0	460.9	460.8	460.7	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.1702	0.1702	0.1702	0.1702	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	9.075e-003	9.075e-003	9.075e-003	9.074e-003	
33	Static Gradient	(kPa/m)	-2.159e-003	-2.159e-003	-2.159e-003	-2.159e-003	
34	Accel. Gradient	(kPa/m)	8.835e-007	8.835e-007	8.835e-007	8.835e-007	
35	Liq. Reynolds		1.831e+005 *	1.831e+005 *	1.830e+005 *	1.830e+005 *	
36	Vap. Reynolds		9.891e+006 *	9.891e+006 *	9.891e+006 *	9.891e+006 *	
37	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
38	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	2.894e+004	2.895e+004	2.896e+004	2.897e+004	
42	Elevation	(m)	-9.400e-002	-9.800e-002	-0.1020	-0.1060	
43	Pressure	(bar)	67.69	67.69	67.69	67.68	
44	Temperature	(C)	47.29	47.29	47.29	47.28	
45	Vapour Fraction		0.8805	0.8805	0.8805	0.8805	
46	Heat Transferred	(kJ/h-m)	460.7	460.6	460.5	460.4	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.1709	0.1709	0.1709	0.1709	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	9.065e-003	9.065e-003	9.065e-003	9.065e-003	
52	Static Gradient	(kPa/m)	-6.187e-004	-6.187e-004	-6.187e-004	-6.187e-004	
53	Accel. Gradient	(kPa/m)	1.079e-006	1.079e-006	1.079e-006	1.079e-006	
54	Liq. Reynolds		1.830e+005 *	1.830e+005 *	1.830e+005 *	1.830e+005 *	
55	Vap. Reynolds		9.891e+006 *	9.891e+006 *	9.891e+006 *	9.891e+006 *	
56	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
57	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	2.898e+004	2.899e+004	2.900e+004	2.901e+004	
61	Elevation	(m)	-0.1100	-0.1180	-0.1260	-0.1340	
62	Pressure	(bar)	67.68	67.68	67.68	67.68	
63	Temperature	(C)	47.28	47.28	47.27	47.27	
64	Vapour Fraction		0.8805	0.8805	0.8805	0.8805	
65	Heat Transferred	(kJ/h-m)	460.3	460.3	460.2	460.1	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.1709	0.1706	0.1706	0.1706	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 463 of 937	

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	9.065e-003	9.069e-003	9.069e-003	9.069e-003
15	Static Gradient (kPa/m)	-6.187e-004	-1.236e-003	-1.236e-003	-1.236e-003
16	Accel. Gradient (kPa/m)	1.079e-006	1.001e-006	1.001e-006	1.001e-006
17	Liq. Reynolds	1.830e+005 *	1.830e+005 *	1.830e+005 *	1.830e+005 *
18	Vap. Reynolds	9.891e+006 *	9.892e+006 *	9.892e+006 *	9.892e+006 *
19	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
20	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	2.902e+004	2.903e+004	2.904e+004	2.904e+004
24	Elevation (m)	-0.1420	-0.1500	-0.1440	-0.1380
25	Pressure (bar)	67.68	67.68	67.68	67.68
26	Temperature (C)	47.27	47.27	47.26	47.26
27	Vapour Fraction	0.8805	0.8805	0.8805	0.8805
28	Heat Transferred (kJ/h-m)	460.0	459.9	459.9	459.9
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.1706	0.1706	0.2360	0.2360
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	9.069e-003	9.069e-003	7.814e-003	7.814e-003
34	Static Gradient (kPa/m)	-1.236e-003	-1.236e-003	3.746e-003	3.746e-003
35	Accel. Gradient (kPa/m)	1.001e-006	1.001e-006	1.477e-006	1.477e-006
36	Liq. Reynolds	1.830e+005 *	1.830e+005 *	1.830e+005 *	1.830e+005 *
37	Vap. Reynolds	9.892e+006 *	9.892e+006 *	9.892e+006 *	9.892e+006 *
38	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
39	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	2.904e+004	2.904e+004	2.905e+004	2.905e+004
43	Elevation (m)	-0.1320	-0.1260	-0.1200	-0.1300
44	Pressure (bar)	67.68	67.68	67.68	67.68
45	Temperature (C)	47.26	47.26	47.26	47.26
46	Vapour Fraction	0.8805	0.8805	0.8805	0.8805
47	Heat Transferred (kJ/h-m)	459.8	459.8	459.8	459.8
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2360	0.2360	0.2360	0.1697
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	7.814e-003	7.814e-003	7.814e-003	9.081e-003
53	Static Gradient (kPa/m)	3.746e-003	3.746e-003	3.746e-003	-3.182e-003
54	Accel. Gradient (kPa/m)	1.477e-006	1.477e-006	1.477e-006	7.538e-007
55	Liq. Reynolds	1.830e+005 *	1.830e+005 *	1.830e+005 *	1.830e+005 *
56	Vap. Reynolds	9.892e+006 *	9.892e+006 *	9.892e+006 *	9.892e+006 *
57	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
58	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	2.906e+004	2.906e+004	2.907e+004	2.907e+004
62	Elevation (m)	-0.1400	-0.1500	-0.1600	-0.1700
63	Pressure (bar)	67.68	67.68	67.68	67.68
64	Temperature (C)	47.26	47.26	47.25	47.25
65	Vapour Fraction	0.8804	0.8804	0.8804	0.8804
66	Heat Transferred (kJ/h-m)	459.7	459.7	459.7	459.6
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 464 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.1697	0.1697	0.1697	0.1697
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	9.081e-003	9.081e-003	9.081e-003	9.081e-003
16	Static Gradient (kPa/m)	-3.182e-003	-3.182e-003	-3.182e-003	-3.182e-003
17	Accel. Gradient (kPa/m)	7.538e-007	7.538e-007	7.538e-007	7.538e-007
18	Liq. Reynolds	1.830e+005 *	1.830e+005 *	1.830e+005 *	1.830e+005 *
19	Vap. Reynolds	9.892e+006 *	9.892e+006 *	9.892e+006 *	9.892e+006 *
20	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
21	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	2.908e+004	2.908e+004	2.909e+004	2.909e+004
25	Elevation (m)	-0.1920	-0.2140	-0.2360	-0.2580
26	Pressure (bar)	67.68	67.68	67.68	67.68
27	Temperature (C)	47.25	47.25	47.25	47.25
28	Vapour Fraction	0.8804	0.8804	0.8804	0.8804
29	Heat Transferred (kJ/h-m)	459.6	459.6	459.5	459.5
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1679	0.1679	0.1679	0.1679
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	9.105e-003	9.105e-003	9.105e-003	9.105e-003
35	Static Gradient (kPa/m)	-6.951e-003	-6.951e-003	-6.951e-003	-6.951e-003
36	Accel. Gradient (kPa/m)	2.753e-007	2.753e-007	2.753e-007	2.753e-007
37	Liq. Reynolds	1.830e+005 *	1.830e+005 *	1.830e+005 *	1.830e+005 *
38	Vap. Reynolds	9.892e+006 *	9.892e+006 *	9.892e+006 *	9.892e+006 *
39	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
40	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	2.910e+004	2.910e+004	2.911e+004	2.911e+004
44	Elevation (m)	-0.2800	-0.2880	-0.2960	-0.3040
45	Pressure (bar)	67.68	67.68	67.68	67.67
46	Temperature (C)	47.25	47.25	47.24	47.24
47	Vapour Fraction	0.8804	0.8804	0.8804	0.8804
48	Heat Transferred (kJ/h-m)	459.5	459.4	459.4	459.3
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1679	0.1700	0.1700	0.1700
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	9.105e-003	9.077e-003	9.077e-003	9.077e-003
54	Static Gradient (kPa/m)	-6.951e-003	-2.548e-003	-2.548e-003	-2.548e-003
55	Accel. Gradient (kPa/m)	2.753e-007	8.342e-007	8.342e-007	8.342e-007
56	Liq. Reynolds	1.830e+005 *	1.830e+005 *	1.830e+005 *	1.830e+005 *
57	Vap. Reynolds	9.892e+006 *	9.892e+006 *	9.892e+006 *	9.892e+006 *
58	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
59	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	2.912e+004	2.912e+004	2.913e+004	2.913e+004
63	Elevation (m)	-0.3120	-0.3200	-0.2840	-0.2480
64	Pressure (bar)	67.67	67.67	67.67	67.67
65	Temperature (C)	47.24	47.24	47.24	47.24
66	Vapour Fraction	0.8804	0.8804	0.8804	0.8804
67	Heat Transferred (kJ/h-m)	459.3	459.3	459.2	459.1
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 465 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1700	0.1700	0.2508	0.2508
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	9.077e-003	9.077e-003	6.096e-003	6.096e-003
17	Static Gradient (kPa/m)	-2.548e-003	-2.548e-003	1.511e-002	1.511e-002
18	Accel. Gradient (kPa/m)	8.342e-007	8.342e-007	2.710e-006	2.711e-006
19	Liq. Reynolds	1.830e+005 *	1.830e+005 *	1.830e+005 *	1.830e+005 *
20	Vap. Reynolds	9.892e+006 *	9.892e+006 *	9.892e+006 *	9.892e+006 *
21	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
22	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	2.914e+004	2.914e+004	2.915e+004	2.915e+004
26	Elevation (m)	-0.2120	-0.1760	-0.1400	-0.1280
27	Pressure (bar)	67.67	67.67	67.67	67.67
28	Temperature (C)	47.23	47.23	47.23	47.23
29	Vapour Fraction	0.8804	0.8804	0.8804	0.8804
30	Heat Transferred (kJ/h-m)	459.1	459.0	459.0	459.0
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2508	0.2508	0.2508	0.2392
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	6.096e-003	6.097e-003	6.097e-003	7.422e-003
36	Static Gradient (kPa/m)	1.511e-002	1.511e-002	1.511e-002	6.079e-003
37	Accel. Gradient (kPa/m)	2.711e-006	2.711e-006	2.711e-006	1.726e-006
38	Liq. Reynolds	1.830e+005 *	1.830e+005 *	1.830e+005 *	1.830e+005 *
39	Vap. Reynolds	9.892e+006 *	9.892e+006 *	9.892e+006 *	9.892e+006 *
40	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
41	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	2.916e+004	2.916e+004	2.916e+004	2.917e+004
45	Elevation (m)	-0.1160	-0.1040	-9.200e-002	-8.000e-002
46	Pressure (bar)	67.67	67.67	67.67	67.67
47	Temperature (C)	47.23	47.23	47.23	47.22
48	Vapour Fraction	0.8804	0.8804	0.8804	0.8804
49	Heat Transferred (kJ/h-m)	458.9	458.9	458.9	458.8
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2392	0.2392	0.2392	0.2392
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	7.422e-003	7.423e-003	7.423e-003	7.423e-003
55	Static Gradient (kPa/m)	6.079e-003	6.079e-003	6.079e-003	6.079e-003
56	Accel. Gradient (kPa/m)	1.726e-006	1.726e-006	1.726e-006	1.726e-006
57	Liq. Reynolds	1.830e+005 *	1.830e+005 *	1.830e+005 *	1.830e+005 *
58	Vap. Reynolds	9.892e+006 *	9.892e+006 *	9.892e+006 *	9.892e+006 *
59	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
60	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	2.917e+004	2.918e+004	2.918e+004	2.919e+004
64	Elevation (m)	-6.000e-002	-4.000e-002	-2.000e-002	-5.551e-016
65	Pressure (bar)	67.67	67.66	67.66	67.66
66	Temperature (C)	47.22	47.22	47.22	47.22
67	Vapour Fraction	0.8804	0.8804	0.8804	0.8804
68	Heat Transferred (kJ/h-m)	458.8	458.7	458.7	458.6
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2409	0.2409	0.2409	0.2409
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	7.209e-003	7.209e-003	7.209e-003	7.210e-003
18	Static Gradient (kPa/m)	7.409e-003	7.409e-003	7.409e-003	7.409e-003
19	Accel. Gradient (kPa/m)	1.868e-006	1.868e-006	1.868e-006	1.868e-006
20	Liq. Reynolds	1.830e+005 *	1.830e+005 *	1.830e+005 *	1.830e+005 *
21	Vap. Reynolds	9.892e+006 *	9.892e+006 *	9.892e+006 *	9.892e+006 *
22	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
23	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	2.920e+004	2.920e+004	2.921e+004	2.921e+004
27	Elevation (m)	2.000e-002	1.600e-002	1.200e-002	8.000e-003
28	Pressure (bar)	67.66	67.66	67.66	67.66
29	Temperature (C)	47.22	47.21	47.21	47.21
30	Vapour Fraction	0.8804	0.8804	0.8804	0.8804
31	Heat Transferred (kJ/h-m)	458.6	458.6	458.6	458.5
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2409	0.1706	0.1706	0.1706
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	7.210e-003	9.069e-003	9.069e-003	9.069e-003
37	Static Gradient (kPa/m)	7.409e-003	-1.161e-003	-1.161e-003	-1.161e-003
38	Accel. Gradient (kPa/m)	1.868e-006	1.011e-006	1.011e-006	1.011e-006
39	Liq. Reynolds	1.830e+005 *	1.830e+005 *	1.829e+005 *	1.829e+005 *
40	Vap. Reynolds	9.892e+006 *	9.893e+006 *	9.893e+006 *	9.893e+006 *
41	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
42	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	2.922e+004	2.922e+004	2.923e+004	2.923e+004
46	Elevation (m)	4.000e-003	-5.655e-016	6.000e-003	1.200e-002
47	Pressure (bar)	67.66	67.66	67.66	67.66
48	Temperature (C)	47.21	47.21	47.21	47.20
49	Vapour Fraction	0.8804	0.8804	0.8804	0.8804
50	Heat Transferred (kJ/h-m)	458.5	458.4	458.3	458.3
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1706	0.1706	0.2339	0.2339
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	9.069e-003	9.069e-003	8.089e-003	8.089e-003
56	Static Gradient (kPa/m)	-1.161e-003	-1.161e-003	2.192e-003	2.192e-003
57	Accel. Gradient (kPa/m)	1.011e-006	1.011e-006	1.314e-006	1.314e-006
58	Liq. Reynolds	1.829e+005 *	1.829e+005 *	1.829e+005 *	1.829e+005 *
59	Vap. Reynolds	9.893e+006 *	9.893e+006 *	9.893e+006 *	9.893e+006 *
60	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
61	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	2.924e+004	2.924e+004	2.925e+004	2.925e+004
65	Elevation (m)	1.800e-002	2.400e-002	3.000e-002	3.800e-002
66	Pressure (bar)	67.66	67.66	67.66	67.66
67	Temperature (C)	47.20	47.20	47.20	47.20
68	Vapour Fraction	0.8804	0.8804	0.8804	0.8804
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Heat Transferred (kJ/h-m)	458.3	458.2	458.2	458.2		
14	Flow Regime	Transition	Transition	Transition	Transition		
15	Water Flow Regime						
16	Liquid HoldUp	0.2339	0.2339	0.2339	0.2419		
17	Water HoldUp	---	---	---	---		
18	Friction Gradient (kPa/m)	8.089e-003	8.089e-003	8.089e-003	7.090e-003		
19	Static Gradient (kPa/m)	2.192e-003	2.192e-003	2.192e-003	8.175e-003		
20	Accel. Gradient (kPa/m)	1.314e-006	1.314e-006	1.314e-006	1.951e-006		
21	Liq. Reynolds	1.829e+005 *	1.829e+005 *	1.829e+005 *	1.829e+005 *		
22	Vap. Reynolds	9.893e+006 *	9.893e+006 *	9.893e+006 *	9.893e+006 *		
23	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
24	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933		
25	Deposit Thickness (mm)	---	---	---	---		
26	Deposit Volume (m3)	---	---	---	---		
27	Length (m)	2.925e+004	2.926e+004	2.926e+004	2.926e+004		
28	Elevation (m)	4.600e-002	5.400e-002	6.200e-002	7.000e-002		
29	Pressure (bar)	67.66	67.66	67.66	67.66		
30	Temperature (C)	47.20	47.20	47.20	47.20		
31	Vapour Fraction	0.8804	0.8804	0.8804	0.8804		
32	Heat Transferred (kJ/h-m)	458.1	458.1	458.1	458.1		
33	Flow Regime	Transition	Transition	Transition	Transition		
34	Water Flow Regime						
35	Liquid HoldUp	0.2419	0.2419	0.2419	0.2419		
36	Water HoldUp	---	---	---	---		
37	Friction Gradient (kPa/m)	7.090e-003	7.090e-003	7.090e-003	7.090e-003		
38	Static Gradient (kPa/m)	8.175e-003	8.175e-003	8.175e-003	8.175e-003		
39	Accel. Gradient (kPa/m)	1.951e-006	1.951e-006	1.951e-006	1.951e-006		
40	Liq. Reynolds	1.829e+005 *	1.829e+005 *	1.829e+005 *	1.829e+005 *		
41	Vap. Reynolds	9.893e+006 *	9.893e+006 *	9.893e+006 *	9.893e+006 *		
42	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
43	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933		
44	Deposit Thickness (mm)	---	---	---	---		
45	Deposit Volume (m3)	---	---	---	---		
46	Length (m)	2.927e+004	2.928e+004	2.929e+004	2.930e+004		
47	Elevation (m)	7.800e-002	8.600e-002	9.400e-002	0.1020		
48	Pressure (bar)	67.65	67.65	67.65	67.65		
49	Temperature (C)	47.19	47.19	47.19	47.19		
50	Vapour Fraction	0.8804	0.8804	0.8804	0.8804		
51	Heat Transferred (kJ/h-m)	458.0	457.9	457.9	457.8		
52	Flow Regime	Transition	Transition	Transition	Transition		
53	Water Flow Regime						
54	Liquid HoldUp	0.2332	0.2332	0.2332	0.2332		
55	Water HoldUp	---	---	---	---		
56	Friction Gradient (kPa/m)	8.183e-003	8.183e-003	8.183e-003	8.183e-003		
57	Static Gradient (kPa/m)	1.675e-003	1.675e-003	1.675e-003	1.675e-003		
58	Accel. Gradient (kPa/m)	1.260e-006	1.260e-006	1.260e-006	1.260e-006		
59	Liq. Reynolds	1.829e+005 *	1.829e+005 *	1.829e+005 *	1.829e+005 *		
60	Vap. Reynolds	9.893e+006 *	9.893e+006 *	9.893e+006 *	9.893e+006 *		
61	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
62	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933		
63	Deposit Thickness (mm)	---	---	---	---		
64	Deposit Volume (m3)	---	---	---	---		
65	Length (m)	2.931e+004	2.932e+004	2.933e+004	2.934e+004		
66	Elevation (m)	0.1100	0.1040	9.800e-002	9.200e-002		
67	Pressure (bar)	67.65	67.65	67.65	67.65		
68	Temperature (C)	47.18	47.18	47.18	47.17		
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Fraction		0.8804	0.8804	0.8804	0.8804	
14	Heat Transferred (kJ/h-m)		457.7	457.7	457.6	457.5	
15	Flow Regime		Transition	Transition	Transition	Transition	
16	Water Flow Regime						
17	Liquid HoldUp		0.2332	0.1707	0.1707	0.1707	
18	Water HoldUp		---	---	---	---	
19	Friction Gradient (kPa/m)		8.183e-003	9.068e-003	9.068e-003	9.068e-003	
20	Static Gradient (kPa/m)		1.675e-003	-1.009e-003	-1.009e-003	-1.009e-003	
21	Accel. Gradient (kPa/m)		1.260e-006	1.030e-006	1.030e-006	1.030e-006	
22	Liq. Reynolds		1.829e+005 *	1.829e+005 *	1.829e+005 *	1.829e+005 *	
23	Vap. Reynolds		9.893e+006 *	9.893e+006 *	9.893e+006 *	9.893e+006 *	
24	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
25	Vapour Velocity (m/s)		2.933	2.933	2.933	2.933	
26	Deposit Thickness (mm)		---	---	---	---	
27	Deposit Volume (m3)		---	---	---	---	
28	Length (m)		2.935e+004	2.936e+004	2.936e+004	2.937e+004	
29	Elevation (m)		8.600e-002	8.000e-002	8.200e-002	8.400e-002	
30	Pressure (bar)		67.65	67.65	67.65	67.65	
31	Temperature (C)		47.17	47.17	47.17	47.16	
32	Vapour Fraction		0.8804	0.8804	0.8804	0.8804	
33	Heat Transferred (kJ/h-m)		457.5	457.4	457.3	457.2	
34	Flow Regime		Transition	Transition	Transition	Transition	
35	Water Flow Regime						
36	Liquid HoldUp		0.1707	0.1707	0.2315	0.2315	
37	Water HoldUp		---	---	---	---	
38	Friction Gradient (kPa/m)		9.068e-003	9.068e-003	8.416e-003	8.416e-003	
39	Static Gradient (kPa/m)		-1.009e-003	-1.009e-003	4.164e-004	4.164e-004	
40	Accel. Gradient (kPa/m)		1.030e-006	1.030e-006	1.129e-006	1.129e-006	
41	Liq. Reynolds		1.829e+005 *	1.829e+005 *	1.829e+005 *	1.829e+005 *	
42	Vap. Reynolds		9.893e+006 *	9.893e+006 *	9.893e+006 *	9.893e+006 *	
43	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
44	Vapour Velocity (m/s)		2.933	2.933	2.933	2.933	
45	Deposit Thickness (mm)		---	---	---	---	
46	Deposit Volume (m3)		---	---	---	---	
47	Length (m)		2.938e+004	2.939e+004	2.940e+004	2.941e+004	
48	Elevation (m)		8.600e-002	8.800e-002	9.000e-002	8.800e-002	
49	Pressure (bar)		67.64	67.64	67.64	67.64	
50	Temperature (C)		47.16	47.16	47.16	47.15	
51	Vapour Fraction		0.8804	0.8804	0.8804	0.8804	
52	Heat Transferred (kJ/h-m)		457.1	457.1	457.0	457.0	
53	Flow Regime		Transition	Transition	Transition	Transition	
54	Water Flow Regime						
55	Liquid HoldUp		0.2315	0.2315	0.2315	0.1710	
56	Water HoldUp		---	---	---	---	
57	Friction Gradient (kPa/m)		8.416e-003	8.416e-003	8.416e-003	9.064e-003	
58	Static Gradient (kPa/m)		4.164e-004	4.164e-004	4.164e-004	-3.367e-004	
59	Accel. Gradient (kPa/m)		1.129e-006	1.129e-006	1.129e-006	1.116e-006	
60	Liq. Reynolds		1.829e+005 *	1.829e+005 *	1.829e+005 *	1.829e+005 *	
61	Vap. Reynolds		9.893e+006 *	9.893e+006 *	9.893e+006 *	9.893e+006 *	
62	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
63	Vapour Velocity (m/s)		2.933	2.933	2.933	2.933	
64	Deposit Thickness (mm)		---	---	---	---	
65	Deposit Volume (m3)		---	---	---	---	
66	Length (m)		2.942e+004	2.943e+004	2.944e+004	2.945e+004	
67	Elevation (m)		8.600e-002	8.400e-002	8.200e-002	8.000e-002	
68	Pressure (bar)		67.64	67.64	67.64	67.64	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 469 of 937	

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	47.15	47.15	47.14	47.14
14	Vapour Fraction	0.8804	0.8804	0.8804	0.8804
15	Heat Transferred (kJ/h-m)	456.9	456.8	456.7	456.7
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.1710	0.1710	0.1710	0.1710
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	9.064e-003	9.064e-003	9.064e-003	9.064e-003
21	Static Gradient (kPa/m)	-3.367e-004	-3.367e-004	-3.367e-004	-3.367e-004
22	Accel. Gradient (kPa/m)	1.116e-006	1.116e-006	1.116e-006	1.116e-006
23	Liq. Reynolds	1.829e+005 *	1.829e+005 *	1.829e+005 *	1.829e+005 *
24	Vap. Reynolds	9.893e+006 *	9.894e+006 *	9.894e+006 *	9.894e+006 *
25	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
26	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	2.946e+004	2.947e+004	2.948e+004	2.949e+004
30	Elevation (m)	6.800e-002	5.600e-002	4.400e-002	3.200e-002
31	Pressure (bar)	67.64	67.64	67.64	67.64
32	Temperature (C)	47.14	47.14	47.13	47.13
33	Vapour Fraction	0.8804	0.8804	0.8804	0.8804
34	Heat Transferred (kJ/h-m)	456.6	456.5	456.4	456.4
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1702	0.1702	0.1702	0.1702
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	9.074e-003	9.074e-003	9.074e-003	9.074e-003
40	Static Gradient (kPa/m)	-2.014e-003	-2.014e-003	-2.014e-003	-2.014e-003
41	Accel. Gradient (kPa/m)	9.028e-007	9.028e-007	9.028e-007	9.028e-007
42	Liq. Reynolds	1.829e+005 *	1.829e+005 *	1.829e+005 *	1.829e+005 *
43	Vap. Reynolds	9.894e+006 *	9.894e+006 *	9.894e+006 *	9.894e+006 *
44	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
45	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	2.950e+004	2.951e+004	2.952e+004	2.953e+004
49	Elevation (m)	2.000e-002	2.000e-002	2.000e-002	2.000e-002
50	Pressure (bar)	67.64	67.63	67.63	67.63
51	Temperature (C)	47.13	47.13	47.12	47.12
52	Vapour Fraction	0.8804	0.8804	0.8804	0.8804
53	Heat Transferred (kJ/h-m)	456.3	456.2	456.1	456.0
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.1702	0.2499	0.2499	0.2499
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	9.074e-003	6.196e-003	6.196e-003	6.196e-003
59	Static Gradient (kPa/m)	-2.014e-003	0.0000	0.0000	0.0000
60	Accel. Gradient (kPa/m)	9.028e-007	7.922e-007	7.922e-007	7.922e-007
61	Liq. Reynolds	1.829e+005 *	1.829e+005 *	1.828e+005 *	1.828e+005 *
62	Vap. Reynolds	9.894e+006 *	9.894e+006 *	9.894e+006 *	9.894e+006 *
63	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
64	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	2.954e+004	2.955e+004	2.955e+004	2.955e+004
68	Elevation (m)	2.000e-002	2.000e-002	2.200e-002	2.400e-002
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Pressure (bar)	67.63	67.63	67.63	67.63
14	Temperature (C)	47.12	47.11	47.11	47.11
15	Vapour Fraction	0.8804	0.8804	0.8804	0.8804
16	Heat Transferred (kJ/h-m)	456.0	455.9	455.9	455.9
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.2499	0.2499	0.2354	0.2354
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	6.196e-003	6.196e-003	7.900e-003	7.900e-003
22	Static Gradient (kPa/m)	0.0000	0.0000	3.256e-003	3.256e-003
23	Accel. Gradient (kPa/m)	7.922e-007	7.922e-007	1.427e-006	1.427e-006
24	Liq. Reynolds	1.828e+005 *	1.828e+005 *	1.828e+005 *	1.828e+005 *
25	Vap. Reynolds	9.894e+006 *	9.894e+006 *	9.894e+006 *	9.894e+006 *
26	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
27	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	2.955e+004	2.955e+004	2.955e+004	2.956e+004
31	Elevation (m)	2.600e-002	2.800e-002	3.000e-002	3.200e-002
32	Pressure (bar)	67.63	67.63	67.63	67.63
33	Temperature (C)	47.11	47.11	47.11	47.11
34	Vapour Fraction	0.8804	0.8804	0.8804	0.8804
35	Heat Transferred (kJ/h-m)	455.9	455.9	455.9	455.8
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.2354	0.2354	0.2354	0.2320
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	7.900e-003	7.900e-003	7.900e-003	8.345e-003
41	Static Gradient (kPa/m)	3.256e-003	3.256e-003	3.256e-003	7.926e-004
42	Accel. Gradient (kPa/m)	1.427e-006	1.427e-006	1.427e-006	1.168e-006
43	Liq. Reynolds	1.828e+005 *	1.828e+005 *	1.828e+005 *	1.828e+005 *
44	Vap. Reynolds	9.894e+006 *	9.894e+006 *	9.894e+006 *	9.894e+006 *
45	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
46	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	2.956e+004	2.957e+004	2.957e+004	2.958e+004
50	Elevation (m)	3.400e-002	3.600e-002	3.800e-002	4.000e-002
51	Pressure (bar)	67.63	67.63	67.63	67.63
52	Temperature (C)	47.11	47.11	47.11	47.11
53	Vapour Fraction	0.8804	0.8804	0.8804	0.8804
54	Heat Transferred (kJ/h-m)	455.8	455.7	455.7	455.7
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.2320	0.2320	0.2320	0.2320
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	8.345e-003	8.345e-003	8.345e-003	8.345e-003
60	Static Gradient (kPa/m)	7.926e-004	7.926e-004	7.926e-004	7.926e-004
61	Accel. Gradient (kPa/m)	1.168e-006	1.169e-006	1.169e-006	1.169e-006
62	Liq. Reynolds	1.828e+005 *	1.828e+005 *	1.828e+005 *	1.828e+005 *
63	Vap. Reynolds	9.894e+006 *	9.894e+006 *	9.894e+006 *	9.894e+006 *
64	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
65	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	2.958e+004	2.959e+004	2.959e+004	2.960e+004
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11										
12										
13	Pipe Table									
13	Elevation (m)	4.400e-002	4.800e-002	5.200e-002	5.600e-002					
14	Pressure (bar)	67.63	67.63	67.63	67.63					
15	Temperature (C)	47.10	47.10	47.10	47.10					
16	Vapour Fraction	0.8804	0.8804	0.8804	0.8804					
17	Heat Transferred (kJ/h-m)	455.6	455.6	455.5	455.5					
18	Flow Regime	Transition	Transition	Transition	Transition					
19	Water Flow Regime									
20	Liquid HoldUp	0.2331	0.2331	0.2331	0.2331					
21	Water HoldUp	---	---	---	---					
22	Friction Gradient (kPa/m)	8.198e-003	8.198e-003	8.198e-003	8.198e-003					
23	Static Gradient (kPa/m)	1.591e-003	1.591e-003	1.591e-003	1.591e-003					
24	Accel. Gradient (kPa/m)	1.252e-006	1.252e-006	1.252e-006	1.252e-006					
25	Liq. Reynolds	1.828e+005 *	1.828e+005 *	1.828e+005 *	1.828e+005 *					
26	Vap. Reynolds	9.894e+006 *	9.894e+006 *	9.894e+006 *	9.894e+006 *					
27	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
28	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933					
29	Deposit Thickness (mm)	---	---	---	---					
30	Deposit Volume (m3)	---	---	---	---					
31	Length (m)	2.960e+004	2.961e+004	2.961e+004	2.962e+004					
32	Elevation (m)	6.000e-002	5.600e-002	5.200e-002	4.800e-002					
33	Pressure (bar)	67.63	67.63	67.63	67.63					
34	Temperature (C)	47.10	47.10	47.10	47.09					
35	Vapour Fraction	0.8804	0.8804	0.8804	0.8804					
36	Heat Transferred (kJ/h-m)	455.5	455.5	455.4	455.4					
37	Flow Regime	Transition	Transition	Transition	Transition					
38	Water Flow Regime									
39	Liquid HoldUp	0.2331	0.1706	0.1706	0.1706					
40	Water HoldUp	---	---	---	---					
41	Friction Gradient (kPa/m)	8.198e-003	9.070e-003	9.070e-003	9.070e-003					
42	Static Gradient (kPa/m)	1.591e-003	-1.277e-003	-1.277e-003	-1.277e-003					
43	Accel. Gradient (kPa/m)	1.252e-006	9.965e-007	9.965e-007	9.965e-007					
44	Liq. Reynolds	1.828e+005 *	1.828e+005 *	1.828e+005 *	1.828e+005 *					
45	Vap. Reynolds	9.894e+006 *	9.894e+006 *	9.894e+006 *	9.894e+006 *					
46	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
47	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933					
48	Deposit Thickness (mm)	---	---	---	---					
49	Deposit Volume (m3)	---	---	---	---					
50	Length (m)	2.962e+004	2.963e+004	2.963e+004	2.964e+004					
51	Elevation (m)	4.400e-002	4.000e-002	3.800e-002	3.600e-002					
52	Pressure (bar)	67.63	67.62	67.62	67.62					
53	Temperature (C)	47.09	47.09	47.09	47.09					
54	Vapour Fraction	0.8804	0.8804	0.8804	0.8804					
55	Heat Transferred (kJ/h-m)	455.4	455.3	455.3	455.2					
56	Flow Regime	Transition	Transition	Transition	Transition					
57	Water Flow Regime									
58	Liquid HoldUp	0.1706	0.1706	0.1709	0.1709					
59	Water HoldUp	---	---	---	---					
60	Friction Gradient (kPa/m)	9.070e-003	9.070e-003	9.066e-003	9.066e-003					
61	Static Gradient (kPa/m)	-1.277e-003	-1.277e-003	-6.393e-004	-6.393e-004					
62	Accel. Gradient (kPa/m)	9.965e-007	9.965e-007	1.078e-006	1.078e-006					
63	Liq. Reynolds	1.828e+005 *	1.828e+005 *	1.828e+005 *	1.828e+005 *					
64	Vap. Reynolds	9.894e+006 *	9.894e+006 *	9.894e+006 *	9.894e+006 *					
65	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
66	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933					
67	Deposit Thickness (mm)	---	---	---	---					
68	Deposit Volume (m3)	---	---	---	---					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length	(m)	2.964e+004	2.965e+004	2.965e+004	2.965e+004	
14	Elevation	(m)	3.400e-002	3.200e-002	3.000e-002	2.400e-002	
15	Pressure	(bar)	67.62	67.62	67.62	67.62	
16	Temperature	(C)	47.09	47.09	47.08	47.08	
17	Vapour Fraction		0.8804	0.8804	0.8804	0.8804	
18	Heat Transferred	(kJ/h-m)	455.2	455.2	455.1	455.1	
19	Flow Regime		Transition	Transition	Transition	Transition	
20	Water Flow Regime						
21	Liquid HoldUp		0.1709	0.1709	0.1709	0.1690	
22	Water HoldUp		---	---	---	---	
23	Friction Gradient	(kPa/m)	9.066e-003	9.066e-003	9.066e-003	9.092e-003	
24	Static Gradient	(kPa/m)	-6.393e-004	-6.393e-004	-6.393e-004	-4.759e-003	
25	Accel. Gradient	(kPa/m)	1.078e-006	1.078e-006	1.078e-006	5.541e-007	
26	Liq. Reynolds		1.828e+005 *	1.828e+005 *	1.828e+005 *	1.828e+005 *	
27	Vap. Reynolds		9.894e+006 *	9.894e+006 *	9.894e+006 *	9.894e+006 *	
28	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
29	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
30	Deposit Thickness	(mm)	---	---	---	---	
31	Deposit Volume	(m3)	---	---	---	---	
32	Length	(m)	2.966e+004	2.966e+004	2.966e+004	2.966e+004	
33	Elevation	(m)	1.800e-002	1.200e-002	6.000e-003	-5.621e-016	
34	Pressure	(bar)	67.62	67.62	67.62	67.62	
35	Temperature	(C)	47.08	47.08	47.08	47.08	
36	Vapour Fraction		0.8804	0.8804	0.8804	0.8804	
37	Heat Transferred	(kJ/h-m)	455.1	455.1	455.1	455.1	
38	Flow Regime		Transition	Transition	Transition	Transition	
39	Water Flow Regime						
40	Liquid HoldUp		0.1690	0.1690	0.1690	0.1690	
41	Water HoldUp		---	---	---	---	
42	Friction Gradient	(kPa/m)	9.092e-003	9.091e-003	9.091e-003	9.091e-003	
43	Static Gradient	(kPa/m)	-4.759e-003	-4.759e-003	-4.759e-003	-4.759e-003	
44	Accel. Gradient	(kPa/m)	5.541e-007	5.541e-007	5.541e-007	5.541e-007	
45	Liq. Reynolds		1.828e+005 *	1.828e+005 *	1.828e+005 *	1.828e+005 *	
46	Vap. Reynolds		9.894e+006 *	9.894e+006 *	9.894e+006 *	9.894e+006 *	
47	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
48	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
49	Deposit Thickness	(mm)	---	---	---	---	
50	Deposit Volume	(m3)	---	---	---	---	
51	Length	(m)	2.966e+004	2.967e+004	2.967e+004	2.967e+004	
52	Elevation	(m)	-6.000e-003	-1.200e-002	-1.800e-002	-2.400e-002	
53	Pressure	(bar)	67.62	67.62	67.62	67.62	
54	Temperature	(C)	47.08	47.08	47.08	47.08	
55	Vapour Fraction		0.8804	0.8804	0.8804	0.8804	
56	Heat Transferred	(kJ/h-m)	455.0	455.0	455.0	455.0	
57	Flow Regime		Transition	Transition	Transition	Transition	
58	Water Flow Regime						
59	Liquid HoldUp		0.1690	0.1690	0.1690	0.1690	
60	Water HoldUp		---	---	---	---	
61	Friction Gradient	(kPa/m)	9.091e-003	9.091e-003	9.091e-003	9.091e-003	
62	Static Gradient	(kPa/m)	-4.759e-003	-4.759e-003	-4.759e-003	-4.759e-003	
63	Accel. Gradient	(kPa/m)	5.541e-007	5.541e-007	5.541e-007	5.541e-007	
64	Liq. Reynolds		1.828e+005 *	1.828e+005 *	1.828e+005 *	1.828e+005 *	
65	Vap. Reynolds		9.894e+006 *	9.894e+006 *	9.894e+006 *	9.894e+006 *	
66	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
67	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
68	Deposit Thickness	(mm)	---	---	---	---	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 473 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	2.967e+004	2.968e+004	2.968e+004	2.968e+004		
15	Elevation (m)	-3.000e-002	-3.200e-002	-3.400e-002	-3.600e-002		
16	Pressure (bar)	67.62	67.62	67.62	67.62		
17	Temperature (C)	47.08	47.08	47.08	47.08		
18	Vapour Fraction	0.8804	0.8804	0.8804	0.8804		
19	Heat Transferred (kJ/h-m)	455.0	454.9	454.9	454.9		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	0.1690	0.1708	0.1708	0.1708		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	9.091e-003	9.066e-003	9.066e-003	9.066e-003		
25	Static Gradient (kPa/m)	-4.759e-003	-7.989e-004	-7.989e-004	-7.989e-004		
26	Accel. Gradient (kPa/m)	5.541e-007	1.057e-006	1.057e-006	1.057e-006		
27	Liq. Reynolds	1.828e+005 *	1.828e+005 *	1.828e+005 *	1.828e+005 *		
28	Vap. Reynolds	9.894e+006 *	9.894e+006 *	9.894e+006 *	9.894e+006 *		
29	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
30	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	2.969e+004	2.969e+004	2.969e+004	2.970e+004		
34	Elevation (m)	-3.800e-002	-4.000e-002	-3.800e-002	-3.600e-002		
35	Pressure (bar)	67.62	67.62	67.62	67.62		
36	Temperature (C)	47.07	47.07	47.07	47.07		
37	Vapour Fraction	0.8804	0.8804	0.8804	0.8804		
38	Heat Transferred (kJ/h-m)	454.9	454.8	454.8	454.7		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.1708	0.1708	0.2327	0.2327		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	9.066e-003	9.066e-003	8.247e-003	8.247e-003		
44	Static Gradient (kPa/m)	-7.989e-004	-7.989e-004	1.324e-003	1.324e-003		
45	Accel. Gradient (kPa/m)	1.057e-006	1.057e-006	1.224e-006	1.224e-006		
46	Liq. Reynolds	1.828e+005 *	1.828e+005 *	1.828e+005 *	1.828e+005 *		
47	Vap. Reynolds	9.894e+006 *	9.894e+006 *	9.895e+006 *	9.895e+006 *		
48	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
49	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	2.970e+004	2.970e+004	2.971e+004	2.971e+004		
53	Elevation (m)	-3.400e-002	-3.200e-002	-3.000e-002	-3.000e-002		
54	Pressure (bar)	67.62	67.62	67.62	67.62		
55	Temperature (C)	47.07	47.07	47.07	47.07		
56	Vapour Fraction	0.8804	0.8804	0.8804	0.8804		
57	Heat Transferred (kJ/h-m)	454.7	454.7	454.7	454.6		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.2327	0.2327	0.2327	0.2499		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	8.247e-003	8.247e-003	8.247e-003	6.196e-003		
63	Static Gradient (kPa/m)	1.324e-003	1.324e-003	1.324e-003	0.0000		
64	Accel. Gradient (kPa/m)	1.224e-006	1.224e-006	1.224e-006	7.924e-007		
65	Liq. Reynolds	1.828e+005 *	1.828e+005 *	1.828e+005 *	1.828e+005 *		
66	Vap. Reynolds	9.895e+006 *	9.895e+006 *	9.895e+006 *	9.895e+006 *		
67	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
68	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933		
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 474 of 937		

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	2.971e+004	2.972e+004	2.972e+004	2.973e+004
16	Elevation (m)	-3.000e-002	-3.000e-002	-3.000e-002	-3.000e-002
17	Pressure (bar)	67.62	67.62	67.62	67.62
18	Temperature (C)	47.07	47.07	47.06	47.06
19	Vapour Fraction	0.8804	0.8804	0.8804	0.8804
20	Heat Transferred (kJ/h-m)	454.6	454.6	454.5	454.5
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2499	0.2499	0.2499	0.2499
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	6.195e-003	6.195e-003	6.195e-003	6.195e-003
26	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
27	Accel. Gradient (kPa/m)	7.924e-007	7.924e-007	7.924e-007	7.924e-007
28	Liq. Reynolds	1.828e+005 *	1.828e+005 *	1.828e+005 *	1.828e+005 *
29	Vap. Reynolds	9.895e+006 *	9.895e+006 *	9.895e+006 *	9.895e+006 *
30	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
31	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	2.973e+004	2.974e+004	2.974e+004	2.975e+004
35	Elevation (m)	-3.200e-002	-3.400e-002	-3.600e-002	-3.800e-002
36	Pressure (bar)	67.62	67.62	67.62	67.62
37	Temperature (C)	47.06	47.06	47.06	47.06
38	Vapour Fraction	0.8804	0.8804	0.8804	0.8804
39	Heat Transferred (kJ/h-m)	454.5	454.5	454.4	454.4
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.1709	0.1709	0.1709	0.1709
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	9.065e-003	9.065e-003	9.065e-003	9.065e-003
45	Static Gradient (kPa/m)	-6.401e-004	-6.401e-004	-6.401e-004	-6.401e-004
46	Accel. Gradient (kPa/m)	1.078e-006	1.078e-006	1.078e-006	1.078e-006
47	Liq. Reynolds	1.828e+005 *	1.828e+005 *	1.828e+005 *	1.828e+005 *
48	Vap. Reynolds	9.895e+006 *	9.895e+006 *	9.895e+006 *	9.895e+006 *
49	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
50	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	2.975e+004	2.976e+004	2.976e+004	2.977e+004
54	Elevation (m)	-4.000e-002	-4.200e-002	-4.400e-002	-4.600e-002
55	Pressure (bar)	67.62	67.62	67.61	67.61
56	Temperature (C)	47.06	47.05	47.05	47.05
57	Vapour Fraction	0.8804	0.8804	0.8804	0.8804
58	Heat Transferred (kJ/h-m)	454.4	454.3	454.3	454.3
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.1709	0.1709	0.1709	0.1709
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	9.065e-003	9.065e-003	9.065e-003	9.065e-003
64	Static Gradient (kPa/m)	-6.401e-004	-6.394e-004	-6.394e-004	-6.394e-004
65	Accel. Gradient (kPa/m)	1.078e-006	1.078e-006	1.078e-006	1.078e-006
66	Liq. Reynolds	1.828e+005 *	1.828e+005 *	1.828e+005 *	1.828e+005 *
67	Vap. Reynolds	9.895e+006 *	9.895e+006 *	9.895e+006 *	9.895e+006 *
68	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	2.977e+004	2.978e+004	2.978e+004	2.979e+004	
17	Elevation	(m)	-4.800e-002	-5.000e-002	-4.600e-002	-4.200e-002	
18	Pressure	(bar)	67.61	67.61	67.61	67.61	
19	Temperature	(C)	47.05	47.05	47.05	47.05	
20	Vapour Fraction		0.8804	0.8804	0.8804	0.8804	
21	Heat Transferred	(kJ/h-m)	454.2	454.2	454.1	454.1	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.1709	0.1709	0.2331	0.2331	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	9.065e-003	9.065e-003	8.198e-003	8.198e-003	
27	Static Gradient	(kPa/m)	-6.394e-004	-6.394e-004	1.591e-003	1.591e-003	
28	Accel. Gradient	(kPa/m)	1.078e-006	1.078e-006	1.252e-006	1.252e-006	
29	Liq. Reynolds		1.828e+005 *	1.828e+005 *	1.828e+005 *	1.828e+005 *	
30	Vap. Reynolds		9.895e+006 *	9.895e+006 *	9.895e+006 *	9.895e+006 *	
31	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
32	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	2.979e+004	2.980e+004	2.980e+004	2.981e+004	
36	Elevation	(m)	-3.800e-002	-3.400e-002	-3.000e-002	-2.200e-002	
37	Pressure	(bar)	67.61	67.61	67.61	67.61	
38	Temperature	(C)	47.04	47.04	47.04	47.04	
39	Vapour Fraction		0.8804	0.8804	0.8804	0.8804	
40	Heat Transferred	(kJ/h-m)	454.0	454.0	453.9	453.9	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2331	0.2331	0.2331	0.2353	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	8.198e-003	8.198e-003	8.198e-003	7.909e-003	
46	Static Gradient	(kPa/m)	1.591e-003	1.591e-003	1.591e-003	3.204e-003	
47	Accel. Gradient	(kPa/m)	1.252e-006	1.252e-006	1.252e-006	1.421e-006	
48	Liq. Reynolds		1.828e+005 *	1.828e+005 *	1.828e+005 *	1.828e+005 *	
49	Vap. Reynolds		9.895e+006 *	9.895e+006 *	9.895e+006 *	9.895e+006 *	
50	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
51	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	2.981e+004	2.982e+004	2.982e+004	2.983e+004	
55	Elevation	(m)	-1.400e-002	-6.000e-003	2.000e-003	1.000e-002	
56	Pressure	(bar)	67.61	67.61	67.61	67.61	
57	Temperature	(C)	47.04	47.04	47.04	47.03	
58	Vapour Fraction		0.8804	0.8804	0.8804	0.8804	
59	Heat Transferred	(kJ/h-m)	453.9	453.8	453.8	453.7	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.2353	0.2353	0.2353	0.2353	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	7.909e-003	7.909e-003	7.909e-003	7.909e-003	
65	Static Gradient	(kPa/m)	3.204e-003	3.204e-003	3.204e-003	3.204e-003	
66	Accel. Gradient	(kPa/m)	1.421e-006	1.421e-006	1.421e-006	1.421e-006	
67	Liq. Reynolds		1.828e+005 *	1.828e+005 *	1.828e+005 *	1.827e+005 *	
68	Vap. Reynolds		9.895e+006 *	9.895e+006 *	9.895e+006 *	9.895e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
14	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	2.983e+004	2.984e+004	2.984e+004	2.985e+004	
18	Elevation	(m)	1.200e-002	1.400e-002	1.600e-002	1.800e-002	
19	Pressure	(bar)	67.61	67.61	67.61	67.61	
20	Temperature	(C)	47.03	47.03	47.03	47.03	
21	Vapour Fraction		0.8804	0.8804	0.8804	0.8804	
22	Heat Transferred	(kJ/h-m)	453.7	453.7	453.6	453.6	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.2320	0.2320	0.2320	0.2320	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	8.346e-003	8.346e-003	8.346e-003	8.346e-003	
28	Static Gradient	(kPa/m)	7.926e-004	7.926e-004	7.926e-004	7.926e-004	
29	Accel. Gradient	(kPa/m)	1.169e-006	1.169e-006	1.169e-006	1.169e-006	
30	Liq. Reynolds		1.827e+005 *	1.827e+005 *	1.827e+005 *	1.827e+005 *	
31	Vap. Reynolds		9.895e+006 *	9.895e+006 *	9.895e+006 *	9.895e+006 *	
32	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
33	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	2.985e+004	2.986e+004	2.986e+004	2.987e+004	
37	Elevation	(m)	2.000e-002	3.000e-002	4.000e-002	5.000e-002	
38	Pressure	(bar)	67.61	67.61	67.60	67.60	
39	Temperature	(C)	47.03	47.02	47.02	47.02	
40	Vapour Fraction		0.8804	0.8804	0.8804	0.8804	
41	Heat Transferred	(kJ/h-m)	453.5	453.5	453.5	453.4	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2320	0.2364	0.2364	0.2364	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	8.346e-003	7.768e-003	7.768e-003	7.768e-003	
47	Static Gradient	(kPa/m)	7.926e-004	4.019e-003	4.019e-003	4.019e-003	
48	Accel. Gradient	(kPa/m)	1.169e-006	1.508e-006	1.508e-006	1.508e-006	
49	Liq. Reynolds		1.827e+005 *	1.827e+005 *	1.827e+005 *	1.827e+005 *	
50	Vap. Reynolds		9.895e+006 *	9.895e+006 *	9.895e+006 *	9.895e+006 *	
51	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
52	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	2.987e+004	2.988e+004	2.988e+004	2.989e+004	
56	Elevation	(m)	6.000e-002	7.000e-002	6.600e-002	6.200e-002	
57	Pressure	(bar)	67.60	67.60	67.60	67.60	
58	Temperature	(C)	47.02	47.02	47.02	47.02	
59	Vapour Fraction		0.8803	0.8803	0.8803	0.8803	
60	Heat Transferred	(kJ/h-m)	453.4	453.3	453.3	453.3	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2364	0.2364	0.1706	0.1706	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	7.768e-003	7.768e-003	9.070e-003	9.070e-003	
66	Static Gradient	(kPa/m)	4.019e-003	4.019e-003	-1.277e-003	-1.277e-003	
67	Accel. Gradient	(kPa/m)	1.508e-006	1.508e-006	9.968e-007	9.968e-007	
68	Liq. Reynolds		1.827e+005 *	1.827e+005 *	1.827e+005 *	1.827e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.895e+006 *	9.895e+006 *	9.895e+006 *	9.895e+006 *
14	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
15	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	2.989e+004	2.990e+004	2.990e+004	2.990e+004
19	Elevation (m)	5.800e-002	5.400e-002	5.000e-002	5.200e-002
20	Pressure (bar)	67.60	67.60	67.60	67.60
21	Temperature (C)	47.01	47.01	47.01	47.01
22	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
23	Heat Transferred (kJ/h-m)	453.3	453.2	453.2	453.1
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1706	0.1706	0.1706	0.2337
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	9.070e-003	9.070e-003	9.070e-003	8.125e-003
29	Static Gradient (kPa/m)	-1.277e-003	-1.277e-003	-1.277e-003	1.992e-003
30	Accel. Gradient (kPa/m)	9.968e-007	9.968e-007	9.968e-007	1.294e-006
31	Liq. Reynolds	1.827e+005 *	1.827e+005 *	1.827e+005 *	1.827e+005 *
32	Vap. Reynolds	9.895e+006 *	9.895e+006 *	9.895e+006 *	9.895e+006 *
33	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
34	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	2.991e+004	2.991e+004	2.991e+004	2.991e+004
38	Elevation (m)	5.400e-002	5.600e-002	5.800e-002	6.000e-002
39	Pressure (bar)	67.60	67.60	67.60	67.60
40	Temperature (C)	47.01	47.01	47.01	47.01
41	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
42	Heat Transferred (kJ/h-m)	453.1	453.1	453.1	453.1
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2337	0.2337	0.2337	0.2337
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	8.125e-003	8.125e-003	8.125e-003	8.125e-003
48	Static Gradient (kPa/m)	1.992e-003	1.992e-003	1.992e-003	1.992e-003
49	Accel. Gradient (kPa/m)	1.294e-006	1.294e-006	1.294e-006	1.294e-006
50	Liq. Reynolds	1.827e+005 *	1.827e+005 *	1.827e+005 *	1.827e+005 *
51	Vap. Reynolds	9.895e+006 *	9.895e+006 *	9.895e+006 *	9.895e+006 *
52	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
53	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	2.992e+004	2.992e+004	2.993e+004	2.993e+004
57	Elevation (m)	5.800e-002	5.600e-002	5.400e-002	5.200e-002
58	Pressure (bar)	67.60	67.60	67.60	67.60
59	Temperature (C)	47.01	47.01	47.00	47.00
60	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
61	Heat Transferred (kJ/h-m)	453.1	453.0	453.0	453.0
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1709	0.1709	0.1709	0.1709
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	9.066e-003	9.066e-003	9.066e-003	9.066e-003
67	Static Gradient (kPa/m)	-6.394e-004	-6.394e-004	-6.394e-004	-6.394e-004
68	Accel. Gradient (kPa/m)	1.078e-006	1.078e-006	1.078e-006	1.078e-006
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	1.827e+005 *	1.827e+005 *	1.827e+005 *	1.827e+005 *
14	Vap. Reynolds	9.895e+006 *	9.895e+006 *	9.896e+006 *	9.896e+006 *
15	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
16	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	2.994e+004	2.994e+004	2.994e+004	2.995e+004
20	Elevation (m)	5.000e-002	5.000e-002	5.000e-002	5.000e-002
21	Pressure (bar)	67.60	67.60	67.60	67.60
22	Temperature (C)	47.00	47.00	47.00	47.00
23	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
24	Heat Transferred (kJ/h-m)	452.9	452.9	452.8	452.8
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.1709	0.2499	0.2499	0.2499
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	9.066e-003	6.196e-003	6.196e-003	6.196e-003
30	Static Gradient (kPa/m)	-6.394e-004	0.0000	0.0000	0.0000
31	Accel. Gradient (kPa/m)	1.078e-006	7.926e-007	7.926e-007	7.926e-007
32	Liq. Reynolds	1.827e+005 *	1.827e+005 *	1.827e+005 *	1.827e+005 *
33	Vap. Reynolds	9.896e+006 *	9.896e+006 *	9.896e+006 *	9.896e+006 *
34	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
35	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	2.995e+004	2.996e+004	2.996e+004	2.997e+004
39	Elevation (m)	5.000e-002	5.000e-002	5.400e-002	5.800e-002
40	Pressure (bar)	67.60	67.60	67.60	67.60
41	Temperature (C)	47.00	47.00	46.99	46.99
42	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
43	Heat Transferred (kJ/h-m)	452.8	452.7	452.7	452.7
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2499	0.2499	0.2331	0.2331
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	6.196e-003	6.196e-003	8.198e-003	8.198e-003
49	Static Gradient (kPa/m)	0.0000	0.0000	1.591e-003	1.591e-003
50	Accel. Gradient (kPa/m)	7.926e-007	7.926e-007	1.252e-006	1.252e-006
51	Liq. Reynolds	1.827e+005 *	1.827e+005 *	1.827e+005 *	1.827e+005 *
52	Vap. Reynolds	9.896e+006 *	9.896e+006 *	9.896e+006 *	9.896e+006 *
53	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
54	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	2.997e+004	2.998e+004	2.998e+004	2.999e+004
58	Elevation (m)	6.200e-002	6.600e-002	7.000e-002	6.800e-002
59	Pressure (bar)	67.60	67.59	67.59	67.59
60	Temperature (C)	46.99	46.99	46.99	46.99
61	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
62	Heat Transferred (kJ/h-m)	452.6	452.6	452.5	452.5
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2331	0.2331	0.2331	0.1709
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	8.198e-003	8.198e-003	8.198e-003	9.066e-003
68	Static Gradient (kPa/m)	1.591e-003	1.591e-003	1.591e-003	-6.394e-004
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	1.252e-006	1.252e-006	1.252e-006	1.078e-006
14	Liq. Reynolds	1.827e+005 *	1.827e+005 *	1.827e+005 *	1.827e+005 *
15	Vap. Reynolds	9.896e+006 *	9.896e+006 *	9.896e+006 *	9.896e+006 *
16	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
17	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	2.999e+004	3.000e+004	3.000e+004	3.001e+004
21	Elevation (m)	6.600e-002	6.400e-002	6.200e-002	6.000e-002
22	Pressure (bar)	67.59	67.59	67.59	67.59
23	Temperature (C)	46.99	46.98	46.98	46.98
24	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
25	Heat Transferred (kJ/h-m)	452.5	452.5	452.4	452.4
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.1709	0.1709	0.1709	0.1709
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	9.066e-003	9.066e-003	9.066e-003	9.066e-003
31	Static Gradient (kPa/m)	-6.394e-004	-6.394e-004	-6.394e-004	-6.394e-004
32	Accel. Gradient (kPa/m)	1.078e-006	1.078e-006	1.078e-006	1.078e-006
33	Liq. Reynolds	1.827e+005 *	1.827e+005 *	1.827e+005 *	1.827e+005 *
34	Vap. Reynolds	9.896e+006 *	9.896e+006 *	9.896e+006 *	9.896e+006 *
35	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
36	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	3.001e+004	3.002e+004	3.002e+004	3.003e+004
40	Elevation (m)	5.800e-002	5.600e-002	5.400e-002	5.200e-002
41	Pressure (bar)	67.59	67.59	67.59	67.59
42	Temperature (C)	46.98	46.98	46.98	46.98
43	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
44	Heat Transferred (kJ/h-m)	452.3	452.3	452.3	452.2
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.1709	0.1709	0.1709	0.1709
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	9.066e-003	9.066e-003	9.066e-003	9.066e-003
50	Static Gradient (kPa/m)	-6.394e-004	-6.394e-004	-6.394e-004	-6.394e-004
51	Accel. Gradient (kPa/m)	1.078e-006	1.078e-006	1.078e-006	1.078e-006
52	Liq. Reynolds	1.827e+005 *	1.827e+005 *	1.827e+005 *	1.827e+005 *
53	Vap. Reynolds	9.896e+006 *	9.896e+006 *	9.896e+006 *	9.896e+006 *
54	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
55	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	3.003e+004	3.004e+004	3.004e+004	3.005e+004
59	Elevation (m)	5.000e-002	4.600e-002	4.200e-002	3.800e-002
60	Pressure (bar)	67.59	67.59	67.59	67.59
61	Temperature (C)	46.97	46.97	46.97	46.97
62	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
63	Heat Transferred (kJ/h-m)	452.2	452.2	452.1	452.1
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.1709	0.1706	0.1706	0.1706
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	9.066e-003	9.070e-003	9.070e-003	9.070e-003
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:13:43 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Static Gradient	(kPa/m)	-6.394e-004	-1.277e-003	-1.277e-003	-1.277e-003	
14	Accel. Gradient	(kPa/m)	1.078e-006	9.970e-007	9.970e-007	9.970e-007	
15	Liq. Reynolds		1.827e+005 *	1.827e+005 *	1.827e+005 *	1.827e+005 *	
16	Vap. Reynolds		9.896e+006 *	9.896e+006 *	9.896e+006 *	9.896e+006 *	
17	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
18	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
19	Deposit Thickness	(mm)	---	---	---	---	
20	Deposit Volume	(m3)	---	---	---	---	
21	Length	(m)	3.005e+004	3.006e+004	3.006e+004	3.007e+004	
22	Elevation	(m)	3.400e-002	3.000e-002	3.400e-002	3.800e-002	
23	Pressure	(bar)	67.59	67.59	67.59	67.59	
24	Temperature	(C)	46.97	46.97	46.97	46.96	
25	Vapour Fraction		0.8803	0.8803	0.8803	0.8803	
26	Heat Transferred	(kJ/h-m)	452.0	452.0	451.9	451.9	
27	Flow Regime		Transition	Transition	Transition	Transition	
28	Water Flow Regime						
29	Liquid HoldUp		0.1706	0.1706	0.2331	0.2331	
30	Water HoldUp		---	---	---	---	
31	Friction Gradient	(kPa/m)	9.070e-003	9.070e-003	8.198e-003	8.198e-003	
32	Static Gradient	(kPa/m)	-1.277e-003	-1.277e-003	1.591e-003	1.591e-003	
33	Accel. Gradient	(kPa/m)	9.970e-007	9.970e-007	1.253e-006	1.253e-006	
34	Liq. Reynolds		1.827e+005 *	1.827e+005 *	1.827e+005 *	1.827e+005 *	
35	Vap. Reynolds		9.896e+006 *	9.896e+006 *	9.896e+006 *	9.896e+006 *	
36	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
37	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
38	Deposit Thickness	(mm)	---	---	---	---	
39	Deposit Volume	(m3)	---	---	---	---	
40	Length	(m)	3.007e+004	3.008e+004	3.008e+004	3.009e+004	
41	Elevation	(m)	4.200e-002	4.600e-002	5.000e-002	4.600e-002	
42	Pressure	(bar)	67.59	67.59	67.59	67.59	
43	Temperature	(C)	46.96	46.96	46.96	46.96	
44	Vapour Fraction		0.8803	0.8803	0.8803	0.8803	
45	Heat Transferred	(kJ/h-m)	451.9	451.8	451.8	451.8	
46	Flow Regime		Transition	Transition	Transition	Transition	
47	Water Flow Regime						
48	Liquid HoldUp		0.2331	0.2331	0.2331	0.1706	
49	Water HoldUp		---	---	---	---	
50	Friction Gradient	(kPa/m)	8.198e-003	8.198e-003	8.198e-003	9.070e-003	
51	Static Gradient	(kPa/m)	1.591e-003	1.591e-003	1.591e-003	-1.277e-003	
52	Accel. Gradient	(kPa/m)	1.253e-006	1.253e-006	1.253e-006	9.971e-007	
53	Liq. Reynolds		1.827e+005 *	1.827e+005 *	1.827e+005 *	1.827e+005 *	
54	Vap. Reynolds		9.896e+006 *	9.896e+006 *	9.896e+006 *	9.896e+006 *	
55	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
56	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
57	Deposit Thickness	(mm)	---	---	---	---	
58	Deposit Volume	(m3)	---	---	---	---	
59	Length	(m)	3.009e+004	3.010e+004	3.010e+004	3.011e+004	
60	Elevation	(m)	4.200e-002	3.800e-002	3.400e-002	3.000e-002	
61	Pressure	(bar)	67.58	67.58	67.58	67.58	
62	Temperature	(C)	46.96	46.96	46.95	46.95	
63	Vapour Fraction		0.8803	0.8803	0.8803	0.8803	
64	Heat Transferred	(kJ/h-m)	451.7	451.7	451.7	451.6	
65	Flow Regime		Transition	Transition	Transition	Transition	
66	Water Flow Regime						
67	Liquid HoldUp		0.1706	0.1706	0.1706	0.1706	
68	Water HoldUp		---	---	---	---	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 481 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	9.070e-003	9.070e-003	9.070e-003	9.070e-003	
14	Static Gradient	(kPa/m)	-1.277e-003	-1.277e-003	-1.277e-003	-1.277e-003	
15	Accel. Gradient	(kPa/m)	9.971e-007	9.971e-007	9.971e-007	9.971e-007	
16	Liq. Reynolds		1.827e+005 *	1.827e+005 *	1.827e+005 *	1.827e+005 *	
17	Vap. Reynolds		9.896e+006 *	9.896e+006 *	9.896e+006 *	9.896e+006 *	
18	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
19	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	3.011e+004	3.012e+004	3.012e+004	3.013e+004	
23	Elevation	(m)	2.600e-002	2.200e-002	1.800e-002	1.400e-002	
24	Pressure	(bar)	67.58	67.58	67.58	67.58	
25	Temperature	(C)	46.95	46.95	46.95	46.95	
26	Vapour Fraction		0.8803	0.8803	0.8803	0.8803	
27	Heat Transferred	(kJ/h-m)	451.6	451.5	451.5	451.5	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.1706	0.1706	0.1706	0.1706	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	9.070e-003	9.070e-003	9.070e-003	9.070e-003	
33	Static Gradient	(kPa/m)	-1.277e-003	-1.277e-003	-1.277e-003	-1.277e-003	
34	Accel. Gradient	(kPa/m)	9.971e-007	9.971e-007	9.971e-007	9.971e-007	
35	Liq. Reynolds		1.827e+005 *	1.827e+005 *	1.827e+005 *	1.827e+005 *	
36	Vap. Reynolds		9.896e+006 *	9.896e+006 *	9.896e+006 *	9.896e+006 *	
37	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
38	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	3.013e+004	3.014e+004	3.014e+004	3.015e+004	
42	Elevation	(m)	1.000e-002	1.400e-002	1.800e-002	2.200e-002	
43	Pressure	(bar)	67.58	67.58	67.58	67.58	
44	Temperature	(C)	46.95	46.94	46.94	46.94	
45	Vapour Fraction		0.8803	0.8803	0.8803	0.8803	
46	Heat Transferred	(kJ/h-m)	451.4	451.4	451.3	451.3	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.1706	0.2331	0.2331	0.2331	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	9.070e-003	8.199e-003	8.199e-003	8.200e-003	
52	Static Gradient	(kPa/m)	-1.277e-003	1.585e-003	1.585e-003	1.585e-003	
53	Accel. Gradient	(kPa/m)	9.971e-007	1.252e-006	1.252e-006	1.252e-006	
54	Liq. Reynolds		1.826e+005 *	1.826e+005 *	1.826e+005 *	1.826e+005 *	
55	Vap. Reynolds		9.896e+006 *	9.896e+006 *	9.896e+006 *	9.896e+006 *	
56	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
57	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	3.015e+004	3.016e+004	3.016e+004	3.017e+004	
61	Elevation	(m)	2.600e-002	3.000e-002	3.200e-002	3.400e-002	
62	Pressure	(bar)	67.58	67.58	67.58	67.58	
63	Temperature	(C)	46.94	46.94	46.94	46.94	
64	Vapour Fraction		0.8803	0.8803	0.8803	0.8803	
65	Heat Transferred	(kJ/h-m)	451.2	451.2	451.2	451.1	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.2331	0.2331	0.2320	0.2320	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 482 of 937	

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	8.200e-003	8.200e-003	8.346e-003	8.346e-003
15	Static Gradient (kPa/m)	1.585e-003	1.585e-003	7.927e-004	7.927e-004
16	Accel. Gradient (kPa/m)	1.252e-006	1.252e-006	1.169e-006	1.169e-006
17	Liq. Reynolds	1.826e+005 *	1.826e+005 *	1.826e+005 *	1.826e+005 *
18	Vap. Reynolds	9.896e+006 *	9.896e+006 *	9.896e+006 *	9.897e+006 *
19	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
20	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	3.017e+004	3.018e+004	3.018e+004	3.019e+004
24	Elevation (m)	3.600e-002	3.800e-002	4.000e-002	4.000e-002
25	Pressure (bar)	67.58	67.58	67.58	67.58
26	Temperature (C)	46.93	46.93	46.93	46.93
27	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
28	Heat Transferred (kJ/h-m)	451.1	451.0	451.0	451.0
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2320	0.2320	0.2320	0.2499
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	8.346e-003	8.346e-003	8.346e-003	6.196e-003
34	Static Gradient (kPa/m)	7.927e-004	7.927e-004	7.927e-004	0.0000
35	Accel. Gradient (kPa/m)	1.169e-006	1.169e-006	1.169e-006	7.929e-007
36	Liq. Reynolds	1.826e+005 *	1.826e+005 *	1.826e+005 *	1.826e+005 *
37	Vap. Reynolds	9.897e+006 *	9.897e+006 *	9.897e+006 *	9.897e+006 *
38	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
39	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	3.019e+004	3.020e+004	3.020e+004	3.021e+004
43	Elevation (m)	4.000e-002	4.000e-002	4.000e-002	4.000e-002
44	Pressure (bar)	67.58	67.58	67.58	67.58
45	Temperature (C)	46.93	46.93	46.93	46.92
46	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
47	Heat Transferred (kJ/h-m)	450.9	450.9	450.9	450.8
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2499	0.2499	0.2499	0.2499
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	6.196e-003	6.196e-003	6.196e-003	6.196e-003
53	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
54	Accel. Gradient (kPa/m)	7.929e-007	7.929e-007	7.929e-007	7.929e-007
55	Liq. Reynolds	1.826e+005 *	1.826e+005 *	1.826e+005 *	1.826e+005 *
56	Vap. Reynolds	9.897e+006 *	9.897e+006 *	9.897e+006 *	9.897e+006 *
57	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
58	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	3.021e+004	3.022e+004	3.022e+004	3.023e+004
62	Elevation (m)	4.400e-002	4.800e-002	5.200e-002	5.600e-002
63	Pressure (bar)	67.57	67.57	67.57	67.57
64	Temperature (C)	46.92	46.92	46.92	46.92
65	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
66	Heat Transferred (kJ/h-m)	450.8	450.7	450.7	450.7
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 483 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.2331	0.2331	0.2331	0.2331
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	8.198e-003	8.198e-003	8.198e-003	8.198e-003
16	Static Gradient (kPa/m)	1.591e-003	1.591e-003	1.591e-003	1.591e-003
17	Accel. Gradient (kPa/m)	1.253e-006	1.253e-006	1.253e-006	1.253e-006
18	Liq. Reynolds	1.826e+005 *	1.826e+005 *	1.826e+005 *	1.826e+005 *
19	Vap. Reynolds	9.897e+006 *	9.897e+006 *	9.897e+006 *	9.897e+006 *
20	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
21	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	3.023e+004	3.024e+004	3.024e+004	3.025e+004
25	Elevation (m)	6.000e-002	5.600e-002	5.200e-002	4.800e-002
26	Pressure (bar)	67.57	67.57	67.57	67.57
27	Temperature (C)	46.92	46.92	46.91	46.91
28	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
29	Heat Transferred (kJ/h-m)	450.6	450.6	450.6	450.6
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2331	0.1706	0.1706	0.1706
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	8.198e-003	9.070e-003	9.070e-003	9.070e-003
35	Static Gradient (kPa/m)	1.591e-003	-1.277e-003	-1.277e-003	-1.277e-003
36	Accel. Gradient (kPa/m)	1.253e-006	9.972e-007	9.972e-007	9.972e-007
37	Liq. Reynolds	1.826e+005 *	1.826e+005 *	1.826e+005 *	1.826e+005 *
38	Vap. Reynolds	9.897e+006 *	9.897e+006 *	9.897e+006 *	9.897e+006 *
39	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
40	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	3.025e+004	3.026e+004	3.026e+004	3.027e+004
44	Elevation (m)	4.400e-002	4.000e-002	4.600e-002	5.200e-002
45	Pressure (bar)	67.57	67.57	67.57	67.57
46	Temperature (C)	46.91	46.91	46.91	46.91
47	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
48	Heat Transferred (kJ/h-m)	450.5	450.5	450.4	450.4
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1706	0.1706	0.2342	0.2342
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	9.070e-003	9.070e-003	8.053e-003	8.053e-003
54	Static Gradient (kPa/m)	-1.277e-003	-1.277e-003	2.395e-003	2.395e-003
55	Accel. Gradient (kPa/m)	9.973e-007	9.973e-007	1.337e-006	1.337e-006
56	Liq. Reynolds	1.826e+005 *	1.826e+005 *	1.826e+005 *	1.826e+005 *
57	Vap. Reynolds	9.897e+006 *	9.897e+006 *	9.897e+006 *	9.897e+006 *
58	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
59	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	3.027e+004	3.028e+004	3.028e+004	3.029e+004
63	Elevation (m)	5.800e-002	6.400e-002	7.000e-002	7.600e-002
64	Pressure (bar)	67.57	67.57	67.57	67.57
65	Temperature (C)	46.91	46.90	46.90	46.90
66	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
67	Heat Transferred (kJ/h-m)	450.3	450.3	450.2	450.2
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 484 of 937

1			Case Name:	Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set:	Mixed		
3	 Epic Atlantic Limited		Date/Time:	Mon Jun 13 15:13:43 2022		
4						
5						
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Water Flow Regime					
14	Liquid HoldUp	0.2342	0.2342	0.2342	0.2342	
15	Water HoldUp	---	---	---	---	
16	Friction Gradient (kPa/m)	8.053e-003	8.053e-003	8.053e-003	8.053e-003	
17	Static Gradient (kPa/m)	2.395e-003	2.395e-003	2.395e-003	2.395e-003	
18	Accel. Gradient (kPa/m)	1.337e-006	1.337e-006	1.337e-006	1.337e-006	
19	Liq. Reynolds	1.826e+005 *	1.826e+005 *	1.826e+005 *	1.826e+005 *	
20	Vap. Reynolds	9.897e+006 *	9.897e+006 *	9.897e+006 *	9.897e+006 *	
21	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	
22	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933	
23	Deposit Thickness (mm)	---	---	---	---	
24	Deposit Volume (m3)	---	---	---	---	
25	Length (m)	3.029e+004	3.030e+004	3.030e+004	3.031e+004	
26	Elevation (m)	8.200e-002	8.800e-002	9.400e-002	1.000e-001	
27	Pressure (bar)	67.57	67.57	67.57	67.57	
28	Temperature (C)	46.90	46.90	46.90	46.89	
29	Vapour Fraction	0.8803	0.8803	0.8803	0.8803	
30	Heat Transferred (kJ/h-m)	450.2	450.1	450.1	450.0	
31	Flow Regime	Transition	Transition	Transition	Transition	
32	Water Flow Regime					
33	Liquid HoldUp	0.2342	0.2342	0.2342	0.2342	
34	Water HoldUp	---	---	---	---	
35	Friction Gradient (kPa/m)	8.053e-003	8.053e-003	8.053e-003	8.053e-003	
36	Static Gradient (kPa/m)	2.395e-003	2.395e-003	2.395e-003	2.395e-003	
37	Accel. Gradient (kPa/m)	1.337e-006	1.337e-006	1.337e-006	1.337e-006	
38	Liq. Reynolds	1.826e+005 *	1.826e+005 *	1.826e+005 *	1.826e+005 *	
39	Vap. Reynolds	9.897e+006 *	9.897e+006 *	9.897e+006 *	9.897e+006 *	
40	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	
41	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933	
42	Deposit Thickness (mm)	---	---	---	---	
43	Deposit Volume (m3)	---	---	---	---	
44	Length (m)	3.031e+004	3.032e+004	3.032e+004	3.033e+004	
45	Elevation (m)	0.1200	0.1400	0.1600	0.1800	
46	Pressure (bar)	67.57	67.56	67.56	67.56	
47	Temperature (C)	46.89	46.89	46.89	46.89	
48	Vapour Fraction	0.8803	0.8803	0.8803	0.8803	
49	Heat Transferred (kJ/h-m)	450.0	450.0	449.9	449.9	
50	Flow Regime	Transition	Transition	Transition	Transition	
51	Water Flow Regime					
52	Liquid HoldUp	0.2420	0.2420	0.2420	0.2420	
53	Water HoldUp	---	---	---	---	
54	Friction Gradient (kPa/m)	7.091e-003	7.091e-003	7.091e-003	7.091e-003	
55	Static Gradient (kPa/m)	8.177e-003	8.177e-003	8.177e-003	8.177e-003	
56	Accel. Gradient (kPa/m)	1.954e-006	1.954e-006	1.954e-006	1.954e-006	
57	Liq. Reynolds	1.826e+005 *	1.826e+005 *	1.826e+005 *	1.826e+005 *	
58	Vap. Reynolds	9.897e+006 *	9.897e+006 *	9.897e+006 *	9.897e+006 *	
59	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	
60	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933	
61	Deposit Thickness (mm)	---	---	---	---	
62	Deposit Volume (m3)	---	---	---	---	
63	Length (m)	3.033e+004	3.034e+004	3.035e+004	3.036e+004	
64	Elevation (m)	0.2000	0.1960	0.1920	0.1880	
65	Pressure (bar)	67.56	67.56	67.56	67.56	
66	Temperature (C)	46.89	46.88	46.88	46.88	
67	Vapour Fraction	0.8803	0.8803	0.8803	0.8803	
68	Heat Transferred (kJ/h-m)	449.8	449.8	449.8	449.7	
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 485 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	
14	Water Flow Regime				
15	Liquid HoldUp	0.2420	0.1708	0.1708	
16	Water HoldUp	---	---	---	
17	Friction Gradient (kPa/m)	7.091e-003	9.067e-003	9.067e-003	
18	Static Gradient (kPa/m)	8.177e-003	-7.990e-004	-7.990e-004	
19	Accel. Gradient (kPa/m)	1.954e-006	1.058e-006	1.058e-006	
20	Liq. Reynolds	1.826e+005 *	1.826e+005 *	1.826e+005 *	
21	Vap. Reynolds	9.897e+006 *	9.897e+006 *	9.897e+006 *	
22	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	
23	Vapour Velocity (m/s)	2.933	2.933	2.933	
24	Deposit Thickness (mm)	---	---	---	
25	Deposit Volume (m3)	---	---	---	
26	Length (m)	3.036e+004	3.037e+004	3.039e+004	
27	Elevation (m)	0.1840	0.1800	0.1820	
28	Pressure (bar)	67.56	67.56	67.56	
29	Temperature (C)	46.88	46.88	46.87	
30	Vapour Fraction	0.8803	0.8803	0.8803	
31	Heat Transferred (kJ/h-m)	449.6	449.6	449.4	
32	Flow Regime	Transition	Transition	Transition	
33	Water Flow Regime				
34	Liquid HoldUp	0.1708	0.1708	0.2313	
35	Water HoldUp	---	---	---	
36	Friction Gradient (kPa/m)	9.067e-003	9.067e-003	8.442e-003	
37	Static Gradient (kPa/m)	-7.990e-004	-7.990e-004	2.825e-004	
38	Accel. Gradient (kPa/m)	1.058e-006	1.058e-006	1.117e-006	
39	Liq. Reynolds	1.826e+005 *	1.826e+005 *	1.826e+005 *	
40	Vap. Reynolds	9.897e+006 *	9.897e+006 *	9.897e+006 *	
41	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	
42	Vapour Velocity (m/s)	2.933	2.933	2.933	
43	Deposit Thickness (mm)	---	---	---	
44	Deposit Volume (m3)	---	---	---	
45	Length (m)	3.041e+004	3.043e+004	3.044e+004	
46	Elevation (m)	0.1860	0.1880	0.1900	
47	Pressure (bar)	67.55	67.55	67.55	
48	Temperature (C)	46.86	46.86	46.86	
49	Vapour Fraction	0.8803	0.8803	0.8803	
50	Heat Transferred (kJ/h-m)	449.2	449.1	449.0	
51	Flow Regime	Transition	Transition	Transition	
52	Water Flow Regime				
53	Liquid HoldUp	0.2313	0.2313	0.2313	
54	Water HoldUp	---	---	---	
55	Friction Gradient (kPa/m)	8.442e-003	8.442e-003	8.442e-003	
56	Static Gradient (kPa/m)	2.825e-004	2.825e-004	2.825e-004	
57	Accel. Gradient (kPa/m)	1.117e-006	1.117e-006	1.117e-006	
58	Liq. Reynolds	1.826e+005 *	1.825e+005 *	1.825e+005 *	
59	Vap. Reynolds	9.898e+006 *	9.898e+006 *	9.898e+006 *	
60	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	
61	Vapour Velocity (m/s)	2.933	2.933	2.933	
62	Deposit Thickness (mm)	---	---	---	
63	Deposit Volume (m3)	---	---	---	
64	Length (m)	3.045e+004	3.045e+004	3.045e+004	
65	Elevation (m)	0.1700	0.1600	0.1500	
66	Pressure (bar)	67.55	67.55	67.55	
67	Temperature (C)	46.85	46.85	46.85	
68	Vapour Fraction	0.8803	0.8803	0.8803	
69	Aspen Technology Inc.		Aspen HYSYS Version 10		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	449.0	449.0	449.0	448.9
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.1687	0.1687	0.1687	0.1687
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	9.095e-003	9.095e-003	9.095e-003	9.095e-003
19	Static Gradient (kPa/m)	-5.283e-003	-5.283e-003	-5.283e-003	-5.283e-003
20	Accel. Gradient (kPa/m)	4.880e-007	4.880e-007	4.880e-007	4.880e-007
21	Liq. Reynolds	1.825e+005 *	1.825e+005 *	1.825e+005 *	1.825e+005 *
22	Vap. Reynolds	9.898e+006 *	9.898e+006 *	9.898e+006 *	9.898e+006 *
23	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
24	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	3.046e+004	3.047e+004	3.047e+004	3.048e+004
28	Elevation (m)	0.1400	0.1400	0.1400	0.1400
29	Pressure (bar)	67.55	67.55	67.55	67.55
30	Temperature (C)	46.85	46.85	46.85	46.85
31	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
32	Heat Transferred (kJ/h-m)	448.9	448.8	448.8	448.7
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.2499	0.2499	0.2499	0.2499
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	6.196e-003	6.196e-003	6.196e-003	6.196e-003
38	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
39	Accel. Gradient (kPa/m)	7.932e-007	7.932e-007	7.932e-007	7.932e-007
40	Liq. Reynolds	1.825e+005 *	1.825e+005 *	1.825e+005 *	1.825e+005 *
41	Vap. Reynolds	9.898e+006 *	9.898e+006 *	9.898e+006 *	9.898e+006 *
42	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
43	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	3.048e+004	3.049e+004	3.049e+004	3.050e+004
47	Elevation (m)	0.1400	0.1380	0.1360	0.1340
48	Pressure (bar)	67.55	67.55	67.55	67.55
49	Temperature (C)	46.84	46.84	46.84	46.84
50	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
51	Heat Transferred (kJ/h-m)	448.7	448.7	448.7	448.6
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.2499	0.1709	0.1709	0.1709
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	6.196e-003	9.066e-003	9.066e-003	9.066e-003
57	Static Gradient (kPa/m)	0.0000	-6.394e-004	-6.394e-004	-6.394e-004
58	Accel. Gradient (kPa/m)	7.932e-007	1.079e-006	1.079e-006	1.079e-006
59	Liq. Reynolds	1.825e+005 *	1.825e+005 *	1.825e+005 *	1.825e+005 *
60	Vap. Reynolds	9.898e+006 *	9.898e+006 *	9.898e+006 *	9.898e+006 *
61	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
62	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	3.050e+004	3.051e+004	3.051e+004	3.052e+004
66	Elevation (m)	0.1320	0.1300	0.1240	0.1180
67	Pressure (bar)	67.55	67.55	67.55	67.55
68	Temperature (C)	46.84	46.84	46.84	46.83
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
14	Heat Transferred (kJ/h-m)	448.6	448.6	448.5	448.5
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.1709	0.1709	0.1703	0.1703
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	9.066e-003	9.066e-003	9.074e-003	9.074e-003
20	Static Gradient (kPa/m)	-6.394e-004	-6.394e-004	-1.914e-003	-1.914e-003
21	Accel. Gradient (kPa/m)	1.079e-006	1.079e-006	9.167e-007	9.167e-007
22	Liq. Reynolds	1.825e+005 *	1.825e+005 *	1.825e+005 *	1.825e+005 *
23	Vap. Reynolds	9.898e+006 *	9.898e+006 *	9.898e+006 *	9.898e+006 *
24	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
25	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	3.052e+004	3.053e+004	3.053e+004	3.054e+004
29	Elevation (m)	0.1120	0.1060	1.000e-001	9.600e-002
30	Pressure (bar)	67.55	67.55	67.55	67.55
31	Temperature (C)	46.83	46.83	46.83	46.83
32	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
33	Heat Transferred (kJ/h-m)	448.4	448.4	448.4	448.3
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1703	0.1703	0.1703	0.1706
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	9.074e-003	9.074e-003	9.074e-003	9.069e-003
39	Static Gradient (kPa/m)	-1.914e-003	-1.914e-003	-1.914e-003	-1.186e-003
40	Accel. Gradient (kPa/m)	9.167e-007	9.167e-007	9.167e-007	1.009e-006
41	Liq. Reynolds	1.825e+005 *	1.825e+005 *	1.825e+005 *	1.825e+005 *
42	Vap. Reynolds	9.898e+006 *	9.898e+006 *	9.898e+006 *	9.898e+006 *
43	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
44	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	3.054e+004	3.055e+004	3.055e+004	3.056e+004
48	Elevation (m)	9.200e-002	8.800e-002	8.400e-002	8.000e-002
49	Pressure (bar)	67.55	67.55	67.54	67.54
50	Temperature (C)	46.83	46.83	46.82	46.82
51	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
52	Heat Transferred (kJ/h-m)	448.3	448.3	448.2	448.2
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.1706	0.1706	0.1706	0.1706
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	9.069e-003	9.069e-003	9.069e-003	9.069e-003
58	Static Gradient (kPa/m)	-1.186e-003	-1.186e-003	-1.186e-003	-1.186e-003
59	Accel. Gradient (kPa/m)	1.009e-006	1.009e-006	1.009e-006	1.009e-006
60	Liq. Reynolds	1.825e+005 *	1.825e+005 *	1.825e+005 *	1.825e+005 *
61	Vap. Reynolds	9.898e+006 *	9.898e+006 *	9.898e+006 *	9.898e+006 *
62	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
63	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	3.056e+004	3.057e+004	3.057e+004	3.058e+004
67	Elevation (m)	7.800e-002	7.600e-002	7.400e-002	7.200e-002
68	Pressure (bar)	67.54	67.54	67.54	67.54
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	46.82	46.82	46.82	46.82
14	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
15	Heat Transferred (kJ/h-m)	448.1	448.1	448.1	448.0
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.1709	0.1709	0.1709	0.1709
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	9.066e-003	9.066e-003	9.066e-003	9.066e-003
21	Static Gradient (kPa/m)	-6.394e-004	-6.394e-004	-6.394e-004	-6.394e-004
22	Accel. Gradient (kPa/m)	1.079e-006	1.079e-006	1.079e-006	1.079e-006
23	Liq. Reynolds	1.825e+005 *	1.825e+005 *	1.825e+005 *	1.825e+005 *
24	Vap. Reynolds	9.898e+006 *	9.898e+006 *	9.898e+006 *	9.898e+006 *
25	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
26	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	3.058e+004	3.059e+004	3.059e+004	3.060e+004
30	Elevation (m)	7.000e-002	7.800e-002	8.600e-002	9.400e-002
31	Pressure (bar)	67.54	67.54	67.54	67.54
32	Temperature (C)	46.82	46.81	46.81	46.81
33	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
34	Heat Transferred (kJ/h-m)	448.0	447.9	447.9	447.8
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1709	0.2353	0.2353	0.2353
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	9.066e-003	7.910e-003	7.910e-003	7.910e-003
40	Static Gradient (kPa/m)	-6.394e-004	3.204e-003	3.204e-003	3.204e-003
41	Accel. Gradient (kPa/m)	1.079e-006	1.423e-006	1.423e-006	1.423e-006
42	Liq. Reynolds	1.825e+005 *	1.825e+005 *	1.825e+005 *	1.825e+005 *
43	Vap. Reynolds	9.898e+006 *	9.898e+006 *	9.898e+006 *	9.898e+006 *
44	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
45	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	3.060e+004	3.061e+004	3.061e+004	3.062e+004
49	Elevation (m)	0.1020	0.1100	0.1160	0.1220
50	Pressure (bar)	67.54	67.54	67.54	67.54
51	Temperature (C)	46.81	46.81	46.81	46.81
52	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
53	Heat Transferred (kJ/h-m)	447.8	447.7	447.7	447.7
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2353	0.2353	0.2342	0.2342
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	7.910e-003	7.910e-003	8.053e-003	8.053e-003
59	Static Gradient (kPa/m)	3.204e-003	3.204e-003	2.395e-003	2.395e-003
60	Accel. Gradient (kPa/m)	1.423e-006	1.423e-006	1.338e-006	1.338e-006
61	Liq. Reynolds	1.825e+005 *	1.825e+005 *	1.825e+005 *	1.825e+005 *
62	Vap. Reynolds	9.898e+006 *	9.898e+006 *	9.898e+006 *	9.898e+006 *
63	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
64	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	3.062e+004	3.063e+004	3.063e+004	3.064e+004
68	Elevation (m)	0.1280	0.1340	0.1400	0.1460
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	67.54	67.54	67.54	67.54		
14	Temperature (C)	46.80	46.80	46.80	46.80		
15	Vapour Fraction	0.8803	0.8803	0.8803	0.8803		
16	Heat Transferred (kJ/h-m)	447.6	447.6	447.6	447.5		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.2342	0.2342	0.2342	0.2342		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	8.053e-003	8.053e-003	8.053e-003	8.054e-003		
22	Static Gradient (kPa/m)	2.395e-003	2.395e-003	2.395e-003	2.395e-003		
23	Accel. Gradient (kPa/m)	1.338e-006	1.338e-006	1.338e-006	1.338e-006		
24	Liq. Reynolds	1.825e+005 *	1.825e+005 *	1.825e+005 *	1.825e+005 *		
25	Vap. Reynolds	9.898e+006 *	9.898e+006 *	9.898e+006 *	9.898e+006 *		
26	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
27	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	3.064e+004	3.065e+004	3.065e+004	3.066e+004		
31	Elevation (m)	0.1520	0.1580	0.1640	0.1700		
32	Pressure (bar)	67.54	67.54	67.53	67.53		
33	Temperature (C)	46.80	46.80	46.80	46.79		
34	Vapour Fraction	0.8803	0.8803	0.8803	0.8803		
35	Heat Transferred (kJ/h-m)	447.5	447.4	447.4	447.4		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.2342	0.2342	0.2342	0.2342		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	8.054e-003	8.054e-003	8.054e-003	8.054e-003		
41	Static Gradient (kPa/m)	2.395e-003	2.395e-003	2.395e-003	2.395e-003		
42	Accel. Gradient (kPa/m)	1.338e-006	1.338e-006	1.338e-006	1.338e-006		
43	Liq. Reynolds	1.825e+005 *	1.825e+005 *	1.825e+005 *	1.825e+005 *		
44	Vap. Reynolds	9.899e+006 *	9.899e+006 *	9.899e+006 *	9.899e+006 *		
45	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
46	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	3.066e+004	3.067e+004	3.067e+004	3.068e+004		
50	Elevation (m)	0.1880	0.2060	0.2240	0.2420		
51	Pressure (bar)	67.53	67.53	67.53	67.53		
52	Temperature (C)	46.79	46.79	46.79	46.79		
53	Vapour Fraction	0.8803	0.8803	0.8803	0.8803		
54	Heat Transferred (kJ/h-m)	447.3	447.3	447.2	447.2		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.2409	0.2409	0.2409	0.2409		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	7.223e-003	7.223e-003	7.223e-003	7.223e-003		
60	Static Gradient (kPa/m)	7.335e-003	7.335e-003	7.335e-003	7.335e-003		
61	Accel. Gradient (kPa/m)	1.864e-006	1.864e-006	1.864e-006	1.864e-006		
62	Liq. Reynolds	1.825e+005 *	1.825e+005 *	1.825e+005 *	1.825e+005 *		
63	Vap. Reynolds	9.899e+006 *	9.899e+006 *	9.899e+006 *	9.899e+006 *		
64	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
65	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	3.068e+004	3.069e+004	3.069e+004	3.070e+004		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Elevation (m)	0.2600	0.2000	0.1400	8.000e-002
14	Pressure (bar)	67.53	67.53	67.53	67.53
15	Temperature (C)	46.79	46.79	46.78	46.78
16	Vapour Fraction	0.8803	0.8803	0.8802	0.8802
17	Heat Transferred (kJ/h-m)	447.2	447.2	447.1	447.1
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2409	0.1623	0.1623	0.1623
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	7.223e-003	9.189e-003	9.189e-003	9.189e-003
23	Static Gradient (kPa/m)	7.335e-003	-1.854e-002	-1.854e-002	-1.854e-002
24	Accel. Gradient (kPa/m)	1.864e-006	-1.197e-006	-1.197e-006	-1.197e-006
25	Liq. Reynolds	1.825e+005 *	1.825e+005 *	1.825e+005 *	1.825e+005 *
26	Vap. Reynolds	9.899e+006 *	9.899e+006 *	9.899e+006 *	9.899e+006 *
27	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
28	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	3.070e+004	3.071e+004	3.071e+004	3.072e+004
32	Elevation (m)	2.000e-002	-4.000e-002	-4.800e-002	-5.600e-002
33	Pressure (bar)	67.53	67.53	67.53	67.53
34	Temperature (C)	46.78	46.78	46.78	46.78
35	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
36	Heat Transferred (kJ/h-m)	447.1	447.1	447.0	447.0
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.1623	0.1623	0.1700	0.1700
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	9.188e-003	9.188e-003	9.078e-003	9.078e-003
42	Static Gradient (kPa/m)	-1.854e-002	-1.854e-002	-2.549e-003	-2.549e-003
43	Accel. Gradient (kPa/m)	-1.197e-006	-1.197e-006	8.360e-007	8.360e-007
44	Liq. Reynolds	1.825e+005 *	1.825e+005 *	1.825e+005 *	1.825e+005 *
45	Vap. Reynolds	9.899e+006 *	9.899e+006 *	9.899e+006 *	9.899e+006 *
46	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
47	Vapour Velocity (m/s)	2.933	2.932	2.932	2.932
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	3.072e+004	3.073e+004	3.073e+004	3.074e+004
51	Elevation (m)	-6.400e-002	-7.200e-002	-8.000e-002	-6.600e-002
52	Pressure (bar)	67.53	67.53	67.53	67.53
53	Temperature (C)	46.78	46.78	46.77	46.77
54	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
55	Heat Transferred (kJ/h-m)	446.9	446.9	446.9	446.8
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.1700	0.1700	0.1700	0.2386
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	9.078e-003	9.078e-003	9.078e-003	7.495e-003
61	Static Gradient (kPa/m)	-2.549e-003	-2.549e-003	-2.549e-003	5.643e-003
62	Accel. Gradient (kPa/m)	8.360e-007	8.360e-007	8.360e-007	1.682e-006
63	Liq. Reynolds	1.825e+005 *	1.825e+005 *	1.825e+005 *	1.825e+005 *
64	Vap. Reynolds	9.899e+006 *	9.899e+006 *	9.899e+006 *	9.899e+006 *
65	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
66	Vapour Velocity (m/s)	2.932	2.932	2.932	2.932
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 491 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:13:43 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length	(m)	3.074e+004	3.075e+004	3.075e+004	3.076e+004	
14	Elevation	(m)	-5.200e-002	-3.800e-002	-2.400e-002	-1.000e-002	
15	Pressure	(bar)	67.53	67.53	67.53	67.53	
16	Temperature	(C)	46.77	46.77	46.77	46.77	
17	Vapour Fraction		0.8802	0.8802	0.8802	0.8802	
18	Heat Transferred	(kJ/h-m)	446.8	446.7	446.7	446.6	
19	Flow Regime		Transition	Transition	Transition	Transition	
20	Water Flow Regime						
21	Liquid HoldUp		0.2386	0.2386	0.2386	0.2386	
22	Water HoldUp		---	---	---	---	
23	Friction Gradient	(kPa/m)	7.495e-003	7.495e-003	7.496e-003	7.496e-003	
24	Static Gradient	(kPa/m)	5.643e-003	5.643e-003	5.643e-003	5.643e-003	
25	Accel. Gradient	(kPa/m)	1.682e-006	1.682e-006	1.682e-006	1.682e-006	
26	Liq. Reynolds		1.825e+005 *	1.825e+005 *	1.824e+005 *	1.824e+005 *	
27	Vap. Reynolds		9.899e+006 *	9.899e+006 *	9.899e+006 *	9.899e+006 *	
28	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
29	Vapour Velocity	(m/s)	2.932	2.933	2.933	2.933	
30	Deposit Thickness	(mm)	---	---	---	---	
31	Deposit Volume	(m3)	---	---	---	---	
32	Length	(m)	3.076e+004	3.077e+004	3.077e+004	3.078e+004	
33	Elevation	(m)	-8.000e-003	-6.000e-003	-4.000e-003	-2.000e-003	
34	Pressure	(bar)	67.53	67.53	67.53	67.53	
35	Temperature	(C)	46.77	46.76	46.76	46.76	
36	Vapour Fraction		0.8802	0.8802	0.8802	0.8802	
37	Heat Transferred	(kJ/h-m)	446.6	446.6	446.5	446.5	
38	Flow Regime		Transition	Transition	Transition	Transition	
39	Water Flow Regime						
40	Liquid HoldUp		0.2320	0.2320	0.2320	0.2320	
41	Water HoldUp		---	---	---	---	
42	Friction Gradient	(kPa/m)	8.346e-003	8.346e-003	8.346e-003	8.346e-003	
43	Static Gradient	(kPa/m)	7.928e-004	7.928e-004	7.928e-004	7.928e-004	
44	Accel. Gradient	(kPa/m)	1.170e-006	1.170e-006	1.170e-006	1.170e-006	
45	Liq. Reynolds		1.824e+005 *	1.824e+005 *	1.824e+005 *	1.824e+005 *	
46	Vap. Reynolds		9.899e+006 *	9.899e+006 *	9.899e+006 *	9.899e+006 *	
47	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
48	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
49	Deposit Thickness	(mm)	---	---	---	---	
50	Deposit Volume	(m3)	---	---	---	---	
51	Length	(m)	3.078e+004	3.079e+004	3.079e+004	3.080e+004	
52	Elevation	(m)	-5.360e-016	-6.000e-003	-1.200e-002	-1.800e-002	
53	Pressure	(bar)	67.53	67.53	67.53	67.52	
54	Temperature	(C)	46.76	46.76	46.76	46.76	
55	Vapour Fraction		0.8802	0.8802	0.8802	0.8802	
56	Heat Transferred	(kJ/h-m)	446.4	446.4	446.4	446.4	
57	Flow Regime		Transition	Transition	Transition	Transition	
58	Water Flow Regime						
59	Liquid HoldUp		0.2320	0.1703	0.1703	0.1703	
60	Water HoldUp		---	---	---	---	
61	Friction Gradient	(kPa/m)	8.346e-003	9.074e-003	9.074e-003	9.074e-003	
62	Static Gradient	(kPa/m)	7.928e-004	-1.914e-003	-1.914e-003	-1.914e-003	
63	Accel. Gradient	(kPa/m)	1.170e-006	9.169e-007	9.169e-007	9.169e-007	
64	Liq. Reynolds		1.824e+005 *	1.824e+005 *	1.824e+005 *	1.824e+005 *	
65	Vap. Reynolds		9.899e+006 *	9.899e+006 *	9.899e+006 *	9.899e+006 *	
66	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
67	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933	
68	Deposit Thickness	(mm)	---	---	---	---	
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	3.080e+004	3.081e+004	3.081e+004	3.082e+004
15	Elevation (m)	-2.400e-002	-3.000e-002	-2.000e-002	-1.000e-002
16	Pressure (bar)	67.52	67.52	67.52	67.52
17	Temperature (C)	46.75	46.75	46.75	46.75
18	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
19	Heat Transferred (kJ/h-m)	446.3	446.3	446.2	446.2
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.1703	0.1703	0.2364	0.2364
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	9.074e-003	9.074e-003	7.769e-003	7.769e-003
25	Static Gradient (kPa/m)	-1.914e-003	-1.914e-003	4.019e-003	4.019e-003
26	Accel. Gradient (kPa/m)	9.169e-007	9.169e-007	1.510e-006	1.510e-006
27	Liq. Reynolds	1.824e+005 *	1.824e+005 *	1.824e+005 *	1.824e+005 *
28	Vap. Reynolds	9.899e+006 *	9.899e+006 *	9.899e+006 *	9.899e+006 *
29	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
30	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	3.082e+004	3.083e+004	3.083e+004	3.084e+004
34	Elevation (m)	-5.343e-016	1.000e-002	2.000e-002	2.600e-002
35	Pressure (bar)	67.52	67.52	67.52	67.52
36	Temperature (C)	46.75	46.75	46.75	46.74
37	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
38	Heat Transferred (kJ/h-m)	446.1	446.1	446.1	446.0
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2364	0.2364	0.2364	0.2342
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	7.769e-003	7.769e-003	7.769e-003	8.053e-003
44	Static Gradient (kPa/m)	4.019e-003	4.019e-003	4.019e-003	2.395e-003
45	Accel. Gradient (kPa/m)	1.510e-006	1.510e-006	1.510e-006	1.338e-006
46	Liq. Reynolds	1.824e+005 *	1.824e+005 *	1.824e+005 *	1.824e+005 *
47	Vap. Reynolds	9.899e+006 *	9.899e+006 *	9.899e+006 *	9.899e+006 *
48	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
49	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	3.084e+004	3.085e+004	3.085e+004	3.086e+004
53	Elevation (m)	3.200e-002	3.800e-002	4.400e-002	5.000e-002
54	Pressure (bar)	67.52	67.52	67.52	67.52
55	Temperature (C)	46.74	46.74	46.74	46.74
56	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
57	Heat Transferred (kJ/h-m)	446.0	445.9	445.9	445.9
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2342	0.2342	0.2342	0.2342
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	8.053e-003	8.053e-003	8.054e-003	8.054e-003
63	Static Gradient (kPa/m)	2.395e-003	2.395e-003	2.395e-003	2.395e-003
64	Accel. Gradient (kPa/m)	1.338e-006	1.338e-006	1.338e-006	1.338e-006
65	Liq. Reynolds	1.824e+005 *	1.824e+005 *	1.824e+005 *	1.824e+005 *
66	Vap. Reynolds	9.899e+006 *	9.899e+006 *	9.899e+006 *	9.899e+006 *
67	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
68	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	3.086e+004	3.087e+004	3.087e+004	3.088e+004
16	Elevation (m)	7.800e-002	0.1060	0.1340	0.1620
17	Pressure (bar)	67.52	67.52	67.52	67.51
18	Temperature (C)	46.74	46.73	46.73	46.73
19	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
20	Heat Transferred (kJ/h-m)	445.8	445.8	445.7	445.7
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2464	0.2464	0.2464	0.2464
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	6.581e-003	6.581e-003	6.581e-003	6.581e-003
26	Static Gradient (kPa/m)	1.160e-002	1.160e-002	1.160e-002	1.160e-002
27	Accel. Gradient (kPa/m)	2.329e-006	2.329e-006	2.329e-006	2.329e-006
28	Liq. Reynolds	1.824e+005 *	1.824e+005 *	1.824e+005 *	1.824e+005 *
29	Vap. Reynolds	9.899e+006 *	9.899e+006 *	9.899e+006 *	9.900e+006 *
30	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
31	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	3.088e+004	3.089e+004	3.089e+004	3.090e+004
35	Elevation (m)	0.1900	0.1940	0.1980	0.2020
36	Pressure (bar)	67.51	67.51	67.51	67.51
37	Temperature (C)	46.73	46.73	46.73	46.73
38	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
39	Heat Transferred (kJ/h-m)	445.6	445.6	445.6	445.5
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2464	0.2331	0.2331	0.2331
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	6.582e-003	8.199e-003	8.199e-003	8.199e-003
45	Static Gradient (kPa/m)	1.160e-002	1.591e-003	1.591e-003	1.591e-003
46	Accel. Gradient (kPa/m)	2.329e-006	1.254e-006	1.254e-006	1.254e-006
47	Liq. Reynolds	1.824e+005 *	1.824e+005 *	1.824e+005 *	1.824e+005 *
48	Vap. Reynolds	9.900e+006 *	9.900e+006 *	9.900e+006 *	9.900e+006 *
49	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
50	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	3.090e+004	3.091e+004	3.091e+004	3.092e+004
54	Elevation (m)	0.2060	0.2100	0.2120	0.2140
55	Pressure (bar)	67.51	67.51	67.51	67.51
56	Temperature (C)	46.72	46.72	46.72	46.72
57	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
58	Heat Transferred (kJ/h-m)	445.5	445.5	445.4	445.4
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2331	0.2331	0.2320	0.2320
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	8.199e-003	8.199e-003	8.347e-003	8.347e-003
64	Static Gradient (kPa/m)	1.591e-003	1.591e-003	7.928e-004	7.928e-004
65	Accel. Gradient (kPa/m)	1.254e-006	1.254e-006	1.171e-006	1.171e-006
66	Liq. Reynolds	1.824e+005 *	1.824e+005 *	1.824e+005 *	1.824e+005 *
67	Vap. Reynolds	9.900e+006 *	9.900e+006 *	9.900e+006 *	9.900e+006 *
68	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:13:43 2022	
6						
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933
14	Deposit Thickness	(mm)	---	---	---	---
15	Deposit Volume	(m3)	---	---	---	---
16	Length	(m)	3.092e+004	3.093e+004	3.093e+004	3.094e+004
17	Elevation	(m)	0.2160	0.2180	0.2200	0.2220
18	Pressure	(bar)	67.51	67.51	67.51	67.51
19	Temperature	(C)	46.72	46.72	46.72	46.71
20	Vapour Fraction		0.8802	0.8802	0.8802	0.8802
21	Heat Transferred	(kJ/h-m)	445.3	445.3	445.3	445.2
22	Flow Regime		Transition	Transition	Transition	Transition
23	Water Flow Regime					
24	Liquid HoldUp		0.2320	0.2320	0.2320	0.2320
25	Water HoldUp		---	---	---	---
26	Friction Gradient	(kPa/m)	8.347e-003	8.347e-003	8.347e-003	8.347e-003
27	Static Gradient	(kPa/m)	7.928e-004	7.928e-004	7.928e-004	7.912e-004
28	Accel. Gradient	(kPa/m)	1.171e-006	1.171e-006	1.171e-006	1.171e-006
29	Liq. Reynolds		1.824e+005 *	1.824e+005 *	1.824e+005 *	1.824e+005 *
30	Vap. Reynolds		9.900e+006 *	9.900e+006 *	9.900e+006 *	9.900e+006 *
31	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
32	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933
33	Deposit Thickness	(mm)	---	---	---	---
34	Deposit Volume	(m3)	---	---	---	---
35	Length	(m)	3.094e+004	3.095e+004	3.095e+004	3.096e+004
36	Elevation	(m)	0.2240	0.2260	0.2280	0.2300
37	Pressure	(bar)	67.51	67.51	67.51	67.51
38	Temperature	(C)	46.71	46.71	46.71	46.71
39	Vapour Fraction		0.8802	0.8802	0.8802	0.8802
40	Heat Transferred	(kJ/h-m)	445.2	445.2	445.1	445.1
41	Flow Regime		Transition	Transition	Transition	Transition
42	Water Flow Regime					
43	Liquid HoldUp		0.2320	0.2320	0.2320	0.2320
44	Water HoldUp		---	---	---	---
45	Friction Gradient	(kPa/m)	8.347e-003	8.347e-003	8.347e-003	8.347e-003
46	Static Gradient	(kPa/m)	7.912e-004	7.912e-004	7.912e-004	7.912e-004
47	Accel. Gradient	(kPa/m)	1.171e-006	1.171e-006	1.171e-006	1.171e-006
48	Liq. Reynolds		1.824e+005 *	1.824e+005 *	1.824e+005 *	1.824e+005 *
49	Vap. Reynolds		9.900e+006 *	9.900e+006 *	9.900e+006 *	9.900e+006 *
50	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
51	Vapour Velocity	(m/s)	2.933	2.933	2.933	2.933
52	Deposit Thickness	(mm)	---	---	---	---
53	Deposit Volume	(m3)	---	---	---	---
54	Length	(m)	3.096e+004	3.097e+004	3.097e+004	3.098e+004
55	Elevation	(m)	0.2300	0.2300	0.2300	0.2300
56	Pressure	(bar)	67.51	67.51	67.51	67.51
57	Temperature	(C)	46.71	46.71	46.70	46.70
58	Vapour Fraction		0.8802	0.8802	0.8802	0.8802
59	Heat Transferred	(kJ/h-m)	445.0	445.0	445.0	444.9
60	Flow Regime		Transition	Transition	Transition	Transition
61	Water Flow Regime					
62	Liquid HoldUp		0.2499	0.2499	0.2499	0.2499
63	Water HoldUp		---	---	---	---
64	Friction Gradient	(kPa/m)	6.197e-003	6.197e-003	6.197e-003	6.197e-003
65	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000
66	Accel. Gradient	(kPa/m)	7.938e-007	7.938e-007	7.938e-007	7.938e-007
67	Liq. Reynolds		1.824e+005 *	1.824e+005 *	1.824e+005 *	1.824e+005 *
68	Vap. Reynolds		9.900e+006 *	9.900e+006 *	9.900e+006 *	9.900e+006 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
14	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
15	Deposit Thickness (mm)	---	---	---	---
16	Deposit Volume (m3)	---	---	---	---
17	Length (m)	3.098e+004	3.099e+004	3.099e+004	3.100e+004
18	Elevation (m)	0.2300	0.2280	0.2260	0.2240
19	Pressure (bar)	67.51	67.50	67.50	67.50
20	Temperature (C)	46.70	46.70	46.70	46.70
21	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
22	Heat Transferred (kJ/h-m)	444.9	444.9	444.9	444.8
23	Flow Regime	Transition	Transition	Transition	Transition
24	Water Flow Regime				
25	Liquid HoldUp	0.2499	0.1709	0.1709	0.1709
26	Water HoldUp	---	---	---	---
27	Friction Gradient (kPa/m)	6.197e-003	9.066e-003	9.066e-003	9.066e-003
28	Static Gradient (kPa/m)	0.0000	-6.395e-004	-6.395e-004	-6.395e-004
29	Accel. Gradient (kPa/m)	7.938e-007	1.080e-006	1.080e-006	1.080e-006
30	Liq. Reynolds	1.824e+005 *	1.824e+005 *	1.824e+005 *	1.824e+005 *
31	Vap. Reynolds	9.900e+006 *	9.900e+006 *	9.900e+006 *	9.900e+006 *
32	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
33	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
34	Deposit Thickness (mm)	---	---	---	---
35	Deposit Volume (m3)	---	---	---	---
36	Length (m)	3.100e+004	3.101e+004	3.103e+004	3.105e+004
37	Elevation (m)	0.2220	0.2200	0.2240	0.2280
38	Pressure (bar)	67.50	67.50	67.50	67.50
39	Temperature (C)	46.70	46.69	46.69	46.68
40	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
41	Heat Transferred (kJ/h-m)	444.8	444.7	444.6	444.4
42	Flow Regime	Transition	Transition	Transition	Transition
43	Water Flow Regime				
44	Liquid HoldUp	0.1709	0.1709	0.2315	0.2315
45	Water HoldUp	---	---	---	---
46	Friction Gradient (kPa/m)	9.066e-003	9.066e-003	8.422e-003	8.422e-003
47	Static Gradient (kPa/m)	-6.395e-004	-6.395e-004	3.957e-004	3.957e-004
48	Accel. Gradient (kPa/m)	1.080e-006	1.080e-006	1.130e-006	1.130e-006
49	Liq. Reynolds	1.824e+005 *	1.824e+005 *	1.824e+005 *	1.823e+005 *
50	Vap. Reynolds	9.900e+006 *	9.900e+006 *	9.900e+006 *	9.900e+006 *
51	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
52	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
53	Deposit Thickness (mm)	---	---	---	---
54	Deposit Volume (m3)	---	---	---	---
55	Length (m)	3.107e+004	3.109e+004	3.111e+004	3.111e+004
56	Elevation (m)	0.2320	0.2360	0.2400	0.2380
57	Pressure (bar)	67.50	67.50	67.49	67.49
58	Temperature (C)	46.68	46.67	46.67	46.66
59	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
60	Heat Transferred (kJ/h-m)	444.2	444.1	443.9	443.9
61	Flow Regime	Transition	Transition	Transition	Transition
62	Water Flow Regime				
63	Liquid HoldUp	0.2315	0.2315	0.2315	0.1709
64	Water HoldUp	---	---	---	---
65	Friction Gradient (kPa/m)	8.422e-003	8.422e-003	8.422e-003	9.067e-003
66	Static Gradient (kPa/m)	3.957e-004	3.957e-004	3.957e-004	-6.395e-004
67	Accel. Gradient (kPa/m)	1.130e-006	1.130e-006	1.130e-006	1.080e-006
68	Liq. Reynolds	1.823e+005 *	1.823e+005 *	1.823e+005 *	1.823e+005 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.900e+006 *	9.900e+006 *	9.900e+006 *	9.901e+006 *
14	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
15	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	3.112e+004	3.112e+004	3.113e+004	3.113e+004
19	Elevation (m)	0.2360	0.2340	0.2320	0.2300
20	Pressure (bar)	67.49	67.49	67.49	67.49
21	Temperature (C)	46.66	46.66	46.66	46.66
22	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
23	Heat Transferred (kJ/h-m)	443.9	443.9	443.8	443.8
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1709	0.1709	0.1709	0.1709
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	9.067e-003	9.067e-003	9.067e-003	9.067e-003
29	Static Gradient (kPa/m)	-6.395e-004	-6.395e-004	-6.395e-004	-6.395e-004
30	Accel. Gradient (kPa/m)	1.080e-006	1.080e-006	1.080e-006	1.080e-006
31	Liq. Reynolds	1.823e+005 *	1.823e+005 *	1.823e+005 *	1.823e+005 *
32	Vap. Reynolds	9.901e+006 *	9.901e+006 *	9.901e+006 *	9.901e+006 *
33	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
34	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	3.114e+004	3.114e+004	3.115e+004	3.115e+004
38	Elevation (m)	0.3460	0.4620	0.5780	0.6940
39	Pressure (bar)	67.49	67.49	67.48	67.48
40	Temperature (C)	46.66	46.65	46.65	46.65
41	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
42	Heat Transferred (kJ/h-m)	443.7	443.6	443.6	443.5
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2949	0.2949	0.2949	0.2949
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	7.942e-003	7.943e-003	7.943e-003	7.943e-003
48	Static Gradient (kPa/m)	5.507e-002	5.507e-002	5.507e-002	5.507e-002
49	Accel. Gradient (kPa/m)	8.075e-006	8.075e-006	8.076e-006	8.077e-006
50	Liq. Reynolds	1.823e+005 *	1.823e+005 *	1.823e+005 *	1.823e+005 *
51	Vap. Reynolds	9.901e+006 *	9.901e+006 *	9.901e+006 *	9.901e+006 *
52	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
53	Vapour Velocity (m/s)	2.933	2.933	2.933	2.933
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	3.116e+004	3.116e+004	3.117e+004	3.117e+004
57	Elevation (m)	0.8100	-1.012	-2.834	-4.656
58	Pressure (bar)	67.48	67.49	67.51	67.52
59	Temperature (C)	46.65	46.66	46.66	46.67
60	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
61	Heat Transferred (kJ/h-m)	443.4	444.6	444.9	445.1
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2949	6.325e-002	6.327e-002	6.328e-002
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	7.943e-003	1.444e-002	1.444e-002	1.443e-002
67	Static Gradient (kPa/m)	5.507e-002	-0.3366	-0.3366	-0.3367
68	Accel. Gradient (kPa/m)	8.077e-006	-4.127e-005	-4.126e-005	-4.125e-005
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		1.823e+005 *	1.823e+005 *	1.823e+005 *	1.824e+005 *	
14	Vap. Reynolds		9.901e+006 *	9.901e+006 *	9.900e+006 *	9.899e+006 *	
15	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	0.1980	
16	Vapour Velocity (m/s)	2.933	2.933	2.932	2.931	2.931	
17	Deposit Thickness (mm)	---	---	---	---	---	
18	Deposit Volume (m3)	---	---	---	---	---	
19	Length (m)	3.118e+004	3.118e+004	3.119e+004	3.119e+004	3.119e+004	
20	Elevation (m)	-6.478	-8.300	-8.220	-8.140	-8.140	
21	Pressure (bar)	67.54	67.56	67.55	67.55	67.55	
22	Temperature (C)	46.68	46.69	46.69	46.69	46.69	
23	Vapour Fraction	0.8802	0.8802	0.8802	0.8802	0.8802	
24	Heat Transferred (kJ/h-m)	445.3	445.6	444.6	444.5	444.5	
25	Flow Regime	Transition	Transition	Transition	Transition	Transition	
26	Water Flow Regime						
27	Liquid HoldUp	6.330e-002	6.331e-002	0.3047	0.3047	0.3047	
28	Water HoldUp	---	---	---	---	---	
29	Friction Gradient (kPa/m)	1.443e-002	1.442e-002	7.918e-003	7.919e-003	7.919e-003	
30	Static Gradient (kPa/m)	-0.3368	-0.3369	6.502e-002	6.502e-002	6.502e-002	
31	Accel. Gradient (kPa/m)	-4.124e-005	-4.123e-005	9.329e-006	9.329e-006	9.329e-006	
32	Liq. Reynolds	1.824e+005 *	1.824e+005 *	1.824e+005 *	1.824e+005 *	1.824e+005 *	
33	Vap. Reynolds	9.899e+006 *	9.898e+006 *	9.898e+006 *	9.898e+006 *	9.898e+006 *	
34	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	0.1980	
35	Vapour Velocity (m/s)	2.931	2.930	2.930	2.930	2.930	
36	Deposit Thickness (mm)	---	---	---	---	---	
37	Deposit Volume (m3)	---	---	---	---	---	
38	Length (m)	3.119e+004	3.120e+004	3.120e+004	3.121e+004	3.121e+004	
39	Elevation (m)	-8.060	-7.980	-7.900	-7.580	-7.580	
40	Pressure (bar)	67.55	67.55	67.55	67.54	67.54	
41	Temperature (C)	46.69	46.69	46.68	46.68	46.68	
42	Vapour Fraction	0.8802	0.8802	0.8802	0.8802	0.8802	
43	Heat Transferred (kJ/h-m)	444.5	444.5	444.4	444.3	444.3	
44	Flow Regime	Transition	Transition	Transition	Transition	Transition	
45	Water Flow Regime						
46	Liquid HoldUp	0.3047	0.3047	0.3047	0.3414	0.3414	
47	Water HoldUp	---	---	---	---	---	
48	Friction Gradient (kPa/m)	7.919e-003	7.919e-003	7.919e-003	7.838e-003	7.838e-003	
49	Static Gradient (kPa/m)	6.502e-002	6.502e-002	6.502e-002	0.1067	0.1067	
50	Accel. Gradient (kPa/m)	9.330e-006	9.330e-006	9.331e-006	1.466e-005	1.466e-005	
51	Liq. Reynolds	1.824e+005 *	1.824e+005 *	1.824e+005 *	1.824e+005 *	1.824e+005 *	
52	Vap. Reynolds	9.898e+006 *	9.899e+006 *	9.899e+006 *	9.899e+006 *	9.899e+006 *	
53	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	0.1980	
54	Vapour Velocity (m/s)	2.930	2.930	2.930	2.931	2.931	
55	Deposit Thickness (mm)	---	---	---	---	---	
56	Deposit Volume (m3)	---	---	---	---	---	
57	Length (m)	3.122e+004	3.122e+004	3.123e+004	3.124e+004	3.124e+004	
58	Elevation (m)	-7.260	-6.940	-6.620	-6.300	-6.300	
59	Pressure (bar)	67.53	67.52	67.51	67.50	67.50	
60	Temperature (C)	46.67	46.67	46.66	46.66	46.66	
61	Vapour Fraction	0.8802	0.8802	0.8802	0.8802	0.8802	
62	Heat Transferred (kJ/h-m)	444.2	444.0	443.9	443.7	443.7	
63	Flow Regime	Transition	Transition	Transition	Transition	Transition	
64	Water Flow Regime						
65	Liquid HoldUp	0.3414	0.3413	0.3413	0.3413	0.3413	
66	Water HoldUp	---	---	---	---	---	
67	Friction Gradient (kPa/m)	7.839e-003	7.840e-003	7.841e-003	7.842e-003	7.842e-003	
68	Static Gradient (kPa/m)	0.1067	0.1067	0.1067	0.1067	0.1067	
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	1.466e-005	1.467e-005	1.467e-005	1.467e-005
14	Liq. Reynolds	1.824e+005 *	1.824e+005 *	1.823e+005 *	1.823e+005 *
15	Vap. Reynolds	9.899e+006 *	9.900e+006 *	9.900e+006 *	9.900e+006 *
16	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
17	Vapour Velocity (m/s)	2.931	2.931	2.932	2.932
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	3.124e+004	3.125e+004	3.125e+004	3.126e+004
21	Elevation (m)	-6.000	-5.700	-5.400	-5.100
22	Pressure (bar)	67.49	67.48	67.47	67.46
23	Temperature (C)	46.65	46.65	46.64	46.64
24	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
25	Heat Transferred (kJ/h-m)	443.3	443.2	443.1	443.0
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.4366	0.4366	0.4365	0.4365
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	7.661e-003	7.662e-003	7.663e-003	7.664e-003
31	Static Gradient (kPa/m)	0.2449	0.2449	0.2449	0.2448
32	Accel. Gradient (kPa/m)	3.236e-005	3.237e-005	3.238e-005	3.238e-005
33	Liq. Reynolds	1.823e+005 *	1.823e+005 *	1.823e+005 *	1.823e+005 *
34	Vap. Reynolds	9.901e+006 *	9.901e+006 *	9.901e+006 *	9.902e+006 *
35	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
36	Vapour Velocity (m/s)	2.933	2.933	2.934	2.934
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	3.126e+004	3.126e+004	3.127e+004	3.127e+004
40	Elevation (m)	-4.800	-3.810	-2.820	-1.830
41	Pressure (bar)	67.45	67.40	67.35	67.29
42	Temperature (C)	46.63	46.62	46.60	46.59
43	Vapour Fraction	0.8802	0.8802	0.8803	0.8803
44	Heat Transferred (kJ/h-m)	442.9	443.7	443.2	442.8
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.4365	0.7476	0.7473	0.7470
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	7.665e-003	7.396e-003	7.401e-003	7.406e-003
50	Static Gradient (kPa/m)	0.2448	1.033	1.032	1.032
51	Accel. Gradient (kPa/m)	3.239e-005	1.337e-004	1.338e-004	1.340e-004
52	Liq. Reynolds	1.823e+005 *	1.822e+005 *	1.821e+005 *	1.820e+005 *
53	Vap. Reynolds	9.902e+006 *	9.904e+006 *	9.906e+006 *	9.908e+006 *
54	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
55	Vapour Velocity (m/s)	2.934	2.937	2.939	2.942
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	3.128e+004	3.128e+004	3.129e+004	3.129e+004
59	Elevation (m)	-0.8400	0.1500	0.1100	7.000e-002
60	Pressure (bar)	67.24	67.19	67.19	67.19
61	Temperature (C)	46.57	46.55	46.55	46.55
62	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
63	Heat Transferred (kJ/h-m)	442.4	441.9	440.9	440.9
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.7467	0.7464	0.1648	0.1648
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	7.412e-003	7.417e-003	9.185e-003	9.185e-003
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1				Case Name:	Simulation - 386MMSCFD_80barg.hsc	
2				Unit Set:	Mixed	
3	 Epic Atlantic Limited			Date/Time:	Mon Jun 13 15:13:43 2022	
4						
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	1.032	1.031	-1.247e-002	-1.247e-002
14	Accel. Gradient	(kPa/m)	1.342e-004	1.343e-004	-4.249e-007	-4.249e-007
15	Liq. Reynolds		1.820e+005 *	1.819e+005 *	1.819e+005 *	1.819e+005 *
16	Vap. Reynolds		9.910e+006 *	9.912e+006 *	9.912e+006 *	9.912e+006 *
17	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
18	Vapour Velocity	(m/s)	2.944	2.946	2.946	2.946
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	3.130e+004	3.130e+004	3.131e+004	3.131e+004
22	Elevation	(m)	3.000e-002	-1.000e-002	-5.000e-002	-3.200e-002
23	Pressure	(bar)	67.19	67.19	67.19	67.19
24	Temperature	(C)	46.55	46.55	46.55	46.54
25	Vapour Fraction		0.8803	0.8803	0.8803	0.8803
26	Heat Transferred	(kJ/h-m)	440.9	440.8	440.8	440.7
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.1648	0.1648	0.1648	0.2402
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	9.185e-003	9.185e-003	9.185e-003	7.269e-003
32	Static Gradient	(kPa/m)	-1.247e-002	-1.247e-002	-1.247e-002	7.315e-003
33	Accel. Gradient	(kPa/m)	-4.249e-007	-4.249e-007	-4.249e-007	1.886e-006
34	Liq. Reynolds		1.819e+005 *	1.819e+005 *	1.819e+005 *	1.819e+005 *
35	Vap. Reynolds		9.912e+006 *	9.912e+006 *	9.912e+006 *	9.912e+006 *
36	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
37	Vapour Velocity	(m/s)	2.946	2.946	2.946	2.946
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	3.132e+004	3.132e+004	3.133e+004	3.133e+004
41	Elevation	(m)	-1.400e-002	4.000e-003	2.200e-002	4.000e-002
42	Pressure	(bar)	67.19	67.19	67.19	67.19
43	Temperature	(C)	46.54	46.54	46.54	46.54
44	Vapour Fraction		0.8803	0.8803	0.8803	0.8803
45	Heat Transferred	(kJ/h-m)	440.7	440.6	440.6	440.6
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2402	0.2402	0.2402	0.2402
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	7.269e-003	7.269e-003	7.269e-003	7.269e-003
51	Static Gradient	(kPa/m)	7.315e-003	7.315e-003	7.315e-003	7.315e-003
52	Accel. Gradient	(kPa/m)	1.886e-006	1.886e-006	1.886e-006	1.886e-006
53	Liq. Reynolds		1.819e+005 *	1.819e+005 *	1.819e+005 *	1.819e+005 *
54	Vap. Reynolds		9.912e+006 *	9.912e+006 *	9.912e+006 *	9.912e+006 *
55	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
56	Vapour Velocity	(m/s)	2.946	2.946	2.946	2.946
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	3.134e+004	3.134e+004	3.135e+004	3.135e+004
60	Elevation	(m)	2.800e-002	1.600e-002	4.000e-003	-8.000e-003
61	Pressure	(bar)	67.19	67.19	67.19	67.19
62	Temperature	(C)	46.54	46.54	46.53	46.53
63	Vapour Fraction		0.8803	0.8803	0.8803	0.8803
64	Heat Transferred	(kJ/h-m)	440.6	440.5	440.5	440.5
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.1689	0.1689	0.1689	0.1689
68	Water HoldUp		---	---	---	---
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1				Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	9.126e-003	9.126e-003	9.126e-003	9.126e-003
14	Static Gradient	(kPa/m)	-3.787e-003	-3.787e-003	-3.787e-003	-3.787e-003
15	Accel. Gradient	(kPa/m)	6.904e-007	6.904e-007	6.904e-007	6.904e-007
16	Liq. Reynolds		1.819e+005 *	1.819e+005 *	1.819e+005 *	1.819e+005 *
17	Vap. Reynolds		9.912e+006 *	9.912e+006 *	9.912e+006 *	9.912e+006 *
18	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
19	Vapour Velocity	(m/s)	2.946	2.946	2.946	2.946
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	3.136e+004	3.136e+004	3.137e+004	3.137e+004
23	Elevation	(m)	-2.000e-002	-1.600e-002	-1.200e-002	-8.000e-003
24	Pressure	(bar)	67.19	67.19	67.18	67.18
25	Temperature	(C)	46.53	46.53	46.53	46.53
26	Vapour Fraction		0.8803	0.8803	0.8803	0.8803
27	Heat Transferred	(kJ/h-m)	440.4	440.3	440.3	440.3
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.1689	0.2324	0.2324	0.2324
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	9.126e-003	8.251e-003	8.251e-003	8.252e-003
33	Static Gradient	(kPa/m)	-3.787e-003	1.587e-003	1.587e-003	1.587e-003
34	Accel. Gradient	(kPa/m)	6.904e-007	1.272e-006	1.272e-006	1.272e-006
35	Liq. Reynolds		1.819e+005 *	1.819e+005 *	1.819e+005 *	1.819e+005 *
36	Vap. Reynolds		9.912e+006 *	9.912e+006 *	9.912e+006 *	9.912e+006 *
37	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
38	Vapour Velocity	(m/s)	2.946	2.946	2.946	2.946
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	3.138e+004	3.138e+004	3.139e+004	3.139e+004
42	Elevation	(m)	-4.000e-003	-5.447e-016	2.400e-002	4.800e-002
43	Pressure	(bar)	67.18	67.18	67.18	67.18
44	Temperature	(C)	46.53	46.52	46.52	46.52
45	Vapour Fraction		0.8803	0.8803	0.8803	0.8803
46	Heat Transferred	(kJ/h-m)	440.2	440.2	440.1	440.1
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2324	0.2324	0.2435	0.2435
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	8.252e-003	8.252e-003	6.877e-003	6.877e-003
52	Static Gradient	(kPa/m)	1.587e-003	1.587e-003	9.853e-003	9.853e-003
53	Accel. Gradient	(kPa/m)	1.272e-006	1.272e-006	2.164e-006	2.164e-006
54	Liq. Reynolds		1.819e+005 *	1.819e+005 *	1.819e+005 *	1.819e+005 *
55	Vap. Reynolds		9.912e+006 *	9.912e+006 *	9.912e+006 *	9.912e+006 *
56	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
57	Vapour Velocity	(m/s)	2.946	2.946	2.946	2.946
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	3.140e+004	3.140e+004	3.141e+004	3.141e+004
61	Elevation	(m)	7.200e-002	9.600e-002	0.1200	0.1080
62	Pressure	(bar)	67.18	67.18	67.18	67.18
63	Temperature	(C)	46.52	46.52	46.52	46.52
64	Vapour Fraction		0.8803	0.8803	0.8803	0.8803
65	Heat Transferred	(kJ/h-m)	440.1	440.0	440.0	440.0
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2435	0.2435	0.2435	0.1689
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	6.877e-003	6.877e-003	6.877e-003	9.126e-003
15	Static Gradient (kPa/m)	9.853e-003	9.853e-003	9.853e-003	-3.803e-003
16	Accel. Gradient (kPa/m)	2.164e-006	2.164e-006	2.164e-006	6.885e-007
17	Liq. Reynolds	1.819e+005 *	1.819e+005 *	1.819e+005 *	1.819e+005 *
18	Vap. Reynolds	9.912e+006 *	9.912e+006 *	9.912e+006 *	9.912e+006 *
19	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
20	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	3.142e+004	3.142e+004	3.143e+004	3.143e+004
24	Elevation (m)	9.600e-002	8.400e-002	7.200e-002	6.000e-002
25	Pressure (bar)	67.18	67.18	67.18	67.18
26	Temperature (C)	46.51	46.51	46.51	46.51
27	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
28	Heat Transferred (kJ/h-m)	439.9	439.9	439.9	439.8
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.1689	0.1689	0.1689	0.1689
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	9.126e-003	9.126e-003	9.126e-003	9.126e-003
34	Static Gradient (kPa/m)	-3.803e-003	-3.803e-003	-3.803e-003	-3.803e-003
35	Accel. Gradient (kPa/m)	6.885e-007	6.884e-007	6.884e-007	6.884e-007
36	Liq. Reynolds	1.819e+005 *	1.819e+005 *	1.819e+005 *	1.819e+005 *
37	Vap. Reynolds	9.912e+006 *	9.912e+006 *	9.912e+006 *	9.912e+006 *
38	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
39	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	3.144e+004	3.144e+004	3.145e+004	3.145e+004
43	Elevation (m)	5.200e-002	4.400e-002	3.600e-002	2.800e-002
44	Pressure (bar)	67.18	67.18	67.18	67.18
45	Temperature (C)	46.51	46.51	46.51	46.50
46	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
47	Heat Transferred (kJ/h-m)	439.8	439.8	439.7	439.7
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.1695	0.1695	0.1695	0.1695
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	9.118e-003	9.118e-003	9.118e-003	9.118e-003
53	Static Gradient (kPa/m)	-2.541e-003	-2.541e-003	-2.541e-003	-2.541e-003
54	Accel. Gradient (kPa/m)	8.506e-007	8.506e-007	8.506e-007	8.506e-007
55	Liq. Reynolds	1.819e+005 *	1.818e+005 *	1.818e+005 *	1.818e+005 *
56	Vap. Reynolds	9.912e+006 *	9.912e+006 *	9.912e+006 *	9.912e+006 *
57	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
58	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	3.146e+004	3.146e+004	3.147e+004	3.147e+004
62	Elevation (m)	2.000e-002	1.800e-002	1.600e-002	1.400e-002
63	Pressure (bar)	67.18	67.18	67.18	67.17
64	Temperature (C)	46.50	46.50	46.50	46.50
65	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
66	Heat Transferred (kJ/h-m)	439.7	439.6	439.6	439.5
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 502 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.1695	0.1704	0.1704	0.1704
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	9.118e-003	9.106e-003	9.106e-003	9.106e-003
16	Static Gradient (kPa/m)	-2.541e-003	-6.376e-004	-6.376e-004	-6.376e-004
17	Accel. Gradient (kPa/m)	8.506e-007	1.095e-006	1.095e-006	1.095e-006
18	Liq. Reynolds	1.818e+005 *	1.818e+005 *	1.818e+005 *	1.818e+005 *
19	Vap. Reynolds	9.912e+006 *	9.912e+006 *	9.912e+006 *	9.912e+006 *
20	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
21	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	3.148e+004	3.148e+004	3.149e+004	3.149e+004
25	Elevation (m)	1.200e-002	1.000e-002	2.000e-002	3.000e-002
26	Pressure (bar)	67.17	67.17	67.17	67.17
27	Temperature (C)	46.50	46.50	46.50	46.49
28	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
29	Heat Transferred (kJ/h-m)	439.5	439.5	439.4	439.4
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1704	0.1704	0.2357	0.2357
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	9.106e-003	9.106e-003	7.819e-003	7.819e-003
35	Static Gradient (kPa/m)	-6.376e-004	-6.376e-004	4.009e-003	4.009e-003
36	Accel. Gradient (kPa/m)	1.095e-006	1.095e-006	1.530e-006	1.530e-006
37	Liq. Reynolds	1.818e+005 *	1.818e+005 *	1.818e+005 *	1.818e+005 *
38	Vap. Reynolds	9.912e+006 *	9.912e+006 *	9.912e+006 *	9.912e+006 *
39	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
40	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	3.150e+004	3.150e+004	3.151e+004	3.151e+004
44	Elevation (m)	4.000e-002	5.000e-002	6.000e-002	5.400e-002
45	Pressure (bar)	67.17	67.17	67.17	67.17
46	Temperature (C)	46.49	46.49	46.49	46.49
47	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
48	Heat Transferred (kJ/h-m)	439.3	439.3	439.2	439.2
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2357	0.2357	0.2357	0.1698
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	7.819e-003	7.819e-003	7.819e-003	9.114e-003
54	Static Gradient (kPa/m)	4.009e-003	4.009e-003	4.009e-003	-1.908e-003
55	Accel. Gradient (kPa/m)	1.530e-006	1.530e-006	1.530e-006	9.321e-007
56	Liq. Reynolds	1.818e+005 *	1.818e+005 *	1.818e+005 *	1.818e+005 *
57	Vap. Reynolds	9.912e+006 *	9.912e+006 *	9.912e+006 *	9.912e+006 *
58	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
59	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	3.152e+004	3.152e+004	3.153e+004	3.153e+004
63	Elevation (m)	4.800e-002	4.200e-002	3.600e-002	3.000e-002
64	Pressure (bar)	67.17	67.17	67.17	67.17
65	Temperature (C)	46.49	46.49	46.48	46.48
66	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
67	Heat Transferred (kJ/h-m)	439.2	439.2	439.1	439.1
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 503 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1698	0.1698	0.1698	0.1698
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	9.114e-003	9.114e-003	9.114e-003	9.114e-003
17	Static Gradient (kPa/m)	-1.908e-003	-1.908e-003	-1.908e-003	-1.908e-003
18	Accel. Gradient (kPa/m)	9.321e-007	9.321e-007	9.321e-007	9.321e-007
19	Liq. Reynolds	1.818e+005 *	1.818e+005 *	1.818e+005 *	1.818e+005 *
20	Vap. Reynolds	9.913e+006 *	9.913e+006 *	9.913e+006 *	9.913e+006 *
21	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
22	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	3.154e+004	3.154e+004	3.155e+004	3.155e+004
26	Elevation (m)	5.000e-002	7.000e-002	9.000e-002	0.1100
27	Pressure (bar)	67.17	67.17	67.17	67.17
28	Temperature (C)	46.48	46.48	46.48	46.48
29	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
30	Heat Transferred (kJ/h-m)	439.0	439.0	438.9	438.9
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2412	0.2412	0.2412	0.2412
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	7.144e-003	7.145e-003	7.145e-003	7.145e-003
36	Static Gradient (kPa/m)	8.109e-003	8.109e-003	8.109e-003	8.109e-003
37	Accel. Gradient (kPa/m)	1.973e-006	1.973e-006	1.973e-006	1.973e-006
38	Liq. Reynolds	1.818e+005 *	1.818e+005 *	1.818e+005 *	1.818e+005 *
39	Vap. Reynolds	9.913e+006 *	9.913e+006 *	9.913e+006 *	9.913e+006 *
40	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
41	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	3.156e+004	3.157e+004	3.157e+004	3.158e+004
45	Elevation (m)	0.1300	0.1060	8.200e-002	5.800e-002
46	Pressure (bar)	67.17	67.17	67.17	67.17
47	Temperature (C)	46.47	46.47	46.47	46.47
48	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
49	Heat Transferred (kJ/h-m)	438.8	438.9	438.8	438.8
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2412	0.1671	0.1671	0.1671
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	7.145e-003	9.151e-003	9.151e-003	9.151e-003
55	Static Gradient (kPa/m)	8.109e-003	-7.553e-003	-7.553e-003	-7.554e-003
56	Accel. Gradient (kPa/m)	1.973e-006	2.067e-007	2.067e-007	2.067e-007
57	Liq. Reynolds	1.818e+005 *	1.818e+005 *	1.818e+005 *	1.818e+005 *
58	Vap. Reynolds	9.913e+006 *	9.913e+006 *	9.913e+006 *	9.913e+006 *
59	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
60	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	3.158e+004	3.159e+004	3.159e+004	3.160e+004
64	Elevation (m)	3.400e-002	1.000e-002	3.400e-002	5.800e-002
65	Pressure (bar)	67.17	67.16	67.16	67.16
66	Temperature (C)	46.47	46.47	46.47	46.47
67	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
68	Heat Transferred (kJ/h-m)	438.8	438.7	438.6	438.6
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 504 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.1671	0.1671	0.2435	0.2435
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	9.151e-003	9.151e-003	6.877e-003	6.877e-003
18	Static Gradient (kPa/m)	-7.554e-003	-7.554e-003	9.853e-003	9.853e-003
19	Accel. Gradient (kPa/m)	2.067e-007	2.067e-007	2.164e-006	2.164e-006
20	Liq. Reynolds	1.818e+005 *	1.818e+005 *	1.818e+005 *	1.818e+005 *
21	Vap. Reynolds	9.913e+006 *	9.913e+006 *	9.913e+006 *	9.913e+006 *
22	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
23	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	3.160e+004	3.161e+004	3.161e+004	3.162e+004
27	Elevation (m)	8.200e-002	0.1060	0.1300	0.1160
28	Pressure (bar)	67.16	67.16	67.16	67.16
29	Temperature (C)	46.46	46.46	46.46	46.46
30	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
31	Heat Transferred (kJ/h-m)	438.6	438.5	438.5	438.5
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2435	0.2435	0.2435	0.1686
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	6.877e-003	6.877e-003	6.878e-003	9.131e-003
37	Static Gradient (kPa/m)	9.853e-003	9.853e-003	9.853e-003	-4.432e-003
38	Accel. Gradient (kPa/m)	2.164e-006	2.164e-006	2.164e-006	6.079e-007
39	Liq. Reynolds	1.818e+005 *	1.818e+005 *	1.818e+005 *	1.818e+005 *
40	Vap. Reynolds	9.913e+006 *	9.913e+006 *	9.913e+006 *	9.913e+006 *
41	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
42	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	3.162e+004	3.163e+004	3.163e+004	3.164e+004
46	Elevation (m)	0.1020	8.800e-002	7.400e-002	6.000e-002
47	Pressure (bar)	67.16	67.16	67.16	67.16
48	Temperature (C)	46.46	46.46	46.46	46.45
49	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
50	Heat Transferred (kJ/h-m)	438.4	438.4	438.4	438.3
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1686	0.1686	0.1686	0.1686
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	9.131e-003	9.131e-003	9.131e-003	9.131e-003
56	Static Gradient (kPa/m)	-4.432e-003	-4.432e-003	-4.432e-003	-4.432e-003
57	Accel. Gradient (kPa/m)	6.079e-007	6.079e-007	6.079e-007	6.079e-007
58	Liq. Reynolds	1.818e+005 *	1.818e+005 *	1.818e+005 *	1.818e+005 *
59	Vap. Reynolds	9.913e+006 *	9.913e+006 *	9.913e+006 *	9.913e+006 *
60	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
61	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	3.164e+004	3.165e+004	3.165e+004	3.166e+004
65	Elevation (m)	6.800e-002	7.600e-002	8.400e-002	9.200e-002
66	Pressure (bar)	67.16	67.16	67.16	67.16
67	Temperature (C)	46.45	46.45	46.45	46.45
68	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Heat Transferred (kJ/h-m)	438.3	438.2	438.2	438.1					
14	Flow Regime	Transition	Transition	Transition	Transition					
15	Water Flow Regime									
16	Liquid HoldUp	0.2346	0.2346	0.2346	0.2346					
17	Water HoldUp	---	---	---	---					
18	Friction Gradient (kPa/m)	7.962e-003	7.962e-003	7.962e-003	7.962e-003					
19	Static Gradient (kPa/m)	3.196e-003	3.196e-003	3.196e-003	3.196e-003					
20	Accel. Gradient (kPa/m)	1.443e-006	1.443e-006	1.443e-006	1.443e-006					
21	Liq. Reynolds	1.818e+005 *	1.818e+005 *	1.818e+005 *	1.818e+005 *					
22	Vap. Reynolds	9.913e+006 *	9.913e+006 *	9.913e+006 *	9.913e+006 *					
23	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
24	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946					
25	Deposit Thickness (mm)	---	---	---	---					
26	Deposit Volume (m3)	---	---	---	---					
27	Length (m)	3.166e+004	3.167e+004	3.167e+004	3.168e+004					
28	Elevation (m)	1.000e-001	0.1080	0.1160	0.1240					
29	Pressure (bar)	67.16	67.16	67.16	67.16					
30	Temperature (C)	46.45	46.45	46.44	46.44					
31	Vapour Fraction	0.8803	0.8803	0.8803	0.8803					
32	Heat Transferred (kJ/h-m)	438.1	438.1	438.0	438.0					
33	Flow Regime	Transition	Transition	Transition	Transition					
34	Water Flow Regime									
35	Liquid HoldUp	0.2346	0.2346	0.2346	0.2346					
36	Water HoldUp	---	---	---	---					
37	Friction Gradient (kPa/m)	7.962e-003	7.962e-003	7.962e-003	7.962e-003					
38	Static Gradient (kPa/m)	3.196e-003	3.196e-003	3.196e-003	3.196e-003					
39	Accel. Gradient (kPa/m)	1.444e-006	1.444e-006	1.444e-006	1.444e-006					
40	Liq. Reynolds	1.818e+005 *	1.818e+005 *	1.818e+005 *	1.818e+005 *					
41	Vap. Reynolds	9.913e+006 *	9.913e+006 *	9.913e+006 *	9.913e+006 *					
42	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
43	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946					
44	Deposit Thickness (mm)	---	---	---	---					
45	Deposit Volume (m3)	---	---	---	---					
46	Length (m)	3.168e+004	3.169e+004	3.169e+004	3.169e+004					
47	Elevation (m)	0.1320	0.1400	0.1360	0.1320					
48	Pressure (bar)	67.15	67.15	67.15	67.15					
49	Temperature (C)	46.44	46.44	46.44	46.44					
50	Vapour Fraction	0.8803	0.8803	0.8803	0.8803					
51	Heat Transferred (kJ/h-m)	438.0	437.9	437.9	437.9					
52	Flow Regime	Transition	Transition	Transition	Transition					
53	Water Flow Regime									
54	Liquid HoldUp	0.2346	0.2346	0.1697	0.1697					
55	Water HoldUp	---	---	---	---					
56	Friction Gradient (kPa/m)	7.962e-003	7.962e-003	9.116e-003	9.116e-003					
57	Static Gradient (kPa/m)	3.196e-003	3.196e-003	-2.120e-003	-2.120e-003					
58	Accel. Gradient (kPa/m)	1.444e-006	1.444e-006	9.052e-007	9.052e-007					
59	Liq. Reynolds	1.818e+005 *	1.818e+005 *	1.818e+005 *	1.818e+005 *					
60	Vap. Reynolds	9.913e+006 *	9.913e+006 *	9.913e+006 *	9.913e+006 *					
61	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
62	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946					
63	Deposit Thickness (mm)	---	---	---	---					
64	Deposit Volume (m3)	---	---	---	---					
65	Length (m)	3.169e+004	3.170e+004	3.170e+004	3.170e+004					
66	Elevation (m)	0.1280	0.1240	0.1200	0.1160					
67	Pressure (bar)	67.15	67.15	67.15	67.15					
68	Temperature (C)	46.44	46.44	46.44	46.43					
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 506 of 937					

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
14	Heat Transferred (kJ/h-m)	437.9	437.9	437.8	437.8
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.1697	0.1697	0.1697	0.1692
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	9.116e-003	9.116e-003	9.116e-003	9.123e-003
20	Static Gradient (kPa/m)	-2.120e-003	-2.120e-003	-2.120e-003	-3.173e-003
21	Accel. Gradient (kPa/m)	9.052e-007	9.052e-007	9.052e-007	7.698e-007
22	Liq. Reynolds	1.818e+005 *	1.818e+005 *	1.818e+005 *	1.818e+005 *
23	Vap. Reynolds	9.913e+006 *	9.913e+006 *	9.913e+006 *	9.913e+006 *
24	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
25	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	3.170e+004	3.171e+004	3.171e+004	3.171e+004
29	Elevation (m)	0.1120	0.1080	0.1040	1.000e-001
30	Pressure (bar)	67.15	67.15	67.15	67.15
31	Temperature (C)	46.43	46.43	46.43	46.43
32	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
33	Heat Transferred (kJ/h-m)	437.8	437.8	437.8	437.8
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1692	0.1692	0.1692	0.1692
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	9.123e-003	9.123e-003	9.123e-003	9.123e-003
39	Static Gradient (kPa/m)	-3.173e-003	-3.173e-003	-3.173e-003	-3.173e-003
40	Accel. Gradient (kPa/m)	7.698e-007	7.698e-007	7.698e-007	7.698e-007
41	Liq. Reynolds	1.818e+005 *	1.818e+005 *	1.818e+005 *	1.818e+005 *
42	Vap. Reynolds	9.913e+006 *	9.913e+006 *	9.913e+006 *	9.913e+006 *
43	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
44	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	3.172e+004	3.172e+004	3.173e+004	3.173e+004
48	Elevation (m)	6.800e-002	3.600e-002	4.000e-003	-2.800e-002
49	Pressure (bar)	67.15	67.15	67.15	67.15
50	Temperature (C)	46.43	46.43	46.43	46.43
51	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
52	Heat Transferred (kJ/h-m)	437.7	437.7	437.7	437.7
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.1660	0.1660	0.1660	0.1660
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	9.168e-003	9.168e-003	9.168e-003	9.168e-003
58	Static Gradient (kPa/m)	-1.002e-002	-1.002e-002	-1.002e-002	-1.002e-002
59	Accel. Gradient (kPa/m)	-1.107e-007	-1.108e-007	-1.108e-007	-1.108e-007
60	Liq. Reynolds	1.818e+005 *	1.818e+005 *	1.818e+005 *	1.818e+005 *
61	Vap. Reynolds	9.913e+006 *	9.913e+006 *	9.913e+006 *	9.913e+006 *
62	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
63	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	3.174e+004	3.174e+004	3.174e+004	3.174e+004
67	Elevation (m)	-6.000e-002	-3.800e-002	-1.600e-002	6.000e-003
68	Pressure (bar)	67.15	67.15	67.15	67.15
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	46.43	46.43	46.42	46.42
14	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
15	Heat Transferred (kJ/h-m)	437.6	437.6	437.5	437.5
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.1660	0.2505	0.2505	0.2505
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	9.168e-003	6.099e-003	6.099e-003	6.099e-003
21	Static Gradient (kPa/m)	-1.002e-002	1.538e-002	1.538e-002	1.538e-002
22	Accel. Gradient (kPa/m)	-1.108e-007	2.778e-006	2.779e-006	2.779e-006
23	Liq. Reynolds	1.818e+005 *	1.818e+005 *	1.818e+005 *	1.818e+005 *
24	Vap. Reynolds	9.913e+006 *	9.913e+006 *	9.913e+006 *	9.913e+006 *
25	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
26	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	3.175e+004	3.175e+004	3.176e+004	3.176e+004
30	Elevation (m)	2.800e-002	5.000e-002	3.800e-002	2.600e-002
31	Pressure (bar)	67.15	67.15	67.15	67.15
32	Temperature (C)	46.42	46.42	46.42	46.42
33	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
34	Heat Transferred (kJ/h-m)	437.5	437.4	437.4	437.4
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2505	0.2505	0.1692	0.1692
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	6.099e-003	6.099e-003	9.123e-003	9.123e-003
40	Static Gradient (kPa/m)	1.538e-002	1.538e-002	-3.173e-003	-3.173e-003
41	Accel. Gradient (kPa/m)	2.779e-006	2.779e-006	7.698e-007	7.698e-007
42	Liq. Reynolds	1.818e+005 *	1.818e+005 *	1.817e+005 *	1.817e+005 *
43	Vap. Reynolds	9.913e+006 *	9.914e+006 *	9.914e+006 *	9.914e+006 *
44	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
45	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	3.177e+004	3.177e+004	3.178e+004	3.179e+004
49	Elevation (m)	1.400e-002	2.000e-003	-1.000e-002	6.000e-003
50	Pressure (bar)	67.15	67.15	67.15	67.15
51	Temperature (C)	46.42	46.42	46.41	46.41
52	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
53	Heat Transferred (kJ/h-m)	437.4	437.3	437.3	437.2
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.1692	0.1692	0.1692	0.2376
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	9.123e-003	9.123e-003	9.123e-003	7.589e-003
59	Static Gradient (kPa/m)	-3.173e-003	-3.173e-003	-3.173e-003	5.365e-003
60	Accel. Gradient (kPa/m)	7.698e-007	7.698e-007	7.698e-007	1.676e-006
61	Liq. Reynolds	1.817e+005 *	1.817e+005 *	1.817e+005 *	1.817e+005 *
62	Vap. Reynolds	9.914e+006 *	9.914e+006 *	9.914e+006 *	9.914e+006 *
63	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
64	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	3.179e+004	3.180e+004	3.180e+004	3.181e+004
68	Elevation (m)	2.200e-002	3.800e-002	5.400e-002	7.000e-002
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Pressure (bar)	67.15	67.15	67.14	67.14
14	Temperature (C)	46.41	46.41	46.41	46.41
15	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
16	Heat Transferred (kJ/h-m)	437.1	437.1	437.0	437.0
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.2376	0.2376	0.2376	0.2376
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	7.589e-003	7.589e-003	7.589e-003	7.589e-003
22	Static Gradient (kPa/m)	5.365e-003	5.365e-003	5.365e-003	5.365e-003
23	Accel. Gradient (kPa/m)	1.676e-006	1.676e-006	1.676e-006	1.676e-006
24	Liq. Reynolds	1.817e+005 *	1.817e+005 *	1.817e+005 *	1.817e+005 *
25	Vap. Reynolds	9.914e+006 *	9.914e+006 *	9.914e+006 *	9.914e+006 *
26	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
27	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	3.182e+004	3.182e+004	3.183e+004	3.183e+004
31	Elevation (m)	6.200e-002	5.400e-002	4.600e-002	3.800e-002
32	Pressure (bar)	67.14	67.14	67.14	67.14
33	Temperature (C)	46.40	46.40	46.40	46.40
34	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
35	Heat Transferred (kJ/h-m)	437.0	437.0	436.9	436.9
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.1695	0.1695	0.1695	0.1695
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	9.119e-003	9.119e-003	9.119e-003	9.119e-003
41	Static Gradient (kPa/m)	-2.542e-003	-2.542e-003	-2.542e-003	-2.542e-003
42	Accel. Gradient (kPa/m)	8.511e-007	8.511e-007	8.511e-007	8.511e-007
43	Liq. Reynolds	1.817e+005 *	1.817e+005 *	1.817e+005 *	1.817e+005 *
44	Vap. Reynolds	9.914e+006 *	9.914e+006 *	9.914e+006 *	9.914e+006 *
45	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
46	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	3.184e+004	3.184e+004	3.185e+004	3.185e+004
50	Elevation (m)	3.000e-002	4.400e-002	5.800e-002	7.200e-002
51	Pressure (bar)	67.14	67.14	67.14	67.14
52	Temperature (C)	46.40	46.40	46.40	46.39
53	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
54	Heat Transferred (kJ/h-m)	436.9	436.8	436.7	436.7
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.1695	0.2379	0.2379	0.2379
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	9.119e-003	7.541e-003	7.541e-003	7.541e-003
60	Static Gradient (kPa/m)	-2.542e-003	5.651e-003	5.651e-003	5.651e-003
61	Accel. Gradient (kPa/m)	8.511e-007	1.707e-006	1.707e-006	1.707e-006
62	Liq. Reynolds	1.817e+005 *	1.817e+005 *	1.817e+005 *	1.817e+005 *
63	Vap. Reynolds	9.914e+006 *	9.914e+006 *	9.914e+006 *	9.914e+006 *
64	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
65	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	3.186e+004	3.186e+004	3.187e+004	3.187e+004
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	8.600e-002	1.000e-001	0.1020	0.1040
14	Pressure (bar)	67.14	67.14	67.14	67.14
15	Temperature (C)	46.39	46.39	46.39	46.39
16	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
17	Heat Transferred (kJ/h-m)	436.7	436.6	436.6	436.5
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2379	0.2379	0.2313	0.2313
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	7.541e-003	7.541e-003	8.401e-003	8.401e-003
23	Static Gradient (kPa/m)	5.651e-003	5.651e-003	7.907e-004	7.907e-004
24	Accel. Gradient (kPa/m)	1.707e-006	1.707e-006	1.190e-006	1.190e-006
25	Liq. Reynolds	1.817e+005 *	1.817e+005 *	1.817e+005 *	1.817e+005 *
26	Vap. Reynolds	9.914e+006 *	9.914e+006 *	9.914e+006 *	9.914e+006 *
27	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
28	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	3.188e+004	3.188e+004	3.189e+004	3.189e+004
32	Elevation (m)	0.1060	0.1080	0.1100	0.1060
33	Pressure (bar)	67.14	67.14	67.14	67.14
34	Temperature (C)	46.39	46.39	46.38	46.38
35	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
36	Heat Transferred (kJ/h-m)	436.5	436.5	436.4	436.4
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2313	0.2313	0.2313	0.1701
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	8.401e-003	8.401e-003	8.401e-003	9.111e-003
42	Static Gradient (kPa/m)	7.907e-004	7.907e-004	7.907e-004	-1.274e-003
43	Accel. Gradient (kPa/m)	1.190e-006	1.190e-006	1.190e-006	1.014e-006
44	Liq. Reynolds	1.817e+005 *	1.817e+005 *	1.817e+005 *	1.817e+005 *
45	Vap. Reynolds	9.914e+006 *	9.914e+006 *	9.914e+006 *	9.914e+006 *
46	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
47	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	3.190e+004	3.190e+004	3.191e+004	3.191e+004
51	Elevation (m)	0.1020	9.800e-002	9.400e-002	9.000e-002
52	Pressure (bar)	67.14	67.14	67.13	67.13
53	Temperature (C)	46.38	46.38	46.38	46.38
54	Vapour Fraction	0.8803	0.8803	0.8803	0.8803
55	Heat Transferred (kJ/h-m)	436.4	436.4	436.3	436.3
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.1701	0.1701	0.1701	0.1701
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	9.111e-003	9.111e-003	9.111e-003	9.111e-003
61	Static Gradient (kPa/m)	-1.274e-003	-1.274e-003	-1.274e-003	-1.274e-003
62	Accel. Gradient (kPa/m)	1.014e-006	1.014e-006	1.014e-006	1.014e-006
63	Liq. Reynolds	1.817e+005 *	1.817e+005 *	1.817e+005 *	1.817e+005 *
64	Vap. Reynolds	9.914e+006 *	9.914e+006 *	9.914e+006 *	9.914e+006 *
65	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
66	Vapour Velocity (m/s)	2.946	2.946	2.946	2.946
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length (m)	3.192e+004	3.192e+004	3.193e+004	3.193e+004		
14	Elevation (m)	0.1480	0.2060	0.2640	0.3220		
15	Pressure (bar)	67.13	67.13	67.13	67.13		
16	Temperature (C)	46.38	46.37	46.37	46.37		
17	Vapour Fraction	0.8803	0.8803	0.8803	0.8803		
18	Heat Transferred (kJ/h-m)	436.2	436.1	436.1	436.0		
19	Flow Regime	Transition	Transition	Transition	Transition		
20	Water Flow Regime						
21	Liquid HoldUp	0.2622	0.2622	0.2622	0.2622		
22	Water HoldUp	---	---	---	---		
23	Friction Gradient (kPa/m)	7.847e-003	7.847e-003	7.848e-003	7.848e-003		
24	Static Gradient (kPa/m)	2.518e-002	2.518e-002	2.518e-002	2.518e-002		
25	Accel. Gradient (kPa/m)	4.274e-006	4.275e-006	4.275e-006	4.275e-006		
26	Liq. Reynolds	1.817e+005 *	1.817e+005 *	1.817e+005 *	1.817e+005 *		
27	Vap. Reynolds	9.914e+006 *	9.914e+006 *	9.914e+006 *	9.914e+006 *		
28	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
29	Vapour Velocity (m/s)	2.946	2.947	2.947	2.947		
30	Deposit Thickness (mm)	---	---	---	---		
31	Deposit Volume (m3)	---	---	---	---		
32	Length (m)	3.194e+004	3.194e+004	3.195e+004	3.195e+004		
33	Elevation (m)	0.3800	0.3520	0.3240	0.2960		
34	Pressure (bar)	67.13	67.13	67.13	67.13		
35	Temperature (C)	46.37	46.37	46.37	46.36		
36	Vapour Fraction	0.8803	0.8803	0.8803	0.8803		
37	Heat Transferred (kJ/h-m)	436.0	436.0	436.0	435.9		
38	Flow Regime	Transition	Transition	Transition	Transition		
39	Water Flow Regime						
40	Liquid HoldUp	0.2622	0.1665	0.1665	0.1665		
41	Water HoldUp	---	---	---	---		
42	Friction Gradient (kPa/m)	7.848e-003	9.161e-003	9.161e-003	9.161e-003		
43	Static Gradient (kPa/m)	2.518e-002	-8.792e-003	-8.792e-003	-8.792e-003		
44	Accel. Gradient (kPa/m)	4.275e-006	4.781e-008	4.780e-008	4.779e-008		
45	Liq. Reynolds	1.817e+005 *	1.817e+005 *	1.817e+005 *	1.817e+005 *		
46	Vap. Reynolds	9.915e+006 *	9.915e+006 *	9.915e+006 *	9.915e+006 *		
47	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
48	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947		
49	Deposit Thickness (mm)	---	---	---	---		
50	Deposit Volume (m3)	---	---	---	---		
51	Length (m)	3.196e+004	3.196e+004	3.197e+004	3.197e+004		
52	Elevation (m)	0.2680	0.2400	0.2400	0.2400		
53	Pressure (bar)	67.13	67.13	67.13	67.13		
54	Temperature (C)	46.36	46.36	46.36	46.36		
55	Vapour Fraction	0.8803	0.8803	0.8803	0.8803		
56	Heat Transferred (kJ/h-m)	435.9	435.9	435.8	435.8		
57	Flow Regime	Transition	Transition	Transition	Transition		
58	Water Flow Regime						
59	Liquid HoldUp	0.1665	0.1665	0.2491	0.2491		
60	Water HoldUp	---	---	---	---		
61	Friction Gradient (kPa/m)	9.161e-003	9.161e-003	6.240e-003	6.240e-003		
62	Static Gradient (kPa/m)	-8.792e-003	-8.792e-003	0.0000	0.0000		
63	Accel. Gradient (kPa/m)	4.778e-008	4.776e-008	8.077e-007	8.077e-007		
64	Liq. Reynolds	1.817e+005 *	1.817e+005 *	1.817e+005 *	1.817e+005 *		
65	Vap. Reynolds	9.915e+006 *	9.915e+006 *	9.915e+006 *	9.915e+006 *		
66	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
67	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947		
68	Deposit Thickness (mm)	---	---	---	---		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	3.198e+004	3.198e+004	3.199e+004	3.199e+004
15	Elevation (m)	0.2400	0.2400	0.2400	0.2340
16	Pressure (bar)	67.13	67.12	67.12	67.12
17	Temperature (C)	46.36	46.36	46.35	46.35
18	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
19	Heat Transferred (kJ/h-m)	435.7	435.7	435.7	435.7
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2491	0.2491	0.2492	0.1698
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	6.240e-003	6.240e-003	6.240e-003	9.115e-003
25	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-1.884e-003
26	Accel. Gradient (kPa/m)	8.077e-007	8.077e-007	8.077e-007	9.360e-007
27	Liq. Reynolds	1.817e+005 *	1.817e+005 *	1.817e+005 *	1.817e+005 *
28	Vap. Reynolds	9.915e+006 *	9.915e+006 *	9.915e+006 *	9.915e+006 *
29	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
30	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	3.200e+004	3.200e+004	3.201e+004	3.201e+004
34	Elevation (m)	0.2280	0.2220	0.2160	0.2100
35	Pressure (bar)	67.12	67.12	67.12	67.12
36	Temperature (C)	46.35	46.35	46.35	46.35
37	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
38	Heat Transferred (kJ/h-m)	435.6	435.6	435.5	435.5
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.1698	0.1698	0.1698	0.1698
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	9.115e-003	9.115e-003	9.115e-003	9.115e-003
44	Static Gradient (kPa/m)	-1.884e-003	-1.884e-003	-1.884e-003	-1.884e-003
45	Accel. Gradient (kPa/m)	9.360e-007	9.360e-007	9.360e-007	9.360e-007
46	Liq. Reynolds	1.817e+005 *	1.817e+005 *	1.817e+005 *	1.817e+005 *
47	Vap. Reynolds	9.915e+006 *	9.915e+006 *	9.915e+006 *	9.915e+006 *
48	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
49	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	3.202e+004	3.202e+004	3.203e+004	3.203e+004
53	Elevation (m)	0.2040	0.1980	0.1920	0.1860
54	Pressure (bar)	67.12	67.12	67.12	67.12
55	Temperature (C)	46.35	46.35	46.34	46.34
56	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
57	Heat Transferred (kJ/h-m)	435.5	435.4	435.4	435.4
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.1698	0.1698	0.1698	0.1698
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	9.115e-003	9.115e-003	9.115e-003	9.115e-003
63	Static Gradient (kPa/m)	-1.908e-003	-1.908e-003	-1.908e-003	-1.908e-003
64	Accel. Gradient (kPa/m)	9.330e-007	9.330e-007	9.330e-007	9.330e-007
65	Liq. Reynolds	1.817e+005 *	1.817e+005 *	1.817e+005 *	1.817e+005 *
66	Vap. Reynolds	9.915e+006 *	9.915e+006 *	9.915e+006 *	9.915e+006 *
67	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
68	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Deposit Thickness (mm)	---	---	---	---					
14	Deposit Volume (m3)	---	---	---	---					
15	Length (m)	3.204e+004	3.204e+004	3.205e+004	3.205e+004					
16	Elevation (m)	0.1800	0.1760	0.1720	0.1680					
17	Pressure (bar)	67.12	67.12	67.12	67.12					
18	Temperature (C)	46.34	46.34	46.34	46.34					
19	Vapour Fraction	0.8802	0.8802	0.8802	0.8802					
20	Heat Transferred (kJ/h-m)	435.3	435.3	435.3	435.2					
21	Flow Regime	Transition	Transition	Transition	Transition					
22	Water Flow Regime									
23	Liquid HoldUp	0.1698	0.1701	0.1701	0.1701					
24	Water HoldUp	---	---	---	---					
25	Friction Gradient (kPa/m)	9.115e-003	9.111e-003	9.111e-003	9.111e-003					
26	Static Gradient (kPa/m)	-1.908e-003	-1.274e-003	-1.274e-003	-1.274e-003					
27	Accel. Gradient (kPa/m)	9.330e-007	1.015e-006	1.015e-006	1.015e-006					
28	Liq. Reynolds	1.817e+005 *	1.817e+005 *	1.817e+005 *	1.816e+005 *					
29	Vap. Reynolds	9.915e+006 *	9.915e+006 *	9.915e+006 *	9.915e+006 *					
30	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
31	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947					
32	Deposit Thickness (mm)	---	---	---	---					
33	Deposit Volume (m3)	---	---	---	---					
34	Length (m)	3.206e+004	3.206e+004	3.207e+004	3.207e+004					
35	Elevation (m)	0.1640	0.1600	0.1520	0.1440					
36	Pressure (bar)	67.12	67.12	67.12	67.12					
37	Temperature (C)	46.34	46.33	46.33	46.33					
38	Vapour Fraction	0.8802	0.8802	0.8802	0.8802					
39	Heat Transferred (kJ/h-m)	435.2	435.1	435.1	435.1					
40	Flow Regime	Transition	Transition	Transition	Transition					
41	Water Flow Regime									
42	Liquid HoldUp	0.1701	0.1701	0.1695	0.1695					
43	Water HoldUp	---	---	---	---					
44	Friction Gradient (kPa/m)	9.111e-003	9.111e-003	9.119e-003	9.119e-003					
45	Static Gradient (kPa/m)	-1.274e-003	-1.274e-003	-2.542e-003	-2.542e-003					
46	Accel. Gradient (kPa/m)	1.015e-006	1.015e-006	8.515e-007	8.516e-007					
47	Liq. Reynolds	1.816e+005 *	1.816e+005 *	1.816e+005 *	1.816e+005 *					
48	Vap. Reynolds	9.915e+006 *	9.915e+006 *	9.915e+006 *	9.915e+006 *					
49	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
50	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947					
51	Deposit Thickness (mm)	---	---	---	---					
52	Deposit Volume (m3)	---	---	---	---					
53	Length (m)	3.208e+004	3.208e+004	3.209e+004	3.209e+004					
54	Elevation (m)	0.1360	0.1280	0.1200	0.1260					
55	Pressure (bar)	67.12	67.12	67.12	67.12					
56	Temperature (C)	46.33	46.33	46.33	46.33					
57	Vapour Fraction	0.8802	0.8802	0.8802	0.8802					
58	Heat Transferred (kJ/h-m)	435.0	435.0	435.0	434.9					
59	Flow Regime	Transition	Transition	Transition	Transition					
60	Water Flow Regime									
61	Liquid HoldUp	0.1695	0.1695	0.1695	0.2335					
62	Water HoldUp	---	---	---	---					
63	Friction Gradient (kPa/m)	9.119e-003	9.119e-003	9.119e-003	8.107e-003					
64	Static Gradient (kPa/m)	-2.542e-003	-2.542e-003	-2.542e-003	2.389e-003					
65	Accel. Gradient (kPa/m)	8.516e-007	8.516e-007	8.516e-007	1.359e-006					
66	Liq. Reynolds	1.816e+005 *	1.816e+005 *	1.816e+005 *	1.816e+005 *					
67	Vap. Reynolds	9.915e+006 *	9.915e+006 *	9.915e+006 *	9.915e+006 *					
68	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	3.210e+004	3.210e+004	3.211e+004	3.211e+004	
17	Elevation	(m)	0.1320	0.1380	0.1440	0.1500	
18	Pressure	(bar)	67.12	67.12	67.12	67.11	
19	Temperature	(C)	46.32	46.32	46.32	46.32	
20	Vapour Fraction		0.8802	0.8802	0.8802	0.8802	
21	Heat Transferred	(kJ/h-m)	434.9	434.8	434.8	434.7	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2335	0.2335	0.2335	0.2335	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	8.107e-003	8.107e-003	8.107e-003	8.107e-003	
27	Static Gradient	(kPa/m)	2.389e-003	2.389e-003	2.389e-003	2.389e-003	
28	Accel. Gradient	(kPa/m)	1.359e-006	1.359e-006	1.359e-006	1.359e-006	
29	Liq. Reynolds		1.816e+005 *	1.816e+005 *	1.816e+005 *	1.816e+005 *	
30	Vap. Reynolds		9.915e+006 *	9.915e+006 *	9.915e+006 *	9.915e+006 *	
31	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
32	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	3.212e+004	3.212e+004	3.213e+004	3.213e+004	
36	Elevation	(m)	0.1540	0.1580	0.1620	0.1660	
37	Pressure	(bar)	67.11	67.11	67.11	67.11	
38	Temperature	(C)	46.32	46.32	46.32	46.31	
39	Vapour Fraction		0.8802	0.8802	0.8802	0.8802	
40	Heat Transferred	(kJ/h-m)	434.7	434.7	434.6	434.6	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2324	0.2324	0.2324	0.2324	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	8.253e-003	8.253e-003	8.254e-003	8.254e-003	
46	Static Gradient	(kPa/m)	1.587e-003	1.587e-003	1.587e-003	1.587e-003	
47	Accel. Gradient	(kPa/m)	1.274e-006	1.274e-006	1.274e-006	1.274e-006	
48	Liq. Reynolds		1.816e+005 *	1.816e+005 *	1.816e+005 *	1.816e+005 *	
49	Vap. Reynolds		9.915e+006 *	9.915e+006 *	9.915e+006 *	9.915e+006 *	
50	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
51	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	3.214e+004	3.214e+004	3.215e+004	3.215e+004	
55	Elevation	(m)	0.1700	0.2160	0.2620	0.3080	
56	Pressure	(bar)	67.11	67.11	67.11	67.11	
57	Temperature	(C)	46.31	46.31	46.31	46.31	
58	Vapour Fraction		0.8802	0.8802	0.8802	0.8802	
59	Heat Transferred	(kJ/h-m)	434.6	434.5	434.5	434.4	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.2324	0.2556	0.2556	0.2556	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	8.254e-003	7.515e-003	7.515e-003	7.515e-003	
65	Static Gradient	(kPa/m)	1.587e-003	1.959e-002	1.959e-002	1.959e-002	
66	Accel. Gradient	(kPa/m)	1.274e-006	3.509e-006	3.509e-006	3.509e-006	
67	Liq. Reynolds		1.816e+005 *	1.816e+005 *	1.816e+005 *	1.816e+005 *	
68	Vap. Reynolds		9.915e+006 *	9.915e+006 *	9.915e+006 *	9.915e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2			Unit Set: Mixed			
3			Date/Time: Mon Jun 13 15:13:43 2022			
4						
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
14	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947
15	Deposit Thickness	(mm)	---	---	---	---
16	Deposit Volume	(m3)	---	---	---	---
17	Length	(m)	3.216e+004	3.216e+004	3.217e+004	3.217e+004
18	Elevation	(m)	0.3540	0.4000	0.4040	0.4080
19	Pressure	(bar)	67.11	67.11	67.10	67.10
20	Temperature	(C)	46.31	46.30	46.30	46.30
21	Vapour Fraction		0.8802	0.8802	0.8802	0.8802
22	Heat Transferred	(kJ/h-m)	434.4	434.3	434.3	434.2
23	Flow Regime		Transition	Transition	Transition	Transition
24	Water Flow Regime					
25	Liquid HoldUp		0.2556	0.2556	0.2324	0.2324
26	Water HoldUp		---	---	---	---
27	Friction Gradient	(kPa/m)	7.515e-003	7.515e-003	8.254e-003	8.254e-003
28	Static Gradient	(kPa/m)	1.959e-002	1.959e-002	1.587e-003	1.587e-003
29	Accel. Gradient	(kPa/m)	3.509e-006	3.509e-006	1.274e-006	1.274e-006
30	Liq. Reynolds		1.816e+005 *	1.816e+005 *	1.816e+005 *	1.816e+005 *
31	Vap. Reynolds		9.915e+006 *	9.916e+006 *	9.916e+006 *	9.916e+006 *
32	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
33	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947
34	Deposit Thickness	(mm)	---	---	---	---
35	Deposit Volume	(m3)	---	---	---	---
36	Length	(m)	3.218e+004	3.218e+004	3.219e+004	3.219e+004
37	Elevation	(m)	0.4120	0.4160	0.4200	0.4880
38	Pressure	(bar)	67.10	67.10	67.10	67.10
39	Temperature	(C)	46.30	46.30	46.30	46.30
40	Vapour Fraction		0.8802	0.8802	0.8802	0.8802
41	Heat Transferred	(kJ/h-m)	434.2	434.2	434.1	434.0
42	Flow Regime		Transition	Transition	Transition	Transition
43	Water Flow Regime					
44	Liquid HoldUp		0.2324	0.2324	0.2324	0.4166
45	Water HoldUp		---	---	---	---
46	Friction Gradient	(kPa/m)	8.254e-003	8.254e-003	8.254e-003	7.726e-003
47	Static Gradient	(kPa/m)	1.587e-003	1.587e-003	1.587e-003	0.2135
48	Accel. Gradient	(kPa/m)	1.274e-006	1.274e-006	1.274e-006	2.865e-005
49	Liq. Reynolds		1.816e+005 *	1.816e+005 *	1.816e+005 *	1.816e+005 *
50	Vap. Reynolds		9.916e+006 *	9.916e+006 *	9.916e+006 *	9.916e+006 *
51	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979
52	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947
53	Deposit Thickness	(mm)	---	---	---	---
54	Deposit Volume	(m3)	---	---	---	---
55	Length	(m)	3.219e+004	3.219e+004	3.219e+004	3.219e+004
56	Elevation	(m)	0.5560	0.6240	0.6920	0.7600
57	Pressure	(bar)	67.10	67.10	67.09	67.09
58	Temperature	(C)	46.30	46.29	46.29	46.29
59	Vapour Fraction		0.8802	0.8802	0.8802	0.8802
60	Heat Transferred	(kJ/h-m)	434.0	433.9	433.9	433.9
61	Flow Regime		Transition	Transition	Transition	Transition
62	Water Flow Regime					
63	Liquid HoldUp		0.4165	0.4165	0.4165	0.4165
64	Water HoldUp		---	---	---	---
65	Friction Gradient	(kPa/m)	7.726e-003	7.727e-003	7.727e-003	7.727e-003
66	Static Gradient	(kPa/m)	0.2135	0.2135	0.2135	0.2135
67	Accel. Gradient	(kPa/m)	2.865e-005	2.865e-005	2.865e-005	2.866e-005
68	Liq. Reynolds		1.816e+005 *	1.816e+005 *	1.816e+005 *	1.816e+005 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.916e+006 *	9.916e+006 *	9.916e+006 *	9.916e+006 *
14	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
15	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	3.219e+004	3.220e+004	3.220e+004	3.221e+004
19	Elevation (m)	0.8460	0.9320	1.018	1.104
20	Pressure (bar)	67.09	67.09	67.08	67.08
21	Temperature (C)	46.29	46.29	46.29	46.29
22	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
23	Heat Transferred (kJ/h-m)	434.0	433.9	433.9	433.8
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2904	0.2904	0.2904	0.2904
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	7.984e-003	7.984e-003	7.984e-003	7.984e-003
29	Static Gradient (kPa/m)	5.126e-002	5.126e-002	5.126e-002	5.126e-002
30	Accel. Gradient (kPa/m)	7.676e-006	7.676e-006	7.676e-006	7.677e-006
31	Liq. Reynolds	1.816e+005 *	1.816e+005 *	1.816e+005 *	1.816e+005 *
32	Vap. Reynolds	9.916e+006 *	9.916e+006 *	9.916e+006 *	9.916e+006 *
33	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
34	Vapour Velocity (m/s)	2.947	2.947	2.948	2.948
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	3.221e+004	3.221e+004	3.222e+004	3.222e+004
38	Elevation (m)	1.190	1.216	1.242	1.268
39	Pressure (bar)	67.08	67.08	67.08	67.08
40	Temperature (C)	46.28	46.28	46.28	46.28
41	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
42	Heat Transferred (kJ/h-m)	433.8	433.7	433.7	433.6
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2904	0.2481	0.2481	0.2481
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	7.985e-003	6.353e-003	6.353e-003	6.353e-003
48	Static Gradient (kPa/m)	5.126e-002	1.353e-002	1.353e-002	1.353e-002
49	Accel. Gradient (kPa/m)	7.677e-006	2.577e-006	2.577e-006	2.577e-006
50	Liq. Reynolds	1.816e+005 *	1.816e+005 *	1.816e+005 *	1.816e+005 *
51	Vap. Reynolds	9.916e+006 *	9.917e+006 *	9.917e+006 *	9.917e+006 *
52	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
53	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	3.223e+004	3.223e+004	3.224e+004	3.226e+004
57	Elevation (m)	1.294	1.320	1.322	1.324
58	Pressure (bar)	67.08	67.08	67.08	67.07
59	Temperature (C)	46.28	46.28	46.27	46.27
60	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
61	Heat Transferred (kJ/h-m)	433.6	433.6	433.5	433.4
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2481	0.2481	0.2306	0.2306
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	6.353e-003	6.353e-003	8.501e-003	8.501e-003
67	Static Gradient (kPa/m)	1.353e-002	1.353e-002	2.943e-004	2.943e-004
68	Accel. Gradient (kPa/m)	2.577e-006	2.577e-006	1.140e-006	1.140e-006
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2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		1.816e+005 *	1.816e+005 *	1.815e+005 *	1.815e+005 *	
14	Vap. Reynolds		9.917e+006 *	9.917e+006 *	9.917e+006 *	9.917e+006 *	
15	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979	0.1979	
16	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948	2.948	
17	Deposit Thickness (mm)	---	---	---	---	---	
18	Deposit Volume (m3)	---	---	---	---	---	
19	Length (m)	3.227e+004	3.228e+004	3.230e+004	3.231e+004		
20	Elevation (m)	1.326	1.328	1.330	1.320		
21	Pressure (bar)	67.07	67.07	67.07	67.07		
22	Temperature (C)	46.27	46.26	46.26	46.25		
23	Vapour Fraction	0.8802	0.8802	0.8802	0.8802		
24	Heat Transferred (kJ/h-m)	433.3	433.2	433.1	433.0		
25	Flow Regime	Transition	Transition	Transition	Transition		
26	Water Flow Regime						
27	Liquid HoldUp	0.2306	0.2306	0.2306	0.1701		
28	Water HoldUp	---	---	---	---		
29	Friction Gradient (kPa/m)	8.501e-003	8.501e-003	8.501e-003	9.115e-003		
30	Static Gradient (kPa/m)	2.943e-004	2.943e-004	2.943e-004	-1.188e-003		
31	Accel. Gradient (kPa/m)	1.140e-006	1.140e-006	1.140e-006	1.027e-006		
32	Liq. Reynolds		1.815e+005 *	1.815e+005 *	1.815e+005 *	1.815e+005 *	
33	Vap. Reynolds		9.917e+006 *	9.917e+006 *	9.917e+006 *	9.917e+006 *	
34	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
35	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948		
36	Deposit Thickness (mm)	---	---	---	---		
37	Deposit Volume (m3)	---	---	---	---		
38	Length (m)	3.232e+004	3.234e+004	3.235e+004	3.236e+004		
39	Elevation (m)	1.310	1.300	1.290	1.280		
40	Pressure (bar)	67.07	67.07	67.07	67.07		
41	Temperature (C)	46.25	46.25	46.24	46.24		
42	Vapour Fraction	0.8802	0.8802	0.8802	0.8802		
43	Heat Transferred (kJ/h-m)	432.9	432.8	432.7	432.6		
44	Flow Regime	Transition	Transition	Transition	Transition		
45	Water Flow Regime						
46	Liquid HoldUp	0.1701	0.1701	0.1701	0.1701		
47	Water HoldUp	---	---	---	---		
48	Friction Gradient (kPa/m)	9.115e-003	9.115e-003	9.115e-003	9.115e-003		
49	Static Gradient (kPa/m)	-1.188e-003	-1.188e-003	-1.188e-003	-1.188e-003		
50	Accel. Gradient (kPa/m)	1.027e-006	1.027e-006	1.027e-006	1.027e-006		
51	Liq. Reynolds		1.815e+005 *	1.815e+005 *	1.815e+005 *	1.815e+005 *	
52	Vap. Reynolds		9.917e+006 *	9.917e+006 *	9.917e+006 *	9.917e+006 *	
53	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979		
54	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948		
55	Deposit Thickness (mm)	---	---	---	---		
56	Deposit Volume (m3)	---	---	---	---		
57	Length (m)	3.238e+004	3.239e+004	3.240e+004	3.242e+004		
58	Elevation (m)	1.292	1.304	1.316	1.328		
59	Pressure (bar)	67.06	67.06	67.06	67.06		
60	Temperature (C)	46.24	46.23	46.23	46.22		
61	Vapour Fraction	0.8802	0.8802	0.8802	0.8802		
62	Heat Transferred (kJ/h-m)	432.5	432.4	432.3	432.2		
63	Flow Regime	Transition	Transition	Transition	Transition		
64	Water Flow Regime						
65	Liquid HoldUp	0.2326	0.2326	0.2326	0.2326		
66	Water HoldUp	---	---	---	---		
67	Friction Gradient (kPa/m)	8.223e-003	8.224e-003	8.224e-003	8.224e-003		
68	Static Gradient (kPa/m)	1.777e-003	1.777e-003	1.777e-003	1.777e-003		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Accel. Gradient	(kPa/m)	1.296e-006	1.296e-006	1.296e-006	1.296e-006				
14	Liq. Reynolds		1.815e+005 *	1.815e+005 *	1.815e+005 *	1.815e+005 *				
15	Vap. Reynolds		9.917e+006 *	9.917e+006 *	9.917e+006 *	9.917e+006 *				
16	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979				
17	Vapour Velocity	(m/s)	2.948	2.948	2.948	2.948				
18	Deposit Thickness	(mm)	---	---	---	---				
19	Deposit Volume	(m3)	---	---	---	---				
20	Length	(m)	3.243e+004	3.244e+004	3.246e+004	3.247e+004				
21	Elevation	(m)	1.340	1.348	1.356	1.364				
22	Pressure	(bar)	67.06	67.06	67.06	67.05				
23	Temperature	(C)	46.22	46.22	46.21	46.21				
24	Vapour Fraction		0.8802	0.8802	0.8802	0.8802				
25	Heat Transferred	(kJ/h-m)	432.1	432.0	431.9	431.8				
26	Flow Regime		Transition	Transition	Transition	Transition				
27	Water Flow Regime									
28	Liquid HoldUp		0.2326	0.2318	0.2318	0.2318				
29	Water HoldUp		---	---	---	---				
30	Friction Gradient	(kPa/m)	8.224e-003	8.334e-003	8.334e-003	8.334e-003				
31	Static Gradient	(kPa/m)	1.777e-003	1.182e-003	1.182e-003	1.182e-003				
32	Accel. Gradient	(kPa/m)	1.296e-006	1.234e-006	1.234e-006	1.234e-006				
33	Liq. Reynolds		1.815e+005 *	1.815e+005 *	1.815e+005 *	1.815e+005 *				
34	Vap. Reynolds		9.917e+006 *	9.918e+006 *	9.918e+006 *	9.918e+006 *				
35	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979				
36	Vapour Velocity	(m/s)	2.948	2.948	2.948	2.948				
37	Deposit Thickness	(mm)	---	---	---	---				
38	Deposit Volume	(m3)	---	---	---	---				
39	Length	(m)	3.248e+004	3.250e+004	3.251e+004	3.252e+004				
40	Elevation	(m)	1.372	1.380	1.340	1.300				
41	Pressure	(bar)	67.05	67.05	67.05	67.05				
42	Temperature	(C)	46.21	46.20	46.20	46.20				
43	Vapour Fraction		0.8802	0.8802	0.8802	0.8802				
44	Heat Transferred	(kJ/h-m)	431.7	431.6	431.5	431.4				
45	Flow Regime		Transition	Transition	Transition	Transition				
46	Water Flow Regime									
47	Liquid HoldUp		0.2318	0.2318	0.1684	0.1684				
48	Water HoldUp		---	---	---	---				
49	Friction Gradient	(kPa/m)	8.334e-003	8.334e-003	9.138e-003	9.138e-003				
50	Static Gradient	(kPa/m)	1.182e-003	1.182e-003	-4.721e-003	-4.722e-003				
51	Accel. Gradient	(kPa/m)	1.234e-006	1.234e-006	5.725e-007	5.725e-007				
52	Liq. Reynolds		1.815e+005 *	1.815e+005 *	1.815e+005 *	1.815e+005 *				
53	Vap. Reynolds		9.918e+006 *	9.918e+006 *	9.918e+006 *	9.918e+006 *				
54	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979				
55	Vapour Velocity	(m/s)	2.948	2.948	2.948	2.948				
56	Deposit Thickness	(mm)	---	---	---	---				
57	Deposit Volume	(m3)	---	---	---	---				
58	Length	(m)	3.254e+004	3.255e+004	3.257e+004	3.257e+004				
59	Elevation	(m)	1.260	1.220	1.180	1.162				
60	Pressure	(bar)	67.05	67.05	67.05	67.05				
61	Temperature	(C)	46.19	46.19	46.19	46.18				
62	Vapour Fraction		0.8802	0.8802	0.8802	0.8802				
63	Heat Transferred	(kJ/h-m)	431.4	431.3	431.2	431.1				
64	Flow Regime		Transition	Transition	Transition	Transition				
65	Water Flow Regime									
66	Liquid HoldUp		0.1684	0.1684	0.1684	0.1684				
67	Water HoldUp		---	---	---	---				
68	Friction Gradient	(kPa/m)	9.138e-003	9.137e-003	9.137e-003	9.138e-003				
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 518 of 937				

1				Case Name:	Simulation - 386MMSCFD_80barg.hsc		
2				Unit Set:	Mixed		
3		Epic Atlantic Limited		Date/Time:	Mon Jun 13 15:13:43 2022		
4							
5							
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Static Gradient	(kPa/m)	-4.722e-003	-4.722e-003	-4.722e-003	-4.751e-003	
14	Accel. Gradient	(kPa/m)	5.725e-007	5.725e-007	5.725e-007	5.687e-007	
15	Liq. Reynolds		1.815e+005 *	1.815e+005 *	1.815e+005 *	1.814e+005 *	
16	Vap. Reynolds		9.918e+006 *	9.918e+006 *	9.918e+006 *	9.918e+006 *	
17	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
18	Vapour Velocity	(m/s)	2.948	2.948	2.948	2.948	
19	Deposit Thickness	(mm)	---	---	---	---	
20	Deposit Volume	(m3)	---	---	---	---	
21	Length	(m)	3.258e+004	3.258e+004	3.259e+004	3.260e+004	
22	Elevation	(m)	1.144	1.126	1.108	1.090	
23	Pressure	(bar)	67.05	67.05	67.05	67.05	
24	Temperature	(C)	46.18	46.18	46.18	46.18	
25	Vapour Fraction		0.8802	0.8802	0.8802	0.8802	
26	Heat Transferred	(kJ/h-m)	431.1	431.1	431.0	431.0	
27	Flow Regime		Transition	Transition	Transition	Transition	
28	Water Flow Regime						
29	Liquid HoldUp		0.1684	0.1684	0.1684	0.1684	
30	Water HoldUp		---	---	---	---	
31	Friction Gradient	(kPa/m)	9.138e-003	9.138e-003	9.138e-003	9.137e-003	
32	Static Gradient	(kPa/m)	-4.751e-003	-4.751e-003	-4.751e-003	-4.751e-003	
33	Accel. Gradient	(kPa/m)	5.686e-007	5.686e-007	5.686e-007	5.686e-007	
34	Liq. Reynolds		1.814e+005 *	1.814e+005 *	1.814e+005 *	1.814e+005 *	
35	Vap. Reynolds		9.918e+006 *	9.918e+006 *	9.918e+006 *	9.918e+006 *	
36	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
37	Vapour Velocity	(m/s)	2.948	2.948	2.948	2.948	
38	Deposit Thickness	(mm)	---	---	---	---	
39	Deposit Volume	(m3)	---	---	---	---	
40	Length	(m)	3.260e+004	3.261e+004	3.261e+004	3.262e+004	
41	Elevation	(m)	1.078	1.066	1.054	1.042	
42	Pressure	(bar)	67.05	67.05	67.05	67.05	
43	Temperature	(C)	46.18	46.17	46.17	46.17	
44	Vapour Fraction		0.8802	0.8802	0.8802	0.8802	
45	Heat Transferred	(kJ/h-m)	430.9	430.9	430.8	430.8	
46	Flow Regime		Transition	Transition	Transition	Transition	
47	Water Flow Regime						
48	Liquid HoldUp		0.1692	0.1692	0.1692	0.1692	
49	Water HoldUp		---	---	---	---	
50	Friction Gradient	(kPa/m)	9.127e-003	9.127e-003	9.127e-003	9.127e-003	
51	Static Gradient	(kPa/m)	-3.173e-003	-3.173e-003	-3.173e-003	-3.173e-003	
52	Accel. Gradient	(kPa/m)	7.720e-007	7.720e-007	7.720e-007	7.720e-007	
53	Liq. Reynolds		1.814e+005 *	1.814e+005 *	1.814e+005 *	1.814e+005 *	
54	Vap. Reynolds		9.918e+006 *	9.918e+006 *	9.918e+006 *	9.918e+006 *	
55	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
56	Vapour Velocity	(m/s)	2.948	2.948	2.948	2.948	
57	Deposit Thickness	(mm)	---	---	---	---	
58	Deposit Volume	(m3)	---	---	---	---	
59	Length	(m)	3.263e+004	3.263e+004	3.263e+004	3.263e+004	
60	Elevation	(m)	1.030	1.026	1.022	1.018	
61	Pressure	(bar)	67.05	67.05	67.05	67.05	
62	Temperature	(C)	46.17	46.17	46.17	46.17	
63	Vapour Fraction		0.8802	0.8802	0.8802	0.8802	
64	Heat Transferred	(kJ/h-m)	430.8	430.7	430.7	430.7	
65	Flow Regime		Transition	Transition	Transition	Transition	
66	Water Flow Regime						
67	Liquid HoldUp		0.1692	0.1692	0.1692	0.1692	
68	Water HoldUp		---	---	---	---	
69	Aspen Technology Inc.		Aspen HYSYS Version 10				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:13:43 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	9.127e-003	9.127e-003	9.127e-003	9.127e-003	
14	Static Gradient	(kPa/m)	-3.173e-003	-3.173e-003	-3.173e-003	-3.173e-003	
15	Accel. Gradient	(kPa/m)	7.720e-007	7.720e-007	7.720e-007	7.720e-007	
16	Liq. Reynolds		1.814e+005 *	1.814e+005 *	1.814e+005 *	1.814e+005 *	
17	Vap. Reynolds		9.918e+006 *	9.918e+006 *	9.918e+006 *	9.918e+006 *	
18	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
19	Vapour Velocity	(m/s)	2.948	2.948	2.948	2.948	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	3.263e+004	3.264e+004	3.264e+004	3.264e+004	
23	Elevation	(m)	1.014	1.010	1.002	0.9940	
24	Pressure	(bar)	67.05	67.05	67.05	67.05	
25	Temperature	(C)	46.17	46.17	46.17	46.17	
26	Vapour Fraction		0.8802	0.8802	0.8802	0.8802	
27	Heat Transferred	(kJ/h-m)	430.7	430.7	430.7	430.7	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.1692	0.1692	0.1677	0.1677	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	9.127e-003	9.127e-003	9.148e-003	9.148e-003	
33	Static Gradient	(kPa/m)	-3.173e-003	-3.173e-003	-6.308e-003	-6.308e-003	
34	Accel. Gradient	(kPa/m)	7.720e-007	7.720e-007	3.681e-007	3.681e-007	
35	Liq. Reynolds		1.814e+005 *	1.814e+005 *	1.814e+005 *	1.814e+005 *	
36	Vap. Reynolds		9.918e+006 *	9.918e+006 *	9.918e+006 *	9.918e+006 *	
37	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
38	Vapour Velocity	(m/s)	2.948	2.948	2.948	2.948	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	3.264e+004	3.264e+004	3.265e+004	3.265e+004	
42	Elevation	(m)	0.9860	0.9780	0.9700	0.9520	
43	Pressure	(bar)	67.05	67.05	67.05	67.04	
44	Temperature	(C)	46.17	46.17	46.16	46.16	
45	Vapour Fraction		0.8802	0.8802	0.8802	0.8802	
46	Heat Transferred	(kJ/h-m)	430.7	430.6	430.6	430.6	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.1677	0.1677	0.1677	0.1680	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	9.148e-003	9.148e-003	9.148e-003	9.143e-003	
52	Static Gradient	(kPa/m)	-6.308e-003	-6.308e-003	-6.308e-003	-5.684e-003	
53	Accel. Gradient	(kPa/m)	3.681e-007	3.681e-007	3.681e-007	4.485e-007	
54	Liq. Reynolds		1.814e+005 *	1.814e+005 *	1.814e+005 *	1.814e+005 *	
55	Vap. Reynolds		9.918e+006 *	9.918e+006 *	9.918e+006 *	9.918e+006 *	
56	Liquid Velocity	(m/s)	0.1979	0.1979	0.1979	0.1979	
57	Vapour Velocity	(m/s)	2.948	2.948	2.948	2.948	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	3.266e+004	3.266e+004	3.267e+004	3.267e+004	
61	Elevation	(m)	0.9340	0.9160	0.8980	0.8800	
62	Pressure	(bar)	67.04	67.04	67.04	67.04	
63	Temperature	(C)	46.16	46.16	46.16	46.16	
64	Vapour Fraction		0.8802	0.8802	0.8802	0.8802	
65	Heat Transferred	(kJ/h-m)	430.6	430.5	430.5	430.5	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.1680	0.1680	0.1680	0.1680	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 520 of 937	

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	9.143e-003	9.143e-003	9.143e-003	9.143e-003
15	Static Gradient (kPa/m)	-5.684e-003	-5.684e-003	-5.684e-003	-5.684e-003
16	Accel. Gradient (kPa/m)	4.485e-007	4.484e-007	4.484e-007	4.484e-007
17	Liq. Reynolds	1.814e+005 *	1.814e+005 *	1.814e+005 *	1.814e+005 *
18	Vap. Reynolds	9.918e+006 *	9.918e+006 *	9.918e+006 *	9.918e+006 *
19	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
20	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	3.268e+004	3.268e+004	3.269e+004	3.269e+004
24	Elevation (m)	0.8680	0.8560	0.8440	0.8320
25	Pressure (bar)	67.04	67.04	67.04	67.04
26	Temperature (C)	46.16	46.16	46.15	46.15
27	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
28	Heat Transferred (kJ/h-m)	430.4	430.4	430.4	430.3
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.1689	0.1689	0.1689	0.1689
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	9.131e-003	9.131e-003	9.131e-003	9.131e-003
34	Static Gradient (kPa/m)	-3.803e-003	-3.803e-003	-3.803e-003	-3.803e-003
35	Accel. Gradient (kPa/m)	6.908e-007	6.908e-007	6.908e-007	6.908e-007
36	Liq. Reynolds	1.814e+005 *	1.814e+005 *	1.814e+005 *	1.814e+005 *
37	Vap. Reynolds	9.918e+006 *	9.918e+006 *	9.918e+006 *	9.918e+006 *
38	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
39	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	3.270e+004	3.270e+004	3.271e+004	3.271e+004
43	Elevation (m)	0.8200	0.8360	0.8520	0.8680
44	Pressure (bar)	67.04	67.04	67.04	67.04
45	Temperature (C)	46.15	46.15	46.15	46.15
46	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
47	Heat Transferred (kJ/h-m)	430.3	430.2	430.2	430.1
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.1689	0.2390	0.2390	0.2390
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	9.131e-003	7.410e-003	7.410e-003	7.410e-003
53	Static Gradient (kPa/m)	-3.803e-003	6.480e-003	6.480e-003	6.480e-003
54	Accel. Gradient (kPa/m)	6.908e-007	1.801e-006	1.801e-006	1.801e-006
55	Liq. Reynolds	1.814e+005 *	1.814e+005 *	1.814e+005 *	1.814e+005 *
56	Vap. Reynolds	9.918e+006 *	9.918e+006 *	9.918e+006 *	9.918e+006 *
57	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
58	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	3.272e+004	3.272e+004	3.273e+004	3.273e+004
62	Elevation (m)	0.8840	0.9000	0.8860	0.8720
63	Pressure (bar)	67.04	67.04	67.04	67.04
64	Temperature (C)	46.15	46.14	46.14	46.14
65	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
66	Heat Transferred (kJ/h-m)	430.1	430.1	430.1	430.0
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 521 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.2390	0.2390	0.1686	0.1686
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	7.410e-003	7.410e-003	9.135e-003	9.135e-003
16	Static Gradient (kPa/m)	6.480e-003	6.480e-003	-4.432e-003	-4.432e-003
17	Accel. Gradient (kPa/m)	1.801e-006	1.801e-006	6.099e-007	6.099e-007
18	Liq. Reynolds	1.814e+005 *	1.814e+005 *	1.814e+005 *	1.814e+005 *
19	Vap. Reynolds	9.918e+006 *	9.918e+006 *	9.918e+006 *	9.918e+006 *
20	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
21	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	3.274e+004	3.274e+004	3.275e+004	3.275e+004
25	Elevation (m)	0.8580	0.8440	0.8300	0.8620
26	Pressure (bar)	67.04	67.04	67.04	67.04
27	Temperature (C)	46.14	46.14	46.14	46.14
28	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
29	Heat Transferred (kJ/h-m)	430.0	430.0	429.9	429.8
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.1686	0.1686	0.1686	0.2478
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	9.135e-003	9.135e-003	9.135e-003	6.383e-003
35	Static Gradient (kPa/m)	-4.432e-003	-4.432e-003	-4.432e-003	1.331e-002
36	Accel. Gradient (kPa/m)	6.099e-007	6.099e-007	6.098e-007	2.554e-006
37	Liq. Reynolds	1.814e+005 *	1.814e+005 *	1.814e+005 *	1.814e+005 *
38	Vap. Reynolds	9.919e+006 *	9.919e+006 *	9.919e+006 *	9.919e+006 *
39	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
40	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	3.276e+004	3.276e+004	3.277e+004	3.277e+004
44	Elevation (m)	0.8940	0.9260	0.9580	0.9900
45	Pressure (bar)	67.04	67.04	67.03	67.03
46	Temperature (C)	46.13	46.13	46.13	46.13
47	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
48	Heat Transferred (kJ/h-m)	429.8	429.8	429.7	429.7
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2478	0.2478	0.2478	0.2478
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.383e-003	6.383e-003	6.384e-003	6.384e-003
54	Static Gradient (kPa/m)	1.331e-002	1.331e-002	1.331e-002	1.331e-002
55	Accel. Gradient (kPa/m)	2.554e-006	2.554e-006	2.554e-006	2.554e-006
56	Liq. Reynolds	1.814e+005 *	1.814e+005 *	1.814e+005 *	1.814e+005 *
57	Vap. Reynolds	9.919e+006 *	9.919e+006 *	9.919e+006 *	9.919e+006 *
58	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
59	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	3.278e+004	3.278e+004	3.279e+004	3.279e+004
63	Elevation (m)	0.9620	0.9340	0.9060	0.8780
64	Pressure (bar)	67.03	67.03	67.03	67.03
65	Temperature (C)	46.13	46.13	46.13	46.13
66	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
67	Heat Transferred (kJ/h-m)	429.7	429.6	429.6	429.6
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 522 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1666	0.1666	0.1666	0.1666
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	9.163e-003	9.163e-003	9.163e-003	9.163e-003
17	Static Gradient (kPa/m)	-8.651e-003	-8.651e-003	-8.651e-003	-8.651e-003
18	Accel. Gradient (kPa/m)	6.645e-008	6.644e-008	6.643e-008	6.642e-008
19	Liq. Reynolds	1.814e+005 *	1.814e+005 *	1.814e+005 *	1.814e+005 *
20	Vap. Reynolds	9.919e+006 *	9.919e+006 *	9.919e+006 *	9.919e+006 *
21	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
22	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	3.280e+004	3.280e+004	3.281e+004	3.281e+004
26	Elevation (m)	0.8500	0.8280	0.8060	0.7840
27	Pressure (bar)	67.03	67.03	67.03	67.03
28	Temperature (C)	46.12	46.12	46.12	46.12
29	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
30	Heat Transferred (kJ/h-m)	429.5	429.5	429.5	429.4
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1666	0.1674	0.1674	0.1674
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	9.163e-003	9.152e-003	9.152e-003	9.152e-003
36	Static Gradient (kPa/m)	-8.651e-003	-6.931e-003	-6.931e-003	-6.931e-003
37	Accel. Gradient (kPa/m)	6.641e-008	2.879e-007	2.879e-007	2.879e-007
38	Liq. Reynolds	1.814e+005 *	1.814e+005 *	1.814e+005 *	1.814e+005 *
39	Vap. Reynolds	9.919e+006 *	9.919e+006 *	9.919e+006 *	9.919e+006 *
40	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
41	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	3.282e+004	3.282e+004	3.283e+004	3.283e+004
45	Elevation (m)	0.7620	0.7400	0.6940	0.6480
46	Pressure (bar)	67.03	67.03	67.03	67.03
47	Temperature (C)	46.12	46.12	46.12	46.12
48	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
49	Heat Transferred (kJ/h-m)	429.4	429.4	429.4	429.3
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.1674	0.1674	0.1639	0.1639
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	9.152e-003	9.152e-003	9.203e-003	9.203e-003
55	Static Gradient (kPa/m)	-6.931e-003	-6.931e-003	-1.429e-002	-1.429e-002
56	Accel. Gradient (kPa/m)	2.879e-007	2.879e-007	-6.595e-007	-6.596e-007
57	Liq. Reynolds	1.814e+005 *	1.814e+005 *	1.814e+005 *	1.814e+005 *
58	Vap. Reynolds	9.919e+006 *	9.919e+006 *	9.919e+006 *	9.919e+006 *
59	Liquid Velocity (m/s)	0.1979	0.1979	0.1979	0.1979
60	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	3.284e+004	3.284e+004	3.285e+004	3.285e+004
64	Elevation (m)	0.6020	0.5560	0.5100	0.4920
65	Pressure (bar)	67.03	67.03	67.03	67.03
66	Temperature (C)	46.11	46.11	46.11	46.11
67	Vapour Fraction	0.8802	0.8802	0.8802	0.8802
68	Heat Transferred (kJ/h-m)	429.3	429.3	429.2	429.2
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc	
2				
3			Unit Set: Mixed	
4			Date/Time: Mon Jun 13 15:13:43 2022	
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Pipe Table			
12				
13	Flow Regime	Transition	Transition	Transition
14	Water Flow Regime			
15	Liquid HoldUp	0.1639	0.1639	0.1639
16	Water HoldUp	---	---	---
17	Friction Gradient (kPa/m)	9.203e-003	9.202e-003	9.202e-003
18	Static Gradient (kPa/m)	-1.429e-002	-1.429e-002	-1.429e-002
19	Accel. Gradient (kPa/m)	-6.596e-007	-6.596e-007	-6.596e-007
20	Liq. Reynolds	1.814e+005 *	1.814e+005 *	1.814e+005 *
21	Vap. Reynolds	9.919e+006 *	9.919e+006 *	9.919e+006 *
22	Liquid Velocity (m/s)	0.1979	0.1979	0.1979
23	Vapour Velocity (m/s)	2.947	2.947	2.947
24	Deposit Thickness (mm)	---	---	---
25	Deposit Volume (m3)	---	---	---
26	Length (m)	3.286e+004	3.286e+004	3.287e+004
27	Elevation (m)	0.4740	0.4560	0.4380
28	Pressure (bar)	67.03	67.03	67.03
29	Temperature (C)	46.11	46.11	46.11
30	Vapour Fraction	0.8802	0.8802	0.8802
31	Heat Transferred (kJ/h-m)	429.2	429.1	429.1
32	Flow Regime	Transition	Transition	Transition
33	Water Flow Regime			
34	Liquid HoldUp	0.1680	0.1680	0.1680
35	Water HoldUp	---	---	---
36	Friction Gradient (kPa/m)	9.143e-003	9.143e-003	9.143e-003
37	Static Gradient (kPa/m)	-5.685e-003	-5.685e-003	-5.685e-003
38	Accel. Gradient (kPa/m)	4.484e-007	4.484e-007	4.483e-007
39	Liq. Reynolds	1.814e+005 *	1.814e+005 *	1.814e+005 *
40	Vap. Reynolds	9.919e+006 *	9.919e+006 *	9.919e+006 *
41	Liquid Velocity (m/s)	0.1979	0.1979	0.1979
42	Vapour Velocity (m/s)	2.947	2.947	2.947
43	Deposit Thickness (mm)	---	---	---
44	Deposit Volume (m3)	---	---	---
45	Length (m)	3.288e+004	3.288e+004	3.289e+004
46	Elevation (m)	0.3780	0.3360	0.2940
47	Pressure (bar)	67.03	67.03	67.03
48	Temperature (C)	46.11	46.10	46.10
49	Vapour Fraction	0.8802	0.8802	0.8802
50	Heat Transferred (kJ/h-m)	429.1	429.0	429.0
51	Flow Regime	Transition	Transition	Transition
52	Water Flow Regime			
53	Liquid HoldUp	0.1645	0.1645	0.1645
54	Water HoldUp	---	---	---
55	Friction Gradient (kPa/m)	9.193e-003	9.193e-003	9.193e-003
56	Static Gradient (kPa/m)	-1.308e-002	-1.308e-002	-1.308e-002
57	Accel. Gradient (kPa/m)	-5.037e-007	-5.037e-007	-5.037e-007
58	Liq. Reynolds	1.814e+005 *	1.814e+005 *	1.814e+005 *
59	Vap. Reynolds	9.919e+006 *	9.919e+006 *	9.919e+006 *
60	Liquid Velocity (m/s)	0.1979	0.1979	0.1979
61	Vapour Velocity (m/s)	2.947	2.947	2.947
62	Deposit Thickness (mm)	---	---	---
63	Deposit Volume (m3)	---	---	---
64	Length (m)	3.290e+004	3.290e+004	3.291e+004
65	Elevation (m)	0.2100	0.2040	0.1980
66	Pressure (bar)	67.03	67.03	67.03
67	Temperature (C)	46.10	46.10	46.10
68	Vapour Fraction	0.8801	0.8801	0.8801
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:13:43 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	428.9	428.9	428.9	428.8
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.1645	0.1698	0.1698	0.1698
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	9.193e-003	9.118e-003	9.118e-003	9.118e-003
19	Static Gradient (kPa/m)	-1.308e-002	-1.908e-003	-1.908e-003	-1.908e-003
20	Accel. Gradient (kPa/m)	-5.038e-007	9.347e-007	9.347e-007	9.347e-007
21	Liq. Reynolds	1.814e+005 *	1.814e+005 *	1.814e+005 *	1.814e+005 *
22	Vap. Reynolds	9.919e+006 *	9.919e+006 *	9.919e+006 *	9.919e+006 *
23	Liquid Velocity (m/s)	0.1979	0.1980	0.1980	0.1980
24	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	3.292e+004	3.292e+004	3.293e+004	3.293e+004
28	Elevation (m)	0.1860	0.1800	0.1800	0.1800
29	Pressure (bar)	67.03	67.03	67.03	67.03
30	Temperature (C)	46.10	46.09	46.09	46.09
31	Vapour Fraction	0.8801	0.8801	0.8801	0.8801
32	Heat Transferred (kJ/h-m)	428.8	428.8	428.7	428.6
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.1698	0.1698	0.2491	0.2491
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	9.118e-003	9.118e-003	6.242e-003	6.242e-003
38	Static Gradient (kPa/m)	-1.908e-003	-1.908e-003	0.0000	0.0000
39	Accel. Gradient (kPa/m)	9.347e-007	9.348e-007	8.093e-007	8.093e-007
40	Liq. Reynolds	1.814e+005 *	1.814e+005 *	1.814e+005 *	1.814e+005 *
41	Vap. Reynolds	9.919e+006 *	9.919e+006 *	9.919e+006 *	9.919e+006 *
42	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
43	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	3.294e+004	3.294e+004	3.295e+004	3.295e+004
47	Elevation (m)	0.1800	0.1800	0.1800	0.1760
48	Pressure (bar)	67.03	67.03	67.03	67.03
49	Temperature (C)	46.09	46.09	46.09	46.09
50	Vapour Fraction	0.8801	0.8801	0.8801	0.8801
51	Heat Transferred (kJ/h-m)	428.6	428.6	428.5	428.5
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.2491	0.2491	0.2491	0.1701
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	6.242e-003	6.242e-003	6.242e-003	9.114e-003
57	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-1.274e-003
58	Accel. Gradient (kPa/m)	8.093e-007	8.093e-007	8.093e-007	1.017e-006
59	Liq. Reynolds	1.814e+005 *	1.814e+005 *	1.814e+005 *	1.814e+005 *
60	Vap. Reynolds	9.919e+006 *	9.919e+006 *	9.919e+006 *	9.919e+006 *
61	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
62	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	3.296e+004	3.296e+004	3.297e+004	3.297e+004
66	Elevation (m)	0.1720	0.1680	0.1640	0.1600
67	Pressure (bar)	67.03	67.03	67.03	67.03
68	Temperature (C)	46.08	46.08	46.08	46.08
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Fraction		0.8801	0.8801	0.8801	0.8801	
14	Heat Transferred (kJ/h-m)		428.5	428.5	428.4	428.4	
15	Flow Regime		Transition	Transition	Transition	Transition	
16	Water Flow Regime						
17	Liquid HoldUp		0.1701	0.1701	0.1701	0.1701	
18	Water HoldUp		---	---	---	---	
19	Friction Gradient (kPa/m)		9.114e-003	9.114e-003	9.114e-003	9.114e-003	
20	Static Gradient (kPa/m)		-1.274e-003	-1.274e-003	-1.274e-003	-1.274e-003	
21	Accel. Gradient (kPa/m)		1.017e-006	1.017e-006	1.017e-006	1.017e-006	
22	Liq. Reynolds		1.813e+005 *	1.813e+005 *	1.813e+005 *	1.813e+005 *	
23	Vap. Reynolds		9.919e+006 *	9.919e+006 *	9.919e+006 *	9.919e+006 *	
24	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
25	Vapour Velocity (m/s)		2.947	2.947	2.947	2.947	
26	Deposit Thickness (mm)		---	---	---	---	
27	Deposit Volume (m3)		---	---	---	---	
28	Length (m)		3.298e+004	3.298e+004	3.299e+004	3.299e+004	
29	Elevation (m)		0.1620	0.1640	0.1660	0.1680	
30	Pressure (bar)		67.03	67.03	67.03	67.03	
31	Temperature (C)		46.08	46.08	46.08	46.08	
32	Vapour Fraction		0.8801	0.8801	0.8801	0.8801	
33	Heat Transferred (kJ/h-m)		428.3	428.3	428.2	428.2	
34	Flow Regime		Transition	Transition	Transition	Transition	
35	Water Flow Regime						
36	Liquid HoldUp		0.2313	0.2313	0.2313	0.2313	
37	Water HoldUp		---	---	---	---	
38	Friction Gradient (kPa/m)		8.408e-003	8.408e-003	8.408e-003	8.408e-003	
39	Static Gradient (kPa/m)		7.770e-004	7.770e-004	7.770e-004	7.770e-004	
40	Accel. Gradient (kPa/m)		1.191e-006	1.191e-006	1.191e-006	1.191e-006	
41	Liq. Reynolds		1.813e+005 *	1.813e+005 *	1.813e+005 *	1.813e+005 *	
42	Vap. Reynolds		9.919e+006 *	9.919e+006 *	9.919e+006 *	9.919e+006 *	
43	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
44	Vapour Velocity (m/s)		2.947	2.947	2.947	2.947	
45	Deposit Thickness (mm)		---	---	---	---	
46	Deposit Volume (m3)		---	---	---	---	
47	Length (m)		3.300e+004	3.300e+004	3.301e+004	3.301e+004	
48	Elevation (m)		0.1700	0.1660	0.1620	0.1580	
49	Pressure (bar)		67.03	67.03	67.03	67.03	
50	Temperature (C)		46.07	46.07	46.07	46.07	
51	Vapour Fraction		0.8801	0.8801	0.8801	0.8801	
52	Heat Transferred (kJ/h-m)		428.2	428.2	428.1	428.1	
53	Flow Regime		Transition	Transition	Transition	Transition	
54	Water Flow Regime						
55	Liquid HoldUp		0.2313	0.1701	0.1701	0.1701	
56	Water HoldUp		---	---	---	---	
57	Friction Gradient (kPa/m)		8.408e-003	9.114e-003	9.114e-003	9.114e-003	
58	Static Gradient (kPa/m)		7.770e-004	-1.274e-003	-1.274e-003	-1.274e-003	
59	Accel. Gradient (kPa/m)		1.191e-006	1.017e-006	1.017e-006	1.017e-006	
60	Liq. Reynolds		1.813e+005 *	1.813e+005 *	1.813e+005 *	1.813e+005 *	
61	Vap. Reynolds		9.919e+006 *	9.919e+006 *	9.919e+006 *	9.919e+006 *	
62	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
63	Vapour Velocity (m/s)		2.947	2.947	2.947	2.947	
64	Deposit Thickness (mm)		---	---	---	---	
65	Deposit Volume (m3)		---	---	---	---	
66	Length (m)		3.302e+004	3.302e+004	3.302e+004	3.302e+004	
67	Elevation (m)		0.1540	0.1500	0.1500	0.1500	
68	Pressure (bar)		67.02	67.02	67.02	67.02	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	46.07	46.07	46.07	46.07
14	Vapour Fraction	0.8801	0.8801	0.8801	0.8801
15	Heat Transferred (kJ/h-m)	428.1	428.0	428.0	428.0
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.1701	0.1701	0.2491	0.2491
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	9.114e-003	9.114e-003	6.242e-003	6.242e-003
21	Static Gradient (kPa/m)	-1.274e-003	-1.274e-003	0.0000	0.0000
22	Accel. Gradient (kPa/m)	1.017e-006	1.017e-006	8.094e-007	8.094e-007
23	Liq. Reynolds	1.813e+005 *	1.813e+005 *	1.813e+005 *	1.813e+005 *
24	Vap. Reynolds	9.919e+006 *	9.919e+006 *	9.919e+006 *	9.919e+006 *
25	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
26	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	3.303e+004	3.303e+004	3.303e+004	3.304e+004
30	Elevation (m)	0.1500	0.1500	0.1500	0.1580
31	Pressure (bar)	67.02	67.02	67.02	67.02
32	Temperature (C)	46.07	46.06	46.06	46.06
33	Vapour Fraction	0.8801	0.8801	0.8801	0.8801
34	Heat Transferred (kJ/h-m)	427.9	427.9	427.9	427.9
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2491	0.2491	0.2491	0.2339
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	6.242e-003	6.242e-003	6.242e-003	8.062e-003
40	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	2.657e-003
41	Accel. Gradient (kPa/m)	8.094e-007	8.094e-007	8.094e-007	1.390e-006
42	Liq. Reynolds	1.813e+005 *	1.813e+005 *	1.813e+005 *	1.813e+005 *
43	Vap. Reynolds	9.919e+006 *	9.919e+006 *	9.919e+006 *	9.919e+006 *
44	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
45	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	3.304e+004	3.305e+004	3.305e+004	3.306e+004
49	Elevation (m)	0.1660	0.1740	0.1820	0.1900
50	Pressure (bar)	67.02	67.02	67.02	67.02
51	Temperature (C)	46.06	46.06	46.06	46.06
52	Vapour Fraction	0.8801	0.8801	0.8801	0.8801
53	Heat Transferred (kJ/h-m)	427.8	427.8	427.7	427.7
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2339	0.2339	0.2339	0.2339
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	8.062e-003	8.062e-003	8.062e-003	8.062e-003
59	Static Gradient (kPa/m)	2.657e-003	2.657e-003	2.657e-003	2.657e-003
60	Accel. Gradient (kPa/m)	1.390e-006	1.390e-006	1.390e-006	1.390e-006
61	Liq. Reynolds	1.813e+005 *	1.813e+005 *	1.813e+005 *	1.813e+005 *
62	Vap. Reynolds	9.919e+006 *	9.919e+006 *	9.919e+006 *	9.919e+006 *
63	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
64	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	3.306e+004	3.306e+004	3.307e+004	3.307e+004
68	Elevation (m)	0.1860	0.1820	0.1780	0.1740
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:13:43 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	67.02	67.02	67.02	67.02		
14	Temperature (C)	46.06	46.05	46.05	46.05		
15	Vapour Fraction	0.8801	0.8801	0.8801	0.8801		
16	Heat Transferred (kJ/h-m)	427.7	427.7	427.7	427.7		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.1692	0.1692	0.1692	0.1692		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	9.126e-003	9.126e-003	9.126e-003	9.126e-003		
22	Static Gradient (kPa/m)	-3.173e-003	-3.173e-003	-3.173e-003	-3.173e-003		
23	Accel. Gradient (kPa/m)	7.719e-007	7.719e-007	7.719e-007	7.719e-007		
24	Liq. Reynolds	1.813e+005 *	1.813e+005 *	1.813e+005 *	1.813e+005 *		
25	Vap. Reynolds	9.919e+006 *	9.920e+006 *	9.920e+006 *	9.920e+006 *		
26	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
27	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	3.307e+004	3.308e+004	3.308e+004	3.309e+004		
31	Elevation (m)	0.1700	0.1680	0.1660	0.1640		
32	Pressure (bar)	67.02	67.02	67.02	67.02		
33	Temperature (C)	46.05	46.05	46.05	46.05		
34	Vapour Fraction	0.8801	0.8801	0.8801	0.8801		
35	Heat Transferred (kJ/h-m)	427.7	427.6	427.6	427.6		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.1692	0.1704	0.1704	0.1704		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	9.126e-003	9.110e-003	9.110e-003	9.110e-003		
41	Static Gradient (kPa/m)	-3.173e-003	-6.376e-004	-6.376e-004	-6.376e-004		
42	Accel. Gradient (kPa/m)	7.719e-007	1.099e-006	1.099e-006	1.099e-006		
43	Liq. Reynolds	1.813e+005 *	1.813e+005 *	1.813e+005 *	1.813e+005 *		
44	Vap. Reynolds	9.920e+006 *	9.920e+006 *	9.920e+006 *	9.920e+006 *		
45	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
46	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	3.309e+004	3.310e+004	3.310e+004	3.311e+004		
50	Elevation (m)	0.1620	0.1600	0.1660	0.1720		
51	Pressure (bar)	67.02	67.02	67.02	67.02		
52	Temperature (C)	46.05	46.05	46.05	46.04		
53	Vapour Fraction	0.8801	0.8801	0.8801	0.8801		
54	Heat Transferred (kJ/h-m)	427.5	427.5	427.4	427.4		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.1704	0.1704	0.2335	0.2335		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	9.110e-003	9.110e-003	8.111e-003	8.111e-003		
60	Static Gradient (kPa/m)	-6.376e-004	-6.376e-004	2.389e-003	2.389e-003		
61	Accel. Gradient (kPa/m)	1.099e-006	1.099e-006	1.362e-006	1.362e-006		
62	Liq. Reynolds	1.813e+005 *	1.813e+005 *	1.813e+005 *	1.813e+005 *		
63	Vap. Reynolds	9.920e+006 *	9.920e+006 *	9.920e+006 *	9.920e+006 *		
64	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
65	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	3.311e+004	3.312e+004	3.312e+004	3.313e+004		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11					
12					
13	Pipe Table				
13	Elevation (m)	0.1780	0.1840	0.1900	0.1840
14	Pressure (bar)	67.02	67.02	67.02	67.01
15	Temperature (C)	46.04	46.04	46.04	46.04
16	Vapour Fraction	0.8801	0.8801	0.8801	0.8801
17	Heat Transferred (kJ/h-m)	427.3	427.3	427.3	427.3
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2335	0.2335	0.2335	0.1698
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	8.111e-003	8.111e-003	8.111e-003	9.118e-003
23	Static Gradient (kPa/m)	2.389e-003	2.389e-003	2.389e-003	-1.908e-003
24	Accel. Gradient (kPa/m)	1.362e-006	1.362e-006	1.362e-006	9.350e-007
25	Liq. Reynolds	1.813e+005 *	1.813e+005 *	1.813e+005 *	1.813e+005 *
26	Vap. Reynolds	9.920e+006 *	9.920e+006 *	9.920e+006 *	9.920e+006 *
27	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
28	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	3.313e+004	3.314e+004	3.314e+004	3.315e+004
32	Elevation (m)	0.1780	0.1720	0.1660	0.1600
33	Pressure (bar)	67.01	67.01	67.01	67.01
34	Temperature (C)	46.04	46.04	46.03	46.03
35	Vapour Fraction	0.8801	0.8801	0.8801	0.8801
36	Heat Transferred (kJ/h-m)	427.2	427.2	427.2	427.1
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.1698	0.1698	0.1698	0.1698
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	9.118e-003	9.118e-003	9.118e-003	9.118e-003
42	Static Gradient (kPa/m)	-1.908e-003	-1.908e-003	-1.908e-003	-1.908e-003
43	Accel. Gradient (kPa/m)	9.350e-007	9.350e-007	9.350e-007	9.350e-007
44	Liq. Reynolds	1.813e+005 *	1.813e+005 *	1.813e+005 *	1.813e+005 *
45	Vap. Reynolds	9.920e+006 *	9.920e+006 *	9.920e+006 *	9.920e+006 *
46	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
47	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	3.315e+004	3.316e+004	3.316e+004	3.317e+004
51	Elevation (m)	0.1560	0.1520	0.1480	0.1440
52	Pressure (bar)	67.01	67.01	67.01	67.01
53	Temperature (C)	46.03	46.03	46.03	46.03
54	Vapour Fraction	0.8801	0.8801	0.8801	0.8801
55	Heat Transferred (kJ/h-m)	427.1	427.0	427.0	427.0
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.1701	0.1701	0.1701	0.1701
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	9.114e-003	9.114e-003	9.114e-003	9.114e-003
61	Static Gradient (kPa/m)	-1.274e-003	-1.274e-003	-1.274e-003	-1.274e-003
62	Accel. Gradient (kPa/m)	1.017e-006	1.017e-006	1.017e-006	1.017e-006
63	Liq. Reynolds	1.813e+005 *	1.813e+005 *	1.813e+005 *	1.813e+005 *
64	Vap. Reynolds	9.920e+006 *	9.920e+006 *	9.920e+006 *	9.920e+006 *
65	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
66	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3						
4						
5			Unit Set: Mixed			
6			Date/Time: Mon Jun 13 15:13:43 2022			
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	PROFILES					
11	Pipe Table					
12						
13	Length (m)	3.317e+004	3.318e+004	3.318e+004	3.319e+004	
14	Elevation (m)	0.1400	0.1520	0.1640	0.1760	
15	Pressure (bar)	67.01	67.01	67.01	67.01	
16	Temperature (C)	46.03	46.02	46.02	46.02	
17	Vapour Fraction	0.8801	0.8801	0.8801	0.8801	
18	Heat Transferred (kJ/h-m)	426.9	426.9	426.8	426.8	
19	Flow Regime	Transition	Transition	Transition	Transition	
20	Water Flow Regime					
21	Liquid HoldUp	0.1701	0.2367	0.2367	0.2367	
22	Water HoldUp	---	---	---	---	
23	Friction Gradient (kPa/m)	9.114e-003	7.696e-003	7.696e-003	7.696e-003	
24	Static Gradient (kPa/m)	-1.274e-003	4.757e-003	4.757e-003	4.757e-003	
25	Accel. Gradient (kPa/m)	1.017e-006	1.615e-006	1.615e-006	1.615e-006	
26	Liq. Reynolds	1.813e+005 *	1.813e+005 *	1.813e+005 *	1.813e+005 *	
27	Vap. Reynolds	9.920e+006 *	9.920e+006 *	9.920e+006 *	9.920e+006 *	
28	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	
29	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947	
30	Deposit Thickness (mm)	---	---	---	---	
31	Deposit Volume (m3)	---	---	---	---	
32	Length (m)	3.319e+004	3.320e+004	3.320e+004	3.321e+004	
33	Elevation (m)	0.1880	0.2000	0.2000	0.2000	
34	Pressure (bar)	67.01	67.01	67.01	67.01	
35	Temperature (C)	46.02	46.02	46.02	46.02	
36	Vapour Fraction	0.8801	0.8801	0.8801	0.8801	
37	Heat Transferred (kJ/h-m)	426.7	426.7	426.7	426.6	
38	Flow Regime	Transition	Transition	Transition	Transition	
39	Water Flow Regime					
40	Liquid HoldUp	0.2367	0.2367	0.2491	0.2491	
41	Water HoldUp	---	---	---	---	
42	Friction Gradient (kPa/m)	7.696e-003	7.696e-003	6.243e-003	6.243e-003	
43	Static Gradient (kPa/m)	4.757e-003	4.757e-003	0.0000	0.0000	
44	Accel. Gradient (kPa/m)	1.615e-006	1.615e-006	8.097e-007	8.097e-007	
45	Liq. Reynolds	1.813e+005 *	1.813e+005 *	1.813e+005 *	1.813e+005 *	
46	Vap. Reynolds	9.920e+006 *	9.920e+006 *	9.920e+006 *	9.920e+006 *	
47	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	
48	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947	
49	Deposit Thickness (mm)	---	---	---	---	
50	Deposit Volume (m3)	---	---	---	---	
51	Length (m)	3.321e+004	3.322e+004	3.322e+004	3.323e+004	
52	Elevation (m)	0.2000	0.2000	0.2000	0.2000	
53	Pressure (bar)	67.01	67.01	67.01	67.01	
54	Temperature (C)	46.01	46.01	46.01	46.01	
55	Vapour Fraction	0.8801	0.8801	0.8801	0.8801	
56	Heat Transferred (kJ/h-m)	426.6	426.6	426.5	426.5	
57	Flow Regime	Transition	Transition	Transition	Transition	
58	Water Flow Regime					
59	Liquid HoldUp	0.2491	0.2491	0.2491	0.2491	
60	Water HoldUp	---	---	---	---	
61	Friction Gradient (kPa/m)	6.243e-003	6.243e-003	6.243e-003	6.243e-003	
62	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000	
63	Accel. Gradient (kPa/m)	8.097e-007	8.097e-007	8.097e-007	8.097e-007	
64	Liq. Reynolds	1.813e+005 *	1.813e+005 *	1.813e+005 *	1.813e+005 *	
65	Vap. Reynolds	9.920e+006 *	9.920e+006 *	9.920e+006 *	9.920e+006 *	
66	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	
67	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947	
68	Deposit Thickness (mm)	---	---	---	---	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	3.323e+004	3.324e+004	3.324e+004	3.325e+004		
15	Elevation (m)	0.2000	0.2000	0.2000	0.2000		
16	Pressure (bar)	67.01	67.01	67.01	67.01		
17	Temperature (C)	46.01	46.01	46.01	46.01		
18	Vapour Fraction	0.8801	0.8801	0.8801	0.8801		
19	Heat Transferred (kJ/h-m)	426.5	426.4	426.4	426.3		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	0.2491	0.2491	0.2491	0.2491		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	6.243e-003	6.243e-003	6.243e-003	6.243e-003		
25	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000		
26	Accel. Gradient (kPa/m)	8.097e-007	8.097e-007	8.097e-007	8.097e-007		
27	Liq. Reynolds	1.813e+005 *	1.813e+005 *	1.813e+005 *	1.813e+005 *		
28	Vap. Reynolds	9.920e+006 *	9.920e+006 *	9.920e+006 *	9.920e+006 *		
29	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
30	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	3.325e+004	3.326e+004	3.326e+004	3.327e+004		
34	Elevation (m)	0.1980	0.1960	0.1940	0.1920		
35	Pressure (bar)	67.00	67.00	67.00	67.00		
36	Temperature (C)	46.00	46.00	46.00	46.00		
37	Vapour Fraction	0.8801	0.8801	0.8801	0.8801		
38	Heat Transferred (kJ/h-m)	426.3	426.3	426.3	426.2		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.1704	0.1704	0.1704	0.1704		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	9.110e-003	9.110e-003	9.110e-003	9.110e-003		
44	Static Gradient (kPa/m)	-6.376e-004	-6.376e-004	-6.376e-004	-6.376e-004		
45	Accel. Gradient (kPa/m)	1.099e-006	1.099e-006	1.099e-006	1.099e-006		
46	Liq. Reynolds	1.813e+005 *	1.813e+005 *	1.813e+005 *	1.813e+005 *		
47	Vap. Reynolds	9.920e+006 *	9.920e+006 *	9.920e+006 *	9.920e+006 *		
48	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
49	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	3.327e+004	3.328e+004	3.328e+004	3.329e+004		
53	Elevation (m)	0.1900	0.1940	0.1980	0.2020		
54	Pressure (bar)	67.00	67.00	67.00	67.00		
55	Temperature (C)	46.00	46.00	46.00	45.99		
56	Vapour Fraction	0.8801	0.8801	0.8801	0.8801		
57	Heat Transferred (kJ/h-m)	426.2	426.1	426.1	426.1		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.1704	0.2324	0.2324	0.2324		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	9.110e-003	8.257e-003	8.257e-003	8.257e-003		
63	Static Gradient (kPa/m)	-6.376e-004	1.587e-003	1.587e-003	1.587e-003		
64	Accel. Gradient (kPa/m)	1.099e-006	1.277e-006	1.277e-006	1.277e-006		
65	Liq. Reynolds	1.813e+005 *	1.812e+005 *	1.812e+005 *	1.812e+005 *		
66	Vap. Reynolds	9.920e+006 *	9.920e+006 *	9.920e+006 *	9.920e+006 *		
67	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
68	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Pipe Segment: ELEP-Akabel Pipeline (continued)		
3					
4					
5			Unit Set: Mixed		
6			Date/Time: Mon Jun 13 15:13:43 2022		
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	3.329e+004	3.330e+004	3.330e+004	3.331e+004
16	Elevation (m)	0.2060	0.2100	0.2080	0.2060
17	Pressure (bar)	67.00	67.00	67.00	67.00
18	Temperature (C)	45.99	45.99	45.99	45.99
19	Vapour Fraction	0.8801	0.8801	0.8801	0.8801
20	Heat Transferred (kJ/h-m)	426.0	426.0	426.0	425.9
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2324	0.2324	0.1704	0.1704
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	8.257e-003	8.257e-003	9.110e-003	9.110e-003
26	Static Gradient (kPa/m)	1.587e-003	1.587e-003	-6.376e-004	-6.376e-004
27	Accel. Gradient (kPa/m)	1.277e-006	1.277e-006	1.099e-006	1.099e-006
28	Liq. Reynolds	1.812e+005 *	1.812e+005 *	1.812e+005 *	1.812e+005 *
29	Vap. Reynolds	9.920e+006 *	9.920e+006 *	9.920e+006 *	9.920e+006 *
30	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
31	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	3.331e+004	3.332e+004	3.332e+004	3.333e+004
35	Elevation (m)	0.2040	0.2020	0.2000	0.2000
36	Pressure (bar)	67.00	67.00	67.00	67.00
37	Temperature (C)	45.99	45.99	45.98	45.98
38	Vapour Fraction	0.8801	0.8801	0.8801	0.8801
39	Heat Transferred (kJ/h-m)	425.9	425.9	425.8	425.8
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.1704	0.1704	0.1704	0.2491
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	9.110e-003	9.110e-003	9.110e-003	6.243e-003
45	Static Gradient (kPa/m)	-6.376e-004	-6.376e-004	-6.376e-004	0.0000
46	Accel. Gradient (kPa/m)	1.099e-006	1.099e-006	1.099e-006	8.098e-007
47	Liq. Reynolds	1.812e+005 *	1.812e+005 *	1.812e+005 *	1.812e+005 *
48	Vap. Reynolds	9.921e+006 *	9.921e+006 *	9.921e+006 *	9.921e+006 *
49	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
50	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	3.333e+004	3.334e+004	3.334e+004	3.335e+004
54	Elevation (m)	0.2000	0.2000	0.2000	0.2000
55	Pressure (bar)	67.00	67.00	67.00	67.00
56	Temperature (C)	45.98	45.98	45.98	45.98
57	Vapour Fraction	0.8801	0.8801	0.8801	0.8801
58	Heat Transferred (kJ/h-m)	425.7	425.7	425.7	425.6
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2491	0.2491	0.2491	0.2491
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	6.243e-003	6.243e-003	6.243e-003	6.243e-003
64	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
65	Accel. Gradient (kPa/m)	8.098e-007	8.098e-007	8.098e-007	8.098e-007
66	Liq. Reynolds	1.812e+005 *	1.812e+005 *	1.812e+005 *	1.812e+005 *
67	Vap. Reynolds	9.921e+006 *	9.921e+006 *	9.921e+006 *	9.921e+006 *
68	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	3.335e+004	3.336e+004	3.336e+004	3.337e+004	
17	Elevation	(m)	0.2060	0.2120	0.2180	0.2240	
18	Pressure	(bar)	67.00	67.00	67.00	66.99	
19	Temperature	(C)	45.98	45.98	45.97	45.97	
20	Vapour Fraction		0.8801	0.8801	0.8801	0.8801	
21	Heat Transferred	(kJ/h-m)	425.6	425.5	425.5	425.5	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2335	0.2335	0.2335	0.2335	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	8.111e-003	8.111e-003	8.111e-003	8.111e-003	
27	Static Gradient	(kPa/m)	2.389e-003	2.389e-003	2.389e-003	2.389e-003	
28	Accel. Gradient	(kPa/m)	1.362e-006	1.362e-006	1.362e-006	1.362e-006	
29	Liq. Reynolds		1.812e+005 *	1.812e+005 *	1.812e+005 *	1.812e+005 *	
30	Vap. Reynolds		9.921e+006 *	9.921e+006 *	9.921e+006 *	9.921e+006 *	
31	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
32	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	3.337e+004	3.338e+004	3.338e+004	3.339e+004	
36	Elevation	(m)	0.2300	0.2260	0.2220	0.2180	
37	Pressure	(bar)	66.99	66.99	66.99	66.99	
38	Temperature	(C)	45.97	45.97	45.97	45.97	
39	Vapour Fraction		0.8801	0.8801	0.8801	0.8801	
40	Heat Transferred	(kJ/h-m)	425.4	425.4	425.4	425.4	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2335	0.1701	0.1701	0.1701	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	8.111e-003	9.114e-003	9.114e-003	9.114e-003	
46	Static Gradient	(kPa/m)	2.389e-003	-1.257e-003	-1.257e-003	-1.257e-003	
47	Accel. Gradient	(kPa/m)	1.362e-006	1.019e-006	1.019e-006	1.019e-006	
48	Liq. Reynolds		1.812e+005 *	1.812e+005 *	1.812e+005 *	1.812e+005 *	
49	Vap. Reynolds		9.921e+006 *	9.921e+006 *	9.921e+006 *	9.921e+006 *	
50	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
51	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	3.339e+004	3.340e+004	3.340e+004	3.341e+004	
55	Elevation	(m)	0.2140	0.2100	0.2080	0.2060	
56	Pressure	(bar)	66.99	66.99	66.99	66.99	
57	Temperature	(C)	45.97	45.96	45.96	45.96	
58	Vapour Fraction		0.8801	0.8801	0.8801	0.8801	
59	Heat Transferred	(kJ/h-m)	425.3	425.3	425.3	425.2	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.1701	0.1701	0.1704	0.1704	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	9.114e-003	9.114e-003	9.110e-003	9.110e-003	
65	Static Gradient	(kPa/m)	-1.257e-003	-1.257e-003	-6.376e-004	-6.376e-004	
66	Accel. Gradient	(kPa/m)	1.019e-006	1.019e-006	1.099e-006	1.099e-006	
67	Liq. Reynolds		1.812e+005 *	1.812e+005 *	1.812e+005 *	1.812e+005 *	
68	Vap. Reynolds		9.921e+006 *	9.921e+006 *	9.921e+006 *	9.921e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
14	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	3.341e+004	3.342e+004	3.342e+004	3.343e+004	
18	Elevation	(m)	0.2040	0.2020	0.2000	0.2040	
19	Pressure	(bar)	66.99	66.99	66.99	66.99	
20	Temperature	(C)	45.96	45.96	45.96	45.96	
21	Vapour Fraction		0.8801	0.8801	0.8801	0.8801	
22	Heat Transferred	(kJ/h-m)	425.2	425.1	425.1	425.0	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.1704	0.1704	0.1704	0.2324	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	9.110e-003	9.110e-003	9.110e-003	8.257e-003	
28	Static Gradient	(kPa/m)	-6.376e-004	-6.376e-004	-6.376e-004	1.587e-003	
29	Accel. Gradient	(kPa/m)	1.099e-006	1.099e-006	1.099e-006	1.277e-006	
30	Liq. Reynolds		1.812e+005 *	1.812e+005 *	1.812e+005 *	1.812e+005 *	
31	Vap. Reynolds		9.921e+006 *	9.921e+006 *	9.921e+006 *	9.921e+006 *	
32	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
33	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	3.343e+004	3.344e+004	3.344e+004	3.345e+004	
37	Elevation	(m)	0.2080	0.2120	0.2160	0.2200	
38	Pressure	(bar)	66.99	66.99	66.99	66.99	
39	Temperature	(C)	45.95	45.95	45.95	45.95	
40	Vapour Fraction		0.8801	0.8801	0.8801	0.8801	
41	Heat Transferred	(kJ/h-m)	425.0	425.0	424.9	424.9	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2324	0.2324	0.2324	0.2324	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	8.257e-003	8.257e-003	8.257e-003	8.258e-003	
47	Static Gradient	(kPa/m)	1.587e-003	1.587e-003	1.587e-003	1.587e-003	
48	Accel. Gradient	(kPa/m)	1.277e-006	1.277e-006	1.277e-006	1.277e-006	
49	Liq. Reynolds		1.812e+005 *	1.812e+005 *	1.812e+005 *	1.812e+005 *	
50	Vap. Reynolds		9.921e+006 *	9.921e+006 *	9.921e+006 *	9.921e+006 *	
51	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
52	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	3.345e+004	3.346e+004	3.346e+004	3.347e+004	
56	Elevation	(m)	0.2180	0.2160	0.2140	0.2120	
57	Pressure	(bar)	66.99	66.99	66.99	66.99	
58	Temperature	(C)	45.95	45.95	45.95	45.95	
59	Vapour Fraction		0.8801	0.8801	0.8801	0.8801	
60	Heat Transferred	(kJ/h-m)	424.9	424.9	424.8	424.8	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.1704	0.1704	0.1704	0.1704	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	9.111e-003	9.111e-003	9.111e-003	9.111e-003	
66	Static Gradient	(kPa/m)	-6.376e-004	-6.376e-004	-6.376e-004	-6.376e-004	
67	Accel. Gradient	(kPa/m)	1.099e-006	1.099e-006	1.099e-006	1.099e-006	
68	Liq. Reynolds		1.812e+005 *	1.812e+005 *	1.812e+005 *	1.812e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vap. Reynolds		9.921e+006 *	9.921e+006 *	9.921e+006 *	9.921e+006 *	
14	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
15	Vapour Velocity (m/s)		2.947	2.947	2.947	2.947	
16	Deposit Thickness (mm)		---	---	---	---	
17	Deposit Volume (m3)		---	---	---	---	
18	Length (m)		3.347e+004	3.348e+004	3.348e+004	3.349e+004	
19	Elevation (m)		0.2100	0.2100	0.2100	0.2100	
20	Pressure (bar)		66.99	66.99	66.98	66.98	
21	Temperature (C)		45.94	45.94	45.94	45.94	
22	Vapour Fraction		0.8801	0.8801	0.8801	0.8801	
23	Heat Transferred (kJ/h-m)		424.7	424.7	424.6	424.6	
24	Flow Regime		Transition	Transition	Transition	Transition	
25	Water Flow Regime						
26	Liquid HoldUp		0.1704	0.2491	0.2491	0.2491	
27	Water HoldUp		---	---	---	---	
28	Friction Gradient (kPa/m)		9.111e-003	6.243e-003	6.243e-003	6.243e-003	
29	Static Gradient (kPa/m)		-6.376e-004	0.0000	0.0000	0.0000	
30	Accel. Gradient (kPa/m)		1.099e-006	8.100e-007	8.100e-007	8.100e-007	
31	Liq. Reynolds		1.812e+005 *	1.812e+005 *	1.812e+005 *	1.812e+005 *	
32	Vap. Reynolds		9.921e+006 *	9.921e+006 *	9.921e+006 *	9.921e+006 *	
33	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
34	Vapour Velocity (m/s)		2.947	2.947	2.947	2.947	
35	Deposit Thickness (mm)		---	---	---	---	
36	Deposit Volume (m3)		---	---	---	---	
37	Length (m)		3.349e+004	3.350e+004	3.350e+004	3.351e+004	
38	Elevation (m)		0.2100	0.2100	0.2120	0.2140	
39	Pressure (bar)		66.98	66.98	66.98	66.98	
40	Temperature (C)		45.94	45.94	45.94	45.93	
41	Vapour Fraction		0.8801	0.8801	0.8801	0.8801	
42	Heat Transferred (kJ/h-m)		424.6	424.5	424.5	424.5	
43	Flow Regime		Transition	Transition	Transition	Transition	
44	Water Flow Regime						
45	Liquid HoldUp		0.2491	0.2491	0.2313	0.2313	
46	Water HoldUp		---	---	---	---	
47	Friction Gradient (kPa/m)		6.243e-003	6.243e-003	8.406e-003	8.406e-003	
48	Static Gradient (kPa/m)		0.0000	0.0000	7.907e-004	7.907e-004	
49	Accel. Gradient (kPa/m)		8.100e-007	8.100e-007	1.193e-006	1.193e-006	
50	Liq. Reynolds		1.812e+005 *	1.812e+005 *	1.812e+005 *	1.812e+005 *	
51	Vap. Reynolds		9.921e+006 *	9.921e+006 *	9.921e+006 *	9.921e+006 *	
52	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
53	Vapour Velocity (m/s)		2.947	2.947	2.947	2.947	
54	Deposit Thickness (mm)		---	---	---	---	
55	Deposit Volume (m3)		---	---	---	---	
56	Length (m)		3.351e+004	3.352e+004	3.352e+004	3.353e+004	
57	Elevation (m)		0.2160	0.2180	0.2200	0.2140	
58	Pressure (bar)		66.98	66.98	66.98	66.98	
59	Temperature (C)		45.93	45.93	45.93	45.93	
60	Vapour Fraction		0.8801	0.8801	0.8801	0.8801	
61	Heat Transferred (kJ/h-m)		424.4	424.4	424.4	424.4	
62	Flow Regime		Transition	Transition	Transition	Transition	
63	Water Flow Regime						
64	Liquid HoldUp		0.2313	0.2313	0.2313	0.1698	
65	Water HoldUp		---	---	---	---	
66	Friction Gradient (kPa/m)		8.406e-003	8.406e-003	8.406e-003	9.118e-003	
67	Static Gradient (kPa/m)		7.908e-004	7.908e-004	7.908e-004	-1.908e-003	
68	Accel. Gradient (kPa/m)		1.193e-006	1.193e-006	1.193e-006	9.355e-007	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	1.812e+005 *	1.812e+005 *	1.812e+005 *	1.812e+005 *
14	Vap. Reynolds	9.921e+006 *	9.921e+006 *	9.921e+006 *	9.921e+006 *
15	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
16	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	3.353e+004	3.354e+004	3.354e+004	3.355e+004
20	Elevation (m)	0.2080	0.2020	0.1960	0.1900
21	Pressure (bar)	66.98	66.98	66.98	66.98
22	Temperature (C)	45.93	45.93	45.93	45.92
23	Vapour Fraction	0.8801	0.8801	0.8801	0.8801
24	Heat Transferred (kJ/h-m)	424.3	424.3	424.2	424.2
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.1698	0.1698	0.1698	0.1698
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	9.118e-003	9.118e-003	9.118e-003	9.118e-003
30	Static Gradient (kPa/m)	-1.908e-003	-1.908e-003	-1.908e-003	-1.908e-003
31	Accel. Gradient (kPa/m)	9.355e-007	9.355e-007	9.355e-007	9.355e-007
32	Liq. Reynolds	1.812e+005 *	1.812e+005 *	1.812e+005 *	1.812e+005 *
33	Vap. Reynolds	9.921e+006 *	9.921e+006 *	9.921e+006 *	9.921e+006 *
34	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
35	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	3.355e+004	3.356e+004	3.356e+004	3.357e+004
39	Elevation (m)	0.1940	0.1980	0.2020	0.2060
40	Pressure (bar)	66.98	66.98	66.98	66.98
41	Temperature (C)	45.92	45.92	45.92	45.92
42	Vapour Fraction	0.8801	0.8801	0.8801	0.8801
43	Heat Transferred (kJ/h-m)	424.1	424.1	424.1	424.0
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2324	0.2324	0.2324	0.2324
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	8.258e-003	8.258e-003	8.258e-003	8.258e-003
49	Static Gradient (kPa/m)	1.587e-003	1.587e-003	1.587e-003	1.587e-003
50	Accel. Gradient (kPa/m)	1.277e-006	1.277e-006	1.277e-006	1.277e-006
51	Liq. Reynolds	1.812e+005 *	1.812e+005 *	1.812e+005 *	1.812e+005 *
52	Vap. Reynolds	9.921e+006 *	9.921e+006 *	9.922e+006 *	9.922e+006 *
53	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
54	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	3.357e+004	3.358e+004	3.358e+004	3.359e+004
58	Elevation (m)	0.2100	0.2000	0.1900	0.1800
59	Pressure (bar)	66.98	66.98	66.98	66.98
60	Temperature (C)	45.92	45.92	45.91	45.91
61	Vapour Fraction	0.8801	0.8801	0.8801	0.8801
62	Heat Transferred (kJ/h-m)	424.0	424.0	424.0	423.9
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2324	0.1692	0.1692	0.1692
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	8.258e-003	9.126e-003	9.126e-003	9.126e-003
68	Static Gradient (kPa/m)	1.587e-003	-3.121e-003	-3.121e-003	-3.121e-003
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 536 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Accel. Gradient	(kPa/m)	1.277e-006	7.792e-007	7.792e-007	7.792e-007				
14	Liq. Reynolds		1.812e+005 *	1.812e+005 *	1.812e+005 *	1.812e+005 *				
15	Vap. Reynolds		9.922e+006 *	9.922e+006 *	9.922e+006 *	9.922e+006 *				
16	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
17	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947				
18	Deposit Thickness	(mm)	---	---	---	---				
19	Deposit Volume	(m3)	---	---	---	---				
20	Length	(m)	3.359e+004	3.360e+004	3.360e+004	3.361e+004				
21	Elevation	(m)	0.1700	0.1600	0.1860	0.2120				
22	Pressure	(bar)	66.98	66.98	66.98	66.97				
23	Temperature	(C)	45.91	45.91	45.91	45.91				
24	Vapour Fraction		0.8801	0.8801	0.8801	0.8801				
25	Heat Transferred	(kJ/h-m)	423.9	423.9	423.8	423.7				
26	Flow Regime		Transition	Transition	Transition	Transition				
27	Water Flow Regime									
28	Liquid HoldUp		0.1692	0.1692	0.2445	0.2445				
29	Water HoldUp		---	---	---	---				
30	Friction Gradient	(kPa/m)	9.126e-003	9.126e-003	6.755e-003	6.755e-003				
31	Static Gradient	(kPa/m)	-3.121e-003	-3.121e-003	1.071e-002	1.071e-002				
32	Accel. Gradient	(kPa/m)	7.792e-007	7.792e-007	2.267e-006	2.267e-006				
33	Liq. Reynolds		1.812e+005 *	1.812e+005 *	1.811e+005 *	1.811e+005 *				
34	Vap. Reynolds		9.922e+006 *	9.922e+006 *	9.922e+006 *	9.922e+006 *				
35	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
36	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947				
37	Deposit Thickness	(mm)	---	---	---	---				
38	Deposit Volume	(m3)	---	---	---	---				
39	Length	(m)	3.361e+004	3.362e+004	3.362e+004	3.363e+004				
40	Elevation	(m)	0.2380	0.2640	0.2900	0.2980				
41	Pressure	(bar)	66.97	66.97	66.97	66.97				
42	Temperature	(C)	45.91	45.90	45.90	45.90				
43	Vapour Fraction		0.8801	0.8801	0.8801	0.8801				
44	Heat Transferred	(kJ/h-m)	423.7	423.6	423.6	423.6				
45	Flow Regime		Transition	Transition	Transition	Transition				
46	Water Flow Regime									
47	Liquid HoldUp		0.2445	0.2445	0.2445	0.2346				
48	Water HoldUp		---	---	---	---				
49	Friction Gradient	(kPa/m)	6.755e-003	6.755e-003	6.755e-003	7.967e-003				
50	Static Gradient	(kPa/m)	1.071e-002	1.071e-002	1.071e-002	3.196e-003				
51	Accel. Gradient	(kPa/m)	2.267e-006	2.267e-006	2.267e-006	1.449e-006				
52	Liq. Reynolds		1.811e+005 *	1.811e+005 *	1.811e+005 *	1.811e+005 *				
53	Vap. Reynolds		9.922e+006 *	9.922e+006 *	9.922e+006 *	9.922e+006 *				
54	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
55	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947				
56	Deposit Thickness	(mm)	---	---	---	---				
57	Deposit Volume	(m3)	---	---	---	---				
58	Length	(m)	3.363e+004	3.364e+004	3.364e+004	3.365e+004				
59	Elevation	(m)	0.3060	0.3140	0.3220	0.3300				
60	Pressure	(bar)	66.97	66.97	66.97	66.97				
61	Temperature	(C)	45.90	45.90	45.90	45.90				
62	Vapour Fraction		0.8801	0.8801	0.8801	0.8801				
63	Heat Transferred	(kJ/h-m)	423.5	423.5	423.5	423.4				
64	Flow Regime		Transition	Transition	Transition	Transition				
65	Water Flow Regime									
66	Liquid HoldUp		0.2346	0.2346	0.2346	0.2346				
67	Water HoldUp		---	---	---	---				
68	Friction Gradient	(kPa/m)	7.967e-003	7.967e-003	7.967e-003	7.967e-003				
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1				Case Name:	Simulation - 386MMSCFD_80barg.hsc	
2				Unit Set:	Mixed	
3	 Epic Atlantic Limited			Date/Time:	Mon Jun 13 15:13:43 2022	
4						
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	3.196e-003	3.196e-003	3.196e-003	3.196e-003
14	Accel. Gradient	(kPa/m)	1.449e-006	1.449e-006	1.449e-006	1.449e-006
15	Liq. Reynolds		1.811e+005 *	1.811e+005 *	1.811e+005 *	1.811e+005 *
16	Vap. Reynolds		9.922e+006 *	9.922e+006 *	9.922e+006 *	9.922e+006 *
17	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
18	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	3.365e+004	3.366e+004	3.366e+004	3.367e+004
22	Elevation	(m)	0.3460	0.3620	0.3780	0.3940
23	Pressure	(bar)	66.97	66.97	66.97	66.97
24	Temperature	(C)	45.89	45.89	45.89	45.89
25	Vapour Fraction		0.8801	0.8801	0.8801	0.8801
26	Heat Transferred	(kJ/h-m)	423.4	423.3	423.3	423.3
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2390	0.2390	0.2390	0.2390
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	7.410e-003	7.410e-003	7.410e-003	7.410e-003
32	Static Gradient	(kPa/m)	6.481e-003	6.481e-003	6.481e-003	6.481e-003
33	Accel. Gradient	(kPa/m)	1.803e-006	1.803e-006	1.803e-006	1.803e-006
34	Liq. Reynolds		1.811e+005 *	1.811e+005 *	1.811e+005 *	1.811e+005 *
35	Vap. Reynolds		9.922e+006 *	9.922e+006 *	9.922e+006 *	9.922e+006 *
36	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
37	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	3.367e+004	3.368e+004	3.368e+004	3.369e+004
41	Elevation	(m)	0.4100	0.3720	0.3340	0.2960
42	Pressure	(bar)	66.97	66.97	66.97	66.97
43	Temperature	(C)	45.89	45.89	45.89	45.88
44	Vapour Fraction		0.8801	0.8801	0.8801	0.8801
45	Heat Transferred	(kJ/h-m)	423.2	423.2	423.2	423.2
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2390	0.1651	0.1651	0.1651
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	7.410e-003	9.185e-003	9.185e-003	9.185e-003
51	Static Gradient	(kPa/m)	6.481e-003	-1.186e-002	-1.186e-002	-1.186e-002
52	Accel. Gradient	(kPa/m)	1.803e-006	-3.474e-007	-3.474e-007	-3.474e-007
53	Liq. Reynolds		1.811e+005 *	1.811e+005 *	1.811e+005 *	1.811e+005 *
54	Vap. Reynolds		9.922e+006 *	9.922e+006 *	9.922e+006 *	9.922e+006 *
55	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
56	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	3.369e+004	3.370e+004	3.370e+004	3.371e+004
60	Elevation	(m)	0.2580	0.2200	0.2300	0.2400
61	Pressure	(bar)	66.97	66.97	66.97	66.96
62	Temperature	(C)	45.88	45.88	45.88	45.88
63	Vapour Fraction		0.8801	0.8801	0.8801	0.8801
64	Heat Transferred	(kJ/h-m)	423.2	423.1	423.0	423.0
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.1651	0.1651	0.2357	0.2357
68	Water HoldUp		---	---	---	---
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1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	9.185e-003	9.185e-003	7.825e-003	7.825e-003
14	Static Gradient	(kPa/m)	-1.186e-002	-1.186e-002	4.009e-003	4.009e-003
15	Accel. Gradient	(kPa/m)	-3.474e-007	-3.474e-007	1.536e-006	1.536e-006
16	Liq. Reynolds		1.811e+005 *	1.811e+005 *	1.811e+005 *	1.811e+005 *
17	Vap. Reynolds		9.922e+006 *	9.922e+006 *	9.922e+006 *	9.922e+006 *
18	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
19	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	3.371e+004	3.372e+004	3.372e+004	3.373e+004
23	Elevation	(m)	0.2500	0.2600	0.2700	0.2800
24	Pressure	(bar)	66.96	66.96	66.96	66.96
25	Temperature	(C)	45.88	45.88	45.88	45.87
26	Vapour Fraction		0.8801	0.8801	0.8801	0.8801
27	Heat Transferred	(kJ/h-m)	423.0	422.9	422.9	422.9
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2357	0.2357	0.2357	0.2357
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	7.825e-003	7.825e-003	7.825e-003	7.825e-003
33	Static Gradient	(kPa/m)	4.009e-003	4.009e-003	4.009e-003	4.009e-003
34	Accel. Gradient	(kPa/m)	1.536e-006	1.536e-006	1.536e-006	1.536e-006
35	Liq. Reynolds		1.811e+005 *	1.811e+005 *	1.811e+005 *	1.811e+005 *
36	Vap. Reynolds		9.922e+006 *	9.922e+006 *	9.922e+006 *	9.922e+006 *
37	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
38	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	3.373e+004	3.374e+004	3.374e+004	3.375e+004
42	Elevation	(m)	0.2900	0.3000	0.3100	0.3200
43	Pressure	(bar)	66.96	66.96	66.96	66.96
44	Temperature	(C)	45.87	45.87	45.87	45.87
45	Vapour Fraction		0.8801	0.8801	0.8801	0.8801
46	Heat Transferred	(kJ/h-m)	422.8	422.8	422.7	422.7
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2357	0.2357	0.2357	0.2357
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	7.825e-003	7.825e-003	7.825e-003	7.825e-003
52	Static Gradient	(kPa/m)	4.009e-003	4.009e-003	4.009e-003	4.009e-003
53	Accel. Gradient	(kPa/m)	1.536e-006	1.536e-006	1.536e-006	1.536e-006
54	Liq. Reynolds		1.811e+005 *	1.811e+005 *	1.811e+005 *	1.811e+005 *
55	Vap. Reynolds		9.922e+006 *	9.922e+006 *	9.922e+006 *	9.922e+006 *
56	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
57	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	3.375e+004	3.376e+004	3.376e+004	3.377e+004
61	Elevation	(m)	0.3280	0.3360	0.3440	0.3520
62	Pressure	(bar)	66.96	66.96	66.96	66.96
63	Temperature	(C)	45.87	45.87	45.86	45.86
64	Vapour Fraction		0.8801	0.8801	0.8801	0.8801
65	Heat Transferred	(kJ/h-m)	422.7	422.6	422.6	422.5
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2346	0.2346	0.2346	0.2346
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 539 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	7.968e-003	7.968e-003	7.968e-003	7.968e-003
15	Static Gradient (kPa/m)	3.196e-003	3.196e-003	3.196e-003	3.196e-003
16	Accel. Gradient (kPa/m)	1.449e-006	1.449e-006	1.449e-006	1.449e-006
17	Liq. Reynolds	1.811e+005 *	1.811e+005 *	1.811e+005 *	1.811e+005 *
18	Vap. Reynolds	9.922e+006 *	9.922e+006 *	9.922e+006 *	9.922e+006 *
19	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
20	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	3.377e+004	3.378e+004	3.378e+004	3.379e+004
24	Elevation (m)	0.3600	0.3220	0.2840	0.2460
25	Pressure (bar)	66.96	66.96	66.96	66.96
26	Temperature (C)	45.86	45.86	45.86	45.86
27	Vapour Fraction	0.8801	0.8800	0.8800	0.8800
28	Heat Transferred (kJ/h-m)	422.5	422.5	422.5	422.5
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2346	0.1651	0.1651	0.1651
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	7.968e-003	9.185e-003	9.185e-003	9.184e-003
34	Static Gradient (kPa/m)	3.196e-003	-1.174e-002	-1.174e-002	-1.174e-002
35	Accel. Gradient (kPa/m)	1.449e-006	-3.321e-007	-3.321e-007	-3.321e-007
36	Liq. Reynolds	1.811e+005 *	1.811e+005 *	1.811e+005 *	1.811e+005 *
37	Vap. Reynolds	9.923e+006 *	9.923e+006 *	9.923e+006 *	9.923e+006 *
38	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
39	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	3.379e+004	3.380e+004	3.380e+004	3.381e+004
43	Elevation (m)	0.2080	0.1700	0.1760	0.1820
44	Pressure (bar)	66.96	66.96	66.96	66.96
45	Temperature (C)	45.86	45.86	45.85	45.85
46	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
47	Heat Transferred (kJ/h-m)	422.4	422.4	422.3	422.3
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.1651	0.1651	0.2335	0.2335
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	9.184e-003	9.184e-003	8.112e-003	8.112e-003
53	Static Gradient (kPa/m)	-1.174e-002	-1.174e-002	2.389e-003	2.389e-003
54	Accel. Gradient (kPa/m)	-3.321e-007	-3.321e-007	1.363e-006	1.363e-006
55	Liq. Reynolds	1.811e+005 *	1.811e+005 *	1.811e+005 *	1.811e+005 *
56	Vap. Reynolds	9.923e+006 *	9.923e+006 *	9.923e+006 *	9.923e+006 *
57	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
58	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	3.381e+004	3.382e+004	3.382e+004	3.383e+004
62	Elevation (m)	0.1880	0.1940	0.2000	0.2100
63	Pressure (bar)	66.96	66.96	66.96	66.95
64	Temperature (C)	45.85	45.85	45.85	45.85
65	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
66	Heat Transferred (kJ/h-m)	422.3	422.2	422.2	422.1
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.2335	0.2335	0.2335	0.2357
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	8.112e-003	8.112e-003	8.112e-003	7.825e-003
16	Static Gradient (kPa/m)	2.389e-003	2.389e-003	2.389e-003	4.009e-003
17	Accel. Gradient (kPa/m)	1.363e-006	1.363e-006	1.363e-006	1.536e-006
18	Liq. Reynolds	1.811e+005 *	1.811e+005 *	1.811e+005 *	1.811e+005 *
19	Vap. Reynolds	9.923e+006 *	9.923e+006 *	9.923e+006 *	9.923e+006 *
20	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
21	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	3.383e+004	3.384e+004	3.384e+004	3.385e+004
25	Elevation (m)	0.2200	0.2300	0.2400	0.2500
26	Pressure (bar)	66.95	66.95	66.95	66.95
27	Temperature (C)	45.85	45.84	45.84	45.84
28	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
29	Heat Transferred (kJ/h-m)	422.1	422.1	422.0	422.0
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2357	0.2357	0.2357	0.2357
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	7.825e-003	7.825e-003	7.825e-003	7.825e-003
35	Static Gradient (kPa/m)	4.009e-003	4.009e-003	4.009e-003	4.009e-003
36	Accel. Gradient (kPa/m)	1.536e-006	1.536e-006	1.536e-006	1.536e-006
37	Liq. Reynolds	1.811e+005 *	1.811e+005 *	1.811e+005 *	1.811e+005 *
38	Vap. Reynolds	9.923e+006 *	9.923e+006 *	9.923e+006 *	9.923e+006 *
39	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
40	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	3.385e+004	3.385e+004	3.385e+004	3.386e+004
44	Elevation (m)	0.2600	0.2700	0.2800	0.2900
45	Pressure (bar)	66.95	66.95	66.95	66.95
46	Temperature (C)	45.84	45.84	45.84	45.84
47	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
48	Heat Transferred (kJ/h-m)	422.0	422.0	421.9	421.9
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2440	0.2440	0.2440	0.2440
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.819e-003	6.819e-003	6.819e-003	6.819e-003
54	Static Gradient (kPa/m)	1.028e-002	1.028e-002	1.028e-002	1.028e-002
55	Accel. Gradient (kPa/m)	2.220e-006	2.220e-006	2.220e-006	2.220e-006
56	Liq. Reynolds	1.811e+005 *	1.811e+005 *	1.811e+005 *	1.811e+005 *
57	Vap. Reynolds	9.923e+006 *	9.923e+006 *	9.923e+006 *	9.923e+006 *
58	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
59	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	3.386e+004	3.386e+004	3.386e+004	3.386e+004
63	Elevation (m)	0.3000	0.3000	0.3000	0.3000
64	Pressure (bar)	66.95	66.95	66.95	66.95
65	Temperature (C)	45.84	45.84	45.84	45.84
66	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
67	Heat Transferred (kJ/h-m)	421.9	421.9	421.9	421.9
68	Flow Regime	Transition	Transition	Transition	Transition
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
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5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2440	0.2491	0.2491	0.2491
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.819e-003	6.244e-003	6.244e-003	6.244e-003
17	Static Gradient (kPa/m)	1.028e-002	0.0000	0.0000	0.0000
18	Accel. Gradient (kPa/m)	2.220e-006	8.105e-007	8.105e-007	8.105e-007
19	Liq. Reynolds	1.811e+005 *	1.811e+005 *	1.811e+005 *	1.811e+005 *
20	Vap. Reynolds	9.923e+006 *	9.923e+006 *	9.923e+006 *	9.923e+006 *
21	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
22	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	3.387e+004	3.387e+004	3.387e+004	3.387e+004
26	Elevation (m)	0.3000	0.3000	0.3000	0.3000
27	Pressure (bar)	66.95	66.95	66.95	66.95
28	Temperature (C)	45.84	45.84	45.84	45.83
29	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
30	Heat Transferred (kJ/h-m)	421.9	421.8	421.8	421.8
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2491	0.2491	0.2491	0.2491
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	6.244e-003	6.244e-003	6.244e-003	6.244e-003
36	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
37	Accel. Gradient (kPa/m)	8.105e-007	8.105e-007	8.105e-007	8.105e-007
38	Liq. Reynolds	1.811e+005 *	1.811e+005 *	1.811e+005 *	1.811e+005 *
39	Vap. Reynolds	9.923e+006 *	9.923e+006 *	9.923e+006 *	9.923e+006 *
40	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
41	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	3.388e+004	3.388e+004	3.388e+004	3.389e+004
45	Elevation (m)	0.3000	0.3000	0.3000	0.3020
46	Pressure (bar)	66.95	66.95	66.95	66.95
47	Temperature (C)	45.83	45.83	45.83	45.83
48	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
49	Heat Transferred (kJ/h-m)	421.8	421.8	421.7	421.7
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2491	0.2491	0.2491	0.2320
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	6.244e-003	6.244e-003	6.244e-003	8.308e-003
55	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	1.321e-003
56	Accel. Gradient (kPa/m)	8.105e-007	8.105e-007	8.105e-007	1.250e-006
57	Liq. Reynolds	1.811e+005 *	1.811e+005 *	1.811e+005 *	1.811e+005 *
58	Vap. Reynolds	9.923e+006 *	9.923e+006 *	9.923e+006 *	9.923e+006 *
59	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
60	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	3.389e+004	3.389e+004	3.389e+004	3.390e+004
64	Elevation (m)	0.3040	0.3060	0.3080	0.3100
65	Pressure (bar)	66.95	66.95	66.95	66.95
66	Temperature (C)	45.83	45.83	45.83	45.83
67	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
68	Heat Transferred (kJ/h-m)	421.7	421.7	421.6	421.6
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3					Unit Set: Mixed
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5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2320	0.2320	0.2320	0.2320
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	8.308e-003	8.308e-003	8.308e-003	8.308e-003
18	Static Gradient (kPa/m)	1.321e-003	1.321e-003	1.321e-003	1.321e-003
19	Accel. Gradient (kPa/m)	1.250e-006	1.250e-006	1.250e-006	1.250e-006
20	Liq. Reynolds	1.811e+005 *	1.811e+005 *	1.811e+005 *	1.811e+005 *
21	Vap. Reynolds	9.923e+006 *	9.923e+006 *	9.923e+006 *	9.923e+006 *
22	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
23	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	3.391e+004	3.392e+004	3.393e+004	3.394e+004
27	Elevation (m)	0.3100	0.3100	0.3100	0.3100
28	Pressure (bar)	66.95	66.95	66.95	66.94
29	Temperature (C)	45.83	45.82	45.82	45.82
30	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
31	Heat Transferred (kJ/h-m)	421.5	421.5	421.4	421.3
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2491	0.2491	0.2491	0.2491
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	6.244e-003	6.244e-003	6.244e-003	6.244e-003
37	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
38	Accel. Gradient (kPa/m)	8.106e-007	8.106e-007	8.106e-007	8.106e-007
39	Liq. Reynolds	1.811e+005 *	1.810e+005 *	1.810e+005 *	1.810e+005 *
40	Vap. Reynolds	9.923e+006 *	9.923e+006 *	9.923e+006 *	9.923e+006 *
41	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
42	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	3.395e+004	3.396e+004	3.398e+004	3.399e+004
46	Elevation (m)	0.3100	0.3080	0.3060	0.3040
47	Pressure (bar)	66.94	66.94	66.94	66.94
48	Temperature (C)	45.81	45.81	45.81	45.80
49	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
50	Heat Transferred (kJ/h-m)	421.2	421.2	421.1	421.0
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2491	0.1705	0.1705	0.1705
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	6.244e-003	9.109e-003	9.109e-003	9.109e-003
56	Static Gradient (kPa/m)	0.0000	-2.863e-004	-2.863e-004	-2.863e-004
57	Accel. Gradient (kPa/m)	8.106e-007	1.145e-006	1.145e-006	1.145e-006
58	Liq. Reynolds	1.810e+005 *	1.810e+005 *	1.810e+005 *	1.810e+005 *
59	Vap. Reynolds	9.923e+006 *	9.923e+006 *	9.923e+006 *	9.923e+006 *
60	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
61	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	3.400e+004	3.401e+004	3.402e+004	3.403e+004
65	Elevation (m)	0.3020	0.3000	0.2980	0.2960
66	Pressure (bar)	66.94	66.94	66.94	66.94
67	Temperature (C)	45.80	45.80	45.80	45.79
68	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
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5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	420.9	420.9	420.8	420.7
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.1705	0.1705	0.1705	0.1705
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	9.109e-003	9.109e-003	9.109e-003	9.109e-003
19	Static Gradient (kPa/m)	-2.863e-004	-2.863e-004	-2.863e-004	-2.863e-004
20	Accel. Gradient (kPa/m)	1.145e-006	1.145e-006	1.146e-006	1.146e-006
21	Liq. Reynolds	1.810e+005 *	1.810e+005 *	1.810e+005 *	1.810e+005 *
22	Vap. Reynolds	9.923e+006 *	9.923e+006 *	9.923e+006 *	9.923e+006 *
23	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
24	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	3.404e+004	3.405e+004	3.406e+004	3.408e+004
28	Elevation (m)	0.2940	0.2920	0.2900	0.2880
29	Pressure (bar)	66.94	66.94	66.93	66.93
30	Temperature (C)	45.79	45.79	45.78	45.78
31	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
32	Heat Transferred (kJ/h-m)	420.6	420.5	420.5	420.4
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.1705	0.1705	0.1705	0.1705
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	9.109e-003	9.109e-003	9.109e-003	9.109e-003
38	Static Gradient (kPa/m)	-2.863e-004	-2.863e-004	-2.863e-004	-2.863e-004
39	Accel. Gradient (kPa/m)	1.146e-006	1.146e-006	1.146e-006	1.146e-006
40	Liq. Reynolds	1.810e+005 *	1.810e+005 *	1.810e+005 *	1.810e+005 *
41	Vap. Reynolds	9.924e+006 *	9.924e+006 *	9.924e+006 *	9.924e+006 *
42	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
43	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	3.409e+004	3.410e+004	3.411e+004	3.412e+004
47	Elevation (m)	0.2860	0.2840	0.2820	0.2800
48	Pressure (bar)	66.93	66.93	66.93	66.93
49	Temperature (C)	45.78	45.77	45.77	45.77
50	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
51	Heat Transferred (kJ/h-m)	420.3	420.2	420.1	420.1
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.1705	0.1705	0.1705	0.1705
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	9.109e-003	9.109e-003	9.109e-003	9.109e-003
57	Static Gradient (kPa/m)	-2.863e-004	-2.863e-004	-2.863e-004	-2.863e-004
58	Accel. Gradient (kPa/m)	1.146e-006	1.146e-006	1.146e-006	1.146e-006
59	Liq. Reynolds	1.810e+005 *	1.810e+005 *	1.810e+005 *	1.810e+005 *
60	Vap. Reynolds	9.924e+006 *	9.924e+006 *	9.924e+006 *	9.924e+006 *
61	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
62	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	3.413e+004	3.414e+004	3.415e+004	3.416e+004
66	Elevation (m)	0.2840	0.2880	0.2920	0.2960
67	Pressure (bar)	66.93	66.93	66.93	66.93
68	Temperature (C)	45.77	45.76	45.76	45.76
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
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3			Unit Set: Mixed		
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5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
14	Heat Transferred (kJ/h-m)	420.0	419.9	419.8	419.7
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2312	0.2312	0.2312	0.2312
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	8.423e-003	8.423e-003	8.423e-003	8.423e-003
20	Static Gradient (kPa/m)	7.094e-004	7.094e-004	7.094e-004	7.095e-004
21	Accel. Gradient (kPa/m)	1.186e-006	1.186e-006	1.186e-006	1.186e-006
22	Liq. Reynolds	1.810e+005 *	1.810e+005 *	1.810e+005 *	1.810e+005 *
23	Vap. Reynolds	9.924e+006 *	9.924e+006 *	9.924e+006 *	9.924e+006 *
24	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
25	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	3.418e+004	3.419e+004	3.420e+004	3.421e+004
29	Elevation (m)	0.3000	0.3300	0.3600	0.3900
30	Pressure (bar)	66.92	66.92	66.92	66.92
31	Temperature (C)	45.75	45.75	45.75	45.74
32	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
33	Heat Transferred (kJ/h-m)	419.6	419.5	419.5	419.4
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2312	0.2376	0.2376	0.2376
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	8.423e-003	7.584e-003	7.584e-003	7.585e-003
39	Static Gradient (kPa/m)	7.095e-004	5.430e-003	5.430e-003	5.430e-003
40	Accel. Gradient (kPa/m)	1.186e-006	1.690e-006	1.690e-006	1.690e-006
41	Liq. Reynolds	1.810e+005 *	1.810e+005 *	1.810e+005 *	1.810e+005 *
42	Vap. Reynolds	9.924e+006 *	9.924e+006 *	9.924e+006 *	9.924e+006 *
43	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
44	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	3.422e+004	3.423e+004	3.424e+004	3.425e+004
48	Elevation (m)	0.4200	0.4500	0.4180	0.3860
49	Pressure (bar)	66.92	66.92	66.92	66.92
50	Temperature (C)	45.74	45.74	45.73	45.73
51	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
52	Heat Transferred (kJ/h-m)	419.3	419.2	419.2	419.1
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2376	0.2376	0.1685	0.1685
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	7.585e-003	7.585e-003	9.137e-003	9.136e-003
58	Static Gradient (kPa/m)	5.430e-003	5.430e-003	-4.545e-003	-4.545e-003
59	Accel. Gradient (kPa/m)	1.690e-006	1.690e-006	5.963e-007	5.963e-007
60	Liq. Reynolds	1.810e+005 *	1.809e+005 *	1.809e+005 *	1.809e+005 *
61	Vap. Reynolds	9.924e+006 *	9.924e+006 *	9.924e+006 *	9.924e+006 *
62	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
63	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	3.426e+004	3.428e+004	3.429e+004	3.429e+004
67	Elevation (m)	0.3540	0.3220	0.2900	0.2960
68	Pressure (bar)	66.92	66.91	66.91	66.91
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	45.73	45.73	45.72	45.72
14	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
15	Heat Transferred (kJ/h-m)	419.0	418.9	418.9	418.8
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.1685	0.1685	0.1685	0.2381
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	9.136e-003	9.136e-003	9.136e-003	7.527e-003
21	Static Gradient (kPa/m)	-4.545e-003	-4.545e-003	-4.545e-003	5.776e-003
22	Accel. Gradient (kPa/m)	5.963e-007	5.963e-007	5.963e-007	1.728e-006
23	Liq. Reynolds	1.809e+005 *	1.809e+005 *	1.809e+005 *	1.809e+005 *
24	Vap. Reynolds	9.925e+006 *	9.925e+006 *	9.925e+006 *	9.925e+006 *
25	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
26	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	3.429e+004	3.429e+004	3.430e+004	3.430e+004
30	Elevation (m)	0.3020	0.3080	0.3140	0.3200
31	Pressure (bar)	66.91	66.91	66.91	66.91
32	Temperature (C)	45.72	45.72	45.72	45.72
33	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
34	Heat Transferred (kJ/h-m)	418.8	418.8	418.8	418.7
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2381	0.2381	0.2381	0.2381
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	7.527e-003	7.527e-003	7.527e-003	7.527e-003
40	Static Gradient (kPa/m)	5.776e-003	5.776e-003	5.776e-003	5.776e-003
41	Accel. Gradient (kPa/m)	1.728e-006	1.728e-006	1.728e-006	1.728e-006
42	Liq. Reynolds	1.809e+005 *	1.809e+005 *	1.809e+005 *	1.809e+005 *
43	Vap. Reynolds	9.925e+006 *	9.925e+006 *	9.925e+006 *	9.925e+006 *
44	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
45	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	3.430e+004	3.431e+004	3.431e+004	3.432e+004
49	Elevation (m)	0.3240	0.3280	0.3320	0.3360
50	Pressure (bar)	66.91	66.91	66.91	66.91
51	Temperature (C)	45.72	45.72	45.72	45.71
52	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
53	Heat Transferred (kJ/h-m)	418.7	418.7	418.6	418.6
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2324	0.2324	0.2324	0.2324
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	8.259e-003	8.259e-003	8.259e-003	8.259e-003
59	Static Gradient (kPa/m)	1.587e-003	1.587e-003	1.587e-003	1.587e-003
60	Accel. Gradient (kPa/m)	1.279e-006	1.279e-006	1.279e-006	1.279e-006
61	Liq. Reynolds	1.809e+005 *	1.809e+005 *	1.809e+005 *	1.809e+005 *
62	Vap. Reynolds	9.925e+006 *	9.925e+006 *	9.925e+006 *	9.925e+006 *
63	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
64	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	3.432e+004	3.433e+004	3.433e+004	3.434e+004
68	Elevation (m)	0.3400	0.3360	0.3320	0.3280
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
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5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
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11	Pipe Table				
12					
13	Pressure (bar)	66.91	66.91	66.91	66.91
14	Temperature (C)	45.71	45.71	45.71	45.71
15	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
16	Heat Transferred (kJ/h-m)	418.6	418.6	418.5	418.5
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.2324	0.1701	0.1701	0.1701
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	8.259e-003	9.116e-003	9.116e-003	9.116e-003
22	Static Gradient (kPa/m)	1.587e-003	-1.274e-003	-1.274e-003	-1.274e-003
23	Accel. Gradient (kPa/m)	1.279e-006	1.019e-006	1.019e-006	1.019e-006
24	Liq. Reynolds	1.809e+005 *	1.809e+005 *	1.809e+005 *	1.809e+005 *
25	Vap. Reynolds	9.925e+006 *	9.925e+006 *	9.925e+006 *	9.925e+006 *
26	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
27	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	3.434e+004	3.435e+004	3.435e+004	3.436e+004
31	Elevation (m)	0.3240	0.3200	0.3220	0.3240
32	Pressure (bar)	66.91	66.91	66.91	66.91
33	Temperature (C)	45.71	45.71	45.71	45.70
34	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
35	Heat Transferred (kJ/h-m)	418.5	418.4	418.4	418.3
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.1701	0.1701	0.2313	0.2313
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	9.116e-003	9.116e-003	8.408e-003	8.408e-003
41	Static Gradient (kPa/m)	-1.274e-003	-1.274e-003	7.908e-004	7.908e-004
42	Accel. Gradient (kPa/m)	1.019e-006	1.019e-006	1.195e-006	1.195e-006
43	Liq. Reynolds	1.809e+005 *	1.809e+005 *	1.809e+005 *	1.809e+005 *
44	Vap. Reynolds	9.925e+006 *	9.925e+006 *	9.925e+006 *	9.925e+006 *
45	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
46	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	3.436e+004	3.437e+004	3.437e+004	3.438e+004
50	Elevation (m)	0.3260	0.3280	0.3300	0.3240
51	Pressure (bar)	66.91	66.91	66.91	66.91
52	Temperature (C)	45.70	45.70	45.70	45.70
53	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
54	Heat Transferred (kJ/h-m)	418.3	418.2	418.2	418.2
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.2313	0.2313	0.2313	0.1698
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	8.408e-003	8.408e-003	8.408e-003	9.120e-003
60	Static Gradient (kPa/m)	7.908e-004	7.908e-004	7.908e-004	-1.909e-003
61	Accel. Gradient (kPa/m)	1.195e-006	1.195e-006	1.195e-006	9.368e-007
62	Liq. Reynolds	1.809e+005 *	1.809e+005 *	1.809e+005 *	1.809e+005 *
63	Vap. Reynolds	9.925e+006 *	9.925e+006 *	9.925e+006 *	9.925e+006 *
64	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
65	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	3.438e+004	3.439e+004	3.439e+004	3.440e+004
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set: Mixed		
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5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
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9	PROFILES				
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11	Pipe Table				
12					
13	Elevation (m)	0.3180	0.3120	0.3060	0.3000
14	Pressure (bar)	66.91	66.91	66.90	66.90
15	Temperature (C)	45.70	45.70	45.69	45.69
16	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
17	Heat Transferred (kJ/h-m)	418.2	418.1	418.1	418.1
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.1698	0.1698	0.1698	0.1698
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	9.120e-003	9.120e-003	9.120e-003	9.120e-003
23	Static Gradient (kPa/m)	-1.909e-003	-1.909e-003	-1.909e-003	-1.909e-003
24	Accel. Gradient (kPa/m)	9.368e-007	9.368e-007	9.368e-007	9.368e-007
25	Liq. Reynolds	1.809e+005 *	1.809e+005 *	1.809e+005 *	1.809e+005 *
26	Vap. Reynolds	9.925e+006 *	9.925e+006 *	9.925e+006 *	9.925e+006 *
27	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
28	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	3.440e+004	3.441e+004	3.441e+004	3.442e+004
32	Elevation (m)	0.3000	0.3000	0.3000	0.3000
33	Pressure (bar)	66.90	66.90	66.90	66.90
34	Temperature (C)	45.69	45.69	45.69	45.69
35	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
36	Heat Transferred (kJ/h-m)	418.0	418.0	417.9	417.9
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2491	0.2491	0.2491	0.2491
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	6.244e-003	6.244e-003	6.244e-003	6.244e-003
42	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
43	Accel. Gradient (kPa/m)	8.112e-007	8.112e-007	8.112e-007	8.112e-007
44	Liq. Reynolds	1.809e+005 *	1.809e+005 *	1.809e+005 *	1.809e+005 *
45	Vap. Reynolds	9.925e+006 *	9.925e+006 *	9.925e+006 *	9.925e+006 *
46	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
47	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	3.442e+004	3.443e+004	3.443e+004	3.444e+004
51	Elevation (m)	0.3000	0.3180	0.3360	0.3540
52	Pressure (bar)	66.90	66.90	66.90	66.90
53	Temperature (C)	45.69	45.69	45.68	45.68
54	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
55	Heat Transferred (kJ/h-m)	417.9	417.8	417.8	417.7
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2491	0.2401	0.2401	0.2401
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	6.244e-003	7.277e-003	7.277e-003	7.277e-003
61	Static Gradient (kPa/m)	0.0000	7.317e-003	7.317e-003	7.317e-003
62	Accel. Gradient (kPa/m)	8.112e-007	1.896e-006	1.896e-006	1.896e-006
63	Liq. Reynolds	1.809e+005 *	1.809e+005 *	1.809e+005 *	1.809e+005 *
64	Vap. Reynolds	9.925e+006 *	9.925e+006 *	9.925e+006 *	9.925e+006 *
65	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
66	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
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5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	3.444e+004	3.445e+004	3.445e+004	3.446e+004
14	Elevation (m)	0.3720	0.3900	0.3720	0.3540
15	Pressure (bar)	66.90	66.90	66.90	66.90
16	Temperature (C)	45.68	45.68	45.68	45.68
17	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
18	Heat Transferred (kJ/h-m)	417.7	417.7	417.7	417.6
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.2401	0.2401	0.1678	0.1678
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	7.277e-003	7.277e-003	9.147e-003	9.147e-003
24	Static Gradient (kPa/m)	7.317e-003	7.317e-003	-6.151e-003	-6.151e-003
25	Accel. Gradient (kPa/m)	1.896e-006	1.896e-006	3.893e-007	3.893e-007
26	Liq. Reynolds	1.809e+005 *	1.809e+005 *	1.809e+005 *	1.809e+005 *
27	Vap. Reynolds	9.925e+006 *	9.925e+006 *	9.925e+006 *	9.925e+006 *
28	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
29	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	3.446e+004	3.447e+004	3.447e+004	3.448e+004
33	Elevation (m)	0.3360	0.3180	0.3000	0.3000
34	Pressure (bar)	66.90	66.90	66.90	66.90
35	Temperature (C)	45.68	45.68	45.67	45.67
36	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
37	Heat Transferred (kJ/h-m)	417.6	417.6	417.6	417.5
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.1678	0.1678	0.1678	0.2491
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	9.147e-003	9.147e-003	9.147e-003	6.244e-003
43	Static Gradient (kPa/m)	-6.151e-003	-6.151e-003	-6.151e-003	0.0000
44	Accel. Gradient (kPa/m)	3.893e-007	3.893e-007	3.893e-007	8.113e-007
45	Liq. Reynolds	1.809e+005 *	1.809e+005 *	1.809e+005 *	1.809e+005 *
46	Vap. Reynolds	9.925e+006 *	9.925e+006 *	9.925e+006 *	9.925e+006 *
47	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
48	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	3.448e+004	3.449e+004	3.449e+004	3.450e+004
52	Elevation (m)	0.3000	0.3000	0.3000	0.3000
53	Pressure (bar)	66.90	66.90	66.90	66.90
54	Temperature (C)	45.67	45.67	45.67	45.67
55	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
56	Heat Transferred (kJ/h-m)	417.4	417.4	417.4	417.3
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.2491	0.2491	0.2491	0.2491
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	6.244e-003	6.244e-003	6.244e-003	6.244e-003
62	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
63	Accel. Gradient (kPa/m)	8.113e-007	8.113e-007	8.113e-007	8.113e-007
64	Liq. Reynolds	1.809e+005 *	1.809e+005 *	1.809e+005 *	1.809e+005 *
65	Vap. Reynolds	9.925e+006 *	9.925e+006 *	9.925e+006 *	9.925e+006 *
66	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
67	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
68	Deposit Thickness (mm)	---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 549 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	3.450e+004	3.451e+004	3.451e+004	3.452e+004		
15	Elevation (m)	0.2980	0.2960	0.2940	0.2920		
16	Pressure (bar)	66.90	66.90	66.90	66.90		
17	Temperature (C)	45.67	45.66	45.66	45.66		
18	Vapour Fraction	0.8800	0.8800	0.8800	0.8800		
19	Heat Transferred (kJ/h-m)	417.3	417.3	417.3	417.2		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	0.1704	0.1704	0.1704	0.1704		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	9.112e-003	9.112e-003	9.112e-003	9.112e-003		
25	Static Gradient (kPa/m)	-6.377e-004	-6.377e-004	-6.377e-004	-6.377e-004		
26	Accel. Gradient (kPa/m)	1.101e-006	1.101e-006	1.101e-006	1.101e-006		
27	Liq. Reynolds	1.809e+005 *	1.809e+005 *	1.809e+005 *	1.809e+005 *		
28	Vap. Reynolds	9.925e+006 *	9.925e+006 *	9.925e+006 *	9.925e+006 *		
29	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
30	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	3.452e+004	3.453e+004	3.453e+004	3.454e+004		
34	Elevation (m)	0.2900	0.3000	0.3100	0.3200		
35	Pressure (bar)	66.89	66.89	66.89	66.89		
36	Temperature (C)	45.66	45.66	45.66	45.66		
37	Vapour Fraction	0.8800	0.8800	0.8800	0.8800		
38	Heat Transferred (kJ/h-m)	417.2	417.1	417.1	417.1		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.1704	0.2357	0.2357	0.2357		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	9.112e-003	7.826e-003	7.826e-003	7.827e-003		
44	Static Gradient (kPa/m)	-6.377e-004	4.010e-003	4.010e-003	4.010e-003		
45	Accel. Gradient (kPa/m)	1.101e-006	1.538e-006	1.538e-006	1.538e-006		
46	Liq. Reynolds	1.809e+005 *	1.809e+005 *	1.809e+005 *	1.809e+005 *		
47	Vap. Reynolds	9.926e+006 *	9.926e+006 *	9.926e+006 *	9.926e+006 *		
48	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
49	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	3.454e+004	3.455e+004	3.455e+004	3.456e+004		
53	Elevation (m)	0.3300	0.3400	0.3320	0.3240		
54	Pressure (bar)	66.89	66.89	66.89	66.89		
55	Temperature (C)	45.66	45.65	45.65	45.65		
56	Vapour Fraction	0.8800	0.8800	0.8800	0.8800		
57	Heat Transferred (kJ/h-m)	417.0	417.0	417.0	416.9		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.2357	0.2357	0.1695	0.1695		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	7.827e-003	7.827e-003	9.124e-003	9.124e-003		
63	Static Gradient (kPa/m)	4.010e-003	4.010e-003	-2.542e-003	-2.542e-003		
64	Accel. Gradient (kPa/m)	1.538e-006	1.538e-006	8.552e-007	8.552e-007		
65	Liq. Reynolds	1.809e+005 *	1.809e+005 *	1.809e+005 *	1.809e+005 *		
66	Vap. Reynolds	9.926e+006 *	9.926e+006 *	9.926e+006 *	9.926e+006 *		
67	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
68	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	3.456e+004	3.457e+004	3.457e+004	3.458e+004
16	Elevation (m)	0.3160	0.3080	0.3000	0.3040
17	Pressure (bar)	66.89	66.89	66.89	66.89
18	Temperature (C)	45.65	45.65	45.65	45.65
19	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
20	Heat Transferred (kJ/h-m)	416.9	416.9	416.8	416.8
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.1695	0.1695	0.1695	0.2324
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	9.124e-003	9.124e-003	9.124e-003	8.260e-003
26	Static Gradient (kPa/m)	-2.542e-003	-2.542e-003	-2.542e-003	1.587e-003
27	Accel. Gradient (kPa/m)	8.552e-007	8.552e-007	8.552e-007	1.279e-006
28	Liq. Reynolds	1.809e+005 *	1.808e+005 *	1.808e+005 *	1.808e+005 *
29	Vap. Reynolds	9.926e+006 *	9.926e+006 *	9.926e+006 *	9.926e+006 *
30	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
31	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	3.458e+004	3.459e+004	3.459e+004	3.460e+004
35	Elevation (m)	0.3080	0.3120	0.3160	0.3200
36	Pressure (bar)	66.89	66.89	66.89	66.89
37	Temperature (C)	45.64	45.64	45.64	45.64
38	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
39	Heat Transferred (kJ/h-m)	416.7	416.7	416.7	416.6
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2324	0.2324	0.2324	0.2324
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	8.260e-003	8.260e-003	8.260e-003	8.260e-003
45	Static Gradient (kPa/m)	1.587e-003	1.587e-003	1.587e-003	1.587e-003
46	Accel. Gradient (kPa/m)	1.279e-006	1.279e-006	1.280e-006	1.280e-006
47	Liq. Reynolds	1.808e+005 *	1.808e+005 *	1.808e+005 *	1.808e+005 *
48	Vap. Reynolds	9.926e+006 *	9.926e+006 *	9.926e+006 *	9.926e+006 *
49	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
50	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	3.460e+004	3.461e+004	3.461e+004	3.462e+004
54	Elevation (m)	0.3180	0.3160	0.3140	0.3120
55	Pressure (bar)	66.89	66.89	66.89	66.89
56	Temperature (C)	45.64	45.64	45.64	45.64
57	Vapour Fraction	0.8800	0.8800	0.8800	0.8800
58	Heat Transferred (kJ/h-m)	416.6	416.6	416.6	416.5
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.1704	0.1704	0.1704	0.1704
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	9.112e-003	9.112e-003	9.112e-003	9.112e-003
64	Static Gradient (kPa/m)	-6.377e-004	-6.377e-004	-6.377e-004	-6.377e-004
65	Accel. Gradient (kPa/m)	1.101e-006	1.101e-006	1.101e-006	1.101e-006
66	Liq. Reynolds	1.808e+005 *	1.808e+005 *	1.808e+005 *	1.808e+005 *
67	Vap. Reynolds	9.926e+006 *	9.926e+006 *	9.926e+006 *	9.926e+006 *
68	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947				
14	Deposit Thickness	(mm)	---	---	---	---				
15	Deposit Volume	(m3)	---	---	---	---				
16	Length	(m)	3.462e+004	3.463e+004	3.463e+004	3.464e+004				
17	Elevation	(m)	0.3100	0.3100	0.3100	0.3100				
18	Pressure	(bar)	66.89	66.89	66.89	66.88				
19	Temperature	(C)	45.63	45.63	45.63	45.63				
20	Vapour Fraction		0.8800	0.8800	0.8800	0.8800				
21	Heat Transferred	(kJ/h-m)	416.5	416.4	416.4	416.3				
22	Flow Regime		Transition	Transition	Transition	Transition				
23	Water Flow Regime									
24	Liquid HoldUp		0.1704	0.2491	0.2491	0.2491				
25	Water HoldUp		---	---	---	---				
26	Friction Gradient	(kPa/m)	9.112e-003	6.245e-003	6.245e-003	6.245e-003				
27	Static Gradient	(kPa/m)	-6.377e-004	0.0000	0.0000	0.0000				
28	Accel. Gradient	(kPa/m)	1.101e-006	8.115e-007	8.115e-007	8.115e-007				
29	Liq. Reynolds		1.808e+005 *	1.808e+005 *	1.808e+005 *	1.808e+005 *				
30	Vap. Reynolds		9.926e+006 *	9.926e+006 *	9.926e+006 *	9.926e+006 *				
31	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
32	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947				
33	Deposit Thickness	(mm)	---	---	---	---				
34	Deposit Volume	(m3)	---	---	---	---				
35	Length	(m)	3.464e+004	3.465e+004	3.465e+004	3.465e+004				
36	Elevation	(m)	0.3100	0.3100	0.3260	0.3420				
37	Pressure	(bar)	66.88	66.88	66.88	66.88				
38	Temperature	(C)	45.63	45.63	45.63	45.62				
39	Vapour Fraction		0.8800	0.8800	0.8800	0.8800				
40	Heat Transferred	(kJ/h-m)	416.3	416.3	416.2	416.2				
41	Flow Regime		Transition	Transition	Transition	Transition				
42	Water Flow Regime									
43	Liquid HoldUp		0.2491	0.2491	0.2410	0.2410				
44	Water HoldUp		---	---	---	---				
45	Friction Gradient	(kPa/m)	6.245e-003	6.245e-003	7.176e-003	7.176e-003				
46	Static Gradient	(kPa/m)	0.0000	0.0000	7.957e-003	7.957e-003				
47	Accel. Gradient	(kPa/m)	8.115e-007	8.115e-007	1.966e-006	1.966e-006				
48	Liq. Reynolds		1.808e+005 *	1.808e+005 *	1.808e+005 *	1.808e+005 *				
49	Vap. Reynolds		9.926e+006 *	9.926e+006 *	9.926e+006 *	9.926e+006 *				
50	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
51	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947				
52	Deposit Thickness	(mm)	---	---	---	---				
53	Deposit Volume	(m3)	---	---	---	---				
54	Length	(m)	3.466e+004	3.466e+004	3.467e+004	3.467e+004				
55	Elevation	(m)	0.3580	0.3740	0.3900	0.3900				
56	Pressure	(bar)	66.88	66.88	66.88	66.88				
57	Temperature	(C)	45.62	45.62	45.62	45.62				
58	Vapour Fraction		0.8800	0.8800	0.8800	0.8800				
59	Heat Transferred	(kJ/h-m)	416.2	416.1	416.1	416.1				
60	Flow Regime		Transition	Transition	Transition	Transition				
61	Water Flow Regime									
62	Liquid HoldUp		0.2410	0.2410	0.2410	0.2491				
63	Water HoldUp		---	---	---	---				
64	Friction Gradient	(kPa/m)	7.176e-003	7.176e-003	7.176e-003	6.245e-003				
65	Static Gradient	(kPa/m)	7.957e-003	7.957e-003	7.957e-003	0.0000				
66	Accel. Gradient	(kPa/m)	1.967e-006	1.967e-006	1.967e-006	8.116e-007				
67	Liq. Reynolds		1.808e+005 *	1.808e+005 *	1.808e+005 *	1.808e+005 *				
68	Vap. Reynolds		9.926e+006 *	9.926e+006 *	9.926e+006 *	9.926e+006 *				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
14	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947				
15	Deposit Thickness	(mm)	---	---	---	---				
16	Deposit Volume	(m3)	---	---	---	---				
17	Length	(m)	3.468e+004	3.468e+004	3.469e+004	3.469e+004				
18	Elevation	(m)	0.3900	0.3900	0.3900	0.3900				
19	Pressure	(bar)	66.88	66.88	66.88	66.88				
20	Temperature	(C)	45.62	45.62	45.62	45.61				
21	Vapour Fraction		0.8799	0.8799	0.8799	0.8799				
22	Heat Transferred	(kJ/h-m)	416.0	416.0	416.0	415.9				
23	Flow Regime		Transition	Transition	Transition	Transition				
24	Water Flow Regime									
25	Liquid HoldUp		0.2491	0.2491	0.2491	0.2491				
26	Water HoldUp		---	---	---	---				
27	Friction Gradient	(kPa/m)	6.245e-003	6.245e-003	6.245e-003	6.245e-003				
28	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000				
29	Accel. Gradient	(kPa/m)	8.116e-007	8.116e-007	8.116e-007	8.116e-007				
30	Liq. Reynolds		1.808e+005 *	1.808e+005 *	1.808e+005 *	1.808e+005 *				
31	Vap. Reynolds		9.926e+006 *	9.926e+006 *	9.926e+006 *	9.926e+006 *				
32	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
33	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947				
34	Deposit Thickness	(mm)	---	---	---	---				
35	Deposit Volume	(m3)	---	---	---	---				
36	Length	(m)	3.470e+004	3.470e+004	3.471e+004	3.471e+004				
37	Elevation	(m)	0.3820	0.3740	0.3660	0.3580				
38	Pressure	(bar)	66.88	66.88	66.88	66.88				
39	Temperature	(C)	45.61	45.61	45.61	45.61				
40	Vapour Fraction		0.8799	0.8799	0.8799	0.8799				
41	Heat Transferred	(kJ/h-m)	415.9	415.9	415.9	415.8				
42	Flow Regime		Transition	Transition	Transition	Transition				
43	Water Flow Regime									
44	Liquid HoldUp		0.1695	0.1695	0.1695	0.1695				
45	Water HoldUp		---	---	---	---				
46	Friction Gradient	(kPa/m)	9.124e-003	9.124e-003	9.124e-003	9.124e-003				
47	Static Gradient	(kPa/m)	-2.542e-003	-2.542e-003	-2.542e-003	-2.542e-003				
48	Accel. Gradient	(kPa/m)	8.554e-007	8.554e-007	8.554e-007	8.554e-007				
49	Liq. Reynolds		1.808e+005 *	1.808e+005 *	1.808e+005 *	1.808e+005 *				
50	Vap. Reynolds		9.926e+006 *	9.926e+006 *	9.926e+006 *	9.926e+006 *				
51	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980				
52	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947				
53	Deposit Thickness	(mm)	---	---	---	---				
54	Deposit Volume	(m3)	---	---	---	---				
55	Length	(m)	3.472e+004	3.472e+004	3.473e+004	3.473e+004				
56	Elevation	(m)	0.3500	0.3400	0.3300	0.3200				
57	Pressure	(bar)	66.88	66.88	66.88	66.88				
58	Temperature	(C)	45.61	45.61	45.61	45.60				
59	Vapour Fraction		0.8799	0.8799	0.8799	0.8799				
60	Heat Transferred	(kJ/h-m)	415.8	415.8	415.7	415.7				
61	Flow Regime		Transition	Transition	Transition	Transition				
62	Water Flow Regime									
63	Liquid HoldUp		0.1695	0.1692	0.1692	0.1692				
64	Water HoldUp		---	---	---	---				
65	Friction Gradient	(kPa/m)	9.124e-003	9.128e-003	9.128e-003	9.128e-003				
66	Static Gradient	(kPa/m)	-2.542e-003	-3.174e-003	-3.174e-003	-3.174e-003				
67	Accel. Gradient	(kPa/m)	8.554e-007	7.738e-007	7.738e-007	7.738e-007				
68	Liq. Reynolds		1.808e+005 *	1.808e+005 *	1.808e+005 *	1.808e+005 *				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.926e+006 *	9.926e+006 *	9.926e+006 *	9.926e+006 *
14	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
15	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	3.474e+004	3.474e+004	3.475e+004	3.475e+004
19	Elevation (m)	0.3100	0.3000	0.2880	0.2760
20	Pressure (bar)	66.88	66.88	66.88	66.88
21	Temperature (C)	45.60	45.60	45.60	45.60
22	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
23	Heat Transferred (kJ/h-m)	415.7	415.6	415.6	415.6
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1692	0.1692	0.1689	0.1689
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	9.128e-003	9.128e-003	9.132e-003	9.132e-003
29	Static Gradient (kPa/m)	-3.174e-003	-3.174e-003	-3.804e-003	-3.804e-003
30	Accel. Gradient (kPa/m)	7.738e-007	7.738e-007	6.924e-007	6.924e-007
31	Liq. Reynolds	1.808e+005 *	1.808e+005 *	1.808e+005 *	1.808e+005 *
32	Vap. Reynolds	9.926e+006 *	9.926e+006 *	9.926e+006 *	9.926e+006 *
33	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
34	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	3.476e+004	3.476e+004	3.477e+004	3.477e+004
38	Elevation (m)	0.2640	0.2520	0.2400	0.2200
39	Pressure (bar)	66.88	66.88	66.88	66.87
40	Temperature (C)	45.60	45.60	45.60	45.59
41	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
42	Heat Transferred (kJ/h-m)	415.5	415.5	415.5	415.4
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1689	0.1689	0.1689	0.1677
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	9.132e-003	9.132e-003	9.132e-003	9.148e-003
48	Static Gradient (kPa/m)	-3.804e-003	-3.804e-003	-3.804e-003	-6.311e-003
49	Accel. Gradient (kPa/m)	6.924e-007	6.924e-007	6.924e-007	3.688e-007
50	Liq. Reynolds	1.808e+005 *	1.808e+005 *	1.808e+005 *	1.808e+005 *
51	Vap. Reynolds	9.926e+006 *	9.926e+006 *	9.926e+006 *	9.926e+006 *
52	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
53	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	3.478e+004	3.478e+004	3.479e+004	3.479e+004
57	Elevation (m)	0.2000	0.1800	0.1600	0.1400
58	Pressure (bar)	66.87	66.87	66.87	66.87
59	Temperature (C)	45.59	45.59	45.59	45.59
60	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
61	Heat Transferred (kJ/h-m)	415.4	415.4	415.3	415.3
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1677	0.1677	0.1677	0.1677
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	9.148e-003	9.148e-003	9.148e-003	9.148e-003
67	Static Gradient (kPa/m)	-6.311e-003	-6.311e-003	-6.311e-003	-6.311e-003
68	Accel. Gradient (kPa/m)	3.688e-007	3.688e-007	3.688e-007	3.688e-007
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		1.808e+005 *	1.808e+005 *	1.808e+005 *	1.808e+005 *	
14	Vap. Reynolds		9.926e+006 *	9.927e+006 *	9.927e+006 *	9.927e+006 *	
15	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
16	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947		
17	Deposit Thickness (mm)	---	---	---	---		
18	Deposit Volume (m3)	---	---	---	---		
19	Length (m)	3.480e+004	3.480e+004	3.481e+004	3.481e+004		
20	Elevation (m)	0.1280	0.1160	0.1040	9.200e-002		
21	Pressure (bar)	66.87	66.87	66.87	66.87		
22	Temperature (C)	45.59	45.59	45.59	45.58		
23	Vapour Fraction	0.8799	0.8799	0.8799	0.8799		
24	Heat Transferred (kJ/h-m)	415.3	415.2	415.2	415.2		
25	Flow Regime	Transition	Transition	Transition	Transition		
26	Water Flow Regime						
27	Liquid HoldUp	0.1689	0.1689	0.1689	0.1689		
28	Water HoldUp	---	---	---	---		
29	Friction Gradient (kPa/m)	9.132e-003	9.132e-003	9.132e-003	9.132e-003		
30	Static Gradient (kPa/m)	-3.804e-003	-3.804e-003	-3.804e-003	-3.804e-003		
31	Accel. Gradient (kPa/m)	6.924e-007	6.924e-007	6.924e-007	6.924e-007		
32	Liq. Reynolds		1.808e+005 *	1.808e+005 *	1.808e+005 *		
33	Vap. Reynolds		9.927e+006 *	9.927e+006 *	9.927e+006 *		
34	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
35	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947		
36	Deposit Thickness (mm)	---	---	---	---		
37	Deposit Volume (m3)	---	---	---	---		
38	Length (m)	3.482e+004	3.482e+004	3.483e+004	3.483e+004		
39	Elevation (m)	8.000e-002	9.200e-002	0.1040	0.1160		
40	Pressure (bar)	66.87	66.87	66.87	66.87		
41	Temperature (C)	45.58	45.58	45.58	45.58		
42	Vapour Fraction	0.8799	0.8799	0.8799	0.8799		
43	Heat Transferred (kJ/h-m)	415.1	415.1	415.0	415.0		
44	Flow Regime	Transition	Transition	Transition	Transition		
45	Water Flow Regime						
46	Liquid HoldUp	0.1689	0.2368	0.2368	0.2368		
47	Water HoldUp	---	---	---	---		
48	Friction Gradient (kPa/m)	9.132e-003	7.686e-003	7.686e-003	7.686e-003		
49	Static Gradient (kPa/m)	-3.804e-003	4.829e-003	4.829e-003	4.829e-003		
50	Accel. Gradient (kPa/m)	6.924e-007	1.626e-006	1.627e-006	1.627e-006		
51	Liq. Reynolds		1.808e+005 *	1.808e+005 *	1.808e+005 *		
52	Vap. Reynolds		9.927e+006 *	9.927e+006 *	9.927e+006 *		
53	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
54	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947		
55	Deposit Thickness (mm)	---	---	---	---		
56	Deposit Volume (m3)	---	---	---	---		
57	Length (m)	3.484e+004	3.484e+004	3.485e+004	3.485e+004		
58	Elevation (m)	0.1280	0.1400	0.1340	0.1280		
59	Pressure (bar)	66.87	66.87	66.87	66.87		
60	Temperature (C)	45.58	45.58	45.58	45.57		
61	Vapour Fraction	0.8799	0.8799	0.8799	0.8799		
62	Heat Transferred (kJ/h-m)	415.0	414.9	414.9	414.9		
63	Flow Regime	Transition	Transition	Transition	Transition		
64	Water Flow Regime						
65	Liquid HoldUp	0.2368	0.2368	0.1698	0.1698		
66	Water HoldUp	---	---	---	---		
67	Friction Gradient (kPa/m)	7.686e-003	7.686e-003	9.120e-003	9.120e-003		
68	Static Gradient (kPa/m)	4.829e-003	4.829e-003	-1.907e-003	-1.907e-003		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	1.627e-006	1.627e-006	9.374e-007	9.374e-007
14	Liq. Reynolds	1.808e+005 *	1.808e+005 *	1.808e+005 *	1.808e+005 *
15	Vap. Reynolds	9.927e+006 *	9.927e+006 *	9.927e+006 *	9.927e+006 *
16	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
17	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	3.486e+004	3.486e+004	3.487e+004	3.487e+004
21	Elevation (m)	0.1220	0.1160	0.1100	0.1680
22	Pressure (bar)	66.87	66.87	66.87	66.87
23	Temperature (C)	45.57	45.57	45.57	45.57
24	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
25	Heat Transferred (kJ/h-m)	414.9	414.8	414.8	414.7
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.1698	0.1698	0.1698	0.2622
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	9.120e-003	9.120e-003	9.120e-003	7.851e-003
31	Static Gradient (kPa/m)	-1.907e-003	-1.907e-003	-1.907e-003	2.519e-002
32	Accel. Gradient (kPa/m)	9.375e-007	9.375e-007	9.375e-007	4.294e-006
33	Liq. Reynolds	1.808e+005 *	1.808e+005 *	1.808e+005 *	1.808e+005 *
34	Vap. Reynolds	9.927e+006 *	9.927e+006 *	9.927e+006 *	9.927e+006 *
35	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
36	Vapour Velocity (m/s)	2.947	2.947	2.947	2.947
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	3.488e+004	3.488e+004	3.489e+004	3.489e+004
40	Elevation (m)	0.2260	0.2840	0.3420	0.4000
41	Pressure (bar)	66.86	66.86	66.86	66.86
42	Temperature (C)	45.57	45.56	45.56	45.56
43	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
44	Heat Transferred (kJ/h-m)	414.6	414.6	414.6	414.5
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.2622	0.2622	0.2622	0.2622
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	7.851e-003	7.851e-003	7.851e-003	7.851e-003
50	Static Gradient (kPa/m)	2.519e-002	2.519e-002	2.519e-002	2.519e-002
51	Accel. Gradient (kPa/m)	4.295e-006	4.295e-006	4.295e-006	4.295e-006
52	Liq. Reynolds	1.808e+005 *	1.808e+005 *	1.807e+005 *	1.807e+005 *
53	Vap. Reynolds	9.927e+006 *	9.927e+006 *	9.927e+006 *	9.927e+006 *
54	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
55	Vapour Velocity (m/s)	2.947	2.948	2.948	2.948
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	3.490e+004	3.490e+004	3.491e+004	3.491e+004
59	Elevation (m)	0.2560	0.1120	-3.200e-002	-0.1760
60	Pressure (bar)	66.86	66.86	66.86	66.87
61	Temperature (C)	45.56	45.56	45.56	45.56
62	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
63	Heat Transferred (kJ/h-m)	414.6	414.5	414.5	414.5
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.1495	0.1495	0.1495	0.1495
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	9.448e-003	9.448e-003	9.447e-003	9.447e-003
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1				Case Name:	Simulation - 386MMSCFD_80barg.hsc		
2				Unit Set:	Mixed		
3	 Epic Atlantic Limited			Date/Time:	Mon Jun 13 15:13:43 2022		
4							
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Static Gradient	(kPa/m)	-4.214e-002	-4.214e-002	-4.214e-002	-4.214e-002	
14	Accel. Gradient	(kPa/m)	-4.250e-006	-4.249e-006	-4.249e-006	-4.249e-006	
15	Liq. Reynolds		1.807e+005 *	1.807e+005 *	1.807e+005 *	1.808e+005 *	
16	Vap. Reynolds		9.927e+006 *	9.927e+006 *	9.927e+006 *	9.927e+006 *	
17	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
18	Vapour Velocity	(m/s)	2.948	2.947	2.947	2.947	
19	Deposit Thickness	(mm)	---	---	---	---	
20	Deposit Volume	(m3)	---	---	---	---	
21	Length	(m)	3.492e+004	3.492e+004	3.493e+004	3.493e+004	
22	Elevation	(m)	-0.3200	-0.4160	-0.5120	-0.6080	
23	Pressure	(bar)	66.87	66.87	66.87	66.87	
24	Temperature	(C)	45.56	45.56	45.56	45.56	
25	Vapour Fraction		0.8799	0.8799	0.8799	0.8799	
26	Heat Transferred	(kJ/h-m)	414.5	414.5	414.5	414.5	
27	Flow Regime		Transition	Transition	Transition	Transition	
28	Water Flow Regime						
29	Liquid HoldUp		0.1495	0.1565	0.1565	0.1566	
30	Water HoldUp		---	---	---	---	
31	Friction Gradient	(kPa/m)	9.447e-003	9.320e-003	9.320e-003	9.320e-003	
32	Static Gradient	(kPa/m)	-4.214e-002	-2.894e-002	-2.894e-002	-2.895e-002	
33	Accel. Gradient	(kPa/m)	-4.249e-006	-2.550e-006	-2.550e-006	-2.550e-006	
34	Liq. Reynolds		1.808e+005 *	1.808e+005 *	1.808e+005 *	1.808e+005 *	
35	Vap. Reynolds		9.927e+006 *	9.927e+006 *	9.927e+006 *	9.927e+006 *	
36	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
37	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947	
38	Deposit Thickness	(mm)	---	---	---	---	
39	Deposit Volume	(m3)	---	---	---	---	
40	Length	(m)	3.494e+004	3.494e+004	3.495e+004	3.495e+004	
41	Elevation	(m)	-0.7040	-0.8000	-0.7000	-0.6000	
42	Pressure	(bar)	66.87	66.87	66.87	66.87	
43	Temperature	(C)	45.56	45.56	45.55	45.55	
44	Vapour Fraction		0.8799	0.8799	0.8799	0.8799	
45	Heat Transferred	(kJ/h-m)	414.4	414.4	414.3	414.2	
46	Flow Regime		Transition	Transition	Transition	Transition	
47	Water Flow Regime						
48	Liquid HoldUp		0.1566	0.1566	0.2762	0.2762	
49	Water HoldUp		---	---	---	---	
50	Friction Gradient	(kPa/m)	9.319e-003	9.319e-003	7.980e-003	7.981e-003	
51	Static Gradient	(kPa/m)	-2.895e-002	-2.895e-002	3.766e-002	3.765e-002	
52	Accel. Gradient	(kPa/m)	-2.550e-006	-2.550e-006	5.931e-006	5.931e-006	
53	Liq. Reynolds		1.808e+005 *	1.808e+005 *	1.808e+005 *	1.807e+005 *	
54	Vap. Reynolds		9.927e+006 *	9.927e+006 *	9.927e+006 *	9.927e+006 *	
55	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980	
56	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947	
57	Deposit Thickness	(mm)	---	---	---	---	
58	Deposit Volume	(m3)	---	---	---	---	
59	Length	(m)	3.496e+004	3.497e+004	3.497e+004	3.497e+004	
60	Elevation	(m)	-0.5000	-0.4000	-0.3000	-0.2440	
61	Pressure	(bar)	66.86	66.86	66.86	66.86	
62	Temperature	(C)	45.55	45.55	45.54	45.54	
63	Vapour Fraction		0.8799	0.8799	0.8799	0.8799	
64	Heat Transferred	(kJ/h-m)	414.2	414.1	414.0	414.0	
65	Flow Regime		Transition	Transition	Transition	Transition	
66	Water Flow Regime						
67	Liquid HoldUp		0.2762	0.2762	0.2761	0.2817	
68	Water HoldUp		---	---	---	---	
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1				Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	7.981e-003	7.981e-003	7.981e-003	7.989e-003
14	Static Gradient	(kPa/m)	3.765e-002	3.765e-002	3.765e-002	4.282e-002
15	Accel. Gradient	(kPa/m)	5.931e-006	5.932e-006	5.932e-006	6.605e-006
16	Liq. Reynolds		1.807e+005 *	1.807e+005 *	1.807e+005 *	1.807e+005 *
17	Vap. Reynolds		9.927e+006 *	9.927e+006 *	9.927e+006 *	9.927e+006 *
18	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
19	Vapour Velocity	(m/s)	2.947	2.947	2.947	2.947
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	3.498e+004	3.498e+004	3.498e+004	3.499e+004
23	Elevation	(m)	-0.1880	-0.1320	-7.600e-002	-2.000e-002
24	Pressure	(bar)	66.86	66.85	66.85	66.85
25	Temperature	(C)	45.54	45.54	45.54	45.54
26	Vapour Fraction		0.8799	0.8799	0.8799	0.8799
27	Heat Transferred	(kJ/h-m)	414.0	413.9	413.9	413.9
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2816	0.2816	0.2816	0.2816
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	7.989e-003	7.989e-003	7.990e-003	7.990e-003
33	Static Gradient	(kPa/m)	4.282e-002	4.282e-002	4.281e-002	4.281e-002
34	Accel. Gradient	(kPa/m)	6.605e-006	6.605e-006	6.605e-006	6.606e-006
35	Liq. Reynolds		1.807e+005 *	1.807e+005 *	1.807e+005 *	1.807e+005 *
36	Vap. Reynolds		9.927e+006 *	9.927e+006 *	9.927e+006 *	9.928e+006 *
37	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
38	Vapour Velocity	(m/s)	2.948	2.948	2.948	2.948
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	3.499e+004	3.500e+004	3.500e+004	3.501e+004
42	Elevation	(m)	1.400e-002	4.800e-002	8.200e-002	0.1160
43	Pressure	(bar)	66.85	66.85	66.85	66.85
44	Temperature	(C)	45.54	45.53	45.53	45.53
45	Vapour Fraction		0.8799	0.8799	0.8799	0.8799
46	Heat Transferred	(kJ/h-m)	413.8	413.8	413.7	413.7
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2458	0.2458	0.2458	0.2458
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	6.612e-003	6.612e-003	6.612e-003	6.612e-003
52	Static Gradient	(kPa/m)	1.172e-002	1.172e-002	1.172e-002	1.172e-002
53	Accel. Gradient	(kPa/m)	2.384e-006	2.384e-006	2.384e-006	2.384e-006
54	Liq. Reynolds		1.807e+005 *	1.807e+005 *	1.807e+005 *	1.807e+005 *
55	Vap. Reynolds		9.928e+006 *	9.928e+006 *	9.928e+006 *	9.928e+006 *
56	Liquid Velocity	(m/s)	0.1980	0.1980	0.1980	0.1980
57	Vapour Velocity	(m/s)	2.948	2.948	2.948	2.948
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	3.502e+004	3.502e+004	3.503e+004	3.503e+004
61	Elevation	(m)	0.1500	0.1560	0.1620	0.1680
62	Pressure	(bar)	66.85	66.85	66.85	66.84
63	Temperature	(C)	45.53	45.53	45.53	45.52
64	Vapour Fraction		0.8799	0.8799	0.8799	0.8799
65	Heat Transferred	(kJ/h-m)	413.6	413.6	413.6	413.5
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2458	0.2335	0.2335	0.2335
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 558 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	6.612e-003	8.115e-003	8.115e-003	8.115e-003
15	Static Gradient (kPa/m)	1.172e-002	2.389e-003	2.389e-003	2.389e-003
16	Accel. Gradient (kPa/m)	2.384e-006	1.366e-006	1.366e-006	1.366e-006
17	Liq. Reynolds	1.807e+005 *	1.807e+005 *	1.807e+005 *	1.807e+005 *
18	Vap. Reynolds	9.928e+006 *	9.928e+006 *	9.928e+006 *	9.928e+006 *
19	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
20	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	3.504e+004	3.504e+004	3.505e+004	3.505e+004
24	Elevation (m)	0.1740	0.1800	0.1760	0.1720
25	Pressure (bar)	66.84	66.84	66.84	66.84
26	Temperature (C)	45.52	45.52	45.52	45.52
27	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
28	Heat Transferred (kJ/h-m)	413.5	413.4	413.4	413.4
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2335	0.2335	0.1701	0.1701
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	8.115e-003	8.115e-003	9.117e-003	9.117e-003
34	Static Gradient (kPa/m)	2.389e-003	2.389e-003	-1.267e-003	-1.267e-003
35	Accel. Gradient (kPa/m)	1.366e-006	1.366e-006	1.021e-006	1.021e-006
36	Liq. Reynolds	1.807e+005 *	1.807e+005 *	1.807e+005 *	1.807e+005 *
37	Vap. Reynolds	9.928e+006 *	9.928e+006 *	9.928e+006 *	9.928e+006 *
38	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
39	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	3.506e+004	3.506e+004	3.507e+004	3.507e+004
43	Elevation (m)	0.1680	0.1640	0.1600	0.1720
44	Pressure (bar)	66.84	66.84	66.84	66.84
45	Temperature (C)	45.52	45.52	45.51	45.51
46	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
47	Heat Transferred (kJ/h-m)	413.4	413.3	413.3	413.2
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.1701	0.1701	0.1701	0.2368
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	9.117e-003	9.117e-003	9.117e-003	7.688e-003
53	Static Gradient (kPa/m)	-1.267e-003	-1.267e-003	-1.267e-003	4.828e-003
54	Accel. Gradient (kPa/m)	1.021e-006	1.021e-006	1.021e-006	1.628e-006
55	Liq. Reynolds	1.807e+005 *	1.807e+005 *	1.807e+005 *	1.807e+005 *
56	Vap. Reynolds	9.928e+006 *	9.928e+006 *	9.928e+006 *	9.928e+006 *
57	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
58	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	3.508e+004	3.508e+004	3.509e+004	3.509e+004
62	Elevation (m)	0.1840	0.1960	0.2080	0.2200
63	Pressure (bar)	66.84	66.84	66.84	66.84
64	Temperature (C)	45.51	45.51	45.51	45.51
65	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
66	Heat Transferred (kJ/h-m)	413.2	413.2	413.1	413.1
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 559 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.2368	0.2368	0.2368	0.2368
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	7.688e-003	7.688e-003	7.688e-003	7.688e-003
16	Static Gradient (kPa/m)	4.828e-003	4.828e-003	4.828e-003	4.828e-003
17	Accel. Gradient (kPa/m)	1.628e-006	1.628e-006	1.628e-006	1.628e-006
18	Liq. Reynolds	1.807e+005 *	1.807e+005 *	1.807e+005 *	1.807e+005 *
19	Vap. Reynolds	9.928e+006 *	9.928e+006 *	9.928e+006 *	9.928e+006 *
20	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
21	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	3.510e+004	3.510e+004	3.511e+004	3.511e+004
25	Elevation (m)	0.2200	0.2200	0.2200	0.2200
26	Pressure (bar)	66.84	66.84	66.84	66.84
27	Temperature (C)	45.51	45.50	45.50	45.50
28	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
29	Heat Transferred (kJ/h-m)	413.0	413.0	413.0	412.9
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2491	0.2491	0.2491	0.2491
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.246e-003	6.246e-003	6.246e-003	6.246e-003
35	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
36	Accel. Gradient (kPa/m)	8.124e-007	8.124e-007	8.124e-007	8.124e-007
37	Liq. Reynolds	1.807e+005 *	1.807e+005 *	1.807e+005 *	1.807e+005 *
38	Vap. Reynolds	9.928e+006 *	9.928e+006 *	9.928e+006 *	9.928e+006 *
39	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
40	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	3.512e+004	3.512e+004	3.513e+004	3.513e+004
44	Elevation (m)	0.2200	0.2180	0.2160	0.2140
45	Pressure (bar)	66.84	66.84	66.84	66.84
46	Temperature (C)	45.50	45.50	45.50	45.50
47	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
48	Heat Transferred (kJ/h-m)	412.9	412.9	412.9	412.8
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2491	0.1703	0.1703	0.1703
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.246e-003	9.114e-003	9.114e-003	9.114e-003
54	Static Gradient (kPa/m)	0.0000	-6.377e-004	-6.377e-004	-6.377e-004
55	Accel. Gradient (kPa/m)	8.124e-007	1.102e-006	1.102e-006	1.102e-006
56	Liq. Reynolds	1.807e+005 *	1.807e+005 *	1.807e+005 *	1.807e+005 *
57	Vap. Reynolds	9.928e+006 *	9.928e+006 *	9.928e+006 *	9.928e+006 *
58	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
59	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	3.514e+004	3.514e+004	3.515e+004	3.515e+004
63	Elevation (m)	0.2120	0.2100	0.2120	0.2140
64	Pressure (bar)	66.84	66.83	66.83	66.83
65	Temperature (C)	45.50	45.49	45.49	45.49
66	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
67	Heat Transferred (kJ/h-m)	412.8	412.8	412.7	412.7
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 560 of 937

1			Case Name:	Simulation - 386MMSCFD_80barg.hsc		
2			Unit Set:	Mixed		
3	 Epic Atlantic Limited		Date/Time:	Mon Jun 13 15:13:43 2022		
4						
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Water Flow Regime					
14	Liquid HoldUp	0.1703	0.1703	0.2313	0.2313	
15	Water HoldUp	---	---	---	---	
16	Friction Gradient (kPa/m)	9.114e-003	9.114e-003	8.410e-003	8.410e-003	
17	Static Gradient (kPa/m)	-6.377e-004	-6.377e-004	7.909e-004	7.909e-004	
18	Accel. Gradient (kPa/m)	1.102e-006	1.102e-006	1.197e-006	1.197e-006	
19	Liq. Reynolds	1.807e+005 *	1.807e+005 *	1.807e+005 *	1.807e+005 *	
20	Vap. Reynolds	9.928e+006 *	9.928e+006 *	9.928e+006 *	9.928e+006 *	
21	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	
22	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948	
23	Deposit Thickness (mm)	---	---	---	---	
24	Deposit Volume (m3)	---	---	---	---	
25	Length (m)	3.516e+004	3.516e+004	3.517e+004	3.517e+004	
26	Elevation (m)	0.2160	0.2180	0.2200	0.2200	
27	Pressure (bar)	66.83	66.83	66.83	66.83	
28	Temperature (C)	45.49	45.49	45.49	45.49	
29	Vapour Fraction	0.8799	0.8799	0.8799	0.8799	
30	Heat Transferred (kJ/h-m)	412.6	412.6	412.6	412.5	
31	Flow Regime	Transition	Transition	Transition	Transition	
32	Water Flow Regime					
33	Liquid HoldUp	0.2313	0.2313	0.2313	0.2491	
34	Water HoldUp	---	---	---	---	
35	Friction Gradient (kPa/m)	8.410e-003	8.410e-003	8.410e-003	6.246e-003	
36	Static Gradient (kPa/m)	7.909e-004	7.909e-004	7.909e-004	0.0000	
37	Accel. Gradient (kPa/m)	1.197e-006	1.197e-006	1.197e-006	8.125e-007	
38	Liq. Reynolds	1.807e+005 *	1.807e+005 *	1.807e+005 *	1.807e+005 *	
39	Vap. Reynolds	9.928e+006 *	9.928e+006 *	9.928e+006 *	9.928e+006 *	
40	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	
41	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948	
42	Deposit Thickness (mm)	---	---	---	---	
43	Deposit Volume (m3)	---	---	---	---	
44	Length (m)	3.518e+004	3.518e+004	3.519e+004	3.519e+004	
45	Elevation (m)	0.2200	0.2200	0.2200	0.2200	
46	Pressure (bar)	66.83	66.83	66.83	66.83	
47	Temperature (C)	45.49	45.48	45.48	45.48	
48	Vapour Fraction	0.8799	0.8799	0.8799	0.8799	
49	Heat Transferred (kJ/h-m)	412.5	412.4	412.4	412.4	
50	Flow Regime	Transition	Transition	Transition	Transition	
51	Water Flow Regime					
52	Liquid HoldUp	0.2491	0.2491	0.2491	0.2491	
53	Water HoldUp	---	---	---	---	
54	Friction Gradient (kPa/m)	6.246e-003	6.246e-003	6.246e-003	6.246e-003	
55	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000	
56	Accel. Gradient (kPa/m)	8.125e-007	8.125e-007	8.125e-007	8.125e-007	
57	Liq. Reynolds	1.807e+005 *	1.807e+005 *	1.807e+005 *	1.806e+005 *	
58	Vap. Reynolds	9.928e+006 *	9.928e+006 *	9.928e+006 *	9.928e+006 *	
59	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	
60	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948	
61	Deposit Thickness (mm)	---	---	---	---	
62	Deposit Volume (m3)	---	---	---	---	
63	Length (m)	3.520e+004	3.520e+004	3.521e+004	3.521e+004	
64	Elevation (m)	0.2240	0.2280	0.2320	0.2360	
65	Pressure (bar)	66.83	66.83	66.83	66.83	
66	Temperature (C)	45.48	45.48	45.48	45.48	
67	Vapour Fraction	0.8799	0.8799	0.8799	0.8799	
68	Heat Transferred (kJ/h-m)	412.3	412.3	412.3	412.2	
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 561 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2324	0.2324	0.2324	0.2324
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	8.262e-003	8.262e-003	8.262e-003	8.262e-003
18	Static Gradient (kPa/m)	1.587e-003	1.587e-003	1.587e-003	1.587e-003
19	Accel. Gradient (kPa/m)	1.281e-006	1.281e-006	1.281e-006	1.281e-006
20	Liq. Reynolds	1.806e+005 *	1.806e+005 *	1.806e+005 *	1.806e+005 *
21	Vap. Reynolds	9.929e+006 *	9.929e+006 *	9.929e+006 *	9.929e+006 *
22	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
23	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	3.522e+004	3.522e+004	3.523e+004	3.523e+004
27	Elevation (m)	0.2400	0.2420	0.2440	0.2460
28	Pressure (bar)	66.83	66.83	66.83	66.83
29	Temperature (C)	45.47	45.47	45.47	45.47
30	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
31	Heat Transferred (kJ/h-m)	412.2	412.2	412.1	412.1
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2324	0.2313	0.2313	0.2313
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	8.262e-003	8.410e-003	8.410e-003	8.410e-003
37	Static Gradient (kPa/m)	1.587e-003	7.909e-004	7.909e-004	7.909e-004
38	Accel. Gradient (kPa/m)	1.281e-006	1.197e-006	1.197e-006	1.197e-006
39	Liq. Reynolds	1.806e+005 *	1.806e+005 *	1.806e+005 *	1.806e+005 *
40	Vap. Reynolds	9.929e+006 *	9.929e+006 *	9.929e+006 *	9.929e+006 *
41	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
42	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	3.524e+004	3.524e+004	3.525e+004	3.525e+004
46	Elevation (m)	0.2480	0.2500	0.2480	0.2460
47	Pressure (bar)	66.83	66.83	66.83	66.83
48	Temperature (C)	45.47	45.47	45.47	45.47
49	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
50	Heat Transferred (kJ/h-m)	412.1	412.0	412.0	412.0
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2313	0.2313	0.1703	0.1703
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	8.411e-003	8.411e-003	9.114e-003	9.114e-003
56	Static Gradient (kPa/m)	7.909e-004	7.909e-004	-6.309e-004	-6.309e-004
57	Accel. Gradient (kPa/m)	1.197e-006	1.197e-006	1.103e-006	1.103e-006
58	Liq. Reynolds	1.806e+005 *	1.806e+005 *	1.806e+005 *	1.806e+005 *
59	Vap. Reynolds	9.929e+006 *	9.929e+006 *	9.929e+006 *	9.929e+006 *
60	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
61	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	3.526e+004	3.526e+004	3.527e+004	3.527e+004
65	Elevation (m)	0.2440	0.2420	0.2400	0.2320
66	Pressure (bar)	66.82	66.82	66.82	66.82
67	Temperature (C)	45.46	45.46	45.46	45.46
68	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	412.0	411.9	411.9	411.8
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.1703	0.1703	0.1703	0.1695
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	9.114e-003	9.114e-003	9.114e-003	9.126e-003
19	Static Gradient (kPa/m)	-6.309e-004	-6.309e-004	-6.309e-004	-2.542e-003
20	Accel. Gradient (kPa/m)	1.103e-006	1.103e-006	1.103e-006	8.565e-007
21	Liq. Reynolds	1.806e+005 *	1.806e+005 *	1.806e+005 *	1.806e+005 *
22	Vap. Reynolds	9.929e+006 *	9.929e+006 *	9.929e+006 *	9.929e+006 *
23	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
24	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	3.528e+004	3.528e+004	3.529e+004	3.529e+004
28	Elevation (m)	0.2240	0.2160	0.2080	0.2000
29	Pressure (bar)	66.82	66.82	66.82	66.82
30	Temperature (C)	45.46	45.46	45.46	45.45
31	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
32	Heat Transferred (kJ/h-m)	411.8	411.8	411.7	411.7
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.1695	0.1695	0.1695	0.1695
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	9.126e-003	9.126e-003	9.126e-003	9.126e-003
38	Static Gradient (kPa/m)	-2.542e-003	-2.542e-003	-2.542e-003	-2.542e-003
39	Accel. Gradient (kPa/m)	8.565e-007	8.565e-007	8.565e-007	8.565e-007
40	Liq. Reynolds	1.806e+005 *	1.806e+005 *	1.806e+005 *	1.806e+005 *
41	Vap. Reynolds	9.929e+006 *	9.929e+006 *	9.929e+006 *	9.929e+006 *
42	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
43	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	3.530e+004	3.530e+004	3.531e+004	3.531e+004
47	Elevation (m)	0.2020	0.2040	0.2060	0.2080
48	Pressure (bar)	66.82	66.82	66.82	66.82
49	Temperature (C)	45.45	45.45	45.45	45.45
50	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
51	Heat Transferred (kJ/h-m)	411.6	411.6	411.6	411.5
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.2313	0.2313	0.2313	0.2313
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	8.411e-003	8.411e-003	8.411e-003	8.411e-003
57	Static Gradient (kPa/m)	7.909e-004	7.909e-004	7.909e-004	7.909e-004
58	Accel. Gradient (kPa/m)	1.197e-006	1.197e-006	1.197e-006	1.197e-006
59	Liq. Reynolds	1.806e+005 *	1.806e+005 *	1.806e+005 *	1.806e+005 *
60	Vap. Reynolds	9.929e+006 *	9.929e+006 *	9.929e+006 *	9.929e+006 *
61	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
62	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	3.532e+004	3.532e+004	3.533e+004	3.533e+004
66	Elevation (m)	0.2100	0.2120	0.2140	0.2160
67	Pressure (bar)	66.82	66.82	66.82	66.82
68	Temperature (C)	45.45	45.45	45.45	45.44
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
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6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Fraction		0.8799	0.8799	0.8799	0.8799	
14	Heat Transferred (kJ/h-m)		411.5	411.5	411.4	411.4	
15	Flow Regime		Transition	Transition	Transition	Transition	
16	Water Flow Regime						
17	Liquid HoldUp		0.2313	0.2313	0.2313	0.2313	
18	Water HoldUp		---	---	---	---	
19	Friction Gradient (kPa/m)		8.411e-003	8.411e-003	8.411e-003	8.411e-003	
20	Static Gradient (kPa/m)		7.909e-004	7.909e-004	7.909e-004	7.909e-004	
21	Accel. Gradient (kPa/m)		1.197e-006	1.197e-006	1.197e-006	1.197e-006	
22	Liq. Reynolds		1.806e+005 *	1.806e+005 *	1.806e+005 *	1.806e+005 *	
23	Vap. Reynolds		9.929e+006 *	9.929e+006 *	9.929e+006 *	9.929e+006 *	
24	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
25	Vapour Velocity (m/s)		2.948	2.948	2.948	2.948	
26	Deposit Thickness (mm)		---	---	---	---	
27	Deposit Volume (m3)		---	---	---	---	
28	Length (m)		3.534e+004	3.534e+004	3.535e+004	3.535e+004	
29	Elevation (m)		0.2180	0.2200	2.600e-002	-0.1680	
30	Pressure (bar)		66.82	66.82	66.82	66.82	
31	Temperature (C)		45.44	45.44	45.44	45.44	
32	Vapour Fraction		0.8799	0.8799	0.8799	0.8799	
33	Heat Transferred (kJ/h-m)		411.4	411.3	411.4	411.4	
34	Flow Regime		Transition	Transition	Transition	Transition	
35	Water Flow Regime						
36	Liquid HoldUp		0.2313	0.2313	0.1422	0.1422	
37	Water HoldUp		---	---	---	---	
38	Friction Gradient (kPa/m)		8.411e-003	8.411e-003	9.598e-003	9.598e-003	
39	Static Gradient (kPa/m)		7.909e-004	7.909e-004	-5.498e-002	-5.499e-002	
40	Accel. Gradient (kPa/m)		1.197e-006	1.197e-006	-5.904e-006	-5.904e-006	
41	Liq. Reynolds		1.806e+005 *	1.806e+005 *	1.806e+005 *	1.806e+005 *	
42	Vap. Reynolds		9.929e+006 *	9.929e+006 *	9.929e+006 *	9.929e+006 *	
43	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
44	Vapour Velocity (m/s)		2.948	2.948	2.948	2.948	
45	Deposit Thickness (mm)		---	---	---	---	
46	Deposit Volume (m3)		---	---	---	---	
47	Length (m)		3.536e+004	3.536e+004	3.537e+004	3.537e+004	
48	Elevation (m)		-0.3620	-0.5560	-0.7500	-0.5520	
49	Pressure (bar)		66.82	66.83	66.83	66.82	
50	Temperature (C)		45.44	45.44	45.44	45.44	
51	Vapour Fraction		0.8799	0.8799	0.8799	0.8799	
52	Heat Transferred (kJ/h-m)		411.4	411.4	411.4	411.3	
53	Flow Regime		Transition	Transition	Transition	Transition	
54	Water Flow Regime						
55	Liquid HoldUp		0.1422	0.1422	0.1422	0.3392	
56	Water HoldUp		---	---	---	---	
57	Friction Gradient (kPa/m)		9.598e-003	9.597e-003	9.597e-003	7.885e-003	
58	Static Gradient (kPa/m)		-5.499e-002	-5.499e-002	-5.499e-002	0.1052	
59	Accel. Gradient (kPa/m)		-5.904e-006	-5.904e-006	-5.903e-006	1.470e-005	
60	Liq. Reynolds		1.806e+005 *	1.806e+005 *	1.806e+005 *	1.806e+005 *	
61	Vap. Reynolds		9.929e+006 *	9.929e+006 *	9.929e+006 *	9.929e+006 *	
62	Liquid Velocity (m/s)		0.1980	0.1980	0.1980	0.1980	
63	Vapour Velocity (m/s)		2.948	2.947	2.947	2.948	
64	Deposit Thickness (mm)		---	---	---	---	
65	Deposit Volume (m3)		---	---	---	---	
66	Length (m)		3.538e+004	3.538e+004	3.539e+004	3.539e+004	
67	Elevation (m)		-0.3540	-0.1560	4.200e-002	0.2400	
68	Pressure (bar)		66.82	66.81	66.81	66.80	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:13:43 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	45.43	45.43	45.43	45.43
14	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
15	Heat Transferred (kJ/h-m)	411.2	411.1	411.0	410.9
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.3391	0.3391	0.3391	0.3391
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	7.885e-003	7.886e-003	7.886e-003	7.887e-003
21	Static Gradient (kPa/m)	0.1051	0.1051	0.1051	0.1051
22	Accel. Gradient (kPa/m)	1.470e-005	1.471e-005	1.471e-005	1.471e-005
23	Liq. Reynolds	1.806e+005 *	1.806e+005 *	1.806e+005 *	1.806e+005 *
24	Vap. Reynolds	9.929e+006 *	9.929e+006 *	9.930e+006 *	9.930e+006 *
25	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
26	Vapour Velocity (m/s)	2.948	2.948	2.948	2.949
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	3.540e+004	3.540e+004	3.541e+004	3.541e+004
30	Elevation (m)	0.2580	0.2760	0.2940	0.3120
31	Pressure (bar)	66.80	66.80	66.80	66.80
32	Temperature (C)	45.42	45.42	45.42	45.42
33	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
34	Heat Transferred (kJ/h-m)	410.9	410.8	410.8	410.7
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2401	0.2401	0.2401	0.2401
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	7.281e-003	7.282e-003	7.282e-003	7.282e-003
40	Static Gradient (kPa/m)	7.317e-003	7.317e-003	7.317e-003	7.317e-003
41	Accel. Gradient (kPa/m)	1.900e-006	1.900e-006	1.900e-006	1.900e-006
42	Liq. Reynolds	1.806e+005 *	1.806e+005 *	1.806e+005 *	1.806e+005 *
43	Vap. Reynolds	9.930e+006 *	9.930e+006 *	9.930e+006 *	9.930e+006 *
44	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
45	Vapour Velocity (m/s)	2.949	2.949	2.949	2.949
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	3.542e+004	3.542e+004	3.543e+004	3.543e+004
49	Elevation (m)	0.3300	0.2940	0.2580	0.2220
50	Pressure (bar)	66.80	66.80	66.80	66.80
51	Temperature (C)	45.42	45.42	45.42	45.41
52	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
53	Heat Transferred (kJ/h-m)	410.7	410.7	410.7	410.7
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2401	0.1653	0.1653	0.1653
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	7.282e-003	9.186e-003	9.186e-003	9.186e-003
59	Static Gradient (kPa/m)	7.317e-003	-1.125e-002	-1.125e-002	-1.125e-002
60	Accel. Gradient (kPa/m)	1.900e-006	-2.687e-007	-2.688e-007	-2.688e-007
61	Liq. Reynolds	1.806e+005 *	1.806e+005 *	1.806e+005 *	1.806e+005 *
62	Vap. Reynolds	9.930e+006 *	9.930e+006 *	9.930e+006 *	9.930e+006 *
63	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
64	Vapour Velocity (m/s)	2.949	2.949	2.949	2.949
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	3.544e+004	3.544e+004	3.545e+004	3.545e+004
68	Elevation (m)	0.1860	0.1500	0.1460	0.1420
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:13:43 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Pressure (bar)	66.80	66.80	66.80	66.80					
14	Temperature (C)	45.41	45.41	45.41	45.41					
15	Vapour Fraction	0.8799	0.8799	0.8799	0.8799					
16	Heat Transferred (kJ/h-m)	410.6	410.6	410.6	410.5					
17	Flow Regime	Transition	Transition	Transition	Transition					
18	Water Flow Regime									
19	Liquid HoldUp	0.1653	0.1653	0.1700	0.1700					
20	Water HoldUp	---	---	---	---					
21	Friction Gradient (kPa/m)	9.186e-003	9.186e-003	9.120e-003	9.120e-003					
22	Static Gradient (kPa/m)	-1.125e-002	-1.125e-002	-1.285e-003	-1.285e-003					
23	Accel. Gradient (kPa/m)	-2.688e-007	-2.688e-007	1.020e-006	1.020e-006					
24	Liq. Reynolds	1.806e+005 *	1.806e+005 *	1.806e+005 *	1.806e+005 *					
25	Vap. Reynolds	9.930e+006 *	9.930e+006 *	9.930e+006 *	9.930e+006 *					
26	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
27	Vapour Velocity (m/s)	2.949	2.948	2.948	2.948					
28	Deposit Thickness (mm)	---	---	---	---					
29	Deposit Volume (m3)	---	---	---	---					
30	Length (m)	3.546e+004	3.546e+004	3.547e+004	3.547e+004					
31	Elevation (m)	0.1380	0.1340	0.1300	0.1320					
32	Pressure (bar)	66.80	66.80	66.80	66.80					
33	Temperature (C)	45.41	45.41	45.41	45.40					
34	Vapour Fraction	0.8799	0.8799	0.8799	0.8799					
35	Heat Transferred (kJ/h-m)	410.5	410.5	410.4	410.3					
36	Flow Regime	Transition	Transition	Transition	Transition					
37	Water Flow Regime									
38	Liquid HoldUp	0.1700	0.1700	0.1700	0.2312					
39	Water HoldUp	---	---	---	---					
40	Friction Gradient (kPa/m)	9.120e-003	9.120e-003	9.120e-003	8.413e-003					
41	Static Gradient (kPa/m)	-1.285e-003	-1.285e-003	-1.285e-003	7.908e-004					
42	Accel. Gradient (kPa/m)	1.020e-006	1.020e-006	1.020e-006	1.198e-006					
43	Liq. Reynolds	1.806e+005 *	1.806e+005 *	1.806e+005 *	1.805e+005 *					
44	Vap. Reynolds	9.930e+006 *	9.930e+006 *	9.930e+006 *	9.930e+006 *					
45	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
46	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948					
47	Deposit Thickness (mm)	---	---	---	---					
48	Deposit Volume (m3)	---	---	---	---					
49	Length (m)	3.548e+004	3.548e+004	3.549e+004	3.549e+004					
50	Elevation (m)	0.1340	0.1360	0.1380	0.1400					
51	Pressure (bar)	66.79	66.79	66.79	66.79					
52	Temperature (C)	45.40	45.40	45.40	45.40					
53	Vapour Fraction	0.8799	0.8799	0.8799	0.8799					
54	Heat Transferred (kJ/h-m)	410.3	410.3	410.2	410.2					
55	Flow Regime	Transition	Transition	Transition	Transition					
56	Water Flow Regime									
57	Liquid HoldUp	0.2312	0.2312	0.2312	0.2312					
58	Water HoldUp	---	---	---	---					
59	Friction Gradient (kPa/m)	8.413e-003	8.413e-003	8.413e-003	8.413e-003					
60	Static Gradient (kPa/m)	7.908e-004	7.908e-004	7.908e-004	7.908e-004					
61	Accel. Gradient (kPa/m)	1.198e-006	1.198e-006	1.198e-006	1.198e-006					
62	Liq. Reynolds	1.805e+005 *	1.805e+005 *	1.805e+005 *	1.805e+005 *					
63	Vap. Reynolds	9.930e+006 *	9.930e+006 *	9.930e+006 *	9.930e+006 *					
64	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980					
65	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948					
66	Deposit Thickness (mm)	---	---	---	---					
67	Deposit Volume (m3)	---	---	---	---					
68	Length (m)	3.550e+004	3.550e+004	3.551e+004	3.551e+004					
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1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	0.1360	0.1320	0.1280	0.1240
14	Pressure (bar)	66.79	66.79	66.79	66.79
15	Temperature (C)	45.40	45.40	45.40	45.39
16	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
17	Heat Transferred (kJ/h-m)	410.2	410.2	410.1	410.1
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.1700	0.1700	0.1700	0.1700
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	9.120e-003	9.120e-003	9.120e-003	9.120e-003
23	Static Gradient (kPa/m)	-1.274e-003	-1.274e-003	-1.274e-003	-1.274e-003
24	Accel. Gradient (kPa/m)	1.021e-006	1.021e-006	1.021e-006	1.021e-006
25	Liq. Reynolds	1.805e+005 *	1.805e+005 *	1.805e+005 *	1.805e+005 *
26	Vap. Reynolds	9.930e+006 *	9.930e+006 *	9.930e+006 *	9.930e+006 *
27	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
28	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	3.552e+004	3.552e+004	3.553e+004	3.553e+004
32	Elevation (m)	0.1200	0.1220	0.1240	0.1260
33	Pressure (bar)	66.79	66.79	66.79	66.79
34	Temperature (C)	45.39	45.39	45.39	45.39
35	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
36	Heat Transferred (kJ/h-m)	410.1	410.0	410.0	409.9
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.1700	0.2312	0.2312	0.2312
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	9.120e-003	8.413e-003	8.413e-003	8.413e-003
42	Static Gradient (kPa/m)	-1.274e-003	7.908e-004	7.908e-004	7.908e-004
43	Accel. Gradient (kPa/m)	1.021e-006	1.198e-006	1.198e-006	1.198e-006
44	Liq. Reynolds	1.805e+005 *	1.805e+005 *	1.805e+005 *	1.805e+005 *
45	Vap. Reynolds	9.930e+006 *	9.930e+006 *	9.930e+006 *	9.930e+006 *
46	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
47	Vapour Velocity (m/s)	2.948	2.948	2.948	2.949
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	3.554e+004	3.554e+004	3.555e+004	3.555e+004
51	Elevation (m)	0.1280	0.1300	0.1260	0.1220
52	Pressure (bar)	66.79	66.79	66.79	66.79
53	Temperature (C)	45.39	45.39	45.39	45.38
54	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
55	Heat Transferred (kJ/h-m)	409.9	409.9	409.9	409.8
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2312	0.2312	0.1700	0.1700
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	8.413e-003	8.413e-003	9.120e-003	9.120e-003
61	Static Gradient (kPa/m)	7.908e-004	7.908e-004	-1.274e-003	-1.274e-003
62	Accel. Gradient (kPa/m)	1.198e-006	1.198e-006	1.021e-006	1.021e-006
63	Liq. Reynolds	1.805e+005 *	1.805e+005 *	1.805e+005 *	1.805e+005 *
64	Vap. Reynolds	9.930e+006 *	9.930e+006 *	9.930e+006 *	9.930e+006 *
65	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
66	Vapour Velocity (m/s)	2.949	2.949	2.949	2.949
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 567 of 937

1			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	3.556e+004	3.556e+004	3.557e+004	3.557e+004
14	Elevation (m)	0.1180	0.1140	0.1100	0.1100
15	Pressure (bar)	66.79	66.79	66.79	66.79
16	Temperature (C)	45.38	45.38	45.38	45.38
17	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
18	Heat Transferred (kJ/h-m)	409.8	409.8	409.7	409.7
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.1700	0.1700	0.1700	0.2491
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	9.120e-003	9.120e-003	9.120e-003	6.249e-003
24	Static Gradient (kPa/m)	-1.274e-003	-1.274e-003	-1.274e-003	0.0000
25	Accel. Gradient (kPa/m)	1.021e-006	1.021e-006	1.021e-006	8.135e-007
26	Liq. Reynolds	1.805e+005 *	1.805e+005 *	1.805e+005 *	1.805e+005 *
27	Vap. Reynolds	9.930e+006 *	9.930e+006 *	9.930e+006 *	9.930e+006 *
28	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
29	Vapour Velocity (m/s)	2.949	2.949	2.949	2.949
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	3.557e+004	3.558e+004	3.558e+004	3.559e+004
33	Elevation (m)	0.1100	0.1100	0.1100	0.1100
34	Pressure (bar)	66.79	66.79	66.79	66.79
35	Temperature (C)	45.38	45.38	45.38	45.37
36	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
37	Heat Transferred (kJ/h-m)	409.6	409.6	409.6	409.5
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.2491	0.2491	0.2491	0.2491
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	6.249e-003	6.249e-003	6.249e-003	6.249e-003
43	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
44	Accel. Gradient (kPa/m)	8.135e-007	8.135e-007	8.135e-007	8.135e-007
45	Liq. Reynolds	1.805e+005 *	1.805e+005 *	1.805e+005 *	1.805e+005 *
46	Vap. Reynolds	9.930e+006 *	9.930e+006 *	9.931e+006 *	9.931e+006 *
47	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
48	Vapour Velocity (m/s)	2.948	2.948	2.948	2.948
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	3.559e+004	3.559e+004	3.560e+004	3.560e+004
52	Elevation (m)	0.1160	0.1220	0.1280	0.1340
53	Pressure (bar)	66.79	66.79	66.78	66.78
54	Temperature (C)	45.37	45.37	45.37	45.37
55	Vapour Fraction	0.8799	0.8799	0.8799	0.8799
56	Heat Transferred (kJ/h-m)	409.5	409.5	409.5	409.5
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.2357	0.2357	0.2357	0.2357
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	7.831e-003	7.832e-003	7.832e-003	7.832e-003
62	Static Gradient (kPa/m)	4.010e-003	4.010e-003	4.010e-003	4.010e-003
63	Accel. Gradient (kPa/m)	1.542e-006	1.542e-006	1.542e-006	1.542e-006
64	Liq. Reynolds	1.805e+005 *	1.805e+005 *	1.805e+005 *	1.805e+005 *
65	Vap. Reynolds	9.931e+006 *	9.931e+006 *	9.931e+006 *	9.931e+006 *
66	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980
67	Vapour Velocity (m/s)	2.948	2.948	2.949	2.949
68	Deposit Thickness (mm)	---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 568 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---	---	---
14	Length (m)	3.560e+004	3.561e+004	3.561e+004	3.562e+004	3.562e+004	3.562e+004
15	Elevation (m)	0.1400	0.1540	0.1680	0.1820	0.1820	0.1820
16	Pressure (bar)	66.78	66.78	66.78	66.78	66.78	66.78
17	Temperature (C)	45.37	45.37	45.37	45.37	45.37	45.37
18	Vapour Fraction	0.8799	0.8799	0.8799	0.8799	0.8799	0.8799
19	Heat Transferred (kJ/h-m)	409.4	409.4	409.4	409.4	409.3	409.3
20	Flow Regime	Transition	Transition	Transition	Transition	Transition	Transition
21	Water Flow Regime						
22	Liquid HoldUp	0.2357	0.2379	0.2379	0.2379	0.2379	0.2379
23	Water HoldUp	---	---	---	---	---	---
24	Friction Gradient (kPa/m)	7.832e-003	7.553e-003	7.553e-003	7.553e-003	7.553e-003	7.553e-003
25	Static Gradient (kPa/m)	4.010e-003	5.652e-003	5.652e-003	5.652e-003	5.652e-003	5.652e-003
26	Accel. Gradient (kPa/m)	1.542e-006	1.719e-006	1.719e-006	1.719e-006	1.719e-006	1.719e-006
27	Liq. Reynolds	1.805e+005 *	1.805e+005 *	1.805e+005 *	1.805e+005 *	1.805e+005 *	1.805e+005 *
28	Vap. Reynolds	9.931e+006 *	9.931e+006 *	9.931e+006 *	9.931e+006 *	9.931e+006 *	9.931e+006 *
29	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	0.1980	0.1980
30	Vapour Velocity (m/s)	2.949	2.949	2.949	2.949	2.949	2.949
31	Deposit Thickness (mm)	---	---	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---	---	---
33	Length (m)	3.562e+004	3.563e+004	3.563e+004	3.564e+004	3.564e+004	3.564e+004
34	Elevation (m)	0.1960	0.2100	0.1940	0.1780	0.1780	0.1780
35	Pressure (bar)	66.78	66.78	66.78	66.78	66.78	66.78
36	Temperature (C)	45.36	45.36	45.36	45.36	45.36	45.36
37	Vapour Fraction	0.8799	0.8799	0.8799	0.8799	0.8799	0.8799
38	Heat Transferred (kJ/h-m)	409.3	409.2	409.3	409.2	409.2	409.2
39	Flow Regime	Transition	Transition	Transition	Transition	Transition	Transition
40	Water Flow Regime						
41	Liquid HoldUp	0.2379	0.2379	0.1683	0.1683	0.1683	0.1683
42	Water HoldUp	---	---	---	---	---	---
43	Friction Gradient (kPa/m)	7.553e-003	7.553e-003	9.144e-003	9.144e-003	9.144e-003	9.144e-003
44	Static Gradient (kPa/m)	5.652e-003	5.652e-003	-5.059e-003	-5.060e-003	-5.060e-003	-5.060e-003
45	Accel. Gradient (kPa/m)	1.719e-006	1.719e-006	5.318e-007	5.318e-007	5.318e-007	5.318e-007
46	Liq. Reynolds	1.805e+005 *	1.805e+005 *	1.805e+005 *	1.805e+005 *	1.805e+005 *	1.805e+005 *
47	Vap. Reynolds	9.931e+006 *	9.931e+006 *	9.931e+006 *	9.931e+006 *	9.931e+006 *	9.931e+006 *
48	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	0.1980	0.1980
49	Vapour Velocity (m/s)	2.949	2.949	2.949	2.949	2.949	2.949
50	Deposit Thickness (mm)	---	---	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---	---	---
52	Length (m)	3.564e+004	3.565e+004	3.565e+004	3.566e+004	3.566e+004	3.566e+004
53	Elevation (m)	0.1620	0.1460	0.1300	0.1840	0.1840	0.1840
54	Pressure (bar)	66.78	66.78	66.78	66.78	66.78	66.78
55	Temperature (C)	45.36	45.36	45.36	45.36	45.36	45.36
56	Vapour Fraction	0.8799	0.8799	0.8799	0.8799	0.8799	0.8799
57	Heat Transferred (kJ/h-m)	409.2	409.2	409.1	409.0	409.0	409.0
58	Flow Regime	Transition	Transition	Transition	Transition	Transition	Transition
59	Water Flow Regime						
60	Liquid HoldUp	0.1683	0.1683	0.1683	0.2597	0.2597	0.2597
61	Water HoldUp	---	---	---	---	---	---
62	Friction Gradient (kPa/m)	9.144e-003	9.144e-003	9.144e-003	7.780e-003	7.780e-003	7.780e-003
63	Static Gradient (kPa/m)	-5.060e-003	-5.060e-003	-5.060e-003	2.310e-002	2.310e-002	2.310e-002
64	Accel. Gradient (kPa/m)	5.318e-007	5.318e-007	5.318e-007	4.020e-006	4.020e-006	4.020e-006
65	Liq. Reynolds	1.805e+005 *	1.805e+005 *	1.805e+005 *	1.805e+005 *	1.805e+005 *	1.805e+005 *
66	Vap. Reynolds	9.931e+006 *	9.931e+006 *	9.931e+006 *	9.931e+006 *	9.931e+006 *	9.931e+006 *
67	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980	0.1980	0.1980
68	Vapour Velocity (m/s)	2.949	2.949	2.949	2.949	2.949	2.949
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 569 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:13:43 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Thickness (mm)	---	---	---	---		
14	Deposit Volume (m3)	---	---	---	---		
15	Length (m)	3.566e+004	3.567e+004	3.567e+004	3.568e+004		
16	Elevation (m)	0.2380	0.2920	0.3460	0.4000		
17	Pressure (bar)	66.78	66.78	66.77	66.77		
18	Temperature (C)	45.35	45.35	45.35	45.35		
19	Vapour Fraction	0.8799	0.8799	0.8799	0.8799		
20	Heat Transferred (kJ/h-m)	409.0	408.9	408.9	408.8		
21	Flow Regime	Transition	Transition	Transition	Transition		
22	Water Flow Regime						
23	Liquid HoldUp	0.2597	0.2597	0.2597	0.2597		
24	Water HoldUp	---	---	---	---		
25	Friction Gradient (kPa/m)	7.780e-003	7.780e-003	7.781e-003	7.781e-003		
26	Static Gradient (kPa/m)	2.310e-002	2.310e-002	2.310e-002	2.310e-002		
27	Accel. Gradient (kPa/m)	4.021e-006	4.021e-006	4.021e-006	4.021e-006		
28	Liq. Reynolds	1.805e+005 *	1.805e+005 *	1.805e+005 *	1.805e+005 *		
29	Vap. Reynolds	9.931e+006 *	9.931e+006 *	9.931e+006 *	9.931e+006 *		
30	Liquid Velocity (m/s)	0.1980	0.1980	0.1980	0.1980		
31	Vapour Velocity (m/s)	2.949	2.949	2.949	2.949		
32	Deposit Thickness (mm)	---	---	---	---		
33	Deposit Volume (m3)	---	---	---	---		
34	Erosion						
35							
36	Model						
37							
38	Erosion Model:	API-RP-14E Continuous service		Empirical Constant:		100.0	
39							
40	Erosion Table						
41	Length (m)	0.0000	5.000	10.00	15.00	20.00	
42	Bulk Density (kg/m3)	104.1	104.1	104.1	104.1	104.1	
43	Bulk Velocity (m/s)	2.832	2.832	2.832	2.832	2.832	
44	Erosion Velocity (m/s)	11.96	11.96	11.96	11.96	11.96	
45	Length (m)	25.00	30.00	35.00	40.00	45.00	
46	Bulk Density (kg/m3)	104.1	104.1	104.1	104.1	104.1	
47	Bulk Velocity (m/s)	2.832	2.832	2.832	2.832	2.832	
48	Erosion Velocity (m/s)	11.96	11.96	11.96	11.96	11.96	
49	Length (m)	50.00	55.00	60.00	65.00	70.00	
50	Bulk Density (kg/m3)	104.1	104.1	104.1	104.1	104.1	
51	Bulk Velocity (m/s)	2.832	2.832	2.832	2.832	2.832	
52	Erosion Velocity (m/s)	11.96	11.96	11.96	11.96	11.96	
53	Length (m)	75.00	80.00	85.00	90.00	95.00	
54	Bulk Density (kg/m3)	104.1	104.1	104.1	104.1	104.1	
55	Bulk Velocity (m/s)	2.832	2.832	2.832	2.832	2.832	
56	Erosion Velocity (m/s)	11.96	11.96	11.96	11.96	11.96	
57	Length (m)	100.0	105.0	110.0	115.0	120.0	
58	Bulk Density (kg/m3)	104.1	104.1	104.1	104.1	104.1	
59	Bulk Velocity (m/s)	2.832	2.832	2.832	2.832	2.832	
60	Erosion Velocity (m/s)	11.96	11.96	11.96	11.96	11.96	
61	Length (m)	125.0	130.0	135.0	140.0	145.0	
62	Bulk Density (kg/m3)	104.1	104.1	104.1	104.1	104.1	
63	Bulk Velocity (m/s)	2.832	2.832	2.832	2.832	2.832	
64	Erosion Velocity (m/s)	11.96	11.96	11.96	11.96	11.96	
65	Length (m)	150.0	155.0	160.0	165.0	170.0	
66	Bulk Density (kg/m3)	104.1	104.1	104.1	104.1	104.1	
67	Bulk Velocity (m/s)	2.832	2.832	2.832	2.832	2.832	
68	Erosion Velocity (m/s)	11.96	11.96	11.96	11.96	11.96	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 570 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	175.0	180.0	185.0	190.0	195.0
14	Bulk Density (kg/m3)	104.1	104.1	104.1	104.1	104.1
15	Bulk Velocity (m/s)	2.832	2.832	2.832	2.832	2.832
16	Erosion Velocity (m/s)	11.96	11.96	11.96	11.96	11.96
17	Length (m)	200.1	205.1	210.1	215.1	220.1
18	Bulk Density (kg/m3)	104.1	104.1	104.1	104.1	104.1
19	Bulk Velocity (m/s)	2.832	2.832	2.832	2.832	2.832
20	Erosion Velocity (m/s)	11.96	11.95	11.95	11.95	11.95
21	Length (m)	225.1	230.1	235.1	240.1	245.1
22	Bulk Density (kg/m3)	104.1	104.1	104.1	104.1	104.1
23	Bulk Velocity (m/s)	2.832	2.832	2.832	2.832	2.832
24	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
25	Length (m)	250.1	255.1	260.1	265.1	270.1
26	Bulk Density (kg/m3)	104.1	104.1	104.1	104.1	104.1
27	Bulk Velocity (m/s)	2.832	2.831	2.831	2.831	2.831
28	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
29	Length (m)	275.1	280.1	285.1	290.1	295.1
30	Bulk Density (kg/m3)	104.1	104.1	104.1	104.1	104.1
31	Bulk Velocity (m/s)	2.831	2.831	2.831	2.831	2.831
32	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
33	Length (m)	300.1	305.1	310.1	315.1	320.1
34	Bulk Density (kg/m3)	104.1	104.1	104.1	104.1	104.1
35	Bulk Velocity (m/s)	2.831	2.831	2.831	2.831	2.831
36	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
37	Length (m)	325.1	330.1	335.1	340.1	345.1
38	Bulk Density (kg/m3)	104.1	104.1	104.1	104.1	104.1
39	Bulk Velocity (m/s)	2.831	2.831	2.831	2.831	2.831
40	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
41	Length (m)	350.1	355.1	360.1	365.1	370.1
42	Bulk Density (kg/m3)	104.2	104.1	104.1	104.1	104.1
43	Bulk Velocity (m/s)	2.831	2.831	2.831	2.831	2.831
44	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
45	Length (m)	375.1	380.0	385.0	390.0	395.0
46	Bulk Density (kg/m3)	104.1	104.1	104.1	104.1	104.1
47	Bulk Velocity (m/s)	2.831	2.831	2.831	2.831	2.831
48	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
49	Length (m)	400.0	405.0	410.0	415.0	420.0
50	Bulk Density (kg/m3)	104.1	104.1	104.1	104.1	104.1
51	Bulk Velocity (m/s)	2.831	2.831	2.831	2.831	2.831
52	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
53	Length (m)	425.0	430.0	435.0	440.0	445.0
54	Bulk Density (kg/m3)	104.1	104.1	104.1	104.1	104.1
55	Bulk Velocity (m/s)	2.831	2.831	2.831	2.831	2.831
56	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
57	Length (m)	450.0	455.0	460.0	465.0	470.0
58	Bulk Density (kg/m3)	104.1	104.1	104.1	104.1	104.1
59	Bulk Velocity (m/s)	2.831	2.831	2.831	2.831	2.831
60	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
61	Length (m)	475.0	480.0	485.0	490.0	495.0
62	Bulk Density (kg/m3)	104.1	104.1	104.1	104.1	104.1
63	Bulk Velocity (m/s)	2.831	2.831	2.831	2.831	2.831
64	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
65	Length (m)	500.0	505.0	510.0	515.0	520.0
66	Bulk Density (kg/m3)	104.1	104.1	104.2	104.2	104.2
67	Bulk Velocity (m/s)	2.831	2.831	2.831	2.831	2.831
68	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 571 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	525.0	530.0	535.0	540.0	545.0
14	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
15	Bulk Velocity (m/s)	2.831	2.831	2.831	2.831	2.831
16	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
17	Length (m)	550.0	555.0	560.0	565.0	570.0
18	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
19	Bulk Velocity (m/s)	2.831	2.831	2.831	2.831	2.831
20	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
21	Length (m)	575.0	580.0	585.0	590.0	595.0
22	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
23	Bulk Velocity (m/s)	2.831	2.831	2.831	2.831	2.831
24	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
25	Length (m)	600.0	605.0	610.0	615.0	620.0
26	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
27	Bulk Velocity (m/s)	2.831	2.831	2.831	2.831	2.831
28	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
29	Length (m)	625.0	630.0	635.0	640.0	645.0
30	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
31	Bulk Velocity (m/s)	2.831	2.831	2.831	2.831	2.831
32	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
33	Length (m)	650.0	655.0	660.0	665.0	670.0
34	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
35	Bulk Velocity (m/s)	2.831	2.831	2.830	2.830	2.830
36	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
37	Length (m)	675.0	682.0	689.0	696.0	703.0
38	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
39	Bulk Velocity (m/s)	2.830	2.830	2.830	2.830	2.830
40	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
41	Length (m)	710.0	712.0	714.0	716.0	718.0
42	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
43	Bulk Velocity (m/s)	2.830	2.830	2.830	2.830	2.830
44	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
45	Length (m)	720.0	722.0	724.0	726.0	728.0
46	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
47	Bulk Velocity (m/s)	2.830	2.830	2.830	2.830	2.830
48	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
49	Length (m)	730.0	732.0	734.0	736.0	738.0
50	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
51	Bulk Velocity (m/s)	2.830	2.830	2.830	2.830	2.830
52	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
53	Length (m)	740.0	742.0	744.0	746.0	748.0
54	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
55	Bulk Velocity (m/s)	2.830	2.830	2.830	2.830	2.830
56	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
57	Length (m)	750.0	755.0	760.0	765.0	770.0
58	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
59	Bulk Velocity (m/s)	2.830	2.830	2.830	2.830	2.830
60	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
61	Length (m)	775.0	780.0	785.0	790.1	795.1
62	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
63	Bulk Velocity (m/s)	2.830	2.830	2.830	2.830	2.830
64	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
65	Length (m)	800.1	805.1	810.1	815.1	820.1
66	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
67	Bulk Velocity (m/s)	2.830	2.830	2.830	2.830	2.830
68	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	825.1	830.1	835.1	840.1	845.1
14	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
15	Bulk Velocity (m/s)	2.830	2.830	2.830	2.830	2.830
16	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
17	Length (m)	850.1	855.1	860.1	865.1	870.1
18	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
19	Bulk Velocity (m/s)	2.830	2.830	2.830	2.830	2.830
20	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
21	Length (m)	875.1	880.1	885.1	890.1	895.1
22	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
23	Bulk Velocity (m/s)	2.830	2.830	2.830	2.830	2.830
24	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
25	Length (m)	900.1	905.1	910.1	915.1	920.1
26	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
27	Bulk Velocity (m/s)	2.830	2.830	2.830	2.830	2.830
28	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
29	Length (m)	925.1	930.1	935.1	940.1	945.1
30	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
31	Bulk Velocity (m/s)	2.830	2.830	2.830	2.830	2.830
32	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
33	Length (m)	950.1	955.1	960.1	965.1	970.1
34	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
35	Bulk Velocity (m/s)	2.830	2.830	2.830	2.830	2.830
36	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
37	Length (m)	975.1	980.1	985.1	990.1	995.1
38	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
39	Bulk Velocity (m/s)	2.829	2.829	2.829	2.829	2.829
40	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
41	Length (m)	1000	1005	1010	1015	1020
42	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
43	Bulk Velocity (m/s)	2.829	2.829	2.829	2.829	2.829
44	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
45	Length (m)	1025	1030	1035	1040	1045
46	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
47	Bulk Velocity (m/s)	2.829	2.829	2.829	2.829	2.829
48	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
49	Length (m)	1050	1055	1060	1065	1070
50	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
51	Bulk Velocity (m/s)	2.829	2.829	2.829	2.829	2.829
52	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
53	Length (m)	1075	1080	1085	1090	1095
54	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
55	Bulk Velocity (m/s)	2.829	2.829	2.829	2.829	2.829
56	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
57	Length (m)	1100	1105	1110	1115	1120
58	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
59	Bulk Velocity (m/s)	2.829	2.829	2.829	2.829	2.829
60	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
61	Length (m)	1125	1130	1135	1140	1145
62	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
63	Bulk Velocity (m/s)	2.829	2.829	2.829	2.829	2.829
64	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
65	Length (m)	1150	1155	1160	1165	1170
66	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
67	Bulk Velocity (m/s)	2.829	2.829	2.829	2.829	2.829
68	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1175	1180	1185	1190	1195
14	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
15	Bulk Velocity (m/s)	2.829	2.829	2.829	2.829	2.829
16	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
17	Length (m)	1200	1205	1210	1215	1220
18	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
19	Bulk Velocity (m/s)	2.829	2.829	2.829	2.829	2.829
20	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
21	Length (m)	1225	1230	1235	1240	1245
22	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
23	Bulk Velocity (m/s)	2.829	2.829	2.829	2.829	2.829
24	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
25	Length (m)	1250	1255	1260	1265	1270
26	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
27	Bulk Velocity (m/s)	2.829	2.829	2.829	2.829	2.829
28	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
29	Length (m)	1275	1280	1285	1290	1295
30	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
31	Bulk Velocity (m/s)	2.829	2.829	2.829	2.829	2.829
32	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
33	Length (m)	1300	1305	1310	1315	1320
34	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
35	Bulk Velocity (m/s)	2.829	2.828	2.828	2.828	2.828
36	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
37	Length (m)	1325	1330	1335	1340	1345
38	Bulk Density (kg/m3)	104.2	104.2	104.2	104.2	104.2
39	Bulk Velocity (m/s)	2.828	2.828	2.828	2.828	2.828
40	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
41	Length (m)	1350	1355	1360	1365	1370
42	Bulk Density (kg/m3)	104.2	104.2	104.2	104.3	104.3
43	Bulk Velocity (m/s)	2.828	2.828	2.828	2.828	2.828
44	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
45	Length (m)	1375	1380	1385	1390	1395
46	Bulk Density (kg/m3)	104.3	104.3	104.3	104.3	104.3
47	Bulk Velocity (m/s)	2.828	2.828	2.828	2.827	2.827
48	Erosion Velocity (m/s)	11.95	11.95	11.95	11.95	11.95
49	Length (m)	1400	1405	1410	1415	1420
50	Bulk Density (kg/m3)	104.3	104.3	104.3	104.3	104.3
51	Bulk Velocity (m/s)	2.827	2.827	2.827	2.826	2.826
52	Erosion Velocity (m/s)	11.94	11.94	11.94	11.94	11.94
53	Length (m)	1425	1430	1435	1440	1445
54	Bulk Density (kg/m3)	104.3	104.3	104.3	104.3	104.3
55	Bulk Velocity (m/s)	2.826	2.826	2.826	2.826	2.826
56	Erosion Velocity (m/s)	11.94	11.94	11.94	11.94	11.94
57	Length (m)	1450	1455	1460	1465	1470
58	Bulk Density (kg/m3)	104.3	104.3	104.3	104.3	104.3
59	Bulk Velocity (m/s)	2.826	2.826	2.826	2.826	2.826
60	Erosion Velocity (m/s)	11.94	11.94	11.94	11.94	11.94
61	Length (m)	1475	1480	1485	1490	1495
62	Bulk Density (kg/m3)	104.3	104.3	104.3	104.3	104.3
63	Bulk Velocity (m/s)	2.826	2.826	2.826	2.826	2.826
64	Erosion Velocity (m/s)	11.94	11.94	11.94	11.94	11.94
65	Length (m)	1500	1505	1510	1515	1520
66	Bulk Density (kg/m3)	104.3	104.4	104.4	104.4	104.4
67	Bulk Velocity (m/s)	2.826	2.825	2.825	2.825	2.825
68	Erosion Velocity (m/s)	11.94	11.94	11.94	11.94	11.94
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10	Erosion Table					
11						
12						
13	Length (m)	1525	1530	1535	1540	1545
14	Bulk Density (kg/m3)	104.4	104.2	104.0	103.8	103.6
15	Bulk Velocity (m/s)	2.825	2.830	2.836	2.842	2.847
16	Erosion Velocity (m/s)	11.94	11.95	11.96	11.98	11.99
17	Length (m)	1550	1555	1560	1565	1570
18	Bulk Density (kg/m3)	103.3	103.3	103.3	103.3	103.3
19	Bulk Velocity (m/s)	2.853	2.853	2.853	2.853	2.853
20	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
21	Length (m)	1575	1580	1585	1590	1595
22	Bulk Density (kg/m3)	103.3	103.3	103.3	103.3	103.3
23	Bulk Velocity (m/s)	2.854	2.853	2.853	2.853	2.853
24	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
25	Length (m)	1600	1605	1610	1615	1620
26	Bulk Density (kg/m3)	103.3	103.3	103.3	103.3	103.3
27	Bulk Velocity (m/s)	2.853	2.853	2.853	2.853	2.853
28	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
29	Length (m)	1625	1630	1635	1640	1645
30	Bulk Density (kg/m3)	103.3	103.3	103.3	103.3	103.3
31	Bulk Velocity (m/s)	2.853	2.853	2.853	2.853	2.853
32	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
33	Length (m)	1650	1655	1660	1665	1670
34	Bulk Density (kg/m3)	103.3	103.3	103.3	103.3	103.3
35	Bulk Velocity (m/s)	2.853	2.853	2.853	2.853	2.853
36	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
37	Length (m)	1675	1680	1685	1690	1695
38	Bulk Density (kg/m3)	103.3	103.4	103.4	103.4	103.4
39	Bulk Velocity (m/s)	2.853	2.853	2.853	2.853	2.853
40	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
41	Length (m)	1700	1705	1710	1715	1720
42	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
43	Bulk Velocity (m/s)	2.853	2.853	2.853	2.853	2.853
44	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
45	Length (m)	1725	1730	1735	1740	1745
46	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
47	Bulk Velocity (m/s)	2.853	2.853	2.852	2.852	2.852
48	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
49	Length (m)	1750	1755	1760	1765	1770
50	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
51	Bulk Velocity (m/s)	2.852	2.852	2.852	2.852	2.852
52	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
53	Length (m)	1775	1780	1785	1790	1795
54	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
55	Bulk Velocity (m/s)	2.852	2.852	2.852	2.852	2.852
56	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
57	Length (m)	1800	1805	1810	1815	1820
58	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
59	Bulk Velocity (m/s)	2.852	2.852	2.852	2.852	2.852
60	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
61	Length (m)	1825	1830	1835	1840	1845
62	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
63	Bulk Velocity (m/s)	2.852	2.852	2.852	2.852	2.852
64	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
65	Length (m)	1850	1855	1860	1865	1870
66	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
67	Bulk Velocity (m/s)	2.852	2.852	2.852	2.852	2.852
68	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 575 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1875	1880	1885	1890	1895
14	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
15	Bulk Velocity (m/s)	2.852	2.852	2.852	2.852	2.852
16	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
17	Length (m)	1900	1905	1910	1915	1920
18	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
19	Bulk Velocity (m/s)	2.852	2.852	2.852	2.852	2.852
20	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
21	Length (m)	1925	1930	1935	1940	1945
22	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
23	Bulk Velocity (m/s)	2.852	2.852	2.852	2.852	2.852
24	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
25	Length (m)	1950	1955	1960	1965	1970
26	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
27	Bulk Velocity (m/s)	2.852	2.852	2.852	2.852	2.852
28	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
29	Length (m)	1975	1980	1985	1990	1995
30	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
31	Bulk Velocity (m/s)	2.852	2.852	2.852	2.852	2.852
32	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
33	Length (m)	2000	2005	2010	2015	2020
34	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
35	Bulk Velocity (m/s)	2.852	2.852	2.852	2.852	2.852
36	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
37	Length (m)	2025	2030	2035	2040	2045
38	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
39	Bulk Velocity (m/s)	2.852	2.852	2.852	2.852	2.852
40	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
41	Length (m)	2050	2055	2060	2065	2070
42	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
43	Bulk Velocity (m/s)	2.852	2.852	2.852	2.852	2.852
44	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
45	Length (m)	2075	2080	2085	2090	2095
46	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
47	Bulk Velocity (m/s)	2.852	2.852	2.852	2.852	2.852
48	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
49	Length (m)	2100	2105	2110	2115	2120
50	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
51	Bulk Velocity (m/s)	2.852	2.852	2.852	2.852	2.852
52	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
53	Length (m)	2125	2130	2135	2140	2145
54	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
55	Bulk Velocity (m/s)	2.852	2.852	2.852	2.852	2.852
56	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
57	Length (m)	2150	2155	2160	2165	2170
58	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
59	Bulk Velocity (m/s)	2.851	2.851	2.851	2.851	2.851
60	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
61	Length (m)	2175	2180	2185	2191	2196
62	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
63	Bulk Velocity (m/s)	2.851	2.851	2.851	2.851	2.851
64	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
65	Length (m)	2201	2206	2211	2216	2221
66	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
67	Bulk Velocity (m/s)	2.851	2.851	2.851	2.851	2.851
68	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 576 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2226	2231	2236	2241	2246
14	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
15	Bulk Velocity (m/s)	2.851	2.851	2.851	2.851	2.851
16	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
17	Length (m)	2251	2256	2261	2266	2271
18	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
19	Bulk Velocity (m/s)	2.851	2.851	2.851	2.851	2.851
20	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
21	Length (m)	2276	2281	2286	2291	2296
22	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
23	Bulk Velocity (m/s)	2.851	2.851	2.851	2.851	2.851
24	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
25	Length (m)	2301	2306	2311	2316	2321
26	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
27	Bulk Velocity (m/s)	2.851	2.851	2.851	2.851	2.851
28	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
29	Length (m)	2326	2331	2336	2341	2346
30	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
31	Bulk Velocity (m/s)	2.851	2.851	2.851	2.851	2.851
32	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
33	Length (m)	2351	2356	2361	2366	2371
34	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
35	Bulk Velocity (m/s)	2.851	2.851	2.851	2.851	2.851
36	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
37	Length (m)	2376	2381	2386	2391	2396
38	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
39	Bulk Velocity (m/s)	2.851	2.851	2.851	2.851	2.851
40	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
41	Length (m)	2401	2406	2411	2416	2421
42	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
43	Bulk Velocity (m/s)	2.851	2.851	2.851	2.851	2.851
44	Erosion Velocity (m/s)	12.00	12.00	12.00	12.00	12.00
45	Length (m)	2426	2431	2436	2441	2446
46	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
47	Bulk Velocity (m/s)	2.851	2.851	2.851	2.851	2.851
48	Erosion Velocity (m/s)	12.00	12.00	11.99	11.99	11.99
49	Length (m)	2451	2456	2461	2466	2471
50	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
51	Bulk Velocity (m/s)	2.851	2.851	2.851	2.851	2.851
52	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
53	Length (m)	2476	2481	2486	2491	2496
54	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
55	Bulk Velocity (m/s)	2.851	2.850	2.850	2.850	2.850
56	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
57	Length (m)	2501	2506	2511	2516	2521
58	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
59	Bulk Velocity (m/s)	2.850	2.850	2.850	2.850	2.850
60	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
61	Length (m)	2526	2531	2536	2541	2546
62	Bulk Density (kg/m3)	103.4	103.4	103.4	103.4	103.4
63	Bulk Velocity (m/s)	2.850	2.850	2.850	2.850	2.850
64	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
65	Length (m)	2551	2556	2561	2566	2571
66	Bulk Density (kg/m3)	103.4	103.4	103.4	103.5	103.5
67	Bulk Velocity (m/s)	2.850	2.850	2.850	2.850	2.850
68	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 577 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2576	2581	2586	2591	2596
14	Bulk Density (kg/m3)	103.5	103.5	103.5	103.5	103.5
15	Bulk Velocity (m/s)	2.850	2.850	2.850	2.850	2.850
16	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
17	Length (m)	2601	2606	2611	2616	2621
18	Bulk Density (kg/m3)	103.5	103.5	103.5	103.5	103.5
19	Bulk Velocity (m/s)	2.850	2.850	2.850	2.850	2.850
20	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
21	Length (m)	2626	2631	2636	2641	2646
22	Bulk Density (kg/m3)	103.5	103.5	103.5	103.5	103.5
23	Bulk Velocity (m/s)	2.850	2.850	2.850	2.850	2.850
24	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
25	Length (m)	2651	2656	2661	2666	2671
26	Bulk Density (kg/m3)	103.5	103.5	103.5	103.5	103.5
27	Bulk Velocity (m/s)	2.850	2.850	2.850	2.850	2.850
28	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
29	Length (m)	2676	2681	2686	2691	2696
30	Bulk Density (kg/m3)	103.5	103.5	103.5	103.5	103.5
31	Bulk Velocity (m/s)	2.850	2.850	2.850	2.850	2.850
32	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
33	Length (m)	2701	2706	2711	2716	2721
34	Bulk Density (kg/m3)	103.5	103.5	103.5	103.5	103.5
35	Bulk Velocity (m/s)	2.850	2.850	2.850	2.850	2.850
36	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
37	Length (m)	2726	2731	2736	2741	2746
38	Bulk Density (kg/m3)	103.5	103.5	103.5	103.5	103.5
39	Bulk Velocity (m/s)	2.850	2.850	2.850	2.850	2.850
40	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
41	Length (m)	2751	2756	2761	2766	2771
42	Bulk Density (kg/m3)	103.5	103.5	103.5	103.5	103.5
43	Bulk Velocity (m/s)	2.850	2.850	2.850	2.850	2.850
44	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
45	Length (m)	2776	2781	2786	2791	2796
46	Bulk Density (kg/m3)	103.5	103.5	103.5	103.5	103.5
47	Bulk Velocity (m/s)	2.850	2.849	2.849	2.849	2.849
48	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
49	Length (m)	2801	2806	2811	2816	2821
50	Bulk Density (kg/m3)	103.5	103.5	103.5	103.5	103.5
51	Bulk Velocity (m/s)	2.849	2.849	2.849	2.849	2.849
52	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
53	Length (m)	2826	2831	2836	2841	2846
54	Bulk Density (kg/m3)	103.5	103.5	103.5	103.5	103.5
55	Bulk Velocity (m/s)	2.849	2.849	2.849	2.849	2.849
56	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
57	Length (m)	2851	2856	2861	2866	2871
58	Bulk Density (kg/m3)	103.5	103.5	103.5	103.5	103.5
59	Bulk Velocity (m/s)	2.849	2.849	2.849	2.849	2.849
60	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
61	Length (m)	2876	2881	2886	2891	2896
62	Bulk Density (kg/m3)	103.5	103.5	103.5	103.5	103.5
63	Bulk Velocity (m/s)	2.849	2.849	2.849	2.849	2.849
64	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
65	Length (m)	2901	2906	2911	2916	2921
66	Bulk Density (kg/m3)	103.5	103.5	103.5	103.5	103.5
67	Bulk Velocity (m/s)	2.849	2.849	2.849	2.849	2.849
68	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 578 of 937	

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2926	2931	2936	2941	2946
14	Bulk Density (kg/m3)	103.5	103.5	103.5	103.5	103.5
15	Bulk Velocity (m/s)	2.849	2.849	2.849	2.849	2.849
16	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
17	Length (m)	2951	2956	2961	2966	2971
18	Bulk Density (kg/m3)	103.5	103.5	103.5	103.5	103.5
19	Bulk Velocity (m/s)	2.849	2.849	2.849	2.849	2.849
20	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
21	Length (m)	2976	2981	2986	2991	2996
22	Bulk Density (kg/m3)	103.5	103.5	103.5	103.5	103.5
23	Bulk Velocity (m/s)	2.848	2.848	2.848	2.848	2.848
24	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
25	Length (m)	3001	3006	3011	3016	3021
26	Bulk Density (kg/m3)	103.5	103.5	103.5	103.6	103.6
27	Bulk Velocity (m/s)	2.848	2.848	2.847	2.847	2.846
28	Erosion Velocity (m/s)	11.99	11.99	11.99	11.99	11.99
29	Length (m)	3026	3031	3036	3041	3046
30	Bulk Density (kg/m3)	103.6	103.6	103.6	103.6	103.6
31	Bulk Velocity (m/s)	2.846	2.846	2.846	2.845	2.845
32	Erosion Velocity (m/s)	11.99	11.98	11.98	11.98	11.98
33	Length (m)	3051	3056	3061	3066	3071
34	Bulk Density (kg/m3)	103.6	103.6	103.6	103.7	103.7
35	Bulk Velocity (m/s)	2.845	2.845	2.845	2.845	2.844
36	Erosion Velocity (m/s)	11.98	11.98	11.98	11.98	11.98
37	Length (m)	3076	3081	3086	3091	3096
38	Bulk Density (kg/m3)	103.7	103.5	103.3	103.1	102.9
39	Bulk Velocity (m/s)	2.844	2.850	2.855	2.861	2.867
40	Erosion Velocity (m/s)	11.98	11.99	12.00	12.02	12.03
41	Length (m)	3101	3106	3111	3116	3121
42	Bulk Density (kg/m3)	102.7	102.7	102.6	102.6	102.6
43	Bulk Velocity (m/s)	2.872	2.872	2.872	2.872	2.873
44	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
45	Length (m)	3126	3131	3136	3141	3146
46	Bulk Density (kg/m3)	102.6	102.6	102.6	102.6	102.6
47	Bulk Velocity (m/s)	2.873	2.873	2.873	2.872	2.872
48	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
49	Length (m)	3151	3156	3161	3166	3171
50	Bulk Density (kg/m3)	102.6	102.6	102.6	102.7	102.7
51	Bulk Velocity (m/s)	2.872	2.872	2.872	2.872	2.872
52	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
53	Length (m)	3176	3181	3186	3192	3197
54	Bulk Density (kg/m3)	102.7	102.7	102.7	102.6	102.6
55	Bulk Velocity (m/s)	2.872	2.872	2.872	2.872	2.872
56	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
57	Length (m)	3202	3207	3212	3217	3222
58	Bulk Density (kg/m3)	102.6	102.6	102.6	102.7	102.7
59	Bulk Velocity (m/s)	2.872	2.872	2.872	2.872	2.872
60	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
61	Length (m)	3227	3232	3237	3242	3247
62	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
63	Bulk Velocity (m/s)	2.872	2.872	2.872	2.872	2.872
64	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
65	Length (m)	3252	3257	3262	3267	3272
66	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
67	Bulk Velocity (m/s)	2.872	2.872	2.872	2.872	2.872
68	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 579 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3277	3282	3287	3292	3297
14	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
15	Bulk Velocity (m/s)	2.872	2.872	2.872	2.872	2.872
16	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
17	Length (m)	3302	3307	3312	3317	3322
18	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
19	Bulk Velocity (m/s)	2.872	2.872	2.872	2.872	2.872
20	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
21	Length (m)	3327	3332	3337	3342	3347
22	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
23	Bulk Velocity (m/s)	2.872	2.872	2.872	2.872	2.872
24	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
25	Length (m)	3352	3357	3362	3367	3372
26	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
27	Bulk Velocity (m/s)	2.872	2.872	2.872	2.872	2.872
28	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
29	Length (m)	3377	3382	3387	3392	3396
30	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
31	Bulk Velocity (m/s)	2.872	2.872	2.872	2.872	2.872
32	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
33	Length (m)	3401	3406	3411	3416	3421
34	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
35	Bulk Velocity (m/s)	2.872	2.872	2.872	2.872	2.872
36	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
37	Length (m)	3426	3431	3436	3441	3446
38	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
39	Bulk Velocity (m/s)	2.872	2.872	2.872	2.872	2.872
40	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
41	Length (m)	3451	3456	3461	3466	3471
42	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
43	Bulk Velocity (m/s)	2.872	2.872	2.872	2.872	2.872
44	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
45	Length (m)	3476	3481	3486	3491	3496
46	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
47	Bulk Velocity (m/s)	2.872	2.872	2.871	2.871	2.871
48	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
49	Length (m)	3501	3506	3511	3516	3521
50	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
51	Bulk Velocity (m/s)	2.871	2.871	2.871	2.871	2.871
52	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
53	Length (m)	3526	3531	3536	3541	3546
54	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
55	Bulk Velocity (m/s)	2.871	2.871	2.871	2.871	2.871
56	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
57	Length (m)	3551	3556	3561	3566	3571
58	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
59	Bulk Velocity (m/s)	2.871	2.871	2.871	2.871	2.871
60	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
61	Length (m)	3576	3581	3586	3591	3596
62	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
63	Bulk Velocity (m/s)	2.871	2.871	2.871	2.871	2.871
64	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
65	Length (m)	3601	3606	3611	3616	3621
66	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
67	Bulk Velocity (m/s)	2.871	2.871	2.871	2.871	2.871
68	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3626	3631	3636	3641	3646
14	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
15	Bulk Velocity (m/s)	2.871	2.871	2.871	2.871	2.871
16	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
17	Length (m)	3651	3656	3661	3666	3671
18	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
19	Bulk Velocity (m/s)	2.871	2.871	2.871	2.871	2.871
20	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
21	Length (m)	3676	3681	3686	3691	3696
22	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
23	Bulk Velocity (m/s)	2.871	2.871	2.871	2.871	2.871
24	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
25	Length (m)	3701	3706	3711	3716	3721
26	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
27	Bulk Velocity (m/s)	2.871	2.871	2.871	2.871	2.871
28	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
29	Length (m)	3726	3731	3736	3741	3746
30	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
31	Bulk Velocity (m/s)	2.871	2.871	2.871	2.871	2.871
32	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
33	Length (m)	3751	3756	3761	3766	3771
34	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
35	Bulk Velocity (m/s)	2.871	2.871	2.871	2.871	2.871
36	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
37	Length (m)	3776	3781	3787	3792	3797
38	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
39	Bulk Velocity (m/s)	2.871	2.870	2.870	2.870	2.870
40	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
41	Length (m)	3802	3807	3812	3817	3822
42	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
43	Bulk Velocity (m/s)	2.870	2.870	2.870	2.870	2.870
44	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
45	Length (m)	3827	3832	3837	3842	3847
46	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
47	Bulk Velocity (m/s)	2.870	2.870	2.870	2.870	2.870
48	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
49	Length (m)	3852	3857	3862	3867	3872
50	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
51	Bulk Velocity (m/s)	2.870	2.870	2.870	2.870	2.870
52	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
53	Length (m)	3877	3882	3887	3892	3897
54	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
55	Bulk Velocity (m/s)	2.870	2.870	2.870	2.870	2.870
56	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
57	Length (m)	3902	3907	3912	3917	3922
58	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
59	Bulk Velocity (m/s)	2.870	2.870	2.870	2.870	2.870
60	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
61	Length (m)	3927	3932	3937	3942	3947
62	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
63	Bulk Velocity (m/s)	2.870	2.870	2.870	2.870	2.870
64	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
65	Length (m)	3952	3957	3962	3967	3972
66	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
67	Bulk Velocity (m/s)	2.870	2.870	2.870	2.870	2.870
68	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 581 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3977	3982	3987	3992	3997
14	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
15	Bulk Velocity (m/s)	2.870	2.870	2.870	2.870	2.870
16	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
17	Length (m)	4002	4007	4012	4017	4022
18	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
19	Bulk Velocity (m/s)	2.870	2.870	2.870	2.870	2.870
20	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
21	Length (m)	4027	4032	4037	4042	4047
22	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
23	Bulk Velocity (m/s)	2.870	2.870	2.870	2.870	2.870
24	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.04
25	Length (m)	4052	4057	4062	4067	4072
26	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
27	Bulk Velocity (m/s)	2.870	2.870	2.870	2.870	2.870
28	Erosion Velocity (m/s)	12.04	12.04	12.04	12.04	12.03
29	Length (m)	4077	4084	4091	4098	4105
30	Bulk Density (kg/m3)	102.7	102.7	102.7	102.7	102.7
31	Bulk Velocity (m/s)	2.870	2.870	2.870	2.870	2.870
32	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
33	Length (m)	4112	4120	4127	4135	4143
34	Bulk Density (kg/m3)	102.7	102.8	102.8	102.8	102.8
35	Bulk Velocity (m/s)	2.870	2.870	2.870	2.869	2.869
36	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
37	Length (m)	4150	4158	4165	4173	4181
38	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
39	Bulk Velocity (m/s)	2.869	2.869	2.869	2.869	2.869
40	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
41	Length (m)	4188	4196	4203	4211	4219
42	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
43	Bulk Velocity (m/s)	2.869	2.869	2.869	2.869	2.869
44	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
45	Length (m)	4226	4234	4241	4249	4257
46	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
47	Bulk Velocity (m/s)	2.869	2.869	2.869	2.869	2.869
48	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
49	Length (m)	4264	4272	4279	4287	4295
50	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
51	Bulk Velocity (m/s)	2.869	2.869	2.869	2.869	2.869
52	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
53	Length (m)	4302	4304	4305	4306	4308
54	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
55	Bulk Velocity (m/s)	2.869	2.869	2.869	2.869	2.869
56	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
57	Length (m)	4309	4314	4319	4324	4329
58	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
59	Bulk Velocity (m/s)	2.869	2.869	2.869	2.869	2.869
60	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
61	Length (m)	4334	4339	4344	4349	4354
62	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
63	Bulk Velocity (m/s)	2.869	2.869	2.869	2.869	2.869
64	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
65	Length (m)	4359	4364	4369	4374	4379
66	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
67	Bulk Velocity (m/s)	2.869	2.869	2.869	2.869	2.869
68	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 582 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	4384	4389	4394	4399	4404
14	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
15	Bulk Velocity (m/s)	2.869	2.869	2.869	2.869	2.869
16	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
17	Length (m)	4409	4414	4419	4424	4429
18	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
19	Bulk Velocity (m/s)	2.869	2.869	2.869	2.869	2.869
20	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
21	Length (m)	4434	4439	4444	4449	4454
22	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
23	Bulk Velocity (m/s)	2.869	2.869	2.869	2.869	2.869
24	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
25	Length (m)	4459	4464	4469	4474	4479
26	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
27	Bulk Velocity (m/s)	2.869	2.869	2.869	2.869	2.869
28	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
29	Length (m)	4484	4489	4494	4499	4504
30	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
31	Bulk Velocity (m/s)	2.869	2.869	2.869	2.869	2.869
32	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
33	Length (m)	4510	4515	4520	4525	4530
34	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
35	Bulk Velocity (m/s)	2.869	2.869	2.869	2.868	2.868
36	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
37	Length (m)	4535	4540	4545	4550	4555
38	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
39	Bulk Velocity (m/s)	2.868	2.868	2.868	2.868	2.868
40	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
41	Length (m)	4560	4565	4570	4575	4580
42	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
43	Bulk Velocity (m/s)	2.868	2.868	2.868	2.868	2.868
44	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
45	Length (m)	4585	4590	4595	4600	4605
46	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
47	Bulk Velocity (m/s)	2.868	2.868	2.868	2.868	2.868
48	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
49	Length (m)	4610	4623	4636	4649	4662
50	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
51	Bulk Velocity (m/s)	2.868	2.868	2.868	2.868	2.868
52	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
53	Length (m)	4675	4677	4679	4681	4683
54	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
55	Bulk Velocity (m/s)	2.868	2.868	2.868	2.868	2.868
56	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
57	Length (m)	4685	4690	4695	4700	4705
58	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
59	Bulk Velocity (m/s)	2.868	2.868	2.868	2.868	2.868
60	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
61	Length (m)	4710	4715	4720	4725	4730
62	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
63	Bulk Velocity (m/s)	2.868	2.868	2.868	2.868	2.868
64	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
65	Length (m)	4735	4740	4745	4750	4755
66	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
67	Bulk Velocity (m/s)	2.868	2.868	2.868	2.868	2.868
68	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 583 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10	Erosion Table					
11						
12						
13	Length (m)	4760	4765	4770	4775	4780
14	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
15	Bulk Velocity (m/s)	2.868	2.868	2.868	2.868	2.868
16	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
17	Length (m)	4785	4790	4795	4800	4805
18	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
19	Bulk Velocity (m/s)	2.868	2.868	2.868	2.868	2.868
20	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
21	Length (m)	4810	4815	4820	4825	4830
22	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
23	Bulk Velocity (m/s)	2.868	2.868	2.868	2.868	2.868
24	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
25	Length (m)	4835	4840	4845	4850	4855
26	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
27	Bulk Velocity (m/s)	2.868	2.868	2.868	2.868	2.868
28	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
29	Length (m)	4860	4865	4870	4875	4880
30	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
31	Bulk Velocity (m/s)	2.868	2.868	2.868	2.867	2.867
32	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
33	Length (m)	4885	4890	4895	4900	4905
34	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
35	Bulk Velocity (m/s)	2.867	2.867	2.867	2.867	2.867
36	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
37	Length (m)	4910	4915	4920	4925	4930
38	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
39	Bulk Velocity (m/s)	2.867	2.867	2.867	2.867	2.867
40	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
41	Length (m)	4935	4940	4945	4950	4955
42	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
43	Bulk Velocity (m/s)	2.867	2.867	2.867	2.867	2.867
44	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
45	Length (m)	4960	4965	4970	4975	4980
46	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
47	Bulk Velocity (m/s)	2.867	2.867	2.867	2.867	2.867
48	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
49	Length (m)	4985	4990	4995	5000	5005
50	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
51	Bulk Velocity (m/s)	2.867	2.867	2.867	2.867	2.867
52	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
53	Length (m)	5010	5015	5020	5025	5030
54	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
55	Bulk Velocity (m/s)	2.867	2.867	2.867	2.867	2.867
56	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
57	Length (m)	5035	5040	5045	5050	5055
58	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
59	Bulk Velocity (m/s)	2.867	2.867	2.867	2.867	2.867
60	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
61	Length (m)	5060	5065	5070	5075	5080
62	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
63	Bulk Velocity (m/s)	2.867	2.867	2.867	2.867	2.867
64	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
65	Length (m)	5085	5090	5095	5100	5105
66	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
67	Bulk Velocity (m/s)	2.867	2.867	2.867	2.867	2.867
68	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
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2						
3				Unit Set: Mixed		
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5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	5110	5115	5120	5125	5130
14	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.8
15	Bulk Velocity (m/s)	2.867	2.867	2.867	2.867	2.867
16	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
17	Length (m)	5135	5140	5145	5150	5155
18	Bulk Density (kg/m3)	102.8	102.8	102.8	102.8	102.9
19	Bulk Velocity (m/s)	2.867	2.867	2.867	2.867	2.867
20	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
21	Length (m)	5160	5165	5170	5175	5180
22	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
23	Bulk Velocity (m/s)	2.867	2.867	2.867	2.867	2.867
24	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
25	Length (m)	5185	5190	5195	5200	5205
26	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
27	Bulk Velocity (m/s)	2.867	2.867	2.867	2.867	2.867
28	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
29	Length (m)	5210	5215	5220	5225	5230
30	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
31	Bulk Velocity (m/s)	2.867	2.867	2.867	2.867	2.866
32	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
33	Length (m)	5235	5240	5245	5250	5255
34	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
35	Bulk Velocity (m/s)	2.866	2.866	2.866	2.866	2.866
36	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
37	Length (m)	5260	5265	5270	5275	5280
38	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
39	Bulk Velocity (m/s)	2.866	2.866	2.866	2.866	2.866
40	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
41	Length (m)	5285	5290	5295	5300	5305
42	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
43	Bulk Velocity (m/s)	2.866	2.866	2.866	2.866	2.866
44	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
45	Length (m)	5310	5315	5320	5325	5330
46	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
47	Bulk Velocity (m/s)	2.866	2.866	2.866	2.866	2.866
48	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
49	Length (m)	5335	5340	5345	5350	5355
50	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
51	Bulk Velocity (m/s)	2.866	2.866	2.866	2.866	2.866
52	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
53	Length (m)	5360	5365	5370	5375	5380
54	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
55	Bulk Velocity (m/s)	2.866	2.866	2.866	2.866	2.866
56	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
57	Length (m)	5385	5390	5395	5400	5405
58	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
59	Bulk Velocity (m/s)	2.866	2.866	2.866	2.866	2.866
60	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
61	Length (m)	5410	5415	5420	5425	5430
62	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
63	Bulk Velocity (m/s)	2.866	2.866	2.866	2.866	2.866
64	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
65	Length (m)	5435	5440	5445	5450	5455
66	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
67	Bulk Velocity (m/s)	2.866	2.866	2.866	2.866	2.866
68	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	5460	5465	5470	5475	5480
14	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
15	Bulk Velocity (m/s)	2.866	2.866	2.866	2.866	2.866
16	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
17	Length (m)	5485	5490	5495	5500	5505
18	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
19	Bulk Velocity (m/s)	2.866	2.866	2.866	2.866	2.866
20	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
21	Length (m)	5510	5515	5520	5525	5530
22	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
23	Bulk Velocity (m/s)	2.866	2.866	2.866	2.866	2.866
24	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
25	Length (m)	5535	5542	5549	5556	5563
26	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
27	Bulk Velocity (m/s)	2.866	2.866	2.866	2.866	2.866
28	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
29	Length (m)	5570	5572	5574	5576	5578
30	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
31	Bulk Velocity (m/s)	2.866	2.865	2.865	2.865	2.865
32	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
33	Length (m)	5580	5585	5590	5595	5600
34	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
35	Bulk Velocity (m/s)	2.865	2.865	2.865	2.865	2.865
36	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
37	Length (m)	5605	5610	5615	5620	5625
38	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
39	Bulk Velocity (m/s)	2.865	2.865	2.865	2.865	2.865
40	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
41	Length (m)	5630	5635	5640	5645	5650
42	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
43	Bulk Velocity (m/s)	2.865	2.865	2.865	2.865	2.865
44	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
45	Length (m)	5655	5660	5665	5670	5675
46	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
47	Bulk Velocity (m/s)	2.865	2.865	2.865	2.865	2.865
48	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
49	Length (m)	5680	5685	5690	5695	5700
50	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
51	Bulk Velocity (m/s)	2.865	2.865	2.865	2.865	2.865
52	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
53	Length (m)	5705	5710	5715	5720	5725
54	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
55	Bulk Velocity (m/s)	2.865	2.865	2.865	2.865	2.865
56	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
57	Length (m)	5730	5735	5740	5745	5750
58	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
59	Bulk Velocity (m/s)	2.865	2.865	2.865	2.865	2.865
60	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
61	Length (m)	5755	5760	5765	5770	5775
62	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
63	Bulk Velocity (m/s)	2.865	2.865	2.865	2.865	2.865
64	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
65	Length (m)	5780	5785	5790	5795	5800
66	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
67	Bulk Velocity (m/s)	2.865	2.865	2.865	2.865	2.865
68	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
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2						
3				Unit Set: Mixed		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	5805	5810	5815	5820	5825
14	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
15	Bulk Velocity (m/s)	2.865	2.865	2.865	2.865	2.865
16	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
17	Length (m)	5830	5835	5840	5845	5850
18	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
19	Bulk Velocity (m/s)	2.865	2.865	2.865	2.865	2.865
20	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
21	Length (m)	5855	5860	5865	5870	5875
22	Bulk Density (kg/m3)	102.9	102.9	102.9	102.9	102.9
23	Bulk Velocity (m/s)	2.865	2.865	2.865	2.865	2.865
24	Erosion Velocity (m/s)	12.03	12.03	12.03	12.03	12.03
25	Length (m)	5880	5885	5890	5895	5900
26	Bulk Density (kg/m3)	102.9	102.9	102.9	103.0	103.0
27	Bulk Velocity (m/s)	2.865	2.865	2.864	2.864	2.863
28	Erosion Velocity (m/s)	12.03	12.02	12.02	12.02	12.02
29	Length (m)	5905	5910	5915	5920	5925
30	Bulk Density (kg/m3)	103.0	103.0	103.0	103.1	103.1
31	Bulk Velocity (m/s)	2.863	2.862	2.862	2.861	2.860
32	Erosion Velocity (m/s)	12.02	12.02	12.02	12.02	12.02
33	Length (m)	5930	5935	5940	5945	5950
34	Bulk Density (kg/m3)	103.1	102.9	102.7	102.5	102.3
35	Bulk Velocity (m/s)	2.860	2.866	2.871	2.877	2.883
36	Erosion Velocity (m/s)	12.01	12.03	12.04	12.05	12.06
37	Length (m)	5955	5960	5965	5970	5975
38	Bulk Density (kg/m3)	102.1	102.0	102.0	101.9	101.9
39	Bulk Velocity (m/s)	2.889	2.890	2.892	2.893	2.894
40	Erosion Velocity (m/s)	12.08	12.08	12.08	12.08	12.09
41	Length (m)	5980	5985	5990	5995	6000
42	Bulk Density (kg/m3)	101.8	101.8	101.9	101.9	101.9
43	Bulk Velocity (m/s)	2.896	2.895	2.894	2.893	2.892
44	Erosion Velocity (m/s)	12.09	12.09	12.09	12.08	12.08
45	Length (m)	6005	6010	6015	6020	6025
46	Bulk Density (kg/m3)	102.0	101.7	101.5	101.3	101.0
47	Bulk Velocity (m/s)	2.891	2.898	2.905	2.912	2.919
48	Erosion Velocity (m/s)	12.08	12.09	12.11	12.12	12.14
49	Length (m)	6030	6035	6040	6045	6050
50	Bulk Density (kg/m3)	100.8	100.8	100.8	100.8	100.8
51	Bulk Velocity (m/s)	2.926	2.926	2.926	2.926	2.926
52	Erosion Velocity (m/s)	12.15	12.15	12.15	12.15	12.15
53	Length (m)	6055	6060	6065	6070	6075
54	Bulk Density (kg/m3)	100.8	100.8	100.8	100.8	100.8
55	Bulk Velocity (m/s)	2.926	2.926	2.926	2.926	2.926
56	Erosion Velocity (m/s)	12.15	12.15	12.15	12.15	12.15
57	Length (m)	6080	6085	6090	6095	6101
58	Bulk Density (kg/m3)	100.8	100.8	100.8	100.8	100.9
59	Bulk Velocity (m/s)	2.926	2.925	2.924	2.924	2.923
60	Erosion Velocity (m/s)	12.15	12.15	12.15	12.15	12.15
61	Length (m)	6106	6111	6116	6121	6126
62	Bulk Density (kg/m3)	100.9	100.7	100.5	100.4	100.2
63	Bulk Velocity (m/s)	2.922	2.928	2.933	2.938	2.943
64	Erosion Velocity (m/s)	12.14	12.16	12.17	12.18	12.19
65	Length (m)	6131	6136	6141	6146	6151
66	Bulk Density (kg/m3)	99.99	99.99	99.99	99.99	99.99
67	Bulk Velocity (m/s)	2.949	2.949	2.949	2.949	2.949
68	Erosion Velocity (m/s)	12.20	12.20	12.20	12.20	12.20
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2						
3				Unit Set: Mixed		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	6156	6161	6166	6171	6176
14	Bulk Density (kg/m3)	99.99	99.99	99.99	99.99	99.99
15	Bulk Velocity (m/s)	2.949	2.949	2.949	2.949	2.949
16	Erosion Velocity (m/s)	12.20	12.20	12.20	12.20	12.20
17	Length (m)	6181	6186	6191	6196	6201
18	Bulk Density (kg/m3)	99.99	99.99	99.99	99.99	99.99
19	Bulk Velocity (m/s)	2.949	2.949	2.949	2.949	2.949
20	Erosion Velocity (m/s)	12.20	12.20	12.20	12.20	12.20
21	Length (m)	6206	6211	6216	6221	6226
22	Bulk Density (kg/m3)	99.99	99.99	99.99	99.99	99.99
23	Bulk Velocity (m/s)	2.949	2.949	2.949	2.949	2.949
24	Erosion Velocity (m/s)	12.20	12.20	12.20	12.20	12.20
25	Length (m)	6231	6236	6241	6246	6251
26	Bulk Density (kg/m3)	99.99	99.99	99.99	99.99	100.0
27	Bulk Velocity (m/s)	2.949	2.949	2.949	2.949	2.949
28	Erosion Velocity (m/s)	12.20	12.20	12.20	12.20	12.20
29	Length (m)	6256	6261	6266	6271	6276
30	Bulk Density (kg/m3)	100.0	100.0	100.0	100.0	100.0
31	Bulk Velocity (m/s)	2.949	2.949	2.949	2.949	2.949
32	Erosion Velocity (m/s)	12.20	12.20	12.20	12.20	12.20
33	Length (m)	6281	6285	6290	6294	6299
34	Bulk Density (kg/m3)	100.0	100.0	100.0	100.1	100.1
35	Bulk Velocity (m/s)	2.949	2.948	2.947	2.946	2.946
36	Erosion Velocity (m/s)	12.20	12.20	12.20	12.19	12.19
37	Length (m)	6303	6308	6313	6318	6323
38	Bulk Density (kg/m3)	100.1	100.1	100.0	99.97	99.92
39	Bulk Velocity (m/s)	2.945	2.946	2.948	2.949	2.951
40	Erosion Velocity (m/s)	12.19	12.19	12.20	12.20	12.20
41	Length (m)	6328	6333	6338	6343	6348
42	Bulk Density (kg/m3)	99.87	99.87	99.88	99.89	99.90
43	Bulk Velocity (m/s)	2.952	2.952	2.952	2.952	2.951
44	Erosion Velocity (m/s)	12.21	12.21	12.21	12.21	12.21
45	Length (m)	6353	6358	6363	6368	6373
46	Bulk Density (kg/m3)	99.91	99.92	99.94	99.96	99.97
47	Bulk Velocity (m/s)	2.951	2.951	2.950	2.950	2.949
48	Erosion Velocity (m/s)	12.20	12.20	12.20	12.20	12.20
49	Length (m)	6378	6383	6388	6393	6398
50	Bulk Density (kg/m3)	99.99	99.95	99.91	99.87	99.83
51	Bulk Velocity (m/s)	2.949	2.950	2.951	2.952	2.954
52	Erosion Velocity (m/s)	12.20	12.20	12.20	12.21	12.21
53	Length (m)	6403	6408	6413	6418	6423
54	Bulk Density (kg/m3)	99.79	99.55	99.32	99.08	98.85
55	Bulk Velocity (m/s)	2.955	2.962	2.969	2.976	2.983
56	Erosion Velocity (m/s)	12.21	12.23	12.24	12.26	12.27
57	Length (m)	6428	6433	6438	6443	6448
58	Bulk Density (kg/m3)	98.62	98.61	98.61	98.61	98.61
59	Bulk Velocity (m/s)	2.990	2.990	2.990	2.990	2.990
60	Erosion Velocity (m/s)	12.28	12.28	12.28	12.28	12.28
61	Length (m)	6453	6459	6465	6471	6477
62	Bulk Density (kg/m3)	98.61	98.61	98.61	98.61	98.61
63	Bulk Velocity (m/s)	2.990	2.990	2.990	2.990	2.990
64	Erosion Velocity (m/s)	12.28	12.28	12.28	12.28	12.28
65	Length (m)	6483	6487	6491	6495	6499
66	Bulk Density (kg/m3)	98.61	98.61	98.61	98.61	98.61
67	Bulk Velocity (m/s)	2.990	2.990	2.990	2.990	2.990
68	Erosion Velocity (m/s)	12.28	12.28	12.28	12.28	12.28
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 588 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10	Erosion Table					
11						
12						
13	Length (m)	6503	6508	6513	6518	6523
14	Bulk Density (kg/m3)	98.61	98.61	98.61	98.61	98.61
15	Bulk Velocity (m/s)	2.990	2.990	2.990	2.990	2.990
16	Erosion Velocity (m/s)	12.28	12.28	12.28	12.28	12.28
17	Length (m)	6528	6533	6538	6543	6548
18	Bulk Density (kg/m3)	98.61	98.61	98.61	98.61	98.61
19	Bulk Velocity (m/s)	2.990	2.990	2.990	2.990	2.990
20	Erosion Velocity (m/s)	12.28	12.28	12.28	12.28	12.28
21	Length (m)	6553	6558	6563	6568	6573
22	Bulk Density (kg/m3)	98.61	98.61	98.61	98.61	98.61
23	Bulk Velocity (m/s)	2.990	2.990	2.990	2.990	2.990
24	Erosion Velocity (m/s)	12.28	12.28	12.28	12.28	12.28
25	Length (m)	6578	6583	6588	6593	6598
26	Bulk Density (kg/m3)	98.61	98.61	98.61	98.61	98.61
27	Bulk Velocity (m/s)	2.990	2.990	2.990	2.990	2.990
28	Erosion Velocity (m/s)	12.28	12.28	12.28	12.28	12.28
29	Length (m)	6603	6608	6613	6618	6623
30	Bulk Density (kg/m3)	98.61	98.61	98.61	98.61	98.62
31	Bulk Velocity (m/s)	2.990	2.990	2.990	2.990	2.990
32	Erosion Velocity (m/s)	12.28	12.28	12.28	12.28	12.28
33	Length (m)	6628	6633	6638	6643	6648
34	Bulk Density (kg/m3)	98.62	98.62	98.62	98.62	98.62
35	Bulk Velocity (m/s)	2.990	2.990	2.990	2.990	2.990
36	Erosion Velocity (m/s)	12.28	12.28	12.28	12.28	12.28
37	Length (m)	6653	6658	6663	6668	6673
38	Bulk Density (kg/m3)	98.62	98.62	98.62	98.62	98.62
39	Bulk Velocity (m/s)	2.990	2.990	2.990	2.990	2.990
40	Erosion Velocity (m/s)	12.28	12.28	12.28	12.28	12.28
41	Length (m)	6678	6683	6688	6693	6698
42	Bulk Density (kg/m3)	98.62	98.62	98.62	98.62	98.62
43	Bulk Velocity (m/s)	2.990	2.990	2.990	2.990	2.990
44	Erosion Velocity (m/s)	12.28	12.28	12.28	12.28	12.28
45	Length (m)	6703	6708	6713	6718	6723
46	Bulk Density (kg/m3)	98.62	98.62	98.63	98.63	98.63
47	Bulk Velocity (m/s)	2.990	2.990	2.990	2.990	2.989
48	Erosion Velocity (m/s)	12.28	12.28	12.28	12.28	12.28
49	Length (m)	6728	6733	6738	6743	6748
50	Bulk Density (kg/m3)	98.63	98.66	98.69	98.72	98.74
51	Bulk Velocity (m/s)	2.989	2.989	2.988	2.987	2.986
52	Erosion Velocity (m/s)	12.28	12.28	12.28	12.28	12.28
53	Length (m)	6754	6759	6764	6769	6774
54	Bulk Density (kg/m3)	98.77	98.71	98.64	98.57	98.50
55	Bulk Velocity (m/s)	2.985	2.987	2.989	2.991	2.993
56	Erosion Velocity (m/s)	12.27	12.28	12.28	12.29	12.29
57	Length (m)	6779	6784	6789	6794	6799
58	Bulk Density (kg/m3)	98.44	98.42	98.41	98.39	98.38
59	Bulk Velocity (m/s)	2.995	2.996	2.996	2.997	2.997
60	Erosion Velocity (m/s)	12.30	12.30	12.30	12.30	12.30
61	Length (m)	6804	6809	6814	6819	6824
62	Bulk Density (kg/m3)	98.36	98.27	98.18	98.08	97.99
63	Bulk Velocity (m/s)	2.998	3.000	3.003	3.006	3.009
64	Erosion Velocity (m/s)	12.30	12.31	12.31	12.32	12.32
65	Length (m)	6829	6834	6839	6844	6849
66	Bulk Density (kg/m3)	97.90	97.93	97.97	98.01	98.05
67	Bulk Velocity (m/s)	3.012	3.011	3.009	3.008	3.007
68	Erosion Velocity (m/s)	12.33	12.33	12.32	12.32	12.32
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 589 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10	Erosion Table					
11						
12						
13	Length (m)	6854	6859	6864	6869	6874
14	Bulk Density (kg/m3)	98.09	97.78	97.47	97.16	96.85
15	Bulk Velocity (m/s)	3.006	3.015	3.025	3.035	3.044
16	Erosion Velocity (m/s)	12.32	12.34	12.36	12.38	12.40
17	Length (m)	6879	6884	6889	6894	6899
18	Bulk Density (kg/m3)	96.54	96.54	96.54	96.54	96.54
19	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
20	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
21	Length (m)	6904	6909	6914	6919	6924
22	Bulk Density (kg/m3)	96.54	96.54	96.54	96.54	96.54
23	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
24	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
25	Length (m)	6929	6934	6939	6944	6949
26	Bulk Density (kg/m3)	96.54	96.54	96.54	96.54	96.54
27	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
28	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
29	Length (m)	6954	6959	6964	6969	6974
30	Bulk Density (kg/m3)	96.54	96.54	96.54	96.54	96.54
31	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
32	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
33	Length (m)	6979	6984	6989	6994	6999
34	Bulk Density (kg/m3)	96.54	96.54	96.54	96.54	96.54
35	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
36	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
37	Length (m)	7004	7009	7014	7019	7024
38	Bulk Density (kg/m3)	96.54	96.54	96.54	96.54	96.54
39	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
40	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
41	Length (m)	7029	7034	7039	7044	7049
42	Bulk Density (kg/m3)	96.54	96.54	96.54	96.55	96.55
43	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
44	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
45	Length (m)	7054	7059	7064	7069	7074
46	Bulk Density (kg/m3)	96.55	96.55	96.55	96.55	96.55
47	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
48	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
49	Length (m)	7079	7081	7083	7085	7087
50	Bulk Density (kg/m3)	96.55	96.55	96.55	96.55	96.55
51	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
52	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
53	Length (m)	7089	7097	7105	7113	7121
54	Bulk Density (kg/m3)	96.55	96.55	96.55	96.55	96.55
55	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
56	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
57	Length (m)	7129	7134	7139	7144	7149
58	Bulk Density (kg/m3)	96.55	96.55	96.55	96.55	96.55
59	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
60	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
61	Length (m)	7154	7159	7164	7169	7174
62	Bulk Density (kg/m3)	96.55	96.55	96.55	96.55	96.55
63	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
64	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.41
65	Length (m)	7179	7184	7189	7194	7199
66	Bulk Density (kg/m3)	96.55	96.55	96.55	96.55	96.55
67	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
68	Erosion Velocity (m/s)	12.41	12.41	12.42	12.42	12.42
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	7204	7209	7214	7219	7224
14	Bulk Density (kg/m3)	96.55	96.55	96.55	96.55	96.55
15	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
16	Erosion Velocity (m/s)	12.42	12.41	12.41	12.41	12.41
17	Length (m)	7229	7234	7239	7244	7249
18	Bulk Density (kg/m3)	96.55	96.55	96.55	96.56	96.56
19	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
20	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
21	Length (m)	7254	7259	7264	7269	7274
22	Bulk Density (kg/m3)	96.56	96.56	96.56	96.56	96.56
23	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
24	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
25	Length (m)	7279	7284	7289	7294	7299
26	Bulk Density (kg/m3)	96.56	96.56	96.56	96.56	96.56
27	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
28	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
29	Length (m)	7304	7309	7314	7319	7324
30	Bulk Density (kg/m3)	96.56	96.56	96.56	96.56	96.56
31	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
32	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
33	Length (m)	7329	7334	7339	7344	7349
34	Bulk Density (kg/m3)	96.56	96.56	96.56	96.56	96.56
35	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
36	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
37	Length (m)	7354	7359	7364	7369	7374
38	Bulk Density (kg/m3)	96.56	96.56	96.56	96.56	96.56
39	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
40	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
41	Length (m)	7379	7384	7389	7394	7399
42	Bulk Density (kg/m3)	96.56	96.56	96.56	96.56	96.56
43	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
44	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
45	Length (m)	7404	7409	7414	7419	7424
46	Bulk Density (kg/m3)	96.56	96.56	96.56	96.56	96.56
47	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
48	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
49	Length (m)	7429	7434	7439	7444	7449
50	Bulk Density (kg/m3)	96.56	96.56	96.56	96.57	96.57
51	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
52	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
53	Length (m)	7454	7459	7464	7469	7474
54	Bulk Density (kg/m3)	96.57	96.57	96.57	96.57	96.57
55	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
56	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
57	Length (m)	7479	7484	7489	7494	7499
58	Bulk Density (kg/m3)	96.57	96.57	96.57	96.57	96.57
59	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
60	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
61	Length (m)	7504	7509	7514	7519	7524
62	Bulk Density (kg/m3)	96.57	96.57	96.57	96.57	96.57
63	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
64	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
65	Length (m)	7529	7533	7538	7543	7548
66	Bulk Density (kg/m3)	96.57	96.57	96.57	96.57	96.57
67	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
68	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 591 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10	Erosion Table					
11						
12						
13	Length (m)	7553	7558	7563	7568	7573
14	Bulk Density (kg/m3)	96.57	96.57	96.57	96.57	96.57
15	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
16	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
17	Length (m)	7578	7583	7588	7593	7598
18	Bulk Density (kg/m3)	96.57	96.58	96.58	96.58	96.58
19	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
20	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
21	Length (m)	7603	7608	7613	7618	7623
22	Bulk Density (kg/m3)	96.58	96.58	96.58	96.58	96.58
23	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
24	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
25	Length (m)	7628	7633	7638	7643	7648
26	Bulk Density (kg/m3)	96.58	96.58	96.58	96.58	96.58
27	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
28	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
29	Length (m)	7653	7658	7663	7668	7673
30	Bulk Density (kg/m3)	96.58	96.58	96.58	96.58	96.58
31	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
32	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
33	Length (m)	7678	7683	7688	7693	7698
34	Bulk Density (kg/m3)	96.58	96.58	96.58	96.58	96.58
35	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
36	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
37	Length (m)	7703	7708	7713	7718	7723
38	Bulk Density (kg/m3)	96.58	96.59	96.59	96.59	96.59
39	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.052
40	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
41	Length (m)	7728	7733	7738	7743	7748
42	Bulk Density (kg/m3)	96.60	96.60	96.60	96.60	96.60
43	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
44	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
45	Length (m)	7753	7758	7763	7768	7773
46	Bulk Density (kg/m3)	96.60	96.60	96.59	96.59	96.59
47	Bulk Velocity (m/s)	3.052	3.052	3.052	3.053	3.053
48	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
49	Length (m)	7778	7783	7788	7793	7798
50	Bulk Density (kg/m3)	96.59	96.58	96.58	96.58	96.58
51	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
52	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
53	Length (m)	7803	7808	7813	7818	7823
54	Bulk Density (kg/m3)	96.58	96.58	96.58	96.59	96.59
55	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
56	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
57	Length (m)	7828	7833	7838	7843	7848
58	Bulk Density (kg/m3)	96.59	96.59	96.59	96.59	96.59
59	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
60	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
61	Length (m)	7853	7858	7863	7868	7873
62	Bulk Density (kg/m3)	96.59	96.59	96.59	96.59	96.59
63	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
64	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
65	Length (m)	7878	7883	7888	7893	7898
66	Bulk Density (kg/m3)	96.59	96.59	96.59	96.59	96.59
67	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
68	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 592 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10	Erosion Table					
11						
12						
13	Length (m)	7903	7908	7913	7918	7923
14	Bulk Density (kg/m3)	96.59	96.59	96.59	96.59	96.59
15	Bulk Velocity (m/s)	3.053	3.053	3.053	3.052	3.052
16	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
17	Length (m)	7928	7933	7938	7943	7948
18	Bulk Density (kg/m3)	96.59	96.59	96.59	96.59	96.59
19	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
20	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
21	Length (m)	7953	7958	7963	7968	7973
22	Bulk Density (kg/m3)	96.59	96.59	96.59	96.59	96.60
23	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
24	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
25	Length (m)	7978	7983	7988	7993	7998
26	Bulk Density (kg/m3)	96.60	96.60	96.60	96.60	96.60
27	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
28	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
29	Length (m)	8003	8008	8013	8018	8023
30	Bulk Density (kg/m3)	96.60	96.60	96.60	96.60	96.60
31	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
32	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
33	Length (m)	8028	8033	8038	8043	8048
34	Bulk Density (kg/m3)	96.60	96.60	96.61	96.61	96.61
35	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
36	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
37	Length (m)	8053	8058	8063	8068	8073
38	Bulk Density (kg/m3)	96.61	96.62	96.63	96.64	96.64
39	Bulk Velocity (m/s)	3.052	3.052	3.051	3.051	3.051
40	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
41	Length (m)	8078	8083	8088	8093	8098
42	Bulk Density (kg/m3)	96.65	96.60	96.56	96.51	96.46
43	Bulk Velocity (m/s)	3.051	3.052	3.054	3.055	3.057
44	Erosion Velocity (m/s)	12.41	12.41	12.41	12.42	12.42
45	Length (m)	8103	8108	8113	8118	8123
46	Bulk Density (kg/m3)	96.42	96.41	96.40	96.40	96.39
47	Bulk Velocity (m/s)	3.058	3.058	3.058	3.059	3.059
48	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.43
49	Length (m)	8128	8133	8138	8143	8148
50	Bulk Density (kg/m3)	96.39	96.39	96.39	96.39	96.39
51	Bulk Velocity (m/s)	3.059	3.059	3.059	3.059	3.059
52	Erosion Velocity (m/s)	12.43	12.43	12.43	12.43	12.43
53	Length (m)	8153	8158	8163	8168	8173
54	Bulk Density (kg/m3)	96.39	96.39	96.39	96.39	96.39
55	Bulk Velocity (m/s)	3.059	3.059	3.059	3.059	3.059
56	Erosion Velocity (m/s)	12.43	12.43	12.43	12.43	12.43
57	Length (m)	8178	8183	8188	8193	8198
58	Bulk Density (kg/m3)	96.39	96.39	96.39	96.39	96.39
59	Bulk Velocity (m/s)	3.059	3.059	3.059	3.059	3.059
60	Erosion Velocity (m/s)	12.43	12.43	12.43	12.43	12.43
61	Length (m)	8203	8208	8213	8218	8223
62	Bulk Density (kg/m3)	96.39	96.39	96.39	96.39	96.39
63	Bulk Velocity (m/s)	3.059	3.059	3.059	3.059	3.059
64	Erosion Velocity (m/s)	12.43	12.43	12.43	12.43	12.43
65	Length (m)	8228	8233	8238	8243	8248
66	Bulk Density (kg/m3)	96.39	96.39	96.39	96.39	96.39
67	Bulk Velocity (m/s)	3.059	3.059	3.059	3.059	3.059
68	Erosion Velocity (m/s)	12.43	12.43	12.43	12.43	12.43
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 593 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	8253	8258	8263	8268	8273
14	Bulk Density (kg/m3)	96.39	96.39	96.39	96.39	96.39
15	Bulk Velocity (m/s)	3.059	3.059	3.059	3.059	3.059
16	Erosion Velocity (m/s)	12.43	12.43	12.43	12.43	12.43
17	Length (m)	8278	8283	8288	8293	8298
18	Bulk Density (kg/m3)	96.39	96.40	96.40	96.40	96.40
19	Bulk Velocity (m/s)	3.059	3.059	3.059	3.059	3.059
20	Erosion Velocity (m/s)	12.43	12.43	12.43	12.42	12.42
21	Length (m)	8303	8308	8313	8318	8323
22	Bulk Density (kg/m3)	96.40	96.40	96.40	96.40	96.40
23	Bulk Velocity (m/s)	3.059	3.059	3.059	3.059	3.059
24	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
25	Length (m)	8328	8333	8338	8343	8348
26	Bulk Density (kg/m3)	96.40	96.40	96.40	96.40	96.40
27	Bulk Velocity (m/s)	3.059	3.059	3.059	3.059	3.058
28	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
29	Length (m)	8353	8358	8363	8368	8373
30	Bulk Density (kg/m3)	96.40	96.41	96.41	96.41	96.41
31	Bulk Velocity (m/s)	3.058	3.058	3.058	3.058	3.058
32	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
33	Length (m)	8378	8383	8388	8393	8398
34	Bulk Density (kg/m3)	96.41	96.41	96.41	96.41	96.41
35	Bulk Velocity (m/s)	3.058	3.058	3.058	3.058	3.058
36	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
37	Length (m)	8403	8408	8413	8418	8423
38	Bulk Density (kg/m3)	96.41	96.41	96.41	96.41	96.41
39	Bulk Velocity (m/s)	3.058	3.058	3.058	3.058	3.058
40	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
41	Length (m)	8428	8433	8438	8443	8448
42	Bulk Density (kg/m3)	96.41	96.41	96.42	96.42	96.42
43	Bulk Velocity (m/s)	3.058	3.058	3.058	3.058	3.058
44	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
45	Length (m)	8453	8458	8463	8468	8473
46	Bulk Density (kg/m3)	96.42	96.42	96.42	96.42	96.41
47	Bulk Velocity (m/s)	3.058	3.058	3.058	3.058	3.058
48	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
49	Length (m)	8478	8483	8488	8493	8498
50	Bulk Density (kg/m3)	96.41	96.41	96.41	96.41	96.42
51	Bulk Velocity (m/s)	3.058	3.058	3.058	3.058	3.058
52	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
53	Length (m)	8503	8508	8513	8518	8523
54	Bulk Density (kg/m3)	96.42	96.42	96.42	96.42	96.42
55	Bulk Velocity (m/s)	3.058	3.058	3.058	3.058	3.058
56	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
57	Length (m)	8528	8533	8538	8543	8548
58	Bulk Density (kg/m3)	96.42	96.42	96.42	96.42	96.42
59	Bulk Velocity (m/s)	3.058	3.058	3.058	3.058	3.058
60	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
61	Length (m)	8553	8558	8563	8568	8573
62	Bulk Density (kg/m3)	96.42	96.42	96.42	96.42	96.42
63	Bulk Velocity (m/s)	3.058	3.058	3.058	3.058	3.058
64	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
65	Length (m)	8578	8583	8588	8593	8598
66	Bulk Density (kg/m3)	96.42	96.42	96.42	96.42	96.42
67	Bulk Velocity (m/s)	3.058	3.058	3.058	3.058	3.058
68	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 594 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	8603	8608	8613	8618	8623
14	Bulk Density (kg/m3)	96.42	96.42	96.42	96.42	96.42
15	Bulk Velocity (m/s)	3.058	3.058	3.058	3.058	3.058
16	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
17	Length (m)	8628	8633	8638	8643	8648
18	Bulk Density (kg/m3)	96.42	96.42	96.42	96.42	96.43
19	Bulk Velocity (m/s)	3.058	3.058	3.058	3.058	3.058
20	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
21	Length (m)	8653	8658	8663	8668	8673
22	Bulk Density (kg/m3)	96.43	96.43	96.43	96.43	96.43
23	Bulk Velocity (m/s)	3.058	3.058	3.058	3.058	3.058
24	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
25	Length (m)	8678	8683	8688	8693	8698
26	Bulk Density (kg/m3)	96.43	96.43	96.43	96.43	96.43
27	Bulk Velocity (m/s)	3.058	3.058	3.058	3.058	3.058
28	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
29	Length (m)	8703	8708	8713	8718	8723
30	Bulk Density (kg/m3)	96.43	96.43	96.43	96.43	96.43
31	Bulk Velocity (m/s)	3.058	3.058	3.058	3.058	3.058
32	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
33	Length (m)	8728	8733	8738	8743	8748
34	Bulk Density (kg/m3)	96.43	96.43	96.43	96.43	96.43
35	Bulk Velocity (m/s)	3.058	3.058	3.058	3.058	3.058
36	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
37	Length (m)	8753	8758	8763	8768	8773
38	Bulk Density (kg/m3)	96.43	96.43	96.43	96.43	96.43
39	Bulk Velocity (m/s)	3.058	3.058	3.057	3.057	3.057
40	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
41	Length (m)	8778	8783	8788	8793	8798
42	Bulk Density (kg/m3)	96.43	96.44	96.44	96.44	96.44
43	Bulk Velocity (m/s)	3.057	3.057	3.057	3.057	3.057
44	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
45	Length (m)	8803	8805	8807	8809	8811
46	Bulk Density (kg/m3)	96.44	96.44	96.44	96.44	96.44
47	Bulk Velocity (m/s)	3.057	3.057	3.057	3.057	3.057
48	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
49	Length (m)	8813	8840	8867	8894	8921
50	Bulk Density (kg/m3)	96.44	96.44	96.44	96.44	96.44
51	Bulk Velocity (m/s)	3.057	3.057	3.057	3.057	3.057
52	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
53	Length (m)	8948	8949	8950	8951	8952
54	Bulk Density (kg/m3)	96.44	96.44	96.44	96.44	96.44
55	Bulk Velocity (m/s)	3.057	3.057	3.057	3.057	3.057
56	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
57	Length (m)	8953	8958	8963	8968	8973
58	Bulk Density (kg/m3)	96.44	96.44	96.45	96.45	96.45
59	Bulk Velocity (m/s)	3.057	3.057	3.057	3.057	3.057
60	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
61	Length (m)	8978	8983	8988	8993	8998
62	Bulk Density (kg/m3)	96.45	96.45	96.45	96.45	96.45
63	Bulk Velocity (m/s)	3.057	3.057	3.057	3.057	3.057
64	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
65	Length (m)	9003	9008	9013	9018	9023
66	Bulk Density (kg/m3)	96.45	96.45	96.45	96.45	96.45
67	Bulk Velocity (m/s)	3.057	3.057	3.057	3.057	3.057
68	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 595 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	9028	9033	9038	9043	9048
14	Bulk Density (kg/m3)	96.45	96.45	96.45	96.45	96.45
15	Bulk Velocity (m/s)	3.057	3.057	3.057	3.057	3.057
16	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
17	Length (m)	9053	9054	9055	9056	9057
18	Bulk Density (kg/m3)	96.45	96.45	96.45	96.45	96.45
19	Bulk Velocity (m/s)	3.057	3.057	3.057	3.057	3.057
20	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
21	Length (m)	9058	9065	9071	9078	9085
22	Bulk Density (kg/m3)	96.45	96.45	96.45	96.45	96.45
23	Bulk Velocity (m/s)	3.057	3.057	3.057	3.057	3.057
24	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
25	Length (m)	9091	9098	9105	9111	9118
26	Bulk Density (kg/m3)	96.45	96.46	96.46	96.46	96.46
27	Bulk Velocity (m/s)	3.057	3.057	3.057	3.057	3.057
28	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
29	Length (m)	9125	9131	9138	9145	9151
30	Bulk Density (kg/m3)	96.46	96.46	96.46	96.46	96.46
31	Bulk Velocity (m/s)	3.057	3.057	3.057	3.057	3.057
32	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
33	Length (m)	9158	9162	9166	9170	9174
34	Bulk Density (kg/m3)	96.46	96.46	96.46	96.46	96.46
35	Bulk Velocity (m/s)	3.057	3.057	3.057	3.057	3.057
36	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
37	Length (m)	9178	9183	9188	9193	9198
38	Bulk Density (kg/m3)	96.46	96.46	96.46	96.47	96.47
39	Bulk Velocity (m/s)	3.057	3.057	3.057	3.056	3.056
40	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
41	Length (m)	9203	9208	9213	9218	9223
42	Bulk Density (kg/m3)	96.47	96.47	96.47	96.47	96.47
43	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
44	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
45	Length (m)	9228	9233	9238	9243	9248
46	Bulk Density (kg/m3)	96.47	96.47	96.47	96.47	96.47
47	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
48	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
49	Length (m)	9253	9258	9263	9268	9273
50	Bulk Density (kg/m3)	96.47	96.47	96.47	96.47	96.47
51	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
52	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
53	Length (m)	9278	9283	9288	9293	9298
54	Bulk Density (kg/m3)	96.47	96.47	96.47	96.47	96.47
55	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
56	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
57	Length (m)	9303	9308	9313	9318	9323
58	Bulk Density (kg/m3)	96.47	96.47	96.47	96.47	96.47
59	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
60	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
61	Length (m)	9328	9333	9338	9343	9348
62	Bulk Density (kg/m3)	96.47	96.47	96.47	96.47	96.47
63	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
64	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
65	Length (m)	9353	9358	9363	9368	9373
66	Bulk Density (kg/m3)	96.47	96.47	96.47	96.47	96.47
67	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
68	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 596 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	9378	9383	9388	9393	9398
14	Bulk Density (kg/m3)	96.47	96.47	96.47	96.47	96.47
15	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
16	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
17	Length (m)	9403	9408	9413	9418	9423
18	Bulk Density (kg/m3)	96.47	96.47	96.47	96.48	96.48
19	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
20	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
21	Length (m)	9428	9433	9438	9443	9448
22	Bulk Density (kg/m3)	96.48	96.48	96.48	96.48	96.48
23	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
24	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
25	Length (m)	9453	9456	9459	9462	9465
26	Bulk Density (kg/m3)	96.48	96.48	96.48	96.47	96.47
27	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
28	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
29	Length (m)	9468	9473	9477	9482	9487
30	Bulk Density (kg/m3)	96.47	96.47	96.47	96.47	96.47
31	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
32	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
33	Length (m)	9491	9496	9501	9505	9510
34	Bulk Density (kg/m3)	96.47	96.47	96.47	96.47	96.48
35	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
36	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
37	Length (m)	9515	9519	9524	9529	9533
38	Bulk Density (kg/m3)	96.48	96.48	96.48	96.48	96.48
39	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
40	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
41	Length (m)	9538	9543	9547	9552	9557
42	Bulk Density (kg/m3)	96.48	96.48	96.48	96.48	96.48
43	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
44	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
45	Length (m)	9561	9566	9571	9575	9580
46	Bulk Density (kg/m3)	96.48	96.47	96.47	96.47	96.47
47	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
48	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
49	Length (m)	9585	9589	9594	9599	9603
50	Bulk Density (kg/m3)	96.47	96.47	96.47	96.47	96.47
51	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
52	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
53	Length (m)	9608	9612	9616	9620	9624
54	Bulk Density (kg/m3)	96.47	96.47	96.47	96.47	96.47
55	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
56	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
57	Length (m)	9628	9633	9638	9643	9648
58	Bulk Density (kg/m3)	96.47	96.47	96.47	96.47	96.47
59	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
60	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
61	Length (m)	9653	9658	9663	9668	9673
62	Bulk Density (kg/m3)	96.47	96.47	96.47	96.47	96.47
63	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
64	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
65	Length (m)	9678	9683	9688	9693	9698
66	Bulk Density (kg/m3)	96.48	96.47	96.47	96.47	96.47
67	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
68	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 597 of 937	

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	9703	9708	9713	9718	9723
14	Bulk Density (kg/m3)	96.47	96.47	96.47	96.47	96.47
15	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
16	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
17	Length (m)	9728	9733	9738	9743	9748
18	Bulk Density (kg/m3)	96.47	96.47	96.47	96.47	96.47
19	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
20	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
21	Length (m)	9753	9758	9763	9768	9773
22	Bulk Density (kg/m3)	96.47	96.47	96.47	96.47	96.47
23	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
24	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
25	Length (m)	9778	9783	9788	9793	9798
26	Bulk Density (kg/m3)	96.48	96.48	96.48	96.48	96.48
27	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
28	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
29	Length (m)	9803	9808	9813	9818	9823
30	Bulk Density (kg/m3)	96.48	96.48	96.48	96.48	96.48
31	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
32	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
33	Length (m)	9828	9833	9838	9843	9848
34	Bulk Density (kg/m3)	96.48	96.48	96.48	96.48	96.48
35	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
36	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
37	Length (m)	9853	9858	9863	9868	9873
38	Bulk Density (kg/m3)	96.48	96.48	96.48	96.48	96.48
39	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
40	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
41	Length (m)	9878	9883	9888	9893	9898
42	Bulk Density (kg/m3)	96.48	96.48	96.48	96.48	96.48
43	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
44	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
45	Length (m)	9903	9905	9907	9909	9911
46	Bulk Density (kg/m3)	96.48	96.48	96.48	96.48	96.48
47	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
48	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
49	Length (m)	9913	9933	9953	9973	9993
50	Bulk Density (kg/m3)	96.48	96.48	96.49	96.49	96.49
51	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
52	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
53	Length (m)	1.001e+004	1.002e+004	1.002e+004	1.002e+004	1.003e+004
54	Bulk Density (kg/m3)	96.49	96.49	96.49	96.50	96.50
55	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.055
56	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
57	Length (m)	1.003e+004	1.003e+004	1.004e+004	1.004e+004	1.005e+004
58	Bulk Density (kg/m3)	96.50	96.50	96.50	96.50	96.50
59	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
60	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
61	Length (m)	1.005e+004	1.006e+004	1.006e+004	1.007e+004	1.007e+004
62	Bulk Density (kg/m3)	96.50	96.50	96.50	96.50	96.49
63	Bulk Velocity (m/s)	3.055	3.055	3.055	3.056	3.056
64	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
65	Length (m)	1.008e+004	1.008e+004	1.009e+004	1.009e+004	1.010e+004
66	Bulk Density (kg/m3)	96.49	96.49	96.49	96.49	96.49
67	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
68	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 598 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.010e+004	1.011e+004	1.011e+004	1.012e+004	1.012e+004
14	Bulk Density (kg/m3)	96.49	96.49	96.49	96.49	96.49
15	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
16	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
17	Length (m)	1.013e+004	1.013e+004	1.014e+004	1.014e+004	1.015e+004
18	Bulk Density (kg/m3)	96.49	96.49	96.49	96.49	96.49
19	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
20	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
21	Length (m)	1.015e+004	1.016e+004	1.016e+004	1.017e+004	1.017e+004
22	Bulk Density (kg/m3)	96.49	96.49	96.49	96.49	96.49
23	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
24	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
25	Length (m)	1.018e+004	1.018e+004	1.019e+004	1.019e+004	1.020e+004
26	Bulk Density (kg/m3)	96.49	96.49	96.49	96.49	96.49
27	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
28	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
29	Length (m)	1.020e+004	1.021e+004	1.021e+004	1.022e+004	1.022e+004
30	Bulk Density (kg/m3)	96.49	96.49	96.49	96.49	96.49
31	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
32	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
33	Length (m)	1.023e+004	1.023e+004	1.024e+004	1.024e+004	1.025e+004
34	Bulk Density (kg/m3)	96.49	96.49	96.49	96.49	96.50
35	Bulk Velocity (m/s)	3.056	3.056	3.056	3.056	3.056
36	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
37	Length (m)	1.025e+004	1.026e+004	1.026e+004	1.027e+004	1.027e+004
38	Bulk Density (kg/m3)	96.50	96.50	96.50	96.50	96.50
39	Bulk Velocity (m/s)	3.056	3.055	3.055	3.055	3.055
40	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
41	Length (m)	1.028e+004	1.028e+004	1.029e+004	1.029e+004	1.030e+004
42	Bulk Density (kg/m3)	96.50	96.50	96.50	96.50	96.50
43	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
44	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
45	Length (m)	1.030e+004	1.031e+004	1.031e+004	1.032e+004	1.032e+004
46	Bulk Density (kg/m3)	96.49	96.50	96.50	96.50	96.50
47	Bulk Velocity (m/s)	3.056	3.056	3.056	3.055	3.055
48	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
49	Length (m)	1.033e+004	1.033e+004	1.034e+004	1.034e+004	1.035e+004
50	Bulk Density (kg/m3)	96.50	96.50	96.50	96.50	96.50
51	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
52	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
53	Length (m)	1.035e+004	1.036e+004	1.036e+004	1.037e+004	1.037e+004
54	Bulk Density (kg/m3)	96.50	96.50	96.50	96.50	96.50
55	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
56	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
57	Length (m)	1.038e+004	1.038e+004	1.039e+004	1.039e+004	1.040e+004
58	Bulk Density (kg/m3)	96.50	96.50	96.50	96.50	96.51
59	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
60	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
61	Length (m)	1.040e+004	1.041e+004	1.041e+004	1.042e+004	1.042e+004
62	Bulk Density (kg/m3)	96.51	96.51	96.51	96.51	96.51
63	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
64	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
65	Length (m)	1.043e+004	1.043e+004	1.044e+004	1.044e+004	1.045e+004
66	Bulk Density (kg/m3)	96.50	96.51	96.51	96.51	96.51
67	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
68	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 599 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.045e+004	1.046e+004	1.046e+004	1.047e+004	1.047e+004
14	Bulk Density (kg/m3)	96.51	96.51	96.51	96.51	96.52
15	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
16	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
17	Length (m)	1.048e+004	1.048e+004	1.049e+004	1.049e+004	1.050e+004
18	Bulk Density (kg/m3)	96.52	96.52	96.52	96.52	96.52
19	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
20	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
21	Length (m)	1.050e+004	1.051e+004	1.051e+004	1.052e+004	1.052e+004
22	Bulk Density (kg/m3)	96.52	96.52	96.51	96.51	96.51
23	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
24	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
25	Length (m)	1.053e+004	1.053e+004	1.054e+004	1.054e+004	1.055e+004
26	Bulk Density (kg/m3)	96.51	96.51	96.51	96.51	96.51
27	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
28	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
29	Length (m)	1.055e+004	1.056e+004	1.056e+004	1.057e+004	1.057e+004
30	Bulk Density (kg/m3)	96.51	96.51	96.51	96.51	96.51
31	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
32	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
33	Length (m)	1.058e+004	1.058e+004	1.059e+004	1.059e+004	1.060e+004
34	Bulk Density (kg/m3)	96.51	96.51	96.51	96.51	96.51
35	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
36	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
37	Length (m)	1.060e+004	1.061e+004	1.061e+004	1.062e+004	1.062e+004
38	Bulk Density (kg/m3)	96.51	96.51	96.51	96.51	96.51
39	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
40	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
41	Length (m)	1.063e+004	1.063e+004	1.064e+004	1.064e+004	1.065e+004
42	Bulk Density (kg/m3)	96.51	96.51	96.51	96.51	96.51
43	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
44	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
45	Length (m)	1.065e+004	1.066e+004	1.066e+004	1.067e+004	1.067e+004
46	Bulk Density (kg/m3)	96.51	96.51	96.51	96.51	96.51
47	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
48	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
49	Length (m)	1.068e+004	1.068e+004	1.069e+004	1.069e+004	1.070e+004
50	Bulk Density (kg/m3)	96.51	96.51	96.51	96.51	96.51
51	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
52	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
53	Length (m)	1.070e+004	1.071e+004	1.071e+004	1.072e+004	1.072e+004
54	Bulk Density (kg/m3)	96.51	96.51	96.51	96.51	96.51
55	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
56	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
57	Length (m)	1.073e+004	1.073e+004	1.074e+004	1.074e+004	1.075e+004
58	Bulk Density (kg/m3)	96.51	96.51	96.51	96.51	96.51
59	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
60	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
61	Length (m)	1.075e+004	1.076e+004	1.076e+004	1.077e+004	1.077e+004
62	Bulk Density (kg/m3)	96.51	96.51	96.51	96.51	96.51
63	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
64	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
65	Length (m)	1.078e+004	1.078e+004	1.079e+004	1.079e+004	1.080e+004
66	Bulk Density (kg/m3)	96.51	96.51	96.51	96.51	96.52
67	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
68	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 600 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.080e+004	1.081e+004	1.081e+004	1.082e+004	1.082e+004
14	Bulk Density (kg/m3)	96.52	96.52	96.52	96.52	96.52
15	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
16	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
17	Length (m)	1.083e+004	1.083e+004	1.084e+004	1.084e+004	1.085e+004
18	Bulk Density (kg/m3)	96.52	96.52	96.52	96.52	96.52
19	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
20	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
21	Length (m)	1.085e+004	1.086e+004	1.086e+004	1.087e+004	1.087e+004
22	Bulk Density (kg/m3)	96.52	96.52	96.52	96.52	96.52
23	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
24	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
25	Length (m)	1.088e+004	1.088e+004	1.089e+004	1.089e+004	1.090e+004
26	Bulk Density (kg/m3)	96.52	96.52	96.52	96.52	96.52
27	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
28	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
29	Length (m)	1.090e+004	1.091e+004	1.091e+004	1.092e+004	1.092e+004
30	Bulk Density (kg/m3)	96.52	96.52	96.52	96.52	96.52
31	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
32	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
33	Length (m)	1.093e+004	1.093e+004	1.094e+004	1.094e+004	1.095e+004
34	Bulk Density (kg/m3)	96.52	96.52	96.52	96.52	96.52
35	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
36	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
37	Length (m)	1.095e+004	1.096e+004	1.096e+004	1.097e+004	1.097e+004
38	Bulk Density (kg/m3)	96.52	96.52	96.52	96.52	96.52
39	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
40	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
41	Length (m)	1.098e+004	1.098e+004	1.099e+004	1.099e+004	1.099e+004
42	Bulk Density (kg/m3)	96.52	96.52	96.52	96.53	96.53
43	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
44	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
45	Length (m)	1.100e+004	1.100e+004	1.101e+004	1.102e+004	1.102e+004
46	Bulk Density (kg/m3)	96.53	96.53	96.53	96.53	96.53
47	Bulk Velocity (m/s)	3.055	3.055	3.055	3.055	3.055
48	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
49	Length (m)	1.103e+004	1.103e+004	1.104e+004	1.104e+004	1.105e+004
50	Bulk Density (kg/m3)	96.53	96.53	96.53	96.53	96.53
51	Bulk Velocity (m/s)	3.055	3.054	3.054	3.054	3.054
52	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
53	Length (m)	1.105e+004	1.106e+004	1.106e+004	1.107e+004	1.107e+004
54	Bulk Density (kg/m3)	96.53	96.53	96.53	96.53	96.53
55	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
56	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
57	Length (m)	1.108e+004	1.108e+004	1.109e+004	1.109e+004	1.110e+004
58	Bulk Density (kg/m3)	96.53	96.53	96.53	96.53	96.53
59	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
60	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
61	Length (m)	1.110e+004	1.111e+004	1.111e+004	1.112e+004	1.112e+004
62	Bulk Density (kg/m3)	96.53	96.53	96.53	96.53	96.53
63	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
64	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
65	Length (m)	1.113e+004	1.113e+004	1.114e+004	1.114e+004	1.115e+004
66	Bulk Density (kg/m3)	96.53	96.53	96.53	96.53	96.53
67	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
68	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 601 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.115e+004	1.116e+004	1.116e+004	1.117e+004	1.117e+004
14	Bulk Density (kg/m3)	96.53	96.54	96.54	96.54	96.54
15	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
16	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
17	Length (m)	1.118e+004	1.118e+004	1.119e+004	1.119e+004	1.120e+004
18	Bulk Density (kg/m3)	96.54	96.54	96.54	96.54	96.54
19	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
20	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
21	Length (m)	1.120e+004	1.121e+004	1.121e+004	1.122e+004	1.122e+004
22	Bulk Density (kg/m3)	96.54	96.54	96.54	96.54	96.54
23	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
24	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
25	Length (m)	1.123e+004	1.123e+004	1.124e+004	1.124e+004	1.125e+004
26	Bulk Density (kg/m3)	96.54	96.54	96.54	96.54	96.54
27	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
28	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
29	Length (m)	1.125e+004	1.126e+004	1.126e+004	1.127e+004	1.127e+004
30	Bulk Density (kg/m3)	96.54	96.54	96.54	96.54	96.54
31	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
32	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
33	Length (m)	1.128e+004	1.128e+004	1.129e+004	1.129e+004	1.130e+004
34	Bulk Density (kg/m3)	96.54	96.54	96.54	96.54	96.54
35	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
36	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
37	Length (m)	1.130e+004	1.131e+004	1.131e+004	1.132e+004	1.132e+004
38	Bulk Density (kg/m3)	96.55	96.55	96.55	96.55	96.55
39	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
40	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
41	Length (m)	1.133e+004	1.133e+004	1.134e+004	1.134e+004	1.135e+004
42	Bulk Density (kg/m3)	96.55	96.55	96.55	96.55	96.55
43	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
44	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
45	Length (m)	1.135e+004	1.136e+004	1.136e+004	1.137e+004	1.137e+004
46	Bulk Density (kg/m3)	96.55	96.55	96.55	96.55	96.55
47	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
48	Erosion Velocity (m/s)	12.42	12.42	12.42	12.41	12.41
49	Length (m)	1.138e+004	1.138e+004	1.139e+004	1.139e+004	1.140e+004
50	Bulk Density (kg/m3)	96.55	96.55	96.55	96.55	96.55
51	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
52	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
53	Length (m)	1.140e+004	1.141e+004	1.141e+004	1.142e+004	1.142e+004
54	Bulk Density (kg/m3)	96.55	96.56	96.56	96.56	96.57
55	Bulk Velocity (m/s)	3.054	3.054	3.053	3.053	3.053
56	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
57	Length (m)	1.143e+004	1.143e+004	1.144e+004	1.144e+004	1.145e+004
58	Bulk Density (kg/m3)	96.57	96.57	96.57	96.57	96.57
59	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
60	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
61	Length (m)	1.145e+004	1.146e+004	1.146e+004	1.147e+004	1.147e+004
62	Bulk Density (kg/m3)	96.56	96.56	96.56	96.55	96.55
63	Bulk Velocity (m/s)	3.053	3.053	3.054	3.054	3.054
64	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
65	Length (m)	1.148e+004	1.148e+004	1.149e+004	1.149e+004	1.150e+004
66	Bulk Density (kg/m3)	96.55	96.55	96.55	96.55	96.55
67	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
68	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 602 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.150e+004	1.151e+004	1.151e+004	1.152e+004	1.152e+004
14	Bulk Density (kg/m3)	96.55	96.55	96.55	96.55	96.55
15	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
16	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
17	Length (m)	1.153e+004	1.153e+004	1.154e+004	1.154e+004	1.155e+004
18	Bulk Density (kg/m3)	96.55	96.55	96.55	96.55	96.55
19	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
20	Erosion Velocity (m/s)	12.42	12.42	12.42	12.42	12.42
21	Length (m)	1.155e+004	1.156e+004	1.156e+004	1.157e+004	1.157e+004
22	Bulk Density (kg/m3)	96.55	96.55	96.55	96.55	96.55
23	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
24	Erosion Velocity (m/s)	12.42	12.41	12.41	12.41	12.41
25	Length (m)	1.158e+004	1.158e+004	1.159e+004	1.159e+004	1.160e+004
26	Bulk Density (kg/m3)	96.55	96.55	96.55	96.55	96.55
27	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
28	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
29	Length (m)	1.160e+004	1.161e+004	1.161e+004	1.162e+004	1.162e+004
30	Bulk Density (kg/m3)	96.55	96.55	96.55	96.56	96.56
31	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
32	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
33	Length (m)	1.163e+004	1.163e+004	1.164e+004	1.164e+004	1.165e+004
34	Bulk Density (kg/m3)	96.56	96.56	96.56	96.56	96.56
35	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
36	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
37	Length (m)	1.165e+004	1.166e+004	1.166e+004	1.167e+004	1.167e+004
38	Bulk Density (kg/m3)	96.56	96.56	96.56	96.56	96.56
39	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
40	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
41	Length (m)	1.168e+004	1.168e+004	1.169e+004	1.169e+004	1.170e+004
42	Bulk Density (kg/m3)	96.56	96.56	96.56	96.56	96.56
43	Bulk Velocity (m/s)	3.054	3.054	3.054	3.054	3.054
44	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
45	Length (m)	1.170e+004	1.171e+004	1.171e+004	1.172e+004	1.172e+004
46	Bulk Density (kg/m3)	96.56	96.56	96.56	96.56	96.56
47	Bulk Velocity (m/s)	3.054	3.053	3.053	3.053	3.053
48	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
49	Length (m)	1.173e+004	1.173e+004	1.174e+004	1.174e+004	1.175e+004
50	Bulk Density (kg/m3)	96.56	96.56	96.56	96.56	96.56
51	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
52	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
53	Length (m)	1.175e+004	1.176e+004	1.176e+004	1.177e+004	1.177e+004
54	Bulk Density (kg/m3)	96.56	96.56	96.56	96.56	96.56
55	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
56	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
57	Length (m)	1.178e+004	1.178e+004	1.179e+004	1.179e+004	1.180e+004
58	Bulk Density (kg/m3)	96.56	96.57	96.57	96.57	96.57
59	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
60	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
61	Length (m)	1.180e+004	1.181e+004	1.181e+004	1.182e+004	1.182e+004
62	Bulk Density (kg/m3)	96.57	96.57	96.57	96.57	96.57
63	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
64	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
65	Length (m)	1.183e+004	1.183e+004	1.184e+004	1.184e+004	1.185e+004
66	Bulk Density (kg/m3)	96.57	96.57	96.57	96.57	96.57
67	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
68	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 603 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.185e+004	1.186e+004	1.186e+004	1.187e+004	1.187e+004
14	Bulk Density (kg/m3)	96.57	96.57	96.57	96.57	96.57
15	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
16	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
17	Length (m)	1.188e+004	1.188e+004	1.189e+004	1.189e+004	1.190e+004
18	Bulk Density (kg/m3)	96.57	96.57	96.57	96.57	96.57
19	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
20	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
21	Length (m)	1.190e+004	1.191e+004	1.191e+004	1.192e+004	1.192e+004
22	Bulk Density (kg/m3)	96.57	96.57	96.57	96.57	96.57
23	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
24	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
25	Length (m)	1.193e+004	1.193e+004	1.194e+004	1.194e+004	1.194e+004
26	Bulk Density (kg/m3)	96.57	96.57	96.57	96.57	96.57
27	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
28	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
29	Length (m)	1.195e+004	1.196e+004	1.196e+004	1.197e+004	1.198e+004
30	Bulk Density (kg/m3)	96.58	96.58	96.58	96.58	96.58
31	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
32	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
33	Length (m)	1.198e+004	1.199e+004	1.200e+004	1.200e+004	1.201e+004
34	Bulk Density (kg/m3)	96.58	96.58	96.58	96.58	96.58
35	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
36	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
37	Length (m)	1.202e+004	1.203e+004	1.203e+004	1.204e+004	1.205e+004
38	Bulk Density (kg/m3)	96.58	96.58	96.58	96.58	96.58
39	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
40	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
41	Length (m)	1.205e+004	1.206e+004	1.207e+004	1.207e+004	1.208e+004
42	Bulk Density (kg/m3)	96.58	96.58	96.58	96.57	96.57
43	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
44	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
45	Length (m)	1.209e+004	1.210e+004	1.210e+004	1.211e+004	1.212e+004
46	Bulk Density (kg/m3)	96.57	96.57	96.57	96.58	96.58
47	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
48	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
49	Length (m)	1.212e+004	1.213e+004	1.214e+004	1.214e+004	1.215e+004
50	Bulk Density (kg/m3)	96.58	96.58	96.58	96.58	96.58
51	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
52	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
53	Length (m)	1.216e+004	1.217e+004	1.217e+004	1.218e+004	1.219e+004
54	Bulk Density (kg/m3)	96.58	96.58	96.58	96.59	96.59
55	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
56	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
57	Length (m)	1.219e+004	1.220e+004	1.221e+004	1.221e+004	1.222e+004
58	Bulk Density (kg/m3)	96.59	96.59	96.59	96.59	96.59
59	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
60	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
61	Length (m)	1.223e+004	1.223e+004	1.224e+004	1.225e+004	1.225e+004
62	Bulk Density (kg/m3)	96.59	96.59	96.59	96.59	96.59
63	Bulk Velocity (m/s)	3.053	3.053	3.053	3.053	3.053
64	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
65	Length (m)	1.226e+004	1.226e+004	1.227e+004	1.227e+004	1.228e+004
66	Bulk Density (kg/m3)	96.59	96.59	96.59	96.59	96.59
67	Bulk Velocity (m/s)	3.053	3.052	3.052	3.052	3.052
68	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 604 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.228e+004	1.229e+004	1.229e+004	1.230e+004	1.230e+004
14	Bulk Density (kg/m3)	96.59	96.59	96.59	96.59	96.59
15	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
16	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
17	Length (m)	1.231e+004	1.231e+004	1.232e+004	1.232e+004	1.233e+004
18	Bulk Density (kg/m3)	96.59	96.59	96.59	96.59	96.59
19	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
20	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
21	Length (m)	1.233e+004	1.234e+004	1.234e+004	1.235e+004	1.235e+004
22	Bulk Density (kg/m3)	96.60	96.60	96.60	96.60	96.60
23	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
24	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
25	Length (m)	1.236e+004	1.237e+004	1.237e+004	1.238e+004	1.239e+004
26	Bulk Density (kg/m3)	96.60	96.60	96.60	96.60	96.60
27	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
28	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
29	Length (m)	1.239e+004	1.240e+004	1.240e+004	1.241e+004	1.241e+004
30	Bulk Density (kg/m3)	96.60	96.60	96.60	96.60	96.60
31	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
32	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
33	Length (m)	1.242e+004	1.242e+004	1.243e+004	1.243e+004	1.244e+004
34	Bulk Density (kg/m3)	96.60	96.60	96.60	96.60	96.60
35	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
36	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
37	Length (m)	1.244e+004	1.245e+004	1.245e+004	1.246e+004	1.246e+004
38	Bulk Density (kg/m3)	96.60	96.60	96.60	96.60	96.60
39	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
40	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
41	Length (m)	1.247e+004	1.247e+004	1.248e+004	1.248e+004	1.249e+004
42	Bulk Density (kg/m3)	96.60	96.60	96.60	96.60	96.60
43	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
44	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
45	Length (m)	1.249e+004	1.250e+004	1.250e+004	1.251e+004	1.251e+004
46	Bulk Density (kg/m3)	96.60	96.61	96.61	96.61	96.61
47	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
48	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
49	Length (m)	1.252e+004	1.252e+004	1.253e+004	1.253e+004	1.254e+004
50	Bulk Density (kg/m3)	96.61	96.61	96.61	96.61	96.61
51	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
52	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
53	Length (m)	1.254e+004	1.255e+004	1.255e+004	1.256e+004	1.256e+004
54	Bulk Density (kg/m3)	96.61	96.61	96.61	96.61	96.61
55	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
56	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
57	Length (m)	1.257e+004	1.257e+004	1.258e+004	1.258e+004	1.259e+004
58	Bulk Density (kg/m3)	96.61	96.61	96.61	96.61	96.61
59	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
60	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
61	Length (m)	1.259e+004	1.260e+004	1.260e+004	1.261e+004	1.261e+004
62	Bulk Density (kg/m3)	96.61	96.61	96.60	96.60	96.60
63	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
64	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
65	Length (m)	1.262e+004	1.262e+004	1.263e+004	1.264e+004	1.265e+004
66	Bulk Density (kg/m3)	96.60	96.60	96.60	96.60	96.60
67	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
68	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 605 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
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11	Erosion Table					
12						
13	Length (m)	1.265e+004	1.266e+004	1.266e+004	1.267e+004	1.267e+004
14	Bulk Density (kg/m3)	96.60	96.60	96.60	96.60	96.60
15	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
16	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
17	Length (m)	1.268e+004	1.268e+004	1.269e+004	1.269e+004	1.270e+004
18	Bulk Density (kg/m3)	96.60	96.60	96.60	96.61	96.61
19	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
20	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
21	Length (m)	1.270e+004	1.271e+004	1.271e+004	1.272e+004	1.272e+004
22	Bulk Density (kg/m3)	96.61	96.61	96.61	96.61	96.61
23	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
24	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
25	Length (m)	1.273e+004	1.273e+004	1.274e+004	1.274e+004	1.275e+004
26	Bulk Density (kg/m3)	96.61	96.61	96.61	96.61	96.61
27	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
28	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
29	Length (m)	1.275e+004	1.276e+004	1.276e+004	1.276e+004	1.277e+004
30	Bulk Density (kg/m3)	96.61	96.61	96.61	96.61	96.62
31	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
32	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
33	Length (m)	1.277e+004	1.278e+004	1.278e+004	1.278e+004	1.278e+004
34	Bulk Density (kg/m3)	96.62	96.62	96.62	96.62	96.62
35	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
36	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
37	Length (m)	1.279e+004	1.279e+004	1.279e+004	1.280e+004	1.280e+004
38	Bulk Density (kg/m3)	96.62	96.62	96.62	96.62	96.61
39	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
40	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
41	Length (m)	1.280e+004	1.281e+004	1.281e+004	1.282e+004	1.282e+004
42	Bulk Density (kg/m3)	96.61	96.62	96.62	96.62	96.62
43	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
44	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
45	Length (m)	1.283e+004	1.283e+004	1.284e+004	1.284e+004	1.285e+004
46	Bulk Density (kg/m3)	96.62	96.62	96.62	96.62	96.62
47	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
48	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
49	Length (m)	1.285e+004	1.286e+004	1.286e+004	1.287e+004	1.287e+004
50	Bulk Density (kg/m3)	96.61	96.61	96.61	96.61	96.61
51	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
52	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
53	Length (m)	1.288e+004	1.288e+004	1.289e+004	1.289e+004	1.290e+004
54	Bulk Density (kg/m3)	96.61	96.61	96.61	96.62	96.62
55	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
56	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
57	Length (m)	1.290e+004	1.291e+004	1.291e+004	1.292e+004	1.292e+004
58	Bulk Density (kg/m3)	96.62	96.62	96.62	96.62	96.62
59	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
60	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
61	Length (m)	1.293e+004	1.293e+004	1.294e+004	1.294e+004	1.295e+004
62	Bulk Density (kg/m3)	96.62	96.62	96.62	96.62	96.62
63	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
64	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
65	Length (m)	1.295e+004	1.296e+004	1.296e+004	1.297e+004	1.297e+004
66	Bulk Density (kg/m3)	96.62	96.62	96.62	96.62	96.62
67	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
68	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 606 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.298e+004	1.298e+004	1.299e+004	1.299e+004	1.300e+004
14	Bulk Density (kg/m3)	96.62	96.62	96.62	96.62	96.62
15	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
16	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
17	Length (m)	1.300e+004	1.301e+004	1.301e+004	1.302e+004	1.302e+004
18	Bulk Density (kg/m3)	96.62	96.62	96.62	96.62	96.62
19	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
20	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
21	Length (m)	1.303e+004	1.303e+004	1.304e+004	1.304e+004	1.305e+004
22	Bulk Density (kg/m3)	96.62	96.62	96.62	96.62	96.62
23	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.052
24	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
25	Length (m)	1.305e+004	1.306e+004	1.306e+004	1.307e+004	1.307e+004
26	Bulk Density (kg/m3)	96.62	96.62	96.62	96.62	96.62
27	Bulk Velocity (m/s)	3.052	3.052	3.052	3.052	3.051
28	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
29	Length (m)	1.308e+004	1.308e+004	1.309e+004	1.309e+004	1.310e+004
30	Bulk Density (kg/m3)	96.62	96.62	96.62	96.62	96.62
31	Bulk Velocity (m/s)	3.051	3.051	3.051	3.051	3.051
32	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
33	Length (m)	1.310e+004	1.311e+004	1.311e+004	1.312e+004	1.312e+004
34	Bulk Density (kg/m3)	96.63	96.63	96.63	96.63	96.63
35	Bulk Velocity (m/s)	3.051	3.051	3.051	3.051	3.051
36	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
37	Length (m)	1.313e+004	1.313e+004	1.314e+004	1.314e+004	1.315e+004
38	Bulk Density (kg/m3)	96.63	96.63	96.63	96.63	96.63
39	Bulk Velocity (m/s)	3.051	3.051	3.051	3.051	3.051
40	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
41	Length (m)	1.315e+004	1.316e+004	1.316e+004	1.317e+004	1.317e+004
42	Bulk Density (kg/m3)	96.63	96.63	96.63	96.63	96.63
43	Bulk Velocity (m/s)	3.051	3.051	3.051	3.051	3.051
44	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
45	Length (m)	1.318e+004	1.318e+004	1.319e+004	1.319e+004	1.320e+004
46	Bulk Density (kg/m3)	96.63	96.63	96.63	96.63	96.63
47	Bulk Velocity (m/s)	3.051	3.051	3.051	3.051	3.051
48	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
49	Length (m)	1.320e+004	1.321e+004	1.321e+004	1.322e+004	1.322e+004
50	Bulk Density (kg/m3)	96.63	96.63	96.64	96.64	96.65
51	Bulk Velocity (m/s)	3.051	3.051	3.051	3.051	3.051
52	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
53	Length (m)	1.323e+004	1.323e+004	1.324e+004	1.324e+004	1.325e+004
54	Bulk Density (kg/m3)	96.65	96.65	96.66	96.66	96.66
55	Bulk Velocity (m/s)	3.051	3.051	3.050	3.050	3.050
56	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
57	Length (m)	1.325e+004	1.326e+004	1.326e+004	1.327e+004	1.327e+004
58	Bulk Density (kg/m3)	96.66	96.67	96.68	96.69	96.70
59	Bulk Velocity (m/s)	3.050	3.050	3.050	3.049	3.049
60	Erosion Velocity (m/s)	12.41	12.41	12.41	12.41	12.41
61	Length (m)	1.328e+004	1.328e+004	1.329e+004	1.329e+004	1.330e+004
62	Bulk Density (kg/m3)	96.70	96.71	96.71	96.72	96.72
63	Bulk Velocity (m/s)	3.049	3.049	3.049	3.049	3.048
64	Erosion Velocity (m/s)	12.41	12.40	12.40	12.40	12.40
65	Length (m)	1.330e+004	1.331e+004	1.331e+004	1.332e+004	1.332e+004
66	Bulk Density (kg/m3)	96.73	96.73	96.74	96.74	96.75
67	Bulk Velocity (m/s)	3.048	3.048	3.048	3.048	3.048
68	Erosion Velocity (m/s)	12.40	12.40	12.40	12.40	12.40
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 607 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.333e+004	1.333e+004	1.334e+004	1.334e+004	1.335e+004
14	Bulk Density (kg/m3)	96.75	96.76	96.77	96.78	96.78
15	Bulk Velocity (m/s)	3.047	3.047	3.047	3.047	3.046
16	Erosion Velocity (m/s)	12.40	12.40	12.40	12.40	12.40
17	Length (m)	1.335e+004	1.336e+004	1.336e+004	1.337e+004	1.337e+004
18	Bulk Density (kg/m3)	96.79	96.80	96.81	96.81	96.82
19	Bulk Velocity (m/s)	3.046	3.046	3.046	3.046	3.045
20	Erosion Velocity (m/s)	12.40	12.40	12.40	12.40	12.40
21	Length (m)	1.338e+004	1.338e+004	1.339e+004	1.339e+004	1.340e+004
22	Bulk Density (kg/m3)	96.83	96.83	96.83	96.83	96.83
23	Bulk Velocity (m/s)	3.045	3.045	3.045	3.045	3.045
24	Erosion Velocity (m/s)	12.40	12.40	12.40	12.40	12.40
25	Length (m)	1.340e+004	1.341e+004	1.341e+004	1.342e+004	1.342e+004
26	Bulk Density (kg/m3)	96.83	96.81	96.79	96.77	96.75
27	Bulk Velocity (m/s)	3.045	3.046	3.046	3.047	3.048
28	Erosion Velocity (m/s)	12.40	12.40	12.40	12.40	12.40
29	Length (m)	1.343e+004	1.343e+004	1.344e+004	1.344e+004	1.345e+004
30	Bulk Density (kg/m3)	96.72	96.71	96.70	96.68	96.67
31	Bulk Velocity (m/s)	3.048	3.049	3.049	3.050	3.050
32	Erosion Velocity (m/s)	12.40	12.40	12.41	12.41	12.41
33	Length (m)	1.345e+004	1.346e+004	1.346e+004	1.347e+004	1.347e+004
34	Bulk Density (kg/m3)	96.66	96.50	96.33	96.17	96.01
35	Bulk Velocity (m/s)	3.050	3.056	3.061	3.066	3.071
36	Erosion Velocity (m/s)	12.41	12.42	12.43	12.44	12.45
37	Length (m)	1.348e+004	1.348e+004	1.349e+004	1.349e+004	1.350e+004
38	Bulk Density (kg/m3)	95.84	95.84	95.83	95.83	95.82
39	Bulk Velocity (m/s)	3.076	3.076	3.077	3.077	3.077
40	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
41	Length (m)	1.350e+004	1.351e+004	1.351e+004	1.352e+004	1.352e+004
42	Bulk Density (kg/m3)	95.82	95.82	95.82	95.82	95.82
43	Bulk Velocity (m/s)	3.077	3.077	3.077	3.077	3.077
44	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
45	Length (m)	1.353e+004	1.353e+004	1.353e+004	1.353e+004	1.354e+004
46	Bulk Density (kg/m3)	95.82	95.82	95.82	95.82	95.82
47	Bulk Velocity (m/s)	3.077	3.077	3.077	3.077	3.077
48	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
49	Length (m)	1.354e+004	1.354e+004	1.354e+004	1.355e+004	1.355e+004
50	Bulk Density (kg/m3)	95.82	95.82	95.82	95.82	95.82
51	Bulk Velocity (m/s)	3.077	3.077	3.077	3.077	3.077
52	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
53	Length (m)	1.355e+004	1.356e+004	1.356e+004	1.356e+004	1.357e+004
54	Bulk Density (kg/m3)	95.82	95.82	95.82	95.82	95.82
55	Bulk Velocity (m/s)	3.077	3.077	3.077	3.077	3.077
56	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
57	Length (m)	1.357e+004	1.357e+004	1.358e+004	1.358e+004	1.358e+004
58	Bulk Density (kg/m3)	95.82	95.82	95.82	95.82	95.82
59	Bulk Velocity (m/s)	3.077	3.077	3.077	3.077	3.077
60	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
61	Length (m)	1.359e+004	1.359e+004	1.359e+004	1.360e+004	1.360e+004
62	Bulk Density (kg/m3)	95.82	95.82	95.82	95.82	95.82
63	Bulk Velocity (m/s)	3.077	3.077	3.077	3.077	3.077
64	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
65	Length (m)	1.360e+004	1.361e+004	1.361e+004	1.362e+004	1.362e+004
66	Bulk Density (kg/m3)	95.82	95.82	95.82	95.82	95.82
67	Bulk Velocity (m/s)	3.077	3.077	3.077	3.077	3.077
68	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 608 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.363e+004	1.363e+004	1.364e+004	1.364e+004	1.365e+004
14	Bulk Density (kg/m3)	95.82	95.82	95.82	95.82	95.82
15	Bulk Velocity (m/s)	3.077	3.077	3.077	3.077	3.077
16	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
17	Length (m)	1.365e+004	1.366e+004	1.366e+004	1.367e+004	1.367e+004
18	Bulk Density (kg/m3)	95.82	95.82	95.82	95.82	95.82
19	Bulk Velocity (m/s)	3.077	3.077	3.077	3.077	3.077
20	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
21	Length (m)	1.368e+004	1.368e+004	1.369e+004	1.369e+004	1.370e+004
22	Bulk Density (kg/m3)	95.82	95.83	95.83	95.83	95.83
23	Bulk Velocity (m/s)	3.077	3.077	3.077	3.077	3.077
24	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
25	Length (m)	1.370e+004	1.371e+004	1.371e+004	1.372e+004	1.372e+004
26	Bulk Density (kg/m3)	95.83	95.83	95.83	95.83	95.83
27	Bulk Velocity (m/s)	3.077	3.077	3.077	3.077	3.077
28	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
29	Length (m)	1.373e+004	1.373e+004	1.374e+004	1.374e+004	1.375e+004
30	Bulk Density (kg/m3)	95.83	95.83	95.83	95.83	95.83
31	Bulk Velocity (m/s)	3.077	3.077	3.077	3.077	3.077
32	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
33	Length (m)	1.375e+004	1.376e+004	1.376e+004	1.377e+004	1.377e+004
34	Bulk Density (kg/m3)	95.83	95.83	95.83	95.83	95.84
35	Bulk Velocity (m/s)	3.077	3.077	3.077	3.077	3.077
36	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
37	Length (m)	1.378e+004	1.378e+004	1.379e+004	1.379e+004	1.380e+004
38	Bulk Density (kg/m3)	95.84	95.84	95.84	95.84	95.84
39	Bulk Velocity (m/s)	3.077	3.077	3.077	3.077	3.077
40	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
41	Length (m)	1.380e+004	1.381e+004	1.381e+004	1.382e+004	1.382e+004
42	Bulk Density (kg/m3)	95.84	95.84	95.83	95.83	95.83
43	Bulk Velocity (m/s)	3.077	3.077	3.077	3.077	3.077
44	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
45	Length (m)	1.383e+004	1.383e+004	1.384e+004	1.384e+004	1.385e+004
46	Bulk Density (kg/m3)	95.83	95.83	95.84	95.84	95.84
47	Bulk Velocity (m/s)	3.077	3.077	3.077	3.077	3.076
48	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
49	Length (m)	1.385e+004	1.386e+004	1.386e+004	1.387e+004	1.387e+004
50	Bulk Density (kg/m3)	95.84	95.84	95.84	95.84	95.84
51	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
52	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
53	Length (m)	1.388e+004	1.388e+004	1.389e+004	1.389e+004	1.390e+004
54	Bulk Density (kg/m3)	95.84	95.84	95.84	95.84	95.84
55	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
56	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
57	Length (m)	1.390e+004	1.391e+004	1.391e+004	1.392e+004	1.392e+004
58	Bulk Density (kg/m3)	95.84	95.84	95.84	95.84	95.84
59	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
60	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
61	Length (m)	1.393e+004	1.393e+004	1.394e+004	1.394e+004	1.395e+004
62	Bulk Density (kg/m3)	95.84	95.84	95.84	95.84	95.84
63	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
64	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
65	Length (m)	1.395e+004	1.396e+004	1.396e+004	1.397e+004	1.397e+004
66	Bulk Density (kg/m3)	95.84	95.84	95.84	95.84	95.84
67	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
68	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 609 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
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11	Erosion Table					
12						
13	Length (m)	1.398e+004	1.398e+004	1.399e+004	1.399e+004	1.400e+004
14	Bulk Density (kg/m3)	95.84	95.84	95.84	95.84	95.84
15	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
16	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
17	Length (m)	1.400e+004	1.401e+004	1.401e+004	1.402e+004	1.402e+004
18	Bulk Density (kg/m3)	95.84	95.84	95.84	95.84	95.84
19	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
20	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
21	Length (m)	1.403e+004	1.403e+004	1.404e+004	1.404e+004	1.405e+004
22	Bulk Density (kg/m3)	95.85	95.84	95.84	95.84	95.84
23	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
24	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
25	Length (m)	1.405e+004	1.405e+004	1.406e+004	1.406e+004	1.406e+004
26	Bulk Density (kg/m3)	95.84	95.84	95.84	95.84	95.84
27	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
28	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
29	Length (m)	1.406e+004	1.406e+004	1.407e+004	1.407e+004	1.408e+004
30	Bulk Density (kg/m3)	95.84	95.84	95.84	95.84	95.84
31	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
32	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
33	Length (m)	1.408e+004	1.408e+004	1.409e+004	1.409e+004	1.409e+004
34	Bulk Density (kg/m3)	95.84	95.84	95.84	95.84	95.84
35	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
36	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
37	Length (m)	1.409e+004	1.409e+004	1.410e+004	1.411e+004	1.411e+004
38	Bulk Density (kg/m3)	95.84	95.84	95.85	95.85	95.85
39	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
40	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
41	Length (m)	1.412e+004	1.413e+004	1.413e+004	1.414e+004	1.415e+004
42	Bulk Density (kg/m3)	95.85	95.85	95.85	95.85	95.85
43	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
44	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
45	Length (m)	1.415e+004	1.416e+004	1.416e+004	1.417e+004	1.417e+004
46	Bulk Density (kg/m3)	95.85	95.85	95.85	95.85	95.84
47	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
48	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
49	Length (m)	1.418e+004	1.418e+004	1.419e+004	1.419e+004	1.420e+004
50	Bulk Density (kg/m3)	95.84	95.84	95.84	95.84	95.84
51	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
52	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
53	Length (m)	1.420e+004	1.421e+004	1.421e+004	1.422e+004	1.422e+004
54	Bulk Density (kg/m3)	95.84	95.84	95.84	95.84	95.84
55	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
56	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
57	Length (m)	1.423e+004	1.423e+004	1.424e+004	1.424e+004	1.425e+004
58	Bulk Density (kg/m3)	95.84	95.84	95.84	95.84	95.84
59	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
60	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
61	Length (m)	1.425e+004	1.426e+004	1.426e+004	1.427e+004	1.427e+004
62	Bulk Density (kg/m3)	95.84	95.84	95.84	95.84	95.84
63	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
64	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
65	Length (m)	1.428e+004	1.428e+004	1.429e+004	1.429e+004	1.430e+004
66	Bulk Density (kg/m3)	95.84	95.84	95.84	95.84	95.84
67	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
68	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 610 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
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11	Erosion Table					
12						
13	Length (m)	1.430e+004	1.431e+004	1.431e+004	1.432e+004	1.432e+004
14	Bulk Density (kg/m3)	95.84	95.84	95.84	95.84	95.84
15	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
16	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
17	Length (m)	1.433e+004	1.433e+004	1.434e+004	1.434e+004	1.435e+004
18	Bulk Density (kg/m3)	95.84	95.84	95.85	95.85	95.85
19	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
20	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
21	Length (m)	1.435e+004	1.436e+004	1.436e+004	1.437e+004	1.437e+004
22	Bulk Density (kg/m3)	95.85	95.85	95.85	95.85	95.85
23	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
24	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
25	Length (m)	1.438e+004	1.438e+004	1.439e+004	1.439e+004	1.440e+004
26	Bulk Density (kg/m3)	95.85	95.85	95.85	95.85	95.85
27	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
28	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
29	Length (m)	1.440e+004	1.441e+004	1.441e+004	1.442e+004	1.442e+004
30	Bulk Density (kg/m3)	95.85	95.85	95.85	95.85	95.85
31	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
32	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
33	Length (m)	1.443e+004	1.443e+004	1.444e+004	1.444e+004	1.445e+004
34	Bulk Density (kg/m3)	95.85	95.85	95.85	95.85	95.85
35	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
36	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
37	Length (m)	1.445e+004	1.446e+004	1.446e+004	1.447e+004	1.447e+004
38	Bulk Density (kg/m3)	95.85	95.85	95.85	95.85	95.85
39	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
40	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
41	Length (m)	1.448e+004	1.448e+004	1.449e+004	1.449e+004	1.450e+004
42	Bulk Density (kg/m3)	95.85	95.85	95.85	95.85	95.86
43	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
44	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
45	Length (m)	1.450e+004	1.451e+004	1.451e+004	1.452e+004	1.452e+004
46	Bulk Density (kg/m3)	95.86	95.86	95.86	95.86	95.86
47	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
48	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
49	Length (m)	1.453e+004	1.453e+004	1.454e+004	1.454e+004	1.455e+004
50	Bulk Density (kg/m3)	95.86	95.86	95.86	95.86	95.86
51	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
52	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
53	Length (m)	1.455e+004	1.456e+004	1.456e+004	1.457e+004	1.457e+004
54	Bulk Density (kg/m3)	95.86	95.86	95.86	95.86	95.86
55	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
56	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
57	Length (m)	1.458e+004	1.458e+004	1.459e+004	1.459e+004	1.460e+004
58	Bulk Density (kg/m3)	95.86	95.86	95.86	95.86	95.86
59	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
60	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
61	Length (m)	1.460e+004	1.461e+004	1.461e+004	1.462e+004	1.462e+004
62	Bulk Density (kg/m3)	95.86	95.86	95.86	95.86	95.86
63	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
64	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
65	Length (m)	1.463e+004	1.463e+004	1.464e+004	1.464e+004	1.465e+004
66	Bulk Density (kg/m3)	95.86	95.86	95.86	95.86	95.86
67	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
68	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 611 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.465e+004	1.466e+004	1.466e+004	1.467e+004	1.467e+004
14	Bulk Density (kg/m3)	95.86	95.86	95.86	95.86	95.86
15	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
16	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
17	Length (m)	1.468e+004	1.468e+004	1.469e+004	1.469e+004	1.470e+004
18	Bulk Density (kg/m3)	95.86	95.86	95.86	95.86	95.87
19	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
20	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
21	Length (m)	1.470e+004	1.471e+004	1.471e+004	1.472e+004	1.472e+004
22	Bulk Density (kg/m3)	95.87	95.87	95.87	95.87	95.87
23	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
24	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
25	Length (m)	1.473e+004	1.473e+004	1.474e+004	1.474e+004	1.475e+004
26	Bulk Density (kg/m3)	95.87	95.87	95.87	95.87	95.87
27	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
28	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
29	Length (m)	1.475e+004	1.476e+004	1.476e+004	1.477e+004	1.477e+004
30	Bulk Density (kg/m3)	95.87	95.87	95.87	95.87	95.87
31	Bulk Velocity (m/s)	3.076	3.076	3.076	3.075	3.075
32	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
33	Length (m)	1.478e+004	1.478e+004	1.479e+004	1.479e+004	1.480e+004
34	Bulk Density (kg/m3)	95.87	95.87	95.87	95.87	95.87
35	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
36	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
37	Length (m)	1.480e+004	1.481e+004	1.481e+004	1.482e+004	1.482e+004
38	Bulk Density (kg/m3)	95.87	95.87	95.87	95.87	95.87
39	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
40	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
41	Length (m)	1.483e+004	1.483e+004	1.484e+004	1.484e+004	1.485e+004
42	Bulk Density (kg/m3)	95.87	95.87	95.87	95.87	95.87
43	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
44	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
45	Length (m)	1.485e+004	1.486e+004	1.486e+004	1.486e+004	1.486e+004
46	Bulk Density (kg/m3)	95.87	95.87	95.87	95.87	95.87
47	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
48	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
49	Length (m)	1.486e+004	1.488e+004	1.490e+004	1.492e+004	1.494e+004
50	Bulk Density (kg/m3)	95.87	95.88	95.88	95.88	95.88
51	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
52	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
53	Length (m)	1.496e+004	1.498e+004	1.500e+004	1.502e+004	1.504e+004
54	Bulk Density (kg/m3)	95.88	95.88	95.88	95.88	95.88
55	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
56	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
57	Length (m)	1.506e+004	1.508e+004	1.510e+004	1.512e+004	1.514e+004
58	Bulk Density (kg/m3)	95.88	95.89	95.89	95.89	95.89
59	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
60	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
61	Length (m)	1.516e+004	1.518e+004	1.520e+004	1.522e+004	1.524e+004
62	Bulk Density (kg/m3)	95.89	95.89	95.89	95.89	95.89
63	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
64	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
65	Length (m)	1.526e+004	1.528e+004	1.530e+004	1.532e+004	1.534e+004
66	Bulk Density (kg/m3)	95.89	95.90	95.90	95.90	95.90
67	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.074
68	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 612 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.536e+004	1.538e+004	1.540e+004	1.542e+004	1.544e+004
14	Bulk Density (kg/m3)	95.90	95.92	95.93	95.94	95.95
15	Bulk Velocity (m/s)	3.074	3.074	3.074	3.073	3.073
16	Erosion Velocity (m/s)	12.46	12.46	12.46	12.45	12.45
17	Length (m)	1.546e+004	1.548e+004	1.550e+004	1.552e+004	1.554e+004
18	Bulk Density (kg/m3)	95.97	95.95	95.93	95.91	95.89
19	Bulk Velocity (m/s)	3.072	3.073	3.074	3.074	3.075
20	Erosion Velocity (m/s)	12.45	12.45	12.46	12.46	12.46
21	Length (m)	1.556e+004	1.558e+004	1.560e+004	1.562e+004	1.564e+004
22	Bulk Density (kg/m3)	95.87	95.86	95.86	95.86	95.86
23	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
24	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
25	Length (m)	1.566e+004	1.568e+004	1.570e+004	1.572e+004	1.574e+004
26	Bulk Density (kg/m3)	95.86	95.86	95.86	95.86	95.87
27	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
28	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
29	Length (m)	1.576e+004	1.577e+004	1.577e+004	1.578e+004	1.578e+004
30	Bulk Density (kg/m3)	95.87	95.87	95.87	95.87	95.87
31	Bulk Velocity (m/s)	3.076	3.076	3.076	3.076	3.076
32	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
33	Length (m)	1.579e+004	1.581e+004	1.583e+004	1.585e+004	1.587e+004
34	Bulk Density (kg/m3)	95.87	95.87	95.87	95.87	95.87
35	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
36	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
37	Length (m)	1.589e+004	1.589e+004	1.590e+004	1.590e+004	1.590e+004
38	Bulk Density (kg/m3)	95.87	95.87	95.87	95.87	95.87
39	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
40	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
41	Length (m)	1.591e+004	1.591e+004	1.592e+004	1.592e+004	1.592e+004
42	Bulk Density (kg/m3)	95.87	95.87	95.87	95.87	95.87
43	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
44	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
45	Length (m)	1.593e+004	1.593e+004	1.593e+004	1.593e+004	1.594e+004
46	Bulk Density (kg/m3)	95.87	95.87	95.87	95.87	95.87
47	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
48	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
49	Length (m)	1.594e+004	1.594e+004	1.595e+004	1.595e+004	1.595e+004
50	Bulk Density (kg/m3)	95.87	95.87	95.87	95.87	95.87
51	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
52	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
53	Length (m)	1.596e+004	1.596e+004	1.597e+004	1.597e+004	1.598e+004
54	Bulk Density (kg/m3)	95.87	95.87	95.87	95.87	95.87
55	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
56	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
57	Length (m)	1.598e+004	1.599e+004	1.599e+004	1.600e+004	1.600e+004
58	Bulk Density (kg/m3)	95.87	95.87	95.87	95.87	95.87
59	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
60	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
61	Length (m)	1.601e+004	1.601e+004	1.602e+004	1.602e+004	1.603e+004
62	Bulk Density (kg/m3)	95.87	95.87	95.87	95.87	95.87
63	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
64	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
65	Length (m)	1.603e+004	1.604e+004	1.604e+004	1.605e+004	1.605e+004
66	Bulk Density (kg/m3)	95.87	95.88	95.88	95.88	95.88
67	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
68	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 613 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.606e+004	1.606e+004	1.607e+004	1.607e+004	1.608e+004
14	Bulk Density (kg/m3)	95.88	95.88	95.88	95.88	95.88
15	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
16	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
17	Length (m)	1.608e+004	1.609e+004	1.609e+004	1.610e+004	1.610e+004
18	Bulk Density (kg/m3)	95.88	95.88	95.88	95.88	95.88
19	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
20	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
21	Length (m)	1.611e+004	1.611e+004	1.612e+004	1.612e+004	1.613e+004
22	Bulk Density (kg/m3)	95.88	95.88	95.88	95.88	95.88
23	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
24	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
25	Length (m)	1.613e+004	1.614e+004	1.614e+004	1.615e+004	1.615e+004
26	Bulk Density (kg/m3)	95.88	95.89	95.89	95.89	95.89
27	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
28	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
29	Length (m)	1.616e+004	1.616e+004	1.617e+004	1.617e+004	1.618e+004
30	Bulk Density (kg/m3)	95.89	95.89	95.89	95.89	95.89
31	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
32	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
33	Length (m)	1.618e+004	1.619e+004	1.619e+004	1.620e+004	1.620e+004
34	Bulk Density (kg/m3)	95.89	95.89	95.89	95.89	95.89
35	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
36	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
37	Length (m)	1.621e+004	1.621e+004	1.622e+004	1.622e+004	1.623e+004
38	Bulk Density (kg/m3)	95.89	95.89	95.89	95.88	95.88
39	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
40	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
41	Length (m)	1.623e+004	1.624e+004	1.624e+004	1.625e+004	1.625e+004
42	Bulk Density (kg/m3)	95.87	95.87	95.88	95.88	95.88
43	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
44	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
45	Length (m)	1.626e+004	1.626e+004	1.627e+004	1.627e+004	1.628e+004
46	Bulk Density (kg/m3)	95.88	95.88	95.88	95.88	95.88
47	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
48	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
49	Length (m)	1.628e+004	1.629e+004	1.629e+004	1.630e+004	1.630e+004
50	Bulk Density (kg/m3)	95.88	95.88	95.88	95.88	95.88
51	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
52	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
53	Length (m)	1.631e+004	1.631e+004	1.631e+004	1.631e+004	1.632e+004
54	Bulk Density (kg/m3)	95.88	95.88	95.88	95.88	95.89
55	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
56	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
57	Length (m)	1.632e+004	1.632e+004	1.632e+004	1.633e+004	1.633e+004
58	Bulk Density (kg/m3)	95.89	95.89	95.88	95.88	95.88
59	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
60	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
61	Length (m)	1.633e+004	1.634e+004	1.634e+004	1.635e+004	1.635e+004
62	Bulk Density (kg/m3)	95.88	95.88	95.88	95.88	95.88
63	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
64	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
65	Length (m)	1.636e+004	1.636e+004	1.637e+004	1.637e+004	1.638e+004
66	Bulk Density (kg/m3)	95.88	95.88	95.88	95.88	95.88
67	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
68	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 614 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.638e+004	1.639e+004	1.639e+004	1.640e+004	1.640e+004
14	Bulk Density (kg/m3)	95.88	95.88	95.88	95.88	95.88
15	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
16	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
17	Length (m)	1.641e+004	1.641e+004	1.642e+004	1.642e+004	1.643e+004
18	Bulk Density (kg/m3)	95.88	95.88	95.88	95.88	95.88
19	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
20	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
21	Length (m)	1.643e+004	1.644e+004	1.644e+004	1.645e+004	1.645e+004
22	Bulk Density (kg/m3)	95.88	95.88	95.88	95.88	95.88
23	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
24	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
25	Length (m)	1.646e+004	1.646e+004	1.647e+004	1.647e+004	1.648e+004
26	Bulk Density (kg/m3)	95.88	95.88	95.88	95.88	95.88
27	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
28	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
29	Length (m)	1.648e+004	1.649e+004	1.649e+004	1.650e+004	1.650e+004
30	Bulk Density (kg/m3)	95.88	95.88	95.88	95.88	95.88
31	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
32	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
33	Length (m)	1.651e+004	1.651e+004	1.652e+004	1.652e+004	1.653e+004
34	Bulk Density (kg/m3)	95.88	95.88	95.88	95.88	95.88
35	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
36	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
37	Length (m)	1.653e+004	1.654e+004	1.654e+004	1.655e+004	1.655e+004
38	Bulk Density (kg/m3)	95.88	95.88	95.88	95.88	95.88
39	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
40	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
41	Length (m)	1.656e+004	1.656e+004	1.657e+004	1.657e+004	1.658e+004
42	Bulk Density (kg/m3)	95.88	95.88	95.88	95.88	95.88
43	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
44	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
45	Length (m)	1.658e+004	1.659e+004	1.659e+004	1.660e+004	1.660e+004
46	Bulk Density (kg/m3)	95.88	95.88	95.88	95.89	95.89
47	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
48	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
49	Length (m)	1.661e+004	1.661e+004	1.662e+004	1.662e+004	1.663e+004
50	Bulk Density (kg/m3)	95.89	95.89	95.89	95.89	95.89
51	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
52	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
53	Length (m)	1.663e+004	1.664e+004	1.664e+004	1.665e+004	1.665e+004
54	Bulk Density (kg/m3)	95.89	95.89	95.89	95.89	95.89
55	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
56	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
57	Length (m)	1.666e+004	1.666e+004	1.667e+004	1.667e+004	1.668e+004
58	Bulk Density (kg/m3)	95.89	95.89	95.89	95.89	95.89
59	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
60	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
61	Length (m)	1.668e+004	1.669e+004	1.669e+004	1.670e+004	1.670e+004
62	Bulk Density (kg/m3)	95.89	95.89	95.89	95.89	95.89
63	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
64	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
65	Length (m)	1.671e+004	1.671e+004	1.672e+004	1.672e+004	1.673e+004
66	Bulk Density (kg/m3)	95.89	95.89	95.89	95.89	95.89
67	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
68	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 615 of 937					

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.673e+004	1.674e+004	1.674e+004	1.675e+004	1.675e+004
14	Bulk Density (kg/m3)	95.89	95.89	95.89	95.89	95.89
15	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
16	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
17	Length (m)	1.676e+004	1.676e+004	1.677e+004	1.677e+004	1.678e+004
18	Bulk Density (kg/m3)	95.89	95.89	95.90	95.90	95.90
19	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
20	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
21	Length (m)	1.678e+004	1.679e+004	1.679e+004	1.680e+004	1.680e+004
22	Bulk Density (kg/m3)	95.90	95.90	95.90	95.90	95.90
23	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
24	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
25	Length (m)	1.681e+004	1.681e+004	1.682e+004	1.682e+004	1.683e+004
26	Bulk Density (kg/m3)	95.90	95.90	95.90	95.90	95.90
27	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
28	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
29	Length (m)	1.683e+004	1.684e+004	1.684e+004	1.685e+004	1.685e+004
30	Bulk Density (kg/m3)	95.90	95.90	95.90	95.90	95.90
31	Bulk Velocity (m/s)	3.075	3.075	3.075	3.075	3.075
32	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
33	Length (m)	1.686e+004	1.686e+004	1.687e+004	1.687e+004	1.688e+004
34	Bulk Density (kg/m3)	95.90	95.90	95.90	95.90	95.90
35	Bulk Velocity (m/s)	3.075	3.074	3.074	3.074	3.074
36	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
37	Length (m)	1.688e+004	1.689e+004	1.689e+004	1.690e+004	1.690e+004
38	Bulk Density (kg/m3)	95.90	95.90	95.90	95.90	95.90
39	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
40	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
41	Length (m)	1.690e+004	1.691e+004	1.691e+004	1.692e+004	1.692e+004
42	Bulk Density (kg/m3)	95.90	95.90	95.91	95.91	95.91
43	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
44	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
45	Length (m)	1.693e+004	1.693e+004	1.694e+004	1.694e+004	1.695e+004
46	Bulk Density (kg/m3)	95.91	95.91	95.91	95.91	95.91
47	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
48	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
49	Length (m)	1.695e+004	1.696e+004	1.696e+004	1.697e+004	1.697e+004
50	Bulk Density (kg/m3)	95.91	95.91	95.91	95.91	95.91
51	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
52	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
53	Length (m)	1.698e+004	1.698e+004	1.699e+004	1.699e+004	1.700e+004
54	Bulk Density (kg/m3)	95.91	95.91	95.91	95.91	95.91
55	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
56	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
57	Length (m)	1.700e+004	1.701e+004	1.701e+004	1.702e+004	1.702e+004
58	Bulk Density (kg/m3)	95.91	95.91	95.91	95.91	95.91
59	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
60	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
61	Length (m)	1.703e+004	1.703e+004	1.704e+004	1.704e+004	1.705e+004
62	Bulk Density (kg/m3)	95.91	95.91	95.91	95.91	95.91
63	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
64	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
65	Length (m)	1.705e+004	1.706e+004	1.706e+004	1.707e+004	1.707e+004
66	Bulk Density (kg/m3)	95.91	95.91	95.91	95.91	95.91
67	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
68	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 616 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.708e+004	1.708e+004	1.709e+004	1.709e+004	1.710e+004
14	Bulk Density (kg/m3)	95.91	95.91	95.91	95.91	95.91
15	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
16	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
17	Length (m)	1.710e+004	1.711e+004	1.711e+004	1.712e+004	1.712e+004
18	Bulk Density (kg/m3)	95.91	95.91	95.91	95.91	95.91
19	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
20	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
21	Length (m)	1.713e+004	1.713e+004	1.714e+004	1.714e+004	1.715e+004
22	Bulk Density (kg/m3)	95.91	95.91	95.91	95.91	95.91
23	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
24	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
25	Length (m)	1.715e+004	1.716e+004	1.716e+004	1.717e+004	1.717e+004
26	Bulk Density (kg/m3)	95.91	95.91	95.91	95.91	95.91
27	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
28	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
29	Length (m)	1.718e+004	1.718e+004	1.719e+004	1.719e+004	1.720e+004
30	Bulk Density (kg/m3)	95.91	95.91	95.91	95.91	95.91
31	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
32	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
33	Length (m)	1.720e+004	1.721e+004	1.721e+004	1.722e+004	1.722e+004
34	Bulk Density (kg/m3)	95.91	95.91	95.91	95.91	95.91
35	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
36	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
37	Length (m)	1.723e+004	1.723e+004	1.724e+004	1.724e+004	1.725e+004
38	Bulk Density (kg/m3)	95.91	95.91	95.91	95.91	95.91
39	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
40	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
41	Length (m)	1.725e+004	1.726e+004	1.726e+004	1.727e+004	1.727e+004
42	Bulk Density (kg/m3)	95.91	95.91	95.91	95.91	95.91
43	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
44	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
45	Length (m)	1.728e+004	1.728e+004	1.729e+004	1.729e+004	1.730e+004
46	Bulk Density (kg/m3)	95.91	95.91	95.91	95.91	95.91
47	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
48	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
49	Length (m)	1.730e+004	1.731e+004	1.731e+004	1.732e+004	1.732e+004
50	Bulk Density (kg/m3)	95.91	95.91	95.91	95.91	95.91
51	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
52	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
53	Length (m)	1.733e+004	1.733e+004	1.734e+004	1.734e+004	1.735e+004
54	Bulk Density (kg/m3)	95.91	95.91	95.91	95.91	95.91
55	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
56	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
57	Length (m)	1.735e+004	1.736e+004	1.736e+004	1.737e+004	1.737e+004
58	Bulk Density (kg/m3)	95.91	95.91	95.92	95.92	95.92
59	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
60	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
61	Length (m)	1.738e+004	1.738e+004	1.739e+004	1.739e+004	1.740e+004
62	Bulk Density (kg/m3)	95.92	95.92	95.92	95.92	95.92
63	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
64	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
65	Length (m)	1.740e+004	1.741e+004	1.741e+004	1.742e+004	1.742e+004
66	Bulk Density (kg/m3)	95.92	95.92	95.92	95.92	95.92
67	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
68	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 617 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.743e+004	1.743e+004	1.744e+004	1.744e+004	1.745e+004
14	Bulk Density (kg/m3)	95.92	95.92	95.92	95.92	95.92
15	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
16	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
17	Length (m)	1.745e+004	1.746e+004	1.746e+004	1.747e+004	1.747e+004
18	Bulk Density (kg/m3)	95.92	95.92	95.92	95.92	95.92
19	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
20	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
21	Length (m)	1.748e+004	1.748e+004	1.749e+004	1.749e+004	1.750e+004
22	Bulk Density (kg/m3)	95.92	95.92	95.92	95.92	95.92
23	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
24	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
25	Length (m)	1.750e+004	1.751e+004	1.751e+004	1.752e+004	1.752e+004
26	Bulk Density (kg/m3)	95.92	95.92	95.92	95.92	95.92
27	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
28	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
29	Length (m)	1.753e+004	1.753e+004	1.754e+004	1.754e+004	1.755e+004
30	Bulk Density (kg/m3)	95.92	95.92	95.92	95.93	95.93
31	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
32	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
33	Length (m)	1.755e+004	1.756e+004	1.756e+004	1.757e+004	1.757e+004
34	Bulk Density (kg/m3)	95.93	95.93	95.93	95.93	95.93
35	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
36	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
37	Length (m)	1.758e+004	1.758e+004	1.759e+004	1.759e+004	1.760e+004
38	Bulk Density (kg/m3)	95.93	95.93	95.93	95.93	95.93
39	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
40	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
41	Length (m)	1.760e+004	1.761e+004	1.761e+004	1.762e+004	1.762e+004
42	Bulk Density (kg/m3)	95.92	95.93	95.93	95.93	95.93
43	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
44	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
45	Length (m)	1.763e+004	1.763e+004	1.764e+004	1.764e+004	1.765e+004
46	Bulk Density (kg/m3)	95.93	95.93	95.93	95.93	95.93
47	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
48	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
49	Length (m)	1.765e+004	1.766e+004	1.766e+004	1.767e+004	1.767e+004
50	Bulk Density (kg/m3)	95.93	95.93	95.93	95.93	95.93
51	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.074
52	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
53	Length (m)	1.768e+004	1.768e+004	1.769e+004	1.769e+004	1.770e+004
54	Bulk Density (kg/m3)	95.93	95.93	95.93	95.93	95.93
55	Bulk Velocity (m/s)	3.074	3.074	3.074	3.074	3.073
56	Erosion Velocity (m/s)	12.46	12.46	12.46	12.46	12.46
57	Length (m)	1.770e+004	1.771e+004	1.771e+004	1.772e+004	1.772e+004
58	Bulk Density (kg/m3)	95.93	95.93	95.93	95.93	95.93
59	Bulk Velocity (m/s)	3.073	3.073	3.073	3.073	3.073
60	Erosion Velocity (m/s)	12.46	12.46	12.45	12.45	12.45
61	Length (m)	1.773e+004	1.773e+004	1.774e+004	1.774e+004	1.775e+004
62	Bulk Density (kg/m3)	95.93	95.93	95.93	95.93	95.94
63	Bulk Velocity (m/s)	3.073	3.073	3.073	3.073	3.073
64	Erosion Velocity (m/s)	12.45	12.45	12.45	12.45	12.45
65	Length (m)	1.775e+004	1.776e+004	1.776e+004	1.777e+004	1.777e+004
66	Bulk Density (kg/m3)	95.94	95.94	95.94	95.94	95.94
67	Bulk Velocity (m/s)	3.073	3.073	3.073	3.073	3.073
68	Erosion Velocity (m/s)	12.45	12.45	12.45	12.45	12.45
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 618 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.778e+004	1.778e+004	1.779e+004	1.779e+004	1.780e+004
14	Bulk Density (kg/m3)	95.94	95.94	95.94	95.94	95.94
15	Bulk Velocity (m/s)	3.073	3.073	3.073	3.073	3.073
16	Erosion Velocity (m/s)	12.45	12.45	12.45	12.45	12.45
17	Length (m)	1.780e+004	1.781e+004	1.781e+004	1.782e+004	1.782e+004
18	Bulk Density (kg/m3)	95.94	95.94	95.94	95.94	95.94
19	Bulk Velocity (m/s)	3.073	3.073	3.073	3.073	3.073
20	Erosion Velocity (m/s)	12.45	12.45	12.45	12.45	12.45
21	Length (m)	1.783e+004	1.783e+004	1.784e+004	1.784e+004	1.785e+004
22	Bulk Density (kg/m3)	95.94	95.94	95.94	95.95	95.95
23	Bulk Velocity (m/s)	3.073	3.073	3.073	3.073	3.073
24	Erosion Velocity (m/s)	12.45	12.45	12.45	12.45	12.45
25	Length (m)	1.785e+004	1.785e+004	1.786e+004	1.786e+004	1.786e+004
26	Bulk Density (kg/m3)	95.95	95.95	95.95	95.95	95.95
27	Bulk Velocity (m/s)	3.073	3.073	3.073	3.073	3.073
28	Erosion Velocity (m/s)	12.45	12.45	12.45	12.45	12.45
29	Length (m)	1.787e+004	1.787e+004	1.787e+004	1.787e+004	1.787e+004
30	Bulk Density (kg/m3)	95.95	95.95	95.96	95.96	95.96
31	Bulk Velocity (m/s)	3.073	3.073	3.073	3.073	3.073
32	Erosion Velocity (m/s)	12.45	12.45	12.45	12.45	12.45
33	Length (m)	1.788e+004	1.788e+004	1.789e+004	1.789e+004	1.790e+004
34	Bulk Density (kg/m3)	95.96	95.97	95.98	95.98	95.99
35	Bulk Velocity (m/s)	3.073	3.072	3.072	3.072	3.072
36	Erosion Velocity (m/s)	12.45	12.45	12.45	12.45	12.45
37	Length (m)	1.790e+004	1.790e+004	1.791e+004	1.791e+004	1.791e+004
38	Bulk Density (kg/m3)	96.00	96.00	96.01	96.01	96.01
39	Bulk Velocity (m/s)	3.071	3.071	3.071	3.071	3.071
40	Erosion Velocity (m/s)	12.45	12.45	12.45	12.45	12.45
41	Length (m)	1.791e+004	1.792e+004	1.792e+004	1.792e+004	1.793e+004
42	Bulk Density (kg/m3)	96.02	96.02	96.02	96.03	96.03
43	Bulk Velocity (m/s)	3.071	3.071	3.071	3.070	3.070
44	Erosion Velocity (m/s)	12.45	12.45	12.45	12.45	12.45
45	Length (m)	1.793e+004	1.793e+004	1.794e+004	1.794e+004	1.794e+004
46	Bulk Density (kg/m3)	96.03	96.04	96.04	96.05	96.05
47	Bulk Velocity (m/s)	3.070	3.070	3.070	3.070	3.070
48	Erosion Velocity (m/s)	12.45	12.45	12.45	12.45	12.45
49	Length (m)	1.795e+004	1.795e+004	1.796e+004	1.796e+004	1.797e+004
50	Bulk Density (kg/m3)	96.06	96.06	96.07	96.07	96.08
51	Bulk Velocity (m/s)	3.069	3.069	3.069	3.069	3.069
52	Erosion Velocity (m/s)	12.45	12.45	12.45	12.45	12.45
53	Length (m)	1.798e+004	1.798e+004	1.799e+004	1.799e+004	1.800e+004
54	Bulk Density (kg/m3)	96.08	96.09	96.10	96.11	96.11
55	Bulk Velocity (m/s)	3.069	3.068	3.068	3.068	3.068
56	Erosion Velocity (m/s)	12.45	12.44	12.44	12.44	12.44
57	Length (m)	1.800e+004	1.801e+004	1.801e+004	1.802e+004	1.802e+004
58	Bulk Density (kg/m3)	96.12	96.13	96.13	96.14	96.15
59	Bulk Velocity (m/s)	3.067	3.067	3.067	3.067	3.067
60	Erosion Velocity (m/s)	12.44	12.44	12.44	12.44	12.44
61	Length (m)	1.803e+004	1.803e+004	1.804e+004	1.804e+004	1.805e+004
62	Bulk Density (kg/m3)	96.16	96.14	96.13	96.12	96.11
63	Bulk Velocity (m/s)	3.066	3.067	3.067	3.067	3.068
64	Erosion Velocity (m/s)	12.44	12.44	12.44	12.44	12.44
65	Length (m)	1.805e+004	1.806e+004	1.806e+004	1.807e+004	1.807e+004
66	Bulk Density (kg/m3)	96.10	95.87	95.64	95.40	95.17
67	Bulk Velocity (m/s)	3.068	3.076	3.083	3.090	3.098
68	Erosion Velocity (m/s)	12.44	12.46	12.47	12.49	12.50
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 619 of 937					

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.808e+004	1.808e+004	1.809e+004	1.809e+004	1.810e+004
14	Bulk Density (kg/m3)	94.94	94.94	94.94	94.94	94.94
15	Bulk Velocity (m/s)	3.105	3.106	3.106	3.106	3.106
16	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
17	Length (m)	1.810e+004	1.811e+004	1.811e+004	1.812e+004	1.812e+004
18	Bulk Density (kg/m3)	94.94	94.94	94.94	94.94	94.94
19	Bulk Velocity (m/s)	3.106	3.106	3.106	3.106	3.106
20	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
21	Length (m)	1.813e+004	1.813e+004	1.814e+004	1.814e+004	1.815e+004
22	Bulk Density (kg/m3)	94.93	94.93	94.93	94.93	94.93
23	Bulk Velocity (m/s)	3.106	3.106	3.106	3.106	3.106
24	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
25	Length (m)	1.815e+004	1.816e+004	1.816e+004	1.817e+004	1.817e+004
26	Bulk Density (kg/m3)	94.93	94.94	94.94	94.94	94.94
27	Bulk Velocity (m/s)	3.106	3.106	3.106	3.106	3.106
28	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
29	Length (m)	1.818e+004	1.818e+004	1.819e+004	1.819e+004	1.820e+004
30	Bulk Density (kg/m3)	94.94	94.94	94.94	94.94	94.94
31	Bulk Velocity (m/s)	3.106	3.106	3.106	3.106	3.106
32	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
33	Length (m)	1.820e+004	1.821e+004	1.821e+004	1.822e+004	1.822e+004
34	Bulk Density (kg/m3)	94.94	94.94	94.94	94.94	94.94
35	Bulk Velocity (m/s)	3.106	3.106	3.106	3.106	3.106
36	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
37	Length (m)	1.823e+004	1.823e+004	1.824e+004	1.824e+004	1.825e+004
38	Bulk Density (kg/m3)	94.94	94.94	94.94	94.94	94.94
39	Bulk Velocity (m/s)	3.106	3.106	3.106	3.106	3.106
40	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
41	Length (m)	1.825e+004	1.826e+004	1.826e+004	1.827e+004	1.827e+004
42	Bulk Density (kg/m3)	94.94	94.94	94.94	94.94	94.94
43	Bulk Velocity (m/s)	3.106	3.106	3.106	3.106	3.106
44	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
45	Length (m)	1.828e+004	1.828e+004	1.829e+004	1.829e+004	1.830e+004
46	Bulk Density (kg/m3)	94.94	94.94	94.94	94.94	94.94
47	Bulk Velocity (m/s)	3.106	3.106	3.106	3.106	3.106
48	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
49	Length (m)	1.830e+004	1.831e+004	1.831e+004	1.832e+004	1.832e+004
50	Bulk Density (kg/m3)	94.94	94.94	94.94	94.94	94.94
51	Bulk Velocity (m/s)	3.106	3.106	3.106	3.106	3.106
52	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
53	Length (m)	1.833e+004	1.833e+004	1.834e+004	1.834e+004	1.835e+004
54	Bulk Density (kg/m3)	94.94	94.94	94.94	94.94	94.94
55	Bulk Velocity (m/s)	3.106	3.106	3.106	3.106	3.106
56	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
57	Length (m)	1.835e+004	1.836e+004	1.836e+004	1.837e+004	1.837e+004
58	Bulk Density (kg/m3)	94.94	94.94	94.94	94.94	94.94
59	Bulk Velocity (m/s)	3.106	3.106	3.106	3.106	3.106
60	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
61	Length (m)	1.838e+004	1.838e+004	1.839e+004	1.839e+004	1.840e+004
62	Bulk Density (kg/m3)	94.94	94.94	94.94	94.94	94.94
63	Bulk Velocity (m/s)	3.106	3.106	3.106	3.106	3.106
64	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
65	Length (m)	1.840e+004	1.841e+004	1.841e+004	1.842e+004	1.842e+004
66	Bulk Density (kg/m3)	94.94	94.94	94.94	94.94	94.94
67	Bulk Velocity (m/s)	3.106	3.106	3.105	3.105	3.105
68	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 620 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.843e+004	1.843e+004	1.844e+004	1.844e+004	1.845e+004
14	Bulk Density (kg/m3)	94.94	94.94	94.94	94.94	94.95
15	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
16	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
17	Length (m)	1.845e+004	1.846e+004	1.846e+004	1.847e+004	1.847e+004
18	Bulk Density (kg/m3)	94.95	94.95	94.95	94.95	94.95
19	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
20	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
21	Length (m)	1.848e+004	1.848e+004	1.849e+004	1.849e+004	1.850e+004
22	Bulk Density (kg/m3)	94.95	94.95	94.95	94.95	94.95
23	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
24	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
25	Length (m)	1.850e+004	1.851e+004	1.851e+004	1.852e+004	1.852e+004
26	Bulk Density (kg/m3)	94.95	94.95	94.95	94.95	94.95
27	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
28	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
29	Length (m)	1.853e+004	1.853e+004	1.854e+004	1.854e+004	1.855e+004
30	Bulk Density (kg/m3)	94.95	94.95	94.95	94.95	94.95
31	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
32	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
33	Length (m)	1.855e+004	1.856e+004	1.856e+004	1.857e+004	1.857e+004
34	Bulk Density (kg/m3)	94.95	94.95	94.95	94.95	94.95
35	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
36	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
37	Length (m)	1.858e+004	1.858e+004	1.859e+004	1.859e+004	1.860e+004
38	Bulk Density (kg/m3)	94.95	94.95	94.95	94.95	94.95
39	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
40	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
41	Length (m)	1.860e+004	1.861e+004	1.861e+004	1.862e+004	1.862e+004
42	Bulk Density (kg/m3)	94.95	94.95	94.95	94.95	94.95
43	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
44	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
45	Length (m)	1.863e+004	1.863e+004	1.864e+004	1.864e+004	1.865e+004
46	Bulk Density (kg/m3)	94.95	94.95	94.95	94.95	94.95
47	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
48	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
49	Length (m)	1.865e+004	1.866e+004	1.866e+004	1.867e+004	1.867e+004
50	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
51	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
52	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
53	Length (m)	1.868e+004	1.868e+004	1.869e+004	1.869e+004	1.870e+004
54	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
55	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
56	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
57	Length (m)	1.870e+004	1.871e+004	1.871e+004	1.872e+004	1.872e+004
58	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
59	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
60	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
61	Length (m)	1.873e+004	1.873e+004	1.874e+004	1.874e+004	1.875e+004
62	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
63	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
64	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
65	Length (m)	1.875e+004	1.876e+004	1.876e+004	1.877e+004	1.877e+004
66	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
67	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
68	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 621 of 937					

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.878e+004	1.878e+004	1.879e+004	1.879e+004	1.880e+004
14	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
15	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
16	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
17	Length (m)	1.880e+004	1.880e+004	1.881e+004	1.881e+004	1.881e+004
18	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
19	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
20	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
21	Length (m)	1.882e+004	1.882e+004	1.882e+004	1.882e+004	1.882e+004
22	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
23	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
24	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
25	Length (m)	1.883e+004	1.883e+004	1.884e+004	1.884e+004	1.885e+004
26	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.95
27	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
28	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
29	Length (m)	1.885e+004	1.886e+004	1.886e+004	1.887e+004	1.887e+004
30	Bulk Density (kg/m3)	94.95	94.95	94.95	94.95	94.95
31	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
32	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
33	Length (m)	1.888e+004	1.888e+004	1.889e+004	1.889e+004	1.890e+004
34	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.95
35	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
36	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
37	Length (m)	1.890e+004	1.890e+004	1.891e+004	1.891e+004	1.891e+004
38	Bulk Density (kg/m3)	94.95	94.95	94.95	94.95	94.96
39	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
40	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
41	Length (m)	1.891e+004	1.891e+004	1.892e+004	1.892e+004	1.892e+004
42	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
43	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
44	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
45	Length (m)	1.893e+004	1.893e+004	1.894e+004	1.894e+004	1.895e+004
46	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
47	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
48	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
49	Length (m)	1.895e+004	1.896e+004	1.896e+004	1.897e+004	1.897e+004
50	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
51	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
52	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
53	Length (m)	1.898e+004	1.898e+004	1.899e+004	1.899e+004	1.900e+004
54	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
55	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
56	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
57	Length (m)	1.900e+004	1.901e+004	1.901e+004	1.902e+004	1.902e+004
58	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
59	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
60	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
61	Length (m)	1.903e+004	1.903e+004	1.904e+004	1.904e+004	1.905e+004
62	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
63	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
64	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
65	Length (m)	1.905e+004	1.906e+004	1.906e+004	1.907e+004	1.907e+004
66	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
67	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
68	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 622 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.908e+004	1.908e+004	1.909e+004	1.909e+004	1.910e+004
14	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
15	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
16	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
17	Length (m)	1.910e+004	1.911e+004	1.911e+004	1.912e+004	1.912e+004
18	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
19	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
20	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
21	Length (m)	1.913e+004	1.913e+004	1.914e+004	1.914e+004	1.915e+004
22	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
23	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
24	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
25	Length (m)	1.915e+004	1.916e+004	1.916e+004	1.917e+004	1.917e+004
26	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
27	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
28	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
29	Length (m)	1.918e+004	1.918e+004	1.919e+004	1.919e+004	1.920e+004
30	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
31	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
32	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
33	Length (m)	1.920e+004	1.921e+004	1.921e+004	1.922e+004	1.922e+004
34	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
35	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
36	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
37	Length (m)	1.923e+004	1.923e+004	1.924e+004	1.924e+004	1.925e+004
38	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
39	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
40	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
41	Length (m)	1.925e+004	1.926e+004	1.926e+004	1.927e+004	1.927e+004
42	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
43	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
44	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
45	Length (m)	1.928e+004	1.928e+004	1.929e+004	1.929e+004	1.930e+004
46	Bulk Density (kg/m3)	94.96	94.96	94.96	94.96	94.96
47	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
48	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
49	Length (m)	1.930e+004	1.931e+004	1.931e+004	1.932e+004	1.932e+004
50	Bulk Density (kg/m3)	94.97	94.97	94.97	94.97	94.97
51	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
52	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
53	Length (m)	1.933e+004	1.933e+004	1.934e+004	1.934e+004	1.935e+004
54	Bulk Density (kg/m3)	94.97	94.97	94.97	94.97	94.97
55	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
56	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
57	Length (m)	1.935e+004	1.936e+004	1.936e+004	1.937e+004	1.937e+004
58	Bulk Density (kg/m3)	94.97	94.97	94.98	94.99	95.00
59	Bulk Velocity (m/s)	3.105	3.104	3.104	3.104	3.104
60	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
61	Length (m)	1.938e+004	1.938e+004	1.939e+004	1.939e+004	1.940e+004
62	Bulk Density (kg/m3)	95.01	95.00	94.99	94.99	94.98
63	Bulk Velocity (m/s)	3.103	3.104	3.104	3.104	3.104
64	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
65	Length (m)	1.940e+004	1.941e+004	1.941e+004	1.942e+004	1.942e+004
66	Bulk Density (kg/m3)	94.97	94.95	94.92	94.90	94.87
67	Bulk Velocity (m/s)	3.105	3.105	3.106	3.107	3.108
68	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 623 of 937					

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.943e+004	1.943e+004	1.944e+004	1.944e+004	1.945e+004
14	Bulk Density (kg/m3)	94.85	94.85	94.85	94.85	94.85
15	Bulk Velocity (m/s)	3.109	3.108	3.108	3.108	3.108
16	Erosion Velocity (m/s)	12.53	12.53	12.53	12.53	12.53
17	Length (m)	1.945e+004	1.946e+004	1.946e+004	1.946e+004	1.947e+004
18	Bulk Density (kg/m3)	94.85	94.85	94.85	94.85	94.84
19	Bulk Velocity (m/s)	3.108	3.109	3.109	3.109	3.109
20	Erosion Velocity (m/s)	12.53	12.53	12.53	12.53	12.53
21	Length (m)	1.947e+004	1.948e+004	1.948e+004	1.949e+004	1.949e+004
22	Bulk Density (kg/m3)	94.84	94.84	94.84	94.84	94.84
23	Bulk Velocity (m/s)	3.109	3.109	3.109	3.109	3.109
24	Erosion Velocity (m/s)	12.53	12.53	12.53	12.53	12.53
25	Length (m)	1.950e+004	1.952e+004	1.954e+004	1.957e+004	1.959e+004
26	Bulk Density (kg/m3)	94.84	94.84	94.84	94.84	94.84
27	Bulk Velocity (m/s)	3.109	3.109	3.109	3.109	3.109
28	Erosion Velocity (m/s)	12.53	12.53	12.53	12.53	12.53
29	Length (m)	1.961e+004	1.964e+004	1.966e+004	1.968e+004	1.971e+004
30	Bulk Density (kg/m3)	94.84	94.84	94.84	94.84	94.84
31	Bulk Velocity (m/s)	3.109	3.109	3.109	3.109	3.109
32	Erosion Velocity (m/s)	12.53	12.53	12.53	12.53	12.53
33	Length (m)	1.973e+004	1.975e+004	1.978e+004	1.980e+004	1.982e+004
34	Bulk Density (kg/m3)	94.84	94.84	94.85	94.85	94.85
35	Bulk Velocity (m/s)	3.109	3.109	3.109	3.109	3.109
36	Erosion Velocity (m/s)	12.53	12.53	12.53	12.53	12.53
37	Length (m)	1.985e+004	1.987e+004	1.989e+004	1.992e+004	1.994e+004
38	Bulk Density (kg/m3)	94.85	94.85	94.85	94.85	94.85
39	Bulk Velocity (m/s)	3.109	3.109	3.109	3.109	3.109
40	Erosion Velocity (m/s)	12.53	12.53	12.53	12.53	12.53
41	Length (m)	1.996e+004	1.999e+004	2.001e+004	2.003e+004	2.006e+004
42	Bulk Density (kg/m3)	94.85	94.85	94.85	94.85	94.85
43	Bulk Velocity (m/s)	3.109	3.109	3.108	3.108	3.108
44	Erosion Velocity (m/s)	12.53	12.53	12.53	12.53	12.53
45	Length (m)	2.008e+004	2.010e+004	2.013e+004	2.015e+004	2.017e+004
46	Bulk Density (kg/m3)	94.85	94.86	94.86	94.86	94.86
47	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
48	Erosion Velocity (m/s)	12.53	12.53	12.53	12.53	12.53
49	Length (m)	2.020e+004	2.022e+004	2.024e+004	2.027e+004	2.029e+004
50	Bulk Density (kg/m3)	94.86	94.86	94.86	94.86	94.86
51	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
52	Erosion Velocity (m/s)	12.53	12.53	12.53	12.53	12.53
53	Length (m)	2.031e+004	2.034e+004	2.036e+004	2.038e+004	2.041e+004
54	Bulk Density (kg/m3)	94.86	94.86	94.86	94.86	94.86
55	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
56	Erosion Velocity (m/s)	12.53	12.53	12.53	12.53	12.53
57	Length (m)	2.043e+004	2.045e+004	2.048e+004	2.050e+004	2.052e+004
58	Bulk Density (kg/m3)	94.86	94.86	94.86	94.86	94.86
59	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
60	Erosion Velocity (m/s)	12.53	12.53	12.53	12.52	12.52
61	Length (m)	2.055e+004	2.058e+004	2.060e+004	2.063e+004	2.066e+004
62	Bulk Density (kg/m3)	94.86	94.87	94.87	94.87	94.87
63	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
64	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
65	Length (m)	2.069e+004	2.069e+004	2.070e+004	2.070e+004	2.070e+004
66	Bulk Density (kg/m3)	94.87	94.87	94.87	94.87	94.87
67	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
68	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 624 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.070e+004	2.071e+004	2.071e+004	2.072e+004	2.072e+004
14	Bulk Density (kg/m3)	94.87	94.87	94.87	94.87	94.87
15	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
16	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
17	Length (m)	2.073e+004	2.073e+004	2.074e+004	2.074e+004	2.075e+004
18	Bulk Density (kg/m3)	94.87	94.87	94.87	94.87	94.87
19	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
20	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
21	Length (m)	2.075e+004	2.076e+004	2.076e+004	2.077e+004	2.077e+004
22	Bulk Density (kg/m3)	94.87	94.87	94.87	94.87	94.87
23	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
24	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
25	Length (m)	2.078e+004	2.078e+004	2.079e+004	2.079e+004	2.080e+004
26	Bulk Density (kg/m3)	94.87	94.87	94.87	94.87	94.87
27	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
28	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
29	Length (m)	2.080e+004	2.081e+004	2.081e+004	2.082e+004	2.082e+004
30	Bulk Density (kg/m3)	94.87	94.87	94.87	94.87	94.88
31	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
32	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
33	Length (m)	2.083e+004	2.083e+004	2.084e+004	2.084e+004	2.085e+004
34	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
35	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
36	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
37	Length (m)	2.085e+004	2.086e+004	2.086e+004	2.087e+004	2.087e+004
38	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
39	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
40	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
41	Length (m)	2.088e+004	2.088e+004	2.089e+004	2.089e+004	2.090e+004
42	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
43	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
44	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
45	Length (m)	2.090e+004	2.091e+004	2.091e+004	2.092e+004	2.092e+004
46	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
47	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
48	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
49	Length (m)	2.093e+004	2.093e+004	2.094e+004	2.094e+004	2.095e+004
50	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
51	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
52	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
53	Length (m)	2.095e+004	2.096e+004	2.096e+004	2.097e+004	2.097e+004
54	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
55	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
56	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
57	Length (m)	2.098e+004	2.098e+004	2.099e+004	2.099e+004	2.100e+004
58	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
59	Bulk Velocity (m/s)	3.108	3.108	3.108	3.107	3.107
60	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
61	Length (m)	2.100e+004	2.101e+004	2.101e+004	2.102e+004	2.102e+004
62	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
63	Bulk Velocity (m/s)	3.107	3.107	3.107	3.107	3.107
64	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
65	Length (m)	2.103e+004	2.103e+004	2.104e+004	2.104e+004	2.105e+004
66	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.89
67	Bulk Velocity (m/s)	3.107	3.107	3.107	3.107	3.107
68	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 625 of 937					

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.105e+004	2.106e+004	2.106e+004	2.107e+004	2.107e+004
14	Bulk Density (kg/m3)	94.89	94.89	94.88	94.88	94.88
15	Bulk Velocity (m/s)	3.107	3.107	3.107	3.107	3.107
16	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
17	Length (m)	2.108e+004	2.108e+004	2.109e+004	2.109e+004	2.110e+004
18	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.89
19	Bulk Velocity (m/s)	3.107	3.107	3.107	3.107	3.107
20	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
21	Length (m)	2.110e+004	2.111e+004	2.111e+004	2.111e+004	2.112e+004
22	Bulk Density (kg/m3)	94.89	94.89	94.89	94.89	94.89
23	Bulk Velocity (m/s)	3.107	3.107	3.107	3.107	3.107
24	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
25	Length (m)	2.112e+004	2.113e+004	2.113e+004	2.113e+004	2.114e+004
26	Bulk Density (kg/m3)	94.90	94.89	94.89	94.89	94.89
27	Bulk Velocity (m/s)	3.107	3.107	3.107	3.107	3.107
28	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
29	Length (m)	2.114e+004	2.114e+004	2.115e+004	2.115e+004	2.115e+004
30	Bulk Density (kg/m3)	94.89	94.89	94.89	94.88	94.88
31	Bulk Velocity (m/s)	3.107	3.107	3.107	3.107	3.107
32	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
33	Length (m)	2.115e+004	2.116e+004	2.116e+004	2.117e+004	2.117e+004
34	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
35	Bulk Velocity (m/s)	3.107	3.107	3.108	3.108	3.108
36	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
37	Length (m)	2.118e+004	2.118e+004	2.119e+004	2.119e+004	2.120e+004
38	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
39	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
40	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
41	Length (m)	2.120e+004	2.121e+004	2.121e+004	2.122e+004	2.122e+004
42	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
43	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
44	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
45	Length (m)	2.123e+004	2.123e+004	2.124e+004	2.124e+004	2.125e+004
46	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
47	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
48	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
49	Length (m)	2.125e+004	2.126e+004	2.126e+004	2.127e+004	2.127e+004
50	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
51	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
52	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
53	Length (m)	2.128e+004	2.128e+004	2.129e+004	2.129e+004	2.130e+004
54	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
55	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
56	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
57	Length (m)	2.130e+004	2.131e+004	2.131e+004	2.132e+004	2.132e+004
58	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
59	Bulk Velocity (m/s)	3.108	3.107	3.107	3.107	3.107
60	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
61	Length (m)	2.133e+004	2.133e+004	2.133e+004	2.133e+004	2.133e+004
62	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
63	Bulk Velocity (m/s)	3.107	3.107	3.107	3.107	3.107
64	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
65	Length (m)	2.134e+004	2.134e+004	2.134e+004	2.134e+004	2.135e+004
66	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
67	Bulk Velocity (m/s)	3.107	3.107	3.107	3.108	3.108
68	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 626 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.135e+004	2.136e+004	2.136e+004	2.137e+004	2.137e+004
14	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
15	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
16	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
17	Length (m)	2.138e+004	2.138e+004	2.139e+004	2.139e+004	2.140e+004
18	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
19	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
20	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
21	Length (m)	2.140e+004	2.141e+004	2.141e+004	2.142e+004	2.142e+004
22	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
23	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
24	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
25	Length (m)	2.143e+004	2.143e+004	2.144e+004	2.144e+004	2.145e+004
26	Bulk Density (kg/m3)	94.87	94.87	94.87	94.88	94.88
27	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
28	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
29	Length (m)	2.145e+004	2.146e+004	2.146e+004	2.147e+004	2.147e+004
30	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
31	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
32	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
33	Length (m)	2.148e+004	2.148e+004	2.149e+004	2.149e+004	2.150e+004
34	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
35	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
36	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
37	Length (m)	2.150e+004	2.150e+004	2.151e+004	2.151e+004	2.152e+004
38	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
39	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
40	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
41	Length (m)	2.152e+004	2.152e+004	2.152e+004	2.152e+004	2.152e+004
42	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
43	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
44	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
45	Length (m)	2.153e+004	2.153e+004	2.154e+004	2.154e+004	2.155e+004
46	Bulk Density (kg/m3)	94.88	94.88	94.88	94.88	94.88
47	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
48	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
49	Length (m)	2.155e+004	2.156e+004	2.156e+004	2.157e+004	2.157e+004
50	Bulk Density (kg/m3)	94.88	94.88	94.87	94.87	94.87
51	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
52	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
53	Length (m)	2.158e+004	2.158e+004	2.159e+004	2.159e+004	2.160e+004
54	Bulk Density (kg/m3)	94.87	94.87	94.88	94.88	94.88
55	Bulk Velocity (m/s)	3.108	3.108	3.108	3.108	3.108
56	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
57	Length (m)	2.160e+004	2.161e+004	2.161e+004	2.162e+004	2.162e+004
58	Bulk Density (kg/m3)	94.88	94.89	94.90	94.90	94.91
59	Bulk Velocity (m/s)	3.107	3.107	3.107	3.107	3.107
60	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
61	Length (m)	2.163e+004	2.163e+004	2.164e+004	2.164e+004	2.165e+004
62	Bulk Density (kg/m3)	94.92	94.92	94.93	94.94	94.94
63	Bulk Velocity (m/s)	3.106	3.106	3.106	3.106	3.105
64	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
65	Length (m)	2.165e+004	2.166e+004	2.166e+004	2.167e+004	2.167e+004
66	Bulk Density (kg/m3)	94.95	94.96	94.96	94.97	94.98
67	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.104
68	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 627 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.168e+004	2.168e+004	2.169e+004	2.169e+004	2.170e+004
14	Bulk Density (kg/m3)	94.98	94.98	94.98	94.98	94.98
15	Bulk Velocity (m/s)	3.104	3.104	3.104	3.104	3.104
16	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
17	Length (m)	2.170e+004	2.171e+004	2.171e+004	2.172e+004	2.172e+004
18	Bulk Density (kg/m3)	94.98	94.98	94.98	94.97	94.97
19	Bulk Velocity (m/s)	3.104	3.104	3.104	3.104	3.105
20	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
21	Length (m)	2.173e+004	2.173e+004	2.174e+004	2.174e+004	2.175e+004
22	Bulk Density (kg/m3)	94.97	94.97	94.96	94.96	94.96
23	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.105
24	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
25	Length (m)	2.175e+004	2.176e+004	2.176e+004	2.177e+004	2.177e+004
26	Bulk Density (kg/m3)	94.95	94.95	94.95	94.95	94.94
27	Bulk Velocity (m/s)	3.105	3.105	3.105	3.105	3.106
28	Erosion Velocity (m/s)	12.52	12.52	12.52	12.52	12.52
29	Length (m)	2.178e+004	2.178e+004	2.179e+004	2.179e+004	2.180e+004
30	Bulk Density (kg/m3)	94.94	94.86	94.78	94.70	94.61
31	Bulk Velocity (m/s)	3.106	3.108	3.111	3.114	3.116
32	Erosion Velocity (m/s)	12.52	12.53	12.53	12.54	12.54
33	Length (m)	2.180e+004	2.181e+004	2.181e+004	2.182e+004	2.182e+004
34	Bulk Density (kg/m3)	94.53	94.53	94.53	94.53	94.53
35	Bulk Velocity (m/s)	3.119	3.119	3.119	3.119	3.119
36	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
37	Length (m)	2.183e+004	2.183e+004	2.184e+004	2.184e+004	2.185e+004
38	Bulk Density (kg/m3)	94.53	94.53	94.53	94.53	94.53
39	Bulk Velocity (m/s)	3.119	3.119	3.119	3.119	3.119
40	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
41	Length (m)	2.185e+004	2.186e+004	2.186e+004	2.187e+004	2.187e+004
42	Bulk Density (kg/m3)	94.53	94.53	94.53	94.53	94.53
43	Bulk Velocity (m/s)	3.119	3.119	3.119	3.119	3.119
44	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
45	Length (m)	2.188e+004	2.188e+004	2.189e+004	2.189e+004	2.190e+004
46	Bulk Density (kg/m3)	94.53	94.53	94.53	94.53	94.53
47	Bulk Velocity (m/s)	3.119	3.119	3.119	3.119	3.119
48	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
49	Length (m)	2.190e+004	2.191e+004	2.191e+004	2.192e+004	2.192e+004
50	Bulk Density (kg/m3)	94.53	94.53	94.53	94.53	94.53
51	Bulk Velocity (m/s)	3.119	3.119	3.119	3.119	3.119
52	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
53	Length (m)	2.193e+004	2.193e+004	2.194e+004	2.194e+004	2.195e+004
54	Bulk Density (kg/m3)	94.53	94.53	94.53	94.53	94.53
55	Bulk Velocity (m/s)	3.119	3.119	3.119	3.119	3.119
56	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
57	Length (m)	2.195e+004	2.196e+004	2.196e+004	2.197e+004	2.197e+004
58	Bulk Density (kg/m3)	94.53	94.53	94.53	94.53	94.53
59	Bulk Velocity (m/s)	3.119	3.119	3.119	3.119	3.119
60	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
61	Length (m)	2.198e+004	2.198e+004	2.199e+004	2.199e+004	2.200e+004
62	Bulk Density (kg/m3)	94.53	94.53	94.53	94.53	94.53
63	Bulk Velocity (m/s)	3.119	3.119	3.119	3.119	3.119
64	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
65	Length (m)	2.200e+004	2.201e+004	2.201e+004	2.202e+004	2.202e+004
66	Bulk Density (kg/m3)	94.53	94.53	94.53	94.53	94.53
67	Bulk Velocity (m/s)	3.119	3.119	3.119	3.119	3.119
68	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 628 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
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11	Erosion Table					
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13	Length (m)	2.203e+004	2.203e+004	2.204e+004	2.204e+004	2.205e+004
14	Bulk Density (kg/m3)	94.53	94.53	94.53	94.54	94.54
15	Bulk Velocity (m/s)	3.119	3.119	3.119	3.119	3.119
16	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
17	Length (m)	2.205e+004	2.206e+004	2.206e+004	2.207e+004	2.207e+004
18	Bulk Density (kg/m3)	94.54	94.54	94.54	94.54	94.54
19	Bulk Velocity (m/s)	3.119	3.119	3.119	3.119	3.119
20	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
21	Length (m)	2.208e+004	2.208e+004	2.209e+004	2.209e+004	2.210e+004
22	Bulk Density (kg/m3)	94.54	94.54	94.54	94.54	94.54
23	Bulk Velocity (m/s)	3.119	3.119	3.119	3.119	3.119
24	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
25	Length (m)	2.210e+004	2.211e+004	2.211e+004	2.212e+004	2.212e+004
26	Bulk Density (kg/m3)	94.54	94.54	94.54	94.54	94.54
27	Bulk Velocity (m/s)	3.119	3.119	3.119	3.119	3.119
28	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
29	Length (m)	2.213e+004	2.213e+004	2.214e+004	2.214e+004	2.215e+004
30	Bulk Density (kg/m3)	94.54	94.54	94.54	94.54	94.54
31	Bulk Velocity (m/s)	3.119	3.119	3.119	3.119	3.119
32	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
33	Length (m)	2.215e+004	2.216e+004	2.216e+004	2.217e+004	2.217e+004
34	Bulk Density (kg/m3)	94.54	94.54	94.54	94.54	94.54
35	Bulk Velocity (m/s)	3.119	3.119	3.119	3.119	3.119
36	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
37	Length (m)	2.218e+004	2.218e+004	2.219e+004	2.219e+004	2.220e+004
38	Bulk Density (kg/m3)	94.54	94.54	94.54	94.54	94.54
39	Bulk Velocity (m/s)	3.119	3.119	3.119	3.119	3.119
40	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
41	Length (m)	2.220e+004	2.221e+004	2.221e+004	2.222e+004	2.222e+004
42	Bulk Density (kg/m3)	94.54	94.54	94.55	94.55	94.55
43	Bulk Velocity (m/s)	3.119	3.119	3.119	3.118	3.118
44	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
45	Length (m)	2.223e+004	2.223e+004	2.224e+004	2.224e+004	2.225e+004
46	Bulk Density (kg/m3)	94.55	94.55	94.55	94.54	94.54
47	Bulk Velocity (m/s)	3.118	3.118	3.118	3.119	3.119
48	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
49	Length (m)	2.225e+004	2.226e+004	2.226e+004	2.227e+004	2.227e+004
50	Bulk Density (kg/m3)	94.54	94.54	94.54	94.54	94.54
51	Bulk Velocity (m/s)	3.119	3.119	3.119	3.119	3.119
52	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
53	Length (m)	2.228e+004	2.229e+004	2.230e+004	2.231e+004	2.232e+004
54	Bulk Density (kg/m3)	94.54	94.54	94.54	94.54	94.54
55	Bulk Velocity (m/s)	3.119	3.119	3.119	3.119	3.119
56	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
57	Length (m)	2.233e+004	2.255e+004	2.278e+004	2.300e+004	2.323e+004
58	Bulk Density (kg/m3)	94.54	94.54	94.55	94.55	94.56
59	Bulk Velocity (m/s)	3.119	3.119	3.118	3.118	3.118
60	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
61	Length (m)	2.345e+004	2.346e+004	2.347e+004	2.348e+004	2.348e+004
62	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
63	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
64	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
65	Length (m)	2.349e+004	2.350e+004	2.350e+004	2.351e+004	2.351e+004
66	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
67	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
68	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
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1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.352e+004	2.352e+004	2.353e+004	2.353e+004	2.354e+004
14	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
15	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
16	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
17	Length (m)	2.354e+004	2.355e+004	2.355e+004	2.356e+004	2.356e+004
18	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
19	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
20	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
21	Length (m)	2.357e+004	2.357e+004	2.358e+004	2.358e+004	2.359e+004
22	Bulk Density (kg/m3)	94.56	94.56	94.55	94.55	94.54
23	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.119
24	Erosion Velocity (m/s)	12.54	12.55	12.55	12.55	12.55
25	Length (m)	2.359e+004	2.360e+004	2.360e+004	2.361e+004	2.361e+004
26	Bulk Density (kg/m3)	94.54	94.54	94.54	94.55	94.55
27	Bulk Velocity (m/s)	3.119	3.119	3.119	3.118	3.118
28	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
29	Length (m)	2.362e+004	2.362e+004	2.363e+004	2.363e+004	2.364e+004
30	Bulk Density (kg/m3)	94.55	94.55	94.55	94.55	94.56
31	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
32	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
33	Length (m)	2.364e+004	2.365e+004	2.365e+004	2.366e+004	2.366e+004
34	Bulk Density (kg/m3)	94.56	94.55	94.55	94.55	94.55
35	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
36	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
37	Length (m)	2.367e+004	2.367e+004	2.368e+004	2.368e+004	2.369e+004
38	Bulk Density (kg/m3)	94.55	94.55	94.55	94.55	94.55
39	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
40	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
41	Length (m)	2.369e+004	2.370e+004	2.370e+004	2.371e+004	2.371e+004
42	Bulk Density (kg/m3)	94.55	94.56	94.56	94.56	94.56
43	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
44	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
45	Length (m)	2.372e+004	2.372e+004	2.373e+004	2.373e+004	2.374e+004
46	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
47	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
48	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
49	Length (m)	2.374e+004	2.375e+004	2.375e+004	2.376e+004	2.376e+004
50	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
51	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
52	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.54
53	Length (m)	2.377e+004	2.377e+004	2.378e+004	2.378e+004	2.379e+004
54	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
55	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
56	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
57	Length (m)	2.379e+004	2.380e+004	2.380e+004	2.381e+004	2.381e+004
58	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
59	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
60	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
61	Length (m)	2.382e+004	2.382e+004	2.383e+004	2.383e+004	2.384e+004
62	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
63	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
64	Erosion Velocity (m/s)	12.54	12.55	12.55	12.55	12.55
65	Length (m)	2.384e+004	2.385e+004	2.385e+004	2.386e+004	2.386e+004
66	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
67	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
68	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 630 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
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11	Erosion Table					
12						
13	Length (m)	2.387e+004	2.387e+004	2.388e+004	2.388e+004	2.389e+004
14	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
15	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
16	Erosion Velocity (m/s)	12.55	12.55	12.54	12.54	12.54
17	Length (m)	2.389e+004	2.390e+004	2.390e+004	2.391e+004	2.391e+004
18	Bulk Density (kg/m3)	94.57	94.56	94.56	94.56	94.56
19	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
20	Erosion Velocity (m/s)	12.54	12.54	12.54	12.55	12.55
21	Length (m)	2.392e+004	2.392e+004	2.393e+004	2.393e+004	2.394e+004
22	Bulk Density (kg/m3)	94.55	94.55	94.55	94.55	94.55
23	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
24	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
25	Length (m)	2.394e+004	2.395e+004	2.395e+004	2.396e+004	2.396e+004
26	Bulk Density (kg/m3)	94.55	94.55	94.55	94.55	94.55
27	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
28	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
29	Length (m)	2.397e+004	2.397e+004	2.398e+004	2.398e+004	2.399e+004
30	Bulk Density (kg/m3)	94.55	94.55	94.55	94.55	94.55
31	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
32	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
33	Length (m)	2.399e+004	2.400e+004	2.400e+004	2.401e+004	2.401e+004
34	Bulk Density (kg/m3)	94.55	94.55	94.56	94.56	94.56
35	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
36	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
37	Length (m)	2.402e+004	2.402e+004	2.403e+004	2.403e+004	2.404e+004
38	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
39	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
40	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.54
41	Length (m)	2.404e+004	2.405e+004	2.405e+004	2.406e+004	2.406e+004
42	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
43	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
44	Erosion Velocity (m/s)	12.54	12.55	12.55	12.55	12.55
45	Length (m)	2.407e+004	2.407e+004	2.408e+004	2.408e+004	2.409e+004
46	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
47	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
48	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
49	Length (m)	2.409e+004	2.410e+004	2.410e+004	2.411e+004	2.411e+004
50	Bulk Density (kg/m3)	94.55	94.55	94.55	94.55	94.55
51	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
52	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
53	Length (m)	2.412e+004	2.412e+004	2.413e+004	2.413e+004	2.414e+004
54	Bulk Density (kg/m3)	94.55	94.55	94.55	94.55	94.55
55	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
56	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
57	Length (m)	2.414e+004	2.415e+004	2.415e+004	2.416e+004	2.416e+004
58	Bulk Density (kg/m3)	94.55	94.55	94.55	94.55	94.55
59	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
60	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
61	Length (m)	2.417e+004	2.417e+004	2.418e+004	2.418e+004	2.419e+004
62	Bulk Density (kg/m3)	94.55	94.55	94.55	94.55	94.55
63	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
64	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
65	Length (m)	2.419e+004	2.420e+004	2.420e+004	2.421e+004	2.421e+004
66	Bulk Density (kg/m3)	94.55	94.56	94.56	94.56	94.56
67	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
68	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 631 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.422e+004	2.422e+004	2.423e+004	2.423e+004	2.424e+004
14	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
15	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
16	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
17	Length (m)	2.424e+004	2.425e+004	2.425e+004	2.426e+004	2.426e+004
18	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
19	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
20	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
21	Length (m)	2.427e+004	2.427e+004	2.428e+004	2.428e+004	2.429e+004
22	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
23	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
24	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
25	Length (m)	2.429e+004	2.430e+004	2.430e+004	2.431e+004	2.431e+004
26	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
27	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
28	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
29	Length (m)	2.432e+004	2.432e+004	2.433e+004	2.433e+004	2.434e+004
30	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
31	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
32	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
33	Length (m)	2.434e+004	2.435e+004	2.435e+004	2.436e+004	2.436e+004
34	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
35	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
36	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
37	Length (m)	2.437e+004	2.437e+004	2.437e+004	2.437e+004	2.437e+004
38	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
39	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
40	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
41	Length (m)	2.438e+004	2.438e+004	2.438e+004	2.438e+004	2.438e+004
42	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
43	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
44	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
45	Length (m)	2.439e+004	2.439e+004	2.440e+004	2.440e+004	2.441e+004
46	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
47	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
48	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
49	Length (m)	2.442e+004	2.442e+004	2.443e+004	2.443e+004	2.444e+004
50	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
51	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
52	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
53	Length (m)	2.444e+004	2.445e+004	2.445e+004	2.446e+004	2.446e+004
54	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
55	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
56	Erosion Velocity (m/s)	12.55	12.55	12.54	12.54	12.54
57	Length (m)	2.447e+004	2.447e+004	2.448e+004	2.448e+004	2.449e+004
58	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
59	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
60	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
61	Length (m)	2.449e+004	2.450e+004	2.450e+004	2.451e+004	2.451e+004
62	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
63	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
64	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
65	Length (m)	2.452e+004	2.452e+004	2.453e+004	2.453e+004	2.454e+004
66	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
67	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
68	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 632 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.454e+004	2.455e+004	2.455e+004	2.456e+004	2.456e+004
14	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
15	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
16	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
17	Length (m)	2.457e+004	2.457e+004	2.458e+004	2.458e+004	2.459e+004
18	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
19	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
20	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
21	Length (m)	2.459e+004	2.460e+004	2.460e+004	2.461e+004	2.461e+004
22	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
23	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
24	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
25	Length (m)	2.462e+004	2.462e+004	2.463e+004	2.463e+004	2.464e+004
26	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
27	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
28	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
29	Length (m)	2.464e+004	2.465e+004	2.465e+004	2.466e+004	2.466e+004
30	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
31	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
32	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
33	Length (m)	2.467e+004	2.467e+004	2.468e+004	2.468e+004	2.469e+004
34	Bulk Density (kg/m3)	94.56	94.57	94.57	94.57	94.57
35	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
36	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
37	Length (m)	2.469e+004	2.470e+004	2.470e+004	2.471e+004	2.471e+004
38	Bulk Density (kg/m3)	94.57	94.57	94.57	94.57	94.56
39	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
40	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
41	Length (m)	2.472e+004	2.472e+004	2.473e+004	2.473e+004	2.474e+004
42	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
43	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
44	Erosion Velocity (m/s)	12.54	12.54	12.54	12.55	12.55
45	Length (m)	2.474e+004	2.475e+004	2.475e+004	2.476e+004	2.476e+004
46	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
47	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
48	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
49	Length (m)	2.477e+004	2.477e+004	2.478e+004	2.478e+004	2.479e+004
50	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
51	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
52	Erosion Velocity (m/s)	12.55	12.55	12.55	12.54	12.54
53	Length (m)	2.479e+004	2.480e+004	2.480e+004	2.481e+004	2.481e+004
54	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
55	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
56	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
57	Length (m)	2.482e+004	2.482e+004	2.483e+004	2.483e+004	2.484e+004
58	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.57
59	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
60	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
61	Length (m)	2.484e+004	2.485e+004	2.485e+004	2.486e+004	2.486e+004
62	Bulk Density (kg/m3)	94.57	94.57	94.57	94.57	94.57
63	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
64	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
65	Length (m)	2.487e+004	2.487e+004	2.488e+004	2.488e+004	2.489e+004
66	Bulk Density (kg/m3)	94.57	94.57	94.57	94.57	94.58
67	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
68	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 633 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.489e+004	2.490e+004	2.490e+004	2.491e+004	2.491e+004
14	Bulk Density (kg/m3)	94.58	94.57	94.57	94.56	94.56
15	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
16	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
17	Length (m)	2.492e+004	2.492e+004	2.493e+004	2.493e+004	2.494e+004
18	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
19	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
20	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
21	Length (m)	2.494e+004	2.495e+004	2.495e+004	2.496e+004	2.496e+004
22	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
23	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
24	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
25	Length (m)	2.497e+004	2.497e+004	2.498e+004	2.498e+004	2.499e+004
26	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
27	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
28	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
29	Length (m)	2.499e+004	2.500e+004	2.500e+004	2.501e+004	2.501e+004
30	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
31	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
32	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
33	Length (m)	2.502e+004	2.502e+004	2.503e+004	2.503e+004	2.504e+004
34	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
35	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
36	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
37	Length (m)	2.504e+004	2.505e+004	2.505e+004	2.506e+004	2.506e+004
38	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
39	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
40	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
41	Length (m)	2.507e+004	2.507e+004	2.508e+004	2.508e+004	2.509e+004
42	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
43	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
44	Erosion Velocity (m/s)	12.54	12.55	12.55	12.55	12.55
45	Length (m)	2.509e+004	2.510e+004	2.510e+004	2.511e+004	2.511e+004
46	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
47	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
48	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
49	Length (m)	2.512e+004	2.512e+004	2.513e+004	2.513e+004	2.514e+004
50	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
51	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
52	Erosion Velocity (m/s)	12.55	12.55	12.54	12.54	12.54
53	Length (m)	2.514e+004	2.515e+004	2.515e+004	2.516e+004	2.516e+004
54	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
55	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
56	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
57	Length (m)	2.517e+004	2.517e+004	2.518e+004	2.518e+004	2.519e+004
58	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.57
59	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
60	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
61	Length (m)	2.519e+004	2.520e+004	2.520e+004	2.521e+004	2.522e+004
62	Bulk Density (kg/m3)	94.57	94.57	94.57	94.57	94.57
63	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
64	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
65	Length (m)	2.522e+004	2.523e+004	2.523e+004	2.524e+004	2.524e+004
66	Bulk Density (kg/m3)	94.57	94.58	94.58	94.59	94.60
67	Bulk Velocity (m/s)	3.118	3.118	3.117	3.117	3.117
68	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 634 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.525e+004	2.525e+004	2.526e+004	2.526e+004	2.527e+004
14	Bulk Density (kg/m3)	94.61	94.61	94.62	94.62	94.63
15	Bulk Velocity (m/s)	3.116	3.116	3.116	3.116	3.116
16	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
17	Length (m)	2.527e+004	2.528e+004	2.528e+004	2.529e+004	2.529e+004
18	Bulk Density (kg/m3)	94.63	94.62	94.61	94.60	94.58
19	Bulk Velocity (m/s)	3.116	3.116	3.116	3.117	3.117
20	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
21	Length (m)	2.529e+004	2.531e+004	2.533e+004	2.535e+004	2.537e+004
22	Bulk Density (kg/m3)	94.57	94.57	94.57	94.57	94.56
23	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
24	Erosion Velocity (m/s)	12.54	12.54	12.54	12.54	12.54
25	Length (m)	2.539e+004	2.539e+004	2.539e+004	2.539e+004	2.540e+004
26	Bulk Density (kg/m3)	94.56	94.56	94.56	94.56	94.56
27	Bulk Velocity (m/s)	3.118	3.118	3.118	3.118	3.118
28	Erosion Velocity (m/s)	12.54	12.55	12.55	12.55	12.55
29	Length (m)	2.540e+004	2.540e+004	2.541e+004	2.541e+004	2.542e+004
30	Bulk Density (kg/m3)	94.55	94.53	94.50	94.48	94.45
31	Bulk Velocity (m/s)	3.118	3.119	3.120	3.121	3.122
32	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
33	Length (m)	2.543e+004	2.543e+004	2.544e+004	2.544e+004	2.545e+004
34	Bulk Density (kg/m3)	94.42	94.42	94.42	94.42	94.42
35	Bulk Velocity (m/s)	3.123	3.123	3.123	3.123	3.123
36	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
37	Length (m)	2.545e+004	2.546e+004	2.546e+004	2.547e+004	2.547e+004
38	Bulk Density (kg/m3)	94.42	94.42	94.42	94.42	94.42
39	Bulk Velocity (m/s)	3.123	3.123	3.123	3.123	3.123
40	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
41	Length (m)	2.548e+004	2.548e+004	2.549e+004	2.549e+004	2.550e+004
42	Bulk Density (kg/m3)	94.42	94.42	94.42	94.42	94.42
43	Bulk Velocity (m/s)	3.123	3.123	3.123	3.123	3.123
44	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
45	Length (m)	2.550e+004	2.551e+004	2.551e+004	2.552e+004	2.552e+004
46	Bulk Density (kg/m3)	94.42	94.42	94.42	94.42	94.42
47	Bulk Velocity (m/s)	3.123	3.123	3.123	3.123	3.123
48	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
49	Length (m)	2.553e+004	2.553e+004	2.554e+004	2.554e+004	2.555e+004
50	Bulk Density (kg/m3)	94.42	94.42	94.42	94.42	94.42
51	Bulk Velocity (m/s)	3.123	3.123	3.123	3.123	3.123
52	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
53	Length (m)	2.555e+004	2.556e+004	2.556e+004	2.557e+004	2.557e+004
54	Bulk Density (kg/m3)	94.42	94.42	94.42	94.42	94.42
55	Bulk Velocity (m/s)	3.123	3.123	3.123	3.123	3.123
56	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
57	Length (m)	2.558e+004	2.558e+004	2.559e+004	2.559e+004	2.560e+004
58	Bulk Density (kg/m3)	94.42	94.42	94.42	94.42	94.42
59	Bulk Velocity (m/s)	3.123	3.123	3.123	3.123	3.123
60	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
61	Length (m)	2.560e+004	2.561e+004	2.561e+004	2.562e+004	2.563e+004
62	Bulk Density (kg/m3)	94.42	94.42	94.42	94.42	94.42
63	Bulk Velocity (m/s)	3.123	3.123	3.123	3.123	3.123
64	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
65	Length (m)	2.563e+004	2.564e+004	2.564e+004	2.565e+004	2.565e+004
66	Bulk Density (kg/m3)	94.42	94.42	94.42	94.42	94.42
67	Bulk Velocity (m/s)	3.123	3.123	3.123	3.123	3.123
68	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 635 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.566e+004	2.566e+004	2.567e+004	2.567e+004	2.568e+004
14	Bulk Density (kg/m3)	94.42	94.42	94.42	94.42	94.42
15	Bulk Velocity (m/s)	3.123	3.123	3.123	3.123	3.123
16	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
17	Length (m)	2.568e+004	2.569e+004	2.569e+004	2.570e+004	2.570e+004
18	Bulk Density (kg/m3)	94.42	94.42	94.42	94.42	94.42
19	Bulk Velocity (m/s)	3.123	3.123	3.123	3.123	3.123
20	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
21	Length (m)	2.571e+004	2.571e+004	2.571e+004	2.571e+004	2.571e+004
22	Bulk Density (kg/m3)	94.42	94.42	94.42	94.42	94.42
23	Bulk Velocity (m/s)	3.123	3.123	3.123	3.123	3.123
24	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
25	Length (m)	2.571e+004	2.572e+004	2.572e+004	2.572e+004	2.572e+004
26	Bulk Density (kg/m3)	94.42	94.42	94.42	94.43	94.43
27	Bulk Velocity (m/s)	3.123	3.123	3.123	3.122	3.122
28	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
29	Length (m)	2.572e+004	2.572e+004	2.573e+004	2.573e+004	2.573e+004
30	Bulk Density (kg/m3)	94.43	94.43	94.44	94.44	94.45
31	Bulk Velocity (m/s)	3.122	3.122	3.122	3.122	3.122
32	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
33	Length (m)	2.573e+004	2.574e+004	2.574e+004	2.575e+004	2.575e+004
34	Bulk Density (kg/m3)	94.45	94.46	94.46	94.47	94.47
35	Bulk Velocity (m/s)	3.122	3.121	3.121	3.121	3.121
36	Erosion Velocity (m/s)	12.55	12.55	12.55	12.55	12.55
37	Length (m)	2.576e+004	2.576e+004	2.577e+004	2.577e+004	2.578e+004
38	Bulk Density (kg/m3)	94.48	94.42	94.37	94.32	94.27
39	Bulk Velocity (m/s)	3.121	3.123	3.124	3.126	3.128
40	Erosion Velocity (m/s)	12.55	12.55	12.56	12.56	12.56
41	Length (m)	2.578e+004	2.579e+004	2.579e+004	2.580e+004	2.580e+004
42	Bulk Density (kg/m3)	94.21	94.21	94.21	94.21	94.20
43	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
44	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
45	Length (m)	2.581e+004	2.581e+004	2.582e+004	2.582e+004	2.583e+004
46	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
47	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
48	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
49	Length (m)	2.583e+004	2.584e+004	2.584e+004	2.585e+004	2.585e+004
50	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
51	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
52	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
53	Length (m)	2.586e+004	2.586e+004	2.587e+004	2.587e+004	2.588e+004
54	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
55	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
56	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
57	Length (m)	2.588e+004	2.589e+004	2.589e+004	2.590e+004	2.590e+004
58	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
59	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
60	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
61	Length (m)	2.591e+004	2.591e+004	2.592e+004	2.592e+004	2.593e+004
62	Bulk Density (kg/m3)	94.20	94.20	94.21	94.21	94.21
63	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
64	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
65	Length (m)	2.593e+004	2.594e+004	2.594e+004	2.595e+004	2.595e+004
66	Bulk Density (kg/m3)	94.21	94.21	94.21	94.21	94.21
67	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
68	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 636 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.596e+004	2.596e+004	2.597e+004	2.597e+004	2.598e+004
14	Bulk Density (kg/m3)	94.21	94.21	94.20	94.20	94.20
15	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
16	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
17	Length (m)	2.598e+004	2.599e+004	2.599e+004	2.600e+004	2.600e+004
18	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
19	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
20	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
21	Length (m)	2.601e+004	2.601e+004	2.602e+004	2.602e+004	2.603e+004
22	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
23	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
24	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
25	Length (m)	2.603e+004	2.604e+004	2.604e+004	2.605e+004	2.605e+004
26	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
27	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
28	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
29	Length (m)	2.606e+004	2.606e+004	2.607e+004	2.607e+004	2.608e+004
30	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
31	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
32	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
33	Length (m)	2.608e+004	2.609e+004	2.609e+004	2.610e+004	2.610e+004
34	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
35	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
36	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
37	Length (m)	2.611e+004	2.611e+004	2.612e+004	2.612e+004	2.613e+004
38	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
39	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
40	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
41	Length (m)	2.613e+004	2.614e+004	2.614e+004	2.615e+004	2.615e+004
42	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
43	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
44	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
45	Length (m)	2.616e+004	2.616e+004	2.617e+004	2.617e+004	2.618e+004
46	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
47	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
48	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
49	Length (m)	2.618e+004	2.619e+004	2.619e+004	2.620e+004	2.620e+004
50	Bulk Density (kg/m3)	94.19	94.20	94.20	94.20	94.20
51	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
52	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
53	Length (m)	2.621e+004	2.621e+004	2.622e+004	2.622e+004	2.623e+004
54	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
55	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
56	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
57	Length (m)	2.623e+004	2.624e+004	2.624e+004	2.625e+004	2.625e+004
58	Bulk Density (kg/m3)	94.19	94.19	94.19	94.20	94.20
59	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
60	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
61	Length (m)	2.626e+004	2.626e+004	2.627e+004	2.627e+004	2.628e+004
62	Bulk Density (kg/m3)	94.20	94.19	94.19	94.19	94.19
63	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
64	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
65	Length (m)	2.628e+004	2.629e+004	2.629e+004	2.630e+004	2.630e+004
66	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
67	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
68	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 637 of 937					

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.631e+004	2.631e+004	2.632e+004	2.632e+004	2.633e+004
14	Bulk Density (kg/m3)	94.19	94.19	94.19	94.20	94.20
15	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
16	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
17	Length (m)	2.633e+004	2.634e+004	2.634e+004	2.635e+004	2.635e+004
18	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
19	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
20	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
21	Length (m)	2.636e+004	2.636e+004	2.637e+004	2.637e+004	2.638e+004
22	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
23	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
24	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
25	Length (m)	2.638e+004	2.639e+004	2.639e+004	2.640e+004	2.640e+004
26	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
27	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
28	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
29	Length (m)	2.641e+004	2.641e+004	2.642e+004	2.642e+004	2.643e+004
30	Bulk Density (kg/m3)	94.20	94.20	94.20	94.19	94.19
31	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
32	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
33	Length (m)	2.644e+004	2.644e+004	2.645e+004	2.645e+004	2.646e+004
34	Bulk Density (kg/m3)	94.19	94.20	94.20	94.20	94.20
35	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
36	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
37	Length (m)	2.646e+004	2.647e+004	2.647e+004	2.648e+004	2.648e+004
38	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
39	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
40	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
41	Length (m)	2.649e+004	2.649e+004	2.650e+004	2.650e+004	2.651e+004
42	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
43	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
44	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
45	Length (m)	2.651e+004	2.652e+004	2.652e+004	2.653e+004	2.653e+004
46	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
47	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
48	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
49	Length (m)	2.654e+004	2.654e+004	2.655e+004	2.655e+004	2.656e+004
50	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
51	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
52	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
53	Length (m)	2.656e+004	2.657e+004	2.657e+004	2.658e+004	2.658e+004
54	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
55	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
56	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
57	Length (m)	2.659e+004	2.659e+004	2.660e+004	2.660e+004	2.661e+004
58	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
59	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
60	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
61	Length (m)	2.661e+004	2.662e+004	2.662e+004	2.663e+004	2.663e+004
62	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.21
63	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
64	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
65	Length (m)	2.664e+004	2.664e+004	2.665e+004	2.665e+004	2.666e+004
66	Bulk Density (kg/m3)	94.21	94.21	94.21	94.21	94.21
67	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
68	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 638 of 937					

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.666e+004	2.667e+004	2.667e+004	2.668e+004	2.668e+004
14	Bulk Density (kg/m3)	94.21	94.20	94.20	94.20	94.20
15	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
16	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
17	Length (m)	2.669e+004	2.669e+004	2.670e+004	2.670e+004	2.671e+004
18	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
19	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
20	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
21	Length (m)	2.671e+004	2.672e+004	2.672e+004	2.673e+004	2.674e+004
22	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
23	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
24	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
25	Length (m)	2.674e+004	2.674e+004	2.675e+004	2.675e+004	2.676e+004
26	Bulk Density (kg/m3)	94.20	94.20	94.20	94.19	94.19
27	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
28	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
29	Length (m)	2.676e+004	2.677e+004	2.677e+004	2.678e+004	2.678e+004
30	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
31	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
32	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
33	Length (m)	2.679e+004	2.679e+004	2.680e+004	2.680e+004	2.681e+004
34	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
35	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
36	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
37	Length (m)	2.681e+004	2.682e+004	2.682e+004	2.683e+004	2.683e+004
38	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
39	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
40	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
41	Length (m)	2.684e+004	2.684e+004	2.685e+004	2.685e+004	2.686e+004
42	Bulk Density (kg/m3)	94.20	94.19	94.19	94.19	94.19
43	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
44	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
45	Length (m)	2.686e+004	2.687e+004	2.687e+004	2.688e+004	2.688e+004
46	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
47	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
48	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
49	Length (m)	2.689e+004	2.689e+004	2.690e+004	2.690e+004	2.691e+004
50	Bulk Density (kg/m3)	94.19	94.19	94.19	94.20	94.20
51	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
52	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
53	Length (m)	2.691e+004	2.692e+004	2.692e+004	2.693e+004	2.693e+004
54	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
55	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
56	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
57	Length (m)	2.694e+004	2.694e+004	2.695e+004	2.695e+004	2.696e+004
58	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
59	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
60	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
61	Length (m)	2.696e+004	2.697e+004	2.697e+004	2.698e+004	2.698e+004
62	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
63	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
64	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
65	Length (m)	2.699e+004	2.699e+004	2.700e+004	2.700e+004	2.701e+004
66	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
67	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
68	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 639 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.701e+004	2.702e+004	2.702e+004	2.703e+004	2.703e+004
14	Bulk Density (kg/m3)	94.20	94.20	94.21	94.21	94.21
15	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
16	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
17	Length (m)	2.704e+004	2.704e+004	2.704e+004	2.704e+004	2.704e+004
18	Bulk Density (kg/m3)	94.21	94.21	94.21	94.20	94.20
19	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
20	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
21	Length (m)	2.705e+004	2.705e+004	2.705e+004	2.705e+004	2.706e+004
22	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
23	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
24	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
25	Length (m)	2.706e+004	2.707e+004	2.707e+004	2.708e+004	2.708e+004
26	Bulk Density (kg/m3)	94.20	94.21	94.21	94.21	94.21
27	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
28	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
29	Length (m)	2.709e+004	2.709e+004	2.710e+004	2.710e+004	2.711e+004
30	Bulk Density (kg/m3)	94.21	94.21	94.21	94.21	94.21
31	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
32	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
33	Length (m)	2.711e+004	2.712e+004	2.712e+004	2.713e+004	2.713e+004
34	Bulk Density (kg/m3)	94.21	94.21	94.20	94.20	94.20
35	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
36	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
37	Length (m)	2.714e+004	2.714e+004	2.715e+004	2.715e+004	2.716e+004
38	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
39	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
40	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
41	Length (m)	2.716e+004	2.717e+004	2.717e+004	2.718e+004	2.718e+004
42	Bulk Density (kg/m3)	94.20	94.19	94.19	94.19	94.19
43	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
44	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
45	Length (m)	2.719e+004	2.719e+004	2.720e+004	2.720e+004	2.721e+004
46	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
47	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
48	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
49	Length (m)	2.721e+004	2.722e+004	2.722e+004	2.723e+004	2.723e+004
50	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
51	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
52	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
53	Length (m)	2.724e+004	2.724e+004	2.725e+004	2.725e+004	2.726e+004
54	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
55	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
56	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
57	Length (m)	2.726e+004	2.727e+004	2.727e+004	2.728e+004	2.728e+004
58	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
59	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
60	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
61	Length (m)	2.729e+004	2.729e+004	2.730e+004	2.730e+004	2.731e+004
62	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
63	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
64	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
65	Length (m)	2.731e+004	2.732e+004	2.732e+004	2.733e+004	2.733e+004
66	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
67	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
68	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 640 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.734e+004	2.734e+004	2.735e+004	2.735e+004	2.736e+004
14	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
15	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
16	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
17	Length (m)	2.736e+004	2.737e+004	2.737e+004	2.738e+004	2.738e+004
18	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
19	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
20	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
21	Length (m)	2.739e+004	2.739e+004	2.740e+004	2.740e+004	2.741e+004
22	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
23	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
24	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
25	Length (m)	2.741e+004	2.742e+004	2.742e+004	2.743e+004	2.743e+004
26	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
27	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
28	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
29	Length (m)	2.744e+004	2.744e+004	2.745e+004	2.745e+004	2.746e+004
30	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
31	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
32	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
33	Length (m)	2.746e+004	2.747e+004	2.747e+004	2.748e+004	2.748e+004
34	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
35	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
36	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
37	Length (m)	2.749e+004	2.749e+004	2.749e+004	2.749e+004	2.749e+004
38	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
39	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
40	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
41	Length (m)	2.750e+004	2.750e+004	2.751e+004	2.751e+004	2.752e+004
42	Bulk Density (kg/m3)	94.20	94.20	94.20	94.20	94.20
43	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
44	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
45	Length (m)	2.753e+004	2.753e+004	2.753e+004	2.753e+004	2.753e+004
46	Bulk Density (kg/m3)	94.20	94.20	94.20	94.19	94.19
47	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
48	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
49	Length (m)	2.754e+004	2.754e+004	2.755e+004	2.755e+004	2.756e+004
50	Bulk Density (kg/m3)	94.19	94.19	94.20	94.20	94.20
51	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
52	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
53	Length (m)	2.756e+004	2.757e+004	2.757e+004	2.758e+004	2.758e+004
54	Bulk Density (kg/m3)	94.20	94.19	94.19	94.18	94.18
55	Bulk Velocity (m/s)	3.130	3.130	3.130	3.131	3.131
56	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
57	Length (m)	2.759e+004	2.759e+004	2.760e+004	2.760e+004	2.761e+004
58	Bulk Density (kg/m3)	94.18	94.18	94.18	94.19	94.19
59	Bulk Velocity (m/s)	3.131	3.131	3.131	3.130	3.130
60	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
61	Length (m)	2.761e+004	2.762e+004	2.762e+004	2.763e+004	2.763e+004
62	Bulk Density (kg/m3)	94.20	94.19	94.19	94.18	94.18
63	Bulk Velocity (m/s)	3.130	3.130	3.130	3.131	3.131
64	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
65	Length (m)	2.764e+004	2.764e+004	2.765e+004	2.765e+004	2.766e+004
66	Bulk Density (kg/m3)	94.17	94.17	94.17	94.17	94.16
67	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
68	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 641 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.766e+004	2.767e+004	2.767e+004	2.768e+004	2.768e+004
14	Bulk Density (kg/m3)	94.16	94.16	94.17	94.17	94.17
15	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
16	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
17	Length (m)	2.769e+004	2.769e+004	2.770e+004	2.770e+004	2.771e+004
18	Bulk Density (kg/m3)	94.17	94.17	94.17	94.17	94.17
19	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
20	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
21	Length (m)	2.771e+004	2.772e+004	2.772e+004	2.773e+004	2.773e+004
22	Bulk Density (kg/m3)	94.17	94.17	94.17	94.17	94.17
23	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
24	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
25	Length (m)	2.774e+004	2.774e+004	2.775e+004	2.775e+004	2.776e+004
26	Bulk Density (kg/m3)	94.17	94.17	94.17	94.17	94.17
27	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
28	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
29	Length (m)	2.776e+004	2.777e+004	2.777e+004	2.778e+004	2.778e+004
30	Bulk Density (kg/m3)	94.17	94.17	94.17	94.17	94.17
31	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
32	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
33	Length (m)	2.779e+004	2.779e+004	2.780e+004	2.780e+004	2.781e+004
34	Bulk Density (kg/m3)	94.17	94.17	94.17	94.18	94.18
35	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
36	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
37	Length (m)	2.781e+004	2.782e+004	2.782e+004	2.783e+004	2.783e+004
38	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
39	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
40	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
41	Length (m)	2.784e+004	2.784e+004	2.785e+004	2.785e+004	2.785e+004
42	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
43	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
44	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
45	Length (m)	2.786e+004	2.788e+004	2.790e+004	2.791e+004	2.793e+004
46	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
47	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
48	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
49	Length (m)	2.795e+004	2.797e+004	2.799e+004	2.801e+004	2.802e+004
50	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
51	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
52	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
53	Length (m)	2.804e+004	2.806e+004	2.808e+004	2.810e+004	2.812e+004
54	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
55	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
56	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
57	Length (m)	2.814e+004	2.815e+004	2.817e+004	2.819e+004	2.821e+004
58	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
59	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
60	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
61	Length (m)	2.823e+004	2.825e+004	2.827e+004	2.828e+004	2.830e+004
62	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
63	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
64	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
65	Length (m)	2.832e+004	2.834e+004	2.836e+004	2.838e+004	2.839e+004
66	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
67	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
68	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 642 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.841e+004	2.842e+004	2.843e+004	2.843e+004	2.844e+004
14	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
15	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
16	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
17	Length (m)	2.845e+004	2.845e+004	2.846e+004	2.846e+004	2.847e+004
18	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
19	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
20	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
21	Length (m)	2.847e+004	2.848e+004	2.848e+004	2.849e+004	2.849e+004
22	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
23	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
24	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
25	Length (m)	2.850e+004	2.850e+004	2.851e+004	2.851e+004	2.852e+004
26	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
27	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
28	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
29	Length (m)	2.852e+004	2.853e+004	2.853e+004	2.854e+004	2.854e+004
30	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
31	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
32	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
33	Length (m)	2.855e+004	2.855e+004	2.856e+004	2.856e+004	2.857e+004
34	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
35	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
36	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
37	Length (m)	2.857e+004	2.858e+004	2.858e+004	2.859e+004	2.859e+004
38	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
39	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.130
40	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
41	Length (m)	2.860e+004	2.860e+004	2.861e+004	2.861e+004	2.862e+004
42	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
43	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
44	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
45	Length (m)	2.862e+004	2.863e+004	2.863e+004	2.864e+004	2.864e+004
46	Bulk Density (kg/m3)	94.19	94.19	94.19	94.19	94.19
47	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
48	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
49	Length (m)	2.865e+004	2.865e+004	2.866e+004	2.866e+004	2.867e+004
50	Bulk Density (kg/m3)	94.20	94.20	94.19	94.19	94.19
51	Bulk Velocity (m/s)	3.130	3.130	3.130	3.130	3.130
52	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
53	Length (m)	2.867e+004	2.868e+004	2.868e+004	2.869e+004	2.869e+004
54	Bulk Density (kg/m3)	94.19	94.19	94.19	94.18	94.18
55	Bulk Velocity (m/s)	3.130	3.130	3.130	3.131	3.131
56	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
57	Length (m)	2.870e+004	2.870e+004	2.871e+004	2.871e+004	2.872e+004
58	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
59	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
60	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
61	Length (m)	2.872e+004	2.873e+004	2.874e+004	2.875e+004	2.876e+004
62	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
63	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
64	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
65	Length (m)	2.877e+004	2.878e+004	2.879e+004	2.880e+004	2.881e+004
66	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
67	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
68	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 643 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.883e+004	2.884e+004	2.885e+004	2.886e+004	2.887e+004
14	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
15	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
16	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
17	Length (m)	2.888e+004	2.889e+004	2.890e+004	2.891e+004	2.892e+004
18	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
19	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
20	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
21	Length (m)	2.893e+004	2.894e+004	2.895e+004	2.896e+004	2.897e+004
22	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
23	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
24	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
25	Length (m)	2.898e+004	2.899e+004	2.900e+004	2.901e+004	2.902e+004
26	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
27	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
28	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
29	Length (m)	2.903e+004	2.904e+004	2.904e+004	2.904e+004	2.904e+004
30	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
31	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
32	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
33	Length (m)	2.905e+004	2.905e+004	2.906e+004	2.906e+004	2.907e+004
34	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
35	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
36	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
37	Length (m)	2.907e+004	2.908e+004	2.908e+004	2.909e+004	2.909e+004
38	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
39	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
40	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
41	Length (m)	2.910e+004	2.910e+004	2.911e+004	2.911e+004	2.912e+004
42	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
43	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
44	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
45	Length (m)	2.912e+004	2.913e+004	2.913e+004	2.914e+004	2.914e+004
46	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
47	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
48	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
49	Length (m)	2.915e+004	2.915e+004	2.916e+004	2.916e+004	2.916e+004
50	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
51	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
52	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
53	Length (m)	2.917e+004	2.917e+004	2.918e+004	2.918e+004	2.919e+004
54	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
55	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
56	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
57	Length (m)	2.920e+004	2.920e+004	2.921e+004	2.921e+004	2.922e+004
58	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
59	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
60	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
61	Length (m)	2.922e+004	2.923e+004	2.923e+004	2.924e+004	2.924e+004
62	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
63	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
64	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
65	Length (m)	2.925e+004	2.925e+004	2.925e+004	2.926e+004	2.926e+004
66	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
67	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
68	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 644 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.926e+004	2.927e+004	2.928e+004	2.929e+004	2.930e+004
14	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
15	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
16	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
17	Length (m)	2.931e+004	2.932e+004	2.933e+004	2.934e+004	2.935e+004
18	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
19	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
20	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
21	Length (m)	2.936e+004	2.936e+004	2.937e+004	2.938e+004	2.939e+004
22	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
23	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
24	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
25	Length (m)	2.940e+004	2.941e+004	2.942e+004	2.943e+004	2.944e+004
26	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
27	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
28	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
29	Length (m)	2.945e+004	2.946e+004	2.947e+004	2.948e+004	2.949e+004
30	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
31	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
32	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
33	Length (m)	2.950e+004	2.951e+004	2.952e+004	2.953e+004	2.954e+004
34	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
35	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
36	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
37	Length (m)	2.955e+004	2.955e+004	2.955e+004	2.955e+004	2.955e+004
38	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
39	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
40	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
41	Length (m)	2.955e+004	2.956e+004	2.956e+004	2.957e+004	2.957e+004
42	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
43	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
44	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
45	Length (m)	2.958e+004	2.958e+004	2.959e+004	2.959e+004	2.960e+004
46	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
47	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
48	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
49	Length (m)	2.960e+004	2.961e+004	2.961e+004	2.962e+004	2.962e+004
50	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
51	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
52	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
53	Length (m)	2.963e+004	2.963e+004	2.964e+004	2.964e+004	2.965e+004
54	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
55	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
56	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
57	Length (m)	2.965e+004	2.965e+004	2.966e+004	2.966e+004	2.966e+004
58	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
59	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
60	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
61	Length (m)	2.966e+004	2.966e+004	2.967e+004	2.967e+004	2.967e+004
62	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
63	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
64	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
65	Length (m)	2.967e+004	2.968e+004	2.968e+004	2.968e+004	2.969e+004
66	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
67	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
68	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 645 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.969e+004	2.969e+004	2.970e+004	2.970e+004	2.970e+004
14	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
15	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
16	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
17	Length (m)	2.971e+004	2.971e+004	2.971e+004	2.972e+004	2.972e+004
18	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
19	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
20	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
21	Length (m)	2.973e+004	2.973e+004	2.974e+004	2.974e+004	2.975e+004
22	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
23	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
24	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
25	Length (m)	2.975e+004	2.976e+004	2.976e+004	2.977e+004	2.977e+004
26	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
27	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
28	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
29	Length (m)	2.978e+004	2.978e+004	2.979e+004	2.979e+004	2.980e+004
30	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
31	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
32	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
33	Length (m)	2.980e+004	2.981e+004	2.981e+004	2.982e+004	2.982e+004
34	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
35	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
36	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
37	Length (m)	2.983e+004	2.983e+004	2.984e+004	2.984e+004	2.985e+004
38	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
39	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
40	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
41	Length (m)	2.985e+004	2.986e+004	2.986e+004	2.987e+004	2.987e+004
42	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
43	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
44	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
45	Length (m)	2.988e+004	2.988e+004	2.989e+004	2.989e+004	2.990e+004
46	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
47	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
48	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
49	Length (m)	2.990e+004	2.990e+004	2.991e+004	2.991e+004	2.991e+004
50	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
51	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
52	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
53	Length (m)	2.991e+004	2.992e+004	2.992e+004	2.993e+004	2.993e+004
54	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
55	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
56	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
57	Length (m)	2.994e+004	2.994e+004	2.994e+004	2.995e+004	2.995e+004
58	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
59	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
60	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
61	Length (m)	2.996e+004	2.996e+004	2.997e+004	2.997e+004	2.998e+004
62	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
63	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
64	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
65	Length (m)	2.998e+004	2.999e+004	2.999e+004	3.000e+004	3.000e+004
66	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
67	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
68	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 646 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.001e+004	3.001e+004	3.002e+004	3.002e+004	3.003e+004
14	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
15	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
16	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
17	Length (m)	3.003e+004	3.004e+004	3.004e+004	3.005e+004	3.005e+004
18	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
19	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
20	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
21	Length (m)	3.006e+004	3.006e+004	3.007e+004	3.007e+004	3.008e+004
22	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
23	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
24	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
25	Length (m)	3.008e+004	3.009e+004	3.009e+004	3.010e+004	3.010e+004
26	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
27	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
28	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
29	Length (m)	3.011e+004	3.011e+004	3.012e+004	3.012e+004	3.013e+004
30	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
31	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
32	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
33	Length (m)	3.013e+004	3.014e+004	3.014e+004	3.015e+004	3.015e+004
34	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
35	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
36	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
37	Length (m)	3.016e+004	3.016e+004	3.017e+004	3.017e+004	3.018e+004
38	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
39	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
40	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
41	Length (m)	3.018e+004	3.019e+004	3.019e+004	3.020e+004	3.020e+004
42	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
43	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
44	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
45	Length (m)	3.021e+004	3.021e+004	3.022e+004	3.022e+004	3.023e+004
46	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
47	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
48	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
49	Length (m)	3.023e+004	3.024e+004	3.024e+004	3.025e+004	3.025e+004
50	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
51	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
52	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
53	Length (m)	3.026e+004	3.026e+004	3.027e+004	3.027e+004	3.028e+004
54	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
55	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
56	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
57	Length (m)	3.028e+004	3.029e+004	3.029e+004	3.030e+004	3.030e+004
58	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
59	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
60	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
61	Length (m)	3.031e+004	3.031e+004	3.032e+004	3.032e+004	3.033e+004
62	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
63	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
64	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
65	Length (m)	3.033e+004	3.034e+004	3.035e+004	3.036e+004	3.036e+004
66	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
67	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
68	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 647 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.037e+004	3.039e+004	3.040e+004	3.041e+004	3.043e+004
14	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
15	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
16	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
17	Length (m)	3.044e+004	3.045e+004	3.045e+004	3.045e+004	3.045e+004
18	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
19	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
20	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
21	Length (m)	3.046e+004	3.046e+004	3.047e+004	3.047e+004	3.048e+004
22	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
23	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
24	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
25	Length (m)	3.048e+004	3.049e+004	3.049e+004	3.050e+004	3.050e+004
26	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
27	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
28	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
29	Length (m)	3.051e+004	3.051e+004	3.052e+004	3.052e+004	3.053e+004
30	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
31	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
32	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
33	Length (m)	3.053e+004	3.054e+004	3.054e+004	3.055e+004	3.055e+004
34	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
35	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
36	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
37	Length (m)	3.056e+004	3.056e+004	3.057e+004	3.057e+004	3.058e+004
38	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
39	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
40	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
41	Length (m)	3.058e+004	3.059e+004	3.059e+004	3.060e+004	3.060e+004
42	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
43	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
44	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
45	Length (m)	3.061e+004	3.061e+004	3.062e+004	3.062e+004	3.063e+004
46	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
47	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
48	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
49	Length (m)	3.063e+004	3.064e+004	3.064e+004	3.065e+004	3.065e+004
50	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
51	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
52	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
53	Length (m)	3.066e+004	3.066e+004	3.067e+004	3.067e+004	3.068e+004
54	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
55	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
56	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
57	Length (m)	3.068e+004	3.069e+004	3.069e+004	3.070e+004	3.070e+004
58	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
59	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
60	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
61	Length (m)	3.071e+004	3.071e+004	3.072e+004	3.072e+004	3.073e+004
62	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.19
63	Bulk Velocity (m/s)	3.131	3.130	3.130	3.130	3.130
64	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
65	Length (m)	3.073e+004	3.074e+004	3.074e+004	3.075e+004	3.075e+004
66	Bulk Density (kg/m3)	94.19	94.18	94.18	94.18	94.18
67	Bulk Velocity (m/s)	3.130	3.130	3.130	3.131	3.131
68	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 648 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.076e+004	3.076e+004	3.077e+004	3.077e+004	3.078e+004
14	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
15	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
16	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
17	Length (m)	3.078e+004	3.079e+004	3.079e+004	3.080e+004	3.080e+004
18	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
19	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
20	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
21	Length (m)	3.081e+004	3.081e+004	3.082e+004	3.082e+004	3.083e+004
22	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
23	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
24	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
25	Length (m)	3.083e+004	3.084e+004	3.084e+004	3.085e+004	3.085e+004
26	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
27	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
28	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
29	Length (m)	3.086e+004	3.086e+004	3.087e+004	3.087e+004	3.088e+004
30	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
31	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
32	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
33	Length (m)	3.088e+004	3.089e+004	3.089e+004	3.090e+004	3.090e+004
34	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
35	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
36	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
37	Length (m)	3.091e+004	3.091e+004	3.092e+004	3.092e+004	3.093e+004
38	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
39	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
40	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
41	Length (m)	3.093e+004	3.094e+004	3.094e+004	3.095e+004	3.095e+004
42	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
43	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
44	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
45	Length (m)	3.096e+004	3.096e+004	3.097e+004	3.097e+004	3.098e+004
46	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
47	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
48	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
49	Length (m)	3.098e+004	3.099e+004	3.099e+004	3.100e+004	3.100e+004
50	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
51	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
52	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
53	Length (m)	3.101e+004	3.103e+004	3.105e+004	3.107e+004	3.109e+004
54	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
55	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
56	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
57	Length (m)	3.111e+004	3.111e+004	3.112e+004	3.112e+004	3.113e+004
58	Bulk Density (kg/m3)	94.18	94.18	94.18	94.18	94.18
59	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
60	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
61	Length (m)	3.113e+004	3.114e+004	3.114e+004	3.115e+004	3.115e+004
62	Bulk Density (kg/m3)	94.18	94.18	94.17	94.17	94.17
63	Bulk Velocity (m/s)	3.131	3.131	3.131	3.131	3.131
64	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
65	Length (m)	3.116e+004	3.116e+004	3.117e+004	3.117e+004	3.118e+004
66	Bulk Density (kg/m3)	94.16	94.18	94.20	94.22	94.24
67	Bulk Velocity (m/s)	3.131	3.131	3.130	3.129	3.129
68	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 649 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.118e+004	3.119e+004	3.119e+004	3.119e+004	3.120e+004
14	Bulk Density (kg/m3)	94.26	94.26	94.26	94.26	94.25
15	Bulk Velocity (m/s)	3.128	3.128	3.128	3.128	3.128
16	Erosion Velocity (m/s)	12.56	12.56	12.57	12.57	12.57
17	Length (m)	3.120e+004	3.121e+004	3.122e+004	3.122e+004	3.123e+004
18	Bulk Density (kg/m3)	94.25	94.24	94.23	94.22	94.20
19	Bulk Velocity (m/s)	3.128	3.129	3.129	3.129	3.130
20	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
21	Length (m)	3.124e+004	3.124e+004	3.125e+004	3.125e+004	3.126e+004
22	Bulk Density (kg/m3)	94.19	94.18	94.17	94.15	94.14
23	Bulk Velocity (m/s)	3.130	3.131	3.131	3.132	3.132
24	Erosion Velocity (m/s)	12.57	12.57	12.57	12.57	12.57
25	Length (m)	3.126e+004	3.126e+004	3.127e+004	3.127e+004	3.128e+004
26	Bulk Density (kg/m3)	94.13	94.06	93.98	93.91	93.84
27	Bulk Velocity (m/s)	3.132	3.135	3.137	3.140	3.142
28	Erosion Velocity (m/s)	12.57	12.58	12.58	12.59	12.59
29	Length (m)	3.128e+004	3.129e+004	3.129e+004	3.130e+004	3.130e+004
30	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
31	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
32	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
33	Length (m)	3.131e+004	3.131e+004	3.132e+004	3.132e+004	3.133e+004
34	Bulk Density (kg/m3)	93.78	93.78	93.77	93.77	93.77
35	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
36	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
37	Length (m)	3.133e+004	3.134e+004	3.134e+004	3.135e+004	3.135e+004
38	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
39	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
40	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
41	Length (m)	3.136e+004	3.136e+004	3.137e+004	3.137e+004	3.138e+004
42	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
43	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
44	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
45	Length (m)	3.138e+004	3.139e+004	3.139e+004	3.140e+004	3.140e+004
46	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
47	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
48	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
49	Length (m)	3.141e+004	3.141e+004	3.142e+004	3.142e+004	3.143e+004
50	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
51	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
52	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
53	Length (m)	3.143e+004	3.144e+004	3.144e+004	3.145e+004	3.145e+004
54	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
55	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
56	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
57	Length (m)	3.146e+004	3.146e+004	3.147e+004	3.147e+004	3.148e+004
58	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
59	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
60	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
61	Length (m)	3.148e+004	3.149e+004	3.149e+004	3.150e+004	3.150e+004
62	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
63	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
64	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
65	Length (m)	3.151e+004	3.151e+004	3.152e+004	3.152e+004	3.153e+004
66	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
67	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
68	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 650 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.153e+004	3.154e+004	3.154e+004	3.155e+004	3.155e+004
14	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
15	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
16	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
17	Length (m)	3.156e+004	3.157e+004	3.157e+004	3.158e+004	3.158e+004
18	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
19	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
20	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
21	Length (m)	3.159e+004	3.159e+004	3.160e+004	3.160e+004	3.161e+004
22	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
23	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
24	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
25	Length (m)	3.161e+004	3.162e+004	3.162e+004	3.163e+004	3.163e+004
26	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
27	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
28	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
29	Length (m)	3.164e+004	3.164e+004	3.165e+004	3.165e+004	3.166e+004
30	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
31	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
32	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
33	Length (m)	3.166e+004	3.167e+004	3.167e+004	3.168e+004	3.168e+004
34	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
35	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
36	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
37	Length (m)	3.169e+004	3.169e+004	3.169e+004	3.169e+004	3.170e+004
38	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
39	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
40	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
41	Length (m)	3.170e+004	3.170e+004	3.170e+004	3.171e+004	3.171e+004
42	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
43	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
44	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
45	Length (m)	3.171e+004	3.172e+004	3.172e+004	3.173e+004	3.173e+004
46	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
47	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
48	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
49	Length (m)	3.174e+004	3.174e+004	3.174e+004	3.174e+004	3.175e+004
50	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
51	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
52	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
53	Length (m)	3.175e+004	3.176e+004	3.176e+004	3.177e+004	3.177e+004
54	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
55	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
56	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
57	Length (m)	3.178e+004	3.179e+004	3.179e+004	3.180e+004	3.180e+004
58	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
59	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
60	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
61	Length (m)	3.181e+004	3.182e+004	3.182e+004	3.183e+004	3.183e+004
62	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
63	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
64	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
65	Length (m)	3.184e+004	3.184e+004	3.185e+004	3.185e+004	3.186e+004
66	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
67	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
68	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 651 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
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11	Erosion Table					
12						
13	Length (m)	3.186e+004	3.187e+004	3.187e+004	3.188e+004	3.188e+004
14	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
15	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
16	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
17	Length (m)	3.189e+004	3.189e+004	3.190e+004	3.190e+004	3.191e+004
18	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
19	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
20	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
21	Length (m)	3.191e+004	3.192e+004	3.192e+004	3.193e+004	3.193e+004
22	Bulk Density (kg/m3)	93.77	93.77	93.77	93.76	93.76
23	Bulk Velocity (m/s)	3.144	3.144	3.144	3.145	3.145
24	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
25	Length (m)	3.194e+004	3.194e+004	3.195e+004	3.195e+004	3.196e+004
26	Bulk Density (kg/m3)	93.76	93.76	93.76	93.76	93.76
27	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
28	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
29	Length (m)	3.196e+004	3.197e+004	3.197e+004	3.198e+004	3.198e+004
30	Bulk Density (kg/m3)	93.76	93.76	93.76	93.76	93.76
31	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
32	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
33	Length (m)	3.199e+004	3.199e+004	3.200e+004	3.200e+004	3.201e+004
34	Bulk Density (kg/m3)	93.76	93.76	93.76	93.76	93.76
35	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
36	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
37	Length (m)	3.201e+004	3.202e+004	3.202e+004	3.203e+004	3.203e+004
38	Bulk Density (kg/m3)	93.76	93.76	93.76	93.76	93.77
39	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
40	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
41	Length (m)	3.204e+004	3.204e+004	3.205e+004	3.205e+004	3.206e+004
42	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
43	Bulk Velocity (m/s)	3.145	3.145	3.145	3.144	3.144
44	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
45	Length (m)	3.206e+004	3.207e+004	3.207e+004	3.208e+004	3.208e+004
46	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
47	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
48	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
49	Length (m)	3.209e+004	3.209e+004	3.210e+004	3.210e+004	3.211e+004
50	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.77
51	Bulk Velocity (m/s)	3.144	3.144	3.144	3.144	3.144
52	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
53	Length (m)	3.211e+004	3.212e+004	3.212e+004	3.213e+004	3.213e+004
54	Bulk Density (kg/m3)	93.77	93.77	93.77	93.77	93.76
55	Bulk Velocity (m/s)	3.144	3.145	3.145	3.145	3.145
56	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
57	Length (m)	3.214e+004	3.214e+004	3.215e+004	3.215e+004	3.216e+004
58	Bulk Density (kg/m3)	93.76	93.76	93.76	93.76	93.76
59	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
60	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
61	Length (m)	3.216e+004	3.217e+004	3.217e+004	3.218e+004	3.218e+004
62	Bulk Density (kg/m3)	93.76	93.76	93.76	93.76	93.76
63	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
64	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
65	Length (m)	3.219e+004	3.219e+004	3.219e+004	3.219e+004	3.219e+004
66	Bulk Density (kg/m3)	93.76	93.76	93.75	93.75	93.75
67	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
68	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 652 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.219e+004	3.219e+004	3.220e+004	3.220e+004	3.221e+004
14	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.73
15	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.146
16	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
17	Length (m)	3.221e+004	3.221e+004	3.222e+004	3.222e+004	3.223e+004
18	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
19	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
20	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
21	Length (m)	3.223e+004	3.224e+004	3.226e+004	3.227e+004	3.228e+004
22	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
23	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
24	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
25	Length (m)	3.230e+004	3.231e+004	3.232e+004	3.234e+004	3.235e+004
26	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
27	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
28	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
29	Length (m)	3.236e+004	3.238e+004	3.239e+004	3.240e+004	3.242e+004
30	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
31	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
32	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
33	Length (m)	3.243e+004	3.244e+004	3.246e+004	3.247e+004	3.248e+004
34	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
35	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
36	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
37	Length (m)	3.250e+004	3.251e+004	3.252e+004	3.254e+004	3.255e+004
38	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
39	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
40	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
41	Length (m)	3.257e+004	3.257e+004	3.258e+004	3.258e+004	3.259e+004
42	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
43	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
44	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
45	Length (m)	3.260e+004	3.260e+004	3.261e+004	3.261e+004	3.262e+004
46	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
47	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
48	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
49	Length (m)	3.263e+004	3.263e+004	3.263e+004	3.263e+004	3.263e+004
50	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
51	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
52	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
53	Length (m)	3.264e+004	3.264e+004	3.264e+004	3.264e+004	3.264e+004
54	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
55	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
56	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
57	Length (m)	3.265e+004	3.265e+004	3.266e+004	3.266e+004	3.267e+004
58	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
59	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
60	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
61	Length (m)	3.267e+004	3.268e+004	3.268e+004	3.269e+004	3.269e+004
62	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
63	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
64	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
65	Length (m)	3.270e+004	3.270e+004	3.271e+004	3.271e+004	3.272e+004
66	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
67	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
68	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 653 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.272e+004	3.273e+004	3.273e+004	3.274e+004	3.274e+004
14	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
15	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
16	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
17	Length (m)	3.275e+004	3.275e+004	3.276e+004	3.276e+004	3.277e+004
18	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
19	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
20	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
21	Length (m)	3.277e+004	3.278e+004	3.278e+004	3.279e+004	3.279e+004
22	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
23	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
24	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
25	Length (m)	3.280e+004	3.280e+004	3.281e+004	3.281e+004	3.282e+004
26	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
27	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
28	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
29	Length (m)	3.282e+004	3.283e+004	3.283e+004	3.284e+004	3.284e+004
30	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
31	Bulk Velocity (m/s)	3.146	3.145	3.145	3.145	3.145
32	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
33	Length (m)	3.285e+004	3.285e+004	3.286e+004	3.286e+004	3.287e+004
34	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
35	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
36	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
37	Length (m)	3.287e+004	3.288e+004	3.288e+004	3.289e+004	3.289e+004
38	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
39	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
40	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
41	Length (m)	3.290e+004	3.290e+004	3.291e+004	3.291e+004	3.292e+004
42	Bulk Density (kg/m3)	93.74	93.75	93.75	93.75	93.75
43	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
44	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
45	Length (m)	3.292e+004	3.293e+004	3.293e+004	3.294e+004	3.294e+004
46	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
47	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
48	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
49	Length (m)	3.295e+004	3.295e+004	3.296e+004	3.296e+004	3.297e+004
50	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
51	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
52	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
53	Length (m)	3.297e+004	3.298e+004	3.298e+004	3.299e+004	3.299e+004
54	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
55	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
56	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
57	Length (m)	3.300e+004	3.300e+004	3.301e+004	3.301e+004	3.302e+004
58	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
59	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
60	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
61	Length (m)	3.302e+004	3.302e+004	3.302e+004	3.303e+004	3.303e+004
62	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
63	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
64	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
65	Length (m)	3.303e+004	3.304e+004	3.304e+004	3.305e+004	3.305e+004
66	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
67	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
68	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 654 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.306e+004	3.306e+004	3.306e+004	3.307e+004	3.307e+004
14	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
15	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
16	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
17	Length (m)	3.307e+004	3.308e+004	3.308e+004	3.309e+004	3.309e+004
18	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
19	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
20	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
21	Length (m)	3.310e+004	3.310e+004	3.311e+004	3.311e+004	3.312e+004
22	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
23	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
24	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
25	Length (m)	3.312e+004	3.313e+004	3.313e+004	3.314e+004	3.314e+004
26	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
27	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
28	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
29	Length (m)	3.315e+004	3.315e+004	3.316e+004	3.316e+004	3.317e+004
30	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
31	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
32	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
33	Length (m)	3.317e+004	3.318e+004	3.318e+004	3.319e+004	3.319e+004
34	Bulk Density (kg/m3)	93.75	93.75	93.75	93.74	93.74
35	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
36	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
37	Length (m)	3.320e+004	3.320e+004	3.321e+004	3.321e+004	3.322e+004
38	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
39	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
40	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
41	Length (m)	3.322e+004	3.323e+004	3.323e+004	3.324e+004	3.324e+004
42	Bulk Density (kg/m3)	93.74	93.75	93.75	93.75	93.75
43	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
44	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
45	Length (m)	3.325e+004	3.325e+004	3.326e+004	3.326e+004	3.327e+004
46	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
47	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
48	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
49	Length (m)	3.327e+004	3.328e+004	3.328e+004	3.329e+004	3.329e+004
50	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
51	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
52	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
53	Length (m)	3.330e+004	3.330e+004	3.331e+004	3.331e+004	3.332e+004
54	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
55	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
56	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
57	Length (m)	3.332e+004	3.333e+004	3.333e+004	3.334e+004	3.334e+004
58	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
59	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
60	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
61	Length (m)	3.335e+004	3.335e+004	3.336e+004	3.336e+004	3.337e+004
62	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
63	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
64	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
65	Length (m)	3.337e+004	3.338e+004	3.338e+004	3.339e+004	3.339e+004
66	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
67	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
68	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 655 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.340e+004	3.340e+004	3.341e+004	3.341e+004	3.342e+004
14	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
15	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
16	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
17	Length (m)	3.342e+004	3.343e+004	3.343e+004	3.344e+004	3.344e+004
18	Bulk Density (kg/m3)	93.75	93.75	93.74	93.74	93.74
19	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
20	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
21	Length (m)	3.345e+004	3.345e+004	3.346e+004	3.346e+004	3.347e+004
22	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
23	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
24	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
25	Length (m)	3.347e+004	3.348e+004	3.348e+004	3.349e+004	3.349e+004
26	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.75
27	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
28	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
29	Length (m)	3.350e+004	3.350e+004	3.351e+004	3.351e+004	3.352e+004
30	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
31	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
32	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
33	Length (m)	3.352e+004	3.353e+004	3.353e+004	3.354e+004	3.354e+004
34	Bulk Density (kg/m3)	93.74	93.75	93.75	93.75	93.75
35	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
36	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
37	Length (m)	3.355e+004	3.355e+004	3.356e+004	3.356e+004	3.357e+004
38	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.74
39	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
40	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
41	Length (m)	3.357e+004	3.358e+004	3.358e+004	3.359e+004	3.359e+004
42	Bulk Density (kg/m3)	93.74	93.74	93.75	93.75	93.75
43	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
44	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
45	Length (m)	3.360e+004	3.360e+004	3.361e+004	3.361e+004	3.362e+004
46	Bulk Density (kg/m3)	93.75	93.74	93.74	93.74	93.74
47	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
48	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
49	Length (m)	3.362e+004	3.363e+004	3.363e+004	3.364e+004	3.364e+004
50	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
51	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
52	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
53	Length (m)	3.365e+004	3.365e+004	3.366e+004	3.366e+004	3.367e+004
54	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
55	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
56	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
57	Length (m)	3.367e+004	3.368e+004	3.368e+004	3.369e+004	3.369e+004
58	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
59	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
60	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
61	Length (m)	3.370e+004	3.370e+004	3.371e+004	3.371e+004	3.372e+004
62	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
63	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
64	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
65	Length (m)	3.372e+004	3.373e+004	3.373e+004	3.374e+004	3.374e+004
66	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
67	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
68	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 656 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.375e+004	3.375e+004	3.376e+004	3.376e+004	3.377e+004
14	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
15	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
16	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
17	Length (m)	3.377e+004	3.378e+004	3.378e+004	3.379e+004	3.379e+004
18	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
19	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
20	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
21	Length (m)	3.380e+004	3.380e+004	3.381e+004	3.381e+004	3.382e+004
22	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
23	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
24	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
25	Length (m)	3.382e+004	3.383e+004	3.383e+004	3.384e+004	3.384e+004
26	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
27	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
28	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
29	Length (m)	3.385e+004	3.385e+004	3.385e+004	3.385e+004	3.386e+004
30	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
31	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
32	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
33	Length (m)	3.386e+004	3.386e+004	3.386e+004	3.386e+004	3.387e+004
34	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
35	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
36	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
37	Length (m)	3.387e+004	3.387e+004	3.387e+004	3.388e+004	3.388e+004
38	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
39	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
40	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
41	Length (m)	3.388e+004	3.389e+004	3.389e+004	3.389e+004	3.389e+004
42	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
43	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
44	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
45	Length (m)	3.390e+004	3.391e+004	3.392e+004	3.393e+004	3.394e+004
46	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
47	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
48	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
49	Length (m)	3.395e+004	3.396e+004	3.398e+004	3.399e+004	3.400e+004
50	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
51	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
52	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
53	Length (m)	3.401e+004	3.402e+004	3.403e+004	3.404e+004	3.405e+004
54	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
55	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
56	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
57	Length (m)	3.406e+004	3.408e+004	3.409e+004	3.410e+004	3.411e+004
58	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
59	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
60	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
61	Length (m)	3.412e+004	3.413e+004	3.414e+004	3.415e+004	3.416e+004
62	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
63	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
64	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
65	Length (m)	3.418e+004	3.419e+004	3.420e+004	3.421e+004	3.422e+004
66	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
67	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
68	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 657 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.423e+004	3.424e+004	3.425e+004	3.426e+004	3.428e+004
14	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
15	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
16	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
17	Length (m)	3.429e+004	3.429e+004	3.429e+004	3.429e+004	3.430e+004
18	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
19	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
20	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
21	Length (m)	3.430e+004	3.430e+004	3.431e+004	3.431e+004	3.432e+004
22	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
23	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
24	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
25	Length (m)	3.432e+004	3.433e+004	3.433e+004	3.434e+004	3.434e+004
26	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
27	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
28	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
29	Length (m)	3.435e+004	3.435e+004	3.436e+004	3.436e+004	3.437e+004
30	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
31	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
32	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
33	Length (m)	3.437e+004	3.438e+004	3.438e+004	3.439e+004	3.439e+004
34	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
35	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
36	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
37	Length (m)	3.440e+004	3.440e+004	3.441e+004	3.441e+004	3.442e+004
38	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
39	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
40	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
41	Length (m)	3.442e+004	3.443e+004	3.443e+004	3.444e+004	3.444e+004
42	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
43	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
44	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
45	Length (m)	3.445e+004	3.445e+004	3.446e+004	3.446e+004	3.447e+004
46	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
47	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
48	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
49	Length (m)	3.447e+004	3.448e+004	3.448e+004	3.449e+004	3.449e+004
50	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
51	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
52	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
53	Length (m)	3.450e+004	3.450e+004	3.451e+004	3.451e+004	3.452e+004
54	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
55	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
56	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
57	Length (m)	3.452e+004	3.453e+004	3.453e+004	3.454e+004	3.454e+004
58	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
59	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
60	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
61	Length (m)	3.455e+004	3.455e+004	3.456e+004	3.456e+004	3.457e+004
62	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
63	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
64	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
65	Length (m)	3.457e+004	3.458e+004	3.458e+004	3.459e+004	3.459e+004
66	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
67	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
68	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 658 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.460e+004	3.460e+004	3.461e+004	3.461e+004	3.462e+004
14	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
15	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
16	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
17	Length (m)	3.462e+004	3.463e+004	3.463e+004	3.464e+004	3.464e+004
18	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
19	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
20	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
21	Length (m)	3.465e+004	3.465e+004	3.465e+004	3.466e+004	3.466e+004
22	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
23	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
24	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
25	Length (m)	3.467e+004	3.467e+004	3.468e+004	3.468e+004	3.469e+004
26	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
27	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
28	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
29	Length (m)	3.469e+004	3.470e+004	3.470e+004	3.471e+004	3.471e+004
30	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
31	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
32	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
33	Length (m)	3.472e+004	3.472e+004	3.473e+004	3.473e+004	3.474e+004
34	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
35	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
36	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
37	Length (m)	3.474e+004	3.475e+004	3.475e+004	3.476e+004	3.476e+004
38	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
39	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
40	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
41	Length (m)	3.477e+004	3.477e+004	3.478e+004	3.478e+004	3.479e+004
42	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
43	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
44	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
45	Length (m)	3.479e+004	3.480e+004	3.480e+004	3.481e+004	3.481e+004
46	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
47	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
48	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
49	Length (m)	3.482e+004	3.482e+004	3.483e+004	3.483e+004	3.484e+004
50	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
51	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
52	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
53	Length (m)	3.484e+004	3.485e+004	3.485e+004	3.486e+004	3.486e+004
54	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.74
55	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
56	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
57	Length (m)	3.487e+004	3.487e+004	3.488e+004	3.488e+004	3.489e+004
58	Bulk Density (kg/m3)	93.74	93.74	93.74	93.74	93.73
59	Bulk Velocity (m/s)	3.145	3.145	3.145	3.146	3.146
60	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
61	Length (m)	3.489e+004	3.490e+004	3.490e+004	3.491e+004	3.491e+004
62	Bulk Density (kg/m3)	93.73	93.73	93.74	93.74	93.74
63	Bulk Velocity (m/s)	3.146	3.146	3.145	3.145	3.145
64	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
65	Length (m)	3.492e+004	3.492e+004	3.493e+004	3.493e+004	3.494e+004
66	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.75
67	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
68	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 659 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.494e+004	3.495e+004	3.495e+004	3.496e+004	3.497e+004
14	Bulk Density (kg/m3)	93.75	93.75	93.75	93.75	93.74
15	Bulk Velocity (m/s)	3.145	3.145	3.145	3.145	3.145
16	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
17	Length (m)	3.497e+004	3.497e+004	3.498e+004	3.498e+004	3.498e+004
18	Bulk Density (kg/m3)	93.74	93.74	93.74	93.73	93.73
19	Bulk Velocity (m/s)	3.145	3.145	3.145	3.146	3.146
20	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
21	Length (m)	3.499e+004	3.499e+004	3.500e+004	3.500e+004	3.501e+004
22	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
23	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
24	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
25	Length (m)	3.502e+004	3.502e+004	3.503e+004	3.503e+004	3.504e+004
26	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
27	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
28	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
29	Length (m)	3.504e+004	3.505e+004	3.505e+004	3.506e+004	3.506e+004
30	Bulk Density (kg/m3)	93.73	93.73	93.73	93.73	93.73
31	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
32	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
33	Length (m)	3.507e+004	3.507e+004	3.508e+004	3.508e+004	3.509e+004
34	Bulk Density (kg/m3)	93.73	93.73	93.73	93.72	93.72
35	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
36	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
37	Length (m)	3.509e+004	3.510e+004	3.510e+004	3.511e+004	3.511e+004
38	Bulk Density (kg/m3)	93.72	93.72	93.72	93.72	93.72
39	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
40	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
41	Length (m)	3.512e+004	3.512e+004	3.513e+004	3.513e+004	3.514e+004
42	Bulk Density (kg/m3)	93.72	93.72	93.72	93.72	93.72
43	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
44	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
45	Length (m)	3.514e+004	3.515e+004	3.515e+004	3.516e+004	3.516e+004
46	Bulk Density (kg/m3)	93.72	93.72	93.72	93.72	93.72
47	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
48	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
49	Length (m)	3.517e+004	3.517e+004	3.518e+004	3.518e+004	3.519e+004
50	Bulk Density (kg/m3)	93.72	93.72	93.72	93.72	93.72
51	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
52	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
53	Length (m)	3.519e+004	3.520e+004	3.520e+004	3.521e+004	3.521e+004
54	Bulk Density (kg/m3)	93.72	93.72	93.72	93.72	93.72
55	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
56	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
57	Length (m)	3.522e+004	3.522e+004	3.523e+004	3.523e+004	3.524e+004
58	Bulk Density (kg/m3)	93.72	93.72	93.72	93.72	93.72
59	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
60	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
61	Length (m)	3.524e+004	3.525e+004	3.525e+004	3.526e+004	3.526e+004
62	Bulk Density (kg/m3)	93.72	93.72	93.72	93.72	93.72
63	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
64	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
65	Length (m)	3.527e+004	3.527e+004	3.528e+004	3.528e+004	3.529e+004
66	Bulk Density (kg/m3)	93.72	93.72	93.72	93.72	93.72
67	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
68	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 660 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.529e+004	3.530e+004	3.530e+004	3.531e+004	3.531e+004
14	Bulk Density (kg/m3)	93.72	93.72	93.72	93.72	93.72
15	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
16	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
17	Length (m)	3.532e+004	3.532e+004	3.533e+004	3.533e+004	3.534e+004
18	Bulk Density (kg/m3)	93.72	93.72	93.72	93.72	93.72
19	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
20	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
21	Length (m)	3.534e+004	3.535e+004	3.535e+004	3.536e+004	3.536e+004
22	Bulk Density (kg/m3)	93.72	93.73	93.73	93.73	93.74
23	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.145
24	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
25	Length (m)	3.537e+004	3.537e+004	3.538e+004	3.538e+004	3.539e+004
26	Bulk Density (kg/m3)	93.74	93.73	93.73	93.72	93.71
27	Bulk Velocity (m/s)	3.145	3.146	3.146	3.146	3.146
28	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
29	Length (m)	3.539e+004	3.540e+004	3.540e+004	3.541e+004	3.541e+004
30	Bulk Density (kg/m3)	93.71	93.70	93.70	93.70	93.70
31	Bulk Velocity (m/s)	3.147	3.147	3.147	3.147	3.147
32	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
33	Length (m)	3.542e+004	3.542e+004	3.543e+004	3.543e+004	3.544e+004
34	Bulk Density (kg/m3)	93.70	93.70	93.70	93.71	93.71
35	Bulk Velocity (m/s)	3.147	3.147	3.147	3.147	3.146
36	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
37	Length (m)	3.544e+004	3.545e+004	3.545e+004	3.546e+004	3.546e+004
38	Bulk Density (kg/m3)	93.71	93.71	93.71	93.71	93.71
39	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
40	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
41	Length (m)	3.547e+004	3.547e+004	3.548e+004	3.548e+004	3.549e+004
42	Bulk Density (kg/m3)	93.71	93.71	93.71	93.71	93.71
43	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
44	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
45	Length (m)	3.549e+004	3.550e+004	3.550e+004	3.551e+004	3.551e+004
46	Bulk Density (kg/m3)	93.71	93.71	93.71	93.71	93.71
47	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
48	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
49	Length (m)	3.552e+004	3.552e+004	3.553e+004	3.553e+004	3.554e+004
50	Bulk Density (kg/m3)	93.71	93.71	93.71	93.71	93.71
51	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
52	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
53	Length (m)	3.554e+004	3.555e+004	3.555e+004	3.556e+004	3.556e+004
54	Bulk Density (kg/m3)	93.71	93.71	93.71	93.71	93.71
55	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
56	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
57	Length (m)	3.557e+004	3.557e+004	3.557e+004	3.558e+004	3.558e+004
58	Bulk Density (kg/m3)	93.71	93.71	93.71	93.71	93.71
59	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.146
60	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
61	Length (m)	3.559e+004	3.559e+004	3.559e+004	3.560e+004	3.560e+004
62	Bulk Density (kg/m3)	93.71	93.71	93.71	93.71	93.71
63	Bulk Velocity (m/s)	3.146	3.146	3.146	3.146	3.147
64	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
65	Length (m)	3.560e+004	3.561e+004	3.561e+004	3.562e+004	3.562e+004
66	Bulk Density (kg/m3)	93.71	93.71	93.70	93.70	93.70
67	Bulk Velocity (m/s)	3.147	3.147	3.147	3.147	3.147
68	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 661 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.563e+004	3.563e+004	3.564e+004	3.564e+004	3.565e+004
14	Bulk Density (kg/m3)	93.70	93.70	93.70	93.70	93.70
15	Bulk Velocity (m/s)	3.147	3.147	3.147	3.147	3.147
16	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
17	Length (m)	3.565e+004	3.566e+004	3.566e+004	3.567e+004	3.567e+004
18	Bulk Density (kg/m3)	93.70	93.70	93.70	93.70	93.70
19	Bulk Velocity (m/s)	3.147	3.147	3.147	3.147	3.147
20	Erosion Velocity (m/s)	12.60	12.60	12.60	12.60	12.60
21	Length (m)	3.568e+004				
22	Bulk Density (kg/m3)	93.70				
23	Bulk Velocity (m/s)	3.147				
24	Erosion Velocity (m/s)	12.60				
25	Hydrates					
26						
27	Model Options					
28						
29	Hydrate Model:			Ng & Robinson		
30	Override Ng and Robinson:			False		
31	Initial Pressure:			From a Correlation		
32	Initial Pressure User Input:			---		
33	Hydrates Table					
34						
35	Distance (m)	0.0000	5.000	10.00	15.00	20.00
36	Pressure (bar)	78.57	78.57	78.57	78.57	78.57
37	Temperature (C)	59.75	59.74	59.74	59.74	59.74
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	25.00	30.00	35.00	40.00	45.00
48	Pressure (bar)	78.57	78.57	78.57	78.57	78.57
49	Temperature (C)	59.73	59.73	59.73	59.73	59.72
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	50.00	55.00	60.00	65.00	70.00
60	Pressure (bar)	78.56	78.56	78.56	78.56	78.56
61	Temperature (C)	59.72	59.72	59.72	59.71	59.71
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 662 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	75.00	80.00	85.00	90.00	95.00
16	Pressure (bar)	78.56	78.56	78.56	78.56	78.56
17	Temperature (C)	59.71	59.71	59.71	59.70	59.70
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	100.0	105.0	110.0	115.0	120.0
28	Pressure (bar)	78.56	78.56	78.56	78.56	78.56
29	Temperature (C)	59.70	59.70	59.69	59.69	59.69
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	125.0	130.0	135.0	140.0	145.0
40	Pressure (bar)	78.56	78.56	78.56	78.56	78.56
41	Temperature (C)	59.69	59.68	59.68	59.68	59.68
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	150.0	155.0	160.0	165.0	170.0
52	Pressure (bar)	78.56	78.56	78.56	78.56	78.56
53	Temperature (C)	59.68	59.67	59.67	59.67	59.67
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	175.0	180.0	185.0	190.0	195.0
64	Pressure (bar)	78.56	78.56	78.56	78.56	78.56
65	Temperature (C)	59.66	59.66	59.66	59.66	59.65
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 663 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	200.1	205.1	210.1	215.1	220.1
20	Pressure (bar)	78.56	78.56	78.55	78.55	78.55
21	Temperature (C)	59.65	59.65	59.65	59.65	59.64
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	225.1	230.1	235.1	240.1	245.1
32	Pressure (bar)	78.55	78.55	78.55	78.55	78.55
33	Temperature (C)	59.64	59.64	59.64	59.63	59.63
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	250.1	255.1	260.1	265.1	270.1
44	Pressure (bar)	78.55	78.55	78.55	78.55	78.55
45	Temperature (C)	59.63	59.63	59.62	59.62	59.62
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	275.1	280.1	285.1	290.1	295.1
56	Pressure (bar)	78.55	78.55	78.55	78.55	78.55
57	Temperature (C)	59.62	59.62	59.61	59.61	59.61
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	300.1	305.1	310.1	315.1	320.1
68	Pressure (bar)	78.55	78.55	78.55	78.55	78.55
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 664 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	59.61	59.60	59.60	59.60	59.60
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	325.1	330.1	335.1	340.1	345.1
24	Pressure (bar)	78.55	78.55	78.55	78.55	78.55
25	Temperature (C)	59.59	59.59	59.59	59.59	59.59
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	350.1	355.1	360.1	365.1	370.1
36	Pressure (bar)	78.55	78.55	78.55	78.55	78.54
37	Temperature (C)	59.59	59.58	59.58	59.58	59.57
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	375.1	380.0	385.0	390.0	395.0
48	Pressure (bar)	78.54	78.54	78.54	78.53	78.53
49	Temperature (C)	59.57	59.57	59.57	59.56	59.56
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	400.0	405.0	410.0	415.0	420.0
60	Pressure (bar)	78.53	78.53	78.53	78.53	78.53
61	Temperature (C)	59.56	59.55	59.55	59.55	59.55
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	425.0	430.0	435.0	440.0	445.0
16	Pressure (bar)	78.53	78.53	78.53	78.53	78.53
17	Temperature (C)	59.55	59.54	59.54	59.54	59.54
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	450.0	455.0	460.0	465.0	470.0
28	Pressure (bar)	78.53	78.53	78.53	78.53	78.53
29	Temperature (C)	59.53	59.53	59.53	59.53	59.52
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	475.0	480.0	485.0	490.0	495.0
40	Pressure (bar)	78.53	78.53	78.53	78.52	78.52
41	Temperature (C)	59.52	59.52	59.52	59.52	59.51
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	500.0	505.0	510.0	515.0	520.0
52	Pressure (bar)	78.52	78.52	78.52	78.52	78.52
53	Temperature (C)	59.51	59.51	59.51	59.50	59.50
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	525.0	530.0	535.0	540.0	545.0
64	Pressure (bar)	78.52	78.52	78.52	78.52	78.52
65	Temperature (C)	59.50	59.50	59.49	59.49	59.49
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	550.0	555.0	560.0	565.0	570.0
20	Pressure (bar)	78.52	78.52	78.52	78.52	78.52
21	Temperature (C)	59.49	59.49	59.48	59.48	59.48
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	575.0	580.0	585.0	590.0	595.0
32	Pressure (bar)	78.52	78.52	78.52	78.52	78.52
33	Temperature (C)	59.48	59.47	59.47	59.47	59.47
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	600.0	605.0	610.0	615.0	620.0
44	Pressure (bar)	78.52	78.52	78.52	78.51	78.51
45	Temperature (C)	59.46	59.46	59.46	59.46	59.45
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	625.0	630.0	635.0	640.0	645.0
56	Pressure (bar)	78.51	78.51	78.51	78.51	78.51
57	Temperature (C)	59.45	59.45	59.45	59.45	59.44
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	650.0	655.0	660.0	665.0	670.0
68	Pressure (bar)	78.51	78.51	78.51	78.51	78.51
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:13:43 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	59.44	59.44	59.44	59.43	59.43
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	675.0	682.0	689.0	696.0	703.0
24	Pressure (bar)	78.51	78.51	78.51	78.51	78.51
25	Temperature (C)	59.43	59.43	59.42	59.42	59.42
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	710.0	712.0	714.0	716.0	718.0
36	Pressure (bar)	78.51	78.51	78.51	78.51	78.51
37	Temperature (C)	59.41	59.41	59.41	59.41	59.41
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	720.0	722.0	724.0	726.0	728.0
48	Pressure (bar)	78.51	78.51	78.51	78.50	78.50
49	Temperature (C)	59.41	59.41	59.41	59.41	59.40
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.06	17.05	17.05	17.05	17.05
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	730.0	732.0	734.0	736.0	738.0
60	Pressure (bar)	78.50	78.50	78.50	78.50	78.50
61	Temperature (C)	59.40	59.40	59.40	59.40	59.40
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 668 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	740.0	742.0	744.0	746.0	748.0
16	Pressure (bar)	78.50	78.50	78.50	78.50	78.50
17	Temperature (C)	59.40	59.40	59.40	59.40	59.40
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	750.0	755.0	760.0	765.0	770.0
28	Pressure (bar)	78.50	78.50	78.50	78.50	78.50
29	Temperature (C)	59.39	59.39	59.39	59.39	59.39
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	775.0	780.0	785.0	790.1	795.1
40	Pressure (bar)	78.50	78.50	78.50	78.50	78.50
41	Temperature (C)	59.38	59.38	59.38	59.38	59.37
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	800.1	805.1	810.1	815.1	820.1
52	Pressure (bar)	78.50	78.50	78.50	78.50	78.50
53	Temperature (C)	59.37	59.37	59.37	59.36	59.36
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	825.1	830.1	835.1	840.1	845.1
64	Pressure (bar)	78.50	78.50	78.50	78.50	78.50
65	Temperature (C)	59.36	59.36	59.36	59.35	59.35
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 669 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	850.1	855.1	860.1	865.1	870.1
20	Pressure (bar)	78.50	78.50	78.50	78.50	78.49
21	Temperature (C)	59.35	59.35	59.35	59.34	59.34
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	875.1	880.1	885.1	890.1	895.1
32	Pressure (bar)	78.49	78.49	78.49	78.49	78.49
33	Temperature (C)	59.34	59.33	59.33	59.33	59.33
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	900.1	905.1	910.1	915.1	920.1
44	Pressure (bar)	78.49	78.49	78.49	78.49	78.49
45	Temperature (C)	59.33	59.32	59.32	59.32	59.32
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	925.1	930.1	935.1	940.1	945.1
56	Pressure (bar)	78.49	78.49	78.49	78.49	78.49
57	Temperature (C)	59.31	59.31	59.31	59.31	59.30
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	950.1	955.1	960.1	965.1	970.1
68	Pressure (bar)	78.49	78.49	78.49	78.49	78.49
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	59.30	59.30	59.30	59.30	59.29
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	975.1	980.1	985.1	990.1	995.1
24	Pressure (bar)	78.49	78.49	78.49	78.48	78.48
25	Temperature (C)	59.29	59.29	59.29	59.28	59.28
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1000	1005	1010	1015	1020
36	Pressure (bar)	78.48	78.48	78.48	78.48	78.48
37	Temperature (C)	59.28	59.28	59.27	59.27	59.27
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1025	1030	1035	1040	1045
48	Pressure (bar)	78.48	78.48	78.48	78.48	78.48
49	Temperature (C)	59.27	59.26	59.26	59.26	59.26
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1050	1055	1060	1065	1070
60	Pressure (bar)	78.48	78.48	78.48	78.48	78.48
61	Temperature (C)	59.26	59.26	59.25	59.25	59.25
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1075	1080	1085	1090	1095
16	Pressure (bar)	78.48	78.48	78.48	78.48	78.47
17	Temperature (C)	59.24	59.24	59.24	59.24	59.24
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1100	1105	1110	1115	1120
28	Pressure (bar)	78.47	78.47	78.47	78.47	78.47
29	Temperature (C)	59.23	59.23	59.23	59.23	59.22
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1125	1130	1135	1140	1145
40	Pressure (bar)	78.47	78.47	78.47	78.47	78.47
41	Temperature (C)	59.22	59.22	59.22	59.21	59.21
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1150	1155	1160	1165	1170
52	Pressure (bar)	78.47	78.47	78.47	78.47	78.47
53	Temperature (C)	59.21	59.21	59.20	59.20	59.20
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1175	1180	1185	1190	1195
64	Pressure (bar)	78.47	78.47	78.47	78.47	78.46
65	Temperature (C)	59.20	59.20	59.19	59.19	59.19
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1200	1205	1210	1215	1220
20	Pressure (bar)	78.46	78.46	78.46	78.46	78.46
21	Temperature (C)	59.19	59.18	59.18	59.18	59.18
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1225	1230	1235	1240	1245
32	Pressure (bar)	78.46	78.46	78.46	78.46	78.46
33	Temperature (C)	59.18	59.17	59.17	59.17	59.17
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1250	1255	1260	1265	1270
44	Pressure (bar)	78.46	78.46	78.46	78.46	78.46
45	Temperature (C)	59.16	59.16	59.16	59.16	59.15
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1275	1280	1285	1290	1295
56	Pressure (bar)	78.46	78.46	78.46	78.46	78.46
57	Temperature (C)	59.15	59.15	59.15	59.15	59.14
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1300	1305	1310	1315	1320
68	Pressure (bar)	78.46	78.46	78.46	78.46	78.46
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 673 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	59.14	59.14	59.14	59.13	59.13
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1325	1330	1335	1340	1345
24	Pressure (bar)	78.46	78.46	78.45	78.45	78.45
25	Temperature (C)	59.13	59.13	59.12	59.12	59.12
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1350	1355	1360	1365	1370
36	Pressure (bar)	78.45	78.45	78.45	78.46	78.46
37	Temperature (C)	59.12	59.12	59.11	59.11	59.11
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1375	1380	1385	1390	1395
48	Pressure (bar)	78.46	78.47	78.47	78.48	78.48
49	Temperature (C)	59.11	59.11	59.11	59.11	59.11
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.05
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1400	1405	1410	1415	1420
60	Pressure (bar)	78.49	78.50	78.50	78.50	78.51
61	Temperature (C)	59.12	59.12	59.12	59.12	59.11
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.05	17.05	17.05	17.05	17.06
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1425	1430	1435	1440	1445
16	Pressure (bar)	78.51	78.51	78.52	78.52	78.52
17	Temperature (C)	59.11	59.11	59.11	59.11	59.11
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1450	1455	1460	1465	1470
28	Pressure (bar)	78.52	78.52	78.51	78.51	78.51
29	Temperature (C)	59.11	59.10	59.10	59.10	59.09
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.05
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1475	1480	1485	1490	1495
40	Pressure (bar)	78.50	78.51	78.51	78.51	78.51
41	Temperature (C)	59.09	59.09	59.09	59.09	59.09
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.05	17.05	17.06	17.06	17.06
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1500	1505	1510	1515	1520
52	Pressure (bar)	78.52	78.52	78.53	78.54	78.54
53	Temperature (C)	59.09	59.09	59.10	59.10	59.11
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.06	17.06	17.06	17.06	17.06
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1525	1530	1535	1540	1545
64	Pressure (bar)	78.55	78.39	78.23	78.07	77.91
65	Temperature (C)	59.11	59.07	59.03	58.99	58.94
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 675 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.06	17.05	17.03	17.02	17.01
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1550	1555	1560	1565	1570
20	Pressure (bar)	77.75	77.74	77.74	77.73	77.73
21	Temperature (C)	58.90	58.90	58.89	58.89	58.89
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.00	16.99	16.99	16.99	16.99
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1575	1580	1585	1590	1595
32	Pressure (bar)	77.73	77.73	77.73	77.73	77.73
33	Temperature (C)	58.88	58.88	58.88	58.88	58.88
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1600	1605	1610	1615	1620
44	Pressure (bar)	77.73	77.73	77.73	77.73	77.73
45	Temperature (C)	58.87	58.87	58.87	58.87	58.87
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1625	1630	1635	1640	1645
56	Pressure (bar)	77.73	77.73	77.73	77.73	77.73
57	Temperature (C)	58.86	58.86	58.86	58.86	58.86
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1650	1655	1660	1665	1670
68	Pressure (bar)	77.73	77.73	77.73	77.73	77.73
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	58.85	58.85	58.85	58.85	58.84
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1675	1680	1685	1690	1695
24	Pressure (bar)	77.73	77.73	77.73	77.73	77.73
25	Temperature (C)	58.84	58.84	58.84	58.84	58.83
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1700	1705	1710	1715	1720
36	Pressure (bar)	77.73	77.73	77.73	77.73	77.73
37	Temperature (C)	58.83	58.83	58.83	58.83	58.82
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1725	1730	1735	1740	1745
48	Pressure (bar)	77.73	77.73	77.73	77.73	77.73
49	Temperature (C)	58.82	58.82	58.82	58.81	58.81
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1750	1755	1760	1765	1770
60	Pressure (bar)	77.73	77.73	77.73	77.73	77.73
61	Temperature (C)	58.81	58.81	58.81	58.80	58.80
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 677 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1775	1780	1785	1790	1795
16	Pressure (bar)	77.72	77.72	77.72	77.72	77.72
17	Temperature (C)	58.80	58.80	58.79	58.79	58.79
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1800	1805	1810	1815	1820
28	Pressure (bar)	77.72	77.72	77.72	77.72	77.72
29	Temperature (C)	58.79	58.78	58.78	58.78	58.78
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1825	1830	1835	1840	1845
40	Pressure (bar)	77.72	77.72	77.72	77.71	77.71
41	Temperature (C)	58.77	58.77	58.77	58.77	58.76
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1850	1855	1860	1865	1870
52	Pressure (bar)	77.71	77.71	77.71	77.71	77.71
53	Temperature (C)	58.76	58.76	58.76	58.76	58.75
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1875	1880	1885	1890	1895
64	Pressure (bar)	77.71	77.71	77.71	77.71	77.71
65	Temperature (C)	58.75	58.75	58.75	58.74	58.74
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1900	1905	1910	1915	1920
20	Pressure (bar)	77.71	77.71	77.71	77.71	77.71
21	Temperature (C)	58.74	58.74	58.73	58.73	58.73
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1925	1930	1935	1940	1945
32	Pressure (bar)	77.71	77.71	77.71	77.70	77.70
33	Temperature (C)	58.73	58.73	58.72	58.72	58.72
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1950	1955	1960	1965	1970
44	Pressure (bar)	77.70	77.70	77.70	77.70	77.70
45	Temperature (C)	58.72	58.71	58.71	58.71	58.71
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1975	1980	1985	1990	1995
56	Pressure (bar)	77.70	77.70	77.70	77.70	77.70
57	Temperature (C)	58.70	58.70	58.70	58.70	58.70
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2000	2005	2010	2015	2020
68	Pressure (bar)	77.70	77.70	77.70	77.70	77.70
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	58.69	58.69	58.69	58.69	58.69
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2025	2030	2035	2040	2045
24	Pressure (bar)	77.70	77.70	77.70	77.70	77.70
25	Temperature (C)	58.68	58.68	58.68	58.68	58.67
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2050	2055	2060	2065	2070
36	Pressure (bar)	77.70	77.70	77.70	77.70	77.70
37	Temperature (C)	58.67	58.67	58.67	58.66	58.66
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2075	2080	2085	2090	2095
48	Pressure (bar)	77.70	77.70	77.69	77.69	77.69
49	Temperature (C)	58.66	58.66	58.66	58.65	58.65
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2100	2105	2110	2115	2120
60	Pressure (bar)	77.69	77.69	77.69	77.69	77.69
61	Temperature (C)	58.65	58.65	58.64	58.64	58.64
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2125	2130	2135	2140	2145
16	Pressure (bar)	77.69	77.69	77.69	77.69	77.69
17	Temperature (C)	58.64	58.63	58.63	58.63	58.63
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2150	2155	2160	2165	2170
28	Pressure (bar)	77.69	77.69	77.69	77.69	77.69
29	Temperature (C)	58.63	58.62	58.62	58.62	58.62
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2175	2180	2185	2191	2196
40	Pressure (bar)	77.69	77.69	77.69	77.69	77.69
41	Temperature (C)	58.61	58.61	58.61	58.61	58.61
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2201	2206	2211	2216	2221
52	Pressure (bar)	77.68	77.68	77.68	77.68	77.68
53	Temperature (C)	58.60	58.60	58.60	58.60	58.59
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2226	2231	2236	2241	2246
64	Pressure (bar)	77.68	77.68	77.68	77.68	77.68
65	Temperature (C)	58.59	58.59	58.59	58.59	58.58
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2251	2256	2261	2266	2271
20	Pressure (bar)	77.68	77.68	77.68	77.68	77.68
21	Temperature (C)	58.58	58.58	58.58	58.57	58.57
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2276	2281	2286	2291	2296
32	Pressure (bar)	77.68	77.68	77.68	77.68	77.68
33	Temperature (C)	58.57	58.57	58.56	58.56	58.56
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2301	2306	2311	2316	2321
44	Pressure (bar)	77.68	77.68	77.68	77.68	77.68
45	Temperature (C)	58.56	58.56	58.55	58.55	58.55
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2326	2331	2336	2341	2346
56	Pressure (bar)	77.68	77.68	77.67	77.67	77.67
57	Temperature (C)	58.55	58.54	58.54	58.54	58.54
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2351	2356	2361	2366	2371
68	Pressure (bar)	77.67	77.67	77.67	77.67	77.67
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 682 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	58.54	58.53	58.53	58.53	58.53
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2376	2381	2386	2391	2396
24	Pressure (bar)	77.67	77.67	77.67	77.67	77.67
25	Temperature (C)	58.52	58.52	58.52	58.52	58.52
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2401	2406	2411	2416	2421
36	Pressure (bar)	77.67	77.67	77.67	77.67	77.67
37	Temperature (C)	58.51	58.51	58.51	58.51	58.50
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2426	2431	2436	2441	2446
48	Pressure (bar)	77.67	77.67	77.67	77.67	77.67
49	Temperature (C)	58.50	58.50	58.50	58.49	58.49
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2451	2456	2461	2466	2471
60	Pressure (bar)	77.67	77.66	77.66	77.66	77.66
61	Temperature (C)	58.49	58.49	58.49	58.48	58.48
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2476	2481	2486	2491	2496
16	Pressure (bar)	77.66	77.66	77.66	77.66	77.66
17	Temperature (C)	58.48	58.48	58.47	58.47	58.47
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2501	2506	2511	2516	2521
28	Pressure (bar)	77.66	77.66	77.66	77.66	77.66
29	Temperature (C)	58.47	58.46	58.46	58.46	58.46
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2526	2531	2536	2541	2546
40	Pressure (bar)	77.66	77.66	77.66	77.66	77.66
41	Temperature (C)	58.46	58.45	58.45	58.45	58.45
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2551	2556	2561	2566	2571
52	Pressure (bar)	77.66	77.66	77.66	77.66	77.66
53	Temperature (C)	58.45	58.44	58.44	58.44	58.44
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2576	2581	2586	2591	2596
64	Pressure (bar)	77.66	77.66	77.66	77.66	77.66
65	Temperature (C)	58.44	58.43	58.43	58.43	58.43
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 684 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2601	2606	2611	2616	2621
20	Pressure (bar)	77.66	77.65	77.65	77.65	77.65
21	Temperature (C)	58.42	58.42	58.42	58.42	58.41
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2626	2631	2636	2641	2646
32	Pressure (bar)	77.65	77.65	77.65	77.65	77.65
33	Temperature (C)	58.41	58.41	58.41	58.41	58.40
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2651	2656	2661	2666	2671
44	Pressure (bar)	77.65	77.65	77.65	77.65	77.65
45	Temperature (C)	58.40	58.40	58.40	58.39	58.39
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2676	2681	2686	2691	2696
56	Pressure (bar)	77.65	77.65	77.65	77.65	77.65
57	Temperature (C)	58.39	58.39	58.39	58.38	58.38
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2701	2706	2711	2716	2721
68	Pressure (bar)	77.65	77.65	77.65	77.65	77.65
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 685 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	58.38	58.38	58.37	58.37	58.37
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2726	2731	2736	2741	2746
24	Pressure (bar)	77.65	77.65	77.64	77.64	77.64
25	Temperature (C)	58.37	58.36	58.36	58.36	58.36
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2751	2756	2761	2766	2771
36	Pressure (bar)	77.64	77.64	77.64	77.64	77.64
37	Temperature (C)	58.36	58.35	58.35	58.35	58.35
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2776	2781	2786	2791	2796
48	Pressure (bar)	77.64	77.64	77.64	77.64	77.64
49	Temperature (C)	58.34	58.34	58.34	58.34	58.34
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2801	2806	2811	2816	2821
60	Pressure (bar)	77.64	77.64	77.64	77.64	77.64
61	Temperature (C)	58.33	58.33	58.33	58.33	58.33
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 686 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2826	2831	2836	2841	2846
16	Pressure (bar)	77.64	77.64	77.64	77.64	77.64
17	Temperature (C)	58.32	58.32	58.32	58.32	58.31
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2851	2856	2861	2866	2871
28	Pressure (bar)	77.64	77.63	77.63	77.63	77.63
29	Temperature (C)	58.31	58.31	58.31	58.30	58.30
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2876	2881	2886	2891	2896
40	Pressure (bar)	77.63	77.63	77.63	77.63	77.63
41	Temperature (C)	58.30	58.30	58.30	58.29	58.29
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2901	2906	2911	2916	2921
52	Pressure (bar)	77.64	77.64	77.64	77.63	77.63
53	Temperature (C)	58.29	58.29	58.29	58.28	58.28
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2926	2931	2936	2941	2946
64	Pressure (bar)	77.63	77.63	77.63	77.63	77.63
65	Temperature (C)	58.28	58.28	58.27	58.27	58.27
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2951	2956	2961	2966	2971
20	Pressure (bar)	77.62	77.63	77.63	77.64	77.64
21	Temperature (C)	58.27	58.27	58.26	58.26	58.26
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2976	2981	2986	2991	2996
32	Pressure (bar)	77.64	77.64	77.64	77.64	77.64
33	Temperature (C)	58.26	58.26	58.26	58.26	58.25
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3001	3006	3011	3016	3021
44	Pressure (bar)	77.64	77.65	77.67	77.68	77.70
45	Temperature (C)	58.25	58.26	58.26	58.27	58.27
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3026	3031	3036	3041	3046
56	Pressure (bar)	77.71	77.72	77.72	77.73	77.73
57	Temperature (C)	58.28	58.28	58.28	58.28	58.28
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.99	16.99	16.99	16.99	16.99
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3051	3056	3061	3066	3071
68	Pressure (bar)	77.74	77.74	77.74	77.75	77.75
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	58.28	58.28	58.28	58.28	58.28
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.99	16.99	16.99	17.00	17.00
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3076	3081	3086	3091	3096
24	Pressure (bar)	77.75	77.60	77.44	77.29	77.13
25	Temperature (C)	58.28	58.24	58.20	58.16	58.12
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.00	16.98	16.97	16.96	16.95
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3101	3106	3111	3116	3121
36	Pressure (bar)	76.97	76.97	76.97	76.97	76.96
37	Temperature (C)	58.08	58.07	58.07	58.07	58.06
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3126	3131	3136	3141	3146
48	Pressure (bar)	76.96	76.96	76.96	76.96	76.96
49	Temperature (C)	58.06	58.06	58.06	58.05	58.05
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3151	3156	3161	3166	3171
60	Pressure (bar)	76.96	76.96	76.96	76.96	76.96
61	Temperature (C)	58.05	58.05	58.05	58.04	58.04
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3176	3181	3186	3192	3197
16	Pressure (bar)	76.96	76.96	76.96	76.96	76.96
17	Temperature (C)	58.04	58.04	58.03	58.03	58.03
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3202	3207	3212	3217	3222
28	Pressure (bar)	76.95	76.95	76.95	76.95	76.95
29	Temperature (C)	58.03	58.02	58.02	58.02	58.02
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3227	3232	3237	3242	3247
40	Pressure (bar)	76.95	76.95	76.95	76.95	76.95
41	Temperature (C)	58.02	58.01	58.01	58.01	58.01
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3252	3257	3262	3267	3272
52	Pressure (bar)	76.95	76.95	76.95	76.95	76.95
53	Temperature (C)	58.01	58.00	58.00	58.00	58.00
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3277	3282	3287	3292	3297
64	Pressure (bar)	76.95	76.95	76.95	76.95	76.95
65	Temperature (C)	57.99	57.99	57.99	57.99	57.99
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 690 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3302	3307	3312	3317	3322
20	Pressure (bar)	76.95	76.95	76.95	76.95	76.95
21	Temperature (C)	57.98	57.98	57.98	57.98	57.97
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3327	3332	3337	3342	3347
32	Pressure (bar)	76.95	76.95	76.95	76.95	76.94
33	Temperature (C)	57.97	57.97	57.97	57.97	57.96
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3352	3357	3362	3367	3372
44	Pressure (bar)	76.94	76.94	76.94	76.94	76.94
45	Temperature (C)	57.96	57.96	57.96	57.95	57.95
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3377	3382	3387	3392	3396
56	Pressure (bar)	76.94	76.94	76.94	76.94	76.94
57	Temperature (C)	57.95	57.95	57.94	57.94	57.94
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3401	3406	3411	3416	3421
68	Pressure (bar)	76.94	76.94	76.94	76.94	76.94
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 691 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	57.94	57.94	57.93	57.93	57.93
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3426	3431	3436	3441	3446
24	Pressure (bar)	76.94	76.94	76.94	76.94	76.94
25	Temperature (C)	57.93	57.93	57.92	57.92	57.92
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3451	3456	3461	3466	3471
36	Pressure (bar)	76.94	76.93	76.93	76.93	76.93
37	Temperature (C)	57.92	57.91	57.91	57.91	57.91
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3476	3481	3486	3491	3496
48	Pressure (bar)	76.93	76.93	76.93	76.93	76.93
49	Temperature (C)	57.91	57.90	57.90	57.90	57.90
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3501	3506	3511	3516	3521
60	Pressure (bar)	76.93	76.93	76.93	76.93	76.93
61	Temperature (C)	57.89	57.89	57.89	57.89	57.89
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3526	3531	3536	3541	3546
16	Pressure (bar)	76.93	76.93	76.93	76.93	76.93
17	Temperature (C)	57.88	57.88	57.88	57.88	57.88
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3551	3556	3561	3566	3571
28	Pressure (bar)	76.93	76.93	76.93	76.93	76.93
29	Temperature (C)	57.87	57.87	57.87	57.87	57.86
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3576	3581	3586	3591	3596
40	Pressure (bar)	76.93	76.93	76.93	76.93	76.93
41	Temperature (C)	57.86	57.86	57.86	57.86	57.85
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3601	3606	3611	3616	3621
52	Pressure (bar)	76.93	76.93	76.93	76.93	76.93
53	Temperature (C)	57.85	57.85	57.85	57.84	57.84
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3626	3631	3636	3641	3646
64	Pressure (bar)	76.93	76.92	76.92	76.92	76.92
65	Temperature (C)	57.84	57.84	57.84	57.83	57.83
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 693 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3651	3656	3661	3666	3671
20	Pressure (bar)	76.92	76.92	76.92	76.92	76.92
21	Temperature (C)	57.83	57.83	57.82	57.82	57.82
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3676	3681	3686	3691	3696
32	Pressure (bar)	76.92	76.92	76.92	76.92	76.92
33	Temperature (C)	57.82	57.82	57.81	57.81	57.81
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3701	3706	3711	3716	3721
44	Pressure (bar)	76.92	76.92	76.92	76.92	76.92
45	Temperature (C)	57.81	57.81	57.80	57.80	57.80
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3726	3731	3736	3741	3746
56	Pressure (bar)	76.92	76.92	76.92	76.92	76.92
57	Temperature (C)	57.80	57.79	57.79	57.79	57.79
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3751	3756	3761	3766	3771
68	Pressure (bar)	76.91	76.91	76.91	76.91	76.91
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 694 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	57.79	57.78	57.78	57.78	57.78
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3776	3781	3787	3792	3797
24	Pressure (bar)	76.91	76.91	76.91	76.91	76.91
25	Temperature (C)	57.77	57.77	57.77	57.77	57.77
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3802	3807	3812	3817	3822
36	Pressure (bar)	76.91	76.91	76.91	76.91	76.91
37	Temperature (C)	57.76	57.76	57.76	57.76	57.75
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3827	3832	3837	3842	3847
48	Pressure (bar)	76.91	76.91	76.91	76.91	76.91
49	Temperature (C)	57.75	57.75	57.75	57.75	57.74
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3852	3857	3862	3867	3872
60	Pressure (bar)	76.91	76.91	76.91	76.91	76.91
61	Temperature (C)	57.74	57.74	57.74	57.73	57.73
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3877	3882	3887	3892	3897
16	Pressure (bar)	76.90	76.90	76.90	76.90	76.90
17	Temperature (C)	57.73	57.73	57.73	57.72	57.72
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3902	3907	3912	3917	3922
28	Pressure (bar)	76.90	76.90	76.90	76.90	76.90
29	Temperature (C)	57.72	57.72	57.71	57.71	57.71
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3927	3932	3937	3942	3947
40	Pressure (bar)	76.90	76.90	76.90	76.90	76.90
41	Temperature (C)	57.71	57.70	57.70	57.70	57.70
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3952	3957	3962	3967	3972
52	Pressure (bar)	76.90	76.90	76.90	76.89	76.89
53	Temperature (C)	57.70	57.69	57.69	57.69	57.69
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3977	3982	3987	3992	3997
64	Pressure (bar)	76.89	76.89	76.89	76.89	76.89
65	Temperature (C)	57.68	57.68	57.68	57.68	57.68
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	4002	4007	4012	4017	4022
20	Pressure (bar)	76.89	76.89	76.89	76.89	76.89
21	Temperature (C)	57.67	57.67	57.67	57.67	57.67
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	4027	4032	4037	4042	4047
32	Pressure (bar)	76.89	76.89	76.89	76.89	76.89
33	Temperature (C)	57.66	57.66	57.66	57.66	57.65
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	4052	4057	4062	4067	4072
44	Pressure (bar)	76.89	76.89	76.89	76.89	76.89
45	Temperature (C)	57.65	57.65	57.65	57.65	57.64
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	4077	4084	4091	4098	4105
56	Pressure (bar)	76.89	76.89	76.89	76.89	76.89
57	Temperature (C)	57.64	57.64	57.64	57.63	57.63
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	4112	4120	4127	4135	4143
68	Pressure (bar)	76.89	76.89	76.89	76.89	76.88
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	57.63	57.62	57.62	57.62	57.61
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	4150	4158	4165	4173	4181
24	Pressure (bar)	76.88	76.88	76.88	76.88	76.88
25	Temperature (C)	57.61	57.61	57.60	57.60	57.60
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	4188	4196	4203	4211	4219
36	Pressure (bar)	76.88	76.88	76.88	76.88	76.88
37	Temperature (C)	57.59	57.59	57.59	57.58	57.58
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	4226	4234	4241	4249	4257
48	Pressure (bar)	76.88	76.88	76.88	76.88	76.88
49	Temperature (C)	57.58	57.58	57.57	57.57	57.56
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	4264	4272	4279	4287	4295
60	Pressure (bar)	76.88	76.87	76.87	76.87	76.87
61	Temperature (C)	57.56	57.56	57.55	57.55	57.55
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	4302	4304	4305	4306	4308
16	Pressure (bar)	76.87	76.87	76.87	76.87	76.87
17	Temperature (C)	57.54	57.54	57.54	57.54	57.54
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	4309	4314	4319	4324	4329
28	Pressure (bar)	76.87	76.87	76.87	76.87	76.87
29	Temperature (C)	57.54	57.54	57.54	57.54	57.53
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	4334	4339	4344	4349	4354
40	Pressure (bar)	76.87	76.87	76.87	76.87	76.87
41	Temperature (C)	57.53	57.53	57.53	57.53	57.52
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	4359	4364	4369	4374	4379
52	Pressure (bar)	76.87	76.87	76.87	76.87	76.87
53	Temperature (C)	57.52	57.52	57.52	57.51	57.51
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.93	16.93	16.93	16.93	16.93
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	4384	4389	4394	4399	4404
64	Pressure (bar)	76.87	76.87	76.87	76.87	76.87
65	Temperature (C)	57.51	57.51	57.50	57.50	57.50
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 699 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.93	16.93	16.92	16.92	16.92
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	4409	4414	4419	4424	4429
20	Pressure (bar)	76.87	76.86	76.86	76.86	76.86
21	Temperature (C)	57.50	57.50	57.49	57.49	57.49
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	4434	4439	4444	4449	4454
32	Pressure (bar)	76.86	76.86	76.86	76.86	76.86
33	Temperature (C)	57.49	57.48	57.48	57.48	57.48
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	4459	4464	4469	4474	4479
44	Pressure (bar)	76.86	76.86	76.86	76.86	76.86
45	Temperature (C)	57.47	57.47	57.47	57.47	57.47
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	4484	4489	4494	4499	4504
56	Pressure (bar)	76.85	76.85	76.85	76.85	76.85
57	Temperature (C)	57.46	57.46	57.46	57.46	57.45
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	4510	4515	4520	4525	4530
68	Pressure (bar)	76.85	76.85	76.85	76.85	76.85
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	57.45	57.45	57.45	57.45	57.44
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	4535	4540	4545	4550	4555
24	Pressure (bar)	76.85	76.85	76.85	76.85	76.85
25	Temperature (C)	57.44	57.44	57.44	57.44	57.43
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	4560	4565	4570	4575	4580
36	Pressure (bar)	76.85	76.85	76.85	76.85	76.85
37	Temperature (C)	57.43	57.43	57.43	57.42	57.42
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	4585	4590	4595	4600	4605
48	Pressure (bar)	76.85	76.85	76.85	76.84	76.84
49	Temperature (C)	57.42	57.42	57.42	57.41	57.41
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	4610	4623	4636	4649	4662
60	Pressure (bar)	76.84	76.84	76.84	76.84	76.84
61	Temperature (C)	57.41	57.40	57.40	57.39	57.39
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	4675	4677	4679	4681	4683
16	Pressure (bar)	76.84	76.84	76.84	76.84	76.84
17	Temperature (C)	57.38	57.38	57.38	57.38	57.38
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	4685	4690	4695	4700	4705
28	Pressure (bar)	76.84	76.84	76.84	76.84	76.84
29	Temperature (C)	57.38	57.37	57.37	57.37	57.37
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	4710	4715	4720	4725	4730
40	Pressure (bar)	76.84	76.83	76.83	76.83	76.83
41	Temperature (C)	57.36	57.36	57.36	57.36	57.36
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	4735	4740	4745	4750	4755
52	Pressure (bar)	76.83	76.83	76.83	76.83	76.83
53	Temperature (C)	57.35	57.35	57.35	57.35	57.34
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	4760	4765	4770	4775	4780
64	Pressure (bar)	76.83	76.83	76.83	76.83	76.83
65	Temperature (C)	57.34	57.34	57.34	57.34	57.33
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	4785	4790	4795	4800	4805
20	Pressure (bar)	76.83	76.83	76.83	76.83	76.83
21	Temperature (C)	57.33	57.33	57.33	57.33	57.32
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	4810	4815	4820	4825	4830
32	Pressure (bar)	76.83	76.83	76.83	76.83	76.83
33	Temperature (C)	57.32	57.32	57.32	57.32	57.31
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	4835	4840	4845	4850	4855
44	Pressure (bar)	76.83	76.83	76.83	76.83	76.83
45	Temperature (C)	57.31	57.31	57.31	57.30	57.30
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	4860	4865	4870	4875	4880
56	Pressure (bar)	76.83	76.82	76.82	76.82	76.82
57	Temperature (C)	57.30	57.30	57.30	57.29	57.29
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	4885	4890	4895	4900	4905
68	Pressure (bar)	76.82	76.82	76.82	76.82	76.82
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	57.29	57.29	57.29	57.28	57.28
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	4910	4915	4920	4925	4930
24	Pressure (bar)	76.82	76.82	76.82	76.82	76.82
25	Temperature (C)	57.28	57.28	57.27	57.27	57.27
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	4935	4940	4945	4950	4955
36	Pressure (bar)	76.82	76.82	76.82	76.82	76.82
37	Temperature (C)	57.27	57.27	57.26	57.26	57.26
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	4960	4965	4970	4975	4980
48	Pressure (bar)	76.82	76.82	76.82	76.82	76.82
49	Temperature (C)	57.26	57.26	57.25	57.25	57.25
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	4985	4990	4995	5000	5005
60	Pressure (bar)	76.82	76.82	76.82	76.82	76.82
61	Temperature (C)	57.25	57.24	57.24	57.24	57.24
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	5010	5015	5020	5025	5030
16	Pressure (bar)	76.81	76.81	76.81	76.81	76.81
17	Temperature (C)	57.24	57.23	57.23	57.23	57.23
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	5035	5040	5045	5050	5055
28	Pressure (bar)	76.81	76.81	76.81	76.81	76.81
29	Temperature (C)	57.23	57.22	57.22	57.22	57.22
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	5060	5065	5070	5075	5080
40	Pressure (bar)	76.81	76.81	76.81	76.81	76.81
41	Temperature (C)	57.21	57.21	57.21	57.21	57.21
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	5085	5090	5095	5100	5105
52	Pressure (bar)	76.81	76.81	76.81	76.81	76.81
53	Temperature (C)	57.20	57.20	57.20	57.20	57.19
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	5110	5115	5120	5125	5130
64	Pressure (bar)	76.80	76.80	76.80	76.80	76.80
65	Temperature (C)	57.19	57.19	57.19	57.19	57.18
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	5135	5140	5145	5150	5155
20	Pressure (bar)	76.80	76.80	76.80	76.80	76.80
21	Temperature (C)	57.18	57.18	57.18	57.17	57.17
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	5160	5165	5170	5175	5180
32	Pressure (bar)	76.80	76.80	76.80	76.80	76.80
33	Temperature (C)	57.17	57.17	57.17	57.16	57.16
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	5185	5190	5195	5200	5205
44	Pressure (bar)	76.80	76.80	76.80	76.80	76.80
45	Temperature (C)	57.16	57.16	57.15	57.15	57.15
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	5210	5215	5220	5225	5230
56	Pressure (bar)	76.80	76.80	76.80	76.80	76.80
57	Temperature (C)	57.15	57.15	57.14	57.14	57.14
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	5235	5240	5245	5250	5255
68	Pressure (bar)	76.80	76.80	76.80	76.79	76.79
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aspentech

Epic Atlantic Limited

Case Name:

Simulation - 386MMSCFD_80barg.hsc

Unit Set:

Mixed

Date/Time:

Mon Jun 13 15:13:43 2022

Pipe Segment: ELEP-Akabel Pipeline (continued)

Hydrates

Hydrates Table

13	Temperature	(C)	57.14	57.14	57.13	57.13	57.13
14	Formation		Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type		No Types	No Types	No Types	No Types	No Types
16	Mode		Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors		Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature	(C)	16.92	16.92	16.92	16.92	16.92
19	Type at Formation Temperature		Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature		Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature		V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature		Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance	(m)	5260	5265	5270	5275	5280
24	Pressure	(bar)	76.79	76.79	76.79	76.79	76.79
25	Temperature	(C)	57.13	57.13	57.12	57.12	57.12
26	Formation		Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type		No Types	No Types	No Types	No Types	No Types
28	Mode		Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors		Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature	(C)	16.92	16.92	16.92	16.92	16.92
31	Type at Formation Temperature		Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature		Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature		V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature		Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance	(m)	5285	5290	5295	5300	5305
36	Pressure	(bar)	76.79	76.79	76.79	76.79	76.79
37	Temperature	(C)	57.12	57.12	57.11	57.11	57.11
38	Formation		Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type		No Types	No Types	No Types	No Types	No Types
40	Mode		Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors		Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature	(C)	16.92	16.92	16.92	16.92	16.92
43	Type at Formation Temperature		Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature		Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature		V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature		Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance	(m)	5310	5315	5320	5325	5330
48	Pressure	(bar)	76.79	76.79	76.79	76.79	76.79
49	Temperature	(C)	57.11	57.10	57.10	57.10	57.10
50	Formation		Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type		No Types	No Types	No Types	No Types	No Types
52	Mode		Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors		Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature	(C)	16.92	16.92	16.92	16.92	16.92
55	Type at Formation Temperature		Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature		Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature		V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature		Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance	(m)	5335	5340	5345	5350	5355
60	Pressure	(bar)	76.79	76.79	76.79	76.79	76.79
61	Temperature	(C)	57.10	57.09	57.09	57.09	57.09
62	Formation		Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type		No Types	No Types	No Types	No Types	No Types
64	Mode		Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors		Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature	(C)	16.92	16.92	16.92	16.92	16.92
67	Type at Formation Temperature		Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature		Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model

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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	5360	5365	5370	5375	5380
16	Pressure (bar)	76.79	76.79	76.79	76.79	76.79
17	Temperature (C)	57.09	57.08	57.08	57.08	57.08
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	5385	5390	5395	5400	5405
28	Pressure (bar)	76.79	76.79	76.79	76.79	76.79
29	Temperature (C)	57.08	57.07	57.07	57.07	57.07
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	5410	5415	5420	5425	5430
40	Pressure (bar)	76.79	76.79	76.79	76.79	76.79
41	Temperature (C)	57.07	57.07	57.06	57.06	57.06
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	5435	5440	5445	5450	5455
52	Pressure (bar)	76.79	76.78	76.78	76.78	76.78
53	Temperature (C)	57.05	57.05	57.05	57.05	57.04
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	5460	5465	5470	5475	5480
64	Pressure (bar)	76.78	76.77	76.77	76.77	76.77
65	Temperature (C)	57.04	57.04	57.04	57.03	57.03
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 708 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	5485	5490	5495	5500	5505
20	Pressure (bar)	76.77	76.77	76.77	76.77	76.77
21	Temperature (C)	57.03	57.03	57.03	57.02	57.02
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	5510	5515	5520	5525	5530
32	Pressure (bar)	76.77	76.77	76.77	76.77	76.77
33	Temperature (C)	57.02	57.02	57.02	57.01	57.01
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	5535	5542	5549	5556	5563
44	Pressure (bar)	76.77	76.77	76.77	76.77	76.77
45	Temperature (C)	57.01	57.01	57.00	57.00	57.00
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	5570	5572	5574	5576	5578
56	Pressure (bar)	76.77	76.77	76.77	76.77	76.77
57	Temperature (C)	57.00	56.99	56.99	56.99	56.99
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	5580	5585	5590	5595	5600
68	Pressure (bar)	76.77	76.77	76.77	76.77	76.77
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 709 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	56.99	56.99	56.99	56.99	56.98
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	5605	5610	5615	5620	5625
24	Pressure (bar)	76.77	76.77	76.77	76.77	76.77
25	Temperature (C)	56.98	56.98	56.98	56.97	56.97
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	5630	5635	5640	5645	5650
36	Pressure (bar)	76.77	76.77	76.77	76.77	76.77
37	Temperature (C)	56.97	56.97	56.97	56.97	56.96
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	5655	5660	5665	5670	5675
48	Pressure (bar)	76.77	76.77	76.77	76.77	76.77
49	Temperature (C)	56.96	56.96	56.96	56.96	56.95
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	5680	5685	5690	5695	5700
60	Pressure (bar)	76.77	76.77	76.76	76.76	76.76
61	Temperature (C)	56.95	56.95	56.95	56.94	56.94
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 710 of 937					

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	5705	5710	5715	5720	5725
16	Pressure (bar)	76.76	76.76	76.76	76.76	76.76
17	Temperature (C)	56.94	56.93	56.93	56.93	56.93
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	5730	5735	5740	5745	5750
28	Pressure (bar)	76.76	76.76	76.75	76.75	76.75
29	Temperature (C)	56.93	56.92	56.92	56.92	56.92
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	5755	5760	5765	5770	5775
40	Pressure (bar)	76.75	76.75	76.75	76.75	76.75
41	Temperature (C)	56.92	56.91	56.91	56.91	56.91
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	5780	5785	5790	5795	5800
52	Pressure (bar)	76.75	76.75	76.75	76.75	76.75
53	Temperature (C)	56.90	56.90	56.90	56.90	56.90
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	5805	5810	5815	5820	5825
64	Pressure (bar)	76.75	76.75	76.75	76.75	76.75
65	Temperature (C)	56.89	56.89	56.89	56.89	56.89
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.92
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	5830	5835	5840	5845	5850
20	Pressure (bar)	76.75	76.74	76.74	76.74	76.74
21	Temperature (C)	56.88	56.88	56.88	56.87	56.87
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.92	16.92	16.91	16.91	16.91
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	5855	5860	5865	5870	5875
32	Pressure (bar)	76.73	76.73	76.73	76.73	76.73
33	Temperature (C)	56.87	56.87	56.86	56.86	56.86
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.91	16.91	16.91	16.91	16.91
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	5880	5885	5890	5895	5900
44	Pressure (bar)	76.73	76.74	76.76	76.78	76.79
45	Temperature (C)	56.86	56.86	56.87	56.88	56.88
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.91	16.92	16.92	16.92	16.92
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	5905	5910	5915	5920	5925
56	Pressure (bar)	76.81	76.82	76.84	76.86	76.87
57	Temperature (C)	56.89	56.90	56.90	56.91	56.92
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.92	16.92	16.92	16.92	16.93
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	5930	5935	5940	5945	5950
68	Pressure (bar)	76.89	76.73	76.57	76.41	76.25
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 712 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	56.93	56.88	56.84	56.80	56.76
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.93	16.91	16.90	16.89	16.88
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	5955	5960	5965	5970	5975
24	Pressure (bar)	76.10	76.06	76.02	75.98	75.94
25	Temperature (C)	56.72	56.70	56.69	56.68	56.67
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.86	16.86	16.86	16.85	16.85
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	5980	5985	5990	5995	6000
36	Pressure (bar)	75.90	75.93	75.96	75.98	76.01
37	Temperature (C)	56.65	56.67	56.68	56.69	56.71
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.85	16.85	16.85	16.85	16.86
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	6005	6010	6015	6020	6025
48	Pressure (bar)	76.04	75.85	75.67	75.49	75.30
49	Temperature (C)	56.72	56.67	56.62	56.57	56.52
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.86	16.84	16.83	16.81	16.80
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	6030	6035	6040	6045	6050
60	Pressure (bar)	75.12	75.12	75.12	75.12	75.11
61	Temperature (C)	56.48	56.47	56.47	56.47	56.47
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.78	16.78	16.78	16.78	16.78
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	6055	6060	6065	6070	6075
16	Pressure (bar)	75.11	75.11	75.11	75.11	75.11
17	Temperature (C)	56.46	56.46	56.46	56.46	56.45
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.78	16.78	16.78	16.78	16.78
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	6080	6085	6090	6095	6101
28	Pressure (bar)	75.11	75.13	75.15	75.17	75.19
29	Temperature (C)	56.45	56.46	56.47	56.48	56.49
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.78	16.78	16.78	16.79	16.79
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	6106	6111	6116	6121	6126
40	Pressure (bar)	75.21	75.07	74.94	74.80	74.66
41	Temperature (C)	56.50	56.46	56.43	56.39	56.35
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.79	16.78	16.77	16.76	16.74
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	6131	6136	6141	6146	6151
52	Pressure (bar)	74.52	74.52	74.52	74.52	74.52
53	Temperature (C)	56.31	56.31	56.31	56.31	56.30
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.73	16.73	16.73	16.73	16.73
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	6156	6161	6166	6171	6176
64	Pressure (bar)	74.52	74.52	74.52	74.51	74.51
65	Temperature (C)	56.30	56.30	56.30	56.30	56.29
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.73	16.73	16.73	16.73	16.73
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	6181	6186	6191	6196	6201
20	Pressure (bar)	74.51	74.51	74.51	74.51	74.51
21	Temperature (C)	56.29	56.29	56.29	56.28	56.28
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.73	16.73	16.73	16.73	16.73
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	6206	6211	6216	6221	6226
32	Pressure (bar)	74.51	74.51	74.51	74.51	74.51
33	Temperature (C)	56.28	56.28	56.28	56.27	56.27
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.73	16.73	16.73	16.73	16.73
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	6231	6236	6241	6246	6251
44	Pressure (bar)	74.51	74.51	74.51	74.51	74.51
45	Temperature (C)	56.27	56.27	56.27	56.26	56.26
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.73	16.73	16.73	16.73	16.73
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	6256	6261	6266	6271	6276
56	Pressure (bar)	74.51	74.51	74.51	74.51	74.51
57	Temperature (C)	56.26	56.26	56.26	56.25	56.25
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.73	16.73	16.73	16.73	16.73
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	6281	6285	6290	6294	6299
68	Pressure (bar)	74.50	74.53	74.55	74.57	74.59
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 715 of 937

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aspentech

Epic Atlantic Limited

Case Name:

Simulation - 386MMSCFD_80barg.hsc

Unit Set:

Mixed

Date/Time:

Mon Jun 13 15:13:43 2022

Pipe Segment: ELEP-Akabel Pipeline (continued)

Hydrates

Hydrates Table

13	Temperature (C)	56.25	56.26	56.27	56.28	56.29
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.73	16.73	16.73	16.74	16.74
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	6303	6308	6313	6318	6323
24	Pressure (bar)	74.61	74.57	74.53	74.49	74.45
25	Temperature (C)	56.30	56.29	56.27	56.26	56.25
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.74	16.74	16.73	16.73	16.73
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	6328	6333	6338	6343	6348
36	Pressure (bar)	74.41	74.42	74.42	74.43	74.43
37	Temperature (C)	56.23	56.24	56.24	56.24	56.24
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.72	16.72	16.72	16.72	16.73
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	6353	6358	6363	6368	6373
48	Pressure (bar)	74.44	74.45	74.47	74.48	74.49
49	Temperature (C)	56.24	56.25	56.25	56.26	56.27
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.73	16.73	16.73	16.73	16.73
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	6378	6383	6388	6393	6398
60	Pressure (bar)	74.50	74.47	74.44	74.41	74.38
61	Temperature (C)	56.27	56.26	56.25	56.24	56.23
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.73	16.73	16.73	16.72	16.72
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 716 of 937					

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	6403	6408	6413	6418	6423
16	Pressure (bar)	74.35	74.17	73.99	73.81	73.64
17	Temperature (C)	56.22	56.17	56.12	56.07	56.03
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.72	16.70	16.69	16.67	16.66
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	6428	6433	6438	6443	6448
28	Pressure (bar)	73.46	73.45	73.45	73.45	73.45
29	Temperature (C)	55.98	55.98	55.97	55.97	55.97
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.64	16.64	16.64	16.64	16.64
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	6453	6459	6465	6471	6477
40	Pressure (bar)	73.45	73.44	73.44	73.44	73.44
41	Temperature (C)	55.96	55.96	55.96	55.96	55.95
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.64	16.64	16.64	16.64	16.64
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	6483	6487	6491	6495	6499
52	Pressure (bar)	73.44	73.44	73.44	73.44	73.44
53	Temperature (C)	55.95	55.95	55.95	55.95	55.94
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.64	16.64	16.64	16.64	16.64
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	6503	6508	6513	6518	6523
64	Pressure (bar)	73.44	73.44	73.44	73.44	73.44
65	Temperature (C)	55.94	55.94	55.94	55.94	55.93
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.64	16.64	16.64	16.64	16.64
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	6528	6533	6538	6543	6548
20	Pressure (bar)	73.44	73.44	73.44	73.44	73.43
21	Temperature (C)	55.93	55.93	55.93	55.93	55.92
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.64	16.64	16.64	16.64	16.64
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	6553	6558	6563	6568	6573
32	Pressure (bar)	73.43	73.43	73.43	73.43	73.43
33	Temperature (C)	55.92	55.92	55.92	55.91	55.91
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.64	16.64	16.64	16.64	16.64
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	6578	6583	6588	6593	6598
44	Pressure (bar)	73.43	73.43	73.43	73.43	73.43
45	Temperature (C)	55.91	55.91	55.91	55.90	55.90
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.64	16.64	16.64	16.64	16.64
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	6603	6608	6613	6618	6623
56	Pressure (bar)	73.43	73.43	73.43	73.43	73.43
57	Temperature (C)	55.90	55.90	55.90	55.89	55.89
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.64	16.64	16.64	16.64	16.64
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	6628	6633	6638	6643	6648
68	Pressure (bar)	73.43	73.43	73.43	73.43	73.43
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	55.89	55.89	55.89	55.88	55.88
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.64	16.64	16.64	16.64	16.64
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	6653	6658	6663	6668	6673
24	Pressure (bar)	73.43	73.43	73.43	73.42	73.42
25	Temperature (C)	55.88	55.88	55.88	55.87	55.87
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.64	16.64	16.64	16.64	16.64
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	6678	6683	6688	6693	6698
36	Pressure (bar)	73.42	73.42	73.42	73.42	73.42
37	Temperature (C)	55.87	55.87	55.87	55.86	55.86
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.64	16.64	16.64	16.64	16.64
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	6703	6708	6713	6718	6723
48	Pressure (bar)	73.42	73.42	73.42	73.42	73.42
49	Temperature (C)	55.86	55.86	55.86	55.85	55.85
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.64	16.64	16.64	16.64	16.64
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	6728	6733	6738	6743	6748
60	Pressure (bar)	73.42	73.45	73.47	73.50	73.52
61	Temperature (C)	55.85	55.86	55.87	55.89	55.90
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.64	16.64	16.64	16.65	16.65
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	6754	6759	6764	6769	6774
16	Pressure (bar)	73.54	73.49	73.44	73.39	73.34
17	Temperature (C)	55.91	55.89	55.88	55.86	55.85
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.65	16.65	16.64	16.64	16.63
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	6779	6784	6789	6794	6799
28	Pressure (bar)	73.28	73.27	73.26	73.25	73.24
29	Temperature (C)	55.83	55.82	55.82	55.81	55.81
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.63	16.63	16.63	16.62	16.62
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	6804	6809	6814	6819	6824
40	Pressure (bar)	73.22	73.15	73.08	73.01	72.94
41	Temperature (C)	55.80	55.78	55.76	55.74	55.72
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.62	16.62	16.61	16.60	16.60
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	6829	6834	6839	6844	6849
52	Pressure (bar)	72.87	72.90	72.93	72.96	73.00
53	Temperature (C)	55.70	55.71	55.73	55.75	55.76
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.59	16.59	16.60	16.60	16.60
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	6854	6859	6864	6869	6874
64	Pressure (bar)	73.03	72.80	72.56	72.33	72.09
65	Temperature (C)	55.78	55.72	55.66	55.59	55.53
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.61	16.59	16.57	16.54	16.52
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	6879	6884	6889	6894	6899
20	Pressure (bar)	71.86	71.85	71.85	71.85	71.85
21	Temperature (C)	55.47	55.47	55.46	55.46	55.46
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	6904	6909	6914	6919	6924
32	Pressure (bar)	71.85	71.85	71.84	71.84	71.84
33	Temperature (C)	55.45	55.45	55.45	55.45	55.45
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	6929	6934	6939	6944	6949
44	Pressure (bar)	71.84	71.84	71.84	71.84	71.84
45	Temperature (C)	55.44	55.44	55.44	55.44	55.44
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	6954	6959	6964	6969	6974
56	Pressure (bar)	71.84	71.84	71.84	71.84	71.84
57	Temperature (C)	55.43	55.43	55.43	55.43	55.43
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	6979	6984	6989	6994	6999
68	Pressure (bar)	71.84	71.84	71.84	71.84	71.84
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	55.42	55.42	55.42	55.42	55.42
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	7004	7009	7014	7019	7024
24	Pressure (bar)	71.84	71.84	71.84	71.84	71.83
25	Temperature (C)	55.41	55.41	55.41	55.41	55.41
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	7029	7034	7039	7044	7049
36	Pressure (bar)	71.83	71.83	71.83	71.83	71.83
37	Temperature (C)	55.40	55.40	55.40	55.40	55.40
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	7054	7059	7064	7069	7074
48	Pressure (bar)	71.83	71.83	71.83	71.83	71.83
49	Temperature (C)	55.39	55.39	55.39	55.39	55.38
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	7079	7081	7083	7085	7087
60	Pressure (bar)	71.83	71.83	71.83	71.83	71.83
61	Temperature (C)	55.38	55.38	55.38	55.38	55.38
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	7089	7097	7105	7113	7121
16	Pressure (bar)	71.83	71.83	71.83	71.83	71.83
17	Temperature (C)	55.38	55.38	55.37	55.37	55.37
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	7129	7134	7139	7144	7149
28	Pressure (bar)	71.83	71.83	71.82	71.82	71.82
29	Temperature (C)	55.36	55.36	55.36	55.36	55.35
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	7154	7159	7164	7169	7174
40	Pressure (bar)	71.82	71.82	71.82	71.82	71.82
41	Temperature (C)	55.35	55.35	55.35	55.35	55.34
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	7179	7184	7189	7194	7199
52	Pressure (bar)	71.82	71.82	71.82	71.82	71.82
53	Temperature (C)	55.34	55.34	55.34	55.33	55.33
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	7204	7209	7214	7219	7224
64	Pressure (bar)	71.82	71.82	71.82	71.82	71.82
65	Temperature (C)	55.33	55.33	55.33	55.32	55.32
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	7229	7234	7239	7244	7249
20	Pressure (bar)	71.82	71.82	71.82	71.81	71.81
21	Temperature (C)	55.32	55.32	55.32	55.31	55.31
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	7254	7259	7264	7269	7274
32	Pressure (bar)	71.81	71.81	71.81	71.81	71.81
33	Temperature (C)	55.31	55.31	55.31	55.30	55.30
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	7279	7284	7289	7294	7299
44	Pressure (bar)	71.81	71.81	71.81	71.81	71.81
45	Temperature (C)	55.30	55.30	55.30	55.29	55.29
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	7304	7309	7314	7319	7324
56	Pressure (bar)	71.81	71.81	71.81	71.81	71.81
57	Temperature (C)	55.29	55.29	55.29	55.28	55.28
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	7329	7334	7339	7344	7349
68	Pressure (bar)	71.81	71.81	71.81	71.81	71.80
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 724 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	55.28	55.28	55.28	55.27	55.27
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	7354	7359	7364	7369	7374
24	Pressure (bar)	71.80	71.80	71.80	71.80	71.80
25	Temperature (C)	55.27	55.27	55.27	55.26	55.26
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	7379	7384	7389	7394	7399
36	Pressure (bar)	71.80	71.80	71.80	71.80	71.80
37	Temperature (C)	55.26	55.26	55.25	55.25	55.25
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	7404	7409	7414	7419	7424
48	Pressure (bar)	71.80	71.80	71.80	71.80	71.80
49	Temperature (C)	55.25	55.25	55.24	55.24	55.24
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	7429	7434	7439	7444	7449
60	Pressure (bar)	71.80	71.80	71.80	71.80	71.79
61	Temperature (C)	55.24	55.24	55.23	55.23	55.23
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 725 of 937					

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	7454	7459	7464	7469	7474
16	Pressure (bar)	71.79	71.79	71.79	71.79	71.79
17	Temperature (C)	55.23	55.23	55.22	55.22	55.22
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	7479	7484	7489	7494	7499
28	Pressure (bar)	71.79	71.79	71.79	71.79	71.79
29	Temperature (C)	55.22	55.22	55.21	55.21	55.21
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	7504	7509	7514	7519	7524
40	Pressure (bar)	71.79	71.79	71.79	71.79	71.79
41	Temperature (C)	55.21	55.21	55.20	55.20	55.20
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	7529	7533	7538	7543	7548
52	Pressure (bar)	71.79	71.79	71.79	71.79	71.79
53	Temperature (C)	55.20	55.20	55.19	55.19	55.19
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	7553	7558	7563	7568	7573
64	Pressure (bar)	71.79	71.79	71.79	71.79	71.79
65	Temperature (C)	55.19	55.19	55.18	55.18	55.18
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	7578	7583	7588	7593	7598
20	Pressure (bar)	71.78	71.78	71.78	71.78	71.78
21	Temperature (C)	55.18	55.18	55.17	55.17	55.17
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	7603	7608	7613	7618	7623
32	Pressure (bar)	71.78	71.78	71.78	71.78	71.78
33	Temperature (C)	55.17	55.17	55.16	55.16	55.16
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	7628	7633	7638	7643	7648
44	Pressure (bar)	71.78	71.78	71.78	71.78	71.78
45	Temperature (C)	55.16	55.16	55.15	55.15	55.15
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	7653	7658	7663	7668	7673
56	Pressure (bar)	71.78	71.78	71.78	71.78	71.78
57	Temperature (C)	55.15	55.15	55.14	55.14	55.14
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	7678	7683	7688	7693	7698
68	Pressure (bar)	71.78	71.78	71.78	71.78	71.78
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	55.14	55.14	55.13	55.13	55.13
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	7703	7708	7713	7718	7723
24	Pressure (bar)	71.77	71.78	71.78	71.78	71.78
25	Temperature (C)	55.13	55.13	55.12	55.12	55.12
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	7728	7733	7738	7743	7748
36	Pressure (bar)	71.78	71.78	71.78	71.78	71.78
37	Temperature (C)	55.12	55.12	55.12	55.11	55.11
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	7753	7758	7763	7768	7773
48	Pressure (bar)	71.78	71.78	71.77	71.77	71.77
49	Temperature (C)	55.11	55.11	55.11	55.10	55.10
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	7778	7783	7788	7793	7798
60	Pressure (bar)	71.77	71.76	71.76	71.76	71.76
61	Temperature (C)	55.10	55.09	55.09	55.09	55.09
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	7803	7808	7813	7818	7823
16	Pressure (bar)	71.76	71.76	71.76	71.76	71.76
17	Temperature (C)	55.09	55.08	55.08	55.08	55.08
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.50
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	7828	7833	7838	7843	7848
28	Pressure (bar)	71.76	71.76	71.76	71.76	71.76
29	Temperature (C)	55.08	55.07	55.07	55.07	55.07
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.50	16.50	16.50	16.50	16.49
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	7853	7858	7863	7868	7873
40	Pressure (bar)	71.76	71.76	71.76	71.76	71.76
41	Temperature (C)	55.07	55.06	55.06	55.06	55.06
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.49	16.49	16.49	16.49	16.49
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	7878	7883	7888	7893	7898
52	Pressure (bar)	71.75	71.75	71.75	71.75	71.75
53	Temperature (C)	55.06	55.05	55.05	55.05	55.05
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.49	16.49	16.49	16.49	16.49
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	7903	7908	7913	7918	7923
64	Pressure (bar)	71.75	71.75	71.75	71.75	71.75
65	Temperature (C)	55.05	55.04	55.04	55.04	55.04
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 729 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.49	16.49	16.49	16.49	16.49
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	7928	7933	7938	7943	7948
20	Pressure (bar)	71.75	71.75	71.75	71.75	71.75
21	Temperature (C)	55.03	55.03	55.03	55.03	55.03
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.49	16.49	16.49	16.49	16.49
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	7953	7958	7963	7968	7973
32	Pressure (bar)	71.75	71.75	71.75	71.75	71.75
33	Temperature (C)	55.02	55.02	55.02	55.02	55.02
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.49	16.49	16.49	16.49	16.49
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	7978	7983	7988	7993	7998
44	Pressure (bar)	71.75	71.75	71.75	71.75	71.75
45	Temperature (C)	55.02	55.01	55.01	55.01	55.01
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.49	16.49	16.49	16.49	16.49
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	8003	8008	8013	8018	8023
56	Pressure (bar)	71.75	71.75	71.75	71.75	71.75
57	Temperature (C)	55.01	55.00	55.00	55.00	55.00
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.49	16.49	16.49	16.49	16.49
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	8028	8033	8038	8043	8048
68	Pressure (bar)	71.75	71.75	71.75	71.75	71.75
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	55.00	54.99	54.99	54.99	54.99
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.49	16.49	16.49	16.49	16.49
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	8053	8058	8063	8068	8073
24	Pressure (bar)	71.75	71.75	71.76	71.77	71.77
25	Temperature (C)	54.99	54.99	54.99	55.00	55.00
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.49	16.49	16.50	16.50	16.50
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	8078	8083	8088	8093	8098
36	Pressure (bar)	71.78	71.74	71.71	71.67	71.63
37	Temperature (C)	55.00	54.99	54.98	54.96	54.95
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.50	16.49	16.49	16.49	16.48
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	8103	8108	8113	8118	8123
48	Pressure (bar)	71.60	71.59	71.59	71.58	71.58
49	Temperature (C)	54.94	54.94	54.93	54.93	54.92
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	8128	8133	8138	8143	8148
60	Pressure (bar)	71.57	71.57	71.57	71.57	71.57
61	Temperature (C)	54.92	54.92	54.92	54.91	54.91
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	8153	8158	8163	8168	8173
16	Pressure (bar)	71.57	71.57	71.57	71.57	71.57
17	Temperature (C)	54.91	54.91	54.91	54.90	54.90
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	8178	8183	8188	8193	8198
28	Pressure (bar)	71.57	71.57	71.57	71.57	71.57
29	Temperature (C)	54.90	54.90	54.90	54.89	54.89
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	8203	8208	8213	8218	8223
40	Pressure (bar)	71.57	71.57	71.56	71.56	71.56
41	Temperature (C)	54.89	54.89	54.89	54.88	54.88
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	8228	8233	8238	8243	8248
52	Pressure (bar)	71.56	71.56	71.56	71.56	71.56
53	Temperature (C)	54.88	54.88	54.88	54.87	54.87
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	8253	8258	8263	8268	8273
64	Pressure (bar)	71.56	71.56	71.56	71.56	71.56
65	Temperature (C)	54.87	54.87	54.87	54.86	54.86
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	8278	8283	8288	8293	8298
20	Pressure (bar)	71.56	71.56	71.56	71.56	71.56
21	Temperature (C)	54.86	54.86	54.86	54.85	54.85
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	8303	8308	8313	8318	8323
32	Pressure (bar)	71.56	71.56	71.56	71.56	71.55
33	Temperature (C)	54.85	54.85	54.85	54.84	54.84
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	8328	8333	8338	8343	8348
44	Pressure (bar)	71.55	71.55	71.55	71.56	71.56
45	Temperature (C)	54.84	54.84	54.84	54.83	54.83
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	8353	8358	8363	8368	8373
56	Pressure (bar)	71.56	71.56	71.56	71.56	71.56
57	Temperature (C)	54.83	54.83	54.83	54.83	54.82
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	8378	8383	8388	8393	8398
68	Pressure (bar)	71.56	71.55	71.55	71.55	71.55
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 733 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	54.82	54.82	54.82	54.82	54.81
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	8403	8408	8413	8418	8423
24	Pressure (bar)	71.55	71.55	71.55	71.55	71.55
25	Temperature (C)	54.81	54.81	54.81	54.81	54.80
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	8428	8433	8438	8443	8448
36	Pressure (bar)	71.55	71.55	71.55	71.55	71.55
37	Temperature (C)	54.80	54.80	54.80	54.80	54.80
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	8453	8458	8463	8468	8473
48	Pressure (bar)	71.55	71.55	71.55	71.55	71.55
49	Temperature (C)	54.79	54.79	54.79	54.79	54.78
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	8478	8483	8488	8493	8498
60	Pressure (bar)	71.55	71.55	71.55	71.55	71.55
61	Temperature (C)	54.78	54.78	54.78	54.78	54.77
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 734 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	8503	8508	8513	8518	8523
16	Pressure (bar)	71.55	71.54	71.54	71.54	71.54
17	Temperature (C)	54.77	54.77	54.77	54.77	54.76
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	8528	8533	8538	8543	8548
28	Pressure (bar)	71.54	71.54	71.54	71.54	71.54
29	Temperature (C)	54.76	54.76	54.76	54.76	54.75
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	8553	8558	8563	8568	8573
40	Pressure (bar)	71.54	71.54	71.54	71.54	71.54
41	Temperature (C)	54.75	54.75	54.75	54.75	54.74
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	8578	8583	8588	8593	8598
52	Pressure (bar)	71.54	71.54	71.54	71.54	71.54
53	Temperature (C)	54.74	54.74	54.74	54.74	54.73
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	8603	8608	8613	8618	8623
64	Pressure (bar)	71.54	71.54	71.54	71.54	71.54
65	Temperature (C)	54.73	54.73	54.73	54.73	54.72
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	8628	8633	8638	8643	8648
20	Pressure (bar)	71.54	71.53	71.53	71.53	71.53
21	Temperature (C)	54.72	54.72	54.72	54.72	54.71
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.48	16.48	16.48	16.48	16.48
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	8653	8658	8663	8668	8673
32	Pressure (bar)	71.53	71.53	71.53	71.53	71.53
33	Temperature (C)	54.71	54.71	54.71	54.71	54.70
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.48	16.48	16.47	16.47	16.47
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	8678	8683	8688	8693	8698
44	Pressure (bar)	71.53	71.53	71.53	71.53	71.53
45	Temperature (C)	54.70	54.70	54.70	54.70	54.69
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	8703	8708	8713	8718	8723
56	Pressure (bar)	71.53	71.53	71.53	71.53	71.53
57	Temperature (C)	54.69	54.69	54.69	54.69	54.68
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	8728	8733	8738	8743	8748
68	Pressure (bar)	71.53	71.53	71.53	71.53	71.53
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	54.68	54.68	54.68	54.68	54.67
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	8753	8758	8763	8768	8773
24	Pressure (bar)	71.52	71.52	71.52	71.52	71.52
25	Temperature (C)	54.67	54.67	54.67	54.67	54.66
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	8778	8783	8788	8793	8798
36	Pressure (bar)	71.52	71.52	71.52	71.52	71.52
37	Temperature (C)	54.66	54.66	54.66	54.66	54.65
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	8803	8805	8807	8809	8811
48	Pressure (bar)	71.52	71.52	71.52	71.52	71.52
49	Temperature (C)	54.65	54.65	54.65	54.65	54.65
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	8813	8840	8867	8894	8921
60	Pressure (bar)	71.52	71.52	71.52	71.51	71.51
61	Temperature (C)	54.65	54.64	54.63	54.62	54.61
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	8948	8949	8950	8951	8952
16	Pressure (bar)	71.51	71.51	71.51	71.51	71.51
17	Temperature (C)	54.59	54.59	54.59	54.59	54.59
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	8953	8958	8963	8968	8973
28	Pressure (bar)	71.51	71.51	71.51	71.51	71.51
29	Temperature (C)	54.59	54.59	54.59	54.59	54.58
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	8978	8983	8988	8993	8998
40	Pressure (bar)	71.51	71.50	71.50	71.50	71.50
41	Temperature (C)	54.58	54.58	54.58	54.58	54.57
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	9003	9008	9013	9018	9023
52	Pressure (bar)	71.50	71.50	71.50	71.50	71.50
53	Temperature (C)	54.57	54.57	54.57	54.57	54.56
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	9028	9033	9038	9043	9048
64	Pressure (bar)	71.50	71.50	71.50	71.50	71.50
65	Temperature (C)	54.56	54.56	54.56	54.56	54.55
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	9053	9054	9055	9056	9057
20	Pressure (bar)	71.50	71.50	71.50	71.50	71.50
21	Temperature (C)	54.55	54.55	54.55	54.55	54.55
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	9058	9065	9071	9078	9085
32	Pressure (bar)	71.50	71.50	71.50	71.50	71.50
33	Temperature (C)	54.55	54.55	54.55	54.54	54.54
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	9091	9098	9105	9111	9118
44	Pressure (bar)	71.50	71.50	71.50	71.50	71.50
45	Temperature (C)	54.54	54.54	54.53	54.53	54.53
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	9125	9131	9138	9145	9151
56	Pressure (bar)	71.50	71.50	71.50	71.50	71.50
57	Temperature (C)	54.53	54.52	54.52	54.52	54.52
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	9158	9162	9166	9170	9174
68	Pressure (bar)	71.49	71.49	71.49	71.49	71.49
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_80barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:13:43 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	54.51	54.51	54.51	54.51	54.51
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	9178	9183	9188	9193	9198
24	Pressure (bar)	71.49	71.49	71.49	71.49	71.49
25	Temperature (C)	54.50	54.50	54.50	54.50	54.50
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	9203	9208	9213	9218	9223
36	Pressure (bar)	71.49	71.49	71.49	71.49	71.49
37	Temperature (C)	54.50	54.49	54.49	54.49	54.49
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	9228	9233	9238	9243	9248
48	Pressure (bar)	71.49	71.49	71.49	71.49	71.49
49	Temperature (C)	54.49	54.48	54.48	54.48	54.48
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	9253	9258	9263	9268	9273
60	Pressure (bar)	71.49	71.49	71.49	71.48	71.48
61	Temperature (C)	54.48	54.47	54.47	54.47	54.47
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	9278	9283	9288	9293	9298
16	Pressure (bar)	71.48	71.48	71.48	71.48	71.48
17	Temperature (C)	54.46	54.46	54.46	54.46	54.46
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	9303	9308	9313	9318	9323
28	Pressure (bar)	71.48	71.48	71.48	71.48	71.48
29	Temperature (C)	54.45	54.45	54.45	54.45	54.45
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	9328	9333	9338	9343	9348
40	Pressure (bar)	71.48	71.48	71.47	71.47	71.47
41	Temperature (C)	54.44	54.44	54.44	54.44	54.44
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	9353	9358	9363	9368	9373
52	Pressure (bar)	71.47	71.47	71.47	71.47	71.47
53	Temperature (C)	54.43	54.43	54.43	54.43	54.43
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	9378	9383	9388	9393	9398
64	Pressure (bar)	71.47	71.47	71.47	71.47	71.47
65	Temperature (C)	54.42	54.42	54.42	54.42	54.42
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	9403	9408	9413	9418	9423
20	Pressure (bar)	71.47	71.47	71.47	71.47	71.47
21	Temperature (C)	54.41	54.41	54.41	54.41	54.41
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	9428	9433	9438	9443	9448
32	Pressure (bar)	71.47	71.47	71.47	71.47	71.47
33	Temperature (C)	54.41	54.40	54.40	54.40	54.40
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	9453	9456	9459	9462	9465
44	Pressure (bar)	71.47	71.47	71.47	71.46	71.46
45	Temperature (C)	54.40	54.40	54.39	54.39	54.39
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	9468	9473	9477	9482	9487
56	Pressure (bar)	71.46	71.46	71.46	71.46	71.46
57	Temperature (C)	54.39	54.39	54.38	54.38	54.38
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	9491	9496	9501	9505	9510
68	Pressure (bar)	71.46	71.46	71.46	71.46	71.46
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 742 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	54.38	54.38	54.38	54.37	54.37
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	9515	9519	9524	9529	9533
24	Pressure (bar)	71.46	71.46	71.46	71.46	71.46
25	Temperature (C)	54.37	54.37	54.37	54.37	54.36
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	9538	9543	9547	9552	9557
36	Pressure (bar)	71.46	71.46	71.46	71.45	71.45
37	Temperature (C)	54.36	54.36	54.36	54.36	54.35
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	9561	9566	9571	9575	9580
48	Pressure (bar)	71.45	71.45	71.45	71.45	71.45
49	Temperature (C)	54.35	54.35	54.35	54.35	54.34
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	9585	9589	9594	9599	9603
60	Pressure (bar)	71.45	71.45	71.45	71.44	71.44
61	Temperature (C)	54.34	54.34	54.34	54.33	54.33
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	9608	9612	9616	9620	9624
16	Pressure (bar)	71.44	71.44	71.44	71.44	71.44
17	Temperature (C)	54.33	54.33	54.33	54.33	54.32
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	9628	9633	9638	9643	9648
28	Pressure (bar)	71.44	71.44	71.44	71.44	71.44
29	Temperature (C)	54.32	54.32	54.32	54.32	54.31
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	9653	9658	9663	9668	9673
40	Pressure (bar)	71.44	71.44	71.44	71.44	71.44
41	Temperature (C)	54.31	54.31	54.31	54.31	54.31
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	9678	9683	9688	9693	9698
52	Pressure (bar)	71.44	71.44	71.43	71.43	71.43
53	Temperature (C)	54.30	54.30	54.30	54.30	54.29
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	9703	9708	9713	9718	9723
64	Pressure (bar)	71.43	71.43	71.43	71.43	71.43
65	Temperature (C)	54.29	54.29	54.29	54.29	54.28
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	9728	9733	9738	9743	9748
20	Pressure (bar)	71.43	71.43	71.43	71.43	71.43
21	Temperature (C)	54.28	54.28	54.28	54.28	54.27
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	9753	9758	9763	9768	9773
32	Pressure (bar)	71.43	71.43	71.42	71.42	71.42
33	Temperature (C)	54.27	54.27	54.27	54.27	54.26
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	9778	9783	9788	9793	9798
44	Pressure (bar)	71.42	71.42	71.42	71.42	71.42
45	Temperature (C)	54.26	54.26	54.26	54.26	54.25
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	9803	9808	9813	9818	9823
56	Pressure (bar)	71.42	71.42	71.42	71.42	71.42
57	Temperature (C)	54.25	54.25	54.25	54.25	54.24
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.47	16.47	16.47	16.47	16.47
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	9828	9833	9838	9843	9848
68	Pressure (bar)	71.42	71.42	71.42	71.42	71.42
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	54.24	54.24	54.24	54.24	54.23
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.47	16.47	16.46	16.46	16.46
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	9853	9858	9863	9868	9873
24	Pressure (bar)	71.42	71.42	71.42	71.42	71.42
25	Temperature (C)	54.23	54.23	54.23	54.23	54.23
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	9878	9883	9888	9893	9898
36	Pressure (bar)	71.42	71.41	71.41	71.41	71.41
37	Temperature (C)	54.22	54.22	54.22	54.22	54.22
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	9903	9905	9907	9909	9911
48	Pressure (bar)	71.41	71.41	71.41	71.41	71.41
49	Temperature (C)	54.21	54.21	54.21	54.21	54.21
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	9913	9933	9953	9973	9993
60	Pressure (bar)	71.41	71.41	71.41	71.41	71.41
61	Temperature (C)	54.21	54.20	54.19	54.19	54.18
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.001e+004	1.002e+004	1.002e+004	1.002e+004	1.003e+004
16	Pressure (bar)	71.40	71.41	71.41	71.41	71.41
17	Temperature (C)	54.17	54.17	54.17	54.17	54.17
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.003e+004	1.003e+004	1.004e+004	1.004e+004	1.005e+004
28	Pressure (bar)	71.41	71.41	71.41	71.41	71.41
29	Temperature (C)	54.17	54.17	54.16	54.16	54.16
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.005e+004	1.006e+004	1.006e+004	1.007e+004	1.007e+004
40	Pressure (bar)	71.41	71.41	71.40	71.40	71.40
41	Temperature (C)	54.16	54.16	54.15	54.15	54.15
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.008e+004	1.008e+004	1.009e+004	1.009e+004	1.010e+004
52	Pressure (bar)	71.40	71.40	71.39	71.39	71.39
53	Temperature (C)	54.15	54.14	54.14	54.14	54.14
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.010e+004	1.011e+004	1.011e+004	1.012e+004	1.012e+004
64	Pressure (bar)	71.39	71.39	71.39	71.39	71.39
65	Temperature (C)	54.13	54.13	54.13	54.13	54.13
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 747 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.013e+004	1.013e+004	1.014e+004	1.014e+004	1.015e+004
20	Pressure (bar)	71.39	71.39	71.39	71.39	71.39
21	Temperature (C)	54.12	54.12	54.12	54.12	54.12
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.015e+004	1.016e+004	1.016e+004	1.017e+004	1.017e+004
32	Pressure (bar)	71.39	71.39	71.39	71.38	71.38
33	Temperature (C)	54.11	54.11	54.11	54.11	54.11
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.018e+004	1.018e+004	1.019e+004	1.019e+004	1.020e+004
44	Pressure (bar)	71.38	71.38	71.38	71.38	71.38
45	Temperature (C)	54.10	54.10	54.10	54.10	54.10
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.020e+004	1.021e+004	1.021e+004	1.022e+004	1.022e+004
56	Pressure (bar)	71.38	71.38	71.38	71.38	71.38
57	Temperature (C)	54.09	54.09	54.09	54.09	54.09
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.023e+004	1.023e+004	1.024e+004	1.024e+004	1.025e+004
68	Pressure (bar)	71.38	71.38	71.38	71.38	71.38
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:13:43 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	54.08	54.08	54.08	54.08	54.08
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.025e+004	1.026e+004	1.026e+004	1.027e+004	1.027e+004
24	Pressure (bar)	71.38	71.38	71.38	71.38	71.38
25	Temperature (C)	54.08	54.07	54.07	54.07	54.07
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.028e+004	1.028e+004	1.029e+004	1.029e+004	1.030e+004
36	Pressure (bar)	71.38	71.38	71.38	71.38	71.37
37	Temperature (C)	54.07	54.07	54.06	54.06	54.06
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.030e+004	1.031e+004	1.031e+004	1.032e+004	1.032e+004
48	Pressure (bar)	71.37	71.37	71.37	71.37	71.37
49	Temperature (C)	54.05	54.05	54.05	54.05	54.05
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.033e+004	1.033e+004	1.034e+004	1.034e+004	1.035e+004
60	Pressure (bar)	71.37	71.37	71.37	71.37	71.37
61	Temperature (C)	54.05	54.04	54.04	54.04	54.04
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.035e+004	1.036e+004	1.036e+004	1.037e+004	1.037e+004
16	Pressure (bar)	71.37	71.37	71.37	71.37	71.37
17	Temperature (C)	54.04	54.03	54.03	54.03	54.03
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.038e+004	1.038e+004	1.039e+004	1.039e+004	1.040e+004
28	Pressure (bar)	71.37	71.37	71.37	71.37	71.37
29	Temperature (C)	54.03	54.02	54.02	54.02	54.02
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.040e+004	1.041e+004	1.041e+004	1.042e+004	1.042e+004
40	Pressure (bar)	71.37	71.37	71.37	71.36	71.36
41	Temperature (C)	54.02	54.02	54.01	54.01	54.01
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.043e+004	1.043e+004	1.044e+004	1.044e+004	1.045e+004
52	Pressure (bar)	71.36	71.36	71.36	71.36	71.36
53	Temperature (C)	54.01	54.01	54.00	54.00	54.00
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.045e+004	1.046e+004	1.046e+004	1.047e+004	1.047e+004
64	Pressure (bar)	71.36	71.37	71.37	71.37	71.37
65	Temperature (C)	54.00	54.00	54.00	53.99	53.99
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 750 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.46	16.46	16.46	16.46
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.048e+004	1.048e+004	1.049e+004	1.050e+004
20	Pressure (bar)	71.37	71.37	71.37	71.37
21	Temperature (C)	53.99	53.99	53.99	53.98
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.46	16.46	16.46	16.46
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.050e+004	1.051e+004	1.051e+004	1.052e+004
32	Pressure (bar)	71.37	71.36	71.36	71.35
33	Temperature (C)	53.98	53.98	53.98	53.97
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.46	16.46	16.46	16.46
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.053e+004	1.053e+004	1.054e+004	1.055e+004
44	Pressure (bar)	71.35	71.35	71.35	71.35
45	Temperature (C)	53.97	53.97	53.96	53.96
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.46	16.46	16.46	16.46
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.055e+004	1.056e+004	1.056e+004	1.057e+004
56	Pressure (bar)	71.35	71.35	71.35	71.35
57	Temperature (C)	53.96	53.96	53.95	53.95
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.46	16.46	16.46	16.46
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.058e+004	1.058e+004	1.059e+004	1.060e+004
68	Pressure (bar)	71.35	71.35	71.34	71.34
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	53.95	53.95	53.94	53.94	53.94
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.060e+004	1.061e+004	1.061e+004	1.062e+004	1.062e+004
24	Pressure (bar)	71.34	71.34	71.34	71.34	71.34
25	Temperature (C)	53.94	53.94	53.93	53.93	53.93
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.063e+004	1.063e+004	1.064e+004	1.064e+004	1.065e+004
36	Pressure (bar)	71.34	71.34	71.34	71.34	71.34
37	Temperature (C)	53.93	53.93	53.92	53.92	53.92
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.065e+004	1.066e+004	1.066e+004	1.067e+004	1.067e+004
48	Pressure (bar)	71.34	71.34	71.34	71.34	71.33
49	Temperature (C)	53.92	53.92	53.91	53.91	53.91
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.068e+004	1.068e+004	1.069e+004	1.069e+004	1.070e+004
60	Pressure (bar)	71.33	71.33	71.33	71.33	71.33
61	Temperature (C)	53.91	53.91	53.90	53.90	53.90
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 752 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.070e+004	1.071e+004	1.071e+004	1.072e+004	1.072e+004
16	Pressure (bar)	71.33	71.33	71.33	71.33	71.33
17	Temperature (C)	53.90	53.90	53.89	53.89	53.89
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.073e+004	1.073e+004	1.074e+004	1.074e+004	1.075e+004
28	Pressure (bar)	71.33	71.33	71.33	71.33	71.33
29	Temperature (C)	53.89	53.89	53.88	53.88	53.88
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.075e+004	1.076e+004	1.076e+004	1.077e+004	1.077e+004
40	Pressure (bar)	71.33	71.33	71.33	71.33	71.32
41	Temperature (C)	53.88	53.88	53.87	53.87	53.87
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.078e+004	1.078e+004	1.079e+004	1.079e+004	1.080e+004
52	Pressure (bar)	71.32	71.32	71.32	71.32	71.32
53	Temperature (C)	53.87	53.87	53.86	53.86	53.86
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.080e+004	1.081e+004	1.081e+004	1.082e+004	1.082e+004
64	Pressure (bar)	71.32	71.32	71.32	71.32	71.32
65	Temperature (C)	53.86	53.86	53.85	53.85	53.85
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 753 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.46	16.46	16.46	16.46
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.083e+004	1.083e+004	1.084e+004	1.085e+004
20	Pressure (bar)	71.32	71.32	71.32	71.32
21	Temperature (C)	53.85	53.85	53.85	53.84
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.46	16.46	16.46	16.46
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.085e+004	1.086e+004	1.086e+004	1.087e+004
32	Pressure (bar)	71.32	71.32	71.32	71.32
33	Temperature (C)	53.84	53.84	53.84	53.83
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.46	16.46	16.46	16.46
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.088e+004	1.088e+004	1.089e+004	1.089e+004
44	Pressure (bar)	71.32	71.32	71.31	71.31
45	Temperature (C)	53.83	53.83	53.83	53.82
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.46	16.46	16.46	16.46
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.090e+004	1.091e+004	1.091e+004	1.092e+004
56	Pressure (bar)	71.31	71.31	71.31	71.31
57	Temperature (C)	53.82	53.82	53.82	53.81
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.46	16.46	16.46	16.46
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.093e+004	1.093e+004	1.094e+004	1.095e+004
68	Pressure (bar)	71.31	71.31	71.31	71.31
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 754 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	53.81	53.81	53.81	53.80	53.80
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.095e+004	1.096e+004	1.096e+004	1.097e+004	1.097e+004
24	Pressure (bar)	71.31	71.31	71.31	71.31	71.31
25	Temperature (C)	53.80	53.80	53.80	53.79	53.79
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.46	16.46	16.46	16.46	16.46
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.098e+004	1.098e+004	1.099e+004	1.099e+004	1.099e+004
36	Pressure (bar)	71.31	71.31	71.31	71.31	71.31
37	Temperature (C)	53.79	53.79	53.79	53.79	53.78
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.46	16.45	16.45	16.45	16.45
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.100e+004	1.100e+004	1.101e+004	1.102e+004	1.102e+004
48	Pressure (bar)	71.31	71.31	71.30	71.30	71.30
49	Temperature (C)	53.78	53.78	53.78	53.78	53.77
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.103e+004	1.103e+004	1.104e+004	1.104e+004	1.105e+004
60	Pressure (bar)	71.30	71.30	71.30	71.30	71.30
61	Temperature (C)	53.77	53.77	53.77	53.77	53.76
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 755 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.105e+004	1.106e+004	1.106e+004	1.107e+004	1.107e+004
16	Pressure (bar)	71.30	71.30	71.30	71.30	71.30
17	Temperature (C)	53.76	53.76	53.76	53.76	53.75
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.108e+004	1.108e+004	1.109e+004	1.109e+004	1.110e+004
28	Pressure (bar)	71.30	71.30	71.30	71.30	71.30
29	Temperature (C)	53.75	53.75	53.75	53.75	53.74
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.110e+004	1.111e+004	1.111e+004	1.112e+004	1.112e+004
40	Pressure (bar)	71.30	71.30	71.30	71.30	71.30
41	Temperature (C)	53.74	53.74	53.74	53.74	53.73
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.113e+004	1.113e+004	1.114e+004	1.114e+004	1.115e+004
52	Pressure (bar)	71.30	71.29	71.29	71.29	71.29
53	Temperature (C)	53.73	53.73	53.73	53.73	53.73
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.115e+004	1.116e+004	1.116e+004	1.117e+004	1.117e+004
64	Pressure (bar)	71.29	71.29	71.29	71.29	71.29
65	Temperature (C)	53.72	53.72	53.72	53.72	53.72
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 756 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.45	16.45	16.45	16.45
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.118e+004	1.118e+004	1.119e+004	1.120e+004
20	Pressure (bar)	71.29	71.29	71.29	71.29
21	Temperature (C)	53.71	53.71	53.71	53.71
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.45	16.45	16.45	16.45
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.120e+004	1.121e+004	1.121e+004	1.122e+004
32	Pressure (bar)	71.29	71.29	71.29	71.29
33	Temperature (C)	53.70	53.70	53.70	53.70
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.45	16.45	16.45	16.45
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.123e+004	1.123e+004	1.124e+004	1.125e+004
44	Pressure (bar)	71.29	71.29	71.29	71.29
45	Temperature (C)	53.69	53.69	53.69	53.69
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.45	16.45	16.45	16.45
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.125e+004	1.126e+004	1.126e+004	1.127e+004
56	Pressure (bar)	71.28	71.28	71.28	71.28
57	Temperature (C)	53.68	53.68	53.68	53.68
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.45	16.45	16.45	16.45
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.128e+004	1.128e+004	1.129e+004	1.130e+004
68	Pressure (bar)	71.28	71.28	71.28	71.28
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 757 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	53.68	53.67	53.67	53.67	53.67
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.130e+004	1.131e+004	1.131e+004	1.132e+004	1.132e+004
24	Pressure (bar)	71.28	71.28	71.28	71.28	71.28
25	Temperature (C)	53.67	53.66	53.66	53.66	53.66
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.133e+004	1.133e+004	1.134e+004	1.134e+004	1.135e+004
36	Pressure (bar)	71.28	71.28	71.28	71.28	71.28
37	Temperature (C)	53.66	53.66	53.65	53.65	53.65
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.135e+004	1.136e+004	1.136e+004	1.137e+004	1.137e+004
48	Pressure (bar)	71.28	71.28	71.28	71.28	71.28
49	Temperature (C)	53.65	53.65	53.64	53.64	53.64
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.138e+004	1.138e+004	1.139e+004	1.139e+004	1.140e+004
60	Pressure (bar)	71.28	71.28	71.28	71.28	71.28
61	Temperature (C)	53.64	53.64	53.63	53.63	53.63
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 758 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.140e+004	1.141e+004	1.141e+004	1.142e+004	1.142e+004
16	Pressure (bar)	71.28	71.28	71.28	71.28	71.28
17	Temperature (C)	53.63	53.63	53.63	53.63	53.62
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.143e+004	1.143e+004	1.144e+004	1.144e+004	1.145e+004
28	Pressure (bar)	71.28	71.28	71.28	71.28	71.28
29	Temperature (C)	53.62	53.62	53.62	53.62	53.61
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.145e+004	1.146e+004	1.146e+004	1.147e+004	1.147e+004
40	Pressure (bar)	71.28	71.27	71.27	71.27	71.27
41	Temperature (C)	53.61	53.61	53.61	53.60	53.60
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.148e+004	1.148e+004	1.149e+004	1.149e+004	1.150e+004
52	Pressure (bar)	71.26	71.26	71.26	71.26	71.26
53	Temperature (C)	53.60	53.60	53.59	53.59	53.59
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.150e+004	1.151e+004	1.151e+004	1.152e+004	1.152e+004
64	Pressure (bar)	71.26	71.26	71.26	71.26	71.26
65	Temperature (C)	53.59	53.59	53.58	53.58	53.58
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 759 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.45	16.45	16.45	16.45
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.153e+004	1.153e+004	1.154e+004	1.155e+004
20	Pressure (bar)	71.26	71.26	71.26	71.25
21	Temperature (C)	53.58	53.58	53.57	53.57
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.45	16.45	16.45	16.45
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.155e+004	1.156e+004	1.156e+004	1.157e+004
32	Pressure (bar)	71.25	71.25	71.25	71.25
33	Temperature (C)	53.57	53.57	53.56	53.56
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.45	16.45	16.45	16.45
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.158e+004	1.158e+004	1.159e+004	1.160e+004
44	Pressure (bar)	71.25	71.25	71.25	71.25
45	Temperature (C)	53.56	53.56	53.56	53.55
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.45	16.45	16.45	16.45
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.160e+004	1.161e+004	1.161e+004	1.162e+004
56	Pressure (bar)	71.25	71.25	71.25	71.25
57	Temperature (C)	53.55	53.55	53.55	53.54
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.45	16.45	16.45	16.45
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.163e+004	1.163e+004	1.164e+004	1.165e+004
68	Pressure (bar)	71.25	71.25	71.25	71.25
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 760 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	53.54	53.54	53.54	53.53	53.53
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.165e+004	1.166e+004	1.166e+004	1.167e+004	1.167e+004
24	Pressure (bar)	71.25	71.25	71.25	71.25	71.25
25	Temperature (C)	53.53	53.53	53.53	53.53	53.52
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.168e+004	1.168e+004	1.169e+004	1.169e+004	1.170e+004
36	Pressure (bar)	71.24	71.24	71.24	71.24	71.24
37	Temperature (C)	53.52	53.52	53.52	53.52	53.51
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.170e+004	1.171e+004	1.171e+004	1.172e+004	1.172e+004
48	Pressure (bar)	71.24	71.24	71.24	71.24	71.24
49	Temperature (C)	53.51	53.51	53.51	53.51	53.50
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.173e+004	1.173e+004	1.174e+004	1.174e+004	1.175e+004
60	Pressure (bar)	71.24	71.24	71.24	71.24	71.24
61	Temperature (C)	53.50	53.50	53.50	53.50	53.49
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 761 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.175e+004	1.176e+004	1.176e+004	1.177e+004	1.177e+004
16	Pressure (bar)	71.24	71.24	71.24	71.24	71.24
17	Temperature (C)	53.49	53.49	53.49	53.49	53.48
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.178e+004	1.178e+004	1.179e+004	1.179e+004	1.180e+004
28	Pressure (bar)	71.24	71.24	71.24	71.24	71.24
29	Temperature (C)	53.48	53.48	53.48	53.48	53.48
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.180e+004	1.181e+004	1.181e+004	1.182e+004	1.182e+004
40	Pressure (bar)	71.23	71.23	71.23	71.23	71.23
41	Temperature (C)	53.47	53.47	53.47	53.47	53.47
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.183e+004	1.183e+004	1.184e+004	1.184e+004	1.185e+004
52	Pressure (bar)	71.23	71.23	71.23	71.23	71.23
53	Temperature (C)	53.46	53.46	53.46	53.46	53.46
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.185e+004	1.186e+004	1.186e+004	1.187e+004	1.187e+004
64	Pressure (bar)	71.23	71.23	71.23	71.23	71.23
65	Temperature (C)	53.45	53.45	53.45	53.45	53.45
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.188e+004	1.188e+004	1.189e+004	1.189e+004	1.190e+004
20	Pressure (bar)	71.23	71.23	71.23	71.23	71.23
21	Temperature (C)	53.44	53.44	53.44	53.44	53.44
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.190e+004	1.191e+004	1.191e+004	1.192e+004	1.192e+004
32	Pressure (bar)	71.23	71.23	71.23	71.22	71.22
33	Temperature (C)	53.44	53.43	53.43	53.43	53.43
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.193e+004	1.193e+004	1.194e+004	1.194e+004	1.194e+004
44	Pressure (bar)	71.22	71.22	71.22	71.22	71.22
45	Temperature (C)	53.43	53.42	53.42	53.42	53.42
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.195e+004	1.196e+004	1.196e+004	1.197e+004	1.198e+004
56	Pressure (bar)	71.22	71.22	71.22	71.22	71.22
57	Temperature (C)	53.42	53.42	53.41	53.41	53.41
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.198e+004	1.199e+004	1.200e+004	1.200e+004	1.201e+004
68	Pressure (bar)	71.22	71.22	71.22	71.22	71.22
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:13:43 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	53.40	53.40	53.40	53.40	53.39
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.202e+004	1.203e+004	1.203e+004	1.204e+004	1.205e+004
24	Pressure (bar)	71.22	71.22	71.22	71.22	71.22
25	Temperature (C)	53.39	53.39	53.39	53.38	53.38
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.205e+004	1.206e+004	1.207e+004	1.207e+004	1.208e+004
36	Pressure (bar)	71.22	71.21	71.21	71.21	71.20
37	Temperature (C)	53.38	53.38	53.37	53.37	53.36
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.209e+004	1.210e+004	1.210e+004	1.211e+004	1.212e+004
48	Pressure (bar)	71.20	71.20	71.20	71.20	71.20
49	Temperature (C)	53.36	53.36	53.36	53.35	53.35
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.212e+004	1.213e+004	1.214e+004	1.214e+004	1.215e+004
60	Pressure (bar)	71.20	71.20	71.20	71.20	71.20
61	Temperature (C)	53.35	53.35	53.35	53.34	53.34
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 764 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.216e+004	1.217e+004	1.217e+004	1.218e+004	1.219e+004
16	Pressure (bar)	71.20	71.20	71.20	71.20	71.20
17	Temperature (C)	53.34	53.34	53.33	53.33	53.33
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.219e+004	1.220e+004	1.221e+004	1.221e+004	1.222e+004
28	Pressure (bar)	71.20	71.20	71.20	71.20	71.20
29	Temperature (C)	53.32	53.32	53.32	53.32	53.31
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.223e+004	1.223e+004	1.224e+004	1.225e+004	1.225e+004
40	Pressure (bar)	71.20	71.20	71.20	71.20	71.20
41	Temperature (C)	53.31	53.31	53.31	53.30	53.30
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.226e+004	1.226e+004	1.227e+004	1.227e+004	1.228e+004
52	Pressure (bar)	71.20	71.20	71.20	71.20	71.20
53	Temperature (C)	53.30	53.30	53.30	53.30	53.29
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.45	16.45	16.45	16.45	16.45
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.228e+004	1.229e+004	1.229e+004	1.230e+004	1.230e+004
64	Pressure (bar)	71.20	71.19	71.19	71.19	71.19
65	Temperature (C)	53.29	53.29	53.29	53.29	53.28
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 765 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.45	16.45	16.45	16.44	16.44
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.231e+004	1.231e+004	1.232e+004	1.232e+004	1.233e+004
20	Pressure (bar)	71.19	71.19	71.19	71.19	71.19
21	Temperature (C)	53.28	53.28	53.28	53.28	53.27
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.233e+004	1.234e+004	1.234e+004	1.235e+004	1.235e+004
32	Pressure (bar)	71.19	71.19	71.19	71.19	71.19
33	Temperature (C)	53.27	53.27	53.27	53.27	53.26
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.236e+004	1.237e+004	1.237e+004	1.238e+004	1.239e+004
44	Pressure (bar)	71.19	71.19	71.19	71.19	71.19
45	Temperature (C)	53.26	53.26	53.26	53.26	53.25
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.239e+004	1.240e+004	1.240e+004	1.241e+004	1.241e+004
56	Pressure (bar)	71.19	71.18	71.18	71.18	71.18
57	Temperature (C)	53.25	53.25	53.25	53.24	53.24
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.242e+004	1.242e+004	1.243e+004	1.243e+004	1.244e+004
68	Pressure (bar)	71.18	71.18	71.18	71.18	71.18
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	53.24	53.24	53.24	53.23	53.23
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.244e+004	1.245e+004	1.245e+004	1.246e+004	1.246e+004
24	Pressure (bar)	71.18	71.18	71.18	71.18	71.18
25	Temperature (C)	53.23	53.23	53.23	53.22	53.22
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.247e+004	1.247e+004	1.248e+004	1.248e+004	1.249e+004
36	Pressure (bar)	71.18	71.18	71.18	71.18	71.18
37	Temperature (C)	53.22	53.22	53.22	53.22	53.21
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.249e+004	1.250e+004	1.250e+004	1.251e+004	1.251e+004
48	Pressure (bar)	71.18	71.18	71.18	71.18	71.18
49	Temperature (C)	53.21	53.21	53.21	53.21	53.21
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.252e+004	1.252e+004	1.253e+004	1.253e+004	1.254e+004
60	Pressure (bar)	71.18	71.18	71.18	71.18	71.18
61	Temperature (C)	53.21	53.20	53.20	53.20	53.20
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.254e+004	1.255e+004	1.255e+004	1.256e+004	1.256e+004
16	Pressure (bar)	71.18	71.18	71.17	71.17	71.17
17	Temperature (C)	53.20	53.19	53.19	53.19	53.19
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.257e+004	1.257e+004	1.258e+004	1.258e+004	1.259e+004
28	Pressure (bar)	71.17	71.17	71.17	71.17	71.17
29	Temperature (C)	53.18	53.18	53.18	53.18	53.18
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.259e+004	1.260e+004	1.260e+004	1.261e+004	1.261e+004
40	Pressure (bar)	71.17	71.16	71.16	71.16	71.16
41	Temperature (C)	53.17	53.17	53.17	53.17	53.16
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.262e+004	1.262e+004	1.263e+004	1.264e+004	1.265e+004
52	Pressure (bar)	71.16	71.16	71.16	71.16	71.15
53	Temperature (C)	53.16	53.16	53.16	53.15	53.15
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.265e+004	1.266e+004	1.266e+004	1.267e+004	1.267e+004
64	Pressure (bar)	71.15	71.15	71.15	71.15	71.15
65	Temperature (C)	53.15	53.15	53.14	53.14	53.14
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 768 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.268e+004	1.268e+004	1.269e+004	1.269e+004	1.270e+004
20	Pressure (bar)	71.15	71.15	71.15	71.15	71.15
21	Temperature (C)	53.14	53.14	53.14	53.13	53.13
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.270e+004	1.271e+004	1.271e+004	1.272e+004	1.272e+004
32	Pressure (bar)	71.15	71.15	71.15	71.15	71.15
33	Temperature (C)	53.13	53.13	53.13	53.13	53.12
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.273e+004	1.273e+004	1.274e+004	1.274e+004	1.275e+004
44	Pressure (bar)	71.15	71.15	71.15	71.15	71.15
45	Temperature (C)	53.12	53.12	53.12	53.12	53.11
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.275e+004	1.276e+004	1.276e+004	1.276e+004	1.277e+004
56	Pressure (bar)	71.15	71.15	71.15	71.15	71.15
57	Temperature (C)	53.11	53.11	53.11	53.11	53.11
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.277e+004	1.278e+004	1.278e+004	1.278e+004	1.278e+004
68	Pressure (bar)	71.15	71.15	71.15	71.15	71.15
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 769 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	53.11	53.11	53.10	53.10	53.10
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.279e+004	1.279e+004	1.279e+004	1.280e+004	1.280e+004
24	Pressure (bar)	71.15	71.15	71.15	71.15	71.15
25	Temperature (C)	53.10	53.10	53.10	53.10	53.10
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.280e+004	1.281e+004	1.281e+004	1.282e+004	1.282e+004
36	Pressure (bar)	71.15	71.15	71.15	71.15	71.15
37	Temperature (C)	53.09	53.09	53.09	53.09	53.09
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.283e+004	1.283e+004	1.284e+004	1.284e+004	1.285e+004
48	Pressure (bar)	71.15	71.15	71.15	71.14	71.14
49	Temperature (C)	53.09	53.08	53.08	53.08	53.08
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.285e+004	1.286e+004	1.286e+004	1.287e+004	1.287e+004
60	Pressure (bar)	71.14	71.14	71.14	71.14	71.14
61	Temperature (C)	53.07	53.07	53.07	53.07	53.07
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 770 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.288e+004	1.288e+004	1.289e+004	1.289e+004	1.290e+004
16	Pressure (bar)	71.14	71.14	71.14	71.14	71.14
17	Temperature (C)	53.07	53.06	53.06	53.06	53.06
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.290e+004	1.291e+004	1.291e+004	1.292e+004	1.292e+004
28	Pressure (bar)	71.13	71.13	71.13	71.13	71.13
29	Temperature (C)	53.06	53.05	53.05	53.05	53.05
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.293e+004	1.293e+004	1.294e+004	1.294e+004	1.295e+004
40	Pressure (bar)	71.13	71.13	71.13	71.13	71.13
41	Temperature (C)	53.05	53.04	53.04	53.04	53.04
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.295e+004	1.296e+004	1.296e+004	1.297e+004	1.297e+004
52	Pressure (bar)	71.13	71.13	71.13	71.13	71.13
53	Temperature (C)	53.04	53.04	53.03	53.03	53.03
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.298e+004	1.298e+004	1.299e+004	1.299e+004	1.300e+004
64	Pressure (bar)	71.13	71.13	71.13	71.13	71.13
65	Temperature (C)	53.03	53.03	53.02	53.02	53.02
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 771 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.44	16.44	16.44	16.44
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.300e+004	1.301e+004	1.301e+004	1.302e+004
20	Pressure (bar)	71.13	71.13	71.12	71.12
21	Temperature (C)	53.02	53.02	53.01	53.01
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.44	16.44	16.44	16.44
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.303e+004	1.303e+004	1.304e+004	1.305e+004
32	Pressure (bar)	71.12	71.12	71.12	71.12
33	Temperature (C)	53.01	53.01	53.00	53.00
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.44	16.44	16.44	16.44
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.305e+004	1.306e+004	1.306e+004	1.307e+004
44	Pressure (bar)	71.12	71.12	71.12	71.12
45	Temperature (C)	53.00	53.00	53.00	52.99
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.44	16.44	16.44	16.44
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.308e+004	1.308e+004	1.309e+004	1.310e+004
56	Pressure (bar)	71.12	71.12	71.12	71.12
57	Temperature (C)	52.99	52.99	52.99	52.98
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.44	16.44	16.44	16.44
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.310e+004	1.311e+004	1.311e+004	1.312e+004
68	Pressure (bar)	71.12	71.12	71.12	71.11
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 772 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	52.98	52.98	52.98	52.97	52.97
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.313e+004	1.313e+004	1.314e+004	1.314e+004	1.315e+004
24	Pressure (bar)	71.11	71.11	71.11	71.11	71.11
25	Temperature (C)	52.97	52.97	52.97	52.96	52.96
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.315e+004	1.316e+004	1.316e+004	1.317e+004	1.317e+004
36	Pressure (bar)	71.11	71.11	71.11	71.11	71.11
37	Temperature (C)	52.96	52.96	52.96	52.96	52.95
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.318e+004	1.318e+004	1.319e+004	1.319e+004	1.320e+004
48	Pressure (bar)	71.11	71.11	71.11	71.11	71.11
49	Temperature (C)	52.95	52.95	52.95	52.95	52.94
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.320e+004	1.321e+004	1.321e+004	1.322e+004	1.322e+004
60	Pressure (bar)	71.11	71.11	71.11	71.12	71.12
61	Temperature (C)	52.94	52.94	52.94	52.94	52.94
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 773 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.323e+004	1.323e+004	1.324e+004	1.324e+004	1.325e+004
16	Pressure (bar)	71.12	71.12	71.12	71.13	71.13
17	Temperature (C)	52.94	52.94	52.94	52.94	52.94
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.325e+004	1.326e+004	1.326e+004	1.327e+004	1.327e+004
28	Pressure (bar)	71.13	71.14	71.14	71.15	71.15
29	Temperature (C)	52.94	52.94	52.94	52.94	52.94
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.328e+004	1.328e+004	1.329e+004	1.329e+004	1.330e+004
40	Pressure (bar)	71.16	71.16	71.17	71.17	71.17
41	Temperature (C)	52.95	52.95	52.95	52.95	52.94
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.330e+004	1.331e+004	1.331e+004	1.332e+004	1.332e+004
52	Pressure (bar)	71.17	71.18	71.18	71.18	71.19
53	Temperature (C)	52.94	52.94	52.94	52.94	52.94
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.44	16.44	16.44	16.44	16.44
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.333e+004	1.333e+004	1.334e+004	1.334e+004	1.335e+004
64	Pressure (bar)	71.19	71.20	71.20	71.21	71.22
65	Temperature (C)	52.94	52.95	52.95	52.95	52.95
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 774 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.44	16.45	16.45	16.45
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.335e+004	1.336e+004	1.336e+004	1.337e+004
20	Pressure (bar)	71.22	71.23	71.23	71.24
21	Temperature (C)	52.95	52.96	52.96	52.96
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.45	16.45	16.45	16.45
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.338e+004	1.338e+004	1.339e+004	1.340e+004
32	Pressure (bar)	71.25	71.25	71.25	71.25
33	Temperature (C)	52.96	52.96	52.96	52.95
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.45	16.45	16.45	16.45
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.340e+004	1.341e+004	1.341e+004	1.342e+004
44	Pressure (bar)	71.25	71.23	71.21	71.20
45	Temperature (C)	52.95	52.94	52.94	52.93
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.45	16.45	16.45	16.44
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.343e+004	1.343e+004	1.344e+004	1.345e+004
56	Pressure (bar)	71.16	71.15	71.14	71.13
57	Temperature (C)	52.92	52.91	52.91	52.90
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.44	16.44	16.44	16.44
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.345e+004	1.346e+004	1.346e+004	1.347e+004
68	Pressure (bar)	71.11	70.99	70.87	70.75
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 775 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	52.89	52.86	52.82	52.79	52.75
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.44	16.43	16.42	16.40	16.39
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.348e+004	1.348e+004	1.349e+004	1.349e+004	1.350e+004
24	Pressure (bar)	70.50	70.50	70.49	70.49	70.48
25	Temperature (C)	52.72	52.72	52.71	52.71	52.71
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.350e+004	1.351e+004	1.351e+004	1.352e+004	1.352e+004
36	Pressure (bar)	70.48	70.48	70.48	70.48	70.48
37	Temperature (C)	52.70	52.70	52.70	52.70	52.70
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.353e+004	1.353e+004	1.353e+004	1.353e+004	1.354e+004
48	Pressure (bar)	70.48	70.48	70.48	70.48	70.47
49	Temperature (C)	52.69	52.69	52.69	52.69	52.69
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.354e+004	1.354e+004	1.354e+004	1.355e+004	1.355e+004
60	Pressure (bar)	70.47	70.47	70.47	70.47	70.47
61	Temperature (C)	52.69	52.69	52.69	52.69	52.68
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 776 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.355e+004	1.356e+004	1.356e+004	1.356e+004	1.357e+004
16	Pressure (bar)	70.47	70.47	70.47	70.47	70.47
17	Temperature (C)	52.68	52.68	52.68	52.68	52.68
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.357e+004	1.357e+004	1.358e+004	1.358e+004	1.358e+004
28	Pressure (bar)	70.47	70.47	70.47	70.47	70.47
29	Temperature (C)	52.68	52.68	52.67	52.67	52.67
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.359e+004	1.359e+004	1.359e+004	1.360e+004	1.360e+004
40	Pressure (bar)	70.47	70.47	70.47	70.47	70.47
41	Temperature (C)	52.67	52.67	52.67	52.67	52.67
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.360e+004	1.361e+004	1.361e+004	1.362e+004	1.362e+004
52	Pressure (bar)	70.47	70.47	70.47	70.47	70.47
53	Temperature (C)	52.67	52.66	52.66	52.66	52.66
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.363e+004	1.363e+004	1.364e+004	1.364e+004	1.365e+004
64	Pressure (bar)	70.47	70.47	70.47	70.47	70.46
65	Temperature (C)	52.66	52.65	52.65	52.65	52.65
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 777 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.365e+004	1.366e+004	1.366e+004	1.367e+004	1.367e+004
20	Pressure (bar)	70.46	70.46	70.46	70.46	70.46
21	Temperature (C)	52.65	52.64	52.64	52.64	52.64
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.368e+004	1.368e+004	1.369e+004	1.369e+004	1.370e+004
32	Pressure (bar)	70.46	70.46	70.46	70.46	70.46
33	Temperature (C)	52.64	52.64	52.63	52.63	52.63
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.370e+004	1.371e+004	1.371e+004	1.372e+004	1.372e+004
44	Pressure (bar)	70.46	70.46	70.46	70.46	70.46
45	Temperature (C)	52.63	52.63	52.62	52.62	52.62
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.373e+004	1.373e+004	1.374e+004	1.374e+004	1.375e+004
56	Pressure (bar)	70.46	70.46	70.46	70.46	70.46
57	Temperature (C)	52.62	52.62	52.62	52.61	52.61
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.375e+004	1.376e+004	1.376e+004	1.377e+004	1.377e+004
68	Pressure (bar)	70.46	70.46	70.46	70.46	70.46
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	52.61	52.61	52.61	52.61	52.60
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.378e+004	1.378e+004	1.379e+004	1.379e+004	1.380e+004
24	Pressure (bar)	70.46	70.46	70.46	70.46	70.46
25	Temperature (C)	52.60	52.60	52.60	52.60	52.60
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.380e+004	1.381e+004	1.381e+004	1.382e+004	1.382e+004
36	Pressure (bar)	70.46	70.45	70.45	70.45	70.45
37	Temperature (C)	52.59	52.59	52.59	52.59	52.58
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.383e+004	1.383e+004	1.384e+004	1.384e+004	1.385e+004
48	Pressure (bar)	70.45	70.45	70.45	70.45	70.45
49	Temperature (C)	52.58	52.58	52.58	52.58	52.58
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.385e+004	1.386e+004	1.386e+004	1.387e+004	1.387e+004
60	Pressure (bar)	70.46	70.45	70.45	70.45	70.45
61	Temperature (C)	52.58	52.57	52.57	52.57	52.57
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.388e+004	1.388e+004	1.389e+004	1.389e+004	1.390e+004
16	Pressure (bar)	70.45	70.45	70.45	70.45	70.45
17	Temperature (C)	52.57	52.56	52.56	52.56	52.56
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.390e+004	1.391e+004	1.391e+004	1.392e+004	1.392e+004
28	Pressure (bar)	70.45	70.45	70.45	70.44	70.44
29	Temperature (C)	52.56	52.55	52.55	52.55	52.55
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.393e+004	1.393e+004	1.394e+004	1.394e+004	1.395e+004
40	Pressure (bar)	70.44	70.44	70.44	70.44	70.44
41	Temperature (C)	52.55	52.54	52.54	52.54	52.54
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.395e+004	1.396e+004	1.396e+004	1.397e+004	1.397e+004
52	Pressure (bar)	70.44	70.44	70.44	70.44	70.44
53	Temperature (C)	52.54	52.54	52.53	52.53	52.53
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.398e+004	1.398e+004	1.399e+004	1.399e+004	1.400e+004
64	Pressure (bar)	70.44	70.44	70.44	70.44	70.43
65	Temperature (C)	52.53	52.53	52.52	52.52	52.52
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.400e+004	1.401e+004	1.401e+004	1.402e+004	1.402e+004
20	Pressure (bar)	70.43	70.43	70.44	70.44	70.44
21	Temperature (C)	52.52	52.52	52.52	52.51	52.51
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.403e+004	1.403e+004	1.404e+004	1.404e+004	1.405e+004
32	Pressure (bar)	70.44	70.44	70.43	70.43	70.43
33	Temperature (C)	52.51	52.51	52.51	52.50	52.50
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.405e+004	1.405e+004	1.406e+004	1.406e+004	1.406e+004
44	Pressure (bar)	70.43	70.43	70.43	70.43	70.43
45	Temperature (C)	52.50	52.50	52.50	52.50	52.50
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.406e+004	1.406e+004	1.407e+004	1.407e+004	1.408e+004
56	Pressure (bar)	70.43	70.43	70.43	70.43	70.43
57	Temperature (C)	52.50	52.50	52.49	52.49	52.49
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.408e+004	1.408e+004	1.409e+004	1.409e+004	1.409e+004
68	Pressure (bar)	70.43	70.43	70.43	70.43	70.43
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:13:43 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	52.49	52.49	52.49	52.49	52.49
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.409e+004	1.409e+004	1.410e+004	1.411e+004	1.411e+004
24	Pressure (bar)	70.43	70.43	70.43	70.43	70.43
25	Temperature (C)	52.49	52.49	52.48	52.48	52.48
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.412e+004	1.413e+004	1.413e+004	1.414e+004	1.415e+004
36	Pressure (bar)	70.43	70.43	70.43	70.43	70.43
37	Temperature (C)	52.48	52.48	52.47	52.47	52.47
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.415e+004	1.416e+004	1.416e+004	1.417e+004	1.417e+004
48	Pressure (bar)	70.43	70.43	70.42	70.42	70.42
49	Temperature (C)	52.47	52.46	52.46	52.46	52.46
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.418e+004	1.418e+004	1.419e+004	1.419e+004	1.420e+004
60	Pressure (bar)	70.42	70.42	70.42	70.42	70.42
61	Temperature (C)	52.45	52.45	52.45	52.45	52.45
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.38	16.38	16.38	16.38	16.38
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 782 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.420e+004	1.421e+004	1.421e+004	1.422e+004	1.422e+004
16	Pressure (bar)	70.41	70.41	70.41	70.41	70.41
17	Temperature (C)	52.44	52.44	52.44	52.44	52.44
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.38	16.38	16.37	16.37	16.37
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.423e+004	1.423e+004	1.424e+004	1.424e+004	1.425e+004
28	Pressure (bar)	70.41	70.41	70.41	70.41	70.41
29	Temperature (C)	52.43	52.43	52.43	52.43	52.43
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.425e+004	1.426e+004	1.426e+004	1.427e+004	1.427e+004
40	Pressure (bar)	70.41	70.41	70.41	70.40	70.40
41	Temperature (C)	52.42	52.42	52.42	52.42	52.42
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.428e+004	1.428e+004	1.429e+004	1.429e+004	1.430e+004
52	Pressure (bar)	70.40	70.40	70.40	70.40	70.40
53	Temperature (C)	52.41	52.41	52.41	52.41	52.41
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.430e+004	1.431e+004	1.431e+004	1.432e+004	1.432e+004
64	Pressure (bar)	70.40	70.40	70.40	70.40	70.40
65	Temperature (C)	52.41	52.40	52.40	52.40	52.40
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.433e+004	1.433e+004	1.434e+004	1.434e+004	1.435e+004
20	Pressure (bar)	70.40	70.40	70.40	70.40	70.40
21	Temperature (C)	52.40	52.39	52.39	52.39	52.39
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.435e+004	1.436e+004	1.436e+004	1.437e+004	1.437e+004
32	Pressure (bar)	70.40	70.40	70.40	70.40	70.40
33	Temperature (C)	52.39	52.39	52.38	52.38	52.38
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.438e+004	1.438e+004	1.439e+004	1.439e+004	1.440e+004
44	Pressure (bar)	70.40	70.40	70.40	70.40	70.39
45	Temperature (C)	52.38	52.38	52.37	52.37	52.37
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.440e+004	1.441e+004	1.441e+004	1.442e+004	1.442e+004
56	Pressure (bar)	70.39	70.39	70.39	70.39	70.39
57	Temperature (C)	52.37	52.37	52.37	52.36	52.36
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.443e+004	1.443e+004	1.444e+004	1.444e+004	1.445e+004
68	Pressure (bar)	70.39	70.39	70.39	70.39	70.39
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:13:43 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	52.36	52.36	52.36	52.35	52.35
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.445e+004	1.446e+004	1.446e+004	1.447e+004	1.447e+004
24	Pressure (bar)	70.39	70.39	70.39	70.39	70.39
25	Temperature (C)	52.35	52.35	52.35	52.35	52.34
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.448e+004	1.448e+004	1.449e+004	1.449e+004	1.450e+004
36	Pressure (bar)	70.39	70.39	70.39	70.39	70.39
37	Temperature (C)	52.34	52.34	52.34	52.34	52.33
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.450e+004	1.451e+004	1.451e+004	1.452e+004	1.452e+004
48	Pressure (bar)	70.39	70.39	70.39	70.39	70.39
49	Temperature (C)	52.33	52.33	52.33	52.33	52.33
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.453e+004	1.453e+004	1.454e+004	1.454e+004	1.455e+004
60	Pressure (bar)	70.38	70.38	70.38	70.38	70.38
61	Temperature (C)	52.32	52.32	52.32	52.32	52.32
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 785 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.455e+004	1.456e+004	1.456e+004	1.457e+004	1.457e+004
16	Pressure (bar)	70.38	70.38	70.38	70.38	70.38
17	Temperature (C)	52.31	52.31	52.31	52.31	52.31
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.458e+004	1.458e+004	1.459e+004	1.459e+004	1.460e+004
28	Pressure (bar)	70.38	70.38	70.38	70.38	70.38
29	Temperature (C)	52.31	52.30	52.30	52.30	52.30
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.460e+004	1.461e+004	1.461e+004	1.462e+004	1.462e+004
40	Pressure (bar)	70.38	70.38	70.38	70.38	70.38
41	Temperature (C)	52.30	52.29	52.29	52.29	52.29
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.463e+004	1.463e+004	1.464e+004	1.464e+004	1.465e+004
52	Pressure (bar)	70.38	70.38	70.38	70.38	70.38
53	Temperature (C)	52.29	52.29	52.28	52.28	52.28
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.465e+004	1.466e+004	1.466e+004	1.467e+004	1.467e+004
64	Pressure (bar)	70.37	70.37	70.37	70.37	70.37
65	Temperature (C)	52.28	52.28	52.27	52.27	52.27
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.468e+004	1.468e+004	1.469e+004	1.469e+004	1.470e+004
20	Pressure (bar)	70.37	70.37	70.37	70.37	70.37
21	Temperature (C)	52.27	52.27	52.27	52.26	52.26
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.470e+004	1.471e+004	1.471e+004	1.472e+004	1.472e+004
32	Pressure (bar)	70.37	70.37	70.37	70.37	70.37
33	Temperature (C)	52.26	52.26	52.26	52.25	52.25
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.473e+004	1.473e+004	1.474e+004	1.474e+004	1.475e+004
44	Pressure (bar)	70.37	70.37	70.37	70.37	70.37
45	Temperature (C)	52.25	52.25	52.25	52.25	52.24
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.475e+004	1.476e+004	1.476e+004	1.477e+004	1.477e+004
56	Pressure (bar)	70.37	70.37	70.37	70.37	70.37
57	Temperature (C)	52.24	52.24	52.24	52.24	52.24
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.478e+004	1.478e+004	1.479e+004	1.479e+004	1.480e+004
68	Pressure (bar)	70.37	70.36	70.36	70.36	70.36
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	52.23	52.23	52.23	52.23	52.23
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.480e+004	1.481e+004	1.481e+004	1.482e+004	1.482e+004
24	Pressure (bar)	70.36	70.36	70.36	70.36	70.36
25	Temperature (C)	52.22	52.22	52.22	52.22	52.22
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.483e+004	1.483e+004	1.484e+004	1.484e+004	1.485e+004
36	Pressure (bar)	70.36	70.36	70.36	70.36	70.36
37	Temperature (C)	52.22	52.21	52.21	52.21	52.21
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.485e+004	1.486e+004	1.486e+004	1.486e+004	1.486e+004
48	Pressure (bar)	70.36	70.36	70.36	70.36	70.36
49	Temperature (C)	52.21	52.21	52.20	52.20	52.20
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.486e+004	1.488e+004	1.490e+004	1.492e+004	1.494e+004
60	Pressure (bar)	70.36	70.36	70.36	70.35	70.35
61	Temperature (C)	52.20	52.20	52.19	52.18	52.17
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.496e+004	1.498e+004	1.500e+004	1.502e+004	1.504e+004
16	Pressure (bar)	70.35	70.35	70.35	70.35	70.34
17	Temperature (C)	52.17	52.16	52.15	52.14	52.14
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.506e+004	1.508e+004	1.510e+004	1.512e+004	1.514e+004
28	Pressure (bar)	70.34	70.34	70.34	70.34	70.34
29	Temperature (C)	52.13	52.12	52.12	52.11	52.10
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.516e+004	1.518e+004	1.520e+004	1.522e+004	1.524e+004
40	Pressure (bar)	70.33	70.33	70.33	70.33	70.33
41	Temperature (C)	52.09	52.09	52.08	52.07	52.07
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.526e+004	1.528e+004	1.530e+004	1.532e+004	1.534e+004
52	Pressure (bar)	70.33	70.33	70.32	70.32	70.32
53	Temperature (C)	52.06	52.05	52.04	52.04	52.03
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.37	16.37	16.37	16.37	16.37
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.536e+004	1.538e+004	1.540e+004	1.542e+004	1.544e+004
64	Pressure (bar)	70.32	70.33	70.34	70.34	70.35
65	Temperature (C)	52.02	52.02	52.02	52.02	52.01
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 789 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.37	16.37	16.37	16.37
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.546e+004	1.548e+004	1.550e+004	1.552e+004
20	Pressure (bar)	70.36	70.34	70.32	70.31
21	Temperature (C)	52.01	52.00	51.98	51.97
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.37	16.37	16.37	16.36
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.556e+004	1.558e+004	1.560e+004	1.562e+004
32	Pressure (bar)	70.27	70.27	70.26	70.26
33	Temperature (C)	51.94	51.94	51.93	51.92
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.36	16.36	16.36	16.36
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.566e+004	1.568e+004	1.570e+004	1.572e+004
44	Pressure (bar)	70.26	70.25	70.25	70.25
45	Temperature (C)	51.91	51.90	51.89	51.88
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.36	16.36	16.36	16.36
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.576e+004	1.577e+004	1.577e+004	1.578e+004
56	Pressure (bar)	70.25	70.25	70.25	70.25
57	Temperature (C)	51.87	51.87	51.87	51.86
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.36	16.36	16.36	16.36
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.579e+004	1.581e+004	1.583e+004	1.587e+004
68	Pressure (bar)	70.25	70.25	70.24	70.24
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 790 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	51.86	51.85	51.85	51.84	51.83
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.589e+004	1.589e+004	1.590e+004	1.590e+004	1.590e+004
24	Pressure (bar)	70.24	70.24	70.24	70.24	70.23
25	Temperature (C)	51.83	51.82	51.82	51.82	51.82
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.591e+004	1.591e+004	1.592e+004	1.592e+004	1.592e+004
36	Pressure (bar)	70.23	70.23	70.23	70.23	70.23
37	Temperature (C)	51.82	51.82	51.82	51.81	51.81
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.593e+004	1.593e+004	1.593e+004	1.593e+004	1.594e+004
48	Pressure (bar)	70.23	70.23	70.23	70.23	70.23
49	Temperature (C)	51.81	51.81	51.81	51.81	51.81
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.594e+004	1.594e+004	1.595e+004	1.595e+004	1.595e+004
60	Pressure (bar)	70.23	70.23	70.23	70.23	70.23
61	Temperature (C)	51.81	51.81	51.80	51.80	51.80
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 791 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.596e+004	1.596e+004	1.597e+004	1.597e+004	1.598e+004
16	Pressure (bar)	70.23	70.23	70.23	70.23	70.23
17	Temperature (C)	51.80	51.80	51.80	51.79	51.79
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.598e+004	1.599e+004	1.599e+004	1.600e+004	1.600e+004
28	Pressure (bar)	70.23	70.23	70.23	70.22	70.22
29	Temperature (C)	51.79	51.79	51.79	51.79	51.78
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.601e+004	1.601e+004	1.602e+004	1.602e+004	1.603e+004
40	Pressure (bar)	70.22	70.22	70.22	70.22	70.22
41	Temperature (C)	51.78	51.78	51.78	51.78	51.77
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.603e+004	1.604e+004	1.604e+004	1.605e+004	1.605e+004
52	Pressure (bar)	70.22	70.22	70.22	70.22	70.22
53	Temperature (C)	51.77	51.77	51.77	51.77	51.77
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.606e+004	1.606e+004	1.607e+004	1.607e+004	1.608e+004
64	Pressure (bar)	70.22	70.22	70.22	70.22	70.22
65	Temperature (C)	51.76	51.76	51.76	51.76	51.76
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 792 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.608e+004	1.609e+004	1.609e+004	1.610e+004	1.610e+004
20	Pressure (bar)	70.22	70.22	70.22	70.22	70.22
21	Temperature (C)	51.76	51.75	51.75	51.75	51.75
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.611e+004	1.611e+004	1.612e+004	1.612e+004	1.613e+004
32	Pressure (bar)	70.22	70.22	70.22	70.22	70.22
33	Temperature (C)	51.75	51.75	51.74	51.74	51.74
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.613e+004	1.614e+004	1.614e+004	1.615e+004	1.615e+004
44	Pressure (bar)	70.22	70.22	70.22	70.22	70.22
45	Temperature (C)	51.74	51.74	51.74	51.73	51.73
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.616e+004	1.616e+004	1.617e+004	1.617e+004	1.618e+004
56	Pressure (bar)	70.22	70.22	70.22	70.22	70.22
57	Temperature (C)	51.73	51.73	51.73	51.73	51.72
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.618e+004	1.619e+004	1.619e+004	1.620e+004	1.620e+004
68	Pressure (bar)	70.22	70.22	70.22	70.22	70.22
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 793 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:13:43 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	51.72	51.72	51.72	51.72	51.72
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.621e+004	1.621e+004	1.622e+004	1.622e+004	1.623e+004
24	Pressure (bar)	70.22	70.21	70.21	70.21	70.20
25	Temperature (C)	51.71	51.71	51.71	51.71	51.70
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.623e+004	1.624e+004	1.624e+004	1.625e+004	1.625e+004
36	Pressure (bar)	70.20	70.20	70.20	70.20	70.20
37	Temperature (C)	51.70	51.70	51.70	51.70	51.69
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.626e+004	1.626e+004	1.627e+004	1.627e+004	1.628e+004
48	Pressure (bar)	70.20	70.20	70.20	70.20	70.20
49	Temperature (C)	51.69	51.69	51.69	51.69	51.68
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.628e+004	1.629e+004	1.629e+004	1.630e+004	1.630e+004
60	Pressure (bar)	70.20	70.20	70.20	70.20	70.20
61	Temperature (C)	51.68	51.68	51.68	51.68	51.68
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 794 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.631e+004	1.631e+004	1.631e+004	1.631e+004	1.632e+004
16	Pressure (bar)	70.20	70.20	70.20	70.20	70.20
17	Temperature (C)	51.67	51.67	51.67	51.67	51.67
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.36
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.632e+004	1.632e+004	1.632e+004	1.633e+004	1.633e+004
28	Pressure (bar)	70.20	70.20	70.20	70.19	70.19
29	Temperature (C)	51.67	51.67	51.67	51.67	51.67
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.36	16.36	16.36	16.36	16.35
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.633e+004	1.634e+004	1.634e+004	1.635e+004	1.635e+004
40	Pressure (bar)	70.19	70.19	70.19	70.19	70.19
41	Temperature (C)	51.67	51.66	51.66	51.66	51.66
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.636e+004	1.636e+004	1.637e+004	1.637e+004	1.638e+004
52	Pressure (bar)	70.19	70.19	70.19	70.19	70.18
53	Temperature (C)	51.66	51.65	51.65	51.65	51.65
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.638e+004	1.639e+004	1.639e+004	1.640e+004	1.640e+004
64	Pressure (bar)	70.18	70.18	70.18	70.18	70.18
65	Temperature (C)	51.65	51.64	51.64	51.64	51.64
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 795 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.641e+004	1.641e+004	1.642e+004	1.642e+004	1.643e+004
20	Pressure (bar)	70.18	70.18	70.18	70.18	70.18
21	Temperature (C)	51.64	51.64	51.63	51.63	51.63
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.643e+004	1.644e+004	1.644e+004	1.645e+004	1.645e+004
32	Pressure (bar)	70.18	70.18	70.18	70.18	70.18
33	Temperature (C)	51.63	51.63	51.63	51.62	51.62
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.646e+004	1.646e+004	1.647e+004	1.647e+004	1.648e+004
44	Pressure (bar)	70.18	70.18	70.18	70.18	70.18
45	Temperature (C)	51.62	51.62	51.62	51.61	51.61
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.648e+004	1.649e+004	1.649e+004	1.650e+004	1.650e+004
56	Pressure (bar)	70.18	70.17	70.17	70.17	70.17
57	Temperature (C)	51.61	51.61	51.61	51.61	51.60
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.651e+004	1.651e+004	1.652e+004	1.652e+004	1.653e+004
68	Pressure (bar)	70.17	70.17	70.17	70.17	70.17
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	51.60	51.60	51.60	51.60	51.59
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.653e+004	1.654e+004	1.654e+004	1.655e+004	1.655e+004
24	Pressure (bar)	70.17	70.17	70.17	70.17	70.17
25	Temperature (C)	51.59	51.59	51.59	51.59	51.59
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.656e+004	1.656e+004	1.657e+004	1.657e+004	1.658e+004
36	Pressure (bar)	70.17	70.17	70.17	70.17	70.17
37	Temperature (C)	51.58	51.58	51.58	51.58	51.58
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.658e+004	1.659e+004	1.659e+004	1.660e+004	1.660e+004
48	Pressure (bar)	70.17	70.17	70.17	70.17	70.17
49	Temperature (C)	51.58	51.57	51.57	51.57	51.57
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.661e+004	1.661e+004	1.662e+004	1.662e+004	1.663e+004
60	Pressure (bar)	70.17	70.17	70.17	70.16	70.16
61	Temperature (C)	51.57	51.57	51.56	51.56	51.56
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 797 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.663e+004	1.664e+004	1.664e+004	1.665e+004	1.665e+004
16	Pressure (bar)	70.16	70.16	70.16	70.16	70.16
17	Temperature (C)	51.56	51.56	51.56	51.55	51.55
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.666e+004	1.666e+004	1.667e+004	1.667e+004	1.668e+004
28	Pressure (bar)	70.16	70.16	70.16	70.16	70.16
29	Temperature (C)	51.55	51.55	51.55	51.54	51.54
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.668e+004	1.669e+004	1.669e+004	1.670e+004	1.670e+004
40	Pressure (bar)	70.16	70.16	70.16	70.16	70.16
41	Temperature (C)	51.54	51.54	51.54	51.54	51.53
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.671e+004	1.671e+004	1.672e+004	1.672e+004	1.673e+004
52	Pressure (bar)	70.16	70.16	70.16	70.16	70.16
53	Temperature (C)	51.53	51.53	51.53	51.53	51.53
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.673e+004	1.674e+004	1.674e+004	1.675e+004	1.675e+004
64	Pressure (bar)	70.16	70.16	70.16	70.16	70.15
65	Temperature (C)	51.52	51.52	51.52	51.52	51.52
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 798 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.676e+004	1.676e+004	1.677e+004	1.677e+004	1.678e+004
20	Pressure (bar)	70.15	70.15	70.15	70.15	70.15
21	Temperature (C)	51.51	51.51	51.51	51.51	51.51
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.678e+004	1.679e+004	1.679e+004	1.680e+004	1.680e+004
32	Pressure (bar)	70.15	70.15	70.15	70.15	70.15
33	Temperature (C)	51.51	51.50	51.50	51.50	51.50
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.681e+004	1.681e+004	1.682e+004	1.682e+004	1.683e+004
44	Pressure (bar)	70.15	70.15	70.15	70.15	70.15
45	Temperature (C)	51.50	51.50	51.49	51.49	51.49
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.683e+004	1.684e+004	1.684e+004	1.685e+004	1.685e+004
56	Pressure (bar)	70.15	70.15	70.15	70.15	70.15
57	Temperature (C)	51.49	51.49	51.49	51.48	51.48
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.686e+004	1.686e+004	1.687e+004	1.687e+004	1.688e+004
68	Pressure (bar)	70.15	70.15	70.15	70.15	70.15
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	51.48	51.48	51.48	51.47	51.47
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.688e+004	1.689e+004	1.689e+004	1.690e+004	1.690e+004
24	Pressure (bar)	70.15	70.15	70.15	70.15	70.15
25	Temperature (C)	51.47	51.47	51.47	51.47	51.47
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.690e+004	1.691e+004	1.691e+004	1.692e+004	1.692e+004
36	Pressure (bar)	70.15	70.15	70.15	70.14	70.14
37	Temperature (C)	51.46	51.46	51.46	51.46	51.46
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.693e+004	1.693e+004	1.694e+004	1.694e+004	1.695e+004
48	Pressure (bar)	70.14	70.14	70.14	70.14	70.14
49	Temperature (C)	51.46	51.45	51.45	51.45	51.45
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.695e+004	1.696e+004	1.696e+004	1.697e+004	1.697e+004
60	Pressure (bar)	70.14	70.14	70.14	70.14	70.14
61	Temperature (C)	51.45	51.45	51.44	51.44	51.44
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.698e+004	1.698e+004	1.699e+004	1.699e+004	1.700e+004
16	Pressure (bar)	70.14	70.14	70.14	70.14	70.14
17	Temperature (C)	51.44	51.44	51.43	51.43	51.43
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.700e+004	1.701e+004	1.701e+004	1.702e+004	1.702e+004
28	Pressure (bar)	70.14	70.14	70.14	70.14	70.14
29	Temperature (C)	51.43	51.43	51.43	51.42	51.42
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.703e+004	1.703e+004	1.704e+004	1.704e+004	1.705e+004
40	Pressure (bar)	70.13	70.13	70.13	70.13	70.13
41	Temperature (C)	51.42	51.42	51.42	51.42	51.41
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.705e+004	1.706e+004	1.706e+004	1.707e+004	1.707e+004
52	Pressure (bar)	70.13	70.13	70.13	70.13	70.13
53	Temperature (C)	51.41	51.41	51.41	51.41	51.40
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.708e+004	1.708e+004	1.709e+004	1.709e+004	1.710e+004
64	Pressure (bar)	70.13	70.13	70.13	70.13	70.13
65	Temperature (C)	51.40	51.40	51.40	51.40	51.40
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 801 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.35	16.35	16.35	16.35
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.710e+004	1.711e+004	1.711e+004	1.712e+004
20	Pressure (bar)	70.13	70.13	70.13	70.12
21	Temperature (C)	51.39	51.39	51.39	51.39
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.35	16.35	16.35	16.35
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.713e+004	1.713e+004	1.714e+004	1.715e+004
32	Pressure (bar)	70.12	70.12	70.12	70.12
33	Temperature (C)	51.38	51.38	51.38	51.38
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.35	16.35	16.35	16.35
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.715e+004	1.716e+004	1.716e+004	1.717e+004
44	Pressure (bar)	70.12	70.12	70.12	70.12
45	Temperature (C)	51.37	51.37	51.37	51.37
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.35	16.35	16.35	16.35
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.718e+004	1.718e+004	1.719e+004	1.720e+004
56	Pressure (bar)	70.12	70.12	70.12	70.12
57	Temperature (C)	51.37	51.36	51.36	51.36
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.35	16.35	16.35	16.35
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.720e+004	1.721e+004	1.721e+004	1.722e+004
68	Pressure (bar)	70.12	70.12	70.11	70.11
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 802 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	51.36	51.36	51.35	51.35	51.35
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.723e+004	1.723e+004	1.724e+004	1.724e+004	1.725e+004
24	Pressure (bar)	70.11	70.11	70.11	70.11	70.11
25	Temperature (C)	51.35	51.35	51.34	51.34	51.34
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.725e+004	1.726e+004	1.726e+004	1.727e+004	1.727e+004
36	Pressure (bar)	70.11	70.11	70.11	70.11	70.11
37	Temperature (C)	51.34	51.34	51.34	51.33	51.33
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.728e+004	1.728e+004	1.729e+004	1.729e+004	1.730e+004
48	Pressure (bar)	70.11	70.11	70.11	70.11	70.11
49	Temperature (C)	51.33	51.33	51.33	51.32	51.32
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.730e+004	1.731e+004	1.731e+004	1.732e+004	1.732e+004
60	Pressure (bar)	70.11	70.11	70.10	70.10	70.10
61	Temperature (C)	51.32	51.32	51.32	51.32	51.31
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 803 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.733e+004	1.733e+004	1.734e+004	1.734e+004	1.735e+004
16	Pressure (bar)	70.10	70.10	70.10	70.10	70.10
17	Temperature (C)	51.31	51.31	51.31	51.31	51.31
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.735e+004	1.736e+004	1.736e+004	1.737e+004	1.737e+004
28	Pressure (bar)	70.10	70.10	70.10	70.10	70.10
29	Temperature (C)	51.30	51.30	51.30	51.30	51.30
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.738e+004	1.738e+004	1.739e+004	1.739e+004	1.740e+004
40	Pressure (bar)	70.10	70.10	70.10	70.10	70.10
41	Temperature (C)	51.29	51.29	51.29	51.29	51.29
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.740e+004	1.741e+004	1.741e+004	1.742e+004	1.742e+004
52	Pressure (bar)	70.10	70.10	70.10	70.10	70.10
53	Temperature (C)	51.29	51.28	51.28	51.28	51.28
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.743e+004	1.743e+004	1.744e+004	1.744e+004	1.745e+004
64	Pressure (bar)	70.10	70.10	70.10	70.10	70.09
65	Temperature (C)	51.28	51.28	51.27	51.27	51.27
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.745e+004	1.746e+004	1.746e+004	1.747e+004	1.747e+004
20	Pressure (bar)	70.09	70.09	70.09	70.09	70.09
21	Temperature (C)	51.27	51.27	51.27	51.26	51.26
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.748e+004	1.748e+004	1.749e+004	1.749e+004	1.750e+004
32	Pressure (bar)	70.09	70.09	70.09	70.09	70.09
33	Temperature (C)	51.26	51.26	51.26	51.26	51.25
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.750e+004	1.751e+004	1.751e+004	1.752e+004	1.752e+004
44	Pressure (bar)	70.09	70.09	70.09	70.09	70.09
45	Temperature (C)	51.25	51.25	51.25	51.25	51.25
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.753e+004	1.753e+004	1.754e+004	1.754e+004	1.755e+004
56	Pressure (bar)	70.09	70.09	70.09	70.09	70.09
57	Temperature (C)	51.24	51.24	51.24	51.24	51.24
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.755e+004	1.756e+004	1.756e+004	1.757e+004	1.757e+004
68	Pressure (bar)	70.09	70.09	70.09	70.09	70.09
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	51.24	51.23	51.23	51.23	51.23
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.758e+004	1.758e+004	1.759e+004	1.759e+004	1.760e+004
24	Pressure (bar)	70.09	70.09	70.09	70.08	70.08
25	Temperature (C)	51.23	51.23	51.22	51.22	51.22
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.34
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.760e+004	1.761e+004	1.761e+004	1.762e+004	1.762e+004
36	Pressure (bar)	70.08	70.08	70.08	70.08	70.08
37	Temperature (C)	51.22	51.22	51.21	51.21	51.21
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.34	16.34	16.34	16.34	16.34
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.763e+004	1.763e+004	1.764e+004	1.764e+004	1.765e+004
48	Pressure (bar)	70.08	70.08	70.08	70.08	70.08
49	Temperature (C)	51.21	51.21	51.21	51.20	51.20
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.34	16.34	16.34	16.34	16.34
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.765e+004	1.766e+004	1.766e+004	1.767e+004	1.767e+004
60	Pressure (bar)	70.08	70.08	70.08	70.08	70.08
61	Temperature (C)	51.20	51.20	51.20	51.20	51.19
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.34	16.34	16.34	16.34	16.34
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 806 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.768e+004	1.768e+004	1.769e+004	1.769e+004	1.770e+004
16	Pressure (bar)	70.08	70.08	70.08	70.08	70.08
17	Temperature (C)	51.19	51.19	51.19	51.19	51.18
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.34	16.34	16.34	16.34	16.34
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.770e+004	1.771e+004	1.771e+004	1.772e+004	1.772e+004
28	Pressure (bar)	70.08	70.07	70.07	70.07	70.07
29	Temperature (C)	51.18	51.18	51.18	51.18	51.18
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.34	16.34	16.34	16.34	16.34
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.773e+004	1.773e+004	1.774e+004	1.774e+004	1.775e+004
40	Pressure (bar)	70.07	70.07	70.07	70.07	70.07
41	Temperature (C)	51.17	51.17	51.17	51.17	51.17
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.34	16.34	16.34	16.34	16.34
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.775e+004	1.776e+004	1.776e+004	1.777e+004	1.777e+004
52	Pressure (bar)	70.07	70.07	70.07	70.07	70.07
53	Temperature (C)	51.17	51.16	51.16	51.16	51.16
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.34	16.34	16.34	16.34	16.34
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.778e+004	1.778e+004	1.779e+004	1.779e+004	1.780e+004
64	Pressure (bar)	70.07	70.07	70.07	70.07	70.07
65	Temperature (C)	51.16	51.16	51.16	51.15	51.15
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 807 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5				Date/Time: Mon Jun 13 15:13:43 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.34	16.34	16.34	16.34	16.34
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.780e+004	1.781e+004	1.781e+004	1.782e+004	1.782e+004
20	Pressure (bar)	70.07	70.07	70.07	70.07	70.07
21	Temperature (C)	51.15	51.15	51.15	51.14	51.14
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.34	16.34	16.34	16.34	16.34
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.783e+004	1.783e+004	1.784e+004	1.784e+004	1.785e+004
32	Pressure (bar)	70.07	70.07	70.07	70.07	70.07
33	Temperature (C)	51.14	51.14	51.14	51.14	51.14
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.34	16.34	16.34	16.34	16.34
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.785e+004	1.785e+004	1.786e+004	1.786e+004	1.786e+004
44	Pressure (bar)	70.07	70.07	70.07	70.07	70.07
45	Temperature (C)	51.14	51.14	51.13	51.13	51.13
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.34	16.34	16.34	16.34	16.34
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.787e+004	1.787e+004	1.787e+004	1.787e+004	1.787e+004
56	Pressure (bar)	70.07	70.07	70.08	70.08	70.08
57	Temperature (C)	51.13	51.13	51.13	51.13	51.13
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.34	16.34	16.34	16.34	16.34
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.788e+004	1.788e+004	1.789e+004	1.789e+004	1.790e+004
68	Pressure (bar)	70.08	70.08	70.09	70.10	70.10
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:13:43 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	51.13	51.13	51.13	51.14	51.14
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.34	16.35	16.35	16.35	16.35
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.790e+004	1.790e+004	1.791e+004	1.791e+004	1.791e+004
24	Pressure (bar)	70.11	70.11	70.11	70.12	70.12
25	Temperature (C)	51.14	51.14	51.14	51.14	51.14
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.791e+004	1.792e+004	1.792e+004	1.792e+004	1.793e+004
36	Pressure (bar)	70.12	70.12	70.12	70.13	70.13
37	Temperature (C)	51.14	51.14	51.14	51.14	51.14
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.793e+004	1.793e+004	1.794e+004	1.794e+004	1.794e+004
48	Pressure (bar)	70.13	70.14	70.14	70.14	70.15
49	Temperature (C)	51.14	51.15	51.15	51.15	51.15
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.795e+004	1.795e+004	1.796e+004	1.796e+004	1.797e+004
60	Pressure (bar)	70.15	70.15	70.16	70.16	70.16
61	Temperature (C)	51.15	51.15	51.15	51.15	51.15
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 809 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.798e+004	1.798e+004	1.799e+004	1.799e+004	1.800e+004
16	Pressure (bar)	70.17	70.17	70.18	70.18	70.19
17	Temperature (C)	51.15	51.15	51.15	51.15	51.15
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.35	16.35	16.35	16.35	16.35
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.800e+004	1.801e+004	1.801e+004	1.802e+004	1.802e+004
28	Pressure (bar)	70.19	70.20	70.20	70.21	70.21
29	Temperature (C)	51.15	51.15	51.16	51.16	51.16
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.35	16.36	16.36	16.36	16.36
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.803e+004	1.803e+004	1.804e+004	1.804e+004	1.805e+004
40	Pressure (bar)	70.22	70.21	70.20	70.19	70.18
41	Temperature (C)	51.16	51.15	51.15	51.15	51.14
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.36	16.36	16.36	16.35	16.35
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.805e+004	1.806e+004	1.806e+004	1.807e+004	1.807e+004
52	Pressure (bar)	70.17	70.00	69.83	69.66	69.49
53	Temperature (C)	51.14	51.09	51.04	50.99	50.95
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.35	16.34	16.32	16.31	16.29
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.808e+004	1.808e+004	1.809e+004	1.809e+004	1.810e+004
64	Pressure (bar)	69.32	69.32	69.32	69.31	69.31
65	Temperature (C)	50.90	50.90	50.89	50.89	50.89
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 810 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.810e+004	1.811e+004	1.811e+004	1.812e+004	1.812e+004
20	Pressure (bar)	69.31	69.31	69.31	69.31	69.31
21	Temperature (C)	50.89	50.89	50.88	50.88	50.88
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.813e+004	1.813e+004	1.814e+004	1.814e+004	1.815e+004
32	Pressure (bar)	69.31	69.30	69.30	69.30	69.30
33	Temperature (C)	50.88	50.88	50.87	50.87	50.87
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.815e+004	1.816e+004	1.816e+004	1.817e+004	1.817e+004
44	Pressure (bar)	69.30	69.30	69.30	69.30	69.30
45	Temperature (C)	50.87	50.87	50.87	50.86	50.86
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.818e+004	1.818e+004	1.819e+004	1.819e+004	1.820e+004
56	Pressure (bar)	69.30	69.30	69.30	69.30	69.30
57	Temperature (C)	50.86	50.86	50.86	50.86	50.85
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.820e+004	1.821e+004	1.821e+004	1.822e+004	1.822e+004
68	Pressure (bar)	69.30	69.30	69.30	69.30	69.30
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 811 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	50.85	50.85	50.85	50.85	50.84
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.823e+004	1.823e+004	1.824e+004	1.824e+004	1.825e+004
24	Pressure (bar)	69.30	69.30	69.29	69.29	69.29
25	Temperature (C)	50.84	50.84	50.84	50.84	50.84
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.825e+004	1.826e+004	1.826e+004	1.827e+004	1.827e+004
36	Pressure (bar)	69.29	69.29	69.29	69.29	69.29
37	Temperature (C)	50.83	50.83	50.83	50.83	50.83
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.828e+004	1.828e+004	1.829e+004	1.829e+004	1.830e+004
48	Pressure (bar)	69.29	69.29	69.29	69.29	69.29
49	Temperature (C)	50.83	50.82	50.82	50.82	50.82
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.830e+004	1.831e+004	1.831e+004	1.832e+004	1.832e+004
60	Pressure (bar)	69.29	69.29	69.29	69.29	69.29
61	Temperature (C)	50.82	50.82	50.81	50.81	50.81
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 812 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.833e+004	1.833e+004	1.834e+004	1.834e+004	1.835e+004
16	Pressure (bar)	69.29	69.29	69.29	69.29	69.29
17	Temperature (C)	50.81	50.81	50.81	50.80	50.80
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.835e+004	1.836e+004	1.836e+004	1.837e+004	1.837e+004
28	Pressure (bar)	69.29	69.29	69.28	69.28	69.28
29	Temperature (C)	50.80	50.80	50.80	50.80	50.79
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.838e+004	1.838e+004	1.839e+004	1.839e+004	1.840e+004
40	Pressure (bar)	69.28	69.28	69.28	69.28	69.28
41	Temperature (C)	50.79	50.79	50.79	50.79	50.79
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.840e+004	1.841e+004	1.841e+004	1.842e+004	1.842e+004
52	Pressure (bar)	69.28	69.28	69.28	69.28	69.28
53	Temperature (C)	50.78	50.78	50.78	50.78	50.78
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.843e+004	1.843e+004	1.844e+004	1.844e+004	1.845e+004
64	Pressure (bar)	69.28	69.28	69.28	69.28	69.28
65	Temperature (C)	50.77	50.77	50.77	50.77	50.77
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 813 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.845e+004	1.846e+004	1.846e+004	1.847e+004	1.847e+004
20	Pressure (bar)	69.28	69.28	69.28	69.28	69.28
21	Temperature (C)	50.77	50.76	50.76	50.76	50.76
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.848e+004	1.848e+004	1.849e+004	1.849e+004	1.850e+004
32	Pressure (bar)	69.28	69.27	69.27	69.27	69.27
33	Temperature (C)	50.76	50.76	50.75	50.75	50.75
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.850e+004	1.851e+004	1.851e+004	1.852e+004	1.852e+004
44	Pressure (bar)	69.27	69.27	69.27	69.27	69.27
45	Temperature (C)	50.75	50.75	50.75	50.74	50.74
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.853e+004	1.853e+004	1.854e+004	1.854e+004	1.855e+004
56	Pressure (bar)	69.27	69.27	69.27	69.27	69.27
57	Temperature (C)	50.74	50.74	50.74	50.74	50.73
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.855e+004	1.856e+004	1.856e+004	1.857e+004	1.857e+004
68	Pressure (bar)	69.27	69.27	69.27	69.27	69.27
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	50.73	50.73	50.73	50.73	50.73
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.858e+004	1.858e+004	1.859e+004	1.859e+004	1.860e+004
24	Pressure (bar)	69.27	69.27	69.27	69.27	69.27
25	Temperature (C)	50.72	50.72	50.72	50.72	50.72
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.860e+004	1.861e+004	1.861e+004	1.862e+004	1.862e+004
36	Pressure (bar)	69.27	69.27	69.27	69.27	69.26
37	Temperature (C)	50.72	50.71	50.71	50.71	50.71
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.863e+004	1.863e+004	1.864e+004	1.864e+004	1.865e+004
48	Pressure (bar)	69.26	69.26	69.26	69.26	69.26
49	Temperature (C)	50.71	50.71	50.70	50.70	50.70
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.865e+004	1.866e+004	1.866e+004	1.867e+004	1.867e+004
60	Pressure (bar)	69.26	69.26	69.26	69.26	69.26
61	Temperature (C)	50.70	50.70	50.70	50.69	50.69
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 815 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.868e+004	1.868e+004	1.869e+004	1.869e+004	1.870e+004
16	Pressure (bar)	69.26	69.26	69.26	69.26	69.26
17	Temperature (C)	50.69	50.69	50.69	50.69	50.69
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.870e+004	1.871e+004	1.871e+004	1.872e+004	1.872e+004
28	Pressure (bar)	69.26	69.26	69.26	69.26	69.26
29	Temperature (C)	50.68	50.68	50.68	50.68	50.68
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.873e+004	1.873e+004	1.874e+004	1.874e+004	1.875e+004
40	Pressure (bar)	69.26	69.26	69.25	69.25	69.25
41	Temperature (C)	50.67	50.67	50.67	50.67	50.67
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.875e+004	1.876e+004	1.876e+004	1.877e+004	1.877e+004
52	Pressure (bar)	69.25	69.25	69.25	69.25	69.25
53	Temperature (C)	50.66	50.66	50.66	50.66	50.66
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.878e+004	1.878e+004	1.879e+004	1.879e+004	1.880e+004
64	Pressure (bar)	69.25	69.25	69.25	69.25	69.25
65	Temperature (C)	50.66	50.66	50.65	50.65	50.65
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 816 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.880e+004	1.880e+004	1.881e+004	1.881e+004	1.881e+004
20	Pressure (bar)	69.25	69.25	69.25	69.25	69.25
21	Temperature (C)	50.65	50.65	50.65	50.65	50.64
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.882e+004	1.882e+004	1.882e+004	1.882e+004	1.882e+004
32	Pressure (bar)	69.25	69.25	69.25	69.25	69.25
33	Temperature (C)	50.64	50.64	50.64	50.64	50.64
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.883e+004	1.883e+004	1.884e+004	1.884e+004	1.885e+004
44	Pressure (bar)	69.25	69.25	69.25	69.24	69.24
45	Temperature (C)	50.64	50.64	50.64	50.63	50.63
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.885e+004	1.886e+004	1.886e+004	1.887e+004	1.887e+004
56	Pressure (bar)	69.24	69.24	69.24	69.24	69.24
57	Temperature (C)	50.63	50.63	50.63	50.62	50.62
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.888e+004	1.888e+004	1.889e+004	1.889e+004	1.890e+004
68	Pressure (bar)	69.24	69.24	69.24	69.24	69.24
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	50.62	50.62	50.62	50.62	50.61
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.890e+004	1.890e+004	1.891e+004	1.891e+004	1.891e+004
24	Pressure (bar)	69.24	69.24	69.24	69.24	69.24
25	Temperature (C)	50.61	50.61	50.61	50.61	50.61
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.891e+004	1.891e+004	1.892e+004	1.892e+004	1.892e+004
36	Pressure (bar)	69.24	69.24	69.23	69.23	69.23
37	Temperature (C)	50.61	50.61	50.61	50.61	50.61
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.893e+004	1.893e+004	1.894e+004	1.894e+004	1.895e+004
48	Pressure (bar)	69.23	69.23	69.23	69.23	69.23
49	Temperature (C)	50.60	50.60	50.60	50.60	50.60
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.895e+004	1.896e+004	1.896e+004	1.897e+004	1.897e+004
60	Pressure (bar)	69.23	69.23	69.23	69.23	69.24
61	Temperature (C)	50.60	50.60	50.59	50.59	50.59
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 818 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.898e+004	1.898e+004	1.899e+004	1.899e+004	1.900e+004
16	Pressure (bar)	69.24	69.23	69.23	69.23	69.23
17	Temperature (C)	50.59	50.59	50.59	50.58	50.58
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.900e+004	1.901e+004	1.901e+004	1.902e+004	1.902e+004
28	Pressure (bar)	69.23	69.23	69.23	69.23	69.22
29	Temperature (C)	50.58	50.58	50.58	50.57	50.57
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.903e+004	1.903e+004	1.904e+004	1.904e+004	1.905e+004
40	Pressure (bar)	69.22	69.22	69.22	69.22	69.22
41	Temperature (C)	50.57	50.57	50.57	50.56	50.56
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.905e+004	1.906e+004	1.906e+004	1.907e+004	1.907e+004
52	Pressure (bar)	69.22	69.22	69.22	69.22	69.22
53	Temperature (C)	50.56	50.56	50.56	50.56	50.55
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.908e+004	1.908e+004	1.909e+004	1.909e+004	1.910e+004
64	Pressure (bar)	69.22	69.22	69.22	69.22	69.22
65	Temperature (C)	50.55	50.55	50.55	50.55	50.55
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 819 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.910e+004	1.911e+004	1.911e+004	1.912e+004	1.912e+004
20	Pressure (bar)	69.22	69.21	69.21	69.21	69.21
21	Temperature (C)	50.54	50.54	50.54	50.54	50.54
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.913e+004	1.913e+004	1.914e+004	1.914e+004	1.915e+004
32	Pressure (bar)	69.21	69.21	69.21	69.21	69.21
33	Temperature (C)	50.53	50.53	50.53	50.53	50.53
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.27	16.27	16.27	16.27	16.27
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.915e+004	1.916e+004	1.916e+004	1.917e+004	1.917e+004
44	Pressure (bar)	69.21	69.21	69.21	69.21	69.21
45	Temperature (C)	50.53	50.52	50.52	50.52	50.52
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.27	16.26	16.26	16.26	16.26
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.918e+004	1.918e+004	1.919e+004	1.919e+004	1.920e+004
56	Pressure (bar)	69.21	69.21	69.21	69.21	69.21
57	Temperature (C)	50.52	50.52	50.51	50.51	50.51
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.26	16.26	16.26	16.26	16.26
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.920e+004	1.921e+004	1.921e+004	1.922e+004	1.922e+004
68	Pressure (bar)	69.21	69.21	69.21	69.21	69.21
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 820 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:13:43 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	50.51	50.51	50.51	50.51	50.50
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.26	16.26	16.26	16.26	16.26
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.923e+004	1.923e+004	1.924e+004	1.924e+004	1.925e+004
24	Pressure (bar)	69.21	69.21	69.20	69.20	69.20
25	Temperature (C)	50.50	50.50	50.50	50.50	50.49
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.26	16.26	16.26	16.26	16.26
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.925e+004	1.926e+004	1.926e+004	1.927e+004	1.927e+004
36	Pressure (bar)	69.20	69.20	69.20	69.20	69.20
37	Temperature (C)	50.49	50.49	50.49	50.49	50.49
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.26	16.26	16.26	16.26	16.26
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.928e+004	1.928e+004	1.929e+004	1.929e+004	1.930e+004
48	Pressure (bar)	69.20	69.20	69.20	69.20	69.20
49	Temperature (C)	50.48	50.48	50.48	50.48	50.48
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.26	16.26	16.26	16.26	16.26
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.930e+004	1.931e+004	1.931e+004	1.932e+004	1.932e+004
60	Pressure (bar)	69.20	69.20	69.20	69.20	69.20
61	Temperature (C)	50.48	50.48	50.47	50.47	50.47
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.26	16.26	16.26	16.26	16.26
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 821 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.933e+004	1.933e+004	1.934e+004	1.934e+004	1.935e+004
16	Pressure (bar)	69.20	69.20	69.20	69.20	69.20
17	Temperature (C)	50.47	50.47	50.47	50.46	50.46
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.26	16.26	16.26	16.26	16.26
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.935e+004	1.936e+004	1.936e+004	1.937e+004	1.937e+004
28	Pressure (bar)	69.20	69.20	69.21	69.22	69.22
29	Temperature (C)	50.46	50.46	50.47	50.47	50.47
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.26	16.26	16.26	16.27	16.27
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.938e+004	1.938e+004	1.939e+004	1.939e+004	1.940e+004
40	Pressure (bar)	69.23	69.22	69.22	69.21	69.20
41	Temperature (C)	50.47	50.47	50.47	50.46	50.46
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.27	16.27	16.27	16.26	16.26
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.940e+004	1.941e+004	1.941e+004	1.942e+004	1.942e+004
52	Pressure (bar)	69.20	69.18	69.16	69.14	69.12
53	Temperature (C)	50.45	50.45	50.44	50.43	50.43
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.26	16.26	16.26	16.26	16.26
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.943e+004	1.943e+004	1.944e+004	1.944e+004	1.945e+004
64	Pressure (bar)	69.11	69.11	69.10	69.10	69.10
65	Temperature (C)	50.42	50.42	50.41	50.41	50.41
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 822 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.26	16.26	16.26	16.26	16.26
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.945e+004	1.946e+004	1.946e+004	1.946e+004	1.947e+004
20	Pressure (bar)	69.10	69.10	69.10	69.10	69.10
21	Temperature (C)	50.41	50.41	50.41	50.40	50.40
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.26	16.25	16.25	16.25	16.25
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.947e+004	1.948e+004	1.948e+004	1.949e+004	1.949e+004
32	Pressure (bar)	69.09	69.09	69.09	69.09	69.09
33	Temperature (C)	50.40	50.40	50.40	50.39	50.39
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.25	16.25	16.25	16.25	16.25
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.950e+004	1.952e+004	1.954e+004	1.957e+004	1.959e+004
44	Pressure (bar)	69.09	69.09	69.08	69.08	69.08
45	Temperature (C)	50.39	50.38	50.37	50.37	50.36
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.25	16.25	16.25	16.25	16.25
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.961e+004	1.964e+004	1.966e+004	1.968e+004	1.971e+004
56	Pressure (bar)	69.08	69.08	69.07	69.07	69.07
57	Temperature (C)	50.35	50.34	50.33	50.33	50.32
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.25	16.25	16.25	16.25	16.25
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.973e+004	1.975e+004	1.978e+004	1.980e+004	1.982e+004
68	Pressure (bar)	69.07	69.07	69.06	69.06	69.06
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 823 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	50.31	50.30	50.29	50.29	50.28
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.25	16.25	16.25	16.25	16.25
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.985e+004	1.987e+004	1.989e+004	1.992e+004	1.994e+004
24	Pressure (bar)	69.06	69.06	69.05	69.05	69.05
25	Temperature (C)	50.27	50.26	50.26	50.25	50.24
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.25	16.25	16.25	16.25	16.25
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.996e+004	1.999e+004	2.001e+004	2.003e+004	2.006e+004
36	Pressure (bar)	69.05	69.05	69.04	69.04	69.04
37	Temperature (C)	50.23	50.22	50.22	50.21	50.20
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.25	16.25	16.25	16.25	16.25
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.008e+004	2.010e+004	2.013e+004	2.015e+004	2.017e+004
48	Pressure (bar)	69.04	69.04	69.03	69.03	69.03
49	Temperature (C)	50.19	50.19	50.18	50.17	50.16
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.25	16.25	16.25	16.25	16.25
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.020e+004	2.022e+004	2.024e+004	2.027e+004	2.029e+004
60	Pressure (bar)	69.03	69.03	69.02	69.02	69.02
61	Temperature (C)	50.16	50.15	50.14	50.13	50.12
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.25	16.25	16.25	16.25	16.25
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.031e+004	2.034e+004	2.036e+004	2.038e+004	2.041e+004
16	Pressure (bar)	69.02	69.01	69.01	69.01	69.01
17	Temperature (C)	50.11	50.11	50.10	50.09	50.08
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.25	16.25	16.25	16.25	16.25
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.043e+004	2.045e+004	2.048e+004	2.050e+004	2.052e+004
28	Pressure (bar)	69.01	69.00	69.00	69.00	69.00
29	Temperature (C)	50.08	50.07	50.06	50.05	50.04
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.25	16.25	16.25	16.25	16.25
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.055e+004	2.058e+004	2.060e+004	2.063e+004	2.066e+004
40	Pressure (bar)	69.00	68.99	68.99	68.99	68.99
41	Temperature (C)	50.04	50.03	50.02	50.01	50.00
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.25	16.24	16.24	16.24	16.24
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.069e+004	2.069e+004	2.070e+004	2.070e+004	2.070e+004
52	Pressure (bar)	68.98	68.98	68.98	68.98	68.98
53	Temperature (C)	49.99	49.99	49.99	49.99	49.99
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.070e+004	2.071e+004	2.071e+004	2.072e+004	2.072e+004
64	Pressure (bar)	68.98	68.98	68.98	68.98	68.98
65	Temperature (C)	49.99	49.99	49.98	49.98	49.98
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.073e+004	2.073e+004	2.074e+004	2.074e+004	2.075e+004
20	Pressure (bar)	68.98	68.98	68.98	68.98	68.98
21	Temperature (C)	49.98	49.98	49.97	49.97	49.97
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.075e+004	2.076e+004	2.076e+004	2.077e+004	2.077e+004
32	Pressure (bar)	68.98	68.98	68.98	68.98	68.98
33	Temperature (C)	49.97	49.97	49.97	49.96	49.96
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.078e+004	2.078e+004	2.079e+004	2.079e+004	2.080e+004
44	Pressure (bar)	68.98	68.98	68.98	68.98	68.98
45	Temperature (C)	49.96	49.96	49.96	49.96	49.96
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.080e+004	2.081e+004	2.081e+004	2.082e+004	2.082e+004
56	Pressure (bar)	68.98	68.98	68.98	68.98	68.97
57	Temperature (C)	49.95	49.95	49.95	49.95	49.95
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.083e+004	2.083e+004	2.084e+004	2.084e+004	2.085e+004
68	Pressure (bar)	68.97	68.97	68.97	68.97	68.97
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	49.95	49.94	49.94	49.94	49.94
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.085e+004	2.086e+004	2.086e+004	2.087e+004	2.087e+004
24	Pressure (bar)	68.97	68.97	68.97	68.97	68.97
25	Temperature (C)	49.94	49.94	49.93	49.93	49.93
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.088e+004	2.088e+004	2.089e+004	2.089e+004	2.090e+004
36	Pressure (bar)	68.97	68.97	68.97	68.97	68.97
37	Temperature (C)	49.93	49.93	49.93	49.92	49.92
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.090e+004	2.091e+004	2.091e+004	2.092e+004	2.092e+004
48	Pressure (bar)	68.97	68.97	68.97	68.97	68.97
49	Temperature (C)	49.92	49.92	49.92	49.92	49.91
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.093e+004	2.093e+004	2.094e+004	2.094e+004	2.095e+004
60	Pressure (bar)	68.97	68.97	68.97	68.96	68.96
61	Temperature (C)	49.91	49.91	49.91	49.91	49.91
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 827 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.095e+004	2.096e+004	2.096e+004	2.097e+004	2.097e+004
16	Pressure (bar)	68.96	68.96	68.96	68.96	68.96
17	Temperature (C)	49.90	49.90	49.90	49.90	49.90
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.098e+004	2.098e+004	2.099e+004	2.099e+004	2.100e+004
28	Pressure (bar)	68.96	68.96	68.96	68.96	68.96
29	Temperature (C)	49.90	49.89	49.89	49.89	49.89
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.100e+004	2.101e+004	2.101e+004	2.102e+004	2.102e+004
40	Pressure (bar)	68.96	68.96	68.96	68.96	68.96
41	Temperature (C)	49.89	49.89	49.89	49.88	49.88
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.103e+004	2.103e+004	2.104e+004	2.104e+004	2.105e+004
52	Pressure (bar)	68.96	68.96	68.96	68.96	68.96
53	Temperature (C)	49.88	49.88	49.88	49.88	49.87
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.105e+004	2.106e+004	2.106e+004	2.107e+004	2.107e+004
64	Pressure (bar)	68.96	68.96	68.96	68.96	68.95
65	Temperature (C)	49.87	49.87	49.87	49.87	49.87
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.24	16.24	16.24	16.24
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.108e+004	2.108e+004	2.109e+004	2.110e+004
20	Pressure (bar)	68.95	68.95	68.95	68.95
21	Temperature (C)	49.86	49.86	49.86	49.86
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.24	16.24	16.24	16.24
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.110e+004	2.111e+004	2.111e+004	2.112e+004
32	Pressure (bar)	68.95	68.95	68.96	68.96
33	Temperature (C)	49.86	49.86	49.85	49.85
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.24	16.24	16.24	16.24
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.112e+004	2.113e+004	2.113e+004	2.114e+004
44	Pressure (bar)	68.96	68.96	68.96	68.95
45	Temperature (C)	49.85	49.85	49.85	49.85
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.24	16.24	16.24	16.24
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.114e+004	2.114e+004	2.115e+004	2.115e+004
56	Pressure (bar)	68.95	68.95	68.95	68.95
57	Temperature (C)	49.84	49.84	49.84	49.84
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.24	16.24	16.24	16.24
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.115e+004	2.116e+004	2.116e+004	2.117e+004
68	Pressure (bar)	68.95	68.95	68.94	68.94
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	49.84	49.84	49.83	49.83	49.83
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.118e+004	2.118e+004	2.119e+004	2.119e+004	2.120e+004
24	Pressure (bar)	68.94	68.94	68.94	68.94	68.94
25	Temperature (C)	49.83	49.83	49.83	49.82	49.82
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.120e+004	2.121e+004	2.121e+004	2.122e+004	2.122e+004
36	Pressure (bar)	68.94	68.94	68.94	68.94	68.93
37	Temperature (C)	49.82	49.82	49.82	49.82	49.81
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.123e+004	2.123e+004	2.124e+004	2.124e+004	2.125e+004
48	Pressure (bar)	68.93	68.93	68.93	68.93	68.93
49	Temperature (C)	49.81	49.81	49.81	49.81	49.80
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.125e+004	2.126e+004	2.126e+004	2.127e+004	2.127e+004
60	Pressure (bar)	68.93	68.93	68.93	68.93	68.93
61	Temperature (C)	49.80	49.80	49.80	49.80	49.80
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.128e+004	2.128e+004	2.129e+004	2.129e+004	2.130e+004
16	Pressure (bar)	68.93	68.93	68.93	68.93	68.93
17	Temperature (C)	49.80	49.79	49.79	49.79	49.79
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.130e+004	2.131e+004	2.131e+004	2.132e+004	2.132e+004
28	Pressure (bar)	68.93	68.93	68.93	68.93	68.93
29	Temperature (C)	49.79	49.79	49.78	49.78	49.78
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.133e+004	2.133e+004	2.133e+004	2.133e+004	2.133e+004
40	Pressure (bar)	68.93	68.93	68.93	68.93	68.93
41	Temperature (C)	49.78	49.78	49.78	49.78	49.78
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.134e+004	2.134e+004	2.134e+004	2.134e+004	2.135e+004
52	Pressure (bar)	68.93	68.93	68.93	68.92	68.92
53	Temperature (C)	49.78	49.78	49.77	49.77	49.77
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.135e+004	2.136e+004	2.136e+004	2.137e+004	2.137e+004
64	Pressure (bar)	68.92	68.92	68.92	68.92	68.92
65	Temperature (C)	49.77	49.77	49.77	49.77	49.76
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.138e+004	2.138e+004	2.139e+004	2.139e+004	2.140e+004
20	Pressure (bar)	68.92	68.92	68.92	68.92	68.92
21	Temperature (C)	49.76	49.76	49.76	49.76	49.76
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.140e+004	2.141e+004	2.141e+004	2.142e+004	2.142e+004
32	Pressure (bar)	68.92	68.92	68.92	68.91	68.91
33	Temperature (C)	49.75	49.75	49.75	49.75	49.75
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.143e+004	2.143e+004	2.144e+004	2.144e+004	2.145e+004
44	Pressure (bar)	68.91	68.91	68.91	68.91	68.91
45	Temperature (C)	49.74	49.74	49.74	49.74	49.74
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.145e+004	2.146e+004	2.146e+004	2.147e+004	2.147e+004
56	Pressure (bar)	68.91	68.91	68.91	68.91	68.91
57	Temperature (C)	49.74	49.73	49.73	49.73	49.73
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.148e+004	2.148e+004	2.149e+004	2.149e+004	2.150e+004
68	Pressure (bar)	68.91	68.91	68.91	68.91	68.91
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	49.73	49.73	49.72	49.72	49.72
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.150e+004	2.150e+004	2.151e+004	2.151e+004	2.152e+004
24	Pressure (bar)	68.90	68.90	68.90	68.90	68.90
25	Temperature (C)	49.72	49.72	49.72	49.72	49.71
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.152e+004	2.152e+004	2.152e+004	2.152e+004	2.152e+004
36	Pressure (bar)	68.90	68.90	68.90	68.91	68.91
37	Temperature (C)	49.71	49.71	49.71	49.71	49.71
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.153e+004	2.153e+004	2.154e+004	2.154e+004	2.155e+004
48	Pressure (bar)	68.91	68.90	68.90	68.90	68.90
49	Temperature (C)	49.71	49.71	49.71	49.71	49.71
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.155e+004	2.156e+004	2.156e+004	2.157e+004	2.157e+004
60	Pressure (bar)	68.90	68.90	68.90	68.90	68.89
61	Temperature (C)	49.70	49.70	49.70	49.70	49.70
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.158e+004	2.158e+004	2.159e+004	2.159e+004	2.160e+004
16	Pressure (bar)	68.89	68.89	68.90	68.90	68.90
17	Temperature (C)	49.69	49.69	49.69	49.69	49.69
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.160e+004	2.161e+004	2.161e+004	2.162e+004	2.162e+004
28	Pressure (bar)	68.90	68.90	68.91	68.91	68.92
29	Temperature (C)	49.69	49.69	49.69	49.69	49.69
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.163e+004	2.163e+004	2.164e+004	2.164e+004	2.165e+004
40	Pressure (bar)	68.92	68.93	68.93	68.94	68.94
41	Temperature (C)	49.69	49.70	49.70	49.70	49.70
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.165e+004	2.166e+004	2.166e+004	2.167e+004	2.167e+004
52	Pressure (bar)	68.95	68.95	68.96	68.96	68.97
53	Temperature (C)	49.70	49.70	49.70	49.70	49.70
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.168e+004	2.168e+004	2.169e+004	2.169e+004	2.170e+004
64	Pressure (bar)	68.97	68.97	68.97	68.97	68.97
65	Temperature (C)	49.71	49.70	49.70	49.70	49.70
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 834 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.170e+004	2.171e+004	2.171e+004	2.172e+004	2.172e+004
20	Pressure (bar)	68.97	68.97	68.96	68.96	68.96
21	Temperature (C)	49.70	49.69	49.69	49.69	49.69
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.173e+004	2.173e+004	2.174e+004	2.174e+004	2.175e+004
32	Pressure (bar)	68.96	68.95	68.95	68.95	68.94
33	Temperature (C)	49.68	49.68	49.68	49.68	49.67
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.175e+004	2.176e+004	2.176e+004	2.177e+004	2.177e+004
44	Pressure (bar)	68.94	68.94	68.94	68.93	68.93
45	Temperature (C)	49.67	49.67	49.67	49.66	49.66
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.24	16.24	16.24	16.24	16.24
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.178e+004	2.178e+004	2.179e+004	2.179e+004	2.180e+004
56	Pressure (bar)	68.93	68.87	68.81	68.75	68.69
57	Temperature (C)	49.66	49.64	49.62	49.60	49.59
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.24	16.23	16.23	16.22	16.22
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.180e+004	2.181e+004	2.181e+004	2.182e+004	2.182e+004
68	Pressure (bar)	68.63	68.63	68.63	68.62	68.62
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 835 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	49.57	49.56	49.56	49.56	49.56
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.183e+004	2.183e+004	2.184e+004	2.184e+004	2.185e+004
24	Pressure (bar)	68.62	68.62	68.62	68.62	68.62
25	Temperature (C)	49.56	49.56	49.55	49.55	49.55
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.185e+004	2.186e+004	2.186e+004	2.187e+004	2.187e+004
36	Pressure (bar)	68.62	68.62	68.62	68.62	68.62
37	Temperature (C)	49.55	49.55	49.55	49.54	49.54
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.188e+004	2.188e+004	2.189e+004	2.189e+004	2.190e+004
48	Pressure (bar)	68.62	68.62	68.62	68.62	68.61
49	Temperature (C)	49.54	49.54	49.54	49.54	49.53
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.190e+004	2.191e+004	2.191e+004	2.192e+004	2.192e+004
60	Pressure (bar)	68.61	68.61	68.61	68.61	68.61
61	Temperature (C)	49.53	49.53	49.53	49.53	49.53
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 836 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.193e+004	2.193e+004	2.194e+004	2.194e+004	2.195e+004
16	Pressure (bar)	68.61	68.61	68.61	68.61	68.61
17	Temperature (C)	49.52	49.52	49.52	49.52	49.52
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.195e+004	2.196e+004	2.196e+004	2.197e+004	2.197e+004
28	Pressure (bar)	68.61	68.61	68.61	68.61	68.61
29	Temperature (C)	49.52	49.51	49.51	49.51	49.51
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.198e+004	2.198e+004	2.199e+004	2.199e+004	2.200e+004
40	Pressure (bar)	68.61	68.61	68.61	68.61	68.61
41	Temperature (C)	49.51	49.51	49.50	49.50	49.50
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.200e+004	2.201e+004	2.201e+004	2.202e+004	2.202e+004
52	Pressure (bar)	68.61	68.61	68.61	68.61	68.61
53	Temperature (C)	49.50	49.50	49.50	49.50	49.49
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.203e+004	2.203e+004	2.204e+004	2.204e+004	2.205e+004
64	Pressure (bar)	68.60	68.60	68.60	68.60	68.60
65	Temperature (C)	49.49	49.49	49.49	49.49	49.49
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.205e+004	2.206e+004	2.206e+004	2.207e+004	2.207e+004
20	Pressure (bar)	68.60	68.60	68.60	68.60	68.60
21	Temperature (C)	49.48	49.48	49.48	49.48	49.48
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.208e+004	2.208e+004	2.209e+004	2.209e+004	2.210e+004
32	Pressure (bar)	68.60	68.60	68.60	68.60	68.60
33	Temperature (C)	49.48	49.48	49.47	49.47	49.47
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.210e+004	2.211e+004	2.211e+004	2.212e+004	2.212e+004
44	Pressure (bar)	68.60	68.60	68.60	68.60	68.60
45	Temperature (C)	49.47	49.47	49.47	49.46	49.46
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.213e+004	2.213e+004	2.214e+004	2.214e+004	2.215e+004
56	Pressure (bar)	68.60	68.60	68.60	68.60	68.60
57	Temperature (C)	49.46	49.46	49.46	49.46	49.45
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.215e+004	2.216e+004	2.216e+004	2.217e+004	2.217e+004
68	Pressure (bar)	68.60	68.60	68.60	68.60	68.59
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:13:43 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	49.45	49.45	49.45	49.45	49.44
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.218e+004	2.218e+004	2.219e+004	2.219e+004	2.220e+004
24	Pressure (bar)	68.59	68.59	68.59	68.59	68.59
25	Temperature (C)	49.44	49.44	49.44	49.44	49.44
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.220e+004	2.221e+004	2.221e+004	2.222e+004	2.222e+004
36	Pressure (bar)	68.59	68.59	68.59	68.60	68.60
37	Temperature (C)	49.44	49.43	49.43	49.43	49.43
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.223e+004	2.223e+004	2.224e+004	2.224e+004	2.225e+004
48	Pressure (bar)	68.60	68.59	68.59	68.59	68.59
49	Temperature (C)	49.43	49.43	49.43	49.42	49.42
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.225e+004	2.226e+004	2.226e+004	2.227e+004	2.227e+004
60	Pressure (bar)	68.59	68.58	68.58	68.58	68.58
61	Temperature (C)	49.42	49.42	49.42	49.41	49.41
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 839 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.228e+004	2.229e+004	2.230e+004	2.231e+004	2.232e+004
16	Pressure (bar)	68.58	68.58	68.58	68.58	68.58
17	Temperature (C)	49.41	49.41	49.40	49.40	49.40
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.21	16.21	16.21	16.21	16.21
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.233e+004	2.255e+004	2.278e+004	2.300e+004	2.323e+004
28	Pressure (bar)	68.58	68.56	68.54	68.52	68.50
29	Temperature (C)	49.39	49.32	49.25	49.18	49.11
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.21	16.20	16.20	16.20	16.20
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.345e+004	2.346e+004	2.347e+004	2.348e+004	2.348e+004
40	Pressure (bar)	68.48	68.48	68.48	68.48	68.48
41	Temperature (C)	49.03	49.03	49.03	49.03	49.02
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.349e+004	2.350e+004	2.350e+004	2.351e+004	2.351e+004
52	Pressure (bar)	68.48	68.48	68.48	68.48	68.48
53	Temperature (C)	49.02	49.02	49.02	49.02	49.02
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.352e+004	2.352e+004	2.353e+004	2.353e+004	2.354e+004
64	Pressure (bar)	68.48	68.47	68.47	68.47	68.47
65	Temperature (C)	49.01	49.01	49.01	49.01	49.01
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 840 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.354e+004	2.355e+004	2.355e+004	2.356e+004	2.356e+004
20	Pressure (bar)	68.47	68.47	68.47	68.47	68.47
21	Temperature (C)	49.01	49.00	49.00	49.00	49.00
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.20	16.20	16.20	16.20	16.20
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.357e+004	2.357e+004	2.358e+004	2.358e+004	2.359e+004
32	Pressure (bar)	68.47	68.47	68.46	68.46	68.45
33	Temperature (C)	49.00	49.00	48.99	48.99	48.99
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.20	16.20	16.20	16.19	16.19
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.359e+004	2.360e+004	2.360e+004	2.361e+004	2.361e+004
44	Pressure (bar)	68.45	68.45	68.45	68.46	68.46
45	Temperature (C)	48.98	48.98	48.98	48.98	48.98
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.362e+004	2.362e+004	2.363e+004	2.363e+004	2.364e+004
56	Pressure (bar)	68.46	68.46	68.46	68.46	68.46
57	Temperature (C)	48.98	48.98	48.98	48.98	48.97
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.20	16.19	16.19	16.19	16.19
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.364e+004	2.365e+004	2.365e+004	2.366e+004	2.366e+004
68	Pressure (bar)	68.46	68.46	68.46	68.46	68.45
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:13:43 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	48.97	48.97	48.97	48.97	48.97
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.367e+004	2.367e+004	2.368e+004	2.368e+004	2.369e+004
24	Pressure (bar)	68.45	68.45	68.45	68.45	68.45
25	Temperature (C)	48.96	48.96	48.96	48.96	48.96
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.369e+004	2.370e+004	2.370e+004	2.371e+004	2.371e+004
36	Pressure (bar)	68.45	68.45	68.45	68.45	68.45
37	Temperature (C)	48.96	48.96	48.95	48.95	48.95
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.372e+004	2.372e+004	2.373e+004	2.373e+004	2.374e+004
48	Pressure (bar)	68.45	68.45	68.45	68.45	68.45
49	Temperature (C)	48.95	48.95	48.95	48.94	48.94
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.374e+004	2.375e+004	2.375e+004	2.376e+004	2.376e+004
60	Pressure (bar)	68.45	68.45	68.45	68.45	68.45
61	Temperature (C)	48.94	48.94	48.94	48.94	48.94
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.377e+004	2.377e+004	2.378e+004	2.378e+004	2.379e+004
16	Pressure (bar)	68.45	68.45	68.45	68.45	68.45
17	Temperature (C)	48.94	48.93	48.93	48.93	48.93
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.379e+004	2.380e+004	2.380e+004	2.381e+004	2.381e+004
28	Pressure (bar)	68.45	68.45	68.45	68.45	68.45
29	Temperature (C)	48.93	48.93	48.92	48.92	48.92
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.382e+004	2.382e+004	2.383e+004	2.383e+004	2.384e+004
40	Pressure (bar)	68.44	68.44	68.44	68.44	68.44
41	Temperature (C)	48.92	48.92	48.92	48.91	48.91
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.384e+004	2.385e+004	2.385e+004	2.386e+004	2.386e+004
52	Pressure (bar)	68.44	68.44	68.44	68.44	68.44
53	Temperature (C)	48.91	48.91	48.91	48.90	48.90
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.387e+004	2.387e+004	2.388e+004	2.388e+004	2.389e+004
64	Pressure (bar)	68.44	68.44	68.44	68.44	68.44
65	Temperature (C)	48.90	48.90	48.90	48.90	48.90
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.389e+004	2.390e+004	2.390e+004	2.391e+004	2.391e+004
20	Pressure (bar)	68.44	68.44	68.44	68.43	68.43
21	Temperature (C)	48.90	48.89	48.89	48.89	48.89
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.392e+004	2.392e+004	2.393e+004	2.393e+004	2.394e+004
32	Pressure (bar)	68.43	68.43	68.43	68.43	68.43
33	Temperature (C)	48.89	48.88	48.88	48.88	48.88
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.394e+004	2.395e+004	2.395e+004	2.396e+004	2.396e+004
44	Pressure (bar)	68.43	68.43	68.42	68.42	68.42
45	Temperature (C)	48.88	48.88	48.87	48.87	48.87
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.397e+004	2.397e+004	2.398e+004	2.398e+004	2.399e+004
56	Pressure (bar)	68.42	68.42	68.42	68.42	68.42
57	Temperature (C)	48.87	48.87	48.87	48.86	48.86
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.399e+004	2.400e+004	2.400e+004	2.401e+004	2.401e+004
68	Pressure (bar)	68.42	68.42	68.42	68.42	68.42
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	48.86	48.86	48.86	48.86	48.86
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.402e+004	2.402e+004	2.403e+004	2.403e+004	2.404e+004
24	Pressure (bar)	68.42	68.42	68.42	68.42	68.42
25	Temperature (C)	48.85	48.85	48.85	48.85	48.85
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.404e+004	2.405e+004	2.405e+004	2.406e+004	2.406e+004
36	Pressure (bar)	68.42	68.42	68.42	68.42	68.42
37	Temperature (C)	48.85	48.85	48.84	48.84	48.84
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.407e+004	2.407e+004	2.408e+004	2.408e+004	2.409e+004
48	Pressure (bar)	68.42	68.42	68.41	68.41	68.41
49	Temperature (C)	48.84	48.84	48.83	48.83	48.83
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.409e+004	2.410e+004	2.410e+004	2.411e+004	2.411e+004
60	Pressure (bar)	68.41	68.41	68.41	68.41	68.41
61	Temperature (C)	48.83	48.83	48.83	48.82	48.82
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.412e+004	2.412e+004	2.413e+004	2.413e+004	2.414e+004
16	Pressure (bar)	68.41	68.41	68.41	68.41	68.41
17	Temperature (C)	48.82	48.82	48.82	48.82	48.81
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.414e+004	2.415e+004	2.415e+004	2.416e+004	2.416e+004
28	Pressure (bar)	68.41	68.41	68.41	68.41	68.40
29	Temperature (C)	48.81	48.81	48.81	48.81	48.81
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.417e+004	2.417e+004	2.418e+004	2.418e+004	2.419e+004
40	Pressure (bar)	68.40	68.40	68.40	68.40	68.40
41	Temperature (C)	48.81	48.80	48.80	48.80	48.80
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.419e+004	2.420e+004	2.420e+004	2.421e+004	2.421e+004
52	Pressure (bar)	68.40	68.40	68.40	68.40	68.40
53	Temperature (C)	48.80	48.80	48.79	48.79	48.79
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.422e+004	2.422e+004	2.423e+004	2.423e+004	2.424e+004
64	Pressure (bar)	68.40	68.40	68.40	68.40	68.40
65	Temperature (C)	48.79	48.79	48.79	48.79	48.78
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.424e+004	2.425e+004	2.425e+004	2.426e+004	2.426e+004
20	Pressure (bar)	68.40	68.40	68.40	68.40	68.40
21	Temperature (C)	48.78	48.78	48.78	48.78	48.78
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.427e+004	2.427e+004	2.428e+004	2.428e+004	2.429e+004
32	Pressure (bar)	68.40	68.40	68.40	68.40	68.39
33	Temperature (C)	48.77	48.77	48.77	48.77	48.77
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.429e+004	2.430e+004	2.430e+004	2.431e+004	2.431e+004
44	Pressure (bar)	68.39	68.39	68.39	68.39	68.39
45	Temperature (C)	48.77	48.76	48.76	48.76	48.76
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.432e+004	2.432e+004	2.433e+004	2.433e+004	2.434e+004
56	Pressure (bar)	68.39	68.39	68.39	68.39	68.39
57	Temperature (C)	48.76	48.76	48.76	48.75	48.75
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.434e+004	2.435e+004	2.435e+004	2.436e+004	2.436e+004
68	Pressure (bar)	68.39	68.39	68.39	68.39	68.39
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	48.75	48.75	48.75	48.75	48.74
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.437e+004	2.437e+004	2.437e+004	2.437e+004	2.437e+004
24	Pressure (bar)	68.39	68.39	68.39	68.39	68.39
25	Temperature (C)	48.74	48.74	48.74	48.74	48.74
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.438e+004	2.438e+004	2.438e+004	2.438e+004	2.438e+004
36	Pressure (bar)	68.39	68.39	68.39	68.39	68.39
37	Temperature (C)	48.74	48.74	48.74	48.74	48.74
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.439e+004	2.439e+004	2.440e+004	2.440e+004	2.441e+004
48	Pressure (bar)	68.39	68.39	68.38	68.38	68.38
49	Temperature (C)	48.74	48.73	48.73	48.73	48.73
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.442e+004	2.442e+004	2.443e+004	2.443e+004	2.444e+004
60	Pressure (bar)	68.38	68.38	68.38	68.38	68.38
61	Temperature (C)	48.73	48.73	48.72	48.72	48.72
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 848 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.444e+004	2.445e+004	2.445e+004	2.446e+004	2.446e+004
16	Pressure (bar)	68.38	68.38	68.38	68.38	68.38
17	Temperature (C)	48.72	48.72	48.72	48.72	48.71
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.447e+004	2.447e+004	2.448e+004	2.448e+004	2.449e+004
28	Pressure (bar)	68.38	68.38	68.38	68.38	68.38
29	Temperature (C)	48.71	48.71	48.71	48.71	48.71
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.449e+004	2.450e+004	2.450e+004	2.451e+004	2.451e+004
40	Pressure (bar)	68.38	68.38	68.38	68.38	68.38
41	Temperature (C)	48.70	48.70	48.70	48.70	48.70
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.452e+004	2.452e+004	2.453e+004	2.453e+004	2.454e+004
52	Pressure (bar)	68.38	68.38	68.38	68.38	68.37
53	Temperature (C)	48.70	48.69	48.69	48.69	48.69
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.454e+004	2.455e+004	2.455e+004	2.456e+004	2.456e+004
64	Pressure (bar)	68.37	68.37	68.37	68.37	68.37
65	Temperature (C)	48.69	48.69	48.69	48.68	48.68
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.457e+004	2.457e+004	2.458e+004	2.458e+004	2.459e+004
20	Pressure (bar)	68.37	68.37	68.37	68.37	68.37
21	Temperature (C)	48.68	48.68	48.68	48.68	48.67
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.459e+004	2.460e+004	2.460e+004	2.461e+004	2.461e+004
32	Pressure (bar)	68.37	68.37	68.37	68.37	68.37
33	Temperature (C)	48.67	48.67	48.67	48.67	48.67
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.462e+004	2.462e+004	2.463e+004	2.463e+004	2.464e+004
44	Pressure (bar)	68.37	68.37	68.37	68.36	68.36
45	Temperature (C)	48.66	48.66	48.66	48.66	48.66
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.464e+004	2.465e+004	2.465e+004	2.466e+004	2.466e+004
56	Pressure (bar)	68.36	68.36	68.36	68.36	68.36
57	Temperature (C)	48.66	48.66	48.65	48.65	48.65
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.467e+004	2.467e+004	2.468e+004	2.468e+004	2.469e+004
68	Pressure (bar)	68.36	68.36	68.36	68.36	68.37
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:13:43 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	48.65	48.65	48.65	48.65	48.65
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.469e+004	2.470e+004	2.470e+004	2.471e+004	2.471e+004
24	Pressure (bar)	68.37	68.36	68.36	68.36	68.36
25	Temperature (C)	48.64	48.64	48.64	48.64	48.64
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.472e+004	2.472e+004	2.473e+004	2.473e+004	2.474e+004
36	Pressure (bar)	68.36	68.36	68.35	68.35	68.35
37	Temperature (C)	48.63	48.63	48.63	48.63	48.63
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.19	16.19	16.19	16.19	16.19
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.474e+004	2.475e+004	2.475e+004	2.476e+004	2.476e+004
48	Pressure (bar)	68.35	68.35	68.35	68.35	68.35
49	Temperature (C)	48.62	48.62	48.62	48.62	48.62
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.477e+004	2.477e+004	2.478e+004	2.478e+004	2.479e+004
60	Pressure (bar)	68.35	68.35	68.35	68.35	68.35
61	Temperature (C)	48.62	48.62	48.61	48.61	48.61
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.479e+004	2.480e+004	2.480e+004	2.481e+004	2.481e+004
16	Pressure (bar)	68.35	68.35	68.35	68.35	68.35
17	Temperature (C)	48.61	48.61	48.61	48.60	48.60
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.482e+004	2.482e+004	2.483e+004	2.483e+004	2.484e+004
28	Pressure (bar)	68.35	68.35	68.35	68.35	68.35
29	Temperature (C)	48.60	48.60	48.60	48.60	48.60
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.484e+004	2.485e+004	2.485e+004	2.486e+004	2.486e+004
40	Pressure (bar)	68.35	68.35	68.35	68.35	68.35
41	Temperature (C)	48.60	48.59	48.59	48.59	48.59
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.487e+004	2.487e+004	2.488e+004	2.488e+004	2.489e+004
52	Pressure (bar)	68.35	68.35	68.35	68.35	68.35
53	Temperature (C)	48.59	48.59	48.59	48.59	48.58
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.489e+004	2.490e+004	2.490e+004	2.491e+004	2.491e+004
64	Pressure (bar)	68.35	68.35	68.34	68.34	68.34
65	Temperature (C)	48.58	48.58	48.58	48.57	48.57
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 852 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.492e+004	2.492e+004	2.493e+004	2.493e+004	2.494e+004
20	Pressure (bar)	68.33	68.33	68.33	68.33	68.33
21	Temperature (C)	48.57	48.57	48.57	48.56	48.56
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.494e+004	2.495e+004	2.495e+004	2.496e+004	2.496e+004
32	Pressure (bar)	68.33	68.33	68.33	68.33	68.33
33	Temperature (C)	48.56	48.56	48.56	48.56	48.56
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.497e+004	2.497e+004	2.498e+004	2.498e+004	2.499e+004
44	Pressure (bar)	68.33	68.33	68.33	68.33	68.33
45	Temperature (C)	48.55	48.55	48.55	48.55	48.55
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.499e+004	2.500e+004	2.500e+004	2.501e+004	2.501e+004
56	Pressure (bar)	68.33	68.33	68.33	68.33	68.33
57	Temperature (C)	48.55	48.54	48.54	48.54	48.54
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.502e+004	2.502e+004	2.503e+004	2.503e+004	2.504e+004
68	Pressure (bar)	68.32	68.32	68.32	68.32	68.32
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	48.54	48.54	48.54	48.53	48.53
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.504e+004	2.505e+004	2.505e+004	2.506e+004	2.506e+004
24	Pressure (bar)	68.32	68.32	68.32	68.32	68.32
25	Temperature (C)	48.53	48.53	48.53	48.53	48.52
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.507e+004	2.507e+004	2.508e+004	2.508e+004	2.509e+004
36	Pressure (bar)	68.32	68.32	68.32	68.32	68.32
37	Temperature (C)	48.52	48.52	48.52	48.52	48.52
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.509e+004	2.510e+004	2.510e+004	2.511e+004	2.511e+004
48	Pressure (bar)	68.32	68.32	68.32	68.32	68.32
49	Temperature (C)	48.52	48.51	48.51	48.51	48.51
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.512e+004	2.512e+004	2.513e+004	2.513e+004	2.514e+004
60	Pressure (bar)	68.32	68.32	68.32	68.32	68.31
61	Temperature (C)	48.51	48.51	48.50	48.50	48.50
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.514e+004	2.515e+004	2.515e+004	2.516e+004	2.516e+004
16	Pressure (bar)	68.31	68.31	68.31	68.31	68.31
17	Temperature (C)	48.50	48.50	48.50	48.50	48.49
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.517e+004	2.517e+004	2.518e+004	2.518e+004	2.519e+004
28	Pressure (bar)	68.31	68.31	68.31	68.31	68.31
29	Temperature (C)	48.49	48.49	48.49	48.49	48.49
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.519e+004	2.520e+004	2.520e+004	2.521e+004	2.522e+004
40	Pressure (bar)	68.31	68.31	68.31	68.31	68.31
41	Temperature (C)	48.49	48.48	48.48	48.48	48.48
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.522e+004	2.523e+004	2.523e+004	2.524e+004	2.524e+004
52	Pressure (bar)	68.31	68.32	68.32	68.33	68.34
53	Temperature (C)	48.48	48.48	48.48	48.48	48.49
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.525e+004	2.525e+004	2.526e+004	2.526e+004	2.527e+004
64	Pressure (bar)	68.34	68.35	68.35	68.35	68.36
65	Temperature (C)	48.49	48.49	48.49	48.49	48.49
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 855 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.19
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.527e+004	2.528e+004	2.528e+004	2.529e+004	2.529e+004
20	Pressure (bar)	68.36	68.35	68.34	68.33	68.32
21	Temperature (C)	48.49	48.49	48.48	48.48	48.47
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.19	16.18	16.18	16.18	16.18
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.529e+004	2.531e+004	2.533e+004	2.535e+004	2.537e+004
32	Pressure (bar)	68.31	68.31	68.31	68.30	68.30
33	Temperature (C)	48.47	48.46	48.46	48.45	48.44
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.539e+004	2.539e+004	2.539e+004	2.539e+004	2.540e+004
44	Pressure (bar)	68.29	68.29	68.29	68.29	68.29
45	Temperature (C)	48.43	48.43	48.43	48.43	48.43
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.18	16.18	16.18	16.18	16.18
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.540e+004	2.540e+004	2.541e+004	2.541e+004	2.542e+004
56	Pressure (bar)	68.29	68.27	68.25	68.23	68.21
57	Temperature (C)	48.43	48.42	48.41	48.41	48.40
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.18	16.18	16.18	16.17	16.17
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.543e+004	2.543e+004	2.544e+004	2.544e+004	2.545e+004
68	Pressure (bar)	68.19	68.19	68.19	68.19	68.18
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 856 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	48.39	48.39	48.39	48.39	48.38
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.17	16.17	16.17	16.17	16.17
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.545e+004	2.546e+004	2.546e+004	2.547e+004	2.547e+004
24	Pressure (bar)	68.18	68.18	68.18	68.18	68.18
25	Temperature (C)	48.38	48.38	48.38	48.38	48.37
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.17	16.17	16.17	16.17	16.17
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.548e+004	2.548e+004	2.549e+004	2.549e+004	2.550e+004
36	Pressure (bar)	68.18	68.18	68.18	68.18	68.18
37	Temperature (C)	48.37	48.37	48.37	48.37	48.37
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.17	16.17	16.17	16.17	16.17
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.550e+004	2.551e+004	2.551e+004	2.552e+004	2.552e+004
48	Pressure (bar)	68.18	68.18	68.18	68.18	68.17
49	Temperature (C)	48.37	48.36	48.36	48.36	48.36
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.17	16.17	16.17	16.17	16.17
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.553e+004	2.553e+004	2.554e+004	2.554e+004	2.555e+004
60	Pressure (bar)	68.17	68.17	68.17	68.17	68.17
61	Temperature (C)	48.36	48.36	48.35	48.35	48.35
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.17	16.17	16.17	16.17	16.17
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.555e+004	2.556e+004	2.556e+004	2.557e+004	2.557e+004
16	Pressure (bar)	68.17	68.17	68.17	68.17	68.17
17	Temperature (C)	48.35	48.35	48.35	48.34	48.34
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.17	16.17	16.17	16.17	16.17
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.558e+004	2.558e+004	2.559e+004	2.559e+004	2.560e+004
28	Pressure (bar)	68.17	68.17	68.17	68.17	68.17
29	Temperature (C)	48.34	48.34	48.34	48.34	48.34
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.17	16.17	16.17	16.17	16.17
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.560e+004	2.561e+004	2.561e+004	2.562e+004	2.563e+004
40	Pressure (bar)	68.17	68.17	68.17	68.17	68.17
41	Temperature (C)	48.33	48.33	48.33	48.33	48.33
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.17	16.17	16.17	16.17	16.17
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.563e+004	2.564e+004	2.564e+004	2.565e+004	2.565e+004
52	Pressure (bar)	68.16	68.16	68.16	68.16	68.16
53	Temperature (C)	48.32	48.32	48.32	48.32	48.32
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.17	16.17	16.17	16.17	16.17
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.566e+004	2.566e+004	2.567e+004	2.567e+004	2.568e+004
64	Pressure (bar)	68.16	68.16	68.16	68.16	68.16
65	Temperature (C)	48.32	48.31	48.31	48.31	48.31
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.17	16.17	16.17	16.17	16.17
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.568e+004	2.569e+004	2.569e+004	2.570e+004	2.570e+004
20	Pressure (bar)	68.16	68.16	68.16	68.16	68.16
21	Temperature (C)	48.31	48.31	48.31	48.30	48.30
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.17	16.17	16.17	16.17	16.17
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.571e+004	2.571e+004	2.571e+004	2.571e+004	2.571e+004
32	Pressure (bar)	68.16	68.16	68.16	68.16	68.16
33	Temperature (C)	48.30	48.30	48.30	48.30	48.30
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.17	16.17	16.17	16.17	16.17
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.571e+004	2.572e+004	2.572e+004	2.572e+004	2.572e+004
44	Pressure (bar)	68.16	68.16	68.16	68.16	68.16
45	Temperature (C)	48.30	48.30	48.30	48.30	48.30
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.17	16.17	16.17	16.17	16.17
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.572e+004	2.572e+004	2.573e+004	2.573e+004	2.573e+004
56	Pressure (bar)	68.17	68.17	68.17	68.18	68.18
57	Temperature (C)	48.30	48.30	48.31	48.31	48.31
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.17	16.17	16.17	16.17	16.17
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.573e+004	2.574e+004	2.574e+004	2.575e+004	2.575e+004
68	Pressure (bar)	68.18	68.19	68.19	68.19	68.20
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	48.31	48.31	48.31	48.31	48.31
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.17	16.17	16.17	16.17	16.17
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.576e+004	2.576e+004	2.577e+004	2.577e+004	2.578e+004
24	Pressure (bar)	68.20	68.16	68.12	68.08	68.04
25	Temperature (C)	48.31	48.30	48.29	48.27	48.26
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.17	16.17	16.16	16.16	16.16
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.578e+004	2.579e+004	2.579e+004	2.580e+004	2.580e+004
36	Pressure (bar)	68.00	68.00	68.00	68.00	68.00
37	Temperature (C)	48.25	48.24	48.24	48.24	48.24
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.581e+004	2.581e+004	2.582e+004	2.582e+004	2.583e+004
48	Pressure (bar)	67.99	67.99	67.99	67.99	67.99
49	Temperature (C)	48.24	48.23	48.23	48.23	48.23
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.583e+004	2.584e+004	2.584e+004	2.585e+004	2.585e+004
60	Pressure (bar)	67.99	67.99	67.99	67.99	67.99
61	Temperature (C)	48.23	48.23	48.22	48.22	48.22
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 860 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.586e+004	2.586e+004	2.587e+004	2.587e+004	2.588e+004
16	Pressure (bar)	67.99	67.99	67.98	67.98	67.98
17	Temperature (C)	48.22	48.22	48.22	48.21	48.21
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.588e+004	2.589e+004	2.589e+004	2.590e+004	2.590e+004
28	Pressure (bar)	67.98	67.98	67.98	67.98	67.98
29	Temperature (C)	48.21	48.21	48.21	48.21	48.21
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.591e+004	2.591e+004	2.592e+004	2.592e+004	2.593e+004
40	Pressure (bar)	67.98	67.98	67.99	67.99	67.99
41	Temperature (C)	48.21	48.20	48.20	48.20	48.20
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.593e+004	2.594e+004	2.594e+004	2.595e+004	2.595e+004
52	Pressure (bar)	67.99	67.99	67.99	67.99	67.99
53	Temperature (C)	48.20	48.20	48.20	48.20	48.19
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.596e+004	2.596e+004	2.597e+004	2.597e+004	2.598e+004
64	Pressure (bar)	67.99	67.98	67.98	67.98	67.97
65	Temperature (C)	48.19	48.19	48.19	48.19	48.18
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 861 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.598e+004	2.599e+004	2.599e+004	2.600e+004	2.600e+004
20	Pressure (bar)	67.97	67.97	67.97	67.97	67.97
21	Temperature (C)	48.18	48.18	48.18	48.18	48.17
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.601e+004	2.601e+004	2.602e+004	2.602e+004	2.603e+004
32	Pressure (bar)	67.97	67.97	67.97	67.97	67.97
33	Temperature (C)	48.17	48.17	48.17	48.17	48.17
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.603e+004	2.604e+004	2.604e+004	2.605e+004	2.605e+004
44	Pressure (bar)	67.97	67.97	67.96	67.96	67.96
45	Temperature (C)	48.17	48.16	48.16	48.16	48.16
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.606e+004	2.606e+004	2.607e+004	2.607e+004	2.608e+004
56	Pressure (bar)	67.96	67.96	67.96	67.96	67.96
57	Temperature (C)	48.16	48.15	48.15	48.15	48.15
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.608e+004	2.609e+004	2.609e+004	2.610e+004	2.610e+004
68	Pressure (bar)	67.96	67.96	67.96	67.96	67.96
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	48.15	48.15	48.15	48.14	48.14
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.611e+004	2.611e+004	2.612e+004	2.612e+004	2.613e+004
24	Pressure (bar)	67.96	67.96	67.96	67.96	67.96
25	Temperature (C)	48.14	48.14	48.14	48.14	48.14
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.613e+004	2.614e+004	2.614e+004	2.615e+004	2.615e+004
36	Pressure (bar)	67.96	67.96	67.95	67.95	67.95
37	Temperature (C)	48.13	48.13	48.13	48.13	48.13
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.616e+004	2.616e+004	2.617e+004	2.617e+004	2.618e+004
48	Pressure (bar)	67.95	67.95	67.95	67.95	67.95
49	Temperature (C)	48.13	48.13	48.12	48.12	48.12
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.618e+004	2.619e+004	2.619e+004	2.620e+004	2.620e+004
60	Pressure (bar)	67.95	67.95	67.95	67.95	67.95
61	Temperature (C)	48.12	48.12	48.12	48.12	48.11
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 863 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.621e+004	2.621e+004	2.622e+004	2.622e+004	2.623e+004
16	Pressure (bar)	67.95	67.95	67.95	67.95	67.95
17	Temperature (C)	48.11	48.11	48.11	48.11	48.11
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.623e+004	2.624e+004	2.624e+004	2.625e+004	2.625e+004
28	Pressure (bar)	67.95	67.95	67.95	67.95	67.95
29	Temperature (C)	48.10	48.10	48.10	48.10	48.10
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.626e+004	2.626e+004	2.627e+004	2.627e+004	2.628e+004
40	Pressure (bar)	67.95	67.95	67.94	67.94	67.94
41	Temperature (C)	48.10	48.10	48.09	48.09	48.09
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.628e+004	2.629e+004	2.629e+004	2.630e+004	2.630e+004
52	Pressure (bar)	67.94	67.94	67.94	67.94	67.94
53	Temperature (C)	48.09	48.09	48.09	48.08	48.08
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.631e+004	2.631e+004	2.632e+004	2.632e+004	2.633e+004
64	Pressure (bar)	67.94	67.94	67.94	67.94	67.94
65	Temperature (C)	48.08	48.08	48.08	48.08	48.08
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 864 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.633e+004	2.634e+004	2.634e+004	2.635e+004	2.635e+004
20	Pressure (bar)	67.94	67.94	67.94	67.94	67.94
21	Temperature (C)	48.07	48.07	48.07	48.07	48.07
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.636e+004	2.636e+004	2.637e+004	2.637e+004	2.638e+004
32	Pressure (bar)	67.94	67.94	67.94	67.94	67.94
33	Temperature (C)	48.07	48.07	48.06	48.06	48.06
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.638e+004	2.639e+004	2.639e+004	2.640e+004	2.640e+004
44	Pressure (bar)	67.94	67.94	67.94	67.93	67.93
45	Temperature (C)	48.06	48.06	48.06	48.05	48.05
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.641e+004	2.641e+004	2.642e+004	2.642e+004	2.643e+004
56	Pressure (bar)	67.93	67.93	67.93	67.93	67.93
57	Temperature (C)	48.05	48.05	48.05	48.05	48.04
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.15
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.644e+004	2.644e+004	2.645e+004	2.645e+004	2.646e+004
68	Pressure (bar)	67.93	67.93	67.93	67.93	67.93
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	48.04	48.04	48.04	48.04	48.04
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.14	16.15	16.15	16.15	16.15
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.646e+004	2.647e+004	2.647e+004	2.648e+004	2.648e+004
24	Pressure (bar)	67.93	67.93	67.93	67.93	67.93
25	Temperature (C)	48.04	48.04	48.03	48.03	48.03
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.15	16.15	16.15	16.15	16.14
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.649e+004	2.649e+004	2.650e+004	2.650e+004	2.651e+004
36	Pressure (bar)	67.93	67.93	67.93	67.93	67.93
37	Temperature (C)	48.03	48.03	48.03	48.02	48.02
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.651e+004	2.652e+004	2.652e+004	2.653e+004	2.653e+004
48	Pressure (bar)	67.93	67.93	67.93	67.93	67.93
49	Temperature (C)	48.02	48.02	48.02	48.02	48.02
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.654e+004	2.654e+004	2.655e+004	2.655e+004	2.656e+004
60	Pressure (bar)	67.92	67.92	67.92	67.92	67.92
61	Temperature (C)	48.01	48.01	48.01	48.01	48.01
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.656e+004	2.657e+004	2.657e+004	2.658e+004	2.658e+004
16	Pressure (bar)	67.92	67.92	67.92	67.92	67.92
17	Temperature (C)	48.01	48.00	48.00	48.00	48.00
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.659e+004	2.659e+004	2.660e+004	2.660e+004	2.661e+004
28	Pressure (bar)	67.92	67.92	67.92	67.92	67.92
29	Temperature (C)	48.00	48.00	48.00	47.99	47.99
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.661e+004	2.662e+004	2.662e+004	2.663e+004	2.663e+004
40	Pressure (bar)	67.92	67.92	67.92	67.92	67.92
41	Temperature (C)	47.99	47.99	47.99	47.99	47.99
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.664e+004	2.664e+004	2.665e+004	2.665e+004	2.666e+004
52	Pressure (bar)	67.92	67.92	67.92	67.92	67.92
53	Temperature (C)	47.99	47.98	47.98	47.98	47.98
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.666e+004	2.667e+004	2.667e+004	2.668e+004	2.668e+004
64	Pressure (bar)	67.92	67.92	67.91	67.91	67.91
65	Temperature (C)	47.98	47.98	47.97	47.97	47.97
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.669e+004	2.669e+004	2.670e+004	2.670e+004	2.671e+004
20	Pressure (bar)	67.91	67.91	67.91	67.91	67.91
21	Temperature (C)	47.97	47.97	47.96	47.96	47.96
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.671e+004	2.672e+004	2.672e+004	2.673e+004	2.674e+004
32	Pressure (bar)	67.90	67.90	67.90	67.90	67.90
33	Temperature (C)	47.96	47.96	47.96	47.95	47.95
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.674e+004	2.674e+004	2.675e+004	2.675e+004	2.676e+004
44	Pressure (bar)	67.90	67.90	67.90	67.90	67.90
45	Temperature (C)	47.95	47.95	47.95	47.95	47.94
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.676e+004	2.677e+004	2.677e+004	2.678e+004	2.678e+004
56	Pressure (bar)	67.90	67.90	67.90	67.90	67.90
57	Temperature (C)	47.94	47.94	47.94	47.94	47.94
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.679e+004	2.679e+004	2.680e+004	2.680e+004	2.681e+004
68	Pressure (bar)	67.90	67.90	67.89	67.89	67.89
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:13:43 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	47.94	47.93	47.93	47.93	47.93
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.681e+004	2.682e+004	2.682e+004	2.683e+004	2.683e+004
24	Pressure (bar)	67.89	67.89	67.89	67.89	67.89
25	Temperature (C)	47.93	47.93	47.92	47.92	47.92
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.684e+004	2.684e+004	2.685e+004	2.685e+004	2.686e+004
36	Pressure (bar)	67.89	67.89	67.89	67.89	67.89
37	Temperature (C)	47.92	47.92	47.92	47.92	47.91
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.686e+004	2.687e+004	2.687e+004	2.688e+004	2.688e+004
48	Pressure (bar)	67.89	67.89	67.89	67.89	67.89
49	Temperature (C)	47.91	47.91	47.91	47.91	47.91
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.689e+004	2.689e+004	2.690e+004	2.690e+004	2.691e+004
60	Pressure (bar)	67.89	67.89	67.89	67.89	67.89
61	Temperature (C)	47.91	47.90	47.90	47.90	47.90
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 869 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.691e+004	2.692e+004	2.692e+004	2.693e+004	2.693e+004
16	Pressure (bar)	67.88	67.88	67.88	67.88	67.88
17	Temperature (C)	47.90	47.90	47.90	47.89	47.89
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.694e+004	2.694e+004	2.695e+004	2.695e+004	2.696e+004
28	Pressure (bar)	67.88	67.88	67.88	67.88	67.88
29	Temperature (C)	47.89	47.89	47.89	47.89	47.89
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.696e+004	2.697e+004	2.697e+004	2.698e+004	2.698e+004
40	Pressure (bar)	67.88	67.88	67.88	67.88	67.88
41	Temperature (C)	47.88	47.88	47.88	47.88	47.88
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.699e+004	2.699e+004	2.700e+004	2.700e+004	2.701e+004
52	Pressure (bar)	67.88	67.88	67.88	67.88	67.88
53	Temperature (C)	47.88	47.88	47.87	47.87	47.87
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.701e+004	2.702e+004	2.702e+004	2.703e+004	2.703e+004
64	Pressure (bar)	67.88	67.88	67.88	67.88	67.88
65	Temperature (C)	47.87	47.87	47.87	47.87	47.87
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 870 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.704e+004	2.704e+004	2.704e+004	2.704e+004	2.704e+004
20	Pressure (bar)	67.88	67.88	67.88	67.88	67.88
21	Temperature (C)	47.86	47.86	47.86	47.86	47.86
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.705e+004	2.705e+004	2.705e+004	2.705e+004	2.706e+004
32	Pressure (bar)	67.88	67.88	67.88	67.88	67.88
33	Temperature (C)	47.86	47.86	47.86	47.86	47.86
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.706e+004	2.707e+004	2.707e+004	2.708e+004	2.708e+004
44	Pressure (bar)	67.88	67.88	67.88	67.88	67.88
45	Temperature (C)	47.86	47.85	47.85	47.85	47.85
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.709e+004	2.709e+004	2.710e+004	2.710e+004	2.711e+004
56	Pressure (bar)	67.88	67.88	67.88	67.88	67.88
57	Temperature (C)	47.85	47.85	47.85	47.84	47.84
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.711e+004	2.712e+004	2.712e+004	2.713e+004	2.713e+004
68	Pressure (bar)	67.88	67.87	67.87	67.87	67.87
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2			Unit Set: Mixed			
3			Date/Time: Mon Jun 13 15:13:43 2022			
4						
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	47.84	47.84	47.84	47.84	47.83
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.714e+004	2.714e+004	2.715e+004	2.715e+004	2.716e+004
24	Pressure (bar)	67.87	67.87	67.86	67.86	67.86
25	Temperature (C)	47.83	47.83	47.83	47.83	47.82
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.716e+004	2.717e+004	2.717e+004	2.718e+004	2.718e+004
36	Pressure (bar)	67.86	67.86	67.86	67.86	67.86
37	Temperature (C)	47.82	47.82	47.82	47.82	47.82
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.719e+004	2.719e+004	2.720e+004	2.720e+004	2.721e+004
48	Pressure (bar)	67.86	67.86	67.86	67.86	67.86
49	Temperature (C)	47.81	47.81	47.81	47.81	47.81
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.721e+004	2.722e+004	2.722e+004	2.723e+004	2.723e+004
60	Pressure (bar)	67.85	67.85	67.85	67.85	67.85
61	Temperature (C)	47.81	47.80	47.80	47.80	47.80
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.724e+004	2.724e+004	2.725e+004	2.725e+004	2.726e+004
16	Pressure (bar)	67.85	67.85	67.85	67.85	67.85
17	Temperature (C)	47.80	47.80	47.80	47.79	47.79
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.726e+004	2.727e+004	2.727e+004	2.728e+004	2.728e+004
28	Pressure (bar)	67.85	67.85	67.85	67.85	67.85
29	Temperature (C)	47.79	47.79	47.79	47.79	47.79
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.729e+004	2.729e+004	2.730e+004	2.730e+004	2.731e+004
40	Pressure (bar)	67.85	67.85	67.85	67.85	67.85
41	Temperature (C)	47.78	47.78	47.78	47.78	47.78
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.731e+004	2.732e+004	2.732e+004	2.733e+004	2.733e+004
52	Pressure (bar)	67.85	67.84	67.84	67.84	67.84
53	Temperature (C)	47.78	47.77	47.77	47.77	47.77
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.734e+004	2.734e+004	2.735e+004	2.735e+004	2.736e+004
64	Pressure (bar)	67.84	67.84	67.84	67.84	67.84
65	Temperature (C)	47.77	47.77	47.77	47.76	47.76
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 873 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.736e+004	2.737e+004	2.737e+004	2.738e+004	2.738e+004
20	Pressure (bar)	67.84	67.84	67.84	67.84	67.84
21	Temperature (C)	47.76	47.76	47.76	47.76	47.76
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.739e+004	2.739e+004	2.740e+004	2.740e+004	2.741e+004
32	Pressure (bar)	67.84	67.84	67.84	67.84	67.84
33	Temperature (C)	47.75	47.75	47.75	47.75	47.75
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.741e+004	2.742e+004	2.742e+004	2.743e+004	2.743e+004
44	Pressure (bar)	67.84	67.84	67.84	67.84	67.83
45	Temperature (C)	47.75	47.74	47.74	47.74	47.74
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.744e+004	2.744e+004	2.745e+004	2.745e+004	2.746e+004
56	Pressure (bar)	67.83	67.83	67.83	67.83	67.83
57	Temperature (C)	47.74	47.74	47.74	47.73	47.73
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.746e+004	2.747e+004	2.747e+004	2.748e+004	2.748e+004
68	Pressure (bar)	67.83	67.83	67.83	67.83	67.83
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	47.73	47.73	47.73	47.73	47.73
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.749e+004	2.749e+004	2.749e+004	2.749e+004	2.749e+004
24	Pressure (bar)	67.83	67.83	67.83	67.83	67.83
25	Temperature (C)	47.72	47.72	47.72	47.72	47.72
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.750e+004	2.750e+004	2.751e+004	2.751e+004	2.752e+004
36	Pressure (bar)	67.83	67.83	67.83	67.83	67.83
37	Temperature (C)	47.72	47.72	47.72	47.72	47.71
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.753e+004	2.753e+004	2.753e+004	2.753e+004	2.753e+004
48	Pressure (bar)	67.83	67.83	67.83	67.83	67.83
49	Temperature (C)	47.71	47.71	47.71	47.71	47.71
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.754e+004	2.754e+004	2.755e+004	2.755e+004	2.756e+004
60	Pressure (bar)	67.83	67.83	67.83	67.82	67.82
61	Temperature (C)	47.71	47.71	47.71	47.70	47.70
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.14	16.14	16.14	16.14	16.14
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.756e+004	2.757e+004	2.757e+004	2.758e+004	2.758e+004
16	Pressure (bar)	67.82	67.82	67.82	67.81	67.81
17	Temperature (C)	47.70	47.70	47.70	47.69	47.69
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.14	16.13	16.13	16.13	16.13
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.759e+004	2.759e+004	2.760e+004	2.760e+004	2.761e+004
28	Pressure (bar)	67.81	67.81	67.81	67.82	67.82
29	Temperature (C)	47.69	47.69	47.69	47.69	47.69
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.761e+004	2.762e+004	2.762e+004	2.763e+004	2.763e+004
40	Pressure (bar)	67.82	67.82	67.81	67.81	67.80
41	Temperature (C)	47.69	47.69	47.68	47.68	47.68
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.764e+004	2.764e+004	2.765e+004	2.765e+004	2.766e+004
52	Pressure (bar)	67.80	67.80	67.80	67.79	67.79
53	Temperature (C)	47.67	47.67	47.67	47.67	47.67
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.766e+004	2.767e+004	2.767e+004	2.768e+004	2.768e+004
64	Pressure (bar)	67.79	67.79	67.79	67.79	67.80
65	Temperature (C)	47.66	47.66	47.66	47.66	47.66
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 876 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.769e+004	2.769e+004	2.770e+004	2.770e+004	2.771e+004
20	Pressure (bar)	67.80	67.80	67.80	67.80	67.80
21	Temperature (C)	47.66	47.66	47.66	47.66	47.65
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.771e+004	2.772e+004	2.772e+004	2.773e+004	2.773e+004
32	Pressure (bar)	67.80	67.79	67.79	67.79	67.79
33	Temperature (C)	47.65	47.65	47.65	47.65	47.65
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.774e+004	2.774e+004	2.775e+004	2.775e+004	2.776e+004
44	Pressure (bar)	67.79	67.79	67.79	67.79	67.79
45	Temperature (C)	47.65	47.64	47.64	47.64	47.64
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.776e+004	2.777e+004	2.777e+004	2.778e+004	2.778e+004
56	Pressure (bar)	67.79	67.79	67.79	67.79	67.79
57	Temperature (C)	47.64	47.64	47.64	47.63	47.63
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.779e+004	2.779e+004	2.780e+004	2.780e+004	2.781e+004
68	Pressure (bar)	67.79	67.79	67.79	67.79	67.79
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 877 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:13:43 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	47.63	47.63	47.63	47.63	47.62
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.781e+004	2.782e+004	2.782e+004	2.783e+004	2.783e+004
24	Pressure (bar)	67.79	67.79	67.79	67.79	67.78
25	Temperature (C)	47.62	47.62	47.62	47.62	47.62
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.784e+004	2.784e+004	2.785e+004	2.785e+004	2.785e+004
36	Pressure (bar)	67.78	67.78	67.78	67.78	67.78
37	Temperature (C)	47.62	47.61	47.61	47.61	47.61
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.786e+004	2.788e+004	2.790e+004	2.791e+004	2.793e+004
48	Pressure (bar)	67.78	67.78	67.78	67.78	67.78
49	Temperature (C)	47.61	47.60	47.60	47.59	47.59
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.795e+004	2.797e+004	2.799e+004	2.801e+004	2.802e+004
60	Pressure (bar)	67.77	67.77	67.77	67.77	67.77
61	Temperature (C)	47.58	47.58	47.57	47.57	47.56
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.804e+004	2.806e+004	2.808e+004	2.810e+004	2.812e+004
16	Pressure (bar)	67.77	67.76	67.76	67.76	67.76
17	Temperature (C)	47.55	47.55	47.54	47.54	47.53
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.814e+004	2.815e+004	2.817e+004	2.819e+004	2.821e+004
28	Pressure (bar)	67.76	67.76	67.76	67.75	67.75
29	Temperature (C)	47.53	47.52	47.52	47.51	47.51
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.823e+004	2.825e+004	2.827e+004	2.828e+004	2.830e+004
40	Pressure (bar)	67.75	67.75	67.75	67.75	67.74
41	Temperature (C)	47.50	47.49	47.49	47.48	47.48
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.832e+004	2.834e+004	2.836e+004	2.838e+004	2.839e+004
52	Pressure (bar)	67.74	67.74	67.74	67.74	67.74
53	Temperature (C)	47.47	47.47	47.46	47.46	47.45
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.841e+004	2.842e+004	2.843e+004	2.843e+004	2.844e+004
64	Pressure (bar)	67.73	67.73	67.73	67.73	67.73
65	Temperature (C)	47.45	47.44	47.44	47.44	47.44
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.845e+004	2.845e+004	2.846e+004	2.846e+004	2.847e+004
20	Pressure (bar)	67.73	67.73	67.73	67.73	67.73
21	Temperature (C)	47.44	47.43	47.43	47.43	47.43
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.847e+004	2.848e+004	2.848e+004	2.849e+004	2.849e+004
32	Pressure (bar)	67.73	67.73	67.73	67.73	67.73
33	Temperature (C)	47.43	47.43	47.43	47.42	47.42
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.850e+004	2.850e+004	2.851e+004	2.851e+004	2.852e+004
44	Pressure (bar)	67.73	67.73	67.73	67.73	67.73
45	Temperature (C)	47.42	47.42	47.42	47.42	47.42
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.852e+004	2.853e+004	2.853e+004	2.854e+004	2.854e+004
56	Pressure (bar)	67.73	67.73	67.73	67.72	67.72
57	Temperature (C)	47.41	47.41	47.41	47.41	47.41
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.855e+004	2.855e+004	2.856e+004	2.856e+004	2.857e+004
68	Pressure (bar)	67.72	67.72	67.72	67.72	67.72
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:13:43 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	47.41	47.41	47.40	47.40	47.40
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.857e+004	2.858e+004	2.858e+004	2.859e+004	2.859e+004
24	Pressure (bar)	67.72	67.72	67.72	67.72	67.72
25	Temperature (C)	47.40	47.40	47.40	47.40	47.39
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.860e+004	2.860e+004	2.861e+004	2.861e+004	2.862e+004
36	Pressure (bar)	67.72	67.72	67.72	67.72	67.72
37	Temperature (C)	47.39	47.39	47.39	47.39	47.39
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.862e+004	2.863e+004	2.863e+004	2.864e+004	2.864e+004
48	Pressure (bar)	67.73	67.73	67.73	67.73	67.73
49	Temperature (C)	47.39	47.39	47.39	47.38	47.38
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.865e+004	2.865e+004	2.866e+004	2.866e+004	2.867e+004
60	Pressure (bar)	67.73	67.72	67.72	67.72	67.72
61	Temperature (C)	47.38	47.38	47.38	47.38	47.38
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.13	16.13	16.13	16.13	16.13
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 881 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.867e+004	2.868e+004	2.868e+004	2.869e+004	2.869e+004
16	Pressure (bar)	67.72	67.72	67.72	67.71	67.71
17	Temperature (C)	47.37	47.37	47.37	47.37	47.36
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.13	16.13	16.12	16.12	16.12
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.870e+004	2.870e+004	2.871e+004	2.871e+004	2.872e+004
28	Pressure (bar)	67.71	67.71	67.71	67.71	67.71
29	Temperature (C)	47.36	47.36	47.36	47.36	47.36
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.872e+004	2.873e+004	2.874e+004	2.875e+004	2.876e+004
40	Pressure (bar)	67.71	67.70	67.70	67.70	67.70
41	Temperature (C)	47.35	47.35	47.35	47.35	47.34
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.877e+004	2.878e+004	2.879e+004	2.880e+004	2.881e+004
52	Pressure (bar)	67.70	67.70	67.70	67.70	67.70
53	Temperature (C)	47.34	47.34	47.33	47.33	47.33
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.883e+004	2.884e+004	2.885e+004	2.886e+004	2.887e+004
64	Pressure (bar)	67.70	67.70	67.69	67.69	67.69
65	Temperature (C)	47.32	47.32	47.32	47.32	47.31
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.888e+004	2.889e+004	2.890e+004	2.891e+004	2.892e+004
20	Pressure (bar)	67.69	67.69	67.69	67.69	67.69
21	Temperature (C)	47.31	47.31	47.30	47.30	47.30
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.893e+004	2.894e+004	2.895e+004	2.896e+004	2.897e+004
32	Pressure (bar)	67.69	67.69	67.69	67.69	67.68
33	Temperature (C)	47.29	47.29	47.29	47.29	47.28
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.898e+004	2.899e+004	2.900e+004	2.901e+004	2.902e+004
44	Pressure (bar)	67.68	67.68	67.68	67.68	67.68
45	Temperature (C)	47.28	47.28	47.27	47.27	47.27
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.903e+004	2.904e+004	2.904e+004	2.904e+004	2.904e+004
56	Pressure (bar)	67.68	67.68	67.68	67.68	67.68
57	Temperature (C)	47.27	47.26	47.26	47.26	47.26
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.905e+004	2.905e+004	2.906e+004	2.906e+004	2.907e+004
68	Pressure (bar)	67.68	67.68	67.68	67.68	67.68
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	47.26	47.26	47.26	47.26	47.25
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.907e+004	2.908e+004	2.908e+004	2.909e+004	2.909e+004
24	Pressure (bar)	67.68	67.68	67.68	67.68	67.68
25	Temperature (C)	47.25	47.25	47.25	47.25	47.25
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.910e+004	2.910e+004	2.911e+004	2.911e+004	2.912e+004
36	Pressure (bar)	67.68	67.68	67.68	67.67	67.67
37	Temperature (C)	47.25	47.25	47.24	47.24	47.24
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.912e+004	2.913e+004	2.913e+004	2.914e+004	2.914e+004
48	Pressure (bar)	67.67	67.67	67.67	67.67	67.67
49	Temperature (C)	47.24	47.24	47.24	47.23	47.23
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.915e+004	2.915e+004	2.916e+004	2.916e+004	2.916e+004
60	Pressure (bar)	67.67	67.67	67.67	67.67	67.67
61	Temperature (C)	47.23	47.23	47.23	47.23	47.23
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 884 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.917e+004	2.917e+004	2.918e+004	2.918e+004	2.919e+004
16	Pressure (bar)	67.67	67.67	67.66	67.66	67.66
17	Temperature (C)	47.22	47.22	47.22	47.22	47.22
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.920e+004	2.920e+004	2.921e+004	2.921e+004	2.922e+004
28	Pressure (bar)	67.66	67.66	67.66	67.66	67.66
29	Temperature (C)	47.22	47.21	47.21	47.21	47.21
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.922e+004	2.923e+004	2.923e+004	2.924e+004	2.924e+004
40	Pressure (bar)	67.66	67.66	67.66	67.66	67.66
41	Temperature (C)	47.21	47.21	47.20	47.20	47.20
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.925e+004	2.925e+004	2.925e+004	2.926e+004	2.926e+004
52	Pressure (bar)	67.66	67.66	67.66	67.66	67.66
53	Temperature (C)	47.20	47.20	47.20	47.20	47.20
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.926e+004	2.927e+004	2.928e+004	2.929e+004	2.930e+004
64	Pressure (bar)	67.66	67.65	67.65	67.65	67.65
65	Temperature (C)	47.20	47.19	47.19	47.19	47.19
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 885 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.931e+004	2.932e+004	2.933e+004	2.934e+004	2.935e+004
20	Pressure (bar)	67.65	67.65	67.65	67.65	67.65
21	Temperature (C)	47.18	47.18	47.18	47.17	47.17
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.936e+004	2.936e+004	2.937e+004	2.938e+004	2.939e+004
32	Pressure (bar)	67.65	67.65	67.65	67.64	67.64
33	Temperature (C)	47.17	47.17	47.16	47.16	47.16
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.940e+004	2.941e+004	2.942e+004	2.943e+004	2.944e+004
44	Pressure (bar)	67.64	67.64	67.64	67.64	67.64
45	Temperature (C)	47.16	47.15	47.15	47.15	47.14
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.945e+004	2.946e+004	2.947e+004	2.948e+004	2.949e+004
56	Pressure (bar)	67.64	67.64	67.64	67.64	67.64
57	Temperature (C)	47.14	47.14	47.14	47.13	47.13
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.950e+004	2.951e+004	2.952e+004	2.953e+004	2.954e+004
68	Pressure (bar)	67.64	67.63	67.63	67.63	67.63
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	47.13	47.13	47.12	47.12	47.12
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.955e+004	2.955e+004	2.955e+004	2.955e+004	2.955e+004
24	Pressure (bar)	67.63	67.63	67.63	67.63	67.63
25	Temperature (C)	47.11	47.11	47.11	47.11	47.11
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.955e+004	2.956e+004	2.956e+004	2.957e+004	2.957e+004
36	Pressure (bar)	67.63	67.63	67.63	67.63	67.63
37	Temperature (C)	47.11	47.11	47.11	47.11	47.11
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.958e+004	2.958e+004	2.959e+004	2.959e+004	2.960e+004
48	Pressure (bar)	67.63	67.63	67.63	67.63	67.63
49	Temperature (C)	47.11	47.10	47.10	47.10	47.10
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.960e+004	2.961e+004	2.961e+004	2.962e+004	2.962e+004
60	Pressure (bar)	67.63	67.63	67.63	67.63	67.63
61	Temperature (C)	47.10	47.10	47.10	47.09	47.09
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.963e+004	2.963e+004	2.964e+004	2.964e+004	2.965e+004
16	Pressure (bar)	67.62	67.62	67.62	67.62	67.62
17	Temperature (C)	47.09	47.09	47.09	47.09	47.09
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.965e+004	2.965e+004	2.966e+004	2.966e+004	2.966e+004
28	Pressure (bar)	67.62	67.62	67.62	67.62	67.62
29	Temperature (C)	47.08	47.08	47.08	47.08	47.08
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.966e+004	2.966e+004	2.967e+004	2.967e+004	2.967e+004
40	Pressure (bar)	67.62	67.62	67.62	67.62	67.62
41	Temperature (C)	47.08	47.08	47.08	47.08	47.08
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.967e+004	2.968e+004	2.968e+004	2.968e+004	2.969e+004
52	Pressure (bar)	67.62	67.62	67.62	67.62	67.62
53	Temperature (C)	47.08	47.08	47.08	47.08	47.07
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.969e+004	2.969e+004	2.970e+004	2.970e+004	2.970e+004
64	Pressure (bar)	67.62	67.62	67.62	67.62	67.62
65	Temperature (C)	47.07	47.07	47.07	47.07	47.07
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 888 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.971e+004	2.971e+004	2.971e+004	2.972e+004	2.972e+004
20	Pressure (bar)	67.62	67.62	67.62	67.62	67.62
21	Temperature (C)	47.07	47.07	47.07	47.07	47.06
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.973e+004	2.973e+004	2.974e+004	2.974e+004	2.975e+004
32	Pressure (bar)	67.62	67.62	67.62	67.62	67.62
33	Temperature (C)	47.06	47.06	47.06	47.06	47.06
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.975e+004	2.976e+004	2.976e+004	2.977e+004	2.977e+004
44	Pressure (bar)	67.62	67.62	67.61	67.61	67.61
45	Temperature (C)	47.06	47.05	47.05	47.05	47.05
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.12	16.12	16.12	16.12	16.12
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.978e+004	2.978e+004	2.979e+004	2.979e+004	2.980e+004
56	Pressure (bar)	67.61	67.61	67.61	67.61	67.61
57	Temperature (C)	47.05	47.05	47.05	47.04	47.04
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.12	16.11	16.11	16.11	16.11
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.980e+004	2.981e+004	2.981e+004	2.982e+004	2.982e+004
68	Pressure (bar)	67.61	67.61	67.61	67.61	67.61
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 889 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	47.04	47.04	47.04	47.04	47.04
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.983e+004	2.983e+004	2.984e+004	2.984e+004	2.985e+004
24	Pressure (bar)	67.61	67.61	67.61	67.61	67.61
25	Temperature (C)	47.03	47.03	47.03	47.03	47.03
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.985e+004	2.986e+004	2.986e+004	2.987e+004	2.987e+004
36	Pressure (bar)	67.61	67.61	67.60	67.60	67.60
37	Temperature (C)	47.03	47.02	47.02	47.02	47.02
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.988e+004	2.988e+004	2.989e+004	2.989e+004	2.990e+004
48	Pressure (bar)	67.60	67.60	67.60	67.60	67.60
49	Temperature (C)	47.02	47.02	47.02	47.01	47.01
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.990e+004	2.990e+004	2.991e+004	2.991e+004	2.991e+004
60	Pressure (bar)	67.60	67.60	67.60	67.60	67.60
61	Temperature (C)	47.01	47.01	47.01	47.01	47.01
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 890 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.991e+004	2.992e+004	2.992e+004	2.993e+004	2.993e+004
16	Pressure (bar)	67.60	67.60	67.60	67.60	67.60
17	Temperature (C)	47.01	47.01	47.01	47.00	47.00
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.994e+004	2.994e+004	2.994e+004	2.995e+004	2.995e+004
28	Pressure (bar)	67.60	67.60	67.60	67.60	67.60
29	Temperature (C)	47.00	47.00	47.00	47.00	47.00
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.996e+004	2.996e+004	2.997e+004	2.997e+004	2.998e+004
40	Pressure (bar)	67.60	67.60	67.60	67.60	67.59
41	Temperature (C)	47.00	46.99	46.99	46.99	46.99
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.998e+004	2.999e+004	2.999e+004	3.000e+004	3.000e+004
52	Pressure (bar)	67.59	67.59	67.59	67.59	67.59
53	Temperature (C)	46.99	46.99	46.99	46.98	46.98
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.001e+004	3.001e+004	3.002e+004	3.002e+004	3.003e+004
64	Pressure (bar)	67.59	67.59	67.59	67.59	67.59
65	Temperature (C)	46.98	46.98	46.98	46.98	46.98
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 891 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.003e+004	3.004e+004	3.004e+004	3.005e+004	3.005e+004
20	Pressure (bar)	67.59	67.59	67.59	67.59	67.59
21	Temperature (C)	46.97	46.97	46.97	46.97	46.97
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.006e+004	3.006e+004	3.007e+004	3.007e+004	3.008e+004
32	Pressure (bar)	67.59	67.59	67.59	67.59	67.59
33	Temperature (C)	46.97	46.97	46.96	46.96	46.96
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.008e+004	3.009e+004	3.009e+004	3.010e+004	3.010e+004
44	Pressure (bar)	67.59	67.59	67.58	67.58	67.58
45	Temperature (C)	46.96	46.96	46.96	46.96	46.95
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.011e+004	3.011e+004	3.012e+004	3.012e+004	3.013e+004
56	Pressure (bar)	67.58	67.58	67.58	67.58	67.58
57	Temperature (C)	46.95	46.95	46.95	46.95	46.95
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.013e+004	3.014e+004	3.014e+004	3.015e+004	3.015e+004
68	Pressure (bar)	67.58	67.58	67.58	67.58	67.58
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:13:43 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	46.95	46.94	46.94	46.94	46.94
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.016e+004	3.016e+004	3.017e+004	3.017e+004	3.018e+004
24	Pressure (bar)	67.58	67.58	67.58	67.58	67.58
25	Temperature (C)	46.94	46.94	46.94	46.93	46.93
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.018e+004	3.019e+004	3.019e+004	3.020e+004	3.020e+004
36	Pressure (bar)	67.58	67.58	67.58	67.58	67.58
37	Temperature (C)	46.93	46.93	46.93	46.93	46.93
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.021e+004	3.021e+004	3.022e+004	3.022e+004	3.023e+004
48	Pressure (bar)	67.58	67.57	67.57	67.57	67.57
49	Temperature (C)	46.92	46.92	46.92	46.92	46.92
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.023e+004	3.024e+004	3.024e+004	3.025e+004	3.025e+004
60	Pressure (bar)	67.57	67.57	67.57	67.57	67.57
61	Temperature (C)	46.92	46.92	46.91	46.91	46.91
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 893 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.026e+004	3.026e+004	3.027e+004	3.027e+004	3.028e+004
16	Pressure (bar)	67.57	67.57	67.57	67.57	67.57
17	Temperature (C)	46.91	46.91	46.91	46.91	46.90
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.028e+004	3.029e+004	3.029e+004	3.030e+004	3.030e+004
28	Pressure (bar)	67.57	67.57	67.57	67.57	67.57
29	Temperature (C)	46.90	46.90	46.90	46.90	46.90
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.031e+004	3.031e+004	3.032e+004	3.032e+004	3.033e+004
40	Pressure (bar)	67.57	67.57	67.56	67.56	67.56
41	Temperature (C)	46.89	46.89	46.89	46.89	46.89
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.033e+004	3.034e+004	3.035e+004	3.036e+004	3.036e+004
52	Pressure (bar)	67.56	67.56	67.56	67.56	67.56
53	Temperature (C)	46.89	46.88	46.88	46.88	46.88
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.037e+004	3.039e+004	3.040e+004	3.041e+004	3.043e+004
64	Pressure (bar)	67.56	67.56	67.56	67.55	67.55
65	Temperature (C)	46.88	46.87	46.87	46.86	46.86
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 894 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.044e+004	3.045e+004	3.045e+004	3.045e+004	3.045e+004
20	Pressure (bar)	67.55	67.55	67.55	67.55	67.55
21	Temperature (C)	46.86	46.85	46.85	46.85	46.85
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.046e+004	3.046e+004	3.047e+004	3.047e+004	3.048e+004
32	Pressure (bar)	67.55	67.55	67.55	67.55	67.55
33	Temperature (C)	46.85	46.85	46.85	46.85	46.85
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.048e+004	3.049e+004	3.049e+004	3.050e+004	3.050e+004
44	Pressure (bar)	67.55	67.55	67.55	67.55	67.55
45	Temperature (C)	46.84	46.84	46.84	46.84	46.84
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.051e+004	3.051e+004	3.052e+004	3.052e+004	3.053e+004
56	Pressure (bar)	67.55	67.55	67.55	67.55	67.55
57	Temperature (C)	46.84	46.84	46.83	46.83	46.83
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.053e+004	3.054e+004	3.054e+004	3.055e+004	3.055e+004
68	Pressure (bar)	67.55	67.55	67.55	67.55	67.54
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	46.83	46.83	46.83	46.83	46.82
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.056e+004	3.056e+004	3.057e+004	3.057e+004	3.058e+004
24	Pressure (bar)	67.54	67.54	67.54	67.54	67.54
25	Temperature (C)	46.82	46.82	46.82	46.82	46.82
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.058e+004	3.059e+004	3.059e+004	3.060e+004	3.060e+004
36	Pressure (bar)	67.54	67.54	67.54	67.54	67.54
37	Temperature (C)	46.82	46.81	46.81	46.81	46.81
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.061e+004	3.061e+004	3.062e+004	3.062e+004	3.063e+004
48	Pressure (bar)	67.54	67.54	67.54	67.54	67.54
49	Temperature (C)	46.81	46.81	46.81	46.80	46.80
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.063e+004	3.064e+004	3.064e+004	3.065e+004	3.065e+004
60	Pressure (bar)	67.54	67.54	67.54	67.54	67.53
61	Temperature (C)	46.80	46.80	46.80	46.80	46.80
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.066e+004	3.066e+004	3.067e+004	3.067e+004	3.068e+004
16	Pressure (bar)	67.53	67.53	67.53	67.53	67.53
17	Temperature (C)	46.79	46.79	46.79	46.79	46.79
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.068e+004	3.069e+004	3.069e+004	3.070e+004	3.070e+004
28	Pressure (bar)	67.53	67.53	67.53	67.53	67.53
29	Temperature (C)	46.79	46.79	46.78	46.78	46.78
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.071e+004	3.071e+004	3.072e+004	3.072e+004	3.073e+004
40	Pressure (bar)	67.53	67.53	67.53	67.53	67.53
41	Temperature (C)	46.78	46.78	46.78	46.78	46.78
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.073e+004	3.074e+004	3.074e+004	3.075e+004	3.075e+004
52	Pressure (bar)	67.53	67.53	67.53	67.53	67.53
53	Temperature (C)	46.77	46.77	46.77	46.77	46.77
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.076e+004	3.076e+004	3.077e+004	3.077e+004	3.078e+004
64	Pressure (bar)	67.53	67.53	67.53	67.53	67.53
65	Temperature (C)	46.77	46.77	46.76	46.76	46.76
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.078e+004	3.079e+004	3.079e+004	3.080e+004	3.080e+004
20	Pressure (bar)	67.53	67.53	67.53	67.52	67.52
21	Temperature (C)	46.76	46.76	46.76	46.76	46.75
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.081e+004	3.081e+004	3.082e+004	3.082e+004	3.083e+004
32	Pressure (bar)	67.52	67.52	67.52	67.52	67.52
33	Temperature (C)	46.75	46.75	46.75	46.75	46.75
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.083e+004	3.084e+004	3.084e+004	3.085e+004	3.085e+004
44	Pressure (bar)	67.52	67.52	67.52	67.52	67.52
45	Temperature (C)	46.75	46.74	46.74	46.74	46.74
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.086e+004	3.086e+004	3.087e+004	3.087e+004	3.088e+004
56	Pressure (bar)	67.52	67.52	67.52	67.52	67.51
57	Temperature (C)	46.74	46.74	46.73	46.73	46.73
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.088e+004	3.089e+004	3.089e+004	3.090e+004	3.090e+004
68	Pressure (bar)	67.51	67.51	67.51	67.51	67.51
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 898 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	46.73	46.73	46.73	46.73	46.72
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.091e+004	3.091e+004	3.092e+004	3.092e+004	3.093e+004
24	Pressure (bar)	67.51	67.51	67.51	67.51	67.51
25	Temperature (C)	46.72	46.72	46.72	46.72	46.72
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.093e+004	3.094e+004	3.094e+004	3.095e+004	3.095e+004
36	Pressure (bar)	67.51	67.51	67.51	67.51	67.51
37	Temperature (C)	46.72	46.71	46.71	46.71	46.71
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.11	16.11	16.10	16.10	16.10
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.096e+004	3.096e+004	3.097e+004	3.097e+004	3.098e+004
48	Pressure (bar)	67.51	67.51	67.51	67.51	67.51
49	Temperature (C)	46.71	46.71	46.71	46.70	46.70
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.10	16.10	16.10	16.10	16.10
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.098e+004	3.099e+004	3.099e+004	3.100e+004	3.100e+004
60	Pressure (bar)	67.51	67.50	67.50	67.50	67.50
61	Temperature (C)	46.70	46.70	46.70	46.70	46.70
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.10	16.10	16.10	16.10	16.10
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 899 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.101e+004	3.103e+004	3.105e+004	3.107e+004	3.109e+004
16	Pressure (bar)	67.50	67.50	67.50	67.50	67.50
17	Temperature (C)	46.69	46.69	46.68	46.68	46.67
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.10	16.10	16.10	16.10	16.10
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.111e+004	3.111e+004	3.112e+004	3.112e+004	3.113e+004
28	Pressure (bar)	67.49	67.49	67.49	67.49	67.49
29	Temperature (C)	46.67	46.66	46.66	46.66	46.66
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.10	16.10	16.10	16.10	16.10
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.113e+004	3.114e+004	3.114e+004	3.115e+004	3.115e+004
40	Pressure (bar)	67.49	67.49	67.49	67.48	67.48
41	Temperature (C)	46.66	46.66	46.65	46.65	46.65
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.10	16.10	16.10	16.10	16.10
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.116e+004	3.116e+004	3.117e+004	3.117e+004	3.118e+004
52	Pressure (bar)	67.48	67.49	67.51	67.52	67.54
53	Temperature (C)	46.65	46.66	46.66	46.67	46.68
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.10	16.10	16.11	16.11	16.11
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.118e+004	3.119e+004	3.119e+004	3.119e+004	3.120e+004
64	Pressure (bar)	67.56	67.55	67.55	67.55	67.55
65	Temperature (C)	46.69	46.69	46.69	46.69	46.69
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.120e+004	3.121e+004	3.122e+004	3.122e+004	3.123e+004
20	Pressure (bar)	67.55	67.54	67.53	67.52	67.51
21	Temperature (C)	46.68	46.68	46.67	46.67	46.66
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.11	16.11	16.11	16.11	16.11
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.124e+004	3.124e+004	3.125e+004	3.125e+004	3.126e+004
32	Pressure (bar)	67.50	67.49	67.48	67.47	67.46
33	Temperature (C)	46.66	46.65	46.65	46.64	46.64
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.10	16.10	16.10	16.10	16.10
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.126e+004	3.126e+004	3.127e+004	3.127e+004	3.128e+004
44	Pressure (bar)	67.45	67.40	67.35	67.29	67.24
45	Temperature (C)	46.63	46.62	46.60	46.59	46.57
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.10	16.09	16.09	16.08	16.08
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.128e+004	3.129e+004	3.129e+004	3.130e+004	3.130e+004
56	Pressure (bar)	67.19	67.19	67.19	67.19	67.19
57	Temperature (C)	46.55	46.55	46.55	46.55	46.55
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.131e+004	3.131e+004	3.132e+004	3.132e+004	3.133e+004
68	Pressure (bar)	67.19	67.19	67.19	67.19	67.19
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:13:43 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	46.55	46.54	46.54	46.54	46.54
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.133e+004	3.134e+004	3.134e+004	3.135e+004	3.135e+004
24	Pressure (bar)	67.19	67.19	67.19	67.19	67.19
25	Temperature (C)	46.54	46.54	46.54	46.53	46.53
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.136e+004	3.136e+004	3.137e+004	3.137e+004	3.138e+004
36	Pressure (bar)	67.19	67.19	67.18	67.18	67.18
37	Temperature (C)	46.53	46.53	46.53	46.53	46.53
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.138e+004	3.139e+004	3.139e+004	3.140e+004	3.140e+004
48	Pressure (bar)	67.18	67.18	67.18	67.18	67.18
49	Temperature (C)	46.52	46.52	46.52	46.52	46.52
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.141e+004	3.141e+004	3.142e+004	3.142e+004	3.143e+004
60	Pressure (bar)	67.18	67.18	67.18	67.18	67.18
61	Temperature (C)	46.52	46.52	46.51	46.51	46.51
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 902 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.143e+004	3.144e+004	3.144e+004	3.145e+004	3.145e+004
16	Pressure (bar)	67.18	67.18	67.18	67.18	67.18
17	Temperature (C)	46.51	46.51	46.51	46.51	46.50
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.146e+004	3.146e+004	3.147e+004	3.147e+004	3.148e+004
28	Pressure (bar)	67.18	67.18	67.18	67.17	67.17
29	Temperature (C)	46.50	46.50	46.50	46.50	46.50
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.148e+004	3.149e+004	3.149e+004	3.150e+004	3.150e+004
40	Pressure (bar)	67.17	67.17	67.17	67.17	67.17
41	Temperature (C)	46.50	46.50	46.49	46.49	46.49
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.151e+004	3.151e+004	3.152e+004	3.152e+004	3.153e+004
52	Pressure (bar)	67.17	67.17	67.17	67.17	67.17
53	Temperature (C)	46.49	46.49	46.49	46.49	46.48
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.153e+004	3.154e+004	3.154e+004	3.155e+004	3.155e+004
64	Pressure (bar)	67.17	67.17	67.17	67.17	67.17
65	Temperature (C)	46.48	46.48	46.48	46.48	46.48
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.156e+004	3.157e+004	3.157e+004	3.158e+004	3.158e+004
20	Pressure (bar)	67.17	67.17	67.17	67.17	67.17
21	Temperature (C)	46.47	46.47	46.47	46.47	46.47
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.159e+004	3.159e+004	3.160e+004	3.160e+004	3.161e+004
32	Pressure (bar)	67.16	67.16	67.16	67.16	67.16
33	Temperature (C)	46.47	46.47	46.47	46.46	46.46
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.161e+004	3.162e+004	3.162e+004	3.163e+004	3.163e+004
44	Pressure (bar)	67.16	67.16	67.16	67.16	67.16
45	Temperature (C)	46.46	46.46	46.46	46.46	46.46
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.164e+004	3.164e+004	3.165e+004	3.165e+004	3.166e+004
56	Pressure (bar)	67.16	67.16	67.16	67.16	67.16
57	Temperature (C)	46.45	46.45	46.45	46.45	46.45
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.166e+004	3.167e+004	3.167e+004	3.168e+004	3.168e+004
68	Pressure (bar)	67.16	67.16	67.16	67.16	67.15
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:13:43 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	46.45	46.45	46.44	46.44	46.44
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.169e+004	3.169e+004	3.169e+004	3.169e+004	3.170e+004
24	Pressure (bar)	67.15	67.15	67.15	67.15	67.15
25	Temperature (C)	46.44	46.44	46.44	46.44	46.44
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.170e+004	3.170e+004	3.170e+004	3.171e+004	3.171e+004
36	Pressure (bar)	67.15	67.15	67.15	67.15	67.15
37	Temperature (C)	46.44	46.43	46.43	46.43	46.43
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.171e+004	3.172e+004	3.172e+004	3.173e+004	3.173e+004
48	Pressure (bar)	67.15	67.15	67.15	67.15	67.15
49	Temperature (C)	46.43	46.43	46.43	46.43	46.43
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.174e+004	3.174e+004	3.174e+004	3.174e+004	3.175e+004
60	Pressure (bar)	67.15	67.15	67.15	67.15	67.15
61	Temperature (C)	46.43	46.43	46.42	46.42	46.42
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.175e+004	3.176e+004	3.176e+004	3.177e+004	3.177e+004
16	Pressure (bar)	67.15	67.15	67.15	67.15	67.15
17	Temperature (C)	46.42	46.42	46.42	46.42	46.42
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.178e+004	3.179e+004	3.179e+004	3.180e+004	3.180e+004
28	Pressure (bar)	67.15	67.15	67.15	67.15	67.14
29	Temperature (C)	46.41	46.41	46.41	46.41	46.41
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.181e+004	3.182e+004	3.182e+004	3.183e+004	3.183e+004
40	Pressure (bar)	67.14	67.14	67.14	67.14	67.14
41	Temperature (C)	46.41	46.40	46.40	46.40	46.40
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.184e+004	3.184e+004	3.185e+004	3.185e+004	3.186e+004
52	Pressure (bar)	67.14	67.14	67.14	67.14	67.14
53	Temperature (C)	46.40	46.40	46.40	46.39	46.39
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.186e+004	3.187e+004	3.187e+004	3.188e+004	3.188e+004
64	Pressure (bar)	67.14	67.14	67.14	67.14	67.14
65	Temperature (C)	46.39	46.39	46.39	46.39	46.39
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 906 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.189e+004	3.189e+004	3.190e+004	3.190e+004	3.191e+004
20	Pressure (bar)	67.14	67.14	67.14	67.14	67.13
21	Temperature (C)	46.38	46.38	46.38	46.38	46.38
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.191e+004	3.192e+004	3.192e+004	3.193e+004	3.193e+004
32	Pressure (bar)	67.13	67.13	67.13	67.13	67.13
33	Temperature (C)	46.38	46.38	46.37	46.37	46.37
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.194e+004	3.194e+004	3.195e+004	3.195e+004	3.196e+004
44	Pressure (bar)	67.13	67.13	67.13	67.13	67.13
45	Temperature (C)	46.37	46.37	46.37	46.36	46.36
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.196e+004	3.197e+004	3.197e+004	3.198e+004	3.198e+004
56	Pressure (bar)	67.13	67.13	67.13	67.13	67.12
57	Temperature (C)	46.36	46.36	46.36	46.36	46.36
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.199e+004	3.199e+004	3.200e+004	3.200e+004	3.201e+004
68	Pressure (bar)	67.12	67.12	67.12	67.12	67.12
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	46.35	46.35	46.35	46.35	46.35
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.201e+004	3.202e+004	3.202e+004	3.203e+004	3.203e+004
24	Pressure (bar)	67.12	67.12	67.12	67.12	67.12
25	Temperature (C)	46.35	46.35	46.35	46.34	46.34
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.204e+004	3.204e+004	3.205e+004	3.205e+004	3.206e+004
36	Pressure (bar)	67.12	67.12	67.12	67.12	67.12
37	Temperature (C)	46.34	46.34	46.34	46.34	46.34
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.206e+004	3.207e+004	3.207e+004	3.208e+004	3.208e+004
48	Pressure (bar)	67.12	67.12	67.12	67.12	67.12
49	Temperature (C)	46.33	46.33	46.33	46.33	46.33
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.209e+004	3.209e+004	3.210e+004	3.210e+004	3.211e+004
60	Pressure (bar)	67.12	67.12	67.12	67.12	67.12
61	Temperature (C)	46.33	46.33	46.32	46.32	46.32
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.211e+004	3.212e+004	3.212e+004	3.213e+004	3.213e+004
16	Pressure (bar)	67.11	67.11	67.11	67.11	67.11
17	Temperature (C)	46.32	46.32	46.32	46.32	46.31
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.214e+004	3.214e+004	3.215e+004	3.215e+004	3.216e+004
28	Pressure (bar)	67.11	67.11	67.11	67.11	67.11
29	Temperature (C)	46.31	46.31	46.31	46.31	46.31
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.216e+004	3.217e+004	3.217e+004	3.218e+004	3.218e+004
40	Pressure (bar)	67.11	67.10	67.10	67.10	67.10
41	Temperature (C)	46.30	46.30	46.30	46.30	46.30
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.219e+004	3.219e+004	3.219e+004	3.219e+004	3.219e+004
52	Pressure (bar)	67.10	67.10	67.10	67.10	67.09
53	Temperature (C)	46.30	46.30	46.30	46.29	46.29
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.07	16.07	16.07	16.07	16.07
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.219e+004	3.219e+004	3.220e+004	3.220e+004	3.221e+004
64	Pressure (bar)	67.09	67.09	67.09	67.08	67.08
65	Temperature (C)	46.29	46.29	46.29	46.29	46.29
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.07	16.06	16.06	16.06	16.06
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.221e+004	3.221e+004	3.222e+004	3.222e+004	3.223e+004
20	Pressure (bar)	67.08	67.08	67.08	67.08	67.08
21	Temperature (C)	46.28	46.28	46.28	46.28	46.28
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.223e+004	3.224e+004	3.226e+004	3.227e+004	3.228e+004
32	Pressure (bar)	67.08	67.08	67.07	67.07	67.07
33	Temperature (C)	46.28	46.27	46.27	46.27	46.26
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.230e+004	3.231e+004	3.232e+004	3.234e+004	3.235e+004
44	Pressure (bar)	67.07	67.07	67.07	67.07	67.07
45	Temperature (C)	46.26	46.25	46.25	46.25	46.24
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.236e+004	3.238e+004	3.239e+004	3.240e+004	3.242e+004
56	Pressure (bar)	67.07	67.06	67.06	67.06	67.06
57	Temperature (C)	46.24	46.24	46.23	46.23	46.22
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.243e+004	3.244e+004	3.246e+004	3.247e+004	3.248e+004
68	Pressure (bar)	67.06	67.06	67.06	67.05	67.05
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	46.22	46.22	46.21	46.21	46.21
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.250e+004	3.251e+004	3.252e+004	3.254e+004	3.255e+004
24	Pressure (bar)	67.05	67.05	67.05	67.05	67.05
25	Temperature (C)	46.20	46.20	46.20	46.19	46.19
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.257e+004	3.257e+004	3.258e+004	3.258e+004	3.259e+004
36	Pressure (bar)	67.05	67.05	67.05	67.05	67.05
37	Temperature (C)	46.19	46.18	46.18	46.18	46.18
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.260e+004	3.260e+004	3.261e+004	3.261e+004	3.262e+004
48	Pressure (bar)	67.05	67.05	67.05	67.05	67.05
49	Temperature (C)	46.18	46.18	46.17	46.17	46.17
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.263e+004	3.263e+004	3.263e+004	3.263e+004	3.263e+004
60	Pressure (bar)	67.05	67.05	67.05	67.05	67.05
61	Temperature (C)	46.17	46.17	46.17	46.17	46.17
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 911 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.264e+004	3.264e+004	3.264e+004	3.264e+004	3.264e+004
16	Pressure (bar)	67.05	67.05	67.05	67.05	67.05
17	Temperature (C)	46.17	46.17	46.17	46.17	46.17
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.265e+004	3.265e+004	3.266e+004	3.266e+004	3.267e+004
28	Pressure (bar)	67.05	67.04	67.04	67.04	67.04
29	Temperature (C)	46.16	46.16	46.16	46.16	46.16
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.267e+004	3.268e+004	3.268e+004	3.269e+004	3.269e+004
40	Pressure (bar)	67.04	67.04	67.04	67.04	67.04
41	Temperature (C)	46.16	46.16	46.16	46.15	46.15
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.270e+004	3.270e+004	3.271e+004	3.271e+004	3.272e+004
52	Pressure (bar)	67.04	67.04	67.04	67.04	67.04
53	Temperature (C)	46.15	46.15	46.15	46.15	46.15
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.272e+004	3.273e+004	3.273e+004	3.274e+004	3.274e+004
64	Pressure (bar)	67.04	67.04	67.04	67.04	67.04
65	Temperature (C)	46.14	46.14	46.14	46.14	46.14
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.275e+004	3.275e+004	3.276e+004	3.276e+004	3.277e+004
20	Pressure (bar)	67.04	67.04	67.04	67.04	67.03
21	Temperature (C)	46.14	46.14	46.13	46.13	46.13
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.277e+004	3.278e+004	3.278e+004	3.279e+004	3.279e+004
32	Pressure (bar)	67.03	67.03	67.03	67.03	67.03
33	Temperature (C)	46.13	46.13	46.13	46.13	46.13
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.280e+004	3.280e+004	3.281e+004	3.281e+004	3.282e+004
44	Pressure (bar)	67.03	67.03	67.03	67.03	67.03
45	Temperature (C)	46.12	46.12	46.12	46.12	46.12
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.282e+004	3.283e+004	3.283e+004	3.284e+004	3.284e+004
56	Pressure (bar)	67.03	67.03	67.03	67.03	67.03
57	Temperature (C)	46.12	46.12	46.12	46.11	46.11
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.285e+004	3.285e+004	3.286e+004	3.286e+004	3.287e+004
68	Pressure (bar)	67.03	67.03	67.03	67.03	67.03
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	46.11	46.11	46.11	46.11	46.11
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.287e+004	3.288e+004	3.288e+004	3.289e+004	3.289e+004
24	Pressure (bar)	67.03	67.03	67.03	67.03	67.03
25	Temperature (C)	46.11	46.11	46.10	46.10	46.10
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.290e+004	3.290e+004	3.291e+004	3.291e+004	3.292e+004
36	Pressure (bar)	67.03	67.03	67.03	67.03	67.03
37	Temperature (C)	46.10	46.10	46.10	46.10	46.10
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.292e+004	3.293e+004	3.293e+004	3.294e+004	3.294e+004
48	Pressure (bar)	67.03	67.03	67.03	67.03	67.03
49	Temperature (C)	46.09	46.09	46.09	46.09	46.09
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.295e+004	3.295e+004	3.296e+004	3.296e+004	3.297e+004
60	Pressure (bar)	67.03	67.03	67.03	67.03	67.03
61	Temperature (C)	46.09	46.09	46.08	46.08	46.08
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 914 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.297e+004	3.298e+004	3.298e+004	3.299e+004	3.299e+004
16	Pressure (bar)	67.03	67.03	67.03	67.03	67.03
17	Temperature (C)	46.08	46.08	46.08	46.08	46.08
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.300e+004	3.300e+004	3.301e+004	3.301e+004	3.302e+004
28	Pressure (bar)	67.03	67.03	67.03	67.03	67.02
29	Temperature (C)	46.07	46.07	46.07	46.07	46.07
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.302e+004	3.302e+004	3.302e+004	3.303e+004	3.303e+004
40	Pressure (bar)	67.02	67.02	67.02	67.02	67.02
41	Temperature (C)	46.07	46.07	46.07	46.07	46.06
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.303e+004	3.304e+004	3.304e+004	3.305e+004	3.305e+004
52	Pressure (bar)	67.02	67.02	67.02	67.02	67.02
53	Temperature (C)	46.06	46.06	46.06	46.06	46.06
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.306e+004	3.306e+004	3.306e+004	3.307e+004	3.307e+004
64	Pressure (bar)	67.02	67.02	67.02	67.02	67.02
65	Temperature (C)	46.06	46.06	46.05	46.05	46.05
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 915 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.307e+004	3.308e+004	3.308e+004	3.309e+004	3.309e+004
20	Pressure (bar)	67.02	67.02	67.02	67.02	67.02
21	Temperature (C)	46.05	46.05	46.05	46.05	46.05
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.310e+004	3.310e+004	3.311e+004	3.311e+004	3.312e+004
32	Pressure (bar)	67.02	67.02	67.02	67.02	67.02
33	Temperature (C)	46.05	46.05	46.04	46.04	46.04
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.312e+004	3.313e+004	3.313e+004	3.314e+004	3.314e+004
44	Pressure (bar)	67.02	67.01	67.01	67.01	67.01
45	Temperature (C)	46.04	46.04	46.04	46.04	46.03
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.315e+004	3.315e+004	3.316e+004	3.316e+004	3.317e+004
56	Pressure (bar)	67.01	67.01	67.01	67.01	67.01
57	Temperature (C)	46.03	46.03	46.03	46.03	46.03
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.317e+004	3.318e+004	3.318e+004	3.319e+004	3.319e+004
68	Pressure (bar)	67.01	67.01	67.01	67.01	67.01
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	46.03	46.02	46.02	46.02	46.02
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.320e+004	3.320e+004	3.321e+004	3.321e+004	3.322e+004
24	Pressure (bar)	67.01	67.01	67.01	67.01	67.01
25	Temperature (C)	46.02	46.02	46.02	46.01	46.01
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.322e+004	3.323e+004	3.323e+004	3.324e+004	3.324e+004
36	Pressure (bar)	67.01	67.01	67.01	67.01	67.01
37	Temperature (C)	46.01	46.01	46.01	46.01	46.01
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.325e+004	3.325e+004	3.326e+004	3.326e+004	3.327e+004
48	Pressure (bar)	67.01	67.00	67.00	67.00	67.00
49	Temperature (C)	46.01	46.00	46.00	46.00	46.00
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.327e+004	3.328e+004	3.328e+004	3.329e+004	3.329e+004
60	Pressure (bar)	67.00	67.00	67.00	67.00	67.00
61	Temperature (C)	46.00	46.00	46.00	45.99	45.99
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 917 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.330e+004	3.330e+004	3.331e+004	3.331e+004	3.332e+004
16	Pressure (bar)	67.00	67.00	67.00	67.00	67.00
17	Temperature (C)	45.99	45.99	45.99	45.99	45.99
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.332e+004	3.333e+004	3.333e+004	3.334e+004	3.334e+004
28	Pressure (bar)	67.00	67.00	67.00	67.00	67.00
29	Temperature (C)	45.98	45.98	45.98	45.98	45.98
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.335e+004	3.335e+004	3.336e+004	3.336e+004	3.337e+004
40	Pressure (bar)	67.00	67.00	67.00	67.00	66.99
41	Temperature (C)	45.98	45.98	45.98	45.97	45.97
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.337e+004	3.338e+004	3.338e+004	3.339e+004	3.339e+004
52	Pressure (bar)	66.99	66.99	66.99	66.99	66.99
53	Temperature (C)	45.97	45.97	45.97	45.97	45.97
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.06	16.06	16.06	16.06	16.06
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.340e+004	3.340e+004	3.341e+004	3.341e+004	3.342e+004
64	Pressure (bar)	66.99	66.99	66.99	66.99	66.99
65	Temperature (C)	45.96	45.96	45.96	45.96	45.96
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 918 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.06	16.06	16.06	16.06
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.342e+004	3.343e+004	3.343e+004	3.344e+004
20	Pressure (bar)	66.99	66.99	66.99	66.99
21	Temperature (C)	45.96	45.96	45.95	45.95
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.06	16.06	16.06	16.06
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.345e+004	3.345e+004	3.346e+004	3.347e+004
32	Pressure (bar)	66.99	66.99	66.99	66.99
33	Temperature (C)	45.95	45.95	45.95	45.95
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.06	16.05	16.05	16.05
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.347e+004	3.348e+004	3.348e+004	3.349e+004
44	Pressure (bar)	66.99	66.99	66.98	66.98
45	Temperature (C)	45.94	45.94	45.94	45.94
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.05	16.05	16.05	16.05
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.350e+004	3.350e+004	3.351e+004	3.352e+004
56	Pressure (bar)	66.98	66.98	66.98	66.98
57	Temperature (C)	45.94	45.94	45.93	45.93
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.05	16.05	16.05	16.05
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.352e+004	3.353e+004	3.353e+004	3.354e+004
68	Pressure (bar)	66.98	66.98	66.98	66.98
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 919 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	45.93	45.93	45.93	45.93	45.93
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.355e+004	3.355e+004	3.356e+004	3.356e+004	3.357e+004
24	Pressure (bar)	66.98	66.98	66.98	66.98	66.98
25	Temperature (C)	45.92	45.92	45.92	45.92	45.92
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.357e+004	3.358e+004	3.358e+004	3.359e+004	3.359e+004
36	Pressure (bar)	66.98	66.98	66.98	66.98	66.98
37	Temperature (C)	45.92	45.92	45.91	45.91	45.91
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.360e+004	3.360e+004	3.361e+004	3.361e+004	3.362e+004
48	Pressure (bar)	66.98	66.98	66.97	66.97	66.97
49	Temperature (C)	45.91	45.91	45.91	45.91	45.90
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.362e+004	3.363e+004	3.363e+004	3.364e+004	3.364e+004
60	Pressure (bar)	66.97	66.97	66.97	66.97	66.97
61	Temperature (C)	45.90	45.90	45.90	45.90	45.90
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.365e+004	3.365e+004	3.366e+004	3.366e+004	3.367e+004
16	Pressure (bar)	66.97	66.97	66.97	66.97	66.97
17	Temperature (C)	45.90	45.89	45.89	45.89	45.89
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.367e+004	3.368e+004	3.368e+004	3.369e+004	3.369e+004
28	Pressure (bar)	66.97	66.97	66.97	66.97	66.97
29	Temperature (C)	45.89	45.89	45.89	45.88	45.88
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.370e+004	3.370e+004	3.371e+004	3.371e+004	3.372e+004
40	Pressure (bar)	66.97	66.97	66.96	66.96	66.96
41	Temperature (C)	45.88	45.88	45.88	45.88	45.88
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.372e+004	3.373e+004	3.373e+004	3.374e+004	3.374e+004
52	Pressure (bar)	66.96	66.96	66.96	66.96	66.96
53	Temperature (C)	45.88	45.87	45.87	45.87	45.87
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.375e+004	3.375e+004	3.376e+004	3.376e+004	3.377e+004
64	Pressure (bar)	66.96	66.96	66.96	66.96	66.96
65	Temperature (C)	45.87	45.87	45.87	45.86	45.86
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.377e+004	3.378e+004	3.378e+004	3.379e+004	3.379e+004
20	Pressure (bar)	66.96	66.96	66.96	66.96	66.96
21	Temperature (C)	45.86	45.86	45.86	45.86	45.86
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.380e+004	3.380e+004	3.381e+004	3.381e+004	3.382e+004
32	Pressure (bar)	66.96	66.96	66.96	66.96	66.96
33	Temperature (C)	45.86	45.85	45.85	45.85	45.85
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.382e+004	3.383e+004	3.383e+004	3.384e+004	3.384e+004
44	Pressure (bar)	66.96	66.95	66.95	66.95	66.95
45	Temperature (C)	45.85	45.85	45.85	45.84	45.84
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.385e+004	3.385e+004	3.385e+004	3.385e+004	3.386e+004
56	Pressure (bar)	66.95	66.95	66.95	66.95	66.95
57	Temperature (C)	45.84	45.84	45.84	45.84	45.84
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.386e+004	3.386e+004	3.386e+004	3.386e+004	3.387e+004
68	Pressure (bar)	66.95	66.95	66.95	66.95	66.95
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	45.84	45.84	45.84	45.84	45.84
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.387e+004	3.387e+004	3.387e+004	3.388e+004	3.388e+004
24	Pressure (bar)	66.95	66.95	66.95	66.95	66.95
25	Temperature (C)	45.84	45.84	45.83	45.83	45.83
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.388e+004	3.389e+004	3.389e+004	3.389e+004	3.389e+004
36	Pressure (bar)	66.95	66.95	66.95	66.95	66.95
37	Temperature (C)	45.83	45.83	45.83	45.83	45.83
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.390e+004	3.391e+004	3.392e+004	3.393e+004	3.394e+004
48	Pressure (bar)	66.95	66.95	66.95	66.95	66.94
49	Temperature (C)	45.83	45.83	45.82	45.82	45.82
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.395e+004	3.396e+004	3.398e+004	3.399e+004	3.400e+004
60	Pressure (bar)	66.94	66.94	66.94	66.94	66.94
61	Temperature (C)	45.81	45.81	45.81	45.80	45.80
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 923 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.401e+004	3.402e+004	3.403e+004	3.404e+004	3.405e+004
16	Pressure (bar)	66.94	66.94	66.94	66.94	66.94
17	Temperature (C)	45.80	45.80	45.79	45.79	45.79
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.406e+004	3.408e+004	3.409e+004	3.410e+004	3.411e+004
28	Pressure (bar)	66.93	66.93	66.93	66.93	66.93
29	Temperature (C)	45.78	45.78	45.78	45.77	45.77
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.412e+004	3.413e+004	3.414e+004	3.415e+004	3.416e+004
40	Pressure (bar)	66.93	66.93	66.93	66.93	66.93
41	Temperature (C)	45.77	45.77	45.76	45.76	45.76
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.418e+004	3.419e+004	3.420e+004	3.421e+004	3.422e+004
52	Pressure (bar)	66.92	66.92	66.92	66.92	66.92
53	Temperature (C)	45.75	45.75	45.75	45.74	45.74
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.423e+004	3.424e+004	3.425e+004	3.426e+004	3.428e+004
64	Pressure (bar)	66.92	66.92	66.92	66.92	66.91
65	Temperature (C)	45.74	45.73	45.73	45.73	45.73
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 924 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:13:43 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.05	16.05	16.05	16.05
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.429e+004	3.429e+004	3.429e+004	3.430e+004
20	Pressure (bar)	66.91	66.91	66.91	66.91
21	Temperature (C)	45.72	45.72	45.72	45.72
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.05	16.05	16.05	16.05
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.430e+004	3.430e+004	3.431e+004	3.432e+004
32	Pressure (bar)	66.91	66.91	66.91	66.91
33	Temperature (C)	45.72	45.72	45.72	45.71
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.05	16.05	16.05	16.05
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.432e+004	3.433e+004	3.433e+004	3.434e+004
44	Pressure (bar)	66.91	66.91	66.91	66.91
45	Temperature (C)	45.71	45.71	45.71	45.71
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.05	16.05	16.05	16.05
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.435e+004	3.435e+004	3.436e+004	3.437e+004
56	Pressure (bar)	66.91	66.91	66.91	66.91
57	Temperature (C)	45.71	45.71	45.70	45.70
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.05	16.05	16.05	16.05
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.437e+004	3.438e+004	3.438e+004	3.439e+004
68	Pressure (bar)	66.91	66.91	66.91	66.90
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	45.70	45.70	45.70	45.70	45.69
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.440e+004	3.440e+004	3.441e+004	3.441e+004	3.442e+004
24	Pressure (bar)	66.90	66.90	66.90	66.90	66.90
25	Temperature (C)	45.69	45.69	45.69	45.69	45.69
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.442e+004	3.443e+004	3.443e+004	3.444e+004	3.444e+004
36	Pressure (bar)	66.90	66.90	66.90	66.90	66.90
37	Temperature (C)	45.69	45.69	45.68	45.68	45.68
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.445e+004	3.445e+004	3.446e+004	3.446e+004	3.447e+004
48	Pressure (bar)	66.90	66.90	66.90	66.90	66.90
49	Temperature (C)	45.68	45.68	45.68	45.68	45.68
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.447e+004	3.448e+004	3.448e+004	3.449e+004	3.449e+004
60	Pressure (bar)	66.90	66.90	66.90	66.90	66.90
61	Temperature (C)	45.67	45.67	45.67	45.67	45.67
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 926 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.450e+004	3.450e+004	3.451e+004	3.451e+004	3.452e+004
16	Pressure (bar)	66.90	66.90	66.90	66.90	66.90
17	Temperature (C)	45.67	45.67	45.66	45.66	45.66
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.452e+004	3.453e+004	3.453e+004	3.454e+004	3.454e+004
28	Pressure (bar)	66.89	66.89	66.89	66.89	66.89
29	Temperature (C)	45.66	45.66	45.66	45.66	45.66
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.455e+004	3.455e+004	3.456e+004	3.456e+004	3.457e+004
40	Pressure (bar)	66.89	66.89	66.89	66.89	66.89
41	Temperature (C)	45.65	45.65	45.65	45.65	45.65
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.457e+004	3.458e+004	3.458e+004	3.459e+004	3.459e+004
52	Pressure (bar)	66.89	66.89	66.89	66.89	66.89
53	Temperature (C)	45.65	45.65	45.64	45.64	45.64
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.460e+004	3.460e+004	3.461e+004	3.461e+004	3.462e+004
64	Pressure (bar)	66.89	66.89	66.89	66.89	66.89
65	Temperature (C)	45.64	45.64	45.64	45.64	45.64
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 927 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.462e+004	3.463e+004	3.463e+004	3.464e+004	3.464e+004
20	Pressure (bar)	66.89	66.89	66.89	66.88	66.88
21	Temperature (C)	45.63	45.63	45.63	45.63	45.63
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.05	16.05	16.05	16.05	16.05
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.465e+004	3.465e+004	3.465e+004	3.466e+004	3.466e+004
32	Pressure (bar)	66.88	66.88	66.88	66.88	66.88
33	Temperature (C)	45.63	45.63	45.62	45.62	45.62
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.05	16.04	16.04	16.04	16.04
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.467e+004	3.467e+004	3.468e+004	3.468e+004	3.469e+004
44	Pressure (bar)	66.88	66.88	66.88	66.88	66.88
45	Temperature (C)	45.62	45.62	45.62	45.62	45.62
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.469e+004	3.470e+004	3.470e+004	3.471e+004	3.471e+004
56	Pressure (bar)	66.88	66.88	66.88	66.88	66.88
57	Temperature (C)	45.61	45.61	45.61	45.61	45.61
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.472e+004	3.472e+004	3.473e+004	3.473e+004	3.474e+004
68	Pressure (bar)	66.88	66.88	66.88	66.88	66.88
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 928 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	45.61	45.61	45.61	45.60	45.60
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.474e+004	3.475e+004	3.475e+004	3.476e+004	3.476e+004
24	Pressure (bar)	66.88	66.88	66.88	66.88	66.88
25	Temperature (C)	45.60	45.60	45.60	45.60	45.60
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.477e+004	3.477e+004	3.478e+004	3.478e+004	3.479e+004
36	Pressure (bar)	66.88	66.87	66.87	66.87	66.87
37	Temperature (C)	45.60	45.59	45.59	45.59	45.59
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.479e+004	3.480e+004	3.480e+004	3.481e+004	3.481e+004
48	Pressure (bar)	66.87	66.87	66.87	66.87	66.87
49	Temperature (C)	45.59	45.59	45.59	45.59	45.58
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.482e+004	3.482e+004	3.483e+004	3.483e+004	3.484e+004
60	Pressure (bar)	66.87	66.87	66.87	66.87	66.87
61	Temperature (C)	45.58	45.58	45.58	45.58	45.58
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.484e+004	3.485e+004	3.485e+004	3.486e+004	3.486e+004
16	Pressure (bar)	66.87	66.87	66.87	66.87	66.87
17	Temperature (C)	45.58	45.58	45.57	45.57	45.57
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.487e+004	3.487e+004	3.488e+004	3.488e+004	3.489e+004
28	Pressure (bar)	66.87	66.87	66.86	66.86	66.86
29	Temperature (C)	45.57	45.57	45.57	45.56	45.56
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.489e+004	3.490e+004	3.490e+004	3.491e+004	3.491e+004
40	Pressure (bar)	66.86	66.86	66.86	66.86	66.87
41	Temperature (C)	45.56	45.56	45.56	45.56	45.56
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.492e+004	3.492e+004	3.493e+004	3.493e+004	3.494e+004
52	Pressure (bar)	66.87	66.87	66.87	66.87	66.87
53	Temperature (C)	45.56	45.56	45.56	45.56	45.56
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.494e+004	3.495e+004	3.495e+004	3.496e+004	3.497e+004
64	Pressure (bar)	66.87	66.87	66.87	66.86	66.86
65	Temperature (C)	45.56	45.55	45.55	45.55	45.55
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.497e+004	3.497e+004	3.498e+004	3.498e+004	3.498e+004
20	Pressure (bar)	66.86	66.86	66.86	66.85	66.85
21	Temperature (C)	45.54	45.54	45.54	45.54	45.54
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.499e+004	3.499e+004	3.500e+004	3.500e+004	3.501e+004
32	Pressure (bar)	66.85	66.85	66.85	66.85	66.85
33	Temperature (C)	45.54	45.54	45.53	45.53	45.53
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.502e+004	3.502e+004	3.503e+004	3.503e+004	3.504e+004
44	Pressure (bar)	66.85	66.85	66.85	66.84	66.84
45	Temperature (C)	45.53	45.53	45.53	45.52	45.52
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.504e+004	3.505e+004	3.505e+004	3.506e+004	3.506e+004
56	Pressure (bar)	66.84	66.84	66.84	66.84	66.84
57	Temperature (C)	45.52	45.52	45.52	45.52	45.52
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.507e+004	3.507e+004	3.508e+004	3.508e+004	3.509e+004
68	Pressure (bar)	66.84	66.84	66.84	66.84	66.84
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 931 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:13:43 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	45.51	45.51	45.51	45.51	45.51
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.509e+004	3.510e+004	3.510e+004	3.511e+004	3.511e+004
24	Pressure (bar)	66.84	66.84	66.84	66.84	66.84
25	Temperature (C)	45.51	45.51	45.50	45.50	45.50
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.512e+004	3.512e+004	3.513e+004	3.513e+004	3.514e+004
36	Pressure (bar)	66.84	66.84	66.84	66.84	66.84
37	Temperature (C)	45.50	45.50	45.50	45.50	45.50
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.514e+004	3.515e+004	3.515e+004	3.516e+004	3.516e+004
48	Pressure (bar)	66.83	66.83	66.83	66.83	66.83
49	Temperature (C)	45.49	45.49	45.49	45.49	45.49
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.517e+004	3.517e+004	3.518e+004	3.518e+004	3.519e+004
60	Pressure (bar)	66.83	66.83	66.83	66.83	66.83
61	Temperature (C)	45.49	45.49	45.49	45.48	45.48
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 932 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.519e+004	3.520e+004	3.520e+004	3.521e+004	3.521e+004
16	Pressure (bar)	66.83	66.83	66.83	66.83	66.83
17	Temperature (C)	45.48	45.48	45.48	45.48	45.48
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.522e+004	3.522e+004	3.523e+004	3.523e+004	3.524e+004
28	Pressure (bar)	66.83	66.83	66.83	66.83	66.83
29	Temperature (C)	45.47	45.47	45.47	45.47	45.47
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.524e+004	3.525e+004	3.525e+004	3.526e+004	3.526e+004
40	Pressure (bar)	66.83	66.83	66.83	66.82	66.82
41	Temperature (C)	45.47	45.47	45.47	45.46	45.46
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.527e+004	3.527e+004	3.528e+004	3.528e+004	3.529e+004
52	Pressure (bar)	66.82	66.82	66.82	66.82	66.82
53	Temperature (C)	45.46	45.46	45.46	45.46	45.46
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.529e+004	3.530e+004	3.530e+004	3.531e+004	3.531e+004
64	Pressure (bar)	66.82	66.82	66.82	66.82	66.82
65	Temperature (C)	45.45	45.45	45.45	45.45	45.45
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 933 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.532e+004	3.532e+004	3.533e+004	3.533e+004	3.534e+004
20	Pressure (bar)	66.82	66.82	66.82	66.82	66.82
21	Temperature (C)	45.45	45.45	45.45	45.44	45.44
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.534e+004	3.535e+004	3.535e+004	3.536e+004	3.536e+004
32	Pressure (bar)	66.82	66.82	66.82	66.82	66.83
33	Temperature (C)	45.44	45.44	45.44	45.44	45.44
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.537e+004	3.537e+004	3.538e+004	3.538e+004	3.539e+004
44	Pressure (bar)	66.83	66.82	66.82	66.81	66.81
45	Temperature (C)	45.44	45.44	45.43	45.43	45.43
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.539e+004	3.540e+004	3.540e+004	3.541e+004	3.541e+004
56	Pressure (bar)	66.80	66.80	66.80	66.80	66.80
57	Temperature (C)	45.43	45.42	45.42	45.42	45.42
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.542e+004	3.542e+004	3.543e+004	3.543e+004	3.544e+004
68	Pressure (bar)	66.80	66.80	66.80	66.80	66.80
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	45.42	45.42	45.42	45.41	45.41
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.544e+004	3.545e+004	3.545e+004	3.546e+004	3.546e+004
24	Pressure (bar)	66.80	66.80	66.80	66.80	66.80
25	Temperature (C)	45.41	45.41	45.41	45.41	45.41
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.547e+004	3.547e+004	3.548e+004	3.548e+004	3.549e+004
36	Pressure (bar)	66.80	66.80	66.79	66.79	66.79
37	Temperature (C)	45.41	45.40	45.40	45.40	45.40
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.549e+004	3.550e+004	3.550e+004	3.551e+004	3.551e+004
48	Pressure (bar)	66.79	66.79	66.79	66.79	66.79
49	Temperature (C)	45.40	45.40	45.40	45.40	45.39
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.552e+004	3.552e+004	3.553e+004	3.553e+004	3.554e+004
60	Pressure (bar)	66.79	66.79	66.79	66.79	66.79
61	Temperature (C)	45.39	45.39	45.39	45.39	45.39
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 935 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.554e+004	3.555e+004	3.555e+004	3.556e+004	3.556e+004
16	Pressure (bar)	66.79	66.79	66.79	66.79	66.79
17	Temperature (C)	45.39	45.39	45.38	45.38	45.38
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.557e+004	3.557e+004	3.557e+004	3.558e+004	3.558e+004
28	Pressure (bar)	66.79	66.79	66.79	66.79	66.79
29	Temperature (C)	45.38	45.38	45.38	45.38	45.38
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.559e+004	3.559e+004	3.559e+004	3.560e+004	3.560e+004
40	Pressure (bar)	66.79	66.79	66.79	66.78	66.78
41	Temperature (C)	45.37	45.37	45.37	45.37	45.37
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.560e+004	3.561e+004	3.561e+004	3.562e+004	3.562e+004
52	Pressure (bar)	66.78	66.78	66.78	66.78	66.78
53	Temperature (C)	45.37	45.37	45.37	45.37	45.36
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	16.04	16.04	16.04	16.04	16.04
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.563e+004	3.563e+004	3.564e+004	3.564e+004	3.565e+004
64	Pressure (bar)	66.78	66.78	66.78	66.78	66.78
65	Temperature (C)	45.36	45.36	45.36	45.36	45.36
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 936 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_80barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:13:43 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	16.04	16.03	16.03	16.03	16.03
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.565e+004	3.566e+004	3.566e+004	3.567e+004	3.567e+004
20	Pressure (bar)	66.78	66.78	66.78	66.78	66.77
21	Temperature (C)	45.36	45.36	45.35	45.35	45.35
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	16.03	16.03	16.03	16.03	16.03
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.568e+004				
32	Pressure (bar)	66.77				
33	Temperature (C)	45.35				
34	Formation	Will NOT Form				
35	Hydrate Type	No Types				
36	Mode	Use 3-Phase Model				
37	Inhibitors	Not Included				
38	Formation Temperature (C)	16.03				
39	Type at Formation Temperature	Type II				
40	Mode at Formation Temperature	Use 3-Phase Model				
41	Phases at Formation Temperature	V - Aq - L - H				
42	Inhibitors at Formation Temperature	Not Included				
43						
44						
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69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 937 of 937	

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline				
8					
9					
10	CONDITIONS				
11	Name	To Elep-Akabel Pipeline	Slug Catcher Tie-In	er Pipeline Energy Stream	
12	Vapour	0.8720	0.8680	---	
13	Temperature (C)	63.1868	47.8947	---	
14	Pressure (bar)	98.6614	87.7668	---	
15	Molar Flow (MMSCFD)	413.8643	413.8643	---	
16	Mass Flow (kg/h)	625694.7736	625694.7736	---	
17	Std Ideal Liq Vol Flow (m3/h)	1439.3914	1439.3914	---	
18	Molar Enthalpy (kJ/kgmole)	-1.003e+005	-1.014e+005	---	
19	Molar Entropy (kJ/kgmole-C)	163.1	160.5	---	
20	Heat Flow (kJ/h)	-2.0676e+09	-2.0909e+09	2.3336e+07	
21					
22	PROPERTIES				
23	Name	To Elep-Akabel Pipeline	Slug Catcher Tie-In		
24	Molecular Weight	30.35	30.35		
25	Molar Density (kgmole/m3)	4.294	4.097		
26	Mass Density (kg/m3)	130.3	124.4		
27	Act. Volume Flow (m3/h)	4801	5032		
28	Mass Enthalpy (kJ/kg)	-3305	-3342		
29	Mass Entropy (kJ/kg-C)	5.374	5.287		
30	Heat Capacity (kJ/kgmole-C)	77.68	76.35		
31	Mass Heat Capacity (kJ/kg-C)	2.559	2.515		
32	LHV Molar Basis (Std) (kJ/kgmole)	1.398e+006	1.398e+006		
33	HHV Molar Basis (Std) (kJ/kgmole)	1.519e+006	1.519e+006		
34	HHV Mass Basis (Std) (kJ/kg)	5.005e+004	5.005e+004		
35	CO2 Loading	---	---		
36	CO2 Apparent Mole Conc. (kgmole/m3)	---	---		
37	CO2 Apparent Wt. Conc. (kgmol/kg)	---	---		
38	LHV Mass Basis (Std) (kJ/kg)	4.606e+004	4.606e+004		
39	Phase Fraction [Vol. Basis]	0.7075	0.7017		
40	Phase Fraction [Mass Basis]	0.5368	0.5278		
41	Phase Fraction [Act. Vol. Basis]	0.9093	0.9137		
42	Mass Exergy (kJ/kg)	335.6	324.3		
43	Partial Pressure of CO2 (bar)	1.008	0.8961		
44	Cost Based on Flow (Cost/s)	0.0000	0.0000		
45	Act. Gas Flow (ACT_m3/h)	4366	4597		
46	Avg. Liq. Density (kgmole/m3)	14.32	14.32		
47	Specific Heat (kJ/kgmole-C)	77.68	76.35		
48	Std. Gas Flow (MMSCFD)	413.1	413.1		
49	Std. Ideal Liq. Mass Density (kg/m3)	434.7	434.7		
50	Act. Liq. Flow (m3/s)	0.1209	0.1207		
51	Z Factor	---	---		
52	Watson K	15.46	15.46		
53	User Property	---	---		
54	Partial Pressure of H2S (bar)	0.0000	0.0000		
55	Cp/(Cp - R)	1.120	1.122		
56	Cp/Cv	1.274	1.281		
57	Heat of Vap. (kJ/kgmole)	3.528e+004	3.559e+004		
58	Kinematic Viscosity (cSt)	---	---		
59	Liq. Mass Density (Std. Cond) (kg/m3)	1.294	1.294		
60	Liq. Vol. Flow (Std. Cond) (m3/h)	4.834e+005	4.834e+005		
61	Liquid Fraction	0.1280	0.1320		
62	Molar Volume (m3/kgmole)	0.2329	0.2441		
63	Mass Heat of Vap. (kJ/kg)	1162	1172		
64	Phase Fraction [Molar Basis]	0.8720	0.8680		
65	Surface Tension (dyne/cm)	---	---		
66	Thermal Conductivity (W/m-K)	---	---		
67	Viscosity (lbm/s-ft)	---	---		
68	Cv (Semi-Ideal) (kJ/kgmole-C)	69.36	68.03		
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 1 of 937				

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROPERTIES					
10						
11	Name	To Elep-Akabel Pipeline	Slug Catcher Tie-In			
12	Mass Cv (Semi-Ideal) (kJ/kg-C)	2.285	2.241			
13	Cv (kJ/kgmole-C)	60.96	59.60			
14	Mass Cv (kJ/kg-C)	2.008	1.964			
15	Cv (Ent. Method) (kJ/kgmole-C)	---	---			
16	Mass Cv (Ent. Method) (kJ/kg-C)	---	---			
17	Cp/Cv (Ent. Method)	---	---			
18	Reid VP at 37.8 C (bar)	---	---			
19	True VP at 37.8 C (bar)	---	---			
20	Liq. Vol. Flow - Sum(Std. Cond) (m3/h)	4.241e+005	4.222e+005			
21	Viscosity Index	---	---			
22	Ideal Gas Cp/Cv	1.156	1.161			
23	Ideal Gas Cp (kJ/kgmole-C)	61.69	60.01			
24	Mass Ideal Gas Cp (kJ/kg-C)	2.032	1.977			
25	Bubble Point Pressure (bar)	---	---			
26						
27	PROFILES					
28						
29	Data					
30	Distance	Elevation		Increments		
31	(m)	(m)				
32	0.0000	0.0000		5		
33	25.00	-0.1600		5		
34	50.00	4.000e-002		5		
35	75.00	-1.000e-002		5		
36	100.0	-8.000e-002		5		
37	125.0	1.000e-002		5		
38	150.0	-7.000e-002		5		
39	175.0	-0.1000		5		
40	200.1	-0.1600		5		
41	225.1	-0.1000		5		
42	250.1	-0.1600		5		
43	275.1	-0.2000		5		
44	300.1	-0.2100		5		
45	325.1	-0.2100		5		
46	350.1	-0.5700		5		
47	375.1	-0.2200		5		
48	400.0	0.2000		5		
49	425.0	8.000e-002		5		
50	450.0	3.000e-002		5		
51	475.0	6.000e-002		5		
52	500.0	6.000e-002		5		
53	525.0	5.000e-002		5		
54	550.0	6.000e-002		5		
55	575.0	8.000e-002		5		
56	600.0	7.000e-002		5		
57	625.0	1.000e-001		5		
58	650.0	0.1300		5		
59	675.0	0.1500		5		
60	710.0	0.1700		5		
61	720.0	9.000e-002		5		
62	730.0	0.2200		5		
63	740.0	0.1500		5		
64	750.0	1.000e-001		5		
65	775.0	8.000e-002		5		
66	800.1	1.000e-001		5		
67	825.1	1.000e-001		5		
68	850.1	-0.2700		5		
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 2 of 937	

1	Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	875.1	5.000e-002	5	
16	900.1	1.000e-002	5	
17	925.1	-2.000e-002	5	
18	950.1	1.000e-002	5	
19	975.1	1.000e-002	5	
20	1000	-1.232e-016	5	
21	1025	0.1800	5	
22	1050	-0.2300	5	
23	1075	-5.000e-002	5	
24	1100	2.000e-002	5	
25	1125	7.000e-002	5	
26	1150	0.1400	5	
27	1175	1.000e-001	5	
28	1200	0.1500	5	
29	1225	7.000e-002	5	
30	1250	-9.714e-017	5	
31	1275	-9.714e-017	5	
32	1300	3.000e-002	5	
33	1325	9.000e-002	5	
34	1350	0.2200	5	
35	1375	-0.3700	5	
36	1400	-3.600	5	
37	1425	-5.300	5	
38	1450	-5.900	5	
39	1475	-5.300	5	
40	1500	-6.300	5	
41	1525	-11.60	5	
42	1550	0.4400	5	
43	1575	1.150	5	
44	1600	0.8400	5	
45	1625	0.5100	5	
46	1650	0.6200	5	
47	1675	0.5900	5	
48	1700	0.2200	5	
49	1725	0.1600	5	
50	1750	0.1700	5	
51	1775	0.2400	5	
52	1800	0.3400	5	
53	1825	0.4800	5	
54	1850	0.5700	5	
55	1875	0.6300	5	
56	1900	0.6700	5	
57	1925	0.6700	5	
58	1950	0.7000	5	
59	1975	0.7000	5	
60	2000	0.5200	5	
61	2025	0.5200	5	
62	2050	0.6800	5	
63	2075	0.6300	5	
64	2100	0.6500	5	
65	2125	0.7000	5	
66	2150	0.7300	5	
67	2175	0.6800	5	
68	2201	0.6700	5	
69	Aspen Technology Inc. Aspen HYSYS Version 10			Page 3 of 937

1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	2226	0.6600	5	
16	2251	0.5900	5	
17	2276	0.6300	5	
18	2301	0.6800	5	
19	2326	0.6500	5	
20	2351	0.6200	5	
21	2376	0.6100	5	
22	2401	0.6600	5	
23	2426	0.7500	5	
24	2451	0.6600	5	
25	2476	0.7100	5	
26	2501	0.7600	5	
27	2526	0.6500	5	
28	2551	0.4700	5	
29	2576	0.5100	5	
30	2601	0.5500	5	
31	2626	0.5400	5	
32	2651	0.5800	5	
33	2676	0.5800	5	
34	2701	0.6000	5	
35	2726	0.6000	5	
36	2751	0.6000	5	
37	2776	0.6500	5	
38	2801	0.4800	5	
39	2826	0.5700	5	
40	2851	0.6500	5	
41	2876	0.7200	5	
42	2901	0.4000	5	
43	2926	0.2900	5	
44	2951	0.6500	5	
45	2976	-0.7100	5	
46	3001	-0.6000	5	
47	3026	-8.000	5	
48	3051	-10.40	5	
49	3076	-11.60	5	
50	3101	2.000e-002	5	
51	3126	0.5200	5	
52	3151	0.3200	5	
53	3176	0.2100	5	
54	3202	0.4800	5	
55	3227	0.4100	5	
56	3252	0.1700	5	
57	3277	0.1600	5	
58	3302	0.1500	5	
59	3327	0.2300	5	
60	3352	0.3200	5	
61	3377	0.3700	5	
62	3401	0.3900	5	
63	3426	0.4000	5	
64	3451	0.3800	5	
65	3476	0.3800	5	
66	3501	0.3200	5	
67	3526	0.3200	5	
68	3551	0.2800	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	3576	0.1300	5	
16	3601	0.1600	5	
17	3626	0.1900	5	
18	3651	0.3100	5	
19	3676	0.2400	5	
20	3701	0.1300	5	
21	3726	0.1600	5	
22	3751	0.2000	5	
23	3776	0.1200	5	
24	3802	7.000e-002	5	
25	3827	6.000e-002	5	
26	3852	0.1600	5	
27	3877	0.1900	5	
28	3902	0.2600	5	
29	3927	0.3000	5	
30	3952	0.3600	5	
31	3977	0.3800	5	
32	4002	0.3800	5	
33	4027	0.2800	5	
34	4052	0.2600	5	
35	4077	0.1600	5	
36	4112	0.1500	5	
37	4150	0.1300	5	
38	4188	5.000e-002	5	
39	4226	-0.1300	5	
40	4264	0.1100	5	
41	4302	0.1900	5	
42	4309	0.1200	5	
43	4334	-0.1300	5	
44	4359	-0.2700	5	
45	4384	-8.000e-002	5	
46	4409	-3.000e-002	5	
47	4434	0.1600	5	
48	4459	0.1300	5	
49	4484	0.1900	5	
50	4510	0.1600	5	
51	4535	0.1800	5	
52	4560	0.1300	5	
53	4585	0.1600	5	
54	4610	0.2700	5	
55	4675	0.1200	5	
56	4685	0.1600	5	
57	4710	0.2600	5	
58	4735	0.2700	5	
59	4760	0.2800	5	
60	4785	0.2700	5	
61	4810	0.2200	5	
62	4835	0.1900	5	
63	4860	0.1400	5	
64	4885	0.1300	5	
65	4910	0.1300	5	
66	4935	2.000e-002	5	
67	4960	-2.000e-002	5	
68	4985	-2.217e-015	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	5010	7.000e-002	5	
16	5035	-2.220e-015	5	
17	5060	0.1300	5	
18	5085	1.000e-001	5	
19	5110	0.1900	5	
20	5135	0.1700	5	
21	5160	0.1400	5	
22	5185	0.1800	5	
23	5210	0.1200	5	
24	5235	9.000e-002	5	
25	5260	8.000e-002	5	
26	5285	2.000e-002	5	
27	5310	-7.000e-002	5	
28	5335	-0.2000	5	
29	5360	-0.3200	5	
30	5385	-0.2600	5	
31	5410	-0.6900	5	
32	5435	-0.3800	5	
33	5460	-2.000e-002	5	
34	5485	-5.000e-002	5	
35	5510	-0.1400	5	
36	5535	-0.2000	5	
37	5570	-0.3200	5	
38	5580	-0.4600	5	
39	5605	-0.3900	5	
40	5630	-0.4300	5	
41	5655	-0.7200	5	
42	5680	-0.7100	5	
43	5705	-0.2900	5	
44	5730	-0.3900	5	
45	5755	-0.2500	5	
46	5780	-5.000e-002	5	
47	5805	-0.3500	5	
48	5830	-0.3900	5	
49	5855	1.000e-001	5	
50	5880	0.2400	5	
51	5905	-7.800	5	
52	5930	-16.10	5	
53	5955	-4.300	5	
54	5980	-0.3100	5	
55	6005	-14.00	5	
56	6030	-0.3600	5	
57	6055	-3.000e-002	5	
58	6080	-0.1400	5	
59	6106	-10.60	5	
60	6131	-0.3100	5	
61	6156	-0.2000	5	
62	6181	-0.1700	5	
63	6206	-7.000e-002	5	
64	6231	-6.000e-002	5	
65	6256	-0.1800	5	
66	6281	-0.1400	5	
67	6303	-10.90	5	
68	6328	-6.800	5	
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 6 of 937			

1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	6353	-10.60	5	
16	6378	-17.30	5	
17	6403	-13.80	5	
18	6428	-0.5700	5	
19	6453	-0.1400	5	
20	6483	-0.1000	5	
21	6503	-9.000e-002	5	
22	6528	4.000e-002	5	
23	6553	3.000e-002	5	
24	6578	2.000e-002	5	
25	6603	8.000e-002	5	
26	6628	4.000e-002	5	
27	6653	-2.000e-002	5	
28	6678	7.000e-002	5	
29	6703	-8.000e-002	5	
30	6728	-0.2500	5	
31	6754	-13.00	5	
32	6779	-8.100	5	
33	6804	-6.300	5	
34	6829	-0.2500	5	
35	6854	-17.80	5	
36	6879	-0.3500	5	
37	6904	-1.000e-002	5	
38	6929	-4.000e-002	5	
39	6954	-8.000e-002	5	
40	6979	-4.000e-002	5	
41	7004	-1.402e-015	5	
42	7029	5.000e-002	5	
43	7054	-0.1200	5	
44	7079	5.000e-002	5	
45	7089	3.000e-002	5	
46	7129	-1.398e-015	5	
47	7154	2.000e-002	5	
48	7179	7.000e-002	5	
49	7204	0.1700	5	
50	7229	8.000e-002	5	
51	7254	6.000e-002	5	
52	7279	0.1600	5	
53	7304	3.000e-002	5	
54	7329	7.000e-002	5	
55	7354	0.1600	5	
56	7379	0.2000	5	
57	7404	0.2200	5	
58	7429	0.2300	5	
59	7454	0.2400	5	
60	7479	0.2100	5	
61	7504	0.1200	5	
62	7529	0.1800	5	
63	7553	0.1200	5	
64	7578	0.1800	5	
65	7603	0.1800	5	
66	7628	0.1100	5	
67	7653	0.1400	5	
68	7678	7.000e-002	5	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	7703	0.1600	5	
16	7728	-0.3700	5	
17	7753	-0.4300	5	
18	7778	7.000e-002	5	
19	7803	0.2000	5	
20	7828	0.2000	5	
21	7853	0.2200	5	
22	7878	0.2600	5	
23	7903	0.2400	5	
24	7928	0.2300	5	
25	7953	0.3400	5	
26	7978	0.1500	5	
27	8003	8.000e-002	5	
28	8028	1.000e-002	5	
29	8053	-0.3100	5	
30	8078	-4.600	5	
31	8103	-0.6800	5	
32	8128	0.2000	5	
33	8153	0.2500	5	
34	8178	0.2400	5	
35	8203	0.2300	5	
36	8228	0.3000	5	
37	8253	0.3000	5	
38	8278	0.3400	5	
39	8303	0.3100	5	
40	8328	0.3700	5	
41	8353	6.000e-002	5	
42	8378	6.000e-002	5	
43	8403	-4.000e-002	5	
44	8428	-0.1600	5	
45	8453	-0.2000	5	
46	8478	-2.000e-002	5	
47	8503	-1.000e-002	5	
48	8528	-1.000e-002	5	
49	8553	-3.000e-002	5	
50	8578	-2.000e-002	5	
51	8603	-3.000e-002	5	
52	8628	-1.000e-002	5	
53	8653	-1.518e-015	5	
54	8678	1.000e-002	5	
55	8703	1.000e-002	5	
56	8728	2.000e-002	5	
57	8753	-1.516e-015	5	
58	8778	-3.000e-002	5	
59	8803	-7.000e-002	5	
60	8813	-2.000e-002	5	
61	8948	-5.000e-002	5	
62	8953	-3.000e-002	5	
63	8978	-1.000e-002	5	
64	9003	8.000e-002	5	
65	9028	-1.513e-015	5	
66	9053	-4.000e-002	5	
67	9058	-8.000e-002	5	
68	9091	-0.1400	5	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	9125	-0.2500	5	
16	9158	-0.3300	5	
17	9178	-0.3000	5	
18	9203	-0.4100	5	
19	9228	-0.4400	5	
20	9253	-0.4500	5	
21	9278	-0.2000	5	
22	9303	-0.1500	5	
23	9328	-0.1600	5	
24	9353	-3.000e-002	5	
25	9378	1.000e-002	5	
26	9403	-0.1500	5	
27	9428	-0.4400	5	
28	9453	-0.6100	5	
29	9468	-9.000e-002	5	
30	9491	-9.000e-002	5	
31	9515	-0.5500	5	
32	9538	-0.4500	5	
33	9561	-0.2800	5	
34	9585	-9.000e-002	5	
35	9608	-2.000e-002	5	
36	9628	0.2300	5	
37	9653	-1.582e-015	5	
38	9678	-7.000e-002	5	
39	9703	0.1700	5	
40	9728	0.1800	5	
41	9753	0.1700	5	
42	9778	0.1500	5	
43	9803	0.1800	5	
44	9828	0.1100	5	
45	9853	0.1800	5	
46	9878	0.1400	5	
47	9903	0.1600	5	
48	9913	0.1400	5	
49	1.001e+004	1.000e-001	5	
50	1.003e+004	-0.5000	5	
51	1.005e+004	-0.4700	5	
52	1.008e+004	-4.000e-002	5	
53	1.010e+004	0.1600	5	
54	1.013e+004	0.1800	5	
55	1.015e+004	0.2100	5	
56	1.018e+004	0.2200	5	
57	1.020e+004	0.1900	5	
58	1.023e+004	0.2200	5	
59	1.025e+004	0.1800	5	
60	1.028e+004	-0.2200	5	
61	1.030e+004	0.1800	5	
62	1.033e+004	1.000e-001	5	
63	1.035e+004	3.000e-002	5	
64	1.038e+004	2.000e-002	5	
65	1.040e+004	-0.1200	5	
66	1.043e+004	-1.554e-015	5	
67	1.045e+004	-0.2900	5	
68	1.048e+004	-0.4800	5	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	1.050e+004	-0.5800		5
16	1.053e+004	-8.000e-002		5
17	1.055e+004	-6.000e-002		5
18	1.058e+004	-1.513e-015		5
19	1.060e+004	9.000e-002		5
20	1.063e+004	0.1600		5
21	1.065e+004	0.1800		5
22	1.068e+004	0.2000		5
23	1.070e+004	0.2500		5
24	1.073e+004	0.2600		5
25	1.075e+004	0.2900		5
26	1.078e+004	0.3100		5
27	1.080e+004	0.2900		5
28	1.083e+004	0.3100		5
29	1.085e+004	0.3300		5
30	1.088e+004	0.3200		5
31	1.090e+004	0.3400		5
32	1.093e+004	0.3600		5
33	1.095e+004	0.4100		5
34	1.098e+004	0.3600		5
35	1.100e+004	0.3600		5
36	1.103e+004	0.3400		5
37	1.105e+004	0.3300		5
38	1.108e+004	0.3400		5
39	1.110e+004	0.3500		5
40	1.113e+004	0.3200		5
41	1.115e+004	0.3400		5
42	1.118e+004	0.3100		5
43	1.120e+004	0.3000		5
44	1.123e+004	0.3000		5
45	1.125e+004	0.3300		5
46	1.128e+004	0.3200		5
47	1.130e+004	0.3200		5
48	1.133e+004	0.1300		5
49	1.135e+004	0.1700		5
50	1.138e+004	0.2000		5
51	1.140e+004	1.000e-001		5
52	1.143e+004	-0.5100		5
53	1.145e+004	-0.3200		5
54	1.148e+004	0.2400		5
55	1.150e+004	0.2800		5
56	1.153e+004	0.3000		5
57	1.155e+004	0.3400		5
58	1.158e+004	0.3300		5
59	1.160e+004	0.3000		5
60	1.163e+004	0.2900		5
61	1.165e+004	0.2200		5
62	1.168e+004	0.2600		5
63	1.170e+004	0.3200		5
64	1.173e+004	0.3200		5
65	1.175e+004	0.3300		5
66	1.178e+004	0.3300		5
67	1.180e+004	0.3000		5
68	1.183e+004	0.2800		5
69	Aspen Technology Inc.		Aspen HYSYS Version 10	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	1.185e+004	0.3000	5	
16	1.188e+004	0.2900	5	
17	1.190e+004	0.3100	5	
18	1.193e+004	0.3000	5	
19	1.195e+004	0.3000	5	
20	1.198e+004	0.2900	5	
21	1.202e+004	0.2900	5	
22	1.205e+004	0.2500	5	
23	1.209e+004	0.8000	5	
24	1.212e+004	0.3000	5	
25	1.216e+004	0.3000	5	
26	1.219e+004	0.3000	5	
27	1.223e+004	0.2800	5	
28	1.226e+004	0.2500	5	
29	1.228e+004	0.1100	5	
30	1.231e+004	0.2200	5	
31	1.233e+004	0.2000	5	
32	1.236e+004	0.2000	5	
33	1.239e+004	0.2100	5	
34	1.242e+004	0.2800	5	
35	1.244e+004	0.2200	5	
36	1.247e+004	0.2800	5	
37	1.249e+004	0.2100	5	
38	1.252e+004	-0.2300	5	
39	1.254e+004	-0.1700	5	
40	1.257e+004	0.1600	5	
41	1.259e+004	0.2300	5	
42	1.262e+004	0.4400	5	
43	1.265e+004	0.6200	5	
44	1.268e+004	0.6000	5	
45	1.270e+004	0.2800	5	
46	1.273e+004	0.2600	5	
47	1.275e+004	0.2500	5	
48	1.277e+004	3.000e-002	5	
49	1.279e+004	1.000e-002	5	
50	1.280e+004	0.2000	5	
51	1.283e+004	-0.1600	5	
52	1.285e+004	0.1800	5	
53	1.288e+004	0.2000	5	
54	1.290e+004	0.2200	5	
55	1.293e+004	0.2400	5	
56	1.295e+004	0.2500	5	
57	1.298e+004	0.2200	5	
58	1.300e+004	0.2700	5	
59	1.303e+004	0.2800	5	
60	1.305e+004	0.2900	5	
61	1.308e+004	0.3000	5	
62	1.310e+004	0.2700	5	
63	1.313e+004	0.3300	5	
64	1.315e+004	0.3400	5	
65	1.318e+004	0.3500	5	
66	1.320e+004	0.3400	5	
67	1.323e+004	-0.8100	5	
68	1.325e+004	-1.440	5	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	1.328e+004	-5.200	5	
16	1.330e+004	-6.300	5	
17	1.333e+004	-7.800	5	
18	1.335e+004	-11.30	5	
19	1.338e+004	-13.55	5	
20	1.340e+004	-13.55	5	
21	1.343e+004	-11.30	5	
22	1.345e+004	-9.650	5	
23	1.348e+004	-0.6500	5	
24	1.350e+004	0.1800	5	
25	1.353e+004	0.2500	5	
26	1.354e+004	0.2500	5	
27	1.355e+004	0.2600	5	
28	1.357e+004	0.2700	5	
29	1.359e+004	0.2800	5	
30	1.360e+004	0.2600	5	
31	1.363e+004	0.2700	5	
32	1.365e+004	0.3100	5	
33	1.368e+004	0.2900	5	
34	1.370e+004	0.3000	5	
35	1.373e+004	0.2000	5	
36	1.375e+004	9.000e-002	5	
37	1.378e+004	-7.000e-002	5	
38	1.380e+004	-2.000e-002	5	
39	1.383e+004	0.2600	5	
40	1.385e+004	-0.3600	5	
41	1.388e+004	-0.2000	5	
42	1.390e+004	-8.000e-002	5	
43	1.393e+004	-4.000e-002	5	
44	1.395e+004	-0.1100	5	
45	1.398e+004	8.000e-002	5	
46	1.400e+004	1.000e-001	5	
47	1.403e+004	-0.2100	5	
48	1.405e+004	1.000e-001	5	
49	1.406e+004	0.1400	5	
50	1.408e+004	9.000e-002	5	
51	1.409e+004	-5.000e-002	5	
52	1.412e+004	-0.3800	5	
53	1.415e+004	-0.4700	5	
54	1.418e+004	-4.000e-002	5	
55	1.420e+004	-0.1000	5	
56	1.423e+004	2.000e-002	5	
57	1.425e+004	0.1500	5	
58	1.428e+004	0.1700	5	
59	1.430e+004	0.1900	5	
60	1.433e+004	0.2000	5	
61	1.435e+004	0.1100	5	
62	1.438e+004	0.1400	5	
63	1.440e+004	0.1400	5	
64	1.443e+004	0.2000	5	
65	1.445e+004	0.2000	5	
66	1.448e+004	0.2000	5	
67	1.450e+004	0.2000	5	
68	1.453e+004	0.2100	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	1.455e+004	0.2000	5	
16	1.458e+004	0.1700	5	
17	1.460e+004	0.1600	5	
18	1.463e+004	0.1700	5	
19	1.465e+004	0.1400	5	
20	1.468e+004	0.1800	5	
21	1.470e+004	0.1100	5	
22	1.473e+004	0.1200	5	
23	1.475e+004	9.000e-002	5	
24	1.478e+004	7.000e-002	5	
25	1.480e+004	7.000e-002	5	
26	1.483e+004	5.000e-002	5	
27	1.485e+004	3.000e-002	5	
28	1.486e+004	2.000e-002	5	
29	1.496e+004	3.000e-002	5	
30	1.506e+004	-4.000e-002	5	
31	1.516e+004	-1.000e-002	5	
32	1.526e+004	-6.000e-002	5	
33	1.536e+004	-0.3800	5	
34	1.546e+004	-3.550	5	
35	1.556e+004	-0.3700	5	
36	1.566e+004	2.000e-002	5	
37	1.576e+004	-9.000e-002	5	
38	1.579e+004	-9.000e-002	5	
39	1.589e+004	2.000e-002	5	
40	1.591e+004	1.000e-001	5	
41	1.593e+004	0.1100	5	
42	1.594e+004	0.1700	5	
43	1.596e+004	0.1800	5	
44	1.598e+004	0.2000	5	
45	1.601e+004	0.2300	5	
46	1.603e+004	0.1400	5	
47	1.606e+004	7.000e-002	5	
48	1.608e+004	-2.000e-002	5	
49	1.611e+004	-9.000e-002	5	
50	1.613e+004	-0.1700	5	
51	1.616e+004	-0.2800	5	
52	1.618e+004	-0.3400	5	
53	1.621e+004	-0.4600	5	
54	1.623e+004	0.1600	5	
55	1.626e+004	0.1400	5	
56	1.628e+004	0.1200	5	
57	1.631e+004	5.000e-002	5	
58	1.632e+004	-0.4100	5	
59	1.633e+004	-1.110e-016	5	
60	1.636e+004	4.000e-002	5	
61	1.638e+004	7.000e-002	5	
62	1.641e+004	1.000e-001	5	
63	1.643e+004	0.1400	5	
64	1.646e+004	1.000e-001	5	
65	1.648e+004	9.000e-002	5	
66	1.651e+004	7.000e-002	5	
67	1.653e+004	0.1200	5	
68	1.656e+004	0.1200	5	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	1.658e+004	0.1500	5	
16	1.661e+004	1.000e-002	5	
17	1.663e+004	2.000e-002	5	
18	1.666e+004	-6.000e-002	5	
19	1.668e+004	-2.000e-002	5	
20	1.671e+004	-1.006e-016	5	
21	1.673e+004	-2.000e-002	5	
22	1.676e+004	-7.000e-002	5	
23	1.678e+004	-0.1400	5	
24	1.681e+004	-0.1000	5	
25	1.683e+004	-0.1300	5	
26	1.686e+004	-0.1700	5	
27	1.688e+004	-0.1700	5	
28	1.690e+004	-0.2600	5	
29	1.693e+004	-0.3500	5	
30	1.695e+004	-0.3600	5	
31	1.698e+004	-0.3500	5	
32	1.700e+004	-0.2400	5	
33	1.703e+004	-0.2700	5	
34	1.705e+004	-0.3400	5	
35	1.708e+004	-0.2500	5	
36	1.710e+004	-0.2100	5	
37	1.713e+004	-0.1500	5	
38	1.715e+004	-8.000e-002	5	
39	1.718e+004	-3.000e-002	5	
40	1.720e+004	-1.000e-002	5	
41	1.723e+004	6.000e-002	5	
42	1.725e+004	8.000e-002	5	
43	1.728e+004	0.1200	5	
44	1.730e+004	0.1200	5	
45	1.733e+004	0.1400	5	
46	1.735e+004	0.1400	5	
47	1.738e+004	0.1600	5	
48	1.740e+004	0.1600	5	
49	1.743e+004	0.1400	5	
50	1.745e+004	0.1100	5	
51	1.748e+004	0.1100	5	
52	1.750e+004	5.000e-002	5	
53	1.753e+004	1.000e-001	5	
54	1.755e+004	-9.714e-017	5	
55	1.758e+004	-0.1700	5	
56	1.760e+004	9.000e-002	5	
57	1.763e+004	1.000e-002	5	
58	1.765e+004	-1.000e-002	5	
59	1.768e+004	-5.000e-002	5	
60	1.770e+004	-5.000e-002	5	
61	1.773e+004	-6.000e-002	5	
62	1.775e+004	-0.1800	5	
63	1.778e+004	-0.1800	5	
64	1.780e+004	-0.3800	5	
65	1.783e+004	-0.1700	5	
66	1.785e+004	-0.8700	5	
67	1.787e+004	-0.8700	5	
68	1.788e+004	-1.300	5	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	1.790e+004	-4.300	5	
16	1.791e+004	-5.800	5	
17	1.793e+004	-6.800	5	
18	1.795e+004	-8.300	5	
19	1.798e+004	-9.800	5	
20	1.800e+004	-12.30	5	
21	1.803e+004	-14.80	5	
22	1.805e+004	-13.30	5	
23	1.808e+004	-0.7000	5	
24	1.810e+004	-0.5000	5	
25	1.813e+004	-0.3000	5	
26	1.815e+004	-0.2600	5	
27	1.818e+004	-0.2800	5	
28	1.820e+004	-0.2400	5	
29	1.823e+004	-0.2100	5	
30	1.825e+004	-0.2200	5	
31	1.828e+004	-0.2300	5	
32	1.830e+004	-0.2500	5	
33	1.833e+004	-0.3000	5	
34	1.835e+004	-0.2800	5	
35	1.838e+004	-0.3000	5	
36	1.840e+004	-0.2800	5	
37	1.843e+004	-0.2700	5	
38	1.845e+004	-0.3500	5	
39	1.848e+004	-0.3000	5	
40	1.850e+004	-0.2900	5	
41	1.853e+004	-0.2600	5	
42	1.855e+004	-0.3400	5	
43	1.858e+004	-0.3300	5	
44	1.860e+004	-0.3800	5	
45	1.863e+004	-0.4800	5	
46	1.865e+004	-0.5400	5	
47	1.868e+004	-0.5200	5	
48	1.870e+004	-0.8000	5	
49	1.873e+004	-0.5500	5	
50	1.875e+004	-0.4800	5	
51	1.878e+004	-0.5900	5	
52	1.880e+004	-0.6800	5	
53	1.882e+004	-0.5400	5	
54	1.883e+004	-0.8600	5	
55	1.885e+004	-0.3700	5	
56	1.888e+004	-0.6800	5	
57	1.890e+004	-0.5500	5	
58	1.891e+004	-0.6000	5	
59	1.893e+004	-0.5600	5	
60	1.895e+004	-0.7000	5	
61	1.898e+004	-0.9800	5	
62	1.900e+004	-0.7100	5	
63	1.903e+004	-0.6100	5	
64	1.905e+004	-0.5400	5	
65	1.908e+004	-0.5700	5	
66	1.910e+004	-0.4500	5	
67	1.913e+004	-0.4400	5	
68	1.915e+004	-0.4500	5	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	1.918e+004	-0.4800	5	
16	1.920e+004	-0.5000	5	
17	1.923e+004	-0.5900	5	
18	1.925e+004	-0.4800	5	
19	1.928e+004	-0.5500	5	
20	1.930e+004	-0.7000	5	
21	1.933e+004	-0.7100	5	
22	1.935e+004	-0.6500	5	
23	1.938e+004	-4.600	5	
24	1.940e+004	-3.550	5	
25	1.943e+004	-1.090	5	
26	1.945e+004	-1.120	5	
27	1.947e+004	-0.7500	5	
28	1.950e+004	-0.6200	5	
29	1.961e+004	-0.4800	5	
30	1.973e+004	-0.4700	5	
31	1.985e+004	-0.4500	5	
32	1.996e+004	-0.4400	5	
33	2.008e+004	-0.4300	5	
34	2.020e+004	-0.7000	5	
35	2.031e+004	-0.3900	5	
36	2.043e+004	-0.4000	5	
37	2.055e+004	-0.3800	5	
38	2.069e+004	-0.4900	5	
39	2.070e+004	-0.5400	5	
40	2.073e+004	-0.5200	5	
41	2.075e+004	-0.4900	5	
42	2.078e+004	-0.4900	5	
43	2.080e+004	-0.4900	5	
44	2.083e+004	-0.5400	5	
45	2.085e+004	-0.5700	5	
46	2.088e+004	-0.5100	5	
47	2.090e+004	-0.6900	5	
48	2.093e+004	-0.6100	5	
49	2.095e+004	-0.5400	5	
50	2.098e+004	-0.5900	5	
51	2.100e+004	-0.7200	5	
52	2.103e+004	-0.7000	5	
53	2.105e+004	-0.8000	5	
54	2.108e+004	-0.6300	5	
55	2.110e+004	-0.7700	5	
56	2.112e+004	-1.230	5	
57	2.114e+004	-0.9300	5	
58	2.115e+004	-0.7900	5	
59	2.118e+004	-0.5700	5	
60	2.120e+004	-0.4900	5	
61	2.123e+004	-0.5000	5	
62	2.125e+004	-0.4100	5	
63	2.128e+004	-0.5100	5	
64	2.130e+004	-0.6100	5	
65	2.133e+004	-0.6500	5	
66	2.134e+004	-0.6600	5	
67	2.135e+004	-0.5100	5	
68	2.138e+004	-0.4900	5	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	2.140e+004	-0.4800	5	
16	2.143e+004	-0.2400	5	
17	2.145e+004	-0.2800	5	
18	2.148e+004	-0.2600	5	
19	2.150e+004	-0.2700	5	
20	2.152e+004	-0.3400	5	
21	2.153e+004	-0.4700	5	
22	2.155e+004	-0.2800	5	
23	2.158e+004	-4.000e-002	5	
24	2.160e+004	-0.5700	5	
25	2.163e+004	-2.920	5	
26	2.165e+004	-5.420	5	
27	2.168e+004	-7.300	5	
28	2.170e+004	-7.300	5	
29	2.173e+004	-6.800	5	
30	2.175e+004	-6.300	5	
31	2.178e+004	-5.800	5	
32	2.180e+004	-0.3600	5	
33	2.183e+004	-0.1800	5	
34	2.185e+004	-0.1700	5	
35	2.188e+004	-0.1400	5	
36	2.190e+004	-9.000e-002	5	
37	2.193e+004	-9.000e-002	5	
38	2.195e+004	-9.000e-002	5	
39	2.198e+004	-8.000e-002	5	
40	2.200e+004	-7.000e-002	5	
41	2.203e+004	-9.000e-002	5	
42	2.205e+004	-0.1000	5	
43	2.208e+004	-0.2300	5	
44	2.210e+004	-0.3600	5	
45	2.213e+004	-0.3600	5	
46	2.215e+004	-0.3300	5	
47	2.218e+004	-0.1900	5	
48	2.220e+004	-0.3000	5	
49	2.223e+004	-0.7500	5	
50	2.225e+004	-0.3200	5	
51	2.228e+004	-0.1600	5	
52	2.233e+004	-0.1200	5	
53	2.345e+004	-0.1000	5	
54	2.349e+004	-0.1200	5	
55	2.352e+004	-0.1100	5	
56	2.354e+004	-0.1300	5	
57	2.357e+004	-0.1300	5	
58	2.359e+004	0.6800	5	
59	2.362e+004	-0.1600	5	
60	2.364e+004	-0.2100	5	
61	2.367e+004	-0.1000	5	
62	2.369e+004	-0.1500	5	
63	2.372e+004	-0.1900	5	
64	2.374e+004	-0.2500	5	
65	2.377e+004	-0.4500	5	
66	2.379e+004	-0.4100	5	
67	2.382e+004	-0.3800	5	
68	2.384e+004	-0.1700	5	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	2.387e+004	-0.2000	5	
16	2.389e+004	-0.6700	5	
17	2.392e+004	-0.2400	5	
18	2.394e+004	-0.1300	5	
19	2.397e+004	-0.1200	5	
20	2.399e+004	-0.1600	5	
21	2.402e+004	-0.3200	5	
22	2.404e+004	-0.5000	5	
23	2.407e+004	-0.2700	5	
24	2.409e+004	-0.2100	5	
25	2.412e+004	-0.1500	5	
26	2.414e+004	-0.1000	5	
27	2.417e+004	-9.000e-002	5	
28	2.419e+004	-0.1300	5	
29	2.422e+004	-0.1700	5	
30	2.424e+004	-0.1100	5	
31	2.427e+004	-0.2200	5	
32	2.429e+004	-0.1400	5	
33	2.432e+004	-9.000e-002	5	
34	2.434e+004	-0.1200	5	
35	2.437e+004	-0.1500	5	
36	2.438e+004	-0.1100	5	
37	2.439e+004	-0.1300	5	
38	2.442e+004	-0.1300	5	
39	2.444e+004	-0.1600	5	
40	2.447e+004	-0.2400	5	
41	2.449e+004	-0.2900	5	
42	2.452e+004	-0.2000	5	
43	2.454e+004	-0.2600	5	
44	2.457e+004	-0.2200	5	
45	2.459e+004	-0.1800	5	
46	2.462e+004	-0.1400	5	
47	2.464e+004	-9.000e-002	5	
48	2.467e+004	-0.2200	5	
49	2.469e+004	-0.5800	5	
50	2.472e+004	-0.2300	5	
51	2.474e+004	-7.000e-002	5	
52	2.477e+004	-7.000e-002	5	
53	2.479e+004	-0.1000	5	
54	2.482e+004	-0.1700	5	
55	2.484e+004	-0.3500	5	
56	2.487e+004	-0.7300	5	
57	2.489e+004	-0.7300	5	
58	2.492e+004	-0.1500	5	
59	2.494e+004	-7.000e-002	5	
60	2.497e+004	-0.1000	5	
61	2.499e+004	-0.1300	5	
62	2.502e+004	-0.1400	5	
63	2.504e+004	-0.1600	5	
64	2.507e+004	-0.2100	5	
65	2.509e+004	-0.1300	5	
66	2.512e+004	-0.1700	5	
67	2.514e+004	-0.2000	5	
68	2.517e+004	-0.2900	5	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	2.519e+004	-0.4200	5	
16	2.522e+004	-0.5200	5	
17	2.525e+004	-3.900	5	
18	2.527e+004	-5.300	5	
19	2.529e+004	-3.900	5	
20	2.539e+004	-3.300	5	
21	2.540e+004	-3.100	5	
22	2.543e+004	-0.4400	5	
23	2.545e+004	-0.2400	5	
24	2.548e+004	-0.1200	5	
25	2.550e+004	-0.1300	5	
26	2.553e+004	-9.000e-002	5	
27	2.555e+004	-9.000e-002	5	
28	2.558e+004	-5.000e-002	5	
29	2.560e+004	-3.000e-002	5	
30	2.563e+004	-5.000e-002	5	
31	2.566e+004	-6.000e-002	5	
32	2.568e+004	-7.000e-002	5	
33	2.571e+004	-0.1000	5	
34	2.571e+004	-0.2000	5	
35	2.572e+004	-0.8000	5	
36	2.573e+004	-3.500	5	
37	2.576e+004	-4.800	5	
38	2.578e+004	-0.7000	5	
39	2.581e+004	-0.3000	5	
40	2.583e+004	-0.2200	5	
41	2.586e+004	-6.000e-002	5	
42	2.588e+004	-7.000e-002	5	
43	2.591e+004	-0.2700	5	
44	2.593e+004	-0.6200	5	
45	2.596e+004	-0.7700	5	
46	2.598e+004	-0.1200	5	
47	2.601e+004	-0.1000	5	
48	2.603e+004	-0.2300	5	
49	2.606e+004	-4.000e-002	5	
50	2.608e+004	7.000e-002	5	
51	2.611e+004	-8.000e-002	5	
52	2.613e+004	-7.000e-002	5	
53	2.616e+004	-0.1400	5	
54	2.618e+004	-0.2000	5	
55	2.621e+004	-0.3800	5	
56	2.623e+004	-0.2000	5	
57	2.626e+004	-0.2300	5	
58	2.628e+004	-0.1000	5	
59	2.631e+004	-0.1700	5	
60	2.633e+004	-0.1700	5	
61	2.636e+004	-0.2400	5	
62	2.638e+004	-0.3100	5	
63	2.641e+004	-0.1800	5	
64	2.644e+004	-8.000e-002	5	
65	2.646e+004	-0.3700	5	
66	2.649e+004	-0.2900	5	
67	2.651e+004	-0.3900	5	
68	2.654e+004	-0.4100	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	2.656e+004	-0.3800	5	
16	2.659e+004	-0.4600	5	
17	2.661e+004	-0.4400	5	
18	2.664e+004	-0.5800	5	
19	2.666e+004	-0.5600	5	
20	2.669e+004	-0.4300	5	
21	2.671e+004	-0.2000	5	
22	2.674e+004	-0.1200	5	
23	2.676e+004	-3.000e-002	5	
24	2.679e+004	-1.000e-002	5	
25	2.681e+004	1.000e-002	5	
26	2.684e+004	-4.000e-002	5	
27	2.686e+004	3.000e-002	5	
28	2.689e+004	1.000e-002	5	
29	2.691e+004	-4.000e-002	5	
30	2.694e+004	-0.1400	5	
31	2.696e+004	-0.2600	5	
32	2.699e+004	-0.3600	5	
33	2.701e+004	-0.4500	5	
34	2.704e+004	-0.5500	5	
35	2.705e+004	-0.4400	5	
36	2.706e+004	-0.4900	5	
37	2.709e+004	-0.6500	5	
38	2.711e+004	-0.6000	5	
39	2.714e+004	-0.3200	5	
40	2.716e+004	-0.1300	5	
41	2.719e+004	-4.000e-002	5	
42	2.721e+004	4.000e-002	5	
43	2.724e+004	3.000e-002	5	
44	2.726e+004	8.000e-002	5	
45	2.729e+004	8.000e-002	5	
46	2.731e+004	0.1100	5	
47	2.734e+004	0.1100	5	
48	2.736e+004	0.1300	5	
49	2.739e+004	0.1500	5	
50	2.741e+004	0.1400	5	
51	2.744e+004	0.1500	5	
52	2.746e+004	0.1600	5	
53	2.749e+004	0.1600	5	
54	2.750e+004	0.1600	5	
55	2.753e+004	0.1500	5	
56	2.754e+004	0.1800	5	
57	2.756e+004	0.1800	5	
58	2.759e+004	0.8000	5	
59	2.761e+004	-0.3300	5	
60	2.764e+004	0.4500	5	
61	2.766e+004	0.7700	5	
62	2.769e+004	0.1800	5	
63	2.771e+004	0.1800	5	
64	2.774e+004	0.1900	5	
65	2.776e+004	0.1600	5	
66	2.779e+004	0.1700	5	
67	2.781e+004	0.1600	5	
68	2.784e+004	0.1500	5	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	2.786e+004	0.1600	5	
16	2.795e+004	0.1800	5	
17	2.804e+004	0.1500	5	
18	2.814e+004	0.1400	5	
19	2.823e+004	1.000e-001	5	
20	2.832e+004	8.000e-002	5	
21	2.841e+004	9.000e-002	5	
22	2.845e+004	8.000e-002	5	
23	2.847e+004	7.000e-002	5	
24	2.850e+004	7.000e-002	5	
25	2.852e+004	2.000e-002	5	
26	2.855e+004	5.000e-002	5	
27	2.857e+004	2.000e-002	5	
28	2.860e+004	-0.1600	5	
29	2.862e+004	-0.5000	5	
30	2.865e+004	-0.6800	5	
31	2.867e+004	-0.5300	5	
32	2.870e+004	-3.000e-002	5	
33	2.872e+004	-1.000e-002	5	
34	2.877e+004	2.000e-002	5	
35	2.883e+004	-1.000e-002	5	
36	2.888e+004	-2.000e-002	5	
37	2.893e+004	-9.000e-002	5	
38	2.898e+004	-0.1100	5	
39	2.903e+004	-0.1500	5	
40	2.905e+004	-0.1200	5	
41	2.907e+004	-0.1700	5	
42	2.910e+004	-0.2800	5	
43	2.912e+004	-0.3200	5	
44	2.915e+004	-0.1400	5	
45	2.917e+004	-8.000e-002	5	
46	2.920e+004	2.000e-002	5	
47	2.922e+004	-5.655e-016	5	
48	2.925e+004	3.000e-002	5	
49	2.926e+004	7.000e-002	5	
50	2.931e+004	0.1100	5	
51	2.936e+004	8.000e-002	5	
52	2.940e+004	9.000e-002	5	
53	2.945e+004	8.000e-002	5	
54	2.950e+004	2.000e-002	5	
55	2.955e+004	2.000e-002	5	
56	2.955e+004	3.000e-002	5	
57	2.958e+004	4.000e-002	5	
58	2.960e+004	6.000e-002	5	
59	2.963e+004	4.000e-002	5	
60	2.965e+004	3.000e-002	5	
61	2.966e+004	-5.621e-016	5	
62	2.967e+004	-3.000e-002	5	
63	2.969e+004	-4.000e-002	5	
64	2.971e+004	-3.000e-002	5	
65	2.973e+004	-3.000e-002	5	
66	2.975e+004	-4.000e-002	5	
67	2.978e+004	-5.000e-002	5	
68	2.980e+004	-3.000e-002	5	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	2.983e+004	1.000e-002	5	
16	2.985e+004	2.000e-002	5	
17	2.988e+004	7.000e-002	5	
18	2.990e+004	5.000e-002	5	
19	2.991e+004	6.000e-002	5	
20	2.994e+004	5.000e-002	5	
21	2.996e+004	5.000e-002	5	
22	2.998e+004	7.000e-002	5	
23	3.001e+004	6.000e-002	5	
24	3.003e+004	5.000e-002	5	
25	3.006e+004	3.000e-002	5	
26	3.008e+004	5.000e-002	5	
27	3.011e+004	3.000e-002	5	
28	3.013e+004	1.000e-002	5	
29	3.016e+004	3.000e-002	5	
30	3.018e+004	4.000e-002	5	
31	3.021e+004	4.000e-002	5	
32	3.023e+004	6.000e-002	5	
33	3.026e+004	4.000e-002	5	
34	3.028e+004	7.000e-002	5	
35	3.031e+004	1.000e-001	5	
36	3.033e+004	0.2000	5	
37	3.037e+004	0.1800	5	
38	3.044e+004	0.1900	5	
39	3.046e+004	0.1400	5	
40	3.048e+004	0.1400	5	
41	3.051e+004	0.1300	5	
42	3.053e+004	1.000e-001	5	
43	3.056e+004	8.000e-002	5	
44	3.058e+004	7.000e-002	5	
45	3.061e+004	0.1100	5	
46	3.063e+004	0.1400	5	
47	3.066e+004	0.1700	5	
48	3.068e+004	0.2600	5	
49	3.071e+004	-4.000e-002	5	
50	3.073e+004	-8.000e-002	5	
51	3.076e+004	-1.000e-002	5	
52	3.078e+004	-5.360e-016	5	
53	3.081e+004	-3.000e-002	5	
54	3.083e+004	2.000e-002	5	
55	3.086e+004	5.000e-002	5	
56	3.088e+004	0.1900	5	
57	3.091e+004	0.2100	5	
58	3.093e+004	0.2200	5	
59	3.096e+004	0.2300	5	
60	3.098e+004	0.2300	5	
61	3.101e+004	0.2200	5	
62	3.111e+004	0.2400	5	
63	3.113e+004	0.2300	5	
64	3.116e+004	0.8100	5	
65	3.118e+004	-8.300	5	
66	3.120e+004	-7.900	5	
67	3.124e+004	-6.300	5	
68	3.126e+004	-4.800	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	3.128e+004	0.1500	5	
16	3.131e+004	-5.000e-002	5	
17	3.133e+004	4.000e-002	5	
18	3.136e+004	-2.000e-002	5	
19	3.138e+004	-5.447e-016	5	
20	3.141e+004	0.1200	5	
21	3.143e+004	6.000e-002	5	
22	3.146e+004	2.000e-002	5	
23	3.148e+004	1.000e-002	5	
24	3.151e+004	6.000e-002	5	
25	3.153e+004	3.000e-002	5	
26	3.156e+004	0.1300	5	
27	3.159e+004	1.000e-002	5	
28	3.161e+004	0.1300	5	
29	3.164e+004	6.000e-002	5	
30	3.166e+004	1.000e-001	5	
31	3.169e+004	0.1400	5	
32	3.170e+004	0.1200	5	
33	3.171e+004	1.000e-001	5	
34	3.174e+004	-6.000e-002	5	
35	3.175e+004	5.000e-002	5	
36	3.178e+004	-1.000e-002	5	
37	3.181e+004	7.000e-002	5	
38	3.184e+004	3.000e-002	5	
39	3.186e+004	1.000e-001	5	
40	3.189e+004	0.1100	5	
41	3.191e+004	9.000e-002	5	
42	3.194e+004	0.3800	5	
43	3.196e+004	0.2400	5	
44	3.199e+004	0.2400	5	
45	3.201e+004	0.2100	5	
46	3.204e+004	0.1800	5	
47	3.206e+004	0.1600	5	
48	3.209e+004	0.1200	5	
49	3.211e+004	0.1500	5	
50	3.214e+004	0.1700	5	
51	3.216e+004	0.4000	5	
52	3.219e+004	0.4200	5	
53	3.219e+004	0.7600	5	
54	3.221e+004	1.190	5	
55	3.223e+004	1.320	5	
56	3.230e+004	1.330	5	
57	3.236e+004	1.280	5	
58	3.243e+004	1.340	5	
59	3.250e+004	1.380	5	
60	3.257e+004	1.180	5	
61	3.260e+004	1.090	5	
62	3.263e+004	1.030	5	
63	3.264e+004	1.010	5	
64	3.265e+004	0.9700	5	
65	3.267e+004	0.8800	5	
66	3.270e+004	0.8200	5	
67	3.272e+004	0.9000	5	
68	3.275e+004	0.8300	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	
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1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	3.277e+004	0.9900	5	
16	3.280e+004	0.8500	5	
17	3.282e+004	0.7400	5	
18	3.285e+004	0.5100	5	
19	3.287e+004	0.4200	5	
20	3.290e+004	0.2100	5	
21	3.292e+004	0.1800	5	
22	3.295e+004	0.1800	5	
23	3.297e+004	0.1600	5	
24	3.300e+004	0.1700	5	
25	3.302e+004	0.1500	5	
26	3.303e+004	0.1500	5	
27	3.306e+004	0.1900	5	
28	3.307e+004	0.1700	5	
29	3.310e+004	0.1600	5	
30	3.312e+004	0.1900	5	
31	3.315e+004	0.1600	5	
32	3.317e+004	0.1400	5	
33	3.320e+004	0.2000	5	
34	3.322e+004	0.2000	5	
35	3.325e+004	0.2000	5	
36	3.327e+004	0.1900	5	
37	3.330e+004	0.2100	5	
38	3.332e+004	0.2000	5	
39	3.335e+004	0.2000	5	
40	3.337e+004	0.2300	5	
41	3.340e+004	0.2100	5	
42	3.342e+004	0.2000	5	
43	3.345e+004	0.2200	5	
44	3.347e+004	0.2100	5	
45	3.350e+004	0.2100	5	
46	3.352e+004	0.2200	5	
47	3.355e+004	0.1900	5	
48	3.357e+004	0.2100	5	
49	3.360e+004	0.1600	5	
50	3.362e+004	0.2900	5	
51	3.365e+004	0.3300	5	
52	3.367e+004	0.4100	5	
53	3.370e+004	0.2200	5	
54	3.372e+004	0.2700	5	
55	3.375e+004	0.3200	5	
56	3.377e+004	0.3600	5	
57	3.380e+004	0.1700	5	
58	3.382e+004	0.2000	5	
59	3.385e+004	0.2500	5	
60	3.386e+004	0.3000	5	
61	3.387e+004	0.3000	5	
62	3.388e+004	0.3000	5	
63	3.390e+004	0.3100	5	
64	3.395e+004	0.3100	5	
65	3.401e+004	0.3000	5	
66	3.406e+004	0.2900	5	
67	3.412e+004	0.2800	5	
68	3.418e+004	0.3000	5	
69	Aspen Technology Inc.		Aspen HYSYS Version 10	Page 24 of 937

1	 Epic Atlantic Limited		Case Name:	Simulation - 386MMSCFD_100barg.hsc
2			Unit Set:	Mixed
3			Date/Time:	Mon Jun 13 15:02:44 2022
4				
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Data			
12				
13	Distance	Elevation	Increments	
14	(m)	(m)		
15	3.423e+004	0.4500	5	
16	3.429e+004	0.2900	5	
17	3.430e+004	0.3200	5	
18	3.432e+004	0.3400	5	
19	3.435e+004	0.3200	5	
20	3.437e+004	0.3300	5	
21	3.440e+004	0.3000	5	
22	3.442e+004	0.3000	5	
23	3.445e+004	0.3900	5	
24	3.447e+004	0.3000	5	
25	3.450e+004	0.3000	5	
26	3.452e+004	0.2900	5	
27	3.455e+004	0.3400	5	
28	3.457e+004	0.3000	5	
29	3.460e+004	0.3200	5	
30	3.462e+004	0.3100	5	
31	3.465e+004	0.3100	5	
32	3.467e+004	0.3900	5	
33	3.469e+004	0.3900	5	
34	3.472e+004	0.3500	5	
35	3.474e+004	0.3000	5	
36	3.477e+004	0.2400	5	
37	3.479e+004	0.1400	5	
38	3.482e+004	8.000e-002	5	
39	3.484e+004	0.1400	5	
40	3.487e+004	0.1100	5	
41	3.489e+004	0.4000	5	
42	3.492e+004	-0.3200	5	
43	3.494e+004	-0.8000	5	
44	3.497e+004	-0.3000	5	
45	3.499e+004	-2.000e-002	5	
46	3.502e+004	0.1500	5	
47	3.504e+004	0.1800	5	
48	3.507e+004	0.1600	5	
49	3.509e+004	0.2200	5	
50	3.512e+004	0.2200	5	
51	3.514e+004	0.2100	5	
52	3.517e+004	0.2200	5	
53	3.519e+004	0.2200	5	
54	3.522e+004	0.2400	5	
55	3.524e+004	0.2500	5	
56	3.527e+004	0.2400	5	
57	3.529e+004	0.2000	5	
58	3.532e+004	0.2100	5	
59	3.534e+004	0.2200	5	
60	3.537e+004	-0.7500	5	
61	3.539e+004	0.2400	5	
62	3.542e+004	0.3300	5	
63	3.544e+004	0.1500	5	
64	3.547e+004	0.1300	5	
65	3.549e+004	0.1400	5	
66	3.552e+004	0.1200	5	
67	3.554e+004	0.1300	5	
68	3.557e+004	0.1100	5	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3							
4			Unit Set: Mixed				
5			Date/Time: Mon Jun 13 15:02:44 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11							
12	Data						
13							
14							
15	Distance (m)		Elevation (m)		Increments		
16	3.559e+004		0.1100		5		
17	3.560e+004		0.1400		5		
18	3.563e+004		0.2100		5		
19	3.565e+004		0.1300		5		
20	3.568e+004		0.4000				
21	Pipe Table						
22	Length (m)	0.0000	5.000	10.00	15.00		
23	Elevation (m)	0.0000	-3.200e-002	-6.400e-002	-9.600e-002		
24	Pressure (bar)	98.66	98.66	98.66	98.66		
25	Temperature (C)	63.19	63.18	63.18	63.18		
26	Vapour Fraction	0.8720	0.8720	0.8720	0.8720		
27	Heat Transferred (kJ/h-m)	---	884.2	884.1	884.0		
28	Flow Regime	Transition	Transition	Transition	Transition		
29	Water Flow Regime						
30	Liquid HoldUp	0.2079	0.2079	0.2079	0.2079		
31	Water HoldUp	---	---	---	---		
32	Friction Gradient (kPa/m)	6.562e-003	6.562e-003	6.562e-003	6.562e-003		
33	Static Gradient (kPa/m)	-1.251e-002	-1.251e-002	-1.251e-002	-1.251e-002		
34	Accel. Gradient (kPa/m)	-3.659e-007	-3.659e-007	-3.659e-007	-3.659e-007		
35	Liq. Reynolds	2.298e+005 *	2.298e+005 *	2.298e+005 *	2.298e+005 *		
36	Vap. Reynolds	8.884e+006 *	8.884e+006 *	8.884e+006 *	8.884e+006 *		
37	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051		
38	Vapour Velocity (m/s)	2.057	2.057	2.057	2.057		
39	Deposit Thickness (mm)	---	---	---	---		
40	Deposit Volume (m3)	---	---	---	---		
41	Length (m)	20.00	25.00	30.00	35.00		
42	Elevation (m)	-0.1280	-0.1600	-0.1200	-8.000e-002		
43	Pressure (bar)	98.66	98.66	98.66	98.66		
44	Temperature (C)	63.18	63.18	63.17	63.17		
45	Vapour Fraction	0.8720	0.8720	0.8720	0.8720		
46	Heat Transferred (kJ/h-m)	884.0	883.9	883.8	883.8		
47	Flow Regime	Transition	Transition	Transition	Transition		
48	Water Flow Regime						
49	Liquid HoldUp	0.2079	0.2079	0.3151	0.3151		
50	Water HoldUp	---	---	---	---		
51	Friction Gradient (kPa/m)	6.562e-003	6.562e-003	5.682e-003	5.682e-003		
52	Static Gradient (kPa/m)	-1.251e-002	-1.251e-002	2.059e-002	2.059e-002		
53	Accel. Gradient (kPa/m)	-3.659e-007	-3.660e-007	1.615e-006	1.615e-006		
54	Liq. Reynolds	2.298e+005 *	2.298e+005 *	2.298e+005 *	2.298e+005 *		
55	Vap. Reynolds	8.884e+006 *	8.884e+006 *	8.884e+006 *	8.884e+006 *		
56	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051		
57	Vapour Velocity (m/s)	2.057	2.057	2.057	2.057		
58	Deposit Thickness (mm)	---	---	---	---		
59	Deposit Volume (m3)	---	---	---	---		
60	Length (m)	40.00	45.00	50.00	55.00		
61	Elevation (m)	-4.000e-002	2.776e-017	4.000e-002	3.000e-002		
62	Pressure (bar)	98.66	98.66	98.66	98.66		
63	Temperature (C)	63.17	63.16	63.16	63.16		
64	Vapour Fraction	0.8720	0.8720	0.8720	0.8720		
65	Heat Transferred (kJ/h-m)	883.7	883.6	883.5	883.5		
66	Flow Regime	Transition	Transition	Transition	Transition		
67	Water Flow Regime						
68	Liquid HoldUp	0.3151	0.3151	0.3151	0.2122		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	5.682e-003	5.682e-003	5.682e-003	6.530e-003
15	Static Gradient (kPa/m)	2.059e-002	2.059e-002	2.059e-002	-3.960e-003
16	Accel. Gradient (kPa/m)	1.615e-006	1.615e-006	1.615e-006	1.580e-007
17	Liq. Reynolds	2.298e+005 *	2.298e+005 *	2.298e+005 *	2.298e+005 *
18	Vap. Reynolds	8.884e+006 *	8.884e+006 *	8.884e+006 *	8.884e+006 *
19	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
20	Vapour Velocity (m/s)	2.057	2.057	2.057	2.057
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	60.00	65.00	70.00	75.00
24	Elevation (m)	2.000e-002	1.000e-002	0.0000	-1.000e-002
25	Pressure (bar)	98.66	98.66	98.66	98.66
26	Temperature (C)	63.16	63.15	63.15	63.15
27	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
28	Heat Transferred (kJ/h-m)	883.4	883.3	883.3	883.2
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2122	0.2122	0.2122	0.2122
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	6.530e-003	6.530e-003	6.530e-003	6.530e-003
34	Static Gradient (kPa/m)	-3.960e-003	-3.960e-003	-3.960e-003	-3.960e-003
35	Accel. Gradient (kPa/m)	1.580e-007	1.580e-007	1.580e-007	1.580e-007
36	Liq. Reynolds	2.298e+005 *	2.298e+005 *	2.297e+005 *	2.297e+005 *
37	Vap. Reynolds	8.884e+006 *	8.884e+006 *	8.884e+006 *	8.884e+006 *
38	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
39	Vapour Velocity (m/s)	2.057	2.057	2.057	2.057
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	80.00	85.00	90.00	95.00
43	Elevation (m)	-2.400e-002	-3.800e-002	-5.200e-002	-6.600e-002
44	Pressure (bar)	98.66	98.66	98.66	98.66
45	Temperature (C)	63.15	63.14	63.14	63.14
46	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
47	Heat Transferred (kJ/h-m)	883.1	883.1	883.0	882.9
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2114	0.2114	0.2114	0.2114
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	6.535e-003	6.535e-003	6.535e-003	6.535e-003
53	Static Gradient (kPa/m)	-5.532e-003	-5.532e-003	-5.532e-003	-5.532e-003
54	Accel. Gradient (kPa/m)	6.169e-008	6.168e-008	6.168e-008	6.167e-008
55	Liq. Reynolds	2.297e+005 *	2.297e+005 *	2.297e+005 *	2.297e+005 *
56	Vap. Reynolds	8.884e+006 *	8.884e+006 *	8.884e+006 *	8.884e+006 *
57	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
58	Vapour Velocity (m/s)	2.057	2.057	2.057	2.057
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	100.0	105.0	110.0	115.0
62	Elevation (m)	-8.000e-002	-6.200e-002	-4.400e-002	-2.600e-002
63	Pressure (bar)	98.66	98.65	98.65	98.65
64	Temperature (C)	63.14	63.13	63.13	63.13
65	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
66	Heat Transferred (kJ/h-m)	882.9	882.8	882.7	882.7
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.2114	0.3007	0.3007	0.3007
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	6.535e-003	4.409e-003	4.409e-003	4.409e-003
16	Static Gradient (kPa/m)	-5.532e-003	8.968e-003	8.968e-003	8.968e-003
17	Accel. Gradient (kPa/m)	6.166e-008	8.223e-007	8.223e-007	8.223e-007
18	Liq. Reynolds	2.297e+005 *	2.297e+005 *	2.297e+005 *	2.297e+005 *
19	Vap. Reynolds	8.884e+006 *	8.884e+006 *	8.884e+006 *	8.884e+006 *
20	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
21	Vapour Velocity (m/s)	2.057	2.057	2.057	2.057
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	120.0	125.0	130.0	135.0
25	Elevation (m)	-8.000e-003	1.000e-002	-6.000e-003	-2.200e-002
26	Pressure (bar)	98.65	98.65	98.65	98.65
27	Temperature (C)	63.13	63.12	63.12	63.12
28	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
29	Heat Transferred (kJ/h-m)	882.6	882.5	882.5	882.4
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.3007	0.3007	0.2111	0.2111
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	4.409e-003	4.409e-003	6.538e-003	6.538e-003
35	Static Gradient (kPa/m)	8.968e-003	8.968e-003	-6.315e-003	-6.315e-003
36	Accel. Gradient (kPa/m)	8.223e-007	8.223e-007	1.366e-008	1.366e-008
37	Liq. Reynolds	2.297e+005 *	2.297e+005 *	2.297e+005 *	2.297e+005 *
38	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
39	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
40	Vapour Velocity (m/s)	2.057	2.057	2.057	2.057
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	140.0	145.0	150.0	155.0
44	Elevation (m)	-3.800e-002	-5.400e-002	-7.000e-002	-7.600e-002
45	Pressure (bar)	98.65	98.65	98.65	98.65
46	Temperature (C)	63.12	63.11	63.11	63.11
47	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
48	Heat Transferred (kJ/h-m)	882.3	882.3	882.2	882.1
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2111	0.2111	0.2111	0.2130
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.538e-003	6.537e-003	6.537e-003	6.523e-003
54	Static Gradient (kPa/m)	-6.316e-003	-6.316e-003	-6.316e-003	-2.382e-003
55	Accel. Gradient (kPa/m)	1.365e-008	1.364e-008	1.363e-008	2.546e-007
56	Liq. Reynolds	2.297e+005 *	2.297e+005 *	2.297e+005 *	2.297e+005 *
57	Vap. Reynolds	8.884e+006 *	8.884e+006 *	8.884e+006 *	8.884e+006 *
58	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
59	Vapour Velocity (m/s)	2.057	2.057	2.057	2.057
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	160.0	165.0	170.0	175.0
63	Elevation (m)	-8.200e-002	-8.800e-002	-9.400e-002	-0.1000
64	Pressure (bar)	98.65	98.65	98.65	98.65
65	Temperature (C)	63.11	63.10	63.10	63.10
66	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
67	Heat Transferred (kJ/h-m)	882.1	882.0	881.9	881.9
68	Flow Regime	Transition	Transition	Transition	Transition
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2130	0.2130	0.2130	0.2130
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.523e-003	6.523e-003	6.523e-003	6.523e-003
17	Static Gradient (kPa/m)	-2.382e-003	-2.382e-003	-2.382e-003	-2.382e-003
18	Accel. Gradient (kPa/m)	2.546e-007	2.546e-007	2.546e-007	2.545e-007
19	Liq. Reynolds	2.297e+005 *	2.297e+005 *	2.297e+005 *	2.297e+005 *
20	Vap. Reynolds	8.884e+006 *	8.884e+006 *	8.884e+006 *	8.884e+006 *
21	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
22	Vapour Velocity (m/s)	2.057	2.057	2.057	2.056
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	180.0	185.0	190.0	195.0
26	Elevation (m)	-0.1120	-0.1240	-0.1360	-0.1480
27	Pressure (bar)	98.65	98.65	98.65	98.65
28	Temperature (C)	63.10	63.09	63.09	63.09
29	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
30	Heat Transferred (kJ/h-m)	881.8	881.7	881.7	881.6
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2119	0.2119	0.2119	0.2119
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	6.531e-003	6.531e-003	6.531e-003	6.531e-003
36	Static Gradient (kPa/m)	-4.738e-003	-4.738e-003	-4.738e-003	-4.738e-003
37	Accel. Gradient (kPa/m)	1.102e-007	1.102e-007	1.102e-007	1.102e-007
38	Liq. Reynolds	2.297e+005 *	2.297e+005 *	2.297e+005 *	2.297e+005 *
39	Vap. Reynolds	8.884e+006 *	8.884e+006 *	8.884e+006 *	8.884e+006 *
40	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
41	Vapour Velocity (m/s)	2.056	2.056	2.056	2.056
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	200.1	205.1	210.1	215.1
45	Elevation (m)	-0.1600	-0.1480	-0.1360	-0.1240
46	Pressure (bar)	98.65	98.65	98.65	98.65
47	Temperature (C)	63.09	63.08	63.08	63.08
48	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
49	Heat Transferred (kJ/h-m)	881.5	881.5	881.4	881.3
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2119	0.2968	0.2968	0.2968
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	6.531e-003	4.665e-003	4.665e-003	4.665e-003
55	Static Gradient (kPa/m)	-4.738e-003	5.925e-003	5.925e-003	5.925e-003
56	Accel. Gradient (kPa/m)	1.102e-007	6.509e-007	6.509e-007	6.509e-007
57	Liq. Reynolds	2.297e+005 *	2.297e+005 *	2.297e+005 *	2.297e+005 *
58	Vap. Reynolds	8.884e+006 *	8.884e+006 *	8.884e+006 *	8.884e+006 *
59	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
60	Vapour Velocity (m/s)	2.056	2.056	2.056	2.056
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	220.1	225.1	230.1	235.1
64	Elevation (m)	-0.1120	-0.1000	-0.1120	-0.1240
65	Pressure (bar)	98.65	98.65	98.65	98.65
66	Temperature (C)	63.08	63.07	63.07	63.07
67	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
68	Heat Transferred (kJ/h-m)	881.3	881.2	881.1	881.1
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc	
2				
3			Unit Set: Mixed	
4			Date/Time: Mon Jun 13 15:02:44 2022	
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Pipe Table			
12				
13	Flow Regime	Transition	Transition	Transition
14	Water Flow Regime			
15	Liquid HoldUp	0.2968	0.2968	0.2119
16	Water HoldUp	---	---	---
17	Friction Gradient (kPa/m)	4.665e-003	4.665e-003	6.531e-003
18	Static Gradient (kPa/m)	5.925e-003	5.925e-003	-4.748e-003
19	Accel. Gradient (kPa/m)	6.509e-007	6.509e-007	1.096e-007
20	Liq. Reynolds	2.297e+005 *	2.297e+005 *	2.297e+005 *
21	Vap. Reynolds	8.884e+006 *	8.885e+006 *	8.885e+006 *
22	Liquid Velocity (m/s)	0.2051	0.2051	0.2051
23	Vapour Velocity (m/s)	2.056	2.056	2.056
24	Deposit Thickness (mm)	---	---	---
25	Deposit Volume (m3)	---	---	---
26	Length (m)	240.1	245.1	250.1
27	Elevation (m)	-0.1360	-0.1480	-0.1600
28	Pressure (bar)	98.65	98.65	98.65
29	Temperature (C)	63.07	63.06	63.06
30	Vapour Fraction	0.8720	0.8719	0.8719
31	Heat Transferred (kJ/h-m)	881.0	880.9	880.8
32	Flow Regime	Transition	Transition	Transition
33	Water Flow Regime			
34	Liquid HoldUp	0.2119	0.2119	0.2119
35	Water HoldUp	---	---	---
36	Friction Gradient (kPa/m)	6.531e-003	6.531e-003	6.531e-003
37	Static Gradient (kPa/m)	-4.748e-003	-4.748e-003	-4.748e-003
38	Accel. Gradient (kPa/m)	1.096e-007	1.095e-007	1.095e-007
39	Liq. Reynolds	2.297e+005 *	2.296e+005 *	2.296e+005 *
40	Vap. Reynolds	8.884e+006 *	8.884e+006 *	8.884e+006 *
41	Liquid Velocity (m/s)	0.2051	0.2051	0.2051
42	Vapour Velocity (m/s)	2.056	2.056	2.056
43	Deposit Thickness (mm)	---	---	---
44	Deposit Volume (m3)	---	---	---
45	Length (m)	260.1	265.1	270.1
46	Elevation (m)	-0.1760	-0.1840	-0.1920
47	Pressure (bar)	98.65	98.65	98.65
48	Temperature (C)	63.06	63.05	63.05
49	Vapour Fraction	0.8719	0.8719	0.8719
50	Heat Transferred (kJ/h-m)	880.7	880.7	880.6
51	Flow Regime	Transition	Transition	Transition
52	Water Flow Regime			
53	Liquid HoldUp	0.2127	0.2127	0.2127
54	Water HoldUp	---	---	---
55	Friction Gradient (kPa/m)	6.525e-003	6.525e-003	6.525e-003
56	Static Gradient (kPa/m)	-3.173e-003	-3.173e-003	-3.173e-003
57	Accel. Gradient (kPa/m)	2.060e-007	2.060e-007	2.060e-007
58	Liq. Reynolds	2.296e+005 *	2.296e+005 *	2.296e+005 *
59	Vap. Reynolds	8.884e+006 *	8.884e+006 *	8.884e+006 *
60	Liquid Velocity (m/s)	0.2051	0.2051	0.2051
61	Vapour Velocity (m/s)	2.056	2.056	2.056
62	Deposit Thickness (mm)	---	---	---
63	Deposit Volume (m3)	---	---	---
64	Length (m)	280.1	285.1	290.1
65	Elevation (m)	-0.2020	-0.2040	-0.2060
66	Pressure (bar)	98.65	98.65	98.65
67	Temperature (C)	63.05	63.04	63.04
68	Vapour Fraction	0.8719	0.8719	0.8719
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
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3							Unit Set: Mixed
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5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
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8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Heat Transferred (kJ/h-m)	880.5	880.4	880.3	880.3		
14	Flow Regime	Transition	Transition	Transition	Transition		
15	Water Flow Regime						
16	Liquid HoldUp	0.2138	0.2138	0.2138	0.2138		
17	Water HoldUp	---	---	---	---		
18	Friction Gradient (kPa/m)	6.516e-003	6.516e-003	6.516e-003	6.516e-003		
19	Static Gradient (kPa/m)	-7.959e-004	-7.959e-004	-7.959e-004	-7.959e-004		
20	Accel. Gradient (kPa/m)	3.516e-007	3.515e-007	3.515e-007	3.515e-007		
21	Liq. Reynolds	2.296e+005 *	2.296e+005 *	2.296e+005 *	2.296e+005 *		
22	Vap. Reynolds	8.884e+006 *	8.884e+006 *	8.884e+006 *	8.884e+006 *		
23	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051		
24	Vapour Velocity (m/s)	2.056	2.056	2.056	2.056		
25	Deposit Thickness (mm)	---	---	---	---		
26	Deposit Volume (m3)	---	---	---	---		
27	Length (m)	300.1	305.1	310.1	315.1		
28	Elevation (m)	-0.2100	-0.2100	-0.2100	-0.2100		
29	Pressure (bar)	98.65	98.64	98.64	98.64		
30	Temperature (C)	63.04	63.03	63.03	63.03		
31	Vapour Fraction	0.8719	0.8719	0.8719	0.8719		
32	Heat Transferred (kJ/h-m)	880.2	880.1	880.1	880.0		
33	Flow Regime	Transition	Transition	Transition	Transition		
34	Water Flow Regime						
35	Liquid HoldUp	0.2138	0.3127	0.3127	0.3127		
36	Water HoldUp	---	---	---	---		
37	Friction Gradient (kPa/m)	6.516e-003	5.641e-003	5.640e-003	5.640e-003		
38	Static Gradient (kPa/m)	-7.959e-004	0.0000	0.0000	0.0000		
39	Accel. Gradient (kPa/m)	3.515e-007	3.466e-007	3.466e-007	3.466e-007		
40	Liq. Reynolds	2.296e+005 *	2.296e+005 *	2.296e+005 *	2.296e+005 *		
41	Vap. Reynolds	8.884e+006 *	8.884e+006 *	8.884e+006 *	8.884e+006 *		
42	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051		
43	Vapour Velocity (m/s)	2.056	2.056	2.056	2.056		
44	Deposit Thickness (mm)	---	---	---	---		
45	Deposit Volume (m3)	---	---	---	---		
46	Length (m)	320.1	325.1	330.1	335.1		
47	Elevation (m)	-0.2100	-0.2100	-0.2820	-0.3540		
48	Pressure (bar)	98.64	98.64	98.64	98.65		
49	Temperature (C)	63.03	63.02	63.02	63.02		
50	Vapour Fraction	0.8719	0.8719	0.8719	0.8719		
51	Heat Transferred (kJ/h-m)	879.9	879.9	879.8	879.8		
52	Flow Regime	Transition	Transition	Transition	Transition		
53	Water Flow Regime						
54	Liquid HoldUp	0.3127	0.3127	0.2003	0.2003		
55	Water HoldUp	---	---	---	---		
56	Friction Gradient (kPa/m)	5.640e-003	5.640e-003	6.622e-003	6.621e-003		
57	Static Gradient (kPa/m)	0.0000	0.0000	-2.753e-002	-2.753e-002		
58	Accel. Gradient (kPa/m)	3.466e-007	3.466e-007	-1.285e-006	-1.285e-006		
59	Liq. Reynolds	2.296e+005 *	2.296e+005 *	2.296e+005 *	2.296e+005 *		
60	Vap. Reynolds	8.884e+006 *	8.884e+006 *	8.884e+006 *	8.884e+006 *		
61	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051		
62	Vapour Velocity (m/s)	2.056	2.056	2.056	2.056		
63	Deposit Thickness (mm)	---	---	---	---		
64	Deposit Volume (m3)	---	---	---	---		
65	Length (m)	340.1	345.1	350.1	355.1		
66	Elevation (m)	-0.4260	-0.4980	-0.5700	-0.5000		
67	Pressure (bar)	98.65	98.65	98.65	98.65		
68	Temperature (C)	63.02	63.02	63.01	63.01		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
14	Heat Transferred (kJ/h-m)	879.7	879.7	879.6	879.5
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2003	0.2003	0.2003	0.3348
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	6.621e-003	6.621e-003	6.621e-003	5.772e-003
20	Static Gradient (kPa/m)	-2.753e-002	-2.753e-002	-2.753e-002	3.764e-002
21	Accel. Gradient (kPa/m)	-1.285e-006	-1.285e-006	-1.285e-006	2.668e-006
22	Liq. Reynolds	2.296e+005 *	2.296e+005 *	2.296e+005 *	2.296e+005 *
23	Vap. Reynolds	8.884e+006 *	8.884e+006 *	8.884e+006 *	8.884e+006 *
24	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
25	Vapour Velocity (m/s)	2.056	2.056	2.056	2.056
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	360.1	365.1	370.1	375.1
29	Elevation (m)	-0.4300	-0.3600	-0.2900	-0.2200
30	Pressure (bar)	98.64	98.64	98.64	98.64
31	Temperature (C)	63.01	63.00	63.00	63.00
32	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
33	Heat Transferred (kJ/h-m)	879.5	879.4	879.3	879.2
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.3348	0.3348	0.3348	0.3348
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	5.772e-003	5.772e-003	5.772e-003	5.772e-003
39	Static Gradient (kPa/m)	3.764e-002	3.764e-002	3.764e-002	3.764e-002
40	Accel. Gradient (kPa/m)	2.668e-006	2.668e-006	2.668e-006	2.668e-006
41	Liq. Reynolds	2.296e+005 *	2.296e+005 *	2.296e+005 *	2.296e+005 *
42	Vap. Reynolds	8.884e+006 *	8.884e+006 *	8.885e+006 *	8.885e+006 *
43	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
44	Vapour Velocity (m/s)	2.056	2.056	2.056	2.056
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	380.0	385.0	390.0	395.0
48	Elevation (m)	-0.1360	-5.200e-002	3.200e-002	0.1160
49	Pressure (bar)	98.64	98.63	98.63	98.63
50	Temperature (C)	63.00	62.99	62.99	62.99
51	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
52	Heat Transferred (kJ/h-m)	879.1	879.0	879.0	878.9
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.3440	0.3440	0.3440	0.3440
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	5.774e-003	5.774e-003	5.774e-003	5.774e-003
58	Static Gradient (kPa/m)	4.610e-002	4.610e-002	4.610e-002	4.610e-002
59	Accel. Gradient (kPa/m)	3.188e-006	3.188e-006	3.188e-006	3.188e-006
60	Liq. Reynolds	2.296e+005 *	2.296e+005 *	2.296e+005 *	2.295e+005 *
61	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
62	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
63	Vapour Velocity (m/s)	2.056	2.056	2.056	2.056
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	400.0	405.0	410.0	415.0
67	Elevation (m)	0.2000	0.1760	0.1520	0.1280
68	Pressure (bar)	98.63	98.63	98.63	98.63
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	62.98	62.98	62.98	62.98
14	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
15	Heat Transferred (kJ/h-m)	878.8	878.7	878.6	878.6
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.3440	0.2096	0.2096	0.2096
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	5.774e-003	6.547e-003	6.547e-003	6.547e-003
21	Static Gradient (kPa/m)	4.610e-002	-9.434e-003	-9.434e-003	-9.434e-003
22	Accel. Gradient (kPa/m)	3.188e-006	-1.774e-007	-1.774e-007	-1.774e-007
23	Liq. Reynolds	2.295e+005 *	2.295e+005 *	2.295e+005 *	2.295e+005 *
24	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
25	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
26	Vapour Velocity (m/s)	2.056	2.056	2.056	2.056
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	420.0	425.0	430.0	435.0
30	Elevation (m)	0.1040	8.000e-002	7.000e-002	6.000e-002
31	Pressure (bar)	98.63	98.63	98.63	98.63
32	Temperature (C)	62.97	62.97	62.97	62.97
33	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
34	Heat Transferred (kJ/h-m)	878.5	878.5	878.4	878.3
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2096	0.2096	0.2123	0.2123
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	6.547e-003	6.547e-003	6.526e-003	6.526e-003
40	Static Gradient (kPa/m)	-9.434e-003	-9.434e-003	-3.962e-003	-3.962e-003
41	Accel. Gradient (kPa/m)	-1.774e-007	-1.775e-007	1.576e-007	1.576e-007
42	Liq. Reynolds	2.295e+005 *	2.295e+005 *	2.295e+005 *	2.295e+005 *
43	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
44	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
45	Vapour Velocity (m/s)	2.056	2.056	2.056	2.056
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	440.0	445.0	450.0	455.0
49	Elevation (m)	5.000e-002	4.000e-002	3.000e-002	3.600e-002
50	Pressure (bar)	98.63	98.63	98.63	98.62
51	Temperature (C)	62.96	62.96	62.96	62.96
52	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
53	Heat Transferred (kJ/h-m)	878.3	878.2	878.1	878.0
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2123	0.2123	0.2123	0.2929
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	6.526e-003	6.526e-003	6.526e-003	4.931e-003
59	Static Gradient (kPa/m)	-3.962e-003	-3.962e-003	-3.962e-003	2.936e-003
60	Accel. Gradient (kPa/m)	1.576e-007	1.576e-007	1.576e-007	4.835e-007
61	Liq. Reynolds	2.295e+005 *	2.295e+005 *	2.295e+005 *	2.295e+005 *
62	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
63	Liquid Velocity (m/s)	0.2051	0.2051	0.2052	0.2052
64	Vapour Velocity (m/s)	2.056	2.056	2.056	2.056
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	460.0	465.0	470.0	475.0
68	Elevation (m)	4.200e-002	4.800e-002	5.400e-002	6.000e-002
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:02:44 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Pressure (bar)	98.62	98.62	98.62	98.62					
14	Temperature (C)	62.95	62.95	62.95	62.95					
15	Vapour Fraction	0.8719	0.8719	0.8719	0.8719					
16	Heat Transferred (kJ/h-m)	878.0	877.9	877.8	877.8					
17	Flow Regime	Transition	Transition	Transition	Transition					
18	Water Flow Regime									
19	Liquid HoldUp	0.2929	0.2929	0.2929	0.2929					
20	Water HoldUp	---	---	---	---					
21	Friction Gradient (kPa/m)	4.931e-003	4.931e-003	4.931e-003	4.931e-003					
22	Static Gradient (kPa/m)	2.936e-003	2.936e-003	2.936e-003	2.936e-003					
23	Accel. Gradient (kPa/m)	4.835e-007	4.835e-007	4.835e-007	4.835e-007					
24	Liq. Reynolds	2.295e+005 *	2.295e+005 *	2.295e+005 *	2.295e+005 *					
25	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *					
26	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052					
27	Vapour Velocity (m/s)	2.055	2.055	2.055	2.055					
28	Deposit Thickness (mm)	---	---	---	---					
29	Deposit Volume (m3)	---	---	---	---					
30	Length (m)	480.0	485.0	490.0	495.0					
31	Elevation (m)	6.000e-002	6.000e-002	6.000e-002	6.000e-002					
32	Pressure (bar)	98.62	98.62	98.62	98.62					
33	Temperature (C)	62.94	62.94	62.94	62.94					
34	Vapour Fraction	0.8719	0.8719	0.8719	0.8719					
35	Heat Transferred (kJ/h-m)	877.7	877.7	877.6	877.5					
36	Flow Regime	Transition	Transition	Transition	Transition					
37	Water Flow Regime									
38	Liquid HoldUp	0.3128	0.3128	0.3128	0.3128					
39	Water HoldUp	---	---	---	---					
40	Friction Gradient (kPa/m)	5.640e-003	5.640e-003	5.640e-003	5.640e-003					
41	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000					
42	Accel. Gradient (kPa/m)	3.466e-007	3.466e-007	3.466e-007	3.466e-007					
43	Liq. Reynolds	2.295e+005 *	2.295e+005 *	2.295e+005 *	2.295e+005 *					
44	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *					
45	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052					
46	Vapour Velocity (m/s)	2.055	2.055	2.055	2.055					
47	Deposit Thickness (mm)	---	---	---	---					
48	Deposit Volume (m3)	---	---	---	---					
49	Length (m)	500.0	505.0	510.0	515.0					
50	Elevation (m)	6.000e-002	5.800e-002	5.600e-002	5.400e-002					
51	Pressure (bar)	98.62	98.62	98.62	98.62					
52	Temperature (C)	62.93	62.93	62.93	62.93					
53	Vapour Fraction	0.8719	0.8719	0.8719	0.8719					
54	Heat Transferred (kJ/h-m)	877.5	877.4	877.3	877.3					
55	Flow Regime	Transition	Transition	Transition	Transition					
56	Water Flow Regime									
57	Liquid HoldUp	0.3128	0.2139	0.2139	0.2139					
58	Water HoldUp	---	---	---	---					
59	Friction Gradient (kPa/m)	5.640e-003	6.515e-003	6.514e-003	6.514e-003					
60	Static Gradient (kPa/m)	0.0000	-7.961e-004	-7.961e-004	-7.961e-004					
61	Accel. Gradient (kPa/m)	3.466e-007	3.514e-007	3.514e-007	3.514e-007					
62	Liq. Reynolds	2.295e+005 *	2.295e+005 *	2.295e+005 *	2.295e+005 *					
63	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *					
64	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052					
65	Vapour Velocity (m/s)	2.055	2.055	2.055	2.055					
66	Deposit Thickness (mm)	---	---	---	---					
67	Deposit Volume (m3)	---	---	---	---					
68	Length (m)	520.0	525.0	530.0	535.0					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	5.200e-002	5.000e-002	5.200e-002	5.400e-002
14	Pressure (bar)	98.62	98.62	98.62	98.62
15	Temperature (C)	62.92	62.92	62.92	62.92
16	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
17	Heat Transferred (kJ/h-m)	877.2	877.1	877.0	877.0
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2139	0.2139	0.2903	0.2903
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	6.514e-003	6.514e-003	5.115e-003	5.115e-003
23	Static Gradient (kPa/m)	-7.961e-004	-7.961e-004	9.728e-004	9.728e-004
24	Accel. Gradient (kPa/m)	3.514e-007	3.514e-007	3.741e-007	3.741e-007
25	Liq. Reynolds	2.295e+005 *	2.295e+005 *	2.295e+005 *	2.295e+005 *
26	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
27	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
28	Vapour Velocity (m/s)	2.055	2.055	2.055	2.055
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	540.0	545.0	550.0	555.0
32	Elevation (m)	5.600e-002	5.800e-002	6.000e-002	6.400e-002
33	Pressure (bar)	98.62	98.62	98.62	98.62
34	Temperature (C)	62.91	62.91	62.91	62.91
35	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
36	Heat Transferred (kJ/h-m)	876.9	876.8	876.8	876.7
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2903	0.2903	0.2903	0.2917
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	5.115e-003	5.115e-003	5.115e-003	5.022e-003
42	Static Gradient (kPa/m)	9.728e-004	9.728e-004	9.728e-004	1.952e-003
43	Accel. Gradient (kPa/m)	3.741e-007	3.741e-007	3.741e-007	4.285e-007
44	Liq. Reynolds	2.295e+005 *	2.295e+005 *	2.295e+005 *	2.295e+005 *
45	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
46	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
47	Vapour Velocity (m/s)	2.055	2.055	2.055	2.055
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	560.0	565.0	570.0	575.0
51	Elevation (m)	6.800e-002	7.200e-002	7.600e-002	8.000e-002
52	Pressure (bar)	98.62	98.62	98.62	98.62
53	Temperature (C)	62.90	62.90	62.90	62.90
54	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
55	Heat Transferred (kJ/h-m)	876.6	876.6	876.5	876.4
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2917	0.2917	0.2917	0.2917
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	5.022e-003	5.021e-003	5.021e-003	5.021e-003
61	Static Gradient (kPa/m)	1.952e-003	1.952e-003	1.952e-003	1.952e-003
62	Accel. Gradient (kPa/m)	4.285e-007	4.285e-007	4.285e-007	4.285e-007
63	Liq. Reynolds	2.295e+005 *	2.294e+005 *	2.294e+005 *	2.294e+005 *
64	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
65	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
66	Vapour Velocity (m/s)	2.055	2.055	2.055	2.055
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length	(m)	580.0	585.0	590.0	595.0	
14	Elevation	(m)	7.800e-002	7.600e-002	7.400e-002	7.200e-002	
15	Pressure	(bar)	98.62	98.62	98.62	98.62	
16	Temperature	(C)	62.89	62.89	62.89	62.89	
17	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
18	Heat Transferred	(kJ/h-m)	876.4	876.3	876.2	876.2	
19	Flow Regime		Transition	Transition	Transition	Transition	
20	Water Flow Regime						
21	Liquid HoldUp		0.2139	0.2139	0.2139	0.2139	
22	Water HoldUp		---	---	---	---	
23	Friction Gradient	(kPa/m)	6.514e-003	6.514e-003	6.514e-003	6.514e-003	
24	Static Gradient	(kPa/m)	-7.981e-004	-7.981e-004	-7.981e-004	-7.981e-004	
25	Accel. Gradient	(kPa/m)	3.512e-007	3.512e-007	3.512e-007	3.512e-007	
26	Liq. Reynolds		2.294e+005 *	2.294e+005 *	2.294e+005 *	2.294e+005 *	
27	Vap. Reynolds		8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *	
28	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052	
29	Vapour Velocity	(m/s)	2.055	2.055	2.055	2.055	
30	Deposit Thickness	(mm)	---	---	---	---	
31	Deposit Volume	(m3)	---	---	---	---	
32	Length	(m)	600.0	605.0	610.0	615.0	
33	Elevation	(m)	7.000e-002	7.600e-002	8.200e-002	8.800e-002	
34	Pressure	(bar)	98.62	98.62	98.62	98.61	
35	Temperature	(C)	62.88	62.88	62.88	62.88	
36	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
37	Heat Transferred	(kJ/h-m)	876.1	876.0	876.0	875.9	
38	Flow Regime		Transition	Transition	Transition	Transition	
39	Water Flow Regime						
40	Liquid HoldUp		0.2139	0.2930	0.2930	0.2930	
41	Water HoldUp		---	---	---	---	
42	Friction Gradient	(kPa/m)	6.514e-003	4.929e-003	4.929e-003	4.929e-003	
43	Static Gradient	(kPa/m)	-7.981e-004	2.937e-003	2.937e-003	2.937e-003	
44	Accel. Gradient	(kPa/m)	3.512e-007	4.833e-007	4.833e-007	4.833e-007	
45	Liq. Reynolds		2.294e+005 *	2.294e+005 *	2.294e+005 *	2.294e+005 *	
46	Vap. Reynolds		8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *	
47	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052	
48	Vapour Velocity	(m/s)	2.055	2.055	2.055	2.055	
49	Deposit Thickness	(mm)	---	---	---	---	
50	Deposit Volume	(m3)	---	---	---	---	
51	Length	(m)	620.0	625.0	630.0	635.0	
52	Elevation	(m)	9.400e-002	1.000e-001	0.1060	0.1120	
53	Pressure	(bar)	98.61	98.61	98.61	98.61	
54	Temperature	(C)	62.87	62.87	62.87	62.86	
55	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
56	Heat Transferred	(kJ/h-m)	875.8	875.8	875.7	875.6	
57	Flow Regime		Transition	Transition	Transition	Transition	
58	Water Flow Regime						
59	Liquid HoldUp		0.2930	0.2930	0.2930	0.2930	
60	Water HoldUp		---	---	---	---	
61	Friction Gradient	(kPa/m)	4.929e-003	4.929e-003	4.929e-003	4.929e-003	
62	Static Gradient	(kPa/m)	2.937e-003	2.937e-003	2.937e-003	2.937e-003	
63	Accel. Gradient	(kPa/m)	4.833e-007	4.833e-007	4.833e-007	4.833e-007	
64	Liq. Reynolds		2.294e+005 *	2.294e+005 *	2.294e+005 *	2.294e+005 *	
65	Vap. Reynolds		8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *	
66	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052	
67	Vapour Velocity	(m/s)	2.055	2.055	2.055	2.055	
68	Deposit Thickness	(mm)	---	---	---	---	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	640.0	645.0	650.0	655.0
15	Elevation (m)	0.1180	0.1240	0.1300	0.1340
16	Pressure (bar)	98.61	98.61	98.61	98.61
17	Temperature (C)	62.86	62.86	62.86	62.85
18	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
19	Heat Transferred (kJ/h-m)	875.6	875.5	875.4	875.4
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2930	0.2930	0.2930	0.2917
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	4.929e-003	4.929e-003	4.929e-003	5.020e-003
25	Static Gradient (kPa/m)	2.937e-003	2.937e-003	2.937e-003	1.952e-003
26	Accel. Gradient (kPa/m)	4.833e-007	4.833e-007	4.833e-007	4.284e-007
27	Liq. Reynolds	2.294e+005 *	2.294e+005 *	2.294e+005 *	2.294e+005 *
28	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
29	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
30	Vapour Velocity (m/s)	2.055	2.055	2.055	2.055
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	660.0	665.0	670.0	675.0
34	Elevation (m)	0.1380	0.1420	0.1460	0.1500
35	Pressure (bar)	98.61	98.61	98.61	98.61
36	Temperature (C)	62.85	62.85	62.85	62.84
37	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
38	Heat Transferred (kJ/h-m)	875.3	875.2	875.2	875.1
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2917	0.2917	0.2917	0.2917
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	5.020e-003	5.020e-003	5.020e-003	5.020e-003
44	Static Gradient (kPa/m)	1.952e-003	1.952e-003	1.952e-003	1.952e-003
45	Accel. Gradient (kPa/m)	4.284e-007	4.284e-007	4.284e-007	4.284e-007
46	Liq. Reynolds	2.294e+005 *	2.294e+005 *	2.294e+005 *	2.294e+005 *
47	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
48	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
49	Vapour Velocity (m/s)	2.055	2.055	2.055	2.055
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	682.0	689.0	696.0	703.0
53	Elevation (m)	0.1540	0.1580	0.1620	0.1660
54	Pressure (bar)	98.61	98.61	98.61	98.61
55	Temperature (C)	62.84	62.84	62.83	62.83
56	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
57	Heat Transferred (kJ/h-m)	875.0	874.9	874.8	874.7
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2910	0.2910	0.2910	0.2910
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	5.073e-003	5.073e-003	5.073e-003	5.073e-003
63	Static Gradient (kPa/m)	1.392e-003	1.392e-003	1.392e-003	1.392e-003
64	Accel. Gradient (kPa/m)	3.972e-007	3.972e-007	3.972e-007	3.972e-007
65	Liq. Reynolds	2.294e+005 *	2.294e+005 *	2.294e+005 *	2.294e+005 *
66	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
67	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
68	Vapour Velocity (m/s)	2.055	2.055	2.055	2.055
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	710.0	712.0	714.0	716.0
16	Elevation (m)	0.1700	0.1540	0.1380	0.1220
17	Pressure (bar)	98.61	98.61	98.61	98.61
18	Temperature (C)	62.83	62.83	62.83	62.82
19	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
20	Heat Transferred (kJ/h-m)	874.6	874.6	874.6	874.6
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2910	0.2066	0.2066	0.2066
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	5.073e-003	6.567e-003	6.567e-003	6.567e-003
26	Static Gradient (kPa/m)	1.392e-003	-1.559e-002	-1.559e-002	-1.559e-002
27	Accel. Gradient (kPa/m)	3.971e-007	-5.541e-007	-5.541e-007	-5.541e-007
28	Liq. Reynolds	2.294e+005 *	2.294e+005 *	2.294e+005 *	2.294e+005 *
29	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
30	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
31	Vapour Velocity (m/s)	2.054	2.054	2.054	2.054
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	718.0	720.0	722.0	724.0
35	Elevation (m)	0.1060	9.000e-002	0.1160	0.1420
36	Pressure (bar)	98.61	98.61	98.61	98.61
37	Temperature (C)	62.82	62.82	62.82	62.82
38	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
39	Heat Transferred (kJ/h-m)	874.5	874.5	874.5	874.5
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2066	0.2066	0.3317	0.3317
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	6.567e-003	6.567e-003	5.766e-003	5.766e-003
45	Static Gradient (kPa/m)	-1.559e-002	-1.559e-002	3.473e-002	3.473e-002
46	Accel. Gradient (kPa/m)	-5.541e-007	-5.541e-007	2.487e-006	2.487e-006
47	Liq. Reynolds	2.294e+005 *	2.294e+005 *	2.294e+005 *	2.294e+005 *
48	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
49	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
50	Vapour Velocity (m/s)	2.054	2.054	2.054	2.054
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	726.0	728.0	730.0	732.0
54	Elevation (m)	0.1680	0.1940	0.2200	0.2060
55	Pressure (bar)	98.61	98.61	98.60	98.60
56	Temperature (C)	62.82	62.82	62.82	62.82
57	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
58	Heat Transferred (kJ/h-m)	874.4	874.4	874.4	874.3
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.3317	0.3317	0.3317	0.2075
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	5.766e-003	5.766e-003	5.766e-003	6.560e-003
64	Static Gradient (kPa/m)	3.473e-002	3.473e-002	3.472e-002	-1.368e-002
65	Accel. Gradient (kPa/m)	2.488e-006	2.488e-006	2.488e-006	-4.373e-007
66	Liq. Reynolds	2.294e+005 *	2.294e+005 *	2.293e+005 *	2.293e+005 *
67	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
68	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
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5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
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9	PROFILES						
10	Pipe Table						
11							
12							
13	Vapour Velocity	(m/s)	2.054	2.054	2.054	2.054	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	734.0	736.0	738.0	740.0	
17	Elevation	(m)	0.1920	0.1780	0.1640	0.1500	
18	Pressure	(bar)	98.61	98.61	98.61	98.61	
19	Temperature	(C)	62.81	62.81	62.81	62.81	
20	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
21	Heat Transferred	(kJ/h-m)	874.3	874.3	874.3	874.2	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2075	0.2075	0.2075	0.2075	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	6.560e-003	6.560e-003	6.560e-003	6.560e-003	
27	Static Gradient	(kPa/m)	-1.368e-002	-1.368e-002	-1.368e-002	-1.368e-002	
28	Accel. Gradient	(kPa/m)	-4.373e-007	-4.373e-007	-4.373e-007	-4.373e-007	
29	Liq. Reynolds		2.293e+005 *	2.293e+005 *	2.293e+005 *	2.293e+005 *	
30	Vap. Reynolds		8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *	
31	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052	
32	Vapour Velocity	(m/s)	2.054	2.054	2.054	2.054	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	742.0	744.0	746.0	748.0	
36	Elevation	(m)	0.1400	0.1300	0.1200	0.1100	
37	Pressure	(bar)	98.61	98.61	98.61	98.61	
38	Temperature	(C)	62.81	62.81	62.81	62.81	
39	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
40	Heat Transferred	(kJ/h-m)	874.2	874.2	874.2	874.1	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2095	0.2095	0.2095	0.2095	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	6.545e-003	6.545e-003	6.545e-003	6.545e-003	
46	Static Gradient	(kPa/m)	-9.826e-003	-9.826e-003	-9.826e-003	-9.826e-003	
47	Accel. Gradient	(kPa/m)	-2.016e-007	-2.016e-007	-2.016e-007	-2.016e-007	
48	Liq. Reynolds		2.293e+005 *	2.293e+005 *	2.293e+005 *	2.293e+005 *	
49	Vap. Reynolds		8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *	
50	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052	
51	Vapour Velocity	(m/s)	2.054	2.054	2.054	2.054	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	750.0	755.0	760.0	765.0	
55	Elevation	(m)	1.000e-001	9.600e-002	9.200e-002	8.800e-002	
56	Pressure	(bar)	98.61	98.61	98.61	98.61	
57	Temperature	(C)	62.81	62.80	62.80	62.80	
58	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
59	Heat Transferred	(kJ/h-m)	874.1	874.0	874.0	873.9	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.2095	0.2136	0.2136	0.2136	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	6.545e-003	6.515e-003	6.515e-003	6.515e-003	
65	Static Gradient	(kPa/m)	-9.826e-003	-1.591e-003	-1.591e-003	-1.591e-003	
66	Accel. Gradient	(kPa/m)	-2.016e-007	3.025e-007	3.025e-007	3.024e-007	
67	Liq. Reynolds		2.293e+005 *	2.293e+005 *	2.293e+005 *	2.293e+005 *	
68	Vap. Reynolds		8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052	
14	Vapour Velocity	(m/s)	2.054	2.054	2.054	2.054	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	770.0	775.0	780.0	785.0	
18	Elevation	(m)	8.400e-002	8.000e-002	8.400e-002	8.800e-002	
19	Pressure	(bar)	98.60	98.60	98.60	98.60	
20	Temperature	(C)	62.80	62.79	62.79	62.79	
21	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
22	Heat Transferred	(kJ/h-m)	873.8	873.8	873.7	873.6	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.2136	0.2136	0.2917	0.2917	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	6.514e-003	6.514e-003	5.020e-003	5.020e-003	
28	Static Gradient	(kPa/m)	-1.591e-003	-1.591e-003	1.938e-003	1.938e-003	
29	Accel. Gradient	(kPa/m)	3.024e-007	3.024e-007	4.274e-007	4.274e-007	
30	Liq. Reynolds		2.293e+005 *	2.293e+005 *	2.293e+005 *	2.293e+005 *	
31	Vap. Reynolds		8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *	
32	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052	
33	Vapour Velocity	(m/s)	2.054	2.054	2.054	2.054	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	790.1	795.1	800.1	805.1	
37	Elevation	(m)	9.200e-002	9.600e-002	1.000e-001	1.000e-001	
38	Pressure	(bar)	98.60	98.60	98.60	98.60	
39	Temperature	(C)	62.79	62.78	62.78	62.78	
40	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
41	Heat Transferred	(kJ/h-m)	873.6	873.5	873.4	873.4	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2917	0.2917	0.2917	0.3129	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	5.020e-003	5.020e-003	5.020e-003	5.637e-003	
47	Static Gradient	(kPa/m)	1.938e-003	1.938e-003	1.938e-003	0.0000	
48	Accel. Gradient	(kPa/m)	4.274e-007	4.274e-007	4.274e-007	3.463e-007	
49	Liq. Reynolds		2.293e+005 *	2.293e+005 *	2.293e+005 *	2.293e+005 *	
50	Vap. Reynolds		8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *	
51	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052	
52	Vapour Velocity	(m/s)	2.054	2.054	2.054	2.054	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	810.1	815.1	820.1	825.1	
56	Elevation	(m)	1.000e-001	1.000e-001	1.000e-001	1.000e-001	
57	Pressure	(bar)	98.60	98.60	98.60	98.60	
58	Temperature	(C)	62.78	62.77	62.77	62.77	
59	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
60	Heat Transferred	(kJ/h-m)	873.3	873.2	873.2	873.1	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.3129	0.3129	0.3129	0.3129	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	5.637e-003	5.637e-003	5.637e-003	5.637e-003	
66	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000	
67	Accel. Gradient	(kPa/m)	3.463e-007	3.463e-007	3.463e-007	3.463e-007	
68	Liq. Reynolds		2.293e+005 *	2.293e+005 *	2.293e+005 *	2.293e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
14	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
15	Vapour Velocity (m/s)	2.054	2.054	2.054	2.054
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	830.1	835.1	840.1	845.1
19	Elevation (m)	2.600e-002	-4.800e-002	-0.1220	-0.1960
20	Pressure (bar)	98.60	98.60	98.60	98.61
21	Temperature (C)	62.77	62.77	62.76	62.76
22	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
23	Heat Transferred (kJ/h-m)	873.1	873.0	873.0	872.9
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2000	0.2000	0.2000	0.2000
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.620e-003	6.620e-003	6.620e-003	6.620e-003
29	Static Gradient (kPa/m)	-2.828e-002	-2.828e-002	-2.828e-002	-2.828e-002
30	Accel. Gradient (kPa/m)	-1.330e-006	-1.330e-006	-1.330e-006	-1.330e-006
31	Liq. Reynolds	2.293e+005 *	2.293e+005 *	2.293e+005 *	2.293e+005 *
32	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
33	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
34	Vapour Velocity (m/s)	2.054	2.054	2.054	2.054
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	850.1	855.1	860.1	865.1
38	Elevation (m)	-0.2700	-0.2060	-0.1420	-7.800e-002
39	Pressure (bar)	98.61	98.60	98.60	98.60
40	Temperature (C)	62.76	62.76	62.75	62.75
41	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
42	Heat Transferred (kJ/h-m)	872.8	872.8	872.7	872.6
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2000	0.3311	0.3311	0.3311
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.620e-003	5.764e-003	5.764e-003	5.764e-003
48	Static Gradient (kPa/m)	-2.828e-002	3.415e-002	3.415e-002	3.415e-002
49	Accel. Gradient (kPa/m)	-1.330e-006	2.451e-006	2.451e-006	2.451e-006
50	Liq. Reynolds	2.293e+005 *	2.293e+005 *	2.293e+005 *	2.293e+005 *
51	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
52	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
53	Vapour Velocity (m/s)	2.054	2.054	2.054	2.054
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	870.1	875.1	880.1	885.1
57	Elevation (m)	-1.400e-002	5.000e-002	4.200e-002	3.400e-002
58	Pressure (bar)	98.60	98.60	98.60	98.60
59	Temperature (C)	62.75	62.74	62.74	62.74
60	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
61	Heat Transferred (kJ/h-m)	872.5	872.4	872.4	872.3
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.3311	0.3311	0.2128	0.2128
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	5.764e-003	5.764e-003	6.519e-003	6.519e-003
67	Static Gradient (kPa/m)	3.415e-002	3.415e-002	-3.175e-003	-3.175e-003
68	Accel. Gradient (kPa/m)	2.452e-006	2.452e-006	2.054e-007	2.054e-007
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 41 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.293e+005 *	2.293e+005 *	2.293e+005 *	2.293e+005 *
14	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
15	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
16	Vapour Velocity (m/s)	2.054	2.054	2.054	2.054
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	890.1	895.1	900.1	905.1
20	Elevation (m)	2.600e-002	1.800e-002	1.000e-002	4.000e-003
21	Pressure (bar)	98.60	98.60	98.60	98.60
22	Temperature (C)	62.74	62.73	62.73	62.73
23	Vapour Fraction	0.8718	0.8718	0.8718	0.8717
24	Heat Transferred (kJ/h-m)	872.2	872.2	872.1	872.0
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2128	0.2128	0.2128	0.2132
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	6.519e-003	6.519e-003	6.519e-003	6.516e-003
30	Static Gradient (kPa/m)	-3.176e-003	-3.176e-003	-3.176e-003	-2.384e-003
31	Accel. Gradient (kPa/m)	2.053e-007	2.053e-007	2.053e-007	2.537e-007
32	Liq. Reynolds	2.293e+005 *	2.293e+005 *	2.293e+005 *	2.292e+005 *
33	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
34	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
35	Vapour Velocity (m/s)	2.054	2.054	2.054	2.054
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	910.1	915.1	920.1	925.1
39	Elevation (m)	-2.000e-003	-8.000e-003	-1.400e-002	-2.000e-002
40	Pressure (bar)	98.60	98.60	98.60	98.59
41	Temperature (C)	62.73	62.72	62.72	62.72
42	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
43	Heat Transferred (kJ/h-m)	872.0	871.9	871.8	871.8
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2132	0.2132	0.2132	0.2132
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	6.516e-003	6.516e-003	6.516e-003	6.516e-003
49	Static Gradient (kPa/m)	-2.384e-003	-2.384e-003	-2.384e-003	-2.384e-003
50	Accel. Gradient (kPa/m)	2.537e-007	2.537e-007	2.537e-007	2.537e-007
51	Liq. Reynolds	2.292e+005 *	2.292e+005 *	2.292e+005 *	2.292e+005 *
52	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
53	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
54	Vapour Velocity (m/s)	2.054	2.054	2.054	2.054
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	930.1	935.1	940.1	945.1
58	Elevation (m)	-1.400e-002	-8.000e-003	-2.000e-003	4.000e-003
59	Pressure (bar)	98.59	98.59	98.59	98.59
60	Temperature (C)	62.72	62.71	62.71	62.71
61	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
62	Heat Transferred (kJ/h-m)	871.7	871.6	871.6	871.5
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2931	0.2931	0.2931	0.2931
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	4.925e-003	4.925e-003	4.925e-003	4.925e-003
68	Static Gradient (kPa/m)	2.938e-003	2.938e-003	2.938e-003	2.938e-003
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	4.830e-007	4.829e-007	4.829e-007	4.829e-007
14	Liq. Reynolds	2.292e+005 *	2.292e+005 *	2.292e+005 *	2.292e+005 *
15	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
16	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
17	Vapour Velocity (m/s)	2.054	2.054	2.054	2.054
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	950.1	955.1	960.1	965.1
21	Elevation (m)	1.000e-002	1.000e-002	1.000e-002	1.000e-002
22	Pressure (bar)	98.59	98.59	98.59	98.59
23	Temperature (C)	62.71	62.70	62.70	62.70
24	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
25	Heat Transferred (kJ/h-m)	871.4	871.4	871.3	871.2
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.2931	0.3130	0.3130	0.3130
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	4.925e-003	5.636e-003	5.636e-003	5.636e-003
31	Static Gradient (kPa/m)	2.938e-003	0.0000	0.0000	0.0000
32	Accel. Gradient (kPa/m)	4.829e-007	3.461e-007	3.461e-007	3.461e-007
33	Liq. Reynolds	2.292e+005 *	2.292e+005 *	2.292e+005 *	2.292e+005 *
34	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
35	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
36	Vapour Velocity (m/s)	2.054	2.053	2.053	2.053
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	970.1	975.1	980.1	985.1
40	Elevation (m)	1.000e-002	1.000e-002	8.000e-003	6.000e-003
41	Pressure (bar)	98.59	98.59	98.59	98.59
42	Temperature (C)	62.70	62.69	62.69	62.69
43	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
44	Heat Transferred (kJ/h-m)	871.2	871.1	871.0	871.0
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.3130	0.3130	0.2140	0.2140
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	5.636e-003	5.636e-003	6.510e-003	6.509e-003
50	Static Gradient (kPa/m)	0.0000	0.0000	-7.976e-004	-7.976e-004
51	Accel. Gradient (kPa/m)	3.461e-007	3.461e-007	3.508e-007	3.508e-007
52	Liq. Reynolds	2.292e+005 *	2.292e+005 *	2.292e+005 *	2.292e+005 *
53	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
54	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
55	Vapour Velocity (m/s)	2.053	2.053	2.053	2.053
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	990.1	995.1	1000	1005
59	Elevation (m)	4.000e-003	2.000e-003	-1.232e-016	3.600e-002
60	Pressure (bar)	98.59	98.59	98.59	98.59
61	Temperature (C)	62.69	62.68	62.68	62.68
62	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
63	Heat Transferred (kJ/h-m)	870.9	870.8	870.8	870.7
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.2140	0.2140	0.2140	0.3128
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	6.509e-003	6.509e-003	6.509e-003	5.632e-003
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 43 of 937

1				Case Name:	Simulation - 386MMSCFD_100barg.hsc	
2				Unit Set:	Mixed	
3		Epic Atlantic Limited				
Date/Time:			Mon Jun 13 15:02:44 2022			
4						
5						
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	-7.976e-004	-7.977e-004	-7.977e-004	1.845e-002
14	Accel. Gradient	(kPa/m)	3.508e-007	3.508e-007	3.508e-007	1.479e-006
15	Liq. Reynolds		2.292e+005 *	2.292e+005 *	2.292e+005 *	2.292e+005 *
16	Vap. Reynolds		8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
17	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052
18	Vapour Velocity	(m/s)	2.053	2.053	2.053	2.053
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	1010	1015	1020	1025
22	Elevation	(m)	7.200e-002	0.1080	0.1440	0.1800
23	Pressure	(bar)	98.59	98.59	98.59	98.58
24	Temperature	(C)	62.68	62.67	62.67	62.67
25	Vapour Fraction		0.8717	0.8717	0.8717	0.8717
26	Heat Transferred	(kJ/h-m)	870.6	870.6	870.5	870.4
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.3128	0.3128	0.3128	0.3128
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	5.632e-003	5.632e-003	5.632e-003	5.632e-003
32	Static Gradient	(kPa/m)	1.845e-002	1.845e-002	1.845e-002	1.845e-002
33	Accel. Gradient	(kPa/m)	1.479e-006	1.479e-006	1.479e-006	1.479e-006
34	Liq. Reynolds		2.292e+005 *	2.292e+005 *	2.292e+005 *	2.292e+005 *
35	Vap. Reynolds		8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
36	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052
37	Vapour Velocity	(m/s)	2.053	2.053	2.053	2.053
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	1030	1035	1040	1045
41	Elevation	(m)	9.800e-002	1.600e-002	-6.600e-002	-0.1480
42	Pressure	(bar)	98.59	98.59	98.59	98.59
43	Temperature	(C)	62.67	62.66	62.66	62.66
44	Vapour Fraction		0.8717	0.8717	0.8717	0.8717
45	Heat Transferred	(kJ/h-m)	870.4	870.3	870.3	870.2
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.1985	0.1985	0.1985	0.1985
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	6.632e-003	6.632e-003	6.632e-003	6.631e-003
51	Static Gradient	(kPa/m)	-3.120e-002	-3.120e-002	-3.120e-002	-3.120e-002
52	Accel. Gradient	(kPa/m)	-1.509e-006	-1.509e-006	-1.508e-006	-1.508e-006
53	Liq. Reynolds		2.292e+005 *	2.292e+005 *	2.292e+005 *	2.292e+005 *
54	Vap. Reynolds		8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
55	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052
56	Vapour Velocity	(m/s)	2.053	2.053	2.053	2.053
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	1050	1055	1060	1065
60	Elevation	(m)	-0.2300	-0.1940	-0.1580	-0.1220
61	Pressure	(bar)	98.59	98.59	98.59	98.59
62	Temperature	(C)	62.66	62.66	62.65	62.65
63	Vapour Fraction		0.8717	0.8717	0.8717	0.8717
64	Heat Transferred	(kJ/h-m)	870.2	870.1	870.0	869.9
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.1985	0.3128	0.3128	0.3128
68	Water HoldUp		---	---	---	---
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1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	6.631e-003	5.632e-003	5.632e-003	5.632e-003
14	Static Gradient	(kPa/m)	-3.120e-002	1.845e-002	1.845e-002	1.845e-002
15	Accel. Gradient	(kPa/m)	-1.508e-006	1.479e-006	1.479e-006	1.479e-006
16	Liq. Reynolds		2.292e+005 *	2.292e+005 *	2.292e+005 *	2.292e+005 *
17	Vap. Reynolds		8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
18	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052
19	Vapour Velocity	(m/s)	2.053	2.053	2.053	2.053
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	1070	1075	1080	1085
23	Elevation	(m)	-8.600e-002	-5.000e-002	-3.600e-002	-2.200e-002
24	Pressure	(bar)	98.59	98.58	98.58	98.58
25	Temperature	(C)	62.65	62.65	62.64	62.64
26	Vapour Fraction		0.8717	0.8717	0.8717	0.8717
27	Heat Transferred	(kJ/h-m)	869.9	869.8	869.7	869.6
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.3128	0.3128	0.2984	0.2984
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	5.632e-003	5.632e-003	4.569e-003	4.569e-003
33	Static Gradient	(kPa/m)	1.845e-002	1.845e-002	6.942e-003	6.943e-003
34	Accel. Gradient	(kPa/m)	1.479e-006	1.479e-006	7.069e-007	7.069e-007
35	Liq. Reynolds		2.292e+005 *	2.292e+005 *	2.291e+005 *	2.291e+005 *
36	Vap. Reynolds		8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
37	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052
38	Vapour Velocity	(m/s)	2.053	2.053	2.053	2.053
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	1090	1095	1100	1105
42	Elevation	(m)	-8.000e-003	6.000e-003	2.000e-002	3.000e-002
43	Pressure	(bar)	98.58	98.58	98.58	98.58
44	Temperature	(C)	62.64	62.63	62.63	62.63
45	Vapour Fraction		0.8717	0.8717	0.8717	0.8717
46	Heat Transferred	(kJ/h-m)	869.6	869.5	869.4	869.4
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2984	0.2984	0.2984	0.2958
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	4.569e-003	4.569e-003	4.569e-003	4.744e-003
52	Static Gradient	(kPa/m)	6.943e-003	6.943e-003	6.943e-003	4.929e-003
53	Accel. Gradient	(kPa/m)	7.069e-007	7.069e-007	7.069e-007	5.939e-007
54	Liq. Reynolds		2.291e+005 *	2.291e+005 *	2.291e+005 *	2.291e+005 *
55	Vap. Reynolds		8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
56	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052
57	Vapour Velocity	(m/s)	2.053	2.053	2.053	2.053
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	1110	1115	1120	1125
61	Elevation	(m)	4.000e-002	5.000e-002	6.000e-002	7.000e-002
62	Pressure	(bar)	98.58	98.58	98.58	98.58
63	Temperature	(C)	62.63	62.62	62.62	62.62
64	Vapour Fraction		0.8717	0.8717	0.8717	0.8717
65	Heat Transferred	(kJ/h-m)	869.3	869.2	869.2	869.1
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2958	0.2958	0.2958	0.2958
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	4.744e-003	4.744e-003	4.744e-003	4.744e-003
15	Static Gradient (kPa/m)	4.929e-003	4.929e-003	4.929e-003	4.929e-003
16	Accel. Gradient (kPa/m)	5.939e-007	5.939e-007	5.939e-007	5.939e-007
17	Liq. Reynolds	2.291e+005 *	2.291e+005 *	2.291e+005 *	2.291e+005 *
18	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
19	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
20	Vapour Velocity (m/s)	2.053	2.053	2.053	2.053
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	1130	1135	1140	1145
24	Elevation (m)	8.400e-002	9.800e-002	0.1120	0.1260
25	Pressure (bar)	98.58	98.58	98.58	98.58
26	Temperature (C)	62.62	62.61	62.61	62.61
27	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
28	Heat Transferred (kJ/h-m)	869.0	868.9	868.9	868.8
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2984	0.2984	0.2984	0.2984
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	4.569e-003	4.569e-003	4.569e-003	4.569e-003
34	Static Gradient (kPa/m)	6.943e-003	6.943e-003	6.943e-003	6.943e-003
35	Accel. Gradient (kPa/m)	7.069e-007	7.069e-007	7.069e-007	7.069e-007
36	Liq. Reynolds	2.291e+005 *	2.291e+005 *	2.291e+005 *	2.291e+005 *
37	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
38	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
39	Vapour Velocity (m/s)	2.053	2.053	2.053	2.053
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	1150	1155	1160	1165
43	Elevation (m)	0.1400	0.1320	0.1240	0.1160
44	Pressure (bar)	98.58	98.58	98.58	98.58
45	Temperature (C)	62.61	62.60	62.60	62.60
46	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
47	Heat Transferred (kJ/h-m)	868.7	868.7	868.6	868.6
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2984	0.2129	0.2129	0.2129
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	4.569e-003	6.516e-003	6.516e-003	6.516e-003
53	Static Gradient (kPa/m)	6.943e-003	-3.177e-003	-3.177e-003	-3.177e-003
54	Accel. Gradient (kPa/m)	7.069e-007	2.051e-007	2.051e-007	2.050e-007
55	Liq. Reynolds	2.291e+005 *	2.291e+005 *	2.291e+005 *	2.291e+005 *
56	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
57	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
58	Vapour Velocity (m/s)	2.053	2.053	2.053	2.053
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	1170	1175	1180	1185
62	Elevation (m)	0.1080	1.000e-001	0.1100	0.1200
63	Pressure (bar)	98.58	98.58	98.57	98.57
64	Temperature (C)	62.60	62.59	62.59	62.59
65	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
66	Heat Transferred (kJ/h-m)	868.5	868.4	868.3	868.3
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.2129	0.2129	0.2958	0.2958
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	6.516e-003	6.516e-003	4.745e-003	4.745e-003
16	Static Gradient (kPa/m)	-3.177e-003	-3.177e-003	4.903e-003	4.903e-003
17	Accel. Gradient (kPa/m)	2.050e-007	2.050e-007	5.924e-007	5.924e-007
18	Liq. Reynolds	2.291e+005 *	2.291e+005 *	2.291e+005 *	2.291e+005 *
19	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
20	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
21	Vapour Velocity (m/s)	2.053	2.053	2.053	2.053
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	1190	1195	1200	1205
25	Elevation (m)	0.1300	0.1400	0.1500	0.1340
26	Pressure (bar)	98.57	98.57	98.57	98.57
27	Temperature (C)	62.59	62.58	62.58	62.58
28	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
29	Heat Transferred (kJ/h-m)	868.2	868.1	868.1	868.0
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2958	0.2958	0.2958	0.2113
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	4.745e-003	4.745e-003	4.745e-003	6.527e-003
35	Static Gradient (kPa/m)	4.903e-003	4.904e-003	4.904e-003	-6.325e-003
36	Accel. Gradient (kPa/m)	5.924e-007	5.924e-007	5.924e-007	1.239e-008
37	Liq. Reynolds	2.291e+005 *	2.291e+005 *	2.291e+005 *	2.291e+005 *
38	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
39	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
40	Vapour Velocity (m/s)	2.053	2.053	2.053	2.053
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	1210	1215	1220	1225
44	Elevation (m)	0.1180	0.1020	8.600e-002	7.000e-002
45	Pressure (bar)	98.57	98.57	98.57	98.57
46	Temperature (C)	62.58	62.57	62.57	62.57
47	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
48	Heat Transferred (kJ/h-m)	868.0	867.9	867.8	867.8
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2113	0.2114	0.2114	0.2114
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.527e-003	6.527e-003	6.527e-003	6.527e-003
54	Static Gradient (kPa/m)	-6.325e-003	-6.325e-003	-6.325e-003	-6.325e-003
55	Accel. Gradient (kPa/m)	1.238e-008	1.238e-008	1.237e-008	1.236e-008
56	Liq. Reynolds	2.291e+005 *	2.291e+005 *	2.291e+005 *	2.291e+005 *
57	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
58	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
59	Vapour Velocity (m/s)	2.053	2.052	2.052	2.052
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	1230	1235	1240	1245
63	Elevation (m)	5.600e-002	4.200e-002	2.800e-002	1.400e-002
64	Pressure (bar)	98.57	98.57	98.57	98.57
65	Temperature (C)	62.57	62.56	62.56	62.56
66	Vapour Fraction	0.8717	0.8716	0.8716	0.8716
67	Heat Transferred (kJ/h-m)	867.7	867.6	867.6	867.5
68	Flow Regime	Transition	Transition	Transition	Transition
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2117	0.2117	0.2117	0.2117
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.524e-003	6.524e-003	6.524e-003	6.524e-003
17	Static Gradient (kPa/m)	-5.541e-003	-5.541e-003	-5.541e-003	-5.541e-003
18	Accel. Gradient (kPa/m)	6.034e-008	6.033e-008	6.032e-008	6.031e-008
19	Liq. Reynolds	2.291e+005 *	2.291e+005 *	2.291e+005 *	2.291e+005 *
20	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
21	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
22	Vapour Velocity (m/s)	2.052	2.052	2.052	2.052
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	1250	1255	1260	1265
26	Elevation (m)	-9.714e-017	-9.714e-017	-9.714e-017	-9.714e-017
27	Pressure (bar)	98.57	98.57	98.57	98.57
28	Temperature (C)	62.56	62.55	62.55	62.55
29	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
30	Heat Transferred (kJ/h-m)	867.4	867.4	867.3	867.2
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2118	0.3131	0.3131	0.3131
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	6.524e-003	5.634e-003	5.634e-003	5.634e-003
36	Static Gradient (kPa/m)	-5.541e-003	0.0000	0.0000	0.0000
37	Accel. Gradient (kPa/m)	6.030e-008	3.459e-007	3.459e-007	3.459e-007
38	Liq. Reynolds	2.290e+005 *	2.290e+005 *	2.290e+005 *	2.290e+005 *
39	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
40	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
41	Vapour Velocity (m/s)	2.052	2.052	2.052	2.052
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	1270	1275	1280	1285
45	Elevation (m)	-9.714e-017	-9.714e-017	6.000e-003	1.200e-002
46	Pressure (bar)	98.57	98.57	98.57	98.57
47	Temperature (C)	62.55	62.55	62.54	62.54
48	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
49	Heat Transferred (kJ/h-m)	867.2	867.1	867.0	867.0
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.3131	0.3131	0.2932	0.2932
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	5.634e-003	5.634e-003	4.921e-003	4.921e-003
55	Static Gradient (kPa/m)	0.0000	0.0000	2.940e-003	2.940e-003
56	Accel. Gradient (kPa/m)	3.459e-007	3.459e-007	4.826e-007	4.826e-007
57	Liq. Reynolds	2.290e+005 *	2.290e+005 *	2.290e+005 *	2.290e+005 *
58	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
59	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
60	Vapour Velocity (m/s)	2.052	2.052	2.052	2.052
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	1290	1295	1300	1305
64	Elevation (m)	1.800e-002	2.400e-002	3.000e-002	4.200e-002
65	Pressure (bar)	98.57	98.57	98.57	98.57
66	Temperature (C)	62.54	62.53	62.53	62.53
67	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
68	Heat Transferred (kJ/h-m)	866.9	866.8	866.8	866.7
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2932	0.2932	0.2932	0.2972
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	4.921e-003	4.921e-003	4.921e-003	4.654e-003
18	Static Gradient (kPa/m)	2.940e-003	2.940e-003	2.940e-003	5.935e-003
19	Accel. Gradient (kPa/m)	4.826e-007	4.826e-007	4.826e-007	6.500e-007
20	Liq. Reynolds	2.290e+005 *	2.290e+005 *	2.290e+005 *	2.290e+005 *
21	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
22	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
23	Vapour Velocity (m/s)	2.052	2.052	2.052	2.052
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	1310	1315	1320	1325
27	Elevation (m)	5.400e-002	6.600e-002	7.800e-002	9.000e-002
28	Pressure (bar)	98.57	98.57	98.57	98.57
29	Temperature (C)	62.53	62.52	62.52	62.52
30	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
31	Heat Transferred (kJ/h-m)	866.6	866.6	866.5	866.4
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2972	0.2972	0.2972	0.2972
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	4.654e-003	4.654e-003	4.653e-003	4.653e-003
37	Static Gradient (kPa/m)	5.935e-003	5.935e-003	5.935e-003	5.935e-003
38	Accel. Gradient (kPa/m)	6.500e-007	6.500e-007	6.500e-007	6.500e-007
39	Liq. Reynolds	2.290e+005 *	2.290e+005 *	2.290e+005 *	2.290e+005 *
40	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *
41	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
42	Vapour Velocity (m/s)	2.052	2.052	2.052	2.052
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	1330	1335	1340	1345
46	Elevation (m)	0.1160	0.1420	0.1680	0.1940
47	Pressure (bar)	98.57	98.56	98.56	98.56
48	Temperature (C)	62.52	62.51	62.51	62.51
49	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
50	Heat Transferred (kJ/h-m)	866.4	866.3	866.2	866.1
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.3064	0.3064	0.3064	0.3064
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	5.372e-003	5.372e-003	5.372e-003	5.372e-003
56	Static Gradient (kPa/m)	1.313e-002	1.313e-002	1.313e-002	1.313e-002
57	Accel. Gradient (kPa/m)	1.136e-006	1.136e-006	1.136e-006	1.136e-006
58	Liq. Reynolds	2.290e+005 *	2.290e+005 *	2.290e+005 *	2.290e+005 *
59	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.886e+006 *
60	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
61	Vapour Velocity (m/s)	2.052	2.052	2.052	2.052
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	1350	1355	1360	1365
65	Elevation (m)	0.2200	0.1020	-1.600e-002	-0.1340
66	Pressure (bar)	98.56	98.56	98.57	98.57
67	Temperature (C)	62.51	62.50	62.50	62.50
68	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022	
6						
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Heat Transferred (kJ/h-m)	866.1	866.1	866.0	866.0	
14	Flow Regime	Transition	Transition	Transition	Transition	
15	Water Flow Regime					
16	Liquid HoldUp	0.3064	0.1916	0.1916	0.1916	
17	Water HoldUp	---	---	---	---	
18	Friction Gradient (kPa/m)	5.372e-003	6.693e-003	6.693e-003	6.692e-003	
19	Static Gradient (kPa/m)	1.313e-002	-4.396e-002	-4.396e-002	-4.396e-002	
20	Accel. Gradient (kPa/m)	1.136e-006	-2.288e-006	-2.288e-006	-2.288e-006	
21	Liq. Reynolds	2.290e+005 *	2.290e+005 *	2.290e+005 *	2.290e+005 *	
22	Vap. Reynolds	8.886e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *	
23	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052	
24	Vapour Velocity (m/s)	2.052	2.052	2.052	2.052	
25	Deposit Thickness (mm)	---	---	---	---	
26	Deposit Volume (m3)	---	---	---	---	
27	Length (m)	1370	1375	1380	1385	
28	Elevation (m)	-0.2520	-0.3700	-1.016	-1.662	
29	Pressure (bar)	98.57	98.57	98.58	98.59	
30	Temperature (C)	62.50	62.50	62.50	62.50	
31	Vapour Fraction	0.8716	0.8716	0.8716	0.8716	
32	Heat Transferred (kJ/h-m)	865.9	865.9	866.3	866.3	
33	Flow Regime	Transition	Transition	Transition	Transition	
34	Water Flow Regime					
35	Liquid HoldUp	0.1916	0.1916	9.226e-002	9.226e-002	
36	Water HoldUp	---	---	---	---	
37	Friction Gradient (kPa/m)	6.692e-003	6.692e-003	9.220e-003	9.219e-003	
38	Static Gradient (kPa/m)	-4.397e-002	-4.397e-002	-0.1664	-0.1664	
39	Accel. Gradient (kPa/m)	-2.288e-006	-2.288e-006	-9.644e-006	-9.643e-006	
40	Liq. Reynolds	2.290e+005 *	2.290e+005 *	2.290e+005 *	2.290e+005 *	
41	Vap. Reynolds	8.885e+006 *	8.885e+006 *	8.885e+006 *	8.885e+006 *	
42	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052	
43	Vapour Velocity (m/s)	2.052	2.052	2.051	2.051	
44	Deposit Thickness (mm)	---	---	---	---	
45	Deposit Volume (m3)	---	---	---	---	
46	Length (m)	1390	1395	1400	1405	
47	Elevation (m)	-2.308	-2.954	-3.600	-3.940	
48	Pressure (bar)	98.59	98.60	98.61	98.62	
49	Temperature (C)	62.50	62.50	62.50	62.50	
50	Vapour Fraction	0.8716	0.8716	0.8716	0.8716	
51	Heat Transferred (kJ/h-m)	866.3	866.4	866.4	866.1	
52	Flow Regime	Transition	Transition	Transition	Transition	
53	Water Flow Regime					
54	Liquid HoldUp	9.227e-002	9.227e-002	9.227e-002	0.1490	
55	Water HoldUp	---	---	---	---	
56	Friction Gradient (kPa/m)	9.218e-003	9.218e-003	9.217e-003	7.264e-003	
57	Static Gradient (kPa/m)	-0.1664	-0.1664	-0.1664	-0.1100	
58	Accel. Gradient (kPa/m)	-9.642e-006	-9.641e-006	-9.641e-006	-6.297e-006	
59	Liq. Reynolds	2.290e+005 *	2.290e+005 *	2.290e+005 *	2.290e+005 *	
60	Vap. Reynolds	8.885e+006 *	8.884e+006 *	8.884e+006 *	8.884e+006 *	
61	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052	
62	Vapour Velocity (m/s)	2.051	2.051	2.051	2.051	
63	Deposit Thickness (mm)	---	---	---	---	
64	Deposit Volume (m3)	---	---	---	---	
65	Length (m)	1410	1415	1420	1425	
66	Elevation (m)	-4.280	-4.620	-4.960	-5.300	
67	Pressure (bar)	98.62	98.63	98.63	98.64	
68	Temperature (C)	62.50	62.50	62.50	62.50	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
14	Heat Transferred (kJ/h-m)	866.1	866.1	866.1	866.1
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.1490	0.1490	0.1490	0.1490
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	7.264e-003	7.263e-003	7.263e-003	7.262e-003
20	Static Gradient (kPa/m)	-0.1100	-0.1100	-0.1100	-0.1100
21	Accel. Gradient (kPa/m)	-6.296e-006	-6.296e-006	-6.296e-006	-6.295e-006
22	Liq. Reynolds	2.290e+005 *	2.291e+005 *	2.291e+005 *	2.291e+005 *
23	Vap. Reynolds	8.884e+006 *	8.884e+006 *	8.883e+006 *	8.883e+006 *
24	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
25	Vapour Velocity (m/s)	2.051	2.050	2.050	2.050
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	1430	1435	1440	1445
29	Elevation (m)	-5.420	-5.540	-5.660	-5.780
30	Pressure (bar)	98.64	98.64	98.64	98.64
31	Temperature (C)	62.50	62.50	62.50	62.49
32	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
33	Heat Transferred (kJ/h-m)	865.9	865.9	865.8	865.8
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1913	0.1913	0.1913	0.1913
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	6.691e-003	6.691e-003	6.691e-003	6.690e-003
39	Static Gradient (kPa/m)	-4.468e-002	-4.468e-002	-4.468e-002	-4.468e-002
40	Accel. Gradient (kPa/m)	-2.328e-006	-2.328e-006	-2.328e-006	-2.328e-006
41	Liq. Reynolds	2.291e+005 *	2.291e+005 *	2.291e+005 *	2.291e+005 *
42	Vap. Reynolds	8.883e+006 *	8.883e+006 *	8.883e+006 *	8.883e+006 *
43	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
44	Vapour Velocity (m/s)	2.050	2.050	2.050	2.050
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	1450	1455	1460	1465
48	Elevation (m)	-5.900	-5.780	-5.660	-5.540
49	Pressure (bar)	98.65	98.64	98.64	98.63
50	Temperature (C)	62.49	62.49	62.49	62.48
51	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
52	Heat Transferred (kJ/h-m)	865.7	865.7	865.6	865.5
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.1914	0.3682	0.3682	0.3682
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	6.690e-003	5.739e-003	5.739e-003	5.739e-003
58	Static Gradient (kPa/m)	-4.468e-002	6.920e-002	6.920e-002	6.920e-002
59	Accel. Gradient (kPa/m)	-2.328e-006	4.592e-006	4.592e-006	4.592e-006
60	Liq. Reynolds	2.291e+005 *	2.291e+005 *	2.290e+005 *	2.290e+005 *
61	Vap. Reynolds	8.883e+006 *	8.883e+006 *	8.883e+006 *	8.883e+006 *
62	Liquid Velocity (m/s)	0.2053	0.2052	0.2052	0.2052
63	Vapour Velocity (m/s)	2.050	2.050	2.050	2.050
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	1470	1475	1480	1485
67	Elevation (m)	-5.420	-5.300	-5.500	-5.700
68	Pressure (bar)	98.63	98.63	98.63	98.63
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	62.48	62.48	62.47	62.47
14	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
15	Heat Transferred (kJ/h-m)	865.4	865.3	865.3	865.2
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.3682	0.3681	0.1759	0.1759
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	5.740e-003	5.740e-003	6.859e-003	6.859e-003
21	Static Gradient (kPa/m)	6.920e-002	6.920e-002	-7.090e-002	-7.090e-002
22	Accel. Gradient (kPa/m)	4.593e-006	4.593e-006	-3.925e-006	-3.924e-006
23	Liq. Reynolds	2.290e+005 *	2.290e+005 *	2.290e+005 *	2.290e+005 *
24	Vap. Reynolds	8.883e+006 *	8.883e+006 *	8.883e+006 *	8.883e+006 *
25	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2053
26	Vapour Velocity (m/s)	2.050	2.050	2.050	2.050
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	1490	1495	1500	1505
30	Elevation (m)	-5.900	-6.100	-6.300	-7.360
31	Pressure (bar)	98.64	98.64	98.64	98.65
32	Temperature (C)	62.47	62.47	62.47	62.47
33	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
34	Heat Transferred (kJ/h-m)	865.2	865.2	865.1	0.0000
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1759	0.1759	0.1759	2.180e-002
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	6.858e-003	6.858e-003	6.858e-003	9.559e-003
40	Static Gradient (kPa/m)	-7.090e-002	-7.090e-002	-7.090e-002	-0.1871
41	Accel. Gradient (kPa/m)	-3.924e-006	-3.924e-006	-3.924e-006	-1.087e-005
42	Liq. Reynolds	2.290e+005 *	2.290e+005 *	2.290e+005 *	2.291e+005 *
43	Vap. Reynolds	8.883e+006 *	8.883e+006 *	8.883e+006 *	8.883e+006 *
44	Liquid Velocity (m/s)	0.2053	0.2053	0.2053	0.2053
45	Vapour Velocity (m/s)	2.050	2.050	2.050	2.049
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	1510	1515	1520	1525
49	Elevation (m)	-8.420	-9.480	-10.54	-11.60
50	Pressure (bar)	98.66	98.67	98.68	98.69
51	Temperature (C)	62.48	62.48	62.49	62.49
52	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
53	Heat Transferred (kJ/h-m)	0.0000	0.0000	0.0000	0.0000
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	2.180e-002	2.180e-002	2.180e-002	2.180e-002
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	9.558e-003	9.557e-003	9.556e-003	9.555e-003
59	Static Gradient (kPa/m)	-0.1871	-0.1871	-0.1871	-0.1871
60	Accel. Gradient (kPa/m)	-1.087e-005	-1.087e-005	-1.087e-005	-1.087e-005
61	Liq. Reynolds	2.291e+005 *	2.291e+005 *	2.291e+005 *	2.291e+005 *
62	Vap. Reynolds	8.882e+006 *	8.882e+006 *	8.882e+006 *	8.882e+006 *
63	Liquid Velocity (m/s)	0.2053	0.2053	0.2053	0.2053
64	Vapour Velocity (m/s)	2.049	2.049	2.049	2.049
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	1530	1535	1540	1545
68	Elevation (m)	-9.192	-6.784	-4.376	-1.968
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Pressure (bar)	98.53	98.37	98.21	98.06
14	Temperature (C)	62.46	62.42	62.38	62.35
15	Vapour Fraction	0.8716	0.8717	0.8717	0.8718
16	Heat Transferred (kJ/h-m)	857.3	856.4	855.4	854.4
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	1.000	1.000	1.000	1.000
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	5.331e-003	5.338e-003	5.346e-003	5.354e-003
22	Static Gradient (kPa/m)	3.147	3.147	3.148	3.148
23	Accel. Gradient (kPa/m)	1.936e-004	1.943e-004	1.949e-004	1.956e-004
24	Liq. Reynolds	2.289e+005 *	2.287e+005 *	2.285e+005 *	2.283e+005 *
25	Vap. Reynolds	8.886e+006 *	8.891e+006 *	8.896e+006 *	8.900e+006 *
26	Liquid Velocity (m/s)	0.2052	0.2052	0.2051	0.2051
27	Vapour Velocity (m/s)	2.052	2.056	2.059	2.063
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	1550	1555	1560	1565
31	Elevation (m)	0.4400	0.5820	0.7240	0.8660
32	Pressure (bar)	97.90	97.89	97.89	97.89
33	Temperature (C)	62.31	62.31	62.30	62.30
34	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
35	Heat Transferred (kJ/h-m)	853.4	860.8	860.7	860.6
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	1.000	0.3808	0.3808	0.3808
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	5.361e-003	5.760e-003	5.760e-003	5.761e-003
41	Static Gradient (kPa/m)	3.149	8.391e-002	8.391e-002	8.391e-002
42	Accel. Gradient (kPa/m)	1.963e-004	5.580e-006	5.581e-006	5.581e-006
43	Liq. Reynolds	2.280e+005 *	2.280e+005 *	2.280e+005 *	2.280e+005 *
44	Vap. Reynolds	8.905e+006 *	8.905e+006 *	8.905e+006 *	8.905e+006 *
45	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
46	Vapour Velocity (m/s)	2.066	2.066	2.066	2.066
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	1570	1575	1580	1585
50	Elevation (m)	1.008	1.150	1.088	1.026
51	Pressure (bar)	97.88	97.88	97.88	97.88
52	Temperature (C)	62.30	62.29	62.29	62.29
53	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
54	Heat Transferred (kJ/h-m)	860.5	860.4	860.3	860.3
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.3808	0.3807	0.2016	0.2016
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	5.761e-003	5.761e-003	6.637e-003	6.637e-003
60	Static Gradient (kPa/m)	8.390e-002	8.390e-002	-2.376e-002	-2.376e-002
61	Accel. Gradient (kPa/m)	5.581e-006	5.582e-006	-1.066e-006	-1.066e-006
62	Liq. Reynolds	2.280e+005 *	2.280e+005 *	2.280e+005 *	2.280e+005 *
63	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
64	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
65	Vapour Velocity (m/s)	2.066	2.066	2.066	2.066
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	1590	1595	1600	1605
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	0.9640	0.9020	0.8400	0.7740
14	Pressure (bar)	97.88	97.88	97.88	97.88
15	Temperature (C)	62.29	62.28	62.28	62.28
16	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
17	Heat Transferred (kJ/h-m)	860.2	860.2	860.1	860.0
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2016	0.2016	0.2016	0.2008
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	6.637e-003	6.637e-003	6.637e-003	6.643e-003
23	Static Gradient (kPa/m)	-2.376e-002	-2.376e-002	-2.376e-002	-2.525e-002
24	Accel. Gradient (kPa/m)	-1.066e-006	-1.066e-006	-1.066e-006	-1.158e-006
25	Liq. Reynolds	2.280e+005 *	2.280e+005 *	2.280e+005 *	2.280e+005 *
26	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.905e+006 *
27	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
28	Vapour Velocity (m/s)	2.066	2.066	2.066	2.066
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	1610	1615	1620	1625
32	Elevation (m)	0.7080	0.6420	0.5760	0.5100
33	Pressure (bar)	97.88	97.88	97.88	97.89
34	Temperature (C)	62.28	62.28	62.27	62.27
35	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
36	Heat Transferred (kJ/h-m)	860.0	859.9	859.9	859.8
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2008	0.2008	0.2008	0.2008
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	6.643e-003	6.643e-003	6.643e-003	6.643e-003
42	Static Gradient (kPa/m)	-2.525e-002	-2.525e-002	-2.525e-002	-2.525e-002
43	Accel. Gradient (kPa/m)	-1.158e-006	-1.158e-006	-1.158e-006	-1.158e-006
44	Liq. Reynolds	2.280e+005 *	2.280e+005 *	2.280e+005 *	2.280e+005 *
45	Vap. Reynolds	8.905e+006 *	8.905e+006 *	8.905e+006 *	8.905e+006 *
46	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
47	Vapour Velocity (m/s)	2.066	2.066	2.066	2.066
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	1630	1635	1640	1645
51	Elevation (m)	0.5320	0.5540	0.5760	0.5980
52	Pressure (bar)	97.88	97.88	97.88	97.88
53	Temperature (C)	62.27	62.27	62.26	62.26
54	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
55	Heat Transferred (kJ/h-m)	859.7	859.7	859.6	859.5
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.3025	0.3025	0.3025	0.3025
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	5.054e-003	5.054e-003	5.054e-003	5.054e-003
61	Static Gradient (kPa/m)	1.101e-002	1.101e-002	1.101e-002	1.101e-002
62	Accel. Gradient (kPa/m)	9.995e-007	9.995e-007	9.995e-007	9.995e-007
63	Liq. Reynolds	2.280e+005 *	2.280e+005 *	2.280e+005 *	2.280e+005 *
64	Vap. Reynolds	8.905e+006 *	8.905e+006 *	8.905e+006 *	8.905e+006 *
65	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
66	Vapour Velocity (m/s)	2.066	2.066	2.066	2.066
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022		
6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length	(m)	1650	1655	1660	1665	
14	Elevation	(m)	0.6200	0.6140	0.6080	0.6020	
15	Pressure	(bar)	97.88	97.88	97.88	97.88	
16	Temperature	(C)	62.26	62.26	62.25	62.25	
17	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
18	Heat Transferred	(kJ/h-m)	859.5	859.4	859.3	859.3	
19	Flow Regime		Transition	Transition	Transition	Transition	
20	Water Flow Regime						
21	Liquid HoldUp		0.3025	0.2124	0.2124	0.2124	
22	Water HoldUp		---	---	---	---	
23	Friction Gradient	(kPa/m)	5.054e-003	6.552e-003	6.552e-003	6.551e-003	
24	Static Gradient	(kPa/m)	1.101e-002	-2.376e-003	-2.376e-003	-2.376e-003	
25	Accel. Gradient	(kPa/m)	9.995e-007	2.598e-007	2.598e-007	2.598e-007	
26	Liq. Reynolds		2.280e+005 *	2.280e+005 *	2.280e+005 *	2.280e+005 *	
27	Vap. Reynolds		8.905e+006 *	8.905e+006 *	8.905e+006 *	8.905e+006 *	
28	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
29	Vapour Velocity	(m/s)	2.066	2.066	2.066	2.066	
30	Deposit Thickness	(mm)	---	---	---	---	
31	Deposit Volume	(m3)	---	---	---	---	
32	Length	(m)	1670	1675	1680	1685	
33	Elevation	(m)	0.5960	0.5900	0.5160	0.4420	
34	Pressure	(bar)	97.88	97.88	97.88	97.88	
35	Temperature	(C)	62.25	62.25	62.24	62.24	
36	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
37	Heat Transferred	(kJ/h-m)	859.2	859.1	859.1	859.1	
38	Flow Regime		Transition	Transition	Transition	Transition	
39	Water Flow Regime						
40	Liquid HoldUp		0.2124	0.2124	0.1993	0.1993	
41	Water HoldUp		---	---	---	---	
42	Friction Gradient	(kPa/m)	6.551e-003	6.551e-003	6.655e-003	6.655e-003	
43	Static Gradient	(kPa/m)	-2.376e-003	-2.376e-003	-2.818e-002	-2.818e-002	
44	Accel. Gradient	(kPa/m)	2.598e-007	2.598e-007	-1.340e-006	-1.340e-006	
45	Liq. Reynolds		2.280e+005 *	2.280e+005 *	2.280e+005 *	2.280e+005 *	
46	Vap. Reynolds		8.905e+006 *	8.905e+006 *	8.905e+006 *	8.905e+006 *	
47	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
48	Vapour Velocity	(m/s)	2.066	2.066	2.066	2.066	
49	Deposit Thickness	(mm)	---	---	---	---	
50	Deposit Volume	(m3)	---	---	---	---	
51	Length	(m)	1690	1695	1700	1705	
52	Elevation	(m)	0.3680	0.2940	0.2200	0.2080	
53	Pressure	(bar)	97.88	97.88	97.89	97.89	
54	Temperature	(C)	62.24	62.24	62.24	62.23	
55	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
56	Heat Transferred	(kJ/h-m)	859.0	858.9	858.9	858.8	
57	Flow Regime		Transition	Transition	Transition	Transition	
58	Water Flow Regime						
59	Liquid HoldUp		0.1993	0.1993	0.1993	0.2113	
60	Water HoldUp		---	---	---	---	
61	Friction Gradient	(kPa/m)	6.655e-003	6.655e-003	6.655e-003	6.559e-003	
62	Static Gradient	(kPa/m)	-2.818e-002	-2.818e-002	-2.819e-002	-4.737e-003	
63	Accel. Gradient	(kPa/m)	-1.340e-006	-1.340e-006	-1.340e-006	1.134e-007	
64	Liq. Reynolds		2.280e+005 *	2.280e+005 *	2.280e+005 *	2.280e+005 *	
65	Vap. Reynolds		8.905e+006 *	8.905e+006 *	8.905e+006 *	8.905e+006 *	
66	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
67	Vapour Velocity	(m/s)	2.066	2.066	2.066	2.065	
68	Deposit Thickness	(mm)	---	---	---	---	
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5			Unit Set: Mixed			
6			Date/Time: Mon Jun 13 15:02:44 2022			
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9						
10	PROFILES					
11	Pipe Table					
12						
13	Deposit Volume (m3)	---	---	---	---	
14	Length (m)	1710	1715	1720	1725	
15	Elevation (m)	0.1960	0.1840	0.1720	0.1600	
16	Pressure (bar)	97.89	97.89	97.89	97.89	
17	Temperature (C)	62.23	62.23	62.23	62.22	
18	Vapour Fraction	0.8718	0.8718	0.8718	0.8718	
19	Heat Transferred (kJ/h-m)	858.8	858.7	858.6	858.6	
20	Flow Regime	Transition	Transition	Transition	Transition	
21	Water Flow Regime					
22	Liquid HoldUp	0.2113	0.2113	0.2113	0.2113	
23	Water HoldUp	---	---	---	---	
24	Friction Gradient (kPa/m)	6.559e-003	6.559e-003	6.559e-003	6.559e-003	
25	Static Gradient (kPa/m)	-4.737e-003	-4.737e-003	-4.737e-003	-4.737e-003	
26	Accel. Gradient (kPa/m)	1.134e-007	1.134e-007	1.134e-007	1.133e-007	
27	Liq. Reynolds	2.280e+005 *	2.280e+005 *	2.279e+005 *	2.279e+005 *	
28	Vap. Reynolds	8.905e+006 *	8.905e+006 *	8.905e+006 *	8.905e+006 *	
29	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051	
30	Vapour Velocity (m/s)	2.065	2.065	2.065	2.065	
31	Deposit Thickness (mm)	---	---	---	---	
32	Deposit Volume (m3)	---	---	---	---	
33	Length (m)	1730	1735	1740	1745	
34	Elevation (m)	0.1620	0.1640	0.1660	0.1680	
35	Pressure (bar)	97.89	97.88	97.88	97.88	
36	Temperature (C)	62.22	62.22	62.22	62.21	
37	Vapour Fraction	0.8718	0.8718	0.8718	0.8718	
38	Heat Transferred (kJ/h-m)	858.5	858.4	858.4	858.3	
39	Flow Regime	Transition	Transition	Transition	Transition	
40	Water Flow Regime					
41	Liquid HoldUp	0.2894	0.2895	0.2895	0.2895	
42	Water HoldUp	---	---	---	---	
43	Friction Gradient (kPa/m)	5.151e-003	5.151e-003	5.151e-003	5.151e-003	
44	Static Gradient (kPa/m)	9.705e-004	9.705e-004	9.705e-004	9.705e-004	
45	Accel. Gradient (kPa/m)	3.809e-007	3.809e-007	3.808e-007	3.808e-007	
46	Liq. Reynolds	2.279e+005 *	2.279e+005 *	2.279e+005 *	2.279e+005 *	
47	Vap. Reynolds	8.905e+006 *	8.905e+006 *	8.905e+006 *	8.905e+006 *	
48	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051	
49	Vapour Velocity (m/s)	2.065	2.065	2.065	2.065	
50	Deposit Thickness (mm)	---	---	---	---	
51	Deposit Volume (m3)	---	---	---	---	
52	Length (m)	1750	1755	1760	1765	
53	Elevation (m)	0.1700	0.1840	0.1980	0.2120	
54	Pressure (bar)	97.88	97.88	97.88	97.88	
55	Temperature (C)	62.21	62.21	62.21	62.20	
56	Vapour Fraction	0.8718	0.8718	0.8718	0.8718	
57	Heat Transferred (kJ/h-m)	858.2	858.2	858.1	858.0	
58	Flow Regime	Transition	Transition	Transition	Transition	
59	Water Flow Regime					
60	Liquid HoldUp	0.2895	0.2973	0.2973	0.2973	
61	Water HoldUp	---	---	---	---	
62	Friction Gradient (kPa/m)	5.151e-003	4.608e-003	4.608e-003	4.608e-003	
63	Static Gradient (kPa/m)	9.705e-004	6.921e-003	6.921e-003	6.921e-003	
64	Accel. Gradient (kPa/m)	3.808e-007	7.172e-007	7.172e-007	7.172e-007	
65	Liq. Reynolds	2.279e+005 *	2.279e+005 *	2.279e+005 *	2.279e+005 *	
66	Vap. Reynolds	8.905e+006 *	8.905e+006 *	8.905e+006 *	8.905e+006 *	
67	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051	
68	Vapour Velocity (m/s)	2.065	2.065	2.065	2.065	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	1770	1775	1780	1785
16	Elevation (m)	0.2260	0.2400	0.2600	0.2800
17	Pressure (bar)	97.88	97.88	97.88	97.88
18	Temperature (C)	62.20	62.20	62.20	62.19
19	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
20	Heat Transferred (kJ/h-m)	857.9	857.9	857.8	857.7
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2973	0.2973	0.3012	0.3012
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	4.608e-003	4.608e-003	4.686e-003	4.686e-003
26	Static Gradient (kPa/m)	6.921e-003	6.921e-003	9.937e-003	9.937e-003
27	Accel. Gradient (kPa/m)	7.172e-007	7.172e-007	9.097e-007	9.097e-007
28	Liq. Reynolds	2.279e+005 *	2.279e+005 *	2.279e+005 *	2.279e+005 *
29	Vap. Reynolds	8.905e+006 *	8.905e+006 *	8.905e+006 *	8.905e+006 *
30	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
31	Vapour Velocity (m/s)	2.065	2.065	2.065	2.065
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	1790	1795	1800	1805
35	Elevation (m)	0.3000	0.3200	0.3400	0.3680
36	Pressure (bar)	97.88	97.88	97.88	97.88
37	Temperature (C)	62.19	62.19	62.19	62.18
38	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
39	Heat Transferred (kJ/h-m)	857.7	857.6	857.5	857.5
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.3012	0.3012	0.3012	0.3065
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	4.686e-003	4.686e-003	4.686e-003	5.474e-003
45	Static Gradient (kPa/m)	9.937e-003	9.937e-003	9.937e-003	1.414e-002
46	Accel. Gradient (kPa/m)	9.097e-007	9.097e-007	9.097e-007	1.220e-006
47	Liq. Reynolds	2.279e+005 *	2.279e+005 *	2.279e+005 *	2.279e+005 *
48	Vap. Reynolds	8.905e+006 *	8.905e+006 *	8.905e+006 *	8.905e+006 *
49	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
50	Vapour Velocity (m/s)	2.065	2.065	2.065	2.065
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	1810	1815	1820	1825
54	Elevation (m)	0.3960	0.4240	0.4520	0.4800
55	Pressure (bar)	97.88	97.87	97.87	97.87
56	Temperature (C)	62.18	62.18	62.18	62.17
57	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
58	Heat Transferred (kJ/h-m)	857.4	857.3	857.3	857.2
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.3065	0.3065	0.3065	0.3065
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	5.474e-003	5.474e-003	5.474e-003	5.474e-003
64	Static Gradient (kPa/m)	1.414e-002	1.414e-002	1.414e-002	1.414e-002
65	Accel. Gradient (kPa/m)	1.220e-006	1.220e-006	1.220e-006	1.220e-006
66	Liq. Reynolds	2.279e+005 *	2.279e+005 *	2.279e+005 *	2.279e+005 *
67	Vap. Reynolds	8.905e+006 *	8.905e+006 *	8.905e+006 *	8.905e+006 *
68	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
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5			Unit Set: Mixed			
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7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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10	PROFILES					
11	Pipe Table					
12						
13	Vapour Velocity (m/s)	2.065	2.065	2.065	2.065	
14	Deposit Thickness (mm)	---	---	---	---	
15	Deposit Volume (m3)	---	---	---	---	
16	Length (m)	1830	1835	1840	1845	
17	Elevation (m)	0.4980	0.5160	0.5340	0.5520	
18	Pressure (bar)	97.87	97.87	97.87	97.87	
19	Temperature (C)	62.17	62.17	62.16	62.16	
20	Vapour Fraction	0.8718	0.8718	0.8718	0.8718	
21	Heat Transferred (kJ/h-m)	857.1	857.0	857.0	856.9	
22	Flow Regime	Transition	Transition	Transition	Transition	
23	Water Flow Regime					
24	Liquid HoldUp	0.2999	0.2999	0.2999	0.2999	
25	Water HoldUp	---	---	---	---	
26	Friction Gradient (kPa/m)	4.436e-003	4.436e-003	4.436e-003	4.436e-003	
27	Static Gradient (kPa/m)	8.953e-003	8.953e-003	8.953e-003	8.953e-003	
28	Accel. Gradient (kPa/m)	8.330e-007	8.330e-007	8.330e-007	8.330e-007	
29	Liq. Reynolds	2.279e+005 *	2.279e+005 *	2.279e+005 *	2.279e+005 *	
30	Vap. Reynolds	8.905e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *	
31	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051	
32	Vapour Velocity (m/s)	2.065	2.065	2.065	2.065	
33	Deposit Thickness (mm)	---	---	---	---	
34	Deposit Volume (m3)	---	---	---	---	
35	Length (m)	1850	1855	1860	1865	
36	Elevation (m)	0.5700	0.5820	0.5940	0.6060	
37	Pressure (bar)	97.87	97.87	97.87	97.87	
38	Temperature (C)	62.16	62.16	62.15	62.15	
39	Vapour Fraction	0.8718	0.8718	0.8718	0.8718	
40	Heat Transferred (kJ/h-m)	856.8	856.8	856.7	856.6	
41	Flow Regime	Transition	Transition	Transition	Transition	
42	Water Flow Regime					
43	Liquid HoldUp	0.2999	0.2960	0.2960	0.2960	
44	Water HoldUp	---	---	---	---	
45	Friction Gradient (kPa/m)	4.436e-003	4.695e-003	4.695e-003	4.695e-003	
46	Static Gradient (kPa/m)	8.953e-003	5.914e-003	5.914e-003	5.915e-003	
47	Accel. Gradient (kPa/m)	8.330e-007	6.601e-007	6.601e-007	6.601e-007	
48	Liq. Reynolds	2.279e+005 *	2.279e+005 *	2.279e+005 *	2.279e+005 *	
49	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *	
50	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051	
51	Vapour Velocity (m/s)	2.065	2.065	2.065	2.065	
52	Deposit Thickness (mm)	---	---	---	---	
53	Deposit Volume (m3)	---	---	---	---	
54	Length (m)	1870	1875	1880	1885	
55	Elevation (m)	0.6180	0.6300	0.6380	0.6460	
56	Pressure (bar)	97.87	97.87	97.87	97.87	
57	Temperature (C)	62.15	62.15	62.14	62.14	
58	Vapour Fraction	0.8718	0.8718	0.8718	0.8718	
59	Heat Transferred (kJ/h-m)	856.6	856.5	856.4	856.4	
60	Flow Regime	Transition	Transition	Transition	Transition	
61	Water Flow Regime					
62	Liquid HoldUp	0.2960	0.2960	0.2934	0.2934	
63	Water HoldUp	---	---	---	---	
64	Friction Gradient (kPa/m)	4.695e-003	4.695e-003	4.873e-003	4.873e-003	
65	Static Gradient (kPa/m)	5.915e-003	5.915e-003	3.919e-003	3.919e-003	
66	Accel. Gradient (kPa/m)	6.601e-007	6.601e-007	5.470e-007	5.470e-007	
67	Liq. Reynolds	2.279e+005 *	2.278e+005 *	2.278e+005 *	2.278e+005 *	
68	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *	
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc				
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3	 Epic Atlantic Limited		Unit Set: Mixed				
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7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
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11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
14	Vapour Velocity	(m/s)	2.065	2.065	2.065	2.065	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	1890	1895	1900	1905	
18	Elevation	(m)	0.6540	0.6620	0.6700	0.6700	
19	Pressure	(bar)	97.87	97.86	97.86	97.86	
20	Temperature	(C)	62.14	62.14	62.13	62.13	
21	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
22	Heat Transferred	(kJ/h-m)	856.3	856.2	856.2	856.1	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.2934	0.2934	0.2934	0.3119	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	4.873e-003	4.873e-003	4.873e-003	5.660e-003	
28	Static Gradient	(kPa/m)	3.919e-003	3.919e-003	3.919e-003	0.0000	
29	Accel. Gradient	(kPa/m)	5.470e-007	5.470e-007	5.470e-007	3.521e-007	
30	Liq. Reynolds		2.278e+005 *	2.278e+005 *	2.278e+005 *	2.278e+005 *	
31	Vap. Reynolds		8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *	
32	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
33	Vapour Velocity	(m/s)	2.065	2.065	2.065	2.065	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	1910	1915	1920	1925	
37	Elevation	(m)	0.6700	0.6700	0.6700	0.6700	
38	Pressure	(bar)	97.86	97.86	97.86	97.86	
39	Temperature	(C)	62.13	62.13	62.12	62.12	
40	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
41	Heat Transferred	(kJ/h-m)	856.0	856.0	855.9	855.8	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.3119	0.3119	0.3119	0.3119	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	5.660e-003	5.660e-003	5.660e-003	5.660e-003	
47	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000	
48	Accel. Gradient	(kPa/m)	3.521e-007	3.521e-007	3.521e-007	3.521e-007	
49	Liq. Reynolds		2.278e+005 *	2.278e+005 *	2.278e+005 *	2.278e+005 *	
50	Vap. Reynolds		8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *	
51	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
52	Vapour Velocity	(m/s)	2.065	2.065	2.065	2.065	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	1930	1935	1940	1945	
56	Elevation	(m)	0.6760	0.6820	0.6880	0.6940	
57	Pressure	(bar)	97.86	97.86	97.86	97.86	
58	Temperature	(C)	62.12	62.12	62.11	62.11	
59	Vapour Fraction		0.8718	0.8718	0.8717	0.8717	
60	Heat Transferred	(kJ/h-m)	855.8	855.7	855.6	855.6	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2921	0.2921	0.2921	0.2921	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	4.964e-003	4.963e-003	4.963e-003	4.963e-003	
66	Static Gradient	(kPa/m)	2.930e-003	2.930e-003	2.930e-003	2.930e-003	
67	Accel. Gradient	(kPa/m)	4.911e-007	4.911e-007	4.911e-007	4.911e-007	
68	Liq. Reynolds		2.278e+005 *	2.278e+005 *	2.278e+005 *	2.278e+005 *	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 59 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
14	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
15	Vapour Velocity (m/s)	2.065	2.065	2.065	2.065
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	1950	1955	1960	1965
19	Elevation (m)	0.7000	0.7000	0.7000	0.7000
20	Pressure (bar)	97.86	97.86	97.86	97.86
21	Temperature (C)	62.11	62.11	62.10	62.10
22	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
23	Heat Transferred (kJ/h-m)	855.5	855.4	855.4	855.3
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2921	0.3119	0.3119	0.3119
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	4.963e-003	5.660e-003	5.660e-003	5.660e-003
29	Static Gradient (kPa/m)	2.930e-003	0.0000	0.0000	0.0000
30	Accel. Gradient (kPa/m)	4.911e-007	3.521e-007	3.521e-007	3.521e-007
31	Liq. Reynolds	2.278e+005 *	2.278e+005 *	2.278e+005 *	2.278e+005 *
32	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
33	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
34	Vapour Velocity (m/s)	2.065	2.065	2.065	2.065
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	1970	1975	1980	1985
38	Elevation (m)	0.7000	0.7000	0.6640	0.6280
39	Pressure (bar)	97.86	97.86	97.86	97.86
40	Temperature (C)	62.10	62.10	62.10	62.09
41	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
42	Heat Transferred (kJ/h-m)	855.2	855.2	855.1	855.1
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.3119	0.3119	0.2067	0.2067
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	5.660e-003	5.660e-003	6.592e-003	6.592e-003
48	Static Gradient (kPa/m)	0.0000	0.0000	-1.403e-002	-1.403e-002
49	Accel. Gradient (kPa/m)	3.521e-007	3.521e-007	-4.629e-007	-4.629e-007
50	Liq. Reynolds	2.278e+005 *	2.278e+005 *	2.278e+005 *	2.278e+005 *
51	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
52	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
53	Vapour Velocity (m/s)	2.065	2.065	2.065	2.065
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1990	1995	2000	2005
57	Elevation (m)	0.5920	0.5560	0.5200	0.5200
58	Pressure (bar)	97.86	97.86	97.86	97.86
59	Temperature (C)	62.09	62.09	62.09	62.08
60	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
61	Heat Transferred (kJ/h-m)	855.0	855.0	854.9	854.8
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2067	0.2067	0.2067	0.3119
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	6.592e-003	6.592e-003	6.592e-003	5.659e-003
67	Static Gradient (kPa/m)	-1.403e-002	-1.403e-002	-1.403e-002	0.0000
68	Accel. Gradient (kPa/m)	-4.629e-007	-4.629e-007	-4.629e-007	3.520e-007
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 60 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		2.278e+005 *	2.278e+005 *	2.278e+005 *	2.278e+005 *	
14	Vap. Reynolds		8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *	
15	Liquid Velocity (m/s)		0.2051	0.2051	0.2051	0.2051	
16	Vapour Velocity (m/s)		2.064	2.064	2.064	2.064	
17	Deposit Thickness (mm)		---	---	---	---	
18	Deposit Volume (m3)		---	---	---	---	
19	Length (m)		2010	2015	2020	2025	
20	Elevation (m)		0.5200	0.5200	0.5200	0.5200	
21	Pressure (bar)		97.86	97.86	97.86	97.86	
22	Temperature (C)		62.08	62.08	62.08	62.07	
23	Vapour Fraction		0.8717	0.8717	0.8717	0.8717	
24	Heat Transferred (kJ/h-m)		854.8	854.7	854.6	854.6	
25	Flow Regime		Transition	Transition	Transition	Transition	
26	Water Flow Regime						
27	Liquid HoldUp		0.3119	0.3119	0.3119	0.3119	
28	Water HoldUp		---	---	---	---	
29	Friction Gradient (kPa/m)		5.659e-003	5.659e-003	5.659e-003	5.659e-003	
30	Static Gradient (kPa/m)		0.0000	0.0000	0.0000	0.0000	
31	Accel. Gradient (kPa/m)		3.520e-007	3.520e-007	3.520e-007	3.520e-007	
32	Liq. Reynolds		2.278e+005 *	2.278e+005 *	2.278e+005 *	2.278e+005 *	
33	Vap. Reynolds		8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *	
34	Liquid Velocity (m/s)		0.2051	0.2051	0.2051	0.2051	
35	Vapour Velocity (m/s)		2.064	2.064	2.064	2.064	
36	Deposit Thickness (mm)		---	---	---	---	
37	Deposit Volume (m3)		---	---	---	---	
38	Length (m)		2030	2035	2040	2045	
39	Elevation (m)		0.5520	0.5840	0.6160	0.6480	
40	Pressure (bar)		97.86	97.86	97.86	97.86	
41	Temperature (C)		62.07	62.07	62.07	62.06	
42	Vapour Fraction		0.8717	0.8717	0.8717	0.8717	
43	Heat Transferred (kJ/h-m)		854.5	854.4	854.3	854.3	
44	Flow Regime		Transition	Transition	Transition	Transition	
45	Water Flow Regime						
46	Liquid HoldUp		0.3092	0.3092	0.3092	0.3092	
47	Water HoldUp		---	---	---	---	
48	Friction Gradient (kPa/m)		5.587e-003	5.587e-003	5.587e-003	5.587e-003	
49	Static Gradient (kPa/m)		1.626e-002	1.626e-002	1.626e-002	1.626e-002	
50	Accel. Gradient (kPa/m)		1.359e-006	1.359e-006	1.359e-006	1.359e-006	
51	Liq. Reynolds		2.278e+005 *	2.278e+005 *	2.278e+005 *	2.277e+005 *	
52	Vap. Reynolds		8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *	
53	Liquid Velocity (m/s)		0.2051	0.2051	0.2051	0.2051	
54	Vapour Velocity (m/s)		2.064	2.064	2.064	2.064	
55	Deposit Thickness (mm)		---	---	---	---	
56	Deposit Volume (m3)		---	---	---	---	
57	Length (m)		2050	2055	2060	2065	
58	Elevation (m)		0.6800	0.6700	0.6600	0.6500	
59	Pressure (bar)		97.85	97.85	97.85	97.85	
60	Temperature (C)		62.06	62.06	62.06	62.05	
61	Vapour Fraction		0.8717	0.8717	0.8717	0.8717	
62	Heat Transferred (kJ/h-m)		854.2	854.1	854.1	854.0	
63	Flow Regime		Transition	Transition	Transition	Transition	
64	Water Flow Regime						
65	Liquid HoldUp		0.3092	0.2118	0.2118	0.2118	
66	Water HoldUp		---	---	---	---	
67	Friction Gradient (kPa/m)		5.587e-003	6.553e-003	6.553e-003	6.553e-003	
68	Static Gradient (kPa/m)		1.626e-002	-3.954e-003	-3.954e-003	-3.954e-003	
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:02:44 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient	(kPa/m)	1.359e-006	1.617e-007	1.617e-007	1.617e-007	
14	Liq. Reynolds		2.277e+005 *	2.277e+005 *	2.277e+005 *	2.277e+005 *	
15	Vap. Reynolds		8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *	
16	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
17	Vapour Velocity	(m/s)	2.064	2.064	2.064	2.064	
18	Deposit Thickness	(mm)	---	---	---	---	
19	Deposit Volume	(m3)	---	---	---	---	
20	Length	(m)	2070	2075	2080	2085	
21	Elevation	(m)	0.6400	0.6300	0.6340	0.6380	
22	Pressure	(bar)	97.85	97.85	97.85	97.85	
23	Temperature	(C)	62.05	62.05	62.05	62.04	
24	Vapour Fraction		0.8717	0.8717	0.8717	0.8717	
25	Heat Transferred	(kJ/h-m)	853.9	853.9	853.8	853.7	
26	Flow Regime		Transition	Transition	Transition	Transition	
27	Water Flow Regime						
28	Liquid HoldUp		0.2118	0.2118	0.2909	0.2909	
29	Water HoldUp		---	---	---	---	
30	Friction Gradient	(kPa/m)	6.553e-003	6.553e-003	5.054e-003	5.054e-003	
31	Static Gradient	(kPa/m)	-3.954e-003	-3.954e-003	1.948e-003	1.948e-003	
32	Accel. Gradient	(kPa/m)	1.617e-007	1.617e-007	4.355e-007	4.355e-007	
33	Liq. Reynolds		2.277e+005 *	2.277e+005 *	2.277e+005 *	2.277e+005 *	
34	Vap. Reynolds		8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *	
35	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
36	Vapour Velocity	(m/s)	2.064	2.064	2.064	2.064	
37	Deposit Thickness	(mm)	---	---	---	---	
38	Deposit Volume	(m3)	---	---	---	---	
39	Length	(m)	2090	2095	2100	2105	
40	Elevation	(m)	0.6420	0.6460	0.6500	0.6600	
41	Pressure	(bar)	97.85	97.85	97.85	97.85	
42	Temperature	(C)	62.04	62.04	62.04	62.03	
43	Vapour Fraction		0.8717	0.8717	0.8717	0.8717	
44	Heat Transferred	(kJ/h-m)	853.7	853.6	853.5	853.5	
45	Flow Regime		Transition	Transition	Transition	Transition	
46	Water Flow Regime						
47	Liquid HoldUp		0.2909	0.2909	0.2909	0.2948	
48	Water HoldUp		---	---	---	---	
49	Friction Gradient	(kPa/m)	5.054e-003	5.054e-003	5.054e-003	4.781e-003	
50	Static Gradient	(kPa/m)	1.948e-003	1.948e-003	1.948e-003	4.915e-003	
51	Accel. Gradient	(kPa/m)	4.355e-007	4.355e-007	4.355e-007	6.031e-007	
52	Liq. Reynolds		2.277e+005 *	2.277e+005 *	2.277e+005 *	2.277e+005 *	
53	Vap. Reynolds		8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *	
54	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
55	Vapour Velocity	(m/s)	2.064	2.064	2.064	2.064	
56	Deposit Thickness	(mm)	---	---	---	---	
57	Deposit Volume	(m3)	---	---	---	---	
58	Length	(m)	2110	2115	2120	2125	
59	Elevation	(m)	0.6700	0.6800	0.6900	0.7000	
60	Pressure	(bar)	97.85	97.85	97.85	97.85	
61	Temperature	(C)	62.03	62.03	62.03	62.02	
62	Vapour Fraction		0.8717	0.8717	0.8717	0.8717	
63	Heat Transferred	(kJ/h-m)	853.4	853.3	853.3	853.2	
64	Flow Regime		Transition	Transition	Transition	Transition	
65	Water Flow Regime						
66	Liquid HoldUp		0.2948	0.2948	0.2948	0.2948	
67	Water HoldUp		---	---	---	---	
68	Friction Gradient	(kPa/m)	4.781e-003	4.781e-003	4.781e-003	4.780e-003	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 62 of 937	

1				Case Name:	Simulation - 386MMSCFD_100barg.hsc	
2				Unit Set:	Mixed	
3		Epic Atlantic Limited				
Date/Time:			Mon Jun 13 15:02:44 2022			
4						
5						
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	4.915e-003	4.915e-003	4.915e-003	4.915e-003
14	Accel. Gradient	(kPa/m)	6.031e-007	6.031e-007	6.031e-007	6.031e-007
15	Liq. Reynolds		2.277e+005 *	2.277e+005 *	2.277e+005 *	2.277e+005 *
16	Vap. Reynolds		8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
17	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051
18	Vapour Velocity	(m/s)	2.064	2.064	2.064	2.064
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	2130	2135	2140	2145
22	Elevation	(m)	0.7060	0.7120	0.7180	0.7240
23	Pressure	(bar)	97.85	97.85	97.85	97.85
24	Temperature	(C)	62.02	62.02	62.02	62.01
25	Vapour Fraction		0.8717	0.8717	0.8717	0.8717
26	Heat Transferred	(kJ/h-m)	853.1	853.1	853.0	852.9
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2922	0.2922	0.2922	0.2922
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	4.961e-003	4.961e-003	4.961e-003	4.961e-003
32	Static Gradient	(kPa/m)	2.931e-003	2.931e-003	2.931e-003	2.931e-003
33	Accel. Gradient	(kPa/m)	4.909e-007	4.909e-007	4.909e-007	4.909e-007
34	Liq. Reynolds		2.277e+005 *	2.277e+005 *	2.277e+005 *	2.277e+005 *
35	Vap. Reynolds		8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
36	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051
37	Vapour Velocity	(m/s)	2.064	2.064	2.064	2.064
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	2150	2155	2160	2165
41	Elevation	(m)	0.7300	0.7200	0.7100	0.7000
42	Pressure	(bar)	97.85	97.85	97.85	97.85
43	Temperature	(C)	62.01	62.01	62.01	62.00
44	Vapour Fraction		0.8717	0.8717	0.8717	0.8717
45	Heat Transferred	(kJ/h-m)	852.9	852.8	852.8	852.7
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2922	0.2118	0.2118	0.2118
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	4.961e-003	6.552e-003	6.552e-003	6.552e-003
51	Static Gradient	(kPa/m)	2.931e-003	-3.954e-003	-3.954e-003	-3.954e-003
52	Accel. Gradient	(kPa/m)	4.909e-007	1.616e-007	1.616e-007	1.616e-007
53	Liq. Reynolds		2.277e+005 *	2.277e+005 *	2.277e+005 *	2.277e+005 *
54	Vap. Reynolds		8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
55	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051
56	Vapour Velocity	(m/s)	2.064	2.064	2.064	2.064
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	2170	2175	2180	2185
60	Elevation	(m)	0.6900	0.6800	0.6780	0.6760
61	Pressure	(bar)	97.85	97.85	97.85	97.85
62	Temperature	(C)	62.00	62.00	62.00	61.99
63	Vapour Fraction		0.8717	0.8717	0.8717	0.8717
64	Heat Transferred	(kJ/h-m)	852.6	852.6	852.5	852.4
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2118	0.2118	0.2133	0.2133
68	Water HoldUp		---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:02:44 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	6.552e-003	6.552e-003	6.541e-003	6.541e-003	
14	Static Gradient	(kPa/m)	-3.954e-003	-3.954e-003	-7.869e-004	-7.869e-004	
15	Accel. Gradient	(kPa/m)	1.616e-007	1.615e-007	3.578e-007	3.578e-007	
16	Liq. Reynolds		2.277e+005 *	2.277e+005 *	2.277e+005 *	2.277e+005 *	
17	Vap. Reynolds		8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *	
18	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
19	Vapour Velocity	(m/s)	2.064	2.064	2.064	2.064	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	2191	2196	2201	2206	
23	Elevation	(m)	0.6740	0.6720	0.6700	0.6680	
24	Pressure	(bar)	97.85	97.85	97.85	97.85	
25	Temperature	(C)	61.99	61.99	61.99	61.98	
26	Vapour Fraction		0.8717	0.8717	0.8717	0.8717	
27	Heat Transferred	(kJ/h-m)	852.4	852.3	852.2	852.2	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.2133	0.2133	0.2133	0.2133	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	6.541e-003	6.540e-003	6.540e-003	6.540e-003	
33	Static Gradient	(kPa/m)	-7.869e-004	-7.869e-004	-7.869e-004	-7.945e-004	
34	Accel. Gradient	(kPa/m)	3.578e-007	3.578e-007	3.578e-007	3.573e-007	
35	Liq. Reynolds		2.277e+005 *	2.277e+005 *	2.277e+005 *	2.277e+005 *	
36	Vap. Reynolds		8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *	
37	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
38	Vapour Velocity	(m/s)	2.064	2.064	2.064	2.064	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	2211	2216	2221	2226	
42	Elevation	(m)	0.6660	0.6640	0.6620	0.6600	
43	Pressure	(bar)	97.84	97.84	97.84	97.84	
44	Temperature	(C)	61.98	61.98	61.98	61.97	
45	Vapour Fraction		0.8717	0.8717	0.8717	0.8717	
46	Heat Transferred	(kJ/h-m)	852.1	852.0	852.0	851.9	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.2133	0.2133	0.2133	0.2133	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	6.540e-003	6.540e-003	6.540e-003	6.540e-003	
52	Static Gradient	(kPa/m)	-7.945e-004	-7.945e-004	-7.945e-004	-7.945e-004	
53	Accel. Gradient	(kPa/m)	3.573e-007	3.573e-007	3.573e-007	3.573e-007	
54	Liq. Reynolds		2.277e+005 *	2.277e+005 *	2.277e+005 *	2.276e+005 *	
55	Vap. Reynolds		8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *	
56	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
57	Vapour Velocity	(m/s)	2.064	2.064	2.064	2.064	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	2231	2236	2241	2246	
61	Elevation	(m)	0.6460	0.6320	0.6180	0.6040	
62	Pressure	(bar)	97.84	97.84	97.84	97.84	
63	Temperature	(C)	61.97	61.97	61.97	61.97	
64	Vapour Fraction		0.8717	0.8717	0.8717	0.8717	
65	Heat Transferred	(kJ/h-m)	851.9	851.8	851.7	851.7	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.2110	0.2110	0.2110	0.2110	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Water HoldUp		---	---	---	---	
14	Friction Gradient (kPa/m)		6.557e-003	6.557e-003	6.557e-003	6.557e-003	
15	Static Gradient (kPa/m)		-5.524e-003	-5.524e-003	-5.524e-003	-5.524e-003	
16	Accel. Gradient (kPa/m)		6.424e-008	6.423e-008	6.422e-008	6.421e-008	
17	Liq. Reynolds		2.276e+005 *	2.276e+005 *	2.276e+005 *	2.276e+005 *	
18	Vap. Reynolds		8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *	
19	Liquid Velocity (m/s)		0.2051	0.2051	0.2051	0.2051	
20	Vapour Velocity (m/s)		2.064	2.064	2.064	2.063	
21	Deposit Thickness (mm)		---	---	---	---	
22	Deposit Volume (m3)		---	---	---	---	
23	Length (m)		2251	2256	2261	2266	
24	Elevation (m)		0.5900	0.5980	0.6060	0.6140	
25	Pressure (bar)		97.84	97.84	97.84	97.84	
26	Temperature (C)		61.96	61.96	61.96	61.96	
27	Vapour Fraction		0.8717	0.8717	0.8717	0.8717	
28	Heat Transferred (kJ/h-m)		851.6	851.5	851.5	851.4	
29	Flow Regime		Transition	Transition	Transition	Transition	
30	Water Flow Regime						
31	Liquid HoldUp		0.2110	0.2936	0.2936	0.2936	
32	Water HoldUp		---	---	---	---	
33	Friction Gradient (kPa/m)		6.557e-003	4.869e-003	4.869e-003	4.869e-003	
34	Static Gradient (kPa/m)		-5.524e-003	3.921e-003	3.921e-003	3.921e-003	
35	Accel. Gradient (kPa/m)		6.420e-008	5.466e-007	5.466e-007	5.466e-007	
36	Liq. Reynolds		2.276e+005 *	2.276e+005 *	2.276e+005 *	2.276e+005 *	
37	Vap. Reynolds		8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *	
38	Liquid Velocity (m/s)		0.2051	0.2051	0.2051	0.2051	
39	Vapour Velocity (m/s)		2.063	2.063	2.063	2.063	
40	Deposit Thickness (mm)		---	---	---	---	
41	Deposit Volume (m3)		---	---	---	---	
42	Length (m)		2271	2276	2281	2286	
43	Elevation (m)		0.6220	0.6300	0.6400	0.6500	
44	Pressure (bar)		97.84	97.84	97.84	97.84	
45	Temperature (C)		61.95	61.95	61.95	61.95	
46	Vapour Fraction		0.8716	0.8716	0.8716	0.8716	
47	Heat Transferred (kJ/h-m)		851.3	851.3	851.2	851.1	
48	Flow Regime		Transition	Transition	Transition	Transition	
49	Water Flow Regime						
50	Liquid HoldUp		0.2936	0.2936	0.2949	0.2949	
51	Water HoldUp		---	---	---	---	
52	Friction Gradient (kPa/m)		4.868e-003	4.868e-003	4.779e-003	4.779e-003	
53	Static Gradient (kPa/m)		3.921e-003	3.921e-003	4.917e-003	4.917e-003	
54	Accel. Gradient (kPa/m)		5.466e-007	5.466e-007	6.029e-007	6.029e-007	
55	Liq. Reynolds		2.276e+005 *	2.276e+005 *	2.276e+005 *	2.276e+005 *	
56	Vap. Reynolds		8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *	
57	Liquid Velocity (m/s)		0.2051	0.2051	0.2051	0.2051	
58	Vapour Velocity (m/s)		2.063	2.063	2.063	2.063	
59	Deposit Thickness (mm)		---	---	---	---	
60	Deposit Volume (m3)		---	---	---	---	
61	Length (m)		2291	2296	2301	2306	
62	Elevation (m)		0.6600	0.6700	0.6800	0.6740	
63	Pressure (bar)		97.84	97.84	97.84	97.84	
64	Temperature (C)		61.94	61.94	61.94	61.94	
65	Vapour Fraction		0.8716	0.8716	0.8716	0.8716	
66	Heat Transferred (kJ/h-m)		851.1	851.0	850.9	850.9	
67	Flow Regime		Transition	Transition	Transition	Transition	
68	Water Flow Regime						
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 65 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.2949	0.2949	0.2949	0.2126
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	4.779e-003	4.778e-003	4.778e-003	6.545e-003
16	Static Gradient (kPa/m)	4.917e-003	4.917e-003	4.917e-003	-2.378e-003
17	Accel. Gradient (kPa/m)	6.029e-007	6.029e-007	6.029e-007	2.591e-007
18	Liq. Reynolds	2.276e+005 *	2.276e+005 *	2.276e+005 *	2.276e+005 *
19	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
20	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
21	Vapour Velocity (m/s)	2.063	2.063	2.063	2.063
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	2311	2316	2321	2326
25	Elevation (m)	0.6680	0.6620	0.6560	0.6500
26	Pressure (bar)	97.84	97.84	97.84	97.84
27	Temperature (C)	61.93	61.93	61.93	61.93
28	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
29	Heat Transferred (kJ/h-m)	850.8	850.7	850.7	850.6
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2126	0.2126	0.2126	0.2126
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.545e-003	6.545e-003	6.545e-003	6.545e-003
35	Static Gradient (kPa/m)	-2.378e-003	-2.378e-003	-2.378e-003	-2.379e-003
36	Accel. Gradient (kPa/m)	2.591e-007	2.591e-007	2.591e-007	2.591e-007
37	Liq. Reynolds	2.276e+005 *	2.276e+005 *	2.276e+005 *	2.276e+005 *
38	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
39	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
40	Vapour Velocity (m/s)	2.063	2.063	2.063	2.063
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	2331	2336	2341	2346
44	Elevation (m)	0.6440	0.6380	0.6320	0.6260
45	Pressure (bar)	97.84	97.84	97.84	97.84
46	Temperature (C)	61.92	61.92	61.92	61.92
47	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
48	Heat Transferred (kJ/h-m)	850.5	850.5	850.4	850.4
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2126	0.2126	0.2126	0.2126
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.545e-003	6.545e-003	6.545e-003	6.544e-003
54	Static Gradient (kPa/m)	-2.379e-003	-2.379e-003	-2.379e-003	-2.379e-003
55	Accel. Gradient (kPa/m)	2.590e-007	2.590e-007	2.590e-007	2.590e-007
56	Liq. Reynolds	2.276e+005 *	2.276e+005 *	2.276e+005 *	2.276e+005 *
57	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
58	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
59	Vapour Velocity (m/s)	2.063	2.063	2.063	2.063
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	2351	2356	2361	2366
63	Elevation (m)	0.6200	0.6180	0.6160	0.6140
64	Pressure (bar)	97.84	97.84	97.84	97.84
65	Temperature (C)	61.91	61.91	61.91	61.91
66	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
67	Heat Transferred (kJ/h-m)	850.3	850.2	850.2	850.1
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 66 of 937

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2126	0.2134	0.2134	0.2134
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.544e-003	6.539e-003	6.539e-003	6.539e-003
17	Static Gradient (kPa/m)	-2.379e-003	-7.947e-004	-7.947e-004	-7.947e-004
18	Accel. Gradient (kPa/m)	2.590e-007	3.572e-007	3.571e-007	3.571e-007
19	Liq. Reynolds	2.276e+005 *	2.276e+005 *	2.276e+005 *	2.276e+005 *
20	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
21	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
22	Vapour Velocity (m/s)	2.063	2.063	2.063	2.063
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	2371	2376	2381	2386
26	Elevation (m)	0.6120	0.6100	0.6200	0.6300
27	Pressure (bar)	97.84	97.84	97.84	97.83
28	Temperature (C)	61.90	61.90	61.90	61.90
29	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
30	Heat Transferred (kJ/h-m)	850.0	850.0	849.9	849.8
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2134	0.2134	0.2949	0.2949
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	6.539e-003	6.539e-003	4.779e-003	4.779e-003
36	Static Gradient (kPa/m)	-7.947e-004	-7.947e-004	4.903e-003	4.903e-003
37	Accel. Gradient (kPa/m)	3.571e-007	3.571e-007	6.020e-007	6.020e-007
38	Liq. Reynolds	2.276e+005 *	2.276e+005 *	2.276e+005 *	2.276e+005 *
39	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
40	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
41	Vapour Velocity (m/s)	2.063	2.063	2.063	2.063
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	2391	2396	2401	2406
45	Elevation (m)	0.6400	0.6500	0.6600	0.6780
46	Pressure (bar)	97.83	97.83	97.83	97.83
47	Temperature (C)	61.89	61.89	61.89	61.89
48	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
49	Heat Transferred (kJ/h-m)	849.8	849.7	849.6	849.6
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2949	0.2949	0.2949	0.3001
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	4.779e-003	4.779e-003	4.779e-003	4.430e-003
55	Static Gradient (kPa/m)	4.904e-003	4.904e-003	4.904e-003	8.961e-003
56	Accel. Gradient (kPa/m)	6.020e-007	6.020e-007	6.020e-007	8.326e-007
57	Liq. Reynolds	2.276e+005 *	2.276e+005 *	2.275e+005 *	2.275e+005 *
58	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
59	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
60	Vapour Velocity (m/s)	2.063	2.063	2.063	2.063
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	2411	2416	2421	2426
64	Elevation (m)	0.6960	0.7140	0.7320	0.7500
65	Pressure (bar)	97.83	97.83	97.83	97.83
66	Temperature (C)	61.88	61.88	61.88	61.88
67	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
68	Heat Transferred (kJ/h-m)	849.5	849.4	849.4	849.3
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.3001	0.3001	0.3001	0.3001
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	4.430e-003	4.430e-003	4.430e-003	4.430e-003
18	Static Gradient (kPa/m)	8.961e-003	8.961e-003	8.961e-003	8.961e-003
19	Accel. Gradient (kPa/m)	8.326e-007	8.326e-007	8.326e-007	8.326e-007
20	Liq. Reynolds	2.275e+005 *	2.275e+005 *	2.275e+005 *	2.275e+005 *
21	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
22	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
23	Vapour Velocity (m/s)	2.063	2.063	2.063	2.063
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	2431	2436	2441	2446
27	Elevation (m)	0.7320	0.7140	0.6960	0.6780
28	Pressure (bar)	97.83	97.83	97.83	97.83
29	Temperature (C)	61.87	61.87	61.87	61.87
30	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
31	Heat Transferred (kJ/h-m)	849.2	849.2	849.1	849.0
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2103	0.2103	0.2103	0.2103
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	6.561e-003	6.561e-003	6.561e-003	6.561e-003
37	Static Gradient (kPa/m)	-7.088e-003	-7.088e-003	-7.088e-003	-7.088e-003
38	Accel. Gradient (kPa/m)	-3.280e-008	-3.280e-008	-3.281e-008	-3.282e-008
39	Liq. Reynolds	2.275e+005 *	2.275e+005 *	2.275e+005 *	2.275e+005 *
40	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
41	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
42	Vapour Velocity (m/s)	2.063	2.063	2.063	2.063
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	2451	2456	2461	2466
46	Elevation (m)	0.6600	0.6700	0.6800	0.6900
47	Pressure (bar)	97.83	97.83	97.83	97.83
48	Temperature (C)	61.86	61.86	61.86	61.86
49	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
50	Heat Transferred (kJ/h-m)	849.0	848.9	848.8	848.8
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2103	0.2949	0.2949	0.2949
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	6.560e-003	4.777e-003	4.777e-003	4.777e-003
56	Static Gradient (kPa/m)	-7.089e-003	4.918e-003	4.918e-003	4.918e-003
57	Accel. Gradient (kPa/m)	-3.283e-008	6.027e-007	6.027e-007	6.027e-007
58	Liq. Reynolds	2.275e+005 *	2.275e+005 *	2.275e+005 *	2.275e+005 *
59	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
60	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
61	Vapour Velocity (m/s)	2.063	2.063	2.063	2.063
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	2471	2476	2481	2486
65	Elevation (m)	0.7000	0.7100	0.7200	0.7300
66	Pressure (bar)	97.83	97.83	97.83	97.83
67	Temperature (C)	61.85	61.85	61.85	61.85
68	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:02:44 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Heat Transferred (kJ/h-m)	848.7	848.6	848.6	848.5					
14	Flow Regime	Transition	Transition	Transition	Transition					
15	Water Flow Regime									
16	Liquid HoldUp	0.2949	0.2949	0.2949	0.2949					
17	Water HoldUp	---	---	---	---					
18	Friction Gradient (kPa/m)	4.777e-003	4.776e-003	4.776e-003	4.776e-003					
19	Static Gradient (kPa/m)	4.918e-003	4.918e-003	4.918e-003	4.918e-003					
20	Accel. Gradient (kPa/m)	6.027e-007	6.027e-007	6.027e-007	6.027e-007					
21	Liq. Reynolds	2.275e+005 *	2.275e+005 *	2.275e+005 *	2.275e+005 *					
22	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *					
23	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051					
24	Vapour Velocity (m/s)	2.063	2.063	2.063	2.063					
25	Deposit Thickness (mm)	---	---	---	---					
26	Deposit Volume (m3)	---	---	---	---					
27	Length (m)	2491	2496	2501	2506					
28	Elevation (m)	0.7400	0.7500	0.7600	0.7380					
29	Pressure (bar)	97.83	97.83	97.83	97.83					
30	Temperature (C)	61.84	61.84	61.84	61.84					
31	Vapour Fraction	0.8716	0.8716	0.8716	0.8716					
32	Heat Transferred (kJ/h-m)	848.4	848.4	848.3	848.3					
33	Flow Regime	Transition	Transition	Transition	Transition					
34	Water Flow Regime									
35	Liquid HoldUp	0.2949	0.2949	0.2949	0.2096					
36	Water HoldUp	---	---	---	---					
37	Friction Gradient (kPa/m)	4.776e-003	4.776e-003	4.776e-003	6.566e-003					
38	Static Gradient (kPa/m)	4.918e-003	4.918e-003	4.918e-003	-8.645e-003					
39	Accel. Gradient (kPa/m)	6.027e-007	6.027e-007	6.027e-007	-1.292e-007					
40	Liq. Reynolds	2.275e+005 *	2.275e+005 *	2.275e+005 *	2.275e+005 *					
41	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *					
42	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051					
43	Vapour Velocity (m/s)	2.063	2.063	2.063	2.062					
44	Deposit Thickness (mm)	---	---	---	---					
45	Deposit Volume (m3)	---	---	---	---					
46	Length (m)	2511	2516	2521	2526					
47	Elevation (m)	0.7160	0.6940	0.6720	0.6500					
48	Pressure (bar)	97.83	97.83	97.83	97.83					
49	Temperature (C)	61.84	61.83	61.83	61.83					
50	Vapour Fraction	0.8716	0.8716	0.8716	0.8716					
51	Heat Transferred (kJ/h-m)	848.2	848.1	848.1	848.0					
52	Flow Regime	Transition	Transition	Transition	Transition					
53	Water Flow Regime									
54	Liquid HoldUp	0.2096	0.2096	0.2096	0.2096					
55	Water HoldUp	---	---	---	---					
56	Friction Gradient (kPa/m)	6.566e-003	6.566e-003	6.566e-003	6.565e-003					
57	Static Gradient (kPa/m)	-8.645e-003	-8.645e-003	-8.645e-003	-8.645e-003					
58	Accel. Gradient (kPa/m)	-1.292e-007	-1.293e-007	-1.293e-007	-1.293e-007					
59	Liq. Reynolds	2.275e+005 *	2.275e+005 *	2.275e+005 *	2.275e+005 *					
60	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *					
61	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051					
62	Vapour Velocity (m/s)	2.062	2.062	2.062	2.062					
63	Deposit Thickness (mm)	---	---	---	---					
64	Deposit Volume (m3)	---	---	---	---					
65	Length (m)	2531	2536	2541	2546					
66	Elevation (m)	0.6140	0.5780	0.5420	0.5060					
67	Pressure (bar)	97.83	97.83	97.83	97.83					
68	Temperature (C)	61.83	61.82	61.82	61.82					
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 69 of 937					

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
14	Heat Transferred (kJ/h-m)	848.0	847.9	847.8	847.8
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2069	0.2069	0.2069	0.2069
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	6.586e-003	6.586e-003	6.586e-003	6.586e-003
20	Static Gradient (kPa/m)	-1.403e-002	-1.403e-002	-1.403e-002	-1.403e-002
21	Accel. Gradient (kPa/m)	-4.630e-007	-4.630e-007	-4.630e-007	-4.630e-007
22	Liq. Reynolds	2.275e+005 *	2.275e+005 *	2.275e+005 *	2.275e+005 *
23	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
24	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
25	Vapour Velocity (m/s)	2.062	2.062	2.062	2.062
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	2551	2556	2561	2566
29	Elevation (m)	0.4700	0.4780	0.4860	0.4940
30	Pressure (bar)	97.83	97.83	97.83	97.83
31	Temperature (C)	61.82	61.81	61.81	61.81
32	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
33	Heat Transferred (kJ/h-m)	847.7	847.6	847.6	847.5
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2069	0.2937	0.2937	0.2937
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	6.586e-003	4.865e-003	4.865e-003	4.865e-003
39	Static Gradient (kPa/m)	-1.403e-002	3.923e-003	3.923e-003	3.923e-003
40	Accel. Gradient (kPa/m)	-4.630e-007	5.462e-007	5.462e-007	5.462e-007
41	Liq. Reynolds	2.275e+005 *	2.275e+005 *	2.275e+005 *	2.275e+005 *
42	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
43	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
44	Vapour Velocity (m/s)	2.062	2.062	2.062	2.062
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	2571	2576	2581	2586
48	Elevation (m)	0.5020	0.5100	0.5180	0.5260
49	Pressure (bar)	97.83	97.83	97.82	97.82
50	Temperature (C)	61.81	61.80	61.80	61.80
51	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
52	Heat Transferred (kJ/h-m)	847.4	847.4	847.3	847.2
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2937	0.2937	0.2937	0.2937
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	4.865e-003	4.865e-003	4.866e-003	4.866e-003
58	Static Gradient (kPa/m)	3.923e-003	3.923e-003	3.909e-003	3.909e-003
59	Accel. Gradient (kPa/m)	5.462e-007	5.462e-007	5.454e-007	5.454e-007
60	Liq. Reynolds	2.275e+005 *	2.275e+005 *	2.275e+005 *	2.274e+005 *
61	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
62	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
63	Vapour Velocity (m/s)	2.062	2.062	2.062	2.062
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	2591	2596	2601	2606
67	Elevation (m)	0.5340	0.5420	0.5500	0.5480
68	Pressure (bar)	97.82	97.82	97.82	97.82
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	61.80	61.79	61.79	61.79
14	Vapour Fraction	0.8716	0.8716	0.8715	0.8715
15	Heat Transferred (kJ/h-m)	847.2	847.1	847.0	847.0
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2937	0.2937	0.2937	0.2135
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	4.866e-003	4.866e-003	4.866e-003	6.536e-003
21	Static Gradient (kPa/m)	3.909e-003	3.909e-003	3.909e-003	-7.950e-004
22	Accel. Gradient (kPa/m)	5.454e-007	5.454e-007	5.454e-007	3.568e-007
23	Liq. Reynolds	2.274e+005 *	2.274e+005 *	2.274e+005 *	2.274e+005 *
24	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
25	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
26	Vapour Velocity (m/s)	2.062	2.062	2.062	2.062
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	2611	2616	2621	2626
30	Elevation (m)	0.5460	0.5440	0.5420	0.5400
31	Pressure (bar)	97.82	97.82	97.82	97.82
32	Temperature (C)	61.79	61.78	61.78	61.78
33	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
34	Heat Transferred (kJ/h-m)	846.9	846.9	846.8	846.7
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2135	0.2135	0.2135	0.2135
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	6.536e-003	6.536e-003	6.536e-003	6.536e-003
40	Static Gradient (kPa/m)	-7.950e-004	-7.950e-004	-7.950e-004	-7.950e-004
41	Accel. Gradient (kPa/m)	3.568e-007	3.568e-007	3.568e-007	3.568e-007
42	Liq. Reynolds	2.274e+005 *	2.274e+005 *	2.274e+005 *	2.274e+005 *
43	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
44	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
45	Vapour Velocity (m/s)	2.062	2.062	2.062	2.062
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	2631	2636	2641	2646
49	Elevation (m)	0.5480	0.5560	0.5640	0.5720
50	Pressure (bar)	97.82	97.82	97.82	97.82
51	Temperature (C)	61.78	61.78	61.77	61.77
52	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
53	Heat Transferred (kJ/h-m)	846.7	846.6	846.5	846.5
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2937	0.2937	0.2937	0.2937
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	4.864e-003	4.864e-003	4.864e-003	4.864e-003
59	Static Gradient (kPa/m)	3.923e-003	3.923e-003	3.923e-003	3.923e-003
60	Accel. Gradient (kPa/m)	5.462e-007	5.462e-007	5.462e-007	5.462e-007
61	Liq. Reynolds	2.274e+005 *	2.274e+005 *	2.274e+005 *	2.274e+005 *
62	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
63	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
64	Vapour Velocity (m/s)	2.062	2.062	2.062	2.062
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	2651	2656	2661	2666
68	Elevation (m)	0.5800	0.5800	0.5800	0.5800
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Pressure (bar)	97.82	97.82	97.82	97.82
14	Temperature (C)	61.77	61.77	61.76	61.76
15	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
16	Heat Transferred (kJ/h-m)	846.4	846.3	846.3	846.2
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.2937	0.3122	0.3122	0.3122
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	4.864e-003	5.655e-003	5.655e-003	5.654e-003
22	Static Gradient (kPa/m)	3.923e-003	0.0000	0.0000	0.0000
23	Accel. Gradient (kPa/m)	5.461e-007	3.514e-007	3.514e-007	3.514e-007
24	Liq. Reynolds	2.274e+005 *	2.274e+005 *	2.274e+005 *	2.274e+005 *
25	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
26	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
27	Vapour Velocity (m/s)	2.062	2.062	2.062	2.062
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	2671	2676	2681	2686
31	Elevation (m)	0.5800	0.5800	0.5840	0.5880
32	Pressure (bar)	97.82	97.82	97.82	97.82
33	Temperature (C)	61.76	61.76	61.75	61.75
34	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
35	Heat Transferred (kJ/h-m)	846.1	846.1	846.0	845.9
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.3122	0.3122	0.2911	0.2911
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	5.654e-003	5.654e-003	5.047e-003	5.047e-003
41	Static Gradient (kPa/m)	0.0000	0.0000	1.950e-003	1.950e-003
42	Accel. Gradient (kPa/m)	3.514e-007	3.514e-007	4.348e-007	4.348e-007
43	Liq. Reynolds	2.274e+005 *	2.274e+005 *	2.274e+005 *	2.274e+005 *
44	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
45	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
46	Vapour Velocity (m/s)	2.062	2.062	2.062	2.062
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	2691	2696	2701	2706
50	Elevation (m)	0.5920	0.5960	0.6000	0.6000
51	Pressure (bar)	97.82	97.82	97.82	97.82
52	Temperature (C)	61.75	61.75	61.74	61.74
53	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
54	Heat Transferred (kJ/h-m)	845.9	845.8	845.7	845.7
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.2911	0.2911	0.2911	0.3122
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	5.046e-003	5.046e-003	5.046e-003	5.654e-003
60	Static Gradient (kPa/m)	1.950e-003	1.950e-003	1.950e-003	0.0000
61	Accel. Gradient (kPa/m)	4.348e-007	4.348e-007	4.348e-007	3.514e-007
62	Liq. Reynolds	2.274e+005 *	2.274e+005 *	2.274e+005 *	2.274e+005 *
63	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
64	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
65	Vapour Velocity (m/s)	2.062	2.062	2.062	2.062
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	2711	2716	2721	2726
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	0.6000	0.6000	0.6000	0.6000
14	Pressure (bar)	97.82	97.82	97.82	97.81
15	Temperature (C)	61.74	61.74	61.73	61.73
16	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
17	Heat Transferred (kJ/h-m)	845.6	845.6	845.5	845.4
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.3122	0.3122	0.3122	0.3122
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	5.654e-003	5.654e-003	5.654e-003	5.654e-003
23	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
24	Accel. Gradient (kPa/m)	3.514e-007	3.514e-007	3.514e-007	3.514e-007
25	Liq. Reynolds	2.274e+005 *	2.274e+005 *	2.274e+005 *	2.274e+005 *
26	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
27	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
28	Vapour Velocity (m/s)	2.062	2.062	2.062	2.062
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	2731	2736	2741	2746
32	Elevation (m)	0.6000	0.6000	0.6000	0.6000
33	Pressure (bar)	97.81	97.81	97.81	97.81
34	Temperature (C)	61.73	61.73	61.72	61.72
35	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
36	Heat Transferred (kJ/h-m)	845.4	845.3	845.2	845.2
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.3122	0.3122	0.3122	0.3122
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	5.654e-003	5.654e-003	5.654e-003	5.654e-003
42	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
43	Accel. Gradient (kPa/m)	3.514e-007	3.514e-007	3.514e-007	3.514e-007
44	Liq. Reynolds	2.274e+005 *	2.274e+005 *	2.274e+005 *	2.274e+005 *
45	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
46	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
47	Vapour Velocity (m/s)	2.062	2.062	2.061	2.061
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	2751	2756	2761	2766
51	Elevation (m)	0.6000	0.6100	0.6200	0.6300
52	Pressure (bar)	97.81	97.81	97.81	97.81
53	Temperature (C)	61.72	61.72	61.71	61.71
54	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
55	Heat Transferred (kJ/h-m)	845.1	845.0	845.0	844.9
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.3122	0.2950	0.2950	0.2950
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	5.654e-003	4.773e-003	4.773e-003	4.773e-003
61	Static Gradient (kPa/m)	0.0000	4.920e-003	4.920e-003	4.920e-003
62	Accel. Gradient (kPa/m)	3.513e-007	6.024e-007	6.024e-007	6.024e-007
63	Liq. Reynolds	2.274e+005 *	2.274e+005 *	2.273e+005 *	2.273e+005 *
64	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
65	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
66	Vapour Velocity (m/s)	2.061	2.061	2.061	2.061
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	2771	2776	2781	2786
14	Elevation (m)	0.6400	0.6500	0.6160	0.5820
15	Pressure (bar)	97.81	97.81	97.81	97.81
16	Temperature (C)	61.71	61.71	61.70	61.70
17	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
18	Heat Transferred (kJ/h-m)	844.8	844.8	844.7	844.7
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.2951	0.2951	0.2072	0.2072
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	4.773e-003	4.773e-003	6.581e-003	6.581e-003
24	Static Gradient (kPa/m)	4.920e-003	4.920e-003	-1.342e-002	-1.342e-002
25	Accel. Gradient (kPa/m)	6.024e-007	6.024e-007	-4.247e-007	-4.248e-007
26	Liq. Reynolds	2.273e+005 *	2.273e+005 *	2.273e+005 *	2.273e+005 *
27	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
28	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2052
29	Vapour Velocity (m/s)	2.061	2.061	2.061	2.061
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	2791	2796	2801	2806
33	Elevation (m)	0.5480	0.5140	0.4800	0.4980
34	Pressure (bar)	97.81	97.81	97.81	97.81
35	Temperature (C)	61.70	61.70	61.70	61.69
36	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
37	Heat Transferred (kJ/h-m)	844.6	844.5	844.5	844.4
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.2072	0.2072	0.2072	0.3003
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	6.581e-003	6.581e-003	6.581e-003	4.426e-003
43	Static Gradient (kPa/m)	-1.342e-002	-1.342e-002	-1.342e-002	8.966e-003
44	Accel. Gradient (kPa/m)	-4.248e-007	-4.248e-007	-4.248e-007	8.321e-007
45	Liq. Reynolds	2.273e+005 *	2.273e+005 *	2.273e+005 *	2.273e+005 *
46	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
47	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
48	Vapour Velocity (m/s)	2.061	2.061	2.061	2.061
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	2811	2816	2821	2826
52	Elevation (m)	0.5160	0.5340	0.5520	0.5700
53	Pressure (bar)	97.81	97.81	97.81	97.81
54	Temperature (C)	61.69	61.69	61.69	61.68
55	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
56	Heat Transferred (kJ/h-m)	844.3	844.3	844.2	844.1
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.3003	0.3003	0.3003	0.3003
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	4.426e-003	4.425e-003	4.425e-003	4.425e-003
62	Static Gradient (kPa/m)	8.966e-003	8.966e-003	8.966e-003	8.966e-003
63	Accel. Gradient (kPa/m)	8.321e-007	8.321e-007	8.321e-007	8.321e-007
64	Liq. Reynolds	2.273e+005 *	2.273e+005 *	2.273e+005 *	2.273e+005 *
65	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
66	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
67	Vapour Velocity (m/s)	2.061	2.061	2.061	2.061
68	Deposit Thickness (mm)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	2831	2836	2841	2846
15	Elevation (m)	0.5860	0.6020	0.6180	0.6340
16	Pressure (bar)	97.81	97.81	97.81	97.81
17	Temperature (C)	61.68	61.68	61.68	61.67
18	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
19	Heat Transferred (kJ/h-m)	844.1	844.0	843.9	843.9
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2990	0.2990	0.2990	0.2990
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	4.510e-003	4.510e-003	4.510e-003	4.510e-003
25	Static Gradient (kPa/m)	7.946e-003	7.946e-003	7.946e-003	7.946e-003
26	Accel. Gradient (kPa/m)	7.740e-007	7.740e-007	7.740e-007	7.740e-007
27	Liq. Reynolds	2.273e+005 *	2.273e+005 *	2.273e+005 *	2.273e+005 *
28	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
29	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
30	Vapour Velocity (m/s)	2.061	2.061	2.061	2.061
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	2851	2856	2861	2866
34	Elevation (m)	0.6500	0.6640	0.6780	0.6920
35	Pressure (bar)	97.81	97.81	97.81	97.80
36	Temperature (C)	61.67	61.67	61.67	61.66
37	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
38	Heat Transferred (kJ/h-m)	843.8	843.7	843.7	843.6
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2990	0.2977	0.2977	0.2977
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	4.510e-003	4.596e-003	4.596e-003	4.596e-003
44	Static Gradient (kPa/m)	7.946e-003	6.932e-003	6.932e-003	6.932e-003
45	Accel. Gradient (kPa/m)	7.740e-007	7.163e-007	7.163e-007	7.163e-007
46	Liq. Reynolds	2.273e+005 *	2.273e+005 *	2.273e+005 *	2.273e+005 *
47	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
48	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
49	Vapour Velocity (m/s)	2.061	2.061	2.061	2.061
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	2871	2876	2881	2886
53	Elevation (m)	0.7060	0.7200	0.6560	0.5920
54	Pressure (bar)	97.80	97.80	97.80	97.81
55	Temperature (C)	61.66	61.66	61.66	61.65
56	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
57	Heat Transferred (kJ/h-m)	843.5	843.5	843.4	843.4
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2977	0.2977	0.2015	0.2015
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	4.596e-003	4.596e-003	6.626e-003	6.626e-003
63	Static Gradient (kPa/m)	6.932e-003	6.932e-003	-2.456e-002	-2.456e-002
64	Accel. Gradient (kPa/m)	7.163e-007	7.163e-007	-1.114e-006	-1.114e-006
65	Liq. Reynolds	2.273e+005 *	2.273e+005 *	2.273e+005 *	2.273e+005 *
66	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
67	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
68	Vapour Velocity (m/s)	2.061	2.061	2.061	2.061
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	2891	2896	2901	2906
16	Elevation (m)	0.5280	0.4640	0.4000	0.3780
17	Pressure (bar)	97.81	97.81	97.81	97.81
18	Temperature (C)	61.65	61.65	61.65	61.65
19	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
20	Heat Transferred (kJ/h-m)	843.3	843.3	843.2	843.1
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2015	0.2015	0.2015	0.2097
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	6.626e-003	6.626e-003	6.626e-003	6.561e-003
26	Static Gradient (kPa/m)	-2.456e-002	-2.456e-002	-2.456e-002	-8.650e-003
27	Accel. Gradient (kPa/m)	-1.114e-006	-1.114e-006	-1.114e-006	-1.298e-007
28	Liq. Reynolds	2.273e+005 *	2.273e+005 *	2.273e+005 *	2.273e+005 *
29	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
30	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
31	Vapour Velocity (m/s)	2.061	2.061	2.061	2.061
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	2911	2916	2921	2926
35	Elevation (m)	0.3560	0.3340	0.3120	0.2900
36	Pressure (bar)	97.81	97.81	97.81	97.81
37	Temperature (C)	61.64	61.64	61.64	61.64
38	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
39	Heat Transferred (kJ/h-m)	843.1	843.0	843.0	842.9
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2097	0.2097	0.2097	0.2097
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	6.561e-003	6.561e-003	6.561e-003	6.561e-003
45	Static Gradient (kPa/m)	-8.650e-003	-8.650e-003	-8.650e-003	-8.650e-003
46	Accel. Gradient (kPa/m)	-1.298e-007	-1.298e-007	-1.298e-007	-1.298e-007
47	Liq. Reynolds	2.273e+005 *	2.273e+005 *	2.273e+005 *	2.273e+005 *
48	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
49	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
50	Vapour Velocity (m/s)	2.061	2.061	2.061	2.061
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	2931	2936	2941	2946
54	Elevation (m)	0.3620	0.4340	0.5060	0.5780
55	Pressure (bar)	97.81	97.80	97.80	97.80
56	Temperature (C)	61.63	61.63	61.63	61.62
57	Vapour Fraction	0.8715	0.8714	0.8714	0.8714
58	Heat Transferred (kJ/h-m)	842.8	842.7	842.7	842.6
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.3357	0.3357	0.3357	0.3357
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	5.788e-003	5.788e-003	5.788e-003	5.788e-003
64	Static Gradient (kPa/m)	3.882e-002	3.882e-002	3.882e-002	3.882e-002
65	Accel. Gradient (kPa/m)	2.771e-006	2.771e-006	2.771e-006	2.771e-006
66	Liq. Reynolds	2.273e+005 *	2.273e+005 *	2.272e+005 *	2.272e+005 *
67	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *
68	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3						
4						
5			Unit Set: Mixed			
6			Date/Time: Mon Jun 13 15:02:44 2022			
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	PROFILES					
11	Pipe Table					
12						
13	Vapour Velocity (m/s)	2.061	2.061	2.061	2.061	
14	Deposit Thickness (mm)	---	---	---	---	
15	Deposit Volume (m3)	---	---	---	---	
16	Length (m)	2951	2956	2961	2966	
17	Elevation (m)	0.6500	0.3780	0.1060	-0.1660	
18	Pressure (bar)	97.80	97.80	97.81	97.81	
19	Temperature (C)	61.62	61.62	61.62	61.62	
20	Vapour Fraction	0.8714	0.8714	0.8714	0.8714	
21	Heat Transferred (kJ/h-m)	842.5	842.6	842.5	842.5	
22	Flow Regime	Transition	Transition	Transition	Transition	
23	Water Flow Regime					
24	Liquid HoldUp	0.3357	0.1615	0.1616	0.1616	
25	Water HoldUp	---	---	---	---	
26	Friction Gradient (kPa/m)	5.788e-003	7.080e-003	7.080e-003	7.080e-003	
27	Static Gradient (kPa/m)	3.882e-002	-9.179e-002	-9.179e-002	-9.180e-002	
28	Accel. Gradient (kPa/m)	2.771e-006	-5.263e-006	-5.262e-006	-5.262e-006	
29	Liq. Reynolds	2.272e+005 *	2.272e+005 *	2.272e+005 *	2.272e+005 *	
30	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *	
31	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052	
32	Vapour Velocity (m/s)	2.061	2.061	2.061	2.060	
33	Deposit Thickness (mm)	---	---	---	---	
34	Deposit Volume (m3)	---	---	---	---	
35	Length (m)	2971	2976	2981	2986	
36	Elevation (m)	-0.4380	-0.7100	-0.6880	-0.6660	
37	Pressure (bar)	97.81	97.82	97.82	97.82	
38	Temperature (C)	61.62	61.62	61.61	61.61	
39	Vapour Fraction	0.8714	0.8714	0.8714	0.8714	
40	Heat Transferred (kJ/h-m)	842.5	842.5	842.3	842.2	
41	Flow Regime	Transition	Transition	Transition	Transition	
42	Water Flow Regime					
43	Liquid HoldUp	0.1616	0.1616	0.3027	0.3027	
44	Water HoldUp	---	---	---	---	
45	Friction Gradient (kPa/m)	7.079e-003	7.079e-003	4.995e-003	4.995e-003	
46	Static Gradient (kPa/m)	-9.180e-002	-9.180e-002	1.078e-002	1.078e-002	
47	Accel. Gradient (kPa/m)	-5.262e-006	-5.261e-006	9.793e-007	9.794e-007	
48	Liq. Reynolds	2.273e+005 *	2.273e+005 *	2.273e+005 *	2.272e+005 *	
49	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.906e+006 *	
50	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052	
51	Vapour Velocity (m/s)	2.060	2.060	2.060	2.060	
52	Deposit Thickness (mm)	---	---	---	---	
53	Deposit Volume (m3)	---	---	---	---	
54	Length (m)	2991	2996	3001	3006	
55	Elevation (m)	-0.6440	-0.6220	-0.6000	-2.080	
56	Pressure (bar)	97.82	97.82	97.81	97.83	
57	Temperature (C)	61.61	61.61	61.60	61.61	
58	Vapour Fraction	0.8714	0.8714	0.8714	0.8714	
59	Heat Transferred (kJ/h-m)	842.2	842.1	842.0	843.5	
60	Flow Regime	Transition	Transition	Transition	Transition	
61	Water Flow Regime					
62	Liquid HoldUp	0.3027	0.3027	0.3027	9.059e-002	
63	Water HoldUp	---	---	---	---	
64	Friction Gradient (kPa/m)	4.995e-003	4.995e-003	4.995e-003	9.348e-003	
65	Static Gradient (kPa/m)	1.078e-002	1.078e-002	1.078e-002	-0.3779	
66	Accel. Gradient (kPa/m)	9.794e-007	9.794e-007	9.794e-007	-2.288e-005	
67	Liq. Reynolds	2.272e+005 *	2.272e+005 *	2.272e+005 *	2.273e+005 *	
68	Vap. Reynolds	8.906e+006 *	8.906e+006 *	8.906e+006 *	8.905e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022		
6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052	
14	Vapour Velocity	(m/s)	2.060	2.060	2.060	2.060	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	3011	3016	3021	3026	
18	Elevation	(m)	-3.560	-5.040	-6.520	-8.000	
19	Pressure	(bar)	97.85	97.87	97.89	97.91	
20	Temperature	(C)	61.62	61.62	61.63	61.63	
21	Vapour Fraction		0.8714	0.8714	0.8714	0.8714	
22	Heat Transferred	(kJ/h-m)	843.6	843.8	843.9	844.1	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		9.061e-002	9.062e-002	9.064e-002	9.066e-002	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	9.346e-003	9.344e-003	9.342e-003	9.340e-003	
28	Static Gradient	(kPa/m)	-0.3779	-0.3780	-0.3781	-0.3781	
29	Accel. Gradient	(kPa/m)	-2.287e-005	-2.287e-005	-2.286e-005	-2.286e-005	
30	Liq. Reynolds		2.273e+005 *	2.273e+005 *	2.273e+005 *	2.274e+005 *	
31	Vap. Reynolds		8.905e+006 *	8.904e+006 *	8.904e+006 *	8.903e+006 *	
32	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052	
33	Vapour Velocity	(m/s)	2.059	2.059	2.059	2.058	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	3031	3036	3041	3046	
37	Elevation	(m)	-8.480	-8.960	-9.440	-9.920	
38	Pressure	(bar)	97.91	97.92	97.93	97.93	
39	Temperature	(C)	61.63	61.63	61.63	61.64	
40	Vapour Fraction		0.8714	0.8714	0.8714	0.8714	
41	Heat Transferred	(kJ/h-m)	843.1	843.1	843.1	843.1	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.1224	0.1224	0.1224	0.1224	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	7.920e-003	7.919e-003	7.919e-003	7.918e-003	
47	Static Gradient	(kPa/m)	-0.1403	-0.1403	-0.1403	-0.1403	
48	Accel. Gradient	(kPa/m)	-8.204e-006	-8.203e-006	-8.203e-006	-8.202e-006	
49	Liq. Reynolds		2.274e+005 *	2.274e+005 *	2.274e+005 *	2.274e+005 *	
50	Vap. Reynolds		8.903e+006 *	8.903e+006 *	8.902e+006 *	8.902e+006 *	
51	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052	
52	Vapour Velocity	(m/s)	2.058	2.058	2.058	2.058	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	3051	3056	3061	3066	
56	Elevation	(m)	-10.40	-10.64	-10.88	-11.12	
57	Pressure	(bar)	97.94	97.94	97.95	97.95	
58	Temperature	(C)	61.64	61.63	61.63	61.63	
59	Vapour Fraction		0.8714	0.8714	0.8714	0.8714	
60	Heat Transferred	(kJ/h-m)	843.1	842.9	842.9	842.9	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.1224	0.1678	0.1678	0.1678	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	7.918e-003	6.982e-003	6.982e-003	6.981e-003	
66	Static Gradient	(kPa/m)	-0.1403	-8.278e-002	-8.278e-002	-8.278e-002	
67	Accel. Gradient	(kPa/m)	-8.201e-006	-4.694e-006	-4.694e-006	-4.694e-006	
68	Liq. Reynolds		2.274e+005 *	2.274e+005 *	2.274e+005 *	2.274e+005 *	
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	8.902e+006 *	8.902e+006 *	8.902e+006 *	8.902e+006 *
14	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
15	Vapour Velocity (m/s)	2.057	2.057	2.057	2.057
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	3071	3076	3081	3086
19	Elevation (m)	-11.36	-11.60	-9.276	-6.952
20	Pressure (bar)	97.95	97.96	97.81	97.65
21	Temperature (C)	61.63	61.63	61.60	61.56
22	Vapour Fraction	0.8714	0.8714	0.8714	0.8715
23	Heat Transferred (kJ/h-m)	842.8	842.8	834.5	833.5
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1678	0.1679	1.000	1.000
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.981e-003	6.981e-003	5.350e-003	5.357e-003
29	Static Gradient (kPa/m)	-8.278e-002	-8.279e-002	3.041	3.042
30	Accel. Gradient (kPa/m)	-4.694e-006	-4.693e-006	1.892e-004	1.899e-004
31	Liq. Reynolds	2.274e+005 *	2.274e+005 *	2.272e+005 *	2.270e+005 *
32	Vap. Reynolds	8.902e+006 *	8.901e+006 *	8.906e+006 *	8.910e+006 *
33	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2051
34	Vapour Velocity (m/s)	2.057	2.057	2.060	2.064
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	3091	3096	3101	3106
38	Elevation (m)	-4.628	-2.304	2.000e-002	0.1200
39	Pressure (bar)	97.50	97.35	97.20	97.19
40	Temperature (C)	61.52	61.49	61.45	61.45
41	Vapour Fraction	0.8716	0.8716	0.8717	0.8717
42	Heat Transferred (kJ/h-m)	832.6	831.6	830.7	837.9
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	1.000	1.000	1.000	0.3526
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	5.365e-003	5.372e-003	5.380e-003	5.814e-003
48	Static Gradient (kPa/m)	3.042	3.043	3.043	5.585e-002
49	Accel. Gradient (kPa/m)	1.905e-004	1.911e-004	1.918e-004	3.879e-006
50	Liq. Reynolds	2.268e+005 *	2.266e+005 *	2.264e+005 *	2.264e+005 *
51	Vap. Reynolds	8.915e+006 *	8.920e+006 *	8.924e+006 *	8.924e+006 *
52	Liquid Velocity (m/s)	0.2051	0.2050	0.2050	0.2050
53	Vapour Velocity (m/s)	2.067	2.070	2.074	2.074
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	3111	3116	3121	3126
57	Elevation (m)	0.2200	0.3200	0.4200	0.5200
58	Pressure (bar)	97.19	97.19	97.18	97.18
59	Temperature (C)	61.44	61.44	61.44	61.43
60	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
61	Heat Transferred (kJ/h-m)	837.8	837.7	837.6	837.6
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.3526	0.3526	0.3526	0.3526
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	5.814e-003	5.815e-003	5.815e-003	5.815e-003
67	Static Gradient (kPa/m)	5.585e-002	5.585e-002	5.585e-002	5.584e-002
68	Accel. Gradient (kPa/m)	3.879e-006	3.880e-006	3.880e-006	3.880e-006
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.264e+005 *	2.264e+005 *	2.264e+005 *	2.264e+005 *
14	Vap. Reynolds	8.924e+006 *	8.924e+006 *	8.924e+006 *	8.925e+006 *
15	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
16	Vapour Velocity (m/s)	2.074	2.074	2.074	2.074
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	3131	3136	3141	3146
20	Elevation (m)	0.4800	0.4400	0.4000	0.3600
21	Pressure (bar)	97.18	97.18	97.18	97.18
22	Temperature (C)	61.43	61.43	61.43	61.43
23	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
24	Heat Transferred (kJ/h-m)	837.5	837.4	837.4	837.3
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2054	0.2054	0.2054	0.2054
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	6.626e-003	6.626e-003	6.626e-003	6.626e-003
30	Static Gradient (kPa/m)	-1.551e-002	-1.551e-002	-1.551e-002	-1.551e-002
31	Accel. Gradient (kPa/m)	-5.587e-007	-5.588e-007	-5.588e-007	-5.588e-007
32	Liq. Reynolds	2.264e+005 *	2.264e+005 *	2.264e+005 *	2.264e+005 *
33	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.924e+006 *	8.924e+006 *
34	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
35	Vapour Velocity (m/s)	2.074	2.074	2.074	2.074
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	3151	3156	3161	3166
39	Elevation (m)	0.3200	0.2980	0.2760	0.2540
40	Pressure (bar)	97.18	97.18	97.18	97.18
41	Temperature (C)	61.42	61.42	61.42	61.42
42	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
43	Heat Transferred (kJ/h-m)	837.2	837.2	837.1	837.1
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2054	0.2088	0.2088	0.2089
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	6.626e-003	6.599e-003	6.599e-003	6.599e-003
49	Static Gradient (kPa/m)	-1.551e-002	-8.617e-003	-8.617e-003	-8.617e-003
50	Accel. Gradient (kPa/m)	-5.588e-007	-1.270e-007	-1.270e-007	-1.270e-007
51	Liq. Reynolds	2.264e+005 *	2.264e+005 *	2.264e+005 *	2.264e+005 *
52	Vap. Reynolds	8.924e+006 *	8.924e+006 *	8.924e+006 *	8.924e+006 *
53	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
54	Vapour Velocity (m/s)	2.074	2.074	2.074	2.074
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	3171	3176	3181	3186
58	Elevation (m)	0.2320	0.2100	0.2640	0.3180
59	Pressure (bar)	97.18	97.18	97.18	97.18
60	Temperature (C)	61.42	61.41	61.41	61.41
61	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
62	Heat Transferred (kJ/h-m)	837.0	836.9	836.9	836.8
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2089	0.2089	0.3214	0.3214
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	6.599e-003	6.598e-003	5.790e-003	5.790e-003
68	Static Gradient (kPa/m)	-8.617e-003	-8.618e-003	2.719e-002	2.719e-002
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	-1.270e-007	-1.270e-007	2.075e-006	2.075e-006
14	Liq. Reynolds	2.263e+005 *	2.263e+005 *	2.263e+005 *	2.263e+005 *
15	Vap. Reynolds	8.924e+006 *	8.924e+006 *	8.924e+006 *	8.924e+006 *
16	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
17	Vapour Velocity (m/s)	2.074	2.074	2.074	2.074
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	3192	3197	3202	3207
21	Elevation (m)	0.3720	0.4260	0.4800	0.4660
22	Pressure (bar)	97.18	97.18	97.18	97.18
23	Temperature (C)	61.40	61.40	61.40	61.40
24	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
25	Heat Transferred (kJ/h-m)	836.7	836.6	836.6	836.5
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.3214	0.3214	0.3214	0.2104
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	5.790e-003	5.790e-003	5.790e-003	6.587e-003
31	Static Gradient (kPa/m)	2.719e-002	2.719e-002	2.719e-002	-5.509e-003
32	Accel. Gradient (kPa/m)	2.075e-006	2.075e-006	2.075e-006	6.783e-008
33	Liq. Reynolds	2.263e+005 *	2.263e+005 *	2.263e+005 *	2.263e+005 *
34	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
35	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
36	Vapour Velocity (m/s)	2.074	2.074	2.074	2.074
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	3212	3217	3222	3227
40	Elevation (m)	0.4520	0.4380	0.4240	0.4100
41	Pressure (bar)	97.18	97.18	97.18	97.18
42	Temperature (C)	61.39	61.39	61.39	61.39
43	Vapour Fraction	0.8717	0.8716	0.8716	0.8716
44	Heat Transferred (kJ/h-m)	836.4	836.4	836.3	836.2
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.2104	0.2104	0.2104	0.2104
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	6.587e-003	6.587e-003	6.587e-003	6.587e-003
50	Static Gradient (kPa/m)	-5.509e-003	-5.509e-003	-5.509e-003	-5.509e-003
51	Accel. Gradient (kPa/m)	6.782e-008	6.781e-008	6.781e-008	6.780e-008
52	Liq. Reynolds	2.263e+005 *	2.263e+005 *	2.263e+005 *	2.263e+005 *
53	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
54	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
55	Vapour Velocity (m/s)	2.074	2.074	2.074	2.074
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	3232	3237	3242	3247
59	Elevation (m)	0.3620	0.3140	0.2660	0.2180
60	Pressure (bar)	97.18	97.18	97.18	97.18
61	Temperature (C)	61.38	61.38	61.38	61.38
62	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
63	Heat Transferred (kJ/h-m)	836.2	836.1	836.1	836.0
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.2039	0.2039	0.2039	0.2039
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	6.638e-003	6.638e-003	6.637e-003	6.637e-003
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1				Case Name:	Simulation - 386MMSCFD_100barg.hsc		
2				Unit Set:	Mixed		
3	 Epic Atlantic Limited			Date/Time:	Mon Jun 13 15:02:44 2022		
4							
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Static Gradient	(kPa/m)	-1.852e-002	-1.852e-002	-1.852e-002	-1.852e-002	
14	Accel. Gradient	(kPa/m)	-7.478e-007	-7.479e-007	-7.479e-007	-7.479e-007	
15	Liq. Reynolds		2.263e+005 *	2.263e+005 *	2.263e+005 *	2.263e+005 *	
16	Vap. Reynolds		8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *	
17	Liquid Velocity	(m/s)	0.2050	0.2050	0.2050	0.2050	
18	Vapour Velocity	(m/s)	2.073	2.073	2.073	2.073	
19	Deposit Thickness	(mm)	---	---	---	---	
20	Deposit Volume	(m3)	---	---	---	---	
21	Length	(m)	3252	3257	3262	3267	
22	Elevation	(m)	0.1700	0.1680	0.1660	0.1640	
23	Pressure	(bar)	97.18	97.18	97.18	97.18	
24	Temperature	(C)	61.38	61.37	61.37	61.37	
25	Vapour Fraction		0.8716	0.8716	0.8716	0.8716	
26	Heat Transferred	(kJ/h-m)	836.0	835.9	835.8	835.8	
27	Flow Regime		Transition	Transition	Transition	Transition	
28	Water Flow Regime						
29	Liquid HoldUp		0.2039	0.2127	0.2127	0.2127	
30	Water HoldUp		---	---	---	---	
31	Friction Gradient	(kPa/m)	6.637e-003	6.569e-003	6.569e-003	6.569e-003	
32	Static Gradient	(kPa/m)	-1.853e-002	-7.924e-004	-7.924e-004	-7.924e-004	
33	Accel. Gradient	(kPa/m)	-7.479e-007	3.634e-007	3.634e-007	3.634e-007	
34	Liq. Reynolds		2.263e+005 *	2.263e+005 *	2.263e+005 *	2.263e+005 *	
35	Vap. Reynolds		8.924e+006 *	8.924e+006 *	8.924e+006 *	8.925e+006 *	
36	Liquid Velocity	(m/s)	0.2050	0.2050	0.2050	0.2050	
37	Vapour Velocity	(m/s)	2.073	2.073	2.073	2.073	
38	Deposit Thickness	(mm)	---	---	---	---	
39	Deposit Volume	(m3)	---	---	---	---	
40	Length	(m)	3272	3277	3282	3287	
41	Elevation	(m)	0.1620	0.1600	0.1580	0.1560	
42	Pressure	(bar)	97.18	97.18	97.18	97.18	
43	Temperature	(C)	61.37	61.36	61.36	61.36	
44	Vapour Fraction		0.8716	0.8716	0.8716	0.8716	
45	Heat Transferred	(kJ/h-m)	835.7	835.6	835.6	835.5	
46	Flow Regime		Transition	Transition	Transition	Transition	
47	Water Flow Regime						
48	Liquid HoldUp		0.2127	0.2127	0.2127	0.2127	
49	Water HoldUp		---	---	---	---	
50	Friction Gradient	(kPa/m)	6.569e-003	6.569e-003	6.569e-003	6.569e-003	
51	Static Gradient	(kPa/m)	-7.924e-004	-7.925e-004	-7.925e-004	-7.925e-004	
52	Accel. Gradient	(kPa/m)	3.634e-007	3.634e-007	3.634e-007	3.634e-007	
53	Liq. Reynolds		2.263e+005 *	2.263e+005 *	2.263e+005 *	2.263e+005 *	
54	Vap. Reynolds		8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *	
55	Liquid Velocity	(m/s)	0.2050	0.2050	0.2050	0.2050	
56	Vapour Velocity	(m/s)	2.073	2.073	2.073	2.073	
57	Deposit Thickness	(mm)	---	---	---	---	
58	Deposit Volume	(m3)	---	---	---	---	
59	Length	(m)	3292	3297	3302	3307	
60	Elevation	(m)	0.1540	0.1520	0.1500	0.1660	
61	Pressure	(bar)	97.18	97.18	97.18	97.17	
62	Temperature	(C)	61.36	61.35	61.35	61.35	
63	Vapour Fraction		0.8716	0.8716	0.8716	0.8716	
64	Heat Transferred	(kJ/h-m)	835.5	835.4	835.3	835.2	
65	Flow Regime		Transition	Transition	Transition	Transition	
66	Water Flow Regime						
67	Liquid HoldUp		0.2127	0.2127	0.2127	0.2979	
68	Water HoldUp		---	---	---	---	
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1				Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
Date/Time: Mon Jun 13 15:02:44 2022						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	6.569e-003	6.569e-003	6.569e-003	4.547e-003
14	Static Gradient	(kPa/m)	-7.925e-004	-7.925e-004	-7.925e-004	7.921e-003
15	Accel. Gradient	(kPa/m)	3.634e-007	3.634e-007	3.634e-007	7.843e-007
16	Liq. Reynolds		2.263e+005 *	2.263e+005 *	2.263e+005 *	2.263e+005 *
17	Vap. Reynolds		8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
18	Liquid Velocity	(m/s)	0.2050	0.2050	0.2050	0.2050
19	Vapour Velocity	(m/s)	2.073	2.073	2.073	2.073
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	3312	3317	3322	3327
23	Elevation	(m)	0.1820	0.1980	0.2140	0.2300
24	Pressure	(bar)	97.17	97.17	97.17	97.17
25	Temperature	(C)	61.35	61.34	61.34	61.34
26	Vapour Fraction		0.8716	0.8716	0.8716	0.8716
27	Heat Transferred	(kJ/h-m)	835.2	835.1	835.0	835.0
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2979	0.2979	0.2979	0.2979
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	4.547e-003	4.547e-003	4.547e-003	4.547e-003
33	Static Gradient	(kPa/m)	7.921e-003	7.921e-003	7.921e-003	7.921e-003
34	Accel. Gradient	(kPa/m)	7.843e-007	7.843e-007	7.843e-007	7.843e-007
35	Liq. Reynolds		2.263e+005 *	2.263e+005 *	2.263e+005 *	2.263e+005 *
36	Vap. Reynolds		8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
37	Liquid Velocity	(m/s)	0.2050	0.2050	0.2050	0.2050
38	Vapour Velocity	(m/s)	2.073	2.073	2.073	2.073
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	3332	3337	3342	3347
42	Elevation	(m)	0.2480	0.2660	0.2840	0.3020
43	Pressure	(bar)	97.17	97.17	97.17	97.17
44	Temperature	(C)	61.34	61.33	61.33	61.33
45	Vapour Fraction		0.8716	0.8716	0.8716	0.8716
46	Heat Transferred	(kJ/h-m)	834.9	834.8	834.8	834.7
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2992	0.2992	0.2992	0.2992
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	4.461e-003	4.461e-003	4.461e-003	4.461e-003
52	Static Gradient	(kPa/m)	8.939e-003	8.939e-003	8.939e-003	8.939e-003
53	Accel. Gradient	(kPa/m)	8.429e-007	8.429e-007	8.430e-007	8.430e-007
54	Liq. Reynolds		2.263e+005 *	2.263e+005 *	2.262e+005 *	2.262e+005 *
55	Vap. Reynolds		8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
56	Liquid Velocity	(m/s)	0.2050	0.2050	0.2050	0.2050
57	Vapour Velocity	(m/s)	2.073	2.073	2.073	2.073
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	3352	3357	3362	3367
61	Elevation	(m)	0.3200	0.3300	0.3400	0.3500
62	Pressure	(bar)	97.17	97.17	97.17	97.17
63	Temperature	(C)	61.33	61.32	61.32	61.32
64	Vapour Fraction		0.8716	0.8716	0.8716	0.8716
65	Heat Transferred	(kJ/h-m)	834.6	834.6	834.5	834.4
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2992	0.2940	0.2940	0.2940
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	4.461e-003	4.811e-003	4.811e-003	4.811e-003
15	Static Gradient (kPa/m)	8.939e-003	4.906e-003	4.906e-003	4.906e-003
16	Accel. Gradient (kPa/m)	8.430e-007	6.112e-007	6.112e-007	6.112e-007
17	Liq. Reynolds	2.262e+005 *	2.262e+005 *	2.262e+005 *	2.262e+005 *
18	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
19	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
20	Vapour Velocity (m/s)	2.073	2.073	2.073	2.073
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	3372	3377	3382	3387
24	Elevation (m)	0.3600	0.3700	0.3740	0.3780
25	Pressure (bar)	97.17	97.17	97.17	97.17
26	Temperature (C)	61.32	61.31	61.31	61.31
27	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
28	Heat Transferred (kJ/h-m)	834.4	834.3	834.2	834.2
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2940	0.2940	0.2902	0.2902
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	4.811e-003	4.810e-003	5.079e-003	5.079e-003
34	Static Gradient (kPa/m)	4.906e-003	4.906e-003	2.007e-003	2.007e-003
35	Accel. Gradient (kPa/m)	6.112e-007	6.112e-007	4.457e-007	4.457e-007
36	Liq. Reynolds	2.262e+005 *	2.262e+005 *	2.262e+005 *	2.262e+005 *
37	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
38	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
39	Vapour Velocity (m/s)	2.073	2.073	2.073	2.073
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	3392	3396	3401	3406
43	Elevation (m)	0.3820	0.3860	0.3900	0.3920
44	Pressure (bar)	97.17	97.17	97.16	97.16
45	Temperature (C)	61.31	61.31	61.30	61.30
46	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
47	Heat Transferred (kJ/h-m)	834.1	834.1	834.0	833.9
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2902	0.2902	0.2902	0.2888
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	5.079e-003	5.079e-003	5.079e-003	5.179e-003
53	Static Gradient (kPa/m)	2.007e-003	2.007e-003	2.007e-003	9.691e-004
54	Accel. Gradient (kPa/m)	4.457e-007	4.457e-007	4.457e-007	3.867e-007
55	Liq. Reynolds	2.262e+005 *	2.262e+005 *	2.262e+005 *	2.262e+005 *
56	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
57	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
58	Vapour Velocity (m/s)	2.073	2.073	2.073	2.073
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	3411	3416	3421	3426
62	Elevation (m)	0.3940	0.3960	0.3980	0.4000
63	Pressure (bar)	97.16	97.16	97.16	97.16
64	Temperature (C)	61.30	61.30	61.29	61.29
65	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
66	Heat Transferred (kJ/h-m)	833.9	833.8	833.7	833.7
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 84 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.2888	0.2888	0.2888	0.2888
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	5.179e-003	5.179e-003	5.178e-003	5.178e-003
16	Static Gradient (kPa/m)	9.691e-004	9.691e-004	9.691e-004	9.691e-004
17	Accel. Gradient (kPa/m)	3.867e-007	3.867e-007	3.867e-007	3.867e-007
18	Liq. Reynolds	2.262e+005 *	2.262e+005 *	2.262e+005 *	2.262e+005 *
19	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
20	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
21	Vapour Velocity (m/s)	2.073	2.073	2.073	2.073
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	3431	3436	3441	3446
25	Elevation (m)	0.3960	0.3920	0.3880	0.3840
26	Pressure (bar)	97.16	97.16	97.16	97.16
27	Temperature (C)	61.29	61.29	61.28	61.28
28	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
29	Heat Transferred (kJ/h-m)	833.6	833.6	833.5	833.4
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2124	0.2124	0.2124	0.2124
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.570e-003	6.570e-003	6.570e-003	6.570e-003
35	Static Gradient (kPa/m)	-1.583e-003	-1.583e-003	-1.583e-003	-1.583e-003
36	Accel. Gradient (kPa/m)	3.137e-007	3.137e-007	3.137e-007	3.137e-007
37	Liq. Reynolds	2.262e+005 *	2.262e+005 *	2.262e+005 *	2.262e+005 *
38	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
39	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
40	Vapour Velocity (m/s)	2.073	2.073	2.073	2.073
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	3451	3456	3461	3466
44	Elevation (m)	0.3800	0.3800	0.3800	0.3800
45	Pressure (bar)	97.16	97.16	97.16	97.16
46	Temperature (C)	61.28	61.28	61.27	61.27
47	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
48	Heat Transferred (kJ/h-m)	833.4	833.3	833.2	833.2
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2124	0.3112	0.3112	0.3112
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.570e-003	5.677e-003	5.677e-003	5.677e-003
54	Static Gradient (kPa/m)	-1.583e-003	0.0000	0.0000	0.0000
55	Accel. Gradient (kPa/m)	3.137e-007	3.571e-007	3.571e-007	3.571e-007
56	Liq. Reynolds	2.262e+005 *	2.262e+005 *	2.262e+005 *	2.262e+005 *
57	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
58	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
59	Vapour Velocity (m/s)	2.073	2.073	2.073	2.073
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	3471	3476	3481	3486
63	Elevation (m)	0.3800	0.3800	0.3680	0.3560
64	Pressure (bar)	97.16	97.16	97.16	97.16
65	Temperature (C)	61.27	61.27	61.26	61.26
66	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
67	Heat Transferred (kJ/h-m)	833.1	833.1	833.0	832.9
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 85 of 937

1			Case Name:	Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set:	Mixed		
3	 Epic Atlantic Limited		Date/Time:	Mon Jun 13 15:02:44 2022		
4						
5						
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Water Flow Regime					
14	Liquid HoldUp		0.3112	0.3112	0.2109	0.2109
15	Water HoldUp		---	---	---	---
16	Friction Gradient (kPa/m)		5.677e-003	5.677e-003	6.581e-003	6.581e-003
17	Static Gradient (kPa/m)		0.0000	0.0000	-4.729e-003	-4.729e-003
18	Accel. Gradient (kPa/m)		3.571e-007	3.571e-007	1.165e-007	1.165e-007
19	Liq. Reynolds		2.262e+005 *	2.262e+005 *	2.262e+005 *	2.262e+005 *
20	Vap. Reynolds		8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
21	Liquid Velocity (m/s)		0.2050	0.2050	0.2050	0.2050
22	Vapour Velocity (m/s)		2.073	2.073	2.072	2.072
23	Deposit Thickness (mm)		---	---	---	---
24	Deposit Volume (m3)		---	---	---	---
25	Length (m)		3491	3496	3501	3506
26	Elevation (m)		0.3440	0.3320	0.3200	0.3200
27	Pressure (bar)		97.16	97.16	97.16	97.16
28	Temperature (C)		61.26	61.26	61.26	61.25
29	Vapour Fraction		0.8716	0.8716	0.8716	0.8716
30	Heat Transferred (kJ/h-m)		832.9	832.8	832.7	832.7
31	Flow Regime		Transition	Transition	Transition	Transition
32	Water Flow Regime					
33	Liquid HoldUp		0.2109	0.2109	0.2109	0.3112
34	Water HoldUp		---	---	---	---
35	Friction Gradient (kPa/m)		6.581e-003	6.581e-003	6.581e-003	5.677e-003
36	Static Gradient (kPa/m)		-4.729e-003	-4.729e-003	-4.729e-003	0.0000
37	Accel. Gradient (kPa/m)		1.165e-007	1.165e-007	1.165e-007	3.570e-007
38	Liq. Reynolds		2.262e+005 *	2.262e+005 *	2.262e+005 *	2.262e+005 *
39	Vap. Reynolds		8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
40	Liquid Velocity (m/s)		0.2050	0.2050	0.2050	0.2050
41	Vapour Velocity (m/s)		2.072	2.072	2.072	2.072
42	Deposit Thickness (mm)		---	---	---	---
43	Deposit Volume (m3)		---	---	---	---
44	Length (m)		3511	3516	3521	3526
45	Elevation (m)		0.3200	0.3200	0.3200	0.3200
46	Pressure (bar)		97.16	97.16	97.16	97.16
47	Temperature (C)		61.25	61.25	61.25	61.24
48	Vapour Fraction		0.8716	0.8716	0.8716	0.8716
49	Heat Transferred (kJ/h-m)		832.6	832.6	832.5	832.4
50	Flow Regime		Transition	Transition	Transition	Transition
51	Water Flow Regime					
52	Liquid HoldUp		0.3112	0.3112	0.3112	0.3112
53	Water HoldUp		---	---	---	---
54	Friction Gradient (kPa/m)		5.677e-003	5.677e-003	5.676e-003	5.676e-003
55	Static Gradient (kPa/m)		0.0000	0.0000	0.0000	0.0000
56	Accel. Gradient (kPa/m)		3.570e-007	3.570e-007	3.570e-007	3.570e-007
57	Liq. Reynolds		2.262e+005 *	2.262e+005 *	2.261e+005 *	2.261e+005 *
58	Vap. Reynolds		8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
59	Liquid Velocity (m/s)		0.2050	0.2050	0.2050	0.2050
60	Vapour Velocity (m/s)		2.072	2.072	2.072	2.072
61	Deposit Thickness (mm)		---	---	---	---
62	Deposit Volume (m3)		---	---	---	---
63	Length (m)		3531	3536	3541	3546
64	Elevation (m)		0.3120	0.3040	0.2960	0.2880
65	Pressure (bar)		97.16	97.16	97.16	97.16
66	Temperature (C)		61.24	61.24	61.24	61.23
67	Vapour Fraction		0.8716	0.8716	0.8716	0.8716
68	Heat Transferred (kJ/h-m)		832.4	832.3	832.2	832.2
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2116	0.2116	0.2116	0.2116
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	6.575e-003	6.575e-003	6.575e-003	6.575e-003
18	Static Gradient (kPa/m)	-3.160e-003	-3.160e-003	-3.160e-003	-3.160e-003
19	Accel. Gradient (kPa/m)	2.148e-007	2.147e-007	2.147e-007	2.147e-007
20	Liq. Reynolds	2.261e+005 *	2.261e+005 *	2.261e+005 *	2.261e+005 *
21	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
22	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
23	Vapour Velocity (m/s)	2.072	2.072	2.072	2.072
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	3551	3556	3561	3566
27	Elevation (m)	0.2800	0.2500	0.2200	0.1900
28	Pressure (bar)	97.16	97.16	97.16	97.16
29	Temperature (C)	61.23	61.23	61.23	61.23
30	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
31	Heat Transferred (kJ/h-m)	832.1	832.1	832.0	831.9
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2116	0.2074	0.2074	0.2074
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	6.575e-003	6.607e-003	6.606e-003	6.606e-003
37	Static Gradient (kPa/m)	-3.160e-003	-1.170e-002	-1.170e-002	-1.170e-002
38	Accel. Gradient (kPa/m)	2.147e-007	-3.205e-007	-3.206e-007	-3.206e-007
39	Liq. Reynolds	2.261e+005 *	2.261e+005 *	2.261e+005 *	2.261e+005 *
40	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
41	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
42	Vapour Velocity (m/s)	2.072	2.072	2.072	2.072
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	3571	3576	3581	3586
46	Elevation (m)	0.1600	0.1300	0.1360	0.1420
47	Pressure (bar)	97.16	97.16	97.16	97.16
48	Temperature (C)	61.22	61.22	61.22	61.22
49	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
50	Heat Transferred (kJ/h-m)	831.9	831.8	831.7	831.7
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2074	0.2074	0.2915	0.2915
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	6.606e-003	6.606e-003	4.991e-003	4.991e-003
56	Static Gradient (kPa/m)	-1.170e-002	-1.170e-002	2.913e-003	2.913e-003
57	Accel. Gradient (kPa/m)	-3.206e-007	-3.206e-007	4.970e-007	4.970e-007
58	Liq. Reynolds	2.261e+005 *	2.261e+005 *	2.261e+005 *	2.261e+005 *
59	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
60	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
61	Vapour Velocity (m/s)	2.072	2.072	2.072	2.072
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	3591	3596	3601	3606
65	Elevation (m)	0.1480	0.1540	0.1600	0.1660
66	Pressure (bar)	97.16	97.16	97.16	97.16
67	Temperature (C)	61.21	61.21	61.21	61.21
68	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
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5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	831.6	831.5	831.5	831.4
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.2915	0.2915	0.2915	0.2915
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	4.991e-003	4.991e-003	4.991e-003	4.989e-003
19	Static Gradient (kPa/m)	2.913e-003	2.913e-003	2.913e-003	2.926e-003
20	Accel. Gradient (kPa/m)	4.970e-007	4.970e-007	4.970e-007	4.977e-007
21	Liq. Reynolds	2.261e+005 *	2.261e+005 *	2.261e+005 *	2.261e+005 *
22	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
23	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
24	Vapour Velocity (m/s)	2.072	2.072	2.072	2.072
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	3611	3616	3621	3626
28	Elevation (m)	0.1720	0.1780	0.1840	0.1900
29	Pressure (bar)	97.16	97.16	97.16	97.16
30	Temperature (C)	61.20	61.20	61.20	61.20
31	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
32	Heat Transferred (kJ/h-m)	831.4	831.3	831.2	831.2
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.2915	0.2915	0.2915	0.2915
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	4.989e-003	4.989e-003	4.989e-003	4.989e-003
38	Static Gradient (kPa/m)	2.926e-003	2.926e-003	2.926e-003	2.926e-003
39	Accel. Gradient (kPa/m)	4.977e-007	4.977e-007	4.977e-007	4.977e-007
40	Liq. Reynolds	2.261e+005 *	2.261e+005 *	2.261e+005 *	2.261e+005 *
41	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
42	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
43	Vapour Velocity (m/s)	2.072	2.072	2.072	2.072
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	3631	3636	3641	3646
47	Elevation (m)	0.2140	0.2380	0.2620	0.2860
48	Pressure (bar)	97.15	97.15	97.15	97.15
49	Temperature (C)	61.19	61.19	61.19	61.19
50	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
51	Heat Transferred (kJ/h-m)	831.1	831.0	831.0	830.9
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.3033	0.3033	0.3033	0.3033
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	5.256e-003	5.256e-003	5.256e-003	5.256e-003
57	Static Gradient (kPa/m)	1.203e-002	1.203e-002	1.203e-002	1.203e-002
58	Accel. Gradient (kPa/m)	1.087e-006	1.087e-006	1.087e-006	1.087e-006
59	Liq. Reynolds	2.261e+005 *	2.261e+005 *	2.261e+005 *	2.261e+005 *
60	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
61	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
62	Vapour Velocity (m/s)	2.072	2.072	2.072	2.072
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	3651	3656	3661	3666
66	Elevation (m)	0.3100	0.2960	0.2820	0.2680
67	Pressure (bar)	97.15	97.15	97.15	97.15
68	Temperature (C)	61.18	61.18	61.18	61.18
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2			Unit Set: Mixed		
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13	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
14	Heat Transferred (kJ/h-m)	830.8	830.8	830.7	830.6
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.3033	0.2105	0.2105	0.2105
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	5.256e-003	6.582e-003	6.582e-003	6.582e-003
20	Static Gradient (kPa/m)	1.203e-002	-5.513e-003	-5.513e-003	-5.513e-003
21	Accel. Gradient (kPa/m)	1.087e-006	6.726e-008	6.725e-008	6.724e-008
22	Liq. Reynolds	2.261e+005 *	2.261e+005 *	2.261e+005 *	2.261e+005 *
23	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
24	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
25	Vapour Velocity (m/s)	2.072	2.072	2.072	2.072
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	3671	3676	3681	3686
29	Elevation (m)	0.2540	0.2400	0.2180	0.1960
30	Pressure (bar)	97.15	97.15	97.15	97.15
31	Temperature (C)	61.17	61.17	61.17	61.17
32	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
33	Heat Transferred (kJ/h-m)	830.6	830.5	830.5	830.4
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2105	0.2105	0.2090	0.2090
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	6.582e-003	6.582e-003	6.593e-003	6.593e-003
39	Static Gradient (kPa/m)	-5.513e-003	-5.513e-003	-8.624e-003	-8.624e-003
40	Accel. Gradient (kPa/m)	6.724e-008	6.723e-008	-1.277e-007	-1.277e-007
41	Liq. Reynolds	2.261e+005 *	2.261e+005 *	2.261e+005 *	2.261e+005 *
42	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
43	Liquid Velocity (m/s)	0.2050	0.2051	0.2051	0.2051
44	Vapour Velocity (m/s)	2.072	2.072	2.072	2.072
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	3691	3696	3701	3706
48	Elevation (m)	0.1740	0.1520	0.1300	0.1360
49	Pressure (bar)	97.15	97.15	97.15	97.15
50	Temperature (C)	61.17	61.16	61.16	61.16
51	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
52	Heat Transferred (kJ/h-m)	830.3	830.3	830.2	830.1
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2090	0.2090	0.2090	0.2915
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	6.593e-003	6.593e-003	6.593e-003	4.988e-003
58	Static Gradient (kPa/m)	-8.624e-003	-8.624e-003	-8.624e-003	2.927e-003
59	Accel. Gradient (kPa/m)	-1.277e-007	-1.277e-007	-1.277e-007	4.976e-007
60	Liq. Reynolds	2.261e+005 *	2.261e+005 *	2.261e+005 *	2.261e+005 *
61	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
62	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
63	Vapour Velocity (m/s)	2.072	2.072	2.072	2.072
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	3711	3716	3721	3726
67	Elevation (m)	0.1420	0.1480	0.1540	0.1600
68	Pressure (bar)	97.15	97.15	97.15	97.15
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
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3					Unit Set: Mixed
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
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11	Pipe Table				
12					
13	Temperature (C)	61.16	61.15	61.15	61.15
14	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
15	Heat Transferred (kJ/h-m)	830.1	830.0	829.9	829.9
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2915	0.2915	0.2915	0.2915
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	4.988e-003	4.988e-003	4.988e-003	4.988e-003
21	Static Gradient (kPa/m)	2.927e-003	2.927e-003	2.927e-003	2.927e-003
22	Accel. Gradient (kPa/m)	4.976e-007	4.976e-007	4.976e-007	4.976e-007
23	Liq. Reynolds	2.260e+005 *	2.260e+005 *	2.260e+005 *	2.260e+005 *
24	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
25	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
26	Vapour Velocity (m/s)	2.072	2.071	2.071	2.071
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	3731	3736	3741	3746
30	Elevation (m)	0.1680	0.1760	0.1840	0.1920
31	Pressure (bar)	97.15	97.15	97.15	97.15
32	Temperature (C)	61.15	61.14	61.14	61.14
33	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
34	Heat Transferred (kJ/h-m)	829.8	829.8	829.7	829.6
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2929	0.2929	0.2929	0.2929
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	4.896e-003	4.896e-003	4.896e-003	4.896e-003
40	Static Gradient (kPa/m)	3.915e-003	3.915e-003	3.915e-003	3.915e-003
41	Accel. Gradient (kPa/m)	5.540e-007	5.540e-007	5.540e-007	5.540e-007
42	Liq. Reynolds	2.260e+005 *	2.260e+005 *	2.260e+005 *	2.260e+005 *
43	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
44	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
45	Vapour Velocity (m/s)	2.071	2.071	2.071	2.071
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	3751	3756	3761	3766
49	Elevation (m)	0.2000	0.1840	0.1680	0.1520
50	Pressure (bar)	97.15	97.15	97.15	97.15
51	Temperature (C)	61.14	61.13	61.13	61.13
52	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
53	Heat Transferred (kJ/h-m)	829.6	829.5	829.5	829.4
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2929	0.2102	0.2102	0.2102
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	4.896e-003	6.584e-003	6.584e-003	6.584e-003
59	Static Gradient (kPa/m)	3.915e-003	-6.294e-003	-6.294e-003	-6.294e-003
60	Accel. Gradient (kPa/m)	5.539e-007	1.823e-008	1.823e-008	1.822e-008
61	Liq. Reynolds	2.260e+005 *	2.260e+005 *	2.260e+005 *	2.260e+005 *
62	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
63	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
64	Vapour Velocity (m/s)	2.071	2.071	2.071	2.071
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	3771	3776	3781	3787
68	Elevation (m)	0.1360	0.1200	0.1100	1.000e-001
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
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4					
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
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13	Pressure (bar)	97.15	97.15	97.15	97.15
14	Temperature (C)	61.13	61.12	61.12	61.12
15	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
16	Heat Transferred (kJ/h-m)	829.3	829.3	829.2	829.1
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.2102	0.2102	0.2114	0.2114
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	6.584e-003	6.584e-003	6.575e-003	6.575e-003
22	Static Gradient (kPa/m)	-6.294e-003	-6.294e-003	-3.834e-003	-3.834e-003
23	Accel. Gradient (kPa/m)	1.821e-008	1.820e-008	1.723e-007	1.723e-007
24	Liq. Reynolds	2.260e+005 *	2.260e+005 *	2.260e+005 *	2.260e+005 *
25	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
26	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
27	Vapour Velocity (m/s)	2.071	2.071	2.071	2.071
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	3792	3797	3802	3807
31	Elevation (m)	9.000e-002	8.000e-002	7.000e-002	6.800e-002
32	Pressure (bar)	97.15	97.15	97.15	97.15
33	Temperature (C)	61.12	61.12	61.11	61.11
34	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
35	Heat Transferred (kJ/h-m)	829.1	829.0	828.9	828.9
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.2114	0.2114	0.2114	0.2129
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	6.574e-003	6.574e-003	6.574e-003	6.564e-003
41	Static Gradient (kPa/m)	-3.834e-003	-3.835e-003	-3.835e-003	-7.931e-004
42	Accel. Gradient (kPa/m)	1.723e-007	1.722e-007	1.722e-007	3.628e-007
43	Liq. Reynolds	2.260e+005 *	2.260e+005 *	2.260e+005 *	2.260e+005 *
44	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
45	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
46	Vapour Velocity (m/s)	2.071	2.071	2.071	2.071
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	3812	3817	3822	3827
50	Elevation (m)	6.600e-002	6.400e-002	6.200e-002	6.000e-002
51	Pressure (bar)	97.15	97.15	97.14	97.14
52	Temperature (C)	61.11	61.11	61.10	61.10
53	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
54	Heat Transferred (kJ/h-m)	828.8	828.8	828.7	828.6
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.2129	0.2129	0.2129	0.2129
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	6.563e-003	6.563e-003	6.563e-003	6.563e-003
60	Static Gradient (kPa/m)	-7.931e-004	-7.931e-004	-7.931e-004	-7.931e-004
61	Accel. Gradient (kPa/m)	3.627e-007	3.627e-007	3.627e-007	3.627e-007
62	Liq. Reynolds	2.260e+005 *	2.260e+005 *	2.260e+005 *	2.260e+005 *
63	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
64	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
65	Vapour Velocity (m/s)	2.071	2.071	2.071	2.071
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	3832	3837	3842	3847
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11	Pipe Table				
12					
13	Elevation (m)	8.000e-002	1.000e-001	0.1200	0.1400
14	Pressure (bar)	97.14	97.14	97.14	97.14
15	Temperature (C)	61.10	61.10	61.09	61.09
16	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
17	Heat Transferred (kJ/h-m)	828.6	828.5	828.4	828.4
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.3007	0.3007	0.3007	0.3007
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	4.674e-003	4.675e-003	4.675e-003	4.675e-003
23	Static Gradient (kPa/m)	9.970e-003	9.970e-003	9.970e-003	9.970e-003
24	Accel. Gradient (kPa/m)	9.206e-007	9.206e-007	9.206e-007	9.206e-007
25	Liq. Reynolds	2.260e+005 *	2.260e+005 *	2.260e+005 *	2.260e+005 *
26	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
27	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
28	Vapour Velocity (m/s)	2.071	2.071	2.071	2.071
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	3852	3857	3862	3867
32	Elevation (m)	0.1600	0.1660	0.1720	0.1780
33	Pressure (bar)	97.14	97.14	97.14	97.14
34	Temperature (C)	61.09	61.09	61.08	61.08
35	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
36	Heat Transferred (kJ/h-m)	828.3	828.2	828.2	828.1
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.3007	0.2916	0.2916	0.2916
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	4.675e-003	4.986e-003	4.986e-003	4.986e-003
42	Static Gradient (kPa/m)	9.970e-003	2.927e-003	2.927e-003	2.927e-003
43	Accel. Gradient (kPa/m)	9.206e-007	4.975e-007	4.975e-007	4.975e-007
44	Liq. Reynolds	2.260e+005 *	2.260e+005 *	2.260e+005 *	2.260e+005 *
45	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
46	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
47	Vapour Velocity (m/s)	2.071	2.071	2.071	2.071
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	3872	3877	3882	3887
51	Elevation (m)	0.1840	0.1900	0.2040	0.2180
52	Pressure (bar)	97.14	97.14	97.14	97.14
53	Temperature (C)	61.08	61.08	61.07	61.07
54	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
55	Heat Transferred (kJ/h-m)	828.0	828.0	827.9	827.8
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2916	0.2916	0.2968	0.2968
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	4.986e-003	4.986e-003	4.628e-003	4.628e-003
61	Static Gradient (kPa/m)	2.927e-003	2.927e-003	6.916e-003	6.916e-003
62	Accel. Gradient (kPa/m)	4.975e-007	4.975e-007	7.256e-007	7.256e-007
63	Liq. Reynolds	2.260e+005 *	2.260e+005 *	2.260e+005 *	2.259e+005 *
64	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
65	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
66	Vapour Velocity (m/s)	2.071	2.071	2.071	2.071
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022	
6						
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Length (m)	3892	3897	3902	3907	
14	Elevation (m)	0.2320	0.2460	0.2600	0.2680	
15	Pressure (bar)	97.14	97.14	97.14	97.14	
16	Temperature (C)	61.07	61.07	61.06	61.06	
17	Vapour Fraction	0.8714	0.8714	0.8714	0.8714	
18	Heat Transferred (kJ/h-m)	827.8	827.7	827.6	827.6	
19	Flow Regime	Transition	Transition	Transition	Transition	
20	Water Flow Regime					
21	Liquid HoldUp	0.2968	0.2968	0.2968	0.2929	
22	Water HoldUp	---	---	---	---	
23	Friction Gradient (kPa/m)	4.628e-003	4.628e-003	4.628e-003	4.895e-003	
24	Static Gradient (kPa/m)	6.916e-003	6.916e-003	6.916e-003	3.915e-003	
25	Accel. Gradient (kPa/m)	7.256e-007	7.256e-007	7.256e-007	5.538e-007	
26	Liq. Reynolds	2.259e+005 *	2.259e+005 *	2.259e+005 *	2.259e+005 *	
27	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *	
28	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051	
29	Vapour Velocity (m/s)	2.071	2.071	2.071	2.071	
30	Deposit Thickness (mm)	---	---	---	---	
31	Deposit Volume (m3)	---	---	---	---	
32	Length (m)	3912	3917	3922	3927	
33	Elevation (m)	0.2760	0.2840	0.2920	0.3000	
34	Pressure (bar)	97.14	97.13	97.13	97.13	
35	Temperature (C)	61.06	61.06	61.05	61.05	
36	Vapour Fraction	0.8714	0.8714	0.8714	0.8714	
37	Heat Transferred (kJ/h-m)	827.5	827.4	827.4	827.3	
38	Flow Regime	Transition	Transition	Transition	Transition	
39	Water Flow Regime					
40	Liquid HoldUp	0.2929	0.2929	0.2929	0.2929	
41	Water HoldUp	---	---	---	---	
42	Friction Gradient (kPa/m)	4.895e-003	4.894e-003	4.894e-003	4.894e-003	
43	Static Gradient (kPa/m)	3.916e-003	3.916e-003	3.916e-003	3.916e-003	
44	Accel. Gradient (kPa/m)	5.538e-007	5.538e-007	5.538e-007	5.538e-007	
45	Liq. Reynolds	2.259e+005 *	2.259e+005 *	2.259e+005 *	2.259e+005 *	
46	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *	
47	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051	
48	Vapour Velocity (m/s)	2.071	2.071	2.071	2.071	
49	Deposit Thickness (mm)	---	---	---	---	
50	Deposit Volume (m3)	---	---	---	---	
51	Length (m)	3932	3937	3942	3947	
52	Elevation (m)	0.3120	0.3240	0.3360	0.3480	
53	Pressure (bar)	97.13	97.13	97.13	97.13	
54	Temperature (C)	61.05	61.05	61.04	61.04	
55	Vapour Fraction	0.8714	0.8714	0.8714	0.8714	
56	Heat Transferred (kJ/h-m)	827.2	827.2	827.1	827.0	
57	Flow Regime	Transition	Transition	Transition	Transition	
58	Water Flow Regime					
59	Liquid HoldUp	0.2955	0.2955	0.2955	0.2955	
60	Water HoldUp	---	---	---	---	
61	Friction Gradient (kPa/m)	4.715e-003	4.715e-003	4.715e-003	4.715e-003	
62	Static Gradient (kPa/m)	5.910e-003	5.910e-003	5.910e-003	5.910e-003	
63	Accel. Gradient (kPa/m)	6.679e-007	6.679e-007	6.679e-007	6.679e-007	
64	Liq. Reynolds	2.259e+005 *	2.259e+005 *	2.259e+005 *	2.259e+005 *	
65	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *	
66	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051	
67	Vapour Velocity (m/s)	2.071	2.071	2.071	2.071	
68	Deposit Thickness (mm)	---	---	---	---	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	3952	3957	3962	3967
15	Elevation (m)	0.3600	0.3640	0.3680	0.3720
16	Pressure (bar)	97.13	97.13	97.13	97.13
17	Temperature (C)	61.04	61.04	61.03	61.03
18	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
19	Heat Transferred (kJ/h-m)	827.0	826.9	826.8	826.8
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2955	0.2903	0.2903	0.2903
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	4.715e-003	5.078e-003	5.078e-003	5.078e-003
25	Static Gradient (kPa/m)	5.910e-003	1.946e-003	1.946e-003	1.946e-003
26	Accel. Gradient (kPa/m)	6.679e-007	4.415e-007	4.415e-007	4.415e-007
27	Liq. Reynolds	2.259e+005 *	2.259e+005 *	2.259e+005 *	2.259e+005 *
28	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
29	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
30	Vapour Velocity (m/s)	2.071	2.071	2.071	2.071
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	3972	3977	3982	3987
34	Elevation (m)	0.3760	0.3800	0.3800	0.3800
35	Pressure (bar)	97.13	97.13	97.13	97.13
36	Temperature (C)	61.03	61.03	61.02	61.02
37	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
38	Heat Transferred (kJ/h-m)	826.7	826.7	826.6	826.5
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2903	0.2903	0.3114	0.3114
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	5.078e-003	5.078e-003	5.673e-003	5.673e-003
44	Static Gradient (kPa/m)	1.946e-003	1.946e-003	0.0000	0.0000
45	Accel. Gradient (kPa/m)	4.415e-007	4.415e-007	3.566e-007	3.566e-007
46	Liq. Reynolds	2.259e+005 *	2.259e+005 *	2.259e+005 *	2.259e+005 *
47	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
48	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
49	Vapour Velocity (m/s)	2.071	2.071	2.071	2.071
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	3992	3997	4002	4007
53	Elevation (m)	0.3800	0.3800	0.3800	0.3600
54	Pressure (bar)	97.13	97.13	97.13	97.13
55	Temperature (C)	61.02	61.02	61.02	61.01
56	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
57	Heat Transferred (kJ/h-m)	826.5	826.4	826.3	826.3
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.3114	0.3114	0.3114	0.2094
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	5.673e-003	5.673e-003	5.673e-003	6.587e-003
63	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-7.852e-003
64	Accel. Gradient (kPa/m)	3.566e-007	3.566e-007	3.566e-007	-7.950e-008
65	Liq. Reynolds	2.259e+005 *	2.259e+005 *	2.259e+005 *	2.259e+005 *
66	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
67	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
68	Vapour Velocity (m/s)	2.071	2.070	2.070	2.070
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	4012	4017	4022	4027
16	Elevation (m)	0.3400	0.3200	0.3000	0.2800
17	Pressure (bar)	97.13	97.13	97.13	97.13
18	Temperature (C)	61.01	61.01	61.01	61.00
19	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
20	Heat Transferred (kJ/h-m)	826.2	826.2	826.1	826.0
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2095	0.2095	0.2095	0.2095
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	6.587e-003	6.587e-003	6.587e-003	6.587e-003
26	Static Gradient (kPa/m)	-7.852e-003	-7.852e-003	-7.852e-003	-7.852e-003
27	Accel. Gradient (kPa/m)	-7.950e-008	-7.951e-008	-7.952e-008	-7.953e-008
28	Liq. Reynolds	2.259e+005 *	2.259e+005 *	2.259e+005 *	2.259e+005 *
29	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
30	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
31	Vapour Velocity (m/s)	2.070	2.070	2.070	2.070
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	4032	4037	4042	4047
35	Elevation (m)	0.2760	0.2720	0.2680	0.2640
36	Pressure (bar)	97.13	97.13	97.13	97.13
37	Temperature (C)	61.00	61.00	61.00	60.99
38	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
39	Heat Transferred (kJ/h-m)	826.0	825.9	825.9	825.8
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2125	0.2125	0.2125	0.2125
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	6.564e-003	6.564e-003	6.564e-003	6.564e-003
45	Static Gradient (kPa/m)	-1.585e-003	-1.585e-003	-1.585e-003	-1.585e-003
46	Accel. Gradient (kPa/m)	3.130e-007	3.130e-007	3.129e-007	3.129e-007
47	Liq. Reynolds	2.259e+005 *	2.259e+005 *	2.259e+005 *	2.259e+005 *
48	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
49	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
50	Vapour Velocity (m/s)	2.070	2.070	2.070	2.070
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	4052	4057	4062	4067
54	Elevation (m)	0.2600	0.2400	0.2200	0.2000
55	Pressure (bar)	97.13	97.13	97.13	97.13
56	Temperature (C)	60.99	60.99	60.99	60.99
57	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
58	Heat Transferred (kJ/h-m)	825.7	825.7	825.6	825.6
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2125	0.2095	0.2095	0.2095
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	6.564e-003	6.587e-003	6.587e-003	6.587e-003
64	Static Gradient (kPa/m)	-1.585e-003	-7.853e-003	-7.853e-003	-7.853e-003
65	Accel. Gradient (kPa/m)	3.129e-007	-7.957e-008	-7.958e-008	-7.959e-008
66	Liq. Reynolds	2.259e+005 *	2.259e+005 *	2.259e+005 *	2.259e+005 *
67	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
68	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.070	2.070	2.070	2.070	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	4072	4077	4084	4091	
17	Elevation	(m)	0.1800	0.1600	0.1580	0.1560	
18	Pressure	(bar)	97.13	97.13	97.13	97.13	
19	Temperature	(C)	60.98	60.98	60.98	60.97	
20	Vapour Fraction		0.8714	0.8714	0.8714	0.8714	
21	Heat Transferred	(kJ/h-m)	825.5	825.4	825.3	825.3	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2095	0.2095	0.2131	0.2131	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	6.587e-003	6.586e-003	6.560e-003	6.560e-003	
27	Static Gradient	(kPa/m)	-7.853e-003	-7.853e-003	-5.669e-004	-5.669e-004	
28	Accel. Gradient	(kPa/m)	-7.959e-008	-7.960e-008	3.766e-007	3.766e-007	
29	Liq. Reynolds		2.258e+005 *	2.258e+005 *	2.258e+005 *	2.258e+005 *	
30	Vap. Reynolds		8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *	
31	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
32	Vapour Velocity	(m/s)	2.070	2.070	2.070	2.070	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	4098	4105	4112	4120	
36	Elevation	(m)	0.1540	0.1520	0.1500	0.1460	
37	Pressure	(bar)	97.13	97.13	97.13	97.12	
38	Temperature	(C)	60.97	60.97	60.96	60.96	
39	Vapour Fraction		0.8714	0.8714	0.8714	0.8714	
40	Heat Transferred	(kJ/h-m)	825.2	825.1	825.0	824.9	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2131	0.2131	0.2131	0.2128	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	6.560e-003	6.560e-003	6.560e-003	6.561e-003	
46	Static Gradient	(kPa/m)	-5.669e-004	-5.669e-004	-5.669e-004	-1.044e-003	
47	Accel. Gradient	(kPa/m)	3.766e-007	3.766e-007	3.766e-007	3.467e-007	
48	Liq. Reynolds		2.258e+005 *	2.258e+005 *	2.258e+005 *	2.258e+005 *	
49	Vap. Reynolds		8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *	
50	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
51	Vapour Velocity	(m/s)	2.070	2.070	2.070	2.070	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	4127	4135	4143	4150	
55	Elevation	(m)	0.1420	0.1380	0.1340	0.1300	
56	Pressure	(bar)	97.12	97.12	97.12	97.12	
57	Temperature	(C)	60.96	60.95	60.95	60.95	
58	Vapour Fraction		0.8714	0.8714	0.8714	0.8714	
59	Heat Transferred	(kJ/h-m)	824.8	824.7	824.6	824.5	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.2128	0.2128	0.2128	0.2128	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	6.561e-003	6.561e-003	6.561e-003	6.561e-003	
65	Static Gradient	(kPa/m)	-1.044e-003	-1.044e-003	-1.044e-003	-1.044e-003	
66	Accel. Gradient	(kPa/m)	3.467e-007	3.467e-007	3.467e-007	3.467e-007	
67	Liq. Reynolds		2.258e+005 *	2.258e+005 *	2.258e+005 *	2.258e+005 *	
68	Vap. Reynolds		8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022		
6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
14	Vapour Velocity	(m/s)	2.070	2.070	2.070	2.070	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	4158	4165	4173	4181	
18	Elevation	(m)	0.1140	9.800e-002	8.200e-002	6.600e-002	
19	Pressure	(bar)	97.12	97.12	97.12	97.12	
20	Temperature	(C)	60.94	60.94	60.94	60.93	
21	Vapour Fraction		0.8714	0.8714	0.8714	0.8714	
22	Heat Transferred	(kJ/h-m)	824.4	824.3	824.2	824.1	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.2113	0.2113	0.2113	0.2113	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	6.572e-003	6.572e-003	6.572e-003	6.572e-003	
28	Static Gradient	(kPa/m)	-4.156e-003	-4.156e-003	-4.156e-003	-4.156e-003	
29	Accel. Gradient	(kPa/m)	1.518e-007	1.518e-007	1.518e-007	1.518e-007	
30	Liq. Reynolds		2.258e+005 *	2.258e+005 *	2.258e+005 *	2.258e+005 *	
31	Vap. Reynolds		8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *	
32	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
33	Vapour Velocity	(m/s)	2.070	2.070	2.070	2.070	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	4188	4196	4203	4211	
37	Elevation	(m)	5.000e-002	1.400e-002	-2.200e-002	-5.800e-002	
38	Pressure	(bar)	97.12	97.12	97.12	97.12	
39	Temperature	(C)	60.93	60.93	60.92	60.92	
40	Vapour Fraction		0.8714	0.8714	0.8714	0.8714	
41	Heat Transferred	(kJ/h-m)	824.0	824.0	823.9	823.8	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2113	0.2088	0.2088	0.2088	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	6.572e-003	6.590e-003	6.590e-003	6.590e-003	
47	Static Gradient	(kPa/m)	-4.156e-003	-9.282e-003	-9.282e-003	-9.282e-003	
48	Accel. Gradient	(kPa/m)	1.518e-007	-1.691e-007	-1.691e-007	-1.691e-007	
49	Liq. Reynolds		2.258e+005 *	2.258e+005 *	2.258e+005 *	2.258e+005 *	
50	Vap. Reynolds		8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *	
51	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
52	Vapour Velocity	(m/s)	2.070	2.070	2.070	2.069	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	4219	4226	4234	4241	
56	Elevation	(m)	-9.400e-002	-0.1300	-8.200e-002	-3.400e-002	
57	Pressure	(bar)	97.12	97.12	97.12	97.12	
58	Temperature	(C)	60.92	60.91	60.91	60.90	
59	Vapour Fraction		0.8714	0.8713	0.8713	0.8713	
60	Heat Transferred	(kJ/h-m)	823.7	823.6	823.5	823.4	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2088	0.2088	0.3084	0.3084	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	6.590e-003	6.590e-003	5.588e-003	5.588e-003	
66	Static Gradient	(kPa/m)	-9.282e-003	-9.282e-003	1.603e-002	1.603e-002	
67	Accel. Gradient	(kPa/m)	-1.691e-007	-1.692e-007	1.358e-006	1.358e-006	
68	Liq. Reynolds		2.258e+005 *	2.258e+005 *	2.258e+005 *	2.258e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
14	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
15	Vapour Velocity (m/s)	2.069	2.069	2.069	2.069
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	4249	4257	4264	4272
19	Elevation (m)	1.400e-002	6.200e-002	0.1100	0.1260
20	Pressure (bar)	97.12	97.12	97.12	97.11
21	Temperature (C)	60.90	60.90	60.89	60.89
22	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
23	Heat Transferred (kJ/h-m)	823.3	823.2	823.1	823.0
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.3084	0.3084	0.3084	0.2947
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	5.588e-003	5.588e-003	5.588e-003	4.777e-003
29	Static Gradient (kPa/m)	1.603e-002	1.603e-002	1.603e-002	5.175e-003
30	Accel. Gradient (kPa/m)	1.358e-006	1.358e-006	1.358e-006	6.253e-007
31	Liq. Reynolds	2.258e+005 *	2.257e+005 *	2.257e+005 *	2.257e+005 *
32	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
33	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
34	Vapour Velocity (m/s)	2.069	2.069	2.069	2.069
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	4279	4287	4295	4302
38	Elevation (m)	0.1420	0.1580	0.1740	0.1900
39	Pressure (bar)	97.11	97.11	97.11	97.11
40	Temperature (C)	60.88	60.88	60.88	60.87
41	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
42	Heat Transferred (kJ/h-m)	822.9	822.8	822.7	822.6
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2947	0.2947	0.2947	0.2947
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	4.777e-003	4.777e-003	4.777e-003	4.777e-003
48	Static Gradient (kPa/m)	5.175e-003	5.175e-003	5.175e-003	5.175e-003
49	Accel. Gradient (kPa/m)	6.253e-007	6.253e-007	6.253e-007	6.253e-007
50	Liq. Reynolds	2.257e+005 *	2.257e+005 *	2.257e+005 *	2.257e+005 *
51	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
52	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
53	Vapour Velocity (m/s)	2.069	2.069	2.069	2.069
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	4304	4305	4306	4308
57	Elevation (m)	0.1760	0.1620	0.1480	0.1340
58	Pressure (bar)	97.11	97.11	97.11	97.11
59	Temperature (C)	60.87	60.87	60.87	60.87
60	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
61	Heat Transferred (kJ/h-m)	822.6	822.5	822.5	822.5
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2039	0.2039	0.2039	0.2039
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	6.628e-003	6.628e-003	6.628e-003	6.628e-003
67	Static Gradient (kPa/m)	-1.896e-002	-1.896e-002	-1.896e-002	-1.896e-002
68	Accel. Gradient (kPa/m)	-7.749e-007	-7.749e-007	-7.749e-007	-7.749e-007
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.257e+005 *	2.257e+005 *	2.257e+005 *	2.257e+005 *
14	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
15	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
16	Vapour Velocity (m/s)	2.069	2.069	2.069	2.069
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	4309	4314	4319	4324
20	Elevation (m)	0.1200	7.000e-002	2.000e-002	-3.000e-002
21	Pressure (bar)	97.11	97.11	97.11	97.11
22	Temperature (C)	60.87	60.87	60.87	60.86
23	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
24	Heat Transferred (kJ/h-m)	822.5	822.4	822.4	822.3
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2039	0.2037	0.2038	0.2038
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	6.628e-003	6.630e-003	6.629e-003	6.629e-003
30	Static Gradient (kPa/m)	-1.896e-002	-1.930e-002	-1.930e-002	-1.930e-002
31	Accel. Gradient (kPa/m)	-7.749e-007	-7.963e-007	-7.963e-007	-7.963e-007
32	Liq. Reynolds	2.257e+005 *	2.257e+005 *	2.257e+005 *	2.257e+005 *
33	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
34	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
35	Vapour Velocity (m/s)	2.069	2.069	2.069	2.069
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	4329	4334	4339	4344
39	Elevation (m)	-8.000e-002	-0.1300	-0.1580	-0.1860
40	Pressure (bar)	97.11	97.12	97.12	97.12
41	Temperature (C)	60.86	60.86	60.86	60.86
42	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
43	Heat Transferred (kJ/h-m)	822.3	822.2	822.2	822.1
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2038	0.2038	0.2080	0.2080
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	6.629e-003	6.629e-003	6.595e-003	6.595e-003
49	Static Gradient (kPa/m)	-1.930e-002	-1.930e-002	-1.095e-002	-1.095e-002
50	Accel. Gradient (kPa/m)	-7.963e-007	-7.963e-007	-2.735e-007	-2.735e-007
51	Liq. Reynolds	2.257e+005 *	2.257e+005 *	2.257e+005 *	2.257e+005 *
52	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
53	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
54	Vapour Velocity (m/s)	2.069	2.069	2.069	2.069
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	4349	4354	4359	4364
58	Elevation (m)	-0.2140	-0.2420	-0.2700	-0.2320
59	Pressure (bar)	97.12	97.12	97.12	97.12
60	Temperature (C)	60.85	60.85	60.85	60.85
61	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
62	Heat Transferred (kJ/h-m)	822.0	822.0	821.9	821.9
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2080	0.2080	0.2080	0.3127
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	6.595e-003	6.595e-003	6.595e-003	5.691e-003
68	Static Gradient (kPa/m)	-1.095e-002	-1.095e-002	-1.095e-002	1.948e-002
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	-2.735e-007	-2.735e-007	-2.735e-007	1.581e-006
14	Liq. Reynolds	2.257e+005 *	2.257e+005 *	2.257e+005 *	2.257e+005 *
15	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
16	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
17	Vapour Velocity (m/s)	2.069	2.069	2.069	2.069
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	4369	4374	4379	4384
21	Elevation (m)	-0.1940	-0.1560	-0.1180	-8.000e-002
22	Pressure (bar)	97.11	97.11	97.11	97.11
23	Temperature (C)	60.84	60.84	60.84	60.84
24	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
25	Heat Transferred (kJ/h-m)	821.8	821.7	821.6	821.6
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.3127	0.3127	0.3127	0.3127
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	5.691e-003	5.691e-003	5.691e-003	5.691e-003
31	Static Gradient (kPa/m)	1.948e-002	1.948e-002	1.948e-002	1.948e-002
32	Accel. Gradient (kPa/m)	1.581e-006	1.581e-006	1.581e-006	1.581e-006
33	Liq. Reynolds	2.257e+005 *	2.257e+005 *	2.257e+005 *	2.257e+005 *
34	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
35	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
36	Vapour Velocity (m/s)	2.069	2.069	2.069	2.069
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	4389	4394	4399	4404
40	Elevation (m)	-7.000e-002	-6.000e-002	-5.000e-002	-4.000e-002
41	Pressure (bar)	97.11	97.11	97.11	97.11
42	Temperature (C)	60.83	60.83	60.83	60.83
43	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
44	Heat Transferred (kJ/h-m)	821.5	821.4	821.4	821.3
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.2944	0.2944	0.2944	0.2944
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	4.799e-003	4.799e-003	4.799e-003	4.799e-003
50	Static Gradient (kPa/m)	4.913e-003	4.913e-003	4.913e-003	4.913e-003
51	Accel. Gradient (kPa/m)	6.101e-007	6.101e-007	6.101e-007	6.101e-007
52	Liq. Reynolds	2.257e+005 *	2.257e+005 *	2.257e+005 *	2.257e+005 *
53	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
54	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
55	Vapour Velocity (m/s)	2.069	2.069	2.069	2.069
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	4409	4414	4419	4424
59	Elevation (m)	-3.000e-002	8.000e-003	4.600e-002	8.400e-002
60	Pressure (bar)	97.11	97.11	97.11	97.10
61	Temperature (C)	60.82	60.82	60.82	60.82
62	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
63	Heat Transferred (kJ/h-m)	821.2	821.2	821.1	821.0
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.2944	0.3127	0.3127	0.3127
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	4.799e-003	5.691e-003	5.691e-003	5.691e-003
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1				Case Name:	Simulation - 386MMSCFD_100barg.hsc	
2				Unit Set:	Mixed	
3		Epic Atlantic Limited				
4				Date/Time:	Mon Jun 13 15:02:44 2022	
5						
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	4.913e-003	1.948e-002	1.948e-002	1.948e-002
14	Accel. Gradient	(kPa/m)	6.101e-007	1.581e-006	1.581e-006	1.581e-006
15	Liq. Reynolds		2.257e+005 *	2.257e+005 *	2.257e+005 *	2.257e+005 *
16	Vap. Reynolds		8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
17	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051
18	Vapour Velocity	(m/s)	2.069	2.069	2.069	2.069
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	4429	4434	4439	4444
22	Elevation	(m)	0.1220	0.1600	0.1540	0.1480
23	Pressure	(bar)	97.10	97.10	97.10	97.10
24	Temperature	(C)	60.81	60.81	60.81	60.81
25	Vapour Fraction		0.8713	0.8713	0.8713	0.8713
26	Heat Transferred	(kJ/h-m)	821.0	820.9	820.8	820.8
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.3127	0.3127	0.2123	0.2123
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	5.691e-003	5.691e-003	6.563e-003	6.563e-003
32	Static Gradient	(kPa/m)	1.948e-002	1.948e-002	-2.376e-003	-2.376e-003
33	Accel. Gradient	(kPa/m)	1.581e-006	1.581e-006	2.630e-007	2.630e-007
34	Liq. Reynolds		2.257e+005 *	2.256e+005 *	2.256e+005 *	2.256e+005 *
35	Vap. Reynolds		8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
36	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051
37	Vapour Velocity	(m/s)	2.069	2.069	2.069	2.069
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	4449	4454	4459	4464
41	Elevation	(m)	0.1420	0.1360	0.1300	0.1420
42	Pressure	(bar)	97.10	97.10	97.10	97.10
43	Temperature	(C)	60.80	60.80	60.80	60.80
44	Vapour Fraction		0.8713	0.8713	0.8713	0.8713
45	Heat Transferred	(kJ/h-m)	820.7	820.6	820.6	820.5
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2123	0.2123	0.2123	0.2957
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	6.563e-003	6.563e-003	6.563e-003	4.709e-003
51	Static Gradient	(kPa/m)	-2.376e-003	-2.376e-003	-2.376e-003	5.914e-003
52	Accel. Gradient	(kPa/m)	2.630e-007	2.630e-007	2.630e-007	6.674e-007
53	Liq. Reynolds		2.256e+005 *	2.256e+005 *	2.256e+005 *	2.256e+005 *
54	Vap. Reynolds		8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
55	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051
56	Vapour Velocity	(m/s)	2.069	2.069	2.069	2.069
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	4469	4474	4479	4484
60	Elevation	(m)	0.1540	0.1660	0.1780	0.1900
61	Pressure	(bar)	97.10	97.10	97.10	97.10
62	Temperature	(C)	60.79	60.79	60.79	60.79
63	Vapour Fraction		0.8713	0.8713	0.8713	0.8713
64	Heat Transferred	(kJ/h-m)	820.4	820.4	820.3	820.2
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2957	0.2957	0.2957	0.2957
68	Water HoldUp		---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 101 of 937

1			Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:02:44 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	4.709e-003	4.709e-003	4.709e-003	4.709e-003	
14	Static Gradient	(kPa/m)	5.914e-003	5.914e-003	5.914e-003	5.914e-003	
15	Accel. Gradient	(kPa/m)	6.674e-007	6.674e-007	6.674e-007	6.674e-007	
16	Liq. Reynolds		2.256e+005 *	2.256e+005 *	2.256e+005 *	2.256e+005 *	
17	Vap. Reynolds		8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *	
18	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
19	Vapour Velocity	(m/s)	2.069	2.069	2.069	2.069	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	4489	4494	4499	4504	
23	Elevation	(m)	0.1840	0.1780	0.1720	0.1660	
24	Pressure	(bar)	97.10	97.10	97.10	97.10	
25	Temperature	(C)	60.78	60.78	60.78	60.78	
26	Vapour Fraction		0.8713	0.8713	0.8713	0.8713	
27	Heat Transferred	(kJ/h-m)	820.2	820.1	820.1	820.0	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.2123	0.2123	0.2123	0.2123	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	6.563e-003	6.563e-003	6.562e-003	6.562e-003	
33	Static Gradient	(kPa/m)	-2.350e-003	-2.350e-003	-2.350e-003	-2.350e-003	
34	Accel. Gradient	(kPa/m)	2.646e-007	2.646e-007	2.646e-007	2.646e-007	
35	Liq. Reynolds		2.256e+005 *	2.256e+005 *	2.256e+005 *	2.256e+005 *	
36	Vap. Reynolds		8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *	
37	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
38	Vapour Velocity	(m/s)	2.069	2.069	2.069	2.069	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	4510	4515	4520	4525	
42	Elevation	(m)	0.1600	0.1640	0.1680	0.1720	
43	Pressure	(bar)	97.10	97.10	97.10	97.10	
44	Temperature	(C)	60.78	60.77	60.77	60.77	
45	Vapour Fraction		0.8713	0.8713	0.8713	0.8713	
46	Heat Transferred	(kJ/h-m)	819.9	819.9	819.8	819.7	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.2123	0.2905	0.2905	0.2905	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	6.562e-003	5.072e-003	5.072e-003	5.071e-003	
52	Static Gradient	(kPa/m)	-2.350e-003	1.947e-003	1.947e-003	1.947e-003	
53	Accel. Gradient	(kPa/m)	2.646e-007	4.409e-007	4.409e-007	4.409e-007	
54	Liq. Reynolds		2.256e+005 *	2.256e+005 *	2.256e+005 *	2.256e+005 *	
55	Vap. Reynolds		8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *	
56	Liquid Velocity	(m/s)	0.2051	0.2051	0.2051	0.2051	
57	Vapour Velocity	(m/s)	2.069	2.068	2.068	2.068	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	4530	4535	4540	4545	
61	Elevation	(m)	0.1760	0.1800	0.1700	0.1600	
62	Pressure	(bar)	97.10	97.10	97.09	97.09	
63	Temperature	(C)	60.77	60.76	60.76	60.76	
64	Vapour Fraction		0.8713	0.8713	0.8713	0.8713	
65	Heat Transferred	(kJ/h-m)	819.7	819.6	819.6	819.5	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.2905	0.2905	0.2115	0.2115	
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	5.071e-003	5.071e-003	6.568e-003	6.568e-003
15	Static Gradient (kPa/m)	1.947e-003	1.947e-003	-3.951e-003	-3.951e-003
16	Accel. Gradient (kPa/m)	4.409e-007	4.409e-007	1.643e-007	1.643e-007
17	Liq. Reynolds	2.256e+005 *	2.256e+005 *	2.256e+005 *	2.256e+005 *
18	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
19	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
20	Vapour Velocity (m/s)	2.068	2.068	2.068	2.068
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	4550	4555	4560	4565
24	Elevation (m)	0.1500	0.1400	0.1300	0.1360
25	Pressure (bar)	97.09	97.09	97.09	97.09
26	Temperature (C)	60.76	60.75	60.75	60.75
27	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
28	Heat Transferred (kJ/h-m)	819.4	819.4	819.3	819.2
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2115	0.2115	0.2115	0.2918
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	6.568e-003	6.568e-003	6.568e-003	4.978e-003
34	Static Gradient (kPa/m)	-3.951e-003	-3.951e-003	-3.952e-003	2.930e-003
35	Accel. Gradient (kPa/m)	1.643e-007	1.643e-007	1.643e-007	4.968e-007
36	Liq. Reynolds	2.256e+005 *	2.256e+005 *	2.256e+005 *	2.256e+005 *
37	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
38	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
39	Vapour Velocity (m/s)	2.068	2.068	2.068	2.068
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	4570	4575	4580	4585
43	Elevation (m)	0.1420	0.1480	0.1540	0.1600
44	Pressure (bar)	97.09	97.09	97.09	97.09
45	Temperature (C)	60.75	60.74	60.74	60.74
46	Vapour Fraction	0.8713	0.8713	0.8713	0.8712
47	Heat Transferred (kJ/h-m)	819.2	819.1	819.1	819.0
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2918	0.2918	0.2918	0.2918
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	4.978e-003	4.978e-003	4.978e-003	4.978e-003
53	Static Gradient (kPa/m)	2.930e-003	2.930e-003	2.930e-003	2.930e-003
54	Accel. Gradient (kPa/m)	4.968e-007	4.968e-007	4.967e-007	4.967e-007
55	Liq. Reynolds	2.256e+005 *	2.256e+005 *	2.256e+005 *	2.256e+005 *
56	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.925e+006 *
57	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
58	Vapour Velocity (m/s)	2.068	2.068	2.068	2.068
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	4590	4595	4600	4605
62	Elevation (m)	0.1820	0.2040	0.2260	0.2480
63	Pressure (bar)	97.09	97.09	97.09	97.09
64	Temperature (C)	60.74	60.73	60.73	60.73
65	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
66	Heat Transferred (kJ/h-m)	818.9	818.9	818.8	818.7
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.3023	0.3023	0.3023	0.3023
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	5.049e-003	5.049e-003	5.049e-003	5.049e-003
16	Static Gradient (kPa/m)	1.101e-002	1.101e-002	1.101e-002	1.101e-002
17	Accel. Gradient (kPa/m)	1.009e-006	1.009e-006	1.009e-006	1.009e-006
18	Liq. Reynolds	2.256e+005 *	2.256e+005 *	2.256e+005 *	2.256e+005 *
19	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.926e+006 *
20	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
21	Vapour Velocity (m/s)	2.068	2.068	2.068	2.068
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	4610	4623	4636	4649
25	Elevation (m)	0.2700	0.2400	0.2100	0.1800
26	Pressure (bar)	97.09	97.09	97.09	97.09
27	Temperature (C)	60.73	60.72	60.72	60.71
28	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
29	Heat Transferred (kJ/h-m)	818.7	818.5	818.3	818.2
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.3023	0.2112	0.2112	0.2112
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	5.049e-003	6.569e-003	6.569e-003	6.569e-003
35	Static Gradient (kPa/m)	1.101e-002	-4.556e-003	-4.556e-003	-4.556e-003
36	Accel. Gradient (kPa/m)	1.009e-006	1.265e-007	1.264e-007	1.264e-007
37	Liq. Reynolds	2.255e+005 *	2.255e+005 *	2.255e+005 *	2.255e+005 *
38	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
39	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
40	Vapour Velocity (m/s)	2.068	2.068	2.068	2.068
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	4662	4675	4677	4679
44	Elevation (m)	0.1500	0.1200	0.1280	0.1360
45	Pressure (bar)	97.09	97.09	97.09	97.09
46	Temperature (C)	60.70	60.70	60.70	60.70
47	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
48	Heat Transferred (kJ/h-m)	818.0	817.9	817.8	817.8
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2112	0.2113	0.3010	0.3010
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.569e-003	6.569e-003	4.687e-003	4.687e-003
54	Static Gradient (kPa/m)	-4.556e-003	-4.556e-003	9.981e-003	9.981e-003
55	Accel. Gradient (kPa/m)	1.264e-007	1.264e-007	9.212e-007	9.212e-007
56	Liq. Reynolds	2.255e+005 *	2.255e+005 *	2.255e+005 *	2.255e+005 *
57	Vap. Reynolds	8.925e+006 *	8.925e+006 *	8.925e+006 *	8.926e+006 *
58	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
59	Vapour Velocity (m/s)	2.068	2.068	2.068	2.068
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	4681	4683	4685	4690
63	Elevation (m)	0.1440	0.1520	0.1600	0.1800
64	Pressure (bar)	97.09	97.09	97.09	97.08
65	Temperature (C)	60.69	60.69	60.69	60.69
66	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
67	Heat Transferred (kJ/h-m)	817.8	817.8	817.7	817.7
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 104 of 937

1			Case Name:	Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set:	Mixed		
3	 Epic Atlantic Limited		Date/Time:	Mon Jun 13 15:02:44 2022		
4						
5						
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Water Flow Regime					
14	Liquid HoldUp	0.3010	0.3010	0.3010	0.3008	
15	Water HoldUp	---	---	---	---	
16	Friction Gradient (kPa/m)	4.687e-003	4.687e-003	4.687e-003	4.599e-003	
17	Static Gradient (kPa/m)	9.981e-003	9.981e-003	9.981e-003	9.812e-003	
18	Accel. Gradient (kPa/m)	9.212e-007	9.212e-007	9.212e-007	9.050e-007	
19	Liq. Reynolds	2.255e+005 *	2.255e+005 *	2.255e+005 *	2.255e+005 *	
20	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *	
21	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051	
22	Vapour Velocity (m/s)	2.068	2.068	2.068	2.068	
23	Deposit Thickness (mm)	---	---	---	---	
24	Deposit Volume (m3)	---	---	---	---	
25	Length (m)	4695	4700	4705	4710	
26	Elevation (m)	0.2000	0.2200	0.2400	0.2600	
27	Pressure (bar)	97.08	97.08	97.08	97.08	
28	Temperature (C)	60.69	60.68	60.68	60.68	
29	Vapour Fraction	0.8712	0.8712	0.8712	0.8712	
30	Heat Transferred (kJ/h-m)	817.6	817.5	817.5	817.4	
31	Flow Regime	Transition	Transition	Transition	Transition	
32	Water Flow Regime					
33	Liquid HoldUp	0.3008	0.3008	0.3008	0.3008	
34	Water HoldUp	---	---	---	---	
35	Friction Gradient (kPa/m)	4.599e-003	4.599e-003	4.599e-003	4.599e-003	
36	Static Gradient (kPa/m)	9.812e-003	9.812e-003	9.812e-003	9.812e-003	
37	Accel. Gradient (kPa/m)	9.050e-007	9.051e-007	9.051e-007	9.051e-007	
38	Liq. Reynolds	2.255e+005 *	2.255e+005 *	2.255e+005 *	2.255e+005 *	
39	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *	
40	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051	
41	Vapour Velocity (m/s)	2.068	2.068	2.068	2.068	
42	Deposit Thickness (mm)	---	---	---	---	
43	Deposit Volume (m3)	---	---	---	---	
44	Length (m)	4715	4720	4725	4730	
45	Elevation (m)	0.2620	0.2640	0.2660	0.2680	
46	Pressure (bar)	97.08	97.08	97.08	97.08	
47	Temperature (C)	60.68	60.67	60.67	60.67	
48	Vapour Fraction	0.8712	0.8712	0.8712	0.8712	
49	Heat Transferred (kJ/h-m)	817.3	817.3	817.2	817.1	
50	Flow Regime	Transition	Transition	Transition	Transition	
51	Water Flow Regime					
52	Liquid HoldUp	0.2893	0.2893	0.2893	0.2893	
53	Water HoldUp	---	---	---	---	
54	Friction Gradient (kPa/m)	5.163e-003	5.163e-003	5.163e-003	5.163e-003	
55	Static Gradient (kPa/m)	9.709e-004	9.709e-004	9.709e-004	9.709e-004	
56	Accel. Gradient (kPa/m)	3.852e-007	3.852e-007	3.852e-007	3.852e-007	
57	Liq. Reynolds	2.255e+005 *	2.255e+005 *	2.255e+005 *	2.255e+005 *	
58	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *	
59	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051	
60	Vapour Velocity (m/s)	2.068	2.068	2.068	2.068	
61	Deposit Thickness (mm)	---	---	---	---	
62	Deposit Volume (m3)	---	---	---	---	
63	Length (m)	4735	4740	4745	4750	
64	Elevation (m)	0.2700	0.2720	0.2740	0.2760	
65	Pressure (bar)	97.08	97.08	97.08	97.08	
66	Temperature (C)	60.67	60.67	60.66	60.66	
67	Vapour Fraction	0.8712	0.8712	0.8712	0.8712	
68	Heat Transferred (kJ/h-m)	817.1	817.0	816.9	816.9	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2893	0.2893	0.2893	0.2893
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	5.163e-003	5.163e-003	5.163e-003	5.163e-003
18	Static Gradient (kPa/m)	9.709e-004	9.709e-004	9.709e-004	9.709e-004
19	Accel. Gradient (kPa/m)	3.852e-007	3.852e-007	3.852e-007	3.852e-007
20	Liq. Reynolds	2.255e+005 *	2.255e+005 *	2.255e+005 *	2.255e+005 *
21	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
22	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
23	Vapour Velocity (m/s)	2.068	2.068	2.068	2.068
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	4755	4760	4765	4770
27	Elevation (m)	0.2780	0.2800	0.2780	0.2760
28	Pressure (bar)	97.08	97.08	97.08	97.08
29	Temperature (C)	60.66	60.66	60.65	60.65
30	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
31	Heat Transferred (kJ/h-m)	816.8	816.8	816.7	816.6
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2893	0.2893	0.2131	0.2131
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	5.163e-003	5.163e-003	6.554e-003	6.554e-003
37	Static Gradient (kPa/m)	9.709e-004	9.710e-004	-7.941e-004	-7.941e-004
38	Accel. Gradient (kPa/m)	3.852e-007	3.852e-007	3.618e-007	3.617e-007
39	Liq. Reynolds	2.255e+005 *	2.255e+005 *	2.255e+005 *	2.255e+005 *
40	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
41	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
42	Vapour Velocity (m/s)	2.068	2.068	2.068	2.068
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	4775	4780	4785	4790
46	Elevation (m)	0.2740	0.2720	0.2700	0.2600
47	Pressure (bar)	97.08	97.08	97.08	97.08
48	Temperature (C)	60.65	60.65	60.64	60.64
49	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
50	Heat Transferred (kJ/h-m)	816.6	816.5	816.5	816.4
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2131	0.2131	0.2131	0.2116
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	6.554e-003	6.554e-003	6.554e-003	6.565e-003
56	Static Gradient (kPa/m)	-7.941e-004	-7.941e-004	-7.941e-004	-3.953e-003
57	Accel. Gradient (kPa/m)	3.617e-007	3.617e-007	3.617e-007	1.641e-007
58	Liq. Reynolds	2.255e+005 *	2.255e+005 *	2.255e+005 *	2.254e+005 *
59	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
60	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
61	Vapour Velocity (m/s)	2.068	2.068	2.067	2.067
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	4795	4800	4805	4810
65	Elevation (m)	0.2500	0.2400	0.2300	0.2200
66	Pressure (bar)	97.08	97.08	97.08	97.08
67	Temperature (C)	60.64	60.64	60.64	60.63
68	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022	
6						
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Heat Transferred (kJ/h-m)	816.3	816.3	816.2	816.2	
14	Flow Regime	Transition	Transition	Transition	Transition	
15	Water Flow Regime					
16	Liquid HoldUp	0.2116	0.2116	0.2116	0.2116	
17	Water HoldUp	---	---	---	---	
18	Friction Gradient (kPa/m)	6.565e-003	6.565e-003	6.565e-003	6.565e-003	
19	Static Gradient (kPa/m)	-3.953e-003	-3.953e-003	-3.953e-003	-3.953e-003	
20	Accel. Gradient (kPa/m)	1.641e-007	1.641e-007	1.640e-007	1.640e-007	
21	Liq. Reynolds	2.254e+005 *	2.254e+005 *	2.254e+005 *	2.254e+005 *	
22	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *	
23	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051	
24	Vapour Velocity (m/s)	2.067	2.067	2.067	2.067	
25	Deposit Thickness (mm)	---	---	---	---	
26	Deposit Volume (m3)	---	---	---	---	
27	Length (m)	4815	4820	4825	4830	
28	Elevation (m)	0.2140	0.2080	0.2020	0.1960	
29	Pressure (bar)	97.08	97.08	97.08	97.08	
30	Temperature (C)	60.63	60.63	60.63	60.62	
31	Vapour Fraction	0.8712	0.8712	0.8712	0.8712	
32	Heat Transferred (kJ/h-m)	816.1	816.0	816.0	815.9	
33	Flow Regime	Transition	Transition	Transition	Transition	
34	Water Flow Regime					
35	Liquid HoldUp	0.2124	0.2124	0.2124	0.2124	
36	Water HoldUp	---	---	---	---	
37	Friction Gradient (kPa/m)	6.559e-003	6.559e-003	6.559e-003	6.559e-003	
38	Static Gradient (kPa/m)	-2.377e-003	-2.377e-003	-2.377e-003	-2.377e-003	
39	Accel. Gradient (kPa/m)	2.626e-007	2.626e-007	2.626e-007	2.626e-007	
40	Liq. Reynolds	2.254e+005 *	2.254e+005 *	2.254e+005 *	2.254e+005 *	
41	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *	
42	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051	
43	Vapour Velocity (m/s)	2.067	2.067	2.067	2.067	
44	Deposit Thickness (mm)	---	---	---	---	
45	Deposit Volume (m3)	---	---	---	---	
46	Length (m)	4835	4840	4845	4850	
47	Elevation (m)	0.1900	0.1800	0.1700	0.1600	
48	Pressure (bar)	97.08	97.08	97.08	97.08	
49	Temperature (C)	60.62	60.62	60.62	60.61	
50	Vapour Fraction	0.8712	0.8712	0.8712	0.8712	
51	Heat Transferred (kJ/h-m)	815.8	815.8	815.7	815.7	
52	Flow Regime	Transition	Transition	Transition	Transition	
53	Water Flow Regime					
54	Liquid HoldUp	0.2124	0.2116	0.2116	0.2116	
55	Water HoldUp	---	---	---	---	
56	Friction Gradient (kPa/m)	6.559e-003	6.565e-003	6.565e-003	6.565e-003	
57	Static Gradient (kPa/m)	-2.377e-003	-3.953e-003	-3.953e-003	-3.953e-003	
58	Accel. Gradient (kPa/m)	2.626e-007	1.640e-007	1.640e-007	1.640e-007	
59	Liq. Reynolds	2.254e+005 *	2.254e+005 *	2.254e+005 *	2.254e+005 *	
60	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *	
61	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051	
62	Vapour Velocity (m/s)	2.067	2.067	2.067	2.067	
63	Deposit Thickness (mm)	---	---	---	---	
64	Deposit Volume (m3)	---	---	---	---	
65	Length (m)	4855	4860	4865	4870	
66	Elevation (m)	0.1500	0.1400	0.1380	0.1360	
67	Pressure (bar)	97.08	97.08	97.07	97.07	
68	Temperature (C)	60.61	60.61	60.61	60.60	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
14	Heat Transferred (kJ/h-m)	815.6	815.5	815.5	815.4
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2116	0.2116	0.2131	0.2131
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	6.565e-003	6.565e-003	6.553e-003	6.553e-003
20	Static Gradient (kPa/m)	-3.953e-003	-3.953e-003	-7.942e-004	-7.942e-004
21	Accel. Gradient (kPa/m)	1.640e-007	1.640e-007	3.616e-007	3.616e-007
22	Liq. Reynolds	2.254e+005 *	2.254e+005 *	2.254e+005 *	2.254e+005 *
23	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
24	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
25	Vapour Velocity (m/s)	2.067	2.067	2.067	2.067
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	4875	4880	4885	4890
29	Elevation (m)	0.1340	0.1320	0.1300	0.1300
30	Pressure (bar)	97.07	97.07	97.07	97.07
31	Temperature (C)	60.60	60.60	60.60	60.60
32	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
33	Heat Transferred (kJ/h-m)	815.3	815.3	815.2	815.2
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2131	0.2131	0.2131	0.3117
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	6.553e-003	6.553e-003	6.553e-003	5.667e-003
39	Static Gradient (kPa/m)	-7.942e-004	-7.942e-004	-7.942e-004	0.0000
40	Accel. Gradient (kPa/m)	3.616e-007	3.616e-007	3.616e-007	3.558e-007
41	Liq. Reynolds	2.254e+005 *	2.254e+005 *	2.254e+005 *	2.254e+005 *
42	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
43	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
44	Vapour Velocity (m/s)	2.067	2.067	2.067	2.067
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	4895	4900	4905	4910
48	Elevation (m)	0.1300	0.1300	0.1300	0.1300
49	Pressure (bar)	97.07	97.07	97.07	97.07
50	Temperature (C)	60.59	60.59	60.59	60.59
51	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
52	Heat Transferred (kJ/h-m)	815.1	815.0	815.0	814.9
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.3117	0.3117	0.3117	0.3117
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	5.667e-003	5.667e-003	5.667e-003	5.667e-003
58	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
59	Accel. Gradient (kPa/m)	3.558e-007	3.558e-007	3.558e-007	3.558e-007
60	Liq. Reynolds	2.254e+005 *	2.254e+005 *	2.254e+005 *	2.254e+005 *
61	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
62	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
63	Vapour Velocity (m/s)	2.067	2.067	2.067	2.067
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	4915	4920	4925	4930
67	Elevation (m)	0.1080	8.600e-002	6.400e-002	4.200e-002
68	Pressure (bar)	97.07	97.07	97.07	97.07
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	60.58	60.58	60.58	60.58
14	Vapour Fraction	0.8712	0.8712	0.8712	0.8711
15	Heat Transferred (kJ/h-m)	814.9	814.8	814.7	814.7
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2093	0.2093	0.2093	0.2093
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	6.581e-003	6.581e-003	6.581e-003	6.581e-003
21	Static Gradient (kPa/m)	-8.639e-003	-8.639e-003	-8.639e-003	-8.639e-003
22	Accel. Gradient (kPa/m)	-1.292e-007	-1.292e-007	-1.292e-007	-1.292e-007
23	Liq. Reynolds	2.254e+005 *	2.254e+005 *	2.254e+005 *	2.254e+005 *
24	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
25	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
26	Vapour Velocity (m/s)	2.067	2.067	2.067	2.067
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	4935	4940	4945	4950
30	Elevation (m)	2.000e-002	1.200e-002	4.000e-003	-4.000e-003
31	Pressure (bar)	97.07	97.07	97.07	97.07
32	Temperature (C)	60.58	60.57	60.57	60.57
33	Vapour Fraction	0.8711	0.8711	0.8711	0.8711
34	Heat Transferred (kJ/h-m)	814.6	814.6	814.5	814.4
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2093	0.2120	0.2120	0.2120
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	6.581e-003	6.561e-003	6.561e-003	6.561e-003
40	Static Gradient (kPa/m)	-8.639e-003	-3.167e-003	-3.167e-003	-3.167e-003
41	Accel. Gradient (kPa/m)	-1.292e-007	2.131e-007	2.131e-007	2.131e-007
42	Liq. Reynolds	2.254e+005 *	2.254e+005 *	2.254e+005 *	2.254e+005 *
43	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
44	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
45	Vapour Velocity (m/s)	2.067	2.067	2.067	2.067
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	4955	4960	4965	4970
49	Elevation (m)	-1.200e-002	-2.000e-002	-1.600e-002	-1.200e-002
50	Pressure (bar)	97.07	97.07	97.07	97.07
51	Temperature (C)	60.57	60.56	60.56	60.56
52	Vapour Fraction	0.8711	0.8711	0.8711	0.8711
53	Heat Transferred (kJ/h-m)	814.4	814.3	814.2	814.2
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2120	0.2120	0.2907	0.2907
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	6.561e-003	6.561e-003	5.066e-003	5.066e-003
59	Static Gradient (kPa/m)	-3.167e-003	-3.167e-003	1.949e-003	1.949e-003
60	Accel. Gradient (kPa/m)	2.131e-007	2.131e-007	4.404e-007	4.404e-007
61	Liq. Reynolds	2.254e+005 *	2.254e+005 *	2.254e+005 *	2.254e+005 *
62	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
63	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2051
64	Vapour Velocity (m/s)	2.067	2.067	2.067	2.067
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	4975	4980	4985	4990
68	Elevation (m)	-8.000e-003	-4.000e-003	-2.217e-015	1.400e-002
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Pressure (bar)	97.07	97.07	97.07	97.07
14	Temperature (C)	60.56	60.55	60.55	60.55
15	Vapour Fraction	0.8711	0.8711	0.8711	0.8711
16	Heat Transferred (kJ/h-m)	814.1	814.0	814.0	813.9
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.2907	0.2907	0.2907	0.2972
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	5.066e-003	5.066e-003	5.066e-003	4.616e-003
22	Static Gradient (kPa/m)	1.949e-003	1.949e-003	1.949e-003	6.926e-003
23	Accel. Gradient (kPa/m)	4.404e-007	4.404e-007	4.403e-007	7.246e-007
24	Liq. Reynolds	2.254e+005 *	2.254e+005 *	2.254e+005 *	2.253e+005 *
25	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
26	Liquid Velocity (m/s)	0.2051	0.2051	0.2051	0.2052
27	Vapour Velocity (m/s)	2.067	2.067	2.067	2.067
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	4995	5000	5005	5010
31	Elevation (m)	2.800e-002	4.200e-002	5.600e-002	7.000e-002
32	Pressure (bar)	97.07	97.07	97.07	97.07
33	Temperature (C)	60.55	60.54	60.54	60.54
34	Vapour Fraction	0.8711	0.8711	0.8711	0.8711
35	Heat Transferred (kJ/h-m)	813.9	813.8	813.7	813.7
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.2972	0.2972	0.2972	0.2972
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	4.616e-003	4.616e-003	4.616e-003	4.616e-003
41	Static Gradient (kPa/m)	6.927e-003	6.927e-003	6.927e-003	6.927e-003
42	Accel. Gradient (kPa/m)	7.246e-007	7.246e-007	7.246e-007	7.246e-007
43	Liq. Reynolds	2.253e+005 *	2.253e+005 *	2.253e+005 *	2.253e+005 *
44	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
45	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
46	Vapour Velocity (m/s)	2.067	2.067	2.067	2.067
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	5015	5020	5025	5030
50	Elevation (m)	5.600e-002	4.200e-002	2.800e-002	1.400e-002
51	Pressure (bar)	97.07	97.07	97.07	97.07
52	Temperature (C)	60.54	60.54	60.53	60.53
53	Vapour Fraction	0.8711	0.8711	0.8711	0.8711
54	Heat Transferred (kJ/h-m)	813.6	813.5	813.5	813.4
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.2109	0.2109	0.2109	0.2109
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	6.569e-003	6.569e-003	6.569e-003	6.568e-003
60	Static Gradient (kPa/m)	-5.523e-003	-5.523e-003	-5.523e-003	-5.523e-003
61	Accel. Gradient (kPa/m)	6.563e-008	6.562e-008	6.561e-008	6.561e-008
62	Liq. Reynolds	2.253e+005 *	2.253e+005 *	2.253e+005 *	2.253e+005 *
63	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
64	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
65	Vapour Velocity (m/s)	2.067	2.067	2.066	2.066
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	5035	5040	5045	5050
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	-2.220e-015	2.600e-002	5.200e-002	7.800e-002
14	Pressure (bar)	97.07	97.07	97.07	97.06
15	Temperature (C)	60.53	60.53	60.52	60.52
16	Vapour Fraction	0.8711	0.8711	0.8711	0.8711
17	Heat Transferred (kJ/h-m)	813.4	813.3	813.2	813.2
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2109	0.3051	0.3051	0.3051
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	6.568e-003	5.386e-003	5.386e-003	5.386e-003
23	Static Gradient (kPa/m)	-5.523e-003	1.310e-002	1.310e-002	1.310e-002
24	Accel. Gradient (kPa/m)	6.560e-008	1.160e-006	1.161e-006	1.161e-006
25	Liq. Reynolds	2.253e+005 *	2.253e+005 *	2.253e+005 *	2.253e+005 *
26	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
27	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
28	Vapour Velocity (m/s)	2.066	2.066	2.066	2.066
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	5055	5060	5065	5070
32	Elevation (m)	0.1040	0.1300	0.1240	0.1180
33	Pressure (bar)	97.06	97.06	97.06	97.06
34	Temperature (C)	60.52	60.52	60.51	60.51
35	Vapour Fraction	0.8711	0.8711	0.8711	0.8711
36	Heat Transferred (kJ/h-m)	813.1	813.0	813.0	812.9
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.3051	0.3051	0.2124	0.2124
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	5.386e-003	5.386e-003	6.557e-003	6.557e-003
42	Static Gradient (kPa/m)	1.310e-002	1.310e-002	-2.378e-003	-2.378e-003
43	Accel. Gradient (kPa/m)	1.161e-006	1.161e-006	2.623e-007	2.623e-007
44	Liq. Reynolds	2.253e+005 *	2.253e+005 *	2.253e+005 *	2.253e+005 *
45	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
46	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
47	Vapour Velocity (m/s)	2.066	2.066	2.066	2.066
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	5075	5080	5085	5090
51	Elevation (m)	0.1120	0.1060	1.000e-001	0.1180
52	Pressure (bar)	97.06	97.06	97.06	97.06
53	Temperature (C)	60.51	60.51	60.50	60.50
54	Vapour Fraction	0.8711	0.8711	0.8711	0.8711
55	Heat Transferred (kJ/h-m)	812.8	812.8	812.7	812.6
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2124	0.2124	0.2124	0.2998
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	6.557e-003	6.557e-003	6.557e-003	4.448e-003
61	Static Gradient (kPa/m)	-2.378e-003	-2.378e-003	-2.378e-003	8.903e-003
62	Accel. Gradient (kPa/m)	2.623e-007	2.623e-007	2.623e-007	8.381e-007
63	Liq. Reynolds	2.253e+005 *	2.253e+005 *	2.253e+005 *	2.253e+005 *
64	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
65	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
66	Vapour Velocity (m/s)	2.066	2.066	2.066	2.066
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022	
6						
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Length (m)	5095	5100	5105	5110	
14	Elevation (m)	0.1360	0.1540	0.1720	0.1900	
15	Pressure (bar)	97.06	97.06	97.06	97.06	
16	Temperature (C)	60.50	60.50	60.49	60.49	
17	Vapour Fraction	0.8711	0.8711	0.8711	0.8711	
18	Heat Transferred (kJ/h-m)	812.6	812.5	812.4	812.4	
19	Flow Regime	Transition	Transition	Transition	Transition	
20	Water Flow Regime					
21	Liquid HoldUp	0.2998	0.2998	0.2998	0.2998	
22	Water HoldUp	---	---	---	---	
23	Friction Gradient (kPa/m)	4.448e-003	4.448e-003	4.448e-003	4.448e-003	
24	Static Gradient (kPa/m)	8.903e-003	8.903e-003	8.903e-003	8.903e-003	
25	Accel. Gradient (kPa/m)	8.381e-007	8.381e-007	8.381e-007	8.381e-007	
26	Liq. Reynolds	2.253e+005 *	2.253e+005 *	2.253e+005 *	2.253e+005 *	
27	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *	
28	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052	
29	Vapour Velocity (m/s)	2.066	2.066	2.066	2.066	
30	Deposit Thickness (mm)	---	---	---	---	
31	Deposit Volume (m3)	---	---	---	---	
32	Length (m)	5115	5120	5125	5130	
33	Elevation (m)	0.1860	0.1820	0.1780	0.1740	
34	Pressure (bar)	97.06	97.06	97.06	97.06	
35	Temperature (C)	60.49	60.49	60.48	60.48	
36	Vapour Fraction	0.8711	0.8711	0.8711	0.8711	
37	Heat Transferred (kJ/h-m)	812.3	812.3	812.2	812.1	
38	Flow Regime	Transition	Transition	Transition	Transition	
39	Water Flow Regime					
40	Liquid HoldUp	0.2128	0.2128	0.2128	0.2128	
41	Water HoldUp	---	---	---	---	
42	Friction Gradient (kPa/m)	6.554e-003	6.554e-003	6.554e-003	6.554e-003	
43	Static Gradient (kPa/m)	-1.587e-003	-1.587e-003	-1.587e-003	-1.587e-003	
44	Accel. Gradient (kPa/m)	3.118e-007	3.118e-007	3.117e-007	3.117e-007	
45	Liq. Reynolds	2.253e+005 *	2.253e+005 *	2.253e+005 *	2.253e+005 *	
46	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *	
47	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052	
48	Vapour Velocity (m/s)	2.066	2.066	2.066	2.066	
49	Deposit Thickness (mm)	---	---	---	---	
50	Deposit Volume (m3)	---	---	---	---	
51	Length (m)	5135	5140	5145	5150	
52	Elevation (m)	0.1700	0.1640	0.1580	0.1520	
53	Pressure (bar)	97.06	97.06	97.06	97.06	
54	Temperature (C)	60.48	60.48	60.48	60.47	
55	Vapour Fraction	0.8711	0.8711	0.8711	0.8711	
56	Heat Transferred (kJ/h-m)	812.1	812.0	812.0	811.9	
57	Flow Regime	Transition	Transition	Transition	Transition	
58	Water Flow Regime					
59	Liquid HoldUp	0.2128	0.2124	0.2124	0.2124	
60	Water HoldUp	---	---	---	---	
61	Friction Gradient (kPa/m)	6.554e-003	6.556e-003	6.556e-003	6.556e-003	
62	Static Gradient (kPa/m)	-1.587e-003	-2.378e-003	-2.378e-003	-2.378e-003	
63	Accel. Gradient (kPa/m)	3.117e-007	2.623e-007	2.623e-007	2.622e-007	
64	Liq. Reynolds	2.253e+005 *	2.253e+005 *	2.253e+005 *	2.253e+005 *	
65	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *	
66	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052	
67	Vapour Velocity (m/s)	2.066	2.066	2.066	2.066	
68	Deposit Thickness (mm)	---	---	---	---	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	5155	5160	5165	5170
15	Elevation (m)	0.1460	0.1400	0.1480	0.1560
16	Pressure (bar)	97.06	97.06	97.06	97.05
17	Temperature (C)	60.47	60.47	60.47	60.46
18	Vapour Fraction	0.8711	0.8711	0.8711	0.8711
19	Heat Transferred (kJ/h-m)	811.8	811.8	811.7	811.6
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2124	0.2124	0.2934	0.2934
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	6.556e-003	6.556e-003	4.881e-003	4.881e-003
25	Static Gradient (kPa/m)	-2.378e-003	-2.378e-003	3.922e-003	3.923e-003
26	Accel. Gradient (kPa/m)	2.622e-007	2.622e-007	5.526e-007	5.525e-007
27	Liq. Reynolds	2.253e+005 *	2.253e+005 *	2.252e+005 *	2.252e+005 *
28	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
29	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
30	Vapour Velocity (m/s)	2.066	2.066	2.066	2.066
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	5175	5180	5185	5190
34	Elevation (m)	0.1640	0.1720	0.1800	0.1680
35	Pressure (bar)	97.05	97.05	97.05	97.05
36	Temperature (C)	60.46	60.46	60.46	60.45
37	Vapour Fraction	0.8711	0.8711	0.8711	0.8711
38	Heat Transferred (kJ/h-m)	811.6	811.5	811.4	811.4
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2934	0.2934	0.2934	0.2113
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	4.881e-003	4.881e-003	4.880e-003	6.564e-003
44	Static Gradient (kPa/m)	3.923e-003	3.923e-003	3.923e-003	-4.741e-003
45	Accel. Gradient (kPa/m)	5.525e-007	5.525e-007	5.525e-007	1.145e-007
46	Liq. Reynolds	2.252e+005 *	2.252e+005 *	2.252e+005 *	2.252e+005 *
47	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
48	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
49	Vapour Velocity (m/s)	2.066	2.066	2.066	2.066
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	5195	5200	5205	5210
53	Elevation (m)	0.1560	0.1440	0.1320	0.1200
54	Pressure (bar)	97.05	97.05	97.05	97.05
55	Temperature (C)	60.45	60.45	60.45	60.45
56	Vapour Fraction	0.8711	0.8711	0.8711	0.8711
57	Heat Transferred (kJ/h-m)	811.3	811.3	811.2	811.1
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2113	0.2113	0.2113	0.2113
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	6.564e-003	6.564e-003	6.564e-003	6.564e-003
63	Static Gradient (kPa/m)	-4.741e-003	-4.741e-003	-4.741e-003	-4.741e-003
64	Accel. Gradient (kPa/m)	1.145e-007	1.144e-007	1.144e-007	1.144e-007
65	Liq. Reynolds	2.252e+005 *	2.252e+005 *	2.252e+005 *	2.252e+005 *
66	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
67	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
68	Vapour Velocity (m/s)	2.066	2.066	2.066	2.066
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	5215	5220	5225	5230
16	Elevation (m)	0.1140	0.1080	0.1020	9.600e-002
17	Pressure (bar)	97.05	97.05	97.05	97.05
18	Temperature (C)	60.44	60.44	60.44	60.44
19	Vapour Fraction	0.8711	0.8711	0.8711	0.8711
20	Heat Transferred (kJ/h-m)	811.1	811.0	811.0	810.9
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2125	0.2125	0.2125	0.2125
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	6.555e-003	6.555e-003	6.555e-003	6.555e-003
26	Static Gradient (kPa/m)	-2.378e-003	-2.378e-003	-2.378e-003	-2.379e-003
27	Accel. Gradient (kPa/m)	2.622e-007	2.622e-007	2.621e-007	2.621e-007
28	Liq. Reynolds	2.252e+005 *	2.252e+005 *	2.252e+005 *	2.252e+005 *
29	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
30	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
31	Vapour Velocity (m/s)	2.066	2.066	2.066	2.066
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	5235	5240	5245	5250
35	Elevation (m)	9.000e-002	8.800e-002	8.600e-002	8.400e-002
36	Pressure (bar)	97.05	97.05	97.05	97.05
37	Temperature (C)	60.43	60.43	60.43	60.43
38	Vapour Fraction	0.8711	0.8711	0.8711	0.8711
39	Heat Transferred (kJ/h-m)	810.8	810.8	810.7	810.7
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2125	0.2132	0.2132	0.2132
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	6.555e-003	6.550e-003	6.550e-003	6.550e-003
45	Static Gradient (kPa/m)	-2.379e-003	-7.946e-004	-7.946e-004	-7.947e-004
46	Accel. Gradient (kPa/m)	2.621e-007	3.612e-007	3.612e-007	3.612e-007
47	Liq. Reynolds	2.252e+005 *	2.252e+005 *	2.252e+005 *	2.252e+005 *
48	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
49	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
50	Vapour Velocity (m/s)	2.066	2.066	2.066	2.066
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	5255	5260	5265	5270
54	Elevation (m)	8.200e-002	8.000e-002	6.800e-002	5.600e-002
55	Pressure (bar)	97.05	97.05	97.05	97.05
56	Temperature (C)	60.42	60.42	60.42	60.42
57	Vapour Fraction	0.8711	0.8711	0.8711	0.8711
58	Heat Transferred (kJ/h-m)	810.6	810.5	810.5	810.4
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2132	0.2132	0.2113	0.2113
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	6.550e-003	6.549e-003	6.563e-003	6.563e-003
64	Static Gradient (kPa/m)	-7.947e-004	-7.947e-004	-4.741e-003	-4.741e-003
65	Accel. Gradient (kPa/m)	3.612e-007	3.612e-007	1.144e-007	1.144e-007
66	Liq. Reynolds	2.252e+005 *	2.252e+005 *	2.252e+005 *	2.252e+005 *
67	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
68	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022		
6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.066	2.066	2.066	2.066	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	5275	5280	5285	5290	
17	Elevation	(m)	4.400e-002	3.200e-002	2.000e-002	2.000e-003	
18	Pressure	(bar)	97.05	97.05	97.05	97.05	
19	Temperature	(C)	60.42	60.41	60.41	60.41	
20	Vapour Fraction		0.8711	0.8710	0.8710	0.8710	
21	Heat Transferred	(kJ/h-m)	810.4	810.3	810.2	810.2	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2113	0.2113	0.2113	0.2102	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	6.563e-003	6.563e-003	6.563e-003	6.572e-003	
27	Static Gradient	(kPa/m)	-4.741e-003	-4.741e-003	-4.741e-003	-7.107e-003	
28	Accel. Gradient	(kPa/m)	1.143e-007	1.143e-007	1.143e-007	-3.362e-008	
29	Liq. Reynolds		2.252e+005 *	2.252e+005 *	2.252e+005 *	2.252e+005 *	
30	Vap. Reynolds		8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *	
31	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052	
32	Vapour Velocity	(m/s)	2.066	2.066	2.066	2.065	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	5295	5300	5305	5310	
36	Elevation	(m)	-1.600e-002	-3.400e-002	-5.200e-002	-7.000e-002	
37	Pressure	(bar)	97.05	97.05	97.05	97.05	
38	Temperature	(C)	60.41	60.40	60.40	60.40	
39	Vapour Fraction		0.8710	0.8710	0.8710	0.8710	
40	Heat Transferred	(kJ/h-m)	810.1	810.1	810.0	809.9	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2102	0.2102	0.2102	0.2102	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	6.572e-003	6.572e-003	6.572e-003	6.571e-003	
46	Static Gradient	(kPa/m)	-7.108e-003	-7.108e-003	-7.108e-003	-7.108e-003	
47	Accel. Gradient	(kPa/m)	-3.363e-008	-3.364e-008	-3.364e-008	-3.365e-008	
48	Liq. Reynolds		2.252e+005 *	2.252e+005 *	2.252e+005 *	2.252e+005 *	
49	Vap. Reynolds		8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *	
50	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052	
51	Vapour Velocity	(m/s)	2.065	2.065	2.065	2.065	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	5315	5320	5325	5330	
55	Elevation	(m)	-9.600e-002	-0.1220	-0.1480	-0.1740	
56	Pressure	(bar)	97.05	97.05	97.05	97.05	
57	Temperature	(C)	60.40	60.40	60.39	60.39	
58	Vapour Fraction		0.8710	0.8710	0.8710	0.8710	
59	Heat Transferred	(kJ/h-m)	809.9	809.8	809.8	809.7	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.2086	0.2086	0.2086	0.2086	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	6.583e-003	6.583e-003	6.583e-003	6.583e-003	
65	Static Gradient	(kPa/m)	-1.019e-002	-1.019e-002	-1.019e-002	-1.019e-002	
66	Accel. Gradient	(kPa/m)	-2.265e-007	-2.265e-007	-2.265e-007	-2.265e-007	
67	Liq. Reynolds		2.252e+005 *	2.252e+005 *	2.252e+005 *	2.252e+005 *	
68	Vap. Reynolds		8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
5			Date/Time: Mon Jun 13 15:02:44 2022				
6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052	
14	Vapour Velocity	(m/s)	2.065	2.065	2.065	2.065	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	5335	5340	5345	5350	
18	Elevation	(m)	-0.2000	-0.2240	-0.2480	-0.2720	
19	Pressure	(bar)	97.05	97.05	97.05	97.05	
20	Temperature	(C)	60.39	60.39	60.38	60.38	
21	Vapour Fraction		0.8710	0.8710	0.8710	0.8710	
22	Heat Transferred	(kJ/h-m)	809.6	809.6	809.5	809.5	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.2086	0.2090	0.2090	0.2090	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	6.583e-003	6.580e-003	6.580e-003	6.580e-003	
28	Static Gradient	(kPa/m)	-1.019e-002	-9.419e-003	-9.419e-003	-9.419e-003	
29	Accel. Gradient	(kPa/m)	-2.265e-007	-1.782e-007	-1.782e-007	-1.782e-007	
30	Liq. Reynolds		2.252e+005 *	2.252e+005 *	2.252e+005 *	2.252e+005 *	
31	Vap. Reynolds		8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *	
32	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052	
33	Vapour Velocity	(m/s)	2.065	2.065	2.065	2.065	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	5355	5360	5365	5370	
37	Elevation	(m)	-0.2960	-0.3200	-0.3080	-0.2960	
38	Pressure	(bar)	97.05	97.05	97.05	97.05	
39	Temperature	(C)	60.38	60.38	60.38	60.37	
40	Vapour Fraction		0.8710	0.8710	0.8710	0.8710	
41	Heat Transferred	(kJ/h-m)	809.4	809.4	809.3	809.2	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2090	0.2090	0.2961	0.2961	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	6.579e-003	6.579e-003	4.699e-003	4.699e-003	
47	Static Gradient	(kPa/m)	-9.419e-003	-9.419e-003	5.922e-003	5.922e-003	
48	Accel. Gradient	(kPa/m)	-1.782e-007	-1.782e-007	6.664e-007	6.664e-007	
49	Liq. Reynolds		2.252e+005 *	2.252e+005 *	2.252e+005 *	2.251e+005 *	
50	Vap. Reynolds		8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *	
51	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052	
52	Vapour Velocity	(m/s)	2.065	2.065	2.065	2.065	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	5375	5380	5385	5390	
56	Elevation	(m)	-0.2840	-0.2720	-0.2600	-0.3460	
57	Pressure	(bar)	97.05	97.05	97.05	97.05	
58	Temperature	(C)	60.37	60.37	60.37	60.36	
59	Vapour Fraction		0.8710	0.8710	0.8710	0.8710	
60	Heat Transferred	(kJ/h-m)	809.2	809.1	809.0	809.0	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2961	0.2961	0.2961	0.1971	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	4.699e-003	4.699e-003	4.699e-003	6.678e-003	
66	Static Gradient	(kPa/m)	5.922e-003	5.922e-003	5.922e-003	-3.256e-002	
67	Accel. Gradient	(kPa/m)	6.664e-007	6.664e-007	6.664e-007	-1.624e-006	
68	Liq. Reynolds		2.251e+005 *	2.251e+005 *	2.251e+005 *	2.251e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
14	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
15	Vapour Velocity (m/s)	2.065	2.065	2.065	2.065
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	5395	5400	5405	5410
19	Elevation (m)	-0.4320	-0.5180	-0.6040	-0.6900
20	Pressure (bar)	97.05	97.05	97.05	97.06
21	Temperature (C)	60.36	60.36	60.36	60.36
22	Vapour Fraction	0.8710	0.8710	0.8710	0.8710
23	Heat Transferred (kJ/h-m)	809.0	808.9	808.9	808.8
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.1971	0.1971	0.1971	0.1971
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.678e-003	6.678e-003	6.678e-003	6.677e-003
29	Static Gradient (kPa/m)	-3.256e-002	-3.256e-002	-3.256e-002	-3.256e-002
30	Accel. Gradient (kPa/m)	-1.624e-006	-1.624e-006	-1.624e-006	-1.624e-006
31	Liq. Reynolds	2.251e+005 *	2.251e+005 *	2.251e+005 *	2.251e+005 *
32	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
33	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
34	Vapour Velocity (m/s)	2.065	2.065	2.065	2.065
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	5415	5420	5425	5430
38	Elevation (m)	-0.6280	-0.5660	-0.5040	-0.4420
39	Pressure (bar)	97.05	97.05	97.05	97.05
40	Temperature (C)	60.35	60.35	60.35	60.35
41	Vapour Fraction	0.8710	0.8710	0.8710	0.8710
42	Heat Transferred (kJ/h-m)	808.7	808.7	808.6	808.5
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.3287	0.3287	0.3287	0.3287
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	5.792e-003	5.792e-003	5.792e-003	5.792e-003
48	Static Gradient (kPa/m)	3.295e-002	3.295e-002	3.295e-002	3.295e-002
49	Accel. Gradient (kPa/m)	2.430e-006	2.430e-006	2.431e-006	2.431e-006
50	Liq. Reynolds	2.251e+005 *	2.251e+005 *	2.251e+005 *	2.251e+005 *
51	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
52	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
53	Vapour Velocity (m/s)	2.065	2.065	2.065	2.065
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	5435	5440	5445	5450
57	Elevation (m)	-0.3800	-0.3080	-0.2360	-0.1640
58	Pressure (bar)	97.05	97.04	97.04	97.04
59	Temperature (C)	60.34	60.34	60.34	60.33
60	Vapour Fraction	0.8710	0.8710	0.8710	0.8710
61	Heat Transferred (kJ/h-m)	808.4	808.4	808.3	808.2
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.3287	0.3353	0.3353	0.3353
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	5.792e-003	5.801e-003	5.801e-003	5.801e-003
67	Static Gradient (kPa/m)	3.295e-002	3.881e-002	3.881e-002	3.881e-002
68	Accel. Gradient (kPa/m)	2.431e-006	2.799e-006	2.799e-006	2.799e-006
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:02:44 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Liq. Reynolds		2.251e+005 *	2.251e+005 *	2.251e+005 *	2.251e+005 *				
14	Vap. Reynolds		8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *				
15	Liquid Velocity (m/s)		0.2052	0.2052	0.2052	0.2052				
16	Vapour Velocity (m/s)		2.065	2.065	2.065	2.065				
17	Deposit Thickness (mm)		---	---	---	---				
18	Deposit Volume (m3)		---	---	---	---				
19	Length (m)		5455	5460	5465	5470				
20	Elevation (m)		-9.200e-002	-2.000e-002	-2.600e-002	-3.200e-002				
21	Pressure (bar)		97.04	97.03	97.03	97.03				
22	Temperature (C)		60.33	60.33	60.33	60.32				
23	Vapour Fraction		0.8710	0.8710	0.8710	0.8710				
24	Heat Transferred (kJ/h-m)		808.1	808.0	808.0	807.9				
25	Flow Regime		Transition	Transition	Transition	Transition				
26	Water Flow Regime									
27	Liquid HoldUp		0.3353	0.3353	0.2125	0.2125				
28	Water HoldUp		---	---	---	---				
29	Friction Gradient (kPa/m)		5.801e-003	5.801e-003	6.553e-003	6.553e-003				
30	Static Gradient (kPa/m)		3.881e-002	3.881e-002	-2.379e-003	-2.379e-003				
31	Accel. Gradient (kPa/m)		2.799e-006	2.799e-006	2.619e-007	2.619e-007				
32	Liq. Reynolds		2.251e+005 *	2.251e+005 *	2.251e+005 *	2.251e+005 *				
33	Vap. Reynolds		8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *				
34	Liquid Velocity (m/s)		0.2052	0.2052	0.2052	0.2052				
35	Vapour Velocity (m/s)		2.065	2.065	2.065	2.065				
36	Deposit Thickness (mm)		---	---	---	---				
37	Deposit Volume (m3)		---	---	---	---				
38	Length (m)		5475	5480	5485	5490				
39	Elevation (m)		-3.800e-002	-4.400e-002	-5.000e-002	-6.800e-002				
40	Pressure (bar)		97.03	97.03	97.03	97.03				
41	Temperature (C)		60.32	60.32	60.32	60.31				
42	Vapour Fraction		0.8710	0.8710	0.8710	0.8710				
43	Heat Transferred (kJ/h-m)		807.8	807.8	807.7	807.7				
44	Flow Regime		Transition	Transition	Transition	Transition				
45	Water Flow Regime									
46	Liquid HoldUp		0.2125	0.2125	0.2125	0.2102				
47	Water HoldUp		---	---	---	---				
48	Friction Gradient (kPa/m)		6.553e-003	6.553e-003	6.553e-003	6.570e-003				
49	Static Gradient (kPa/m)		-2.379e-003	-2.379e-003	-2.379e-003	-7.028e-003				
50	Accel. Gradient (kPa/m)		2.619e-007	2.619e-007	2.619e-007	-2.877e-008				
51	Liq. Reynolds		2.251e+005 *	2.251e+005 *	2.251e+005 *	2.251e+005 *				
52	Vap. Reynolds		8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *				
53	Liquid Velocity (m/s)		0.2052	0.2052	0.2052	0.2052				
54	Vapour Velocity (m/s)		2.065	2.065	2.065	2.065				
55	Deposit Thickness (mm)		---	---	---	---				
56	Deposit Volume (m3)		---	---	---	---				
57	Length (m)		5495	5500	5505	5510				
58	Elevation (m)		-8.600e-002	-0.1040	-0.1220	-0.1400				
59	Pressure (bar)		97.03	97.03	97.03	97.03				
60	Temperature (C)		60.31	60.31	60.31	60.31				
61	Vapour Fraction		0.8710	0.8710	0.8710	0.8710				
62	Heat Transferred (kJ/h-m)		807.6	807.5	807.5	807.4				
63	Flow Regime		Transition	Transition	Transition	Transition				
64	Water Flow Regime									
65	Liquid HoldUp		0.2102	0.2102	0.2102	0.2102				
66	Water HoldUp		---	---	---	---				
67	Friction Gradient (kPa/m)		6.570e-003	6.570e-003	6.570e-003	6.569e-003				
68	Static Gradient (kPa/m)		-7.028e-003	-7.028e-003	-7.029e-003	-7.029e-003				
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:02:44 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient	(kPa/m)	-2.878e-008	-2.879e-008	-2.880e-008	-2.881e-008	
14	Liq. Reynolds		2.251e+005 *	2.251e+005 *	2.251e+005 *	2.251e+005 *	
15	Vap. Reynolds		8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *	
16	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052	
17	Vapour Velocity	(m/s)	2.065	2.065	2.065	2.065	
18	Deposit Thickness	(mm)	---	---	---	---	
19	Deposit Volume	(m3)	---	---	---	---	
20	Length	(m)	5515	5520	5525	5530	
21	Elevation	(m)	-0.1520	-0.1640	-0.1760	-0.1880	
22	Pressure	(bar)	97.03	97.03	97.03	97.03	
23	Temperature	(C)	60.30	60.30	60.30	60.30	
24	Vapour Fraction		0.8710	0.8710	0.8710	0.8710	
25	Heat Transferred	(kJ/h-m)	807.4	807.3	807.2	807.2	
26	Flow Regime		Transition	Transition	Transition	Transition	
27	Water Flow Regime						
28	Liquid HoldUp		0.2114	0.2114	0.2114	0.2114	
29	Water HoldUp		---	---	---	---	
30	Friction Gradient	(kPa/m)	6.561e-003	6.561e-003	6.561e-003	6.561e-003	
31	Static Gradient	(kPa/m)	-4.743e-003	-4.743e-003	-4.743e-003	-4.743e-003	
32	Accel. Gradient	(kPa/m)	1.141e-007	1.141e-007	1.141e-007	1.141e-007	
33	Liq. Reynolds		2.251e+005 *	2.251e+005 *	2.251e+005 *	2.251e+005 *	
34	Vap. Reynolds		8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *	
35	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052	
36	Vapour Velocity	(m/s)	2.065	2.065	2.065	2.065	
37	Deposit Thickness	(mm)	---	---	---	---	
38	Deposit Volume	(m3)	---	---	---	---	
39	Length	(m)	5535	5542	5549	5556	
40	Elevation	(m)	-0.2000	-0.2240	-0.2480	-0.2720	
41	Pressure	(bar)	97.03	97.03	97.03	97.03	
42	Temperature	(C)	60.29	60.29	60.29	60.28	
43	Vapour Fraction		0.8710	0.8710	0.8710	0.8710	
44	Heat Transferred	(kJ/h-m)	807.1	807.0	807.0	806.9	
45	Flow Regime		Transition	Transition	Transition	Transition	
46	Water Flow Regime						
47	Liquid HoldUp		0.2114	0.2104	0.2104	0.2104	
48	Water HoldUp		---	---	---	---	
49	Friction Gradient	(kPa/m)	6.561e-003	6.568e-003	6.568e-003	6.568e-003	
50	Static Gradient	(kPa/m)	-4.743e-003	-6.756e-003	-6.756e-003	-6.756e-003	
51	Accel. Gradient	(kPa/m)	1.141e-007	-1.178e-008	-1.179e-008	-1.180e-008	
52	Liq. Reynolds		2.251e+005 *	2.250e+005 *	2.250e+005 *	2.250e+005 *	
53	Vap. Reynolds		8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *	
54	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052	
55	Vapour Velocity	(m/s)	2.065	2.065	2.065	2.064	
56	Deposit Thickness	(mm)	---	---	---	---	
57	Deposit Volume	(m3)	---	---	---	---	
58	Length	(m)	5563	5570	5572	5574	
59	Elevation	(m)	-0.2960	-0.3200	-0.3480	-0.3760	
60	Pressure	(bar)	97.03	97.03	97.03	97.03	
61	Temperature	(C)	60.28	60.28	60.28	60.28	
62	Vapour Fraction		0.8710	0.8710	0.8710	0.8710	
63	Heat Transferred	(kJ/h-m)	806.8	806.7	806.7	806.7	
64	Flow Regime		Transition	Transition	Transition	Transition	
65	Water Flow Regime						
66	Liquid HoldUp		0.2104	0.2104	0.2002	0.2002	
67	Water HoldUp		---	---	---	---	
68	Friction Gradient	(kPa/m)	6.568e-003	6.568e-003	6.649e-003	6.649e-003	
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1				Case Name:	Simulation - 386MMSCFD_100barg.hsc	
2				Unit Set:	Mixed	
3	 Epic Atlantic Limited			Date/Time:	Mon Jun 13 15:02:44 2022	
4						
5						
6						
7						
8						
9						
10						
11						
12						
13	Static Gradient	(kPa/m)	-6.756e-003	-6.756e-003	-2.676e-002	-2.676e-002
14	Accel. Gradient	(kPa/m)	-1.182e-008	-1.183e-008	-1.261e-006	-1.261e-006
15	Liq. Reynolds		2.250e+005 *	2.250e+005 *	2.250e+005 *	2.250e+005 *
16	Vap. Reynolds		8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
17	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052
18	Vapour Velocity	(m/s)	2.064	2.064	2.064	2.064
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	5576	5578	5580	5585
22	Elevation	(m)	-0.4040	-0.4320	-0.4600	-0.4460
23	Pressure	(bar)	97.03	97.04	97.04	97.03
24	Temperature	(C)	60.28	60.28	60.27	60.27
25	Vapour Fraction		0.8710	0.8710	0.8710	0.8710
26	Heat Transferred	(kJ/h-m)	806.7	806.6	806.6	806.5
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2002	0.2002	0.2002	0.2974
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	6.649e-003	6.649e-003	6.649e-003	4.610e-003
32	Static Gradient	(kPa/m)	-2.676e-002	-2.676e-002	-2.676e-002	6.932e-003
33	Accel. Gradient	(kPa/m)	-1.261e-006	-1.261e-006	-1.261e-006	7.240e-007
34	Liq. Reynolds		2.250e+005 *	2.250e+005 *	2.250e+005 *	2.250e+005 *
35	Vap. Reynolds		8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
36	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052
37	Vapour Velocity	(m/s)	2.064	2.064	2.064	2.064
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	5590	5595	5600	5605
41	Elevation	(m)	-0.4320	-0.4180	-0.4040	-0.3900
42	Pressure	(bar)	97.03	97.03	97.03	97.03
43	Temperature	(C)	60.27	60.27	60.27	60.26
44	Vapour Fraction		0.8710	0.8710	0.8710	0.8710
45	Heat Transferred	(kJ/h-m)	806.5	806.4	806.3	806.3
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2974	0.2974	0.2974	0.2974
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	4.610e-003	4.610e-003	4.610e-003	4.610e-003
51	Static Gradient	(kPa/m)	6.932e-003	6.932e-003	6.932e-003	6.933e-003
52	Accel. Gradient	(kPa/m)	7.240e-007	7.240e-007	7.240e-007	7.240e-007
53	Liq. Reynolds		2.250e+005 *	2.250e+005 *	2.250e+005 *	2.250e+005 *
54	Vap. Reynolds		8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
55	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052
56	Vapour Velocity	(m/s)	2.064	2.064	2.064	2.064
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	5610	5615	5620	5625
60	Elevation	(m)	-0.3980	-0.4060	-0.4140	-0.4220
61	Pressure	(bar)	97.03	97.03	97.03	97.03
62	Temperature	(C)	60.26	60.26	60.26	60.25
63	Vapour Fraction		0.8710	0.8710	0.8710	0.8710
64	Heat Transferred	(kJ/h-m)	806.2	806.2	806.1	806.0
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2122	0.2122	0.2122	0.2122
68	Water HoldUp		---	---	---	---
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1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	6.554e-003	6.554e-003	6.554e-003	6.554e-003
14	Static Gradient	(kPa/m)	-3.170e-003	-3.170e-003	-3.170e-003	-3.170e-003
15	Accel. Gradient	(kPa/m)	2.123e-007	2.123e-007	2.123e-007	2.123e-007
16	Liq. Reynolds		2.250e+005 *	2.250e+005 *	2.250e+005 *	2.250e+005 *
17	Vap. Reynolds		8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
18	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052
19	Vapour Velocity	(m/s)	2.064	2.064	2.064	2.064
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	5630	5635	5640	5645
23	Elevation	(m)	-0.4300	-0.4880	-0.5460	-0.6040
24	Pressure	(bar)	97.03	97.03	97.03	97.03
25	Temperature	(C)	60.25	60.25	60.25	60.25
26	Vapour Fraction		0.8709	0.8709	0.8709	0.8709
27	Heat Transferred	(kJ/h-m)	806.0	805.9	805.9	805.8
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2122	0.2025	0.2025	0.2025
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	6.554e-003	6.629e-003	6.629e-003	6.629e-003
33	Static Gradient	(kPa/m)	-3.170e-003	-2.233e-002	-2.233e-002	-2.233e-002
34	Accel. Gradient	(kPa/m)	2.123e-007	-9.848e-007	-9.848e-007	-9.848e-007
35	Liq. Reynolds		2.250e+005 *	2.250e+005 *	2.250e+005 *	2.250e+005 *
36	Vap. Reynolds		8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
37	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052
38	Vapour Velocity	(m/s)	2.064	2.064	2.064	2.064
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	5650	5655	5660	5665
42	Elevation	(m)	-0.6620	-0.7200	-0.7180	-0.7160
43	Pressure	(bar)	97.03	97.04	97.04	97.04
44	Temperature	(C)	60.24	60.24	60.24	60.24
45	Vapour Fraction		0.8709	0.8709	0.8709	0.8709
46	Heat Transferred	(kJ/h-m)	805.8	805.7	805.7	805.6
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2025	0.2025	0.2896	0.2896
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	6.629e-003	6.629e-003	5.151e-003	5.151e-003
52	Static Gradient	(kPa/m)	-2.233e-002	-2.233e-002	9.723e-004	9.723e-004
53	Accel. Gradient	(kPa/m)	-9.848e-007	-9.848e-007	3.840e-007	3.840e-007
54	Liq. Reynolds		2.250e+005 *	2.250e+005 *	2.250e+005 *	2.250e+005 *
55	Vap. Reynolds		8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
56	Liquid Velocity	(m/s)	0.2052	0.2052	0.2052	0.2052
57	Vapour Velocity	(m/s)	2.064	2.064	2.064	2.064
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	5670	5675	5680	5685
61	Elevation	(m)	-0.7140	-0.7120	-0.7100	-0.6260
62	Pressure	(bar)	97.03	97.03	97.03	97.03
63	Temperature	(C)	60.23	60.23	60.23	60.23
64	Vapour Fraction		0.8709	0.8709	0.8709	0.8709
65	Heat Transferred	(kJ/h-m)	805.5	805.5	805.4	805.4
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2896	0.2896	0.2896	0.3436
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	5.151e-003	5.151e-003	5.151e-003	5.799e-003
15	Static Gradient (kPa/m)	9.723e-004	9.723e-004	9.723e-004	4.645e-002
16	Accel. Gradient (kPa/m)	3.840e-007	3.840e-007	3.840e-007	3.277e-006
17	Liq. Reynolds	2.250e+005 *	2.250e+005 *	2.250e+005 *	2.250e+005 *
18	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
19	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
20	Vapour Velocity (m/s)	2.064	2.064	2.064	2.064
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	5690	5695	5700	5705
24	Elevation (m)	-0.5420	-0.4580	-0.3740	-0.2900
25	Pressure (bar)	97.03	97.03	97.02	97.02
26	Temperature (C)	60.22	60.22	60.22	60.22
27	Vapour Fraction	0.8709	0.8709	0.8709	0.8709
28	Heat Transferred (kJ/h-m)	805.3	805.2	805.1	805.0
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.3436	0.3436	0.3436	0.3436
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	5.799e-003	5.799e-003	5.799e-003	5.800e-003
34	Static Gradient (kPa/m)	4.645e-002	4.645e-002	4.645e-002	4.645e-002
35	Accel. Gradient (kPa/m)	3.277e-006	3.277e-006	3.278e-006	3.278e-006
36	Liq. Reynolds	2.250e+005 *	2.250e+005 *	2.250e+005 *	2.250e+005 *
37	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
38	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
39	Vapour Velocity (m/s)	2.064	2.064	2.064	2.064
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	5710	5715	5720	5725
43	Elevation (m)	-0.3100	-0.3300	-0.3500	-0.3700
44	Pressure (bar)	97.02	97.02	97.02	97.02
45	Temperature (C)	60.21	60.21	60.21	60.21
46	Vapour Fraction	0.8709	0.8709	0.8709	0.8709
47	Heat Transferred (kJ/h-m)	805.0	804.9	804.8	804.8
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2099	0.2099	0.2099	0.2099
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	6.571e-003	6.571e-003	6.571e-003	6.570e-003
53	Static Gradient (kPa/m)	-7.871e-003	-7.871e-003	-7.871e-003	-7.871e-003
54	Accel. Gradient (kPa/m)	-8.155e-008	-8.156e-008	-8.157e-008	-8.157e-008
55	Liq. Reynolds	2.250e+005 *	2.250e+005 *	2.250e+005 *	2.250e+005 *
56	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
57	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
58	Vapour Velocity (m/s)	2.064	2.064	2.064	2.064
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	5730	5735	5740	5745
62	Elevation (m)	-0.3900	-0.3620	-0.3340	-0.3060
63	Pressure (bar)	97.02	97.02	97.02	97.02
64	Temperature (C)	60.20	60.20	60.20	60.20
65	Vapour Fraction	0.8709	0.8709	0.8709	0.8709
66	Heat Transferred (kJ/h-m)	804.7	804.7	804.6	804.5
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.2099	0.3066	0.3066	0.3066
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	6.570e-003	5.474e-003	5.474e-003	5.474e-003
16	Static Gradient (kPa/m)	-7.871e-003	1.416e-002	1.416e-002	1.416e-002
17	Accel. Gradient (kPa/m)	-8.158e-008	1.232e-006	1.232e-006	1.232e-006
18	Liq. Reynolds	2.249e+005 *	2.249e+005 *	2.249e+005 *	2.249e+005 *
19	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
20	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
21	Vapour Velocity (m/s)	2.064	2.064	2.064	2.064
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	5750	5755	5760	5765
25	Elevation (m)	-0.2780	-0.2500	-0.2100	-0.1700
26	Pressure (bar)	97.02	97.02	97.02	97.01
27	Temperature (C)	60.19	60.19	60.19	60.19
28	Vapour Fraction	0.8709	0.8709	0.8709	0.8709
29	Heat Transferred (kJ/h-m)	804.5	804.4	804.3	804.2
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.3066	0.3066	0.3145	0.3145
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	5.474e-003	5.474e-003	5.702e-003	5.702e-003
35	Static Gradient (kPa/m)	1.416e-002	1.416e-002	2.060e-002	2.060e-002
36	Accel. Gradient (kPa/m)	1.232e-006	1.232e-006	1.650e-006	1.650e-006
37	Liq. Reynolds	2.249e+005 *	2.249e+005 *	2.249e+005 *	2.249e+005 *
38	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
39	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
40	Vapour Velocity (m/s)	2.064	2.064	2.064	2.064
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	5770	5775	5780	5785
44	Elevation (m)	-0.1300	-9.000e-002	-5.000e-002	-0.1100
45	Pressure (bar)	97.01	97.01	97.01	97.01
46	Temperature (C)	60.18	60.18	60.18	60.18
47	Vapour Fraction	0.8709	0.8709	0.8709	0.8709
48	Heat Transferred (kJ/h-m)	804.2	804.1	804.0	804.0
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.3145	0.3145	0.3145	0.2022
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	5.702e-003	5.702e-003	5.702e-003	6.632e-003
54	Static Gradient (kPa/m)	2.060e-002	2.060e-002	2.060e-002	-2.308e-002
55	Accel. Gradient (kPa/m)	1.650e-006	1.650e-006	1.650e-006	-1.032e-006
56	Liq. Reynolds	2.249e+005 *	2.249e+005 *	2.249e+005 *	2.249e+005 *
57	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.927e+006 *	8.926e+006 *
58	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
59	Vapour Velocity (m/s)	2.064	2.064	2.064	2.064
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	5790	5795	5800	5805
63	Elevation (m)	-0.1700	-0.2300	-0.2900	-0.3500
64	Pressure (bar)	97.01	97.01	97.01	97.01
65	Temperature (C)	60.17	60.17	60.17	60.17
66	Vapour Fraction	0.8709	0.8709	0.8709	0.8709
67	Heat Transferred (kJ/h-m)	803.9	803.9	803.8	803.8
68	Flow Regime	Transition	Transition	Transition	Transition
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2022	0.2022	0.2022	0.2022
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.631e-003	6.631e-003	6.631e-003	6.631e-003
17	Static Gradient (kPa/m)	-2.308e-002	-2.308e-002	-2.308e-002	-2.308e-002
18	Accel. Gradient (kPa/m)	-1.032e-006	-1.032e-006	-1.032e-006	-1.032e-006
19	Liq. Reynolds	2.249e+005 *	2.249e+005 *	2.249e+005 *	2.249e+005 *
20	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
21	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
22	Vapour Velocity (m/s)	2.064	2.064	2.064	2.064
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	5810	5815	5820	5825
26	Elevation (m)	-0.3580	-0.3660	-0.3740	-0.3820
27	Pressure (bar)	97.01	97.01	97.01	97.01
28	Temperature (C)	60.17	60.16	60.16	60.16
29	Vapour Fraction	0.8709	0.8709	0.8709	0.8709
30	Heat Transferred (kJ/h-m)	803.7	803.7	803.6	803.5
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2122	0.2122	0.2122	0.2122
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	6.553e-003	6.553e-003	6.553e-003	6.552e-003
36	Static Gradient (kPa/m)	-3.170e-003	-3.170e-003	-3.170e-003	-3.170e-003
37	Accel. Gradient (kPa/m)	2.122e-007	2.121e-007	2.121e-007	2.121e-007
38	Liq. Reynolds	2.249e+005 *	2.249e+005 *	2.249e+005 *	2.249e+005 *
39	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.926e+006 *	8.926e+006 *
40	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
41	Vapour Velocity (m/s)	2.064	2.064	2.064	2.064
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	5830	5835	5840	5845
45	Elevation (m)	-0.3900	-0.2920	-0.1940	-9.600e-002
46	Pressure (bar)	97.01	97.01	97.01	97.00
47	Temperature (C)	60.16	60.15	60.15	60.15
48	Vapour Fraction	0.8709	0.8709	0.8709	0.8709
49	Heat Transferred (kJ/h-m)	803.5	803.4	803.3	803.3
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2122	0.3524	0.3524	0.3524
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	6.552e-003	5.793e-003	5.793e-003	5.793e-003
55	Static Gradient (kPa/m)	-3.170e-003	5.478e-002	5.478e-002	5.478e-002
56	Accel. Gradient (kPa/m)	2.121e-007	3.800e-006	3.800e-006	3.800e-006
57	Liq. Reynolds	2.249e+005 *	2.249e+005 *	2.249e+005 *	2.249e+005 *
58	Vap. Reynolds	8.926e+006 *	8.926e+006 *	8.927e+006 *	8.927e+006 *
59	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
60	Vapour Velocity (m/s)	2.064	2.064	2.064	2.064
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	5850	5855	5860	5865
64	Elevation (m)	2.000e-003	1.000e-001	0.1280	0.1560
65	Pressure (bar)	97.00	97.00	97.00	97.00
66	Temperature (C)	60.14	60.14	60.14	60.14
67	Vapour Fraction	0.8709	0.8709	0.8709	0.8709
68	Heat Transferred (kJ/h-m)	803.2	803.1	803.0	802.9
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
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9	PROFILES				
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11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.3524	0.3524	0.3066	0.3066
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	5.793e-003	5.793e-003	5.474e-003	5.474e-003
18	Static Gradient (kPa/m)	5.478e-002	5.478e-002	1.417e-002	1.417e-002
19	Accel. Gradient (kPa/m)	3.800e-006	3.800e-006	1.232e-006	1.232e-006
20	Liq. Reynolds	2.249e+005 *	2.249e+005 *	2.249e+005 *	2.249e+005 *
21	Vap. Reynolds	8.927e+006 *	8.927e+006 *	8.927e+006 *	8.927e+006 *
22	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
23	Vapour Velocity (m/s)	2.064	2.064	2.064	2.064
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	5870	5875	5880	5885
27	Elevation (m)	0.1840	0.2120	0.2400	-1.368
28	Pressure (bar)	97.00	96.99	96.99	97.01
29	Temperature (C)	60.13	60.13	60.13	60.14
30	Vapour Fraction	0.8709	0.8709	0.8709	0.8709
31	Heat Transferred (kJ/h-m)	802.9	802.8	802.7	804.1
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.3066	0.3066	0.3066	9.046e-002
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	5.474e-003	5.474e-003	5.474e-003	9.371e-003
37	Static Gradient (kPa/m)	1.417e-002	1.417e-002	1.417e-002	-0.4071
38	Accel. Gradient (kPa/m)	1.232e-006	1.232e-006	1.232e-006	-2.495e-005
39	Liq. Reynolds	2.248e+005 *	2.248e+005 *	2.248e+005 *	2.249e+005 *
40	Vap. Reynolds	8.927e+006 *	8.927e+006 *	8.927e+006 *	8.926e+006 *
41	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
42	Vapour Velocity (m/s)	2.064	2.064	2.064	2.063
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	5890	5895	5900	5905
46	Elevation (m)	-2.976	-4.584	-6.192	-7.800
47	Pressure (bar)	97.03	97.05	97.07	97.09
48	Temperature (C)	60.14	60.15	60.16	60.16
49	Vapour Fraction	0.8709	0.8709	0.8709	0.8709
50	Heat Transferred (kJ/h-m)	804.3	804.5	804.7	804.8
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	9.048e-002	9.050e-002	9.052e-002	9.054e-002
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	9.369e-003	9.366e-003	9.364e-003	9.362e-003
56	Static Gradient (kPa/m)	-0.4072	-0.4073	-0.4074	-0.4074
57	Accel. Gradient (kPa/m)	-2.494e-005	-2.493e-005	-2.493e-005	-2.492e-005
58	Liq. Reynolds	2.249e+005 *	2.249e+005 *	2.250e+005 *	2.250e+005 *
59	Vap. Reynolds	8.926e+006 *	8.925e+006 *	8.925e+006 *	8.924e+006 *
60	Liquid Velocity (m/s)	0.2052	0.2052	0.2052	0.2052
61	Vapour Velocity (m/s)	2.063	2.062	2.062	2.062
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	5910	5915	5920	5925
65	Elevation (m)	-9.460	-11.12	-12.78	-14.44
66	Pressure (bar)	97.11	97.13	97.16	97.18
67	Temperature (C)	60.17	60.18	60.18	60.19
68	Vapour Fraction	0.8708	0.8708	0.8708	0.8708
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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11	Pipe Table				
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13	Heat Transferred (kJ/h-m)	805.0	805.2	805.4	805.6
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	9.056e-002	9.058e-002	9.060e-002	9.062e-002
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	9.359e-003	9.357e-003	9.355e-003	9.352e-003
19	Static Gradient (kPa/m)	-0.4236	-0.4236	-0.4237	-0.4238
20	Accel. Gradient (kPa/m)	-2.592e-005	-2.592e-005	-2.591e-005	-2.590e-005
21	Liq. Reynolds	2.250e+005 *	2.250e+005 *	2.251e+005 *	2.251e+005 *
22	Vap. Reynolds	8.923e+006 *	8.923e+006 *	8.922e+006 *	8.921e+006 *
23	Liquid Velocity (m/s)	0.2052	0.2052	0.2053	0.2053
24	Vapour Velocity (m/s)	2.061	2.061	2.060	2.060
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	5930	5935	5940	5945
28	Elevation (m)	-16.10	-13.74	-11.38	-9.020
29	Pressure (bar)	97.20	97.04	96.89	96.73
30	Temperature (C)	60.20	60.16	60.12	60.09
31	Vapour Fraction	0.8708	0.8709	0.8709	0.8710
32	Heat Transferred (kJ/h-m)	805.8	796.5	795.6	794.6
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	9.064e-002	1.000	1.000	1.000
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	9.350e-003	5.358e-003	5.366e-003	5.374e-003
38	Static Gradient (kPa/m)	-0.4239	3.094	3.094	3.095
39	Accel. Gradient (kPa/m)	-2.590e-005	1.943e-004	1.949e-004	1.956e-004
40	Liq. Reynolds	2.251e+005 *	2.249e+005 *	2.247e+005 *	2.245e+005 *
41	Vap. Reynolds	8.921e+006 *	8.925e+006 *	8.930e+006 *	8.935e+006 *
42	Liquid Velocity (m/s)	0.2053	0.2052	0.2052	0.2051
43	Vapour Velocity (m/s)	2.059	2.063	2.066	2.070
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	5950	5955	5960	5965
47	Elevation (m)	-6.660	-4.300	-3.502	-2.704
48	Pressure (bar)	96.58	96.42	96.38	96.34
49	Temperature (C)	60.05	60.01	60.00	59.99
50	Vapour Fraction	0.8711	0.8711	0.8712	0.8712
51	Heat Transferred (kJ/h-m)	793.6	792.6	800.8	800.5
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	1.000	1.000	0.7881	0.7879
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	5.381e-003	5.389e-003	5.455e-003	5.458e-003
57	Static Gradient (kPa/m)	3.096	3.096	0.8503	0.8501
58	Accel. Gradient (kPa/m)	1.963e-004	1.970e-004	5.439e-005	5.443e-005
59	Liq. Reynolds	2.243e+005 *	2.241e+005 *	2.240e+005 *	2.240e+005 *
60	Vap. Reynolds	8.940e+006 *	8.944e+006 *	8.945e+006 *	8.947e+006 *
61	Liquid Velocity (m/s)	0.2051	0.2051	0.2050	0.2050
62	Vapour Velocity (m/s)	2.073	2.077	2.078	2.079
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	5970	5975	5980	5985
66	Elevation (m)	-1.906	-1.108	-0.3100	-3.048
67	Pressure (bar)	96.29	96.25	96.21	96.24
68	Temperature (C)	59.98	59.96	59.95	59.96
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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9	PROFILES				
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11	Pipe Table				
12					
13	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
14	Heat Transferred (kJ/h-m)	800.2	799.8	799.5	799.6
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.7877	0.7875	0.7874	8.969e-002
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	5.460e-003	5.462e-003	5.464e-003	9.466e-003
20	Static Gradient (kPa/m)	0.8500	0.8498	0.8497	-0.6927
21	Accel. Gradient (kPa/m)	5.447e-005	5.451e-005	5.455e-005	-4.356e-005
22	Liq. Reynolds	2.239e+005 *	2.239e+005 *	2.238e+005 *	2.238e+005 *
23	Vap. Reynolds	8.948e+006 *	8.949e+006 *	8.951e+006 *	8.950e+006 *
24	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
25	Vapour Velocity (m/s)	2.080	2.080	2.081	2.081
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	5990	5995	6000	6005
29	Elevation (m)	-5.786	-8.524	-11.26	-14.00
30	Pressure (bar)	96.28	96.31	96.34	96.38
31	Temperature (C)	59.98	59.99	60.00	60.02
32	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
33	Heat Transferred (kJ/h-m)	799.9	800.3	800.6	801.0
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	8.972e-002	8.975e-002	8.978e-002	8.982e-002
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	9.462e-003	9.458e-003	9.454e-003	9.450e-003
39	Static Gradient (kPa/m)	-0.6929	-0.6931	-0.6934	-0.6936
40	Accel. Gradient (kPa/m)	-4.354e-005	-4.352e-005	-4.351e-005	-4.349e-005
41	Liq. Reynolds	2.239e+005 *	2.239e+005 *	2.240e+005 *	2.240e+005 *
42	Vap. Reynolds	8.949e+006 *	8.948e+006 *	8.947e+006 *	8.946e+006 *
43	Liquid Velocity (m/s)	0.2050	0.2050	0.2050	0.2050
44	Vapour Velocity (m/s)	2.080	2.079	2.079	2.078
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	6010	6015	6020	6025
48	Elevation (m)	-11.27	-8.544	-5.816	-3.088
49	Pressure (bar)	96.20	96.02	95.84	95.66
50	Temperature (C)	59.97	59.93	59.89	59.85
51	Vapour Fraction	0.8712	0.8713	0.8714	0.8715
52	Heat Transferred (kJ/h-m)	791.6	790.5	789.4	788.2
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	1.000	1.000	1.000	1.000
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	5.400e-003	5.410e-003	5.419e-003	5.428e-003
58	Static Gradient (kPa/m)	3.580	3.580	3.581	3.582
59	Accel. Gradient (kPa/m)	2.288e-004	2.297e-004	2.306e-004	2.315e-004
60	Liq. Reynolds	2.238e+005 *	2.236e+005 *	2.233e+005 *	2.231e+005 *
61	Vap. Reynolds	8.951e+006 *	8.956e+006 *	8.962e+006 *	8.967e+006 *
62	Liquid Velocity (m/s)	0.2050	0.2049	0.2049	0.2048
63	Vapour Velocity (m/s)	2.082	2.086	2.090	2.094
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	6030	6035	6040	6045
67	Elevation (m)	-0.3600	-0.2940	-0.2280	-0.1620
68	Pressure (bar)	95.48	95.48	95.48	95.48
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
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9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	59.80	59.80	59.80	59.80
14	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
15	Heat Transferred (kJ/h-m)	787.1	794.0	793.9	793.9
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	1.000	0.3281	0.3281	0.3281
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	5.437e-003	5.879e-003	5.879e-003	5.879e-003
21	Static Gradient (kPa/m)	3.582	3.498e-002	3.498e-002	3.498e-002
22	Accel. Gradient (kPa/m)	2.325e-004	2.647e-006	2.647e-006	2.648e-006
23	Liq. Reynolds	2.228e+005 *	2.228e+005 *	2.228e+005 *	2.228e+005 *
24	Vap. Reynolds	8.973e+006 *	8.973e+006 *	8.973e+006 *	8.973e+006 *
25	Liquid Velocity (m/s)	0.2048	0.2048	0.2048	0.2048
26	Vapour Velocity (m/s)	2.098	2.098	2.098	2.098
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	6050	6055	6060	6065
30	Elevation (m)	-9.600e-002	-3.000e-002	-5.200e-002	-7.400e-002
31	Pressure (bar)	95.47	95.47	95.47	95.47
32	Temperature (C)	59.79	59.79	59.79	59.79
33	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
34	Heat Transferred (kJ/h-m)	793.8	793.7	793.7	793.6
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.3281	0.3281	0.2073	0.2073
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	5.879e-003	5.879e-003	6.673e-003	6.673e-003
40	Static Gradient (kPa/m)	3.498e-002	3.498e-002	-8.561e-003	-8.561e-003
41	Accel. Gradient (kPa/m)	2.648e-006	2.648e-006	-1.223e-007	-1.223e-007
42	Liq. Reynolds	2.228e+005 *	2.228e+005 *	2.228e+005 *	2.228e+005 *
43	Vap. Reynolds	8.973e+006 *	8.973e+006 *	8.973e+006 *	8.973e+006 *
44	Liquid Velocity (m/s)	0.2048	0.2048	0.2048	0.2048
45	Vapour Velocity (m/s)	2.098	2.098	2.098	2.098
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	6070	6075	6080	6085
49	Elevation (m)	-9.600e-002	-0.1180	-0.1400	-2.232
50	Pressure (bar)	95.47	95.47	95.47	95.50
51	Temperature (C)	59.78	59.78	59.78	59.79
52	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
53	Heat Transferred (kJ/h-m)	793.5	793.5	793.4	794.9
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2073	0.2073	0.2073	8.895e-002
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	6.673e-003	6.673e-003	6.673e-003	9.558e-003
59	Static Gradient (kPa/m)	-8.561e-003	-8.561e-003	-8.561e-003	-0.5196
60	Accel. Gradient (kPa/m)	-1.223e-007	-1.224e-007	-1.224e-007	-3.303e-005
61	Liq. Reynolds	2.228e+005 *	2.228e+005 *	2.228e+005 *	2.228e+005 *
62	Vap. Reynolds	8.973e+006 *	8.973e+006 *	8.973e+006 *	8.972e+006 *
63	Liquid Velocity (m/s)	0.2048	0.2048	0.2048	0.2048
64	Vapour Velocity (m/s)	2.098	2.098	2.098	2.098
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	6090	6095	6101	6106
68	Elevation (m)	-4.324	-6.416	-8.508	-10.60
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Pressure (bar)	95.52	95.55	95.58	95.60
14	Temperature (C)	59.80	59.81	59.82	59.83
15	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
16	Heat Transferred (kJ/h-m)	795.2	795.4	795.7	795.9
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	8.897e-002	8.900e-002	8.902e-002	8.905e-002
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	9.555e-003	9.552e-003	9.549e-003	9.546e-003
22	Static Gradient (kPa/m)	-0.5197	-0.5198	-0.5200	-0.5201
23	Accel. Gradient (kPa/m)	-3.303e-005	-3.302e-005	-3.301e-005	-3.300e-005
24	Liq. Reynolds	2.229e+005 *	2.229e+005 *	2.230e+005 *	2.230e+005 *
25	Vap. Reynolds	8.971e+006 *	8.971e+006 *	8.970e+006 *	8.969e+006 *
26	Liquid Velocity (m/s)	0.2048	0.2048	0.2048	0.2048
27	Vapour Velocity (m/s)	2.097	2.097	2.096	2.095
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	6111	6116	6121	6126
31	Elevation (m)	-8.542	-6.484	-4.426	-2.368
32	Pressure (bar)	95.47	95.33	95.20	95.06
33	Temperature (C)	59.79	59.76	59.73	59.70
34	Vapour Fraction	0.8715	0.8716	0.8717	0.8717
35	Heat Transferred (kJ/h-m)	786.8	786.0	785.1	784.2
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	1.000	1.000	1.000	1.000
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	5.438e-003	5.444e-003	5.451e-003	5.458e-003
41	Static Gradient (kPa/m)	2.703	2.703	2.703	2.704
42	Accel. Gradient (kPa/m)	1.755e-004	1.761e-004	1.766e-004	1.772e-004
43	Liq. Reynolds	2.228e+005 *	2.226e+005 *	2.225e+005 *	2.223e+005 *
44	Vap. Reynolds	8.973e+006 *	8.977e+006 *	8.981e+006 *	8.985e+006 *
45	Liquid Velocity (m/s)	0.2048	0.2048	0.2047	0.2047
46	Vapour Velocity (m/s)	2.099	2.102	2.105	2.108
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	6131	6136	6141	6146
50	Elevation (m)	-0.3100	-0.2880	-0.2660	-0.2440
51	Pressure (bar)	94.92	94.92	94.92	94.92
52	Temperature (C)	59.66	59.66	59.66	59.66
53	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
54	Heat Transferred (kJ/h-m)	783.4	790.2	790.2	790.1
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	1.000	0.2985	0.2985	0.2985
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	5.465e-003	4.966e-003	4.966e-003	4.966e-003
60	Static Gradient (kPa/m)	2.704	1.089e-002	1.089e-002	1.089e-002
61	Accel. Gradient (kPa/m)	1.777e-004	1.040e-006	1.040e-006	1.040e-006
62	Liq. Reynolds	2.221e+005 *	2.221e+005 *	2.221e+005 *	2.221e+005 *
63	Vap. Reynolds	8.990e+006 *	8.990e+006 *	8.990e+006 *	8.990e+006 *
64	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
65	Vapour Velocity (m/s)	2.111	2.111	2.111	2.111
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	6151	6156	6161	6166
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3			Unit Set: Mixed		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
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9	PROFILES				
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11	Pipe Table				
12					
13	Elevation (m)	-0.2220	-0.2000	-0.1940	-0.1880
14	Pressure (bar)	94.92	94.92	94.92	94.92
15	Temperature (C)	59.65	59.65	59.65	59.65
16	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
17	Heat Transferred (kJ/h-m)	790.1	790.0	789.9	789.9
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2985	0.2985	0.2881	0.2881
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	4.966e-003	4.966e-003	5.122e-003	5.122e-003
23	Static Gradient (kPa/m)	1.089e-002	1.089e-002	2.898e-003	2.898e-003
24	Accel. Gradient (kPa/m)	1.040e-006	1.040e-006	5.259e-007	5.259e-007
25	Liq. Reynolds	2.221e+005 *	2.221e+005 *	2.221e+005 *	2.221e+005 *
26	Vap. Reynolds	8.990e+006 *	8.990e+006 *	8.990e+006 *	8.990e+006 *
27	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
28	Vapour Velocity (m/s)	2.111	2.111	2.111	2.111
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	6171	6176	6181	6186
32	Elevation (m)	-0.1820	-0.1760	-0.1700	-0.1500
33	Pressure (bar)	94.92	94.92	94.92	94.92
34	Temperature (C)	59.64	59.64	59.64	59.64
35	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
36	Heat Transferred (kJ/h-m)	789.8	789.7	789.7	789.6
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2881	0.2881	0.2881	0.2972
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	5.122e-003	5.122e-003	5.122e-003	4.492e-003
42	Static Gradient (kPa/m)	2.898e-003	2.898e-003	2.898e-003	9.870e-003
43	Accel. Gradient (kPa/m)	5.259e-007	5.259e-007	5.259e-007	9.418e-007
44	Liq. Reynolds	2.221e+005 *	2.221e+005 *	2.221e+005 *	2.221e+005 *
45	Vap. Reynolds	8.990e+006 *	8.990e+006 *	8.990e+006 *	8.990e+006 *
46	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
47	Vapour Velocity (m/s)	2.111	2.111	2.111	2.111
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	6191	6196	6201	6206
51	Elevation (m)	-0.1300	-0.1100	-9.000e-002	-7.000e-002
52	Pressure (bar)	94.92	94.92	94.92	94.91
53	Temperature (C)	59.63	59.63	59.63	59.63
54	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
55	Heat Transferred (kJ/h-m)	789.5	789.5	789.4	789.4
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2972	0.2972	0.2972	0.2972
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	4.492e-003	4.492e-003	4.492e-003	4.492e-003
61	Static Gradient (kPa/m)	9.870e-003	9.870e-003	9.870e-003	9.870e-003
62	Accel. Gradient (kPa/m)	9.418e-007	9.418e-007	9.418e-007	9.418e-007
63	Liq. Reynolds	2.220e+005 *	2.220e+005 *	2.220e+005 *	2.220e+005 *
64	Vap. Reynolds	8.990e+006 *	8.990e+006 *	8.990e+006 *	8.990e+006 *
65	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
66	Vapour Velocity (m/s)	2.111	2.111	2.111	2.111
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
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5			Unit Set: Mixed			
6			Date/Time: Mon Jun 13 15:02:44 2022			
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	PROFILES					
11	Pipe Table					
12						
13	Length (m)	6211	6216	6221	6226	
14	Elevation (m)	-6.800e-002	-6.600e-002	-6.400e-002	-6.200e-002	
15	Pressure (bar)	94.91	94.91	94.91	94.91	
16	Temperature (C)	59.63	59.62	59.62	59.62	
17	Vapour Fraction	0.8718	0.8718	0.8718	0.8718	
18	Heat Transferred (kJ/h-m)	789.3	789.2	789.2	789.1	
19	Flow Regime	Transition	Transition	Transition	Transition	
20	Water Flow Regime					
21	Liquid HoldUp	0.2855	0.2855	0.2855	0.2855	
22	Water HoldUp	---	---	---	---	
23	Friction Gradient (kPa/m)	5.313e-003	5.313e-003	5.313e-003	5.313e-003	
24	Static Gradient (kPa/m)	9.598e-004	9.599e-004	9.599e-004	9.599e-004	
25	Accel. Gradient (kPa/m)	4.114e-007	4.114e-007	4.114e-007	4.113e-007	
26	Liq. Reynolds	2.220e+005 *	2.220e+005 *	2.220e+005 *	2.220e+005 *	
27	Vap. Reynolds	8.990e+006 *	8.990e+006 *	8.990e+006 *	8.990e+006 *	
28	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047	
29	Vapour Velocity (m/s)	2.111	2.111	2.111	2.111	
30	Deposit Thickness (mm)	---	---	---	---	
31	Deposit Volume (m3)	---	---	---	---	
32	Length (m)	6231	6236	6241	6246	
33	Elevation (m)	-6.000e-002	-8.400e-002	-0.1080	-0.1320	
34	Pressure (bar)	94.91	94.91	94.91	94.91	
35	Temperature (C)	59.62	59.61	59.61	59.61	
36	Vapour Fraction	0.8718	0.8717	0.8717	0.8717	
37	Heat Transferred (kJ/h-m)	789.0	789.0	788.9	788.9	
38	Flow Regime	Transition	Transition	Transition	Transition	
39	Water Flow Regime					
40	Liquid HoldUp	0.2855	0.2062	0.2062	0.2062	
41	Water HoldUp	---	---	---	---	
42	Friction Gradient (kPa/m)	5.313e-003	6.712e-003	6.712e-003	6.712e-003	
43	Static Gradient (kPa/m)	9.599e-004	-9.294e-003	-9.294e-003	-9.295e-003	
44	Accel. Gradient (kPa/m)	4.113e-007	-1.693e-007	-1.693e-007	-1.694e-007	
45	Liq. Reynolds	2.220e+005 *	2.220e+005 *	2.220e+005 *	2.220e+005 *	
46	Vap. Reynolds	8.990e+006 *	8.990e+006 *	8.990e+006 *	8.990e+006 *	
47	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047	
48	Vapour Velocity (m/s)	2.111	2.111	2.111	2.111	
49	Deposit Thickness (mm)	---	---	---	---	
50	Deposit Volume (m3)	---	---	---	---	
51	Length (m)	6251	6256	6261	6266	
52	Elevation (m)	-0.1560	-0.1800	-0.1720	-0.1640	
53	Pressure (bar)	94.91	94.91	94.91	94.91	
54	Temperature (C)	59.61	59.61	59.60	59.60	
55	Vapour Fraction	0.8717	0.8717	0.8717	0.8717	
56	Heat Transferred (kJ/h-m)	788.8	788.8	788.7	788.6	
57	Flow Regime	Transition	Transition	Transition	Transition	
58	Water Flow Regime					
59	Liquid HoldUp	0.2062	0.2062	0.2894	0.2894	
60	Water HoldUp	---	---	---	---	
61	Friction Gradient (kPa/m)	6.712e-003	6.712e-003	5.028e-003	5.028e-003	
62	Static Gradient (kPa/m)	-9.295e-003	-9.295e-003	3.876e-003	3.876e-003	
63	Accel. Gradient (kPa/m)	-1.694e-007	-1.694e-007	5.838e-007	5.838e-007	
64	Liq. Reynolds	2.220e+005 *	2.220e+005 *	2.220e+005 *	2.220e+005 *	
65	Vap. Reynolds	8.990e+006 *	8.990e+006 *	8.990e+006 *	8.990e+006 *	
66	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047	
67	Vapour Velocity (m/s)	2.111	2.111	2.111	2.111	
68	Deposit Thickness (mm)	---	---	---	---	
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
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5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	6271	6276	6281	6285
15	Elevation (m)	-0.1560	-0.1480	-0.1400	-2.292
16	Pressure (bar)	94.91	94.91	94.91	94.94
17	Temperature (C)	59.60	59.60	59.59	59.60
18	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
19	Heat Transferred (kJ/h-m)	788.6	788.5	788.4	790.0
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2894	0.2894	0.2894	8.842e-002
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	5.027e-003	5.027e-003	5.027e-003	9.626e-003
25	Static Gradient (kPa/m)	3.876e-003	3.876e-003	3.876e-003	-0.5919
26	Accel. Gradient (kPa/m)	5.838e-007	5.838e-007	5.838e-007	-3.816e-005
27	Liq. Reynolds	2.220e+005 *	2.220e+005 *	2.220e+005 *	2.220e+005 *
28	Vap. Reynolds	8.990e+006 *	8.990e+006 *	8.990e+006 *	8.989e+006 *
29	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
30	Vapour Velocity (m/s)	2.111	2.111	2.111	2.110
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	6290	6294	6299	6303
34	Elevation (m)	-4.444	-6.596	-8.748	-10.90
35	Pressure (bar)	94.96	94.99	95.02	95.04
36	Temperature (C)	59.61	59.62	59.63	59.64
37	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
38	Heat Transferred (kJ/h-m)	790.2	790.5	790.8	791.0
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	8.845e-002	8.847e-002	8.849e-002	8.852e-002
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	9.623e-003	9.619e-003	9.616e-003	9.613e-003
44	Static Gradient (kPa/m)	-0.5920	-0.5921	-0.5923	-0.5924
45	Accel. Gradient (kPa/m)	-3.814e-005	-3.813e-005	-3.812e-005	-3.811e-005
46	Liq. Reynolds	2.221e+005 *	2.221e+005 *	2.222e+005 *	2.222e+005 *
47	Vap. Reynolds	8.988e+006 *	8.987e+006 *	8.987e+006 *	8.986e+006 *
48	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
49	Vapour Velocity (m/s)	2.109	2.109	2.108	2.108
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	6308	6313	6318	6323
53	Elevation (m)	-10.08	-9.260	-8.440	-7.620
54	Pressure (bar)	95.00	94.96	94.91	94.87
55	Temperature (C)	59.63	59.62	59.61	59.59
56	Vapour Fraction	0.8717	0.8717	0.8717	0.8718
57	Heat Transferred (kJ/h-m)	791.0	790.6	790.3	790.0
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.7946	0.7944	0.7942	0.7940
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	5.522e-003	5.524e-003	5.527e-003	5.529e-003
63	Static Gradient (kPa/m)	0.8808	0.8807	0.8805	0.8804
64	Accel. Gradient (kPa/m)	5.801e-005	5.806e-005	5.810e-005	5.815e-005
65	Liq. Reynolds	2.221e+005 *	2.221e+005 *	2.220e+005 *	2.220e+005 *
66	Vap. Reynolds	8.987e+006 *	8.989e+006 *	8.990e+006 *	8.991e+006 *
67	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2046
68	Vapour Velocity (m/s)	2.109	2.110	2.111	2.112
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	6328	6333	6338	6343
16	Elevation (m)	-6.800	-7.560	-8.320	-9.080
17	Pressure (bar)	94.82	94.83	94.84	94.85
18	Temperature (C)	59.58	59.58	59.58	59.59
19	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
20	Heat Transferred (kJ/h-m)	789.6	788.7	788.7	788.8
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.7938	7.104e-002	7.105e-002	7.105e-002
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	5.531e-003	1.113e-002	1.113e-002	1.113e-002
26	Static Gradient (kPa/m)	0.8802	-0.1743	-0.1743	-0.1744
27	Accel. Gradient (kPa/m)	5.819e-005	-1.072e-005	-1.072e-005	-1.072e-005
28	Liq. Reynolds	2.219e+005 *	2.219e+005 *	2.219e+005 *	2.219e+005 *
29	Vap. Reynolds	8.993e+006 *	8.992e+006 *	8.992e+006 *	8.992e+006 *
30	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
31	Vapour Velocity (m/s)	2.113	2.113	2.112	2.112
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	6348	6353	6358	6363
35	Elevation (m)	-9.840	-10.60	-11.94	-13.28
36	Pressure (bar)	94.86	94.86	94.88	94.90
37	Temperature (C)	59.59	59.59	59.59	59.60
38	Vapour Fraction	0.8718	0.8718	0.8718	0.8717
39	Heat Transferred (kJ/h-m)	788.8	788.9	789.7	789.9
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	7.105e-002	7.105e-002	8.836e-002	8.838e-002
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	1.113e-002	1.113e-002	9.633e-003	9.631e-003
45	Static Gradient (kPa/m)	-0.1744	-0.1744	-0.3346	-0.3346
46	Accel. Gradient (kPa/m)	-1.072e-005	-1.072e-005	-2.132e-005	-2.132e-005
47	Liq. Reynolds	2.219e+005 *	2.219e+005 *	2.220e+005 *	2.220e+005 *
48	Vap. Reynolds	8.992e+006 *	8.991e+006 *	8.991e+006 *	8.990e+006 *
49	Liquid Velocity (m/s)	0.2046	0.2046	0.2047	0.2047
50	Vapour Velocity (m/s)	2.112	2.112	2.111	2.111
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	6368	6373	6378	6383
54	Elevation (m)	-14.62	-15.96	-17.30	-16.60
55	Pressure (bar)	94.91	94.93	94.94	94.91
56	Temperature (C)	59.60	59.61	59.62	59.60
57	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
58	Heat Transferred (kJ/h-m)	790.0	790.1	790.3	790.2
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	8.839e-002	8.841e-002	8.842e-002	0.7242
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	9.630e-003	9.628e-003	9.626e-003	5.557e-003
64	Static Gradient (kPa/m)	-0.3347	-0.3347	-0.3348	0.6945
65	Accel. Gradient (kPa/m)	-2.131e-005	-2.131e-005	-2.131e-005	4.591e-005
66	Liq. Reynolds	2.220e+005 *	2.220e+005 *	2.221e+005 *	2.220e+005 *
67	Vap. Reynolds	8.990e+006 *	8.989e+006 *	8.989e+006 *	8.990e+006 *
68	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
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4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
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9	PROFILES						
10	Pipe Table						
11							
12							
13	Vapour Velocity	(m/s)	2.111	2.110	2.110	2.111	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	6388	6393	6398	6403	
17	Elevation	(m)	-15.90	-15.20	-14.50	-13.80	
18	Pressure	(bar)	94.87	94.84	94.80	94.77	
19	Temperature	(C)	59.59	59.58	59.57	59.56	
20	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
21	Heat Transferred	(kJ/h-m)	789.9	789.6	789.3	789.1	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.7240	0.7239	0.7238	0.7236	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	5.559e-003	5.561e-003	5.563e-003	5.565e-003	
27	Static Gradient	(kPa/m)	0.6944	0.6943	0.6941	0.6940	
28	Accel. Gradient	(kPa/m)	4.593e-005	4.596e-005	4.599e-005	4.602e-005	
29	Liq. Reynolds		2.220e+005 *	2.219e+005 *	2.219e+005 *	2.218e+005 *	
30	Vap. Reynolds		8.991e+006 *	8.992e+006 *	8.993e+006 *	8.994e+006 *	
31	Liquid Velocity	(m/s)	0.2047	0.2046	0.2046	0.2046	
32	Vapour Velocity	(m/s)	2.112	2.112	2.113	2.114	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	6408	6413	6418	6423	
36	Elevation	(m)	-11.15	-8.508	-5.862	-3.216	
37	Pressure	(bar)	94.60	94.42	94.25	94.07	
38	Temperature	(C)	59.52	59.48	59.44	59.40	
39	Vapour Fraction		0.8719	0.8719	0.8720	0.8721	
40	Heat Transferred	(kJ/h-m)	779.6	778.5	777.4	776.3	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		1.000	1.000	1.000	1.000	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	5.481e-003	5.490e-003	5.499e-003	5.508e-003	
46	Static Gradient	(kPa/m)	3.478	3.479	3.480	3.480	
47	Accel. Gradient	(kPa/m)	2.300e-004	2.309e-004	2.318e-004	2.328e-004	
48	Liq. Reynolds		2.216e+005 *	2.213e+005 *	2.211e+005 *	2.209e+005 *	
49	Vap. Reynolds		8.999e+006 *	9.005e+006 *	9.010e+006 *	9.015e+006 *	
50	Liquid Velocity	(m/s)	0.2046	0.2045	0.2045	0.2044	
51	Vapour Velocity	(m/s)	2.118	2.122	2.126	2.130	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	6428	6433	6438	6443	
55	Elevation	(m)	-0.5700	-0.4840	-0.3980	-0.3120	
56	Pressure	(bar)	93.90	93.90	93.89	93.89	
57	Temperature	(C)	59.35	59.35	59.35	59.34	
58	Vapour Fraction		0.8722	0.8722	0.8722	0.8722	
59	Heat Transferred	(kJ/h-m)	775.2	782.0	781.9	781.8	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		1.000	0.3375	0.3375	0.3375	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	5.518e-003	5.974e-003	5.974e-003	5.974e-003	
65	Static Gradient	(kPa/m)	3.481	4.645e-002	4.645e-002	4.645e-002	
66	Accel. Gradient	(kPa/m)	2.337e-004	3.514e-006	3.514e-006	3.514e-006	
67	Liq. Reynolds		2.206e+005 *	2.206e+005 *	2.206e+005 *	2.206e+005 *	
68	Vap. Reynolds		9.021e+006 *	9.021e+006 *	9.021e+006 *	9.021e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3							
4			Unit Set: Mixed				
5			Date/Time: Mon Jun 13 15:02:44 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
14	Vapour Velocity	(m/s)	2.135	2.135	2.135	2.135	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	6448	6453	6459	6465	
18	Elevation	(m)	-0.2260	-0.1400	-0.1320	-0.1240	
19	Pressure	(bar)	93.89	93.89	93.88	93.88	
20	Temperature	(C)	59.34	59.34	59.34	59.33	
21	Vapour Fraction		0.8722	0.8722	0.8722	0.8722	
22	Heat Transferred	(kJ/h-m)	781.8	781.7	781.6	781.5	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.3375	0.3375	0.2865	0.2865	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	5.974e-003	5.974e-003	5.170e-003	5.170e-003	
28	Static Gradient	(kPa/m)	4.645e-002	4.645e-002	3.202e-003	3.202e-003	
29	Accel. Gradient	(kPa/m)	3.514e-006	3.514e-006	5.613e-007	5.613e-007	
30	Liq. Reynolds		2.206e+005 *	2.206e+005 *	2.206e+005 *	2.206e+005 *	
31	Vap. Reynolds		9.021e+006 *	9.021e+006 *	9.021e+006 *	9.021e+006 *	
32	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
33	Vapour Velocity	(m/s)	2.135	2.135	2.135	2.135	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	6471	6477	6483	6487	
37	Elevation	(m)	-0.1160	-0.1080	-0.1000	-9.800e-002	
38	Pressure	(bar)	93.88	93.88	93.88	93.88	
39	Temperature	(C)	59.33	59.33	59.32	59.32	
40	Vapour Fraction		0.8722	0.8722	0.8722	0.8722	
41	Heat Transferred	(kJ/h-m)	781.4	781.4	781.3	781.2	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2865	0.2865	0.2865	0.2839	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	5.170e-003	5.169e-003	5.169e-003	5.371e-003	
47	Static Gradient	(kPa/m)	3.202e-003	3.203e-003	3.203e-003	1.193e-003	
48	Accel. Gradient	(kPa/m)	5.613e-007	5.613e-007	5.613e-007	4.400e-007	
49	Liq. Reynolds		2.206e+005 *	2.206e+005 *	2.206e+005 *	2.206e+005 *	
50	Vap. Reynolds		9.021e+006 *	9.021e+006 *	9.021e+006 *	9.021e+006 *	
51	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
52	Vapour Velocity	(m/s)	2.135	2.135	2.135	2.135	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	6491	6495	6499	6503	
56	Elevation	(m)	-9.600e-002	-9.400e-002	-9.200e-002	-9.000e-002	
57	Pressure	(bar)	93.88	93.88	93.88	93.88	
58	Temperature	(C)	59.32	59.32	59.32	59.32	
59	Vapour Fraction		0.8722	0.8722	0.8722	0.8722	
60	Heat Transferred	(kJ/h-m)	781.2	781.1	781.1	781.0	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2839	0.2839	0.2839	0.2839	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	5.370e-003	5.370e-003	5.370e-003	5.370e-003	
66	Static Gradient	(kPa/m)	1.193e-003	1.193e-003	1.193e-003	1.193e-003	
67	Accel. Gradient	(kPa/m)	4.400e-007	4.400e-007	4.400e-007	4.400e-007	
68	Liq. Reynolds		2.206e+005 *	2.206e+005 *	2.206e+005 *	2.206e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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8					
9	PROFILES				
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11	Pipe Table				
12					
13	Vap. Reynolds	9.021e+006 *	9.021e+006 *	9.021e+006 *	9.021e+006 *
14	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
15	Vapour Velocity (m/s)	2.135	2.135	2.135	2.135
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	6508	6513	6518	6523
19	Elevation (m)	-6.400e-002	-3.800e-002	-1.200e-002	1.400e-002
20	Pressure (bar)	93.88	93.88	93.88	93.88
21	Temperature (C)	59.31	59.31	59.31	59.31
22	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
23	Heat Transferred (kJ/h-m)	781.0	780.9	780.8	780.8
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2990	0.2990	0.2990	0.2990
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	5.429e-003	5.429e-003	5.429e-003	5.429e-003
29	Static Gradient (kPa/m)	1.287e-002	1.287e-002	1.287e-002	1.287e-002
30	Accel. Gradient (kPa/m)	1.227e-006	1.227e-006	1.227e-006	1.227e-006
31	Liq. Reynolds	2.206e+005 *	2.206e+005 *	2.206e+005 *	2.206e+005 *
32	Vap. Reynolds	9.021e+006 *	9.021e+006 *	9.021e+006 *	9.021e+006 *
33	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
34	Vapour Velocity (m/s)	2.135	2.135	2.135	2.134
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	6528	6533	6538	6543
38	Elevation (m)	4.000e-002	3.800e-002	3.600e-002	3.400e-002
39	Pressure (bar)	93.88	93.88	93.88	93.88
40	Temperature (C)	59.30	59.30	59.30	59.30
41	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
42	Heat Transferred (kJ/h-m)	780.7	780.7	780.6	780.5
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2990	0.2089	0.2089	0.2089
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	5.429e-003	6.748e-003	6.748e-003	6.748e-003
48	Static Gradient (kPa/m)	1.287e-002	-7.808e-004	-7.808e-004	-7.808e-004
49	Accel. Gradient (kPa/m)	1.227e-006	4.001e-007	4.001e-007	4.001e-007
50	Liq. Reynolds	2.206e+005 *	2.206e+005 *	2.206e+005 *	2.206e+005 *
51	Vap. Reynolds	9.021e+006 *	9.021e+006 *	9.021e+006 *	9.021e+006 *
52	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
53	Vapour Velocity (m/s)	2.134	2.134	2.134	2.134
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	6548	6553	6558	6563
57	Elevation (m)	3.200e-002	3.000e-002	2.800e-002	2.600e-002
58	Pressure (bar)	93.88	93.88	93.87	93.87
59	Temperature (C)	59.29	59.29	59.29	59.29
60	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
61	Heat Transferred (kJ/h-m)	780.5	780.4	780.4	780.3
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2089	0.2089	0.2089	0.2089
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	6.748e-003	6.748e-003	6.748e-003	6.748e-003
67	Static Gradient (kPa/m)	-7.808e-004	-7.808e-004	-7.790e-004	-7.790e-004
68	Accel. Gradient (kPa/m)	4.001e-007	4.001e-007	4.002e-007	4.002e-007
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.206e+005 *	2.206e+005 *	2.206e+005 *	2.205e+005 *
14	Vap. Reynolds	9.021e+006 *	9.021e+006 *	9.021e+006 *	9.021e+006 *
15	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
16	Vapour Velocity (m/s)	2.134	2.134	2.134	2.134
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	6568	6573	6578	6583
20	Elevation (m)	2.400e-002	2.200e-002	2.000e-002	3.200e-002
21	Pressure (bar)	93.87	93.87	93.87	93.87
22	Temperature (C)	59.29	59.28	59.28	59.28
23	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
24	Heat Transferred (kJ/h-m)	780.2	780.2	780.1	780.0
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2089	0.2089	0.2089	0.2900
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	6.748e-003	6.748e-003	6.748e-003	4.919e-003
30	Static Gradient (kPa/m)	-7.790e-004	-7.790e-004	-7.790e-004	5.813e-003
31	Accel. Gradient (kPa/m)	4.002e-007	4.002e-007	4.002e-007	7.195e-007
32	Liq. Reynolds	2.205e+005 *	2.205e+005 *	2.205e+005 *	2.205e+005 *
33	Vap. Reynolds	9.021e+006 *	9.021e+006 *	9.021e+006 *	9.021e+006 *
34	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
35	Vapour Velocity (m/s)	2.134	2.134	2.134	2.134
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	6588	6593	6598	6603
39	Elevation (m)	4.400e-002	5.600e-002	6.800e-002	8.000e-002
40	Pressure (bar)	93.87	93.87	93.87	93.87
41	Temperature (C)	59.28	59.27	59.27	59.27
42	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
43	Heat Transferred (kJ/h-m)	780.0	779.9	779.9	779.8
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2900	0.2900	0.2900	0.2900
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	4.919e-003	4.919e-003	4.919e-003	4.919e-003
49	Static Gradient (kPa/m)	5.813e-003	5.813e-003	5.814e-003	5.814e-003
50	Accel. Gradient (kPa/m)	7.195e-007	7.195e-007	7.195e-007	7.195e-007
51	Liq. Reynolds	2.205e+005 *	2.205e+005 *	2.205e+005 *	2.205e+005 *
52	Vap. Reynolds	9.021e+006 *	9.021e+006 *	9.021e+006 *	9.021e+006 *
53	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
54	Vapour Velocity (m/s)	2.134	2.134	2.134	2.134
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	6608	6613	6618	6623
58	Elevation (m)	7.200e-002	6.400e-002	5.600e-002	4.800e-002
59	Pressure (bar)	93.87	93.87	93.87	93.87
60	Temperature (C)	59.27	59.26	59.26	59.26
61	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
62	Heat Transferred (kJ/h-m)	779.7	779.7	779.6	779.6
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2078	0.2078	0.2078	0.2078
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	6.756e-003	6.756e-003	6.756e-003	6.756e-003
68	Static Gradient (kPa/m)	-3.105e-003	-3.105e-003	-3.106e-003	-3.106e-003
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3							Unit Set: Mixed
4							Date/Time: Mon Jun 13 15:02:44 2022
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Accel. Gradient (kPa/m)	2.447e-007	2.447e-007	2.447e-007	2.447e-007		
14	Liq. Reynolds	2.205e+005 *	2.205e+005 *	2.205e+005 *	2.205e+005 *		
15	Vap. Reynolds	9.021e+006 *	9.021e+006 *	9.021e+006 *	9.021e+006 *		
16	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
17	Vapour Velocity (m/s)	2.134	2.134	2.134	2.134		
18	Deposit Thickness (mm)	---	---	---	---		
19	Deposit Volume (m3)	---	---	---	---		
20	Length (m)	6628	6633	6638	6643		
21	Elevation (m)	4.000e-002	2.800e-002	1.600e-002	4.000e-003		
22	Pressure (bar)	93.87	93.87	93.87	93.87		
23	Temperature (C)	59.26	59.26	59.25	59.25		
24	Vapour Fraction	0.8721	0.8721	0.8721	0.8721		
25	Heat Transferred (kJ/h-m)	779.5	779.5	779.4	779.3		
26	Flow Regime	Transition	Transition	Transition	Transition		
27	Water Flow Regime						
28	Liquid HoldUp	0.2078	0.2070	0.2070	0.2070		
29	Water HoldUp	---	---	---	---		
30	Friction Gradient (kPa/m)	6.756e-003	6.762e-003	6.762e-003	6.762e-003		
31	Static Gradient (kPa/m)	-3.106e-003	-4.648e-003	-4.648e-003	-4.648e-003		
32	Accel. Gradient (kPa/m)	2.447e-007	1.417e-007	1.417e-007	1.417e-007		
33	Liq. Reynolds	2.205e+005 *	2.205e+005 *	2.205e+005 *	2.205e+005 *		
34	Vap. Reynolds	9.021e+006 *	9.021e+006 *	9.021e+006 *	9.021e+006 *		
35	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
36	Vapour Velocity (m/s)	2.134	2.134	2.134	2.134		
37	Deposit Thickness (mm)	---	---	---	---		
38	Deposit Volume (m3)	---	---	---	---		
39	Length (m)	6648	6653	6658	6663		
40	Elevation (m)	-8.000e-003	-2.000e-002	-2.000e-003	1.600e-002		
41	Pressure (bar)	93.87	93.87	93.87	93.87		
42	Temperature (C)	59.25	59.25	59.24	59.24		
43	Vapour Fraction	0.8721	0.8721	0.8721	0.8721		
44	Heat Transferred (kJ/h-m)	779.3	779.2	779.1	779.1		
45	Flow Regime	Transition	Transition	Transition	Transition		
46	Water Flow Regime						
47	Liquid HoldUp	0.2070	0.2070	0.2939	0.2939		
48	Water HoldUp	---	---	---	---		
49	Friction Gradient (kPa/m)	6.762e-003	6.762e-003	4.648e-003	4.648e-003		
50	Static Gradient (kPa/m)	-4.648e-003	-4.648e-003	8.802e-003	8.802e-003		
51	Accel. Gradient (kPa/m)	1.417e-007	1.417e-007	9.017e-007	9.017e-007		
52	Liq. Reynolds	2.205e+005 *	2.205e+005 *	2.205e+005 *	2.205e+005 *		
53	Vap. Reynolds	9.021e+006 *	9.021e+006 *	9.021e+006 *	9.021e+006 *		
54	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
55	Vapour Velocity (m/s)	2.134	2.134	2.134	2.134		
56	Deposit Thickness (mm)	---	---	---	---		
57	Deposit Volume (m3)	---	---	---	---		
58	Length (m)	6668	6673	6678	6683		
59	Elevation (m)	3.400e-002	5.200e-002	7.000e-002	4.000e-002		
60	Pressure (bar)	93.87	93.87	93.87	93.87		
61	Temperature (C)	59.24	59.24	59.24	59.23		
62	Vapour Fraction	0.8721	0.8721	0.8721	0.8721		
63	Heat Transferred (kJ/h-m)	779.0	779.0	778.9	778.9		
64	Flow Regime	Transition	Transition	Transition	Transition		
65	Water Flow Regime						
66	Liquid HoldUp	0.2939	0.2939	0.2939	0.2036		
67	Water HoldUp	---	---	---	---		
68	Friction Gradient (kPa/m)	4.648e-003	4.648e-003	4.648e-003	6.788e-003		
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3	 Epic Atlantic Limited		Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	8.802e-003	8.802e-003	8.803e-003	-1.150e-002
14	Accel. Gradient	(kPa/m)	9.017e-007	9.017e-007	9.017e-007	-3.159e-007
15	Liq. Reynolds		2.205e+005 *	2.205e+005 *	2.205e+005 *	2.205e+005 *
16	Vap. Reynolds		9.021e+006 *	9.021e+006 *	9.022e+006 *	9.021e+006 *
17	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
18	Vapour Velocity	(m/s)	2.134	2.134	2.134	2.134
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	6688	6693	6698	6703
22	Elevation	(m)	1.000e-002	-2.000e-002	-5.000e-002	-8.000e-002
23	Pressure	(bar)	93.87	93.87	93.87	93.87
24	Temperature	(C)	59.23	59.23	59.23	59.22
25	Vapour Fraction		0.8721	0.8721	0.8721	0.8721
26	Heat Transferred	(kJ/h-m)	778.8	778.7	778.7	778.6
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2036	0.2036	0.2036	0.2036
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	6.788e-003	6.788e-003	6.788e-003	6.788e-003
32	Static Gradient	(kPa/m)	-1.150e-002	-1.150e-002	-1.150e-002	-1.150e-002
33	Accel. Gradient	(kPa/m)	-3.159e-007	-3.159e-007	-3.159e-007	-3.159e-007
34	Liq. Reynolds		2.205e+005 *	2.205e+005 *	2.205e+005 *	2.205e+005 *
35	Vap. Reynolds		9.021e+006 *	9.021e+006 *	9.021e+006 *	9.021e+006 *
36	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
37	Vapour Velocity	(m/s)	2.134	2.134	2.134	2.134
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	6708	6713	6718	6723
41	Elevation	(m)	-0.1140	-0.1480	-0.1820	-0.2160
42	Pressure	(bar)	93.87	93.87	93.87	93.87
43	Temperature	(C)	59.22	59.22	59.22	59.22
44	Vapour Fraction		0.8721	0.8721	0.8721	0.8721
45	Heat Transferred	(kJ/h-m)	778.6	778.5	778.5	778.4
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2029	0.2029	0.2029	0.2029
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	6.794e-003	6.794e-003	6.794e-003	6.794e-003
51	Static Gradient	(kPa/m)	-1.301e-002	-1.301e-002	-1.301e-002	-1.301e-002
52	Accel. Gradient	(kPa/m)	-4.163e-007	-4.163e-007	-4.163e-007	-4.163e-007
53	Liq. Reynolds		2.205e+005 *	2.205e+005 *	2.205e+005 *	2.205e+005 *
54	Vap. Reynolds		9.021e+006 *	9.021e+006 *	9.021e+006 *	9.021e+006 *
55	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
56	Vapour Velocity	(m/s)	2.134	2.134	2.134	2.134
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	6728	6733	6738	6743
60	Elevation	(m)	-0.2500	-2.800	-5.350	-7.900
61	Pressure	(bar)	93.87	93.90	93.93	93.96
62	Temperature	(C)	59.21	59.23	59.24	59.25
63	Vapour Fraction		0.8721	0.8721	0.8721	0.8721
64	Heat Transferred	(kJ/h-m)	778.4	779.9	780.2	780.6
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2029	8.746e-002	8.748e-002	8.751e-002
68	Water HoldUp		---	---	---	---
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1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	6.794e-003	9.752e-003	9.748e-003	9.744e-003
14	Static Gradient	(kPa/m)	-1.301e-002	-0.6244	-0.6246	-0.6248
15	Accel. Gradient	(kPa/m)	-4.163e-007	-4.117e-005	-4.115e-005	-4.114e-005
16	Liq. Reynolds		2.205e+005 *	2.205e+005 *	2.206e+005 *	2.206e+005 *
17	Vap. Reynolds		9.021e+006 *	9.020e+006 *	9.019e+006 *	9.019e+006 *
18	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
19	Vapour Velocity	(m/s)	2.134	2.133	2.132	2.132
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	6748	6754	6759	6764
23	Elevation	(m)	-10.45	-13.00	-12.02	-11.04
24	Pressure	(bar)	93.99	94.02	93.97	93.91
25	Temperature	(C)	59.26	59.27	59.26	59.24
26	Vapour Fraction		0.8720	0.8720	0.8721	0.8721
27	Heat Transferred	(kJ/h-m)	780.9	781.2	781.0	780.6
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		8.754e-002	8.757e-002	0.8791	0.8788
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	9.740e-003	9.737e-003	5.544e-003	5.547e-003
33	Static Gradient	(kPa/m)	-0.6249	-0.6251	1.151	1.150
34	Accel. Gradient	(kPa/m)	-4.113e-005	-4.111e-005	7.733e-005	7.741e-005
35	Liq. Reynolds		2.207e+005 *	2.207e+005 *	2.206e+005 *	2.205e+005 *
36	Vap. Reynolds		9.018e+006 *	9.017e+006 *	9.018e+006 *	9.020e+006 *
37	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
38	Vapour Velocity	(m/s)	2.131	2.130	2.132	2.133
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	6769	6774	6779	6784
42	Elevation	(m)	-10.06	-9.080	-8.100	-7.740
43	Pressure	(bar)	93.85	93.79	93.74	93.72
44	Temperature	(C)	59.23	59.21	59.20	59.19
45	Vapour Fraction		0.8721	0.8721	0.8722	0.8722
46	Heat Transferred	(kJ/h-m)	780.2	779.7	779.3	777.9
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.8785	0.8782	0.8779	0.5117
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	5.550e-003	5.553e-003	5.556e-003	5.765e-003
52	Static Gradient	(kPa/m)	1.150	1.150	1.149	0.2679
53	Accel. Gradient	(kPa/m)	7.749e-005	7.757e-005	7.764e-005	1.840e-005
54	Liq. Reynolds		2.205e+005 *	2.204e+005 *	2.203e+005 *	2.203e+005 *
55	Vap. Reynolds		9.022e+006 *	9.024e+006 *	9.025e+006 *	9.026e+006 *
56	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
57	Vapour Velocity	(m/s)	2.134	2.136	2.137	2.137
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	6789	6794	6799	6804
61	Elevation	(m)	-7.380	-7.020	-6.660	-6.300
62	Pressure	(bar)	93.71	93.69	93.68	93.67
63	Temperature	(C)	59.18	59.18	59.17	59.17
64	Vapour Fraction		0.8722	0.8722	0.8722	0.8722
65	Heat Transferred	(kJ/h-m)	777.8	777.6	777.5	777.3
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.5117	0.5116	0.5116	0.5115
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4					
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp		---	---	---
14	Friction Gradient (kPa/m)	5.765e-003	5.766e-003	5.767e-003	5.767e-003
15	Static Gradient (kPa/m)	0.2678	0.2678	0.2678	0.2678
16	Accel. Gradient (kPa/m)	1.840e-005	1.841e-005	1.841e-005	1.842e-005
17	Liq. Reynolds	2.203e+005 *	2.202e+005 *	2.202e+005 *	2.202e+005 *
18	Vap. Reynolds	9.026e+006 *	9.027e+006 *	9.027e+006 *	9.028e+006 *
19	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
20	Vapour Velocity (m/s)	2.138	2.138	2.138	2.138
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	6809	6814	6819	6824
24	Elevation (m)	-5.090	-3.880	-2.670	-1.460
25	Pressure (bar)	93.59	93.51	93.43	93.35
26	Temperature (C)	59.15	59.13	59.11	59.09
27	Vapour Fraction	0.8722	0.8722	0.8723	0.8723
28	Heat Transferred (kJ/h-m)	778.0	777.5	776.9	776.4
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.9958	0.9953	0.9949	0.9944
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	5.532e-003	5.536e-003	5.540e-003	5.545e-003
34	Static Gradient (kPa/m)	1.587	1.586	1.586	1.585
35	Accel. Gradient (kPa/m)	1.074e-004	1.075e-004	1.077e-004	1.078e-004
36	Liq. Reynolds	2.201e+005 *	2.200e+005 *	2.199e+005 *	2.198e+005 *
37	Vap. Reynolds	9.030e+006 *	9.032e+006 *	9.035e+006 *	9.037e+006 *
38	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
39	Vapour Velocity (m/s)	2.140	2.142	2.144	2.146
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	6829	6834	6839	6844
43	Elevation (m)	-0.2500	-3.760	-7.270	-10.78
44	Pressure (bar)	93.27	93.31	93.35	93.40
45	Temperature (C)	59.07	59.08	59.10	59.12
46	Vapour Fraction	0.8723	0.8723	0.8723	0.8723
47	Heat Transferred (kJ/h-m)	775.8	776.1	776.5	777.0
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.9940	8.687e-002	8.691e-002	8.695e-002
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	5.549e-003	9.829e-003	9.824e-003	9.819e-003
53	Static Gradient (kPa/m)	1.585	-0.8633	-0.8636	-0.8639
54	Accel. Gradient (kPa/m)	1.080e-004	-5.790e-005	-5.787e-005	-5.785e-005
55	Liq. Reynolds	2.197e+005 *	2.197e+005 *	2.198e+005 *	2.199e+005 *
56	Vap. Reynolds	9.040e+006 *	9.038e+006 *	9.037e+006 *	9.036e+006 *
57	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
58	Vapour Velocity (m/s)	2.148	2.147	2.146	2.145
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	6849	6854	6859	6864
62	Elevation (m)	-14.29	-17.80	-14.31	-10.82
63	Pressure (bar)	93.44	93.48	93.25	93.02
64	Temperature (C)	59.14	59.15	59.10	59.04
65	Vapour Fraction	0.8723	0.8723	0.8724	0.8725
66	Heat Transferred (kJ/h-m)	777.5	777.9	768.4	766.9
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	8.699e-002	8.703e-002	1.000	1.000
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	9.813e-003	9.808e-003	5.549e-003	5.562e-003
16	Static Gradient (kPa/m)	-0.8643	-0.8646	4.595	4.596
17	Accel. Gradient (kPa/m)	-5.782e-005	-5.779e-005	3.126e-004	3.142e-004
18	Liq. Reynolds	2.199e+005 *	2.200e+005 *	2.197e+005 *	2.194e+005 *
19	Vap. Reynolds	9.035e+006 *	9.033e+006 *	9.040e+006 *	9.047e+006 *
20	Liquid Velocity (m/s)	0.2043	0.2043	0.2042	0.2042
21	Vapour Velocity (m/s)	2.144	2.143	2.149	2.154
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	6869	6874	6879	6884
25	Elevation (m)	-7.330	-3.840	-0.3500	-0.2820
26	Pressure (bar)	92.79	92.56	92.33	92.33
27	Temperature (C)	58.99	58.93	58.88	58.88
28	Vapour Fraction	0.8726	0.8727	0.8728	0.8728
29	Heat Transferred (kJ/h-m)	765.5	764.0	762.6	769.3
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	1.000	1.000	1.000	0.3226
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	5.574e-003	5.587e-003	5.599e-003	6.059e-003
35	Static Gradient (kPa/m)	4.597	4.598	4.599	3.547e-002
36	Accel. Gradient (kPa/m)	3.159e-004	3.176e-004	3.193e-004	2.880e-006
37	Liq. Reynolds	2.191e+005 *	2.187e+005 *	2.184e+005 *	2.184e+005 *
38	Vap. Reynolds	9.054e+006 *	9.062e+006 *	9.069e+006 *	9.069e+006 *
39	Liquid Velocity (m/s)	0.2041	0.2041	0.2040	0.2040
40	Vapour Velocity (m/s)	2.160	2.166	2.171	2.171
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	6889	6894	6899	6904
44	Elevation (m)	-0.2140	-0.1460	-7.800e-002	-1.000e-002
45	Pressure (bar)	92.33	92.33	92.32	92.32
46	Temperature (C)	58.87	58.87	58.87	58.87
47	Vapour Fraction	0.8728	0.8728	0.8728	0.8728
48	Heat Transferred (kJ/h-m)	769.3	769.2	769.1	769.0
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.3226	0.3226	0.3226	0.3226
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.059e-003	6.059e-003	6.060e-003	6.060e-003
54	Static Gradient (kPa/m)	3.547e-002	3.547e-002	3.547e-002	3.547e-002
55	Accel. Gradient (kPa/m)	2.880e-006	2.880e-006	2.880e-006	2.881e-006
56	Liq. Reynolds	2.184e+005 *	2.184e+005 *	2.184e+005 *	2.184e+005 *
57	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *
58	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
59	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	6909	6914	6919	6924
63	Elevation (m)	-1.600e-002	-2.200e-002	-2.800e-002	-3.400e-002
64	Pressure (bar)	92.32	92.32	92.32	92.32
65	Temperature (C)	58.86	58.86	58.86	58.86
66	Vapour Fraction	0.8728	0.8728	0.8728	0.8728
67	Heat Transferred (kJ/h-m)	769.0	768.9	768.9	768.8
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 142 of 937

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2059	0.2059	0.2059	0.2059
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.861e-003	6.860e-003	6.860e-003	6.860e-003
17	Static Gradient (kPa/m)	-2.307e-003	-2.307e-003	-2.307e-003	-2.307e-003
18	Accel. Gradient (kPa/m)	3.158e-007	3.158e-007	3.158e-007	3.158e-007
19	Liq. Reynolds	2.184e+005 *	2.184e+005 *	2.184e+005 *	2.184e+005 *
20	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *
21	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
22	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	6929	6934	6939	6944
26	Elevation (m)	-4.000e-002	-4.800e-002	-5.600e-002	-6.400e-002
27	Pressure (bar)	92.32	92.32	92.32	92.32
28	Temperature (C)	58.85	58.85	58.85	58.85
29	Vapour Fraction	0.8728	0.8728	0.8728	0.8728
30	Heat Transferred (kJ/h-m)	768.7	768.7	768.6	768.6
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2059	0.2055	0.2055	0.2055
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	6.860e-003	6.863e-003	6.863e-003	6.863e-003
36	Static Gradient (kPa/m)	-2.307e-003	-3.064e-003	-3.064e-003	-3.064e-003
37	Accel. Gradient (kPa/m)	3.158e-007	2.635e-007	2.635e-007	2.635e-007
38	Liq. Reynolds	2.184e+005 *	2.184e+005 *	2.184e+005 *	2.184e+005 *
39	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *
40	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
41	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	6949	6954	6959	6964
45	Elevation (m)	-7.200e-002	-8.000e-002	-7.200e-002	-6.400e-002
46	Pressure (bar)	92.32	92.32	92.32	92.32
47	Temperature (C)	58.85	58.84	58.84	58.84
48	Vapour Fraction	0.8727	0.8727	0.8727	0.8727
49	Heat Transferred (kJ/h-m)	768.5	768.5	768.4	768.3
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2055	0.2055	0.2843	0.2843
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	6.863e-003	6.863e-003	5.228e-003	5.228e-003
55	Static Gradient (kPa/m)	-3.064e-003	-3.064e-003	3.814e-003	3.814e-003
56	Accel. Gradient (kPa/m)	2.635e-007	2.634e-007	6.270e-007	6.270e-007
57	Liq. Reynolds	2.184e+005 *	2.184e+005 *	2.184e+005 *	2.184e+005 *
58	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *
59	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
60	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	6969	6974	6979	6984
64	Elevation (m)	-5.600e-002	-4.800e-002	-4.000e-002	-3.200e-002
65	Pressure (bar)	92.32	92.32	92.32	92.32
66	Temperature (C)	58.84	58.83	58.83	58.83
67	Vapour Fraction	0.8727	0.8727	0.8727	0.8727
68	Heat Transferred (kJ/h-m)	768.3	768.2	768.1	768.1
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2843	0.2843	0.2843	0.2843
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	5.227e-003	5.227e-003	5.227e-003	5.227e-003
18	Static Gradient (kPa/m)	3.814e-003	3.814e-003	3.814e-003	3.814e-003
19	Accel. Gradient (kPa/m)	6.270e-007	6.270e-007	6.270e-007	6.270e-007
20	Liq. Reynolds	2.184e+005 *	2.184e+005 *	2.184e+005 *	2.184e+005 *
21	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *
22	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
23	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	6989	6994	6999	7004
27	Elevation (m)	-2.400e-002	-1.600e-002	-8.000e-003	-1.402e-015
28	Pressure (bar)	92.32	92.32	92.32	92.32
29	Temperature (C)	58.83	58.82	58.82	58.82
30	Vapour Fraction	0.8727	0.8727	0.8727	0.8727
31	Heat Transferred (kJ/h-m)	768.0	767.9	767.9	767.8
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2843	0.2843	0.2843	0.2843
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	5.227e-003	5.227e-003	5.227e-003	5.227e-003
37	Static Gradient (kPa/m)	3.814e-003	3.814e-003	3.814e-003	3.814e-003
38	Accel. Gradient (kPa/m)	6.270e-007	6.270e-007	6.270e-007	6.270e-007
39	Liq. Reynolds	2.184e+005 *	2.184e+005 *	2.184e+005 *	2.184e+005 *
40	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *
41	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
42	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	7009	7014	7019	7024
46	Elevation (m)	1.000e-002	2.000e-002	3.000e-002	4.000e-002
47	Pressure (bar)	92.31	92.31	92.31	92.31
48	Temperature (C)	58.82	58.82	58.81	58.81
49	Vapour Fraction	0.8727	0.8727	0.8727	0.8727
50	Heat Transferred (kJ/h-m)	767.8	767.7	767.6	767.6
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2856	0.2856	0.2856	0.2856
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	5.131e-003	5.131e-003	5.131e-003	5.131e-003
56	Static Gradient (kPa/m)	4.783e-003	4.783e-003	4.783e-003	4.783e-003
57	Accel. Gradient (kPa/m)	6.875e-007	6.875e-007	6.875e-007	6.875e-007
58	Liq. Reynolds	2.184e+005 *	2.184e+005 *	2.184e+005 *	2.184e+005 *
59	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *
60	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
61	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	7029	7034	7039	7044
65	Elevation (m)	5.000e-002	1.600e-002	-1.800e-002	-5.200e-002
66	Pressure (bar)	92.31	92.31	92.31	92.31
67	Temperature (C)	58.81	58.81	58.80	58.80
68	Vapour Fraction	0.8727	0.8727	0.8727	0.8727
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	767.5	767.5	767.4	767.4
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.2856	0.2007	0.2007	0.2007
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	5.131e-003	6.902e-003	6.902e-003	6.902e-003
19	Static Gradient (kPa/m)	4.783e-003	-1.287e-002	-1.287e-002	-1.287e-002
20	Accel. Gradient (kPa/m)	6.875e-007	-4.136e-007	-4.136e-007	-4.136e-007
21	Liq. Reynolds	2.183e+005 *	2.183e+005 *	2.183e+005 *	2.183e+005 *
22	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *
23	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
24	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	7049	7054	7059	7064
28	Elevation (m)	-8.600e-002	-0.1200	-8.600e-002	-5.200e-002
29	Pressure (bar)	92.31	92.31	92.31	92.31
30	Temperature (C)	58.80	58.80	58.80	58.79
31	Vapour Fraction	0.8727	0.8727	0.8727	0.8727
32	Heat Transferred (kJ/h-m)	767.3	767.3	767.2	767.1
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.2007	0.2007	0.3010	0.3010
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	6.902e-003	6.902e-003	5.827e-003	5.827e-003
38	Static Gradient (kPa/m)	-1.287e-002	-1.287e-002	1.687e-002	1.687e-002
39	Accel. Gradient (kPa/m)	-4.136e-007	-4.136e-007	1.574e-006	1.574e-006
40	Liq. Reynolds	2.183e+005 *	2.183e+005 *	2.183e+005 *	2.183e+005 *
41	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *
42	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
43	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	7069	7074	7079	7081
47	Elevation (m)	-1.800e-002	1.600e-002	5.000e-002	4.600e-002
48	Pressure (bar)	92.31	92.31	92.31	92.31
49	Temperature (C)	58.79	58.79	58.79	58.79
50	Vapour Fraction	0.8727	0.8727	0.8727	0.8727
51	Heat Transferred (kJ/h-m)	767.1	767.0	766.9	766.9
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.3010	0.3010	0.3010	0.2052
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	5.827e-003	5.827e-003	5.827e-003	6.865e-003
57	Static Gradient (kPa/m)	1.687e-002	1.687e-002	1.687e-002	-3.837e-003
58	Accel. Gradient (kPa/m)	1.574e-006	1.574e-006	1.574e-006	2.100e-007
59	Liq. Reynolds	2.183e+005 *	2.183e+005 *	2.183e+005 *	2.183e+005 *
60	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *
61	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
62	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	7083	7085	7087	7089
66	Elevation (m)	4.200e-002	3.800e-002	3.400e-002	3.000e-002
67	Pressure (bar)	92.31	92.31	92.31	92.31
68	Temperature (C)	58.78	58.78	58.78	58.78
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8727	0.8727	0.8727	0.8727
14	Heat Transferred (kJ/h-m)	766.9	766.9	766.8	766.8
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2052	0.2052	0.2052	0.2052
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	6.865e-003	6.865e-003	6.865e-003	6.865e-003
20	Static Gradient (kPa/m)	-3.837e-003	-3.837e-003	-3.837e-003	-3.837e-003
21	Accel. Gradient (kPa/m)	2.100e-007	2.100e-007	2.100e-007	2.100e-007
22	Liq. Reynolds	2.183e+005 *	2.183e+005 *	2.183e+005 *	2.183e+005 *
23	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *
24	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
25	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	7097	7105	7113	7121
29	Elevation (m)	2.400e-002	1.800e-002	1.200e-002	6.000e-003
30	Pressure (bar)	92.31	92.31	92.31	92.31
31	Temperature (C)	58.78	58.77	58.77	58.77
32	Vapour Fraction	0.8727	0.8727	0.8727	0.8727
33	Heat Transferred (kJ/h-m)	766.7	766.6	766.5	766.4
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2063	0.2063	0.2063	0.2063
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	6.856e-003	6.856e-003	6.855e-003	6.855e-003
39	Static Gradient (kPa/m)	-1.444e-003	-1.444e-003	-1.444e-003	-1.444e-003
40	Accel. Gradient (kPa/m)	3.753e-007	3.752e-007	3.752e-007	3.752e-007
41	Liq. Reynolds	2.183e+005 *	2.183e+005 *	2.183e+005 *	2.183e+005 *
42	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *
43	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
44	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	7129	7134	7139	7144
48	Elevation (m)	-1.398e-015	4.000e-003	8.000e-003	1.200e-002
49	Pressure (bar)	92.31	92.31	92.31	92.30
50	Temperature (C)	58.76	58.76	58.76	58.76
51	Vapour Fraction	0.8727	0.8727	0.8727	0.8727
52	Heat Transferred (kJ/h-m)	766.3	766.3	766.2	766.2
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2063	0.2818	0.2818	0.2818
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	6.855e-003	5.421e-003	5.421e-003	5.421e-003
58	Static Gradient (kPa/m)	-1.444e-003	1.895e-003	1.895e-003	1.895e-003
59	Accel. Gradient (kPa/m)	3.752e-007	5.073e-007	5.073e-007	5.073e-007
60	Liq. Reynolds	2.183e+005 *	2.183e+005 *	2.183e+005 *	2.183e+005 *
61	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *
62	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
63	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	7149	7154	7159	7164
67	Elevation (m)	1.600e-002	2.000e-002	3.000e-002	4.000e-002
68	Pressure (bar)	92.30	92.30	92.30	92.30
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	58.76	58.75	58.75	58.75
14	Vapour Fraction	0.8727	0.8727	0.8727	0.8727
15	Heat Transferred (kJ/h-m)	766.1	766.0	766.0	765.9
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2818	0.2818	0.2857	0.2857
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	5.421e-003	5.421e-003	5.130e-003	5.130e-003
21	Static Gradient (kPa/m)	1.895e-003	1.895e-003	4.783e-003	4.783e-003
22	Accel. Gradient (kPa/m)	5.073e-007	5.073e-007	6.874e-007	6.874e-007
23	Liq. Reynolds	2.183e+005 *	2.183e+005 *	2.183e+005 *	2.183e+005 *
24	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *
25	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
26	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	7169	7174	7179	7184
30	Elevation (m)	5.000e-002	6.000e-002	7.000e-002	9.000e-002
31	Pressure (bar)	92.30	92.30	92.30	92.30
32	Temperature (C)	58.75	58.74	58.74	58.74
33	Vapour Fraction	0.8727	0.8727	0.8727	0.8727
34	Heat Transferred (kJ/h-m)	765.8	765.8	765.7	765.7
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2857	0.2857	0.2857	0.2921
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	5.130e-003	5.130e-003	5.130e-003	4.670e-003
40	Static Gradient (kPa/m)	4.784e-003	4.784e-003	4.784e-003	9.717e-003
41	Accel. Gradient (kPa/m)	6.874e-007	6.874e-007	6.874e-007	9.976e-007
42	Liq. Reynolds	2.183e+005 *	2.183e+005 *	2.183e+005 *	2.183e+005 *
43	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *
44	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
45	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	7189	7194	7199	7204
49	Elevation (m)	0.1100	0.1300	0.1500	0.1700
50	Pressure (bar)	92.30	92.30	92.30	92.30
51	Temperature (C)	58.74	58.73	58.73	58.73
52	Vapour Fraction	0.8727	0.8727	0.8727	0.8727
53	Heat Transferred (kJ/h-m)	765.6	765.5	765.5	765.4
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2921	0.2921	0.2921	0.2921
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	4.670e-003	4.670e-003	4.670e-003	4.670e-003
59	Static Gradient (kPa/m)	9.717e-003	9.717e-003	9.717e-003	9.717e-003
60	Accel. Gradient (kPa/m)	9.976e-007	9.976e-007	9.976e-007	9.976e-007
61	Liq. Reynolds	2.183e+005 *	2.183e+005 *	2.183e+005 *	2.183e+005 *
62	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *
63	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
64	Vapour Velocity (m/s)	2.171	2.171	2.170	2.170
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	7209	7214	7219	7224
68	Elevation (m)	0.1520	0.1340	0.1160	9.800e-002
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:02:44 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Pressure (bar)	92.30	92.30	92.30	92.30					
14	Temperature (C)	58.73	58.73	58.72	58.72					
15	Vapour Fraction	0.8727	0.8727	0.8727	0.8727					
16	Heat Transferred (kJ/h-m)	765.4	765.3	765.3	765.2					
17	Flow Regime	Transition	Transition	Transition	Transition					
18	Water Flow Regime									
19	Liquid HoldUp	0.2037	0.2037	0.2037	0.2037					
20	Water HoldUp	---	---	---	---					
21	Friction Gradient (kPa/m)	6.876e-003	6.876e-003	6.876e-003	6.876e-003					
22	Static Gradient (kPa/m)	-6.876e-003	-6.876e-003	-6.876e-003	-6.876e-003					
23	Accel. Gradient (kPa/m)	-8.226e-012	-1.690e-011	-2.556e-011	-3.423e-011					
24	Liq. Reynolds	2.183e+005 *	2.182e+005 *	2.182e+005 *	2.182e+005 *					
25	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *					
26	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040					
27	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170					
28	Deposit Thickness (mm)	---	---	---	---					
29	Deposit Volume (m3)	---	---	---	---					
30	Length (m)	7229	7234	7239	7244					
31	Elevation (m)	8.000e-002	7.600e-002	7.200e-002	6.800e-002					
32	Pressure (bar)	92.30	92.30	92.30	92.30					
33	Temperature (C)	58.72	58.72	58.71	58.71					
34	Vapour Fraction	0.8727	0.8727	0.8727	0.8727					
35	Heat Transferred (kJ/h-m)	765.1	765.1	765.0	765.0					
36	Flow Regime	Transition	Transition	Transition	Transition					
37	Water Flow Regime									
38	Liquid HoldUp	0.2037	0.2063	0.2063	0.2063					
39	Water HoldUp	---	---	---	---					
40	Friction Gradient (kPa/m)	6.876e-003	6.855e-003	6.855e-003	6.855e-003					
41	Static Gradient (kPa/m)	-6.876e-003	-1.540e-003	-1.540e-003	-1.540e-003					
42	Accel. Gradient (kPa/m)	-4.290e-011	3.685e-007	3.685e-007	3.685e-007					
43	Liq. Reynolds	2.182e+005 *	2.182e+005 *	2.182e+005 *	2.182e+005 *					
44	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *					
45	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040					
46	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170					
47	Deposit Thickness (mm)	---	---	---	---					
48	Deposit Volume (m3)	---	---	---	---					
49	Length (m)	7249	7254	7259	7264					
50	Elevation (m)	6.400e-002	6.000e-002	8.000e-002	1.000e-001					
51	Pressure (bar)	92.30	92.30	92.30	92.30					
52	Temperature (C)	58.71	58.71	58.71	58.70					
53	Vapour Fraction	0.8727	0.8727	0.8727	0.8727					
54	Heat Transferred (kJ/h-m)	764.9	764.8	764.8	764.7					
55	Flow Regime	Transition	Transition	Transition	Transition					
56	Water Flow Regime									
57	Liquid HoldUp	0.2063	0.2063	0.2921	0.2921					
58	Water HoldUp	---	---	---	---					
59	Friction Gradient (kPa/m)	6.855e-003	6.855e-003	4.669e-003	4.669e-003					
60	Static Gradient (kPa/m)	-1.540e-003	-1.540e-003	9.718e-003	9.718e-003					
61	Accel. Gradient (kPa/m)	3.685e-007	3.685e-007	9.975e-007	9.975e-007					
62	Liq. Reynolds	2.182e+005 *	2.182e+005 *	2.182e+005 *	2.182e+005 *					
63	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *					
64	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040					
65	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170					
66	Deposit Thickness (mm)	---	---	---	---					
67	Deposit Volume (m3)	---	---	---	---					
68	Length (m)	7269	7274	7279	7284					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	0.1200	0.1400	0.1600	0.1340
14	Pressure (bar)	92.29	92.29	92.29	92.29
15	Temperature (C)	58.70	58.70	58.70	58.69
16	Vapour Fraction	0.8727	0.8727	0.8727	0.8727
17	Heat Transferred (kJ/h-m)	764.6	764.6	764.5	764.5
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2921	0.2921	0.2921	0.2022
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	4.669e-003	4.669e-003	4.669e-003	6.888e-003
23	Static Gradient (kPa/m)	9.718e-003	9.718e-003	9.718e-003	-9.888e-003
24	Accel. Gradient (kPa/m)	9.975e-007	9.975e-007	9.975e-007	-2.080e-007
25	Liq. Reynolds	2.182e+005 *	2.182e+005 *	2.182e+005 *	2.182e+005 *
26	Vap. Reynolds	9.069e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
27	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
28	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	7289	7294	7299	7304
32	Elevation (m)	0.1080	8.200e-002	5.600e-002	3.000e-002
33	Pressure (bar)	92.29	92.29	92.29	92.29
34	Temperature (C)	58.69	58.69	58.69	58.69
35	Vapour Fraction	0.8727	0.8727	0.8727	0.8727
36	Heat Transferred (kJ/h-m)	764.4	764.4	764.3	764.3
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2022	0.2022	0.2022	0.2022
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	6.888e-003	6.887e-003	6.887e-003	6.887e-003
42	Static Gradient (kPa/m)	-9.888e-003	-9.888e-003	-9.888e-003	-9.888e-003
43	Accel. Gradient (kPa/m)	-2.080e-007	-2.080e-007	-2.080e-007	-2.080e-007
44	Liq. Reynolds	2.182e+005 *	2.182e+005 *	2.182e+005 *	2.182e+005 *
45	Vap. Reynolds	9.070e+006 *	9.069e+006 *	9.069e+006 *	9.069e+006 *
46	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
47	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	7309	7314	7319	7324
51	Elevation (m)	3.800e-002	4.600e-002	5.400e-002	6.200e-002
52	Pressure (bar)	92.29	92.29	92.29	92.29
53	Temperature (C)	58.68	58.68	58.68	58.68
54	Vapour Fraction	0.8727	0.8727	0.8727	0.8727
55	Heat Transferred (kJ/h-m)	764.2	764.1	764.1	764.0
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2844	0.2844	0.2844	0.2844
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	5.224e-003	5.224e-003	5.224e-003	5.224e-003
61	Static Gradient (kPa/m)	3.815e-003	3.816e-003	3.816e-003	3.816e-003
62	Accel. Gradient (kPa/m)	6.267e-007	6.267e-007	6.267e-007	6.267e-007
63	Liq. Reynolds	2.182e+005 *	2.182e+005 *	2.182e+005 *	2.182e+005 *
64	Vap. Reynolds	9.069e+006 *	9.069e+006 *	9.070e+006 *	9.070e+006 *
65	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
66	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	7329	7334	7339	7344
14	Elevation (m)	7.000e-002	8.800e-002	0.1060	0.1240
15	Pressure (bar)	92.29	92.29	92.29	92.29
16	Temperature (C)	58.67	58.67	58.67	58.67
17	Vapour Fraction	0.8727	0.8726	0.8726	0.8726
18	Heat Transferred (kJ/h-m)	763.9	763.9	763.8	763.8
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.2844	0.2908	0.2908	0.2908
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	5.224e-003	4.756e-003	4.756e-003	4.756e-003
24	Static Gradient (kPa/m)	3.816e-003	8.738e-003	8.738e-003	8.738e-003
25	Accel. Gradient (kPa/m)	6.267e-007	9.356e-007	9.356e-007	9.356e-007
26	Liq. Reynolds	2.182e+005 *	2.182e+005 *	2.182e+005 *	2.182e+005 *
27	Vap. Reynolds	9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
28	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
29	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	7349	7354	7359	7364
33	Elevation (m)	0.1420	0.1600	0.1680	0.1760
34	Pressure (bar)	92.29	92.29	92.29	92.29
35	Temperature (C)	58.66	58.66	58.66	58.66
36	Vapour Fraction	0.8726	0.8726	0.8726	0.8726
37	Heat Transferred (kJ/h-m)	763.7	763.6	763.6	763.5
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.2908	0.2908	0.2844	0.2844
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	4.756e-003	4.756e-003	5.224e-003	5.224e-003
43	Static Gradient (kPa/m)	8.738e-003	8.738e-003	3.816e-003	3.816e-003
44	Accel. Gradient (kPa/m)	9.356e-007	9.356e-007	6.267e-007	6.267e-007
45	Liq. Reynolds	2.182e+005 *	2.182e+005 *	2.182e+005 *	2.182e+005 *
46	Vap. Reynolds	9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
47	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
48	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	7369	7374	7379	7384
52	Elevation (m)	0.1840	0.1920	0.2000	0.2040
53	Pressure (bar)	92.29	92.29	92.29	92.29
54	Temperature (C)	58.66	58.65	58.65	58.65
55	Vapour Fraction	0.8726	0.8726	0.8726	0.8726
56	Heat Transferred (kJ/h-m)	763.4	763.4	763.3	763.3
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.2844	0.2844	0.2844	0.2819
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	5.224e-003	5.224e-003	5.224e-003	5.419e-003
62	Static Gradient (kPa/m)	3.816e-003	3.816e-003	3.816e-003	1.896e-003
63	Accel. Gradient (kPa/m)	6.267e-007	6.267e-007	6.267e-007	5.071e-007
64	Liq. Reynolds	2.182e+005 *	2.182e+005 *	2.182e+005 *	2.182e+005 *
65	Vap. Reynolds	9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
66	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
67	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
68	Deposit Thickness (mm)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	7389	7394	7399	7404
15	Elevation (m)	0.2080	0.2120	0.2160	0.2200
16	Pressure (bar)	92.29	92.28	92.28	92.28
17	Temperature (C)	58.65	58.64	58.64	58.64
18	Vapour Fraction	0.8726	0.8726	0.8726	0.8726
19	Heat Transferred (kJ/h-m)	763.2	763.1	763.1	763.0
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2819	0.2819	0.2819	0.2819
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	5.419e-003	5.419e-003	5.419e-003	5.419e-003
25	Static Gradient (kPa/m)	1.896e-003	1.896e-003	1.896e-003	1.896e-003
26	Accel. Gradient (kPa/m)	5.071e-007	5.071e-007	5.071e-007	5.071e-007
27	Liq. Reynolds	2.182e+005 *	2.182e+005 *	2.182e+005 *	2.181e+005 *
28	Vap. Reynolds	9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
29	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
30	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	7409	7414	7419	7424
34	Elevation (m)	0.2220	0.2240	0.2260	0.2280
35	Pressure (bar)	92.28	92.28	92.28	92.28
36	Temperature (C)	58.64	58.64	58.63	58.63
37	Vapour Fraction	0.8726	0.8726	0.8726	0.8726
38	Heat Transferred (kJ/h-m)	763.0	762.9	762.8	762.8
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2806	0.2806	0.2806	0.2806
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	5.518e-003	5.518e-003	5.518e-003	5.518e-003
44	Static Gradient (kPa/m)	9.450e-004	9.450e-004	9.450e-004	9.450e-004
45	Accel. Gradient (kPa/m)	4.481e-007	4.481e-007	4.481e-007	4.481e-007
46	Liq. Reynolds	2.181e+005 *	2.181e+005 *	2.181e+005 *	2.181e+005 *
47	Vap. Reynolds	9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
48	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
49	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	7429	7434	7439	7444
53	Elevation (m)	0.2300	0.2320	0.2340	0.2360
54	Pressure (bar)	92.28	92.28	92.28	92.28
55	Temperature (C)	58.63	58.63	58.62	58.62
56	Vapour Fraction	0.8726	0.8726	0.8726	0.8726
57	Heat Transferred (kJ/h-m)	762.7	762.7	762.6	762.5
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2806	0.2806	0.2806	0.2806
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	5.518e-003	5.518e-003	5.518e-003	5.518e-003
63	Static Gradient (kPa/m)	9.450e-004	9.450e-004	9.450e-004	9.450e-004
64	Accel. Gradient (kPa/m)	4.481e-007	4.481e-007	4.481e-007	4.481e-007
65	Liq. Reynolds	2.181e+005 *	2.181e+005 *	2.181e+005 *	2.181e+005 *
66	Vap. Reynolds	9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
67	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
68	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	7449	7454	7459	7464
16	Elevation (m)	0.2380	0.2400	0.2340	0.2280
17	Pressure (bar)	92.28	92.28	92.28	92.28
18	Temperature (C)	58.62	58.62	58.62	58.61
19	Vapour Fraction	0.8726	0.8726	0.8726	0.8726
20	Heat Transferred (kJ/h-m)	762.5	762.4	762.4	762.3
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2806	0.2806	0.2060	0.2060
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	5.518e-003	5.518e-003	6.856e-003	6.856e-003
26	Static Gradient (kPa/m)	9.450e-004	9.450e-004	-2.308e-003	-2.309e-003
27	Accel. Gradient (kPa/m)	4.481e-007	4.480e-007	3.152e-007	3.152e-007
28	Liq. Reynolds	2.181e+005 *	2.181e+005 *	2.181e+005 *	2.181e+005 *
29	Vap. Reynolds	9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
30	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
31	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	7469	7474	7479	7484
35	Elevation (m)	0.2220	0.2160	0.2100	0.1920
36	Pressure (bar)	92.28	92.28	92.28	92.28
37	Temperature (C)	58.61	58.61	58.61	58.60
38	Vapour Fraction	0.8726	0.8726	0.8726	0.8726
39	Heat Transferred (kJ/h-m)	762.3	762.2	762.2	762.1
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2060	0.2060	0.2060	0.2038
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	6.856e-003	6.856e-003	6.856e-003	6.874e-003
45	Static Gradient (kPa/m)	-2.309e-003	-2.309e-003	-2.309e-003	-6.878e-003
46	Accel. Gradient (kPa/m)	3.152e-007	3.152e-007	3.152e-007	-3.422e-010
47	Liq. Reynolds	2.181e+005 *	2.181e+005 *	2.181e+005 *	2.181e+005 *
48	Vap. Reynolds	9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
49	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
50	Vapour Velocity (m/s)	2.170	2.170	2.170	2.169
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	7489	7494	7499	7504
54	Elevation (m)	0.1740	0.1560	0.1380	0.1200
55	Pressure (bar)	92.28	92.28	92.28	92.28
56	Temperature (C)	58.60	58.60	58.60	58.60
57	Vapour Fraction	0.8726	0.8726	0.8726	0.8726
58	Heat Transferred (kJ/h-m)	762.0	762.0	761.9	761.9
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2038	0.2038	0.2038	0.2038
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	6.873e-003	6.873e-003	6.873e-003	6.873e-003
64	Static Gradient (kPa/m)	-6.879e-003	-6.879e-003	-6.879e-003	-6.879e-003
65	Accel. Gradient (kPa/m)	-3.509e-010	-3.595e-010	-3.681e-010	-3.768e-010
66	Liq. Reynolds	2.181e+005 *	2.181e+005 *	2.181e+005 *	2.181e+005 *
67	Vap. Reynolds	9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
68	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3							
4							
5			Unit Set: Mixed				
6			Date/Time: Mon Jun 13 15:02:44 2022				
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9							
10	PROFILES						
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	7509	7514	7519	7524	
17	Elevation	(m)	0.1320	0.1440	0.1560	0.1680	
18	Pressure	(bar)	92.28	92.28	92.28	92.28	
19	Temperature	(C)	58.59	58.59	58.59	58.59	
20	Vapour Fraction		0.8726	0.8726	0.8726	0.8726	
21	Heat Transferred	(kJ/h-m)	761.8	761.7	761.7	761.6	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2870	0.2870	0.2870	0.2870	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	5.032e-003	5.032e-003	5.032e-003	5.032e-003	
27	Static Gradient	(kPa/m)	5.761e-003	5.761e-003	5.761e-003	5.761e-003	
28	Accel. Gradient	(kPa/m)	7.481e-007	7.481e-007	7.481e-007	7.481e-007	
29	Liq. Reynolds		2.181e+005 *	2.181e+005 *	2.181e+005 *	2.181e+005 *	
30	Vap. Reynolds		9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *	
31	Liquid Velocity	(m/s)	0.2040	0.2040	0.2040	0.2040	
32	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	7529	7533	7538	7543	
36	Elevation	(m)	0.1800	0.1680	0.1560	0.1440	
37	Pressure	(bar)	92.28	92.28	92.28	92.28	
38	Temperature	(C)	58.58	58.58	58.58	58.58	
39	Vapour Fraction		0.8726	0.8726	0.8726	0.8726	
40	Heat Transferred	(kJ/h-m)	761.6	761.5	761.5	761.4	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2870	0.2049	0.2049	0.2049	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	5.032e-003	6.864e-003	6.864e-003	6.864e-003	
46	Static Gradient	(kPa/m)	5.761e-003	-4.680e-003	-4.680e-003	-4.680e-003	
47	Accel. Gradient	(kPa/m)	7.481e-007	1.514e-007	1.514e-007	1.514e-007	
48	Liq. Reynolds		2.181e+005 *	2.181e+005 *	2.181e+005 *	2.181e+005 *	
49	Vap. Reynolds		9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *	
50	Liquid Velocity	(m/s)	0.2040	0.2040	0.2040	0.2040	
51	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	7548	7553	7558	7563	
55	Elevation	(m)	0.1320	0.1200	0.1320	0.1440	
56	Pressure	(bar)	92.28	92.28	92.28	92.28	
57	Temperature	(C)	58.58	58.57	58.57	58.57	
58	Vapour Fraction		0.8726	0.8726	0.8726	0.8726	
59	Heat Transferred	(kJ/h-m)	761.3	761.3	761.2	761.1	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.2049	0.2049	0.2871	0.2871	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	6.864e-003	6.864e-003	5.032e-003	5.032e-003	
65	Static Gradient	(kPa/m)	-4.680e-003	-4.680e-003	5.761e-003	5.761e-003	
66	Accel. Gradient	(kPa/m)	1.514e-007	1.514e-007	7.481e-007	7.481e-007	
67	Liq. Reynolds		2.181e+005 *	2.181e+005 *	2.181e+005 *	2.181e+005 *	
68	Vap. Reynolds		9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *	
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:02:44 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2040	0.2040	0.2040	0.2040	
14	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	7568	7573	7578	7583	
18	Elevation	(m)	0.1560	0.1680	0.1800	0.1800	
19	Pressure	(bar)	92.27	92.27	92.27	92.27	
20	Temperature	(C)	58.57	58.56	58.56	58.56	
21	Vapour Fraction		0.8726	0.8726	0.8726	0.8726	
22	Heat Transferred	(kJ/h-m)	761.1	761.0	761.0	760.9	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.2871	0.2871	0.2871	0.3024	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	5.031e-003	5.031e-003	5.031e-003	5.868e-003	
28	Static Gradient	(kPa/m)	5.761e-003	5.761e-003	5.761e-003	0.0000	
29	Accel. Gradient	(kPa/m)	7.481e-007	7.481e-007	7.481e-007	4.067e-007	
30	Liq. Reynolds		2.181e+005 *	2.181e+005 *	2.181e+005 *	2.181e+005 *	
31	Vap. Reynolds		9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *	
32	Liquid Velocity	(m/s)	0.2040	0.2040	0.2040	0.2040	
33	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	7588	7593	7598	7603	
37	Elevation	(m)	0.1800	0.1800	0.1800	0.1800	
38	Pressure	(bar)	92.27	92.27	92.27	92.27	
39	Temperature	(C)	58.56	58.56	58.55	58.55	
40	Vapour Fraction		0.8726	0.8726	0.8726	0.8726	
41	Heat Transferred	(kJ/h-m)	760.9	760.8	760.7	760.7	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.3024	0.3024	0.3024	0.3024	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	5.868e-003	5.868e-003	5.868e-003	5.868e-003	
47	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000	
48	Accel. Gradient	(kPa/m)	4.067e-007	4.067e-007	4.067e-007	4.067e-007	
49	Liq. Reynolds		2.181e+005 *	2.181e+005 *	2.181e+005 *	2.180e+005 *	
50	Vap. Reynolds		9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *	
51	Liquid Velocity	(m/s)	0.2040	0.2040	0.2040	0.2040	
52	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	7608	7613	7618	7623	
56	Elevation	(m)	0.1660	0.1520	0.1380	0.1240	
57	Pressure	(bar)	92.27	92.27	92.27	92.27	
58	Temperature	(C)	58.55	58.55	58.55	58.54	
59	Vapour Fraction		0.8726	0.8726	0.8726	0.8726	
60	Heat Transferred	(kJ/h-m)	760.6	760.6	760.5	760.5	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2045	0.2045	0.2045	0.2045	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	6.866e-003	6.866e-003	6.866e-003	6.866e-003	
66	Static Gradient	(kPa/m)	-5.363e-003	-5.363e-003	-5.363e-003	-5.363e-003	
67	Accel. Gradient	(kPa/m)	1.042e-007	1.042e-007	1.042e-007	1.042e-007	
68	Liq. Reynolds		2.180e+005 *	2.180e+005 *	2.180e+005 *	2.180e+005 *	
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
14	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
15	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	7628	7633	7638	7643
19	Elevation (m)	0.1100	0.1160	0.1220	0.1280
20	Pressure (bar)	92.27	92.27	92.27	92.27
21	Temperature (C)	58.54	58.54	58.54	58.53
22	Vapour Fraction	0.8726	0.8726	0.8726	0.8726
23	Heat Transferred (kJ/h-m)	760.4	760.3	760.3	760.2
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2045	0.2833	0.2833	0.2833
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.866e-003	5.318e-003	5.318e-003	5.318e-003
29	Static Gradient (kPa/m)	-5.363e-003	2.854e-003	2.854e-003	2.854e-003
30	Accel. Gradient (kPa/m)	1.042e-007	5.664e-007	5.664e-007	5.664e-007
31	Liq. Reynolds	2.180e+005 *	2.180e+005 *	2.180e+005 *	2.180e+005 *
32	Vap. Reynolds	9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
33	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
34	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	7648	7653	7658	7663
38	Elevation (m)	0.1340	0.1400	0.1260	0.1120
39	Pressure (bar)	92.27	92.27	92.27	92.27
40	Temperature (C)	58.53	58.53	58.53	58.53
41	Vapour Fraction	0.8726	0.8726	0.8726	0.8726
42	Heat Transferred (kJ/h-m)	760.1	760.1	760.0	760.0
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2833	0.2833	0.2045	0.2045
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	5.318e-003	5.318e-003	6.866e-003	6.866e-003
48	Static Gradient (kPa/m)	2.854e-003	2.854e-003	-5.363e-003	-5.363e-003
49	Accel. Gradient (kPa/m)	5.664e-007	5.664e-007	1.041e-007	1.041e-007
50	Liq. Reynolds	2.180e+005 *	2.180e+005 *	2.180e+005 *	2.180e+005 *
51	Vap. Reynolds	9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
52	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
53	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	7668	7673	7678	7683
57	Elevation (m)	9.800e-002	8.400e-002	7.000e-002	8.800e-002
58	Pressure (bar)	92.27	92.27	92.27	92.27
59	Temperature (C)	58.52	58.52	58.52	58.52
60	Vapour Fraction	0.8726	0.8726	0.8726	0.8726
61	Heat Transferred (kJ/h-m)	759.9	759.9	759.8	759.7
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2045	0.2046	0.2046	0.2909
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	6.866e-003	6.866e-003	6.866e-003	4.755e-003
67	Static Gradient (kPa/m)	-5.363e-003	-5.364e-003	-5.364e-003	8.724e-003
68	Accel. Gradient (kPa/m)	1.041e-007	1.041e-007	1.041e-007	9.342e-007
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.180e+005 *	2.180e+005 *	2.180e+005 *	2.180e+005 *
14	Vap. Reynolds	9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
15	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
16	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	7688	7693	7698	7703
20	Elevation (m)	0.1060	0.1240	0.1420	0.1600
21	Pressure (bar)	92.27	92.27	92.27	92.27
22	Temperature (C)	58.51	58.51	58.51	58.51
23	Vapour Fraction	0.8726	0.8726	0.8726	0.8726
24	Heat Transferred (kJ/h-m)	759.7	759.6	759.6	759.5
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2909	0.2909	0.2909	0.2909
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	4.754e-003	4.754e-003	4.754e-003	4.754e-003
30	Static Gradient (kPa/m)	8.724e-003	8.724e-003	8.724e-003	8.724e-003
31	Accel. Gradient (kPa/m)	9.342e-007	9.342e-007	9.342e-007	9.342e-007
32	Liq. Reynolds	2.180e+005 *	2.180e+005 *	2.180e+005 *	2.180e+005 *
33	Vap. Reynolds	9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
34	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
35	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	7708	7713	7718	7723
39	Elevation (m)	5.400e-002	-5.200e-002	-0.1580	-0.2640
40	Pressure (bar)	92.27	92.27	92.27	92.27
41	Temperature (C)	58.51	58.50	58.50	58.50
42	Vapour Fraction	0.8726	0.8725	0.8725	0.8725
43	Heat Transferred (kJ/h-m)	759.5	759.4	759.4	759.4
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.1874	0.1874	0.1874	0.1874
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	7.022e-003	7.021e-003	7.021e-003	7.021e-003
49	Static Gradient (kPa/m)	-3.848e-002	-3.848e-002	-3.848e-002	-3.848e-002
50	Accel. Gradient (kPa/m)	-2.180e-006	-2.180e-006	-2.180e-006	-2.180e-006
51	Liq. Reynolds	2.180e+005 *	2.180e+005 *	2.180e+005 *	2.180e+005 *
52	Vap. Reynolds	9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
53	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
54	Vapour Velocity (m/s)	2.169	2.169	2.169	2.168
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	7728	7733	7738	7743
58	Elevation (m)	-0.3700	-0.3820	-0.3940	-0.4060
59	Pressure (bar)	92.27	92.27	92.27	92.27
60	Temperature (C)	58.50	58.50	58.50	58.49
61	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
62	Heat Transferred (kJ/h-m)	759.3	759.2	759.2	759.1
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.1874	0.2049	0.2049	0.2049
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	7.021e-003	6.862e-003	6.862e-003	6.861e-003
68	Static Gradient (kPa/m)	-3.848e-002	-4.610e-003	-4.611e-003	-4.611e-003
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 156 of 937				

1			Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:02:44 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient	(kPa/m)	-2.180e-006	1.560e-007	1.560e-007	1.560e-007	
14	Liq. Reynolds		2.180e+005 *	2.180e+005 *	2.180e+005 *	2.180e+005 *	
15	Vap. Reynolds		9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *	
16	Liquid Velocity	(m/s)	0.2040	0.2040	0.2040	0.2040	
17	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168	
18	Deposit Thickness	(mm)	---	---	---	---	
19	Deposit Volume	(m3)	---	---	---	---	
20	Length	(m)	7748	7753	7758	7763	
21	Elevation	(m)	-0.4180	-0.4300	-0.3300	-0.2300	
22	Pressure	(bar)	92.27	92.27	92.27	92.27	
23	Temperature	(C)	58.49	58.49	58.49	58.48	
24	Vapour Fraction		0.8725	0.8725	0.8725	0.8725	
25	Heat Transferred	(kJ/h-m)	759.1	759.0	759.0	758.9	
26	Flow Regime		Transition	Transition	Transition	Transition	
27	Water Flow Regime						
28	Liquid HoldUp		0.2049	0.2049	0.3433	0.3433	
29	Water HoldUp		---	---	---	---	
30	Friction Gradient	(kPa/m)	6.861e-003	6.861e-003	6.053e-003	6.053e-003	
31	Static Gradient	(kPa/m)	-4.611e-003	-4.611e-003	5.462e-002	5.462e-002	
32	Accel. Gradient	(kPa/m)	1.560e-007	1.560e-007	4.204e-006	4.205e-006	
33	Liq. Reynolds		2.180e+005 *	2.180e+005 *	2.180e+005 *	2.180e+005 *	
34	Vap. Reynolds		9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *	
35	Liquid Velocity	(m/s)	0.2040	0.2040	0.2040	0.2040	
36	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168	
37	Deposit Thickness	(mm)	---	---	---	---	
38	Deposit Volume	(m3)	---	---	---	---	
39	Length	(m)	7768	7773	7778	7783	
40	Elevation	(m)	-0.1300	-3.000e-002	7.000e-002	9.600e-002	
41	Pressure	(bar)	92.26	92.26	92.26	92.26	
42	Temperature	(C)	58.48	58.48	58.47	58.47	
43	Vapour Fraction		0.8725	0.8725	0.8725	0.8725	
44	Heat Transferred	(kJ/h-m)	758.8	758.7	758.6	758.5	
45	Flow Regime		Transition	Transition	Transition	Transition	
46	Water Flow Regime						
47	Liquid HoldUp		0.3433	0.3433	0.3433	0.2961	
48	Water HoldUp		---	---	---	---	
49	Friction Gradient	(kPa/m)	6.053e-003	6.053e-003	6.053e-003	5.432e-003	
50	Static Gradient	(kPa/m)	5.462e-002	5.462e-002	5.462e-002	1.276e-002	
51	Accel. Gradient	(kPa/m)	4.205e-006	4.205e-006	4.205e-006	1.261e-006	
52	Liq. Reynolds		2.180e+005 *	2.180e+005 *	2.180e+005 *	2.180e+005 *	
53	Vap. Reynolds		9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *	
54	Liquid Velocity	(m/s)	0.2040	0.2040	0.2040	0.2040	
55	Vapour Velocity	(m/s)	2.168	2.168	2.169	2.169	
56	Deposit Thickness	(mm)	---	---	---	---	
57	Deposit Volume	(m3)	---	---	---	---	
58	Length	(m)	7788	7793	7798	7803	
59	Elevation	(m)	0.1220	0.1480	0.1740	0.2000	
60	Pressure	(bar)	92.26	92.26	92.25	92.25	
61	Temperature	(C)	58.47	58.47	58.46	58.46	
62	Vapour Fraction		0.8725	0.8725	0.8725	0.8725	
63	Heat Transferred	(kJ/h-m)	758.5	758.4	758.3	758.3	
64	Flow Regime		Transition	Transition	Transition	Transition	
65	Water Flow Regime						
66	Liquid HoldUp		0.2961	0.2961	0.2961	0.2961	
67	Water HoldUp		---	---	---	---	
68	Friction Gradient	(kPa/m)	5.432e-003	5.432e-003	5.432e-003	5.432e-003	
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
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5			Date/Time: Mon Jun 13 15:02:44 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	1.276e-002	1.276e-002	1.276e-002	1.276e-002
14	Accel. Gradient	(kPa/m)	1.261e-006	1.261e-006	1.261e-006	1.261e-006
15	Liq. Reynolds		2.179e+005 *	2.179e+005 *	2.179e+005 *	2.179e+005 *
16	Vap. Reynolds		9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
17	Liquid Velocity	(m/s)	0.2040	0.2040	0.2040	0.2040
18	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	7808	7813	7818	7823
22	Elevation	(m)	0.2000	0.2000	0.2000	0.2000
23	Pressure	(bar)	92.25	92.25	92.25	92.25
24	Temperature	(C)	58.46	58.46	58.46	58.45
25	Vapour Fraction		0.8725	0.8725	0.8725	0.8725
26	Heat Transferred	(kJ/h-m)	758.2	758.2	758.1	758.0
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.3024	0.3025	0.3025	0.3025
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	5.867e-003	5.867e-003	5.867e-003	5.867e-003
32	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000
33	Accel. Gradient	(kPa/m)	4.066e-007	4.066e-007	4.066e-007	4.066e-007
34	Liq. Reynolds		2.179e+005 *	2.179e+005 *	2.179e+005 *	2.179e+005 *
35	Vap. Reynolds		9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
36	Liquid Velocity	(m/s)	0.2040	0.2040	0.2040	0.2040
37	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	7828	7833	7838	7843
41	Elevation	(m)	0.2000	0.2040	0.2080	0.2120
42	Pressure	(bar)	92.25	92.25	92.25	92.25
43	Temperature	(C)	58.45	58.45	58.45	58.44
44	Vapour Fraction		0.8725	0.8725	0.8725	0.8725
45	Heat Transferred	(kJ/h-m)	758.0	757.9	757.9	757.8
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.3025	0.2820	0.2820	0.2820
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	5.867e-003	5.415e-003	5.414e-003	5.414e-003
51	Static Gradient	(kPa/m)	0.0000	1.897e-003	1.897e-003	1.897e-003
52	Accel. Gradient	(kPa/m)	4.066e-007	5.067e-007	5.067e-007	5.067e-007
53	Liq. Reynolds		2.179e+005 *	2.179e+005 *	2.179e+005 *	2.179e+005 *
54	Vap. Reynolds		9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
55	Liquid Velocity	(m/s)	0.2040	0.2040	0.2040	0.2041
56	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	7848	7853	7858	7863
60	Elevation	(m)	0.2160	0.2200	0.2280	0.2360
61	Pressure	(bar)	92.25	92.25	92.25	92.25
62	Temperature	(C)	58.44	58.44	58.44	58.43
63	Vapour Fraction		0.8725	0.8725	0.8725	0.8725
64	Heat Transferred	(kJ/h-m)	757.7	757.7	757.6	757.6
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2820	0.2820	0.2846	0.2846
68	Water HoldUp		---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	5.414e-003	5.414e-003	5.219e-003	5.219e-003
14	Static Gradient	(kPa/m)	1.897e-003	1.897e-003	3.818e-003	3.818e-003
15	Accel. Gradient	(kPa/m)	5.067e-007	5.067e-007	6.263e-007	6.263e-007
16	Liq. Reynolds		2.179e+005 *	2.179e+005 *	2.179e+005 *	2.179e+005 *
17	Vap. Reynolds		9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
18	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041
19	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	7868	7873	7878	7883
23	Elevation	(m)	0.2440	0.2520	0.2600	0.2560
24	Pressure	(bar)	92.25	92.25	92.25	92.25
25	Temperature	(C)	58.43	58.43	58.43	58.43
26	Vapour Fraction		0.8725	0.8725	0.8725	0.8725
27	Heat Transferred	(kJ/h-m)	757.5	757.4	757.4	757.3
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2846	0.2846	0.2846	0.2064
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	5.219e-003	5.219e-003	5.219e-003	6.850e-003
33	Static Gradient	(kPa/m)	3.818e-003	3.818e-003	3.818e-003	-1.542e-003
34	Accel. Gradient	(kPa/m)	6.263e-007	6.263e-007	6.263e-007	3.679e-007
35	Liq. Reynolds		2.179e+005 *	2.179e+005 *	2.179e+005 *	2.179e+005 *
36	Vap. Reynolds		9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
37	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041
38	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	7888	7893	7898	7903
42	Elevation	(m)	0.2520	0.2480	0.2440	0.2400
43	Pressure	(bar)	92.25	92.25	92.25	92.25
44	Temperature	(C)	58.42	58.42	58.42	58.42
45	Vapour Fraction		0.8725	0.8725	0.8725	0.8725
46	Heat Transferred	(kJ/h-m)	757.3	757.2	757.2	757.1
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2064	0.2065	0.2065	0.2065
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	6.849e-003	6.849e-003	6.849e-003	6.849e-003
52	Static Gradient	(kPa/m)	-1.542e-003	-1.542e-003	-1.542e-003	-1.542e-003
53	Accel. Gradient	(kPa/m)	3.679e-007	3.679e-007	3.679e-007	3.679e-007
54	Liq. Reynolds		2.179e+005 *	2.179e+005 *	2.179e+005 *	2.179e+005 *
55	Vap. Reynolds		9.070e+006 *	9.070e+006 *	9.070e+006 *	9.071e+006 *
56	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041
57	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	7908	7913	7918	7923
61	Elevation	(m)	0.2380	0.2360	0.2340	0.2320
62	Pressure	(bar)	92.25	92.25	92.25	92.25
63	Temperature	(C)	58.42	58.41	58.41	58.41
64	Vapour Fraction		0.8725	0.8725	0.8725	0.8725
65	Heat Transferred	(kJ/h-m)	757.0	757.0	756.9	756.9
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2068	0.2068	0.2068	0.2068
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	6.846e-003	6.846e-003	6.846e-003	6.846e-003
15	Static Gradient (kPa/m)	-7.717e-004	-7.717e-004	-7.717e-004	-7.717e-004
16	Accel. Gradient (kPa/m)	4.210e-007	4.210e-007	4.210e-007	4.210e-007
17	Liq. Reynolds	2.179e+005 *	2.179e+005 *	2.179e+005 *	2.179e+005 *
18	Vap. Reynolds	9.071e+006 *	9.071e+006 *	9.071e+006 *	9.071e+006 *
19	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
20	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	7928	7933	7938	7943
24	Elevation (m)	0.2300	0.2520	0.2740	0.2960
25	Pressure (bar)	92.25	92.24	92.24	92.24
26	Temperature (C)	58.41	58.40	58.40	58.40
27	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
28	Heat Transferred (kJ/h-m)	756.8	756.7	756.7	756.6
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2068	0.2936	0.2936	0.2936
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	6.846e-003	4.773e-003	4.773e-003	4.773e-003
34	Static Gradient (kPa/m)	-7.717e-004	1.080e-002	1.080e-002	1.080e-002
35	Accel. Gradient (kPa/m)	4.210e-007	1.079e-006	1.079e-006	1.079e-006
36	Liq. Reynolds	2.179e+005 *	2.179e+005 *	2.179e+005 *	2.179e+005 *
37	Vap. Reynolds	9.071e+006 *	9.071e+006 *	9.071e+006 *	9.071e+006 *
38	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
39	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	7948	7953	7958	7963
43	Elevation (m)	0.3180	0.3400	0.3020	0.2640
44	Pressure (bar)	92.24	92.24	92.24	92.24
45	Temperature (C)	58.40	58.39	58.39	58.39
46	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
47	Heat Transferred (kJ/h-m)	756.5	756.5	756.5	756.4
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2936	0.2936	0.2001	0.2001
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	4.773e-003	4.773e-003	6.901e-003	6.901e-003
53	Static Gradient (kPa/m)	1.080e-002	1.080e-002	-1.436e-002	-1.436e-002
54	Accel. Gradient (kPa/m)	1.079e-006	1.080e-006	-5.171e-007	-5.171e-007
55	Liq. Reynolds	2.179e+005 *	2.179e+005 *	2.179e+005 *	2.179e+005 *
56	Vap. Reynolds	9.071e+006 *	9.071e+006 *	9.071e+006 *	9.071e+006 *
57	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
58	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	7968	7973	7978	7983
62	Elevation (m)	0.2260	0.1880	0.1500	0.1360
63	Pressure (bar)	92.24	92.24	92.24	92.24
64	Temperature (C)	58.39	58.39	58.38	58.38
65	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
66	Heat Transferred (kJ/h-m)	756.4	756.3	756.2	756.2
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.2001	0.2001	0.2001	0.2046
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	6.901e-003	6.901e-003	6.901e-003	6.863e-003
16	Static Gradient (kPa/m)	-1.436e-002	-1.436e-002	-1.436e-002	-5.365e-003
17	Accel. Gradient (kPa/m)	-5.171e-007	-5.171e-007	-5.171e-007	1.038e-007
18	Liq. Reynolds	2.179e+005 *	2.179e+005 *	2.179e+005 *	2.178e+005 *
19	Vap. Reynolds	9.071e+006 *	9.071e+006 *	9.071e+006 *	9.071e+006 *
20	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
21	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	7988	7993	7998	8003
25	Elevation (m)	0.1220	0.1080	9.400e-002	8.000e-002
26	Pressure (bar)	92.24	92.24	92.24	92.24
27	Temperature (C)	58.38	58.38	58.38	58.37
28	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
29	Heat Transferred (kJ/h-m)	756.1	756.1	756.0	756.0
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2046	0.2046	0.2046	0.2046
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.863e-003	6.863e-003	6.863e-003	6.863e-003
35	Static Gradient (kPa/m)	-5.365e-003	-5.365e-003	-5.365e-003	-5.366e-003
36	Accel. Gradient (kPa/m)	1.038e-007	1.038e-007	1.038e-007	1.038e-007
37	Liq. Reynolds	2.178e+005 *	2.178e+005 *	2.178e+005 *	2.178e+005 *
38	Vap. Reynolds	9.071e+006 *	9.071e+006 *	9.071e+006 *	9.071e+006 *
39	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
40	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	8008	8013	8018	8023
44	Elevation (m)	6.600e-002	5.200e-002	3.800e-002	2.400e-002
45	Pressure (bar)	92.24	92.24	92.24	92.24
46	Temperature (C)	58.37	58.37	58.37	58.37
47	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
48	Heat Transferred (kJ/h-m)	755.9	755.8	755.8	755.7
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2046	0.2046	0.2046	0.2046
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.863e-003	6.863e-003	6.863e-003	6.863e-003
54	Static Gradient (kPa/m)	-5.366e-003	-5.366e-003	-5.366e-003	-5.366e-003
55	Accel. Gradient (kPa/m)	1.038e-007	1.038e-007	1.037e-007	1.037e-007
56	Liq. Reynolds	2.178e+005 *	2.178e+005 *	2.178e+005 *	2.178e+005 *
57	Vap. Reynolds	9.071e+006 *	9.071e+006 *	9.071e+006 *	9.071e+006 *
58	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
59	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	8028	8033	8038	8043
63	Elevation (m)	1.000e-002	-5.400e-002	-0.1180	-0.1820
64	Pressure (bar)	92.24	92.24	92.24	92.25
65	Temperature (C)	58.36	58.36	58.36	58.36
66	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
67	Heat Transferred (kJ/h-m)	755.7	755.6	755.6	755.5
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 161 of 937				

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
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3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:02:44 2022		
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6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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11	Pipe Table				
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13	Water Flow Regime				
14	Liquid HoldUp	0.2046	0.1953	0.1953	0.1953
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.863e-003	6.943e-003	6.943e-003	6.943e-003
17	Static Gradient (kPa/m)	-5.366e-003	-2.383e-002	-2.383e-002	-2.383e-002
18	Accel. Gradient (kPa/m)	1.037e-007	-1.170e-006	-1.170e-006	-1.170e-006
19	Liq. Reynolds	2.178e+005 *	2.178e+005 *	2.178e+005 *	2.178e+005 *
20	Vap. Reynolds	9.071e+006 *	9.071e+006 *	9.070e+006 *	9.070e+006 *
21	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
22	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	8048	8053	8058	8063
26	Elevation (m)	-0.2460	-0.3100	-1.168	-2.026
27	Pressure (bar)	92.25	92.25	92.26	92.26
28	Temperature (C)	58.36	58.35	58.36	58.36
29	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
30	Heat Transferred (kJ/h-m)	755.5	755.5	756.2	756.2
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1953	0.1953	5.405e-002	5.405e-002
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	6.942e-003	6.942e-003	1.342e-002	1.342e-002
36	Static Gradient (kPa/m)	-2.383e-002	-2.383e-002	-0.1769	-0.1770
37	Accel. Gradient (kPa/m)	-1.170e-006	-1.170e-006	-1.133e-005	-1.133e-005
38	Liq. Reynolds	2.178e+005 *	2.178e+005 *	2.178e+005 *	2.178e+005 *
39	Vap. Reynolds	9.070e+006 *	9.070e+006 *	9.070e+006 *	9.070e+006 *
40	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
41	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	8068	8073	8078	8083
45	Elevation (m)	-2.884	-3.742	-4.600	-3.816
46	Pressure (bar)	92.27	92.28	92.29	92.25
47	Temperature (C)	58.36	58.36	58.37	58.35
48	Vapour Fraction	0.8725	0.8724	0.8724	0.8725
49	Heat Transferred (kJ/h-m)	756.3	756.3	756.4	756.9
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	5.405e-002	5.405e-002	5.405e-002	0.7626
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	1.342e-002	1.342e-002	1.341e-002	5.667e-003
55	Static Gradient (kPa/m)	-0.1770	-0.1770	-0.1770	0.8149
56	Accel. Gradient (kPa/m)	-1.132e-005	-1.132e-005	-1.132e-005	5.685e-005
57	Liq. Reynolds	2.179e+005 *	2.179e+005 *	2.179e+005 *	2.178e+005 *
58	Vap. Reynolds	9.070e+006 *	9.069e+006 *	9.069e+006 *	9.070e+006 *
59	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
60	Vapour Velocity (m/s)	2.167	2.167	2.166	2.167
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	8088	8093	8098	8103
64	Elevation (m)	-3.032	-2.248	-1.464	-0.6800
65	Pressure (bar)	92.21	92.16	92.12	92.08
66	Temperature (C)	58.34	58.33	58.32	58.31
67	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
68	Heat Transferred (kJ/h-m)	756.5	756.2	755.9	755.6
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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9	PROFILES				
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11	Pipe Table				
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13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.7624	0.7622	0.7620	0.7618
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	5.670e-003	5.672e-003	5.674e-003	5.676e-003
18	Static Gradient (kPa/m)	0.8147	0.8146	0.8144	0.8143
19	Accel. Gradient (kPa/m)	5.689e-005	5.693e-005	5.697e-005	5.701e-005
20	Liq. Reynolds	2.178e+005 *	2.177e+005 *	2.177e+005 *	2.176e+005 *
21	Vap. Reynolds	9.072e+006 *	9.073e+006 *	9.074e+006 *	9.075e+006 *
22	Liquid Velocity (m/s)	0.2041	0.2040	0.2040	0.2040
23	Vapour Velocity (m/s)	2.168	2.169	2.170	2.171
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	8108	8113	8118	8123
27	Elevation (m)	-0.5040	-0.3280	-0.1520	2.400e-002
28	Pressure (bar)	92.08	92.07	92.07	92.06
29	Temperature (C)	58.30	58.30	58.29	58.29
30	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
31	Heat Transferred (kJ/h-m)	754.1	754.0	753.9	753.8
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.3913	0.3913	0.3913	0.3913
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	5.994e-003	5.994e-003	5.994e-003	5.995e-003
37	Static Gradient (kPa/m)	0.1061	0.1061	0.1061	0.1061
38	Accel. Gradient (kPa/m)	7.793e-006	7.794e-006	7.795e-006	7.796e-006
39	Liq. Reynolds	2.176e+005 *	2.176e+005 *	2.176e+005 *	2.176e+005 *
40	Vap. Reynolds	9.076e+006 *	9.076e+006 *	9.076e+006 *	9.076e+006 *
41	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
42	Vapour Velocity (m/s)	2.171	2.172	2.172	2.172
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	8128	8133	8138	8143
46	Elevation (m)	0.2000	0.2100	0.2200	0.2300
47	Pressure (bar)	92.05	92.05	92.05	92.05
48	Temperature (C)	58.29	58.28	58.28	58.28
49	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
50	Heat Transferred (kJ/h-m)	753.7	753.5	753.5	753.4
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.3912	0.2856	0.2856	0.2856
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	5.995e-003	5.134e-003	5.133e-003	5.133e-003
56	Static Gradient (kPa/m)	0.1061	4.803e-003	4.803e-003	4.803e-003
57	Accel. Gradient (kPa/m)	7.796e-006	6.913e-007	6.913e-007	6.913e-007
58	Liq. Reynolds	2.175e+005 *	2.175e+005 *	2.175e+005 *	2.175e+005 *
59	Vap. Reynolds	9.076e+006 *	9.076e+006 *	9.076e+006 *	9.076e+006 *
60	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
61	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	8148	8153	8158	8163
65	Elevation (m)	0.2400	0.2500	0.2480	0.2460
66	Pressure (bar)	92.05	92.05	92.05	92.05
67	Temperature (C)	58.28	58.28	58.27	58.27
68	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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13	Heat Transferred (kJ/h-m)	753.4	753.3	753.3	753.2
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.2856	0.2856	0.2066	0.2066
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	5.133e-003	5.133e-003	6.857e-003	6.857e-003
19	Static Gradient (kPa/m)	4.803e-003	4.803e-003	-7.709e-004	-7.709e-004
20	Accel. Gradient (kPa/m)	6.913e-007	6.913e-007	4.234e-007	4.234e-007
21	Liq. Reynolds	2.175e+005 *	2.175e+005 *	2.175e+005 *	2.175e+005 *
22	Vap. Reynolds	9.076e+006 *	9.076e+006 *	9.076e+006 *	9.076e+006 *
23	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
24	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	8168	8173	8178	8183
28	Elevation (m)	0.2440	0.2420	0.2400	0.2380
29	Pressure (bar)	92.05	92.05	92.05	92.05
30	Temperature (C)	58.27	58.27	58.26	58.26
31	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
32	Heat Transferred (kJ/h-m)	753.1	753.1	753.0	753.0
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.2066	0.2066	0.2066	0.2066
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	6.857e-003	6.857e-003	6.857e-003	6.857e-003
38	Static Gradient (kPa/m)	-7.709e-004	-7.709e-004	-7.709e-004	-7.709e-004
39	Accel. Gradient (kPa/m)	4.234e-007	4.234e-007	4.234e-007	4.234e-007
40	Liq. Reynolds	2.175e+005 *	2.175e+005 *	2.175e+005 *	2.175e+005 *
41	Vap. Reynolds	9.076e+006 *	9.076e+006 *	9.076e+006 *	9.076e+006 *
42	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
43	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	8188	8193	8198	8203
47	Elevation (m)	0.2360	0.2340	0.2320	0.2300
48	Pressure (bar)	92.05	92.05	92.05	92.05
49	Temperature (C)	58.26	58.26	58.26	58.25
50	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
51	Heat Transferred (kJ/h-m)	752.9	752.9	752.8	752.7
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.2066	0.2066	0.2066	0.2066
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	6.857e-003	6.857e-003	6.857e-003	6.857e-003
57	Static Gradient (kPa/m)	-7.709e-004	-7.709e-004	-7.709e-004	-7.709e-004
58	Accel. Gradient (kPa/m)	4.234e-007	4.234e-007	4.234e-007	4.234e-007
59	Liq. Reynolds	2.175e+005 *	2.175e+005 *	2.175e+005 *	2.175e+005 *
60	Vap. Reynolds	9.076e+006 *	9.076e+006 *	9.076e+006 *	9.076e+006 *
61	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
62	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	8208	8213	8218	8223
66	Elevation (m)	0.2440	0.2580	0.2720	0.2860
67	Pressure (bar)	92.05	92.05	92.05	92.05
68	Temperature (C)	58.25	58.25	58.25	58.24
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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13	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
14	Heat Transferred (kJ/h-m)	752.7	752.6	752.5	752.5
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2881	0.2881	0.2881	0.2881
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	4.947e-003	4.947e-003	4.947e-003	4.947e-003
20	Static Gradient (kPa/m)	6.739e-003	6.739e-003	6.739e-003	6.739e-003
21	Accel. Gradient (kPa/m)	8.130e-007	8.130e-007	8.130e-007	8.130e-007
22	Liq. Reynolds	2.175e+005 *	2.175e+005 *	2.175e+005 *	2.175e+005 *
23	Vap. Reynolds	9.076e+006 *	9.076e+006 *	9.076e+006 *	9.076e+006 *
24	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
25	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	8228	8233	8238	8243
29	Elevation (m)	0.3000	0.3000	0.3000	0.3000
30	Pressure (bar)	92.05	92.05	92.05	92.05
31	Temperature (C)	58.24	58.24	58.24	58.24
32	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
33	Heat Transferred (kJ/h-m)	752.4	752.4	752.3	752.2
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2881	0.3022	0.3022	0.3022
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	4.947e-003	5.873e-003	5.873e-003	5.873e-003
39	Static Gradient (kPa/m)	6.739e-003	0.0000	0.0000	0.0000
40	Accel. Gradient (kPa/m)	8.130e-007	4.086e-007	4.086e-007	4.086e-007
41	Liq. Reynolds	2.175e+005 *	2.175e+005 *	2.175e+005 *	2.175e+005 *
42	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
43	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
44	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	8248	8253	8258	8263
48	Elevation (m)	0.3000	0.3000	0.3080	0.3160
49	Pressure (bar)	92.04	92.04	92.04	92.04
50	Temperature (C)	58.23	58.23	58.23	58.23
51	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
52	Heat Transferred (kJ/h-m)	752.2	752.1	752.1	752.0
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.3022	0.3022	0.2843	0.2843
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	5.873e-003	5.873e-003	5.230e-003	5.230e-003
58	Static Gradient (kPa/m)	0.0000	0.0000	3.815e-003	3.815e-003
59	Accel. Gradient (kPa/m)	4.086e-007	4.086e-007	6.292e-007	6.292e-007
60	Liq. Reynolds	2.175e+005 *	2.175e+005 *	2.175e+005 *	2.175e+005 *
61	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
62	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
63	Vapour Velocity (m/s)	2.172	2.172	2.172	2.171
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	8268	8273	8278	8283
67	Elevation (m)	0.3240	0.3320	0.3400	0.3340
68	Pressure (bar)	92.04	92.04	92.04	92.04
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 165 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	58.22	58.22	58.22	58.22
14	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
15	Heat Transferred (kJ/h-m)	751.9	751.9	751.8	751.8
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2843	0.2843	0.2843	0.2059
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	5.230e-003	5.230e-003	5.230e-003	6.862e-003
21	Static Gradient (kPa/m)	3.815e-003	3.815e-003	3.815e-003	-2.308e-003
22	Accel. Gradient (kPa/m)	6.292e-007	6.292e-007	6.292e-007	3.168e-007
23	Liq. Reynolds	2.175e+005 *	2.175e+005 *	2.175e+005 *	2.175e+005 *
24	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
25	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
26	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	8288	8293	8298	8303
30	Elevation (m)	0.3280	0.3220	0.3160	0.3100
31	Pressure (bar)	92.04	92.04	92.04	92.04
32	Temperature (C)	58.22	58.21	58.21	58.21
33	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
34	Heat Transferred (kJ/h-m)	751.7	751.7	751.6	751.6
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2059	0.2059	0.2059	0.2059
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	6.862e-003	6.862e-003	6.862e-003	6.862e-003
40	Static Gradient (kPa/m)	-2.308e-003	-2.308e-003	-2.308e-003	-2.308e-003
41	Accel. Gradient (kPa/m)	3.168e-007	3.168e-007	3.168e-007	3.168e-007
42	Liq. Reynolds	2.175e+005 *	2.175e+005 *	2.175e+005 *	2.175e+005 *
43	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
44	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
45	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	8308	8313	8318	8323
49	Elevation (m)	0.3220	0.3340	0.3460	0.3580
50	Pressure (bar)	92.04	92.04	92.04	92.04
51	Temperature (C)	58.21	58.20	58.20	58.20
52	Vapour Fraction	0.8725	0.8725	0.8725	0.8725
53	Heat Transferred (kJ/h-m)	751.5	751.4	751.4	751.3
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2869	0.2869	0.2869	0.2869
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	5.039e-003	5.039e-003	5.039e-003	5.039e-003
59	Static Gradient (kPa/m)	5.759e-003	5.759e-003	5.759e-003	5.759e-003
60	Accel. Gradient (kPa/m)	7.512e-007	7.512e-007	7.512e-007	7.512e-007
61	Liq. Reynolds	2.175e+005 *	2.174e+005 *	2.174e+005 *	2.174e+005 *
62	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
63	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
64	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	8328	8333	8338	8343
68	Elevation (m)	0.3700	0.3080	0.2460	0.1840
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Pressure (bar)	92.04	92.04	92.04	92.04		
14	Temperature (C)	58.20	58.20	58.19	58.19		
15	Vapour Fraction	0.8725	0.8725	0.8725	0.8725		
16	Heat Transferred (kJ/h-m)	751.2	751.2	751.2	751.1		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.2869	0.1955	0.1955	0.1955		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	5.039e-003	6.951e-003	6.950e-003	6.950e-003		
22	Static Gradient (kPa/m)	5.759e-003	-2.311e-002	-2.312e-002	-2.312e-002		
23	Accel. Gradient (kPa/m)	7.512e-007	-1.124e-006	-1.124e-006	-1.124e-006		
24	Liq. Reynolds	2.174e+005 *	2.174e+005 *	2.174e+005 *	2.174e+005 *		
25	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *		
26	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040		
27	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	8348	8353	8358	8363		
31	Elevation (m)	0.1220	6.000e-002	6.000e-002	6.000e-002		
32	Pressure (bar)	92.04	92.04	92.04	92.04		
33	Temperature (C)	58.19	58.19	58.19	58.18		
34	Vapour Fraction	0.8725	0.8725	0.8725	0.8725		
35	Heat Transferred (kJ/h-m)	751.1	751.0	750.9	750.9		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.1955	0.1955	0.3022	0.3022		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	6.950e-003	6.950e-003	5.872e-003	5.872e-003		
41	Static Gradient (kPa/m)	-2.312e-002	-2.312e-002	0.0000	0.0000		
42	Accel. Gradient (kPa/m)	-1.124e-006	-1.124e-006	4.084e-007	4.084e-007		
43	Liq. Reynolds	2.174e+005 *	2.174e+005 *	2.174e+005 *	2.174e+005 *		
44	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *		
45	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040		
46	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	8368	8373	8378	8383		
50	Elevation (m)	6.000e-002	6.000e-002	6.000e-002	4.000e-002		
51	Pressure (bar)	92.04	92.04	92.04	92.04		
52	Temperature (C)	58.18	58.18	58.18	58.18		
53	Vapour Fraction	0.8725	0.8725	0.8725	0.8725		
54	Heat Transferred (kJ/h-m)	750.8	750.8	750.7	750.7		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.3022	0.3022	0.3022	0.2033		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	5.872e-003	5.872e-003	5.872e-003	6.882e-003		
60	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-7.632e-003		
61	Accel. Gradient (kPa/m)	4.084e-007	4.084e-007	4.084e-007	-5.221e-008		
62	Liq. Reynolds	2.174e+005 *	2.174e+005 *	2.174e+005 *	2.174e+005 *		
63	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *		
64	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040		
65	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	8388	8393	8398	8403		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Elevation (m)	2.000e-002	-1.499e-015	-2.000e-002	-4.000e-002		
14	Pressure (bar)	92.04	92.04	92.04	92.04		
15	Temperature (C)	58.17	58.17	58.17	58.17		
16	Vapour Fraction	0.8725	0.8725	0.8725	0.8725		
17	Heat Transferred (kJ/h-m)	750.6	750.6	750.5	750.5		
18	Flow Regime	Transition	Transition	Transition	Transition		
19	Water Flow Regime						
20	Liquid HoldUp	0.2033	0.2033	0.2033	0.2033		
21	Water HoldUp	---	---	---	---		
22	Friction Gradient (kPa/m)	6.882e-003	6.882e-003	6.882e-003	6.881e-003		
23	Static Gradient (kPa/m)	-7.632e-003	-7.633e-003	-7.633e-003	-7.633e-003		
24	Accel. Gradient (kPa/m)	-5.221e-008	-5.222e-008	-5.223e-008	-5.224e-008		
25	Liq. Reynolds	2.174e+005 *	2.174e+005 *	2.174e+005 *	2.174e+005 *		
26	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *		
27	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040		
28	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
29	Deposit Thickness (mm)	---	---	---	---		
30	Deposit Volume (m3)	---	---	---	---		
31	Length (m)	8408	8413	8418	8423		
32	Elevation (m)	-6.400e-002	-8.800e-002	-0.1120	-0.1360		
33	Pressure (bar)	92.04	92.04	92.04	92.04		
34	Temperature (C)	58.17	58.16	58.16	58.16		
35	Vapour Fraction	0.8725	0.8725	0.8725	0.8724		
36	Heat Transferred (kJ/h-m)	750.4	750.3	750.3	750.2		
37	Flow Regime	Transition	Transition	Transition	Transition		
38	Water Flow Regime						
39	Liquid HoldUp	0.2026	0.2026	0.2026	0.2026		
40	Water HoldUp	---	---	---	---		
41	Friction Gradient (kPa/m)	6.888e-003	6.887e-003	6.887e-003	6.887e-003		
42	Static Gradient (kPa/m)	-9.138e-003	-9.138e-003	-9.138e-003	-9.138e-003		
43	Accel. Gradient (kPa/m)	-1.565e-007	-1.565e-007	-1.565e-007	-1.565e-007		
44	Liq. Reynolds	2.174e+005 *	2.174e+005 *	2.174e+005 *	2.174e+005 *		
45	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *		
46	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040		
47	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
48	Deposit Thickness (mm)	---	---	---	---		
49	Deposit Volume (m3)	---	---	---	---		
50	Length (m)	8428	8433	8438	8443		
51	Elevation (m)	-0.1600	-0.1680	-0.1760	-0.1840		
52	Pressure (bar)	92.04	92.04	92.04	92.04		
53	Temperature (C)	58.16	58.16	58.15	58.15		
54	Vapour Fraction	0.8724	0.8724	0.8724	0.8724		
55	Heat Transferred (kJ/h-m)	750.2	750.1	750.1	750.0		
56	Flow Regime	Transition	Transition	Transition	Transition		
57	Water Flow Regime						
58	Liquid HoldUp	0.2026	0.2056	0.2056	0.2056		
59	Water HoldUp	---	---	---	---		
60	Friction Gradient (kPa/m)	6.887e-003	6.863e-003	6.863e-003	6.863e-003		
61	Static Gradient (kPa/m)	-9.139e-003	-3.074e-003	-3.074e-003	-3.074e-003		
62	Accel. Gradient (kPa/m)	-1.566e-007	2.635e-007	2.635e-007	2.635e-007		
63	Liq. Reynolds	2.174e+005 *	2.174e+005 *	2.174e+005 *	2.174e+005 *		
64	Vap. Reynolds	9.076e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *		
65	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040		
66	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
67	Deposit Thickness (mm)	---	---	---	---		
68	Deposit Volume (m3)	---	---	---	---		
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	8448	8453	8458	8463
14	Elevation (m)	-0.1920	-0.2000	-0.1640	-0.1280
15	Pressure (bar)	92.04	92.04	92.04	92.04
16	Temperature (C)	58.15	58.15	58.14	58.14
17	Vapour Fraction	0.8724	0.8724	0.8724	0.8724
18	Heat Transferred (kJ/h-m)	750.0	749.9	749.8	749.8
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.2056	0.2056	0.3023	0.3023
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	6.863e-003	6.863e-003	5.871e-003	5.871e-003
24	Static Gradient (kPa/m)	-3.074e-003	-3.074e-003	1.793e-002	1.793e-002
25	Accel. Gradient (kPa/m)	2.635e-007	2.634e-007	1.655e-006	1.655e-006
26	Liq. Reynolds	2.174e+005 *	2.174e+005 *	2.174e+005 *	2.174e+005 *
27	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
28	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
29	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	8468	8473	8478	8483
33	Elevation (m)	-9.200e-002	-5.600e-002	-2.000e-002	-1.800e-002
34	Pressure (bar)	92.04	92.04	92.04	92.03
35	Temperature (C)	58.14	58.14	58.13	58.13
36	Vapour Fraction	0.8724	0.8724	0.8724	0.8724
37	Heat Transferred (kJ/h-m)	749.7	749.6	749.6	749.5
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.3023	0.3023	0.3023	0.2806
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	5.871e-003	5.871e-003	5.871e-003	5.523e-003
43	Static Gradient (kPa/m)	1.793e-002	1.793e-002	1.793e-002	9.451e-004
44	Accel. Gradient (kPa/m)	1.655e-006	1.655e-006	1.655e-006	4.498e-007
45	Liq. Reynolds	2.174e+005 *	2.174e+005 *	2.174e+005 *	2.174e+005 *
46	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
47	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
48	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	8488	8493	8498	8503
52	Elevation (m)	-1.600e-002	-1.400e-002	-1.200e-002	-1.000e-002
53	Pressure (bar)	92.03	92.03	92.03	92.03
54	Temperature (C)	58.13	58.13	58.13	58.12
55	Vapour Fraction	0.8724	0.8724	0.8724	0.8724
56	Heat Transferred (kJ/h-m)	749.4	749.4	749.3	749.3
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.2806	0.2806	0.2806	0.2806
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	5.523e-003	5.523e-003	5.522e-003	5.522e-003
62	Static Gradient (kPa/m)	9.451e-004	9.451e-004	9.451e-004	9.451e-004
63	Accel. Gradient (kPa/m)	4.498e-007	4.498e-007	4.498e-007	4.498e-007
64	Liq. Reynolds	2.174e+005 *	2.174e+005 *	2.174e+005 *	2.174e+005 *
65	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
66	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
67	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
68	Deposit Thickness (mm)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	8508	8513	8518	8523
15	Elevation (m)	-1.000e-002	-1.000e-002	-1.000e-002	-1.000e-002
16	Pressure (bar)	92.03	92.03	92.03	92.03
17	Temperature (C)	58.12	58.12	58.12	58.11
18	Vapour Fraction	0.8724	0.8724	0.8724	0.8724
19	Heat Transferred (kJ/h-m)	749.2	749.1	749.1	749.0
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.3023	0.3023	0.3023	0.3023
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	5.871e-003	5.871e-003	5.871e-003	5.871e-003
25	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
26	Accel. Gradient (kPa/m)	4.083e-007	4.083e-007	4.083e-007	4.083e-007
27	Liq. Reynolds	2.174e+005 *	2.174e+005 *	2.174e+005 *	2.173e+005 *
28	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
29	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
30	Vapour Velocity (m/s)	2.171	2.171	2.170	2.170
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	8528	8533	8538	8543
34	Elevation (m)	-1.000e-002	-1.400e-002	-1.800e-002	-2.200e-002
35	Pressure (bar)	92.03	92.03	92.03	92.03
36	Temperature (C)	58.11	58.11	58.11	58.11
37	Vapour Fraction	0.8724	0.8724	0.8724	0.8724
38	Heat Transferred (kJ/h-m)	749.0	748.9	748.9	748.8
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.3023	0.2063	0.2063	0.2063
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	5.871e-003	6.857e-003	6.856e-003	6.856e-003
44	Static Gradient (kPa/m)	0.0000	-1.541e-003	-1.541e-003	-1.541e-003
45	Accel. Gradient (kPa/m)	4.083e-007	3.697e-007	3.696e-007	3.696e-007
46	Liq. Reynolds	2.173e+005 *	2.173e+005 *	2.173e+005 *	2.173e+005 *
47	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
48	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
49	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	8548	8553	8558	8563
53	Elevation (m)	-2.600e-002	-3.000e-002	-2.800e-002	-2.600e-002
54	Pressure (bar)	92.03	92.03	92.03	92.03
55	Temperature (C)	58.10	58.10	58.10	58.10
56	Vapour Fraction	0.8724	0.8724	0.8724	0.8724
57	Heat Transferred (kJ/h-m)	748.7	748.7	748.6	748.6
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2063	0.2063	0.2806	0.2806
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	6.856e-003	6.856e-003	5.522e-003	5.522e-003
63	Static Gradient (kPa/m)	-1.541e-003	-1.541e-003	9.418e-004	9.418e-004
64	Accel. Gradient (kPa/m)	3.696e-007	3.696e-007	4.495e-007	4.495e-007
65	Liq. Reynolds	2.173e+005 *	2.173e+005 *	2.173e+005 *	2.173e+005 *
66	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
67	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
68	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	8568	8573	8578	8583
16	Elevation (m)	-2.400e-002	-2.200e-002	-2.000e-002	-2.200e-002
17	Pressure (bar)	92.03	92.03	92.03	92.03
18	Temperature (C)	58.09	58.09	58.09	58.09
19	Vapour Fraction	0.8724	0.8724	0.8724	0.8724
20	Heat Transferred (kJ/h-m)	748.5	748.4	748.4	748.3
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2806	0.2806	0.2806	0.2067
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	5.522e-003	5.522e-003	5.522e-003	6.853e-003
26	Static Gradient (kPa/m)	9.418e-004	9.418e-004	9.418e-004	-7.713e-004
27	Accel. Gradient (kPa/m)	4.495e-007	4.495e-007	4.495e-007	4.229e-007
28	Liq. Reynolds	2.173e+005 *	2.173e+005 *	2.173e+005 *	2.173e+005 *
29	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
30	Liquid Velocity (m/s)	0.2040	0.2040	0.2040	0.2040
31	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	8588	8593	8598	8603
35	Elevation (m)	-2.400e-002	-2.600e-002	-2.800e-002	-3.000e-002
36	Pressure (bar)	92.03	92.03	92.03	92.03
37	Temperature (C)	58.09	58.08	58.08	58.08
38	Vapour Fraction	0.8724	0.8724	0.8724	0.8724
39	Heat Transferred (kJ/h-m)	748.3	748.2	748.2	748.1
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2067	0.2067	0.2067	0.2067
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	6.853e-003	6.853e-003	6.853e-003	6.853e-003
45	Static Gradient (kPa/m)	-7.713e-004	-7.713e-004	-7.713e-004	-7.713e-004
46	Accel. Gradient (kPa/m)	4.229e-007	4.229e-007	4.229e-007	4.229e-007
47	Liq. Reynolds	2.173e+005 *	2.173e+005 *	2.173e+005 *	2.173e+005 *
48	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
49	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
50	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	8608	8613	8618	8623
54	Elevation (m)	-2.600e-002	-2.200e-002	-1.800e-002	-1.400e-002
55	Pressure (bar)	92.03	92.03	92.03	92.03
56	Temperature (C)	58.08	58.08	58.07	58.07
57	Vapour Fraction	0.8724	0.8724	0.8724	0.8724
58	Heat Transferred (kJ/h-m)	748.0	748.0	747.9	747.9
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2819	0.2819	0.2819	0.2819
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	5.422e-003	5.422e-003	5.421e-003	5.421e-003
64	Static Gradient (kPa/m)	1.896e-003	1.897e-003	1.897e-003	1.897e-003
65	Accel. Gradient (kPa/m)	5.089e-007	5.089e-007	5.089e-007	5.088e-007
66	Liq. Reynolds	2.173e+005 *	2.173e+005 *	2.173e+005 *	2.173e+005 *
67	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
68	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
14	Deposit Thickness (mm)	---	---	---	---
15	Deposit Volume (m3)	---	---	---	---
16	Length (m)	8628	8633	8638	8643
17	Elevation (m)	-1.000e-002	-8.000e-003	-6.000e-003	-4.000e-003
18	Pressure (bar)	92.03	92.03	92.02	92.02
19	Temperature (C)	58.07	58.07	58.06	58.06
20	Vapour Fraction	0.8724	0.8724	0.8724	0.8724
21	Heat Transferred (kJ/h-m)	747.8	747.7	747.7	747.6
22	Flow Regime	Transition	Transition	Transition	Transition
23	Water Flow Regime				
24	Liquid HoldUp	0.2819	0.2806	0.2806	0.2806
25	Water HoldUp	---	---	---	---
26	Friction Gradient (kPa/m)	5.421e-003	5.521e-003	5.521e-003	5.521e-003
27	Static Gradient (kPa/m)	1.897e-003	9.453e-004	9.453e-004	9.453e-004
28	Accel. Gradient (kPa/m)	5.088e-007	4.496e-007	4.496e-007	4.496e-007
29	Liq. Reynolds	2.173e+005 *	2.173e+005 *	2.173e+005 *	2.173e+005 *
30	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
31	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
32	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
33	Deposit Thickness (mm)	---	---	---	---
34	Deposit Volume (m3)	---	---	---	---
35	Length (m)	8648	8653	8658	8663
36	Elevation (m)	-2.000e-003	-1.518e-015	2.000e-003	4.000e-003
37	Pressure (bar)	92.02	92.02	92.02	92.02
38	Temperature (C)	58.06	58.06	58.06	58.05
39	Vapour Fraction	0.8724	0.8724	0.8724	0.8724
40	Heat Transferred (kJ/h-m)	747.6	747.5	747.4	747.4
41	Flow Regime	Transition	Transition	Transition	Transition
42	Water Flow Regime				
43	Liquid HoldUp	0.2806	0.2806	0.2806	0.2806
44	Water HoldUp	---	---	---	---
45	Friction Gradient (kPa/m)	5.521e-003	5.521e-003	5.521e-003	5.521e-003
46	Static Gradient (kPa/m)	9.453e-004	9.453e-004	9.453e-004	9.453e-004
47	Accel. Gradient (kPa/m)	4.496e-007	4.496e-007	4.496e-007	4.496e-007
48	Liq. Reynolds	2.173e+005 *	2.173e+005 *	2.173e+005 *	2.173e+005 *
49	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
50	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
51	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
52	Deposit Thickness (mm)	---	---	---	---
53	Deposit Volume (m3)	---	---	---	---
54	Length (m)	8668	8673	8678	8683
55	Elevation (m)	6.000e-003	8.000e-003	1.000e-002	1.000e-002
56	Pressure (bar)	92.02	92.02	92.02	92.02
57	Temperature (C)	58.05	58.05	58.05	58.04
58	Vapour Fraction	0.8724	0.8724	0.8724	0.8724
59	Heat Transferred (kJ/h-m)	747.3	747.3	747.2	747.2
60	Flow Regime	Transition	Transition	Transition	Transition
61	Water Flow Regime				
62	Liquid HoldUp	0.2806	0.2806	0.2806	0.3023
63	Water HoldUp	---	---	---	---
64	Friction Gradient (kPa/m)	5.521e-003	5.521e-003	5.521e-003	5.870e-003
65	Static Gradient (kPa/m)	9.453e-004	9.453e-004	9.453e-004	0.0000
66	Accel. Gradient (kPa/m)	4.496e-007	4.496e-007	4.496e-007	4.082e-007
67	Liq. Reynolds	2.173e+005 *	2.173e+005 *	2.173e+005 *	2.173e+005 *
68	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:02:44 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041	
14	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	8688	8693	8698	8703	
18	Elevation	(m)	1.000e-002	1.000e-002	1.000e-002	1.000e-002	
19	Pressure	(bar)	92.02	92.02	92.02	92.02	
20	Temperature	(C)	58.04	58.04	58.04	58.04	
21	Vapour Fraction		0.8724	0.8724	0.8724	0.8724	
22	Heat Transferred	(kJ/h-m)	747.1	747.0	747.0	746.9	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.3023	0.3023	0.3023	0.3023	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	5.870e-003	5.870e-003	5.870e-003	5.870e-003	
28	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000	
29	Accel. Gradient	(kPa/m)	4.082e-007	4.082e-007	4.081e-007	4.081e-007	
30	Liq. Reynolds		2.173e+005 *	2.173e+005 *	2.173e+005 *	2.173e+005 *	
31	Vap. Reynolds		9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *	
32	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041	
33	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	8708	8713	8718	8723	
37	Elevation	(m)	1.200e-002	1.400e-002	1.600e-002	1.800e-002	
38	Pressure	(bar)	92.02	92.02	92.02	92.02	
39	Temperature	(C)	58.03	58.03	58.03	58.03	
40	Vapour Fraction		0.8724	0.8724	0.8724	0.8724	
41	Heat Transferred	(kJ/h-m)	746.9	746.8	746.7	746.7	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2806	0.2806	0.2806	0.2806	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	5.520e-003	5.520e-003	5.520e-003	5.520e-003	
47	Static Gradient	(kPa/m)	9.454e-004	9.454e-004	9.454e-004	9.454e-004	
48	Accel. Gradient	(kPa/m)	4.495e-007	4.495e-007	4.495e-007	4.495e-007	
49	Liq. Reynolds		2.173e+005 *	2.173e+005 *	2.173e+005 *	2.172e+005 *	
50	Vap. Reynolds		9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *	
51	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041	
52	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	8728	8733	8738	8743	
56	Elevation	(m)	2.000e-002	1.600e-002	1.200e-002	8.000e-003	
57	Pressure	(bar)	92.02	92.02	92.02	92.02	
58	Temperature	(C)	58.02	58.02	58.02	58.02	
59	Vapour Fraction		0.8724	0.8724	0.8724	0.8724	
60	Heat Transferred	(kJ/h-m)	746.6	746.6	746.5	746.5	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2806	0.2064	0.2064	0.2064	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	5.520e-003	6.855e-003	6.855e-003	6.855e-003	
66	Static Gradient	(kPa/m)	9.454e-004	-1.541e-003	-1.541e-003	-1.541e-003	
67	Accel. Gradient	(kPa/m)	4.495e-007	3.694e-007	3.694e-007	3.694e-007	
68	Liq. Reynolds		2.172e+005 *	2.172e+005 *	2.172e+005 *	2.172e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
14	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
15	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	8748	8753	8758	8763
19	Elevation (m)	4.000e-003	-1.516e-015	-6.000e-003	-1.200e-002
20	Pressure (bar)	92.02	92.02	92.02	92.02
21	Temperature (C)	58.02	58.01	58.01	58.01
22	Vapour Fraction	0.8724	0.8724	0.8724	0.8724
23	Heat Transferred (kJ/h-m)	746.4	746.4	746.3	746.2
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2064	0.2064	0.2060	0.2060
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.855e-003	6.855e-003	6.857e-003	6.857e-003
29	Static Gradient (kPa/m)	-1.541e-003	-1.541e-003	-2.310e-003	-2.310e-003
30	Accel. Gradient (kPa/m)	3.694e-007	3.694e-007	3.162e-007	3.161e-007
31	Liq. Reynolds	2.172e+005 *	2.172e+005 *	2.172e+005 *	2.172e+005 *
32	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
33	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
34	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	8768	8773	8778	8783
38	Elevation (m)	-1.800e-002	-2.400e-002	-3.000e-002	-3.800e-002
39	Pressure (bar)	92.02	92.02	92.02	92.02
40	Temperature (C)	58.01	58.01	58.00	58.00
41	Vapour Fraction	0.8724	0.8724	0.8724	0.8724
42	Heat Transferred (kJ/h-m)	746.2	746.1	746.1	746.0
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2060	0.2060	0.2060	0.2056
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.857e-003	6.857e-003	6.857e-003	6.860e-003
48	Static Gradient (kPa/m)	-2.310e-003	-2.310e-003	-2.310e-003	-3.076e-003
49	Accel. Gradient (kPa/m)	3.161e-007	3.161e-007	3.161e-007	2.631e-007
50	Liq. Reynolds	2.172e+005 *	2.172e+005 *	2.172e+005 *	2.172e+005 *
51	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
52	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
53	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	8788	8793	8798	8803
57	Elevation (m)	-4.600e-002	-5.400e-002	-6.200e-002	-7.000e-002
58	Pressure (bar)	92.02	92.02	92.02	92.02
59	Temperature (C)	58.00	58.00	57.99	57.99
60	Vapour Fraction	0.8724	0.8724	0.8724	0.8724
61	Heat Transferred (kJ/h-m)	746.0	745.9	745.8	745.8
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2056	0.2056	0.2056	0.2056
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	6.860e-003	6.860e-003	6.860e-003	6.860e-003
67	Static Gradient (kPa/m)	-3.076e-003	-3.076e-003	-3.076e-003	-3.076e-003
68	Accel. Gradient (kPa/m)	2.631e-007	2.631e-007	2.631e-007	2.631e-007
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.172e+005 *	2.172e+005 *	2.172e+005 *	2.172e+005 *
14	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
15	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
16	Vapour Velocity (m/s)	2.170	2.170	2.169	2.169
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	8805	8807	8809	8811
20	Elevation (m)	-6.000e-002	-5.000e-002	-4.000e-002	-3.000e-002
21	Pressure (bar)	92.02	92.02	92.01	92.01
22	Temperature (C)	57.99	57.99	57.99	57.99
23	Vapour Fraction	0.8724	0.8724	0.8723	0.8723
24	Heat Transferred (kJ/h-m)	745.8	745.7	745.7	745.7
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2953	0.2953	0.2953	0.2953
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	5.323e-003	5.323e-003	5.323e-003	5.323e-003
30	Static Gradient (kPa/m)	1.225e-002	1.225e-002	1.225e-002	1.225e-002
31	Accel. Gradient (kPa/m)	1.222e-006	1.222e-006	1.222e-006	1.222e-006
32	Liq. Reynolds	2.172e+005 *	2.172e+005 *	2.172e+005 *	2.172e+005 *
33	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
34	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
35	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	8813	8840	8867	8894
39	Elevation (m)	-2.000e-002	-2.600e-002	-3.200e-002	-3.800e-002
40	Pressure (bar)	92.01	92.01	92.01	92.01
41	Temperature (C)	57.99	57.98	57.96	57.95
42	Vapour Fraction	0.8723	0.8723	0.8723	0.8723
43	Heat Transferred (kJ/h-m)	745.7	745.3	745.0	744.7
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2953	0.2069	0.2069	0.2069
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	5.323e-003	6.850e-003	6.849e-003	6.849e-003
49	Static Gradient (kPa/m)	1.225e-002	-4.289e-004	-4.289e-004	-4.289e-004
50	Accel. Gradient (kPa/m)	1.222e-006	4.464e-007	4.463e-007	4.463e-007
51	Liq. Reynolds	2.172e+005 *	2.172e+005 *	2.172e+005 *	2.172e+005 *
52	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
53	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
54	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	8921	8948	8949	8950
58	Elevation (m)	-4.400e-002	-5.000e-002	-4.600e-002	-4.200e-002
59	Pressure (bar)	92.01	92.01	92.01	92.01
60	Temperature (C)	57.94	57.93	57.93	57.93
61	Vapour Fraction	0.8723	0.8723	0.8723	0.8723
62	Heat Transferred (kJ/h-m)	744.4	744.1	744.1	744.1
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2069	0.2070	0.2922	0.2922
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	6.849e-003	6.849e-003	4.667e-003	4.667e-003
68	Static Gradient (kPa/m)	-4.289e-004	-4.289e-004	9.727e-003	9.727e-003
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:02:44 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Accel. Gradient	(kPa/m)	4.463e-007	4.463e-007	1.001e-006	1.001e-006				
14	Liq. Reynolds		2.171e+005 *	2.171e+005 *	2.171e+005 *	2.171e+005 *				
15	Vap. Reynolds		9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *				
16	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041				
17	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169				
18	Deposit Thickness	(mm)	---	---	---	---				
19	Deposit Volume	(m3)	---	---	---	---				
20	Length	(m)	8951	8952	8953	8958				
21	Elevation	(m)	-3.800e-002	-3.400e-002	-3.000e-002	-2.600e-002				
22	Pressure	(bar)	92.01	92.00	92.00	92.00				
23	Temperature	(C)	57.93	57.93	57.93	57.92				
24	Vapour Fraction		0.8723	0.8723	0.8723	0.8723				
25	Heat Transferred	(kJ/h-m)	744.0	744.0	744.0	744.0				
26	Flow Regime		Transition	Transition	Transition	Transition				
27	Water Flow Regime									
28	Liquid HoldUp		0.2922	0.2922	0.2922	0.2820				
29	Water HoldUp		---	---	---	---				
30	Friction Gradient	(kPa/m)	4.667e-003	4.667e-003	4.667e-003	5.418e-003				
31	Static Gradient	(kPa/m)	9.727e-003	9.727e-003	9.727e-003	1.898e-003				
32	Accel. Gradient	(kPa/m)	1.001e-006	1.001e-006	1.001e-006	5.086e-007				
33	Liq. Reynolds		2.171e+005 *	2.171e+005 *	2.171e+005 *	2.171e+005 *				
34	Vap. Reynolds		9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *				
35	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041				
36	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169				
37	Deposit Thickness	(mm)	---	---	---	---				
38	Deposit Volume	(m3)	---	---	---	---				
39	Length	(m)	8963	8968	8973	8978				
40	Elevation	(m)	-2.200e-002	-1.800e-002	-1.400e-002	-1.000e-002				
41	Pressure	(bar)	92.00	92.00	92.00	92.00				
42	Temperature	(C)	57.92	57.92	57.92	57.92				
43	Vapour Fraction		0.8723	0.8723	0.8723	0.8723				
44	Heat Transferred	(kJ/h-m)	743.9	743.8	743.8	743.7				
45	Flow Regime		Transition	Transition	Transition	Transition				
46	Water Flow Regime									
47	Liquid HoldUp		0.2820	0.2820	0.2820	0.2820				
48	Water HoldUp		---	---	---	---				
49	Friction Gradient	(kPa/m)	5.418e-003	5.418e-003	5.418e-003	5.418e-003				
50	Static Gradient	(kPa/m)	1.898e-003	1.898e-003	1.898e-003	1.898e-003				
51	Accel. Gradient	(kPa/m)	5.085e-007	5.085e-007	5.085e-007	5.085e-007				
52	Liq. Reynolds		2.171e+005 *	2.171e+005 *	2.171e+005 *	2.171e+005 *				
53	Vap. Reynolds		9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *				
54	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041				
55	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169				
56	Deposit Thickness	(mm)	---	---	---	---				
57	Deposit Volume	(m3)	---	---	---	---				
58	Length	(m)	8983	8988	8993	8998				
59	Elevation	(m)	8.000e-003	2.600e-002	4.400e-002	6.200e-002				
60	Pressure	(bar)	92.00	92.00	92.00	92.00				
61	Temperature	(C)	57.91	57.91	57.91	57.91				
62	Vapour Fraction		0.8723	0.8723	0.8723	0.8723				
63	Heat Transferred	(kJ/h-m)	743.7	743.6	743.5	743.5				
64	Flow Regime		Transition	Transition	Transition	Transition				
65	Water Flow Regime									
66	Liquid HoldUp		0.2909	0.2909	0.2909	0.2909				
67	Water HoldUp		---	---	---	---				
68	Friction Gradient	(kPa/m)	4.757e-003	4.757e-003	4.757e-003	4.757e-003				
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	8.727e-003	8.727e-003	8.728e-003	8.728e-003
14	Accel. Gradient	(kPa/m)	9.373e-007	9.373e-007	9.373e-007	9.373e-007
15	Liq. Reynolds		2.171e+005 *	2.171e+005 *	2.171e+005 *	2.171e+005 *
16	Vap. Reynolds		9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
17	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041
18	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	9003	9008	9013	9018
22	Elevation	(m)	8.000e-002	6.400e-002	4.800e-002	3.200e-002
23	Pressure	(bar)	92.00	92.00	92.00	92.00
24	Temperature	(C)	57.90	57.90	57.90	57.90
25	Vapour Fraction		0.8723	0.8723	0.8723	0.8723
26	Heat Transferred	(kJ/h-m)	743.4	743.4	743.3	743.3
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2909	0.2042	0.2042	0.2042
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	4.757e-003	6.870e-003	6.870e-003	6.870e-003
32	Static Gradient	(kPa/m)	8.728e-003	-6.125e-003	-6.125e-003	-6.125e-003
33	Accel. Gradient	(kPa/m)	9.373e-007	5.184e-008	5.183e-008	5.182e-008
34	Liq. Reynolds		2.171e+005 *	2.171e+005 *	2.171e+005 *	2.171e+005 *
35	Vap. Reynolds		9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
36	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041
37	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	9023	9028	9033	9038
41	Elevation	(m)	1.600e-002	-1.513e-015	-8.000e-003	-1.600e-002
42	Pressure	(bar)	92.00	92.00	92.00	92.00
43	Temperature	(C)	57.90	57.89	57.89	57.89
44	Vapour Fraction		0.8723	0.8723	0.8723	0.8723
45	Heat Transferred	(kJ/h-m)	743.2	743.2	743.1	743.0
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2042	0.2042	0.2057	0.2057
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	6.870e-003	6.870e-003	6.858e-003	6.858e-003
51	Static Gradient	(kPa/m)	-6.125e-003	-6.125e-003	-3.076e-003	-3.076e-003
52	Accel. Gradient	(kPa/m)	5.181e-008	5.180e-008	2.629e-007	2.628e-007
53	Liq. Reynolds		2.171e+005 *	2.171e+005 *	2.171e+005 *	2.171e+005 *
54	Vap. Reynolds		9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
55	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041
56	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	9043	9048	9053	9054
60	Elevation	(m)	-2.400e-002	-3.200e-002	-4.000e-002	-4.800e-002
61	Pressure	(bar)	92.00	92.00	92.00	92.00
62	Temperature	(C)	57.89	57.89	57.88	57.88
63	Vapour Fraction		0.8723	0.8723	0.8723	0.8723
64	Heat Transferred	(kJ/h-m)	743.0	742.9	742.9	742.9
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2057	0.2057	0.2057	0.1997
68	Water HoldUp		---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:02:44 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	6.858e-003	6.858e-003	6.858e-003	6.907e-003	
14	Static Gradient	(kPa/m)	-3.076e-003	-3.076e-003	-3.076e-003	-1.510e-002	
15	Accel. Gradient	(kPa/m)	2.628e-007	2.628e-007	2.628e-007	-5.696e-007	
16	Liq. Reynolds		2.171e+005 *	2.171e+005 *	2.171e+005 *	2.171e+005 *	
17	Vap. Reynolds		9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *	
18	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041	
19	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	9055	9056	9057	9058	
23	Elevation	(m)	-5.600e-002	-6.400e-002	-7.200e-002	-8.000e-002	
24	Pressure	(bar)	92.00	92.00	92.00	92.00	
25	Temperature	(C)	57.88	57.88	57.88	57.88	
26	Vapour Fraction		0.8723	0.8723	0.8723	0.8723	
27	Heat Transferred	(kJ/h-m)	742.9	742.9	742.8	742.8	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.1997	0.1997	0.1997	0.1997	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	6.907e-003	6.907e-003	6.907e-003	6.907e-003	
33	Static Gradient	(kPa/m)	-1.510e-002	-1.510e-002	-1.510e-002	-1.510e-002	
34	Accel. Gradient	(kPa/m)	-5.696e-007	-5.696e-007	-5.696e-007	-5.696e-007	
35	Liq. Reynolds		2.171e+005 *	2.171e+005 *	2.171e+005 *	2.171e+005 *	
36	Vap. Reynolds		9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *	
37	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041	
38	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	9065	9071	9078	9085	
42	Elevation	(m)	-9.200e-002	-0.1040	-0.1160	-0.1280	
43	Pressure	(bar)	92.00	92.00	92.00	92.00	
44	Temperature	(C)	57.88	57.88	57.87	57.87	
45	Vapour Fraction		0.8723	0.8723	0.8723	0.8723	
46	Heat Transferred	(kJ/h-m)	742.8	742.7	742.6	742.5	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.2055	0.2055	0.2055	0.2055	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	6.859e-003	6.859e-003	6.859e-003	6.859e-003	
52	Static Gradient	(kPa/m)	-3.459e-003	-3.460e-003	-3.460e-003	-3.460e-003	
53	Accel. Gradient	(kPa/m)	2.363e-007	2.363e-007	2.363e-007	2.363e-007	
54	Liq. Reynolds		2.171e+005 *	2.171e+005 *	2.171e+005 *	2.171e+005 *	
55	Vap. Reynolds		9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *	
56	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041	
57	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.168	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	9091	9098	9105	9111	
61	Elevation	(m)	-0.1400	-0.1620	-0.1840	-0.2060	
62	Pressure	(bar)	92.00	92.00	92.00	92.00	
63	Temperature	(C)	57.87	57.86	57.86	57.86	
64	Vapour Fraction		0.8723	0.8723	0.8723	0.8723	
65	Heat Transferred	(kJ/h-m)	742.5	742.4	742.3	742.2	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.2055	0.2041	0.2041	0.2041	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Water HoldUp		---	---	---	---	
14	Friction Gradient (kPa/m)		6.859e-003	6.870e-003	6.870e-003	6.870e-003	
15	Static Gradient (kPa/m)		-3.460e-003	-6.316e-003	-6.316e-003	-6.316e-003	
16	Accel. Gradient (kPa/m)		2.363e-007	3.853e-008	3.852e-008	3.851e-008	
17	Liq. Reynolds		2.171e+005 *	2.171e+005 *	2.171e+005 *	2.171e+005 *	
18	Vap. Reynolds		9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *	
19	Liquid Velocity (m/s)		0.2041	0.2041	0.2041	0.2041	
20	Vapour Velocity (m/s)		2.168	2.168	2.168	2.168	
21	Deposit Thickness (mm)		---	---	---	---	
22	Deposit Volume (m3)		---	---	---	---	
23	Length (m)		9118	9125	9131	9138	
24	Elevation (m)		-0.2280	-0.2500	-0.2660	-0.2820	
25	Pressure (bar)		92.00	92.00	92.00	92.00	
26	Temperature (C)		57.86	57.85	57.85	57.85	
27	Vapour Fraction		0.8723	0.8723	0.8723	0.8723	
28	Heat Transferred (kJ/h-m)		742.2	742.1	742.0	741.9	
29	Flow Regime		Transition	Transition	Transition	Transition	
30	Water Flow Regime						
31	Liquid HoldUp		0.2041	0.2041	0.2050	0.2050	
32	Water HoldUp		---	---	---	---	
33	Friction Gradient (kPa/m)		6.870e-003	6.870e-003	6.863e-003	6.863e-003	
34	Static Gradient (kPa/m)		-6.316e-003	-6.316e-003	-4.604e-003	-4.604e-003	
35	Accel. Gradient (kPa/m)		3.850e-008	3.848e-008	1.570e-007	1.570e-007	
36	Liq. Reynolds		2.171e+005 *	2.171e+005 *	2.170e+005 *	2.170e+005 *	
37	Vap. Reynolds		9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *	
38	Liquid Velocity (m/s)		0.2041	0.2041	0.2041	0.2041	
39	Vapour Velocity (m/s)		2.168	2.168	2.168	2.168	
40	Deposit Thickness (mm)		---	---	---	---	
41	Deposit Volume (m3)		---	---	---	---	
42	Length (m)		9145	9151	9158	9162	
43	Elevation (m)		-0.2980	-0.3140	-0.3300	-0.3240	
44	Pressure (bar)		92.00	92.00	92.00	92.00	
45	Temperature (C)		57.85	57.84	57.84	57.84	
46	Vapour Fraction		0.8723	0.8723	0.8723	0.8723	
47	Heat Transferred (kJ/h-m)		741.9	741.8	741.7	741.7	
48	Flow Regime		Transition	Transition	Transition	Transition	
49	Water Flow Regime						
50	Liquid HoldUp		0.2050	0.2050	0.2050	0.2843	
51	Water HoldUp		---	---	---	---	
52	Friction Gradient (kPa/m)		6.863e-003	6.863e-003	6.862e-003	5.245e-003	
53	Static Gradient (kPa/m)		-4.604e-003	-4.604e-003	-4.604e-003	3.571e-003	
54	Accel. Gradient (kPa/m)		1.570e-007	1.570e-007	1.569e-007	6.127e-007	
55	Liq. Reynolds		2.170e+005 *	2.170e+005 *	2.170e+005 *	2.170e+005 *	
56	Vap. Reynolds		9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *	
57	Liquid Velocity (m/s)		0.2041	0.2041	0.2041	0.2041	
58	Vapour Velocity (m/s)		2.168	2.168	2.168	2.168	
59	Deposit Thickness (mm)		---	---	---	---	
60	Deposit Volume (m3)		---	---	---	---	
61	Length (m)		9166	9170	9174	9178	
62	Elevation (m)		-0.3180	-0.3120	-0.3060	-0.3000	
63	Pressure (bar)		92.00	92.00	92.00	92.00	
64	Temperature (C)		57.84	57.83	57.83	57.83	
65	Vapour Fraction		0.8723	0.8723	0.8723	0.8723	
66	Heat Transferred (kJ/h-m)		741.6	741.6	741.5	741.5	
67	Flow Regime		Transition	Transition	Transition	Transition	
68	Water Flow Regime						
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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9	PROFILES				
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11	Pipe Table				
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13	Liquid HoldUp	0.2843	0.2843	0.2843	0.2843
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	5.245e-003	5.245e-003	5.245e-003	5.245e-003
16	Static Gradient (kPa/m)	3.571e-003	3.571e-003	3.571e-003	3.571e-003
17	Accel. Gradient (kPa/m)	6.127e-007	6.126e-007	6.126e-007	6.126e-007
18	Liq. Reynolds	2.170e+005 *	2.170e+005 *	2.170e+005 *	2.170e+005 *
19	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
20	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
21	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	9183	9188	9193	9198
25	Elevation (m)	-0.3220	-0.3440	-0.3660	-0.3880
26	Pressure (bar)	92.00	92.00	92.00	92.00
27	Temperature (C)	57.83	57.83	57.82	57.82
28	Vapour Fraction	0.8723	0.8723	0.8722	0.8722
29	Heat Transferred (kJ/h-m)	741.4	741.4	741.3	741.3
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2031	0.2031	0.2031	0.2031
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.877e-003	6.877e-003	6.877e-003	6.877e-003
35	Static Gradient (kPa/m)	-8.394e-003	-8.395e-003	-8.395e-003	-8.395e-003
36	Accel. Gradient (kPa/m)	-1.054e-007	-1.054e-007	-1.054e-007	-1.055e-007
37	Liq. Reynolds	2.170e+005 *	2.170e+005 *	2.170e+005 *	2.170e+005 *
38	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
39	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
40	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	9203	9208	9213	9218
44	Elevation (m)	-0.4100	-0.4160	-0.4220	-0.4280
45	Pressure (bar)	92.00	92.00	91.99	91.99
46	Temperature (C)	57.82	57.82	57.82	57.81
47	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
48	Heat Transferred (kJ/h-m)	741.2	741.2	741.1	741.0
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2031	0.2061	0.2061	0.2061
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.877e-003	6.853e-003	6.853e-003	6.853e-003
54	Static Gradient (kPa/m)	-8.395e-003	-2.311e-003	-2.311e-003	-2.311e-003
55	Accel. Gradient (kPa/m)	-1.055e-007	3.156e-007	3.156e-007	3.156e-007
56	Liq. Reynolds	2.170e+005 *	2.170e+005 *	2.170e+005 *	2.170e+005 *
57	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
58	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
59	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	9223	9228	9233	9238
63	Elevation (m)	-0.4340	-0.4400	-0.4420	-0.4440
64	Pressure (bar)	91.99	91.99	91.99	91.99
65	Temperature (C)	57.81	57.81	57.81	57.81
66	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
67	Heat Transferred (kJ/h-m)	741.0	740.9	740.9	740.8
68	Flow Regime	Transition	Transition	Transition	Transition
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
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5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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8					
9	PROFILES				
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11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2061	0.2061	0.2069	0.2069
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.853e-003	6.853e-003	6.847e-003	6.847e-003
17	Static Gradient (kPa/m)	-2.311e-003	-2.311e-003	-7.720e-004	-7.720e-004
18	Accel. Gradient (kPa/m)	3.156e-007	3.156e-007	4.221e-007	4.221e-007
19	Liq. Reynolds	2.170e+005 *	2.170e+005 *	2.170e+005 *	2.170e+005 *
20	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.077e+006 *
21	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
22	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	9243	9248	9253	9258
26	Elevation (m)	-0.4460	-0.4480	-0.4500	-0.4000
27	Pressure (bar)	91.99	91.99	91.99	91.99
28	Temperature (C)	57.80	57.80	57.80	57.80
29	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
30	Heat Transferred (kJ/h-m)	740.8	740.7	740.6	740.6
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2069	0.2069	0.2069	0.3114
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	6.847e-003	6.847e-003	6.847e-003	6.008e-003
36	Static Gradient (kPa/m)	-7.720e-004	-7.720e-004	-7.720e-004	2.545e-002
37	Accel. Gradient (kPa/m)	4.221e-007	4.221e-007	4.221e-007	2.186e-006
38	Liq. Reynolds	2.170e+005 *	2.170e+005 *	2.170e+005 *	2.170e+005 *
39	Vap. Reynolds	9.077e+006 *	9.077e+006 *	9.077e+006 *	9.078e+006 *
40	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
41	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	9263	9268	9273	9278
45	Elevation (m)	-0.3500	-0.3000	-0.2500	-0.2000
46	Pressure (bar)	91.99	91.99	91.99	91.98
47	Temperature (C)	57.79	57.79	57.79	57.79
48	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
49	Heat Transferred (kJ/h-m)	740.5	740.4	740.4	740.3
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.3114	0.3114	0.3114	0.3114
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	6.008e-003	6.008e-003	6.008e-003	6.008e-003
55	Static Gradient (kPa/m)	2.545e-002	2.545e-002	2.545e-002	2.545e-002
56	Accel. Gradient (kPa/m)	2.186e-006	2.186e-006	2.186e-006	2.186e-006
57	Liq. Reynolds	2.170e+005 *	2.170e+005 *	2.170e+005 *	2.170e+005 *
58	Vap. Reynolds	9.078e+006 *	9.078e+006 *	9.078e+006 *	9.078e+006 *
59	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
60	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	9283	9288	9293	9298
64	Elevation (m)	-0.1900	-0.1800	-0.1700	-0.1600
65	Pressure (bar)	91.98	91.98	91.98	91.98
66	Temperature (C)	57.78	57.78	57.78	57.78
67	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
68	Heat Transferred (kJ/h-m)	740.2	740.2	740.1	740.0
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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9	PROFILES				
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11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2859	0.2859	0.2859	0.2859
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	5.124e-003	5.124e-003	5.124e-003	5.124e-003
18	Static Gradient (kPa/m)	4.790e-003	4.790e-003	4.790e-003	4.790e-003
19	Accel. Gradient (kPa/m)	6.889e-007	6.889e-007	6.889e-007	6.889e-007
20	Liq. Reynolds	2.170e+005 *	2.170e+005 *	2.170e+005 *	2.170e+005 *
21	Vap. Reynolds	9.078e+006 *	9.078e+006 *	9.078e+006 *	9.078e+006 *
22	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
23	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	9303	9308	9313	9318
27	Elevation (m)	-0.1500	-0.1520	-0.1540	-0.1560
28	Pressure (bar)	91.98	91.98	91.98	91.98
29	Temperature (C)	57.78	57.77	57.77	57.77
30	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
31	Heat Transferred (kJ/h-m)	740.0	739.9	739.9	739.8
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2859	0.2069	0.2069	0.2069
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	5.124e-003	6.847e-003	6.847e-003	6.847e-003
37	Static Gradient (kPa/m)	4.790e-003	-7.720e-004	-7.720e-004	-7.720e-004
38	Accel. Gradient (kPa/m)	6.889e-007	4.221e-007	4.221e-007	4.221e-007
39	Liq. Reynolds	2.170e+005 *	2.170e+005 *	2.170e+005 *	2.170e+005 *
40	Vap. Reynolds	9.078e+006 *	9.078e+006 *	9.078e+006 *	9.078e+006 *
41	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
42	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	9323	9328	9333	9338
46	Elevation (m)	-0.1580	-0.1600	-0.1340	-0.1080
47	Pressure (bar)	91.98	91.98	91.98	91.98
48	Temperature (C)	57.77	57.76	57.76	57.76
49	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
50	Heat Transferred (kJ/h-m)	739.8	739.7	739.6	739.6
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2069	0.2069	0.2961	0.2961
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	6.847e-003	6.847e-003	5.432e-003	5.432e-003
56	Static Gradient (kPa/m)	-7.720e-004	-7.720e-004	1.277e-002	1.277e-002
57	Accel. Gradient (kPa/m)	4.221e-007	4.221e-007	1.265e-006	1.265e-006
58	Liq. Reynolds	2.169e+005 *	2.169e+005 *	2.169e+005 *	2.169e+005 *
59	Vap. Reynolds	9.078e+006 *	9.078e+006 *	9.078e+006 *	9.078e+006 *
60	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
61	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	9343	9348	9353	9358
65	Elevation (m)	-8.200e-002	-5.600e-002	-3.000e-002	-2.200e-002
66	Pressure (bar)	91.98	91.98	91.98	91.98
67	Temperature (C)	57.76	57.76	57.75	57.75
68	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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9	PROFILES				
10					
11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	739.5	739.4	739.4	739.3
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.2961	0.2961	0.2961	0.2846
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	5.432e-003	5.432e-003	5.432e-003	5.219e-003
19	Static Gradient (kPa/m)	1.277e-002	1.277e-002	1.277e-002	3.816e-003
20	Accel. Gradient (kPa/m)	1.265e-006	1.265e-006	1.265e-006	6.279e-007
21	Liq. Reynolds	2.169e+005 *	2.169e+005 *	2.169e+005 *	2.169e+005 *
22	Vap. Reynolds	9.078e+006 *	9.078e+006 *	9.078e+006 *	9.078e+006 *
23	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
24	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	9363	9368	9373	9378
28	Elevation (m)	-1.400e-002	-6.000e-003	2.000e-003	1.000e-002
29	Pressure (bar)	91.98	91.97	91.97	91.97
30	Temperature (C)	57.75	57.75	57.74	57.74
31	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
32	Heat Transferred (kJ/h-m)	739.3	739.2	739.1	739.1
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.2846	0.2847	0.2847	0.2847
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	5.219e-003	5.219e-003	5.219e-003	5.219e-003
38	Static Gradient (kPa/m)	3.816e-003	3.816e-003	3.816e-003	3.816e-003
39	Accel. Gradient (kPa/m)	6.279e-007	6.279e-007	6.279e-007	6.279e-007
40	Liq. Reynolds	2.169e+005 *	2.169e+005 *	2.169e+005 *	2.169e+005 *
41	Vap. Reynolds	9.078e+006 *	9.078e+006 *	9.078e+006 *	9.078e+006 *
42	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
43	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	9383	9388	9393	9398
47	Elevation (m)	-2.200e-002	-5.400e-002	-8.600e-002	-0.1180
48	Pressure (bar)	91.97	91.97	91.97	91.98
49	Temperature (C)	57.74	57.74	57.74	57.73
50	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
51	Heat Transferred (kJ/h-m)	739.1	739.0	739.0	738.9
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.2013	0.2013	0.2013	0.2013
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	6.892e-003	6.892e-003	6.892e-003	6.891e-003
57	Static Gradient (kPa/m)	-1.214e-002	-1.214e-002	-1.214e-002	-1.214e-002
58	Accel. Gradient (kPa/m)	-3.649e-007	-3.649e-007	-3.649e-007	-3.649e-007
59	Liq. Reynolds	2.169e+005 *	2.169e+005 *	2.169e+005 *	2.169e+005 *
60	Vap. Reynolds	9.078e+006 *	9.078e+006 *	9.078e+006 *	9.078e+006 *
61	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
62	Vapour Velocity (m/s)	2.168	2.168	2.168	2.167
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	9403	9408	9413	9418
66	Elevation (m)	-0.1500	-0.2080	-0.2660	-0.3240
67	Pressure (bar)	91.98	91.98	91.98	91.98
68	Temperature (C)	57.73	57.73	57.73	57.73
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
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5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
14	Heat Transferred (kJ/h-m)	738.8	738.8	738.8	738.7
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2013	0.1964	0.1965	0.1965
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	6.891e-003	6.934e-003	6.934e-003	6.933e-003
20	Static Gradient (kPa/m)	-1.214e-002	-2.168e-002	-2.168e-002	-2.168e-002
21	Accel. Gradient (kPa/m)	-3.649e-007	-1.025e-006	-1.025e-006	-1.025e-006
22	Liq. Reynolds	2.169e+005 *	2.169e+005 *	2.169e+005 *	2.169e+005 *
23	Vap. Reynolds	9.078e+006 *	9.078e+006 *	9.078e+006 *	9.078e+006 *
24	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
25	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	9423	9428	9433	9438
29	Elevation (m)	-0.3820	-0.4400	-0.4740	-0.5080
30	Pressure (bar)	91.98	91.98	91.98	91.98
31	Temperature (C)	57.72	57.72	57.72	57.72
32	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
33	Heat Transferred (kJ/h-m)	738.7	738.6	738.6	738.5
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1965	0.1965	0.2009	0.2009
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	6.933e-003	6.933e-003	6.894e-003	6.894e-003
39	Static Gradient (kPa/m)	-2.168e-002	-2.168e-002	-1.289e-002	-1.289e-002
40	Accel. Gradient (kPa/m)	-1.025e-006	-1.025e-006	-4.164e-007	-4.164e-007
41	Liq. Reynolds	2.169e+005 *	2.169e+005 *	2.169e+005 *	2.169e+005 *
42	Vap. Reynolds	9.078e+006 *	9.078e+006 *	9.078e+006 *	9.078e+006 *
43	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
44	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	9443	9448	9453	9456
48	Elevation (m)	-0.5420	-0.5760	-0.6100	-0.5060
49	Pressure (bar)	91.98	91.98	91.98	91.98
50	Temperature (C)	57.72	57.72	57.71	57.71
51	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
52	Heat Transferred (kJ/h-m)	738.5	738.4	738.4	738.3
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2009	0.2009	0.2009	0.3901
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	6.894e-003	6.894e-003	6.894e-003	5.987e-003
58	Static Gradient (kPa/m)	-1.289e-002	-1.289e-002	-1.289e-002	0.1043
59	Accel. Gradient (kPa/m)	-4.164e-007	-4.165e-007	-4.165e-007	7.661e-006
60	Liq. Reynolds	2.169e+005 *	2.169e+005 *	2.169e+005 *	2.169e+005 *
61	Vap. Reynolds	9.078e+006 *	9.078e+006 *	9.078e+006 *	9.078e+006 *
62	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
63	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	9459	9462	9465	9468
67	Elevation (m)	-0.4020	-0.2980	-0.1940	-9.000e-002
68	Pressure (bar)	91.97	91.97	91.97	91.96
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2			Unit Set: Mixed		
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12					
13	Temperature (C)	57.71	57.71	57.70	57.70
14	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
15	Heat Transferred (kJ/h-m)	738.3	738.2	738.2	738.1
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.3900	0.3900	0.3900	0.3900
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	5.987e-003	5.987e-003	5.988e-003	5.988e-003
21	Static Gradient (kPa/m)	0.1043	0.1043	0.1043	0.1043
22	Accel. Gradient (kPa/m)	7.662e-006	7.662e-006	7.663e-006	7.663e-006
23	Liq. Reynolds	2.169e+005 *	2.169e+005 *	2.169e+005 *	2.169e+005 *
24	Vap. Reynolds	9.078e+006 *	9.078e+006 *	9.078e+006 *	9.078e+006 *
25	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
26	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	9473	9477	9482	9487
30	Elevation (m)	-9.000e-002	-9.000e-002	-9.000e-002	-9.000e-002
31	Pressure (bar)	91.96	91.96	91.96	91.96
32	Temperature (C)	57.70	57.70	57.70	57.69
33	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
34	Heat Transferred (kJ/h-m)	738.0	737.9	737.9	737.8
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.3026	0.3026	0.3026	0.3026
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	5.866e-003	5.866e-003	5.866e-003	5.866e-003
40	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
41	Accel. Gradient (kPa/m)	4.077e-007	4.077e-007	4.077e-007	4.077e-007
42	Liq. Reynolds	2.169e+005 *	2.169e+005 *	2.169e+005 *	2.169e+005 *
43	Vap. Reynolds	9.078e+006 *	9.078e+006 *	9.078e+006 *	9.078e+006 *
44	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
45	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	9491	9496	9501	9505
49	Elevation (m)	-9.000e-002	-0.1820	-0.2740	-0.3660
50	Pressure (bar)	91.96	91.96	91.97	91.97
51	Temperature (C)	57.69	57.69	57.69	57.69
52	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
53	Heat Transferred (kJ/h-m)	737.8	737.8	737.7	737.7
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.3026	0.1889	0.1889	0.1889
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	5.866e-003	7.005e-003	7.005e-003	7.005e-003
59	Static Gradient (kPa/m)	0.0000	-3.596e-002	-3.596e-002	-3.596e-002
60	Accel. Gradient (kPa/m)	4.076e-007	-2.012e-006	-2.012e-006	-2.012e-006
61	Liq. Reynolds	2.169e+005 *	2.169e+005 *	2.169e+005 *	2.169e+005 *
62	Vap. Reynolds	9.078e+006 *	9.078e+006 *	9.078e+006 *	9.078e+006 *
63	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
64	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	9510	9515	9519	9524
68	Elevation (m)	-0.4580	-0.5500	-0.5300	-0.5100
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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12					
13	Pressure (bar)	91.97	91.97	91.97	91.97
14	Temperature (C)	57.69	57.69	57.68	57.68
15	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
16	Heat Transferred (kJ/h-m)	737.7	737.6	737.5	737.5
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.1889	0.1889	0.2933	0.2933
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	7.005e-003	7.005e-003	4.599e-003	4.599e-003
22	Static Gradient (kPa/m)	-3.596e-002	-3.596e-002	1.045e-002	1.045e-002
23	Accel. Gradient (kPa/m)	-2.012e-006	-2.012e-006	1.046e-006	1.046e-006
24	Liq. Reynolds	2.169e+005 *	2.169e+005 *	2.169e+005 *	2.168e+005 *
25	Vap. Reynolds	9.078e+006 *	9.078e+006 *	9.078e+006 *	9.078e+006 *
26	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
27	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	9529	9533	9538	9543
31	Elevation (m)	-0.4900	-0.4700	-0.4500	-0.4160
32	Pressure (bar)	91.97	91.97	91.97	91.96
33	Temperature (C)	57.68	57.68	57.67	57.67
34	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
35	Heat Transferred (kJ/h-m)	737.4	737.4	737.3	737.2
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.2933	0.2933	0.2933	0.3029
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	4.599e-003	4.599e-003	4.599e-003	5.873e-003
41	Static Gradient (kPa/m)	1.045e-002	1.045e-002	1.045e-002	1.818e-002
42	Accel. Gradient (kPa/m)	1.046e-006	1.046e-006	1.046e-006	1.671e-006
43	Liq. Reynolds	2.168e+005 *	2.168e+005 *	2.168e+005 *	2.168e+005 *
44	Vap. Reynolds	9.078e+006 *	9.078e+006 *	9.078e+006 *	9.078e+006 *
45	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
46	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	9547	9552	9557	9561
50	Elevation (m)	-0.3820	-0.3480	-0.3140	-0.2800
51	Pressure (bar)	91.96	91.96	91.96	91.96
52	Temperature (C)	57.67	57.67	57.67	57.66
53	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
54	Heat Transferred (kJ/h-m)	737.2	737.1	737.1	737.0
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.3029	0.3029	0.3029	0.3029
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	5.873e-003	5.873e-003	5.873e-003	5.873e-003
60	Static Gradient (kPa/m)	1.818e-002	1.818e-002	1.818e-002	1.818e-002
61	Accel. Gradient (kPa/m)	1.671e-006	1.671e-006	1.671e-006	1.671e-006
62	Liq. Reynolds	2.168e+005 *	2.168e+005 *	2.168e+005 *	2.168e+005 *
63	Vap. Reynolds	9.078e+006 *	9.078e+006 *	9.078e+006 *	9.078e+006 *
64	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
65	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	9566	9571	9575	9580
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
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3			Unit Set: Mixed		
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5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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10					
11	Pipe Table				
12					
13	Elevation (m)	-0.2420	-0.2040	-0.1660	-0.1280
14	Pressure (bar)	91.96	91.96	91.96	91.96
15	Temperature (C)	57.66	57.66	57.66	57.65
16	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
17	Heat Transferred (kJ/h-m)	736.9	736.9	736.8	736.8
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.3056	0.3056	0.3056	0.3056
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	5.935e-003	5.935e-003	5.935e-003	5.935e-003
23	Static Gradient (kPa/m)	2.044e-002	2.044e-002	2.044e-002	2.044e-002
24	Accel. Gradient (kPa/m)	1.833e-006	1.833e-006	1.833e-006	1.833e-006
25	Liq. Reynolds	2.168e+005 *	2.168e+005 *	2.168e+005 *	2.168e+005 *
26	Vap. Reynolds	9.078e+006 *	9.078e+006 *	9.078e+006 *	9.078e+006 *
27	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
28	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	9585	9589	9594	9599
32	Elevation (m)	-9.000e-002	-7.600e-002	-6.200e-002	-4.800e-002
33	Pressure (bar)	91.95	91.95	91.95	91.95
34	Temperature (C)	57.65	57.65	57.65	57.65
35	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
36	Heat Transferred (kJ/h-m)	736.7	736.6	736.6	736.5
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.3056	0.2892	0.2892	0.2892
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	5.935e-003	4.888e-003	4.888e-003	4.888e-003
42	Static Gradient (kPa/m)	2.044e-002	7.245e-003	7.245e-003	7.245e-003
43	Accel. Gradient (kPa/m)	1.833e-006	8.431e-007	8.432e-007	8.432e-007
44	Liq. Reynolds	2.168e+005 *	2.168e+005 *	2.168e+005 *	2.168e+005 *
45	Vap. Reynolds	9.078e+006 *	9.078e+006 *	9.078e+006 *	9.078e+006 *
46	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
47	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	9603	9608	9612	9616
51	Elevation (m)	-3.400e-002	-2.000e-002	3.000e-002	8.000e-002
52	Pressure (bar)	91.95	91.95	91.95	91.95
53	Temperature (C)	57.64	57.64	57.64	57.64
54	Vapour Fraction	0.8722	0.8722	0.8722	0.8722
55	Heat Transferred (kJ/h-m)	736.5	736.4	736.4	736.3
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2892	0.2892	0.3193	0.3193
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	4.888e-003	4.888e-003	6.044e-003	6.044e-003
61	Static Gradient (kPa/m)	7.245e-003	7.245e-003	3.227e-002	3.227e-002
62	Accel. Gradient (kPa/m)	8.432e-007	8.432e-007	2.663e-006	2.663e-006
63	Liq. Reynolds	2.168e+005 *	2.168e+005 *	2.168e+005 *	2.168e+005 *
64	Vap. Reynolds	9.078e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
65	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
66	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	9620	9624	9628	9633
14	Elevation (m)	0.1300	0.1800	0.2300	0.1840
15	Pressure (bar)	91.95	91.95	91.94	91.94
16	Temperature (C)	57.63	57.63	57.63	57.63
17	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
18	Heat Transferred (kJ/h-m)	736.2	736.2	736.1	736.1
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.3193	0.3193	0.3193	0.1987
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	6.044e-003	6.044e-003	6.044e-003	6.913e-003
24	Static Gradient (kPa/m)	3.227e-002	3.227e-002	3.227e-002	-1.732e-002
25	Accel. Gradient (kPa/m)	2.663e-006	2.663e-006	2.663e-006	-7.230e-007
26	Liq. Reynolds	2.168e+005 *	2.168e+005 *	2.168e+005 *	2.168e+005 *
27	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
28	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
29	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	9638	9643	9648	9653
33	Elevation (m)	0.1380	9.200e-002	4.600e-002	-1.582e-015
34	Pressure (bar)	91.94	91.95	91.95	91.95
35	Temperature (C)	57.63	57.62	57.62	57.62
36	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
37	Heat Transferred (kJ/h-m)	736.0	736.0	735.9	735.9
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.1987	0.1987	0.1987	0.1987
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	6.913e-003	6.913e-003	6.913e-003	6.913e-003
43	Static Gradient (kPa/m)	-1.732e-002	-1.732e-002	-1.732e-002	-1.732e-002
44	Accel. Gradient (kPa/m)	-7.230e-007	-7.231e-007	-7.231e-007	-7.231e-007
45	Liq. Reynolds	2.168e+005 *	2.168e+005 *	2.168e+005 *	2.168e+005 *
46	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
47	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
48	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	9658	9663	9668	9673
52	Elevation (m)	-1.400e-002	-2.800e-002	-4.200e-002	-5.600e-002
53	Pressure (bar)	91.95	91.95	91.95	91.95
54	Temperature (C)	57.62	57.62	57.61	57.61
55	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
56	Heat Transferred (kJ/h-m)	735.8	735.8	735.7	735.7
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.2047	0.2047	0.2047	0.2047
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	6.862e-003	6.862e-003	6.862e-003	6.862e-003
62	Static Gradient (kPa/m)	-5.369e-003	-5.369e-003	-5.369e-003	-5.369e-003
63	Accel. Gradient (kPa/m)	1.038e-007	1.038e-007	1.038e-007	1.038e-007
64	Liq. Reynolds	2.168e+005 *	2.168e+005 *	2.168e+005 *	2.168e+005 *
65	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
66	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
67	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
68	Deposit Thickness (mm)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	9678	9683	9688	9693
15	Elevation (m)	-7.000e-002	-2.200e-002	2.600e-002	7.400e-002
16	Pressure (bar)	91.95	91.94	91.94	91.94
17	Temperature (C)	57.61	57.61	57.61	57.60
18	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
19	Heat Transferred (kJ/h-m)	735.6	735.5	735.5	735.4
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2047	0.3103	0.3103	0.3103
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	6.862e-003	5.995e-003	5.995e-003	5.995e-003
25	Static Gradient (kPa/m)	-5.369e-003	2.437e-002	2.437e-002	2.437e-002
26	Accel. Gradient (kPa/m)	1.038e-007	2.110e-006	2.110e-006	2.110e-006
27	Liq. Reynolds	2.168e+005 *	2.168e+005 *	2.167e+005 *	2.167e+005 *
28	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
29	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
30	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	9698	9703	9708	9713
34	Elevation (m)	0.1220	0.1700	0.1720	0.1740
35	Pressure (bar)	91.94	91.94	91.94	91.94
36	Temperature (C)	57.60	57.60	57.60	57.59
37	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
38	Heat Transferred (kJ/h-m)	735.3	735.3	735.2	735.1
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.3103	0.3103	0.2809	0.2809
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	5.995e-003	5.995e-003	5.511e-003	5.511e-003
44	Static Gradient (kPa/m)	2.437e-002	2.437e-002	9.464e-004	9.464e-004
45	Accel. Gradient (kPa/m)	2.110e-006	2.110e-006	4.488e-007	4.488e-007
46	Liq. Reynolds	2.167e+005 *	2.167e+005 *	2.167e+005 *	2.167e+005 *
47	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
48	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
49	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	9718	9723	9728	9733
53	Elevation (m)	0.1760	0.1780	0.1800	0.1780
54	Pressure (bar)	91.94	91.94	91.94	91.94
55	Temperature (C)	57.59	57.59	57.59	57.59
56	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
57	Heat Transferred (kJ/h-m)	735.1	735.0	735.0	734.9
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2809	0.2809	0.2809	0.2069
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	5.511e-003	5.511e-003	5.511e-003	6.844e-003
63	Static Gradient (kPa/m)	9.464e-004	9.464e-004	9.464e-004	-7.723e-004
64	Accel. Gradient (kPa/m)	4.488e-007	4.488e-007	4.488e-007	4.220e-007
65	Liq. Reynolds	2.167e+005 *	2.167e+005 *	2.167e+005 *	2.167e+005 *
66	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
67	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
68	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 189 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	9738	9743	9748	9753
16	Elevation (m)	0.1760	0.1740	0.1720	0.1700
17	Pressure (bar)	91.94	91.94	91.94	91.94
18	Temperature (C)	57.58	57.58	57.58	57.58
19	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
20	Heat Transferred (kJ/h-m)	734.9	734.8	734.8	734.7
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2069	0.2069	0.2069	0.2069
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	6.844e-003	6.844e-003	6.844e-003	6.844e-003
26	Static Gradient (kPa/m)	-7.723e-004	-7.723e-004	-7.723e-004	-7.723e-004
27	Accel. Gradient (kPa/m)	4.220e-007	4.220e-007	4.220e-007	4.220e-007
28	Liq. Reynolds	2.167e+005 *	2.167e+005 *	2.167e+005 *	2.167e+005 *
29	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
30	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
31	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	9758	9763	9768	9773
35	Elevation (m)	0.1660	0.1620	0.1580	0.1540
36	Pressure (bar)	91.93	91.93	91.93	91.93
37	Temperature (C)	57.57	57.57	57.57	57.57
38	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
39	Heat Transferred (kJ/h-m)	734.6	734.6	734.5	734.5
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2066	0.2066	0.2066	0.2066
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	6.847e-003	6.847e-003	6.847e-003	6.847e-003
45	Static Gradient (kPa/m)	-1.543e-003	-1.543e-003	-1.543e-003	-1.543e-003
46	Accel. Gradient (kPa/m)	3.686e-007	3.686e-007	3.686e-007	3.686e-007
47	Liq. Reynolds	2.167e+005 *	2.167e+005 *	2.167e+005 *	2.167e+005 *
48	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
49	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
50	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	9778	9783	9788	9793
54	Elevation (m)	0.1500	0.1560	0.1620	0.1680
55	Pressure (bar)	91.93	91.93	91.93	91.93
56	Temperature (C)	57.57	57.56	57.56	57.56
57	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
58	Heat Transferred (kJ/h-m)	734.4	734.3	734.3	734.2
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2066	0.2835	0.2835	0.2835
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	6.847e-003	5.313e-003	5.313e-003	5.313e-003
64	Static Gradient (kPa/m)	-1.543e-003	2.858e-003	2.858e-003	2.858e-003
65	Accel. Gradient (kPa/m)	3.686e-007	5.678e-007	5.678e-007	5.678e-007
66	Liq. Reynolds	2.167e+005 *	2.167e+005 *	2.167e+005 *	2.167e+005 *
67	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
68	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
14	Deposit Thickness (mm)	---	---	---	---
15	Deposit Volume (m3)	---	---	---	---
16	Length (m)	9798	9803	9808	9813
17	Elevation (m)	0.1740	0.1800	0.1660	0.1520
18	Pressure (bar)	91.93	91.93	91.93	91.93
19	Temperature (C)	57.56	57.55	57.55	57.55
20	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
21	Heat Transferred (kJ/h-m)	734.2	734.1	734.1	734.0
22	Flow Regime	Transition	Transition	Transition	Transition
23	Water Flow Regime				
24	Liquid HoldUp	0.2835	0.2835	0.2047	0.2047
25	Water HoldUp	---	---	---	---
26	Friction Gradient (kPa/m)	5.313e-003	5.313e-003	6.862e-003	6.861e-003
27	Static Gradient (kPa/m)	2.858e-003	2.858e-003	-5.385e-003	-5.385e-003
28	Accel. Gradient (kPa/m)	5.678e-007	5.678e-007	1.026e-007	1.026e-007
29	Liq. Reynolds	2.167e+005 *	2.167e+005 *	2.167e+005 *	2.167e+005 *
30	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
31	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
32	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
33	Deposit Thickness (mm)	---	---	---	---
34	Deposit Volume (m3)	---	---	---	---
35	Length (m)	9818	9823	9828	9833
36	Elevation (m)	0.1380	0.1240	0.1100	0.1240
37	Pressure (bar)	91.93	91.93	91.93	91.93
38	Temperature (C)	57.55	57.55	57.54	57.54
39	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
40	Heat Transferred (kJ/h-m)	734.0	733.9	733.9	733.8
41	Flow Regime	Transition	Transition	Transition	Transition
42	Water Flow Regime				
43	Liquid HoldUp	0.2047	0.2047	0.2047	0.2886
44	Water HoldUp	---	---	---	---
45	Friction Gradient (kPa/m)	6.861e-003	6.861e-003	6.861e-003	4.933e-003
46	Static Gradient (kPa/m)	-5.385e-003	-5.385e-003	-5.385e-003	6.752e-003
47	Accel. Gradient (kPa/m)	1.026e-007	1.026e-007	1.026e-007	8.119e-007
48	Liq. Reynolds	2.167e+005 *	2.167e+005 *	2.167e+005 *	2.167e+005 *
49	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
50	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
51	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
52	Deposit Thickness (mm)	---	---	---	---
53	Deposit Volume (m3)	---	---	---	---
54	Length (m)	9838	9843	9848	9853
55	Elevation (m)	0.1380	0.1520	0.1660	0.1800
56	Pressure (bar)	91.93	91.93	91.93	91.93
57	Temperature (C)	57.54	57.54	57.54	57.53
58	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
59	Heat Transferred (kJ/h-m)	733.7	733.7	733.6	733.5
60	Flow Regime	Transition	Transition	Transition	Transition
61	Water Flow Regime				
62	Liquid HoldUp	0.2886	0.2886	0.2886	0.2886
63	Water HoldUp	---	---	---	---
64	Friction Gradient (kPa/m)	4.933e-003	4.932e-003	4.932e-003	4.932e-003
65	Static Gradient (kPa/m)	6.752e-003	6.752e-003	6.752e-003	6.752e-003
66	Accel. Gradient (kPa/m)	8.119e-007	8.120e-007	8.120e-007	8.120e-007
67	Liq. Reynolds	2.167e+005 *	2.167e+005 *	2.167e+005 *	2.167e+005 *
68	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041	
14	Vapour Velocity	(m/s)	2.166	2.166	2.166	2.166	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	9858	9863	9868	9873	
18	Elevation	(m)	0.1720	0.1640	0.1560	0.1480	
19	Pressure	(bar)	91.93	91.93	91.93	91.93	
20	Temperature	(C)	57.53	57.53	57.53	57.53	
21	Vapour Fraction		0.8721	0.8721	0.8721	0.8721	
22	Heat Transferred	(kJ/h-m)	733.5	733.4	733.4	733.3	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.2058	0.2058	0.2058	0.2058	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	6.852e-003	6.852e-003	6.852e-003	6.852e-003	
28	Static Gradient	(kPa/m)	-3.079e-003	-3.079e-003	-3.079e-003	-3.079e-003	
29	Accel. Gradient	(kPa/m)	2.622e-007	2.622e-007	2.622e-007	2.622e-007	
30	Liq. Reynolds		2.167e+005 *	2.167e+005 *	2.167e+005 *	2.167e+005 *	
31	Vap. Reynolds		9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *	
32	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041	
33	Vapour Velocity	(m/s)	2.166	2.166	2.166	2.166	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	9878	9883	9888	9893	
37	Elevation	(m)	0.1400	0.1440	0.1480	0.1520	
38	Pressure	(bar)	91.93	91.93	91.93	91.93	
39	Temperature	(C)	57.52	57.52	57.52	57.52	
40	Vapour Fraction		0.8721	0.8721	0.8721	0.8721	
41	Heat Transferred	(kJ/h-m)	733.3	733.2	733.1	733.1	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2058	0.2822	0.2822	0.2822	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	6.852e-003	5.410e-003	5.410e-003	5.410e-003	
47	Static Gradient	(kPa/m)	-3.079e-003	1.899e-003	1.899e-003	1.899e-003	
48	Accel. Gradient	(kPa/m)	2.622e-007	5.079e-007	5.079e-007	5.079e-007	
49	Liq. Reynolds		2.166e+005 *	2.166e+005 *	2.166e+005 *	2.166e+005 *	
50	Vap. Reynolds		9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *	
51	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041	
52	Vapour Velocity	(m/s)	2.166	2.166	2.166	2.166	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	9898	9903	9905	9907	
56	Elevation	(m)	0.1560	0.1600	0.1560	0.1520	
57	Pressure	(bar)	91.93	91.93	91.93	91.93	
58	Temperature	(C)	57.51	57.51	57.51	57.51	
59	Vapour Fraction		0.8721	0.8721	0.8721	0.8721	
60	Heat Transferred	(kJ/h-m)	733.0	733.0	733.0	732.9	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2822	0.2822	0.2055	0.2055	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	5.410e-003	5.410e-003	6.855e-003	6.855e-003	
66	Static Gradient	(kPa/m)	1.899e-003	1.899e-003	-3.845e-003	-3.845e-003	
67	Accel. Gradient	(kPa/m)	5.079e-007	5.079e-007	2.091e-007	2.091e-007	
68	Liq. Reynolds		2.166e+005 *	2.166e+005 *	2.166e+005 *	2.166e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
14	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
15	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	9909	9911	9913	9933
19	Elevation (m)	0.1480	0.1440	0.1400	0.1320
20	Pressure (bar)	91.93	91.93	91.93	91.92
21	Temperature (C)	57.51	57.51	57.51	57.50
22	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
23	Heat Transferred (kJ/h-m)	732.9	732.9	732.9	732.6
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2055	0.2055	0.2055	0.2070
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.855e-003	6.855e-003	6.855e-003	6.843e-003
29	Static Gradient (kPa/m)	-3.845e-003	-3.845e-003	-3.845e-003	-7.725e-004
30	Accel. Gradient (kPa/m)	2.091e-007	2.091e-007	2.091e-007	4.218e-007
31	Liq. Reynolds	2.166e+005 *	2.166e+005 *	2.166e+005 *	2.166e+005 *
32	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
33	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
34	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	9953	9973	9993	1.001e+004
38	Elevation (m)	0.1240	0.1160	0.1080	1.000e-001
39	Pressure (bar)	91.92	91.92	91.92	91.92
40	Temperature (C)	57.49	57.48	57.47	57.47
41	Vapour Fraction	0.8721	0.8721	0.8721	0.8721
42	Heat Transferred (kJ/h-m)	732.4	732.2	732.0	731.7
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2070	0.2070	0.2070	0.2070
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.842e-003	6.842e-003	6.842e-003	6.842e-003
48	Static Gradient (kPa/m)	-7.725e-004	-7.725e-004	-7.725e-004	-7.726e-004
49	Accel. Gradient (kPa/m)	4.217e-007	4.217e-007	4.217e-007	4.217e-007
50	Liq. Reynolds	2.166e+005 *	2.166e+005 *	2.166e+005 *	2.166e+005 *
51	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
52	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
53	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1.002e+004	1.002e+004	1.002e+004	1.003e+004
57	Elevation (m)	-2.000e-002	-0.1400	-0.2600	-0.3800
58	Pressure (bar)	91.92	91.92	91.92	91.93
59	Temperature (C)	57.46	57.46	57.46	57.46
60	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
61	Heat Transferred (kJ/h-m)	731.8	731.8	731.7	731.7
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1704	0.1704	0.1704	0.1704
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	7.217e-003	7.217e-003	7.217e-003	7.216e-003
67	Static Gradient (kPa/m)	-6.823e-002	-6.823e-002	-6.823e-002	-6.823e-002
68	Accel. Gradient (kPa/m)	-4.239e-006	-4.238e-006	-4.238e-006	-4.238e-006
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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11	Pipe Table				
12					
13	Liq. Reynolds	2.166e+005 *	2.166e+005 *	2.166e+005 *	2.166e+005 *
14	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
15	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
16	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	1.003e+004	1.003e+004	1.004e+004	1.004e+004
20	Elevation (m)	-0.5000	-0.4940	-0.4880	-0.4820
21	Pressure (bar)	91.93	91.93	91.93	91.93
22	Temperature (C)	57.46	57.46	57.46	57.46
23	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
24	Heat Transferred (kJ/h-m)	731.7	731.6	731.5	731.5
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.1704	0.2836	0.2836	0.2836
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	7.216e-003	5.309e-003	5.309e-003	5.309e-003
30	Static Gradient (kPa/m)	-6.823e-002	2.859e-003	2.859e-003	2.859e-003
31	Accel. Gradient (kPa/m)	-4.238e-006	5.674e-007	5.673e-007	5.673e-007
32	Liq. Reynolds	2.166e+005 *	2.166e+005 *	2.166e+005 *	2.166e+005 *
33	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
34	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
35	Vapour Velocity (m/s)	2.166	2.166	2.166	2.165
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	1.005e+004	1.005e+004	1.006e+004	1.006e+004
39	Elevation (m)	-0.4760	-0.4700	-0.3840	-0.2980
40	Pressure (bar)	91.93	91.93	91.92	91.92
41	Temperature (C)	57.45	57.45	57.45	57.45
42	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
43	Heat Transferred (kJ/h-m)	731.4	731.4	731.3	731.2
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2836	0.2836	0.3346	0.3346
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	5.309e-003	5.309e-003	6.053e-003	6.053e-003
49	Static Gradient (kPa/m)	2.859e-003	2.859e-003	4.614e-002	4.614e-002
50	Accel. Gradient (kPa/m)	5.673e-007	5.673e-007	3.626e-006	3.626e-006
51	Liq. Reynolds	2.166e+005 *	2.166e+005 *	2.166e+005 *	2.166e+005 *
52	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
53	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
54	Vapour Velocity (m/s)	2.165	2.165	2.166	2.166
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	1.007e+004	1.007e+004	1.008e+004	1.008e+004
58	Elevation (m)	-0.2120	-0.1260	-4.000e-002	-1.527e-015
59	Pressure (bar)	91.92	91.92	91.91	91.91
60	Temperature (C)	57.44	57.44	57.44	57.44
61	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
62	Heat Transferred (kJ/h-m)	731.2	731.1	731.0	730.9
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.3346	0.3346	0.3346	0.3053
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	6.054e-003	6.054e-003	6.054e-003	5.924e-003
68	Static Gradient (kPa/m)	4.614e-002	4.614e-002	4.614e-002	2.008e-002
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	3.626e-006	3.626e-006	3.626e-006	1.807e-006
14	Liq. Reynolds	2.166e+005 *	2.166e+005 *	2.165e+005 *	2.165e+005 *
15	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
16	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
17	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	1.009e+004	1.009e+004	1.010e+004	1.010e+004
21	Elevation (m)	4.000e-002	8.000e-002	0.1200	0.1600
22	Pressure (bar)	91.91	91.91	91.91	91.91
23	Temperature (C)	57.43	57.43	57.43	57.43
24	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
25	Heat Transferred (kJ/h-m)	730.9	730.8	730.7	730.7
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.3053	0.3053	0.3053	0.3053
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	5.924e-003	5.924e-003	5.924e-003	5.924e-003
31	Static Gradient (kPa/m)	2.008e-002	2.008e-002	2.008e-002	2.008e-002
32	Accel. Gradient (kPa/m)	1.807e-006	1.807e-006	1.807e-006	1.807e-006
33	Liq. Reynolds	2.165e+005 *	2.165e+005 *	2.165e+005 *	2.165e+005 *
34	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.080e+006 *	9.080e+006 *
35	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
36	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	1.011e+004	1.011e+004	1.012e+004	1.012e+004
40	Elevation (m)	0.1640	0.1680	0.1720	0.1760
41	Pressure (bar)	91.91	91.91	91.91	91.91
42	Temperature (C)	57.42	57.42	57.42	57.42
43	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
44	Heat Transferred (kJ/h-m)	730.6	730.5	730.5	730.4
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.2823	0.2823	0.2823	0.2823
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	5.408e-003	5.408e-003	5.408e-003	5.408e-003
50	Static Gradient (kPa/m)	1.900e-003	1.900e-003	1.900e-003	1.900e-003
51	Accel. Gradient (kPa/m)	5.078e-007	5.077e-007	5.077e-007	5.077e-007
52	Liq. Reynolds	2.165e+005 *	2.165e+005 *	2.165e+005 *	2.165e+005 *
53	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
54	Liquid Velocity (m/s)	0.2041	0.2041	0.2041	0.2041
55	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	1.013e+004	1.013e+004	1.014e+004	1.014e+004
59	Elevation (m)	0.1800	0.1860	0.1920	0.1980
60	Pressure (bar)	91.91	91.90	91.90	91.90
61	Temperature (C)	57.41	57.41	57.41	57.41
62	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
63	Heat Transferred (kJ/h-m)	730.4	730.3	730.3	730.2
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.2823	0.2836	0.2836	0.2836
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	5.408e-003	5.310e-003	5.310e-003	5.310e-003
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1			Case Name:	Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set:	Mixed		
3	 Epic Atlantic Limited		Date/Time:	Mon Jun 13 15:02:44 2022		
4						
5						
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	1.900e-003	2.859e-003	2.859e-003	2.859e-003
14	Accel. Gradient	(kPa/m)	5.077e-007	5.675e-007	5.675e-007	5.675e-007
15	Liq. Reynolds		2.165e+005 *	2.165e+005 *	2.165e+005 *	2.165e+005 *
16	Vap. Reynolds		9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
17	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041
18	Vapour Velocity	(m/s)	2.166	2.166	2.166	2.166
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	1.015e+004	1.015e+004	1.016e+004	1.016e+004
22	Elevation	(m)	0.2040	0.2100	0.2120	0.2140
23	Pressure	(bar)	91.90	91.90	91.90	91.90
24	Temperature	(C)	57.41	57.40	57.40	57.40
25	Vapour Fraction		0.8720	0.8720	0.8720	0.8720
26	Heat Transferred	(kJ/h-m)	730.1	730.1	730.0	730.0
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2836	0.2836	0.2810	0.2810
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	5.310e-003	5.310e-003	5.507e-003	5.507e-003
32	Static Gradient	(kPa/m)	2.859e-003	2.859e-003	9.469e-004	9.469e-004
33	Accel. Gradient	(kPa/m)	5.675e-007	5.675e-007	4.484e-007	4.484e-007
34	Liq. Reynolds		2.165e+005 *	2.165e+005 *	2.165e+005 *	2.165e+005 *
35	Vap. Reynolds		9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
36	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041
37	Vapour Velocity	(m/s)	2.166	2.166	2.166	2.165
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	1.017e+004	1.017e+004	1.018e+004	1.018e+004
41	Elevation	(m)	0.2160	0.2180	0.2200	0.2140
42	Pressure	(bar)	91.90	91.90	91.90	91.90
43	Temperature	(C)	57.40	57.40	57.39	57.39
44	Vapour Fraction		0.8720	0.8720	0.8720	0.8720
45	Heat Transferred	(kJ/h-m)	729.9	729.8	729.8	729.8
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2810	0.2810	0.2810	0.2063
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	5.507e-003	5.507e-003	5.507e-003	6.847e-003
51	Static Gradient	(kPa/m)	9.469e-004	9.469e-004	9.469e-004	-2.313e-003
52	Accel. Gradient	(kPa/m)	4.484e-007	4.484e-007	4.484e-007	3.150e-007
53	Liq. Reynolds		2.165e+005 *	2.165e+005 *	2.165e+005 *	2.165e+005 *
54	Vap. Reynolds		9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
55	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041
56	Vapour Velocity	(m/s)	2.165	2.165	2.165	2.165
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	1.019e+004	1.019e+004	1.020e+004	1.020e+004
60	Elevation	(m)	0.2080	0.2020	0.1960	0.1900
61	Pressure	(bar)	91.90	91.90	91.90	91.90
62	Temperature	(C)	57.39	57.39	57.38	57.38
63	Vapour Fraction		0.8720	0.8720	0.8720	0.8720
64	Heat Transferred	(kJ/h-m)	729.7	729.6	729.6	729.5
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2063	0.2063	0.2063	0.2063
68	Water HoldUp		---	---	---	---
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
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7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	6.847e-003	6.847e-003	6.847e-003	6.847e-003
14	Static Gradient	(kPa/m)	-2.313e-003	-2.313e-003	-2.313e-003	-2.313e-003
15	Accel. Gradient	(kPa/m)	3.150e-007	3.150e-007	3.150e-007	3.150e-007
16	Liq. Reynolds		2.165e+005 *	2.165e+005 *	2.165e+005 *	2.165e+005 *
17	Vap. Reynolds		9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
18	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2041
19	Vapour Velocity	(m/s)	2.165	2.165	2.165	2.165
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	1.021e+004	1.021e+004	1.022e+004	1.022e+004
23	Elevation	(m)	0.1960	0.2020	0.2080	0.2140
24	Pressure	(bar)	91.90	91.90	91.90	91.90
25	Temperature	(C)	57.38	57.38	57.38	57.37
26	Vapour Fraction		0.8720	0.8720	0.8720	0.8720
27	Heat Transferred	(kJ/h-m)	729.5	729.4	729.3	729.3
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2836	0.2836	0.2836	0.2836
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	5.308e-003	5.308e-003	5.308e-003	5.308e-003
33	Static Gradient	(kPa/m)	2.867e-003	2.867e-003	2.867e-003	2.867e-003
34	Accel. Gradient	(kPa/m)	5.680e-007	5.680e-007	5.680e-007	5.680e-007
35	Liq. Reynolds		2.165e+005 *	2.165e+005 *	2.165e+005 *	2.165e+005 *
36	Vap. Reynolds		9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
37	Liquid Velocity	(m/s)	0.2041	0.2041	0.2041	0.2042
38	Vapour Velocity	(m/s)	2.165	2.165	2.165	2.165
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	1.023e+004	1.023e+004	1.024e+004	1.024e+004
42	Elevation	(m)	0.2200	0.2120	0.2040	0.1960
43	Pressure	(bar)	91.90	91.90	91.90	91.90
44	Temperature	(C)	57.37	57.37	57.37	57.37
45	Vapour Fraction		0.8720	0.8720	0.8720	0.8720
46	Heat Transferred	(kJ/h-m)	729.2	729.2	729.1	729.1
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2836	0.2059	0.2059	0.2059
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	5.308e-003	6.849e-003	6.849e-003	6.849e-003
52	Static Gradient	(kPa/m)	2.867e-003	-3.080e-003	-3.080e-003	-3.080e-003
53	Accel. Gradient	(kPa/m)	5.680e-007	2.619e-007	2.618e-007	2.618e-007
54	Liq. Reynolds		2.165e+005 *	2.165e+005 *	2.165e+005 *	2.165e+005 *
55	Vap. Reynolds		9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
56	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042
57	Vapour Velocity	(m/s)	2.165	2.165	2.165	2.165
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	1.025e+004	1.025e+004	1.026e+004	1.026e+004
61	Elevation	(m)	0.1880	0.1800	1.000e-001	2.000e-002
62	Pressure	(bar)	91.90	91.90	91.90	91.90
63	Temperature	(C)	57.36	57.36	57.36	57.36
64	Vapour Fraction		0.8720	0.8720	0.8720	0.8720
65	Heat Transferred	(kJ/h-m)	729.0	729.0	728.9	728.9
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2059	0.2059	0.1925	0.1925
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	6.849e-003	6.849e-003	6.966e-003	6.966e-003
15	Static Gradient (kPa/m)	-3.080e-003	-3.080e-003	-2.954e-002	-2.954e-002
16	Accel. Gradient (kPa/m)	2.618e-007	2.618e-007	-1.568e-006	-1.568e-006
17	Liq. Reynolds	2.165e+005 *	2.165e+005 *	2.165e+005 *	2.165e+005 *
18	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
19	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
20	Vapour Velocity (m/s)	2.165	2.165	2.165	2.165
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	1.027e+004	1.027e+004	1.028e+004	1.028e+004
24	Elevation (m)	-6.000e-002	-0.1400	-0.2200	-0.1400
25	Pressure (bar)	91.90	91.90	91.90	91.90
26	Temperature (C)	57.36	57.35	57.35	57.35
27	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
28	Heat Transferred (kJ/h-m)	728.8	728.8	728.8	728.7
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.1925	0.1925	0.1925	0.3309
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	6.966e-003	6.966e-003	6.965e-003	6.053e-003
34	Static Gradient (kPa/m)	-2.954e-002	-2.954e-002	-2.954e-002	4.257e-002
35	Accel. Gradient (kPa/m)	-1.568e-006	-1.568e-006	-1.568e-006	3.378e-006
36	Liq. Reynolds	2.165e+005 *	2.165e+005 *	2.165e+005 *	2.164e+005 *
37	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
38	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
39	Vapour Velocity (m/s)	2.165	2.165	2.165	2.165
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	1.029e+004	1.029e+004	1.030e+004	1.030e+004
43	Elevation (m)	-6.000e-002	2.000e-002	1.000e-001	0.1800
44	Pressure (bar)	91.90	91.90	91.89	91.89
45	Temperature (C)	57.35	57.34	57.34	57.34
46	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
47	Heat Transferred (kJ/h-m)	728.6	728.5	728.5	728.4
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.3309	0.3308	0.3308	0.3308
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	6.053e-003	6.053e-003	6.053e-003	6.053e-003
53	Static Gradient (kPa/m)	4.257e-002	4.257e-002	4.257e-002	4.257e-002
54	Accel. Gradient (kPa/m)	3.378e-006	3.378e-006	3.378e-006	3.378e-006
55	Liq. Reynolds	2.164e+005 *	2.164e+005 *	2.164e+005 *	2.164e+005 *
56	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
57	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
58	Vapour Velocity (m/s)	2.165	2.165	2.165	2.165
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	1.031e+004	1.031e+004	1.032e+004	1.032e+004
62	Elevation (m)	0.1640	0.1480	0.1320	0.1160
63	Pressure (bar)	91.89	91.89	91.89	91.89
64	Temperature (C)	57.34	57.34	57.33	57.33
65	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
66	Heat Transferred (kJ/h-m)	728.3	728.3	728.2	728.2
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 198 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.2044	0.2044	0.2044	0.2044
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	6.861e-003	6.861e-003	6.861e-003	6.861e-003
16	Static Gradient (kPa/m)	-6.133e-003	-6.133e-003	-6.133e-003	-6.133e-003
17	Accel. Gradient (kPa/m)	5.056e-008	5.056e-008	5.055e-008	5.054e-008
18	Liq. Reynolds	2.164e+005 *	2.164e+005 *	2.164e+005 *	2.164e+005 *
19	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
20	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
21	Vapour Velocity (m/s)	2.165	2.165	2.165	2.165
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	1.033e+004	1.033e+004	1.034e+004	1.034e+004
25	Elevation (m)	1.000e-001	8.600e-002	7.200e-002	5.800e-002
26	Pressure (bar)	91.89	91.89	91.89	91.89
27	Temperature (C)	57.33	57.33	57.32	57.32
28	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
29	Heat Transferred (kJ/h-m)	728.1	728.1	728.0	727.9
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2044	0.2048	0.2048	0.2048
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.861e-003	6.857e-003	6.857e-003	6.857e-003
35	Static Gradient (kPa/m)	-6.133e-003	-5.373e-003	-5.373e-003	-5.373e-003
36	Accel. Gradient (kPa/m)	5.053e-008	1.031e-007	1.031e-007	1.031e-007
37	Liq. Reynolds	2.164e+005 *	2.164e+005 *	2.164e+005 *	2.164e+005 *
38	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
39	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
40	Vapour Velocity (m/s)	2.165	2.165	2.165	2.165
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	1.035e+004	1.035e+004	1.036e+004	1.036e+004
44	Elevation (m)	4.400e-002	3.000e-002	2.800e-002	2.600e-002
45	Pressure (bar)	91.89	91.89	91.89	91.89
46	Temperature (C)	57.32	57.32	57.32	57.31
47	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
48	Heat Transferred (kJ/h-m)	727.9	727.8	727.8	727.7
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2048	0.2048	0.2070	0.2070
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.857e-003	6.857e-003	6.839e-003	6.839e-003
54	Static Gradient (kPa/m)	-5.373e-003	-5.373e-003	-7.728e-004	-7.728e-004
55	Accel. Gradient (kPa/m)	1.031e-007	1.031e-007	4.214e-007	4.214e-007
56	Liq. Reynolds	2.164e+005 *	2.164e+005 *	2.164e+005 *	2.164e+005 *
57	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
58	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
59	Vapour Velocity (m/s)	2.165	2.165	2.165	2.165
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	1.037e+004	1.037e+004	1.038e+004	1.038e+004
63	Elevation (m)	2.400e-002	2.200e-002	2.000e-002	-8.000e-003
64	Pressure (bar)	91.89	91.89	91.89	91.89
65	Temperature (C)	57.31	57.31	57.31	57.31
66	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
67	Heat Transferred (kJ/h-m)	727.7	727.6	727.6	727.5
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 199 of 937

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2070	0.2071	0.2071	0.2022
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.839e-003	6.839e-003	6.839e-003	6.878e-003
17	Static Gradient (kPa/m)	-7.728e-004	-7.728e-004	-7.728e-004	-1.066e-002
18	Accel. Gradient (kPa/m)	4.214e-007	4.214e-007	4.214e-007	-2.627e-007
19	Liq. Reynolds	2.164e+005 *	2.164e+005 *	2.164e+005 *	2.164e+005 *
20	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
21	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
22	Vapour Velocity (m/s)	2.165	2.165	2.165	2.165
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	1.039e+004	1.039e+004	1.040e+004	1.040e+004
26	Elevation (m)	-3.600e-002	-6.400e-002	-9.200e-002	-0.1200
27	Pressure (bar)	91.89	91.89	91.89	91.89
28	Temperature (C)	57.30	57.30	57.30	57.30
29	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
30	Heat Transferred (kJ/h-m)	727.5	727.4	727.3	727.3
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2022	0.2022	0.2022	0.2022
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	6.878e-003	6.878e-003	6.878e-003	6.878e-003
36	Static Gradient (kPa/m)	-1.066e-002	-1.066e-002	-1.066e-002	-1.066e-002
37	Accel. Gradient (kPa/m)	-2.627e-007	-2.627e-007	-2.627e-007	-2.627e-007
38	Liq. Reynolds	2.164e+005 *	2.164e+005 *	2.164e+005 *	2.164e+005 *
39	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
40	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
41	Vapour Velocity (m/s)	2.165	2.165	2.165	2.165
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	1.041e+004	1.041e+004	1.042e+004	1.042e+004
45	Elevation (m)	-9.600e-002	-7.200e-002	-4.800e-002	-2.400e-002
46	Pressure (bar)	91.89	91.89	91.89	91.89
47	Temperature (C)	57.30	57.29	57.29	57.29
48	Vapour Fraction	0.8720	0.8720	0.8720	0.8720
49	Heat Transferred (kJ/h-m)	727.2	727.2	727.1	727.0
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2951	0.2951	0.2951	0.2951
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	5.189e-003	5.189e-003	5.189e-003	5.189e-003
55	Static Gradient (kPa/m)	1.177e-002	1.177e-002	1.177e-002	1.177e-002
56	Accel. Gradient (kPa/m)	1.178e-006	1.178e-006	1.178e-006	1.178e-006
57	Liq. Reynolds	2.164e+005 *	2.164e+005 *	2.164e+005 *	2.164e+005 *
58	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
59	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
60	Vapour Velocity (m/s)	2.165	2.165	2.165	2.165
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	1.043e+004	1.043e+004	1.044e+004	1.044e+004
64	Elevation (m)	-1.554e-015	-5.800e-002	-0.1160	-0.1740
65	Pressure (bar)	91.89	91.89	91.89	91.89
66	Temperature (C)	57.29	57.29	57.28	57.28
67	Vapour Fraction	0.8720	0.8719	0.8719	0.8719
68	Heat Transferred (kJ/h-m)	727.0	727.0	726.9	726.9
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 200 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc	
2				
3			Unit Set: Mixed	
4			Date/Time: Mon Jun 13 15:02:44 2022	
5				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)			
7				
8				
9	PROFILES			
10				
11	Pipe Table			
12				
13	Flow Regime	Transition	Transition	Transition
14	Water Flow Regime			
15	Liquid HoldUp	0.2951	0.1966	0.1966
16	Water HoldUp	---	---	---
17	Friction Gradient (kPa/m)	5.189e-003	6.926e-003	6.926e-003
18	Static Gradient (kPa/m)	1.177e-002	-2.170e-002	-2.170e-002
19	Accel. Gradient (kPa/m)	1.178e-006	-1.026e-006	-1.026e-006
20	Liq. Reynolds	2.164e+005 *	2.164e+005 *	2.164e+005 *
21	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *
22	Liquid Velocity (m/s)	0.2042	0.2042	0.2042
23	Vapour Velocity (m/s)	2.165	2.165	2.165
24	Deposit Thickness (mm)	---	---	---
25	Deposit Volume (m3)	---	---	---
26	Length (m)	1.045e+004	1.045e+004	1.046e+004
27	Elevation (m)	-0.2320	-0.2900	-0.3280
28	Pressure (bar)	91.89	91.89	91.89
29	Temperature (C)	57.28	57.28	57.28
30	Vapour Fraction	0.8719	0.8719	0.8719
31	Heat Transferred (kJ/h-m)	726.8	726.8	726.7
32	Flow Regime	Transition	Transition	Transition
33	Water Flow Regime			
34	Liquid HoldUp	0.1966	0.1966	0.2004
35	Water HoldUp	---	---	---
36	Friction Gradient (kPa/m)	6.926e-003	6.926e-003	6.893e-003
37	Static Gradient (kPa/m)	-2.170e-002	-2.170e-002	-1.439e-002
38	Accel. Gradient (kPa/m)	-1.026e-006	-1.026e-006	-5.204e-007
39	Liq. Reynolds	2.164e+005 *	2.164e+005 *	2.164e+005 *
40	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *
41	Liquid Velocity (m/s)	0.2042	0.2042	0.2042
42	Vapour Velocity (m/s)	2.164	2.164	2.164
43	Deposit Thickness (mm)	---	---	---
44	Deposit Volume (m3)	---	---	---
45	Length (m)	1.047e+004	1.047e+004	1.048e+004
46	Elevation (m)	-0.4040	-0.4420	-0.4800
47	Pressure (bar)	91.89	91.89	91.89
48	Temperature (C)	57.27	57.27	57.27
49	Vapour Fraction	0.8719	0.8719	0.8719
50	Heat Transferred (kJ/h-m)	726.6	726.6	726.5
51	Flow Regime	Transition	Transition	Transition
52	Water Flow Regime			
53	Liquid HoldUp	0.2004	0.2004	0.2004
54	Water HoldUp	---	---	---
55	Friction Gradient (kPa/m)	6.893e-003	6.893e-003	6.893e-003
56	Static Gradient (kPa/m)	-1.439e-002	-1.439e-002	-1.439e-002
57	Accel. Gradient (kPa/m)	-5.204e-007	-5.204e-007	-5.204e-007
58	Liq. Reynolds	2.164e+005 *	2.164e+005 *	2.164e+005 *
59	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *
60	Liquid Velocity (m/s)	0.2042	0.2042	0.2042
61	Vapour Velocity (m/s)	2.164	2.164	2.164
62	Deposit Thickness (mm)	---	---	---
63	Deposit Volume (m3)	---	---	---
64	Length (m)	1.049e+004	1.049e+004	1.050e+004
65	Elevation (m)	-0.5200	-0.5400	-0.5600
66	Pressure (bar)	91.89	91.89	91.89
67	Temperature (C)	57.26	57.26	57.26
68	Vapour Fraction	0.8719	0.8719	0.8719
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Heat Transferred (kJ/h-m)	726.4	726.3	726.3	726.2		
14	Flow Regime	Transition	Transition	Transition	Transition		
15	Water Flow Regime						
16	Liquid HoldUp	0.2037	0.2037	0.2037	0.2037		
17	Water HoldUp	---	---	---	---		
18	Friction Gradient (kPa/m)	6.865e-003	6.864e-003	6.864e-003	6.864e-003		
19	Static Gradient (kPa/m)	-7.651e-003	-7.651e-003	-7.651e-003	-7.651e-003		
20	Accel. Gradient (kPa/m)	-5.461e-008	-5.462e-008	-5.463e-008	-5.464e-008		
21	Liq. Reynolds	2.164e+005 *	2.164e+005 *	2.164e+005 *	2.163e+005 *		
22	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *		
23	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
24	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164		
25	Deposit Thickness (mm)	---	---	---	---		
26	Deposit Volume (m3)	---	---	---	---		
27	Length (m)	1.051e+004	1.051e+004	1.052e+004	1.052e+004		
28	Elevation (m)	-0.4800	-0.3800	-0.2800	-0.1800		
29	Pressure (bar)	91.89	91.89	91.88	91.88		
30	Temperature (C)	57.26	57.25	57.25	57.25		
31	Vapour Fraction	0.8719	0.8719	0.8719	0.8719		
32	Heat Transferred (kJ/h-m)	726.2	726.1	726.0	725.9		
33	Flow Regime	Transition	Transition	Transition	Transition		
34	Water Flow Regime						
35	Liquid HoldUp	0.3437	0.3437	0.3437	0.3437		
36	Water HoldUp	---	---	---	---		
37	Friction Gradient (kPa/m)	6.045e-003	6.045e-003	6.045e-003	6.045e-003		
38	Static Gradient (kPa/m)	5.473e-002	5.473e-002	5.473e-002	5.473e-002		
39	Accel. Gradient (kPa/m)	4.221e-006	4.221e-006	4.221e-006	4.221e-006		
40	Liq. Reynolds	2.163e+005 *	2.163e+005 *	2.163e+005 *	2.163e+005 *		
41	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *		
42	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
43	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164		
44	Deposit Thickness (mm)	---	---	---	---		
45	Deposit Volume (m3)	---	---	---	---		
46	Length (m)	1.053e+004	1.053e+004	1.054e+004	1.054e+004		
47	Elevation (m)	-8.000e-002	-7.600e-002	-7.200e-002	-6.800e-002		
48	Pressure (bar)	91.88	91.88	91.88	91.87		
49	Temperature (C)	57.24	57.24	57.24	57.24		
50	Vapour Fraction	0.8719	0.8719	0.8719	0.8719		
51	Heat Transferred (kJ/h-m)	725.9	725.8	725.7	725.7		
52	Flow Regime	Transition	Transition	Transition	Transition		
53	Water Flow Regime						
54	Liquid HoldUp	0.3437	0.2824	0.2824	0.2824		
55	Water HoldUp	---	---	---	---		
56	Friction Gradient (kPa/m)	6.045e-003	5.404e-003	5.404e-003	5.404e-003		
57	Static Gradient (kPa/m)	5.473e-002	1.901e-003	1.901e-003	1.901e-003		
58	Accel. Gradient (kPa/m)	4.222e-006	5.074e-007	5.074e-007	5.074e-007		
59	Liq. Reynolds	2.163e+005 *	2.163e+005 *	2.163e+005 *	2.163e+005 *		
60	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *		
61	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
62	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164		
63	Deposit Thickness (mm)	---	---	---	---		
64	Deposit Volume (m3)	---	---	---	---		
65	Length (m)	1.055e+004	1.055e+004	1.056e+004	1.056e+004		
66	Elevation (m)	-6.400e-002	-6.000e-002	-4.800e-002	-3.600e-002		
67	Pressure (bar)	91.87	91.87	91.87	91.87		
68	Temperature (C)	57.24	57.23	57.23	57.23		
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
14	Heat Transferred (kJ/h-m)	725.6	725.5	725.5	725.4
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2824	0.2824	0.2875	0.2875
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	5.404e-003	5.404e-003	5.020e-003	5.020e-003
20	Static Gradient (kPa/m)	1.901e-003	1.901e-003	5.774e-003	5.774e-003
21	Accel. Gradient (kPa/m)	5.074e-007	5.074e-007	7.498e-007	7.498e-007
22	Liq. Reynolds	2.163e+005 *	2.163e+005 *	2.163e+005 *	2.163e+005 *
23	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
24	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
25	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	1.057e+004	1.057e+004	1.058e+004	1.058e+004
29	Elevation (m)	-2.400e-002	-1.200e-002	-1.513e-015	1.800e-002
30	Pressure (bar)	91.87	91.87	91.87	91.87
31	Temperature (C)	57.23	57.22	57.22	57.22
32	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
33	Heat Transferred (kJ/h-m)	725.4	725.3	725.2	725.2
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2875	0.2875	0.2875	0.2913
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	5.019e-003	5.019e-003	5.019e-003	4.744e-003
39	Static Gradient (kPa/m)	5.774e-003	5.774e-003	5.774e-003	8.743e-003
40	Accel. Gradient (kPa/m)	7.498e-007	7.498e-007	7.498e-007	9.368e-007
41	Liq. Reynolds	2.163e+005 *	2.163e+005 *	2.163e+005 *	2.163e+005 *
42	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
43	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
44	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	1.059e+004	1.059e+004	1.060e+004	1.060e+004
48	Elevation (m)	3.600e-002	5.400e-002	7.200e-002	9.000e-002
49	Pressure (bar)	91.87	91.87	91.87	91.87
50	Temperature (C)	57.22	57.22	57.21	57.21
51	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
52	Heat Transferred (kJ/h-m)	725.1	725.1	725.0	725.0
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2913	0.2913	0.2913	0.2913
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	4.744e-003	4.744e-003	4.744e-003	4.744e-003
58	Static Gradient (kPa/m)	8.743e-003	8.743e-003	8.743e-003	8.743e-003
59	Accel. Gradient (kPa/m)	9.369e-007	9.369e-007	9.369e-007	9.369e-007
60	Liq. Reynolds	2.163e+005 *	2.163e+005 *	2.163e+005 *	2.163e+005 *
61	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
62	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
63	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	1.061e+004	1.061e+004	1.062e+004	1.062e+004
67	Elevation (m)	0.1040	0.1180	0.1320	0.1460
68	Pressure (bar)	91.87	91.87	91.87	91.87
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	57.21	57.21	57.20	57.20
14	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
15	Heat Transferred (kJ/h-m)	724.9	724.8	724.8	724.7
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2888	0.2888	0.2888	0.2888
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	4.928e-003	4.928e-003	4.928e-003	4.928e-003
21	Static Gradient (kPa/m)	6.736e-003	6.736e-003	6.736e-003	6.736e-003
22	Accel. Gradient (kPa/m)	8.102e-007	8.102e-007	8.103e-007	8.103e-007
23	Liq. Reynolds	2.163e+005 *	2.163e+005 *	2.163e+005 *	2.163e+005 *
24	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *
25	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
26	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	1.063e+004	1.063e+004	1.064e+004	1.064e+004
30	Elevation (m)	0.1600	0.1640	0.1680	0.1720
31	Pressure (bar)	91.87	91.86	91.86	91.86
32	Temperature (C)	57.20	57.20	57.20	57.19
33	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
34	Heat Transferred (kJ/h-m)	724.7	724.6	724.5	724.5
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2888	0.2824	0.2824	0.2824
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	4.928e-003	5.403e-003	5.403e-003	5.403e-003
40	Static Gradient (kPa/m)	6.736e-003	1.901e-003	1.901e-003	1.901e-003
41	Accel. Gradient (kPa/m)	8.103e-007	5.074e-007	5.074e-007	5.074e-007
42	Liq. Reynolds	2.163e+005 *	2.163e+005 *	2.163e+005 *	2.163e+005 *
43	Vap. Reynolds	9.080e+006 *	9.080e+006 *	9.081e+006 *	9.081e+006 *
44	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
45	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	1.065e+004	1.065e+004	1.066e+004	1.066e+004
49	Elevation (m)	0.1760	0.1800	0.1840	0.1880
50	Pressure (bar)	91.86	91.86	91.86	91.86
51	Temperature (C)	57.19	57.19	57.19	57.18
52	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
53	Heat Transferred (kJ/h-m)	724.4	724.4	724.3	724.3
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2824	0.2824	0.2824	0.2824
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	5.403e-003	5.403e-003	5.403e-003	5.403e-003
59	Static Gradient (kPa/m)	1.901e-003	1.901e-003	1.901e-003	1.901e-003
60	Accel. Gradient (kPa/m)	5.074e-007	5.074e-007	5.074e-007	5.073e-007
61	Liq. Reynolds	2.163e+005 *	2.162e+005 *	2.162e+005 *	2.162e+005 *
62	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
63	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
64	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	1.067e+004	1.067e+004	1.068e+004	1.068e+004
68	Elevation (m)	0.1920	0.1960	0.2000	0.2100
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:02:44 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Pressure (bar)	91.86	91.86	91.86	91.86					
14	Temperature (C)	57.18	57.18	57.18	57.18					
15	Vapour Fraction	0.8719	0.8719	0.8719	0.8719					
16	Heat Transferred (kJ/h-m)	724.2	724.1	724.1	724.0					
17	Flow Regime	Transition	Transition	Transition	Transition					
18	Water Flow Regime									
19	Liquid HoldUp	0.2824	0.2824	0.2824	0.2863					
20	Water HoldUp	---	---	---	---					
21	Friction Gradient (kPa/m)	5.403e-003	5.403e-003	5.403e-003	5.113e-003					
22	Static Gradient (kPa/m)	1.901e-003	1.901e-003	1.901e-003	4.797e-003					
23	Accel. Gradient (kPa/m)	5.073e-007	5.073e-007	5.073e-007	6.884e-007					
24	Liq. Reynolds	2.162e+005 *	2.162e+005 *	2.162e+005 *	2.162e+005 *					
25	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *					
26	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042					
27	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164					
28	Deposit Thickness (mm)	---	---	---	---					
29	Deposit Volume (m3)	---	---	---	---					
30	Length (m)	1.069e+004	1.069e+004	1.070e+004	1.070e+004					
31	Elevation (m)	0.2200	0.2300	0.2400	0.2500					
32	Pressure (bar)	91.86	91.86	91.86	91.86					
33	Temperature (C)	57.17	57.17	57.17	57.17					
34	Vapour Fraction	0.8719	0.8719	0.8719	0.8719					
35	Heat Transferred (kJ/h-m)	724.0	723.9	723.8	723.8					
36	Flow Regime	Transition	Transition	Transition	Transition					
37	Water Flow Regime									
38	Liquid HoldUp	0.2863	0.2863	0.2863	0.2863					
39	Water HoldUp	---	---	---	---					
40	Friction Gradient (kPa/m)	5.113e-003	5.113e-003	5.113e-003	5.113e-003					
41	Static Gradient (kPa/m)	4.797e-003	4.797e-003	4.798e-003	4.798e-003					
42	Accel. Gradient (kPa/m)	6.884e-007	6.884e-007	6.884e-007	6.884e-007					
43	Liq. Reynolds	2.162e+005 *	2.162e+005 *	2.162e+005 *	2.162e+005 *					
44	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *					
45	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042					
46	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164					
47	Deposit Thickness (mm)	---	---	---	---					
48	Deposit Volume (m3)	---	---	---	---					
49	Length (m)	1.071e+004	1.071e+004	1.072e+004	1.072e+004					
50	Elevation (m)	0.2520	0.2540	0.2560	0.2580					
51	Pressure (bar)	91.86	91.86	91.86	91.86					
52	Temperature (C)	57.17	57.16	57.16	57.16					
53	Vapour Fraction	0.8719	0.8719	0.8719	0.8719					
54	Heat Transferred (kJ/h-m)	723.7	723.7	723.6	723.6					
55	Flow Regime	Transition	Transition	Transition	Transition					
56	Water Flow Regime									
57	Liquid HoldUp	0.2812	0.2812	0.2812	0.2812					
58	Water HoldUp	---	---	---	---					
59	Friction Gradient (kPa/m)	5.502e-003	5.502e-003	5.502e-003	5.502e-003					
60	Static Gradient (kPa/m)	9.475e-004	9.475e-004	9.475e-004	9.475e-004					
61	Accel. Gradient (kPa/m)	4.480e-007	4.480e-007	4.480e-007	4.480e-007					
62	Liq. Reynolds	2.162e+005 *	2.162e+005 *	2.162e+005 *	2.162e+005 *					
63	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *					
64	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042					
65	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164					
66	Deposit Thickness (mm)	---	---	---	---					
67	Deposit Volume (m3)	---	---	---	---					
68	Length (m)	1.073e+004	1.073e+004	1.074e+004	1.074e+004					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	0.2600	0.2660	0.2720	0.2780
14	Pressure (bar)	91.86	91.86	91.86	91.86
15	Temperature (C)	57.16	57.15	57.15	57.15
16	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
17	Heat Transferred (kJ/h-m)	723.5	723.4	723.4	723.3
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2812	0.2837	0.2837	0.2837
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	5.502e-003	5.304e-003	5.304e-003	5.304e-003
23	Static Gradient (kPa/m)	9.475e-004	2.861e-003	2.861e-003	2.861e-003
24	Accel. Gradient (kPa/m)	4.480e-007	5.671e-007	5.671e-007	5.671e-007
25	Liq. Reynolds	2.162e+005 *	2.162e+005 *	2.162e+005 *	2.162e+005 *
26	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
27	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
28	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	1.075e+004	1.075e+004	1.076e+004	1.076e+004
32	Elevation (m)	0.2840	0.2900	0.2940	0.2980
33	Pressure (bar)	91.86	91.86	91.86	91.85
34	Temperature (C)	57.15	57.15	57.14	57.14
35	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
36	Heat Transferred (kJ/h-m)	723.3	723.2	723.2	723.1
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2837	0.2837	0.2824	0.2824
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	5.304e-003	5.304e-003	5.402e-003	5.402e-003
42	Static Gradient (kPa/m)	2.861e-003	2.861e-003	1.901e-003	1.901e-003
43	Accel. Gradient (kPa/m)	5.671e-007	5.671e-007	5.073e-007	5.073e-007
44	Liq. Reynolds	2.162e+005 *	2.162e+005 *	2.162e+005 *	2.162e+005 *
45	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
46	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
47	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	1.077e+004	1.077e+004	1.078e+004	1.078e+004
51	Elevation (m)	0.3020	0.3060	0.3100	0.3060
52	Pressure (bar)	91.85	91.85	91.85	91.85
53	Temperature (C)	57.14	57.14	57.14	57.13
54	Vapour Fraction	0.8719	0.8719	0.8719	0.8719
55	Heat Transferred (kJ/h-m)	723.0	723.0	722.9	722.9
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2824	0.2824	0.2824	0.2068
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	5.402e-003	5.402e-003	5.402e-003	6.839e-003
61	Static Gradient (kPa/m)	1.901e-003	1.901e-003	1.901e-003	-1.545e-003
62	Accel. Gradient (kPa/m)	5.073e-007	5.073e-007	5.073e-007	3.678e-007
63	Liq. Reynolds	2.162e+005 *	2.162e+005 *	2.162e+005 *	2.162e+005 *
64	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
65	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
66	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length (m)	1.079e+004	1.079e+004	1.080e+004	1.080e+004		
14	Elevation (m)	0.3020	0.2980	0.2940	0.2900		
15	Pressure (bar)	91.85	91.85	91.85	91.85		
16	Temperature (C)	57.13	57.13	57.13	57.12		
17	Vapour Fraction	0.8719	0.8719	0.8719	0.8719		
18	Heat Transferred (kJ/h-m)	722.8	722.8	722.7	722.7		
19	Flow Regime	Transition	Transition	Transition	Transition		
20	Water Flow Regime						
21	Liquid HoldUp	0.2068	0.2068	0.2068	0.2068		
22	Water HoldUp	---	---	---	---		
23	Friction Gradient (kPa/m)	6.839e-003	6.839e-003	6.839e-003	6.839e-003		
24	Static Gradient (kPa/m)	-1.545e-003	-1.545e-003	-1.545e-003	-1.545e-003		
25	Accel. Gradient (kPa/m)	3.678e-007	3.678e-007	3.678e-007	3.677e-007		
26	Liq. Reynolds	2.162e+005 *	2.162e+005 *	2.162e+005 *	2.162e+005 *		
27	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *		
28	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
29	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164		
30	Deposit Thickness (mm)	---	---	---	---		
31	Deposit Volume (m3)	---	---	---	---		
32	Length (m)	1.081e+004	1.081e+004	1.082e+004	1.082e+004		
33	Elevation (m)	0.2940	0.2980	0.3020	0.3060		
34	Pressure (bar)	91.85	91.85	91.85	91.85		
35	Temperature (C)	57.12	57.12	57.12	57.12		
36	Vapour Fraction	0.8719	0.8719	0.8719	0.8719		
37	Heat Transferred (kJ/h-m)	722.6	722.5	722.5	722.4		
38	Flow Regime	Transition	Transition	Transition	Transition		
39	Water Flow Regime						
40	Liquid HoldUp	0.2825	0.2825	0.2825	0.2825		
41	Water HoldUp	---	---	---	---		
42	Friction Gradient (kPa/m)	5.402e-003	5.402e-003	5.402e-003	5.401e-003		
43	Static Gradient (kPa/m)	1.903e-003	1.903e-003	1.903e-003	1.903e-003		
44	Accel. Gradient (kPa/m)	5.073e-007	5.073e-007	5.073e-007	5.073e-007		
45	Liq. Reynolds	2.162e+005 *	2.162e+005 *	2.162e+005 *	2.162e+005 *		
46	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *		
47	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
48	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164		
49	Deposit Thickness (mm)	---	---	---	---		
50	Deposit Volume (m3)	---	---	---	---		
51	Length (m)	1.083e+004	1.083e+004	1.084e+004	1.084e+004		
52	Elevation (m)	0.3100	0.3140	0.3180	0.3220		
53	Pressure (bar)	91.85	91.85	91.85	91.85		
54	Temperature (C)	57.11	57.11	57.11	57.11		
55	Vapour Fraction	0.8719	0.8719	0.8719	0.8719		
56	Heat Transferred (kJ/h-m)	722.4	722.3	722.3	722.2		
57	Flow Regime	Transition	Transition	Transition	Transition		
58	Water Flow Regime						
59	Liquid HoldUp	0.2825	0.2825	0.2825	0.2825		
60	Water HoldUp	---	---	---	---		
61	Friction Gradient (kPa/m)	5.401e-003	5.402e-003	5.401e-003	5.401e-003		
62	Static Gradient (kPa/m)	1.903e-003	1.901e-003	1.901e-003	1.901e-003		
63	Accel. Gradient (kPa/m)	5.073e-007	5.072e-007	5.072e-007	5.072e-007		
64	Liq. Reynolds	2.162e+005 *	2.162e+005 *	2.162e+005 *	2.162e+005 *		
65	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *		
66	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
67	Vapour Velocity (m/s)	2.164	2.163	2.163	2.163		
68	Deposit Thickness (mm)	---	---	---	---		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	1.085e+004	1.085e+004	1.086e+004	1.086e+004
15	Elevation (m)	0.3260	0.3300	0.3280	0.3260
16	Pressure (bar)	91.85	91.85	91.85	91.85
17	Temperature (C)	57.11	57.10	57.10	57.10
18	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
19	Heat Transferred (kJ/h-m)	722.1	722.1	722.0	722.0
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2825	0.2825	0.2071	0.2071
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	5.401e-003	5.401e-003	6.836e-003	6.836e-003
25	Static Gradient (kPa/m)	1.901e-003	1.901e-003	-7.732e-004	-7.732e-004
26	Accel. Gradient (kPa/m)	5.072e-007	5.072e-007	4.211e-007	4.211e-007
27	Liq. Reynolds	2.162e+005 *	2.161e+005 *	2.161e+005 *	2.161e+005 *
28	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
29	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
30	Vapour Velocity (m/s)	2.163	2.163	2.163	2.163
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	1.087e+004	1.087e+004	1.088e+004	1.088e+004
34	Elevation (m)	0.3240	0.3220	0.3200	0.3240
35	Pressure (bar)	91.85	91.85	91.85	91.85
36	Temperature (C)	57.10	57.10	57.09	57.09
37	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
38	Heat Transferred (kJ/h-m)	721.9	721.9	721.8	721.7
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2071	0.2071	0.2071	0.2825
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	6.836e-003	6.836e-003	6.836e-003	5.401e-003
44	Static Gradient (kPa/m)	-7.732e-004	-7.732e-004	-7.733e-004	1.901e-003
45	Accel. Gradient (kPa/m)	4.211e-007	4.211e-007	4.211e-007	5.072e-007
46	Liq. Reynolds	2.161e+005 *	2.161e+005 *	2.161e+005 *	2.161e+005 *
47	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
48	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
49	Vapour Velocity (m/s)	2.163	2.163	2.163	2.163
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	1.089e+004	1.089e+004	1.090e+004	1.090e+004
53	Elevation (m)	0.3280	0.3320	0.3360	0.3400
54	Pressure (bar)	91.85	91.85	91.85	91.85
55	Temperature (C)	57.09	57.09	57.08	57.08
56	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
57	Heat Transferred (kJ/h-m)	721.7	721.6	721.6	721.5
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2825	0.2825	0.2825	0.2825
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	5.401e-003	5.401e-003	5.401e-003	5.401e-003
63	Static Gradient (kPa/m)	1.901e-003	1.901e-003	1.901e-003	1.901e-003
64	Accel. Gradient (kPa/m)	5.072e-007	5.072e-007	5.072e-007	5.071e-007
65	Liq. Reynolds	2.161e+005 *	2.161e+005 *	2.161e+005 *	2.161e+005 *
66	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
67	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
68	Vapour Velocity (m/s)	2.163	2.163	2.163	2.163
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	1.091e+004	1.091e+004	1.092e+004	1.092e+004
16	Elevation (m)	0.3440	0.3480	0.3520	0.3560
17	Pressure (bar)	91.84	91.84	91.84	91.84
18	Temperature (C)	57.08	57.08	57.08	57.07
19	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
20	Heat Transferred (kJ/h-m)	721.5	721.4	721.3	721.3
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2825	0.2825	0.2825	0.2825
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	5.401e-003	5.401e-003	5.401e-003	5.401e-003
26	Static Gradient (kPa/m)	1.901e-003	1.901e-003	1.901e-003	1.901e-003
27	Accel. Gradient (kPa/m)	5.071e-007	5.071e-007	5.071e-007	5.071e-007
28	Liq. Reynolds	2.161e+005 *	2.161e+005 *	2.161e+005 *	2.161e+005 *
29	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
30	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
31	Vapour Velocity (m/s)	2.163	2.163	2.163	2.163
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	1.093e+004	1.093e+004	1.094e+004	1.094e+004
35	Elevation (m)	0.3600	0.3700	0.3800	0.3900
36	Pressure (bar)	91.84	91.84	91.84	91.84
37	Temperature (C)	57.07	57.07	57.07	57.07
38	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
39	Heat Transferred (kJ/h-m)	721.2	721.2	721.1	721.1
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2825	0.2863	0.2863	0.2863
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	5.401e-003	5.111e-003	5.111e-003	5.111e-003
45	Static Gradient (kPa/m)	1.901e-003	4.799e-003	4.799e-003	4.799e-003
46	Accel. Gradient (kPa/m)	5.071e-007	6.882e-007	6.882e-007	6.882e-007
47	Liq. Reynolds	2.161e+005 *	2.161e+005 *	2.161e+005 *	2.161e+005 *
48	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
49	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
50	Vapour Velocity (m/s)	2.163	2.163	2.163	2.163
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	1.095e+004	1.095e+004	1.096e+004	1.096e+004
54	Elevation (m)	0.4000	0.4100	0.4000	0.3900
55	Pressure (bar)	91.84	91.84	91.84	91.84
56	Temperature (C)	57.06	57.06	57.06	57.06
57	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
58	Heat Transferred (kJ/h-m)	721.0	720.9	720.9	720.8
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2863	0.2863	0.2057	0.2057
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	5.111e-003	5.110e-003	6.847e-003	6.847e-003
64	Static Gradient (kPa/m)	4.799e-003	4.799e-003	-3.849e-003	-3.849e-003
65	Accel. Gradient (kPa/m)	6.882e-007	6.882e-007	2.082e-007	2.082e-007
66	Liq. Reynolds	2.161e+005 *	2.161e+005 *	2.161e+005 *	2.161e+005 *
67	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
68	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	1.097e+004	1.097e+004	1.098e+004	1.098e+004	
17	Elevation	(m)	0.3800	0.3700	0.3600	0.3600	
18	Pressure	(bar)	91.84	91.84	91.84	91.84	
19	Temperature	(C)	57.05	57.05	57.05	57.05	
20	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
21	Heat Transferred	(kJ/h-m)	720.8	720.7	720.7	720.6	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2057	0.2057	0.2057	0.3030	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	6.847e-003	6.847e-003	6.847e-003	5.859e-003	
27	Static Gradient	(kPa/m)	-3.849e-003	-3.849e-003	-3.849e-003	0.0000	
28	Accel. Gradient	(kPa/m)	2.082e-007	2.082e-007	2.082e-007	4.069e-007	
29	Liq. Reynolds		2.161e+005 *	2.161e+005 *	2.161e+005 *	2.161e+005 *	
30	Vap. Reynolds		9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *	
31	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
32	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	1.099e+004	1.099e+004	1.099e+004	1.100e+004	
36	Elevation	(m)	0.3600	0.3600	0.3600	0.3600	
37	Pressure	(bar)	91.84	91.84	91.84	91.84	
38	Temperature	(C)	57.05	57.05	57.04	57.04	
39	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
40	Heat Transferred	(kJ/h-m)	720.6	720.5	720.5	720.4	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.3030	0.3030	0.3030	0.3030	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	5.859e-003	5.859e-003	5.859e-003	5.859e-003	
46	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000	
47	Accel. Gradient	(kPa/m)	4.069e-007	4.069e-007	4.069e-007	4.069e-007	
48	Liq. Reynolds		2.161e+005 *	2.161e+005 *	2.161e+005 *	2.161e+005 *	
49	Vap. Reynolds		9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *	
50	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
51	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	1.100e+004	1.101e+004	1.102e+004	1.102e+004	
55	Elevation	(m)	0.3560	0.3520	0.3480	0.3440	
56	Pressure	(bar)	91.84	91.84	91.84	91.84	
57	Temperature	(C)	57.04	57.04	57.03	57.03	
58	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
59	Heat Transferred	(kJ/h-m)	720.4	720.3	720.3	720.2	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.2069	0.2069	0.2069	0.2069	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	6.837e-003	6.837e-003	6.837e-003	6.836e-003	
65	Static Gradient	(kPa/m)	-1.285e-003	-1.285e-003	-1.285e-003	-1.285e-003	
66	Accel. Gradient	(kPa/m)	3.855e-007	3.855e-007	3.855e-007	3.855e-007	
67	Liq. Reynolds		2.161e+005 *	2.161e+005 *	2.161e+005 *	2.161e+005 *	
68	Vap. Reynolds		9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
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9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
14	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	1.103e+004	1.103e+004	1.104e+004	1.104e+004	
18	Elevation	(m)	0.3400	0.3380	0.3360	0.3340	
19	Pressure	(bar)	91.84	91.84	91.84	91.84	
20	Temperature	(C)	57.03	57.03	57.03	57.02	
21	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
22	Heat Transferred	(kJ/h-m)	720.1	720.1	720.0	720.0	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.2069	0.2072	0.2072	0.2072	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	6.836e-003	6.834e-003	6.834e-003	6.834e-003	
28	Static Gradient	(kPa/m)	-1.285e-003	-7.734e-004	-7.734e-004	-7.734e-004	
29	Accel. Gradient	(kPa/m)	3.855e-007	4.209e-007	4.209e-007	4.209e-007	
30	Liq. Reynolds		2.161e+005 *	2.161e+005 *	2.161e+005 *	2.161e+005 *	
31	Vap. Reynolds		9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *	
32	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
33	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	1.105e+004	1.105e+004	1.106e+004	1.106e+004	
37	Elevation	(m)	0.3320	0.3300	0.3320	0.3340	
38	Pressure	(bar)	91.84	91.84	91.84	91.84	
39	Temperature	(C)	57.02	57.02	57.02	57.01	
40	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
41	Heat Transferred	(kJ/h-m)	719.9	719.8	719.8	719.7	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2072	0.2072	0.2812	0.2812	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	6.834e-003	6.834e-003	5.499e-003	5.498e-003	
47	Static Gradient	(kPa/m)	-7.734e-004	-7.734e-004	9.479e-004	9.479e-004	
48	Accel. Gradient	(kPa/m)	4.209e-007	4.209e-007	4.476e-007	4.476e-007	
49	Liq. Reynolds		2.161e+005 *	2.161e+005 *	2.160e+005 *	2.160e+005 *	
50	Vap. Reynolds		9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *	
51	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
52	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	1.107e+004	1.107e+004	1.108e+004	1.108e+004	
56	Elevation	(m)	0.3360	0.3380	0.3400	0.3420	
57	Pressure	(bar)	91.83	91.83	91.83	91.83	
58	Temperature	(C)	57.01	57.01	57.01	57.01	
59	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
60	Heat Transferred	(kJ/h-m)	719.7	719.6	719.5	719.5	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2813	0.2813	0.2813	0.2813	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	5.498e-003	5.498e-003	5.498e-003	5.498e-003	
66	Static Gradient	(kPa/m)	9.479e-004	9.479e-004	9.479e-004	9.479e-004	
67	Accel. Gradient	(kPa/m)	4.476e-007	4.476e-007	4.476e-007	4.476e-007	
68	Liq. Reynolds		2.160e+005 *	2.160e+005 *	2.160e+005 *	2.160e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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11	Pipe Table				
12					
13	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
14	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
15	Vapour Velocity (m/s)	2.163	2.163	2.163	2.163
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	1.109e+004	1.109e+004	1.110e+004	1.110e+004
19	Elevation (m)	0.3440	0.3460	0.3480	0.3500
20	Pressure (bar)	91.83	91.83	91.83	91.83
21	Temperature (C)	57.00	57.00	57.00	57.00
22	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
23	Heat Transferred (kJ/h-m)	719.4	719.4	719.3	719.3
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2813	0.2813	0.2813	0.2813
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	5.498e-003	5.498e-003	5.498e-003	5.498e-003
29	Static Gradient (kPa/m)	9.479e-004	9.479e-004	9.479e-004	9.479e-004
30	Accel. Gradient (kPa/m)	4.476e-007	4.476e-007	4.476e-007	4.476e-007
31	Liq. Reynolds	2.160e+005 *	2.160e+005 *	2.160e+005 *	2.160e+005 *
32	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
33	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
34	Vapour Velocity (m/s)	2.163	2.163	2.163	2.163
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	1.111e+004	1.111e+004	1.112e+004	1.112e+004
38	Elevation (m)	0.3440	0.3380	0.3320	0.3260
39	Pressure (bar)	91.83	91.83	91.83	91.83
40	Temperature (C)	57.00	56.99	56.99	56.99
41	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
42	Heat Transferred (kJ/h-m)	719.2	719.2	719.1	719.1
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2064	0.2064	0.2064	0.2064
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.840e-003	6.840e-003	6.840e-003	6.839e-003
48	Static Gradient (kPa/m)	-2.315e-003	-2.315e-003	-2.315e-003	-2.315e-003
49	Accel. Gradient (kPa/m)	3.142e-007	3.142e-007	3.142e-007	3.142e-007
50	Liq. Reynolds	2.160e+005 *	2.160e+005 *	2.160e+005 *	2.160e+005 *
51	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
52	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
53	Vapour Velocity (m/s)	2.163	2.163	2.163	2.163
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1.113e+004	1.113e+004	1.114e+004	1.114e+004
57	Elevation (m)	0.3200	0.3240	0.3280	0.3320
58	Pressure (bar)	91.83	91.83	91.83	91.83
59	Temperature (C)	56.99	56.99	56.98	56.98
60	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
61	Heat Transferred (kJ/h-m)	719.0	718.9	718.9	718.8
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2064	0.2825	0.2825	0.2826
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	6.839e-003	5.398e-003	5.398e-003	5.398e-003
67	Static Gradient (kPa/m)	-2.315e-003	1.902e-003	1.902e-003	1.902e-003
68	Accel. Gradient (kPa/m)	3.142e-007	5.069e-007	5.069e-007	5.069e-007
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
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11	Pipe Table				
12					
13	Liq. Reynolds	2.160e+005 *	2.160e+005 *	2.160e+005 *	2.160e+005 *
14	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
15	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
16	Vapour Velocity (m/s)	2.163	2.163	2.162	2.162
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	1.115e+004	1.115e+004	1.116e+004	1.116e+004
20	Elevation (m)	0.3360	0.3400	0.3340	0.3280
21	Pressure (bar)	91.83	91.83	91.83	91.83
22	Temperature (C)	56.98	56.98	56.97	56.97
23	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
24	Heat Transferred (kJ/h-m)	718.8	718.7	718.7	718.6
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2826	0.2826	0.2065	0.2065
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	5.398e-003	5.398e-003	6.839e-003	6.839e-003
30	Static Gradient (kPa/m)	1.902e-003	1.902e-003	-2.315e-003	-2.315e-003
31	Accel. Gradient (kPa/m)	5.069e-007	5.069e-007	3.141e-007	3.141e-007
32	Liq. Reynolds	2.160e+005 *	2.160e+005 *	2.160e+005 *	2.160e+005 *
33	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
34	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
35	Vapour Velocity (m/s)	2.162	2.162	2.162	2.162
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	1.117e+004	1.117e+004	1.118e+004	1.118e+004
39	Elevation (m)	0.3220	0.3160	0.3100	0.3080
40	Pressure (bar)	91.83	91.83	91.83	91.83
41	Temperature (C)	56.97	56.97	56.97	56.96
42	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
43	Heat Transferred (kJ/h-m)	718.6	718.5	718.4	718.4
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2065	0.2065	0.2065	0.2072
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	6.839e-003	6.839e-003	6.839e-003	6.833e-003
49	Static Gradient (kPa/m)	-2.315e-003	-2.315e-003	-2.315e-003	-7.735e-004
50	Accel. Gradient (kPa/m)	3.141e-007	3.141e-007	3.141e-007	4.207e-007
51	Liq. Reynolds	2.160e+005 *	2.160e+005 *	2.160e+005 *	2.160e+005 *
52	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
53	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
54	Vapour Velocity (m/s)	2.162	2.162	2.162	2.162
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	1.119e+004	1.119e+004	1.120e+004	1.120e+004
58	Elevation (m)	0.3060	0.3040	0.3020	0.3000
59	Pressure (bar)	91.83	91.83	91.83	91.83
60	Temperature (C)	56.96	56.96	56.96	56.96
61	Vapour Fraction	0.8718	0.8718	0.8718	0.8718
62	Heat Transferred (kJ/h-m)	718.3	718.3	718.2	718.2
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2072	0.2072	0.2072	0.2072
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	6.833e-003	6.833e-003	6.833e-003	6.833e-003
68	Static Gradient (kPa/m)	-7.735e-004	-7.735e-004	-7.735e-004	-7.736e-004
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
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12							
13	Accel. Gradient	(kPa/m)	4.207e-007	4.207e-007	4.207e-007	4.207e-007	
14	Liq. Reynolds		2.160e+005 *	2.160e+005 *	2.160e+005 *	2.160e+005 *	
15	Vap. Reynolds		9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *	
16	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
17	Vapour Velocity	(m/s)	2.162	2.162	2.162	2.162	
18	Deposit Thickness	(mm)	---	---	---	---	
19	Deposit Volume	(m3)	---	---	---	---	
20	Length	(m)	1.121e+004	1.121e+004	1.122e+004	1.122e+004	
21	Elevation	(m)	0.3000	0.3000	0.3000	0.3000	
22	Pressure	(bar)	91.83	91.83	91.83	91.83	
23	Temperature	(C)	56.95	56.95	56.95	56.95	
24	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
25	Heat Transferred	(kJ/h-m)	718.1	718.0	718.0	717.9	
26	Flow Regime		Transition	Transition	Transition	Transition	
27	Water Flow Regime						
28	Liquid HoldUp		0.3030	0.3030	0.3030	0.3031	
29	Water HoldUp		---	---	---	---	
30	Friction Gradient	(kPa/m)	5.858e-003	5.858e-003	5.858e-003	5.858e-003	
31	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000	
32	Accel. Gradient	(kPa/m)	4.067e-007	4.067e-007	4.067e-007	4.067e-007	
33	Liq. Reynolds		2.160e+005 *	2.160e+005 *	2.160e+005 *	2.160e+005 *	
34	Vap. Reynolds		9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *	
35	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
36	Vapour Velocity	(m/s)	2.162	2.162	2.162	2.162	
37	Deposit Thickness	(mm)	---	---	---	---	
38	Deposit Volume	(m3)	---	---	---	---	
39	Length	(m)	1.123e+004	1.123e+004	1.124e+004	1.124e+004	
40	Elevation	(m)	0.3000	0.3060	0.3120	0.3180	
41	Pressure	(bar)	91.83	91.83	91.82	91.82	
42	Temperature	(C)	56.95	56.94	56.94	56.94	
43	Vapour Fraction		0.8718	0.8718	0.8718	0.8718	
44	Heat Transferred	(kJ/h-m)	717.9	717.8	717.8	717.7	
45	Flow Regime		Transition	Transition	Transition	Transition	
46	Water Flow Regime						
47	Liquid HoldUp		0.3031	0.2839	0.2839	0.2839	
48	Water HoldUp		---	---	---	---	
49	Friction Gradient	(kPa/m)	5.858e-003	5.300e-003	5.299e-003	5.299e-003	
50	Static Gradient	(kPa/m)	0.0000	2.862e-003	2.862e-003	2.862e-003	
51	Accel. Gradient	(kPa/m)	4.067e-007	5.667e-007	5.667e-007	5.667e-007	
52	Liq. Reynolds		2.160e+005 *	2.160e+005 *	2.160e+005 *	2.160e+005 *	
53	Vap. Reynolds		9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *	
54	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
55	Vapour Velocity	(m/s)	2.162	2.162	2.162	2.162	
56	Deposit Thickness	(mm)	---	---	---	---	
57	Deposit Volume	(m3)	---	---	---	---	
58	Length	(m)	1.125e+004	1.125e+004	1.126e+004	1.126e+004	
59	Elevation	(m)	0.3240	0.3300	0.3280	0.3260	
60	Pressure	(bar)	91.82	91.82	91.82	91.82	
61	Temperature	(C)	56.94	56.93	56.93	56.93	
62	Vapour Fraction		0.8717	0.8717	0.8717	0.8717	
63	Heat Transferred	(kJ/h-m)	717.6	717.6	717.5	717.5	
64	Flow Regime		Transition	Transition	Transition	Transition	
65	Water Flow Regime						
66	Liquid HoldUp		0.2839	0.2839	0.2072	0.2072	
67	Water HoldUp		---	---	---	---	
68	Friction Gradient	(kPa/m)	5.299e-003	5.299e-003	6.833e-003	6.832e-003	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
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9	PROFILES						
10							
11	Pipe Table						
12							
13	Static Gradient	(kPa/m)	2.862e-003	2.862e-003	-7.736e-004	-7.736e-004	
14	Accel. Gradient	(kPa/m)	5.667e-007	5.667e-007	4.207e-007	4.207e-007	
15	Liq. Reynolds		2.160e+005 *	2.160e+005 *	2.160e+005 *	2.160e+005 *	
16	Vap. Reynolds		9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *	
17	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
18	Vapour Velocity	(m/s)	2.162	2.162	2.162	2.162	
19	Deposit Thickness	(mm)	---	---	---	---	
20	Deposit Volume	(m3)	---	---	---	---	
21	Length	(m)	1.127e+004	1.127e+004	1.128e+004	1.128e+004	
22	Elevation	(m)	0.3240	0.3220	0.3200	0.3200	
23	Pressure	(bar)	91.82	91.82	91.82	91.82	
24	Temperature	(C)	56.93	56.93	56.92	56.92	
25	Vapour Fraction		0.8717	0.8717	0.8717	0.8717	
26	Heat Transferred	(kJ/h-m)	717.4	717.4	717.3	717.3	
27	Flow Regime		Transition	Transition	Transition	Transition	
28	Water Flow Regime						
29	Liquid HoldUp		0.2072	0.2072	0.2072	0.3031	
30	Water HoldUp		---	---	---	---	
31	Friction Gradient	(kPa/m)	6.832e-003	6.832e-003	6.832e-003	5.857e-003	
32	Static Gradient	(kPa/m)	-7.736e-004	-7.736e-004	-7.736e-004	0.0000	
33	Accel. Gradient	(kPa/m)	4.207e-007	4.207e-007	4.206e-007	4.067e-007	
34	Liq. Reynolds		2.159e+005 *	2.159e+005 *	2.159e+005 *	2.159e+005 *	
35	Vap. Reynolds		9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *	
36	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
37	Vapour Velocity	(m/s)	2.162	2.162	2.162	2.162	
38	Deposit Thickness	(mm)	---	---	---	---	
39	Deposit Volume	(m3)	---	---	---	---	
40	Length	(m)	1.129e+004	1.129e+004	1.130e+004	1.130e+004	
41	Elevation	(m)	0.3200	0.3200	0.3200	0.3200	
42	Pressure	(bar)	91.82	91.82	91.82	91.82	
43	Temperature	(C)	56.92	56.92	56.92	56.91	
44	Vapour Fraction		0.8717	0.8717	0.8717	0.8717	
45	Heat Transferred	(kJ/h-m)	717.2	717.1	717.1	717.0	
46	Flow Regime		Transition	Transition	Transition	Transition	
47	Water Flow Regime						
48	Liquid HoldUp		0.3031	0.3031	0.3031	0.3031	
49	Water HoldUp		---	---	---	---	
50	Friction Gradient	(kPa/m)	5.857e-003	5.857e-003	5.857e-003	5.857e-003	
51	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000	
52	Accel. Gradient	(kPa/m)	4.067e-007	4.067e-007	4.066e-007	4.066e-007	
53	Liq. Reynolds		2.159e+005 *	2.159e+005 *	2.159e+005 *	2.159e+005 *	
54	Vap. Reynolds		9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *	
55	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
56	Vapour Velocity	(m/s)	2.162	2.162	2.162	2.162	
57	Deposit Thickness	(mm)	---	---	---	---	
58	Deposit Volume	(m3)	---	---	---	---	
59	Length	(m)	1.131e+004	1.131e+004	1.132e+004	1.132e+004	
60	Elevation	(m)	0.2820	0.2440	0.2060	0.1680	
61	Pressure	(bar)	91.82	91.82	91.82	91.82	
62	Temperature	(C)	56.91	56.91	56.91	56.91	
63	Vapour Fraction		0.8717	0.8717	0.8717	0.8717	
64	Heat Transferred	(kJ/h-m)	717.0	716.9	716.9	716.8	
65	Flow Regime		Transition	Transition	Transition	Transition	
66	Water Flow Regime						
67	Liquid HoldUp		0.2005	0.2005	0.2005	0.2005	
68	Water HoldUp		---	---	---	---	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 215 of 937	

1			Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3	 Epic Atlantic Limited		Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	6.887e-003	6.887e-003	6.887e-003	6.887e-003
14	Static Gradient	(kPa/m)	-1.440e-002	-1.440e-002	-1.440e-002	-1.440e-002
15	Accel. Gradient	(kPa/m)	-5.215e-007	-5.215e-007	-5.215e-007	-5.215e-007
16	Liq. Reynolds		2.159e+005 *	2.159e+005 *	2.159e+005 *	2.159e+005 *
17	Vap. Reynolds		9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
18	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042
19	Vapour Velocity	(m/s)	2.162	2.162	2.162	2.162
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	1.133e+004	1.133e+004	1.134e+004	1.134e+004
23	Elevation	(m)	0.1300	0.1380	0.1460	0.1540
24	Pressure	(bar)	91.82	91.82	91.82	91.82
25	Temperature	(C)	56.90	56.90	56.90	56.90
26	Vapour Fraction		0.8717	0.8717	0.8717	0.8717
27	Heat Transferred	(kJ/h-m)	716.8	716.7	716.7	716.6
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2005	0.2852	0.2852	0.2852
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	6.886e-003	5.202e-003	5.202e-003	5.202e-003
33	Static Gradient	(kPa/m)	-1.440e-002	3.829e-003	3.829e-003	3.829e-003
34	Accel. Gradient	(kPa/m)	-5.215e-007	6.269e-007	6.269e-007	6.269e-007
35	Liq. Reynolds		2.159e+005 *	2.159e+005 *	2.159e+005 *	2.159e+005 *
36	Vap. Reynolds		9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
37	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042
38	Vapour Velocity	(m/s)	2.162	2.162	2.162	2.162
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	1.135e+004	1.135e+004	1.136e+004	1.136e+004
42	Elevation	(m)	0.1620	0.1700	0.1760	0.1820
43	Pressure	(bar)	91.82	91.82	91.82	91.82
44	Temperature	(C)	56.90	56.89	56.89	56.89
45	Vapour Fraction		0.8717	0.8717	0.8717	0.8717
46	Heat Transferred	(kJ/h-m)	716.5	716.5	716.4	716.4
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2852	0.2852	0.2839	0.2839
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	5.202e-003	5.202e-003	5.298e-003	5.298e-003
52	Static Gradient	(kPa/m)	3.829e-003	3.829e-003	2.863e-003	2.863e-003
53	Accel. Gradient	(kPa/m)	6.269e-007	6.269e-007	5.665e-007	5.665e-007
54	Liq. Reynolds		2.159e+005 *	2.159e+005 *	2.159e+005 *	2.159e+005 *
55	Vap. Reynolds		9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
56	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042
57	Vapour Velocity	(m/s)	2.162	2.162	2.162	2.162
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	1.137e+004	1.137e+004	1.138e+004	1.138e+004
61	Elevation	(m)	0.1880	0.1940	0.2000	0.1800
62	Pressure	(bar)	91.82	91.82	91.82	91.82
63	Temperature	(C)	56.89	56.88	56.88	56.88
64	Vapour Fraction		0.8717	0.8717	0.8717	0.8717
65	Heat Transferred	(kJ/h-m)	716.3	716.3	716.2	716.2
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2839	0.2839	0.2839	0.2039
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 216 of 937	

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	5.298e-003	5.298e-003	5.298e-003	6.858e-003
15	Static Gradient (kPa/m)	2.863e-003	2.863e-003	2.863e-003	-7.658e-003
16	Accel. Gradient (kPa/m)	5.665e-007	5.665e-007	5.665e-007	-5.554e-008
17	Liq. Reynolds	2.159e+005 *	2.159e+005 *	2.159e+005 *	2.159e+005 *
18	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
19	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
20	Vapour Velocity (m/s)	2.162	2.162	2.162	2.162
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	1.139e+004	1.139e+004	1.140e+004	1.140e+004
24	Elevation (m)	0.1600	0.1400	0.1200	1.000e-001
25	Pressure (bar)	91.82	91.82	91.82	91.82
26	Temperature (C)	56.88	56.88	56.87	56.87
27	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
28	Heat Transferred (kJ/h-m)	716.1	716.1	716.0	716.0
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2039	0.2039	0.2039	0.2039
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	6.858e-003	6.858e-003	6.858e-003	6.858e-003
34	Static Gradient (kPa/m)	-7.658e-003	-7.658e-003	-7.658e-003	-7.658e-003
35	Accel. Gradient (kPa/m)	-5.555e-008	-5.556e-008	-5.557e-008	-5.558e-008
36	Liq. Reynolds	2.159e+005 *	2.159e+005 *	2.159e+005 *	2.159e+005 *
37	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
38	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
39	Vapour Velocity (m/s)	2.162	2.162	2.162	2.162
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	1.141e+004	1.141e+004	1.142e+004	1.142e+004
43	Elevation (m)	-2.200e-002	-0.1440	-0.2660	-0.3880
44	Pressure (bar)	91.82	91.82	91.82	91.83
45	Temperature (C)	56.87	56.87	56.87	56.87
46	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
47	Heat Transferred (kJ/h-m)	715.9	715.9	715.9	715.8
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.1849	0.1849	0.1849	0.1849
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	7.035e-003	7.035e-003	7.035e-003	7.035e-003
53	Static Gradient (kPa/m)	-4.392e-002	-4.392e-002	-4.392e-002	-4.392e-002
54	Accel. Gradient (kPa/m)	-2.560e-006	-2.560e-006	-2.560e-006	-2.560e-006
55	Liq. Reynolds	2.159e+005 *	2.159e+005 *	2.159e+005 *	2.159e+005 *
56	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
57	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
58	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	1.143e+004	1.143e+004	1.144e+004	1.144e+004
62	Elevation (m)	-0.5100	-0.4720	-0.4340	-0.3960
63	Pressure (bar)	91.83	91.83	91.83	91.82
64	Temperature (C)	56.87	56.86	56.86	56.86
65	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
66	Heat Transferred (kJ/h-m)	715.8	715.7	715.6	715.6
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 217 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.1849	0.3044	0.3044	0.3044
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	7.034e-003	5.889e-003	5.889e-003	5.889e-003
16	Static Gradient (kPa/m)	-4.392e-002	1.905e-002	1.905e-002	1.905e-002
17	Accel. Gradient (kPa/m)	-2.560e-006	1.731e-006	1.731e-006	1.731e-006
18	Liq. Reynolds	2.159e+005 *	2.159e+005 *	2.159e+005 *	2.159e+005 *
19	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *
20	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
21	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	1.145e+004	1.145e+004	1.146e+004	1.146e+004
25	Elevation (m)	-0.3580	-0.3200	-0.2080	-9.600e-002
26	Pressure (bar)	91.82	91.82	91.82	91.81
27	Temperature (C)	56.86	56.85	56.85	56.85
28	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
29	Heat Transferred (kJ/h-m)	715.5	715.5	715.4	715.3
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.3044	0.3044	0.3516	0.3516
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	5.889e-003	5.889e-003	6.030e-003	6.030e-003
35	Static Gradient (kPa/m)	1.905e-002	1.905e-002	6.238e-002	6.237e-002
36	Accel. Gradient (kPa/m)	1.731e-006	1.731e-006	4.748e-006	4.748e-006
37	Liq. Reynolds	2.159e+005 *	2.159e+005 *	2.159e+005 *	2.159e+005 *
38	Vap. Reynolds	9.081e+006 *	9.081e+006 *	9.081e+006 *	9.082e+006 *
39	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
40	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	1.147e+004	1.147e+004	1.148e+004	1.148e+004
44	Elevation (m)	1.600e-002	0.1280	0.2400	0.2480
45	Pressure (bar)	91.81	91.81	91.80	91.80
46	Temperature (C)	56.85	56.84	56.84	56.84
47	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
48	Heat Transferred (kJ/h-m)	715.2	715.2	715.1	715.0
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.3516	0.3516	0.3516	0.2852
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.031e-003	6.031e-003	6.031e-003	5.201e-003
54	Static Gradient (kPa/m)	6.237e-002	6.237e-002	6.237e-002	3.829e-003
55	Accel. Gradient (kPa/m)	4.748e-006	4.749e-006	4.749e-006	6.269e-007
56	Liq. Reynolds	2.159e+005 *	2.158e+005 *	2.158e+005 *	2.158e+005 *
57	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *
58	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
59	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	1.149e+004	1.149e+004	1.150e+004	1.150e+004
63	Elevation (m)	0.2560	0.2640	0.2720	0.2800
64	Pressure (bar)	91.80	91.80	91.80	91.80
65	Temperature (C)	56.84	56.83	56.83	56.83
66	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
67	Heat Transferred (kJ/h-m)	714.9	714.9	714.8	714.8
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 218 of 937

1			Case Name:	Simulation - 386MMSCFD_100barg.hsc								
2			Unit Set:	Mixed								
3	 Epic Atlantic Limited		Date/Time:	Mon Jun 13 15:02:44 2022								
4												
5												
6	Pipe Segment: ELEP-Akabel Pipeline (continued)											
7												
8												
9	PROFILES											
10												
11	Pipe Table											
12												
13	Water Flow Regime											
14	Liquid HoldUp		0.2852	0.2852	0.2852	0.2852						
15	Water HoldUp		---	---	---	---						
16	Friction Gradient (kPa/m)		5.201e-003	5.201e-003	5.201e-003	5.201e-003						
17	Static Gradient (kPa/m)		3.829e-003	3.829e-003	3.829e-003	3.830e-003						
18	Accel. Gradient (kPa/m)		6.269e-007	6.269e-007	6.269e-007	6.269e-007						
19	Liq. Reynolds		2.158e+005 *	2.158e+005 *	2.158e+005 *	2.158e+005 *						
20	Vap. Reynolds		9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *						
21	Liquid Velocity (m/s)		0.2042	0.2042	0.2042	0.2042						
22	Vapour Velocity (m/s)		2.161	2.161	2.161	2.161						
23	Deposit Thickness (mm)		---	---	---	---						
24	Deposit Volume (m3)		---	---	---	---						
25	Length (m)		1.151e+004	1.151e+004	1.152e+004	1.152e+004						
26	Elevation (m)		0.2840	0.2880	0.2920	0.2960						
27	Pressure (bar)		91.80	91.80	91.80	91.80						
28	Temperature (C)		56.83	56.82	56.82	56.82						
29	Vapour Fraction		0.8717	0.8717	0.8717	0.8717						
30	Heat Transferred (kJ/h-m)		714.7	714.6	714.6	714.5						
31	Flow Regime		Transition	Transition	Transition	Transition						
32	Water Flow Regime											
33	Liquid HoldUp		0.2826	0.2826	0.2826	0.2826						
34	Water HoldUp		---	---	---	---						
35	Friction Gradient (kPa/m)		5.395e-003	5.395e-003	5.395e-003	5.395e-003						
36	Static Gradient (kPa/m)		1.903e-003	1.903e-003	1.903e-003	1.903e-003						
37	Accel. Gradient (kPa/m)		5.067e-007	5.066e-007	5.066e-007	5.066e-007						
38	Liq. Reynolds		2.158e+005 *	2.158e+005 *	2.158e+005 *	2.158e+005 *						
39	Vap. Reynolds		9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *						
40	Liquid Velocity (m/s)		0.2042	0.2042	0.2042	0.2042						
41	Vapour Velocity (m/s)		2.161	2.161	2.161	2.161						
42	Deposit Thickness (mm)		---	---	---	---						
43	Deposit Volume (m3)		---	---	---	---						
44	Length (m)		1.153e+004	1.153e+004	1.154e+004	1.154e+004						
45	Elevation (m)		0.3000	0.3080	0.3160	0.3240						
46	Pressure (bar)		91.80	91.80	91.80	91.80						
47	Temperature (C)		56.82	56.82	56.81	56.81						
48	Vapour Fraction		0.8717	0.8717	0.8717	0.8717						
49	Heat Transferred (kJ/h-m)		714.5	714.4	714.4	714.3						
50	Flow Regime		Transition	Transition	Transition	Transition						
51	Water Flow Regime											
52	Liquid HoldUp		0.2826	0.2852	0.2852	0.2852						
53	Water HoldUp		---	---	---	---						
54	Friction Gradient (kPa/m)		5.395e-003	5.201e-003	5.201e-003	5.201e-003						
55	Static Gradient (kPa/m)		1.903e-003	3.830e-003	3.830e-003	3.830e-003						
56	Accel. Gradient (kPa/m)		5.066e-007	6.269e-007	6.269e-007	6.269e-007						
57	Liq. Reynolds		2.158e+005 *	2.158e+005 *	2.158e+005 *	2.158e+005 *						
58	Vap. Reynolds		9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *						
59	Liquid Velocity (m/s)		0.2042	0.2042	0.2042	0.2042						
60	Vapour Velocity (m/s)		2.161	2.161	2.161	2.161						
61	Deposit Thickness (mm)		---	---	---	---						
62	Deposit Volume (m3)		---	---	---	---						
63	Length (m)		1.155e+004	1.155e+004	1.156e+004	1.156e+004						
64	Elevation (m)		0.3320	0.3400	0.3380	0.3360						
65	Pressure (bar)		91.80	91.80	91.80	91.80						
66	Temperature (C)		56.81	56.81	56.81	56.80						
67	Vapour Fraction		0.8717	0.8717	0.8717	0.8717						
68	Heat Transferred (kJ/h-m)		714.2	714.2	714.1	714.1						
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
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5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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9	PROFILES				
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11	Pipe Table				
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13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2852	0.2852	0.2073	0.2073
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	5.200e-003	5.200e-003	6.830e-003	6.830e-003
18	Static Gradient (kPa/m)	3.830e-003	3.830e-003	-7.738e-004	-7.738e-004
19	Accel. Gradient (kPa/m)	6.269e-007	6.269e-007	4.205e-007	4.205e-007
20	Liq. Reynolds	2.158e+005 *	2.158e+005 *	2.158e+005 *	2.158e+005 *
21	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *
22	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
23	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	1.157e+004	1.157e+004	1.158e+004	1.158e+004
27	Elevation (m)	0.3340	0.3320	0.3300	0.3240
28	Pressure (bar)	91.80	91.80	91.80	91.80
29	Temperature (C)	56.80	56.80	56.80	56.79
30	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
31	Heat Transferred (kJ/h-m)	714.0	714.0	713.9	713.9
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2073	0.2073	0.2073	0.2065
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	6.830e-003	6.830e-003	6.830e-003	6.836e-003
37	Static Gradient (kPa/m)	-7.738e-004	-7.738e-004	-7.739e-004	-2.316e-003
38	Accel. Gradient (kPa/m)	4.205e-007	4.205e-007	4.204e-007	3.138e-007
39	Liq. Reynolds	2.158e+005 *	2.158e+005 *	2.158e+005 *	2.158e+005 *
40	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *
41	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
42	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	1.159e+004	1.159e+004	1.160e+004	1.160e+004
46	Elevation (m)	0.3180	0.3120	0.3060	0.3000
47	Pressure (bar)	91.80	91.80	91.80	91.80
48	Temperature (C)	56.79	56.79	56.79	56.79
49	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
50	Heat Transferred (kJ/h-m)	713.8	713.8	713.7	713.7
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2065	0.2065	0.2065	0.2065
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	6.836e-003	6.836e-003	6.836e-003	6.836e-003
56	Static Gradient (kPa/m)	-2.316e-003	-2.316e-003	-2.316e-003	-2.316e-003
57	Accel. Gradient (kPa/m)	3.138e-007	3.138e-007	3.137e-007	3.137e-007
58	Liq. Reynolds	2.158e+005 *	2.158e+005 *	2.158e+005 *	2.158e+005 *
59	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *
60	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
61	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	1.161e+004	1.161e+004	1.162e+004	1.162e+004
65	Elevation (m)	0.2980	0.2960	0.2940	0.2920
66	Pressure (bar)	91.79	91.79	91.79	91.79
67	Temperature (C)	56.78	56.78	56.78	56.78
68	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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13	Heat Transferred (kJ/h-m)	713.6	713.5	713.5	713.4
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.2073	0.2073	0.2073	0.2073
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	6.830e-003	6.830e-003	6.830e-003	6.830e-003
19	Static Gradient (kPa/m)	-7.754e-004	-7.754e-004	-7.754e-004	-7.754e-004
20	Accel. Gradient (kPa/m)	4.203e-007	4.203e-007	4.203e-007	4.203e-007
21	Liq. Reynolds	2.158e+005 *	2.158e+005 *	2.158e+005 *	2.158e+005 *
22	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *
23	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
24	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	1.163e+004	1.163e+004	1.164e+004	1.164e+004
28	Elevation (m)	0.2900	0.2760	0.2620	0.2480
29	Pressure (bar)	91.79	91.79	91.79	91.79
30	Temperature (C)	56.78	56.77	56.77	56.77
31	Vapour Fraction	0.8717	0.8717	0.8717	0.8717
32	Heat Transferred (kJ/h-m)	713.4	713.3	713.3	713.2
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.2073	0.2051	0.2051	0.2051
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	6.830e-003	6.847e-003	6.847e-003	6.847e-003
38	Static Gradient (kPa/m)	-7.755e-004	-5.380e-003	-5.380e-003	-5.380e-003
39	Accel. Gradient (kPa/m)	4.203e-007	1.018e-007	1.018e-007	1.018e-007
40	Liq. Reynolds	2.158e+005 *	2.158e+005 *	2.158e+005 *	2.158e+005 *
41	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *
42	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
43	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	1.165e+004	1.165e+004	1.166e+004	1.166e+004
47	Elevation (m)	0.2340	0.2200	0.2280	0.2360
48	Pressure (bar)	91.79	91.79	91.79	91.79
49	Temperature (C)	56.77	56.77	56.76	56.76
50	Vapour Fraction	0.8717	0.8717	0.8717	0.8716
51	Heat Transferred (kJ/h-m)	713.2	713.1	713.0	713.0
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.2051	0.2051	0.2852	0.2852
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	6.847e-003	6.847e-003	5.199e-003	5.199e-003
57	Static Gradient (kPa/m)	-5.381e-003	-5.381e-003	3.830e-003	3.830e-003
58	Accel. Gradient (kPa/m)	1.018e-007	1.018e-007	6.268e-007	6.268e-007
59	Liq. Reynolds	2.158e+005 *	2.158e+005 *	2.158e+005 *	2.158e+005 *
60	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *
61	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
62	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	1.167e+004	1.167e+004	1.168e+004	1.168e+004
66	Elevation (m)	0.2440	0.2520	0.2600	0.2720
67	Pressure (bar)	91.79	91.79	91.79	91.79
68	Temperature (C)	56.76	56.76	56.76	56.75
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13	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
14	Heat Transferred (kJ/h-m)	712.9	712.9	712.8	712.7
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2852	0.2852	0.2852	0.2878
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	5.199e-003	5.199e-003	5.199e-003	5.010e-003
20	Static Gradient (kPa/m)	3.830e-003	3.830e-003	3.830e-003	5.782e-003
21	Accel. Gradient (kPa/m)	6.268e-007	6.268e-007	6.268e-007	7.491e-007
22	Liq. Reynolds	2.157e+005 *	2.157e+005 *	2.157e+005 *	2.157e+005 *
23	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *
24	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
25	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	1.169e+004	1.169e+004	1.170e+004	1.170e+004
29	Elevation (m)	0.2840	0.2960	0.3080	0.3200
30	Pressure (bar)	91.79	91.79	91.79	91.79
31	Temperature (C)	56.75	56.75	56.75	56.74
32	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
33	Heat Transferred (kJ/h-m)	712.7	712.6	712.6	712.5
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2878	0.2878	0.2878	0.2878
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	5.010e-003	5.010e-003	5.010e-003	5.010e-003
39	Static Gradient (kPa/m)	5.782e-003	5.782e-003	5.782e-003	5.782e-003
40	Accel. Gradient (kPa/m)	7.491e-007	7.491e-007	7.491e-007	7.491e-007
41	Liq. Reynolds	2.157e+005 *	2.157e+005 *	2.157e+005 *	2.157e+005 *
42	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *
43	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
44	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	1.171e+004	1.171e+004	1.172e+004	1.172e+004
48	Elevation (m)	0.3200	0.3200	0.3200	0.3200
49	Pressure (bar)	91.79	91.79	91.79	91.79
50	Temperature (C)	56.74	56.74	56.74	56.74
51	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
52	Heat Transferred (kJ/h-m)	712.5	712.4	712.4	712.3
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.3032	0.3032	0.3032	0.3032
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	5.855e-003	5.855e-003	5.855e-003	5.855e-003
58	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
59	Accel. Gradient (kPa/m)	4.064e-007	4.064e-007	4.064e-007	4.064e-007
60	Liq. Reynolds	2.157e+005 *	2.157e+005 *	2.157e+005 *	2.157e+005 *
61	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *
62	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
63	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	1.173e+004	1.173e+004	1.174e+004	1.174e+004
67	Elevation (m)	0.3200	0.3220	0.3240	0.3260
68	Pressure (bar)	91.79	91.79	91.79	91.79
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12					
13	Temperature (C)	56.73	56.73	56.73	56.73
14	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
15	Heat Transferred (kJ/h-m)	712.2	712.2	712.1	712.1
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.3032	0.2814	0.2814	0.2814
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	5.855e-003	5.492e-003	5.492e-003	5.492e-003
21	Static Gradient (kPa/m)	0.0000	9.486e-004	9.486e-004	9.486e-004
22	Accel. Gradient (kPa/m)	4.064e-007	4.471e-007	4.471e-007	4.471e-007
23	Liq. Reynolds	2.157e+005 *	2.157e+005 *	2.157e+005 *	2.157e+005 *
24	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *
25	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
26	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	1.175e+004	1.175e+004	1.176e+004	1.176e+004
30	Elevation (m)	0.3280	0.3300	0.3300	0.3300
31	Pressure (bar)	91.79	91.79	91.79	91.78
32	Temperature (C)	56.73	56.72	56.72	56.72
33	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
34	Heat Transferred (kJ/h-m)	712.0	712.0	711.9	711.9
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2814	0.2814	0.3032	0.3032
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	5.492e-003	5.492e-003	5.855e-003	5.855e-003
40	Static Gradient (kPa/m)	9.486e-004	9.486e-004	0.0000	0.0000
41	Accel. Gradient (kPa/m)	4.471e-007	4.470e-007	4.064e-007	4.064e-007
42	Liq. Reynolds	2.157e+005 *	2.157e+005 *	2.157e+005 *	2.157e+005 *
43	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *
44	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
45	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	1.177e+004	1.177e+004	1.178e+004	1.178e+004
49	Elevation (m)	0.3300	0.3300	0.3300	0.3240
50	Pressure (bar)	91.78	91.78	91.78	91.78
51	Temperature (C)	56.72	56.72	56.71	56.71
52	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
53	Heat Transferred (kJ/h-m)	711.8	711.7	711.7	711.6
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.3032	0.3032	0.3032	0.2066
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	5.855e-003	5.855e-003	5.855e-003	6.834e-003
59	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-2.317e-003
60	Accel. Gradient (kPa/m)	4.064e-007	4.064e-007	4.064e-007	3.136e-007
61	Liq. Reynolds	2.157e+005 *	2.157e+005 *	2.157e+005 *	2.157e+005 *
62	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *
63	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
64	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	1.179e+004	1.179e+004	1.180e+004	1.180e+004
68	Elevation (m)	0.3180	0.3120	0.3060	0.3000
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3			Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:02:44 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	91.78	91.78	91.78	91.78		
14	Temperature (C)	56.71	56.71	56.70	56.70		
15	Vapour Fraction	0.8716	0.8716	0.8716	0.8716		
16	Heat Transferred (kJ/h-m)	711.6	711.5	711.5	711.4		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.2066	0.2066	0.2066	0.2066		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	6.834e-003	6.834e-003	6.834e-003	6.834e-003		
22	Static Gradient (kPa/m)	-2.317e-003	-2.317e-003	-2.317e-003	-2.317e-003		
23	Accel. Gradient (kPa/m)	3.136e-007	3.135e-007	3.135e-007	3.135e-007		
24	Liq. Reynolds	2.157e+005 *	2.157e+005 *	2.157e+005 *	2.157e+005 *		
25	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *		
26	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
27	Vapour Velocity (m/s)	2.161	2.160	2.160	2.160		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	1.181e+004	1.181e+004	1.182e+004	1.182e+004		
31	Elevation (m)	0.2960	0.2920	0.2880	0.2840		
32	Pressure (bar)	91.78	91.78	91.78	91.78		
33	Temperature (C)	56.70	56.70	56.70	56.69		
34	Vapour Fraction	0.8716	0.8716	0.8716	0.8716		
35	Heat Transferred (kJ/h-m)	711.4	711.3	711.3	711.2		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.2070	0.2070	0.2070	0.2070		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	6.831e-003	6.831e-003	6.831e-003	6.831e-003		
41	Static Gradient (kPa/m)	-1.550e-003	-1.550e-003	-1.550e-003	-1.550e-003		
42	Accel. Gradient (kPa/m)	3.666e-007	3.665e-007	3.665e-007	3.665e-007		
43	Liq. Reynolds	2.157e+005 *	2.157e+005 *	2.157e+005 *	2.157e+005 *		
44	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *		
45	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
46	Vapour Velocity (m/s)	2.160	2.160	2.160	2.160		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	1.183e+004	1.183e+004	1.184e+004	1.184e+004		
50	Elevation (m)	0.2800	0.2840	0.2880	0.2920		
51	Pressure (bar)	91.78	91.78	91.78	91.78		
52	Temperature (C)	56.69	56.69	56.69	56.69		
53	Vapour Fraction	0.8716	0.8716	0.8716	0.8716		
54	Heat Transferred (kJ/h-m)	711.1	711.1	711.0	711.0		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.2070	0.2827	0.2827	0.2827		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	6.831e-003	5.392e-003	5.392e-003	5.392e-003		
60	Static Gradient (kPa/m)	-1.550e-003	1.903e-003	1.903e-003	1.903e-003		
61	Accel. Gradient (kPa/m)	3.665e-007	5.063e-007	5.063e-007	5.063e-007		
62	Liq. Reynolds	2.157e+005 *	2.157e+005 *	2.157e+005 *	2.157e+005 *		
63	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *		
64	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
65	Vapour Velocity (m/s)	2.160	2.160	2.160	2.160		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	1.185e+004	1.185e+004	1.186e+004	1.186e+004		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	0.2960	0.3000	0.2980	0.2960
14	Pressure (bar)	91.78	91.78	91.78	91.78
15	Temperature (C)	56.68	56.68	56.68	56.68
16	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
17	Heat Transferred (kJ/h-m)	710.9	710.9	710.8	710.8
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2827	0.2827	0.2073	0.2073
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	5.392e-003	5.392e-003	6.828e-003	6.828e-003
23	Static Gradient (kPa/m)	1.903e-003	1.903e-003	-7.741e-004	-7.741e-004
24	Accel. Gradient (kPa/m)	5.063e-007	5.063e-007	4.202e-007	4.202e-007
25	Liq. Reynolds	2.157e+005 *	2.157e+005 *	2.157e+005 *	2.157e+005 *
26	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *
27	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
28	Vapour Velocity (m/s)	2.160	2.160	2.160	2.160
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	1.187e+004	1.187e+004	1.188e+004	1.188e+004
32	Elevation (m)	0.2940	0.2920	0.2900	0.2940
33	Pressure (bar)	91.78	91.78	91.78	91.78
34	Temperature (C)	56.68	56.67	56.67	56.67
35	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
36	Heat Transferred (kJ/h-m)	710.7	710.6	710.6	710.5
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2073	0.2073	0.2073	0.2827
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	6.828e-003	6.828e-003	6.828e-003	5.391e-003
42	Static Gradient (kPa/m)	-7.741e-004	-7.741e-004	-7.741e-004	1.904e-003
43	Accel. Gradient (kPa/m)	4.201e-007	4.201e-007	4.201e-007	5.063e-007
44	Liq. Reynolds	2.157e+005 *	2.157e+005 *	2.156e+005 *	2.156e+005 *
45	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *
46	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
47	Vapour Velocity (m/s)	2.160	2.160	2.160	2.160
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	1.189e+004	1.189e+004	1.190e+004	1.190e+004
51	Elevation (m)	0.2980	0.3020	0.3060	0.3100
52	Pressure (bar)	91.78	91.78	91.78	91.78
53	Temperature (C)	56.67	56.67	56.66	56.66
54	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
55	Heat Transferred (kJ/h-m)	710.5	710.4	710.3	710.3
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2827	0.2828	0.2828	0.2828
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	5.391e-003	5.391e-003	5.391e-003	5.391e-003
61	Static Gradient (kPa/m)	1.904e-003	1.904e-003	1.904e-003	1.904e-003
62	Accel. Gradient (kPa/m)	5.063e-007	5.063e-007	5.063e-007	5.063e-007
63	Liq. Reynolds	2.156e+005 *	2.156e+005 *	2.156e+005 *	2.156e+005 *
64	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *
65	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
66	Vapour Velocity (m/s)	2.160	2.160	2.160	2.160
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	1.191e+004	1.191e+004	1.192e+004	1.192e+004
14	Elevation (m)	0.3080	0.3060	0.3040	0.3020
15	Pressure (bar)	91.78	91.78	91.78	91.78
16	Temperature (C)	56.66	56.66	56.65	56.65
17	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
18	Heat Transferred (kJ/h-m)	710.3	710.2	710.1	710.1
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.2074	0.2074	0.2074	0.2074
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	6.827e-003	6.827e-003	6.827e-003	6.827e-003
24	Static Gradient (kPa/m)	-7.695e-004	-7.695e-004	-7.696e-004	-7.696e-004
25	Accel. Gradient (kPa/m)	4.204e-007	4.204e-007	4.204e-007	4.204e-007
26	Liq. Reynolds	2.156e+005 *	2.156e+005 *	2.156e+005 *	2.156e+005 *
27	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *
28	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
29	Vapour Velocity (m/s)	2.160	2.160	2.160	2.160
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	1.193e+004	1.193e+004	1.194e+004	1.194e+004
33	Elevation (m)	0.3000	0.3000	0.3000	0.3000
34	Pressure (bar)	91.77	91.77	91.77	91.77
35	Temperature (C)	56.65	56.65	56.65	56.65
36	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
37	Heat Transferred (kJ/h-m)	710.0	710.0	709.9	709.9
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.2074	0.3033	0.3033	0.3033
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	6.827e-003	5.854e-003	5.854e-003	5.854e-003
43	Static Gradient (kPa/m)	-7.696e-004	0.0000	0.0000	0.0000
44	Accel. Gradient (kPa/m)	4.204e-007	4.063e-007	4.063e-007	4.063e-007
45	Liq. Reynolds	2.156e+005 *	2.156e+005 *	2.156e+005 *	2.156e+005 *
46	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *
47	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
48	Vapour Velocity (m/s)	2.160	2.160	2.160	2.160
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	1.194e+004	1.195e+004	1.196e+004	1.196e+004
52	Elevation (m)	0.3000	0.3000	0.2980	0.2960
53	Pressure (bar)	91.77	91.77	91.77	91.77
54	Temperature (C)	56.64	56.64	56.64	56.64
55	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
56	Heat Transferred (kJ/h-m)	709.8	709.8	709.7	709.7
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.3033	0.3033	0.2075	0.2075
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	5.854e-003	5.854e-003	6.826e-003	6.826e-003
62	Static Gradient (kPa/m)	0.0000	0.0000	-5.532e-004	-5.532e-004
63	Accel. Gradient (kPa/m)	4.063e-007	4.063e-007	4.353e-007	4.353e-007
64	Liq. Reynolds	2.156e+005 *	2.156e+005 *	2.156e+005 *	2.156e+005 *
65	Vap. Reynolds	9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *
66	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
67	Vapour Velocity (m/s)	2.160	2.160	2.160	2.160
68	Deposit Thickness (mm)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:02:44 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Deposit Volume (m3)	---	---	---	---					
14	Length (m)	1.197e+004	1.198e+004	1.198e+004	1.199e+004					
15	Elevation (m)	0.2940	0.2920	0.2900	0.2900					
16	Pressure (bar)	91.77	91.77	91.77	91.77					
17	Temperature (C)	56.63	56.63	56.63	56.62					
18	Vapour Fraction	0.8716	0.8716	0.8716	0.8716					
19	Heat Transferred (kJ/h-m)	709.6	709.5	709.4	709.3					
20	Flow Regime	Transition	Transition	Transition	Transition					
21	Water Flow Regime									
22	Liquid HoldUp	0.2075	0.2075	0.2075	0.3033					
23	Water HoldUp	---	---	---	---					
24	Friction Gradient (kPa/m)	6.826e-003	6.826e-003	6.826e-003	5.854e-003					
25	Static Gradient (kPa/m)	-5.532e-004	-5.532e-004	-5.532e-004	0.0000					
26	Accel. Gradient (kPa/m)	4.353e-007	4.353e-007	4.353e-007	4.062e-007					
27	Liq. Reynolds	2.156e+005 *	2.156e+005 *	2.156e+005 *	2.156e+005 *					
28	Vap. Reynolds	9.082e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *					
29	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043					
30	Vapour Velocity (m/s)	2.160	2.160	2.160	2.160					
31	Deposit Thickness (mm)	---	---	---	---					
32	Deposit Volume (m3)	---	---	---	---					
33	Length (m)	1.200e+004	1.200e+004	1.201e+004	1.202e+004					
34	Elevation (m)	0.2900	0.2900	0.2900	0.2900					
35	Pressure (bar)	91.77	91.77	91.77	91.77					
36	Temperature (C)	56.62	56.62	56.62	56.61					
37	Vapour Fraction	0.8716	0.8716	0.8716	0.8716					
38	Heat Transferred (kJ/h-m)	709.3	709.2	709.1	709.0					
39	Flow Regime	Transition	Transition	Transition	Transition					
40	Water Flow Regime									
41	Liquid HoldUp	0.3033	0.3033	0.3033	0.3033					
42	Water HoldUp	---	---	---	---					
43	Friction Gradient (kPa/m)	5.854e-003	5.854e-003	5.854e-003	5.854e-003					
44	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000					
45	Accel. Gradient (kPa/m)	4.062e-007	4.062e-007	4.062e-007	4.062e-007					
46	Liq. Reynolds	2.156e+005 *	2.156e+005 *	2.156e+005 *	2.156e+005 *					
47	Vap. Reynolds	9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *					
48	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043					
49	Vapour Velocity (m/s)	2.160	2.160	2.160	2.160					
50	Deposit Thickness (mm)	---	---	---	---					
51	Deposit Volume (m3)	---	---	---	---					
52	Length (m)	1.203e+004	1.203e+004	1.204e+004	1.205e+004					
53	Elevation (m)	0.2820	0.2740	0.2660	0.2580					
54	Pressure (bar)	91.77	91.77	91.77	91.77					
55	Temperature (C)	56.61	56.61	56.60	56.60					
56	Vapour Fraction	0.8716	0.8716	0.8716	0.8716					
57	Heat Transferred (kJ/h-m)	709.0	708.9	708.8	708.7					
58	Flow Regime	Transition	Transition	Transition	Transition					
59	Water Flow Regime									
60	Liquid HoldUp	0.2067	0.2067	0.2067	0.2067					
61	Water HoldUp	---	---	---	---					
62	Friction Gradient (kPa/m)	6.832e-003	6.832e-003	6.832e-003	6.832e-003					
63	Static Gradient (kPa/m)	-2.208e-003	-2.208e-003	-2.208e-003	-2.208e-003					
64	Accel. Gradient (kPa/m)	3.209e-007	3.209e-007	3.209e-007	3.209e-007					
65	Liq. Reynolds	2.156e+005 *	2.156e+005 *	2.156e+005 *	2.156e+005 *					
66	Vap. Reynolds	9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *					
67	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043					
68	Vapour Velocity (m/s)	2.160	2.160	2.160	2.160					
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 227 of 937					

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	1.205e+004	1.206e+004	1.207e+004	1.207e+004
16	Elevation (m)	0.2500	0.3600	0.4700	0.5800
17	Pressure (bar)	91.77	91.76	91.76	91.76
18	Temperature (C)	56.60	56.60	56.59	56.59
19	Vapour Fraction	0.8716	0.8716	0.8716	0.8716
20	Heat Transferred (kJ/h-m)	708.7	708.6	708.5	708.4
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2067	0.3304	0.3304	0.3304
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	6.832e-003	6.041e-003	6.041e-003	6.041e-003
26	Static Gradient (kPa/m)	-2.208e-003	4.181e-002	4.181e-002	4.181e-002
27	Accel. Gradient (kPa/m)	3.209e-007	3.321e-006	3.321e-006	3.321e-006
28	Liq. Reynolds	2.156e+005 *	2.156e+005 *	2.156e+005 *	2.155e+005 *
29	Vap. Reynolds	9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *
30	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
31	Vapour Velocity (m/s)	2.160	2.160	2.160	2.160
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	1.208e+004	1.209e+004	1.210e+004	1.210e+004
35	Elevation (m)	0.6900	0.8000	0.7000	0.6000
36	Pressure (bar)	91.75	91.75	91.75	91.75
37	Temperature (C)	56.58	56.58	56.58	56.58
38	Vapour Fraction	0.8716	0.8716	0.8715	0.8715
39	Heat Transferred (kJ/h-m)	708.3	708.2	708.1	708.0
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.3304	0.3304	0.1944	0.1944
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	6.041e-003	6.042e-003	6.937e-003	6.937e-003
45	Static Gradient (kPa/m)	4.181e-002	4.181e-002	-2.656e-002	-2.656e-002
46	Accel. Gradient (kPa/m)	3.321e-006	3.321e-006	-1.362e-006	-1.362e-006
47	Liq. Reynolds	2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *
48	Vap. Reynolds	9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *
49	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
50	Vapour Velocity (m/s)	2.160	2.160	2.160	2.160
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	1.211e+004	1.212e+004	1.212e+004	1.213e+004
54	Elevation (m)	0.5000	0.4000	0.3000	0.3000
55	Pressure (bar)	91.76	91.76	91.76	91.76
56	Temperature (C)	56.57	56.57	56.57	56.57
57	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
58	Heat Transferred (kJ/h-m)	708.0	707.9	707.9	707.8
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.1944	0.1944	0.1944	0.3033
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	6.937e-003	6.937e-003	6.936e-003	5.853e-003
64	Static Gradient (kPa/m)	-2.656e-002	-2.656e-002	-2.656e-002	0.0000
65	Accel. Gradient (kPa/m)	-1.362e-006	-1.362e-006	-1.362e-006	4.062e-007
66	Liq. Reynolds	2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *
67	Vap. Reynolds	9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *
68	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.160	2.160	2.160	2.160	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	1.214e+004	1.214e+004	1.215e+004	1.216e+004	
17	Elevation	(m)	0.3000	0.3000	0.3000	0.3000	
18	Pressure	(bar)	91.76	91.76	91.76	91.76	
19	Temperature	(C)	56.56	56.56	56.56	56.55	
20	Vapour Fraction		0.8715	0.8715	0.8715	0.8715	
21	Heat Transferred	(kJ/h-m)	707.7	707.6	707.5	707.5	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.3033	0.3033	0.3033	0.3033	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	5.853e-003	5.853e-003	5.853e-003	5.853e-003	
27	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000	
28	Accel. Gradient	(kPa/m)	4.062e-007	4.062e-007	4.062e-007	4.062e-007	
29	Liq. Reynolds		2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *	
30	Vap. Reynolds		9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *	
31	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
32	Vapour Velocity	(m/s)	2.159	2.159	2.159	2.159	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	1.217e+004	1.217e+004	1.218e+004	1.219e+004	
36	Elevation	(m)	0.3000	0.3000	0.3000	0.3000	
37	Pressure	(bar)	91.76	91.75	91.75	91.75	
38	Temperature	(C)	56.55	56.55	56.55	56.54	
39	Vapour Fraction		0.8715	0.8715	0.8715	0.8715	
40	Heat Transferred	(kJ/h-m)	707.4	707.3	707.2	707.1	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.3033	0.3033	0.3033	0.3033	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	5.853e-003	5.853e-003	5.853e-003	5.853e-003	
46	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000	
47	Accel. Gradient	(kPa/m)	4.062e-007	4.062e-007	4.062e-007	4.061e-007	
48	Liq. Reynolds		2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *	
49	Vap. Reynolds		9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *	
50	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
51	Vapour Velocity	(m/s)	2.159	2.159	2.159	2.159	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	1.219e+004	1.220e+004	1.221e+004	1.221e+004	
55	Elevation	(m)	0.3000	0.2960	0.2920	0.2880	
56	Pressure	(bar)	91.75	91.75	91.75	91.75	
57	Temperature	(C)	56.54	56.54	56.53	56.53	
58	Vapour Fraction		0.8715	0.8715	0.8715	0.8715	
59	Heat Transferred	(kJ/h-m)	707.1	707.0	706.9	706.8	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.3033	0.2072	0.2072	0.2072	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	5.853e-003	6.827e-003	6.827e-003	6.826e-003	
65	Static Gradient	(kPa/m)	0.0000	-1.106e-003	-1.106e-003	-1.106e-003	
66	Accel. Gradient	(kPa/m)	4.061e-007	3.970e-007	3.970e-007	3.970e-007	
67	Liq. Reynolds		2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *	
68	Vap. Reynolds		9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
14	Vapour Velocity	(m/s)	2.159	2.159	2.159	2.159	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	1.222e+004	1.223e+004	1.223e+004	1.224e+004	
18	Elevation	(m)	0.2840	0.2800	0.2740	0.2680	
19	Pressure	(bar)	91.75	91.75	91.75	91.75	
20	Temperature	(C)	56.53	56.53	56.52	56.52	
21	Vapour Fraction		0.8715	0.8715	0.8715	0.8715	
22	Heat Transferred	(kJ/h-m)	706.8	706.7	706.6	706.6	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.2072	0.2072	0.2068	0.2068	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	6.826e-003	6.826e-003	6.829e-003	6.829e-003	
28	Static Gradient	(kPa/m)	-1.106e-003	-1.106e-003	-1.939e-003	-1.939e-003	
29	Accel. Gradient	(kPa/m)	3.970e-007	3.970e-007	3.394e-007	3.394e-007	
30	Liq. Reynolds		2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *	
31	Vap. Reynolds		9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *	
32	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
33	Vapour Velocity	(m/s)	2.159	2.159	2.159	2.159	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	1.225e+004	1.225e+004	1.226e+004	1.226e+004	
37	Elevation	(m)	0.2620	0.2560	0.2500	0.2220	
38	Pressure	(bar)	91.75	91.75	91.75	91.75	
39	Temperature	(C)	56.52	56.52	56.51	56.51	
40	Vapour Fraction		0.8715	0.8715	0.8715	0.8715	
41	Heat Transferred	(kJ/h-m)	706.5	706.4	706.4	706.3	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2068	0.2069	0.2069	0.2026	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	6.829e-003	6.829e-003	6.829e-003	6.864e-003	
47	Static Gradient	(kPa/m)	-1.939e-003	-1.939e-003	-1.939e-003	-1.068e-002	
48	Accel. Gradient	(kPa/m)	3.394e-007	3.394e-007	3.393e-007	-2.649e-007	
49	Liq. Reynolds		2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *	
50	Vap. Reynolds		9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *	
51	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
52	Vapour Velocity	(m/s)	2.159	2.159	2.159	2.159	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	1.227e+004	1.227e+004	1.228e+004	1.228e+004	
56	Elevation	(m)	0.1940	0.1660	0.1380	0.1100	
57	Pressure	(bar)	91.75	91.75	91.75	91.75	
58	Temperature	(C)	56.51	56.51	56.51	56.50	
59	Vapour Fraction		0.8715	0.8715	0.8715	0.8715	
60	Heat Transferred	(kJ/h-m)	706.3	706.2	706.2	706.1	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2026	0.2026	0.2026	0.2026	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	6.864e-003	6.864e-003	6.864e-003	6.863e-003	
66	Static Gradient	(kPa/m)	-1.068e-002	-1.068e-002	-1.068e-002	-1.068e-002	
67	Accel. Gradient	(kPa/m)	-2.649e-007	-2.650e-007	-2.650e-007	-2.650e-007	
68	Liq. Reynolds		2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *	
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *
14	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
15	Vapour Velocity (m/s)	2.159	2.159	2.159	2.159
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	1.229e+004	1.229e+004	1.230e+004	1.230e+004
19	Elevation (m)	0.1320	0.1540	0.1760	0.1980
20	Pressure (bar)	91.75	91.75	91.75	91.75
21	Temperature (C)	56.50	56.50	56.50	56.49
22	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
23	Heat Transferred (kJ/h-m)	706.0	706.0	705.9	705.9
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2944	0.2944	0.2944	0.2944
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	4.767e-003	4.767e-003	4.767e-003	4.767e-003
29	Static Gradient (kPa/m)	1.077e-002	1.077e-002	1.077e-002	1.077e-002
30	Accel. Gradient (kPa/m)	1.078e-006	1.078e-006	1.078e-006	1.078e-006
31	Liq. Reynolds	2.155e+005 *	2.154e+005 *	2.154e+005 *	2.154e+005 *
32	Vap. Reynolds	9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *
33	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
34	Vapour Velocity (m/s)	2.159	2.159	2.159	2.159
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	1.231e+004	1.231e+004	1.232e+004	1.232e+004
38	Elevation (m)	0.2200	0.2160	0.2120	0.2080
39	Pressure (bar)	91.75	91.75	91.75	91.75
40	Temperature (C)	56.49	56.49	56.49	56.49
41	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
42	Heat Transferred (kJ/h-m)	705.8	705.8	705.7	705.6
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2944	0.2071	0.2071	0.2071
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	4.767e-003	6.827e-003	6.827e-003	6.827e-003
48	Static Gradient (kPa/m)	1.077e-002	-1.547e-003	-1.547e-003	-1.547e-003
49	Accel. Gradient (kPa/m)	1.078e-006	3.664e-007	3.663e-007	3.663e-007
50	Liq. Reynolds	2.154e+005 *	2.154e+005 *	2.154e+005 *	2.154e+005 *
51	Vap. Reynolds	9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *
52	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
53	Vapour Velocity (m/s)	2.159	2.159	2.159	2.159
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1.233e+004	1.233e+004	1.234e+004	1.234e+004
57	Elevation (m)	0.2040	0.2000	0.2000	0.2000
58	Pressure (bar)	91.75	91.75	91.75	91.75
59	Temperature (C)	56.48	56.48	56.48	56.48
60	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
61	Heat Transferred (kJ/h-m)	705.6	705.5	705.5	705.4
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2071	0.2071	0.3034	0.3034
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	6.827e-003	6.827e-003	5.852e-003	5.852e-003
67	Static Gradient (kPa/m)	-1.547e-003	-1.547e-003	0.0000	0.0000
68	Accel. Gradient (kPa/m)	3.663e-007	3.663e-007	4.060e-007	4.060e-007
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.154e+005 *	2.154e+005 *	2.154e+005 *	2.154e+005 *
14	Vap. Reynolds	9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *
15	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
16	Vapour Velocity (m/s)	2.159	2.159	2.159	2.159
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	1.235e+004	1.235e+004	1.236e+004	1.237e+004
20	Elevation (m)	0.2000	0.2000	0.2000	0.2020
21	Pressure (bar)	91.75	91.74	91.74	91.74
22	Temperature (C)	56.48	56.47	56.47	56.47
23	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
24	Heat Transferred (kJ/h-m)	705.4	705.3	705.3	705.2
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.3034	0.3034	0.3034	0.2813
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	5.852e-003	5.852e-003	5.852e-003	5.512e-003
30	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	6.972e-004
31	Accel. Gradient (kPa/m)	4.060e-007	4.060e-007	4.060e-007	4.308e-007
32	Liq. Reynolds	2.154e+005 *	2.154e+005 *	2.154e+005 *	2.154e+005 *
33	Vap. Reynolds	9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *
34	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
35	Vapour Velocity (m/s)	2.159	2.159	2.159	2.159
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	1.237e+004	1.238e+004	1.239e+004	1.239e+004
39	Elevation (m)	0.2040	0.2060	0.2080	0.2100
40	Pressure (bar)	91.74	91.74	91.74	91.74
41	Temperature (C)	56.47	56.46	56.46	56.46
42	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
43	Heat Transferred (kJ/h-m)	705.1	705.0	704.9	704.9
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2813	0.2813	0.2813	0.2813
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	5.512e-003	5.512e-003	5.512e-003	5.512e-003
49	Static Gradient (kPa/m)	6.972e-004	6.972e-004	6.972e-004	6.972e-004
50	Accel. Gradient (kPa/m)	4.308e-007	4.308e-007	4.308e-007	4.308e-007
51	Liq. Reynolds	2.154e+005 *	2.154e+005 *	2.154e+005 *	2.154e+005 *
52	Vap. Reynolds	9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *
53	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
54	Vapour Velocity (m/s)	2.159	2.159	2.159	2.159
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	1.240e+004	1.240e+004	1.241e+004	1.241e+004
58	Elevation (m)	0.2240	0.2380	0.2520	0.2660
59	Pressure (bar)	91.74	91.74	91.74	91.74
60	Temperature (C)	56.46	56.45	56.45	56.45
61	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
62	Heat Transferred (kJ/h-m)	704.8	704.8	704.7	704.6
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2893	0.2893	0.2893	0.2893
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	4.911e-003	4.911e-003	4.911e-003	4.911e-003
68	Static Gradient (kPa/m)	6.772e-003	6.772e-003	6.772e-003	6.772e-003
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:02:44 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Accel. Gradient	(kPa/m)	8.106e-007	8.106e-007	8.106e-007	8.106e-007				
14	Liq. Reynolds		2.154e+005 *	2.154e+005 *	2.154e+005 *	2.154e+005 *				
15	Vap. Reynolds		9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *				
16	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043				
17	Vapour Velocity	(m/s)	2.159	2.159	2.159	2.159				
18	Deposit Thickness	(mm)	---	---	---	---				
19	Deposit Volume	(m3)	---	---	---	---				
20	Length	(m)	1.242e+004	1.242e+004	1.243e+004	1.243e+004				
21	Elevation	(m)	0.2800	0.2680	0.2560	0.2440				
22	Pressure	(bar)	91.74	91.74	91.74	91.74				
23	Temperature	(C)	56.45	56.44	56.44	56.44				
24	Vapour Fraction		0.8715	0.8715	0.8715	0.8715				
25	Heat Transferred	(kJ/h-m)	704.6	704.5	704.5	704.4				
26	Flow Regime		Transition	Transition	Transition	Transition				
27	Water Flow Regime									
28	Liquid HoldUp		0.2893	0.2056	0.2056	0.2056				
29	Water HoldUp		---	---	---	---				
30	Friction Gradient	(kPa/m)	4.911e-003	6.838e-003	6.838e-003	6.838e-003				
31	Static Gradient	(kPa/m)	6.772e-003	-4.621e-003	-4.621e-003	-4.621e-003				
32	Accel. Gradient	(kPa/m)	8.106e-007	1.538e-007	1.538e-007	1.538e-007				
33	Liq. Reynolds		2.154e+005 *	2.154e+005 *	2.154e+005 *	2.154e+005 *				
34	Vap. Reynolds		9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *				
35	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043				
36	Vapour Velocity	(m/s)	2.159	2.159	2.159	2.159				
37	Deposit Thickness	(mm)	---	---	---	---				
38	Deposit Volume	(m3)	---	---	---	---				
39	Length	(m)	1.244e+004	1.244e+004	1.245e+004	1.245e+004				
40	Elevation	(m)	0.2320	0.2200	0.2320	0.2440				
41	Pressure	(bar)	91.74	91.74	91.74	91.74				
42	Temperature	(C)	56.44	56.44	56.43	56.43				
43	Vapour Fraction		0.8715	0.8715	0.8715	0.8715				
44	Heat Transferred	(kJ/h-m)	704.4	704.3	704.3	704.2				
45	Flow Regime		Transition	Transition	Transition	Transition				
46	Water Flow Regime									
47	Liquid HoldUp		0.2056	0.2056	0.2880	0.2880				
48	Water HoldUp		---	---	---	---				
49	Friction Gradient	(kPa/m)	6.838e-003	6.838e-003	5.003e-003	5.003e-003				
50	Static Gradient	(kPa/m)	-4.621e-003	-4.621e-003	5.787e-003	5.787e-003				
51	Accel. Gradient	(kPa/m)	1.538e-007	1.538e-007	7.486e-007	7.486e-007				
52	Liq. Reynolds		2.154e+005 *	2.154e+005 *	2.154e+005 *	2.154e+005 *				
53	Vap. Reynolds		9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *				
54	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043				
55	Vapour Velocity	(m/s)	2.158	2.158	2.158	2.158				
56	Deposit Thickness	(mm)	---	---	---	---				
57	Deposit Volume	(m3)	---	---	---	---				
58	Length	(m)	1.246e+004	1.246e+004	1.247e+004	1.247e+004				
59	Elevation	(m)	0.2560	0.2680	0.2800	0.2660				
60	Pressure	(bar)	91.74	91.74	91.74	91.74				
61	Temperature	(C)	56.43	56.43	56.43	56.42				
62	Vapour Fraction		0.8715	0.8715	0.8715	0.8715				
63	Heat Transferred	(kJ/h-m)	704.1	704.1	704.0	704.0				
64	Flow Regime		Transition	Transition	Transition	Transition				
65	Water Flow Regime									
66	Liquid HoldUp		0.2880	0.2880	0.2880	0.2052				
67	Water HoldUp		---	---	---	---				
68	Friction Gradient	(kPa/m)	5.003e-003	5.003e-003	5.003e-003	6.841e-003				
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	5.787e-003	5.787e-003	5.787e-003	-5.385e-003
14	Accel. Gradient	(kPa/m)	7.486e-007	7.486e-007	7.486e-007	1.010e-007
15	Liq. Reynolds		2.154e+005 *	2.154e+005 *	2.154e+005 *	2.154e+005 *
16	Vap. Reynolds		9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *
17	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
18	Vapour Velocity	(m/s)	2.158	2.158	2.158	2.158
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	1.248e+004	1.248e+004	1.249e+004	1.249e+004
22	Elevation	(m)	0.2520	0.2380	0.2240	0.2100
23	Pressure	(bar)	91.74	91.74	91.74	91.74
24	Temperature	(C)	56.42	56.42	56.42	56.42
25	Vapour Fraction		0.8715	0.8715	0.8714	0.8714
26	Heat Transferred	(kJ/h-m)	703.9	703.9	703.8	703.8
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2052	0.2052	0.2052	0.2052
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	6.841e-003	6.841e-003	6.841e-003	6.841e-003
32	Static Gradient	(kPa/m)	-5.385e-003	-5.386e-003	-5.386e-003	-5.386e-003
33	Accel. Gradient	(kPa/m)	1.010e-007	1.010e-007	1.009e-007	1.009e-007
34	Liq. Reynolds		2.154e+005 *	2.154e+005 *	2.154e+005 *	2.154e+005 *
35	Vap. Reynolds		9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *
36	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
37	Vapour Velocity	(m/s)	2.158	2.158	2.158	2.158
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	1.250e+004	1.250e+004	1.251e+004	1.251e+004
41	Elevation	(m)	0.1220	3.400e-002	-5.400e-002	-0.1420
42	Pressure	(bar)	91.74	91.74	91.74	91.74
43	Temperature	(C)	56.41	56.41	56.41	56.41
44	Vapour Fraction		0.8714	0.8714	0.8714	0.8714
45	Heat Transferred	(kJ/h-m)	703.8	703.7	703.7	703.6
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.1914	0.1914	0.1914	0.1914
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	6.963e-003	6.962e-003	6.962e-003	6.962e-003
51	Static Gradient	(kPa/m)	-3.242e-002	-3.242e-002	-3.242e-002	-3.242e-002
52	Accel. Gradient	(kPa/m)	-1.766e-006	-1.766e-006	-1.766e-006	-1.766e-006
53	Liq. Reynolds		2.154e+005 *	2.154e+005 *	2.154e+005 *	2.154e+005 *
54	Vap. Reynolds		9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *
55	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
56	Vapour Velocity	(m/s)	2.158	2.158	2.158	2.158
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	1.252e+004	1.252e+004	1.253e+004	1.253e+004
60	Elevation	(m)	-0.2300	-0.2180	-0.2060	-0.1940
61	Pressure	(bar)	91.74	91.74	91.74	91.74
62	Temperature	(C)	56.41	56.41	56.40	56.40
63	Vapour Fraction		0.8714	0.8714	0.8714	0.8714
64	Heat Transferred	(kJ/h-m)	703.6	703.5	703.5	703.4
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.1914	0.2880	0.2881	0.2881
68	Water HoldUp		---	---	---	---
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1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	6.962e-003	5.002e-003	5.002e-003	5.002e-003
14	Static Gradient	(kPa/m)	-3.242e-002	5.787e-003	5.787e-003	5.788e-003
15	Accel. Gradient	(kPa/m)	-1.766e-006	7.484e-007	7.484e-007	7.484e-007
16	Liq. Reynolds		2.154e+005 *	2.153e+005 *	2.153e+005 *	2.153e+005 *
17	Vap. Reynolds		9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *
18	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
19	Vapour Velocity	(m/s)	2.158	2.158	2.158	2.158
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	1.254e+004	1.254e+004	1.255e+004	1.255e+004
23	Elevation	(m)	-0.1820	-0.1700	-0.1040	-3.800e-002
24	Pressure	(bar)	91.74	91.74	91.74	91.74
25	Temperature	(C)	56.40	56.40	56.40	56.39
26	Vapour Fraction		0.8714	0.8714	0.8714	0.8714
27	Heat Transferred	(kJ/h-m)	703.3	703.3	703.2	703.2
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2881	0.2881	0.3226	0.3226
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	5.002e-003	5.002e-003	6.030e-003	6.030e-003
33	Static Gradient	(kPa/m)	5.788e-003	5.788e-003	3.452e-002	3.452e-002
34	Accel. Gradient	(kPa/m)	7.484e-007	7.484e-007	2.812e-006	2.812e-006
35	Liq. Reynolds		2.153e+005 *	2.153e+005 *	2.153e+005 *	2.153e+005 *
36	Vap. Reynolds		9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *
37	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
38	Vapour Velocity	(m/s)	2.158	2.158	2.158	2.158
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	1.256e+004	1.256e+004	1.257e+004	1.257e+004
42	Elevation	(m)	2.800e-002	9.400e-002	0.1600	0.1740
43	Pressure	(bar)	91.73	91.73	91.73	91.73
44	Temperature	(C)	56.39	56.39	56.39	56.38
45	Vapour Fraction		0.8714	0.8714	0.8714	0.8714
46	Heat Transferred	(kJ/h-m)	703.1	703.0	703.0	702.9
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.3226	0.3226	0.3226	0.2893
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	6.030e-003	6.030e-003	6.030e-003	4.910e-003
52	Static Gradient	(kPa/m)	3.452e-002	3.452e-002	3.452e-002	6.773e-003
53	Accel. Gradient	(kPa/m)	2.813e-006	2.813e-006	2.813e-006	8.105e-007
54	Liq. Reynolds		2.153e+005 *	2.153e+005 *	2.153e+005 *	2.153e+005 *
55	Vap. Reynolds		9.083e+006 *	9.083e+006 *	9.083e+006 *	9.083e+006 *
56	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
57	Vapour Velocity	(m/s)	2.158	2.158	2.158	2.158
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	1.258e+004	1.258e+004	1.259e+004	1.259e+004
61	Elevation	(m)	0.1880	0.2020	0.2160	0.2300
62	Pressure	(bar)	91.73	91.73	91.73	91.73
63	Temperature	(C)	56.38	56.38	56.38	56.37
64	Vapour Fraction		0.8714	0.8714	0.8714	0.8714
65	Heat Transferred	(kJ/h-m)	702.8	702.8	702.7	702.7
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2893	0.2893	0.2893	0.2893
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Pipe Segment: ELEP-Akabel Pipeline (continued)		
3					
4					
5			Unit Set: Mixed		
6			Date/Time: Mon Jun 13 15:02:44 2022		
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	4.910e-003	4.910e-003	4.910e-003	4.910e-003
15	Static Gradient (kPa/m)	6.773e-003	6.773e-003	6.773e-003	6.773e-003
16	Accel. Gradient (kPa/m)	8.105e-007	8.105e-007	8.105e-007	8.105e-007
17	Liq. Reynolds	2.153e+005 *	2.153e+005 *	2.153e+005 *	2.153e+005 *
18	Vap. Reynolds	9.083e+006 *	9.083e+006 *	9.083e+006 *	9.084e+006 *
19	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
20	Vapour Velocity (m/s)	2.158	2.158	2.158	2.158
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	1.260e+004	1.260e+004	1.261e+004	1.261e+004
24	Elevation (m)	0.2720	0.3140	0.3560	0.3980
25	Pressure (bar)	91.73	91.72	91.72	91.72
26	Temperature (C)	56.37	56.37	56.37	56.36
27	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
28	Heat Transferred (kJ/h-m)	702.6	702.5	702.5	702.4
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.3072	0.3072	0.3072	0.3072
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	5.932e-003	5.932e-003	5.932e-003	5.932e-003
34	Static Gradient (kPa/m)	2.121e-002	2.121e-002	2.121e-002	2.121e-002
35	Accel. Gradient (kPa/m)	1.883e-006	1.883e-006	1.883e-006	1.883e-006
36	Liq. Reynolds	2.153e+005 *	2.153e+005 *	2.153e+005 *	2.153e+005 *
37	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
38	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
39	Vapour Velocity (m/s)	2.158	2.158	2.158	2.158
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	1.262e+004	1.262e+004	1.263e+004	1.264e+004
43	Elevation (m)	0.4400	0.4760	0.5120	0.5480
44	Pressure (bar)	91.72	91.72	91.72	91.72
45	Temperature (C)	56.36	56.36	56.36	56.35
46	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
47	Heat Transferred (kJ/h-m)	702.3	702.3	702.2	702.1
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.3072	0.2968	0.2968	0.2968
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	5.932e-003	5.397e-003	5.397e-003	5.397e-003
53	Static Gradient (kPa/m)	2.121e-002	1.263e-002	1.263e-002	1.263e-002
54	Accel. Gradient (kPa/m)	1.883e-006	1.251e-006	1.251e-006	1.251e-006
55	Liq. Reynolds	2.153e+005 *	2.153e+005 *	2.153e+005 *	2.153e+005 *
56	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
57	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
58	Vapour Velocity (m/s)	2.158	2.158	2.158	2.158
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	1.265e+004	1.265e+004	1.266e+004	1.266e+004
62	Elevation (m)	0.5840	0.6200	0.6160	0.6120
63	Pressure (bar)	91.71	91.71	91.71	91.71
64	Temperature (C)	56.35	56.35	56.34	56.34
65	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
66	Heat Transferred (kJ/h-m)	702.0	701.9	701.9	701.8
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 236 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.2968	0.2968	0.2071	0.2071
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	5.397e-003	5.397e-003	6.825e-003	6.825e-003
16	Static Gradient (kPa/m)	1.263e-002	1.263e-002	-1.548e-003	-1.548e-003
17	Accel. Gradient (kPa/m)	1.251e-006	1.251e-006	3.662e-007	3.662e-007
18	Liq. Reynolds	2.153e+005 *	2.153e+005 *	2.153e+005 *	2.153e+005 *
19	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
20	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
21	Vapour Velocity (m/s)	2.158	2.158	2.158	2.158
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	1.267e+004	1.267e+004	1.268e+004	1.268e+004
25	Elevation (m)	0.6080	0.6040	0.6000	0.5360
26	Pressure (bar)	91.71	91.71	91.71	91.71
27	Temperature (C)	56.34	56.34	56.34	56.33
28	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
29	Heat Transferred (kJ/h-m)	701.8	701.7	701.7	701.6
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2071	0.2071	0.2071	0.1959
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.825e-003	6.825e-003	6.825e-003	6.920e-003
35	Static Gradient (kPa/m)	-1.548e-003	-1.548e-003	-1.548e-003	-2.392e-002
36	Accel. Gradient (kPa/m)	3.662e-007	3.662e-007	3.662e-007	-1.179e-006
37	Liq. Reynolds	2.153e+005 *	2.152e+005 *	2.152e+005 *	2.152e+005 *
38	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
39	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
40	Vapour Velocity (m/s)	2.158	2.158	2.158	2.158
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	1.269e+004	1.269e+004	1.270e+004	1.270e+004
44	Elevation (m)	0.4720	0.4080	0.3440	0.2800
45	Pressure (bar)	91.71	91.71	91.72	91.72
46	Temperature (C)	56.33	56.33	56.33	56.33
47	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
48	Heat Transferred (kJ/h-m)	701.6	701.5	701.5	701.4
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.1959	0.1959	0.1959	0.1959
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.920e-003	6.919e-003	6.919e-003	6.919e-003
54	Static Gradient (kPa/m)	-2.392e-002	-2.392e-002	-2.392e-002	-2.392e-002
55	Accel. Gradient (kPa/m)	-1.179e-006	-1.179e-006	-1.179e-006	-1.179e-006
56	Liq. Reynolds	2.152e+005 *	2.152e+005 *	2.152e+005 *	2.152e+005 *
57	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
58	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
59	Vapour Velocity (m/s)	2.158	2.158	2.158	2.158
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	1.271e+004	1.271e+004	1.272e+004	1.272e+004
63	Elevation (m)	0.2760	0.2720	0.2680	0.2640
64	Pressure (bar)	91.72	91.72	91.72	91.72
65	Temperature (C)	56.33	56.32	56.32	56.32
66	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
67	Heat Transferred (kJ/h-m)	701.4	701.3	701.3	701.2
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 237 of 937

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2071	0.2071	0.2071	0.2071
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.824e-003	6.824e-003	6.824e-003	6.824e-003
17	Static Gradient (kPa/m)	-1.548e-003	-1.548e-003	-1.548e-003	-1.548e-003
18	Accel. Gradient (kPa/m)	3.660e-007	3.660e-007	3.660e-007	3.660e-007
19	Liq. Reynolds	2.152e+005 *	2.152e+005 *	2.152e+005 *	2.152e+005 *
20	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
21	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
22	Vapour Velocity (m/s)	2.158	2.158	2.158	2.158
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	1.273e+004	1.273e+004	1.274e+004	1.274e+004
26	Elevation (m)	0.2600	0.2580	0.2560	0.2540
27	Pressure (bar)	91.71	91.71	91.71	91.71
28	Temperature (C)	56.32	56.32	56.31	56.31
29	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
30	Heat Transferred (kJ/h-m)	701.2	701.1	701.1	701.0
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2071	0.2075	0.2075	0.2075
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	6.824e-003	6.821e-003	6.821e-003	6.821e-003
36	Static Gradient (kPa/m)	-1.548e-003	-7.748e-004	-7.748e-004	-7.748e-004
37	Accel. Gradient (kPa/m)	3.660e-007	4.194e-007	4.194e-007	4.194e-007
38	Liq. Reynolds	2.152e+005 *	2.153e+005 *	2.153e+005 *	2.153e+005 *
39	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
40	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
41	Vapour Velocity (m/s)	2.158	2.158	2.158	2.158
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	1.275e+004	1.275e+004	1.276e+004	1.276e+004
45	Elevation (m)	0.2520	0.2500	0.2060	0.1620
46	Pressure (bar)	91.71	91.71	91.71	91.71
47	Temperature (C)	56.31	56.31	56.31	56.30
48	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
49	Heat Transferred (kJ/h-m)	700.9	700.9	700.9	700.8
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2075	0.2075	0.1976	0.1976
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	6.821e-003	6.821e-003	6.903e-003	6.903e-003
55	Static Gradient (kPa/m)	-7.749e-004	-7.749e-004	-2.067e-002	-2.067e-002
56	Accel. Gradient (kPa/m)	4.194e-007	4.194e-007	-9.547e-007	-9.547e-007
57	Liq. Reynolds	2.153e+005 *	2.153e+005 *	2.153e+005 *	2.153e+005 *
58	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
59	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
60	Vapour Velocity (m/s)	2.158	2.158	2.158	2.158
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	1.276e+004	1.277e+004	1.277e+004	1.278e+004
64	Elevation (m)	0.1180	7.400e-002	3.000e-002	2.600e-002
65	Pressure (bar)	91.72	91.72	91.72	91.72
66	Temperature (C)	56.30	56.30	56.30	56.30
67	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
68	Heat Transferred (kJ/h-m)	700.8	700.8	700.7	700.7
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.1976	0.1976	0.1976	0.2066
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	6.903e-003	6.903e-003	6.903e-003	6.827e-003
18	Static Gradient (kPa/m)	-2.067e-002	-2.067e-002	-2.067e-002	-2.576e-003
19	Accel. Gradient (kPa/m)	-9.547e-007	-9.547e-007	-9.547e-007	2.949e-007
20	Liq. Reynolds	2.153e+005 *	2.153e+005 *	2.153e+005 *	2.153e+005 *
21	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
22	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
23	Vapour Velocity (m/s)	2.158	2.157	2.157	2.157
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	1.278e+004	1.278e+004	1.278e+004	1.279e+004
27	Elevation (m)	2.200e-002	1.800e-002	1.400e-002	1.000e-002
28	Pressure (bar)	91.72	91.72	91.72	91.72
29	Temperature (C)	56.30	56.30	56.30	56.29
30	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
31	Heat Transferred (kJ/h-m)	700.6	700.6	700.6	700.5
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2066	0.2066	0.2066	0.2066
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	6.827e-003	6.827e-003	6.827e-003	6.827e-003
37	Static Gradient (kPa/m)	-2.576e-003	-2.576e-003	-2.576e-003	-2.576e-003
38	Accel. Gradient (kPa/m)	2.949e-007	2.949e-007	2.949e-007	2.949e-007
39	Liq. Reynolds	2.153e+005 *	2.153e+005 *	2.153e+005 *	2.153e+005 *
40	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
41	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
42	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	1.279e+004	1.279e+004	1.280e+004	1.280e+004
46	Elevation (m)	4.800e-002	8.600e-002	0.1240	0.1620
47	Pressure (bar)	91.71	91.71	91.71	91.71
48	Temperature (C)	56.29	56.29	56.29	56.29
49	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
50	Heat Transferred (kJ/h-m)	700.5	700.5	700.4	700.4
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.3209	0.3209	0.3209	0.3209
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	6.025e-003	6.025e-003	6.025e-003	6.025e-003
56	Static Gradient (kPa/m)	3.300e-002	3.300e-002	3.300e-002	3.300e-002
57	Accel. Gradient (kPa/m)	2.707e-006	2.707e-006	2.707e-006	2.707e-006
58	Liq. Reynolds	2.153e+005 *	2.153e+005 *	2.153e+005 *	2.153e+005 *
59	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
60	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
61	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	1.280e+004	1.281e+004	1.281e+004	1.282e+004
65	Elevation (m)	0.2000	0.1280	5.600e-002	-1.600e-002
66	Pressure (bar)	91.71	91.71	91.71	91.71
67	Temperature (C)	56.29	56.29	56.28	56.28
68	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	700.3	700.3	700.3	700.2
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.3209	0.1944	0.1944	0.1944
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	6.025e-003	6.932e-003	6.932e-003	6.932e-003
19	Static Gradient (kPa/m)	3.300e-002	-2.679e-002	-2.679e-002	-2.679e-002
20	Accel. Gradient (kPa/m)	2.707e-006	-1.377e-006	-1.377e-006	-1.377e-006
21	Liq. Reynolds	2.153e+005 *	2.154e+005 *	2.154e+005 *	2.154e+005 *
22	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
23	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
24	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	1.282e+004	1.283e+004	1.283e+004	1.284e+004
28	Elevation (m)	-8.800e-002	-0.1600	-9.200e-002	-2.400e-002
29	Pressure (bar)	91.71	91.71	91.71	91.71
30	Temperature (C)	56.28	56.28	56.28	56.27
31	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
32	Heat Transferred (kJ/h-m)	700.2	700.1	700.1	700.0
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.1944	0.1944	0.3237	0.3237
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	6.931e-003	6.931e-003	6.030e-003	6.030e-003
38	Static Gradient (kPa/m)	-2.679e-002	-2.679e-002	3.544e-002	3.544e-002
39	Accel. Gradient (kPa/m)	-1.377e-006	-1.377e-006	2.876e-006	2.876e-006
40	Liq. Reynolds	2.154e+005 *	2.154e+005 *	2.154e+005 *	2.154e+005 *
41	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
42	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
43	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	1.284e+004	1.285e+004	1.285e+004	1.286e+004
47	Elevation (m)	4.400e-002	0.1120	0.1800	0.1840
48	Pressure (bar)	91.71	91.71	91.70	91.70
49	Temperature (C)	56.27	56.27	56.27	56.26
50	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
51	Heat Transferred (kJ/h-m)	699.9	699.9	699.8	699.7
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.3237	0.3237	0.3237	0.2830
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	6.030e-003	6.030e-003	6.030e-003	5.383e-003
57	Static Gradient (kPa/m)	3.544e-002	3.544e-002	3.544e-002	1.906e-003
58	Accel. Gradient (kPa/m)	2.876e-006	2.877e-006	2.877e-006	5.056e-007
59	Liq. Reynolds	2.154e+005 *	2.154e+005 *	2.154e+005 *	2.154e+005 *
60	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
61	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
62	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	1.286e+004	1.287e+004	1.287e+004	1.288e+004
66	Elevation (m)	0.1880	0.1920	0.1960	0.2000
67	Pressure (bar)	91.70	91.70	91.70	91.70
68	Temperature (C)	56.26	56.26	56.26	56.26
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
14	Heat Transferred (kJ/h-m)	699.7	699.6	699.5	699.5
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2830	0.2830	0.2830	0.2830
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	5.383e-003	5.383e-003	5.383e-003	5.383e-003
20	Static Gradient (kPa/m)	1.906e-003	1.906e-003	1.906e-003	1.906e-003
21	Accel. Gradient (kPa/m)	5.056e-007	5.056e-007	5.056e-007	5.056e-007
22	Liq. Reynolds	2.154e+005 *	2.154e+005 *	2.154e+005 *	2.154e+005 *
23	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
24	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
25	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	1.288e+004	1.289e+004	1.289e+004	1.290e+004
29	Elevation (m)	0.2040	0.2080	0.2120	0.2160
30	Pressure (bar)	91.70	91.70	91.70	91.70
31	Temperature (C)	56.25	56.25	56.25	56.25
32	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
33	Heat Transferred (kJ/h-m)	699.4	699.4	699.3	699.3
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2830	0.2830	0.2830	0.2830
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	5.382e-003	5.382e-003	5.382e-003	5.382e-003
39	Static Gradient (kPa/m)	1.906e-003	1.906e-003	1.906e-003	1.906e-003
40	Accel. Gradient (kPa/m)	5.056e-007	5.055e-007	5.055e-007	5.055e-007
41	Liq. Reynolds	2.154e+005 *	2.154e+005 *	2.154e+005 *	2.154e+005 *
42	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
43	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
44	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	1.290e+004	1.291e+004	1.291e+004	1.292e+004
48	Elevation (m)	0.2200	0.2240	0.2280	0.2320
49	Pressure (bar)	91.70	91.70	91.70	91.70
50	Temperature (C)	56.25	56.24	56.24	56.24
51	Vapour Fraction	0.8714	0.8714	0.8714	0.8713
52	Heat Transferred (kJ/h-m)	699.2	699.2	699.1	699.0
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2830	0.2830	0.2830	0.2830
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	5.382e-003	5.382e-003	5.382e-003	5.382e-003
58	Static Gradient (kPa/m)	1.906e-003	1.906e-003	1.906e-003	1.906e-003
59	Accel. Gradient (kPa/m)	5.055e-007	5.055e-007	5.055e-007	5.055e-007
60	Liq. Reynolds	2.154e+005 *	2.154e+005 *	2.154e+005 *	2.154e+005 *
61	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
62	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
63	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	1.292e+004	1.293e+004	1.293e+004	1.294e+004
67	Elevation (m)	0.2360	0.2400	0.2420	0.2440
68	Pressure (bar)	91.70	91.70	91.70	91.70
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022	
6						
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Temperature (C)	56.24	56.23	56.23	56.23	
14	Vapour Fraction	0.8713	0.8713	0.8713	0.8713	
15	Heat Transferred (kJ/h-m)	699.0	698.9	698.9	698.8	
16	Flow Regime	Transition	Transition	Transition	Transition	
17	Water Flow Regime					
18	Liquid HoldUp	0.2830	0.2830	0.2817	0.2817	
19	Water HoldUp	---	---	---	---	
20	Friction Gradient (kPa/m)	5.382e-003	5.382e-003	5.481e-003	5.481e-003	
21	Static Gradient (kPa/m)	1.906e-003	1.906e-003	9.499e-004	9.499e-004	
22	Accel. Gradient (kPa/m)	5.055e-007	5.055e-007	4.461e-007	4.461e-007	
23	Liq. Reynolds	2.154e+005 *	2.154e+005 *	2.154e+005 *	2.154e+005 *	
24	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *	
25	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043	
26	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157	
27	Deposit Thickness (mm)	---	---	---	---	
28	Deposit Volume (m3)	---	---	---	---	
29	Length (m)	1.294e+004	1.295e+004	1.295e+004	1.296e+004	
30	Elevation (m)	0.2460	0.2480	0.2500	0.2440	
31	Pressure (bar)	91.70	91.70	91.70	91.70	
32	Temperature (C)	56.23	56.23	56.22	56.22	
33	Vapour Fraction	0.8713	0.8713	0.8713	0.8713	
34	Heat Transferred (kJ/h-m)	698.8	698.7	698.7	698.6	
35	Flow Regime	Transition	Transition	Transition	Transition	
36	Water Flow Regime					
37	Liquid HoldUp	0.2817	0.2817	0.2817	0.2068	
38	Water HoldUp	---	---	---	---	
39	Friction Gradient (kPa/m)	5.481e-003	5.481e-003	5.481e-003	6.825e-003	
40	Static Gradient (kPa/m)	9.499e-004	9.499e-004	9.499e-004	-2.320e-003	
41	Accel. Gradient (kPa/m)	4.461e-007	4.461e-007	4.460e-007	3.125e-007	
42	Liq. Reynolds	2.154e+005 *	2.154e+005 *	2.155e+005 *	2.155e+005 *	
43	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *	
44	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043	
45	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157	
46	Deposit Thickness (mm)	---	---	---	---	
47	Deposit Volume (m3)	---	---	---	---	
48	Length (m)	1.296e+004	1.297e+004	1.297e+004	1.298e+004	
49	Elevation (m)	0.2380	0.2320	0.2260	0.2200	
50	Pressure (bar)	91.70	91.70	91.70	91.70	
51	Temperature (C)	56.22	56.22	56.22	56.21	
52	Vapour Fraction	0.8713	0.8713	0.8713	0.8713	
53	Heat Transferred (kJ/h-m)	698.6	698.5	698.5	698.4	
54	Flow Regime	Transition	Transition	Transition	Transition	
55	Water Flow Regime					
56	Liquid HoldUp	0.2068	0.2068	0.2068	0.2068	
57	Water HoldUp	---	---	---	---	
58	Friction Gradient (kPa/m)	6.825e-003	6.825e-003	6.825e-003	6.825e-003	
59	Static Gradient (kPa/m)	-2.320e-003	-2.320e-003	-2.320e-003	-2.320e-003	
60	Accel. Gradient (kPa/m)	3.125e-007	3.125e-007	3.125e-007	3.125e-007	
61	Liq. Reynolds	2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *	
62	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *	
63	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043	
64	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157	
65	Deposit Thickness (mm)	---	---	---	---	
66	Deposit Volume (m3)	---	---	---	---	
67	Length (m)	1.298e+004	1.299e+004	1.299e+004	1.300e+004	
68	Elevation (m)	0.2300	0.2400	0.2500	0.2600	
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Pressure (bar)	91.70	91.70	91.69	91.69
14	Temperature (C)	56.21	56.21	56.21	56.21
15	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
16	Heat Transferred (kJ/h-m)	698.3	698.3	698.2	698.2
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.2869	0.2869	0.2869	0.2869
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	5.093e-003	5.092e-003	5.092e-003	5.092e-003
22	Static Gradient (kPa/m)	4.810e-003	4.810e-003	4.810e-003	4.810e-003
23	Accel. Gradient (kPa/m)	6.868e-007	6.868e-007	6.868e-007	6.868e-007
24	Liq. Reynolds	2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *
25	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
26	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
27	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	1.300e+004	1.301e+004	1.301e+004	1.302e+004
31	Elevation (m)	0.2700	0.2720	0.2740	0.2760
32	Pressure (bar)	91.69	91.69	91.69	91.69
33	Temperature (C)	56.20	56.20	56.20	56.20
34	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
35	Heat Transferred (kJ/h-m)	698.1	698.1	698.0	697.9
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.2869	0.2818	0.2818	0.2818
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	5.092e-003	5.480e-003	5.480e-003	5.480e-003
41	Static Gradient (kPa/m)	4.810e-003	9.499e-004	9.499e-004	9.499e-004
42	Accel. Gradient (kPa/m)	6.868e-007	4.460e-007	4.460e-007	4.460e-007
43	Liq. Reynolds	2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *
44	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
45	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
46	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	1.302e+004	1.303e+004	1.303e+004	1.304e+004
50	Elevation (m)	0.2780	0.2800	0.2820	0.2840
51	Pressure (bar)	91.69	91.69	91.69	91.69
52	Temperature (C)	56.20	56.19	56.19	56.19
53	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
54	Heat Transferred (kJ/h-m)	697.9	697.8	697.8	697.7
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.2818	0.2818	0.2818	0.2818
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	5.480e-003	5.480e-003	5.480e-003	5.480e-003
60	Static Gradient (kPa/m)	9.500e-004	9.500e-004	9.439e-004	9.439e-004
61	Accel. Gradient (kPa/m)	4.460e-007	4.460e-007	4.456e-007	4.456e-007
62	Liq. Reynolds	2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *
63	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
64	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
65	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	1.304e+004	1.305e+004	1.305e+004	1.306e+004
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Elevation (m)	0.2860	0.2880	0.2900	0.2920
14	Pressure (bar)	91.69	91.69	91.69	91.69
15	Temperature (C)	56.19	56.19	56.18	56.18
16	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
17	Heat Transferred (kJ/h-m)	697.7	697.6	697.6	697.5
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2818	0.2818	0.2818	0.2818
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	5.480e-003	5.480e-003	5.480e-003	5.480e-003
23	Static Gradient (kPa/m)	9.439e-004	9.439e-004	9.439e-004	9.500e-004
24	Accel. Gradient (kPa/m)	4.456e-007	4.456e-007	4.456e-007	4.459e-007
25	Liq. Reynolds	2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *
26	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
27	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
28	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	1.306e+004	1.307e+004	1.307e+004	1.308e+004
32	Elevation (m)	0.2940	0.2960	0.2980	0.3000
33	Pressure (bar)	91.69	91.69	91.69	91.69
34	Temperature (C)	56.18	56.18	56.18	56.17
35	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
36	Heat Transferred (kJ/h-m)	697.5	697.4	697.3	697.3
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2818	0.2818	0.2818	0.2818
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	5.480e-003	5.479e-003	5.479e-003	5.479e-003
42	Static Gradient (kPa/m)	9.500e-004	9.500e-004	9.500e-004	9.500e-004
43	Accel. Gradient (kPa/m)	4.459e-007	4.459e-007	4.459e-007	4.459e-007
44	Liq. Reynolds	2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *
45	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
46	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
47	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	1.308e+004	1.309e+004	1.309e+004	1.310e+004
51	Elevation (m)	0.2940	0.2880	0.2820	0.2760
52	Pressure (bar)	91.69	91.69	91.69	91.69
53	Temperature (C)	56.17	56.17	56.17	56.16
54	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
55	Heat Transferred (kJ/h-m)	697.2	697.2	697.1	697.1
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2068	0.2068	0.2068	0.2068
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	6.824e-003	6.824e-003	6.824e-003	6.824e-003
61	Static Gradient (kPa/m)	-2.320e-003	-2.320e-003	-2.320e-003	-2.320e-003
62	Accel. Gradient (kPa/m)	3.124e-007	3.124e-007	3.124e-007	3.124e-007
63	Liq. Reynolds	2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *
64	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
65	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
66	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:02:44 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Length (m)	1.310e+004	1.311e+004	1.311e+004	1.312e+004					
14	Elevation (m)	0.2700	0.2820	0.2940	0.3060					
15	Pressure (bar)	91.69	91.69	91.69	91.69					
16	Temperature (C)	56.16	56.16	56.16	56.16					
17	Vapour Fraction	0.8713	0.8713	0.8713	0.8713					
18	Heat Transferred (kJ/h-m)	697.0	697.0	696.9	696.9					
19	Flow Regime	Transition	Transition	Transition	Transition					
20	Water Flow Regime									
21	Liquid HoldUp	0.2068	0.2882	0.2882	0.2882					
22	Water HoldUp	---	---	---	---					
23	Friction Gradient (kPa/m)	6.824e-003	4.998e-003	4.998e-003	4.998e-003					
24	Static Gradient (kPa/m)	-2.320e-003	5.791e-003	5.791e-003	5.791e-003					
25	Accel. Gradient (kPa/m)	3.124e-007	7.482e-007	7.482e-007	7.482e-007					
26	Liq. Reynolds	2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *					
27	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *					
28	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043					
29	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157					
30	Deposit Thickness (mm)	---	---	---	---					
31	Deposit Volume (m3)	---	---	---	---					
32	Length (m)	1.312e+004	1.313e+004	1.313e+004	1.314e+004					
33	Elevation (m)	0.3180	0.3300	0.3320	0.3340					
34	Pressure (bar)	91.69	91.68	91.68	91.68					
35	Temperature (C)	56.15	56.15	56.15	56.15					
36	Vapour Fraction	0.8713	0.8713	0.8713	0.8713					
37	Heat Transferred (kJ/h-m)	696.8	696.7	696.7	696.6					
38	Flow Regime	Transition	Transition	Transition	Transition					
39	Water Flow Regime									
40	Liquid HoldUp	0.2882	0.2882	0.2818	0.2818					
41	Water HoldUp	---	---	---	---					
42	Friction Gradient (kPa/m)	4.998e-003	4.997e-003	5.479e-003	5.479e-003					
43	Static Gradient (kPa/m)	5.791e-003	5.791e-003	9.501e-004	9.501e-004					
44	Accel. Gradient (kPa/m)	7.482e-007	7.482e-007	4.459e-007	4.459e-007					
45	Liq. Reynolds	2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *					
46	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *					
47	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043					
48	Vapour Velocity (m/s)	2.157	2.157	2.157	2.156					
49	Deposit Thickness (mm)	---	---	---	---					
50	Deposit Volume (m3)	---	---	---	---					
51	Length (m)	1.314e+004	1.315e+004	1.315e+004	1.316e+004					
52	Elevation (m)	0.3360	0.3380	0.3400	0.3420					
53	Pressure (bar)	91.68	91.68	91.68	91.68					
54	Temperature (C)	56.15	56.14	56.14	56.14					
55	Vapour Fraction	0.8713	0.8713	0.8713	0.8713					
56	Heat Transferred (kJ/h-m)	696.6	696.5	696.5	696.4					
57	Flow Regime	Transition	Transition	Transition	Transition					
58	Water Flow Regime									
59	Liquid HoldUp	0.2818	0.2818	0.2818	0.2818					
60	Water HoldUp	---	---	---	---					
61	Friction Gradient (kPa/m)	5.479e-003	5.479e-003	5.479e-003	5.479e-003					
62	Static Gradient (kPa/m)	9.501e-004	9.501e-004	9.501e-004	9.501e-004					
63	Accel. Gradient (kPa/m)	4.459e-007	4.459e-007	4.459e-007	4.459e-007					
64	Liq. Reynolds	2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *					
65	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *					
66	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043					
67	Vapour Velocity (m/s)	2.156	2.156	2.156	2.156					
68	Deposit Thickness (mm)	---	---	---	---					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---	---	
14	Length (m)	1.316e+004	1.317e+004	1.317e+004	1.318e+004	1.318e+004	
15	Elevation (m)	0.3440	0.3460	0.3480	0.3500	0.3500	
16	Pressure (bar)	91.68	91.68	91.68	91.68	91.68	
17	Temperature (C)	56.14	56.14	56.13	56.13	56.13	
18	Vapour Fraction	0.8713	0.8713	0.8713	0.8713	0.8713	
19	Heat Transferred (kJ/h-m)	696.4	696.3	696.2	696.2	696.2	
20	Flow Regime	Transition	Transition	Transition	Transition	Transition	
21	Water Flow Regime						
22	Liquid HoldUp	0.2818	0.2818	0.2818	0.2818	0.2818	
23	Water HoldUp	---	---	---	---	---	
24	Friction Gradient (kPa/m)	5.479e-003	5.478e-003	5.478e-003	5.478e-003	5.478e-003	
25	Static Gradient (kPa/m)	9.501e-004	9.501e-004	9.501e-004	9.501e-004	9.501e-004	
26	Accel. Gradient (kPa/m)	4.458e-007	4.458e-007	4.458e-007	4.458e-007	4.458e-007	
27	Liq. Reynolds	2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *	
28	Vap. Reynolds	9.085e+006 *	9.085e+006 *	9.085e+006 *	9.085e+006 *	9.085e+006 *	
29	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043	0.2043	
30	Vapour Velocity (m/s)	2.156	2.156	2.156	2.156	2.156	
31	Deposit Thickness (mm)	---	---	---	---	---	
32	Deposit Volume (m3)	---	---	---	---	---	
33	Length (m)	1.318e+004	1.319e+004	1.319e+004	1.320e+004	1.320e+004	
34	Elevation (m)	0.3480	0.3460	0.3440	0.3420	0.3420	
35	Pressure (bar)	91.68	91.68	91.68	91.68	91.68	
36	Temperature (C)	56.13	56.13	56.13	56.12	56.12	
37	Vapour Fraction	0.8713	0.8713	0.8713	0.8713	0.8713	
38	Heat Transferred (kJ/h-m)	696.1	696.1	696.0	696.0	696.0	
39	Flow Regime	Transition	Transition	Transition	Transition	Transition	
40	Water Flow Regime						
41	Liquid HoldUp	0.2076	0.2076	0.2076	0.2076	0.2076	
42	Water HoldUp	---	---	---	---	---	
43	Friction Gradient (kPa/m)	6.817e-003	6.817e-003	6.817e-003	6.817e-003	6.817e-003	
44	Static Gradient (kPa/m)	-7.752e-004	-7.752e-004	-7.752e-004	-7.752e-004	-7.752e-004	
45	Accel. Gradient (kPa/m)	4.190e-007	4.190e-007	4.190e-007	4.190e-007	4.190e-007	
46	Liq. Reynolds	2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *	
47	Vap. Reynolds	9.085e+006 *	9.085e+006 *	9.085e+006 *	9.085e+006 *	9.085e+006 *	
48	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043	0.2043	
49	Vapour Velocity (m/s)	2.156	2.156	2.156	2.156	2.156	
50	Deposit Thickness (mm)	---	---	---	---	---	
51	Deposit Volume (m3)	---	---	---	---	---	
52	Length (m)	1.320e+004	1.321e+004	1.321e+004	1.322e+004	1.322e+004	
53	Elevation (m)	0.3400	0.1100	-0.1200	-0.3500	-0.3500	
54	Pressure (bar)	91.68	91.68	91.69	91.69	91.69	
55	Temperature (C)	56.12	56.12	56.12	56.12	56.12	
56	Vapour Fraction	0.8713	0.8713	0.8713	0.8713	0.8713	
57	Heat Transferred (kJ/h-m)	695.9	696.0	696.0	695.9	695.9	
58	Flow Regime	Transition	Transition	Transition	Transition	Transition	
59	Water Flow Regime						
60	Liquid HoldUp	0.2076	0.1651	0.1651	0.1651	0.1651	
61	Water HoldUp	---	---	---	---	---	
62	Friction Gradient (kPa/m)	6.817e-003	7.270e-003	7.270e-003	7.270e-003	7.270e-003	
63	Static Gradient (kPa/m)	-7.752e-004	-7.763e-002	-7.763e-002	-7.763e-002	-7.763e-002	
64	Accel. Gradient (kPa/m)	4.190e-007	-4.879e-006	-4.879e-006	-4.879e-006	-4.879e-006	
65	Liq. Reynolds	2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *	2.155e+005 *	
66	Vap. Reynolds	9.085e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *	
67	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043	0.2043	
68	Vapour Velocity (m/s)	2.156	2.156	2.156	2.156	2.156	
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	1.322e+004	1.323e+004	1.323e+004	1.324e+004
16	Elevation (m)	-0.5800	-0.8100	-0.9360	-1.062
17	Pressure (bar)	91.69	91.70	91.70	91.70
18	Temperature (C)	56.12	56.12	56.12	56.12
19	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
20	Heat Transferred (kJ/h-m)	695.9	695.9	695.9	695.8
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.1651	0.1651	0.1846	0.1846
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	7.269e-003	7.269e-003	7.026e-003	7.026e-003
26	Static Gradient (kPa/m)	-7.764e-002	-7.764e-002	-4.517e-002	-4.517e-002
27	Accel. Gradient (kPa/m)	-4.878e-006	-4.878e-006	-2.644e-006	-2.644e-006
28	Liq. Reynolds	2.155e+005 *	2.156e+005 *	2.156e+005 *	2.156e+005 *
29	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.084e+006 *
30	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
31	Vapour Velocity (m/s)	2.156	2.156	2.156	2.156
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	1.324e+004	1.325e+004	1.325e+004	1.326e+004
35	Elevation (m)	-1.188	-1.314	-1.440	-2.192
36	Pressure (bar)	91.70	91.71	91.71	91.72
37	Temperature (C)	56.12	56.11	56.11	56.11
38	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
39	Heat Transferred (kJ/h-m)	695.8	695.8	695.7	696.2
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.1846	0.1846	0.1846	7.185e-002
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	7.026e-003	7.026e-003	7.026e-003	1.118e-002
45	Static Gradient (kPa/m)	-4.517e-002	-4.518e-002	-4.518e-002	-0.1712
46	Accel. Gradient (kPa/m)	-2.644e-006	-2.644e-006	-2.644e-006	-1.109e-005
47	Liq. Reynolds	2.156e+005 *	2.156e+005 *	2.156e+005 *	2.156e+005 *
48	Vap. Reynolds	9.084e+006 *	9.084e+006 *	9.084e+006 *	9.083e+006 *
49	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
50	Vapour Velocity (m/s)	2.156	2.156	2.155	2.155
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	1.326e+004	1.327e+004	1.327e+004	1.328e+004
54	Elevation (m)	-2.944	-3.696	-4.448	-5.200
55	Pressure (bar)	91.72	91.73	91.74	91.75
56	Temperature (C)	56.12	56.12	56.12	56.12
57	Vapour Fraction	0.8713	0.8712	0.8712	0.8712
58	Heat Transferred (kJ/h-m)	696.3	696.3	696.4	696.4
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	7.185e-002	7.186e-002	7.186e-002	7.186e-002
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	1.118e-002	1.117e-002	1.117e-002	1.117e-002
64	Static Gradient (kPa/m)	-0.1712	-0.1713	-0.1713	-0.1713
65	Accel. Gradient (kPa/m)	-1.109e-005	-1.109e-005	-1.109e-005	-1.109e-005
66	Liq. Reynolds	2.156e+005 *	2.156e+005 *	2.156e+005 *	2.156e+005 *
67	Vap. Reynolds	9.083e+006 *	9.083e+006 *	9.083e+006 *	9.082e+006 *
68	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2044
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022		
6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.155	2.155	2.155	2.154	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	1.328e+004	1.329e+004	1.329e+004	1.330e+004	
17	Elevation	(m)	-5.420	-5.640	-5.860	-6.080	
18	Pressure	(bar)	91.75	91.75	91.76	91.76	
19	Temperature	(C)	56.12	56.12	56.12	56.12	
20	Vapour Fraction		0.8712	0.8712	0.8712	0.8712	
21	Heat Transferred	(kJ/h-m)	696.0	696.0	696.0	696.0	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.1670	0.1670	0.1670	0.1670	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	7.238e-003	7.238e-003	7.238e-003	7.238e-003	
27	Static Gradient	(kPa/m)	-7.477e-002	-7.478e-002	-7.478e-002	-7.478e-002	
28	Accel. Gradient	(kPa/m)	-4.676e-006	-4.676e-006	-4.675e-006	-4.675e-006	
29	Liq. Reynolds		2.156e+005 *	2.156e+005 *	2.156e+005 *	2.156e+005 *	
30	Vap. Reynolds		9.082e+006 *	9.082e+006 *	9.082e+006 *	9.082e+006 *	
31	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
32	Vapour Velocity	(m/s)	2.154	2.154	2.154	2.154	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	1.330e+004	1.331e+004	1.331e+004	1.332e+004	
36	Elevation	(m)	-6.300	-6.600	-6.900	-7.200	
37	Pressure	(bar)	91.76	91.77	91.77	91.78	
38	Temperature	(C)	56.12	56.12	56.12	56.12	
39	Vapour Fraction		0.8712	0.8712	0.8712	0.8712	
40	Heat Transferred	(kJ/h-m)	695.9	696.0	696.0	696.0	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.1670	0.1522	0.1522	0.1523	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	7.237e-003	7.469e-003	7.469e-003	7.469e-003	
46	Static Gradient	(kPa/m)	-7.478e-002	-9.675e-002	-9.675e-002	-9.676e-002	
47	Accel. Gradient	(kPa/m)	-4.675e-006	-6.179e-006	-6.178e-006	-6.178e-006	
48	Liq. Reynolds		2.156e+005 *	2.156e+005 *	2.156e+005 *	2.156e+005 *	
49	Vap. Reynolds		9.082e+006 *	9.082e+006 *	9.082e+006 *	9.081e+006 *	
50	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
51	Vapour Velocity	(m/s)	2.154	2.154	2.154	2.154	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	1.332e+004	1.333e+004	1.333e+004	1.334e+004	
55	Elevation	(m)	-7.500	-7.800	-8.500	-9.200	
56	Pressure	(bar)	91.78	91.79	91.79	91.80	
57	Temperature	(C)	56.12	56.12	56.12	56.12	
58	Vapour Fraction		0.8712	0.8712	0.8712	0.8712	
59	Heat Transferred	(kJ/h-m)	696.0	696.0	696.3	696.4	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.1523	0.1523	8.075e-002	8.075e-002	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	7.468e-003	7.468e-003	1.033e-002	1.033e-002	
65	Static Gradient	(kPa/m)	-9.676e-002	-9.676e-002	-0.1668	-0.1668	
66	Accel. Gradient	(kPa/m)	-6.178e-006	-6.177e-006	-1.082e-005	-1.082e-005	
67	Liq. Reynolds		2.156e+005 *	2.156e+005 *	2.157e+005 *	2.157e+005 *	
68	Vap. Reynolds		9.081e+006 *	9.081e+006 *	9.081e+006 *	9.081e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
14	Vapour Velocity	(m/s)	2.153	2.153	2.153	2.153	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	1.334e+004	1.335e+004	1.335e+004	1.336e+004	
18	Elevation	(m)	-9.900	-10.60	-11.30	-11.75	
19	Pressure	(bar)	91.81	91.82	91.83	91.83	
20	Temperature	(C)	56.13	56.13	56.13	56.13	
21	Vapour Fraction		0.8712	0.8712	0.8712	0.8712	
22	Heat Transferred	(kJ/h-m)	696.4	696.5	696.5	696.3	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		8.075e-002	8.076e-002	8.076e-002	0.1249	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	1.033e-002	1.033e-002	1.033e-002	8.089e-003	
28	Static Gradient	(kPa/m)	-0.1668	-0.1668	-0.1668	-0.1307	
29	Accel. Gradient	(kPa/m)	-1.082e-005	-1.082e-005	-1.082e-005	-8.471e-006	
30	Liq. Reynolds		2.157e+005 *	2.157e+005 *	2.157e+005 *	2.157e+005 *	
31	Vap. Reynolds		9.080e+006 *	9.080e+006 *	9.080e+006 *	9.080e+006 *	
32	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
33	Vapour Velocity	(m/s)	2.153	2.153	2.152	2.152	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	1.336e+004	1.337e+004	1.337e+004	1.338e+004	
37	Elevation	(m)	-12.20	-12.65	-13.10	-13.55	
38	Pressure	(bar)	91.84	91.84	91.85	91.86	
39	Temperature	(C)	56.13	56.13	56.13	56.13	
40	Vapour Fraction		0.8712	0.8712	0.8712	0.8712	
41	Heat Transferred	(kJ/h-m)	696.3	696.4	696.4	696.4	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.1249	0.1249	0.1249	0.1249	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	8.089e-003	8.088e-003	8.088e-003	8.087e-003	
47	Static Gradient	(kPa/m)	-0.1307	-0.1307	-0.1307	-0.1307	
48	Accel. Gradient	(kPa/m)	-8.470e-006	-8.469e-006	-8.469e-006	-8.468e-006	
49	Liq. Reynolds		2.157e+005 *	2.157e+005 *	2.157e+005 *	2.157e+005 *	
50	Vap. Reynolds		9.080e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *	
51	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
52	Vapour Velocity	(m/s)	2.152	2.152	2.152	2.152	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	1.338e+004	1.339e+004	1.339e+004	1.340e+004	
56	Elevation	(m)	-13.55	-13.55	-13.55	-13.55	
57	Pressure	(bar)	91.86	91.86	91.86	91.86	
58	Temperature	(C)	56.13	56.13	56.13	56.12	
59	Vapour Fraction		0.8712	0.8712	0.8712	0.8712	
60	Heat Transferred	(kJ/h-m)	696.2	696.1	696.0	696.0	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.3040	0.3040	0.3040	0.3040	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	5.838e-003	5.838e-003	5.838e-003	5.838e-003	
66	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000	
67	Accel. Gradient	(kPa/m)	4.032e-007	4.032e-007	4.032e-007	4.032e-007	
68	Liq. Reynolds		2.157e+005 *	2.157e+005 *	2.157e+005 *	2.157e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.079e+006 *	9.079e+006 *	9.079e+006 *	9.079e+006 *
14	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
15	Vapour Velocity (m/s)	2.152	2.152	2.152	2.151
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	1.340e+004	1.341e+004	1.341e+004	1.342e+004
19	Elevation (m)	-13.55	-13.10	-12.65	-12.20
20	Pressure (bar)	91.85	91.84	91.82	91.80
21	Temperature (C)	56.12	56.12	56.11	56.10
22	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
23	Heat Transferred (kJ/h-m)	695.9	696.8	696.6	696.4
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.3040	0.5656	0.5656	0.5655
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	5.838e-003	5.754e-003	5.755e-003	5.756e-003
29	Static Gradient (kPa/m)	0.0000	0.3643	0.3643	0.3642
30	Accel. Gradient (kPa/m)	4.032e-007	2.557e-005	2.558e-005	2.558e-005
31	Liq. Reynolds	2.157e+005 *	2.157e+005 *	2.157e+005 *	2.157e+005 *
32	Vap. Reynolds	9.079e+006 *	9.080e+006 *	9.080e+006 *	9.081e+006 *
33	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
34	Vapour Velocity (m/s)	2.151	2.152	2.152	2.153
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	1.342e+004	1.343e+004	1.343e+004	1.344e+004
38	Elevation (m)	-11.75	-11.30	-10.97	-10.64
39	Pressure (bar)	91.78	91.76	91.75	91.74
40	Temperature (C)	56.09	56.09	56.08	56.08
41	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
42	Heat Transferred (kJ/h-m)	696.2	696.0	694.7	694.6
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.5654	0.5654	0.4889	0.4888
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	5.757e-003	5.758e-003	5.831e-003	5.832e-003
48	Static Gradient (kPa/m)	0.3642	0.3642	0.2353	0.2353
49	Accel. Gradient (kPa/m)	2.559e-005	2.560e-005	1.669e-005	1.670e-005
50	Liq. Reynolds	2.157e+005 *	2.156e+005 *	2.156e+005 *	2.156e+005 *
51	Vap. Reynolds	9.081e+006 *	9.082e+006 *	9.082e+006 *	9.083e+006 *
52	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
53	Vapour Velocity (m/s)	2.153	2.154	2.154	2.154
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1.344e+004	1.345e+004	1.345e+004	1.346e+004
57	Elevation (m)	-10.31	-9.980	-9.650	-7.850
58	Pressure (bar)	91.73	91.71	91.70	91.58
59	Temperature (C)	56.07	56.07	56.06	56.03
60	Vapour Fraction	0.8712	0.8712	0.8712	0.8713
61	Heat Transferred (kJ/h-m)	694.4	694.3	694.1	687.3
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.4888	0.4888	0.4887	1.000
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	5.832e-003	5.833e-003	5.834e-003	5.575e-003
67	Static Gradient (kPa/m)	0.2353	0.2353	0.2353	2.379
68	Accel. Gradient (kPa/m)	1.670e-005	1.670e-005	1.671e-005	1.657e-004
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 250 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.156e+005 *	2.156e+005 *	2.156e+005 *	2.155e+005 *
14	Vap. Reynolds	9.083e+006 *	9.083e+006 *	9.084e+006 *	9.088e+006 *
15	Liquid Velocity (m/s)	0.2044	0.2044	0.2043	0.2043
16	Vapour Velocity (m/s)	2.154	2.155	2.155	2.158
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	1.346e+004	1.347e+004	1.347e+004	1.348e+004
20	Elevation (m)	-6.050	-4.250	-2.450	-0.6500
21	Pressure (bar)	91.46	91.34	91.22	91.11
22	Temperature (C)	56.00	55.97	55.94	55.91
23	Vapour Fraction	0.8713	0.8714	0.8714	0.8715
24	Heat Transferred (kJ/h-m)	686.5	685.7	684.9	684.1
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	1.000	1.000	1.000	1.000
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	5.581e-003	5.588e-003	5.594e-003	5.601e-003
30	Static Gradient (kPa/m)	2.379	2.380	2.380	2.380
31	Accel. Gradient (kPa/m)	1.661e-004	1.666e-004	1.671e-004	1.675e-004
32	Liq. Reynolds	2.153e+005 *	2.152e+005 *	2.151e+005 *	2.150e+005 *
33	Vap. Reynolds	9.091e+006 *	9.095e+006 *	9.099e+006 *	9.103e+006 *
34	Liquid Velocity (m/s)	0.2043	0.2043	0.2042	0.2042
35	Vapour Velocity (m/s)	2.161	2.164	2.167	2.170
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	1.348e+004	1.349e+004	1.349e+004	1.350e+004
39	Elevation (m)	-0.4840	-0.3180	-0.1520	1.400e-002
40	Pressure (bar)	91.10	91.09	91.09	91.08
41	Temperature (C)	55.91	55.90	55.90	55.90
42	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
43	Heat Transferred (kJ/h-m)	690.3	690.2	690.1	690.0
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.3851	0.3851	0.3851	0.3851
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	6.004e-003	6.004e-003	6.004e-003	6.004e-003
49	Static Gradient (kPa/m)	9.904e-002	9.904e-002	9.903e-002	9.903e-002
50	Accel. Gradient (kPa/m)	7.377e-006	7.378e-006	7.379e-006	7.379e-006
51	Liq. Reynolds	2.150e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
52	Vap. Reynolds	9.103e+006 *	9.103e+006 *	9.103e+006 *	9.103e+006 *
53	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
54	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	1.350e+004	1.351e+004	1.351e+004	1.352e+004
58	Elevation (m)	0.1800	0.1940	0.2080	0.2220
59	Pressure (bar)	91.08	91.08	91.08	91.08
60	Temperature (C)	55.89	55.89	55.89	55.89
61	Vapour Fraction	0.8715	0.8715	0.8715	0.8715
62	Heat Transferred (kJ/h-m)	689.9	689.8	689.7	689.7
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.3851	0.2883	0.2883	0.2883
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	6.005e-003	4.948e-003	4.948e-003	4.948e-003
68	Static Gradient (kPa/m)	9.903e-002	6.753e-003	6.753e-003	6.753e-003
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient	(kPa/m)	7.380e-006	8.222e-007	8.222e-007	8.222e-007	
14	Liq. Reynolds		2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *	
15	Vap. Reynolds		9.103e+006 *	9.103e+006 *	9.103e+006 *	9.103e+006 *	
16	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
17	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
18	Deposit Thickness	(mm)	---	---	---	---	
19	Deposit Volume	(m3)	---	---	---	---	
20	Length	(m)	1.352e+004	1.353e+004	1.353e+004	1.353e+004	
21	Elevation	(m)	0.2360	0.2500	0.2500	0.2500	
22	Pressure	(bar)	91.08	91.08	91.08	91.08	
23	Temperature	(C)	55.89	55.88	55.88	55.88	
24	Vapour Fraction		0.8715	0.8715	0.8715	0.8715	
25	Heat Transferred	(kJ/h-m)	689.6	689.6	689.6	689.5	
26	Flow Regime		Transition	Transition	Transition	Transition	
27	Water Flow Regime						
28	Liquid HoldUp		0.2883	0.2883	0.3024	0.3024	
29	Water HoldUp		---	---	---	---	
30	Friction Gradient	(kPa/m)	4.948e-003	4.948e-003	5.873e-003	5.873e-003	
31	Static Gradient	(kPa/m)	6.753e-003	6.753e-003	0.0000	0.0000	
32	Accel. Gradient	(kPa/m)	8.222e-007	8.222e-007	4.127e-007	4.127e-007	
33	Liq. Reynolds		2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *	
34	Vap. Reynolds		9.103e+006 *	9.103e+006 *	9.103e+006 *	9.103e+006 *	
35	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
36	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
37	Deposit Thickness	(mm)	---	---	---	---	
38	Deposit Volume	(m3)	---	---	---	---	
39	Length	(m)	1.353e+004	1.354e+004	1.354e+004	1.354e+004	
40	Elevation	(m)	0.2500	0.2500	0.2500	0.2520	
41	Pressure	(bar)	91.08	91.08	91.08	91.08	
42	Temperature	(C)	55.88	55.88	55.88	55.88	
43	Vapour Fraction		0.8715	0.8715	0.8715	0.8715	
44	Heat Transferred	(kJ/h-m)	689.5	689.5	689.5	689.4	
45	Flow Regime		Transition	Transition	Transition	Transition	
46	Water Flow Regime						
47	Liquid HoldUp		0.3024	0.3024	0.3024	0.2813	
48	Water HoldUp		---	---	---	---	
49	Friction Gradient	(kPa/m)	5.873e-003	5.873e-003	5.873e-003	5.477e-003	
50	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	1.422e-003	
51	Accel. Gradient	(kPa/m)	4.127e-007	4.127e-007	4.127e-007	4.847e-007	
52	Liq. Reynolds		2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *	
53	Vap. Reynolds		9.103e+006 *	9.103e+006 *	9.103e+006 *	9.103e+006 *	
54	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
55	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
56	Deposit Thickness	(mm)	---	---	---	---	
57	Deposit Volume	(m3)	---	---	---	---	
58	Length	(m)	1.354e+004	1.355e+004	1.355e+004	1.355e+004	
59	Elevation	(m)	0.2540	0.2560	0.2580	0.2600	
60	Pressure	(bar)	91.08	91.07	91.07	91.07	
61	Temperature	(C)	55.88	55.88	55.87	55.87	
62	Vapour Fraction		0.8715	0.8715	0.8715	0.8715	
63	Heat Transferred	(kJ/h-m)	689.4	689.4	689.3	689.3	
64	Flow Regime		Transition	Transition	Transition	Transition	
65	Water Flow Regime						
66	Liquid HoldUp		0.2813	0.2813	0.2813	0.2813	
67	Water HoldUp		---	---	---	---	
68	Friction Gradient	(kPa/m)	5.477e-003	5.477e-003	5.477e-003	5.476e-003	
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	1.422e-003	1.422e-003	1.422e-003	1.422e-003
14	Accel. Gradient	(kPa/m)	4.847e-007	4.847e-007	4.847e-007	4.847e-007
15	Liq. Reynolds		2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
16	Vap. Reynolds		9.103e+006 *	9.103e+006 *	9.103e+006 *	9.103e+006 *
17	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042
18	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	1.356e+004	1.356e+004	1.356e+004	1.357e+004
22	Elevation	(m)	0.2620	0.2640	0.2660	0.2680
23	Pressure	(bar)	91.07	91.07	91.07	91.07
24	Temperature	(C)	55.87	55.87	55.87	55.87
25	Vapour Fraction		0.8715	0.8715	0.8715	0.8715
26	Heat Transferred	(kJ/h-m)	689.2	689.2	689.2	689.1
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2813	0.2813	0.2813	0.2813
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	5.476e-003	5.476e-003	5.476e-003	5.476e-003
32	Static Gradient	(kPa/m)	1.423e-003	1.423e-003	1.423e-003	1.423e-003
33	Accel. Gradient	(kPa/m)	4.848e-007	4.848e-007	4.847e-007	4.847e-007
34	Liq. Reynolds		2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
35	Vap. Reynolds		9.103e+006 *	9.103e+006 *	9.103e+006 *	9.103e+006 *
36	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042
37	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	1.357e+004	1.357e+004	1.358e+004	1.358e+004
41	Elevation	(m)	0.2700	0.2720	0.2740	0.2760
42	Pressure	(bar)	91.07	91.07	91.07	91.07
43	Temperature	(C)	55.87	55.86	55.86	55.86
44	Vapour Fraction		0.8715	0.8715	0.8715	0.8715
45	Heat Transferred	(kJ/h-m)	689.1	689.1	689.0	689.0
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2813	0.2813	0.2813	0.2813
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	5.476e-003	5.476e-003	5.476e-003	5.476e-003
51	Static Gradient	(kPa/m)	1.423e-003	1.422e-003	1.422e-003	1.422e-003
52	Accel. Gradient	(kPa/m)	4.847e-007	4.847e-007	4.847e-007	4.847e-007
53	Liq. Reynolds		2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
54	Vap. Reynolds		9.103e+006 *	9.103e+006 *	9.103e+006 *	9.104e+006 *
55	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042
56	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	1.358e+004	1.359e+004	1.359e+004	1.359e+004
60	Elevation	(m)	0.2780	0.2800	0.2760	0.2720
61	Pressure	(bar)	91.07	91.07	91.07	91.07
62	Temperature	(C)	55.86	55.86	55.86	55.86
63	Vapour Fraction		0.8715	0.8715	0.8715	0.8715
64	Heat Transferred	(kJ/h-m)	689.0	688.9	688.9	688.9
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2813	0.2813	0.2059	0.2059
68	Water HoldUp		---	---	---	---
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1				Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	5.476e-003	5.476e-003	6.864e-003	6.863e-003
14	Static Gradient	(kPa/m)	1.422e-003	1.422e-003	-2.567e-003	-2.567e-003
15	Accel. Gradient	(kPa/m)	4.847e-007	4.847e-007	3.019e-007	3.019e-007
16	Liq. Reynolds		2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
17	Vap. Reynolds		9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
18	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042
19	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	1.360e+004	1.360e+004	1.360e+004	1.361e+004
23	Elevation	(m)	0.2680	0.2640	0.2600	0.2620
24	Pressure	(bar)	91.07	91.07	91.07	91.07
25	Temperature	(C)	55.86	55.85	55.85	55.85
26	Vapour Fraction		0.8715	0.8715	0.8715	0.8715
27	Heat Transferred	(kJ/h-m)	688.8	688.8	688.8	688.7
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2059	0.2059	0.2059	0.2807
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	6.863e-003	6.863e-003	6.863e-003	5.526e-003
33	Static Gradient	(kPa/m)	-2.567e-003	-2.567e-003	-2.567e-003	9.468e-004
34	Accel. Gradient	(kPa/m)	3.018e-007	3.018e-007	3.018e-007	4.548e-007
35	Liq. Reynolds		2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
36	Vap. Reynolds		9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
37	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042
38	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	1.361e+004	1.362e+004	1.362e+004	1.363e+004
42	Elevation	(m)	0.2640	0.2660	0.2680	0.2700
43	Pressure	(bar)	91.07	91.07	91.07	91.07
44	Temperature	(C)	55.85	55.85	55.84	55.84
45	Vapour Fraction		0.8715	0.8715	0.8715	0.8715
46	Heat Transferred	(kJ/h-m)	688.7	688.6	688.5	688.5
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2807	0.2807	0.2807	0.2807
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	5.526e-003	5.526e-003	5.526e-003	5.526e-003
52	Static Gradient	(kPa/m)	9.468e-004	9.468e-004	9.468e-004	9.468e-004
53	Accel. Gradient	(kPa/m)	4.548e-007	4.548e-007	4.547e-007	4.547e-007
54	Liq. Reynolds		2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
55	Vap. Reynolds		9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
56	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042
57	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	1.363e+004	1.364e+004	1.364e+004	1.365e+004
61	Elevation	(m)	0.2780	0.2860	0.2940	0.3020
62	Pressure	(bar)	91.07	91.07	91.07	91.07
63	Temperature	(C)	55.84	55.84	55.84	55.83
64	Vapour Fraction		0.8715	0.8715	0.8715	0.8715
65	Heat Transferred	(kJ/h-m)	688.4	688.4	688.3	688.3
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2845	0.2845	0.2845	0.2845
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp		---	---	---
14	Friction Gradient (kPa/m)	5.234e-003	5.234e-003	5.234e-003	5.234e-003
15	Static Gradient (kPa/m)	3.791e-003	3.791e-003	3.791e-003	3.791e-003
16	Accel. Gradient (kPa/m)	6.341e-007	6.341e-007	6.341e-007	6.341e-007
17	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
18	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
19	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
20	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	1.365e+004	1.366e+004	1.366e+004	1.367e+004
24	Elevation (m)	0.3100	0.3060	0.3020	0.2980
25	Pressure (bar)	91.07	91.07	91.07	91.07
26	Temperature (C)	55.83	55.83	55.83	55.83
27	Vapour Fraction	0.8715	0.8714	0.8714	0.8714
28	Heat Transferred (kJ/h-m)	688.2	688.2	688.1	688.1
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2845	0.2064	0.2064	0.2064
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	5.234e-003	6.859e-003	6.859e-003	6.859e-003
34	Static Gradient (kPa/m)	3.791e-003	-1.543e-003	-1.543e-003	-1.543e-003
35	Accel. Gradient (kPa/m)	6.341e-007	3.735e-007	3.735e-007	3.735e-007
36	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
37	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
38	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
39	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	1.367e+004	1.368e+004	1.368e+004	1.369e+004
43	Elevation (m)	0.2940	0.2900	0.2920	0.2940
44	Pressure (bar)	91.07	91.07	91.07	91.07
45	Temperature (C)	55.82	55.82	55.82	55.82
46	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
47	Heat Transferred (kJ/h-m)	688.0	688.0	687.9	687.8
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2064	0.2064	0.2807	0.2807
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	6.859e-003	6.859e-003	5.525e-003	5.525e-003
53	Static Gradient (kPa/m)	-1.543e-003	-1.543e-003	9.468e-004	9.468e-004
54	Accel. Gradient (kPa/m)	3.735e-007	3.735e-007	4.547e-007	4.547e-007
55	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
56	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
57	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
58	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	1.369e+004	1.370e+004	1.370e+004	1.371e+004
62	Elevation (m)	0.2960	0.2980	0.3000	0.2800
63	Pressure (bar)	91.07	91.06	91.06	91.06
64	Temperature (C)	55.82	55.81	55.81	55.81
65	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
66	Heat Transferred (kJ/h-m)	687.8	687.7	687.7	687.6
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 255 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.2807	0.2807	0.2807	0.2034
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	5.525e-003	5.525e-003	5.525e-003	6.883e-003
16	Static Gradient (kPa/m)	9.468e-004	9.468e-004	9.469e-004	-7.644e-003
17	Accel. Gradient (kPa/m)	4.547e-007	4.547e-007	4.547e-007	-5.351e-008
18	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
19	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
20	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
21	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	1.371e+004	1.372e+004	1.372e+004	1.373e+004
25	Elevation (m)	0.2600	0.2400	0.2200	0.2000
26	Pressure (bar)	91.06	91.06	91.06	91.06
27	Temperature (C)	55.81	55.81	55.80	55.80
28	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
29	Heat Transferred (kJ/h-m)	687.6	687.5	687.5	687.4
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2034	0.2034	0.2034	0.2034
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.882e-003	6.882e-003	6.882e-003	6.882e-003
35	Static Gradient (kPa/m)	-7.644e-003	-7.644e-003	-7.644e-003	-7.644e-003
36	Accel. Gradient (kPa/m)	-5.352e-008	-5.353e-008	-5.353e-008	-5.354e-008
37	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
38	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
39	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
40	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	1.373e+004	1.374e+004	1.374e+004	1.375e+004
44	Elevation (m)	0.1780	0.1560	0.1340	0.1120
45	Pressure (bar)	91.06	91.06	91.07	91.07
46	Temperature (C)	55.80	55.80	55.80	55.80
47	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
48	Heat Transferred (kJ/h-m)	687.4	687.3	687.3	687.2
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2031	0.2031	0.2031	0.2031
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.885e-003	6.885e-003	6.885e-003	6.885e-003
54	Static Gradient (kPa/m)	-8.399e-003	-8.399e-003	-8.399e-003	-8.399e-003
55	Accel. Gradient (kPa/m)	-1.064e-007	-1.064e-007	-1.064e-007	-1.064e-007
56	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
57	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
58	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
59	Vapour Velocity (m/s)	2.170	2.170	2.170	2.169
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	1.375e+004	1.376e+004	1.376e+004	1.377e+004
63	Elevation (m)	9.000e-002	5.800e-002	2.600e-002	-6.000e-003
64	Pressure (bar)	91.07	91.07	91.07	91.07
65	Temperature (C)	55.79	55.79	55.79	55.79
66	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
67	Heat Transferred (kJ/h-m)	687.2	687.1	687.1	687.0
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 256 of 937				

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2	 Epic Atlantic Limited		Unit Set: Mixed		
Date/Time: Mon Jun 13 15:02:44 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
9	PROFILES				
11	Pipe Table				
13	Water Flow Regime				
14	Liquid HoldUp	0.2031	0.2012	0.2012	0.2012
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.885e-003	6.900e-003	6.900e-003	6.900e-003
17	Static Gradient (kPa/m)	-8.400e-003	-1.215e-002	-1.215e-002	-1.215e-002
18	Accel. Gradient (kPa/m)	-1.064e-007	-3.686e-007	-3.686e-007	-3.686e-007
19	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
20	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
21	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
22	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	1.377e+004	1.378e+004	1.378e+004	1.379e+004
26	Elevation (m)	-3.800e-002	-7.000e-002	-6.000e-002	-5.000e-002
27	Pressure (bar)	91.07	91.07	91.07	91.07
28	Temperature (C)	55.79	55.78	55.78	55.78
29	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
30	Heat Transferred (kJ/h-m)	687.0	687.0	686.9	686.8
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2012	0.2012	0.2858	0.2858
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	6.900e-003	6.900e-003	5.133e-003	5.133e-003
36	Static Gradient (kPa/m)	-1.215e-002	-1.215e-002	4.795e-003	4.795e-003
37	Accel. Gradient (kPa/m)	-3.686e-007	-3.686e-007	6.974e-007	6.974e-007
38	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
39	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
40	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
41	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	1.379e+004	1.380e+004	1.380e+004	1.381e+004
45	Elevation (m)	-4.000e-002	-3.000e-002	-2.000e-002	3.600e-002
46	Pressure (bar)	91.07	91.06	91.06	91.06
47	Temperature (C)	55.78	55.78	55.77	55.77
48	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
49	Heat Transferred (kJ/h-m)	686.8	686.7	686.7	686.6
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2858	0.2858	0.2858	0.3152
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	5.133e-003	5.133e-003	5.133e-003	6.037e-003
55	Static Gradient (kPa/m)	4.795e-003	4.795e-003	4.795e-003	2.879e-002
56	Accel. Gradient (kPa/m)	6.974e-007	6.974e-007	6.974e-007	2.446e-006
57	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
58	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
59	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
60	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	1.381e+004	1.382e+004	1.382e+004	1.383e+004
64	Elevation (m)	9.200e-002	0.1480	0.2040	0.2600
65	Pressure (bar)	91.06	91.06	91.06	91.06
66	Temperature (C)	55.77	55.77	55.76	55.76
67	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
68	Heat Transferred (kJ/h-m)	686.5	686.5	686.4	686.3
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.3152	0.3152	0.3152	0.3152
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	6.037e-003	6.037e-003	6.037e-003	6.037e-003
18	Static Gradient (kPa/m)	2.879e-002	2.879e-002	2.879e-002	2.879e-002
19	Accel. Gradient (kPa/m)	2.447e-006	2.447e-006	2.447e-006	2.447e-006
20	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
21	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
22	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
23	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	1.383e+004	1.384e+004	1.384e+004	1.385e+004
27	Elevation (m)	0.1360	1.200e-002	-0.1120	-0.2360
28	Pressure (bar)	91.06	91.06	91.06	91.06
29	Temperature (C)	55.76	55.76	55.76	55.76
30	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
31	Heat Transferred (kJ/h-m)	686.3	686.3	686.3	686.2
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1843	0.1843	0.1843	0.1843
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	7.062e-003	7.062e-003	7.062e-003	7.061e-003
37	Static Gradient (kPa/m)	-4.419e-002	-4.419e-002	-4.419e-002	-4.420e-002
38	Accel. Gradient (kPa/m)	-2.608e-006	-2.608e-006	-2.608e-006	-2.608e-006
39	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
40	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
41	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
42	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	1.385e+004	1.386e+004	1.386e+004	1.387e+004
46	Elevation (m)	-0.3600	-0.3280	-0.2960	-0.2640
47	Pressure (bar)	91.06	91.06	91.06	91.06
48	Temperature (C)	55.76	55.75	55.75	55.75
49	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
50	Heat Transferred (kJ/h-m)	686.2	686.1	686.0	686.0
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1843	0.2999	0.2999	0.2999
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	7.061e-003	5.769e-003	5.769e-003	5.769e-003
56	Static Gradient (kPa/m)	-4.420e-002	1.588e-002	1.588e-002	1.588e-002
57	Accel. Gradient (kPa/m)	-2.608e-006	1.520e-006	1.520e-006	1.520e-006
58	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
59	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
60	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
61	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	1.387e+004	1.388e+004	1.388e+004	1.389e+004
65	Elevation (m)	-0.2320	-0.2000	-0.1760	-0.1520
66	Pressure (bar)	91.06	91.06	91.06	91.06
67	Temperature (C)	55.75	55.74	55.74	55.74
68	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Pipe Segment: ELEP-Akabel Pipeline (continued)		
3					
4					
5			Unit Set: Mixed		
6			Date/Time: Mon Jun 13 15:02:44 2022		
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	685.9	685.9	685.8	685.7
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.2999	0.2999	0.2948	0.2948
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	5.769e-003	5.769e-003	5.175e-003	5.175e-003
19	Static Gradient (kPa/m)	1.588e-002	1.588e-002	1.176e-002	1.176e-002
20	Accel. Gradient (kPa/m)	1.520e-006	1.520e-006	1.190e-006	1.190e-006
21	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
22	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
23	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
24	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	1.389e+004	1.390e+004	1.390e+004	1.391e+004
28	Elevation (m)	-0.1280	-0.1040	-8.000e-002	-7.200e-002
29	Pressure (bar)	91.06	91.06	91.06	91.05
30	Temperature (C)	55.74	55.74	55.73	55.73
31	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
32	Heat Transferred (kJ/h-m)	685.7	685.6	685.6	685.5
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.2948	0.2948	0.2948	0.2846
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	5.175e-003	5.175e-003	5.175e-003	5.228e-003
38	Static Gradient (kPa/m)	1.176e-002	1.176e-002	1.176e-002	3.825e-003
39	Accel. Gradient (kPa/m)	1.190e-006	1.190e-006	1.190e-006	6.358e-007
40	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
41	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
42	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
43	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	1.391e+004	1.392e+004	1.392e+004	1.393e+004
47	Elevation (m)	-6.400e-002	-5.600e-002	-4.800e-002	-4.000e-002
48	Pressure (bar)	91.05	91.05	91.05	91.05
49	Temperature (C)	55.73	55.73	55.72	55.72
50	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
51	Heat Transferred (kJ/h-m)	685.5	685.4	685.3	685.3
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.2846	0.2846	0.2846	0.2846
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	5.228e-003	5.228e-003	5.228e-003	5.228e-003
57	Static Gradient (kPa/m)	3.825e-003	3.825e-003	3.825e-003	3.825e-003
58	Accel. Gradient (kPa/m)	6.358e-007	6.358e-007	6.358e-007	6.358e-007
59	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
60	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
61	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
62	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	1.393e+004	1.394e+004	1.394e+004	1.395e+004
66	Elevation (m)	-5.400e-002	-6.800e-002	-8.200e-002	-9.600e-002
67	Pressure (bar)	91.05	91.05	91.05	91.05
68	Temperature (C)	55.72	55.72	55.72	55.72
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 259 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
14	Heat Transferred (kJ/h-m)	685.3	685.2	685.2	685.1
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2046	0.2046	0.2046	0.2046
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	6.871e-003	6.871e-003	6.871e-003	6.871e-003
20	Static Gradient (kPa/m)	-5.371e-003	-5.371e-003	-5.371e-003	-5.371e-003
21	Accel. Gradient (kPa/m)	1.054e-007	1.054e-007	1.054e-007	1.054e-007
22	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
23	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
24	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
25	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	1.395e+004	1.396e+004	1.396e+004	1.397e+004
29	Elevation (m)	-0.1100	-7.200e-002	-3.400e-002	4.000e-003
30	Pressure (bar)	91.05	91.05	91.05	91.05
31	Temperature (C)	55.71	55.71	55.71	55.71
32	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
33	Heat Transferred (kJ/h-m)	685.1	685.0	684.9	684.9
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2046	0.3037	0.3037	0.3037
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	6.871e-003	5.905e-003	5.905e-003	5.905e-003
39	Static Gradient (kPa/m)	-5.371e-003	1.903e-002	1.903e-002	1.903e-002
40	Accel. Gradient (kPa/m)	1.054e-007	1.751e-006	1.751e-006	1.751e-006
41	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
42	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
43	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
44	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	1.397e+004	1.398e+004	1.398e+004	1.399e+004
48	Elevation (m)	4.200e-002	8.000e-002	8.400e-002	8.800e-002
49	Pressure (bar)	91.05	91.05	91.05	91.05
50	Temperature (C)	55.70	55.70	55.70	55.70
51	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
52	Heat Transferred (kJ/h-m)	684.8	684.7	684.7	684.6
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.3037	0.3037	0.2821	0.2821
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	5.905e-003	5.905e-003	5.423e-003	5.423e-003
58	Static Gradient (kPa/m)	1.903e-002	1.903e-002	1.900e-003	1.900e-003
59	Accel. Gradient (kPa/m)	1.751e-006	1.751e-006	5.144e-007	5.144e-007
60	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
61	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
62	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
63	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	1.399e+004	1.400e+004	1.400e+004	1.401e+004
67	Elevation (m)	9.200e-002	9.600e-002	1.000e-001	3.800e-002
68	Pressure (bar)	91.05	91.04	91.04	91.05
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	55.70	55.69	55.69	55.69
14	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
15	Heat Transferred (kJ/h-m)	684.6	684.5	684.5	684.4
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2821	0.2821	0.2821	0.1957
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	5.423e-003	5.423e-003	5.423e-003	6.948e-003
21	Static Gradient (kPa/m)	1.900e-003	1.900e-003	1.900e-003	-2.313e-002
22	Accel. Gradient (kPa/m)	5.144e-007	5.144e-007	5.144e-007	-1.137e-006
23	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
24	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
25	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
26	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	1.401e+004	1.402e+004	1.402e+004	1.403e+004
30	Elevation (m)	-2.400e-002	-8.600e-002	-0.1480	-0.2100
31	Pressure (bar)	91.05	91.05	91.05	91.05
32	Temperature (C)	55.69	55.69	55.68	55.68
33	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
34	Heat Transferred (kJ/h-m)	684.4	684.4	684.3	684.3
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1957	0.1957	0.1957	0.1957
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	6.948e-003	6.948e-003	6.948e-003	6.947e-003
40	Static Gradient (kPa/m)	-2.313e-002	-2.313e-002	-2.313e-002	-2.313e-002
41	Accel. Gradient (kPa/m)	-1.137e-006	-1.137e-006	-1.137e-006	-1.137e-006
42	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
43	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
44	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
45	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	1.403e+004	1.404e+004	1.404e+004	1.405e+004
49	Elevation (m)	-0.1480	-8.600e-002	-2.400e-002	3.800e-002
50	Pressure (bar)	91.05	91.04	91.04	91.04
51	Temperature (C)	55.68	55.68	55.68	55.67
52	Vapour Fraction	0.8714	0.8714	0.8714	0.8714
53	Heat Transferred (kJ/h-m)	684.2	684.1	684.1	684.0
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.3188	0.3188	0.3188	0.3188
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	6.049e-003	6.049e-003	6.049e-003	6.049e-003
59	Static Gradient (kPa/m)	3.197e-002	3.197e-002	3.197e-002	3.197e-002
60	Accel. Gradient (kPa/m)	2.670e-006	2.670e-006	2.670e-006	2.670e-006
61	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
62	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
63	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
64	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	1.405e+004	1.405e+004	1.406e+004	1.406e+004
68	Elevation (m)	1.000e-001	0.1080	0.1160	0.1240
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	91.04	91.04	91.04	91.04		
14	Temperature (C)	55.67	55.67	55.67	55.67		
15	Vapour Fraction	0.8714	0.8714	0.8714	0.8714		
16	Heat Transferred (kJ/h-m)	683.9	683.9	683.9	683.9		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.3188	0.3050	0.3050	0.3050		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	6.049e-003	5.932e-003	5.932e-003	5.932e-003		
22	Static Gradient (kPa/m)	3.197e-002	2.009e-002	2.009e-002	2.009e-002		
23	Accel. Gradient (kPa/m)	2.670e-006	1.828e-006	1.828e-006	1.828e-006		
24	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *		
25	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *		
26	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
27	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	1.406e+004	1.406e+004	1.406e+004	1.407e+004		
31	Elevation (m)	0.1320	0.1400	0.1300	0.1200		
32	Pressure (bar)	91.04	91.04	91.04	91.04		
33	Temperature (C)	55.67	55.67	55.67	55.66		
34	Vapour Fraction	0.8714	0.8714	0.8714	0.8714		
35	Heat Transferred (kJ/h-m)	683.9	683.8	683.8	683.8		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.3050	0.3050	0.2054	0.2054		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	5.932e-003	5.932e-003	6.865e-003	6.865e-003		
41	Static Gradient (kPa/m)	2.009e-002	2.009e-002	-3.845e-003	-3.845e-003		
42	Accel. Gradient (kPa/m)	1.828e-006	1.828e-006	2.121e-007	2.121e-007		
43	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *		
44	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *		
45	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
46	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	1.407e+004	1.408e+004	1.408e+004	1.408e+004		
50	Elevation (m)	0.1100	1.000e-001	9.000e-002	6.200e-002		
51	Pressure (bar)	91.04	91.04	91.04	91.04		
52	Temperature (C)	55.66	55.66	55.66	55.66		
53	Vapour Fraction	0.8714	0.8714	0.8714	0.8714		
54	Heat Transferred (kJ/h-m)	683.7	683.6	683.6	683.6		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.2054	0.2054	0.2054	0.1812		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	6.865e-003	6.865e-003	6.865e-003	7.095e-003		
60	Static Gradient (kPa/m)	-3.845e-003	-3.846e-003	-3.846e-003	-4.984e-002		
61	Accel. Gradient (kPa/m)	2.121e-007	2.121e-007	2.121e-007	-3.002e-006		
62	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *		
63	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *		
64	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
65	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	1.409e+004	1.409e+004	1.409e+004	1.409e+004		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Elevation (m)	3.400e-002	6.000e-003	-2.200e-002	-5.000e-002		
14	Pressure (bar)	91.04	91.04	91.04	91.04		
15	Temperature (C)	55.66	55.66	55.66	55.66		
16	Vapour Fraction	0.8714	0.8714	0.8713	0.8713		
17	Heat Transferred (kJ/h-m)	683.6	683.6	683.6	683.6		
18	Flow Regime	Transition	Transition	Transition	Transition		
19	Water Flow Regime						
20	Liquid HoldUp	0.1812	0.1812	0.1812	0.1812		
21	Water HoldUp	---	---	---	---		
22	Friction Gradient (kPa/m)	7.095e-003	7.095e-003	7.095e-003	7.095e-003		
23	Static Gradient (kPa/m)	-4.984e-002	-4.984e-002	-4.984e-002	-4.984e-002		
24	Accel. Gradient (kPa/m)	-3.002e-006	-3.002e-006	-3.002e-006	-3.002e-006		
25	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *		
26	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *		
27	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
28	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169		
29	Deposit Thickness (mm)	---	---	---	---		
30	Deposit Volume (m3)	---	---	---	---		
31	Length (m)	1.409e+004	1.410e+004	1.411e+004	1.411e+004		
32	Elevation (m)	-0.1160	-0.1820	-0.2480	-0.3140		
33	Pressure (bar)	91.04	91.04	91.04	91.04		
34	Temperature (C)	55.66	55.65	55.65	55.65		
35	Vapour Fraction	0.8713	0.8713	0.8713	0.8713		
36	Heat Transferred (kJ/h-m)	683.5	683.5	683.4	683.4		
37	Flow Regime	Transition	Transition	Transition	Transition		
38	Water Flow Regime						
39	Liquid HoldUp	0.1970	0.1970	0.1970	0.1970		
40	Water HoldUp	---	---	---	---		
41	Friction Gradient (kPa/m)	6.935e-003	6.935e-003	6.935e-003	6.935e-003		
42	Static Gradient (kPa/m)	-2.061e-002	-2.061e-002	-2.061e-002	-2.061e-002		
43	Accel. Gradient (kPa/m)	-9.602e-007	-9.602e-007	-9.602e-007	-9.602e-007		
44	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *		
45	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *		
46	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
47	Vapour Velocity (m/s)	2.169	2.168	2.168	2.168		
48	Deposit Thickness (mm)	---	---	---	---		
49	Deposit Volume (m3)	---	---	---	---		
50	Length (m)	1.412e+004	1.413e+004	1.413e+004	1.414e+004		
51	Elevation (m)	-0.3800	-0.3980	-0.4160	-0.4340		
52	Pressure (bar)	91.04	91.04	91.04	91.04		
53	Temperature (C)	55.65	55.65	55.64	55.64		
54	Vapour Fraction	0.8713	0.8713	0.8713	0.8713		
55	Heat Transferred (kJ/h-m)	683.3	683.2	683.2	683.1		
56	Flow Regime	Transition	Transition	Transition	Transition		
57	Water Flow Regime						
58	Liquid HoldUp	0.1970	0.2048	0.2049	0.2049		
59	Water HoldUp	---	---	---	---		
60	Friction Gradient (kPa/m)	6.935e-003	6.868e-003	6.868e-003	6.868e-003		
61	Static Gradient (kPa/m)	-2.061e-002	-4.937e-003	-4.937e-003	-4.937e-003		
62	Accel. Gradient (kPa/m)	-9.602e-007	1.356e-007	1.356e-007	1.356e-007		
63	Liq. Reynolds	2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *		
64	Vap. Reynolds	9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *		
65	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
66	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
67	Deposit Thickness (mm)	---	---	---	---		
68	Deposit Volume (m3)	---	---	---	---		
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 263 of 937		

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022	
6						
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Length	(m)	1.415e+004	1.415e+004	1.416e+004	1.416e+004
14	Elevation	(m)	-0.4520	-0.4700	-0.3840	-0.2980
15	Pressure	(bar)	91.04	91.04	91.04	91.04
16	Temperature	(C)	55.64	55.63	55.63	55.63
17	Vapour Fraction		0.8713	0.8713	0.8713	0.8713
18	Heat Transferred	(kJ/h-m)	683.0	682.9	682.9	682.8
19	Flow Regime		Transition	Transition	Transition	Transition
20	Water Flow Regime					
21	Liquid HoldUp		0.2049	0.2049	0.3344	0.3344
22	Water HoldUp		---	---	---	---
23	Friction Gradient	(kPa/m)	6.868e-003	6.868e-003	6.063e-003	6.063e-003
24	Static Gradient	(kPa/m)	-4.937e-003	-4.937e-003	4.617e-002	4.617e-002
25	Accel. Gradient	(kPa/m)	1.356e-007	1.356e-007	3.668e-006	3.669e-006
26	Liq. Reynolds		2.149e+005 *	2.149e+005 *	2.149e+005 *	2.149e+005 *
27	Vap. Reynolds		9.104e+006 *	9.104e+006 *	9.104e+006 *	9.104e+006 *
28	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042
29	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168
30	Deposit Thickness	(mm)	---	---	---	---
31	Deposit Volume	(m3)	---	---	---	---
32	Length	(m)	1.417e+004	1.417e+004	1.418e+004	1.418e+004
33	Elevation	(m)	-0.2120	-0.1260	-4.000e-002	-5.200e-002
34	Pressure	(bar)	91.03	91.03	91.03	91.03
35	Temperature	(C)	55.63	55.62	55.62	55.62
36	Vapour Fraction		0.8713	0.8713	0.8713	0.8713
37	Heat Transferred	(kJ/h-m)	682.7	682.7	682.6	682.5
38	Flow Regime		Transition	Transition	Transition	Transition
39	Water Flow Regime					
40	Liquid HoldUp		0.3344	0.3344	0.3344	0.2050
41	Water HoldUp		---	---	---	---
42	Friction Gradient	(kPa/m)	6.063e-003	6.063e-003	6.063e-003	6.867e-003
43	Static Gradient	(kPa/m)	4.617e-002	4.617e-002	4.617e-002	-4.610e-003
44	Accel. Gradient	(kPa/m)	3.669e-006	3.669e-006	3.669e-006	1.585e-007
45	Liq. Reynolds		2.149e+005 *	2.149e+005 *	2.148e+005 *	2.148e+005 *
46	Vap. Reynolds		9.104e+006 *	9.104e+006 *	9.105e+006 *	9.105e+006 *
47	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042
48	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168
49	Deposit Thickness	(mm)	---	---	---	---
50	Deposit Volume	(m3)	---	---	---	---
51	Length	(m)	1.419e+004	1.419e+004	1.420e+004	1.420e+004
52	Elevation	(m)	-6.400e-002	-7.600e-002	-8.800e-002	-0.1000
53	Pressure	(bar)	91.03	91.03	91.03	91.03
54	Temperature	(C)	55.62	55.61	55.61	55.61
55	Vapour Fraction		0.8713	0.8713	0.8713	0.8713
56	Heat Transferred	(kJ/h-m)	682.5	682.4	682.4	682.3
57	Flow Regime		Transition	Transition	Transition	Transition
58	Water Flow Regime					
59	Liquid HoldUp		0.2050	0.2050	0.2050	0.2050
60	Water HoldUp		---	---	---	---
61	Friction Gradient	(kPa/m)	6.867e-003	6.867e-003	6.867e-003	6.867e-003
62	Static Gradient	(kPa/m)	-4.610e-003	-4.610e-003	-4.610e-003	-4.610e-003
63	Accel. Gradient	(kPa/m)	1.585e-007	1.585e-007	1.585e-007	1.585e-007
64	Liq. Reynolds		2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *
65	Vap. Reynolds		9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
66	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042
67	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168
68	Deposit Thickness	(mm)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	1.421e+004	1.421e+004	1.422e+004	1.422e+004		
15	Elevation (m)	-7.600e-002	-5.200e-002	-2.800e-002	-4.000e-003		
16	Pressure (bar)	91.03	91.03	91.03	91.03		
17	Temperature (C)	55.61	55.61	55.60	55.60		
18	Vapour Fraction	0.8713	0.8713	0.8713	0.8713		
19	Heat Transferred (kJ/h-m)	682.3	682.2	682.1	682.1		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	0.2949	0.2949	0.2949	0.2949		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	5.176e-003	5.176e-003	5.176e-003	5.176e-003		
25	Static Gradient (kPa/m)	1.177e-002	1.177e-002	1.177e-002	1.177e-002		
26	Accel. Gradient (kPa/m)	1.190e-006	1.190e-006	1.190e-006	1.190e-006		
27	Liq. Reynolds	2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *		
28	Vap. Reynolds	9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *		
29	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
30	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	1.423e+004	1.423e+004	1.424e+004	1.424e+004		
34	Elevation (m)	2.000e-002	4.600e-002	7.200e-002	9.800e-002		
35	Pressure (bar)	91.02	91.02	91.02	91.02		
36	Temperature (C)	55.60	55.60	55.60	55.59		
37	Vapour Fraction	0.8713	0.8713	0.8713	0.8713		
38	Heat Transferred (kJ/h-m)	682.0	682.0	681.9	681.8		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.2949	0.2961	0.2961	0.2961		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	5.176e-003	5.419e-003	5.419e-003	5.419e-003		
44	Static Gradient (kPa/m)	1.177e-002	1.273e-002	1.273e-002	1.273e-002		
45	Accel. Gradient (kPa/m)	1.190e-006	1.275e-006	1.275e-006	1.275e-006		
46	Liq. Reynolds	2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *		
47	Vap. Reynolds	9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *		
48	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
49	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	1.425e+004	1.425e+004	1.426e+004	1.426e+004		
53	Elevation (m)	0.1240	0.1500	0.1540	0.1580		
54	Pressure (bar)	91.02	91.02	91.02	91.02		
55	Temperature (C)	55.59	55.59	55.59	55.58		
56	Vapour Fraction	0.8713	0.8713	0.8713	0.8713		
57	Heat Transferred (kJ/h-m)	681.8	681.7	681.7	681.6		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.2961	0.2961	0.2821	0.2821		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	5.419e-003	5.419e-003	5.421e-003	5.421e-003		
63	Static Gradient (kPa/m)	1.273e-002	1.273e-002	1.901e-003	1.901e-003		
64	Accel. Gradient (kPa/m)	1.275e-006	1.275e-006	5.143e-007	5.143e-007		
65	Liq. Reynolds	2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *		
66	Vap. Reynolds	9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *		
67	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
68	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	1.427e+004	1.427e+004	1.428e+004	1.428e+004
16	Elevation (m)	0.1620	0.1660	0.1700	0.1740
17	Pressure (bar)	91.02	91.02	91.02	91.02
18	Temperature (C)	55.58	55.58	55.58	55.58
19	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
20	Heat Transferred (kJ/h-m)	681.6	681.5	681.5	681.4
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2821	0.2821	0.2821	0.2821
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	5.421e-003	5.421e-003	5.421e-003	5.421e-003
26	Static Gradient (kPa/m)	1.901e-003	1.901e-003	1.901e-003	1.901e-003
27	Accel. Gradient (kPa/m)	5.143e-007	5.143e-007	5.143e-007	5.143e-007
28	Liq. Reynolds	2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *
29	Vap. Reynolds	9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
30	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
31	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	1.429e+004	1.429e+004	1.430e+004	1.430e+004
35	Elevation (m)	0.1780	0.1820	0.1860	0.1900
36	Pressure (bar)	91.02	91.02	91.02	91.02
37	Temperature (C)	55.57	55.57	55.57	55.57
38	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
39	Heat Transferred (kJ/h-m)	681.3	681.3	681.2	681.2
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2821	0.2821	0.2821	0.2821
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	5.420e-003	5.420e-003	5.420e-003	5.420e-003
45	Static Gradient (kPa/m)	1.901e-003	1.901e-003	1.901e-003	1.901e-003
46	Accel. Gradient (kPa/m)	5.143e-007	5.143e-007	5.142e-007	5.142e-007
47	Liq. Reynolds	2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *
48	Vap. Reynolds	9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
49	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
50	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	1.431e+004	1.431e+004	1.432e+004	1.432e+004
54	Elevation (m)	0.1920	0.1940	0.1960	0.1980
55	Pressure (bar)	91.02	91.02	91.02	91.01
56	Temperature (C)	55.57	55.56	55.56	55.56
57	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
58	Heat Transferred (kJ/h-m)	681.1	681.1	681.0	681.0
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2808	0.2808	0.2808	0.2809
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	5.520e-003	5.520e-003	5.520e-003	5.520e-003
64	Static Gradient (kPa/m)	9.474e-004	9.474e-004	9.474e-004	9.474e-004
65	Accel. Gradient (kPa/m)	4.543e-007	4.543e-007	4.543e-007	4.543e-007
66	Liq. Reynolds	2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *
67	Vap. Reynolds	9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
68	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	1.433e+004	1.433e+004	1.434e+004	1.434e+004	
17	Elevation	(m)	0.2000	0.1820	0.1640	0.1460	
18	Pressure	(bar)	91.01	91.01	91.01	91.01	
19	Temperature	(C)	55.56	55.56	55.55	55.55	
20	Vapour Fraction		0.8713	0.8713	0.8713	0.8713	
21	Heat Transferred	(kJ/h-m)	680.9	680.9	680.8	680.8	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2809	0.2039	0.2039	0.2039	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	5.520e-003	6.875e-003	6.875e-003	6.875e-003	
27	Static Gradient	(kPa/m)	9.475e-004	-6.892e-003	-6.892e-003	-6.892e-003	
28	Accel. Gradient	(kPa/m)	4.542e-007	-1.186e-009	-1.194e-009	-1.202e-009	
29	Liq. Reynolds		2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *	
30	Vap. Reynolds		9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *	
31	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
32	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	1.435e+004	1.435e+004	1.436e+004	1.436e+004	
36	Elevation	(m)	0.1280	0.1100	0.1160	0.1220	
37	Pressure	(bar)	91.01	91.01	91.01	91.01	
38	Temperature	(C)	55.55	55.55	55.55	55.54	
39	Vapour Fraction		0.8713	0.8713	0.8713	0.8713	
40	Heat Transferred	(kJ/h-m)	680.7	680.7	680.6	680.6	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2039	0.2039	0.2834	0.2834	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	6.875e-003	6.875e-003	5.321e-003	5.321e-003	
46	Static Gradient	(kPa/m)	-6.892e-003	-6.892e-003	2.861e-003	2.861e-003	
47	Accel. Gradient	(kPa/m)	-1.210e-009	-1.218e-009	5.747e-007	5.747e-007	
48	Liq. Reynolds		2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *	
49	Vap. Reynolds		9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *	
50	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
51	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	1.437e+004	1.437e+004	1.438e+004	1.438e+004	
55	Elevation	(m)	0.1280	0.1340	0.1400	0.1400	
56	Pressure	(bar)	91.01	91.01	91.01	91.01	
57	Temperature	(C)	55.54	55.54	55.54	55.54	
58	Vapour Fraction		0.8713	0.8713	0.8713	0.8713	
59	Heat Transferred	(kJ/h-m)	680.5	680.4	680.4	680.3	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.2834	0.2834	0.2834	0.3026	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	5.321e-003	5.321e-003	5.321e-003	5.869e-003	
65	Static Gradient	(kPa/m)	2.861e-003	2.861e-003	2.861e-003	0.0000	
66	Accel. Gradient	(kPa/m)	5.747e-007	5.747e-007	5.747e-007	4.122e-007	
67	Liq. Reynolds		2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *	
68	Vap. Reynolds		9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
14	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	1.439e+004	1.439e+004	1.440e+004	1.440e+004	
18	Elevation	(m)	0.1400	0.1400	0.1400	0.1400	
19	Pressure	(bar)	91.01	91.01	91.01	91.01	
20	Temperature	(C)	55.53	55.53	55.53	55.53	
21	Vapour Fraction		0.8713	0.8713	0.8713	0.8713	
22	Heat Transferred	(kJ/h-m)	680.3	680.2	680.2	680.1	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.3026	0.3026	0.3026	0.3026	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	5.869e-003	5.869e-003	5.869e-003	5.869e-003	
28	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000	
29	Accel. Gradient	(kPa/m)	4.122e-007	4.122e-007	4.122e-007	4.122e-007	
30	Liq. Reynolds		2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *	
31	Vap. Reynolds		9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *	
32	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
33	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	1.441e+004	1.441e+004	1.442e+004	1.442e+004	
37	Elevation	(m)	0.1520	0.1640	0.1760	0.1880	
38	Pressure	(bar)	91.01	91.01	91.01	91.01	
39	Temperature	(C)	55.53	55.52	55.52	55.52	
40	Vapour Fraction		0.8713	0.8713	0.8713	0.8713	
41	Heat Transferred	(kJ/h-m)	680.1	680.0	680.0	679.9	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2873	0.2873	0.2873	0.2873	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	5.034e-003	5.034e-003	5.034e-003	5.034e-003	
47	Static Gradient	(kPa/m)	5.776e-003	5.776e-003	5.776e-003	5.776e-003	
48	Accel. Gradient	(kPa/m)	7.592e-007	7.592e-007	7.592e-007	7.592e-007	
49	Liq. Reynolds		2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *	
50	Vap. Reynolds		9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *	
51	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
52	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	1.443e+004	1.443e+004	1.444e+004	1.444e+004	
56	Elevation	(m)	0.2000	0.2000	0.2000	0.2000	
57	Pressure	(bar)	91.01	91.01	91.01	91.01	
58	Temperature	(C)	55.52	55.52	55.51	55.51	
59	Vapour Fraction		0.8713	0.8713	0.8713	0.8713	
60	Heat Transferred	(kJ/h-m)	679.8	679.8	679.7	679.7	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2873	0.3026	0.3026	0.3026	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	5.034e-003	5.869e-003	5.869e-003	5.869e-003	
66	Static Gradient	(kPa/m)	5.776e-003	0.0000	0.0000	0.0000	
67	Accel. Gradient	(kPa/m)	7.592e-007	4.122e-007	4.122e-007	4.122e-007	
68	Liq. Reynolds		2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 268 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
14	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
15	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	1.445e+004	1.445e+004	1.446e+004	1.446e+004
19	Elevation (m)	0.2000	0.2000	0.2000	0.2000
20	Pressure (bar)	91.01	91.01	91.01	91.01
21	Temperature (C)	55.51	55.51	55.51	55.50
22	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
23	Heat Transferred (kJ/h-m)	679.6	679.6	679.5	679.5
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.3026	0.3026	0.3026	0.3026
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	5.869e-003	5.869e-003	5.869e-003	5.869e-003
29	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
30	Accel. Gradient (kPa/m)	4.122e-007	4.122e-007	4.122e-007	4.122e-007
31	Liq. Reynolds	2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *
32	Vap. Reynolds	9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
33	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
34	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	1.447e+004	1.447e+004	1.448e+004	1.448e+004
38	Elevation (m)	0.2000	0.2000	0.2000	0.2000
39	Pressure (bar)	91.01	91.01	91.01	91.01
40	Temperature (C)	55.50	55.50	55.50	55.50
41	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
42	Heat Transferred (kJ/h-m)	679.4	679.4	679.3	679.3
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.3026	0.3026	0.3026	0.3026
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	5.869e-003	5.869e-003	5.869e-003	5.869e-003
48	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
49	Accel. Gradient (kPa/m)	4.122e-007	4.122e-007	4.122e-007	4.122e-007
50	Liq. Reynolds	2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *
51	Vap. Reynolds	9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
52	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
53	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1.449e+004	1.449e+004	1.450e+004	1.450e+004
57	Elevation (m)	0.2000	0.2000	0.2000	0.2000
58	Pressure (bar)	91.00	91.00	91.00	91.00
59	Temperature (C)	55.49	55.49	55.49	55.49
60	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
61	Heat Transferred (kJ/h-m)	679.2	679.1	679.1	679.0
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.3026	0.3026	0.3026	0.3026
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	5.869e-003	5.869e-003	5.869e-003	5.869e-003
67	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
68	Accel. Gradient (kPa/m)	4.122e-007	4.121e-007	4.121e-007	4.121e-007
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *
14	Vap. Reynolds	9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
15	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
16	Vapour Velocity (m/s)	2.168	2.168	2.167	2.167
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	1.451e+004	1.451e+004	1.452e+004	1.452e+004
20	Elevation (m)	0.2020	0.2040	0.2060	0.2080
21	Pressure (bar)	91.00	91.00	91.00	91.00
22	Temperature (C)	55.49	55.48	55.48	55.48
23	Vapour Fraction	0.8713	0.8713	0.8713	0.8713
24	Heat Transferred (kJ/h-m)	679.0	678.9	678.9	678.8
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2809	0.2809	0.2809	0.2809
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	5.518e-003	5.518e-003	5.518e-003	5.518e-003
30	Static Gradient (kPa/m)	9.477e-004	9.477e-004	9.477e-004	9.477e-004
31	Accel. Gradient (kPa/m)	4.541e-007	4.541e-007	4.541e-007	4.541e-007
32	Liq. Reynolds	2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *
33	Vap. Reynolds	9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
34	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
35	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	1.453e+004	1.453e+004	1.454e+004	1.454e+004
39	Elevation (m)	0.2100	0.2080	0.2060	0.2040
40	Pressure (bar)	91.00	91.00	91.00	91.00
41	Temperature (C)	55.48	55.48	55.47	55.47
42	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
43	Heat Transferred (kJ/h-m)	678.8	678.7	678.7	678.6
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2809	0.2069	0.2069	0.2069
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	5.518e-003	6.850e-003	6.850e-003	6.849e-003
49	Static Gradient (kPa/m)	9.477e-004	-7.730e-004	-7.730e-004	-7.730e-004
50	Accel. Gradient (kPa/m)	4.540e-007	4.267e-007	4.267e-007	4.267e-007
51	Liq. Reynolds	2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *
52	Vap. Reynolds	9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
53	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
54	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	1.455e+004	1.455e+004	1.456e+004	1.456e+004
58	Elevation (m)	0.2020	0.2000	0.1940	0.1880
59	Pressure (bar)	91.00	91.00	91.00	91.00
60	Temperature (C)	55.47	55.47	55.47	55.46
61	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
62	Heat Transferred (kJ/h-m)	678.6	678.5	678.5	678.4
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2069	0.2069	0.2062	0.2062
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	6.849e-003	6.849e-003	6.855e-003	6.855e-003
68	Static Gradient (kPa/m)	-7.730e-004	-7.730e-004	-2.314e-003	-2.314e-003
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient	(kPa/m)	4.267e-007	4.267e-007	3.189e-007	3.189e-007	
14	Liq. Reynolds		2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *	
15	Vap. Reynolds		9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *	
16	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
17	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167	
18	Deposit Thickness	(mm)	---	---	---	---	
19	Deposit Volume	(m3)	---	---	---	---	
20	Length	(m)	1.457e+004	1.457e+004	1.458e+004	1.458e+004	
21	Elevation	(m)	0.1820	0.1760	0.1700	0.1680	
22	Pressure	(bar)	91.00	91.00	91.00	91.00	
23	Temperature	(C)	55.46	55.46	55.46	55.46	
24	Vapour Fraction		0.8712	0.8712	0.8712	0.8712	
25	Heat Transferred	(kJ/h-m)	678.4	678.3	678.3	678.2	
26	Flow Regime		Transition	Transition	Transition	Transition	
27	Water Flow Regime						
28	Liquid HoldUp		0.2062	0.2062	0.2062	0.2069	
29	Water HoldUp		---	---	---	---	
30	Friction Gradient	(kPa/m)	6.855e-003	6.855e-003	6.855e-003	6.849e-003	
31	Static Gradient	(kPa/m)	-2.314e-003	-2.314e-003	-2.314e-003	-7.730e-004	
32	Accel. Gradient	(kPa/m)	3.189e-007	3.189e-007	3.189e-007	4.267e-007	
33	Liq. Reynolds		2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *	
34	Vap. Reynolds		9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *	
35	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
36	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167	
37	Deposit Thickness	(mm)	---	---	---	---	
38	Deposit Volume	(m3)	---	---	---	---	
39	Length	(m)	1.459e+004	1.459e+004	1.460e+004	1.460e+004	
40	Elevation	(m)	0.1660	0.1640	0.1620	0.1600	
41	Pressure	(bar)	91.00	91.00	91.00	91.00	
42	Temperature	(C)	55.45	55.45	55.45	55.45	
43	Vapour Fraction		0.8712	0.8712	0.8712	0.8712	
44	Heat Transferred	(kJ/h-m)	678.1	678.1	678.0	678.0	
45	Flow Regime		Transition	Transition	Transition	Transition	
46	Water Flow Regime						
47	Liquid HoldUp		0.2069	0.2069	0.2069	0.2069	
48	Water HoldUp		---	---	---	---	
49	Friction Gradient	(kPa/m)	6.849e-003	6.849e-003	6.849e-003	6.849e-003	
50	Static Gradient	(kPa/m)	-7.730e-004	-7.730e-004	-7.730e-004	-7.730e-004	
51	Accel. Gradient	(kPa/m)	4.267e-007	4.267e-007	4.267e-007	4.267e-007	
52	Liq. Reynolds		2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *	
53	Vap. Reynolds		9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *	
54	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
55	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167	
56	Deposit Thickness	(mm)	---	---	---	---	
57	Deposit Volume	(m3)	---	---	---	---	
58	Length	(m)	1.461e+004	1.461e+004	1.462e+004	1.462e+004	
59	Elevation	(m)	0.1620	0.1640	0.1660	0.1680	
60	Pressure	(bar)	91.00	91.00	91.00	91.00	
61	Temperature	(C)	55.45	55.44	55.44	55.44	
62	Vapour Fraction		0.8712	0.8712	0.8712	0.8712	
63	Heat Transferred	(kJ/h-m)	677.9	677.9	677.8	677.8	
64	Flow Regime		Transition	Transition	Transition	Transition	
65	Water Flow Regime						
66	Liquid HoldUp		0.2809	0.2809	0.2809	0.2809	
67	Water HoldUp		---	---	---	---	
68	Friction Gradient	(kPa/m)	5.517e-003	5.517e-003	5.517e-003	5.517e-003	
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1				Case Name:	Simulation - 386MMSCFD_100barg.hsc	
2				Unit Set:	Mixed	
3		Epic Atlantic Limited				
Date/Time:			Mon Jun 13 15:02:44 2022			
4						
5						
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	9.478e-004	9.478e-004	9.478e-004	9.478e-004
14	Accel. Gradient	(kPa/m)	4.540e-007	4.540e-007	4.540e-007	4.539e-007
15	Liq. Reynolds		2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *
16	Vap. Reynolds		9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
17	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042
18	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	1.463e+004	1.463e+004	1.464e+004	1.464e+004
22	Elevation	(m)	0.1700	0.1640	0.1580	0.1520
23	Pressure	(bar)	91.00	91.00	91.00	91.00
24	Temperature	(C)	55.44	55.44	55.43	55.43
25	Vapour Fraction		0.8712	0.8712	0.8712	0.8712
26	Heat Transferred	(kJ/h-m)	677.7	677.7	677.6	677.6
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2809	0.2062	0.2062	0.2062
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	5.517e-003	6.855e-003	6.855e-003	6.855e-003
32	Static Gradient	(kPa/m)	9.478e-004	-2.358e-003	-2.358e-003	-2.358e-003
33	Accel. Gradient	(kPa/m)	4.539e-007	3.158e-007	3.158e-007	3.157e-007
34	Liq. Reynolds		2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *
35	Vap. Reynolds		9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
36	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042
37	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	1.465e+004	1.465e+004	1.466e+004	1.466e+004
41	Elevation	(m)	0.1460	0.1400	0.1480	0.1560
42	Pressure	(bar)	91.00	91.00	91.00	90.99
43	Temperature	(C)	55.43	55.43	55.43	55.42
44	Vapour Fraction		0.8712	0.8712	0.8712	0.8712
45	Heat Transferred	(kJ/h-m)	677.5	677.5	677.4	677.3
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2062	0.2062	0.2848	0.2848
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	6.855e-003	6.855e-003	5.222e-003	5.222e-003
51	Static Gradient	(kPa/m)	-2.358e-003	-2.358e-003	3.827e-003	3.827e-003
52	Accel. Gradient	(kPa/m)	3.157e-007	3.157e-007	6.354e-007	6.354e-007
53	Liq. Reynolds		2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *
54	Vap. Reynolds		9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
55	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042
56	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	1.467e+004	1.467e+004	1.468e+004	1.468e+004
60	Elevation	(m)	0.1640	0.1720	0.1800	0.1660
61	Pressure	(bar)	90.99	90.99	90.99	90.99
62	Temperature	(C)	55.42	55.42	55.42	55.42
63	Vapour Fraction		0.8712	0.8712	0.8712	0.8712
64	Heat Transferred	(kJ/h-m)	677.3	677.2	677.2	677.1
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2848	0.2848	0.2848	0.2047
68	Water HoldUp		---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	5.222e-003	5.222e-003	5.222e-003	6.866e-003	
14	Static Gradient	(kPa/m)	3.827e-003	3.827e-003	3.827e-003	-5.375e-003	
15	Accel. Gradient	(kPa/m)	6.354e-007	6.354e-007	6.354e-007	1.047e-007	
16	Liq. Reynolds		2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *	
17	Vap. Reynolds		9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *	
18	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
19	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	1.469e+004	1.469e+004	1.470e+004	1.470e+004	
23	Elevation	(m)	0.1520	0.1380	0.1240	0.1100	
24	Pressure	(bar)	90.99	90.99	90.99	90.99	
25	Temperature	(C)	55.41	55.41	55.41	55.41	
26	Vapour Fraction		0.8712	0.8712	0.8712	0.8712	
27	Heat Transferred	(kJ/h-m)	677.1	677.0	677.0	676.9	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.2047	0.2047	0.2047	0.2047	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	6.866e-003	6.866e-003	6.866e-003	6.866e-003	
33	Static Gradient	(kPa/m)	-5.375e-003	-5.375e-003	-5.375e-003	-5.375e-003	
34	Accel. Gradient	(kPa/m)	1.047e-007	1.047e-007	1.047e-007	1.047e-007	
35	Liq. Reynolds		2.148e+005 *	2.148e+005 *	2.148e+005 *	2.148e+005 *	
36	Vap. Reynolds		9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *	
37	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2043	
38	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	1.471e+004	1.471e+004	1.472e+004	1.472e+004	
42	Elevation	(m)	0.1120	0.1140	0.1160	0.1180	
43	Pressure	(bar)	90.99	90.99	90.99	90.99	
44	Temperature	(C)	55.41	55.40	55.40	55.40	
45	Vapour Fraction		0.8712	0.8712	0.8712	0.8712	
46	Heat Transferred	(kJ/h-m)	676.9	676.8	676.8	676.7	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.2810	0.2810	0.2810	0.2810	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	5.516e-003	5.516e-003	5.516e-003	5.516e-003	
52	Static Gradient	(kPa/m)	9.479e-004	9.479e-004	9.479e-004	9.479e-004	
53	Accel. Gradient	(kPa/m)	4.539e-007	4.539e-007	4.538e-007	4.538e-007	
54	Liq. Reynolds		2.148e+005 *	2.147e+005 *	2.147e+005 *	2.147e+005 *	
55	Vap. Reynolds		9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *	
56	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
57	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	1.473e+004	1.473e+004	1.474e+004	1.474e+004	
61	Elevation	(m)	0.1200	0.1140	0.1080	0.1020	
62	Pressure	(bar)	90.99	90.99	90.99	90.99	
63	Temperature	(C)	55.40	55.40	55.39	55.39	
64	Vapour Fraction		0.8712	0.8712	0.8712	0.8712	
65	Heat Transferred	(kJ/h-m)	676.7	676.6	676.6	676.5	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.2810	0.2062	0.2062	0.2062	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 273 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp		---	---	---
14	Friction Gradient (kPa/m)	5.516e-003	6.854e-003	6.854e-003	6.854e-003
15	Static Gradient (kPa/m)	9.479e-004	-2.314e-003	-2.314e-003	-2.314e-003
16	Accel. Gradient (kPa/m)	4.538e-007	3.187e-007	3.187e-007	3.187e-007
17	Liq. Reynolds	2.147e+005 *	2.147e+005 *	2.147e+005 *	2.147e+005 *
18	Vap. Reynolds	9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
19	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
20	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	1.475e+004	1.475e+004	1.476e+004	1.476e+004
24	Elevation (m)	9.600e-002	9.000e-002	8.600e-002	8.200e-002
25	Pressure (bar)	90.99	90.99	90.99	90.99
26	Temperature (C)	55.39	55.39	55.39	55.39
27	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
28	Heat Transferred (kJ/h-m)	676.5	676.4	676.4	676.3
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2062	0.2062	0.2066	0.2066
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	6.854e-003	6.853e-003	6.851e-003	6.850e-003
34	Static Gradient (kPa/m)	-2.314e-003	-2.314e-003	-1.545e-003	-1.545e-003
35	Accel. Gradient (kPa/m)	3.187e-007	3.187e-007	3.725e-007	3.725e-007
36	Liq. Reynolds	2.147e+005 *	2.147e+005 *	2.147e+005 *	2.147e+005 *
37	Vap. Reynolds	9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
38	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
39	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	1.477e+004	1.477e+004	1.478e+004	1.478e+004
43	Elevation (m)	7.800e-002	7.400e-002	7.000e-002	7.000e-002
44	Pressure (bar)	90.99	90.99	90.99	90.99
45	Temperature (C)	55.38	55.38	55.38	55.38
46	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
47	Heat Transferred (kJ/h-m)	676.3	676.2	676.1	676.1
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2066	0.2066	0.2066	0.3027
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	6.850e-003	6.850e-003	6.850e-003	5.867e-003
53	Static Gradient (kPa/m)	-1.545e-003	-1.545e-003	-1.545e-003	0.0000
54	Accel. Gradient (kPa/m)	3.725e-007	3.725e-007	3.725e-007	4.119e-007
55	Liq. Reynolds	2.147e+005 *	2.147e+005 *	2.147e+005 *	2.147e+005 *
56	Vap. Reynolds	9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
57	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
58	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	1.479e+004	1.479e+004	1.480e+004	1.480e+004
62	Elevation (m)	7.000e-002	7.000e-002	7.000e-002	7.000e-002
63	Pressure (bar)	90.99	90.99	90.99	90.99
64	Temperature (C)	55.38	55.37	55.37	55.37
65	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
66	Heat Transferred (kJ/h-m)	676.0	676.0	675.9	675.9
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 274 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.3027	0.3027	0.3027	0.3027
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	5.867e-003	5.867e-003	5.867e-003	5.867e-003
16	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
17	Accel. Gradient (kPa/m)	4.119e-007	4.119e-007	4.119e-007	4.119e-007
18	Liq. Reynolds	2.147e+005 *	2.147e+005 *	2.147e+005 *	2.147e+005 *
19	Vap. Reynolds	9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
20	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
21	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	1.481e+004	1.481e+004	1.482e+004	1.482e+004
25	Elevation (m)	6.600e-002	6.200e-002	5.800e-002	5.400e-002
26	Pressure (bar)	90.99	90.99	90.99	90.99
27	Temperature (C)	55.37	55.37	55.36	55.36
28	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
29	Heat Transferred (kJ/h-m)	675.8	675.8	675.7	675.7
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2066	0.2066	0.2066	0.2066
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.850e-003	6.850e-003	6.850e-003	6.850e-003
35	Static Gradient (kPa/m)	-1.545e-003	-1.545e-003	-1.545e-003	-1.545e-003
36	Accel. Gradient (kPa/m)	3.725e-007	3.725e-007	3.725e-007	3.725e-007
37	Liq. Reynolds	2.147e+005 *	2.147e+005 *	2.147e+005 *	2.147e+005 *
38	Vap. Reynolds	9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
39	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
40	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	1.483e+004	1.483e+004	1.484e+004	1.484e+004
44	Elevation (m)	5.000e-002	4.600e-002	4.200e-002	3.800e-002
45	Pressure (bar)	90.99	90.99	90.99	90.99
46	Temperature (C)	55.36	55.36	55.36	55.35
47	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
48	Heat Transferred (kJ/h-m)	675.6	675.6	675.5	675.5
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2066	0.2066	0.2066	0.2066
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.850e-003	6.850e-003	6.850e-003	6.850e-003
54	Static Gradient (kPa/m)	-1.545e-003	-1.534e-003	-1.534e-003	-1.534e-003
55	Accel. Gradient (kPa/m)	3.725e-007	3.732e-007	3.732e-007	3.732e-007
56	Liq. Reynolds	2.147e+005 *	2.147e+005 *	2.147e+005 *	2.147e+005 *
57	Vap. Reynolds	9.105e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
58	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
59	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	1.485e+004	1.485e+004	1.486e+004	1.486e+004
63	Elevation (m)	3.400e-002	3.000e-002	2.800e-002	2.600e-002
64	Pressure (bar)	90.98	90.98	90.98	90.98
65	Temperature (C)	55.35	55.35	55.35	55.35
66	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
67	Heat Transferred (kJ/h-m)	675.4	675.4	675.3	675.3
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 275 of 937

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2066	0.2066	0.2064	0.2064
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.850e-003	6.850e-003	6.851e-003	6.851e-003
17	Static Gradient (kPa/m)	-1.534e-003	-1.534e-003	-1.930e-003	-1.930e-003
18	Accel. Gradient (kPa/m)	3.732e-007	3.732e-007	3.455e-007	3.455e-007
19	Liq. Reynolds	2.147e+005 *	2.147e+005 *	2.147e+005 *	2.147e+005 *
20	Vap. Reynolds	9.106e+006 *	9.106e+006 *	9.106e+006 *	9.106e+006 *
21	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
22	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	1.486e+004	1.486e+004	1.486e+004	1.488e+004
26	Elevation (m)	2.400e-002	2.200e-002	2.000e-002	2.200e-002
27	Pressure (bar)	90.98	90.98	90.98	90.98
28	Temperature (C)	55.35	55.35	55.35	55.34
29	Vapour Fraction	0.8712	0.8712	0.8712	0.8712
30	Heat Transferred (kJ/h-m)	675.3	675.3	675.3	675.0
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2064	0.2064	0.2064	0.2800
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	6.851e-003	6.851e-003	6.851e-003	5.589e-003
36	Static Gradient (kPa/m)	-1.930e-003	-1.930e-003	-1.930e-003	2.365e-004
37	Accel. Gradient (kPa/m)	3.455e-007	3.455e-007	3.455e-007	4.090e-007
38	Liq. Reynolds	2.147e+005 *	2.147e+005 *	2.147e+005 *	2.147e+005 *
39	Vap. Reynolds	9.106e+006 *	9.106e+006 *	9.106e+006 *	9.106e+006 *
40	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
41	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	1.490e+004	1.492e+004	1.494e+004	1.496e+004
45	Elevation (m)	2.400e-002	2.600e-002	2.800e-002	3.000e-002
46	Pressure (bar)	90.98	90.98	90.98	90.98
47	Temperature (C)	55.33	55.32	55.31	55.31
48	Vapour Fraction	0.8712	0.8712	0.8712	0.8711
49	Heat Transferred (kJ/h-m)	674.8	674.6	674.4	674.2
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2801	0.2801	0.2801	0.2801
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	5.589e-003	5.589e-003	5.589e-003	5.589e-003
55	Static Gradient (kPa/m)	2.365e-004	2.365e-004	2.365e-004	2.365e-004
56	Accel. Gradient (kPa/m)	4.090e-007	4.090e-007	4.089e-007	4.089e-007
57	Liq. Reynolds	2.147e+005 *	2.147e+005 *	2.147e+005 *	2.147e+005 *
58	Vap. Reynolds	9.106e+006 *	9.106e+006 *	9.106e+006 *	9.106e+006 *
59	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
60	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	1.498e+004	1.500e+004	1.502e+004	1.504e+004
64	Elevation (m)	1.600e-002	2.000e-003	-1.200e-002	-2.600e-002
65	Pressure (bar)	90.98	90.98	90.97	90.97
66	Temperature (C)	55.30	55.29	55.28	55.27
67	Vapour Fraction	0.8711	0.8711	0.8711	0.8711
68	Heat Transferred (kJ/h-m)	674.0	673.8	673.6	673.4
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 276 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Pipe Segment: ELEP-Akabel Pipeline (continued)		
3					
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2067	0.2067	0.2068	0.2068
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	6.848e-003	6.848e-003	6.848e-003	6.847e-003
18	Static Gradient (kPa/m)	-1.352e-003	-1.352e-003	-1.352e-003	-1.352e-003
19	Accel. Gradient (kPa/m)	3.858e-007	3.858e-007	3.857e-007	3.857e-007
20	Liq. Reynolds	2.147e+005 *	2.147e+005 *	2.147e+005 *	2.147e+005 *
21	Vap. Reynolds	9.106e+006 *	9.106e+006 *	9.106e+006 *	9.106e+006 *
22	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
23	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	1.506e+004	1.508e+004	1.510e+004	1.512e+004
27	Elevation (m)	-4.000e-002	-3.400e-002	-2.800e-002	-2.200e-002
28	Pressure (bar)	90.97	90.97	90.97	90.97
29	Temperature (C)	55.27	55.26	55.25	55.24
30	Vapour Fraction	0.8711	0.8711	0.8711	0.8711
31	Heat Transferred (kJ/h-m)	673.1	672.9	672.7	672.5
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2068	0.2807	0.2807	0.2808
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	6.847e-003	5.537e-003	5.537e-003	5.537e-003
37	Static Gradient (kPa/m)	-1.352e-003	7.106e-004	7.107e-004	7.107e-004
38	Accel. Gradient (kPa/m)	3.857e-007	4.385e-007	4.385e-007	4.385e-007
39	Liq. Reynolds	2.147e+005 *	2.147e+005 *	2.147e+005 *	2.147e+005 *
40	Vap. Reynolds	9.106e+006 *	9.106e+006 *	9.106e+006 *	9.106e+006 *
41	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
42	Vapour Velocity (m/s)	2.166	2.166	2.166	2.165
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	1.514e+004	1.516e+004	1.518e+004	1.520e+004
46	Elevation (m)	-1.600e-002	-1.000e-002	-2.000e-002	-3.000e-002
47	Pressure (bar)	90.97	90.97	90.97	90.96
48	Temperature (C)	55.23	55.23	55.22	55.21
49	Vapour Fraction	0.8711	0.8711	0.8711	0.8711
50	Heat Transferred (kJ/h-m)	672.3	672.1	671.9	671.7
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2808	0.2808	0.2070	0.2070
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	5.536e-003	5.536e-003	6.845e-003	6.845e-003
56	Static Gradient (kPa/m)	7.107e-004	7.107e-004	-9.667e-004	-9.667e-004
57	Accel. Gradient (kPa/m)	4.385e-007	4.385e-007	4.126e-007	4.126e-007
58	Liq. Reynolds	2.147e+005 *	2.147e+005 *	2.147e+005 *	2.146e+005 *
59	Vap. Reynolds	9.106e+006 *	9.106e+006 *	9.106e+006 *	9.106e+006 *
60	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
61	Vapour Velocity (m/s)	2.165	2.165	2.165	2.165
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	1.522e+004	1.524e+004	1.526e+004	1.528e+004
65	Elevation (m)	-4.000e-002	-5.000e-002	-6.000e-002	-0.1240
66	Pressure (bar)	90.96	90.96	90.96	90.96
67	Temperature (C)	55.20	55.20	55.19	55.18
68	Vapour Fraction	0.8711	0.8711	0.8711	0.8711
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	671.5	671.2	671.0	670.8
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.2070	0.2070	0.2070	0.2045
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	6.844e-003	6.844e-003	6.844e-003	6.864e-003
19	Static Gradient (kPa/m)	-9.667e-004	-9.667e-004	-9.667e-004	-6.140e-003
20	Accel. Gradient (kPa/m)	4.125e-007	4.125e-007	4.125e-007	5.082e-008
21	Liq. Reynolds	2.146e+005 *	2.146e+005 *	2.146e+005 *	2.146e+005 *
22	Vap. Reynolds	9.106e+006 *	9.106e+006 *	9.106e+006 *	9.106e+006 *
23	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
24	Vapour Velocity (m/s)	2.165	2.165	2.165	2.165
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	1.530e+004	1.532e+004	1.534e+004	1.536e+004
28	Elevation (m)	-0.1880	-0.2520	-0.3160	-0.3800
29	Pressure (bar)	90.96	90.96	90.96	90.96
30	Temperature (C)	55.17	55.16	55.16	55.15
31	Vapour Fraction	0.8711	0.8711	0.8711	0.8711
32	Heat Transferred (kJ/h-m)	670.6	670.4	670.2	670.0
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.2045	0.2045	0.2045	0.2045
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	6.864e-003	6.863e-003	6.863e-003	6.863e-003
38	Static Gradient (kPa/m)	-6.140e-003	-6.140e-003	-6.140e-003	-6.140e-003
39	Accel. Gradient (kPa/m)	5.078e-008	5.075e-008	5.072e-008	5.069e-008
40	Liq. Reynolds	2.146e+005 *	2.146e+005 *	2.146e+005 *	2.146e+005 *
41	Vap. Reynolds	9.106e+006 *	9.106e+006 *	9.106e+006 *	9.106e+006 *
42	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
43	Vapour Velocity (m/s)	2.165	2.165	2.165	2.165
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	1.538e+004	1.540e+004	1.542e+004	1.544e+004
47	Elevation (m)	-1.014	-1.648	-2.282	-2.916
48	Pressure (bar)	90.97	90.98	90.99	91.00
49	Temperature (C)	55.15	55.14	55.14	55.14
50	Vapour Fraction	0.8710	0.8710	0.8710	0.8710
51	Heat Transferred (kJ/h-m)	670.0	669.9	669.8	669.7
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.1780	0.1780	0.1780	0.1780
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	7.123e-003	7.122e-003	7.121e-003	7.120e-003
57	Static Gradient (kPa/m)	-5.587e-002	-5.587e-002	-5.588e-002	-5.588e-002
58	Accel. Gradient (kPa/m)	-3.419e-006	-3.419e-006	-3.419e-006	-3.418e-006
59	Liq. Reynolds	2.146e+005 *	2.146e+005 *	2.146e+005 *	2.147e+005 *
60	Vap. Reynolds	9.106e+006 *	9.105e+006 *	9.105e+006 *	9.105e+006 *
61	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
62	Vapour Velocity (m/s)	2.164	2.164	2.164	2.163
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	1.546e+004	1.548e+004	1.550e+004	1.552e+004
66	Elevation (m)	-3.550	-2.914	-2.278	-1.642
67	Pressure (bar)	91.01	90.99	90.97	90.95
68	Temperature (C)	55.13	55.12	55.10	55.09
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8710	0.8710	0.8710	0.8710
14	Heat Transferred (kJ/h-m)	669.6	669.2	668.9	668.5
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.1780	0.3813	0.3813	0.3813
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	7.119e-003	5.996e-003	5.997e-003	5.998e-003
20	Static Gradient (kPa/m)	-5.589e-002	9.424e-002	9.423e-002	9.422e-002
21	Accel. Gradient (kPa/m)	-3.418e-006	7.027e-006	7.029e-006	7.032e-006
22	Liq. Reynolds	2.147e+005 *	2.146e+005 *	2.146e+005 *	2.146e+005 *
23	Vap. Reynolds	9.104e+006 *	9.105e+006 *	9.106e+006 *	9.106e+006 *
24	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
25	Vapour Velocity (m/s)	2.163	2.163	2.164	2.164
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	1.554e+004	1.556e+004	1.558e+004	1.560e+004
29	Elevation (m)	-1.006	-0.3700	-0.2920	-0.2140
30	Pressure (bar)	90.93	90.91	90.91	90.90
31	Temperature (C)	55.08	55.06	55.06	55.05
32	Vapour Fraction	0.8710	0.8710	0.8710	0.8710
33	Heat Transferred (kJ/h-m)	668.1	667.8	667.5	667.3
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.3812	0.3812	0.2923	0.2923
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	5.999e-003	6.000e-003	4.684e-003	4.684e-003
39	Static Gradient (kPa/m)	9.421e-002	9.420e-002	9.506e-003	9.506e-003
40	Accel. Gradient (kPa/m)	7.034e-006	7.036e-006	9.965e-007	9.965e-007
41	Liq. Reynolds	2.146e+005 *	2.145e+005 *	2.145e+005 *	2.145e+005 *
42	Vap. Reynolds	9.107e+006 *	9.107e+006 *	9.108e+006 *	9.108e+006 *
43	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
44	Vapour Velocity (m/s)	2.165	2.165	2.165	2.165
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	1.562e+004	1.564e+004	1.566e+004	1.568e+004
48	Elevation (m)	-0.1360	-5.800e-002	2.000e-002	-2.000e-003
49	Pressure (bar)	90.90	90.90	90.89	90.89
50	Temperature (C)	55.04	55.03	55.02	55.01
51	Vapour Fraction	0.8710	0.8710	0.8710	0.8710
52	Heat Transferred (kJ/h-m)	667.1	666.8	666.6	666.4
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2923	0.2923	0.2923	0.2064
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	4.684e-003	4.684e-003	4.684e-003	6.848e-003
58	Static Gradient (kPa/m)	9.506e-003	9.506e-003	9.506e-003	-2.124e-003
59	Accel. Gradient (kPa/m)	9.965e-007	9.966e-007	9.966e-007	3.318e-007
60	Liq. Reynolds	2.145e+005 *	2.145e+005 *	2.145e+005 *	2.145e+005 *
61	Vap. Reynolds	9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *
62	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
63	Vapour Velocity (m/s)	2.165	2.165	2.165	2.165
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	1.570e+004	1.572e+004	1.574e+004	1.576e+004
67	Elevation (m)	-2.400e-002	-4.600e-002	-6.800e-002	-9.000e-002
68	Pressure (bar)	90.89	90.89	90.89	90.89
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	55.01	55.00	54.99	54.98
14	Vapour Fraction	0.8710	0.8710	0.8710	0.8710
15	Heat Transferred (kJ/h-m)	666.2	666.0	665.8	665.6
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2064	0.2064	0.2065	0.2065
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	6.848e-003	6.848e-003	6.848e-003	6.847e-003
21	Static Gradient (kPa/m)	-2.124e-003	-2.124e-003	-2.124e-003	-2.124e-003
22	Accel. Gradient (kPa/m)	3.318e-007	3.318e-007	3.317e-007	3.317e-007
23	Liq. Reynolds	2.145e+005 *	2.145e+005 *	2.145e+005 *	2.145e+005 *
24	Vap. Reynolds	9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *
25	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
26	Vapour Velocity (m/s)	2.165	2.165	2.165	2.165
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	1.577e+004	1.577e+004	1.578e+004	1.578e+004
30	Elevation (m)	-9.000e-002	-9.000e-002	-9.000e-002	-9.000e-002
31	Pressure (bar)	90.89	90.89	90.89	90.89
32	Temperature (C)	54.98	54.98	54.98	54.97
33	Vapour Fraction	0.8710	0.8710	0.8710	0.8710
34	Heat Transferred (kJ/h-m)	665.5	665.5	665.4	665.4
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.3029	0.3029	0.3029	0.3029
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	5.864e-003	5.864e-003	5.864e-003	5.863e-003
40	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
41	Accel. Gradient (kPa/m)	4.118e-007	4.118e-007	4.117e-007	4.117e-007
42	Liq. Reynolds	2.145e+005 *	2.145e+005 *	2.145e+005 *	2.145e+005 *
43	Vap. Reynolds	9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *
44	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
45	Vapour Velocity (m/s)	2.165	2.165	2.165	2.164
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	1.579e+004	1.581e+004	1.583e+004	1.585e+004
49	Elevation (m)	-9.000e-002	-6.800e-002	-4.600e-002	-2.400e-002
50	Pressure (bar)	90.89	90.89	90.88	90.88
51	Temperature (C)	54.97	54.96	54.96	54.95
52	Vapour Fraction	0.8710	0.8710	0.8710	0.8710
53	Heat Transferred (kJ/h-m)	665.3	665.1	664.9	664.7
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.3029	0.2834	0.2834	0.2834
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	5.863e-003	5.335e-003	5.335e-003	5.335e-003
59	Static Gradient (kPa/m)	0.0000	2.624e-003	2.624e-003	2.624e-003
60	Accel. Gradient (kPa/m)	4.117e-007	5.589e-007	5.589e-007	5.589e-007
61	Liq. Reynolds	2.145e+005 *	2.145e+005 *	2.145e+005 *	2.144e+005 *
62	Vap. Reynolds	9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *
63	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
64	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	1.587e+004	1.589e+004	1.589e+004	1.590e+004
68	Elevation (m)	-2.000e-003	2.000e-002	3.600e-002	5.200e-002
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Pressure (bar)	90.88	90.88	90.88	90.88
14	Temperature (C)	54.94	54.93	54.93	54.93
15	Vapour Fraction	0.8710	0.8710	0.8710	0.8710
16	Heat Transferred (kJ/h-m)	664.5	664.2	664.2	664.2
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.2834	0.2834	0.2942	0.2942
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	5.335e-003	5.335e-003	4.833e-003	4.833e-003
22	Static Gradient (kPa/m)	2.624e-003	2.624e-003	1.095e-002	1.095e-002
23	Accel. Gradient (kPa/m)	5.589e-007	5.589e-007	1.108e-006	1.108e-006
24	Liq. Reynolds	2.144e+005 *	2.144e+005 *	2.144e+005 *	2.144e+005 *
25	Vap. Reynolds	9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *
26	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
27	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	1.590e+004	1.590e+004	1.591e+004	1.591e+004
31	Elevation (m)	6.800e-002	8.400e-002	1.000e-001	0.1020
32	Pressure (bar)	90.88	90.88	90.88	90.88
33	Temperature (C)	54.93	54.93	54.93	54.92
34	Vapour Fraction	0.8710	0.8709	0.8709	0.8709
35	Heat Transferred (kJ/h-m)	664.1	664.1	664.0	664.0
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.2942	0.2942	0.2942	0.2815
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	4.833e-003	4.833e-003	4.833e-003	5.483e-003
41	Static Gradient (kPa/m)	1.095e-002	1.095e-002	1.095e-002	1.187e-003
42	Accel. Gradient (kPa/m)	1.108e-006	1.108e-006	1.108e-006	4.684e-007
43	Liq. Reynolds	2.144e+005 *	2.144e+005 *	2.144e+005 *	2.144e+005 *
44	Vap. Reynolds	9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *
45	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
46	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	1.592e+004	1.592e+004	1.592e+004	1.593e+004
50	Elevation (m)	0.1040	0.1060	0.1080	0.1100
51	Pressure (bar)	90.88	90.88	90.88	90.88
52	Temperature (C)	54.92	54.92	54.92	54.92
53	Vapour Fraction	0.8709	0.8709	0.8709	0.8709
54	Heat Transferred (kJ/h-m)	664.0	663.9	663.9	663.8
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.2815	0.2815	0.2815	0.2815
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	5.483e-003	5.483e-003	5.483e-003	5.483e-003
60	Static Gradient (kPa/m)	1.187e-003	1.187e-003	1.187e-003	1.187e-003
61	Accel. Gradient (kPa/m)	4.684e-007	4.684e-007	4.683e-007	4.683e-007
62	Liq. Reynolds	2.144e+005 *	2.144e+005 *	2.144e+005 *	2.144e+005 *
63	Vap. Reynolds	9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *
64	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
65	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	1.593e+004	1.593e+004	1.593e+004	1.594e+004
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	0.1220	0.1340	0.1460	0.1580
14	Pressure (bar)	90.88	90.88	90.87	90.87
15	Temperature (C)	54.92	54.92	54.91	54.91
16	Vapour Fraction	0.8709	0.8709	0.8709	0.8709
17	Heat Transferred (kJ/h-m)	663.8	663.8	663.8	663.7
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2991	0.2991	0.2991	0.2991
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	5.687e-003	5.687e-003	5.687e-003	5.687e-003
23	Static Gradient (kPa/m)	1.487e-002	1.487e-002	1.487e-002	1.487e-002
24	Accel. Gradient (kPa/m)	1.443e-006	1.443e-006	1.443e-006	1.443e-006
25	Liq. Reynolds	2.144e+005 *	2.144e+005 *	2.144e+005 *	2.144e+005 *
26	Vap. Reynolds	9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *
27	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
28	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	1.594e+004	1.594e+004	1.595e+004	1.595e+004
32	Elevation (m)	0.1700	0.1720	0.1740	0.1760
33	Pressure (bar)	90.87	90.87	90.87	90.87
34	Temperature (C)	54.91	54.91	54.91	54.91
35	Vapour Fraction	0.8709	0.8709	0.8709	0.8709
36	Heat Transferred (kJ/h-m)	663.7	663.7	663.6	663.6
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2991	0.2815	0.2815	0.2815
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	5.687e-003	5.483e-003	5.483e-003	5.483e-003
42	Static Gradient (kPa/m)	1.487e-002	1.187e-003	1.187e-003	1.187e-003
43	Accel. Gradient (kPa/m)	1.443e-006	4.684e-007	4.684e-007	4.683e-007
44	Liq. Reynolds	2.144e+005 *	2.144e+005 *	2.144e+005 *	2.144e+005 *
45	Vap. Reynolds	9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *
46	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
47	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	1.595e+004	1.596e+004	1.596e+004	1.597e+004
51	Elevation (m)	0.1780	0.1800	0.1840	0.1880
52	Pressure (bar)	90.87	90.87	90.87	90.87
53	Temperature (C)	54.91	54.91	54.90	54.90
54	Vapour Fraction	0.8709	0.8709	0.8709	0.8709
55	Heat Transferred (kJ/h-m)	663.5	663.5	663.5	663.4
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2815	0.2815	0.2825	0.2825
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	5.483e-003	5.482e-003	5.408e-003	5.408e-003
61	Static Gradient (kPa/m)	1.187e-003	1.187e-003	1.904e-003	1.904e-003
62	Accel. Gradient (kPa/m)	4.683e-007	4.683e-007	5.134e-007	5.134e-007
63	Liq. Reynolds	2.144e+005 *	2.144e+005 *	2.144e+005 *	2.144e+005 *
64	Vap. Reynolds	9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *
65	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
66	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3							Unit Set: Mixed
4							Date/Time: Mon Jun 13 15:02:44 2022
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Length (m)	1.597e+004	1.598e+004	1.598e+004	1.599e+004		
14	Elevation (m)	0.1920	0.1960	0.2000	0.2060		
15	Pressure (bar)	90.87	90.87	90.87	90.87		
16	Temperature (C)	54.90	54.90	54.90	54.89		
17	Vapour Fraction	0.8709	0.8709	0.8709	0.8709		
18	Heat Transferred (kJ/h-m)	663.3	663.3	663.2	663.2		
19	Flow Regime	Transition	Transition	Transition	Transition		
20	Water Flow Regime						
21	Liquid HoldUp	0.2825	0.2825	0.2825	0.2838		
22	Water HoldUp	---	---	---	---		
23	Friction Gradient (kPa/m)	5.408e-003	5.408e-003	5.408e-003	5.310e-003		
24	Static Gradient (kPa/m)	1.904e-003	1.904e-003	1.904e-003	2.865e-003		
25	Accel. Gradient (kPa/m)	5.134e-007	5.134e-007	5.134e-007	5.740e-007		
26	Liq. Reynolds	2.144e+005 *	2.144e+005 *	2.144e+005 *	2.144e+005 *		
27	Vap. Reynolds	9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *		
28	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043		
29	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164		
30	Deposit Thickness (mm)	---	---	---	---		
31	Deposit Volume (m3)	---	---	---	---		
32	Length (m)	1.599e+004	1.600e+004	1.600e+004	1.601e+004		
33	Elevation (m)	0.2120	0.2180	0.2240	0.2300		
34	Pressure (bar)	90.87	90.87	90.87	90.87		
35	Temperature (C)	54.89	54.89	54.89	54.89		
36	Vapour Fraction	0.8709	0.8709	0.8709	0.8709		
37	Heat Transferred (kJ/h-m)	663.1	663.1	663.0	663.0		
38	Flow Regime	Transition	Transition	Transition	Transition		
39	Water Flow Regime						
40	Liquid HoldUp	0.2838	0.2838	0.2838	0.2838		
41	Water HoldUp	---	---	---	---		
42	Friction Gradient (kPa/m)	5.310e-003	5.310e-003	5.310e-003	5.310e-003		
43	Static Gradient (kPa/m)	2.865e-003	2.865e-003	2.865e-003	2.865e-003		
44	Accel. Gradient (kPa/m)	5.740e-007	5.740e-007	5.740e-007	5.740e-007		
45	Liq. Reynolds	2.144e+005 *	2.144e+005 *	2.144e+005 *	2.144e+005 *		
46	Vap. Reynolds	9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *		
47	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043		
48	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164		
49	Deposit Thickness (mm)	---	---	---	---		
50	Deposit Volume (m3)	---	---	---	---		
51	Length (m)	1.601e+004	1.602e+004	1.602e+004	1.603e+004		
52	Elevation (m)	0.2120	0.1940	0.1760	0.1580		
53	Pressure (bar)	90.87	90.87	90.87	90.87		
54	Temperature (C)	54.88	54.88	54.88	54.88		
55	Vapour Fraction	0.8709	0.8709	0.8709	0.8709		
56	Heat Transferred (kJ/h-m)	662.9	662.9	662.8	662.8		
57	Flow Regime	Transition	Transition	Transition	Transition		
58	Water Flow Regime						
59	Liquid HoldUp	0.2042	0.2042	0.2042	0.2042		
60	Water HoldUp	---	---	---	---		
61	Friction Gradient (kPa/m)	6.864e-003	6.864e-003	6.864e-003	6.864e-003		
62	Static Gradient (kPa/m)	-6.902e-003	-6.902e-003	-6.902e-003	-6.902e-003		
63	Accel. Gradient (kPa/m)	-2.664e-009	-2.672e-009	-2.680e-009	-2.688e-009		
64	Liq. Reynolds	2.144e+005 *	2.144e+005 *	2.144e+005 *	2.144e+005 *		
65	Vap. Reynolds	9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *		
66	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043		
67	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164		
68	Deposit Thickness (mm)	---	---	---	---		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	1.603e+004	1.604e+004	1.604e+004	1.605e+004		
15	Elevation (m)	0.1400	0.1260	0.1120	9.800e-002		
16	Pressure (bar)	90.87	90.87	90.87	90.87		
17	Temperature (C)	54.88	54.87	54.87	54.87		
18	Vapour Fraction	0.8709	0.8709	0.8709	0.8709		
19	Heat Transferred (kJ/h-m)	662.7	662.7	662.6	662.6		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	0.2042	0.2049	0.2049	0.2049		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	6.864e-003	6.858e-003	6.858e-003	6.858e-003		
25	Static Gradient (kPa/m)	-6.903e-003	-5.381e-003	-5.381e-003	-5.381e-003		
26	Accel. Gradient (kPa/m)	-2.696e-009	1.037e-007	1.037e-007	1.037e-007		
27	Liq. Reynolds	2.144e+005 *	2.144e+005 *	2.144e+005 *	2.144e+005 *		
28	Vap. Reynolds	9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *		
29	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043		
30	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	1.605e+004	1.606e+004	1.606e+004	1.607e+004		
34	Elevation (m)	8.400e-002	7.000e-002	5.200e-002	3.400e-002		
35	Pressure (bar)	90.87	90.87	90.87	90.87		
36	Temperature (C)	54.87	54.87	54.86	54.86		
37	Vapour Fraction	0.8709	0.8709	0.8709	0.8709		
38	Heat Transferred (kJ/h-m)	662.5	662.5	662.5	662.4		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.2049	0.2049	0.2042	0.2042		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	6.858e-003	6.858e-003	6.864e-003	6.864e-003		
44	Static Gradient (kPa/m)	-5.381e-003	-5.381e-003	-6.903e-003	-6.903e-003		
45	Accel. Gradient (kPa/m)	1.037e-007	1.037e-007	-2.740e-009	-2.748e-009		
46	Liq. Reynolds	2.144e+005 *	2.144e+005 *	2.144e+005 *	2.144e+005 *		
47	Vap. Reynolds	9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *		
48	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043		
49	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	1.607e+004	1.608e+004	1.608e+004	1.609e+004		
53	Elevation (m)	1.600e-002	-2.000e-003	-2.000e-002	-3.400e-002		
54	Pressure (bar)	90.87	90.87	90.87	90.87		
55	Temperature (C)	54.86	54.86	54.86	54.86		
56	Vapour Fraction	0.8709	0.8709	0.8709	0.8709		
57	Heat Transferred (kJ/h-m)	662.4	662.3	662.3	662.2		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.2042	0.2042	0.2042	0.2049		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	6.864e-003	6.864e-003	6.864e-003	6.857e-003		
63	Static Gradient (kPa/m)	-6.903e-003	-6.903e-003	-6.903e-003	-5.364e-003		
64	Accel. Gradient (kPa/m)	-2.756e-009	-2.763e-009	-2.771e-009	1.048e-007		
65	Liq. Reynolds	2.144e+005 *	2.144e+005 *	2.144e+005 *	2.144e+005 *		
66	Vap. Reynolds	9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *		
67	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043		
68	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022	
6						
7						
8						
9			PROFILES			
10						
11			Pipe Table			
12						
13	Deposit Thickness	(mm)	---	---	---	---
14	Deposit Volume	(m3)	---	---	---	---
15	Length	(m)	1.609e+004	1.610e+004	1.610e+004	1.611e+004
16	Elevation	(m)	-4.800e-002	-6.200e-002	-7.600e-002	-9.000e-002
17	Pressure	(bar)	90.87	90.87	90.87	90.87
18	Temperature	(C)	54.85	54.85	54.85	54.85
19	Vapour Fraction		0.8709	0.8709	0.8709	0.8709
20	Heat Transferred	(kJ/h-m)	662.2	662.1	662.1	662.0
21	Flow Regime		Transition	Transition	Transition	Transition
22	Water Flow Regime					
23	Liquid HoldUp		0.2049	0.2049	0.2049	0.2049
24	Water HoldUp		---	---	---	---
25	Friction Gradient	(kPa/m)	6.857e-003	6.857e-003	6.857e-003	6.857e-003
26	Static Gradient	(kPa/m)	-5.364e-003	-5.364e-003	-5.364e-003	-5.365e-003
27	Accel. Gradient	(kPa/m)	1.048e-007	1.048e-007	1.048e-007	1.048e-007
28	Liq. Reynolds		2.144e+005 *	2.144e+005 *	2.144e+005 *	2.144e+005 *
29	Vap. Reynolds		9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *
30	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
31	Vapour Velocity	(m/s)	2.164	2.164	2.164	2.164
32	Deposit Thickness	(mm)	---	---	---	---
33	Deposit Volume	(m3)	---	---	---	---
34	Length	(m)	1.611e+004	1.612e+004	1.612e+004	1.613e+004
35	Elevation	(m)	-0.1060	-0.1220	-0.1380	-0.1540
36	Pressure	(bar)	90.87	90.87	90.87	90.87
37	Temperature	(C)	54.85	54.84	54.84	54.84
38	Vapour Fraction		0.8709	0.8709	0.8709	0.8709
39	Heat Transferred	(kJ/h-m)	662.0	661.9	661.9	661.8
40	Flow Regime		Transition	Transition	Transition	Transition
41	Water Flow Regime					
42	Liquid HoldUp		0.2046	0.2046	0.2046	0.2046
43	Water HoldUp		---	---	---	---
44	Friction Gradient	(kPa/m)	6.860e-003	6.860e-003	6.860e-003	6.860e-003
45	Static Gradient	(kPa/m)	-6.143e-003	-6.143e-003	-6.143e-003	-6.143e-003
46	Accel. Gradient	(kPa/m)	5.033e-008	5.032e-008	5.031e-008	5.030e-008
47	Liq. Reynolds		2.144e+005 *	2.144e+005 *	2.144e+005 *	2.144e+005 *
48	Vap. Reynolds		9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *
49	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
50	Vapour Velocity	(m/s)	2.164	2.163	2.163	2.163
51	Deposit Thickness	(mm)	---	---	---	---
52	Deposit Volume	(m3)	---	---	---	---
53	Length	(m)	1.613e+004	1.614e+004	1.614e+004	1.615e+004
54	Elevation	(m)	-0.1700	-0.1920	-0.2140	-0.2360
55	Pressure	(bar)	90.87	90.87	90.87	90.87
56	Temperature	(C)	54.84	54.84	54.84	54.83
57	Vapour Fraction		0.8709	0.8709	0.8709	0.8709
58	Heat Transferred	(kJ/h-m)	661.8	661.7	661.7	661.6
59	Flow Regime		Transition	Transition	Transition	Transition
60	Water Flow Regime					
61	Liquid HoldUp		0.2046	0.2035	0.2035	0.2035
62	Water HoldUp		---	---	---	---
63	Friction Gradient	(kPa/m)	6.860e-003	6.869e-003	6.869e-003	6.869e-003
64	Static Gradient	(kPa/m)	-6.144e-003	-8.418e-003	-8.418e-003	-8.418e-003
65	Accel. Gradient	(kPa/m)	5.030e-008	-1.088e-007	-1.088e-007	-1.088e-007
66	Liq. Reynolds		2.144e+005 *	2.144e+005 *	2.144e+005 *	2.144e+005 *
67	Vap. Reynolds		9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *
68	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	1.615e+004	1.616e+004	1.616e+004	1.617e+004	
17	Elevation	(m)	-0.2580	-0.2800	-0.2920	-0.3040	
18	Pressure	(bar)	90.87	90.87	90.87	90.87	
19	Temperature	(C)	54.83	54.83	54.83	54.83	
20	Vapour Fraction		0.8709	0.8709	0.8709	0.8709	
21	Heat Transferred	(kJ/h-m)	661.6	661.5	661.5	661.4	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2035	0.2035	0.2053	0.2053	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	6.869e-003	6.869e-003	6.854e-003	6.854e-003	
27	Static Gradient	(kPa/m)	-8.419e-003	-8.419e-003	-4.618e-003	-4.618e-003	
28	Accel. Gradient	(kPa/m)	-1.088e-007	-1.088e-007	1.569e-007	1.569e-007	
29	Liq. Reynolds		2.144e+005 *	2.144e+005 *	2.144e+005 *	2.144e+005 *	
30	Vap. Reynolds		9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *	
31	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
32	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	1.617e+004	1.618e+004	1.618e+004	1.619e+004	
36	Elevation	(m)	-0.3160	-0.3280	-0.3400	-0.3640	
37	Pressure	(bar)	90.87	90.87	90.87	90.87	
38	Temperature	(C)	54.82	54.82	54.82	54.82	
39	Vapour Fraction		0.8709	0.8709	0.8709	0.8709	
40	Heat Transferred	(kJ/h-m)	661.4	661.3	661.3	661.2	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2053	0.2053	0.2053	0.2031	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	6.853e-003	6.853e-003	6.853e-003	6.871e-003	
46	Static Gradient	(kPa/m)	-4.619e-003	-4.619e-003	-4.619e-003	-9.174e-003	
47	Accel. Gradient	(kPa/m)	1.569e-007	1.569e-007	1.569e-007	-1.616e-007	
48	Liq. Reynolds		2.144e+005 *	2.144e+005 *	2.144e+005 *	2.144e+005 *	
49	Vap. Reynolds		9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *	
50	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
51	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	1.619e+004	1.620e+004	1.620e+004	1.621e+004	
55	Elevation	(m)	-0.3880	-0.4120	-0.4360	-0.4600	
56	Pressure	(bar)	90.87	90.87	90.87	90.87	
57	Temperature	(C)	54.82	54.82	54.81	54.81	
58	Vapour Fraction		0.8709	0.8709	0.8709	0.8709	
59	Heat Transferred	(kJ/h-m)	661.2	661.1	661.1	661.0	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.2031	0.2031	0.2031	0.2031	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	6.871e-003	6.871e-003	6.871e-003	6.871e-003	
65	Static Gradient	(kPa/m)	-9.174e-003	-9.174e-003	-9.174e-003	-9.174e-003	
66	Accel. Gradient	(kPa/m)	-1.616e-007	-1.616e-007	-1.616e-007	-1.616e-007	
67	Liq. Reynolds		2.144e+005 *	2.144e+005 *	2.144e+005 *	2.144e+005 *	
68	Vap. Reynolds		9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *	
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
14	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163
15	Deposit Thickness	(mm)	---	---	---	---
16	Deposit Volume	(m3)	---	---	---	---
17	Length	(m)	1.621e+004	1.622e+004	1.622e+004	1.623e+004
18	Elevation	(m)	-0.3360	-0.2120	-8.800e-002	3.600e-002
19	Pressure	(bar)	90.86	90.86	90.86	90.85
20	Temperature	(C)	54.81	54.81	54.80	54.80
21	Vapour Fraction		0.8709	0.8709	0.8709	0.8709
22	Heat Transferred	(kJ/h-m)	661.0	660.9	660.8	660.7
23	Flow Regime		Transition	Transition	Transition	Transition
24	Water Flow Regime					
25	Liquid HoldUp		0.3591	0.3591	0.3591	0.3591
26	Water HoldUp		---	---	---	---
27	Friction Gradient	(kPa/m)	6.028e-003	6.028e-003	6.028e-003	6.028e-003
28	Static Gradient	(kPa/m)	7.025e-002	7.025e-002	7.025e-002	7.025e-002
29	Accel. Gradient	(kPa/m)	5.355e-006	5.355e-006	5.355e-006	5.356e-006
30	Liq. Reynolds		2.144e+005 *	2.144e+005 *	2.143e+005 *	2.143e+005 *
31	Vap. Reynolds		9.109e+006 *	9.109e+006 *	9.109e+006 *	9.109e+006 *
32	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
33	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163
34	Deposit Thickness	(mm)	---	---	---	---
35	Deposit Volume	(m3)	---	---	---	---
36	Length	(m)	1.623e+004	1.624e+004	1.624e+004	1.625e+004
37	Elevation	(m)	0.1600	0.1560	0.1520	0.1480
38	Pressure	(bar)	90.85	90.85	90.85	90.85
39	Temperature	(C)	54.80	54.79	54.79	54.79
40	Vapour Fraction		0.8709	0.8709	0.8709	0.8709
41	Heat Transferred	(kJ/h-m)	660.6	660.6	660.5	660.5
42	Flow Regime		Transition	Transition	Transition	Transition
43	Water Flow Regime					
44	Liquid HoldUp		0.3591	0.2068	0.2068	0.2068
45	Water HoldUp		---	---	---	---
46	Friction Gradient	(kPa/m)	6.028e-003	6.842e-003	6.842e-003	6.842e-003
47	Static Gradient	(kPa/m)	7.025e-002	-1.547e-003	-1.547e-003	-1.547e-003
48	Accel. Gradient	(kPa/m)	5.356e-006	3.719e-007	3.718e-007	3.718e-007
49	Liq. Reynolds		2.143e+005 *	2.143e+005 *	2.143e+005 *	2.143e+005 *
50	Vap. Reynolds		9.109e+006 *	9.109e+006 *	9.109e+006 *	9.109e+006 *
51	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
52	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163
53	Deposit Thickness	(mm)	---	---	---	---
54	Deposit Volume	(m3)	---	---	---	---
55	Length	(m)	1.625e+004	1.626e+004	1.626e+004	1.627e+004
56	Elevation	(m)	0.1440	0.1400	0.1360	0.1320
57	Pressure	(bar)	90.85	90.85	90.85	90.85
58	Temperature	(C)	54.79	54.79	54.78	54.78
59	Vapour Fraction		0.8709	0.8709	0.8709	0.8709
60	Heat Transferred	(kJ/h-m)	660.4	660.4	660.3	660.3
61	Flow Regime		Transition	Transition	Transition	Transition
62	Water Flow Regime					
63	Liquid HoldUp		0.2068	0.2068	0.2068	0.2068
64	Water HoldUp		---	---	---	---
65	Friction Gradient	(kPa/m)	6.842e-003	6.842e-003	6.842e-003	6.842e-003
66	Static Gradient	(kPa/m)	-1.547e-003	-1.547e-003	-1.547e-003	-1.547e-003
67	Accel. Gradient	(kPa/m)	3.718e-007	3.718e-007	3.718e-007	3.718e-007
68	Liq. Reynolds		2.143e+005 *	2.143e+005 *	2.143e+005 *	2.143e+005 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.109e+006 *	9.109e+006 *	9.109e+006 *	9.109e+006 *
14	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
15	Vapour Velocity (m/s)	2.163	2.163	2.163	2.163
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	1.627e+004	1.628e+004	1.628e+004	1.629e+004
19	Elevation (m)	0.1280	0.1240	0.1200	0.1060
20	Pressure (bar)	90.85	90.85	90.85	90.85
21	Temperature (C)	54.78	54.78	54.78	54.78
22	Vapour Fraction	0.8709	0.8709	0.8709	0.8709
23	Heat Transferred (kJ/h-m)	660.2	660.2	660.1	660.1
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2068	0.2068	0.2068	0.2050
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.842e-003	6.842e-003	6.842e-003	6.857e-003
29	Static Gradient (kPa/m)	-1.547e-003	-1.547e-003	-1.547e-003	-5.382e-003
30	Accel. Gradient (kPa/m)	3.718e-007	3.718e-007	3.718e-007	1.035e-007
31	Liq. Reynolds	2.143e+005 *	2.143e+005 *	2.143e+005 *	2.143e+005 *
32	Vap. Reynolds	9.109e+006 *	9.109e+006 *	9.109e+006 *	9.109e+006 *
33	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
34	Vapour Velocity (m/s)	2.163	2.163	2.163	2.163
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	1.629e+004	1.630e+004	1.630e+004	1.631e+004
38	Elevation (m)	9.200e-002	7.800e-002	6.400e-002	5.000e-002
39	Pressure (bar)	90.85	90.85	90.85	90.85
40	Temperature (C)	54.77	54.77	54.77	54.77
41	Vapour Fraction	0.8709	0.8709	0.8709	0.8709
42	Heat Transferred (kJ/h-m)	660.0	660.0	659.9	659.9
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2050	0.2050	0.2050	0.2050
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.857e-003	6.856e-003	6.856e-003	6.856e-003
48	Static Gradient (kPa/m)	-5.382e-003	-5.382e-003	-5.382e-003	-5.382e-003
49	Accel. Gradient (kPa/m)	1.035e-007	1.035e-007	1.035e-007	1.035e-007
50	Liq. Reynolds	2.143e+005 *	2.143e+005 *	2.143e+005 *	2.143e+005 *
51	Vap. Reynolds	9.109e+006 *	9.109e+006 *	9.109e+006 *	9.109e+006 *
52	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
53	Vapour Velocity (m/s)	2.163	2.163	2.163	2.163
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1.631e+004	1.631e+004	1.631e+004	1.632e+004
57	Elevation (m)	-4.200e-002	-0.1340	-0.2260	-0.3180
58	Pressure (bar)	90.85	90.85	90.85	90.85
59	Temperature (C)	54.77	54.77	54.77	54.77
60	Vapour Fraction	0.8709	0.8709	0.8709	0.8709
61	Heat Transferred (kJ/h-m)	659.9	659.9	659.9	659.9
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1648	0.1648	0.1648	0.1648
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	7.293e-003	7.293e-003	7.293e-003	7.293e-003
67	Static Gradient (kPa/m)	-7.751e-002	-7.751e-002	-7.751e-002	-7.751e-002
68	Accel. Gradient (kPa/m)	-4.930e-006	-4.929e-006	-4.929e-006	-4.929e-006
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		2.143e+005 *	2.143e+005 *	2.143e+005 *	2.143e+005 *	
14	Vap. Reynolds		9.109e+006 *	9.109e+006 *	9.109e+006 *	9.109e+006 *	
15	Liquid Velocity (m/s)		0.2043	0.2043	0.2043	0.2043	
16	Vapour Velocity (m/s)		2.163	2.163	2.163	2.163	
17	Deposit Thickness (mm)		---	---	---	---	
18	Deposit Volume (m3)		---	---	---	---	
19	Length (m)		1.632e+004	1.632e+004	1.632e+004	1.633e+004	
20	Elevation (m)		-0.4100	-0.3280	-0.2460	-0.1640	
21	Pressure (bar)		90.85	90.85	90.85	90.85	
22	Temperature (C)		54.77	54.77	54.76	54.76	
23	Vapour Fraction		0.8709	0.8709	0.8709	0.8709	
24	Heat Transferred (kJ/h-m)		659.9	659.8	659.8	659.7	
25	Flow Regime		Transition	Transition	Transition	Transition	
26	Water Flow Regime						
27	Liquid HoldUp		0.1648	0.3672	0.3672	0.3672	
28	Water HoldUp		---	---	---	---	
29	Friction Gradient (kPa/m)		7.293e-003	6.016e-003	6.016e-003	6.016e-003	
30	Static Gradient (kPa/m)		-7.751e-002	7.874e-002	7.874e-002	7.873e-002	
31	Accel. Gradient (kPa/m)		-4.929e-006	5.950e-006	5.950e-006	5.950e-006	
32	Liq. Reynolds		2.143e+005 *	2.143e+005 *	2.143e+005 *	2.143e+005 *	
33	Vap. Reynolds		9.109e+006 *	9.109e+006 *	9.109e+006 *	9.109e+006 *	
34	Liquid Velocity (m/s)		0.2043	0.2043	0.2043	0.2043	
35	Vapour Velocity (m/s)		2.163	2.163	2.163	2.163	
36	Deposit Thickness (mm)		---	---	---	---	
37	Deposit Volume (m3)		---	---	---	---	
38	Length (m)		1.633e+004	1.633e+004	1.634e+004	1.634e+004	
39	Elevation (m)		-8.200e-002	-1.110e-016	8.000e-003	1.600e-002	
40	Pressure (bar)		90.84	90.84	90.84	90.84	
41	Temperature (C)		54.76	54.76	54.76	54.75	
42	Vapour Fraction		0.8709	0.8709	0.8709	0.8709	
43	Heat Transferred (kJ/h-m)		659.7	659.6	659.5	659.5	
44	Flow Regime		Transition	Transition	Transition	Transition	
45	Water Flow Regime						
46	Liquid HoldUp		0.3672	0.3672	0.2851	0.2851	
47	Water HoldUp		---	---	---	---	
48	Friction Gradient (kPa/m)		6.016e-003	6.017e-003	5.211e-003	5.211e-003	
49	Static Gradient (kPa/m)		7.873e-002	7.873e-002	3.833e-003	3.833e-003	
50	Accel. Gradient (kPa/m)		5.951e-006	5.951e-006	6.350e-007	6.350e-007	
51	Liq. Reynolds		2.143e+005 *	2.143e+005 *	2.143e+005 *	2.143e+005 *	
52	Vap. Reynolds		9.109e+006 *	9.109e+006 *	9.109e+006 *	9.109e+006 *	
53	Liquid Velocity (m/s)		0.2043	0.2043	0.2043	0.2043	
54	Vapour Velocity (m/s)		2.163	2.163	2.163	2.163	
55	Deposit Thickness (mm)		---	---	---	---	
56	Deposit Volume (m3)		---	---	---	---	
57	Length (m)		1.635e+004	1.635e+004	1.636e+004	1.636e+004	
58	Elevation (m)		2.400e-002	3.200e-002	4.000e-002	4.600e-002	
59	Pressure (bar)		90.84	90.84	90.84	90.84	
60	Temperature (C)		54.75	54.75	54.75	54.75	
61	Vapour Fraction		0.8709	0.8709	0.8709	0.8709	
62	Heat Transferred (kJ/h-m)		659.4	659.4	659.3	659.3	
63	Flow Regime		Transition	Transition	Transition	Transition	
64	Water Flow Regime						
65	Liquid HoldUp		0.2851	0.2851	0.2851	0.2838	
66	Water HoldUp		---	---	---	---	
67	Friction Gradient (kPa/m)		5.211e-003	5.211e-003	5.211e-003	5.307e-003	
68	Static Gradient (kPa/m)		3.833e-003	3.833e-003	3.833e-003	2.866e-003	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient	(kPa/m)	6.350e-007	6.350e-007	6.350e-007	5.739e-007	
14	Liq. Reynolds		2.143e+005 *	2.143e+005 *	2.143e+005 *	2.143e+005 *	
15	Vap. Reynolds		9.109e+006 *	9.109e+006 *	9.109e+006 *	9.109e+006 *	
16	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
17	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163	
18	Deposit Thickness	(mm)	---	---	---	---	
19	Deposit Volume	(m3)	---	---	---	---	
20	Length	(m)	1.637e+004	1.637e+004	1.638e+004	1.638e+004	
21	Elevation	(m)	5.200e-002	5.800e-002	6.400e-002	7.000e-002	
22	Pressure	(bar)	90.84	90.84	90.84	90.84	
23	Temperature	(C)	54.74	54.74	54.74	54.74	
24	Vapour Fraction		0.8708	0.8708	0.8708	0.8708	
25	Heat Transferred	(kJ/h-m)	659.2	659.1	659.1	659.0	
26	Flow Regime		Transition	Transition	Transition	Transition	
27	Water Flow Regime						
28	Liquid HoldUp		0.2838	0.2838	0.2838	0.2838	
29	Water HoldUp		---	---	---	---	
30	Friction Gradient	(kPa/m)	5.307e-003	5.307e-003	5.307e-003	5.307e-003	
31	Static Gradient	(kPa/m)	2.866e-003	2.866e-003	2.866e-003	2.866e-003	
32	Accel. Gradient	(kPa/m)	5.739e-007	5.739e-007	5.739e-007	5.739e-007	
33	Liq. Reynolds		2.143e+005 *	2.143e+005 *	2.143e+005 *	2.143e+005 *	
34	Vap. Reynolds		9.109e+006 *	9.109e+006 *	9.109e+006 *	9.109e+006 *	
35	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
36	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163	
37	Deposit Thickness	(mm)	---	---	---	---	
38	Deposit Volume	(m3)	---	---	---	---	
39	Length	(m)	1.639e+004	1.639e+004	1.640e+004	1.640e+004	
40	Elevation	(m)	7.600e-002	8.200e-002	8.800e-002	9.400e-002	
41	Pressure	(bar)	90.84	90.84	90.83	90.83	
42	Temperature	(C)	54.74	54.73	54.73	54.73	
43	Vapour Fraction		0.8708	0.8708	0.8708	0.8708	
44	Heat Transferred	(kJ/h-m)	659.0	658.9	658.9	658.8	
45	Flow Regime		Transition	Transition	Transition	Transition	
46	Water Flow Regime						
47	Liquid HoldUp		0.2838	0.2838	0.2838	0.2838	
48	Water HoldUp		---	---	---	---	
49	Friction Gradient	(kPa/m)	5.307e-003	5.307e-003	5.307e-003	5.307e-003	
50	Static Gradient	(kPa/m)	2.866e-003	2.866e-003	2.866e-003	2.866e-003	
51	Accel. Gradient	(kPa/m)	5.739e-007	5.739e-007	5.739e-007	5.739e-007	
52	Liq. Reynolds		2.143e+005 *	2.143e+005 *	2.143e+005 *	2.143e+005 *	
53	Vap. Reynolds		9.109e+006 *	9.109e+006 *	9.109e+006 *	9.109e+006 *	
54	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
55	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163	
56	Deposit Thickness	(mm)	---	---	---	---	
57	Deposit Volume	(m3)	---	---	---	---	
58	Length	(m)	1.641e+004	1.641e+004	1.642e+004	1.642e+004	
59	Elevation	(m)	1.000e-001	0.1080	0.1160	0.1240	
60	Pressure	(bar)	90.83	90.83	90.83	90.83	
61	Temperature	(C)	54.73	54.73	54.72	54.72	
62	Vapour Fraction		0.8708	0.8708	0.8708	0.8708	
63	Heat Transferred	(kJ/h-m)	658.8	658.7	658.7	658.6	
64	Flow Regime		Transition	Transition	Transition	Transition	
65	Water Flow Regime						
66	Liquid HoldUp		0.2838	0.2851	0.2851	0.2851	
67	Water HoldUp		---	---	---	---	
68	Friction Gradient	(kPa/m)	5.307e-003	5.210e-003	5.210e-003	5.210e-003	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 290 of 937	

1				Case Name:	Simulation - 386MMSCFD_100barg.hsc	
2				Unit Set:	Mixed	
3		Epic Atlantic Limited				
Date/Time:			Mon Jun 13 15:02:44 2022			
4						
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	2.866e-003	3.833e-003	3.833e-003	3.833e-003
14	Accel. Gradient	(kPa/m)	5.739e-007	6.350e-007	6.350e-007	6.350e-007
15	Liq. Reynolds		2.143e+005 *	2.143e+005 *	2.143e+005 *	2.143e+005 *
16	Vap. Reynolds		9.109e+006 *	9.109e+006 *	9.109e+006 *	9.109e+006 *
17	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
18	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	1.643e+004	1.643e+004	1.644e+004	1.644e+004
22	Elevation	(m)	0.1320	0.1400	0.1320	0.1240
23	Pressure	(bar)	90.83	90.83	90.83	90.83
24	Temperature	(C)	54.72	54.72	54.72	54.71
25	Vapour Fraction		0.8708	0.8708	0.8708	0.8708
26	Heat Transferred	(kJ/h-m)	658.6	658.5	658.5	658.4
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2851	0.2851	0.2061	0.2061
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	5.210e-003	5.210e-003	6.847e-003	6.847e-003
32	Static Gradient	(kPa/m)	3.833e-003	3.834e-003	-3.086e-003	-3.087e-003
33	Accel. Gradient	(kPa/m)	6.350e-007	6.350e-007	2.640e-007	2.640e-007
34	Liq. Reynolds		2.143e+005 *	2.143e+005 *	2.143e+005 *	2.143e+005 *
35	Vap. Reynolds		9.109e+006 *	9.109e+006 *	9.109e+006 *	9.109e+006 *
36	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
37	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	1.645e+004	1.645e+004	1.646e+004	1.646e+004
41	Elevation	(m)	0.1160	0.1080	1.000e-001	9.800e-002
42	Pressure	(bar)	90.83	90.83	90.83	90.83
43	Temperature	(C)	54.71	54.71	54.71	54.71
44	Vapour Fraction		0.8708	0.8708	0.8708	0.8708
45	Heat Transferred	(kJ/h-m)	658.4	658.3	658.3	658.2
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2061	0.2061	0.2061	0.2072
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	6.847e-003	6.847e-003	6.847e-003	6.838e-003
51	Static Gradient	(kPa/m)	-3.087e-003	-3.087e-003	-3.087e-003	-7.743e-004
52	Accel. Gradient	(kPa/m)	2.640e-007	2.640e-007	2.640e-007	4.257e-007
53	Liq. Reynolds		2.143e+005 *	2.143e+005 *	2.143e+005 *	2.143e+005 *
54	Vap. Reynolds		9.109e+006 *	9.109e+006 *	9.109e+006 *	9.109e+006 *
55	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
56	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	1.647e+004	1.647e+004	1.648e+004	1.648e+004
60	Elevation	(m)	9.600e-002	9.400e-002	9.200e-002	9.000e-002
61	Pressure	(bar)	90.83	90.83	90.83	90.83
62	Temperature	(C)	54.70	54.70	54.70	54.70
63	Vapour Fraction		0.8708	0.8708	0.8708	0.8708
64	Heat Transferred	(kJ/h-m)	658.2	658.1	658.1	658.0
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2072	0.2072	0.2072	0.2072
68	Water HoldUp		---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 291 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	6.838e-003	6.838e-003	6.838e-003	6.838e-003	
14	Static Gradient	(kPa/m)	-7.743e-004	-7.743e-004	-7.743e-004	-7.743e-004	
15	Accel. Gradient	(kPa/m)	4.257e-007	4.257e-007	4.257e-007	4.257e-007	
16	Liq. Reynolds		2.143e+005 *	2.143e+005 *	2.143e+005 *	2.143e+005 *	
17	Vap. Reynolds		9.109e+006 *	9.109e+006 *	9.109e+006 *	9.109e+006 *	
18	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
19	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	1.649e+004	1.649e+004	1.650e+004	1.650e+004	
23	Elevation	(m)	8.600e-002	8.200e-002	7.800e-002	7.400e-002	
24	Pressure	(bar)	90.83	90.83	90.83	90.83	
25	Temperature	(C)	54.70	54.69	54.69	54.69	
26	Vapour Fraction		0.8708	0.8708	0.8708	0.8708	
27	Heat Transferred	(kJ/h-m)	658.0	657.9	657.9	657.8	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.2068	0.2068	0.2068	0.2069	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	6.841e-003	6.841e-003	6.841e-003	6.841e-003	
33	Static Gradient	(kPa/m)	-1.556e-003	-1.556e-003	-1.556e-003	-1.556e-003	
34	Accel. Gradient	(kPa/m)	3.711e-007	3.711e-007	3.711e-007	3.711e-007	
35	Liq. Reynolds		2.143e+005 *	2.143e+005 *	2.143e+005 *	2.143e+005 *	
36	Vap. Reynolds		9.109e+006 *	9.109e+006 *	9.109e+006 *	9.109e+006 *	
37	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
38	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	1.651e+004	1.651e+004	1.652e+004	1.652e+004	
42	Elevation	(m)	7.000e-002	8.000e-002	9.000e-002	1.000e-001	
43	Pressure	(bar)	90.83	90.83	90.83	90.83	
44	Temperature	(C)	54.69	54.69	54.68	54.68	
45	Vapour Fraction		0.8708	0.8708	0.8708	0.8708	
46	Heat Transferred	(kJ/h-m)	657.8	657.7	657.6	657.6	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.2069	0.2864	0.2864	0.2864	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	6.840e-003	5.114e-003	5.114e-003	5.114e-003	
52	Static Gradient	(kPa/m)	-1.556e-003	4.807e-003	4.807e-003	4.807e-003	
53	Accel. Gradient	(kPa/m)	3.711e-007	6.966e-007	6.966e-007	6.966e-007	
54	Liq. Reynolds		2.143e+005 *	2.143e+005 *	2.143e+005 *	2.143e+005 *	
55	Vap. Reynolds		9.109e+006 *	9.109e+006 *	9.110e+006 *	9.110e+006 *	
56	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
57	Vapour Velocity	(m/s)	2.163	2.163	2.163	2.163	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	1.653e+004	1.653e+004	1.654e+004	1.654e+004	
61	Elevation	(m)	0.1100	0.1200	0.1200	0.1200	
62	Pressure	(bar)	90.83	90.83	90.83	90.82	
63	Temperature	(C)	54.68	54.68	54.68	54.68	
64	Vapour Fraction		0.8708	0.8708	0.8708	0.8708	
65	Heat Transferred	(kJ/h-m)	657.5	657.5	657.4	657.4	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.2864	0.2864	0.3031	0.3031	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 292 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	5.114e-003	5.114e-003	5.860e-003	5.860e-003
15	Static Gradient (kPa/m)	4.807e-003	4.807e-003	0.0000	0.0000
16	Accel. Gradient (kPa/m)	6.966e-007	6.966e-007	4.115e-007	4.115e-007
17	Liq. Reynolds	2.143e+005 *	2.143e+005 *	2.142e+005 *	2.142e+005 *
18	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
19	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
20	Vapour Velocity (m/s)	2.163	2.163	2.163	2.163
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	1.655e+004	1.655e+004	1.656e+004	1.656e+004
24	Elevation (m)	0.1200	0.1200	0.1200	0.1260
25	Pressure (bar)	90.82	90.82	90.82	90.82
26	Temperature (C)	54.67	54.67	54.67	54.67
27	Vapour Fraction	0.8708	0.8708	0.8708	0.8708
28	Heat Transferred (kJ/h-m)	657.3	657.3	657.2	657.2
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.3031	0.3031	0.3031	0.2839
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	5.860e-003	5.860e-003	5.860e-003	5.306e-003
34	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	2.866e-003
35	Accel. Gradient (kPa/m)	4.115e-007	4.114e-007	4.114e-007	5.738e-007
36	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *
37	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
38	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
39	Vapour Velocity (m/s)	2.163	2.163	2.163	2.163
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	1.657e+004	1.657e+004	1.658e+004	1.658e+004
43	Elevation (m)	0.1320	0.1380	0.1440	0.1500
44	Pressure (bar)	90.82	90.82	90.82	90.82
45	Temperature (C)	54.67	54.66	54.66	54.66
46	Vapour Fraction	0.8708	0.8708	0.8708	0.8708
47	Heat Transferred (kJ/h-m)	657.1	657.1	657.0	657.0
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2839	0.2839	0.2839	0.2839
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	5.306e-003	5.305e-003	5.305e-003	5.305e-003
53	Static Gradient (kPa/m)	2.866e-003	2.867e-003	2.867e-003	2.867e-003
54	Accel. Gradient (kPa/m)	5.738e-007	5.737e-007	5.737e-007	5.737e-007
55	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *
56	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
57	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
58	Vapour Velocity (m/s)	2.163	2.163	2.163	2.163
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	1.659e+004	1.659e+004	1.660e+004	1.660e+004
62	Elevation (m)	0.1220	9.400e-002	6.600e-002	3.800e-002
63	Pressure (bar)	90.82	90.82	90.82	90.82
64	Temperature (C)	54.66	54.66	54.65	54.65
65	Vapour Fraction	0.8708	0.8708	0.8708	0.8708
66	Heat Transferred (kJ/h-m)	656.9	656.9	656.8	656.8
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 293 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.2024	0.2024	0.2024	0.2024
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	6.876e-003	6.876e-003	6.876e-003	6.876e-003
16	Static Gradient (kPa/m)	-1.068e-002	-1.068e-002	-1.068e-002	-1.068e-002
17	Accel. Gradient (kPa/m)	-2.671e-007	-2.671e-007	-2.672e-007	-2.672e-007
18	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *
19	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
20	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
21	Vapour Velocity (m/s)	2.162	2.162	2.162	2.162
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	1.661e+004	1.661e+004	1.662e+004	1.662e+004
25	Elevation (m)	1.000e-002	1.200e-002	1.400e-002	1.600e-002
26	Pressure (bar)	90.82	90.82	90.82	90.82
27	Temperature (C)	54.65	54.65	54.65	54.64
28	Vapour Fraction	0.8708	0.8708	0.8708	0.8708
29	Heat Transferred (kJ/h-m)	656.7	656.7	656.6	656.6
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2024	0.2813	0.2813	0.2813
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.876e-003	5.502e-003	5.502e-003	5.502e-003
35	Static Gradient (kPa/m)	-1.068e-002	9.495e-004	9.495e-004	9.495e-004
36	Accel. Gradient (kPa/m)	-2.672e-007	4.529e-007	4.529e-007	4.529e-007
37	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *
38	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
39	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
40	Vapour Velocity (m/s)	2.162	2.162	2.162	2.162
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	1.663e+004	1.663e+004	1.664e+004	1.664e+004
44	Elevation (m)	1.800e-002	2.000e-002	4.000e-003	-1.200e-002
45	Pressure (bar)	90.82	90.82	90.82	90.82
46	Temperature (C)	54.64	54.64	54.64	54.64
47	Vapour Fraction	0.8708	0.8708	0.8708	0.8708
48	Heat Transferred (kJ/h-m)	656.5	656.5	656.4	656.4
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2813	0.2813	0.2046	0.2046
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	5.502e-003	5.502e-003	6.857e-003	6.857e-003
54	Static Gradient (kPa/m)	9.495e-004	9.495e-004	-6.146e-003	-6.146e-003
55	Accel. Gradient (kPa/m)	4.529e-007	4.529e-007	4.991e-008	4.990e-008
56	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *
57	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
58	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
59	Vapour Velocity (m/s)	2.162	2.162	2.162	2.162
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	1.665e+004	1.665e+004	1.666e+004	1.666e+004
63	Elevation (m)	-2.800e-002	-4.400e-002	-6.000e-002	-5.200e-002
64	Pressure (bar)	90.82	90.82	90.82	90.82
65	Temperature (C)	54.64	54.63	54.63	54.63
66	Vapour Fraction	0.8708	0.8708	0.8708	0.8708
67	Heat Transferred (kJ/h-m)	656.3	656.3	656.2	656.2
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 294 of 937

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2046	0.2046	0.2046	0.2852
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.857e-003	6.857e-003	6.857e-003	5.208e-003
17	Static Gradient (kPa/m)	-6.146e-003	-6.146e-003	-6.146e-003	3.835e-003
18	Accel. Gradient (kPa/m)	4.990e-008	4.989e-008	4.988e-008	6.348e-007
19	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *
20	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
21	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
22	Vapour Velocity (m/s)	2.162	2.162	2.162	2.162
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	1.667e+004	1.667e+004	1.668e+004	1.668e+004
26	Elevation (m)	-4.400e-002	-3.600e-002	-2.800e-002	-2.000e-002
27	Pressure (bar)	90.82	90.82	90.82	90.82
28	Temperature (C)	54.63	54.63	54.62	54.62
29	Vapour Fraction	0.8708	0.8708	0.8708	0.8708
30	Heat Transferred (kJ/h-m)	656.1	656.1	656.0	656.0
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2852	0.2852	0.2852	0.2852
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	5.208e-003	5.208e-003	5.208e-003	5.208e-003
36	Static Gradient (kPa/m)	3.835e-003	3.835e-003	3.835e-003	3.835e-003
37	Accel. Gradient (kPa/m)	6.348e-007	6.348e-007	6.348e-007	6.348e-007
38	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *
39	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
40	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
41	Vapour Velocity (m/s)	2.162	2.162	2.162	2.162
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	1.669e+004	1.669e+004	1.670e+004	1.670e+004
45	Elevation (m)	-1.600e-002	-1.200e-002	-8.000e-003	-4.000e-003
46	Pressure (bar)	90.82	90.82	90.82	90.82
47	Temperature (C)	54.62	54.62	54.62	54.61
48	Vapour Fraction	0.8708	0.8708	0.8708	0.8708
49	Heat Transferred (kJ/h-m)	655.9	655.8	655.8	655.7
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2826	0.2826	0.2826	0.2826
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	5.403e-003	5.403e-003	5.403e-003	5.403e-003
55	Static Gradient (kPa/m)	1.900e-003	1.900e-003	1.900e-003	1.900e-003
56	Accel. Gradient (kPa/m)	5.126e-007	5.126e-007	5.126e-007	5.126e-007
57	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *
58	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
59	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
60	Vapour Velocity (m/s)	2.162	2.162	2.162	2.162
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	1.671e+004	1.671e+004	1.672e+004	1.672e+004
64	Elevation (m)	-1.006e-016	-4.000e-003	-8.000e-003	-1.200e-002
65	Pressure (bar)	90.82	90.82	90.82	90.82
66	Temperature (C)	54.61	54.61	54.61	54.61
67	Vapour Fraction	0.8708	0.8708	0.8708	0.8708
68	Heat Transferred (kJ/h-m)	655.7	655.7	655.6	655.6
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 295 of 937

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2826	0.2069	0.2069	0.2069
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	5.402e-003	6.839e-003	6.839e-003	6.839e-003
18	Static Gradient (kPa/m)	1.900e-003	-1.547e-003	-1.547e-003	-1.547e-003
19	Accel. Gradient (kPa/m)	5.126e-007	3.715e-007	3.714e-007	3.714e-007
20	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *
21	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
22	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
23	Vapour Velocity (m/s)	2.162	2.162	2.162	2.162
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	1.673e+004	1.673e+004	1.674e+004	1.674e+004
27	Elevation (m)	-1.600e-002	-2.000e-002	-3.000e-002	-4.000e-002
28	Pressure (bar)	90.82	90.82	90.82	90.82
29	Temperature (C)	54.60	54.60	54.60	54.60
30	Vapour Fraction	0.8708	0.8708	0.8708	0.8708
31	Heat Transferred (kJ/h-m)	655.5	655.4	655.4	655.3
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2069	0.2069	0.2058	0.2058
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	6.839e-003	6.839e-003	6.847e-003	6.847e-003
37	Static Gradient (kPa/m)	-1.547e-003	-1.547e-003	-3.855e-003	-3.855e-003
38	Accel. Gradient (kPa/m)	3.714e-007	3.714e-007	2.101e-007	2.100e-007
39	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *
40	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
41	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
42	Vapour Velocity (m/s)	2.162	2.162	2.162	2.162
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	1.675e+004	1.675e+004	1.676e+004	1.676e+004
46	Elevation (m)	-5.000e-002	-6.000e-002	-7.000e-002	-8.400e-002
47	Pressure (bar)	90.82	90.82	90.82	90.81
48	Temperature (C)	54.60	54.59	54.59	54.59
49	Vapour Fraction	0.8708	0.8708	0.8708	0.8708
50	Heat Transferred (kJ/h-m)	655.3	655.3	655.2	655.2
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2058	0.2058	0.2058	0.2050
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	6.847e-003	6.847e-003	6.847e-003	6.853e-003
56	Static Gradient (kPa/m)	-3.855e-003	-3.855e-003	-3.855e-003	-5.385e-003
57	Accel. Gradient (kPa/m)	2.100e-007	2.100e-007	2.100e-007	1.031e-007
58	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *
59	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
60	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
61	Vapour Velocity (m/s)	2.162	2.162	2.162	2.162
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	1.677e+004	1.677e+004	1.678e+004	1.678e+004
65	Elevation (m)	-9.800e-002	-0.1120	-0.1260	-0.1400
66	Pressure (bar)	90.81	90.81	90.81	90.81
67	Temperature (C)	54.59	54.59	54.59	54.58
68	Vapour Fraction	0.8708	0.8708	0.8708	0.8708
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022		
6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Heat Transferred (kJ/h-m)	655.1	655.1	655.0	655.0		
14	Flow Regime	Transition	Transition	Transition	Transition		
15	Water Flow Regime						
16	Liquid HoldUp	0.2050	0.2050	0.2050	0.2050		
17	Water HoldUp	---	---	---	---		
18	Friction Gradient (kPa/m)	6.853e-003	6.853e-003	6.853e-003	6.853e-003		
19	Static Gradient (kPa/m)	-5.385e-003	-5.385e-003	-5.385e-003	-5.385e-003		
20	Accel. Gradient (kPa/m)	1.031e-007	1.030e-007	1.030e-007	1.030e-007		
21	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *		
22	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *		
23	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
24	Vapour Velocity (m/s)	2.162	2.162	2.162	2.162		
25	Deposit Thickness (mm)	---	---	---	---		
26	Deposit Volume (m3)	---	---	---	---		
27	Length (m)	1.679e+004	1.679e+004	1.680e+004	1.680e+004		
28	Elevation (m)	-0.1320	-0.1240	-0.1160	-0.1080		
29	Pressure (bar)	90.81	90.81	90.81	90.81		
30	Temperature (C)	54.58	54.58	54.58	54.58		
31	Vapour Fraction	0.8708	0.8708	0.8708	0.8707		
32	Heat Transferred (kJ/h-m)	654.9	654.8	654.8	654.7		
33	Flow Regime	Transition	Transition	Transition	Transition		
34	Water Flow Regime						
35	Liquid HoldUp	0.2852	0.2852	0.2852	0.2852		
36	Water HoldUp	---	---	---	---		
37	Friction Gradient (kPa/m)	5.207e-003	5.207e-003	5.207e-003	5.207e-003		
38	Static Gradient (kPa/m)	3.835e-003	3.835e-003	3.835e-003	3.835e-003		
39	Accel. Gradient (kPa/m)	6.347e-007	6.346e-007	6.346e-007	6.346e-007		
40	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *		
41	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *		
42	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
43	Vapour Velocity (m/s)	2.162	2.162	2.162	2.162		
44	Deposit Thickness (mm)	---	---	---	---		
45	Deposit Volume (m3)	---	---	---	---		
46	Length (m)	1.681e+004	1.681e+004	1.682e+004	1.682e+004		
47	Elevation (m)	-0.1000	-0.1060	-0.1120	-0.1180		
48	Pressure (bar)	90.81	90.81	90.81	90.81		
49	Temperature (C)	54.57	54.57	54.57	54.57		
50	Vapour Fraction	0.8707	0.8707	0.8707	0.8707		
51	Heat Transferred (kJ/h-m)	654.7	654.6	654.6	654.5		
52	Flow Regime	Transition	Transition	Transition	Transition		
53	Water Flow Regime						
54	Liquid HoldUp	0.2852	0.2065	0.2065	0.2065		
55	Water HoldUp	---	---	---	---		
56	Friction Gradient (kPa/m)	5.206e-003	6.841e-003	6.841e-003	6.841e-003		
57	Static Gradient (kPa/m)	3.835e-003	-2.318e-003	-2.319e-003	-2.319e-003		
58	Accel. Gradient (kPa/m)	6.346e-007	3.174e-007	3.174e-007	3.174e-007		
59	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *		
60	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *		
61	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
62	Vapour Velocity (m/s)	2.162	2.162	2.162	2.162		
63	Deposit Thickness (mm)	---	---	---	---		
64	Deposit Volume (m3)	---	---	---	---		
65	Length (m)	1.683e+004	1.683e+004	1.684e+004	1.684e+004		
66	Elevation (m)	-0.1240	-0.1300	-0.1380	-0.1460		
67	Pressure (bar)	90.81	90.81	90.81	90.81		
68	Temperature (C)	54.57	54.56	54.56	54.56		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
14	Heat Transferred (kJ/h-m)	654.5	654.4	654.4	654.3
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2065	0.2065	0.2062	0.2062
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	6.841e-003	6.841e-003	6.843e-003	6.843e-003
20	Static Gradient (kPa/m)	-2.319e-003	-2.319e-003	-3.088e-003	-3.088e-003
21	Accel. Gradient (kPa/m)	3.174e-007	3.174e-007	2.636e-007	2.636e-007
22	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *
23	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
24	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
25	Vapour Velocity (m/s)	2.162	2.162	2.162	2.162
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	1.685e+004	1.685e+004	1.686e+004	1.686e+004
29	Elevation (m)	-0.1540	-0.1620	-0.1700	-0.1700
30	Pressure (bar)	90.81	90.81	90.81	90.81
31	Temperature (C)	54.56	54.56	54.55	54.55
32	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
33	Heat Transferred (kJ/h-m)	654.3	654.2	654.2	654.1
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2062	0.2062	0.2062	0.3032
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	6.843e-003	6.843e-003	6.843e-003	5.858e-003
39	Static Gradient (kPa/m)	-3.088e-003	-3.088e-003	-3.088e-003	0.0000
40	Accel. Gradient (kPa/m)	2.636e-007	2.636e-007	2.636e-007	4.112e-007
41	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *
42	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
43	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
44	Vapour Velocity (m/s)	2.162	2.162	2.162	2.162
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	1.687e+004	1.687e+004	1.688e+004	1.688e+004
48	Elevation (m)	-0.1700	-0.1700	-0.1700	-0.1700
49	Pressure (bar)	90.81	90.81	90.81	90.81
50	Temperature (C)	54.55	54.55	54.55	54.55
51	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
52	Heat Transferred (kJ/h-m)	654.1	654.0	654.0	653.9
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.3032	0.3032	0.3032	0.3032
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	5.858e-003	5.858e-003	5.858e-003	5.858e-003
58	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
59	Accel. Gradient (kPa/m)	4.112e-007	4.112e-007	4.112e-007	4.112e-007
60	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *
61	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
62	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
63	Vapour Velocity (m/s)	2.162	2.162	2.162	2.161
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	1.689e+004	1.689e+004	1.690e+004	1.690e+004
67	Elevation (m)	-0.1880	-0.2060	-0.2240	-0.2420
68	Pressure (bar)	90.81	90.81	90.81	90.81
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Pipe Segment: ELEP-Akabel Pipeline (continued)		
3					
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Temperature (C)	54.54	54.54	54.54	54.54
14	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
15	Heat Transferred (kJ/h-m)	653.9	653.8	653.8	653.8
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2039	0.2039	0.2039	0.2039
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	6.861e-003	6.861e-003	6.861e-003	6.861e-003
21	Static Gradient (kPa/m)	-7.725e-003	-7.725e-003	-7.725e-003	-7.725e-003
22	Accel. Gradient (kPa/m)	-6.062e-008	-6.062e-008	-6.063e-008	-6.064e-008
23	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *
24	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
25	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
26	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	1.690e+004	1.691e+004	1.691e+004	1.692e+004
30	Elevation (m)	-0.2600	-0.2780	-0.2960	-0.3140
31	Pressure (bar)	90.81	90.81	90.81	90.81
32	Temperature (C)	54.54	54.54	54.53	54.53
33	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
34	Heat Transferred (kJ/h-m)	653.7	653.7	653.6	653.6
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2039	0.2043	0.2043	0.2043
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	6.861e-003	6.858e-003	6.858e-003	6.858e-003
40	Static Gradient (kPa/m)	-7.725e-003	-6.909e-003	-6.909e-003	-6.909e-003
41	Accel. Gradient (kPa/m)	-6.065e-008	-3.579e-009	-3.586e-009	-3.594e-009
42	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *
43	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
44	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
45	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	1.692e+004	1.693e+004	1.693e+004	1.694e+004
49	Elevation (m)	-0.3320	-0.3500	-0.3520	-0.3540
50	Pressure (bar)	90.81	90.81	90.81	90.81
51	Temperature (C)	54.53	54.53	54.53	54.52
52	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
53	Heat Transferred (kJ/h-m)	653.5	653.5	653.4	653.4
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2043	0.2043	0.2073	0.2073
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	6.857e-003	6.857e-003	6.834e-003	6.834e-003
59	Static Gradient (kPa/m)	-6.909e-003	-6.909e-003	-7.747e-004	-7.747e-004
60	Accel. Gradient (kPa/m)	-3.602e-009	-3.610e-009	4.252e-007	4.252e-007
61	Liq. Reynolds	2.142e+005 *	2.142e+005 *	2.142e+005 *	2.142e+005 *
62	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
63	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
64	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	1.694e+004	1.695e+004	1.695e+004	1.696e+004
68	Elevation (m)	-0.3560	-0.3580	-0.3600	-0.3580
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	90.81	90.81	90.81	90.81		
14	Temperature (C)	54.52	54.52	54.52	54.52		
15	Vapour Fraction	0.8707	0.8707	0.8707	0.8707		
16	Heat Transferred (kJ/h-m)	653.3	653.3	653.2	653.1		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.2073	0.2073	0.2073	0.2814		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	6.834e-003	6.834e-003	6.834e-003	5.498e-003		
22	Static Gradient (kPa/m)	-7.747e-004	-7.747e-004	-7.747e-004	9.499e-004		
23	Accel. Gradient (kPa/m)	4.252e-007	4.252e-007	4.252e-007	4.525e-007		
24	Liq. Reynolds	2.141e+005 *	2.141e+005 *	2.141e+005 *	2.141e+005 *		
25	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *		
26	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
27	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	1.696e+004	1.697e+004	1.697e+004	1.698e+004		
31	Elevation (m)	-0.3560	-0.3540	-0.3520	-0.3500		
32	Pressure (bar)	90.81	90.81	90.81	90.81		
33	Temperature (C)	54.51	54.51	54.51	54.51		
34	Vapour Fraction	0.8707	0.8707	0.8707	0.8707		
35	Heat Transferred (kJ/h-m)	653.1	653.0	653.0	652.9		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.2814	0.2814	0.2814	0.2814		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	5.498e-003	5.498e-003	5.498e-003	5.498e-003		
41	Static Gradient (kPa/m)	9.499e-004	9.499e-004	9.499e-004	9.499e-004		
42	Accel. Gradient (kPa/m)	4.525e-007	4.525e-007	4.525e-007	4.525e-007		
43	Liq. Reynolds	2.141e+005 *	2.141e+005 *	2.141e+005 *	2.141e+005 *		
44	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *		
45	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
46	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	1.698e+004	1.699e+004	1.699e+004	1.700e+004		
50	Elevation (m)	-0.3280	-0.3060	-0.2840	-0.2620		
51	Pressure (bar)	90.81	90.80	90.80	90.80		
52	Temperature (C)	54.51	54.50	54.50	54.50		
53	Vapour Fraction	0.8707	0.8707	0.8707	0.8707		
54	Heat Transferred (kJ/h-m)	652.9	652.8	652.8	652.7		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.2942	0.2942	0.2942	0.2942		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	4.740e-003	4.740e-003	4.740e-003	4.740e-003		
60	Static Gradient (kPa/m)	1.078e-002	1.078e-002	1.078e-002	1.078e-002		
61	Accel. Gradient (kPa/m)	1.089e-006	1.089e-006	1.089e-006	1.089e-006		
62	Liq. Reynolds	2.141e+005 *	2.141e+005 *	2.141e+005 *	2.141e+005 *		
63	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *		
64	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
65	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	1.700e+004	1.701e+004	1.701e+004	1.702e+004		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3						
4						
5			Unit Set: Mixed			
6			Date/Time: Mon Jun 13 15:02:44 2022			
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	PROFILES					
11	Pipe Table					
12						
13	Elevation (m)	-0.2400	-0.2460	-0.2520	-0.2580	
14	Pressure (bar)	90.80	90.80	90.80	90.80	
15	Temperature (C)	54.50	54.50	54.49	54.49	
16	Vapour Fraction	0.8707	0.8707	0.8707	0.8707	
17	Heat Transferred (kJ/h-m)	652.7	652.6	652.6	652.5	
18	Flow Regime	Transition	Transition	Transition	Transition	
19	Water Flow Regime					
20	Liquid HoldUp	0.2942	0.2066	0.2066	0.2066	
21	Water HoldUp	---	---	---	---	
22	Friction Gradient (kPa/m)	4.740e-003	6.839e-003	6.839e-003	6.839e-003	
23	Static Gradient (kPa/m)	1.078e-002	-2.319e-003	-2.319e-003	-2.319e-003	
24	Accel. Gradient (kPa/m)	1.089e-006	3.172e-007	3.172e-007	3.172e-007	
25	Liq. Reynolds	2.141e+005 *	2.141e+005 *	2.141e+005 *	2.141e+005 *	
26	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *	
27	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044	
28	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161	
29	Deposit Thickness (mm)	---	---	---	---	
30	Deposit Volume (m3)	---	---	---	---	
31	Length (m)	1.702e+004	1.703e+004	1.703e+004	1.704e+004	
32	Elevation (m)	-0.2640	-0.2700	-0.2840	-0.2980	
33	Pressure (bar)	90.80	90.80	90.80	90.80	
34	Temperature (C)	54.49	54.49	54.49	54.49	
35	Vapour Fraction	0.8707	0.8707	0.8707	0.8707	
36	Heat Transferred (kJ/h-m)	652.5	652.4	652.4	652.3	
37	Flow Regime	Transition	Transition	Transition	Transition	
38	Water Flow Regime					
39	Liquid HoldUp	0.2066	0.2066	0.2051	0.2051	
40	Water HoldUp	---	---	---	---	
41	Friction Gradient (kPa/m)	6.839e-003	6.839e-003	6.851e-003	6.851e-003	
42	Static Gradient (kPa/m)	-2.319e-003	-2.319e-003	-5.386e-003	-5.386e-003	
43	Accel. Gradient (kPa/m)	3.172e-007	3.172e-007	1.028e-007	1.027e-007	
44	Liq. Reynolds	2.141e+005 *	2.141e+005 *	2.141e+005 *	2.141e+005 *	
45	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *	
46	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044	
47	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161	
48	Deposit Thickness (mm)	---	---	---	---	
49	Deposit Volume (m3)	---	---	---	---	
50	Length (m)	1.704e+004	1.705e+004	1.705e+004	1.706e+004	
51	Elevation (m)	-0.3120	-0.3260	-0.3400	-0.3220	
52	Pressure (bar)	90.80	90.80	90.80	90.80	
53	Temperature (C)	54.48	54.48	54.48	54.48	
54	Vapour Fraction	0.8707	0.8707	0.8707	0.8707	
55	Heat Transferred (kJ/h-m)	652.3	652.2	652.2	652.1	
56	Flow Regime	Transition	Transition	Transition	Transition	
57	Water Flow Regime					
58	Liquid HoldUp	0.2051	0.2051	0.2051	0.2917	
59	Water HoldUp	---	---	---	---	
60	Friction Gradient (kPa/m)	6.851e-003	6.851e-003	6.851e-003	4.740e-003	
61	Static Gradient (kPa/m)	-5.387e-003	-5.387e-003	-5.387e-003	8.767e-003	
62	Accel. Gradient (kPa/m)	1.027e-007	1.027e-007	1.027e-007	9.479e-007	
63	Liq. Reynolds	2.141e+005 *	2.141e+005 *	2.141e+005 *	2.141e+005 *	
64	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *	
65	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044	
66	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161	
67	Deposit Thickness (mm)	---	---	---	---	
68	Deposit Volume (m3)	---	---	---	---	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length (m)	1.706e+004	1.707e+004	1.707e+004	1.708e+004		
14	Elevation (m)	-0.3040	-0.2860	-0.2680	-0.2500		
15	Pressure (bar)	90.80	90.80	90.80	90.80		
16	Temperature (C)	54.48	54.47	54.47	54.47		
17	Vapour Fraction	0.8707	0.8707	0.8707	0.8707		
18	Heat Transferred (kJ/h-m)	652.1	652.0	651.9	651.9		
19	Flow Regime	Transition	Transition	Transition	Transition		
20	Water Flow Regime						
21	Liquid HoldUp	0.2917	0.2917	0.2917	0.2917		
22	Water HoldUp	---	---	---	---		
23	Friction Gradient (kPa/m)	4.740e-003	4.740e-003	4.740e-003	4.740e-003		
24	Static Gradient (kPa/m)	8.767e-003	8.768e-003	8.768e-003	8.768e-003		
25	Accel. Gradient (kPa/m)	9.479e-007	9.479e-007	9.479e-007	9.479e-007		
26	Liq. Reynolds	2.141e+005 *	2.141e+005 *	2.141e+005 *	2.141e+005 *		
27	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *		
28	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
29	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161		
30	Deposit Thickness (mm)	---	---	---	---		
31	Deposit Volume (m3)	---	---	---	---		
32	Length (m)	1.708e+004	1.709e+004	1.709e+004	1.710e+004		
33	Elevation (m)	-0.2420	-0.2340	-0.2260	-0.2180		
34	Pressure (bar)	90.80	90.80	90.80	90.80		
35	Temperature (C)	54.47	54.47	54.46	54.46		
36	Vapour Fraction	0.8707	0.8707	0.8707	0.8707		
37	Heat Transferred (kJ/h-m)	651.8	651.8	651.7	651.7		
38	Flow Regime	Transition	Transition	Transition	Transition		
39	Water Flow Regime						
40	Liquid HoldUp	0.2855	0.2855	0.2855	0.2855		
41	Water HoldUp	---	---	---	---		
42	Friction Gradient (kPa/m)	5.191e-003	5.191e-003	5.191e-003	5.191e-003		
43	Static Gradient (kPa/m)	3.963e-003	3.963e-003	3.963e-003	3.963e-003		
44	Accel. Gradient (kPa/m)	6.424e-007	6.424e-007	6.424e-007	6.424e-007		
45	Liq. Reynolds	2.141e+005 *	2.141e+005 *	2.141e+005 *	2.141e+005 *		
46	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *		
47	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
48	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161		
49	Deposit Thickness (mm)	---	---	---	---		
50	Deposit Volume (m3)	---	---	---	---		
51	Length (m)	1.710e+004	1.711e+004	1.711e+004	1.712e+004		
52	Elevation (m)	-0.2100	-0.1980	-0.1860	-0.1740		
53	Pressure (bar)	90.80	90.79	90.79	90.79		
54	Temperature (C)	54.46	54.46	54.46	54.45		
55	Vapour Fraction	0.8707	0.8707	0.8707	0.8707		
56	Heat Transferred (kJ/h-m)	651.6	651.6	651.5	651.5		
57	Flow Regime	Transition	Transition	Transition	Transition		
58	Water Flow Regime						
59	Liquid HoldUp	0.2855	0.2879	0.2879	0.2879		
60	Water HoldUp	---	---	---	---		
61	Friction Gradient (kPa/m)	5.191e-003	5.015e-003	5.015e-003	5.015e-003		
62	Static Gradient (kPa/m)	3.963e-003	5.791e-003	5.791e-003	5.791e-003		
63	Accel. Gradient (kPa/m)	6.424e-007	7.583e-007	7.583e-007	7.583e-007		
64	Liq. Reynolds	2.141e+005 *	2.141e+005 *	2.141e+005 *	2.141e+005 *		
65	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *		
66	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
67	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161		
68	Deposit Thickness (mm)	---	---	---	---		
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	1.712e+004	1.713e+004	1.713e+004	1.714e+004
15	Elevation (m)	-0.1620	-0.1500	-0.1360	-0.1220
16	Pressure (bar)	90.79	90.79	90.79	90.79
17	Temperature (C)	54.45	54.45	54.45	54.45
18	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
19	Heat Transferred (kJ/h-m)	651.4	651.4	651.3	651.3
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2879	0.2879	0.2891	0.2891
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	5.015e-003	5.015e-003	4.922e-003	4.922e-003
25	Static Gradient (kPa/m)	5.791e-003	5.791e-003	6.777e-003	6.777e-003
26	Accel. Gradient (kPa/m)	7.583e-007	7.583e-007	8.210e-007	8.210e-007
27	Liq. Reynolds	2.141e+005 *	2.141e+005 *	2.141e+005 *	2.141e+005 *
28	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
29	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
30	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	1.714e+004	1.715e+004	1.715e+004	1.716e+004
34	Elevation (m)	-0.1080	-9.400e-002	-8.000e-002	-7.000e-002
35	Pressure (bar)	90.79	90.79	90.79	90.79
36	Temperature (C)	54.44	54.44	54.44	54.44
37	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
38	Heat Transferred (kJ/h-m)	651.2	651.2	651.1	651.0
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2891	0.2891	0.2891	0.2866
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	4.922e-003	4.922e-003	4.922e-003	5.108e-003
44	Static Gradient (kPa/m)	6.777e-003	6.777e-003	6.777e-003	4.811e-003
45	Accel. Gradient (kPa/m)	8.210e-007	8.210e-007	8.210e-007	6.961e-007
46	Liq. Reynolds	2.141e+005 *	2.141e+005 *	2.141e+005 *	2.141e+005 *
47	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
48	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
49	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	1.716e+004	1.717e+004	1.717e+004	1.718e+004
53	Elevation (m)	-6.000e-002	-5.000e-002	-4.000e-002	-3.000e-002
54	Pressure (bar)	90.79	90.79	90.79	90.79
55	Temperature (C)	54.44	54.43	54.43	54.43
56	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
57	Heat Transferred (kJ/h-m)	651.0	650.9	650.9	650.8
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2866	0.2866	0.2866	0.2866
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	5.108e-003	5.108e-003	5.108e-003	5.108e-003
63	Static Gradient (kPa/m)	4.811e-003	4.811e-003	4.811e-003	4.811e-003
64	Accel. Gradient (kPa/m)	6.961e-007	6.961e-007	6.961e-007	6.961e-007
65	Liq. Reynolds	2.141e+005 *	2.141e+005 *	2.141e+005 *	2.141e+005 *
66	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
67	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
68	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	1.718e+004	1.719e+004	1.719e+004	1.720e+004
16	Elevation (m)	-2.600e-002	-2.200e-002	-1.800e-002	-1.400e-002
17	Pressure (bar)	90.79	90.79	90.79	90.79
18	Temperature (C)	54.43	54.43	54.42	54.42
19	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
20	Heat Transferred (kJ/h-m)	650.8	650.7	650.7	650.6
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2828	0.2828	0.2828	0.2828
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	5.398e-003	5.398e-003	5.398e-003	5.398e-003
26	Static Gradient (kPa/m)	1.906e-003	1.906e-003	1.906e-003	1.906e-003
27	Accel. Gradient (kPa/m)	5.126e-007	5.126e-007	5.126e-007	5.126e-007
28	Liq. Reynolds	2.141e+005 *	2.141e+005 *	2.141e+005 *	2.141e+005 *
29	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
30	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
31	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	1.720e+004	1.721e+004	1.721e+004	1.722e+004
35	Elevation (m)	-1.000e-002	4.000e-003	1.800e-002	3.200e-002
36	Pressure (bar)	90.79	90.78	90.78	90.78
37	Temperature (C)	54.42	54.42	54.42	54.41
38	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
39	Heat Transferred (kJ/h-m)	650.6	650.5	650.5	650.4
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2828	0.2892	0.2892	0.2892
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	5.398e-003	4.921e-003	4.921e-003	4.921e-003
45	Static Gradient (kPa/m)	1.906e-003	6.778e-003	6.778e-003	6.778e-003
46	Accel. Gradient (kPa/m)	5.126e-007	8.210e-007	8.210e-007	8.210e-007
47	Liq. Reynolds	2.141e+005 *	2.141e+005 *	2.141e+005 *	2.141e+005 *
48	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.110e+006 *	9.110e+006 *
49	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
50	Vapour Velocity (m/s)	2.161	2.161	2.161	2.161
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	1.722e+004	1.723e+004	1.723e+004	1.724e+004
54	Elevation (m)	4.600e-002	6.000e-002	6.400e-002	6.800e-002
55	Pressure (bar)	90.78	90.78	90.78	90.78
56	Temperature (C)	54.41	54.41	54.41	54.41
57	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
58	Heat Transferred (kJ/h-m)	650.4	650.3	650.3	650.2
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2892	0.2892	0.2828	0.2828
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	4.921e-003	4.921e-003	5.397e-003	5.397e-003
64	Static Gradient (kPa/m)	6.778e-003	6.778e-003	1.906e-003	1.906e-003
65	Accel. Gradient (kPa/m)	8.210e-007	8.210e-007	5.126e-007	5.125e-007
66	Liq. Reynolds	2.141e+005 *	2.141e+005 *	2.141e+005 *	2.141e+005 *
67	Vap. Reynolds	9.110e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
68	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3							
4			Unit Set: Mixed				
5			Date/Time: Mon Jun 13 15:02:44 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11							
12	Pipe Table						
13	Vapour Velocity	(m/s)	2.161	2.161	2.161	2.161	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	1.724e+004	1.725e+004	1.725e+004	1.726e+004	
17	Elevation	(m)	7.200e-002	7.600e-002	8.000e-002	8.800e-002	
18	Pressure	(bar)	90.78	90.78	90.78	90.78	
19	Temperature	(C)	54.40	54.40	54.40	54.40	
20	Vapour Fraction		0.8707	0.8707	0.8706	0.8706	
21	Heat Transferred	(kJ/h-m)	650.1	650.1	650.0	650.0	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2828	0.2828	0.2828	0.2853	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	5.397e-003	5.397e-003	5.397e-003	5.203e-003	
27	Static Gradient	(kPa/m)	1.906e-003	1.906e-003	1.906e-003	3.837e-003	
28	Accel. Gradient	(kPa/m)	5.125e-007	5.125e-007	5.125e-007	6.344e-007	
29	Liq. Reynolds		2.141e+005 *	2.141e+005 *	2.141e+005 *	2.140e+005 *	
30	Vap. Reynolds		9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *	
31	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
32	Vapour Velocity	(m/s)	2.161	2.161	2.161	2.161	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	1.726e+004	1.727e+004	1.727e+004	1.728e+004	
36	Elevation	(m)	9.600e-002	0.1040	0.1120	0.1200	
37	Pressure	(bar)	90.78	90.78	90.78	90.78	
38	Temperature	(C)	54.40	54.39	54.39	54.39	
39	Vapour Fraction		0.8706	0.8706	0.8706	0.8706	
40	Heat Transferred	(kJ/h-m)	649.9	649.9	649.8	649.8	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2853	0.2853	0.2853	0.2853	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	5.203e-003	5.203e-003	5.203e-003	5.203e-003	
46	Static Gradient	(kPa/m)	3.837e-003	3.837e-003	3.837e-003	3.837e-003	
47	Accel. Gradient	(kPa/m)	6.344e-007	6.344e-007	6.344e-007	6.344e-007	
48	Liq. Reynolds		2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *	
49	Vap. Reynolds		9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *	
50	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
51	Vapour Velocity	(m/s)	2.161	2.160	2.160	2.160	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	1.728e+004	1.729e+004	1.729e+004	1.730e+004	
55	Elevation	(m)	0.1200	0.1200	0.1200	0.1200	
56	Pressure	(bar)	90.78	90.78	90.78	90.78	
57	Temperature	(C)	54.39	54.39	54.38	54.38	
58	Vapour Fraction		0.8706	0.8706	0.8706	0.8706	
59	Heat Transferred	(kJ/h-m)	649.7	649.7	649.6	649.6	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.3033	0.3033	0.3033	0.3033	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	5.857e-003	5.857e-003	5.857e-003	5.857e-003	
65	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000	
66	Accel. Gradient	(kPa/m)	4.110e-007	4.110e-007	4.110e-007	4.110e-007	
67	Liq. Reynolds		2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *	
68	Vap. Reynolds		9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
14	Vapour Velocity	(m/s)	2.160	2.160	2.160	2.160
15	Deposit Thickness	(mm)	---	---	---	---
16	Deposit Volume	(m3)	---	---	---	---
17	Length	(m)	1.730e+004	1.731e+004	1.731e+004	1.732e+004
18	Elevation	(m)	0.1200	0.1240	0.1280	0.1320
19	Pressure	(bar)	90.78	90.78	90.78	90.78
20	Temperature	(C)	54.38	54.38	54.38	54.37
21	Vapour Fraction		0.8706	0.8706	0.8706	0.8706
22	Heat Transferred	(kJ/h-m)	649.5	649.5	649.4	649.4
23	Flow Regime		Transition	Transition	Transition	Transition
24	Water Flow Regime					
25	Liquid HoldUp		0.3033	0.2828	0.2828	0.2828
26	Water HoldUp		---	---	---	---
27	Friction Gradient	(kPa/m)	5.857e-003	5.397e-003	5.397e-003	5.397e-003
28	Static Gradient	(kPa/m)	0.0000	1.906e-003	1.906e-003	1.906e-003
29	Accel. Gradient	(kPa/m)	4.110e-007	5.125e-007	5.125e-007	5.125e-007
30	Liq. Reynolds		2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *
31	Vap. Reynolds		9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
32	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
33	Vapour Velocity	(m/s)	2.160	2.160	2.160	2.160
34	Deposit Thickness	(mm)	---	---	---	---
35	Deposit Volume	(m3)	---	---	---	---
36	Length	(m)	1.732e+004	1.733e+004	1.733e+004	1.734e+004
37	Elevation	(m)	0.1360	0.1400	0.1400	0.1400
38	Pressure	(bar)	90.78	90.77	90.77	90.77
39	Temperature	(C)	54.37	54.37	54.37	54.37
40	Vapour Fraction		0.8706	0.8706	0.8706	0.8706
41	Heat Transferred	(kJ/h-m)	649.3	649.3	649.2	649.2
42	Flow Regime		Transition	Transition	Transition	Transition
43	Water Flow Regime					
44	Liquid HoldUp		0.2828	0.2828	0.3033	0.3033
45	Water HoldUp		---	---	---	---
46	Friction Gradient	(kPa/m)	5.397e-003	5.397e-003	5.856e-003	5.856e-003
47	Static Gradient	(kPa/m)	1.906e-003	1.907e-003	0.0000	0.0000
48	Accel. Gradient	(kPa/m)	5.125e-007	5.125e-007	4.110e-007	4.110e-007
49	Liq. Reynolds		2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *
50	Vap. Reynolds		9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
51	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
52	Vapour Velocity	(m/s)	2.160	2.160	2.160	2.160
53	Deposit Thickness	(mm)	---	---	---	---
54	Deposit Volume	(m3)	---	---	---	---
55	Length	(m)	1.734e+004	1.735e+004	1.735e+004	1.736e+004
56	Elevation	(m)	0.1400	0.1400	0.1400	0.1440
57	Pressure	(bar)	90.77	90.77	90.77	90.77
58	Temperature	(C)	54.37	54.36	54.36	54.36
59	Vapour Fraction		0.8706	0.8706	0.8706	0.8706
60	Heat Transferred	(kJ/h-m)	649.1	649.1	649.0	649.0
61	Flow Regime		Transition	Transition	Transition	Transition
62	Water Flow Regime					
63	Liquid HoldUp		0.3033	0.3033	0.3033	0.2828
64	Water HoldUp		---	---	---	---
65	Friction Gradient	(kPa/m)	5.856e-003	5.856e-003	5.856e-003	5.396e-003
66	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	1.907e-003
67	Accel. Gradient	(kPa/m)	4.110e-007	4.110e-007	4.110e-007	5.125e-007
68	Liq. Reynolds		2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
14	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
15	Vapour Velocity (m/s)	2.160	2.160	2.160	2.160
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	1.736e+004	1.737e+004	1.737e+004	1.738e+004
19	Elevation (m)	0.1480	0.1520	0.1560	0.1600
20	Pressure (bar)	90.77	90.77	90.77	90.77
21	Temperature (C)	54.36	54.36	54.35	54.35
22	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
23	Heat Transferred (kJ/h-m)	648.9	648.9	648.8	648.8
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2828	0.2828	0.2828	0.2828
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	5.396e-003	5.396e-003	5.396e-003	5.396e-003
29	Static Gradient (kPa/m)	1.907e-003	1.907e-003	1.907e-003	1.907e-003
30	Accel. Gradient (kPa/m)	5.125e-007	5.125e-007	5.125e-007	5.125e-007
31	Liq. Reynolds	2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *
32	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
33	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
34	Vapour Velocity (m/s)	2.160	2.160	2.160	2.160
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	1.738e+004	1.739e+004	1.739e+004	1.740e+004
38	Elevation (m)	0.1600	0.1600	0.1600	0.1600
39	Pressure (bar)	90.77	90.77	90.77	90.77
40	Temperature (C)	54.35	54.35	54.35	54.34
41	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
42	Heat Transferred (kJ/h-m)	648.7	648.7	648.6	648.6
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.3033	0.3033	0.3033	0.3033
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	5.856e-003	5.856e-003	5.856e-003	5.856e-003
48	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
49	Accel. Gradient (kPa/m)	4.109e-007	4.109e-007	4.109e-007	4.109e-007
50	Liq. Reynolds	2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *
51	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
52	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
53	Vapour Velocity (m/s)	2.160	2.160	2.160	2.160
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1.740e+004	1.741e+004	1.741e+004	1.742e+004
57	Elevation (m)	0.1600	0.1560	0.1520	0.1480
58	Pressure (bar)	90.77	90.77	90.77	90.77
59	Temperature (C)	54.34	54.34	54.34	54.34
60	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
61	Heat Transferred (kJ/h-m)	648.5	648.5	648.4	648.4
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.3033	0.2070	0.2070	0.2070
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	5.856e-003	6.834e-003	6.834e-003	6.834e-003
67	Static Gradient (kPa/m)	0.0000	-1.548e-003	-1.548e-003	-1.548e-003
68	Accel. Gradient (kPa/m)	4.109e-007	3.709e-007	3.709e-007	3.709e-007
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *
14	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
15	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
16	Vapour Velocity (m/s)	2.160	2.160	2.160	2.160
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	1.742e+004	1.743e+004	1.743e+004	1.744e+004
20	Elevation (m)	0.1440	0.1400	0.1340	0.1280
21	Pressure (bar)	90.77	90.77	90.77	90.77
22	Temperature (C)	54.33	54.33	54.33	54.33
23	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
24	Heat Transferred (kJ/h-m)	648.3	648.3	648.2	648.2
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2070	0.2070	0.2067	0.2067
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	6.833e-003	6.833e-003	6.836e-003	6.836e-003
30	Static Gradient (kPa/m)	-1.548e-003	-1.548e-003	-2.320e-003	-2.320e-003
31	Accel. Gradient (kPa/m)	3.709e-007	3.709e-007	3.169e-007	3.169e-007
32	Liq. Reynolds	2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *
33	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
34	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
35	Vapour Velocity (m/s)	2.160	2.160	2.160	2.160
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	1.744e+004	1.745e+004	1.745e+004	1.746e+004
39	Elevation (m)	0.1220	0.1160	0.1100	0.1100
40	Pressure (bar)	90.77	90.77	90.77	90.77
41	Temperature (C)	54.33	54.32	54.32	54.32
42	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
43	Heat Transferred (kJ/h-m)	648.1	648.1	648.0	647.9
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2067	0.2067	0.2067	0.3033
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	6.836e-003	6.836e-003	6.836e-003	5.856e-003
49	Static Gradient (kPa/m)	-2.320e-003	-2.320e-003	-2.320e-003	0.0000
50	Accel. Gradient (kPa/m)	3.169e-007	3.169e-007	3.169e-007	4.109e-007
51	Liq. Reynolds	2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *
52	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
53	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
54	Vapour Velocity (m/s)	2.160	2.160	2.160	2.160
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	1.746e+004	1.747e+004	1.747e+004	1.748e+004
58	Elevation (m)	0.1100	0.1100	0.1100	0.1100
59	Pressure (bar)	90.77	90.77	90.77	90.77
60	Temperature (C)	54.32	54.32	54.32	54.31
61	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
62	Heat Transferred (kJ/h-m)	647.9	647.8	647.8	647.7
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.3033	0.3033	0.3033	0.3033
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	5.856e-003	5.856e-003	5.856e-003	5.856e-003
68	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:02:44 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Accel. Gradient	(kPa/m)	4.109e-007	4.109e-007	4.109e-007	4.109e-007				
14	Liq. Reynolds		2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *				
15	Vap. Reynolds		9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *				
16	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044				
17	Vapour Velocity	(m/s)	2.160	2.160	2.160	2.160				
18	Deposit Thickness	(mm)	---	---	---	---				
19	Deposit Volume	(m3)	---	---	---	---				
20	Length	(m)	1.748e+004	1.749e+004	1.749e+004	1.750e+004				
21	Elevation	(m)	9.800e-002	8.600e-002	7.400e-002	6.200e-002				
22	Pressure	(bar)	90.77	90.77	90.77	90.77				
23	Temperature	(C)	54.31	54.31	54.31	54.31				
24	Vapour Fraction		0.8706	0.8706	0.8706	0.8706				
25	Heat Transferred	(kJ/h-m)	647.7	647.7	647.6	647.6				
26	Flow Regime		Transition	Transition	Transition	Transition				
27	Water Flow Regime									
28	Liquid HoldUp		0.2053	0.2053	0.2054	0.2054				
29	Water HoldUp		---	---	---	---				
30	Friction Gradient	(kPa/m)	6.846e-003	6.846e-003	6.846e-003	6.846e-003				
31	Static Gradient	(kPa/m)	-5.019e-003	-5.019e-003	-5.019e-003	-5.019e-003				
32	Accel. Gradient	(kPa/m)	1.282e-007	1.282e-007	1.282e-007	1.282e-007				
33	Liq. Reynolds		2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *				
34	Vap. Reynolds		9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *				
35	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044				
36	Vapour Velocity	(m/s)	2.160	2.160	2.160	2.160				
37	Deposit Thickness	(mm)	---	---	---	---				
38	Deposit Volume	(m3)	---	---	---	---				
39	Length	(m)	1.750e+004	1.751e+004	1.751e+004	1.752e+004				
40	Elevation	(m)	5.000e-002	6.000e-002	7.000e-002	8.000e-002				
41	Pressure	(bar)	90.77	90.77	90.76	90.76				
42	Temperature	(C)	54.31	54.30	54.30	54.30				
43	Vapour Fraction		0.8706	0.8706	0.8706	0.8706				
44	Heat Transferred	(kJ/h-m)	647.5	647.5	647.4	647.4				
45	Flow Regime		Transition	Transition	Transition	Transition				
46	Water Flow Regime									
47	Liquid HoldUp		0.2054	0.2867	0.2867	0.2867				
48	Water HoldUp		---	---	---	---				
49	Friction Gradient	(kPa/m)	6.846e-003	5.106e-003	5.105e-003	5.105e-003				
50	Static Gradient	(kPa/m)	-5.019e-003	4.813e-003	4.813e-003	4.813e-003				
51	Accel. Gradient	(kPa/m)	1.282e-007	6.959e-007	6.959e-007	6.959e-007				
52	Liq. Reynolds		2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *				
53	Vap. Reynolds		9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *				
54	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044				
55	Vapour Velocity	(m/s)	2.160	2.160	2.160	2.160				
56	Deposit Thickness	(mm)	---	---	---	---				
57	Deposit Volume	(m3)	---	---	---	---				
58	Length	(m)	1.752e+004	1.753e+004	1.753e+004	1.754e+004				
59	Elevation	(m)	9.000e-002	1.000e-001	8.000e-002	6.000e-002				
60	Pressure	(bar)	90.76	90.76	90.76	90.76				
61	Temperature	(C)	54.30	54.30	54.29	54.29				
62	Vapour Fraction		0.8706	0.8706	0.8706	0.8706				
63	Heat Transferred	(kJ/h-m)	647.3	647.2	647.2	647.2				
64	Flow Regime		Transition	Transition	Transition	Transition				
65	Water Flow Regime									
66	Liquid HoldUp		0.2867	0.2867	0.2041	0.2041				
67	Water HoldUp		---	---	---	---				
68	Friction Gradient	(kPa/m)	5.105e-003	5.105e-003	6.856e-003	6.856e-003				
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1				Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	4.813e-003	4.813e-003	-7.672e-003	-7.672e-003
14	Accel. Gradient	(kPa/m)	6.959e-007	6.959e-007	-5.723e-008	-5.724e-008
15	Liq. Reynolds		2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *
16	Vap. Reynolds		9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
17	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
18	Vapour Velocity	(m/s)	2.160	2.160	2.160	2.160
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	1.754e+004	1.755e+004	1.755e+004	1.756e+004
22	Elevation	(m)	4.000e-002	2.000e-002	-9.714e-017	-3.400e-002
23	Pressure	(bar)	90.76	90.76	90.76	90.76
24	Temperature	(C)	54.29	54.29	54.29	54.28
25	Vapour Fraction		0.8706	0.8706	0.8706	0.8706
26	Heat Transferred	(kJ/h-m)	647.1	647.1	647.0	647.0
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2041	0.2041	0.2041	0.2014
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	6.856e-003	6.856e-003	6.856e-003	6.878e-003
32	Static Gradient	(kPa/m)	-7.672e-003	-7.672e-003	-7.672e-003	-1.294e-002
33	Accel. Gradient	(kPa/m)	-5.725e-008	-5.725e-008	-5.726e-008	-4.252e-007
34	Liq. Reynolds		2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *
35	Vap. Reynolds		9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
36	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
37	Vapour Velocity	(m/s)	2.160	2.160	2.160	2.160
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	1.756e+004	1.757e+004	1.757e+004	1.758e+004
41	Elevation	(m)	-6.800e-002	-0.1020	-0.1360	-0.1700
42	Pressure	(bar)	90.76	90.76	90.76	90.76
43	Temperature	(C)	54.28	54.28	54.28	54.28
44	Vapour Fraction		0.8706	0.8706	0.8706	0.8706
45	Heat Transferred	(kJ/h-m)	646.9	646.9	646.9	646.8
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2014	0.2015	0.2015	0.2015
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	6.878e-003	6.878e-003	6.878e-003	6.877e-003
51	Static Gradient	(kPa/m)	-1.294e-002	-1.294e-002	-1.294e-002	-1.294e-002
52	Accel. Gradient	(kPa/m)	-4.252e-007	-4.252e-007	-4.252e-007	-4.252e-007
53	Liq. Reynolds		2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *
54	Vap. Reynolds		9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
55	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
56	Vapour Velocity	(m/s)	2.160	2.160	2.160	2.159
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	1.758e+004	1.759e+004	1.759e+004	1.760e+004
60	Elevation	(m)	-0.1180	-6.600e-002	-1.400e-002	3.800e-002
61	Pressure	(bar)	90.76	90.76	90.76	90.76
62	Temperature	(C)	54.28	54.27	54.27	54.27
63	Vapour Fraction		0.8706	0.8706	0.8706	0.8706
64	Heat Transferred	(kJ/h-m)	646.7	646.7	646.6	646.5
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.3135	0.3135	0.3135	0.3135
68	Water HoldUp		---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:02:44 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	6.002e-003	6.002e-003	6.002e-003	6.002e-003	
14	Static Gradient	(kPa/m)	2.668e-002	2.668e-002	2.668e-002	2.668e-002	
15	Accel. Gradient	(kPa/m)	2.293e-006	2.293e-006	2.293e-006	2.293e-006	
16	Liq. Reynolds		2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *	
17	Vap. Reynolds		9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *	
18	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
19	Vapour Velocity	(m/s)	2.159	2.160	2.160	2.160	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	1.760e+004	1.761e+004	1.761e+004	1.762e+004	
23	Elevation	(m)	9.000e-002	7.400e-002	5.800e-002	4.200e-002	
24	Pressure	(bar)	90.76	90.76	90.76	90.76	
25	Temperature	(C)	54.27	54.26	54.26	54.26	
26	Vapour Fraction		0.8706	0.8706	0.8706	0.8706	
27	Heat Transferred	(kJ/h-m)	646.5	646.4	646.4	646.3	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.3135	0.2048	0.2048	0.2048	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	6.002e-003	6.850e-003	6.850e-003	6.850e-003	
33	Static Gradient	(kPa/m)	2.668e-002	-6.152e-003	-6.152e-003	-6.152e-003	
34	Accel. Gradient	(kPa/m)	2.293e-006	4.895e-008	4.895e-008	4.894e-008	
35	Liq. Reynolds		2.140e+005 *	2.139e+005 *	2.139e+005 *	2.139e+005 *	
36	Vap. Reynolds		9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *	
37	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
38	Vapour Velocity	(m/s)	2.160	2.160	2.160	2.160	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	1.762e+004	1.763e+004	1.763e+004	1.764e+004	
42	Elevation	(m)	2.600e-002	1.000e-002	6.000e-003	2.000e-003	
43	Pressure	(bar)	90.76	90.76	90.76	90.76	
44	Temperature	(C)	54.26	54.26	54.26	54.25	
45	Vapour Fraction		0.8706	0.8706	0.8706	0.8706	
46	Heat Transferred	(kJ/h-m)	646.3	646.3	646.2	646.1	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.2048	0.2048	0.2071	0.2071	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	6.850e-003	6.850e-003	6.832e-003	6.832e-003	
52	Static Gradient	(kPa/m)	-6.152e-003	-6.152e-003	-1.549e-003	-1.549e-003	
53	Accel. Gradient	(kPa/m)	4.893e-008	4.892e-008	3.707e-007	3.707e-007	
54	Liq. Reynolds		2.139e+005 *	2.139e+005 *	2.139e+005 *	2.139e+005 *	
55	Vap. Reynolds		9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *	
56	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
57	Vapour Velocity	(m/s)	2.159	2.159	2.159	2.159	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	1.764e+004	1.765e+004	1.765e+004	1.766e+004	
61	Elevation	(m)	-2.000e-003	-6.000e-003	-1.000e-002	-1.800e-002	
62	Pressure	(bar)	90.76	90.76	90.76	90.76	
63	Temperature	(C)	54.25	54.25	54.25	54.25	
64	Vapour Fraction		0.8706	0.8706	0.8706	0.8706	
65	Heat Transferred	(kJ/h-m)	646.1	646.0	646.0	645.9	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.2071	0.2071	0.2071	0.2063	
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	6.832e-003	6.832e-003	6.832e-003	6.838e-003
15	Static Gradient (kPa/m)	-1.549e-003	-1.549e-003	-1.549e-003	-3.090e-003
16	Accel. Gradient (kPa/m)	3.707e-007	3.707e-007	3.706e-007	2.629e-007
17	Liq. Reynolds	2.139e+005 *	2.139e+005 *	2.139e+005 *	2.139e+005 *
18	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
19	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
20	Vapour Velocity (m/s)	2.159	2.159	2.159	2.159
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	1.766e+004	1.767e+004	1.767e+004	1.768e+004
24	Elevation (m)	-2.600e-002	-3.400e-002	-4.200e-002	-5.000e-002
25	Pressure (bar)	90.75	90.75	90.75	90.75
26	Temperature (C)	54.24	54.24	54.24	54.24
27	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
28	Heat Transferred (kJ/h-m)	645.9	645.8	645.8	645.7
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2063	0.2063	0.2063	0.2063
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	6.837e-003	6.837e-003	6.837e-003	6.837e-003
34	Static Gradient (kPa/m)	-3.090e-003	-3.090e-003	-3.090e-003	-3.090e-003
35	Accel. Gradient (kPa/m)	2.629e-007	2.629e-007	2.629e-007	2.629e-007
36	Liq. Reynolds	2.139e+005 *	2.139e+005 *	2.139e+005 *	2.139e+005 *
37	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
38	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
39	Vapour Velocity (m/s)	2.159	2.159	2.159	2.159
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	1.768e+004	1.769e+004	1.769e+004	1.770e+004
43	Elevation (m)	-5.000e-002	-5.000e-002	-5.000e-002	-5.000e-002
44	Pressure (bar)	90.75	90.75	90.75	90.75
45	Temperature (C)	54.24	54.23	54.23	54.23
46	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
47	Heat Transferred (kJ/h-m)	645.7	645.6	645.6	645.5
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.3034	0.3034	0.3034	0.3034
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	5.855e-003	5.855e-003	5.855e-003	5.855e-003
53	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
54	Accel. Gradient (kPa/m)	4.107e-007	4.107e-007	4.107e-007	4.107e-007
55	Liq. Reynolds	2.139e+005 *	2.139e+005 *	2.139e+005 *	2.139e+005 *
56	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
57	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
58	Vapour Velocity (m/s)	2.159	2.159	2.159	2.159
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	1.770e+004	1.771e+004	1.771e+004	1.772e+004
62	Elevation (m)	-5.000e-002	-5.200e-002	-5.400e-002	-5.600e-002
63	Pressure (bar)	90.75	90.75	90.75	90.75
64	Temperature (C)	54.23	54.23	54.22	54.22
65	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
66	Heat Transferred (kJ/h-m)	645.5	645.4	645.4	645.3
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 312 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.3034	0.2075	0.2075	0.2075
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	5.854e-003	6.828e-003	6.828e-003	6.828e-003
16	Static Gradient (kPa/m)	0.0000	-7.753e-004	-7.753e-004	-7.753e-004
17	Accel. Gradient (kPa/m)	4.107e-007	4.247e-007	4.246e-007	4.246e-007
18	Liq. Reynolds	2.139e+005 *	2.139e+005 *	2.139e+005 *	2.139e+005 *
19	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
20	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
21	Vapour Velocity (m/s)	2.159	2.159	2.159	2.159
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	1.772e+004	1.773e+004	1.773e+004	1.774e+004
25	Elevation (m)	-5.800e-002	-6.000e-002	-8.400e-002	-0.1080
26	Pressure (bar)	90.75	90.75	90.75	90.75
27	Temperature (C)	54.22	54.22	54.22	54.22
28	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
29	Heat Transferred (kJ/h-m)	645.3	645.2	645.2	645.2
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2075	0.2075	0.2033	0.2033
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.828e-003	6.828e-003	6.861e-003	6.861e-003
35	Static Gradient (kPa/m)	-7.753e-004	-7.753e-004	-9.187e-003	-9.187e-003
36	Accel. Gradient (kPa/m)	4.246e-007	4.246e-007	-1.632e-007	-1.632e-007
37	Liq. Reynolds	2.139e+005 *	2.139e+005 *	2.139e+005 *	2.139e+005 *
38	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
39	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
40	Vapour Velocity (m/s)	2.159	2.159	2.159	2.159
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	1.774e+004	1.775e+004	1.775e+004	1.776e+004
44	Elevation (m)	-0.1320	-0.1560	-0.1800	-0.1800
45	Pressure (bar)	90.75	90.75	90.75	90.75
46	Temperature (C)	54.21	54.21	54.21	54.21
47	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
48	Heat Transferred (kJ/h-m)	645.1	645.1	645.0	645.0
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2033	0.2033	0.2034	0.3034
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.861e-003	6.861e-003	6.861e-003	5.854e-003
54	Static Gradient (kPa/m)	-9.187e-003	-9.187e-003	-9.188e-003	0.0000
55	Accel. Gradient (kPa/m)	-1.632e-007	-1.632e-007	-1.632e-007	4.107e-007
56	Liq. Reynolds	2.139e+005 *	2.139e+005 *	2.139e+005 *	2.139e+005 *
57	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
58	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
59	Vapour Velocity (m/s)	2.159	2.159	2.159	2.159
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	1.776e+004	1.777e+004	1.777e+004	1.778e+004
63	Elevation (m)	-0.1800	-0.1800	-0.1800	-0.1800
64	Pressure (bar)	90.75	90.75	90.75	90.75
65	Temperature (C)	54.21	54.20	54.20	54.20
66	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
67	Heat Transferred (kJ/h-m)	644.9	644.8	644.8	644.7
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 313 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.3034	0.3034	0.3034	0.3034
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	5.854e-003	5.854e-003	5.854e-003	5.854e-003
17	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
18	Accel. Gradient (kPa/m)	4.107e-007	4.107e-007	4.107e-007	4.106e-007
19	Liq. Reynolds	2.139e+005 *	2.139e+005 *	2.139e+005 *	2.139e+005 *
20	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
21	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
22	Vapour Velocity (m/s)	2.159	2.159	2.159	2.159
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	1.778e+004	1.779e+004	1.779e+004	1.780e+004
26	Elevation (m)	-0.2200	-0.2600	-0.3000	-0.3400
27	Pressure (bar)	90.75	90.75	90.75	90.75
28	Temperature (C)	54.20	54.20	54.20	54.19
29	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
30	Heat Transferred (kJ/h-m)	644.7	644.7	644.6	644.6
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2004	0.2004	0.2004	0.2004
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	6.886e-003	6.885e-003	6.885e-003	6.885e-003
36	Static Gradient (kPa/m)	-1.517e-002	-1.517e-002	-1.517e-002	-1.517e-002
37	Accel. Gradient (kPa/m)	-5.813e-007	-5.813e-007	-5.813e-007	-5.813e-007
38	Liq. Reynolds	2.139e+005 *	2.139e+005 *	2.139e+005 *	2.139e+005 *
39	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
40	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
41	Vapour Velocity (m/s)	2.159	2.159	2.159	2.159
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	1.780e+004	1.781e+004	1.781e+004	1.782e+004
45	Elevation (m)	-0.3800	-0.3380	-0.2960	-0.2540
46	Pressure (bar)	90.75	90.75	90.75	90.75
47	Temperature (C)	54.19	54.19	54.19	54.19
48	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
49	Heat Transferred (kJ/h-m)	644.5	644.5	644.4	644.4
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2004	0.3072	0.3072	0.3072
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	6.885e-003	5.935e-003	5.935e-003	5.935e-003
55	Static Gradient (kPa/m)	-1.517e-002	2.123e-002	2.123e-002	2.123e-002
56	Accel. Gradient (kPa/m)	-5.813e-007	1.906e-006	1.906e-006	1.906e-006
57	Liq. Reynolds	2.139e+005 *	2.139e+005 *	2.139e+005 *	2.139e+005 *
58	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
59	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
60	Vapour Velocity (m/s)	2.159	2.159	2.159	2.159
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	1.782e+004	1.783e+004	1.783e+004	1.784e+004
64	Elevation (m)	-0.2120	-0.1700	-0.3100	-0.4500
65	Pressure (bar)	90.75	90.75	90.75	90.75
66	Temperature (C)	54.18	54.18	54.18	54.18
67	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
68	Heat Transferred (kJ/h-m)	644.3	644.2	644.2	644.2
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 314 of 937

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.3072	0.3072	0.1817	0.1817
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	5.935e-003	5.935e-003	7.069e-003	7.068e-003
18	Static Gradient (kPa/m)	2.123e-002	2.123e-002	-5.002e-002	-5.002e-002
19	Accel. Gradient (kPa/m)	1.906e-006	1.906e-006	-3.012e-006	-3.012e-006
20	Liq. Reynolds	2.139e+005 *	2.139e+005 *	2.139e+005 *	2.139e+005 *
21	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
22	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
23	Vapour Velocity (m/s)	2.159	2.159	2.159	2.159
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	1.784e+004	1.785e+004	1.785e+004	1.785e+004
27	Elevation (m)	-0.5900	-0.7300	-0.8700	-0.8700
28	Pressure (bar)	90.75	90.75	90.76	90.76
29	Temperature (C)	54.18	54.18	54.18	54.18
30	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
31	Heat Transferred (kJ/h-m)	644.2	644.2	644.1	644.1
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1817	0.1817	0.1817	0.3035
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	7.068e-003	7.068e-003	7.068e-003	5.853e-003
37	Static Gradient (kPa/m)	-5.002e-002	-5.002e-002	-5.002e-002	0.0000
38	Accel. Gradient (kPa/m)	-3.012e-006	-3.012e-006	-3.012e-006	4.105e-007
39	Liq. Reynolds	2.139e+005 *	2.139e+005 *	2.139e+005 *	2.139e+005 *
40	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
41	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
42	Vapour Velocity (m/s)	2.159	2.159	2.158	2.158
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	1.786e+004	1.786e+004	1.786e+004	1.787e+004
46	Elevation (m)	-0.8700	-0.8700	-0.8700	-0.8700
47	Pressure (bar)	90.76	90.76	90.76	90.76
48	Temperature (C)	54.17	54.17	54.17	54.17
49	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
50	Heat Transferred (kJ/h-m)	644.0	644.0	644.0	644.0
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.3035	0.3035	0.3035	0.3035
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	5.853e-003	5.853e-003	5.853e-003	5.853e-003
56	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
57	Accel. Gradient (kPa/m)	4.105e-007	4.105e-007	4.105e-007	4.105e-007
58	Liq. Reynolds	2.139e+005 *	2.139e+005 *	2.139e+005 *	2.139e+005 *
59	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *
60	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
61	Vapour Velocity (m/s)	2.158	2.158	2.158	2.158
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	1.787e+004	1.787e+004	1.787e+004	1.787e+004
65	Elevation (m)	-0.9560	-1.042	-1.128	-1.214
66	Pressure (bar)	90.76	90.76	90.76	90.76
67	Temperature (C)	54.17	54.17	54.17	54.17
68	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed			
6			Date/Time: Mon Jun 13 15:02:44 2022			
7						
8						
9			PROFILES			
10						
11			Pipe Table			
12						
13	Heat Transferred (kJ/h-m)	644.0	644.0	644.0	644.0	
14	Flow Regime	Transition	Transition	Transition	Transition	
15	Water Flow Regime					
16	Liquid HoldUp	0.1678	0.1678	0.1678	0.1678	
17	Water HoldUp	---	---	---	---	
18	Friction Gradient (kPa/m)	7.241e-003	7.241e-003	7.241e-003	7.241e-003	
19	Static Gradient (kPa/m)	-7.327e-002	-7.328e-002	-7.328e-002	-7.328e-002	
20	Accel. Gradient (kPa/m)	-4.631e-006	-4.631e-006	-4.630e-006	-4.630e-006	
21	Liq. Reynolds	2.139e+005 *	2.139e+005 *	2.139e+005 *	2.139e+005 *	
22	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.111e+006 *	9.111e+006 *	
23	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044	
24	Vapour Velocity (m/s)	2.158	2.158	2.158	2.158	
25	Deposit Thickness (mm)	---	---	---	---	
26	Deposit Volume (m3)	---	---	---	---	
27	Length (m)	1.788e+004	1.788e+004	1.789e+004	1.789e+004	
28	Elevation (m)	-1.300	-1.900	-2.500	-3.100	
29	Pressure (bar)	90.76	90.77	90.78	90.78	
30	Temperature (C)	54.17	54.17	54.17	54.17	
31	Vapour Fraction	0.8705	0.8705	0.8705	0.8705	
32	Heat Transferred (kJ/h-m)	644.0	644.2	644.3	644.3	
33	Flow Regime	Transition	Transition	Transition	Transition	
34	Water Flow Regime					
35	Liquid HoldUp	0.1678	9.804e-002	9.805e-002	9.805e-002	
36	Water HoldUp	---	---	---	---	
37	Friction Gradient (kPa/m)	7.240e-003	9.180e-003	9.179e-003	9.178e-003	
38	Static Gradient (kPa/m)	-7.328e-002	-0.1550	-0.1550	-0.1550	
39	Accel. Gradient (kPa/m)	-4.630e-006	-1.022e-005	-1.022e-005	-1.022e-005	
40	Liq. Reynolds	2.139e+005 *	2.139e+005 *	2.139e+005 *	2.139e+005 *	
41	Vap. Reynolds	9.111e+006 *	9.111e+006 *	9.110e+006 *	9.110e+006 *	
42	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044	
43	Vapour Velocity (m/s)	2.158	2.158	2.158	2.158	
44	Deposit Thickness (mm)	---	---	---	---	
45	Deposit Volume (m3)	---	---	---	---	
46	Length (m)	1.790e+004	1.790e+004	1.790e+004	1.791e+004	
47	Elevation (m)	-3.700	-4.300	-4.600	-4.900	
48	Pressure (bar)	90.79	90.80	90.80	90.81	
49	Temperature (C)	54.18	54.18	54.18	54.18	
50	Vapour Fraction	0.8705	0.8705	0.8705	0.8705	
51	Heat Transferred (kJ/h-m)	644.4	644.4	644.6	644.6	
52	Flow Regime	Transition	Transition	Transition	Transition	
53	Water Flow Regime					
54	Liquid HoldUp	9.805e-002	9.806e-002	7.219e-002	7.219e-002	
55	Water HoldUp	---	---	---	---	
56	Friction Gradient (kPa/m)	9.178e-003	9.177e-003	1.115e-002	1.115e-002	
57	Static Gradient (kPa/m)	-0.1551	-0.1551	-0.1709	-0.1709	
58	Accel. Gradient (kPa/m)	-1.022e-005	-1.022e-005	-1.119e-005	-1.119e-005	
59	Liq. Reynolds	2.139e+005 *	2.139e+005 *	2.140e+005 *	2.140e+005 *	
60	Vap. Reynolds	9.110e+006 *	9.110e+006 *	9.109e+006 *	9.109e+006 *	
61	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044	
62	Vapour Velocity (m/s)	2.158	2.157	2.157	2.157	
63	Deposit Thickness (mm)	---	---	---	---	
64	Deposit Volume (m3)	---	---	---	---	
65	Length (m)	1.791e+004	1.791e+004	1.791e+004	1.792e+004	
66	Elevation (m)	-5.200	-5.500	-5.800	-6.000	
67	Pressure (bar)	90.81	90.81	90.81	90.82	
68	Temperature (C)	54.18	54.18	54.18	54.18	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
14	Heat Transferred (kJ/h-m)	644.6	644.7	644.7	644.3
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	7.220e-002	7.220e-002	7.220e-002	0.1548
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	1.115e-002	1.115e-002	1.115e-002	7.436e-003
20	Static Gradient (kPa/m)	-0.1709	-0.1709	-0.1709	-9.300e-002
21	Accel. Gradient (kPa/m)	-1.119e-005	-1.119e-005	-1.119e-005	-5.992e-006
22	Liq. Reynolds	2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *
23	Vap. Reynolds	9.109e+006 *	9.109e+006 *	9.109e+006 *	9.109e+006 *
24	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
25	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	1.792e+004	1.792e+004	1.793e+004	1.793e+004
29	Elevation (m)	-6.200	-6.400	-6.600	-6.800
30	Pressure (bar)	90.82	90.82	90.83	90.83
31	Temperature (C)	54.18	54.18	54.18	54.18
32	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
33	Heat Transferred (kJ/h-m)	644.3	644.3	644.3	644.3
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.1548	0.1548	0.1548	0.1548
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	7.436e-003	7.436e-003	7.436e-003	7.435e-003
39	Static Gradient (kPa/m)	-9.301e-002	-9.301e-002	-9.301e-002	-9.301e-002
40	Accel. Gradient (kPa/m)	-5.992e-006	-5.992e-006	-5.991e-006	-5.991e-006
41	Liq. Reynolds	2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *
42	Vap. Reynolds	9.109e+006 *	9.109e+006 *	9.109e+006 *	9.109e+006 *
43	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
44	Vapour Velocity (m/s)	2.157	2.157	2.157	2.157
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	1.793e+004	1.794e+004	1.794e+004	1.794e+004
48	Elevation (m)	-7.100	-7.400	-7.700	-8.000
49	Pressure (bar)	90.83	90.84	90.84	90.85
50	Temperature (C)	54.18	54.18	54.18	54.18
51	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
52	Heat Transferred (kJ/h-m)	644.4	644.4	644.4	644.5
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.1287	0.1287	0.1287	0.1287
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	7.997e-003	7.997e-003	7.996e-003	7.996e-003
58	Static Gradient (kPa/m)	-0.1263	-0.1263	-0.1263	-0.1263
59	Accel. Gradient (kPa/m)	-8.281e-006	-8.280e-006	-8.280e-006	-8.279e-006
60	Liq. Reynolds	2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *
61	Vap. Reynolds	9.108e+006 *	9.108e+006 *	9.108e+006 *	9.108e+006 *
62	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
63	Vapour Velocity (m/s)	2.156	2.156	2.156	2.156
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	1.795e+004	1.795e+004	1.796e+004	1.796e+004
67	Elevation (m)	-8.300	-8.600	-8.900	-9.200
68	Pressure (bar)	90.85	90.85	90.86	90.86
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2					
3		Epic Atlantic Limited	Unit Set: Mixed		
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5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	54.18	54.18	54.18	54.18
14	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
15	Heat Transferred (kJ/h-m)	644.5	644.4	644.4	644.4
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.1287	0.1614	0.1614	0.1614
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	7.996e-003	7.327e-003	7.327e-003	7.326e-003
21	Static Gradient (kPa/m)	-0.1263	-8.335e-002	-8.336e-002	-8.336e-002
22	Accel. Gradient (kPa/m)	-8.279e-006	-5.319e-006	-5.319e-006	-5.318e-006
23	Liq. Reynolds	2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *
24	Vap. Reynolds	9.108e+006 *	9.108e+006 *	9.108e+006 *	9.107e+006 *
25	Liquid Velocity (m/s)	0.2044	0.2044	0.2045	0.2045
26	Vapour Velocity (m/s)	2.156	2.156	2.156	2.156
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	1.797e+004	1.798e+004	1.798e+004	1.799e+004
30	Elevation (m)	-9.500	-9.800	-10.30	-10.80
31	Pressure (bar)	90.87	90.87	90.88	90.89
32	Temperature (C)	54.18	54.18	54.18	54.19
33	Vapour Fraction	0.8704	0.8704	0.8704	0.8704
34	Heat Transferred (kJ/h-m)	644.4	644.3	644.5	644.5
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.1614	0.1614	0.1158	0.1159
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	7.326e-003	7.325e-003	8.393e-003	8.392e-003
40	Static Gradient (kPa/m)	-8.336e-002	-8.336e-002	-0.1398	-0.1398
41	Accel. Gradient (kPa/m)	-5.318e-006	-5.318e-006	-9.188e-006	-9.187e-006
42	Liq. Reynolds	2.140e+005 *	2.140e+005 *	2.140e+005 *	2.140e+005 *
43	Vap. Reynolds	9.107e+006 *	9.107e+006 *	9.107e+006 *	9.107e+006 *
44	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
45	Vapour Velocity (m/s)	2.155	2.155	2.155	2.155
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	1.799e+004	1.800e+004	1.800e+004	1.801e+004
49	Elevation (m)	-11.30	-11.80	-12.30	-12.80
50	Pressure (bar)	90.89	90.90	90.91	90.91
51	Temperature (C)	54.19	54.19	54.19	54.19
52	Vapour Fraction	0.8704	0.8704	0.8704	0.8704
53	Heat Transferred (kJ/h-m)	644.6	644.6	644.6	644.7
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.1159	0.1159	0.1159	0.1159
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	8.391e-003	8.391e-003	8.390e-003	8.390e-003
59	Static Gradient (kPa/m)	-0.1398	-0.1398	-0.1398	-0.1398
60	Accel. Gradient (kPa/m)	-9.186e-006	-9.186e-006	-9.185e-006	-9.184e-006
61	Liq. Reynolds	2.141e+005 *	2.141e+005 *	2.141e+005 *	2.141e+005 *
62	Vap. Reynolds	9.107e+006 *	9.106e+006 *	9.106e+006 *	9.106e+006 *
63	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
64	Vapour Velocity (m/s)	2.155	2.155	2.154	2.154
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	1.801e+004	1.802e+004	1.802e+004	1.803e+004
68	Elevation (m)	-13.30	-13.80	-14.30	-14.80
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3		Epic Atlantic Limited	Unit Set: Mixed		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
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11	Pipe Table				
12					
13	Pressure (bar)	90.92	90.93	90.93	90.94
14	Temperature (C)	54.19	54.19	54.19	54.19
15	Vapour Fraction	0.8704	0.8704	0.8704	0.8704
16	Heat Transferred (kJ/h-m)	644.7	644.7	644.7	644.8
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.1159	0.1159	0.1159	0.1159
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	8.389e-003	8.388e-003	8.388e-003	8.387e-003
22	Static Gradient (kPa/m)	-0.1398	-0.1398	-0.1399	-0.1399
23	Accel. Gradient (kPa/m)	-9.183e-006	-9.182e-006	-9.181e-006	-9.180e-006
24	Liq. Reynolds	2.141e+005 *	2.141e+005 *	2.141e+005 *	2.141e+005 *
25	Vap. Reynolds	9.106e+006 *	9.106e+006 *	9.105e+006 *	9.105e+006 *
26	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
27	Vapour Velocity (m/s)	2.154	2.154	2.154	2.154
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	1.803e+004	1.804e+004	1.804e+004	1.805e+004
31	Elevation (m)	-14.50	-14.20	-13.90	-13.60
32	Pressure (bar)	90.93	90.92	90.91	90.90
33	Temperature (C)	54.19	54.18	54.18	54.17
34	Vapour Fraction	0.8704	0.8704	0.8704	0.8704
35	Heat Transferred (kJ/h-m)	644.2	644.1	644.0	643.8
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.4717	0.4716	0.4716	0.4716
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	5.853e-003	5.853e-003	5.854e-003	5.854e-003
41	Static Gradient (kPa/m)	0.2100	0.2100	0.2100	0.2100
42	Accel. Gradient (kPa/m)	1.508e-005	1.508e-005	1.508e-005	1.509e-005
43	Liq. Reynolds	2.141e+005 *	2.141e+005 *	2.141e+005 *	2.141e+005 *
44	Vap. Reynolds	9.105e+006 *	9.106e+006 *	9.106e+006 *	9.106e+006 *
45	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
46	Vapour Velocity (m/s)	2.154	2.154	2.154	2.155
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	1.805e+004	1.806e+004	1.806e+004	1.807e+004
50	Elevation (m)	-13.30	-10.78	-8.260	-5.740
51	Pressure (bar)	90.88	90.72	90.55	90.38
52	Temperature (C)	54.17	54.13	54.09	54.05
53	Vapour Fraction	0.8704	0.8705	0.8706	0.8706
54	Heat Transferred (kJ/h-m)	643.7	637.0	635.9	634.8
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.4716	1.000	1.000	1.000
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	5.855e-003	5.579e-003	5.589e-003	5.598e-003
60	Static Gradient (kPa/m)	0.2100	3.338	3.339	3.339
61	Accel. Gradient (kPa/m)	1.509e-005	2.346e-004	2.356e-004	2.365e-004
62	Liq. Reynolds	2.140e+005 *	2.138e+005 *	2.136e+005 *	2.134e+005 *
63	Vap. Reynolds	9.107e+006 *	9.112e+006 *	9.117e+006 *	9.123e+006 *
64	Liquid Velocity (m/s)	0.2045	0.2044	0.2044	0.2043
65	Vapour Velocity (m/s)	2.155	2.159	2.163	2.167
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	1.807e+004	1.808e+004	1.808e+004	1.809e+004
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
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10							
11	Pipe Table						
12							
13	Elevation	(m)	-3.220	-0.7000	-0.6600	-0.6200	
14	Pressure	(bar)	90.22	90.05	90.05	90.05	
15	Temperature	(C)	54.00	53.96	53.96	53.96	
16	Vapour Fraction		0.8707	0.8708	0.8708	0.8708	
17	Heat Transferred	(kJ/h-m)	633.7	632.6	638.3	638.3	
18	Flow Regime		Transition	Transition	Transition	Transition	
19	Water Flow Regime						
20	Liquid HoldUp		1.000	1.000	0.3044	0.3044	
21	Water HoldUp		---	---	---	---	
22	Friction Gradient	(kPa/m)	5.607e-003	5.617e-003	5.949e-003	5.949e-003	
23	Static Gradient	(kPa/m)	3.340	3.340	2.007e-002	2.007e-002	
24	Accel. Gradient	(kPa/m)	2.375e-004	2.384e-004	1.854e-006	1.854e-006	
25	Liq. Reynolds		2.132e+005 *	2.130e+005 *	2.130e+005 *	2.130e+005 *	
26	Vap. Reynolds		9.128e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *	
27	Liquid Velocity	(m/s)	0.2043	0.2042	0.2042	0.2042	
28	Vapour Velocity	(m/s)	2.172	2.176	2.176	2.176	
29	Deposit Thickness	(mm)	---	---	---	---	
30	Deposit Volume	(m3)	---	---	---	---	
31	Length	(m)	1.809e+004	1.810e+004	1.810e+004	1.811e+004	
32	Elevation	(m)	-0.5800	-0.5400	-0.5000	-0.4600	
33	Pressure	(bar)	90.04	90.04	90.04	90.04	
34	Temperature	(C)	53.96	53.95	53.95	53.95	
35	Vapour Fraction		0.8708	0.8708	0.8708	0.8708	
36	Heat Transferred	(kJ/h-m)	638.2	638.2	638.1	638.1	
37	Flow Regime		Transition	Transition	Transition	Transition	
38	Water Flow Regime						
39	Liquid HoldUp		0.3044	0.3044	0.3044	0.3044	
40	Water HoldUp		---	---	---	---	
41	Friction Gradient	(kPa/m)	5.949e-003	5.949e-003	5.949e-003	5.949e-003	
42	Static Gradient	(kPa/m)	2.007e-002	2.007e-002	2.007e-002	2.007e-002	
43	Accel. Gradient	(kPa/m)	1.854e-006	1.854e-006	1.854e-006	1.854e-006	
44	Liq. Reynolds		2.130e+005 *	2.130e+005 *	2.130e+005 *	2.130e+005 *	
45	Vap. Reynolds		9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *	
46	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
47	Vapour Velocity	(m/s)	2.176	2.176	2.176	2.176	
48	Deposit Thickness	(mm)	---	---	---	---	
49	Deposit Volume	(m3)	---	---	---	---	
50	Length	(m)	1.811e+004	1.812e+004	1.812e+004	1.813e+004	
51	Elevation	(m)	-0.4200	-0.3800	-0.3400	-0.3000	
52	Pressure	(bar)	90.04	90.04	90.04	90.04	
53	Temperature	(C)	53.95	53.95	53.94	53.94	
54	Vapour Fraction		0.8708	0.8708	0.8708	0.8708	
55	Heat Transferred	(kJ/h-m)	638.0	637.9	637.9	637.8	
56	Flow Regime		Transition	Transition	Transition	Transition	
57	Water Flow Regime						
58	Liquid HoldUp		0.3044	0.3044	0.3044	0.3044	
59	Water HoldUp		---	---	---	---	
60	Friction Gradient	(kPa/m)	5.949e-003	5.949e-003	5.949e-003	5.949e-003	
61	Static Gradient	(kPa/m)	2.007e-002	2.007e-002	2.007e-002	2.007e-002	
62	Accel. Gradient	(kPa/m)	1.854e-006	1.854e-006	1.854e-006	1.855e-006	
63	Liq. Reynolds		2.130e+005 *	2.130e+005 *	2.130e+005 *	2.130e+005 *	
64	Vap. Reynolds		9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *	
65	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
66	Vapour Velocity	(m/s)	2.176	2.176	2.176	2.176	
67	Deposit Thickness	(mm)	---	---	---	---	
68	Deposit Volume	(m3)	---	---	---	---	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
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5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length (m)	1.813e+004	1.814e+004	1.814e+004	1.815e+004		
14	Elevation (m)	-0.2920	-0.2840	-0.2760	-0.2680		
15	Pressure (bar)	90.04	90.03	90.03	90.03		
16	Temperature (C)	53.94	53.94	53.94	53.93		
17	Vapour Fraction	0.8708	0.8708	0.8708	0.8708		
18	Heat Transferred (kJ/h-m)	637.8	637.7	637.7	637.6		
19	Flow Regime	Transition	Transition	Transition	Transition		
20	Water Flow Regime						
21	Liquid HoldUp	0.2841	0.2841	0.2841	0.2841		
22	Water HoldUp	---	---	---	---		
23	Friction Gradient (kPa/m)	5.255e-003	5.255e-003	5.255e-003	5.255e-003		
24	Static Gradient (kPa/m)	3.822e-003	3.822e-003	3.822e-003	3.822e-003		
25	Accel. Gradient (kPa/m)	6.468e-007	6.468e-007	6.468e-007	6.468e-007		
26	Liq. Reynolds	2.130e+005 *	2.129e+005 *	2.129e+005 *	2.129e+005 *		
27	Vap. Reynolds	9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *		
28	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
29	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176		
30	Deposit Thickness (mm)	---	---	---	---		
31	Deposit Volume (m3)	---	---	---	---		
32	Length (m)	1.815e+004	1.816e+004	1.816e+004	1.817e+004		
33	Elevation (m)	-0.2600	-0.2640	-0.2680	-0.2720		
34	Pressure (bar)	90.03	90.03	90.03	90.03		
35	Temperature (C)	53.93	53.93	53.93	53.93		
36	Vapour Fraction	0.8708	0.8708	0.8708	0.8708		
37	Heat Transferred (kJ/h-m)	637.6	637.5	637.5	637.4		
38	Flow Regime	Transition	Transition	Transition	Transition		
39	Water Flow Regime						
40	Liquid HoldUp	0.2841	0.2061	0.2061	0.2061		
41	Water HoldUp	---	---	---	---		
42	Friction Gradient (kPa/m)	5.255e-003	6.880e-003	6.880e-003	6.880e-003		
43	Static Gradient (kPa/m)	3.822e-003	-1.542e-003	-1.542e-003	-1.542e-003		
44	Accel. Gradient (kPa/m)	6.468e-007	3.804e-007	3.804e-007	3.804e-007		
45	Liq. Reynolds	2.129e+005 *	2.129e+005 *	2.129e+005 *	2.129e+005 *		
46	Vap. Reynolds	9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *		
47	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
48	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176		
49	Deposit Thickness (mm)	---	---	---	---		
50	Deposit Volume (m3)	---	---	---	---		
51	Length (m)	1.817e+004	1.818e+004	1.818e+004	1.819e+004		
52	Elevation (m)	-0.2760	-0.2800	-0.2720	-0.2640		
53	Pressure (bar)	90.03	90.03	90.03	90.03		
54	Temperature (C)	53.92	53.92	53.92	53.92		
55	Vapour Fraction	0.8708	0.8708	0.8708	0.8708		
56	Heat Transferred (kJ/h-m)	637.4	637.3	637.3	637.2		
57	Flow Regime	Transition	Transition	Transition	Transition		
58	Water Flow Regime						
59	Liquid HoldUp	0.2061	0.2061	0.2841	0.2841		
60	Water HoldUp	---	---	---	---		
61	Friction Gradient (kPa/m)	6.879e-003	6.879e-003	5.254e-003	5.254e-003		
62	Static Gradient (kPa/m)	-1.542e-003	-1.542e-003	3.822e-003	3.822e-003		
63	Accel. Gradient (kPa/m)	3.804e-007	3.804e-007	6.468e-007	6.468e-007		
64	Liq. Reynolds	2.129e+005 *	2.129e+005 *	2.129e+005 *	2.129e+005 *		
65	Vap. Reynolds	9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *		
66	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042		
67	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176		
68	Deposit Thickness (mm)	---	---	---	---		
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 321 of 937		

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	1.819e+004	1.820e+004	1.820e+004	1.821e+004
15	Elevation (m)	-0.2560	-0.2480	-0.2400	-0.2340
16	Pressure (bar)	90.03	90.03	90.03	90.03
17	Temperature (C)	53.92	53.91	53.91	53.91
18	Vapour Fraction	0.8708	0.8708	0.8708	0.8708
19	Heat Transferred (kJ/h-m)	637.1	637.1	637.0	637.0
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2841	0.2841	0.2841	0.2828
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	5.254e-003	5.254e-003	5.254e-003	5.352e-003
25	Static Gradient (kPa/m)	3.822e-003	3.822e-003	3.822e-003	2.858e-003
26	Accel. Gradient (kPa/m)	6.468e-007	6.468e-007	6.468e-007	5.850e-007
27	Liq. Reynolds	2.129e+005 *	2.129e+005 *	2.129e+005 *	2.129e+005 *
28	Vap. Reynolds	9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *
29	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
30	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	1.821e+004	1.822e+004	1.822e+004	1.823e+004
34	Elevation (m)	-0.2280	-0.2220	-0.2160	-0.2100
35	Pressure (bar)	90.03	90.03	90.03	90.03
36	Temperature (C)	53.91	53.91	53.90	53.90
37	Vapour Fraction	0.8708	0.8708	0.8708	0.8708
38	Heat Transferred (kJ/h-m)	636.9	636.9	636.8	636.8
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2828	0.2828	0.2828	0.2828
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	5.352e-003	5.352e-003	5.352e-003	5.352e-003
44	Static Gradient (kPa/m)	2.858e-003	2.858e-003	2.858e-003	2.858e-003
45	Accel. Gradient (kPa/m)	5.850e-007	5.850e-007	5.850e-007	5.850e-007
46	Liq. Reynolds	2.129e+005 *	2.129e+005 *	2.129e+005 *	2.129e+005 *
47	Vap. Reynolds	9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *
48	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
49	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	1.823e+004	1.824e+004	1.824e+004	1.825e+004
53	Elevation (m)	-0.2120	-0.2140	-0.2160	-0.2180
54	Pressure (bar)	90.03	90.03	90.03	90.03
55	Temperature (C)	53.90	53.90	53.90	53.89
56	Vapour Fraction	0.8708	0.8708	0.8708	0.8708
57	Heat Transferred (kJ/h-m)	636.8	636.7	636.7	636.6
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2065	0.2065	0.2065	0.2065
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	6.876e-003	6.876e-003	6.876e-003	6.876e-003
63	Static Gradient (kPa/m)	-7.717e-004	-7.717e-004	-7.717e-004	-7.717e-004
64	Accel. Gradient (kPa/m)	4.350e-007	4.350e-007	4.350e-007	4.350e-007
65	Liq. Reynolds	2.129e+005 *	2.129e+005 *	2.129e+005 *	2.129e+005 *
66	Vap. Reynolds	9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *
67	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
68	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:02:44 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Deposit Thickness (mm)	---	---	---	---					
14	Deposit Volume (m3)	---	---	---	---					
15	Length (m)	1.825e+004	1.826e+004	1.826e+004	1.827e+004					
16	Elevation (m)	-0.2200	-0.2220	-0.2240	-0.2260					
17	Pressure (bar)	90.03	90.03	90.03	90.03					
18	Temperature (C)	53.89	53.89	53.89	53.89					
19	Vapour Fraction	0.8708	0.8708	0.8708	0.8708					
20	Heat Transferred (kJ/h-m)	636.6	636.5	636.5	636.4					
21	Flow Regime	Transition	Transition	Transition	Transition					
22	Water Flow Regime									
23	Liquid HoldUp	0.2065	0.2065	0.2065	0.2065					
24	Water HoldUp	---	---	---	---					
25	Friction Gradient (kPa/m)	6.876e-003	6.876e-003	6.876e-003	6.876e-003					
26	Static Gradient (kPa/m)	-7.717e-004	-7.717e-004	-7.717e-004	-7.717e-004					
27	Accel. Gradient (kPa/m)	4.350e-007	4.350e-007	4.350e-007	4.350e-007					
28	Liq. Reynolds	2.129e+005 *	2.129e+005 *	2.129e+005 *	2.129e+005 *					
29	Vap. Reynolds	9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *					
30	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042					
31	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176					
32	Deposit Thickness (mm)	---	---	---	---					
33	Deposit Volume (m3)	---	---	---	---					
34	Length (m)	1.827e+004	1.828e+004	1.828e+004	1.829e+004					
35	Elevation (m)	-0.2280	-0.2300	-0.2340	-0.2380					
36	Pressure (bar)	90.03	90.02	90.02	90.02					
37	Temperature (C)	53.89	53.88	53.88	53.88					
38	Vapour Fraction	0.8708	0.8708	0.8708	0.8708					
39	Heat Transferred (kJ/h-m)	636.4	636.3	636.3	636.2					
40	Flow Regime	Transition	Transition	Transition	Transition					
41	Water Flow Regime									
42	Liquid HoldUp	0.2065	0.2065	0.2061	0.2061					
43	Water HoldUp	---	---	---	---					
44	Friction Gradient (kPa/m)	6.876e-003	6.876e-003	6.879e-003	6.879e-003					
45	Static Gradient (kPa/m)	-7.717e-004	-7.717e-004	-1.551e-003	-1.551e-003					
46	Accel. Gradient (kPa/m)	4.350e-007	4.350e-007	3.796e-007	3.796e-007					
47	Liq. Reynolds	2.129e+005 *	2.129e+005 *	2.129e+005 *	2.129e+005 *					
48	Vap. Reynolds	9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *					
49	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042					
50	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176					
51	Deposit Thickness (mm)	---	---	---	---					
52	Deposit Volume (m3)	---	---	---	---					
53	Length (m)	1.829e+004	1.830e+004	1.830e+004	1.831e+004					
54	Elevation (m)	-0.2420	-0.2460	-0.2500	-0.2600					
55	Pressure (bar)	90.02	90.02	90.02	90.02					
56	Temperature (C)	53.88	53.88	53.87	53.87					
57	Vapour Fraction	0.8708	0.8708	0.8707	0.8707					
58	Heat Transferred (kJ/h-m)	636.2	636.1	636.1	636.0					
59	Flow Regime	Transition	Transition	Transition	Transition					
60	Water Flow Regime									
61	Liquid HoldUp	0.2061	0.2061	0.2061	0.2050					
62	Water HoldUp	---	---	---	---					
63	Friction Gradient (kPa/m)	6.879e-003	6.879e-003	6.879e-003	6.887e-003					
64	Static Gradient (kPa/m)	-1.551e-003	-1.551e-003	-1.551e-003	-3.841e-003					
65	Accel. Gradient (kPa/m)	3.796e-007	3.796e-007	3.796e-007	2.171e-007					
66	Liq. Reynolds	2.129e+005 *	2.129e+005 *	2.129e+005 *	2.129e+005 *					
67	Vap. Reynolds	9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *					
68	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3							
4			Unit Set: Mixed				
5			Date/Time: Mon Jun 13 15:02:44 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11							
12	Pipe Table						
13	Vapour Velocity	(m/s)	2.176	2.176	2.176	2.176	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	1.831e+004	1.832e+004	1.832e+004	1.833e+004	
17	Elevation	(m)	-0.2700	-0.2800	-0.2900	-0.3000	
18	Pressure	(bar)	90.02	90.02	90.02	90.02	
19	Temperature	(C)	53.87	53.87	53.87	53.87	
20	Vapour Fraction		0.8707	0.8707	0.8707	0.8707	
21	Heat Transferred	(kJ/h-m)	636.0	635.9	635.9	635.8	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2050	0.2050	0.2050	0.2050	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	6.887e-003	6.887e-003	6.887e-003	6.887e-003	
27	Static Gradient	(kPa/m)	-3.841e-003	-3.841e-003	-3.841e-003	-3.841e-003	
28	Accel. Gradient	(kPa/m)	2.171e-007	2.170e-007	2.170e-007	2.170e-007	
29	Liq. Reynolds		2.129e+005 *	2.129e+005 *	2.129e+005 *	2.129e+005 *	
30	Vap. Reynolds		9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *	
31	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
32	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	1.833e+004	1.834e+004	1.834e+004	1.835e+004	
36	Elevation	(m)	-0.2960	-0.2920	-0.2880	-0.2840	
37	Pressure	(bar)	90.02	90.02	90.02	90.02	
38	Temperature	(C)	53.86	53.86	53.86	53.86	
39	Vapour Fraction		0.8707	0.8707	0.8707	0.8707	
40	Heat Transferred	(kJ/h-m)	635.7	635.7	635.6	635.6	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2816	0.2816	0.2816	0.2816	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	5.449e-003	5.449e-003	5.449e-003	5.449e-003	
46	Static Gradient	(kPa/m)	1.899e-003	1.899e-003	1.899e-003	1.899e-003	
47	Accel. Gradient	(kPa/m)	5.236e-007	5.236e-007	5.236e-007	5.236e-007	
48	Liq. Reynolds		2.129e+005 *	2.129e+005 *	2.129e+005 *	2.129e+005 *	
49	Vap. Reynolds		9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *	
50	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042	
51	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	1.835e+004	1.836e+004	1.836e+004	1.837e+004	
55	Elevation	(m)	-0.2800	-0.2840	-0.2880	-0.2920	
56	Pressure	(bar)	90.02	90.02	90.02	90.02	
57	Temperature	(C)	53.86	53.85	53.85	53.85	
58	Vapour Fraction		0.8707	0.8707	0.8707	0.8707	
59	Heat Transferred	(kJ/h-m)	635.5	635.5	635.5	635.4	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.2816	0.2061	0.2061	0.2061	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	5.449e-003	6.878e-003	6.878e-003	6.878e-003	
65	Static Gradient	(kPa/m)	1.899e-003	-1.542e-003	-1.542e-003	-1.542e-003	
66	Accel. Gradient	(kPa/m)	5.236e-007	3.802e-007	3.802e-007	3.802e-007	
67	Liq. Reynolds		2.129e+005 *	2.129e+005 *	2.129e+005 *	2.129e+005 *	
68	Vap. Reynolds		9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Unit Set: Mixed			
3			Date/Time: Mon Jun 13 15:02:44 2022			
4						
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042
14	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175
15	Deposit Thickness	(mm)	---	---	---	---
16	Deposit Volume	(m3)	---	---	---	---
17	Length	(m)	1.837e+004	1.838e+004	1.838e+004	1.839e+004
18	Elevation	(m)	-0.2960	-0.3000	-0.2960	-0.2920
19	Pressure	(bar)	90.02	90.02	90.02	90.02
20	Temperature	(C)	53.85	53.85	53.84	53.84
21	Vapour Fraction		0.8707	0.8707	0.8707	0.8707
22	Heat Transferred	(kJ/h-m)	635.4	635.3	635.2	635.2
23	Flow Regime		Transition	Transition	Transition	Transition
24	Water Flow Regime					
25	Liquid HoldUp		0.2061	0.2061	0.2816	0.2816
26	Water HoldUp		---	---	---	---
27	Friction Gradient	(kPa/m)	6.878e-003	6.878e-003	5.449e-003	5.449e-003
28	Static Gradient	(kPa/m)	-1.542e-003	-1.542e-003	1.899e-003	1.899e-003
29	Accel. Gradient	(kPa/m)	3.802e-007	3.802e-007	5.236e-007	5.236e-007
30	Liq. Reynolds		2.129e+005 *	2.129e+005 *	2.129e+005 *	2.129e+005 *
31	Vap. Reynolds		9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *
32	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042
33	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175
34	Deposit Thickness	(mm)	---	---	---	---
35	Deposit Volume	(m3)	---	---	---	---
36	Length	(m)	1.839e+004	1.840e+004	1.840e+004	1.841e+004
37	Elevation	(m)	-0.2880	-0.2840	-0.2800	-0.2780
38	Pressure	(bar)	90.02	90.02	90.02	90.02
39	Temperature	(C)	53.84	53.84	53.84	53.84
40	Vapour Fraction		0.8707	0.8707	0.8707	0.8707
41	Heat Transferred	(kJ/h-m)	635.1	635.1	635.0	635.0
42	Flow Regime		Transition	Transition	Transition	Transition
43	Water Flow Regime					
44	Liquid HoldUp		0.2816	0.2816	0.2816	0.2803
45	Water HoldUp		---	---	---	---
46	Friction Gradient	(kPa/m)	5.449e-003	5.449e-003	5.449e-003	5.549e-003
47	Static Gradient	(kPa/m)	1.899e-003	1.899e-003	1.899e-003	9.467e-004
48	Accel. Gradient	(kPa/m)	5.236e-007	5.236e-007	5.236e-007	4.628e-007
49	Liq. Reynolds		2.129e+005 *	2.129e+005 *	2.129e+005 *	2.129e+005 *
50	Vap. Reynolds		9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *
51	Liquid Velocity	(m/s)	0.2042	0.2042	0.2042	0.2042
52	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175
53	Deposit Thickness	(mm)	---	---	---	---
54	Deposit Volume	(m3)	---	---	---	---
55	Length	(m)	1.841e+004	1.842e+004	1.842e+004	1.843e+004
56	Elevation	(m)	-0.2760	-0.2740	-0.2720	-0.2700
57	Pressure	(bar)	90.02	90.02	90.02	90.02
58	Temperature	(C)	53.83	53.83	53.83	53.83
59	Vapour Fraction		0.8707	0.8707	0.8707	0.8707
60	Heat Transferred	(kJ/h-m)	634.9	634.9	634.8	634.8
61	Flow Regime		Transition	Transition	Transition	Transition
62	Water Flow Regime					
63	Liquid HoldUp		0.2803	0.2803	0.2803	0.2803
64	Water HoldUp		---	---	---	---
65	Friction Gradient	(kPa/m)	5.549e-003	5.548e-003	5.548e-003	5.548e-003
66	Static Gradient	(kPa/m)	9.467e-004	9.467e-004	9.467e-004	9.467e-004
67	Accel. Gradient	(kPa/m)	4.628e-007	4.628e-007	4.628e-007	4.628e-007
68	Liq. Reynolds		2.129e+005 *	2.129e+005 *	2.129e+005 *	2.129e+005 *
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *
14	Liquid Velocity (m/s)	0.2042	0.2042	0.2042	0.2042
15	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	1.843e+004	1.844e+004	1.844e+004	1.845e+004
19	Elevation (m)	-0.2860	-0.3020	-0.3180	-0.3340
20	Pressure (bar)	90.02	90.02	90.02	90.02
21	Temperature (C)	53.83	53.82	53.82	53.82
22	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
23	Heat Transferred (kJ/h-m)	634.8	634.7	634.7	634.6
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2039	0.2039	0.2039	0.2039
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.895e-003	6.895e-003	6.895e-003	6.895e-003
29	Static Gradient (kPa/m)	-6.126e-003	-6.126e-003	-6.126e-003	-6.126e-003
30	Accel. Gradient (kPa/m)	5.485e-008	5.484e-008	5.483e-008	5.483e-008
31	Liq. Reynolds	2.129e+005 *	2.129e+005 *	2.129e+005 *	2.129e+005 *
32	Vap. Reynolds	9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *
33	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
34	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	1.845e+004	1.846e+004	1.846e+004	1.847e+004
38	Elevation (m)	-0.3500	-0.3400	-0.3300	-0.3200
39	Pressure (bar)	90.02	90.02	90.01	90.01
40	Temperature (C)	53.82	53.82	53.81	53.81
41	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
42	Heat Transferred (kJ/h-m)	634.6	634.5	634.4	634.4
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2039	0.2854	0.2854	0.2854
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.895e-003	5.156e-003	5.156e-003	5.156e-003
48	Static Gradient (kPa/m)	-6.126e-003	4.794e-003	4.794e-003	4.794e-003
49	Accel. Gradient (kPa/m)	5.482e-008	7.089e-007	7.089e-007	7.089e-007
50	Liq. Reynolds	2.129e+005 *	2.129e+005 *	2.129e+005 *	2.129e+005 *
51	Vap. Reynolds	9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *
52	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
53	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1.847e+004	1.848e+004	1.848e+004	1.849e+004
57	Elevation (m)	-0.3100	-0.3000	-0.2980	-0.2960
58	Pressure (bar)	90.01	90.01	90.01	90.01
59	Temperature (C)	53.81	53.81	53.81	53.81
60	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
61	Heat Transferred (kJ/h-m)	634.3	634.3	634.2	634.2
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2854	0.2854	0.2803	0.2803
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	5.156e-003	5.156e-003	5.548e-003	5.548e-003
67	Static Gradient (kPa/m)	4.794e-003	4.794e-003	9.453e-004	9.453e-004
68	Accel. Gradient (kPa/m)	7.089e-007	7.089e-007	4.626e-007	4.626e-007
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.128e+005 *	2.128e+005 *	2.128e+005 *	2.128e+005 *
14	Vap. Reynolds	9.134e+006 *	9.134e+006 *	9.134e+006 *	9.134e+006 *
15	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
16	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	1.849e+004	1.850e+004	1.850e+004	1.851e+004
20	Elevation (m)	-0.2940	-0.2920	-0.2900	-0.2840
21	Pressure (bar)	90.01	90.01	90.01	90.01
22	Temperature (C)	53.80	53.80	53.80	53.80
23	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
24	Heat Transferred (kJ/h-m)	634.1	634.1	634.0	634.0
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2803	0.2803	0.2803	0.2829
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	5.548e-003	5.548e-003	5.548e-003	5.349e-003
30	Static Gradient (kPa/m)	9.453e-004	9.453e-004	9.453e-004	2.859e-003
31	Accel. Gradient (kPa/m)	4.626e-007	4.626e-007	4.626e-007	5.848e-007
32	Liq. Reynolds	2.128e+005 *	2.128e+005 *	2.128e+005 *	2.128e+005 *
33	Vap. Reynolds	9.134e+006 *	9.134e+006 *	9.134e+006 *	9.135e+006 *
34	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
35	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	1.851e+004	1.852e+004	1.852e+004	1.853e+004
39	Elevation (m)	-0.2780	-0.2720	-0.2660	-0.2600
40	Pressure (bar)	90.01	90.01	90.01	90.01
41	Temperature (C)	53.80	53.79	53.79	53.79
42	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
43	Heat Transferred (kJ/h-m)	633.9	633.9	633.8	633.8
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2829	0.2829	0.2829	0.2829
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	5.349e-003	5.349e-003	5.349e-003	5.349e-003
49	Static Gradient (kPa/m)	2.859e-003	2.859e-003	2.859e-003	2.859e-003
50	Accel. Gradient (kPa/m)	5.848e-007	5.848e-007	5.848e-007	5.848e-007
51	Liq. Reynolds	2.128e+005 *	2.128e+005 *	2.128e+005 *	2.128e+005 *
52	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
53	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
54	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	1.853e+004	1.854e+004	1.854e+004	1.855e+004
58	Elevation (m)	-0.2760	-0.2920	-0.3080	-0.3240
59	Pressure (bar)	90.01	90.01	90.01	90.01
60	Temperature (C)	53.79	53.79	53.78	53.78
61	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
62	Heat Transferred (kJ/h-m)	633.8	633.7	633.7	633.6
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2039	0.2039	0.2039	0.2039
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	6.895e-003	6.895e-003	6.895e-003	6.895e-003
68	Static Gradient (kPa/m)	-6.126e-003	-6.126e-003	-6.126e-003	-6.126e-003
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	5.475e-008	5.474e-008	5.473e-008	5.473e-008
14	Liq. Reynolds	2.128e+005 *	2.128e+005 *	2.128e+005 *	2.128e+005 *
15	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
16	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
17	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	1.855e+004	1.856e+004	1.856e+004	1.857e+004
21	Elevation (m)	-0.3400	-0.3380	-0.3360	-0.3340
22	Pressure (bar)	90.01	90.01	90.01	90.01
23	Temperature (C)	53.78	53.78	53.78	53.78
24	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
25	Heat Transferred (kJ/h-m)	633.6	633.5	633.4	633.4
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.2039	0.2803	0.2803	0.2803
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	6.894e-003	5.547e-003	5.547e-003	5.547e-003
31	Static Gradient (kPa/m)	-6.126e-003	9.469e-004	9.469e-004	9.469e-004
32	Accel. Gradient (kPa/m)	5.472e-008	4.627e-007	4.627e-007	4.626e-007
33	Liq. Reynolds	2.128e+005 *	2.128e+005 *	2.128e+005 *	2.128e+005 *
34	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
35	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
36	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	1.857e+004	1.858e+004	1.858e+004	1.859e+004
40	Elevation (m)	-0.3320	-0.3300	-0.3400	-0.3500
41	Pressure (bar)	90.01	90.01	90.01	90.01
42	Temperature (C)	53.77	53.77	53.77	53.77
43	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
44	Heat Transferred (kJ/h-m)	633.3	633.3	633.3	633.2
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.2804	0.2804	0.2050	0.2050
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	5.547e-003	5.547e-003	6.885e-003	6.885e-003
50	Static Gradient (kPa/m)	9.469e-004	9.469e-004	-3.842e-003	-3.842e-003
51	Accel. Gradient (kPa/m)	4.626e-007	4.626e-007	2.168e-007	2.168e-007
52	Liq. Reynolds	2.128e+005 *	2.128e+005 *	2.128e+005 *	2.128e+005 *
53	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
54	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
55	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	1.859e+004	1.860e+004	1.860e+004	1.861e+004
59	Elevation (m)	-0.3600	-0.3700	-0.3800	-0.4000
60	Pressure (bar)	90.01	90.01	90.01	90.01
61	Temperature (C)	53.77	53.76	53.76	53.76
62	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
63	Heat Transferred (kJ/h-m)	633.2	633.1	633.1	633.0
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.2050	0.2050	0.2050	0.2032
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	6.885e-003	6.885e-003	6.885e-003	6.900e-003
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1				Case Name:	Simulation - 386MMSCFD_100barg.hsc		
2				Unit Set:	Mixed		
3	 Epic Atlantic Limited			Date/Time:	Mon Jun 13 15:02:44 2022		
4							
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Static Gradient	(kPa/m)	-3.842e-003	-3.842e-003	-3.842e-003	-7.641e-003	
14	Accel. Gradient	(kPa/m)	2.168e-007	2.168e-007	2.168e-007	-5.278e-008	
15	Liq. Reynolds		2.128e+005 *	2.128e+005 *	2.128e+005 *	2.128e+005 *	
16	Vap. Reynolds		9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *	
17	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
18	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175	
19	Deposit Thickness	(mm)	---	---	---	---	
20	Deposit Volume	(m3)	---	---	---	---	
21	Length	(m)	1.861e+004	1.862e+004	1.862e+004	1.863e+004	
22	Elevation	(m)	-0.4200	-0.4400	-0.4600	-0.4800	
23	Pressure	(bar)	90.01	90.01	90.01	90.01	
24	Temperature	(C)	53.76	53.76	53.76	53.75	
25	Vapour Fraction		0.8707	0.8707	0.8707	0.8707	
26	Heat Transferred	(kJ/h-m)	633.0	632.9	632.9	632.8	
27	Flow Regime		Transition	Transition	Transition	Transition	
28	Water Flow Regime						
29	Liquid HoldUp		0.2032	0.2032	0.2032	0.2032	
30	Water HoldUp		---	---	---	---	
31	Friction Gradient	(kPa/m)	6.900e-003	6.900e-003	6.900e-003	6.900e-003	
32	Static Gradient	(kPa/m)	-7.641e-003	-7.641e-003	-7.641e-003	-7.641e-003	
33	Accel. Gradient	(kPa/m)	-5.278e-008	-5.279e-008	-5.280e-008	-5.281e-008	
34	Liq. Reynolds		2.128e+005 *	2.128e+005 *	2.128e+005 *	2.128e+005 *	
35	Vap. Reynolds		9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *	
36	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
37	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175	
38	Deposit Thickness	(mm)	---	---	---	---	
39	Deposit Volume	(m3)	---	---	---	---	
40	Length	(m)	1.863e+004	1.864e+004	1.864e+004	1.865e+004	
41	Elevation	(m)	-0.4920	-0.5040	-0.5160	-0.5280	
42	Pressure	(bar)	90.01	90.01	90.01	90.01	
43	Temperature	(C)	53.75	53.75	53.75	53.75	
44	Vapour Fraction		0.8707	0.8707	0.8707	0.8707	
45	Heat Transferred	(kJ/h-m)	632.8	632.7	632.7	632.6	
46	Flow Regime		Transition	Transition	Transition	Transition	
47	Water Flow Regime						
48	Liquid HoldUp		0.2047	0.2047	0.2047	0.2047	
49	Water HoldUp		---	---	---	---	
50	Friction Gradient	(kPa/m)	6.888e-003	6.888e-003	6.888e-003	6.888e-003	
51	Static Gradient	(kPa/m)	-4.606e-003	-4.606e-003	-4.606e-003	-4.606e-003	
52	Accel. Gradient	(kPa/m)	1.626e-007	1.625e-007	1.625e-007	1.625e-007	
53	Liq. Reynolds		2.128e+005 *	2.128e+005 *	2.128e+005 *	2.128e+005 *	
54	Vap. Reynolds		9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *	
55	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
56	Vapour Velocity	(m/s)	2.174	2.174	2.174	2.174	
57	Deposit Thickness	(mm)	---	---	---	---	
58	Deposit Volume	(m3)	---	---	---	---	
59	Length	(m)	1.865e+004	1.866e+004	1.866e+004	1.867e+004	
60	Elevation	(m)	-0.5400	-0.5360	-0.5320	-0.5280	
61	Pressure	(bar)	90.01	90.01	90.01	90.01	
62	Temperature	(C)	53.74	53.74	53.74	53.74	
63	Vapour Fraction		0.8707	0.8707	0.8707	0.8707	
64	Heat Transferred	(kJ/h-m)	632.6	632.5	632.5	632.4	
65	Flow Regime		Transition	Transition	Transition	Transition	
66	Water Flow Regime						
67	Liquid HoldUp		0.2047	0.2817	0.2817	0.2817	
68	Water HoldUp		---	---	---	---	
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1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc			
2							
3				Unit Set: Mixed			
4							
5				Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	6.887e-003	5.446e-003	5.446e-003	5.446e-003	
14	Static Gradient	(kPa/m)	-4.606e-003	1.900e-003	1.900e-003	1.900e-003	
15	Accel. Gradient	(kPa/m)	1.625e-007	5.233e-007	5.233e-007	5.233e-007	
16	Liq. Reynolds		2.128e+005 *	2.128e+005 *	2.128e+005 *	2.128e+005 *	
17	Vap. Reynolds		9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *	
18	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
19	Vapour Velocity	(m/s)	2.174	2.174	2.174	2.174	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	1.867e+004	1.868e+004	1.868e+004	1.869e+004	
23	Elevation	(m)	-0.5240	-0.5200	-0.5760	-0.6320	
24	Pressure	(bar)	90.01	90.00	90.01	90.01	
25	Temperature	(C)	53.74	53.74	53.73	53.73	
26	Vapour Fraction		0.8707	0.8707	0.8707	0.8707	
27	Heat Transferred	(kJ/h-m)	632.4	632.3	632.3	632.3	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.2817	0.2817	0.1965	0.1965	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	5.446e-003	5.446e-003	6.957e-003	6.957e-003	
33	Static Gradient	(kPa/m)	1.900e-003	1.900e-003	-2.094e-002	-2.094e-002	
34	Accel. Gradient	(kPa/m)	5.233e-007	5.233e-007	-9.964e-007	-9.964e-007	
35	Liq. Reynolds		2.128e+005 *	2.128e+005 *	2.128e+005 *	2.128e+005 *	
36	Vap. Reynolds		9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *	
37	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
38	Vapour Velocity	(m/s)	2.174	2.174	2.174	2.174	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	1.869e+004	1.870e+004	1.870e+004	1.871e+004	
42	Elevation	(m)	-0.6880	-0.7440	-0.8000	-0.7500	
43	Pressure	(bar)	90.01	90.01	90.01	90.01	
44	Temperature	(C)	53.73	53.73	53.73	53.73	
45	Vapour Fraction		0.8707	0.8707	0.8707	0.8707	
46	Heat Transferred	(kJ/h-m)	632.2	632.2	632.2	632.1	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.1965	0.1965	0.1965	0.3110	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	6.957e-003	6.957e-003	6.957e-003	6.027e-003	
52	Static Gradient	(kPa/m)	-2.095e-002	-2.095e-002	-2.095e-002	2.549e-002	
53	Accel. Gradient	(kPa/m)	-9.964e-007	-9.964e-007	-9.964e-007	2.245e-006	
54	Liq. Reynolds		2.128e+005 *	2.128e+005 *	2.128e+005 *	2.128e+005 *	
55	Vap. Reynolds		9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *	
56	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
57	Vapour Velocity	(m/s)	2.174	2.174	2.174	2.174	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	1.871e+004	1.872e+004	1.872e+004	1.873e+004	
61	Elevation	(m)	-0.7000	-0.6500	-0.6000	-0.5500	
62	Pressure	(bar)	90.01	90.00	90.00	90.00	
63	Temperature	(C)	53.72	53.72	53.72	53.72	
64	Vapour Fraction		0.8707	0.8707	0.8707	0.8707	
65	Heat Transferred	(kJ/h-m)	632.0	632.0	631.9	631.8	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.3110	0.3110	0.3110	0.3110	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	6.027e-003	6.027e-003	6.028e-003	6.028e-003
15	Static Gradient (kPa/m)	2.549e-002	2.549e-002	2.549e-002	2.549e-002
16	Accel. Gradient (kPa/m)	2.245e-006	2.245e-006	2.245e-006	2.245e-006
17	Liq. Reynolds	2.128e+005 *	2.128e+005 *	2.128e+005 *	2.128e+005 *
18	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
19	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
20	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	1.873e+004	1.874e+004	1.874e+004	1.875e+004
24	Elevation (m)	-0.5360	-0.5220	-0.5080	-0.4940
25	Pressure (bar)	90.00	90.00	90.00	90.00
26	Temperature (C)	53.71	53.71	53.71	53.71
27	Vapour Fraction	0.8707	0.8707	0.8707	0.8707
28	Heat Transferred (kJ/h-m)	631.8	631.7	631.7	631.6
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2880	0.2880	0.2880	0.2880
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	4.966e-003	4.966e-003	4.966e-003	4.966e-003
34	Static Gradient (kPa/m)	6.756e-003	6.756e-003	6.756e-003	6.756e-003
35	Accel. Gradient (kPa/m)	8.350e-007	8.350e-007	8.350e-007	8.350e-007
36	Liq. Reynolds	2.128e+005 *	2.128e+005 *	2.128e+005 *	2.128e+005 *
37	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
38	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
39	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	1.875e+004	1.876e+004	1.876e+004	1.877e+004
43	Elevation (m)	-0.4800	-0.5020	-0.5240	-0.5460
44	Pressure (bar)	90.00	90.00	90.00	90.00
45	Temperature (C)	53.71	53.70	53.70	53.70
46	Vapour Fraction	0.8707	0.8706	0.8706	0.8706
47	Heat Transferred (kJ/h-m)	631.6	631.5	631.5	631.4
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2880	0.2028	0.2028	0.2028
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	4.965e-003	6.902e-003	6.902e-003	6.902e-003
53	Static Gradient (kPa/m)	6.756e-003	-8.397e-003	-8.397e-003	-8.397e-003
54	Accel. Gradient (kPa/m)	8.350e-007	-1.065e-007	-1.065e-007	-1.065e-007
55	Liq. Reynolds	2.128e+005 *	2.128e+005 *	2.128e+005 *	2.128e+005 *
56	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
57	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
58	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	1.877e+004	1.878e+004	1.878e+004	1.879e+004
62	Elevation (m)	-0.5680	-0.5900	-0.6080	-0.6260
63	Pressure (bar)	90.00	90.00	90.00	90.00
64	Temperature (C)	53.70	53.70	53.70	53.69
65	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
66	Heat Transferred (kJ/h-m)	631.4	631.3	631.3	631.3
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
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9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.2028	0.2028	0.2036	0.2036
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	6.902e-003	6.902e-003	6.896e-003	6.896e-003
16	Static Gradient (kPa/m)	-8.397e-003	-8.397e-003	-6.886e-003	-6.886e-003
17	Accel. Gradient (kPa/m)	-1.065e-007	-1.065e-007	6.858e-010	6.782e-010
18	Liq. Reynolds	2.128e+005 *	2.128e+005 *	2.128e+005 *	2.128e+005 *
19	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
20	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
21	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	1.879e+004	1.880e+004	1.880e+004	1.880e+004
25	Elevation (m)	-0.6440	-0.6620	-0.6800	-0.6520
26	Pressure (bar)	90.00	90.00	90.00	90.00
27	Temperature (C)	53.69	53.69	53.69	53.69
28	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
29	Heat Transferred (kJ/h-m)	631.2	631.2	631.1	631.1
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2036	0.2036	0.2036	0.3089
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.896e-003	6.895e-003	6.895e-003	6.007e-003
35	Static Gradient (kPa/m)	-6.886e-003	-6.886e-003	-6.886e-003	2.367e-002
36	Accel. Gradient (kPa/m)	6.706e-010	6.629e-010	6.553e-010	2.114e-006
37	Liq. Reynolds	2.128e+005 *	2.128e+005 *	2.128e+005 *	2.128e+005 *
38	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
39	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
40	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	1.881e+004	1.881e+004	1.881e+004	1.882e+004
44	Elevation (m)	-0.6240	-0.5960	-0.5680	-0.5400
45	Pressure (bar)	90.00	90.00	89.99	89.99
46	Temperature (C)	53.69	53.68	53.68	53.68
47	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
48	Heat Transferred (kJ/h-m)	631.0	631.0	631.0	630.9
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.3089	0.3089	0.3088	0.3088
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.008e-003	6.008e-003	6.008e-003	6.008e-003
54	Static Gradient (kPa/m)	2.367e-002	2.367e-002	2.367e-002	2.367e-002
55	Accel. Gradient (kPa/m)	2.114e-006	2.114e-006	2.114e-006	2.114e-006
56	Liq. Reynolds	2.128e+005 *	2.128e+005 *	2.127e+005 *	2.127e+005 *
57	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
58	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
59	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	1.882e+004	1.882e+004	1.882e+004	1.882e+004
63	Elevation (m)	-0.6040	-0.6680	-0.7320	-0.7960
64	Pressure (bar)	89.99	90.00	90.00	90.00
65	Temperature (C)	53.68	53.68	53.68	53.68
66	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
67	Heat Transferred (kJ/h-m)	631.0	630.9	630.9	630.9
68	Flow Regime	Transition	Transition	Transition	Transition
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Pipe Segment: ELEP-Akabel Pipeline (continued)		
3					
4					
5			Unit Set: Mixed		
6			Date/Time: Mon Jun 13 15:02:44 2022		
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.1772	0.1772	0.1772	0.1772
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	7.158e-003	7.158e-003	7.158e-003	7.158e-003
17	Static Gradient (kPa/m)	-5.622e-002	-5.622e-002	-5.622e-002	-5.622e-002
18	Accel. Gradient (kPa/m)	-3.494e-006	-3.494e-006	-3.494e-006	-3.494e-006
19	Liq. Reynolds	2.127e+005 *	2.127e+005 *	2.128e+005 *	2.128e+005 *
20	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
21	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
22	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	1.883e+004	1.883e+004	1.884e+004	1.884e+004
26	Elevation (m)	-0.8600	-0.7620	-0.6640	-0.5660
27	Pressure (bar)	90.00	90.00	89.99	89.99
28	Temperature (C)	53.68	53.68	53.68	53.67
29	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
30	Heat Transferred (kJ/h-m)	630.9	630.8	630.8	630.7
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1772	0.3415	0.3415	0.3415
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	7.157e-003	6.075e-003	6.075e-003	6.075e-003
36	Static Gradient (kPa/m)	-5.622e-002	5.350e-002	5.350e-002	5.350e-002
37	Accel. Gradient (kPa/m)	-3.494e-006	4.244e-006	4.244e-006	4.244e-006
38	Liq. Reynolds	2.128e+005 *	2.127e+005 *	2.127e+005 *	2.127e+005 *
39	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
40	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
41	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	1.885e+004	1.885e+004	1.886e+004	1.886e+004
45	Elevation (m)	-0.4680	-0.3700	-0.4320	-0.4940
46	Pressure (bar)	89.99	89.98	89.98	89.99
47	Temperature (C)	53.67	53.67	53.67	53.66
48	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
49	Heat Transferred (kJ/h-m)	630.6	630.5	630.5	630.5
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.3415	0.3415	0.1954	0.1954
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	6.075e-003	6.075e-003	6.967e-003	6.967e-003
55	Static Gradient (kPa/m)	5.350e-002	5.350e-002	-2.312e-002	-2.312e-002
56	Accel. Gradient (kPa/m)	4.244e-006	4.244e-006	-1.151e-006	-1.151e-006
57	Liq. Reynolds	2.127e+005 *	2.127e+005 *	2.127e+005 *	2.127e+005 *
58	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
59	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
60	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	1.887e+004	1.887e+004	1.888e+004	1.888e+004
64	Elevation (m)	-0.5560	-0.6180	-0.6800	-0.6540
65	Pressure (bar)	89.99	89.99	89.99	89.99
66	Temperature (C)	53.66	53.66	53.66	53.66
67	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
68	Heat Transferred (kJ/h-m)	630.4	630.4	630.3	630.3
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.1954	0.1954	0.1954	0.2957
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	6.967e-003	6.966e-003	6.966e-003	5.427e-003
18	Static Gradient (kPa/m)	-2.312e-002	-2.312e-002	-2.312e-002	1.279e-002
19	Accel. Gradient (kPa/m)	-1.151e-006	-1.151e-006	-1.151e-006	1.298e-006
20	Liq. Reynolds	2.127e+005 *	2.127e+005 *	2.127e+005 *	2.127e+005 *
21	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
22	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
23	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	1.889e+004	1.889e+004	1.890e+004	1.890e+004
27	Elevation (m)	-0.6280	-0.6020	-0.5760	-0.5500
28	Pressure (bar)	89.99	89.98	89.98	89.98
29	Temperature (C)	53.66	53.65	53.65	53.65
30	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
31	Heat Transferred (kJ/h-m)	630.2	630.2	630.1	630.0
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2957	0.2957	0.2957	0.2957
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	5.427e-003	5.427e-003	5.427e-003	5.427e-003
37	Static Gradient (kPa/m)	1.279e-002	1.279e-002	1.279e-002	1.279e-002
38	Accel. Gradient (kPa/m)	1.298e-006	1.298e-006	1.298e-006	1.298e-006
39	Liq. Reynolds	2.127e+005 *	2.127e+005 *	2.127e+005 *	2.127e+005 *
40	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
41	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
42	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	1.890e+004	1.891e+004	1.891e+004	1.891e+004
46	Elevation (m)	-0.5600	-0.5700	-0.5800	-0.5900
47	Pressure (bar)	89.98	89.98	89.98	89.98
48	Temperature (C)	53.65	53.65	53.65	53.65
49	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
50	Heat Transferred (kJ/h-m)	630.0	630.0	630.0	630.0
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2023	0.2023	0.2023	0.2023
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	6.906e-003	6.906e-003	6.906e-003	6.906e-003
56	Static Gradient (kPa/m)	-9.526e-003	-9.526e-003	-9.526e-003	-9.526e-003
57	Accel. Gradient (kPa/m)	-1.866e-007	-1.866e-007	-1.866e-007	-1.866e-007
58	Liq. Reynolds	2.127e+005 *	2.127e+005 *	2.127e+005 *	2.127e+005 *
59	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
60	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
61	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	1.891e+004	1.891e+004	1.892e+004	1.892e+004
65	Elevation (m)	-0.6000	-0.5920	-0.5840	-0.5760
66	Pressure (bar)	89.98	89.98	89.98	89.98
67	Temperature (C)	53.65	53.64	53.64	53.64
68	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:02:44 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Heat Transferred (kJ/h-m)	630.0	629.9	629.9	629.9					
14	Flow Regime	Transition	Transition	Transition	Transition					
15	Water Flow Regime									
16	Liquid HoldUp	0.2023	0.2876	0.2876	0.2876					
17	Water HoldUp	---	---	---	---					
18	Friction Gradient (kPa/m)	6.906e-003	4.996e-003	4.996e-003	4.996e-003					
19	Static Gradient (kPa/m)	-9.526e-003	6.428e-003	6.429e-003	6.429e-003					
20	Accel. Gradient (kPa/m)	-1.867e-007	8.138e-007	8.138e-007	8.138e-007					
21	Liq. Reynolds	2.127e+005 *	2.127e+005 *	2.127e+005 *	2.127e+005 *					
22	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *					
23	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043					
24	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174					
25	Deposit Thickness (mm)	---	---	---	---					
26	Deposit Volume (m3)	---	---	---	---					
27	Length (m)	1.892e+004	1.893e+004	1.893e+004	1.894e+004					
28	Elevation (m)	-0.5680	-0.5600	-0.5880	-0.6160					
29	Pressure (bar)	89.98	89.98	89.98	89.98					
30	Temperature (C)	53.64	53.64	53.64	53.64					
31	Vapour Fraction	0.8706	0.8706	0.8706	0.8706					
32	Heat Transferred (kJ/h-m)	629.8	629.8	629.8	629.7					
33	Flow Regime	Transition	Transition	Transition	Transition					
34	Water Flow Regime									
35	Liquid HoldUp	0.2876	0.2876	0.2017	0.2017					
36	Water HoldUp	---	---	---	---					
37	Friction Gradient (kPa/m)	4.996e-003	4.996e-003	6.910e-003	6.910e-003					
38	Static Gradient (kPa/m)	6.429e-003	6.429e-003	-1.065e-002	-1.065e-002					
39	Accel. Gradient (kPa/m)	8.138e-007	8.138e-007	-2.665e-007	-2.665e-007					
40	Liq. Reynolds	2.127e+005 *	2.127e+005 *	2.127e+005 *	2.127e+005 *					
41	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *					
42	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043					
43	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174					
44	Deposit Thickness (mm)	---	---	---	---					
45	Deposit Volume (m3)	---	---	---	---					
46	Length (m)	1.894e+004	1.895e+004	1.895e+004	1.896e+004					
47	Elevation (m)	-0.6440	-0.6720	-0.7000	-0.7560					
48	Pressure (bar)	89.98	89.98	89.98	89.98					
49	Temperature (C)	53.64	53.63	53.63	53.63					
50	Vapour Fraction	0.8706	0.8706	0.8706	0.8706					
51	Heat Transferred (kJ/h-m)	629.7	629.6	629.6	629.6					
52	Flow Regime	Transition	Transition	Transition	Transition					
53	Water Flow Regime									
54	Liquid HoldUp	0.2017	0.2017	0.2017	0.1965					
55	Water HoldUp	---	---	---	---					
56	Friction Gradient (kPa/m)	6.910e-003	6.910e-003	6.910e-003	6.956e-003					
57	Static Gradient (kPa/m)	-1.065e-002	-1.065e-002	-1.065e-002	-2.096e-002					
58	Accel. Gradient (kPa/m)	-2.665e-007	-2.665e-007	-2.665e-007	-9.974e-007					
59	Liq. Reynolds	2.127e+005 *	2.127e+005 *	2.127e+005 *	2.127e+005 *					
60	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *					
61	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043					
62	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174					
63	Deposit Thickness (mm)	---	---	---	---					
64	Deposit Volume (m3)	---	---	---	---					
65	Length (m)	1.896e+004	1.897e+004	1.897e+004	1.898e+004					
66	Elevation (m)	-0.8120	-0.8680	-0.9240	-0.9800					
67	Pressure (bar)	89.98	89.98	89.99	89.99					
68	Temperature (C)	53.63	53.63	53.63	53.62					
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 335 of 937					

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
14	Heat Transferred (kJ/h-m)	629.5	629.5	629.4	629.4
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.1965	0.1965	0.1966	0.1966
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	6.955e-003	6.955e-003	6.955e-003	6.955e-003
20	Static Gradient (kPa/m)	-2.096e-002	-2.096e-002	-2.096e-002	-2.096e-002
21	Accel. Gradient (kPa/m)	-9.974e-007	-9.974e-007	-9.974e-007	-9.974e-007
22	Liq. Reynolds	2.127e+005 *	2.127e+005 *	2.127e+005 *	2.127e+005 *
23	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
24	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
25	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	1.898e+004	1.899e+004	1.899e+004	1.900e+004
29	Elevation (m)	-0.9260	-0.8720	-0.8180	-0.7640
30	Pressure (bar)	89.98	89.98	89.98	89.98
31	Temperature (C)	53.62	53.62	53.62	53.62
32	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
33	Heat Transferred (kJ/h-m)	629.3	629.3	629.2	629.1
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.3136	0.3136	0.3136	0.3136
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	6.043e-003	6.043e-003	6.043e-003	6.043e-003
39	Static Gradient (kPa/m)	2.769e-002	2.769e-002	2.769e-002	2.769e-002
40	Accel. Gradient (kPa/m)	2.403e-006	2.403e-006	2.403e-006	2.403e-006
41	Liq. Reynolds	2.127e+005 *	2.127e+005 *	2.127e+005 *	2.127e+005 *
42	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
43	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
44	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	1.900e+004	1.901e+004	1.901e+004	1.902e+004
48	Elevation (m)	-0.7100	-0.6900	-0.6700	-0.6500
49	Pressure (bar)	89.98	89.98	89.98	89.98
50	Temperature (C)	53.61	53.61	53.61	53.61
51	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
52	Heat Transferred (kJ/h-m)	629.1	629.0	629.0	628.9
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.3136	0.2919	0.2919	0.2919
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	6.043e-003	4.691e-003	4.691e-003	4.691e-003
58	Static Gradient (kPa/m)	2.769e-002	9.744e-003	9.744e-003	9.744e-003
59	Accel. Gradient (kPa/m)	2.403e-006	1.028e-006	1.028e-006	1.028e-006
60	Liq. Reynolds	2.127e+005 *	2.127e+005 *	2.127e+005 *	2.127e+005 *
61	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
62	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
63	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	1.902e+004	1.903e+004	1.903e+004	1.904e+004
67	Elevation (m)	-0.6300	-0.6100	-0.5960	-0.5820
68	Pressure (bar)	89.97	89.97	89.97	89.97
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	53.60	53.60	53.60	53.60
14	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
15	Heat Transferred (kJ/h-m)	628.9	628.8	628.7	628.7
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2919	0.2919	0.2881	0.2881
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	4.691e-003	4.691e-003	4.964e-003	4.964e-003
21	Static Gradient (kPa/m)	9.744e-003	9.744e-003	6.758e-003	6.758e-003
22	Accel. Gradient (kPa/m)	1.028e-006	1.028e-006	8.350e-007	8.350e-007
23	Liq. Reynolds	2.127e+005 *	2.127e+005 *	2.127e+005 *	2.127e+005 *
24	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
25	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
26	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	1.904e+004	1.905e+004	1.905e+004	1.906e+004
30	Elevation (m)	-0.5680	-0.5540	-0.5400	-0.5460
31	Pressure (bar)	89.97	89.97	89.97	89.97
32	Temperature (C)	53.60	53.59	53.59	53.59
33	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
34	Heat Transferred (kJ/h-m)	628.6	628.6	628.5	628.5
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2881	0.2881	0.2881	0.2058
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	4.964e-003	4.964e-003	4.964e-003	6.876e-003
40	Static Gradient (kPa/m)	6.758e-003	6.758e-003	6.758e-003	-2.312e-003
41	Accel. Gradient (kPa/m)	8.350e-007	8.350e-007	8.350e-007	3.252e-007
42	Liq. Reynolds	2.127e+005 *	2.127e+005 *	2.127e+005 *	2.127e+005 *
43	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.136e+006 *	9.136e+006 *
44	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
45	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	1.906e+004	1.907e+004	1.907e+004	1.908e+004
49	Elevation (m)	-0.5520	-0.5580	-0.5640	-0.5700
50	Pressure (bar)	89.97	89.97	89.97	89.97
51	Temperature (C)	53.59	53.59	53.59	53.58
52	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
53	Heat Transferred (kJ/h-m)	628.5	628.4	628.4	628.3
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2059	0.2059	0.2059	0.2059
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	6.876e-003	6.876e-003	6.876e-003	6.876e-003
59	Static Gradient (kPa/m)	-2.312e-003	-2.312e-003	-2.312e-003	-2.312e-003
60	Accel. Gradient (kPa/m)	3.252e-007	3.252e-007	3.252e-007	3.251e-007
61	Liq. Reynolds	2.127e+005 *	2.127e+005 *	2.127e+005 *	2.127e+005 *
62	Vap. Reynolds	9.136e+006 *	9.136e+006 *	9.136e+006 *	9.136e+006 *
63	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
64	Vapour Velocity (m/s)	2.174	2.174	2.174	2.173
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	1.908e+004	1.909e+004	1.909e+004	1.910e+004
68	Elevation (m)	-0.5460	-0.5220	-0.4980	-0.4740
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:02:44 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	89.97	89.97	89.97	89.97		
14	Temperature (C)	53.58	53.58	53.58	53.58		
15	Vapour Fraction	0.8706	0.8706	0.8706	0.8706		
16	Heat Transferred (kJ/h-m)	628.2	628.2	628.1	628.1		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.2946	0.2946	0.2946	0.2946		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	5.185e-003	5.185e-003	5.185e-003	5.185e-003		
22	Static Gradient (kPa/m)	1.184e-002	1.184e-002	1.184e-002	1.184e-002		
23	Accel. Gradient (kPa/m)	1.213e-006	1.213e-006	1.213e-006	1.213e-006		
24	Liq. Reynolds	2.127e+005 *	2.127e+005 *	2.127e+005 *	2.127e+005 *		
25	Vap. Reynolds	9.136e+006 *	9.136e+006 *	9.136e+006 *	9.136e+006 *		
26	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043		
27	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	1.910e+004	1.911e+004	1.911e+004	1.912e+004		
31	Elevation (m)	-0.4500	-0.4480	-0.4460	-0.4440		
32	Pressure (bar)	89.97	89.97	89.96	89.96		
33	Temperature (C)	53.57	53.57	53.57	53.57		
34	Vapour Fraction	0.8706	0.8706	0.8706	0.8706		
35	Heat Transferred (kJ/h-m)	628.0	628.0	627.9	627.9		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.2946	0.2805	0.2805	0.2805		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	5.185e-003	5.543e-003	5.543e-003	5.543e-003		
41	Static Gradient (kPa/m)	1.184e-002	9.473e-004	9.473e-004	9.473e-004		
42	Accel. Gradient (kPa/m)	1.213e-006	4.624e-007	4.624e-007	4.623e-007		
43	Liq. Reynolds	2.126e+005 *	2.126e+005 *	2.126e+005 *	2.126e+005 *		
44	Vap. Reynolds	9.136e+006 *	9.136e+006 *	9.136e+006 *	9.136e+006 *		
45	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043		
46	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	1.912e+004	1.913e+004	1.913e+004	1.914e+004		
50	Elevation (m)	-0.4420	-0.4400	-0.4420	-0.4440		
51	Pressure (bar)	89.96	89.96	89.96	89.96		
52	Temperature (C)	53.57	53.56	53.56	53.56		
53	Vapour Fraction	0.8706	0.8706	0.8706	0.8706		
54	Heat Transferred (kJ/h-m)	627.8	627.8	627.7	627.7		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.2805	0.2805	0.2066	0.2066		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	5.543e-003	5.543e-003	6.870e-003	6.870e-003		
60	Static Gradient (kPa/m)	9.473e-004	9.473e-004	-7.724e-004	-7.724e-004		
61	Accel. Gradient (kPa/m)	4.623e-007	4.623e-007	4.344e-007	4.344e-007		
62	Liq. Reynolds	2.126e+005 *	2.126e+005 *	2.126e+005 *	2.126e+005 *		
63	Vap. Reynolds	9.136e+006 *	9.136e+006 *	9.136e+006 *	9.136e+006 *		
64	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043		
65	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	1.914e+004	1.915e+004	1.915e+004	1.916e+004		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	-0.4460	-0.4480	-0.4500	-0.4560
14	Pressure (bar)	89.96	89.96	89.96	89.96
15	Temperature (C)	53.56	53.56	53.55	53.55
16	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
17	Heat Transferred (kJ/h-m)	627.6	627.6	627.5	627.5
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2066	0.2066	0.2066	0.2059
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	6.870e-003	6.870e-003	6.870e-003	6.876e-003
23	Static Gradient (kPa/m)	-7.724e-004	-7.724e-004	-7.724e-004	-2.312e-003
24	Accel. Gradient (kPa/m)	4.344e-007	4.344e-007	4.344e-007	3.251e-007
25	Liq. Reynolds	2.126e+005 *	2.126e+005 *	2.126e+005 *	2.126e+005 *
26	Vap. Reynolds	9.136e+006 *	9.136e+006 *	9.136e+006 *	9.136e+006 *
27	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
28	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	1.916e+004	1.917e+004	1.917e+004	1.918e+004
32	Elevation (m)	-0.4620	-0.4680	-0.4740	-0.4800
33	Pressure (bar)	89.96	89.96	89.96	89.96
34	Temperature (C)	53.55	53.55	53.55	53.55
35	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
36	Heat Transferred (kJ/h-m)	627.4	627.4	627.3	627.3
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2059	0.2059	0.2059	0.2059
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	6.876e-003	6.876e-003	6.876e-003	6.876e-003
42	Static Gradient (kPa/m)	-2.312e-003	-2.312e-003	-2.312e-003	-2.312e-003
43	Accel. Gradient (kPa/m)	3.251e-007	3.251e-007	3.251e-007	3.251e-007
44	Liq. Reynolds	2.126e+005 *	2.126e+005 *	2.126e+005 *	2.126e+005 *
45	Vap. Reynolds	9.136e+006 *	9.136e+006 *	9.136e+006 *	9.136e+006 *
46	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
47	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	1.918e+004	1.919e+004	1.919e+004	1.920e+004
51	Elevation (m)	-0.4840	-0.4880	-0.4920	-0.4960
52	Pressure (bar)	89.96	89.96	89.96	89.96
53	Temperature (C)	53.54	53.54	53.54	53.54
54	Vapour Fraction	0.8706	0.8706	0.8706	0.8706
55	Heat Transferred (kJ/h-m)	627.2	627.2	627.2	627.1
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2062	0.2062	0.2062	0.2062
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	6.873e-003	6.873e-003	6.873e-003	6.873e-003
61	Static Gradient (kPa/m)	-1.543e-003	-1.543e-003	-1.543e-003	-1.543e-003
62	Accel. Gradient (kPa/m)	3.796e-007	3.796e-007	3.796e-007	3.796e-007
63	Liq. Reynolds	2.126e+005 *	2.126e+005 *	2.126e+005 *	2.126e+005 *
64	Vap. Reynolds	9.136e+006 *	9.136e+006 *	9.136e+006 *	9.136e+006 *
65	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
66	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length (m)	1.920e+004	1.921e+004	1.921e+004	1.922e+004		
14	Elevation (m)	-0.5000	-0.5180	-0.5360	-0.5540		
15	Pressure (bar)	89.96	89.96	89.96	89.96		
16	Temperature (C)	53.54	53.53	53.53	53.53		
17	Vapour Fraction	0.8706	0.8706	0.8706	0.8706		
18	Heat Transferred (kJ/h-m)	627.1	627.0	627.0	626.9		
19	Flow Regime	Transition	Transition	Transition	Transition		
20	Water Flow Regime						
21	Liquid HoldUp	0.2062	0.2036	0.2036	0.2036		
22	Water HoldUp	---	---	---	---		
23	Friction Gradient (kPa/m)	6.873e-003	6.893e-003	6.893e-003	6.893e-003		
24	Static Gradient (kPa/m)	-1.543e-003	-6.888e-003	-6.888e-003	-6.888e-003		
25	Accel. Gradient (kPa/m)	3.796e-007	3.594e-010	3.518e-010	3.443e-010		
26	Liq. Reynolds	2.126e+005 *	2.126e+005 *	2.126e+005 *	2.126e+005 *		
27	Vap. Reynolds	9.136e+006 *	9.136e+006 *	9.136e+006 *	9.136e+006 *		
28	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043		
29	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173		
30	Deposit Thickness (mm)	---	---	---	---		
31	Deposit Volume (m3)	---	---	---	---		
32	Length (m)	1.922e+004	1.923e+004	1.923e+004	1.924e+004		
33	Elevation (m)	-0.5720	-0.5900	-0.5680	-0.5460		
34	Pressure (bar)	89.96	89.96	89.96	89.96		
35	Temperature (C)	53.53	53.53	53.53	53.52		
36	Vapour Fraction	0.8706	0.8706	0.8706	0.8705		
37	Heat Transferred (kJ/h-m)	626.9	626.8	626.8	626.7		
38	Flow Regime	Transition	Transition	Transition	Transition		
39	Water Flow Regime						
40	Liquid HoldUp	0.2036	0.2036	0.2932	0.2932		
41	Water HoldUp	---	---	---	---		
42	Friction Gradient (kPa/m)	6.893e-003	6.893e-003	4.658e-003	4.658e-003		
43	Static Gradient (kPa/m)	-6.888e-003	-6.889e-003	1.075e-002	1.075e-002		
44	Accel. Gradient (kPa/m)	3.367e-010	3.291e-010	1.098e-006	1.098e-006		
45	Liq. Reynolds	2.126e+005 *	2.126e+005 *	2.126e+005 *	2.126e+005 *		
46	Vap. Reynolds	9.136e+006 *	9.136e+006 *	9.136e+006 *	9.136e+006 *		
47	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043		
48	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173		
49	Deposit Thickness (mm)	---	---	---	---		
50	Deposit Volume (m3)	---	---	---	---		
51	Length (m)	1.924e+004	1.925e+004	1.925e+004	1.926e+004		
52	Elevation (m)	-0.5240	-0.5020	-0.4800	-0.4940		
53	Pressure (bar)	89.96	89.96	89.96	89.96		
54	Temperature (C)	53.52	53.52	53.52	53.52		
55	Vapour Fraction	0.8705	0.8705	0.8705	0.8705		
56	Heat Transferred (kJ/h-m)	626.6	626.6	626.5	626.5		
57	Flow Regime	Transition	Transition	Transition	Transition		
58	Water Flow Regime						
59	Liquid HoldUp	0.2932	0.2932	0.2932	0.2044		
60	Water HoldUp	---	---	---	---		
61	Friction Gradient (kPa/m)	4.658e-003	4.658e-003	4.658e-003	6.887e-003		
62	Static Gradient (kPa/m)	1.075e-002	1.075e-002	1.075e-002	-5.370e-003		
63	Accel. Gradient (kPa/m)	1.098e-006	1.098e-006	1.098e-006	1.080e-007		
64	Liq. Reynolds	2.126e+005 *	2.126e+005 *	2.126e+005 *	2.126e+005 *		
65	Vap. Reynolds	9.136e+006 *	9.136e+006 *	9.136e+006 *	9.136e+006 *		
66	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043		
67	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173		
68	Deposit Thickness (mm)	---	---	---	---		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:02:44 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	1.926e+004	1.927e+004	1.927e+004	1.928e+004		
15	Elevation (m)	-0.5080	-0.5220	-0.5360	-0.5500		
16	Pressure (bar)	89.96	89.96	89.96	89.96		
17	Temperature (C)	53.51	53.51	53.51	53.51		
18	Vapour Fraction	0.8705	0.8705	0.8705	0.8705		
19	Heat Transferred (kJ/h-m)	626.5	626.4	626.4	626.3		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	0.2044	0.2044	0.2044	0.2044		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	6.887e-003	6.887e-003	6.887e-003	6.887e-003		
25	Static Gradient (kPa/m)	-5.370e-003	-5.370e-003	-5.370e-003	-5.370e-003		
26	Accel. Gradient (kPa/m)	1.080e-007	1.080e-007	1.080e-007	1.080e-007		
27	Liq. Reynolds	2.126e+005 *	2.126e+005 *	2.126e+005 *	2.126e+005 *		
28	Vap. Reynolds	9.136e+006 *	9.136e+006 *	9.136e+006 *	9.136e+006 *		
29	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043		
30	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	1.928e+004	1.929e+004	1.929e+004	1.930e+004		
34	Elevation (m)	-0.5800	-0.6100	-0.6400	-0.6700		
35	Pressure (bar)	89.96	89.96	89.96	89.96		
36	Temperature (C)	53.51	53.51	53.50	53.50		
37	Vapour Fraction	0.8705	0.8705	0.8705	0.8705		
38	Heat Transferred (kJ/h-m)	626.3	626.2	626.2	626.1		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.2015	0.2015	0.2015	0.2015		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	6.911e-003	6.911e-003	6.911e-003	6.911e-003		
44	Static Gradient (kPa/m)	-1.132e-002	-1.132e-002	-1.132e-002	-1.132e-002		
45	Accel. Gradient (kPa/m)	-3.142e-007	-3.142e-007	-3.142e-007	-3.142e-007		
46	Liq. Reynolds	2.126e+005 *	2.126e+005 *	2.126e+005 *	2.126e+005 *		
47	Vap. Reynolds	9.136e+006 *	9.136e+006 *	9.136e+006 *	9.136e+006 *		
48	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043		
49	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	1.930e+004	1.931e+004	1.931e+004	1.932e+004		
53	Elevation (m)	-0.7000	-0.7020	-0.7040	-0.7060		
54	Pressure (bar)	89.96	89.96	89.96	89.96		
55	Temperature (C)	53.50	53.50	53.50	53.50		
56	Vapour Fraction	0.8705	0.8705	0.8705	0.8705		
57	Heat Transferred (kJ/h-m)	626.1	626.0	626.0	625.9		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.2015	0.2066	0.2066	0.2066		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	6.911e-003	6.869e-003	6.869e-003	6.869e-003		
63	Static Gradient (kPa/m)	-1.132e-002	-7.725e-004	-7.725e-004	-7.725e-004		
64	Accel. Gradient (kPa/m)	-3.142e-007	4.342e-007	4.342e-007	4.342e-007		
65	Liq. Reynolds	2.126e+005 *	2.126e+005 *	2.126e+005 *	2.126e+005 *		
66	Vap. Reynolds	9.136e+006 *	9.136e+006 *	9.136e+006 *	9.136e+006 *		
67	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043		
68	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	1.932e+004	1.933e+004	1.933e+004	1.934e+004
16	Elevation (m)	-0.7080	-0.7100	-0.6980	-0.6860
17	Pressure (bar)	89.96	89.96	89.95	89.95
18	Temperature (C)	53.49	53.49	53.49	53.49
19	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
20	Heat Transferred (kJ/h-m)	625.9	625.8	625.8	625.7
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2066	0.2066	0.2869	0.2869
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	6.869e-003	6.868e-003	5.055e-003	5.055e-003
26	Static Gradient (kPa/m)	-7.725e-004	-7.725e-004	5.776e-003	5.776e-003
27	Accel. Gradient (kPa/m)	4.342e-007	4.342e-007	7.714e-007	7.714e-007
28	Liq. Reynolds	2.126e+005 *	2.126e+005 *	2.126e+005 *	2.126e+005 *
29	Vap. Reynolds	9.136e+006 *	9.136e+006 *	9.136e+006 *	9.136e+006 *
30	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
31	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	1.934e+004	1.935e+004	1.935e+004	1.936e+004
35	Elevation (m)	-0.6740	-0.6620	-0.6500	-1.440
36	Pressure (bar)	89.95	89.95	89.95	89.96
37	Temperature (C)	53.49	53.48	53.48	53.48
38	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
39	Heat Transferred (kJ/h-m)	625.7	625.6	625.6	626.1
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2869	0.2869	0.2869	6.529e-002
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	5.055e-003	5.055e-003	5.055e-003	1.198e-002
45	Static Gradient (kPa/m)	5.776e-003	5.776e-003	5.776e-003	-0.1729
46	Accel. Gradient (kPa/m)	7.714e-007	7.714e-007	7.714e-007	-1.146e-005
47	Liq. Reynolds	2.126e+005 *	2.126e+005 *	2.126e+005 *	2.126e+005 *
48	Vap. Reynolds	9.136e+006 *	9.136e+006 *	9.136e+006 *	9.136e+006 *
49	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
50	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	1.936e+004	1.937e+004	1.937e+004	1.938e+004
54	Elevation (m)	-2.230	-3.020	-3.810	-4.600
55	Pressure (bar)	89.97	89.98	89.98	89.99
56	Temperature (C)	53.49	53.49	53.49	53.49
57	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
58	Heat Transferred (kJ/h-m)	626.2	626.2	626.3	626.4
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	6.529e-002	6.530e-002	6.530e-002	6.530e-002
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	1.198e-002	1.198e-002	1.198e-002	1.198e-002
64	Static Gradient (kPa/m)	-0.1729	-0.1729	-0.1729	-0.1729
65	Accel. Gradient (kPa/m)	-1.146e-005	-1.145e-005	-1.145e-005	-1.145e-005
66	Liq. Reynolds	2.126e+005 *	2.126e+005 *	2.126e+005 *	2.126e+005 *
67	Vap. Reynolds	9.135e+006 *	9.135e+006 *	9.135e+006 *	9.135e+006 *
68	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022	
6						
7						
8						
9			PROFILES			
10						
11			Pipe Table			
12						
13	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172
14	Deposit Thickness	(mm)	---	---	---	---
15	Deposit Volume	(m3)	---	---	---	---
16	Length	(m)	1.938e+004	1.939e+004	1.939e+004	1.940e+004
17	Elevation	(m)	-4.390	-4.180	-3.970	-3.760
18	Pressure	(bar)	89.99	89.98	89.97	89.97
19	Temperature	(C)	53.49	53.49	53.48	53.48
20	Vapour Fraction		0.8705	0.8705	0.8705	0.8705
21	Heat Transferred	(kJ/h-m)	625.8	625.7	625.6	625.5
22	Flow Regime		Transition	Transition	Transition	Transition
23	Water Flow Regime					
24	Liquid HoldUp		0.4128	0.4127	0.4127	0.4127
25	Water HoldUp		---	---	---	---
26	Friction Gradient	(kPa/m)	5.971e-003	5.971e-003	5.972e-003	5.972e-003
27	Static Gradient	(kPa/m)	0.1324	0.1324	0.1324	0.1324
28	Accel. Gradient	(kPa/m)	9.847e-006	9.848e-006	9.850e-006	9.851e-006
29	Liq. Reynolds		2.126e+005 *	2.126e+005 *	2.126e+005 *	2.126e+005 *
30	Vap. Reynolds		9.135e+006 *	9.135e+006 *	9.135e+006 *	9.136e+006 *
31	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
32	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172
33	Deposit Thickness	(mm)	---	---	---	---
34	Deposit Volume	(m3)	---	---	---	---
35	Length	(m)	1.940e+004	1.941e+004	1.941e+004	1.942e+004
36	Elevation	(m)	-3.550	-3.058	-2.566	-2.074
37	Pressure	(bar)	89.96	89.94	89.92	89.90
38	Temperature	(C)	53.47	53.47	53.46	53.45
39	Vapour Fraction		0.8705	0.8705	0.8705	0.8705
40	Heat Transferred	(kJ/h-m)	625.4	626.2	626.0	625.8
41	Flow Regime		Transition	Transition	Transition	Transition
42	Water Flow Regime					
43	Liquid HoldUp		0.4127	0.5882	0.5881	0.5880
44	Water HoldUp		---	---	---	---
45	Friction Gradient	(kPa/m)	5.972e-003	5.787e-003	5.788e-003	5.789e-003
46	Static Gradient	(kPa/m)	0.1324	0.4124	0.4124	0.4124
47	Accel. Gradient	(kPa/m)	9.852e-006	2.980e-005	2.981e-005	2.982e-005
48	Liq. Reynolds		2.126e+005 *	2.126e+005 *	2.125e+005 *	2.125e+005 *
49	Vap. Reynolds		9.136e+006 *	9.136e+006 *	9.137e+006 *	9.138e+006 *
50	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
51	Vapour Velocity	(m/s)	2.172	2.173	2.173	2.174
52	Deposit Thickness	(mm)	---	---	---	---
53	Deposit Volume	(m3)	---	---	---	---
54	Length	(m)	1.942e+004	1.943e+004	1.943e+004	1.944e+004
55	Elevation	(m)	-1.582	-1.090	-1.096	-1.102
56	Pressure	(bar)	89.87	89.85	89.85	89.85
57	Temperature	(C)	53.45	53.44	53.44	53.43
58	Vapour Fraction		0.8705	0.8706	0.8706	0.8706
59	Heat Transferred	(kJ/h-m)	625.6	625.4	624.4	624.3
60	Flow Regime		Transition	Transition	Transition	Transition
61	Water Flow Regime					
62	Liquid HoldUp		0.5880	0.5879	0.2058	0.2058
63	Water HoldUp		---	---	---	---
64	Friction Gradient	(kPa/m)	5.791e-003	5.792e-003	6.881e-003	6.881e-003
65	Static Gradient	(kPa/m)	0.4123	0.4123	-2.311e-003	-2.311e-003
66	Accel. Gradient	(kPa/m)	2.983e-005	2.984e-005	3.262e-007	3.262e-007
67	Liq. Reynolds		2.125e+005 *	2.124e+005 *	2.124e+005 *	2.124e+005 *
68	Vap. Reynolds		9.138e+006 *	9.139e+006 *	9.139e+006 *	9.139e+006 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
14	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	1.944e+004	1.945e+004	1.945e+004	1.946e+004	
18	Elevation	(m)	-1.108	-1.114	-1.120	-1.046	
19	Pressure	(bar)	89.85	89.85	89.85	89.85	
20	Temperature	(C)	53.43	53.43	53.43	53.43	
21	Vapour Fraction		0.8706	0.8706	0.8705	0.8705	
22	Heat Transferred	(kJ/h-m)	624.3	624.2	624.2	624.1	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.2058	0.2058	0.2058	0.3377	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	6.881e-003	6.881e-003	6.881e-003	6.081e-003	
28	Static Gradient	(kPa/m)	-2.311e-003	-2.311e-003	-2.311e-003	4.991e-002	
29	Accel. Gradient	(kPa/m)	3.262e-007	3.262e-007	3.262e-007	3.997e-006	
30	Liq. Reynolds		2.124e+005 *	2.124e+005 *	2.124e+005 *	2.124e+005 *	
31	Vap. Reynolds		9.139e+006 *	9.139e+006 *	9.139e+006 *	9.139e+006 *	
32	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
33	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	1.946e+004	1.946e+004	1.947e+004	1.947e+004	
37	Elevation	(m)	-0.9720	-0.8980	-0.8240	-0.7500	
38	Pressure	(bar)	89.85	89.85	89.84	89.84	
39	Temperature	(C)	53.42	53.42	53.42	53.42	
40	Vapour Fraction		0.8705	0.8705	0.8705	0.8705	
41	Heat Transferred	(kJ/h-m)	624.1	624.0	624.0	623.9	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.3377	0.3377	0.3377	0.3377	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	6.081e-003	6.081e-003	6.081e-003	6.081e-003	
47	Static Gradient	(kPa/m)	4.991e-002	4.991e-002	4.991e-002	4.991e-002	
48	Accel. Gradient	(kPa/m)	3.997e-006	3.997e-006	3.997e-006	3.997e-006	
49	Liq. Reynolds		2.124e+005 *	2.124e+005 *	2.124e+005 *	2.124e+005 *	
50	Vap. Reynolds		9.139e+006 *	9.139e+006 *	9.139e+006 *	9.139e+006 *	
51	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
52	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	1.948e+004	1.948e+004	1.949e+004	1.949e+004	
56	Elevation	(m)	-0.7240	-0.6980	-0.6720	-0.6460	
57	Pressure	(bar)	89.84	89.84	89.84	89.84	
58	Temperature	(C)	53.42	53.41	53.41	53.41	
59	Vapour Fraction		0.8705	0.8705	0.8705	0.8705	
60	Heat Transferred	(kJ/h-m)	623.8	623.8	623.7	623.7	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2956	0.2956	0.2956	0.2956	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	5.424e-003	5.424e-003	5.424e-003	5.424e-003	
66	Static Gradient	(kPa/m)	1.278e-002	1.278e-002	1.278e-002	1.278e-002	
67	Accel. Gradient	(kPa/m)	1.300e-006	1.300e-006	1.300e-006	1.300e-006	
68	Liq. Reynolds		2.124e+005 *	2.124e+005 *	2.124e+005 *	2.124e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.139e+006 *	9.140e+006 *	9.140e+006 *	9.140e+006 *
14	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
15	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	1.950e+004	1.952e+004	1.954e+004	1.957e+004
19	Elevation (m)	-0.6200	-0.5920	-0.5640	-0.5360
20	Pressure (bar)	89.84	89.83	89.83	89.83
21	Temperature (C)	53.41	53.40	53.39	53.38
22	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
23	Heat Transferred (kJ/h-m)	623.6	623.4	623.1	622.9
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2956	0.2829	0.2829	0.2829
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	5.424e-003	5.351e-003	5.350e-003	5.350e-003
29	Static Gradient (kPa/m)	1.278e-002	2.859e-003	2.859e-003	2.859e-003
30	Accel. Gradient (kPa/m)	1.300e-006	5.861e-007	5.861e-007	5.861e-007
31	Liq. Reynolds	2.124e+005 *	2.124e+005 *	2.124e+005 *	2.124e+005 *
32	Vap. Reynolds	9.140e+006 *	9.140e+006 *	9.140e+006 *	9.140e+006 *
33	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
34	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	1.959e+004	1.961e+004	1.964e+004	1.966e+004
38	Elevation (m)	-0.5080	-0.4800	-0.4780	-0.4760
39	Pressure (bar)	89.83	89.83	89.83	89.82
40	Temperature (C)	53.37	53.36	53.36	53.35
41	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
42	Heat Transferred (kJ/h-m)	622.7	622.4	622.2	622.0
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2829	0.2829	0.2794	0.2794
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	5.350e-003	5.350e-003	5.628e-003	5.628e-003
48	Static Gradient (kPa/m)	2.859e-003	2.859e-003	2.024e-004	2.025e-004
49	Accel. Gradient (kPa/m)	5.861e-007	5.861e-007	4.162e-007	4.162e-007
50	Liq. Reynolds	2.124e+005 *	2.124e+005 *	2.124e+005 *	2.123e+005 *
51	Vap. Reynolds	9.140e+006 *	9.140e+006 *	9.140e+006 *	9.140e+006 *
52	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
53	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	1.968e+004	1.971e+004	1.973e+004	1.975e+004
57	Elevation (m)	-0.4740	-0.4720	-0.4700	-0.4660
58	Pressure (bar)	89.82	89.82	89.82	89.82
59	Temperature (C)	53.34	53.33	53.32	53.31
60	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
61	Heat Transferred (kJ/h-m)	621.7	621.5	621.3	621.1
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2794	0.2794	0.2794	0.2797
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	5.627e-003	5.627e-003	5.627e-003	5.605e-003
67	Static Gradient (kPa/m)	2.025e-004	2.025e-004	2.025e-004	4.052e-004
68	Accel. Gradient (kPa/m)	4.162e-007	4.162e-007	4.161e-007	4.290e-007
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.123e+005 *	2.123e+005 *	2.123e+005 *	2.123e+005 *
14	Vap. Reynolds	9.140e+006 *	9.140e+006 *	9.140e+006 *	9.140e+006 *
15	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
16	Vapour Velocity (m/s)	2.175	2.175	2.175	2.174
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	1.978e+004	1.980e+004	1.982e+004	1.985e+004
20	Elevation (m)	-0.4620	-0.4580	-0.4540	-0.4500
21	Pressure (bar)	89.82	89.82	89.81	89.81
22	Temperature (C)	53.30	53.29	53.29	53.28
23	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
24	Heat Transferred (kJ/h-m)	620.8	620.6	620.4	620.1
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2797	0.2797	0.2797	0.2797
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	5.605e-003	5.605e-003	5.604e-003	5.604e-003
30	Static Gradient (kPa/m)	4.052e-004	4.052e-004	4.052e-004	4.053e-004
31	Accel. Gradient (kPa/m)	4.290e-007	4.290e-007	4.290e-007	4.290e-007
32	Liq. Reynolds	2.123e+005 *	2.123e+005 *	2.123e+005 *	2.123e+005 *
33	Vap. Reynolds	9.140e+006 *	9.140e+006 *	9.140e+006 *	9.140e+006 *
34	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
35	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	1.987e+004	1.989e+004	1.992e+004	1.994e+004
39	Elevation (m)	-0.4480	-0.4460	-0.4440	-0.4420
40	Pressure (bar)	89.81	89.81	89.81	89.81
41	Temperature (C)	53.27	53.26	53.25	53.24
42	Vapour Fraction	0.8705	0.8705	0.8705	0.8705
43	Heat Transferred (kJ/h-m)	619.9	619.7	619.5	619.2
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2794	0.2794	0.2794	0.2794
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	5.626e-003	5.625e-003	5.625e-003	5.625e-003
49	Static Gradient (kPa/m)	2.025e-004	2.025e-004	2.025e-004	2.025e-004
50	Accel. Gradient (kPa/m)	4.160e-007	4.160e-007	4.160e-007	4.159e-007
51	Liq. Reynolds	2.123e+005 *	2.123e+005 *	2.123e+005 *	2.123e+005 *
52	Vap. Reynolds	9.140e+006 *	9.140e+006 *	9.140e+006 *	9.140e+006 *
53	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
54	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	1.996e+004	1.999e+004	2.001e+004	2.003e+004
58	Elevation (m)	-0.4400	-0.4380	-0.4360	-0.4340
59	Pressure (bar)	89.81	89.81	89.80	89.80
60	Temperature (C)	53.23	53.23	53.22	53.21
61	Vapour Fraction	0.8704	0.8704	0.8704	0.8704
62	Heat Transferred (kJ/h-m)	619.0	618.8	618.5	618.3
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2794	0.2794	0.2794	0.2795
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	5.625e-003	5.625e-003	5.624e-003	5.624e-003
68	Static Gradient (kPa/m)	2.025e-004	2.025e-004	2.025e-004	2.025e-004
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient	(kPa/m)	4.159e-007	4.159e-007	4.159e-007	4.159e-007	
14	Liq. Reynolds		2.123e+005 *	2.122e+005 *	2.122e+005 *	2.122e+005 *	
15	Vap. Reynolds		9.140e+006 *	9.140e+006 *	9.140e+006 *	9.140e+006 *	
16	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
17	Vapour Velocity	(m/s)	2.174	2.174	2.174	2.174	
18	Deposit Thickness	(mm)	---	---	---	---	
19	Deposit Volume	(m3)	---	---	---	---	
20	Length	(m)	2.006e+004	2.008e+004	2.010e+004	2.013e+004	
21	Elevation	(m)	-0.4320	-0.4300	-0.4840	-0.5380	
22	Pressure	(bar)	89.80	89.80	89.80	89.80	
23	Temperature	(C)	53.20	53.19	53.18	53.18	
24	Vapour Fraction		0.8704	0.8704	0.8704	0.8704	
25	Heat Transferred	(kJ/h-m)	618.1	617.9	617.7	617.4	
26	Flow Regime		Transition	Transition	Transition	Transition	
27	Water Flow Regime						
28	Liquid HoldUp		0.2795	0.2795	0.2048	0.2048	
29	Water HoldUp		---	---	---	---	
30	Friction Gradient	(kPa/m)	5.624e-003	5.624e-003	6.885e-003	6.885e-003	
31	Static Gradient	(kPa/m)	2.025e-004	2.025e-004	-4.445e-003	-4.445e-003	
32	Accel. Gradient	(kPa/m)	4.158e-007	4.158e-007	1.741e-007	1.741e-007	
33	Liq. Reynolds		2.122e+005 *	2.122e+005 *	2.122e+005 *	2.122e+005 *	
34	Vap. Reynolds		9.140e+006 *	9.140e+006 *	9.140e+006 *	9.141e+006 *	
35	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
36	Vapour Velocity	(m/s)	2.174	2.174	2.173	2.173	
37	Deposit Thickness	(mm)	---	---	---	---	
38	Deposit Volume	(m3)	---	---	---	---	
39	Length	(m)	2.015e+004	2.017e+004	2.020e+004	2.022e+004	
40	Elevation	(m)	-0.5920	-0.6460	-0.7000	-0.6380	
41	Pressure	(bar)	89.80	89.80	89.80	89.79	
42	Temperature	(C)	53.17	53.16	53.15	53.14	
43	Vapour Fraction		0.8704	0.8704	0.8704	0.8704	
44	Heat Transferred	(kJ/h-m)	617.2	617.0	616.8	616.5	
45	Flow Regime		Transition	Transition	Transition	Transition	
46	Water Flow Regime						
47	Liquid HoldUp		0.2048	0.2048	0.2048	0.2877	
48	Water HoldUp		---	---	---	---	
49	Friction Gradient	(kPa/m)	6.885e-003	6.884e-003	6.884e-003	4.997e-003	
50	Static Gradient	(kPa/m)	-4.445e-003	-4.445e-003	-4.446e-003	6.409e-003	
51	Accel. Gradient	(kPa/m)	1.741e-007	1.741e-007	1.740e-007	8.139e-007	
52	Liq. Reynolds		2.122e+005 *	2.122e+005 *	2.122e+005 *	2.122e+005 *	
53	Vap. Reynolds		9.141e+006 *	9.141e+006 *	9.141e+006 *	9.141e+006 *	
54	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043	
55	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173	
56	Deposit Thickness	(mm)	---	---	---	---	
57	Deposit Volume	(m3)	---	---	---	---	
58	Length	(m)	2.024e+004	2.027e+004	2.029e+004	2.031e+004	
59	Elevation	(m)	-0.5760	-0.5140	-0.4520	-0.3900	
60	Pressure	(bar)	89.79	89.79	89.79	89.78	
61	Temperature	(C)	53.13	53.12	53.11	53.11	
62	Vapour Fraction		0.8704	0.8704	0.8704	0.8704	
63	Heat Transferred	(kJ/h-m)	616.3	616.0	615.8	615.6	
64	Flow Regime		Transition	Transition	Transition	Transition	
65	Water Flow Regime						
66	Liquid HoldUp		0.2877	0.2877	0.2877	0.2877	
67	Water HoldUp		---	---	---	---	
68	Friction Gradient	(kPa/m)	4.996e-003	4.996e-003	4.996e-003	4.996e-003	
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1				Case Name:	Simulation - 386MMSCFD_100barg.hsc	
2				Unit Set:	Mixed	
3		Epic Atlantic Limited				
4				Date/Time:	Mon Jun 13 15:02:44 2022	
5						
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	6.409e-003	6.409e-003	6.409e-003	6.409e-003
14	Accel. Gradient	(kPa/m)	8.139e-007	8.139e-007	8.139e-007	8.139e-007
15	Liq. Reynolds		2.122e+005 *	2.122e+005 *	2.122e+005 *	2.121e+005 *
16	Vap. Reynolds		9.141e+006 *	9.141e+006 *	9.141e+006 *	9.141e+006 *
17	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
18	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	2.034e+004	2.036e+004	2.038e+004	2.041e+004
22	Elevation	(m)	-0.3920	-0.3940	-0.3960	-0.3980
23	Pressure	(bar)	89.78	89.78	89.78	89.78
24	Temperature	(C)	53.10	53.09	53.08	53.07
25	Vapour Fraction		0.8704	0.8704	0.8704	0.8704
26	Heat Transferred	(kJ/h-m)	615.3	615.1	614.9	614.7
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2069	0.2069	0.2069	0.2069
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	6.867e-003	6.867e-003	6.867e-003	6.867e-003
32	Static Gradient	(kPa/m)	-1.657e-004	-1.657e-004	-1.657e-004	-1.657e-004
33	Accel. Gradient	(kPa/m)	4.782e-007	4.782e-007	4.782e-007	4.782e-007
34	Liq. Reynolds		2.121e+005 *	2.121e+005 *	2.121e+005 *	2.121e+005 *
35	Vap. Reynolds		9.141e+006 *	9.141e+006 *	9.141e+006 *	9.141e+006 *
36	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
37	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	2.043e+004	2.045e+004	2.048e+004	2.050e+004
41	Elevation	(m)	-0.4000	-0.3960	-0.3920	-0.3880
42	Pressure	(bar)	89.78	89.77	89.77	89.77
43	Temperature	(C)	53.06	53.05	53.05	53.04
44	Vapour Fraction		0.8703	0.8703	0.8703	0.8703
45	Heat Transferred	(kJ/h-m)	614.4	614.2	614.0	613.7
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2070	0.2798	0.2798	0.2798
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	6.867e-003	5.599e-003	5.599e-003	5.599e-003
51	Static Gradient	(kPa/m)	-1.657e-004	4.055e-004	4.055e-004	4.055e-004
52	Accel. Gradient	(kPa/m)	4.782e-007	4.285e-007	4.284e-007	4.284e-007
53	Liq. Reynolds		2.121e+005 *	2.121e+005 *	2.121e+005 *	2.121e+005 *
54	Vap. Reynolds		9.141e+006 *	9.141e+006 *	9.141e+006 *	9.141e+006 *
55	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
56	Vapour Velocity	(m/s)	2.173	2.173	2.172	2.172
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	2.052e+004	2.055e+004	2.058e+004	2.060e+004
60	Elevation	(m)	-0.3840	-0.3800	-0.4020	-0.4240
61	Pressure	(bar)	89.77	89.77	89.77	89.77
62	Temperature	(C)	53.03	53.02	53.01	53.00
63	Vapour Fraction		0.8703	0.8703	0.8703	0.8703
64	Heat Transferred	(kJ/h-m)	613.5	613.3	613.0	612.7
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2798	0.2798	0.2064	0.2064
68	Water HoldUp		---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 348 of 937

1				Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	5.598e-003	5.598e-003	6.871e-003	6.870e-003
14	Static Gradient	(kPa/m)	4.055e-004	4.055e-004	-1.464e-003	-1.464e-003
15	Accel. Gradient	(kPa/m)	4.284e-007	4.284e-007	3.858e-007	3.857e-007
16	Liq. Reynolds		2.121e+005 *	2.121e+005 *	2.121e+005 *	2.121e+005 *
17	Vap. Reynolds		9.141e+006 *	9.141e+006 *	9.141e+006 *	9.141e+006 *
18	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
19	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	2.063e+004	2.066e+004	2.069e+004	2.069e+004
23	Elevation	(m)	-0.4460	-0.4680	-0.4900	-0.5000
24	Pressure	(bar)	89.76	89.76	89.76	89.76
25	Temperature	(C)	52.99	52.98	52.97	52.97
26	Vapour Fraction		0.8703	0.8703	0.8703	0.8703
27	Heat Transferred	(kJ/h-m)	612.5	612.2	611.9	611.9
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2064	0.2064	0.2064	0.2018
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	6.870e-003	6.870e-003	6.870e-003	6.907e-003
33	Static Gradient	(kPa/m)	-1.464e-003	-1.464e-003	-1.464e-003	-1.081e-002
34	Accel. Gradient	(kPa/m)	3.857e-007	3.857e-007	3.856e-007	-2.787e-007
35	Liq. Reynolds		2.120e+005 *	2.120e+005 *	2.120e+005 *	2.120e+005 *
36	Vap. Reynolds		9.141e+006 *	9.141e+006 *	9.141e+006 *	9.141e+006 *
37	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
38	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	2.070e+004	2.070e+004	2.070e+004	2.070e+004
42	Elevation	(m)	-0.5100	-0.5200	-0.5300	-0.5400
43	Pressure	(bar)	89.76	89.76	89.76	89.76
44	Temperature	(C)	52.97	52.97	52.97	52.97
45	Vapour Fraction		0.8703	0.8703	0.8703	0.8703
46	Heat Transferred	(kJ/h-m)	611.9	611.9	611.9	611.8
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2018	0.2018	0.2018	0.2018
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	6.907e-003	6.907e-003	6.907e-003	6.907e-003
52	Static Gradient	(kPa/m)	-1.081e-002	-1.081e-002	-1.081e-002	-1.081e-002
53	Accel. Gradient	(kPa/m)	-2.787e-007	-2.787e-007	-2.787e-007	-2.787e-007
54	Liq. Reynolds		2.120e+005 *	2.120e+005 *	2.120e+005 *	2.120e+005 *
55	Vap. Reynolds		9.141e+006 *	9.141e+006 *	9.141e+006 *	9.141e+006 *
56	Liquid Velocity	(m/s)	0.2043	0.2043	0.2043	0.2043
57	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	2.071e+004	2.071e+004	2.072e+004	2.072e+004
61	Elevation	(m)	-0.5360	-0.5320	-0.5280	-0.5240
62	Pressure	(bar)	89.76	89.76	89.76	89.76
63	Temperature	(C)	52.96	52.96	52.96	52.96
64	Vapour Fraction		0.8703	0.8703	0.8703	0.8703
65	Heat Transferred	(kJ/h-m)	611.8	611.7	611.7	611.6
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2819	0.2819	0.2819	0.2819
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1			Case Name:	Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set:	Mixed		
3	 Epic Atlantic Limited		Date/Time:	Mon Jun 13 15:02:44 2022		
4						
5						
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Water HoldUp		---	---	---	---
14	Friction Gradient (kPa/m)		5.439e-003	5.439e-003	5.439e-003	5.439e-003
15	Static Gradient (kPa/m)		1.902e-003	1.902e-003	1.902e-003	1.902e-003
16	Accel. Gradient (kPa/m)		5.238e-007	5.238e-007	5.238e-007	5.238e-007
17	Liq. Reynolds		2.120e+005 *	2.120e+005 *	2.120e+005 *	2.120e+005 *
18	Vap. Reynolds		9.141e+006 *	9.141e+006 *	9.141e+006 *	9.141e+006 *
19	Liquid Velocity (m/s)		0.2043	0.2043	0.2043	0.2043
20	Vapour Velocity (m/s)		2.172	2.172	2.172	2.172
21	Deposit Thickness (mm)		---	---	---	---
22	Deposit Volume (m3)		---	---	---	---
23	Length (m)		2.073e+004	2.073e+004	2.074e+004	2.074e+004
24	Elevation (m)		-0.5200	-0.5140	-0.5080	-0.5020
25	Pressure (bar)		89.76	89.76	89.76	89.76
26	Temperature (C)		52.96	52.95	52.95	52.95
27	Vapour Fraction		0.8703	0.8703	0.8703	0.8703
28	Heat Transferred (kJ/h-m)		611.6	611.5	611.5	611.4
29	Flow Regime		Transition	Transition	Transition	Transition
30	Water Flow Regime					
31	Liquid HoldUp		0.2819	0.2832	0.2832	0.2832
32	Water HoldUp		---	---	---	---
33	Friction Gradient (kPa/m)		5.439e-003	5.341e-003	5.340e-003	5.340e-003
34	Static Gradient (kPa/m)		1.902e-003	2.863e-003	2.863e-003	2.863e-003
35	Accel. Gradient (kPa/m)		5.238e-007	5.852e-007	5.852e-007	5.852e-007
36	Liq. Reynolds		2.120e+005 *	2.120e+005 *	2.120e+005 *	2.120e+005 *
37	Vap. Reynolds		9.141e+006 *	9.141e+006 *	9.141e+006 *	9.141e+006 *
38	Liquid Velocity (m/s)		0.2043	0.2043	0.2043	0.2043
39	Vapour Velocity (m/s)		2.172	2.172	2.172	2.172
40	Deposit Thickness (mm)		---	---	---	---
41	Deposit Volume (m3)		---	---	---	---
42	Length (m)		2.075e+004	2.075e+004	2.076e+004	2.076e+004
43	Elevation (m)		-0.4960	-0.4900	-0.4900	-0.4900
44	Pressure (bar)		89.76	89.76	89.76	89.76
45	Temperature (C)		52.95	52.95	52.95	52.94
46	Vapour Fraction		0.8703	0.8703	0.8703	0.8703
47	Heat Transferred (kJ/h-m)		611.4	611.3	611.3	611.2
48	Flow Regime		Transition	Transition	Transition	Transition
49	Water Flow Regime					
50	Liquid HoldUp		0.2832	0.2832	0.3023	0.3023
51	Water HoldUp		---	---	---	---
52	Friction Gradient (kPa/m)		5.340e-003	5.340e-003	5.879e-003	5.879e-003
53	Static Gradient (kPa/m)		2.863e-003	2.863e-003	0.0000	0.0000
54	Accel. Gradient (kPa/m)		5.852e-007	5.852e-007	4.194e-007	4.194e-007
55	Liq. Reynolds		2.120e+005 *	2.120e+005 *	2.120e+005 *	2.120e+005 *
56	Vap. Reynolds		9.141e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
57	Liquid Velocity (m/s)		0.2043	0.2043	0.2043	0.2043
58	Vapour Velocity (m/s)		2.172	2.172	2.172	2.172
59	Deposit Thickness (mm)		---	---	---	---
60	Deposit Volume (m3)		---	---	---	---
61	Length (m)		2.077e+004	2.077e+004	2.078e+004	2.078e+004
62	Elevation (m)		-0.4900	-0.4900	-0.4900	-0.4900
63	Pressure (bar)		89.76	89.76	89.76	89.76
64	Temperature (C)		52.94	52.94	52.94	52.94
65	Vapour Fraction		0.8703	0.8703	0.8703	0.8703
66	Heat Transferred (kJ/h-m)		611.2	611.1	611.1	611.0
67	Flow Regime		Transition	Transition	Transition	Transition
68	Water Flow Regime					
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 350 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.3023	0.3023	0.3023	0.3023
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	5.879e-003	5.879e-003	5.879e-003	5.879e-003
16	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
17	Accel. Gradient (kPa/m)	4.194e-007	4.194e-007	4.194e-007	4.194e-007
18	Liq. Reynolds	2.120e+005 *	2.120e+005 *	2.120e+005 *	2.120e+005 *
19	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
20	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
21	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	2.079e+004	2.079e+004	2.080e+004	2.080e+004
25	Elevation (m)	-0.4900	-0.4900	-0.4900	-0.4900
26	Pressure (bar)	89.76	89.76	89.75	89.75
27	Temperature (C)	52.93	52.93	52.93	52.93
28	Vapour Fraction	0.8703	0.8703	0.8703	0.8703
29	Heat Transferred (kJ/h-m)	611.0	610.9	610.9	610.9
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.3023	0.3023	0.3023	0.3023
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	5.879e-003	5.879e-003	5.879e-003	5.879e-003
35	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
36	Accel. Gradient (kPa/m)	4.194e-007	4.194e-007	4.194e-007	4.194e-007
37	Liq. Reynolds	2.120e+005 *	2.120e+005 *	2.120e+005 *	2.120e+005 *
38	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
39	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
40	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	2.081e+004	2.081e+004	2.082e+004	2.082e+004
44	Elevation (m)	-0.5000	-0.5100	-0.5200	-0.5300
45	Pressure (bar)	89.75	89.75	89.75	89.75
46	Temperature (C)	52.93	52.93	52.92	52.92
47	Vapour Fraction	0.8703	0.8703	0.8703	0.8703
48	Heat Transferred (kJ/h-m)	610.8	610.8	610.7	610.7
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2052	0.2052	0.2052	0.2052
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.878e-003	6.878e-003	6.878e-003	6.878e-003
54	Static Gradient (kPa/m)	-3.847e-003	-3.847e-003	-3.847e-003	-3.847e-003
55	Accel. Gradient (kPa/m)	2.162e-007	2.162e-007	2.162e-007	2.162e-007
56	Liq. Reynolds	2.120e+005 *	2.120e+005 *	2.120e+005 *	2.120e+005 *
57	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
58	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
59	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	2.083e+004	2.083e+004	2.084e+004	2.084e+004
63	Elevation (m)	-0.5400	-0.5460	-0.5520	-0.5580
64	Pressure (bar)	89.75	89.75	89.75	89.75
65	Temperature (C)	52.92	52.92	52.92	52.91
66	Vapour Fraction	0.8703	0.8703	0.8703	0.8703
67	Heat Transferred (kJ/h-m)	610.6	610.6	610.5	610.5
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 351 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2052	0.2060	0.2060	0.2060
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.878e-003	6.872e-003	6.872e-003	6.872e-003
17	Static Gradient (kPa/m)	-3.848e-003	-2.314e-003	-2.314e-003	-2.314e-003
18	Accel. Gradient (kPa/m)	2.162e-007	3.252e-007	3.252e-007	3.252e-007
19	Liq. Reynolds	2.120e+005 *	2.120e+005 *	2.120e+005 *	2.120e+005 *
20	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
21	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
22	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	2.085e+004	2.085e+004	2.086e+004	2.086e+004
26	Elevation (m)	-0.5640	-0.5700	-0.5580	-0.5460
27	Pressure (bar)	89.75	89.75	89.75	89.75
28	Temperature (C)	52.91	52.91	52.91	52.91
29	Vapour Fraction	0.8703	0.8703	0.8703	0.8703
30	Heat Transferred (kJ/h-m)	610.4	610.4	610.3	610.3
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2060	0.2060	0.2870	0.2870
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	6.872e-003	6.872e-003	5.052e-003	5.052e-003
36	Static Gradient (kPa/m)	-2.314e-003	-2.314e-003	5.780e-003	5.780e-003
37	Accel. Gradient (kPa/m)	3.252e-007	3.251e-007	7.728e-007	7.728e-007
38	Liq. Reynolds	2.120e+005 *	2.120e+005 *	2.120e+005 *	2.120e+005 *
39	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
40	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
41	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	2.087e+004	2.087e+004	2.088e+004	2.088e+004
45	Elevation (m)	-0.5340	-0.5220	-0.5100	-0.5460
46	Pressure (bar)	89.75	89.75	89.75	89.75
47	Temperature (C)	52.91	52.90	52.90	52.90
48	Vapour Fraction	0.8703	0.8703	0.8703	0.8703
49	Heat Transferred (kJ/h-m)	610.2	610.2	610.1	610.1
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2870	0.2870	0.2870	0.2005
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	5.052e-003	5.052e-003	5.052e-003	6.917e-003
55	Static Gradient (kPa/m)	5.780e-003	5.780e-003	5.780e-003	-1.351e-002
56	Accel. Gradient (kPa/m)	7.728e-007	7.728e-007	7.728e-007	-4.705e-007
57	Liq. Reynolds	2.120e+005 *	2.120e+005 *	2.120e+005 *	2.120e+005 *
58	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
59	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
60	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	2.089e+004	2.089e+004	2.090e+004	2.090e+004
64	Elevation (m)	-0.5820	-0.6180	-0.6540	-0.6900
65	Pressure (bar)	89.75	89.75	89.75	89.75
66	Temperature (C)	52.90	52.90	52.90	52.89
67	Vapour Fraction	0.8703	0.8702	0.8702	0.8702
68	Heat Transferred (kJ/h-m)	610.1	610.0	610.0	609.9
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2005	0.2005	0.2005	0.2005
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	6.917e-003	6.917e-003	6.917e-003	6.917e-003
18	Static Gradient (kPa/m)	-1.351e-002	-1.351e-002	-1.351e-002	-1.351e-002
19	Accel. Gradient (kPa/m)	-4.706e-007	-4.706e-007	-4.706e-007	-4.706e-007
20	Liq. Reynolds	2.120e+005 *	2.120e+005 *	2.120e+005 *	2.120e+005 *
21	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
22	Liquid Velocity (m/s)	0.2043	0.2043	0.2044	0.2044
23	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	2.091e+004	2.091e+004	2.092e+004	2.092e+004
27	Elevation (m)	-0.6740	-0.6580	-0.6420	-0.6260
28	Pressure (bar)	89.75	89.75	89.75	89.75
29	Temperature (C)	52.89	52.89	52.89	52.89
30	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
31	Heat Transferred (kJ/h-m)	609.9	609.8	609.8	609.7
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2896	0.2896	0.2896	0.2896
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	4.866e-003	4.866e-003	4.866e-003	4.866e-003
37	Static Gradient (kPa/m)	7.756e-003	7.756e-003	7.756e-003	7.756e-003
38	Accel. Gradient (kPa/m)	9.004e-007	9.004e-007	9.004e-007	9.004e-007
39	Liq. Reynolds	2.120e+005 *	2.120e+005 *	2.120e+005 *	2.120e+005 *
40	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
41	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
42	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	2.093e+004	2.093e+004	2.094e+004	2.094e+004
46	Elevation (m)	-0.6100	-0.5960	-0.5820	-0.5680
47	Pressure (bar)	89.75	89.75	89.75	89.75
48	Temperature (C)	52.88	52.88	52.88	52.88
49	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
50	Heat Transferred (kJ/h-m)	609.7	609.6	609.6	609.5
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2896	0.2883	0.2883	0.2883
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	4.866e-003	4.958e-003	4.958e-003	4.958e-003
56	Static Gradient (kPa/m)	7.756e-003	6.765e-003	6.765e-003	6.765e-003
57	Accel. Gradient (kPa/m)	9.004e-007	8.363e-007	8.363e-007	8.363e-007
58	Liq. Reynolds	2.120e+005 *	2.120e+005 *	2.120e+005 *	2.120e+005 *
59	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
60	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
61	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	2.095e+004	2.095e+004	2.096e+004	2.096e+004
65	Elevation (m)	-0.5540	-0.5400	-0.5500	-0.5600
66	Pressure (bar)	89.75	89.75	89.75	89.75
67	Temperature (C)	52.88	52.87	52.87	52.87
68	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	609.5	609.4	609.4	609.3
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.2883	0.2883	0.2053	0.2053
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	4.958e-003	4.958e-003	6.877e-003	6.877e-003
19	Static Gradient (kPa/m)	6.765e-003	6.765e-003	-3.848e-003	-3.848e-003
20	Accel. Gradient (kPa/m)	8.363e-007	8.363e-007	2.161e-007	2.161e-007
21	Liq. Reynolds	2.120e+005 *	2.119e+005 *	2.119e+005 *	2.119e+005 *
22	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
23	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
24	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	2.097e+004	2.097e+004	2.098e+004	2.098e+004
28	Elevation (m)	-0.5700	-0.5800	-0.5900	-0.6160
29	Pressure (bar)	89.75	89.74	89.74	89.74
30	Temperature (C)	52.87	52.87	52.87	52.86
31	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
32	Heat Transferred (kJ/h-m)	609.3	609.2	609.2	609.1
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.2053	0.2053	0.2053	0.2023
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	6.877e-003	6.877e-003	6.877e-003	6.901e-003
38	Static Gradient (kPa/m)	-3.848e-003	-3.848e-003	-3.848e-003	-9.913e-003
39	Accel. Gradient (kPa/m)	2.161e-007	2.161e-007	2.160e-007	-2.149e-007
40	Liq. Reynolds	2.119e+005 *	2.119e+005 *	2.119e+005 *	2.119e+005 *
41	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
42	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
43	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	2.099e+004	2.099e+004	2.100e+004	2.100e+004
47	Elevation (m)	-0.6420	-0.6680	-0.6940	-0.7200
48	Pressure (bar)	89.75	89.75	89.75	89.75
49	Temperature (C)	52.86	52.86	52.86	52.86
50	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
51	Heat Transferred (kJ/h-m)	609.1	609.1	609.0	609.0
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.2023	0.2023	0.2023	0.2023
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	6.901e-003	6.901e-003	6.901e-003	6.901e-003
57	Static Gradient (kPa/m)	-9.913e-003	-9.913e-003	-9.913e-003	-9.914e-003
58	Accel. Gradient (kPa/m)	-2.149e-007	-2.149e-007	-2.149e-007	-2.149e-007
59	Liq. Reynolds	2.119e+005 *	2.119e+005 *	2.119e+005 *	2.119e+005 *
60	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
61	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
62	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	2.101e+004	2.101e+004	2.102e+004	2.102e+004
66	Elevation (m)	-0.7160	-0.7120	-0.7080	-0.7040
67	Pressure (bar)	89.75	89.74	89.74	89.74
68	Temperature (C)	52.86	52.85	52.85	52.85
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
14	Heat Transferred (kJ/h-m)	608.9	608.8	608.8	608.8
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2820	0.2820	0.2820	0.2820
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	5.436e-003	5.436e-003	5.436e-003	5.436e-003
20	Static Gradient (kPa/m)	1.903e-003	1.903e-003	1.903e-003	1.903e-003
21	Accel. Gradient (kPa/m)	5.235e-007	5.235e-007	5.235e-007	5.235e-007
22	Liq. Reynolds	2.119e+005 *	2.119e+005 *	2.119e+005 *	2.119e+005 *
23	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
24	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
25	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	2.103e+004	2.103e+004	2.104e+004	2.104e+004
29	Elevation (m)	-0.7000	-0.7200	-0.7400	-0.7600
30	Pressure (bar)	89.74	89.74	89.74	89.74
31	Temperature (C)	52.85	52.85	52.85	52.84
32	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
33	Heat Transferred (kJ/h-m)	608.7	608.7	608.6	608.6
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2820	0.2034	0.2034	0.2034
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	5.436e-003	6.891e-003	6.891e-003	6.891e-003
39	Static Gradient (kPa/m)	1.903e-003	-7.653e-003	-7.653e-003	-7.653e-003
40	Accel. Gradient (kPa/m)	5.235e-007	-5.430e-008	-5.431e-008	-5.431e-008
41	Liq. Reynolds	2.119e+005 *	2.119e+005 *	2.119e+005 *	2.119e+005 *
42	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
43	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
44	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	2.105e+004	2.105e+004	2.106e+004	2.106e+004
48	Elevation (m)	-0.7800	-0.8000	-0.7660	-0.7320
49	Pressure (bar)	89.74	89.74	89.74	89.74
50	Temperature (C)	52.84	52.84	52.84	52.84
51	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
52	Heat Transferred (kJ/h-m)	608.5	608.5	608.4	608.4
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2034	0.2034	0.3011	0.3011
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	6.891e-003	6.891e-003	5.832e-003	5.832e-003
58	Static Gradient (kPa/m)	-7.653e-003	-7.653e-003	1.695e-002	1.695e-002
59	Accel. Gradient (kPa/m)	-5.432e-008	-5.433e-008	1.625e-006	1.625e-006
60	Liq. Reynolds	2.119e+005 *	2.119e+005 *	2.119e+005 *	2.119e+005 *
61	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
62	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
63	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	2.107e+004	2.107e+004	2.108e+004	2.108e+004
67	Elevation (m)	-0.6980	-0.6640	-0.6300	-0.6580
68	Pressure (bar)	89.74	89.74	89.74	89.74
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Temperature (C)	52.83	52.83	52.83	52.83		
14	Vapour Fraction	0.8702	0.8702	0.8702	0.8702		
15	Heat Transferred (kJ/h-m)	608.3	608.3	608.2	608.2		
16	Flow Regime	Transition	Transition	Transition	Transition		
17	Water Flow Regime						
18	Liquid HoldUp	0.3011	0.3011	0.3011	0.2020		
19	Water HoldUp	---	---	---	---		
20	Friction Gradient (kPa/m)	5.832e-003	5.832e-003	5.832e-003	6.903e-003		
21	Static Gradient (kPa/m)	1.695e-002	1.695e-002	1.695e-002	-1.050e-002		
22	Accel. Gradient (kPa/m)	1.625e-006	1.625e-006	1.625e-006	-2.568e-007		
23	Liq. Reynolds	2.119e+005 *	2.119e+005 *	2.119e+005 *	2.119e+005 *		
24	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *		
25	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
26	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
27	Deposit Thickness (mm)	---	---	---	---		
28	Deposit Volume (m3)	---	---	---	---		
29	Length (m)	2.109e+004	2.109e+004	2.110e+004	2.110e+004		
30	Elevation (m)	-0.6860	-0.7140	-0.7420	-0.7700		
31	Pressure (bar)	89.74	89.74	89.74	89.74		
32	Temperature (C)	52.83	52.82	52.82	52.82		
33	Vapour Fraction	0.8702	0.8702	0.8702	0.8702		
34	Heat Transferred (kJ/h-m)	608.1	608.1	608.0	608.0		
35	Flow Regime	Transition	Transition	Transition	Transition		
36	Water Flow Regime						
37	Liquid HoldUp	0.2020	0.2020	0.2020	0.2020		
38	Water HoldUp	---	---	---	---		
39	Friction Gradient (kPa/m)	6.903e-003	6.903e-003	6.903e-003	6.902e-003		
40	Static Gradient (kPa/m)	-1.050e-002	-1.050e-002	-1.050e-002	-1.050e-002		
41	Accel. Gradient (kPa/m)	-2.568e-007	-2.568e-007	-2.568e-007	-2.568e-007		
42	Liq. Reynolds	2.119e+005 *	2.119e+005 *	2.119e+005 *	2.119e+005 *		
43	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *		
44	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
45	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
46	Deposit Thickness (mm)	---	---	---	---		
47	Deposit Volume (m3)	---	---	---	---		
48	Length (m)	2.111e+004	2.111e+004	2.111e+004	2.112e+004		
49	Elevation (m)	-0.8620	-0.9540	-1.046	-1.138		
50	Pressure (bar)	89.74	89.74	89.74	89.74		
51	Temperature (C)	52.82	52.82	52.82	52.82		
52	Vapour Fraction	0.8702	0.8702	0.8702	0.8702		
53	Heat Transferred (kJ/h-m)	608.0	608.0	607.9	607.9		
54	Flow Regime	Transition	Transition	Transition	Transition		
55	Water Flow Regime						
56	Liquid HoldUp	0.1858	0.1858	0.1858	0.1858		
57	Water HoldUp	---	---	---	---		
58	Friction Gradient (kPa/m)	7.056e-003	7.056e-003	7.055e-003	7.055e-003		
59	Static Gradient (kPa/m)	-4.160e-002	-4.160e-002	-4.160e-002	-4.160e-002		
60	Accel. Gradient (kPa/m)	-2.464e-006	-2.463e-006	-2.463e-006	-2.463e-006		
61	Liq. Reynolds	2.119e+005 *	2.119e+005 *	2.119e+005 *	2.119e+005 *		
62	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *		
63	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
64	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
65	Deposit Thickness (mm)	---	---	---	---		
66	Deposit Volume (m3)	---	---	---	---		
67	Length (m)	2.112e+004	2.113e+004	2.113e+004	2.113e+004		
68	Elevation (m)	-1.230	-1.170	-1.110	-1.050		
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Pressure (bar)	89.75	89.74	89.74	89.74
14	Temperature (C)	52.82	52.82	52.81	52.81
15	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
16	Heat Transferred (kJ/h-m)	607.9	607.8	607.8	607.7
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.1858	0.3272	0.3272	0.3272
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	7.055e-003	6.072e-003	6.072e-003	6.072e-003
22	Static Gradient (kPa/m)	-4.160e-002	3.971e-002	3.971e-002	3.971e-002
23	Accel. Gradient (kPa/m)	-2.463e-006	3.265e-006	3.265e-006	3.265e-006
24	Liq. Reynolds	2.119e+005 *	2.119e+005 *	2.119e+005 *	2.119e+005 *
25	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
26	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
27	Vapour Velocity (m/s)	2.170	2.170	2.171	2.171
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	2.114e+004	2.114e+004	2.114e+004	2.115e+004
31	Elevation (m)	-0.9900	-0.9300	-0.9020	-0.8740
32	Pressure (bar)	89.74	89.74	89.74	89.74
33	Temperature (C)	52.81	52.81	52.81	52.81
34	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
35	Heat Transferred (kJ/h-m)	607.7	607.6	607.6	607.6
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.3272	0.3272	0.3240	0.3240
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	6.072e-003	6.072e-003	6.069e-003	6.069e-003
41	Static Gradient (kPa/m)	3.971e-002	3.971e-002	3.679e-002	3.679e-002
42	Accel. Gradient (kPa/m)	3.265e-006	3.265e-006	3.057e-006	3.057e-006
43	Liq. Reynolds	2.119e+005 *	2.119e+005 *	2.119e+005 *	2.119e+005 *
44	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
45	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
46	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	2.115e+004	2.115e+004	2.115e+004	2.116e+004
50	Elevation (m)	-0.8460	-0.8180	-0.7900	-0.7460
51	Pressure (bar)	89.73	89.73	89.73	89.73
52	Temperature (C)	52.80	52.80	52.80	52.80
53	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
54	Heat Transferred (kJ/h-m)	607.6	607.5	607.5	607.4
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.3240	0.3240	0.3240	0.3075
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	6.069e-003	6.069e-003	6.069e-003	5.981e-003
60	Static Gradient (kPa/m)	3.679e-002	3.679e-002	3.679e-002	2.226e-002
61	Accel. Gradient (kPa/m)	3.057e-006	3.057e-006	3.057e-006	2.015e-006
62	Liq. Reynolds	2.119e+005 *	2.119e+005 *	2.119e+005 *	2.119e+005 *
63	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
64	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
65	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	2.116e+004	2.117e+004	2.117e+004	2.118e+004
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	-0.7020	-0.6580	-0.6140	-0.5700
14	Pressure (bar)	89.73	89.73	89.73	89.73
15	Temperature (C)	52.80	52.80	52.79	52.79
16	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
17	Heat Transferred (kJ/h-m)	607.4	607.3	607.3	607.2
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.3075	0.3075	0.3075	0.3075
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	5.981e-003	5.981e-003	5.981e-003	5.981e-003
23	Static Gradient (kPa/m)	2.226e-002	2.226e-002	2.226e-002	2.226e-002
24	Accel. Gradient (kPa/m)	2.015e-006	2.015e-006	2.015e-006	2.015e-006
25	Liq. Reynolds	2.119e+005 *	2.119e+005 *	2.119e+005 *	2.119e+005 *
26	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
27	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
28	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	2.118e+004	2.119e+004	2.119e+004	2.120e+004
32	Elevation (m)	-0.5540	-0.5380	-0.5220	-0.5060
33	Pressure (bar)	89.72	89.72	89.72	89.72
34	Temperature (C)	52.79	52.79	52.79	52.78
35	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
36	Heat Transferred (kJ/h-m)	607.1	607.1	607.0	607.0
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2896	0.2896	0.2896	0.2896
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	4.865e-003	4.865e-003	4.865e-003	4.865e-003
42	Static Gradient (kPa/m)	7.758e-003	7.758e-003	7.758e-003	7.758e-003
43	Accel. Gradient (kPa/m)	9.004e-007	9.004e-007	9.005e-007	9.005e-007
44	Liq. Reynolds	2.119e+005 *	2.119e+005 *	2.119e+005 *	2.119e+005 *
45	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
46	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
47	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	2.120e+004	2.121e+004	2.121e+004	2.122e+004
51	Elevation (m)	-0.4900	-0.4920	-0.4940	-0.4960
52	Pressure (bar)	89.72	89.72	89.72	89.72
53	Temperature (C)	52.78	52.78	52.78	52.78
54	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
55	Heat Transferred (kJ/h-m)	606.9	606.9	606.9	606.8
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2896	0.2068	0.2068	0.2068
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	4.865e-003	6.864e-003	6.864e-003	6.864e-003
61	Static Gradient (kPa/m)	7.758e-003	-7.733e-004	-7.733e-004	-7.733e-004
62	Accel. Gradient (kPa/m)	9.005e-007	4.345e-007	4.345e-007	4.345e-007
63	Liq. Reynolds	2.119e+005 *	2.119e+005 *	2.119e+005 *	2.119e+005 *
64	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *
65	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
66	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length (m)	2.122e+004	2.123e+004	2.123e+004	2.124e+004		
14	Elevation (m)	-0.4980	-0.5000	-0.4820	-0.4640		
15	Pressure (bar)	89.72	89.72	89.72	89.72		
16	Temperature (C)	52.78	52.77	52.77	52.77		
17	Vapour Fraction	0.8702	0.8702	0.8702	0.8702		
18	Heat Transferred (kJ/h-m)	606.8	606.7	606.7	606.6		
19	Flow Regime	Transition	Transition	Transition	Transition		
20	Water Flow Regime						
21	Liquid HoldUp	0.2068	0.2068	0.2909	0.2909		
22	Water HoldUp	---	---	---	---		
23	Friction Gradient (kPa/m)	6.864e-003	6.864e-003	4.774e-003	4.774e-003		
24	Static Gradient (kPa/m)	-7.733e-004	-7.733e-004	8.755e-003	8.755e-003		
25	Accel. Gradient (kPa/m)	4.345e-007	4.345e-007	9.651e-007	9.651e-007		
26	Liq. Reynolds	2.119e+005 *	2.119e+005 *	2.119e+005 *	2.119e+005 *		
27	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.142e+006 *	9.142e+006 *		
28	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
29	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
30	Deposit Thickness (mm)	---	---	---	---		
31	Deposit Volume (m3)	---	---	---	---		
32	Length (m)	2.124e+004	2.125e+004	2.125e+004	2.126e+004		
33	Elevation (m)	-0.4460	-0.4280	-0.4100	-0.4300		
34	Pressure (bar)	89.72	89.72	89.72	89.72		
35	Temperature (C)	52.77	52.77	52.76	52.76		
36	Vapour Fraction	0.8702	0.8702	0.8702	0.8702		
37	Heat Transferred (kJ/h-m)	606.5	606.5	606.4	606.4		
38	Flow Regime	Transition	Transition	Transition	Transition		
39	Water Flow Regime						
40	Liquid HoldUp	0.2909	0.2909	0.2909	0.2034		
41	Water HoldUp	---	---	---	---		
42	Friction Gradient (kPa/m)	4.774e-003	4.774e-003	4.774e-003	6.891e-003		
43	Static Gradient (kPa/m)	8.755e-003	8.755e-003	8.755e-003	-7.653e-003		
44	Accel. Gradient (kPa/m)	9.651e-007	9.651e-007	9.651e-007	-5.442e-008		
45	Liq. Reynolds	2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *		
46	Vap. Reynolds	9.142e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *		
47	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
48	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
49	Deposit Thickness (mm)	---	---	---	---		
50	Deposit Volume (m3)	---	---	---	---		
51	Length (m)	2.126e+004	2.127e+004	2.127e+004	2.128e+004		
52	Elevation (m)	-0.4500	-0.4700	-0.4900	-0.5100		
53	Pressure (bar)	89.72	89.72	89.72	89.72		
54	Temperature (C)	52.76	52.76	52.76	52.76		
55	Vapour Fraction	0.8702	0.8702	0.8702	0.8702		
56	Heat Transferred (kJ/h-m)	606.4	606.3	606.3	606.2		
57	Flow Regime	Transition	Transition	Transition	Transition		
58	Water Flow Regime						
59	Liquid HoldUp	0.2034	0.2034	0.2034	0.2034		
60	Water HoldUp	---	---	---	---		
61	Friction Gradient (kPa/m)	6.891e-003	6.890e-003	6.890e-003	6.890e-003		
62	Static Gradient (kPa/m)	-7.654e-003	-7.654e-003	-7.654e-003	-7.654e-003		
63	Accel. Gradient (kPa/m)	-5.443e-008	-5.444e-008	-5.445e-008	-5.445e-008		
64	Liq. Reynolds	2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *		
65	Vap. Reynolds	9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *		
66	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
67	Vapour Velocity (m/s)	2.171	2.171	2.171	2.170		
68	Deposit Thickness (mm)	---	---	---	---		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---	---	---
14	Length (m)	2.128e+004	2.129e+004	2.129e+004	2.130e+004	2.130e+004	2.130e+004
15	Elevation (m)	-0.5300	-0.5500	-0.5700	-0.5900	-0.5900	-0.5900
16	Pressure (bar)	89.72	89.72	89.72	89.72	89.72	89.72
17	Temperature (C)	52.75	52.75	52.75	52.75	52.75	52.75
18	Vapour Fraction	0.8702	0.8702	0.8702	0.8702	0.8702	0.8702
19	Heat Transferred (kJ/h-m)	606.2	606.1	606.1	606.1	606.1	606.1
20	Flow Regime	Transition	Transition	Transition	Transition	Transition	Transition
21	Water Flow Regime						
22	Liquid HoldUp	0.2034	0.2034	0.2034	0.2034	0.2034	0.2034
23	Water HoldUp	---	---	---	---	---	---
24	Friction Gradient (kPa/m)	6.891e-003	6.890e-003	6.890e-003	6.890e-003	6.890e-003	6.890e-003
25	Static Gradient (kPa/m)	-7.712e-003	-7.712e-003	-7.712e-003	-7.712e-003	-7.712e-003	-7.712e-003
26	Accel. Gradient (kPa/m)	-5.858e-008	-5.859e-008	-5.859e-008	-5.859e-008	-5.860e-008	-5.860e-008
27	Liq. Reynolds	2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *
28	Vap. Reynolds	9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *
29	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044	0.2044	0.2044
30	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170	2.170	2.170
31	Deposit Thickness (mm)	---	---	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---	---	---
33	Length (m)	2.130e+004	2.131e+004	2.131e+004	2.132e+004	2.132e+004	2.132e+004
34	Elevation (m)	-0.6100	-0.6180	-0.6260	-0.6340	-0.6340	-0.6340
35	Pressure (bar)	89.72	89.72	89.72	89.72	89.72	89.72
36	Temperature (C)	52.75	52.75	52.74	52.74	52.74	52.74
37	Vapour Fraction	0.8702	0.8702	0.8702	0.8702	0.8702	0.8702
38	Heat Transferred (kJ/h-m)	606.0	606.0	605.9	605.9	605.9	605.9
39	Flow Regime	Transition	Transition	Transition	Transition	Transition	Transition
40	Water Flow Regime						
41	Liquid HoldUp	0.2034	0.2057	0.2057	0.2057	0.2057	0.2057
42	Water HoldUp	---	---	---	---	---	---
43	Friction Gradient (kPa/m)	6.890e-003	6.872e-003	6.872e-003	6.872e-003	6.872e-003	6.872e-003
44	Static Gradient (kPa/m)	-7.712e-003	-3.083e-003	-3.083e-003	-3.083e-003	-3.083e-003	-3.083e-003
45	Accel. Gradient (kPa/m)	-5.861e-008	2.703e-007	2.703e-007	2.703e-007	2.703e-007	2.703e-007
46	Liq. Reynolds	2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *
47	Vap. Reynolds	9.142e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *
48	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044	0.2044	0.2044
49	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170	2.170	2.170
50	Deposit Thickness (mm)	---	---	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---	---	---
52	Length (m)	2.132e+004	2.133e+004	2.133e+004	2.133e+004	2.133e+004	2.133e+004
53	Elevation (m)	-0.6420	-0.6500	-0.6520	-0.6540	-0.6540	-0.6540
54	Pressure (bar)	89.72	89.72	89.72	89.72	89.72	89.72
55	Temperature (C)	52.74	52.74	52.74	52.74	52.74	52.74
56	Vapour Fraction	0.8702	0.8702	0.8702	0.8702	0.8702	0.8702
57	Heat Transferred (kJ/h-m)	605.8	605.8	605.8	605.8	605.7	605.7
58	Flow Regime	Transition	Transition	Transition	Transition	Transition	Transition
59	Water Flow Regime						
60	Liquid HoldUp	0.2057	0.2057	0.2062	0.2062	0.2062	0.2062
61	Water HoldUp	---	---	---	---	---	---
62	Friction Gradient (kPa/m)	6.872e-003	6.872e-003	6.867e-003	6.867e-003	6.867e-003	6.867e-003
63	Static Gradient (kPa/m)	-3.083e-003	-3.083e-003	-1.930e-003	-1.930e-003	-1.930e-003	-1.930e-003
64	Accel. Gradient (kPa/m)	2.703e-007	2.703e-007	3.522e-007	3.522e-007	3.522e-007	3.522e-007
65	Liq. Reynolds	2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *
66	Vap. Reynolds	9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *
67	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044	0.2044	0.2044
68	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170	2.170	2.170
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 360 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:02:44 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Deposit Thickness	(mm)	---	---	---	---				
14	Deposit Volume	(m3)	---	---	---	---				
15	Length	(m)	2.133e+004	2.133e+004	2.134e+004	2.134e+004				
16	Elevation	(m)	-0.6560	-0.6580	-0.6600	-0.6300				
17	Pressure	(bar)	89.72	89.72	89.72	89.72				
18	Temperature	(C)	52.74	52.74	52.73	52.73				
19	Vapour Fraction		0.8702	0.8702	0.8702	0.8702				
20	Heat Transferred	(kJ/h-m)	605.7	605.7	605.7	605.6				
21	Flow Regime		Transition	Transition	Transition	Transition				
22	Water Flow Regime									
23	Liquid HoldUp		0.2062	0.2062	0.2062	0.3113				
24	Water HoldUp		---	---	---	---				
25	Friction Gradient	(kPa/m)	6.867e-003	6.867e-003	6.867e-003	6.020e-003				
26	Static Gradient	(kPa/m)	-1.930e-003	-1.930e-003	-1.930e-003	2.553e-002				
27	Accel. Gradient	(kPa/m)	3.522e-007	3.522e-007	3.522e-007	2.250e-006				
28	Liq. Reynolds		2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *				
29	Vap. Reynolds		9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *				
30	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044				
31	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170				
32	Deposit Thickness	(mm)	---	---	---	---				
33	Deposit Volume	(m3)	---	---	---	---				
34	Length	(m)	2.134e+004	2.134e+004	2.135e+004	2.135e+004				
35	Elevation	(m)	-0.6000	-0.5700	-0.5400	-0.5100				
36	Pressure	(bar)	89.71	89.71	89.71	89.71				
37	Temperature	(C)	52.73	52.73	52.73	52.73				
38	Vapour Fraction		0.8702	0.8702	0.8702	0.8702				
39	Heat Transferred	(kJ/h-m)	605.6	605.6	605.5	605.5				
40	Flow Regime		Transition	Transition	Transition	Transition				
41	Water Flow Regime									
42	Liquid HoldUp		0.3113	0.3113	0.3113	0.3113				
43	Water HoldUp		---	---	---	---				
44	Friction Gradient	(kPa/m)	6.020e-003	6.020e-003	6.020e-003	6.020e-003				
45	Static Gradient	(kPa/m)	2.553e-002	2.553e-002	2.553e-002	2.553e-002				
46	Accel. Gradient	(kPa/m)	2.251e-006	2.251e-006	2.251e-006	2.251e-006				
47	Liq. Reynolds		2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *				
48	Vap. Reynolds		9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *				
49	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044				
50	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170				
51	Deposit Thickness	(mm)	---	---	---	---				
52	Deposit Volume	(m3)	---	---	---	---				
53	Length	(m)	2.136e+004	2.136e+004	2.137e+004	2.137e+004				
54	Elevation	(m)	-0.5060	-0.5020	-0.4980	-0.4940				
55	Pressure	(bar)	89.71	89.71	89.71	89.71				
56	Temperature	(C)	52.73	52.72	52.72	52.72				
57	Vapour Fraction		0.8702	0.8702	0.8702	0.8702				
58	Heat Transferred	(kJ/h-m)	605.4	605.4	605.3	605.3				
59	Flow Regime		Transition	Transition	Transition	Transition				
60	Water Flow Regime									
61	Liquid HoldUp		0.2820	0.2820	0.2820	0.2820				
62	Water HoldUp		---	---	---	---				
63	Friction Gradient	(kPa/m)	5.435e-003	5.435e-003	5.435e-003	5.435e-003				
64	Static Gradient	(kPa/m)	1.903e-003	1.903e-003	1.903e-003	1.903e-003				
65	Accel. Gradient	(kPa/m)	5.235e-007	5.235e-007	5.235e-007	5.234e-007				
66	Liq. Reynolds		2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *				
67	Vap. Reynolds		9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *				
68	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	2.138e+004	2.138e+004	2.139e+004	2.139e+004	
17	Elevation	(m)	-0.4900	-0.4880	-0.4860	-0.4840	
18	Pressure	(bar)	89.71	89.71	89.71	89.71	
19	Temperature	(C)	52.72	52.72	52.72	52.71	
20	Vapour Fraction		0.8702	0.8702	0.8702	0.8701	
21	Heat Transferred	(kJ/h-m)	605.2	605.2	605.2	605.1	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2820	0.2807	0.2807	0.2807	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	5.434e-003	5.534e-003	5.534e-003	5.534e-003	
27	Static Gradient	(kPa/m)	1.903e-003	9.487e-004	9.487e-004	9.487e-004	
28	Accel. Gradient	(kPa/m)	5.234e-007	4.624e-007	4.624e-007	4.624e-007	
29	Liq. Reynolds		2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *	
30	Vap. Reynolds		9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *	
31	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
32	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	2.140e+004	2.140e+004	2.141e+004	2.141e+004	
36	Elevation	(m)	-0.4820	-0.4800	-0.4320	-0.3840	
37	Pressure	(bar)	89.71	89.71	89.71	89.71	
38	Temperature	(C)	52.71	52.71	52.71	52.71	
39	Vapour Fraction		0.8701	0.8701	0.8701	0.8701	
40	Heat Transferred	(kJ/h-m)	605.1	605.0	605.0	604.9	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2807	0.2807	0.3100	0.3100	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	5.534e-003	5.534e-003	6.009e-003	6.009e-003	
46	Static Gradient	(kPa/m)	9.487e-004	9.487e-004	2.444e-002	2.444e-002	
47	Accel. Gradient	(kPa/m)	4.624e-007	4.624e-007	2.172e-006	2.172e-006	
48	Liq. Reynolds		2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *	
49	Vap. Reynolds		9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *	
50	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
51	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	2.142e+004	2.142e+004	2.143e+004	2.143e+004	
55	Elevation	(m)	-0.3360	-0.2880	-0.2400	-0.2480	
56	Pressure	(bar)	89.70	89.70	89.70	89.70	
57	Temperature	(C)	52.70	52.70	52.70	52.70	
58	Vapour Fraction		0.8701	0.8701	0.8701	0.8701	
59	Heat Transferred	(kJ/h-m)	604.8	604.8	604.7	604.7	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.3100	0.3100	0.3100	0.2057	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	6.009e-003	6.009e-003	6.009e-003	6.872e-003	
65	Static Gradient	(kPa/m)	2.444e-002	2.444e-002	2.444e-002	-3.083e-003	
66	Accel. Gradient	(kPa/m)	2.172e-006	2.172e-006	2.172e-006	2.703e-007	
67	Liq. Reynolds		2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *	
68	Vap. Reynolds		9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *	
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
14	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170
15	Deposit Thickness	(mm)	---	---	---	---
16	Deposit Volume	(m3)	---	---	---	---
17	Length	(m)	2.144e+004	2.144e+004	2.145e+004	2.145e+004
18	Elevation	(m)	-0.2560	-0.2640	-0.2720	-0.2800
19	Pressure	(bar)	89.70	89.70	89.70	89.70
20	Temperature	(C)	52.70	52.69	52.69	52.69
21	Vapour Fraction		0.8701	0.8701	0.8701	0.8701
22	Heat Transferred	(kJ/h-m)	604.6	604.6	604.5	604.5
23	Flow Regime		Transition	Transition	Transition	Transition
24	Water Flow Regime					
25	Liquid HoldUp		0.2057	0.2057	0.2057	0.2057
26	Water HoldUp		---	---	---	---
27	Friction Gradient	(kPa/m)	6.872e-003	6.872e-003	6.872e-003	6.872e-003
28	Static Gradient	(kPa/m)	-3.083e-003	-3.083e-003	-3.083e-003	-3.083e-003
29	Accel. Gradient	(kPa/m)	2.703e-007	2.703e-007	2.703e-007	2.703e-007
30	Liq. Reynolds		2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *
31	Vap. Reynolds		9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *
32	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
33	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170
34	Deposit Thickness	(mm)	---	---	---	---
35	Deposit Volume	(m3)	---	---	---	---
36	Length	(m)	2.146e+004	2.146e+004	2.147e+004	2.147e+004
37	Elevation	(m)	-0.2760	-0.2720	-0.2680	-0.2640
38	Pressure	(bar)	89.70	89.70	89.70	89.70
39	Temperature	(C)	52.69	52.69	52.68	52.68
40	Vapour Fraction		0.8701	0.8701	0.8701	0.8701
41	Heat Transferred	(kJ/h-m)	604.4	604.4	604.3	604.3
42	Flow Regime		Transition	Transition	Transition	Transition
43	Water Flow Regime					
44	Liquid HoldUp		0.2820	0.2820	0.2820	0.2820
45	Water HoldUp		---	---	---	---
46	Friction Gradient	(kPa/m)	5.434e-003	5.434e-003	5.434e-003	5.434e-003
47	Static Gradient	(kPa/m)	1.904e-003	1.904e-003	1.904e-003	1.904e-003
48	Accel. Gradient	(kPa/m)	5.235e-007	5.235e-007	5.235e-007	5.235e-007
49	Liq. Reynolds		2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *
50	Vap. Reynolds		9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *
51	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
52	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170
53	Deposit Thickness	(mm)	---	---	---	---
54	Deposit Volume	(m3)	---	---	---	---
55	Length	(m)	2.148e+004	2.148e+004	2.149e+004	2.149e+004
56	Elevation	(m)	-0.2600	-0.2620	-0.2640	-0.2660
57	Pressure	(bar)	89.70	89.70	89.70	89.70
58	Temperature	(C)	52.68	52.68	52.68	52.68
59	Vapour Fraction		0.8701	0.8701	0.8701	0.8701
60	Heat Transferred	(kJ/h-m)	604.2	604.2	604.2	604.1
61	Flow Regime		Transition	Transition	Transition	Transition
62	Water Flow Regime					
63	Liquid HoldUp		0.2820	0.2068	0.2068	0.2068
64	Water HoldUp		---	---	---	---
65	Friction Gradient	(kPa/m)	5.434e-003	6.863e-003	6.863e-003	6.863e-003
66	Static Gradient	(kPa/m)	1.904e-003	-7.734e-004	-7.734e-004	-7.734e-004
67	Accel. Gradient	(kPa/m)	5.235e-007	4.344e-007	4.344e-007	4.344e-007
68	Liq. Reynolds		2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *
14	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
15	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	2.150e+004	2.150e+004	2.150e+004	2.151e+004
19	Elevation (m)	-0.2680	-0.2700	-0.2840	-0.2980
20	Pressure (bar)	89.70	89.70	89.70	89.70
21	Temperature (C)	52.67	52.67	52.67	52.67
22	Vapour Fraction	0.8701	0.8701	0.8701	0.8701
23	Heat Transferred (kJ/h-m)	604.1	604.0	604.0	603.9
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2068	0.2068	0.2039	0.2039
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.863e-003	6.862e-003	6.885e-003	6.885e-003
29	Static Gradient (kPa/m)	-7.734e-004	-7.734e-004	-6.708e-003	-6.708e-003
30	Accel. Gradient (kPa/m)	4.344e-007	4.344e-007	1.268e-008	1.267e-008
31	Liq. Reynolds	2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *
32	Vap. Reynolds	9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *
33	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
34	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	2.151e+004	2.152e+004	2.152e+004	2.152e+004
38	Elevation (m)	-0.3120	-0.3260	-0.3400	-0.3660
39	Pressure (bar)	89.70	89.70	89.70	89.70
40	Temperature (C)	52.67	52.67	52.67	52.67
41	Vapour Fraction	0.8701	0.8701	0.8701	0.8701
42	Heat Transferred (kJ/h-m)	603.9	603.9	603.8	603.9
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2039	0.2039	0.2039	0.1825
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.885e-003	6.885e-003	6.885e-003	7.090e-003
48	Static Gradient (kPa/m)	-6.708e-003	-6.708e-003	-6.708e-003	-4.748e-002
49	Accel. Gradient (kPa/m)	1.267e-008	1.266e-008	1.265e-008	-2.881e-006
50	Liq. Reynolds	2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *
51	Vap. Reynolds	9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *
52	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
53	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	2.152e+004	2.152e+004	2.152e+004	2.153e+004
57	Elevation (m)	-0.3920	-0.4180	-0.4440	-0.4700
58	Pressure (bar)	89.70	89.70	89.70	89.70
59	Temperature (C)	52.66	52.66	52.66	52.66
60	Vapour Fraction	0.8701	0.8701	0.8701	0.8701
61	Heat Transferred (kJ/h-m)	603.9	603.8	603.8	603.8
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1825	0.1825	0.1825	0.1825
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	7.090e-003	7.090e-003	7.090e-003	7.089e-003
67	Static Gradient (kPa/m)	-4.748e-002	-4.748e-002	-4.748e-002	-4.748e-002
68	Accel. Gradient (kPa/m)	-2.881e-006	-2.881e-006	-2.881e-006	-2.881e-006
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *
14	Vap. Reynolds	9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *
15	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
16	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	2.153e+004	2.154e+004	2.154e+004	2.155e+004
20	Elevation (m)	-0.4320	-0.3940	-0.3560	-0.3180
21	Pressure (bar)	89.70	89.70	89.69	89.69
22	Temperature (C)	52.66	52.66	52.66	52.66
23	Vapour Fraction	0.8701	0.8701	0.8701	0.8701
24	Heat Transferred (kJ/h-m)	603.7	603.7	603.6	603.6
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.3037	0.3037	0.3037	0.3037
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	5.910e-003	5.910e-003	5.910e-003	5.910e-003
30	Static Gradient (kPa/m)	1.906e-002	1.906e-002	1.906e-002	1.906e-002
31	Accel. Gradient (kPa/m)	1.781e-006	1.781e-006	1.781e-006	1.781e-006
32	Liq. Reynolds	2.118e+005 *	2.118e+005 *	2.117e+005 *	2.117e+005 *
33	Vap. Reynolds	9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *
34	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
35	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	2.155e+004	2.156e+004	2.156e+004	2.157e+004
39	Elevation (m)	-0.2800	-0.2320	-0.1840	-0.1360
40	Pressure (bar)	89.69	89.69	89.69	89.69
41	Temperature (C)	52.65	52.65	52.65	52.65
42	Vapour Fraction	0.8701	0.8701	0.8701	0.8701
43	Heat Transferred (kJ/h-m)	603.5	603.5	603.4	603.4
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.3037	0.3101	0.3101	0.3101
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	5.910e-003	6.008e-003	6.008e-003	6.009e-003
49	Static Gradient (kPa/m)	1.906e-002	2.444e-002	2.444e-002	2.444e-002
50	Accel. Gradient (kPa/m)	1.781e-006	2.172e-006	2.172e-006	2.172e-006
51	Liq. Reynolds	2.117e+005 *	2.117e+005 *	2.117e+005 *	2.117e+005 *
52	Vap. Reynolds	9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *
53	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
54	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	2.157e+004	2.158e+004	2.158e+004	2.159e+004
58	Elevation (m)	-8.800e-002	-4.000e-002	-0.1460	-0.2520
59	Pressure (bar)	89.69	89.68	89.69	89.69
60	Temperature (C)	52.65	52.64	52.64	52.64
61	Vapour Fraction	0.8701	0.8701	0.8701	0.8701
62	Heat Transferred (kJ/h-m)	603.3	603.2	603.2	603.2
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.3101	0.3101	0.1875	0.1875
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	6.009e-003	6.009e-003	7.037e-003	7.037e-003
68	Static Gradient (kPa/m)	2.444e-002	2.444e-002	-3.856e-002	-3.856e-002
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3							
4			Unit Set: Mixed				
5			Date/Time: Mon Jun 13 15:02:44 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Accel. Gradient (kPa/m)	2.172e-006	2.172e-006	-2.249e-006	-2.249e-006		
14	Liq. Reynolds	2.117e+005 *	2.117e+005 *	2.117e+005 *	2.117e+005 *		
15	Vap. Reynolds	9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *		
16	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
17	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170		
18	Deposit Thickness (mm)	---	---	---	---		
19	Deposit Volume (m3)	---	---	---	---		
20	Length (m)	2.159e+004	2.160e+004	2.160e+004	2.161e+004		
21	Elevation (m)	-0.3580	-0.4640	-0.5700	-1.040		
22	Pressure (bar)	89.69	89.69	89.69	89.70		
23	Temperature (C)	52.64	52.64	52.64	52.64		
24	Vapour Fraction	0.8701	0.8701	0.8701	0.8701		
25	Heat Transferred (kJ/h-m)	603.2	603.1	603.1	603.3		
26	Flow Regime	Transition	Transition	Transition	Transition		
27	Water Flow Regime						
28	Liquid HoldUp	0.1875	0.1875	0.1875	0.1208		
29	Water HoldUp	---	---	---	---		
30	Friction Gradient (kPa/m)	7.036e-003	7.036e-003	7.036e-003	8.265e-003		
31	Static Gradient (kPa/m)	-3.856e-002	-3.856e-002	-3.856e-002	-0.1338		
32	Accel. Gradient (kPa/m)	-2.249e-006	-2.249e-006	-2.249e-006	-8.956e-006		
33	Liq. Reynolds	2.117e+005 *	2.117e+005 *	2.117e+005 *	2.117e+005 *		
34	Vap. Reynolds	9.143e+006 *	9.143e+006 *	9.143e+006 *	9.143e+006 *		
35	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
36	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170		
37	Deposit Thickness (mm)	---	---	---	---		
38	Deposit Volume (m3)	---	---	---	---		
39	Length (m)	2.161e+004	2.162e+004	2.162e+004	2.163e+004		
40	Elevation (m)	-1.510	-1.980	-2.450	-2.920		
41	Pressure (bar)	89.70	89.71	89.72	89.72		
42	Temperature (C)	52.64	52.64	52.64	52.64		
43	Vapour Fraction	0.8701	0.8701	0.8701	0.8701		
44	Heat Transferred (kJ/h-m)	603.3	603.3	603.4	603.4		
45	Flow Regime	Transition	Transition	Transition	Transition		
46	Water Flow Regime						
47	Liquid HoldUp	0.1208	0.1208	0.1208	0.1208		
48	Water HoldUp	---	---	---	---		
49	Friction Gradient (kPa/m)	8.264e-003	8.264e-003	8.263e-003	8.263e-003		
50	Static Gradient (kPa/m)	-0.1338	-0.1339	-0.1339	-0.1339		
51	Accel. Gradient (kPa/m)	-8.955e-006	-8.954e-006	-8.954e-006	-8.953e-006		
52	Liq. Reynolds	2.117e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *		
53	Vap. Reynolds	9.143e+006 *	9.143e+006 *	9.142e+006 *	9.142e+006 *		
54	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
55	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169		
56	Deposit Thickness (mm)	---	---	---	---		
57	Deposit Volume (m3)	---	---	---	---		
58	Length (m)	2.163e+004	2.164e+004	2.164e+004	2.165e+004		
59	Elevation (m)	-3.420	-3.920	-4.420	-4.920		
60	Pressure (bar)	89.73	89.74	89.74	89.75		
61	Temperature (C)	52.64	52.65	52.65	52.65		
62	Vapour Fraction	0.8701	0.8701	0.8701	0.8701		
63	Heat Transferred (kJ/h-m)	603.5	603.5	603.5	603.6		
64	Flow Regime	Transition	Transition	Transition	Transition		
65	Water Flow Regime						
66	Liquid HoldUp	0.1155	0.1155	0.1155	0.1155		
67	Water HoldUp	---	---	---	---		
68	Friction Gradient (kPa/m)	8.443e-003	8.443e-003	8.442e-003	8.442e-003		
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Unit Set: Mixed			
3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:02:44 2022			
4						
5						
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	-0.1392	-0.1392	-0.1393	-0.1393
14	Accel. Gradient	(kPa/m)	-9.322e-006	-9.321e-006	-9.320e-006	-9.319e-006
15	Liq. Reynolds		2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *
16	Vap. Reynolds		9.142e+006 *	9.142e+006 *	9.142e+006 *	9.141e+006 *
17	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
18	Vapour Velocity	(m/s)	2.169	2.169	2.168	2.168
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	2.165e+004	2.166e+004	2.166e+004	2.167e+004
22	Elevation	(m)	-5.420	-5.796	-6.172	-6.548
23	Pressure	(bar)	89.76	89.76	89.77	89.77
24	Temperature	(C)	52.65	52.65	52.65	52.65
25	Vapour Fraction		0.8701	0.8701	0.8701	0.8701
26	Heat Transferred	(kJ/h-m)	603.6	603.5	603.5	603.6
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.1155	0.1378	0.1378	0.1378
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	8.441e-003	7.801e-003	7.800e-003	7.800e-003
32	Static Gradient	(kPa/m)	-0.1393	-0.1147	-0.1147	-0.1147
33	Accel. Gradient	(kPa/m)	-9.318e-006	-7.612e-006	-7.612e-006	-7.611e-006
34	Liq. Reynolds		2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *
35	Vap. Reynolds		9.141e+006 *	9.141e+006 *	9.141e+006 *	9.141e+006 *
36	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
37	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	2.167e+004	2.168e+004	2.168e+004	2.169e+004
41	Elevation	(m)	-6.924	-7.300	-7.300	-7.300
42	Pressure	(bar)	89.78	89.78	89.78	89.78
43	Temperature	(C)	52.65	52.65	52.65	52.65
44	Vapour Fraction		0.8701	0.8701	0.8701	0.8701
45	Heat Transferred	(kJ/h-m)	603.6	603.6	603.4	603.4
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.1379	0.1379	0.3027	0.3027
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	7.799e-003	7.799e-003	5.872e-003	5.872e-003
51	Static Gradient	(kPa/m)	-0.1147	-0.1147	0.0000	0.0000
52	Accel. Gradient	(kPa/m)	-7.611e-006	-7.610e-006	4.179e-007	4.179e-007
53	Liq. Reynolds		2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *
54	Vap. Reynolds		9.140e+006 *	9.140e+006 *	9.140e+006 *	9.140e+006 *
55	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
56	Vapour Velocity	(m/s)	2.168	2.167	2.167	2.167
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	2.169e+004	2.170e+004	2.170e+004	2.171e+004
60	Elevation	(m)	-7.300	-7.300	-7.300	-7.200
61	Pressure	(bar)	89.78	89.78	89.78	89.78
62	Temperature	(C)	52.65	52.65	52.64	52.64
63	Vapour Fraction		0.8701	0.8701	0.8701	0.8701
64	Heat Transferred	(kJ/h-m)	603.3	603.3	603.2	603.2
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.3027	0.3027	0.3027	0.3433
68	Water HoldUp		---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	5.872e-003	5.872e-003	5.871e-003	6.060e-003
14	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	5.478e-002
15	Accel. Gradient	(kPa/m)	4.179e-007	4.179e-007	4.179e-007	4.331e-006
16	Liq. Reynolds		2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *
17	Vap. Reynolds		9.140e+006 *	9.140e+006 *	9.140e+006 *	9.140e+006 *
18	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
19	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	2.171e+004	2.172e+004	2.172e+004	2.173e+004
23	Elevation	(m)	-7.100	-7.000	-6.900	-6.800
24	Pressure	(bar)	89.78	89.77	89.77	89.77
25	Temperature	(C)	52.64	52.64	52.63	52.63
26	Vapour Fraction		0.8701	0.8701	0.8701	0.8701
27	Heat Transferred	(kJ/h-m)	603.1	603.0	603.0	602.9
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.3433	0.3433	0.3433	0.3433
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	6.060e-003	6.060e-003	6.060e-003	6.060e-003
33	Static Gradient	(kPa/m)	5.478e-002	5.478e-002	5.478e-002	5.478e-002
34	Accel. Gradient	(kPa/m)	4.331e-006	4.331e-006	4.331e-006	4.332e-006
35	Liq. Reynolds		2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *
36	Vap. Reynolds		9.140e+006 *	9.141e+006 *	9.141e+006 *	9.141e+006 *
37	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
38	Vapour Velocity	(m/s)	2.167	2.167	2.168	2.168
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	2.173e+004	2.174e+004	2.174e+004	2.175e+004
42	Elevation	(m)	-6.700	-6.600	-6.500	-6.400
43	Pressure	(bar)	89.76	89.76	89.76	89.75
44	Temperature	(C)	52.63	52.62	52.62	52.62
45	Vapour Fraction		0.8701	0.8701	0.8701	0.8701
46	Heat Transferred	(kJ/h-m)	602.8	602.8	602.7	602.6
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.3434	0.3434	0.3434	0.3434
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	6.060e-003	6.060e-003	6.060e-003	6.061e-003
52	Static Gradient	(kPa/m)	5.487e-002	5.487e-002	5.487e-002	5.487e-002
53	Accel. Gradient	(kPa/m)	4.339e-006	4.339e-006	4.339e-006	4.340e-006
54	Liq. Reynolds		2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *
55	Vap. Reynolds		9.141e+006 *	9.141e+006 *	9.141e+006 *	9.141e+006 *
56	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
57	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	2.175e+004	2.176e+004	2.176e+004	2.177e+004
61	Elevation	(m)	-6.300	-6.200	-6.100	-6.000
62	Pressure	(bar)	89.75	89.75	89.74	89.74
63	Temperature	(C)	52.62	52.61	52.61	52.61
64	Vapour Fraction		0.8701	0.8701	0.8701	0.8701
65	Heat Transferred	(kJ/h-m)	602.6	602.5	602.4	602.3
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.3434	0.3434	0.3434	0.3434
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	6.061e-003	6.061e-003	6.061e-003	6.061e-003
15	Static Gradient (kPa/m)	5.487e-002	5.487e-002	5.487e-002	5.487e-002
16	Accel. Gradient (kPa/m)	4.340e-006	4.340e-006	4.340e-006	4.340e-006
17	Liq. Reynolds	2.118e+005 *	2.118e+005 *	2.118e+005 *	2.118e+005 *
18	Vap. Reynolds	9.141e+006 *	9.141e+006 *	9.141e+006 *	9.141e+006 *
19	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
20	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	2.177e+004	2.178e+004	2.178e+004	2.179e+004
24	Elevation (m)	-5.900	-5.800	-4.712	-3.624
25	Pressure (bar)	89.74	89.74	89.67	89.60
26	Temperature (C)	52.61	52.60	52.59	52.57
27	Vapour Fraction	0.8701	0.8701	0.8701	0.8701
28	Heat Transferred (kJ/h-m)	602.3	602.2	602.9	602.4
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.3434	0.3434	0.9267	0.9264
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	6.061e-003	6.061e-003	5.624e-003	5.628e-003
34	Static Gradient (kPa/m)	5.487e-002	5.487e-002	1.350	1.349
35	Accel. Gradient (kPa/m)	4.341e-006	4.341e-006	9.670e-005	9.682e-005
36	Liq. Reynolds	2.118e+005 *	2.118e+005 *	2.117e+005 *	2.116e+005 *
37	Vap. Reynolds	9.142e+006 *	9.142e+006 *	9.144e+006 *	9.146e+006 *
38	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
39	Vapour Velocity (m/s)	2.168	2.168	2.170	2.172
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	2.179e+004	2.180e+004	2.180e+004	2.181e+004
43	Elevation (m)	-2.536	-1.448	-0.3600	-0.3240
44	Pressure (bar)	89.53	89.46	89.40	89.40
45	Temperature (C)	52.55	52.53	52.51	52.51
46	Vapour Fraction	0.8701	0.8702	0.8702	0.8702
47	Heat Transferred (kJ/h-m)	601.9	601.4	600.9	599.7
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.9260	0.9256	0.9252	0.3019
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	5.632e-003	5.636e-003	5.640e-003	5.888e-003
53	Static Gradient (kPa/m)	1.349	1.348	1.348	1.797e-002
54	Accel. Gradient (kPa/m)	9.694e-005	9.707e-005	9.719e-005	1.713e-006
55	Liq. Reynolds	2.115e+005 *	2.114e+005 *	2.113e+005 *	2.113e+005 *
56	Vap. Reynolds	9.148e+006 *	9.150e+006 *	9.153e+006 *	9.153e+006 *
57	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
58	Vapour Velocity (m/s)	2.173	2.175	2.177	2.177
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	2.181e+004	2.182e+004	2.182e+004	2.183e+004
62	Elevation (m)	-0.2880	-0.2520	-0.2160	-0.1800
63	Pressure (bar)	89.39	89.39	89.39	89.39
64	Temperature (C)	52.51	52.51	52.50	52.50
65	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
66	Heat Transferred (kJ/h-m)	599.6	599.6	599.5	599.5
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.3018	0.3018	0.3018	0.3018
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	5.888e-003	5.888e-003	5.888e-003	5.888e-003
16	Static Gradient (kPa/m)	1.797e-002	1.797e-002	1.797e-002	1.797e-002
17	Accel. Gradient (kPa/m)	1.713e-006	1.713e-006	1.713e-006	1.713e-006
18	Liq. Reynolds	2.113e+005 *	2.113e+005 *	2.113e+005 *	2.113e+005 *
19	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *
20	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
21	Vapour Velocity (m/s)	2.177	2.177	2.177	2.177
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	2.183e+004	2.184e+004	2.184e+004	2.185e+004
25	Elevation (m)	-0.1780	-0.1760	-0.1740	-0.1720
26	Pressure (bar)	89.39	89.39	89.39	89.39
27	Temperature (C)	52.50	52.50	52.50	52.49
28	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
29	Heat Transferred (kJ/h-m)	599.4	599.4	599.3	599.3
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2802	0.2802	0.2802	0.2802
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	5.557e-003	5.557e-003	5.557e-003	5.557e-003
35	Static Gradient (kPa/m)	9.472e-004	9.472e-004	9.472e-004	9.472e-004
36	Accel. Gradient (kPa/m)	4.670e-007	4.670e-007	4.670e-007	4.670e-007
37	Liq. Reynolds	2.113e+005 *	2.113e+005 *	2.113e+005 *	2.113e+005 *
38	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *
39	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
40	Vapour Velocity (m/s)	2.177	2.177	2.177	2.177
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	2.185e+004	2.186e+004	2.186e+004	2.187e+004
44	Elevation (m)	-0.1700	-0.1640	-0.1580	-0.1520
45	Pressure (bar)	89.39	89.39	89.39	89.39
46	Temperature (C)	52.49	52.49	52.49	52.49
47	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
48	Heat Transferred (kJ/h-m)	599.2	599.2	599.1	599.1
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2802	0.2828	0.2828	0.2828
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	5.557e-003	5.358e-003	5.358e-003	5.358e-003
54	Static Gradient (kPa/m)	9.472e-004	2.860e-003	2.860e-003	2.860e-003
55	Accel. Gradient (kPa/m)	4.670e-007	5.900e-007	5.900e-007	5.900e-007
56	Liq. Reynolds	2.113e+005 *	2.113e+005 *	2.113e+005 *	2.113e+005 *
57	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *
58	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
59	Vapour Velocity (m/s)	2.177	2.177	2.177	2.177
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	2.187e+004	2.188e+004	2.188e+004	2.189e+004
63	Elevation (m)	-0.1460	-0.1400	-0.1300	-0.1200
64	Pressure (bar)	89.39	89.39	89.39	89.39
65	Temperature (C)	52.49	52.48	52.48	52.48
66	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
67	Heat Transferred (kJ/h-m)	599.0	599.0	598.9	598.9
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 370 of 937

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2828	0.2828	0.2853	0.2853
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	5.358e-003	5.358e-003	5.164e-003	5.164e-003
17	Static Gradient (kPa/m)	2.860e-003	2.860e-003	4.796e-003	4.796e-003
18	Accel. Gradient (kPa/m)	5.900e-007	5.900e-007	7.152e-007	7.152e-007
19	Liq. Reynolds	2.113e+005 *	2.113e+005 *	2.113e+005 *	2.113e+005 *
20	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *
21	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
22	Vapour Velocity (m/s)	2.177	2.177	2.177	2.177
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	2.189e+004	2.190e+004	2.190e+004	2.191e+004
26	Elevation (m)	-0.1100	-0.1000	-9.000e-002	-9.000e-002
27	Pressure (bar)	89.39	89.39	89.39	89.38
28	Temperature (C)	52.48	52.48	52.47	52.47
29	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
30	Heat Transferred (kJ/h-m)	598.8	598.8	598.7	598.7
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2853	0.2853	0.2853	0.3019
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	5.164e-003	5.164e-003	5.164e-003	5.889e-003
36	Static Gradient (kPa/m)	4.796e-003	4.797e-003	4.797e-003	0.0000
37	Accel. Gradient (kPa/m)	7.152e-007	7.152e-007	7.152e-007	4.228e-007
38	Liq. Reynolds	2.113e+005 *	2.113e+005 *	2.113e+005 *	2.113e+005 *
39	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *
40	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
41	Vapour Velocity (m/s)	2.177	2.177	2.177	2.177
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	2.191e+004	2.192e+004	2.192e+004	2.193e+004
45	Elevation (m)	-9.000e-002	-9.000e-002	-9.000e-002	-9.000e-002
46	Pressure (bar)	89.38	89.38	89.38	89.38
47	Temperature (C)	52.47	52.47	52.47	52.47
48	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
49	Heat Transferred (kJ/h-m)	598.7	598.6	598.6	598.5
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.3019	0.3019	0.3019	0.3019
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	5.889e-003	5.889e-003	5.889e-003	5.889e-003
55	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
56	Accel. Gradient (kPa/m)	4.228e-007	4.228e-007	4.228e-007	4.228e-007
57	Liq. Reynolds	2.113e+005 *	2.113e+005 *	2.113e+005 *	2.113e+005 *
58	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *
59	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
60	Vapour Velocity (m/s)	2.177	2.177	2.177	2.177
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	2.193e+004	2.194e+004	2.194e+004	2.195e+004
64	Elevation (m)	-9.000e-002	-9.000e-002	-9.000e-002	-9.000e-002
65	Pressure (bar)	89.38	89.38	89.38	89.38
66	Temperature (C)	52.46	52.46	52.46	52.46
67	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
68	Heat Transferred (kJ/h-m)	598.5	598.4	598.4	598.3
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.3019	0.3019	0.3019	0.3019
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	5.889e-003	5.889e-003	5.889e-003	5.889e-003
18	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
19	Accel. Gradient (kPa/m)	4.228e-007	4.228e-007	4.228e-007	4.228e-007
20	Liq. Reynolds	2.113e+005 *	2.113e+005 *	2.113e+005 *	2.113e+005 *
21	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *
22	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
23	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	2.195e+004	2.196e+004	2.196e+004	2.197e+004
27	Elevation (m)	-9.000e-002	-8.800e-002	-8.600e-002	-8.400e-002
28	Pressure (bar)	89.38	89.38	89.38	89.38
29	Temperature (C)	52.46	52.46	52.45	52.45
30	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
31	Heat Transferred (kJ/h-m)	598.3	598.2	598.2	598.1
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.3019	0.2802	0.2802	0.2802
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	5.889e-003	5.556e-003	5.556e-003	5.556e-003
37	Static Gradient (kPa/m)	0.0000	9.473e-004	9.473e-004	9.473e-004
38	Accel. Gradient (kPa/m)	4.228e-007	4.669e-007	4.669e-007	4.669e-007
39	Liq. Reynolds	2.113e+005 *	2.113e+005 *	2.113e+005 *	2.113e+005 *
40	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *
41	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
42	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	2.197e+004	2.198e+004	2.198e+004	2.199e+004
46	Elevation (m)	-8.200e-002	-8.000e-002	-7.800e-002	-7.600e-002
47	Pressure (bar)	89.38	89.38	89.38	89.38
48	Temperature (C)	52.45	52.45	52.45	52.44
49	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
50	Heat Transferred (kJ/h-m)	598.1	598.0	598.0	597.9
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2802	0.2802	0.2803	0.2803
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	5.556e-003	5.556e-003	5.556e-003	5.556e-003
56	Static Gradient (kPa/m)	9.473e-004	9.473e-004	9.473e-004	9.473e-004
57	Accel. Gradient (kPa/m)	4.669e-007	4.669e-007	4.669e-007	4.669e-007
58	Liq. Reynolds	2.113e+005 *	2.113e+005 *	2.113e+005 *	2.112e+005 *
59	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *
60	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
61	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	2.199e+004	2.200e+004	2.200e+004	2.201e+004
65	Elevation (m)	-7.400e-002	-7.200e-002	-7.000e-002	-7.400e-002
66	Pressure (bar)	89.38	89.38	89.38	89.38
67	Temperature (C)	52.44	52.44	52.44	52.44
68	Vapour Fraction	0.8702	0.8702	0.8702	0.8702
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022	
6						
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Heat Transferred (kJ/h-m)	597.9	597.8	597.8	597.8	
14	Flow Regime	Transition	Transition	Transition	Transition	
15	Water Flow Regime					
16	Liquid HoldUp	0.2803	0.2803	0.2803	0.2061	
17	Water HoldUp	---	---	---	---	
18	Friction Gradient (kPa/m)	5.556e-003	5.556e-003	5.556e-003	6.884e-003	
19	Static Gradient (kPa/m)	9.473e-004	9.473e-004	9.473e-004	-1.543e-003	
20	Accel. Gradient (kPa/m)	4.669e-007	4.669e-007	4.669e-007	3.835e-007	
21	Liq. Reynolds	2.112e+005 *	2.112e+005 *	2.112e+005 *	2.112e+005 *	
22	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *	
23	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043	
24	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176	
25	Deposit Thickness (mm)	---	---	---	---	
26	Deposit Volume (m3)	---	---	---	---	
27	Length (m)	2.201e+004	2.202e+004	2.202e+004	2.203e+004	
28	Elevation (m)	-7.800e-002	-8.200e-002	-8.600e-002	-9.000e-002	
29	Pressure (bar)	89.38	89.38	89.38	89.38	
30	Temperature (C)	52.44	52.43	52.43	52.43	
31	Vapour Fraction	0.8702	0.8702	0.8702	0.8702	
32	Heat Transferred (kJ/h-m)	597.7	597.7	597.6	597.6	
33	Flow Regime	Transition	Transition	Transition	Transition	
34	Water Flow Regime					
35	Liquid HoldUp	0.2061	0.2061	0.2061	0.2061	
36	Water HoldUp	---	---	---	---	
37	Friction Gradient (kPa/m)	6.884e-003	6.884e-003	6.884e-003	6.883e-003	
38	Static Gradient (kPa/m)	-1.543e-003	-1.543e-003	-1.543e-003	-1.543e-003	
39	Accel. Gradient (kPa/m)	3.835e-007	3.835e-007	3.835e-007	3.835e-007	
40	Liq. Reynolds	2.112e+005 *	2.112e+005 *	2.112e+005 *	2.112e+005 *	
41	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *	
42	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043	
43	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176	
44	Deposit Thickness (mm)	---	---	---	---	
45	Deposit Volume (m3)	---	---	---	---	
46	Length (m)	2.203e+004	2.204e+004	2.204e+004	2.205e+004	
47	Elevation (m)	-9.200e-002	-9.400e-002	-9.600e-002	-9.800e-002	
48	Pressure (bar)	89.38	89.38	89.38	89.38	
49	Temperature (C)	52.43	52.43	52.42	52.42	
50	Vapour Fraction	0.8702	0.8702	0.8702	0.8702	
51	Heat Transferred (kJ/h-m)	597.5	597.5	597.4	597.4	
52	Flow Regime	Transition	Transition	Transition	Transition	
53	Water Flow Regime					
54	Liquid HoldUp	0.2065	0.2065	0.2065	0.2065	
55	Water HoldUp	---	---	---	---	
56	Friction Gradient (kPa/m)	6.881e-003	6.880e-003	6.880e-003	6.880e-003	
57	Static Gradient (kPa/m)	-7.722e-004	-7.722e-004	-7.722e-004	-7.722e-004	
58	Accel. Gradient (kPa/m)	4.386e-007	4.386e-007	4.386e-007	4.385e-007	
59	Liq. Reynolds	2.112e+005 *	2.112e+005 *	2.112e+005 *	2.112e+005 *	
60	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *	
61	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043	
62	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176	
63	Deposit Thickness (mm)	---	---	---	---	
64	Deposit Volume (m3)	---	---	---	---	
65	Length (m)	2.205e+004	2.206e+004	2.206e+004	2.207e+004	
66	Elevation (m)	-0.1000	-0.1260	-0.1520	-0.1780	
67	Pressure (bar)	89.38	89.38	89.38	89.38	
68	Temperature (C)	52.42	52.42	52.42	52.42	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8702	0.8702	0.8702	0.8701
14	Heat Transferred (kJ/h-m)	597.3	597.3	597.2	597.2
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2065	0.2020	0.2020	0.2020
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	6.880e-003	6.917e-003	6.916e-003	6.916e-003
20	Static Gradient (kPa/m)	-7.722e-004	-9.901e-003	-9.901e-003	-9.901e-003
21	Accel. Gradient (kPa/m)	4.385e-007	-2.142e-007	-2.143e-007	-2.143e-007
22	Liq. Reynolds	2.112e+005 *	2.112e+005 *	2.112e+005 *	2.112e+005 *
23	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *
24	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
25	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	2.207e+004	2.208e+004	2.208e+004	2.209e+004
29	Elevation (m)	-0.2040	-0.2300	-0.2560	-0.2820
30	Pressure (bar)	89.38	89.38	89.38	89.38
31	Temperature (C)	52.41	52.41	52.41	52.41
32	Vapour Fraction	0.8701	0.8701	0.8701	0.8701
33	Heat Transferred (kJ/h-m)	597.2	597.1	597.1	597.0
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2020	0.2020	0.2020	0.2020
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	6.916e-003	6.916e-003	6.916e-003	6.916e-003
39	Static Gradient (kPa/m)	-9.901e-003	-9.901e-003	-9.901e-003	-9.901e-003
40	Accel. Gradient (kPa/m)	-2.143e-007	-2.143e-007	-2.143e-007	-2.143e-007
41	Liq. Reynolds	2.112e+005 *	2.112e+005 *	2.112e+005 *	2.112e+005 *
42	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *
43	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
44	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	2.209e+004	2.210e+004	2.210e+004	2.211e+004
48	Elevation (m)	-0.3080	-0.3340	-0.3600	-0.3600
49	Pressure (bar)	89.38	89.38	89.38	89.38
50	Temperature (C)	52.41	52.41	52.41	52.40
51	Vapour Fraction	0.8701	0.8701	0.8701	0.8701
52	Heat Transferred (kJ/h-m)	597.0	596.9	596.9	596.8
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2020	0.2020	0.2020	0.3020
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	6.916e-003	6.916e-003	6.916e-003	5.888e-003
58	Static Gradient (kPa/m)	-9.901e-003	-9.901e-003	-9.901e-003	0.0000
59	Accel. Gradient (kPa/m)	-2.143e-007	-2.143e-007	-2.143e-007	4.227e-007
60	Liq. Reynolds	2.112e+005 *	2.112e+005 *	2.112e+005 *	2.112e+005 *
61	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *
62	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
63	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	2.211e+004	2.212e+004	2.212e+004	2.213e+004
67	Elevation (m)	-0.3600	-0.3600	-0.3600	-0.3600
68	Pressure (bar)	89.38	89.38	89.38	89.38
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	52.40	52.40	52.40	52.40
14	Vapour Fraction	0.8701	0.8701	0.8701	0.8701
15	Heat Transferred (kJ/h-m)	596.8	596.7	596.7	596.6
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.3020	0.3020	0.3020	0.3020
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	5.888e-003	5.888e-003	5.888e-003	5.888e-003
21	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
22	Accel. Gradient (kPa/m)	4.227e-007	4.227e-007	4.226e-007	4.226e-007
23	Liq. Reynolds	2.112e+005 *	2.112e+005 *	2.112e+005 *	2.112e+005 *
24	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *
25	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
26	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	2.213e+004	2.214e+004	2.214e+004	2.215e+004
30	Elevation (m)	-0.3540	-0.3480	-0.3420	-0.3360
31	Pressure (bar)	89.38	89.37	89.37	89.37
32	Temperature (C)	52.39	52.39	52.39	52.39
33	Vapour Fraction	0.8701	0.8701	0.8701	0.8701
34	Heat Transferred (kJ/h-m)	596.6	596.5	596.5	596.4
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2828	0.2828	0.2828	0.2828
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	5.356e-003	5.356e-003	5.356e-003	5.356e-003
40	Static Gradient (kPa/m)	2.861e-003	2.861e-003	2.861e-003	2.861e-003
41	Accel. Gradient (kPa/m)	5.898e-007	5.898e-007	5.898e-007	5.898e-007
42	Liq. Reynolds	2.112e+005 *	2.112e+005 *	2.112e+005 *	2.112e+005 *
43	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *
44	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
45	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	2.215e+004	2.216e+004	2.216e+004	2.217e+004
49	Elevation (m)	-0.3300	-0.3020	-0.2740	-0.2460
50	Pressure (bar)	89.37	89.37	89.37	89.37
51	Temperature (C)	52.39	52.38	52.38	52.38
52	Vapour Fraction	0.8701	0.8701	0.8701	0.8701
53	Heat Transferred (kJ/h-m)	596.4	596.3	596.3	596.2
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2828	0.2968	0.2968	0.2968
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	5.356e-003	5.589e-003	5.589e-003	5.589e-003
59	Static Gradient (kPa/m)	2.861e-003	1.381e-002	1.381e-002	1.381e-002
60	Accel. Gradient (kPa/m)	5.898e-007	1.393e-006	1.393e-006	1.393e-006
61	Liq. Reynolds	2.112e+005 *	2.112e+005 *	2.112e+005 *	2.112e+005 *
62	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *
63	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
64	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	2.217e+004	2.218e+004	2.218e+004	2.219e+004
68	Elevation (m)	-0.2180	-0.1900	-0.2120	-0.2340
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:02:44 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Pressure (bar)	89.37	89.37	89.37	89.37					
14	Temperature (C)	52.38	52.38	52.37	52.37					
15	Vapour Fraction	0.8701	0.8701	0.8701	0.8701					
16	Heat Transferred (kJ/h-m)	596.2	596.1	596.1	596.0					
17	Flow Regime	Transition	Transition	Transition	Transition					
18	Water Flow Regime									
19	Liquid HoldUp	0.2968	0.2968	0.2028	0.2028					
20	Water HoldUp	---	---	---	---					
21	Friction Gradient (kPa/m)	5.589e-003	5.589e-003	6.909e-003	6.909e-003					
22	Static Gradient (kPa/m)	1.381e-002	1.381e-002	-8.398e-003	-8.398e-003					
23	Accel. Gradient (kPa/m)	1.393e-006	1.393e-006	-1.069e-007	-1.069e-007					
24	Liq. Reynolds	2.112e+005 *	2.112e+005 *	2.112e+005 *	2.112e+005 *					
25	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *					
26	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043					
27	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176					
28	Deposit Thickness (mm)	---	---	---	---					
29	Deposit Volume (m3)	---	---	---	---					
30	Length (m)	2.219e+004	2.220e+004	2.220e+004	2.221e+004					
31	Elevation (m)	-0.2560	-0.2780	-0.3000	-0.3900					
32	Pressure (bar)	89.37	89.37	89.37	89.37					
33	Temperature (C)	52.37	52.37	52.37	52.37					
34	Vapour Fraction	0.8701	0.8701	0.8701	0.8701					
35	Heat Transferred (kJ/h-m)	596.0	596.0	595.9	595.9					
36	Flow Regime	Transition	Transition	Transition	Transition					
37	Water Flow Regime									
38	Liquid HoldUp	0.2028	0.2028	0.2028	0.1901					
39	Water HoldUp	---	---	---	---					
40	Friction Gradient (kPa/m)	6.909e-003	6.909e-003	6.909e-003	7.024e-003					
41	Static Gradient (kPa/m)	-8.398e-003	-8.398e-003	-8.398e-003	-3.301e-002					
42	Accel. Gradient (kPa/m)	-1.069e-007	-1.069e-007	-1.069e-007	-1.865e-006					
43	Liq. Reynolds	2.112e+005 *	2.112e+005 *	2.112e+005 *	2.112e+005 *					
44	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *					
45	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043					
46	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176					
47	Deposit Thickness (mm)	---	---	---	---					
48	Deposit Volume (m3)	---	---	---	---					
49	Length (m)	2.221e+004	2.222e+004	2.222e+004	2.223e+004					
50	Elevation (m)	-0.4800	-0.5700	-0.6600	-0.7500					
51	Pressure (bar)	89.37	89.37	89.37	89.38					
52	Temperature (C)	52.37	52.36	52.36	52.36					
53	Vapour Fraction	0.8701	0.8701	0.8701	0.8701					
54	Heat Transferred (kJ/h-m)	595.9	595.8	595.8	595.8					
55	Flow Regime	Transition	Transition	Transition	Transition					
56	Water Flow Regime									
57	Liquid HoldUp	0.1901	0.1901	0.1902	0.1902					
58	Water HoldUp	---	---	---	---					
59	Friction Gradient (kPa/m)	7.024e-003	7.024e-003	7.023e-003	7.023e-003					
60	Static Gradient (kPa/m)	-3.301e-002	-3.301e-002	-3.301e-002	-3.301e-002					
61	Accel. Gradient (kPa/m)	-1.865e-006	-1.865e-006	-1.865e-006	-1.865e-006					
62	Liq. Reynolds	2.112e+005 *	2.112e+005 *	2.112e+005 *	2.112e+005 *					
63	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *					
64	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043					
65	Vapour Velocity (m/s)	2.176	2.176	2.176	2.175					
66	Deposit Thickness (mm)	---	---	---	---					
67	Deposit Volume (m3)	---	---	---	---					
68	Length (m)	2.223e+004	2.224e+004	2.224e+004	2.225e+004					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	-0.6640	-0.5780	-0.4920	-0.4060
14	Pressure (bar)	89.37	89.37	89.37	89.37
15	Temperature (C)	52.36	52.36	52.35	52.35
16	Vapour Fraction	0.8701	0.8701	0.8701	0.8701
17	Heat Transferred (kJ/h-m)	595.7	595.6	595.6	595.5
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.3338	0.3337	0.3337	0.3337
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	6.086e-003	6.086e-003	6.086e-003	6.086e-003
23	Static Gradient (kPa/m)	4.620e-002	4.620e-002	4.620e-002	4.620e-002
24	Accel. Gradient (kPa/m)	3.753e-006	3.753e-006	3.753e-006	3.753e-006
25	Liq. Reynolds	2.112e+005 *	2.112e+005 *	2.112e+005 *	2.112e+005 *
26	Vap. Reynolds	9.153e+006 *	9.153e+006 *	9.153e+006 *	9.153e+006 *
27	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
28	Vapour Velocity (m/s)	2.175	2.176	2.176	2.176
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	2.225e+004	2.226e+004	2.226e+004	2.227e+004
32	Elevation (m)	-0.3200	-0.2880	-0.2560	-0.2240
33	Pressure (bar)	89.36	89.36	89.36	89.36
34	Temperature (C)	52.35	52.35	52.35	52.34
35	Vapour Fraction	0.8701	0.8701	0.8701	0.8701
36	Heat Transferred (kJ/h-m)	595.4	595.4	595.3	595.3
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.3337	0.2995	0.2995	0.2995
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	6.086e-003	5.786e-003	5.786e-003	5.786e-003
42	Static Gradient (kPa/m)	4.620e-002	1.596e-002	1.596e-002	1.596e-002
43	Accel. Gradient (kPa/m)	3.753e-006	1.561e-006	1.561e-006	1.561e-006
44	Liq. Reynolds	2.112e+005 *	2.112e+005 *	2.112e+005 *	2.112e+005 *
45	Vap. Reynolds	9.154e+006 *	9.154e+006 *	9.154e+006 *	9.154e+006 *
46	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
47	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	2.227e+004	2.228e+004	2.229e+004	2.230e+004
51	Elevation (m)	-0.1920	-0.1600	-0.1520	-0.1440
52	Pressure (bar)	89.36	89.36	89.36	89.36
53	Temperature (C)	52.34	52.34	52.34	52.33
54	Vapour Fraction	0.8701	0.8701	0.8701	0.8701
55	Heat Transferred (kJ/h-m)	595.2	595.1	595.0	594.9
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2995	0.2995	0.2816	0.2816
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	5.786e-003	5.786e-003	5.454e-003	5.454e-003
61	Static Gradient (kPa/m)	1.596e-002	1.596e-002	1.901e-003	1.901e-003
62	Accel. Gradient (kPa/m)	1.561e-006	1.561e-006	5.281e-007	5.280e-007
63	Liq. Reynolds	2.112e+005 *	2.112e+005 *	2.111e+005 *	2.111e+005 *
64	Vap. Reynolds	9.154e+006 *	9.154e+006 *	9.154e+006 *	9.154e+006 *
65	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043
66	Vapour Velocity (m/s)	2.176	2.176	2.176	2.176
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Length (m)	2.231e+004	2.232e+004	2.233e+004	2.255e+004		
14	Elevation (m)	-0.1360	-0.1280	-0.1200	-0.1160		
15	Pressure (bar)	89.36	89.35	89.35	89.34		
16	Temperature (C)	52.33	52.33	52.32	52.24		
17	Vapour Fraction	0.8701	0.8701	0.8701	0.8701		
18	Heat Transferred (kJ/h-m)	594.9	594.8	594.7	592.5		
19	Flow Regime	Transition	Transition	Transition	Transition		
20	Water Flow Regime						
21	Liquid HoldUp	0.2816	0.2816	0.2816	0.2792		
22	Water HoldUp	---	---	---	---		
23	Friction Gradient (kPa/m)	5.454e-003	5.454e-003	5.454e-003	5.649e-003		
24	Static Gradient (kPa/m)	1.901e-003	1.901e-003	1.901e-003	4.200e-005		
25	Accel. Gradient (kPa/m)	5.280e-007	5.280e-007	5.280e-007	4.085e-007		
26	Liq. Reynolds	2.111e+005 *	2.111e+005 *	2.111e+005 *	2.111e+005 *		
27	Vap. Reynolds	9.154e+006 *	9.154e+006 *	9.154e+006 *	9.154e+006 *		
28	Liquid Velocity (m/s)	0.2043	0.2043	0.2043	0.2043		
29	Vapour Velocity (m/s)	2.176	2.176	2.176	2.175		
30	Deposit Thickness (mm)	---	---	---	---		
31	Deposit Volume (m3)	---	---	---	---		
32	Length (m)	2.278e+004	2.300e+004	2.323e+004	2.345e+004		
33	Elevation (m)	-0.1120	-0.1080	-0.1040	-0.1000		
34	Pressure (bar)	89.33	89.32	89.30	89.29		
35	Temperature (C)	52.16	52.08	52.01	51.93		
36	Vapour Fraction	0.8700	0.8700	0.8699	0.8699		
37	Heat Transferred (kJ/h-m)	590.4	588.3	586.2	584.2		
38	Flow Regime	Transition	Transition	Transition	Transition		
39	Water Flow Regime						
40	Liquid HoldUp	0.2792	0.2793	0.2793	0.2794		
41	Water HoldUp	---	---	---	---		
42	Friction Gradient (kPa/m)	5.647e-003	5.645e-003	5.643e-003	5.641e-003		
43	Static Gradient (kPa/m)	4.200e-005	4.201e-005	4.202e-005	4.203e-005		
44	Accel. Gradient (kPa/m)	4.083e-007	4.081e-007	4.079e-007	4.077e-007		
45	Liq. Reynolds	2.110e+005 *	2.109e+005 *	2.109e+005 *	2.108e+005 *		
46	Vap. Reynolds	9.154e+006 *	9.155e+006 *	9.155e+006 *	9.155e+006 *		
47	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
48	Vapour Velocity (m/s)	2.174	2.174	2.173	2.173		
49	Deposit Thickness (mm)	---	---	---	---		
50	Deposit Volume (m3)	---	---	---	---		
51	Length (m)	2.346e+004	2.347e+004	2.348e+004	2.348e+004		
52	Elevation (m)	-0.1040	-0.1080	-0.1120	-0.1160		
53	Pressure (bar)	89.29	89.29	89.29	89.29		
54	Temperature (C)	51.92	51.92	51.92	51.92		
55	Vapour Fraction	0.8699	0.8699	0.8699	0.8699		
56	Heat Transferred (kJ/h-m)	584.1	584.0	584.0	583.9		
57	Flow Regime	Transition	Transition	Transition	Transition		
58	Water Flow Regime						
59	Liquid HoldUp	0.2066	0.2066	0.2066	0.2066		
60	Water HoldUp	---	---	---	---		
61	Friction Gradient (kPa/m)	6.872e-003	6.872e-003	6.872e-003	6.872e-003		
62	Static Gradient (kPa/m)	-1.051e-003	-1.051e-003	-1.051e-003	-1.051e-003		
63	Accel. Gradient (kPa/m)	4.176e-007	4.176e-007	4.176e-007	4.176e-007		
64	Liq. Reynolds	2.108e+005 *	2.108e+005 *	2.108e+005 *	2.108e+005 *		
65	Vap. Reynolds	9.155e+006 *	9.155e+006 *	9.155e+006 *	9.155e+006 *		
66	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
67	Vapour Velocity (m/s)	2.173	2.173	2.172	2.172		
68	Deposit Thickness (mm)	---	---	---	---		
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 378 of 937		

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	2.349e+004	2.350e+004	2.350e+004	2.351e+004		
15	Elevation (m)	-0.1200	-0.1180	-0.1160	-0.1140		
16	Pressure (bar)	89.29	89.29	89.29	89.29		
17	Temperature (C)	51.91	51.91	51.91	51.91		
18	Vapour Fraction	0.8699	0.8699	0.8699	0.8699		
19	Heat Transferred (kJ/h-m)	583.8	583.8	583.7	583.7		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	0.2066	0.2806	0.2806	0.2806		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	6.872e-003	5.544e-003	5.544e-003	5.544e-003		
25	Static Gradient (kPa/m)	-1.051e-003	9.487e-004	9.487e-004	9.487e-004		
26	Accel. Gradient (kPa/m)	4.176e-007	4.658e-007	4.658e-007	4.658e-007		
27	Liq. Reynolds	2.108e+005 *	2.108e+005 *	2.108e+005 *	2.108e+005 *		
28	Vap. Reynolds	9.155e+006 *	9.155e+006 *	9.155e+006 *	9.155e+006 *		
29	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
30	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	2.351e+004	2.352e+004	2.352e+004	2.353e+004		
34	Elevation (m)	-0.1120	-0.1100	-0.1140	-0.1180		
35	Pressure (bar)	89.29	89.29	89.29	89.29		
36	Temperature (C)	51.91	51.91	51.90	51.90		
37	Vapour Fraction	0.8699	0.8699	0.8699	0.8699		
38	Heat Transferred (kJ/h-m)	583.6	583.6	583.6	583.5		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.2806	0.2806	0.2063	0.2063		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	5.544e-003	5.544e-003	6.874e-003	6.873e-003		
44	Static Gradient (kPa/m)	9.487e-004	9.487e-004	-1.545e-003	-1.545e-003		
45	Accel. Gradient (kPa/m)	4.658e-007	4.658e-007	3.823e-007	3.823e-007		
46	Liq. Reynolds	2.108e+005 *	2.108e+005 *	2.108e+005 *	2.108e+005 *		
47	Vap. Reynolds	9.155e+006 *	9.155e+006 *	9.155e+006 *	9.155e+006 *		
48	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
49	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	2.353e+004	2.354e+004	2.354e+004	2.355e+004		
53	Elevation (m)	-0.1220	-0.1260	-0.1300	-0.1300		
54	Pressure (bar)	89.29	89.28	89.28	89.28		
55	Temperature (C)	51.90	51.90	51.90	51.90		
56	Vapour Fraction	0.8699	0.8699	0.8699	0.8699		
57	Heat Transferred (kJ/h-m)	583.5	583.4	583.4	583.3		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.2063	0.2063	0.2063	0.3023		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	6.873e-003	6.873e-003	6.873e-003	5.882e-003		
63	Static Gradient (kPa/m)	-1.545e-003	-1.545e-003	-1.545e-003	0.0000		
64	Accel. Gradient (kPa/m)	3.823e-007	3.823e-007	3.823e-007	4.220e-007		
65	Liq. Reynolds	2.108e+005 *	2.108e+005 *	2.108e+005 *	2.108e+005 *		
66	Vap. Reynolds	9.155e+006 *	9.155e+006 *	9.155e+006 *	9.155e+006 *		
67	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
68	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172		
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 379 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:02:44 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Deposit Thickness	(mm)	---	---	---	---				
14	Deposit Volume	(m3)	---	---	---	---				
15	Length	(m)	2.355e+004	2.356e+004	2.356e+004	2.357e+004				
16	Elevation	(m)	-0.1300	-0.1300	-0.1300	-0.1300				
17	Pressure	(bar)	89.28	89.28	89.28	89.28				
18	Temperature	(C)	51.89	51.89	51.89	51.89				
19	Vapour Fraction		0.8698	0.8698	0.8698	0.8698				
20	Heat Transferred	(kJ/h-m)	583.3	583.2	583.2	583.1				
21	Flow Regime		Transition	Transition	Transition	Transition				
22	Water Flow Regime									
23	Liquid HoldUp		0.3023	0.3023	0.3023	0.3023				
24	Water HoldUp		---	---	---	---				
25	Friction Gradient	(kPa/m)	5.882e-003	5.882e-003	5.882e-003	5.882e-003				
26	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000				
27	Accel. Gradient	(kPa/m)	4.220e-007	4.220e-007	4.220e-007	4.220e-007				
28	Liq. Reynolds		2.108e+005 *	2.108e+005 *	2.108e+005 *	2.108e+005 *				
29	Vap. Reynolds		9.155e+006 *	9.155e+006 *	9.155e+006 *	9.155e+006 *				
30	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044				
31	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172				
32	Deposit Thickness	(mm)	---	---	---	---				
33	Deposit Volume	(m3)	---	---	---	---				
34	Length	(m)	2.357e+004	2.358e+004	2.358e+004	2.359e+004				
35	Elevation	(m)	3.200e-002	0.1940	0.3560	0.5180				
36	Pressure	(bar)	89.28	89.27	89.27	89.26				
37	Temperature	(C)	51.89	51.88	51.88	51.88				
38	Vapour Fraction		0.8698	0.8698	0.8698	0.8698				
39	Heat Transferred	(kJ/h-m)	583.1	583.0	582.9	582.8				
40	Flow Regime		Transition	Transition	Transition	Transition				
41	Water Flow Regime									
42	Liquid HoldUp		0.3823	0.3823	0.3823	0.3823				
43	Water HoldUp		---	---	---	---				
44	Friction Gradient	(kPa/m)	6.020e-003	6.020e-003	6.021e-003	6.021e-003				
45	Static Gradient	(kPa/m)	9.641e-002	9.641e-002	9.640e-002	9.640e-002				
46	Accel. Gradient	(kPa/m)	7.349e-006	7.350e-006	7.350e-006	7.351e-006				
47	Liq. Reynolds		2.107e+005 *	2.107e+005 *	2.107e+005 *	2.107e+005 *				
48	Vap. Reynolds		9.156e+006 *	9.156e+006 *	9.156e+006 *	9.156e+006 *				
49	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044				
50	Vapour Velocity	(m/s)	2.172	2.172	2.173	2.173				
51	Deposit Thickness	(mm)	---	---	---	---				
52	Deposit Volume	(m3)	---	---	---	---				
53	Length	(m)	2.359e+004	2.360e+004	2.360e+004	2.361e+004				
54	Elevation	(m)	0.6800	0.5120	0.3440	0.1760				
55	Pressure	(bar)	89.26	89.26	89.26	89.27				
56	Temperature	(C)	51.87	51.87	51.87	51.87				
57	Vapour Fraction		0.8699	0.8698	0.8698	0.8698				
58	Heat Transferred	(kJ/h-m)	582.7	582.7	582.7	582.7				
59	Flow Regime		Transition	Transition	Transition	Transition				
60	Water Flow Regime									
61	Liquid HoldUp		0.3823	0.1758	0.1759	0.1759				
62	Water HoldUp		---	---	---	---				
63	Friction Gradient	(kPa/m)	6.021e-003	7.176e-003	7.175e-003	7.175e-003				
64	Static Gradient	(kPa/m)	9.640e-002	-5.880e-002	-5.880e-002	-5.880e-002				
65	Accel. Gradient	(kPa/m)	7.352e-006	-3.705e-006	-3.705e-006	-3.705e-006				
66	Liq. Reynolds		2.107e+005 *	2.107e+005 *	2.107e+005 *	2.107e+005 *				
67	Vap. Reynolds		9.156e+006 *	9.156e+006 *	9.156e+006 *	9.156e+006 *				
68	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	2.361e+004	2.362e+004	2.362e+004	2.363e+004	
17	Elevation	(m)	8.000e-003	-0.1600	-0.1700	-0.1800	
18	Pressure	(bar)	89.27	89.27	89.27	89.27	
19	Temperature	(C)	51.87	51.87	51.87	51.87	
20	Vapour Fraction		0.8698	0.8698	0.8698	0.8698	
21	Heat Transferred	(kJ/h-m)	582.7	582.7	582.6	582.5	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.1759	0.1759	0.2052	0.2052	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	7.175e-003	7.175e-003	6.882e-003	6.882e-003	
27	Static Gradient	(kPa/m)	-5.880e-002	-5.880e-002	-3.848e-003	-3.849e-003	
28	Accel. Gradient	(kPa/m)	-3.705e-006	-3.704e-006	2.177e-007	2.177e-007	
29	Liq. Reynolds		2.107e+005 *	2.107e+005 *	2.107e+005 *	2.107e+005 *	
30	Vap. Reynolds		9.156e+006 *	9.156e+006 *	9.156e+006 *	9.156e+006 *	
31	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
32	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	2.363e+004	2.364e+004	2.364e+004	2.365e+004	
36	Elevation	(m)	-0.1900	-0.2000	-0.2100	-0.1880	
37	Pressure	(bar)	89.27	89.27	89.27	89.27	
38	Temperature	(C)	51.86	51.86	51.86	51.86	
39	Vapour Fraction		0.8698	0.8698	0.8698	0.8698	
40	Heat Transferred	(kJ/h-m)	582.5	582.5	582.4	582.3	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2052	0.2052	0.2052	0.2933	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	6.882e-003	6.882e-003	6.882e-003	4.649e-003	
46	Static Gradient	(kPa/m)	-3.849e-003	-3.849e-003	-3.849e-003	1.077e-002	
47	Accel. Gradient	(kPa/m)	2.177e-007	2.177e-007	2.177e-007	1.106e-006	
48	Liq. Reynolds		2.107e+005 *	2.107e+005 *	2.107e+005 *	2.107e+005 *	
49	Vap. Reynolds		9.156e+006 *	9.156e+006 *	9.156e+006 *	9.156e+006 *	
50	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
51	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	2.365e+004	2.366e+004	2.366e+004	2.367e+004	
55	Elevation	(m)	-0.1660	-0.1440	-0.1220	-0.1000	
56	Pressure	(bar)	89.27	89.27	89.27	89.27	
57	Temperature	(C)	51.86	51.86	51.85	51.85	
58	Vapour Fraction		0.8698	0.8698	0.8698	0.8698	
59	Heat Transferred	(kJ/h-m)	582.3	582.2	582.2	582.1	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.2933	0.2933	0.2933	0.2933	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	4.649e-003	4.649e-003	4.649e-003	4.649e-003	
65	Static Gradient	(kPa/m)	1.077e-002	1.077e-002	1.077e-002	1.077e-002	
66	Accel. Gradient	(kPa/m)	1.106e-006	1.106e-006	1.106e-006	1.106e-006	
67	Liq. Reynolds		2.107e+005 *	2.107e+005 *	2.107e+005 *	2.107e+005 *	
68	Vap. Reynolds		9.156e+006 *	9.156e+006 *	9.156e+006 *	9.156e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
14	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	2.367e+004	2.368e+004	2.368e+004	2.369e+004	
18	Elevation	(m)	-0.1100	-0.1200	-0.1300	-0.1400	
19	Pressure	(bar)	89.27	89.27	89.27	89.27	
20	Temperature	(C)	51.85	51.85	51.85	51.84	
21	Vapour Fraction		0.8698	0.8698	0.8698	0.8698	
22	Heat Transferred	(kJ/h-m)	582.1	582.1	582.0	582.0	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.2052	0.2052	0.2052	0.2052	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	6.882e-003	6.882e-003	6.882e-003	6.882e-003	
28	Static Gradient	(kPa/m)	-3.846e-003	-3.846e-003	-3.846e-003	-3.846e-003	
29	Accel. Gradient	(kPa/m)	2.179e-007	2.179e-007	2.179e-007	2.179e-007	
30	Liq. Reynolds		2.107e+005 *	2.107e+005 *	2.107e+005 *	2.107e+005 *	
31	Vap. Reynolds		9.156e+006 *	9.156e+006 *	9.156e+006 *	9.156e+006 *	
32	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
33	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	2.369e+004	2.370e+004	2.370e+004	2.371e+004	
37	Elevation	(m)	-0.1500	-0.1580	-0.1660	-0.1740	
38	Pressure	(bar)	89.26	89.26	89.26	89.26	
39	Temperature	(C)	51.84	51.84	51.84	51.84	
40	Vapour Fraction		0.8698	0.8698	0.8698	0.8698	
41	Heat Transferred	(kJ/h-m)	581.9	581.9	581.8	581.8	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2052	0.2056	0.2056	0.2056	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	6.882e-003	6.879e-003	6.879e-003	6.879e-003	
47	Static Gradient	(kPa/m)	-3.846e-003	-3.083e-003	-3.083e-003	-3.083e-003	
48	Accel. Gradient	(kPa/m)	2.179e-007	2.724e-007	2.724e-007	2.724e-007	
49	Liq. Reynolds		2.107e+005 *	2.107e+005 *	2.107e+005 *	2.107e+005 *	
50	Vap. Reynolds		9.156e+006 *	9.156e+006 *	9.156e+006 *	9.156e+006 *	
51	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
52	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	2.371e+004	2.372e+004	2.372e+004	2.373e+004	
56	Elevation	(m)	-0.1820	-0.1900	-0.2020	-0.2140	
57	Pressure	(bar)	89.26	89.26	89.26	89.26	
58	Temperature	(C)	51.84	51.83	51.83	51.83	
59	Vapour Fraction		0.8698	0.8698	0.8698	0.8698	
60	Heat Transferred	(kJ/h-m)	581.8	581.7	581.7	581.6	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2056	0.2056	0.2049	0.2049	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	6.879e-003	6.879e-003	6.885e-003	6.885e-003	
66	Static Gradient	(kPa/m)	-3.083e-003	-3.083e-003	-4.613e-003	-4.613e-003	
67	Accel. Gradient	(kPa/m)	2.724e-007	2.724e-007	1.630e-007	1.630e-007	
68	Liq. Reynolds		2.107e+005 *	2.107e+005 *	2.107e+005 *	2.107e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.156e+006 *	9.156e+006 *	9.156e+006 *	9.156e+006 *
14	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
15	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	2.373e+004	2.374e+004	2.374e+004	2.375e+004
19	Elevation (m)	-0.2260	-0.2380	-0.2500	-0.2900
20	Pressure (bar)	89.26	89.26	89.26	89.26
21	Temperature (C)	51.83	51.83	51.83	51.83
22	Vapour Fraction	0.8698	0.8698	0.8698	0.8698
23	Heat Transferred (kJ/h-m)	581.6	581.5	581.5	581.5
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2049	0.2049	0.2049	0.1996
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.885e-003	6.885e-003	6.885e-003	6.928e-003
29	Static Gradient (kPa/m)	-4.613e-003	-4.613e-003	-4.613e-003	-1.513e-002
30	Accel. Gradient (kPa/m)	1.630e-007	1.630e-007	1.630e-007	-5.885e-007
31	Liq. Reynolds	2.107e+005 *	2.107e+005 *	2.107e+005 *	2.107e+005 *
32	Vap. Reynolds	9.156e+006 *	9.156e+006 *	9.156e+006 *	9.156e+006 *
33	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
34	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	2.375e+004	2.376e+004	2.376e+004	2.377e+004
38	Elevation (m)	-0.3300	-0.3700	-0.4100	-0.4500
39	Pressure (bar)	89.26	89.26	89.27	89.27
40	Temperature (C)	51.82	51.82	51.82	51.82
41	Vapour Fraction	0.8698	0.8698	0.8698	0.8698
42	Heat Transferred (kJ/h-m)	581.4	581.4	581.3	581.3
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1997	0.1997	0.1997	0.1997
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.928e-003	6.928e-003	6.928e-003	6.928e-003
48	Static Gradient (kPa/m)	-1.513e-002	-1.513e-002	-1.513e-002	-1.513e-002
49	Accel. Gradient (kPa/m)	-5.885e-007	-5.885e-007	-5.885e-007	-5.886e-007
50	Liq. Reynolds	2.107e+005 *	2.107e+005 *	2.107e+005 *	2.107e+005 *
51	Vap. Reynolds	9.156e+006 *	9.156e+006 *	9.156e+006 *	9.156e+006 *
52	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
53	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	2.377e+004	2.378e+004	2.378e+004	2.379e+004
57	Elevation (m)	-0.4420	-0.4340	-0.4260	-0.4180
58	Pressure (bar)	89.27	89.26	89.26	89.26
59	Temperature (C)	51.82	51.82	51.81	51.81
60	Vapour Fraction	0.8698	0.8698	0.8698	0.8698
61	Heat Transferred (kJ/h-m)	581.2	581.2	581.1	581.1
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2845	0.2845	0.2845	0.2845
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	5.247e-003	5.247e-003	5.247e-003	5.247e-003
67	Static Gradient (kPa/m)	3.832e-003	3.832e-003	3.832e-003	3.832e-003
68	Accel. Gradient (kPa/m)	6.513e-007	6.513e-007	6.513e-007	6.513e-007
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 383 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.107e+005 *	2.107e+005 *	2.107e+005 *	2.107e+005 *
14	Vap. Reynolds	9.156e+006 *	9.156e+006 *	9.156e+006 *	9.156e+006 *
15	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
16	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	2.379e+004	2.380e+004	2.380e+004	2.381e+004
20	Elevation (m)	-0.4100	-0.4040	-0.3980	-0.3920
21	Pressure (bar)	89.26	89.26	89.26	89.26
22	Temperature (C)	51.81	51.81	51.81	51.80
23	Vapour Fraction	0.8698	0.8698	0.8698	0.8698
24	Heat Transferred (kJ/h-m)	581.0	581.0	580.9	580.9
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2845	0.2832	0.2832	0.2832
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	5.247e-003	5.344e-003	5.344e-003	5.344e-003
30	Static Gradient (kPa/m)	3.832e-003	2.865e-003	2.865e-003	2.865e-003
31	Accel. Gradient (kPa/m)	6.513e-007	5.889e-007	5.889e-007	5.889e-007
32	Liq. Reynolds	2.107e+005 *	2.107e+005 *	2.107e+005 *	2.107e+005 *
33	Vap. Reynolds	9.156e+006 *	9.156e+006 *	9.156e+006 *	9.156e+006 *
34	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
35	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	2.381e+004	2.382e+004	2.382e+004	2.383e+004
39	Elevation (m)	-0.3860	-0.3800	-0.3380	-0.2960
40	Pressure (bar)	89.26	89.26	89.26	89.26
41	Temperature (C)	51.80	51.80	51.80	51.80
42	Vapour Fraction	0.8698	0.8698	0.8698	0.8698
43	Heat Transferred (kJ/h-m)	580.8	580.8	580.8	580.7
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2832	0.2832	0.3061	0.3061
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	5.344e-003	5.344e-003	5.966e-003	5.966e-003
49	Static Gradient (kPa/m)	2.865e-003	2.865e-003	2.120e-002	2.120e-002
50	Accel. Gradient (kPa/m)	5.889e-007	5.889e-007	1.949e-006	1.949e-006
51	Liq. Reynolds	2.107e+005 *	2.107e+005 *	2.107e+005 *	2.107e+005 *
52	Vap. Reynolds	9.156e+006 *	9.156e+006 *	9.156e+006 *	9.156e+006 *
53	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
54	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	2.383e+004	2.384e+004	2.384e+004	2.385e+004
58	Elevation (m)	-0.2540	-0.2120	-0.1700	-0.1760
59	Pressure (bar)	89.26	89.26	89.25	89.25
60	Temperature (C)	51.80	51.79	51.79	51.79
61	Vapour Fraction	0.8698	0.8698	0.8698	0.8698
62	Heat Transferred (kJ/h-m)	580.6	580.6	580.5	580.5
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.3061	0.3061	0.3061	0.2060
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	5.966e-003	5.966e-003	5.966e-003	6.875e-003
68	Static Gradient (kPa/m)	2.120e-002	2.120e-002	2.120e-002	-2.315e-003
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 384 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:02:44 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Accel. Gradient	(kPa/m)	1.949e-006	1.949e-006	1.949e-006	3.272e-007				
14	Liq. Reynolds		2.107e+005 *	2.107e+005 *	2.107e+005 *	2.107e+005 *				
15	Vap. Reynolds		9.156e+006 *	9.156e+006 *	9.156e+006 *	9.156e+006 *				
16	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044				
17	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172				
18	Deposit Thickness	(mm)	---	---	---	---				
19	Deposit Volume	(m3)	---	---	---	---				
20	Length	(m)	2.385e+004	2.386e+004	2.386e+004	2.387e+004				
21	Elevation	(m)	-0.1820	-0.1880	-0.1940	-0.2000				
22	Pressure	(bar)	89.25	89.25	89.25	89.25				
23	Temperature	(C)	51.79	51.79	51.78	51.78				
24	Vapour Fraction		0.8698	0.8698	0.8698	0.8698				
25	Heat Transferred	(kJ/h-m)	580.4	580.4	580.4	580.3				
26	Flow Regime		Transition	Transition	Transition	Transition				
27	Water Flow Regime									
28	Liquid HoldUp		0.2060	0.2060	0.2060	0.2060				
29	Water HoldUp		---	---	---	---				
30	Friction Gradient	(kPa/m)	6.875e-003	6.875e-003	6.875e-003	6.875e-003				
31	Static Gradient	(kPa/m)	-2.315e-003	-2.315e-003	-2.315e-003	-2.315e-003				
32	Accel. Gradient	(kPa/m)	3.272e-007	3.272e-007	3.272e-007	3.272e-007				
33	Liq. Reynolds		2.107e+005 *	2.107e+005 *	2.107e+005 *	2.107e+005 *				
34	Vap. Reynolds		9.156e+006 *	9.156e+006 *	9.156e+006 *	9.156e+006 *				
35	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044				
36	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172				
37	Deposit Thickness	(mm)	---	---	---	---				
38	Deposit Volume	(m3)	---	---	---	---				
39	Length	(m)	2.387e+004	2.388e+004	2.388e+004	2.389e+004				
40	Elevation	(m)	-0.2940	-0.3880	-0.4820	-0.5760				
41	Pressure	(bar)	89.25	89.26	89.26	89.26				
42	Temperature	(C)	51.78	51.78	51.78	51.78				
43	Vapour Fraction		0.8698	0.8698	0.8698	0.8698				
44	Heat Transferred	(kJ/h-m)	580.3	580.3	580.2	580.2				
45	Flow Regime		Transition	Transition	Transition	Transition				
46	Water Flow Regime									
47	Liquid HoldUp		0.1896	0.1896	0.1896	0.1896				
48	Water HoldUp		---	---	---	---				
49	Friction Gradient	(kPa/m)	7.021e-003	7.021e-003	7.021e-003	7.020e-003				
50	Static Gradient	(kPa/m)	-3.443e-002	-3.443e-002	-3.443e-002	-3.443e-002				
51	Accel. Gradient	(kPa/m)	-1.966e-006	-1.966e-006	-1.966e-006	-1.966e-006				
52	Liq. Reynolds		2.107e+005 *	2.107e+005 *	2.107e+005 *	2.107e+005 *				
53	Vap. Reynolds		9.156e+006 *	9.156e+006 *	9.156e+006 *	9.156e+006 *				
54	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044				
55	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172				
56	Deposit Thickness	(mm)	---	---	---	---				
57	Deposit Volume	(m3)	---	---	---	---				
58	Length	(m)	2.389e+004	2.390e+004	2.390e+004	2.391e+004				
59	Elevation	(m)	-0.6700	-0.5840	-0.4980	-0.4120				
60	Pressure	(bar)	89.26	89.26	89.25	89.25				
61	Temperature	(C)	51.78	51.77	51.77	51.77				
62	Vapour Fraction		0.8698	0.8698	0.8698	0.8698				
63	Heat Transferred	(kJ/h-m)	580.2	580.1	580.0	580.0				
64	Flow Regime		Transition	Transition	Transition	Transition				
65	Water Flow Regime									
66	Liquid HoldUp		0.1896	0.3341	0.3341	0.3341				
67	Water HoldUp		---	---	---	---				
68	Friction Gradient	(kPa/m)	7.020e-003	6.077e-003	6.077e-003	6.077e-003				
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 385 of 937				

1				Case Name:	Simulation - 386MMSCFD_100barg.hsc		
2				Unit Set:	Mixed		
3	 Epic Atlantic Limited			Date/Time:	Mon Jun 13 15:02:44 2022		
4							
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Static Gradient	(kPa/m)	-3.443e-002	4.627e-002	4.626e-002	4.626e-002	
14	Accel. Gradient	(kPa/m)	-1.966e-006	3.755e-006	3.755e-006	3.755e-006	
15	Liq. Reynolds		2.107e+005 *	2.106e+005 *	2.106e+005 *	2.106e+005 *	
16	Vap. Reynolds		9.156e+006 *	9.156e+006 *	9.156e+006 *	9.156e+006 *	
17	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
18	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172	
19	Deposit Thickness	(mm)	---	---	---	---	
20	Deposit Volume	(m3)	---	---	---	---	
21	Length	(m)	2.391e+004	2.392e+004	2.392e+004	2.393e+004	
22	Elevation	(m)	-0.3260	-0.2400	-0.2180	-0.1960	
23	Pressure	(bar)	89.25	89.25	89.25	89.25	
24	Temperature	(C)	51.77	51.76	51.76	51.76	
25	Vapour Fraction		0.8698	0.8698	0.8698	0.8698	
26	Heat Transferred	(kJ/h-m)	579.9	579.8	579.8	579.7	
27	Flow Regime		Transition	Transition	Transition	Transition	
28	Water Flow Regime						
29	Liquid HoldUp		0.3341	0.3341	0.2934	0.2934	
30	Water HoldUp		---	---	---	---	
31	Friction Gradient	(kPa/m)	6.077e-003	6.077e-003	4.652e-003	4.652e-003	
32	Static Gradient	(kPa/m)	4.626e-002	4.626e-002	1.077e-002	1.077e-002	
33	Accel. Gradient	(kPa/m)	3.755e-006	3.756e-006	1.107e-006	1.107e-006	
34	Liq. Reynolds		2.106e+005 *	2.106e+005 *	2.106e+005 *	2.106e+005 *	
35	Vap. Reynolds		9.156e+006 *	9.156e+006 *	9.157e+006 *	9.157e+006 *	
36	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
37	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172	
38	Deposit Thickness	(mm)	---	---	---	---	
39	Deposit Volume	(m3)	---	---	---	---	
40	Length	(m)	2.393e+004	2.394e+004	2.394e+004	2.395e+004	
41	Elevation	(m)	-0.1740	-0.1520	-0.1300	-0.1280	
42	Pressure	(bar)	89.24	89.24	89.24	89.24	
43	Temperature	(C)	51.76	51.76	51.76	51.75	
44	Vapour Fraction		0.8698	0.8698	0.8698	0.8698	
45	Heat Transferred	(kJ/h-m)	579.7	579.6	579.6	579.5	
46	Flow Regime		Transition	Transition	Transition	Transition	
47	Water Flow Regime						
48	Liquid HoldUp		0.2934	0.2934	0.2934	0.2806	
49	Water HoldUp		---	---	---	---	
50	Friction Gradient	(kPa/m)	4.652e-003	4.652e-003	4.652e-003	5.542e-003	
51	Static Gradient	(kPa/m)	1.077e-002	1.077e-002	1.077e-002	9.489e-004	
52	Accel. Gradient	(kPa/m)	1.107e-006	1.107e-006	1.107e-006	4.657e-007	
53	Liq. Reynolds		2.106e+005 *	2.106e+005 *	2.106e+005 *	2.106e+005 *	
54	Vap. Reynolds		9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *	
55	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
56	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172	
57	Deposit Thickness	(mm)	---	---	---	---	
58	Deposit Volume	(m3)	---	---	---	---	
59	Length	(m)	2.395e+004	2.396e+004	2.396e+004	2.397e+004	
60	Elevation	(m)	-0.1260	-0.1240	-0.1220	-0.1200	
61	Pressure	(bar)	89.24	89.24	89.24	89.24	
62	Temperature	(C)	51.75	51.75	51.75	51.75	
63	Vapour Fraction		0.8698	0.8698	0.8698	0.8698	
64	Heat Transferred	(kJ/h-m)	579.5	579.4	579.4	579.3	
65	Flow Regime		Transition	Transition	Transition	Transition	
66	Water Flow Regime						
67	Liquid HoldUp		0.2806	0.2806	0.2806	0.2807	
68	Water HoldUp		---	---	---	---	
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1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	5.542e-003	5.542e-003	5.542e-003	5.542e-003
14	Static Gradient	(kPa/m)	9.489e-004	9.489e-004	9.489e-004	9.490e-004
15	Accel. Gradient	(kPa/m)	4.657e-007	4.657e-007	4.657e-007	4.657e-007
16	Liq. Reynolds		2.106e+005 *	2.106e+005 *	2.106e+005 *	2.106e+005 *
17	Vap. Reynolds		9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
18	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
19	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	2.397e+004	2.398e+004	2.398e+004	2.399e+004
23	Elevation	(m)	-0.1280	-0.1360	-0.1440	-0.1520
24	Pressure	(bar)	89.24	89.24	89.24	89.24
25	Temperature	(C)	51.74	51.74	51.74	51.74
26	Vapour Fraction		0.8698	0.8698	0.8698	0.8698
27	Heat Transferred	(kJ/h-m)	579.3	579.3	579.2	579.2
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2056	0.2056	0.2056	0.2056
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	6.878e-003	6.878e-003	6.878e-003	6.878e-003
33	Static Gradient	(kPa/m)	-3.083e-003	-3.083e-003	-3.083e-003	-3.083e-003
34	Accel. Gradient	(kPa/m)	2.723e-007	2.723e-007	2.723e-007	2.723e-007
35	Liq. Reynolds		2.106e+005 *	2.106e+005 *	2.106e+005 *	2.106e+005 *
36	Vap. Reynolds		9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
37	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
38	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	2.399e+004	2.400e+004	2.400e+004	2.401e+004
42	Elevation	(m)	-0.1600	-0.1920	-0.2240	-0.2560
43	Pressure	(bar)	89.24	89.24	89.24	89.24
44	Temperature	(C)	51.74	51.74	51.74	51.73
45	Vapour Fraction		0.8698	0.8698	0.8698	0.8698
46	Heat Transferred	(kJ/h-m)	579.1	579.1	579.1	579.0
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2056	0.2012	0.2012	0.2012
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	6.878e-003	6.914e-003	6.914e-003	6.914e-003
52	Static Gradient	(kPa/m)	-3.083e-003	-1.216e-002	-1.216e-002	-1.216e-002
53	Accel. Gradient	(kPa/m)	2.723e-007	-3.766e-007	-3.766e-007	-3.766e-007
54	Liq. Reynolds		2.106e+005 *	2.106e+005 *	2.106e+005 *	2.106e+005 *
55	Vap. Reynolds		9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
56	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
57	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	2.401e+004	2.402e+004	2.402e+004	2.403e+004
61	Elevation	(m)	-0.2880	-0.3200	-0.3560	-0.3920
62	Pressure	(bar)	89.24	89.24	89.24	89.24
63	Temperature	(C)	51.73	51.73	51.73	51.73
64	Vapour Fraction		0.8698	0.8698	0.8698	0.8698
65	Heat Transferred	(kJ/h-m)	579.0	578.9	578.9	578.9
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2012	0.2012	0.2004	0.2004
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	6.914e-003	6.914e-003	6.920e-003	6.920e-003
15	Static Gradient (kPa/m)	-1.216e-002	-1.216e-002	-1.365e-002	-1.365e-002
16	Accel. Gradient (kPa/m)	-3.766e-007	-3.767e-007	-4.830e-007	-4.830e-007
17	Liq. Reynolds	2.106e+005 *	2.106e+005 *	2.106e+005 *	2.106e+005 *
18	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
19	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
20	Vapour Velocity (m/s)	2.172	2.171	2.171	2.171
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	2.403e+004	2.404e+004	2.404e+004	2.405e+004
24	Elevation (m)	-0.4280	-0.4640	-0.5000	-0.4540
25	Pressure (bar)	89.24	89.24	89.24	89.24
26	Temperature (C)	51.73	51.72	51.72	51.72
27	Vapour Fraction	0.8698	0.8698	0.8698	0.8698
28	Heat Transferred (kJ/h-m)	578.8	578.8	578.7	578.7
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2004	0.2004	0.2004	0.3087
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	6.920e-003	6.920e-003	6.920e-003	6.000e-003
34	Static Gradient (kPa/m)	-1.365e-002	-1.365e-002	-1.365e-002	2.336e-002
35	Accel. Gradient (kPa/m)	-4.830e-007	-4.830e-007	-4.830e-007	2.106e-006
36	Liq. Reynolds	2.106e+005 *	2.106e+005 *	2.106e+005 *	2.106e+005 *
37	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
38	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
39	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	2.405e+004	2.406e+004	2.406e+004	2.407e+004
43	Elevation (m)	-0.4080	-0.3620	-0.3160	-0.2700
44	Pressure (bar)	89.24	89.24	89.24	89.24
45	Temperature (C)	51.72	51.72	51.71	51.71
46	Vapour Fraction	0.8698	0.8698	0.8698	0.8698
47	Heat Transferred (kJ/h-m)	578.6	578.6	578.5	578.4
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.3087	0.3087	0.3087	0.3087
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	6.000e-003	6.000e-003	6.000e-003	6.000e-003
53	Static Gradient (kPa/m)	2.336e-002	2.336e-002	2.336e-002	2.336e-002
54	Accel. Gradient (kPa/m)	2.106e-006	2.106e-006	2.106e-006	2.106e-006
55	Liq. Reynolds	2.106e+005 *	2.106e+005 *	2.106e+005 *	2.106e+005 *
56	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
57	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
58	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	2.407e+004	2.408e+004	2.408e+004	2.409e+004
62	Elevation (m)	-0.2580	-0.2460	-0.2340	-0.2220
63	Pressure (bar)	89.24	89.24	89.23	89.23
64	Temperature (C)	51.71	51.71	51.71	51.71
65	Vapour Fraction	0.8698	0.8698	0.8698	0.8698
66	Heat Transferred (kJ/h-m)	578.4	578.3	578.3	578.2
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.2871	0.2871	0.2871	0.2871
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	5.054e-003	5.054e-003	5.054e-003	5.054e-003
16	Static Gradient (kPa/m)	5.794e-003	5.794e-003	5.795e-003	5.795e-003
17	Accel. Gradient (kPa/m)	7.784e-007	7.784e-007	7.784e-007	7.784e-007
18	Liq. Reynolds	2.106e+005 *	2.106e+005 *	2.106e+005 *	2.106e+005 *
19	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
20	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
21	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	2.409e+004	2.410e+004	2.410e+004	2.411e+004
25	Elevation (m)	-0.2100	-0.1980	-0.1860	-0.1740
26	Pressure (bar)	89.23	89.23	89.23	89.23
27	Temperature (C)	51.70	51.70	51.70	51.70
28	Vapour Fraction	0.8697	0.8697	0.8697	0.8697
29	Heat Transferred (kJ/h-m)	578.2	578.1	578.1	578.1
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2871	0.2870	0.2870	0.2870
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	5.054e-003	5.055e-003	5.055e-003	5.055e-003
35	Static Gradient (kPa/m)	5.795e-003	5.785e-003	5.785e-003	5.785e-003
36	Accel. Gradient (kPa/m)	7.784e-007	7.778e-007	7.778e-007	7.778e-007
37	Liq. Reynolds	2.106e+005 *	2.106e+005 *	2.106e+005 *	2.106e+005 *
38	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
39	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
40	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	2.411e+004	2.412e+004	2.412e+004	2.413e+004
44	Elevation (m)	-0.1620	-0.1500	-0.1400	-0.1300
45	Pressure (bar)	89.23	89.23	89.23	89.23
46	Temperature (C)	51.70	51.69	51.69	51.69
47	Vapour Fraction	0.8697	0.8697	0.8697	0.8697
48	Heat Transferred (kJ/h-m)	578.0	578.0	577.9	577.9
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2870	0.2870	0.2858	0.2858
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	5.055e-003	5.055e-003	5.149e-003	5.149e-003
54	Static Gradient (kPa/m)	5.785e-003	5.785e-003	4.806e-003	4.806e-003
55	Accel. Gradient (kPa/m)	7.778e-007	7.778e-007	7.143e-007	7.143e-007
56	Liq. Reynolds	2.106e+005 *	2.106e+005 *	2.106e+005 *	2.106e+005 *
57	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
58	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
59	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	2.413e+004	2.414e+004	2.414e+004	2.415e+004
63	Elevation (m)	-0.1200	-0.1100	-0.1000	-9.800e-002
64	Pressure (bar)	89.23	89.23	89.23	89.23
65	Temperature (C)	51.69	51.69	51.69	51.68
66	Vapour Fraction	0.8697	0.8697	0.8697	0.8697
67	Heat Transferred (kJ/h-m)	577.8	577.8	577.7	577.7
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 389 of 937

1			Case Name:	Simulation - 386MMSCFD_100barg.hsc	
2			Unit Set:	Mixed	
3	 Epic Atlantic Limited		Date/Time:	Mon Jun 13 15:02:44 2022	
4					
5					
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2858	0.2858	0.2858	0.2807
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	5.149e-003	5.149e-003	5.149e-003	5.540e-003
17	Static Gradient (kPa/m)	4.806e-003	4.806e-003	4.806e-003	9.491e-004
18	Accel. Gradient (kPa/m)	7.143e-007	7.143e-007	7.143e-007	4.656e-007
19	Liq. Reynolds	2.106e+005 *	2.106e+005 *	2.106e+005 *	2.106e+005 *
20	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
21	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
22	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	2.415e+004	2.416e+004	2.416e+004	2.417e+004
26	Elevation (m)	-9.600e-002	-9.400e-002	-9.200e-002	-9.000e-002
27	Pressure (bar)	89.23	89.23	89.23	89.23
28	Temperature (C)	51.68	51.68	51.68	51.68
29	Vapour Fraction	0.8697	0.8697	0.8697	0.8697
30	Heat Transferred (kJ/h-m)	577.6	577.6	577.5	577.5
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2807	0.2807	0.2807	0.2807
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	5.540e-003	5.540e-003	5.540e-003	5.540e-003
36	Static Gradient (kPa/m)	9.491e-004	9.491e-004	9.491e-004	9.491e-004
37	Accel. Gradient (kPa/m)	4.656e-007	4.656e-007	4.656e-007	4.656e-007
38	Liq. Reynolds	2.106e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *
39	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
40	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
41	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	2.417e+004	2.418e+004	2.418e+004	2.419e+004
45	Elevation (m)	-9.800e-002	-0.1060	-0.1140	-0.1220
46	Pressure (bar)	89.23	89.23	89.23	89.23
47	Temperature (C)	51.68	51.67	51.67	51.67
48	Vapour Fraction	0.8697	0.8697	0.8697	0.8697
49	Heat Transferred (kJ/h-m)	577.5	577.4	577.4	577.3
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2057	0.2057	0.2057	0.2057
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	6.877e-003	6.877e-003	6.877e-003	6.877e-003
55	Static Gradient (kPa/m)	-3.084e-003	-3.084e-003	-3.084e-003	-3.084e-003
56	Accel. Gradient (kPa/m)	2.722e-007	2.722e-007	2.721e-007	2.721e-007
57	Liq. Reynolds	2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *
58	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
59	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
60	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	2.419e+004	2.420e+004	2.420e+004	2.421e+004
64	Elevation (m)	-0.1300	-0.1380	-0.1460	-0.1540
65	Pressure (bar)	89.23	89.23	89.23	89.23
66	Temperature (C)	51.67	51.67	51.67	51.66
67	Vapour Fraction	0.8697	0.8697	0.8697	0.8697
68	Heat Transferred (kJ/h-m)	577.3	577.2	577.2	577.1
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 390 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2057	0.2057	0.2057	0.2057
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	6.876e-003	6.876e-003	6.876e-003	6.876e-003
18	Static Gradient (kPa/m)	-3.084e-003	-3.084e-003	-3.084e-003	-3.084e-003
19	Accel. Gradient (kPa/m)	2.721e-007	2.721e-007	2.721e-007	2.721e-007
20	Liq. Reynolds	2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *
21	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
22	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
23	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	2.421e+004	2.422e+004	2.422e+004	2.423e+004
27	Elevation (m)	-0.1620	-0.1700	-0.1580	-0.1460
28	Pressure (bar)	89.23	89.22	89.22	89.22
29	Temperature (C)	51.66	51.66	51.66	51.66
30	Vapour Fraction	0.8697	0.8697	0.8697	0.8697
31	Heat Transferred (kJ/h-m)	577.1	577.0	577.0	576.9
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2057	0.2057	0.2871	0.2871
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	6.876e-003	6.876e-003	5.054e-003	5.054e-003
37	Static Gradient (kPa/m)	-3.084e-003	-3.084e-003	5.786e-003	5.786e-003
38	Accel. Gradient (kPa/m)	2.721e-007	2.721e-007	7.778e-007	7.778e-007
39	Liq. Reynolds	2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *
40	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
41	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
42	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	2.423e+004	2.424e+004	2.424e+004	2.425e+004
46	Elevation (m)	-0.1340	-0.1220	-0.1100	-0.1320
47	Pressure (bar)	89.22	89.22	89.22	89.22
48	Temperature (C)	51.65	51.65	51.65	51.65
49	Vapour Fraction	0.8697	0.8697	0.8697	0.8697
50	Heat Transferred (kJ/h-m)	576.9	576.8	576.8	576.8
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2871	0.2871	0.2871	0.2031
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	5.054e-003	5.054e-003	5.054e-003	6.897e-003
56	Static Gradient (kPa/m)	5.786e-003	5.786e-003	5.786e-003	-8.412e-003
57	Accel. Gradient (kPa/m)	7.778e-007	7.778e-007	7.778e-007	-1.087e-007
58	Liq. Reynolds	2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *
59	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
60	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
61	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	2.425e+004	2.426e+004	2.426e+004	2.427e+004
65	Elevation (m)	-0.1540	-0.1760	-0.1980	-0.2200
66	Pressure (bar)	89.22	89.22	89.22	89.22
67	Temperature (C)	51.65	51.65	51.64	51.64
68	Vapour Fraction	0.8697	0.8697	0.8697	0.8697
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 391 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022	
6						
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Heat Transferred (kJ/h-m)	576.7	576.7	576.6	576.6	
14	Flow Regime	Transition	Transition	Transition	Transition	
15	Water Flow Regime					
16	Liquid HoldUp	0.2031	0.2031	0.2031	0.2031	
17	Water HoldUp	---	---	---	---	
18	Friction Gradient (kPa/m)	6.897e-003	6.897e-003	6.897e-003	6.897e-003	
19	Static Gradient (kPa/m)	-8.413e-003	-8.413e-003	-8.413e-003	-8.413e-003	
20	Accel. Gradient (kPa/m)	-1.087e-007	-1.087e-007	-1.087e-007	-1.087e-007	
21	Liq. Reynolds	2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *	
22	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *	
23	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044	
24	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171	
25	Deposit Thickness (mm)	---	---	---	---	
26	Deposit Volume (m3)	---	---	---	---	
27	Length (m)	2.427e+004	2.428e+004	2.428e+004	2.429e+004	
28	Elevation (m)	-0.2040	-0.1880	-0.1720	-0.1560	
29	Pressure (bar)	89.22	89.22	89.22	89.22	
30	Temperature (C)	51.64	51.64	51.64	51.64	
31	Vapour Fraction	0.8697	0.8697	0.8697	0.8697	
32	Heat Transferred (kJ/h-m)	576.5	576.5	576.4	576.4	
33	Flow Regime	Transition	Transition	Transition	Transition	
34	Water Flow Regime					
35	Liquid HoldUp	0.2896	0.2896	0.2896	0.2896	
36	Water HoldUp	---	---	---	---	
37	Friction Gradient (kPa/m)	4.872e-003	4.872e-003	4.872e-003	4.872e-003	
38	Static Gradient (kPa/m)	7.728e-003	7.728e-003	7.728e-003	7.728e-003	
39	Accel. Gradient (kPa/m)	9.040e-007	9.040e-007	9.040e-007	9.040e-007	
40	Liq. Reynolds	2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *	
41	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *	
42	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044	
43	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171	
44	Deposit Thickness (mm)	---	---	---	---	
45	Deposit Volume (m3)	---	---	---	---	
46	Length (m)	2.429e+004	2.430e+004	2.430e+004	2.431e+004	
47	Elevation (m)	-0.1400	-0.1300	-0.1200	-0.1100	
48	Pressure (bar)	89.22	89.22	89.22	89.22	
49	Temperature (C)	51.63	51.63	51.63	51.63	
50	Vapour Fraction	0.8697	0.8697	0.8697	0.8697	
51	Heat Transferred (kJ/h-m)	576.3	576.3	576.2	576.2	
52	Flow Regime	Transition	Transition	Transition	Transition	
53	Water Flow Regime					
54	Liquid HoldUp	0.2896	0.2858	0.2858	0.2858	
55	Water HoldUp	---	---	---	---	
56	Friction Gradient (kPa/m)	4.872e-003	5.148e-003	5.148e-003	5.148e-003	
57	Static Gradient (kPa/m)	7.728e-003	4.807e-003	4.807e-003	4.807e-003	
58	Accel. Gradient (kPa/m)	9.040e-007	7.142e-007	7.142e-007	7.142e-007	
59	Liq. Reynolds	2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *	
60	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *	
61	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044	
62	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171	
63	Deposit Thickness (mm)	---	---	---	---	
64	Deposit Volume (m3)	---	---	---	---	
65	Length (m)	2.431e+004	2.432e+004	2.432e+004	2.433e+004	
66	Elevation (m)	-0.1000	-9.000e-002	-9.600e-002	-0.1020	
67	Pressure (bar)	89.22	89.22	89.22	89.22	
68	Temperature (C)	51.63	51.62	51.62	51.62	
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 392 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8697	0.8697	0.8697	0.8697
14	Heat Transferred (kJ/h-m)	576.1	576.1	576.1	576.0
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2858	0.2858	0.2060	0.2060
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	5.148e-003	5.148e-003	6.873e-003	6.873e-003
20	Static Gradient (kPa/m)	4.807e-003	4.807e-003	-2.316e-003	-2.316e-003
21	Accel. Gradient (kPa/m)	7.143e-007	7.143e-007	3.270e-007	3.270e-007
22	Liq. Reynolds	2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *
23	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
24	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
25	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	2.433e+004	2.434e+004	2.434e+004	2.435e+004
29	Elevation (m)	-0.1080	-0.1140	-0.1200	-0.1260
30	Pressure (bar)	89.22	89.22	89.22	89.22
31	Temperature (C)	51.62	51.62	51.62	51.61
32	Vapour Fraction	0.8697	0.8697	0.8697	0.8697
33	Heat Transferred (kJ/h-m)	576.0	575.9	575.9	575.8
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2060	0.2061	0.2061	0.2061
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	6.873e-003	6.873e-003	6.873e-003	6.873e-003
39	Static Gradient (kPa/m)	-2.316e-003	-2.316e-003	-2.316e-003	-2.316e-003
40	Accel. Gradient (kPa/m)	3.270e-007	3.270e-007	3.269e-007	3.269e-007
41	Liq. Reynolds	2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *
42	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
43	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
44	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	2.435e+004	2.436e+004	2.436e+004	2.437e+004
48	Elevation (m)	-0.1320	-0.1380	-0.1440	-0.1500
49	Pressure (bar)	89.22	89.22	89.21	89.21
50	Temperature (C)	51.61	51.61	51.61	51.61
51	Vapour Fraction	0.8697	0.8697	0.8697	0.8697
52	Heat Transferred (kJ/h-m)	575.8	575.8	575.7	575.7
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2061	0.2061	0.2061	0.2061
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	6.873e-003	6.873e-003	6.872e-003	6.872e-003
58	Static Gradient (kPa/m)	-2.316e-003	-2.316e-003	-2.316e-003	-2.316e-003
59	Accel. Gradient (kPa/m)	3.269e-007	3.269e-007	3.269e-007	3.269e-007
60	Liq. Reynolds	2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *
61	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
62	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
63	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	2.437e+004	2.437e+004	2.437e+004	2.437e+004
67	Elevation (m)	-0.1420	-0.1340	-0.1260	-0.1180
68	Pressure (bar)	89.21	89.21	89.21	89.21
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	51.61	51.61	51.61	51.61
14	Vapour Fraction	0.8697	0.8697	0.8697	0.8697
15	Heat Transferred (kJ/h-m)	575.6	575.6	575.6	575.6
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2922	0.2922	0.2922	0.2922
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	4.687e-003	4.687e-003	4.687e-003	4.687e-003
21	Static Gradient (kPa/m)	9.765e-003	9.765e-003	9.765e-003	9.765e-003
22	Accel. Gradient (kPa/m)	1.037e-006	1.037e-006	1.037e-006	1.037e-006
23	Liq. Reynolds	2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *
24	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
25	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
26	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	2.438e+004	2.438e+004	2.438e+004	2.438e+004
30	Elevation (m)	-0.1100	-0.1140	-0.1180	-0.1220
31	Pressure (bar)	89.21	89.21	89.21	89.21
32	Temperature (C)	51.60	51.60	51.60	51.60
33	Vapour Fraction	0.8697	0.8697	0.8697	0.8697
34	Heat Transferred (kJ/h-m)	575.6	575.5	575.5	575.5
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2922	0.2053	0.2053	0.2053
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	4.687e-003	6.878e-003	6.878e-003	6.878e-003
40	Static Gradient (kPa/m)	9.765e-003	-3.851e-003	-3.851e-003	-3.851e-003
41	Accel. Gradient (kPa/m)	1.037e-006	2.172e-007	2.172e-007	2.172e-007
42	Liq. Reynolds	2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *
43	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
44	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
45	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	2.438e+004	2.439e+004	2.439e+004	2.440e+004
49	Elevation (m)	-0.1260	-0.1300	-0.1300	-0.1300
50	Pressure (bar)	89.21	89.21	89.21	89.21
51	Temperature (C)	51.60	51.60	51.60	51.60
52	Vapour Fraction	0.8697	0.8697	0.8697	0.8697
53	Heat Transferred (kJ/h-m)	575.5	575.5	575.4	575.4
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2053	0.2053	0.3024	0.3024
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	6.878e-003	6.878e-003	5.879e-003	5.879e-003
59	Static Gradient (kPa/m)	-3.851e-003	-3.851e-003	0.0000	0.0000
60	Accel. Gradient (kPa/m)	2.172e-007	2.172e-007	4.218e-007	4.218e-007
61	Liq. Reynolds	2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *
62	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
63	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
64	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	2.440e+004	2.441e+004	2.442e+004	2.442e+004
68	Elevation (m)	-0.1300	-0.1300	-0.1300	-0.1360
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022		
6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	89.21	89.21	89.21	89.21	89.21	
14	Temperature (C)	51.59	51.59	51.59	51.59	51.59	
15	Vapour Fraction	0.8697	0.8697	0.8697	0.8697	0.8697	
16	Heat Transferred (kJ/h-m)	575.3	575.2	575.2	575.2	575.2	
17	Flow Regime	Transition	Transition	Transition	Transition	Transition	
18	Water Flow Regime						
19	Liquid HoldUp	0.3025	0.3025	0.3025	0.3025	0.2061	
20	Water HoldUp	---	---	---	---	---	
21	Friction Gradient (kPa/m)	5.879e-003	5.879e-003	5.879e-003	5.879e-003	6.872e-003	
22	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000	-2.316e-003	
23	Accel. Gradient (kPa/m)	4.218e-007	4.218e-007	4.218e-007	4.218e-007	3.269e-007	
24	Liq. Reynolds	2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *	
25	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *	
26	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044	0.2044	
27	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171	2.171	
28	Deposit Thickness (mm)	---	---	---	---	---	
29	Deposit Volume (m3)	---	---	---	---	---	
30	Length (m)	2.443e+004	2.443e+004	2.444e+004	2.444e+004	2.444e+004	
31	Elevation (m)	-0.1420	-0.1480	-0.1540	-0.1540	-0.1600	
32	Pressure (bar)	89.21	89.21	89.21	89.21	89.21	
33	Temperature (C)	51.59	51.59	51.58	51.58	51.58	
34	Vapour Fraction	0.8697	0.8697	0.8697	0.8697	0.8697	
35	Heat Transferred (kJ/h-m)	575.1	575.1	575.0	575.0	575.0	
36	Flow Regime	Transition	Transition	Transition	Transition	Transition	
37	Water Flow Regime						
38	Liquid HoldUp	0.2061	0.2061	0.2061	0.2061	0.2061	
39	Water HoldUp	---	---	---	---	---	
40	Friction Gradient (kPa/m)	6.872e-003	6.872e-003	6.872e-003	6.872e-003	6.872e-003	
41	Static Gradient (kPa/m)	-2.316e-003	-2.316e-003	-2.316e-003	-2.316e-003	-2.316e-003	
42	Accel. Gradient (kPa/m)	3.269e-007	3.269e-007	3.269e-007	3.269e-007	3.269e-007	
43	Liq. Reynolds	2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *	
44	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *	
45	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044	0.2044	
46	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171	2.171	
47	Deposit Thickness (mm)	---	---	---	---	---	
48	Deposit Volume (m3)	---	---	---	---	---	
49	Length (m)	2.445e+004	2.445e+004	2.446e+004	2.446e+004	2.446e+004	
50	Elevation (m)	-0.1760	-0.1920	-0.2080	-0.2080	-0.2240	
51	Pressure (bar)	89.21	89.21	89.21	89.21	89.21	
52	Temperature (C)	51.58	51.58	51.58	51.58	51.58	
53	Vapour Fraction	0.8697	0.8697	0.8697	0.8697	0.8697	
54	Heat Transferred (kJ/h-m)	574.9	574.9	574.8	574.8	574.8	
55	Flow Regime	Transition	Transition	Transition	Transition	Transition	
56	Water Flow Regime						
57	Liquid HoldUp	0.2042	0.2042	0.2042	0.2042	0.2042	
58	Water HoldUp	---	---	---	---	---	
59	Friction Gradient (kPa/m)	6.887e-003	6.887e-003	6.887e-003	6.887e-003	6.887e-003	
60	Static Gradient (kPa/m)	-6.140e-003	-6.141e-003	-6.141e-003	-6.141e-003	-6.141e-003	
61	Accel. Gradient (kPa/m)	5.355e-008	5.354e-008	5.353e-008	5.353e-008	5.353e-008	
62	Liq. Reynolds	2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *	
63	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *	
64	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044	0.2044	
65	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171	2.170	
66	Deposit Thickness (mm)	---	---	---	---	---	
67	Deposit Volume (m3)	---	---	---	---	---	
68	Length (m)	2.447e+004	2.447e+004	2.448e+004	2.448e+004	2.448e+004	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Elevation	(m)	-0.2400	-0.2500	-0.2600	-0.2700	
14	Pressure	(bar)	89.21	89.21	89.21	89.21	
15	Temperature	(C)	51.57	51.57	51.57	51.57	
16	Vapour Fraction		0.8697	0.8697	0.8697	0.8697	
17	Heat Transferred	(kJ/h-m)	574.8	574.7	574.7	574.6	
18	Flow Regime		Transition	Transition	Transition	Transition	
19	Water Flow Regime						
20	Liquid HoldUp		0.2042	0.2053	0.2053	0.2053	
21	Water HoldUp		---	---	---	---	
22	Friction Gradient	(kPa/m)	6.887e-003	6.878e-003	6.878e-003	6.877e-003	
23	Static Gradient	(kPa/m)	-6.141e-003	-3.848e-003	-3.848e-003	-3.848e-003	
24	Accel. Gradient	(kPa/m)	5.352e-008	2.173e-007	2.173e-007	2.173e-007	
25	Liq. Reynolds		2.105e+005 *	2.105e+005 *	2.105e+005 *	2.105e+005 *	
26	Vap. Reynolds		9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *	
27	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
28	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
29	Deposit Thickness	(mm)	---	---	---	---	
30	Deposit Volume	(m3)	---	---	---	---	
31	Length	(m)	2.449e+004	2.449e+004	2.450e+004	2.450e+004	
32	Elevation	(m)	-0.2800	-0.2900	-0.2720	-0.2540	
33	Pressure	(bar)	89.21	89.21	89.21	89.21	
34	Temperature	(C)	51.57	51.57	51.56	51.56	
35	Vapour Fraction		0.8697	0.8697	0.8697	0.8697	
36	Heat Transferred	(kJ/h-m)	574.6	574.5	574.5	574.4	
37	Flow Regime		Transition	Transition	Transition	Transition	
38	Water Flow Regime						
39	Liquid HoldUp		0.2053	0.2053	0.2910	0.2910	
40	Water HoldUp		---	---	---	---	
41	Friction Gradient	(kPa/m)	6.877e-003	6.877e-003	4.776e-003	4.776e-003	
42	Static Gradient	(kPa/m)	-3.848e-003	-3.848e-003	8.763e-003	8.763e-003	
43	Accel. Gradient	(kPa/m)	2.173e-007	2.173e-007	9.713e-007	9.713e-007	
44	Liq. Reynolds		2.105e+005 *	2.104e+005 *	2.104e+005 *	2.104e+005 *	
45	Vap. Reynolds		9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *	
46	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
47	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
48	Deposit Thickness	(mm)	---	---	---	---	
49	Deposit Volume	(m3)	---	---	---	---	
50	Length	(m)	2.451e+004	2.451e+004	2.452e+004	2.452e+004	
51	Elevation	(m)	-0.2360	-0.2180	-0.2000	-0.2120	
52	Pressure	(bar)	89.21	89.21	89.21	89.21	
53	Temperature	(C)	51.56	51.56	51.56	51.56	
54	Vapour Fraction		0.8697	0.8697	0.8697	0.8697	
55	Heat Transferred	(kJ/h-m)	574.4	574.3	574.3	574.3	
56	Flow Regime		Transition	Transition	Transition	Transition	
57	Water Flow Regime						
58	Liquid HoldUp		0.2910	0.2910	0.2910	0.2050	
59	Water HoldUp		---	---	---	---	
60	Friction Gradient	(kPa/m)	4.776e-003	4.776e-003	4.776e-003	6.880e-003	
61	Static Gradient	(kPa/m)	8.763e-003	8.763e-003	8.763e-003	-4.616e-003	
62	Accel. Gradient	(kPa/m)	9.713e-007	9.713e-007	9.713e-007	1.624e-007	
63	Liq. Reynolds		2.104e+005 *	2.104e+005 *	2.104e+005 *	2.104e+005 *	
64	Vap. Reynolds		9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *	
65	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
66	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
67	Deposit Thickness	(mm)	---	---	---	---	
68	Deposit Volume	(m3)	---	---	---	---	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length	(m)	2.453e+004	2.453e+004	2.454e+004	2.454e+004	
14	Elevation	(m)	-0.2240	-0.2360	-0.2480	-0.2600	
15	Pressure	(bar)	89.21	89.21	89.21	89.21	
16	Temperature	(C)	51.55	51.55	51.55	51.55	
17	Vapour Fraction		0.8697	0.8697	0.8697	0.8697	
18	Heat Transferred	(kJ/h-m)	574.2	574.2	574.1	574.1	
19	Flow Regime		Transition	Transition	Transition	Transition	
20	Water Flow Regime						
21	Liquid HoldUp		0.2050	0.2050	0.2050	0.2050	
22	Water HoldUp		---	---	---	---	
23	Friction Gradient	(kPa/m)	6.880e-003	6.880e-003	6.880e-003	6.880e-003	
24	Static Gradient	(kPa/m)	-4.616e-003	-4.616e-003	-4.616e-003	-4.616e-003	
25	Accel. Gradient	(kPa/m)	1.624e-007	1.624e-007	1.624e-007	1.624e-007	
26	Liq. Reynolds		2.104e+005 *	2.104e+005 *	2.104e+005 *	2.104e+005 *	
27	Vap. Reynolds		9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *	
28	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
29	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
30	Deposit Thickness	(mm)	---	---	---	---	
31	Deposit Volume	(m3)	---	---	---	---	
32	Length	(m)	2.455e+004	2.455e+004	2.456e+004	2.456e+004	
33	Elevation	(m)	-0.2520	-0.2440	-0.2360	-0.2280	
34	Pressure	(bar)	89.20	89.20	89.20	89.20	
35	Temperature	(C)	51.55	51.54	51.54	51.54	
36	Vapour Fraction		0.8697	0.8697	0.8697	0.8697	
37	Heat Transferred	(kJ/h-m)	574.0	574.0	573.9	573.9	
38	Flow Regime		Transition	Transition	Transition	Transition	
39	Water Flow Regime						
40	Liquid HoldUp		0.2846	0.2846	0.2846	0.2846	
41	Water HoldUp		---	---	---	---	
42	Friction Gradient	(kPa/m)	5.242e-003	5.242e-003	5.242e-003	5.242e-003	
43	Static Gradient	(kPa/m)	3.834e-003	3.834e-003	3.834e-003	3.834e-003	
44	Accel. Gradient	(kPa/m)	6.511e-007	6.511e-007	6.511e-007	6.511e-007	
45	Liq. Reynolds		2.104e+005 *	2.104e+005 *	2.104e+005 *	2.104e+005 *	
46	Vap. Reynolds		9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *	
47	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
48	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
49	Deposit Thickness	(mm)	---	---	---	---	
50	Deposit Volume	(m3)	---	---	---	---	
51	Length	(m)	2.457e+004	2.457e+004	2.458e+004	2.458e+004	
52	Elevation	(m)	-0.2200	-0.2120	-0.2040	-0.1960	
53	Pressure	(bar)	89.20	89.20	89.20	89.20	
54	Temperature	(C)	51.54	51.54	51.54	51.53	
55	Vapour Fraction		0.8697	0.8697	0.8697	0.8697	
56	Heat Transferred	(kJ/h-m)	573.8	573.8	573.7	573.7	
57	Flow Regime		Transition	Transition	Transition	Transition	
58	Water Flow Regime						
59	Liquid HoldUp		0.2846	0.2846	0.2846	0.2846	
60	Water HoldUp		---	---	---	---	
61	Friction Gradient	(kPa/m)	5.242e-003	5.242e-003	5.242e-003	5.242e-003	
62	Static Gradient	(kPa/m)	3.834e-003	3.834e-003	3.834e-003	3.834e-003	
63	Accel. Gradient	(kPa/m)	6.511e-007	6.511e-007	6.511e-007	6.511e-007	
64	Liq. Reynolds		2.104e+005 *	2.104e+005 *	2.104e+005 *	2.104e+005 *	
65	Vap. Reynolds		9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *	
66	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
67	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
68	Deposit Thickness	(mm)	---	---	---	---	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:02:44 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	2.459e+004	2.459e+004	2.460e+004	2.460e+004		
15	Elevation (m)	-0.1880	-0.1800	-0.1720	-0.1640		
16	Pressure (bar)	89.20	89.20	89.20	89.20		
17	Temperature (C)	51.53	51.53	51.53	51.53		
18	Vapour Fraction	0.8697	0.8697	0.8697	0.8697		
19	Heat Transferred (kJ/h-m)	573.6	573.6	573.5	573.5		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	0.2846	0.2846	0.2846	0.2846		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	5.242e-003	5.242e-003	5.242e-003	5.242e-003		
25	Static Gradient (kPa/m)	3.834e-003	3.834e-003	3.834e-003	3.834e-003		
26	Accel. Gradient (kPa/m)	6.511e-007	6.511e-007	6.511e-007	6.511e-007		
27	Liq. Reynolds	2.104e+005 *	2.104e+005 *	2.104e+005 *	2.104e+005 *		
28	Vap. Reynolds	9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *		
29	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
30	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	2.461e+004	2.461e+004	2.462e+004	2.462e+004		
34	Elevation (m)	-0.1560	-0.1480	-0.1400	-0.1300		
35	Pressure (bar)	89.20	89.20	89.20	89.20		
36	Temperature (C)	51.53	51.52	51.52	51.52		
37	Vapour Fraction	0.8696	0.8696	0.8696	0.8696		
38	Heat Transferred (kJ/h-m)	573.5	573.4	573.4	573.3		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.2846	0.2846	0.2846	0.2859		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	5.242e-003	5.242e-003	5.242e-003	5.146e-003		
44	Static Gradient (kPa/m)	3.834e-003	3.834e-003	3.834e-003	4.808e-003		
45	Accel. Gradient (kPa/m)	6.511e-007	6.511e-007	6.511e-007	7.141e-007		
46	Liq. Reynolds	2.104e+005 *	2.104e+005 *	2.104e+005 *	2.104e+005 *		
47	Vap. Reynolds	9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *		
48	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
49	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	2.463e+004	2.463e+004	2.464e+004	2.464e+004		
53	Elevation (m)	-0.1200	-0.1100	-0.1000	-9.000e-002		
54	Pressure (bar)	89.20	89.20	89.20	89.20		
55	Temperature (C)	51.52	51.52	51.51	51.51		
56	Vapour Fraction	0.8696	0.8696	0.8696	0.8696		
57	Heat Transferred (kJ/h-m)	573.3	573.2	573.2	573.1		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.2859	0.2859	0.2859	0.2859		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	5.146e-003	5.146e-003	5.146e-003	5.146e-003		
63	Static Gradient (kPa/m)	4.808e-003	4.808e-003	4.808e-003	4.808e-003		
64	Accel. Gradient (kPa/m)	7.141e-007	7.141e-007	7.141e-007	7.141e-007		
65	Liq. Reynolds	2.104e+005 *	2.104e+005 *	2.104e+005 *	2.104e+005 *		
66	Vap. Reynolds	9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *		
67	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044		
68	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Thickness	(mm)	---	---	---	---	
14	Deposit Volume	(m3)	---	---	---	---	
15	Length	(m)	2.465e+004	2.465e+004	2.466e+004	2.466e+004	
16	Elevation	(m)	-0.1160	-0.1420	-0.1680	-0.1940	
17	Pressure	(bar)	89.20	89.20	89.20	89.20	
18	Temperature	(C)	51.51	51.51	51.51	51.51	
19	Vapour Fraction		0.8696	0.8696	0.8696	0.8696	
20	Heat Transferred	(kJ/h-m)	573.1	573.1	573.0	573.0	
21	Flow Regime		Transition	Transition	Transition	Transition	
22	Water Flow Regime						
23	Liquid HoldUp		0.2024	0.2024	0.2024	0.2024	
24	Water HoldUp		---	---	---	---	
25	Friction Gradient	(kPa/m)	6.901e-003	6.901e-003	6.901e-003	6.901e-003	
26	Static Gradient	(kPa/m)	-9.922e-003	-9.922e-003	-9.923e-003	-9.923e-003	
27	Accel. Gradient	(kPa/m)	-2.168e-007	-2.168e-007	-2.168e-007	-2.168e-007	
28	Liq. Reynolds		2.104e+005 *	2.104e+005 *	2.104e+005 *	2.104e+005 *	
29	Vap. Reynolds		9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *	
30	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
31	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
32	Deposit Thickness	(mm)	---	---	---	---	
33	Deposit Volume	(m3)	---	---	---	---	
34	Length	(m)	2.467e+004	2.467e+004	2.468e+004	2.468e+004	
35	Elevation	(m)	-0.2200	-0.2920	-0.3640	-0.4360	
36	Pressure	(bar)	89.20	89.20	89.20	89.20	
37	Temperature	(C)	51.51	51.50	51.50	51.50	
38	Vapour Fraction		0.8696	0.8696	0.8696	0.8696	
39	Heat Transferred	(kJ/h-m)	572.9	572.9	572.9	572.8	
40	Flow Regime		Transition	Transition	Transition	Transition	
41	Water Flow Regime						
42	Liquid HoldUp		0.2024	0.1938	0.1938	0.1938	
43	Water HoldUp		---	---	---	---	
44	Friction Gradient	(kPa/m)	6.901e-003	6.977e-003	6.977e-003	6.976e-003	
45	Static Gradient	(kPa/m)	-9.923e-003	-2.675e-002	-2.675e-002	-2.675e-002	
46	Accel. Gradient	(kPa/m)	-2.168e-007	-1.418e-006	-1.418e-006	-1.418e-006	
47	Liq. Reynolds		2.104e+005 *	2.104e+005 *	2.104e+005 *	2.104e+005 *	
48	Vap. Reynolds		9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *	
49	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
50	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
51	Deposit Thickness	(mm)	---	---	---	---	
52	Deposit Volume	(m3)	---	---	---	---	
53	Length	(m)	2.469e+004	2.469e+004	2.470e+004	2.470e+004	
54	Elevation	(m)	-0.5080	-0.5800	-0.5100	-0.4400	
55	Pressure	(bar)	89.20	89.20	89.20	89.20	
56	Temperature	(C)	51.50	51.50	51.50	51.49	
57	Vapour Fraction		0.8696	0.8696	0.8696	0.8696	
58	Heat Transferred	(kJ/h-m)	572.8	572.8	572.7	572.6	
59	Flow Regime		Transition	Transition	Transition	Transition	
60	Water Flow Regime						
61	Liquid HoldUp		0.1938	0.1938	0.3241	0.3241	
62	Water HoldUp		---	---	---	---	
63	Friction Gradient	(kPa/m)	6.976e-003	6.976e-003	6.069e-003	6.069e-003	
64	Static Gradient	(kPa/m)	-2.675e-002	-2.675e-002	3.684e-002	3.684e-002	
65	Accel. Gradient	(kPa/m)	-1.418e-006	-1.418e-006	3.077e-006	3.078e-006	
66	Liq. Reynolds		2.104e+005 *	2.104e+005 *	2.104e+005 *	2.104e+005 *	
67	Vap. Reynolds		9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *	
68	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
14	Deposit Thickness (mm)	---	---	---	---
15	Deposit Volume (m3)	---	---	---	---
16	Length (m)	2.471e+004	2.471e+004	2.472e+004	2.472e+004
17	Elevation (m)	-0.3700	-0.3000	-0.2300	-0.1980
18	Pressure (bar)	89.19	89.19	89.19	89.19
19	Temperature (C)	51.49	51.49	51.49	51.49
20	Vapour Fraction	0.8696	0.8696	0.8696	0.8696
21	Heat Transferred (kJ/h-m)	572.6	572.5	572.5	572.4
22	Flow Regime	Transition	Transition	Transition	Transition
23	Water Flow Regime				
24	Liquid HoldUp	0.3241	0.3241	0.3241	0.2999
25	Water HoldUp	---	---	---	---
26	Friction Gradient (kPa/m)	6.069e-003	6.069e-003	6.069e-003	5.773e-003
27	Static Gradient (kPa/m)	3.684e-002	3.684e-002	3.683e-002	1.592e-002
28	Accel. Gradient (kPa/m)	3.078e-006	3.078e-006	3.078e-006	1.556e-006
29	Liq. Reynolds	2.104e+005 *	2.104e+005 *	2.104e+005 *	2.104e+005 *
30	Vap. Reynolds	9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *
31	Liquid Velocity (m/s)	0.2045	0.2044	0.2044	0.2044
32	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
33	Deposit Thickness (mm)	---	---	---	---
34	Deposit Volume (m3)	---	---	---	---
35	Length (m)	2.473e+004	2.473e+004	2.474e+004	2.474e+004
36	Elevation (m)	-0.1660	-0.1340	-0.1020	-7.000e-002
37	Pressure (bar)	89.19	89.19	89.19	89.19
38	Temperature (C)	51.48	51.48	51.48	51.48
39	Vapour Fraction	0.8696	0.8696	0.8696	0.8696
40	Heat Transferred (kJ/h-m)	572.3	572.3	572.2	572.2
41	Flow Regime	Transition	Transition	Transition	Transition
42	Water Flow Regime				
43	Liquid HoldUp	0.2999	0.2999	0.2999	0.2999
44	Water HoldUp	---	---	---	---
45	Friction Gradient (kPa/m)	5.773e-003	5.773e-003	5.773e-003	5.773e-003
46	Static Gradient (kPa/m)	1.592e-002	1.592e-002	1.592e-002	1.592e-002
47	Accel. Gradient (kPa/m)	1.556e-006	1.556e-006	1.556e-006	1.556e-006
48	Liq. Reynolds	2.104e+005 *	2.104e+005 *	2.104e+005 *	2.104e+005 *
49	Vap. Reynolds	9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *
50	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
51	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
52	Deposit Thickness (mm)	---	---	---	---
53	Deposit Volume (m3)	---	---	---	---
54	Length (m)	2.475e+004	2.475e+004	2.476e+004	2.476e+004
55	Elevation (m)	-7.000e-002	-7.000e-002	-7.000e-002	-7.000e-002
56	Pressure (bar)	89.18	89.18	89.18	89.18
57	Temperature (C)	51.48	51.47	51.47	51.47
58	Vapour Fraction	0.8696	0.8696	0.8696	0.8696
59	Heat Transferred (kJ/h-m)	572.1	572.1	572.1	572.0
60	Flow Regime	Transition	Transition	Transition	Transition
61	Water Flow Regime				
62	Liquid HoldUp	0.3025	0.3025	0.3025	0.3025
63	Water HoldUp	---	---	---	---
64	Friction Gradient (kPa/m)	5.878e-003	5.878e-003	5.878e-003	5.878e-003
65	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
66	Accel. Gradient (kPa/m)	4.217e-007	4.217e-007	4.217e-007	4.217e-007
67	Liq. Reynolds	2.104e+005 *	2.104e+005 *	2.104e+005 *	2.104e+005 *
68	Vap. Reynolds	9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
14	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	2.477e+004	2.477e+004	2.478e+004	2.478e+004	
18	Elevation	(m)	-7.000e-002	-7.600e-002	-8.200e-002	-8.800e-002	
19	Pressure	(bar)	89.18	89.18	89.18	89.18	
20	Temperature	(C)	51.47	51.47	51.47	51.46	
21	Vapour Fraction		0.8696	0.8696	0.8696	0.8696	
22	Heat Transferred	(kJ/h-m)	572.0	571.9	571.9	571.8	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.3025	0.2061	0.2061	0.2061	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	5.878e-003	6.870e-003	6.870e-003	6.870e-003	
28	Static Gradient	(kPa/m)	0.0000	-2.316e-003	-2.317e-003	-2.317e-003	
29	Accel. Gradient	(kPa/m)	4.217e-007	3.267e-007	3.267e-007	3.267e-007	
30	Liq. Reynolds		2.104e+005 *	2.104e+005 *	2.103e+005 *	2.103e+005 *	
31	Vap. Reynolds		9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *	
32	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
33	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	2.479e+004	2.479e+004	2.480e+004	2.480e+004	
37	Elevation	(m)	-9.400e-002	-0.1000	-0.1140	-0.1280	
38	Pressure	(bar)	89.18	89.18	89.18	89.18	
39	Temperature	(C)	51.46	51.46	51.46	51.46	
40	Vapour Fraction		0.8696	0.8696	0.8696	0.8696	
41	Heat Transferred	(kJ/h-m)	571.8	571.7	571.7	571.7	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2061	0.2061	0.2046	0.2046	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	6.870e-003	6.870e-003	6.882e-003	6.882e-003	
47	Static Gradient	(kPa/m)	-2.317e-003	-2.317e-003	-5.381e-003	-5.381e-003	
48	Accel. Gradient	(kPa/m)	3.267e-007	3.267e-007	1.077e-007	1.077e-007	
49	Liq. Reynolds		2.103e+005 *	2.103e+005 *	2.103e+005 *	2.103e+005 *	
50	Vap. Reynolds		9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *	
51	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
52	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	2.481e+004	2.481e+004	2.482e+004	2.482e+004	
56	Elevation	(m)	-0.1420	-0.1560	-0.1700	-0.2060	
57	Pressure	(bar)	89.18	89.18	89.18	89.18	
58	Temperature	(C)	51.46	51.45	51.45	51.45	
59	Vapour Fraction		0.8696	0.8696	0.8696	0.8696	
60	Heat Transferred	(kJ/h-m)	571.6	571.6	571.5	571.5	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2046	0.2046	0.2046	0.2005	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	6.882e-003	6.882e-003	6.882e-003	6.916e-003	
66	Static Gradient	(kPa/m)	-5.381e-003	-5.381e-003	-5.381e-003	-1.366e-002	
67	Accel. Gradient	(kPa/m)	1.077e-007	1.077e-007	1.077e-007	-4.839e-007	
68	Liq. Reynolds		2.103e+005 *	2.103e+005 *	2.103e+005 *	2.103e+005 *	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 401 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *
14	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
15	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	2.483e+004	2.483e+004	2.484e+004	2.484e+004
19	Elevation (m)	-0.2420	-0.2780	-0.3140	-0.3500
20	Pressure (bar)	89.18	89.18	89.18	89.18
21	Temperature (C)	51.45	51.45	51.45	51.45
22	Vapour Fraction	0.8696	0.8696	0.8696	0.8696
23	Heat Transferred (kJ/h-m)	571.5	571.4	571.4	571.3
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2005	0.2005	0.2005	0.2005
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.916e-003	6.916e-003	6.916e-003	6.916e-003
29	Static Gradient (kPa/m)	-1.366e-002	-1.366e-002	-1.366e-002	-1.366e-002
30	Accel. Gradient (kPa/m)	-4.839e-007	-4.839e-007	-4.839e-007	-4.839e-007
31	Liq. Reynolds	2.103e+005 *	2.103e+005 *	2.103e+005 *	2.103e+005 *
32	Vap. Reynolds	9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *
33	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
34	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	2.485e+004	2.485e+004	2.486e+004	2.486e+004
38	Elevation (m)	-0.4260	-0.5020	-0.5780	-0.6540
39	Pressure (bar)	89.19	89.19	89.19	89.19
40	Temperature (C)	51.44	51.44	51.44	51.44
41	Vapour Fraction	0.8696	0.8696	0.8696	0.8696
42	Heat Transferred (kJ/h-m)	571.3	571.3	571.3	571.2
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.1931	0.1931	0.1931	0.1931
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.983e-003	6.983e-003	6.982e-003	6.982e-003
48	Static Gradient (kPa/m)	-2.817e-002	-2.817e-002	-2.817e-002	-2.817e-002
49	Accel. Gradient (kPa/m)	-1.520e-006	-1.520e-006	-1.520e-006	-1.520e-006
50	Liq. Reynolds	2.103e+005 *	2.103e+005 *	2.103e+005 *	2.103e+005 *
51	Vap. Reynolds	9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *
52	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
53	Vapour Velocity (m/s)	2.170	2.170	2.169	2.169
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	2.487e+004	2.487e+004	2.488e+004	2.488e+004
57	Elevation (m)	-0.7300	-0.7300	-0.7300	-0.7300
58	Pressure (bar)	89.19	89.19	89.19	89.19
59	Temperature (C)	51.44	51.44	51.44	51.43
60	Vapour Fraction	0.8696	0.8696	0.8696	0.8696
61	Heat Transferred (kJ/h-m)	571.2	571.1	571.1	571.0
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1931	0.3026	0.3026	0.3026
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	6.982e-003	5.877e-003	5.877e-003	5.877e-003
67	Static Gradient (kPa/m)	-2.817e-002	0.0000	0.0000	0.0000
68	Accel. Gradient (kPa/m)	-1.520e-006	4.215e-007	4.215e-007	4.215e-007
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.103e+005 *	2.103e+005 *	2.103e+005 *	2.103e+005 *
14	Vap. Reynolds	9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *
15	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
16	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	2.489e+004	2.489e+004	2.490e+004	2.490e+004
20	Elevation (m)	-0.7300	-0.7300	-0.6140	-0.4980
21	Pressure (bar)	89.19	89.19	89.18	89.18
22	Temperature (C)	51.43	51.43	51.43	51.43
23	Vapour Fraction	0.8696	0.8696	0.8696	0.8696
24	Heat Transferred (kJ/h-m)	571.0	570.9	570.9	570.8
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.3026	0.3026	0.3534	0.3534
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	5.877e-003	5.877e-003	6.056e-003	6.056e-003
30	Static Gradient (kPa/m)	0.0000	0.0000	6.508e-002	6.508e-002
31	Accel. Gradient (kPa/m)	4.215e-007	4.215e-007	5.103e-006	5.103e-006
32	Liq. Reynolds	2.103e+005 *	2.103e+005 *	2.103e+005 *	2.103e+005 *
33	Vap. Reynolds	9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *
34	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
35	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	2.491e+004	2.491e+004	2.492e+004	2.492e+004
39	Elevation (m)	-0.3820	-0.2660	-0.1500	-0.1340
40	Pressure (bar)	89.18	89.17	89.17	89.17
41	Temperature (C)	51.42	51.42	51.42	51.42
42	Vapour Fraction	0.8696	0.8696	0.8696	0.8696
43	Heat Transferred (kJ/h-m)	570.7	570.7	570.6	570.5
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.3534	0.3534	0.3534	0.2897
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	6.056e-003	6.056e-003	6.056e-003	4.865e-003
49	Static Gradient (kPa/m)	6.508e-002	6.508e-002	6.508e-002	7.767e-003
50	Accel. Gradient (kPa/m)	5.103e-006	5.103e-006	5.104e-006	9.063e-007
51	Liq. Reynolds	2.103e+005 *	2.103e+005 *	2.103e+005 *	2.103e+005 *
52	Vap. Reynolds	9.158e+006 *	9.158e+006 *	9.159e+006 *	9.159e+006 *
53	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
54	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	2.493e+004	2.493e+004	2.494e+004	2.494e+004
58	Elevation (m)	-0.1180	-0.1020	-8.600e-002	-7.000e-002
59	Pressure (bar)	89.17	89.17	89.17	89.17
60	Temperature (C)	51.41	51.41	51.41	51.41
61	Vapour Fraction	0.8696	0.8696	0.8696	0.8696
62	Heat Transferred (kJ/h-m)	570.5	570.4	570.4	570.3
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2897	0.2897	0.2897	0.2897
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	4.865e-003	4.865e-003	4.865e-003	4.865e-003
68	Static Gradient (kPa/m)	7.767e-003	7.767e-003	7.767e-003	7.767e-003
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	9.063e-007	9.063e-007	9.063e-007	9.063e-007
14	Liq. Reynolds	2.103e+005 *	2.103e+005 *	2.103e+005 *	2.103e+005 *
15	Vap. Reynolds	9.159e+006 *	9.159e+006 *	9.159e+006 *	9.159e+006 *
16	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
17	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	2.495e+004	2.495e+004	2.496e+004	2.496e+004
21	Elevation (m)	-7.600e-002	-8.200e-002	-8.800e-002	-9.400e-002
22	Pressure (bar)	89.17	89.17	89.17	89.17
23	Temperature (C)	51.41	51.40	51.40	51.40
24	Vapour Fraction	0.8696	0.8696	0.8696	0.8696
25	Heat Transferred (kJ/h-m)	570.3	570.3	570.2	570.2
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.2061	0.2061	0.2061	0.2061
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	6.870e-003	6.870e-003	6.870e-003	6.870e-003
31	Static Gradient (kPa/m)	-2.317e-003	-2.317e-003	-2.317e-003	-2.317e-003
32	Accel. Gradient (kPa/m)	3.267e-007	3.267e-007	3.267e-007	3.266e-007
33	Liq. Reynolds	2.103e+005 *	2.103e+005 *	2.103e+005 *	2.103e+005 *
34	Vap. Reynolds	9.159e+006 *	9.159e+006 *	9.159e+006 *	9.159e+006 *
35	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
36	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	2.497e+004	2.497e+004	2.498e+004	2.498e+004
40	Elevation (m)	-0.1000	-0.1060	-0.1120	-0.1180
41	Pressure (bar)	89.17	89.17	89.17	89.17
42	Temperature (C)	51.40	51.40	51.40	51.39
43	Vapour Fraction	0.8696	0.8696	0.8696	0.8696
44	Heat Transferred (kJ/h-m)	570.1	570.1	570.0	570.0
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.2061	0.2061	0.2061	0.2061
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	6.870e-003	6.869e-003	6.869e-003	6.869e-003
50	Static Gradient (kPa/m)	-2.317e-003	-2.317e-003	-2.317e-003	-2.317e-003
51	Accel. Gradient (kPa/m)	3.266e-007	3.266e-007	3.266e-007	3.266e-007
52	Liq. Reynolds	2.103e+005 *	2.103e+005 *	2.103e+005 *	2.103e+005 *
53	Vap. Reynolds	9.159e+006 *	9.159e+006 *	9.159e+006 *	9.159e+006 *
54	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
55	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	2.499e+004	2.499e+004	2.500e+004	2.500e+004
59	Elevation (m)	-0.1240	-0.1300	-0.1320	-0.1340
60	Pressure (bar)	89.16	89.16	89.16	89.16
61	Temperature (C)	51.39	51.39	51.39	51.39
62	Vapour Fraction	0.8696	0.8696	0.8696	0.8696
63	Heat Transferred (kJ/h-m)	569.9	569.9	569.9	569.8
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.2061	0.2061	0.2069	0.2069
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	6.869e-003	6.869e-003	6.863e-003	6.863e-003
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 404 of 937

1				Case Name:	Simulation - 386MMSCFD_100barg.hsc	
2				Unit Set:	Mixed	
3	 Epic Atlantic Limited			Date/Time:	Mon Jun 13 15:02:44 2022	
4						
5						
6						
7						
8						
9						
10						
11						
12						
13	Static Gradient	(kPa/m)	-2.317e-003	-2.317e-003	-7.741e-004	-7.741e-004
14	Accel. Gradient	(kPa/m)	3.266e-007	3.266e-007	4.369e-007	4.369e-007
15	Liq. Reynolds		2.103e+005 *	2.103e+005 *	2.103e+005 *	2.103e+005 *
16	Vap. Reynolds		9.159e+006 *	9.159e+006 *	9.159e+006 *	9.159e+006 *
17	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045
18	Vapour Velocity	(m/s)	2.170	2.170	2.169	2.169
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	2.501e+004	2.501e+004	2.502e+004	2.502e+004
22	Elevation	(m)	-0.1360	-0.1380	-0.1400	-0.1440
23	Pressure	(bar)	89.16	89.16	89.16	89.16
24	Temperature	(C)	51.39	51.38	51.38	51.38
25	Vapour Fraction		0.8696	0.8696	0.8696	0.8696
26	Heat Transferred	(kJ/h-m)	569.8	569.7	569.7	569.6
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2069	0.2069	0.2069	0.2065
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	6.863e-003	6.863e-003	6.863e-003	6.866e-003
32	Static Gradient	(kPa/m)	-7.741e-004	-7.741e-004	-7.741e-004	-1.546e-003
33	Accel. Gradient	(kPa/m)	4.369e-007	4.369e-007	4.369e-007	3.817e-007
34	Liq. Reynolds		2.103e+005 *	2.103e+005 *	2.103e+005 *	2.103e+005 *
35	Vap. Reynolds		9.159e+006 *	9.159e+006 *	9.159e+006 *	9.159e+006 *
36	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045
37	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	2.503e+004	2.503e+004	2.504e+004	2.504e+004
41	Elevation	(m)	-0.1480	-0.1520	-0.1560	-0.1600
42	Pressure	(bar)	89.16	89.16	89.16	89.16
43	Temperature	(C)	51.38	51.38	51.38	51.37
44	Vapour Fraction		0.8696	0.8696	0.8696	0.8696
45	Heat Transferred	(kJ/h-m)	569.6	569.5	569.5	569.4
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2065	0.2065	0.2065	0.2065
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	6.866e-003	6.866e-003	6.866e-003	6.866e-003
51	Static Gradient	(kPa/m)	-1.546e-003	-1.546e-003	-1.546e-003	-1.546e-003
52	Accel. Gradient	(kPa/m)	3.817e-007	3.817e-007	3.817e-007	3.817e-007
53	Liq. Reynolds		2.103e+005 *	2.103e+005 *	2.103e+005 *	2.103e+005 *
54	Vap. Reynolds		9.159e+006 *	9.159e+006 *	9.159e+006 *	9.159e+006 *
55	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045
56	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	2.505e+004	2.505e+004	2.506e+004	2.506e+004
60	Elevation	(m)	-0.1700	-0.1800	-0.1900	-0.2000
61	Pressure	(bar)	89.16	89.16	89.16	89.16
62	Temperature	(C)	51.37	51.37	51.37	51.37
63	Vapour Fraction		0.8696	0.8696	0.8696	0.8696
64	Heat Transferred	(kJ/h-m)	569.4	569.4	569.3	569.3
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2054	0.2054	0.2054	0.2054
68	Water HoldUp		---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	6.875e-003	6.875e-003	6.875e-003	6.875e-003	
14	Static Gradient	(kPa/m)	-3.853e-003	-3.853e-003	-3.853e-003	-3.853e-003	
15	Accel. Gradient	(kPa/m)	2.168e-007	2.168e-007	2.168e-007	2.168e-007	
16	Liq. Reynolds		2.103e+005 *	2.103e+005 *	2.103e+005 *	2.103e+005 *	
17	Vap. Reynolds		9.159e+006 *	9.159e+006 *	9.159e+006 *	9.159e+006 *	
18	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
19	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	2.507e+004	2.507e+004	2.508e+004	2.508e+004	
23	Elevation	(m)	-0.2100	-0.1940	-0.1780	-0.1620	
24	Pressure	(bar)	89.16	89.16	89.16	89.16	
25	Temperature	(C)	51.37	51.36	51.36	51.36	
26	Vapour Fraction		0.8696	0.8696	0.8696	0.8696	
27	Heat Transferred	(kJ/h-m)	569.2	569.2	569.1	569.1	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.2054	0.2898	0.2898	0.2898	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	6.875e-003	4.865e-003	4.865e-003	4.865e-003	
33	Static Gradient	(kPa/m)	-3.853e-003	7.755e-003	7.755e-003	7.755e-003	
34	Accel. Gradient	(kPa/m)	2.168e-007	9.054e-007	9.054e-007	9.054e-007	
35	Liq. Reynolds		2.103e+005 *	2.103e+005 *	2.103e+005 *	2.102e+005 *	
36	Vap. Reynolds		9.159e+006 *	9.159e+006 *	9.159e+006 *	9.159e+006 *	
37	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
38	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	2.509e+004	2.509e+004	2.510e+004	2.510e+004	
42	Elevation	(m)	-0.1460	-0.1300	-0.1380	-0.1460	
43	Pressure	(bar)	89.16	89.16	89.16	89.16	
44	Temperature	(C)	51.36	51.36	51.36	51.35	
45	Vapour Fraction		0.8696	0.8696	0.8696	0.8696	
46	Heat Transferred	(kJ/h-m)	569.0	569.0	568.9	568.9	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.2898	0.2898	0.2058	0.2058	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	4.865e-003	4.865e-003	6.872e-003	6.872e-003	
52	Static Gradient	(kPa/m)	7.755e-003	7.755e-003	-3.086e-003	-3.086e-003	
53	Accel. Gradient	(kPa/m)	9.054e-007	9.054e-007	2.716e-007	2.716e-007	
54	Liq. Reynolds		2.102e+005 *	2.102e+005 *	2.102e+005 *	2.102e+005 *	
55	Vap. Reynolds		9.159e+006 *	9.159e+006 *	9.159e+006 *	9.159e+006 *	
56	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
57	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	2.511e+004	2.511e+004	2.512e+004	2.512e+004	
61	Elevation	(m)	-0.1540	-0.1620	-0.1700	-0.1760	
62	Pressure	(bar)	89.16	89.16	89.16	89.16	
63	Temperature	(C)	51.35	51.35	51.35	51.35	
64	Vapour Fraction		0.8696	0.8696	0.8696	0.8696	
65	Heat Transferred	(kJ/h-m)	568.9	568.8	568.8	568.7	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.2058	0.2058	0.2058	0.2062	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	6.872e-003	6.872e-003	6.872e-003	6.869e-003
15	Static Gradient (kPa/m)	-3.086e-003	-3.086e-003	-3.086e-003	-2.317e-003
16	Accel. Gradient (kPa/m)	2.716e-007	2.716e-007	2.716e-007	3.265e-007
17	Liq. Reynolds	2.102e+005 *	2.102e+005 *	2.102e+005 *	2.102e+005 *
18	Vap. Reynolds	9.159e+006 *	9.159e+006 *	9.159e+006 *	9.159e+006 *
19	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
20	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	2.513e+004	2.513e+004	2.514e+004	2.514e+004
24	Elevation (m)	-0.1820	-0.1880	-0.1940	-0.2000
25	Pressure (bar)	89.16	89.16	89.16	89.16
26	Temperature (C)	51.35	51.34	51.34	51.34
27	Vapour Fraction	0.8696	0.8696	0.8696	0.8695
28	Heat Transferred (kJ/h-m)	568.7	568.6	568.6	568.5
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2062	0.2062	0.2062	0.2062
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	6.868e-003	6.868e-003	6.868e-003	6.868e-003
34	Static Gradient (kPa/m)	-2.317e-003	-2.317e-003	-2.317e-003	-2.317e-003
35	Accel. Gradient (kPa/m)	3.265e-007	3.265e-007	3.265e-007	3.265e-007
36	Liq. Reynolds	2.102e+005 *	2.102e+005 *	2.102e+005 *	2.102e+005 *
37	Vap. Reynolds	9.159e+006 *	9.159e+006 *	9.159e+006 *	9.159e+006 *
38	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
39	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	2.515e+004	2.515e+004	2.516e+004	2.516e+004
43	Elevation (m)	-0.2180	-0.2360	-0.2540	-0.2720
44	Pressure (bar)	89.16	89.16	89.16	89.16
45	Temperature (C)	51.34	51.34	51.34	51.33
46	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
47	Heat Transferred (kJ/h-m)	568.5	568.5	568.4	568.4
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2039	0.2039	0.2039	0.2039
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	6.886e-003	6.886e-003	6.886e-003	6.886e-003
53	Static Gradient (kPa/m)	-6.904e-003	-6.904e-003	-6.904e-003	-6.904e-003
54	Accel. Gradient (kPa/m)	-1.254e-009	-1.261e-009	-1.268e-009	-1.275e-009
55	Liq. Reynolds	2.102e+005 *	2.102e+005 *	2.102e+005 *	2.102e+005 *
56	Vap. Reynolds	9.159e+006 *	9.159e+006 *	9.159e+006 *	9.159e+006 *
57	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
58	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	2.517e+004	2.517e+004	2.518e+004	2.518e+004
62	Elevation (m)	-0.2900	-0.3160	-0.3420	-0.3680
63	Pressure (bar)	89.16	89.16	89.16	89.16
64	Temperature (C)	51.33	51.33	51.33	51.33
65	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
66	Heat Transferred (kJ/h-m)	568.3	568.3	568.3	568.2
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.2039	0.2024	0.2024	0.2024
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	6.886e-003	6.898e-003	6.898e-003	6.898e-003
16	Static Gradient (kPa/m)	-6.904e-003	-9.926e-003	-9.926e-003	-9.927e-003
17	Accel. Gradient (kPa/m)	-1.282e-009	-2.172e-007	-2.172e-007	-2.172e-007
18	Liq. Reynolds	2.102e+005 *	2.102e+005 *	2.102e+005 *	2.102e+005 *
19	Vap. Reynolds	9.159e+006 *	9.159e+006 *	9.159e+006 *	9.159e+006 *
20	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
21	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	2.519e+004	2.519e+004	2.520e+004	2.520e+004
25	Elevation (m)	-0.3940	-0.4200	-0.4400	-0.4600
26	Pressure (bar)	89.16	89.16	89.16	89.16
27	Temperature (C)	51.33	51.33	51.32	51.32
28	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
29	Heat Transferred (kJ/h-m)	568.2	568.1	568.1	568.0
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2024	0.2024	0.2044	0.2044
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.898e-003	6.898e-003	6.882e-003	6.882e-003
35	Static Gradient (kPa/m)	-9.927e-003	-9.927e-003	-5.885e-003	-5.885e-003
36	Accel. Gradient (kPa/m)	-2.172e-007	-2.173e-007	7.150e-008	7.149e-008
37	Liq. Reynolds	2.102e+005 *	2.102e+005 *	2.102e+005 *	2.102e+005 *
38	Vap. Reynolds	9.159e+006 *	9.159e+006 *	9.159e+006 *	9.159e+006 *
39	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
40	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	2.521e+004	2.522e+004	2.522e+004	2.523e+004
44	Elevation (m)	-0.4800	-0.5000	-0.5200	-1.196
45	Pressure (bar)	89.16	89.16	89.16	89.16
46	Temperature (C)	51.32	51.32	51.31	51.32
47	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
48	Heat Transferred (kJ/h-m)	568.0	567.9	567.9	568.2
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2044	0.2044	0.2044	8.467e-002
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.881e-003	6.881e-003	6.881e-003	1.007e-002
54	Static Gradient (kPa/m)	-5.885e-003	-5.885e-003	-5.885e-003	-0.1635
55	Accel. Gradient (kPa/m)	7.148e-008	7.147e-008	7.147e-008	-1.101e-005
56	Liq. Reynolds	2.102e+005 *	2.102e+005 *	2.102e+005 *	2.102e+005 *
57	Vap. Reynolds	9.159e+006 *	9.159e+006 *	9.159e+006 *	9.159e+006 *
58	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
59	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	2.523e+004	2.524e+004	2.524e+004	2.525e+004
63	Elevation (m)	-1.872	-2.548	-3.224	-3.900
64	Pressure (bar)	89.17	89.18	89.19	89.19
65	Temperature (C)	51.32	51.32	51.32	51.33
66	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
67	Heat Transferred (kJ/h-m)	568.3	568.3	568.4	568.4
68	Flow Regime	Transition	Transition	Transition	Transition
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	8.468e-002	8.468e-002	8.468e-002	8.469e-002
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	1.007e-002	1.007e-002	1.007e-002	1.007e-002
17	Static Gradient (kPa/m)	-0.1635	-0.1636	-0.1636	-0.1636
18	Accel. Gradient (kPa/m)	-1.100e-005	-1.100e-005	-1.100e-005	-1.100e-005
19	Liq. Reynolds	2.102e+005 *	2.102e+005 *	2.103e+005 *	2.103e+005 *
20	Vap. Reynolds	9.158e+006 *	9.158e+006 *	9.158e+006 *	9.158e+006 *
21	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
22	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	2.525e+004	2.526e+004	2.526e+004	2.527e+004
26	Elevation (m)	-4.180	-4.460	-4.740	-5.020
27	Pressure (bar)	89.20	89.20	89.21	89.21
28	Temperature (C)	51.33	51.33	51.33	51.33
29	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
30	Heat Transferred (kJ/h-m)	568.2	568.2	568.2	568.2
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.1554	0.1554	0.1554	0.1554
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	7.459e-003	7.458e-003	7.458e-003	7.457e-003
36	Static Gradient (kPa/m)	-9.128e-002	-9.128e-002	-9.128e-002	-9.129e-002
37	Accel. Gradient (kPa/m)	-6.006e-006	-6.006e-006	-6.006e-006	-6.005e-006
38	Liq. Reynolds	2.103e+005 *	2.103e+005 *	2.103e+005 *	2.103e+005 *
39	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.157e+006 *	9.157e+006 *
40	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
41	Vapour Velocity (m/s)	2.168	2.168	2.167	2.167
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	2.527e+004	2.528e+004	2.528e+004	2.529e+004
45	Elevation (m)	-5.300	-5.020	-4.740	-4.460
46	Pressure (bar)	89.22	89.20	89.19	89.18
47	Temperature (C)	51.33	51.32	51.32	51.31
48	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
49	Heat Transferred (kJ/h-m)	568.2	568.1	567.9	567.8
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.1554	0.5012	0.5012	0.5011
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	7.457e-003	5.856e-003	5.856e-003	5.857e-003
55	Static Gradient (kPa/m)	-9.129e-002	0.2578	0.2578	0.2578
56	Accel. Gradient (kPa/m)	-6.005e-006	1.889e-005	1.890e-005	1.890e-005
57	Liq. Reynolds	2.103e+005 *	2.103e+005 *	2.103e+005 *	2.102e+005 *
58	Vap. Reynolds	9.157e+006 *	9.157e+006 *	9.158e+006 *	9.158e+006 *
59	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
60	Vapour Velocity (m/s)	2.167	2.167	2.168	2.168
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	2.529e+004	2.529e+004	2.531e+004	2.533e+004
64	Elevation (m)	-4.180	-3.900	-3.780	-3.660
65	Pressure (bar)	89.17	89.16	89.16	89.15
66	Temperature (C)	51.31	51.31	51.30	51.29
67	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
68	Heat Transferred (kJ/h-m)	567.7	567.6	567.4	567.2
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.5011	0.5011	0.2998	0.2998
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	5.858e-003	5.858e-003	5.757e-003	5.757e-003
18	Static Gradient (kPa/m)	0.2578	0.2578	1.571e-002	1.571e-002
19	Accel. Gradient (kPa/m)	1.890e-005	1.891e-005	1.539e-006	1.539e-006
20	Liq. Reynolds	2.102e+005 *	2.102e+005 *	2.102e+005 *	2.102e+005 *
21	Vap. Reynolds	9.158e+006 *	9.159e+006 *	9.159e+006 *	9.159e+006 *
22	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
23	Vapour Velocity (m/s)	2.168	2.168	2.169	2.169
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	2.535e+004	2.537e+004	2.539e+004	2.539e+004
27	Elevation (m)	-3.540	-3.420	-3.300	-3.260
28	Pressure (bar)	89.15	89.15	89.14	89.14
29	Temperature (C)	51.28	51.28	51.27	51.27
30	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
31	Heat Transferred (kJ/h-m)	567.0	566.8	566.6	566.6
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2998	0.2998	0.2998	0.3433
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	5.757e-003	5.757e-003	5.757e-003	6.065e-003
37	Static Gradient (kPa/m)	1.571e-002	1.571e-002	1.571e-002	5.492e-002
38	Accel. Gradient (kPa/m)	1.540e-006	1.540e-006	1.540e-006	4.374e-006
39	Liq. Reynolds	2.102e+005 *	2.102e+005 *	2.102e+005 *	2.102e+005 *
40	Vap. Reynolds	9.159e+006 *	9.159e+006 *	9.159e+006 *	9.159e+006 *
41	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
42	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	2.539e+004	2.539e+004	2.540e+004	2.540e+004
46	Elevation (m)	-3.220	-3.180	-3.140	-3.100
47	Pressure (bar)	89.14	89.14	89.14	89.14
48	Temperature (C)	51.27	51.26	51.26	51.26
49	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
50	Heat Transferred (kJ/h-m)	566.6	566.5	566.5	566.5
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.3433	0.3433	0.3433	0.3433
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	6.065e-003	6.065e-003	6.065e-003	6.065e-003
56	Static Gradient (kPa/m)	5.491e-002	5.491e-002	5.491e-002	5.491e-002
57	Accel. Gradient (kPa/m)	4.375e-006	4.375e-006	4.375e-006	4.375e-006
58	Liq. Reynolds	2.102e+005 *	2.102e+005 *	2.102e+005 *	2.102e+005 *
59	Vap. Reynolds	9.159e+006 *	9.159e+006 *	9.159e+006 *	9.159e+006 *
60	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
61	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	2.540e+004	2.541e+004	2.541e+004	2.542e+004
65	Elevation (m)	-2.568	-2.036	-1.504	-0.9720
66	Pressure (bar)	89.11	89.09	89.07	89.05
67	Temperature (C)	51.25	51.25	51.24	51.23
68	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	567.2	567.0	566.8	566.6
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.5908	0.5907	0.5906	0.5905
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	5.781e-003	5.782e-003	5.783e-003	5.785e-003
19	Static Gradient (kPa/m)	0.4178	0.4177	0.4177	0.4177
20	Accel. Gradient (kPa/m)	3.040e-005	3.042e-005	3.043e-005	3.044e-005
21	Liq. Reynolds	2.101e+005 *	2.101e+005 *	2.101e+005 *	2.100e+005 *
22	Vap. Reynolds	9.160e+006 *	9.161e+006 *	9.162e+006 *	9.162e+006 *
23	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
24	Vapour Velocity (m/s)	2.169	2.170	2.170	2.171
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	2.543e+004	2.543e+004	2.544e+004	2.544e+004
28	Elevation (m)	-0.4400	-0.4000	-0.3600	-0.3200
29	Pressure (bar)	89.02	89.02	89.02	89.02
30	Temperature (C)	51.22	51.22	51.22	51.22
31	Vapour Fraction	0.8696	0.8696	0.8695	0.8695
32	Heat Transferred (kJ/h-m)	566.3	565.4	565.3	565.2
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.5904	0.3049	0.3049	0.3049
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	5.786e-003	5.943e-003	5.943e-003	5.943e-003
38	Static Gradient (kPa/m)	0.4176	2.014e-002	2.014e-002	2.014e-002
39	Accel. Gradient (kPa/m)	3.045e-005	1.876e-006	1.876e-006	1.876e-006
40	Liq. Reynolds	2.100e+005 *	2.100e+005 *	2.100e+005 *	2.100e+005 *
41	Vap. Reynolds	9.163e+006 *	9.163e+006 *	9.163e+006 *	9.163e+006 *
42	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
43	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	2.545e+004	2.545e+004	2.546e+004	2.546e+004
47	Elevation (m)	-0.2800	-0.2400	-0.2160	-0.1920
48	Pressure (bar)	89.02	89.02	89.02	89.01
49	Temperature (C)	51.22	51.21	51.21	51.21
50	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
51	Heat Transferred (kJ/h-m)	565.2	565.1	565.1	565.0
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.3049	0.3049	0.2921	0.2921
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	5.943e-003	5.943e-003	4.691e-003	4.691e-003
57	Static Gradient (kPa/m)	2.014e-002	2.014e-002	9.766e-003	9.766e-003
58	Accel. Gradient (kPa/m)	1.876e-006	1.876e-006	1.040e-006	1.040e-006
59	Liq. Reynolds	2.100e+005 *	2.100e+005 *	2.100e+005 *	2.100e+005 *
60	Vap. Reynolds	9.163e+006 *	9.163e+006 *	9.163e+006 *	9.163e+006 *
61	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
62	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	2.547e+004	2.547e+004	2.548e+004	2.548e+004
66	Elevation (m)	-0.1680	-0.1440	-0.1200	-0.1220
67	Pressure (bar)	89.01	89.01	89.01	89.01
68	Temperature (C)	51.21	51.20	51.20	51.20
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
14	Heat Transferred (kJ/h-m)	565.0	564.9	564.8	564.8
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2921	0.2921	0.2921	0.2067
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	4.691e-003	4.691e-003	4.691e-003	6.870e-003
20	Static Gradient (kPa/m)	9.766e-003	9.766e-003	9.766e-003	-9.668e-004
21	Accel. Gradient (kPa/m)	1.040e-006	1.040e-006	1.040e-006	4.247e-007
22	Liq. Reynolds	2.100e+005 *	2.100e+005 *	2.100e+005 *	2.100e+005 *
23	Vap. Reynolds	9.163e+006 *	9.163e+006 *	9.163e+006 *	9.163e+006 *
24	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
25	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	2.549e+004	2.549e+004	2.550e+004	2.550e+004
29	Elevation (m)	-0.1240	-0.1260	-0.1280	-0.1300
30	Pressure (bar)	89.01	89.01	89.01	89.01
31	Temperature (C)	51.20	51.20	51.20	51.20
32	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
33	Heat Transferred (kJ/h-m)	564.8	564.7	564.7	564.7
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2067	0.2067	0.2067	0.2067
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	6.870e-003	6.870e-003	6.870e-003	6.870e-003
39	Static Gradient (kPa/m)	-9.668e-004	-9.668e-004	-9.668e-004	-9.668e-004
40	Accel. Gradient (kPa/m)	4.247e-007	4.247e-007	4.247e-007	4.247e-007
41	Liq. Reynolds	2.100e+005 *	2.100e+005 *	2.100e+005 *	2.100e+005 *
42	Vap. Reynolds	9.163e+006 *	9.163e+006 *	9.163e+006 *	9.163e+006 *
43	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
44	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	2.551e+004	2.551e+004	2.552e+004	2.552e+004
48	Elevation (m)	-0.1220	-0.1140	-0.1060	-9.800e-002
49	Pressure (bar)	89.01	89.01	89.01	89.01
50	Temperature (C)	51.19	51.19	51.19	51.19
51	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
52	Heat Transferred (kJ/h-m)	564.6	564.6	564.5	564.5
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2845	0.2845	0.2845	0.2845
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	5.247e-003	5.247e-003	5.247e-003	5.247e-003
58	Static Gradient (kPa/m)	3.834e-003	3.834e-003	3.834e-003	3.834e-003
59	Accel. Gradient (kPa/m)	6.532e-007	6.532e-007	6.532e-007	6.532e-007
60	Liq. Reynolds	2.100e+005 *	2.100e+005 *	2.100e+005 *	2.100e+005 *
61	Vap. Reynolds	9.163e+006 *	9.163e+006 *	9.164e+006 *	9.164e+006 *
62	Liquid Velocity (m/s)	0.2044	0.2045	0.2045	0.2045
63	Vapour Velocity (m/s)	2.172	2.172	2.171	2.171
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	2.553e+004	2.553e+004	2.554e+004	2.554e+004
67	Elevation (m)	-9.000e-002	-9.000e-002	-9.000e-002	-9.000e-002
68	Pressure (bar)	89.01	89.01	89.01	89.01
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	51.19	51.18	51.18	51.18
14	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
15	Heat Transferred (kJ/h-m)	564.4	564.4	564.3	564.3
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2845	0.3024	0.3024	0.3024
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	5.247e-003	5.881e-003	5.881e-003	5.881e-003
21	Static Gradient (kPa/m)	3.834e-003	0.0000	0.0000	0.0000
22	Accel. Gradient (kPa/m)	6.532e-007	4.231e-007	4.231e-007	4.231e-007
23	Liq. Reynolds	2.100e+005 *	2.100e+005 *	2.100e+005 *	2.099e+005 *
24	Vap. Reynolds	9.164e+006 *	9.164e+006 *	9.164e+006 *	9.164e+006 *
25	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
26	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	2.555e+004	2.555e+004	2.556e+004	2.556e+004
30	Elevation (m)	-9.000e-002	-9.000e-002	-8.200e-002	-7.400e-002
31	Pressure (bar)	89.01	89.01	89.01	89.01
32	Temperature (C)	51.18	51.18	51.18	51.17
33	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
34	Heat Transferred (kJ/h-m)	564.2	564.2	564.2	564.1
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.3024	0.3024	0.2845	0.2845
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	5.881e-003	5.881e-003	5.247e-003	5.247e-003
40	Static Gradient (kPa/m)	0.0000	0.0000	3.834e-003	3.834e-003
41	Accel. Gradient (kPa/m)	4.231e-007	4.231e-007	6.532e-007	6.532e-007
42	Liq. Reynolds	2.099e+005 *	2.099e+005 *	2.099e+005 *	2.099e+005 *
43	Vap. Reynolds	9.164e+006 *	9.164e+006 *	9.164e+006 *	9.164e+006 *
44	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
45	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	2.557e+004	2.557e+004	2.558e+004	2.558e+004
49	Elevation (m)	-6.600e-002	-5.800e-002	-5.000e-002	-4.600e-002
50	Pressure (bar)	89.01	89.00	89.00	89.00
51	Temperature (C)	51.17	51.17	51.17	51.17
52	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
53	Heat Transferred (kJ/h-m)	564.1	564.0	564.0	563.9
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2845	0.2845	0.2845	0.2820
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	5.247e-003	5.247e-003	5.247e-003	5.442e-003
59	Static Gradient (kPa/m)	3.834e-003	3.834e-003	3.834e-003	1.905e-003
60	Accel. Gradient (kPa/m)	6.532e-007	6.532e-007	6.532e-007	5.285e-007
61	Liq. Reynolds	2.099e+005 *	2.099e+005 *	2.099e+005 *	2.099e+005 *
62	Vap. Reynolds	9.164e+006 *	9.164e+006 *	9.164e+006 *	9.164e+006 *
63	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
64	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	2.559e+004	2.559e+004	2.560e+004	2.560e+004
68	Elevation (m)	-4.200e-002	-3.800e-002	-3.400e-002	-3.000e-002
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	89.00	89.00	89.00	89.00		
14	Temperature (C)	51.17	51.16	51.16	51.16		
15	Vapour Fraction	0.8695	0.8695	0.8695	0.8695		
16	Heat Transferred (kJ/h-m)	563.9	563.8	563.8	563.7		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.2820	0.2820	0.2820	0.2820		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	5.442e-003	5.442e-003	5.442e-003	5.442e-003		
22	Static Gradient (kPa/m)	1.905e-003	1.905e-003	1.905e-003	1.905e-003		
23	Accel. Gradient (kPa/m)	5.285e-007	5.285e-007	5.285e-007	5.285e-007		
24	Liq. Reynolds	2.099e+005 *	2.099e+005 *	2.099e+005 *	2.099e+005 *		
25	Vap. Reynolds	9.164e+006 *	9.164e+006 *	9.164e+006 *	9.164e+006 *		
26	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045		
27	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	2.561e+004	2.561e+004	2.562e+004	2.563e+004		
31	Elevation (m)	-3.400e-002	-3.800e-002	-4.200e-002	-4.600e-002		
32	Pressure (bar)	89.00	89.00	89.00	89.00		
33	Temperature (C)	51.16	51.16	51.15	51.15		
34	Vapour Fraction	0.8695	0.8695	0.8695	0.8695		
35	Heat Transferred (kJ/h-m)	563.7	563.6	563.6	563.5		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.2066	0.2066	0.2066	0.2066		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	6.870e-003	6.870e-003	6.870e-003	6.870e-003		
41	Static Gradient (kPa/m)	-1.122e-003	-1.122e-003	-1.122e-003	-1.122e-003		
42	Accel. Gradient (kPa/m)	4.135e-007	4.135e-007	4.135e-007	4.135e-007		
43	Liq. Reynolds	2.099e+005 *	2.099e+005 *	2.099e+005 *	2.099e+005 *		
44	Vap. Reynolds	9.164e+006 *	9.164e+006 *	9.164e+006 *	9.164e+006 *		
45	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045		
46	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	2.563e+004	2.564e+004	2.564e+004	2.565e+004		
50	Elevation (m)	-5.000e-002	-5.200e-002	-5.400e-002	-5.600e-002		
51	Pressure (bar)	89.00	89.00	89.00	89.00		
52	Temperature (C)	51.15	51.15	51.15	51.14		
53	Vapour Fraction	0.8695	0.8695	0.8695	0.8695		
54	Heat Transferred (kJ/h-m)	563.4	563.4	563.4	563.3		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.2066	0.2068	0.2068	0.2068		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	6.870e-003	6.869e-003	6.869e-003	6.869e-003		
60	Static Gradient (kPa/m)	-1.122e-003	-7.738e-004	-7.738e-004	-7.738e-004		
61	Accel. Gradient (kPa/m)	4.135e-007	4.384e-007	4.384e-007	4.384e-007		
62	Liq. Reynolds	2.099e+005 *	2.099e+005 *	2.099e+005 *	2.099e+005 *		
63	Vap. Reynolds	9.164e+006 *	9.164e+006 *	9.164e+006 *	9.164e+006 *		
64	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045		
65	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	2.565e+004	2.566e+004	2.566e+004	2.567e+004		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Elevation (m)	-5.800e-002	-6.000e-002	-6.200e-002	-6.400e-002		
14	Pressure (bar)	89.00	89.00	89.00	89.00		
15	Temperature (C)	51.14	51.14	51.14	51.14		
16	Vapour Fraction	0.8695	0.8695	0.8695	0.8695		
17	Heat Transferred (kJ/h-m)	563.3	563.2	563.2	563.1		
18	Flow Regime	Transition	Transition	Transition	Transition		
19	Water Flow Regime						
20	Liquid HoldUp	0.2068	0.2068	0.2068	0.2068		
21	Water HoldUp	---	---	---	---		
22	Friction Gradient (kPa/m)	6.869e-003	6.869e-003	6.869e-003	6.869e-003		
23	Static Gradient (kPa/m)	-7.738e-004	-7.738e-004	-7.738e-004	-7.738e-004		
24	Accel. Gradient (kPa/m)	4.384e-007	4.384e-007	4.384e-007	4.384e-007		
25	Liq. Reynolds	2.099e+005 *	2.099e+005 *	2.099e+005 *	2.099e+005 *		
26	Vap. Reynolds	9.164e+006 *	9.164e+006 *	9.164e+006 *	9.164e+006 *		
27	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045		
28	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
29	Deposit Thickness (mm)	---	---	---	---		
30	Deposit Volume (m3)	---	---	---	---		
31	Length (m)	2.567e+004	2.568e+004	2.568e+004	2.569e+004		
32	Elevation (m)	-6.600e-002	-6.800e-002	-7.000e-002	-7.600e-002		
33	Pressure (bar)	89.00	89.00	89.00	89.00		
34	Temperature (C)	51.14	51.13	51.13	51.13		
35	Vapour Fraction	0.8695	0.8695	0.8695	0.8695		
36	Heat Transferred (kJ/h-m)	563.1	563.0	563.0	563.0		
37	Flow Regime	Transition	Transition	Transition	Transition		
38	Water Flow Regime						
39	Liquid HoldUp	0.2068	0.2068	0.2068	0.2060		
40	Water HoldUp	---	---	---	---		
41	Friction Gradient (kPa/m)	6.869e-003	6.869e-003	6.869e-003	6.874e-003		
42	Static Gradient (kPa/m)	-7.738e-004	-7.738e-004	-7.738e-004	-2.316e-003		
43	Accel. Gradient (kPa/m)	4.384e-007	4.384e-007	4.384e-007	3.279e-007		
44	Liq. Reynolds	2.099e+005 *	2.099e+005 *	2.099e+005 *	2.099e+005 *		
45	Vap. Reynolds	9.164e+006 *	9.164e+006 *	9.164e+006 *	9.164e+006 *		
46	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045		
47	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
48	Deposit Thickness (mm)	---	---	---	---		
49	Deposit Volume (m3)	---	---	---	---		
50	Length (m)	2.569e+004	2.570e+004	2.570e+004	2.571e+004		
51	Elevation (m)	-8.200e-002	-8.800e-002	-9.400e-002	-0.1000		
52	Pressure (bar)	89.00	89.00	89.00	89.00		
53	Temperature (C)	51.13	51.13	51.13	51.12		
54	Vapour Fraction	0.8695	0.8695	0.8695	0.8695		
55	Heat Transferred (kJ/h-m)	562.9	562.9	562.8	562.8		
56	Flow Regime	Transition	Transition	Transition	Transition		
57	Water Flow Regime						
58	Liquid HoldUp	0.2060	0.2060	0.2060	0.2060		
59	Water HoldUp	---	---	---	---		
60	Friction Gradient (kPa/m)	6.874e-003	6.874e-003	6.874e-003	6.874e-003		
61	Static Gradient (kPa/m)	-2.316e-003	-2.316e-003	-2.316e-003	-2.316e-003		
62	Accel. Gradient (kPa/m)	3.279e-007	3.279e-007	3.279e-007	3.279e-007		
63	Liq. Reynolds	2.099e+005 *	2.099e+005 *	2.099e+005 *	2.099e+005 *		
64	Vap. Reynolds	9.164e+006 *	9.164e+006 *	9.164e+006 *	9.164e+006 *		
65	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045		
66	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
67	Deposit Thickness (mm)	---	---	---	---		
68	Deposit Volume (m3)	---	---	---	---		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length (m)	2.571e+004	2.571e+004	2.571e+004	2.571e+004		
14	Elevation (m)	-0.1200	-0.1400	-0.1600	-0.1800		
15	Pressure (bar)	89.00	89.00	89.00	89.00		
16	Temperature (C)	51.12	51.12	51.12	51.12		
17	Vapour Fraction	0.8695	0.8695	0.8695	0.8695		
18	Heat Transferred (kJ/h-m)	562.8	562.8	562.8	562.8		
19	Flow Regime	Transition	Transition	Transition	Transition		
20	Water Flow Regime						
21	Liquid HoldUp	0.1886	0.1886	0.1886	0.1886		
22	Water HoldUp	---	---	---	---		
23	Friction Gradient (kPa/m)	7.032e-003	7.032e-003	7.032e-003	7.031e-003		
24	Static Gradient (kPa/m)	-3.652e-002	-3.652e-002	-3.652e-002	-3.652e-002		
25	Accel. Gradient (kPa/m)	-2.121e-006	-2.121e-006	-2.121e-006	-2.121e-006		
26	Liq. Reynolds	2.099e+005 *	2.099e+005 *	2.099e+005 *	2.099e+005 *		
27	Vap. Reynolds	9.164e+006 *	9.164e+006 *	9.164e+006 *	9.164e+006 *		
28	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045		
29	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
30	Deposit Thickness (mm)	---	---	---	---		
31	Deposit Volume (m3)	---	---	---	---		
32	Length (m)	2.571e+004	2.572e+004	2.572e+004	2.572e+004		
33	Elevation (m)	-0.2000	-0.3200	-0.4400	-0.5600		
34	Pressure (bar)	89.00	89.00	89.00	89.00		
35	Temperature (C)	51.12	51.12	51.12	51.12		
36	Vapour Fraction	0.8695	0.8695	0.8695	0.8695		
37	Heat Transferred (kJ/h-m)	562.8	563.0	563.0	563.0		
38	Flow Regime	Transition	Transition	Transition	Transition		
39	Water Flow Regime						
40	Liquid HoldUp	0.1886	9.780e-002	9.780e-002	9.780e-002		
41	Water HoldUp	---	---	---	---		
42	Friction Gradient (kPa/m)	7.031e-003	9.235e-003	9.235e-003	9.235e-003		
43	Static Gradient (kPa/m)	-3.652e-002	-0.1544	-0.1544	-0.1544		
44	Accel. Gradient (kPa/m)	-2.121e-006	-1.044e-005	-1.044e-005	-1.044e-005		
45	Liq. Reynolds	2.099e+005 *	2.099e+005 *	2.099e+005 *	2.099e+005 *		
46	Vap. Reynolds	9.164e+006 *	9.164e+006 *	9.164e+006 *	9.164e+006 *		
47	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045		
48	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
49	Deposit Thickness (mm)	---	---	---	---		
50	Deposit Volume (m3)	---	---	---	---		
51	Length (m)	2.572e+004	2.572e+004	2.572e+004	2.573e+004		
52	Elevation (m)	-0.6800	-0.8000	-1.340	-1.880		
53	Pressure (bar)	89.00	89.01	89.01	89.02		
54	Temperature (C)	51.12	51.13	51.13	51.13		
55	Vapour Fraction	0.8695	0.8695	0.8695	0.8695		
56	Heat Transferred (kJ/h-m)	563.0	563.0	0.0000	0.0000		
57	Flow Regime	Transition	Transition	Transition	Transition		
58	Water Flow Regime						
59	Liquid HoldUp	9.780e-002	9.780e-002	4.734e-002	4.734e-002		
60	Water HoldUp	---	---	---	---		
61	Friction Gradient (kPa/m)	9.234e-003	9.234e-003	1.427e-002	1.427e-002		
62	Static Gradient (kPa/m)	-0.1544	-0.1544	-0.1777	-0.1777		
63	Accel. Gradient (kPa/m)	-1.044e-005	-1.044e-005	-1.175e-005	-1.175e-005		
64	Liq. Reynolds	2.099e+005 *	2.099e+005 *	2.099e+005 *	2.099e+005 *		
65	Vap. Reynolds	9.164e+006 *	9.164e+006 *	9.163e+006 *	9.163e+006 *		
66	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045		
67	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171		
68	Deposit Thickness (mm)	---	---	---	---		
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	2.573e+004	2.573e+004	2.573e+004	2.574e+004
15	Elevation (m)	-2.420	-2.960	-3.500	-3.760
16	Pressure (bar)	89.02	89.02	89.03	89.03
17	Temperature (C)	51.13	51.14	51.14	51.14
18	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
19	Heat Transferred (kJ/h-m)	0.0000	0.0000	0.0000	563.2
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	4.734e-002	4.734e-002	4.734e-002	0.1590
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	1.427e-002	1.427e-002	1.427e-002	7.406e-003
25	Static Gradient (kPa/m)	-0.1777	-0.1777	-0.1777	-8.584e-002
26	Accel. Gradient (kPa/m)	-1.175e-005	-1.175e-005	-1.175e-005	-5.637e-006
27	Liq. Reynolds	2.099e+005 *	2.099e+005 *	2.099e+005 *	2.099e+005 *
28	Vap. Reynolds	9.163e+006 *	9.163e+006 *	9.163e+006 *	9.163e+006 *
29	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
30	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	2.574e+004	2.575e+004	2.575e+004	2.576e+004
34	Elevation (m)	-4.020	-4.280	-4.540	-4.800
35	Pressure (bar)	89.04	89.04	89.05	89.05
36	Temperature (C)	51.14	51.14	51.14	51.14
37	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
38	Heat Transferred (kJ/h-m)	563.2	563.2	563.2	563.2
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.1590	0.1590	0.1590	0.1590
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	7.406e-003	7.406e-003	7.405e-003	7.405e-003
44	Static Gradient (kPa/m)	-8.584e-002	-8.584e-002	-8.585e-002	-8.585e-002
45	Accel. Gradient (kPa/m)	-5.637e-006	-5.636e-006	-5.636e-006	-5.636e-006
46	Liq. Reynolds	2.100e+005 *	2.100e+005 *	2.100e+005 *	2.100e+005 *
47	Vap. Reynolds	9.162e+006 *	9.162e+006 *	9.162e+006 *	9.162e+006 *
48	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
49	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	2.576e+004	2.577e+004	2.577e+004	2.578e+004
53	Elevation (m)	-3.980	-3.160	-2.340	-1.520
54	Pressure (bar)	89.01	88.96	88.92	88.87
55	Temperature (C)	51.13	51.11	51.10	51.09
56	Vapour Fraction	0.8695	0.8695	0.8695	0.8696
57	Heat Transferred (kJ/h-m)	563.9	563.6	563.3	562.9
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.7816	0.7814	0.7812	0.7810
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	5.678e-003	5.681e-003	5.683e-003	5.686e-003
63	Static Gradient (kPa/m)	0.8774	0.8773	0.8771	0.8769
64	Accel. Gradient (kPa/m)	6.351e-005	6.356e-005	6.361e-005	6.367e-005
65	Liq. Reynolds	2.099e+005 *	2.098e+005 *	2.098e+005 *	2.097e+005 *
66	Vap. Reynolds	9.164e+006 *	9.165e+006 *	9.166e+006 *	9.168e+006 *
67	Liquid Velocity (m/s)	0.2045	0.2044	0.2044	0.2044
68	Vapour Velocity (m/s)	2.171	2.172	2.173	2.174
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	2.578e+004	2.579e+004	2.579e+004	2.580e+004
16	Elevation (m)	-0.7000	-0.6200	-0.5400	-0.4600
17	Pressure (bar)	88.83	88.83	88.82	88.82
18	Temperature (C)	51.08	51.07	51.07	51.07
19	Vapour Fraction	0.8696	0.8696	0.8696	0.8696
20	Heat Transferred (kJ/h-m)	562.6	561.4	561.3	561.3
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.7808	0.3300	0.3300	0.3300
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	5.688e-003	6.088e-003	6.088e-003	6.088e-003
26	Static Gradient (kPa/m)	0.8767	4.266e-002	4.265e-002	4.265e-002
27	Accel. Gradient (kPa/m)	6.372e-005	3.520e-006	3.520e-006	3.520e-006
28	Liq. Reynolds	2.097e+005 *	2.097e+005 *	2.097e+005 *	2.097e+005 *
29	Vap. Reynolds	9.169e+006 *	9.169e+006 *	9.169e+006 *	9.170e+006 *
30	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
31	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	2.580e+004	2.581e+004	2.581e+004	2.582e+004
35	Elevation (m)	-0.3800	-0.3000	-0.2840	-0.2680
36	Pressure (bar)	88.82	88.82	88.82	88.82
37	Temperature (C)	51.07	51.06	51.06	51.06
38	Vapour Fraction	0.8696	0.8696	0.8696	0.8696
39	Heat Transferred (kJ/h-m)	561.2	561.2	561.1	561.0
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.3300	0.3300	0.2894	0.2894
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	6.088e-003	6.088e-003	4.873e-003	4.873e-003
45	Static Gradient (kPa/m)	4.265e-002	4.265e-002	7.876e-003	7.876e-003
46	Accel. Gradient (kPa/m)	3.520e-006	3.520e-006	9.208e-007	9.208e-007
47	Liq. Reynolds	2.097e+005 *	2.097e+005 *	2.096e+005 *	2.096e+005 *
48	Vap. Reynolds	9.170e+006 *	9.170e+006 *	9.170e+006 *	9.170e+006 *
49	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
50	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	2.582e+004	2.583e+004	2.583e+004	2.584e+004
54	Elevation (m)	-0.2520	-0.2360	-0.2200	-0.1880
55	Pressure (bar)	88.81	88.81	88.81	88.81
56	Temperature (C)	51.06	51.06	51.05	51.05
57	Vapour Fraction	0.8696	0.8696	0.8696	0.8696
58	Heat Transferred (kJ/h-m)	561.0	561.0	560.9	560.9
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2894	0.2894	0.2894	0.2994
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	4.873e-003	4.873e-003	4.873e-003	5.782e-003
64	Static Gradient (kPa/m)	7.876e-003	7.876e-003	7.876e-003	1.590e-002
65	Accel. Gradient (kPa/m)	9.208e-007	9.208e-007	9.208e-007	1.566e-006
66	Liq. Reynolds	2.096e+005 *	2.096e+005 *	2.096e+005 *	2.096e+005 *
67	Vap. Reynolds	9.170e+006 *	9.170e+006 *	9.170e+006 *	9.170e+006 *
68	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
14	Deposit Thickness (mm)	---	---	---	---
15	Deposit Volume (m3)	---	---	---	---
16	Length (m)	2.584e+004	2.585e+004	2.585e+004	2.586e+004
17	Elevation (m)	-0.1560	-0.1240	-9.200e-002	-6.000e-002
18	Pressure (bar)	88.81	88.81	88.81	88.81
19	Temperature (C)	51.05	51.05	51.05	51.04
20	Vapour Fraction	0.8696	0.8696	0.8696	0.8696
21	Heat Transferred (kJ/h-m)	560.8	560.8	560.7	560.7
22	Flow Regime	Transition	Transition	Transition	Transition
23	Water Flow Regime				
24	Liquid HoldUp	0.2994	0.2994	0.2994	0.2994
25	Water HoldUp	---	---	---	---
26	Friction Gradient (kPa/m)	5.782e-003	5.782e-003	5.782e-003	5.782e-003
27	Static Gradient (kPa/m)	1.590e-002	1.590e-002	1.590e-002	1.590e-002
28	Accel. Gradient (kPa/m)	1.566e-006	1.566e-006	1.566e-006	1.566e-006
29	Liq. Reynolds	2.096e+005 *	2.096e+005 *	2.096e+005 *	2.096e+005 *
30	Vap. Reynolds	9.170e+006 *	9.170e+006 *	9.170e+006 *	9.170e+006 *
31	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
32	Vapour Velocity (m/s)	2.175	2.175	2.176	2.176
33	Deposit Thickness (mm)	---	---	---	---
34	Deposit Volume (m3)	---	---	---	---
35	Length (m)	2.586e+004	2.587e+004	2.587e+004	2.588e+004
36	Elevation (m)	-6.200e-002	-6.400e-002	-6.600e-002	-6.800e-002
37	Pressure (bar)	88.81	88.81	88.81	88.81
38	Temperature (C)	51.04	51.04	51.04	51.04
39	Vapour Fraction	0.8696	0.8696	0.8696	0.8696
40	Heat Transferred (kJ/h-m)	560.6	560.6	560.5	560.5
41	Flow Regime	Transition	Transition	Transition	Transition
42	Water Flow Regime				
43	Liquid HoldUp	0.2065	0.2065	0.2065	0.2065
44	Water HoldUp	---	---	---	---
45	Friction Gradient (kPa/m)	6.881e-003	6.881e-003	6.881e-003	6.881e-003
46	Static Gradient (kPa/m)	-7.728e-004	-7.728e-004	-7.728e-004	-7.728e-004
47	Accel. Gradient (kPa/m)	4.412e-007	4.412e-007	4.412e-007	4.412e-007
48	Liq. Reynolds	2.096e+005 *	2.096e+005 *	2.096e+005 *	2.096e+005 *
49	Vap. Reynolds	9.170e+006 *	9.170e+006 *	9.170e+006 *	9.170e+006 *
50	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
51	Vapour Velocity (m/s)	2.176	2.175	2.175	2.175
52	Deposit Thickness (mm)	---	---	---	---
53	Deposit Volume (m3)	---	---	---	---
54	Length (m)	2.588e+004	2.589e+004	2.589e+004	2.590e+004
55	Elevation (m)	-7.000e-002	-0.1100	-0.1500	-0.1900
56	Pressure (bar)	88.81	88.81	88.81	88.81
57	Temperature (C)	51.04	51.04	51.03	51.03
58	Vapour Fraction	0.8696	0.8696	0.8696	0.8696
59	Heat Transferred (kJ/h-m)	560.4	560.4	560.4	560.3
60	Flow Regime	Transition	Transition	Transition	Transition
61	Water Flow Regime				
62	Liquid HoldUp	0.2065	0.1995	0.1995	0.1995
63	Water HoldUp	---	---	---	---
64	Friction Gradient (kPa/m)	6.881e-003	6.940e-003	6.939e-003	6.939e-003
65	Static Gradient (kPa/m)	-7.729e-004	-1.512e-002	-1.512e-002	-1.512e-002
66	Accel. Gradient (kPa/m)	4.412e-007	-5.910e-007	-5.910e-007	-5.910e-007
67	Liq. Reynolds	2.096e+005 *	2.096e+005 *	2.096e+005 *	2.096e+005 *
68	Vap. Reynolds	9.170e+006 *	9.170e+006 *	9.170e+006 *	9.170e+006 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
14	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175
15	Deposit Thickness	(mm)	---	---	---	---
16	Deposit Volume	(m3)	---	---	---	---
17	Length	(m)	2.590e+004	2.591e+004	2.591e+004	2.592e+004
18	Elevation	(m)	-0.2300	-0.2700	-0.3400	-0.4100
19	Pressure	(bar)	88.81	88.81	88.81	88.81
20	Temperature	(C)	51.03	51.03	51.03	51.03
21	Vapour Fraction		0.8696	0.8695	0.8695	0.8695
22	Heat Transferred	(kJ/h-m)	560.3	560.3	560.2	560.2
23	Flow Regime		Transition	Transition	Transition	Transition
24	Water Flow Regime					
25	Liquid HoldUp		0.1995	0.1995	0.1939	0.1939
26	Water HoldUp		---	---	---	---
27	Friction Gradient	(kPa/m)	6.939e-003	6.939e-003	6.990e-003	6.989e-003
28	Static Gradient	(kPa/m)	-1.512e-002	-1.512e-002	-2.600e-002	-2.600e-002
29	Accel. Gradient	(kPa/m)	-5.910e-007	-5.910e-007	-1.373e-006	-1.373e-006
30	Liq. Reynolds		2.096e+005 *	2.096e+005 *	2.096e+005 *	2.096e+005 *
31	Vap. Reynolds		9.170e+006 *	9.170e+006 *	9.170e+006 *	9.170e+006 *
32	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
33	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175
34	Deposit Thickness	(mm)	---	---	---	---
35	Deposit Volume	(m3)	---	---	---	---
36	Length	(m)	2.592e+004	2.593e+004	2.593e+004	2.594e+004
37	Elevation	(m)	-0.4800	-0.5500	-0.6200	-0.6500
38	Pressure	(bar)	88.81	88.81	88.81	88.81
39	Temperature	(C)	51.03	51.02	51.02	51.02
40	Vapour Fraction		0.8695	0.8695	0.8695	0.8695
41	Heat Transferred	(kJ/h-m)	560.2	560.1	560.1	560.1
42	Flow Regime		Transition	Transition	Transition	Transition
43	Water Flow Regime					
44	Liquid HoldUp		0.1939	0.1939	0.1939	0.2013
45	Water HoldUp		---	---	---	---
46	Friction Gradient	(kPa/m)	6.989e-003	6.989e-003	6.989e-003	6.923e-003
47	Static Gradient	(kPa/m)	-2.600e-002	-2.600e-002	-2.600e-002	-1.141e-002
48	Accel. Gradient	(kPa/m)	-1.373e-006	-1.373e-006	-1.373e-006	-3.239e-007
49	Liq. Reynolds		2.096e+005 *	2.096e+005 *	2.096e+005 *	2.096e+005 *
50	Vap. Reynolds		9.170e+006 *	9.170e+006 *	9.170e+006 *	9.170e+006 *
51	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
52	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175
53	Deposit Thickness	(mm)	---	---	---	---
54	Deposit Volume	(m3)	---	---	---	---
55	Length	(m)	2.594e+004	2.595e+004	2.595e+004	2.596e+004
56	Elevation	(m)	-0.6800	-0.7100	-0.7400	-0.7700
57	Pressure	(bar)	88.81	88.81	88.81	88.81
58	Temperature	(C)	51.02	51.02	51.02	51.02
59	Vapour Fraction		0.8695	0.8695	0.8695	0.8695
60	Heat Transferred	(kJ/h-m)	560.0	560.0	559.9	559.9
61	Flow Regime		Transition	Transition	Transition	Transition
62	Water Flow Regime					
63	Liquid HoldUp		0.2013	0.2014	0.2014	0.2014
64	Water HoldUp		---	---	---	---
65	Friction Gradient	(kPa/m)	6.923e-003	6.922e-003	6.922e-003	6.922e-003
66	Static Gradient	(kPa/m)	-1.141e-002	-1.141e-002	-1.141e-002	-1.141e-002
67	Accel. Gradient	(kPa/m)	-3.240e-007	-3.240e-007	-3.240e-007	-3.240e-007
68	Liq. Reynolds		2.096e+005 *	2.096e+005 *	2.096e+005 *	2.096e+005 *
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vap. Reynolds		9.170e+006 *	9.170e+006 *	9.170e+006 *	9.170e+006 *	
14	Liquid Velocity (m/s)		0.2044	0.2044	0.2044	0.2044	
15	Vapour Velocity (m/s)		2.175	2.175	2.175	2.175	
16	Deposit Thickness (mm)		---	---	---	---	
17	Deposit Volume (m3)		---	---	---	---	
18	Length (m)		2.596e+004	2.597e+004	2.597e+004	2.598e+004	
19	Elevation (m)		-0.6400	-0.5100	-0.3800	-0.2500	
20	Pressure (bar)		88.81	88.81	88.80	88.80	
21	Temperature (C)		51.01	51.01	51.01	51.00	
22	Vapour Fraction		0.8695	0.8695	0.8695	0.8695	
23	Heat Transferred (kJ/h-m)		559.8	559.8	559.7	559.6	
24	Flow Regime		Transition	Transition	Transition	Transition	
25	Water Flow Regime						
26	Liquid HoldUp		0.3617	0.3617	0.3617	0.3617	
27	Water HoldUp		---	---	---	---	
28	Friction Gradient (kPa/m)		6.058e-003	6.059e-003	6.059e-003	6.059e-003	
29	Static Gradient (kPa/m)		7.423e-002	7.423e-002	7.423e-002	7.423e-002	
30	Accel. Gradient (kPa/m)		5.798e-006	5.798e-006	5.799e-006	5.799e-006	
31	Liq. Reynolds		2.096e+005 *	2.096e+005 *	2.096e+005 *	2.096e+005 *	
32	Vap. Reynolds		9.170e+006 *	9.170e+006 *	9.170e+006 *	9.170e+006 *	
33	Liquid Velocity (m/s)		0.2044	0.2044	0.2044	0.2044	
34	Vapour Velocity (m/s)		2.175	2.175	2.175	2.175	
35	Deposit Thickness (mm)		---	---	---	---	
36	Deposit Volume (m3)		---	---	---	---	
37	Length (m)		2.598e+004	2.599e+004	2.599e+004	2.600e+004	
38	Elevation (m)		-0.1200	-0.1160	-0.1120	-0.1080	
39	Pressure (bar)		88.79	88.79	88.79	88.79	
40	Temperature (C)		51.00	51.00	51.00	51.00	
41	Vapour Fraction		0.8695	0.8695	0.8695	0.8695	
42	Heat Transferred (kJ/h-m)		559.5	559.5	559.4	559.4	
43	Flow Regime		Transition	Transition	Transition	Transition	
44	Water Flow Regime						
45	Liquid HoldUp		0.3617	0.2816	0.2816	0.2816	
46	Water HoldUp		---	---	---	---	
47	Friction Gradient (kPa/m)		6.059e-003	5.456e-003	5.456e-003	5.456e-003	
48	Static Gradient (kPa/m)		7.422e-002	1.903e-003	1.903e-003	1.903e-003	
49	Accel. Gradient (kPa/m)		5.800e-006	5.316e-007	5.316e-007	5.316e-007	
50	Liq. Reynolds		2.096e+005 *	2.096e+005 *	2.096e+005 *	2.096e+005 *	
51	Vap. Reynolds		9.170e+006 *	9.170e+006 *	9.170e+006 *	9.170e+006 *	
52	Liquid Velocity (m/s)		0.2044	0.2044	0.2044	0.2044	
53	Vapour Velocity (m/s)		2.175	2.175	2.175	2.175	
54	Deposit Thickness (mm)		---	---	---	---	
55	Deposit Volume (m3)		---	---	---	---	
56	Length (m)		2.600e+004	2.601e+004	2.601e+004	2.602e+004	
57	Elevation (m)		-0.1040	-0.1000	-0.1260	-0.1520	
58	Pressure (bar)		88.79	88.79	88.79	88.79	
59	Temperature (C)		51.00	50.99	50.99	50.99	
60	Vapour Fraction		0.8695	0.8695	0.8695	0.8695	
61	Heat Transferred (kJ/h-m)		559.3	559.3	559.3	559.2	
62	Flow Regime		Transition	Transition	Transition	Transition	
63	Water Flow Regime						
64	Liquid HoldUp		0.2816	0.2816	0.2020	0.2020	
65	Water HoldUp		---	---	---	---	
66	Friction Gradient (kPa/m)		5.456e-003	5.456e-003	6.918e-003	6.918e-003	
67	Static Gradient (kPa/m)		1.903e-003	1.903e-003	-1.007e-002	-1.007e-002	
68	Accel. Gradient (kPa/m)		5.316e-007	5.316e-007	-2.276e-007	-2.276e-007	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.096e+005 *	2.096e+005 *	2.096e+005 *	2.096e+005 *
14	Vap. Reynolds	9.170e+006 *	9.170e+006 *	9.170e+006 *	9.170e+006 *
15	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
16	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	2.602e+004	2.603e+004	2.603e+004	2.604e+004
20	Elevation (m)	-0.1780	-0.2040	-0.2300	-0.1920
21	Pressure (bar)	88.79	88.79	88.79	88.79
22	Temperature (C)	50.99	50.99	50.99	50.98
23	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
24	Heat Transferred (kJ/h-m)	559.2	559.1	559.1	559.0
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2020	0.2020	0.2020	0.3033
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	6.918e-003	6.918e-003	6.917e-003	5.922e-003
30	Static Gradient (kPa/m)	-1.007e-002	-1.007e-002	-1.007e-002	1.906e-002
31	Accel. Gradient (kPa/m)	-2.276e-007	-2.276e-007	-2.276e-007	1.804e-006
32	Liq. Reynolds	2.096e+005 *	2.096e+005 *	2.096e+005 *	2.096e+005 *
33	Vap. Reynolds	9.170e+006 *	9.170e+006 *	9.170e+006 *	9.170e+006 *
34	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
35	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	2.604e+004	2.605e+004	2.605e+004	2.606e+004
39	Elevation (m)	-0.1540	-0.1160	-7.800e-002	-4.000e-002
40	Pressure (bar)	88.79	88.79	88.79	88.79
41	Temperature (C)	50.98	50.98	50.98	50.98
42	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
43	Heat Transferred (kJ/h-m)	559.0	558.9	558.9	558.8
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.3033	0.3033	0.3033	0.3033
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	5.922e-003	5.922e-003	5.923e-003	5.923e-003
49	Static Gradient (kPa/m)	1.906e-002	1.906e-002	1.906e-002	1.906e-002
50	Accel. Gradient (kPa/m)	1.804e-006	1.804e-006	1.804e-006	1.805e-006
51	Liq. Reynolds	2.096e+005 *	2.096e+005 *	2.096e+005 *	2.096e+005 *
52	Vap. Reynolds	9.170e+006 *	9.170e+006 *	9.171e+006 *	9.171e+006 *
53	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
54	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	2.606e+004	2.607e+004	2.607e+004	2.608e+004
58	Elevation (m)	-1.800e-002	4.000e-003	2.600e-002	4.800e-002
59	Pressure (bar)	88.79	88.79	88.78	88.78
60	Temperature (C)	50.97	50.97	50.97	50.97
61	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
62	Heat Transferred (kJ/h-m)	558.8	558.7	558.7	558.6
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2931	0.2931	0.2931	0.2931
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	4.620e-003	4.620e-003	4.620e-003	4.620e-003
68	Static Gradient (kPa/m)	1.077e-002	1.077e-002	1.077e-002	1.077e-002
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient	(kPa/m)	1.112e-006	1.112e-006	1.112e-006	1.112e-006	
14	Liq. Reynolds		2.096e+005 *	2.096e+005 *	2.095e+005 *	2.095e+005 *	
15	Vap. Reynolds		9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *	
16	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
17	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175	
18	Deposit Thickness	(mm)	---	---	---	---	
19	Deposit Volume	(m3)	---	---	---	---	
20	Length	(m)	2.608e+004	2.609e+004	2.609e+004	2.610e+004	
21	Elevation	(m)	7.000e-002	4.000e-002	1.000e-002	-2.000e-002	
22	Pressure	(bar)	88.78	88.78	88.78	88.78	
23	Temperature	(C)	50.97	50.97	50.96	50.96	
24	Vapour Fraction		0.8695	0.8695	0.8695	0.8695	
25	Heat Transferred	(kJ/h-m)	558.6	558.5	558.5	558.5	
26	Flow Regime		Transition	Transition	Transition	Transition	
27	Water Flow Regime						
28	Liquid HoldUp		0.2931	0.2013	0.2013	0.2013	
29	Water HoldUp		---	---	---	---	
30	Friction Gradient	(kPa/m)	4.620e-003	6.923e-003	6.923e-003	6.923e-003	
31	Static Gradient	(kPa/m)	1.077e-002	-1.141e-002	-1.141e-002	-1.141e-002	
32	Accel. Gradient	(kPa/m)	1.112e-006	-3.240e-007	-3.240e-007	-3.240e-007	
33	Liq. Reynolds		2.095e+005 *	2.095e+005 *	2.095e+005 *	2.095e+005 *	
34	Vap. Reynolds		9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *	
35	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
36	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175	
37	Deposit Thickness	(mm)	---	---	---	---	
38	Deposit Volume	(m3)	---	---	---	---	
39	Length	(m)	2.610e+004	2.611e+004	2.611e+004	2.612e+004	
40	Elevation	(m)	-5.000e-002	-8.000e-002	-7.800e-002	-7.600e-002	
41	Pressure	(bar)	88.78	88.78	88.78	88.78	
42	Temperature	(C)	50.96	50.96	50.96	50.96	
43	Vapour Fraction		0.8695	0.8695	0.8695	0.8695	
44	Heat Transferred	(kJ/h-m)	558.4	558.4	558.3	558.3	
45	Flow Regime		Transition	Transition	Transition	Transition	
46	Water Flow Regime						
47	Liquid HoldUp		0.2013	0.2013	0.2804	0.2804	
48	Water HoldUp		---	---	---	---	
49	Friction Gradient	(kPa/m)	6.923e-003	6.923e-003	5.556e-003	5.556e-003	
50	Static Gradient	(kPa/m)	-1.141e-002	-1.141e-002	9.486e-004	9.486e-004	
51	Accel. Gradient	(kPa/m)	-3.240e-007	-3.240e-007	4.699e-007	4.699e-007	
52	Liq. Reynolds		2.095e+005 *	2.095e+005 *	2.095e+005 *	2.095e+005 *	
53	Vap. Reynolds		9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *	
54	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044	
55	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175	
56	Deposit Thickness	(mm)	---	---	---	---	
57	Deposit Volume	(m3)	---	---	---	---	
58	Length	(m)	2.612e+004	2.613e+004	2.613e+004	2.614e+004	
59	Elevation	(m)	-7.400e-002	-7.200e-002	-7.000e-002	-8.400e-002	
60	Pressure	(bar)	88.78	88.78	88.78	88.78	
61	Temperature	(C)	50.95	50.95	50.95	50.95	
62	Vapour Fraction		0.8695	0.8695	0.8695	0.8695	
63	Heat Transferred	(kJ/h-m)	558.2	558.2	558.1	558.1	
64	Flow Regime		Transition	Transition	Transition	Transition	
65	Water Flow Regime						
66	Liquid HoldUp		0.2804	0.2804	0.2804	0.2043	
67	Water HoldUp		---	---	---	---	
68	Friction Gradient	(kPa/m)	5.555e-003	5.555e-003	5.555e-003	6.898e-003	
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3	 Epic Atlantic Limited		Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	9.486e-004	9.486e-004	9.486e-004	-5.374e-003
14	Accel. Gradient	(kPa/m)	4.698e-007	4.698e-007	4.698e-007	1.101e-007
15	Liq. Reynolds		2.095e+005 *	2.095e+005 *	2.095e+005 *	2.095e+005 *
16	Vap. Reynolds		9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
17	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
18	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	2.614e+004	2.615e+004	2.615e+004	2.616e+004
22	Elevation	(m)	-9.800e-002	-0.1120	-0.1260	-0.1400
23	Pressure	(bar)	88.78	88.78	88.78	88.78
24	Temperature	(C)	50.95	50.95	50.94	50.94
25	Vapour Fraction		0.8695	0.8695	0.8695	0.8695
26	Heat Transferred	(kJ/h-m)	558.1	558.0	558.0	558.0
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2043	0.2043	0.2043	0.2043
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	6.898e-003	6.898e-003	6.898e-003	6.898e-003
32	Static Gradient	(kPa/m)	-5.374e-003	-5.374e-003	-5.374e-003	-5.374e-003
33	Accel. Gradient	(kPa/m)	1.101e-007	1.101e-007	1.101e-007	1.101e-007
34	Liq. Reynolds		2.095e+005 *	2.095e+005 *	2.095e+005 *	2.095e+005 *
35	Vap. Reynolds		9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
36	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
37	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	2.616e+004	2.617e+004	2.617e+004	2.618e+004
41	Elevation	(m)	-0.1520	-0.1640	-0.1760	-0.1880
42	Pressure	(bar)	88.78	88.78	88.78	88.78
43	Temperature	(C)	50.94	50.94	50.94	50.94
44	Vapour Fraction		0.8695	0.8695	0.8695	0.8695
45	Heat Transferred	(kJ/h-m)	557.9	557.9	557.8	557.8
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2047	0.2047	0.2047	0.2047
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	6.895e-003	6.895e-003	6.895e-003	6.895e-003
51	Static Gradient	(kPa/m)	-4.611e-003	-4.611e-003	-4.611e-003	-4.612e-003
52	Accel. Gradient	(kPa/m)	1.649e-007	1.649e-007	1.649e-007	1.649e-007
53	Liq. Reynolds		2.095e+005 *	2.095e+005 *	2.095e+005 *	2.095e+005 *
54	Vap. Reynolds		9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
55	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
56	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	2.618e+004	2.619e+004	2.619e+004	2.620e+004
60	Elevation	(m)	-0.2000	-0.2360	-0.2720	-0.3080
61	Pressure	(bar)	88.78	88.78	88.78	88.78
62	Temperature	(C)	50.94	50.93	50.93	50.93
63	Vapour Fraction		0.8695	0.8695	0.8695	0.8695
64	Heat Transferred	(kJ/h-m)	557.7	557.7	557.7	557.6
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2047	0.2002	0.2002	0.2002
68	Water HoldUp		---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3						
4						
5	Unit Set: Mixed					
6	Date/Time: Mon Jun 13 15:02:44 2022					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	PROFILES					
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	6.895e-003	6.932e-003	6.932e-003	6.932e-003
14	Static Gradient	(kPa/m)	-4.612e-003	-1.364e-002	-1.364e-002	-1.364e-002
15	Accel. Gradient	(kPa/m)	1.649e-007	-4.848e-007	-4.848e-007	-4.848e-007
16	Liq. Reynolds		2.095e+005 *	2.095e+005 *	2.095e+005 *	2.095e+005 *
17	Vap. Reynolds		9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
18	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
19	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	2.620e+004	2.621e+004	2.621e+004	2.622e+004
23	Elevation	(m)	-0.3440	-0.3800	-0.3440	-0.3080
24	Pressure	(bar)	88.78	88.78	88.78	88.78
25	Temperature	(C)	50.93	50.93	50.93	50.92
26	Vapour Fraction		0.8695	0.8695	0.8695	0.8695
27	Heat Transferred	(kJ/h-m)	557.6	557.6	557.5	557.4
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2002	0.2003	0.3024	0.3024
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	6.932e-003	6.931e-003	5.895e-003	5.895e-003
33	Static Gradient	(kPa/m)	-1.364e-002	-1.364e-002	1.826e-002	1.826e-002
34	Accel. Gradient	(kPa/m)	-4.848e-007	-4.849e-007	1.744e-006	1.744e-006
35	Liq. Reynolds		2.095e+005 *	2.095e+005 *	2.095e+005 *	2.095e+005 *
36	Vap. Reynolds		9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
37	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
38	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	2.622e+004	2.623e+004	2.623e+004	2.624e+004
42	Elevation	(m)	-0.2720	-0.2360	-0.2000	-0.2060
43	Pressure	(bar)	88.78	88.78	88.78	88.78
44	Temperature	(C)	50.92	50.92	50.92	50.92
45	Vapour Fraction		0.8695	0.8695	0.8695	0.8695
46	Heat Transferred	(kJ/h-m)	557.4	557.3	557.3	557.2
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.3024	0.3024	0.3024	0.2058
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	5.895e-003	5.895e-003	5.895e-003	6.885e-003
52	Static Gradient	(kPa/m)	1.826e-002	1.826e-002	1.826e-002	-2.314e-003
53	Accel. Gradient	(kPa/m)	1.745e-006	1.745e-006	1.745e-006	3.302e-007
54	Liq. Reynolds		2.095e+005 *	2.095e+005 *	2.095e+005 *	2.095e+005 *
55	Vap. Reynolds		9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
56	Liquid Velocity	(m/s)	0.2044	0.2044	0.2044	0.2044
57	Vapour Velocity	(m/s)	2.175	2.175	2.175	2.175
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	2.624e+004	2.625e+004	2.625e+004	2.626e+004
61	Elevation	(m)	-0.2120	-0.2180	-0.2240	-0.2300
62	Pressure	(bar)	88.78	88.78	88.78	88.78
63	Temperature	(C)	50.92	50.91	50.91	50.91
64	Vapour Fraction		0.8695	0.8695	0.8695	0.8695
65	Heat Transferred	(kJ/h-m)	557.2	557.2	557.1	557.1
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2058	0.2058	0.2058	0.2058
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	6.885e-003	6.885e-003	6.885e-003	6.885e-003
15	Static Gradient (kPa/m)	-2.314e-003	-2.314e-003	-2.314e-003	-2.314e-003
16	Accel. Gradient (kPa/m)	3.302e-007	3.302e-007	3.302e-007	3.302e-007
17	Liq. Reynolds	2.095e+005 *	2.095e+005 *	2.095e+005 *	2.095e+005 *
18	Vap. Reynolds	9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
19	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
20	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	2.626e+004	2.627e+004	2.627e+004	2.628e+004
24	Elevation (m)	-0.2040	-0.1780	-0.1520	-0.1260
25	Pressure (bar)	88.78	88.77	88.77	88.77
26	Temperature (C)	50.91	50.91	50.90	50.90
27	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
28	Heat Transferred (kJ/h-m)	557.0	557.0	556.9	556.9
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2957	0.2957	0.2957	0.2957
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	5.421e-003	5.421e-003	5.421e-003	5.421e-003
34	Static Gradient (kPa/m)	1.280e-002	1.281e-002	1.281e-002	1.281e-002
35	Accel. Gradient (kPa/m)	1.317e-006	1.317e-006	1.317e-006	1.317e-006
36	Liq. Reynolds	2.095e+005 *	2.095e+005 *	2.095e+005 *	2.095e+005 *
37	Vap. Reynolds	9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
38	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
39	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	2.628e+004	2.629e+004	2.629e+004	2.630e+004
43	Elevation (m)	-0.1000	-0.1140	-0.1280	-0.1420
44	Pressure (bar)	88.77	88.77	88.77	88.77
45	Temperature (C)	50.90	50.90	50.90	50.90
46	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
47	Heat Transferred (kJ/h-m)	556.8	556.8	556.7	556.7
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2957	0.2043	0.2043	0.2043
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	5.421e-003	6.897e-003	6.897e-003	6.897e-003
53	Static Gradient (kPa/m)	1.281e-002	-5.374e-003	-5.374e-003	-5.374e-003
54	Accel. Gradient (kPa/m)	1.317e-006	1.100e-007	1.100e-007	1.100e-007
55	Liq. Reynolds	2.095e+005 *	2.095e+005 *	2.095e+005 *	2.095e+005 *
56	Vap. Reynolds	9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
57	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
58	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	2.630e+004	2.631e+004	2.631e+004	2.632e+004
62	Elevation (m)	-0.1560	-0.1700	-0.1700	-0.1700
63	Pressure (bar)	88.77	88.77	88.77	88.77
64	Temperature (C)	50.89	50.89	50.89	50.89
65	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
66	Heat Transferred (kJ/h-m)	556.7	556.6	556.6	556.5
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.2043	0.2043	0.3021	0.3021
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	6.897e-003	6.897e-003	5.887e-003	5.887e-003
16	Static Gradient (kPa/m)	-5.374e-003	-5.374e-003	0.0000	0.0000
17	Accel. Gradient (kPa/m)	1.100e-007	1.100e-007	4.253e-007	4.253e-007
18	Liq. Reynolds	2.095e+005 *	2.095e+005 *	2.095e+005 *	2.095e+005 *
19	Vap. Reynolds	9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
20	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
21	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	2.632e+004	2.633e+004	2.633e+004	2.634e+004
25	Elevation (m)	-0.1700	-0.1700	-0.1700	-0.1840
26	Pressure (bar)	88.77	88.77	88.77	88.77
27	Temperature (C)	50.89	50.89	50.88	50.88
28	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
29	Heat Transferred (kJ/h-m)	556.5	556.4	556.4	556.4
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.3021	0.3021	0.3021	0.2044
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	5.887e-003	5.887e-003	5.887e-003	6.897e-003
35	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-5.375e-003
36	Accel. Gradient (kPa/m)	4.253e-007	4.253e-007	4.253e-007	1.100e-007
37	Liq. Reynolds	2.095e+005 *	2.095e+005 *	2.095e+005 *	2.095e+005 *
38	Vap. Reynolds	9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
39	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
40	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	2.634e+004	2.635e+004	2.635e+004	2.636e+004
44	Elevation (m)	-0.1980	-0.2120	-0.2260	-0.2400
45	Pressure (bar)	88.77	88.77	88.77	88.77
46	Temperature (C)	50.88	50.88	50.88	50.88
47	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
48	Heat Transferred (kJ/h-m)	556.3	556.3	556.2	556.2
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2044	0.2044	0.2044	0.2044
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.897e-003	6.897e-003	6.897e-003	6.897e-003
54	Static Gradient (kPa/m)	-5.375e-003	-5.375e-003	-5.375e-003	-5.375e-003
55	Accel. Gradient (kPa/m)	1.100e-007	1.100e-007	1.099e-007	1.099e-007
56	Liq. Reynolds	2.095e+005 *	2.095e+005 *	2.095e+005 *	2.095e+005 *
57	Vap. Reynolds	9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
58	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
59	Vapour Velocity (m/s)	2.175	2.175	2.175	2.175
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	2.636e+004	2.637e+004	2.637e+004	2.638e+004
63	Elevation (m)	-0.2540	-0.2680	-0.2820	-0.2960
64	Pressure (bar)	88.77	88.77	88.77	88.77
65	Temperature (C)	50.88	50.87	50.87	50.87
66	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
67	Heat Transferred (kJ/h-m)	556.1	556.1	556.1	556.0
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 427 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2044	0.2044	0.2044	0.2044
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.897e-003	6.897e-003	6.897e-003	6.897e-003
17	Static Gradient (kPa/m)	-5.375e-003	-5.375e-003	-5.375e-003	-5.375e-003
18	Accel. Gradient (kPa/m)	1.099e-007	1.099e-007	1.099e-007	1.099e-007
19	Liq. Reynolds	2.095e+005 *	2.095e+005 *	2.095e+005 *	2.095e+005 *
20	Vap. Reynolds	9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
21	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
22	Vapour Velocity (m/s)	2.175	2.175	2.174	2.174
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	2.638e+004	2.639e+004	2.639e+004	2.640e+004
26	Elevation (m)	-0.3100	-0.2840	-0.2580	-0.2320
27	Pressure (bar)	88.77	88.77	88.77	88.77
28	Temperature (C)	50.87	50.87	50.87	50.86
29	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
30	Heat Transferred (kJ/h-m)	556.0	555.9	555.9	555.8
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2044	0.2957	0.2957	0.2957
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	6.896e-003	5.421e-003	5.421e-003	5.421e-003
36	Static Gradient (kPa/m)	-5.375e-003	1.281e-002	1.281e-002	1.281e-002
37	Accel. Gradient (kPa/m)	1.099e-007	1.317e-006	1.317e-006	1.317e-006
38	Liq. Reynolds	2.095e+005 *	2.095e+005 *	2.095e+005 *	2.095e+005 *
39	Vap. Reynolds	9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
40	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
41	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	2.640e+004	2.641e+004	2.641e+004	2.642e+004
45	Elevation (m)	-0.2060	-0.1800	-0.1600	-0.1400
46	Pressure (bar)	88.77	88.76	88.76	88.76
47	Temperature (C)	50.86	50.86	50.86	50.86
48	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
49	Heat Transferred (kJ/h-m)	555.8	555.7	555.7	555.6
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2957	0.2957	0.2910	0.2910
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	5.421e-003	5.421e-003	4.763e-003	4.763e-003
55	Static Gradient (kPa/m)	1.281e-002	1.281e-002	9.061e-003	9.061e-003
56	Accel. Gradient (kPa/m)	1.317e-006	1.317e-006	9.986e-007	9.986e-007
57	Liq. Reynolds	2.094e+005 *	2.094e+005 *	2.094e+005 *	2.094e+005 *
58	Vap. Reynolds	9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
59	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
60	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	2.642e+004	2.643e+004	2.644e+004	2.644e+004
64	Elevation (m)	-0.1200	-0.1000	-8.000e-002	-0.1380
65	Pressure (bar)	88.76	88.76	88.76	88.76
66	Temperature (C)	50.85	50.85	50.85	50.85
67	Vapour Fraction	0.8695	0.8695	0.8695	0.8695
68	Heat Transferred (kJ/h-m)	555.6	555.5	555.5	555.4
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 428 of 937

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2910	0.2910	0.2910	0.1962
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	4.763e-003	4.763e-003	4.763e-003	6.967e-003
18	Static Gradient (kPa/m)	9.062e-003	9.062e-003	9.062e-003	-2.170e-002
19	Accel. Gradient (kPa/m)	9.986e-007	9.986e-007	9.986e-007	-1.064e-006
20	Liq. Reynolds	2.094e+005 *	2.094e+005 *	2.094e+005 *	2.094e+005 *
21	Vap. Reynolds	9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
22	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
23	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	2.645e+004	2.645e+004	2.646e+004	2.646e+004
27	Elevation (m)	-0.1960	-0.2540	-0.3120	-0.3700
28	Pressure (bar)	88.76	88.76	88.76	88.76
29	Temperature (C)	50.85	50.85	50.84	50.84
30	Vapour Fraction	0.8695	0.8695	0.8695	0.8694
31	Heat Transferred (kJ/h-m)	555.4	555.4	555.3	555.3
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.1962	0.1962	0.1962	0.1962
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	6.967e-003	6.967e-003	6.967e-003	6.966e-003
37	Static Gradient (kPa/m)	-2.170e-002	-2.170e-002	-2.170e-002	-2.170e-002
38	Accel. Gradient (kPa/m)	-1.064e-006	-1.064e-006	-1.064e-006	-1.064e-006
39	Liq. Reynolds	2.094e+005 *	2.094e+005 *	2.094e+005 *	2.094e+005 *
40	Vap. Reynolds	9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
41	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
42	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	2.647e+004	2.647e+004	2.648e+004	2.648e+004
46	Elevation (m)	-0.3540	-0.3380	-0.3220	-0.3060
47	Pressure (bar)	88.76	88.76	88.76	88.76
48	Temperature (C)	50.84	50.84	50.84	50.84
49	Vapour Fraction	0.8694	0.8694	0.8694	0.8694
50	Heat Transferred (kJ/h-m)	555.2	555.2	555.1	555.1
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2894	0.2894	0.2894	0.2894
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	4.881e-003	4.881e-003	4.881e-003	4.881e-003
56	Static Gradient (kPa/m)	7.760e-003	7.760e-003	7.760e-003	7.760e-003
57	Accel. Gradient (kPa/m)	9.130e-007	9.130e-007	9.130e-007	9.130e-007
58	Liq. Reynolds	2.094e+005 *	2.094e+005 *	2.094e+005 *	2.094e+005 *
59	Vap. Reynolds	9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
60	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
61	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	2.649e+004	2.649e+004	2.650e+004	2.650e+004
65	Elevation (m)	-0.2900	-0.3100	-0.3300	-0.3500
66	Pressure (bar)	88.76	88.76	88.76	88.76
67	Temperature (C)	50.83	50.83	50.83	50.83
68	Vapour Fraction	0.8694	0.8694	0.8694	0.8694
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 429 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022	
6						
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Heat Transferred (kJ/h-m)	555.0	555.0	555.0	554.9	
14	Flow Regime	Transition	Transition	Transition	Transition	
15	Water Flow Regime					
16	Liquid HoldUp	0.2894	0.2033	0.2033	0.2033	
17	Water HoldUp	---	---	---	---	
18	Friction Gradient (kPa/m)	4.881e-003	6.905e-003	6.905e-003	6.905e-003	
19	Static Gradient (kPa/m)	7.760e-003	-7.652e-003	-7.652e-003	-7.653e-003	
20	Accel. Gradient (kPa/m)	9.131e-007	-5.398e-008	-5.399e-008	-5.400e-008	
21	Liq. Reynolds	2.094e+005 *	2.094e+005 *	2.094e+005 *	2.094e+005 *	
22	Vap. Reynolds	9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *	
23	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044	
24	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174	
25	Deposit Thickness (mm)	---	---	---	---	
26	Deposit Volume (m3)	---	---	---	---	
27	Length (m)	2.651e+004	2.651e+004	2.652e+004	2.652e+004	
28	Elevation (m)	-0.3700	-0.3900	-0.3940	-0.3980	
29	Pressure (bar)	88.76	88.76	88.76	88.76	
30	Temperature (C)	50.83	50.83	50.83	50.82	
31	Vapour Fraction	0.8694	0.8694	0.8694	0.8694	
32	Heat Transferred (kJ/h-m)	554.9	554.9	554.8	554.8	
33	Flow Regime	Transition	Transition	Transition	Transition	
34	Water Flow Regime					
35	Liquid HoldUp	0.2033	0.2033	0.2062	0.2062	
36	Water HoldUp	---	---	---	---	
37	Friction Gradient (kPa/m)	6.905e-003	6.905e-003	6.881e-003	6.881e-003	
38	Static Gradient (kPa/m)	-7.653e-003	-7.653e-003	-1.545e-003	-1.545e-003	
39	Accel. Gradient (kPa/m)	-5.400e-008	-5.401e-008	3.854e-007	3.854e-007	
40	Liq. Reynolds	2.094e+005 *	2.094e+005 *	2.094e+005 *	2.094e+005 *	
41	Vap. Reynolds	9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *	
42	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044	
43	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174	
44	Deposit Thickness (mm)	---	---	---	---	
45	Deposit Volume (m3)	---	---	---	---	
46	Length (m)	2.653e+004	2.653e+004	2.654e+004	2.654e+004	
47	Elevation (m)	-0.4020	-0.4060	-0.4100	-0.4040	
48	Pressure (bar)	88.76	88.76	88.76	88.76	
49	Temperature (C)	50.82	50.82	50.82	50.82	
50	Vapour Fraction	0.8694	0.8694	0.8694	0.8694	
51	Heat Transferred (kJ/h-m)	554.7	554.7	554.6	554.6	
52	Flow Regime	Transition	Transition	Transition	Transition	
53	Water Flow Regime					
54	Liquid HoldUp	0.2062	0.2062	0.2062	0.2830	
55	Water HoldUp	---	---	---	---	
56	Friction Gradient (kPa/m)	6.881e-003	6.881e-003	6.881e-003	5.354e-003	
57	Static Gradient (kPa/m)	-1.545e-003	-1.545e-003	-1.545e-003	2.865e-003	
58	Accel. Gradient (kPa/m)	3.854e-007	3.854e-007	3.854e-007	5.936e-007	
59	Liq. Reynolds	2.094e+005 *	2.094e+005 *	2.094e+005 *	2.094e+005 *	
60	Vap. Reynolds	9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *	
61	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044	
62	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174	
63	Deposit Thickness (mm)	---	---	---	---	
64	Deposit Volume (m3)	---	---	---	---	
65	Length (m)	2.655e+004	2.655e+004	2.656e+004	2.656e+004	
66	Elevation (m)	-0.3980	-0.3920	-0.3860	-0.3800	
67	Pressure (bar)	88.76	88.76	88.76	88.76	
68	Temperature (C)	50.82	50.81	50.81	50.81	
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 430 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Fraction		0.8694	0.8694	0.8694	0.8694	
14	Heat Transferred (kJ/h-m)		554.5	554.5	554.4	554.4	
15	Flow Regime		Transition	Transition	Transition	Transition	
16	Water Flow Regime						
17	Liquid HoldUp		0.2830	0.2830	0.2830	0.2830	
18	Water HoldUp		---	---	---	---	
19	Friction Gradient (kPa/m)		5.354e-003	5.354e-003	5.354e-003	5.354e-003	
20	Static Gradient (kPa/m)		2.865e-003	2.865e-003	2.865e-003	2.865e-003	
21	Accel. Gradient (kPa/m)		5.936e-007	5.936e-007	5.936e-007	5.936e-007	
22	Liq. Reynolds		2.094e+005 *	2.094e+005 *	2.094e+005 *	2.094e+005 *	
23	Vap. Reynolds		9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *	
24	Liquid Velocity (m/s)		0.2044	0.2044	0.2044	0.2044	
25	Vapour Velocity (m/s)		2.174	2.174	2.174	2.174	
26	Deposit Thickness (mm)		---	---	---	---	
27	Deposit Volume (m3)		---	---	---	---	
28	Length (m)		2.657e+004	2.657e+004	2.658e+004	2.658e+004	
29	Elevation (m)		-0.3960	-0.4120	-0.4280	-0.4440	
30	Pressure (bar)		88.76	88.76	88.76	88.76	
31	Temperature (C)		50.81	50.81	50.81	50.80	
32	Vapour Fraction		0.8694	0.8694	0.8694	0.8694	
33	Heat Transferred (kJ/h-m)		554.4	554.3	554.3	554.2	
34	Flow Regime		Transition	Transition	Transition	Transition	
35	Water Flow Regime						
36	Liquid HoldUp		0.2040	0.2040	0.2040	0.2040	
37	Water HoldUp		---	---	---	---	
38	Friction Gradient (kPa/m)		6.898e-003	6.898e-003	6.898e-003	6.898e-003	
39	Static Gradient (kPa/m)		-6.137e-003	-6.137e-003	-6.137e-003	-6.137e-003	
40	Accel. Gradient (kPa/m)		5.503e-008	5.502e-008	5.501e-008	5.501e-008	
41	Liq. Reynolds		2.094e+005 *	2.094e+005 *	2.094e+005 *	2.094e+005 *	
42	Vap. Reynolds		9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *	
43	Liquid Velocity (m/s)		0.2044	0.2044	0.2044	0.2044	
44	Vapour Velocity (m/s)		2.174	2.174	2.174	2.174	
45	Deposit Thickness (mm)		---	---	---	---	
46	Deposit Volume (m3)		---	---	---	---	
47	Length (m)		2.659e+004	2.659e+004	2.660e+004	2.660e+004	
48	Elevation (m)		-0.4600	-0.4560	-0.4520	-0.4480	
49	Pressure (bar)		88.76	88.76	88.76	88.76	
50	Temperature (C)		50.80	50.80	50.80	50.80	
51	Vapour Fraction		0.8694	0.8694	0.8694	0.8694	
52	Heat Transferred (kJ/h-m)		554.2	554.1	554.1	554.1	
53	Flow Regime		Transition	Transition	Transition	Transition	
54	Water Flow Regime						
55	Liquid HoldUp		0.2040	0.2818	0.2818	0.2818	
56	Water HoldUp		---	---	---	---	
57	Friction Gradient (kPa/m)		6.898e-003	5.452e-003	5.452e-003	5.452e-003	
58	Static Gradient (kPa/m)		-6.137e-003	1.904e-003	1.904e-003	1.904e-003	
59	Accel. Gradient (kPa/m)		5.500e-008	5.313e-007	5.313e-007	5.312e-007	
60	Liq. Reynolds		2.094e+005 *	2.094e+005 *	2.094e+005 *	2.094e+005 *	
61	Vap. Reynolds		9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *	
62	Liquid Velocity (m/s)		0.2044	0.2044	0.2044	0.2044	
63	Vapour Velocity (m/s)		2.174	2.174	2.174	2.174	
64	Deposit Thickness (mm)		---	---	---	---	
65	Deposit Volume (m3)		---	---	---	---	
66	Length (m)		2.661e+004	2.661e+004	2.662e+004	2.662e+004	
67	Elevation (m)		-0.4440	-0.4400	-0.4680	-0.4960	
68	Pressure (bar)		88.76	88.76	88.76	88.76	
69	Aspen Technology Inc.		Aspen HYSYS Version 10				Page 431 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Pipe Segment: ELEP-Akabel Pipeline (continued)		
3					
4					
5			Unit Set: Mixed		
6			Date/Time: Mon Jun 13 15:02:44 2022		
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	50.80	50.79	50.79	50.79
14	Vapour Fraction	0.8694	0.8694	0.8694	0.8694
15	Heat Transferred (kJ/h-m)	554.0	554.0	553.9	553.9
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2818	0.2818	0.2018	0.2018
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	5.452e-003	5.452e-003	6.917e-003	6.916e-003
21	Static Gradient (kPa/m)	1.904e-003	1.904e-003	-1.067e-002	-1.067e-002
22	Accel. Gradient (kPa/m)	5.312e-007	5.312e-007	-2.707e-007	-2.707e-007
23	Liq. Reynolds	2.094e+005 *	2.094e+005 *	2.094e+005 *	2.094e+005 *
24	Vap. Reynolds	9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
25	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
26	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	2.663e+004	2.663e+004	2.664e+004	2.664e+004
30	Elevation (m)	-0.5240	-0.5520	-0.5800	-0.5760
31	Pressure (bar)	88.76	88.76	88.76	88.76
32	Temperature (C)	50.79	50.79	50.79	50.79
33	Vapour Fraction	0.8694	0.8694	0.8694	0.8694
34	Heat Transferred (kJ/h-m)	553.9	553.8	553.8	553.7
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2018	0.2018	0.2018	0.2818
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	6.916e-003	6.916e-003	6.916e-003	5.451e-003
40	Static Gradient (kPa/m)	-1.067e-002	-1.067e-002	-1.067e-002	1.904e-003
41	Accel. Gradient (kPa/m)	-2.707e-007	-2.708e-007	-2.708e-007	5.312e-007
42	Liq. Reynolds	2.094e+005 *	2.094e+005 *	2.094e+005 *	2.094e+005 *
43	Vap. Reynolds	9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
44	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
45	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	2.665e+004	2.665e+004	2.666e+004	2.666e+004
49	Elevation (m)	-0.5720	-0.5680	-0.5640	-0.5600
50	Pressure (bar)	88.76	88.76	88.76	88.76
51	Temperature (C)	50.78	50.78	50.78	50.78
52	Vapour Fraction	0.8694	0.8694	0.8694	0.8694
53	Heat Transferred (kJ/h-m)	553.7	553.6	553.6	553.5
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2818	0.2818	0.2818	0.2818
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	5.451e-003	5.451e-003	5.451e-003	5.451e-003
59	Static Gradient (kPa/m)	1.904e-003	1.904e-003	1.904e-003	1.904e-003
60	Accel. Gradient (kPa/m)	5.312e-007	5.312e-007	5.312e-007	5.312e-007
61	Liq. Reynolds	2.094e+005 *	2.094e+005 *	2.094e+005 *	2.094e+005 *
62	Vap. Reynolds	9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *
63	Liquid Velocity (m/s)	0.2044	0.2044	0.2044	0.2044
64	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	2.667e+004	2.667e+004	2.668e+004	2.668e+004
68	Elevation (m)	-0.5340	-0.5080	-0.4820	-0.4560
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 432 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	88.75	88.75	88.75	88.75		
14	Temperature (C)	50.78	50.77	50.77	50.77		
15	Vapour Fraction	0.8694	0.8694	0.8694	0.8694		
16	Heat Transferred (kJ/h-m)	553.5	553.4	553.4	553.4		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.2958	0.2958	0.2958	0.2958		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	5.421e-003	5.421e-003	5.421e-003	5.421e-003		
22	Static Gradient (kPa/m)	1.281e-002	1.281e-002	1.281e-002	1.281e-002		
23	Accel. Gradient (kPa/m)	1.317e-006	1.317e-006	1.317e-006	1.317e-006		
24	Liq. Reynolds	2.094e+005 *	2.094e+005 *	2.094e+005 *	2.094e+005 *		
25	Vap. Reynolds	9.171e+006 *	9.171e+006 *	9.171e+006 *	9.171e+006 *		
26	Liquid Velocity (m/s)	0.2044	0.2045	0.2045	0.2045		
27	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	2.669e+004	2.669e+004	2.670e+004	2.670e+004		
31	Elevation (m)	-0.4300	-0.3840	-0.3380	-0.2920		
32	Pressure (bar)	88.75	88.75	88.75	88.75		
33	Temperature (C)	50.77	50.77	50.76	50.76		
34	Vapour Fraction	0.8694	0.8694	0.8694	0.8694		
35	Heat Transferred (kJ/h-m)	553.3	553.2	553.2	553.1		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.2958	0.3058	0.3058	0.3058		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	5.421e-003	5.970e-003	5.970e-003	5.970e-003		
41	Static Gradient (kPa/m)	1.281e-002	2.110e-002	2.110e-002	2.110e-002		
42	Accel. Gradient (kPa/m)	1.317e-006	1.955e-006	1.955e-006	1.955e-006		
43	Liq. Reynolds	2.094e+005 *	2.094e+005 *	2.094e+005 *	2.094e+005 *		
44	Vap. Reynolds	9.171e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *		
45	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045		
46	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	2.671e+004	2.671e+004	2.672e+004	2.672e+004		
50	Elevation (m)	-0.2460	-0.2000	-0.1840	-0.1680		
51	Pressure (bar)	88.74	88.74	88.74	88.74		
52	Temperature (C)	50.76	50.76	50.76	50.75		
53	Vapour Fraction	0.8694	0.8694	0.8694	0.8694		
54	Heat Transferred (kJ/h-m)	553.1	553.0	553.0	552.9		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.3058	0.3058	0.2885	0.2885		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	5.970e-003	5.970e-003	4.946e-003	4.946e-003		
60	Static Gradient (kPa/m)	2.110e-002	2.110e-002	7.040e-003	7.040e-003		
61	Accel. Gradient (kPa/m)	1.955e-006	1.955e-006	8.658e-007	8.658e-007		
62	Liq. Reynolds	2.094e+005 *	2.093e+005 *	2.093e+005 *	2.093e+005 *		
63	Vap. Reynolds	9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *		
64	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045		
65	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	2.673e+004	2.674e+004	2.674e+004	2.674e+004		
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 433 of 937		

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Elevation	(m)	-0.1520	-0.1360	-0.1200	-0.1020	
14	Pressure	(bar)	88.74	88.74	88.74	88.74	
15	Temperature	(C)	50.75	50.75	50.75	50.75	
16	Vapour Fraction		0.8694	0.8694	0.8694	0.8694	
17	Heat Transferred	(kJ/h-m)	552.8	552.8	552.7	552.7	
18	Flow Regime		Transition	Transition	Transition	Transition	
19	Water Flow Regime						
20	Liquid HoldUp		0.2885	0.2885	0.2885	0.2935	
21	Water HoldUp		---	---	---	---	
22	Friction Gradient	(kPa/m)	4.946e-003	4.946e-003	4.946e-003	4.798e-003	
23	Static Gradient	(kPa/m)	7.040e-003	7.040e-003	7.040e-003	1.103e-002	
24	Accel. Gradient	(kPa/m)	8.658e-007	8.658e-007	8.658e-007	1.143e-006	
25	Liq. Reynolds		2.093e+005 *	2.093e+005 *	2.093e+005 *	2.093e+005 *	
26	Vap. Reynolds		9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *	
27	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
28	Vapour Velocity	(m/s)	2.174	2.174	2.174	2.174	
29	Deposit Thickness	(mm)	---	---	---	---	
30	Deposit Volume	(m3)	---	---	---	---	
31	Length	(m)	2.675e+004	2.675e+004	2.676e+004	2.676e+004	
32	Elevation	(m)	-8.400e-002	-6.600e-002	-4.800e-002	-3.000e-002	
33	Pressure	(bar)	88.74	88.74	88.74	88.74	
34	Temperature	(C)	50.75	50.74	50.74	50.74	
35	Vapour Fraction		0.8694	0.8694	0.8694	0.8694	
36	Heat Transferred	(kJ/h-m)	552.7	552.6	552.6	552.6	
37	Flow Regime		Transition	Transition	Transition	Transition	
38	Water Flow Regime						
39	Liquid HoldUp		0.2935	0.2935	0.2935	0.2935	
40	Water HoldUp		---	---	---	---	
41	Friction Gradient	(kPa/m)	4.798e-003	4.798e-003	4.798e-003	4.798e-003	
42	Static Gradient	(kPa/m)	1.103e-002	1.103e-002	1.103e-002	1.103e-002	
43	Accel. Gradient	(kPa/m)	1.143e-006	1.143e-006	1.143e-006	1.143e-006	
44	Liq. Reynolds		2.093e+005 *	2.093e+005 *	2.093e+005 *	2.093e+005 *	
45	Vap. Reynolds		9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *	
46	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
47	Vapour Velocity	(m/s)	2.174	2.174	2.174	2.174	
48	Deposit Thickness	(mm)	---	---	---	---	
49	Deposit Volume	(m3)	---	---	---	---	
50	Length	(m)	2.677e+004	2.677e+004	2.678e+004	2.678e+004	
51	Elevation	(m)	-2.600e-002	-2.200e-002	-1.800e-002	-1.400e-002	
52	Pressure	(bar)	88.74	88.74	88.74	88.74	
53	Temperature	(C)	50.74	50.74	50.74	50.73	
54	Vapour Fraction		0.8694	0.8694	0.8694	0.8694	
55	Heat Transferred	(kJ/h-m)	552.5	552.5	552.4	552.4	
56	Flow Regime		Transition	Transition	Transition	Transition	
57	Water Flow Regime						
58	Liquid HoldUp		0.2818	0.2818	0.2818	0.2818	
59	Water HoldUp		---	---	---	---	
60	Friction Gradient	(kPa/m)	5.452e-003	5.451e-003	5.451e-003	5.451e-003	
61	Static Gradient	(kPa/m)	1.904e-003	1.904e-003	1.904e-003	1.904e-003	
62	Accel. Gradient	(kPa/m)	5.313e-007	5.313e-007	5.313e-007	5.313e-007	
63	Liq. Reynolds		2.093e+005 *	2.093e+005 *	2.093e+005 *	2.093e+005 *	
64	Vap. Reynolds		9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *	
65	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
66	Vapour Velocity	(m/s)	2.174	2.174	2.174	2.174	
67	Deposit Thickness	(mm)	---	---	---	---	
68	Deposit Volume	(m3)	---	---	---	---	
69	Aspen Technology Inc.		Aspen HYSYS Version 10				Page 434 of 937

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	2.679e+004	2.679e+004	2.680e+004	2.680e+004
14	Elevation (m)	-1.000e-002	-6.000e-003	-2.000e-003	2.000e-003
15	Pressure (bar)	88.73	88.73	88.73	88.73
16	Temperature (C)	50.73	50.73	50.73	50.73
17	Vapour Fraction	0.8694	0.8694	0.8694	0.8694
18	Heat Transferred (kJ/h-m)	552.3	552.3	552.2	552.2
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.2818	0.2818	0.2818	0.2818
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	5.451e-003	5.451e-003	5.451e-003	5.451e-003
24	Static Gradient (kPa/m)	1.904e-003	1.904e-003	1.904e-003	1.904e-003
25	Accel. Gradient (kPa/m)	5.313e-007	5.313e-007	5.313e-007	5.313e-007
26	Liq. Reynolds	2.093e+005 *	2.093e+005 *	2.093e+005 *	2.093e+005 *
27	Vap. Reynolds	9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *
28	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
29	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	2.681e+004	2.681e+004	2.682e+004	2.682e+004
33	Elevation (m)	6.000e-003	1.000e-002	-9.298e-016	-1.000e-002
34	Pressure (bar)	88.73	88.73	88.73	88.73
35	Temperature (C)	50.73	50.72	50.72	50.72
36	Vapour Fraction	0.8694	0.8694	0.8694	0.8694
37	Heat Transferred (kJ/h-m)	552.1	552.1	552.1	552.0
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.2818	0.2818	0.2052	0.2052
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	5.451e-003	5.451e-003	6.888e-003	6.888e-003
43	Static Gradient (kPa/m)	1.904e-003	1.904e-003	-3.805e-003	-3.805e-003
44	Accel. Gradient (kPa/m)	5.313e-007	5.313e-007	2.227e-007	2.227e-007
45	Liq. Reynolds	2.093e+005 *	2.093e+005 *	2.093e+005 *	2.093e+005 *
46	Vap. Reynolds	9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *
47	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
48	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	2.683e+004	2.683e+004	2.684e+004	2.684e+004
52	Elevation (m)	-2.000e-002	-3.000e-002	-4.000e-002	-2.600e-002
53	Pressure (bar)	88.73	88.73	88.73	88.73
54	Temperature (C)	50.72	50.72	50.72	50.71
55	Vapour Fraction	0.8694	0.8694	0.8694	0.8694
56	Heat Transferred (kJ/h-m)	552.0	551.9	551.9	551.8
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.2052	0.2052	0.2052	0.2882
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	6.888e-003	6.888e-003	6.888e-003	4.971e-003
62	Static Gradient (kPa/m)	-3.805e-003	-3.805e-003	-3.805e-003	6.771e-003
63	Accel. Gradient (kPa/m)	2.227e-007	2.227e-007	2.227e-007	8.482e-007
64	Liq. Reynolds	2.093e+005 *	2.093e+005 *	2.093e+005 *	2.093e+005 *
65	Vap. Reynolds	9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *
66	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
67	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174
68	Deposit Thickness (mm)	---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 435 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	2.685e+004	2.685e+004	2.686e+004	2.686e+004		
15	Elevation (m)	-1.200e-002	2.000e-003	1.600e-002	3.000e-002		
16	Pressure (bar)	88.73	88.73	88.73	88.73		
17	Temperature (C)	50.71	50.71	50.71	50.71		
18	Vapour Fraction	0.8694	0.8694	0.8694	0.8694		
19	Heat Transferred (kJ/h-m)	551.8	551.7	551.7	551.7		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	0.2882	0.2882	0.2882	0.2882		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	4.971e-003	4.971e-003	4.971e-003	4.971e-003		
25	Static Gradient (kPa/m)	6.771e-003	6.771e-003	6.771e-003	6.771e-003		
26	Accel. Gradient (kPa/m)	8.482e-007	8.482e-007	8.482e-007	8.482e-007		
27	Liq. Reynolds	2.093e+005 *	2.093e+005 *	2.093e+005 *	2.093e+005 *		
28	Vap. Reynolds	9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *		
29	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045		
30	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	2.687e+004	2.687e+004	2.688e+004	2.688e+004		
34	Elevation (m)	2.600e-002	2.200e-002	1.800e-002	1.400e-002		
35	Pressure (bar)	88.73	88.73	88.73	88.73		
36	Temperature (C)	50.71	50.70	50.70	50.70		
37	Vapour Fraction	0.8694	0.8694	0.8694	0.8694		
38	Heat Transferred (kJ/h-m)	551.6	551.6	551.5	551.5		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.2063	0.2063	0.2063	0.2063		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	6.879e-003	6.879e-003	6.879e-003	6.879e-003		
44	Static Gradient (kPa/m)	-1.545e-003	-1.545e-003	-1.545e-003	-1.545e-003		
45	Accel. Gradient (kPa/m)	3.853e-007	3.853e-007	3.853e-007	3.853e-007		
46	Liq. Reynolds	2.093e+005 *	2.093e+005 *	2.093e+005 *	2.093e+005 *		
47	Vap. Reynolds	9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *		
48	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045		
49	Vapour Velocity (m/s)	2.174	2.174	2.174	2.174		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	2.689e+004	2.689e+004	2.690e+004	2.690e+004		
53	Elevation (m)	1.000e-002	-9.263e-016	-1.000e-002	-2.000e-002		
54	Pressure (bar)	88.73	88.73	88.73	88.73		
55	Temperature (C)	50.70	50.70	50.70	50.69		
56	Vapour Fraction	0.8694	0.8694	0.8694	0.8694		
57	Heat Transferred (kJ/h-m)	551.4	551.4	551.4	551.3		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.2063	0.2052	0.2052	0.2052		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	6.879e-003	6.888e-003	6.888e-003	6.888e-003		
63	Static Gradient (kPa/m)	-1.545e-003	-3.849e-003	-3.849e-003	-3.849e-003		
64	Accel. Gradient (kPa/m)	3.853e-007	2.195e-007	2.195e-007	2.195e-007		
65	Liq. Reynolds	2.093e+005 *	2.093e+005 *	2.093e+005 *	2.093e+005 *		
66	Vap. Reynolds	9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *		
67	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045		
68	Vapour Velocity (m/s)	2.174	2.174	2.173	2.173		
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 436 of 937		

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	2.691e+004	2.691e+004	2.692e+004	2.692e+004
16	Elevation (m)	-3.000e-002	-4.000e-002	-6.000e-002	-8.000e-002
17	Pressure (bar)	88.73	88.73	88.73	88.73
18	Temperature (C)	50.69	50.69	50.69	50.69
19	Vapour Fraction	0.8694	0.8694	0.8694	0.8694
20	Heat Transferred (kJ/h-m)	551.3	551.2	551.2	551.2
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2052	0.2052	0.2033	0.2033
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	6.888e-003	6.888e-003	6.903e-003	6.903e-003
26	Static Gradient (kPa/m)	-3.849e-003	-3.849e-003	-7.655e-003	-7.655e-003
27	Accel. Gradient (kPa/m)	2.195e-007	2.195e-007	-5.429e-008	-5.430e-008
28	Liq. Reynolds	2.093e+005 *	2.093e+005 *	2.093e+005 *	2.093e+005 *
29	Vap. Reynolds	9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *
30	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
31	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	2.693e+004	2.693e+004	2.694e+004	2.694e+004
35	Elevation (m)	-0.1000	-0.1200	-0.1400	-0.1640
36	Pressure (bar)	88.73	88.73	88.73	88.73
37	Temperature (C)	50.69	50.69	50.68	50.68
38	Vapour Fraction	0.8694	0.8694	0.8694	0.8694
39	Heat Transferred (kJ/h-m)	551.1	551.1	551.0	551.0
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2033	0.2033	0.2033	0.2026
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	6.903e-003	6.903e-003	6.903e-003	6.909e-003
45	Static Gradient (kPa/m)	-7.655e-003	-7.655e-003	-7.655e-003	-9.165e-003
46	Accel. Gradient (kPa/m)	-5.431e-008	-5.431e-008	-5.432e-008	-1.629e-007
47	Liq. Reynolds	2.093e+005 *	2.093e+005 *	2.093e+005 *	2.093e+005 *
48	Vap. Reynolds	9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *
49	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
50	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	2.695e+004	2.695e+004	2.696e+004	2.696e+004
54	Elevation (m)	-0.1880	-0.2120	-0.2360	-0.2600
55	Pressure (bar)	88.73	88.73	88.73	88.73
56	Temperature (C)	50.68	50.68	50.68	50.68
57	Vapour Fraction	0.8694	0.8694	0.8694	0.8694
58	Heat Transferred (kJ/h-m)	551.0	550.9	550.9	550.8
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2026	0.2026	0.2026	0.2026
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	6.909e-003	6.909e-003	6.909e-003	6.909e-003
64	Static Gradient (kPa/m)	-9.165e-003	-9.165e-003	-9.165e-003	-9.165e-003
65	Accel. Gradient (kPa/m)	-1.629e-007	-1.629e-007	-1.629e-007	-1.630e-007
66	Liq. Reynolds	2.093e+005 *	2.093e+005 *	2.093e+005 *	2.093e+005 *
67	Vap. Reynolds	9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *
68	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
14	Deposit Thickness (mm)	---	---	---	---
15	Deposit Volume (m3)	---	---	---	---
16	Length (m)	2.697e+004	2.697e+004	2.698e+004	2.698e+004
17	Elevation (m)	-0.2800	-0.3000	-0.3200	-0.3400
18	Pressure (bar)	88.73	88.73	88.73	88.73
19	Temperature (C)	50.67	50.67	50.67	50.67
20	Vapour Fraction	0.8694	0.8694	0.8694	0.8694
21	Heat Transferred (kJ/h-m)	550.8	550.8	550.7	550.7
22	Flow Regime	Transition	Transition	Transition	Transition
23	Water Flow Regime				
24	Liquid HoldUp	0.2033	0.2033	0.2033	0.2033
25	Water HoldUp	---	---	---	---
26	Friction Gradient (kPa/m)	6.903e-003	6.903e-003	6.902e-003	6.902e-003
27	Static Gradient (kPa/m)	-7.673e-003	-7.674e-003	-7.674e-003	-7.674e-003
28	Accel. Gradient (kPa/m)	-5.567e-008	-5.568e-008	-5.569e-008	-5.570e-008
29	Liq. Reynolds	2.093e+005 *	2.093e+005 *	2.093e+005 *	2.093e+005 *
30	Vap. Reynolds	9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *
31	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
32	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
33	Deposit Thickness (mm)	---	---	---	---
34	Deposit Volume (m3)	---	---	---	---
35	Length (m)	2.699e+004	2.699e+004	2.700e+004	2.700e+004
36	Elevation (m)	-0.3600	-0.3780	-0.3960	-0.4140
37	Pressure (bar)	88.73	88.73	88.73	88.73
38	Temperature (C)	50.67	50.67	50.67	50.66
39	Vapour Fraction	0.8694	0.8694	0.8693	0.8693
40	Heat Transferred (kJ/h-m)	550.6	550.6	550.6	550.5
41	Flow Regime	Transition	Transition	Transition	Transition
42	Water Flow Regime				
43	Liquid HoldUp	0.2033	0.2037	0.2037	0.2037
44	Water HoldUp	---	---	---	---
45	Friction Gradient (kPa/m)	6.902e-003	6.899e-003	6.899e-003	6.899e-003
46	Static Gradient (kPa/m)	-7.674e-003	-6.898e-003	-6.898e-003	-6.898e-003
47	Accel. Gradient (kPa/m)	-5.570e-008	9.090e-011	8.406e-011	7.722e-011
48	Liq. Reynolds	2.093e+005 *	2.093e+005 *	2.093e+005 *	2.093e+005 *
49	Vap. Reynolds	9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *
50	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
51	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
52	Deposit Thickness (mm)	---	---	---	---
53	Deposit Volume (m3)	---	---	---	---
54	Length (m)	2.701e+004	2.701e+004	2.702e+004	2.702e+004
55	Elevation (m)	-0.4320	-0.4500	-0.4700	-0.4900
56	Pressure (bar)	88.73	88.73	88.73	88.73
57	Temperature (C)	50.66	50.66	50.66	50.66
58	Vapour Fraction	0.8693	0.8693	0.8693	0.8693
59	Heat Transferred (kJ/h-m)	550.5	550.4	550.4	550.4
60	Flow Regime	Transition	Transition	Transition	Transition
61	Water Flow Regime				
62	Liquid HoldUp	0.2037	0.2037	0.2033	0.2033
63	Water HoldUp	---	---	---	---
64	Friction Gradient (kPa/m)	6.899e-003	6.899e-003	6.902e-003	6.902e-003
65	Static Gradient (kPa/m)	-6.898e-003	-6.898e-003	-7.656e-003	-7.656e-003
66	Accel. Gradient (kPa/m)	7.038e-011	6.355e-011	-5.443e-008	-5.444e-008
67	Liq. Reynolds	2.093e+005 *	2.093e+005 *	2.093e+005 *	2.093e+005 *
68	Vap. Reynolds	9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045
14	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173
15	Deposit Thickness	(mm)	---	---	---	---
16	Deposit Volume	(m3)	---	---	---	---
17	Length	(m)	2.703e+004	2.703e+004	2.704e+004	2.704e+004
18	Elevation	(m)	-0.5100	-0.5300	-0.5500	-0.5280
19	Pressure	(bar)	88.73	88.73	88.73	88.73
20	Temperature	(C)	50.66	50.65	50.65	50.65
21	Vapour Fraction		0.8693	0.8693	0.8693	0.8693
22	Heat Transferred	(kJ/h-m)	550.3	550.3	550.2	550.2
23	Flow Regime		Transition	Transition	Transition	Transition
24	Water Flow Regime					
25	Liquid HoldUp		0.2033	0.2033	0.2033	0.3143
26	Water HoldUp		---	---	---	---
27	Friction Gradient	(kPa/m)	6.902e-003	6.902e-003	6.902e-003	6.050e-003
28	Static Gradient	(kPa/m)	-7.656e-003	-7.656e-003	-7.656e-003	2.831e-002
29	Accel. Gradient	(kPa/m)	-5.445e-008	-5.445e-008	-5.446e-008	2.481e-006
30	Liq. Reynolds		2.093e+005 *	2.093e+005 *	2.093e+005 *	2.093e+005 *
31	Vap. Reynolds		9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *
32	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045
33	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173
34	Deposit Thickness	(mm)	---	---	---	---
35	Deposit Volume	(m3)	---	---	---	---
36	Length	(m)	2.704e+004	2.704e+004	2.704e+004	2.705e+004
37	Elevation	(m)	-0.5060	-0.4840	-0.4620	-0.4400
38	Pressure	(bar)	88.73	88.73	88.73	88.72
39	Temperature	(C)	50.65	50.65	50.65	50.65
40	Vapour Fraction		0.8693	0.8693	0.8693	0.8693
41	Heat Transferred	(kJ/h-m)	550.2	550.2	550.1	550.1
42	Flow Regime		Transition	Transition	Transition	Transition
43	Water Flow Regime					
44	Liquid HoldUp		0.3143	0.3143	0.3143	0.3143
45	Water HoldUp		---	---	---	---
46	Friction Gradient	(kPa/m)	6.050e-003	6.050e-003	6.050e-003	6.050e-003
47	Static Gradient	(kPa/m)	2.831e-002	2.831e-002	2.831e-002	2.831e-002
48	Accel. Gradient	(kPa/m)	2.481e-006	2.481e-006	2.482e-006	2.482e-006
49	Liq. Reynolds		2.093e+005 *	2.093e+005 *	2.093e+005 *	2.093e+005 *
50	Vap. Reynolds		9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *
51	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045
52	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173
53	Deposit Thickness	(mm)	---	---	---	---
54	Deposit Volume	(m3)	---	---	---	---
55	Length	(m)	2.705e+004	2.705e+004	2.705e+004	2.706e+004
56	Elevation	(m)	-0.4500	-0.4600	-0.4700	-0.4800
57	Pressure	(bar)	88.72	88.72	88.72	88.72
58	Temperature	(C)	50.65	50.65	50.65	50.65
59	Vapour Fraction		0.8693	0.8693	0.8693	0.8693
60	Heat Transferred	(kJ/h-m)	550.1	550.1	550.0	550.0
61	Flow Regime		Transition	Transition	Transition	Transition
62	Water Flow Regime					
63	Liquid HoldUp		0.2040	0.2040	0.2040	0.2040
64	Water HoldUp		---	---	---	---
65	Friction Gradient	(kPa/m)	6.897e-003	6.897e-003	6.897e-003	6.897e-003
66	Static Gradient	(kPa/m)	-6.392e-003	-6.392e-003	-6.392e-003	-6.392e-003
67	Accel. Gradient	(kPa/m)	3.644e-008	3.644e-008	3.643e-008	3.643e-008
68	Liq. Reynolds		2.092e+005 *	2.092e+005 *	2.092e+005 *	2.092e+005 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *
14	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
15	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	2.706e+004	2.707e+004	2.707e+004	2.708e+004
19	Elevation (m)	-0.4900	-0.5220	-0.5540	-0.5860
20	Pressure (bar)	88.72	88.73	88.73	88.73
21	Temperature (C)	50.64	50.64	50.64	50.64
22	Vapour Fraction	0.8693	0.8693	0.8693	0.8693
23	Heat Transferred (kJ/h-m)	550.0	550.0	549.9	549.9
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2040	0.2011	0.2011	0.2011
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.897e-003	6.920e-003	6.920e-003	6.920e-003
29	Static Gradient (kPa/m)	-6.392e-003	-1.216e-002	-1.216e-002	-1.216e-002
30	Accel. Gradient (kPa/m)	3.642e-008	-3.787e-007	-3.787e-007	-3.787e-007
31	Liq. Reynolds	2.092e+005 *	2.092e+005 *	2.092e+005 *	2.092e+005 *
32	Vap. Reynolds	9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *
33	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
34	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	2.708e+004	2.709e+004	2.709e+004	2.710e+004
38	Elevation (m)	-0.6180	-0.6500	-0.6400	-0.6300
39	Pressure (bar)	88.73	88.73	88.73	88.73
40	Temperature (C)	50.64	50.64	50.64	50.63
41	Vapour Fraction	0.8693	0.8693	0.8693	0.8693
42	Heat Transferred (kJ/h-m)	549.8	549.8	549.7	549.7
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2011	0.2011	0.2857	0.2857
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.920e-003	6.920e-003	5.157e-003	5.157e-003
48	Static Gradient (kPa/m)	-1.217e-002	-1.217e-002	4.807e-003	4.807e-003
49	Accel. Gradient (kPa/m)	-3.787e-007	-3.787e-007	7.195e-007	7.195e-007
50	Liq. Reynolds	2.092e+005 *	2.092e+005 *	2.092e+005 *	2.092e+005 *
51	Vap. Reynolds	9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *
52	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
53	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	2.710e+004	2.711e+004	2.711e+004	2.712e+004
57	Elevation (m)	-0.6200	-0.6100	-0.6000	-0.5440
58	Pressure (bar)	88.72	88.72	88.72	88.72
59	Temperature (C)	50.63	50.63	50.63	50.63
60	Vapour Fraction	0.8693	0.8693	0.8693	0.8693
61	Heat Transferred (kJ/h-m)	549.7	549.6	549.6	549.5
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2857	0.2857	0.2857	0.3149
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	5.157e-003	5.157e-003	5.157e-003	6.053e-003
67	Static Gradient (kPa/m)	4.807e-003	4.807e-003	4.807e-003	2.887e-002
68	Accel. Gradient (kPa/m)	7.195e-007	7.195e-007	7.195e-007	2.522e-006
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		2.092e+005 *	2.092e+005 *	2.092e+005 *	2.092e+005 *	
14	Vap. Reynolds		9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *	
15	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045	0.2045	
16	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173	2.173	
17	Deposit Thickness (mm)	---	---	---	---	---	
18	Deposit Volume (m3)	---	---	---	---	---	
19	Length (m)	2.712e+004	2.713e+004	2.713e+004	2.714e+004		
20	Elevation (m)	-0.4880	-0.4320	-0.3760	-0.3200		
21	Pressure (bar)	88.72	88.72	88.72	88.71		
22	Temperature (C)	50.62	50.62	50.62	50.62		
23	Vapour Fraction	0.8693	0.8693	0.8693	0.8693		
24	Heat Transferred (kJ/h-m)	549.5	549.4	549.3	549.3		
25	Flow Regime	Transition	Transition	Transition	Transition		
26	Water Flow Regime						
27	Liquid HoldUp	0.3149	0.3149	0.3149	0.3149		
28	Water HoldUp	---	---	---	---		
29	Friction Gradient (kPa/m)	6.053e-003	6.053e-003	6.053e-003	6.053e-003		
30	Static Gradient (kPa/m)	2.887e-002	2.887e-002	2.887e-002	2.887e-002		
31	Accel. Gradient (kPa/m)	2.522e-006	2.522e-006	2.522e-006	2.522e-006		
32	Liq. Reynolds		2.092e+005 *	2.092e+005 *	2.092e+005 *	2.092e+005 *	
33	Vap. Reynolds		9.172e+006 *	9.172e+006 *	9.172e+006 *	9.172e+006 *	
34	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045	0.2045	
35	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173	2.173	
36	Deposit Thickness (mm)	---	---	---	---	---	
37	Deposit Volume (m3)	---	---	---	---	---	
38	Length (m)	2.714e+004	2.715e+004	2.715e+004	2.716e+004		
39	Elevation (m)	-0.2820	-0.2440	-0.2060	-0.1680		
40	Pressure (bar)	88.71	88.71	88.71	88.71		
41	Temperature (C)	50.62	50.61	50.61	50.61		
42	Vapour Fraction	0.8693	0.8693	0.8693	0.8693		
43	Heat Transferred (kJ/h-m)	549.2	549.2	549.1	549.1		
44	Flow Regime	Transition	Transition	Transition	Transition		
45	Water Flow Regime						
46	Liquid HoldUp	0.3035	0.3035	0.3035	0.3035		
47	Water HoldUp	---	---	---	---		
48	Friction Gradient (kPa/m)	5.918e-003	5.918e-003	5.918e-003	5.918e-003		
49	Static Gradient (kPa/m)	1.907e-002	1.907e-002	1.907e-002	1.907e-002		
50	Accel. Gradient (kPa/m)	1.805e-006	1.805e-006	1.805e-006	1.805e-006		
51	Liq. Reynolds		2.092e+005 *	2.092e+005 *	2.092e+005 *	2.092e+005 *	
52	Vap. Reynolds		9.172e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *	
53	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045	0.2045	
54	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173	2.173	
55	Deposit Thickness (mm)	---	---	---	---	---	
56	Deposit Volume (m3)	---	---	---	---	---	
57	Length (m)	2.716e+004	2.717e+004	2.717e+004	2.718e+004		
58	Elevation (m)	-0.1300	-0.1120	-9.400e-002	-7.600e-002		
59	Pressure (bar)	88.71	88.71	88.71	88.71		
60	Temperature (C)	50.61	50.61	50.60	50.60		
61	Vapour Fraction	0.8693	0.8693	0.8693	0.8693		
62	Heat Transferred (kJ/h-m)	549.0	549.0	548.9	548.9		
63	Flow Regime	Transition	Transition	Transition	Transition		
64	Water Flow Regime						
65	Liquid HoldUp	0.3035	0.2908	0.2908	0.2908		
66	Water HoldUp	---	---	---	---		
67	Friction Gradient (kPa/m)	5.918e-003	4.786e-003	4.786e-003	4.786e-003		
68	Static Gradient (kPa/m)	1.907e-002	8.762e-003	8.762e-003	8.762e-003		
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 441 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient	(kPa/m)	1.805e-006	9.786e-007	9.786e-007	9.786e-007	
14	Liq. Reynolds		2.092e+005 *	2.092e+005 *	2.092e+005 *	2.092e+005 *	
15	Vap. Reynolds		9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *	
16	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
17	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173	
18	Deposit Thickness	(mm)	---	---	---	---	
19	Deposit Volume	(m3)	---	---	---	---	
20	Length	(m)	2.718e+004	2.719e+004	2.719e+004	2.720e+004	
21	Elevation	(m)	-5.800e-002	-4.000e-002	-2.400e-002	-8.000e-003	
22	Pressure	(bar)	88.71	88.71	88.70	88.70	
23	Temperature	(C)	50.60	50.60	50.60	50.60	
24	Vapour Fraction		0.8693	0.8693	0.8693	0.8693	
25	Heat Transferred	(kJ/h-m)	548.8	548.8	548.7	548.7	
26	Flow Regime		Transition	Transition	Transition	Transition	
27	Water Flow Regime						
28	Liquid HoldUp		0.2908	0.2908	0.2895	0.2895	
29	Water HoldUp		---	---	---	---	
30	Friction Gradient	(kPa/m)	4.786e-003	4.786e-003	4.877e-003	4.877e-003	
31	Static Gradient	(kPa/m)	8.762e-003	8.762e-003	7.764e-003	7.764e-003	
32	Accel. Gradient	(kPa/m)	9.786e-007	9.786e-007	9.131e-007	9.131e-007	
33	Liq. Reynolds		2.092e+005 *	2.092e+005 *	2.092e+005 *	2.092e+005 *	
34	Vap. Reynolds		9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *	
35	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
36	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173	
37	Deposit Thickness	(mm)	---	---	---	---	
38	Deposit Volume	(m3)	---	---	---	---	
39	Length	(m)	2.720e+004	2.721e+004	2.721e+004	2.722e+004	
40	Elevation	(m)	8.000e-003	2.400e-002	4.000e-002	3.800e-002	
41	Pressure	(bar)	88.70	88.70	88.70	88.70	
42	Temperature	(C)	50.59	50.59	50.59	50.59	
43	Vapour Fraction		0.8693	0.8693	0.8693	0.8693	
44	Heat Transferred	(kJ/h-m)	548.6	548.6	548.5	548.5	
45	Flow Regime		Transition	Transition	Transition	Transition	
46	Water Flow Regime						
47	Liquid HoldUp		0.2895	0.2895	0.2895	0.2067	
48	Water HoldUp		---	---	---	---	
49	Friction Gradient	(kPa/m)	4.877e-003	4.877e-003	4.877e-003	6.875e-003	
50	Static Gradient	(kPa/m)	7.764e-003	7.764e-003	7.764e-003	-7.705e-004	
51	Accel. Gradient	(kPa/m)	9.131e-007	9.131e-007	9.131e-007	4.409e-007	
52	Liq. Reynolds		2.092e+005 *	2.092e+005 *	2.092e+005 *	2.092e+005 *	
53	Vap. Reynolds		9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *	
54	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
55	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173	
56	Deposit Thickness	(mm)	---	---	---	---	
57	Deposit Volume	(m3)	---	---	---	---	
58	Length	(m)	2.722e+004	2.723e+004	2.723e+004	2.724e+004	
59	Elevation	(m)	3.600e-002	3.400e-002	3.200e-002	3.000e-002	
60	Pressure	(bar)	88.70	88.70	88.70	88.70	
61	Temperature	(C)	50.59	50.59	50.58	50.58	
62	Vapour Fraction		0.8693	0.8693	0.8693	0.8693	
63	Heat Transferred	(kJ/h-m)	548.5	548.4	548.4	548.3	
64	Flow Regime		Transition	Transition	Transition	Transition	
65	Water Flow Regime						
66	Liquid HoldUp		0.2067	0.2067	0.2067	0.2067	
67	Water HoldUp		---	---	---	---	
68	Friction Gradient	(kPa/m)	6.875e-003	6.875e-003	6.875e-003	6.875e-003	
69	Aspen Technology Inc.		Aspen HYSYS Version 10				
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:02:44 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Static Gradient	(kPa/m)	-7.705e-004	-7.705e-004	-7.705e-004	-7.705e-004	
14	Accel. Gradient	(kPa/m)	4.409e-007	4.409e-007	4.409e-007	4.409e-007	
15	Liq. Reynolds		2.092e+005 *	2.092e+005 *	2.092e+005 *	2.092e+005 *	
16	Vap. Reynolds		9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *	
17	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
18	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173	
19	Deposit Thickness	(mm)	---	---	---	---	
20	Deposit Volume	(m3)	---	---	---	---	
21	Length	(m)	2.724e+004	2.725e+004	2.725e+004	2.726e+004	
22	Elevation	(m)	4.000e-002	5.000e-002	6.000e-002	7.000e-002	
23	Pressure	(bar)	88.70	88.70	88.70	88.70	
24	Temperature	(C)	50.58	50.58	50.58	50.58	
25	Vapour Fraction		0.8693	0.8693	0.8693	0.8693	
26	Heat Transferred	(kJ/h-m)	548.3	548.2	548.2	548.1	
27	Flow Regime		Transition	Transition	Transition	Transition	
28	Water Flow Regime						
29	Liquid HoldUp		0.2857	0.2857	0.2857	0.2857	
30	Water HoldUp		---	---	---	---	
31	Friction Gradient	(kPa/m)	5.157e-003	5.157e-003	5.157e-003	5.157e-003	
32	Static Gradient	(kPa/m)	4.808e-003	4.808e-003	4.808e-003	4.808e-003	
33	Accel. Gradient	(kPa/m)	7.198e-007	7.198e-007	7.198e-007	7.198e-007	
34	Liq. Reynolds		2.092e+005 *	2.092e+005 *	2.092e+005 *	2.092e+005 *	
35	Vap. Reynolds		9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *	
36	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
37	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173	
38	Deposit Thickness	(mm)	---	---	---	---	
39	Deposit Volume	(m3)	---	---	---	---	
40	Length	(m)	2.726e+004	2.727e+004	2.727e+004	2.728e+004	
41	Elevation	(m)	8.000e-002	8.000e-002	8.000e-002	8.000e-002	
42	Pressure	(bar)	88.70	88.70	88.70	88.70	
43	Temperature	(C)	50.57	50.57	50.57	50.57	
44	Vapour Fraction		0.8693	0.8693	0.8693	0.8693	
45	Heat Transferred	(kJ/h-m)	548.1	548.1	548.0	548.0	
46	Flow Regime		Transition	Transition	Transition	Transition	
47	Water Flow Regime						
48	Liquid HoldUp		0.2857	0.3023	0.3023	0.3023	
49	Water HoldUp		---	---	---	---	
50	Friction Gradient	(kPa/m)	5.157e-003	5.884e-003	5.884e-003	5.884e-003	
51	Static Gradient	(kPa/m)	4.808e-003	0.0000	0.0000	0.0000	
52	Accel. Gradient	(kPa/m)	7.198e-007	4.251e-007	4.251e-007	4.250e-007	
53	Liq. Reynolds		2.092e+005 *	2.092e+005 *	2.092e+005 *	2.092e+005 *	
54	Vap. Reynolds		9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *	
55	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
56	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173	
57	Deposit Thickness	(mm)	---	---	---	---	
58	Deposit Volume	(m3)	---	---	---	---	
59	Length	(m)	2.728e+004	2.729e+004	2.729e+004	2.730e+004	
60	Elevation	(m)	8.000e-002	8.000e-002	8.600e-002	9.200e-002	
61	Pressure	(bar)	88.70	88.70	88.70	88.70	
62	Temperature	(C)	50.57	50.57	50.56	50.56	
63	Vapour Fraction		0.8693	0.8693	0.8693	0.8693	
64	Heat Transferred	(kJ/h-m)	547.9	547.9	547.8	547.8	
65	Flow Regime		Transition	Transition	Transition	Transition	
66	Water Flow Regime						
67	Liquid HoldUp		0.3023	0.3023	0.2831	0.2831	
68	Water HoldUp		---	---	---	---	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 443 of 937	

1				Case Name:	Simulation - 386MMSCFD_100barg.hsc	
2				Unit Set:	Mixed	
3		Epic Atlantic Limited				
Date/Time:			Mon Jun 13 15:02:44 2022			
4						
5						
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	5.884e-003	5.884e-003	5.350e-003	5.350e-003
14	Static Gradient	(kPa/m)	0.0000	0.0000	2.866e-003	2.866e-003
15	Accel. Gradient	(kPa/m)	4.250e-007	4.250e-007	5.935e-007	5.935e-007
16	Liq. Reynolds		2.092e+005 *	2.092e+005 *	2.092e+005 *	2.092e+005 *
17	Vap. Reynolds		9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *
18	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045
19	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	2.730e+004	2.731e+004	2.731e+004	2.732e+004
23	Elevation	(m)	9.800e-002	0.1040	0.1100	0.1100
24	Pressure	(bar)	88.70	88.69	88.69	88.69
25	Temperature	(C)	50.56	50.56	50.56	50.56
26	Vapour Fraction		0.8693	0.8693	0.8693	0.8693
27	Heat Transferred	(kJ/h-m)	547.7	547.7	547.7	547.6
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2831	0.2831	0.2831	0.3023
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	5.350e-003	5.350e-003	5.350e-003	5.884e-003
33	Static Gradient	(kPa/m)	2.866e-003	2.866e-003	2.866e-003	0.0000
34	Accel. Gradient	(kPa/m)	5.935e-007	5.935e-007	5.935e-007	4.250e-007
35	Liq. Reynolds		2.092e+005 *	2.092e+005 *	2.091e+005 *	2.091e+005 *
36	Vap. Reynolds		9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *
37	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045
38	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	2.732e+004	2.733e+004	2.733e+004	2.734e+004
42	Elevation	(m)	0.1100	0.1100	0.1100	0.1100
43	Pressure	(bar)	88.69	88.69	88.69	88.69
44	Temperature	(C)	50.55	50.55	50.55	50.55
45	Vapour Fraction		0.8693	0.8693	0.8693	0.8693
46	Heat Transferred	(kJ/h-m)	547.6	547.5	547.5	547.4
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.3023	0.3023	0.3023	0.3023
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	5.884e-003	5.884e-003	5.884e-003	5.884e-003
52	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000
53	Accel. Gradient	(kPa/m)	4.250e-007	4.250e-007	4.250e-007	4.250e-007
54	Liq. Reynolds		2.091e+005 *	2.091e+005 *	2.091e+005 *	2.091e+005 *
55	Vap. Reynolds		9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *
56	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045
57	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	2.734e+004	2.735e+004	2.735e+004	2.736e+004
61	Elevation	(m)	0.1140	0.1180	0.1220	0.1260
62	Pressure	(bar)	88.69	88.69	88.69	88.69
63	Temperature	(C)	50.55	50.55	50.54	50.54
64	Vapour Fraction		0.8693	0.8693	0.8693	0.8693
65	Heat Transferred	(kJ/h-m)	547.4	547.3	547.3	547.3
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2819	0.2819	0.2819	0.2819
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	5.448e-003	5.448e-003	5.448e-003	5.448e-003
15	Static Gradient (kPa/m)	1.905e-003	1.905e-003	1.905e-003	1.905e-003
16	Accel. Gradient (kPa/m)	5.311e-007	5.311e-007	5.311e-007	5.311e-007
17	Liq. Reynolds	2.091e+005 *	2.091e+005 *	2.091e+005 *	2.091e+005 *
18	Vap. Reynolds	9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *
19	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
20	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	2.736e+004	2.737e+004	2.737e+004	2.738e+004
24	Elevation (m)	0.1300	0.1340	0.1380	0.1420
25	Pressure (bar)	88.69	88.69	88.69	88.69
26	Temperature (C)	50.54	50.54	50.54	50.54
27	Vapour Fraction	0.8693	0.8693	0.8693	0.8693
28	Heat Transferred (kJ/h-m)	547.2	547.2	547.1	547.1
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2819	0.2819	0.2819	0.2819
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	5.448e-003	5.448e-003	5.448e-003	5.448e-003
34	Static Gradient (kPa/m)	1.905e-003	1.905e-003	1.905e-003	1.905e-003
35	Accel. Gradient (kPa/m)	5.311e-007	5.311e-007	5.311e-007	5.311e-007
36	Liq. Reynolds	2.091e+005 *	2.091e+005 *	2.091e+005 *	2.091e+005 *
37	Vap. Reynolds	9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *
38	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
39	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	2.738e+004	2.739e+004	2.739e+004	2.740e+004
43	Elevation (m)	0.1460	0.1500	0.1480	0.1460
44	Pressure (bar)	88.69	88.69	88.69	88.69
45	Temperature (C)	50.53	50.53	50.53	50.53
46	Vapour Fraction	0.8693	0.8693	0.8693	0.8693
47	Heat Transferred (kJ/h-m)	547.0	547.0	547.0	546.9
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2819	0.2819	0.2067	0.2067
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	5.448e-003	5.448e-003	6.874e-003	6.874e-003
53	Static Gradient (kPa/m)	1.905e-003	1.905e-003	-7.737e-004	-7.737e-004
54	Accel. Gradient (kPa/m)	5.311e-007	5.311e-007	4.406e-007	4.406e-007
55	Liq. Reynolds	2.091e+005 *	2.091e+005 *	2.091e+005 *	2.091e+005 *
56	Vap. Reynolds	9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *
57	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
58	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	2.740e+004	2.741e+004	2.741e+004	2.742e+004
62	Elevation (m)	0.1440	0.1420	0.1400	0.1420
63	Pressure (bar)	88.69	88.69	88.69	88.69
64	Temperature (C)	50.53	50.53	50.52	50.52
65	Vapour Fraction	0.8693	0.8693	0.8693	0.8693
66	Heat Transferred (kJ/h-m)	546.9	546.8	546.8	546.7
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 445 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.2067	0.2067	0.2067	0.2806
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	6.874e-003	6.874e-003	6.874e-003	5.549e-003
16	Static Gradient (kPa/m)	-7.737e-004	-7.737e-004	-7.737e-004	9.308e-004
17	Accel. Gradient (kPa/m)	4.406e-007	4.406e-007	4.406e-007	4.681e-007
18	Liq. Reynolds	2.091e+005 *	2.091e+005 *	2.091e+005 *	2.091e+005 *
19	Vap. Reynolds	9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *
20	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
21	Vapour Velocity (m/s)	2.173	2.173	2.173	2.172
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	2.742e+004	2.743e+004	2.743e+004	2.744e+004
25	Elevation (m)	0.1440	0.1460	0.1480	0.1500
26	Pressure (bar)	88.69	88.69	88.69	88.69
27	Temperature (C)	50.52	50.52	50.52	50.52
28	Vapour Fraction	0.8693	0.8693	0.8693	0.8693
29	Heat Transferred (kJ/h-m)	546.7	546.6	546.6	546.5
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2806	0.2806	0.2806	0.2806
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	5.549e-003	5.549e-003	5.549e-003	5.549e-003
35	Static Gradient (kPa/m)	9.308e-004	9.309e-004	9.309e-004	9.309e-004
36	Accel. Gradient (kPa/m)	4.681e-007	4.681e-007	4.681e-007	4.681e-007
37	Liq. Reynolds	2.091e+005 *	2.091e+005 *	2.091e+005 *	2.091e+005 *
38	Vap. Reynolds	9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *
39	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
40	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	2.744e+004	2.745e+004	2.745e+004	2.746e+004
44	Elevation (m)	0.1520	0.1540	0.1560	0.1580
45	Pressure (bar)	88.69	88.69	88.69	88.68
46	Temperature (C)	50.51	50.51	50.51	50.51
47	Vapour Fraction	0.8693	0.8693	0.8693	0.8693
48	Heat Transferred (kJ/h-m)	546.5	546.5	546.4	546.4
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2806	0.2806	0.2806	0.2806
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	5.547e-003	5.547e-003	5.547e-003	5.547e-003
54	Static Gradient (kPa/m)	9.495e-004	9.495e-004	9.495e-004	9.496e-004
55	Accel. Gradient (kPa/m)	4.693e-007	4.693e-007	4.693e-007	4.693e-007
56	Liq. Reynolds	2.091e+005 *	2.091e+005 *	2.091e+005 *	2.091e+005 *
57	Vap. Reynolds	9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *
58	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
59	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	2.746e+004	2.747e+004	2.747e+004	2.748e+004
63	Elevation (m)	0.1600	0.1600	0.1600	0.1600
64	Pressure (bar)	88.68	88.68	88.68	88.68
65	Temperature (C)	50.51	50.51	50.50	50.50
66	Vapour Fraction	0.8693	0.8693	0.8693	0.8693
67	Heat Transferred (kJ/h-m)	546.3	546.3	546.2	546.2
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 446 of 937

1			Case Name:	Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set:	Mixed		
3	 Epic Atlantic Limited		Date/Time:	Mon Jun 13 15:02:44 2022		
4						
5						
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Water Flow Regime					
14	Liquid HoldUp	0.2806	0.3023	0.3023	0.3023	
15	Water HoldUp	---	---	---	---	
16	Friction Gradient (kPa/m)	5.547e-003	5.884e-003	5.884e-003	5.884e-003	
17	Static Gradient (kPa/m)	9.496e-004	0.0000	0.0000	0.0000	
18	Accel. Gradient (kPa/m)	4.693e-007	4.250e-007	4.250e-007	4.250e-007	
19	Liq. Reynolds	2.091e+005 *	2.091e+005 *	2.091e+005 *	2.091e+005 *	
20	Vap. Reynolds	9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *	
21	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045	
22	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172	
23	Deposit Thickness (mm)	---	---	---	---	
24	Deposit Volume (m3)	---	---	---	---	
25	Length (m)	2.748e+004	2.749e+004	2.749e+004	2.749e+004	
26	Elevation (m)	0.1600	0.1600	0.1600	0.1600	
27	Pressure (bar)	88.68	88.68	88.68	88.68	
28	Temperature (C)	50.50	50.50	50.50	50.50	
29	Vapour Fraction	0.8693	0.8693	0.8693	0.8693	
30	Heat Transferred (kJ/h-m)	546.2	546.1	546.1	546.1	
31	Flow Regime	Transition	Transition	Transition	Transition	
32	Water Flow Regime					
33	Liquid HoldUp	0.3023	0.3023	0.3023	0.3023	
34	Water HoldUp	---	---	---	---	
35	Friction Gradient (kPa/m)	5.884e-003	5.884e-003	5.884e-003	5.884e-003	
36	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000	
37	Accel. Gradient (kPa/m)	4.250e-007	4.250e-007	4.250e-007	4.250e-007	
38	Liq. Reynolds	2.091e+005 *	2.091e+005 *	2.091e+005 *	2.091e+005 *	
39	Vap. Reynolds	9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *	
40	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045	
41	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172	
42	Deposit Thickness (mm)	---	---	---	---	
43	Deposit Volume (m3)	---	---	---	---	
44	Length (m)	2.749e+004	2.749e+004	2.750e+004	2.750e+004	
45	Elevation (m)	0.1600	0.1600	0.1600	0.1580	
46	Pressure (bar)	88.68	88.68	88.68	88.68	
47	Temperature (C)	50.50	50.50	50.50	50.49	
48	Vapour Fraction	0.8693	0.8693	0.8693	0.8693	
49	Heat Transferred (kJ/h-m)	546.1	546.0	546.0	546.0	
50	Flow Regime	Transition	Transition	Transition	Transition	
51	Water Flow Regime					
52	Liquid HoldUp	0.3023	0.3023	0.3023	0.2068	
53	Water HoldUp	---	---	---	---	
54	Friction Gradient (kPa/m)	5.884e-003	5.884e-003	5.884e-003	6.873e-003	
55	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-6.449e-004	
56	Accel. Gradient (kPa/m)	4.250e-007	4.250e-007	4.250e-007	4.498e-007	
57	Liq. Reynolds	2.091e+005 *	2.091e+005 *	2.091e+005 *	2.091e+005 *	
58	Vap. Reynolds	9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *	
59	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045	
60	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172	
61	Deposit Thickness (mm)	---	---	---	---	
62	Deposit Volume (m3)	---	---	---	---	
63	Length (m)	2.751e+004	2.751e+004	2.752e+004	2.753e+004	
64	Elevation (m)	0.1560	0.1540	0.1520	0.1500	
65	Pressure (bar)	88.68	88.68	88.68	88.68	
66	Temperature (C)	50.49	50.49	50.49	50.49	
67	Vapour Fraction	0.8693	0.8693	0.8693	0.8693	
68	Heat Transferred (kJ/h-m)	545.9	545.9	545.8	545.8	
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 447 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2068	0.2068	0.2068	0.2068
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	6.873e-003	6.873e-003	6.873e-003	6.873e-003
18	Static Gradient (kPa/m)	-6.449e-004	-6.449e-004	-6.449e-004	-6.449e-004
19	Accel. Gradient (kPa/m)	4.498e-007	4.498e-007	4.498e-007	4.498e-007
20	Liq. Reynolds	2.091e+005 *	2.091e+005 *	2.091e+005 *	2.091e+005 *
21	Vap. Reynolds	9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *
22	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
23	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	2.753e+004	2.753e+004	2.753e+004	2.753e+004
27	Elevation (m)	0.1560	0.1620	0.1680	0.1740
28	Pressure (bar)	88.68	88.68	88.68	88.68
29	Temperature (C)	50.49	50.48	50.48	50.48
30	Vapour Fraction	0.8693	0.8693	0.8693	0.8693
31	Heat Transferred (kJ/h-m)	545.7	545.7	545.7	545.7
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2889	0.2889	0.2889	0.2889
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	4.921e-003	4.921e-003	4.921e-003	4.921e-003
37	Static Gradient (kPa/m)	7.270e-003	7.270e-003	7.270e-003	7.270e-003
38	Accel. Gradient (kPa/m)	8.805e-007	8.805e-007	8.805e-007	8.805e-007
39	Liq. Reynolds	2.091e+005 *	2.091e+005 *	2.091e+005 *	2.091e+005 *
40	Vap. Reynolds	9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *
41	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
42	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	2.754e+004	2.754e+004	2.755e+004	2.755e+004
46	Elevation (m)	0.1800	0.1800	0.1800	0.1800
47	Pressure (bar)	88.68	88.68	88.68	88.68
48	Temperature (C)	50.48	50.48	50.48	50.48
49	Vapour Fraction	0.8693	0.8693	0.8693	0.8693
50	Heat Transferred (kJ/h-m)	545.7	545.6	545.6	545.5
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2889	0.3023	0.3023	0.3023
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	4.921e-003	5.884e-003	5.883e-003	5.883e-003
56	Static Gradient (kPa/m)	7.270e-003	0.0000	0.0000	0.0000
57	Accel. Gradient (kPa/m)	8.805e-007	4.250e-007	4.250e-007	4.250e-007
58	Liq. Reynolds	2.091e+005 *	2.091e+005 *	2.091e+005 *	2.091e+005 *
59	Vap. Reynolds	9.173e+006 *	9.173e+006 *	9.173e+006 *	9.173e+006 *
60	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
61	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	2.756e+004	2.756e+004	2.757e+004	2.757e+004
65	Elevation (m)	0.1800	0.1800	0.3040	0.4280
66	Pressure (bar)	88.68	88.68	88.67	88.67
67	Temperature (C)	50.48	50.47	50.47	50.47
68	Vapour Fraction	0.8693	0.8692	0.8693	0.8693
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 448 of 937

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	545.5	545.5	545.4	545.3
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.3023	0.3023	0.3582	0.3582
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	5.883e-003	5.883e-003	6.058e-003	6.058e-003
19	Static Gradient (kPa/m)	0.0000	0.0000	7.032e-002	7.032e-002
20	Accel. Gradient (kPa/m)	4.249e-007	4.249e-007	5.517e-006	5.517e-006
21	Liq. Reynolds	2.091e+005 *	2.091e+005 *	2.091e+005 *	2.091e+005 *
22	Vap. Reynolds	9.173e+006 *	9.173e+006 *	9.174e+006 *	9.174e+006 *
23	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
24	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	2.758e+004	2.758e+004	2.759e+004	2.759e+004
28	Elevation (m)	0.5520	0.6760	0.8000	0.5740
29	Pressure (bar)	88.67	88.66	88.66	88.66
30	Temperature (C)	50.47	50.46	50.46	50.46
31	Vapour Fraction	0.8693	0.8693	0.8693	0.8693
32	Heat Transferred (kJ/h-m)	545.3	545.2	545.1	545.1
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.3582	0.3582	0.3582	0.1652
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	6.058e-003	6.058e-003	6.058e-003	7.320e-003
38	Static Gradient (kPa/m)	7.032e-002	7.032e-002	7.032e-002	-7.626e-002
39	Accel. Gradient (kPa/m)	5.518e-006	5.518e-006	5.518e-006	-4.981e-006
40	Liq. Reynolds	2.091e+005 *	2.090e+005 *	2.090e+005 *	2.090e+005 *
41	Vap. Reynolds	9.174e+006 *	9.174e+006 *	9.174e+006 *	9.174e+006 *
42	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
43	Vapour Velocity (m/s)	2.172	2.172	2.173	2.172
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	2.760e+004	2.760e+004	2.761e+004	2.761e+004
47	Elevation (m)	0.3480	0.1220	-0.1040	-0.3300
48	Pressure (bar)	88.67	88.67	88.67	88.68
49	Temperature (C)	50.46	50.46	50.46	50.46
50	Vapour Fraction	0.8692	0.8692	0.8692	0.8692
51	Heat Transferred (kJ/h-m)	545.1	545.1	545.1	545.1
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.1652	0.1652	0.1652	0.1652
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	7.320e-003	7.320e-003	7.319e-003	7.319e-003
57	Static Gradient (kPa/m)	-7.626e-002	-7.627e-002	-7.627e-002	-7.627e-002
58	Accel. Gradient (kPa/m)	-4.981e-006	-4.981e-006	-4.980e-006	-4.980e-006
59	Liq. Reynolds	2.090e+005 *	2.091e+005 *	2.091e+005 *	2.091e+005 *
60	Vap. Reynolds	9.174e+006 *	9.174e+006 *	9.174e+006 *	9.174e+006 *
61	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
62	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	2.762e+004	2.762e+004	2.763e+004	2.763e+004
66	Elevation (m)	-0.1740	-1.800e-002	0.1380	0.2940
67	Pressure (bar)	88.67	88.67	88.66	88.66
68	Temperature (C)	50.46	50.45	50.45	50.45
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 449 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3			Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8692	0.8692	0.8692	0.8692
14	Heat Transferred (kJ/h-m)	545.0	545.0	544.9	544.8
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.3805	0.3805	0.3805	0.3805
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	6.025e-003	6.025e-003	6.026e-003	6.026e-003
20	Static Gradient (kPa/m)	9.446e-002	9.445e-002	9.445e-002	9.445e-002
21	Accel. Gradient (kPa/m)	7.258e-006	7.259e-006	7.259e-006	7.260e-006
22	Liq. Reynolds	2.091e+005 *	2.090e+005 *	2.090e+005 *	2.090e+005 *
23	Vap. Reynolds	9.174e+006 *	9.174e+006 *	9.174e+006 *	9.174e+006 *
24	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
25	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	2.764e+004	2.764e+004	2.765e+004	2.765e+004
29	Elevation (m)	0.4500	0.5140	0.5780	0.6420
30	Pressure (bar)	88.65	88.65	88.65	88.65
31	Temperature (C)	50.45	50.44	50.44	50.44
32	Vapour Fraction	0.8692	0.8692	0.8692	0.8692
33	Heat Transferred (kJ/h-m)	544.7	544.6	544.6	544.5
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.3805	0.3201	0.3201	0.3201
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	6.026e-003	6.071e-003	6.071e-003	6.071e-003
39	Static Gradient (kPa/m)	9.445e-002	3.339e-002	3.339e-002	3.339e-002
40	Accel. Gradient (kPa/m)	7.260e-006	2.852e-006	2.852e-006	2.852e-006
41	Liq. Reynolds	2.090e+005 *	2.090e+005 *	2.090e+005 *	2.090e+005 *
42	Vap. Reynolds	9.174e+006 *	9.174e+006 *	9.174e+006 *	9.174e+006 *
43	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
44	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	2.766e+004	2.766e+004	2.767e+004	2.767e+004
48	Elevation (m)	0.7060	0.7700	0.6520	0.5340
49	Pressure (bar)	88.64	88.64	88.64	88.65
50	Temperature (C)	50.44	50.43	50.43	50.43
51	Vapour Fraction	0.8692	0.8692	0.8692	0.8692
52	Heat Transferred (kJ/h-m)	544.4	544.4	544.4	544.4
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.3201	0.3200	0.1851	0.1851
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	6.071e-003	6.071e-003	7.073e-003	7.073e-003
58	Static Gradient (kPa/m)	3.339e-002	3.339e-002	-4.262e-002	-4.262e-002
59	Accel. Gradient (kPa/m)	2.852e-006	2.852e-006	-2.569e-006	-2.569e-006
60	Liq. Reynolds	2.090e+005 *	2.090e+005 *	2.090e+005 *	2.090e+005 *
61	Vap. Reynolds	9.175e+006 *	9.175e+006 *	9.175e+006 *	9.174e+006 *
62	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
63	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	2.768e+004	2.768e+004	2.769e+004	2.769e+004
67	Elevation (m)	0.4160	0.2980	0.1800	0.1800
68	Pressure (bar)	88.65	88.65	88.65	88.65
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 450 of 937				

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	50.43	50.43	50.43	50.43
14	Vapour Fraction	0.8692	0.8692	0.8692	0.8692
15	Heat Transferred (kJ/h-m)	544.3	544.3	544.3	544.2
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.1852	0.1852	0.1852	0.3023
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	7.072e-003	7.072e-003	7.072e-003	5.884e-003
21	Static Gradient (kPa/m)	-4.262e-002	-4.262e-002	-4.262e-002	0.0000
22	Accel. Gradient (kPa/m)	-2.569e-006	-2.569e-006	-2.569e-006	4.252e-007
23	Liq. Reynolds	2.090e+005 *	2.090e+005 *	2.090e+005 *	2.090e+005 *
24	Vap. Reynolds	9.174e+006 *	9.174e+006 *	9.174e+006 *	9.174e+006 *
25	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
26	Vapour Velocity (m/s)	2.173	2.172	2.172	2.172
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	2.770e+004	2.770e+004	2.771e+004	2.771e+004
30	Elevation (m)	0.1800	0.1800	0.1800	0.1800
31	Pressure (bar)	88.65	88.65	88.65	88.65
32	Temperature (C)	50.43	50.43	50.42	50.42
33	Vapour Fraction	0.8692	0.8692	0.8692	0.8692
34	Heat Transferred (kJ/h-m)	544.2	544.1	544.1	544.1
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.3023	0.3023	0.3023	0.3023
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	5.884e-003	5.884e-003	5.884e-003	5.884e-003
40	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
41	Accel. Gradient (kPa/m)	4.252e-007	4.252e-007	4.252e-007	4.252e-007
42	Liq. Reynolds	2.090e+005 *	2.090e+005 *	2.090e+005 *	2.090e+005 *
43	Vap. Reynolds	9.174e+006 *	9.174e+006 *	9.174e+006 *	9.174e+006 *
44	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
45	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	2.772e+004	2.772e+004	2.773e+004	2.773e+004
49	Elevation (m)	0.1820	0.1840	0.1860	0.1880
50	Pressure (bar)	88.65	88.65	88.65	88.65
51	Temperature (C)	50.42	50.42	50.42	50.42
52	Vapour Fraction	0.8692	0.8692	0.8692	0.8692
53	Heat Transferred (kJ/h-m)	544.0	544.0	543.9	543.9
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2806	0.2806	0.2806	0.2806
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	5.547e-003	5.547e-003	5.547e-003	5.547e-003
59	Static Gradient (kPa/m)	9.496e-004	9.496e-004	9.496e-004	9.496e-004
60	Accel. Gradient (kPa/m)	4.694e-007	4.694e-007	4.694e-007	4.694e-007
61	Liq. Reynolds	2.090e+005 *	2.090e+005 *	2.090e+005 *	2.090e+005 *
62	Vap. Reynolds	9.174e+006 *	9.174e+006 *	9.174e+006 *	9.174e+006 *
63	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
64	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	2.774e+004	2.774e+004	2.775e+004	2.775e+004
68	Elevation (m)	0.1900	0.1840	0.1780	0.1720
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:02:44 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Pressure (bar)	88.65	88.65	88.65	88.65					
14	Temperature (C)	50.41	50.41	50.41	50.41					
15	Vapour Fraction	0.8692	0.8692	0.8692	0.8692					
16	Heat Transferred (kJ/h-m)	543.8	543.8	543.8	543.7					
17	Flow Regime	Transition	Transition	Transition	Transition					
18	Water Flow Regime									
19	Liquid HoldUp	0.2806	0.2060	0.2060	0.2060					
20	Water HoldUp	---	---	---	---					
21	Friction Gradient (kPa/m)	5.547e-003	6.879e-003	6.879e-003	6.879e-003					
22	Static Gradient (kPa/m)	9.496e-004	-2.316e-003	-2.316e-003	-2.316e-003					
23	Accel. Gradient (kPa/m)	4.694e-007	3.297e-007	3.297e-007	3.297e-007					
24	Liq. Reynolds	2.090e+005 *	2.090e+005 *	2.090e+005 *	2.090e+005 *					
25	Vap. Reynolds	9.174e+006 *	9.174e+006 *	9.174e+006 *	9.174e+006 *					
26	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045					
27	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172					
28	Deposit Thickness (mm)	---	---	---	---					
29	Deposit Volume (m3)	---	---	---	---					
30	Length (m)	2.776e+004	2.776e+004	2.777e+004	2.777e+004					
31	Elevation (m)	0.1660	0.1600	0.1620	0.1640					
32	Pressure (bar)	88.65	88.65	88.65	88.65					
33	Temperature (C)	50.41	50.41	50.40	50.40					
34	Vapour Fraction	0.8692	0.8692	0.8692	0.8692					
35	Heat Transferred (kJ/h-m)	543.7	543.6	543.6	543.5					
36	Flow Regime	Transition	Transition	Transition	Transition					
37	Water Flow Regime									
38	Liquid HoldUp	0.2060	0.2060	0.2806	0.2806					
39	Water HoldUp	---	---	---	---					
40	Friction Gradient (kPa/m)	6.879e-003	6.879e-003	5.547e-003	5.547e-003					
41	Static Gradient (kPa/m)	-2.316e-003	-2.316e-003	9.497e-004	9.497e-004					
42	Accel. Gradient (kPa/m)	3.297e-007	3.297e-007	4.694e-007	4.694e-007					
43	Liq. Reynolds	2.090e+005 *	2.090e+005 *	2.090e+005 *	2.090e+005 *					
44	Vap. Reynolds	9.174e+006 *	9.174e+006 *	9.174e+006 *	9.174e+006 *					
45	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045					
46	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172					
47	Deposit Thickness (mm)	---	---	---	---					
48	Deposit Volume (m3)	---	---	---	---					
49	Length (m)	2.778e+004	2.778e+004	2.779e+004	2.779e+004					
50	Elevation (m)	0.1660	0.1680	0.1700	0.1680					
51	Pressure (bar)	88.65	88.64	88.64	88.64					
52	Temperature (C)	50.40	50.40	50.40	50.40					
53	Vapour Fraction	0.8692	0.8692	0.8692	0.8692					
54	Heat Transferred (kJ/h-m)	543.5	543.4	543.4	543.4					
55	Flow Regime	Transition	Transition	Transition	Transition					
56	Water Flow Regime									
57	Liquid HoldUp	0.2806	0.2806	0.2806	0.2067					
58	Water HoldUp	---	---	---	---					
59	Friction Gradient (kPa/m)	5.547e-003	5.547e-003	5.547e-003	6.873e-003					
60	Static Gradient (kPa/m)	9.497e-004	9.497e-004	9.497e-004	-7.738e-004					
61	Accel. Gradient (kPa/m)	4.694e-007	4.694e-007	4.694e-007	4.407e-007					
62	Liq. Reynolds	2.090e+005 *	2.090e+005 *	2.090e+005 *	2.090e+005 *					
63	Vap. Reynolds	9.174e+006 *	9.174e+006 *	9.174e+006 *	9.174e+006 *					
64	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045					
65	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172					
66	Deposit Thickness (mm)	---	---	---	---					
67	Deposit Volume (m3)	---	---	---	---					
68	Length (m)	2.780e+004	2.780e+004	2.781e+004	2.781e+004					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	0.1660	0.1640	0.1620	0.1600
14	Pressure (bar)	88.64	88.64	88.64	88.64
15	Temperature (C)	50.39	50.39	50.39	50.39
16	Vapour Fraction	0.8692	0.8692	0.8692	0.8692
17	Heat Transferred (kJ/h-m)	543.3	543.3	543.2	543.2
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2067	0.2067	0.2067	0.2067
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	6.873e-003	6.873e-003	6.873e-003	6.873e-003
23	Static Gradient (kPa/m)	-7.738e-004	-7.738e-004	-7.738e-004	-7.738e-004
24	Accel. Gradient (kPa/m)	4.407e-007	4.407e-007	4.407e-007	4.407e-007
25	Liq. Reynolds	2.090e+005 *	2.090e+005 *	2.090e+005 *	2.090e+005 *
26	Vap. Reynolds	9.174e+006 *	9.174e+006 *	9.174e+006 *	9.174e+006 *
27	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
28	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	2.782e+004	2.782e+004	2.783e+004	2.783e+004
32	Elevation (m)	0.1580	0.1560	0.1540	0.1520
33	Pressure (bar)	88.64	88.64	88.64	88.64
34	Temperature (C)	50.39	50.39	50.38	50.38
35	Vapour Fraction	0.8692	0.8692	0.8692	0.8692
36	Heat Transferred (kJ/h-m)	543.1	543.1	543.0	543.0
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2068	0.2068	0.2068	0.2068
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	6.873e-003	6.873e-003	6.873e-003	6.873e-003
42	Static Gradient (kPa/m)	-7.095e-004	-7.095e-004	-7.095e-004	-7.095e-004
43	Accel. Gradient (kPa/m)	4.453e-007	4.453e-007	4.453e-007	4.453e-007
44	Liq. Reynolds	2.090e+005 *	2.090e+005 *	2.090e+005 *	2.090e+005 *
45	Vap. Reynolds	9.175e+006 *	9.175e+006 *	9.175e+006 *	9.175e+006 *
46	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
47	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	2.784e+004	2.784e+004	2.785e+004	2.785e+004
51	Elevation (m)	0.1500	0.1520	0.1540	0.1560
52	Pressure (bar)	88.64	88.64	88.64	88.64
53	Temperature (C)	50.38	50.38	50.38	50.38
54	Vapour Fraction	0.8692	0.8692	0.8692	0.8692
55	Heat Transferred (kJ/h-m)	543.0	542.9	542.9	542.8
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2068	0.2810	0.2810	0.2810
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	6.873e-003	5.521e-003	5.521e-003	5.521e-003
61	Static Gradient (kPa/m)	-7.095e-004	1.188e-003	1.188e-003	1.188e-003
62	Accel. Gradient (kPa/m)	4.453e-007	4.848e-007	4.848e-007	4.848e-007
63	Liq. Reynolds	2.090e+005 *	2.090e+005 *	2.090e+005 *	2.090e+005 *
64	Vap. Reynolds	9.175e+006 *	9.175e+006 *	9.175e+006 *	9.175e+006 *
65	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
66	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 453 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:02:44 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Length	(m)	2.785e+004	2.786e+004	2.788e+004	2.790e+004				
14	Elevation	(m)	0.1580	0.1600	0.1640	0.1680				
15	Pressure	(bar)	88.64	88.64	88.64	88.64				
16	Temperature	(C)	50.38	50.37	50.37	50.36				
17	Vapour Fraction		0.8692	0.8692	0.8692	0.8692				
18	Heat Transferred	(kJ/h-m)	542.8	542.8	542.6	542.4				
19	Flow Regime		Transition	Transition	Transition	Transition				
20	Water Flow Regime									
21	Liquid HoldUp		0.2810	0.2810	0.2801	0.2801				
22	Water HoldUp		---	---	---	---				
23	Friction Gradient	(kPa/m)	5.521e-003	5.521e-003	5.592e-003	5.592e-003				
24	Static Gradient	(kPa/m)	1.188e-003	1.188e-003	5.126e-004	5.126e-004				
25	Accel. Gradient	(kPa/m)	4.848e-007	4.848e-007	4.411e-007	4.411e-007				
26	Liq. Reynolds		2.090e+005 *	2.090e+005 *	2.090e+005 *	2.089e+005 *				
27	Vap. Reynolds		9.175e+006 *	9.175e+006 *	9.175e+006 *	9.175e+006 *				
28	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045				
29	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172				
30	Deposit Thickness	(mm)	---	---	---	---				
31	Deposit Volume	(m3)	---	---	---	---				
32	Length	(m)	2.791e+004	2.793e+004	2.795e+004	2.797e+004				
33	Elevation	(m)	0.1720	0.1760	0.1800	0.1740				
34	Pressure	(bar)	88.64	88.64	88.63	88.63				
35	Temperature	(C)	50.36	50.35	50.34	50.34				
36	Vapour Fraction		0.8692	0.8692	0.8692	0.8692				
37	Heat Transferred	(kJ/h-m)	542.3	542.1	542.0	541.8				
38	Flow Regime		Transition	Transition	Transition	Transition				
39	Water Flow Regime									
40	Liquid HoldUp		0.2801	0.2801	0.2801	0.2068				
41	Water HoldUp		---	---	---	---				
42	Friction Gradient	(kPa/m)	5.592e-003	5.592e-003	5.592e-003	6.872e-003				
43	Static Gradient	(kPa/m)	5.126e-004	5.127e-004	5.127e-004	-6.277e-004				
44	Accel. Gradient	(kPa/m)	4.411e-007	4.411e-007	4.411e-007	4.511e-007				
45	Liq. Reynolds		2.089e+005 *	2.089e+005 *	2.089e+005 *	2.089e+005 *				
46	Vap. Reynolds		9.175e+006 *	9.175e+006 *	9.175e+006 *	9.175e+006 *				
47	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045				
48	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172				
49	Deposit Thickness	(mm)	---	---	---	---				
50	Deposit Volume	(m3)	---	---	---	---				
51	Length	(m)	2.799e+004	2.801e+004	2.802e+004	2.804e+004				
52	Elevation	(m)	0.1680	0.1620	0.1560	0.1500				
53	Pressure	(bar)	88.63	88.63	88.63	88.63				
54	Temperature	(C)	50.33	50.33	50.32	50.31				
55	Vapour Fraction		0.8692	0.8692	0.8692	0.8692				
56	Heat Transferred	(kJ/h-m)	541.7	541.5	541.3	541.2				
57	Flow Regime		Transition	Transition	Transition	Transition				
58	Water Flow Regime									
59	Liquid HoldUp		0.2068	0.2068	0.2068	0.2068				
60	Water HoldUp		---	---	---	---				
61	Friction Gradient	(kPa/m)	6.871e-003	6.871e-003	6.871e-003	6.871e-003				
62	Static Gradient	(kPa/m)	-6.277e-004	-6.277e-004	-6.277e-004	-6.277e-004				
63	Accel. Gradient	(kPa/m)	4.511e-007	4.511e-007	4.511e-007	4.511e-007				
64	Liq. Reynolds		2.089e+005 *	2.089e+005 *	2.089e+005 *	2.089e+005 *				
65	Vap. Reynolds		9.175e+006 *	9.175e+006 *	9.175e+006 *	9.175e+006 *				
66	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045				
67	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172				
68	Deposit Thickness	(mm)	---	---	---	---				
69	Aspen Technology Inc.		Aspen HYSYS Version 10							
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	2.806e+004	2.808e+004	2.810e+004	2.812e+004
15	Elevation (m)	0.1480	0.1460	0.1440	0.1420
16	Pressure (bar)	88.63	88.63	88.62	88.62
17	Temperature (C)	50.31	50.30	50.30	50.29
18	Vapour Fraction	0.8692	0.8692	0.8692	0.8692
19	Heat Transferred (kJ/h-m)	541.0	540.9	540.7	540.5
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2070	0.2071	0.2071	0.2071
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	6.870e-003	6.869e-003	6.869e-003	6.869e-003
25	Static Gradient (kPa/m)	-2.094e-004	-2.094e-004	-2.094e-004	-2.094e-004
26	Accel. Gradient (kPa/m)	4.812e-007	4.812e-007	4.812e-007	4.812e-007
27	Liq. Reynolds	2.089e+005 *	2.089e+005 *	2.089e+005 *	2.089e+005 *
28	Vap. Reynolds	9.175e+006 *	9.175e+006 *	9.175e+006 *	9.175e+006 *
29	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
30	Vapour Velocity (m/s)	2.172	2.172	2.171	2.171
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	2.814e+004	2.815e+004	2.817e+004	2.819e+004
34	Elevation (m)	0.1400	0.1320	0.1240	0.1160
35	Pressure (bar)	88.62	88.62	88.62	88.62
36	Temperature (C)	50.28	50.28	50.27	50.27
37	Vapour Fraction	0.8692	0.8691	0.8691	0.8691
38	Heat Transferred (kJ/h-m)	540.4	540.2	540.1	539.9
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2071	0.2068	0.2068	0.2068
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	6.869e-003	6.871e-003	6.871e-003	6.871e-003
44	Static Gradient (kPa/m)	-2.094e-004	-8.367e-004	-8.368e-004	-8.368e-004
45	Accel. Gradient (kPa/m)	4.811e-007	4.360e-007	4.360e-007	4.360e-007
46	Liq. Reynolds	2.089e+005 *	2.089e+005 *	2.089e+005 *	2.089e+005 *
47	Vap. Reynolds	9.175e+006 *	9.175e+006 *	9.175e+006 *	9.175e+006 *
48	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
49	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	2.821e+004	2.823e+004	2.825e+004	2.827e+004
53	Elevation (m)	0.1080	1.000e-001	9.600e-002	9.200e-002
54	Pressure (bar)	88.62	88.62	88.62	88.61
55	Temperature (C)	50.26	50.25	50.25	50.24
56	Vapour Fraction	0.8691	0.8691	0.8691	0.8691
57	Heat Transferred (kJ/h-m)	539.7	539.6	539.4	539.3
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2068	0.2068	0.2070	0.2070
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	6.871e-003	6.871e-003	6.869e-003	6.869e-003
63	Static Gradient (kPa/m)	-8.368e-004	-8.368e-004	-4.187e-004	-4.187e-004
64	Accel. Gradient (kPa/m)	4.359e-007	4.359e-007	4.660e-007	4.660e-007
65	Liq. Reynolds	2.088e+005 *	2.088e+005 *	2.088e+005 *	2.088e+005 *
66	Vap. Reynolds	9.175e+006 *	9.175e+006 *	9.175e+006 *	9.175e+006 *
67	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
68	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	2.828e+004	2.830e+004	2.832e+004	2.834e+004
16	Elevation (m)	8.800e-002	8.400e-002	8.000e-002	8.200e-002
17	Pressure (bar)	88.61	88.61	88.61	88.61
18	Temperature (C)	50.24	50.23	50.22	50.22
19	Vapour Fraction	0.8691	0.8691	0.8691	0.8691
20	Heat Transferred (kJ/h-m)	539.1	538.9	538.8	538.6
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2070	0.2070	0.2070	0.2798
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	6.869e-003	6.869e-003	6.869e-003	5.617e-003
26	Static Gradient (kPa/m)	-4.187e-004	-4.187e-004	-4.187e-004	2.562e-004
27	Accel. Gradient (kPa/m)	4.660e-007	4.660e-007	4.660e-007	4.243e-007
28	Liq. Reynolds	2.088e+005 *	2.088e+005 *	2.088e+005 *	2.088e+005 *
29	Vap. Reynolds	9.175e+006 *	9.175e+006 *	9.175e+006 *	9.175e+006 *
30	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
31	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	2.836e+004	2.838e+004	2.839e+004	2.841e+004
35	Elevation (m)	8.400e-002	8.600e-002	8.800e-002	9.000e-002
36	Pressure (bar)	88.61	88.61	88.61	88.61
37	Temperature (C)	50.21	50.21	50.20	50.19
38	Vapour Fraction	0.8691	0.8691	0.8691	0.8691
39	Heat Transferred (kJ/h-m)	538.4	538.3	538.1	538.0
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2798	0.2798	0.2798	0.2798
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	5.616e-003	5.616e-003	5.616e-003	5.616e-003
45	Static Gradient (kPa/m)	2.562e-004	2.562e-004	2.562e-004	2.562e-004
46	Accel. Gradient (kPa/m)	4.242e-007	4.242e-007	4.242e-007	4.242e-007
47	Liq. Reynolds	2.088e+005 *	2.088e+005 *	2.088e+005 *	2.088e+005 *
48	Vap. Reynolds	9.175e+006 *	9.175e+006 *	9.175e+006 *	9.175e+006 *
49	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
50	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	2.842e+004	2.843e+004	2.843e+004	2.844e+004
54	Elevation (m)	8.800e-002	8.600e-002	8.400e-002	8.200e-002
55	Pressure (bar)	88.61	88.60	88.60	88.60
56	Temperature (C)	50.19	50.19	50.19	50.19
57	Vapour Fraction	0.8691	0.8691	0.8691	0.8691
58	Heat Transferred (kJ/h-m)	537.9	537.9	537.8	537.8
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2069	0.2069	0.2069	0.2069
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	6.869e-003	6.869e-003	6.869e-003	6.869e-003
64	Static Gradient (kPa/m)	-5.781e-004	-5.781e-004	-5.781e-004	-5.781e-004
65	Accel. Gradient (kPa/m)	4.544e-007	4.544e-007	4.544e-007	4.544e-007
66	Liq. Reynolds	2.088e+005 *	2.088e+005 *	2.088e+005 *	2.088e+005 *
67	Vap. Reynolds	9.175e+006 *	9.175e+006 *	9.175e+006 *	9.176e+006 *
68	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.171	2.171	2.171	2.171	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	2.845e+004	2.845e+004	2.846e+004	2.846e+004	
17	Elevation	(m)	8.000e-002	7.800e-002	7.600e-002	7.400e-002	
18	Pressure	(bar)	88.60	88.60	88.60	88.60	
19	Temperature	(C)	50.18	50.18	50.18	50.18	
20	Vapour Fraction		0.8691	0.8691	0.8691	0.8691	
21	Heat Transferred	(kJ/h-m)	537.7	537.7	537.6	537.6	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2069	0.2068	0.2068	0.2068	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	6.869e-003	6.869e-003	6.869e-003	6.869e-003	
27	Static Gradient	(kPa/m)	-5.781e-004	-7.742e-004	-7.742e-004	-7.742e-004	
28	Accel. Gradient	(kPa/m)	4.544e-007	4.403e-007	4.403e-007	4.403e-007	
29	Liq. Reynolds		2.088e+005 *	2.088e+005 *	2.088e+005 *	2.088e+005 *	
30	Vap. Reynolds		9.176e+006 *	9.176e+006 *	9.176e+006 *	9.176e+006 *	
31	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
32	Vapour Velocity	(m/s)	2.171	2.171	2.171	2.171	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	2.847e+004	2.847e+004	2.848e+004	2.848e+004	
36	Elevation	(m)	7.200e-002	7.000e-002	7.000e-002	7.000e-002	
37	Pressure	(bar)	88.60	88.60	88.60	88.60	
38	Temperature	(C)	50.18	50.18	50.17	50.17	
39	Vapour Fraction		0.8691	0.8691	0.8691	0.8691	
40	Heat Transferred	(kJ/h-m)	537.5	537.5	537.4	537.4	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2068	0.2068	0.3025	0.3025	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	6.869e-003	6.869e-003	5.881e-003	5.881e-003	
46	Static Gradient	(kPa/m)	-7.743e-004	-7.743e-004	0.0000	0.0000	
47	Accel. Gradient	(kPa/m)	4.403e-007	4.403e-007	4.248e-007	4.248e-007	
48	Liq. Reynolds		2.088e+005 *	2.088e+005 *	2.088e+005 *	2.088e+005 *	
49	Vap. Reynolds		9.176e+006 *	9.176e+006 *	9.176e+006 *	9.176e+006 *	
50	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
51	Vapour Velocity	(m/s)	2.171	2.171	2.171	2.171	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	2.849e+004	2.849e+004	2.850e+004	2.850e+004	
55	Elevation	(m)	7.000e-002	7.000e-002	7.000e-002	6.000e-002	
56	Pressure	(bar)	88.60	88.60	88.60	88.60	
57	Temperature	(C)	50.17	50.17	50.17	50.17	
58	Vapour Fraction		0.8691	0.8691	0.8691	0.8691	
59	Heat Transferred	(kJ/h-m)	537.3	537.3	537.3	537.2	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.3025	0.3025	0.3025	0.2054	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	5.881e-003	5.881e-003	5.881e-003	6.881e-003	
65	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	-3.854e-003	
66	Accel. Gradient	(kPa/m)	4.248e-007	4.248e-007	4.248e-007	2.187e-007	
67	Liq. Reynolds		2.088e+005 *	2.088e+005 *	2.088e+005 *	2.088e+005 *	
68	Vap. Reynolds		9.176e+006 *	9.176e+006 *	9.176e+006 *	9.176e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
14	Vapour Velocity	(m/s)	2.171	2.171	2.171	2.171	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	2.851e+004	2.851e+004	2.852e+004	2.852e+004	
18	Elevation	(m)	5.000e-002	4.000e-002	3.000e-002	2.000e-002	
19	Pressure	(bar)	88.60	88.60	88.60	88.60	
20	Temperature	(C)	50.16	50.16	50.16	50.16	
21	Vapour Fraction		0.8691	0.8691	0.8691	0.8691	
22	Heat Transferred	(kJ/h-m)	537.2	537.1	537.1	537.1	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.2054	0.2054	0.2054	0.2054	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	6.881e-003	6.881e-003	6.881e-003	6.881e-003	
28	Static Gradient	(kPa/m)	-3.854e-003	-3.854e-003	-3.854e-003	-3.854e-003	
29	Accel. Gradient	(kPa/m)	2.187e-007	2.187e-007	2.187e-007	2.187e-007	
30	Liq. Reynolds		2.088e+005 *	2.088e+005 *	2.088e+005 *	2.088e+005 *	
31	Vap. Reynolds		9.176e+006 *	9.176e+006 *	9.176e+006 *	9.176e+006 *	
32	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
33	Vapour Velocity	(m/s)	2.171	2.171	2.171	2.171	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	2.853e+004	2.853e+004	2.854e+004	2.854e+004	
37	Elevation	(m)	2.600e-002	3.200e-002	3.800e-002	4.400e-002	
38	Pressure	(bar)	88.60	88.60	88.60	88.60	
39	Temperature	(C)	50.16	50.16	50.15	50.15	
40	Vapour Fraction		0.8691	0.8691	0.8691	0.8691	
41	Heat Transferred	(kJ/h-m)	537.0	537.0	536.9	536.9	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2833	0.2833	0.2833	0.2833	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	5.343e-003	5.343e-003	5.343e-003	5.343e-003	
47	Static Gradient	(kPa/m)	2.869e-003	2.869e-003	2.869e-003	2.869e-003	
48	Accel. Gradient	(kPa/m)	5.932e-007	5.932e-007	5.932e-007	5.932e-007	
49	Liq. Reynolds		2.088e+005 *	2.088e+005 *	2.088e+005 *	2.087e+005 *	
50	Vap. Reynolds		9.176e+006 *	9.176e+006 *	9.176e+006 *	9.176e+006 *	
51	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
52	Vapour Velocity	(m/s)	2.171	2.170	2.170	2.170	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	2.855e+004	2.855e+004	2.856e+004	2.856e+004	
56	Elevation	(m)	5.000e-002	4.400e-002	3.800e-002	3.200e-002	
57	Pressure	(bar)	88.60	88.60	88.60	88.60	
58	Temperature	(C)	50.15	50.15	50.15	50.15	
59	Vapour Fraction		0.8691	0.8691	0.8691	0.8691	
60	Heat Transferred	(kJ/h-m)	536.8	536.8	536.8	536.7	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2833	0.2061	0.2061	0.2061	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	5.343e-003	6.875e-003	6.875e-003	6.875e-003	
66	Static Gradient	(kPa/m)	2.869e-003	-2.318e-003	-2.318e-003	-2.318e-003	
67	Accel. Gradient	(kPa/m)	5.932e-007	3.292e-007	3.292e-007	3.292e-007	
68	Liq. Reynolds		2.087e+005 *	2.087e+005 *	2.087e+005 *	2.087e+005 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.176e+006 *	9.176e+006 *	9.176e+006 *	9.176e+006 *
14	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
15	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	2.857e+004	2.857e+004	2.858e+004	2.858e+004
19	Elevation (m)	2.600e-002	2.000e-002	-1.600e-002	-5.200e-002
20	Pressure (bar)	88.60	88.60	88.60	88.60
21	Temperature (C)	50.14	50.14	50.14	50.14
22	Vapour Fraction	0.8691	0.8691	0.8691	0.8691
23	Heat Transferred (kJ/h-m)	536.7	536.6	536.6	536.6
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2061	0.2061	0.2005	0.2005
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.875e-003	6.875e-003	6.920e-003	6.920e-003
29	Static Gradient (kPa/m)	-2.318e-003	-2.318e-003	-1.367e-002	-1.367e-002
30	Accel. Gradient (kPa/m)	3.292e-007	3.292e-007	-4.873e-007	-4.873e-007
31	Liq. Reynolds	2.087e+005 *	2.087e+005 *	2.087e+005 *	2.087e+005 *
32	Vap. Reynolds	9.176e+006 *	9.176e+006 *	9.176e+006 *	9.176e+006 *
33	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
34	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	2.859e+004	2.859e+004	2.860e+004	2.860e+004
38	Elevation (m)	-8.800e-002	-0.1240	-0.1600	-0.2280
39	Pressure (bar)	88.60	88.60	88.60	88.60
40	Temperature (C)	50.14	50.14	50.14	50.14
41	Vapour Fraction	0.8691	0.8691	0.8691	0.8691
42	Heat Transferred (kJ/h-m)	536.5	536.5	536.5	536.4
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2005	0.2005	0.2005	0.1946
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.920e-003	6.920e-003	6.920e-003	6.973e-003
48	Static Gradient (kPa/m)	-1.367e-002	-1.367e-002	-1.367e-002	-2.533e-002
49	Accel. Gradient (kPa/m)	-4.873e-007	-4.873e-007	-4.873e-007	-1.326e-006
50	Liq. Reynolds	2.087e+005 *	2.087e+005 *	2.087e+005 *	2.087e+005 *
51	Vap. Reynolds	9.176e+006 *	9.176e+006 *	9.176e+006 *	9.176e+006 *
52	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
53	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	2.861e+004	2.861e+004	2.862e+004	2.862e+004
57	Elevation (m)	-0.2960	-0.3640	-0.4320	-0.5000
58	Pressure (bar)	88.60	88.60	88.60	88.60
59	Temperature (C)	50.13	50.13	50.13	50.13
60	Vapour Fraction	0.8691	0.8691	0.8691	0.8691
61	Heat Transferred (kJ/h-m)	536.4	536.4	536.3	536.3
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.1946	0.1946	0.1946	0.1946
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	6.973e-003	6.973e-003	6.973e-003	6.973e-003
67	Static Gradient (kPa/m)	-2.533e-002	-2.533e-002	-2.534e-002	-2.534e-002
68	Accel. Gradient (kPa/m)	-1.326e-006	-1.326e-006	-1.326e-006	-1.326e-006
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.087e+005 *	2.087e+005 *	2.087e+005 *	2.087e+005 *
14	Vap. Reynolds	9.176e+006 *	9.176e+006 *	9.175e+006 *	9.175e+006 *
15	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
16	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	2.863e+004	2.863e+004	2.864e+004	2.864e+004
20	Elevation (m)	-0.5360	-0.5720	-0.6080	-0.6440
21	Pressure (bar)	88.60	88.60	88.60	88.60
22	Temperature (C)	50.13	50.13	50.13	50.13
23	Vapour Fraction	0.8691	0.8691	0.8691	0.8691
24	Heat Transferred (kJ/h-m)	536.3	536.2	536.2	536.2
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2005	0.2005	0.2005	0.2005
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	6.920e-003	6.920e-003	6.919e-003	6.919e-003
30	Static Gradient (kPa/m)	-1.368e-002	-1.368e-002	-1.368e-002	-1.368e-002
31	Accel. Gradient (kPa/m)	-4.881e-007	-4.881e-007	-4.882e-007	-4.882e-007
32	Liq. Reynolds	2.087e+005 *	2.087e+005 *	2.087e+005 *	2.087e+005 *
33	Vap. Reynolds	9.175e+006 *	9.175e+006 *	9.175e+006 *	9.175e+006 *
34	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
35	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	2.865e+004	2.865e+004	2.866e+004	2.866e+004
39	Elevation (m)	-0.6800	-0.6500	-0.6200	-0.5900
40	Pressure (bar)	88.60	88.60	88.60	88.60
41	Temperature (C)	50.12	50.12	50.12	50.12
42	Vapour Fraction	0.8691	0.8691	0.8691	0.8691
43	Heat Transferred (kJ/h-m)	536.1	536.1	536.0	536.0
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2005	0.2987	0.2987	0.2987
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	6.919e-003	5.695e-003	5.695e-003	5.695e-003
49	Static Gradient (kPa/m)	-1.368e-002	1.489e-002	1.489e-002	1.489e-002
50	Accel. Gradient (kPa/m)	-4.882e-007	1.487e-006	1.487e-006	1.487e-006
51	Liq. Reynolds	2.087e+005 *	2.087e+005 *	2.087e+005 *	2.087e+005 *
52	Vap. Reynolds	9.175e+006 *	9.175e+006 *	9.175e+006 *	9.175e+006 *
53	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
54	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	2.867e+004	2.867e+004	2.868e+004	2.868e+004
58	Elevation (m)	-0.5600	-0.5300	-0.4300	-0.3300
59	Pressure (bar)	88.60	88.60	88.60	88.59
60	Temperature (C)	50.12	50.11	50.11	50.11
61	Vapour Fraction	0.8691	0.8691	0.8691	0.8691
62	Heat Transferred (kJ/h-m)	535.9	535.9	535.8	535.7
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2987	0.2987	0.3432	0.3432
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	5.695e-003	5.695e-003	6.071e-003	6.071e-003
68	Static Gradient (kPa/m)	1.489e-002	1.489e-002	5.494e-002	5.494e-002
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	1.487e-006	1.487e-006	4.406e-006	4.407e-006
14	Liq. Reynolds	2.087e+005 *	2.087e+005 *	2.087e+005 *	2.087e+005 *
15	Vap. Reynolds	9.176e+006 *	9.176e+006 *	9.176e+006 *	9.176e+006 *
16	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
17	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	2.869e+004	2.869e+004	2.870e+004	2.870e+004
21	Elevation (m)	-0.2300	-0.1300	-3.000e-002	-2.600e-002
22	Pressure (bar)	88.59	88.59	88.58	88.58
23	Temperature (C)	50.11	50.10	50.10	50.10
24	Vapour Fraction	0.8691	0.8691	0.8691	0.8691
25	Heat Transferred (kJ/h-m)	535.7	535.6	535.5	535.5
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.3432	0.3432	0.3432	0.2821
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	6.071e-003	6.071e-003	6.071e-003	5.441e-003
31	Static Gradient (kPa/m)	5.494e-002	5.494e-002	5.494e-002	1.907e-003
32	Accel. Gradient (kPa/m)	4.407e-006	4.407e-006	4.407e-006	5.308e-007
33	Liq. Reynolds	2.087e+005 *	2.087e+005 *	2.087e+005 *	2.087e+005 *
34	Vap. Reynolds	9.176e+006 *	9.176e+006 *	9.176e+006 *	9.176e+006 *
35	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
36	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	2.871e+004	2.871e+004	2.872e+004	2.872e+004
40	Elevation (m)	-2.200e-002	-1.800e-002	-1.400e-002	-1.000e-002
41	Pressure (bar)	88.58	88.58	88.58	88.58
42	Temperature (C)	50.10	50.10	50.10	50.09
43	Vapour Fraction	0.8691	0.8690	0.8690	0.8690
44	Heat Transferred (kJ/h-m)	535.4	535.4	535.3	535.3
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.2821	0.2821	0.2821	0.2821
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	5.441e-003	5.441e-003	5.441e-003	5.441e-003
50	Static Gradient (kPa/m)	1.907e-003	1.907e-003	1.907e-003	1.907e-003
51	Accel. Gradient (kPa/m)	5.308e-007	5.308e-007	5.308e-007	5.308e-007
52	Liq. Reynolds	2.087e+005 *	2.087e+005 *	2.087e+005 *	2.087e+005 *
53	Vap. Reynolds	9.176e+006 *	9.176e+006 *	9.176e+006 *	9.176e+006 *
54	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
55	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	2.873e+004	2.874e+004	2.875e+004	2.876e+004
59	Elevation (m)	-4.000e-003	2.000e-003	8.000e-003	1.400e-002
60	Pressure (bar)	88.58	88.58	88.58	88.58
61	Temperature (C)	50.09	50.09	50.08	50.08
62	Vapour Fraction	0.8690	0.8690	0.8690	0.8690
63	Heat Transferred (kJ/h-m)	535.2	535.1	535.0	534.9
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.2814	0.2814	0.2814	0.2814
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	5.495e-003	5.495e-003	5.495e-003	5.495e-003
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Static Gradient	(kPa/m)	1.382e-003	1.382e-003	1.382e-003
14	Accel. Gradient	(kPa/m)	4.968e-007	4.968e-007	4.968e-007
15	Liq. Reynolds		2.087e+005 *	2.087e+005 *	2.087e+005 *
16	Vap. Reynolds		9.176e+006 *	9.176e+006 *	9.176e+006 *
17	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045
18	Vapour Velocity	(m/s)	2.170	2.170	2.170
19	Deposit Thickness	(mm)	---	---	---
20	Deposit Volume	(m3)	---	---	---
21	Length	(m)	2.877e+004	2.878e+004	2.879e+004
22	Elevation	(m)	2.000e-002	1.400e-002	8.000e-003
23	Pressure	(bar)	88.58	88.58	88.58
24	Temperature	(C)	50.08	50.07	50.07
25	Vapour Fraction		0.8690	0.8690	0.8690
26	Heat Transferred	(kJ/h-m)	534.9	534.8	534.7
27	Flow Regime		Transition	Transition	Transition
28	Water Flow Regime				
29	Liquid HoldUp		0.2814	0.2067	0.2067
30	Water HoldUp		---	---	---
31	Friction Gradient	(kPa/m)	5.495e-003	6.869e-003	6.869e-003
32	Static Gradient	(kPa/m)	1.382e-003	-1.124e-003	-1.124e-003
33	Accel. Gradient	(kPa/m)	4.968e-007	4.151e-007	4.150e-007
34	Liq. Reynolds		2.087e+005 *	2.087e+005 *	2.087e+005 *
35	Vap. Reynolds		9.176e+006 *	9.176e+006 *	9.176e+006 *
36	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045
37	Vapour Velocity	(m/s)	2.170	2.170	2.170
38	Deposit Thickness	(mm)	---	---	---
39	Deposit Volume	(m3)	---	---	---
40	Length	(m)	2.881e+004	2.883e+004	2.884e+004
41	Elevation	(m)	-4.000e-003	-1.000e-002	-1.200e-002
42	Pressure	(bar)	88.58	88.58	88.58
43	Temperature	(C)	50.06	50.06	50.06
44	Vapour Fraction		0.8690	0.8690	0.8690
45	Heat Transferred	(kJ/h-m)	534.5	534.4	534.3
46	Flow Regime		Transition	Transition	Transition
47	Water Flow Regime				
48	Liquid HoldUp		0.2067	0.2067	0.2071
49	Water HoldUp		---	---	---
50	Friction Gradient	(kPa/m)	6.869e-003	6.869e-003	6.866e-003
51	Static Gradient	(kPa/m)	-1.124e-003	-1.124e-003	-3.750e-004
52	Accel. Gradient	(kPa/m)	4.150e-007	4.150e-007	4.689e-007
53	Liq. Reynolds		2.087e+005 *	2.087e+005 *	2.087e+005 *
54	Vap. Reynolds		9.176e+006 *	9.176e+006 *	9.176e+006 *
55	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045
56	Vapour Velocity	(m/s)	2.170	2.170	2.170
57	Deposit Thickness	(mm)	---	---	---
58	Deposit Volume	(m3)	---	---	---
59	Length	(m)	2.886e+004	2.887e+004	2.888e+004
60	Elevation	(m)	-1.600e-002	-1.800e-002	-2.000e-002
61	Pressure	(bar)	88.57	88.57	88.57
62	Temperature	(C)	50.05	50.05	50.04
63	Vapour Fraction		0.8690	0.8690	0.8690
64	Heat Transferred	(kJ/h-m)	534.2	534.1	534.0
65	Flow Regime		Transition	Transition	Transition
66	Water Flow Regime				
67	Liquid HoldUp		0.2071	0.2071	0.2060
68	Water HoldUp		---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3							
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5	Date/Time: Mon Jun 13 15:02:44 2022						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Friction Gradient	(kPa/m)	6.866e-003	6.866e-003	6.866e-003	6.874e-003	
14	Static Gradient	(kPa/m)	-3.750e-004	-3.750e-004	-3.750e-004	-2.616e-003	
15	Accel. Gradient	(kPa/m)	4.689e-007	4.689e-007	4.689e-007	3.076e-007	
16	Liq. Reynolds		2.086e+005 *	2.086e+005 *	2.086e+005 *	2.086e+005 *	
17	Vap. Reynolds		9.176e+006 *	9.176e+006 *	9.176e+006 *	9.176e+006 *	
18	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
19	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	2.890e+004	2.891e+004	2.892e+004	2.893e+004	
23	Elevation	(m)	-4.800e-002	-6.200e-002	-7.600e-002	-9.000e-002	
24	Pressure	(bar)	88.57	88.57	88.57	88.57	
25	Temperature	(C)	50.04	50.03	50.03	50.03	
26	Vapour Fraction		0.8690	0.8690	0.8690	0.8690	
27	Heat Transferred	(kJ/h-m)	533.8	533.7	533.6	533.6	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.2060	0.2060	0.2060	0.2060	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	6.874e-003	6.874e-003	6.874e-003	6.874e-003	
33	Static Gradient	(kPa/m)	-2.616e-003	-2.616e-003	-2.616e-003	-2.616e-003	
34	Accel. Gradient	(kPa/m)	3.076e-007	3.076e-007	3.076e-007	3.076e-007	
35	Liq. Reynolds		2.086e+005 *	2.086e+005 *	2.086e+005 *	2.086e+005 *	
36	Vap. Reynolds		9.176e+006 *	9.176e+006 *	9.176e+006 *	9.176e+006 *	
37	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
38	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	2.894e+004	2.895e+004	2.896e+004	2.897e+004	
42	Elevation	(m)	-9.400e-002	-9.800e-002	-0.1020	-0.1060	
43	Pressure	(bar)	88.57	88.57	88.57	88.57	
44	Temperature	(C)	50.02	50.02	50.02	50.01	
45	Vapour Fraction		0.8690	0.8690	0.8690	0.8690	
46	Heat Transferred	(kJ/h-m)	533.5	533.4	533.3	533.2	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.2069	0.2069	0.2069	0.2069	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	6.867e-003	6.867e-003	6.867e-003	6.867e-003	
52	Static Gradient	(kPa/m)	-7.496e-004	-7.496e-004	-7.496e-004	-7.496e-004	
53	Accel. Gradient	(kPa/m)	4.419e-007	4.419e-007	4.419e-007	4.419e-007	
54	Liq. Reynolds		2.086e+005 *	2.086e+005 *	2.086e+005 *	2.086e+005 *	
55	Vap. Reynolds		9.176e+006 *	9.176e+006 *	9.176e+006 *	9.176e+006 *	
56	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2045	
57	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	2.898e+004	2.899e+004	2.900e+004	2.901e+004	
61	Elevation	(m)	-0.1100	-0.1180	-0.1260	-0.1340	
62	Pressure	(bar)	88.57	88.57	88.57	88.57	
63	Temperature	(C)	50.01	50.01	50.00	50.00	
64	Vapour Fraction		0.8690	0.8690	0.8690	0.8690	
65	Heat Transferred	(kJ/h-m)	533.1	533.0	532.9	532.9	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.2069	0.2066	0.2066	0.2066	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 463 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	6.867e-003	6.869e-003	6.869e-003	6.869e-003
15	Static Gradient (kPa/m)	-7.496e-004	-1.498e-003	-1.498e-003	-1.498e-003
16	Accel. Gradient (kPa/m)	4.418e-007	3.880e-007	3.880e-007	3.880e-007
17	Liq. Reynolds	2.086e+005 *	2.086e+005 *	2.086e+005 *	2.086e+005 *
18	Vap. Reynolds	9.176e+006 *	9.176e+006 *	9.177e+006 *	9.177e+006 *
19	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
20	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	2.902e+004	2.903e+004	2.904e+004	2.904e+004
24	Elevation (m)	-0.1420	-0.1500	-0.1440	-0.1380
25	Pressure (bar)	88.56	88.56	88.56	88.56
26	Temperature (C)	50.00	50.00	49.99	49.99
27	Vapour Fraction	0.8690	0.8690	0.8690	0.8690
28	Heat Transferred (kJ/h-m)	532.8	532.7	532.6	532.6
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2066	0.2066	0.2855	0.2855
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	6.869e-003	6.869e-003	5.179e-003	5.179e-003
34	Static Gradient (kPa/m)	-1.498e-003	-1.498e-003	4.489e-003	4.489e-003
35	Accel. Gradient (kPa/m)	3.880e-007	3.880e-007	6.983e-007	6.983e-007
36	Liq. Reynolds	2.086e+005 *	2.086e+005 *	2.086e+005 *	2.086e+005 *
37	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
38	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
39	Vapour Velocity (m/s)	2.170	2.169	2.169	2.169
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	2.904e+004	2.904e+004	2.905e+004	2.905e+004
43	Elevation (m)	-0.1320	-0.1260	-0.1200	-0.1300
44	Pressure (bar)	88.56	88.56	88.56	88.56
45	Temperature (C)	49.99	49.99	49.99	49.99
46	Vapour Fraction	0.8690	0.8690	0.8690	0.8690
47	Heat Transferred (kJ/h-m)	532.6	532.6	532.5	532.5
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2855	0.2855	0.2855	0.2054
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	5.179e-003	5.179e-003	5.179e-003	6.878e-003
53	Static Gradient (kPa/m)	4.489e-003	4.489e-003	4.489e-003	-3.855e-003
54	Accel. Gradient (kPa/m)	6.983e-007	6.983e-007	6.983e-007	2.183e-007
55	Liq. Reynolds	2.086e+005 *	2.086e+005 *	2.086e+005 *	2.086e+005 *
56	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
57	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
58	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	2.906e+004	2.906e+004	2.907e+004	2.907e+004
62	Elevation (m)	-0.1400	-0.1500	-0.1600	-0.1700
63	Pressure (bar)	88.56	88.56	88.56	88.56
64	Temperature (C)	49.99	49.99	49.98	49.98
65	Vapour Fraction	0.8690	0.8690	0.8690	0.8690
66	Heat Transferred (kJ/h-m)	532.5	532.4	532.4	532.3
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 464 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.2054	0.2054	0.2054	0.2054
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	6.878e-003	6.878e-003	6.878e-003	6.878e-003
16	Static Gradient (kPa/m)	-3.855e-003	-3.855e-003	-3.855e-003	-3.855e-003
17	Accel. Gradient (kPa/m)	2.183e-007	2.183e-007	2.183e-007	2.183e-007
18	Liq. Reynolds	2.086e+005 *	2.086e+005 *	2.086e+005 *	2.086e+005 *
19	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
20	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
21	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	2.908e+004	2.908e+004	2.909e+004	2.909e+004
25	Elevation (m)	-0.1920	-0.2140	-0.2360	-0.2580
26	Pressure (bar)	88.56	88.56	88.56	88.56
27	Temperature (C)	49.98	49.98	49.98	49.98
28	Vapour Fraction	0.8690	0.8690	0.8690	0.8690
29	Heat Transferred (kJ/h-m)	532.3	532.3	532.2	532.2
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2032	0.2032	0.2032	0.2032
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.896e-003	6.896e-003	6.896e-003	6.896e-003
35	Static Gradient (kPa/m)	-8.423e-003	-8.423e-003	-8.423e-003	-8.423e-003
36	Accel. Gradient (kPa/m)	-1.103e-007	-1.103e-007	-1.103e-007	-1.103e-007
37	Liq. Reynolds	2.086e+005 *	2.086e+005 *	2.086e+005 *	2.086e+005 *
38	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
39	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
40	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	2.910e+004	2.910e+004	2.911e+004	2.911e+004
44	Elevation (m)	-0.2800	-0.2880	-0.2960	-0.3040
45	Pressure (bar)	88.56	88.56	88.56	88.56
46	Temperature (C)	49.97	49.97	49.97	49.97
47	Vapour Fraction	0.8690	0.8690	0.8690	0.8690
48	Heat Transferred (kJ/h-m)	532.1	532.1	532.1	532.0
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2032	0.2058	0.2058	0.2058
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.896e-003	6.875e-003	6.875e-003	6.875e-003
54	Static Gradient (kPa/m)	-8.423e-003	-3.088e-003	-3.088e-003	-3.088e-003
55	Accel. Gradient (kPa/m)	-1.103e-007	2.735e-007	2.735e-007	2.735e-007
56	Liq. Reynolds	2.086e+005 *	2.086e+005 *	2.086e+005 *	2.086e+005 *
57	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
58	Liquid Velocity (m/s)	0.2045	0.2045	0.2045	0.2045
59	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	2.912e+004	2.912e+004	2.913e+004	2.913e+004
63	Elevation (m)	-0.3120	-0.3200	-0.2840	-0.2480
64	Pressure (bar)	88.56	88.56	88.56	88.56
65	Temperature (C)	49.97	49.97	49.97	49.96
66	Vapour Fraction	0.8690	0.8690	0.8690	0.8690
67	Heat Transferred (kJ/h-m)	532.0	531.9	531.9	531.8
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 465 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Pipe Segment: ELEP-Akabel Pipeline (continued)		
3					
4					
5			Unit Set: Mixed		
6			Date/Time: Mon Jun 13 15:02:44 2022		
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2058	0.2058	0.3025	0.3025
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.875e-003	6.874e-003	5.876e-003	5.876e-003
17	Static Gradient (kPa/m)	-3.088e-003	-3.088e-003	1.804e-002	1.804e-002
18	Accel. Gradient (kPa/m)	2.735e-007	2.735e-007	1.727e-006	1.727e-006
19	Liq. Reynolds	2.086e+005 *	2.086e+005 *	2.086e+005 *	2.086e+005 *
20	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
21	Liquid Velocity (m/s)	0.2045	0.2046	0.2046	0.2046
22	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	2.914e+004	2.914e+004	2.915e+004	2.915e+004
26	Elevation (m)	-0.2120	-0.1760	-0.1400	-0.1280
27	Pressure (bar)	88.56	88.56	88.56	88.55
28	Temperature (C)	49.96	49.96	49.96	49.96
29	Vapour Fraction	0.8690	0.8690	0.8690	0.8690
30	Heat Transferred (kJ/h-m)	531.8	531.7	531.7	531.6
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.3025	0.3025	0.3025	0.2892
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	5.876e-003	5.876e-003	5.876e-003	4.913e-003
36	Static Gradient (kPa/m)	1.804e-002	1.804e-002	1.804e-002	7.278e-003
37	Accel. Gradient (kPa/m)	1.727e-006	1.728e-006	1.728e-006	8.806e-007
38	Liq. Reynolds	2.086e+005 *	2.086e+005 *	2.086e+005 *	2.086e+005 *
39	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
40	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
41	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	2.916e+004	2.916e+004	2.916e+004	2.917e+004
45	Elevation (m)	-0.1160	-0.1040	-9.200e-002	-8.000e-002
46	Pressure (bar)	88.55	88.55	88.55	88.55
47	Temperature (C)	49.95	49.95	49.95	49.95
48	Vapour Fraction	0.8690	0.8690	0.8690	0.8690
49	Heat Transferred (kJ/h-m)	531.6	531.6	531.5	531.5
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2892	0.2892	0.2892	0.2892
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	4.913e-003	4.913e-003	4.913e-003	4.913e-003
55	Static Gradient (kPa/m)	7.278e-003	7.278e-003	7.278e-003	7.278e-003
56	Accel. Gradient (kPa/m)	8.806e-007	8.806e-007	8.806e-007	8.806e-007
57	Liq. Reynolds	2.086e+005 *	2.086e+005 *	2.086e+005 *	2.086e+005 *
58	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
59	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
60	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	2.917e+004	2.918e+004	2.918e+004	2.919e+004
64	Elevation (m)	-6.000e-002	-4.000e-002	-2.000e-002	-5.551e-016
65	Pressure (bar)	88.55	88.55	88.55	88.55
66	Temperature (C)	49.95	49.95	49.94	49.94
67	Vapour Fraction	0.8690	0.8690	0.8690	0.8690
68	Heat Transferred (kJ/h-m)	531.4	531.4	531.3	531.3
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 466 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2912	0.2912	0.2912	0.2912
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	4.769e-003	4.769e-003	4.769e-003	4.769e-003
18	Static Gradient (kPa/m)	8.866e-003	8.866e-003	8.866e-003	8.866e-003
19	Accel. Gradient (kPa/m)	9.849e-007	9.849e-007	9.849e-007	9.849e-007
20	Liq. Reynolds	2.086e+005 *	2.085e+005 *	2.085e+005 *	2.085e+005 *
21	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
22	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
23	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	2.920e+004	2.920e+004	2.921e+004	2.921e+004
27	Elevation (m)	2.000e-002	1.600e-002	1.200e-002	8.000e-003
28	Pressure (bar)	88.55	88.55	88.55	88.55
29	Temperature (C)	49.94	49.94	49.94	49.94
30	Vapour Fraction	0.8690	0.8690	0.8690	0.8690
31	Heat Transferred (kJ/h-m)	531.2	531.2	531.2	531.1
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2912	0.2066	0.2066	0.2066
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	4.769e-003	6.868e-003	6.868e-003	6.868e-003
37	Static Gradient (kPa/m)	8.866e-003	-1.407e-003	-1.407e-003	-1.407e-003
38	Accel. Gradient (kPa/m)	9.849e-007	3.945e-007	3.945e-007	3.945e-007
39	Liq. Reynolds	2.085e+005 *	2.085e+005 *	2.085e+005 *	2.085e+005 *
40	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
41	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
42	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	2.922e+004	2.922e+004	2.923e+004	2.923e+004
46	Elevation (m)	4.000e-003	-5.655e-016	6.000e-003	1.200e-002
47	Pressure (bar)	88.55	88.55	88.55	88.55
48	Temperature (C)	49.93	49.93	49.93	49.93
49	Vapour Fraction	0.8690	0.8690	0.8690	0.8690
50	Heat Transferred (kJ/h-m)	531.1	531.0	531.0	530.9
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2066	0.2066	0.2831	0.2831
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	6.868e-003	6.868e-003	5.364e-003	5.364e-003
56	Static Gradient (kPa/m)	-1.407e-003	-1.407e-003	2.628e-003	2.628e-003
57	Accel. Gradient (kPa/m)	3.945e-007	3.945e-007	5.773e-007	5.773e-007
58	Liq. Reynolds	2.085e+005 *	2.085e+005 *	2.085e+005 *	2.085e+005 *
59	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
60	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
61	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	2.924e+004	2.924e+004	2.925e+004	2.925e+004
65	Elevation (m)	1.800e-002	2.400e-002	3.000e-002	3.800e-002
66	Pressure (bar)	88.55	88.55	88.55	88.55
67	Temperature (C)	49.93	49.93	49.92	49.92
68	Vapour Fraction	0.8690	0.8690	0.8690	0.8690
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 467 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	530.9	530.8	530.8	530.8
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.2831	0.2831	0.2831	0.2924
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	5.364e-003	5.364e-003	5.364e-003	4.687e-003
19	Static Gradient (kPa/m)	2.628e-003	2.628e-003	2.629e-003	9.781e-003
20	Accel. Gradient (kPa/m)	5.773e-007	5.773e-007	5.773e-007	1.045e-006
21	Liq. Reynolds	2.085e+005 *	2.085e+005 *	2.085e+005 *	2.085e+005 *
22	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
23	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
24	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	2.925e+004	2.926e+004	2.926e+004	2.926e+004
28	Elevation (m)	4.600e-002	5.400e-002	6.200e-002	7.000e-002
29	Pressure (bar)	88.54	88.54	88.54	88.54
30	Temperature (C)	49.92	49.92	49.92	49.92
31	Vapour Fraction	0.8690	0.8690	0.8690	0.8690
32	Heat Transferred (kJ/h-m)	530.7	530.7	530.7	530.7
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.2924	0.2924	0.2924	0.2924
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	4.687e-003	4.687e-003	4.687e-003	4.687e-003
38	Static Gradient (kPa/m)	9.781e-003	9.781e-003	9.781e-003	9.781e-003
39	Accel. Gradient (kPa/m)	1.045e-006	1.045e-006	1.045e-006	1.045e-006
40	Liq. Reynolds	2.085e+005 *	2.085e+005 *	2.085e+005 *	2.085e+005 *
41	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
42	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
43	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	2.927e+004	2.928e+004	2.929e+004	2.930e+004
47	Elevation (m)	7.800e-002	8.600e-002	9.400e-002	0.1020
48	Pressure (bar)	88.54	88.54	88.54	88.54
49	Temperature (C)	49.92	49.91	49.91	49.91
50	Vapour Fraction	0.8690	0.8689	0.8689	0.8689
51	Heat Transferred (kJ/h-m)	530.6	530.5	530.4	530.3
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.2823	0.2823	0.2823	0.2823
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	5.427e-003	5.427e-003	5.427e-003	5.427e-003
57	Static Gradient (kPa/m)	2.009e-003	2.009e-003	2.009e-003	2.009e-003
58	Accel. Gradient (kPa/m)	5.371e-007	5.371e-007	5.371e-007	5.371e-007
59	Liq. Reynolds	2.085e+005 *	2.085e+005 *	2.085e+005 *	2.085e+005 *
60	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
61	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
62	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	2.931e+004	2.932e+004	2.933e+004	2.934e+004
66	Elevation (m)	0.1100	0.1040	9.800e-002	9.200e-002
67	Pressure (bar)	88.54	88.54	88.54	88.54
68	Temperature (C)	49.90	49.90	49.90	49.90
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8689	0.8689	0.8689	0.8689
14	Heat Transferred (kJ/h-m)	530.3	530.2	530.1	530.0
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2823	0.2067	0.2067	0.2067
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	5.427e-003	6.867e-003	6.867e-003	6.867e-003
20	Static Gradient (kPa/m)	2.009e-003	-1.222e-003	-1.222e-003	-1.222e-003
21	Accel. Gradient (kPa/m)	5.371e-007	4.077e-007	4.077e-007	4.077e-007
22	Liq. Reynolds	2.085e+005 *	2.085e+005 *	2.085e+005 *	2.085e+005 *
23	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
24	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
25	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	2.935e+004	2.936e+004	2.936e+004	2.937e+004
29	Elevation (m)	8.600e-002	8.000e-002	8.200e-002	8.400e-002
30	Pressure (bar)	88.54	88.54	88.54	88.54
31	Temperature (C)	49.89	49.89	49.89	49.88
32	Vapour Fraction	0.8689	0.8689	0.8689	0.8689
33	Heat Transferred (kJ/h-m)	530.0	529.9	529.8	529.7
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2067	0.2067	0.2803	0.2803
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	6.867e-003	6.867e-003	5.584e-003	5.584e-003
39	Static Gradient (kPa/m)	-1.222e-003	-1.222e-003	4.997e-004	4.997e-004
40	Accel. Gradient (kPa/m)	4.077e-007	4.077e-007	4.395e-007	4.395e-007
41	Liq. Reynolds	2.085e+005 *	2.085e+005 *	2.085e+005 *	2.085e+005 *
42	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
43	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
44	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	2.938e+004	2.939e+004	2.940e+004	2.941e+004
48	Elevation (m)	8.600e-002	8.800e-002	9.000e-002	8.800e-002
49	Pressure (bar)	88.54	88.54	88.53	88.53
50	Temperature (C)	49.88	49.88	49.87	49.87
51	Vapour Fraction	0.8689	0.8689	0.8689	0.8689
52	Heat Transferred (kJ/h-m)	529.6	529.5	529.5	529.4
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2803	0.2803	0.2803	0.2071
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	5.584e-003	5.584e-003	5.584e-003	6.863e-003
58	Static Gradient (kPa/m)	4.997e-004	4.997e-004	4.997e-004	-4.080e-004
59	Accel. Gradient (kPa/m)	4.395e-007	4.395e-007	4.395e-007	4.663e-007
60	Liq. Reynolds	2.085e+005 *	2.085e+005 *	2.085e+005 *	2.085e+005 *
61	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
62	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
63	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	2.942e+004	2.943e+004	2.944e+004	2.945e+004
67	Elevation (m)	8.600e-002	8.400e-002	8.200e-002	8.000e-002
68	Pressure (bar)	88.53	88.53	88.53	88.53
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Temperature (C)	49.87	49.87	49.86	49.86
14	Vapour Fraction	0.8689	0.8689	0.8689	0.8689
15	Heat Transferred (kJ/h-m)	529.3	529.2	529.1	529.1
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2071	0.2071	0.2071	0.2071
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	6.863e-003	6.863e-003	6.863e-003	6.863e-003
21	Static Gradient (kPa/m)	-4.080e-004	-4.080e-004	-4.080e-004	-4.080e-004
22	Accel. Gradient (kPa/m)	4.663e-007	4.663e-007	4.663e-007	4.663e-007
23	Liq. Reynolds	2.085e+005 *	2.085e+005 *	2.085e+005 *	2.085e+005 *
24	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
25	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
26	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	2.946e+004	2.947e+004	2.948e+004	2.949e+004
30	Elevation (m)	6.800e-002	5.600e-002	4.400e-002	3.200e-002
31	Pressure (bar)	88.53	88.53	88.53	88.53
32	Temperature (C)	49.86	49.85	49.85	49.85
33	Vapour Fraction	0.8689	0.8689	0.8689	0.8689
34	Heat Transferred (kJ/h-m)	529.0	528.9	528.8	528.8
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2062	0.2062	0.2062	0.2062
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	6.871e-003	6.871e-003	6.871e-003	6.871e-003
40	Static Gradient (kPa/m)	-2.441e-003	-2.441e-003	-2.441e-003	-2.441e-003
41	Accel. Gradient (kPa/m)	3.200e-007	3.200e-007	3.200e-007	3.200e-007
42	Liq. Reynolds	2.085e+005 *	2.085e+005 *	2.085e+005 *	2.085e+005 *
43	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
44	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
45	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	2.950e+004	2.951e+004	2.952e+004	2.953e+004
49	Elevation (m)	2.000e-002	2.000e-002	2.000e-002	2.000e-002
50	Pressure (bar)	88.53	88.53	88.53	88.53
51	Temperature (C)	49.84	49.84	49.84	49.84
52	Vapour Fraction	0.8689	0.8689	0.8689	0.8689
53	Heat Transferred (kJ/h-m)	528.7	528.6	528.5	528.4
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2062	0.3027	0.3027	0.3027
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	6.871e-003	5.878e-003	5.878e-003	5.878e-003
59	Static Gradient (kPa/m)	-2.441e-003	0.0000	0.0000	0.0000
60	Accel. Gradient (kPa/m)	3.200e-007	4.245e-007	4.245e-007	4.245e-007
61	Liq. Reynolds	2.084e+005 *	2.084e+005 *	2.084e+005 *	2.084e+005 *
62	Vap. Reynolds	9.177e+006 *	9.177e+006 *	9.178e+006 *	9.178e+006 *
63	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
64	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	2.954e+004	2.955e+004	2.955e+004	2.955e+004
68	Elevation (m)	2.000e-002	2.000e-002	2.200e-002	2.400e-002
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	88.53	88.53	88.53	88.53		
14	Temperature (C)	49.83	49.83	49.83	49.83		
15	Vapour Fraction	0.8689	0.8689	0.8689	0.8689		
16	Heat Transferred (kJ/h-m)	528.3	528.3	528.3	528.2		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.3027	0.3027	0.2849	0.2849		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	5.878e-003	5.878e-003	5.234e-003	5.234e-003		
22	Static Gradient (kPa/m)	0.0000	0.0000	3.904e-003	3.904e-003		
23	Accel. Gradient (kPa/m)	4.245e-007	4.245e-007	6.600e-007	6.600e-007		
24	Liq. Reynolds	2.084e+005 *	2.084e+005 *	2.084e+005 *	2.084e+005 *		
25	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *		
26	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
27	Vapour Velocity (m/s)	2.169	2.168	2.168	2.168		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	2.955e+004	2.955e+004	2.955e+004	2.956e+004		
31	Elevation (m)	2.600e-002	2.800e-002	3.000e-002	3.200e-002		
32	Pressure (bar)	88.53	88.53	88.53	88.53		
33	Temperature (C)	49.83	49.83	49.83	49.83		
34	Vapour Fraction	0.8689	0.8689	0.8689	0.8689		
35	Heat Transferred (kJ/h-m)	528.2	528.2	528.2	528.2		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.2849	0.2849	0.2849	0.2810		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	5.234e-003	5.234e-003	5.234e-003	5.535e-003		
41	Static Gradient (kPa/m)	3.904e-003	3.904e-003	3.904e-003	9.510e-004		
42	Accel. Gradient (kPa/m)	6.600e-007	6.600e-007	6.600e-007	4.685e-007		
43	Liq. Reynolds	2.084e+005 *	2.084e+005 *	2.084e+005 *	2.084e+005 *		
44	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *		
45	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
46	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	2.956e+004	2.957e+004	2.957e+004	2.958e+004		
50	Elevation (m)	3.400e-002	3.600e-002	3.800e-002	4.000e-002		
51	Pressure (bar)	88.53	88.53	88.53	88.52		
52	Temperature (C)	49.82	49.82	49.82	49.82		
53	Vapour Fraction	0.8689	0.8689	0.8689	0.8689		
54	Heat Transferred (kJ/h-m)	528.1	528.1	528.0	528.0		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.2810	0.2810	0.2810	0.2810		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	5.535e-003	5.535e-003	5.535e-003	5.535e-003		
60	Static Gradient (kPa/m)	9.510e-004	9.510e-004	9.510e-004	9.510e-004		
61	Accel. Gradient (kPa/m)	4.685e-007	4.685e-007	4.685e-007	4.685e-007		
62	Liq. Reynolds	2.084e+005 *	2.084e+005 *	2.084e+005 *	2.084e+005 *		
63	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *		
64	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
65	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	2.958e+004	2.959e+004	2.959e+004	2.960e+004		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Elevation (m)	4.400e-002	4.800e-002	5.200e-002	5.600e-002		
14	Pressure (bar)	88.52	88.52	88.52	88.52		
15	Temperature (C)	49.82	49.82	49.81	49.81		
16	Vapour Fraction	0.8689	0.8689	0.8689	0.8689		
17	Heat Transferred (kJ/h-m)	528.0	527.9	527.9	527.8		
18	Flow Regime	Transition	Transition	Transition	Transition		
19	Water Flow Regime						
20	Liquid HoldUp	0.2822	0.2822	0.2822	0.2822		
21	Water HoldUp	---	---	---	---		
22	Friction Gradient (kPa/m)	5.436e-003	5.436e-003	5.436e-003	5.436e-003		
23	Static Gradient (kPa/m)	1.908e-003	1.908e-003	1.908e-003	1.908e-003		
24	Accel. Gradient (kPa/m)	5.304e-007	5.304e-007	5.304e-007	5.304e-007		
25	Liq. Reynolds	2.084e+005 *	2.084e+005 *	2.084e+005 *	2.084e+005 *		
26	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *		
27	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
28	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
29	Deposit Thickness (mm)	---	---	---	---		
30	Deposit Volume (m3)	---	---	---	---		
31	Length (m)	2.960e+004	2.961e+004	2.961e+004	2.962e+004		
32	Elevation (m)	6.000e-002	5.600e-002	5.200e-002	4.800e-002		
33	Pressure (bar)	88.52	88.52	88.52	88.52		
34	Temperature (C)	49.81	49.81	49.81	49.81		
35	Vapour Fraction	0.8689	0.8689	0.8689	0.8689		
36	Heat Transferred (kJ/h-m)	527.8	527.7	527.7	527.7		
37	Flow Regime	Transition	Transition	Transition	Transition		
38	Water Flow Regime						
39	Liquid HoldUp	0.2822	0.2066	0.2066	0.2066		
40	Water HoldUp	---	---	---	---		
41	Friction Gradient (kPa/m)	5.436e-003	6.867e-003	6.866e-003	6.866e-003		
42	Static Gradient (kPa/m)	1.908e-003	-1.548e-003	-1.548e-003	-1.548e-003		
43	Accel. Gradient (kPa/m)	5.304e-007	3.841e-007	3.841e-007	3.841e-007		
44	Liq. Reynolds	2.084e+005 *	2.084e+005 *	2.084e+005 *	2.084e+005 *		
45	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *		
46	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
47	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
48	Deposit Thickness (mm)	---	---	---	---		
49	Deposit Volume (m3)	---	---	---	---		
50	Length (m)	2.962e+004	2.963e+004	2.963e+004	2.964e+004		
51	Elevation (m)	4.400e-002	4.000e-002	3.800e-002	3.600e-002		
52	Pressure (bar)	88.52	88.52	88.52	88.52		
53	Temperature (C)	49.81	49.80	49.80	49.80		
54	Vapour Fraction	0.8689	0.8689	0.8689	0.8689		
55	Heat Transferred (kJ/h-m)	527.6	527.6	527.5	527.5		
56	Flow Regime	Transition	Transition	Transition	Transition		
57	Water Flow Regime						
58	Liquid HoldUp	0.2066	0.2066	0.2070	0.2070		
59	Water HoldUp	---	---	---	---		
60	Friction Gradient (kPa/m)	6.866e-003	6.866e-003	6.863e-003	6.863e-003		
61	Static Gradient (kPa/m)	-1.548e-003	-1.548e-003	-7.749e-004	-7.749e-004		
62	Accel. Gradient (kPa/m)	3.841e-007	3.841e-007	4.398e-007	4.398e-007		
63	Liq. Reynolds	2.084e+005 *	2.084e+005 *	2.084e+005 *	2.084e+005 *		
64	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *		
65	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
66	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
67	Deposit Thickness (mm)	---	---	---	---		
68	Deposit Volume (m3)	---	---	---	---		
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 472 of 937		

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length (m)	2.964e+004	2.965e+004	2.965e+004	2.965e+004		
14	Elevation (m)	3.400e-002	3.200e-002	3.000e-002	2.400e-002		
15	Pressure (bar)	88.52	88.52	88.52	88.52		
16	Temperature (C)	49.80	49.80	49.80	49.80		
17	Vapour Fraction	0.8689	0.8689	0.8689	0.8689		
18	Heat Transferred (kJ/h-m)	527.5	527.4	527.4	527.4		
19	Flow Regime	Transition	Transition	Transition	Transition		
20	Water Flow Regime						
21	Liquid HoldUp	0.2070	0.2070	0.2070	0.2046		
22	Water HoldUp	---	---	---	---		
23	Friction Gradient (kPa/m)	6.863e-003	6.863e-003	6.863e-003	6.883e-003		
24	Static Gradient (kPa/m)	-7.749e-004	-7.749e-004	-7.749e-004	-5.769e-003		
25	Accel. Gradient (kPa/m)	4.398e-007	4.397e-007	4.397e-007	8.045e-008		
26	Liq. Reynolds	2.084e+005 *	2.084e+005 *	2.084e+005 *	2.084e+005 *		
27	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *		
28	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
29	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
30	Deposit Thickness (mm)	---	---	---	---		
31	Deposit Volume (m3)	---	---	---	---		
32	Length (m)	2.966e+004	2.966e+004	2.966e+004	2.966e+004		
33	Elevation (m)	1.800e-002	1.200e-002	6.000e-003	-5.621e-016		
34	Pressure (bar)	88.52	88.52	88.52	88.52		
35	Temperature (C)	49.79	49.79	49.79	49.79		
36	Vapour Fraction	0.8689	0.8689	0.8689	0.8689		
37	Heat Transferred (kJ/h-m)	527.3	527.3	527.3	527.3		
38	Flow Regime	Transition	Transition	Transition	Transition		
39	Water Flow Regime						
40	Liquid HoldUp	0.2046	0.2046	0.2046	0.2046		
41	Water HoldUp	---	---	---	---		
42	Friction Gradient (kPa/m)	6.883e-003	6.883e-003	6.883e-003	6.883e-003		
43	Static Gradient (kPa/m)	-5.769e-003	-5.769e-003	-5.769e-003	-5.769e-003		
44	Accel. Gradient (kPa/m)	8.045e-008	8.044e-008	8.044e-008	8.044e-008		
45	Liq. Reynolds	2.084e+005 *	2.084e+005 *	2.084e+005 *	2.084e+005 *		
46	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *		
47	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
48	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
49	Deposit Thickness (mm)	---	---	---	---		
50	Deposit Volume (m3)	---	---	---	---		
51	Length (m)	2.966e+004	2.967e+004	2.967e+004	2.967e+004		
52	Elevation (m)	-6.000e-003	-1.200e-002	-1.800e-002	-2.400e-002		
53	Pressure (bar)	88.52	88.52	88.52	88.52		
54	Temperature (C)	49.79	49.79	49.79	49.79		
55	Vapour Fraction	0.8689	0.8689	0.8689	0.8689		
56	Heat Transferred (kJ/h-m)	527.3	527.3	527.2	527.2		
57	Flow Regime	Transition	Transition	Transition	Transition		
58	Water Flow Regime						
59	Liquid HoldUp	0.2046	0.2046	0.2046	0.2046		
60	Water HoldUp	---	---	---	---		
61	Friction Gradient (kPa/m)	6.882e-003	6.882e-003	6.882e-003	6.882e-003		
62	Static Gradient (kPa/m)	-5.769e-003	-5.769e-003	-5.769e-003	-5.769e-003		
63	Accel. Gradient (kPa/m)	8.044e-008	8.043e-008	8.043e-008	8.043e-008		
64	Liq. Reynolds	2.084e+005 *	2.084e+005 *	2.084e+005 *	2.084e+005 *		
65	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *		
66	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
67	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
68	Deposit Thickness (mm)	---	---	---	---		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	2.967e+004	2.968e+004	2.968e+004	2.968e+004		
15	Elevation (m)	-3.000e-002	-3.200e-002	-3.400e-002	-3.600e-002		
16	Pressure (bar)	88.52	88.52	88.52	88.52		
17	Temperature (C)	49.79	49.79	49.79	49.79		
18	Vapour Fraction	0.8689	0.8689	0.8689	0.8689		
19	Heat Transferred (kJ/h-m)	527.2	527.2	527.1	527.1		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	0.2046	0.2069	0.2069	0.2069		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	6.882e-003	6.864e-003	6.864e-003	6.864e-003		
25	Static Gradient (kPa/m)	-5.769e-003	-9.684e-004	-9.684e-004	-9.684e-004		
26	Accel. Gradient (kPa/m)	8.043e-008	4.258e-007	4.258e-007	4.258e-007		
27	Liq. Reynolds	2.084e+005 *	2.084e+005 *	2.084e+005 *	2.084e+005 *		
28	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *		
29	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
30	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	2.969e+004	2.969e+004	2.969e+004	2.970e+004		
34	Elevation (m)	-3.800e-002	-4.000e-002	-3.800e-002	-3.600e-002		
35	Pressure (bar)	88.52	88.52	88.52	88.52		
36	Temperature (C)	49.78	49.78	49.78	49.78		
37	Vapour Fraction	0.8689	0.8689	0.8689	0.8689		
38	Heat Transferred (kJ/h-m)	527.1	527.0	527.0	527.0		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.2069	0.2069	0.2818	0.2818		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	6.864e-003	6.864e-003	5.468e-003	5.468e-003		
44	Static Gradient (kPa/m)	-9.684e-004	-9.684e-004	1.589e-003	1.589e-003		
45	Accel. Gradient (kPa/m)	4.258e-007	4.258e-007	5.096e-007	5.096e-007		
46	Liq. Reynolds	2.084e+005 *	2.084e+005 *	2.084e+005 *	2.084e+005 *		
47	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *		
48	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
49	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	2.970e+004	2.970e+004	2.971e+004	2.971e+004		
53	Elevation (m)	-3.400e-002	-3.200e-002	-3.000e-002	-3.000e-002		
54	Pressure (bar)	88.52	88.52	88.52	88.52		
55	Temperature (C)	49.78	49.78	49.78	49.78		
56	Vapour Fraction	0.8689	0.8689	0.8689	0.8689		
57	Heat Transferred (kJ/h-m)	527.0	526.9	526.9	526.9		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.2818	0.2818	0.2818	0.3027		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	5.468e-003	5.468e-003	5.468e-003	5.877e-003		
63	Static Gradient (kPa/m)	1.589e-003	1.589e-003	1.589e-003	0.0000		
64	Accel. Gradient (kPa/m)	5.096e-007	5.096e-007	5.096e-007	4.245e-007		
65	Liq. Reynolds	2.084e+005 *	2.084e+005 *	2.084e+005 *	2.084e+005 *		
66	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *		
67	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
68	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
4							Date/Time: Mon Jun 13 15:02:44 2022			
5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10	Pipe Table									
11										
12										
13	Deposit Thickness (mm)	---	---	---	---					
14	Deposit Volume (m3)	---	---	---	---					
15	Length (m)	2.971e+004	2.972e+004	2.972e+004	2.973e+004					
16	Elevation (m)	-3.000e-002	-3.000e-002	-3.000e-002	-3.000e-002					
17	Pressure (bar)	88.52	88.52	88.52	88.52					
18	Temperature (C)	49.78	49.77	49.77	49.77					
19	Vapour Fraction	0.8689	0.8689	0.8689	0.8689					
20	Heat Transferred (kJ/h-m)	526.8	526.8	526.8	526.7					
21	Flow Regime	Transition	Transition	Transition	Transition					
22	Water Flow Regime									
23	Liquid HoldUp	0.3027	0.3027	0.3027	0.3027					
24	Water HoldUp	---	---	---	---					
25	Friction Gradient (kPa/m)	5.877e-003	5.877e-003	5.877e-003	5.877e-003					
26	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000					
27	Accel. Gradient (kPa/m)	4.245e-007	4.245e-007	4.245e-007	4.244e-007					
28	Liq. Reynolds	2.084e+005 *	2.084e+005 *	2.084e+005 *	2.084e+005 *					
29	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *					
30	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046					
31	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168					
32	Deposit Thickness (mm)	---	---	---	---					
33	Deposit Volume (m3)	---	---	---	---					
34	Length (m)	2.973e+004	2.974e+004	2.974e+004	2.975e+004					
35	Elevation (m)	-3.200e-002	-3.400e-002	-3.600e-002	-3.800e-002					
36	Pressure (bar)	88.52	88.52	88.52	88.52					
37	Temperature (C)	49.77	49.77	49.77	49.77					
38	Vapour Fraction	0.8689	0.8689	0.8689	0.8689					
39	Heat Transferred (kJ/h-m)	526.7	526.7	526.6	526.6					
40	Flow Regime	Transition	Transition	Transition	Transition					
41	Water Flow Regime									
42	Liquid HoldUp	0.2070	0.2070	0.2070	0.2070					
43	Water HoldUp	---	---	---	---					
44	Friction Gradient (kPa/m)	6.863e-003	6.863e-003	6.863e-003	6.863e-003					
45	Static Gradient (kPa/m)	-7.759e-004	-7.759e-004	-7.759e-004	-7.759e-004					
46	Accel. Gradient (kPa/m)	4.396e-007	4.396e-007	4.396e-007	4.396e-007					
47	Liq. Reynolds	2.084e+005 *	2.084e+005 *	2.084e+005 *	2.084e+005 *					
48	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *					
49	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046					
50	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168					
51	Deposit Thickness (mm)	---	---	---	---					
52	Deposit Volume (m3)	---	---	---	---					
53	Length (m)	2.975e+004	2.976e+004	2.976e+004	2.977e+004					
54	Elevation (m)	-4.000e-002	-4.200e-002	-4.400e-002	-4.600e-002					
55	Pressure (bar)	88.51	88.51	88.51	88.51					
56	Temperature (C)	49.76	49.76	49.76	49.76					
57	Vapour Fraction	0.8689	0.8689	0.8689	0.8689					
58	Heat Transferred (kJ/h-m)	526.5	526.5	526.5	526.4					
59	Flow Regime	Transition	Transition	Transition	Transition					
60	Water Flow Regime									
61	Liquid HoldUp	0.2070	0.2070	0.2070	0.2070					
62	Water HoldUp	---	---	---	---					
63	Friction Gradient (kPa/m)	6.863e-003	6.863e-003	6.863e-003	6.863e-003					
64	Static Gradient (kPa/m)	-7.759e-004	-7.750e-004	-7.750e-004	-7.750e-004					
65	Accel. Gradient (kPa/m)	4.396e-007	4.397e-007	4.397e-007	4.397e-007					
66	Liq. Reynolds	2.084e+005 *	2.084e+005 *	2.084e+005 *	2.084e+005 *					
67	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *					
68	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046					
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168	
14	Deposit Thickness	(mm)	---	---	---	---	
15	Deposit Volume	(m3)	---	---	---	---	
16	Length	(m)	2.977e+004	2.978e+004	2.978e+004	2.979e+004	
17	Elevation	(m)	-4.800e-002	-5.000e-002	-4.600e-002	-4.200e-002	
18	Pressure	(bar)	88.51	88.51	88.51	88.51	
19	Temperature	(C)	49.76	49.76	49.75	49.75	
20	Vapour Fraction		0.8689	0.8689	0.8689	0.8689	
21	Heat Transferred	(kJ/h-m)	526.4	526.3	526.3	526.2	
22	Flow Regime		Transition	Transition	Transition	Transition	
23	Water Flow Regime						
24	Liquid HoldUp		0.2070	0.2070	0.2823	0.2823	
25	Water HoldUp		---	---	---	---	
26	Friction Gradient	(kPa/m)	6.863e-003	6.863e-003	5.434e-003	5.434e-003	
27	Static Gradient	(kPa/m)	-7.750e-004	-7.750e-004	1.908e-003	1.908e-003	
28	Accel. Gradient	(kPa/m)	4.397e-007	4.397e-007	5.303e-007	5.303e-007	
29	Liq. Reynolds		2.084e+005 *	2.084e+005 *	2.084e+005 *	2.084e+005 *	
30	Vap. Reynolds		9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *	
31	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
32	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168	
33	Deposit Thickness	(mm)	---	---	---	---	
34	Deposit Volume	(m3)	---	---	---	---	
35	Length	(m)	2.979e+004	2.980e+004	2.980e+004	2.981e+004	
36	Elevation	(m)	-3.800e-002	-3.400e-002	-3.000e-002	-2.200e-002	
37	Pressure	(bar)	88.51	88.51	88.51	88.51	
38	Temperature	(C)	49.75	49.75	49.75	49.75	
39	Vapour Fraction		0.8689	0.8689	0.8689	0.8689	
40	Heat Transferred	(kJ/h-m)	526.2	526.1	526.1	526.1	
41	Flow Regime		Transition	Transition	Transition	Transition	
42	Water Flow Regime						
43	Liquid HoldUp		0.2823	0.2823	0.2823	0.2848	
44	Water HoldUp		---	---	---	---	
45	Friction Gradient	(kPa/m)	5.434e-003	5.434e-003	5.434e-003	5.239e-003	
46	Static Gradient	(kPa/m)	1.908e-003	1.908e-003	1.908e-003	3.841e-003	
47	Accel. Gradient	(kPa/m)	5.303e-007	5.303e-007	5.303e-007	6.558e-007	
48	Liq. Reynolds		2.084e+005 *	2.084e+005 *	2.084e+005 *	2.084e+005 *	
49	Vap. Reynolds		9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *	
50	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
51	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168	
52	Deposit Thickness	(mm)	---	---	---	---	
53	Deposit Volume	(m3)	---	---	---	---	
54	Length	(m)	2.981e+004	2.982e+004	2.982e+004	2.983e+004	
55	Elevation	(m)	-1.400e-002	-6.000e-003	2.000e-003	1.000e-002	
56	Pressure	(bar)	88.51	88.51	88.51	88.51	
57	Temperature	(C)	49.75	49.74	49.74	49.74	
58	Vapour Fraction		0.8689	0.8689	0.8689	0.8688	
59	Heat Transferred	(kJ/h-m)	526.0	526.0	525.9	525.9	
60	Flow Regime		Transition	Transition	Transition	Transition	
61	Water Flow Regime						
62	Liquid HoldUp		0.2848	0.2848	0.2848	0.2848	
63	Water HoldUp		---	---	---	---	
64	Friction Gradient	(kPa/m)	5.239e-003	5.239e-003	5.239e-003	5.239e-003	
65	Static Gradient	(kPa/m)	3.841e-003	3.841e-003	3.841e-003	3.841e-003	
66	Accel. Gradient	(kPa/m)	6.558e-007	6.558e-007	6.558e-007	6.558e-007	
67	Liq. Reynolds		2.084e+005 *	2.084e+005 *	2.084e+005 *	2.084e+005 *	
68	Vap. Reynolds		9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
14	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168	
15	Deposit Thickness	(mm)	---	---	---	---	
16	Deposit Volume	(m3)	---	---	---	---	
17	Length	(m)	2.983e+004	2.984e+004	2.984e+004	2.985e+004	
18	Elevation	(m)	1.200e-002	1.400e-002	1.600e-002	1.800e-002	
19	Pressure	(bar)	88.51	88.51	88.51	88.51	
20	Temperature	(C)	49.74	49.74	49.74	49.73	
21	Vapour Fraction		0.8688	0.8688	0.8688	0.8688	
22	Heat Transferred	(kJ/h-m)	525.8	525.8	525.8	525.7	
23	Flow Regime		Transition	Transition	Transition	Transition	
24	Water Flow Regime						
25	Liquid HoldUp		0.2810	0.2810	0.2810	0.2810	
26	Water HoldUp		---	---	---	---	
27	Friction Gradient	(kPa/m)	5.534e-003	5.534e-003	5.533e-003	5.533e-003	
28	Static Gradient	(kPa/m)	9.512e-004	9.512e-004	9.512e-004	9.512e-004	
29	Accel. Gradient	(kPa/m)	4.683e-007	4.683e-007	4.683e-007	4.683e-007	
30	Liq. Reynolds		2.083e+005 *	2.083e+005 *	2.083e+005 *	2.083e+005 *	
31	Vap. Reynolds		9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *	
32	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
33	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168	
34	Deposit Thickness	(mm)	---	---	---	---	
35	Deposit Volume	(m3)	---	---	---	---	
36	Length	(m)	2.985e+004	2.986e+004	2.986e+004	2.987e+004	
37	Elevation	(m)	2.000e-002	3.000e-002	4.000e-002	5.000e-002	
38	Pressure	(bar)	88.51	88.51	88.51	88.51	
39	Temperature	(C)	49.73	49.73	49.73	49.73	
40	Vapour Fraction		0.8688	0.8688	0.8688	0.8688	
41	Heat Transferred	(kJ/h-m)	525.7	525.6	525.6	525.5	
42	Flow Regime		Transition	Transition	Transition	Transition	
43	Water Flow Regime						
44	Liquid HoldUp		0.2810	0.2861	0.2861	0.2861	
45	Water HoldUp		---	---	---	---	
46	Friction Gradient	(kPa/m)	5.533e-003	5.143e-003	5.143e-003	5.143e-003	
47	Static Gradient	(kPa/m)	9.512e-004	4.817e-003	4.817e-003	4.817e-003	
48	Accel. Gradient	(kPa/m)	4.683e-007	7.193e-007	7.193e-007	7.193e-007	
49	Liq. Reynolds		2.083e+005 *	2.083e+005 *	2.083e+005 *	2.083e+005 *	
50	Vap. Reynolds		9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *	
51	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
52	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168	
53	Deposit Thickness	(mm)	---	---	---	---	
54	Deposit Volume	(m3)	---	---	---	---	
55	Length	(m)	2.987e+004	2.988e+004	2.988e+004	2.989e+004	
56	Elevation	(m)	6.000e-002	7.000e-002	6.600e-002	6.200e-002	
57	Pressure	(bar)	88.51	88.51	88.50	88.50	
58	Temperature	(C)	49.73	49.72	49.72	49.72	
59	Vapour Fraction		0.8688	0.8688	0.8688	0.8688	
60	Heat Transferred	(kJ/h-m)	525.5	525.5	525.4	525.4	
61	Flow Regime		Transition	Transition	Transition	Transition	
62	Water Flow Regime						
63	Liquid HoldUp		0.2861	0.2861	0.2066	0.2066	
64	Water HoldUp		---	---	---	---	
65	Friction Gradient	(kPa/m)	5.143e-003	5.143e-003	6.865e-003	6.865e-003	
66	Static Gradient	(kPa/m)	4.817e-003	4.817e-003	-1.548e-003	-1.548e-003	
67	Accel. Gradient	(kPa/m)	7.193e-007	7.193e-007	3.840e-007	3.840e-007	
68	Liq. Reynolds		2.083e+005 *	2.083e+005 *	2.083e+005 *	2.083e+005 *	
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *
14	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
15	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	2.989e+004	2.990e+004	2.990e+004	2.990e+004
19	Elevation (m)	5.800e-002	5.400e-002	5.000e-002	5.200e-002
20	Pressure (bar)	88.50	88.50	88.50	88.50
21	Temperature (C)	49.72	49.72	49.72	49.72
22	Vapour Fraction	0.8688	0.8688	0.8688	0.8688
23	Heat Transferred (kJ/h-m)	525.3	525.3	525.3	525.2
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2066	0.2066	0.2066	0.2829
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.865e-003	6.865e-003	6.865e-003	5.384e-003
29	Static Gradient (kPa/m)	-1.548e-003	-1.548e-003	-1.548e-003	2.390e-003
30	Accel. Gradient (kPa/m)	3.840e-007	3.840e-007	3.840e-007	5.614e-007
31	Liq. Reynolds	2.083e+005 *	2.083e+005 *	2.083e+005 *	2.083e+005 *
32	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *
33	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
34	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	2.991e+004	2.991e+004	2.991e+004	2.991e+004
38	Elevation (m)	5.400e-002	5.600e-002	5.800e-002	6.000e-002
39	Pressure (bar)	88.50	88.50	88.50	88.50
40	Temperature (C)	49.71	49.71	49.71	49.71
41	Vapour Fraction	0.8688	0.8688	0.8688	0.8688
42	Heat Transferred (kJ/h-m)	525.2	525.2	525.2	525.2
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2829	0.2829	0.2829	0.2829
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	5.384e-003	5.384e-003	5.384e-003	5.384e-003
48	Static Gradient (kPa/m)	2.390e-003	2.390e-003	2.390e-003	2.390e-003
49	Accel. Gradient (kPa/m)	5.614e-007	5.614e-007	5.614e-007	5.614e-007
50	Liq. Reynolds	2.083e+005 *	2.083e+005 *	2.083e+005 *	2.083e+005 *
51	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *
52	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
53	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	2.992e+004	2.992e+004	2.993e+004	2.993e+004
57	Elevation (m)	5.800e-002	5.600e-002	5.400e-002	5.200e-002
58	Pressure (bar)	88.50	88.50	88.50	88.50
59	Temperature (C)	49.71	49.71	49.71	49.71
60	Vapour Fraction	0.8688	0.8688	0.8688	0.8688
61	Heat Transferred (kJ/h-m)	525.1	525.1	525.0	525.0
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2070	0.2070	0.2070	0.2070
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	6.862e-003	6.862e-003	6.862e-003	6.862e-003
67	Static Gradient (kPa/m)	-7.751e-004	-7.751e-004	-7.751e-004	-7.751e-004
68	Accel. Gradient (kPa/m)	4.396e-007	4.396e-007	4.396e-007	4.396e-007
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.083e+005 *	2.083e+005 *	2.083e+005 *	2.083e+005 *
14	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *
15	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
16	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	2.994e+004	2.994e+004	2.994e+004	2.995e+004
20	Elevation (m)	5.000e-002	5.000e-002	5.000e-002	5.000e-002
21	Pressure (bar)	88.50	88.50	88.50	88.50
22	Temperature (C)	49.71	49.70	49.70	49.70
23	Vapour Fraction	0.8688	0.8688	0.8688	0.8688
24	Heat Transferred (kJ/h-m)	525.0	524.9	524.9	524.8
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2070	0.3028	0.3028	0.3028
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	6.862e-003	5.876e-003	5.876e-003	5.876e-003
30	Static Gradient (kPa/m)	-7.751e-004	0.0000	0.0000	0.0000
31	Accel. Gradient (kPa/m)	4.396e-007	4.244e-007	4.244e-007	4.244e-007
32	Liq. Reynolds	2.083e+005 *	2.083e+005 *	2.083e+005 *	2.083e+005 *
33	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *
34	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
35	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	2.995e+004	2.996e+004	2.996e+004	2.997e+004
39	Elevation (m)	5.000e-002	5.000e-002	5.400e-002	5.800e-002
40	Pressure (bar)	88.50	88.50	88.50	88.50
41	Temperature (C)	49.70	49.70	49.70	49.70
42	Vapour Fraction	0.8688	0.8688	0.8688	0.8688
43	Heat Transferred (kJ/h-m)	524.8	524.8	524.7	524.7
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.3028	0.3028	0.2823	0.2823
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	5.876e-003	5.876e-003	5.433e-003	5.433e-003
49	Static Gradient (kPa/m)	0.0000	0.0000	1.909e-003	1.909e-003
50	Accel. Gradient (kPa/m)	4.244e-007	4.244e-007	5.302e-007	5.302e-007
51	Liq. Reynolds	2.083e+005 *	2.083e+005 *	2.083e+005 *	2.083e+005 *
52	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *
53	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
54	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	2.997e+004	2.998e+004	2.998e+004	2.999e+004
58	Elevation (m)	6.200e-002	6.600e-002	7.000e-002	6.800e-002
59	Pressure (bar)	88.50	88.50	88.50	88.50
60	Temperature (C)	49.69	49.69	49.69	49.69
61	Vapour Fraction	0.8688	0.8688	0.8688	0.8688
62	Heat Transferred (kJ/h-m)	524.6	524.6	524.6	524.5
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2823	0.2823	0.2823	0.2070
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	5.433e-003	5.433e-003	5.433e-003	6.862e-003
68	Static Gradient (kPa/m)	1.909e-003	1.909e-003	1.909e-003	-7.751e-004
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	5.302e-007	5.302e-007	5.302e-007	4.396e-007
14	Liq. Reynolds	2.083e+005 *	2.083e+005 *	2.083e+005 *	2.083e+005 *
15	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *
16	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
17	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	2.999e+004	3.000e+004	3.000e+004	3.001e+004
21	Elevation (m)	6.600e-002	6.400e-002	6.200e-002	6.000e-002
22	Pressure (bar)	88.50	88.50	88.50	88.50
23	Temperature (C)	49.69	49.69	49.68	49.68
24	Vapour Fraction	0.8688	0.8688	0.8688	0.8688
25	Heat Transferred (kJ/h-m)	524.5	524.4	524.4	524.4
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.2070	0.2070	0.2070	0.2070
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	6.862e-003	6.862e-003	6.861e-003	6.861e-003
31	Static Gradient (kPa/m)	-7.751e-004	-7.751e-004	-7.751e-004	-7.751e-004
32	Accel. Gradient (kPa/m)	4.396e-007	4.396e-007	4.396e-007	4.396e-007
33	Liq. Reynolds	2.083e+005 *	2.083e+005 *	2.083e+005 *	2.083e+005 *
34	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *
35	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
36	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	3.001e+004	3.002e+004	3.002e+004	3.003e+004
40	Elevation (m)	5.800e-002	5.600e-002	5.400e-002	5.200e-002
41	Pressure (bar)	88.50	88.50	88.50	88.50
42	Temperature (C)	49.68	49.68	49.68	49.68
43	Vapour Fraction	0.8688	0.8688	0.8688	0.8688
44	Heat Transferred (kJ/h-m)	524.3	524.3	524.2	524.2
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.2070	0.2070	0.2070	0.2070
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	6.861e-003	6.861e-003	6.861e-003	6.861e-003
50	Static Gradient (kPa/m)	-7.751e-004	-7.751e-004	-7.751e-004	-7.752e-004
51	Accel. Gradient (kPa/m)	4.396e-007	4.396e-007	4.395e-007	4.395e-007
52	Liq. Reynolds	2.083e+005 *	2.083e+005 *	2.083e+005 *	2.083e+005 *
53	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *
54	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
55	Vapour Velocity (m/s)	2.168	2.168	2.167	2.167
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	3.003e+004	3.004e+004	3.004e+004	3.005e+004
59	Elevation (m)	5.000e-002	4.600e-002	4.200e-002	3.800e-002
60	Pressure (bar)	88.50	88.50	88.49	88.49
61	Temperature (C)	49.67	49.67	49.67	49.67
62	Vapour Fraction	0.8688	0.8688	0.8688	0.8688
63	Heat Transferred (kJ/h-m)	524.1	524.1	524.1	524.0
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.2070	0.2067	0.2067	0.2067
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	6.861e-003	6.864e-003	6.864e-003	6.864e-003
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1				Case Name:	Simulation - 386MMSCFD_100barg.hsc	
2				Unit Set:	Mixed	
3	 Epic Atlantic Limited			Date/Time:	Mon Jun 13 15:02:44 2022	
4						
5						
6						
7						
8						
9						
10						
11						
12						
13	Static Gradient	(kPa/m)	-7.752e-004	-1.549e-003	-1.549e-003	-1.549e-003
14	Accel. Gradient	(kPa/m)	4.395e-007	3.839e-007	3.839e-007	3.839e-007
15	Liq. Reynolds		2.083e+005 *	2.083e+005 *	2.083e+005 *	2.083e+005 *
16	Vap. Reynolds		9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *
17	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
18	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	3.005e+004	3.006e+004	3.006e+004	3.007e+004
22	Elevation	(m)	3.400e-002	3.000e-002	3.400e-002	3.800e-002
23	Pressure	(bar)	88.49	88.49	88.49	88.49
24	Temperature	(C)	49.67	49.67	49.67	49.66
25	Vapour Fraction		0.8688	0.8688	0.8688	0.8688
26	Heat Transferred	(kJ/h-m)	524.0	523.9	523.9	523.8
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2067	0.2067	0.2823	0.2823
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	6.864e-003	6.864e-003	5.433e-003	5.433e-003
32	Static Gradient	(kPa/m)	-1.549e-003	-1.549e-003	1.909e-003	1.909e-003
33	Accel. Gradient	(kPa/m)	3.839e-007	3.839e-007	5.302e-007	5.302e-007
34	Liq. Reynolds		2.083e+005 *	2.083e+005 *	2.083e+005 *	2.083e+005 *
35	Vap. Reynolds		9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *
36	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
37	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	3.007e+004	3.008e+004	3.008e+004	3.009e+004
41	Elevation	(m)	4.200e-002	4.600e-002	5.000e-002	4.600e-002
42	Pressure	(bar)	88.49	88.49	88.49	88.49
43	Temperature	(C)	49.66	49.66	49.66	49.66
44	Vapour Fraction		0.8688	0.8688	0.8688	0.8688
45	Heat Transferred	(kJ/h-m)	523.8	523.8	523.7	523.7
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2823	0.2823	0.2823	0.2067
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	5.433e-003	5.432e-003	5.432e-003	6.864e-003
51	Static Gradient	(kPa/m)	1.909e-003	1.909e-003	1.909e-003	-1.549e-003
52	Accel. Gradient	(kPa/m)	5.302e-007	5.302e-007	5.302e-007	3.839e-007
53	Liq. Reynolds		2.083e+005 *	2.083e+005 *	2.083e+005 *	2.083e+005 *
54	Vap. Reynolds		9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *
55	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
56	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	3.009e+004	3.010e+004	3.010e+004	3.011e+004
60	Elevation	(m)	4.200e-002	3.800e-002	3.400e-002	3.000e-002
61	Pressure	(bar)	88.49	88.49	88.49	88.49
62	Temperature	(C)	49.66	49.65	49.65	49.65
63	Vapour Fraction		0.8688	0.8688	0.8688	0.8688
64	Heat Transferred	(kJ/h-m)	523.6	523.6	523.6	523.5
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2067	0.2067	0.2067	0.2067
68	Water HoldUp		---	---	---	---
69	Aspen Technology Inc.			Aspen HYSYS Version 10		Page 481 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	6.864e-003	6.864e-003	6.864e-003	6.864e-003	
14	Static Gradient	(kPa/m)	-1.549e-003	-1.549e-003	-1.549e-003	-1.549e-003	
15	Accel. Gradient	(kPa/m)	3.839e-007	3.839e-007	3.839e-007	3.839e-007	
16	Liq. Reynolds		2.083e+005 *	2.083e+005 *	2.083e+005 *	2.083e+005 *	
17	Vap. Reynolds		9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *	
18	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
19	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	3.011e+004	3.012e+004	3.012e+004	3.013e+004	
23	Elevation	(m)	2.600e-002	2.200e-002	1.800e-002	1.400e-002	
24	Pressure	(bar)	88.49	88.49	88.49	88.49	
25	Temperature	(C)	49.65	49.65	49.65	49.64	
26	Vapour Fraction		0.8688	0.8688	0.8688	0.8688	
27	Heat Transferred	(kJ/h-m)	523.5	523.4	523.4	523.4	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.2067	0.2067	0.2067	0.2067	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	6.864e-003	6.864e-003	6.864e-003	6.864e-003	
33	Static Gradient	(kPa/m)	-1.549e-003	-1.549e-003	-1.549e-003	-1.549e-003	
34	Accel. Gradient	(kPa/m)	3.839e-007	3.838e-007	3.838e-007	3.838e-007	
35	Liq. Reynolds		2.083e+005 *	2.083e+005 *	2.083e+005 *	2.083e+005 *	
36	Vap. Reynolds		9.178e+006 *	9.178e+006 *	9.179e+006 *	9.179e+006 *	
37	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
38	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	3.013e+004	3.014e+004	3.014e+004	3.015e+004	
42	Elevation	(m)	1.000e-002	1.400e-002	1.800e-002	2.200e-002	
43	Pressure	(bar)	88.49	88.49	88.49	88.49	
44	Temperature	(C)	49.64	49.64	49.64	49.64	
45	Vapour Fraction		0.8688	0.8688	0.8688	0.8688	
46	Heat Transferred	(kJ/h-m)	523.3	523.3	523.2	523.2	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.2067	0.2823	0.2823	0.2823	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	6.864e-003	5.433e-003	5.433e-003	5.433e-003	
52	Static Gradient	(kPa/m)	-1.549e-003	1.901e-003	1.901e-003	1.901e-003	
53	Accel. Gradient	(kPa/m)	3.838e-007	5.297e-007	5.297e-007	5.297e-007	
54	Liq. Reynolds		2.083e+005 *	2.083e+005 *	2.083e+005 *	2.083e+005 *	
55	Vap. Reynolds		9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *	
56	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
57	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	3.015e+004	3.016e+004	3.016e+004	3.017e+004	
61	Elevation	(m)	2.600e-002	3.000e-002	3.200e-002	3.400e-002	
62	Pressure	(bar)	88.49	88.49	88.49	88.49	
63	Temperature	(C)	49.64	49.64	49.63	49.63	
64	Vapour Fraction		0.8688	0.8688	0.8688	0.8688	
65	Heat Transferred	(kJ/h-m)	523.1	523.1	523.0	523.0	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.2823	0.2823	0.2811	0.2811	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 482 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	5.433e-003	5.433e-003	5.531e-003	5.531e-003
15	Static Gradient (kPa/m)	1.901e-003	1.901e-003	9.515e-004	9.515e-004
16	Accel. Gradient (kPa/m)	5.296e-007	5.296e-007	4.682e-007	4.682e-007
17	Liq. Reynolds	2.083e+005 *	2.083e+005 *	2.082e+005 *	2.082e+005 *
18	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
19	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
20	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	3.017e+004	3.018e+004	3.018e+004	3.019e+004
24	Elevation (m)	3.600e-002	3.800e-002	4.000e-002	4.000e-002
25	Pressure (bar)	88.49	88.49	88.49	88.49
26	Temperature (C)	49.63	49.63	49.63	49.63
27	Vapour Fraction	0.8688	0.8688	0.8688	0.8688
28	Heat Transferred (kJ/h-m)	523.0	522.9	522.9	522.8
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2811	0.2811	0.2811	0.3028
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	5.531e-003	5.531e-003	5.531e-003	5.875e-003
34	Static Gradient (kPa/m)	9.515e-004	9.515e-004	9.515e-004	0.0000
35	Accel. Gradient (kPa/m)	4.682e-007	4.682e-007	4.682e-007	4.243e-007
36	Liq. Reynolds	2.082e+005 *	2.082e+005 *	2.082e+005 *	2.082e+005 *
37	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
38	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
39	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	3.019e+004	3.020e+004	3.020e+004	3.021e+004
43	Elevation (m)	4.000e-002	4.000e-002	4.000e-002	4.000e-002
44	Pressure (bar)	88.49	88.49	88.48	88.48
45	Temperature (C)	49.62	49.62	49.62	49.62
46	Vapour Fraction	0.8688	0.8688	0.8688	0.8688
47	Heat Transferred (kJ/h-m)	522.8	522.8	522.7	522.7
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.3028	0.3028	0.3028	0.3028
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	5.875e-003	5.875e-003	5.875e-003	5.875e-003
53	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
54	Accel. Gradient (kPa/m)	4.243e-007	4.243e-007	4.243e-007	4.243e-007
55	Liq. Reynolds	2.082e+005 *	2.082e+005 *	2.082e+005 *	2.082e+005 *
56	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
57	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
58	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	3.021e+004	3.022e+004	3.022e+004	3.023e+004
62	Elevation (m)	4.400e-002	4.800e-002	5.200e-002	5.600e-002
63	Pressure (bar)	88.48	88.48	88.48	88.48
64	Temperature (C)	49.62	49.62	49.61	49.61
65	Vapour Fraction	0.8688	0.8688	0.8688	0.8688
66	Heat Transferred (kJ/h-m)	522.6	522.6	522.5	522.5
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 483 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.2824	0.2824	0.2824	0.2824
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	5.432e-003	5.432e-003	5.432e-003	5.432e-003
16	Static Gradient (kPa/m)	1.909e-003	1.909e-003	1.909e-003	1.909e-003
17	Accel. Gradient (kPa/m)	5.301e-007	5.301e-007	5.301e-007	5.301e-007
18	Liq. Reynolds	2.082e+005 *	2.082e+005 *	2.082e+005 *	2.082e+005 *
19	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
20	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
21	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	3.023e+004	3.024e+004	3.024e+004	3.025e+004
25	Elevation (m)	6.000e-002	5.600e-002	5.200e-002	4.800e-002
26	Pressure (bar)	88.48	88.48	88.48	88.48
27	Temperature (C)	49.61	49.61	49.61	49.61
28	Vapour Fraction	0.8688	0.8688	0.8688	0.8688
29	Heat Transferred (kJ/h-m)	522.5	522.4	522.4	522.3
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2824	0.2067	0.2067	0.2067
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	5.431e-003	6.863e-003	6.863e-003	6.863e-003
35	Static Gradient (kPa/m)	1.909e-003	-1.549e-003	-1.549e-003	-1.549e-003
36	Accel. Gradient (kPa/m)	5.301e-007	3.838e-007	3.838e-007	3.838e-007
37	Liq. Reynolds	2.082e+005 *	2.082e+005 *	2.082e+005 *	2.082e+005 *
38	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
39	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
40	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	3.025e+004	3.026e+004	3.026e+004	3.027e+004
44	Elevation (m)	4.400e-002	4.000e-002	4.600e-002	5.200e-002
45	Pressure (bar)	88.48	88.48	88.48	88.48
46	Temperature (C)	49.61	49.60	49.60	49.60
47	Vapour Fraction	0.8688	0.8688	0.8688	0.8688
48	Heat Transferred (kJ/h-m)	522.3	522.3	522.2	522.2
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2067	0.2067	0.2836	0.2836
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	6.863e-003	6.863e-003	5.333e-003	5.333e-003
54	Static Gradient (kPa/m)	-1.549e-003	-1.549e-003	2.873e-003	2.873e-003
55	Accel. Gradient (kPa/m)	3.838e-007	3.838e-007	5.926e-007	5.926e-007
56	Liq. Reynolds	2.082e+005 *	2.082e+005 *	2.082e+005 *	2.082e+005 *
57	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
58	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
59	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	3.027e+004	3.028e+004	3.028e+004	3.029e+004
63	Elevation (m)	5.800e-002	6.400e-002	7.000e-002	7.600e-002
64	Pressure (bar)	88.48	88.48	88.48	88.48
65	Temperature (C)	49.60	49.60	49.60	49.59
66	Vapour Fraction	0.8688	0.8688	0.8688	0.8688
67	Heat Transferred (kJ/h-m)	522.1	522.1	522.0	522.0
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 484 of 937

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2836	0.2836	0.2836	0.2836
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	5.333e-003	5.333e-003	5.333e-003	5.333e-003
17	Static Gradient (kPa/m)	2.873e-003	2.873e-003	2.873e-003	2.873e-003
18	Accel. Gradient (kPa/m)	5.926e-007	5.926e-007	5.926e-007	5.926e-007
19	Liq. Reynolds	2.082e+005 *	2.082e+005 *	2.082e+005 *	2.082e+005 *
20	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
21	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
22	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	3.029e+004	3.030e+004	3.030e+004	3.031e+004
26	Elevation (m)	8.200e-002	8.800e-002	9.400e-002	1.000e-001
27	Pressure (bar)	88.48	88.48	88.48	88.48
28	Temperature (C)	49.59	49.59	49.59	49.59
29	Vapour Fraction	0.8688	0.8688	0.8688	0.8688
30	Heat Transferred (kJ/h-m)	521.9	521.9	521.9	521.8
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2836	0.2836	0.2836	0.2836
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	5.333e-003	5.333e-003	5.333e-003	5.333e-003
36	Static Gradient (kPa/m)	2.873e-003	2.873e-003	2.873e-003	2.873e-003
37	Accel. Gradient (kPa/m)	5.926e-007	5.926e-007	5.926e-007	5.926e-007
38	Liq. Reynolds	2.082e+005 *	2.082e+005 *	2.082e+005 *	2.082e+005 *
39	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
40	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
41	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	3.031e+004	3.032e+004	3.032e+004	3.033e+004
45	Elevation (m)	0.1200	0.1400	0.1600	0.1800
46	Pressure (bar)	88.48	88.48	88.48	88.47
47	Temperature (C)	49.59	49.58	49.58	49.58
48	Vapour Fraction	0.8688	0.8688	0.8688	0.8688
49	Heat Transferred (kJ/h-m)	521.8	521.7	521.7	521.6
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2926	0.2926	0.2926	0.2926
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	4.681e-003	4.681e-003	4.681e-003	4.681e-003
55	Static Gradient (kPa/m)	9.789e-003	9.789e-003	9.789e-003	9.789e-003
56	Accel. Gradient (kPa/m)	1.045e-006	1.045e-006	1.045e-006	1.045e-006
57	Liq. Reynolds	2.082e+005 *	2.082e+005 *	2.082e+005 *	2.082e+005 *
58	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
59	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
60	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	3.033e+004	3.034e+004	3.035e+004	3.036e+004
64	Elevation (m)	0.2000	0.1960	0.1920	0.1880
65	Pressure (bar)	88.47	88.47	88.47	88.47
66	Temperature (C)	49.58	49.58	49.57	49.57
67	Vapour Fraction	0.8688	0.8688	0.8688	0.8688
68	Heat Transferred (kJ/h-m)	521.6	521.5	521.5	521.4
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2926	0.2070	0.2070	0.2070
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	4.681e-003	6.861e-003	6.861e-003	6.860e-003
18	Static Gradient (kPa/m)	9.789e-003	-9.689e-004	-9.689e-004	-9.689e-004
19	Accel. Gradient (kPa/m)	1.045e-006	4.255e-007	4.255e-007	4.255e-007
20	Liq. Reynolds	2.082e+005 *	2.082e+005 *	2.082e+005 *	2.082e+005 *
21	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
22	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
23	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	3.036e+004	3.037e+004	3.039e+004	3.040e+004
27	Elevation (m)	0.1840	0.1800	0.1820	0.1840
28	Pressure (bar)	88.47	88.47	88.47	88.47
29	Temperature (C)	49.57	49.57	49.56	49.56
30	Vapour Fraction	0.8688	0.8688	0.8688	0.8687
31	Heat Transferred (kJ/h-m)	521.3	521.3	521.1	521.0
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2070	0.2070	0.2803	0.2803
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	6.860e-003	6.860e-003	5.595e-003	5.595e-003
37	Static Gradient (kPa/m)	-9.689e-004	-9.689e-004	3.392e-004	3.392e-004
38	Accel. Gradient (kPa/m)	4.255e-007	4.255e-007	4.285e-007	4.285e-007
39	Liq. Reynolds	2.082e+005 *	2.082e+005 *	2.082e+005 *	2.082e+005 *
40	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
41	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
42	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	3.041e+004	3.043e+004	3.044e+004	3.045e+004
46	Elevation (m)	0.1860	0.1880	0.1900	0.1800
47	Pressure (bar)	88.47	88.47	88.47	88.47
48	Temperature (C)	49.55	49.55	49.54	49.54
49	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
50	Heat Transferred (kJ/h-m)	520.9	520.8	520.7	520.7
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2803	0.2803	0.2803	0.2044
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	5.595e-003	5.595e-003	5.595e-003	6.881e-003
56	Static Gradient (kPa/m)	3.392e-004	3.392e-004	3.392e-004	-6.407e-003
57	Accel. Gradient (kPa/m)	4.285e-007	4.285e-007	4.285e-007	3.421e-008
58	Liq. Reynolds	2.082e+005 *	2.082e+005 *	2.082e+005 *	2.082e+005 *
59	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
60	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
61	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	3.045e+004	3.045e+004	3.045e+004	3.046e+004
65	Elevation (m)	0.1700	0.1600	0.1500	0.1400
66	Pressure (bar)	88.47	88.47	88.47	88.47
67	Temperature (C)	49.54	49.54	49.54	49.54
68	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022	
6						
7						
8						
9			PROFILES			
10						
11			Pipe Table			
12						
13	Heat Transferred (kJ/h-m)	520.6	520.6	520.6	520.6	
14	Flow Regime	Transition	Transition	Transition	Transition	
15	Water Flow Regime					
16	Liquid HoldUp	0.2044	0.2044	0.2044	0.2044	
17	Water HoldUp	---	---	---	---	
18	Friction Gradient (kPa/m)	6.881e-003	6.881e-003	6.881e-003	6.881e-003	
19	Static Gradient (kPa/m)	-6.407e-003	-6.407e-003	-6.407e-003	-6.407e-003	
20	Accel. Gradient (kPa/m)	3.421e-008	3.420e-008	3.420e-008	3.419e-008	
21	Liq. Reynolds	2.082e+005 *	2.082e+005 *	2.082e+005 *	2.082e+005 *	
22	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *	
23	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046	
24	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167	
25	Deposit Thickness (mm)	---	---	---	---	
26	Deposit Volume (m3)	---	---	---	---	
27	Length (m)	3.046e+004	3.047e+004	3.047e+004	3.048e+004	
28	Elevation (m)	0.1400	0.1400	0.1400	0.1400	
29	Pressure (bar)	88.47	88.47	88.47	88.47	
30	Temperature (C)	49.54	49.54	49.54	49.53	
31	Vapour Fraction	0.8687	0.8687	0.8687	0.8687	
32	Heat Transferred (kJ/h-m)	520.5	520.5	520.4	520.4	
33	Flow Regime	Transition	Transition	Transition	Transition	
34	Water Flow Regime					
35	Liquid HoldUp	0.3029	0.3029	0.3029	0.3029	
36	Water HoldUp	---	---	---	---	
37	Friction Gradient (kPa/m)	5.874e-003	5.874e-003	5.874e-003	5.874e-003	
38	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000	
39	Accel. Gradient (kPa/m)	4.242e-007	4.242e-007	4.242e-007	4.242e-007	
40	Liq. Reynolds	2.082e+005 *	2.082e+005 *	2.082e+005 *	2.082e+005 *	
41	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *	
42	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046	
43	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167	
44	Deposit Thickness (mm)	---	---	---	---	
45	Deposit Volume (m3)	---	---	---	---	
46	Length (m)	3.048e+004	3.049e+004	3.049e+004	3.050e+004	
47	Elevation (m)	0.1400	0.1380	0.1360	0.1340	
48	Pressure (bar)	88.47	88.47	88.47	88.46	
49	Temperature (C)	49.53	49.53	49.53	49.53	
50	Vapour Fraction	0.8687	0.8687	0.8687	0.8687	
51	Heat Transferred (kJ/h-m)	520.3	520.3	520.3	520.2	
52	Flow Regime	Transition	Transition	Transition	Transition	
53	Water Flow Regime					
54	Liquid HoldUp	0.3029	0.2071	0.2071	0.2071	
55	Water HoldUp	---	---	---	---	
56	Friction Gradient (kPa/m)	5.874e-003	6.859e-003	6.859e-003	6.859e-003	
57	Static Gradient (kPa/m)	0.0000	-7.754e-004	-7.754e-004	-7.754e-004	
58	Accel. Gradient (kPa/m)	4.242e-007	4.393e-007	4.393e-007	4.393e-007	
59	Liq. Reynolds	2.082e+005 *	2.081e+005 *	2.081e+005 *	2.081e+005 *	
60	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *	
61	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046	
62	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167	
63	Deposit Thickness (mm)	---	---	---	---	
64	Deposit Volume (m3)	---	---	---	---	
65	Length (m)	3.050e+004	3.051e+004	3.051e+004	3.052e+004	
66	Elevation (m)	0.1320	0.1300	0.1240	0.1180	
67	Pressure (bar)	88.46	88.46	88.46	88.46	
68	Temperature (C)	49.53	49.52	49.52	49.52	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Pipe Segment: ELEP-Akabel Pipeline (continued)		
3					
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
14	Heat Transferred (kJ/h-m)	520.2	520.1	520.1	520.1
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2071	0.2071	0.2064	0.2064
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	6.859e-003	6.859e-003	6.865e-003	6.865e-003
20	Static Gradient (kPa/m)	-7.754e-004	-7.754e-004	-2.321e-003	-2.321e-003
21	Accel. Gradient (kPa/m)	4.393e-007	4.393e-007	3.281e-007	3.281e-007
22	Liq. Reynolds	2.081e+005 *	2.081e+005 *	2.081e+005 *	2.081e+005 *
23	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
24	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
25	Vapour Velocity (m/s)	2.167	2.167	2.166	2.166
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	3.052e+004	3.053e+004	3.053e+004	3.054e+004
29	Elevation (m)	0.1120	0.1060	1.000e-001	9.600e-002
30	Pressure (bar)	88.46	88.46	88.46	88.46
31	Temperature (C)	49.52	49.52	49.52	49.52
32	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
33	Heat Transferred (kJ/h-m)	520.0	520.0	519.9	519.9
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2064	0.2064	0.2064	0.2068
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	6.865e-003	6.865e-003	6.865e-003	6.861e-003
39	Static Gradient (kPa/m)	-2.321e-003	-2.321e-003	-2.321e-003	-1.439e-003
40	Accel. Gradient (kPa/m)	3.281e-007	3.281e-007	3.281e-007	3.916e-007
41	Liq. Reynolds	2.081e+005 *	2.081e+005 *	2.081e+005 *	2.081e+005 *
42	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
43	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
44	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	3.054e+004	3.055e+004	3.055e+004	3.056e+004
48	Elevation (m)	9.200e-002	8.800e-002	8.400e-002	8.000e-002
49	Pressure (bar)	88.46	88.46	88.46	88.46
50	Temperature (C)	49.51	49.51	49.51	49.51
51	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
52	Heat Transferred (kJ/h-m)	519.9	519.8	519.8	519.7
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2068	0.2068	0.2068	0.2068
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	6.861e-003	6.861e-003	6.861e-003	6.861e-003
58	Static Gradient (kPa/m)	-1.439e-003	-1.439e-003	-1.439e-003	-1.439e-003
59	Accel. Gradient (kPa/m)	3.916e-007	3.915e-007	3.915e-007	3.915e-007
60	Liq. Reynolds	2.081e+005 *	2.081e+005 *	2.081e+005 *	2.081e+005 *
61	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
62	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
63	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	3.056e+004	3.057e+004	3.057e+004	3.058e+004
67	Elevation (m)	7.800e-002	7.600e-002	7.400e-002	7.200e-002
68	Pressure (bar)	88.46	88.46	88.46	88.46
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	49.51	49.51	49.50	49.50
14	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
15	Heat Transferred (kJ/h-m)	519.7	519.6	519.6	519.6
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2071	0.2071	0.2071	0.2071
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	6.859e-003	6.858e-003	6.858e-003	6.858e-003
21	Static Gradient (kPa/m)	-7.755e-004	-7.755e-004	-7.755e-004	-7.755e-004
22	Accel. Gradient (kPa/m)	4.393e-007	4.393e-007	4.393e-007	4.393e-007
23	Liq. Reynolds	2.081e+005 *	2.081e+005 *	2.081e+005 *	2.081e+005 *
24	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
25	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
26	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	3.058e+004	3.059e+004	3.059e+004	3.060e+004
30	Elevation (m)	7.000e-002	7.800e-002	8.600e-002	9.400e-002
31	Pressure (bar)	88.46	88.46	88.46	88.46
32	Temperature (C)	49.50	49.50	49.50	49.50
33	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
34	Heat Transferred (kJ/h-m)	519.5	519.5	519.4	519.4
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2071	0.2850	0.2850	0.2850
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	6.858e-003	5.234e-003	5.234e-003	5.234e-003
40	Static Gradient (kPa/m)	-7.755e-004	3.844e-003	3.844e-003	3.844e-003
41	Accel. Gradient (kPa/m)	4.393e-007	6.555e-007	6.555e-007	6.555e-007
42	Liq. Reynolds	2.081e+005 *	2.081e+005 *	2.081e+005 *	2.081e+005 *
43	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
44	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
45	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	3.060e+004	3.061e+004	3.061e+004	3.062e+004
49	Elevation (m)	0.1020	0.1100	0.1160	0.1220
50	Pressure (bar)	88.46	88.46	88.46	88.46
51	Temperature (C)	49.49	49.49	49.49	49.49
52	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
53	Heat Transferred (kJ/h-m)	519.3	519.3	519.2	519.2
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2850	0.2850	0.2837	0.2837
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	5.234e-003	5.234e-003	5.331e-003	5.331e-003
59	Static Gradient (kPa/m)	3.844e-003	3.844e-003	2.874e-003	2.874e-003
60	Accel. Gradient (kPa/m)	6.555e-007	6.555e-007	5.925e-007	5.925e-007
61	Liq. Reynolds	2.081e+005 *	2.081e+005 *	2.081e+005 *	2.081e+005 *
62	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
63	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
64	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	3.062e+004	3.063e+004	3.063e+004	3.064e+004
68	Elevation (m)	0.1280	0.1340	0.1400	0.1460
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 489 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:02:44 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	88.46	88.46	88.46	88.46		
14	Temperature (C)	49.49	49.49	49.48	49.48		
15	Vapour Fraction	0.8687	0.8687	0.8687	0.8687		
16	Heat Transferred (kJ/h-m)	519.2	519.1	519.1	519.0		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.2837	0.2837	0.2837	0.2837		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	5.331e-003	5.331e-003	5.331e-003	5.331e-003		
22	Static Gradient (kPa/m)	2.874e-003	2.874e-003	2.874e-003	2.874e-003		
23	Accel. Gradient (kPa/m)	5.925e-007	5.925e-007	5.925e-007	5.925e-007		
24	Liq. Reynolds	2.081e+005 *	2.081e+005 *	2.081e+005 *	2.081e+005 *		
25	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *		
26	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
27	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	3.064e+004	3.065e+004	3.065e+004	3.066e+004		
31	Elevation (m)	0.1520	0.1580	0.1640	0.1700		
32	Pressure (bar)	88.45	88.45	88.45	88.45		
33	Temperature (C)	49.48	49.48	49.48	49.48		
34	Vapour Fraction	0.8687	0.8687	0.8687	0.8687		
35	Heat Transferred (kJ/h-m)	519.0	518.9	518.9	518.9		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.2837	0.2837	0.2837	0.2837		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	5.331e-003	5.331e-003	5.331e-003	5.331e-003		
41	Static Gradient (kPa/m)	2.874e-003	2.874e-003	2.874e-003	2.874e-003		
42	Accel. Gradient (kPa/m)	5.925e-007	5.925e-007	5.925e-007	5.925e-007		
43	Liq. Reynolds	2.081e+005 *	2.081e+005 *	2.081e+005 *	2.081e+005 *		
44	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *		
45	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
46	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	3.066e+004	3.067e+004	3.067e+004	3.068e+004		
50	Elevation (m)	0.1880	0.2060	0.2240	0.2420		
51	Pressure (bar)	88.45	88.45	88.45	88.45		
52	Temperature (C)	49.47	49.47	49.47	49.47		
53	Vapour Fraction	0.8687	0.8687	0.8687	0.8687		
54	Heat Transferred (kJ/h-m)	518.8	518.8	518.7	518.7		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.2914	0.2914	0.2914	0.2914		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	4.769e-003	4.769e-003	4.769e-003	4.769e-003		
60	Static Gradient (kPa/m)	8.785e-003	8.785e-003	8.785e-003	8.785e-003		
61	Accel. Gradient (kPa/m)	9.787e-007	9.788e-007	9.788e-007	9.788e-007		
62	Liq. Reynolds	2.081e+005 *	2.081e+005 *	2.081e+005 *	2.081e+005 *		
63	Vap. Reynolds	9.179e+006 *	9.180e+006 *	9.180e+006 *	9.180e+006 *		
64	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
65	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	3.068e+004	3.069e+004	3.069e+004	3.070e+004		
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	0.2600	0.2000	0.1400	8.000e-002
14	Pressure (bar)	88.45	88.45	88.45	88.45
15	Temperature (C)	49.47	49.47	49.47	49.46
16	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
17	Heat Transferred (kJ/h-m)	518.6	518.6	518.6	518.6
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2914	0.1963	0.1963	0.1963
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	4.769e-003	6.949e-003	6.949e-003	6.949e-003
23	Static Gradient (kPa/m)	8.785e-003	-2.249e-002	-2.249e-002	-2.249e-002
24	Accel. Gradient (kPa/m)	9.788e-007	-1.122e-006	-1.122e-006	-1.122e-006
25	Liq. Reynolds	2.081e+005 *	2.081e+005 *	2.081e+005 *	2.081e+005 *
26	Vap. Reynolds	9.180e+006 *	9.180e+006 *	9.180e+006 *	9.179e+006 *
27	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
28	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	3.070e+004	3.071e+004	3.071e+004	3.072e+004
32	Elevation (m)	2.000e-002	-4.000e-002	-4.800e-002	-5.600e-002
33	Pressure (bar)	88.45	88.45	88.45	88.45
34	Temperature (C)	49.46	49.46	49.46	49.46
35	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
36	Heat Transferred (kJ/h-m)	518.5	518.5	518.5	518.4
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.1963	0.1963	0.2060	0.2060
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	6.949e-003	6.949e-003	6.866e-003	6.866e-003
42	Static Gradient (kPa/m)	-2.249e-002	-2.249e-002	-3.092e-003	-3.092e-003
43	Accel. Gradient (kPa/m)	-1.122e-006	-1.122e-006	2.726e-007	2.726e-007
44	Liq. Reynolds	2.081e+005 *	2.081e+005 *	2.081e+005 *	2.081e+005 *
45	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
46	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
47	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	3.072e+004	3.073e+004	3.073e+004	3.074e+004
51	Elevation (m)	-6.400e-002	-7.200e-002	-8.000e-002	-6.600e-002
52	Pressure (bar)	88.45	88.45	88.45	88.45
53	Temperature (C)	49.46	49.46	49.45	49.45
54	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
55	Heat Transferred (kJ/h-m)	518.4	518.3	518.3	518.2
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2060	0.2060	0.2060	0.2888
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	6.866e-003	6.866e-003	6.866e-003	4.953e-003
61	Static Gradient (kPa/m)	-3.092e-003	-3.092e-003	-3.092e-003	6.763e-003
62	Accel. Gradient (kPa/m)	2.725e-007	2.725e-007	2.725e-007	8.460e-007
63	Liq. Reynolds	2.081e+005 *	2.081e+005 *	2.081e+005 *	2.081e+005 *
64	Vap. Reynolds	9.179e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
65	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
66	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 491 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Length	(m)	3.074e+004	3.075e+004	3.075e+004	3.076e+004	
14	Elevation	(m)	-5.200e-002	-3.800e-002	-2.400e-002	-1.000e-002	
15	Pressure	(bar)	88.45	88.45	88.45	88.45	
16	Temperature	(C)	49.45	49.45	49.45	49.45	
17	Vapour Fraction		0.8687	0.8687	0.8687	0.8687	
18	Heat Transferred	(kJ/h-m)	518.2	518.1	518.1	518.1	
19	Flow Regime		Transition	Transition	Transition	Transition	
20	Water Flow Regime						
21	Liquid HoldUp		0.2888	0.2888	0.2888	0.2888	
22	Water HoldUp		---	---	---	---	
23	Friction Gradient	(kPa/m)	4.953e-003	4.953e-003	4.953e-003	4.953e-003	
24	Static Gradient	(kPa/m)	6.763e-003	6.763e-003	6.763e-003	6.763e-003	
25	Accel. Gradient	(kPa/m)	8.460e-007	8.460e-007	8.460e-007	8.460e-007	
26	Liq. Reynolds		2.081e+005 *	2.081e+005 *	2.081e+005 *	2.081e+005 *	
27	Vap. Reynolds		9.179e+006 *	9.179e+006 *	9.180e+006 *	9.180e+006 *	
28	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
29	Vapour Velocity	(m/s)	2.166	2.166	2.166	2.166	
30	Deposit Thickness	(mm)	---	---	---	---	
31	Deposit Volume	(m3)	---	---	---	---	
32	Length	(m)	3.076e+004	3.077e+004	3.077e+004	3.078e+004	
33	Elevation	(m)	-8.000e-003	-6.000e-003	-4.000e-003	-2.000e-003	
34	Pressure	(bar)	88.45	88.45	88.45	88.45	
35	Temperature	(C)	49.44	49.44	49.44	49.44	
36	Vapour Fraction		0.8687	0.8687	0.8687	0.8687	
37	Heat Transferred	(kJ/h-m)	518.0	518.0	517.9	517.9	
38	Flow Regime		Transition	Transition	Transition	Transition	
39	Water Flow Regime						
40	Liquid HoldUp		0.2812	0.2812	0.2812	0.2812	
41	Water HoldUp		---	---	---	---	
42	Friction Gradient	(kPa/m)	5.528e-003	5.527e-003	5.527e-003	5.527e-003	
43	Static Gradient	(kPa/m)	9.519e-004	9.519e-004	9.519e-004	9.519e-004	
44	Accel. Gradient	(kPa/m)	4.678e-007	4.678e-007	4.678e-007	4.678e-007	
45	Liq. Reynolds		2.081e+005 *	2.081e+005 *	2.081e+005 *	2.081e+005 *	
46	Vap. Reynolds		9.180e+006 *	9.180e+006 *	9.180e+006 *	9.180e+006 *	
47	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
48	Vapour Velocity	(m/s)	2.166	2.166	2.166	2.166	
49	Deposit Thickness	(mm)	---	---	---	---	
50	Deposit Volume	(m3)	---	---	---	---	
51	Length	(m)	3.078e+004	3.079e+004	3.079e+004	3.080e+004	
52	Elevation	(m)	-5.360e-016	-6.000e-003	-1.200e-002	-1.800e-002	
53	Pressure	(bar)	88.45	88.45	88.45	88.45	
54	Temperature	(C)	49.44	49.44	49.44	49.43	
55	Vapour Fraction		0.8687	0.8687	0.8687	0.8687	
56	Heat Transferred	(kJ/h-m)	517.8	517.8	517.8	517.7	
57	Flow Regime		Transition	Transition	Transition	Transition	
58	Water Flow Regime						
59	Liquid HoldUp		0.2812	0.2064	0.2064	0.2064	
60	Water HoldUp		---	---	---	---	
61	Friction Gradient	(kPa/m)	5.527e-003	6.863e-003	6.863e-003	6.863e-003	
62	Static Gradient	(kPa/m)	9.519e-004	-2.321e-003	-2.321e-003	-2.322e-003	
63	Accel. Gradient	(kPa/m)	4.678e-007	3.279e-007	3.279e-007	3.279e-007	
64	Liq. Reynolds		2.081e+005 *	2.081e+005 *	2.081e+005 *	2.081e+005 *	
65	Vap. Reynolds		9.180e+006 *	9.180e+006 *	9.180e+006 *	9.180e+006 *	
66	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
67	Vapour Velocity	(m/s)	2.166	2.166	2.166	2.166	
68	Deposit Thickness	(mm)	---	---	---	---	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 492 of 937	

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	3.080e+004	3.081e+004	3.081e+004	3.082e+004
15	Elevation (m)	-2.400e-002	-3.000e-002	-2.000e-002	-1.000e-002
16	Pressure (bar)	88.45	88.45	88.45	88.45
17	Temperature (C)	49.43	49.43	49.43	49.43
18	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
19	Heat Transferred (kJ/h-m)	517.7	517.7	517.6	517.6
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2064	0.2064	0.2863	0.2863
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	6.863e-003	6.863e-003	5.137e-003	5.137e-003
25	Static Gradient (kPa/m)	-2.322e-003	-2.322e-003	4.820e-003	4.820e-003
26	Accel. Gradient (kPa/m)	3.279e-007	3.279e-007	7.190e-007	7.190e-007
27	Liq. Reynolds	2.081e+005 *	2.081e+005 *	2.081e+005 *	2.081e+005 *
28	Vap. Reynolds	9.180e+006 *	9.180e+006 *	9.180e+006 *	9.180e+006 *
29	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
30	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	3.082e+004	3.083e+004	3.083e+004	3.084e+004
34	Elevation (m)	-5.343e-016	1.000e-002	2.000e-002	2.600e-002
35	Pressure (bar)	88.45	88.45	88.45	88.44
36	Temperature (C)	49.43	49.42	49.42	49.42
37	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
38	Heat Transferred (kJ/h-m)	517.5	517.5	517.4	517.4
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2863	0.2863	0.2863	0.2837
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	5.137e-003	5.137e-003	5.137e-003	5.330e-003
44	Static Gradient (kPa/m)	4.820e-003	4.820e-003	4.821e-003	2.874e-003
45	Accel. Gradient (kPa/m)	7.190e-007	7.190e-007	7.190e-007	5.923e-007
46	Liq. Reynolds	2.081e+005 *	2.081e+005 *	2.080e+005 *	2.080e+005 *
47	Vap. Reynolds	9.180e+006 *	9.180e+006 *	9.180e+006 *	9.180e+006 *
48	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
49	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	3.084e+004	3.085e+004	3.085e+004	3.086e+004
53	Elevation (m)	3.200e-002	3.800e-002	4.400e-002	5.000e-002
54	Pressure (bar)	88.44	88.44	88.44	88.44
55	Temperature (C)	49.42	49.42	49.42	49.41
56	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
57	Heat Transferred (kJ/h-m)	517.3	517.3	517.3	517.2
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2837	0.2837	0.2837	0.2837
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	5.330e-003	5.329e-003	5.329e-003	5.329e-003
63	Static Gradient (kPa/m)	2.874e-003	2.874e-003	2.874e-003	2.874e-003
64	Accel. Gradient (kPa/m)	5.923e-007	5.923e-007	5.923e-007	5.923e-007
65	Liq. Reynolds	2.080e+005 *	2.080e+005 *	2.080e+005 *	2.080e+005 *
66	Vap. Reynolds	9.180e+006 *	9.180e+006 *	9.180e+006 *	9.180e+006 *
67	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
68	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 493 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	3.086e+004	3.087e+004	3.087e+004	3.088e+004
16	Elevation (m)	7.800e-002	0.1060	0.1340	0.1620
17	Pressure (bar)	88.44	88.44	88.44	88.44
18	Temperature (C)	49.41	49.41	49.41	49.41
19	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
20	Heat Transferred (kJ/h-m)	517.2	517.1	517.1	517.0
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2978	0.2978	0.2978	0.2978
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	5.582e-003	5.582e-003	5.582e-003	5.582e-003
26	Static Gradient (kPa/m)	1.388e-002	1.388e-002	1.388e-002	1.388e-002
27	Accel. Gradient (kPa/m)	1.405e-006	1.405e-006	1.405e-006	1.405e-006
28	Liq. Reynolds	2.080e+005 *	2.080e+005 *	2.080e+005 *	2.080e+005 *
29	Vap. Reynolds	9.180e+006 *	9.180e+006 *	9.180e+006 *	9.180e+006 *
30	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
31	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	3.088e+004	3.089e+004	3.089e+004	3.090e+004
35	Elevation (m)	0.1900	0.1940	0.1980	0.2020
36	Pressure (bar)	88.44	88.44	88.44	88.44
37	Temperature (C)	49.41	49.40	49.40	49.40
38	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
39	Heat Transferred (kJ/h-m)	517.0	516.9	516.9	516.8
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2978	0.2825	0.2825	0.2825
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	5.582e-003	5.428e-003	5.428e-003	5.428e-003
45	Static Gradient (kPa/m)	1.388e-002	1.910e-003	1.910e-003	1.910e-003
46	Accel. Gradient (kPa/m)	1.405e-006	5.299e-007	5.299e-007	5.299e-007
47	Liq. Reynolds	2.080e+005 *	2.080e+005 *	2.080e+005 *	2.080e+005 *
48	Vap. Reynolds	9.180e+006 *	9.180e+006 *	9.180e+006 *	9.180e+006 *
49	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
50	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	3.090e+004	3.091e+004	3.091e+004	3.092e+004
54	Elevation (m)	0.2060	0.2100	0.2120	0.2140
55	Pressure (bar)	88.44	88.44	88.44	88.44
56	Temperature (C)	49.40	49.40	49.40	49.39
57	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
58	Heat Transferred (kJ/h-m)	516.8	516.8	516.7	516.7
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2825	0.2825	0.2812	0.2812
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	5.428e-003	5.428e-003	5.527e-003	5.527e-003
64	Static Gradient (kPa/m)	1.910e-003	1.910e-003	9.520e-004	9.520e-004
65	Accel. Gradient (kPa/m)	5.298e-007	5.298e-007	4.678e-007	4.678e-007
66	Liq. Reynolds	2.080e+005 *	2.080e+005 *	2.080e+005 *	2.080e+005 *
67	Vap. Reynolds	9.180e+006 *	9.180e+006 *	9.180e+006 *	9.180e+006 *
68	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Unit Set: Mixed			
3			Date/Time: Mon Jun 13 15:02:44 2022			
4						
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Vapour Velocity	(m/s)	2.166	2.166	2.166	2.166
14	Deposit Thickness	(mm)	---	---	---	---
15	Deposit Volume	(m3)	---	---	---	---
16	Length	(m)	3.092e+004	3.093e+004	3.093e+004	3.094e+004
17	Elevation	(m)	0.2160	0.2180	0.2200	0.2220
18	Pressure	(bar)	88.44	88.44	88.43	88.43
19	Temperature	(C)	49.39	49.39	49.39	49.39
20	Vapour Fraction		0.8687	0.8687	0.8687	0.8687
21	Heat Transferred	(kJ/h-m)	516.6	516.6	516.6	516.5
22	Flow Regime		Transition	Transition	Transition	Transition
23	Water Flow Regime					
24	Liquid HoldUp		0.2812	0.2812	0.2812	0.2812
25	Water HoldUp		---	---	---	---
26	Friction Gradient	(kPa/m)	5.527e-003	5.527e-003	5.527e-003	5.527e-003
27	Static Gradient	(kPa/m)	9.520e-004	9.520e-004	9.520e-004	9.501e-004
28	Accel. Gradient	(kPa/m)	4.678e-007	4.678e-007	4.678e-007	4.677e-007
29	Liq. Reynolds		2.080e+005 *	2.080e+005 *	2.080e+005 *	2.080e+005 *
30	Vap. Reynolds		9.180e+006 *	9.180e+006 *	9.180e+006 *	9.180e+006 *
31	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
32	Vapour Velocity	(m/s)	2.166	2.166	2.166	2.166
33	Deposit Thickness	(mm)	---	---	---	---
34	Deposit Volume	(m3)	---	---	---	---
35	Length	(m)	3.094e+004	3.095e+004	3.095e+004	3.096e+004
36	Elevation	(m)	0.2240	0.2260	0.2280	0.2300
37	Pressure	(bar)	88.43	88.43	88.43	88.43
38	Temperature	(C)	49.39	49.39	49.38	49.38
39	Vapour Fraction		0.8687	0.8687	0.8687	0.8687
40	Heat Transferred	(kJ/h-m)	516.5	516.4	516.4	516.3
41	Flow Regime		Transition	Transition	Transition	Transition
42	Water Flow Regime					
43	Liquid HoldUp		0.2812	0.2812	0.2812	0.2812
44	Water HoldUp		---	---	---	---
45	Friction Gradient	(kPa/m)	5.527e-003	5.527e-003	5.527e-003	5.527e-003
46	Static Gradient	(kPa/m)	9.501e-004	9.501e-004	9.501e-004	9.501e-004
47	Accel. Gradient	(kPa/m)	4.677e-007	4.677e-007	4.677e-007	4.677e-007
48	Liq. Reynolds		2.080e+005 *	2.080e+005 *	2.080e+005 *	2.080e+005 *
49	Vap. Reynolds		9.180e+006 *	9.180e+006 *	9.180e+006 *	9.180e+006 *
50	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
51	Vapour Velocity	(m/s)	2.166	2.166	2.166	2.166
52	Deposit Thickness	(mm)	---	---	---	---
53	Deposit Volume	(m3)	---	---	---	---
54	Length	(m)	3.096e+004	3.097e+004	3.097e+004	3.098e+004
55	Elevation	(m)	0.2300	0.2300	0.2300	0.2300
56	Pressure	(bar)	88.43	88.43	88.43	88.43
57	Temperature	(C)	49.38	49.38	49.38	49.38
58	Vapour Fraction		0.8686	0.8686	0.8686	0.8686
59	Heat Transferred	(kJ/h-m)	516.3	516.3	516.2	516.2
60	Flow Regime		Transition	Transition	Transition	Transition
61	Water Flow Regime					
62	Liquid HoldUp		0.3030	0.3030	0.3030	0.3030
63	Water HoldUp		---	---	---	---
64	Friction Gradient	(kPa/m)	5.873e-003	5.873e-003	5.873e-003	5.873e-003
65	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000
66	Accel. Gradient	(kPa/m)	4.241e-007	4.241e-007	4.241e-007	4.241e-007
67	Liq. Reynolds		2.080e+005 *	2.080e+005 *	2.080e+005 *	2.080e+005 *
68	Vap. Reynolds		9.180e+006 *	9.180e+006 *	9.180e+006 *	9.180e+006 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022	
6						
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8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
14	Vapour Velocity	(m/s)	2.166	2.166	2.166	2.166
15	Deposit Thickness	(mm)	---	---	---	---
16	Deposit Volume	(m3)	---	---	---	---
17	Length	(m)	3.098e+004	3.099e+004	3.099e+004	3.100e+004
18	Elevation	(m)	0.2300	0.2280	0.2260	0.2240
19	Pressure	(bar)	88.43	88.43	88.43	88.43
20	Temperature	(C)	49.37	49.37	49.37	49.37
21	Vapour Fraction		0.8686	0.8686	0.8686	0.8686
22	Heat Transferred	(kJ/h-m)	516.1	516.1	516.1	516.0
23	Flow Regime		Transition	Transition	Transition	Transition
24	Water Flow Regime					
25	Liquid HoldUp		0.3030	0.2072	0.2072	0.2072
26	Water HoldUp		---	---	---	---
27	Friction Gradient	(kPa/m)	5.873e-003	6.856e-003	6.856e-003	6.856e-003
28	Static Gradient	(kPa/m)	0.0000	-7.757e-004	-7.757e-004	-7.757e-004
29	Accel. Gradient	(kPa/m)	4.241e-007	4.391e-007	4.391e-007	4.391e-007
30	Liq. Reynolds		2.080e+005 *	2.080e+005 *	2.080e+005 *	2.080e+005 *
31	Vap. Reynolds		9.180e+006 *	9.180e+006 *	9.180e+006 *	9.180e+006 *
32	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
33	Vapour Velocity	(m/s)	2.166	2.166	2.166	2.166
34	Deposit Thickness	(mm)	---	---	---	---
35	Deposit Volume	(m3)	---	---	---	---
36	Length	(m)	3.100e+004	3.101e+004	3.103e+004	3.105e+004
37	Elevation	(m)	0.2220	0.2200	0.2240	0.2280
38	Pressure	(bar)	88.43	88.43	88.43	88.43
39	Temperature	(C)	49.37	49.37	49.36	49.35
40	Vapour Fraction		0.8686	0.8686	0.8686	0.8686
41	Heat Transferred	(kJ/h-m)	516.0	515.9	515.8	515.6
42	Flow Regime		Transition	Transition	Transition	Transition
43	Water Flow Regime					
44	Liquid HoldUp		0.2072	0.2072	0.2806	0.2806
45	Water HoldUp		---	---	---	---
46	Friction Gradient	(kPa/m)	6.856e-003	6.856e-003	5.576e-003	5.576e-003
47	Static Gradient	(kPa/m)	-7.757e-004	-7.757e-004	4.753e-004	4.753e-004
48	Accel. Gradient	(kPa/m)	4.391e-007	4.391e-007	4.370e-007	4.370e-007
49	Liq. Reynolds		2.080e+005 *	2.080e+005 *	2.080e+005 *	2.080e+005 *
50	Vap. Reynolds		9.180e+006 *	9.180e+006 *	9.180e+006 *	9.180e+006 *
51	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
52	Vapour Velocity	(m/s)	2.166	2.166	2.165	2.165
53	Deposit Thickness	(mm)	---	---	---	---
54	Deposit Volume	(m3)	---	---	---	---
55	Length	(m)	3.107e+004	3.109e+004	3.111e+004	3.111e+004
56	Elevation	(m)	0.2320	0.2360	0.2400	0.2380
57	Pressure	(bar)	88.43	88.43	88.42	88.42
58	Temperature	(C)	49.35	49.34	49.34	49.33
59	Vapour Fraction		0.8686	0.8686	0.8686	0.8686
60	Heat Transferred	(kJ/h-m)	515.4	515.3	515.1	515.1
61	Flow Regime		Transition	Transition	Transition	Transition
62	Water Flow Regime					
63	Liquid HoldUp		0.2806	0.2806	0.2806	0.2072
64	Water HoldUp		---	---	---	---
65	Friction Gradient	(kPa/m)	5.576e-003	5.576e-003	5.576e-003	6.856e-003
66	Static Gradient	(kPa/m)	4.753e-004	4.753e-004	4.753e-004	-7.758e-004
67	Accel. Gradient	(kPa/m)	4.370e-007	4.369e-007	4.369e-007	4.390e-007
68	Liq. Reynolds		2.080e+005 *	2.080e+005 *	2.080e+005 *	2.080e+005 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.180e+006 *	9.180e+006 *	9.180e+006 *	9.180e+006 *
14	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
15	Vapour Velocity (m/s)	2.165	2.165	2.165	2.165
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	3.112e+004	3.112e+004	3.113e+004	3.113e+004
19	Elevation (m)	0.2360	0.2340	0.2320	0.2300
20	Pressure (bar)	88.42	88.42	88.42	88.42
21	Temperature (C)	49.33	49.33	49.33	49.33
22	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
23	Heat Transferred (kJ/h-m)	515.0	515.0	514.9	514.9
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2072	0.2072	0.2072	0.2072
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.856e-003	6.856e-003	6.856e-003	6.856e-003
29	Static Gradient (kPa/m)	-7.758e-004	-7.758e-004	-7.758e-004	-7.758e-004
30	Accel. Gradient (kPa/m)	4.390e-007	4.390e-007	4.390e-007	4.390e-007
31	Liq. Reynolds	2.080e+005 *	2.080e+005 *	2.080e+005 *	2.080e+005 *
32	Vap. Reynolds	9.180e+006 *	9.180e+006 *	9.180e+006 *	9.180e+006 *
33	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
34	Vapour Velocity (m/s)	2.165	2.165	2.165	2.165
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	3.114e+004	3.114e+004	3.115e+004	3.115e+004
38	Elevation (m)	0.3460	0.4620	0.5780	0.6940
39	Pressure (bar)	88.42	88.42	88.41	88.41
40	Temperature (C)	49.33	49.32	49.32	49.32
41	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
42	Heat Transferred (kJ/h-m)	514.9	514.8	514.7	514.6
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.3537	0.3537	0.3537	0.3537
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.050e-003	6.050e-003	6.050e-003	6.050e-003
48	Static Gradient (kPa/m)	6.516e-002	6.516e-002	6.516e-002	6.516e-002
49	Accel. Gradient (kPa/m)	5.142e-006	5.142e-006	5.143e-006	5.143e-006
50	Liq. Reynolds	2.079e+005 *	2.079e+005 *	2.079e+005 *	2.079e+005 *
51	Vap. Reynolds	9.180e+006 *	9.181e+006 *	9.181e+006 *	9.181e+006 *
52	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
53	Vapour Velocity (m/s)	2.165	2.165	2.165	2.166
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	3.116e+004	3.116e+004	3.117e+004	3.117e+004
57	Elevation (m)	0.8100	-1.012	-2.834	-4.656
58	Pressure (bar)	88.40	88.43	88.45	88.47
59	Temperature (C)	49.31	49.32	49.33	49.34
60	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
61	Heat Transferred (kJ/h-m)	514.6	515.6	515.9	516.1
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.3537	8.635e-002	8.638e-002	8.640e-002
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	6.050e-003	9.947e-003	9.944e-003	9.941e-003
67	Static Gradient (kPa/m)	6.516e-002	-0.4446	-0.4447	-0.4448
68	Accel. Gradient (kPa/m)	5.143e-006	-3.138e-005	-3.137e-005	-3.137e-005
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.079e+005 *	2.080e+005 *	2.080e+005 *	2.080e+005 *
14	Vap. Reynolds	9.181e+006 *	9.180e+006 *	9.179e+006 *	9.179e+006 *
15	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
16	Vapour Velocity (m/s)	2.166	2.165	2.165	2.164
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	3.118e+004	3.118e+004	3.119e+004	3.119e+004
20	Elevation (m)	-6.478	-8.300	-8.220	-8.140
21	Pressure (bar)	88.49	88.51	88.51	88.51
22	Temperature (C)	49.35	49.36	49.36	49.36
23	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
24	Heat Transferred (kJ/h-m)	516.4	516.6	515.8	515.7
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	8.642e-002	8.644e-002	0.3651	0.3651
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	9.938e-003	9.935e-003	6.029e-003	6.029e-003
30	Static Gradient (kPa/m)	-0.4449	-0.4450	7.679e-002	7.679e-002
31	Accel. Gradient (kPa/m)	-3.136e-005	-3.135e-005	5.968e-006	5.968e-006
32	Liq. Reynolds	2.080e+005 *	2.081e+005 *	2.081e+005 *	2.081e+005 *
33	Vap. Reynolds	9.178e+006 *	9.177e+006 *	9.177e+006 *	9.177e+006 *
34	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
35	Vapour Velocity (m/s)	2.164	2.163	2.163	2.163
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	3.119e+004	3.120e+004	3.120e+004	3.121e+004
39	Elevation (m)	-8.060	-7.980	-7.900	-7.580
40	Pressure (bar)	88.51	88.50	88.50	88.49
41	Temperature (C)	49.36	49.35	49.35	49.35
42	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
43	Heat Transferred (kJ/h-m)	515.7	515.6	515.6	515.5
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.3651	0.3650	0.3650	0.4073
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	6.029e-003	6.029e-003	6.030e-003	5.967e-003
49	Static Gradient (kPa/m)	7.679e-002	7.679e-002	7.679e-002	0.1252
50	Accel. Gradient (kPa/m)	5.969e-006	5.969e-006	5.969e-006	9.459e-006
51	Liq. Reynolds	2.081e+005 *	2.081e+005 *	2.081e+005 *	2.080e+005 *
52	Vap. Reynolds	9.178e+006 *	9.178e+006 *	9.178e+006 *	9.178e+006 *
53	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
54	Vapour Velocity (m/s)	2.163	2.163	2.163	2.164
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	3.122e+004	3.122e+004	3.123e+004	3.124e+004
58	Elevation (m)	-7.260	-6.940	-6.620	-6.300
59	Pressure (bar)	88.48	88.47	88.46	88.45
60	Temperature (C)	49.34	49.34	49.33	49.32
61	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
62	Heat Transferred (kJ/h-m)	515.3	515.2	515.0	514.9
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.4073	0.4073	0.4072	0.4072
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	5.968e-003	5.969e-003	5.969e-003	5.970e-003
68	Static Gradient (kPa/m)	0.1252	0.1252	0.1252	0.1252
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 498 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	9.461e-006	9.463e-006	9.464e-006	9.466e-006
14	Liq. Reynolds	2.080e+005 *	2.080e+005 *	2.080e+005 *	2.080e+005 *
15	Vap. Reynolds	9.178e+006 *	9.179e+006 *	9.179e+006 *	9.179e+006 *
16	Liquid Velocity (m/s)	0.2047	0.2046	0.2046	0.2046
17	Vapour Velocity (m/s)	2.164	2.164	2.164	2.164
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	3.124e+004	3.125e+004	3.125e+004	3.126e+004
21	Elevation (m)	-6.000	-5.700	-5.400	-5.100
22	Pressure (bar)	88.44	88.43	88.41	88.40
23	Temperature (C)	49.32	49.32	49.31	49.31
24	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
25	Heat Transferred (kJ/h-m)	514.9	514.8	514.7	514.6
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.5171	0.5170	0.5170	0.5169
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	5.837e-003	5.838e-003	5.839e-003	5.839e-003
31	Static Gradient (kPa/m)	0.2838	0.2838	0.2838	0.2838
32	Accel. Gradient (kPa/m)	2.091e-005	2.091e-005	2.092e-005	2.092e-005
33	Liq. Reynolds	2.080e+005 *	2.079e+005 *	2.079e+005 *	2.079e+005 *
34	Vap. Reynolds	9.180e+006 *	9.180e+006 *	9.181e+006 *	9.181e+006 *
35	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
36	Vapour Velocity (m/s)	2.165	2.165	2.165	2.166
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	3.126e+004	3.126e+004	3.127e+004	3.127e+004
40	Elevation (m)	-4.800	-3.810	-2.820	-1.830
41	Pressure (bar)	88.39	88.33	88.27	88.21
42	Temperature (C)	49.30	49.29	49.27	49.25
43	Vapour Fraction	0.8686	0.8686	0.8687	0.8687
44	Heat Transferred (kJ/h-m)	514.4	514.8	514.4	513.9
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.5169	0.8755	0.8752	0.8749
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	5.840e-003	5.640e-003	5.643e-003	5.647e-003
50	Static Gradient (kPa/m)	0.2838	1.172	1.172	1.171
51	Accel. Gradient (kPa/m)	2.092e-005	8.520e-005	8.529e-005	8.539e-005
52	Liq. Reynolds	2.079e+005 *	2.078e+005 *	2.077e+005 *	2.077e+005 *
53	Vap. Reynolds	9.181e+006 *	9.183e+006 *	9.185e+006 *	9.187e+006 *
54	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
55	Vapour Velocity (m/s)	2.166	2.167	2.169	2.170
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	3.128e+004	3.128e+004	3.129e+004	3.129e+004
59	Elevation (m)	-0.8400	0.1500	0.1100	7.000e-002
60	Pressure (bar)	88.16	88.10	88.10	88.10
61	Temperature (C)	49.24	49.22	49.22	49.22
62	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
63	Heat Transferred (kJ/h-m)	513.5	513.1	512.0	512.0
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.8746	0.8742	0.1996	0.1996
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	5.650e-003	5.654e-003	6.938e-003	6.938e-003
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1				Case Name:	Simulation - 386MMSCFD_100barg.hsc	
2				Unit Set:	Mixed	
3		Epic Atlantic Limited				
Date/Time:			Mon Jun 13 15:02:44 2022			
4						
5						
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	1.171	1.171	-1.514e-002	-1.514e-002
14	Accel. Gradient	(kPa/m)	8.548e-005	8.558e-005	-5.969e-007	-5.969e-007
15	Liq. Reynolds		2.076e+005 *	2.075e+005 *	2.075e+005 *	2.075e+005 *
16	Vap. Reynolds		9.189e+006 *	9.191e+006 *	9.191e+006 *	9.191e+006 *
17	Liquid Velocity	(m/s)	0.2046	0.2045	0.2045	0.2045
18	Vapour Velocity	(m/s)	2.172	2.173	2.173	2.173
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	3.130e+004	3.130e+004	3.131e+004	3.131e+004
22	Elevation	(m)	3.000e-002	-1.000e-002	-5.000e-002	-3.200e-002
23	Pressure	(bar)	88.10	88.10	88.10	88.10
24	Temperature	(C)	49.22	49.22	49.21	49.21
25	Vapour Fraction		0.8687	0.8687	0.8687	0.8687
26	Heat Transferred	(kJ/h-m)	512.0	511.9	511.9	511.8
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.1996	0.1996	0.1996	0.2908
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	6.937e-003	6.937e-003	6.937e-003	4.791e-003
32	Static Gradient	(kPa/m)	-1.514e-002	-1.514e-002	-1.514e-002	8.770e-003
33	Accel. Gradient	(kPa/m)	-5.969e-007	-5.969e-007	-5.969e-007	9.864e-007
34	Liq. Reynolds		2.075e+005 *	2.075e+005 *	2.075e+005 *	2.075e+005 *
35	Vap. Reynolds		9.191e+006 *	9.191e+006 *	9.191e+006 *	9.191e+006 *
36	Liquid Velocity	(m/s)	0.2045	0.2045	0.2045	0.2046
37	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	3.132e+004	3.132e+004	3.133e+004	3.133e+004
41	Elevation	(m)	-1.400e-002	4.000e-003	2.200e-002	4.000e-002
42	Pressure	(bar)	88.10	88.10	88.10	88.09
43	Temperature	(C)	49.21	49.21	49.21	49.21
44	Vapour Fraction		0.8687	0.8687	0.8687	0.8687
45	Heat Transferred	(kJ/h-m)	511.8	511.7	511.7	511.7
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2908	0.2908	0.2908	0.2908
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	4.791e-003	4.791e-003	4.791e-003	4.791e-003
51	Static Gradient	(kPa/m)	8.770e-003	8.770e-003	8.770e-003	8.770e-003
52	Accel. Gradient	(kPa/m)	9.864e-007	9.864e-007	9.864e-007	9.864e-007
53	Liq. Reynolds		2.075e+005 *	2.075e+005 *	2.075e+005 *	2.075e+005 *
54	Vap. Reynolds		9.191e+006 *	9.191e+006 *	9.191e+006 *	9.191e+006 *
55	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
56	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	3.134e+004	3.134e+004	3.135e+004	3.135e+004
60	Elevation	(m)	2.800e-002	1.600e-002	4.000e-003	-8.000e-003
61	Pressure	(bar)	88.09	88.09	88.09	88.09
62	Temperature	(C)	49.20	49.20	49.20	49.20
63	Vapour Fraction		0.8687	0.8687	0.8687	0.8687
64	Heat Transferred	(kJ/h-m)	511.6	511.6	511.5	511.5
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2048	0.2048	0.2048	0.2048
68	Water HoldUp		---	---	---	---
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 500 of 937

1			Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3	 Epic Atlantic Limited		Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:02:44 2022				
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	6.894e-003	6.894e-003	6.894e-003	6.893e-003	
14	Static Gradient	(kPa/m)	-4.597e-003	-4.597e-003	-4.598e-003	-4.598e-003	
15	Accel. Gradient	(kPa/m)	1.670e-007	1.670e-007	1.670e-007	1.670e-007	
16	Liq. Reynolds		2.075e+005 *	2.075e+005 *	2.075e+005 *	2.075e+005 *	
17	Vap. Reynolds		9.191e+006 *	9.191e+006 *	9.191e+006 *	9.191e+006 *	
18	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
19	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	3.136e+004	3.136e+004	3.137e+004	3.137e+004	
23	Elevation	(m)	-2.000e-002	-1.600e-002	-1.200e-002	-8.000e-003	
24	Pressure	(bar)	88.09	88.09	88.09	88.09	
25	Temperature	(C)	49.20	49.20	49.20	49.19	
26	Vapour Fraction		0.8687	0.8687	0.8687	0.8687	
27	Heat Transferred	(kJ/h-m)	511.5	511.4	511.4	511.3	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.2048	0.2819	0.2819	0.2819	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	6.893e-003	5.454e-003	5.454e-003	5.454e-003	
33	Static Gradient	(kPa/m)	-4.598e-003	1.906e-003	1.906e-003	1.906e-003	
34	Accel. Gradient	(kPa/m)	1.670e-007	5.354e-007	5.354e-007	5.354e-007	
35	Liq. Reynolds		2.075e+005 *	2.075e+005 *	2.075e+005 *	2.075e+005 *	
36	Vap. Reynolds		9.191e+006 *	9.191e+006 *	9.191e+006 *	9.191e+006 *	
37	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
38	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	3.138e+004	3.138e+004	3.139e+004	3.139e+004	
42	Elevation	(m)	-4.000e-003	-5.447e-016	2.400e-002	4.800e-002	
43	Pressure	(bar)	88.09	88.09	88.09	88.09	
44	Temperature	(C)	49.19	49.19	49.19	49.19	
45	Vapour Fraction		0.8687	0.8687	0.8687	0.8687	
46	Heat Transferred	(kJ/h-m)	511.3	511.2	511.2	511.2	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.2819	0.2819	0.2946	0.2946	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	5.454e-003	5.453e-003	5.144e-003	5.144e-003	
52	Static Gradient	(kPa/m)	1.906e-003	1.906e-003	1.180e-002	1.180e-002	
53	Accel. Gradient	(kPa/m)	5.354e-007	5.354e-007	1.233e-006	1.233e-006	
54	Liq. Reynolds		2.075e+005 *	2.075e+005 *	2.075e+005 *	2.075e+005 *	
55	Vap. Reynolds		9.191e+006 *	9.191e+006 *	9.191e+006 *	9.191e+006 *	
56	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
57	Vapour Velocity	(m/s)	2.173	2.173	2.173	2.173	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	3.140e+004	3.140e+004	3.141e+004	3.141e+004	
61	Elevation	(m)	7.200e-002	9.600e-002	0.1200	0.1080	
62	Pressure	(bar)	88.09	88.09	88.09	88.09	
63	Temperature	(C)	49.19	49.18	49.18	49.18	
64	Vapour Fraction		0.8687	0.8687	0.8687	0.8687	
65	Heat Transferred	(kJ/h-m)	511.1	511.1	511.0	511.0	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.2946	0.2946	0.2946	0.2048	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 501 of 937	

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	5.144e-003	5.144e-003	5.144e-003	6.893e-003
15	Static Gradient (kPa/m)	1.180e-002	1.180e-002	1.180e-002	-4.618e-003
16	Accel. Gradient (kPa/m)	1.233e-006	1.233e-006	1.233e-006	1.655e-007
17	Liq. Reynolds	2.075e+005 *	2.075e+005 *	2.075e+005 *	2.075e+005 *
18	Vap. Reynolds	9.191e+006 *	9.191e+006 *	9.191e+006 *	9.191e+006 *
19	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
20	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	3.142e+004	3.142e+004	3.143e+004	3.143e+004
24	Elevation (m)	9.600e-002	8.400e-002	7.200e-002	6.000e-002
25	Pressure (bar)	88.09	88.09	88.09	88.09
26	Temperature (C)	49.18	49.18	49.18	49.17
27	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
28	Heat Transferred (kJ/h-m)	511.0	510.9	510.9	510.8
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2048	0.2048	0.2048	0.2048
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	6.893e-003	6.893e-003	6.893e-003	6.893e-003
34	Static Gradient (kPa/m)	-4.618e-003	-4.618e-003	-4.618e-003	-4.618e-003
35	Accel. Gradient (kPa/m)	1.655e-007	1.655e-007	1.655e-007	1.655e-007
36	Liq. Reynolds	2.075e+005 *	2.075e+005 *	2.075e+005 *	2.075e+005 *
37	Vap. Reynolds	9.191e+006 *	9.191e+006 *	9.191e+006 *	9.191e+006 *
38	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
39	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	3.144e+004	3.144e+004	3.145e+004	3.145e+004
43	Elevation (m)	5.200e-002	4.400e-002	3.600e-002	2.800e-002
44	Pressure (bar)	88.09	88.09	88.09	88.09
45	Temperature (C)	49.17	49.17	49.17	49.17
46	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
47	Heat Transferred (kJ/h-m)	510.8	510.8	510.7	510.7
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2056	0.2056	0.2056	0.2056
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	6.887e-003	6.887e-003	6.887e-003	6.887e-003
53	Static Gradient (kPa/m)	-3.086e-003	-3.086e-003	-3.086e-003	-3.086e-003
54	Accel. Gradient (kPa/m)	2.765e-007	2.765e-007	2.765e-007	2.765e-007
55	Liq. Reynolds	2.075e+005 *	2.075e+005 *	2.075e+005 *	2.075e+005 *
56	Vap. Reynolds	9.191e+006 *	9.191e+006 *	9.191e+006 *	9.191e+006 *
57	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
58	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	3.146e+004	3.146e+004	3.147e+004	3.147e+004
62	Elevation (m)	2.000e-002	1.800e-002	1.600e-002	1.400e-002
63	Pressure (bar)	88.09	88.09	88.09	88.09
64	Temperature (C)	49.17	49.17	49.16	49.16
65	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
66	Heat Transferred (kJ/h-m)	510.6	510.6	510.6	510.5
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 502 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.2056	0.2067	0.2067	0.2067
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	6.887e-003	6.878e-003	6.878e-003	6.878e-003
16	Static Gradient (kPa/m)	-3.086e-003	-7.741e-004	-7.741e-004	-7.741e-004
17	Accel. Gradient (kPa/m)	2.765e-007	4.440e-007	4.440e-007	4.440e-007
18	Liq. Reynolds	2.075e+005 *	2.075e+005 *	2.075e+005 *	2.075e+005 *
19	Vap. Reynolds	9.191e+006 *	9.191e+006 *	9.191e+006 *	9.191e+006 *
20	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
21	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	3.148e+004	3.148e+004	3.149e+004	3.149e+004
25	Elevation (m)	1.200e-002	1.000e-002	2.000e-002	3.000e-002
26	Pressure (bar)	88.09	88.09	88.08	88.08
27	Temperature (C)	49.16	49.16	49.16	49.16
28	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
29	Heat Transferred (kJ/h-m)	510.5	510.4	510.4	510.3
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2067	0.2067	0.2857	0.2857
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.878e-003	6.878e-003	5.161e-003	5.161e-003
35	Static Gradient (kPa/m)	-7.741e-004	-7.741e-004	4.812e-003	4.812e-003
36	Accel. Gradient (kPa/m)	4.440e-007	4.440e-007	7.255e-007	7.255e-007
37	Liq. Reynolds	2.075e+005 *	2.074e+005 *	2.074e+005 *	2.074e+005 *
38	Vap. Reynolds	9.191e+006 *	9.191e+006 *	9.191e+006 *	9.191e+006 *
39	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
40	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	3.150e+004	3.150e+004	3.151e+004	3.151e+004
44	Elevation (m)	4.000e-002	5.000e-002	6.000e-002	5.400e-002
45	Pressure (bar)	88.08	88.08	88.08	88.08
46	Temperature (C)	49.15	49.15	49.15	49.15
47	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
48	Heat Transferred (kJ/h-m)	510.3	510.2	510.2	510.2
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2857	0.2857	0.2857	0.2060
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	5.161e-003	5.161e-003	5.161e-003	6.884e-003
54	Static Gradient (kPa/m)	4.812e-003	4.812e-003	4.812e-003	-2.317e-003
55	Accel. Gradient (kPa/m)	7.255e-007	7.255e-007	7.255e-007	3.322e-007
56	Liq. Reynolds	2.074e+005 *	2.074e+005 *	2.074e+005 *	2.074e+005 *
57	Vap. Reynolds	9.191e+006 *	9.191e+006 *	9.191e+006 *	9.191e+006 *
58	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
59	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	3.152e+004	3.152e+004	3.153e+004	3.153e+004
63	Elevation (m)	4.800e-002	4.200e-002	3.600e-002	3.000e-002
64	Pressure (bar)	88.08	88.08	88.08	88.08
65	Temperature (C)	49.15	49.15	49.15	49.14
66	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
67	Heat Transferred (kJ/h-m)	510.1	510.1	510.1	510.0
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 503 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Pipe Segment: ELEP-Akabel Pipeline (continued)		
3					
4					
5			Unit Set: Mixed		
6			Date/Time: Mon Jun 13 15:02:44 2022		
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2060	0.2060	0.2060	0.2060
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.884e-003	6.884e-003	6.884e-003	6.884e-003
17	Static Gradient (kPa/m)	-2.317e-003	-2.317e-003	-2.317e-003	-2.317e-003
18	Accel. Gradient (kPa/m)	3.322e-007	3.322e-007	3.322e-007	3.322e-007
19	Liq. Reynolds	2.074e+005 *	2.074e+005 *	2.074e+005 *	2.074e+005 *
20	Vap. Reynolds	9.191e+006 *	9.191e+006 *	9.191e+006 *	9.191e+006 *
21	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
22	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	3.154e+004	3.154e+004	3.155e+004	3.155e+004
26	Elevation (m)	5.000e-002	7.000e-002	9.000e-002	0.1100
27	Pressure (bar)	88.08	88.08	88.08	88.08
28	Temperature (C)	49.14	49.14	49.14	49.14
29	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
30	Heat Transferred (kJ/h-m)	510.0	509.9	509.9	509.8
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2920	0.2920	0.2920	0.2920
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	4.705e-003	4.705e-003	4.705e-003	4.705e-003
36	Static Gradient (kPa/m)	9.720e-003	9.720e-003	9.720e-003	9.720e-003
37	Accel. Gradient (kPa/m)	1.049e-006	1.049e-006	1.049e-006	1.049e-006
38	Liq. Reynolds	2.074e+005 *	2.074e+005 *	2.074e+005 *	2.074e+005 *
39	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *
40	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
41	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	3.156e+004	3.157e+004	3.157e+004	3.158e+004
45	Elevation (m)	0.1300	0.1060	8.200e-002	5.800e-002
46	Pressure (bar)	88.08	88.08	88.08	88.08
47	Temperature (C)	49.14	49.13	49.13	49.13
48	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
49	Heat Transferred (kJ/h-m)	509.8	509.8	509.7	509.7
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2920	0.2026	0.2026	0.2026
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	4.705e-003	6.911e-003	6.911e-003	6.911e-003
55	Static Gradient (kPa/m)	9.720e-003	-9.173e-003	-9.173e-003	-9.173e-003
56	Accel. Gradient (kPa/m)	1.049e-006	-1.645e-007	-1.645e-007	-1.645e-007
57	Liq. Reynolds	2.074e+005 *	2.074e+005 *	2.074e+005 *	2.074e+005 *
58	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *
59	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
60	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	3.158e+004	3.159e+004	3.159e+004	3.160e+004
64	Elevation (m)	3.400e-002	1.000e-002	3.400e-002	5.800e-002
65	Pressure (bar)	88.08	88.08	88.08	88.08
66	Temperature (C)	49.13	49.13	49.13	49.13
67	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
68	Heat Transferred (kJ/h-m)	509.6	509.6	509.6	509.5
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2026	0.2026	0.2946	0.2946
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	6.911e-003	6.911e-003	5.145e-003	5.145e-003
18	Static Gradient (kPa/m)	-9.173e-003	-9.173e-003	1.180e-002	1.180e-002
19	Accel. Gradient (kPa/m)	-1.645e-007	-1.645e-007	1.233e-006	1.233e-006
20	Liq. Reynolds	2.074e+005 *	2.074e+005 *	2.074e+005 *	2.074e+005 *
21	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *
22	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
23	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	3.160e+004	3.161e+004	3.161e+004	3.162e+004
27	Elevation (m)	8.200e-002	0.1060	0.1300	0.1160
28	Pressure (bar)	88.08	88.08	88.07	88.07
29	Temperature (C)	49.12	49.12	49.12	49.12
30	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
31	Heat Transferred (kJ/h-m)	509.5	509.4	509.4	509.3
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2946	0.2946	0.2946	0.2045
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	5.145e-003	5.145e-003	5.145e-003	6.896e-003
37	Static Gradient (kPa/m)	1.180e-002	1.180e-002	1.180e-002	-5.382e-003
38	Accel. Gradient (kPa/m)	1.233e-006	1.233e-006	1.233e-006	1.101e-007
39	Liq. Reynolds	2.074e+005 *	2.074e+005 *	2.074e+005 *	2.074e+005 *
40	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *
41	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
42	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	3.162e+004	3.163e+004	3.163e+004	3.164e+004
46	Elevation (m)	0.1020	8.800e-002	7.400e-002	6.000e-002
47	Pressure (bar)	88.07	88.07	88.07	88.07
48	Temperature (C)	49.12	49.12	49.11	49.11
49	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
50	Heat Transferred (kJ/h-m)	509.3	509.3	509.2	509.2
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2045	0.2045	0.2045	0.2045
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	6.895e-003	6.895e-003	6.895e-003	6.895e-003
56	Static Gradient (kPa/m)	-5.382e-003	-5.382e-003	-5.382e-003	-5.382e-003
57	Accel. Gradient (kPa/m)	1.101e-007	1.101e-007	1.101e-007	1.101e-007
58	Liq. Reynolds	2.074e+005 *	2.074e+005 *	2.074e+005 *	2.074e+005 *
59	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *
60	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
61	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	3.164e+004	3.165e+004	3.165e+004	3.166e+004
65	Elevation (m)	6.800e-002	7.600e-002	8.400e-002	9.200e-002
66	Pressure (bar)	88.07	88.07	88.07	88.07
67	Temperature (C)	49.11	49.11	49.11	49.11
68	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Pipe Segment: ELEP-Akabel Pipeline (continued)				
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022		
6							
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Heat Transferred (kJ/h-m)	509.1	509.1	509.0	509.0		
14	Flow Regime	Transition	Transition	Transition	Transition		
15	Water Flow Regime						
16	Liquid HoldUp	0.2845	0.2845	0.2845	0.2845		
17	Water HoldUp	---	---	---	---		
18	Friction Gradient (kPa/m)	5.256e-003	5.256e-003	5.256e-003	5.256e-003		
19	Static Gradient (kPa/m)	3.838e-003	3.838e-003	3.838e-003	3.838e-003		
20	Accel. Gradient (kPa/m)	6.615e-007	6.615e-007	6.615e-007	6.615e-007		
21	Liq. Reynolds	2.074e+005 *	2.074e+005 *	2.074e+005 *	2.074e+005 *		
22	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *		
23	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
24	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173		
25	Deposit Thickness (mm)	---	---	---	---		
26	Deposit Volume (m3)	---	---	---	---		
27	Length (m)	3.166e+004	3.167e+004	3.167e+004	3.168e+004		
28	Elevation (m)	1.000e-001	0.1080	0.1160	0.1240		
29	Pressure (bar)	88.07	88.07	88.07	88.07		
30	Temperature (C)	49.11	49.10	49.10	49.10		
31	Vapour Fraction	0.8687	0.8687	0.8687	0.8687		
32	Heat Transferred (kJ/h-m)	509.0	508.9	508.9	508.8		
33	Flow Regime	Transition	Transition	Transition	Transition		
34	Water Flow Regime						
35	Liquid HoldUp	0.2845	0.2845	0.2845	0.2845		
36	Water HoldUp	---	---	---	---		
37	Friction Gradient (kPa/m)	5.256e-003	5.256e-003	5.256e-003	5.256e-003		
38	Static Gradient (kPa/m)	3.838e-003	3.838e-003	3.838e-003	3.838e-003		
39	Accel. Gradient (kPa/m)	6.615e-007	6.615e-007	6.615e-007	6.615e-007		
40	Liq. Reynolds	2.074e+005 *	2.074e+005 *	2.074e+005 *	2.074e+005 *		
41	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *		
42	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
43	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173		
44	Deposit Thickness (mm)	---	---	---	---		
45	Deposit Volume (m3)	---	---	---	---		
46	Length (m)	3.168e+004	3.169e+004	3.169e+004	3.169e+004		
47	Elevation (m)	0.1320	0.1400	0.1360	0.1320		
48	Pressure (bar)	88.07	88.07	88.07	88.07		
49	Temperature (C)	49.10	49.10	49.10	49.10		
50	Vapour Fraction	0.8687	0.8687	0.8687	0.8687		
51	Heat Transferred (kJ/h-m)	508.8	508.7	508.7	508.7		
52	Flow Regime	Transition	Transition	Transition	Transition		
53	Water Flow Regime						
54	Liquid HoldUp	0.2845	0.2845	0.2059	0.2059		
55	Water HoldUp	---	---	---	---		
56	Friction Gradient (kPa/m)	5.256e-003	5.256e-003	6.884e-003	6.884e-003		
57	Static Gradient (kPa/m)	3.838e-003	3.838e-003	-2.574e-003	-2.574e-003		
58	Accel. Gradient (kPa/m)	6.615e-007	6.615e-007	3.136e-007	3.136e-007		
59	Liq. Reynolds	2.074e+005 *	2.074e+005 *	2.074e+005 *	2.074e+005 *		
60	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *		
61	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
62	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173		
63	Deposit Thickness (mm)	---	---	---	---		
64	Deposit Volume (m3)	---	---	---	---		
65	Length (m)	3.169e+004	3.170e+004	3.170e+004	3.170e+004		
66	Elevation (m)	0.1280	0.1240	0.1200	0.1160		
67	Pressure (bar)	88.07	88.07	88.07	88.07		
68	Temperature (C)	49.09	49.09	49.09	49.09		
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 506 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
14	Heat Transferred (kJ/h-m)	508.7	508.7	508.6	508.6
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2059	0.2059	0.2059	0.2052
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	6.884e-003	6.884e-003	6.884e-003	6.889e-003
20	Static Gradient (kPa/m)	-2.574e-003	-2.574e-003	-2.574e-003	-3.853e-003
21	Accel. Gradient (kPa/m)	3.136e-007	3.135e-007	3.135e-007	2.208e-007
22	Liq. Reynolds	2.074e+005 *	2.074e+005 *	2.074e+005 *	2.074e+005 *
23	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *
24	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
25	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	3.170e+004	3.171e+004	3.171e+004	3.171e+004
29	Elevation (m)	0.1120	0.1080	0.1040	1.000e-001
30	Pressure (bar)	88.07	88.07	88.07	88.07
31	Temperature (C)	49.09	49.09	49.09	49.09
32	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
33	Heat Transferred (kJ/h-m)	508.6	508.6	508.6	508.6
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2052	0.2052	0.2053	0.2053
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	6.889e-003	6.889e-003	6.889e-003	6.889e-003
39	Static Gradient (kPa/m)	-3.853e-003	-3.853e-003	-3.853e-003	-3.853e-003
40	Accel. Gradient (kPa/m)	2.208e-007	2.208e-007	2.208e-007	2.208e-007
41	Liq. Reynolds	2.074e+005 *	2.074e+005 *	2.074e+005 *	2.074e+005 *
42	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *
43	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
44	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	3.172e+004	3.172e+004	3.173e+004	3.173e+004
48	Elevation (m)	6.800e-002	3.600e-002	4.000e-003	-2.800e-002
49	Pressure (bar)	88.07	88.07	88.07	88.07
50	Temperature (C)	49.09	49.09	49.09	49.08
51	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
52	Heat Transferred (kJ/h-m)	508.5	508.5	508.5	508.4
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2012	0.2012	0.2012	0.2012
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	6.923e-003	6.923e-003	6.923e-003	6.923e-003
58	Static Gradient (kPa/m)	-1.217e-002	-1.217e-002	-1.217e-002	-1.217e-002
59	Accel. Gradient (kPa/m)	-3.820e-007	-3.820e-007	-3.820e-007	-3.821e-007
60	Liq. Reynolds	2.074e+005 *	2.074e+005 *	2.074e+005 *	2.074e+005 *
61	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *
62	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
63	Vapour Velocity (m/s)	2.173	2.173	2.173	2.173
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	3.174e+004	3.174e+004	3.174e+004	3.174e+004
67	Elevation (m)	-6.000e-002	-3.800e-002	-1.600e-002	6.000e-003
68	Pressure (bar)	88.07	88.07	88.07	88.07
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Temperature (C)	49.08	49.08	49.08	49.08		
14	Vapour Fraction	0.8687	0.8687	0.8687	0.8687		
15	Heat Transferred (kJ/h-m)	508.4	508.4	508.3	508.3		
16	Flow Regime	Transition	Transition	Transition	Transition		
17	Water Flow Regime						
18	Liquid HoldUp	0.2012	0.3027	0.3027	0.3027		
19	Water HoldUp	---	---	---	---		
20	Friction Gradient (kPa/m)	6.922e-003	5.896e-003	5.896e-003	5.896e-003		
21	Static Gradient (kPa/m)	-1.218e-002	1.839e-002	1.839e-002	1.839e-002		
22	Accel. Gradient (kPa/m)	-3.821e-007	1.766e-006	1.766e-006	1.766e-006		
23	Liq. Reynolds	2.074e+005 *	2.074e+005 *	2.074e+005 *	2.074e+005 *		
24	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *		
25	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
26	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172		
27	Deposit Thickness (mm)	---	---	---	---		
28	Deposit Volume (m3)	---	---	---	---		
29	Length (m)	3.175e+004	3.175e+004	3.176e+004	3.176e+004		
30	Elevation (m)	2.800e-002	5.000e-002	3.800e-002	2.600e-002		
31	Pressure (bar)	88.07	88.07	88.07	88.07		
32	Temperature (C)	49.08	49.08	49.08	49.07		
33	Vapour Fraction	0.8687	0.8687	0.8687	0.8687		
34	Heat Transferred (kJ/h-m)	508.3	508.2	508.2	508.1		
35	Flow Regime	Transition	Transition	Transition	Transition		
36	Water Flow Regime						
37	Liquid HoldUp	0.3027	0.3027	0.2053	0.2053		
38	Water HoldUp	---	---	---	---		
39	Friction Gradient (kPa/m)	5.896e-003	5.896e-003	6.889e-003	6.889e-003		
40	Static Gradient (kPa/m)	1.839e-002	1.839e-002	-3.853e-003	-3.854e-003		
41	Accel. Gradient (kPa/m)	1.766e-006	1.766e-006	2.208e-007	2.208e-007		
42	Liq. Reynolds	2.074e+005 *	2.074e+005 *	2.074e+005 *	2.074e+005 *		
43	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *		
44	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
45	Vapour Velocity (m/s)	2.173	2.173	2.172	2.172		
46	Deposit Thickness (mm)	---	---	---	---		
47	Deposit Volume (m3)	---	---	---	---		
48	Length (m)	3.177e+004	3.177e+004	3.178e+004	3.179e+004		
49	Elevation (m)	1.400e-002	2.000e-003	-1.000e-002	6.000e-003		
50	Pressure (bar)	88.07	88.07	88.07	88.06		
51	Temperature (C)	49.07	49.07	49.07	49.07		
52	Vapour Fraction	0.8687	0.8687	0.8687	0.8687		
53	Heat Transferred (kJ/h-m)	508.1	508.1	508.0	507.9		
54	Flow Regime	Transition	Transition	Transition	Transition		
55	Water Flow Regime						
56	Liquid HoldUp	0.2053	0.2053	0.2053	0.2879		
57	Water HoldUp	---	---	---	---		
58	Friction Gradient (kPa/m)	6.889e-003	6.889e-003	6.889e-003	5.004e-003		
59	Static Gradient (kPa/m)	-3.854e-003	-3.854e-003	-3.854e-003	6.438e-003		
60	Accel. Gradient (kPa/m)	2.208e-007	2.208e-007	2.208e-007	8.323e-007		
61	Liq. Reynolds	2.074e+005 *	2.074e+005 *	2.074e+005 *	2.074e+005 *		
62	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *		
63	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
64	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172		
65	Deposit Thickness (mm)	---	---	---	---		
66	Deposit Volume (m3)	---	---	---	---		
67	Length (m)	3.179e+004	3.180e+004	3.180e+004	3.181e+004		
68	Elevation (m)	2.200e-002	3.800e-002	5.400e-002	7.000e-002		
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Pressure (bar)	88.06	88.06	88.06	88.06
14	Temperature (C)	49.06	49.06	49.06	49.06
15	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
16	Heat Transferred (kJ/h-m)	507.9	507.8	507.8	507.7
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.2879	0.2879	0.2879	0.2879
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	5.004e-003	5.004e-003	5.004e-003	5.004e-003
22	Static Gradient (kPa/m)	6.438e-003	6.438e-003	6.438e-003	6.438e-003
23	Accel. Gradient (kPa/m)	8.323e-007	8.323e-007	8.323e-007	8.323e-007
24	Liq. Reynolds	2.074e+005 *	2.074e+005 *	2.074e+005 *	2.073e+005 *
25	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *
26	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
27	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	3.182e+004	3.182e+004	3.183e+004	3.183e+004
31	Elevation (m)	6.200e-002	5.400e-002	4.600e-002	3.800e-002
32	Pressure (bar)	88.06	88.06	88.06	88.06
33	Temperature (C)	49.06	49.06	49.05	49.05
34	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
35	Heat Transferred (kJ/h-m)	507.7	507.7	507.6	507.6
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.2056	0.2056	0.2056	0.2056
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	6.886e-003	6.886e-003	6.885e-003	6.885e-003
41	Static Gradient (kPa/m)	-3.086e-003	-3.086e-003	-3.086e-003	-3.086e-003
42	Accel. Gradient (kPa/m)	2.763e-007	2.763e-007	2.763e-007	2.763e-007
43	Liq. Reynolds	2.073e+005 *	2.073e+005 *	2.073e+005 *	2.073e+005 *
44	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *
45	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
46	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	3.184e+004	3.184e+004	3.185e+004	3.185e+004
50	Elevation (m)	3.000e-002	4.400e-002	5.800e-002	7.200e-002
51	Pressure (bar)	88.06	88.06	88.06	88.06
52	Temperature (C)	49.05	49.05	49.05	49.05
53	Vapour Fraction	0.8687	0.8687	0.8687	0.8687
54	Heat Transferred (kJ/h-m)	507.5	507.5	507.4	507.4
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.2056	0.2883	0.2883	0.2883
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	6.885e-003	4.972e-003	4.972e-003	4.972e-003
60	Static Gradient (kPa/m)	-3.087e-003	6.781e-003	6.781e-003	6.781e-003
61	Accel. Gradient (kPa/m)	2.763e-007	8.549e-007	8.549e-007	8.549e-007
62	Liq. Reynolds	2.073e+005 *	2.073e+005 *	2.073e+005 *	2.073e+005 *
63	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *
64	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
65	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	3.186e+004	3.186e+004	3.187e+004	3.187e+004
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	8.600e-002	1.000e-001	0.1020	0.1040
14	Pressure (bar)	88.06	88.06	88.06	88.06
15	Temperature (C)	49.04	49.04	49.04	49.04
16	Vapour Fraction	0.8687	0.8687	0.8686	0.8686
17	Heat Transferred (kJ/h-m)	507.4	507.3	507.3	507.2
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2883	0.2883	0.2807	0.2807
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	4.972e-003	4.972e-003	5.551e-003	5.551e-003
23	Static Gradient (kPa/m)	6.781e-003	6.781e-003	9.505e-004	9.505e-004
24	Accel. Gradient (kPa/m)	8.549e-007	8.549e-007	4.729e-007	4.729e-007
25	Liq. Reynolds	2.073e+005 *	2.073e+005 *	2.073e+005 *	2.073e+005 *
26	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *
27	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
28	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	3.188e+004	3.188e+004	3.189e+004	3.189e+004
32	Elevation (m)	0.1060	0.1080	0.1100	0.1060
33	Pressure (bar)	88.06	88.06	88.06	88.06
34	Temperature (C)	49.04	49.04	49.04	49.03
35	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
36	Heat Transferred (kJ/h-m)	507.2	507.2	507.1	507.1
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2807	0.2807	0.2807	0.2064
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	5.551e-003	5.551e-003	5.551e-003	6.879e-003
42	Static Gradient (kPa/m)	9.505e-004	9.505e-004	9.505e-004	-1.547e-003
43	Accel. Gradient (kPa/m)	4.729e-007	4.729e-007	4.729e-007	3.879e-007
44	Liq. Reynolds	2.073e+005 *	2.073e+005 *	2.073e+005 *	2.073e+005 *
45	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *
46	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
47	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	3.190e+004	3.190e+004	3.191e+004	3.191e+004
51	Elevation (m)	0.1020	9.800e-002	9.400e-002	9.000e-002
52	Pressure (bar)	88.06	88.06	88.06	88.05
53	Temperature (C)	49.03	49.03	49.03	49.03
54	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
55	Heat Transferred (kJ/h-m)	507.0	507.0	507.0	506.9
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2064	0.2064	0.2064	0.2064
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	6.879e-003	6.879e-003	6.879e-003	6.879e-003
61	Static Gradient (kPa/m)	-1.547e-003	-1.547e-003	-1.547e-003	-1.547e-003
62	Accel. Gradient (kPa/m)	3.879e-007	3.879e-007	3.879e-007	3.879e-007
63	Liq. Reynolds	2.073e+005 *	2.073e+005 *	2.073e+005 *	2.073e+005 *
64	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *
65	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
66	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	3.192e+004	3.192e+004	3.193e+004	3.193e+004
14	Elevation (m)	0.1480	0.2060	0.2640	0.3220
15	Pressure (bar)	88.05	88.05	88.05	88.05
16	Temperature (C)	49.03	49.02	49.02	49.02
17	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
18	Heat Transferred (kJ/h-m)	506.9	506.8	506.8	506.7
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.3163	0.3163	0.3163	0.3163
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	6.060e-003	6.060e-003	6.060e-003	6.060e-003
24	Static Gradient (kPa/m)	3.003e-002	3.003e-002	3.003e-002	3.003e-002
25	Accel. Gradient (kPa/m)	2.625e-006	2.625e-006	2.625e-006	2.625e-006
26	Liq. Reynolds	2.073e+005 *	2.073e+005 *	2.073e+005 *	2.073e+005 *
27	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *
28	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
29	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	3.194e+004	3.194e+004	3.195e+004	3.195e+004
33	Elevation (m)	0.3800	0.3520	0.3240	0.2960
34	Pressure (bar)	88.05	88.05	88.05	88.05
35	Temperature (C)	49.02	49.02	49.02	49.01
36	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
37	Heat Transferred (kJ/h-m)	506.6	506.6	506.6	506.5
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.3163	0.2019	0.2019	0.2019
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	6.060e-003	6.916e-003	6.916e-003	6.916e-003
43	Static Gradient (kPa/m)	3.003e-002	-1.068e-002	-1.068e-002	-1.068e-002
44	Accel. Gradient (kPa/m)	2.625e-006	-2.738e-007	-2.738e-007	-2.738e-007
45	Liq. Reynolds	2.073e+005 *	2.073e+005 *	2.073e+005 *	2.073e+005 *
46	Vap. Reynolds	9.193e+006 *	9.193e+006 *	9.192e+006 *	9.192e+006 *
47	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
48	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	3.196e+004	3.196e+004	3.197e+004	3.197e+004
52	Elevation (m)	0.2680	0.2400	0.2400	0.2400
53	Pressure (bar)	88.05	88.05	88.05	88.05
54	Temperature (C)	49.01	49.01	49.01	49.01
55	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
56	Heat Transferred (kJ/h-m)	506.5	506.5	506.4	506.4
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.2019	0.2019	0.3024	0.3024
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	6.916e-003	6.916e-003	5.886e-003	5.886e-003
62	Static Gradient (kPa/m)	-1.068e-002	-1.068e-002	0.0000	0.0000
63	Accel. Gradient (kPa/m)	-2.738e-007	-2.738e-007	4.282e-007	4.282e-007
64	Liq. Reynolds	2.073e+005 *	2.073e+005 *	2.073e+005 *	2.073e+005 *
65	Vap. Reynolds	9.192e+006 *	9.192e+006 *	9.192e+006 *	9.192e+006 *
66	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
67	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
68	Deposit Thickness (mm)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4							
5			Date/Time: Mon Jun 13 15:02:44 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Volume (m3)	---	---	---	---		
14	Length (m)	3.198e+004	3.198e+004	3.199e+004	3.199e+004		
15	Elevation (m)	0.2400	0.2400	0.2400	0.2340		
16	Pressure (bar)	88.05	88.05	88.05	88.05		
17	Temperature (C)	49.01	49.01	49.00	49.00		
18	Vapour Fraction	0.8686	0.8686	0.8686	0.8686		
19	Heat Transferred (kJ/h-m)	506.3	506.3	506.3	506.2		
20	Flow Regime	Transition	Transition	Transition	Transition		
21	Water Flow Regime						
22	Liquid HoldUp	0.3024	0.3024	0.3024	0.2060		
23	Water HoldUp	---	---	---	---		
24	Friction Gradient (kPa/m)	5.886e-003	5.886e-003	5.886e-003	6.882e-003		
25	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-2.289e-003		
26	Accel. Gradient (kPa/m)	4.282e-007	4.282e-007	4.282e-007	3.342e-007		
27	Liq. Reynolds	2.073e+005 *	2.073e+005 *	2.073e+005 *	2.073e+005 *		
28	Vap. Reynolds	9.192e+006 *	9.193e+006 *	9.193e+006 *	9.193e+006 *		
29	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
30	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172		
31	Deposit Thickness (mm)	---	---	---	---		
32	Deposit Volume (m3)	---	---	---	---		
33	Length (m)	3.200e+004	3.200e+004	3.201e+004	3.201e+004		
34	Elevation (m)	0.2280	0.2220	0.2160	0.2100		
35	Pressure (bar)	88.04	88.04	88.04	88.04		
36	Temperature (C)	49.00	49.00	49.00	49.00		
37	Vapour Fraction	0.8686	0.8686	0.8686	0.8686		
38	Heat Transferred (kJ/h-m)	506.2	506.1	506.1	506.1		
39	Flow Regime	Transition	Transition	Transition	Transition		
40	Water Flow Regime						
41	Liquid HoldUp	0.2060	0.2060	0.2060	0.2060		
42	Water HoldUp	---	---	---	---		
43	Friction Gradient (kPa/m)	6.882e-003	6.882e-003	6.882e-003	6.882e-003		
44	Static Gradient (kPa/m)	-2.289e-003	-2.289e-003	-2.289e-003	-2.289e-003		
45	Accel. Gradient (kPa/m)	3.342e-007	3.342e-007	3.341e-007	3.341e-007		
46	Liq. Reynolds	2.073e+005 *	2.073e+005 *	2.073e+005 *	2.073e+005 *		
47	Vap. Reynolds	9.193e+006 *	9.193e+006 *	9.193e+006 *	9.193e+006 *		
48	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
49	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172		
50	Deposit Thickness (mm)	---	---	---	---		
51	Deposit Volume (m3)	---	---	---	---		
52	Length (m)	3.202e+004	3.202e+004	3.203e+004	3.203e+004		
53	Elevation (m)	0.2040	0.1980	0.1920	0.1860		
54	Pressure (bar)	88.04	88.04	88.04	88.04		
55	Temperature (C)	48.99	48.99	48.99	48.99		
56	Vapour Fraction	0.8686	0.8686	0.8686	0.8686		
57	Heat Transferred (kJ/h-m)	506.0	506.0	505.9	505.9		
58	Flow Regime	Transition	Transition	Transition	Transition		
59	Water Flow Regime						
60	Liquid HoldUp	0.2060	0.2060	0.2060	0.2060		
61	Water HoldUp	---	---	---	---		
62	Friction Gradient (kPa/m)	6.882e-003	6.882e-003	6.882e-003	6.882e-003		
63	Static Gradient (kPa/m)	-2.318e-003	-2.318e-003	-2.318e-003	-2.318e-003		
64	Accel. Gradient (kPa/m)	3.320e-007	3.320e-007	3.320e-007	3.320e-007		
65	Liq. Reynolds	2.073e+005 *	2.073e+005 *	2.073e+005 *	2.073e+005 *		
66	Vap. Reynolds	9.193e+006 *	9.193e+006 *	9.193e+006 *	9.193e+006 *		
67	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
68	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	3.204e+004	3.204e+004	3.205e+004	3.205e+004
16	Elevation (m)	0.1800	0.1760	0.1720	0.1680
17	Pressure (bar)	88.04	88.04	88.04	88.04
18	Temperature (C)	48.99	48.99	48.99	48.98
19	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
20	Heat Transferred (kJ/h-m)	505.9	505.8	505.8	505.7
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2060	0.2064	0.2064	0.2064
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	6.882e-003	6.879e-003	6.879e-003	6.879e-003
26	Static Gradient (kPa/m)	-2.318e-003	-1.547e-003	-1.547e-003	-1.547e-003
27	Accel. Gradient (kPa/m)	3.320e-007	3.879e-007	3.879e-007	3.879e-007
28	Liq. Reynolds	2.073e+005 *	2.073e+005 *	2.073e+005 *	2.073e+005 *
29	Vap. Reynolds	9.193e+006 *	9.193e+006 *	9.193e+006 *	9.193e+006 *
30	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
31	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	3.206e+004	3.206e+004	3.207e+004	3.207e+004
35	Elevation (m)	0.1640	0.1600	0.1520	0.1440
36	Pressure (bar)	88.04	88.04	88.04	88.04
37	Temperature (C)	48.98	48.98	48.98	48.98
38	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
39	Heat Transferred (kJ/h-m)	505.7	505.7	505.6	505.6
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2064	0.2064	0.2057	0.2057
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	6.879e-003	6.879e-003	6.885e-003	6.885e-003
45	Static Gradient (kPa/m)	-1.547e-003	-1.547e-003	-3.087e-003	-3.087e-003
46	Accel. Gradient (kPa/m)	3.879e-007	3.879e-007	2.763e-007	2.763e-007
47	Liq. Reynolds	2.073e+005 *	2.073e+005 *	2.073e+005 *	2.073e+005 *
48	Vap. Reynolds	9.193e+006 *	9.193e+006 *	9.193e+006 *	9.193e+006 *
49	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
50	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	3.208e+004	3.208e+004	3.209e+004	3.209e+004
54	Elevation (m)	0.1360	0.1280	0.1200	0.1260
55	Pressure (bar)	88.04	88.04	88.04	88.04
56	Temperature (C)	48.98	48.98	48.97	48.97
57	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
58	Heat Transferred (kJ/h-m)	505.6	505.5	505.5	505.4
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2057	0.2057	0.2057	0.2832
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	6.885e-003	6.884e-003	6.884e-003	5.351e-003
64	Static Gradient (kPa/m)	-3.087e-003	-3.087e-003	-3.087e-003	2.870e-003
65	Accel. Gradient (kPa/m)	2.762e-007	2.762e-007	2.762e-007	5.981e-007
66	Liq. Reynolds	2.073e+005 *	2.073e+005 *	2.073e+005 *	2.073e+005 *
67	Vap. Reynolds	9.193e+006 *	9.193e+006 *	9.193e+006 *	9.193e+006 *
68	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
14	Deposit Thickness (mm)	---	---	---	---
15	Deposit Volume (m3)	---	---	---	---
16	Length (m)	3.210e+004	3.210e+004	3.211e+004	3.211e+004
17	Elevation (m)	0.1320	0.1380	0.1440	0.1500
18	Pressure (bar)	88.04	88.04	88.04	88.04
19	Temperature (C)	48.97	48.97	48.97	48.97
20	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
21	Heat Transferred (kJ/h-m)	505.4	505.3	505.3	505.3
22	Flow Regime	Transition	Transition	Transition	Transition
23	Water Flow Regime				
24	Liquid HoldUp	0.2832	0.2832	0.2832	0.2833
25	Water HoldUp	---	---	---	---
26	Friction Gradient (kPa/m)	5.351e-003	5.351e-003	5.351e-003	5.351e-003
27	Static Gradient (kPa/m)	2.870e-003	2.870e-003	2.870e-003	2.870e-003
28	Accel. Gradient (kPa/m)	5.981e-007	5.981e-007	5.981e-007	5.981e-007
29	Liq. Reynolds	2.073e+005 *	2.073e+005 *	2.073e+005 *	2.073e+005 *
30	Vap. Reynolds	9.193e+006 *	9.193e+006 *	9.193e+006 *	9.193e+006 *
31	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
32	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
33	Deposit Thickness (mm)	---	---	---	---
34	Deposit Volume (m3)	---	---	---	---
35	Length (m)	3.212e+004	3.212e+004	3.213e+004	3.213e+004
36	Elevation (m)	0.1540	0.1580	0.1620	0.1660
37	Pressure (bar)	88.04	88.04	88.04	88.04
38	Temperature (C)	48.96	48.96	48.96	48.96
39	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
40	Heat Transferred (kJ/h-m)	505.2	505.2	505.1	505.1
41	Flow Regime	Transition	Transition	Transition	Transition
42	Water Flow Regime				
43	Liquid HoldUp	0.2820	0.2820	0.2820	0.2820
44	Water HoldUp	---	---	---	---
45	Friction Gradient (kPa/m)	5.450e-003	5.450e-003	5.450e-003	5.450e-003
46	Static Gradient (kPa/m)	1.907e-003	1.907e-003	1.907e-003	1.907e-003
47	Accel. Gradient (kPa/m)	5.352e-007	5.352e-007	5.352e-007	5.352e-007
48	Liq. Reynolds	2.073e+005 *	2.073e+005 *	2.073e+005 *	2.072e+005 *
49	Vap. Reynolds	9.193e+006 *	9.193e+006 *	9.193e+006 *	9.193e+006 *
50	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
51	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
52	Deposit Thickness (mm)	---	---	---	---
53	Deposit Volume (m3)	---	---	---	---
54	Length (m)	3.214e+004	3.214e+004	3.215e+004	3.215e+004
55	Elevation (m)	0.1700	0.2160	0.2620	0.3080
56	Pressure (bar)	88.04	88.04	88.03	88.03
57	Temperature (C)	48.96	48.96	48.95	48.95
58	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
59	Heat Transferred (kJ/h-m)	505.0	505.0	505.0	504.9
60	Flow Regime	Transition	Transition	Transition	Transition
61	Water Flow Regime				
62	Liquid HoldUp	0.2820	0.3087	0.3087	0.3087
63	Water HoldUp	---	---	---	---
64	Friction Gradient (kPa/m)	5.450e-003	6.005e-003	6.005e-003	6.005e-003
65	Static Gradient (kPa/m)	1.907e-003	2.340e-002	2.340e-002	2.340e-002
66	Accel. Gradient (kPa/m)	5.352e-007	2.139e-006	2.139e-006	2.139e-006
67	Liq. Reynolds	2.072e+005 *	2.072e+005 *	2.072e+005 *	2.072e+005 *
68	Vap. Reynolds	9.193e+006 *	9.193e+006 *	9.193e+006 *	9.193e+006 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc							
2										
3							Unit Set: Mixed			
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5										
6	Pipe Segment: ELEP-Akabel Pipeline (continued)									
7										
8										
9	PROFILES									
10										
11	Pipe Table									
12										
13	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046				
14	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172				
15	Deposit Thickness	(mm)	---	---	---	---				
16	Deposit Volume	(m3)	---	---	---	---				
17	Length	(m)	3.216e+004	3.216e+004	3.217e+004	3.217e+004				
18	Elevation	(m)	0.3540	0.4000	0.4040	0.4080				
19	Pressure	(bar)	88.03	88.03	88.03	88.03				
20	Temperature	(C)	48.95	48.95	48.95	48.95				
21	Vapour Fraction		0.8686	0.8686	0.8686	0.8686				
22	Heat Transferred	(kJ/h-m)	504.8	504.8	504.7	504.7				
23	Flow Regime		Transition	Transition	Transition	Transition				
24	Water Flow Regime									
25	Liquid HoldUp		0.3087	0.3087	0.2820	0.2820				
26	Water HoldUp		---	---	---	---				
27	Friction Gradient	(kPa/m)	6.005e-003	6.005e-003	5.450e-003	5.450e-003				
28	Static Gradient	(kPa/m)	2.340e-002	2.340e-002	1.907e-003	1.907e-003				
29	Accel. Gradient	(kPa/m)	2.139e-006	2.139e-006	5.353e-007	5.353e-007				
30	Liq. Reynolds		2.072e+005 *	2.072e+005 *	2.072e+005 *	2.072e+005 *				
31	Vap. Reynolds		9.193e+006 *	9.193e+006 *	9.193e+006 *	9.193e+006 *				
32	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046				
33	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172				
34	Deposit Thickness	(mm)	---	---	---	---				
35	Deposit Volume	(m3)	---	---	---	---				
36	Length	(m)	3.218e+004	3.218e+004	3.219e+004	3.219e+004				
37	Elevation	(m)	0.4120	0.4160	0.4200	0.4880				
38	Pressure	(bar)	88.03	88.03	88.03	88.03				
39	Temperature	(C)	48.94	48.94	48.94	48.94				
40	Vapour Fraction		0.8686	0.8686	0.8686	0.8686				
41	Heat Transferred	(kJ/h-m)	504.7	504.6	504.6	504.5				
42	Flow Regime		Transition	Transition	Transition	Transition				
43	Water Flow Regime									
44	Liquid HoldUp		0.2820	0.2820	0.2820	0.4943				
45	Water HoldUp		---	---	---	---				
46	Friction Gradient	(kPa/m)	5.450e-003	5.450e-003	5.450e-003	5.878e-003				
47	Static Gradient	(kPa/m)	1.907e-003	1.907e-003	1.907e-003	0.2482				
48	Accel. Gradient	(kPa/m)	5.353e-007	5.353e-007	5.353e-007	1.848e-005				
49	Liq. Reynolds		2.072e+005 *	2.072e+005 *	2.072e+005 *	2.072e+005 *				
50	Vap. Reynolds		9.193e+006 *	9.193e+006 *	9.193e+006 *	9.193e+006 *				
51	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046				
52	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172				
53	Deposit Thickness	(mm)	---	---	---	---				
54	Deposit Volume	(m3)	---	---	---	---				
55	Length	(m)	3.219e+004	3.219e+004	3.219e+004	3.219e+004				
56	Elevation	(m)	0.5560	0.6240	0.6920	0.7600				
57	Pressure	(bar)	88.02	88.02	88.02	88.01				
58	Temperature	(C)	48.94	48.94	48.94	48.94				
59	Vapour Fraction		0.8686	0.8686	0.8686	0.8686				
60	Heat Transferred	(kJ/h-m)	504.5	504.4	504.4	504.4				
61	Flow Regime		Transition	Transition	Transition	Transition				
62	Water Flow Regime									
63	Liquid HoldUp		0.4943	0.4943	0.4943	0.4943				
64	Water HoldUp		---	---	---	---				
65	Friction Gradient	(kPa/m)	5.878e-003	5.878e-003	5.878e-003	5.878e-003				
66	Static Gradient	(kPa/m)	0.2482	0.2482	0.2482	0.2482				
67	Accel. Gradient	(kPa/m)	1.849e-005	1.849e-005	1.849e-005	1.849e-005				
68	Liq. Reynolds		2.072e+005 *	2.072e+005 *	2.072e+005 *	2.072e+005 *				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.193e+006 *	9.193e+006 *	9.193e+006 *	9.193e+006 *
14	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
15	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	3.219e+004	3.220e+004	3.220e+004	3.221e+004
19	Elevation (m)	0.8460	0.9320	1.018	1.104
20	Pressure (bar)	88.01	88.01	88.01	88.00
21	Temperature (C)	48.93	48.93	48.93	48.93
22	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
23	Heat Transferred (kJ/h-m)	504.4	504.4	504.3	504.3
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.3488	0.3488	0.3488	0.3488
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.072e-003	6.073e-003	6.073e-003	6.073e-003
29	Static Gradient (kPa/m)	6.077e-002	6.077e-002	6.077e-002	6.077e-002
30	Accel. Gradient (kPa/m)	4.865e-006	4.865e-006	4.865e-006	4.865e-006
31	Liq. Reynolds	2.072e+005 *	2.072e+005 *	2.072e+005 *	2.072e+005 *
32	Vap. Reynolds	9.194e+006 *	9.194e+006 *	9.194e+006 *	9.194e+006 *
33	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
34	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	3.221e+004	3.221e+004	3.222e+004	3.222e+004
38	Elevation (m)	1.190	1.216	1.242	1.268
39	Pressure (bar)	88.00	88.00	88.00	88.00
40	Temperature (C)	48.93	48.92	48.92	48.92
41	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
42	Heat Transferred (kJ/h-m)	504.2	504.2	504.1	504.1
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.3488	0.3001	0.3001	0.3001
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.073e-003	5.795e-003	5.795e-003	5.795e-003
48	Static Gradient (kPa/m)	6.077e-002	1.620e-002	1.620e-002	1.620e-002
49	Accel. Gradient (kPa/m)	4.865e-006	1.601e-006	1.601e-006	1.601e-006
50	Liq. Reynolds	2.072e+005 *	2.072e+005 *	2.072e+005 *	2.072e+005 *
51	Vap. Reynolds	9.194e+006 *	9.194e+006 *	9.194e+006 *	9.194e+006 *
52	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
53	Vapour Velocity (m/s)	2.172	2.172	2.173	2.173
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	3.223e+004	3.223e+004	3.224e+004	3.226e+004
57	Elevation (m)	1.294	1.320	1.322	1.324
58	Pressure (bar)	88.00	88.00	88.00	88.00
59	Temperature (C)	48.92	48.92	48.91	48.91
60	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
61	Heat Transferred (kJ/h-m)	504.0	504.0	503.9	503.8
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.3001	0.3001	0.2799	0.2799
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	5.795e-003	5.795e-003	5.615e-003	5.615e-003
67	Static Gradient (kPa/m)	1.620e-002	1.620e-002	3.540e-004	3.540e-004
68	Accel. Gradient (kPa/m)	1.601e-006	1.601e-006	4.345e-007	4.345e-007
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 516 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liq. Reynolds	2.072e+005 *	2.072e+005 *	2.072e+005 *	2.072e+005 *
14	Vap. Reynolds	9.194e+006 *	9.194e+006 *	9.194e+006 *	9.194e+006 *
15	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
16	Vapour Velocity (m/s)	2.173	2.173	2.172	2.172
17	Deposit Thickness (mm)	---	---	---	---
18	Deposit Volume (m3)	---	---	---	---
19	Length (m)	3.227e+004	3.228e+004	3.230e+004	3.231e+004
20	Elevation (m)	1.326	1.328	1.330	1.320
21	Pressure (bar)	88.00	87.99	87.99	87.99
22	Temperature (C)	48.91	48.90	48.90	48.89
23	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
24	Heat Transferred (kJ/h-m)	503.7	503.6	503.4	503.4
25	Flow Regime	Transition	Transition	Transition	Transition
26	Water Flow Regime				
27	Liquid HoldUp	0.2799	0.2799	0.2799	0.2064
28	Water HoldUp	---	---	---	---
29	Friction Gradient (kPa/m)	5.615e-003	5.615e-003	5.615e-003	6.880e-003
30	Static Gradient (kPa/m)	3.540e-004	3.540e-004	3.540e-004	-1.443e-003
31	Accel. Gradient (kPa/m)	4.345e-007	4.345e-007	4.345e-007	3.957e-007
32	Liq. Reynolds	2.072e+005 *	2.072e+005 *	2.072e+005 *	2.072e+005 *
33	Vap. Reynolds	9.194e+006 *	9.194e+006 *	9.194e+006 *	9.194e+006 *
34	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
35	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
36	Deposit Thickness (mm)	---	---	---	---
37	Deposit Volume (m3)	---	---	---	---
38	Length (m)	3.232e+004	3.234e+004	3.235e+004	3.236e+004
39	Elevation (m)	1.310	1.300	1.290	1.280
40	Pressure (bar)	87.99	87.99	87.99	87.99
41	Temperature (C)	48.89	48.89	48.88	48.88
42	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
43	Heat Transferred (kJ/h-m)	503.2	503.1	503.0	502.9
44	Flow Regime	Transition	Transition	Transition	Transition
45	Water Flow Regime				
46	Liquid HoldUp	0.2064	0.2064	0.2064	0.2064
47	Water HoldUp	---	---	---	---
48	Friction Gradient (kPa/m)	6.879e-003	6.879e-003	6.879e-003	6.879e-003
49	Static Gradient (kPa/m)	-1.443e-003	-1.443e-003	-1.443e-003	-1.443e-003
50	Accel. Gradient (kPa/m)	3.957e-007	3.957e-007	3.957e-007	3.957e-007
51	Liq. Reynolds	2.071e+005 *	2.071e+005 *	2.071e+005 *	2.071e+005 *
52	Vap. Reynolds	9.194e+006 *	9.194e+006 *	9.194e+006 *	9.194e+006 *
53	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
54	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
55	Deposit Thickness (mm)	---	---	---	---
56	Deposit Volume (m3)	---	---	---	---
57	Length (m)	3.238e+004	3.239e+004	3.240e+004	3.242e+004
58	Elevation (m)	1.292	1.304	1.316	1.328
59	Pressure (bar)	87.99	87.99	87.99	87.99
60	Temperature (C)	48.87	48.87	48.87	48.86
61	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
62	Heat Transferred (kJ/h-m)	502.8	502.7	502.6	502.5
63	Flow Regime	Transition	Transition	Transition	Transition
64	Water Flow Regime				
65	Liquid HoldUp	0.2823	0.2823	0.2823	0.2823
66	Water HoldUp	---	---	---	---
67	Friction Gradient (kPa/m)	5.427e-003	5.427e-003	5.427e-003	5.427e-003
68	Static Gradient (kPa/m)	2.137e-003	2.137e-003	2.137e-003	2.137e-003
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 517 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Accel. Gradient (kPa/m)	5.506e-007	5.506e-007	5.506e-007	5.506e-007
14	Liq. Reynolds	2.071e+005 *	2.071e+005 *	2.071e+005 *	2.071e+005 *
15	Vap. Reynolds	9.194e+006 *	9.194e+006 *	9.194e+006 *	9.194e+006 *
16	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
17	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	3.243e+004	3.244e+004	3.246e+004	3.247e+004
21	Elevation (m)	1.340	1.348	1.356	1.364
22	Pressure (bar)	87.98	87.98	87.98	87.98
23	Temperature (C)	48.86	48.85	48.85	48.85
24	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
25	Heat Transferred (kJ/h-m)	502.4	502.3	502.1	502.0
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.2823	0.2813	0.2813	0.2813
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	5.427e-003	5.501e-003	5.501e-003	5.501e-003
31	Static Gradient (kPa/m)	2.137e-003	1.421e-003	1.421e-003	1.421e-003
32	Accel. Gradient (kPa/m)	5.506e-007	5.039e-007	5.039e-007	5.039e-007
33	Liq. Reynolds	2.071e+005 *	2.071e+005 *	2.071e+005 *	2.071e+005 *
34	Vap. Reynolds	9.194e+006 *	9.194e+006 *	9.194e+006 *	9.194e+006 *
35	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
36	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	3.248e+004	3.250e+004	3.251e+004	3.252e+004
40	Elevation (m)	1.372	1.380	1.340	1.300
41	Pressure (bar)	87.98	87.98	87.98	87.98
42	Temperature (C)	48.84	48.84	48.83	48.83
43	Vapour Fraction	0.8686	0.8686	0.8686	0.8686
44	Heat Transferred (kJ/h-m)	501.9	501.8	501.7	501.6
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.2813	0.2813	0.2044	0.2044
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	5.501e-003	5.501e-003	6.895e-003	6.895e-003
50	Static Gradient (kPa/m)	1.421e-003	1.421e-003	-5.737e-003	-5.737e-003
51	Accel. Gradient (kPa/m)	5.039e-007	5.039e-007	8.433e-008	8.431e-008
52	Liq. Reynolds	2.071e+005 *	2.071e+005 *	2.071e+005 *	2.071e+005 *
53	Vap. Reynolds	9.194e+006 *	9.194e+006 *	9.194e+006 *	9.194e+006 *
54	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
55	Vapour Velocity (m/s)	2.172	2.172	2.172	2.172
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	3.254e+004	3.255e+004	3.257e+004	3.257e+004
59	Elevation (m)	1.260	1.220	1.180	1.162
60	Pressure (bar)	87.98	87.98	87.98	87.98
61	Temperature (C)	48.83	48.82	48.82	48.82
62	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
63	Heat Transferred (kJ/h-m)	501.5	501.4	501.3	501.3
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.2044	0.2044	0.2044	0.2044
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	6.895e-003	6.895e-003	6.895e-003	6.895e-003
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 518 of 937

1				Case Name:	Simulation - 386MMSCFD_100barg.hsc		
2				Unit Set:	Mixed		
3		Epic Atlantic Limited					
4				Date/Time:	Mon Jun 13 15:02:44 2022		
5							
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Static Gradient	(kPa/m)	-5.737e-003	-5.737e-003	-5.737e-003	-5.773e-003	
14	Accel. Gradient	(kPa/m)	8.430e-008	8.428e-008	8.426e-008	8.164e-008	
15	Liq. Reynolds		2.071e+005 *	2.071e+005 *	2.071e+005 *	2.071e+005 *	
16	Vap. Reynolds		9.194e+006 *	9.194e+006 *	9.194e+006 *	9.194e+006 *	
17	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
18	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172	
19	Deposit Thickness	(mm)	---	---	---	---	
20	Deposit Volume	(m3)	---	---	---	---	
21	Length	(m)	3.258e+004	3.258e+004	3.259e+004	3.260e+004	
22	Elevation	(m)	1.144	1.126	1.108	1.090	
23	Pressure	(bar)	87.98	87.98	87.98	87.98	
24	Temperature	(C)	48.81	48.81	48.81	48.81	
25	Vapour Fraction		0.8685	0.8685	0.8685	0.8685	
26	Heat Transferred	(kJ/h-m)	501.2	501.2	501.1	501.1	
27	Flow Regime		Transition	Transition	Transition	Transition	
28	Water Flow Regime						
29	Liquid HoldUp		0.2044	0.2044	0.2044	0.2044	
30	Water HoldUp		---	---	---	---	
31	Friction Gradient	(kPa/m)	6.895e-003	6.895e-003	6.895e-003	6.895e-003	
32	Static Gradient	(kPa/m)	-5.773e-003	-5.773e-003	-5.773e-003	-5.773e-003	
33	Accel. Gradient	(kPa/m)	8.163e-008	8.163e-008	8.162e-008	8.161e-008	
34	Liq. Reynolds		2.071e+005 *	2.071e+005 *	2.071e+005 *	2.071e+005 *	
35	Vap. Reynolds		9.194e+006 *	9.194e+006 *	9.194e+006 *	9.194e+006 *	
36	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
37	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172	
38	Deposit Thickness	(mm)	---	---	---	---	
39	Deposit Volume	(m3)	---	---	---	---	
40	Length	(m)	3.260e+004	3.261e+004	3.261e+004	3.262e+004	
41	Elevation	(m)	1.078	1.066	1.054	1.042	
42	Pressure	(bar)	87.98	87.98	87.98	87.98	
43	Temperature	(C)	48.81	48.81	48.80	48.80	
44	Vapour Fraction		0.8685	0.8685	0.8685	0.8685	
45	Heat Transferred	(kJ/h-m)	501.0	501.0	501.0	500.9	
46	Flow Regime		Transition	Transition	Transition	Transition	
47	Water Flow Regime						
48	Liquid HoldUp		0.2053	0.2053	0.2053	0.2053	
49	Water HoldUp		---	---	---	---	
50	Friction Gradient	(kPa/m)	6.887e-003	6.887e-003	6.887e-003	6.887e-003	
51	Static Gradient	(kPa/m)	-3.855e-003	-3.855e-003	-3.855e-003	-3.855e-003	
52	Accel. Gradient	(kPa/m)	2.207e-007	2.207e-007	2.207e-007	2.207e-007	
53	Liq. Reynolds		2.071e+005 *	2.071e+005 *	2.071e+005 *	2.071e+005 *	
54	Vap. Reynolds		9.194e+006 *	9.194e+006 *	9.194e+006 *	9.194e+006 *	
55	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
56	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172	
57	Deposit Thickness	(mm)	---	---	---	---	
58	Deposit Volume	(m3)	---	---	---	---	
59	Length	(m)	3.263e+004	3.263e+004	3.263e+004	3.263e+004	
60	Elevation	(m)	1.030	1.026	1.022	1.018	
61	Pressure	(bar)	87.98	87.98	87.98	87.98	
62	Temperature	(C)	48.80	48.80	48.80	48.80	
63	Vapour Fraction		0.8685	0.8685	0.8685	0.8685	
64	Heat Transferred	(kJ/h-m)	500.9	500.8	500.8	500.8	
65	Flow Regime		Transition	Transition	Transition	Transition	
66	Water Flow Regime						
67	Liquid HoldUp		0.2053	0.2053	0.2053	0.2053	
68	Water HoldUp		---	---	---	---	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3							
4			Unit Set: Mixed				
5			Date/Time: Mon Jun 13 15:02:44 2022				
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Friction Gradient	(kPa/m)	6.887e-003	6.887e-003	6.887e-003	6.887e-003	
14	Static Gradient	(kPa/m)	-3.855e-003	-3.855e-003	-3.855e-003	-3.855e-003	
15	Accel. Gradient	(kPa/m)	2.207e-007	2.207e-007	2.207e-007	2.207e-007	
16	Liq. Reynolds		2.071e+005 *	2.071e+005 *	2.071e+005 *	2.071e+005 *	
17	Vap. Reynolds		9.194e+006 *	9.194e+006 *	9.194e+006 *	9.194e+006 *	
18	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
19	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172	
20	Deposit Thickness	(mm)	---	---	---	---	
21	Deposit Volume	(m3)	---	---	---	---	
22	Length	(m)	3.263e+004	3.264e+004	3.264e+004	3.264e+004	
23	Elevation	(m)	1.014	1.010	1.002	0.9940	
24	Pressure	(bar)	87.98	87.98	87.98	87.98	
25	Temperature	(C)	48.80	48.80	48.80	48.80	
26	Vapour Fraction		0.8685	0.8685	0.8685	0.8685	
27	Heat Transferred	(kJ/h-m)	500.8	500.8	500.8	500.8	
28	Flow Regime		Transition	Transition	Transition	Transition	
29	Water Flow Regime						
30	Liquid HoldUp		0.2053	0.2053	0.2035	0.2035	
31	Water HoldUp		---	---	---	---	
32	Friction Gradient	(kPa/m)	6.887e-003	6.887e-003	6.902e-003	6.902e-003	
33	Static Gradient	(kPa/m)	-3.855e-003	-3.855e-003	-7.666e-003	-7.666e-003	
34	Accel. Gradient	(kPa/m)	2.207e-007	2.207e-007	-5.559e-008	-5.560e-008	
35	Liq. Reynolds		2.071e+005 *	2.071e+005 *	2.071e+005 *	2.071e+005 *	
36	Vap. Reynolds		9.194e+006 *	9.194e+006 *	9.194e+006 *	9.194e+006 *	
37	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
38	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172	
39	Deposit Thickness	(mm)	---	---	---	---	
40	Deposit Volume	(m3)	---	---	---	---	
41	Length	(m)	3.264e+004	3.264e+004	3.265e+004	3.265e+004	
42	Elevation	(m)	0.9860	0.9780	0.9700	0.9520	
43	Pressure	(bar)	87.98	87.98	87.98	87.98	
44	Temperature	(C)	48.80	48.80	48.80	48.79	
45	Vapour Fraction		0.8685	0.8685	0.8685	0.8685	
46	Heat Transferred	(kJ/h-m)	500.7	500.7	500.7	500.7	
47	Flow Regime		Transition	Transition	Transition	Transition	
48	Water Flow Regime						
49	Liquid HoldUp		0.2035	0.2035	0.2035	0.2038	
50	Water HoldUp		---	---	---	---	
51	Friction Gradient	(kPa/m)	6.902e-003	6.902e-003	6.902e-003	6.899e-003	
52	Static Gradient	(kPa/m)	-7.666e-003	-7.666e-003	-7.666e-003	-6.907e-003	
53	Accel. Gradient	(kPa/m)	-5.560e-008	-5.560e-008	-5.560e-008	-6.213e-010	
54	Liq. Reynolds		2.071e+005 *	2.071e+005 *	2.071e+005 *	2.071e+005 *	
55	Vap. Reynolds		9.194e+006 *	9.194e+006 *	9.194e+006 *	9.194e+006 *	
56	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
57	Vapour Velocity	(m/s)	2.172	2.172	2.172	2.172	
58	Deposit Thickness	(mm)	---	---	---	---	
59	Deposit Volume	(m3)	---	---	---	---	
60	Length	(m)	3.266e+004	3.266e+004	3.267e+004	3.267e+004	
61	Elevation	(m)	0.9340	0.9160	0.8980	0.8800	
62	Pressure	(bar)	87.98	87.98	87.98	87.98	
63	Temperature	(C)	48.79	48.79	48.79	48.79	
64	Vapour Fraction		0.8685	0.8685	0.8685	0.8685	
65	Heat Transferred	(kJ/h-m)	500.6	500.6	500.6	500.5	
66	Flow Regime		Transition	Transition	Transition	Transition	
67	Water Flow Regime						
68	Liquid HoldUp		0.2038	0.2038	0.2038	0.2038	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 520 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	6.899e-003	6.899e-003	6.899e-003	6.898e-003
15	Static Gradient (kPa/m)	-6.907e-003	-6.907e-003	-6.907e-003	-6.907e-003
16	Accel. Gradient (kPa/m)	-6.276e-010	-6.339e-010	-6.402e-010	-6.465e-010
17	Liq. Reynolds	2.071e+005 *	2.071e+005 *	2.071e+005 *	2.071e+005 *
18	Vap. Reynolds	9.194e+006 *	9.194e+006 *	9.194e+006 *	9.194e+006 *
19	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
20	Vapour Velocity (m/s)	2.172	2.171	2.171	2.171
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	3.268e+004	3.268e+004	3.269e+004	3.269e+004
24	Elevation (m)	0.8680	0.8560	0.8440	0.8320
25	Pressure (bar)	87.98	87.98	87.98	87.98
26	Temperature (C)	48.79	48.79	48.78	48.78
27	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
28	Heat Transferred (kJ/h-m)	500.5	500.5	500.4	500.4
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2050	0.2050	0.2050	0.2050
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	6.889e-003	6.889e-003	6.889e-003	6.889e-003
34	Static Gradient (kPa/m)	-4.621e-003	-4.621e-003	-4.621e-003	-4.621e-003
35	Accel. Gradient (kPa/m)	1.651e-007	1.651e-007	1.651e-007	1.651e-007
36	Liq. Reynolds	2.071e+005 *	2.071e+005 *	2.071e+005 *	2.071e+005 *
37	Vap. Reynolds	9.194e+006 *	9.194e+006 *	9.194e+006 *	9.194e+006 *
38	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
39	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	3.270e+004	3.270e+004	3.271e+004	3.271e+004
43	Elevation (m)	0.8200	0.8360	0.8520	0.8680
44	Pressure (bar)	87.98	87.98	87.98	87.98
45	Temperature (C)	48.78	48.78	48.78	48.78
46	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
47	Heat Transferred (kJ/h-m)	500.3	500.3	500.2	500.2
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2050	0.2897	0.2897	0.2897
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	6.889e-003	4.877e-003	4.877e-003	4.877e-003
53	Static Gradient (kPa/m)	-4.621e-003	7.777e-003	7.777e-003	7.777e-003
54	Accel. Gradient (kPa/m)	1.651e-007	9.209e-007	9.210e-007	9.210e-007
55	Liq. Reynolds	2.071e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *
56	Vap. Reynolds	9.194e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
57	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
58	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	3.272e+004	3.272e+004	3.273e+004	3.273e+004
62	Elevation (m)	0.8840	0.9000	0.8860	0.8720
63	Pressure (bar)	87.97	87.97	87.97	87.97
64	Temperature (C)	48.77	48.77	48.77	48.77
65	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
66	Heat Transferred (kJ/h-m)	500.2	500.1	500.1	500.0
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 521 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Liquid HoldUp	0.2897	0.2897	0.2046	0.2046
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	4.877e-003	4.877e-003	6.892e-003	6.892e-003
16	Static Gradient (kPa/m)	7.777e-003	7.777e-003	-5.385e-003	-5.385e-003
17	Accel. Gradient (kPa/m)	9.210e-007	9.210e-007	1.097e-007	1.097e-007
18	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *
19	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
20	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
21	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	3.274e+004	3.274e+004	3.275e+004	3.275e+004
25	Elevation (m)	0.8580	0.8440	0.8300	0.8620
26	Pressure (bar)	87.97	87.97	87.97	87.97
27	Temperature (C)	48.77	48.77	48.77	48.76
28	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
29	Heat Transferred (kJ/h-m)	500.0	500.0	499.9	499.9
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2046	0.2046	0.2046	0.2998
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	6.892e-003	6.892e-003	6.892e-003	5.777e-003
35	Static Gradient (kPa/m)	-5.385e-003	-5.385e-003	-5.385e-003	1.594e-002
36	Accel. Gradient (kPa/m)	1.097e-007	1.097e-007	1.097e-007	1.581e-006
37	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *
38	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
39	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
40	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	3.276e+004	3.276e+004	3.277e+004	3.277e+004
44	Elevation (m)	0.8940	0.9260	0.9580	0.9900
45	Pressure (bar)	87.97	87.97	87.97	87.97
46	Temperature (C)	48.76	48.76	48.76	48.76
47	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
48	Heat Transferred (kJ/h-m)	499.8	499.8	499.7	499.7
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2998	0.2998	0.2998	0.2998
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	5.777e-003	5.777e-003	5.777e-003	5.777e-003
54	Static Gradient (kPa/m)	1.594e-002	1.594e-002	1.594e-002	1.594e-002
55	Accel. Gradient (kPa/m)	1.581e-006	1.581e-006	1.581e-006	1.581e-006
56	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *
57	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
58	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
59	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	3.278e+004	3.278e+004	3.279e+004	3.279e+004
63	Elevation (m)	0.9620	0.9340	0.9060	0.8780
64	Pressure (bar)	87.97	87.97	87.97	87.97
65	Temperature (C)	48.76	48.75	48.75	48.75
66	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
67	Heat Transferred (kJ/h-m)	499.7	499.6	499.6	499.6
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 522 of 937

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Unit Set: Mixed		
3	 Epic Atlantic Limited		Date/Time: Mon Jun 13 15:02:44 2022		
4					
5					
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2021	0.2021	0.2021	0.2021
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.913e-003	6.913e-003	6.913e-003	6.913e-003
17	Static Gradient (kPa/m)	-1.051e-002	-1.051e-002	-1.051e-002	-1.051e-002
18	Accel. Gradient (kPa/m)	-2.621e-007	-2.621e-007	-2.621e-007	-2.621e-007
19	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *
20	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
21	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
22	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	3.280e+004	3.280e+004	3.281e+004	3.281e+004
26	Elevation (m)	0.8500	0.8280	0.8060	0.7840
27	Pressure (bar)	87.97	87.97	87.97	87.97
28	Temperature (C)	48.75	48.75	48.75	48.75
29	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
30	Heat Transferred (kJ/h-m)	499.5	499.5	499.4	499.4
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2021	0.2031	0.2031	0.2031
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	6.913e-003	6.904e-003	6.904e-003	6.904e-003
36	Static Gradient (kPa/m)	-1.051e-002	-8.424e-003	-8.424e-003	-8.424e-003
37	Accel. Gradient (kPa/m)	-2.621e-007	-1.106e-007	-1.106e-007	-1.106e-007
38	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *
39	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
40	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
41	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	3.282e+004	3.282e+004	3.283e+004	3.283e+004
45	Elevation (m)	0.7620	0.7400	0.6940	0.6480
46	Pressure (bar)	87.97	87.97	87.97	87.97
47	Temperature (C)	48.74	48.74	48.74	48.74
48	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
49	Heat Transferred (kJ/h-m)	499.4	499.3	499.3	499.3
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.2031	0.2031	0.1986	0.1986
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	6.904e-003	6.904e-003	6.942e-003	6.942e-003
55	Static Gradient (kPa/m)	-8.424e-003	-8.424e-003	-1.737e-002	-1.737e-002
56	Accel. Gradient (kPa/m)	-1.106e-007	-1.106e-007	-7.588e-007	-7.588e-007
57	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *
58	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
59	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
60	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	3.284e+004	3.284e+004	3.285e+004	3.285e+004
64	Elevation (m)	0.6020	0.5560	0.5100	0.4920
65	Pressure (bar)	87.97	87.97	87.97	87.97
66	Temperature (C)	48.74	48.74	48.74	48.74
67	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
68	Heat Transferred (kJ/h-m)	499.2	499.2	499.2	499.1
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.1987	0.1987	0.1987	0.2039
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	6.942e-003	6.942e-003	6.941e-003	6.897e-003
18	Static Gradient (kPa/m)	-1.737e-002	-1.737e-002	-1.737e-002	-6.909e-003
19	Accel. Gradient (kPa/m)	-7.588e-007	-7.588e-007	-7.588e-007	-8.247e-010
20	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *
21	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
22	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
23	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	3.286e+004	3.286e+004	3.287e+004	3.287e+004
27	Elevation (m)	0.4740	0.4560	0.4380	0.4200
28	Pressure (bar)	87.97	87.97	87.97	87.97
29	Temperature (C)	48.73	48.73	48.73	48.73
30	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
31	Heat Transferred (kJ/h-m)	499.1	499.1	499.0	499.0
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2039	0.2039	0.2039	0.2039
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	6.897e-003	6.897e-003	6.897e-003	6.897e-003
37	Static Gradient (kPa/m)	-6.909e-003	-6.909e-003	-6.909e-003	-6.909e-003
38	Accel. Gradient (kPa/m)	-8.310e-010	-8.373e-010	-8.436e-010	-8.499e-010
39	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *
40	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
41	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
42	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	3.288e+004	3.288e+004	3.289e+004	3.289e+004
46	Elevation (m)	0.3780	0.3360	0.2940	0.2520
47	Pressure (bar)	87.97	87.97	87.97	87.97
48	Temperature (C)	48.73	48.73	48.73	48.73
49	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
50	Heat Transferred (kJ/h-m)	499.0	498.9	498.9	498.9
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.1994	0.1994	0.1994	0.1994
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	6.935e-003	6.935e-003	6.934e-003	6.934e-003
56	Static Gradient (kPa/m)	-1.590e-002	-1.590e-002	-1.590e-002	-1.590e-002
57	Accel. Gradient (kPa/m)	-6.521e-007	-6.521e-007	-6.521e-007	-6.521e-007
58	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *
59	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
60	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
61	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	3.290e+004	3.290e+004	3.291e+004	3.291e+004
65	Elevation (m)	0.2100	0.2040	0.1980	0.1920
66	Pressure (bar)	87.97	87.97	87.97	87.97
67	Temperature (C)	48.72	48.72	48.72	48.72
68	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	498.8	498.8	498.8	498.7
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.1994	0.2061	0.2061	0.2061
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	6.934e-003	6.879e-003	6.879e-003	6.879e-003
19	Static Gradient (kPa/m)	-1.590e-002	-2.319e-003	-2.319e-003	-2.319e-003
20	Accel. Gradient (kPa/m)	-6.521e-007	3.317e-007	3.317e-007	3.317e-007
21	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *
22	Vap. Reynolds	9.194e+006 *	9.194e+006 *	9.195e+006 *	9.195e+006 *
23	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
24	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	3.292e+004	3.292e+004	3.293e+004	3.293e+004
28	Elevation (m)	0.1860	0.1800	0.1800	0.1800
29	Pressure (bar)	87.97	87.97	87.97	87.97
30	Temperature (C)	48.72	48.72	48.72	48.71
31	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
32	Heat Transferred (kJ/h-m)	498.7	498.6	498.6	498.5
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.2061	0.2061	0.3025	0.3025
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	6.878e-003	6.878e-003	5.883e-003	5.883e-003
38	Static Gradient (kPa/m)	-2.319e-003	-2.319e-003	0.0000	0.0000
39	Accel. Gradient (kPa/m)	3.317e-007	3.317e-007	4.280e-007	4.280e-007
40	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *
41	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
42	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
43	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	3.294e+004	3.294e+004	3.295e+004	3.295e+004
47	Elevation (m)	0.1800	0.1800	0.1800	0.1760
48	Pressure (bar)	87.97	87.97	87.97	87.97
49	Temperature (C)	48.71	48.71	48.71	48.71
50	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
51	Heat Transferred (kJ/h-m)	498.5	498.5	498.4	498.4
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.3025	0.3025	0.3025	0.2065
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	5.883e-003	5.883e-003	5.883e-003	6.875e-003
57	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	-1.548e-003
58	Accel. Gradient (kPa/m)	4.280e-007	4.280e-007	4.280e-007	3.876e-007
59	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *
60	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
61	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
62	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	3.296e+004	3.296e+004	3.297e+004	3.297e+004
66	Elevation (m)	0.1720	0.1680	0.1640	0.1600
67	Pressure (bar)	87.97	87.97	87.97	87.97
68	Temperature (C)	48.71	48.71	48.70	48.70
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
14	Heat Transferred (kJ/h-m)	498.4	498.3	498.3	498.2
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2065	0.2065	0.2065	0.2065
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	6.875e-003	6.875e-003	6.875e-003	6.875e-003
20	Static Gradient (kPa/m)	-1.548e-003	-1.548e-003	-1.548e-003	-1.548e-003
21	Accel. Gradient (kPa/m)	3.876e-007	3.876e-007	3.876e-007	3.876e-007
22	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *
23	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
24	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
25	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	3.298e+004	3.298e+004	3.299e+004	3.299e+004
29	Elevation (m)	0.1620	0.1640	0.1660	0.1680
30	Pressure (bar)	87.97	87.97	87.97	87.97
31	Temperature (C)	48.70	48.70	48.70	48.70
32	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
33	Heat Transferred (kJ/h-m)	498.2	498.1	498.1	498.1
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2808	0.2808	0.2808	0.2808
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	5.547e-003	5.547e-003	5.547e-003	5.547e-003
39	Static Gradient (kPa/m)	9.347e-004	9.347e-004	9.347e-004	9.347e-004
40	Accel. Gradient (kPa/m)	4.716e-007	4.716e-007	4.716e-007	4.716e-007
41	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *
42	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
43	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
44	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	3.300e+004	3.300e+004	3.301e+004	3.301e+004
48	Elevation (m)	0.1700	0.1660	0.1620	0.1580
49	Pressure (bar)	87.97	87.97	87.97	87.97
50	Temperature (C)	48.69	48.69	48.69	48.69
51	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
52	Heat Transferred (kJ/h-m)	498.0	498.0	497.9	497.9
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2808	0.2065	0.2065	0.2065
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	5.547e-003	6.875e-003	6.875e-003	6.875e-003
58	Static Gradient (kPa/m)	9.347e-004	-1.548e-003	-1.548e-003	-1.548e-003
59	Accel. Gradient (kPa/m)	4.716e-007	3.876e-007	3.876e-007	3.876e-007
60	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *
61	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
62	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
63	Vapour Velocity (m/s)	2.171	2.171	2.171	2.171
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	3.302e+004	3.302e+004	3.302e+004	3.302e+004
67	Elevation (m)	0.1540	0.1500	0.1500	0.1500
68	Pressure (bar)	87.97	87.97	87.97	87.97
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	48.69	48.69	48.69	48.69
14	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
15	Heat Transferred (kJ/h-m)	497.9	497.8	497.8	497.8
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2065	0.2065	0.3026	0.3026
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	6.875e-003	6.875e-003	5.883e-003	5.883e-003
21	Static Gradient (kPa/m)	-1.548e-003	-1.548e-003	0.0000	0.0000
22	Accel. Gradient (kPa/m)	3.876e-007	3.876e-007	4.280e-007	4.280e-007
23	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *
24	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
25	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
26	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	3.303e+004	3.303e+004	3.303e+004	3.304e+004
30	Elevation (m)	0.1500	0.1500	0.1500	0.1580
31	Pressure (bar)	87.97	87.97	87.97	87.97
32	Temperature (C)	48.69	48.68	48.68	48.68
33	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
34	Heat Transferred (kJ/h-m)	497.8	497.8	497.7	497.7
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.3026	0.3026	0.3026	0.2838
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	5.883e-003	5.883e-003	5.883e-003	5.315e-003
40	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	3.194e-003
41	Accel. Gradient (kPa/m)	4.280e-007	4.280e-007	4.280e-007	6.191e-007
42	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *
43	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
44	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
45	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	3.304e+004	3.305e+004	3.305e+004	3.306e+004
49	Elevation (m)	0.1660	0.1740	0.1820	0.1900
50	Pressure (bar)	87.97	87.97	87.96	87.96
51	Temperature (C)	48.68	48.68	48.68	48.67
52	Vapour Fraction	0.8685	0.8685	0.8685	0.8685
53	Heat Transferred (kJ/h-m)	497.6	497.6	497.5	497.5
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2838	0.2838	0.2838	0.2838
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	5.315e-003	5.315e-003	5.315e-003	5.315e-003
59	Static Gradient (kPa/m)	3.194e-003	3.194e-003	3.194e-003	3.194e-003
60	Accel. Gradient (kPa/m)	6.191e-007	6.191e-007	6.191e-007	6.191e-007
61	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *
62	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
63	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
64	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	3.306e+004	3.306e+004	3.307e+004	3.307e+004
68	Elevation (m)	0.1860	0.1820	0.1780	0.1740
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	87.96	87.96	87.96	87.96		
14	Temperature (C)	48.67	48.67	48.67	48.67		
15	Vapour Fraction	0.8685	0.8685	0.8685	0.8685		
16	Heat Transferred (kJ/h-m)	497.5	497.5	497.5	497.4		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.2054	0.2054	0.2054	0.2054		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	6.884e-003	6.884e-003	6.884e-003	6.884e-003		
22	Static Gradient (kPa/m)	-3.857e-003	-3.857e-003	-3.857e-003	-3.857e-003		
23	Accel. Gradient (kPa/m)	2.202e-007	2.202e-007	2.202e-007	2.202e-007		
24	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *		
25	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *		
26	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
27	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	3.307e+004	3.308e+004	3.308e+004	3.309e+004		
31	Elevation (m)	0.1700	0.1680	0.1660	0.1640		
32	Pressure (bar)	87.96	87.96	87.96	87.96		
33	Temperature (C)	48.67	48.67	48.67	48.67		
34	Vapour Fraction	0.8685	0.8685	0.8685	0.8685		
35	Heat Transferred (kJ/h-m)	497.4	497.4	497.3	497.3		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.2054	0.2069	0.2069	0.2069		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	6.884e-003	6.872e-003	6.872e-003	6.872e-003		
41	Static Gradient (kPa/m)	-3.857e-003	-7.749e-004	-7.749e-004	-7.749e-004		
42	Accel. Gradient (kPa/m)	2.202e-007	4.436e-007	4.436e-007	4.436e-007		
43	Liq. Reynolds	2.070e+005 *	2.070e+005 *	2.070e+005 *	2.070e+005 *		
44	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *		
45	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
46	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	3.309e+004	3.310e+004	3.310e+004	3.311e+004		
50	Elevation (m)	0.1620	0.1600	0.1660	0.1720		
51	Pressure (bar)	87.96	87.96	87.96	87.96		
52	Temperature (C)	48.67	48.66	48.66	48.66		
53	Vapour Fraction	0.8684	0.8684	0.8684	0.8684		
54	Heat Transferred (kJ/h-m)	497.3	497.2	497.2	497.1		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.2069	0.2069	0.2834	0.2834		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	6.872e-003	6.872e-003	5.347e-003	5.347e-003		
60	Static Gradient (kPa/m)	-7.749e-004	-7.749e-004	2.872e-003	2.872e-003		
61	Accel. Gradient (kPa/m)	4.436e-007	4.436e-007	5.980e-007	5.980e-007		
62	Liq. Reynolds	2.070e+005 *	2.069e+005 *	2.069e+005 *	2.069e+005 *		
63	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *		
64	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046		
65	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	3.311e+004	3.312e+004	3.312e+004	3.313e+004		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	0.1780	0.1840	0.1900	0.1840
14	Pressure (bar)	87.96	87.96	87.96	87.96
15	Temperature (C)	48.66	48.66	48.66	48.66
16	Vapour Fraction	0.8684	0.8684	0.8684	0.8684
17	Heat Transferred (kJ/h-m)	497.1	497.1	497.0	497.0
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2834	0.2834	0.2834	0.2061
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	5.347e-003	5.347e-003	5.347e-003	6.877e-003
23	Static Gradient (kPa/m)	2.872e-003	2.872e-003	2.872e-003	-2.319e-003
24	Accel. Gradient (kPa/m)	5.980e-007	5.980e-007	5.980e-007	3.316e-007
25	Liq. Reynolds	2.069e+005 *	2.069e+005 *	2.069e+005 *	2.069e+005 *
26	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
27	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
28	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	3.313e+004	3.314e+004	3.314e+004	3.315e+004
32	Elevation (m)	0.1780	0.1720	0.1660	0.1600
33	Pressure (bar)	87.96	87.96	87.96	87.96
34	Temperature (C)	48.65	48.65	48.65	48.65
35	Vapour Fraction	0.8684	0.8684	0.8684	0.8684
36	Heat Transferred (kJ/h-m)	496.9	496.9	496.9	496.8
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2061	0.2061	0.2061	0.2061
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	6.877e-003	6.877e-003	6.877e-003	6.877e-003
42	Static Gradient (kPa/m)	-2.319e-003	-2.319e-003	-2.319e-003	-2.319e-003
43	Accel. Gradient (kPa/m)	3.316e-007	3.316e-007	3.316e-007	3.316e-007
44	Liq. Reynolds	2.069e+005 *	2.069e+005 *	2.069e+005 *	2.069e+005 *
45	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
46	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
47	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	3.315e+004	3.316e+004	3.316e+004	3.317e+004
51	Elevation (m)	0.1560	0.1520	0.1480	0.1440
52	Pressure (bar)	87.96	87.96	87.96	87.96
53	Temperature (C)	48.65	48.65	48.64	48.64
54	Vapour Fraction	0.8684	0.8684	0.8684	0.8684
55	Heat Transferred (kJ/h-m)	496.8	496.7	496.7	496.7
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2065	0.2065	0.2065	0.2065
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	6.874e-003	6.874e-003	6.874e-003	6.874e-003
61	Static Gradient (kPa/m)	-1.548e-003	-1.548e-003	-1.548e-003	-1.548e-003
62	Accel. Gradient (kPa/m)	3.875e-007	3.875e-007	3.875e-007	3.875e-007
63	Liq. Reynolds	2.069e+005 *	2.069e+005 *	2.069e+005 *	2.069e+005 *
64	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
65	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
66	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022	
6						
7						
8						
9			PROFILES			
10						
11			Pipe Table			
12						
13	Length	(m)	3.317e+004	3.318e+004	3.318e+004	3.319e+004
14	Elevation	(m)	0.1400	0.1520	0.1640	0.1760
15	Pressure	(bar)	87.96	87.96	87.96	87.96
16	Temperature	(C)	48.64	48.64	48.64	48.64
17	Vapour Fraction		0.8684	0.8684	0.8684	0.8684
18	Heat Transferred	(kJ/h-m)	496.6	496.6	496.5	496.5
19	Flow Regime		Transition	Transition	Transition	Transition
20	Water Flow Regime					
21	Liquid HoldUp		0.2065	0.2871	0.2871	0.2871
22	Water HoldUp		---	---	---	---
23	Friction Gradient	(kPa/m)	6.874e-003	5.067e-003	5.067e-003	5.067e-003
24	Static Gradient	(kPa/m)	-1.548e-003	5.715e-003	5.715e-003	5.715e-003
25	Accel. Gradient	(kPa/m)	3.875e-007	7.844e-007	7.844e-007	7.844e-007
26	Liq. Reynolds		2.069e+005 *	2.069e+005 *	2.069e+005 *	2.069e+005 *
27	Vap. Reynolds		9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
28	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
29	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170
30	Deposit Thickness	(mm)	---	---	---	---
31	Deposit Volume	(m3)	---	---	---	---
32	Length	(m)	3.319e+004	3.320e+004	3.320e+004	3.321e+004
33	Elevation	(m)	0.1880	0.2000	0.2000	0.2000
34	Pressure	(bar)	87.96	87.96	87.96	87.95
35	Temperature	(C)	48.64	48.63	48.63	48.63
36	Vapour Fraction		0.8684	0.8684	0.8684	0.8684
37	Heat Transferred	(kJ/h-m)	496.4	496.4	496.4	496.3
38	Flow Regime		Transition	Transition	Transition	Transition
39	Water Flow Regime					
40	Liquid HoldUp		0.2871	0.2871	0.3026	0.3026
41	Water HoldUp		---	---	---	---
42	Friction Gradient	(kPa/m)	5.067e-003	5.067e-003	5.882e-003	5.882e-003
43	Static Gradient	(kPa/m)	5.715e-003	5.715e-003	0.0000	0.0000
44	Accel. Gradient	(kPa/m)	7.844e-007	7.844e-007	4.280e-007	4.280e-007
45	Liq. Reynolds		2.069e+005 *	2.069e+005 *	2.069e+005 *	2.069e+005 *
46	Vap. Reynolds		9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
47	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
48	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170
49	Deposit Thickness	(mm)	---	---	---	---
50	Deposit Volume	(m3)	---	---	---	---
51	Length	(m)	3.321e+004	3.322e+004	3.322e+004	3.323e+004
52	Elevation	(m)	0.2000	0.2000	0.2000	0.2000
53	Pressure	(bar)	87.95	87.95	87.95	87.95
54	Temperature	(C)	48.63	48.63	48.63	48.62
55	Vapour Fraction		0.8684	0.8684	0.8684	0.8684
56	Heat Transferred	(kJ/h-m)	496.3	496.2	496.2	496.2
57	Flow Regime		Transition	Transition	Transition	Transition
58	Water Flow Regime					
59	Liquid HoldUp		0.3026	0.3026	0.3026	0.3026
60	Water HoldUp		---	---	---	---
61	Friction Gradient	(kPa/m)	5.882e-003	5.882e-003	5.882e-003	5.882e-003
62	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000
63	Accel. Gradient	(kPa/m)	4.280e-007	4.280e-007	4.280e-007	4.280e-007
64	Liq. Reynolds		2.069e+005 *	2.069e+005 *	2.069e+005 *	2.069e+005 *
65	Vap. Reynolds		9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
66	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
67	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170
68	Deposit Thickness	(mm)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
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9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	3.323e+004	3.324e+004	3.324e+004	3.325e+004
15	Elevation (m)	0.2000	0.2000	0.2000	0.2000
16	Pressure (bar)	87.95	87.95	87.95	87.95
17	Temperature (C)	48.62	48.62	48.62	48.62
18	Vapour Fraction	0.8684	0.8684	0.8684	0.8684
19	Heat Transferred (kJ/h-m)	496.1	496.1	496.0	496.0
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.3026	0.3026	0.3026	0.3026
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	5.882e-003	5.882e-003	5.882e-003	5.882e-003
25	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
26	Accel. Gradient (kPa/m)	4.280e-007	4.280e-007	4.279e-007	4.279e-007
27	Liq. Reynolds	2.069e+005 *	2.069e+005 *	2.069e+005 *	2.069e+005 *
28	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
29	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
30	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	3.325e+004	3.326e+004	3.326e+004	3.327e+004
34	Elevation (m)	0.1980	0.1960	0.1940	0.1920
35	Pressure (bar)	87.95	87.95	87.95	87.95
36	Temperature (C)	48.62	48.62	48.61	48.61
37	Vapour Fraction	0.8684	0.8684	0.8684	0.8684
38	Heat Transferred (kJ/h-m)	496.0	495.9	495.9	495.9
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2069	0.2069	0.2069	0.2069
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	6.871e-003	6.871e-003	6.871e-003	6.871e-003
44	Static Gradient (kPa/m)	-7.750e-004	-7.750e-004	-7.750e-004	-7.750e-004
45	Accel. Gradient (kPa/m)	4.435e-007	4.435e-007	4.435e-007	4.435e-007
46	Liq. Reynolds	2.069e+005 *	2.069e+005 *	2.069e+005 *	2.069e+005 *
47	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
48	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
49	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	3.327e+004	3.328e+004	3.328e+004	3.329e+004
53	Elevation (m)	0.1900	0.1940	0.1980	0.2020
54	Pressure (bar)	87.95	87.95	87.95	87.95
55	Temperature (C)	48.61	48.61	48.61	48.61
56	Vapour Fraction	0.8684	0.8684	0.8684	0.8684
57	Heat Transferred (kJ/h-m)	495.8	495.8	495.7	495.7
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2069	0.2821	0.2821	0.2821
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	6.871e-003	5.444e-003	5.444e-003	5.444e-003
63	Static Gradient (kPa/m)	-7.750e-004	1.909e-003	1.909e-003	1.909e-003
64	Accel. Gradient (kPa/m)	4.435e-007	5.350e-007	5.350e-007	5.350e-007
65	Liq. Reynolds	2.069e+005 *	2.069e+005 *	2.069e+005 *	2.069e+005 *
66	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
67	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
68	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
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9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	3.329e+004	3.330e+004	3.330e+004	3.331e+004
16	Elevation (m)	0.2060	0.2100	0.2080	0.2060
17	Pressure (bar)	87.95	87.95	87.95	87.95
18	Temperature (C)	48.61	48.60	48.60	48.60
19	Vapour Fraction	0.8684	0.8684	0.8684	0.8684
20	Heat Transferred (kJ/h-m)	495.6	495.6	495.6	495.5
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2821	0.2821	0.2069	0.2069
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	5.444e-003	5.444e-003	6.871e-003	6.871e-003
26	Static Gradient (kPa/m)	1.909e-003	1.909e-003	-7.750e-004	-7.750e-004
27	Accel. Gradient (kPa/m)	5.350e-007	5.350e-007	4.435e-007	4.435e-007
28	Liq. Reynolds	2.069e+005 *	2.069e+005 *	2.069e+005 *	2.069e+005 *
29	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
30	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
31	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	3.331e+004	3.332e+004	3.332e+004	3.333e+004
35	Elevation (m)	0.2040	0.2020	0.2000	0.2000
36	Pressure (bar)	87.95	87.95	87.95	87.95
37	Temperature (C)	48.60	48.60	48.60	48.59
38	Vapour Fraction	0.8684	0.8684	0.8684	0.8684
39	Heat Transferred (kJ/h-m)	495.5	495.5	495.4	495.4
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2069	0.2069	0.2069	0.3026
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	6.871e-003	6.871e-003	6.871e-003	5.882e-003
45	Static Gradient (kPa/m)	-7.750e-004	-7.750e-004	-7.750e-004	0.0000
46	Accel. Gradient (kPa/m)	4.435e-007	4.435e-007	4.435e-007	4.279e-007
47	Liq. Reynolds	2.069e+005 *	2.069e+005 *	2.069e+005 *	2.069e+005 *
48	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
49	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
50	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	3.333e+004	3.334e+004	3.334e+004	3.335e+004
54	Elevation (m)	0.2000	0.2000	0.2000	0.2000
55	Pressure (bar)	87.95	87.95	87.95	87.95
56	Temperature (C)	48.59	48.59	48.59	48.59
57	Vapour Fraction	0.8684	0.8684	0.8684	0.8684
58	Heat Transferred (kJ/h-m)	495.3	495.3	495.2	495.2
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.3026	0.3026	0.3026	0.3026
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	5.882e-003	5.882e-003	5.882e-003	5.882e-003
64	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
65	Accel. Gradient (kPa/m)	4.279e-007	4.279e-007	4.279e-007	4.279e-007
66	Liq. Reynolds	2.069e+005 *	2.069e+005 *	2.069e+005 *	2.069e+005 *
67	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
68	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022	
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10	Pipe Table					
11						
12						
13	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170	
14	Deposit Thickness (mm)	---	---	---	---	
15	Deposit Volume (m3)	---	---	---	---	
16	Length (m)	3.335e+004	3.336e+004	3.336e+004	3.337e+004	
17	Elevation (m)	0.2060	0.2120	0.2180	0.2240	
18	Pressure (bar)	87.95	87.95	87.94	87.94	
19	Temperature (C)	48.59	48.59	48.58	48.58	
20	Vapour Fraction	0.8684	0.8684	0.8684	0.8684	
21	Heat Transferred (kJ/h-m)	495.2	495.1	495.1	495.0	
22	Flow Regime	Transition	Transition	Transition	Transition	
23	Water Flow Regime					
24	Liquid HoldUp	0.2834	0.2834	0.2834	0.2834	
25	Water HoldUp	---	---	---	---	
26	Friction Gradient (kPa/m)	5.345e-003	5.345e-003	5.345e-003	5.345e-003	
27	Static Gradient (kPa/m)	2.872e-003	2.872e-003	2.872e-003	2.872e-003	
28	Accel. Gradient (kPa/m)	5.979e-007	5.979e-007	5.979e-007	5.979e-007	
29	Liq. Reynolds	2.069e+005 *	2.069e+005 *	2.069e+005 *	2.069e+005 *	
30	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *	
31	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046	
32	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170	
33	Deposit Thickness (mm)	---	---	---	---	
34	Deposit Volume (m3)	---	---	---	---	
35	Length (m)	3.337e+004	3.338e+004	3.338e+004	3.339e+004	
36	Elevation (m)	0.2300	0.2260	0.2220	0.2180	
37	Pressure (bar)	87.94	87.94	87.94	87.94	
38	Temperature (C)	48.58	48.58	48.58	48.58	
39	Vapour Fraction	0.8684	0.8684	0.8684	0.8684	
40	Heat Transferred (kJ/h-m)	495.0	495.0	494.9	494.9	
41	Flow Regime	Transition	Transition	Transition	Transition	
42	Water Flow Regime					
43	Liquid HoldUp	0.2834	0.2066	0.2066	0.2066	
44	Water HoldUp	---	---	---	---	
45	Friction Gradient (kPa/m)	5.345e-003	6.873e-003	6.873e-003	6.873e-003	
46	Static Gradient (kPa/m)	2.873e-003	-1.528e-003	-1.528e-003	-1.528e-003	
47	Accel. Gradient (kPa/m)	5.979e-007	3.889e-007	3.889e-007	3.889e-007	
48	Liq. Reynolds	2.069e+005 *	2.069e+005 *	2.069e+005 *	2.069e+005 *	
49	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *	
50	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046	
51	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170	
52	Deposit Thickness (mm)	---	---	---	---	
53	Deposit Volume (m3)	---	---	---	---	
54	Length (m)	3.339e+004	3.340e+004	3.340e+004	3.341e+004	
55	Elevation (m)	0.2140	0.2100	0.2080	0.2060	
56	Pressure (bar)	87.94	87.94	87.94	87.94	
57	Temperature (C)	48.58	48.57	48.57	48.57	
58	Vapour Fraction	0.8684	0.8684	0.8684	0.8684	
59	Heat Transferred (kJ/h-m)	494.8	494.8	494.8	494.7	
60	Flow Regime	Transition	Transition	Transition	Transition	
61	Water Flow Regime					
62	Liquid HoldUp	0.2066	0.2066	0.2069	0.2069	
63	Water HoldUp	---	---	---	---	
64	Friction Gradient (kPa/m)	6.873e-003	6.873e-003	6.870e-003	6.870e-003	
65	Static Gradient (kPa/m)	-1.528e-003	-1.528e-003	-7.750e-004	-7.750e-004	
66	Accel. Gradient (kPa/m)	3.889e-007	3.889e-007	4.434e-007	4.434e-007	
67	Liq. Reynolds	2.069e+005 *	2.069e+005 *	2.069e+005 *	2.069e+005 *	
68	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *	
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
14	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170
15	Deposit Thickness	(mm)	---	---	---	---
16	Deposit Volume	(m3)	---	---	---	---
17	Length	(m)	3.341e+004	3.342e+004	3.342e+004	3.343e+004
18	Elevation	(m)	0.2040	0.2020	0.2000	0.2040
19	Pressure	(bar)	87.94	87.94	87.94	87.94
20	Temperature	(C)	48.57	48.57	48.57	48.56
21	Vapour Fraction		0.8684	0.8684	0.8684	0.8684
22	Heat Transferred	(kJ/h-m)	494.7	494.7	494.6	494.6
23	Flow Regime		Transition	Transition	Transition	Transition
24	Water Flow Regime					
25	Liquid HoldUp		0.2069	0.2069	0.2069	0.2822
26	Water HoldUp		---	---	---	---
27	Friction Gradient	(kPa/m)	6.870e-003	6.870e-003	6.870e-003	5.443e-003
28	Static Gradient	(kPa/m)	-7.750e-004	-7.751e-004	-7.751e-004	1.909e-003
29	Accel. Gradient	(kPa/m)	4.434e-007	4.434e-007	4.434e-007	5.349e-007
30	Liq. Reynolds		2.069e+005 *	2.069e+005 *	2.069e+005 *	2.069e+005 *
31	Vap. Reynolds		9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
32	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
33	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170
34	Deposit Thickness	(mm)	---	---	---	---
35	Deposit Volume	(m3)	---	---	---	---
36	Length	(m)	3.343e+004	3.344e+004	3.344e+004	3.345e+004
37	Elevation	(m)	0.2080	0.2120	0.2160	0.2200
38	Pressure	(bar)	87.94	87.94	87.94	87.94
39	Temperature	(C)	48.56	48.56	48.56	48.56
40	Vapour Fraction		0.8684	0.8684	0.8684	0.8684
41	Heat Transferred	(kJ/h-m)	494.5	494.5	494.4	494.4
42	Flow Regime		Transition	Transition	Transition	Transition
43	Water Flow Regime					
44	Liquid HoldUp		0.2822	0.2822	0.2822	0.2822
45	Water HoldUp		---	---	---	---
46	Friction Gradient	(kPa/m)	5.443e-003	5.443e-003	5.443e-003	5.443e-003
47	Static Gradient	(kPa/m)	1.909e-003	1.909e-003	1.909e-003	1.909e-003
48	Accel. Gradient	(kPa/m)	5.349e-007	5.349e-007	5.349e-007	5.349e-007
49	Liq. Reynolds		2.068e+005 *	2.068e+005 *	2.068e+005 *	2.068e+005 *
50	Vap. Reynolds		9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
51	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
52	Vapour Velocity	(m/s)	2.170	2.170	2.170	2.170
53	Deposit Thickness	(mm)	---	---	---	---
54	Deposit Volume	(m3)	---	---	---	---
55	Length	(m)	3.345e+004	3.346e+004	3.346e+004	3.347e+004
56	Elevation	(m)	0.2180	0.2160	0.2140	0.2120
57	Pressure	(bar)	87.94	87.94	87.94	87.94
58	Temperature	(C)	48.56	48.56	48.55	48.55
59	Vapour Fraction		0.8684	0.8684	0.8684	0.8684
60	Heat Transferred	(kJ/h-m)	494.4	494.3	494.3	494.2
61	Flow Regime		Transition	Transition	Transition	Transition
62	Water Flow Regime					
63	Liquid HoldUp		0.2069	0.2069	0.2069	0.2069
64	Water HoldUp		---	---	---	---
65	Friction Gradient	(kPa/m)	6.870e-003	6.870e-003	6.870e-003	6.870e-003
66	Static Gradient	(kPa/m)	-7.751e-004	-7.751e-004	-7.751e-004	-7.751e-004
67	Accel. Gradient	(kPa/m)	4.434e-007	4.434e-007	4.434e-007	4.434e-007
68	Liq. Reynolds		2.068e+005 *	2.068e+005 *	2.068e+005 *	2.068e+005 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.195e+006 *
14	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
15	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	3.347e+004	3.348e+004	3.348e+004	3.349e+004
19	Elevation (m)	0.2100	0.2100	0.2100	0.2100
20	Pressure (bar)	87.94	87.94	87.94	87.94
21	Temperature (C)	48.55	48.55	48.55	48.55
22	Vapour Fraction	0.8684	0.8684	0.8684	0.8684
23	Heat Transferred (kJ/h-m)	494.2	494.2	494.1	494.1
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2069	0.3026	0.3026	0.3026
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.870e-003	5.881e-003	5.881e-003	5.881e-003
29	Static Gradient (kPa/m)	-7.751e-004	0.0000	0.0000	0.0000
30	Accel. Gradient (kPa/m)	4.434e-007	4.279e-007	4.279e-007	4.279e-007
31	Liq. Reynolds	2.068e+005 *	2.068e+005 *	2.068e+005 *	2.068e+005 *
32	Vap. Reynolds	9.195e+006 *	9.195e+006 *	9.195e+006 *	9.196e+006 *
33	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
34	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	3.349e+004	3.350e+004	3.350e+004	3.351e+004
38	Elevation (m)	0.2100	0.2100	0.2120	0.2140
39	Pressure (bar)	87.94	87.94	87.94	87.94
40	Temperature (C)	48.55	48.54	48.54	48.54
41	Vapour Fraction	0.8684	0.8684	0.8684	0.8684
42	Heat Transferred (kJ/h-m)	494.0	494.0	494.0	493.9
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.3026	0.3026	0.2809	0.2809
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	5.881e-003	5.881e-003	5.543e-003	5.543e-003
48	Static Gradient (kPa/m)	0.0000	0.0000	9.515e-004	9.515e-004
49	Accel. Gradient (kPa/m)	4.279e-007	4.279e-007	4.725e-007	4.725e-007
50	Liq. Reynolds	2.068e+005 *	2.068e+005 *	2.068e+005 *	2.068e+005 *
51	Vap. Reynolds	9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *
52	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
53	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	3.351e+004	3.352e+004	3.352e+004	3.353e+004
57	Elevation (m)	0.2160	0.2180	0.2200	0.2140
58	Pressure (bar)	87.94	87.93	87.93	87.93
59	Temperature (C)	48.54	48.54	48.54	48.53
60	Vapour Fraction	0.8684	0.8684	0.8684	0.8684
61	Heat Transferred (kJ/h-m)	493.9	493.8	493.8	493.8
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2809	0.2809	0.2809	0.2062
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	5.543e-003	5.543e-003	5.543e-003	6.876e-003
67	Static Gradient (kPa/m)	9.515e-004	9.515e-004	9.515e-004	-2.320e-003
68	Accel. Gradient (kPa/m)	4.724e-007	4.724e-007	4.724e-007	3.314e-007
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		2.068e+005 *	2.068e+005 *	2.068e+005 *	2.068e+005 *	
14	Vap. Reynolds		9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *	
15	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046	0.2046	
16	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170	2.170	
17	Deposit Thickness (mm)	---	---	---	---	---	
18	Deposit Volume (m3)	---	---	---	---	---	
19	Length (m)	3.353e+004	3.354e+004	3.354e+004	3.355e+004		
20	Elevation (m)	0.2080	0.2020	0.1960	0.1900		
21	Pressure (bar)	87.93	87.93	87.93	87.93		
22	Temperature (C)	48.53	48.53	48.53	48.53		
23	Vapour Fraction	0.8684	0.8684	0.8684	0.8684		
24	Heat Transferred (kJ/h-m)	493.7	493.7	493.7	493.6		
25	Flow Regime	Transition	Transition	Transition	Transition		
26	Water Flow Regime						
27	Liquid HoldUp	0.2062	0.2062	0.2062	0.2062		
28	Water HoldUp	---	---	---	---		
29	Friction Gradient (kPa/m)	6.876e-003	6.875e-003	6.875e-003	6.875e-003		
30	Static Gradient (kPa/m)	-2.320e-003	-2.320e-003	-2.320e-003	-2.320e-003		
31	Accel. Gradient (kPa/m)	3.314e-007	3.314e-007	3.314e-007	3.314e-007		
32	Liq. Reynolds		2.068e+005 *	2.068e+005 *	2.068e+005 *	2.068e+005 *	
33	Vap. Reynolds		9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *	
34	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046	0.2046	
35	Vapour Velocity (m/s)	2.170	2.170	2.170	2.170	2.169	
36	Deposit Thickness (mm)	---	---	---	---	---	
37	Deposit Volume (m3)	---	---	---	---	---	
38	Length (m)	3.355e+004	3.356e+004	3.356e+004	3.357e+004		
39	Elevation (m)	0.1940	0.1980	0.2020	0.2060		
40	Pressure (bar)	87.93	87.93	87.93	87.93		
41	Temperature (C)	48.53	48.53	48.52	48.52		
42	Vapour Fraction	0.8684	0.8684	0.8684	0.8684		
43	Heat Transferred (kJ/h-m)	493.6	493.5	493.5	493.4		
44	Flow Regime	Transition	Transition	Transition	Transition		
45	Water Flow Regime						
46	Liquid HoldUp	0.2822	0.2822	0.2822	0.2822		
47	Water HoldUp	---	---	---	---		
48	Friction Gradient (kPa/m)	5.443e-003	5.443e-003	5.443e-003	5.443e-003		
49	Static Gradient (kPa/m)	1.909e-003	1.909e-003	1.909e-003	1.909e-003		
50	Accel. Gradient (kPa/m)	5.348e-007	5.348e-007	5.348e-007	5.348e-007		
51	Liq. Reynolds		2.068e+005 *	2.068e+005 *	2.068e+005 *	2.068e+005 *	
52	Vap. Reynolds		9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *	
53	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046	0.2046	
54	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169	2.169	
55	Deposit Thickness (mm)	---	---	---	---	---	
56	Deposit Volume (m3)	---	---	---	---	---	
57	Length (m)	3.357e+004	3.358e+004	3.358e+004	3.359e+004		
58	Elevation (m)	0.2100	0.2000	0.1900	0.1800		
59	Pressure (bar)	87.93	87.93	87.93	87.93		
60	Temperature (C)	48.52	48.52	48.52	48.52		
61	Vapour Fraction	0.8684	0.8684	0.8684	0.8684		
62	Heat Transferred (kJ/h-m)	493.4	493.4	493.3	493.3		
63	Flow Regime	Transition	Transition	Transition	Transition		
64	Water Flow Regime						
65	Liquid HoldUp	0.2822	0.2055	0.2055	0.2055		
66	Water HoldUp	---	---	---	---		
67	Friction Gradient (kPa/m)	5.443e-003	6.881e-003	6.881e-003	6.881e-003		
68	Static Gradient (kPa/m)	1.909e-003	-3.795e-003	-3.795e-003	-3.795e-003		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Accel. Gradient	(kPa/m)	5.348e-007	2.245e-007	2.245e-007	2.245e-007	
14	Liq. Reynolds		2.068e+005 *	2.068e+005 *	2.068e+005 *	2.068e+005 *	
15	Vap. Reynolds		9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *	
16	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
17	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169	
18	Deposit Thickness	(mm)	---	---	---	---	
19	Deposit Volume	(m3)	---	---	---	---	
20	Length	(m)	3.359e+004	3.360e+004	3.360e+004	3.361e+004	
21	Elevation	(m)	0.1700	0.1600	0.1860	0.2120	
22	Pressure	(bar)	87.93	87.93	87.93	87.93	
23	Temperature	(C)	48.52	48.51	48.51	48.51	
24	Vapour Fraction		0.8684	0.8684	0.8684	0.8684	
25	Heat Transferred	(kJ/h-m)	493.3	493.2	493.2	493.1	
26	Flow Regime		Transition	Transition	Transition	Transition	
27	Water Flow Regime						
28	Liquid HoldUp		0.2055	0.2055	0.2962	0.2962	
29	Water HoldUp		---	---	---	---	
30	Friction Gradient	(kPa/m)	6.881e-003	6.881e-003	5.419e-003	5.419e-003	
31	Static Gradient	(kPa/m)	-3.795e-003	-3.795e-003	1.284e-002	1.284e-002	
32	Accel. Gradient	(kPa/m)	2.245e-007	2.245e-007	1.329e-006	1.329e-006	
33	Liq. Reynolds		2.068e+005 *	2.068e+005 *	2.068e+005 *	2.068e+005 *	
34	Vap. Reynolds		9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *	
35	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
36	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169	
37	Deposit Thickness	(mm)	---	---	---	---	
38	Deposit Volume	(m3)	---	---	---	---	
39	Length	(m)	3.361e+004	3.362e+004	3.362e+004	3.363e+004	
40	Elevation	(m)	0.2380	0.2640	0.2900	0.2980	
41	Pressure	(bar)	87.93	87.93	87.93	87.93	
42	Temperature	(C)	48.51	48.51	48.51	48.50	
43	Vapour Fraction		0.8684	0.8684	0.8684	0.8684	
44	Heat Transferred	(kJ/h-m)	493.1	493.0	493.0	492.9	
45	Flow Regime		Transition	Transition	Transition	Transition	
46	Water Flow Regime						
47	Liquid HoldUp		0.2962	0.2962	0.2962	0.2847	
48	Water HoldUp		---	---	---	---	
49	Friction Gradient	(kPa/m)	5.419e-003	5.419e-003	5.419e-003	5.247e-003	
50	Static Gradient	(kPa/m)	1.284e-002	1.284e-002	1.284e-002	3.843e-003	
51	Accel. Gradient	(kPa/m)	1.329e-006	1.329e-006	1.329e-006	6.613e-007	
52	Liq. Reynolds		2.068e+005 *	2.068e+005 *	2.068e+005 *	2.068e+005 *	
53	Vap. Reynolds		9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *	
54	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046	
55	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169	
56	Deposit Thickness	(mm)	---	---	---	---	
57	Deposit Volume	(m3)	---	---	---	---	
58	Length	(m)	3.363e+004	3.364e+004	3.364e+004	3.365e+004	
59	Elevation	(m)	0.3060	0.3140	0.3220	0.3300	
60	Pressure	(bar)	87.93	87.92	87.92	87.92	
61	Temperature	(C)	48.50	48.50	48.50	48.50	
62	Vapour Fraction		0.8684	0.8684	0.8684	0.8684	
63	Heat Transferred	(kJ/h-m)	492.9	492.9	492.8	492.8	
64	Flow Regime		Transition	Transition	Transition	Transition	
65	Water Flow Regime						
66	Liquid HoldUp		0.2847	0.2847	0.2847	0.2847	
67	Water HoldUp		---	---	---	---	
68	Friction Gradient	(kPa/m)	5.247e-003	5.247e-003	5.247e-003	5.247e-003	
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3		Epic Atlantic Limited	Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Static Gradient	(kPa/m)	3.843e-003	3.843e-003	3.843e-003	3.843e-003
14	Accel. Gradient	(kPa/m)	6.613e-007	6.613e-007	6.613e-007	6.613e-007
15	Liq. Reynolds		2.068e+005 *	2.068e+005 *	2.068e+005 *	2.068e+005 *
16	Vap. Reynolds		9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *
17	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
18	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169
19	Deposit Thickness	(mm)	---	---	---	---
20	Deposit Volume	(m3)	---	---	---	---
21	Length	(m)	3.365e+004	3.366e+004	3.366e+004	3.367e+004
22	Elevation	(m)	0.3460	0.3620	0.3780	0.3940
23	Pressure	(bar)	87.92	87.92	87.92	87.92
24	Temperature	(C)	48.50	48.49	48.49	48.49
25	Vapour Fraction		0.8684	0.8684	0.8684	0.8684
26	Heat Transferred	(kJ/h-m)	492.7	492.7	492.6	492.6
27	Flow Regime		Transition	Transition	Transition	Transition
28	Water Flow Regime					
29	Liquid HoldUp		0.2898	0.2898	0.2898	0.2898
30	Water HoldUp		---	---	---	---
31	Friction Gradient	(kPa/m)	4.872e-003	4.872e-003	4.872e-003	4.872e-003
32	Static Gradient	(kPa/m)	7.783e-003	7.783e-003	7.783e-003	7.783e-003
33	Accel. Gradient	(kPa/m)	9.207e-007	9.207e-007	9.207e-007	9.207e-007
34	Liq. Reynolds		2.068e+005 *	2.068e+005 *	2.068e+005 *	2.068e+005 *
35	Vap. Reynolds		9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *
36	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
37	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169
38	Deposit Thickness	(mm)	---	---	---	---
39	Deposit Volume	(m3)	---	---	---	---
40	Length	(m)	3.367e+004	3.368e+004	3.368e+004	3.369e+004
41	Elevation	(m)	0.4100	0.3720	0.3340	0.2960
42	Pressure	(bar)	87.92	87.92	87.92	87.92
43	Temperature	(C)	48.49	48.49	48.49	48.49
44	Vapour Fraction		0.8684	0.8684	0.8684	0.8684
45	Heat Transferred	(kJ/h-m)	492.6	492.5	492.5	492.5
46	Flow Regime		Transition	Transition	Transition	Transition
47	Water Flow Regime					
48	Liquid HoldUp		0.2898	0.2002	0.2002	0.2002
49	Water HoldUp		---	---	---	---
50	Friction Gradient	(kPa/m)	4.872e-003	6.924e-003	6.924e-003	6.924e-003
51	Static Gradient	(kPa/m)	7.783e-003	-1.442e-002	-1.442e-002	-1.442e-002
52	Accel. Gradient	(kPa/m)	9.207e-007	-5.457e-007	-5.457e-007	-5.457e-007
53	Liq. Reynolds		2.068e+005 *	2.068e+005 *	2.068e+005 *	2.068e+005 *
54	Vap. Reynolds		9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *
55	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
56	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169
57	Deposit Thickness	(mm)	---	---	---	---
58	Deposit Volume	(m3)	---	---	---	---
59	Length	(m)	3.369e+004	3.370e+004	3.370e+004	3.371e+004
60	Elevation	(m)	0.2580	0.2200	0.2300	0.2400
61	Pressure	(bar)	87.92	87.92	87.92	87.92
62	Temperature	(C)	48.48	48.48	48.48	48.48
63	Vapour Fraction		0.8683	0.8683	0.8683	0.8683
64	Heat Transferred	(kJ/h-m)	492.4	492.4	492.4	492.3
65	Flow Regime		Transition	Transition	Transition	Transition
66	Water Flow Regime					
67	Liquid HoldUp		0.2002	0.2002	0.2860	0.2860
68	Water HoldUp		---	---	---	---
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1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3						Unit Set: Mixed
4						Date/Time: Mon Jun 13 15:02:44 2022
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	6.924e-003	6.924e-003	5.151e-003	5.151e-003
14	Static Gradient	(kPa/m)	-1.442e-002	-1.442e-002	4.819e-003	4.819e-003
15	Accel. Gradient	(kPa/m)	-5.457e-007	-5.457e-007	7.253e-007	7.253e-007
16	Liq. Reynolds		2.068e+005 *	2.068e+005 *	2.068e+005 *	2.068e+005 *
17	Vap. Reynolds		9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *
18	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
19	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	3.371e+004	3.372e+004	3.372e+004	3.373e+004
23	Elevation	(m)	0.2500	0.2600	0.2700	0.2800
24	Pressure	(bar)	87.92	87.92	87.92	87.92
25	Temperature	(C)	48.48	48.48	48.48	48.47
26	Vapour Fraction		0.8683	0.8683	0.8683	0.8683
27	Heat Transferred	(kJ/h-m)	492.3	492.2	492.2	492.1
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.2860	0.2860	0.2860	0.2860
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	5.151e-003	5.151e-003	5.151e-003	5.151e-003
33	Static Gradient	(kPa/m)	4.819e-003	4.819e-003	4.819e-003	4.819e-003
34	Accel. Gradient	(kPa/m)	7.253e-007	7.253e-007	7.253e-007	7.253e-007
35	Liq. Reynolds		2.068e+005 *	2.068e+005 *	2.068e+005 *	2.068e+005 *
36	Vap. Reynolds		9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *
37	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
38	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	3.373e+004	3.374e+004	3.374e+004	3.375e+004
42	Elevation	(m)	0.2900	0.3000	0.3100	0.3200
43	Pressure	(bar)	87.92	87.92	87.92	87.92
44	Temperature	(C)	48.47	48.47	48.47	48.47
45	Vapour Fraction		0.8683	0.8683	0.8683	0.8683
46	Heat Transferred	(kJ/h-m)	492.1	492.1	492.0	492.0
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2860	0.2860	0.2860	0.2860
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	5.151e-003	5.151e-003	5.151e-003	5.151e-003
52	Static Gradient	(kPa/m)	4.819e-003	4.819e-003	4.819e-003	4.819e-003
53	Accel. Gradient	(kPa/m)	7.253e-007	7.253e-007	7.253e-007	7.253e-007
54	Liq. Reynolds		2.068e+005 *	2.068e+005 *	2.068e+005 *	2.068e+005 *
55	Vap. Reynolds		9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *
56	Liquid Velocity	(m/s)	0.2046	0.2046	0.2046	0.2046
57	Vapour Velocity	(m/s)	2.169	2.169	2.169	2.169
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	3.375e+004	3.376e+004	3.376e+004	3.377e+004
61	Elevation	(m)	0.3280	0.3360	0.3440	0.3520
62	Pressure	(bar)	87.92	87.92	87.92	87.92
63	Temperature	(C)	48.47	48.46	48.46	48.46
64	Vapour Fraction		0.8683	0.8683	0.8683	0.8683
65	Heat Transferred	(kJ/h-m)	491.9	491.9	491.9	491.8
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2848	0.2848	0.2848	0.2848
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	5.246e-003	5.246e-003	5.246e-003	5.246e-003
15	Static Gradient (kPa/m)	3.843e-003	3.843e-003	3.843e-003	3.843e-003
16	Accel. Gradient (kPa/m)	6.613e-007	6.613e-007	6.613e-007	6.613e-007
17	Liq. Reynolds	2.068e+005 *	2.068e+005 *	2.067e+005 *	2.067e+005 *
18	Vap. Reynolds	9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *
19	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
20	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	3.377e+004	3.378e+004	3.378e+004	3.379e+004
24	Elevation (m)	0.3600	0.3220	0.2840	0.2460
25	Pressure (bar)	87.92	87.92	87.92	87.92
26	Temperature (C)	48.46	48.46	48.46	48.46
27	Vapour Fraction	0.8683	0.8683	0.8683	0.8683
28	Heat Transferred (kJ/h-m)	491.8	491.8	491.7	491.7
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2848	0.2003	0.2003	0.2003
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	5.246e-003	6.923e-003	6.923e-003	6.923e-003
34	Static Gradient (kPa/m)	3.843e-003	-1.428e-002	-1.428e-002	-1.428e-002
35	Accel. Gradient (kPa/m)	6.613e-007	-5.352e-007	-5.353e-007	-5.353e-007
36	Liq. Reynolds	2.067e+005 *	2.067e+005 *	2.067e+005 *	2.067e+005 *
37	Vap. Reynolds	9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *
38	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
39	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	3.379e+004	3.380e+004	3.380e+004	3.381e+004
43	Elevation (m)	0.2080	0.1700	0.1760	0.1820
44	Pressure (bar)	87.92	87.92	87.92	87.92
45	Temperature (C)	48.46	48.45	48.45	48.45
46	Vapour Fraction	0.8683	0.8683	0.8683	0.8683
47	Heat Transferred (kJ/h-m)	491.7	491.6	491.6	491.5
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2003	0.2003	0.2835	0.2835
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	6.923e-003	6.923e-003	5.343e-003	5.343e-003
53	Static Gradient (kPa/m)	-1.428e-002	-1.428e-002	2.873e-003	2.873e-003
54	Accel. Gradient (kPa/m)	-5.353e-007	-5.353e-007	5.977e-007	5.977e-007
55	Liq. Reynolds	2.067e+005 *	2.067e+005 *	2.067e+005 *	2.067e+005 *
56	Vap. Reynolds	9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *
57	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
58	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	3.381e+004	3.382e+004	3.382e+004	3.383e+004
62	Elevation (m)	0.1880	0.1940	0.2000	0.2100
63	Pressure (bar)	87.92	87.92	87.92	87.91
64	Temperature (C)	48.45	48.45	48.45	48.44
65	Vapour Fraction	0.8683	0.8683	0.8683	0.8683
66	Heat Transferred (kJ/h-m)	491.5	491.4	491.4	491.4
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.2835	0.2835	0.2835	0.2861
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	5.343e-003	5.343e-003	5.343e-003	5.150e-003
16	Static Gradient (kPa/m)	2.873e-003	2.873e-003	2.873e-003	4.819e-003
17	Accel. Gradient (kPa/m)	5.977e-007	5.977e-007	5.977e-007	7.253e-007
18	Liq. Reynolds	2.067e+005 *	2.067e+005 *	2.067e+005 *	2.067e+005 *
19	Vap. Reynolds	9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *
20	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
21	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	3.383e+004	3.384e+004	3.384e+004	3.385e+004
25	Elevation (m)	0.2200	0.2300	0.2400	0.2500
26	Pressure (bar)	87.91	87.91	87.91	87.91
27	Temperature (C)	48.44	48.44	48.44	48.44
28	Vapour Fraction	0.8683	0.8683	0.8683	0.8683
29	Heat Transferred (kJ/h-m)	491.3	491.3	491.2	491.2
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.2861	0.2861	0.2861	0.2861
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	5.150e-003	5.150e-003	5.150e-003	5.150e-003
35	Static Gradient (kPa/m)	4.819e-003	4.819e-003	4.819e-003	4.819e-003
36	Accel. Gradient (kPa/m)	7.253e-007	7.253e-007	7.253e-007	7.253e-007
37	Liq. Reynolds	2.067e+005 *	2.067e+005 *	2.067e+005 *	2.067e+005 *
38	Vap. Reynolds	9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *
39	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
40	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	3.385e+004	3.385e+004	3.385e+004	3.386e+004
44	Elevation (m)	0.2600	0.2700	0.2800	0.2900
45	Pressure (bar)	87.91	87.91	87.91	87.91
46	Temperature (C)	48.44	48.44	48.44	48.44
47	Vapour Fraction	0.8683	0.8683	0.8683	0.8683
48	Heat Transferred (kJ/h-m)	491.2	491.2	491.1	491.1
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.2956	0.2956	0.2956	0.2956
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	5.303e-003	5.303e-003	5.303e-003	5.303e-003
54	Static Gradient (kPa/m)	1.233e-002	1.233e-002	1.233e-002	1.233e-002
55	Accel. Gradient (kPa/m)	1.283e-006	1.283e-006	1.283e-006	1.283e-006
56	Liq. Reynolds	2.067e+005 *	2.067e+005 *	2.067e+005 *	2.067e+005 *
57	Vap. Reynolds	9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *
58	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
59	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	3.386e+004	3.386e+004	3.386e+004	3.386e+004
63	Elevation (m)	0.3000	0.3000	0.3000	0.3000
64	Pressure (bar)	87.91	87.91	87.91	87.91
65	Temperature (C)	48.44	48.43	48.43	48.43
66	Vapour Fraction	0.8683	0.8683	0.8683	0.8683
67	Heat Transferred (kJ/h-m)	491.1	491.1	491.1	491.1
68	Flow Regime	Transition	Transition	Transition	Transition
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 541 of 937

1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2956	0.3027	0.3027	0.3027
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	5.303e-003	5.880e-003	5.880e-003	5.880e-003
17	Static Gradient (kPa/m)	1.233e-002	0.0000	0.0000	0.0000
18	Accel. Gradient (kPa/m)	1.283e-006	4.278e-007	4.278e-007	4.278e-007
19	Liq. Reynolds	2.067e+005 *	2.067e+005 *	2.067e+005 *	2.067e+005 *
20	Vap. Reynolds	9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *
21	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
22	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	3.387e+004	3.387e+004	3.387e+004	3.387e+004
26	Elevation (m)	0.3000	0.3000	0.3000	0.3000
27	Pressure (bar)	87.91	87.91	87.91	87.91
28	Temperature (C)	48.43	48.43	48.43	48.43
29	Vapour Fraction	0.8683	0.8683	0.8683	0.8683
30	Heat Transferred (kJ/h-m)	491.1	491.0	491.0	491.0
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.3027	0.3027	0.3027	0.3027
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	5.880e-003	5.880e-003	5.880e-003	5.880e-003
36	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
37	Accel. Gradient (kPa/m)	4.278e-007	4.278e-007	4.278e-007	4.278e-007
38	Liq. Reynolds	2.067e+005 *	2.067e+005 *	2.067e+005 *	2.067e+005 *
39	Vap. Reynolds	9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *
40	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
41	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	3.388e+004	3.388e+004	3.388e+004	3.389e+004
45	Elevation (m)	0.3000	0.3000	0.3000	0.3020
46	Pressure (bar)	87.91	87.91	87.91	87.91
47	Temperature (C)	48.43	48.43	48.43	48.43
48	Vapour Fraction	0.8683	0.8683	0.8683	0.8683
49	Heat Transferred (kJ/h-m)	491.0	490.9	490.9	490.9
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.3027	0.3027	0.3027	0.2818
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	5.880e-003	5.880e-003	5.880e-003	5.474e-003
55	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	1.590e-003
56	Accel. Gradient (kPa/m)	4.278e-007	4.278e-007	4.278e-007	5.139e-007
57	Liq. Reynolds	2.067e+005 *	2.067e+005 *	2.067e+005 *	2.067e+005 *
58	Vap. Reynolds	9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *
59	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
60	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	3.389e+004	3.389e+004	3.389e+004	3.390e+004
64	Elevation (m)	0.3040	0.3060	0.3080	0.3100
65	Pressure (bar)	87.91	87.91	87.91	87.91
66	Temperature (C)	48.43	48.42	48.42	48.42
67	Vapour Fraction	0.8683	0.8683	0.8683	0.8683
68	Heat Transferred (kJ/h-m)	490.9	490.8	490.8	490.8
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 542 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2818	0.2818	0.2818	0.2818
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	5.474e-003	5.474e-003	5.474e-003	5.474e-003
18	Static Gradient (kPa/m)	1.590e-003	1.590e-003	1.590e-003	1.590e-003
19	Accel. Gradient (kPa/m)	5.139e-007	5.139e-007	5.139e-007	5.139e-007
20	Liq. Reynolds	2.067e+005 *	2.067e+005 *	2.067e+005 *	2.067e+005 *
21	Vap. Reynolds	9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *
22	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2046
23	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	3.391e+004	3.392e+004	3.393e+004	3.394e+004
27	Elevation (m)	0.3100	0.3100	0.3100	0.3100
28	Pressure (bar)	87.91	87.91	87.91	87.91
29	Temperature (C)	48.42	48.42	48.41	48.41
30	Vapour Fraction	0.8683	0.8683	0.8683	0.8683
31	Heat Transferred (kJ/h-m)	490.7	490.6	490.5	490.4
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.3027	0.3027	0.3027	0.3027
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	5.880e-003	5.880e-003	5.880e-003	5.880e-003
37	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
38	Accel. Gradient (kPa/m)	4.278e-007	4.278e-007	4.278e-007	4.278e-007
39	Liq. Reynolds	2.067e+005 *	2.067e+005 *	2.067e+005 *	2.067e+005 *
40	Vap. Reynolds	9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *
41	Liquid Velocity (m/s)	0.2046	0.2046	0.2046	0.2047
42	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	3.395e+004	3.396e+004	3.398e+004	3.399e+004
46	Elevation (m)	0.3100	0.3080	0.3060	0.3040
47	Pressure (bar)	87.91	87.90	87.90	87.90
48	Temperature (C)	48.41	48.40	48.40	48.40
49	Vapour Fraction	0.8683	0.8683	0.8683	0.8683
50	Heat Transferred (kJ/h-m)	490.4	490.3	490.2	490.1
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.3027	0.2072	0.2072	0.2072
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	5.880e-003	6.866e-003	6.866e-003	6.866e-003
56	Static Gradient (kPa/m)	0.0000	-3.481e-004	-3.481e-004	-3.481e-004
57	Accel. Gradient (kPa/m)	4.278e-007	4.742e-007	4.742e-007	4.742e-007
58	Liq. Reynolds	2.067e+005 *	2.067e+005 *	2.067e+005 *	2.067e+005 *
59	Vap. Reynolds	9.196e+006 *	9.196e+006 *	9.196e+006 *	9.196e+006 *
60	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
61	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	3.400e+004	3.401e+004	3.402e+004	3.403e+004
65	Elevation (m)	0.3020	0.3000	0.2980	0.2960
66	Pressure (bar)	87.90	87.90	87.90	87.90
67	Temperature (C)	48.39	48.39	48.39	48.38
68	Vapour Fraction	0.8683	0.8683	0.8683	0.8683
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022	
6						
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Heat Transferred (kJ/h-m)	490.0	489.9	489.8	489.7	
14	Flow Regime	Transition	Transition	Transition	Transition	
15	Water Flow Regime					
16	Liquid HoldUp	0.2072	0.2072	0.2072	0.2072	
17	Water HoldUp	---	---	---	---	
18	Friction Gradient (kPa/m)	6.866e-003	6.866e-003	6.866e-003	6.866e-003	
19	Static Gradient (kPa/m)	-3.481e-004	-3.481e-004	-3.481e-004	-3.481e-004	
20	Accel. Gradient (kPa/m)	4.742e-007	4.742e-007	4.742e-007	4.742e-007	
21	Liq. Reynolds	2.067e+005 *	2.067e+005 *	2.067e+005 *	2.067e+005 *	
22	Vap. Reynolds	9.196e+006 *	9.196e+006 *	9.196e+006 *	9.197e+006 *	
23	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047	
24	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169	
25	Deposit Thickness (mm)	---	---	---	---	
26	Deposit Volume (m3)	---	---	---	---	
27	Length (m)	3.404e+004	3.405e+004	3.406e+004	3.408e+004	
28	Elevation (m)	0.2940	0.2920	0.2900	0.2880	
29	Pressure (bar)	87.90	87.90	87.90	87.90	
30	Temperature (C)	48.38	48.38	48.37	48.37	
31	Vapour Fraction	0.8683	0.8683	0.8683	0.8683	
32	Heat Transferred (kJ/h-m)	489.7	489.6	489.5	489.4	
33	Flow Regime	Transition	Transition	Transition	Transition	
34	Water Flow Regime					
35	Liquid HoldUp	0.2072	0.2072	0.2072	0.2072	
36	Water HoldUp	---	---	---	---	
37	Friction Gradient (kPa/m)	6.866e-003	6.866e-003	6.866e-003	6.866e-003	
38	Static Gradient (kPa/m)	-3.481e-004	-3.481e-004	-3.481e-004	-3.482e-004	
39	Accel. Gradient (kPa/m)	4.741e-007	4.741e-007	4.741e-007	4.741e-007	
40	Liq. Reynolds	2.067e+005 *	2.067e+005 *	2.067e+005 *	2.067e+005 *	
41	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *	
42	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047	
43	Vapour Velocity (m/s)	2.169	2.169	2.169	2.169	
44	Deposit Thickness (mm)	---	---	---	---	
45	Deposit Volume (m3)	---	---	---	---	
46	Length (m)	3.409e+004	3.410e+004	3.411e+004	3.412e+004	
47	Elevation (m)	0.2860	0.2840	0.2820	0.2800	
48	Pressure (bar)	87.90	87.90	87.90	87.89	
49	Temperature (C)	48.37	48.36	48.36	48.36	
50	Vapour Fraction	0.8683	0.8683	0.8683	0.8683	
51	Heat Transferred (kJ/h-m)	489.3	489.2	489.1	489.0	
52	Flow Regime	Transition	Transition	Transition	Transition	
53	Water Flow Regime					
54	Liquid HoldUp	0.2072	0.2072	0.2072	0.2072	
55	Water HoldUp	---	---	---	---	
56	Friction Gradient (kPa/m)	6.866e-003	6.866e-003	6.865e-003	6.865e-003	
57	Static Gradient (kPa/m)	-3.482e-004	-3.482e-004	-3.482e-004	-3.482e-004	
58	Accel. Gradient (kPa/m)	4.741e-007	4.741e-007	4.741e-007	4.741e-007	
59	Liq. Reynolds	2.067e+005 *	2.067e+005 *	2.067e+005 *	2.066e+005 *	
60	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *	
61	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047	
62	Vapour Velocity (m/s)	2.169	2.169	2.168	2.168	
63	Deposit Thickness (mm)	---	---	---	---	
64	Deposit Volume (m3)	---	---	---	---	
65	Length (m)	3.413e+004	3.414e+004	3.415e+004	3.416e+004	
66	Elevation (m)	0.2840	0.2880	0.2920	0.2960	
67	Pressure (bar)	87.89	87.89	87.89	87.89	
68	Temperature (C)	48.35	48.35	48.35	48.34	
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8683	0.8683	0.8683	0.8683
14	Heat Transferred (kJ/h-m)	488.9	488.9	488.8	488.7
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2809	0.2809	0.2809	0.2809
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	5.550e-003	5.549e-003	5.549e-003	5.549e-003
20	Static Gradient (kPa/m)	8.540e-004	8.540e-004	8.540e-004	8.540e-004
21	Accel. Gradient (kPa/m)	4.658e-007	4.658e-007	4.658e-007	4.658e-007
22	Liq. Reynolds	2.066e+005 *	2.066e+005 *	2.066e+005 *	2.066e+005 *
23	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *
24	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
25	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	3.418e+004	3.419e+004	3.420e+004	3.421e+004
29	Elevation (m)	0.3000	0.3300	0.3600	0.3900
30	Pressure (bar)	87.89	87.89	87.89	87.89
31	Temperature (C)	48.34	48.34	48.33	48.33
32	Vapour Fraction	0.8683	0.8683	0.8683	0.8683
33	Heat Transferred (kJ/h-m)	488.6	488.5	488.4	488.3
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2809	0.2883	0.2883	0.2883
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	5.549e-003	4.986e-003	4.986e-003	4.986e-003
39	Static Gradient (kPa/m)	8.540e-004	6.525e-003	6.525e-003	6.525e-003
40	Accel. Gradient (kPa/m)	4.658e-007	8.374e-007	8.374e-007	8.374e-007
41	Liq. Reynolds	2.066e+005 *	2.066e+005 *	2.066e+005 *	2.066e+005 *
42	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *
43	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
44	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	3.422e+004	3.423e+004	3.424e+004	3.425e+004
48	Elevation (m)	0.4200	0.4500	0.4180	0.3860
49	Pressure (bar)	87.89	87.88	87.88	87.88
50	Temperature (C)	48.33	48.32	48.32	48.32
51	Vapour Fraction	0.8683	0.8683	0.8683	0.8683
52	Heat Transferred (kJ/h-m)	488.2	488.1	488.0	488.0
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.2883	0.2883	0.2047	0.2047
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	4.986e-003	4.986e-003	6.885e-003	6.885e-003
58	Static Gradient (kPa/m)	6.525e-003	6.525e-003	-5.528e-003	-5.528e-003
59	Accel. Gradient (kPa/m)	8.374e-007	8.374e-007	9.871e-008	9.870e-008
60	Liq. Reynolds	2.066e+005 *	2.066e+005 *	2.066e+005 *	2.066e+005 *
61	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *
62	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
63	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	3.426e+004	3.428e+004	3.429e+004	3.429e+004
67	Elevation (m)	0.3540	0.3220	0.2900	0.2960
68	Pressure (bar)	87.88	87.88	87.88	87.88
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	48.31	48.31	48.31	48.31
14	Vapour Fraction	0.8683	0.8683	0.8683	0.8683
15	Heat Transferred (kJ/h-m)	487.9	487.8	487.7	487.7
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.2047	0.2047	0.2047	0.2889
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	6.885e-003	6.885e-003	6.885e-003	4.947e-003
21	Static Gradient (kPa/m)	-5.528e-003	-5.528e-003	-5.528e-003	6.940e-003
22	Accel. Gradient (kPa/m)	9.869e-008	9.868e-008	9.866e-008	8.647e-007
23	Liq. Reynolds	2.066e+005 *	2.066e+005 *	2.066e+005 *	2.066e+005 *
24	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *
25	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
26	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	3.429e+004	3.429e+004	3.430e+004	3.430e+004
30	Elevation (m)	0.3020	0.3080	0.3140	0.3200
31	Pressure (bar)	87.88	87.88	87.88	87.88
32	Temperature (C)	48.31	48.31	48.30	48.30
33	Vapour Fraction	0.8683	0.8683	0.8683	0.8682
34	Heat Transferred (kJ/h-m)	487.7	487.7	487.6	487.6
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2889	0.2889	0.2889	0.2889
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	4.947e-003	4.947e-003	4.947e-003	4.947e-003
40	Static Gradient (kPa/m)	6.940e-003	6.940e-003	6.940e-003	6.940e-003
41	Accel. Gradient (kPa/m)	8.647e-007	8.647e-007	8.647e-007	8.647e-007
42	Liq. Reynolds	2.066e+005 *	2.066e+005 *	2.066e+005 *	2.066e+005 *
43	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *
44	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
45	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	3.430e+004	3.431e+004	3.431e+004	3.432e+004
49	Elevation (m)	0.3240	0.3280	0.3320	0.3360
50	Pressure (bar)	87.88	87.88	87.88	87.88
51	Temperature (C)	48.30	48.30	48.30	48.30
52	Vapour Fraction	0.8682	0.8682	0.8682	0.8682
53	Heat Transferred (kJ/h-m)	487.6	487.5	487.5	487.5
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2823	0.2823	0.2823	0.2823
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	5.439e-003	5.439e-003	5.439e-003	5.439e-003
59	Static Gradient (kPa/m)	1.910e-003	1.910e-003	1.910e-003	1.910e-003
60	Accel. Gradient (kPa/m)	5.346e-007	5.346e-007	5.346e-007	5.346e-007
61	Liq. Reynolds	2.066e+005 *	2.066e+005 *	2.066e+005 *	2.066e+005 *
62	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *
63	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
64	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	3.432e+004	3.433e+004	3.433e+004	3.434e+004
68	Elevation (m)	0.3400	0.3360	0.3320	0.3280
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Pressure (bar)	87.88	87.88	87.88	87.88		
14	Temperature (C)	48.30	48.30	48.29	48.29		
15	Vapour Fraction	0.8682	0.8682	0.8682	0.8682		
16	Heat Transferred (kJ/h-m)	487.4	487.4	487.4	487.3		
17	Flow Regime	Transition	Transition	Transition	Transition		
18	Water Flow Regime						
19	Liquid HoldUp	0.2823	0.2067	0.2067	0.2067		
20	Water HoldUp	---	---	---	---		
21	Friction Gradient (kPa/m)	5.439e-003	6.869e-003	6.869e-003	6.869e-003		
22	Static Gradient (kPa/m)	1.910e-003	-1.549e-003	-1.549e-003	-1.549e-003		
23	Accel. Gradient (kPa/m)	5.346e-007	3.870e-007	3.870e-007	3.870e-007		
24	Liq. Reynolds	2.066e+005 *	2.066e+005 *	2.066e+005 *	2.066e+005 *		
25	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *		
26	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047		
27	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
28	Deposit Thickness (mm)	---	---	---	---		
29	Deposit Volume (m3)	---	---	---	---		
30	Length (m)	3.434e+004	3.435e+004	3.435e+004	3.436e+004		
31	Elevation (m)	0.3240	0.3200	0.3220	0.3240		
32	Pressure (bar)	87.88	87.88	87.88	87.88		
33	Temperature (C)	48.29	48.29	48.29	48.29		
34	Vapour Fraction	0.8682	0.8682	0.8682	0.8682		
35	Heat Transferred (kJ/h-m)	487.3	487.2	487.2	487.1		
36	Flow Regime	Transition	Transition	Transition	Transition		
37	Water Flow Regime						
38	Liquid HoldUp	0.2067	0.2067	0.2810	0.2810		
39	Water HoldUp	---	---	---	---		
40	Friction Gradient (kPa/m)	6.869e-003	6.869e-003	5.538e-003	5.538e-003		
41	Static Gradient (kPa/m)	-1.549e-003	-1.549e-003	9.521e-004	9.521e-004		
42	Accel. Gradient (kPa/m)	3.870e-007	3.870e-007	4.721e-007	4.721e-007		
43	Liq. Reynolds	2.066e+005 *	2.066e+005 *	2.066e+005 *	2.066e+005 *		
44	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *		
45	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047		
46	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
47	Deposit Thickness (mm)	---	---	---	---		
48	Deposit Volume (m3)	---	---	---	---		
49	Length (m)	3.436e+004	3.437e+004	3.437e+004	3.438e+004		
50	Elevation (m)	0.3260	0.3280	0.3300	0.3240		
51	Pressure (bar)	87.88	87.88	87.88	87.88		
52	Temperature (C)	48.28	48.28	48.28	48.28		
53	Vapour Fraction	0.8682	0.8682	0.8682	0.8682		
54	Heat Transferred (kJ/h-m)	487.1	487.1	487.0	487.0		
55	Flow Regime	Transition	Transition	Transition	Transition		
56	Water Flow Regime						
57	Liquid HoldUp	0.2810	0.2810	0.2810	0.2063		
58	Water HoldUp	---	---	---	---		
59	Friction Gradient (kPa/m)	5.538e-003	5.538e-003	5.538e-003	6.872e-003		
60	Static Gradient (kPa/m)	9.521e-004	9.521e-004	9.521e-004	-2.321e-003		
61	Accel. Gradient (kPa/m)	4.721e-007	4.721e-007	4.721e-007	3.310e-007		
62	Liq. Reynolds	2.066e+005 *	2.066e+005 *	2.066e+005 *	2.066e+005 *		
63	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *		
64	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047		
65	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
66	Deposit Thickness (mm)	---	---	---	---		
67	Deposit Volume (m3)	---	---	---	---		
68	Length (m)	3.438e+004	3.439e+004	3.439e+004	3.440e+004		
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	0.3180	0.3120	0.3060	0.3000
14	Pressure (bar)	87.88	87.88	87.88	87.88
15	Temperature (C)	48.28	48.28	48.28	48.27
16	Vapour Fraction	0.8682	0.8682	0.8682	0.8682
17	Heat Transferred (kJ/h-m)	487.0	486.9	486.9	486.8
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2063	0.2063	0.2063	0.2063
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	6.872e-003	6.872e-003	6.872e-003	6.872e-003
23	Static Gradient (kPa/m)	-2.321e-003	-2.321e-003	-2.321e-003	-2.321e-003
24	Accel. Gradient (kPa/m)	3.310e-007	3.310e-007	3.310e-007	3.310e-007
25	Liq. Reynolds	2.066e+005 *	2.066e+005 *	2.066e+005 *	2.066e+005 *
26	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *
27	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
28	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	3.440e+004	3.441e+004	3.441e+004	3.442e+004
32	Elevation (m)	0.3000	0.3000	0.3000	0.3000
33	Pressure (bar)	87.88	87.88	87.88	87.88
34	Temperature (C)	48.27	48.27	48.27	48.27
35	Vapour Fraction	0.8682	0.8682	0.8682	0.8682
36	Heat Transferred (kJ/h-m)	486.8	486.8	486.7	486.7
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.3028	0.3028	0.3028	0.3028
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	5.879e-003	5.879e-003	5.879e-003	5.879e-003
42	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
43	Accel. Gradient (kPa/m)	4.277e-007	4.276e-007	4.276e-007	4.276e-007
44	Liq. Reynolds	2.066e+005 *	2.066e+005 *	2.066e+005 *	2.066e+005 *
45	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *
46	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
47	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	3.442e+004	3.443e+004	3.443e+004	3.444e+004
51	Elevation (m)	0.3000	0.3180	0.3360	0.3540
52	Pressure (bar)	87.87	87.87	87.87	87.87
53	Temperature (C)	48.27	48.27	48.26	48.26
54	Vapour Fraction	0.8682	0.8682	0.8682	0.8682
55	Heat Transferred (kJ/h-m)	486.6	486.6	486.6	486.5
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.3028	0.2912	0.2912	0.2912
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	5.879e-003	4.777e-003	4.777e-003	4.777e-003
61	Static Gradient (kPa/m)	0.0000	8.788e-003	8.788e-003	8.788e-003
62	Accel. Gradient (kPa/m)	4.276e-007	9.868e-007	9.868e-007	9.868e-007
63	Liq. Reynolds	2.066e+005 *	2.066e+005 *	2.066e+005 *	2.066e+005 *
64	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *
65	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
66	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3							
4							
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10	Pipe Table						
11							
12							
13	Length (m)	3.444e+004	3.445e+004	3.445e+004	3.446e+004		
14	Elevation (m)	0.3720	0.3900	0.3720	0.3540		
15	Pressure (bar)	87.87	87.87	87.87	87.87		
16	Temperature (C)	48.26	48.26	48.26	48.26		
17	Vapour Fraction	0.8682	0.8682	0.8682	0.8682		
18	Heat Transferred (kJ/h-m)	486.5	486.4	486.4	486.4		
19	Flow Regime	Transition	Transition	Transition	Transition		
20	Water Flow Regime						
21	Liquid HoldUp	0.2912	0.2912	0.2038	0.2038		
22	Water HoldUp	---	---	---	---		
23	Friction Gradient (kPa/m)	4.777e-003	4.777e-003	6.892e-003	6.892e-003		
24	Static Gradient (kPa/m)	8.788e-003	8.788e-003	-7.482e-003	-7.482e-003		
25	Accel. Gradient (kPa/m)	9.868e-007	9.868e-007	-4.294e-008	-4.294e-008		
26	Liq. Reynolds	2.066e+005 *	2.065e+005 *	2.065e+005 *	2.065e+005 *		
27	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *		
28	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047		
29	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
30	Deposit Thickness (mm)	---	---	---	---		
31	Deposit Volume (m3)	---	---	---	---		
32	Length (m)	3.446e+004	3.447e+004	3.447e+004	3.448e+004		
33	Elevation (m)	0.3360	0.3180	0.3000	0.3000		
34	Pressure (bar)	87.87	87.87	87.87	87.87		
35	Temperature (C)	48.26	48.25	48.25	48.25		
36	Vapour Fraction	0.8682	0.8682	0.8682	0.8682		
37	Heat Transferred (kJ/h-m)	486.3	486.3	486.3	486.2		
38	Flow Regime	Transition	Transition	Transition	Transition		
39	Water Flow Regime						
40	Liquid HoldUp	0.2038	0.2038	0.2038	0.3028		
41	Water HoldUp	---	---	---	---		
42	Friction Gradient (kPa/m)	6.892e-003	6.892e-003	6.891e-003	5.879e-003		
43	Static Gradient (kPa/m)	-7.482e-003	-7.482e-003	-7.482e-003	0.0000		
44	Accel. Gradient (kPa/m)	-4.295e-008	-4.295e-008	-4.296e-008	4.276e-007		
45	Liq. Reynolds	2.065e+005 *	2.065e+005 *	2.065e+005 *	2.065e+005 *		
46	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *		
47	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047		
48	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
49	Deposit Thickness (mm)	---	---	---	---		
50	Deposit Volume (m3)	---	---	---	---		
51	Length (m)	3.448e+004	3.449e+004	3.449e+004	3.450e+004		
52	Elevation (m)	0.3000	0.3000	0.3000	0.3000		
53	Pressure (bar)	87.87	87.87	87.87	87.87		
54	Temperature (C)	48.25	48.25	48.25	48.25		
55	Vapour Fraction	0.8682	0.8682	0.8682	0.8682		
56	Heat Transferred (kJ/h-m)	486.2	486.2	486.1	486.1		
57	Flow Regime	Transition	Transition	Transition	Transition		
58	Water Flow Regime						
59	Liquid HoldUp	0.3028	0.3028	0.3028	0.3028		
60	Water HoldUp	---	---	---	---		
61	Friction Gradient (kPa/m)	5.879e-003	5.879e-003	5.879e-003	5.878e-003		
62	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000		
63	Accel. Gradient (kPa/m)	4.276e-007	4.276e-007	4.276e-007	4.276e-007		
64	Liq. Reynolds	2.065e+005 *	2.065e+005 *	2.065e+005 *	2.065e+005 *		
65	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *		
66	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047		
67	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168		
68	Deposit Thickness (mm)	---	---	---	---		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	3.450e+004	3.451e+004	3.451e+004	3.452e+004
15	Elevation (m)	0.2980	0.2960	0.2940	0.2920
16	Pressure (bar)	87.87	87.87	87.87	87.87
17	Temperature (C)	48.24	48.24	48.24	48.24
18	Vapour Fraction	0.8682	0.8682	0.8682	0.8682
19	Heat Transferred (kJ/h-m)	486.0	486.0	486.0	485.9
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2070	0.2070	0.2070	0.2070
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	6.865e-003	6.865e-003	6.865e-003	6.865e-003
25	Static Gradient (kPa/m)	-7.756e-004	-7.756e-004	-7.756e-004	-7.756e-004
26	Accel. Gradient (kPa/m)	4.430e-007	4.430e-007	4.430e-007	4.430e-007
27	Liq. Reynolds	2.065e+005 *	2.065e+005 *	2.065e+005 *	2.065e+005 *
28	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *
29	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
30	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	3.452e+004	3.453e+004	3.453e+004	3.454e+004
34	Elevation (m)	0.2900	0.3000	0.3100	0.3200
35	Pressure (bar)	87.87	87.87	87.87	87.87
36	Temperature (C)	48.24	48.24	48.24	48.23
37	Vapour Fraction	0.8682	0.8682	0.8682	0.8682
38	Heat Transferred (kJ/h-m)	485.9	485.8	485.8	485.7
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2071	0.2862	0.2862	0.2862
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	6.865e-003	5.147e-003	5.147e-003	5.147e-003
44	Static Gradient (kPa/m)	-7.756e-004	4.822e-003	4.822e-003	4.822e-003
45	Accel. Gradient (kPa/m)	4.430e-007	7.251e-007	7.251e-007	7.251e-007
46	Liq. Reynolds	2.065e+005 *	2.065e+005 *	2.065e+005 *	2.065e+005 *
47	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *
48	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
49	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	3.454e+004	3.455e+004	3.455e+004	3.456e+004
53	Elevation (m)	0.3300	0.3400	0.3320	0.3240
54	Pressure (bar)	87.87	87.87	87.87	87.87
55	Temperature (C)	48.23	48.23	48.23	48.23
56	Vapour Fraction	0.8682	0.8682	0.8682	0.8682
57	Heat Transferred (kJ/h-m)	485.7	485.7	485.6	485.6
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2862	0.2862	0.2059	0.2059
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	5.147e-003	5.147e-003	6.874e-003	6.874e-003
63	Static Gradient (kPa/m)	4.822e-003	4.822e-003	-3.092e-003	-3.092e-003
64	Accel. Gradient (kPa/m)	7.251e-007	7.252e-007	2.751e-007	2.751e-007
65	Liq. Reynolds	2.065e+005 *	2.065e+005 *	2.065e+005 *	2.065e+005 *
66	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *
67	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
68	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Thickness (mm)	---	---	---	---
14	Deposit Volume (m3)	---	---	---	---
15	Length (m)	3.456e+004	3.457e+004	3.457e+004	3.458e+004
16	Elevation (m)	0.3160	0.3080	0.3000	0.3040
17	Pressure (bar)	87.87	87.87	87.87	87.86
18	Temperature (C)	48.23	48.23	48.22	48.22
19	Vapour Fraction	0.8682	0.8682	0.8682	0.8682
20	Heat Transferred (kJ/h-m)	485.6	485.5	485.5	485.4
21	Flow Regime	Transition	Transition	Transition	Transition
22	Water Flow Regime				
23	Liquid HoldUp	0.2059	0.2059	0.2059	0.2823
24	Water HoldUp	---	---	---	---
25	Friction Gradient (kPa/m)	6.874e-003	6.874e-003	6.874e-003	5.437e-003
26	Static Gradient (kPa/m)	-3.092e-003	-3.092e-003	-3.092e-003	1.911e-003
27	Accel. Gradient (kPa/m)	2.751e-007	2.751e-007	2.751e-007	5.345e-007
28	Liq. Reynolds	2.065e+005 *	2.065e+005 *	2.065e+005 *	2.065e+005 *
29	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *
30	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
31	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
32	Deposit Thickness (mm)	---	---	---	---
33	Deposit Volume (m3)	---	---	---	---
34	Length (m)	3.458e+004	3.459e+004	3.459e+004	3.460e+004
35	Elevation (m)	0.3080	0.3120	0.3160	0.3200
36	Pressure (bar)	87.86	87.86	87.86	87.86
37	Temperature (C)	48.22	48.22	48.22	48.22
38	Vapour Fraction	0.8682	0.8682	0.8682	0.8682
39	Heat Transferred (kJ/h-m)	485.4	485.4	485.3	485.3
40	Flow Regime	Transition	Transition	Transition	Transition
41	Water Flow Regime				
42	Liquid HoldUp	0.2823	0.2824	0.2824	0.2824
43	Water HoldUp	---	---	---	---
44	Friction Gradient (kPa/m)	5.437e-003	5.437e-003	5.437e-003	5.437e-003
45	Static Gradient (kPa/m)	1.911e-003	1.911e-003	1.911e-003	1.911e-003
46	Accel. Gradient (kPa/m)	5.345e-007	5.345e-007	5.345e-007	5.345e-007
47	Liq. Reynolds	2.065e+005 *	2.065e+005 *	2.065e+005 *	2.065e+005 *
48	Vap. Reynolds	9.197e+006 *	9.197e+006 *	9.197e+006 *	9.197e+006 *
49	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
50	Vapour Velocity (m/s)	2.168	2.168	2.168	2.168
51	Deposit Thickness (mm)	---	---	---	---
52	Deposit Volume (m3)	---	---	---	---
53	Length (m)	3.460e+004	3.461e+004	3.461e+004	3.462e+004
54	Elevation (m)	0.3180	0.3160	0.3140	0.3120
55	Pressure (bar)	87.86	87.86	87.86	87.86
56	Temperature (C)	48.21	48.21	48.21	48.21
57	Vapour Fraction	0.8682	0.8682	0.8682	0.8682
58	Heat Transferred (kJ/h-m)	485.2	485.2	485.2	485.1
59	Flow Regime	Transition	Transition	Transition	Transition
60	Water Flow Regime				
61	Liquid HoldUp	0.2071	0.2071	0.2071	0.2071
62	Water HoldUp	---	---	---	---
63	Friction Gradient (kPa/m)	6.865e-003	6.865e-003	6.865e-003	6.865e-003
64	Static Gradient (kPa/m)	-7.757e-004	-7.757e-004	-7.757e-004	-7.757e-004
65	Accel. Gradient (kPa/m)	4.429e-007	4.429e-007	4.429e-007	4.429e-007
66	Liq. Reynolds	2.065e+005 *	2.065e+005 *	2.065e+005 *	2.065e+005 *
67	Vap. Reynolds	9.198e+006 *	9.198e+006 *	9.198e+006 *	9.198e+006 *
68	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2			Pipe Segment: ELEP-Akabel Pipeline (continued)			
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022	
6						
7						
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168
14	Deposit Thickness	(mm)	---	---	---	---
15	Deposit Volume	(m3)	---	---	---	---
16	Length	(m)	3.462e+004	3.463e+004	3.463e+004	3.464e+004
17	Elevation	(m)	0.3100	0.3100	0.3100	0.3100
18	Pressure	(bar)	87.86	87.86	87.86	87.86
19	Temperature	(C)	48.21	48.21	48.21	48.20
20	Vapour Fraction		0.8682	0.8682	0.8682	0.8682
21	Heat Transferred	(kJ/h-m)	485.1	485.0	485.0	485.0
22	Flow Regime		Transition	Transition	Transition	Transition
23	Water Flow Regime					
24	Liquid HoldUp		0.2071	0.3028	0.3028	0.3028
25	Water HoldUp		---	---	---	---
26	Friction Gradient	(kPa/m)	6.865e-003	5.878e-003	5.878e-003	5.878e-003
27	Static Gradient	(kPa/m)	-7.757e-004	0.0000	0.0000	0.0000
28	Accel. Gradient	(kPa/m)	4.429e-007	4.276e-007	4.276e-007	4.276e-007
29	Liq. Reynolds		2.065e+005 *	2.065e+005 *	2.065e+005 *	2.065e+005 *
30	Vap. Reynolds		9.198e+006 *	9.198e+006 *	9.198e+006 *	9.198e+006 *
31	Liquid Velocity	(m/s)	0.2047	0.2047	0.2047	0.2047
32	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168
33	Deposit Thickness	(mm)	---	---	---	---
34	Deposit Volume	(m3)	---	---	---	---
35	Length	(m)	3.464e+004	3.465e+004	3.465e+004	3.465e+004
36	Elevation	(m)	0.3100	0.3100	0.3260	0.3420
37	Pressure	(bar)	87.86	87.86	87.86	87.86
38	Temperature	(C)	48.20	48.20	48.20	48.20
39	Vapour Fraction		0.8682	0.8682	0.8682	0.8682
40	Heat Transferred	(kJ/h-m)	484.9	484.9	484.9	484.8
41	Flow Regime		Transition	Transition	Transition	Transition
42	Water Flow Regime					
43	Liquid HoldUp		0.3028	0.3028	0.2922	0.2922
44	Water HoldUp		---	---	---	---
45	Friction Gradient	(kPa/m)	5.878e-003	5.878e-003	4.708e-003	4.708e-003
46	Static Gradient	(kPa/m)	0.0000	0.0000	9.555e-003	9.555e-003
47	Accel. Gradient	(kPa/m)	4.276e-007	4.276e-007	1.038e-006	1.038e-006
48	Liq. Reynolds		2.065e+005 *	2.065e+005 *	2.065e+005 *	2.065e+005 *
49	Vap. Reynolds		9.198e+006 *	9.198e+006 *	9.198e+006 *	9.198e+006 *
50	Liquid Velocity	(m/s)	0.2047	0.2047	0.2047	0.2047
51	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.168
52	Deposit Thickness	(mm)	---	---	---	---
53	Deposit Volume	(m3)	---	---	---	---
54	Length	(m)	3.466e+004	3.466e+004	3.467e+004	3.467e+004
55	Elevation	(m)	0.3580	0.3740	0.3900	0.3900
56	Pressure	(bar)	87.86	87.86	87.86	87.86
57	Temperature	(C)	48.20	48.20	48.19	48.19
58	Vapour Fraction		0.8682	0.8682	0.8682	0.8682
59	Heat Transferred	(kJ/h-m)	484.8	484.7	484.7	484.7
60	Flow Regime		Transition	Transition	Transition	Transition
61	Water Flow Regime					
62	Liquid HoldUp		0.2922	0.2922	0.2922	0.3028
63	Water HoldUp		---	---	---	---
64	Friction Gradient	(kPa/m)	4.708e-003	4.708e-003	4.708e-003	5.878e-003
65	Static Gradient	(kPa/m)	9.555e-003	9.555e-003	9.555e-003	0.0000
66	Accel. Gradient	(kPa/m)	1.038e-006	1.038e-006	1.038e-006	4.276e-007
67	Liq. Reynolds		2.065e+005 *	2.065e+005 *	2.065e+005 *	2.065e+005 *
68	Vap. Reynolds		9.198e+006 *	9.198e+006 *	9.198e+006 *	9.198e+006 *
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3						
4						
5			Unit Set: Mixed		Date/Time: Mon Jun 13 15:02:44 2022	
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	PROFILES					
10	Pipe Table					
11						
12						
13	Liquid Velocity	(m/s)	0.2047	0.2047	0.2047	0.2047
14	Vapour Velocity	(m/s)	2.168	2.168	2.168	2.167
15	Deposit Thickness	(mm)	---	---	---	---
16	Deposit Volume	(m3)	---	---	---	---
17	Length	(m)	3.468e+004	3.468e+004	3.469e+004	3.469e+004
18	Elevation	(m)	0.3900	0.3900	0.3900	0.3900
19	Pressure	(bar)	87.86	87.86	87.86	87.86
20	Temperature	(C)	48.19	48.19	48.19	48.19
21	Vapour Fraction		0.8682	0.8682	0.8682	0.8682
22	Heat Transferred	(kJ/h-m)	484.6	484.6	484.6	484.5
23	Flow Regime		Transition	Transition	Transition	Transition
24	Water Flow Regime					
25	Liquid HoldUp		0.3028	0.3028	0.3028	0.3028
26	Water HoldUp		---	---	---	---
27	Friction Gradient	(kPa/m)	5.878e-003	5.878e-003	5.878e-003	5.878e-003
28	Static Gradient	(kPa/m)	0.0000	0.0000	0.0000	0.0000
29	Accel. Gradient	(kPa/m)	4.276e-007	4.276e-007	4.276e-007	4.276e-007
30	Liq. Reynolds		2.065e+005 *	2.065e+005 *	2.065e+005 *	2.065e+005 *
31	Vap. Reynolds		9.198e+006 *	9.198e+006 *	9.198e+006 *	9.198e+006 *
32	Liquid Velocity	(m/s)	0.2047	0.2047	0.2047	0.2047
33	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167
34	Deposit Thickness	(mm)	---	---	---	---
35	Deposit Volume	(m3)	---	---	---	---
36	Length	(m)	3.470e+004	3.470e+004	3.471e+004	3.471e+004
37	Elevation	(m)	0.3820	0.3740	0.3660	0.3580
38	Pressure	(bar)	87.86	87.86	87.86	87.86
39	Temperature	(C)	48.19	48.18	48.18	48.18
40	Vapour Fraction		0.8682	0.8682	0.8682	0.8682
41	Heat Transferred	(kJ/h-m)	484.5	484.5	484.4	484.4
42	Flow Regime		Transition	Transition	Transition	Transition
43	Water Flow Regime					
44	Liquid HoldUp		0.2060	0.2060	0.2060	0.2060
45	Water HoldUp		---	---	---	---
46	Friction Gradient	(kPa/m)	6.873e-003	6.873e-003	6.873e-003	6.873e-003
47	Static Gradient	(kPa/m)	-3.092e-003	-3.092e-003	-3.092e-003	-3.092e-003
48	Accel. Gradient	(kPa/m)	2.751e-007	2.751e-007	2.750e-007	2.750e-007
49	Liq. Reynolds		2.065e+005 *	2.065e+005 *	2.065e+005 *	2.065e+005 *
50	Vap. Reynolds		9.198e+006 *	9.198e+006 *	9.198e+006 *	9.198e+006 *
51	Liquid Velocity	(m/s)	0.2047	0.2047	0.2047	0.2047
52	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167
53	Deposit Thickness	(mm)	---	---	---	---
54	Deposit Volume	(m3)	---	---	---	---
55	Length	(m)	3.472e+004	3.472e+004	3.473e+004	3.473e+004
56	Elevation	(m)	0.3500	0.3400	0.3300	0.3200
57	Pressure	(bar)	87.86	87.85	87.85	87.85
58	Temperature	(C)	48.18	48.18	48.18	48.18
59	Vapour Fraction		0.8682	0.8682	0.8682	0.8682
60	Heat Transferred	(kJ/h-m)	484.3	484.3	484.3	484.2
61	Flow Regime		Transition	Transition	Transition	Transition
62	Water Flow Regime					
63	Liquid HoldUp		0.2060	0.2056	0.2056	0.2056
64	Water HoldUp		---	---	---	---
65	Friction Gradient	(kPa/m)	6.873e-003	6.876e-003	6.876e-003	6.876e-003
66	Static Gradient	(kPa/m)	-3.092e-003	-3.861e-003	-3.861e-003	-3.861e-003
67	Accel. Gradient	(kPa/m)	2.750e-007	2.193e-007	2.193e-007	2.193e-007
68	Liq. Reynolds		2.065e+005 *	2.065e+005 *	2.065e+005 *	2.065e+005 *
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vap. Reynolds	9.198e+006 *	9.198e+006 *	9.198e+006 *	9.198e+006 *
14	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
15	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
16	Deposit Thickness (mm)	---	---	---	---
17	Deposit Volume (m3)	---	---	---	---
18	Length (m)	3.474e+004	3.474e+004	3.475e+004	3.475e+004
19	Elevation (m)	0.3100	0.3000	0.2880	0.2760
20	Pressure (bar)	87.85	87.85	87.85	87.85
21	Temperature (C)	48.17	48.17	48.17	48.17
22	Vapour Fraction	0.8682	0.8682	0.8682	0.8682
23	Heat Transferred (kJ/h-m)	484.2	484.2	484.1	484.1
24	Flow Regime	Transition	Transition	Transition	Transition
25	Water Flow Regime				
26	Liquid HoldUp	0.2056	0.2056	0.2052	0.2052
27	Water HoldUp	---	---	---	---
28	Friction Gradient (kPa/m)	6.876e-003	6.876e-003	6.879e-003	6.879e-003
29	Static Gradient (kPa/m)	-3.861e-003	-3.861e-003	-4.628e-003	-4.628e-003
30	Accel. Gradient (kPa/m)	2.193e-007	2.193e-007	1.637e-007	1.637e-007
31	Liq. Reynolds	2.065e+005 *	2.065e+005 *	2.065e+005 *	2.065e+005 *
32	Vap. Reynolds	9.198e+006 *	9.198e+006 *	9.198e+006 *	9.198e+006 *
33	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
34	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
35	Deposit Thickness (mm)	---	---	---	---
36	Deposit Volume (m3)	---	---	---	---
37	Length (m)	3.476e+004	3.476e+004	3.477e+004	3.477e+004
38	Elevation (m)	0.2640	0.2520	0.2400	0.2200
39	Pressure (bar)	87.85	87.85	87.85	87.85
40	Temperature (C)	48.17	48.17	48.17	48.17
41	Vapour Fraction	0.8682	0.8682	0.8682	0.8682
42	Heat Transferred (kJ/h-m)	484.0	484.0	484.0	483.9
43	Flow Regime	Transition	Transition	Transition	Transition
44	Water Flow Regime				
45	Liquid HoldUp	0.2052	0.2052	0.2052	0.2037
46	Water HoldUp	---	---	---	---
47	Friction Gradient (kPa/m)	6.879e-003	6.879e-003	6.879e-003	6.891e-003
48	Static Gradient (kPa/m)	-4.628e-003	-4.628e-003	-4.628e-003	-7.678e-003
49	Accel. Gradient (kPa/m)	1.637e-007	1.637e-007	1.637e-007	-5.723e-008
50	Liq. Reynolds	2.065e+005 *	2.065e+005 *	2.065e+005 *	2.065e+005 *
51	Vap. Reynolds	9.198e+006 *	9.198e+006 *	9.198e+006 *	9.198e+006 *
52	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
53	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
54	Deposit Thickness (mm)	---	---	---	---
55	Deposit Volume (m3)	---	---	---	---
56	Length (m)	3.478e+004	3.478e+004	3.479e+004	3.479e+004
57	Elevation (m)	0.2000	0.1800	0.1600	0.1400
58	Pressure (bar)	87.85	87.85	87.85	87.85
59	Temperature (C)	48.16	48.16	48.16	48.16
60	Vapour Fraction	0.8682	0.8682	0.8682	0.8682
61	Heat Transferred (kJ/h-m)	483.9	483.9	483.8	483.8
62	Flow Regime	Transition	Transition	Transition	Transition
63	Water Flow Regime				
64	Liquid HoldUp	0.2037	0.2037	0.2037	0.2037
65	Water HoldUp	---	---	---	---
66	Friction Gradient (kPa/m)	6.891e-003	6.891e-003	6.891e-003	6.891e-003
67	Static Gradient (kPa/m)	-7.678e-003	-7.678e-003	-7.678e-003	-7.678e-003
68	Accel. Gradient (kPa/m)	-5.724e-008	-5.725e-008	-5.725e-008	-5.726e-008
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Liq. Reynolds		2.065e+005 *	2.065e+005 *	2.065e+005 *	2.065e+005 *	
14	Vap. Reynolds		9.198e+006 *	9.198e+006 *	9.198e+006 *	9.198e+006 *	
15	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047		
16	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167		
17	Deposit Thickness (mm)	---	---	---	---		
18	Deposit Volume (m3)	---	---	---	---		
19	Length (m)	3.480e+004	3.480e+004	3.481e+004	3.481e+004		
20	Elevation (m)	0.1280	0.1160	0.1040	9.200e-002		
21	Pressure (bar)	87.85	87.85	87.85	87.85		
22	Temperature (C)	48.16	48.16	48.16	48.15		
23	Vapour Fraction	0.8682	0.8682	0.8682	0.8682		
24	Heat Transferred (kJ/h-m)	483.8	483.7	483.7	483.6		
25	Flow Regime	Transition	Transition	Transition	Transition		
26	Water Flow Regime						
27	Liquid HoldUp	0.2052	0.2052	0.2052	0.2052		
28	Water HoldUp	---	---	---	---		
29	Friction Gradient (kPa/m)	6.878e-003	6.878e-003	6.878e-003	6.878e-003		
30	Static Gradient (kPa/m)	-4.628e-003	-4.628e-003	-4.628e-003	-4.628e-003		
31	Accel. Gradient (kPa/m)	1.637e-007	1.637e-007	1.637e-007	1.637e-007		
32	Liq. Reynolds		2.065e+005 *	2.065e+005 *	2.065e+005 *		
33	Vap. Reynolds		9.198e+006 *	9.198e+006 *	9.198e+006 *		
34	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047		
35	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167		
36	Deposit Thickness (mm)	---	---	---	---		
37	Deposit Volume (m3)	---	---	---	---		
38	Length (m)	3.482e+004	3.482e+004	3.483e+004	3.483e+004		
39	Elevation (m)	8.000e-002	9.200e-002	0.1040	0.1160		
40	Pressure (bar)	87.85	87.85	87.85	87.85		
41	Temperature (C)	48.15	48.15	48.15	48.15		
42	Vapour Fraction	0.8682	0.8682	0.8682	0.8682		
43	Heat Transferred (kJ/h-m)	483.6	483.6	483.5	483.5		
44	Flow Regime	Transition	Transition	Transition	Transition		
45	Water Flow Regime						
46	Liquid HoldUp	0.2052	0.2875	0.2875	0.2875		
47	Water HoldUp	---	---	---	---		
48	Friction Gradient (kPa/m)	6.878e-003	5.051e-003	5.051e-003	5.051e-003		
49	Static Gradient (kPa/m)	-4.628e-003	5.806e-003	5.806e-003	5.806e-003		
50	Accel. Gradient (kPa/m)	1.637e-007	7.896e-007	7.896e-007	7.896e-007		
51	Liq. Reynolds		2.065e+005 *	2.064e+005 *	2.064e+005 *		
52	Vap. Reynolds		9.198e+006 *	9.198e+006 *	9.198e+006 *		
53	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047		
54	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167		
55	Deposit Thickness (mm)	---	---	---	---		
56	Deposit Volume (m3)	---	---	---	---		
57	Length (m)	3.484e+004	3.484e+004	3.485e+004	3.485e+004		
58	Elevation (m)	0.1280	0.1400	0.1340	0.1280		
59	Pressure (bar)	87.85	87.85	87.85	87.85		
60	Temperature (C)	48.15	48.15	48.14	48.14		
61	Vapour Fraction	0.8682	0.8682	0.8682	0.8682		
62	Heat Transferred (kJ/h-m)	483.4	483.4	483.4	483.3		
63	Flow Regime	Transition	Transition	Transition	Transition		
64	Water Flow Regime						
65	Liquid HoldUp	0.2875	0.2875	0.2063	0.2063		
66	Water HoldUp	---	---	---	---		
67	Friction Gradient (kPa/m)	5.051e-003	5.051e-003	6.869e-003	6.869e-003		
68	Static Gradient (kPa/m)	5.806e-003	5.806e-003	-2.320e-003	-2.320e-003		
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Accel. Gradient (kPa/m)	7.896e-007	7.896e-007	3.309e-007	3.309e-007
14	Liq. Reynolds	2.064e+005 *	2.064e+005 *	2.064e+005 *	2.064e+005 *
15	Vap. Reynolds	9.198e+006 *	9.198e+006 *	9.198e+006 *	9.198e+006 *
16	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
17	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
18	Deposit Thickness (mm)	---	---	---	---
19	Deposit Volume (m3)	---	---	---	---
20	Length (m)	3.486e+004	3.486e+004	3.487e+004	3.487e+004
21	Elevation (m)	0.1220	0.1160	0.1100	0.1680
22	Pressure (bar)	87.85	87.85	87.85	87.85
23	Temperature (C)	48.14	48.14	48.14	48.14
24	Vapour Fraction	0.8682	0.8682	0.8682	0.8682
25	Heat Transferred (kJ/h-m)	483.3	483.2	483.2	483.2
26	Flow Regime	Transition	Transition	Transition	Transition
27	Water Flow Regime				
28	Liquid HoldUp	0.2063	0.2063	0.2064	0.3168
29	Water HoldUp	---	---	---	---
30	Friction Gradient (kPa/m)	6.869e-003	6.869e-003	6.869e-003	6.049e-003
31	Static Gradient (kPa/m)	-2.320e-003	-2.320e-003	-2.320e-003	3.009e-002
32	Accel. Gradient (kPa/m)	3.309e-007	3.309e-007	3.309e-007	2.629e-006
33	Liq. Reynolds	2.064e+005 *	2.064e+005 *	2.064e+005 *	2.064e+005 *
34	Vap. Reynolds	9.198e+006 *	9.198e+006 *	9.198e+006 *	9.198e+006 *
35	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
36	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
37	Deposit Thickness (mm)	---	---	---	---
38	Deposit Volume (m3)	---	---	---	---
39	Length (m)	3.488e+004	3.488e+004	3.489e+004	3.489e+004
40	Elevation (m)	0.2260	0.2840	0.3420	0.4000
41	Pressure (bar)	87.85	87.84	87.84	87.84
42	Temperature (C)	48.13	48.13	48.13	48.13
43	Vapour Fraction	0.8682	0.8682	0.8682	0.8682
44	Heat Transferred (kJ/h-m)	483.1	483.0	483.0	482.9
45	Flow Regime	Transition	Transition	Transition	Transition
46	Water Flow Regime				
47	Liquid HoldUp	0.3168	0.3168	0.3168	0.3168
48	Water HoldUp	---	---	---	---
49	Friction Gradient (kPa/m)	6.049e-003	6.050e-003	6.050e-003	6.050e-003
50	Static Gradient (kPa/m)	3.009e-002	3.009e-002	3.009e-002	3.009e-002
51	Accel. Gradient (kPa/m)	2.629e-006	2.629e-006	2.629e-006	2.629e-006
52	Liq. Reynolds	2.064e+005 *	2.064e+005 *	2.064e+005 *	2.064e+005 *
53	Vap. Reynolds	9.198e+006 *	9.198e+006 *	9.198e+006 *	9.198e+006 *
54	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
55	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
56	Deposit Thickness (mm)	---	---	---	---
57	Deposit Volume (m3)	---	---	---	---
58	Length (m)	3.490e+004	3.490e+004	3.491e+004	3.491e+004
59	Elevation (m)	0.2560	0.1120	-3.200e-002	-0.1760
60	Pressure (bar)	87.84	87.84	87.85	87.85
61	Temperature (C)	48.13	48.13	48.13	48.13
62	Vapour Fraction	0.8682	0.8682	0.8681	0.8681
63	Heat Transferred (kJ/h-m)	483.0	482.9	482.9	482.9
64	Flow Regime	Transition	Transition	Transition	Transition
65	Water Flow Regime				
66	Liquid HoldUp	0.1807	0.1807	0.1807	0.1807
67	Water HoldUp	---	---	---	---
68	Friction Gradient (kPa/m)	7.115e-003	7.114e-003	7.114e-003	7.114e-003
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1				Case Name:	Simulation - 386MMSCFD_100barg.hsc		
2				Unit Set:	Mixed		
3		Epic Atlantic Limited					
4				Date/Time:	Mon Jun 13 15:02:44 2022		
5							
6							
7	Pipe Segment: ELEP-Akabel Pipeline (continued)						
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Static Gradient	(kPa/m)	-5.132e-002	-5.132e-002	-5.132e-002	-5.132e-002	
14	Accel. Gradient	(kPa/m)	-3.215e-006	-3.215e-006	-3.215e-006	-3.215e-006	
15	Liq. Reynolds		2.064e+005 *	2.064e+005 *	2.064e+005 *	2.064e+005 *	
16	Vap. Reynolds		9.198e+006 *	9.198e+006 *	9.198e+006 *	9.198e+006 *	
17	Liquid Velocity	(m/s)	0.2047	0.2047	0.2047	0.2047	
18	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167	
19	Deposit Thickness	(mm)	---	---	---	---	
20	Deposit Volume	(m3)	---	---	---	---	
21	Length	(m)	3.492e+004	3.492e+004	3.493e+004	3.493e+004	
22	Elevation	(m)	-0.3200	-0.4160	-0.5120	-0.6080	
23	Pressure	(bar)	87.85	87.85	87.85	87.86	
24	Temperature	(C)	48.13	48.13	48.12	48.12	
25	Vapour Fraction		0.8681	0.8681	0.8681	0.8681	
26	Heat Transferred	(kJ/h-m)	482.9	482.9	482.9	482.8	
27	Flow Regime		Transition	Transition	Transition	Transition	
28	Water Flow Regime						
29	Liquid HoldUp		0.1807	0.1896	0.1896	0.1896	
30	Water HoldUp		---	---	---	---	
31	Friction Gradient	(kPa/m)	7.114e-003	7.018e-003	7.018e-003	7.018e-003	
32	Static Gradient	(kPa/m)	-5.132e-002	-3.524e-002	-3.524e-002	-3.524e-002	
33	Accel. Gradient	(kPa/m)	-3.215e-006	-2.052e-006	-2.052e-006	-2.052e-006	
34	Liq. Reynolds		2.064e+005 *	2.064e+005 *	2.064e+005 *	2.064e+005 *	
35	Vap. Reynolds		9.198e+006 *	9.198e+006 *	9.198e+006 *	9.198e+006 *	
36	Liquid Velocity	(m/s)	0.2047	0.2047	0.2047	0.2047	
37	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167	
38	Deposit Thickness	(mm)	---	---	---	---	
39	Deposit Volume	(m3)	---	---	---	---	
40	Length	(m)	3.494e+004	3.494e+004	3.495e+004	3.495e+004	
41	Elevation	(m)	-0.7040	-0.8000	-0.7000	-0.6000	
42	Pressure	(bar)	87.86	87.86	87.86	87.85	
43	Temperature	(C)	48.12	48.12	48.12	48.12	
44	Vapour Fraction		0.8681	0.8681	0.8681	0.8681	
45	Heat Transferred	(kJ/h-m)	482.8	482.8	482.7	482.6	
46	Flow Regime		Transition	Transition	Transition	Transition	
47	Water Flow Regime						
48	Liquid HoldUp		0.1896	0.1896	0.3329	0.3329	
49	Water HoldUp		---	---	---	---	
50	Friction Gradient	(kPa/m)	7.018e-003	7.018e-003	6.072e-003	6.072e-003	
51	Static Gradient	(kPa/m)	-3.524e-002	-3.524e-002	4.484e-002	4.484e-002	
52	Accel. Gradient	(kPa/m)	-2.052e-006	-2.052e-006	3.702e-006	3.702e-006	
53	Liq. Reynolds		2.064e+005 *	2.064e+005 *	2.064e+005 *	2.064e+005 *	
54	Vap. Reynolds		9.198e+006 *	9.198e+006 *	9.198e+006 *	9.198e+006 *	
55	Liquid Velocity	(m/s)	0.2047	0.2047	0.2047	0.2047	
56	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167	
57	Deposit Thickness	(mm)	---	---	---	---	
58	Deposit Volume	(m3)	---	---	---	---	
59	Length	(m)	3.496e+004	3.497e+004	3.497e+004	3.497e+004	
60	Elevation	(m)	-0.5000	-0.4000	-0.3000	-0.2440	
61	Pressure	(bar)	87.85	87.85	87.84	87.84	
62	Temperature	(C)	48.11	48.11	48.11	48.11	
63	Vapour Fraction		0.8681	0.8681	0.8681	0.8681	
64	Heat Transferred	(kJ/h-m)	482.6	482.5	482.4	482.4	
65	Flow Regime		Transition	Transition	Transition	Transition	
66	Water Flow Regime						
67	Liquid HoldUp		0.3329	0.3329	0.3329	0.3393	
68	Water HoldUp		---	---	---	---	
69	Aspen Technology Inc.		Aspen HYSYS Version 10				
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1				Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3	 Epic Atlantic Limited			Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	PROFILES					
10						
11	Pipe Table					
12						
13	Friction Gradient	(kPa/m)	6.072e-003	6.072e-003	6.072e-003	6.070e-003
14	Static Gradient	(kPa/m)	4.484e-002	4.484e-002	4.483e-002	5.092e-002
15	Accel. Gradient	(kPa/m)	3.702e-006	3.703e-006	3.703e-006	4.145e-006
16	Liq. Reynolds		2.064e+005 *	2.064e+005 *	2.064e+005 *	2.064e+005 *
17	Vap. Reynolds		9.198e+006 *	9.198e+006 *	9.198e+006 *	9.198e+006 *
18	Liquid Velocity	(m/s)	0.2047	0.2047	0.2047	0.2047
19	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167
20	Deposit Thickness	(mm)	---	---	---	---
21	Deposit Volume	(m3)	---	---	---	---
22	Length	(m)	3.498e+004	3.498e+004	3.498e+004	3.499e+004
23	Elevation	(m)	-0.1880	-0.1320	-7.600e-002	-2.000e-002
24	Pressure	(bar)	87.84	87.84	87.84	87.83
25	Temperature	(C)	48.11	48.10	48.10	48.10
26	Vapour Fraction		0.8681	0.8681	0.8681	0.8681
27	Heat Transferred	(kJ/h-m)	482.4	482.3	482.3	482.3
28	Flow Regime		Transition	Transition	Transition	Transition
29	Water Flow Regime					
30	Liquid HoldUp		0.3393	0.3393	0.3393	0.3393
31	Water HoldUp		---	---	---	---
32	Friction Gradient	(kPa/m)	6.070e-003	6.070e-003	6.070e-003	6.070e-003
33	Static Gradient	(kPa/m)	5.092e-002	5.092e-002	5.092e-002	5.092e-002
34	Accel. Gradient	(kPa/m)	4.146e-006	4.146e-006	4.146e-006	4.146e-006
35	Liq. Reynolds		2.064e+005 *	2.064e+005 *	2.064e+005 *	2.064e+005 *
36	Vap. Reynolds		9.198e+006 *	9.198e+006 *	9.198e+006 *	9.198e+006 *
37	Liquid Velocity	(m/s)	0.2047	0.2047	0.2047	0.2047
38	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167
39	Deposit Thickness	(mm)	---	---	---	---
40	Deposit Volume	(m3)	---	---	---	---
41	Length	(m)	3.499e+004	3.500e+004	3.500e+004	3.501e+004
42	Elevation	(m)	1.400e-002	4.800e-002	8.200e-002	0.1160
43	Pressure	(bar)	87.83	87.83	87.83	87.83
44	Temperature	(C)	48.10	48.10	48.10	48.09
45	Vapour Fraction		0.8681	0.8681	0.8681	0.8681
46	Heat Transferred	(kJ/h-m)	482.2	482.1	482.1	482.0
47	Flow Regime		Transition	Transition	Transition	Transition
48	Water Flow Regime					
49	Liquid HoldUp		0.2979	0.2979	0.2979	0.2979
50	Water HoldUp		---	---	---	---
51	Friction Gradient	(kPa/m)	5.604e-003	5.604e-003	5.604e-003	5.604e-003
52	Static Gradient	(kPa/m)	1.406e-002	1.406e-002	1.406e-002	1.406e-002
53	Accel. Gradient	(kPa/m)	1.431e-006	1.431e-006	1.431e-006	1.431e-006
54	Liq. Reynolds		2.064e+005 *	2.064e+005 *	2.064e+005 *	2.064e+005 *
55	Vap. Reynolds		9.198e+006 *	9.19e+006 *	9.198e+006 *	9.198e+006 *
56	Liquid Velocity	(m/s)	0.2047	0.2047	0.2047	0.2047
57	Vapour Velocity	(m/s)	2.167	2.167	2.167	2.167
58	Deposit Thickness	(mm)	---	---	---	---
59	Deposit Volume	(m3)	---	---	---	---
60	Length	(m)	3.502e+004	3.502e+004	3.503e+004	3.503e+004
61	Elevation	(m)	0.1500	0.1560	0.1620	0.1680
62	Pressure	(bar)	87.83	87.83	87.83	87.83
63	Temperature	(C)	48.09	48.09	48.09	48.09
64	Vapour Fraction		0.8681	0.8681	0.8681	0.8681
65	Heat Transferred	(kJ/h-m)	482.0	481.9	481.9	481.8
66	Flow Regime		Transition	Transition	Transition	Transition
67	Water Flow Regime					
68	Liquid HoldUp		0.2979	0.2837	0.2837	0.2837
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Water HoldUp	---	---	---	---
14	Friction Gradient (kPa/m)	5.604e-003	5.337e-003	5.337e-003	5.337e-003
15	Static Gradient (kPa/m)	1.406e-002	2.876e-003	2.876e-003	2.876e-003
16	Accel. Gradient (kPa/m)	1.431e-006	5.975e-007	5.975e-007	5.975e-007
17	Liq. Reynolds	2.064e+005 *	2.064e+005 *	2.064e+005 *	2.064e+005 *
18	Vap. Reynolds	9.198e+006 *	9.198e+006 *	9.199e+006 *	9.199e+006 *
19	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
20	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
21	Deposit Thickness (mm)	---	---	---	---
22	Deposit Volume (m3)	---	---	---	---
23	Length (m)	3.504e+004	3.504e+004	3.505e+004	3.505e+004
24	Elevation (m)	0.1740	0.1800	0.1760	0.1720
25	Pressure (bar)	87.83	87.83	87.83	87.83
26	Temperature (C)	48.09	48.08	48.08	48.08
27	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
28	Heat Transferred (kJ/h-m)	481.8	481.8	481.7	481.7
29	Flow Regime	Transition	Transition	Transition	Transition
30	Water Flow Regime				
31	Liquid HoldUp	0.2837	0.2837	0.2067	0.2067
32	Water HoldUp	---	---	---	---
33	Friction Gradient (kPa/m)	5.337e-003	5.337e-003	6.866e-003	6.866e-003
34	Static Gradient (kPa/m)	2.876e-003	2.876e-003	-1.541e-003	-1.541e-003
35	Accel. Gradient (kPa/m)	5.975e-007	5.975e-007	3.874e-007	3.874e-007
36	Liq. Reynolds	2.064e+005 *	2.064e+005 *	2.064e+005 *	2.064e+005 *
37	Vap. Reynolds	9.199e+006 *	9.199e+006 *	9.199e+006 *	9.199e+006 *
38	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
39	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
40	Deposit Thickness (mm)	---	---	---	---
41	Deposit Volume (m3)	---	---	---	---
42	Length (m)	3.506e+004	3.506e+004	3.507e+004	3.507e+004
43	Elevation (m)	0.1680	0.1640	0.1600	0.1720
44	Pressure (bar)	87.83	87.83	87.83	87.82
45	Temperature (C)	48.08	48.08	48.08	48.08
46	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
47	Heat Transferred (kJ/h-m)	481.7	481.6	481.6	481.5
48	Flow Regime	Transition	Transition	Transition	Transition
49	Water Flow Regime				
50	Liquid HoldUp	0.2067	0.2067	0.2067	0.2875
51	Water HoldUp	---	---	---	---
52	Friction Gradient (kPa/m)	6.866e-003	6.866e-003	6.866e-003	5.050e-003
53	Static Gradient (kPa/m)	-1.541e-003	-1.541e-003	-1.541e-003	5.806e-003
54	Accel. Gradient (kPa/m)	3.874e-007	3.874e-007	3.874e-007	7.898e-007
55	Liq. Reynolds	2.064e+005 *	2.064e+005 *	2.064e+005 *	2.064e+005 *
56	Vap. Reynolds	9.199e+006 *	9.199e+006 *	9.199e+006 *	9.199e+006 *
57	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
58	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
59	Deposit Thickness (mm)	---	---	---	---
60	Deposit Volume (m3)	---	---	---	---
61	Length (m)	3.508e+004	3.508e+004	3.509e+004	3.509e+004
62	Elevation (m)	0.1840	0.1960	0.2080	0.2200
63	Pressure (bar)	87.82	87.82	87.82	87.82
64	Temperature (C)	48.07	48.07	48.07	48.07
65	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
66	Heat Transferred (kJ/h-m)	481.5	481.4	481.4	481.4
67	Flow Regime	Transition	Transition	Transition	Transition
68	Water Flow Regime				
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Liquid HoldUp	0.2875	0.2875	0.2875	0.2875
14	Water HoldUp	---	---	---	---
15	Friction Gradient (kPa/m)	5.050e-003	5.050e-003	5.050e-003	5.050e-003
16	Static Gradient (kPa/m)	5.806e-003	5.806e-003	5.806e-003	5.806e-003
17	Accel. Gradient (kPa/m)	7.899e-007	7.899e-007	7.899e-007	7.899e-007
18	Liq. Reynolds	2.064e+005 *	2.064e+005 *	2.064e+005 *	2.064e+005 *
19	Vap. Reynolds	9.199e+006 *	9.199e+006 *	9.199e+006 *	9.199e+006 *
20	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
21	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
22	Deposit Thickness (mm)	---	---	---	---
23	Deposit Volume (m3)	---	---	---	---
24	Length (m)	3.510e+004	3.510e+004	3.511e+004	3.511e+004
25	Elevation (m)	0.2200	0.2200	0.2200	0.2200
26	Pressure (bar)	87.82	87.82	87.82	87.82
27	Temperature (C)	48.07	48.07	48.06	48.06
28	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
29	Heat Transferred (kJ/h-m)	481.3	481.3	481.3	481.2
30	Flow Regime	Transition	Transition	Transition	Transition
31	Water Flow Regime				
32	Liquid HoldUp	0.3029	0.3029	0.3029	0.3029
33	Water HoldUp	---	---	---	---
34	Friction Gradient (kPa/m)	5.877e-003	5.877e-003	5.877e-003	5.877e-003
35	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
36	Accel. Gradient (kPa/m)	4.276e-007	4.276e-007	4.276e-007	4.276e-007
37	Liq. Reynolds	2.064e+005 *	2.064e+005 *	2.064e+005 *	2.063e+005 *
38	Vap. Reynolds	9.199e+006 *	9.199e+006 *	9.199e+006 *	9.199e+006 *
39	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
40	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
41	Deposit Thickness (mm)	---	---	---	---
42	Deposit Volume (m3)	---	---	---	---
43	Length (m)	3.512e+004	3.512e+004	3.513e+004	3.513e+004
44	Elevation (m)	0.2200	0.2180	0.2160	0.2140
45	Pressure (bar)	87.82	87.82	87.82	87.82
46	Temperature (C)	48.06	48.06	48.06	48.06
47	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
48	Heat Transferred (kJ/h-m)	481.2	481.1	481.1	481.1
49	Flow Regime	Transition	Transition	Transition	Transition
50	Water Flow Regime				
51	Liquid HoldUp	0.3029	0.2071	0.2071	0.2071
52	Water HoldUp	---	---	---	---
53	Friction Gradient (kPa/m)	5.877e-003	6.863e-003	6.863e-003	6.863e-003
54	Static Gradient (kPa/m)	0.0000	-7.759e-004	-7.759e-004	-7.759e-004
55	Accel. Gradient (kPa/m)	4.276e-007	4.429e-007	4.429e-007	4.429e-007
56	Liq. Reynolds	2.063e+005 *	2.063e+005 *	2.063e+005 *	2.063e+005 *
57	Vap. Reynolds	9.199e+006 *	9.199e+006 *	9.199e+006 *	9.199e+006 *
58	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
59	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
60	Deposit Thickness (mm)	---	---	---	---
61	Deposit Volume (m3)	---	---	---	---
62	Length (m)	3.514e+004	3.514e+004	3.515e+004	3.515e+004
63	Elevation (m)	0.2120	0.2100	0.2120	0.2140
64	Pressure (bar)	87.82	87.82	87.82	87.82
65	Temperature (C)	48.06	48.05	48.05	48.05
66	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
67	Heat Transferred (kJ/h-m)	481.0	481.0	480.9	480.9
68	Flow Regime	Transition	Transition	Transition	Transition
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10	Pipe Table				
11					
12					
13	Water Flow Regime				
14	Liquid HoldUp	0.2071	0.2071	0.2811	0.2811
15	Water HoldUp	---	---	---	---
16	Friction Gradient (kPa/m)	6.863e-003	6.863e-003	5.534e-003	5.534e-003
17	Static Gradient (kPa/m)	-7.759e-004	-7.759e-004	9.525e-004	9.525e-004
18	Accel. Gradient (kPa/m)	4.429e-007	4.429e-007	4.720e-007	4.720e-007
19	Liq. Reynolds	2.063e+005 *	2.063e+005 *	2.063e+005 *	2.063e+005 *
20	Vap. Reynolds	9.199e+006 *	9.199e+006 *	9.199e+006 *	9.199e+006 *
21	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
22	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
23	Deposit Thickness (mm)	---	---	---	---
24	Deposit Volume (m3)	---	---	---	---
25	Length (m)	3.516e+004	3.516e+004	3.517e+004	3.517e+004
26	Elevation (m)	0.2160	0.2180	0.2200	0.2200
27	Pressure (bar)	87.82	87.82	87.82	87.82
28	Temperature (C)	48.05	48.05	48.05	48.05
29	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
30	Heat Transferred (kJ/h-m)	480.9	480.8	480.8	480.7
31	Flow Regime	Transition	Transition	Transition	Transition
32	Water Flow Regime				
33	Liquid HoldUp	0.2811	0.2811	0.2811	0.3029
34	Water HoldUp	---	---	---	---
35	Friction Gradient (kPa/m)	5.534e-003	5.534e-003	5.534e-003	5.877e-003
36	Static Gradient (kPa/m)	9.526e-004	9.526e-004	9.526e-004	0.0000
37	Accel. Gradient (kPa/m)	4.719e-007	4.719e-007	4.719e-007	4.276e-007
38	Liq. Reynolds	2.063e+005 *	2.063e+005 *	2.063e+005 *	2.063e+005 *
39	Vap. Reynolds	9.199e+006 *	9.199e+006 *	9.199e+006 *	9.199e+006 *
40	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
41	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
42	Deposit Thickness (mm)	---	---	---	---
43	Deposit Volume (m3)	---	---	---	---
44	Length (m)	3.518e+004	3.518e+004	3.519e+004	3.519e+004
45	Elevation (m)	0.2200	0.2200	0.2200	0.2200
46	Pressure (bar)	87.82	87.82	87.82	87.82
47	Temperature (C)	48.04	48.04	48.04	48.04
48	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
49	Heat Transferred (kJ/h-m)	480.7	480.7	480.6	480.6
50	Flow Regime	Transition	Transition	Transition	Transition
51	Water Flow Regime				
52	Liquid HoldUp	0.3029	0.3029	0.3029	0.3029
53	Water HoldUp	---	---	---	---
54	Friction Gradient (kPa/m)	5.877e-003	5.877e-003	5.877e-003	5.877e-003
55	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
56	Accel. Gradient (kPa/m)	4.276e-007	4.276e-007	4.276e-007	4.276e-007
57	Liq. Reynolds	2.063e+005 *	2.063e+005 *	2.063e+005 *	2.063e+005 *
58	Vap. Reynolds	9.199e+006 *	9.199e+006 *	9.199e+006 *	9.199e+006 *
59	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
60	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
61	Deposit Thickness (mm)	---	---	---	---
62	Deposit Volume (m3)	---	---	---	---
63	Length (m)	3.520e+004	3.520e+004	3.521e+004	3.521e+004
64	Elevation (m)	0.2240	0.2280	0.2320	0.2360
65	Pressure (bar)	87.82	87.82	87.82	87.82
66	Temperature (C)	48.04	48.04	48.04	48.03
67	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
68	Heat Transferred (kJ/h-m)	480.5	480.5	480.5	480.4
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3					Unit Set: Mixed
4					Date/Time: Mon Jun 13 15:02:44 2022
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Flow Regime	Transition	Transition	Transition	Transition
14	Water Flow Regime				
15	Liquid HoldUp	0.2824	0.2824	0.2824	0.2824
16	Water HoldUp	---	---	---	---
17	Friction Gradient (kPa/m)	5.435e-003	5.435e-003	5.435e-003	5.435e-003
18	Static Gradient (kPa/m)	1.911e-003	1.911e-003	1.911e-003	1.911e-003
19	Accel. Gradient (kPa/m)	5.344e-007	5.344e-007	5.344e-007	5.344e-007
20	Liq. Reynolds	2.063e+005 *	2.063e+005 *	2.063e+005 *	2.063e+005 *
21	Vap. Reynolds	9.199e+006 *	9.199e+006 *	9.199e+006 *	9.199e+006 *
22	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
23	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
24	Deposit Thickness (mm)	---	---	---	---
25	Deposit Volume (m3)	---	---	---	---
26	Length (m)	3.522e+004	3.522e+004	3.523e+004	3.523e+004
27	Elevation (m)	0.2400	0.2420	0.2440	0.2460
28	Pressure (bar)	87.81	87.81	87.81	87.81
29	Temperature (C)	48.03	48.03	48.03	48.03
30	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
31	Heat Transferred (kJ/h-m)	480.4	480.4	480.3	480.3
32	Flow Regime	Transition	Transition	Transition	Transition
33	Water Flow Regime				
34	Liquid HoldUp	0.2824	0.2812	0.2812	0.2812
35	Water HoldUp	---	---	---	---
36	Friction Gradient (kPa/m)	5.435e-003	5.534e-003	5.534e-003	5.534e-003
37	Static Gradient (kPa/m)	1.911e-003	9.526e-004	9.526e-004	9.526e-004
38	Accel. Gradient (kPa/m)	5.344e-007	4.719e-007	4.719e-007	4.719e-007
39	Liq. Reynolds	2.063e+005 *	2.063e+005 *	2.063e+005 *	2.063e+005 *
40	Vap. Reynolds	9.199e+006 *	9.199e+006 *	9.199e+006 *	9.199e+006 *
41	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
42	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
43	Deposit Thickness (mm)	---	---	---	---
44	Deposit Volume (m3)	---	---	---	---
45	Length (m)	3.524e+004	3.524e+004	3.525e+004	3.525e+004
46	Elevation (m)	0.2480	0.2500	0.2480	0.2460
47	Pressure (bar)	87.81	87.81	87.81	87.81
48	Temperature (C)	48.03	48.03	48.02	48.02
49	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
50	Heat Transferred (kJ/h-m)	480.2	480.2	480.2	480.1
51	Flow Regime	Transition	Transition	Transition	Transition
52	Water Flow Regime				
53	Liquid HoldUp	0.2812	0.2812	0.2071	0.2071
54	Water HoldUp	---	---	---	---
55	Friction Gradient (kPa/m)	5.534e-003	5.534e-003	6.863e-003	6.863e-003
56	Static Gradient (kPa/m)	9.526e-004	9.526e-004	-7.677e-004	-7.677e-004
57	Accel. Gradient (kPa/m)	4.719e-007	4.719e-007	4.434e-007	4.434e-007
58	Liq. Reynolds	2.063e+005 *	2.063e+005 *	2.063e+005 *	2.063e+005 *
59	Vap. Reynolds	9.199e+006 *	9.199e+006 *	9.199e+006 *	9.199e+006 *
60	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
61	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
62	Deposit Thickness (mm)	---	---	---	---
63	Deposit Volume (m3)	---	---	---	---
64	Length (m)	3.526e+004	3.526e+004	3.527e+004	3.527e+004
65	Elevation (m)	0.2440	0.2420	0.2400	0.2320
66	Pressure (bar)	87.81	87.81	87.81	87.81
67	Temperature (C)	48.02	48.02	48.02	48.02
68	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3		Epic Atlantic Limited	Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Heat Transferred (kJ/h-m)	480.1	480.0	480.0	480.0
14	Flow Regime	Transition	Transition	Transition	Transition
15	Water Flow Regime				
16	Liquid HoldUp	0.2071	0.2071	0.2071	0.2060
17	Water HoldUp	---	---	---	---
18	Friction Gradient (kPa/m)	6.862e-003	6.862e-003	6.862e-003	6.871e-003
19	Static Gradient (kPa/m)	-7.677e-004	-7.677e-004	-7.677e-004	-3.093e-003
20	Accel. Gradient (kPa/m)	4.434e-007	4.434e-007	4.434e-007	2.749e-007
21	Liq. Reynolds	2.063e+005 *	2.063e+005 *	2.063e+005 *	2.063e+005 *
22	Vap. Reynolds	9.199e+006 *	9.199e+006 *	9.199e+006 *	9.199e+006 *
23	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
24	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
25	Deposit Thickness (mm)	---	---	---	---
26	Deposit Volume (m3)	---	---	---	---
27	Length (m)	3.528e+004	3.528e+004	3.529e+004	3.529e+004
28	Elevation (m)	0.2240	0.2160	0.2080	0.2000
29	Pressure (bar)	87.81	87.81	87.81	87.81
30	Temperature (C)	48.02	48.01	48.01	48.01
31	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
32	Heat Transferred (kJ/h-m)	479.9	479.9	479.9	479.8
33	Flow Regime	Transition	Transition	Transition	Transition
34	Water Flow Regime				
35	Liquid HoldUp	0.2060	0.2060	0.2060	0.2060
36	Water HoldUp	---	---	---	---
37	Friction Gradient (kPa/m)	6.871e-003	6.871e-003	6.871e-003	6.871e-003
38	Static Gradient (kPa/m)	-3.093e-003	-3.093e-003	-3.093e-003	-3.093e-003
39	Accel. Gradient (kPa/m)	2.749e-007	2.749e-007	2.749e-007	2.749e-007
40	Liq. Reynolds	2.063e+005 *	2.063e+005 *	2.063e+005 *	2.063e+005 *
41	Vap. Reynolds	9.199e+006 *	9.199e+006 *	9.199e+006 *	9.199e+006 *
42	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
43	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
44	Deposit Thickness (mm)	---	---	---	---
45	Deposit Volume (m3)	---	---	---	---
46	Length (m)	3.530e+004	3.530e+004	3.531e+004	3.531e+004
47	Elevation (m)	0.2020	0.2040	0.2060	0.2080
48	Pressure (bar)	87.81	87.81	87.81	87.81
49	Temperature (C)	48.01	48.01	48.01	48.01
50	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
51	Heat Transferred (kJ/h-m)	479.8	479.7	479.7	479.7
52	Flow Regime	Transition	Transition	Transition	Transition
53	Water Flow Regime				
54	Liquid HoldUp	0.2812	0.2812	0.2812	0.2812
55	Water HoldUp	---	---	---	---
56	Friction Gradient (kPa/m)	5.534e-003	5.533e-003	5.533e-003	5.533e-003
57	Static Gradient (kPa/m)	9.526e-004	9.526e-004	9.527e-004	9.527e-004
58	Accel. Gradient (kPa/m)	4.719e-007	4.719e-007	4.719e-007	4.719e-007
59	Liq. Reynolds	2.063e+005 *	2.063e+005 *	2.063e+005 *	2.063e+005 *
60	Vap. Reynolds	9.199e+006 *	9.199e+006 *	9.199e+006 *	9.199e+006 *
61	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
62	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
63	Deposit Thickness (mm)	---	---	---	---
64	Deposit Volume (m3)	---	---	---	---
65	Length (m)	3.532e+004	3.532e+004	3.533e+004	3.533e+004
66	Elevation (m)	0.2100	0.2120	0.2140	0.2160
67	Pressure (bar)	87.81	87.81	87.81	87.81
68	Temperature (C)	48.00	48.00	48.00	48.00
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Pipe Segment: ELEP-Akabel Pipeline (continued)		
3					
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
14	Heat Transferred (kJ/h-m)	479.6	479.6	479.5	479.5
15	Flow Regime	Transition	Transition	Transition	Transition
16	Water Flow Regime				
17	Liquid HoldUp	0.2812	0.2812	0.2812	0.2812
18	Water HoldUp	---	---	---	---
19	Friction Gradient (kPa/m)	5.533e-003	5.533e-003	5.533e-003	5.533e-003
20	Static Gradient (kPa/m)	9.527e-004	9.527e-004	9.527e-004	9.527e-004
21	Accel. Gradient (kPa/m)	4.719e-007	4.719e-007	4.719e-007	4.719e-007
22	Liq. Reynolds	2.063e+005 *	2.063e+005 *	2.063e+005 *	2.063e+005 *
23	Vap. Reynolds	9.199e+006 *	9.199e+006 *	9.199e+006 *	9.199e+006 *
24	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
25	Vapour Velocity (m/s)	2.167	2.167	2.166	2.166
26	Deposit Thickness (mm)	---	---	---	---
27	Deposit Volume (m3)	---	---	---	---
28	Length (m)	3.534e+004	3.534e+004	3.535e+004	3.535e+004
29	Elevation (m)	0.2180	0.2200	2.600e-002	-0.1680
30	Pressure (bar)	87.81	87.81	87.81	87.81
31	Temperature (C)	48.00	48.00	48.00	48.00
32	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
33	Heat Transferred (kJ/h-m)	479.5	479.4	479.5	479.5
34	Flow Regime	Transition	Transition	Transition	Transition
35	Water Flow Regime				
36	Liquid HoldUp	0.2812	0.2812	0.1714	0.1714
37	Water HoldUp	---	---	---	---
38	Friction Gradient (kPa/m)	5.533e-003	5.533e-003	7.226e-003	7.226e-003
39	Static Gradient (kPa/m)	9.527e-004	9.527e-004	-6.701e-002	-6.701e-002
40	Accel. Gradient (kPa/m)	4.719e-007	4.719e-007	-4.349e-006	-4.349e-006
41	Liq. Reynolds	2.063e+005 *	2.063e+005 *	2.063e+005 *	2.063e+005 *
42	Vap. Reynolds	9.199e+006 *	9.199e+006 *	9.199e+006 *	9.199e+006 *
43	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
44	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
45	Deposit Thickness (mm)	---	---	---	---
46	Deposit Volume (m3)	---	---	---	---
47	Length (m)	3.536e+004	3.536e+004	3.537e+004	3.537e+004
48	Elevation (m)	-0.3620	-0.5560	-0.7500	-0.5520
49	Pressure (bar)	87.82	87.82	87.82	87.82
50	Temperature (C)	48.00	48.00	48.00	47.99
51	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
52	Heat Transferred (kJ/h-m)	479.5	479.5	479.4	479.4
53	Flow Regime	Transition	Transition	Transition	Transition
54	Water Flow Regime				
55	Liquid HoldUp	0.1714	0.1714	0.1714	0.4057
56	Water HoldUp	---	---	---	---
57	Friction Gradient (kPa/m)	7.226e-003	7.225e-003	7.225e-003	5.978e-003
58	Static Gradient (kPa/m)	-6.701e-002	-6.701e-002	-6.701e-002	0.1237
59	Accel. Gradient (kPa/m)	-4.349e-006	-4.348e-006	-4.348e-006	9.435e-006
60	Liq. Reynolds	2.063e+005 *	2.063e+005 *	2.063e+005 *	2.063e+005 *
61	Vap. Reynolds	9.199e+006 *	9.199e+006 *	9.199e+006 *	9.199e+006 *
62	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
63	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
64	Deposit Thickness (mm)	---	---	---	---
65	Deposit Volume (m3)	---	---	---	---
66	Length (m)	3.538e+004	3.538e+004	3.539e+004	3.539e+004
67	Elevation (m)	-0.3540	-0.1560	4.200e-002	0.2400
68	Pressure (bar)	87.81	87.80	87.80	87.79
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Temperature (C)	47.99	47.99	47.98	47.98
14	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
15	Heat Transferred (kJ/h-m)	479.3	479.2	479.1	479.0
16	Flow Regime	Transition	Transition	Transition	Transition
17	Water Flow Regime				
18	Liquid HoldUp	0.4057	0.4057	0.4057	0.4057
19	Water HoldUp	---	---	---	---
20	Friction Gradient (kPa/m)	5.979e-003	5.979e-003	5.979e-003	5.980e-003
21	Static Gradient (kPa/m)	0.1237	0.1237	0.1237	0.1237
22	Accel. Gradient (kPa/m)	9.436e-006	9.437e-006	9.438e-006	9.439e-006
23	Liq. Reynolds	2.063e+005 *	2.063e+005 *	2.063e+005 *	2.063e+005 *
24	Vap. Reynolds	9.199e+006 *	9.199e+006 *	9.199e+006 *	9.200e+006 *
25	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
26	Vapour Velocity (m/s)	2.166	2.166	2.167	2.167
27	Deposit Thickness (mm)	---	---	---	---
28	Deposit Volume (m3)	---	---	---	---
29	Length (m)	3.540e+004	3.540e+004	3.541e+004	3.541e+004
30	Elevation (m)	0.2580	0.2760	0.2940	0.3120
31	Pressure (bar)	87.79	87.79	87.79	87.79
32	Temperature (C)	47.98	47.98	47.97	47.97
33	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
34	Heat Transferred (kJ/h-m)	478.9	478.9	478.8	478.8
35	Flow Regime	Transition	Transition	Transition	Transition
36	Water Flow Regime				
37	Liquid HoldUp	0.2913	0.2913	0.2913	0.2913
38	Water HoldUp	---	---	---	---
39	Friction Gradient (kPa/m)	4.774e-003	4.774e-003	4.774e-003	4.774e-003
40	Static Gradient (kPa/m)	8.792e-003	8.792e-003	8.792e-003	8.792e-003
41	Accel. Gradient (kPa/m)	9.873e-007	9.874e-007	9.874e-007	9.874e-007
42	Liq. Reynolds	2.062e+005 *	2.062e+005 *	2.062e+005 *	2.062e+005 *
43	Vap. Reynolds	9.200e+006 *	9.200e+006 *	9.200e+006 *	9.200e+006 *
44	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
45	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
46	Deposit Thickness (mm)	---	---	---	---
47	Deposit Volume (m3)	---	---	---	---
48	Length (m)	3.542e+004	3.542e+004	3.543e+004	3.543e+004
49	Elevation (m)	0.3300	0.2940	0.2580	0.2220
50	Pressure (bar)	87.79	87.79	87.79	87.79
51	Temperature (C)	47.97	47.97	47.97	47.97
52	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
53	Heat Transferred (kJ/h-m)	478.7	478.7	478.7	478.7
54	Flow Regime	Transition	Transition	Transition	Transition
55	Water Flow Regime				
56	Liquid HoldUp	0.2913	0.2008	0.2008	0.2008
57	Water HoldUp	---	---	---	---
58	Friction Gradient (kPa/m)	4.774e-003	6.915e-003	6.915e-003	6.915e-003
59	Static Gradient (kPa/m)	8.792e-003	-1.369e-002	-1.369e-002	-1.370e-002
60	Accel. Gradient (kPa/m)	9.874e-007	-4.934e-007	-4.935e-007	-4.935e-007
61	Liq. Reynolds	2.062e+005 *	2.062e+005 *	2.062e+005 *	2.062e+005 *
62	Vap. Reynolds	9.200e+006 *	9.200e+006 *	9.200e+006 *	9.200e+006 *
63	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
64	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
65	Deposit Thickness (mm)	---	---	---	---
66	Deposit Volume (m3)	---	---	---	---
67	Length (m)	3.544e+004	3.544e+004	3.545e+004	3.545e+004
68	Elevation (m)	0.1860	0.1500	0.1460	0.1420
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Pressure (bar)	87.79	87.79	87.79	87.79
14	Temperature (C)	47.97	47.97	47.96	47.96
15	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
16	Heat Transferred (kJ/h-m)	478.6	478.6	478.6	478.5
17	Flow Regime	Transition	Transition	Transition	Transition
18	Water Flow Regime				
19	Liquid HoldUp	0.2008	0.2008	0.2067	0.2067
20	Water HoldUp	---	---	---	---
21	Friction Gradient (kPa/m)	6.915e-003	6.914e-003	6.866e-003	6.866e-003
22	Static Gradient (kPa/m)	-1.370e-002	-1.370e-002	-1.564e-003	-1.564e-003
23	Accel. Gradient (kPa/m)	-4.935e-007	-4.935e-007	3.858e-007	3.858e-007
24	Liq. Reynolds	2.062e+005 *	2.062e+005 *	2.062e+005 *	2.062e+005 *
25	Vap. Reynolds	9.200e+006 *	9.200e+006 *	9.200e+006 *	9.200e+006 *
26	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
27	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
28	Deposit Thickness (mm)	---	---	---	---
29	Deposit Volume (m3)	---	---	---	---
30	Length (m)	3.546e+004	3.546e+004	3.547e+004	3.547e+004
31	Elevation (m)	0.1380	0.1340	0.1300	0.1320
32	Pressure (bar)	87.79	87.79	87.79	87.79
33	Temperature (C)	47.96	47.96	47.96	47.96
34	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
35	Heat Transferred (kJ/h-m)	478.5	478.4	478.4	478.4
36	Flow Regime	Transition	Transition	Transition	Transition
37	Water Flow Regime				
38	Liquid HoldUp	0.2067	0.2067	0.2067	0.2812
39	Water HoldUp	---	---	---	---
40	Friction Gradient (kPa/m)	6.866e-003	6.866e-003	6.865e-003	5.534e-003
41	Static Gradient (kPa/m)	-1.564e-003	-1.564e-003	-1.564e-003	9.527e-004
42	Accel. Gradient (kPa/m)	3.858e-007	3.858e-007	3.858e-007	4.720e-007
43	Liq. Reynolds	2.062e+005 *	2.062e+005 *	2.062e+005 *	2.062e+005 *
44	Vap. Reynolds	9.200e+006 *	9.200e+006 *	9.200e+006 *	9.200e+006 *
45	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
46	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
47	Deposit Thickness (mm)	---	---	---	---
48	Deposit Volume (m3)	---	---	---	---
49	Length (m)	3.548e+004	3.548e+004	3.549e+004	3.549e+004
50	Elevation (m)	0.1340	0.1360	0.1380	0.1400
51	Pressure (bar)	87.79	87.79	87.79	87.79
52	Temperature (C)	47.96	47.95	47.95	47.95
53	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
54	Heat Transferred (kJ/h-m)	478.3	478.3	478.2	478.2
55	Flow Regime	Transition	Transition	Transition	Transition
56	Water Flow Regime				
57	Liquid HoldUp	0.2812	0.2812	0.2812	0.2812
58	Water HoldUp	---	---	---	---
59	Friction Gradient (kPa/m)	5.534e-003	5.534e-003	5.534e-003	5.534e-003
60	Static Gradient (kPa/m)	9.527e-004	9.527e-004	9.527e-004	9.527e-004
61	Accel. Gradient (kPa/m)	4.720e-007	4.720e-007	4.720e-007	4.720e-007
62	Liq. Reynolds	2.062e+005 *	2.062e+005 *	2.062e+005 *	2.062e+005 *
63	Vap. Reynolds	9.200e+006 *	9.200e+006 *	9.200e+006 *	9.200e+006 *
64	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
65	Vapour Velocity (m/s)	2.167	2.167	2.167	2.167
66	Deposit Thickness (mm)	---	---	---	---
67	Deposit Volume (m3)	---	---	---	---
68	Length (m)	3.550e+004	3.550e+004	3.551e+004	3.551e+004
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Elevation (m)	0.1360	0.1320	0.1280	0.1240
14	Pressure (bar)	87.79	87.78	87.78	87.78
15	Temperature (C)	47.95	47.95	47.95	47.94
16	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
17	Heat Transferred (kJ/h-m)	478.2	478.1	478.1	478.1
18	Flow Regime	Transition	Transition	Transition	Transition
19	Water Flow Regime				
20	Liquid HoldUp	0.2068	0.2068	0.2068	0.2068
21	Water HoldUp	---	---	---	---
22	Friction Gradient (kPa/m)	6.865e-003	6.865e-003	6.865e-003	6.865e-003
23	Static Gradient (kPa/m)	-1.550e-003	-1.550e-003	-1.550e-003	-1.550e-003
24	Accel. Gradient (kPa/m)	3.868e-007	3.868e-007	3.868e-007	3.868e-007
25	Liq. Reynolds	2.062e+005 *	2.062e+005 *	2.062e+005 *	2.062e+005 *
26	Vap. Reynolds	9.200e+006 *	9.200e+006 *	9.200e+006 *	9.200e+006 *
27	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
28	Vapour Velocity (m/s)	2.167	2.167	2.167	2.166
29	Deposit Thickness (mm)	---	---	---	---
30	Deposit Volume (m3)	---	---	---	---
31	Length (m)	3.552e+004	3.552e+004	3.553e+004	3.553e+004
32	Elevation (m)	0.1200	0.1220	0.1240	0.1260
33	Pressure (bar)	87.78	87.78	87.78	87.78
34	Temperature (C)	47.94	47.94	47.94	47.94
35	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
36	Heat Transferred (kJ/h-m)	478.0	478.0	477.9	477.9
37	Flow Regime	Transition	Transition	Transition	Transition
38	Water Flow Regime				
39	Liquid HoldUp	0.2068	0.2812	0.2812	0.2812
40	Water HoldUp	---	---	---	---
41	Friction Gradient (kPa/m)	6.865e-003	5.533e-003	5.533e-003	5.533e-003
42	Static Gradient (kPa/m)	-1.550e-003	9.527e-004	9.527e-004	9.527e-004
43	Accel. Gradient (kPa/m)	3.868e-007	4.720e-007	4.720e-007	4.720e-007
44	Liq. Reynolds	2.062e+005 *	2.062e+005 *	2.062e+005 *	2.062e+005 *
45	Vap. Reynolds	9.200e+006 *	9.200e+006 *	9.200e+006 *	9.200e+006 *
46	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
47	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
48	Deposit Thickness (mm)	---	---	---	---
49	Deposit Volume (m3)	---	---	---	---
50	Length (m)	3.554e+004	3.554e+004	3.555e+004	3.555e+004
51	Elevation (m)	0.1280	0.1300	0.1260	0.1220
52	Pressure (bar)	87.78	87.78	87.78	87.78
53	Temperature (C)	47.94	47.94	47.93	47.93
54	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
55	Heat Transferred (kJ/h-m)	477.9	477.8	477.8	477.8
56	Flow Regime	Transition	Transition	Transition	Transition
57	Water Flow Regime				
58	Liquid HoldUp	0.2812	0.2812	0.2068	0.2068
59	Water HoldUp	---	---	---	---
60	Friction Gradient (kPa/m)	5.533e-003	5.533e-003	6.865e-003	6.865e-003
61	Static Gradient (kPa/m)	9.527e-004	9.527e-004	-1.550e-003	-1.550e-003
62	Accel. Gradient (kPa/m)	4.720e-007	4.720e-007	3.868e-007	3.868e-007
63	Liq. Reynolds	2.062e+005 *	2.062e+005 *	2.062e+005 *	2.062e+005 *
64	Vap. Reynolds	9.200e+006 *	9.200e+006 *	9.200e+006 *	9.200e+006 *
65	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
66	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
67	Deposit Thickness (mm)	---	---	---	---
68	Deposit Volume (m3)	---	---	---	---
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3	 Epic Atlantic Limited		Unit Set: Mixed		
4					
5			Date/Time: Mon Jun 13 15:02:44 2022		
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Length (m)	3.556e+004	3.556e+004	3.557e+004	3.557e+004
14	Elevation (m)	0.1180	0.1140	0.1100	0.1100
15	Pressure (bar)	87.78	87.78	87.78	87.78
16	Temperature (C)	47.93	47.93	47.93	47.93
17	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
18	Heat Transferred (kJ/h-m)	477.7	477.7	477.6	477.6
19	Flow Regime	Transition	Transition	Transition	Transition
20	Water Flow Regime				
21	Liquid HoldUp	0.2068	0.2068	0.2068	0.3029
22	Water HoldUp	---	---	---	---
23	Friction Gradient (kPa/m)	6.865e-003	6.865e-003	6.865e-003	5.876e-003
24	Static Gradient (kPa/m)	-1.550e-003	-1.550e-003	-1.550e-003	0.0000
25	Accel. Gradient (kPa/m)	3.868e-007	3.868e-007	3.868e-007	4.276e-007
26	Liq. Reynolds	2.062e+005 *	2.062e+005 *	2.062e+005 *	2.062e+005 *
27	Vap. Reynolds	9.200e+006 *	9.200e+006 *	9.200e+006 *	9.200e+006 *
28	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
29	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
30	Deposit Thickness (mm)	---	---	---	---
31	Deposit Volume (m3)	---	---	---	---
32	Length (m)	3.557e+004	3.558e+004	3.558e+004	3.559e+004
33	Elevation (m)	0.1100	0.1100	0.1100	0.1100
34	Pressure (bar)	87.78	87.78	87.78	87.78
35	Temperature (C)	47.93	47.93	47.92	47.92
36	Vapour Fraction	0.8681	0.8681	0.8681	0.8681
37	Heat Transferred (kJ/h-m)	477.6	477.5	477.5	477.5
38	Flow Regime	Transition	Transition	Transition	Transition
39	Water Flow Regime				
40	Liquid HoldUp	0.3029	0.3029	0.3029	0.3029
41	Water HoldUp	---	---	---	---
42	Friction Gradient (kPa/m)	5.876e-003	5.876e-003	5.876e-003	5.876e-003
43	Static Gradient (kPa/m)	0.0000	0.0000	0.0000	0.0000
44	Accel. Gradient (kPa/m)	4.276e-007	4.276e-007	4.276e-007	4.276e-007
45	Liq. Reynolds	2.062e+005 *	2.062e+005 *	2.062e+005 *	2.062e+005 *
46	Vap. Reynolds	9.200e+006 *	9.200e+006 *	9.200e+006 *	9.200e+006 *
47	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
48	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
49	Deposit Thickness (mm)	---	---	---	---
50	Deposit Volume (m3)	---	---	---	---
51	Length (m)	3.559e+004	3.559e+004	3.560e+004	3.560e+004
52	Elevation (m)	0.1160	0.1220	0.1280	0.1340
53	Pressure (bar)	87.78	87.78	87.78	87.78
54	Temperature (C)	47.92	47.92	47.92	47.92
55	Vapour Fraction	0.8681	0.8680	0.8680	0.8680
56	Heat Transferred (kJ/h-m)	477.5	477.4	477.4	477.4
57	Flow Regime	Transition	Transition	Transition	Transition
58	Water Flow Regime				
59	Liquid HoldUp	0.2863	0.2863	0.2863	0.2863
60	Water HoldUp	---	---	---	---
61	Friction Gradient (kPa/m)	5.143e-003	5.143e-003	5.143e-003	5.143e-003
62	Static Gradient (kPa/m)	4.825e-003	4.825e-003	4.825e-003	4.825e-003
63	Accel. Gradient (kPa/m)	7.254e-007	7.254e-007	7.254e-007	7.254e-007
64	Liq. Reynolds	2.062e+005 *	2.062e+005 *	2.062e+005 *	2.062e+005 *
65	Vap. Reynolds	9.200e+006 *	9.200e+006 *	9.200e+006 *	9.200e+006 *
66	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
67	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
68	Deposit Thickness (mm)	---	---	---	---
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2			Pipe Segment: ELEP-Akabel Pipeline (continued)		
3					
4					
5			Unit Set: Mixed		
6			Date/Time: Mon Jun 13 15:02:44 2022		
7					
8					
9	PROFILES				
10					
11	Pipe Table				
12					
13	Deposit Volume (m3)	---	---	---	---
14	Length (m)	3.560e+004	3.561e+004	3.561e+004	3.562e+004
15	Elevation (m)	0.1400	0.1540	0.1680	0.1820
16	Pressure (bar)	87.78	87.78	87.78	87.78
17	Temperature (C)	47.92	47.92	47.92	47.91
18	Vapour Fraction	0.8680	0.8680	0.8680	0.8680
19	Heat Transferred (kJ/h-m)	477.4	477.3	477.3	477.2
20	Flow Regime	Transition	Transition	Transition	Transition
21	Water Flow Regime				
22	Liquid HoldUp	0.2863	0.2888	0.2888	0.2888
23	Water HoldUp	---	---	---	---
24	Friction Gradient (kPa/m)	5.143e-003	4.956e-003	4.956e-003	4.956e-003
25	Static Gradient (kPa/m)	4.825e-003	6.797e-003	6.797e-003	6.797e-003
26	Accel. Gradient (kPa/m)	7.254e-007	8.553e-007	8.553e-007	8.553e-007
27	Liq. Reynolds	2.062e+005 *	2.062e+005 *	2.062e+005 *	2.062e+005 *
28	Vap. Reynolds	9.200e+006 *	9.200e+006 *	9.200e+006 *	9.200e+006 *
29	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
30	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
31	Deposit Thickness (mm)	---	---	---	---
32	Deposit Volume (m3)	---	---	---	---
33	Length (m)	3.562e+004	3.563e+004	3.563e+004	3.564e+004
34	Elevation (m)	0.1960	0.2100	0.1940	0.1780
35	Pressure (bar)	87.78	87.78	87.78	87.78
36	Temperature (C)	47.91	47.91	47.91	47.91
37	Vapour Fraction	0.8680	0.8680	0.8680	0.8680
38	Heat Transferred (kJ/h-m)	477.2	477.1	477.1	477.1
39	Flow Regime	Transition	Transition	Transition	Transition
40	Water Flow Regime				
41	Liquid HoldUp	0.2888	0.2888	0.2045	0.2045
42	Water HoldUp	---	---	---	---
43	Friction Gradient (kPa/m)	4.956e-003	4.956e-003	6.883e-003	6.883e-003
44	Static Gradient (kPa/m)	6.797e-003	6.797e-003	-6.159e-003	-6.159e-003
45	Accel. Gradient (kPa/m)	8.553e-007	8.553e-007	5.268e-008	5.267e-008
46	Liq. Reynolds	2.062e+005 *	2.062e+005 *	2.062e+005 *	2.062e+005 *
47	Vap. Reynolds	9.200e+006 *	9.200e+006 *	9.200e+006 *	9.200e+006 *
48	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
49	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
50	Deposit Thickness (mm)	---	---	---	---
51	Deposit Volume (m3)	---	---	---	---
52	Length (m)	3.564e+004	3.565e+004	3.565e+004	3.566e+004
53	Elevation (m)	0.1620	0.1460	0.1300	0.1840
54	Pressure (bar)	87.78	87.78	87.78	87.77
55	Temperature (C)	47.91	47.91	47.90	47.90
56	Vapour Fraction	0.8680	0.8680	0.8680	0.8680
57	Heat Transferred (kJ/h-m)	477.1	477.0	477.0	476.9
58	Flow Regime	Transition	Transition	Transition	Transition
59	Water Flow Regime				
60	Liquid HoldUp	0.2045	0.2045	0.2045	0.3140
61	Water HoldUp	---	---	---	---
62	Friction Gradient (kPa/m)	6.883e-003	6.882e-003	6.882e-003	6.034e-003
63	Static Gradient (kPa/m)	-6.159e-003	-6.159e-003	-6.159e-003	2.762e-002
64	Accel. Gradient (kPa/m)	5.267e-008	5.266e-008	5.266e-008	2.449e-006
65	Liq. Reynolds	2.062e+005 *	2.062e+005 *	2.062e+005 *	2.062e+005 *
66	Vap. Reynolds	9.200e+006 *	9.200e+006 *	9.200e+006 *	9.200e+006 *
67	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047
68	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 569 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2			Unit Set: Mixed				
3							
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	PROFILES						
10							
11	Pipe Table						
12							
13	Deposit Thickness (mm)	---	---	---	---	---	
14	Deposit Volume (m3)	---	---	---	---	---	
15	Length (m)	3.566e+004	3.567e+004	3.567e+004	3.568e+004	3.568e+004	
16	Elevation (m)	0.2380	0.2920	0.3460	0.4000	0.4000	
17	Pressure (bar)	87.77	87.77	87.77	87.77	87.77	
18	Temperature (C)	47.90	47.90	47.90	47.89	47.89	
19	Vapour Fraction	0.8680	0.8680	0.8680	0.8680	0.8680	
20	Heat Transferred (kJ/h-m)	476.9	476.8	476.8	476.7	476.7	
21	Flow Regime	Transition	Transition	Transition	Transition	Transition	
22	Water Flow Regime						
23	Liquid HoldUp	0.3140	0.3140	0.3140	0.3140	0.3140	
24	Water HoldUp	---	---	---	---	---	
25	Friction Gradient (kPa/m)	6.034e-003	6.034e-003	6.034e-003	6.034e-003	6.034e-003	
26	Static Gradient (kPa/m)	2.762e-002	2.762e-002	2.762e-002	2.762e-002	2.762e-002	
27	Accel. Gradient (kPa/m)	2.449e-006	2.449e-006	2.449e-006	2.449e-006	2.449e-006	
28	Liq. Reynolds	2.062e+005 *	2.062e+005 *	2.062e+005 *	2.062e+005 *	2.062e+005 *	
29	Vap. Reynolds	9.200e+006 *	9.200e+006 *	9.200e+006 *	9.200e+006 *	9.200e+006 *	
30	Liquid Velocity (m/s)	0.2047	0.2047	0.2047	0.2047	0.2047	
31	Vapour Velocity (m/s)	2.166	2.166	2.166	2.166	2.166	
32	Deposit Thickness (mm)	---	---	---	---	---	
33	Deposit Volume (m3)	---	---	---	---	---	
34	Erosion						
35							
36	Model						
37							
38	Erosion Model:	API-RP-14E Continuous service		Empirical Constant:	100.0		
39							
40	Erosion Table						
41	Length (m)	0.0000	5.000	10.00	15.00	20.00	
42	Bulk Density (kg/m3)	130.3	130.3	130.3	130.3	130.3	
43	Bulk Velocity (m/s)	2.262	2.262	2.262	2.262	2.262	
44	Erosion Velocity (m/s)	10.69	10.69	10.69	10.69	10.69	
45	Length (m)	25.00	30.00	35.00	40.00	45.00	
46	Bulk Density (kg/m3)	130.3	130.3	130.3	130.3	130.3	
47	Bulk Velocity (m/s)	2.262	2.262	2.262	2.262	2.262	
48	Erosion Velocity (m/s)	10.69	10.69	10.69	10.69	10.69	
49	Length (m)	50.00	55.00	60.00	65.00	70.00	
50	Bulk Density (kg/m3)	130.3	130.3	130.3	130.3	130.3	
51	Bulk Velocity (m/s)	2.262	2.262	2.262	2.262	2.262	
52	Erosion Velocity (m/s)	10.69	10.69	10.69	10.69	10.69	
53	Length (m)	75.00	80.00	85.00	90.00	95.00	
54	Bulk Density (kg/m3)	130.3	130.3	130.3	130.3	130.3	
55	Bulk Velocity (m/s)	2.262	2.262	2.262	2.262	2.262	
56	Erosion Velocity (m/s)	10.69	10.69	10.69	10.68	10.68	
57	Length (m)	100.0	105.0	110.0	115.0	120.0	
58	Bulk Density (kg/m3)	130.4	130.4	130.4	130.4	130.4	
59	Bulk Velocity (m/s)	2.262	2.262	2.262	2.262	2.262	
60	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68	
61	Length (m)	125.0	130.0	135.0	140.0	145.0	
62	Bulk Density (kg/m3)	130.4	130.4	130.4	130.4	130.4	
63	Bulk Velocity (m/s)	2.262	2.262	2.262	2.262	2.262	
64	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68	
65	Length (m)	150.0	155.0	160.0	165.0	170.0	
66	Bulk Density (kg/m3)	130.4	130.4	130.4	130.4	130.4	
67	Bulk Velocity (m/s)	2.262	2.262	2.262	2.262	2.262	
68	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 570 of 937	

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	175.0	180.0	185.0	190.0	195.0
14	Bulk Density (kg/m3)	130.4	130.4	130.4	130.4	130.4
15	Bulk Velocity (m/s)	2.262	2.262	2.262	2.262	2.262
16	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
17	Length (m)	200.1	205.1	210.1	215.1	220.1
18	Bulk Density (kg/m3)	130.4	130.4	130.4	130.4	130.4
19	Bulk Velocity (m/s)	2.261	2.261	2.261	2.261	2.261
20	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
21	Length (m)	225.1	230.1	235.1	240.1	245.1
22	Bulk Density (kg/m3)	130.4	130.4	130.4	130.4	130.4
23	Bulk Velocity (m/s)	2.261	2.261	2.261	2.261	2.261
24	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
25	Length (m)	250.1	255.1	260.1	265.1	270.1
26	Bulk Density (kg/m3)	130.4	130.4	130.4	130.4	130.4
27	Bulk Velocity (m/s)	2.261	2.261	2.261	2.261	2.261
28	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
29	Length (m)	275.1	280.1	285.1	290.1	295.1
30	Bulk Density (kg/m3)	130.4	130.4	130.4	130.4	130.4
31	Bulk Velocity (m/s)	2.261	2.261	2.261	2.261	2.261
32	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
33	Length (m)	300.1	305.1	310.1	315.1	320.1
34	Bulk Density (kg/m3)	130.4	130.4	130.4	130.4	130.4
35	Bulk Velocity (m/s)	2.261	2.261	2.261	2.261	2.261
36	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
37	Length (m)	325.1	330.1	335.1	340.1	345.1
38	Bulk Density (kg/m3)	130.4	130.4	130.4	130.4	130.4
39	Bulk Velocity (m/s)	2.261	2.261	2.261	2.261	2.261
40	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
41	Length (m)	350.1	355.1	360.1	365.1	370.1
42	Bulk Density (kg/m3)	130.4	130.4	130.4	130.4	130.4
43	Bulk Velocity (m/s)	2.261	2.261	2.261	2.261	2.261
44	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
45	Length (m)	375.1	380.0	385.0	390.0	395.0
46	Bulk Density (kg/m3)	130.4	130.4	130.4	130.4	130.4
47	Bulk Velocity (m/s)	2.261	2.261	2.261	2.261	2.261
48	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
49	Length (m)	400.0	405.0	410.0	415.0	420.0
50	Bulk Density (kg/m3)	130.4	130.4	130.4	130.4	130.4
51	Bulk Velocity (m/s)	2.261	2.261	2.261	2.261	2.261
52	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
53	Length (m)	425.0	430.0	435.0	440.0	445.0
54	Bulk Density (kg/m3)	130.4	130.4	130.4	130.4	130.4
55	Bulk Velocity (m/s)	2.261	2.261	2.261	2.261	2.261
56	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
57	Length (m)	450.0	455.0	460.0	465.0	470.0
58	Bulk Density (kg/m3)	130.4	130.4	130.4	130.4	130.4
59	Bulk Velocity (m/s)	2.261	2.261	2.261	2.261	2.261
60	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
61	Length (m)	475.0	480.0	485.0	490.0	495.0
62	Bulk Density (kg/m3)	130.4	130.4	130.4	130.4	130.4
63	Bulk Velocity (m/s)	2.261	2.261	2.261	2.261	2.261
64	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
65	Length (m)	500.0	505.0	510.0	515.0	520.0
66	Bulk Density (kg/m3)	130.4	130.4	130.4	130.4	130.4
67	Bulk Velocity (m/s)	2.260	2.260	2.260	2.260	2.260
68	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 571 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	525.0	530.0	535.0	540.0	545.0
14	Bulk Density (kg/m3)	130.4	130.4	130.4	130.4	130.4
15	Bulk Velocity (m/s)	2.260	2.260	2.260	2.260	2.260
16	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
17	Length (m)	550.0	555.0	560.0	565.0	570.0
18	Bulk Density (kg/m3)	130.4	130.4	130.4	130.4	130.5
19	Bulk Velocity (m/s)	2.260	2.260	2.260	2.260	2.260
20	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
21	Length (m)	575.0	580.0	585.0	590.0	595.0
22	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.5
23	Bulk Velocity (m/s)	2.260	2.260	2.260	2.260	2.260
24	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
25	Length (m)	600.0	605.0	610.0	615.0	620.0
26	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.5
27	Bulk Velocity (m/s)	2.260	2.260	2.260	2.260	2.260
28	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
29	Length (m)	625.0	630.0	635.0	640.0	645.0
30	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.5
31	Bulk Velocity (m/s)	2.260	2.260	2.260	2.260	2.260
32	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
33	Length (m)	650.0	655.0	660.0	665.0	670.0
34	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.5
35	Bulk Velocity (m/s)	2.260	2.260	2.260	2.260	2.260
36	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
37	Length (m)	675.0	682.0	689.0	696.0	703.0
38	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.5
39	Bulk Velocity (m/s)	2.260	2.260	2.260	2.260	2.260
40	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
41	Length (m)	710.0	712.0	714.0	716.0	718.0
42	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.5
43	Bulk Velocity (m/s)	2.260	2.260	2.260	2.260	2.260
44	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
45	Length (m)	720.0	722.0	724.0	726.0	728.0
46	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.5
47	Bulk Velocity (m/s)	2.260	2.260	2.260	2.260	2.260
48	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
49	Length (m)	730.0	732.0	734.0	736.0	738.0
50	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.5
51	Bulk Velocity (m/s)	2.260	2.260	2.260	2.260	2.260
52	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
53	Length (m)	740.0	742.0	744.0	746.0	748.0
54	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.5
55	Bulk Velocity (m/s)	2.260	2.260	2.260	2.260	2.260
56	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
57	Length (m)	750.0	755.0	760.0	765.0	770.0
58	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.5
59	Bulk Velocity (m/s)	2.259	2.259	2.259	2.259	2.259
60	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
61	Length (m)	775.0	780.0	785.0	790.1	795.1
62	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.5
63	Bulk Velocity (m/s)	2.259	2.259	2.259	2.259	2.259
64	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
65	Length (m)	800.1	805.1	810.1	815.1	820.1
66	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.5
67	Bulk Velocity (m/s)	2.259	2.259	2.259	2.259	2.259
68	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 572 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	825.1	830.1	835.1	840.1	845.1
14	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.5
15	Bulk Velocity (m/s)	2.259	2.259	2.259	2.259	2.259
16	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
17	Length (m)	850.1	855.1	860.1	865.1	870.1
18	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.5
19	Bulk Velocity (m/s)	2.259	2.259	2.259	2.259	2.259
20	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
21	Length (m)	875.1	880.1	885.1	890.1	895.1
22	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.5
23	Bulk Velocity (m/s)	2.259	2.259	2.259	2.259	2.259
24	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
25	Length (m)	900.1	905.1	910.1	915.1	920.1
26	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.5
27	Bulk Velocity (m/s)	2.259	2.259	2.259	2.259	2.259
28	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
29	Length (m)	925.1	930.1	935.1	940.1	945.1
30	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.5
31	Bulk Velocity (m/s)	2.259	2.259	2.259	2.259	2.259
32	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
33	Length (m)	950.1	955.1	960.1	965.1	970.1
34	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.5
35	Bulk Velocity (m/s)	2.259	2.259	2.259	2.259	2.259
36	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
37	Length (m)	975.1	980.1	985.1	990.1	995.1
38	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.5
39	Bulk Velocity (m/s)	2.259	2.259	2.259	2.259	2.259
40	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
41	Length (m)	1000	1005	1010	1015	1020
42	Bulk Density (kg/m3)	130.5	130.5	130.5	130.5	130.6
43	Bulk Velocity (m/s)	2.258	2.258	2.258	2.258	2.258
44	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
45	Length (m)	1025	1030	1035	1040	1045
46	Bulk Density (kg/m3)	130.6	130.6	130.6	130.6	130.6
47	Bulk Velocity (m/s)	2.258	2.258	2.258	2.258	2.258
48	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
49	Length (m)	1050	1055	1060	1065	1070
50	Bulk Density (kg/m3)	130.6	130.6	130.6	130.6	130.6
51	Bulk Velocity (m/s)	2.258	2.258	2.258	2.258	2.258
52	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
53	Length (m)	1075	1080	1085	1090	1095
54	Bulk Density (kg/m3)	130.6	130.6	130.6	130.6	130.6
55	Bulk Velocity (m/s)	2.258	2.258	2.258	2.258	2.258
56	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
57	Length (m)	1100	1105	1110	1115	1120
58	Bulk Density (kg/m3)	130.6	130.6	130.6	130.6	130.6
59	Bulk Velocity (m/s)	2.258	2.258	2.258	2.258	2.258
60	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
61	Length (m)	1125	1130	1135	1140	1145
62	Bulk Density (kg/m3)	130.6	130.6	130.6	130.6	130.6
63	Bulk Velocity (m/s)	2.258	2.258	2.258	2.258	2.258
64	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
65	Length (m)	1150	1155	1160	1165	1170
66	Bulk Density (kg/m3)	130.6	130.6	130.6	130.6	130.6
67	Bulk Velocity (m/s)	2.258	2.258	2.258	2.258	2.258
68	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10	Erosion Table					
11						
12						
13	Length (m)	1175	1180	1185	1190	1195
14	Bulk Density (kg/m3)	130.6	130.6	130.6	130.6	130.6
15	Bulk Velocity (m/s)	2.258	2.258	2.258	2.258	2.258
16	Erosion Velocity (m/s)	10.68	10.68	10.68	10.68	10.68
17	Length (m)	1200	1205	1210	1215	1220
18	Bulk Density (kg/m3)	130.6	130.6	130.6	130.6	130.6
19	Bulk Velocity (m/s)	2.258	2.258	2.258	2.258	2.258
20	Erosion Velocity (m/s)	10.68	10.67	10.67	10.67	10.67
21	Length (m)	1225	1230	1235	1240	1245
22	Bulk Density (kg/m3)	130.6	130.6	130.6	130.6	130.6
23	Bulk Velocity (m/s)	2.258	2.258	2.258	2.258	2.258
24	Erosion Velocity (m/s)	10.67	10.67	10.67	10.67	10.67
25	Length (m)	1250	1255	1260	1265	1270
26	Bulk Density (kg/m3)	130.6	130.6	130.6	130.6	130.6
27	Bulk Velocity (m/s)	2.258	2.258	2.257	2.257	2.257
28	Erosion Velocity (m/s)	10.67	10.67	10.67	10.67	10.67
29	Length (m)	1275	1280	1285	1290	1295
30	Bulk Density (kg/m3)	130.6	130.6	130.6	130.6	130.6
31	Bulk Velocity (m/s)	2.257	2.257	2.257	2.257	2.257
32	Erosion Velocity (m/s)	10.67	10.67	10.67	10.67	10.67
33	Length (m)	1300	1305	1310	1315	1320
34	Bulk Density (kg/m3)	130.6	130.6	130.6	130.6	130.6
35	Bulk Velocity (m/s)	2.257	2.257	2.257	2.257	2.257
36	Erosion Velocity (m/s)	10.67	10.67	10.67	10.67	10.67
37	Length (m)	1325	1330	1335	1340	1345
38	Bulk Density (kg/m3)	130.6	130.6	130.6	130.6	130.6
39	Bulk Velocity (m/s)	2.257	2.257	2.257	2.257	2.257
40	Erosion Velocity (m/s)	10.67	10.67	10.67	10.67	10.67
41	Length (m)	1350	1355	1360	1365	1370
42	Bulk Density (kg/m3)	130.6	130.6	130.6	130.6	130.6
43	Bulk Velocity (m/s)	2.257	2.257	2.257	2.257	2.257
44	Erosion Velocity (m/s)	10.67	10.67	10.67	10.67	10.67
45	Length (m)	1375	1380	1385	1390	1395
46	Bulk Density (kg/m3)	130.6	130.7	130.7	130.7	130.7
47	Bulk Velocity (m/s)	2.257	2.257	2.257	2.256	2.256
48	Erosion Velocity (m/s)	10.67	10.67	10.67	10.67	10.67
49	Length (m)	1400	1405	1410	1415	1420
50	Bulk Density (kg/m3)	130.7	130.7	130.7	130.7	130.7
51	Bulk Velocity (m/s)	2.256	2.256	2.256	2.256	2.256
52	Erosion Velocity (m/s)	10.67	10.67	10.67	10.67	10.67
53	Length (m)	1425	1430	1435	1440	1445
54	Bulk Density (kg/m3)	130.7	130.7	130.7	130.7	130.7
55	Bulk Velocity (m/s)	2.255	2.255	2.255	2.255	2.255
56	Erosion Velocity (m/s)	10.67	10.67	10.67	10.67	10.67
57	Length (m)	1450	1455	1460	1465	1470
58	Bulk Density (kg/m3)	130.7	130.7	130.7	130.7	130.7
59	Bulk Velocity (m/s)	2.255	2.255	2.255	2.255	2.255
60	Erosion Velocity (m/s)	10.67	10.67	10.67	10.67	10.67
61	Length (m)	1475	1480	1485	1490	1495
62	Bulk Density (kg/m3)	130.7	130.7	130.7	130.7	130.8
63	Bulk Velocity (m/s)	2.255	2.255	2.255	2.255	2.255
64	Erosion Velocity (m/s)	10.67	10.67	10.67	10.67	10.67
65	Length (m)	1500	1505	1510	1515	1520
66	Bulk Density (kg/m3)	130.8	130.8	130.8	130.8	130.8
67	Bulk Velocity (m/s)	2.255	2.255	2.255	2.254	2.254
68	Erosion Velocity (m/s)	10.67	10.67	10.67	10.67	10.67
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10	Erosion Table					
11						
12						
13	Length (m)	1525	1530	1535	1540	1545
14	Bulk Density (kg/m3)	130.8	130.6	130.4	130.2	130.0
15	Bulk Velocity (m/s)	2.254	2.257	2.261	2.264	2.268
16	Erosion Velocity (m/s)	10.67	10.67	10.68	10.69	10.70
17	Length (m)	1550	1555	1560	1565	1570
18	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
19	Bulk Velocity (m/s)	2.271	2.271	2.271	2.271	2.271
20	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
21	Length (m)	1575	1580	1585	1590	1595
22	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
23	Bulk Velocity (m/s)	2.271	2.271	2.271	2.271	2.271
24	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
25	Length (m)	1600	1605	1610	1615	1620
26	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
27	Bulk Velocity (m/s)	2.271	2.271	2.271	2.271	2.271
28	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
29	Length (m)	1625	1630	1635	1640	1645
30	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
31	Bulk Velocity (m/s)	2.271	2.271	2.271	2.271	2.271
32	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
33	Length (m)	1650	1655	1660	1665	1670
34	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
35	Bulk Velocity (m/s)	2.271	2.271	2.271	2.271	2.271
36	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
37	Length (m)	1675	1680	1685	1690	1695
38	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.9
39	Bulk Velocity (m/s)	2.271	2.271	2.271	2.271	2.271
40	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
41	Length (m)	1700	1705	1710	1715	1720
42	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
43	Bulk Velocity (m/s)	2.271	2.271	2.271	2.271	2.270
44	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
45	Length (m)	1725	1730	1735	1740	1745
46	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
47	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
48	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
49	Length (m)	1750	1755	1760	1765	1770
50	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
51	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
52	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
53	Length (m)	1775	1780	1785	1790	1795
54	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
55	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
56	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
57	Length (m)	1800	1805	1810	1815	1820
58	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
59	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
60	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
61	Length (m)	1825	1830	1835	1840	1845
62	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
63	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
64	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
65	Length (m)	1850	1855	1860	1865	1870
66	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
67	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
68	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 575 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10	Erosion Table					
11						
12						
13	Length (m)	1875	1880	1885	1890	1895
14	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
15	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
16	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
17	Length (m)	1900	1905	1910	1915	1920
18	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
19	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
20	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
21	Length (m)	1925	1930	1935	1940	1945
22	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
23	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
24	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
25	Length (m)	1950	1955	1960	1965	1970
26	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
27	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
28	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
29	Length (m)	1975	1980	1985	1990	1995
30	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
31	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
32	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
33	Length (m)	2000	2005	2010	2015	2020
34	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
35	Bulk Velocity (m/s)	2.270	2.269	2.269	2.269	2.269
36	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
37	Length (m)	2025	2030	2035	2040	2045
38	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
39	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
40	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
41	Length (m)	2050	2055	2060	2065	2070
42	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
43	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
44	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
45	Length (m)	2075	2080	2085	2090	2095
46	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
47	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
48	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
49	Length (m)	2100	2105	2110	2115	2120
50	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
51	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
52	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
53	Length (m)	2125	2130	2135	2140	2145
54	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
55	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
56	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
57	Length (m)	2150	2155	2160	2165	2170
58	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
59	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
60	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
61	Length (m)	2175	2180	2185	2191	2196
62	Bulk Density (kg/m3)	130.0	130.0	130.0	130.0	130.0
63	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
64	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
65	Length (m)	2201	2206	2211	2216	2221
66	Bulk Density (kg/m3)	130.0	130.0	130.0	130.0	130.0
67	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
68	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 576 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2226	2231	2236	2241	2246
14	Bulk Density (kg/m3)	130.0	130.0	130.0	130.0	130.0
15	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
16	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
17	Length (m)	2251	2256	2261	2266	2271
18	Bulk Density (kg/m3)	130.0	130.0	130.0	130.0	130.0
19	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
20	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
21	Length (m)	2276	2281	2286	2291	2296
22	Bulk Density (kg/m3)	130.0	130.0	130.0	130.0	130.0
23	Bulk Velocity (m/s)	2.268	2.268	2.268	2.268	2.268
24	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
25	Length (m)	2301	2306	2311	2316	2321
26	Bulk Density (kg/m3)	130.0	130.0	130.0	130.0	130.0
27	Bulk Velocity (m/s)	2.268	2.268	2.268	2.268	2.268
28	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
29	Length (m)	2326	2331	2336	2341	2346
30	Bulk Density (kg/m3)	130.0	130.0	130.0	130.0	130.0
31	Bulk Velocity (m/s)	2.268	2.268	2.268	2.268	2.268
32	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
33	Length (m)	2351	2356	2361	2366	2371
34	Bulk Density (kg/m3)	130.0	130.0	130.0	130.0	130.0
35	Bulk Velocity (m/s)	2.268	2.268	2.268	2.268	2.268
36	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
37	Length (m)	2376	2381	2386	2391	2396
38	Bulk Density (kg/m3)	130.0	130.0	130.0	130.0	130.0
39	Bulk Velocity (m/s)	2.268	2.268	2.268	2.268	2.268
40	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
41	Length (m)	2401	2406	2411	2416	2421
42	Bulk Density (kg/m3)	130.0	130.0	130.0	130.0	130.0
43	Bulk Velocity (m/s)	2.268	2.268	2.268	2.268	2.268
44	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
45	Length (m)	2426	2431	2436	2441	2446
46	Bulk Density (kg/m3)	130.0	130.0	130.0	130.0	130.0
47	Bulk Velocity (m/s)	2.268	2.268	2.268	2.268	2.268
48	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
49	Length (m)	2451	2456	2461	2466	2471
50	Bulk Density (kg/m3)	130.0	130.0	130.0	130.0	130.0
51	Bulk Velocity (m/s)	2.268	2.268	2.268	2.268	2.268
52	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
53	Length (m)	2476	2481	2486	2491	2496
54	Bulk Density (kg/m3)	130.0	130.0	130.0	130.0	130.0
55	Bulk Velocity (m/s)	2.268	2.268	2.268	2.268	2.268
56	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
57	Length (m)	2501	2506	2511	2516	2521
58	Bulk Density (kg/m3)	130.0	130.0	130.0	130.0	130.0
59	Bulk Velocity (m/s)	2.268	2.268	2.268	2.268	2.268
60	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
61	Length (m)	2526	2531	2536	2541	2546
62	Bulk Density (kg/m3)	130.0	130.0	130.0	130.0	130.0
63	Bulk Velocity (m/s)	2.268	2.267	2.267	2.267	2.267
64	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
65	Length (m)	2551	2556	2561	2566	2571
66	Bulk Density (kg/m3)	130.0	130.0	130.0	130.0	130.0
67	Bulk Velocity (m/s)	2.267	2.267	2.267	2.267	2.267
68	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 577 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2576	2581	2586	2591	2596
14	Bulk Density (kg/m3)	130.0	130.0	130.0	130.0	130.0
15	Bulk Velocity (m/s)	2.267	2.267	2.267	2.267	2.267
16	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
17	Length (m)	2601	2606	2611	2616	2621
18	Bulk Density (kg/m3)	130.0	130.1	130.1	130.1	130.1
19	Bulk Velocity (m/s)	2.267	2.267	2.267	2.267	2.267
20	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
21	Length (m)	2626	2631	2636	2641	2646
22	Bulk Density (kg/m3)	130.1	130.1	130.1	130.1	130.1
23	Bulk Velocity (m/s)	2.267	2.267	2.267	2.267	2.267
24	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
25	Length (m)	2651	2656	2661	2666	2671
26	Bulk Density (kg/m3)	130.1	130.1	130.1	130.1	130.1
27	Bulk Velocity (m/s)	2.267	2.267	2.267	2.267	2.267
28	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
29	Length (m)	2676	2681	2686	2691	2696
30	Bulk Density (kg/m3)	130.1	130.1	130.1	130.1	130.1
31	Bulk Velocity (m/s)	2.267	2.267	2.267	2.267	2.267
32	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
33	Length (m)	2701	2706	2711	2716	2721
34	Bulk Density (kg/m3)	130.1	130.1	130.1	130.1	130.1
35	Bulk Velocity (m/s)	2.267	2.267	2.267	2.267	2.267
36	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
37	Length (m)	2726	2731	2736	2741	2746
38	Bulk Density (kg/m3)	130.1	130.1	130.1	130.1	130.1
39	Bulk Velocity (m/s)	2.267	2.267	2.267	2.267	2.267
40	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
41	Length (m)	2751	2756	2761	2766	2771
42	Bulk Density (kg/m3)	130.1	130.1	130.1	130.1	130.1
43	Bulk Velocity (m/s)	2.267	2.267	2.267	2.267	2.267
44	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
45	Length (m)	2776	2781	2786	2791	2796
46	Bulk Density (kg/m3)	130.1	130.1	130.1	130.1	130.1
47	Bulk Velocity (m/s)	2.267	2.266	2.266	2.266	2.266
48	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
49	Length (m)	2801	2806	2811	2816	2821
50	Bulk Density (kg/m3)	130.1	130.1	130.1	130.1	130.1
51	Bulk Velocity (m/s)	2.266	2.266	2.266	2.266	2.266
52	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
53	Length (m)	2826	2831	2836	2841	2846
54	Bulk Density (kg/m3)	130.1	130.1	130.1	130.1	130.1
55	Bulk Velocity (m/s)	2.266	2.266	2.266	2.266	2.266
56	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
57	Length (m)	2851	2856	2861	2866	2871
58	Bulk Density (kg/m3)	130.1	130.1	130.1	130.1	130.1
59	Bulk Velocity (m/s)	2.266	2.266	2.266	2.266	2.266
60	Erosion Velocity (m/s)	10.69	10.69	10.69	10.69	10.69
61	Length (m)	2876	2881	2886	2891	2896
62	Bulk Density (kg/m3)	130.1	130.1	130.1	130.1	130.1
63	Bulk Velocity (m/s)	2.266	2.266	2.266	2.266	2.266
64	Erosion Velocity (m/s)	10.69	10.69	10.69	10.69	10.69
65	Length (m)	2901	2906	2911	2916	2921
66	Bulk Density (kg/m3)	130.1	130.1	130.1	130.1	130.1
67	Bulk Velocity (m/s)	2.266	2.266	2.266	2.266	2.266
68	Erosion Velocity (m/s)	10.69	10.69	10.69	10.69	10.69
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 578 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2926	2931	2936	2941	2946
14	Bulk Density (kg/m3)	130.1	130.1	130.1	130.1	130.1
15	Bulk Velocity (m/s)	2.266	2.266	2.266	2.266	2.266
16	Erosion Velocity (m/s)	10.69	10.69	10.69	10.69	10.69
17	Length (m)	2951	2956	2961	2966	2971
18	Bulk Density (kg/m3)	130.1	130.1	130.1	130.1	130.1
19	Bulk Velocity (m/s)	2.266	2.266	2.266	2.266	2.265
20	Erosion Velocity (m/s)	10.69	10.69	10.69	10.69	10.69
21	Length (m)	2976	2981	2986	2991	2996
22	Bulk Density (kg/m3)	130.2	130.2	130.2	130.2	130.2
23	Bulk Velocity (m/s)	2.265	2.265	2.265	2.265	2.265
24	Erosion Velocity (m/s)	10.69	10.69	10.69	10.69	10.69
25	Length (m)	3001	3006	3011	3016	3021
26	Bulk Density (kg/m3)	130.2	130.2	130.2	130.2	130.2
27	Bulk Velocity (m/s)	2.265	2.265	2.265	2.264	2.264
28	Erosion Velocity (m/s)	10.69	10.69	10.69	10.69	10.69
29	Length (m)	3026	3031	3036	3041	3046
30	Bulk Density (kg/m3)	130.3	130.3	130.3	130.3	130.3
31	Bulk Velocity (m/s)	2.263	2.263	2.263	2.263	2.263
32	Erosion Velocity (m/s)	10.69	10.69	10.69	10.69	10.69
33	Length (m)	3051	3056	3061	3066	3071
34	Bulk Density (kg/m3)	130.3	130.3	130.3	130.3	130.3
35	Bulk Velocity (m/s)	2.263	2.262	2.262	2.262	2.262
36	Erosion Velocity (m/s)	10.69	10.69	10.69	10.69	10.69
37	Length (m)	3076	3081	3086	3091	3096
38	Bulk Density (kg/m3)	130.3	130.2	130.0	129.8	129.6
39	Bulk Velocity (m/s)	2.262	2.265	2.269	2.272	2.275
40	Erosion Velocity (m/s)	10.69	10.69	10.70	10.71	10.72
41	Length (m)	3101	3106	3111	3116	3121
42	Bulk Density (kg/m3)	129.4	129.4	129.4	129.4	129.4
43	Bulk Velocity (m/s)	2.279	2.279	2.279	2.279	2.279
44	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
45	Length (m)	3126	3131	3136	3141	3146
46	Bulk Density (kg/m3)	129.4	129.4	129.4	129.4	129.4
47	Bulk Velocity (m/s)	2.279	2.279	2.279	2.279	2.279
48	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
49	Length (m)	3151	3156	3161	3166	3171
50	Bulk Density (kg/m3)	129.4	129.4	129.4	129.4	129.4
51	Bulk Velocity (m/s)	2.279	2.279	2.279	2.279	2.279
52	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
53	Length (m)	3176	3181	3186	3192	3197
54	Bulk Density (kg/m3)	129.4	129.4	129.4	129.4	129.4
55	Bulk Velocity (m/s)	2.279	2.279	2.279	2.279	2.279
56	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
57	Length (m)	3202	3207	3212	3217	3222
58	Bulk Density (kg/m3)	129.4	129.4	129.4	129.4	129.4
59	Bulk Velocity (m/s)	2.279	2.279	2.279	2.279	2.279
60	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
61	Length (m)	3227	3232	3237	3242	3247
62	Bulk Density (kg/m3)	129.4	129.4	129.4	129.4	129.4
63	Bulk Velocity (m/s)	2.279	2.278	2.278	2.278	2.278
64	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
65	Length (m)	3252	3257	3262	3267	3272
66	Bulk Density (kg/m3)	129.4	129.4	129.4	129.4	129.4
67	Bulk Velocity (m/s)	2.278	2.278	2.278	2.278	2.278
68	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 579 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3277	3282	3287	3292	3297
14	Bulk Density (kg/m3)	129.4	129.4	129.4	129.4	129.4
15	Bulk Velocity (m/s)	2.278	2.278	2.278	2.278	2.278
16	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
17	Length (m)	3302	3307	3312	3317	3322
18	Bulk Density (kg/m3)	129.4	129.4	129.4	129.4	129.4
19	Bulk Velocity (m/s)	2.278	2.278	2.278	2.278	2.278
20	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
21	Length (m)	3327	3332	3337	3342	3347
22	Bulk Density (kg/m3)	129.4	129.4	129.4	129.4	129.4
23	Bulk Velocity (m/s)	2.278	2.278	2.278	2.278	2.278
24	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
25	Length (m)	3352	3357	3362	3367	3372
26	Bulk Density (kg/m3)	129.4	129.4	129.4	129.4	129.4
27	Bulk Velocity (m/s)	2.278	2.278	2.278	2.278	2.278
28	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
29	Length (m)	3377	3382	3387	3392	3396
30	Bulk Density (kg/m3)	129.4	129.4	129.4	129.4	129.4
31	Bulk Velocity (m/s)	2.278	2.278	2.278	2.278	2.278
32	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
33	Length (m)	3401	3406	3411	3416	3421
34	Bulk Density (kg/m3)	129.4	129.4	129.4	129.4	129.4
35	Bulk Velocity (m/s)	2.278	2.278	2.278	2.278	2.278
36	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
37	Length (m)	3426	3431	3436	3441	3446
38	Bulk Density (kg/m3)	129.4	129.4	129.4	129.4	129.5
39	Bulk Velocity (m/s)	2.278	2.278	2.278	2.278	2.278
40	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
41	Length (m)	3451	3456	3461	3466	3471
42	Bulk Density (kg/m3)	129.5	129.5	129.5	129.5	129.5
43	Bulk Velocity (m/s)	2.278	2.278	2.278	2.278	2.278
44	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
45	Length (m)	3476	3481	3486	3491	3496
46	Bulk Density (kg/m3)	129.5	129.5	129.5	129.5	129.5
47	Bulk Velocity (m/s)	2.278	2.278	2.278	2.277	2.277
48	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
49	Length (m)	3501	3506	3511	3516	3521
50	Bulk Density (kg/m3)	129.5	129.5	129.5	129.5	129.5
51	Bulk Velocity (m/s)	2.277	2.277	2.277	2.277	2.277
52	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
53	Length (m)	3526	3531	3536	3541	3546
54	Bulk Density (kg/m3)	129.5	129.5	129.5	129.5	129.5
55	Bulk Velocity (m/s)	2.277	2.277	2.277	2.277	2.277
56	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
57	Length (m)	3551	3556	3561	3566	3571
58	Bulk Density (kg/m3)	129.5	129.5	129.5	129.5	129.5
59	Bulk Velocity (m/s)	2.277	2.277	2.277	2.277	2.277
60	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
61	Length (m)	3576	3581	3586	3591	3596
62	Bulk Density (kg/m3)	129.5	129.5	129.5	129.5	129.5
63	Bulk Velocity (m/s)	2.277	2.277	2.277	2.277	2.277
64	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
65	Length (m)	3601	3606	3611	3616	3621
66	Bulk Density (kg/m3)	129.5	129.5	129.5	129.5	129.5
67	Bulk Velocity (m/s)	2.277	2.277	2.277	2.277	2.277
68	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3626	3631	3636	3641	3646
14	Bulk Density (kg/m3)	129.5	129.5	129.5	129.5	129.5
15	Bulk Velocity (m/s)	2.277	2.277	2.277	2.277	2.277
16	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
17	Length (m)	3651	3656	3661	3666	3671
18	Bulk Density (kg/m3)	129.5	129.5	129.5	129.5	129.5
19	Bulk Velocity (m/s)	2.277	2.277	2.277	2.277	2.277
20	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
21	Length (m)	3676	3681	3686	3691	3696
22	Bulk Density (kg/m3)	129.5	129.5	129.5	129.5	129.5
23	Bulk Velocity (m/s)	2.277	2.277	2.277	2.277	2.277
24	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
25	Length (m)	3701	3706	3711	3716	3721
26	Bulk Density (kg/m3)	129.5	129.5	129.5	129.5	129.5
27	Bulk Velocity (m/s)	2.277	2.277	2.277	2.277	2.277
28	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
29	Length (m)	3726	3731	3736	3741	3746
30	Bulk Density (kg/m3)	129.5	129.5	129.5	129.5	129.5
31	Bulk Velocity (m/s)	2.277	2.276	2.276	2.276	2.276
32	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
33	Length (m)	3751	3756	3761	3766	3771
34	Bulk Density (kg/m3)	129.5	129.5	129.5	129.5	129.5
35	Bulk Velocity (m/s)	2.276	2.276	2.276	2.276	2.276
36	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
37	Length (m)	3776	3781	3787	3792	3797
38	Bulk Density (kg/m3)	129.5	129.5	129.5	129.5	129.5
39	Bulk Velocity (m/s)	2.276	2.276	2.276	2.276	2.276
40	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
41	Length (m)	3802	3807	3812	3817	3822
42	Bulk Density (kg/m3)	129.5	129.5	129.5	129.5	129.5
43	Bulk Velocity (m/s)	2.276	2.276	2.276	2.276	2.276
44	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
45	Length (m)	3827	3832	3837	3842	3847
46	Bulk Density (kg/m3)	129.5	129.5	129.5	129.5	129.5
47	Bulk Velocity (m/s)	2.276	2.276	2.276	2.276	2.276
48	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
49	Length (m)	3852	3857	3862	3867	3872
50	Bulk Density (kg/m3)	129.5	129.5	129.5	129.5	129.5
51	Bulk Velocity (m/s)	2.276	2.276	2.276	2.276	2.276
52	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
53	Length (m)	3877	3882	3887	3892	3897
54	Bulk Density (kg/m3)	129.5	129.5	129.5	129.6	129.6
55	Bulk Velocity (m/s)	2.276	2.276	2.276	2.276	2.276
56	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
57	Length (m)	3902	3907	3912	3917	3922
58	Bulk Density (kg/m3)	129.6	129.6	129.6	129.6	129.6
59	Bulk Velocity (m/s)	2.276	2.276	2.276	2.276	2.276
60	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
61	Length (m)	3927	3932	3937	3942	3947
62	Bulk Density (kg/m3)	129.6	129.6	129.6	129.6	129.6
63	Bulk Velocity (m/s)	2.276	2.276	2.276	2.276	2.276
64	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
65	Length (m)	3952	3957	3962	3967	3972
66	Bulk Density (kg/m3)	129.6	129.6	129.6	129.6	129.6
67	Bulk Velocity (m/s)	2.276	2.276	2.276	2.276	2.276
68	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 581 of 937	

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10	Erosion Table					
11						
12						
13	Length (m)	3977	3982	3987	3992	3997
14	Bulk Density (kg/m3)	129.6	129.6	129.6	129.6	129.6
15	Bulk Velocity (m/s)	2.276	2.276	2.276	2.276	2.276
16	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
17	Length (m)	4002	4007	4012	4017	4022
18	Bulk Density (kg/m3)	129.6	129.6	129.6	129.6	129.6
19	Bulk Velocity (m/s)	2.276	2.276	2.275	2.275	2.275
20	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
21	Length (m)	4027	4032	4037	4042	4047
22	Bulk Density (kg/m3)	129.6	129.6	129.6	129.6	129.6
23	Bulk Velocity (m/s)	2.275	2.275	2.275	2.275	2.275
24	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
25	Length (m)	4052	4057	4062	4067	4072
26	Bulk Density (kg/m3)	129.6	129.6	129.6	129.6	129.6
27	Bulk Velocity (m/s)	2.275	2.275	2.275	2.275	2.275
28	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
29	Length (m)	4077	4084	4091	4098	4105
30	Bulk Density (kg/m3)	129.6	129.6	129.6	129.6	129.6
31	Bulk Velocity (m/s)	2.275	2.275	2.275	2.275	2.275
32	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
33	Length (m)	4112	4120	4127	4135	4143
34	Bulk Density (kg/m3)	129.6	129.6	129.6	129.6	129.6
35	Bulk Velocity (m/s)	2.275	2.275	2.275	2.275	2.275
36	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
37	Length (m)	4150	4158	4165	4173	4181
38	Bulk Density (kg/m3)	129.6	129.6	129.6	129.6	129.6
39	Bulk Velocity (m/s)	2.275	2.275	2.275	2.275	2.275
40	Erosion Velocity (m/s)	10.72	10.72	10.72	10.72	10.72
41	Length (m)	4188	4196	4203	4211	4219
42	Bulk Density (kg/m3)	129.6	129.6	129.6	129.6	129.6
43	Bulk Velocity (m/s)	2.275	2.275	2.275	2.275	2.275
44	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
45	Length (m)	4226	4234	4241	4249	4257
46	Bulk Density (kg/m3)	129.6	129.6	129.6	129.6	129.6
47	Bulk Velocity (m/s)	2.274	2.274	2.274	2.274	2.274
48	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
49	Length (m)	4264	4272	4279	4287	4295
50	Bulk Density (kg/m3)	129.6	129.6	129.6	129.6	129.6
51	Bulk Velocity (m/s)	2.274	2.274	2.274	2.274	2.274
52	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
53	Length (m)	4302	4304	4305	4306	4308
54	Bulk Density (kg/m3)	129.6	129.6	129.6	129.6	129.6
55	Bulk Velocity (m/s)	2.274	2.274	2.274	2.274	2.274
56	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
57	Length (m)	4309	4314	4319	4324	4329
58	Bulk Density (kg/m3)	129.6	129.6	129.6	129.6	129.7
59	Bulk Velocity (m/s)	2.274	2.274	2.274	2.274	2.274
60	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
61	Length (m)	4334	4339	4344	4349	4354
62	Bulk Density (kg/m3)	129.7	129.7	129.7	129.7	129.7
63	Bulk Velocity (m/s)	2.274	2.274	2.274	2.274	2.274
64	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
65	Length (m)	4359	4364	4369	4374	4379
66	Bulk Density (kg/m3)	129.7	129.7	129.7	129.7	129.7
67	Bulk Velocity (m/s)	2.274	2.274	2.274	2.274	2.274
68	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
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1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	4384	4389	4394	4399	4404
14	Bulk Density (kg/m3)	129.7	129.7	129.7	129.7	129.7
15	Bulk Velocity (m/s)	2.274	2.274	2.274	2.274	2.274
16	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
17	Length (m)	4409	4414	4419	4424	4429
18	Bulk Density (kg/m3)	129.7	129.7	129.7	129.7	129.7
19	Bulk Velocity (m/s)	2.274	2.274	2.274	2.274	2.274
20	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
21	Length (m)	4434	4439	4444	4449	4454
22	Bulk Density (kg/m3)	129.7	129.7	129.7	129.7	129.7
23	Bulk Velocity (m/s)	2.274	2.274	2.274	2.274	2.274
24	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
25	Length (m)	4459	4464	4469	4474	4479
26	Bulk Density (kg/m3)	129.7	129.7	129.7	129.7	129.7
27	Bulk Velocity (m/s)	2.274	2.274	2.274	2.274	2.274
28	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
29	Length (m)	4484	4489	4494	4499	4504
30	Bulk Density (kg/m3)	129.7	129.7	129.7	129.7	129.7
31	Bulk Velocity (m/s)	2.274	2.274	2.274	2.274	2.274
32	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
33	Length (m)	4510	4515	4520	4525	4530
34	Bulk Density (kg/m3)	129.7	129.7	129.7	129.7	129.7
35	Bulk Velocity (m/s)	2.274	2.274	2.274	2.274	2.274
36	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
37	Length (m)	4535	4540	4545	4550	4555
38	Bulk Density (kg/m3)	129.7	129.7	129.7	129.7	129.7
39	Bulk Velocity (m/s)	2.274	2.274	2.273	2.273	2.273
40	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
41	Length (m)	4560	4565	4570	4575	4580
42	Bulk Density (kg/m3)	129.7	129.7	129.7	129.7	129.7
43	Bulk Velocity (m/s)	2.273	2.273	2.273	2.273	2.273
44	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
45	Length (m)	4585	4590	4595	4600	4605
46	Bulk Density (kg/m3)	129.7	129.7	129.7	129.7	129.7
47	Bulk Velocity (m/s)	2.273	2.273	2.273	2.273	2.273
48	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
49	Length (m)	4610	4623	4636	4649	4662
50	Bulk Density (kg/m3)	129.7	129.7	129.7	129.7	129.7
51	Bulk Velocity (m/s)	2.273	2.273	2.273	2.273	2.273
52	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
53	Length (m)	4675	4677	4679	4681	4683
54	Bulk Density (kg/m3)	129.7	129.7	129.7	129.7	129.7
55	Bulk Velocity (m/s)	2.273	2.273	2.273	2.273	2.273
56	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
57	Length (m)	4685	4690	4695	4700	4705
58	Bulk Density (kg/m3)	129.7	129.7	129.7	129.7	129.7
59	Bulk Velocity (m/s)	2.273	2.273	2.273	2.273	2.273
60	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
61	Length (m)	4710	4715	4720	4725	4730
62	Bulk Density (kg/m3)	129.7	129.7	129.7	129.7	129.7
63	Bulk Velocity (m/s)	2.273	2.273	2.273	2.273	2.273
64	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
65	Length (m)	4735	4740	4745	4750	4755
66	Bulk Density (kg/m3)	129.7	129.7	129.7	129.7	129.7
67	Bulk Velocity (m/s)	2.273	2.273	2.273	2.273	2.273
68	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 583 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	4760	4765	4770	4775	4780
14	Bulk Density (kg/m3)	129.7	129.7	129.7	129.7	129.7
15	Bulk Velocity (m/s)	2.273	2.273	2.273	2.273	2.273
16	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
17	Length (m)	4785	4790	4795	4800	4805
18	Bulk Density (kg/m3)	129.7	129.7	129.7	129.7	129.7
19	Bulk Velocity (m/s)	2.273	2.273	2.273	2.273	2.273
20	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
21	Length (m)	4810	4815	4820	4825	4830
22	Bulk Density (kg/m3)	129.7	129.7	129.7	129.7	129.7
23	Bulk Velocity (m/s)	2.273	2.272	2.272	2.272	2.272
24	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
25	Length (m)	4835	4840	4845	4850	4855
26	Bulk Density (kg/m3)	129.7	129.8	129.8	129.8	129.8
27	Bulk Velocity (m/s)	2.272	2.272	2.272	2.272	2.272
28	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
29	Length (m)	4860	4865	4870	4875	4880
30	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
31	Bulk Velocity (m/s)	2.272	2.272	2.272	2.272	2.272
32	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
33	Length (m)	4885	4890	4895	4900	4905
34	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
35	Bulk Velocity (m/s)	2.272	2.272	2.272	2.272	2.272
36	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
37	Length (m)	4910	4915	4920	4925	4930
38	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
39	Bulk Velocity (m/s)	2.272	2.272	2.272	2.272	2.272
40	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
41	Length (m)	4935	4940	4945	4950	4955
42	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
43	Bulk Velocity (m/s)	2.272	2.272	2.272	2.272	2.272
44	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
45	Length (m)	4960	4965	4970	4975	4980
46	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
47	Bulk Velocity (m/s)	2.272	2.272	2.272	2.272	2.272
48	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
49	Length (m)	4985	4990	4995	5000	5005
50	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
51	Bulk Velocity (m/s)	2.272	2.272	2.272	2.272	2.272
52	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
53	Length (m)	5010	5015	5020	5025	5030
54	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
55	Bulk Velocity (m/s)	2.272	2.272	2.272	2.272	2.272
56	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
57	Length (m)	5035	5040	5045	5050	5055
58	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
59	Bulk Velocity (m/s)	2.272	2.272	2.272	2.272	2.272
60	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
61	Length (m)	5060	5065	5070	5075	5080
62	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
63	Bulk Velocity (m/s)	2.272	2.272	2.272	2.271	2.271
64	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
65	Length (m)	5085	5090	5095	5100	5105
66	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
67	Bulk Velocity (m/s)	2.271	2.271	2.271	2.271	2.271
68	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
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1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	5110	5115	5120	5125	5130
14	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
15	Bulk Velocity (m/s)	2.271	2.271	2.271	2.271	2.271
16	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
17	Length (m)	5135	5140	5145	5150	5155
18	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
19	Bulk Velocity (m/s)	2.271	2.271	2.271	2.271	2.271
20	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
21	Length (m)	5160	5165	5170	5175	5180
22	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
23	Bulk Velocity (m/s)	2.271	2.271	2.271	2.271	2.271
24	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
25	Length (m)	5185	5190	5195	5200	5205
26	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
27	Bulk Velocity (m/s)	2.271	2.271	2.271	2.271	2.271
28	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
29	Length (m)	5210	5215	5220	5225	5230
30	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
31	Bulk Velocity (m/s)	2.271	2.271	2.271	2.271	2.271
32	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
33	Length (m)	5235	5240	5245	5250	5255
34	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
35	Bulk Velocity (m/s)	2.271	2.271	2.271	2.271	2.271
36	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
37	Length (m)	5260	5265	5270	5275	5280
38	Bulk Density (kg/m3)	129.8	129.8	129.8	129.8	129.8
39	Bulk Velocity (m/s)	2.271	2.271	2.271	2.271	2.271
40	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
41	Length (m)	5285	5290	5295	5300	5305
42	Bulk Density (kg/m3)	129.8	129.9	129.9	129.9	129.9
43	Bulk Velocity (m/s)	2.271	2.271	2.271	2.271	2.271
44	Erosion Velocity (m/s)	10.71	10.71	10.71	10.71	10.71
45	Length (m)	5310	5315	5320	5325	5330
46	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
47	Bulk Velocity (m/s)	2.271	2.271	2.270	2.270	2.270
48	Erosion Velocity (m/s)	10.71	10.71	10.71	10.70	10.70
49	Length (m)	5335	5340	5345	5350	5355
50	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
51	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
52	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
53	Length (m)	5360	5365	5370	5375	5380
54	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
55	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
56	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
57	Length (m)	5385	5390	5395	5400	5405
58	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
59	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
60	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
61	Length (m)	5410	5415	5420	5425	5430
62	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
63	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
64	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
65	Length (m)	5435	5440	5445	5450	5455
66	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
67	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
68	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
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1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	5460	5465	5470	5475	5480
14	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
15	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
16	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
17	Length (m)	5485	5490	5495	5500	5505
18	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
19	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
20	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
21	Length (m)	5510	5515	5520	5525	5530
22	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
23	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
24	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
25	Length (m)	5535	5542	5549	5556	5563
26	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
27	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
28	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
29	Length (m)	5570	5572	5574	5576	5578
30	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
31	Bulk Velocity (m/s)	2.270	2.270	2.270	2.270	2.270
32	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
33	Length (m)	5580	5585	5590	5595	5600
34	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
35	Bulk Velocity (m/s)	2.270	2.270	2.269	2.269	2.269
36	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
37	Length (m)	5605	5610	5615	5620	5625
38	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
39	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
40	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
41	Length (m)	5630	5635	5640	5645	5650
42	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
43	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
44	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
45	Length (m)	5655	5660	5665	5670	5675
46	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
47	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
48	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
49	Length (m)	5680	5685	5690	5695	5700
50	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
51	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
52	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
53	Length (m)	5705	5710	5715	5720	5725
54	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
55	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
56	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
57	Length (m)	5730	5735	5740	5745	5750
58	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
59	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
60	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
61	Length (m)	5755	5760	5765	5770	5775
62	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
63	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
64	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
65	Length (m)	5780	5785	5790	5795	5800
66	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	130.0
67	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
68	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	5805	5810	5815	5820	5825
14	Bulk Density (kg/m3)	130.0	130.0	130.0	130.0	130.0
15	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
16	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
17	Length (m)	5830	5835	5840	5845	5850
18	Bulk Density (kg/m3)	130.0	130.0	130.0	130.0	130.0
19	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
20	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
21	Length (m)	5855	5860	5865	5870	5875
22	Bulk Density (kg/m3)	129.9	129.9	129.9	129.9	129.9
23	Bulk Velocity (m/s)	2.269	2.269	2.269	2.269	2.269
24	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
25	Length (m)	5880	5885	5890	5895	5900
26	Bulk Density (kg/m3)	129.9	130.0	130.0	130.0	130.0
27	Bulk Velocity (m/s)	2.269	2.269	2.268	2.268	2.267
28	Erosion Velocity (m/s)	10.70	10.70	10.70	10.70	10.70
29	Length (m)	5905	5910	5915	5920	5925
30	Bulk Density (kg/m3)	130.1	130.1	130.1	130.1	130.2
31	Bulk Velocity (m/s)	2.267	2.266	2.266	2.266	2.265
32	Erosion Velocity (m/s)	10.70	10.70	10.69	10.69	10.69
33	Length (m)	5930	5935	5940	5945	5950
34	Bulk Density (kg/m3)	130.2	130.0	129.8	129.6	129.4
35	Bulk Velocity (m/s)	2.265	2.268	2.271	2.275	2.278
36	Erosion Velocity (m/s)	10.69	10.70	10.71	10.72	10.72
37	Length (m)	5955	5960	5965	5970	5975
38	Bulk Density (kg/m3)	129.2	129.2	129.1	129.1	129.0
39	Bulk Velocity (m/s)	2.282	2.283	2.284	2.285	2.285
40	Erosion Velocity (m/s)	10.73	10.73	10.74	10.74	10.74
41	Length (m)	5980	5985	5990	5995	6000
42	Bulk Density (kg/m3)	129.0	129.0	129.0	129.1	129.1
43	Bulk Velocity (m/s)	2.286	2.286	2.285	2.284	2.284
44	Erosion Velocity (m/s)	10.74	10.74	10.74	10.74	10.74
45	Length (m)	6005	6010	6015	6020	6025
46	Bulk Density (kg/m3)	129.2	128.9	128.7	128.5	128.2
47	Bulk Velocity (m/s)	2.283	2.287	2.291	2.295	2.299
48	Erosion Velocity (m/s)	10.73	10.74	10.75	10.76	10.77
49	Length (m)	6030	6035	6040	6045	6050
50	Bulk Density (kg/m3)	128.0	128.0	128.0	128.0	128.0
51	Bulk Velocity (m/s)	2.303	2.303	2.303	2.303	2.303
52	Erosion Velocity (m/s)	10.78	10.78	10.78	10.78	10.78
53	Length (m)	6055	6060	6065	6070	6075
54	Bulk Density (kg/m3)	128.0	128.0	128.0	128.0	128.0
55	Bulk Velocity (m/s)	2.303	2.303	2.303	2.303	2.303
56	Erosion Velocity (m/s)	10.78	10.78	10.78	10.78	10.78
57	Length (m)	6080	6085	6090	6095	6101
58	Bulk Density (kg/m3)	128.0	128.1	128.1	128.1	128.1
59	Bulk Velocity (m/s)	2.303	2.302	2.302	2.301	2.301
60	Erosion Velocity (m/s)	10.78	10.78	10.78	10.78	10.78
61	Length (m)	6106	6111	6116	6121	6126
62	Bulk Density (kg/m3)	128.2	128.0	127.8	127.7	127.5
63	Bulk Velocity (m/s)	2.300	2.303	2.306	2.310	2.313
64	Erosion Velocity (m/s)	10.78	10.78	10.79	10.80	10.80
65	Length (m)	6131	6136	6141	6146	6151
66	Bulk Density (kg/m3)	127.3	127.3	127.3	127.3	127.3
67	Bulk Velocity (m/s)	2.316	2.316	2.316	2.316	2.316
68	Erosion Velocity (m/s)	10.81	10.81	10.81	10.81	10.81
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	6156	6161	6166	6171	6176
14	Bulk Density (kg/m3)	127.3	127.3	127.3	127.3	127.3
15	Bulk Velocity (m/s)	2.316	2.316	2.316	2.316	2.316
16	Erosion Velocity (m/s)	10.81	10.81	10.81	10.81	10.81
17	Length (m)	6181	6186	6191	6196	6201
18	Bulk Density (kg/m3)	127.3	127.3	127.3	127.3	127.3
19	Bulk Velocity (m/s)	2.316	2.316	2.316	2.316	2.316
20	Erosion Velocity (m/s)	10.81	10.81	10.81	10.81	10.81
21	Length (m)	6206	6211	6216	6221	6226
22	Bulk Density (kg/m3)	127.3	127.3	127.3	127.3	127.3
23	Bulk Velocity (m/s)	2.316	2.316	2.316	2.316	2.316
24	Erosion Velocity (m/s)	10.81	10.81	10.81	10.81	10.81
25	Length (m)	6231	6236	6241	6246	6251
26	Bulk Density (kg/m3)	127.3	127.3	127.3	127.3	127.3
27	Bulk Velocity (m/s)	2.315	2.315	2.315	2.315	2.315
28	Erosion Velocity (m/s)	10.81	10.81	10.81	10.81	10.81
29	Length (m)	6256	6261	6266	6271	6276
30	Bulk Density (kg/m3)	127.3	127.3	127.3	127.3	127.3
31	Bulk Velocity (m/s)	2.315	2.315	2.315	2.315	2.315
32	Erosion Velocity (m/s)	10.81	10.81	10.81	10.81	10.81
33	Length (m)	6281	6285	6290	6294	6299
34	Bulk Density (kg/m3)	127.3	127.4	127.4	127.4	127.5
35	Bulk Velocity (m/s)	2.315	2.315	2.314	2.314	2.313
36	Erosion Velocity (m/s)	10.81	10.81	10.81	10.81	10.80
37	Length (m)	6303	6308	6313	6318	6323
38	Bulk Density (kg/m3)	127.5	127.4	127.4	127.3	127.3
39	Bulk Velocity (m/s)	2.312	2.313	2.314	2.315	2.316
40	Erosion Velocity (m/s)	10.80	10.81	10.81	10.81	10.81
41	Length (m)	6328	6333	6338	6343	6348
42	Bulk Density (kg/m3)	127.2	127.2	127.3	127.3	127.3
43	Bulk Velocity (m/s)	2.317	2.317	2.317	2.317	2.317
44	Erosion Velocity (m/s)	10.82	10.81	10.81	10.81	10.81
45	Length (m)	6353	6358	6363	6368	6373
46	Bulk Density (kg/m3)	127.3	127.3	127.3	127.3	127.4
47	Bulk Velocity (m/s)	2.316	2.316	2.316	2.315	2.315
48	Erosion Velocity (m/s)	10.81	10.81	10.81	10.81	10.81
49	Length (m)	6378	6383	6388	6393	6398
50	Bulk Density (kg/m3)	127.4	127.3	127.3	127.3	127.2
51	Bulk Velocity (m/s)	2.315	2.315	2.316	2.317	2.318
52	Erosion Velocity (m/s)	10.81	10.81	10.81	10.81	10.82
53	Length (m)	6403	6408	6413	6418	6423
54	Bulk Density (kg/m3)	127.2	126.9	126.7	126.5	126.3
55	Bulk Velocity (m/s)	2.319	2.323	2.327	2.331	2.335
56	Erosion Velocity (m/s)	10.82	10.83	10.84	10.85	10.86
57	Length (m)	6428	6433	6438	6443	6448
58	Bulk Density (kg/m3)	126.1	126.1	126.1	126.1	126.1
59	Bulk Velocity (m/s)	2.339	2.339	2.339	2.339	2.339
60	Erosion Velocity (m/s)	10.87	10.87	10.87	10.87	10.87
61	Length (m)	6453	6459	6465	6471	6477
62	Bulk Density (kg/m3)	126.1	126.1	126.1	126.1	126.1
63	Bulk Velocity (m/s)	2.339	2.339	2.339	2.339	2.339
64	Erosion Velocity (m/s)	10.87	10.87	10.87	10.87	10.87
65	Length (m)	6483	6487	6491	6495	6499
66	Bulk Density (kg/m3)	126.1	126.1	126.1	126.1	126.1
67	Bulk Velocity (m/s)	2.339	2.339	2.339	2.339	2.339
68	Erosion Velocity (m/s)	10.87	10.87	10.87	10.87	10.87
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 588 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	6503	6508	6513	6518	6523
14	Bulk Density (kg/m3)	126.1	126.1	126.1	126.1	126.1
15	Bulk Velocity (m/s)	2.339	2.339	2.339	2.339	2.339
16	Erosion Velocity (m/s)	10.87	10.87	10.87	10.87	10.87
17	Length (m)	6528	6533	6538	6543	6548
18	Bulk Density (kg/m3)	126.1	126.1	126.1	126.1	126.1
19	Bulk Velocity (m/s)	2.339	2.339	2.339	2.339	2.339
20	Erosion Velocity (m/s)	10.87	10.87	10.87	10.86	10.86
21	Length (m)	6553	6558	6563	6568	6573
22	Bulk Density (kg/m3)	126.1	126.1	126.1	126.1	126.1
23	Bulk Velocity (m/s)	2.339	2.339	2.339	2.339	2.339
24	Erosion Velocity (m/s)	10.86	10.86	10.86	10.86	10.86
25	Length (m)	6578	6583	6588	6593	6598
26	Bulk Density (kg/m3)	126.1	126.1	126.1	126.1	126.1
27	Bulk Velocity (m/s)	2.339	2.339	2.339	2.339	2.339
28	Erosion Velocity (m/s)	10.86	10.86	10.86	10.86	10.86
29	Length (m)	6603	6608	6613	6618	6623
30	Bulk Density (kg/m3)	126.1	126.1	126.1	126.1	126.1
31	Bulk Velocity (m/s)	2.339	2.339	2.339	2.339	2.339
32	Erosion Velocity (m/s)	10.86	10.86	10.86	10.86	10.86
33	Length (m)	6628	6633	6638	6643	6648
34	Bulk Density (kg/m3)	126.1	126.1	126.1	126.1	126.1
35	Bulk Velocity (m/s)	2.339	2.339	2.338	2.338	2.338
36	Erosion Velocity (m/s)	10.86	10.86	10.86	10.86	10.86
37	Length (m)	6653	6658	6663	6668	6673
38	Bulk Density (kg/m3)	126.1	126.1	126.1	126.1	126.1
39	Bulk Velocity (m/s)	2.338	2.338	2.338	2.338	2.338
40	Erosion Velocity (m/s)	10.86	10.86	10.86	10.86	10.86
41	Length (m)	6678	6683	6688	6693	6698
42	Bulk Density (kg/m3)	126.1	126.1	126.1	126.1	126.1
43	Bulk Velocity (m/s)	2.338	2.338	2.338	2.338	2.338
44	Erosion Velocity (m/s)	10.86	10.86	10.86	10.86	10.86
45	Length (m)	6703	6708	6713	6718	6723
46	Bulk Density (kg/m3)	126.1	126.1	126.1	126.1	126.1
47	Bulk Velocity (m/s)	2.338	2.338	2.338	2.338	2.338
48	Erosion Velocity (m/s)	10.86	10.86	10.86	10.86	10.86
49	Length (m)	6728	6733	6738	6743	6748
50	Bulk Density (kg/m3)	126.1	126.1	126.2	126.2	126.3
51	Bulk Velocity (m/s)	2.338	2.337	2.337	2.336	2.335
52	Erosion Velocity (m/s)	10.86	10.86	10.86	10.86	10.86
53	Length (m)	6754	6759	6764	6769	6774
54	Bulk Density (kg/m3)	126.3	126.2	126.1	126.1	126.0
55	Bulk Velocity (m/s)	2.335	2.336	2.337	2.339	2.340
56	Erosion Velocity (m/s)	10.86	10.86	10.86	10.86	10.87
57	Length (m)	6779	6784	6789	6794	6799
58	Bulk Density (kg/m3)	125.9	125.9	125.9	125.9	125.9
59	Bulk Velocity (m/s)	2.341	2.342	2.342	2.342	2.343
60	Erosion Velocity (m/s)	10.87	10.87	10.87	10.87	10.87
61	Length (m)	6804	6809	6814	6819	6824
62	Bulk Density (kg/m3)	125.8	125.7	125.6	125.5	125.4
63	Bulk Velocity (m/s)	2.343	2.345	2.347	2.348	2.350
64	Erosion Velocity (m/s)	10.87	10.88	10.88	10.89	10.89
65	Length (m)	6829	6834	6839	6844	6849
66	Bulk Density (kg/m3)	125.3	125.4	125.4	125.5	125.5
67	Bulk Velocity (m/s)	2.352	2.351	2.350	2.349	2.349
68	Erosion Velocity (m/s)	10.90	10.89	10.89	10.89	10.89
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10	Erosion Table					
11						
12						
13	Length (m)	6854	6859	6864	6869	6874
14	Bulk Density (kg/m3)	125.6	125.3	125.0	124.7	124.4
15	Bulk Velocity (m/s)	2.348	2.353	2.359	2.364	2.370
16	Erosion Velocity (m/s)	10.89	10.90	10.91	10.92	10.94
17	Length (m)	6879	6884	6889	6894	6899
18	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
19	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	6904	6909	6914	6919	6924
22	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
23	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	6929	6934	6939	6944	6949
26	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
27	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	6954	6959	6964	6969	6974
30	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
31	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	6979	6984	6989	6994	6999
34	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
35	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	7004	7009	7014	7019	7024
38	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
39	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	7029	7034	7039	7044	7049
42	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.2
43	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	7054	7059	7064	7069	7074
46	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
47	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	7079	7081	7083	7085	7087
50	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
51	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	7089	7097	7105	7113	7121
54	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
55	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	7129	7134	7139	7144	7149
58	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
59	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	7154	7159	7164	7169	7174
62	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
63	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	7179	7184	7189	7194	7199
66	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
67	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 590 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	Erosion						
10							
11	Erosion Table						
12							
13	Length (m)	7204	7209	7214	7219	7224	
14	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2	
15	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374	
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95	
17	Length (m)	7229	7234	7239	7244	7249	
18	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2	
19	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374	
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95	
21	Length (m)	7254	7259	7264	7269	7274	
22	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2	
23	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374	
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95	
25	Length (m)	7279	7284	7289	7294	7299	
26	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2	
27	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374	
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95	
29	Length (m)	7304	7309	7314	7319	7324	
30	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2	
31	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374	
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95	
33	Length (m)	7329	7334	7339	7344	7349	
34	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2	
35	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374	
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95	
37	Length (m)	7354	7359	7364	7369	7374	
38	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2	
39	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374	
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95	
41	Length (m)	7379	7384	7389	7394	7399	
42	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2	
43	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374	
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95	
45	Length (m)	7404	7409	7414	7419	7424	
46	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2	
47	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374	
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95	
49	Length (m)	7429	7434	7439	7444	7449	
50	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2	
51	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374	
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95	
53	Length (m)	7454	7459	7464	7469	7474	
54	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2	
55	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374	
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95	
57	Length (m)	7479	7484	7489	7494	7499	
58	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2	
59	Bulk Velocity (m/s)	2.374	2.374	2.374	2.373	2.373	
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95	
61	Length (m)	7504	7509	7514	7519	7524	
62	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2	
63	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373	
64	Erosion Velocity (m/s)	10.95	10.94	10.94	10.94	10.94	
65	Length (m)	7529	7533	7538	7543	7548	
66	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2	
67	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373	
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 591 of 937	

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10	Erosion Table					
11						
12						
13	Length (m)	7553	7558	7563	7568	7573
14	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
15	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
17	Length (m)	7578	7583	7588	7593	7598
18	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
19	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
21	Length (m)	7603	7608	7613	7618	7623
22	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
23	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
25	Length (m)	7628	7633	7638	7643	7648
26	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
27	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
29	Length (m)	7653	7658	7663	7668	7673
30	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
31	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
32	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
33	Length (m)	7678	7683	7688	7693	7698
34	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
35	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
36	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
37	Length (m)	7703	7708	7713	7718	7723
38	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
39	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
40	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
41	Length (m)	7728	7733	7738	7743	7748
42	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
43	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
44	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
45	Length (m)	7753	7758	7763	7768	7773
46	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
47	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.373
48	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
49	Length (m)	7778	7783	7788	7793	7798
50	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
51	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
52	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
53	Length (m)	7803	7808	7813	7818	7823
54	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
55	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.372
56	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
57	Length (m)	7828	7833	7838	7843	7848
58	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
59	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
60	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
61	Length (m)	7853	7858	7863	7868	7873
62	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
63	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
65	Length (m)	7878	7883	7888	7893	7898
66	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
67	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 592 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	7903	7908	7913	7918	7923
14	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
15	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
17	Length (m)	7928	7933	7938	7943	7948
18	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
19	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
21	Length (m)	7953	7958	7963	7968	7973
22	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
23	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
25	Length (m)	7978	7983	7988	7993	7998
26	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
27	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
29	Length (m)	8003	8008	8013	8018	8023
30	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
31	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
32	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
33	Length (m)	8028	8033	8038	8043	8048
34	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
35	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
36	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
37	Length (m)	8053	8058	8063	8068	8073
38	Bulk Density (kg/m3)	124.3	124.3	124.3	124.4	124.4
39	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
40	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
41	Length (m)	8078	8083	8088	8093	8098
42	Bulk Density (kg/m3)	124.4	124.3	124.3	124.2	124.2
43	Bulk Velocity (m/s)	2.371	2.371	2.372	2.373	2.374
44	Erosion Velocity (m/s)	10.94	10.94	10.94	10.95	10.95
45	Length (m)	8103	8108	8113	8118	8123
46	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
47	Bulk Velocity (m/s)	2.375	2.375	2.376	2.376	2.376
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	8128	8133	8138	8143	8148
50	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
51	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	8153	8158	8163	8168	8173
54	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
55	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	8178	8183	8188	8193	8198
58	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
59	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	8203	8208	8213	8218	8223
62	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
63	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	8228	8233	8238	8243	8248
66	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
67	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 593 of 937

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	8253	8258	8263	8268	8273
14	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
15	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.375
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	8278	8283	8288	8293	8298
18	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
19	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	8303	8308	8313	8318	8323
22	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
23	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	8328	8333	8338	8343	8348
26	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
27	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	8353	8358	8363	8368	8373
30	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
31	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	8378	8383	8388	8393	8398
34	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
35	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	8403	8408	8413	8418	8423
38	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
39	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	8428	8433	8438	8443	8448
42	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
43	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	8453	8458	8463	8468	8473
46	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
47	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	8478	8483	8488	8493	8498
50	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
51	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	8503	8508	8513	8518	8523
54	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
55	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	8528	8533	8538	8543	8548
58	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
59	Bulk Velocity (m/s)	2.375	2.374	2.374	2.374	2.374
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	8553	8558	8563	8568	8573
62	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
63	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	8578	8583	8588	8593	8598
66	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
67	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 594 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	8603	8608	8613	8618	8623
14	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
15	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	8628	8633	8638	8643	8648
18	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
19	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	8653	8658	8663	8668	8673
22	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
23	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	8678	8683	8688	8693	8698
26	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
27	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	8703	8708	8713	8718	8723
30	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
31	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	8728	8733	8738	8743	8748
34	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
35	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	8753	8758	8763	8768	8773
38	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
39	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	8778	8783	8788	8793	8798
42	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
43	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	8803	8805	8807	8809	8811
46	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
47	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	8813	8840	8867	8894	8921
50	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
51	Bulk Velocity (m/s)	2.374	2.373	2.373	2.373	2.373
52	Erosion Velocity (m/s)	10.95	10.95	10.94	10.94	10.94
53	Length (m)	8948	8949	8950	8951	8952
54	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
55	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
56	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
57	Length (m)	8953	8958	8963	8968	8973
58	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
59	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
60	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
61	Length (m)	8978	8983	8988	8993	8998
62	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
63	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
65	Length (m)	9003	9008	9013	9018	9023
66	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
67	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 595 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	9028	9033	9038	9043	9048
14	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
15	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
17	Length (m)	9053	9054	9055	9056	9057
18	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
19	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
21	Length (m)	9058	9065	9071	9078	9085
22	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
23	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
25	Length (m)	9091	9098	9105	9111	9118
26	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
27	Bulk Velocity (m/s)	2.373	2.373	2.372	2.372	2.372
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
29	Length (m)	9125	9131	9138	9145	9151
30	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
31	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
32	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
33	Length (m)	9158	9162	9166	9170	9174
34	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
35	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
36	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
37	Length (m)	9178	9183	9188	9193	9198
38	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
39	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
40	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
41	Length (m)	9203	9208	9213	9218	9223
42	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
43	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
44	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
45	Length (m)	9228	9233	9238	9243	9248
46	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
47	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
48	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
49	Length (m)	9253	9258	9263	9268	9273
50	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
51	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
52	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
53	Length (m)	9278	9283	9288	9293	9298
54	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
55	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
56	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
57	Length (m)	9303	9308	9313	9318	9323
58	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
59	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
60	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
61	Length (m)	9328	9333	9338	9343	9348
62	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
63	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
65	Length (m)	9353	9358	9363	9368	9373
66	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
67	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 596 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10	Erosion Table					
11						
12						
13	Length (m)	9378	9383	9388	9393	9398
14	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
15	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
17	Length (m)	9403	9408	9413	9418	9423
18	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
19	Bulk Velocity (m/s)	2.372	2.372	2.371	2.371	2.371
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
21	Length (m)	9428	9433	9438	9443	9448
22	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
23	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
25	Length (m)	9453	9456	9459	9462	9465
26	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
27	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
29	Length (m)	9468	9473	9477	9482	9487
30	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
31	Bulk Velocity (m/s)	2.372	2.372	2.371	2.371	2.371
32	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
33	Length (m)	9491	9496	9501	9505	9510
34	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
35	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
36	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
37	Length (m)	9515	9519	9524	9529	9533
38	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
39	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
40	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
41	Length (m)	9538	9543	9547	9552	9557
42	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
43	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
44	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
45	Length (m)	9561	9566	9571	9575	9580
46	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
47	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
48	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
49	Length (m)	9585	9589	9594	9599	9603
50	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
51	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
52	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
53	Length (m)	9608	9612	9616	9620	9624
54	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
55	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
56	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
57	Length (m)	9628	9633	9638	9643	9648
58	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
59	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
60	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
61	Length (m)	9653	9658	9663	9668	9673
62	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
63	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
65	Length (m)	9678	9683	9688	9693	9698
66	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
67	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 597 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3			Unit Set: Mixed				
4			Date/Time: Mon Jun 13 15:02:44 2022				
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	Erosion						
10							
11	Erosion Table						
12							
13	Length (m)	9703	9708	9713	9718	9723	
14	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4	
15	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371	
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
17	Length (m)	9728	9733	9738	9743	9748	
18	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4	
19	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371	
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
21	Length (m)	9753	9758	9763	9768	9773	
22	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4	
23	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371	
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
25	Length (m)	9778	9783	9788	9793	9798	
26	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4	
27	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371	
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
29	Length (m)	9803	9808	9813	9818	9823	
30	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4	
31	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371	
32	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
33	Length (m)	9828	9833	9838	9843	9848	
34	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4	
35	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371	
36	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
37	Length (m)	9853	9858	9863	9868	9873	
38	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4	
39	Bulk Velocity (m/s)	2.371	2.370	2.370	2.370	2.370	
40	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
41	Length (m)	9878	9883	9888	9893	9898	
42	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4	
43	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370	
44	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
45	Length (m)	9903	9905	9907	9909	9911	
46	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4	
47	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370	
48	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
49	Length (m)	9913	9933	9953	9973	9993	
50	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4	
51	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370	
52	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
53	Length (m)	1.001e+004	1.002e+004	1.002e+004	1.002e+004	1.003e+004	
54	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4	
55	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370	
56	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
57	Length (m)	1.003e+004	1.003e+004	1.004e+004	1.004e+004	1.005e+004	
58	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4	
59	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370	
60	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
61	Length (m)	1.005e+004	1.006e+004	1.006e+004	1.007e+004	1.007e+004	
62	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4	
63	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370	
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
65	Length (m)	1.008e+004	1.008e+004	1.009e+004	1.009e+004	1.010e+004	
66	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4	
67	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370	
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 598 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.010e+004	1.011e+004	1.011e+004	1.012e+004	1.012e+004
14	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
15	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
17	Length (m)	1.013e+004	1.013e+004	1.014e+004	1.014e+004	1.015e+004
18	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
19	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
21	Length (m)	1.015e+004	1.016e+004	1.016e+004	1.017e+004	1.017e+004
22	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
23	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
25	Length (m)	1.018e+004	1.018e+004	1.019e+004	1.019e+004	1.020e+004
26	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
27	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
29	Length (m)	1.020e+004	1.021e+004	1.021e+004	1.022e+004	1.022e+004
30	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
31	Bulk Velocity (m/s)	2.370	2.369	2.369	2.369	2.369
32	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
33	Length (m)	1.023e+004	1.023e+004	1.024e+004	1.024e+004	1.025e+004
34	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
35	Bulk Velocity (m/s)	2.369	2.369	2.369	2.369	2.369
36	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
37	Length (m)	1.025e+004	1.026e+004	1.026e+004	1.027e+004	1.027e+004
38	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.5
39	Bulk Velocity (m/s)	2.369	2.369	2.369	2.369	2.369
40	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
41	Length (m)	1.028e+004	1.028e+004	1.029e+004	1.029e+004	1.030e+004
42	Bulk Density (kg/m3)	124.5	124.5	124.5	124.4	124.4
43	Bulk Velocity (m/s)	2.369	2.369	2.369	2.369	2.369
44	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
45	Length (m)	1.030e+004	1.031e+004	1.031e+004	1.032e+004	1.032e+004
46	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.5
47	Bulk Velocity (m/s)	2.369	2.369	2.369	2.369	2.369
48	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
49	Length (m)	1.033e+004	1.033e+004	1.034e+004	1.034e+004	1.035e+004
50	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
51	Bulk Velocity (m/s)	2.369	2.369	2.369	2.369	2.369
52	Erosion Velocity (m/s)	10.94	10.94	10.94	10.93	10.93
53	Length (m)	1.035e+004	1.036e+004	1.036e+004	1.037e+004	1.037e+004
54	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
55	Bulk Velocity (m/s)	2.369	2.369	2.369	2.369	2.369
56	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
57	Length (m)	1.038e+004	1.038e+004	1.039e+004	1.039e+004	1.040e+004
58	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
59	Bulk Velocity (m/s)	2.369	2.369	2.369	2.369	2.369
60	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
61	Length (m)	1.040e+004	1.041e+004	1.041e+004	1.042e+004	1.042e+004
62	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
63	Bulk Velocity (m/s)	2.369	2.369	2.369	2.369	2.369
64	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
65	Length (m)	1.043e+004	1.043e+004	1.044e+004	1.044e+004	1.045e+004
66	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
67	Bulk Velocity (m/s)	2.369	2.369	2.369	2.369	2.369
68	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 599 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.045e+004	1.046e+004	1.046e+004	1.047e+004	1.047e+004
14	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
15	Bulk Velocity (m/s)	2.369	2.369	2.369	2.368	2.368
16	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
17	Length (m)	1.048e+004	1.048e+004	1.049e+004	1.049e+004	1.050e+004
18	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
19	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
20	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
21	Length (m)	1.050e+004	1.051e+004	1.051e+004	1.052e+004	1.052e+004
22	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
23	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
24	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
25	Length (m)	1.053e+004	1.053e+004	1.054e+004	1.054e+004	1.055e+004
26	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
27	Bulk Velocity (m/s)	2.369	2.369	2.369	2.368	2.368
28	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
29	Length (m)	1.055e+004	1.056e+004	1.056e+004	1.057e+004	1.057e+004
30	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
31	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
32	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
33	Length (m)	1.058e+004	1.058e+004	1.059e+004	1.059e+004	1.060e+004
34	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
35	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
36	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
37	Length (m)	1.060e+004	1.061e+004	1.061e+004	1.062e+004	1.062e+004
38	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
39	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
40	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
41	Length (m)	1.063e+004	1.063e+004	1.064e+004	1.064e+004	1.065e+004
42	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
43	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
44	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
45	Length (m)	1.065e+004	1.066e+004	1.066e+004	1.067e+004	1.067e+004
46	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
47	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
48	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
49	Length (m)	1.068e+004	1.068e+004	1.069e+004	1.069e+004	1.070e+004
50	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
51	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
52	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
53	Length (m)	1.070e+004	1.071e+004	1.071e+004	1.072e+004	1.072e+004
54	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
55	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
56	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
57	Length (m)	1.073e+004	1.073e+004	1.074e+004	1.074e+004	1.075e+004
58	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
59	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
60	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
61	Length (m)	1.075e+004	1.076e+004	1.076e+004	1.077e+004	1.077e+004
62	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
63	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
64	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
65	Length (m)	1.078e+004	1.078e+004	1.079e+004	1.079e+004	1.080e+004
66	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
67	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
68	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 600 of 937	

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.080e+004	1.081e+004	1.081e+004	1.082e+004	1.082e+004
14	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
15	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
16	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
17	Length (m)	1.083e+004	1.083e+004	1.084e+004	1.084e+004	1.085e+004
18	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
19	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
20	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
21	Length (m)	1.085e+004	1.086e+004	1.086e+004	1.087e+004	1.087e+004
22	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
23	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
24	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
25	Length (m)	1.088e+004	1.088e+004	1.089e+004	1.089e+004	1.090e+004
26	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
27	Bulk Velocity (m/s)	2.368	2.368	2.368	2.367	2.367
28	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
29	Length (m)	1.090e+004	1.091e+004	1.091e+004	1.092e+004	1.092e+004
30	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
31	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
32	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
33	Length (m)	1.093e+004	1.093e+004	1.094e+004	1.094e+004	1.095e+004
34	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
35	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
36	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
37	Length (m)	1.095e+004	1.096e+004	1.096e+004	1.097e+004	1.097e+004
38	Bulk Density (kg/m3)	124.5	124.5	124.5	124.6	124.6
39	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
40	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
41	Length (m)	1.098e+004	1.098e+004	1.099e+004	1.099e+004	1.099e+004
42	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
43	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
44	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
45	Length (m)	1.100e+004	1.100e+004	1.101e+004	1.102e+004	1.102e+004
46	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
47	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
48	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
49	Length (m)	1.103e+004	1.103e+004	1.104e+004	1.104e+004	1.105e+004
50	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
51	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
52	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
53	Length (m)	1.105e+004	1.106e+004	1.106e+004	1.107e+004	1.107e+004
54	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
55	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
56	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
57	Length (m)	1.108e+004	1.108e+004	1.109e+004	1.109e+004	1.110e+004
58	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
59	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
60	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
61	Length (m)	1.110e+004	1.111e+004	1.111e+004	1.112e+004	1.112e+004
62	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
63	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
64	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
65	Length (m)	1.113e+004	1.113e+004	1.114e+004	1.114e+004	1.115e+004
66	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
67	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
68	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 601 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.115e+004	1.116e+004	1.116e+004	1.117e+004	1.117e+004
14	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
15	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
16	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
17	Length (m)	1.118e+004	1.118e+004	1.119e+004	1.119e+004	1.120e+004
18	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
19	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.366
20	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
21	Length (m)	1.120e+004	1.121e+004	1.121e+004	1.122e+004	1.122e+004
22	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
23	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
24	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
25	Length (m)	1.123e+004	1.123e+004	1.124e+004	1.124e+004	1.125e+004
26	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
27	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
28	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
29	Length (m)	1.125e+004	1.126e+004	1.126e+004	1.127e+004	1.127e+004
30	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
31	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
32	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
33	Length (m)	1.128e+004	1.128e+004	1.129e+004	1.129e+004	1.130e+004
34	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
35	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
36	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
37	Length (m)	1.130e+004	1.131e+004	1.131e+004	1.132e+004	1.132e+004
38	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
39	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
40	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
41	Length (m)	1.133e+004	1.133e+004	1.134e+004	1.134e+004	1.135e+004
42	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
43	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
44	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
45	Length (m)	1.135e+004	1.136e+004	1.136e+004	1.137e+004	1.137e+004
46	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
47	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
48	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
49	Length (m)	1.138e+004	1.138e+004	1.139e+004	1.139e+004	1.140e+004
50	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
51	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
52	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
53	Length (m)	1.140e+004	1.141e+004	1.141e+004	1.142e+004	1.142e+004
54	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
55	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.365
56	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
57	Length (m)	1.143e+004	1.143e+004	1.144e+004	1.144e+004	1.145e+004
58	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
59	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
60	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
61	Length (m)	1.145e+004	1.146e+004	1.146e+004	1.147e+004	1.147e+004
62	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
63	Bulk Velocity (m/s)	2.365	2.365	2.366	2.366	2.366
64	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
65	Length (m)	1.148e+004	1.148e+004	1.149e+004	1.149e+004	1.150e+004
66	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
67	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
68	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 602 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.150e+004	1.151e+004	1.151e+004	1.152e+004	1.152e+004
14	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
15	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
16	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
17	Length (m)	1.153e+004	1.153e+004	1.154e+004	1.154e+004	1.155e+004
18	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
19	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
20	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
21	Length (m)	1.155e+004	1.156e+004	1.156e+004	1.157e+004	1.157e+004
22	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
23	Bulk Velocity (m/s)	2.366	2.365	2.365	2.365	2.365
24	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
25	Length (m)	1.158e+004	1.158e+004	1.159e+004	1.159e+004	1.160e+004
26	Bulk Density (kg/m3)	124.6	124.6	124.6	124.7	124.7
27	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
28	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
29	Length (m)	1.160e+004	1.161e+004	1.161e+004	1.162e+004	1.162e+004
30	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
31	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
32	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
33	Length (m)	1.163e+004	1.163e+004	1.164e+004	1.164e+004	1.165e+004
34	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
35	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
36	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
37	Length (m)	1.165e+004	1.166e+004	1.166e+004	1.167e+004	1.167e+004
38	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
39	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
40	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
41	Length (m)	1.168e+004	1.168e+004	1.169e+004	1.169e+004	1.170e+004
42	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
43	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
44	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
45	Length (m)	1.170e+004	1.171e+004	1.171e+004	1.172e+004	1.172e+004
46	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
47	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
48	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
49	Length (m)	1.173e+004	1.173e+004	1.174e+004	1.174e+004	1.175e+004
50	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
51	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
52	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
53	Length (m)	1.175e+004	1.176e+004	1.176e+004	1.177e+004	1.177e+004
54	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
55	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
56	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
57	Length (m)	1.178e+004	1.178e+004	1.179e+004	1.179e+004	1.180e+004
58	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
59	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
60	Erosion Velocity (m/s)	10.93	10.93	10.92	10.92	10.92
61	Length (m)	1.180e+004	1.181e+004	1.181e+004	1.182e+004	1.182e+004
62	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
63	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
64	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
65	Length (m)	1.183e+004	1.183e+004	1.184e+004	1.184e+004	1.185e+004
66	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
67	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
68	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 603 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.185e+004	1.186e+004	1.186e+004	1.187e+004	1.187e+004
14	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
15	Bulk Velocity (m/s)	2.365	2.365	2.365	2.364	2.364
16	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
17	Length (m)	1.188e+004	1.188e+004	1.189e+004	1.189e+004	1.190e+004
18	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
19	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
20	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
21	Length (m)	1.190e+004	1.191e+004	1.191e+004	1.192e+004	1.192e+004
22	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
23	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
24	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
25	Length (m)	1.193e+004	1.193e+004	1.194e+004	1.194e+004	1.194e+004
26	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
27	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
28	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
29	Length (m)	1.195e+004	1.196e+004	1.196e+004	1.197e+004	1.198e+004
30	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
31	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
32	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
33	Length (m)	1.198e+004	1.199e+004	1.200e+004	1.200e+004	1.201e+004
34	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
35	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
36	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
37	Length (m)	1.202e+004	1.203e+004	1.203e+004	1.204e+004	1.205e+004
38	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
39	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
40	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
41	Length (m)	1.205e+004	1.206e+004	1.207e+004	1.207e+004	1.208e+004
42	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
43	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
44	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
45	Length (m)	1.209e+004	1.210e+004	1.210e+004	1.211e+004	1.212e+004
46	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
47	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
48	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
49	Length (m)	1.212e+004	1.213e+004	1.214e+004	1.214e+004	1.215e+004
50	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
51	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
52	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
53	Length (m)	1.216e+004	1.217e+004	1.217e+004	1.218e+004	1.219e+004
54	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
55	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
56	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
57	Length (m)	1.219e+004	1.220e+004	1.221e+004	1.221e+004	1.222e+004
58	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.8
59	Bulk Velocity (m/s)	2.364	2.364	2.364	2.363	2.363
60	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
61	Length (m)	1.223e+004	1.223e+004	1.224e+004	1.225e+004	1.225e+004
62	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
63	Bulk Velocity (m/s)	2.363	2.363	2.363	2.363	2.363
64	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
65	Length (m)	1.226e+004	1.226e+004	1.227e+004	1.227e+004	1.228e+004
66	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
67	Bulk Velocity (m/s)	2.363	2.363	2.363	2.363	2.363
68	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 604 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.228e+004	1.229e+004	1.229e+004	1.230e+004	1.230e+004
14	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
15	Bulk Velocity (m/s)	2.363	2.363	2.363	2.363	2.363
16	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
17	Length (m)	1.231e+004	1.231e+004	1.232e+004	1.232e+004	1.233e+004
18	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
19	Bulk Velocity (m/s)	2.363	2.363	2.363	2.363	2.363
20	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
21	Length (m)	1.233e+004	1.234e+004	1.234e+004	1.235e+004	1.235e+004
22	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
23	Bulk Velocity (m/s)	2.363	2.363	2.363	2.363	2.363
24	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
25	Length (m)	1.236e+004	1.237e+004	1.237e+004	1.238e+004	1.239e+004
26	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
27	Bulk Velocity (m/s)	2.363	2.363	2.363	2.363	2.363
28	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
29	Length (m)	1.239e+004	1.240e+004	1.240e+004	1.241e+004	1.241e+004
30	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
31	Bulk Velocity (m/s)	2.363	2.363	2.363	2.363	2.363
32	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
33	Length (m)	1.242e+004	1.242e+004	1.243e+004	1.243e+004	1.244e+004
34	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
35	Bulk Velocity (m/s)	2.363	2.363	2.363	2.363	2.363
36	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
37	Length (m)	1.244e+004	1.245e+004	1.245e+004	1.246e+004	1.246e+004
38	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
39	Bulk Velocity (m/s)	2.363	2.363	2.363	2.363	2.363
40	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
41	Length (m)	1.247e+004	1.247e+004	1.248e+004	1.248e+004	1.249e+004
42	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
43	Bulk Velocity (m/s)	2.363	2.363	2.363	2.363	2.363
44	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
45	Length (m)	1.249e+004	1.250e+004	1.250e+004	1.251e+004	1.251e+004
46	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
47	Bulk Velocity (m/s)	2.363	2.363	2.362	2.362	2.362
48	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
49	Length (m)	1.252e+004	1.252e+004	1.253e+004	1.253e+004	1.254e+004
50	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
51	Bulk Velocity (m/s)	2.362	2.362	2.362	2.362	2.362
52	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
53	Length (m)	1.254e+004	1.255e+004	1.255e+004	1.256e+004	1.256e+004
54	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
55	Bulk Velocity (m/s)	2.362	2.362	2.362	2.362	2.362
56	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
57	Length (m)	1.257e+004	1.257e+004	1.258e+004	1.258e+004	1.259e+004
58	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
59	Bulk Velocity (m/s)	2.362	2.362	2.362	2.362	2.362
60	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
61	Length (m)	1.259e+004	1.260e+004	1.260e+004	1.261e+004	1.261e+004
62	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
63	Bulk Velocity (m/s)	2.362	2.362	2.362	2.362	2.362
64	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
65	Length (m)	1.262e+004	1.262e+004	1.263e+004	1.264e+004	1.265e+004
66	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
67	Bulk Velocity (m/s)	2.362	2.362	2.362	2.362	2.362
68	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 605 of 937	

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
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11	Erosion Table					
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13	Length (m)	1.265e+004	1.266e+004	1.266e+004	1.267e+004	1.267e+004
14	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
15	Bulk Velocity (m/s)	2.362	2.362	2.362	2.362	2.362
16	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
17	Length (m)	1.268e+004	1.268e+004	1.269e+004	1.269e+004	1.270e+004
18	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
19	Bulk Velocity (m/s)	2.362	2.362	2.362	2.362	2.362
20	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
21	Length (m)	1.270e+004	1.271e+004	1.271e+004	1.272e+004	1.272e+004
22	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
23	Bulk Velocity (m/s)	2.362	2.362	2.362	2.362	2.362
24	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
25	Length (m)	1.273e+004	1.273e+004	1.274e+004	1.274e+004	1.275e+004
26	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
27	Bulk Velocity (m/s)	2.362	2.362	2.362	2.362	2.362
28	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
29	Length (m)	1.275e+004	1.276e+004	1.276e+004	1.276e+004	1.277e+004
30	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
31	Bulk Velocity (m/s)	2.362	2.362	2.362	2.362	2.362
32	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
33	Length (m)	1.277e+004	1.278e+004	1.278e+004	1.278e+004	1.278e+004
34	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
35	Bulk Velocity (m/s)	2.362	2.362	2.362	2.362	2.362
36	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
37	Length (m)	1.279e+004	1.279e+004	1.279e+004	1.280e+004	1.280e+004
38	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
39	Bulk Velocity (m/s)	2.362	2.362	2.362	2.362	2.362
40	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
41	Length (m)	1.280e+004	1.281e+004	1.281e+004	1.282e+004	1.282e+004
42	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.9
43	Bulk Velocity (m/s)	2.362	2.362	2.362	2.362	2.362
44	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
45	Length (m)	1.283e+004	1.283e+004	1.284e+004	1.284e+004	1.285e+004
46	Bulk Density (kg/m3)	124.9	124.9	124.8	124.8	124.8
47	Bulk Velocity (m/s)	2.362	2.362	2.362	2.362	2.362
48	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
49	Length (m)	1.285e+004	1.286e+004	1.286e+004	1.287e+004	1.287e+004
50	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
51	Bulk Velocity (m/s)	2.362	2.362	2.362	2.362	2.362
52	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
53	Length (m)	1.288e+004	1.288e+004	1.289e+004	1.289e+004	1.290e+004
54	Bulk Density (kg/m3)	124.8	124.9	124.9	124.9	124.9
55	Bulk Velocity (m/s)	2.362	2.362	2.362	2.362	2.362
56	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
57	Length (m)	1.290e+004	1.291e+004	1.291e+004	1.292e+004	1.292e+004
58	Bulk Density (kg/m3)	124.9	124.9	124.9	124.9	124.9
59	Bulk Velocity (m/s)	2.362	2.362	2.361	2.361	2.361
60	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
61	Length (m)	1.293e+004	1.293e+004	1.294e+004	1.294e+004	1.295e+004
62	Bulk Density (kg/m3)	124.9	124.9	124.9	124.9	124.9
63	Bulk Velocity (m/s)	2.361	2.361	2.361	2.361	2.361
64	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
65	Length (m)	1.295e+004	1.296e+004	1.296e+004	1.297e+004	1.297e+004
66	Bulk Density (kg/m3)	124.9	124.9	124.9	124.9	124.9
67	Bulk Velocity (m/s)	2.361	2.361	2.361	2.361	2.361
68	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.298e+004	1.298e+004	1.299e+004	1.299e+004	1.300e+004
14	Bulk Density (kg/m3)	124.9	124.9	124.9	124.9	124.9
15	Bulk Velocity (m/s)	2.361	2.361	2.361	2.361	2.361
16	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
17	Length (m)	1.300e+004	1.301e+004	1.301e+004	1.302e+004	1.302e+004
18	Bulk Density (kg/m3)	124.9	124.9	124.9	124.9	124.9
19	Bulk Velocity (m/s)	2.361	2.361	2.361	2.361	2.361
20	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
21	Length (m)	1.303e+004	1.303e+004	1.304e+004	1.304e+004	1.305e+004
22	Bulk Density (kg/m3)	124.9	124.9	124.9	124.9	124.9
23	Bulk Velocity (m/s)	2.361	2.361	2.361	2.361	2.361
24	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
25	Length (m)	1.305e+004	1.306e+004	1.306e+004	1.307e+004	1.307e+004
26	Bulk Density (kg/m3)	124.9	124.9	124.9	124.9	124.9
27	Bulk Velocity (m/s)	2.361	2.361	2.361	2.361	2.361
28	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
29	Length (m)	1.308e+004	1.308e+004	1.309e+004	1.309e+004	1.310e+004
30	Bulk Density (kg/m3)	124.9	124.9	124.9	124.9	124.9
31	Bulk Velocity (m/s)	2.361	2.361	2.361	2.361	2.361
32	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
33	Length (m)	1.310e+004	1.311e+004	1.311e+004	1.312e+004	1.312e+004
34	Bulk Density (kg/m3)	124.9	124.9	124.9	124.9	124.9
35	Bulk Velocity (m/s)	2.361	2.361	2.361	2.361	2.361
36	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
37	Length (m)	1.313e+004	1.313e+004	1.314e+004	1.314e+004	1.315e+004
38	Bulk Density (kg/m3)	124.9	124.9	124.9	124.9	124.9
39	Bulk Velocity (m/s)	2.361	2.361	2.361	2.361	2.361
40	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
41	Length (m)	1.315e+004	1.316e+004	1.316e+004	1.317e+004	1.317e+004
42	Bulk Density (kg/m3)	124.9	124.9	124.9	124.9	124.9
43	Bulk Velocity (m/s)	2.361	2.361	2.361	2.361	2.361
44	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
45	Length (m)	1.318e+004	1.318e+004	1.319e+004	1.319e+004	1.320e+004
46	Bulk Density (kg/m3)	124.9	124.9	124.9	124.9	124.9
47	Bulk Velocity (m/s)	2.361	2.361	2.361	2.361	2.361
48	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
49	Length (m)	1.320e+004	1.321e+004	1.321e+004	1.322e+004	1.322e+004
50	Bulk Density (kg/m3)	124.9	124.9	124.9	124.9	124.9
51	Bulk Velocity (m/s)	2.361	2.361	2.360	2.360	2.360
52	Erosion Velocity (m/s)	10.92	10.92	10.91	10.91	10.91
53	Length (m)	1.323e+004	1.323e+004	1.324e+004	1.324e+004	1.325e+004
54	Bulk Density (kg/m3)	124.9	124.9	124.9	124.9	124.9
55	Bulk Velocity (m/s)	2.360	2.360	2.360	2.360	2.360
56	Erosion Velocity (m/s)	10.91	10.91	10.91	10.91	10.91
57	Length (m)	1.325e+004	1.326e+004	1.326e+004	1.327e+004	1.327e+004
58	Bulk Density (kg/m3)	124.9	125.0	125.0	125.0	125.0
59	Bulk Velocity (m/s)	2.360	2.360	2.359	2.359	2.359
60	Erosion Velocity (m/s)	10.91	10.91	10.91	10.91	10.91
61	Length (m)	1.328e+004	1.328e+004	1.329e+004	1.329e+004	1.330e+004
62	Bulk Density (kg/m3)	125.0	125.0	125.0	125.0	125.0
63	Bulk Velocity (m/s)	2.359	2.359	2.359	2.359	2.358
64	Erosion Velocity (m/s)	10.91	10.91	10.91	10.91	10.91
65	Length (m)	1.330e+004	1.331e+004	1.331e+004	1.332e+004	1.332e+004
66	Bulk Density (kg/m3)	125.0	125.0	125.0	125.0	125.1
67	Bulk Velocity (m/s)	2.358	2.358	2.358	2.358	2.358
68	Erosion Velocity (m/s)	10.91	10.91	10.91	10.91	10.91
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 607 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.333e+004	1.333e+004	1.334e+004	1.334e+004	1.335e+004
14	Bulk Density (kg/m3)	125.1	125.1	125.1	125.1	125.1
15	Bulk Velocity (m/s)	2.358	2.357	2.357	2.357	2.357
16	Erosion Velocity (m/s)	10.91	10.91	10.91	10.91	10.91
17	Length (m)	1.335e+004	1.336e+004	1.336e+004	1.337e+004	1.337e+004
18	Bulk Density (kg/m3)	125.1	125.1	125.1	125.1	125.1
19	Bulk Velocity (m/s)	2.357	2.357	2.356	2.356	2.356
20	Erosion Velocity (m/s)	10.91	10.91	10.91	10.91	10.90
21	Length (m)	1.338e+004	1.338e+004	1.339e+004	1.339e+004	1.340e+004
22	Bulk Density (kg/m3)	125.1	125.2	125.2	125.2	125.2
23	Bulk Velocity (m/s)	2.356	2.356	2.356	2.356	2.356
24	Erosion Velocity (m/s)	10.90	10.90	10.90	10.90	10.90
25	Length (m)	1.340e+004	1.341e+004	1.341e+004	1.342e+004	1.342e+004
26	Bulk Density (kg/m3)	125.2	125.1	125.1	125.1	125.1
27	Bulk Velocity (m/s)	2.356	2.356	2.357	2.357	2.358
28	Erosion Velocity (m/s)	10.90	10.91	10.91	10.91	10.91
29	Length (m)	1.343e+004	1.343e+004	1.344e+004	1.344e+004	1.345e+004
30	Bulk Density (kg/m3)	125.0	125.0	125.0	125.0	125.0
31	Bulk Velocity (m/s)	2.358	2.358	2.359	2.359	2.359
32	Erosion Velocity (m/s)	10.91	10.91	10.91	10.91	10.91
33	Length (m)	1.345e+004	1.346e+004	1.346e+004	1.347e+004	1.347e+004
34	Bulk Density (kg/m3)	125.0	124.8	124.7	124.5	124.4
35	Bulk Velocity (m/s)	2.359	2.362	2.365	2.368	2.371
36	Erosion Velocity (m/s)	10.91	10.92	10.93	10.93	10.94
37	Length (m)	1.348e+004	1.348e+004	1.349e+004	1.349e+004	1.350e+004
38	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
39	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	1.350e+004	1.351e+004	1.351e+004	1.352e+004	1.352e+004
42	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
43	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	1.353e+004	1.353e+004	1.353e+004	1.353e+004	1.354e+004
46	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
47	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	1.354e+004	1.354e+004	1.354e+004	1.355e+004	1.355e+004
50	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
51	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	1.355e+004	1.356e+004	1.356e+004	1.356e+004	1.357e+004
54	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
55	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	1.357e+004	1.357e+004	1.358e+004	1.358e+004	1.358e+004
58	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
59	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	1.359e+004	1.359e+004	1.359e+004	1.360e+004	1.360e+004
62	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
63	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	1.360e+004	1.361e+004	1.361e+004	1.362e+004	1.362e+004
66	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
67	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 608 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.363e+004	1.363e+004	1.364e+004	1.364e+004	1.365e+004
14	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
15	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	1.365e+004	1.366e+004	1.366e+004	1.367e+004	1.367e+004
18	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
19	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	1.368e+004	1.368e+004	1.369e+004	1.369e+004	1.370e+004
22	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
23	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	1.370e+004	1.371e+004	1.371e+004	1.372e+004	1.372e+004
26	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
27	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	1.373e+004	1.373e+004	1.374e+004	1.374e+004	1.375e+004
30	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
31	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	1.375e+004	1.376e+004	1.376e+004	1.377e+004	1.377e+004
34	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
35	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	1.378e+004	1.378e+004	1.379e+004	1.379e+004	1.380e+004
38	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
39	Bulk Velocity (m/s)	2.374	2.374	2.373	2.373	2.373
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	1.380e+004	1.381e+004	1.381e+004	1.382e+004	1.382e+004
42	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
43	Bulk Velocity (m/s)	2.373	2.373	2.373	2.374	2.374
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	1.383e+004	1.383e+004	1.384e+004	1.384e+004	1.385e+004
46	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
47	Bulk Velocity (m/s)	2.374	2.373	2.373	2.373	2.373
48	Erosion Velocity (m/s)	10.95	10.95	10.94	10.94	10.94
49	Length (m)	1.385e+004	1.386e+004	1.386e+004	1.387e+004	1.387e+004
50	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
51	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
52	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
53	Length (m)	1.388e+004	1.388e+004	1.389e+004	1.389e+004	1.390e+004
54	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
55	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
56	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
57	Length (m)	1.390e+004	1.391e+004	1.391e+004	1.392e+004	1.392e+004
58	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
59	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
60	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
61	Length (m)	1.393e+004	1.393e+004	1.394e+004	1.394e+004	1.395e+004
62	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
63	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
65	Length (m)	1.395e+004	1.396e+004	1.396e+004	1.397e+004	1.397e+004
66	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
67	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 609 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.398e+004	1.398e+004	1.399e+004	1.399e+004	1.400e+004
14	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
15	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
17	Length (m)	1.400e+004	1.401e+004	1.401e+004	1.402e+004	1.402e+004
18	Bulk Density (kg/m3)	124.2	124.3	124.3	124.3	124.3
19	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
21	Length (m)	1.403e+004	1.403e+004	1.404e+004	1.404e+004	1.405e+004
22	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
23	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
25	Length (m)	1.405e+004	1.405e+004	1.406e+004	1.406e+004	1.406e+004
26	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
27	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
29	Length (m)	1.406e+004	1.406e+004	1.407e+004	1.407e+004	1.408e+004
30	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
31	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
32	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
33	Length (m)	1.408e+004	1.408e+004	1.409e+004	1.409e+004	1.409e+004
34	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
35	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
36	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
37	Length (m)	1.409e+004	1.409e+004	1.410e+004	1.411e+004	1.411e+004
38	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
39	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
40	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
41	Length (m)	1.412e+004	1.413e+004	1.413e+004	1.414e+004	1.415e+004
42	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
43	Bulk Velocity (m/s)	2.373	2.373	2.372	2.372	2.372
44	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
45	Length (m)	1.415e+004	1.416e+004	1.416e+004	1.417e+004	1.417e+004
46	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
47	Bulk Velocity (m/s)	2.372	2.372	2.372	2.373	2.373
48	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
49	Length (m)	1.418e+004	1.418e+004	1.419e+004	1.419e+004	1.420e+004
50	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
51	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
52	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
53	Length (m)	1.420e+004	1.421e+004	1.421e+004	1.422e+004	1.422e+004
54	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
55	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
56	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
57	Length (m)	1.423e+004	1.423e+004	1.424e+004	1.424e+004	1.425e+004
58	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
59	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
60	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
61	Length (m)	1.425e+004	1.426e+004	1.426e+004	1.427e+004	1.427e+004
62	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
63	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
65	Length (m)	1.428e+004	1.428e+004	1.429e+004	1.429e+004	1.430e+004
66	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
67	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 610 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.430e+004	1.431e+004	1.431e+004	1.432e+004	1.432e+004
14	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
15	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
17	Length (m)	1.433e+004	1.433e+004	1.434e+004	1.434e+004	1.435e+004
18	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
19	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
21	Length (m)	1.435e+004	1.436e+004	1.436e+004	1.437e+004	1.437e+004
22	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
23	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
25	Length (m)	1.438e+004	1.438e+004	1.439e+004	1.439e+004	1.440e+004
26	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
27	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
29	Length (m)	1.440e+004	1.441e+004	1.441e+004	1.442e+004	1.442e+004
30	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
31	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
32	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
33	Length (m)	1.443e+004	1.443e+004	1.444e+004	1.444e+004	1.445e+004
34	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
35	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
36	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
37	Length (m)	1.445e+004	1.446e+004	1.446e+004	1.447e+004	1.447e+004
38	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
39	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
40	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
41	Length (m)	1.448e+004	1.448e+004	1.449e+004	1.449e+004	1.450e+004
42	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
43	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
44	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
45	Length (m)	1.450e+004	1.451e+004	1.451e+004	1.452e+004	1.452e+004
46	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
47	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
48	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
49	Length (m)	1.453e+004	1.453e+004	1.454e+004	1.454e+004	1.455e+004
50	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
51	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
52	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
53	Length (m)	1.455e+004	1.456e+004	1.456e+004	1.457e+004	1.457e+004
54	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
55	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.371
56	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
57	Length (m)	1.458e+004	1.458e+004	1.459e+004	1.459e+004	1.460e+004
58	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
59	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
60	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
61	Length (m)	1.460e+004	1.461e+004	1.461e+004	1.462e+004	1.462e+004
62	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
63	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
65	Length (m)	1.463e+004	1.463e+004	1.464e+004	1.464e+004	1.465e+004
66	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
67	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 611 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.465e+004	1.466e+004	1.466e+004	1.467e+004	1.467e+004
14	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
15	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
17	Length (m)	1.468e+004	1.468e+004	1.469e+004	1.469e+004	1.470e+004
18	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
19	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
21	Length (m)	1.470e+004	1.471e+004	1.471e+004	1.472e+004	1.472e+004
22	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
23	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
25	Length (m)	1.473e+004	1.473e+004	1.474e+004	1.474e+004	1.475e+004
26	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
27	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
29	Length (m)	1.475e+004	1.476e+004	1.476e+004	1.477e+004	1.477e+004
30	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
31	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
32	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
33	Length (m)	1.478e+004	1.478e+004	1.479e+004	1.479e+004	1.480e+004
34	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
35	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
36	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
37	Length (m)	1.480e+004	1.481e+004	1.481e+004	1.482e+004	1.482e+004
38	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
39	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
40	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
41	Length (m)	1.483e+004	1.483e+004	1.484e+004	1.484e+004	1.485e+004
42	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
43	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
44	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
45	Length (m)	1.485e+004	1.486e+004	1.486e+004	1.486e+004	1.486e+004
46	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
47	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
48	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
49	Length (m)	1.486e+004	1.488e+004	1.490e+004	1.492e+004	1.494e+004
50	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
51	Bulk Velocity (m/s)	2.371	2.370	2.370	2.370	2.370
52	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
53	Length (m)	1.496e+004	1.498e+004	1.500e+004	1.502e+004	1.504e+004
54	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
55	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370
56	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
57	Length (m)	1.506e+004	1.508e+004	1.510e+004	1.512e+004	1.514e+004
58	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
59	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370
60	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
61	Length (m)	1.516e+004	1.518e+004	1.520e+004	1.522e+004	1.524e+004
62	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
63	Bulk Velocity (m/s)	2.370	2.370	2.370	2.369	2.369
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
65	Length (m)	1.526e+004	1.528e+004	1.530e+004	1.532e+004	1.534e+004
66	Bulk Density (kg/m3)	124.4	124.4	124.5	124.5	124.5
67	Bulk Velocity (m/s)	2.369	2.369	2.369	2.369	2.369
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.93	10.93
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 612 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.536e+004	1.538e+004	1.540e+004	1.542e+004	1.544e+004
14	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
15	Bulk Velocity (m/s)	2.369	2.369	2.368	2.368	2.368
16	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
17	Length (m)	1.546e+004	1.548e+004	1.550e+004	1.552e+004	1.554e+004
18	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
19	Bulk Velocity (m/s)	2.367	2.368	2.368	2.369	2.369
20	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
21	Length (m)	1.556e+004	1.558e+004	1.560e+004	1.562e+004	1.564e+004
22	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
23	Bulk Velocity (m/s)	2.369	2.369	2.369	2.369	2.369
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
25	Length (m)	1.566e+004	1.568e+004	1.570e+004	1.572e+004	1.574e+004
26	Bulk Density (kg/m3)	124.4	124.5	124.5	124.5	124.5
27	Bulk Velocity (m/s)	2.369	2.369	2.369	2.369	2.369
28	Erosion Velocity (m/s)	10.94	10.94	10.93	10.93	10.93
29	Length (m)	1.576e+004	1.577e+004	1.577e+004	1.578e+004	1.578e+004
30	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
31	Bulk Velocity (m/s)	2.369	2.369	2.369	2.369	2.369
32	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
33	Length (m)	1.579e+004	1.581e+004	1.583e+004	1.585e+004	1.587e+004
34	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
35	Bulk Velocity (m/s)	2.369	2.369	2.369	2.369	2.369
36	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
37	Length (m)	1.589e+004	1.589e+004	1.590e+004	1.590e+004	1.590e+004
38	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
39	Bulk Velocity (m/s)	2.369	2.369	2.369	2.369	2.369
40	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
41	Length (m)	1.591e+004	1.591e+004	1.592e+004	1.592e+004	1.592e+004
42	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
43	Bulk Velocity (m/s)	2.369	2.368	2.368	2.368	2.368
44	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
45	Length (m)	1.593e+004	1.593e+004	1.593e+004	1.593e+004	1.594e+004
46	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
47	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
48	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
49	Length (m)	1.594e+004	1.594e+004	1.595e+004	1.595e+004	1.595e+004
50	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
51	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
52	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
53	Length (m)	1.596e+004	1.596e+004	1.597e+004	1.597e+004	1.598e+004
54	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
55	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
56	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
57	Length (m)	1.598e+004	1.599e+004	1.599e+004	1.600e+004	1.600e+004
58	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
59	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
60	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
61	Length (m)	1.601e+004	1.601e+004	1.602e+004	1.602e+004	1.603e+004
62	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
63	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
64	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
65	Length (m)	1.603e+004	1.604e+004	1.604e+004	1.605e+004	1.605e+004
66	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
67	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
68	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 613 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.606e+004	1.606e+004	1.607e+004	1.607e+004	1.608e+004
14	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
15	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
16	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
17	Length (m)	1.608e+004	1.609e+004	1.609e+004	1.610e+004	1.610e+004
18	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
19	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
20	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
21	Length (m)	1.611e+004	1.611e+004	1.612e+004	1.612e+004	1.613e+004
22	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
23	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
24	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
25	Length (m)	1.613e+004	1.614e+004	1.614e+004	1.615e+004	1.615e+004
26	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
27	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
28	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
29	Length (m)	1.616e+004	1.616e+004	1.617e+004	1.617e+004	1.618e+004
30	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
31	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
32	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
33	Length (m)	1.618e+004	1.619e+004	1.619e+004	1.620e+004	1.620e+004
34	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
35	Bulk Velocity (m/s)	2.368	2.367	2.367	2.367	2.367
36	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
37	Length (m)	1.621e+004	1.621e+004	1.622e+004	1.622e+004	1.623e+004
38	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
39	Bulk Velocity (m/s)	2.367	2.367	2.368	2.368	2.368
40	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
41	Length (m)	1.623e+004	1.624e+004	1.624e+004	1.625e+004	1.625e+004
42	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
43	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
44	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
45	Length (m)	1.626e+004	1.626e+004	1.627e+004	1.627e+004	1.628e+004
46	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
47	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
48	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
49	Length (m)	1.628e+004	1.629e+004	1.629e+004	1.630e+004	1.630e+004
50	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
51	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
52	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
53	Length (m)	1.631e+004	1.631e+004	1.631e+004	1.631e+004	1.632e+004
54	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
55	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
56	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
57	Length (m)	1.632e+004	1.632e+004	1.632e+004	1.633e+004	1.633e+004
58	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
59	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
60	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
61	Length (m)	1.633e+004	1.634e+004	1.634e+004	1.635e+004	1.635e+004
62	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
63	Bulk Velocity (m/s)	2.368	2.368	2.368	2.367	2.367
64	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
65	Length (m)	1.636e+004	1.636e+004	1.637e+004	1.637e+004	1.638e+004
66	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
67	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
68	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 614 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.638e+004	1.639e+004	1.639e+004	1.640e+004	1.640e+004
14	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
15	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
16	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
17	Length (m)	1.641e+004	1.641e+004	1.642e+004	1.642e+004	1.643e+004
18	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
19	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
20	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
21	Length (m)	1.643e+004	1.644e+004	1.644e+004	1.645e+004	1.645e+004
22	Bulk Density (kg/m3)	124.5	124.6	124.6	124.6	124.6
23	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
24	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
25	Length (m)	1.646e+004	1.646e+004	1.647e+004	1.647e+004	1.648e+004
26	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
27	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
28	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
29	Length (m)	1.648e+004	1.649e+004	1.649e+004	1.650e+004	1.650e+004
30	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
31	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
32	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
33	Length (m)	1.651e+004	1.651e+004	1.652e+004	1.652e+004	1.653e+004
34	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
35	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
36	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
37	Length (m)	1.653e+004	1.654e+004	1.654e+004	1.655e+004	1.655e+004
38	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
39	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
40	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
41	Length (m)	1.656e+004	1.656e+004	1.657e+004	1.657e+004	1.658e+004
42	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
43	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
44	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
45	Length (m)	1.658e+004	1.659e+004	1.659e+004	1.660e+004	1.660e+004
46	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
47	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
48	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
49	Length (m)	1.661e+004	1.661e+004	1.662e+004	1.662e+004	1.663e+004
50	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
51	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
52	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
53	Length (m)	1.663e+004	1.664e+004	1.664e+004	1.665e+004	1.665e+004
54	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
55	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
56	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
57	Length (m)	1.666e+004	1.666e+004	1.667e+004	1.667e+004	1.668e+004
58	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
59	Bulk Velocity (m/s)	2.367	2.367	2.367	2.367	2.367
60	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
61	Length (m)	1.668e+004	1.669e+004	1.669e+004	1.670e+004	1.670e+004
62	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
63	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
64	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
65	Length (m)	1.671e+004	1.671e+004	1.672e+004	1.672e+004	1.673e+004
66	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
67	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
68	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 615 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.673e+004	1.674e+004	1.674e+004	1.675e+004	1.675e+004
14	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
15	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
16	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
17	Length (m)	1.676e+004	1.676e+004	1.677e+004	1.677e+004	1.678e+004
18	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
19	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
20	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
21	Length (m)	1.678e+004	1.679e+004	1.679e+004	1.680e+004	1.680e+004
22	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
23	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
24	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
25	Length (m)	1.681e+004	1.681e+004	1.682e+004	1.682e+004	1.683e+004
26	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
27	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
28	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
29	Length (m)	1.683e+004	1.684e+004	1.684e+004	1.685e+004	1.685e+004
30	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
31	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
32	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
33	Length (m)	1.686e+004	1.686e+004	1.687e+004	1.687e+004	1.688e+004
34	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
35	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
36	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
37	Length (m)	1.688e+004	1.689e+004	1.689e+004	1.690e+004	1.690e+004
38	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
39	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
40	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
41	Length (m)	1.690e+004	1.691e+004	1.691e+004	1.692e+004	1.692e+004
42	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
43	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
44	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
45	Length (m)	1.693e+004	1.693e+004	1.694e+004	1.694e+004	1.695e+004
46	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
47	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
48	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
49	Length (m)	1.695e+004	1.696e+004	1.696e+004	1.697e+004	1.697e+004
50	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
51	Bulk Velocity (m/s)	2.366	2.366	2.366	2.366	2.366
52	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
53	Length (m)	1.698e+004	1.698e+004	1.699e+004	1.699e+004	1.700e+004
54	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
55	Bulk Velocity (m/s)	2.366	2.366	2.365	2.365	2.365
56	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
57	Length (m)	1.700e+004	1.701e+004	1.701e+004	1.702e+004	1.702e+004
58	Bulk Density (kg/m3)	124.6	124.6	124.6	124.6	124.6
59	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
60	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
61	Length (m)	1.703e+004	1.703e+004	1.704e+004	1.704e+004	1.705e+004
62	Bulk Density (kg/m3)	124.6	124.6	124.7	124.7	124.7
63	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
64	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
65	Length (m)	1.705e+004	1.706e+004	1.706e+004	1.707e+004	1.707e+004
66	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
67	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
68	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 616 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.708e+004	1.708e+004	1.709e+004	1.709e+004	1.710e+004
14	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
15	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
16	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
17	Length (m)	1.710e+004	1.711e+004	1.711e+004	1.712e+004	1.712e+004
18	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
19	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
20	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
21	Length (m)	1.713e+004	1.713e+004	1.714e+004	1.714e+004	1.715e+004
22	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
23	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
24	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
25	Length (m)	1.715e+004	1.716e+004	1.716e+004	1.717e+004	1.717e+004
26	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
27	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
28	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
29	Length (m)	1.718e+004	1.718e+004	1.719e+004	1.719e+004	1.720e+004
30	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
31	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
32	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
33	Length (m)	1.720e+004	1.721e+004	1.721e+004	1.722e+004	1.722e+004
34	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
35	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
36	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
37	Length (m)	1.723e+004	1.723e+004	1.724e+004	1.724e+004	1.725e+004
38	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
39	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
40	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
41	Length (m)	1.725e+004	1.726e+004	1.726e+004	1.727e+004	1.727e+004
42	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
43	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
44	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
45	Length (m)	1.728e+004	1.728e+004	1.729e+004	1.729e+004	1.730e+004
46	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
47	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
48	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
49	Length (m)	1.730e+004	1.731e+004	1.731e+004	1.732e+004	1.732e+004
50	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
51	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
52	Erosion Velocity (m/s)	10.93	10.93	10.93	10.92	10.92
53	Length (m)	1.733e+004	1.733e+004	1.734e+004	1.734e+004	1.735e+004
54	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
55	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
56	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
57	Length (m)	1.735e+004	1.736e+004	1.736e+004	1.737e+004	1.737e+004
58	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
59	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
60	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
61	Length (m)	1.738e+004	1.738e+004	1.739e+004	1.739e+004	1.740e+004
62	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
63	Bulk Velocity (m/s)	2.365	2.365	2.365	2.365	2.365
64	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
65	Length (m)	1.740e+004	1.741e+004	1.741e+004	1.742e+004	1.742e+004
66	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
67	Bulk Velocity (m/s)	2.365	2.364	2.364	2.364	2.364
68	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 617 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.743e+004	1.743e+004	1.744e+004	1.744e+004	1.745e+004
14	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
15	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
16	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
17	Length (m)	1.745e+004	1.746e+004	1.746e+004	1.747e+004	1.747e+004
18	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
19	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
20	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
21	Length (m)	1.748e+004	1.748e+004	1.749e+004	1.749e+004	1.750e+004
22	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
23	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
24	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
25	Length (m)	1.750e+004	1.751e+004	1.751e+004	1.752e+004	1.752e+004
26	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
27	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
28	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
29	Length (m)	1.753e+004	1.753e+004	1.754e+004	1.754e+004	1.755e+004
30	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
31	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
32	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
33	Length (m)	1.755e+004	1.756e+004	1.756e+004	1.757e+004	1.757e+004
34	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
35	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
36	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
37	Length (m)	1.758e+004	1.758e+004	1.759e+004	1.759e+004	1.760e+004
38	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
39	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
40	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
41	Length (m)	1.760e+004	1.761e+004	1.761e+004	1.762e+004	1.762e+004
42	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
43	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
44	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
45	Length (m)	1.763e+004	1.763e+004	1.764e+004	1.764e+004	1.765e+004
46	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
47	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
48	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
49	Length (m)	1.765e+004	1.766e+004	1.766e+004	1.767e+004	1.767e+004
50	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
51	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
52	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
53	Length (m)	1.768e+004	1.768e+004	1.769e+004	1.769e+004	1.770e+004
54	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
55	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
56	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
57	Length (m)	1.770e+004	1.771e+004	1.771e+004	1.772e+004	1.772e+004
58	Bulk Density (kg/m3)	124.7	124.7	124.7	124.7	124.7
59	Bulk Velocity (m/s)	2.364	2.364	2.364	2.364	2.364
60	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
61	Length (m)	1.773e+004	1.773e+004	1.774e+004	1.774e+004	1.775e+004
62	Bulk Density (kg/m3)	124.7	124.7	124.7	124.8	124.8
63	Bulk Velocity (m/s)	2.364	2.364	2.364	2.363	2.363
64	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
65	Length (m)	1.775e+004	1.776e+004	1.776e+004	1.777e+004	1.777e+004
66	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
67	Bulk Velocity (m/s)	2.363	2.363	2.363	2.363	2.363
68	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 618 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.778e+004	1.778e+004	1.779e+004	1.779e+004	1.780e+004
14	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
15	Bulk Velocity (m/s)	2.363	2.363	2.363	2.363	2.363
16	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
17	Length (m)	1.780e+004	1.781e+004	1.781e+004	1.782e+004	1.782e+004
18	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
19	Bulk Velocity (m/s)	2.363	2.363	2.363	2.363	2.363
20	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
21	Length (m)	1.783e+004	1.783e+004	1.784e+004	1.784e+004	1.785e+004
22	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
23	Bulk Velocity (m/s)	2.363	2.363	2.363	2.363	2.363
24	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
25	Length (m)	1.785e+004	1.785e+004	1.786e+004	1.786e+004	1.786e+004
26	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
27	Bulk Velocity (m/s)	2.363	2.363	2.363	2.363	2.363
28	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
29	Length (m)	1.787e+004	1.787e+004	1.787e+004	1.787e+004	1.787e+004
30	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
31	Bulk Velocity (m/s)	2.363	2.363	2.363	2.363	2.363
32	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
33	Length (m)	1.788e+004	1.788e+004	1.789e+004	1.789e+004	1.790e+004
34	Bulk Density (kg/m3)	124.8	124.8	124.8	124.8	124.8
35	Bulk Velocity (m/s)	2.363	2.362	2.362	2.362	2.362
36	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
37	Length (m)	1.790e+004	1.790e+004	1.791e+004	1.791e+004	1.791e+004
38	Bulk Density (kg/m3)	124.8	124.8	124.8	124.9	124.9
39	Bulk Velocity (m/s)	2.362	2.362	2.362	2.362	2.361
40	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
41	Length (m)	1.791e+004	1.792e+004	1.792e+004	1.792e+004	1.793e+004
42	Bulk Density (kg/m3)	124.9	124.9	124.9	124.9	124.9
43	Bulk Velocity (m/s)	2.361	2.361	2.361	2.361	2.361
44	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
45	Length (m)	1.793e+004	1.793e+004	1.794e+004	1.794e+004	1.794e+004
46	Bulk Density (kg/m3)	124.9	124.9	124.9	124.9	124.9
47	Bulk Velocity (m/s)	2.361	2.361	2.361	2.361	2.361
48	Erosion Velocity (m/s)	10.92	10.92	10.92	10.92	10.92
49	Length (m)	1.795e+004	1.795e+004	1.796e+004	1.796e+004	1.797e+004
50	Bulk Density (kg/m3)	124.9	124.9	124.9	124.9	124.9
51	Bulk Velocity (m/s)	2.360	2.360	2.360	2.360	2.360
52	Erosion Velocity (m/s)	10.91	10.91	10.91	10.91	10.91
53	Length (m)	1.798e+004	1.798e+004	1.799e+004	1.799e+004	1.800e+004
54	Bulk Density (kg/m3)	124.9	125.0	125.0	125.0	125.0
55	Bulk Velocity (m/s)	2.360	2.360	2.359	2.359	2.359
56	Erosion Velocity (m/s)	10.91	10.91	10.91	10.91	10.91
57	Length (m)	1.800e+004	1.801e+004	1.801e+004	1.802e+004	1.802e+004
58	Bulk Density (kg/m3)	125.0	125.0	125.0	125.0	125.0
59	Bulk Velocity (m/s)	2.359	2.359	2.359	2.358	2.358
60	Erosion Velocity (m/s)	10.91	10.91	10.91	10.91	10.91
61	Length (m)	1.803e+004	1.803e+004	1.804e+004	1.804e+004	1.805e+004
62	Bulk Density (kg/m3)	125.0	125.0	125.0	125.0	125.0
63	Bulk Velocity (m/s)	2.358	2.358	2.359	2.359	2.359
64	Erosion Velocity (m/s)	10.91	10.91	10.91	10.91	10.91
65	Length (m)	1.805e+004	1.806e+004	1.806e+004	1.807e+004	1.807e+004
66	Bulk Density (kg/m3)	125.0	124.8	124.5	124.3	124.1
67	Bulk Velocity (m/s)	2.359	2.363	2.368	2.372	2.376
68	Erosion Velocity (m/s)	10.91	10.92	10.93	10.94	10.95
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 619 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.808e+004	1.808e+004	1.809e+004	1.809e+004	1.810e+004
14	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
15	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
16	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
17	Length (m)	1.810e+004	1.811e+004	1.811e+004	1.812e+004	1.812e+004
18	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
19	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
20	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
21	Length (m)	1.813e+004	1.813e+004	1.814e+004	1.814e+004	1.815e+004
22	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
23	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
24	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
25	Length (m)	1.815e+004	1.816e+004	1.816e+004	1.817e+004	1.817e+004
26	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
27	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
28	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
29	Length (m)	1.818e+004	1.818e+004	1.819e+004	1.819e+004	1.820e+004
30	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
31	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
32	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
33	Length (m)	1.820e+004	1.821e+004	1.821e+004	1.822e+004	1.822e+004
34	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
35	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
36	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
37	Length (m)	1.823e+004	1.823e+004	1.824e+004	1.824e+004	1.825e+004
38	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
39	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
40	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
41	Length (m)	1.825e+004	1.826e+004	1.826e+004	1.827e+004	1.827e+004
42	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
43	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
44	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
45	Length (m)	1.828e+004	1.828e+004	1.829e+004	1.829e+004	1.830e+004
46	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
47	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
48	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
49	Length (m)	1.830e+004	1.831e+004	1.831e+004	1.832e+004	1.832e+004
50	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
51	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
52	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
53	Length (m)	1.833e+004	1.833e+004	1.834e+004	1.834e+004	1.835e+004
54	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
55	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
56	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
57	Length (m)	1.835e+004	1.836e+004	1.836e+004	1.837e+004	1.837e+004
58	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
59	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
60	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
61	Length (m)	1.838e+004	1.838e+004	1.839e+004	1.839e+004	1.840e+004
62	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
63	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.379
64	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
65	Length (m)	1.840e+004	1.841e+004	1.841e+004	1.842e+004	1.842e+004
66	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
67	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
68	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 620 of 937					

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.843e+004	1.843e+004	1.844e+004	1.844e+004	1.845e+004
14	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
15	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
16	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
17	Length (m)	1.845e+004	1.846e+004	1.846e+004	1.847e+004	1.847e+004
18	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
19	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
20	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
21	Length (m)	1.848e+004	1.848e+004	1.849e+004	1.849e+004	1.850e+004
22	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
23	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
24	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
25	Length (m)	1.850e+004	1.851e+004	1.851e+004	1.852e+004	1.852e+004
26	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
27	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
28	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
29	Length (m)	1.853e+004	1.853e+004	1.854e+004	1.854e+004	1.855e+004
30	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
31	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
32	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
33	Length (m)	1.855e+004	1.856e+004	1.856e+004	1.857e+004	1.857e+004
34	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
35	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
36	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
37	Length (m)	1.858e+004	1.858e+004	1.859e+004	1.859e+004	1.860e+004
38	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
39	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
40	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
41	Length (m)	1.860e+004	1.861e+004	1.861e+004	1.862e+004	1.862e+004
42	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
43	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
44	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
45	Length (m)	1.863e+004	1.863e+004	1.864e+004	1.864e+004	1.865e+004
46	Bulk Density (kg/m3)	123.9	123.9	124.0	124.0	124.0
47	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
48	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
49	Length (m)	1.865e+004	1.866e+004	1.866e+004	1.867e+004	1.867e+004
50	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
51	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
52	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
53	Length (m)	1.868e+004	1.868e+004	1.869e+004	1.869e+004	1.870e+004
54	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
55	Bulk Velocity (m/s)	2.379	2.379	2.379	2.378	2.378
56	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
57	Length (m)	1.870e+004	1.871e+004	1.871e+004	1.872e+004	1.872e+004
58	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
59	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
60	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
61	Length (m)	1.873e+004	1.873e+004	1.874e+004	1.874e+004	1.875e+004
62	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
63	Bulk Velocity (m/s)	2.379	2.378	2.378	2.378	2.378
64	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
65	Length (m)	1.875e+004	1.876e+004	1.876e+004	1.877e+004	1.877e+004
66	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
67	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
68	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 621 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.878e+004	1.878e+004	1.879e+004	1.879e+004	1.880e+004
14	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
15	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
16	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
17	Length (m)	1.880e+004	1.880e+004	1.881e+004	1.881e+004	1.881e+004
18	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
19	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
20	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
21	Length (m)	1.882e+004	1.882e+004	1.882e+004	1.882e+004	1.882e+004
22	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
23	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
24	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
25	Length (m)	1.883e+004	1.883e+004	1.884e+004	1.884e+004	1.885e+004
26	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
27	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
28	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
29	Length (m)	1.885e+004	1.886e+004	1.886e+004	1.887e+004	1.887e+004
30	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
31	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
32	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
33	Length (m)	1.888e+004	1.888e+004	1.889e+004	1.889e+004	1.890e+004
34	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
35	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
36	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
37	Length (m)	1.890e+004	1.890e+004	1.891e+004	1.891e+004	1.891e+004
38	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
39	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
40	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
41	Length (m)	1.891e+004	1.891e+004	1.892e+004	1.892e+004	1.892e+004
42	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
43	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
44	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
45	Length (m)	1.893e+004	1.893e+004	1.894e+004	1.894e+004	1.895e+004
46	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
47	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
48	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
49	Length (m)	1.895e+004	1.896e+004	1.896e+004	1.897e+004	1.897e+004
50	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
51	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
52	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
53	Length (m)	1.898e+004	1.898e+004	1.899e+004	1.899e+004	1.900e+004
54	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
55	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
56	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
57	Length (m)	1.900e+004	1.901e+004	1.901e+004	1.902e+004	1.902e+004
58	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
59	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
60	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
61	Length (m)	1.903e+004	1.903e+004	1.904e+004	1.904e+004	1.905e+004
62	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
63	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
64	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
65	Length (m)	1.905e+004	1.906e+004	1.906e+004	1.907e+004	1.907e+004
66	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
67	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
68	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 622 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.908e+004	1.908e+004	1.909e+004	1.909e+004	1.910e+004
14	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
15	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
16	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
17	Length (m)	1.910e+004	1.911e+004	1.911e+004	1.912e+004	1.912e+004
18	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
19	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
20	Erosion Velocity (m/s)	10.96	10.95	10.95	10.95	10.95
21	Length (m)	1.913e+004	1.913e+004	1.914e+004	1.914e+004	1.915e+004
22	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
23	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	1.915e+004	1.916e+004	1.916e+004	1.917e+004	1.917e+004
26	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
27	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	1.918e+004	1.918e+004	1.919e+004	1.919e+004	1.920e+004
30	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
31	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.377
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	1.920e+004	1.921e+004	1.921e+004	1.922e+004	1.922e+004
34	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
35	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	1.923e+004	1.923e+004	1.924e+004	1.924e+004	1.925e+004
38	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
39	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	1.925e+004	1.926e+004	1.926e+004	1.927e+004	1.927e+004
42	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
43	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	1.928e+004	1.928e+004	1.929e+004	1.929e+004	1.930e+004
46	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
47	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	1.930e+004	1.931e+004	1.931e+004	1.932e+004	1.932e+004
50	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
51	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	1.933e+004	1.933e+004	1.934e+004	1.934e+004	1.935e+004
54	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
55	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	1.935e+004	1.936e+004	1.936e+004	1.937e+004	1.937e+004
58	Bulk Density (kg/m3)	124.0	124.0	124.1	124.1	124.1
59	Bulk Velocity (m/s)	2.377	2.377	2.377	2.376	2.376
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	1.938e+004	1.938e+004	1.939e+004	1.939e+004	1.940e+004
62	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
63	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.377
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	1.940e+004	1.941e+004	1.941e+004	1.942e+004	1.942e+004
66	Bulk Density (kg/m3)	124.1	124.0	124.0	124.0	123.9
67	Bulk Velocity (m/s)	2.377	2.377	2.378	2.378	2.379
68	Erosion Velocity (m/s)	10.95	10.95	10.96	10.96	10.96
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 623 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	1.943e+004	1.943e+004	1.944e+004	1.944e+004	1.945e+004
14	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
15	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
16	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
17	Length (m)	1.945e+004	1.946e+004	1.946e+004	1.946e+004	1.947e+004
18	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
19	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
20	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
21	Length (m)	1.947e+004	1.948e+004	1.948e+004	1.949e+004	1.949e+004
22	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
23	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
24	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
25	Length (m)	1.950e+004	1.952e+004	1.954e+004	1.957e+004	1.959e+004
26	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
27	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
28	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
29	Length (m)	1.961e+004	1.964e+004	1.966e+004	1.968e+004	1.971e+004
30	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
31	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
32	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
33	Length (m)	1.973e+004	1.975e+004	1.978e+004	1.980e+004	1.982e+004
34	Bulk Density (kg/m3)	123.9	123.9	124.0	124.0	124.0
35	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
36	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
37	Length (m)	1.985e+004	1.987e+004	1.989e+004	1.992e+004	1.994e+004
38	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
39	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
40	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
41	Length (m)	1.996e+004	1.999e+004	2.001e+004	2.003e+004	2.006e+004
42	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
43	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
44	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
45	Length (m)	2.008e+004	2.010e+004	2.013e+004	2.015e+004	2.017e+004
46	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
47	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.377
48	Erosion Velocity (m/s)	10.96	10.95	10.95	10.95	10.95
49	Length (m)	2.020e+004	2.022e+004	2.024e+004	2.027e+004	2.029e+004
50	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
51	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	2.031e+004	2.034e+004	2.036e+004	2.038e+004	2.041e+004
54	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
55	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	2.043e+004	2.045e+004	2.048e+004	2.050e+004	2.052e+004
58	Bulk Density (kg/m3)	124.0	124.0	124.0	124.1	124.1
59	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	2.055e+004	2.058e+004	2.060e+004	2.063e+004	2.066e+004
62	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
63	Bulk Velocity (m/s)	2.377	2.377	2.376	2.376	2.376
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	2.069e+004	2.069e+004	2.070e+004	2.070e+004	2.070e+004
66	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
67	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 624 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.070e+004	2.071e+004	2.071e+004	2.072e+004	2.072e+004
14	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
15	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	2.073e+004	2.073e+004	2.074e+004	2.074e+004	2.075e+004
18	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
19	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	2.075e+004	2.076e+004	2.076e+004	2.077e+004	2.077e+004
22	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
23	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	2.078e+004	2.078e+004	2.079e+004	2.079e+004	2.080e+004
26	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
27	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	2.080e+004	2.081e+004	2.081e+004	2.082e+004	2.082e+004
30	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
31	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	2.083e+004	2.083e+004	2.084e+004	2.084e+004	2.085e+004
34	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
35	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	2.085e+004	2.086e+004	2.086e+004	2.087e+004	2.087e+004
38	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
39	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	2.088e+004	2.088e+004	2.089e+004	2.089e+004	2.090e+004
42	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
43	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	2.090e+004	2.091e+004	2.091e+004	2.092e+004	2.092e+004
46	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
47	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	2.093e+004	2.093e+004	2.094e+004	2.094e+004	2.095e+004
50	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
51	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	2.095e+004	2.096e+004	2.096e+004	2.097e+004	2.097e+004
54	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
55	Bulk Velocity (m/s)	2.376	2.376	2.375	2.375	2.375
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	2.098e+004	2.098e+004	2.099e+004	2.099e+004	2.100e+004
58	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
59	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	2.100e+004	2.101e+004	2.101e+004	2.102e+004	2.102e+004
62	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
63	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	2.103e+004	2.103e+004	2.104e+004	2.104e+004	2.105e+004
66	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
67	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 625 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.105e+004	2.106e+004	2.106e+004	2.107e+004	2.107e+004
14	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
15	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	2.108e+004	2.108e+004	2.109e+004	2.109e+004	2.110e+004
18	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
19	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	2.110e+004	2.111e+004	2.111e+004	2.111e+004	2.112e+004
22	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.2
23	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	2.112e+004	2.113e+004	2.113e+004	2.113e+004	2.114e+004
26	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.1
27	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	2.114e+004	2.114e+004	2.115e+004	2.115e+004	2.115e+004
30	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
31	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	2.115e+004	2.116e+004	2.116e+004	2.117e+004	2.117e+004
34	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
35	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	2.118e+004	2.118e+004	2.119e+004	2.119e+004	2.120e+004
38	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
39	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	2.120e+004	2.121e+004	2.121e+004	2.122e+004	2.122e+004
42	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
43	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	2.123e+004	2.123e+004	2.124e+004	2.124e+004	2.125e+004
46	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
47	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	2.125e+004	2.126e+004	2.126e+004	2.127e+004	2.127e+004
50	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.2
51	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	2.128e+004	2.128e+004	2.129e+004	2.129e+004	2.130e+004
54	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
55	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	2.130e+004	2.131e+004	2.131e+004	2.132e+004	2.132e+004
58	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
59	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	2.133e+004	2.133e+004	2.133e+004	2.133e+004	2.133e+004
62	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
63	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	2.134e+004	2.134e+004	2.134e+004	2.134e+004	2.135e+004
66	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
67	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.135e+004	2.136e+004	2.136e+004	2.137e+004	2.137e+004
14	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
15	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	2.138e+004	2.138e+004	2.139e+004	2.139e+004	2.140e+004
18	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
19	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	2.140e+004	2.141e+004	2.141e+004	2.142e+004	2.142e+004
22	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
23	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	2.143e+004	2.143e+004	2.144e+004	2.144e+004	2.145e+004
26	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
27	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	2.145e+004	2.146e+004	2.146e+004	2.147e+004	2.147e+004
30	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
31	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	2.148e+004	2.148e+004	2.149e+004	2.149e+004	2.150e+004
34	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
35	Bulk Velocity (m/s)	2.375	2.375	2.375	2.374	2.374
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	2.150e+004	2.150e+004	2.151e+004	2.151e+004	2.152e+004
38	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
39	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	2.152e+004	2.152e+004	2.152e+004	2.152e+004	2.152e+004
42	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
43	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	2.153e+004	2.153e+004	2.154e+004	2.154e+004	2.155e+004
46	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
47	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	2.155e+004	2.156e+004	2.156e+004	2.157e+004	2.157e+004
50	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
51	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	2.158e+004	2.158e+004	2.159e+004	2.159e+004	2.160e+004
54	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
55	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	2.160e+004	2.161e+004	2.161e+004	2.162e+004	2.162e+004
58	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
59	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	2.163e+004	2.163e+004	2.164e+004	2.164e+004	2.165e+004
62	Bulk Density (kg/m3)	124.2	124.2	124.2	124.3	124.3
63	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
65	Length (m)	2.165e+004	2.166e+004	2.166e+004	2.167e+004	2.167e+004
66	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
67	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 627 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.168e+004	2.168e+004	2.169e+004	2.169e+004	2.170e+004
14	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
15	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
17	Length (m)	2.170e+004	2.171e+004	2.171e+004	2.172e+004	2.172e+004
18	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
19	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
21	Length (m)	2.173e+004	2.173e+004	2.174e+004	2.174e+004	2.175e+004
22	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
23	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
25	Length (m)	2.175e+004	2.176e+004	2.176e+004	2.177e+004	2.177e+004
26	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
27	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
29	Length (m)	2.178e+004	2.178e+004	2.179e+004	2.179e+004	2.180e+004
30	Bulk Density (kg/m3)	124.3	124.2	124.1	124.0	123.9
31	Bulk Velocity (m/s)	2.373	2.374	2.376	2.378	2.379
32	Erosion Velocity (m/s)	10.94	10.95	10.95	10.95	10.96
33	Length (m)	2.180e+004	2.181e+004	2.181e+004	2.182e+004	2.182e+004
34	Bulk Density (kg/m3)	123.8	123.8	123.8	123.8	123.8
35	Bulk Velocity (m/s)	2.381	2.381	2.381	2.381	2.381
36	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
37	Length (m)	2.183e+004	2.183e+004	2.184e+004	2.184e+004	2.185e+004
38	Bulk Density (kg/m3)	123.8	123.8	123.8	123.8	123.8
39	Bulk Velocity (m/s)	2.381	2.381	2.381	2.381	2.381
40	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
41	Length (m)	2.185e+004	2.186e+004	2.186e+004	2.187e+004	2.187e+004
42	Bulk Density (kg/m3)	123.8	123.8	123.8	123.8	123.8
43	Bulk Velocity (m/s)	2.381	2.381	2.381	2.381	2.381
44	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
45	Length (m)	2.188e+004	2.188e+004	2.189e+004	2.189e+004	2.190e+004
46	Bulk Density (kg/m3)	123.8	123.8	123.8	123.8	123.8
47	Bulk Velocity (m/s)	2.381	2.381	2.381	2.381	2.381
48	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
49	Length (m)	2.190e+004	2.191e+004	2.191e+004	2.192e+004	2.192e+004
50	Bulk Density (kg/m3)	123.8	123.8	123.8	123.8	123.8
51	Bulk Velocity (m/s)	2.381	2.381	2.381	2.381	2.381
52	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
53	Length (m)	2.193e+004	2.193e+004	2.194e+004	2.194e+004	2.195e+004
54	Bulk Density (kg/m3)	123.8	123.8	123.8	123.8	123.8
55	Bulk Velocity (m/s)	2.381	2.381	2.381	2.381	2.381
56	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
57	Length (m)	2.195e+004	2.196e+004	2.196e+004	2.197e+004	2.197e+004
58	Bulk Density (kg/m3)	123.8	123.8	123.8	123.8	123.8
59	Bulk Velocity (m/s)	2.381	2.381	2.381	2.381	2.381
60	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
61	Length (m)	2.198e+004	2.198e+004	2.199e+004	2.199e+004	2.200e+004
62	Bulk Density (kg/m3)	123.8	123.8	123.8	123.9	123.9
63	Bulk Velocity (m/s)	2.381	2.381	2.381	2.381	2.381
64	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
65	Length (m)	2.200e+004	2.201e+004	2.201e+004	2.202e+004	2.202e+004
66	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
67	Bulk Velocity (m/s)	2.381	2.381	2.381	2.381	2.381
68	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 628 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.203e+004	2.203e+004	2.204e+004	2.204e+004	2.205e+004
14	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
15	Bulk Velocity (m/s)	2.381	2.381	2.381	2.381	2.381
16	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
17	Length (m)	2.205e+004	2.206e+004	2.206e+004	2.207e+004	2.207e+004
18	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
19	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
20	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
21	Length (m)	2.208e+004	2.208e+004	2.209e+004	2.209e+004	2.210e+004
22	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
23	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
24	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
25	Length (m)	2.210e+004	2.211e+004	2.211e+004	2.212e+004	2.212e+004
26	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
27	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
28	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
29	Length (m)	2.213e+004	2.213e+004	2.214e+004	2.214e+004	2.215e+004
30	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
31	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
32	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
33	Length (m)	2.215e+004	2.216e+004	2.216e+004	2.217e+004	2.217e+004
34	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
35	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
36	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
37	Length (m)	2.218e+004	2.218e+004	2.219e+004	2.219e+004	2.220e+004
38	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
39	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
40	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
41	Length (m)	2.220e+004	2.221e+004	2.221e+004	2.222e+004	2.222e+004
42	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
43	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
44	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
45	Length (m)	2.223e+004	2.223e+004	2.224e+004	2.224e+004	2.225e+004
46	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
47	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
48	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
49	Length (m)	2.225e+004	2.226e+004	2.226e+004	2.227e+004	2.227e+004
50	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
51	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
52	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
53	Length (m)	2.228e+004	2.229e+004	2.230e+004	2.231e+004	2.232e+004
54	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
55	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
56	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
57	Length (m)	2.233e+004	2.255e+004	2.278e+004	2.300e+004	2.323e+004
58	Bulk Density (kg/m3)	123.9	123.9	124.0	124.0	124.0
59	Bulk Velocity (m/s)	2.380	2.379	2.379	2.378	2.378
60	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.95
61	Length (m)	2.345e+004	2.346e+004	2.347e+004	2.348e+004	2.348e+004
62	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
63	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	2.349e+004	2.350e+004	2.350e+004	2.351e+004	2.351e+004
66	Bulk Density (kg/m3)	124.0	124.0	124.0	124.1	124.1
67	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 629 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.352e+004	2.352e+004	2.353e+004	2.353e+004	2.354e+004
14	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
15	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	2.354e+004	2.355e+004	2.355e+004	2.356e+004	2.356e+004
18	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
19	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	2.357e+004	2.357e+004	2.358e+004	2.358e+004	2.359e+004
22	Bulk Density (kg/m3)	124.1	124.1	124.0	124.0	124.0
23	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	2.359e+004	2.360e+004	2.360e+004	2.361e+004	2.361e+004
26	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
27	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	2.362e+004	2.362e+004	2.363e+004	2.363e+004	2.364e+004
30	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
31	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	2.364e+004	2.365e+004	2.365e+004	2.366e+004	2.366e+004
34	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
35	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	2.367e+004	2.367e+004	2.368e+004	2.368e+004	2.369e+004
38	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
39	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	2.369e+004	2.370e+004	2.370e+004	2.371e+004	2.371e+004
42	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
43	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	2.372e+004	2.372e+004	2.373e+004	2.373e+004	2.374e+004
46	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
47	Bulk Velocity (m/s)	2.377	2.377	2.377	2.376	2.376
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	2.374e+004	2.375e+004	2.375e+004	2.376e+004	2.376e+004
50	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
51	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	2.377e+004	2.377e+004	2.378e+004	2.378e+004	2.379e+004
54	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
55	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	2.379e+004	2.380e+004	2.380e+004	2.381e+004	2.381e+004
58	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
59	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	2.382e+004	2.382e+004	2.383e+004	2.383e+004	2.384e+004
62	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
63	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	2.384e+004	2.385e+004	2.385e+004	2.386e+004	2.386e+004
66	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
67	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 630 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.387e+004	2.387e+004	2.388e+004	2.388e+004	2.389e+004
14	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
15	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	2.389e+004	2.390e+004	2.390e+004	2.391e+004	2.391e+004
18	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
19	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	2.392e+004	2.392e+004	2.393e+004	2.393e+004	2.394e+004
22	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
23	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	2.394e+004	2.395e+004	2.395e+004	2.396e+004	2.396e+004
26	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
27	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	2.397e+004	2.397e+004	2.398e+004	2.398e+004	2.399e+004
30	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
31	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	2.399e+004	2.400e+004	2.400e+004	2.401e+004	2.401e+004
34	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
35	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	2.402e+004	2.402e+004	2.403e+004	2.403e+004	2.404e+004
38	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
39	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	2.404e+004	2.405e+004	2.405e+004	2.406e+004	2.406e+004
42	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
43	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	2.407e+004	2.407e+004	2.408e+004	2.408e+004	2.409e+004
46	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
47	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	2.409e+004	2.410e+004	2.410e+004	2.411e+004	2.411e+004
50	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
51	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	2.412e+004	2.412e+004	2.413e+004	2.413e+004	2.414e+004
54	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
55	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	2.414e+004	2.415e+004	2.415e+004	2.416e+004	2.416e+004
58	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
59	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	2.417e+004	2.417e+004	2.418e+004	2.418e+004	2.419e+004
62	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
63	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	2.419e+004	2.420e+004	2.420e+004	2.421e+004	2.421e+004
66	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
67	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 631 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.422e+004	2.422e+004	2.423e+004	2.423e+004	2.424e+004
14	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
15	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	2.424e+004	2.425e+004	2.425e+004	2.426e+004	2.426e+004
18	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
19	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	2.427e+004	2.427e+004	2.428e+004	2.428e+004	2.429e+004
22	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
23	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	2.429e+004	2.430e+004	2.430e+004	2.431e+004	2.431e+004
26	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
27	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	2.432e+004	2.432e+004	2.433e+004	2.433e+004	2.434e+004
30	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
31	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	2.434e+004	2.435e+004	2.435e+004	2.436e+004	2.436e+004
34	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
35	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	2.437e+004	2.437e+004	2.437e+004	2.437e+004	2.437e+004
38	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
39	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	2.438e+004	2.438e+004	2.438e+004	2.438e+004	2.438e+004
42	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
43	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	2.439e+004	2.439e+004	2.440e+004	2.440e+004	2.441e+004
46	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
47	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	2.442e+004	2.442e+004	2.443e+004	2.443e+004	2.444e+004
50	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
51	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	2.444e+004	2.445e+004	2.445e+004	2.446e+004	2.446e+004
54	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
55	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	2.447e+004	2.447e+004	2.448e+004	2.448e+004	2.449e+004
58	Bulk Density (kg/m3)	124.1	124.2	124.2	124.2	124.2
59	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	2.449e+004	2.450e+004	2.450e+004	2.451e+004	2.451e+004
62	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
63	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	2.452e+004	2.452e+004	2.453e+004	2.453e+004	2.454e+004
66	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
67	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 632 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.454e+004	2.455e+004	2.455e+004	2.456e+004	2.456e+004
14	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
15	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	2.457e+004	2.457e+004	2.458e+004	2.458e+004	2.459e+004
18	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
19	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	2.459e+004	2.460e+004	2.460e+004	2.461e+004	2.461e+004
22	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
23	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	2.462e+004	2.462e+004	2.463e+004	2.463e+004	2.464e+004
26	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
27	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	2.464e+004	2.465e+004	2.465e+004	2.466e+004	2.466e+004
30	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
31	Bulk Velocity (m/s)	2.375	2.375	2.375	2.374	2.374
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	2.467e+004	2.467e+004	2.468e+004	2.468e+004	2.469e+004
34	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
35	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	2.469e+004	2.470e+004	2.470e+004	2.471e+004	2.471e+004
38	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
39	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	2.472e+004	2.472e+004	2.473e+004	2.473e+004	2.474e+004
42	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
43	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	2.474e+004	2.475e+004	2.475e+004	2.476e+004	2.476e+004
46	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
47	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	2.477e+004	2.477e+004	2.478e+004	2.478e+004	2.479e+004
50	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
51	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	2.479e+004	2.480e+004	2.480e+004	2.481e+004	2.481e+004
54	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
55	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	2.482e+004	2.482e+004	2.483e+004	2.483e+004	2.484e+004
58	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
59	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	2.484e+004	2.485e+004	2.485e+004	2.486e+004	2.486e+004
62	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
63	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	2.487e+004	2.487e+004	2.488e+004	2.488e+004	2.489e+004
66	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
67	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 633 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.489e+004	2.490e+004	2.490e+004	2.491e+004	2.491e+004
14	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
15	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	2.492e+004	2.492e+004	2.493e+004	2.493e+004	2.494e+004
18	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
19	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	2.494e+004	2.495e+004	2.495e+004	2.496e+004	2.496e+004
22	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
23	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	2.497e+004	2.497e+004	2.498e+004	2.498e+004	2.499e+004
26	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
27	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	2.499e+004	2.500e+004	2.500e+004	2.501e+004	2.501e+004
30	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
31	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	2.502e+004	2.502e+004	2.503e+004	2.503e+004	2.504e+004
34	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
35	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	2.504e+004	2.505e+004	2.505e+004	2.506e+004	2.506e+004
38	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
39	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	2.507e+004	2.507e+004	2.508e+004	2.508e+004	2.509e+004
42	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
43	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	2.509e+004	2.510e+004	2.510e+004	2.511e+004	2.511e+004
46	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
47	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	2.512e+004	2.512e+004	2.513e+004	2.513e+004	2.514e+004
50	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
51	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	2.514e+004	2.515e+004	2.515e+004	2.516e+004	2.516e+004
54	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
55	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	2.517e+004	2.517e+004	2.518e+004	2.518e+004	2.519e+004
58	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
59	Bulk Velocity (m/s)	2.374	2.373	2.373	2.373	2.373
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.94
61	Length (m)	2.519e+004	2.520e+004	2.520e+004	2.521e+004	2.522e+004
62	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
63	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
65	Length (m)	2.522e+004	2.523e+004	2.523e+004	2.524e+004	2.524e+004
66	Bulk Density (kg/m3)	124.2	124.2	124.3	124.3	124.3
67	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 634 of 937					

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.525e+004	2.525e+004	2.526e+004	2.526e+004	2.527e+004
14	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
15	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
17	Length (m)	2.527e+004	2.528e+004	2.528e+004	2.529e+004	2.529e+004
18	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
19	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.373
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
21	Length (m)	2.529e+004	2.531e+004	2.533e+004	2.535e+004	2.537e+004
22	Bulk Density (kg/m3)	124.3	124.3	124.2	124.2	124.2
23	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
25	Length (m)	2.539e+004	2.539e+004	2.539e+004	2.539e+004	2.540e+004
26	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
27	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
29	Length (m)	2.540e+004	2.540e+004	2.541e+004	2.541e+004	2.542e+004
30	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.1
31	Bulk Velocity (m/s)	2.373	2.374	2.374	2.375	2.375
32	Erosion Velocity (m/s)	10.94	10.95	10.95	10.95	10.95
33	Length (m)	2.543e+004	2.543e+004	2.544e+004	2.544e+004	2.545e+004
34	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
35	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	2.545e+004	2.546e+004	2.546e+004	2.547e+004	2.547e+004
38	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
39	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	2.548e+004	2.548e+004	2.549e+004	2.549e+004	2.550e+004
42	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
43	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	2.550e+004	2.551e+004	2.551e+004	2.552e+004	2.552e+004
46	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
47	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	2.553e+004	2.553e+004	2.554e+004	2.554e+004	2.555e+004
50	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
51	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	2.555e+004	2.556e+004	2.556e+004	2.557e+004	2.557e+004
54	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
55	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	2.558e+004	2.558e+004	2.559e+004	2.559e+004	2.560e+004
58	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
59	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	2.560e+004	2.561e+004	2.561e+004	2.562e+004	2.563e+004
62	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
63	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	2.563e+004	2.564e+004	2.564e+004	2.565e+004	2.565e+004
66	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
67	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 635 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.566e+004	2.566e+004	2.567e+004	2.567e+004	2.568e+004
14	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
15	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	2.568e+004	2.569e+004	2.569e+004	2.570e+004	2.570e+004
18	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
19	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	2.571e+004	2.571e+004	2.571e+004	2.571e+004	2.571e+004
22	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
23	Bulk Velocity (m/s)	2.376	2.375	2.375	2.375	2.375
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	2.571e+004	2.572e+004	2.572e+004	2.572e+004	2.572e+004
26	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
27	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	2.572e+004	2.572e+004	2.573e+004	2.573e+004	2.573e+004
30	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.2
31	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	2.573e+004	2.574e+004	2.574e+004	2.575e+004	2.575e+004
34	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
35	Bulk Velocity (m/s)	2.375	2.375	2.375	2.374	2.374
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	2.576e+004	2.576e+004	2.577e+004	2.577e+004	2.578e+004
38	Bulk Density (kg/m3)	124.2	124.1	124.1	124.0	124.0
39	Bulk Velocity (m/s)	2.374	2.375	2.376	2.377	2.379
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.96
41	Length (m)	2.578e+004	2.579e+004	2.579e+004	2.580e+004	2.580e+004
42	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
43	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
44	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
45	Length (m)	2.581e+004	2.581e+004	2.582e+004	2.582e+004	2.583e+004
46	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
47	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
48	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
49	Length (m)	2.583e+004	2.584e+004	2.584e+004	2.585e+004	2.585e+004
50	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
51	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
52	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
53	Length (m)	2.586e+004	2.586e+004	2.587e+004	2.587e+004	2.588e+004
54	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
55	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
56	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
57	Length (m)	2.588e+004	2.589e+004	2.589e+004	2.590e+004	2.590e+004
58	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
59	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
60	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
61	Length (m)	2.591e+004	2.591e+004	2.592e+004	2.592e+004	2.593e+004
62	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
63	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
64	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
65	Length (m)	2.593e+004	2.594e+004	2.594e+004	2.595e+004	2.595e+004
66	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
67	Bulk Velocity (m/s)	2.380	2.379	2.379	2.379	2.379
68	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 636 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.596e+004	2.596e+004	2.597e+004	2.597e+004	2.598e+004
14	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
15	Bulk Velocity (m/s)	2.379	2.379	2.380	2.380	2.380
16	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
17	Length (m)	2.598e+004	2.599e+004	2.599e+004	2.600e+004	2.600e+004
18	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
19	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
20	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
21	Length (m)	2.601e+004	2.601e+004	2.602e+004	2.602e+004	2.603e+004
22	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
23	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
24	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
25	Length (m)	2.603e+004	2.604e+004	2.604e+004	2.605e+004	2.605e+004
26	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
27	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
28	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
29	Length (m)	2.606e+004	2.606e+004	2.607e+004	2.607e+004	2.608e+004
30	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
31	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
32	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
33	Length (m)	2.608e+004	2.609e+004	2.609e+004	2.610e+004	2.610e+004
34	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
35	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
36	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
37	Length (m)	2.611e+004	2.611e+004	2.612e+004	2.612e+004	2.613e+004
38	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
39	Bulk Velocity (m/s)	2.380	2.380	2.380	2.380	2.380
40	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
41	Length (m)	2.613e+004	2.614e+004	2.614e+004	2.615e+004	2.615e+004
42	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
43	Bulk Velocity (m/s)	2.380	2.379	2.379	2.379	2.379
44	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
45	Length (m)	2.616e+004	2.616e+004	2.617e+004	2.617e+004	2.618e+004
46	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
47	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
48	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
49	Length (m)	2.618e+004	2.619e+004	2.619e+004	2.620e+004	2.620e+004
50	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
51	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
52	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
53	Length (m)	2.621e+004	2.621e+004	2.622e+004	2.622e+004	2.623e+004
54	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
55	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
56	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
57	Length (m)	2.623e+004	2.624e+004	2.624e+004	2.625e+004	2.625e+004
58	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
59	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
60	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
61	Length (m)	2.626e+004	2.626e+004	2.627e+004	2.627e+004	2.628e+004
62	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
63	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
64	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
65	Length (m)	2.628e+004	2.629e+004	2.629e+004	2.630e+004	2.630e+004
66	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
67	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
68	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 637 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.631e+004	2.631e+004	2.632e+004	2.632e+004	2.633e+004
14	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
15	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
16	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
17	Length (m)	2.633e+004	2.634e+004	2.634e+004	2.635e+004	2.635e+004
18	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
19	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
20	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
21	Length (m)	2.636e+004	2.636e+004	2.637e+004	2.637e+004	2.638e+004
22	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
23	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
24	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
25	Length (m)	2.638e+004	2.639e+004	2.639e+004	2.640e+004	2.640e+004
26	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
27	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
28	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
29	Length (m)	2.641e+004	2.641e+004	2.642e+004	2.642e+004	2.643e+004
30	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
31	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
32	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
33	Length (m)	2.644e+004	2.644e+004	2.645e+004	2.645e+004	2.646e+004
34	Bulk Density (kg/m3)	123.9	123.9	123.9	123.9	123.9
35	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
36	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
37	Length (m)	2.646e+004	2.647e+004	2.647e+004	2.648e+004	2.648e+004
38	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
39	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
40	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
41	Length (m)	2.649e+004	2.649e+004	2.650e+004	2.650e+004	2.651e+004
42	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
43	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
44	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
45	Length (m)	2.651e+004	2.652e+004	2.652e+004	2.653e+004	2.653e+004
46	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
47	Bulk Velocity (m/s)	2.379	2.379	2.379	2.379	2.379
48	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
49	Length (m)	2.654e+004	2.654e+004	2.655e+004	2.655e+004	2.656e+004
50	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
51	Bulk Velocity (m/s)	2.379	2.379	2.379	2.378	2.378
52	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
53	Length (m)	2.656e+004	2.657e+004	2.657e+004	2.658e+004	2.658e+004
54	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
55	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
56	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
57	Length (m)	2.659e+004	2.659e+004	2.660e+004	2.660e+004	2.661e+004
58	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
59	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
60	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
61	Length (m)	2.661e+004	2.662e+004	2.662e+004	2.663e+004	2.663e+004
62	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
63	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
64	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
65	Length (m)	2.664e+004	2.664e+004	2.665e+004	2.665e+004	2.666e+004
66	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
67	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
68	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 638 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.666e+004	2.667e+004	2.667e+004	2.668e+004	2.668e+004
14	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
15	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
16	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
17	Length (m)	2.669e+004	2.669e+004	2.670e+004	2.670e+004	2.671e+004
18	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
19	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
20	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
21	Length (m)	2.671e+004	2.672e+004	2.672e+004	2.673e+004	2.674e+004
22	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
23	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
24	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
25	Length (m)	2.674e+004	2.674e+004	2.675e+004	2.675e+004	2.676e+004
26	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
27	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
28	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
29	Length (m)	2.676e+004	2.677e+004	2.677e+004	2.678e+004	2.678e+004
30	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
31	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
32	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
33	Length (m)	2.679e+004	2.679e+004	2.680e+004	2.680e+004	2.681e+004
34	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
35	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
36	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
37	Length (m)	2.681e+004	2.682e+004	2.682e+004	2.683e+004	2.683e+004
38	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
39	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
40	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
41	Length (m)	2.684e+004	2.684e+004	2.685e+004	2.685e+004	2.686e+004
42	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
43	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
44	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
45	Length (m)	2.686e+004	2.687e+004	2.687e+004	2.688e+004	2.688e+004
46	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
47	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
48	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
49	Length (m)	2.689e+004	2.689e+004	2.690e+004	2.690e+004	2.691e+004
50	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
51	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
52	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
53	Length (m)	2.691e+004	2.692e+004	2.692e+004	2.693e+004	2.693e+004
54	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
55	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
56	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
57	Length (m)	2.694e+004	2.694e+004	2.695e+004	2.695e+004	2.696e+004
58	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
59	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
60	Erosion Velocity (m/s)	10.96	10.96	10.96	10.95	10.95
61	Length (m)	2.696e+004	2.697e+004	2.697e+004	2.698e+004	2.698e+004
62	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
63	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	2.699e+004	2.699e+004	2.700e+004	2.700e+004	2.701e+004
66	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
67	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 639 of 937					

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.701e+004	2.702e+004	2.702e+004	2.703e+004	2.703e+004
14	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
15	Bulk Velocity (m/s)	2.378	2.377	2.377	2.377	2.377
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	2.704e+004	2.704e+004	2.704e+004	2.704e+004	2.704e+004
18	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
19	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	2.705e+004	2.705e+004	2.705e+004	2.705e+004	2.706e+004
22	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
23	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	2.706e+004	2.707e+004	2.707e+004	2.708e+004	2.708e+004
26	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
27	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	2.709e+004	2.709e+004	2.710e+004	2.710e+004	2.711e+004
30	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
31	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	2.711e+004	2.712e+004	2.712e+004	2.713e+004	2.713e+004
34	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
35	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	2.714e+004	2.714e+004	2.715e+004	2.715e+004	2.716e+004
38	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
39	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	2.716e+004	2.717e+004	2.717e+004	2.718e+004	2.718e+004
42	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
43	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	2.719e+004	2.719e+004	2.720e+004	2.720e+004	2.721e+004
46	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
47	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	2.721e+004	2.722e+004	2.722e+004	2.723e+004	2.723e+004
50	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
51	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	2.724e+004	2.724e+004	2.725e+004	2.725e+004	2.726e+004
54	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
55	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	2.726e+004	2.727e+004	2.727e+004	2.728e+004	2.728e+004
58	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
59	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	2.729e+004	2.729e+004	2.730e+004	2.730e+004	2.731e+004
62	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
63	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	2.731e+004	2.732e+004	2.732e+004	2.733e+004	2.733e+004
66	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
67	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 640 of 937	

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.734e+004	2.734e+004	2.735e+004	2.735e+004	2.736e+004
14	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
15	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	2.736e+004	2.737e+004	2.737e+004	2.738e+004	2.738e+004
18	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
19	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	2.739e+004	2.739e+004	2.740e+004	2.740e+004	2.741e+004
22	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
23	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	2.741e+004	2.742e+004	2.742e+004	2.743e+004	2.743e+004
26	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
27	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	2.744e+004	2.744e+004	2.745e+004	2.745e+004	2.746e+004
30	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
31	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	2.746e+004	2.747e+004	2.747e+004	2.748e+004	2.748e+004
34	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
35	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	2.749e+004	2.749e+004	2.749e+004	2.749e+004	2.749e+004
38	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
39	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	2.750e+004	2.750e+004	2.751e+004	2.751e+004	2.752e+004
42	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
43	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	2.753e+004	2.753e+004	2.753e+004	2.753e+004	2.753e+004
46	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
47	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	2.754e+004	2.754e+004	2.755e+004	2.755e+004	2.756e+004
50	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
51	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	2.756e+004	2.757e+004	2.757e+004	2.758e+004	2.758e+004
54	Bulk Density (kg/m3)	124.1	124.1	124.1	124.0	124.0
55	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	2.759e+004	2.759e+004	2.760e+004	2.760e+004	2.761e+004
58	Bulk Density (kg/m3)	124.0	124.0	124.0	124.1	124.1
59	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	2.761e+004	2.762e+004	2.762e+004	2.763e+004	2.763e+004
62	Bulk Density (kg/m3)	124.1	124.1	124.1	124.0	124.0
63	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	2.764e+004	2.764e+004	2.765e+004	2.765e+004	2.766e+004
66	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
67	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 641 of 937					

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.766e+004	2.767e+004	2.767e+004	2.768e+004	2.768e+004
14	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
15	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	2.769e+004	2.769e+004	2.770e+004	2.770e+004	2.771e+004
18	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
19	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	2.771e+004	2.772e+004	2.772e+004	2.773e+004	2.773e+004
22	Bulk Density (kg/m3)	124.0	124.0	124.1	124.1	124.1
23	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	2.774e+004	2.774e+004	2.775e+004	2.775e+004	2.776e+004
26	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
27	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	2.776e+004	2.777e+004	2.777e+004	2.778e+004	2.778e+004
30	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
31	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	2.779e+004	2.779e+004	2.780e+004	2.780e+004	2.781e+004
34	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
35	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	2.781e+004	2.782e+004	2.782e+004	2.783e+004	2.783e+004
38	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
39	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	2.784e+004	2.784e+004	2.785e+004	2.785e+004	2.785e+004
42	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
43	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	2.786e+004	2.788e+004	2.790e+004	2.791e+004	2.793e+004
46	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
47	Bulk Velocity (m/s)	2.377	2.376	2.376	2.376	2.376
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	2.795e+004	2.797e+004	2.799e+004	2.801e+004	2.802e+004
50	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
51	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	2.804e+004	2.806e+004	2.808e+004	2.810e+004	2.812e+004
54	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
55	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	2.814e+004	2.815e+004	2.817e+004	2.819e+004	2.821e+004
58	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
59	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	2.823e+004	2.825e+004	2.827e+004	2.828e+004	2.830e+004
62	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
63	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	2.832e+004	2.834e+004	2.836e+004	2.838e+004	2.839e+004
66	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
67	Bulk Velocity (m/s)	2.376	2.375	2.375	2.375	2.375
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 642 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.841e+004	2.842e+004	2.843e+004	2.843e+004	2.844e+004
14	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
15	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	2.845e+004	2.845e+004	2.846e+004	2.846e+004	2.847e+004
18	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
19	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	2.847e+004	2.848e+004	2.848e+004	2.849e+004	2.849e+004
22	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
23	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	2.850e+004	2.850e+004	2.851e+004	2.851e+004	2.852e+004
26	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
27	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	2.852e+004	2.853e+004	2.853e+004	2.854e+004	2.854e+004
30	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
31	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	2.855e+004	2.855e+004	2.856e+004	2.856e+004	2.857e+004
34	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
35	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	2.857e+004	2.858e+004	2.858e+004	2.859e+004	2.859e+004
38	Bulk Density (kg/m3)	124.1	124.2	124.2	124.2	124.2
39	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	2.860e+004	2.860e+004	2.861e+004	2.861e+004	2.862e+004
42	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
43	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	2.862e+004	2.863e+004	2.863e+004	2.864e+004	2.864e+004
46	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
47	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.374
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	2.865e+004	2.865e+004	2.866e+004	2.866e+004	2.867e+004
50	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
51	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	2.867e+004	2.868e+004	2.868e+004	2.869e+004	2.869e+004
54	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
55	Bulk Velocity (m/s)	2.374	2.375	2.375	2.375	2.375
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	2.870e+004	2.870e+004	2.871e+004	2.871e+004	2.872e+004
58	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
59	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	2.872e+004	2.873e+004	2.874e+004	2.875e+004	2.876e+004
62	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
63	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	2.877e+004	2.878e+004	2.879e+004	2.880e+004	2.881e+004
66	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
67	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 643 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.883e+004	2.884e+004	2.885e+004	2.886e+004	2.887e+004
14	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
15	Bulk Velocity (m/s)	2.375	2.374	2.374	2.374	2.374
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	2.888e+004	2.889e+004	2.890e+004	2.891e+004	2.892e+004
18	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
19	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	2.893e+004	2.894e+004	2.895e+004	2.896e+004	2.897e+004
22	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
23	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	2.898e+004	2.899e+004	2.900e+004	2.901e+004	2.902e+004
26	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
27	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	2.903e+004	2.904e+004	2.904e+004	2.904e+004	2.904e+004
30	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
31	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	2.905e+004	2.905e+004	2.906e+004	2.906e+004	2.907e+004
34	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
35	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	2.907e+004	2.908e+004	2.908e+004	2.909e+004	2.909e+004
38	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
39	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	2.910e+004	2.910e+004	2.911e+004	2.911e+004	2.912e+004
42	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
43	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	2.912e+004	2.913e+004	2.913e+004	2.914e+004	2.914e+004
46	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
47	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	2.915e+004	2.915e+004	2.916e+004	2.916e+004	2.916e+004
50	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
51	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	2.917e+004	2.917e+004	2.918e+004	2.918e+004	2.919e+004
54	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
55	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	2.920e+004	2.920e+004	2.921e+004	2.921e+004	2.922e+004
58	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
59	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	2.922e+004	2.923e+004	2.923e+004	2.924e+004	2.924e+004
62	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
63	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	2.925e+004	2.925e+004	2.925e+004	2.926e+004	2.926e+004
66	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
67	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 644 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.926e+004	2.927e+004	2.928e+004	2.929e+004	2.930e+004
14	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
15	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	2.931e+004	2.932e+004	2.933e+004	2.934e+004	2.935e+004
18	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
19	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	2.936e+004	2.936e+004	2.937e+004	2.938e+004	2.939e+004
22	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
23	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.94
25	Length (m)	2.940e+004	2.941e+004	2.942e+004	2.943e+004	2.944e+004
26	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
27	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
29	Length (m)	2.945e+004	2.946e+004	2.947e+004	2.948e+004	2.949e+004
30	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
31	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
32	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
33	Length (m)	2.950e+004	2.951e+004	2.952e+004	2.953e+004	2.954e+004
34	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
35	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
36	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
37	Length (m)	2.955e+004	2.955e+004	2.955e+004	2.955e+004	2.955e+004
38	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
39	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
40	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
41	Length (m)	2.955e+004	2.956e+004	2.956e+004	2.957e+004	2.957e+004
42	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
43	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
44	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
45	Length (m)	2.958e+004	2.958e+004	2.959e+004	2.959e+004	2.960e+004
46	Bulk Density (kg/m3)	124.2	124.3	124.3	124.3	124.3
47	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
48	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
49	Length (m)	2.960e+004	2.961e+004	2.961e+004	2.962e+004	2.962e+004
50	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
51	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
52	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
53	Length (m)	2.963e+004	2.963e+004	2.964e+004	2.964e+004	2.965e+004
54	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
55	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
56	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
57	Length (m)	2.965e+004	2.965e+004	2.966e+004	2.966e+004	2.966e+004
58	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
59	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
60	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
61	Length (m)	2.966e+004	2.966e+004	2.967e+004	2.967e+004	2.967e+004
62	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
63	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
65	Length (m)	2.967e+004	2.968e+004	2.968e+004	2.968e+004	2.969e+004
66	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
67	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 645 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	2.969e+004	2.969e+004	2.970e+004	2.970e+004	2.970e+004
14	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
15	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
17	Length (m)	2.971e+004	2.971e+004	2.971e+004	2.972e+004	2.972e+004
18	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
19	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
21	Length (m)	2.973e+004	2.973e+004	2.974e+004	2.974e+004	2.975e+004
22	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
23	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
25	Length (m)	2.975e+004	2.976e+004	2.976e+004	2.977e+004	2.977e+004
26	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
27	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
29	Length (m)	2.978e+004	2.978e+004	2.979e+004	2.979e+004	2.980e+004
30	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
31	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
32	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
33	Length (m)	2.980e+004	2.981e+004	2.981e+004	2.982e+004	2.982e+004
34	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
35	Bulk Velocity (m/s)	2.373	2.373	2.372	2.372	2.372
36	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
37	Length (m)	2.983e+004	2.983e+004	2.984e+004	2.984e+004	2.985e+004
38	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
39	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
40	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
41	Length (m)	2.985e+004	2.986e+004	2.986e+004	2.987e+004	2.987e+004
42	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
43	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
44	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
45	Length (m)	2.988e+004	2.988e+004	2.989e+004	2.989e+004	2.990e+004
46	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
47	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
48	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
49	Length (m)	2.990e+004	2.990e+004	2.991e+004	2.991e+004	2.991e+004
50	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
51	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
52	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
53	Length (m)	2.991e+004	2.992e+004	2.992e+004	2.993e+004	2.993e+004
54	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
55	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
56	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
57	Length (m)	2.994e+004	2.994e+004	2.994e+004	2.995e+004	2.995e+004
58	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
59	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
60	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
61	Length (m)	2.996e+004	2.996e+004	2.997e+004	2.997e+004	2.998e+004
62	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
63	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
65	Length (m)	2.998e+004	2.999e+004	2.999e+004	3.000e+004	3.000e+004
66	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
67	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 646 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.001e+004	3.001e+004	3.002e+004	3.002e+004	3.003e+004
14	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
15	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
17	Length (m)	3.003e+004	3.004e+004	3.004e+004	3.005e+004	3.005e+004
18	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
19	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
21	Length (m)	3.006e+004	3.006e+004	3.007e+004	3.007e+004	3.008e+004
22	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
23	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
25	Length (m)	3.008e+004	3.009e+004	3.009e+004	3.010e+004	3.010e+004
26	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
27	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
29	Length (m)	3.011e+004	3.011e+004	3.012e+004	3.012e+004	3.013e+004
30	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
31	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
32	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
33	Length (m)	3.013e+004	3.014e+004	3.014e+004	3.015e+004	3.015e+004
34	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
35	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
36	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
37	Length (m)	3.016e+004	3.016e+004	3.017e+004	3.017e+004	3.018e+004
38	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
39	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
40	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
41	Length (m)	3.018e+004	3.019e+004	3.019e+004	3.020e+004	3.020e+004
42	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
43	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
44	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
45	Length (m)	3.021e+004	3.021e+004	3.022e+004	3.022e+004	3.023e+004
46	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
47	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
48	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
49	Length (m)	3.023e+004	3.024e+004	3.024e+004	3.025e+004	3.025e+004
50	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
51	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
52	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
53	Length (m)	3.026e+004	3.026e+004	3.027e+004	3.027e+004	3.028e+004
54	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
55	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
56	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
57	Length (m)	3.028e+004	3.029e+004	3.029e+004	3.030e+004	3.030e+004
58	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
59	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
60	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
61	Length (m)	3.031e+004	3.031e+004	3.032e+004	3.032e+004	3.033e+004
62	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
63	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
65	Length (m)	3.033e+004	3.034e+004	3.035e+004	3.036e+004	3.036e+004
66	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
67	Bulk Velocity (m/s)	2.372	2.371	2.371	2.371	2.371
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 647 of 937					

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.037e+004	3.039e+004	3.040e+004	3.041e+004	3.043e+004
14	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
15	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
17	Length (m)	3.044e+004	3.045e+004	3.045e+004	3.045e+004	3.045e+004
18	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
19	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
21	Length (m)	3.046e+004	3.046e+004	3.047e+004	3.047e+004	3.048e+004
22	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
23	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
25	Length (m)	3.048e+004	3.049e+004	3.049e+004	3.050e+004	3.050e+004
26	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
27	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
29	Length (m)	3.051e+004	3.051e+004	3.052e+004	3.052e+004	3.053e+004
30	Bulk Density (kg/m3)	124.3	124.3	124.4	124.4	124.4
31	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
32	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
33	Length (m)	3.053e+004	3.054e+004	3.054e+004	3.055e+004	3.055e+004
34	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
35	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
36	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
37	Length (m)	3.056e+004	3.056e+004	3.057e+004	3.057e+004	3.058e+004
38	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
39	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
40	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
41	Length (m)	3.058e+004	3.059e+004	3.059e+004	3.060e+004	3.060e+004
42	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
43	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
44	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
45	Length (m)	3.061e+004	3.061e+004	3.062e+004	3.062e+004	3.063e+004
46	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
47	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
48	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
49	Length (m)	3.063e+004	3.064e+004	3.064e+004	3.065e+004	3.065e+004
50	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
51	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
52	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
53	Length (m)	3.066e+004	3.066e+004	3.067e+004	3.067e+004	3.068e+004
54	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
55	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
56	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
57	Length (m)	3.068e+004	3.069e+004	3.069e+004	3.070e+004	3.070e+004
58	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
59	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
60	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
61	Length (m)	3.071e+004	3.071e+004	3.072e+004	3.072e+004	3.073e+004
62	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
63	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
65	Length (m)	3.073e+004	3.074e+004	3.074e+004	3.075e+004	3.075e+004
66	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
67	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 648 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.076e+004	3.076e+004	3.077e+004	3.077e+004	3.078e+004
14	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
15	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
17	Length (m)	3.078e+004	3.079e+004	3.079e+004	3.080e+004	3.080e+004
18	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
19	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
21	Length (m)	3.081e+004	3.081e+004	3.082e+004	3.082e+004	3.083e+004
22	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
23	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
25	Length (m)	3.083e+004	3.084e+004	3.084e+004	3.085e+004	3.085e+004
26	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
27	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
29	Length (m)	3.086e+004	3.086e+004	3.087e+004	3.087e+004	3.088e+004
30	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
31	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370
32	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
33	Length (m)	3.088e+004	3.089e+004	3.089e+004	3.090e+004	3.090e+004
34	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
35	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370
36	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
37	Length (m)	3.091e+004	3.091e+004	3.092e+004	3.092e+004	3.093e+004
38	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
39	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370
40	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
41	Length (m)	3.093e+004	3.094e+004	3.094e+004	3.095e+004	3.095e+004
42	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
43	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370
44	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
45	Length (m)	3.096e+004	3.096e+004	3.097e+004	3.097e+004	3.098e+004
46	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
47	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370
48	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
49	Length (m)	3.098e+004	3.099e+004	3.099e+004	3.100e+004	3.100e+004
50	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
51	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370
52	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
53	Length (m)	3.101e+004	3.103e+004	3.105e+004	3.107e+004	3.109e+004
54	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
55	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370
56	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
57	Length (m)	3.111e+004	3.111e+004	3.112e+004	3.112e+004	3.113e+004
58	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
59	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370
60	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
61	Length (m)	3.113e+004	3.114e+004	3.114e+004	3.115e+004	3.115e+004
62	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
63	Bulk Velocity (m/s)	2.370	2.370	2.370	2.370	2.370
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
65	Length (m)	3.116e+004	3.116e+004	3.117e+004	3.117e+004	3.118e+004
66	Bulk Density (kg/m3)	124.4	124.4	124.4	124.5	124.5
67	Bulk Velocity (m/s)	2.370	2.370	2.369	2.369	2.368
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.93	10.93
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.118e+004	3.119e+004	3.119e+004	3.119e+004	3.120e+004
14	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
15	Bulk Velocity (m/s)	2.368	2.368	2.368	2.368	2.368
16	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
17	Length (m)	3.120e+004	3.121e+004	3.122e+004	3.122e+004	3.123e+004
18	Bulk Density (kg/m3)	124.5	124.5	124.5	124.5	124.5
19	Bulk Velocity (m/s)	2.368	2.368	2.368	2.369	2.369
20	Erosion Velocity (m/s)	10.93	10.93	10.93	10.93	10.93
21	Length (m)	3.124e+004	3.124e+004	3.125e+004	3.125e+004	3.126e+004
22	Bulk Density (kg/m3)	124.5	124.4	124.4	124.4	124.4
23	Bulk Velocity (m/s)	2.369	2.369	2.370	2.370	2.370
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
25	Length (m)	3.126e+004	3.126e+004	3.127e+004	3.127e+004	3.128e+004
26	Bulk Density (kg/m3)	124.4	124.3	124.2	124.1	124.1
27	Bulk Velocity (m/s)	2.370	2.372	2.373	2.375	2.376
28	Erosion Velocity (m/s)	10.94	10.94	10.95	10.95	10.95
29	Length (m)	3.128e+004	3.129e+004	3.129e+004	3.130e+004	3.130e+004
30	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
31	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
32	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
33	Length (m)	3.131e+004	3.131e+004	3.132e+004	3.132e+004	3.133e+004
34	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
35	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
36	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
37	Length (m)	3.133e+004	3.134e+004	3.134e+004	3.135e+004	3.135e+004
38	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
39	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
40	Erosion Velocity (m/s)	10.96	10.96	10.96	10.96	10.96
41	Length (m)	3.136e+004	3.136e+004	3.137e+004	3.137e+004	3.138e+004
42	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
43	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
44	Erosion Velocity (m/s)	10.96	10.95	10.95	10.95	10.95
45	Length (m)	3.138e+004	3.139e+004	3.139e+004	3.140e+004	3.140e+004
46	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
47	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	3.141e+004	3.141e+004	3.142e+004	3.142e+004	3.143e+004
50	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
51	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	3.143e+004	3.144e+004	3.144e+004	3.145e+004	3.145e+004
54	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
55	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	3.146e+004	3.146e+004	3.147e+004	3.147e+004	3.148e+004
58	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
59	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	3.148e+004	3.149e+004	3.149e+004	3.150e+004	3.150e+004
62	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
63	Bulk Velocity (m/s)	2.378	2.378	2.378	2.378	2.378
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	3.151e+004	3.151e+004	3.152e+004	3.152e+004	3.153e+004
66	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
67	Bulk Velocity (m/s)	2.378	2.378	2.377	2.377	2.377
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
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1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.153e+004	3.154e+004	3.154e+004	3.155e+004	3.155e+004
14	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
15	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	3.156e+004	3.157e+004	3.157e+004	3.158e+004	3.158e+004
18	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
19	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	3.159e+004	3.159e+004	3.160e+004	3.160e+004	3.161e+004
22	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
23	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	3.161e+004	3.162e+004	3.162e+004	3.163e+004	3.163e+004
26	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
27	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	3.164e+004	3.164e+004	3.165e+004	3.165e+004	3.166e+004
30	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
31	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	3.166e+004	3.167e+004	3.167e+004	3.168e+004	3.168e+004
34	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
35	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	3.169e+004	3.169e+004	3.169e+004	3.169e+004	3.170e+004
38	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
39	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	3.170e+004	3.170e+004	3.170e+004	3.171e+004	3.171e+004
42	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
43	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	3.171e+004	3.172e+004	3.172e+004	3.173e+004	3.173e+004
46	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
47	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	3.174e+004	3.174e+004	3.174e+004	3.174e+004	3.175e+004
50	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
51	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	3.175e+004	3.176e+004	3.176e+004	3.177e+004	3.177e+004
54	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
55	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	3.178e+004	3.179e+004	3.179e+004	3.180e+004	3.180e+004
58	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
59	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	3.181e+004	3.182e+004	3.182e+004	3.183e+004	3.183e+004
62	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
63	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	3.184e+004	3.184e+004	3.185e+004	3.185e+004	3.186e+004
66	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
67	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 651 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.186e+004	3.187e+004	3.187e+004	3.188e+004	3.188e+004
14	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
15	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	3.189e+004	3.189e+004	3.190e+004	3.190e+004	3.191e+004
18	Bulk Density (kg/m3)	124.0	124.0	124.0	124.1	124.1
19	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	3.191e+004	3.192e+004	3.192e+004	3.193e+004	3.193e+004
22	Bulk Density (kg/m3)	124.1	124.1	124.0	124.0	124.0
23	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	3.194e+004	3.194e+004	3.195e+004	3.195e+004	3.196e+004
26	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
27	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	3.196e+004	3.197e+004	3.197e+004	3.198e+004	3.198e+004
30	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
31	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	3.199e+004	3.199e+004	3.200e+004	3.200e+004	3.201e+004
34	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
35	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	3.201e+004	3.202e+004	3.202e+004	3.203e+004	3.203e+004
38	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
39	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	3.204e+004	3.204e+004	3.205e+004	3.205e+004	3.206e+004
42	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
43	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	3.206e+004	3.207e+004	3.207e+004	3.208e+004	3.208e+004
46	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
47	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	3.209e+004	3.209e+004	3.210e+004	3.210e+004	3.211e+004
50	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
51	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.376
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	3.211e+004	3.212e+004	3.212e+004	3.213e+004	3.213e+004
54	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
55	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	3.214e+004	3.214e+004	3.215e+004	3.215e+004	3.216e+004
58	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
59	Bulk Velocity (m/s)	2.376	2.376	2.376	2.377	2.377
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	3.216e+004	3.217e+004	3.217e+004	3.218e+004	3.218e+004
62	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
63	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	3.219e+004	3.219e+004	3.219e+004	3.219e+004	3.219e+004
66	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
67	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 652 of 937

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.219e+004	3.219e+004	3.220e+004	3.220e+004	3.221e+004
14	Bulk Density (kg/m3)	124.1	124.0	124.0	124.0	124.0
15	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	3.221e+004	3.221e+004	3.222e+004	3.222e+004	3.223e+004
18	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
19	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	3.223e+004	3.224e+004	3.226e+004	3.227e+004	3.228e+004
22	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
23	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	3.230e+004	3.231e+004	3.232e+004	3.234e+004	3.235e+004
26	Bulk Density (kg/m3)	124.0	124.0	124.0	124.0	124.0
27	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	3.236e+004	3.238e+004	3.239e+004	3.240e+004	3.242e+004
30	Bulk Density (kg/m3)	124.0	124.1	124.1	124.1	124.1
31	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	3.243e+004	3.244e+004	3.246e+004	3.247e+004	3.248e+004
34	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
35	Bulk Velocity (m/s)	2.377	2.377	2.377	2.377	2.377
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	3.250e+004	3.251e+004	3.252e+004	3.254e+004	3.255e+004
38	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
39	Bulk Velocity (m/s)	2.377	2.377	2.377	2.376	2.376
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	3.257e+004	3.257e+004	3.258e+004	3.258e+004	3.259e+004
42	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
43	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	3.260e+004	3.260e+004	3.261e+004	3.261e+004	3.262e+004
46	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
47	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	3.263e+004	3.263e+004	3.263e+004	3.263e+004	3.263e+004
50	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
51	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	3.264e+004	3.264e+004	3.264e+004	3.264e+004	3.264e+004
54	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
55	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	3.265e+004	3.265e+004	3.266e+004	3.266e+004	3.267e+004
58	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
59	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	3.267e+004	3.268e+004	3.268e+004	3.269e+004	3.269e+004
62	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
63	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	3.270e+004	3.270e+004	3.271e+004	3.271e+004	3.272e+004
66	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
67	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 653 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.272e+004	3.273e+004	3.273e+004	3.274e+004	3.274e+004
14	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
15	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	3.275e+004	3.275e+004	3.276e+004	3.276e+004	3.277e+004
18	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
19	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	3.277e+004	3.278e+004	3.278e+004	3.279e+004	3.279e+004
22	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
23	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	3.280e+004	3.280e+004	3.281e+004	3.281e+004	3.282e+004
26	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
27	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	3.282e+004	3.283e+004	3.283e+004	3.284e+004	3.284e+004
30	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
31	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	3.285e+004	3.285e+004	3.286e+004	3.286e+004	3.287e+004
34	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
35	Bulk Velocity (m/s)	2.376	2.376	2.376	2.376	2.376
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	3.287e+004	3.288e+004	3.288e+004	3.289e+004	3.289e+004
38	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
39	Bulk Velocity (m/s)	2.376	2.375	2.375	2.375	2.375
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	3.290e+004	3.290e+004	3.291e+004	3.291e+004	3.292e+004
42	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
43	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	3.292e+004	3.293e+004	3.293e+004	3.294e+004	3.294e+004
46	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
47	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	3.295e+004	3.295e+004	3.296e+004	3.296e+004	3.297e+004
50	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
51	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	3.297e+004	3.298e+004	3.298e+004	3.299e+004	3.299e+004
54	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
55	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	3.300e+004	3.300e+004	3.301e+004	3.301e+004	3.302e+004
58	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
59	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	3.302e+004	3.302e+004	3.302e+004	3.303e+004	3.303e+004
62	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
63	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	3.303e+004	3.304e+004	3.304e+004	3.305e+004	3.305e+004
66	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
67	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 654 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
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9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.306e+004	3.306e+004	3.306e+004	3.307e+004	3.307e+004
14	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
15	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	3.307e+004	3.308e+004	3.308e+004	3.309e+004	3.309e+004
18	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
19	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	3.310e+004	3.310e+004	3.311e+004	3.311e+004	3.312e+004
22	Bulk Density (kg/m3)	124.1	124.1	124.1	124.1	124.1
23	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	3.312e+004	3.313e+004	3.313e+004	3.314e+004	3.314e+004
26	Bulk Density (kg/m3)	124.1	124.1	124.2	124.2	124.2
27	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	3.315e+004	3.315e+004	3.316e+004	3.316e+004	3.317e+004
30	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
31	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	3.317e+004	3.318e+004	3.318e+004	3.319e+004	3.319e+004
34	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
35	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	3.320e+004	3.320e+004	3.321e+004	3.321e+004	3.322e+004
38	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
39	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	3.322e+004	3.323e+004	3.323e+004	3.324e+004	3.324e+004
42	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
43	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	3.325e+004	3.325e+004	3.326e+004	3.326e+004	3.327e+004
46	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
47	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	3.327e+004	3.328e+004	3.328e+004	3.329e+004	3.329e+004
50	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
51	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	3.330e+004	3.330e+004	3.331e+004	3.331e+004	3.332e+004
54	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
55	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	3.332e+004	3.333e+004	3.333e+004	3.334e+004	3.334e+004
58	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
59	Bulk Velocity (m/s)	2.375	2.375	2.375	2.375	2.375
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	3.335e+004	3.335e+004	3.336e+004	3.336e+004	3.337e+004
62	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
63	Bulk Velocity (m/s)	2.375	2.374	2.374	2.374	2.374
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	3.337e+004	3.338e+004	3.338e+004	3.339e+004	3.339e+004
66	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
67	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 655 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.340e+004	3.340e+004	3.341e+004	3.341e+004	3.342e+004
14	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
15	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	3.342e+004	3.343e+004	3.343e+004	3.344e+004	3.344e+004
18	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
19	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	3.345e+004	3.345e+004	3.346e+004	3.346e+004	3.347e+004
22	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
23	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	3.347e+004	3.348e+004	3.348e+004	3.349e+004	3.349e+004
26	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
27	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	3.350e+004	3.350e+004	3.351e+004	3.351e+004	3.352e+004
30	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
31	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	3.352e+004	3.353e+004	3.353e+004	3.354e+004	3.354e+004
34	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
35	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	3.355e+004	3.355e+004	3.356e+004	3.356e+004	3.357e+004
38	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
39	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	3.357e+004	3.358e+004	3.358e+004	3.359e+004	3.359e+004
42	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
43	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	3.360e+004	3.360e+004	3.361e+004	3.361e+004	3.362e+004
46	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
47	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	3.362e+004	3.363e+004	3.363e+004	3.364e+004	3.364e+004
50	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
51	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
52	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
53	Length (m)	3.365e+004	3.365e+004	3.366e+004	3.366e+004	3.367e+004
54	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
55	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
56	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
57	Length (m)	3.367e+004	3.368e+004	3.368e+004	3.369e+004	3.369e+004
58	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
59	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
60	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
61	Length (m)	3.370e+004	3.370e+004	3.371e+004	3.371e+004	3.372e+004
62	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
63	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
64	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
65	Length (m)	3.372e+004	3.373e+004	3.373e+004	3.374e+004	3.374e+004
66	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
67	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
68	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 656 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
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8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.375e+004	3.375e+004	3.376e+004	3.376e+004	3.377e+004
14	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
15	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
16	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
17	Length (m)	3.377e+004	3.378e+004	3.378e+004	3.379e+004	3.379e+004
18	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
19	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
20	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
21	Length (m)	3.380e+004	3.380e+004	3.381e+004	3.381e+004	3.382e+004
22	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
23	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
24	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
25	Length (m)	3.382e+004	3.383e+004	3.383e+004	3.384e+004	3.384e+004
26	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
27	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
28	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
29	Length (m)	3.385e+004	3.385e+004	3.385e+004	3.385e+004	3.386e+004
30	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
31	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
32	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
33	Length (m)	3.386e+004	3.386e+004	3.386e+004	3.386e+004	3.387e+004
34	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
35	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
36	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
37	Length (m)	3.387e+004	3.387e+004	3.387e+004	3.388e+004	3.388e+004
38	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
39	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
40	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
41	Length (m)	3.388e+004	3.389e+004	3.389e+004	3.389e+004	3.389e+004
42	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
43	Bulk Velocity (m/s)	2.374	2.374	2.374	2.374	2.374
44	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
45	Length (m)	3.390e+004	3.391e+004	3.392e+004	3.393e+004	3.394e+004
46	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
47	Bulk Velocity (m/s)	2.374	2.374	2.373	2.373	2.373
48	Erosion Velocity (m/s)	10.95	10.95	10.95	10.95	10.95
49	Length (m)	3.395e+004	3.396e+004	3.398e+004	3.399e+004	3.400e+004
50	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
51	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
52	Erosion Velocity (m/s)	10.95	10.94	10.94	10.94	10.94
53	Length (m)	3.401e+004	3.402e+004	3.403e+004	3.404e+004	3.405e+004
54	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
55	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
56	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
57	Length (m)	3.406e+004	3.408e+004	3.409e+004	3.410e+004	3.411e+004
58	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
59	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
60	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
61	Length (m)	3.412e+004	3.413e+004	3.414e+004	3.415e+004	3.416e+004
62	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
63	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
65	Length (m)	3.418e+004	3.419e+004	3.420e+004	3.421e+004	3.422e+004
66	Bulk Density (kg/m3)	124.2	124.2	124.2	124.2	124.2
67	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 657 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.423e+004	3.424e+004	3.425e+004	3.426e+004	3.428e+004
14	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
15	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
17	Length (m)	3.429e+004	3.429e+004	3.429e+004	3.429e+004	3.430e+004
18	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
19	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
21	Length (m)	3.430e+004	3.430e+004	3.431e+004	3.431e+004	3.432e+004
22	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
23	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
25	Length (m)	3.432e+004	3.433e+004	3.433e+004	3.434e+004	3.434e+004
26	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
27	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
29	Length (m)	3.435e+004	3.435e+004	3.436e+004	3.436e+004	3.437e+004
30	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
31	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
32	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
33	Length (m)	3.437e+004	3.438e+004	3.438e+004	3.439e+004	3.439e+004
34	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
35	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
36	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
37	Length (m)	3.440e+004	3.440e+004	3.441e+004	3.441e+004	3.442e+004
38	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
39	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
40	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
41	Length (m)	3.442e+004	3.443e+004	3.443e+004	3.444e+004	3.444e+004
42	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
43	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
44	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
45	Length (m)	3.445e+004	3.445e+004	3.446e+004	3.446e+004	3.447e+004
46	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
47	Bulk Velocity (m/s)	2.373	2.373	2.373	2.373	2.373
48	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
49	Length (m)	3.447e+004	3.448e+004	3.448e+004	3.449e+004	3.449e+004
50	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
51	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
52	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
53	Length (m)	3.450e+004	3.450e+004	3.451e+004	3.451e+004	3.452e+004
54	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
55	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
56	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
57	Length (m)	3.452e+004	3.453e+004	3.453e+004	3.454e+004	3.454e+004
58	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
59	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
60	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
61	Length (m)	3.455e+004	3.455e+004	3.456e+004	3.456e+004	3.457e+004
62	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
63	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
65	Length (m)	3.457e+004	3.458e+004	3.458e+004	3.459e+004	3.459e+004
66	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
67	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 658 of 937					

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc			
2							
3				Unit Set: Mixed			
4							
5				Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	Erosion						
10							
11	Erosion Table						
12							
13	Length (m)	3.460e+004	3.460e+004	3.461e+004	3.461e+004	3.462e+004	
14	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3	
15	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372	
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
17	Length (m)	3.462e+004	3.463e+004	3.463e+004	3.464e+004	3.464e+004	
18	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3	
19	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372	
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
21	Length (m)	3.465e+004	3.465e+004	3.465e+004	3.466e+004	3.466e+004	
22	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3	
23	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372	
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
25	Length (m)	3.467e+004	3.467e+004	3.468e+004	3.468e+004	3.469e+004	
26	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3	
27	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372	
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
29	Length (m)	3.469e+004	3.470e+004	3.470e+004	3.471e+004	3.471e+004	
30	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3	
31	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372	
32	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
33	Length (m)	3.472e+004	3.472e+004	3.473e+004	3.473e+004	3.474e+004	
34	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3	
35	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372	
36	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
37	Length (m)	3.474e+004	3.475e+004	3.475e+004	3.476e+004	3.476e+004	
38	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3	
39	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372	
40	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
41	Length (m)	3.477e+004	3.477e+004	3.478e+004	3.478e+004	3.479e+004	
42	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3	
43	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372	
44	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
45	Length (m)	3.479e+004	3.480e+004	3.480e+004	3.481e+004	3.481e+004	
46	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3	
47	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372	
48	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
49	Length (m)	3.482e+004	3.482e+004	3.483e+004	3.483e+004	3.484e+004	
50	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3	
51	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372	
52	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
53	Length (m)	3.484e+004	3.485e+004	3.485e+004	3.486e+004	3.486e+004	
54	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3	
55	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372	
56	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
57	Length (m)	3.487e+004	3.487e+004	3.488e+004	3.488e+004	3.489e+004	
58	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3	
59	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372	
60	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
61	Length (m)	3.489e+004	3.490e+004	3.490e+004	3.491e+004	3.491e+004	
62	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3	
63	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372	
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
65	Length (m)	3.492e+004	3.492e+004	3.493e+004	3.493e+004	3.494e+004	
66	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3	
67	Bulk Velocity (m/s)	2.372	2.371	2.371	2.371	2.371	
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94	
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 659 of 937						

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.494e+004	3.495e+004	3.495e+004	3.496e+004	3.497e+004
14	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
15	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
17	Length (m)	3.497e+004	3.497e+004	3.498e+004	3.498e+004	3.498e+004
18	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
19	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
21	Length (m)	3.499e+004	3.499e+004	3.500e+004	3.500e+004	3.501e+004
22	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
23	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
25	Length (m)	3.502e+004	3.502e+004	3.503e+004	3.503e+004	3.504e+004
26	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
27	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
29	Length (m)	3.504e+004	3.505e+004	3.505e+004	3.506e+004	3.506e+004
30	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
31	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
32	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
33	Length (m)	3.507e+004	3.507e+004	3.508e+004	3.508e+004	3.509e+004
34	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
35	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
36	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
37	Length (m)	3.509e+004	3.510e+004	3.510e+004	3.511e+004	3.511e+004
38	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
39	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
40	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
41	Length (m)	3.512e+004	3.512e+004	3.513e+004	3.513e+004	3.514e+004
42	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
43	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
44	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
45	Length (m)	3.514e+004	3.515e+004	3.515e+004	3.516e+004	3.516e+004
46	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
47	Bulk Velocity (m/s)	2.372	2.372	2.372	2.372	2.372
48	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
49	Length (m)	3.517e+004	3.517e+004	3.518e+004	3.518e+004	3.519e+004
50	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
51	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
52	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
53	Length (m)	3.519e+004	3.520e+004	3.520e+004	3.521e+004	3.521e+004
54	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
55	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
56	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
57	Length (m)	3.522e+004	3.522e+004	3.523e+004	3.523e+004	3.524e+004
58	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
59	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
60	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
61	Length (m)	3.524e+004	3.525e+004	3.525e+004	3.526e+004	3.526e+004
62	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
63	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
65	Length (m)	3.527e+004	3.527e+004	3.528e+004	3.528e+004	3.529e+004
66	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
67	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 660 of 937					

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.529e+004	3.530e+004	3.530e+004	3.531e+004	3.531e+004
14	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
15	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
17	Length (m)	3.532e+004	3.532e+004	3.533e+004	3.533e+004	3.534e+004
18	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
19	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
21	Length (m)	3.534e+004	3.535e+004	3.535e+004	3.536e+004	3.536e+004
22	Bulk Density (kg/m3)	124.3	124.4	124.4	124.4	124.4
23	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
24	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
25	Length (m)	3.537e+004	3.537e+004	3.538e+004	3.538e+004	3.539e+004
26	Bulk Density (kg/m3)	124.4	124.4	124.4	124.3	124.3
27	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
28	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
29	Length (m)	3.539e+004	3.540e+004	3.540e+004	3.541e+004	3.541e+004
30	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
31	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
32	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
33	Length (m)	3.542e+004	3.542e+004	3.543e+004	3.543e+004	3.544e+004
34	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
35	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
36	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
37	Length (m)	3.544e+004	3.545e+004	3.545e+004	3.546e+004	3.546e+004
38	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
39	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
40	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
41	Length (m)	3.547e+004	3.547e+004	3.548e+004	3.548e+004	3.549e+004
42	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
43	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
44	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
45	Length (m)	3.549e+004	3.550e+004	3.550e+004	3.551e+004	3.551e+004
46	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
47	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
48	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
49	Length (m)	3.552e+004	3.552e+004	3.553e+004	3.553e+004	3.554e+004
50	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
51	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
52	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
53	Length (m)	3.554e+004	3.555e+004	3.555e+004	3.556e+004	3.556e+004
54	Bulk Density (kg/m3)	124.3	124.3	124.3	124.3	124.3
55	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
56	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
57	Length (m)	3.557e+004	3.557e+004	3.557e+004	3.558e+004	3.558e+004
58	Bulk Density (kg/m3)	124.3	124.3	124.4	124.4	124.4
59	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
60	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
61	Length (m)	3.559e+004	3.559e+004	3.559e+004	3.560e+004	3.560e+004
62	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
63	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
64	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
65	Length (m)	3.560e+004	3.561e+004	3.561e+004	3.562e+004	3.562e+004
66	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
67	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
68	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 661 of 937					

1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Erosion					
10						
11	Erosion Table					
12						
13	Length (m)	3.563e+004	3.563e+004	3.564e+004	3.564e+004	3.565e+004
14	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
15	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
16	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
17	Length (m)	3.565e+004	3.566e+004	3.566e+004	3.567e+004	3.567e+004
18	Bulk Density (kg/m3)	124.4	124.4	124.4	124.4	124.4
19	Bulk Velocity (m/s)	2.371	2.371	2.371	2.371	2.371
20	Erosion Velocity (m/s)	10.94	10.94	10.94	10.94	10.94
21	Length (m)	3.568e+004				
22	Bulk Density (kg/m3)	124.4				
23	Bulk Velocity (m/s)	2.371				
24	Erosion Velocity (m/s)	10.94				
25	Hydrates					
26						
27	Model Options					
28						
29	Hydrate Model:			Ng & Robinson		
30	Override Ng and Robinson:			False		
31	Initial Pressure:			From a Correlation		
32	Initial Pressure User Input:			---		
33						
34	Hydrates Table					
35	Distance (m)	0.0000	5.000	10.00	15.00	20.00
36	Pressure (bar)	98.66	98.66	98.66	98.66	98.66
37	Temperature (C)	63.19	63.18	63.18	63.18	63.18
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	25.00	30.00	35.00	40.00	45.00
48	Pressure (bar)	98.66	98.66	98.66	98.66	98.66
49	Temperature (C)	63.18	63.17	63.17	63.17	63.16
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	50.00	55.00	60.00	65.00	70.00
60	Pressure (bar)	98.66	98.66	98.66	98.66	98.66
61	Temperature (C)	63.16	63.16	63.16	63.15	63.15
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 662 of 937					

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	75.00	80.00	85.00	90.00	95.00
16	Pressure (bar)	98.66	98.66	98.66	98.66	98.66
17	Temperature (C)	63.15	63.15	63.14	63.14	63.14
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	100.0	105.0	110.0	115.0	120.0
28	Pressure (bar)	98.66	98.65	98.65	98.65	98.65
29	Temperature (C)	63.14	63.13	63.13	63.13	63.13
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	125.0	130.0	135.0	140.0	145.0
40	Pressure (bar)	98.65	98.65	98.65	98.65	98.65
41	Temperature (C)	63.12	63.12	63.12	63.12	63.11
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	150.0	155.0	160.0	165.0	170.0
52	Pressure (bar)	98.65	98.65	98.65	98.65	98.65
53	Temperature (C)	63.11	63.11	63.11	63.10	63.10
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	175.0	180.0	185.0	190.0	195.0
64	Pressure (bar)	98.65	98.65	98.65	98.65	98.65
65	Temperature (C)	63.10	63.10	63.09	63.09	63.09
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 663 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	200.1	205.1	210.1	215.1	220.1
20	Pressure (bar)	98.65	98.65	98.65	98.65	98.65
21	Temperature (C)	63.09	63.08	63.08	63.08	63.08
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	225.1	230.1	235.1	240.1	245.1
32	Pressure (bar)	98.65	98.65	98.65	98.65	98.65
33	Temperature (C)	63.07	63.07	63.07	63.07	63.06
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	250.1	255.1	260.1	265.1	270.1
44	Pressure (bar)	98.65	98.65	98.65	98.65	98.65
45	Temperature (C)	63.06	63.06	63.06	63.05	63.05
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	275.1	280.1	285.1	290.1	295.1
56	Pressure (bar)	98.65	98.65	98.65	98.65	98.65
57	Temperature (C)	63.05	63.05	63.04	63.04	63.04
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	300.1	305.1	310.1	315.1	320.1
68	Pressure (bar)	98.65	98.64	98.64	98.64	98.64
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	63.04	63.03	63.03	63.03	63.03
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	325.1	330.1	335.1	340.1	345.1
24	Pressure (bar)	98.64	98.64	98.65	98.65	98.65
25	Temperature (C)	63.02	63.02	63.02	63.02	63.02
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	350.1	355.1	360.1	365.1	370.1
36	Pressure (bar)	98.65	98.65	98.64	98.64	98.64
37	Temperature (C)	63.01	63.01	63.01	63.00	63.00
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	375.1	380.0	385.0	390.0	395.0
48	Pressure (bar)	98.64	98.64	98.63	98.63	98.63
49	Temperature (C)	63.00	63.00	62.99	62.99	62.99
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	400.0	405.0	410.0	415.0	420.0
60	Pressure (bar)	98.63	98.63	98.63	98.63	98.63
61	Temperature (C)	62.98	62.98	62.98	62.98	62.97
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	425.0	430.0	435.0	440.0	445.0
16	Pressure (bar)	98.63	98.63	98.63	98.63	98.63
17	Temperature (C)	62.97	62.97	62.97	62.96	62.96
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	450.0	455.0	460.0	465.0	470.0
28	Pressure (bar)	98.63	98.62	98.62	98.62	98.62
29	Temperature (C)	62.96	62.96	62.95	62.95	62.95
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	475.0	480.0	485.0	490.0	495.0
40	Pressure (bar)	98.62	98.62	98.62	98.62	98.62
41	Temperature (C)	62.95	62.94	62.94	62.94	62.94
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	500.0	505.0	510.0	515.0	520.0
52	Pressure (bar)	98.62	98.62	98.62	98.62	98.62
53	Temperature (C)	62.93	62.93	62.93	62.93	62.92
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	525.0	530.0	535.0	540.0	545.0
64	Pressure (bar)	98.62	98.62	98.62	98.62	98.62
65	Temperature (C)	62.92	62.92	62.92	62.91	62.91
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 666 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	550.0	555.0	560.0	565.0	570.0
20	Pressure (bar)	98.62	98.62	98.62	98.62	98.62
21	Temperature (C)	62.91	62.91	62.90	62.90	62.90
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	575.0	580.0	585.0	590.0	595.0
32	Pressure (bar)	98.62	98.62	98.62	98.62	98.62
33	Temperature (C)	62.90	62.89	62.89	62.89	62.89
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	600.0	605.0	610.0	615.0	620.0
44	Pressure (bar)	98.62	98.62	98.62	98.61	98.61
45	Temperature (C)	62.88	62.88	62.88	62.88	62.87
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	625.0	630.0	635.0	640.0	645.0
56	Pressure (bar)	98.61	98.61	98.61	98.61	98.61
57	Temperature (C)	62.87	62.87	62.86	62.86	62.86
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	650.0	655.0	660.0	665.0	670.0
68	Pressure (bar)	98.61	98.61	98.61	98.61	98.61
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1	 Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	62.86	62.85	62.85	62.85	62.85
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	675.0	682.0	689.0	696.0	703.0
24	Pressure (bar)	98.61	98.61	98.61	98.61	98.61
25	Temperature (C)	62.84	62.84	62.84	62.83	62.83
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	710.0	712.0	714.0	716.0	718.0
36	Pressure (bar)	98.61	98.61	98.61	98.61	98.61
37	Temperature (C)	62.83	62.83	62.83	62.82	62.82
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	720.0	722.0	724.0	726.0	728.0
48	Pressure (bar)	98.61	98.61	98.61	98.61	98.61
49	Temperature (C)	62.82	62.82	62.82	62.82	62.82
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	730.0	732.0	734.0	736.0	738.0
60	Pressure (bar)	98.60	98.60	98.61	98.61	98.61
61	Temperature (C)	62.82	62.82	62.81	62.81	62.81
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 668 of 937					

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	740.0	742.0	744.0	746.0	748.0
16	Pressure (bar)	98.61	98.61	98.61	98.61	98.61
17	Temperature (C)	62.81	62.81	62.81	62.81	62.81
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	750.0	755.0	760.0	765.0	770.0
28	Pressure (bar)	98.61	98.61	98.61	98.61	98.60
29	Temperature (C)	62.81	62.80	62.80	62.80	62.80
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	775.0	780.0	785.0	790.1	795.1
40	Pressure (bar)	98.60	98.60	98.60	98.60	98.60
41	Temperature (C)	62.79	62.79	62.79	62.79	62.78
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	800.1	805.1	810.1	815.1	820.1
52	Pressure (bar)	98.60	98.60	98.60	98.60	98.60
53	Temperature (C)	62.78	62.78	62.78	62.77	62.77
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	825.1	830.1	835.1	840.1	845.1
64	Pressure (bar)	98.60	98.60	98.60	98.60	98.61
65	Temperature (C)	62.77	62.77	62.77	62.76	62.76
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	850.1	855.1	860.1	865.1	870.1
20	Pressure (bar)	98.61	98.60	98.60	98.60	98.60
21	Temperature (C)	62.76	62.76	62.75	62.75	62.75
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	875.1	880.1	885.1	890.1	895.1
32	Pressure (bar)	98.60	98.60	98.60	98.60	98.60
33	Temperature (C)	62.74	62.74	62.74	62.74	62.73
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	900.1	905.1	910.1	915.1	920.1
44	Pressure (bar)	98.60	98.60	98.60	98.60	98.60
45	Temperature (C)	62.73	62.73	62.73	62.72	62.72
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	925.1	930.1	935.1	940.1	945.1
56	Pressure (bar)	98.59	98.59	98.59	98.59	98.59
57	Temperature (C)	62.72	62.72	62.71	62.71	62.71
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	950.1	955.1	960.1	965.1	970.1
68	Pressure (bar)	98.59	98.59	98.59	98.59	98.59
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	62.71	62.70	62.70	62.70	62.70
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	975.1	980.1	985.1	990.1	995.1
24	Pressure (bar)	98.59	98.59	98.59	98.59	98.59
25	Temperature (C)	62.69	62.69	62.69	62.69	62.68
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1000	1005	1010	1015	1020
36	Pressure (bar)	98.59	98.59	98.59	98.59	98.59
37	Temperature (C)	62.68	62.68	62.68	62.67	62.67
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1025	1030	1035	1040	1045
48	Pressure (bar)	98.58	98.59	98.59	98.59	98.59
49	Temperature (C)	62.67	62.67	62.66	62.66	62.66
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1050	1055	1060	1065	1070
60	Pressure (bar)	98.59	98.59	98.59	98.59	98.59
61	Temperature (C)	62.66	62.66	62.65	62.65	62.65
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1075	1080	1085	1090	1095
16	Pressure (bar)	98.58	98.58	98.58	98.58	98.58
17	Temperature (C)	62.65	62.64	62.64	62.64	62.63
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1100	1105	1110	1115	1120
28	Pressure (bar)	98.58	98.58	98.58	98.58	98.58
29	Temperature (C)	62.63	62.63	62.63	62.62	62.62
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1125	1130	1135	1140	1145
40	Pressure (bar)	98.58	98.58	98.58	98.58	98.58
41	Temperature (C)	62.62	62.62	62.61	62.61	62.61
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1150	1155	1160	1165	1170
52	Pressure (bar)	98.58	98.58	98.58	98.58	98.58
53	Temperature (C)	62.61	62.60	62.60	62.60	62.60
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1175	1180	1185	1190	1195
64	Pressure (bar)	98.58	98.57	98.57	98.57	98.57
65	Temperature (C)	62.59	62.59	62.59	62.59	62.58
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 672 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1200	1205	1210	1215	1220
20	Pressure (bar)	98.57	98.57	98.57	98.57	98.57
21	Temperature (C)	62.58	62.58	62.58	62.57	62.57
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1225	1230	1235	1240	1245
32	Pressure (bar)	98.57	98.57	98.57	98.57	98.57
33	Temperature (C)	62.57	62.57	62.56	62.56	62.56
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1250	1255	1260	1265	1270
44	Pressure (bar)	98.57	98.57	98.57	98.57	98.57
45	Temperature (C)	62.56	62.55	62.55	62.55	62.55
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1275	1280	1285	1290	1295
56	Pressure (bar)	98.57	98.57	98.57	98.57	98.57
57	Temperature (C)	62.55	62.54	62.54	62.54	62.53
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1300	1305	1310	1315	1320
68	Pressure (bar)	98.57	98.57	98.57	98.57	98.57
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	62.53	62.53	62.53	62.52	62.52
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1325	1330	1335	1340	1345
24	Pressure (bar)	98.57	98.57	98.56	98.56	98.56
25	Temperature (C)	62.52	62.52	62.51	62.51	62.51
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1350	1355	1360	1365	1370
36	Pressure (bar)	98.56	98.56	98.57	98.57	98.57
37	Temperature (C)	62.51	62.50	62.50	62.50	62.50
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1375	1380	1385	1390	1395
48	Pressure (bar)	98.57	98.58	98.59	98.59	98.60
49	Temperature (C)	62.50	62.50	62.50	62.50	62.50
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1400	1405	1410	1415	1420
60	Pressure (bar)	98.61	98.62	98.62	98.63	98.63
61	Temperature (C)	62.50	62.50	62.50	62.50	62.50
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1425	1430	1435	1440	1445
16	Pressure (bar)	98.64	98.64	98.64	98.64	98.64
17	Temperature (C)	62.50	62.50	62.50	62.50	62.49
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1450	1455	1460	1465	1470
28	Pressure (bar)	98.65	98.64	98.64	98.63	98.63
29	Temperature (C)	62.49	62.49	62.49	62.48	62.48
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1475	1480	1485	1490	1495
40	Pressure (bar)	98.63	98.63	98.63	98.64	98.64
41	Temperature (C)	62.48	62.47	62.47	62.47	62.47
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1500	1505	1510	1515	1520
52	Pressure (bar)	98.64	98.65	98.66	98.67	98.68
53	Temperature (C)	62.47	62.47	62.48	62.48	62.49
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.40	18.40	18.40	18.40	18.40
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1525	1530	1535	1540	1545
64	Pressure (bar)	98.69	98.53	98.37	98.21	98.06
65	Temperature (C)	62.49	62.46	62.42	62.38	62.35
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.40	18.39	18.38	18.38	18.37
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1550	1555	1560	1565	1570
20	Pressure (bar)	97.90	97.89	97.89	97.89	97.88
21	Temperature (C)	62.31	62.31	62.30	62.30	62.30
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.36	18.36	18.36	18.36	18.36
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1575	1580	1585	1590	1595
32	Pressure (bar)	97.88	97.88	97.88	97.88	97.88
33	Temperature (C)	62.29	62.29	62.29	62.29	62.28
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.36	18.36	18.36	18.36	18.36
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1600	1605	1610	1615	1620
44	Pressure (bar)	97.88	97.88	97.88	97.88	97.88
45	Temperature (C)	62.28	62.28	62.28	62.28	62.27
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.36	18.36	18.36	18.36	18.36
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1625	1630	1635	1640	1645
56	Pressure (bar)	97.89	97.88	97.88	97.88	97.88
57	Temperature (C)	62.27	62.27	62.27	62.26	62.26
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.36	18.36	18.36	18.36	18.36
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1650	1655	1660	1665	1670
68	Pressure (bar)	97.88	97.88	97.88	97.88	97.88
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 676 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	62.26	62.26	62.25	62.25	62.25
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.36	18.36	18.36	18.36	18.36
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1675	1680	1685	1690	1695
24	Pressure (bar)	97.88	97.88	97.88	97.88	97.88
25	Temperature (C)	62.25	62.24	62.24	62.24	62.24
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.36	18.36	18.36	18.36	18.36
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1700	1705	1710	1715	1720
36	Pressure (bar)	97.89	97.89	97.89	97.89	97.89
37	Temperature (C)	62.24	62.23	62.23	62.23	62.23
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.36	18.36	18.36	18.36	18.36
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1725	1730	1735	1740	1745
48	Pressure (bar)	97.89	97.89	97.88	97.88	97.88
49	Temperature (C)	62.22	62.22	62.22	62.22	62.21
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.36	18.36	18.36	18.36	18.36
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1750	1755	1760	1765	1770
60	Pressure (bar)	97.88	97.88	97.88	97.88	97.88
61	Temperature (C)	62.21	62.21	62.21	62.20	62.20
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.36	18.36	18.36	18.36	18.36
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1775	1780	1785	1790	1795
16	Pressure (bar)	97.88	97.88	97.88	97.88	97.88
17	Temperature (C)	62.20	62.20	62.19	62.19	62.19
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.36	18.36	18.36	18.36	18.36
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1800	1805	1810	1815	1820
28	Pressure (bar)	97.88	97.88	97.88	97.87	97.87
29	Temperature (C)	62.19	62.18	62.18	62.18	62.18
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.36	18.36	18.36	18.36	18.36
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1825	1830	1835	1840	1845
40	Pressure (bar)	97.87	97.87	97.87	97.87	97.87
41	Temperature (C)	62.17	62.17	62.17	62.16	62.16
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.36	18.36	18.36	18.36	18.36
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1850	1855	1860	1865	1870
52	Pressure (bar)	97.87	97.87	97.87	97.87	97.87
53	Temperature (C)	62.16	62.16	62.15	62.15	62.15
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.36	18.36	18.36	18.36	18.36
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1875	1880	1885	1890	1895
64	Pressure (bar)	97.87	97.87	97.87	97.87	97.86
65	Temperature (C)	62.15	62.14	62.14	62.14	62.14
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.36	18.36	18.36	18.36	18.36
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1900	1905	1910	1915	1920
20	Pressure (bar)	97.86	97.86	97.86	97.86	97.86
21	Temperature (C)	62.13	62.13	62.13	62.13	62.12
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.36	18.36	18.36	18.36	18.36
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1925	1930	1935	1940	1945
32	Pressure (bar)	97.86	97.86	97.86	97.86	97.86
33	Temperature (C)	62.12	62.12	62.12	62.11	62.11
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.36	18.36	18.36	18.36	18.36
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1950	1955	1960	1965	1970
44	Pressure (bar)	97.86	97.86	97.86	97.86	97.86
45	Temperature (C)	62.11	62.11	62.10	62.10	62.10
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.36	18.36	18.35	18.35	18.35
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1975	1980	1985	1990	1995
56	Pressure (bar)	97.86	97.86	97.86	97.86	97.86
57	Temperature (C)	62.10	62.10	62.09	62.09	62.09
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.35	18.35	18.35	18.36	18.36
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2000	2005	2010	2015	2020
68	Pressure (bar)	97.86	97.86	97.86	97.86	97.86
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	62.09	62.08	62.08	62.08	62.08
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.36	18.36	18.36	18.35	18.35
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2025	2030	2035	2040	2045
24	Pressure (bar)	97.86	97.86	97.86	97.86	97.86
25	Temperature (C)	62.07	62.07	62.07	62.07	62.06
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2050	2055	2060	2065	2070
36	Pressure (bar)	97.85	97.85	97.85	97.85	97.85
37	Temperature (C)	62.06	62.06	62.06	62.05	62.05
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2075	2080	2085	2090	2095
48	Pressure (bar)	97.85	97.85	97.85	97.85	97.85
49	Temperature (C)	62.05	62.05	62.04	62.04	62.04
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2100	2105	2110	2115	2120
60	Pressure (bar)	97.85	97.85	97.85	97.85	97.85
61	Temperature (C)	62.04	62.03	62.03	62.03	62.03
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2125	2130	2135	2140	2145
16	Pressure (bar)	97.85	97.85	97.85	97.85	97.85
17	Temperature (C)	62.02	62.02	62.02	62.02	62.01
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2150	2155	2160	2165	2170
28	Pressure (bar)	97.85	97.85	97.85	97.85	97.85
29	Temperature (C)	62.01	62.01	62.01	62.00	62.00
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2175	2180	2185	2191	2196
40	Pressure (bar)	97.85	97.85	97.85	97.85	97.85
41	Temperature (C)	62.00	62.00	61.99	61.99	61.99
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2201	2206	2211	2216	2221
52	Pressure (bar)	97.85	97.85	97.84	97.84	97.84
53	Temperature (C)	61.99	61.98	61.98	61.98	61.98
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2226	2231	2236	2241	2246
64	Pressure (bar)	97.84	97.84	97.84	97.84	97.84
65	Temperature (C)	61.97	61.97	61.97	61.97	61.97
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2251	2256	2261	2266	2271
20	Pressure (bar)	97.84	97.84	97.84	97.84	97.84
21	Temperature (C)	61.96	61.96	61.96	61.96	61.95
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2276	2281	2286	2291	2296
32	Pressure (bar)	97.84	97.84	97.84	97.84	97.84
33	Temperature (C)	61.95	61.95	61.95	61.94	61.94
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2301	2306	2311	2316	2321
44	Pressure (bar)	97.84	97.84	97.84	97.84	97.84
45	Temperature (C)	61.94	61.94	61.93	61.93	61.93
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2326	2331	2336	2341	2346
56	Pressure (bar)	97.84	97.84	97.84	97.84	97.84
57	Temperature (C)	61.93	61.92	61.92	61.92	61.92
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2351	2356	2361	2366	2371
68	Pressure (bar)	97.84	97.84	97.84	97.84	97.84
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4				Date/Time: Mon Jun 13 15:02:44 2022		
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	61.91	61.91	61.91	61.91	61.90
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2376	2381	2386	2391	2396
24	Pressure (bar)	97.84	97.84	97.83	97.83	97.83
25	Temperature (C)	61.90	61.90	61.90	61.89	61.89
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2401	2406	2411	2416	2421
36	Pressure (bar)	97.83	97.83	97.83	97.83	97.83
37	Temperature (C)	61.89	61.89	61.88	61.88	61.88
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2426	2431	2436	2441	2446
48	Pressure (bar)	97.83	97.83	97.83	97.83	97.83
49	Temperature (C)	61.88	61.87	61.87	61.87	61.87
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2451	2456	2461	2466	2471
60	Pressure (bar)	97.83	97.83	97.83	97.83	97.83
61	Temperature (C)	61.86	61.86	61.86	61.86	61.85
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2476	2481	2486	2491	2496
16	Pressure (bar)	97.83	97.83	97.83	97.83	97.83
17	Temperature (C)	61.85	61.85	61.85	61.84	61.84
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2501	2506	2511	2516	2521
28	Pressure (bar)	97.83	97.83	97.83	97.83	97.83
29	Temperature (C)	61.84	61.84	61.84	61.83	61.83
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2526	2531	2536	2541	2546
40	Pressure (bar)	97.83	97.83	97.83	97.83	97.83
41	Temperature (C)	61.83	61.83	61.82	61.82	61.82
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2551	2556	2561	2566	2571
52	Pressure (bar)	97.83	97.83	97.83	97.83	97.83
53	Temperature (C)	61.82	61.81	61.81	61.81	61.81
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2576	2581	2586	2591	2596
64	Pressure (bar)	97.83	97.82	97.82	97.82	97.82
65	Temperature (C)	61.80	61.80	61.80	61.80	61.79
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 684 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2601	2606	2611	2616	2621
20	Pressure (bar)	97.82	97.82	97.82	97.82	97.82
21	Temperature (C)	61.79	61.79	61.79	61.78	61.78
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2626	2631	2636	2641	2646
32	Pressure (bar)	97.82	97.82	97.82	97.82	97.82
33	Temperature (C)	61.78	61.78	61.78	61.77	61.77
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2651	2656	2661	2666	2671
44	Pressure (bar)	97.82	97.82	97.82	97.82	97.82
45	Temperature (C)	61.77	61.77	61.76	61.76	61.76
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2676	2681	2686	2691	2696
56	Pressure (bar)	97.82	97.82	97.82	97.82	97.82
57	Temperature (C)	61.76	61.75	61.75	61.75	61.75
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2701	2706	2711	2716	2721
68	Pressure (bar)	97.82	97.82	97.82	97.82	97.82
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 685 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	61.74	61.74	61.74	61.74	61.73
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2726	2731	2736	2741	2746
24	Pressure (bar)	97.81	97.81	97.81	97.81	97.81
25	Temperature (C)	61.73	61.73	61.73	61.72	61.72
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2751	2756	2761	2766	2771
36	Pressure (bar)	97.81	97.81	97.81	97.81	97.81
37	Temperature (C)	61.72	61.72	61.71	61.71	61.71
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2776	2781	2786	2791	2796
48	Pressure (bar)	97.81	97.81	97.81	97.81	97.81
49	Temperature (C)	61.71	61.70	61.70	61.70	61.70
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2801	2806	2811	2816	2821
60	Pressure (bar)	97.81	97.81	97.81	97.81	97.81
61	Temperature (C)	61.70	61.69	61.69	61.69	61.69
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2826	2831	2836	2841	2846
16	Pressure (bar)	97.81	97.81	97.81	97.81	97.81
17	Temperature (C)	61.68	61.68	61.68	61.68	61.67
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2851	2856	2861	2866	2871
28	Pressure (bar)	97.81	97.81	97.81	97.80	97.80
29	Temperature (C)	61.67	61.67	61.67	61.66	61.66
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2876	2881	2886	2891	2896
40	Pressure (bar)	97.80	97.80	97.81	97.81	97.81
41	Temperature (C)	61.66	61.66	61.65	61.65	61.65
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2901	2906	2911	2916	2921
52	Pressure (bar)	97.81	97.81	97.81	97.81	97.81
53	Temperature (C)	61.65	61.65	61.64	61.64	61.64
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2926	2931	2936	2941	2946
64	Pressure (bar)	97.81	97.81	97.80	97.80	97.80
65	Temperature (C)	61.64	61.63	61.63	61.63	61.62
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2951	2956	2961	2966	2971
20	Pressure (bar)	97.80	97.80	97.81	97.81	97.81
21	Temperature (C)	61.62	61.62	61.62	61.62	61.62
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2976	2981	2986	2991	2996
32	Pressure (bar)	97.82	97.82	97.82	97.82	97.82
33	Temperature (C)	61.62	61.61	61.61	61.61	61.61
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.35	18.35	18.35	18.35	18.35
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3001	3006	3011	3016	3021
44	Pressure (bar)	97.81	97.83	97.85	97.87	97.89
45	Temperature (C)	61.60	61.61	61.62	61.62	61.63
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.35	18.35	18.35	18.36	18.36
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3026	3031	3036	3041	3046
56	Pressure (bar)	97.91	97.91	97.92	97.93	97.93
57	Temperature (C)	61.63	61.63	61.63	61.63	61.64
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.36	18.36	18.36	18.36	18.36
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3051	3056	3061	3066	3071
68	Pressure (bar)	97.94	97.94	97.95	97.95	97.95
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	61.64	61.63	61.63	61.63	61.63
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.36	18.36	18.36	18.36	18.36
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3076	3081	3086	3091	3096
24	Pressure (bar)	97.96	97.81	97.65	97.50	97.35
25	Temperature (C)	61.63	61.60	61.56	61.52	61.49
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.36	18.35	18.34	18.33	18.33
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3101	3106	3111	3116	3121
36	Pressure (bar)	97.20	97.19	97.19	97.19	97.18
37	Temperature (C)	61.45	61.45	61.44	61.44	61.44
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.32	18.32	18.32	18.32	18.32
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3126	3131	3136	3141	3146
48	Pressure (bar)	97.18	97.18	97.18	97.18	97.18
49	Temperature (C)	61.43	61.43	61.43	61.43	61.43
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.32	18.32	18.32	18.32	18.32
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3151	3156	3161	3166	3171
60	Pressure (bar)	97.18	97.18	97.18	97.18	97.18
61	Temperature (C)	61.42	61.42	61.42	61.42	61.42
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.32	18.32	18.32	18.32	18.32
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3176	3181	3186	3192	3197
16	Pressure (bar)	97.18	97.18	97.18	97.18	97.18
17	Temperature (C)	61.41	61.41	61.41	61.40	61.40
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.32	18.32	18.32	18.32	18.32
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3202	3207	3212	3217	3222
28	Pressure (bar)	97.18	97.18	97.18	97.18	97.18
29	Temperature (C)	61.40	61.40	61.39	61.39	61.39
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.32	18.32	18.32	18.32	18.32
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3227	3232	3237	3242	3247
40	Pressure (bar)	97.18	97.18	97.18	97.18	97.18
41	Temperature (C)	61.39	61.38	61.38	61.38	61.38
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.32	18.32	18.32	18.32	18.32
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3252	3257	3262	3267	3272
52	Pressure (bar)	97.18	97.18	97.18	97.18	97.18
53	Temperature (C)	61.38	61.37	61.37	61.37	61.37
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.32	18.32	18.32	18.32	18.32
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3277	3282	3287	3292	3297
64	Pressure (bar)	97.18	97.18	97.18	97.18	97.18
65	Temperature (C)	61.36	61.36	61.36	61.36	61.35
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 690 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.32	18.32	18.32	18.32	18.32
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3302	3307	3312	3317	3322
20	Pressure (bar)	97.18	97.17	97.17	97.17	97.17
21	Temperature (C)	61.35	61.35	61.35	61.34	61.34
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.32	18.32	18.32	18.32	18.31
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3327	3332	3337	3342	3347
32	Pressure (bar)	97.17	97.17	97.17	97.17	97.17
33	Temperature (C)	61.34	61.34	61.33	61.33	61.33
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3352	3357	3362	3367	3372
44	Pressure (bar)	97.17	97.17	97.17	97.17	97.17
45	Temperature (C)	61.33	61.32	61.32	61.32	61.32
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3377	3382	3387	3392	3396
56	Pressure (bar)	97.17	97.17	97.17	97.17	97.17
57	Temperature (C)	61.31	61.31	61.31	61.31	61.31
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3401	3406	3411	3416	3421
68	Pressure (bar)	97.16	97.16	97.16	97.16	97.16
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	61.30	61.30	61.30	61.30	61.29
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3426	3431	3436	3441	3446
24	Pressure (bar)	97.16	97.16	97.16	97.16	97.16
25	Temperature (C)	61.29	61.29	61.29	61.28	61.28
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3451	3456	3461	3466	3471
36	Pressure (bar)	97.16	97.16	97.16	97.16	97.16
37	Temperature (C)	61.28	61.28	61.27	61.27	61.27
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3476	3481	3486	3491	3496
48	Pressure (bar)	97.16	97.16	97.16	97.16	97.16
49	Temperature (C)	61.27	61.26	61.26	61.26	61.26
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3501	3506	3511	3516	3521
60	Pressure (bar)	97.16	97.16	97.16	97.16	97.16
61	Temperature (C)	61.26	61.25	61.25	61.25	61.25
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3526	3531	3536	3541	3546
16	Pressure (bar)	97.16	97.16	97.16	97.16	97.16
17	Temperature (C)	61.24	61.24	61.24	61.24	61.23
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3551	3556	3561	3566	3571
28	Pressure (bar)	97.16	97.16	97.16	97.16	97.16
29	Temperature (C)	61.23	61.23	61.23	61.23	61.22
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3576	3581	3586	3591	3596
40	Pressure (bar)	97.16	97.16	97.16	97.16	97.16
41	Temperature (C)	61.22	61.22	61.22	61.21	61.21
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3601	3606	3611	3616	3621
52	Pressure (bar)	97.16	97.16	97.16	97.16	97.16
53	Temperature (C)	61.21	61.21	61.20	61.20	61.20
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3626	3631	3636	3641	3646
64	Pressure (bar)	97.16	97.15	97.15	97.15	97.15
65	Temperature (C)	61.20	61.19	61.19	61.19	61.19
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3651	3656	3661	3666	3671
20	Pressure (bar)	97.15	97.15	97.15	97.15	97.15
21	Temperature (C)	61.18	61.18	61.18	61.18	61.17
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3676	3681	3686	3691	3696
32	Pressure (bar)	97.15	97.15	97.15	97.15	97.15
33	Temperature (C)	61.17	61.17	61.17	61.17	61.16
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3701	3706	3711	3716	3721
44	Pressure (bar)	97.15	97.15	97.15	97.15	97.15
45	Temperature (C)	61.16	61.16	61.16	61.15	61.15
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3726	3731	3736	3741	3746
56	Pressure (bar)	97.15	97.15	97.15	97.15	97.15
57	Temperature (C)	61.15	61.15	61.14	61.14	61.14
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3751	3756	3761	3766	3771
68	Pressure (bar)	97.15	97.15	97.15	97.15	97.15
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	61.14	61.13	61.13	61.13	61.13
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3776	3781	3787	3792	3797
24	Pressure (bar)	97.15	97.15	97.15	97.15	97.15
25	Temperature (C)	61.12	61.12	61.12	61.12	61.12
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3802	3807	3812	3817	3822
36	Pressure (bar)	97.15	97.15	97.15	97.15	97.14
37	Temperature (C)	61.11	61.11	61.11	61.11	61.10
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3827	3832	3837	3842	3847
48	Pressure (bar)	97.14	97.14	97.14	97.14	97.14
49	Temperature (C)	61.10	61.10	61.10	61.09	61.09
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3852	3857	3862	3867	3872
60	Pressure (bar)	97.14	97.14	97.14	97.14	97.14
61	Temperature (C)	61.09	61.09	61.08	61.08	61.08
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3877	3882	3887	3892	3897
16	Pressure (bar)	97.14	97.14	97.14	97.14	97.14
17	Temperature (C)	61.08	61.07	61.07	61.07	61.07
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3902	3907	3912	3917	3922
28	Pressure (bar)	97.14	97.14	97.14	97.13	97.13
29	Temperature (C)	61.06	61.06	61.06	61.06	61.05
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3927	3932	3937	3942	3947
40	Pressure (bar)	97.13	97.13	97.13	97.13	97.13
41	Temperature (C)	61.05	61.05	61.05	61.04	61.04
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3952	3957	3962	3967	3972
52	Pressure (bar)	97.13	97.13	97.13	97.13	97.13
53	Temperature (C)	61.04	61.04	61.03	61.03	61.03
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3977	3982	3987	3992	3997
64	Pressure (bar)	97.13	97.13	97.13	97.13	97.13
65	Temperature (C)	61.03	61.02	61.02	61.02	61.02
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	4002	4007	4012	4017	4022
20	Pressure (bar)	97.13	97.13	97.13	97.13	97.13
21	Temperature (C)	61.02	61.01	61.01	61.01	61.01
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	4027	4032	4037	4042	4047
32	Pressure (bar)	97.13	97.13	97.13	97.13	97.13
33	Temperature (C)	61.00	61.00	61.00	61.00	60.99
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	4052	4057	4062	4067	4072
44	Pressure (bar)	97.13	97.13	97.13	97.13	97.13
45	Temperature (C)	60.99	60.99	60.99	60.99	60.98
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	4077	4084	4091	4098	4105
56	Pressure (bar)	97.13	97.13	97.13	97.13	97.13
57	Temperature (C)	60.98	60.98	60.97	60.97	60.97
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	4112	4120	4127	4135	4143
68	Pressure (bar)	97.13	97.12	97.12	97.12	97.12
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	60.96	60.96	60.96	60.95	60.95
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	4150	4158	4165	4173	4181
24	Pressure (bar)	97.12	97.12	97.12	97.12	97.12
25	Temperature (C)	60.95	60.94	60.94	60.94	60.93
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	4188	4196	4203	4211	4219
36	Pressure (bar)	97.12	97.12	97.12	97.12	97.12
37	Temperature (C)	60.93	60.93	60.92	60.92	60.92
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	4226	4234	4241	4249	4257
48	Pressure (bar)	97.12	97.12	97.12	97.12	97.12
49	Temperature (C)	60.91	60.91	60.90	60.90	60.90
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	4264	4272	4279	4287	4295
60	Pressure (bar)	97.12	97.11	97.11	97.11	97.11
61	Temperature (C)	60.89	60.89	60.88	60.88	60.88
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	4302	4304	4305	4306	4308
16	Pressure (bar)	97.11	97.11	97.11	97.11	97.11
17	Temperature (C)	60.87	60.87	60.87	60.87	60.87
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	4309	4314	4319	4324	4329
28	Pressure (bar)	97.11	97.11	97.11	97.11	97.11
29	Temperature (C)	60.87	60.87	60.87	60.86	60.86
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	4334	4339	4344	4349	4354
40	Pressure (bar)	97.12	97.12	97.12	97.12	97.12
41	Temperature (C)	60.86	60.86	60.86	60.85	60.85
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	4359	4364	4369	4374	4379
52	Pressure (bar)	97.12	97.12	97.11	97.11	97.11
53	Temperature (C)	60.85	60.85	60.84	60.84	60.84
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	4384	4389	4394	4399	4404
64	Pressure (bar)	97.11	97.11	97.11	97.11	97.11
65	Temperature (C)	60.84	60.83	60.83	60.83	60.83
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	4409	4414	4419	4424	4429
20	Pressure (bar)	97.11	97.11	97.11	97.10	97.10
21	Temperature (C)	60.82	60.82	60.82	60.82	60.81
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	4434	4439	4444	4449	4454
32	Pressure (bar)	97.10	97.10	97.10	97.10	97.10
33	Temperature (C)	60.81	60.81	60.81	60.80	60.80
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	4459	4464	4469	4474	4479
44	Pressure (bar)	97.10	97.10	97.10	97.10	97.10
45	Temperature (C)	60.80	60.80	60.79	60.79	60.79
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	4484	4489	4494	4499	4504
56	Pressure (bar)	97.10	97.10	97.10	97.10	97.10
57	Temperature (C)	60.79	60.78	60.78	60.78	60.78
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	4510	4515	4520	4525	4530
68	Pressure (bar)	97.10	97.10	97.10	97.10	97.10
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	60.78	60.77	60.77	60.77	60.77
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	4535	4540	4545	4550	4555
24	Pressure (bar)	97.10	97.09	97.09	97.09	97.09
25	Temperature (C)	60.76	60.76	60.76	60.76	60.75
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	4560	4565	4570	4575	4580
36	Pressure (bar)	97.09	97.09	97.09	97.09	97.09
37	Temperature (C)	60.75	60.75	60.75	60.74	60.74
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	4585	4590	4595	4600	4605
48	Pressure (bar)	97.09	97.09	97.09	97.09	97.09
49	Temperature (C)	60.74	60.74	60.73	60.73	60.73
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	4610	4623	4636	4649	4662
60	Pressure (bar)	97.09	97.09	97.09	97.09	97.09
61	Temperature (C)	60.73	60.72	60.72	60.71	60.70
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
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6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	4675	4677	4679	4681	4683
16	Pressure (bar)	97.09	97.09	97.09	97.09	97.09
17	Temperature (C)	60.70	60.70	60.70	60.69	60.69
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	4685	4690	4695	4700	4705
28	Pressure (bar)	97.09	97.08	97.08	97.08	97.08
29	Temperature (C)	60.69	60.69	60.69	60.68	60.68
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	4710	4715	4720	4725	4730
40	Pressure (bar)	97.08	97.08	97.08	97.08	97.08
41	Temperature (C)	60.68	60.68	60.67	60.67	60.67
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	4735	4740	4745	4750	4755
52	Pressure (bar)	97.08	97.08	97.08	97.08	97.08
53	Temperature (C)	60.67	60.67	60.66	60.66	60.66
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	4760	4765	4770	4775	4780
64	Pressure (bar)	97.08	97.08	97.08	97.08	97.08
65	Temperature (C)	60.66	60.65	60.65	60.65	60.65
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	4785	4790	4795	4800	4805
20	Pressure (bar)	97.08	97.08	97.08	97.08	97.08
21	Temperature (C)	60.64	60.64	60.64	60.64	60.64
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	4810	4815	4820	4825	4830
32	Pressure (bar)	97.08	97.08	97.08	97.08	97.08
33	Temperature (C)	60.63	60.63	60.63	60.63	60.62
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	4835	4840	4845	4850	4855
44	Pressure (bar)	97.08	97.08	97.08	97.08	97.08
45	Temperature (C)	60.62	60.62	60.62	60.61	60.61
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	4860	4865	4870	4875	4880
56	Pressure (bar)	97.08	97.07	97.07	97.07	97.07
57	Temperature (C)	60.61	60.61	60.60	60.60	60.60
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	4885	4890	4895	4900	4905
68	Pressure (bar)	97.07	97.07	97.07	97.07	97.07
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	60.60	60.60	60.59	60.59	60.59
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	4910	4915	4920	4925	4930
24	Pressure (bar)	97.07	97.07	97.07	97.07	97.07
25	Temperature (C)	60.59	60.58	60.58	60.58	60.58
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	4935	4940	4945	4950	4955
36	Pressure (bar)	97.07	97.07	97.07	97.07	97.07
37	Temperature (C)	60.58	60.57	60.57	60.57	60.57
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	4960	4965	4970	4975	4980
48	Pressure (bar)	97.07	97.07	97.07	97.07	97.07
49	Temperature (C)	60.56	60.56	60.56	60.56	60.55
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	4985	4990	4995	5000	5005
60	Pressure (bar)	97.07	97.07	97.07	97.07	97.07
61	Temperature (C)	60.55	60.55	60.55	60.54	60.54
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	5010	5015	5020	5025	5030
16	Pressure (bar)	97.07	97.07	97.07	97.07	97.07
17	Temperature (C)	60.54	60.54	60.54	60.53	60.53
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	5035	5040	5045	5050	5055
28	Pressure (bar)	97.07	97.07	97.07	97.06	97.06
29	Temperature (C)	60.53	60.53	60.52	60.52	60.52
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	5060	5065	5070	5075	5080
40	Pressure (bar)	97.06	97.06	97.06	97.06	97.06
41	Temperature (C)	60.52	60.51	60.51	60.51	60.51
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	5085	5090	5095	5100	5105
52	Pressure (bar)	97.06	97.06	97.06	97.06	97.06
53	Temperature (C)	60.50	60.50	60.50	60.50	60.49
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	5110	5115	5120	5125	5130
64	Pressure (bar)	97.06	97.06	97.06	97.06	97.06
65	Temperature (C)	60.49	60.49	60.49	60.48	60.48
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	5135	5140	5145	5150	5155
20	Pressure (bar)	97.06	97.06	97.06	97.06	97.06
21	Temperature (C)	60.48	60.48	60.48	60.47	60.47
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	5160	5165	5170	5175	5180
32	Pressure (bar)	97.06	97.06	97.05	97.05	97.05
33	Temperature (C)	60.47	60.47	60.46	60.46	60.46
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	5185	5190	5195	5200	5205
44	Pressure (bar)	97.05	97.05	97.05	97.05	97.05
45	Temperature (C)	60.46	60.45	60.45	60.45	60.45
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	5210	5215	5220	5225	5230
56	Pressure (bar)	97.05	97.05	97.05	97.05	97.05
57	Temperature (C)	60.45	60.44	60.44	60.44	60.44
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	5235	5240	5245	5250	5255
68	Pressure (bar)	97.05	97.05	97.05	97.05	97.05
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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	60.43	60.43	60.43	60.43	60.42
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	5260	5265	5270	5275	5280
24	Pressure (bar)	97.05	97.05	97.05	97.05	97.05
25	Temperature (C)	60.42	60.42	60.42	60.42	60.41
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	5285	5290	5295	5300	5305
36	Pressure (bar)	97.05	97.05	97.05	97.05	97.05
37	Temperature (C)	60.41	60.41	60.41	60.40	60.40
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	5310	5315	5320	5325	5330
48	Pressure (bar)	97.05	97.05	97.05	97.05	97.05
49	Temperature (C)	60.40	60.40	60.40	60.39	60.39
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	5335	5340	5345	5350	5355
60	Pressure (bar)	97.05	97.05	97.05	97.05	97.05
61	Temperature (C)	60.39	60.39	60.38	60.38	60.38
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 707 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	5360	5365	5370	5375	5380
16	Pressure (bar)	97.05	97.05	97.05	97.05	97.05
17	Temperature (C)	60.38	60.38	60.37	60.37	60.37
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	5385	5390	5395	5400	5405
28	Pressure (bar)	97.05	97.05	97.05	97.05	97.05
29	Temperature (C)	60.37	60.36	60.36	60.36	60.36
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	5410	5415	5420	5425	5430
40	Pressure (bar)	97.06	97.05	97.05	97.05	97.05
41	Temperature (C)	60.36	60.35	60.35	60.35	60.35
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	5435	5440	5445	5450	5455
52	Pressure (bar)	97.05	97.04	97.04	97.04	97.04
53	Temperature (C)	60.34	60.34	60.34	60.33	60.33
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	5460	5465	5470	5475	5480
64	Pressure (bar)	97.03	97.03	97.03	97.03	97.03
65	Temperature (C)	60.33	60.33	60.32	60.32	60.32
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	5485	5490	5495	5500	5505
20	Pressure (bar)	97.03	97.03	97.03	97.03	97.03
21	Temperature (C)	60.32	60.31	60.31	60.31	60.31
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	5510	5515	5520	5525	5530
32	Pressure (bar)	97.03	97.03	97.03	97.03	97.03
33	Temperature (C)	60.31	60.30	60.30	60.30	60.30
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	5535	5542	5549	5556	5563
44	Pressure (bar)	97.03	97.03	97.03	97.03	97.03
45	Temperature (C)	60.29	60.29	60.29	60.28	60.28
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	5570	5572	5574	5576	5578
56	Pressure (bar)	97.03	97.03	97.03	97.03	97.04
57	Temperature (C)	60.28	60.28	60.28	60.28	60.28
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	5580	5585	5590	5595	5600
68	Pressure (bar)	97.04	97.03	97.03	97.03	97.03
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3						
4						
5	Unit Set: Mixed					
6	Date/Time: Mon Jun 13 15:02:44 2022					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	Hydrates					
11	Hydrates Table					
12						
13	Temperature (C)	60.27	60.27	60.27	60.27	60.27
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	5605	5610	5615	5620	5625
24	Pressure (bar)	97.03	97.03	97.03	97.03	97.03
25	Temperature (C)	60.26	60.26	60.26	60.26	60.25
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	5630	5635	5640	5645	5650
36	Pressure (bar)	97.03	97.03	97.03	97.03	97.03
37	Temperature (C)	60.25	60.25	60.25	60.25	60.24
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	5655	5660	5665	5670	5675
48	Pressure (bar)	97.04	97.04	97.04	97.03	97.03
49	Temperature (C)	60.24	60.24	60.24	60.23	60.23
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	5680	5685	5690	5695	5700
60	Pressure (bar)	97.03	97.03	97.03	97.03	97.02
61	Temperature (C)	60.23	60.23	60.22	60.22	60.22
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	5705	5710	5715	5720	5725
16	Pressure (bar)	97.02	97.02	97.02	97.02	97.02
17	Temperature (C)	60.22	60.21	60.21	60.21	60.21
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	5730	5735	5740	5745	5750
28	Pressure (bar)	97.02	97.02	97.02	97.02	97.02
29	Temperature (C)	60.20	60.20	60.20	60.20	60.19
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	5755	5760	5765	5770	5775
40	Pressure (bar)	97.02	97.02	97.01	97.01	97.01
41	Temperature (C)	60.19	60.19	60.19	60.18	60.18
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	5780	5785	5790	5795	5800
52	Pressure (bar)	97.01	97.01	97.01	97.01	97.01
53	Temperature (C)	60.18	60.18	60.17	60.17	60.17
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	5805	5810	5815	5820	5825
64	Pressure (bar)	97.01	97.01	97.01	97.01	97.01
65	Temperature (C)	60.17	60.17	60.16	60.16	60.16
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 711 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.31
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	5830	5835	5840	5845	5850
20	Pressure (bar)	97.01	97.01	97.01	97.00	97.00
21	Temperature (C)	60.16	60.15	60.15	60.15	60.14
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.30
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	5855	5860	5865	5870	5875
32	Pressure (bar)	97.00	97.00	97.00	97.00	96.99
33	Temperature (C)	60.14	60.14	60.14	60.13	60.13
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.30	18.30	18.30	18.30	18.30
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	5880	5885	5890	5895	5900
44	Pressure (bar)	96.99	97.01	97.03	97.05	97.07
45	Temperature (C)	60.13	60.14	60.14	60.15	60.16
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.30	18.31	18.31	18.31	18.31
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	5905	5910	5915	5920	5925
56	Pressure (bar)	97.09	97.11	97.13	97.16	97.18
57	Temperature (C)	60.16	60.17	60.18	60.18	60.19
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.31	18.31	18.31	18.31	18.32
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	5930	5935	5940	5945	5950
68	Pressure (bar)	97.20	97.04	96.89	96.73	96.58
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	60.20	60.16	60.12	60.09	60.05
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.32	18.31	18.30	18.29	18.28
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	5955	5960	5965	5970	5975
24	Pressure (bar)	96.42	96.38	96.34	96.29	96.25
25	Temperature (C)	60.01	60.00	59.99	59.98	59.96
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.27	18.27	18.27	18.26	18.26
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	5980	5985	5990	5995	6000
36	Pressure (bar)	96.21	96.24	96.28	96.31	96.34
37	Temperature (C)	59.95	59.96	59.98	59.99	60.00
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.26	18.26	18.26	18.26	18.27
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	6005	6010	6015	6020	6025
48	Pressure (bar)	96.38	96.20	96.02	95.84	95.66
49	Temperature (C)	60.02	59.97	59.93	59.89	59.85
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.27	18.26	18.25	18.24	18.23
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	6030	6035	6040	6045	6050
60	Pressure (bar)	95.48	95.48	95.48	95.48	95.47
61	Temperature (C)	59.80	59.80	59.80	59.80	59.79
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.21	18.21	18.21	18.21	18.21
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	6055	6060	6065	6070	6075
16	Pressure (bar)	95.47	95.47	95.47	95.47	95.47
17	Temperature (C)	59.79	59.79	59.79	59.78	59.78
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.21	18.21	18.21	18.21	18.21
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	6080	6085	6090	6095	6101
28	Pressure (bar)	95.47	95.50	95.52	95.55	95.58
29	Temperature (C)	59.78	59.79	59.80	59.81	59.82
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.21	18.22	18.22	18.22	18.22
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	6106	6111	6116	6121	6126
40	Pressure (bar)	95.60	95.47	95.33	95.20	95.06
41	Temperature (C)	59.83	59.79	59.76	59.73	59.70
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.22	18.21	18.21	18.20	18.19
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	6131	6136	6141	6146	6151
52	Pressure (bar)	94.92	94.92	94.92	94.92	94.92
53	Temperature (C)	59.66	59.66	59.66	59.66	59.65
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.18	18.18	18.18	18.18	18.18
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	6156	6161	6166	6171	6176
64	Pressure (bar)	94.92	94.92	94.92	94.92	94.92
65	Temperature (C)	59.65	59.65	59.65	59.64	59.64
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.18	18.18	18.18	18.18	18.18
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	6181	6186	6191	6196	6201
20	Pressure (bar)	94.92	94.92	94.92	94.92	94.92
21	Temperature (C)	59.64	59.64	59.63	59.63	59.63
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.18	18.18	18.18	18.18	18.18
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	6206	6211	6216	6221	6226
32	Pressure (bar)	94.91	94.91	94.91	94.91	94.91
33	Temperature (C)	59.63	59.63	59.62	59.62	59.62
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.18	18.18	18.18	18.18	18.18
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	6231	6236	6241	6246	6251
44	Pressure (bar)	94.91	94.91	94.91	94.91	94.91
45	Temperature (C)	59.62	59.61	59.61	59.61	59.61
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.18	18.18	18.18	18.18	18.18
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	6256	6261	6266	6271	6276
56	Pressure (bar)	94.91	94.91	94.91	94.91	94.91
57	Temperature (C)	59.61	59.60	59.60	59.60	59.60
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.18	18.18	18.18	18.18	18.18
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	6281	6285	6290	6294	6299
68	Pressure (bar)	94.91	94.94	94.96	94.99	95.02
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	59.59	59.60	59.61	59.62	59.63
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.18	18.18	18.18	18.19	18.19
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	6303	6308	6313	6318	6323
24	Pressure (bar)	95.04	95.00	94.96	94.91	94.87
25	Temperature (C)	59.64	59.63	59.62	59.61	59.59
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.19	18.19	18.18	18.18	18.18
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	6328	6333	6338	6343	6348
36	Pressure (bar)	94.82	94.83	94.84	94.85	94.86
37	Temperature (C)	59.58	59.58	59.58	59.59	59.59
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.18	18.18	18.18	18.18	18.18
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	6353	6358	6363	6368	6373
48	Pressure (bar)	94.86	94.88	94.90	94.91	94.93
49	Temperature (C)	59.59	59.59	59.60	59.60	59.61
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.18	18.18	18.18	18.18	18.18
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	6378	6383	6388	6393	6398
60	Pressure (bar)	94.94	94.91	94.87	94.84	94.80
61	Temperature (C)	59.62	59.60	59.59	59.58	59.57
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.18	18.18	18.18	18.18	18.17
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 716 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	6403	6408	6413	6418	6423
16	Pressure (bar)	94.77	94.60	94.42	94.25	94.07
17	Temperature (C)	59.56	59.52	59.48	59.44	59.40
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.17	18.16	18.15	18.14	18.13
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	6428	6433	6438	6443	6448
28	Pressure (bar)	93.90	93.90	93.89	93.89	93.89
29	Temperature (C)	59.35	59.35	59.35	59.34	59.34
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.12	18.12	18.12	18.12	18.12
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	6453	6459	6465	6471	6477
40	Pressure (bar)	93.89	93.88	93.88	93.88	93.88
41	Temperature (C)	59.34	59.34	59.33	59.33	59.33
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.12	18.12	18.12	18.12	18.12
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	6483	6487	6491	6495	6499
52	Pressure (bar)	93.88	93.88	93.88	93.88	93.88
53	Temperature (C)	59.32	59.32	59.32	59.32	59.32
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.12	18.12	18.12	18.12	18.12
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	6503	6508	6513	6518	6523
64	Pressure (bar)	93.88	93.88	93.88	93.88	93.88
65	Temperature (C)	59.32	59.31	59.31	59.31	59.31
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.12	18.12	18.12	18.12	18.12
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	6528	6533	6538	6543	6548
20	Pressure (bar)	93.88	93.88	93.88	93.88	93.88
21	Temperature (C)	59.30	59.30	59.30	59.30	59.29
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.12	18.12	18.12	18.12	18.12
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	6553	6558	6563	6568	6573
32	Pressure (bar)	93.88	93.87	93.87	93.87	93.87
33	Temperature (C)	59.29	59.29	59.29	59.29	59.28
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.12	18.12	18.12	18.12	18.12
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	6578	6583	6588	6593	6598
44	Pressure (bar)	93.87	93.87	93.87	93.87	93.87
45	Temperature (C)	59.28	59.28	59.28	59.27	59.27
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.12	18.12	18.12	18.12	18.12
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	6603	6608	6613	6618	6623
56	Pressure (bar)	93.87	93.87	93.87	93.87	93.87
57	Temperature (C)	59.27	59.27	59.26	59.26	59.26
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.12	18.12	18.12	18.12	18.12
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	6628	6633	6638	6643	6648
68	Pressure (bar)	93.87	93.87	93.87	93.87	93.87
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	59.26	59.26	59.25	59.25	59.25
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.12	18.12	18.12	18.12	18.12
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	6653	6658	6663	6668	6673
24	Pressure (bar)	93.87	93.87	93.87	93.87	93.87
25	Temperature (C)	59.25	59.24	59.24	59.24	59.24
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.12	18.12	18.12	18.12	18.12
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	6678	6683	6688	6693	6698
36	Pressure (bar)	93.87	93.87	93.87	93.87	93.87
37	Temperature (C)	59.24	59.23	59.23	59.23	59.23
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.12	18.12	18.12	18.12	18.12
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	6703	6708	6713	6718	6723
48	Pressure (bar)	93.87	93.87	93.87	93.87	93.87
49	Temperature (C)	59.22	59.22	59.22	59.22	59.22
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.12	18.12	18.12	18.12	18.12
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	6728	6733	6738	6743	6748
60	Pressure (bar)	93.87	93.90	93.93	93.96	93.99
61	Temperature (C)	59.21	59.23	59.24	59.25	59.26
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.12	18.12	18.12	18.12	18.12
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	6754	6759	6764	6769	6774
16	Pressure (bar)	94.02	93.97	93.91	93.85	93.79
17	Temperature (C)	59.27	59.26	59.24	59.23	59.21
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.13	18.12	18.12	18.12	18.11
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	6779	6784	6789	6794	6799
28	Pressure (bar)	93.74	93.72	93.71	93.69	93.68
29	Temperature (C)	59.20	59.19	59.18	59.18	59.17
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.11	18.11	18.11	18.11	18.11
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	6804	6809	6814	6819	6824
40	Pressure (bar)	93.67	93.59	93.51	93.43	93.35
41	Temperature (C)	59.17	59.15	59.13	59.11	59.09
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.10	18.10	18.10	18.09	18.09
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	6829	6834	6839	6844	6849
52	Pressure (bar)	93.27	93.31	93.35	93.40	93.44
53	Temperature (C)	59.07	59.08	59.10	59.12	59.14
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.08	18.08	18.09	18.09	18.09
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	6854	6859	6864	6869	6874
64	Pressure (bar)	93.48	93.25	93.02	92.79	92.56
65	Temperature (C)	59.15	59.10	59.04	58.99	58.93
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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aspentech

Epic Atlantic Limited

Case Name:

Simulation - 386MMSCFD_100barg.hsc

Unit Set:

Mixed

Date/Time:

Mon Jun 13 15:02:44 2022

Pipe Segment: ELEP-Akabel Pipeline (continued)

Hydrates

Hydrates Table

Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
Formation Temperature (C)	18.09	18.08	18.07	18.05	18.04
Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
Distance (m)	6879	6884	6889	6894	6899
Pressure (bar)	92.33	92.33	92.33	92.33	92.32
Temperature (C)	58.88	58.88	58.87	58.87	58.87
Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
Hydrate Type	No Types	No Types	No Types	No Types	No Types
Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
Distance (m)	6904	6909	6914	6919	6924
Pressure (bar)	92.32	92.32	92.32	92.32	92.32
Temperature (C)	58.87	58.86	58.86	58.86	58.86
Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
Hydrate Type	No Types	No Types	No Types	No Types	No Types
Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
Distance (m)	6929	6934	6939	6944	6949
Pressure (bar)	92.32	92.32	92.32	92.32	92.32
Temperature (C)	58.85	58.85	58.85	58.85	58.85
Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
Hydrate Type	No Types	No Types	No Types	No Types	No Types
Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
Distance (m)	6954	6959	6964	6969	6974
Pressure (bar)	92.32	92.32	92.32	92.32	92.32
Temperature (C)	58.84	58.84	58.84	58.84	58.83
Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
Hydrate Type	No Types	No Types	No Types	No Types	No Types
Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
Distance (m)	6979	6984	6989	6994	6999
Pressure (bar)	92.32	92.32	92.32	92.32	92.32

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1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	58.83	58.83	58.83	58.82	58.82
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	7004	7009	7014	7019	7024
24	Pressure (bar)	92.32	92.31	92.31	92.31	92.31
25	Temperature (C)	58.82	58.82	58.82	58.81	58.81
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	7029	7034	7039	7044	7049
36	Pressure (bar)	92.31	92.31	92.31	92.31	92.31
37	Temperature (C)	58.81	58.81	58.80	58.80	58.80
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	7054	7059	7064	7069	7074
48	Pressure (bar)	92.31	92.31	92.31	92.31	92.31
49	Temperature (C)	58.80	58.80	58.79	58.79	58.79
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	7079	7081	7083	7085	7087
60	Pressure (bar)	92.31	92.31	92.31	92.31	92.31
61	Temperature (C)	58.79	58.79	58.78	58.78	58.78
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	7089	7097	7105	7113	7121
16	Pressure (bar)	92.31	92.31	92.31	92.31	92.31
17	Temperature (C)	58.78	58.78	58.77	58.77	58.77
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	7129	7134	7139	7144	7149
28	Pressure (bar)	92.31	92.31	92.31	92.30	92.30
29	Temperature (C)	58.76	58.76	58.76	58.76	58.76
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	7154	7159	7164	7169	7174
40	Pressure (bar)	92.30	92.30	92.30	92.30	92.30
41	Temperature (C)	58.75	58.75	58.75	58.75	58.74
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	7179	7184	7189	7194	7199
52	Pressure (bar)	92.30	92.30	92.30	92.30	92.30
53	Temperature (C)	58.74	58.74	58.74	58.73	58.73
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	7204	7209	7214	7219	7224
64	Pressure (bar)	92.30	92.30	92.30	92.30	92.30
65	Temperature (C)	58.73	58.73	58.73	58.72	58.72
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3						
4						
5	Unit Set: Mixed					
6	Date/Time: Mon Jun 13 15:02:44 2022					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	Hydrates					
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	7229	7234	7239	7244	7249
20	Pressure (bar)	92.30	92.30	92.30	92.30	92.30
21	Temperature (C)	58.72	58.72	58.71	58.71	58.71
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	7254	7259	7264	7269	7274
32	Pressure (bar)	92.30	92.30	92.30	92.29	92.29
33	Temperature (C)	58.71	58.71	58.70	58.70	58.70
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	7279	7284	7289	7294	7299
44	Pressure (bar)	92.29	92.29	92.29	92.29	92.29
45	Temperature (C)	58.70	58.69	58.69	58.69	58.69
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	7304	7309	7314	7319	7324
56	Pressure (bar)	92.29	92.29	92.29	92.29	92.29
57	Temperature (C)	58.69	58.68	58.68	58.68	58.68
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	7329	7334	7339	7344	7349
68	Pressure (bar)	92.29	92.29	92.29	92.29	92.29
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 724 of 937

1	Epic Atlantic Limited			Case Name: Simulation - 386MMSCFD_100barg.hsc		
2						
3				Unit Set: Mixed		
4						
5				Date/Time: Mon Jun 13 15:02:44 2022		
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	58.67	58.67	58.67	58.67	58.66
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	7354	7359	7364	7369	7374
24	Pressure (bar)	92.29	92.29	92.29	92.29	92.29
25	Temperature (C)	58.66	58.66	58.66	58.66	58.65
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	7379	7384	7389	7394	7399
36	Pressure (bar)	92.29	92.29	92.29	92.28	92.28
37	Temperature (C)	58.65	58.65	58.65	58.64	58.64
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	7404	7409	7414	7419	7424
48	Pressure (bar)	92.28	92.28	92.28	92.28	92.28
49	Temperature (C)	58.64	58.64	58.64	58.63	58.63
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	7429	7434	7439	7444	7449
60	Pressure (bar)	92.28	92.28	92.28	92.28	92.28
61	Temperature (C)	58.63	58.63	58.62	58.62	58.62
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	7454	7459	7464	7469	7474
16	Pressure (bar)	92.28	92.28	92.28	92.28	92.28
17	Temperature (C)	58.62	58.62	58.61	58.61	58.61
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	7479	7484	7489	7494	7499
28	Pressure (bar)	92.28	92.28	92.28	92.28	92.28
29	Temperature (C)	58.61	58.60	58.60	58.60	58.60
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	7504	7509	7514	7519	7524
40	Pressure (bar)	92.28	92.28	92.28	92.28	92.28
41	Temperature (C)	58.60	58.59	58.59	58.59	58.59
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	7529	7533	7538	7543	7548
52	Pressure (bar)	92.28	92.28	92.28	92.28	92.28
53	Temperature (C)	58.58	58.58	58.58	58.58	58.58
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	7553	7558	7563	7568	7573
64	Pressure (bar)	92.28	92.28	92.28	92.27	92.27
65	Temperature (C)	58.57	58.57	58.57	58.57	58.56
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	7578	7583	7588	7593	7598
20	Pressure (bar)	92.27	92.27	92.27	92.27	92.27
21	Temperature (C)	58.56	58.56	58.56	58.56	58.55
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	7603	7608	7613	7618	7623
32	Pressure (bar)	92.27	92.27	92.27	92.27	92.27
33	Temperature (C)	58.55	58.55	58.55	58.55	58.54
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	7628	7633	7638	7643	7648
44	Pressure (bar)	92.27	92.27	92.27	92.27	92.27
45	Temperature (C)	58.54	58.54	58.54	58.53	58.53
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	7653	7658	7663	7668	7673
56	Pressure (bar)	92.27	92.27	92.27	92.27	92.27
57	Temperature (C)	58.53	58.53	58.53	58.52	58.52
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	7678	7683	7688	7693	7698
68	Pressure (bar)	92.27	92.27	92.27	92.27	92.27
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	58.52	58.52	58.51	58.51	58.51
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	7703	7708	7713	7718	7723
24	Pressure (bar)	92.27	92.27	92.27	92.27	92.27
25	Temperature (C)	58.51	58.51	58.50	58.50	58.50
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	7728	7733	7738	7743	7748
36	Pressure (bar)	92.27	92.27	92.27	92.27	92.27
37	Temperature (C)	58.50	58.50	58.50	58.49	58.49
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	7753	7758	7763	7768	7773
48	Pressure (bar)	92.27	92.27	92.27	92.26	92.26
49	Temperature (C)	58.49	58.49	58.48	58.48	58.48
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	7778	7783	7788	7793	7798
60	Pressure (bar)	92.26	92.26	92.26	92.26	92.25
61	Temperature (C)	58.47	58.47	58.47	58.47	58.46
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	7803	7808	7813	7818	7823
16	Pressure (bar)	92.25	92.25	92.25	92.25	92.25
17	Temperature (C)	58.46	58.46	58.46	58.46	58.45
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	7828	7833	7838	7843	7848
28	Pressure (bar)	92.25	92.25	92.25	92.25	92.25
29	Temperature (C)	58.45	58.45	58.45	58.44	58.44
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	7853	7858	7863	7868	7873
40	Pressure (bar)	92.25	92.25	92.25	92.25	92.25
41	Temperature (C)	58.44	58.44	58.43	58.43	58.43
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	7878	7883	7888	7893	7898
52	Pressure (bar)	92.25	92.25	92.25	92.25	92.25
53	Temperature (C)	58.43	58.43	58.42	58.42	58.42
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	7903	7908	7913	7918	7923
64	Pressure (bar)	92.25	92.25	92.25	92.25	92.25
65	Temperature (C)	58.42	58.42	58.41	58.41	58.41
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 729 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	7928	7933	7938	7943	7948
20	Pressure (bar)	92.25	92.24	92.24	92.24	92.24
21	Temperature (C)	58.41	58.40	58.40	58.40	58.40
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	7953	7958	7963	7968	7973
32	Pressure (bar)	92.24	92.24	92.24	92.24	92.24
33	Temperature (C)	58.39	58.39	58.39	58.39	58.39
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	7978	7983	7988	7993	7998
44	Pressure (bar)	92.24	92.24	92.24	92.24	92.24
45	Temperature (C)	58.38	58.38	58.38	58.38	58.38
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	8003	8008	8013	8018	8023
56	Pressure (bar)	92.24	92.24	92.24	92.24	92.24
57	Temperature (C)	58.37	58.37	58.37	58.37	58.37
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	8028	8033	8038	8043	8048
68	Pressure (bar)	92.24	92.24	92.24	92.25	92.25
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	58.36	58.36	58.36	58.36	58.36
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	8053	8058	8063	8068	8073
24	Pressure (bar)	92.25	92.26	92.26	92.27	92.28
25	Temperature (C)	58.35	58.36	58.36	58.36	58.36
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.02	18.02	18.02	18.02	18.02
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	8078	8083	8088	8093	8098
36	Pressure (bar)	92.29	92.25	92.21	92.16	92.12
37	Temperature (C)	58.37	58.35	58.34	58.33	58.32
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.02	18.02	18.01	18.01	18.01
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	8103	8108	8113	8118	8123
48	Pressure (bar)	92.08	92.08	92.07	92.07	92.06
49	Temperature (C)	58.31	58.30	58.30	58.29	58.29
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.01	18.01	18.01	18.01	18.00
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	8128	8133	8138	8143	8148
60	Pressure (bar)	92.05	92.05	92.05	92.05	92.05
61	Temperature (C)	58.29	58.28	58.28	58.28	58.28
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3						
4						
5	Unit Set: Mixed					
6	Date/Time: Mon Jun 13 15:02:44 2022					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	Hydrates					
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	8153	8158	8163	8168	8173
16	Pressure (bar)	92.05	92.05	92.05	92.05	92.05
17	Temperature (C)	58.28	58.27	58.27	58.27	58.27
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	8178	8183	8188	8193	8198
28	Pressure (bar)	92.05	92.05	92.05	92.05	92.05
29	Temperature (C)	58.26	58.26	58.26	58.26	58.26
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	8203	8208	8213	8218	8223
40	Pressure (bar)	92.05	92.05	92.05	92.05	92.05
41	Temperature (C)	58.25	58.25	58.25	58.25	58.24
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	8228	8233	8238	8243	8248
52	Pressure (bar)	92.05	92.05	92.05	92.05	92.04
53	Temperature (C)	58.24	58.24	58.24	58.24	58.23
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	8253	8258	8263	8268	8273
64	Pressure (bar)	92.04	92.04	92.04	92.04	92.04
65	Temperature (C)	58.23	58.23	58.23	58.22	58.22
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	8278	8283	8288	8293	8298
20	Pressure (bar)	92.04	92.04	92.04	92.04	92.04
21	Temperature (C)	58.22	58.22	58.22	58.21	58.21
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	8303	8308	8313	8318	8323
32	Pressure (bar)	92.04	92.04	92.04	92.04	92.04
33	Temperature (C)	58.21	58.21	58.20	58.20	58.20
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	8328	8333	8338	8343	8348
44	Pressure (bar)	92.04	92.04	92.04	92.04	92.04
45	Temperature (C)	58.20	58.20	58.19	58.19	58.19
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	8353	8358	8363	8368	8373
56	Pressure (bar)	92.04	92.04	92.04	92.04	92.04
57	Temperature (C)	58.19	58.19	58.18	58.18	58.18
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	8378	8383	8388	8393	8398
68	Pressure (bar)	92.04	92.04	92.04	92.04	92.04
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	58.18	58.18	58.17	58.17	58.17
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	8403	8408	8413	8418	8423
24	Pressure (bar)	92.04	92.04	92.04	92.04	92.04
25	Temperature (C)	58.17	58.17	58.16	58.16	58.16
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	8428	8433	8438	8443	8448
36	Pressure (bar)	92.04	92.04	92.04	92.04	92.04
37	Temperature (C)	58.16	58.16	58.15	58.15	58.15
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	8453	8458	8463	8468	8473
48	Pressure (bar)	92.04	92.04	92.04	92.04	92.04
49	Temperature (C)	58.15	58.14	58.14	58.14	58.14
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	8478	8483	8488	8493	8498
60	Pressure (bar)	92.04	92.03	92.03	92.03	92.03
61	Temperature (C)	58.13	58.13	58.13	58.13	58.13
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc				
2							
3							Unit Set: Mixed
4							Date/Time: Mon Jun 13 15:02:44 2022
5							
6	Pipe Segment: ELEP-Akabel Pipeline (continued)						
7							
8							
9	Hydrates						
10	Hydrates Table						
11							
12							
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included	
15	Distance (m)	8503	8508	8513	8518	8523	
16	Pressure (bar)	92.03	92.03	92.03	92.03	92.03	
17	Temperature (C)	58.12	58.12	58.12	58.12	58.11	
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types	
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included	
22	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00	
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II	
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included	
27	Distance (m)	8528	8533	8538	8543	8548	
28	Pressure (bar)	92.03	92.03	92.03	92.03	92.03	
29	Temperature (C)	58.11	58.11	58.11	58.11	58.10	
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types	
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included	
34	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00	
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II	
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included	
39	Distance (m)	8553	8558	8563	8568	8573	
40	Pressure (bar)	92.03	92.03	92.03	92.03	92.03	
41	Temperature (C)	58.10	58.10	58.10	58.09	58.09	
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types	
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included	
46	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00	
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II	
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included	
51	Distance (m)	8578	8583	8588	8593	8598	
52	Pressure (bar)	92.03	92.03	92.03	92.03	92.03	
53	Temperature (C)	58.09	58.09	58.09	58.08	58.08	
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types	
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included	
58	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00	
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II	
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included	
63	Distance (m)	8603	8608	8613	8618	8623	
64	Pressure (bar)	92.03	92.03	92.03	92.03	92.03	
65	Temperature (C)	58.08	58.08	58.08	58.07	58.07	
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types	
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 735 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	8628	8633	8638	8643	8648
20	Pressure (bar)	92.03	92.03	92.02	92.02	92.02
21	Temperature (C)	58.07	58.07	58.06	58.06	58.06
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	8653	8658	8663	8668	8673
32	Pressure (bar)	92.02	92.02	92.02	92.02	92.02
33	Temperature (C)	58.06	58.06	58.05	58.05	58.05
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	8678	8683	8688	8693	8698
44	Pressure (bar)	92.02	92.02	92.02	92.02	92.02
45	Temperature (C)	58.05	58.04	58.04	58.04	58.04
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	8703	8708	8713	8718	8723
56	Pressure (bar)	92.02	92.02	92.02	92.02	92.02
57	Temperature (C)	58.04	58.03	58.03	58.03	58.03
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	8728	8733	8738	8743	8748
68	Pressure (bar)	92.02	92.02	92.02	92.02	92.02
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	58.02	58.02	58.02	58.02	58.02
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	8753	8758	8763	8768	8773
24	Pressure (bar)	92.02	92.02	92.02	92.02	92.02
25	Temperature (C)	58.01	58.01	58.01	58.01	58.01
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	8778	8783	8788	8793	8798
36	Pressure (bar)	92.02	92.02	92.02	92.02	92.02
37	Temperature (C)	58.00	58.00	58.00	58.00	57.99
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	8803	8805	8807	8809	8811
48	Pressure (bar)	92.02	92.02	92.02	92.01	92.01
49	Temperature (C)	57.99	57.99	57.99	57.99	57.99
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	8813	8840	8867	8894	8921
60	Pressure (bar)	92.01	92.01	92.01	92.01	92.01
61	Temperature (C)	57.99	57.98	57.96	57.95	57.94
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	8948	8949	8950	8951	8952
16	Pressure (bar)	92.01	92.01	92.01	92.01	92.00
17	Temperature (C)	57.93	57.93	57.93	57.93	57.93
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	8953	8958	8963	8968	8973
28	Pressure (bar)	92.00	92.00	92.00	92.00	92.00
29	Temperature (C)	57.93	57.92	57.92	57.92	57.92
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	8978	8983	8988	8993	8998
40	Pressure (bar)	92.00	92.00	92.00	92.00	92.00
41	Temperature (C)	57.92	57.91	57.91	57.91	57.91
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	9003	9008	9013	9018	9023
52	Pressure (bar)	92.00	92.00	92.00	92.00	92.00
53	Temperature (C)	57.90	57.90	57.90	57.90	57.90
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	9028	9033	9038	9043	9048
64	Pressure (bar)	92.00	92.00	92.00	92.00	92.00
65	Temperature (C)	57.89	57.89	57.89	57.89	57.89
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 738 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	9053	9054	9055	9056	9057
20	Pressure (bar)	92.00	92.00	92.00	92.00	92.00
21	Temperature (C)	57.88	57.88	57.88	57.88	57.88
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	9058	9065	9071	9078	9085
32	Pressure (bar)	92.00	92.00	92.00	92.00	92.00
33	Temperature (C)	57.88	57.88	57.88	57.87	57.87
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	9091	9098	9105	9111	9118
44	Pressure (bar)	92.00	92.00	92.00	92.00	92.00
45	Temperature (C)	57.87	57.86	57.86	57.86	57.86
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	9125	9131	9138	9145	9151
56	Pressure (bar)	92.00	92.00	92.00	92.00	92.00
57	Temperature (C)	57.85	57.85	57.85	57.85	57.84
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	9158	9162	9166	9170	9174
68	Pressure (bar)	92.00	92.00	92.00	92.00	92.00
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	57.84	57.84	57.84	57.83	57.83
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	9178	9183	9188	9193	9198
24	Pressure (bar)	92.00	92.00	92.00	92.00	92.00
25	Temperature (C)	57.83	57.83	57.83	57.82	57.82
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	9203	9208	9213	9218	9223
36	Pressure (bar)	92.00	92.00	91.99	91.99	91.99
37	Temperature (C)	57.82	57.82	57.82	57.81	57.81
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	9228	9233	9238	9243	9248
48	Pressure (bar)	91.99	91.99	91.99	91.99	91.99
49	Temperature (C)	57.81	57.81	57.81	57.80	57.80
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	9253	9258	9263	9268	9273
60	Pressure (bar)	91.99	91.99	91.99	91.99	91.99
61	Temperature (C)	57.80	57.80	57.79	57.79	57.79
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	9278	9283	9288	9293	9298
16	Pressure (bar)	91.98	91.98	91.98	91.98	91.98
17	Temperature (C)	57.79	57.78	57.78	57.78	57.78
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	9303	9308	9313	9318	9323
28	Pressure (bar)	91.98	91.98	91.98	91.98	91.98
29	Temperature (C)	57.78	57.77	57.77	57.77	57.77
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	9328	9333	9338	9343	9348
40	Pressure (bar)	91.98	91.98	91.98	91.98	91.98
41	Temperature (C)	57.76	57.76	57.76	57.76	57.76
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	9353	9358	9363	9368	9373
52	Pressure (bar)	91.98	91.98	91.98	91.97	91.97
53	Temperature (C)	57.75	57.75	57.75	57.75	57.74
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	9378	9383	9388	9393	9398
64	Pressure (bar)	91.97	91.97	91.97	91.97	91.98
65	Temperature (C)	57.74	57.74	57.74	57.74	57.73
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	9403	9408	9413	9418	9423
20	Pressure (bar)	91.98	91.98	91.98	91.98	91.98
21	Temperature (C)	57.73	57.73	57.73	57.73	57.72
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	9428	9433	9438	9443	9448
32	Pressure (bar)	91.98	91.98	91.98	91.98	91.98
33	Temperature (C)	57.72	57.72	57.72	57.72	57.72
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	9453	9456	9459	9462	9465
44	Pressure (bar)	91.98	91.98	91.97	91.97	91.97
45	Temperature (C)	57.71	57.71	57.71	57.71	57.70
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	9468	9473	9477	9482	9487
56	Pressure (bar)	91.96	91.96	91.96	91.96	91.96
57	Temperature (C)	57.70	57.70	57.70	57.70	57.69
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	9491	9496	9501	9505	9510
68	Pressure (bar)	91.96	91.96	91.97	91.97	91.97
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	57.69	57.69	57.69	57.69	57.69
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	9515	9519	9524	9529	9533
24	Pressure (bar)	91.97	91.97	91.97	91.97	91.97
25	Temperature (C)	57.69	57.68	57.68	57.68	57.68
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	9538	9543	9547	9552	9557
36	Pressure (bar)	91.97	91.96	91.96	91.96	91.96
37	Temperature (C)	57.67	57.67	57.67	57.67	57.67
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	9561	9566	9571	9575	9580
48	Pressure (bar)	91.96	91.96	91.96	91.96	91.96
49	Temperature (C)	57.66	57.66	57.66	57.66	57.65
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	9585	9589	9594	9599	9603
60	Pressure (bar)	91.95	91.95	91.95	91.95	91.95
61	Temperature (C)	57.65	57.65	57.65	57.65	57.64
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	9608	9612	9616	9620	9624
16	Pressure (bar)	91.95	91.95	91.95	91.95	91.95
17	Temperature (C)	57.64	57.64	57.64	57.63	57.63
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	9628	9633	9638	9643	9648
28	Pressure (bar)	91.94	91.94	91.94	91.95	91.95
29	Temperature (C)	57.63	57.63	57.63	57.62	57.62
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	9653	9658	9663	9668	9673
40	Pressure (bar)	91.95	91.95	91.95	91.95	91.95
41	Temperature (C)	57.62	57.62	57.62	57.61	57.61
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	9678	9683	9688	9693	9698
52	Pressure (bar)	91.95	91.94	91.94	91.94	91.94
53	Temperature (C)	57.61	57.61	57.61	57.60	57.60
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	9703	9708	9713	9718	9723
64	Pressure (bar)	91.94	91.94	91.94	91.94	91.94
65	Temperature (C)	57.60	57.60	57.59	57.59	57.59
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3						
4						
5	Unit Set: Mixed					
6	Date/Time: Mon Jun 13 15:02:44 2022					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	Hydrates					
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	9728	9733	9738	9743	9748
20	Pressure (bar)	91.94	91.94	91.94	91.94	91.94
21	Temperature (C)	57.59	57.59	57.58	57.58	57.58
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	9753	9758	9763	9768	9773
32	Pressure (bar)	91.94	91.93	91.93	91.93	91.93
33	Temperature (C)	57.58	57.57	57.57	57.57	57.57
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	9778	9783	9788	9793	9798
44	Pressure (bar)	91.93	91.93	91.93	91.93	91.93
45	Temperature (C)	57.57	57.56	57.56	57.56	57.56
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	9803	9808	9813	9818	9823
56	Pressure (bar)	91.93	91.93	91.93	91.93	91.93
57	Temperature (C)	57.55	57.55	57.55	57.55	57.55
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	9828	9833	9838	9843	9848
68	Pressure (bar)	91.93	91.93	91.93	91.93	91.93
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	57.54	57.54	57.54	57.54	57.54
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	9853	9858	9863	9868	9873
24	Pressure (bar)	91.93	91.93	91.93	91.93	91.93
25	Temperature (C)	57.53	57.53	57.53	57.53	57.53
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	9878	9883	9888	9893	9898
36	Pressure (bar)	91.93	91.93	91.93	91.93	91.93
37	Temperature (C)	57.52	57.52	57.52	57.52	57.51
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	9903	9905	9907	9909	9911
48	Pressure (bar)	91.93	91.93	91.93	91.93	91.93
49	Temperature (C)	57.51	57.51	57.51	57.51	57.51
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	9913	9933	9953	9973	9993
60	Pressure (bar)	91.93	91.92	91.92	91.92	91.92
61	Temperature (C)	57.51	57.50	57.49	57.48	57.47
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.001e+004	1.002e+004	1.002e+004	1.002e+004	1.003e+004
16	Pressure (bar)	91.92	91.92	91.92	91.92	91.93
17	Temperature (C)	57.47	57.46	57.46	57.46	57.46
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.003e+004	1.003e+004	1.004e+004	1.004e+004	1.005e+004
28	Pressure (bar)	91.93	91.93	91.93	91.93	91.93
29	Temperature (C)	57.46	57.46	57.46	57.46	57.45
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.005e+004	1.006e+004	1.006e+004	1.007e+004	1.007e+004
40	Pressure (bar)	91.93	91.92	91.92	91.92	91.92
41	Temperature (C)	57.45	57.45	57.45	57.44	57.44
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.008e+004	1.008e+004	1.009e+004	1.009e+004	1.010e+004
52	Pressure (bar)	91.91	91.91	91.91	91.91	91.91
53	Temperature (C)	57.44	57.44	57.43	57.43	57.43
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.010e+004	1.011e+004	1.011e+004	1.012e+004	1.012e+004
64	Pressure (bar)	91.91	91.91	91.91	91.91	91.91
65	Temperature (C)	57.43	57.42	57.42	57.42	57.42
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.013e+004	1.013e+004	1.014e+004	1.014e+004	1.015e+004
20	Pressure (bar)	91.91	91.90	91.90	91.90	91.90
21	Temperature (C)	57.41	57.41	57.41	57.41	57.41
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	18.00	18.00	18.00	18.00	18.00
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.015e+004	1.016e+004	1.016e+004	1.017e+004	1.017e+004
32	Pressure (bar)	91.90	91.90	91.90	91.90	91.90
33	Temperature (C)	57.40	57.40	57.40	57.40	57.40
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.018e+004	1.018e+004	1.019e+004	1.019e+004	1.020e+004
44	Pressure (bar)	91.90	91.90	91.90	91.90	91.90
45	Temperature (C)	57.39	57.39	57.39	57.39	57.38
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.020e+004	1.021e+004	1.021e+004	1.022e+004	1.022e+004
56	Pressure (bar)	91.90	91.90	91.90	91.90	91.90
57	Temperature (C)	57.38	57.38	57.38	57.38	57.37
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.023e+004	1.023e+004	1.024e+004	1.024e+004	1.025e+004
68	Pressure (bar)	91.90	91.90	91.90	91.90	91.90
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	57.37	57.37	57.37	57.37	57.36
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.025e+004	1.026e+004	1.026e+004	1.027e+004	1.027e+004
24	Pressure (bar)	91.90	91.90	91.90	91.90	91.90
25	Temperature (C)	57.36	57.36	57.36	57.36	57.35
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.028e+004	1.028e+004	1.029e+004	1.029e+004	1.030e+004
36	Pressure (bar)	91.90	91.90	91.90	91.90	91.89
37	Temperature (C)	57.35	57.35	57.35	57.34	57.34
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.030e+004	1.031e+004	1.031e+004	1.032e+004	1.032e+004
48	Pressure (bar)	91.89	91.89	91.89	91.89	91.89
49	Temperature (C)	57.34	57.34	57.34	57.33	57.33
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.033e+004	1.033e+004	1.034e+004	1.034e+004	1.035e+004
60	Pressure (bar)	91.89	91.89	91.89	91.89	91.89
61	Temperature (C)	57.33	57.33	57.32	57.32	57.32
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.035e+004	1.036e+004	1.036e+004	1.037e+004	1.037e+004
16	Pressure (bar)	91.89	91.89	91.89	91.89	91.89
17	Temperature (C)	57.32	57.32	57.31	57.31	57.31
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.038e+004	1.038e+004	1.039e+004	1.039e+004	1.040e+004
28	Pressure (bar)	91.89	91.89	91.89	91.89	91.89
29	Temperature (C)	57.31	57.31	57.30	57.30	57.30
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.040e+004	1.041e+004	1.041e+004	1.042e+004	1.042e+004
40	Pressure (bar)	91.89	91.89	91.89	91.89	91.89
41	Temperature (C)	57.30	57.30	57.29	57.29	57.29
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.043e+004	1.043e+004	1.044e+004	1.044e+004	1.045e+004
52	Pressure (bar)	91.89	91.89	91.89	91.89	91.89
53	Temperature (C)	57.29	57.29	57.28	57.28	57.28
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.045e+004	1.046e+004	1.046e+004	1.047e+004	1.047e+004
64	Pressure (bar)	91.89	91.89	91.89	91.89	91.89
65	Temperature (C)	57.28	57.28	57.27	57.27	57.27
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 750 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.048e+004	1.048e+004	1.049e+004	1.049e+004	1.050e+004
20	Pressure (bar)	91.89	91.89	91.89	91.89	91.89
21	Temperature (C)	57.27	57.27	57.26	57.26	57.26
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.050e+004	1.051e+004	1.051e+004	1.052e+004	1.052e+004
32	Pressure (bar)	91.89	91.89	91.89	91.88	91.88
33	Temperature (C)	57.26	57.26	57.25	57.25	57.25
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.053e+004	1.053e+004	1.054e+004	1.054e+004	1.055e+004
44	Pressure (bar)	91.88	91.88	91.88	91.87	91.87
45	Temperature (C)	57.24	57.24	57.24	57.24	57.24
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.055e+004	1.056e+004	1.056e+004	1.057e+004	1.057e+004
56	Pressure (bar)	91.87	91.87	91.87	91.87	91.87
57	Temperature (C)	57.23	57.23	57.23	57.23	57.22
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.058e+004	1.058e+004	1.059e+004	1.059e+004	1.060e+004
68	Pressure (bar)	91.87	91.87	91.87	91.87	91.87
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 751 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	57.22	57.22	57.22	57.22	57.21
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.060e+004	1.061e+004	1.061e+004	1.062e+004	1.062e+004
24	Pressure (bar)	91.87	91.87	91.87	91.87	91.87
25	Temperature (C)	57.21	57.21	57.21	57.20	57.20
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.063e+004	1.063e+004	1.064e+004	1.064e+004	1.065e+004
36	Pressure (bar)	91.87	91.86	91.86	91.86	91.86
37	Temperature (C)	57.20	57.20	57.20	57.19	57.19
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.065e+004	1.066e+004	1.066e+004	1.067e+004	1.067e+004
48	Pressure (bar)	91.86	91.86	91.86	91.86	91.86
49	Temperature (C)	57.19	57.19	57.18	57.18	57.18
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.068e+004	1.068e+004	1.069e+004	1.069e+004	1.070e+004
60	Pressure (bar)	91.86	91.86	91.86	91.86	91.86
61	Temperature (C)	57.18	57.18	57.17	57.17	57.17
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.070e+004	1.071e+004	1.071e+004	1.072e+004	1.072e+004
16	Pressure (bar)	91.86	91.86	91.86	91.86	91.86
17	Temperature (C)	57.17	57.17	57.16	57.16	57.16
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.073e+004	1.073e+004	1.074e+004	1.074e+004	1.075e+004
28	Pressure (bar)	91.86	91.86	91.86	91.86	91.86
29	Temperature (C)	57.16	57.15	57.15	57.15	57.15
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.075e+004	1.076e+004	1.076e+004	1.077e+004	1.077e+004
40	Pressure (bar)	91.86	91.86	91.85	91.85	91.85
41	Temperature (C)	57.15	57.14	57.14	57.14	57.14
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.078e+004	1.078e+004	1.079e+004	1.079e+004	1.080e+004
52	Pressure (bar)	91.85	91.85	91.85	91.85	91.85
53	Temperature (C)	57.14	57.13	57.13	57.13	57.13
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.080e+004	1.081e+004	1.081e+004	1.082e+004	1.082e+004
64	Pressure (bar)	91.85	91.85	91.85	91.85	91.85
65	Temperature (C)	57.12	57.12	57.12	57.12	57.12
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 753 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.083e+004	1.083e+004	1.084e+004	1.084e+004	1.085e+004
20	Pressure (bar)	91.85	91.85	91.85	91.85	91.85
21	Temperature (C)	57.11	57.11	57.11	57.11	57.11
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.085e+004	1.086e+004	1.086e+004	1.087e+004	1.087e+004
32	Pressure (bar)	91.85	91.85	91.85	91.85	91.85
33	Temperature (C)	57.10	57.10	57.10	57.10	57.10
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.088e+004	1.088e+004	1.089e+004	1.089e+004	1.090e+004
44	Pressure (bar)	91.85	91.85	91.85	91.85	91.85
45	Temperature (C)	57.09	57.09	57.09	57.09	57.08
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.090e+004	1.091e+004	1.091e+004	1.092e+004	1.092e+004
56	Pressure (bar)	91.85	91.84	91.84	91.84	91.84
57	Temperature (C)	57.08	57.08	57.08	57.08	57.07
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.093e+004	1.093e+004	1.094e+004	1.094e+004	1.095e+004
68	Pressure (bar)	91.84	91.84	91.84	91.84	91.84
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	57.07	57.07	57.07	57.07	57.06
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.095e+004	1.096e+004	1.096e+004	1.097e+004	1.097e+004
24	Pressure (bar)	91.84	91.84	91.84	91.84	91.84
25	Temperature (C)	57.06	57.06	57.06	57.05	57.05
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.098e+004	1.098e+004	1.099e+004	1.099e+004	1.099e+004
36	Pressure (bar)	91.84	91.84	91.84	91.84	91.84
37	Temperature (C)	57.05	57.05	57.05	57.05	57.04
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.100e+004	1.100e+004	1.101e+004	1.102e+004	1.102e+004
48	Pressure (bar)	91.84	91.84	91.84	91.84	91.84
49	Temperature (C)	57.04	57.04	57.04	57.03	57.03
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.103e+004	1.103e+004	1.104e+004	1.104e+004	1.105e+004
60	Pressure (bar)	91.84	91.84	91.84	91.84	91.84
61	Temperature (C)	57.03	57.03	57.03	57.02	57.02
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 755 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.105e+004	1.106e+004	1.106e+004	1.107e+004	1.107e+004
16	Pressure (bar)	91.84	91.84	91.84	91.83	91.83
17	Temperature (C)	57.02	57.02	57.01	57.01	57.01
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.108e+004	1.108e+004	1.109e+004	1.109e+004	1.110e+004
28	Pressure (bar)	91.83	91.83	91.83	91.83	91.83
29	Temperature (C)	57.01	57.01	57.00	57.00	57.00
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.110e+004	1.111e+004	1.111e+004	1.112e+004	1.112e+004
40	Pressure (bar)	91.83	91.83	91.83	91.83	91.83
41	Temperature (C)	57.00	57.00	56.99	56.99	56.99
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.113e+004	1.113e+004	1.114e+004	1.114e+004	1.115e+004
52	Pressure (bar)	91.83	91.83	91.83	91.83	91.83
53	Temperature (C)	56.99	56.99	56.98	56.98	56.98
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.115e+004	1.116e+004	1.116e+004	1.117e+004	1.117e+004
64	Pressure (bar)	91.83	91.83	91.83	91.83	91.83
65	Temperature (C)	56.98	56.97	56.97	56.97	56.97
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.99	17.99	17.99	17.99
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.118e+004	1.118e+004	1.119e+004	1.120e+004
20	Pressure (bar)	91.83	91.83	91.83	91.83
21	Temperature (C)	56.97	56.96	56.96	56.96
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.99	17.99	17.99	17.99
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.120e+004	1.121e+004	1.121e+004	1.122e+004
32	Pressure (bar)	91.83	91.83	91.83	91.83
33	Temperature (C)	56.96	56.95	56.95	56.95
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.99	17.99	17.99	17.99
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.123e+004	1.123e+004	1.124e+004	1.125e+004
44	Pressure (bar)	91.83	91.83	91.82	91.82
45	Temperature (C)	56.95	56.94	56.94	56.94
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.99	17.99	17.99	17.99
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.125e+004	1.126e+004	1.126e+004	1.127e+004
56	Pressure (bar)	91.82	91.82	91.82	91.82
57	Temperature (C)	56.93	56.93	56.93	56.93
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.99	17.99	17.99	17.99
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.128e+004	1.128e+004	1.129e+004	1.130e+004
68	Pressure (bar)	91.82	91.82	91.82	91.82
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	56.92	56.92	56.92	56.92	56.92
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.130e+004	1.131e+004	1.131e+004	1.132e+004	1.132e+004
24	Pressure (bar)	91.82	91.82	91.82	91.82	91.82
25	Temperature (C)	56.91	56.91	56.91	56.91	56.91
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.133e+004	1.133e+004	1.134e+004	1.134e+004	1.135e+004
36	Pressure (bar)	91.82	91.82	91.82	91.82	91.82
37	Temperature (C)	56.90	56.90	56.90	56.90	56.90
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.135e+004	1.136e+004	1.136e+004	1.137e+004	1.137e+004
48	Pressure (bar)	91.82	91.82	91.82	91.82	91.82
49	Temperature (C)	56.89	56.89	56.89	56.89	56.88
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.138e+004	1.138e+004	1.139e+004	1.139e+004	1.140e+004
60	Pressure (bar)	91.82	91.82	91.82	91.82	91.82
61	Temperature (C)	56.88	56.88	56.88	56.88	56.87
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.140e+004	1.141e+004	1.141e+004	1.142e+004	1.142e+004
16	Pressure (bar)	91.82	91.82	91.82	91.82	91.83
17	Temperature (C)	56.87	56.87	56.87	56.87	56.87
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.143e+004	1.143e+004	1.144e+004	1.144e+004	1.145e+004
28	Pressure (bar)	91.83	91.83	91.83	91.82	91.82
29	Temperature (C)	56.87	56.86	56.86	56.86	56.86
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.145e+004	1.146e+004	1.146e+004	1.147e+004	1.147e+004
40	Pressure (bar)	91.82	91.82	91.81	91.81	91.81
41	Temperature (C)	56.85	56.85	56.85	56.85	56.84
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.148e+004	1.148e+004	1.149e+004	1.149e+004	1.150e+004
52	Pressure (bar)	91.80	91.80	91.80	91.80	91.80
53	Temperature (C)	56.84	56.84	56.84	56.83	56.83
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.150e+004	1.151e+004	1.151e+004	1.152e+004	1.152e+004
64	Pressure (bar)	91.80	91.80	91.80	91.80	91.80
65	Temperature (C)	56.83	56.83	56.82	56.82	56.82
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 759 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.99	17.99	17.99	17.99
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.153e+004	1.153e+004	1.154e+004	1.155e+004
20	Pressure (bar)	91.80	91.80	91.80	91.80
21	Temperature (C)	56.82	56.82	56.81	56.81
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.99	17.99	17.99	17.99
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.155e+004	1.156e+004	1.156e+004	1.157e+004
32	Pressure (bar)	91.80	91.80	91.80	91.80
33	Temperature (C)	56.81	56.81	56.80	56.80
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.99	17.99	17.99	17.99
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.158e+004	1.158e+004	1.159e+004	1.160e+004
44	Pressure (bar)	91.80	91.80	91.80	91.80
45	Temperature (C)	56.80	56.79	56.79	56.79
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.99	17.99	17.99	17.99
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.160e+004	1.161e+004	1.161e+004	1.162e+004
56	Pressure (bar)	91.80	91.79	91.79	91.79
57	Temperature (C)	56.79	56.78	56.78	56.78
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.99	17.99	17.99	17.99
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.163e+004	1.163e+004	1.164e+004	1.165e+004
68	Pressure (bar)	91.79	91.79	91.79	91.79
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 760 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	56.78	56.77	56.77	56.77	56.77
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.165e+004	1.166e+004	1.166e+004	1.167e+004	1.167e+004
24	Pressure (bar)	91.79	91.79	91.79	91.79	91.79
25	Temperature (C)	56.77	56.76	56.76	56.76	56.76
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.168e+004	1.168e+004	1.169e+004	1.169e+004	1.170e+004
36	Pressure (bar)	91.79	91.79	91.79	91.79	91.79
37	Temperature (C)	56.76	56.75	56.75	56.75	56.75
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.170e+004	1.171e+004	1.171e+004	1.172e+004	1.172e+004
48	Pressure (bar)	91.79	91.79	91.79	91.79	91.79
49	Temperature (C)	56.74	56.74	56.74	56.74	56.74
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.173e+004	1.173e+004	1.174e+004	1.174e+004	1.175e+004
60	Pressure (bar)	91.79	91.79	91.79	91.79	91.79
61	Temperature (C)	56.73	56.73	56.73	56.73	56.73
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 761 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.175e+004	1.176e+004	1.176e+004	1.177e+004	1.177e+004
16	Pressure (bar)	91.79	91.79	91.78	91.78	91.78
17	Temperature (C)	56.72	56.72	56.72	56.72	56.72
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.178e+004	1.178e+004	1.179e+004	1.179e+004	1.180e+004
28	Pressure (bar)	91.78	91.78	91.78	91.78	91.78
29	Temperature (C)	56.71	56.71	56.71	56.71	56.70
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.180e+004	1.181e+004	1.181e+004	1.182e+004	1.182e+004
40	Pressure (bar)	91.78	91.78	91.78	91.78	91.78
41	Temperature (C)	56.70	56.70	56.70	56.70	56.69
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.183e+004	1.183e+004	1.184e+004	1.184e+004	1.185e+004
52	Pressure (bar)	91.78	91.78	91.78	91.78	91.78
53	Temperature (C)	56.69	56.69	56.69	56.69	56.68
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.185e+004	1.186e+004	1.186e+004	1.187e+004	1.187e+004
64	Pressure (bar)	91.78	91.78	91.78	91.78	91.78
65	Temperature (C)	56.68	56.68	56.68	56.68	56.67
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 762 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.188e+004	1.188e+004	1.189e+004	1.189e+004	1.190e+004
20	Pressure (bar)	91.78	91.78	91.78	91.78	91.78
21	Temperature (C)	56.67	56.67	56.67	56.67	56.66
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.190e+004	1.191e+004	1.191e+004	1.192e+004	1.192e+004
32	Pressure (bar)	91.78	91.78	91.78	91.78	91.78
33	Temperature (C)	56.66	56.66	56.66	56.65	56.65
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.193e+004	1.193e+004	1.194e+004	1.194e+004	1.194e+004
44	Pressure (bar)	91.77	91.77	91.77	91.77	91.77
45	Temperature (C)	56.65	56.65	56.65	56.65	56.64
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.195e+004	1.196e+004	1.196e+004	1.197e+004	1.198e+004
56	Pressure (bar)	91.77	91.77	91.77	91.77	91.77
57	Temperature (C)	56.64	56.64	56.64	56.63	56.63
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.198e+004	1.199e+004	1.200e+004	1.200e+004	1.201e+004
68	Pressure (bar)	91.77	91.77	91.77	91.77	91.77
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 763 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	56.63	56.62	56.62	56.62	56.62
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.202e+004	1.203e+004	1.203e+004	1.204e+004	1.205e+004
24	Pressure (bar)	91.77	91.77	91.77	91.77	91.77
25	Temperature (C)	56.61	56.61	56.61	56.60	56.60
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.205e+004	1.206e+004	1.207e+004	1.207e+004	1.208e+004
36	Pressure (bar)	91.77	91.76	91.76	91.76	91.75
37	Temperature (C)	56.60	56.60	56.59	56.59	56.58
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.209e+004	1.210e+004	1.210e+004	1.211e+004	1.212e+004
48	Pressure (bar)	91.75	91.75	91.75	91.76	91.76
49	Temperature (C)	56.58	56.58	56.58	56.57	56.57
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.212e+004	1.213e+004	1.214e+004	1.214e+004	1.215e+004
60	Pressure (bar)	91.76	91.76	91.76	91.76	91.76
61	Temperature (C)	56.57	56.57	56.56	56.56	56.56
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.216e+004	1.217e+004	1.217e+004	1.218e+004	1.219e+004
16	Pressure (bar)	91.76	91.76	91.75	91.75	91.75
17	Temperature (C)	56.55	56.55	56.55	56.55	56.54
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.219e+004	1.220e+004	1.221e+004	1.221e+004	1.222e+004
28	Pressure (bar)	91.75	91.75	91.75	91.75	91.75
29	Temperature (C)	56.54	56.54	56.53	56.53	56.53
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.223e+004	1.223e+004	1.224e+004	1.225e+004	1.225e+004
40	Pressure (bar)	91.75	91.75	91.75	91.75	91.75
41	Temperature (C)	56.53	56.52	56.52	56.52	56.52
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.226e+004	1.226e+004	1.227e+004	1.227e+004	1.228e+004
52	Pressure (bar)	91.75	91.75	91.75	91.75	91.75
53	Temperature (C)	56.51	56.51	56.51	56.51	56.51
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.228e+004	1.229e+004	1.229e+004	1.230e+004	1.230e+004
64	Pressure (bar)	91.75	91.75	91.75	91.75	91.75
65	Temperature (C)	56.50	56.50	56.50	56.50	56.49
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.231e+004	1.231e+004	1.232e+004	1.232e+004	1.233e+004
20	Pressure (bar)	91.75	91.75	91.75	91.75	91.75
21	Temperature (C)	56.49	56.49	56.49	56.49	56.48
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.233e+004	1.234e+004	1.234e+004	1.235e+004	1.235e+004
32	Pressure (bar)	91.75	91.75	91.75	91.75	91.74
33	Temperature (C)	56.48	56.48	56.48	56.48	56.47
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.98
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.236e+004	1.237e+004	1.237e+004	1.238e+004	1.239e+004
44	Pressure (bar)	91.74	91.74	91.74	91.74	91.74
45	Temperature (C)	56.47	56.47	56.47	56.46	56.46
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.239e+004	1.240e+004	1.240e+004	1.241e+004	1.241e+004
56	Pressure (bar)	91.74	91.74	91.74	91.74	91.74
57	Temperature (C)	56.46	56.46	56.45	56.45	56.45
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.242e+004	1.242e+004	1.243e+004	1.243e+004	1.244e+004
68	Pressure (bar)	91.74	91.74	91.74	91.74	91.74
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	56.45	56.44	56.44	56.44	56.44
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.244e+004	1.245e+004	1.245e+004	1.246e+004	1.246e+004
24	Pressure (bar)	91.74	91.74	91.74	91.74	91.74
25	Temperature (C)	56.44	56.43	56.43	56.43	56.43
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.247e+004	1.247e+004	1.248e+004	1.248e+004	1.249e+004
36	Pressure (bar)	91.74	91.74	91.74	91.74	91.74
37	Temperature (C)	56.43	56.42	56.42	56.42	56.42
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.249e+004	1.250e+004	1.250e+004	1.251e+004	1.251e+004
48	Pressure (bar)	91.74	91.74	91.74	91.74	91.74
49	Temperature (C)	56.42	56.41	56.41	56.41	56.41
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.252e+004	1.252e+004	1.253e+004	1.253e+004	1.254e+004
60	Pressure (bar)	91.74	91.74	91.74	91.74	91.74
61	Temperature (C)	56.41	56.41	56.40	56.40	56.40
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 767 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.254e+004	1.255e+004	1.255e+004	1.256e+004	1.256e+004
16	Pressure (bar)	91.74	91.74	91.74	91.73	91.73
17	Temperature (C)	56.40	56.40	56.39	56.39	56.39
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.257e+004	1.257e+004	1.258e+004	1.258e+004	1.259e+004
28	Pressure (bar)	91.73	91.73	91.73	91.73	91.73
29	Temperature (C)	56.39	56.38	56.38	56.38	56.38
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.259e+004	1.260e+004	1.260e+004	1.261e+004	1.261e+004
40	Pressure (bar)	91.73	91.73	91.72	91.72	91.72
41	Temperature (C)	56.37	56.37	56.37	56.37	56.36
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.262e+004	1.262e+004	1.263e+004	1.264e+004	1.265e+004
52	Pressure (bar)	91.72	91.72	91.72	91.72	91.71
53	Temperature (C)	56.36	56.36	56.36	56.35	56.35
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.265e+004	1.266e+004	1.266e+004	1.267e+004	1.267e+004
64	Pressure (bar)	91.71	91.71	91.71	91.71	91.71
65	Temperature (C)	56.35	56.34	56.34	56.34	56.34
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 768 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.98	17.98	17.98	17.98
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.268e+004	1.268e+004	1.269e+004	1.270e+004
20	Pressure (bar)	91.71	91.71	91.71	91.72
21	Temperature (C)	56.34	56.33	56.33	56.33
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.98	17.98	17.98	17.98
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.270e+004	1.271e+004	1.271e+004	1.272e+004
32	Pressure (bar)	91.72	91.72	91.72	91.72
33	Temperature (C)	56.33	56.33	56.32	56.32
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.98	17.98	17.98	17.98
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.273e+004	1.273e+004	1.274e+004	1.275e+004
44	Pressure (bar)	91.71	91.71	91.71	91.71
45	Temperature (C)	56.32	56.32	56.31	56.31
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.98	17.98	17.98	17.98
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.275e+004	1.276e+004	1.276e+004	1.277e+004
56	Pressure (bar)	91.71	91.71	91.71	91.72
57	Temperature (C)	56.31	56.31	56.30	56.30
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.98	17.98	17.98	17.98
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.277e+004	1.278e+004	1.278e+004	1.278e+004
68	Pressure (bar)	91.72	91.72	91.72	91.72
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 769 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	56.30	56.30	56.30	56.30	56.30
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.279e+004	1.279e+004	1.279e+004	1.280e+004	1.280e+004
24	Pressure (bar)	91.72	91.71	91.71	91.71	91.71
25	Temperature (C)	56.29	56.29	56.29	56.29	56.29
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.280e+004	1.281e+004	1.281e+004	1.282e+004	1.282e+004
36	Pressure (bar)	91.71	91.71	91.71	91.71	91.71
37	Temperature (C)	56.29	56.29	56.28	56.28	56.28
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.283e+004	1.283e+004	1.284e+004	1.284e+004	1.285e+004
48	Pressure (bar)	91.71	91.71	91.71	91.71	91.71
49	Temperature (C)	56.28	56.28	56.27	56.27	56.27
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.285e+004	1.286e+004	1.286e+004	1.287e+004	1.287e+004
60	Pressure (bar)	91.70	91.70	91.70	91.70	91.70
61	Temperature (C)	56.27	56.26	56.26	56.26	56.26
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 770 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.288e+004	1.288e+004	1.289e+004	1.289e+004	1.290e+004
16	Pressure (bar)	91.70	91.70	91.70	91.70	91.70
17	Temperature (C)	56.26	56.25	56.25	56.25	56.25
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.290e+004	1.291e+004	1.291e+004	1.292e+004	1.292e+004
28	Pressure (bar)	91.70	91.70	91.70	91.70	91.70
29	Temperature (C)	56.25	56.24	56.24	56.24	56.24
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.293e+004	1.293e+004	1.294e+004	1.294e+004	1.295e+004
40	Pressure (bar)	91.70	91.70	91.70	91.70	91.70
41	Temperature (C)	56.23	56.23	56.23	56.23	56.23
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.295e+004	1.296e+004	1.296e+004	1.297e+004	1.297e+004
52	Pressure (bar)	91.70	91.70	91.70	91.70	91.70
53	Temperature (C)	56.22	56.22	56.22	56.22	56.22
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.298e+004	1.298e+004	1.299e+004	1.299e+004	1.300e+004
64	Pressure (bar)	91.70	91.70	91.70	91.69	91.69
65	Temperature (C)	56.21	56.21	56.21	56.21	56.21
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 771 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.98	17.98	17.98	17.98
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.300e+004	1.301e+004	1.301e+004	1.302e+004
20	Pressure (bar)	91.69	91.69	91.69	91.69
21	Temperature (C)	56.20	56.20	56.20	56.20
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.98	17.98	17.98	17.98
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.303e+004	1.303e+004	1.304e+004	1.305e+004
32	Pressure (bar)	91.69	91.69	91.69	91.69
33	Temperature (C)	56.19	56.19	56.19	56.19
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.98	17.98	17.98	17.98
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.305e+004	1.306e+004	1.306e+004	1.307e+004
44	Pressure (bar)	91.69	91.69	91.69	91.69
45	Temperature (C)	56.18	56.18	56.18	56.18
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.98	17.98	17.98	17.98
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.308e+004	1.308e+004	1.309e+004	1.310e+004
56	Pressure (bar)	91.69	91.69	91.69	91.69
57	Temperature (C)	56.17	56.17	56.17	56.16
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.98	17.98	17.98	17.98
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.310e+004	1.311e+004	1.311e+004	1.312e+004
68	Pressure (bar)	91.69	91.69	91.69	91.69
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 772 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	56.16	56.16	56.16	56.16	56.15
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.313e+004	1.313e+004	1.314e+004	1.314e+004	1.315e+004
24	Pressure (bar)	91.68	91.68	91.68	91.68	91.68
25	Temperature (C)	56.15	56.15	56.15	56.15	56.14
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.315e+004	1.316e+004	1.316e+004	1.317e+004	1.317e+004
36	Pressure (bar)	91.68	91.68	91.68	91.68	91.68
37	Temperature (C)	56.14	56.14	56.14	56.14	56.13
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.318e+004	1.318e+004	1.319e+004	1.319e+004	1.320e+004
48	Pressure (bar)	91.68	91.68	91.68	91.68	91.68
49	Temperature (C)	56.13	56.13	56.13	56.13	56.12
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.320e+004	1.321e+004	1.321e+004	1.322e+004	1.322e+004
60	Pressure (bar)	91.68	91.68	91.69	91.69	91.69
61	Temperature (C)	56.12	56.12	56.12	56.12	56.12
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.323e+004	1.323e+004	1.324e+004	1.324e+004	1.325e+004
16	Pressure (bar)	91.70	91.70	91.70	91.70	91.71
17	Temperature (C)	56.12	56.12	56.12	56.12	56.11
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.325e+004	1.326e+004	1.326e+004	1.327e+004	1.327e+004
28	Pressure (bar)	91.71	91.72	91.72	91.73	91.74
29	Temperature (C)	56.11	56.11	56.12	56.12	56.12
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.98	17.98	17.98	17.98	17.98
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.328e+004	1.328e+004	1.329e+004	1.329e+004	1.330e+004
40	Pressure (bar)	91.75	91.75	91.75	91.76	91.76
41	Temperature (C)	56.12	56.12	56.12	56.12	56.12
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.330e+004	1.331e+004	1.331e+004	1.332e+004	1.332e+004
52	Pressure (bar)	91.76	91.77	91.77	91.78	91.78
53	Temperature (C)	56.12	56.12	56.12	56.12	56.12
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.333e+004	1.333e+004	1.334e+004	1.334e+004	1.335e+004
64	Pressure (bar)	91.79	91.79	91.80	91.81	91.82
65	Temperature (C)	56.12	56.12	56.12	56.13	56.13
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.335e+004	1.336e+004	1.336e+004	1.337e+004	1.337e+004
20	Pressure (bar)	91.83	91.83	91.84	91.84	91.85
21	Temperature (C)	56.13	56.13	56.13	56.13	56.13
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.338e+004	1.338e+004	1.339e+004	1.339e+004	1.340e+004
32	Pressure (bar)	91.86	91.86	91.86	91.86	91.86
33	Temperature (C)	56.13	56.13	56.13	56.13	56.12
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.340e+004	1.341e+004	1.341e+004	1.342e+004	1.342e+004
44	Pressure (bar)	91.85	91.84	91.82	91.80	91.78
45	Temperature (C)	56.12	56.12	56.11	56.10	56.09
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.99	17.99	17.99	17.99	17.99
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.343e+004	1.343e+004	1.344e+004	1.344e+004	1.345e+004
56	Pressure (bar)	91.76	91.75	91.74	91.73	91.71
57	Temperature (C)	56.09	56.08	56.08	56.07	56.07
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.99	17.99	17.98	17.98	17.98
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.345e+004	1.346e+004	1.346e+004	1.347e+004	1.347e+004
68	Pressure (bar)	91.70	91.58	91.46	91.34	91.22
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	56.06	56.03	56.00	55.97	55.94
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.98	17.97	17.97	17.96	17.95
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.348e+004	1.348e+004	1.349e+004	1.349e+004	1.350e+004
24	Pressure (bar)	91.11	91.10	91.09	91.09	91.08
25	Temperature (C)	55.91	55.91	55.90	55.90	55.90
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.350e+004	1.351e+004	1.351e+004	1.352e+004	1.352e+004
36	Pressure (bar)	91.08	91.08	91.08	91.08	91.08
37	Temperature (C)	55.89	55.89	55.89	55.89	55.89
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.353e+004	1.353e+004	1.353e+004	1.353e+004	1.354e+004
48	Pressure (bar)	91.08	91.08	91.08	91.08	91.08
49	Temperature (C)	55.88	55.88	55.88	55.88	55.88
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.354e+004	1.354e+004	1.354e+004	1.355e+004	1.355e+004
60	Pressure (bar)	91.08	91.08	91.08	91.07	91.07
61	Temperature (C)	55.88	55.88	55.88	55.88	55.87
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 776 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3						
4						
5	Unit Set: Mixed					
6	Date/Time: Mon Jun 13 15:02:44 2022					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	Hydrates					
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.355e+004	1.356e+004	1.356e+004	1.356e+004	1.357e+004
16	Pressure (bar)	91.07	91.07	91.07	91.07	91.07
17	Temperature (C)	55.87	55.87	55.87	55.87	55.87
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.357e+004	1.357e+004	1.358e+004	1.358e+004	1.358e+004
28	Pressure (bar)	91.07	91.07	91.07	91.07	91.07
29	Temperature (C)	55.87	55.86	55.86	55.86	55.86
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.359e+004	1.359e+004	1.359e+004	1.360e+004	1.360e+004
40	Pressure (bar)	91.07	91.07	91.07	91.07	91.07
41	Temperature (C)	55.86	55.86	55.86	55.86	55.85
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.360e+004	1.361e+004	1.361e+004	1.362e+004	1.362e+004
52	Pressure (bar)	91.07	91.07	91.07	91.07	91.07
53	Temperature (C)	55.85	55.85	55.85	55.85	55.84
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.363e+004	1.363e+004	1.364e+004	1.364e+004	1.365e+004
64	Pressure (bar)	91.07	91.07	91.07	91.07	91.07
65	Temperature (C)	55.84	55.84	55.84	55.84	55.83
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 777 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3						
4						
5	Unit Set: Mixed					
6	Date/Time: Mon Jun 13 15:02:44 2022					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	Hydrates					
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.365e+004	1.366e+004	1.366e+004	1.367e+004	1.367e+004
20	Pressure (bar)	91.07	91.07	91.07	91.07	91.07
21	Temperature (C)	55.83	55.83	55.83	55.83	55.82
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.368e+004	1.368e+004	1.369e+004	1.369e+004	1.370e+004
32	Pressure (bar)	91.07	91.07	91.07	91.07	91.06
33	Temperature (C)	55.82	55.82	55.82	55.82	55.81
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.370e+004	1.371e+004	1.371e+004	1.372e+004	1.372e+004
44	Pressure (bar)	91.06	91.06	91.06	91.06	91.06
45	Temperature (C)	55.81	55.81	55.81	55.81	55.80
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.373e+004	1.373e+004	1.374e+004	1.374e+004	1.375e+004
56	Pressure (bar)	91.06	91.06	91.06	91.07	91.07
57	Temperature (C)	55.80	55.80	55.80	55.80	55.80
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.375e+004	1.376e+004	1.376e+004	1.377e+004	1.377e+004
68	Pressure (bar)	91.07	91.07	91.07	91.07	91.07
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	55.79	55.79	55.79	55.79	55.79
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.378e+004	1.378e+004	1.379e+004	1.379e+004	1.380e+004
24	Pressure (bar)	91.07	91.07	91.07	91.07	91.06
25	Temperature (C)	55.78	55.78	55.78	55.78	55.78
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.380e+004	1.381e+004	1.381e+004	1.382e+004	1.382e+004
36	Pressure (bar)	91.06	91.06	91.06	91.06	91.06
37	Temperature (C)	55.77	55.77	55.77	55.77	55.76
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.383e+004	1.383e+004	1.384e+004	1.384e+004	1.385e+004
48	Pressure (bar)	91.06	91.06	91.06	91.06	91.06
49	Temperature (C)	55.76	55.76	55.76	55.76	55.76
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.385e+004	1.386e+004	1.386e+004	1.387e+004	1.387e+004
60	Pressure (bar)	91.06	91.06	91.06	91.06	91.06
61	Temperature (C)	55.76	55.75	55.75	55.75	55.75
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.388e+004	1.388e+004	1.389e+004	1.389e+004	1.390e+004
16	Pressure (bar)	91.06	91.06	91.06	91.06	91.06
17	Temperature (C)	55.74	55.74	55.74	55.74	55.74
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.390e+004	1.391e+004	1.391e+004	1.392e+004	1.392e+004
28	Pressure (bar)	91.06	91.05	91.05	91.05	91.05
29	Temperature (C)	55.73	55.73	55.73	55.73	55.72
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.393e+004	1.393e+004	1.394e+004	1.394e+004	1.395e+004
40	Pressure (bar)	91.05	91.05	91.05	91.05	91.05
41	Temperature (C)	55.72	55.72	55.72	55.72	55.72
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.395e+004	1.396e+004	1.396e+004	1.397e+004	1.397e+004
52	Pressure (bar)	91.05	91.05	91.05	91.05	91.05
53	Temperature (C)	55.71	55.71	55.71	55.71	55.70
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.398e+004	1.398e+004	1.399e+004	1.399e+004	1.400e+004
64	Pressure (bar)	91.05	91.05	91.05	91.05	91.04
65	Temperature (C)	55.70	55.70	55.70	55.70	55.69
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 780 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.400e+004	1.401e+004	1.401e+004	1.402e+004	1.402e+004
20	Pressure (bar)	91.04	91.05	91.05	91.05	91.05
21	Temperature (C)	55.69	55.69	55.69	55.69	55.68
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.403e+004	1.403e+004	1.404e+004	1.404e+004	1.405e+004
32	Pressure (bar)	91.05	91.05	91.04	91.04	91.04
33	Temperature (C)	55.68	55.68	55.68	55.68	55.67
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.405e+004	1.405e+004	1.406e+004	1.406e+004	1.406e+004
44	Pressure (bar)	91.04	91.04	91.04	91.04	91.04
45	Temperature (C)	55.67	55.67	55.67	55.67	55.67
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.406e+004	1.406e+004	1.407e+004	1.407e+004	1.408e+004
56	Pressure (bar)	91.04	91.04	91.04	91.04	91.04
57	Temperature (C)	55.67	55.67	55.66	55.66	55.66
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.408e+004	1.408e+004	1.409e+004	1.409e+004	1.409e+004
68	Pressure (bar)	91.04	91.04	91.04	91.04	91.04
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	55.66	55.66	55.66	55.66	55.66
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.409e+004	1.409e+004	1.410e+004	1.411e+004	1.411e+004
24	Pressure (bar)	91.04	91.04	91.04	91.04	91.04
25	Temperature (C)	55.66	55.66	55.65	55.65	55.65
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.412e+004	1.413e+004	1.413e+004	1.414e+004	1.415e+004
36	Pressure (bar)	91.04	91.04	91.04	91.04	91.04
37	Temperature (C)	55.65	55.65	55.64	55.64	55.64
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.415e+004	1.416e+004	1.416e+004	1.417e+004	1.417e+004
48	Pressure (bar)	91.04	91.04	91.04	91.03	91.03
49	Temperature (C)	55.63	55.63	55.63	55.63	55.62
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.418e+004	1.418e+004	1.419e+004	1.419e+004	1.420e+004
60	Pressure (bar)	91.03	91.03	91.03	91.03	91.03
61	Temperature (C)	55.62	55.62	55.62	55.61	55.61
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.420e+004	1.421e+004	1.421e+004	1.422e+004	1.422e+004
16	Pressure (bar)	91.03	91.03	91.03	91.03	91.03
17	Temperature (C)	55.61	55.61	55.61	55.60	55.60
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.423e+004	1.423e+004	1.424e+004	1.424e+004	1.425e+004
28	Pressure (bar)	91.02	91.02	91.02	91.02	91.02
29	Temperature (C)	55.60	55.60	55.60	55.59	55.59
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.425e+004	1.426e+004	1.426e+004	1.427e+004	1.427e+004
40	Pressure (bar)	91.02	91.02	91.02	91.02	91.02
41	Temperature (C)	55.59	55.59	55.58	55.58	55.58
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.428e+004	1.428e+004	1.429e+004	1.429e+004	1.430e+004
52	Pressure (bar)	91.02	91.02	91.02	91.02	91.02
53	Temperature (C)	55.58	55.58	55.57	55.57	55.57
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.430e+004	1.431e+004	1.431e+004	1.432e+004	1.432e+004
64	Pressure (bar)	91.02	91.02	91.02	91.02	91.01
65	Temperature (C)	55.57	55.57	55.56	55.56	55.56
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 783 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.433e+004	1.433e+004	1.434e+004	1.434e+004	1.435e+004
20	Pressure (bar)	91.01	91.01	91.01	91.01	91.01
21	Temperature (C)	55.56	55.56	55.55	55.55	55.55
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.435e+004	1.436e+004	1.436e+004	1.437e+004	1.437e+004
32	Pressure (bar)	91.01	91.01	91.01	91.01	91.01
33	Temperature (C)	55.55	55.55	55.54	55.54	55.54
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.438e+004	1.438e+004	1.439e+004	1.439e+004	1.440e+004
44	Pressure (bar)	91.01	91.01	91.01	91.01	91.01
45	Temperature (C)	55.54	55.54	55.53	55.53	55.53
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.440e+004	1.441e+004	1.441e+004	1.442e+004	1.442e+004
56	Pressure (bar)	91.01	91.01	91.01	91.01	91.01
57	Temperature (C)	55.53	55.53	55.52	55.52	55.52
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.443e+004	1.443e+004	1.444e+004	1.444e+004	1.445e+004
68	Pressure (bar)	91.01	91.01	91.01	91.01	91.01
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 784 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	55.52	55.52	55.51	55.51	55.51
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.445e+004	1.446e+004	1.446e+004	1.447e+004	1.447e+004
24	Pressure (bar)	91.01	91.01	91.01	91.01	91.01
25	Temperature (C)	55.51	55.51	55.50	55.50	55.50
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.448e+004	1.448e+004	1.449e+004	1.449e+004	1.450e+004
36	Pressure (bar)	91.01	91.01	91.00	91.00	91.00
37	Temperature (C)	55.50	55.50	55.49	55.49	55.49
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.450e+004	1.451e+004	1.451e+004	1.452e+004	1.452e+004
48	Pressure (bar)	91.00	91.00	91.00	91.00	91.00
49	Temperature (C)	55.49	55.49	55.48	55.48	55.48
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.453e+004	1.453e+004	1.454e+004	1.454e+004	1.455e+004
60	Pressure (bar)	91.00	91.00	91.00	91.00	91.00
61	Temperature (C)	55.48	55.48	55.47	55.47	55.47
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 785 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.455e+004	1.456e+004	1.456e+004	1.457e+004	1.457e+004
16	Pressure (bar)	91.00	91.00	91.00	91.00	91.00
17	Temperature (C)	55.47	55.47	55.46	55.46	55.46
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.458e+004	1.458e+004	1.459e+004	1.459e+004	1.460e+004
28	Pressure (bar)	91.00	91.00	91.00	91.00	91.00
29	Temperature (C)	55.46	55.46	55.45	55.45	55.45
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.460e+004	1.461e+004	1.461e+004	1.462e+004	1.462e+004
40	Pressure (bar)	91.00	91.00	91.00	91.00	91.00
41	Temperature (C)	55.45	55.45	55.44	55.44	55.44
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.463e+004	1.463e+004	1.464e+004	1.464e+004	1.465e+004
52	Pressure (bar)	91.00	91.00	91.00	91.00	91.00
53	Temperature (C)	55.44	55.44	55.43	55.43	55.43
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.465e+004	1.466e+004	1.466e+004	1.467e+004	1.467e+004
64	Pressure (bar)	91.00	91.00	90.99	90.99	90.99
65	Temperature (C)	55.43	55.43	55.42	55.42	55.42
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 786 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.468e+004	1.468e+004	1.469e+004	1.469e+004	1.470e+004
20	Pressure (bar)	90.99	90.99	90.99	90.99	90.99
21	Temperature (C)	55.42	55.42	55.41	55.41	55.41
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.470e+004	1.471e+004	1.471e+004	1.472e+004	1.472e+004
32	Pressure (bar)	90.99	90.99	90.99	90.99	90.99
33	Temperature (C)	55.41	55.41	55.40	55.40	55.40
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.473e+004	1.473e+004	1.474e+004	1.474e+004	1.475e+004
44	Pressure (bar)	90.99	90.99	90.99	90.99	90.99
45	Temperature (C)	55.40	55.40	55.39	55.39	55.39
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.475e+004	1.476e+004	1.476e+004	1.477e+004	1.477e+004
56	Pressure (bar)	90.99	90.99	90.99	90.99	90.99
57	Temperature (C)	55.39	55.39	55.39	55.38	55.38
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.478e+004	1.478e+004	1.479e+004	1.479e+004	1.480e+004
68	Pressure (bar)	90.99	90.99	90.99	90.99	90.99
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3						
4						
5	Date/Time: Mon Jun 13 15:02:44 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10	Hydrates Table					
11						
12						
13	Temperature (C)	55.38	55.38	55.38	55.37	55.37
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.480e+004	1.481e+004	1.481e+004	1.482e+004	1.482e+004
24	Pressure (bar)	90.99	90.99	90.99	90.99	90.99
25	Temperature (C)	55.37	55.37	55.37	55.36	55.36
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.483e+004	1.483e+004	1.484e+004	1.484e+004	1.485e+004
36	Pressure (bar)	90.99	90.99	90.99	90.99	90.98
37	Temperature (C)	55.36	55.36	55.36	55.35	55.35
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.485e+004	1.486e+004	1.486e+004	1.486e+004	1.486e+004
48	Pressure (bar)	90.98	90.98	90.98	90.98	90.98
49	Temperature (C)	55.35	55.35	55.35	55.35	55.35
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.486e+004	1.488e+004	1.490e+004	1.492e+004	1.494e+004
60	Pressure (bar)	90.98	90.98	90.98	90.98	90.98
61	Temperature (C)	55.35	55.34	55.33	55.32	55.31
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 788 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.496e+004	1.498e+004	1.500e+004	1.502e+004	1.504e+004
16	Pressure (bar)	90.98	90.98	90.98	90.97	90.97
17	Temperature (C)	55.31	55.30	55.29	55.28	55.27
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.506e+004	1.508e+004	1.510e+004	1.512e+004	1.514e+004
28	Pressure (bar)	90.97	90.97	90.97	90.97	90.97
29	Temperature (C)	55.27	55.26	55.25	55.24	55.23
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.516e+004	1.518e+004	1.520e+004	1.522e+004	1.524e+004
40	Pressure (bar)	90.97	90.97	90.96	90.96	90.96
41	Temperature (C)	55.23	55.22	55.21	55.20	55.20
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.526e+004	1.528e+004	1.530e+004	1.532e+004	1.534e+004
52	Pressure (bar)	90.96	90.96	90.96	90.96	90.96
53	Temperature (C)	55.19	55.18	55.17	55.16	55.16
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.536e+004	1.538e+004	1.540e+004	1.542e+004	1.544e+004
64	Pressure (bar)	90.96	90.97	90.98	90.99	91.00
65	Temperature (C)	55.15	55.15	55.14	55.14	55.14
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 789 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.94	17.94	17.94	17.94	17.94
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.546e+004	1.548e+004	1.550e+004	1.552e+004	1.554e+004
20	Pressure (bar)	91.01	90.99	90.97	90.95	90.93
21	Temperature (C)	55.13	55.12	55.10	55.09	55.08
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.94	17.94	17.94	17.93	17.93
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.556e+004	1.558e+004	1.560e+004	1.562e+004	1.564e+004
32	Pressure (bar)	90.91	90.91	90.90	90.90	90.90
33	Temperature (C)	55.06	55.06	55.05	55.04	55.03
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.566e+004	1.568e+004	1.570e+004	1.572e+004	1.574e+004
44	Pressure (bar)	90.89	90.89	90.89	90.89	90.89
45	Temperature (C)	55.02	55.01	55.01	55.00	54.99
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.576e+004	1.577e+004	1.577e+004	1.578e+004	1.578e+004
56	Pressure (bar)	90.89	90.89	90.89	90.89	90.89
57	Temperature (C)	54.98	54.98	54.98	54.98	54.97
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.579e+004	1.581e+004	1.583e+004	1.585e+004	1.587e+004
68	Pressure (bar)	90.89	90.89	90.88	90.88	90.88
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	54.97	54.96	54.96	54.95	54.94
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.589e+004	1.589e+004	1.590e+004	1.590e+004	1.590e+004
24	Pressure (bar)	90.88	90.88	90.88	90.88	90.88
25	Temperature (C)	54.93	54.93	54.93	54.93	54.93
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.591e+004	1.591e+004	1.592e+004	1.592e+004	1.592e+004
36	Pressure (bar)	90.88	90.88	90.88	90.88	90.88
37	Temperature (C)	54.93	54.92	54.92	54.92	54.92
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.593e+004	1.593e+004	1.593e+004	1.593e+004	1.594e+004
48	Pressure (bar)	90.88	90.88	90.88	90.87	90.87
49	Temperature (C)	54.92	54.92	54.92	54.91	54.91
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.594e+004	1.594e+004	1.595e+004	1.595e+004	1.595e+004
60	Pressure (bar)	90.87	90.87	90.87	90.87	90.87
61	Temperature (C)	54.91	54.91	54.91	54.91	54.91
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 791 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.596e+004	1.596e+004	1.597e+004	1.597e+004	1.598e+004
16	Pressure (bar)	90.87	90.87	90.87	90.87	90.87
17	Temperature (C)	54.91	54.90	54.90	54.90	54.90
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.598e+004	1.599e+004	1.599e+004	1.600e+004	1.600e+004
28	Pressure (bar)	90.87	90.87	90.87	90.87	90.87
29	Temperature (C)	54.90	54.89	54.89	54.89	54.89
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.601e+004	1.601e+004	1.602e+004	1.602e+004	1.603e+004
40	Pressure (bar)	90.87	90.87	90.87	90.87	90.87
41	Temperature (C)	54.89	54.88	54.88	54.88	54.88
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.603e+004	1.604e+004	1.604e+004	1.605e+004	1.605e+004
52	Pressure (bar)	90.87	90.87	90.87	90.87	90.87
53	Temperature (C)	54.88	54.87	54.87	54.87	54.87
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.606e+004	1.606e+004	1.607e+004	1.607e+004	1.608e+004
64	Pressure (bar)	90.87	90.87	90.87	90.87	90.87
65	Temperature (C)	54.87	54.86	54.86	54.86	54.86
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.608e+004	1.609e+004	1.609e+004	1.610e+004	1.610e+004
20	Pressure (bar)	90.87	90.87	90.87	90.87	90.87
21	Temperature (C)	54.86	54.86	54.85	54.85	54.85
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.611e+004	1.611e+004	1.612e+004	1.612e+004	1.613e+004
32	Pressure (bar)	90.87	90.87	90.87	90.87	90.87
33	Temperature (C)	54.85	54.85	54.84	54.84	54.84
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.613e+004	1.614e+004	1.614e+004	1.615e+004	1.615e+004
44	Pressure (bar)	90.87	90.87	90.87	90.87	90.87
45	Temperature (C)	54.84	54.84	54.84	54.83	54.83
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.616e+004	1.616e+004	1.617e+004	1.617e+004	1.618e+004
56	Pressure (bar)	90.87	90.87	90.87	90.87	90.87
57	Temperature (C)	54.83	54.83	54.83	54.82	54.82
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.618e+004	1.619e+004	1.619e+004	1.620e+004	1.620e+004
68	Pressure (bar)	90.87	90.87	90.87	90.87	90.87
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 793 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	54.82	54.82	54.82	54.82	54.81
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.621e+004	1.621e+004	1.622e+004	1.622e+004	1.623e+004
24	Pressure (bar)	90.87	90.86	90.86	90.86	90.85
25	Temperature (C)	54.81	54.81	54.81	54.80	54.80
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.623e+004	1.624e+004	1.624e+004	1.625e+004	1.625e+004
36	Pressure (bar)	90.85	90.85	90.85	90.85	90.85
37	Temperature (C)	54.80	54.79	54.79	54.79	54.79
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.626e+004	1.626e+004	1.627e+004	1.627e+004	1.628e+004
48	Pressure (bar)	90.85	90.85	90.85	90.85	90.85
49	Temperature (C)	54.79	54.78	54.78	54.78	54.78
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.628e+004	1.629e+004	1.629e+004	1.630e+004	1.630e+004
60	Pressure (bar)	90.85	90.85	90.85	90.85	90.85
61	Temperature (C)	54.78	54.78	54.77	54.77	54.77
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.631e+004	1.631e+004	1.631e+004	1.631e+004	1.632e+004
16	Pressure (bar)	90.85	90.85	90.85	90.85	90.85
17	Temperature (C)	54.77	54.77	54.77	54.77	54.77
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.632e+004	1.632e+004	1.632e+004	1.633e+004	1.633e+004
28	Pressure (bar)	90.85	90.85	90.85	90.85	90.84
29	Temperature (C)	54.77	54.77	54.76	54.76	54.76
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.633e+004	1.634e+004	1.634e+004	1.635e+004	1.635e+004
40	Pressure (bar)	90.84	90.84	90.84	90.84	90.84
41	Temperature (C)	54.76	54.76	54.75	54.75	54.75
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.636e+004	1.636e+004	1.637e+004	1.637e+004	1.638e+004
52	Pressure (bar)	90.84	90.84	90.84	90.84	90.84
53	Temperature (C)	54.75	54.75	54.74	54.74	54.74
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.638e+004	1.639e+004	1.639e+004	1.640e+004	1.640e+004
64	Pressure (bar)	90.84	90.84	90.84	90.83	90.83
65	Temperature (C)	54.74	54.74	54.73	54.73	54.73
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.93	17.93	17.93	17.93
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.641e+004	1.641e+004	1.642e+004	1.643e+004
20	Pressure (bar)	90.83	90.83	90.83	90.83
21	Temperature (C)	54.73	54.73	54.72	54.72
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.93	17.93	17.93	17.93
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.643e+004	1.644e+004	1.644e+004	1.645e+004
32	Pressure (bar)	90.83	90.83	90.83	90.83
33	Temperature (C)	54.72	54.72	54.71	54.71
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.93	17.93	17.93	17.93
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.646e+004	1.646e+004	1.647e+004	1.648e+004
44	Pressure (bar)	90.83	90.83	90.83	90.83
45	Temperature (C)	54.71	54.71	54.70	54.70
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.93	17.93	17.93	17.93
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.648e+004	1.649e+004	1.649e+004	1.650e+004
56	Pressure (bar)	90.83	90.83	90.83	90.83
57	Temperature (C)	54.70	54.70	54.69	54.69
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.93	17.93	17.93	17.93
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.651e+004	1.651e+004	1.652e+004	1.653e+004
68	Pressure (bar)	90.83	90.83	90.83	90.83
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3						
4			Unit Set: Mixed			
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11						
12	Hydrates Table					
13	Temperature (C)	54.69	54.69	54.68	54.68	54.68
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.653e+004	1.654e+004	1.654e+004	1.655e+004	1.655e+004
24	Pressure (bar)	90.83	90.83	90.82	90.82	90.82
25	Temperature (C)	54.68	54.68	54.68	54.67	54.67
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.656e+004	1.656e+004	1.657e+004	1.657e+004	1.658e+004
36	Pressure (bar)	90.82	90.82	90.82	90.82	90.82
37	Temperature (C)	54.67	54.67	54.67	54.66	54.66
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.658e+004	1.659e+004	1.659e+004	1.660e+004	1.660e+004
48	Pressure (bar)	90.82	90.82	90.82	90.82	90.82
49	Temperature (C)	54.66	54.66	54.66	54.65	54.65
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.661e+004	1.661e+004	1.662e+004	1.662e+004	1.663e+004
60	Pressure (bar)	90.82	90.82	90.82	90.82	90.82
61	Temperature (C)	54.65	54.65	54.65	54.64	54.64
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.663e+004	1.664e+004	1.664e+004	1.665e+004	1.665e+004
16	Pressure (bar)	90.82	90.82	90.82	90.82	90.82
17	Temperature (C)	54.64	54.64	54.64	54.64	54.63
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.666e+004	1.666e+004	1.667e+004	1.667e+004	1.668e+004
28	Pressure (bar)	90.82	90.82	90.82	90.82	90.82
29	Temperature (C)	54.63	54.63	54.63	54.63	54.62
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.668e+004	1.669e+004	1.669e+004	1.670e+004	1.670e+004
40	Pressure (bar)	90.82	90.82	90.82	90.82	90.82
41	Temperature (C)	54.62	54.62	54.62	54.62	54.61
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.671e+004	1.671e+004	1.672e+004	1.672e+004	1.673e+004
52	Pressure (bar)	90.82	90.82	90.82	90.82	90.82
53	Temperature (C)	54.61	54.61	54.61	54.61	54.60
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.673e+004	1.674e+004	1.674e+004	1.675e+004	1.675e+004
64	Pressure (bar)	90.82	90.82	90.82	90.82	90.82
65	Temperature (C)	54.60	54.60	54.60	54.60	54.59
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 798 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.93	17.93	17.93	17.93
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.676e+004	1.676e+004	1.677e+004	1.678e+004
20	Pressure (bar)	90.82	90.81	90.81	90.81
21	Temperature (C)	54.59	54.59	54.59	54.59
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.93	17.93	17.93	17.93
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.678e+004	1.679e+004	1.679e+004	1.680e+004
32	Pressure (bar)	90.81	90.81	90.81	90.81
33	Temperature (C)	54.58	54.58	54.58	54.58
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.93	17.93	17.93	17.93
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.681e+004	1.681e+004	1.682e+004	1.682e+004
44	Pressure (bar)	90.81	90.81	90.81	90.81
45	Temperature (C)	54.57	54.57	54.57	54.57
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.93	17.93	17.93	17.93
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.683e+004	1.684e+004	1.684e+004	1.685e+004
56	Pressure (bar)	90.81	90.81	90.81	90.81
57	Temperature (C)	54.56	54.56	54.56	54.56
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.93	17.93	17.93	17.93
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.686e+004	1.686e+004	1.687e+004	1.688e+004
68	Pressure (bar)	90.81	90.81	90.81	90.81
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	54.55	54.55	54.55	54.55	54.55
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.688e+004	1.689e+004	1.689e+004	1.690e+004	1.690e+004
24	Pressure (bar)	90.81	90.81	90.81	90.81	90.81
25	Temperature (C)	54.55	54.54	54.54	54.54	54.54
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.690e+004	1.691e+004	1.691e+004	1.692e+004	1.692e+004
36	Pressure (bar)	90.81	90.81	90.81	90.81	90.81
37	Temperature (C)	54.54	54.54	54.53	54.53	54.53
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.693e+004	1.693e+004	1.694e+004	1.694e+004	1.695e+004
48	Pressure (bar)	90.81	90.81	90.81	90.81	90.81
49	Temperature (C)	54.53	54.53	54.52	54.52	54.52
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.695e+004	1.696e+004	1.696e+004	1.697e+004	1.697e+004
60	Pressure (bar)	90.81	90.81	90.81	90.81	90.81
61	Temperature (C)	54.52	54.52	54.51	54.51	54.51
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.698e+004	1.698e+004	1.699e+004	1.699e+004	1.700e+004
16	Pressure (bar)	90.81	90.81	90.80	90.80	90.80
17	Temperature (C)	54.51	54.51	54.50	54.50	54.50
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.700e+004	1.701e+004	1.701e+004	1.702e+004	1.702e+004
28	Pressure (bar)	90.80	90.80	90.80	90.80	90.80
29	Temperature (C)	54.50	54.50	54.49	54.49	54.49
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.703e+004	1.703e+004	1.704e+004	1.704e+004	1.705e+004
40	Pressure (bar)	90.80	90.80	90.80	90.80	90.80
41	Temperature (C)	54.49	54.49	54.49	54.48	54.48
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.705e+004	1.706e+004	1.706e+004	1.707e+004	1.707e+004
52	Pressure (bar)	90.80	90.80	90.80	90.80	90.80
53	Temperature (C)	54.48	54.48	54.48	54.47	54.47
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.708e+004	1.708e+004	1.709e+004	1.709e+004	1.710e+004
64	Pressure (bar)	90.80	90.80	90.80	90.80	90.80
65	Temperature (C)	54.47	54.47	54.47	54.46	54.46
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 801 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.92	17.92	17.92	17.92
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.710e+004	1.711e+004	1.711e+004	1.712e+004
20	Pressure (bar)	90.80	90.79	90.79	90.79
21	Temperature (C)	54.46	54.46	54.46	54.45
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.92	17.92	17.92	17.92
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.713e+004	1.713e+004	1.714e+004	1.715e+004
32	Pressure (bar)	90.79	90.79	90.79	90.79
33	Temperature (C)	54.45	54.45	54.45	54.44
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.92	17.92	17.92	17.92
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.715e+004	1.716e+004	1.716e+004	1.717e+004
44	Pressure (bar)	90.79	90.79	90.79	90.79
45	Temperature (C)	54.44	54.44	54.44	54.43
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.92	17.92	17.92	17.92
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.718e+004	1.718e+004	1.719e+004	1.720e+004
56	Pressure (bar)	90.79	90.79	90.79	90.79
57	Temperature (C)	54.43	54.43	54.43	54.42
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.92	17.92	17.92	17.92
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.720e+004	1.721e+004	1.721e+004	1.722e+004
68	Pressure (bar)	90.79	90.78	90.78	90.78
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 802 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	54.42	54.42	54.42	54.41	54.41
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.723e+004	1.723e+004	1.724e+004	1.724e+004	1.725e+004
24	Pressure (bar)	90.78	90.78	90.78	90.78	90.78
25	Temperature (C)	54.41	54.41	54.41	54.40	54.40
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.725e+004	1.726e+004	1.726e+004	1.727e+004	1.727e+004
36	Pressure (bar)	90.78	90.78	90.78	90.78	90.78
37	Temperature (C)	54.40	54.40	54.40	54.39	54.39
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.728e+004	1.728e+004	1.729e+004	1.729e+004	1.730e+004
48	Pressure (bar)	90.78	90.78	90.78	90.78	90.78
49	Temperature (C)	54.39	54.39	54.39	54.38	54.38
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.730e+004	1.731e+004	1.731e+004	1.732e+004	1.732e+004
60	Pressure (bar)	90.78	90.78	90.78	90.78	90.78
61	Temperature (C)	54.38	54.38	54.38	54.37	54.37
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 803 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.733e+004	1.733e+004	1.734e+004	1.734e+004	1.735e+004
16	Pressure (bar)	90.77	90.77	90.77	90.77	90.77
17	Temperature (C)	54.37	54.37	54.37	54.37	54.36
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.735e+004	1.736e+004	1.736e+004	1.737e+004	1.737e+004
28	Pressure (bar)	90.77	90.77	90.77	90.77	90.77
29	Temperature (C)	54.36	54.36	54.36	54.36	54.35
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.738e+004	1.738e+004	1.739e+004	1.739e+004	1.740e+004
40	Pressure (bar)	90.77	90.77	90.77	90.77	90.77
41	Temperature (C)	54.35	54.35	54.35	54.35	54.34
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.740e+004	1.741e+004	1.741e+004	1.742e+004	1.742e+004
52	Pressure (bar)	90.77	90.77	90.77	90.77	90.77
53	Temperature (C)	54.34	54.34	54.34	54.34	54.33
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.743e+004	1.743e+004	1.744e+004	1.744e+004	1.745e+004
64	Pressure (bar)	90.77	90.77	90.77	90.77	90.77
65	Temperature (C)	54.33	54.33	54.33	54.33	54.32
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 804 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.92	17.92	17.92	17.92
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.745e+004	1.746e+004	1.746e+004	1.747e+004
20	Pressure (bar)	90.77	90.77	90.77	90.77
21	Temperature (C)	54.32	54.32	54.32	54.32
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.92	17.92	17.92	17.92
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.748e+004	1.748e+004	1.749e+004	1.750e+004
32	Pressure (bar)	90.77	90.77	90.77	90.77
33	Temperature (C)	54.31	54.31	54.31	54.31
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.92	17.92	17.92	17.92
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.750e+004	1.751e+004	1.751e+004	1.752e+004
44	Pressure (bar)	90.77	90.77	90.76	90.76
45	Temperature (C)	54.31	54.30	54.30	54.30
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.92	17.92	17.92	17.92
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.753e+004	1.753e+004	1.754e+004	1.755e+004
56	Pressure (bar)	90.76	90.76	90.76	90.76
57	Temperature (C)	54.30	54.29	54.29	54.29
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.92	17.92	17.92	17.92
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.755e+004	1.756e+004	1.756e+004	1.757e+004
68	Pressure (bar)	90.76	90.76	90.76	90.76
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 805 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	54.29	54.28	54.28	54.28	54.28
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.758e+004	1.758e+004	1.759e+004	1.759e+004	1.760e+004
24	Pressure (bar)	90.76	90.76	90.76	90.76	90.76
25	Temperature (C)	54.28	54.28	54.27	54.27	54.27
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.760e+004	1.761e+004	1.761e+004	1.762e+004	1.762e+004
36	Pressure (bar)	90.76	90.76	90.76	90.76	90.76
37	Temperature (C)	54.27	54.26	54.26	54.26	54.26
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.763e+004	1.763e+004	1.764e+004	1.764e+004	1.765e+004
48	Pressure (bar)	90.76	90.76	90.76	90.76	90.76
49	Temperature (C)	54.26	54.26	54.25	54.25	54.25
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.765e+004	1.766e+004	1.766e+004	1.767e+004	1.767e+004
60	Pressure (bar)	90.76	90.76	90.75	90.75	90.75
61	Temperature (C)	54.25	54.25	54.24	54.24	54.24
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 806 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.768e+004	1.768e+004	1.769e+004	1.769e+004	1.770e+004
16	Pressure (bar)	90.75	90.75	90.75	90.75	90.75
17	Temperature (C)	54.24	54.24	54.23	54.23	54.23
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.770e+004	1.771e+004	1.771e+004	1.772e+004	1.772e+004
28	Pressure (bar)	90.75	90.75	90.75	90.75	90.75
29	Temperature (C)	54.23	54.23	54.22	54.22	54.22
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.773e+004	1.773e+004	1.774e+004	1.774e+004	1.775e+004
40	Pressure (bar)	90.75	90.75	90.75	90.75	90.75
41	Temperature (C)	54.22	54.22	54.22	54.21	54.21
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.775e+004	1.776e+004	1.776e+004	1.777e+004	1.777e+004
52	Pressure (bar)	90.75	90.75	90.75	90.75	90.75
53	Temperature (C)	54.21	54.21	54.21	54.20	54.20
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.778e+004	1.778e+004	1.779e+004	1.779e+004	1.780e+004
64	Pressure (bar)	90.75	90.75	90.75	90.75	90.75
65	Temperature (C)	54.20	54.20	54.20	54.20	54.19
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 807 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.780e+004	1.781e+004	1.781e+004	1.782e+004	1.782e+004
20	Pressure (bar)	90.75	90.75	90.75	90.75	90.75
21	Temperature (C)	54.19	54.19	54.19	54.19	54.18
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.783e+004	1.783e+004	1.784e+004	1.784e+004	1.785e+004
32	Pressure (bar)	90.75	90.75	90.75	90.75	90.75
33	Temperature (C)	54.18	54.18	54.18	54.18	54.18
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.785e+004	1.785e+004	1.786e+004	1.786e+004	1.786e+004
44	Pressure (bar)	90.76	90.76	90.76	90.76	90.76
45	Temperature (C)	54.18	54.18	54.17	54.17	54.17
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.787e+004	1.787e+004	1.787e+004	1.787e+004	1.787e+004
56	Pressure (bar)	90.76	90.76	90.76	90.76	90.76
57	Temperature (C)	54.17	54.17	54.17	54.17	54.17
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.788e+004	1.788e+004	1.789e+004	1.789e+004	1.790e+004
68	Pressure (bar)	90.76	90.77	90.78	90.78	90.79
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	54.17	54.17	54.17	54.17	54.18
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.92	17.92	17.92	17.92	17.92
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.790e+004	1.790e+004	1.791e+004	1.791e+004	1.791e+004
24	Pressure (bar)	90.80	90.80	90.81	90.81	90.81
25	Temperature (C)	54.18	54.18	54.18	54.18	54.18
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.92	17.92	17.93	17.93	17.93
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.791e+004	1.792e+004	1.792e+004	1.792e+004	1.793e+004
36	Pressure (bar)	90.81	90.82	90.82	90.82	90.83
37	Temperature (C)	54.18	54.18	54.18	54.18	54.18
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.793e+004	1.793e+004	1.794e+004	1.794e+004	1.794e+004
48	Pressure (bar)	90.83	90.83	90.84	90.84	90.85
49	Temperature (C)	54.18	54.18	54.18	54.18	54.18
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.795e+004	1.795e+004	1.796e+004	1.796e+004	1.797e+004
60	Pressure (bar)	90.85	90.85	90.86	90.86	90.87
61	Temperature (C)	54.18	54.18	54.18	54.18	54.18
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 809 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.798e+004	1.798e+004	1.799e+004	1.799e+004	1.800e+004
16	Pressure (bar)	90.87	90.88	90.89	90.89	90.90
17	Temperature (C)	54.18	54.18	54.19	54.19	54.19
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.800e+004	1.801e+004	1.801e+004	1.802e+004	1.802e+004
28	Pressure (bar)	90.91	90.91	90.92	90.93	90.93
29	Temperature (C)	54.19	54.19	54.19	54.19	54.19
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.803e+004	1.803e+004	1.804e+004	1.804e+004	1.805e+004
40	Pressure (bar)	90.94	90.93	90.92	90.91	90.90
41	Temperature (C)	54.19	54.19	54.18	54.18	54.17
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.93	17.93	17.93	17.93	17.93
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.805e+004	1.806e+004	1.806e+004	1.807e+004	1.807e+004
52	Pressure (bar)	90.88	90.72	90.55	90.38	90.22
53	Temperature (C)	54.17	54.13	54.09	54.05	54.00
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.93	17.92	17.91	17.90	17.89
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.808e+004	1.808e+004	1.809e+004	1.809e+004	1.810e+004
64	Pressure (bar)	90.05	90.05	90.05	90.04	90.04
65	Temperature (C)	53.96	53.96	53.96	53.96	53.95
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 810 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.88	17.88	17.88	17.88	17.88
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.810e+004	1.811e+004	1.811e+004	1.812e+004	1.812e+004
20	Pressure (bar)	90.04	90.04	90.04	90.04	90.04
21	Temperature (C)	53.95	53.95	53.95	53.95	53.94
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.88	17.88	17.88	17.88	17.88
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.813e+004	1.813e+004	1.814e+004	1.814e+004	1.815e+004
32	Pressure (bar)	90.04	90.04	90.03	90.03	90.03
33	Temperature (C)	53.94	53.94	53.94	53.94	53.93
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.88	17.88	17.88	17.88	17.88
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.815e+004	1.816e+004	1.816e+004	1.817e+004	1.817e+004
44	Pressure (bar)	90.03	90.03	90.03	90.03	90.03
45	Temperature (C)	53.93	53.93	53.93	53.93	53.92
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.88	17.88	17.88	17.88	17.88
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.818e+004	1.818e+004	1.819e+004	1.819e+004	1.820e+004
56	Pressure (bar)	90.03	90.03	90.03	90.03	90.03
57	Temperature (C)	53.92	53.92	53.92	53.92	53.91
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.88	17.88	17.88	17.88	17.88
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.820e+004	1.821e+004	1.821e+004	1.822e+004	1.822e+004
68	Pressure (bar)	90.03	90.03	90.03	90.03	90.03
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 811 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	53.91	53.91	53.91	53.91	53.90
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.88	17.88	17.88	17.88	17.88
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.823e+004	1.823e+004	1.824e+004	1.824e+004	1.825e+004
24	Pressure (bar)	90.03	90.03	90.03	90.03	90.03
25	Temperature (C)	53.90	53.90	53.90	53.90	53.89
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.825e+004	1.826e+004	1.826e+004	1.827e+004	1.827e+004
36	Pressure (bar)	90.03	90.03	90.03	90.03	90.03
37	Temperature (C)	53.89	53.89	53.89	53.89	53.89
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.828e+004	1.828e+004	1.829e+004	1.829e+004	1.830e+004
48	Pressure (bar)	90.02	90.02	90.02	90.02	90.02
49	Temperature (C)	53.88	53.88	53.88	53.88	53.88
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.830e+004	1.831e+004	1.831e+004	1.832e+004	1.832e+004
60	Pressure (bar)	90.02	90.02	90.02	90.02	90.02
61	Temperature (C)	53.87	53.87	53.87	53.87	53.87
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.833e+004	1.833e+004	1.834e+004	1.834e+004	1.835e+004
16	Pressure (bar)	90.02	90.02	90.02	90.02	90.02
17	Temperature (C)	53.87	53.86	53.86	53.86	53.86
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.835e+004	1.836e+004	1.836e+004	1.837e+004	1.837e+004
28	Pressure (bar)	90.02	90.02	90.02	90.02	90.02
29	Temperature (C)	53.86	53.85	53.85	53.85	53.85
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.838e+004	1.838e+004	1.839e+004	1.839e+004	1.840e+004
40	Pressure (bar)	90.02	90.02	90.02	90.02	90.02
41	Temperature (C)	53.85	53.84	53.84	53.84	53.84
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.840e+004	1.841e+004	1.841e+004	1.842e+004	1.842e+004
52	Pressure (bar)	90.02	90.02	90.02	90.02	90.02
53	Temperature (C)	53.84	53.84	53.83	53.83	53.83
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.843e+004	1.843e+004	1.844e+004	1.844e+004	1.845e+004
64	Pressure (bar)	90.02	90.02	90.02	90.02	90.02
65	Temperature (C)	53.83	53.83	53.82	53.82	53.82
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.845e+004	1.846e+004	1.846e+004	1.847e+004	1.847e+004
20	Pressure (bar)	90.02	90.02	90.01	90.01	90.01
21	Temperature (C)	53.82	53.82	53.81	53.81	53.81
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.848e+004	1.848e+004	1.849e+004	1.849e+004	1.850e+004
32	Pressure (bar)	90.01	90.01	90.01	90.01	90.01
33	Temperature (C)	53.81	53.81	53.81	53.80	53.80
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.850e+004	1.851e+004	1.851e+004	1.852e+004	1.852e+004
44	Pressure (bar)	90.01	90.01	90.01	90.01	90.01
45	Temperature (C)	53.80	53.80	53.80	53.79	53.79
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.853e+004	1.853e+004	1.854e+004	1.854e+004	1.855e+004
56	Pressure (bar)	90.01	90.01	90.01	90.01	90.01
57	Temperature (C)	53.79	53.79	53.79	53.78	53.78
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.855e+004	1.856e+004	1.856e+004	1.857e+004	1.857e+004
68	Pressure (bar)	90.01	90.01	90.01	90.01	90.01
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 814 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	53.78	53.78	53.78	53.78	53.77
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.858e+004	1.858e+004	1.859e+004	1.859e+004	1.860e+004
24	Pressure (bar)	90.01	90.01	90.01	90.01	90.01
25	Temperature (C)	53.77	53.77	53.77	53.77	53.76
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.860e+004	1.861e+004	1.861e+004	1.862e+004	1.862e+004
36	Pressure (bar)	90.01	90.01	90.01	90.01	90.01
37	Temperature (C)	53.76	53.76	53.76	53.76	53.76
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.863e+004	1.863e+004	1.864e+004	1.864e+004	1.865e+004
48	Pressure (bar)	90.01	90.01	90.01	90.01	90.01
49	Temperature (C)	53.75	53.75	53.75	53.75	53.75
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.865e+004	1.866e+004	1.866e+004	1.867e+004	1.867e+004
60	Pressure (bar)	90.01	90.01	90.01	90.01	90.01
61	Temperature (C)	53.74	53.74	53.74	53.74	53.74
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3						
4						
5	Unit Set: Mixed					
6	Date/Time: Mon Jun 13 15:02:44 2022					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	Hydrates					
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.868e+004	1.868e+004	1.869e+004	1.869e+004	1.870e+004
16	Pressure (bar)	90.00	90.01	90.01	90.01	90.01
17	Temperature (C)	53.74	53.73	53.73	53.73	53.73
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.870e+004	1.871e+004	1.871e+004	1.872e+004	1.872e+004
28	Pressure (bar)	90.01	90.01	90.01	90.00	90.00
29	Temperature (C)	53.73	53.73	53.72	53.72	53.72
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.873e+004	1.873e+004	1.874e+004	1.874e+004	1.875e+004
40	Pressure (bar)	90.00	90.00	90.00	90.00	90.00
41	Temperature (C)	53.72	53.71	53.71	53.71	53.71
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.875e+004	1.876e+004	1.876e+004	1.877e+004	1.877e+004
52	Pressure (bar)	90.00	90.00	90.00	90.00	90.00
53	Temperature (C)	53.71	53.70	53.70	53.70	53.70
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.878e+004	1.878e+004	1.879e+004	1.879e+004	1.880e+004
64	Pressure (bar)	90.00	90.00	90.00	90.00	90.00
65	Temperature (C)	53.70	53.70	53.69	53.69	53.69
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.880e+004	1.880e+004	1.881e+004	1.881e+004	1.881e+004
20	Pressure (bar)	90.00	90.00	90.00	90.00	89.99
21	Temperature (C)	53.69	53.69	53.69	53.68	53.68
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.882e+004	1.882e+004	1.882e+004	1.882e+004	1.882e+004
32	Pressure (bar)	89.99	89.99	90.00	90.00	90.00
33	Temperature (C)	53.68	53.68	53.68	53.68	53.68
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.883e+004	1.883e+004	1.884e+004	1.884e+004	1.885e+004
44	Pressure (bar)	90.00	90.00	89.99	89.99	89.99
45	Temperature (C)	53.68	53.68	53.68	53.67	53.67
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.885e+004	1.886e+004	1.886e+004	1.887e+004	1.887e+004
56	Pressure (bar)	89.98	89.98	89.99	89.99	89.99
57	Temperature (C)	53.67	53.67	53.66	53.66	53.66
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.888e+004	1.888e+004	1.889e+004	1.889e+004	1.890e+004
68	Pressure (bar)	89.99	89.99	89.99	89.98	89.98
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	53.66	53.66	53.66	53.65	53.65
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.890e+004	1.890e+004	1.891e+004	1.891e+004	1.891e+004
24	Pressure (bar)	89.98	89.98	89.98	89.98	89.98
25	Temperature (C)	53.65	53.65	53.65	53.65	53.65
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.891e+004	1.891e+004	1.892e+004	1.892e+004	1.892e+004
36	Pressure (bar)	89.98	89.98	89.98	89.98	89.98
37	Temperature (C)	53.65	53.64	53.64	53.64	53.64
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.893e+004	1.893e+004	1.894e+004	1.894e+004	1.895e+004
48	Pressure (bar)	89.98	89.98	89.98	89.98	89.98
49	Temperature (C)	53.64	53.64	53.64	53.64	53.63
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.895e+004	1.896e+004	1.896e+004	1.897e+004	1.897e+004
60	Pressure (bar)	89.98	89.98	89.98	89.98	89.99
61	Temperature (C)	53.63	53.63	53.63	53.63	53.63
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 818 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.898e+004	1.898e+004	1.899e+004	1.899e+004	1.900e+004
16	Pressure (bar)	89.99	89.98	89.98	89.98	89.98
17	Temperature (C)	53.62	53.62	53.62	53.62	53.62
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.900e+004	1.901e+004	1.901e+004	1.902e+004	1.902e+004
28	Pressure (bar)	89.98	89.98	89.98	89.98	89.97
29	Temperature (C)	53.61	53.61	53.61	53.61	53.60
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.903e+004	1.903e+004	1.904e+004	1.904e+004	1.905e+004
40	Pressure (bar)	89.97	89.97	89.97	89.97	89.97
41	Temperature (C)	53.60	53.60	53.60	53.60	53.59
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.905e+004	1.906e+004	1.906e+004	1.907e+004	1.907e+004
52	Pressure (bar)	89.97	89.97	89.97	89.97	89.97
53	Temperature (C)	53.59	53.59	53.59	53.59	53.59
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.908e+004	1.908e+004	1.909e+004	1.909e+004	1.910e+004
64	Pressure (bar)	89.97	89.97	89.97	89.97	89.97
65	Temperature (C)	53.58	53.58	53.58	53.58	53.58
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 819 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.87	17.87	17.87	17.87
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.910e+004	1.911e+004	1.911e+004	1.912e+004
20	Pressure (bar)	89.97	89.97	89.96	89.96
21	Temperature (C)	53.57	53.57	53.57	53.57
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.87	17.87	17.87	17.87
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.913e+004	1.913e+004	1.914e+004	1.915e+004
32	Pressure (bar)	89.96	89.96	89.96	89.96
33	Temperature (C)	53.56	53.56	53.56	53.56
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.87	17.87	17.87	17.87
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.915e+004	1.916e+004	1.916e+004	1.917e+004
44	Pressure (bar)	89.96	89.96	89.96	89.96
45	Temperature (C)	53.55	53.55	53.55	53.55
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.87	17.87	17.87	17.87
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.918e+004	1.918e+004	1.919e+004	1.920e+004
56	Pressure (bar)	89.96	89.96	89.96	89.96
57	Temperature (C)	53.55	53.54	53.54	53.54
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.87	17.87	17.87	17.87
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.920e+004	1.921e+004	1.921e+004	1.922e+004
68	Pressure (bar)	89.96	89.96	89.96	89.96
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 820 of 937				

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	53.54	53.53	53.53	53.53	53.53
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.923e+004	1.923e+004	1.924e+004	1.924e+004	1.925e+004
24	Pressure (bar)	89.96	89.96	89.96	89.96	89.96
25	Temperature (C)	53.53	53.53	53.52	53.52	53.52
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.925e+004	1.926e+004	1.926e+004	1.927e+004	1.927e+004
36	Pressure (bar)	89.96	89.96	89.96	89.96	89.96
37	Temperature (C)	53.52	53.52	53.51	53.51	53.51
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	1.928e+004	1.928e+004	1.929e+004	1.929e+004	1.930e+004
48	Pressure (bar)	89.96	89.96	89.96	89.96	89.96
49	Temperature (C)	53.51	53.51	53.51	53.50	53.50
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	1.930e+004	1.931e+004	1.931e+004	1.932e+004	1.932e+004
60	Pressure (bar)	89.96	89.96	89.96	89.96	89.96
61	Temperature (C)	53.50	53.50	53.50	53.50	53.49
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 821 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	1.933e+004	1.933e+004	1.934e+004	1.934e+004	1.935e+004
16	Pressure (bar)	89.96	89.95	89.95	89.95	89.95
17	Temperature (C)	53.49	53.49	53.49	53.49	53.48
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	1.935e+004	1.936e+004	1.936e+004	1.937e+004	1.937e+004
28	Pressure (bar)	89.95	89.96	89.97	89.98	89.98
29	Temperature (C)	53.48	53.48	53.49	53.49	53.49
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	1.938e+004	1.938e+004	1.939e+004	1.939e+004	1.940e+004
40	Pressure (bar)	89.99	89.99	89.98	89.97	89.97
41	Temperature (C)	53.49	53.49	53.49	53.48	53.48
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	1.940e+004	1.941e+004	1.941e+004	1.942e+004	1.942e+004
52	Pressure (bar)	89.96	89.94	89.92	89.90	89.87
53	Temperature (C)	53.47	53.47	53.46	53.45	53.45
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.87	17.87	17.87	17.87	17.87
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	1.943e+004	1.943e+004	1.944e+004	1.944e+004	1.945e+004
64	Pressure (bar)	89.85	89.85	89.85	89.85	89.85
65	Temperature (C)	53.44	53.44	53.43	53.43	53.43
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 822 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	1.945e+004	1.946e+004	1.946e+004	1.946e+004	1.947e+004
20	Pressure (bar)	89.85	89.85	89.85	89.85	89.84
21	Temperature (C)	53.43	53.43	53.42	53.42	53.42
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	1.947e+004	1.948e+004	1.948e+004	1.949e+004	1.949e+004
32	Pressure (bar)	89.84	89.84	89.84	89.84	89.84
33	Temperature (C)	53.42	53.42	53.41	53.41	53.41
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	1.950e+004	1.952e+004	1.954e+004	1.957e+004	1.959e+004
44	Pressure (bar)	89.84	89.83	89.83	89.83	89.83
45	Temperature (C)	53.41	53.40	53.39	53.38	53.37
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	1.961e+004	1.964e+004	1.966e+004	1.968e+004	1.971e+004
56	Pressure (bar)	89.83	89.83	89.82	89.82	89.82
57	Temperature (C)	53.36	53.36	53.35	53.34	53.33
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	1.973e+004	1.975e+004	1.978e+004	1.980e+004	1.982e+004
68	Pressure (bar)	89.82	89.82	89.82	89.82	89.81
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 823 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	53.32	53.31	53.30	53.29	53.29
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	1.985e+004	1.987e+004	1.989e+004	1.992e+004	1.994e+004
24	Pressure (bar)	89.81	89.81	89.81	89.81	89.81
25	Temperature (C)	53.28	53.27	53.26	53.25	53.24
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	1.996e+004	1.999e+004	2.001e+004	2.003e+004	2.006e+004
36	Pressure (bar)	89.81	89.81	89.80	89.80	89.80
37	Temperature (C)	53.23	53.23	53.22	53.21	53.20
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.008e+004	2.010e+004	2.013e+004	2.015e+004	2.017e+004
48	Pressure (bar)	89.80	89.80	89.80	89.80	89.80
49	Temperature (C)	53.19	53.18	53.18	53.17	53.16
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.020e+004	2.022e+004	2.024e+004	2.027e+004	2.029e+004
60	Pressure (bar)	89.80	89.79	89.79	89.79	89.79
61	Temperature (C)	53.15	53.14	53.13	53.12	53.11
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 824 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.031e+004	2.034e+004	2.036e+004	2.038e+004	2.041e+004
16	Pressure (bar)	89.78	89.78	89.78	89.78	89.78
17	Temperature (C)	53.11	53.10	53.09	53.08	53.07
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.043e+004	2.045e+004	2.048e+004	2.050e+004	2.052e+004
28	Pressure (bar)	89.78	89.77	89.77	89.77	89.77
29	Temperature (C)	53.06	53.05	53.05	53.04	53.03
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.055e+004	2.058e+004	2.060e+004	2.063e+004	2.066e+004
40	Pressure (bar)	89.77	89.77	89.77	89.76	89.76
41	Temperature (C)	53.02	53.01	53.00	52.99	52.98
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.069e+004	2.069e+004	2.070e+004	2.070e+004	2.070e+004
52	Pressure (bar)	89.76	89.76	89.76	89.76	89.76
53	Temperature (C)	52.97	52.97	52.97	52.97	52.97
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.070e+004	2.071e+004	2.071e+004	2.072e+004	2.072e+004
64	Pressure (bar)	89.76	89.76	89.76	89.76	89.76
65	Temperature (C)	52.97	52.96	52.96	52.96	52.96
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 825 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.073e+004	2.073e+004	2.074e+004	2.074e+004	2.075e+004
20	Pressure (bar)	89.76	89.76	89.76	89.76	89.76
21	Temperature (C)	52.96	52.95	52.95	52.95	52.95
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.075e+004	2.076e+004	2.076e+004	2.077e+004	2.077e+004
32	Pressure (bar)	89.76	89.76	89.76	89.76	89.76
33	Temperature (C)	52.95	52.95	52.94	52.94	52.94
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.078e+004	2.078e+004	2.079e+004	2.079e+004	2.080e+004
44	Pressure (bar)	89.76	89.76	89.76	89.76	89.75
45	Temperature (C)	52.94	52.94	52.93	52.93	52.93
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.080e+004	2.081e+004	2.081e+004	2.082e+004	2.082e+004
56	Pressure (bar)	89.75	89.75	89.75	89.75	89.75
57	Temperature (C)	52.93	52.93	52.93	52.92	52.92
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.083e+004	2.083e+004	2.084e+004	2.084e+004	2.085e+004
68	Pressure (bar)	89.75	89.75	89.75	89.75	89.75
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 826 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	52.92	52.92	52.92	52.91	52.91
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.085e+004	2.086e+004	2.086e+004	2.087e+004	2.087e+004
24	Pressure (bar)	89.75	89.75	89.75	89.75	89.75
25	Temperature (C)	52.91	52.91	52.91	52.91	52.90
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.088e+004	2.088e+004	2.089e+004	2.089e+004	2.090e+004
36	Pressure (bar)	89.75	89.75	89.75	89.75	89.75
37	Temperature (C)	52.90	52.90	52.90	52.90	52.90
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.090e+004	2.091e+004	2.091e+004	2.092e+004	2.092e+004
48	Pressure (bar)	89.75	89.75	89.75	89.75	89.75
49	Temperature (C)	52.89	52.89	52.89	52.89	52.89
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.093e+004	2.093e+004	2.094e+004	2.094e+004	2.095e+004
60	Pressure (bar)	89.75	89.75	89.75	89.75	89.75
61	Temperature (C)	52.88	52.88	52.88	52.88	52.88
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 827 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.095e+004	2.096e+004	2.096e+004	2.097e+004	2.097e+004
16	Pressure (bar)	89.75	89.75	89.75	89.75	89.74
17	Temperature (C)	52.87	52.87	52.87	52.87	52.87
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.098e+004	2.098e+004	2.099e+004	2.099e+004	2.100e+004
28	Pressure (bar)	89.74	89.74	89.75	89.75	89.75
29	Temperature (C)	52.87	52.86	52.86	52.86	52.86
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.100e+004	2.101e+004	2.101e+004	2.102e+004	2.102e+004
40	Pressure (bar)	89.75	89.75	89.74	89.74	89.74
41	Temperature (C)	52.86	52.86	52.85	52.85	52.85
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.103e+004	2.103e+004	2.104e+004	2.104e+004	2.105e+004
52	Pressure (bar)	89.74	89.74	89.74	89.74	89.74
53	Temperature (C)	52.85	52.85	52.85	52.84	52.84
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.105e+004	2.106e+004	2.106e+004	2.107e+004	2.107e+004
64	Pressure (bar)	89.74	89.74	89.74	89.74	89.74
65	Temperature (C)	52.84	52.84	52.84	52.83	52.83
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.86	17.86	17.86	17.86
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.108e+004	2.108e+004	2.109e+004	2.110e+004
20	Pressure (bar)	89.74	89.74	89.74	89.74
21	Temperature (C)	52.83	52.83	52.83	52.82
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.86	17.86	17.86	17.86
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.110e+004	2.111e+004	2.111e+004	2.112e+004
32	Pressure (bar)	89.74	89.74	89.74	89.74
33	Temperature (C)	52.82	52.82	52.82	52.82
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.86	17.86	17.86	17.86
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.112e+004	2.113e+004	2.113e+004	2.114e+004
44	Pressure (bar)	89.75	89.74	89.74	89.74
45	Temperature (C)	52.82	52.82	52.81	52.81
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.86	17.86	17.86	17.86
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.114e+004	2.114e+004	2.115e+004	2.115e+004
56	Pressure (bar)	89.74	89.74	89.74	89.73
57	Temperature (C)	52.81	52.81	52.81	52.80
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.86	17.86	17.86	17.86
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.115e+004	2.116e+004	2.116e+004	2.117e+004
68	Pressure (bar)	89.73	89.73	89.73	89.73
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 829 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	52.80	52.80	52.80	52.80	52.79
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.118e+004	2.118e+004	2.119e+004	2.119e+004	2.120e+004
24	Pressure (bar)	89.73	89.72	89.72	89.72	89.72
25	Temperature (C)	52.79	52.79	52.79	52.79	52.78
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.120e+004	2.121e+004	2.121e+004	2.122e+004	2.122e+004
36	Pressure (bar)	89.72	89.72	89.72	89.72	89.72
37	Temperature (C)	52.78	52.78	52.78	52.78	52.78
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.123e+004	2.123e+004	2.124e+004	2.124e+004	2.125e+004
48	Pressure (bar)	89.72	89.72	89.72	89.72	89.72
49	Temperature (C)	52.77	52.77	52.77	52.77	52.77
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.86	17.85	17.85	17.85	17.85
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.125e+004	2.126e+004	2.126e+004	2.127e+004	2.127e+004
60	Pressure (bar)	89.72	89.72	89.72	89.72	89.72
61	Temperature (C)	52.76	52.76	52.76	52.76	52.76
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.85	17.85	17.85	17.85	17.85
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 830 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.128e+004	2.128e+004	2.129e+004	2.129e+004	2.130e+004
16	Pressure (bar)	89.72	89.72	89.72	89.72	89.72
17	Temperature (C)	52.76	52.75	52.75	52.75	52.75
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.85	17.85	17.85	17.85	17.85
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.130e+004	2.131e+004	2.131e+004	2.132e+004	2.132e+004
28	Pressure (bar)	89.72	89.72	89.72	89.72	89.72
29	Temperature (C)	52.75	52.75	52.74	52.74	52.74
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.85	17.85	17.85	17.85	17.85
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.133e+004	2.133e+004	2.133e+004	2.133e+004	2.133e+004
40	Pressure (bar)	89.72	89.72	89.72	89.72	89.72
41	Temperature (C)	52.74	52.74	52.74	52.74	52.74
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.85	17.85	17.85	17.85	17.85
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.134e+004	2.134e+004	2.134e+004	2.134e+004	2.135e+004
52	Pressure (bar)	89.72	89.72	89.71	89.71	89.71
53	Temperature (C)	52.73	52.73	52.73	52.73	52.73
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.85	17.85	17.85	17.85	17.85
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.135e+004	2.136e+004	2.136e+004	2.137e+004	2.137e+004
64	Pressure (bar)	89.71	89.71	89.71	89.71	89.71
65	Temperature (C)	52.73	52.73	52.72	52.72	52.72
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 831 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.85	17.85	17.85	17.85	17.85
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.138e+004	2.138e+004	2.139e+004	2.139e+004	2.140e+004
20	Pressure (bar)	89.71	89.71	89.71	89.71	89.71
21	Temperature (C)	52.72	52.72	52.72	52.71	52.71
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.85	17.85	17.85	17.85	17.85
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.140e+004	2.141e+004	2.141e+004	2.142e+004	2.142e+004
32	Pressure (bar)	89.71	89.71	89.71	89.70	89.70
33	Temperature (C)	52.71	52.71	52.71	52.70	52.70
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.85	17.85	17.85	17.85	17.85
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.143e+004	2.143e+004	2.144e+004	2.144e+004	2.145e+004
44	Pressure (bar)	89.70	89.70	89.70	89.70	89.70
45	Temperature (C)	52.70	52.70	52.70	52.69	52.69
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.85	17.85	17.85	17.85	17.85
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.145e+004	2.146e+004	2.146e+004	2.147e+004	2.147e+004
56	Pressure (bar)	89.70	89.70	89.70	89.70	89.70
57	Temperature (C)	52.69	52.69	52.69	52.68	52.68
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.85	17.85	17.85	17.85	17.85
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.148e+004	2.148e+004	2.149e+004	2.149e+004	2.150e+004
68	Pressure (bar)	89.70	89.70	89.70	89.70	89.70
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	52.68	52.68	52.68	52.68	52.67
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.85	17.85	17.85	17.85	17.85
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.150e+004	2.150e+004	2.151e+004	2.151e+004	2.152e+004
24	Pressure (bar)	89.70	89.70	89.70	89.70	89.70
25	Temperature (C)	52.67	52.67	52.67	52.67	52.67
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.85	17.85	17.85	17.85	17.85
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.152e+004	2.152e+004	2.152e+004	2.152e+004	2.152e+004
36	Pressure (bar)	89.70	89.70	89.70	89.70	89.70
37	Temperature (C)	52.67	52.67	52.66	52.66	52.66
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.85	17.85	17.85	17.85	17.85
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.153e+004	2.153e+004	2.154e+004	2.154e+004	2.155e+004
48	Pressure (bar)	89.70	89.70	89.70	89.69	89.69
49	Temperature (C)	52.66	52.66	52.66	52.66	52.66
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.85	17.85	17.85	17.85	17.85
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.155e+004	2.156e+004	2.156e+004	2.157e+004	2.157e+004
60	Pressure (bar)	89.69	89.69	89.69	89.69	89.69
61	Temperature (C)	52.65	52.65	52.65	52.65	52.65
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.85	17.85	17.85	17.85	17.85
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 833 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.158e+004	2.158e+004	2.159e+004	2.159e+004	2.160e+004
16	Pressure (bar)	89.68	89.69	89.69	89.69	89.69
17	Temperature (C)	52.64	52.64	52.64	52.64	52.64
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.85	17.85	17.85	17.85	17.85
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.160e+004	2.161e+004	2.161e+004	2.162e+004	2.162e+004
28	Pressure (bar)	89.69	89.70	89.70	89.71	89.72
29	Temperature (C)	52.64	52.64	52.64	52.64	52.64
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.85	17.85	17.85	17.85	17.85
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.163e+004	2.163e+004	2.164e+004	2.164e+004	2.165e+004
40	Pressure (bar)	89.72	89.73	89.74	89.74	89.75
41	Temperature (C)	52.64	52.64	52.65	52.65	52.65
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.165e+004	2.166e+004	2.166e+004	2.167e+004	2.167e+004
52	Pressure (bar)	89.76	89.76	89.77	89.77	89.78
53	Temperature (C)	52.65	52.65	52.65	52.65	52.65
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.86	17.86	17.86	17.86	17.86
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.168e+004	2.168e+004	2.169e+004	2.169e+004	2.170e+004
64	Pressure (bar)	89.78	89.78	89.78	89.78	89.78
65	Temperature (C)	52.65	52.65	52.65	52.65	52.65
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 834 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.86	17.86	17.86	17.86
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.170e+004	2.171e+004	2.171e+004	2.172e+004
20	Pressure (bar)	89.78	89.78	89.78	89.77
21	Temperature (C)	52.64	52.64	52.64	52.63
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.86	17.86	17.86	17.86
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.173e+004	2.173e+004	2.174e+004	2.175e+004
32	Pressure (bar)	89.77	89.76	89.76	89.75
33	Temperature (C)	52.63	52.63	52.62	52.62
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.86	17.86	17.86	17.86
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.175e+004	2.176e+004	2.176e+004	2.177e+004
44	Pressure (bar)	89.75	89.75	89.74	89.74
45	Temperature (C)	52.62	52.61	52.61	52.61
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.86	17.86	17.86	17.86
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.178e+004	2.178e+004	2.179e+004	2.180e+004
56	Pressure (bar)	89.74	89.67	89.60	89.53
57	Temperature (C)	52.60	52.59	52.57	52.55
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.86	17.85	17.85	17.84
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.180e+004	2.181e+004	2.181e+004	2.182e+004
68	Pressure (bar)	89.40	89.40	89.39	89.39
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 835 of 937				

1				Case Name: Simulation - 386MMSCFD_100barg.hsc		
2	 Epic Atlantic Limited			Unit Set: Mixed		
Date/Time: Mon Jun 13 15:02:44 2022						
3						
4						
5						
6						
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	52.51	52.51	52.51	52.51	52.50
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.183e+004	2.183e+004	2.184e+004	2.184e+004	2.185e+004
24	Pressure (bar)	89.39	89.39	89.39	89.39	89.39
25	Temperature (C)	52.50	52.50	52.50	52.50	52.49
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.185e+004	2.186e+004	2.186e+004	2.187e+004	2.187e+004
36	Pressure (bar)	89.39	89.39	89.39	89.39	89.39
37	Temperature (C)	52.49	52.49	52.49	52.49	52.49
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.188e+004	2.188e+004	2.189e+004	2.189e+004	2.190e+004
48	Pressure (bar)	89.39	89.39	89.39	89.39	89.39
49	Temperature (C)	52.48	52.48	52.48	52.48	52.48
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.190e+004	2.191e+004	2.191e+004	2.192e+004	2.192e+004
60	Pressure (bar)	89.39	89.38	89.38	89.38	89.38
61	Temperature (C)	52.47	52.47	52.47	52.47	52.47
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 836 of 937					

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.193e+004	2.193e+004	2.194e+004	2.194e+004	2.195e+004
16	Pressure (bar)	89.38	89.38	89.38	89.38	89.38
17	Temperature (C)	52.47	52.46	52.46	52.46	52.46
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.195e+004	2.196e+004	2.196e+004	2.197e+004	2.197e+004
28	Pressure (bar)	89.38	89.38	89.38	89.38	89.38
29	Temperature (C)	52.46	52.46	52.45	52.45	52.45
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.198e+004	2.198e+004	2.199e+004	2.199e+004	2.200e+004
40	Pressure (bar)	89.38	89.38	89.38	89.38	89.38
41	Temperature (C)	52.45	52.45	52.44	52.44	52.44
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.200e+004	2.201e+004	2.201e+004	2.202e+004	2.202e+004
52	Pressure (bar)	89.38	89.38	89.38	89.38	89.38
53	Temperature (C)	52.44	52.44	52.44	52.43	52.43
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.203e+004	2.203e+004	2.204e+004	2.204e+004	2.205e+004
64	Pressure (bar)	89.38	89.38	89.38	89.38	89.38
65	Temperature (C)	52.43	52.43	52.43	52.42	52.42
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.205e+004	2.206e+004	2.206e+004	2.207e+004	2.207e+004
20	Pressure (bar)	89.38	89.38	89.38	89.38	89.38
21	Temperature (C)	52.42	52.42	52.42	52.42	52.41
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.208e+004	2.208e+004	2.209e+004	2.209e+004	2.210e+004
32	Pressure (bar)	89.38	89.38	89.38	89.38	89.38
33	Temperature (C)	52.41	52.41	52.41	52.41	52.41
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.210e+004	2.211e+004	2.211e+004	2.212e+004	2.212e+004
44	Pressure (bar)	89.38	89.38	89.38	89.38	89.38
45	Temperature (C)	52.41	52.40	52.40	52.40	52.40
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.213e+004	2.213e+004	2.214e+004	2.214e+004	2.215e+004
56	Pressure (bar)	89.38	89.38	89.37	89.37	89.37
57	Temperature (C)	52.40	52.39	52.39	52.39	52.39
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.215e+004	2.216e+004	2.216e+004	2.217e+004	2.217e+004
68	Pressure (bar)	89.37	89.37	89.37	89.37	89.37
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5	Date/Time: Mon Jun 13 15:02:44 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	52.39	52.38	52.38	52.38	52.38
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.218e+004	2.218e+004	2.219e+004	2.219e+004	2.220e+004
24	Pressure (bar)	89.37	89.37	89.37	89.37	89.37
25	Temperature (C)	52.38	52.37	52.37	52.37	52.37
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.220e+004	2.221e+004	2.221e+004	2.222e+004	2.222e+004
36	Pressure (bar)	89.37	89.37	89.37	89.37	89.37
37	Temperature (C)	52.37	52.37	52.37	52.36	52.36
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.223e+004	2.223e+004	2.224e+004	2.224e+004	2.225e+004
48	Pressure (bar)	89.38	89.37	89.37	89.37	89.37
49	Temperature (C)	52.36	52.36	52.36	52.35	52.35
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.225e+004	2.226e+004	2.226e+004	2.227e+004	2.227e+004
60	Pressure (bar)	89.36	89.36	89.36	89.36	89.36
61	Temperature (C)	52.35	52.35	52.35	52.34	52.34
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 839 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3						
4						
5	Unit Set: Mixed					
6	Date/Time: Mon Jun 13 15:02:44 2022					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	Hydrates					
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.228e+004	2.229e+004	2.230e+004	2.231e+004	2.232e+004
16	Pressure (bar)	89.36	89.36	89.36	89.36	89.35
17	Temperature (C)	52.34	52.34	52.33	52.33	52.33
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.233e+004	2.255e+004	2.278e+004	2.300e+004	2.323e+004
28	Pressure (bar)	89.35	89.34	89.33	89.32	89.30
29	Temperature (C)	52.32	52.24	52.16	52.08	52.01
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.345e+004	2.346e+004	2.347e+004	2.348e+004	2.348e+004
40	Pressure (bar)	89.29	89.29	89.29	89.29	89.29
41	Temperature (C)	51.93	51.92	51.92	51.92	51.92
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.349e+004	2.350e+004	2.350e+004	2.351e+004	2.351e+004
52	Pressure (bar)	89.29	89.29	89.29	89.29	89.29
53	Temperature (C)	51.91	51.91	51.91	51.91	51.91
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.352e+004	2.352e+004	2.353e+004	2.353e+004	2.354e+004
64	Pressure (bar)	89.29	89.29	89.29	89.29	89.28
65	Temperature (C)	51.91	51.90	51.90	51.90	51.90
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 840 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)				
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.83	17.83	17.83	17.83
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.354e+004	2.355e+004	2.355e+004	2.356e+004
20	Pressure (bar)	89.28	89.28	89.28	89.28
21	Temperature (C)	51.90	51.90	51.89	51.89
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.83	17.83	17.83	17.83
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.357e+004	2.357e+004	2.358e+004	2.359e+004
32	Pressure (bar)	89.28	89.28	89.27	89.26
33	Temperature (C)	51.89	51.89	51.88	51.88
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.83	17.83	17.83	17.83
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.359e+004	2.360e+004	2.360e+004	2.361e+004
44	Pressure (bar)	89.26	89.26	89.26	89.27
45	Temperature (C)	51.87	51.87	51.87	51.87
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.82	17.82	17.83	17.83
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.362e+004	2.362e+004	2.363e+004	2.364e+004
56	Pressure (bar)	89.27	89.27	89.27	89.27
57	Temperature (C)	51.87	51.87	51.87	51.86
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.83	17.83	17.83	17.83
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.364e+004	2.365e+004	2.365e+004	2.366e+004
68	Pressure (bar)	89.27	89.27	89.27	89.27
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 841 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3						
4						
5	Date/Time: Mon Jun 13 15:02:44 2022					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10	Hydrates Table					
11						
12						
13	Temperature (C)	51.86	51.86	51.86	51.86	51.85
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.367e+004	2.367e+004	2.368e+004	2.368e+004	2.369e+004
24	Pressure (bar)	89.27	89.27	89.27	89.27	89.27
25	Temperature (C)	51.85	51.85	51.85	51.85	51.84
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.369e+004	2.370e+004	2.370e+004	2.371e+004	2.371e+004
36	Pressure (bar)	89.26	89.26	89.26	89.26	89.26
37	Temperature (C)	51.84	51.84	51.84	51.84	51.84
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.372e+004	2.372e+004	2.373e+004	2.373e+004	2.374e+004
48	Pressure (bar)	89.26	89.26	89.26	89.26	89.26
49	Temperature (C)	51.83	51.83	51.83	51.83	51.83
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.374e+004	2.375e+004	2.375e+004	2.376e+004	2.376e+004
60	Pressure (bar)	89.26	89.26	89.26	89.26	89.27
61	Temperature (C)	51.83	51.83	51.82	51.82	51.82
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 842 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.377e+004	2.377e+004	2.378e+004	2.378e+004	2.379e+004
16	Pressure (bar)	89.27	89.27	89.26	89.26	89.26
17	Temperature (C)	51.82	51.82	51.82	51.81	51.81
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.379e+004	2.380e+004	2.380e+004	2.381e+004	2.381e+004
28	Pressure (bar)	89.26	89.26	89.26	89.26	89.26
29	Temperature (C)	51.81	51.81	51.81	51.80	51.80
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.83	17.83	17.83	17.83	17.83
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.382e+004	2.382e+004	2.383e+004	2.383e+004	2.384e+004
40	Pressure (bar)	89.26	89.26	89.26	89.26	89.26
41	Temperature (C)	51.80	51.80	51.80	51.80	51.79
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.384e+004	2.385e+004	2.385e+004	2.386e+004	2.386e+004
52	Pressure (bar)	89.25	89.25	89.25	89.25	89.25
53	Temperature (C)	51.79	51.79	51.79	51.79	51.78
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.387e+004	2.387e+004	2.388e+004	2.388e+004	2.389e+004
64	Pressure (bar)	89.25	89.25	89.26	89.26	89.26
65	Temperature (C)	51.78	51.78	51.78	51.78	51.78
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 843 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.389e+004	2.390e+004	2.390e+004	2.391e+004	2.391e+004
20	Pressure (bar)	89.26	89.26	89.25	89.25	89.25
21	Temperature (C)	51.78	51.77	51.77	51.77	51.77
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.392e+004	2.392e+004	2.393e+004	2.393e+004	2.394e+004
32	Pressure (bar)	89.25	89.25	89.25	89.24	89.24
33	Temperature (C)	51.76	51.76	51.76	51.76	51.76
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.394e+004	2.395e+004	2.395e+004	2.396e+004	2.396e+004
44	Pressure (bar)	89.24	89.24	89.24	89.24	89.24
45	Temperature (C)	51.76	51.75	51.75	51.75	51.75
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.397e+004	2.397e+004	2.398e+004	2.398e+004	2.399e+004
56	Pressure (bar)	89.24	89.24	89.24	89.24	89.24
57	Temperature (C)	51.75	51.74	51.74	51.74	51.74
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.399e+004	2.400e+004	2.400e+004	2.401e+004	2.401e+004
68	Pressure (bar)	89.24	89.24	89.24	89.24	89.24
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 844 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	51.74	51.74	51.74	51.73	51.73
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.402e+004	2.402e+004	2.403e+004	2.403e+004	2.404e+004
24	Pressure (bar)	89.24	89.24	89.24	89.24	89.24
25	Temperature (C)	51.73	51.73	51.73	51.73	51.72
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.404e+004	2.405e+004	2.405e+004	2.406e+004	2.406e+004
36	Pressure (bar)	89.24	89.24	89.24	89.24	89.24
37	Temperature (C)	51.72	51.72	51.72	51.72	51.71
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.407e+004	2.407e+004	2.408e+004	2.408e+004	2.409e+004
48	Pressure (bar)	89.24	89.24	89.24	89.23	89.23
49	Temperature (C)	51.71	51.71	51.71	51.71	51.71
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.409e+004	2.410e+004	2.410e+004	2.411e+004	2.411e+004
60	Pressure (bar)	89.23	89.23	89.23	89.23	89.23
61	Temperature (C)	51.70	51.70	51.70	51.70	51.70
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.412e+004	2.412e+004	2.413e+004	2.413e+004	2.414e+004
16	Pressure (bar)	89.23	89.23	89.23	89.23	89.23
17	Temperature (C)	51.69	51.69	51.69	51.69	51.69
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.414e+004	2.415e+004	2.415e+004	2.416e+004	2.416e+004
28	Pressure (bar)	89.23	89.23	89.23	89.23	89.23
29	Temperature (C)	51.69	51.68	51.68	51.68	51.68
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.417e+004	2.417e+004	2.418e+004	2.418e+004	2.419e+004
40	Pressure (bar)	89.23	89.23	89.23	89.23	89.23
41	Temperature (C)	51.68	51.68	51.67	51.67	51.67
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.419e+004	2.420e+004	2.420e+004	2.421e+004	2.421e+004
52	Pressure (bar)	89.23	89.23	89.23	89.23	89.23
53	Temperature (C)	51.67	51.67	51.67	51.66	51.66
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.422e+004	2.422e+004	2.423e+004	2.423e+004	2.424e+004
64	Pressure (bar)	89.22	89.22	89.22	89.22	89.22
65	Temperature (C)	51.66	51.66	51.66	51.65	51.65
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.424e+004	2.425e+004	2.425e+004	2.426e+004	2.426e+004
20	Pressure (bar)	89.22	89.22	89.22	89.22	89.22
21	Temperature (C)	51.65	51.65	51.65	51.65	51.64
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.427e+004	2.427e+004	2.428e+004	2.428e+004	2.429e+004
32	Pressure (bar)	89.22	89.22	89.22	89.22	89.22
33	Temperature (C)	51.64	51.64	51.64	51.64	51.64
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.429e+004	2.430e+004	2.430e+004	2.431e+004	2.431e+004
44	Pressure (bar)	89.22	89.22	89.22	89.22	89.22
45	Temperature (C)	51.63	51.63	51.63	51.63	51.63
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.432e+004	2.432e+004	2.433e+004	2.433e+004	2.434e+004
56	Pressure (bar)	89.22	89.22	89.22	89.22	89.22
57	Temperature (C)	51.62	51.62	51.62	51.62	51.62
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.434e+004	2.435e+004	2.435e+004	2.436e+004	2.436e+004
68	Pressure (bar)	89.22	89.22	89.22	89.22	89.21
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	51.62	51.61	51.61	51.61	51.61
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.437e+004	2.437e+004	2.437e+004	2.437e+004	2.437e+004
24	Pressure (bar)	89.21	89.21	89.21	89.21	89.21
25	Temperature (C)	51.61	51.61	51.61	51.61	51.61
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.438e+004	2.438e+004	2.438e+004	2.438e+004	2.438e+004
36	Pressure (bar)	89.21	89.21	89.21	89.21	89.21
37	Temperature (C)	51.60	51.60	51.60	51.60	51.60
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.439e+004	2.439e+004	2.440e+004	2.440e+004	2.441e+004
48	Pressure (bar)	89.21	89.21	89.21	89.21	89.21
49	Temperature (C)	51.60	51.60	51.60	51.59	51.59
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.442e+004	2.442e+004	2.443e+004	2.443e+004	2.444e+004
60	Pressure (bar)	89.21	89.21	89.21	89.21	89.21
61	Temperature (C)	51.59	51.59	51.59	51.59	51.58
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 848 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.444e+004	2.445e+004	2.445e+004	2.446e+004	2.446e+004
16	Pressure (bar)	89.21	89.21	89.21	89.21	89.21
17	Temperature (C)	51.58	51.58	51.58	51.58	51.58
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.447e+004	2.447e+004	2.448e+004	2.448e+004	2.449e+004
28	Pressure (bar)	89.21	89.21	89.21	89.21	89.21
29	Temperature (C)	51.57	51.57	51.57	51.57	51.57
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.449e+004	2.450e+004	2.450e+004	2.451e+004	2.451e+004
40	Pressure (bar)	89.21	89.21	89.21	89.21	89.21
41	Temperature (C)	51.57	51.56	51.56	51.56	51.56
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.452e+004	2.452e+004	2.453e+004	2.453e+004	2.454e+004
52	Pressure (bar)	89.21	89.21	89.21	89.21	89.21
53	Temperature (C)	51.56	51.56	51.55	51.55	51.55
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.454e+004	2.455e+004	2.455e+004	2.456e+004	2.456e+004
64	Pressure (bar)	89.21	89.20	89.20	89.20	89.20
65	Temperature (C)	51.55	51.55	51.54	51.54	51.54
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 849 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.457e+004	2.457e+004	2.458e+004	2.458e+004	2.459e+004
20	Pressure (bar)	89.20	89.20	89.20	89.20	89.20
21	Temperature (C)	51.54	51.54	51.54	51.53	51.53
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.459e+004	2.460e+004	2.460e+004	2.461e+004	2.461e+004
32	Pressure (bar)	89.20	89.20	89.20	89.20	89.20
33	Temperature (C)	51.53	51.53	51.53	51.53	51.52
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.462e+004	2.462e+004	2.463e+004	2.463e+004	2.464e+004
44	Pressure (bar)	89.20	89.20	89.20	89.20	89.20
45	Temperature (C)	51.52	51.52	51.52	51.52	51.51
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.464e+004	2.465e+004	2.465e+004	2.466e+004	2.466e+004
56	Pressure (bar)	89.20	89.20	89.20	89.20	89.20
57	Temperature (C)	51.51	51.51	51.51	51.51	51.51
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.467e+004	2.467e+004	2.468e+004	2.468e+004	2.469e+004
68	Pressure (bar)	89.20	89.20	89.20	89.20	89.20
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	51.51	51.50	51.50	51.50	51.50
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.469e+004	2.470e+004	2.470e+004	2.471e+004	2.471e+004
24	Pressure (bar)	89.20	89.20	89.20	89.19	89.19
25	Temperature (C)	51.50	51.50	51.49	51.49	51.49
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.472e+004	2.472e+004	2.473e+004	2.473e+004	2.474e+004
36	Pressure (bar)	89.19	89.19	89.19	89.19	89.19
37	Temperature (C)	51.49	51.49	51.48	51.48	51.48
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.474e+004	2.475e+004	2.475e+004	2.476e+004	2.476e+004
48	Pressure (bar)	89.19	89.18	89.18	89.18	89.18
49	Temperature (C)	51.48	51.48	51.47	51.47	51.47
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.477e+004	2.477e+004	2.478e+004	2.478e+004	2.479e+004
60	Pressure (bar)	89.18	89.18	89.18	89.18	89.18
61	Temperature (C)	51.47	51.47	51.47	51.46	51.46
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.479e+004	2.480e+004	2.480e+004	2.481e+004	2.481e+004
16	Pressure (bar)	89.18	89.18	89.18	89.18	89.18
17	Temperature (C)	51.46	51.46	51.46	51.46	51.45
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.482e+004	2.482e+004	2.483e+004	2.483e+004	2.484e+004
28	Pressure (bar)	89.18	89.18	89.18	89.18	89.18
29	Temperature (C)	51.45	51.45	51.45	51.45	51.45
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.484e+004	2.485e+004	2.485e+004	2.486e+004	2.486e+004
40	Pressure (bar)	89.18	89.19	89.19	89.19	89.19
41	Temperature (C)	51.45	51.44	51.44	51.44	51.44
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.487e+004	2.487e+004	2.488e+004	2.488e+004	2.489e+004
52	Pressure (bar)	89.19	89.19	89.19	89.19	89.19
53	Temperature (C)	51.44	51.44	51.44	51.43	51.43
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.489e+004	2.490e+004	2.490e+004	2.491e+004	2.491e+004
64	Pressure (bar)	89.19	89.18	89.18	89.18	89.17
65	Temperature (C)	51.43	51.43	51.43	51.42	51.42
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 852 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.492e+004	2.492e+004	2.493e+004	2.493e+004	2.494e+004
20	Pressure (bar)	89.17	89.17	89.17	89.17	89.17
21	Temperature (C)	51.42	51.42	51.41	51.41	51.41
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.494e+004	2.495e+004	2.495e+004	2.496e+004	2.496e+004
32	Pressure (bar)	89.17	89.17	89.17	89.17	89.17
33	Temperature (C)	51.41	51.41	51.40	51.40	51.40
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.497e+004	2.497e+004	2.498e+004	2.498e+004	2.499e+004
44	Pressure (bar)	89.17	89.17	89.17	89.17	89.16
45	Temperature (C)	51.40	51.40	51.40	51.39	51.39
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.499e+004	2.500e+004	2.500e+004	2.501e+004	2.501e+004
56	Pressure (bar)	89.16	89.16	89.16	89.16	89.16
57	Temperature (C)	51.39	51.39	51.39	51.39	51.38
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.502e+004	2.502e+004	2.503e+004	2.503e+004	2.504e+004
68	Pressure (bar)	89.16	89.16	89.16	89.16	89.16
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	51.38	51.38	51.38	51.38	51.38
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.504e+004	2.505e+004	2.505e+004	2.506e+004	2.506e+004
24	Pressure (bar)	89.16	89.16	89.16	89.16	89.16
25	Temperature (C)	51.37	51.37	51.37	51.37	51.37
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.507e+004	2.507e+004	2.508e+004	2.508e+004	2.509e+004
36	Pressure (bar)	89.16	89.16	89.16	89.16	89.16
37	Temperature (C)	51.37	51.36	51.36	51.36	51.36
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.509e+004	2.510e+004	2.510e+004	2.511e+004	2.511e+004
48	Pressure (bar)	89.16	89.16	89.16	89.16	89.16
49	Temperature (C)	51.36	51.36	51.35	51.35	51.35
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.512e+004	2.512e+004	2.513e+004	2.513e+004	2.514e+004
60	Pressure (bar)	89.16	89.16	89.16	89.16	89.16
61	Temperature (C)	51.35	51.35	51.35	51.34	51.34
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.514e+004	2.515e+004	2.515e+004	2.516e+004	2.516e+004
16	Pressure (bar)	89.16	89.16	89.16	89.16	89.16
17	Temperature (C)	51.34	51.34	51.34	51.34	51.33
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.517e+004	2.517e+004	2.518e+004	2.518e+004	2.519e+004
28	Pressure (bar)	89.16	89.16	89.16	89.16	89.16
29	Temperature (C)	51.33	51.33	51.33	51.33	51.33
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.519e+004	2.520e+004	2.520e+004	2.521e+004	2.522e+004
40	Pressure (bar)	89.16	89.16	89.16	89.16	89.16
41	Temperature (C)	51.33	51.32	51.32	51.32	51.32
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.522e+004	2.523e+004	2.523e+004	2.524e+004	2.524e+004
52	Pressure (bar)	89.16	89.16	89.17	89.18	89.19
53	Temperature (C)	51.31	51.32	51.32	51.32	51.32
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.525e+004	2.525e+004	2.526e+004	2.526e+004	2.527e+004
64	Pressure (bar)	89.19	89.20	89.20	89.21	89.21
65	Temperature (C)	51.33	51.33	51.33	51.33	51.33
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.527e+004	2.528e+004	2.528e+004	2.529e+004	2.529e+004
20	Pressure (bar)	89.22	89.20	89.19	89.18	89.17
21	Temperature (C)	51.33	51.32	51.32	51.31	51.31
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.529e+004	2.531e+004	2.533e+004	2.535e+004	2.537e+004
32	Pressure (bar)	89.16	89.16	89.15	89.15	89.15
33	Temperature (C)	51.31	51.30	51.29	51.28	51.28
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.539e+004	2.539e+004	2.539e+004	2.539e+004	2.540e+004
44	Pressure (bar)	89.14	89.14	89.14	89.14	89.14
45	Temperature (C)	51.27	51.27	51.27	51.26	51.26
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.82	17.82	17.82	17.82	17.82
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.540e+004	2.540e+004	2.541e+004	2.541e+004	2.542e+004
56	Pressure (bar)	89.14	89.11	89.09	89.07	89.05
57	Temperature (C)	51.26	51.25	51.25	51.24	51.23
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.82	17.82	17.81	17.81	17.81
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.543e+004	2.543e+004	2.544e+004	2.544e+004	2.545e+004
68	Pressure (bar)	89.02	89.02	89.02	89.02	89.02
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 856 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	51.22	51.22	51.22	51.22	51.22
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.81	17.81	17.81	17.81	17.81
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.545e+004	2.546e+004	2.546e+004	2.547e+004	2.547e+004
24	Pressure (bar)	89.02	89.02	89.01	89.01	89.01
25	Temperature (C)	51.21	51.21	51.21	51.21	51.20
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.81	17.81	17.81	17.81	17.81
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.548e+004	2.548e+004	2.549e+004	2.549e+004	2.550e+004
36	Pressure (bar)	89.01	89.01	89.01	89.01	89.01
37	Temperature (C)	51.20	51.20	51.20	51.20	51.20
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.81	17.81	17.81	17.81	17.81
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.550e+004	2.551e+004	2.551e+004	2.552e+004	2.552e+004
48	Pressure (bar)	89.01	89.01	89.01	89.01	89.01
49	Temperature (C)	51.20	51.19	51.19	51.19	51.19
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.81	17.81	17.81	17.81	17.81
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.553e+004	2.553e+004	2.554e+004	2.554e+004	2.555e+004
60	Pressure (bar)	89.01	89.01	89.01	89.01	89.01
61	Temperature (C)	51.19	51.18	51.18	51.18	51.18
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.81	17.81	17.81	17.81	17.81
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.555e+004	2.556e+004	2.556e+004	2.557e+004	2.557e+004
16	Pressure (bar)	89.01	89.01	89.01	89.01	89.00
17	Temperature (C)	51.18	51.18	51.17	51.17	51.17
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.81	17.81	17.81	17.81	17.81
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.558e+004	2.558e+004	2.559e+004	2.559e+004	2.560e+004
28	Pressure (bar)	89.00	89.00	89.00	89.00	89.00
29	Temperature (C)	51.17	51.17	51.17	51.16	51.16
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.81	17.81	17.81	17.81	17.81
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.560e+004	2.561e+004	2.561e+004	2.562e+004	2.563e+004
40	Pressure (bar)	89.00	89.00	89.00	89.00	89.00
41	Temperature (C)	51.16	51.16	51.16	51.15	51.15
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.81	17.81	17.81	17.81	17.81
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.563e+004	2.564e+004	2.564e+004	2.565e+004	2.565e+004
52	Pressure (bar)	89.00	89.00	89.00	89.00	89.00
53	Temperature (C)	51.15	51.15	51.15	51.14	51.14
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.81	17.81	17.81	17.81	17.81
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.566e+004	2.566e+004	2.567e+004	2.567e+004	2.568e+004
64	Pressure (bar)	89.00	89.00	89.00	89.00	89.00
65	Temperature (C)	51.14	51.14	51.14	51.14	51.13
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.81	17.81	17.81	17.81	17.81
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.568e+004	2.569e+004	2.569e+004	2.570e+004	2.570e+004
20	Pressure (bar)	89.00	89.00	89.00	89.00	89.00
21	Temperature (C)	51.13	51.13	51.13	51.13	51.13
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.81	17.81	17.81	17.81	17.81
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.571e+004	2.571e+004	2.571e+004	2.571e+004	2.571e+004
32	Pressure (bar)	89.00	89.00	89.00	89.00	89.00
33	Temperature (C)	51.12	51.12	51.12	51.12	51.12
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.81	17.81	17.81	17.81	17.81
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.571e+004	2.572e+004	2.572e+004	2.572e+004	2.572e+004
44	Pressure (bar)	89.00	89.00	89.00	89.00	89.00
45	Temperature (C)	51.12	51.12	51.12	51.12	51.12
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.81	17.81	17.81	17.81	17.81
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.572e+004	2.572e+004	2.573e+004	2.573e+004	2.573e+004
56	Pressure (bar)	89.01	89.01	89.02	89.02	89.02
57	Temperature (C)	51.13	51.13	51.13	51.13	51.14
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.81	17.81	17.81	17.81	17.81
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.573e+004	2.574e+004	2.574e+004	2.575e+004	2.575e+004
68	Pressure (bar)	89.03	89.03	89.04	89.04	89.05
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	51.14	51.14	51.14	51.14	51.14
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.81	17.81	17.81	17.81	17.81
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.576e+004	2.576e+004	2.577e+004	2.577e+004	2.578e+004
24	Pressure (bar)	89.05	89.01	88.96	88.92	88.87
25	Temperature (C)	51.14	51.13	51.11	51.10	51.09
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.81	17.81	17.81	17.80	17.80
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.578e+004	2.579e+004	2.579e+004	2.580e+004	2.580e+004
36	Pressure (bar)	88.83	88.83	88.82	88.82	88.82
37	Temperature (C)	51.08	51.07	51.07	51.07	51.07
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.80	17.80	17.80	17.80	17.80
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.581e+004	2.581e+004	2.582e+004	2.582e+004	2.583e+004
48	Pressure (bar)	88.82	88.82	88.82	88.81	88.81
49	Temperature (C)	51.06	51.06	51.06	51.06	51.06
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.80	17.80	17.80	17.80	17.80
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.583e+004	2.584e+004	2.584e+004	2.585e+004	2.585e+004
60	Pressure (bar)	88.81	88.81	88.81	88.81	88.81
61	Temperature (C)	51.05	51.05	51.05	51.05	51.05
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.80	17.80	17.80	17.80	17.80
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 860 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.586e+004	2.586e+004	2.587e+004	2.587e+004	2.588e+004
16	Pressure (bar)	88.81	88.81	88.81	88.81	88.81
17	Temperature (C)	51.04	51.04	51.04	51.04	51.04
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.80	17.80	17.80	17.80	17.80
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.588e+004	2.589e+004	2.589e+004	2.590e+004	2.590e+004
28	Pressure (bar)	88.81	88.81	88.81	88.81	88.81
29	Temperature (C)	51.04	51.04	51.03	51.03	51.03
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.80	17.80	17.80	17.80	17.80
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.591e+004	2.591e+004	2.592e+004	2.592e+004	2.593e+004
40	Pressure (bar)	88.81	88.81	88.81	88.81	88.81
41	Temperature (C)	51.03	51.03	51.03	51.03	51.02
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.80	17.80	17.80	17.80	17.80
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.593e+004	2.594e+004	2.594e+004	2.595e+004	2.595e+004
52	Pressure (bar)	88.81	88.81	88.81	88.81	88.81
53	Temperature (C)	51.02	51.02	51.02	51.02	51.02
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.80	17.80	17.80	17.80	17.80
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.596e+004	2.596e+004	2.597e+004	2.597e+004	2.598e+004
64	Pressure (bar)	88.81	88.81	88.81	88.80	88.80
65	Temperature (C)	51.02	51.01	51.01	51.01	51.00
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 861 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.80	17.80	17.80	17.79
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.598e+004	2.599e+004	2.599e+004	2.600e+004
20	Pressure (bar)	88.79	88.79	88.79	88.79
21	Temperature (C)	51.00	51.00	51.00	51.00
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.79	17.79	17.79	17.79
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.601e+004	2.601e+004	2.602e+004	2.603e+004
32	Pressure (bar)	88.79	88.79	88.79	88.79
33	Temperature (C)	50.99	50.99	50.99	50.99
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.79	17.79	17.79	17.79
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.603e+004	2.604e+004	2.604e+004	2.605e+004
44	Pressure (bar)	88.79	88.79	88.79	88.79
45	Temperature (C)	50.99	50.98	50.98	50.98
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.79	17.79	17.79	17.79
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.606e+004	2.606e+004	2.607e+004	2.608e+004
56	Pressure (bar)	88.79	88.79	88.79	88.78
57	Temperature (C)	50.98	50.97	50.97	50.97
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.79	17.79	17.79	17.79
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.608e+004	2.609e+004	2.609e+004	2.610e+004
68	Pressure (bar)	88.78	88.78	88.78	88.78
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 862 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	50.97	50.97	50.96	50.96	50.96
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.611e+004	2.611e+004	2.612e+004	2.612e+004	2.613e+004
24	Pressure (bar)	88.78	88.78	88.78	88.78	88.78
25	Temperature (C)	50.96	50.96	50.96	50.95	50.95
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.613e+004	2.614e+004	2.614e+004	2.615e+004	2.615e+004
36	Pressure (bar)	88.78	88.78	88.78	88.78	88.78
37	Temperature (C)	50.95	50.95	50.95	50.95	50.94
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.616e+004	2.616e+004	2.617e+004	2.617e+004	2.618e+004
48	Pressure (bar)	88.78	88.78	88.78	88.78	88.78
49	Temperature (C)	50.94	50.94	50.94	50.94	50.94
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.618e+004	2.619e+004	2.619e+004	2.620e+004	2.620e+004
60	Pressure (bar)	88.78	88.78	88.78	88.78	88.78
61	Temperature (C)	50.94	50.93	50.93	50.93	50.93
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 863 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.621e+004	2.621e+004	2.622e+004	2.622e+004	2.623e+004
16	Pressure (bar)	88.78	88.78	88.78	88.78	88.78
17	Temperature (C)	50.93	50.93	50.92	50.92	50.92
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.623e+004	2.624e+004	2.624e+004	2.625e+004	2.625e+004
28	Pressure (bar)	88.78	88.78	88.78	88.78	88.78
29	Temperature (C)	50.92	50.92	50.92	50.91	50.91
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.626e+004	2.626e+004	2.627e+004	2.627e+004	2.628e+004
40	Pressure (bar)	88.78	88.78	88.77	88.77	88.77
41	Temperature (C)	50.91	50.91	50.91	50.90	50.90
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.628e+004	2.629e+004	2.629e+004	2.630e+004	2.630e+004
52	Pressure (bar)	88.77	88.77	88.77	88.77	88.77
53	Temperature (C)	50.90	50.90	50.90	50.90	50.89
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.631e+004	2.631e+004	2.632e+004	2.632e+004	2.633e+004
64	Pressure (bar)	88.77	88.77	88.77	88.77	88.77
65	Temperature (C)	50.89	50.89	50.89	50.89	50.89
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 864 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.633e+004	2.634e+004	2.634e+004	2.635e+004	2.635e+004
20	Pressure (bar)	88.77	88.77	88.77	88.77	88.77
21	Temperature (C)	50.88	50.88	50.88	50.88	50.88
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.636e+004	2.636e+004	2.637e+004	2.637e+004	2.638e+004
32	Pressure (bar)	88.77	88.77	88.77	88.77	88.77
33	Temperature (C)	50.88	50.88	50.87	50.87	50.87
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.638e+004	2.639e+004	2.639e+004	2.640e+004	2.640e+004
44	Pressure (bar)	88.77	88.77	88.77	88.77	88.77
45	Temperature (C)	50.87	50.87	50.87	50.86	50.86
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.641e+004	2.641e+004	2.642e+004	2.642e+004	2.643e+004
56	Pressure (bar)	88.76	88.76	88.76	88.76	88.76
57	Temperature (C)	50.86	50.86	50.86	50.85	50.85
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.644e+004	2.644e+004	2.645e+004	2.645e+004	2.646e+004
68	Pressure (bar)	88.76	88.76	88.76	88.76	88.76
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 865 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	50.85	50.85	50.85	50.85	50.84
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.646e+004	2.647e+004	2.647e+004	2.648e+004	2.648e+004
24	Pressure (bar)	88.76	88.76	88.76	88.76	88.76
25	Temperature (C)	50.84	50.84	50.84	50.84	50.84
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.649e+004	2.649e+004	2.650e+004	2.650e+004	2.651e+004
36	Pressure (bar)	88.76	88.76	88.76	88.76	88.76
37	Temperature (C)	50.83	50.83	50.83	50.83	50.83
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.651e+004	2.652e+004	2.652e+004	2.653e+004	2.653e+004
48	Pressure (bar)	88.76	88.76	88.76	88.76	88.76
49	Temperature (C)	50.83	50.83	50.82	50.82	50.82
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.654e+004	2.654e+004	2.655e+004	2.655e+004	2.656e+004
60	Pressure (bar)	88.76	88.76	88.76	88.76	88.76
61	Temperature (C)	50.82	50.82	50.82	50.81	50.81
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 866 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.656e+004	2.657e+004	2.657e+004	2.658e+004	2.658e+004
16	Pressure (bar)	88.76	88.76	88.76	88.76	88.76
17	Temperature (C)	50.81	50.81	50.81	50.81	50.80
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.659e+004	2.659e+004	2.660e+004	2.660e+004	2.661e+004
28	Pressure (bar)	88.76	88.76	88.76	88.76	88.76
29	Temperature (C)	50.80	50.80	50.80	50.80	50.80
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.661e+004	2.662e+004	2.662e+004	2.663e+004	2.663e+004
40	Pressure (bar)	88.76	88.76	88.76	88.76	88.76
41	Temperature (C)	50.79	50.79	50.79	50.79	50.79
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.664e+004	2.664e+004	2.665e+004	2.665e+004	2.666e+004
52	Pressure (bar)	88.76	88.76	88.76	88.76	88.76
53	Temperature (C)	50.79	50.79	50.78	50.78	50.78
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.666e+004	2.667e+004	2.667e+004	2.668e+004	2.668e+004
64	Pressure (bar)	88.76	88.75	88.75	88.75	88.75
65	Temperature (C)	50.78	50.78	50.77	50.77	50.77
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 867 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.669e+004	2.669e+004	2.670e+004	2.670e+004	2.671e+004
20	Pressure (bar)	88.75	88.75	88.75	88.75	88.74
21	Temperature (C)	50.77	50.77	50.76	50.76	50.76
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.671e+004	2.672e+004	2.672e+004	2.673e+004	2.674e+004
32	Pressure (bar)	88.74	88.74	88.74	88.74	88.74
33	Temperature (C)	50.76	50.76	50.75	50.75	50.75
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.674e+004	2.674e+004	2.675e+004	2.675e+004	2.676e+004
44	Pressure (bar)	88.74	88.74	88.74	88.74	88.74
45	Temperature (C)	50.75	50.75	50.75	50.74	50.74
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.676e+004	2.677e+004	2.677e+004	2.678e+004	2.678e+004
56	Pressure (bar)	88.74	88.74	88.74	88.74	88.74
57	Temperature (C)	50.74	50.74	50.74	50.74	50.73
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.679e+004	2.679e+004	2.680e+004	2.680e+004	2.681e+004
68	Pressure (bar)	88.73	88.73	88.73	88.73	88.73
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	50.73	50.73	50.73	50.73	50.73
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.681e+004	2.682e+004	2.682e+004	2.683e+004	2.683e+004
24	Pressure (bar)	88.73	88.73	88.73	88.73	88.73
25	Temperature (C)	50.72	50.72	50.72	50.72	50.72
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.684e+004	2.684e+004	2.685e+004	2.685e+004	2.686e+004
36	Pressure (bar)	88.73	88.73	88.73	88.73	88.73
37	Temperature (C)	50.72	50.71	50.71	50.71	50.71
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.686e+004	2.687e+004	2.687e+004	2.688e+004	2.688e+004
48	Pressure (bar)	88.73	88.73	88.73	88.73	88.73
49	Temperature (C)	50.71	50.71	50.70	50.70	50.70
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.689e+004	2.689e+004	2.690e+004	2.690e+004	2.691e+004
60	Pressure (bar)	88.73	88.73	88.73	88.73	88.73
61	Temperature (C)	50.70	50.70	50.70	50.69	50.69
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 869 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.691e+004	2.692e+004	2.692e+004	2.693e+004	2.693e+004
16	Pressure (bar)	88.73	88.73	88.73	88.73	88.73
17	Temperature (C)	50.69	50.69	50.69	50.69	50.69
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.694e+004	2.694e+004	2.695e+004	2.695e+004	2.696e+004
28	Pressure (bar)	88.73	88.73	88.73	88.73	88.73
29	Temperature (C)	50.68	50.68	50.68	50.68	50.68
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.696e+004	2.697e+004	2.697e+004	2.698e+004	2.698e+004
40	Pressure (bar)	88.73	88.73	88.73	88.73	88.73
41	Temperature (C)	50.68	50.67	50.67	50.67	50.67
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.699e+004	2.699e+004	2.700e+004	2.700e+004	2.701e+004
52	Pressure (bar)	88.73	88.73	88.73	88.73	88.73
53	Temperature (C)	50.67	50.67	50.67	50.66	50.66
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.701e+004	2.702e+004	2.702e+004	2.703e+004	2.703e+004
64	Pressure (bar)	88.73	88.73	88.73	88.73	88.73
65	Temperature (C)	50.66	50.66	50.66	50.66	50.65
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 870 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.704e+004	2.704e+004	2.704e+004	2.704e+004	2.704e+004
20	Pressure (bar)	88.73	88.73	88.73	88.73	88.73
21	Temperature (C)	50.65	50.65	50.65	50.65	50.65
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.705e+004	2.705e+004	2.705e+004	2.705e+004	2.706e+004
32	Pressure (bar)	88.72	88.72	88.72	88.72	88.72
33	Temperature (C)	50.65	50.65	50.65	50.65	50.65
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.706e+004	2.707e+004	2.707e+004	2.708e+004	2.708e+004
44	Pressure (bar)	88.72	88.73	88.73	88.73	88.73
45	Temperature (C)	50.64	50.64	50.64	50.64	50.64
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.709e+004	2.709e+004	2.710e+004	2.710e+004	2.711e+004
56	Pressure (bar)	88.73	88.73	88.73	88.72	88.72
57	Temperature (C)	50.64	50.64	50.63	50.63	50.63
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.711e+004	2.712e+004	2.712e+004	2.713e+004	2.713e+004
68	Pressure (bar)	88.72	88.72	88.72	88.72	88.72
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	50.63	50.63	50.62	50.62	50.62
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.714e+004	2.714e+004	2.715e+004	2.715e+004	2.716e+004
24	Pressure (bar)	88.71	88.71	88.71	88.71	88.71
25	Temperature (C)	50.62	50.62	50.61	50.61	50.61
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.716e+004	2.717e+004	2.717e+004	2.718e+004	2.718e+004
36	Pressure (bar)	88.71	88.71	88.71	88.71	88.71
37	Temperature (C)	50.61	50.61	50.60	50.60	50.60
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.719e+004	2.719e+004	2.720e+004	2.720e+004	2.721e+004
48	Pressure (bar)	88.71	88.70	88.70	88.70	88.70
49	Temperature (C)	50.60	50.60	50.60	50.59	50.59
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.721e+004	2.722e+004	2.722e+004	2.723e+004	2.723e+004
60	Pressure (bar)	88.70	88.70	88.70	88.70	88.70
61	Temperature (C)	50.59	50.59	50.59	50.59	50.58
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 872 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.724e+004	2.724e+004	2.725e+004	2.725e+004	2.726e+004
16	Pressure (bar)	88.70	88.70	88.70	88.70	88.70
17	Temperature (C)	50.58	50.58	50.58	50.58	50.58
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.726e+004	2.727e+004	2.727e+004	2.728e+004	2.728e+004
28	Pressure (bar)	88.70	88.70	88.70	88.70	88.70
29	Temperature (C)	50.57	50.57	50.57	50.57	50.57
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.729e+004	2.729e+004	2.730e+004	2.730e+004	2.731e+004
40	Pressure (bar)	88.70	88.70	88.70	88.70	88.69
41	Temperature (C)	50.57	50.56	50.56	50.56	50.56
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.731e+004	2.732e+004	2.732e+004	2.733e+004	2.733e+004
52	Pressure (bar)	88.69	88.69	88.69	88.69	88.69
53	Temperature (C)	50.56	50.56	50.55	50.55	50.55
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.734e+004	2.734e+004	2.735e+004	2.735e+004	2.736e+004
64	Pressure (bar)	88.69	88.69	88.69	88.69	88.69
65	Temperature (C)	50.55	50.55	50.55	50.54	50.54
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 873 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.736e+004	2.737e+004	2.737e+004	2.738e+004	2.738e+004
20	Pressure (bar)	88.69	88.69	88.69	88.69	88.69
21	Temperature (C)	50.54	50.54	50.54	50.54	50.53
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.739e+004	2.739e+004	2.740e+004	2.740e+004	2.741e+004
32	Pressure (bar)	88.69	88.69	88.69	88.69	88.69
33	Temperature (C)	50.53	50.53	50.53	50.53	50.53
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.741e+004	2.742e+004	2.742e+004	2.743e+004	2.743e+004
44	Pressure (bar)	88.69	88.69	88.69	88.69	88.69
45	Temperature (C)	50.52	50.52	50.52	50.52	50.52
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.744e+004	2.744e+004	2.745e+004	2.745e+004	2.746e+004
56	Pressure (bar)	88.69	88.69	88.69	88.69	88.68
57	Temperature (C)	50.52	50.51	50.51	50.51	50.51
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.746e+004	2.747e+004	2.747e+004	2.748e+004	2.748e+004
68	Pressure (bar)	88.68	88.68	88.68	88.68	88.68
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 874 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	50.51	50.51	50.50	50.50	50.50
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.749e+004	2.749e+004	2.749e+004	2.749e+004	2.749e+004
24	Pressure (bar)	88.68	88.68	88.68	88.68	88.68
25	Temperature (C)	50.50	50.50	50.50	50.50	50.50
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.750e+004	2.750e+004	2.751e+004	2.751e+004	2.752e+004
36	Pressure (bar)	88.68	88.68	88.68	88.68	88.68
37	Temperature (C)	50.50	50.49	50.49	50.49	50.49
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.753e+004	2.753e+004	2.753e+004	2.753e+004	2.753e+004
48	Pressure (bar)	88.68	88.68	88.68	88.68	88.68
49	Temperature (C)	50.49	50.49	50.48	50.48	50.48
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.754e+004	2.754e+004	2.755e+004	2.755e+004	2.756e+004
60	Pressure (bar)	88.68	88.68	88.68	88.68	88.68
61	Temperature (C)	50.48	50.48	50.48	50.48	50.48
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 875 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.756e+004	2.757e+004	2.757e+004	2.758e+004	2.758e+004
16	Pressure (bar)	88.68	88.67	88.67	88.67	88.66
17	Temperature (C)	50.47	50.47	50.47	50.47	50.46
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.759e+004	2.759e+004	2.760e+004	2.760e+004	2.761e+004
28	Pressure (bar)	88.66	88.66	88.67	88.67	88.67
29	Temperature (C)	50.46	50.46	50.46	50.46	50.46
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.761e+004	2.762e+004	2.762e+004	2.763e+004	2.763e+004
40	Pressure (bar)	88.68	88.67	88.67	88.66	88.66
41	Temperature (C)	50.46	50.46	50.45	50.45	50.45
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.79	17.79	17.79	17.79	17.79
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.764e+004	2.764e+004	2.765e+004	2.765e+004	2.766e+004
52	Pressure (bar)	88.65	88.65	88.65	88.65	88.64
53	Temperature (C)	50.45	50.44	50.44	50.44	50.44
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.766e+004	2.767e+004	2.767e+004	2.768e+004	2.768e+004
64	Pressure (bar)	88.64	88.64	88.65	88.65	88.65
65	Temperature (C)	50.43	50.43	50.43	50.43	50.43
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.769e+004	2.769e+004	2.770e+004	2.770e+004	2.771e+004
20	Pressure (bar)	88.65	88.65	88.65	88.65	88.65
21	Temperature (C)	50.43	50.43	50.43	50.43	50.42
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.771e+004	2.772e+004	2.772e+004	2.773e+004	2.773e+004
32	Pressure (bar)	88.65	88.65	88.65	88.65	88.65
33	Temperature (C)	50.42	50.42	50.42	50.42	50.42
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.774e+004	2.774e+004	2.775e+004	2.775e+004	2.776e+004
44	Pressure (bar)	88.65	88.65	88.65	88.65	88.65
45	Temperature (C)	50.41	50.41	50.41	50.41	50.41
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.776e+004	2.777e+004	2.777e+004	2.778e+004	2.778e+004
56	Pressure (bar)	88.65	88.65	88.65	88.65	88.64
57	Temperature (C)	50.41	50.40	50.40	50.40	50.40
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.779e+004	2.779e+004	2.780e+004	2.780e+004	2.781e+004
68	Pressure (bar)	88.64	88.64	88.64	88.64	88.64
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 877 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	50.40	50.40	50.39	50.39	50.39
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.781e+004	2.782e+004	2.782e+004	2.783e+004	2.783e+004
24	Pressure (bar)	88.64	88.64	88.64	88.64	88.64
25	Temperature (C)	50.39	50.39	50.39	50.38	50.38
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.784e+004	2.784e+004	2.785e+004	2.785e+004	2.785e+004
36	Pressure (bar)	88.64	88.64	88.64	88.64	88.64
37	Temperature (C)	50.38	50.38	50.38	50.38	50.38
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.786e+004	2.788e+004	2.790e+004	2.791e+004	2.793e+004
48	Pressure (bar)	88.64	88.64	88.64	88.64	88.64
49	Temperature (C)	50.37	50.37	50.36	50.36	50.35
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.795e+004	2.797e+004	2.799e+004	2.801e+004	2.802e+004
60	Pressure (bar)	88.63	88.63	88.63	88.63	88.63
61	Temperature (C)	50.34	50.34	50.33	50.33	50.32
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.804e+004	2.806e+004	2.808e+004	2.810e+004	2.812e+004
16	Pressure (bar)	88.63	88.63	88.63	88.62	88.62
17	Temperature (C)	50.31	50.31	50.30	50.30	50.29
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.814e+004	2.815e+004	2.817e+004	2.819e+004	2.821e+004
28	Pressure (bar)	88.62	88.62	88.62	88.62	88.62
29	Temperature (C)	50.28	50.28	50.27	50.27	50.26
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.823e+004	2.825e+004	2.827e+004	2.828e+004	2.830e+004
40	Pressure (bar)	88.62	88.62	88.61	88.61	88.61
41	Temperature (C)	50.25	50.25	50.24	50.24	50.23
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.832e+004	2.834e+004	2.836e+004	2.838e+004	2.839e+004
52	Pressure (bar)	88.61	88.61	88.61	88.61	88.61
53	Temperature (C)	50.22	50.22	50.21	50.21	50.20
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.841e+004	2.842e+004	2.843e+004	2.843e+004	2.844e+004
64	Pressure (bar)	88.61	88.61	88.60	88.60	88.60
65	Temperature (C)	50.19	50.19	50.19	50.19	50.19
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 879 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.845e+004	2.845e+004	2.846e+004	2.846e+004	2.847e+004
20	Pressure (bar)	88.60	88.60	88.60	88.60	88.60
21	Temperature (C)	50.18	50.18	50.18	50.18	50.18
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.847e+004	2.848e+004	2.848e+004	2.849e+004	2.849e+004
32	Pressure (bar)	88.60	88.60	88.60	88.60	88.60
33	Temperature (C)	50.18	50.17	50.17	50.17	50.17
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.850e+004	2.850e+004	2.851e+004	2.851e+004	2.852e+004
44	Pressure (bar)	88.60	88.60	88.60	88.60	88.60
45	Temperature (C)	50.17	50.17	50.16	50.16	50.16
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.852e+004	2.853e+004	2.853e+004	2.854e+004	2.854e+004
56	Pressure (bar)	88.60	88.60	88.60	88.60	88.60
57	Temperature (C)	50.16	50.16	50.16	50.15	50.15
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.855e+004	2.855e+004	2.856e+004	2.856e+004	2.857e+004
68	Pressure (bar)	88.60	88.60	88.60	88.60	88.60
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	50.15	50.15	50.15	50.15	50.14
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.857e+004	2.858e+004	2.858e+004	2.859e+004	2.859e+004
24	Pressure (bar)	88.60	88.60	88.60	88.60	88.60
25	Temperature (C)	50.14	50.14	50.14	50.14	50.14
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.860e+004	2.860e+004	2.861e+004	2.861e+004	2.862e+004
36	Pressure (bar)	88.60	88.60	88.60	88.60	88.60
37	Temperature (C)	50.14	50.14	50.13	50.13	50.13
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.862e+004	2.863e+004	2.863e+004	2.864e+004	2.864e+004
48	Pressure (bar)	88.60	88.60	88.60	88.60	88.60
49	Temperature (C)	50.13	50.13	50.13	50.13	50.13
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.865e+004	2.865e+004	2.866e+004	2.866e+004	2.867e+004
60	Pressure (bar)	88.60	88.60	88.60	88.60	88.60
61	Temperature (C)	50.12	50.12	50.12	50.12	50.12
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 881 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.867e+004	2.868e+004	2.868e+004	2.869e+004	2.869e+004
16	Pressure (bar)	88.60	88.60	88.59	88.59	88.59
17	Temperature (C)	50.11	50.11	50.11	50.11	50.10
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.870e+004	2.870e+004	2.871e+004	2.871e+004	2.872e+004
28	Pressure (bar)	88.58	88.58	88.58	88.58	88.58
29	Temperature (C)	50.10	50.10	50.10	50.10	50.10
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.872e+004	2.873e+004	2.874e+004	2.875e+004	2.876e+004
40	Pressure (bar)	88.58	88.58	88.58	88.58	88.58
41	Temperature (C)	50.09	50.09	50.09	50.08	50.08
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.877e+004	2.878e+004	2.879e+004	2.880e+004	2.881e+004
52	Pressure (bar)	88.58	88.58	88.58	88.58	88.58
53	Temperature (C)	50.08	50.07	50.07	50.07	50.06
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.883e+004	2.884e+004	2.885e+004	2.886e+004	2.887e+004
64	Pressure (bar)	88.58	88.58	88.57	88.57	88.57
65	Temperature (C)	50.06	50.06	50.05	50.05	50.05
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 882 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.888e+004	2.889e+004	2.890e+004	2.891e+004	2.892e+004
20	Pressure (bar)	88.57	88.57	88.57	88.57	88.57
21	Temperature (C)	50.04	50.04	50.04	50.03	50.03
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.893e+004	2.894e+004	2.895e+004	2.896e+004	2.897e+004
32	Pressure (bar)	88.57	88.57	88.57	88.57	88.57
33	Temperature (C)	50.03	50.02	50.02	50.02	50.01
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.898e+004	2.899e+004	2.900e+004	2.901e+004	2.902e+004
44	Pressure (bar)	88.57	88.57	88.57	88.57	88.56
45	Temperature (C)	50.01	50.01	50.00	50.00	50.00
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.903e+004	2.904e+004	2.904e+004	2.904e+004	2.904e+004
56	Pressure (bar)	88.56	88.56	88.56	88.56	88.56
57	Temperature (C)	50.00	49.99	49.99	49.99	49.99
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.905e+004	2.905e+004	2.906e+004	2.906e+004	2.907e+004
68	Pressure (bar)	88.56	88.56	88.56	88.56	88.56
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	49.99	49.99	49.99	49.99	49.98
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.907e+004	2.908e+004	2.908e+004	2.909e+004	2.909e+004
24	Pressure (bar)	88.56	88.56	88.56	88.56	88.56
25	Temperature (C)	49.98	49.98	49.98	49.98	49.98
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.910e+004	2.910e+004	2.911e+004	2.911e+004	2.912e+004
36	Pressure (bar)	88.56	88.56	88.56	88.56	88.56
37	Temperature (C)	49.97	49.97	49.97	49.97	49.97
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.912e+004	2.913e+004	2.913e+004	2.914e+004	2.914e+004
48	Pressure (bar)	88.56	88.56	88.56	88.56	88.56
49	Temperature (C)	49.97	49.97	49.96	49.96	49.96
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.915e+004	2.915e+004	2.916e+004	2.916e+004	2.916e+004
60	Pressure (bar)	88.56	88.55	88.55	88.55	88.55
61	Temperature (C)	49.96	49.96	49.95	49.95	49.95
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 884 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.917e+004	2.917e+004	2.918e+004	2.918e+004	2.919e+004
16	Pressure (bar)	88.55	88.55	88.55	88.55	88.55
17	Temperature (C)	49.95	49.95	49.95	49.94	49.94
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.920e+004	2.920e+004	2.921e+004	2.921e+004	2.922e+004
28	Pressure (bar)	88.55	88.55	88.55	88.55	88.55
29	Temperature (C)	49.94	49.94	49.94	49.94	49.93
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.922e+004	2.923e+004	2.923e+004	2.924e+004	2.924e+004
40	Pressure (bar)	88.55	88.55	88.55	88.55	88.55
41	Temperature (C)	49.93	49.93	49.93	49.93	49.93
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.925e+004	2.925e+004	2.925e+004	2.926e+004	2.926e+004
52	Pressure (bar)	88.55	88.55	88.54	88.54	88.54
53	Temperature (C)	49.92	49.92	49.92	49.92	49.92
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.926e+004	2.927e+004	2.928e+004	2.929e+004	2.930e+004
64	Pressure (bar)	88.54	88.54	88.54	88.54	88.54
65	Temperature (C)	49.92	49.92	49.91	49.91	49.91
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 885 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.931e+004	2.932e+004	2.933e+004	2.934e+004	2.935e+004
20	Pressure (bar)	88.54	88.54	88.54	88.54	88.54
21	Temperature (C)	49.90	49.90	49.90	49.90	49.89
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.936e+004	2.936e+004	2.937e+004	2.938e+004	2.939e+004
32	Pressure (bar)	88.54	88.54	88.54	88.54	88.54
33	Temperature (C)	49.89	49.89	49.88	49.88	49.88
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.940e+004	2.941e+004	2.942e+004	2.943e+004	2.944e+004
44	Pressure (bar)	88.53	88.53	88.53	88.53	88.53
45	Temperature (C)	49.87	49.87	49.87	49.87	49.86
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.945e+004	2.946e+004	2.947e+004	2.948e+004	2.949e+004
56	Pressure (bar)	88.53	88.53	88.53	88.53	88.53
57	Temperature (C)	49.86	49.86	49.85	49.85	49.85
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.950e+004	2.951e+004	2.952e+004	2.953e+004	2.954e+004
68	Pressure (bar)	88.53	88.53	88.53	88.53	88.53
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	49.84	49.84	49.84	49.84	49.83
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.955e+004	2.955e+004	2.955e+004	2.955e+004	2.955e+004
24	Pressure (bar)	88.53	88.53	88.53	88.53	88.53
25	Temperature (C)	49.83	49.83	49.83	49.83	49.83
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.955e+004	2.956e+004	2.956e+004	2.957e+004	2.957e+004
36	Pressure (bar)	88.53	88.53	88.53	88.53	88.53
37	Temperature (C)	49.83	49.83	49.82	49.82	49.82
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.958e+004	2.958e+004	2.959e+004	2.959e+004	2.960e+004
48	Pressure (bar)	88.52	88.52	88.52	88.52	88.52
49	Temperature (C)	49.82	49.82	49.82	49.81	49.81
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.960e+004	2.961e+004	2.961e+004	2.962e+004	2.962e+004
60	Pressure (bar)	88.52	88.52	88.52	88.52	88.52
61	Temperature (C)	49.81	49.81	49.81	49.81	49.81
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.963e+004	2.963e+004	2.964e+004	2.964e+004	2.965e+004
16	Pressure (bar)	88.52	88.52	88.52	88.52	88.52
17	Temperature (C)	49.80	49.80	49.80	49.80	49.80
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.965e+004	2.965e+004	2.966e+004	2.966e+004	2.966e+004
28	Pressure (bar)	88.52	88.52	88.52	88.52	88.52
29	Temperature (C)	49.80	49.80	49.79	49.79	49.79
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.966e+004	2.966e+004	2.967e+004	2.967e+004	2.967e+004
40	Pressure (bar)	88.52	88.52	88.52	88.52	88.52
41	Temperature (C)	49.79	49.79	49.79	49.79	49.79
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.967e+004	2.968e+004	2.968e+004	2.968e+004	2.969e+004
52	Pressure (bar)	88.52	88.52	88.52	88.52	88.52
53	Temperature (C)	49.79	49.79	49.79	49.79	49.78
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	2.969e+004	2.969e+004	2.970e+004	2.970e+004	2.970e+004
64	Pressure (bar)	88.52	88.52	88.52	88.52	88.52
65	Temperature (C)	49.78	49.78	49.78	49.78	49.78
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 888 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	2.971e+004	2.971e+004	2.971e+004	2.972e+004	2.972e+004
20	Pressure (bar)	88.52	88.52	88.52	88.52	88.52
21	Temperature (C)	49.78	49.78	49.78	49.77	49.77
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	2.973e+004	2.973e+004	2.974e+004	2.974e+004	2.975e+004
32	Pressure (bar)	88.52	88.52	88.52	88.52	88.52
33	Temperature (C)	49.77	49.77	49.77	49.77	49.77
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	2.975e+004	2.976e+004	2.976e+004	2.977e+004	2.977e+004
44	Pressure (bar)	88.51	88.51	88.51	88.51	88.51
45	Temperature (C)	49.76	49.76	49.76	49.76	49.76
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	2.978e+004	2.978e+004	2.979e+004	2.979e+004	2.980e+004
56	Pressure (bar)	88.51	88.51	88.51	88.51	88.51
57	Temperature (C)	49.76	49.75	49.75	49.75	49.75
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	2.980e+004	2.981e+004	2.981e+004	2.982e+004	2.982e+004
68	Pressure (bar)	88.51	88.51	88.51	88.51	88.51
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	49.75	49.75	49.75	49.74	49.74
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	2.983e+004	2.983e+004	2.984e+004	2.984e+004	2.985e+004
24	Pressure (bar)	88.51	88.51	88.51	88.51	88.51
25	Temperature (C)	49.74	49.74	49.74	49.74	49.73
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	2.985e+004	2.986e+004	2.986e+004	2.987e+004	2.987e+004
36	Pressure (bar)	88.51	88.51	88.51	88.51	88.51
37	Temperature (C)	49.73	49.73	49.73	49.73	49.73
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.78
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	2.988e+004	2.988e+004	2.989e+004	2.989e+004	2.990e+004
48	Pressure (bar)	88.51	88.50	88.50	88.50	88.50
49	Temperature (C)	49.72	49.72	49.72	49.72	49.72
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.77
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	2.990e+004	2.990e+004	2.991e+004	2.991e+004	2.991e+004
60	Pressure (bar)	88.50	88.50	88.50	88.50	88.50
61	Temperature (C)	49.72	49.72	49.71	49.71	49.71
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 890 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	2.991e+004	2.992e+004	2.992e+004	2.993e+004	2.993e+004
16	Pressure (bar)	88.50	88.50	88.50	88.50	88.50
17	Temperature (C)	49.71	49.71	49.71	49.71	49.71
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	2.994e+004	2.994e+004	2.994e+004	2.995e+004	2.995e+004
28	Pressure (bar)	88.50	88.50	88.50	88.50	88.50
29	Temperature (C)	49.71	49.70	49.70	49.70	49.70
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	2.996e+004	2.996e+004	2.997e+004	2.997e+004	2.998e+004
40	Pressure (bar)	88.50	88.50	88.50	88.50	88.50
41	Temperature (C)	49.70	49.70	49.70	49.69	49.69
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	2.998e+004	2.999e+004	2.999e+004	3.000e+004	3.000e+004
52	Pressure (bar)	88.50	88.50	88.50	88.50	88.50
53	Temperature (C)	49.69	49.69	49.69	49.69	49.68
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.001e+004	3.001e+004	3.002e+004	3.002e+004	3.003e+004
64	Pressure (bar)	88.50	88.50	88.50	88.50	88.50
65	Temperature (C)	49.68	49.68	49.68	49.68	49.68
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 891 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.003e+004	3.004e+004	3.004e+004	3.005e+004	3.005e+004
20	Pressure (bar)	88.50	88.50	88.49	88.49	88.49
21	Temperature (C)	49.67	49.67	49.67	49.67	49.67
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.006e+004	3.006e+004	3.007e+004	3.007e+004	3.008e+004
32	Pressure (bar)	88.49	88.49	88.49	88.49	88.49
33	Temperature (C)	49.67	49.67	49.66	49.66	49.66
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.008e+004	3.009e+004	3.009e+004	3.010e+004	3.010e+004
44	Pressure (bar)	88.49	88.49	88.49	88.49	88.49
45	Temperature (C)	49.66	49.66	49.66	49.65	49.65
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.011e+004	3.011e+004	3.012e+004	3.012e+004	3.013e+004
56	Pressure (bar)	88.49	88.49	88.49	88.49	88.49
57	Temperature (C)	49.65	49.65	49.65	49.65	49.64
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.013e+004	3.014e+004	3.014e+004	3.015e+004	3.015e+004
68	Pressure (bar)	88.49	88.49	88.49	88.49	88.49
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	49.64	49.64	49.64	49.64	49.64
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.016e+004	3.016e+004	3.017e+004	3.017e+004	3.018e+004
24	Pressure (bar)	88.49	88.49	88.49	88.49	88.49
25	Temperature (C)	49.64	49.63	49.63	49.63	49.63
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.018e+004	3.019e+004	3.019e+004	3.020e+004	3.020e+004
36	Pressure (bar)	88.49	88.49	88.49	88.49	88.48
37	Temperature (C)	49.63	49.63	49.62	49.62	49.62
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.021e+004	3.021e+004	3.022e+004	3.022e+004	3.023e+004
48	Pressure (bar)	88.48	88.48	88.48	88.48	88.48
49	Temperature (C)	49.62	49.62	49.62	49.61	49.61
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.023e+004	3.024e+004	3.024e+004	3.025e+004	3.025e+004
60	Pressure (bar)	88.48	88.48	88.48	88.48	88.48
61	Temperature (C)	49.61	49.61	49.61	49.61	49.61
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 893 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3						
4						
5	Unit Set: Mixed					
6	Date/Time: Mon Jun 13 15:02:44 2022					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	Hydrates					
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.026e+004	3.026e+004	3.027e+004	3.027e+004	3.028e+004
16	Pressure (bar)	88.48	88.48	88.48	88.48	88.48
17	Temperature (C)	49.60	49.60	49.60	49.60	49.60
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.028e+004	3.029e+004	3.029e+004	3.030e+004	3.030e+004
28	Pressure (bar)	88.48	88.48	88.48	88.48	88.48
29	Temperature (C)	49.60	49.59	49.59	49.59	49.59
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.031e+004	3.031e+004	3.032e+004	3.032e+004	3.033e+004
40	Pressure (bar)	88.48	88.48	88.48	88.48	88.47
41	Temperature (C)	49.59	49.59	49.58	49.58	49.58
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.033e+004	3.034e+004	3.035e+004	3.036e+004	3.036e+004
52	Pressure (bar)	88.47	88.47	88.47	88.47	88.47
53	Temperature (C)	49.58	49.58	49.57	49.57	49.57
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.037e+004	3.039e+004	3.040e+004	3.041e+004	3.043e+004
64	Pressure (bar)	88.47	88.47	88.47	88.47	88.47
65	Temperature (C)	49.57	49.56	49.56	49.55	49.55
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 894 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc		
2					
3			Unit Set: Mixed		
4			Date/Time: Mon Jun 13 15:02:44 2022		
5					
6	Pipe Segment: ELEP-Akabel Pipeline (continued)				
7					
8					
9	Hydrates				
10					
11	Hydrates Table				
12					
13	Inhibitors	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.77	17.77	17.77	17.77
15	Type at Formation Temperature	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.044e+004	3.045e+004	3.045e+004	3.045e+004
20	Pressure (bar)	88.47	88.47	88.47	88.47
21	Temperature (C)	49.54	49.54	49.54	49.54
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.77	17.77	17.77	17.77
27	Type at Formation Temperature	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.046e+004	3.046e+004	3.047e+004	3.048e+004
32	Pressure (bar)	88.47	88.47	88.47	88.47
33	Temperature (C)	49.54	49.54	49.54	49.53
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.77	17.77	17.77	17.77
39	Type at Formation Temperature	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.048e+004	3.049e+004	3.049e+004	3.050e+004
44	Pressure (bar)	88.47	88.47	88.47	88.46
45	Temperature (C)	49.53	49.53	49.53	49.53
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.77	17.77	17.77	17.77
51	Type at Formation Temperature	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.051e+004	3.051e+004	3.052e+004	3.053e+004
56	Pressure (bar)	88.46	88.46	88.46	88.46
57	Temperature (C)	49.52	49.52	49.52	49.52
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.77	17.77	17.77	17.77
63	Type at Formation Temperature	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.053e+004	3.054e+004	3.054e+004	3.055e+004
68	Pressure (bar)	88.46	88.46	88.46	88.46
69	Aspen Technology Inc. Aspen HYSYS Version 10 Page 895 of 937				

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	49.52	49.52	49.51	49.51	49.51
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.056e+004	3.056e+004	3.057e+004	3.057e+004	3.058e+004
24	Pressure (bar)	88.46	88.46	88.46	88.46	88.46
25	Temperature (C)	49.51	49.51	49.51	49.50	49.50
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.058e+004	3.059e+004	3.059e+004	3.060e+004	3.060e+004
36	Pressure (bar)	88.46	88.46	88.46	88.46	88.46
37	Temperature (C)	49.50	49.50	49.50	49.50	49.49
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.061e+004	3.061e+004	3.062e+004	3.062e+004	3.063e+004
48	Pressure (bar)	88.46	88.46	88.46	88.46	88.46
49	Temperature (C)	49.49	49.49	49.49	49.49	49.49
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.063e+004	3.064e+004	3.064e+004	3.065e+004	3.065e+004
60	Pressure (bar)	88.46	88.46	88.45	88.45	88.45
61	Temperature (C)	49.48	49.48	49.48	49.48	49.48
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 896 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.066e+004	3.066e+004	3.067e+004	3.067e+004	3.068e+004
16	Pressure (bar)	88.45	88.45	88.45	88.45	88.45
17	Temperature (C)	49.48	49.47	49.47	49.47	49.47
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.068e+004	3.069e+004	3.069e+004	3.070e+004	3.070e+004
28	Pressure (bar)	88.45	88.45	88.45	88.45	88.45
29	Temperature (C)	49.47	49.47	49.47	49.46	49.46
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.071e+004	3.071e+004	3.072e+004	3.072e+004	3.073e+004
40	Pressure (bar)	88.45	88.45	88.45	88.45	88.45
41	Temperature (C)	49.46	49.46	49.46	49.46	49.46
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.073e+004	3.074e+004	3.074e+004	3.075e+004	3.075e+004
52	Pressure (bar)	88.45	88.45	88.45	88.45	88.45
53	Temperature (C)	49.45	49.45	49.45	49.45	49.45
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.076e+004	3.076e+004	3.077e+004	3.077e+004	3.078e+004
64	Pressure (bar)	88.45	88.45	88.45	88.45	88.45
65	Temperature (C)	49.45	49.44	49.44	49.44	49.44
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 897 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.078e+004	3.079e+004	3.079e+004	3.080e+004	3.080e+004
20	Pressure (bar)	88.45	88.45	88.45	88.45	88.45
21	Temperature (C)	49.44	49.44	49.44	49.43	49.43
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.081e+004	3.081e+004	3.082e+004	3.082e+004	3.083e+004
32	Pressure (bar)	88.45	88.45	88.45	88.45	88.45
33	Temperature (C)	49.43	49.43	49.43	49.43	49.42
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.083e+004	3.084e+004	3.084e+004	3.085e+004	3.085e+004
44	Pressure (bar)	88.45	88.44	88.44	88.44	88.44
45	Temperature (C)	49.42	49.42	49.42	49.42	49.42
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.086e+004	3.086e+004	3.087e+004	3.087e+004	3.088e+004
56	Pressure (bar)	88.44	88.44	88.44	88.44	88.44
57	Temperature (C)	49.41	49.41	49.41	49.41	49.41
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.088e+004	3.089e+004	3.089e+004	3.090e+004	3.090e+004
68	Pressure (bar)	88.44	88.44	88.44	88.44	88.44
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 898 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	49.41	49.40	49.40	49.40	49.40
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.091e+004	3.091e+004	3.092e+004	3.092e+004	3.093e+004
24	Pressure (bar)	88.44	88.44	88.44	88.44	88.44
25	Temperature (C)	49.40	49.40	49.39	49.39	49.39
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.093e+004	3.094e+004	3.094e+004	3.095e+004	3.095e+004
36	Pressure (bar)	88.43	88.43	88.43	88.43	88.43
37	Temperature (C)	49.39	49.39	49.39	49.39	49.38
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.096e+004	3.096e+004	3.097e+004	3.097e+004	3.098e+004
48	Pressure (bar)	88.43	88.43	88.43	88.43	88.43
49	Temperature (C)	49.38	49.38	49.38	49.38	49.38
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.098e+004	3.099e+004	3.099e+004	3.100e+004	3.100e+004
60	Pressure (bar)	88.43	88.43	88.43	88.43	88.43
61	Temperature (C)	49.37	49.37	49.37	49.37	49.37
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.101e+004	3.103e+004	3.105e+004	3.107e+004	3.109e+004
16	Pressure (bar)	88.43	88.43	88.43	88.43	88.43
17	Temperature (C)	49.37	49.36	49.35	49.35	49.34
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.111e+004	3.111e+004	3.112e+004	3.112e+004	3.113e+004
28	Pressure (bar)	88.42	88.42	88.42	88.42	88.42
29	Temperature (C)	49.34	49.33	49.33	49.33	49.33
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.113e+004	3.114e+004	3.114e+004	3.115e+004	3.115e+004
40	Pressure (bar)	88.42	88.42	88.42	88.41	88.41
41	Temperature (C)	49.33	49.33	49.32	49.32	49.32
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.116e+004	3.116e+004	3.117e+004	3.117e+004	3.118e+004
52	Pressure (bar)	88.40	88.43	88.45	88.47	88.49
53	Temperature (C)	49.31	49.32	49.33	49.34	49.35
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.118e+004	3.119e+004	3.119e+004	3.119e+004	3.120e+004
64	Pressure (bar)	88.51	88.51	88.51	88.51	88.50
65	Temperature (C)	49.36	49.36	49.36	49.36	49.35
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.78	17.78	17.78	17.78	17.77
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.120e+004	3.121e+004	3.122e+004	3.122e+004	3.123e+004
20	Pressure (bar)	88.50	88.49	88.48	88.47	88.46
21	Temperature (C)	49.35	49.35	49.34	49.34	49.33
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.124e+004	3.124e+004	3.125e+004	3.125e+004	3.126e+004
32	Pressure (bar)	88.45	88.44	88.43	88.41	88.40
33	Temperature (C)	49.32	49.32	49.32	49.31	49.31
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.77	17.77	17.77	17.77	17.77
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.126e+004	3.126e+004	3.127e+004	3.127e+004	3.128e+004
44	Pressure (bar)	88.39	88.33	88.27	88.21	88.16
45	Temperature (C)	49.30	49.29	49.27	49.25	49.24
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.77	17.76	17.76	17.76	17.75
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.128e+004	3.129e+004	3.129e+004	3.130e+004	3.130e+004
56	Pressure (bar)	88.10	88.10	88.10	88.10	88.10
57	Temperature (C)	49.22	49.22	49.22	49.22	49.22
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.131e+004	3.131e+004	3.132e+004	3.132e+004	3.133e+004
68	Pressure (bar)	88.10	88.10	88.10	88.10	88.10
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	49.21	49.21	49.21	49.21	49.21
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.133e+004	3.134e+004	3.134e+004	3.135e+004	3.135e+004
24	Pressure (bar)	88.09	88.09	88.09	88.09	88.09
25	Temperature (C)	49.21	49.20	49.20	49.20	49.20
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.136e+004	3.136e+004	3.137e+004	3.137e+004	3.138e+004
36	Pressure (bar)	88.09	88.09	88.09	88.09	88.09
37	Temperature (C)	49.20	49.20	49.20	49.19	49.19
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.138e+004	3.139e+004	3.139e+004	3.140e+004	3.140e+004
48	Pressure (bar)	88.09	88.09	88.09	88.09	88.09
49	Temperature (C)	49.19	49.19	49.19	49.19	49.18
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.141e+004	3.141e+004	3.142e+004	3.142e+004	3.143e+004
60	Pressure (bar)	88.09	88.09	88.09	88.09	88.09
61	Temperature (C)	49.18	49.18	49.18	49.18	49.18
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 902 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.143e+004	3.144e+004	3.144e+004	3.145e+004	3.145e+004
16	Pressure (bar)	88.09	88.09	88.09	88.09	88.09
17	Temperature (C)	49.17	49.17	49.17	49.17	49.17
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.146e+004	3.146e+004	3.147e+004	3.147e+004	3.148e+004
28	Pressure (bar)	88.09	88.09	88.09	88.09	88.09
29	Temperature (C)	49.17	49.17	49.16	49.16	49.16
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.148e+004	3.149e+004	3.149e+004	3.150e+004	3.150e+004
40	Pressure (bar)	88.09	88.08	88.08	88.08	88.08
41	Temperature (C)	49.16	49.16	49.16	49.15	49.15
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.151e+004	3.151e+004	3.152e+004	3.152e+004	3.153e+004
52	Pressure (bar)	88.08	88.08	88.08	88.08	88.08
53	Temperature (C)	49.15	49.15	49.15	49.15	49.15
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.153e+004	3.154e+004	3.154e+004	3.155e+004	3.155e+004
64	Pressure (bar)	88.08	88.08	88.08	88.08	88.08
65	Temperature (C)	49.14	49.14	49.14	49.14	49.14
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 903 of 937	

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.156e+004	3.157e+004	3.157e+004	3.158e+004	3.158e+004
20	Pressure (bar)	88.08	88.08	88.08	88.08	88.08
21	Temperature (C)	49.14	49.13	49.13	49.13	49.13
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.159e+004	3.159e+004	3.160e+004	3.160e+004	3.161e+004
32	Pressure (bar)	88.08	88.08	88.08	88.08	88.08
33	Temperature (C)	49.13	49.13	49.13	49.12	49.12
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.161e+004	3.162e+004	3.162e+004	3.163e+004	3.163e+004
44	Pressure (bar)	88.07	88.07	88.07	88.07	88.07
45	Temperature (C)	49.12	49.12	49.12	49.12	49.11
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.164e+004	3.164e+004	3.165e+004	3.165e+004	3.166e+004
56	Pressure (bar)	88.07	88.07	88.07	88.07	88.07
57	Temperature (C)	49.11	49.11	49.11	49.11	49.11
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.166e+004	3.167e+004	3.167e+004	3.168e+004	3.168e+004
68	Pressure (bar)	88.07	88.07	88.07	88.07	88.07
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	49.11	49.10	49.10	49.10	49.10
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.169e+004	3.169e+004	3.169e+004	3.169e+004	3.170e+004
24	Pressure (bar)	88.07	88.07	88.07	88.07	88.07
25	Temperature (C)	49.10	49.10	49.10	49.09	49.09
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.170e+004	3.170e+004	3.170e+004	3.171e+004	3.171e+004
36	Pressure (bar)	88.07	88.07	88.07	88.07	88.07
37	Temperature (C)	49.09	49.09	49.09	49.09	49.09
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.171e+004	3.172e+004	3.172e+004	3.173e+004	3.173e+004
48	Pressure (bar)	88.07	88.07	88.07	88.07	88.07
49	Temperature (C)	49.09	49.09	49.09	49.09	49.08
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.174e+004	3.174e+004	3.174e+004	3.174e+004	3.175e+004
60	Pressure (bar)	88.07	88.07	88.07	88.07	88.07
61	Temperature (C)	49.08	49.08	49.08	49.08	49.08
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 905 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.175e+004	3.176e+004	3.176e+004	3.177e+004	3.177e+004
16	Pressure (bar)	88.07	88.07	88.07	88.07	88.07
17	Temperature (C)	49.08	49.08	49.07	49.07	49.07
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.178e+004	3.179e+004	3.179e+004	3.180e+004	3.180e+004
28	Pressure (bar)	88.07	88.06	88.06	88.06	88.06
29	Temperature (C)	49.07	49.07	49.06	49.06	49.06
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.181e+004	3.182e+004	3.182e+004	3.183e+004	3.183e+004
40	Pressure (bar)	88.06	88.06	88.06	88.06	88.06
41	Temperature (C)	49.06	49.06	49.06	49.05	49.05
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.184e+004	3.184e+004	3.185e+004	3.185e+004	3.186e+004
52	Pressure (bar)	88.06	88.06	88.06	88.06	88.06
53	Temperature (C)	49.05	49.05	49.05	49.05	49.04
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.186e+004	3.187e+004	3.187e+004	3.188e+004	3.188e+004
64	Pressure (bar)	88.06	88.06	88.06	88.06	88.06
65	Temperature (C)	49.04	49.04	49.04	49.04	49.04
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 906 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.189e+004	3.189e+004	3.190e+004	3.190e+004	3.191e+004
20	Pressure (bar)	88.06	88.06	88.06	88.06	88.06
21	Temperature (C)	49.04	49.03	49.03	49.03	49.03
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.75	17.75	17.75	17.75	17.75
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.191e+004	3.192e+004	3.192e+004	3.193e+004	3.193e+004
32	Pressure (bar)	88.05	88.05	88.05	88.05	88.05
33	Temperature (C)	49.03	49.03	49.02	49.02	49.02
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.75	17.74	17.74	17.74	17.74
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.194e+004	3.194e+004	3.195e+004	3.195e+004	3.196e+004
44	Pressure (bar)	88.05	88.05	88.05	88.05	88.05
45	Temperature (C)	49.02	49.02	49.02	49.01	49.01
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.196e+004	3.197e+004	3.197e+004	3.198e+004	3.198e+004
56	Pressure (bar)	88.05	88.05	88.05	88.05	88.05
57	Temperature (C)	49.01	49.01	49.01	49.01	49.01
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.199e+004	3.199e+004	3.200e+004	3.200e+004	3.201e+004
68	Pressure (bar)	88.05	88.05	88.04	88.04	88.04
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	49.00	49.00	49.00	49.00	49.00
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.201e+004	3.202e+004	3.202e+004	3.203e+004	3.203e+004
24	Pressure (bar)	88.04	88.04	88.04	88.04	88.04
25	Temperature (C)	49.00	48.99	48.99	48.99	48.99
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.204e+004	3.204e+004	3.205e+004	3.205e+004	3.206e+004
36	Pressure (bar)	88.04	88.04	88.04	88.04	88.04
37	Temperature (C)	48.99	48.99	48.99	48.98	48.98
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.206e+004	3.207e+004	3.207e+004	3.208e+004	3.208e+004
48	Pressure (bar)	88.04	88.04	88.04	88.04	88.04
49	Temperature (C)	48.98	48.98	48.98	48.98	48.98
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.209e+004	3.209e+004	3.210e+004	3.210e+004	3.211e+004
60	Pressure (bar)	88.04	88.04	88.04	88.04	88.04
61	Temperature (C)	48.97	48.97	48.97	48.97	48.97
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 908 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3						
4						
5	Unit Set: Mixed					
6	Date/Time: Mon Jun 13 15:02:44 2022					
7	Pipe Segment: ELEP-Akabel Pipeline (continued)					
8						
9						
10	Hydrates					
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.211e+004	3.212e+004	3.212e+004	3.213e+004	3.213e+004
16	Pressure (bar)	88.04	88.04	88.04	88.04	88.04
17	Temperature (C)	48.97	48.96	48.96	48.96	48.96
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.214e+004	3.214e+004	3.215e+004	3.215e+004	3.216e+004
28	Pressure (bar)	88.04	88.04	88.03	88.03	88.03
29	Temperature (C)	48.96	48.96	48.95	48.95	48.95
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.216e+004	3.217e+004	3.217e+004	3.218e+004	3.218e+004
40	Pressure (bar)	88.03	88.03	88.03	88.03	88.03
41	Temperature (C)	48.95	48.95	48.95	48.94	48.94
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.219e+004	3.219e+004	3.219e+004	3.219e+004	3.219e+004
52	Pressure (bar)	88.03	88.03	88.02	88.02	88.02
53	Temperature (C)	48.94	48.94	48.94	48.94	48.94
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.219e+004	3.219e+004	3.220e+004	3.220e+004	3.221e+004
64	Pressure (bar)	88.01	88.01	88.01	88.01	88.00
65	Temperature (C)	48.94	48.93	48.93	48.93	48.93
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1			Case Name: Simulation - 386MMSCFD_100barg.hsc			
2	 Epic Atlantic Limited		Unit Set: Mixed			
Date/Time: Mon Jun 13 15:02:44 2022						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.221e+004	3.221e+004	3.222e+004	3.222e+004	3.223e+004
20	Pressure (bar)	88.00	88.00	88.00	88.00	88.00
21	Temperature (C)	48.93	48.92	48.92	48.92	48.92
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.223e+004	3.224e+004	3.226e+004	3.227e+004	3.228e+004
32	Pressure (bar)	88.00	88.00	88.00	88.00	87.99
33	Temperature (C)	48.92	48.91	48.91	48.91	48.90
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.230e+004	3.231e+004	3.232e+004	3.234e+004	3.235e+004
44	Pressure (bar)	87.99	87.99	87.99	87.99	87.99
45	Temperature (C)	48.90	48.89	48.89	48.89	48.88
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.236e+004	3.238e+004	3.239e+004	3.240e+004	3.242e+004
56	Pressure (bar)	87.99	87.99	87.99	87.99	87.99
57	Temperature (C)	48.88	48.87	48.87	48.87	48.86
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.243e+004	3.244e+004	3.246e+004	3.247e+004	3.248e+004
68	Pressure (bar)	87.98	87.98	87.98	87.98	87.98
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 910 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	48.86	48.85	48.85	48.85	48.84
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.250e+004	3.251e+004	3.252e+004	3.254e+004	3.255e+004
24	Pressure (bar)	87.98	87.98	87.98	87.98	87.98
25	Temperature (C)	48.84	48.83	48.83	48.83	48.82
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.257e+004	3.257e+004	3.258e+004	3.258e+004	3.259e+004
36	Pressure (bar)	87.98	87.98	87.98	87.98	87.98
37	Temperature (C)	48.82	48.82	48.81	48.81	48.81
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.260e+004	3.260e+004	3.261e+004	3.261e+004	3.262e+004
48	Pressure (bar)	87.98	87.98	87.98	87.98	87.98
49	Temperature (C)	48.81	48.81	48.81	48.80	48.80
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.263e+004	3.263e+004	3.263e+004	3.263e+004	3.263e+004
60	Pressure (bar)	87.98	87.98	87.98	87.98	87.98
61	Temperature (C)	48.80	48.80	48.80	48.80	48.80
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 911 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.264e+004	3.264e+004	3.264e+004	3.264e+004	3.264e+004
16	Pressure (bar)	87.98	87.98	87.98	87.98	87.98
17	Temperature (C)	48.80	48.80	48.80	48.80	48.80
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.265e+004	3.265e+004	3.266e+004	3.266e+004	3.267e+004
28	Pressure (bar)	87.98	87.98	87.98	87.98	87.98
29	Temperature (C)	48.80	48.79	48.79	48.79	48.79
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.267e+004	3.268e+004	3.268e+004	3.269e+004	3.269e+004
40	Pressure (bar)	87.98	87.98	87.98	87.98	87.98
41	Temperature (C)	48.79	48.79	48.79	48.78	48.78
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.270e+004	3.270e+004	3.271e+004	3.271e+004	3.272e+004
52	Pressure (bar)	87.98	87.98	87.98	87.98	87.97
53	Temperature (C)	48.78	48.78	48.78	48.78	48.77
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.272e+004	3.273e+004	3.273e+004	3.274e+004	3.274e+004
64	Pressure (bar)	87.97	87.97	87.97	87.97	87.97
65	Temperature (C)	48.77	48.77	48.77	48.77	48.77
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 912 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.275e+004	3.275e+004	3.276e+004	3.276e+004	3.277e+004
20	Pressure (bar)	87.97	87.97	87.97	87.97	87.97
21	Temperature (C)	48.77	48.76	48.76	48.76	48.76
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.277e+004	3.278e+004	3.278e+004	3.279e+004	3.279e+004
32	Pressure (bar)	87.97	87.97	87.97	87.97	87.97
33	Temperature (C)	48.76	48.76	48.75	48.75	48.75
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.280e+004	3.280e+004	3.281e+004	3.281e+004	3.282e+004
44	Pressure (bar)	87.97	87.97	87.97	87.97	87.97
45	Temperature (C)	48.75	48.75	48.75	48.75	48.74
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.282e+004	3.283e+004	3.283e+004	3.284e+004	3.284e+004
56	Pressure (bar)	87.97	87.97	87.97	87.97	87.97
57	Temperature (C)	48.74	48.74	48.74	48.74	48.74
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.285e+004	3.285e+004	3.286e+004	3.286e+004	3.287e+004
68	Pressure (bar)	87.97	87.97	87.97	87.97	87.97
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 913 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	48.74	48.74	48.73	48.73	48.73
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.287e+004	3.288e+004	3.288e+004	3.289e+004	3.289e+004
24	Pressure (bar)	87.97	87.97	87.97	87.97	87.97
25	Temperature (C)	48.73	48.73	48.73	48.73	48.73
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.290e+004	3.290e+004	3.291e+004	3.291e+004	3.292e+004
36	Pressure (bar)	87.97	87.97	87.97	87.97	87.97
37	Temperature (C)	48.72	48.72	48.72	48.72	48.72
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.292e+004	3.293e+004	3.293e+004	3.294e+004	3.294e+004
48	Pressure (bar)	87.97	87.97	87.97	87.97	87.97
49	Temperature (C)	48.72	48.72	48.71	48.71	48.71
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.295e+004	3.295e+004	3.296e+004	3.296e+004	3.297e+004
60	Pressure (bar)	87.97	87.97	87.97	87.97	87.97
61	Temperature (C)	48.71	48.71	48.71	48.71	48.70
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 914 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.297e+004	3.298e+004	3.298e+004	3.299e+004	3.299e+004
16	Pressure (bar)	87.97	87.97	87.97	87.97	87.97
17	Temperature (C)	48.70	48.70	48.70	48.70	48.70
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.300e+004	3.300e+004	3.301e+004	3.301e+004	3.302e+004
28	Pressure (bar)	87.97	87.97	87.97	87.97	87.97
29	Temperature (C)	48.69	48.69	48.69	48.69	48.69
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.302e+004	3.302e+004	3.302e+004	3.303e+004	3.303e+004
40	Pressure (bar)	87.97	87.97	87.97	87.97	87.97
41	Temperature (C)	48.69	48.69	48.69	48.69	48.68
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.303e+004	3.304e+004	3.304e+004	3.305e+004	3.305e+004
52	Pressure (bar)	87.97	87.97	87.97	87.97	87.96
53	Temperature (C)	48.68	48.68	48.68	48.68	48.68
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.306e+004	3.306e+004	3.306e+004	3.307e+004	3.307e+004
64	Pressure (bar)	87.96	87.96	87.96	87.96	87.96
65	Temperature (C)	48.67	48.67	48.67	48.67	48.67
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 915 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.307e+004	3.308e+004	3.308e+004	3.309e+004	3.309e+004
20	Pressure (bar)	87.96	87.96	87.96	87.96	87.96
21	Temperature (C)	48.67	48.67	48.67	48.67	48.67
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.310e+004	3.310e+004	3.311e+004	3.311e+004	3.312e+004
32	Pressure (bar)	87.96	87.96	87.96	87.96	87.96
33	Temperature (C)	48.66	48.66	48.66	48.66	48.66
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.312e+004	3.313e+004	3.313e+004	3.314e+004	3.314e+004
44	Pressure (bar)	87.96	87.96	87.96	87.96	87.96
45	Temperature (C)	48.66	48.66	48.65	48.65	48.65
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.315e+004	3.315e+004	3.316e+004	3.316e+004	3.317e+004
56	Pressure (bar)	87.96	87.96	87.96	87.96	87.96
57	Temperature (C)	48.65	48.65	48.65	48.64	48.64
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.317e+004	3.318e+004	3.318e+004	3.319e+004	3.319e+004
68	Pressure (bar)	87.96	87.96	87.96	87.96	87.96
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	48.64	48.64	48.64	48.64	48.64
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.320e+004	3.320e+004	3.321e+004	3.321e+004	3.322e+004
24	Pressure (bar)	87.96	87.96	87.95	87.95	87.95
25	Temperature (C)	48.63	48.63	48.63	48.63	48.63
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.322e+004	3.323e+004	3.323e+004	3.324e+004	3.324e+004
36	Pressure (bar)	87.95	87.95	87.95	87.95	87.95
37	Temperature (C)	48.63	48.62	48.62	48.62	48.62
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.325e+004	3.325e+004	3.326e+004	3.326e+004	3.327e+004
48	Pressure (bar)	87.95	87.95	87.95	87.95	87.95
49	Temperature (C)	48.62	48.62	48.62	48.61	48.61
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.327e+004	3.328e+004	3.328e+004	3.329e+004	3.329e+004
60	Pressure (bar)	87.95	87.95	87.95	87.95	87.95
61	Temperature (C)	48.61	48.61	48.61	48.61	48.61
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.330e+004	3.330e+004	3.331e+004	3.331e+004	3.332e+004
16	Pressure (bar)	87.95	87.95	87.95	87.95	87.95
17	Temperature (C)	48.60	48.60	48.60	48.60	48.60
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.332e+004	3.333e+004	3.333e+004	3.334e+004	3.334e+004
28	Pressure (bar)	87.95	87.95	87.95	87.95	87.95
29	Temperature (C)	48.60	48.59	48.59	48.59	48.59
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.335e+004	3.335e+004	3.336e+004	3.336e+004	3.337e+004
40	Pressure (bar)	87.95	87.95	87.95	87.94	87.94
41	Temperature (C)	48.59	48.59	48.59	48.58	48.58
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.337e+004	3.338e+004	3.338e+004	3.339e+004	3.339e+004
52	Pressure (bar)	87.94	87.94	87.94	87.94	87.94
53	Temperature (C)	48.58	48.58	48.58	48.58	48.58
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.340e+004	3.340e+004	3.341e+004	3.341e+004	3.342e+004
64	Pressure (bar)	87.94	87.94	87.94	87.94	87.94
65	Temperature (C)	48.57	48.57	48.57	48.57	48.57
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.342e+004	3.343e+004	3.343e+004	3.344e+004	3.344e+004
20	Pressure (bar)	87.94	87.94	87.94	87.94	87.94
21	Temperature (C)	48.57	48.56	48.56	48.56	48.56
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.345e+004	3.345e+004	3.346e+004	3.346e+004	3.347e+004
32	Pressure (bar)	87.94	87.94	87.94	87.94	87.94
33	Temperature (C)	48.56	48.56	48.56	48.55	48.55
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.347e+004	3.348e+004	3.348e+004	3.349e+004	3.349e+004
44	Pressure (bar)	87.94	87.94	87.94	87.94	87.94
45	Temperature (C)	48.55	48.55	48.55	48.55	48.55
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.350e+004	3.350e+004	3.351e+004	3.351e+004	3.352e+004
56	Pressure (bar)	87.94	87.94	87.94	87.94	87.93
57	Temperature (C)	48.54	48.54	48.54	48.54	48.54
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.352e+004	3.353e+004	3.353e+004	3.354e+004	3.354e+004
68	Pressure (bar)	87.93	87.93	87.93	87.93	87.93
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 919 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	48.54	48.53	48.53	48.53	48.53
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.355e+004	3.355e+004	3.356e+004	3.356e+004	3.357e+004
24	Pressure (bar)	87.93	87.93	87.93	87.93	87.93
25	Temperature (C)	48.53	48.53	48.53	48.52	48.52
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.357e+004	3.358e+004	3.358e+004	3.359e+004	3.359e+004
36	Pressure (bar)	87.93	87.93	87.93	87.93	87.93
37	Temperature (C)	48.52	48.52	48.52	48.52	48.52
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.360e+004	3.360e+004	3.361e+004	3.361e+004	3.362e+004
48	Pressure (bar)	87.93	87.93	87.93	87.93	87.93
49	Temperature (C)	48.51	48.51	48.51	48.51	48.51
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.362e+004	3.363e+004	3.363e+004	3.364e+004	3.364e+004
60	Pressure (bar)	87.93	87.93	87.93	87.92	87.92
61	Temperature (C)	48.51	48.50	48.50	48.50	48.50
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.365e+004	3.365e+004	3.366e+004	3.366e+004	3.367e+004
16	Pressure (bar)	87.92	87.92	87.92	87.92	87.92
17	Temperature (C)	48.50	48.50	48.49	48.49	48.49
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.367e+004	3.368e+004	3.368e+004	3.369e+004	3.369e+004
28	Pressure (bar)	87.92	87.92	87.92	87.92	87.92
29	Temperature (C)	48.49	48.49	48.49	48.49	48.48
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.370e+004	3.370e+004	3.371e+004	3.371e+004	3.372e+004
40	Pressure (bar)	87.92	87.92	87.92	87.92	87.92
41	Temperature (C)	48.48	48.48	48.48	48.48	48.48
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.372e+004	3.373e+004	3.373e+004	3.374e+004	3.374e+004
52	Pressure (bar)	87.92	87.92	87.92	87.92	87.92
53	Temperature (C)	48.48	48.47	48.47	48.47	48.47
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.375e+004	3.375e+004	3.376e+004	3.376e+004	3.377e+004
64	Pressure (bar)	87.92	87.92	87.92	87.92	87.92
65	Temperature (C)	48.47	48.47	48.46	48.46	48.46
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.377e+004	3.378e+004	3.378e+004	3.379e+004	3.379e+004
20	Pressure (bar)	87.92	87.92	87.92	87.92	87.92
21	Temperature (C)	48.46	48.46	48.46	48.46	48.46
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.380e+004	3.380e+004	3.381e+004	3.381e+004	3.382e+004
32	Pressure (bar)	87.92	87.92	87.92	87.92	87.92
33	Temperature (C)	48.45	48.45	48.45	48.45	48.45
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.382e+004	3.383e+004	3.383e+004	3.384e+004	3.384e+004
44	Pressure (bar)	87.92	87.91	87.91	87.91	87.91
45	Temperature (C)	48.45	48.44	48.44	48.44	48.44
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.385e+004	3.385e+004	3.385e+004	3.385e+004	3.386e+004
56	Pressure (bar)	87.91	87.91	87.91	87.91	87.91
57	Temperature (C)	48.44	48.44	48.44	48.44	48.44
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.386e+004	3.386e+004	3.386e+004	3.386e+004	3.387e+004
68	Pressure (bar)	87.91	87.91	87.91	87.91	87.91
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	48.44	48.43	48.43	48.43	48.43
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.387e+004	3.387e+004	3.387e+004	3.388e+004	3.388e+004
24	Pressure (bar)	87.91	87.91	87.91	87.91	87.91
25	Temperature (C)	48.43	48.43	48.43	48.43	48.43
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.388e+004	3.389e+004	3.389e+004	3.389e+004	3.389e+004
36	Pressure (bar)	87.91	87.91	87.91	87.91	87.91
37	Temperature (C)	48.43	48.43	48.43	48.42	48.42
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.390e+004	3.391e+004	3.392e+004	3.393e+004	3.394e+004
48	Pressure (bar)	87.91	87.91	87.91	87.91	87.91
49	Temperature (C)	48.42	48.42	48.42	48.41	48.41
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.74	17.74	17.74	17.74	17.74
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.395e+004	3.396e+004	3.398e+004	3.399e+004	3.400e+004
60	Pressure (bar)	87.91	87.90	87.90	87.90	87.90
61	Temperature (C)	48.41	48.40	48.40	48.40	48.39
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.74	17.73	17.73	17.73	17.73
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 923 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.401e+004	3.402e+004	3.403e+004	3.404e+004	3.405e+004
16	Pressure (bar)	87.90	87.90	87.90	87.90	87.90
17	Temperature (C)	48.39	48.39	48.38	48.38	48.38
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.406e+004	3.408e+004	3.409e+004	3.410e+004	3.411e+004
28	Pressure (bar)	87.90	87.90	87.90	87.90	87.90
29	Temperature (C)	48.37	48.37	48.37	48.36	48.36
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.412e+004	3.413e+004	3.414e+004	3.415e+004	3.416e+004
40	Pressure (bar)	87.89	87.89	87.89	87.89	87.89
41	Temperature (C)	48.36	48.35	48.35	48.35	48.34
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.418e+004	3.419e+004	3.420e+004	3.421e+004	3.422e+004
52	Pressure (bar)	87.89	87.89	87.89	87.89	87.89
53	Temperature (C)	48.34	48.34	48.33	48.33	48.33
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.423e+004	3.424e+004	3.425e+004	3.426e+004	3.428e+004
64	Pressure (bar)	87.88	87.88	87.88	87.88	87.88
65	Temperature (C)	48.32	48.32	48.32	48.31	48.31
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10		Page 924 of 937	

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.429e+004	3.429e+004	3.429e+004	3.429e+004	3.430e+004
20	Pressure (bar)	87.88	87.88	87.88	87.88	87.88
21	Temperature (C)	48.31	48.31	48.31	48.31	48.30
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.430e+004	3.430e+004	3.431e+004	3.431e+004	3.432e+004
32	Pressure (bar)	87.88	87.88	87.88	87.88	87.88
33	Temperature (C)	48.30	48.30	48.30	48.30	48.30
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.432e+004	3.433e+004	3.433e+004	3.434e+004	3.434e+004
44	Pressure (bar)	87.88	87.88	87.88	87.88	87.88
45	Temperature (C)	48.30	48.30	48.29	48.29	48.29
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.435e+004	3.435e+004	3.436e+004	3.436e+004	3.437e+004
56	Pressure (bar)	87.88	87.88	87.88	87.88	87.88
57	Temperature (C)	48.29	48.29	48.29	48.28	48.28
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.437e+004	3.438e+004	3.438e+004	3.439e+004	3.439e+004
68	Pressure (bar)	87.88	87.88	87.88	87.88	87.88
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	48.28	48.28	48.28	48.28	48.28
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.440e+004	3.440e+004	3.441e+004	3.441e+004	3.442e+004
24	Pressure (bar)	87.88	87.88	87.88	87.88	87.88
25	Temperature (C)	48.27	48.27	48.27	48.27	48.27
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.442e+004	3.443e+004	3.443e+004	3.444e+004	3.444e+004
36	Pressure (bar)	87.87	87.87	87.87	87.87	87.87
37	Temperature (C)	48.27	48.27	48.26	48.26	48.26
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.445e+004	3.445e+004	3.446e+004	3.446e+004	3.447e+004
48	Pressure (bar)	87.87	87.87	87.87	87.87	87.87
49	Temperature (C)	48.26	48.26	48.26	48.26	48.25
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.447e+004	3.448e+004	3.448e+004	3.449e+004	3.449e+004
60	Pressure (bar)	87.87	87.87	87.87	87.87	87.87
61	Temperature (C)	48.25	48.25	48.25	48.25	48.25
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.450e+004	3.450e+004	3.451e+004	3.451e+004	3.452e+004
16	Pressure (bar)	87.87	87.87	87.87	87.87	87.87
17	Temperature (C)	48.25	48.24	48.24	48.24	48.24
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.452e+004	3.453e+004	3.453e+004	3.454e+004	3.454e+004
28	Pressure (bar)	87.87	87.87	87.87	87.87	87.87
29	Temperature (C)	48.24	48.24	48.24	48.23	48.23
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.455e+004	3.455e+004	3.456e+004	3.456e+004	3.457e+004
40	Pressure (bar)	87.87	87.87	87.87	87.87	87.87
41	Temperature (C)	48.23	48.23	48.23	48.23	48.23
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.457e+004	3.458e+004	3.458e+004	3.459e+004	3.459e+004
52	Pressure (bar)	87.87	87.86	87.86	87.86	87.86
53	Temperature (C)	48.22	48.22	48.22	48.22	48.22
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.460e+004	3.460e+004	3.461e+004	3.461e+004	3.462e+004
64	Pressure (bar)	87.86	87.86	87.86	87.86	87.86
65	Temperature (C)	48.22	48.21	48.21	48.21	48.21
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 927 of 937

1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.462e+004	3.463e+004	3.463e+004	3.464e+004	3.464e+004
20	Pressure (bar)	87.86	87.86	87.86	87.86	87.86
21	Temperature (C)	48.21	48.21	48.21	48.20	48.20
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.465e+004	3.465e+004	3.465e+004	3.466e+004	3.466e+004
32	Pressure (bar)	87.86	87.86	87.86	87.86	87.86
33	Temperature (C)	48.20	48.20	48.20	48.20	48.20
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.467e+004	3.467e+004	3.468e+004	3.468e+004	3.469e+004
44	Pressure (bar)	87.86	87.86	87.86	87.86	87.86
45	Temperature (C)	48.19	48.19	48.19	48.19	48.19
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.469e+004	3.470e+004	3.470e+004	3.471e+004	3.471e+004
56	Pressure (bar)	87.86	87.86	87.86	87.86	87.86
57	Temperature (C)	48.19	48.19	48.18	48.18	48.18
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.472e+004	3.472e+004	3.473e+004	3.473e+004	3.474e+004
68	Pressure (bar)	87.86	87.85	87.85	87.85	87.85
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	48.18	48.18	48.18	48.18	48.17
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.474e+004	3.475e+004	3.475e+004	3.476e+004	3.476e+004
24	Pressure (bar)	87.85	87.85	87.85	87.85	87.85
25	Temperature (C)	48.17	48.17	48.17	48.17	48.17
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.477e+004	3.477e+004	3.478e+004	3.478e+004	3.479e+004
36	Pressure (bar)	87.85	87.85	87.85	87.85	87.85
37	Temperature (C)	48.17	48.17	48.16	48.16	48.16
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.479e+004	3.480e+004	3.480e+004	3.481e+004	3.481e+004
48	Pressure (bar)	87.85	87.85	87.85	87.85	87.85
49	Temperature (C)	48.16	48.16	48.16	48.16	48.15
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.482e+004	3.482e+004	3.483e+004	3.483e+004	3.484e+004
60	Pressure (bar)	87.85	87.85	87.85	87.85	87.85
61	Temperature (C)	48.15	48.15	48.15	48.15	48.15
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.484e+004	3.485e+004	3.485e+004	3.486e+004	3.486e+004
16	Pressure (bar)	87.85	87.85	87.85	87.85	87.85
17	Temperature (C)	48.15	48.14	48.14	48.14	48.14
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.487e+004	3.487e+004	3.488e+004	3.488e+004	3.489e+004
28	Pressure (bar)	87.85	87.85	87.85	87.84	87.84
29	Temperature (C)	48.14	48.14	48.13	48.13	48.13
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.489e+004	3.490e+004	3.490e+004	3.491e+004	3.491e+004
40	Pressure (bar)	87.84	87.84	87.84	87.85	87.85
41	Temperature (C)	48.13	48.13	48.13	48.13	48.13
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.492e+004	3.492e+004	3.493e+004	3.493e+004	3.494e+004
52	Pressure (bar)	87.85	87.85	87.85	87.86	87.86
53	Temperature (C)	48.13	48.13	48.12	48.12	48.12
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.494e+004	3.495e+004	3.495e+004	3.496e+004	3.497e+004
64	Pressure (bar)	87.86	87.86	87.85	87.85	87.85
65	Temperature (C)	48.12	48.12	48.12	48.11	48.11
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.497e+004	3.497e+004	3.498e+004	3.498e+004	3.498e+004
20	Pressure (bar)	87.84	87.84	87.84	87.84	87.84
21	Temperature (C)	48.11	48.11	48.11	48.10	48.10
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.499e+004	3.499e+004	3.500e+004	3.500e+004	3.501e+004
32	Pressure (bar)	87.83	87.83	87.83	87.83	87.83
33	Temperature (C)	48.10	48.10	48.10	48.10	48.09
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.502e+004	3.502e+004	3.503e+004	3.503e+004	3.504e+004
44	Pressure (bar)	87.83	87.83	87.83	87.83	87.83
45	Temperature (C)	48.09	48.09	48.09	48.09	48.09
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.504e+004	3.505e+004	3.505e+004	3.506e+004	3.506e+004
56	Pressure (bar)	87.83	87.83	87.83	87.83	87.83
57	Temperature (C)	48.08	48.08	48.08	48.08	48.08
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.507e+004	3.507e+004	3.508e+004	3.508e+004	3.509e+004
68	Pressure (bar)	87.83	87.82	87.82	87.82	87.82
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 931 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	48.08	48.08	48.07	48.07	48.07
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.509e+004	3.510e+004	3.510e+004	3.511e+004	3.511e+004
24	Pressure (bar)	87.82	87.82	87.82	87.82	87.82
25	Temperature (C)	48.07	48.07	48.07	48.06	48.06
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.512e+004	3.512e+004	3.513e+004	3.513e+004	3.514e+004
36	Pressure (bar)	87.82	87.82	87.82	87.82	87.82
37	Temperature (C)	48.06	48.06	48.06	48.06	48.06
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.514e+004	3.515e+004	3.515e+004	3.516e+004	3.516e+004
48	Pressure (bar)	87.82	87.82	87.82	87.82	87.82
49	Temperature (C)	48.05	48.05	48.05	48.05	48.05
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.517e+004	3.517e+004	3.518e+004	3.518e+004	3.519e+004
60	Pressure (bar)	87.82	87.82	87.82	87.82	87.82
61	Temperature (C)	48.05	48.05	48.04	48.04	48.04
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 932 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.519e+004	3.520e+004	3.520e+004	3.521e+004	3.521e+004
16	Pressure (bar)	87.82	87.82	87.82	87.82	87.82
17	Temperature (C)	48.04	48.04	48.04	48.04	48.03
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.522e+004	3.522e+004	3.523e+004	3.523e+004	3.524e+004
28	Pressure (bar)	87.81	87.81	87.81	87.81	87.81
29	Temperature (C)	48.03	48.03	48.03	48.03	48.03
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.524e+004	3.525e+004	3.525e+004	3.526e+004	3.526e+004
40	Pressure (bar)	87.81	87.81	87.81	87.81	87.81
41	Temperature (C)	48.03	48.02	48.02	48.02	48.02
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.527e+004	3.527e+004	3.528e+004	3.528e+004	3.529e+004
52	Pressure (bar)	87.81	87.81	87.81	87.81	87.81
53	Temperature (C)	48.02	48.02	48.02	48.01	48.01
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.529e+004	3.530e+004	3.530e+004	3.531e+004	3.531e+004
64	Pressure (bar)	87.81	87.81	87.81	87.81	87.81
65	Temperature (C)	48.01	48.01	48.01	48.01	48.01
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.532e+004	3.532e+004	3.533e+004	3.533e+004	3.534e+004
20	Pressure (bar)	87.81	87.81	87.81	87.81	87.81
21	Temperature (C)	48.00	48.00	48.00	48.00	48.00
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.534e+004	3.535e+004	3.535e+004	3.536e+004	3.536e+004
32	Pressure (bar)	87.81	87.81	87.81	87.82	87.82
33	Temperature (C)	48.00	48.00	48.00	48.00	48.00
34	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
35	Hydrate Type	No Types	No Types	No Types	No Types	No Types
36	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
38	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
39	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
40	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
42	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
43	Distance (m)	3.537e+004	3.537e+004	3.538e+004	3.538e+004	3.539e+004
44	Pressure (bar)	87.82	87.82	87.81	87.80	87.80
45	Temperature (C)	48.00	47.99	47.99	47.99	47.98
46	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
47	Hydrate Type	No Types	No Types	No Types	No Types	No Types
48	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
50	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
51	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
52	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
54	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
55	Distance (m)	3.539e+004	3.540e+004	3.540e+004	3.541e+004	3.541e+004
56	Pressure (bar)	87.79	87.79	87.79	87.79	87.79
57	Temperature (C)	47.98	47.98	47.98	47.97	47.97
58	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
59	Hydrate Type	No Types	No Types	No Types	No Types	No Types
60	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
62	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
63	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
64	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
66	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
67	Distance (m)	3.542e+004	3.542e+004	3.543e+004	3.543e+004	3.544e+004
68	Pressure (bar)	87.79	87.79	87.79	87.79	87.79
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4						
5			Date/Time: Mon Jun 13 15:02:44 2022			
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Temperature (C)	47.97	47.97	47.97	47.97	47.97
14	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
15	Hydrate Type	No Types	No Types	No Types	No Types	No Types
16	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
18	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
19	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
20	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
22	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
23	Distance (m)	3.544e+004	3.545e+004	3.545e+004	3.546e+004	3.546e+004
24	Pressure (bar)	87.79	87.79	87.79	87.79	87.79
25	Temperature (C)	47.97	47.96	47.96	47.96	47.96
26	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
27	Hydrate Type	No Types	No Types	No Types	No Types	No Types
28	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
30	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
31	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
32	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
34	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
35	Distance (m)	3.547e+004	3.547e+004	3.548e+004	3.548e+004	3.549e+004
36	Pressure (bar)	87.79	87.79	87.79	87.79	87.79
37	Temperature (C)	47.96	47.96	47.96	47.95	47.95
38	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
39	Hydrate Type	No Types	No Types	No Types	No Types	No Types
40	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
41	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
42	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
43	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
44	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
46	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
47	Distance (m)	3.549e+004	3.550e+004	3.550e+004	3.551e+004	3.551e+004
48	Pressure (bar)	87.79	87.79	87.78	87.78	87.78
49	Temperature (C)	47.95	47.95	47.95	47.95	47.94
50	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
51	Hydrate Type	No Types	No Types	No Types	No Types	No Types
52	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
53	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
54	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
55	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
56	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
58	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
59	Distance (m)	3.552e+004	3.552e+004	3.553e+004	3.553e+004	3.554e+004
60	Pressure (bar)	87.78	87.78	87.78	87.78	87.78
61	Temperature (C)	47.94	47.94	47.94	47.94	47.94
62	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
63	Hydrate Type	No Types	No Types	No Types	No Types	No Types
64	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
65	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
66	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
67	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
68	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
69	Aspen Technology Inc.		Aspen HYSYS Version 10			Page 935 of 937

1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
14	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
15	Distance (m)	3.554e+004	3.555e+004	3.555e+004	3.556e+004	3.556e+004
16	Pressure (bar)	87.78	87.78	87.78	87.78	87.78
17	Temperature (C)	47.94	47.93	47.93	47.93	47.93
18	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
19	Hydrate Type	No Types	No Types	No Types	No Types	No Types
20	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
21	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
22	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
23	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
24	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
26	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
27	Distance (m)	3.557e+004	3.557e+004	3.557e+004	3.558e+004	3.558e+004
28	Pressure (bar)	87.78	87.78	87.78	87.78	87.78
29	Temperature (C)	47.93	47.93	47.93	47.93	47.92
30	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
31	Hydrate Type	No Types	No Types	No Types	No Types	No Types
32	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
33	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
34	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
35	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
36	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
37	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
38	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
39	Distance (m)	3.559e+004	3.559e+004	3.559e+004	3.560e+004	3.560e+004
40	Pressure (bar)	87.78	87.78	87.78	87.78	87.78
41	Temperature (C)	47.92	47.92	47.92	47.92	47.92
42	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
43	Hydrate Type	No Types	No Types	No Types	No Types	No Types
44	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
45	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
46	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
47	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
48	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
49	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
50	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
51	Distance (m)	3.560e+004	3.561e+004	3.561e+004	3.562e+004	3.562e+004
52	Pressure (bar)	87.78	87.78	87.78	87.78	87.78
53	Temperature (C)	47.92	47.92	47.92	47.91	47.91
54	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
55	Hydrate Type	No Types	No Types	No Types	No Types	No Types
56	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
57	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
58	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
59	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
60	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
61	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
62	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
63	Distance (m)	3.563e+004	3.563e+004	3.564e+004	3.564e+004	3.565e+004
64	Pressure (bar)	87.78	87.78	87.78	87.78	87.78
65	Temperature (C)	47.91	47.91	47.91	47.91	47.91
66	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
67	Hydrate Type	No Types	No Types	No Types	No Types	No Types
68	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
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1	 Epic Atlantic Limited		Case Name: Simulation - 386MMSCFD_100barg.hsc			
2						
3			Unit Set: Mixed			
4			Date/Time: Mon Jun 13 15:02:44 2022			
5						
6	Pipe Segment: ELEP-Akabel Pipeline (continued)					
7						
8						
9	Hydrates					
10						
11	Hydrates Table					
12						
13	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
14	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
15	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
16	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
17	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
18	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
19	Distance (m)	3.565e+004	3.566e+004	3.566e+004	3.567e+004	3.567e+004
20	Pressure (bar)	87.78	87.77	87.77	87.77	87.77
21	Temperature (C)	47.90	47.90	47.90	47.90	47.90
22	Formation	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form	Will NOT Form
23	Hydrate Type	No Types	No Types	No Types	No Types	No Types
24	Mode	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
25	Inhibitors	Not Included	Not Included	Not Included	Not Included	Not Included
26	Formation Temperature (C)	17.73	17.73	17.73	17.73	17.73
27	Type at Formation Temperature	Type II	Type II	Type II	Type II	Type II
28	Mode at Formation Temperature	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model	Use 3-Phase Model
29	Phases at Formation Temperature	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H	V - Aq - L - H
30	Inhibitors at Formation Temperature	Not Included	Not Included	Not Included	Not Included	Not Included
31	Distance (m)	3.568e+004				
32	Pressure (bar)	87.77				
33	Temperature (C)	47.89				
34	Formation	Will NOT Form				
35	Hydrate Type	No Types				
36	Mode	Use 3-Phase Model				
37	Inhibitors	Not Included				
38	Formation Temperature (C)	17.73				
39	Type at Formation Temperature	Type II				
40	Mode at Formation Temperature	Use 3-Phase Model				
41	Phases at Formation Temperature	V - Aq - L - H				
42	Inhibitors at Formation Temperature	Not Included				
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